Mathematical Methods for Physicists

Problems and Worked Solutions — Arfken, Weber & Harris, 7th ed.

Dark red text marks published errata (from the Instructor's Manual, *Errata and Revision Status*, 6 April 2012) and additional corrections or necessary qualifications explained in the corresponding solutions.

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1 Chapter 1. Mathematical Preliminaries

Problems

1.1 Infinite Series

1.1.1

(a) Prove that if $\lim_{n\to\infty} n^p u_n = A < \infty$, $p > 1$, the series $\sum_{n=1}^{\infty} u_n$ converges.

(b) Prove that if $\lim_{n\to\infty} n u_n = A > 0$, the series diverges. (The test fails for $A = 0$.)

These two tests, known as **limit tests**, are often convenient for establishing the convergence of a series. They may be treated as comparison tests, comparing with

$$\sum_n n^{-q}, \quad 1 \le q < p.$$

Solution.

(a) Interpret A as a finite real or complex number. By the definition of limit there is N such that $|n^p u_n - A| < 1$ for $n \ge N$. Hence $|n^p u_n| < |A| + 1$, and

$$|u_n| \le \frac{|A|+1}{n^p}, \qquad \sum_{n=N}^{\infty} |u_n| \le (|A|+1) \sum_{n=N}^{\infty} n^{-p} < \infty.$$

The last series converges by the integral test, because $\int_N^{\infty} x^{-p} dx = N^{1-p}/(p-1) < \infty$ for $p > 1$. Adding finitely many initial terms proves absolute convergence. This also covers $A = 0$.

(b) If $n u_n \to A > 0$, choose N so that $n u_n > A/2$ for $n \ge N$ (if $A = +\infty$, choose N so that $n u_n > 1$, and read 1 for $A/2$ below). Then $\sum_{n=N}^{M} u_n \ge (A/2) \sum_{n=N}^{M} n^{-1} \to +\infty$, because the harmonic series diverges. When $A = 0$, both $u_n = n^{-2}$ (convergent) and $u_n = 1/(n \ln n)$, $n \ge 2$ (divergent, Exercise 1.1.6(b)), meet the limit condition, since $n \cdot n^{-2} = 1/n \to 0$ and $n/(n \ln n) = 1/\ln n \to 0$; the test therefore cannot decide that case.

1.1.2 For eventually positive sequences, if $\lim_{n\to\infty} \dfrac{b_n}{a_n} = K$, a constant with $0 < K < \infty$, show that $\sum_n b_n$ converges or diverges with $\sum a_n$.

Hint. If $\sum a_n$ converges, rescale b_n to $b_n' = \dfrac{b_n}{2K}$. If $\sum_n a_n$ diverges, rescale to $b_n'' = \dfrac{2b_n}{K}$.

Solution.

The limit-comparison statement requires eventually positive terms (as in the positive-series tests of this section). Take $\epsilon = K/2$ in the definition of the limit. For sufficiently large n, $a_n > 0$ and $|b_n/a_n - K| < K/2$, which is the first inequality below; multiplying it by $a_n > 0$ gives the second:

$$\frac{K}{2} < \frac{b_n}{a_n} < \frac{3K}{2}, \qquad \frac{K}{2} a_n < b_n < \frac{3K}{2} a_n.$$

If $\sum a_n$ converges, the upper bound proves convergence of $\sum b_n$ by comparison. If $\sum a_n$ diverges, the lower bound proves divergence of $\sum b_n$. Finite initial segments do not change either conclusion. (These are the hint's rescalings: $b_n' = b_n/(2K) < \frac{3}{4} a_n$ and $b_n'' = 2b_n/K > a_n$.) The sign hypothesis matters: with $a_n = (-1)^n/\sqrt{n}$ and $b_n = a_n + 1/n$, one has $b_n/a_n = 1 + (-1)^n/\sqrt{n} \to 1$, but $\sum a_n$ converges whereas the partial sums of $\sum b_n$ tend to $+\infty$. Thus the unrestricted statement is false.

1.1.3

(a) Show that the series $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges.

(b) By direct addition $\sum_{n=2}^{100,000}[n(\ln n)^2]^{-1} = 2.02288$. Use Eq. (1.9) to make a five-significant-figure estimate of the sum of this series.

Solution.

(a) Set $f(x) = 1/[x(\ln x)^2]$. For $x > 1$,

$$f'(x) = -\frac{\ln x + 2}{x^2(\ln x)^3} < 0, \qquad \int_2^{\infty} f(x)\,dx = \left[-\frac{1}{\ln x}\right]_2^{\infty} = \frac{1}{\ln 2} < \infty.$$

(The antiderivative follows from $t = \ln x$, $dt = dx/x$: $\int dt/t^2 = -1/t$.) The integral test proves convergence.

(b) Let S be the sum and $S_N = \sum_{n=2}^{N} f(n)$. The tail $S - S_N = \sum_{n=N+1}^{\infty} f(n)$ of the decreasing f obeys Eq. (1.9) in the form

$$\int_{N+1}^{\infty} f(x)dx < S - S_N < \int_{N+1}^{\infty} f(x)dx + f(N+1) < \int_N^{\infty} f(x)dx,$$

the last step because $f(N+1) < \int_N^{N+1} f(x)dx$. With the antiderivative $-1/\ln x$ from (a),

$$S_N + \frac{1}{\ln(N+1)} < S < S_N + \frac{1}{\ln N}.$$

Summing the 99,999 terms $n = 2, \ldots, 100{,}000$ directly in double precision, adding the smallest terms ($n = 100{,}000$ downward) first, gives

$$S_{100000} = 2.022883942578507\ldots.$$

(Summing upward from $n = 2$ instead changes the last two digits shown here and the last digit of each bound below.) Consequently, with $1/\ln 100{,}001 = 0.0868588209\ldots$ and $1/\ln 100{,}000 = 0.0868588963\ldots$,

$$2.10974276351492 < S < 2.10974283895916.$$

Both bounds round to $\boxed{2.1097}$ (five significant figures). Using instead the rounded value 2.02288 ± 0.000005 given in the problem yields $2.1097338 < S < 2.1097439$, which also rounds to 2.1097.

1.1.4 Gauss' test is often given in the form of a test of the ratio

$$\frac{u_n}{u_{n+1}} = \frac{n^2 + a_1 n + a_0}{n^2 + b_1 n + b_0}.$$

Assume $u_n \neq 0$ for all sufficiently large n, so that the ratio is defined. For what values of the parameters a_1 and b_1 is there convergence? divergence?

ANS. Convergent for $a_1 - b_1 > 1$, divergent for $a_1 - b_1 \leq 1$.

Solution.

The right side tends to 1, so it is positive for large n: the u_n eventually have one sign, and replacing u_n by $-u_n$ if necessary we may take $u_n > 0$ for large n. Put $d = a_1 - b_1$. Dividing numerator and denominator by n^2 and using $(1+z)^{-1} = 1 - z + O(z^2)$ with $z = b_1/n + b_0/n^2$ gives

$$\frac{u_n}{u_{n+1}} = \left(1 + \frac{a_1}{n} + \frac{a_0}{n^2}\right)\left(1 - \frac{b_1}{n} + O(n^{-2})\right) = 1 + \frac{d}{n} + O(n^{-2}).$$

This is the form required by Gauss' test with $h = d$; indeed,

$$\frac{u_n}{u_{n+1}} - 1 = \frac{dn + a_0 - b_0}{n^2 + b_1 n + b_0},$$

so the exact remainder is $B(n)/n^2$ with

$$B(n) = n^2\Big(\frac{u_n}{u_{n+1}} - 1 - \frac{d}{n}\Big) = \frac{n[(a_0 - b_0 - b_1 d)n - b_0 d]}{n^2 + b_1 n + b_0} \longrightarrow a_0 - b_0 - b_1 d,$$

a bounded function. To see directly why d decides, use $\log(1 + y) = y + O(y^2)$:

$$\log \frac{u_{n+1}}{u_n} = -\log \frac{u_n}{u_{n+1}} = -\frac{d}{n} + O(n^{-2}).$$

Summing from a fixed large N to $n - 1$,

$$\log u_n = \log u_N - d \sum_{k=N}^{n-1} \frac{1}{k} + \sum_{k=N}^{n-1} O(k^{-2}) = -d \log n + C + o(1).$$

Here the last error sum converges (compare $\sum k^{-2}$), and $\sum_{k=N}^{n-1} k^{-1} - \log n$ converges because $H_{n-1} - \log n \to \gamma$, Eq. (1.13). Exponentiating, $u_n = e^{C+o(1)} n^{-d}$, so $u_n \sim C_1 n^{-d}$ with $C_1 = e^C > 0$. Limit comparison with a p-series proves convergence for $d > 1$ and divergence for $d \le 1$, including the borderline $d = 1$. The parameters a_0, b_0 affect the constant, but not this classification.

1.1.5 Test for convergence

(a) $\displaystyle\sum_{n=2}^{\infty} (\ln n)^{-1}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{10^n}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2n(2n+1)}$

(d) $\displaystyle\sum_{n=1}^{\infty} [n(n+1)]^{-1/2}$

(e) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{2n+1}$

Solution.

(a) For $n \ge 2$, $\ln n < n$, so $1/\ln n > 1/n$. Comparison with the harmonic series proves divergence.

(b) For $u_n = n!/10^n$, $u_{n+1}/u_n = (n+1)/10$. For $n \ge 19$ this is at least 2, so the terms grow instead of tending to zero. The series diverges.

(c) Since $0 < 1/[2n(2n+1)] < 1/(4n^2)$, the series converges. As a check, partial fractions give $1/[2n(2n+1)] = 1/(2n) - 1/(2n+1)$, so the sum is $\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \cdots = 1 - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) = 1 - \ln 2$, by the alternating harmonic series.

(d) With $u_n = [n(n+1)]^{-1/2}$, $nu_n = (1 + 1/n)^{-1/2} \to 1$. Limit comparison with $1/n$ proves divergence.

(e) For $n \geq 1$, $1/(2n+1) \geq 1/(3n)$, so the series diverges. Its first term $n = 0$ has no effect on this conclusion.

1.1.6 Test for convergence

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

(b) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n\, 2^n}$

(d) $\displaystyle\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right)$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n \cdot n^{1/n}}$

Solution.

(a) $1/[n(n+1)] = 1/n - 1/(n+1)$, so the partial sum through N is $1 - 1/(N+1) \to 1$. The series converges.

(b) The positive function $1/(x \ln x)$ decreases for $x > 1$, but $\int_2^M dx/(x \ln x) = \ln\ln M - \ln\ln 2 \to \infty$. The series diverges.

(c) $0 < 1/(n2^n) \leq 2^{-n}$ proves convergence. Integrating $\sum_{n=0}^{\infty} x^n = (1-x)^{-1}$ termwise from 0 to 1/2 (inside the interval of convergence) gives $\sum_{n=0}^{\infty} (1/2)^{n+1}/(n+1) = \sum_{n=1}^{\infty} 1/(n2^n) = [-\ln(1-x)]_0^{1/2} = \ln 2$.

(d) The partial sum is $\sum_{n=1}^{N} \ln[(n+1)/n] = \ln \prod_{n=1}^{N} (n+1)/n = \ln(N+1)$, which tends to infinity.

(e) $n^{1/n} = \exp[(\ln n)/n] \to 1$, so $n/[nn^{1/n}] \to 1$. Limit comparison with the harmonic series proves divergence.

1.1.7 For what values of p and q will $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^p(\ln n)^q}$ converge?

ANS. Convergent for $\begin{cases} p>1, \text{ all } q, \\ p=1,\ q>1, \end{cases}$ divergent for $\begin{cases} p<1, \text{ all } q, \\ p=1,\ q\le 1. \end{cases}$

Solution.

Let p, q be real. We use the elementary fact that, for every $\epsilon > 0$ and fixed real q, $(\ln n)^{|q|} < n^{\epsilon}$ for all sufficiently large n; taking logarithms reduces it to $|q| \ln\ln n < \epsilon \ln n$, which holds for all large n because $(\ln y)/y \to 0$ as $y = \ln n \to \infty$. If $p > 1$, take $\epsilon = (p-1)/2$. Eventually

$$\frac{1}{n^p(\ln n)^q} \le \frac{(\ln n)^{|q|}}{n^p} \le n^{-(p+1)/2},$$

and the comparison series converges. If $p < 1$, take $\epsilon = (1-p)/2$. Eventually $n^p(\ln n)^q \le n^{(p+1)/2}$, so the terms dominate a divergent p-series. For $p = 1$ make the substitution $t = \ln x$:

$$\int_2^M \frac{dx}{x(\ln x)^q} = \int_{\ln 2}^{\ln M} t^{-q}dt = \begin{cases} \dfrac{(\ln M)^{1-q} - (\ln 2)^{1-q}}{1-q}, & q \ne 1, \\ \ln\ln M - \ln\ln 2, & q = 1. \end{cases}$$

As $M \to \infty$ this stays bounded exactly when $q > 1$. The integrand $f(x) = 1/[x(\ln x)^q]$ has $f'(x) = -(\ln x + q)/[x^2(\ln x)^{q+1}] < 0$ once $\ln x > -q$, so it is eventually decreasing and the integral test applies. These three cases give all the stated ranges.

1.1.8 Given $\sum_{n=1}^{1,000} n^{-1} = 7.485\,470\ldots$ set upper and lower bounds on the Euler-Mascheroni constant.

ANS. $0.5767 < \gamma < 0.5778$.

Solution.

Write $H_N = \sum_{n=1}^{N} 1/n$. Since $1/t > 1/(N+1)$ on $(N, N+1)$ and $1/t < 1/(N+1)$ on $(N+1, N+2)$,

$$[H_{N+1} - \ln(N+1)] - [H_N - \ln N] = \frac{1}{N+1} - \int_N^{N+1} \frac{dt}{t} < 0,$$
$$[H_{N+1} - \ln(N+2)] - [H_N - \ln(N+1)] = \frac{1}{N+1} - \int_{N+1}^{N+2} \frac{dt}{t} > 0.$$

Thus $H_N - \ln N$ decreases and $H_N - \ln(N+1)$ increases. Moreover $H_N - \ln N > H_N - \ln(N+1) \ge 1 - \ln 2$ (its value at $N = 1$) and $H_N - \ln(N+1) < H_N - \ln N \le 1$, so the monotone sequences have finite limits. Their difference is $\ln(1 + 1/N) \to 0$, so both approach the same constant γ. Consequently

$$H_N - \ln(N+1) < \gamma < H_N - \ln N.$$

For $N = 1000$, direct summation gives $H_{1000} = 7.48547086055034\ldots$, and the bounds are

$$0.57671608123512 < \gamma < 0.57771558156821.$$

Rounding the lower bound downward and the upper bound upward gives $0.5767 < \gamma < 0.5778$.

1.1.9 (From **Olbers' paradox**.) Assume a static universe in which the stars are uniformly distributed. Divide all space into shells of constant thickness; the stars in any one shell by themselves subtend a solid angle of ω_0. **Allowing for the blocking out of distant stars by nearer stars**, show that the total net solid angle subtended by all stars, shells extending to infinity, is **exactly** 4π with probability one in the model of independent uniform angular coverage, where $0 < \omega_0 \leq 4\pi$. [Therefore the night sky should be ablaze with light. For more details, see E. Harrison, *Darkness at Night: A Riddle of the Universe*. Cambridge, MA: Harvard University Press (1987).]

Solution.

Use the intended model of uniform, independent angular coverage by successive shells, with $0 < \omega_0 \leq 4\pi$. Let $f = \omega_0/(4\pi)$. For a fixed direction the probability of remaining unobscured after $n-1$ independent shells is $(1-f)^{n-1}$. The probability that shell n supplies its first visible star is therefore $f(1-f)^{n-1}$. The solid angle $\Delta\Omega_n$ in which shell n supplies the first visible star is the integral over all directions of the indicator of this event; taking the expectation inside the integral replaces the indicator by this probability, so

$$\mathbb{E}[\Delta\Omega_n] = 4\pi f(1-f)^{n-1}.$$

For $0 < f < 1$, the solid angle covered by the first N shells, $\Omega_N = \sum_{n=1}^{N} \Delta\Omega_n$, has expectation given by the finite geometric sum

$$\mathbb{E}[\Omega_N] = 4\pi f \frac{1-(1-f)^N}{1-(1-f)} = 4\pi[1-(1-f)^N] \longrightarrow 4\pi.$$

This also proves the almost-sure statement about the limiting solid angle. Let $U_N = 4\pi - \Omega_N$ be the uncovered area. It is nonnegative and decreases to U_∞, so for every $\epsilon > 0$, by Markov's inequality $\mathbb{P}(U_N > \epsilon) \leq \mathbb{E}[U_N]/\epsilon$,

$$\mathbb{P}(U_\infty > \epsilon) \leq \mathbb{P}(U_N > \epsilon) \leq \frac{\mathbb{E}[U_N]}{\epsilon} = \frac{4\pi(1-f)^N}{\epsilon} \longrightarrow 0.$$

Thus $U_\infty = 0$ with probability one (take $\epsilon = 1/k$, $k = 1, 2, \ldots$) and $\Omega_\infty = 4\pi$. In the deterministic idealization where each shell covers exactly fraction f of every remaining open region, the same geometric formulas hold without the expectation symbols: shell n adds $\omega_0(1-f)^{n-1}$, and

$$\sum_{n=1}^{\infty} \omega_0(1-f)^{n-1} = \frac{\omega_0}{f} = 4\pi.$$

If $f = 1$, the first shell already covers the sky. Independent angular coverage, or this equivalent idealization, is essential: placing every shell behind the same patch would not establish the conclusion merely from the value of ω_0.

1.1.10 Test for convergence

$$\sum_{n=1}^{\infty} \left[\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}\right]^2 = \frac{1}{4} + \frac{9}{64} + \frac{25}{256} + \cdots.$$

Solution.

Let u_n denote the squared product. The product for u_{n+1} has the extra factor $(2n+1)/(2n+2)$, so cancelling the common factors gives

$$\frac{u_n}{u_{n+1}} = \left(\frac{2n+2}{2n+1}\right)^2 = \frac{4n^2+8n+4}{4n^2+4n+1} = \frac{n^2+2n+1}{n^2+n+1/4}.$$

Thus $a_1 - b_1 = 2 - 1 = 1$, the divergent borderline case of Exercise 1.1.4. For an independent check, use $2 \cdot 4 \cdots (2n) = 2^n n!$ and $1 \cdot 3 \cdots (2n-1) = (2n)!/(2^n n!)$ to write

$$u_n = \left[\frac{(2n)!}{2^{2n}(n!)^2}\right]^2.$$

Stirling's formula $m! \sim \sqrt{2\pi m}\,(m/e)^m$ gives

$$\frac{(2n)!}{2^{2n}(n!)^2} \sim \frac{\sqrt{4\pi n}\,(2n/e)^{2n}}{2^{2n}\,2\pi n\,(n/e)^{2n}} = \frac{1}{\sqrt{\pi n}},$$

so $u_n \sim 1/(\pi n)$ and the harmonic comparison again proves divergence. The Instructor's Manual, p. 8, prints $u_{n+1}/u_n = 1 + 1/n + O(n^{-2})$. For that orientation of the ratio the sign is minus: $u_{n+1}/u_n = [(2n+1)/(2n+2)]^2 = 1 - 1/n + O(n^{-2})$. The reciprocal used above has the plus sign.

1.1.11 Determine whether each of these series is convergent, and if so, whether it is absolutely convergent:

(a) $\dfrac{\ln 2}{2} - \dfrac{\ln 3}{3} + \dfrac{\ln 4}{4} - \dfrac{\ln 5}{5} + \dfrac{\ln 6}{6} - \cdots,$

(b) $\dfrac{1}{1} + \dfrac{1}{2} - \dfrac{1}{3} - \dfrac{1}{4} + \dfrac{1}{5} + \dfrac{1}{6} - \dfrac{1}{7} - \dfrac{1}{8} + \cdots,$

(c) $1 - \dfrac{1}{2} - \dfrac{1}{3} + \dfrac{1}{4} + \dfrac{1}{5} + \dfrac{1}{6} - \dfrac{1}{7} - \dfrac{1}{8} - \dfrac{1}{9} - \dfrac{1}{10} + \dfrac{1}{11} + \cdots + \dfrac{1}{15} - \dfrac{1}{16} - \cdots - \dfrac{1}{21} + \cdots.$

Solution.

(a) For $a(x) = \ln x/x$, $a'(x) = (1 - \ln x)/x^2 < 0$ when $x > e$, and $a(x) \to 0$. The alternating-series test, ignoring finitely many initial terms, proves convergence. The absolute series diverges by the integral test, since $a(x)$ decreases for $x > e$ and $\int_e^M (\ln x)/x\,dx = [(\ln x)^2/2]_e^M = [(\ln M)^2 - 1]/2 \to \infty$.

(b) Combine consecutive pairs to obtain

$$\sum_{k=0}^{\infty} (-1)^k b_k, \qquad b_k = \frac{1}{2k+1} + \frac{1}{2k+2}.$$

Both summands decrease, so b_k decreases, and $b_k \to 0$; the grouped series therefore converges by the alternating-series (Leibniz) test. An unpaired last term is at most $1/(2k+1) \to 0$; hence the original partial sums have the same limit. Removing signs gives the harmonic series, so convergence is conditional.

(c) Put $T_k = k(k+1)/2$. The kth block has sign $(-1)^{k+1}$, length k, and magnitude

$$B_k = \sum_{j=1}^{k} \frac{1}{T_{k-1} + j}.$$

We must prove its decrease, not merely bound its size. Write $a_j = T_{k-1} + j$. Since $T_k = T_{k-1} + k$, the first k terms of B_{k+1} are $1/(T_k + j) = 1/(a_j + k)$, $j = 1, \ldots, k$, and its last term is $1/(T_k + k + 1) = 1/T_{k+1}$. Hence, using $1/a_j - 1/(a_j + k) = k/[a_j(a_j + k)]$,

$$B_k - B_{k+1} = \sum_{j=1}^{k} \frac{k}{a_j(a_j + k)} - \frac{1}{T_{k+1}} \geq \frac{k^2/T_k - 1}{T_{k+1}} = \frac{k-1}{(k+1)T_{k+1}} > 0 \quad (k > 1).$$

Here $a_j \le T_k$ and $a_j + k < T_{k+1}$, so each of the k terms of the sum exceeds $k/(T_k T_{k+1})$; and $k^2/T_k - 1 = 2k/(k+1) - 1 = (k-1)/(k+1)$. Also $B_1 = 1 > B_2 = 5/6$, and $0 < B_k \le k/(T_{k-1}+1) \to 0$. The block series thus converges by the alternating-series test. Every incomplete block has magnitude at most $B_k \to 0$, proving convergence of all original partial sums. The absolute series is again harmonic. All three series are therefore conditionally, but not absolutely, convergent.

1.1.12 Catalan's constant $\beta(2)$ is defined by

$$\beta(2) = \sum_{k=0}^{\infty} (-1)^k (2k+1)^{-2} = \frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} \cdots .$$

Calculate $\beta(2)$ to six-digit accuracy.

Hint. The rate of convergence is enhanced by pairing the terms,

$$(4k-1)^{-2} - (4k+1)^{-2} = \frac{16k}{(16k^2-1)^2}.$$

If you have carried enough digits in your summation, $\sum_{1 \le k \le N} 16k/(16k^2-1)^2$, additional significant figures may be obtained by setting upper and lower bounds on the tail of the series, $\sum_{k=N+1}^{\infty}$. These bounds may be set by comparison with integrals, as in the Maclaurin integral test.

ANS. $\beta(2) = 0.9159\,6559\,4177 \cdots$.

Solution.

Write $G = \beta(2)$ and pair the terms after the first:

$$G = 1 - \sum_{k=1}^{\infty} \left[\frac{1}{(4k-1)^2} - \frac{1}{(4k+1)^2} \right] = 1 - \sum_{k=1}^{\infty} f(k), \quad f(x) = \frac{16x}{(16x^2-1)^2}.$$

For $x \ge 1$, $f'(x) = -16(48x^2+1)/(16x^2-1)^3 < 0$, and the substitution $u = 16x^2 - 1$, $du = 32x\,dx$ gives

$$\int_N^{\infty} f(x)dx = \int_{16N^2-1}^{\infty} \frac{du}{2u^2} = \frac{1}{2(16N^2-1)}.$$

Since f decreases, $\int_k^{k+1} f\,dx < f(k) < \int_{k-1}^{k} f\,dx$, so the tail $\sum_{k=N+1}^{\infty} f(k)$ lies between $\int_{N+1}^{\infty} f\,dx$ and $\int_N^{\infty} f\,dx$. If $G_N = 1 - \sum_{k=1}^{N} f(k)$, then $G = G_N - \sum_{k>N} f(k)$ and

$$G_N - \frac{1}{2(16N^2-1)} < G < G_N - \frac{1}{2(16(N+1)^2-1)}.$$

With $N = 100$, direct summation gives $G_{100} = 0.9159686881\ldots$, and the interval is $0.91596556308 < G < 0.91596562466$. Thus to six decimal places (and also to six significant figures), $\boxed{G = 0.915966}$. As an independent check, integrating $\arctan t/t = \sum_{k \ge 0} (-1)^k t^{2k}/(2k+1)$ termwise gives $G = \int_0^1 (\arctan t/t)\,dt$; numerical quadrature of this integral (e.g., Simpson's rule) gives $0.9159655941\ldots$, inside the interval.

1.1.13 Show how to combine $\zeta(2) = \sum_{n=1}^{\infty} n^{-2}$ with α_1 and α_2 to obtain a series converging as n^{-4}.

Note. $\zeta(2)$ has the known value $\pi^2/6$. See Eq. (12.66).

Solution.

Use $\alpha_1 = \sum[n(n+1)]^{-1} = 1$ and $\alpha_2 = \sum[n(n+1)(n+2)]^{-1} = 1/4$, the cases $p = 1, 2$ of Eq. (1.26), $\alpha_p = 1/(p\,p!)$. With undetermined c_1, c_2,

$$\frac{1}{n^2} - \frac{c_1}{n(n+1)} - \frac{c_2}{n(n+1)(n+2)} = \frac{(1-c_1)n^2 + (3-2c_1-c_2)n + 2}{n^2(n+1)(n+2)}.$$

(The numerator is $(n+1)(n+2) - c_1 n(n+2) - c_2 n$.) Set $1 - c_1 = 0$ and $3 - 2c_1 - c_2 = 0$, giving $c_1 = c_2 = 1$, i.e., the exact identity

$$\frac{1}{n^2} = \frac{1}{n(n+1)} + \frac{1}{n(n+1)(n+2)} + \frac{2}{n^2(n+1)(n+2)}.$$

Summing it over $n \geq 1$ with $\alpha_1 = 1$, $\alpha_2 = 1/4$ yields

$$\boxed{\zeta(2) = \frac{5}{4} + \sum_{n=1}^{\infty} \frac{2}{n^2(n+1)(n+2)}}.$$

The new summand is asymptotic to $2n^{-4}$, as requested.

1.1.14 Give a method of computing

$$\lambda(3) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3}$$

that converges at least as fast as n^{-8} and obtain a result good to six decimal places.

ANS. $\lambda(3) = 1.051800$.

Solution.

Separate the even terms of $\zeta(3)$: since $\sum_{n\geq 1}(2n)^{-3} = 2^{-3}\zeta(3)$, $\lambda(3) = \zeta(3) - 2^{-3}\zeta(3) = 7\zeta(3)/8$. To accelerate the evaluation, define (for $n \geq 2$, $n \geq 3$, and $n \geq 4$, respectively, where the denominators are positive)

$$A_2(n) = \frac{1}{n(n^2-1)}, \quad A_4(n) = \frac{1}{n(n^2-1)(n^2-4)}, \quad A_6(n) = \frac{1}{n(n^2-1)(n^2-4)(n^2-9)}.$$

Use the common denominator $D = n^3(n^2-1)(n^2-4)(n^2-9)$. With $z = n^2$,

$$\frac{D}{n^3} = (z-1)(z-4)(z-9), \quad DA_2 = z(z-4)(z-9), \quad DA_4 = z(z-9), \quad DA_6 = z,$$

so the numerator of $n^{-3} - A_2(n) + A_4(n) - 4A_6(n)$ becomes

$$\begin{aligned}
&(z-1)(z-4)(z-9) - z(z-4)(z-9) + z(z-9) - 4z \\
&\quad = -(z-4)(z-9) + z(z-9) - 4z \\
&\quad = (-z^2 + 13z - 36) + (z^2 - 9z) - 4z = -36.
\end{aligned}$$

Rearranging therefore gives the exact identity, valid for $n \geq 4$,

$$\frac{1}{n^3} = A_2(n) - A_4(n) + 4A_6(n) - \frac{36}{n^3(n^2-1)(n^2-4)(n^2-9)}.$$

To spell out the shifted sums, $A_2(n)$ is $[(n-1)n(n+1)]^{-1}$, $A_4(n)$ is the reciprocal of the five consecutive factors from $n-2$ to $n+2$, and $A_6(n)$ is the reciprocal of the seven from $n-3$ to $n+3$. Their complete sums therefore start at $n=2,3,4$ and equal $\alpha_2,\alpha_4,\alpha_6$, respectively. The telescoping formula $\alpha_p = 1/(p\,p!)$, Eq. (1.26) (proved in Exercise 1.5.3), gives $\alpha_2 = 1/4$, $\alpha_4 = 1/96$, $\alpha_6 = 1/4320$. Removing the terms below $n=4$, namely $A_2(2)=1/6$, $A_2(3)=1/24$, and $A_4(3)=1/120$,

$$\sum_{n=4}^{\infty} A_2(n) = \frac{1}{4}-\frac{1}{6}-\frac{1}{24}=\frac{1}{24}, \quad \sum_{n=4}^{\infty} A_4(n) = \frac{1}{96}-\frac{1}{120}=\frac{1}{480}, \quad \sum_{n=4}^{\infty} A_6(n) = \frac{1}{4320}.$$

Summing the identity over $n \ge 4$ therefore gives

$$\sum_{n=4}^{\infty}\frac{1}{n^3} = \frac{1}{24}-\frac{1}{480}+\frac{4}{4320}-36\sum_{n=4}^{\infty}\frac{1}{D} = \frac{35}{864}-36\sum_{n=4}^{\infty}\frac{1}{D}.$$

Adding $n=1,2,3$, i.e., $1+\frac{1}{8}+\frac{1}{27}=\frac{251}{216}=\frac{1004}{864}$, and multiplying $\zeta(3)$ by $7/8$ gives

$$\boxed{\lambda(3) = \frac{7}{8}\left[\frac{1039}{864}-36\sum_{n=4}^{\infty}\frac{1}{n^3(n^2-1)(n^2-4)(n^2-9)}\right]}.$$

The summand is $O(n^{-9})$, faster than required. Let L_N denote the boxed expression with the sum stopped at $n=N$. The omitted terms are positive and enter with a minus sign, so $L_N > \lambda(3)$. Summing the seven terms $n=4,\dots,10$ gives $L_{10} = 1.051799819028417\ldots$. For the error bound put $C_N = \prod_{j=1}^{3}[1-j^2/(N+1)^2]$. For $n>N$ each factor $1-j^2/n^2$ is at least $1-j^2/(N+1)^2$, so $D = n^9\prod_{j=1}^{3}(1-j^2/n^2) \ge C_N n^9$. Since $\frac{7}{8}\cdot 36 = \frac{63}{2}$ and $\sum_{n>N} n^{-9} < \int_N^\infty x^{-9}dx$,

$$0 < L_N - \lambda(3) = \frac{63}{2}\sum_{n=N+1}^{\infty}\frac{1}{D} < \frac{63}{2C_N}\int_N^\infty x^{-9}dx = \frac{63}{16C_N N^8}.$$

At $N=10$, $C_{10} = 120\cdot 117\cdot 112/121^3 = 0.8876\ldots$, and the bound is $4.44\times10^{-8} < 4.5\times10^{-8}$. Hence $1.05179977 < \lambda(3) < 1.05179982$, and the whole interval rounds to $\boxed{1.051800}$ at six decimal places.

1.1.15 Show that (a) $\sum_{n=2}^{\infty}[\zeta(n)-1] = 1$, (b) $\sum_{n=2}^{\infty}(-1)^n[\zeta(n)-1] = \frac{1}{2}$, where $\zeta(n)$ is the Riemann zeta function.

Solution.

(a) Since $\zeta(n)-1 = \sum_{k=2}^{\infty}k^{-n}$, the nonnegative double series can first be summed over n:

$$\sum_{n=2}^{\infty}[\zeta(n)-1] = \sum_{k=2}^{\infty}\sum_{n=2}^{\infty}k^{-n} = \sum_{k=2}^{\infty}\frac{k^{-2}}{1-k^{-1}} = \sum_{k=2}^{\infty}\frac{1}{k(k-1)} = \sum_{k=2}^{\infty}\left(\frac{1}{k-1}-\frac{1}{k}\right) = 1.$$

The inner sum is geometric with first term k^{-2} and ratio $1/k$.

(b) The absolute values k^{-n} of the terms of the signed double series are the terms summed in (a), so it converges absolutely and its order may be exchanged:

$$\sum_{n=2}^{\infty}(-1)^n[\zeta(n)-1] = \sum_{k=2}^{\infty}\sum_{n=2}^{\infty}(-k^{-1})^n = \sum_{k=2}^{\infty}\frac{k^{-2}}{1+k^{-1}} = \sum_{k=2}^{\infty}\frac{1}{k(k+1)} = \sum_{k=2}^{\infty}\left(\frac{1}{k}-\frac{1}{k+1}\right) = \frac{1}{2}.$$

Both final sums telescope; their partial sums are respectively $1-1/K$ and $1/2-1/(K+1)$.

1.1.16 The convergence improvement of Example 1.1.11 may be carried out more expediently (in this special case) by putting α_2, from Eq. (1.26), into a more symmetric form: Replacing n by $n-1$, we have

$$\alpha_2' = \sum_{n=2}^{\infty} \frac{1}{(n-1)n(n+1)} = \frac{1}{4}.$$

(a) Combine $\zeta(3)$ and α_2' to obtain convergence as n^{-5}.

(b) Let α_4' be α_4 with $n \to n-2$. Combine $\zeta(3)$, α_2', and α_4' to obtain convergence as n^{-7}.

(c) If $\zeta(3)$ is to be calculated to six-decimal place accuracy (error 5×10^{-7}), how many terms are required for $\zeta(3)$ alone? combined as in part (a)? combined as in part (b)?

Note. The error may be estimated using the corresponding integral.

ANS. (a) $\zeta(3) = \dfrac{5}{4} - \displaystyle\sum_{n=2}^{\infty} \frac{1}{n^3(n^2-1)}$.

Solution.

The opening reference is to Example 1.1.11 (the acceleration of $\zeta(3)$), not Exercise 1.1.11. (a) For $n \geq 2$, over the common denominator $n^3(n^2-1)$ the numerator is $(n^2-1) - n^2 = -1$:

$$\frac{1}{n^3} - \frac{1}{n(n^2-1)} = -\frac{1}{n^3(n^2-1)}.$$

Separate $n=1$ and use $\alpha_2' = \sum_{n\geq 2}[n(n^2-1)]^{-1} = 1/4$:

$$\zeta(3) = 1 + \sum_{n=2}^{\infty}\left[\frac{1}{n(n^2-1)} - \frac{1}{n^3(n^2-1)}\right] = \frac{5}{4} - \sum_{n=2}^{\infty}\frac{1}{n^3(n^2-1)},$$

whose summand is $O(n^{-5})$.

(b) Separate the $n=2$ term, $1/(8\cdot 3) = 1/24$, in this last sum and add/subtract $\alpha_4' = \sum_{n=3}^{\infty}[n(n^2-1)(n^2-4)]^{-1} = \alpha_4 = 1/96$ (Eq. (1.26): $\alpha_4 = 1/(4\cdot 4!)$). Over the common denominator $n^3(n^2-1)(n^2-4)$ the numerator is $-(n^2-4)+n^2 = 4$:

$$-\frac{1}{n^3(n^2-1)} + \frac{1}{n(n^2-1)(n^2-4)} = \frac{4}{n^3(n^2-1)(n^2-4)}.$$

Summing over $n \geq 3$,

$$-\sum_{n=3}^{\infty}\frac{1}{n^3(n^2-1)} = -\alpha_4' + \sum_{n=3}^{\infty}\frac{4}{n^3(n^2-1)(n^2-4)},$$

and inserting this and the $n=2$ term into (a),

$$\boxed{\zeta(3) = \frac{5}{4} - \frac{1}{24} - \frac{1}{96} + \sum_{n=3}^{\infty}\frac{4}{n^3(n^2-1)(n^2-4)} = \frac{115}{96} + \sum_{n=3}^{\infty}\frac{4}{n^3(n^2-1)(n^2-4)}}.$$

The summand is $O(n^{-7})$. In the Instructor's Manual, p. 10, the final formula of part (b) has n instead of n^3 in the denominator. The factor n^3 is required by the displayed subtraction and by the claimed n^{-7} rate.

(c) Specify the cutoff by the largest included index N, avoiding ambiguity about the separately summed initial terms. The summands behave as n^{-3}, n^{-5}, and $4n^{-7}$, so the integral estimates of the three tails are

$$\int_N^\infty \frac{dx}{x^3} = \frac{1}{2N^2}, \qquad \int_N^\infty \frac{dx}{x^5} = \frac{1}{4N^4}, \qquad \int_N^\infty \frac{4\,dx}{x^7} = \frac{2}{3N^6}.$$

Requiring each to be at most 5×10^{-7} gives $N^2 \geq 10^6$, $N^4 \geq 5 \times 10^5$, and $N^6 \geq \frac{4}{3} \times 10^6$, i.e., $N \geq 1000$, $N \geq 27$, and $N \geq 11$ (since $26^4 = 456{,}976 < 5 \times 10^5 < 27^4 = 531{,}441$ and $10^6 < \frac{4}{3} \times 10^6 < 11^6 = 1{,}771{,}561$). The exact positive tails, obtained as the difference between each truncated sum and $\zeta(3) = 1.2020569031595942\ldots$, confirm these cutoffs:

	$N_{\min} - 1$	error	$N_{\min}$	error
$\sum n^{-3}$	999	5.0050025×10^{-7}	1000	4.9950025×10^{-7}
(a)	26	5.0681999×10^{-7}	27	4.3703254×10^{-7}
(b)	10	5.0671099×10^{-7}	11	2.9266119×10^{-7}

These cutoffs correspond to 1000 summands ($n = 1, \ldots, 1000$), 26 summands ($n = 2, \ldots, 27$), and 9 summands ($n = 3, \ldots, 11$), respectively, with constants 5/4 or 115/96 evaluated exactly. Positivity makes the error decrease with N, so checking the two neighboring values identifies the minimum. The Instructor's Manual's "999" for the unaccelerated sum is one term too small for an error below 5×10^{-7}.

1.2 Series of Functions

1.2.1 Find the range of uniform convergence of the series

(a) $\eta(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^x}$,

(b) $\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}$.

ANS. (a) $0 < s \leq x < \infty$. (b) $1 < s \leq x < \infty$.

Solution.

(a) For fixed $x > 0$, n^{-x} decreases to zero. By the Leibniz criterion, the alternating-series remainder is bounded by the first omitted term:

$$|\eta(x) - S_N(x)| \leq (N+1)^{-x} \leq (N+1)^{-s}, \qquad x \geq s > 0.$$

The last expression is independent of x and tends to zero, proving uniform convergence on every $[s, \infty)$ with $s > 0$. On $(0, \infty)$ it is not uniform: a necessary condition is uniform convergence of the terms $u_n = S_n - S_{n-1}$ to zero, whereas $\sup_{x>0} n^{-x} = 1$ for every n (let $x \to 0^+$). At $x \leq 0$ the terms do not even tend to zero.

(b) If $x \geq s > 1$, then $|n^{-x}| \leq n^{-s}$, and the convergent p-series supplies the Weierstrass bound. Thus convergence is uniform on every $[s, \infty)$, $s > 1$. It is not uniform on $(1, \infty)$: for every N,

$$\sup_{x>1} \sum_{n=N+1}^{2N} n^{-x} = \sum_{n=N+1}^{2N} \frac{1}{n} \geq N \cdot \frac{1}{2N} = \frac{1}{2}$$

(the supremum is the limit $x \to 1^+$ of this finite sum, and each of its N terms is at least $1/(2N)$), contradicting the uniform Cauchy criterion. For $x \leq 1$ the positive series diverges. The lower bounds $s > 0$ and $s > 1$ cannot be omitted.

1.2.2 For what range of x is the geometric series $\sum_{n=0}^{\infty} x^n$ uniformly convergent?

ANS. $-1 < -s \le x \le s < 1$.

Solution.

For $|x| < 1$ the finite-sum identity gives the exact remainder

$$\sum_{n=N+1}^{\infty} x^n = \frac{x^{N+1}}{1-x}.$$

If $|x| \le s < 1$, then $|1-x| \ge 1-|x| \ge 1-s$, so its absolute value is at most $s^{N+1}/(1-s)$, which tends to zero independently of x. Thus the series is uniformly convergent on every $[-s, s]$, $s < 1$, or more generally any subset of $(-1, 1)$ bounded away from both endpoints ± 1. If a set has points with $|x|$ arbitrarily close to 1, then $\sup |x^n| = 1$ and the necessary uniform term test fails. The full interval $(-1, 1)$ is therefore not a range of uniform convergence, though it is the range of pointwise convergence.

1.2.3 For what range of positive values of x is $\sum_{n=0}^{\infty} 1/(1+x^n)$

(a) convergent? (b) uniformly convergent?

Solution.

The term $n = 0$ is $1/2$. If $0 < x < 1$, the terms approach 1; if $x = 1$ they all equal $1/2$. Both cases diverge. If $x > 1$,

$$0 < \frac{1}{1+x^n} \le x^{-n},$$

so geometric comparison proves convergence. For $x \ge 1 + \epsilon$, $\epsilon > 0$, the terms are bounded by $(1+\epsilon)^{-n}$, the terms of a convergent geometric series independent of x; the Weierstrass M test proves uniform convergence on $[1+\epsilon, \infty)$. Uniform convergence fails on $(1, \infty)$, since $\sup_{x>1}(1+x^n)^{-1} = 1/2$ for each n (let $x \to 1^+$), so the terms do not tend to zero uniformly. Thus a range extending arbitrarily close to 1 from above must be excluded. In summary: (a) convergent for $x > 1$; (b) uniformly convergent for $1 < s \le x < \infty$.

1.2.4 If the series of the coefficients $\sum a_n$ and $\sum b_n$ are absolutely convergent, show that the Fourier series

$$\sum (a_n \cos nx + b_n \sin nx)$$

is **uniformly** convergent for $-\infty < x < \infty$.

Solution.

For every real x,

$$|a_n \cos nx + b_n \sin nx| \le |a_n|\,|\cos nx| + |b_n|\,|\sin nx| \le |a_n| + |b_n|.$$

The hypotheses give $\sum_n (|a_n| + |b_n|) < \infty$. In particular, for any $M > N$,

$$\sup_{x \in \mathbb{R}} \left| \sum_{n=N+1}^{M} (a_n \cos nx + b_n \sin nx) \right| \le \sum_{n=N+1}^{M} (|a_n| + |b_n|) \longrightarrow 0.$$

This is the uniform Cauchy criterion on the entire real line, and proves both absolute pointwise convergence and uniform convergence.

1.2.5 Assume a nonzero initial term and real l. The Legendre series $\sum_{j\ \text{even}} u_j(x)$ satisfies the recurrence relations

$$u_{j+2}(x) = \frac{(j+1)(j+2) - l(l+1)}{(j+2)(j+3)}\, x^2 u_j(x),$$

in which the index j is even and l is some constant (but, in this problem, **not** a nonnegative odd integer, nor a negative even integer if negative l is allowed). Find the range of values of x for which this Legendre series is convergent. Test the endpoints.

ANS. $-1 < x < 1$.

Solution.

The quoted answer concerns a nonzero, nonterminating series. If arbitrary real l is allowed, also exclude $l = -2, -4, -6, \ldots$; these values terminate just as $l = 1, 3, 5, \ldots$ do. Write $j = 2r$. For a nonterminating series the magnitude of the ratio of successive nonzero terms is $|(j+1)(j+2) - l(l+1)|\, x^2/[(j+2)(j+3)] \to x^2$ as $j \to \infty$. Hence the ratio test proves absolute convergence for $|x| < 1$ and divergence for $|x| > 1$. At $x = \pm 1$, the ratio is eventually positive, and its reciprocal is

$$\begin{aligned}\frac{(2r+2)(2r+3)}{(2r+1)(2r+2) - l(l+1)} &= \frac{4r^2 + 10r + 6}{4r^2 + 6r + 2 - l(l+1)} \\ &= 1 + \frac{4r + 4 + l(l+1)}{4r^2 + 6r + 2 - l(l+1)} = 1 + \frac{1}{r} + O(r^{-2}).\end{aligned}$$

This is Exercise 1.1.4 with $a_1 - b_1 = \frac{5}{2} - \frac{3}{2} = 1$ (divide numerator and denominator by 4), or Gauss' test with $h = 1$: the terms have one eventual sign and magnitude asymptotic to C/r, so both endpoints diverge. To check the exception, the numerator vanishes exactly when $l(l+1) = (j+1)(j+2)$; since $l^2 + l - (j+1)(j+2) = (l - j - 1)(l + j + 2)$, this means $l = j + 1$ or $l = -j - 2$. For even $j \geq 0$ these are precisely the excluded parameter values. A terminated series, or the identically zero series, converges for all x.

1.2.6 A series solution of the Chebyshev equation leads to successive terms having the ratio

$$\frac{u_{j+2}(x)}{u_j(x)} = \frac{(k+j)^2 - n^2}{(k+j+1)(k+j+2)}\, x^2,$$

with $k = 0$ and $k = 1$. Test for convergence at $x = \pm 1$.

ANS. Convergent.

Solution.

Within either parity branch put $j = 2r$ and $m = k + 2r$. At $x = \pm 1$ the reciprocal ratio of successive terms is

$$\frac{(m+1)(m+2)}{m^2 - n^2} = \frac{m^2 + 3m + 2}{m^2 - n^2} = 1 + \frac{3m + 2 + n^2}{m^2 - n^2} = 1 + \frac{3}{m} + O(m^{-2}) = 1 + \frac{3}{2r} + O(r^{-2}).$$

(Here $3/m = 3/(2r + k) = 3/(2r) + O(r^{-2})$.) This is Gauss' test with $h = 3/2 > 1$, the ratio being positive once $m > |n|$. For a nonterminating branch the terms have magnitude asymptotic to $Cr^{-3/2}$ by the product/logarithm argument of Exercise 1.1.4. The series therefore converges absolutely at both endpoints. If $m^2 = n^2$ at an index of the branch, the recurrence terminates; that polynomial also converges at both endpoints. This covers $k = 0$ and $k = 1$, including terminating cases.

1.2.7 A series solution for the ultraspherical (Gegenbauer) function $C_n^\alpha(x)$ leads to the recurrence

$$a_{j+2} = a_j \frac{(k+j)(k+j+2\alpha) - n(n+2\alpha)}{(k+j+1)(k+j+2)}.$$

Take real n, α, $k = 0$ or 1, and even $j = 0, 2, \ldots$ in each parity branch. Investigate the convergence of each of these series at $x = \pm 1$ as a function of the parameter α.

ANS. For a nonterminating branch: convergent for $\alpha < 1/2$, divergent for $\alpha \geq 1/2$. A terminated branch converges at both endpoints.

Solution.

For the stated recurrence the nonterminating series converges at $x = \pm 1$ for $\alpha < 1/2$, and diverges for $\alpha \geq 1/2$. The book's threshold 1 is incorrect; it is what Gauss' test gives if j, which advances by 2 from term to term, is wrongly used as the term counter ($h = 3 - 2\alpha$ instead of $3/2 - \alpha$). Terminated series converge for all α. Here α, n are real and $k = 0$ or 1. Indeed, put $j = 2r$ and $m = k + 2r$. For large r the ratio is positive, and at either endpoint its reciprocal is

$$\begin{aligned}\frac{(m+1)(m+2)}{m(m+2\alpha) - n(n+2\alpha)} &= 1 + \frac{(3-2\alpha)m + 2 + n(n+2\alpha)}{m^2 + 2\alpha m - n(n+2\alpha)} \\ &= 1 + \frac{3-2\alpha}{m} + O(m^{-2}) = 1 + \frac{3/2 - \alpha}{r} + O(r^{-2}),\end{aligned}$$

since $1/m = 1/(2r + k) = 1/(2r) + O(r^{-2})$. This is Gauss' test with $h = 3/2 - \alpha$. By the product/logarithm argument of Exercise 1.1.4 (with $d = 3/2 - \alpha$), $|a_{2r}| \sim C r^{\alpha - 3/2}$ in a nonzero nonterminating branch. The p-series criterion requires $3/2 - \alpha > 1$, or $\alpha < 1/2$. At equality the tail is harmonic, and larger α also diverges. Since the parity is fixed, replacing 1 by -1 only multiplies the whole branch by $(-1)^k$; it cannot produce alternating convergence. Finally,

$$m(m+2\alpha) - n(n+2\alpha) = (m-n)(m+n+2\alpha),$$

so the branch terminates if an allowed $m = k + 2r$ equals n or $-n - 2\alpha$. Such branches are the explicit exception to the endpoint test. As a check, $\alpha = 1/2$ gives Legendre's recurrence and its nonterminating logarithmic endpoint divergence (Exercise 1.2.5, the case $k = 1$), while $\alpha = 0$ gives the Chebyshev ratio of Exercise 1.2.6, convergent at the endpoints.

1.2.8 Show that

(a) $\sin x = \displaystyle\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$,

(b) $\cos x = \displaystyle\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$.

Solution.

Successive derivatives of sine repeat as $\sin x, \cos x, -\sin x, -\cos x$, while those of cosine repeat as $\cos x, -\sin x, -\cos x, \sin x$. At $x = 0$ this gives

$$\sin^{(2n)}(0) = 0, \quad \sin^{(2n+1)}(0) = (-1)^n, \qquad \cos^{(2n)}(0) = (-1)^n, \quad \cos^{(2n+1)}(0) = 0.$$

Insert these values into Taylor's formula $f(x) = \sum_{j=0}^{N} f^{(j)}(0) x^j/j! + R_N(x)$, with the Lagrange remainder $R_N(x) = f^{(N+1)}(\xi) x^{N+1}/(N+1)!$, ξ between 0 and x. For real x all derivatives have magnitude at most one on the segment from 0 to x, so $|R_N(x)| \leq |x|^{N+1}/(N+1)! \to 0$; this is the general term of the convergent series for $e^{|x|}$. The two stated series therefore equal sine and cosine for every real x. Their term ratios tend to zero for every complex x as well, giving entire power-series extensions.

1.2.9 Derive a series expansion of $\cot x$ in increasing powers of x by dividing the power series for $\cos x$ by that for $\sin x$.

Note. The resultant series that starts with $1/x$ is known as a Laurent series ($\cot x$ does not have a Taylor expansion about $x = 0$, although $\cot(x) - x^{-1}$ does). Although the two series for $\sin x$ and $\cos x$ were valid for all x, the convergence of the series for $\cot x$ is limited by the zeros of the denominator, $\sin x$.

Solution.

Sine is odd and cosine is even, so write $\cot x = x^{-1} + a_1 x + a_3 x^3 + a_5 x^5 + \cdots$. Multiply by $\sin x = x - x^3/6 + x^5/120 - x^7/5040 + x^9/362880 - \cdots$ and equate the result to $\cos x = 1 - x^2/2 + x^4/24 - x^6/720 + x^8/40320 - \cdots$:

$$\begin{aligned}
x^2: &\quad a_1 - \tfrac{1}{6} = -\tfrac{1}{2} && \Rightarrow a_1 = -\tfrac{1}{3},\\
x^4: &\quad a_3 - \tfrac{a_1}{6} + \tfrac{1}{120} = \tfrac{1}{24} && \Rightarrow a_3 = -\tfrac{1}{45},\\
x^6: &\quad a_5 - \tfrac{a_3}{6} + \tfrac{a_1}{120} - \tfrac{1}{5040} = -\tfrac{1}{720} && \Rightarrow a_5 = -\tfrac{2}{945},\\
x^8: &\quad a_7 - \tfrac{a_5}{6} + \tfrac{a_3}{120} - \tfrac{a_1}{5040} + \tfrac{1}{362880} = \tfrac{1}{40320} && \Rightarrow a_7 = -\tfrac{1}{4725}.
\end{aligned}$$

Solving each line with the previous values, over common denominators,

$$\begin{aligned}
a_3 &= \tfrac{1}{24} - \tfrac{1}{120} - \tfrac{1}{18} = \tfrac{15-3-20}{360} = -\tfrac{1}{45},\\
a_5 &= -\tfrac{1}{720} + \tfrac{1}{5040} - \tfrac{1}{270} + \tfrac{1}{360} = \tfrac{-21+3-56+42}{15120} = -\tfrac{2}{945},\\
a_7 &= \left(\tfrac{1}{40320} - \tfrac{1}{362880}\right) - \tfrac{1}{2835} + \tfrac{1}{5400} - \tfrac{1}{15120} = \tfrac{5-80+42-15}{226800} = -\tfrac{1}{4725},
\end{aligned}$$

where $\frac{1}{40320} - \frac{1}{362880} = \frac{8}{362880} = \frac{1}{45360}$. Consequently

$$\cot x = \frac{1}{x} - \frac{x}{3} - \frac{x^3}{45} - \frac{2x^5}{945} - \frac{x^7}{4725} - \cdots.$$

More generally, if $\cot x = \sum_{m\ge 0} c_m x^{2m-1}$ and $c_0 = 1$, the coefficient of x^{2m} in $\cot x \sin x$ is $\sum_{j=0}^{m} (-1)^j c_{m-j}/(2j+1)!$; equating it to that of $\cos x$, $(-1)^m/(2m)!$, gives the recursion

$$c_m = \frac{(-1)^m}{(2m)!} - \sum_{j=1}^{m} \frac{(-1)^j c_{m-j}}{(2j+1)!}, \qquad m \ge 1.$$

The nearest nonzero zeros of $\sin x$ are $\pm\pi$. Since the quotient has poles there and no other singularity closer to zero, its Laurent series converges for $0 < |x| < \pi$.

1.2.10 Show by series expansion that

$$\frac{1}{2}\ln\frac{\eta_0+1}{\eta_0-1} = \coth^{-1}\eta_0, \qquad |\eta_0| > 1.$$

This identity may be used to obtain a second solution for Legendre's equation.

Solution.

Put $t = 1/\eta_0$, so $|t| < 1$. Dividing numerator and denominator by η_0 and subtracting the two logarithm series $\ln(1 \pm t) = \pm t - t^2/2 \pm t^3/3 - \cdots$ gives

$$\frac{1}{2}\ln\frac{\eta_0+1}{\eta_0-1} = \frac{1}{2}[\ln(1+t) - \ln(1-t)] = t + \frac{t^3}{3} + \frac{t^5}{5} + \cdots.$$

The even powers cancel. The same series represents $\coth^{-1}\eta_0$. Since $d\coth y/dy = 1-\coth^2 y$, for $|\eta|>1$

$$\frac{d}{d\eta}\coth^{-1}\eta = \frac{1}{1-\eta^2} = -\frac{\eta^{-2}}{1-\eta^{-2}} = -\sum_{n\ge 0}\eta^{-2n-2}.$$

Integrating termwise from $\eta_0>1$ to ∞, where $\coth^{-1}$ vanishes, gives

$$\coth^{-1}\eta_0 = \sum_{n\ge 0}\frac{\eta_0^{-2n-1}}{2n+1} = t+\frac{t^3}{3}+\frac{t^5}{5}+\cdots;$$

both sides are odd in η_0, which covers $\eta_0<-1$. The two series agree term by term, proving the identity. As a direct check without series, set the left side equal to y. Then $e^{2y}=(1+t)/(1-t)$ and hence

$$\coth y = \frac{e^{2y}+1}{e^{2y}-1} = \frac{(1+t)+(1-t)}{(1+t)-(1-t)} = \frac{1}{t} = \eta_0.$$

For real $|\eta_0|>1$ the logarithm's argument is positive and y has the same sign as η_0, giving the real inverse $\coth^{-1}\eta_0$. For complex arguments a consistent logarithm branch is required.

1.2.11 Show that $f(x)=x^{1/2}$ (a) has no Maclaurin expansion but (b) has a Taylor expansion about any point $x_0\neq 0$ (for the real-valued square root, $x_0>0$; at a complex $x_0\neq 0$, with a chosen local branch). Find the range of convergence of the Taylor expansion about $x=x_0$.

Solution.

(a) A Maclaurin expansion requires the coefficient $f'(0)$. But $f(0)=0$, and the difference quotient $[f(x)-f(0)]/x = x^{-1/2}$ tends to ∞ as $x\downarrow 0$, so $f'(0)$ does not exist. More strongly, no power series $P(x)=\sum_n a_n x^n$ can equal $\sqrt{x}$ for $0<x<\epsilon$: its derivative $P'(x)=\sum_n n a_n x^{n-1}$ is again a power series convergent near 0 (Exercise 1.2.16), hence continuous at 0, so $P'(x)\to a_1$ as $x\downarrow 0$, whereas $(\sqrt{x})' = 1/(2\sqrt{x})\to\infty$.

(b) For a real-valued square root, take $x_0>0$. At a complex $x_0\neq 0$, choose a local branch. Set $h=(x-x_0)/x_0$. The binomial expansion gives

$$\sqrt{x} = \sqrt{x_0}(1+h)^{1/2} = \sqrt{x_0}\left[1+\frac{h}{2}-\frac{h^2}{8}+\frac{h^3}{16}-\frac{5h^4}{128}+\cdots\right],$$

with general coefficient $\binom{1/2}{m} = (1/2)(1/2-1)\cdots(1/2-m+1)/m!$. The ratio test gives radius $|h|<1$, or $|x-x_0|<|x_0|$; the branch point at zero prevents a larger radius. For real $x_0>0$, the two endpoints $h=\pm 1$ also converge absolutely: the absolute coefficient ratio is $|\binom{1/2}{m+1}/\binom{1/2}{m}| = (m-1/2)/(m+1)$, whose reciprocal is $(m+1)/(m-1/2) = 1+3/(2m)+O(m^{-2})$, so Gauss' test with parameter $3/2>1$ proves that $\sum_m|\binom{1/2}{m}|$ converges. (As in Exercise 1.2.6, the argument of Exercise 1.1.4 shows more precisely that the coefficients are of order $m^{-3/2}$.) By Abel's theorem (a power series is continuous up to an endpoint at which it converges) their sums are 0 at $x=0$ and $\sqrt{2x_0}$ at $x=2x_0$. Thus the real convergence interval is $[0,2x_0]$, with analyticity only in its interior.

1.2.12 Prove l'Hôpital's rule, Eq. (1.58). For the 0/0 version, assume real f,g are continuous at x_0, differentiable in a punctured one-sided neighborhood, $f(x_0)=g(x_0)=0$, and $g'\neq 0$ there. Assume that $f'/g'\to L$, where L may be an extended real limit. Prove $f/g\to L$.

Solution.

The rule needs hypotheses: assume f,g are continuous at x_0, differentiable in a punctured one-sided neighborhood, $f(x_0)=g(x_0)=0$, $g'(x)\neq 0$ there, and $\lim_{x\to x_0} f'(x)/g'(x)=L$ exists

(possibly infinite). For $x \neq x_0$ sufficiently close to x_0, Cauchy's mean value theorem supplies ξ strictly between x and x_0 such that

$$[f(x) - f(x_0)]g'(\xi) = [g(x) - g(x_0)]f'(\xi).$$

To prove it, apply Rolle's theorem to $\phi(t) = [f(t) - f(x_0)][g(x) - g(x_0)] - [g(t) - g(x_0)][f(x) - f(x_0)]$, which is continuous on the closed interval between x_0 and x, differentiable inside, and vanishes at both ends; $\phi'(\xi) = 0$ is the displayed equality. Rolle's theorem applied to g itself and $g' \neq 0$ ensure $g(x) \neq g(x_0) = 0$. Dividing the equality gives

$$\frac{f(x)}{g(x)} = \frac{f'(\xi)}{g'(\xi)}.$$

As $x \to x_0$, $\xi \to x_0$, and the right side tends to L. This proves the one-sided $0/0$ rule in Eq. (1.58). If both one-sided limits agree, the two-sided version follows. This proof does not require f, g to have convergent Taylor series.

1.2.13 With $n > 1$, show that

$$\text{(a)}\ \frac{1}{n} - \ln\left(\frac{n}{n-1}\right) < 0, \qquad \text{(b)}\ \frac{1}{n} - \ln\left(\frac{n+1}{n}\right) > 0.$$

Use these inequalities to show that the limit defining the Euler-Mascheroni constant, Eq. (1.13), is finite.

Solution.

Since $1/t > 1/n$ for $n - 1 < t < n$ and $1/t < 1/n$ for $n < t < n + 1$,

$$\ln\frac{n}{n-1} = \int_{n-1}^{n} \frac{dt}{t} > \frac{1}{n}, \qquad \ln\frac{n+1}{n} = \int_{n}^{n+1} \frac{dt}{t} < \frac{1}{n}.$$

These are the two requested strict inequalities. Let $a_n = H_n - \ln n$ and $b_n = H_n - \ln(n+1)$. Then

$$a_n - a_{n-1} = \frac{1}{n} - \ln\frac{n}{n-1} < 0, \quad b_n - b_{n-1} = \frac{1}{n} - \ln\frac{n+1}{n} > 0.$$

Moreover $a_n > b_n$ and $a_n - b_n = \ln(1 + 1/n) \to 0$. Thus a_n is decreasing and bounded below by $b_1 = 1 - \ln 2$ (since $a_n > b_n \geq b_1$); b_n is increasing and bounded above by $a_1 = 1$ (since $b_n < a_n \leq a_1$). Both have finite limits, and their vanishing difference proves that the limits agree. That common limit is γ.

1.2.14 In numerical analysis it is often convenient to approximate $d^2\psi(x)/dx^2$ by

$$\frac{d^2}{dx^2}\psi(x) \approx \frac{1}{h^2}[\psi(x+h) - 2\psi(x) + \psi(x-h)].$$

Find the error in this approximation. Assume $\psi \in C^6$ near x, and define error as approximation minus exact value.

ANS. Error $= \dfrac{h^2}{12}\psi^{(4)}(x) + O(h^4)$.

Solution.

With the stated C^6 regularity, Taylor's remainder through degree five is $O(h^6)$. Taylor expansion on both sides gives

$$\psi(x \pm h) = \psi(x) \pm h\psi'(x) + \frac{h^2}{2}\psi''(x) \pm \frac{h^3}{6}\psi^{(3)}(x) + \frac{h^4}{24}\psi^{(4)}(x) \pm \frac{h^5}{120}\psi^{(5)}(x) + O(h^6).$$

Adding cancels the odd powers and doubles the even ones ($2 \cdot \frac{1}{24} = \frac{1}{12}$):

$$\psi(x+h) + \psi(x-h) = 2\psi(x) + h^2\psi''(x) + \frac{h^4}{12}\psi^{(4)}(x) + O(h^6).$$

Subtracting $2\psi(x)$ and dividing by h^2 yields

$$D_h\psi = \frac{\psi(x+h) - 2\psi(x) + \psi(x-h)}{h^2} = \psi''(x) + \frac{h^2}{12}\psi^{(4)}(x) + O(h^4).$$

Thus the printed error means approximation minus exact value, $D_h\psi - \psi''(x) = h^2\psi^{(4)}(x)/12 + O(h^4)$. If error is defined as exact minus approximation, its sign is reversed. For $\psi(x) = x^4$, $(x \pm h)^4$ summed is $2x^4 + 12x^2h^2 + 2h^4$, so $D_h\psi = 12x^2 + 2h^2$ and the difference from $\psi'' = 12x^2$ is exactly $2h^2$, confirming the coefficient $24/12 = 2$ and the stated sign convention. The printed answer is the leading error, not generally the whole error. For $\psi(x) = x^6$ at $x = 0$, the displayed fourth derivative vanishes, but $D_h\psi - \psi''(0) = 2h^4 \neq 0$. This shows why the remainder term must be retained.

1.2.15 Evaluate $\lim_{x\to 0}\left[\dfrac{\sin(\tan x) - \tan(\sin x)}{x^7}\right]$.

ANS. $-\dfrac{1}{30}$.

Solution.

Put $t = \tan x = x + x^3/3 + 2x^5/15 + 17x^7/315 + O(x^9)$ and $s = \sin x = x - x^3/6 + x^5/120 - x^7/5040 + O(x^9)$. The tangent series follows from Exercise 1.2.9 and $\tan x = \cot x - 2\cot 2x$ (because $\cot x - \tan x = (\cos^2 x - \sin^2 x)/(\sin x \cos x) = 2\cot 2x$): the $1/x$ terms cancel, and its coefficients of x, x^3, x^5, x^7 are $-\frac{1}{3} + \frac{4}{3} = 1$, $-\frac{1}{45} + \frac{16}{45} = \frac{1}{3}$, $-\frac{2}{945} + \frac{128}{945} = \frac{2}{15}$, and $-\frac{1}{4725} + \frac{256}{4725} = \frac{17}{315}$. Multiplication gives

$$t^3 = x^3 + x^5 + \tfrac{11}{15}x^7 + O(x^9), \qquad t^5 = x^5 + \tfrac{5}{3}x^7 + O(x^9),$$
$$s^3 = x^3 - \tfrac{1}{2}x^5 + \tfrac{13}{120}x^7 + O(x^9), \quad s^5 = x^5 - \tfrac{5}{6}x^7 + O(x^9).$$

(For instance, the x^7 coefficient of t^3 collects $3 \cdot \frac{2}{15}$ from the three orderings of $x \cdot x \cdot \frac{2x^5}{15}$ and $3 \cdot \frac{1}{9}$ from those of $x \cdot \frac{x^3}{3} \cdot \frac{x^3}{3}$, total $\frac{11}{15}$; that of s^3 is $3 \cdot \frac{1}{120} + 3 \cdot \frac{1}{36} = \frac{13}{120}$; those of t^5, s^5 are $5 \cdot \frac{1}{3}$ and $5 \cdot (-\frac{1}{6})$, from the five places of the single cubic factor.) Also $t^7 = s^7 = x^7 + O(x^9)$. Substituting into the outer series,

$$\sin t = t - \frac{t^3}{6} + \frac{t^5}{120} - \frac{t^7}{5040} + O(x^9) = x + \frac{x^3}{6} - \frac{x^5}{40} - \frac{55x^7}{1008} + O(x^9),$$
$$\tan s = s + \frac{s^3}{3} + \frac{2s^5}{15} + \frac{17s^7}{315} + O(x^9) = x + \frac{x^3}{6} - \frac{x^5}{40} - \frac{107x^7}{5040} + O(x^9).$$

For example, the x^5 coefficients of $\sin t$ and $\tan s$ are $\frac{2}{15} - \frac{1}{6} + \frac{1}{120} = -\frac{1}{40}$ and $\frac{1}{120} - \frac{1}{6} + \frac{2}{15} = -\frac{1}{40}$, while over the common denominator 5040 the x^7 coefficients are

$$\sin t: \quad \frac{17}{315} - \frac{1}{6}\cdot\frac{11}{15} + \frac{1}{120}\cdot\frac{5}{3} - \frac{1}{5040} = \frac{272 - 616 + 70 - 1}{5040} = -\frac{275}{5040},$$
$$\tan s: \quad -\frac{1}{5040} + \frac{1}{3}\cdot\frac{13}{120} - \frac{2}{15}\cdot\frac{5}{6} + \frac{17}{315} = \frac{-1 + 182 - 560 + 272}{5040} = -\frac{107}{5040}.$$

Their difference is $[-275/5040 + 107/5040]x^7 + O(x^9) = -168x^7/5040 + O(x^9) = -x^7/30 + O(x^9)$. Division by x^7 therefore gives $\boxed{-1/30}$ in the limit.

1.2.16 A power series has actual radius of convergence $R > 0$ (possibly $R = \infty$) and converges for $-R < x < R$. Show that the differentiated series and the integrated series have the same interval of convergence. (Do not bother about the endpoints $x = \pm R$.)

Solution.

Write $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Fix r, ρ with $0 < r < \rho < R$. Convergence at ρ implies $|a_n|\rho^n \le M$ for some finite M. Consequently, on $|x| \le r$,

$$\sum_{n=1}^{\infty} n|a_n|r^{n-1} \le \frac{M}{\rho}\sum_{n=1}^{\infty} n(r/\rho)^{n-1} < \infty,$$

since $n|a_n|r^{n-1} = n|a_n|\rho^n r^{n-1}/\rho^n \le (M/\rho)\, n(r/\rho)^{n-1}$ and $\sum_n nq^{n-1}$ converges for $0 \le q < 1$ (ratio test). As $r < R$ is arbitrary, the differentiated series converges for $|x| < R$, and by the Weierstrass M test uniformly on each $|x| \le r$, so termwise differentiation is justified there. Similarly the integral series $C + \sum_{n=0}^{\infty} a_n x^{n+1}/(n+1)$ converges for $|x| < R$, because $|a_n|r^{n+1}/(n+1) \le rM(r/\rho)^n$. Neither resulting radius can exceed R. The argument above uses only that the terms of a series are bounded at one point x_1. If the differentiated series converged at some x_1 with $|x_1| > R$, the same argument applied to it would make its integrated series, which is the original series minus a_0, converge for all $|x| < |x_1|$. If the integrated series converged at such an x_1, the same argument would make its differentiated series, the original one, converge for all $|x| < |x_1|$. Both contradict the definition of R. Thus both radii equal R. Here R denotes the actual radius of convergence, not merely an arbitrary smaller interval on which the original series converges. Endpoint behavior may change, as the question explicitly permits.

1.3 Binomial Theorem

1.3.1 The classical Langevin theory of paramagnetism leads to an expression for the magnetic polarization,

$$P(x) = c\left(\frac{\cosh x}{\sinh x} - \frac{1}{x}\right).$$

Expand $P(x)$ as a power series for small x (low fields, high temperature).

Solution.

Since $x\coth x = x\cosh x/\sinh x$ is even and regular at $x = 0$ (it tends to 1), write $\coth x = x^{-1} + ax + bx^3 + dx^5 + \cdots$ and multiply by $\sinh x = x + x^3/6 + x^5/120 + x^7/5040 + \cdots$:

$$\begin{aligned}\coth x \sinh x = 1 &+ \Big(a + \frac{1}{6}\Big)x^2 + \Big(b + \frac{a}{6} + \frac{1}{120}\Big)x^4 \\ &+ \Big(d + \frac{b}{6} + \frac{a}{120} + \frac{1}{5040}\Big)x^6 + O(x^8).\end{aligned}$$

Equating to $\cosh x = 1 + x^2/2 + x^4/24 + x^6/720 + \cdots$ gives

$$a + \frac{1}{6} = \frac{1}{2}, \qquad b + \frac{a}{6} + \frac{1}{120} = \frac{1}{24}, \qquad d + \frac{b}{6} + \frac{a}{120} + \frac{1}{5040} = \frac{1}{720}.$$

Solving successively yields $a = 1/3$, $b = \frac{1}{24} - \frac{1}{18} - \frac{1}{120} = \frac{15-20-3}{360} = -\frac{1}{45}$, and

$$d = \frac{1}{720} + \frac{1}{270} - \frac{1}{360} - \frac{1}{5040} = \frac{21 + 56 - 42 - 3}{15120} = \frac{2}{945}.$$

The apparent pole cancels in P, leaving

$$\boxed{P(x) = c\left(\frac{x}{3} - \frac{x^3}{45} + \frac{2x^5}{945} + O(x^7)\right).}$$

In particular $P(0) = 0$ by continuous extension, and the leading small-field response is $cx/3$. The series converges for $|x| < \pi$: the singularity of $\coth x - 1/x$ at $x = 0$ is removable, and the nearest other singularities are the poles of $\coth x$ at the zeros $x = \pm i\pi$ of $\sinh x$ (compare $\cot x$ in Exercise 1.2.9).

1.3.2 Given that

$$\int_0^1 \frac{dx}{1+x^2} = \tan^{-1} x \Big|_0^1 = \frac{\pi}{4},$$

expand the integrand into a series and integrate term by term obtaining[1]

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots + (-1)^n \frac{1}{2n+1} + \cdots,$$

which is Leibniz's formula for π. Compare the convergence of the integrand series and the integrated series at $x = 1$. Leibniz's formula converges so slowly that it is quite useless for numerical work.

Solution.

For $|x| < 1$, the geometric expansion is $(1+x^2)^{-1} = \sum_{n=0}^{\infty}(-1)^n x^{2n}$. To justify integrating all the way to 1, use the finite geometric sum $\sum_{n=0}^{N} r^n = (1 - r^{N+1})/(1-r)$ with $r = -x^2$, which gives the identity

$$\frac{1}{1+x^2} = \sum_{n=0}^{N}(-1)^n x^{2n} + \frac{(-1)^{N+1}x^{2N+2}}{1+x^2}.$$

Integrating from 0 to 1, with $\int_0^1 x^{2n}dx = 1/(2n+1)$, gives

$$\frac{\pi}{4} = \sum_{n=0}^{N}\frac{(-1)^n}{2n+1} + R_N, \qquad R_N = (-1)^{N+1}\int_0^1 \frac{x^{2N+2}}{1+x^2}\,dx.$$

Since $1 + x^2 \geq 1$, $|R_N| \leq \int_0^1 x^{2N+2}dx = 1/(2N+3) \to 0$. Thus the stated Leibniz series follows. At $x = 1$ the integrand series is $1 - 1 + 1 - 1 + \cdots$, whose terms fail to tend to zero. The integrated series has decreasing term magnitudes $1/(2n+1) \to 0$ and converges conditionally; its absolute series diverges by comparison with the harmonic series, since $1/(2n+1) \geq \frac{1}{2}\cdot 1/(n+1)$.

1.3.3 Expand the incomplete gamma function $\gamma(n+1, x) \equiv \int_0^x e^{-t}t^n\,dt$ in a series of powers of x, initially for a nonnegative integer n. What is the range of convergence of the resulting series?

ANS. $$\int_0^x e^{-t}t^n\,dt = x^{n+1}\left[\frac{1}{n+1} - \frac{x}{n+2} + \frac{x^2}{2!(n+3)} - \cdots + \frac{(-1)^p x^p}{p!(n+p+1)} + \cdots\right].$$

[1]The series expansion of $\tan^{-1} x$ (upper limit 1 replaced by x) was discovered by James Gregory in 1671, 3 years before Leibniz. See Peter Beckmann's entertaining book, *A History of Pi*, 2nd ed., Boulder, CO: Golem Press (1971), and L. Berggren, J. Borwein, and P. Borwein, *Pi: A Source Book*, New York: Springer (1997).

Solution.

For a nonnegative integer n, expand e^{-t} and integrate:

$$\gamma(n+1,x) = \int_0^x \sum_{p=0}^{\infty} \frac{(-1)^p t^{n+p}}{p!}\,dt$$
$$= \sum_{p=0}^{\infty} \frac{(-1)^p x^{n+p+1}}{p!(n+p+1)} = x^{n+1} \sum_{p=0}^{\infty} \frac{(-1)^p x^p}{p!(n+p+1)}.$$

Here $\int_0^x t^{n+p}dt = x^{n+p+1}/(n+p+1)$. On every bounded integration segment the exponential series is uniformly absolutely convergent, which justifies the interchange. For $x \neq 0$ the absolute ratio of successive terms of the bracketed series is

$$\frac{|x|^{p+1}}{(p+1)!\,(n+p+2)} \cdot \frac{p!\,(n+p+1)}{|x|^p} = |x| \frac{n+p+1}{(p+1)(n+p+2)} \longrightarrow 0 \quad (p \to \infty).$$

Thus, by the ratio test, the resulting power series converges for every real or complex x. For nonintegral n with $\Re n > -1$ the same coefficient formula holds with a consistent branch of x^{n+1}; it is then a power series times that prefactor rather than an ordinary Maclaurin series.

1.3.4 Develop a series expansion of $y = \sinh^{-1} x$ (that is, $\sinh y = x$) in powers of x by

(a) inversion of the series for $\sinh y$,

(b) a direct Maclaurin expansion.

Solution.

(a) Because sinh is odd, so is its inverse, and $\sinh y = y + O(y^3)$ gives $y = x + O(x^3)$. Hence seek $y = x + ax^3 + bx^5 + dx^7 + O(x^9)$ in $x = \sinh y = y + y^3/6 + y^5/120 + y^7/5040 + O(y^9)$. Since

$$y^3 = x^3 + 3ax^5 + (3b + 3a^2)x^7 + O(x^9), \quad y^5 = x^5 + 5ax^7 + O(x^9), \quad y^7 = x^7 + O(x^9),$$

the coefficients of x^3, x^5, x^7 in $y + y^3/6 + y^5/120 + y^7/5040$ give

$$a + \frac{1}{6} = 0, \quad b + \frac{a}{2} + \frac{1}{120} = 0, \quad d + \frac{b + a^2}{2} + \frac{a}{24} + \frac{1}{5040} = 0.$$

Hence $a = -1/6$, $b = -a/2 - 1/120 = 1/12 - 1/120 = 3/40$, and, with $b + a^2 = 3/40 + 1/36 = 37/360$,

$$d = -\frac{37}{720} + \frac{1}{144} - \frac{1}{5040} = \frac{-259 + 35 - 1}{5040} = -\frac{225}{5040} = -\frac{5}{112}.$$

Thus $y = x - x^3/6 + 3x^5/40 - 5x^7/112 + O(x^9)$.

(b) The Maclaurin coefficients are $y^{(k)}(0)/k!$. Differentiating $\sinh y = x$ gives $\cosh y\, y' = 1$. As $\cosh y = \sqrt{1 + \sinh^2 y} = \sqrt{1 + x^2}$ on the branch through zero, the binomial theorem gives

$$y' = (1 + x^2)^{-1/2} = \sum_{r=0}^{\infty} \binom{-1/2}{r} x^{2r} = \sum_{r=0}^{\infty} (-1)^r \frac{\binom{2r}{r}}{4^r} x^{2r},$$

because, using $(2r-1)!! = (2r)!/(2^r r!)$,

$$\binom{-1/2}{r} = \frac{(-\frac{1}{2})(-\frac{3}{2}) \cdots (-\frac{2r-1}{2})}{r!} = (-1)^r \frac{(2r-1)!!}{2^r r!} = (-1)^r \frac{(2r)!}{4^r (r!)^2}.$$

Hence $y^{(2r+1)}(0) = (2r)!\,(-1)^r\binom{2r}{r}/4^r = (-1)^r[(2r-1)!!]^2 = 1, -1, 9, -225, \ldots$, while the even-order derivatives vanish because y is odd. Dividing by $(2r+1)!$ (equivalently, integrating the series for y' termwise with $y(0) = 0$) produces the full Maclaurin series

$$\boxed{\operatorname{arsinh} x = \sum_{r=0}^{\infty}(-1)^r\frac{\binom{2r}{r}}{4^r(2r+1)}x^{2r+1} = x - \frac{x^3}{6} + \frac{3x^5}{40} - \frac{5x^7}{112} + \cdots},$$

in agreement with (a): $9/5! = 3/40$ and $225/7! = 5/112$. Its radius is 1 (branch points at $\pm i$): the coefficient magnitudes are $[(2r-1)!!]^2/(2r+1)!$, so the ratio of successive ones is $(2r+1)^2\,(2r+1)!/(2r+3)! = (2r+1)^2/[(2r+2)(2r+3)] \to 1$. At the real endpoints $x = \pm 1$ the coefficient magnitudes are $O(r^{-3/2})$, because $\binom{2r}{r}/4^r \sim 1/\sqrt{\pi r}$ (Stirling's formula, as in Exercise 1.1.10), so the series also converges absolutely there.

1.3.5 Show that for integral $n \geq 0$ and $|x| < 1$, $\dfrac{1}{(1-x)^{n+1}} = \displaystyle\sum_{m=n}^{\infty}\binom{m}{n}x^{m-n}$.

Solution.

For $|x| < 1$, differentiate the geometric series n times. Each differentiation turns $k!(1-x)^{-k-1}$ into $(k+1)!(1-x)^{-k-2}$, and $d^n x^m/dx^n = m(m-1)\cdots(m-n+1)x^{m-n} = \frac{m!}{(m-n)!}x^{m-n}$ for $m \geq n$, while it vanishes for $m < n$:

$$\frac{d^n}{dx^n}\frac{1}{1-x} = \frac{n!}{(1-x)^{n+1}}, \qquad \frac{d^n}{dx^n}\sum_{m=0}^{\infty}x^m = \sum_{m=n}^{\infty}\frac{m!}{(m-n)!}x^{m-n}.$$

Termwise differentiation is valid inside the radius of convergence, as established in Exercise 1.2.16. Divide by $n!$, using $m!/[(m-n)!\,n!] = \binom{m}{n}$, to obtain

$$\frac{1}{(1-x)^{n+1}} = \sum_{m=n}^{\infty}\binom{m}{n}x^{m-n}.$$

The case $n = 0$ is the geometric series itself. The expansion's ordinary convergence domain is $|x| < 1$.

1.3.6 Show that $(1+x)^{-m/2} = \displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{(m+2n-2)!!}{2^n\,n!\,(m-2)!!}\,x^n$, for $m = 1, 2, 3, \ldots$ and $|x| < 1$, with $(-1)!! = 0!! = 1$.

Solution.

The n factors in the numerator of the generalized binomial coefficient are $-m/2-k = -(m+2k)/2$, $k = 0, \ldots, n-1$, so

$$\binom{-m/2}{n} = \frac{(-m/2)(-m/2-1)\cdots(-m/2-n+1)}{n!} = (-1)^n\frac{m(m+2)\cdots(m+2n-2)}{2^n n!}.$$

Since $(m+2n-2)!! = (m+2n-2)(m+2n-4)\cdots m\cdot(m-2)!!$, canceling the final factors $(m-2)!!$ of the double factorial gives $m(m+2)\cdots(m+2n-2) = (m+2n-2)!!/(m-2)!!$. Substitute into $(1+x)^{-m/2} = \sum_{n\geq 0}\binom{-m/2}{n}x^n$ to obtain the requested formula for $|x| < 1$. The conventions $(-1)!! = 0!! = 1$ cover $m = 1, 2$, and an empty product is 1, so the $n = 0$ term is correctly equal to 1 in every case.

1.3.7 Using binomial expansions, compare the three Doppler shift formulas:

(a) $\nu' = \nu\left(1 \mp \dfrac{v}{c}\right)^{-1}$ moving source;

(b) $\nu' = \nu\left(1 \pm \dfrac{v}{c}\right)$ moving observer;

(c) $\nu' = \nu\left(1 \pm \dfrac{v}{c}\right)\left(1 - \dfrac{v^2}{c^2}\right)^{-1/2}$ relativistic.

Note. The relativistic formula agrees with the classical formulas if terms of order v^2/c^2 can be neglected.

Solution.

Put $\beta = v/c$ and use a single sign $\sigma = +1$ for the upper signs and $\sigma = -1$ for the lower signs, so $\sigma^2 = 1$. With the geometric series in (a) and the binomial series $(1-\beta^2)^{-1/2} = 1 + \frac{1}{2}\beta^2 + \frac{3}{8}\beta^4 + \cdots$ in (c),

$$\begin{aligned}
&\text{(a)} \quad \nu'/\nu = (1-\sigma\beta)^{-1} = 1 + \sigma\beta + \beta^2 + \sigma\beta^3 + O(\beta^4),\\
&\text{(b)} \quad \nu'/\nu = 1 + \sigma\beta,\\
&\text{(c)} \quad \nu'/\nu = (1+\sigma\beta)(1-\beta^2)^{-1/2}\\
&\qquad\qquad = (1+\sigma\beta)(1 + \tfrac{1}{2}\beta^2 + O(\beta^4)) = 1 + \sigma\beta + \tfrac{1}{2}\beta^2 + \tfrac{1}{2}\sigma\beta^3 + O(\beta^4).
\end{aligned}$$

All three share $1 + \sigma\beta$ at first order. Their second-order coefficients are respectively $1, 0, 1/2$, so distinguishing the formulas requires retaining terms of order v^2/c^2.

1.3.8 In the theory of general relativity there are various ways of relating (defining) a velocity of recession of a galaxy to its red shift, δ. Milne's model (kinematic relativity) gives

(a) $v_1 = c\delta\left(1 + \dfrac{\delta}{2}\right)$,

(b) $v_2 = c\delta\left(1 + \dfrac{\delta}{2}\right)(1+\delta)^{-2}$,

(c) $1 + \delta = \left[\dfrac{1 + v_3/c}{1 - v_3/c}\right]^{1/2}$.

1. Show that for $\delta \ll 1$ (and $v_3/c \ll 1$), all three formulas reduce to $v = c\delta$.
2. Compare the three velocities through terms of order δ^2.

Note. In special relativity (with δ replaced by z), the ratio of observed wavelength λ to emitted wavelength λ_0 is given by

$$\frac{\lambda}{\lambda_0} = 1 + z = \left(\frac{c+v}{c-v}\right)^{1/2}.$$

Solution.

(a) The first formula already reads $v_1/c = \delta + \delta^2/2$. (b) Since $(1+\delta)^{-2} = 1 - 2\delta + 3\delta^2 + O(\delta^3)$ (binomial series),

$$\frac{v_2}{c} = (\delta + \tfrac{1}{2}\delta^2)(1 - 2\delta + 3\delta^2 + \cdots) = \delta - 2\delta^2 + \tfrac{1}{2}\delta^2 + O(\delta^3) = \delta - \frac{3}{2}\delta^2 + O(\delta^3).$$

(c) Squaring the third formula and multiplying by $1-\beta_3$, where $\beta_3 = v_3/c$, gives $(1+\delta)^2(1-\beta_3) = 1+\beta_3$, i.e., $(1+\delta)^2 - 1 = \beta_3[(1+\delta)^2+1]$. Hence

$$\beta_3 = \frac{2\delta+\delta^2}{2+2\delta+\delta^2} = \frac{\delta+\delta^2/2}{1+\delta+\delta^2/2} = (\delta+\tfrac{1}{2}\delta^2)(1-\delta+O(\delta^2)) = \delta - \frac{1}{2}\delta^2 + O(\delta^3).$$

1. All three therefore reduce to $v = c\delta$ at first order. *2.* Through second order their coefficients of $c\delta^2$ are $+1/2, -3/2, -1/2$; for small positive nonzero δ, $v_1 > v_3 > v_2$.

1.3.9 The relativistic sum w of two velocities u and v in the same direction is given by

$$\frac{w}{c} = \frac{u/c + v/c}{1 + uv/c^2}.$$

If

$$\frac{v}{c} = \frac{u}{c} = 1 - \alpha,$$

where $0 \le \alpha \le 1$, find w/c in powers of α through terms in α^3.

Solution.

Substitution first gives an exact rational function:

$$\frac{w}{c} = \frac{2(1-\alpha)}{1+(1-\alpha)^2} = 1 - \frac{\alpha^2}{2-2\alpha+\alpha^2},$$

because $1+(1-\alpha)^2 - 2(1-\alpha) = [1-(1-\alpha)]^2 = \alpha^2$ and $1+(1-\alpha)^2 = 2-2\alpha+\alpha^2$. Near $\alpha = 0$, with $1/(1-\eta) = 1+\eta+O(\eta^2)$ for $\eta = \alpha - \alpha^2/2$,

$$\frac{1}{2-2\alpha+\alpha^2} = \frac{1}{2}\frac{1}{1-\alpha+\alpha^2/2} = \frac{1}{2}[1+\alpha+O(\alpha^2)].$$

Multiplying by α^2 gives $\alpha^2/2 + \alpha^3/2 + O(\alpha^4)$. Therefore

$$\boxed{w/c = 1 - \frac{\alpha^2}{2} - \frac{\alpha^3}{2} + O(\alpha^4)}.$$

There is no linear term. This is an expansion about $\alpha = 0$, not an assertion that truncation is accurate throughout $0 \le \alpha \le 1$. The unexpanded expression gives $w/c = 1$ at $\alpha = 0$ and 0 at $\alpha = 1$, consistent with the limiting input velocities.

1.3.10 The displacement x of a particle of rest mass m_0, resulting from a constant force $m_0 g$ along the x-axis, is

$$x = \frac{c^2}{g}\left\{\left[1+\left(g\frac{t}{c}\right)^2\right]^{1/2} - 1\right\},$$

including relativistic effects. Find the displacement x as a power series in time t. Compare with the classical result,

$$x = \frac{1}{2}gt^2.$$

Solution.

Set $q = (gt/c)^2$ and use the binomial series $(1+q)^{1/2} = 1 + q/2 - q^2/8 + q^3/16 - 5q^4/128 + \cdots$, whose coefficients are

$$\binom{1/2}{2} = \frac{\frac{1}{2}(-\frac{1}{2})}{2!} = -\frac{1}{8}, \quad \binom{1/2}{3} = \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{3!} = \frac{1}{16}, \quad \binom{1/2}{4} = \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{4!} = -\frac{5}{128}.$$

After subtraction of 1 and multiplication by c^2/g, each term $\binom{1/2}{k} q^k$ becomes $\binom{1/2}{k} g^{2k-1} t^{2k}/c^{2k-2}$:

$$\boxed{x = \frac{gt^2}{2} - \frac{g^3 t^4}{8c^2} + \frac{g^5 t^6}{16c^4} - \frac{5g^7 t^8}{128 c^6} + \cdots}.$$

The first term is the classical displacement. For $|gt/c| \ll 1$ the leading relative correction is $[-g^3t^4/(8c^2)]/(gt^2/2) = -g^2t^2/(4c^2)$. The binomial series converges for $|gt/c| < 1$, and also at the real endpoints $|gt/c| = 1$, where its coefficients are absolutely summable. To see this, note that for $k \geq 1$ the k numerator factors $\frac{1}{2} - j$ $(j = 0, \ldots, k-1)$ of $\binom{1/2}{k}$ are $\frac{1}{2}$ followed by $k-1$ negative ones, so, with $(2k-3)!! = (2k-1)!!/(2k-1)$ (where $(-1)!! = 1$) and $(2k-1)!! = (2k)!/(2^k k!)$ as in Exercise 1.3.4,

$$\binom{1/2}{k} = \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})\cdots(-\frac{2k-3}{2})}{k!} = (-1)^{k-1}\frac{(2k-3)!!}{2^k k!} = (-1)^{k-1}\frac{\binom{2k}{k}}{4^k(2k-1)}.$$

Since $\binom{2k}{k}/4^k \sim 1/\sqrt{\pi k}$ (Exercise 1.1.10), these coefficients are $O(k^{-3/2})$, hence absolutely summable by comparison with $\sum k^{-3/2}$. The original expression, unlike the small-time truncation, also describes larger times.

1.3.11 By use of Dirac's relativistic theory, the fine structure formula of atomic spectroscopy is given by

$$E = mc^2\left[1 + \frac{\gamma^2}{(s+n-|k|)^2}\right]^{-1/2},$$

where

$$s = (|k|^2 - \gamma^2)^{1/2}, \quad k = \pm 1, \pm 2, \pm 3, \ldots.$$

Take the principal quantum number $n > 0$ and $|\gamma| < |k|$. Expand in powers of γ^2 through order γ^4 ($\gamma = Ze^2/(4\pi\epsilon_0\hbar c)$, with Z the atomic number). This expansion is useful in comparing the predictions of the Dirac electron theory with those of a relativistic Schrödinger electron theory. Experimental results support the Dirac theory.

Solution.

In the physical definition the parameter is $\gamma = Ze^2/(4\pi\epsilon_0\hbar c) = Z\alpha$, not $\gamma^2 = Ze^2/(4\pi\epsilon_0\hbar c)$. Here $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$ is the fine-structure constant. The correction follows by matching the given denominator with the standard Dirac formula containing $(Z\alpha)^2$; see Kramida's NIST compilation, Sec. 2, Eq. (3). For the algebra set $K = |k| > 0$ and $q = \gamma^2$. With $\sqrt{1-w} = 1 - w/2 - w^2/8 + O(w^3)$ for $w = q/K^2$,

$$s = K\sqrt{1 - q/K^2} = K - \frac{q}{2K} - \frac{q^2}{8K^3} + O(q^3), \quad s + n - K = n - \frac{q}{2K} + O(q^2).$$

Consequently, using $(1-\varepsilon)^{-2} = 1 + 2\varepsilon + O(\varepsilon^2)$ with $\varepsilon = q/(2Kn) + O(q^2)$,

$$\frac{q}{(s+n-K)^2} = \frac{q}{n^2}\left(1 - \frac{q}{2Kn} + O(q^2)\right)^{-2} = \frac{q}{n^2} + \frac{q^2}{Kn^3} + O(q^3).$$

Now $E/mc^2 = (1+u)^{-1/2}$ with $u = q/(s+n-K)^2$ as just expanded, so $u^2 = q^2/n^4 + O(q^3)$, and $(1+u)^{-1/2} = 1 - u/2 + 3u^2/8 + O(u^3)$ gives

$$\frac{E}{mc^2} = 1 - \frac{q}{2n^2} - \frac{q^2}{2Kn^3} + \frac{3q^2}{8n^4} + O(q^3),$$

and $-q^2/(2Kn^3) + 3q^2/(8n^4) = -[q^2/(2n^4)](n/K - 3/4)$. Thus

$$\boxed{E = mc^2\left[1 - \frac{\gamma^2}{2n^2} - \frac{\gamma^4}{2n^4}\left(\frac{n}{|k|} - \frac{3}{4}\right) + O(\gamma^6)\right].}$$

The expansion is for small γ at fixed n. The qualification in the statement is needed because $|\gamma| < |k|$ keeps s real and $n > 0$ keeps the denominator $s + n - |k| \to n$ away from zero as $\gamma \to 0$; both hold for the physical states, which have $n \geq |k| \geq 1$ and $|\gamma| = Z\alpha < 1$ for every known element. For $n = |k| = 1$ the exact expression reduces to $mc^2[1 + \gamma^2/(1-\gamma^2)]^{-1/2} = mc^2\sqrt{1-\gamma^2}$, whose expansion $mc^2(1 - \gamma^2/2 - \gamma^4/8 + \cdots)$ checks the result. The γ^2 term $-mc^2\gamma^2/(2n^2)$ is the Bohr energy $-mc^2(Z\alpha)^2/(2n^2)$ exactly when $\gamma = Z\alpha$, which confirms the corrected definition of γ.

1.3.12 In a head-on proton-proton collision, the ratio of the kinetic energy in the center of mass system to the incident kinetic energy is

$$R = \left[\sqrt{2mc^2(E_k + 2mc^2)} - 2mc^2\right]/E_k.$$

Find the value of this ratio of kinetic energies for

(a) $E_k \ll mc^2$ (nonrelativistic),

(b) $E_k \gg mc^2$ (extreme-relativistic).

ANS. (a) $\frac{1}{2}$, (b) 0. The latter answer is a sort of law of diminishing returns for high-energy particle accelerators (with stationary targets).

Solution.

Let $q = E_k/(mc^2) > 0$. Since $\sqrt{2mc^2(E_k + 2mc^2)} = mc^2\sqrt{4+2q}$, dividing numerator and denominator by mc^2 gives the first form below; rationalizing (multiplying above and below by $\sqrt{4+2q}+2$) removes the cancellation at $q = 0$:

$$R = \frac{\sqrt{4+2q} - 2}{q} = \frac{2}{\sqrt{4+2q}+2} = \frac{1}{\sqrt{1+q/2}+1}.$$

(a) For $q \ll 1$, $\sqrt{1+q/2} = 1 + q/4 + O(q^2)$, giving $R = 1/[2 + q/4 + O(q^2)] = \frac{1}{2}(1 + q/8 + O(q^2))^{-1} = 1/2 - q/16 + O(q^2)$. Thus the nonrelativistic limit is 1/2.

(b) For $q \gg 1$, write $\sqrt{4+2q} = \sqrt{2q}\sqrt{1+2/q}$ in the original form and use $\sqrt{1+2/q} = 1 + 1/q + O(q^{-2})$:

$$R = \sqrt{\frac{2}{q}}\sqrt{1 + \frac{2}{q}} - \frac{2}{q} = \sqrt{\frac{2}{q}} - \frac{2}{q} + O(q^{-3/2}) \longrightarrow 0.$$

The positive leading term also specifies how the ratio approaches zero.

1.3.13 With binomial expansions

$$\frac{x}{1-x} = \sum_{n=1}^{\infty} x^n, \qquad \frac{x}{x-1} = \frac{1}{1-x^{-1}} = \sum_{n=0}^{\infty} x^{-n}.$$

Adding these two series yields $\sum_{n=-\infty}^{\infty} x^n = 0$.

Hopefully, we can agree that this is nonsense, but what has gone wrong?

Solution.

The first equality uses a geometric series with ratio x, and thus requires $|x| < 1$. The second uses ratio x^{-1} and requires $|x| > 1$. There is no common point where the two infinite series both converge, so adding their sums termwise is unjustified. Indeed, for any nonzero x, the positive-power tail fails to converge when $|x| \geq 1$, while the negative-power tail fails when $|x| \leq 1$. On $|x| = 1$ neither tail's terms tend to zero. At $x = 0$ negative powers are undefined. Thus the proposed bilateral series has no ordinary convergence domain, although the two rational functions algebraically cancel wherever they are defined.

1.3.14

(a) Planck's theory of quantized oscillators leads to an average energy

$$\langle \varepsilon \rangle = \frac{\displaystyle\sum_{n=1}^{\infty} n\varepsilon_0 \exp(-n\varepsilon_0/kT)}{\displaystyle\sum_{n=0}^{\infty} \exp(-n\varepsilon_0/kT)},$$

where ε_0 is a fixed energy. Identify the numerator and denominator as binomial expansions and show that the ratio is

$$\langle \varepsilon \rangle = \frac{\varepsilon_0}{\exp(\varepsilon_0/kT) - 1}.$$

(b) Show that the $\langle \varepsilon \rangle$ of part (a) reduces to kT, the classical result, for $kT \gg \varepsilon_0$.

Solution.

(a) Put $q = e^{-\varepsilon_0/(kT)}$, so $0 < q < 1$ for $\varepsilon_0, kT > 0$. Both sums are binomial expansions: the denominator is $(1-q)^{-1} = \sum_{n\geq 0} q^n$, and the numerator is $\varepsilon_0 q$ times $(1-q)^{-2} = \sum_{n\geq 0}(n+1)q^n$ (Exercise 1.3.5 with $n = 1$; equivalently, apply $q\,d/dq$ to the geometric series):

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}, \qquad \sum_{n=1}^{\infty} nq^n = q\sum_{n=0}^{\infty}(n+1)q^n = \frac{q}{(1-q)^2}.$$

Their ratio, including the energy factor, is

$$\langle \varepsilon \rangle = \varepsilon_0 \frac{q/(1-q)^2}{1/(1-q)} = \frac{\varepsilon_0 q}{1-q} = \frac{\varepsilon_0}{e^{\varepsilon_0/(kT)} - 1}.$$

(b) With $z = \varepsilon_0/(kT) \ll 1$, write $\langle \varepsilon \rangle = kT\, z/(e^z - 1)$ and $e^z - 1 = z(1 + z/2 + z^2/6 + z^3/24 + \cdots)$. The reciprocal series $1 + az + bz^2 + cz^3 + \cdots$ of the bracket follows from setting $(1 + z/2 + z^2/6 +$

$z^3/24+\cdots)(1+az+bz^2+cz^3+\cdots)=1$: $a+1/2=0$, $b+a/2+1/6=0$, and $c+b/2+a/6+1/24=0$, so $a=-1/2$, $b=1/12$, and $c=0$. Hence

$$\langle\varepsilon\rangle = kT\left(1-\frac{z}{2}+\frac{z^2}{12}+O(z^4)\right) = kT-\frac{\varepsilon_0}{2}+O(\varepsilon_0^2/(kT)).$$

Its leading high-temperature value is kT. The Instructor's Manual, p. 14, gives a plus sign in the linear term of $z/(e^z-1)$ and a positive constant correction to the energy. The expansion above fixes the first correction as $-\varepsilon_0/2$.

1.3.15 Expand by the binomial theorem and integrate term by term to obtain the Gregory series for $y=\tan^{-1}x$ (note $\tan y = x$):

$$\tan^{-1}x = \int_0^x \frac{dt}{1+t^2} = \int_0^x \left\{1-t^2+t^4-t^6+\cdots\right\}dt$$
$$= \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1}, \quad -1\le x\le 1.$$

Solution.

For $|x|<1$ the expansion $(1+t^2)^{-1}=\sum_{n=0}^{\infty}(-1)^n t^{2n}$ is uniformly convergent on the segment from 0 to x (Weierstrass M test with $M_n=|x|^{2n}$). Termwise integration, with $\int_0^x t^{2n}dt = x^{2n+1}/(2n+1)$, therefore gives

$$\arctan x = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1}.$$

At $x=1$, the finite geometric remainder after integration is bounded by $1/(2N+3)$ as in Exercise 1.3.2, proving convergence to $\pi/4$. Oddness gives convergence to $-\pi/4$ at $x=-1$. At either endpoint convergence is conditional; in the interior it is absolute. For $|x|>1$, $|x|^{2n+1}/(2n+1)$ does not tend to zero, so no further real points belong to the convergence interval.

1.3.16 The Klein-Nishina formula for the scattering of photons by electrons contains a term of the form

$$f(\varepsilon)=\frac{(1+\varepsilon)}{\varepsilon^2}\left[\frac{2+2\varepsilon}{1+2\varepsilon}-\frac{\ln(1+2\varepsilon)}{\varepsilon}\right].$$

Here $\varepsilon = h\nu/mc^2$, the ratio of the photon energy to the electron rest mass energy. Find $\lim_{\varepsilon\to 0} f(\varepsilon)$.

ANS. $\dfrac{4}{3}$.

Solution.

The prefactor contains ε^{-2}, so expand the bracket at least through ε^2. Use the geometric series for $1/(1+2\varepsilon)$ and $\ln(1+u)=u-u^2/2+u^3/3-u^4/4+\cdots$ with $u=2\varepsilon$:

$$\frac{2+2\varepsilon}{1+2\varepsilon} = 2(1+\varepsilon)(1-2\varepsilon+4\varepsilon^2-8\varepsilon^3+\cdots) = 2(1-\varepsilon+2\varepsilon^2-4\varepsilon^3+\cdots)$$
$$= 2-2\varepsilon+4\varepsilon^2-8\varepsilon^3+O(\varepsilon^4),$$
$$\frac{\ln(1+2\varepsilon)}{\varepsilon} = \frac{2\varepsilon-2\varepsilon^2+\frac{8}{3}\varepsilon^3-4\varepsilon^4+\cdots}{\varepsilon} = 2-2\varepsilon+\frac{8}{3}\varepsilon^2-4\varepsilon^3+O(\varepsilon^4).$$

The constant and linear terms cancel, leaving $(4-\frac{8}{3})\varepsilon^2+(-8+4)\varepsilon^3+O(\varepsilon^4)=\frac{4}{3}\varepsilon^2-4\varepsilon^3+O(\varepsilon^4)$. Multiplication by $(1+\varepsilon)/\varepsilon^2$ now gives

$$f(\varepsilon)=(1+\varepsilon)\Big(\frac{4}{3}-4\varepsilon\Big)+O(\varepsilon^2)=\frac{4}{3}-\frac{8}{3}\varepsilon+O(\varepsilon^2), \qquad \boxed{\lim_{\varepsilon\to 0} f(\varepsilon)=\frac{4}{3}}.$$

1.3.17 The behavior of a neutron losing energy by colliding elastically with nuclei of mass A is described by a parameter ξ_1,

$$\xi_1=1+\frac{(A-1)^2}{2A}\ln\frac{A-1}{A+1}.$$

An approximation, good for large A, is

$$\xi_2=\frac{2}{A+\frac{2}{3}}.$$

Expand ξ_1 and ξ_2 in powers of A^{-1}. Show that ξ_2 agrees with ξ_1 through $(A^{-1})^2$. Find the difference in the coefficients of the $(A^{-1})^3$ term.

Solution.

Set $x=1/A$, with $|x|<1$. Then

$$\ln\frac{A-1}{A+1}=\ln(1-x)-\ln(1+x)=-2\left(x+\frac{x^3}{3}+\frac{x^5}{5}+\cdots\right), \qquad \frac{(A-1)^2}{2A}=\frac{1-2x+x^2}{2x}.$$

The logarithm series is that of Exercise 1.6.1 with $x\to -x$, so the product of the two factors is

$$\frac{1-2x+x^2}{2x}\cdot(-2x)\Big(1+\frac{x^2}{3}+\frac{x^4}{5}+\cdots\Big)=-(1-2x+x^2)\Big(1+\frac{x^2}{3}+\frac{x^4}{5}+\cdots\Big).$$

Retain through x^4 in this product (the x^{-1} prefactor requires the logarithm through x^5):

$$(1-2x+x^2)\Big(1+\frac{x^2}{3}+\frac{x^4}{5}+\cdots\Big)=1-2x+\frac{4}{3}x^2-\frac{2}{3}x^3+\Big(\frac{1}{3}+\frac{1}{5}\Big)x^4+O(x^5),$$

with $\frac{1}{3}+\frac{1}{5}=\frac{8}{15}$. Expanding $\xi_2=2x/(1+2x/3)$ by the geometric series,

$$\xi_1=1-\Big(1-2x+\frac{4}{3}x^2-\frac{2}{3}x^3+\frac{8}{15}x^4+O(x^5)\Big)=2x-\frac{4}{3}x^2+\frac{2}{3}x^3-\frac{8}{15}x^4+O(x^5),$$
$$\xi_2=2x\Big(1-\frac{2x}{3}+\frac{4x^2}{9}-\frac{8x^3}{27}+\cdots\Big)=2x-\frac{4}{3}x^2+\frac{8}{9}x^3-\frac{16}{27}x^4+O(x^5).$$

Thus they agree through A^{-2}, and $\xi_2-\xi_1=(8/9-2/3)A^{-3}+O(A^{-4})=\boxed{(2/9)A^{-3}+O(A^{-4})}$. The Instructor's Manual, p. 14, prints $4x^3/3$ in its intermediate expansion of ξ_1. The coefficient is $2/3$, as follows from the product above; its final coefficient difference $2/9$ is correct.

1.3.18 Show that each of these two integrals equals Catalan's constant:

(a) $\displaystyle\int_0^1 \arctan t\,\frac{dt}{t}$, (b) $\displaystyle-\int_0^1 \ln x\,\frac{dx}{1+x^2}$.

Note. The definition and numerical computation of Catalan's constant was addressed in Exercise 1.1.12.

Solution.

(a) Divide the Gregory series (Exercise 1.3.15) by t and integrate:

$$\int_0^1 \frac{\arctan t}{t} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \int_0^1 t^{2n} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} = G,$$

Catalan's constant $G = \beta(2)$ of Exercise 1.1.12. The apparent singularity at zero is removable since $\arctan t/t \to 1$. To justify the termwise integration, integrate the finite identity of Exercise 1.3.2 from 0 to t and divide by t: $\arctan t/t = \sum_{n=0}^{N}(-1)^n t^{2n}/(2n+1) + r_N(t)$ with $|r_N(t)| \le t^{-1}\int_0^t s^{2N+2} ds = t^{2N+2}/(2N+3)$. The integrated remainder is at most $1/(2N+3)^2 \to 0$.

(b) Similarly expand $1/(1+x^2)$ and use integration by parts with $u = \ln x$, $dv = x^{2n}dx$:

$$\int_0^1 x^{2n} \ln x \, dx = \left[\frac{x^{2n+1}\ln x}{2n+1}\right]_0^1 - \frac{1}{2n+1}\int_0^1 x^{2n} dx = -\frac{1}{(2n+1)^2}.$$

The boundary term vanishes: $\ln 1 = 0$, and $x^{2n+1}\ln x \to 0$ as $x \to 0^+$ (l'Hôpital's rule applied to $\ln x / x^{-(2n+1)}$). Therefore $-\int_0^1 \ln x/(1+x^2)dx = \sum_{n\ge 0}(-1)^n/(2n+1)^2 = G$. This termwise integration is justified in the same way: multiplying the finite identity of Exercise 1.3.2 by $-\ln x \ge 0$ and integrating, the remainder is at most $\int_0^1 x^{2N+2}(-\ln x)\,dx = 1/(2N+3)^2 \to 0$, by the integral just computed.

1.4 Mathematical Induction

1.4.1 Show that $\displaystyle\sum_{j=1}^{n} j^4 = \frac{n}{30}(2n+1)(n+1)(3n^2+3n-1)$.

Solution.

Let $F(n) = n(n+1)(2n+1)(3n^2+3n-1)/30$. Since $n(n+1)(2n+1) = 2n^3+3n^2+n$, multiplying the factors gives

$$F(n) = \frac{6n^5+15n^4+10n^3-n}{30} = \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n.$$

The base case is $F(1) = \frac{1}{5}+\frac{1}{2}+\frac{1}{3}-\frac{1}{30} = \frac{6+15+10-1}{30} = 1$. For the inductive step use the binomial expansions of $(n+1)^j$ in $(n+1)^j - n^j$:

$$\begin{aligned} F(n+1) - F(n) &= \tfrac{1}{5}(5n^4+10n^3+10n^2+5n+1) \\ &\quad + \tfrac{1}{2}(4n^3+6n^2+4n+1) + \tfrac{1}{3}(3n^2+3n+1) - \tfrac{1}{30} \\ &= n^4 + (2+2)n^3 + (2+3+1)n^2 + (1+2+1)n + \left(\tfrac{1}{5}+\tfrac{1}{2}+\tfrac{1}{3}-\tfrac{1}{30}\right) \\ &= n^4 + 4n^3 + 6n^2 + 4n + 1 = (n+1)^4. \end{aligned}$$

If $\sum_{j=1}^{n} j^4 = F(n)$, adding $(n+1)^4$ consequently gives $F(n+1)$. The formula follows for every positive integer n.

1.4.2 Prove the Leibniz formula for the repeated differentiation of a product:

$$\left(\frac{d}{dx}\right)^n \left[f(x)g(x)\right] = \sum_{j=0}^{n} \binom{n}{j} \left[\left(\frac{d}{dx}\right)^j f(x)\right] \left[\left(\frac{d}{dx}\right)^{n-j} g(x)\right].$$

Solution.

For $n = 0$ the identity is $fg = fg$, and for $n = 1$ it is the product rule. Assume it holds at order n, with the needed derivatives existing. Differentiating each term gives

$$(fg)^{(n+1)} = \sum_{j=0}^{n} \binom{n}{j} f^{(j+1)} g^{(n-j)} + \sum_{j=0}^{n} \binom{n}{j} f^{(j)} g^{(n+1-j)}.$$

In the first sum replace $j + 1$ by r, and in the second write r for j:

$$(fg)^{(n+1)} = \sum_{r=1}^{n+1} \binom{n}{r-1} f^{(r)} g^{(n+1-r)} + \sum_{r=0}^{n} \binom{n}{r} f^{(r)} g^{(n+1-r)}.$$

The boundary terms $f^{(n+1)}g$ ($r = n + 1$, first sum only) and $fg^{(n+1)}$ ($r = 0$, second sum only) have coefficient 1, and for $1 \le r \le n$ the coefficient of $f^{(r)}g^{(n+1-r)}$ is, over the common denominator $r!(n + 1 - r)!$,

$$\binom{n}{r-1} + \binom{n}{r} = \frac{n!\, r + n!\, (n + 1 - r)}{r!(n + 1 - r)!} = \frac{(n + 1)!}{r!(n + 1 - r)!} = \binom{n+1}{r}.$$

The boundary coefficients also equal the corresponding binomial coefficients, $\binom{n+1}{n+1} = \binom{n+1}{0} = 1$. This is exactly the formula at order $n + 1$, completing induction.

1.5 Operations on Series Expansions of Functions

1.5.1 Using a partial fraction expansion, show that for $0 < x < 1$,

$$\int_{-x}^{x} \frac{dt}{1 - t^2} = \ln\left(\frac{1 + x}{1 - x}\right).$$

Solution.

Solve $1/(1 - t^2) = A/(1 - t) + B/(1 + t)$ by multiplying by $1 - t^2$: $1 = A(1 + t) + B(1 - t)$. Constant and linear coefficients give $A + B = 1$ and $A - B = 0$, hence $A = B = 1/2$. Therefore

$$\begin{aligned}\int_{-x}^{x} \frac{dt}{1 - t^2} &= \frac{1}{2}[-\ln(1 - t) + \ln(1 + t)]_{-x}^{x} \\ &= \frac{1}{2}\left[\ln\frac{1 + x}{1 - x} - \ln\frac{1 - x}{1 + x}\right] = \ln\frac{1 + x}{1 - x}.\end{aligned}$$

For $0 < x < 1$ the denominators stay nonzero and all logarithms have positive arguments, so no principal-value or branch convention is needed.

1.5.2 Prove the partial fraction expansion

$$\frac{1}{n(n+1)\cdots(n+p)} = \frac{1}{p!}\left[\binom{p}{0}\frac{1}{n} - \binom{p}{1}\frac{1}{n+1} + \binom{p}{2}\frac{1}{n+2} - \cdots + (-1)^p\binom{p}{p}\frac{1}{n+p}\right],$$

where p is a positive integer.

Hint. Use mathematical induction. Two binomial coefficient formulas of use here are

$$\frac{p+1}{p+1-j}\binom{p}{j} = \binom{p+1}{j}, \qquad \sum_{j=1}^{p+1}(-1)^{j-1}\binom{p+1}{j} = 1.$$

Solution.

For an induction convenient here, let $F_p(n) = 1/[n(n+1)\cdots(n+p)]$. The formula holds trivially for $p = 0$, where it reads $F_0(n) = 1/n$; this is the base case. Directly bringing two fractions to a common denominator gives

$$F_p(n) - F_p(n+1) = \frac{(n+p+1)-n}{n(n+1)\cdots(n+p+1)} = (p+1)F_{p+1}(n).$$

Assume $F_p(n) = \frac{1}{p!}\sum_{j=0}^{p}(-1)^j\binom{p}{j}/(n+j)$. Apply it at both n and $n+1$, subtract, and divide by $p+1$; in the second sum, $\sum_{j=0}^{p}(-1)^j\binom{p}{j}/(n+1+j)$, shift the index $j \to j-1$:

$$F_{p+1}(n) = \frac{1}{(p+1)!}\left[\sum_{j=0}^{p}\frac{(-1)^j\binom{p}{j}}{n+j} - \sum_{j=1}^{p+1}\frac{(-1)^{j-1}\binom{p}{j-1}}{n+j}\right].$$

For $1 \le j \le p$ the numerator of $1/(n+j)$ becomes $(-1)^j\binom{p}{j} - (-1)^{j-1}\binom{p}{j-1} = (-1)^j[\binom{p}{j} + \binom{p}{j-1}] = (-1)^j\binom{p+1}{j}$, by Pascal's rule (proved in Exercise 1.4.2). The endpoint numerators are $\binom{p}{0} = 1 = \binom{p+1}{0}$ for $j = 0$ and $-(-1)^p\binom{p}{p} = (-1)^{p+1}\binom{p+1}{p+1}$ for $j = p+1$, as required. Thus the formula holds at $p+1$ and for all p by induction. All formulas are rational identities away from the poles $n = 0, -1, \ldots$.

1.5.3 The formula for α_p, Eq. (1.26), is a summation of the form $\sum_{n=1}^{\infty} u_n(p)$, with

$$u_n(p) = \frac{1}{n(n+1)\cdots(n+p)}.$$

Applying a partial fraction decomposition to the first and last factors of the denominator, i.e.,

$$\frac{1}{n(n+p)} = \frac{1}{p}\left[\frac{1}{n} - \frac{1}{n+p}\right],$$

show that $u_n(p) = \dfrac{u_n(p-1) - u_{n+1}(p-1)}{p}$ and that $\sum_{n=1}^{\infty} u_n(p) = \dfrac{1}{p\,p!}$.

Hint. It is useful to note that $u_1(p-1) = 1/p!$.

Solution.

Multiplying the supplied partial fraction decomposition by $1/[(n+1)\cdots(n+p-1)]$ gives

$$u_n(p) = \frac{1}{p}\left[\frac{1}{n(n+1)\cdots(n+p-1)} - \frac{1}{(n+1)\cdots(n+p)}\right] = \frac{u_n(p-1) - u_{n+1}(p-1)}{p}.$$

Now sum only to a finite N, so the cancellation is unambiguous: the sum telescopes, since each $u_n(p-1)$ with $2 \le n \le N$ appears once with each sign. With $u_1(p-1) = 1/(1 \cdot 2 \cdots p) = 1/p!$ (the hint) and $u_{N+1}(p-1) = 1/[(N+1)(N+2)\cdots(N+p)]$,

$$\sum_{n=1}^{N} u_n(p) = \frac{1}{p}[u_1(p-1) - u_{N+1}(p-1)] = \frac{1}{p\,p!} - \frac{1}{p(N+1)(N+2)\cdots(N+p)}.$$

For every integer $p \ge 1$ the last term tends to zero. Taking the limit gives $\alpha_p = 1/(p\,p!)$, and the same expression supplies the exact finite-sum remainder.

1.5.4 Proof of Euler transformation: By substituting Eq. (1.88) into Eq. (1.87), verify that Eq. (1.86) is recovered.

Hint. It may help to rearrange the resultant double series so that both indices are summed on the range $(0, \infty)$. Then the summation not containing the coefficients c_j can be recognized as a binomial expansion.

Solution.

In the book's notation put $y = x/(1+x)$ and $a_n = \sum_{j=0}^{n}(-1)^j \binom{n}{j} c_{n-j}$, Eq. (1.88). Substitute this in the transformed series, Eq. (1.87), and set $k = n - j$. Then $n = k + j$, $(-1)^n(-1)^j = (-1)^{k+2j} = (-1)^k$, and $\binom{n}{j} = \binom{k+j}{j}$; as n runs over $0, 1, 2, \ldots$ and j over $0, \ldots, n$, the pair (k, j) runs over all pairs of nonnegative integers:

$$\begin{aligned}\frac{1}{1+x}\sum_{n=0}^{\infty}(-1)^n a_n y^n &= \frac{1}{1+x}\sum_{k=0}^{\infty}\sum_{j=0}^{\infty}(-1)^k c_k \binom{k+j}{j} y^{k+j} \\ &= \frac{1}{1+x}\sum_{k=0}^{\infty}(-1)^k c_k y^k \sum_{j=0}^{\infty}\binom{k+j}{j} y^j.\end{aligned}$$

Exercise 1.3.5, with n replaced by k and $m = k + j$, reads $\sum_{j\ge 0}\binom{k+j}{k} y^j = (1-y)^{-k-1}$ for $|y| < 1$; since $\binom{k+j}{k} = \binom{k+j}{j}$, this identifies the inner sum. Because $1 - y = 1/(1+x)$, its complete prefactor simplifies to

$$\frac{y^k}{(1+x)(1-y)^{k+1}} = \frac{x^k}{(1+x)^k} \cdot \frac{(1+x)^{k+1}}{1+x} = x^k.$$

Thus the result is $\sum_{k\ge 0}(-1)^k c_k x^k$, precisely Eq. (1.86). The exchanges hold wherever the double series converges absolutely; if the original radius is $R > 0$, it suffices that $|y| < 1$ and $|y|/(1-|y|) < R$, because by the same identity the double series of absolute values sums to

$$\sum_{k=0}^{\infty}|c_k|\,|y|^k \sum_{j=0}^{\infty}\binom{k+j}{j}|y|^j = \frac{1}{1-|y|}\sum_{k=0}^{\infty}|c_k|\Big(\frac{|y|}{1-|y|}\Big)^k,$$

and $\sum_k |c_k| t^k$, which has the same radius R as $\sum_k c_k x^k$, converges at $t = |y|/(1-|y|) < R$. Both conditions hold near $x = 0$, since $y = x/(1+x) \to 0$ as $x \to 0$. By the identity theorem for analytic functions (uniqueness of analytic continuation), the equality then persists on any connected open set containing $x = 0$ on which both series converge. It is not a license to rearrange an arbitrary divergent double series.

1.5.5 Carry out the Euler transformation on the series for $\arctan(x)$:

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots .$$

Check your work by computing $\arctan(1) = \pi/4$ and $\arctan(3^{-1/2}) = \pi/6$.

Solution.

Apply Exercise 1.5.4 to $\arctan x/x = \sum_{j\ge0}(-1)^j c_j u^j$ with variable $u = x^2$ and $c_j = 1/(2j+1) = \int_0^1 t^{2j}dt$. By the binomial theorem its difference coefficient is

$$a_n = \int_0^1 \sum_{j=0}^{n}(-1)^j\binom{n}{j}t^{2(n-j)}dt = \int_0^1 (t^2-1)^n dt = (-1)^n I_n, \quad I_n = \int_0^1 (1-t^2)^n dt.$$

For $n \ge 1$, integration by parts (differentiating $(1-t^2)^n$ and integrating 1 to t; the boundary term $[t(1-t^2)^n]_0^1$ vanishes), followed by $t^2 = 1-(1-t^2)$, gives

$$I_n = 2n\int_0^1 t^2(1-t^2)^{n-1}dt = 2n(I_{n-1} - I_n).$$

Thus $(2n+1)I_n = 2nI_{n-1}$, and since $I_0 = 1$,

$$I_n = \frac{2n}{2n+1}I_{n-1} = \frac{2n(2n-2)\cdots 2}{(2n+1)(2n-1)\cdots 3} = \frac{(2n)!!}{(2n+1)!!}.$$

In Eq. (1.87) with x replaced by $u = x^2$, the prefactor is $1/(1+x^2)$, the variable is $q = x^2/(1+x^2)$, and the coefficients are $(-1)^n a_n = I_n$. Multiplying by x, the transformed series is therefore

$$\boxed{\arctan x = \frac{x}{1+x^2}\sum_{n=0}^{\infty}\frac{(2n)!!}{(2n+1)!!}\left(\frac{x^2}{1+x^2}\right)^n}.$$

For every finite real x, $0 \le q < 1$ and $0 < I_n \le 1$ (the integrand $(1-t^2)^n$ lies between 0 and 1), so this series converges. Summing the geometric series under the integral sign (its ratio $q(1-t^2)$ is at most $q < 1$, so it converges uniformly) gives $\sum_n I_n q^n = \int_0^1 [1-q(1-t^2)]^{-1}dt$; since $1-q(1-t^2) = (1+x^2t^2)/(1+x^2)$,

$$\frac{x}{1+x^2}\int_0^1 \frac{dt}{1-q(1-t^2)} = \int_0^1 \frac{x\,dt}{1+x^2t^2} = \arctan x,$$

confirming its full real domain. Keeping $n = 0, \ldots, 39$, with I_n from the recurrence and the sums evaluated to 30 significant digits, gives (rounded to 16 digits)

x	q	computed value
1	1/2	0.7853981633973235
$1/\sqrt{3}$	1/4	0.5235987755982989

These agree with $\pi/4 = 0.78539816339744830\ldots$ and $\pi/6 = 0.52359877559829887\ldots$ within the truncation error. For $x > 0$, since $I_n \le 1$ and $1/(1-q) = 1+x^2$, the omitted terms satisfy

$$0 < \frac{x}{1+x^2}\sum_{n=40}^{\infty} I_n q^n \le \frac{x}{1+x^2}\cdot\frac{q^{40}}{1-q} = xq^{40},$$

that is, $2^{-40} \approx 9.09\times10^{-13}$ for $x = 1$ and $4^{-40}/\sqrt{3} \approx 4.78\times10^{-25}$ for $x = 1/\sqrt{3}$. (The actual differences from $\pi/4$ and $\pi/6$ are 1.25×10^{-13} and 6.6×10^{-26}.)

1.6 Some Important Series

1.6.1 Show that $\ln\left(\dfrac{1+x}{1-x}\right) = 2\left(x + \dfrac{x^3}{3} + \dfrac{x^5}{5} + \cdots\right), \quad -1 < x < 1.$

Solution.

For $|x| < 1$,

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}x^n}{n}, \qquad \ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}.$$

Subtracting cancels every even power and doubles every odd power:

$$\ln\frac{1+x}{1-x} = \sum_{n=1}^{\infty} \frac{[(-1)^{n-1}+1]x^n}{n} = 2\sum_{r=0}^{\infty} \frac{x^{2r+1}}{2r+1}.$$

The subtraction is justified by absolute convergence in $|x| < 1$. At $x = \pm 1$ the odd harmonic series diverges, agreeing with the logarithmic singularities at the interval boundaries.

1.7 Vectors

1.7.1 The vector $\mathbf{A}$ whose magnitude is 1.732 units makes equal angles with the positive coordinate axes. Find A_x, A_y, and A_z.

Solution.

If the common angle with the positive coordinate axes is θ, then the direction cosines are equal: $A_x = A_y = A_z = A\cos\theta$. Taking the squared magnitude gives $A^2 = A_x^2 + A_y^2 + A_z^2 = 3A^2\cos^2\theta$, so $\cos\theta = \pm 1/\sqrt{3}$. Thus

$$(A_x, A_y, A_z) = \pm\frac{1.732}{\sqrt{3}}(1,1,1) \simeq \pm(0.999971, 0.999971, 0.999971).$$

The usual first-octant interpretation selects approximately $(1,1,1)$. With the axes read as undirected lines, equal angles would give only $|A_x| = |A_y| = |A_z|$, i.e., all eight vectors $(1.732/\sqrt{3})(\pm 1, \pm 1, \pm 1)$; hence "positive" in the statement. Even then the opposite vector is allowed; a first-octant or acute-angle condition is needed to select the plus sign. The value 1.732 is a rounded magnitude, so components exactly equal to 1 correspond to the unrounded magnitude $\sqrt{3}$.

1.7.2 A triangle is defined by the vertices of three vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ that extend from the origin. In terms of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ show that the vector sum of the successive sides of the triangle $(\mathbf{AB} + \mathbf{BC} + \mathbf{CA})$ is zero, where the side $\mathbf{AB}$ is from $\mathbf{A}$ to $\mathbf{B}$, etc.

Solution.

A displacement from a point to another is final position minus initial position. Therefore

$$\mathbf{AB} = \mathbf{B} - \mathbf{A}, \quad \mathbf{BC} = \mathbf{C} - \mathbf{B}, \quad \mathbf{CA} = \mathbf{A} - \mathbf{C}.$$

Adding cancels each position vector once with its negative: $\mathbf{AB} + \mathbf{BC} + \mathbf{CA} = (\mathbf{B} - \mathbf{A}) + (\mathbf{C} - \mathbf{B}) + (\mathbf{A} - \mathbf{C}) = \mathbf{0}$. This is the zero net displacement on traversing the closed triangle.

1.7.3 A sphere of radius a is centered at a point $\mathbf{r}_1$.

(a) Write out the algebraic equation for the sphere.

(b) Write out a vector equation for the sphere.

ANS. (a) $(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=a^2$.
(b) $\mathbf{r}=\mathbf{r}_1+\mathbf{a}$, where $\mathbf{a}$ takes on all directions but has a fixed magnitude a.

Solution.

A point $\mathbf{r}=(x,y,z)$ is on the sphere precisely when its distance from $\mathbf{r}_1=(x_1,y_1,z_1)$ equals a. Squaring that condition yields

$$|\mathbf{r}-\mathbf{r}_1|^2=(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=a^2.$$

This is the Cartesian equation in (a). In vector form one may use $(\mathbf{r}-\mathbf{r}_1)\cdot(\mathbf{r}-\mathbf{r}_1)=a^2$, or $\mathbf{r}=\mathbf{r}_1+\mathbf{a}$ with $|\mathbf{a}|=a$ and all orientations of $\mathbf{a}$ allowed. A fixed direction for $\mathbf{a}$ would describe only one point, not the whole sphere.

1.7.4 Hubble's law. Hubble found that distant galaxies are receding with a velocity proportional to their distance from where we are on Earth. For the ith galaxy,

$$\mathbf{v}_i=H_0\mathbf{r}_i$$

with us at the origin. Show that this recession of the galaxies from us does not imply that we are at the center of the universe. Specifically, take the galaxy at $\mathbf{r}_1$ as a new origin and show that Hubble's law is still obeyed.

Solution.

An observer moving with galaxy 1 measures the simultaneous relative position $\mathbf{r}_i'=\mathbf{r}_i-\mathbf{r}_1$ and relative velocity $\mathbf{v}_i'=\mathbf{v}_i-\mathbf{v}_1$. Substitution gives

$$\mathbf{v}_i'=H_0\mathbf{r}_i-H_0\mathbf{r}_1=H_0(\mathbf{r}_i-\mathbf{r}_1)=H_0\mathbf{r}_i'.$$

Thus the same instantaneous linear recession law holds with every comoving galaxy as origin. The equation does not select our position as a distinguished center. This argument uses the relative-position and relative-velocity description assumed by the exercise.

1.7.5 Find the diagonal vectors of a unit cube with one corner at the origin and its three sides lying along Cartesian coordinates axes. Show that there are four diagonals with length $\sqrt{3}$. Representing these as vectors, what are their components? Show that the diagonals of the cube's faces have length $\sqrt{2}$ and determine their components.

Solution.

Take the cube to have vertices (a,b,c) with each coordinate 0 or 1. Opposite vertices are (a,b,c) and $(1-a,1-b,1-c)$; the connecting vector is $(1-2a,1-2b,1-2c)$, with each component ±1. Opposite orientations describe the same geometric diagonal, leaving four representatives:

$$(1,1,1),\quad (1,1,-1),\quad (1,-1,1),\quad (1,-1,-1),$$

running from $(0,0,0)$ to $(1,1,1)$, from $(0,0,1)$ to $(1,1,0)$, from $(0,1,0)$ to $(1,0,1)$, and from $(0,1,1)$ to $(1,0,0)$. Each has squared length $1+1+1=3$, hence length $\sqrt{3}$. A face diagonal joins two

vertices of one face, which share one coordinate and differ in the other two; this gives the six unoriented direction representatives

$$(1, \pm 1, 0), \qquad (1, 0, \pm 1), \qquad (0, 1, \pm 1),$$

all of squared length $1 + 1 = 2$, hence length $\sqrt{2}$. Each direction occurs on two opposite faces ($(1, \pm 1, 0)$ on $z = 0$ and $z = 1$, $(1, 0, \pm 1)$ on $y = 0$ and $y = 1$, $(0, 1, \pm 1)$ on $x = 0$ and $x = 1$), so there are twelve face-diagonal segments, two on each of six faces. Reversing any listed vector supplies its opposite orientation.

1.7.6 The vector $\mathbf{r}$, starting at the origin, terminates at and specifies the point in space (x, y, z). Find the surface swept out by the tip of $\mathbf{r}$ if

(a) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{a} = 0$. Characterize $\mathbf{a}$ geometrically.

(b) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{r} = 0$. Describe the geometric role of $\mathbf{a}$.

The vector $\mathbf{a}$ is constant (in magnitude and direction).

Solution.

(a) Expanding gives $\mathbf{r} \cdot \mathbf{a} = |\mathbf{a}|^2$. For $\mathbf{a} \neq 0$ this is a plane perpendicular to $\mathbf{a}$ and passing through its tip. Its signed unit-normal equation is $\mathbf{r} \cdot \widehat{\mathbf{a}} = |\mathbf{a}|$, so $\mathbf{a}$ is the perpendicular displacement from the origin to the plane. If $\mathbf{a} = 0$, the equation is $0 = 0$ and the set is all space.

(b) Expanding gives $|\mathbf{r}|^2 - \mathbf{a} \cdot \mathbf{r} = 0$. Since $|\mathbf{r} - \mathbf{a}/2|^2 = |\mathbf{r}|^2 - \mathbf{a} \cdot \mathbf{r} + |\mathbf{a}|^2/4$, completing the square gives

$$|\mathbf{r}|^2 - \mathbf{a} \cdot \mathbf{r} = 0 \quad \Longleftrightarrow \quad \left|\mathbf{r} - \frac{\mathbf{a}}{2}\right|^2 = \frac{|\mathbf{a}|^2}{4}.$$

This is the sphere centered at $\mathbf{a}/2$ of radius $|\mathbf{a}|/2$; the segment from 0 to $\mathbf{a}$ is a diameter. For $\mathbf{a} = 0$ it degenerates to the single point at the origin.

1.7.7 A pipe comes diagonally down the south wall of a building, making an angle of 45° with the horizontal. Coming into a corner, the pipe turns and continues diagonally down a west-facing wall, still making an angle of 45° with the horizontal. What is the angle between the south-wall and west-wall sections of the pipe?

ANS. 120°.

Solution.

Choose x east, y north, and z upward. From the southwest corner, a ray back along the south-wall pipe can be represented by $\mathbf{a} = (1, 0, 1)$, while the ray forward down the west-wall pipe is $\mathbf{b} = (0, 1, -1)$. Each has equal horizontal and vertical component magnitudes, as required by the 45° slopes. With $\mathbf{a} \cdot \mathbf{b} = 1 \cdot 0 + 0 \cdot 1 + 1 \cdot (-1) = -1$ and $|\mathbf{a}| = |\mathbf{b}| = \sqrt{2}$, their included angle is

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{-1}{\sqrt{2}\sqrt{2}} = -\frac{1}{2}, \qquad \boxed{\theta = 120^\circ}.$$

The change of direction of travel is the supplementary angle 60°, since the incoming travel direction is $-\mathbf{a}$. The quoted answer is the angle between the two rays of pipe extending from the corner.

1.7.8 Find the shortest distance of an observer at the point $(2,1,3)$ from a rocket in free flight with velocity $(1,2,3)$ km/s. The rocket was launched at time $t=0$ from $(1,1,1)$. Lengths are in kilometers.

Solution.

At time $t \geq 0$ seconds the rocket is at $\mathbf{r}(t) = (1+t, 1+2t, 1+3t)$, so $\mathbf{r}(t)-(2,1,3) = (t-1, 2t, 3t-2)$. Its squared distance from the observer is

$$d^2(t) = (t-1)^2 + (2t)^2 + (3t-2)^2 = (t^2-2t+1) + 4t^2 + (9t^2-12t+4)$$
$$= 14t^2 - 14t + 5 = 14\big(t^2 - t + \tfrac{1}{4}\big) - \tfrac{14}{4} + 5 = 14(t-\tfrac{1}{2})^2 + \frac{3}{2}.$$

The minimum is attained at the permitted time $t = 1/2$ s, with

$$\boxed{d_{\min} = \sqrt{3/2} \text{ km} \simeq 1.224745 \text{ km}}.$$

At that time the displacement to the rocket is $(-1/2, 1, -1/2)$. Its dot product with $(1,2,3)$ is $-1/2+2-3/2 = 0$, independently confirming that the shortest connecting segment is perpendicular to the trajectory.

1.7.9 Show that the medians of a triangle intersect in the center which is $2/3$ of the median's length from each vertex. Construct a numerical example and plot it.

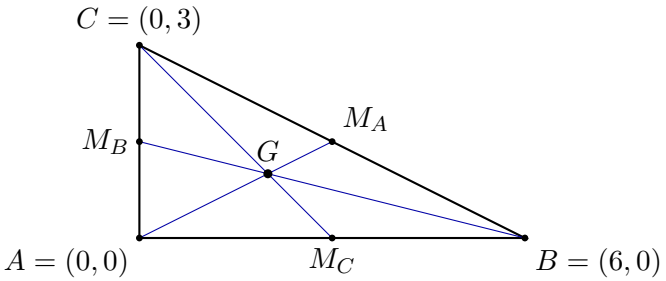
Solution.

Let the position vectors of the vertices be $\mathbf{a}, \mathbf{b}, \mathbf{c}$. The midpoint opposite $\mathbf{a}$ is $\mathbf{m}_a = (\mathbf{b}+\mathbf{c})/2$. At fraction $2/3$ of this median from $\mathbf{a}$ the position is

$$\mathbf{a} + \frac{2}{3}(\mathbf{m}_a - \mathbf{a}) = \frac{1}{3}\mathbf{a} + \frac{2}{3}\cdot\frac{\mathbf{b}+\mathbf{c}}{2} = \frac{\mathbf{a}+\mathbf{b}+\mathbf{c}}{3} = \mathbf{g}.$$

Cyclically interchanging the vertex labels gives exactly the same $\mathbf{g}$ on each of the other two medians, because $\mathbf{g}$ is symmetric in $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Thus all three intersect there, and each median is divided in the length ratio $2:1$. For a concrete example use $A=(0,0)$, $B=(6,0)$, $C=(0,3)$. The midpoints are $M_A = (3, 3/2)$, $M_B = (0, 3/2)$, $M_C = (3,0)$, and $G = (A+B+C)/3 = (2,1)$. Substitution in the three parametric medians gives G at parameter $2/3$ in every case:

$$A + \tfrac{2}{3}(M_A - A) = \tfrac{2}{3}(3, \tfrac{3}{2}) = (2,1),$$
$$B + \tfrac{2}{3}(M_B - B) = (6,0) + \tfrac{2}{3}(-6, \tfrac{3}{2}) = (2,1),$$
$$C + \tfrac{2}{3}(M_C - C) = (0,3) + \tfrac{2}{3}(3,-3) = (2,1).$$



1.7.10 Prove the law of cosines starting from $A^2 = (\mathbf{B} - \mathbf{C})^2$.

Solution.

Here $(\mathbf{B} - \mathbf{C})^2$ means $(\mathbf{B} - \mathbf{C}) \cdot (\mathbf{B} - \mathbf{C}) = |\mathbf{B} - \mathbf{C}|^2$, and A, B, C denote the side lengths $|\mathbf{A}|, |\mathbf{B}|, |\mathbf{C}|$. Place the tails of $\mathbf{B}$ and $\mathbf{C}$ at the same triangle vertex, and let θ be the included angle. The opposite side is $\mathbf{A} = \mathbf{B} - \mathbf{C}$ up to orientation. Therefore

$$\begin{aligned} A^2 = (\mathbf{B} - \mathbf{C}) \cdot (\mathbf{B} - \mathbf{C}) &= \mathbf{B} \cdot \mathbf{B} - \mathbf{B} \cdot \mathbf{C} - \mathbf{C} \cdot \mathbf{B} + \mathbf{C} \cdot \mathbf{C} \\ &= \mathbf{B} \cdot \mathbf{B} + \mathbf{C} \cdot \mathbf{C} - 2\mathbf{B} \cdot \mathbf{C} = B^2 + C^2 - 2BC\cos\theta, \end{aligned}$$

using $\mathbf{C} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{C}$ and $\mathbf{B} \cdot \mathbf{C} = BC\cos\theta$. This is the law of cosines, reducing to Pythagoras when $\theta = \pi/2$.

1.7.11 Given the three vectors,

$$\mathbf{P} = 3\hat{\mathbf{e}}_x + 2\hat{\mathbf{e}}_y - \hat{\mathbf{e}}_z,$$
$$\mathbf{Q} = -6\hat{\mathbf{e}}_x - 4\hat{\mathbf{e}}_y + 2\hat{\mathbf{e}}_z,$$
$$\mathbf{R} = \hat{\mathbf{e}}_x - 2\hat{\mathbf{e}}_y - \hat{\mathbf{e}}_z,$$

find two that are perpendicular and two that are parallel or antiparallel.

Solution.

The component vectors are $\mathbf{P} = (3, 2, -1)$, $\mathbf{Q} = (-6, -4, 2)$, and $\mathbf{R} = (1, -2, -1)$. First $\mathbf{Q} = -2\mathbf{P}$, so $\mathbf{P}$ and $\mathbf{Q}$ are antiparallel. Next

$$\mathbf{P} \cdot \mathbf{R} = 3(1) + 2(-2) + (-1)(-1) = 3 - 4 + 1 = 0,$$

so $\mathbf{P}$ and $\mathbf{R}$ are perpendicular. Also $\mathbf{Q} \cdot \mathbf{R} = -2(\mathbf{P} \cdot \mathbf{R}) = 0$, so $\mathbf{Q}$ and $\mathbf{R}$ form a second perpendicular pair.

1.8 Complex Numbers and Functions

1.8.1 Find the reciprocal of $x+iy$, working in polar form but expressing the final result in Cartesian form.

Solution.

For $z = x + iy \neq 0$, set $r = \sqrt{x^2 + y^2}$ and $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$, so that $\cos\theta = x/r$ and $\sin\theta = y/r$. Since $re^{i\theta} \cdot r^{-1}e^{-i\theta} = e^0 = 1$, the reciprocal has magnitude $1/r$ and angle $-\theta$:

$$\frac{1}{z} = r^{-1}e^{-i\theta} = \frac{1}{r}(\cos\theta - i\sin\theta) = \frac{1}{r}\left(\frac{x}{r} - i\frac{y}{r}\right) = \boxed{\frac{x - iy}{x^2 + y^2}}.$$

Multiplication by $x + iy$ gives $(x - iy)(x + iy)/(x^2 + y^2) = (x^2 + y^2)/(x^2 + y^2) = 1$, checking the result. No reciprocal exists at $x = y = 0$.

1.8.2 Show that complex numbers have square roots and that the square roots are contained in the complex plane. What are the square roots of i?

Solution.

Write a nonzero complex number as $z = re^{i\theta}$ with $r > 0$. The two numbers $w_k = \sqrt{r}\, e^{i(\theta/2 + k\pi)}$, $k = 0, 1$, satisfy $w_k^2 = re^{i\theta}e^{2\pi ik} = z$. They are opposites, $w_1 = e^{i\pi}w_0 = -w_0$, and are distinct

because $w_0 \neq 0$. Conversely, if $w = \rho e^{i\phi}$ with $\rho > 0$ satisfies $w^2 = z$, then $\rho^2 e^{2i\phi} = re^{i\theta}$. Equal moduli give $\rho = \sqrt{r}$, and equal phases give $2\phi = \theta + 2\pi k$ with k an integer, that is, $\phi = \theta/2 + k\pi$; even k gives w_0 and odd k gives w_1. So there are precisely these two roots, and both are again complex numbers: the square roots lie in the complex plane. Zero has the single square root zero, since $w^2 = 0$ forces $w = 0$. For $i = e^{i\pi/2}$ ($r = 1$, $\theta = \pi/2$) this gives, with $\cos(\pi/4) = \sin(\pi/4) = 1/\sqrt{2}$,

$$\boxed{\sqrt{i} = \pm e^{i\pi/4} = \pm\frac{1+i}{\sqrt{2}}}.$$

Squaring either value gives $(1 + 2i - 1)/2 = i$.

1.8.3 For a nonnegative integer n, show that

(a) $\cos n\theta = \cos^n\theta - \binom{n}{2}\cos^{n-2}\theta\sin^2\theta + \binom{n}{4}\cos^{n-4}\theta\sin^4\theta - \cdots$,

(b) $\sin n\theta = \binom{n}{1}\cos^{n-1}\theta\sin\theta - \binom{n}{3}\cos^{n-3}\theta\sin^3\theta + \cdots$.

Solution.

The added hypothesis is needed: only for an integer $n \geq 0$ is the binomial expansion below a finite sum; for other n the expansions (a) and (b) become infinite series, which diverge when $|\tan\theta| > 1$. For a nonnegative integer n, de Moivre's formula and the binomial expansion give

$$\cos n\theta + i\sin n\theta = (\cos\theta + i\sin\theta)^n = \sum_{j=0}^{n}\binom{n}{j}\cos^{n-j}\theta\, i^j \sin^j\theta.$$

For $j = 2r$, $i^{2r} = (-1)^r$, while for $j = 2r+1$, $i^{2r+1} = i(-1)^r$. For real θ the terms with even j are therefore real and those with odd j are imaginary; the conditions $2r \leq n$ and $2r + 1 \leq n$ give the upper limits $\lfloor n/2\rfloor$ and $\lfloor (n-1)/2\rfloor$. Equating real and imaginary parts yields

$$\cos n\theta = \sum_{r=0}^{\lfloor n/2\rfloor}(-1)^r\binom{n}{2r}\cos^{n-2r}\theta\sin^{2r}\theta,$$

$$\sin n\theta = \sum_{r=0}^{\lfloor (n-1)/2\rfloor}(-1)^r\binom{n}{2r+1}\cos^{n-2r-1}\theta\sin^{2r+1}\theta.$$

Their terms $r = 0, 1, 2, \ldots$ are exactly those displayed in (a) and (b), so these are the two requested finite expansions. For $n = 0$ the second sum is empty and gives zero.

The identities hold for complex θ as well. Euler's formula $e^{iz} = \cos z + i\sin z$, obtained by comparing power series, holds for every complex z, and cos is even and sin odd; hence $(\cos\theta \pm i\sin\theta)^n = e^{\pm in\theta} = \cos n\theta \pm i\sin n\theta$ for complex θ. The binomial expansion of $(\cos\theta - i\sin\theta)^n$ has $(-i)^j$ in place of i^j. Since $[i^j + (-i)^j]/2$ is $(-1)^r$ for $j = 2r$ and 0 for odd j, while $[i^j - (-i)^j]/(2i)$ is $(-1)^r$ for $j = 2r+1$ and 0 for even j, half the sum and $1/(2i)$ times the difference of the two expansions give the same two sums.

1.8.4 Prove that

(a) $$\sum_{n=0}^{N-1} \cos nx = \frac{\sin(Nx/2)}{\sin(x/2)} \cos\left(\frac{(N-1)x}{2}\right),$$

(b) $$\sum_{n=0}^{N-1} \sin nx = \frac{\sin(Nx/2)}{\sin(x/2)} \sin\left(\frac{(N-1)x}{2}\right).$$

Here N is a positive integer; at $x = 2\pi m$, use the continuous limits N and 0, respectively. These series occur in the analysis of the multiple-slit diffraction pattern.

Solution.

The parentheses in the statement fix the argument of the last factor as $(N-1)x/2$, as derived below; the textbook's typesetting of this argument is ambiguous. For $e^{ix} \neq 1$, that is, $x \neq 2\pi m$ with m an integer, sum the finite geometric progression with ratio e^{ix}:

$$\sum_{n=0}^{N-1} e^{inx} = \frac{1 - e^{iNx}}{1 - e^{ix}}.$$

Factoring out the half angle, $1 - e^{it} = e^{it/2}(e^{-it/2} - e^{it/2}) = -2ie^{it/2}\sin(t/2)$. Using this with $t = Nx$ in the numerator and $t = x$ in the denominator gives

$$\sum_{n=0}^{N-1} e^{inx} = \frac{-2ie^{iNx/2}\sin(Nx/2)}{-2ie^{ix/2}\sin(x/2)} = e^{i(N-1)x/2}\frac{\sin(Nx/2)}{\sin(x/2)}.$$

For real x the sine quotient is real, while $e^{inx} = \cos nx + i\sin nx$ and $e^{i(N-1)x/2} = \cos[(N-1)x/2] + i\sin[(N-1)x/2]$; taking real and imaginary parts gives (a) and (b). At $x = 2\pi m$ the quotient is $0/0$, but every $e^{inx} = 1$, so the sums are N and 0. These values are also the limits of the right sides: with $x = 2\pi m + h$, $\sin(Nx/2) = (-1)^{Nm}\sin(Nh/2)$ and $\sin(x/2) = (-1)^m \sin(h/2)$, so the quotient tends to $(-1)^{(N-1)m}N$ as $h \to 0$, while $\cos[(N-1)x/2] \to (-1)^{(N-1)m}$ and $\sin[(N-1)x/2] \to 0$. The products tend to N and 0, so the exception at $x = 2\pi m$ is removable.

The formulas hold for complex $x \neq 2\pi m$ as well. The sine quotient is unchanged when x is replaced by $-x$, so

$$\sum_{n=0}^{N-1} e^{-inx} = e^{-i(N-1)x/2}\frac{\sin(Nx/2)}{\sin(x/2)},$$

and since $\cos z = (e^{iz} + e^{-iz})/2$ and $\sin z = (e^{iz} - e^{-iz})/(2i)$ for complex z, half the sum and $1/(2i)$ times the difference of the two geometric sums give (a) and (b).

1.8.5 Assume that the trigonometric functions and the hyperbolic functions are defined for complex argument by the appropriate power series. Show that

$$i\sin z = \sinh iz, \quad \sin iz = i\sinh z,$$

$$\cos z = \cosh iz, \quad \cos iz = \cosh z.$$

Solution.

Direct substitution of iz into the entire power series, with $i^{2n} = (-1)^n$, $i^{2n+1} = i(-1)^n$ and $(-1)^n(-1)^n = 1$, gives

$$\begin{aligned}
\sinh(iz) &= \sum_{n=0}^{\infty} \frac{i^{2n+1} z^{2n+1}}{(2n+1)!} = i \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!} = i \sin z, \\
\sin(iz) &= \sum_{n=0}^{\infty} \frac{(-1)^n i^{2n+1} z^{2n+1}}{(2n+1)!} = i \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = i \sinh z, \\
\cosh(iz) &= \sum_{n=0}^{\infty} \frac{i^{2n} z^{2n}}{(2n)!} = \cos z, \\
\cos(iz) &= \sum_{n=0}^{\infty} \frac{(-1)^n i^{2n} z^{2n}}{(2n)!} = \cosh z.
\end{aligned}$$

No restriction on complex z is required because all four power series converge absolutely everywhere.

1.8.6 Using the identities

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i},$$

established from comparison of power series, show that

(a) $\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$,
$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$,

(b) $|\sin z|^2 = \sin^2 x + \sinh^2 y, \quad |\cos z|^2 = \cos^2 x + \sinh^2 y.$

This demonstrates that we may have $|\sin z|, |\cos z| > 1$ in the complex plane.

Solution.

(a) For real x, y, $e^{i(x+iy)} = e^{-y}(\cos x + i \sin x)$ and $e^{-i(x+iy)} = e^{y}(\cos x - i \sin x)$. With $z = x + iy$, collecting the terms in $\cos x$ and $\sin x$ gives

$$\begin{aligned}
e^{iz} - e^{-iz} &= -(e^{y} - e^{-y}) \cos x + i(e^{y} + e^{-y}) \sin x, \\
e^{iz} + e^{-iz} &= (e^{y} + e^{-y}) \cos x - i(e^{y} - e^{-y}) \sin x.
\end{aligned}$$

Dividing the first line by $2i$ (note $-1/i = i$) and the second by 2,

$$\begin{aligned}
\sin(x + iy) &= \sin x \frac{e^{y} + e^{-y}}{2} + i \cos x \frac{e^{y} - e^{-y}}{2} = \sin x \cosh y + i \cos x \sinh y, \\
\cos(x + iy) &= \cos x \frac{e^{y} + e^{-y}}{2} - i \sin x \frac{e^{y} - e^{-y}}{2} = \cos x \cosh y - i \sin x \sinh y.
\end{aligned}$$

(b) Add the squares of each expression's real and imaginary parts:

$$\begin{aligned}
|\sin z|^2 &= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sin^2 x + (\sin^2 x + \cos^2 x) \sinh^2 y = \sin^2 x + \sinh^2 y, \\
|\cos z|^2 &= \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y = \cos^2 x + (\cos^2 x + \sin^2 x) \sinh^2 y = \cos^2 x + \sinh^2 y.
\end{aligned}$$

Here $\cosh^2 y = 1 + \sinh^2 y$ was used. For example, for $y \neq 0$, (b) gives $|\cos(iy)|^2 = 1 + \sinh^2 y = \cosh^2 y > 1$ and $|\sin(\pi/2 + iy)|^2 = 1 + \sinh^2 y = \cosh^2 y > 1$.

1.8.7 From the identities in Exercises 1.8.5 and 1.8.6 show that

(a) $\sinh(x+iy) = \sinh x \cos y + i \cosh x \sin y$,
$\cosh(x+iy) = \cosh x \cos y + i \sinh x \sin y$,

(b) $|\sinh z|^2 = \sinh^2 x + \sin^2 y$, $\quad |\cosh z|^2 = \cosh^2 x - \sin^2 y$.

Solution.

(a) By Exercise 1.8.5, $\sin iz = i \sinh z$ and $\cos iz = \cosh z$, so $\sinh z = -i \sin(iz)$ and $\cosh z = \cos(iz)$. For $z = x + iy$, $iz = -y + ix$ has real part $-y$ and imaginary part x. Exercise 1.8.6(a), with $\sin(-y) = -\sin y$ and $\cos(-y) = \cos y$, then gives

$$\sinh(x+iy) = -i[\sin(-y)\cosh x + i\cos(-y)\sinh x] = \sinh x \cos y + i \cosh x \sin y,$$
$$\cosh(x+iy) = \cos(-y)\cosh x - i\sin(-y)\sinh x = \cosh x \cos y + i \sinh x \sin y.$$

(b) Consequently, using $\cosh^2 x - \sinh^2 x = 1$ and $\cos^2 y + \sin^2 y = 1$,

$$|\sinh z|^2 = \sinh^2 x \cos^2 y + \cosh^2 x \sin^2 y = \sinh^2 x \cos^2 y + (1 + \sinh^2 x)\sin^2 y = \sinh^2 x + \sin^2 y,$$
$$|\cosh z|^2 = \cosh^2 x \cos^2 y + \sinh^2 x \sin^2 y = \cosh^2 x \cos^2 y + (\cosh^2 x - 1)\sin^2 y = \cosh^2 x - \sin^2 y.$$

The textbook (p. 61) prints a plus sign in the last identity; the correct sign is minus, as just derived. As a check, at $x = 0$, $y = \pi/2$, $\cosh(i\pi/2) = \cos(\pi/2) = 0$; the corrected right side is $1 - 1 = 0$, whereas the printed plus sign would give 2.

1.8.8 Put $z = x + iy$ with real x, y, and exclude the poles of each side. Show that

(a) $\tanh \dfrac{z}{2} = \dfrac{\sinh x + i \sin y}{\cosh x + \cos y}$,

(b) $\coth \dfrac{z}{2} = \dfrac{\sinh x - i \sin y}{\cosh x - \cos y}$.

Solution.

Let $z = x + iy$ with x, y real, and $\bar{z} = x - iy$. Multiplying numerator and denominator of $\tanh(z/2) = (e^{z/2} - e^{-z/2})/(e^{z/2} + e^{-z/2})$ by $e^{z/2}$ gives $\tanh(z/2) = (e^z - 1)/(e^z + 1)$. Now multiply numerator and denominator by $e^{\bar{z}} + 1$, the conjugate of $e^z + 1$, and use

$$e^z e^{\bar{z}} = e^{2x}, \qquad e^z - e^{\bar{z}} = e^x(e^{iy} - e^{-iy}) = 2ie^x \sin y, \qquad e^z + e^{\bar{z}} = 2e^x \cos y.$$

This gives

$$\tanh \frac{z}{2} = \frac{e^{2x} - 1 + (e^z - e^{\bar{z}})}{e^{2x} + 1 + (e^z + e^{\bar{z}})} = \frac{e^{2x} - 1 + 2ie^x \sin y}{e^{2x} + 1 + 2e^x \cos y} = \frac{\sinh x + i \sin y}{\cosh x + \cos y},$$

the last step dividing numerator and denominator by $2e^x$. Likewise $\coth(z/2) = (e^z + 1)/(e^z - 1)$. Multiplying numerator and denominator by $e^{\bar{z}} - 1$ gives

$$\coth \frac{z}{2} = \frac{e^{2x} - 1 - (e^z - e^{\bar{z}})}{e^{2x} + 1 - (e^z + e^{\bar{z}})} = \frac{e^{2x} - 1 - 2ie^x \sin y}{e^{2x} + 1 - 2e^x \cos y} = \frac{\sinh x - i \sin y}{\cosh x - \cos y}.$$

The divisions exclude the poles. Since $\cosh x \geq 1 \geq |\cos y|$, the denominator $\cosh x + \cos y$ vanishes only for $x = 0$, $\cos y = -1$, that is, at $z = (2m+1)\pi i$, the poles of $\tanh(z/2)$; likewise $\cosh x - \cos y$ vanishes only at $z = 2m\pi i$, the poles of $\coth(z/2)$ ($m \in \mathbb{Z}$). Elsewhere the factors $e^{\bar{z}} \pm 1$ used above are nonzero.

1.8.9 By comparing series expansions, show that $\tan^{-1} x = \dfrac{i}{2}\mathrm{Log}\left(\dfrac{1-ix}{1+ix}\right)$, for real x, with the principal inverse tangent and principal logarithm.

Solution.

For $|x| < 1$, use the series $\log(1+w) = \sum_{k\geq 1}(-1)^{k+1}w^k/k$ ($|w| < 1$), on the branch with $\log 1 = 0$, at $w = -ix$ and $w = ix$: $\log(1-ix) = -\sum_k (ix)^k/k$ and $\log(1+ix) = \sum_k (-1)^{k+1}(ix)^k/k$. Subtracting, the even powers cancel and the odd ones double; with $(ix)^{2r+1} = i(-1)^r x^{2r+1}$,

$$\log(1-ix) - \log(1+ix) = -2\left(ix + \frac{(ix)^3}{3} + \frac{(ix)^5}{5} + \cdots\right) = -2i\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right).$$

Multiplication by $i/2$, with $(i/2)(-2i) = 1$, gives exactly the Gregory series for $\arctan x$. This branch is the principal one: the series is a logarithm of $1+w$ that varies continuously on $|w| < 1$ and vanishes at $w = 0$, and since $\mathrm{Re}(1+w) > 0$ there, its imaginary part, an argument of $1+w$, stays in $(-\pi/2, \pi/2)$. So the left side above is $\mathrm{Log}(1-ix) - \mathrm{Log}(1+ix) = -2i\arctan x$, a logarithm of $(1-ix)/(1+ix)$ whose imaginary part $-2\arctan x$ lies in $(-\pi/2, \pi/2)$; hence it is $\mathrm{Log}[(1-ix)/(1+ix)]$, which proves the identity for $|x| < 1$.

For all real x, write $1+ix = \sqrt{1+x^2}\,e^{i\alpha}$ with $\alpha = \arctan x \in (-\pi/2, \pi/2)$; then $1-ix = \sqrt{1+x^2}\,e^{-i\alpha}$, so $(1-ix)/(1+ix) = e^{-2i\alpha}$ has unit modulus and argument $-2\alpha \in (-\pi, \pi)$. Hence its principal logarithm is $-2i\alpha$, and $(i/2)(-2i\alpha) = \alpha$:

$$\boxed{\arctan x = \frac{i}{2}\,\mathrm{Log}\,\frac{1-ix}{1+ix}}.$$

Other logarithm branches differ by $2\pi i m$ and change the right side by $-\pi m$, the corresponding multivalued inverse-tangent branches. For complex x, branch choices and the singularities $x = \pm i$ must be retained.

1.8.10 Find the Cartesian form for all values of

(a) $(-8)^{1/3}$,

(b) $i^{1/4}$,

(c) $e^{i\pi/4}$.

Solution.

(a) Write $-8 = 8e^{i(\pi+2\pi k)}$. The cube roots have magnitude $8^{1/3} = 2$ and arguments $(\pi+2\pi k)/3 = \pi/3, \pi, 5\pi/3$ for $k = 0, 1, 2$. Since $2e^{\pm i\pi/3} = 2(1/2 \pm i\sqrt{3}/2)$ and $2e^{5i\pi/3} = 2e^{-i\pi/3}$, they are

$$\boxed{1+i\sqrt{3}, \quad -2, \quad 1-i\sqrt{3}}.$$

(b) The fourth roots of $i = e^{i(\pi/2+2\pi k)}$ have unit modulus and arguments $\pi/8 + k\pi/2$, $k = 0, 1, 2, 3$. Set $A = \sqrt{2+\sqrt{2}}/2$ and $B = \sqrt{2-\sqrt{2}}/2$; the half-angle identities $\cos^2(\pi/8) = [1+\cos(\pi/4)]/2 = (2+\sqrt{2})/4$ and $\sin^2(\pi/8) = [1-\cos(\pi/4)]/2 = (2-\sqrt{2})/4$ give $\cos(\pi/8) = A$, $\sin(\pi/8) = B$. Each further root is the previous one times $e^{i\pi/2} = i$, and $i(A+iB) = -B+iA$, so the four Cartesian values are

$$\boxed{A+iB, \quad -B+iA, \quad -A-iB, \quad B-iA}.$$

(c) The exponential is single-valued: $\boxed{e^{i\pi/4} = \cos(\pi/4) + i\sin(\pi/4) = (1+i)/\sqrt{2}}$. Adding 2π to its angle reproduces this same value; it does not create additional roots. Cubing the values in (a) or taking fourth powers in (b) returns the specified original number.

1.8.11 Find the polar form for all values of

(a) $(1+i)^3$,

(b) $(-1)^{1/5}$.

Solution.

(a) Since $1+i=\sqrt{2}e^{i\pi/4}$ and $(\sqrt{2})^3=2\sqrt{2}$,

$$\boxed{(1+i)^3=2\sqrt{2}e^{3\pi i/4}}.$$

This is one value, equal to $2\sqrt{2}(-1+i)/\sqrt{2}=-2+2i$ in Cartesian form; directly, $(1+i)^2=2i$ and $2i(1+i)=-2+2i$. Its argument may be written $3\pi/4+2\pi m$, but all such choices represent the same complex number.

(b) Write $-1=e^{i(\pi+2\pi k)}$. Its fifth roots are

$$\boxed{e^{i(2k+1)\pi/5}, \qquad k=0,1,2,3,4}.$$

They all have unit modulus. Their arguments are $\pi/5, 3\pi/5, \pi, 7\pi/5, 9\pi/5$, and raising any one to the fifth power gives $e^{i(2k+1)\pi}=-1$. Values of k outside this range repeat the roots.

1.9 Derivatives and Extrema

1.9.1 Derive the following formula for the Maclaurin expansion of a function of two variables analytic near $(0,0)$, for (x,y) within the convergence neighborhood of its Maclaurin series:

$$\begin{aligned} f(x,y) &= f(0,0)+x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y} \\ &+\frac{1}{2!}\left[\binom{2}{0}x^2\frac{\partial^2 f}{\partial x^2}+\binom{2}{1}xy\frac{\partial^2 f}{\partial x\,\partial y}+\binom{2}{2}y^2\frac{\partial^2 f}{\partial y^2}\right] \\ &+\frac{1}{3!}\left[\binom{3}{0}x^3\frac{\partial^3 f}{\partial x^3}+\binom{3}{1}x^2y\frac{\partial^3 f}{\partial x^2\,\partial y}+\binom{3}{2}xy^2\frac{\partial^3 f}{\partial x\,\partial y^2}+\binom{3}{3}y^3\frac{\partial^3 f}{\partial y^3}\right]+\cdots, \end{aligned}$$

where all the partial derivatives are to be evaluated at the point $(0,0)$.

Solution.

An infinite Taylor expansion represents the function only where its remainder tends to zero; analyticity near the origin is a sufficient hypothesis. Smoothness alone is not sufficient. For example, $f(x,y)=e^{-1/x^2}$ (with $f=0$ at $x=0$) is infinitely differentiable, but all its partial derivatives vanish at the origin, so the series sums to $0\neq f(x,y)$ for $x\neq 0$. Indeed, f does not depend on y, and for $x\neq 0$ each x-derivative has the form $P(1/x)\,e^{-1/x^2}$ with P a polynomial; since e^{-1/x^2} tends to zero faster than any power of x, each derivative, and each difference quotient $P(1/h)\,e^{-1/h^2}/h$ at $x=0$, tends to zero, so by induction every x-derivative exists at $x=0$ and is zero there.

Fix x,y and define the one-variable function $F(t)=f(tx,ty)$. The chain rule gives $F'(t)=(x\partial_x+y\partial_y)f(tx,ty)$: differentiation in t acts as the operator $x\partial_x+y\partial_y$ followed by evaluation at (tx,ty). Applying this to $f, (x\partial_x+y\partial_y)f, \ldots$ in turn gives $F^{(n)}(t)=[(x\partial_x+y\partial_y)^n f](tx,ty)$. Since

x, y are constants and the mixed partial derivatives commute, the binomial theorem expands the operator power, and at $t = 0$

$$F^{(n)}(0) = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j \frac{\partial^n f}{\partial x^{n-j} \partial y^j}(0,0).$$

Taylor-expand F about $t = 0$ and then set $t = 1$. The hypothesis ensures that this series converges to $F(1) = f(x, y)$: if the Maclaurin series $\sum_{j,k} c_{jk} X^j Y^k$ converges absolutely to f on a disk about the origin containing (x, y), then $F(t) = \sum_{j,k} c_{jk} x^j y^k t^{j+k}$ for $|t| \le 1$; grouping the terms by total degree $j + k = n$ gives a power series in t that converges to $F(t)$ there, and a convergent power series is the Taylor series of its sum. With $\binom{n}{j}/n! = 1/[(n-j)!\, j!]$,

$$f(x,y) = \sum_{n=0}^{\infty} \frac{F^{(n)}(0)}{n!} = \sum_{n=0}^{\infty} \sum_{j=0}^{n} \frac{x^{n-j} y^j}{(n-j)!\, j!} \frac{\partial^n f}{\partial x^{n-j} \partial y^j}(0,0).$$

For $n = 1$ this is $x f_x + y f_y$; for $n = 2$ it is $(x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy})/2!$; for $n = 3$ it is $(x^3 f_{xxx} + 3x^2 y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy})/3!$. These are exactly the displayed terms. For a finite order N, ordinary Taylor's theorem also provides a remainder $F^{(N+1)}(\theta)/(N+1)!$ for some $0 < \theta < 1$, under its usual real-variable hypotheses.

1.9.2 The result in Exercise 1.9.1 can be generalized to larger numbers of independent variables. Prove that for an m-variable system analytic near the origin, at points sufficiently close to it, the Maclaurin expansion can be written in the symbolic form

$$f(x_1, \ldots, x_m) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \left[\left(\sum_{i=1}^{m} \alpha_i \frac{\partial}{\partial x_i} \right)^n f(\mathbf{x}) \right]_{\mathbf{x}=0},$$

where in the right-hand side we have made the substitutions $x_j = \alpha_j t$.

Solution.

Define $F(t) = f(\alpha_1 t, \ldots, \alpha_m t)$ with fixed α_i, and write $D = \sum_{i=1}^{m} \alpha_i \partial_{x_i}$. For any differentiable g the chain rule gives $\frac{d}{dt} g(\alpha_1 t, \ldots, \alpha_m t) = \sum_i \alpha_i (\partial_{x_i} g)(\alpha_1 t, \ldots, \alpha_m t) = (Dg)(\alpha_1 t, \ldots, \alpha_m t)$. Applying this to $g = f, Df, D^2 f, \ldots$ in turn gives, by induction,

$$F^{(n)}(t) = \left[\left(\sum_{i=1}^{m} \alpha_i \partial_{x_i} \right)^n f(\mathbf{x}) \right]_{\mathbf{x}=(\alpha_1 t, \ldots, \alpha_m t)}.$$

Therefore Taylor's formula in t about $t = 0$, $F(t) = \sum_n F^{(n)}(0)\, t^n/n!$, is

$$F(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \left[\left(\sum_{i=1}^{m} \alpha_i \partial_{x_i} \right)^n f(\mathbf{x}) \right]_{\mathbf{x}=0}.$$

The book's symbolic $D^n f(0)$ means that derivatives act on f before evaluation at zero; the brackets in the corrected statement make this order explicit. Differentiating the constant $f(0)$ would instead give zero for $n > 0$. Since $x_i = \alpha_i t$, $F(t) = f(x_1, \ldots, x_m)$, and this is the required symbolic form. Explicitly, since the ∂_{x_i} commute, the multinomial theorem gives

$$D^n = \sum_{r_1 + \cdots + r_m = n} \frac{n!}{r_1! \cdots r_m!}\, \alpha_1^{r_1} \cdots \alpha_m^{r_m}\, \partial_{x_1}^{r_1} \cdots \partial_{x_m}^{r_m},$$

and with $(\alpha_i t)^{r_i} = x_i^{r_i}$ the nth term $t^n[D^n f]_{\mathbf{x}=0}/n!$ becomes the sum of the terms with $r_1+\cdots+r_m = n$ in

$$f(\mathbf{x}) = \sum_{r_1,\ldots,r_m \geq 0} \frac{x_1^{r_1}\cdots x_m^{r_m}}{r_1!\cdots r_m!}(\partial_{x_1}^{r_1}\cdots\partial_{x_m}^{r_m} f)(0).$$

As in Exercise 1.9.1, equality to the infinite expansion requires convergence to f; assume analyticity on the neighborhood under consideration, or retain a finite Taylor remainder.

1.10 Evaluation of Integrals

1.10.1 Use a recursive method to show that, for all positive integers n, $\Gamma(n) = (n-1)!$.

Solution.

Start from $\Gamma(n) = \int_0^\infty t^{n-1}e^{-t}dt$ for $n > 0$. For an integer $n \geq 1$, integration by parts with $u = t^n$, $dv = e^{-t}dt$, $v = -e^{-t}$ gives

$$\Gamma(n+1) = \int_0^\infty t^n e^{-t}dt = [-t^n e^{-t}]_0^\infty + n\int_0^\infty t^{n-1}e^{-t}dt = n\Gamma(n).$$

The boundary term vanishes because the exponential dominates any fixed power at infinity and $t^n \to 0$ at zero. The initial value is $\Gamma(1) = \int_0^\infty e^{-t}dt = 1$. Repeated application of the recurrence, $\Gamma(n) = (n-1)\Gamma(n-1) = (n-1)(n-2)\Gamma(n-2) = \cdots$, yields (by induction on n) $\Gamma(n) = (n-1)(n-2)\cdots 1\,\Gamma(1) = \boxed{(n-1)!}$, including $n = 1$ with $0! = 1$.

Evaluate the integrals in Exercises 1.10.2 through 1.10.9.

1.10.2 $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx.$

Hint. Multiply integrand by e^{-ax} and take the limit $a \downarrow 0$.

Solution.

The limit in the hint is one-sided because the damped integral diverges for $a < 0$. For $a > 0$ define $I(a) = \int_0^\infty e^{-ax}\sin x\,dx/x$. For $a \geq a_0 > 0$ the a-derivative of the integrand, $-e^{-ax}\sin x$, is bounded by the integrable function $e^{-a_0 x}$, which permits differentiation under the integral:

$$\begin{aligned} I'(a) &= -\int_0^\infty e^{-ax}\sin x\,dx = -\operatorname{Im}\int_0^\infty e^{-(a-i)x}dx \\ &= -\operatorname{Im}\frac{1}{a-i} = -\operatorname{Im}\frac{a+i}{a^2+1} = -\frac{1}{1+a^2}. \end{aligned}$$

Also $|\sin x/x| \leq 1$ gives $|I(a)| \leq \int_0^\infty e^{-ax}dx = 1/a$, so $I(a) \to 0$ as $a \to \infty$. Integrating the derivative from a to infinity therefore gives $0 - I(a) = \int_a^\infty I'(t)\,dt$, that is,

$$I(a) = \int_a^\infty \frac{dt}{1+t^2} = \frac{\pi}{2} - \arctan a.$$

It remains to justify removing the damping. For $a \geq 0$ the function $g(x) = e^{-ax}/x$ is decreasing on $x > 0$, so $g' \leq 0$. Integrating by parts on $[R,T]$, $0 < R < T$,

$$\int_R^T g(x)\sin x\,dx = [-g(x)\cos x]_R^T + \int_R^T g'(x)\cos x\,dx.$$

Its absolute value is at most $g(R)+g(T)+\int_R^T |g'(x)|dx = g(R)+g(T)+g(R)-g(T) = 2g(R) \le 2/R$. Applied on any $[T_1, T_2]$, the same bound $2/T_1 \to 0$ shows by the Cauchy criterion that $\int_R^\infty g(x)\sin x\,dx$ converges, also for the undamped $a = 0$; letting $T \to \infty$ above, its absolute value is at most $2/R$ for every $a \ge 0$. Let $I(0) = \int_0^\infty \sin x\,dx/x$. Splitting at R and using $|\sin x/x| \le 1$ and $0 \le 1-e^{-ax} \le ax$ on $[0, R]$,

$$|I(0) - I(a)| \le \int_0^R (1 - e^{-ax})\,dx + \frac{2}{R} + \frac{2}{R} \le \frac{aR^2}{2} + \frac{4}{R}.$$

Choosing R large and then a small makes this arbitrarily small, so the ordinary improper integral equals $\lim_{a\downarrow 0} I(a) = \lim_{a\downarrow 0}(\pi/2 - \arctan a)$:

$$\boxed{\int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}}.$$

The tail argument also proves convergence, rather than merely assigning an Abel-summed value to the expression.

1.10.3 $\displaystyle\int_0^\infty \frac{dx}{\cosh x}$.

Hint. Expand the denominator in a way that converges for all relevant x.

Solution.

Since $\cosh x = e^x(1+e^{-2x})/2$, and the finite geometric sum $1/(1+q) = \sum_{n=0}^N (-q)^n + (-q)^{N+1}/(1+q)$ holds for $q = e^{-2x}$, for $x > 0$

$$\frac{1}{\cosh x} = \frac{2e^{-x}}{1+e^{-2x}} = 2\sum_{n=0}^{N}(-1)^n e^{-(2n+1)x} + \frac{2(-1)^{N+1}e^{-(2N+3)x}}{1+e^{-2x}}.$$

Each term integrates to $\int_0^\infty e^{-(2n+1)x}dx = 1/(2n+1)$, and since $1 + e^{-2x} \ge 1$, the integral of the last term has absolute value at most $2\int_0^\infty e^{-(2N+3)x}dx = 2/(2N+3) \to 0$. The expansion converges for every $x > 0$, as the hint requires, and diverges only at the single point $x = 0$, where it reads $2(1-1+1-\cdots)$; that point does not affect the integral, and the remainder bound justifies integrating term by term and taking the limit $N \to \infty$. Therefore, by Leibniz's series $1 - \frac{1}{3} + \frac{1}{5} - \cdots = \pi/4$ (the Gregory series at $x = 1$),

$$\int_0^\infty \frac{dx}{\cosh x} = 2\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1} = \boxed{\frac{\pi}{2}}.$$

An independent substitution $t = e^{-x}$, $dx = -dt/t$, turns the integrand $2e^{-x}dx/(1+e^{-2x})$ into $-2\,dt/(1+t^2)$, with limits $t = 1$ to $t = 0$; this gives $2\int_0^1 dt/(1+t^2) = 2[\arctan t]_0^1 = \pi/2$.

1.10.4 $\displaystyle\int_0^\infty \frac{dx}{e^{ax}+1}$, for $a > 0$.

Solution.

Set $u = e^{-ax}$ for $a > 0$. Then $du = -au\,dx$, so $dx = -du/(au)$; $e^{ax} = 1/u$, so $1/(e^{ax}+1) = u/(1+u)$; and the limits $x = 0, \infty$ become $u = 1, 0$. The integrand becomes $[u/(1+u)][-du/(au)] = -du/[a(1+u)]$, and reversing the limits,

$$\int_0^\infty \frac{dx}{e^{ax}+1} = \frac{1}{a}\int_0^1 \frac{du}{1+u} = \frac{1}{a}[\ln(1+u)]_0^1 = \boxed{\frac{\ln 2}{a}}.$$

As a series check, for $x > 0$ write $1/(e^{ax}+1) = e^{-ax}/(1+e^{-ax}) = \sum_{n=1}^{N}(-1)^{n-1}e^{-nax} + (-1)^N e^{-(N+1)ax}/(1+e^{-ax})$. The terms integrate to $(-1)^{n-1}/(na)$, and the integral of the remainder is at most $\int_0^\infty e^{-(N+1)ax}dx = 1/[a(N+1)] \to 0$ in absolute value. Hence the integral is $(1/a)\sum_{n\geq 1}(-1)^{n-1}/n = (\ln 2)/a$ as well, by the alternating harmonic series.

1.10.5 $\displaystyle\int_\pi^\infty \frac{\sin x}{x^2}\,dx.$

Solution.

Differentiating $\sin x/x$ gives $d(\sin x/x)/dx = \cos x/x - \sin x/x^2$, so $\sin x/x^2 = \cos x/x - d(\sin x/x)/dx$. On a finite interval $[\pi, R]$,

$$\int_\pi^R \frac{\sin x}{x^2}dx = \int_\pi^R \frac{\cos x}{x}dx - \left[\frac{\sin x}{x}\right]_\pi^R.$$

The boundary term $\sin R/R - \sin\pi/\pi = \sin R/R$ tends to zero because $|\sin R/R| \leq 1/R$. With the convention $\mathrm{Ci}(z) = -\int_z^\infty \cos t\,dt/t$ for $z > 0$, the limit $R \to \infty$ gives

$$\boxed{\int_\pi^\infty \frac{\sin x}{x^2}dx = -\,\mathrm{Ci}(\pi) = -0.073667912046425\ldots}.$$

The negative sign is consistent with the negative first lobe on $(\pi, 2\pi)$ and the decreasing weights of subsequent lobes. The number follows from the convergent series $\mathrm{Ci}(x) = \gamma + \ln x + \sum_{k=1}^\infty (-1)^k x^{2k}/[2k\,(2k)!]$ of Exercise 13.6.13(b), where $\gamma = 0.57721566490153286\ldots$ is the Euler-Mascheroni constant of Eq. (1.13). Its derivative is $\cos x/x$, as it must be; this check fixes the series only up to an additive constant, which the derivation in Exercise 13.6.13(b) shows to be γ. At $x = \pi$ the terms of the sum alternate in sign and decrease in magnitude from $k = 1$ on, so stopping after $k = 13$ leaves an error smaller than the first omitted term, $\pi^{28}/(28\cdot 28!) < 10^{-17}$.

1.10.6 $\displaystyle\int_0^\infty \frac{e^{-x}\sin x}{x}\,dx.$

Solution.

Use $I(a)$ derived in Exercise 1.10.2:

$$I(a) = \int_0^\infty \frac{e^{-ax}\sin x}{x}dx = \frac{\pi}{2} - \arctan a \quad (a > 0).$$

At $a = 1$, this gives $\boxed{I(1) = \pi/2 - \pi/4 = \pi/4}$. For a direct check, write $\sin x/x = \int_0^1 \cos(tx)\,dt$ (a t-antiderivative of $\cos(tx)$ is $\sin(tx)/x$). Absolute integrability of the exponentially damped double integral, $\int_0^1 dt \int_0^\infty e^{-x}|\cos tx|\,dx \leq 1$, permits interchanging the integrals. Since

$$\int_0^\infty e^{-x}\cos(tx)\,dx = \mathrm{Re}\int_0^\infty e^{-(1-it)x}dx = \mathrm{Re}\,\frac{1}{1-it} = \frac{1}{1+t^2},$$

the interchanged integral is

$$\int_0^1 dt\int_0^\infty e^{-x}\cos(tx)dx = \int_0^1 \frac{dt}{1+t^2} = \arctan 1 = \frac{\pi}{4}.$$

1.10.7 $\displaystyle\int_0^x \operatorname{erf}(t)\,dt.$

The result can be expressed in terms of special functions in Table 1.2.

Solution.

From the definition $\operatorname{erf}(t) = (2/\sqrt{\pi})\int_0^t e^{-u^2}du$, $\operatorname{erf}'(t) = 2e^{-t^2}/\sqrt{\pi}$. Integration by parts, differentiating $\operatorname{erf}(t)$ and integrating dt to t, and then $\int_0^x te^{-t^2}dt = [-e^{-t^2}/2]_0^x = (1-e^{-x^2})/2$ give

$$\int_0^x \operatorname{erf}(t)dt = [t\operatorname{erf}(t)]_0^x - \frac{2}{\sqrt{\pi}}\int_0^x te^{-t^2}dt = x\operatorname{erf}(x) - \frac{1-e^{-x^2}}{\sqrt{\pi}}$$
$$= \boxed{x\operatorname{erf}(x) + \frac{e^{-x^2}-1}{\sqrt{\pi}}}.$$

Differentiating the answer cancels $2xe^{-x^2}/\sqrt{\pi}$ against $-2xe^{-x^2}/\sqrt{\pi}$, leaving $\operatorname{erf}(x)$; at $x = 0$ the answer is zero. Both checks fix the additive constant.

1.10.8 $\displaystyle\int_1^x E_1(t)\,dt.$

Obtain a result in which the only special function is E_1, for real $x > 0$.

Solution.

The restriction $x > 0$ keeps the interval between 1 and x inside $t > 0$, where $E_1(t)$ is defined by the integral below; E_1 has a logarithmic singularity at $t = 0$. For $t > 0$, $E_1(t) = \int_1^\infty e^{-ts}\,ds/s = \int_t^\infty e^{-u}\,du/u$ (substitute $u = ts$, so $ds/s = du/u$ and $s = 1$ becomes $u = t$), and hence $E_1'(t) = -e^{-t}/t$. Integration by parts, differentiating $E_1(t)$ and integrating dt to t, then $tE_1'(t) = -e^{-t}$ and $\int_1^x e^{-t}dt = e^{-1} - e^{-x}$, give

$$\int_1^x E_1(t)dt = [tE_1(t)]_1^x - \int_1^x tE_1'(t)dt$$
$$= xE_1(x) - E_1(1) + \int_1^x e^{-t}dt$$
$$= \boxed{xE_1(x) - e^{-x} - E_1(1) + e^{-1}}, \qquad x > 0.$$

Its derivative is $E_1(x) + x(-e^{-x}/x) + e^{-x} = E_1(x)$, and its value at $x = 1$ is zero. Thus it has both the correct derivative and the prescribed lower-limit constant.

1.10.9 $\displaystyle\int_0^\infty \frac{e^{-x}}{x+1}\,dx.$

Solution.

Set $u = x + 1$, so $e^{-x} = e^{1-u} = e\,e^{-u}$ and $dx = du$. The limits are $u = 1$ and $u = \infty$, and therefore, by the definition $E_1(z) = \int_z^\infty e^{-u}du/u$ used in Exercise 1.10.8,

$$\int_0^\infty \frac{e^{-x}}{x+1}dx = e\int_1^\infty \frac{e^{-u}}{u}du = \boxed{eE_1(1) = 0.596347362323194\ldots}.$$

The integral is convergent because its integrand is bounded at zero and bounded above by e^{-x} for $x \geq 0$. The number follows from the convergent series $E_1(x) = -\gamma - \ln x - \sum_{k=1}^\infty (-x)^k/(k\,k!)$ of Exercise 13.6.9 (its derivative is $-e^{-x}/x$, as it must be; this check fixes the series only up to an additive constant, and that exercise derives the constant $-\gamma$ from $\Gamma'(1) = -\gamma$), which at $x = 1$ gives $E_1(1) = -\gamma + 1 - \frac{1}{4} + \frac{1}{18} - \frac{1}{96} + \cdots = 0.2193839343955202\ldots$, with $\gamma = 0.57721566490153286\ldots$ as in Exercise 1.10.5. The terms alternate and decrease, so stopping after $k = 17$ leaves an error below $1/(18\cdot 18!) < 10^{-17}$; multiplying by $e = 2.718281828459045\ldots$ gives the boxed value.

1.10.10 Show that $\displaystyle\int_0^\infty \left(\frac{\tan^{-1} x}{x}\right)^2 dx = \pi \ln 2.$

Hint. Integrate by parts, to linearize in $\tan^{-1}$. Then replace $\tan^{-1} x$ by $\tan^{-1} ax$ and evaluate for $a = 1$.

Solution.

Integrate by parts with $u = (\arctan x)^2$, $du = 2 \arctan x\, dx/(1+x^2)$, and $dv = dx/x^2$, $v = -1/x$:

$$I = \left[-\frac{(\arctan x)^2}{x}\right]_0^\infty + 2\int_0^\infty \frac{\arctan x}{x(1+x^2)} dx.$$

The boundary term vanishes: near zero $|\arctan x| \le |x|$ makes its magnitude at most x, and at infinity it is at most $(\pi/2)^2/x$. Define, for $a \ge 0$,

$$J(a) = 2\int_0^\infty \frac{\arctan(ax)}{x(1+x^2)} dx, \qquad I = J(1), \qquad J(0) = 0.$$

Since $\partial_a \arctan(ax) = x/(1+a^2x^2)$, the a-derivative of the integrand is $2/[(1+x^2)(1+a^2x^2)] \le 2/(1+x^2)$, an integrable bound independent of a; differentiation under the integral is therefore allowed and yields

$$J'(a) = 2\int_0^\infty \frac{dx}{(1+x^2)(1+a^2x^2)}.$$

For $a > 0$, $a \ne 1$, partial fractions give

$$\frac{1}{(1+x^2)(1+a^2x^2)} = \frac{1}{1-a^2}\left(\frac{1}{1+x^2} - \frac{a^2}{1+a^2x^2}\right),$$

as recombining the bracket over the common denominator confirms $(1 + a^2x^2 - a^2(1+x^2) = 1 - a^2)$. Here $\int_0^\infty dx/(1+x^2) = \pi/2$, and the substitution $y = ax$ gives $\int_0^\infty a^2 dx/(1+a^2x^2) = a\int_0^\infty dy/(1+y^2) = a\pi/2$, so

$$J'(a) = \frac{2}{1-a^2}\left(\frac{\pi}{2} - \frac{a\pi}{2}\right) = \frac{\pi(1-a)}{(1-a)(1+a)} = \frac{\pi}{1+a}.$$

The integral $J'(a)$ is continuous in a (by the same dominating bound), which extends this to $a = 0, 1$. Integrating from 0 to a with $J(0) = 0$ gives $J(a) = \int_0^a \pi\, db/(1+b) = \pi \ln(1+a)$, and hence $\boxed{I = \pi \ln 2}$.

1.10.11 By direct integration in Cartesian coordinates, find the area of the ellipse with positive semiaxes a, b, defined by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Solution.

Take $a, b > 0$ (the equation involves only a^2, b^2; for semiaxes of either sign the area is $\pi|ab|$). In the first quadrant the ellipse has upper boundary $y = b\sqrt{1 - x^2/a^2}$, $0 \le x \le a$. Reflection symmetry in both axes gives

$$A = 4b\int_0^a \sqrt{1 - x^2/a^2}\, dx.$$

Use $x = a\sin\theta$, $dx = a\cos\theta\, d\theta$, with $0 \le \theta \le \pi/2$, where $\sqrt{1-\sin^2\theta} = \cos\theta \ge 0$; then $\cos^2\theta = (1 + \cos 2\theta)/2$ gives

$$A = 4ab\int_0^{\pi/2} \cos^2\theta\, d\theta = 2ab\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{\pi/2} = \boxed{\pi ab}.$$

For $a = b = r$ this reduces to the circle area πr^2, checking the scale and numerical coefficient.

1.10.12 A unit circle is divided into two pieces by a straight line whose distance of closest approach to the center is $1/2$ unit. By evaluating a suitable integral, find the area of the smaller piece thereby produced. Then use simple geometric considerations to verify your answer.

Solution.

Rotate coordinates so that the cutting line is $x = 1/2$ in the unit circle $x^2 + y^2 = 1$. The smaller cap, $1/2 \le x \le 1$, has vertical extent $2\sqrt{1-x^2}$. An antiderivative of $2\sqrt{1-x^2}$ is $x\sqrt{1-x^2} + \arcsin x$, since its derivative is $\sqrt{1-x^2} - x^2/\sqrt{1-x^2} + 1/\sqrt{1-x^2} = 2\sqrt{1-x^2}$ (it also follows from $x = \sin\theta$ as in Exercise 1.10.11). With $\arcsin 1 = \pi/2$ and $\arcsin \frac{1}{2} = \pi/6$,

$$\begin{aligned} A &= 2\int_{1/2}^{1} \sqrt{1-x^2}\,dx \\ &= \Big[x\sqrt{1-x^2} + \arcsin x\Big]_{1/2}^{1} = \frac{\pi}{2} - \left(\frac{\sqrt{3}}{4} + \frac{\pi}{6}\right) = \boxed{\frac{\pi}{3} - \frac{\sqrt{3}}{4}} \approx 0.614, \end{aligned}$$

less than half of the area π of the circle, as it should be. For the geometric check, the chord's endpoints $(1/2, \pm\sqrt{3}/2)$ make angles $\pm\pi/3$ with the positive x axis. The sector between them has angle $2\pi/3$ and, the radius being 1, area $\frac{1}{2}\cdot 1^2 \cdot \frac{2\pi}{3} = \frac{\pi}{3}$. The triangle formed by the center and the chord has base $\sqrt{3}$ and altitude $1/2$, hence area $\sqrt{3}/4$. Subtracting it from the sector leaves the cap, with the same area $\pi/3 - \sqrt{3}/4$.

1.11 Dirac Delta Function

1.11.1 Let

$$\delta_n(x) = \begin{cases} 0, & x < -\dfrac{1}{2n}, \\ n, & -\dfrac{1}{2n} < x < \dfrac{1}{2n}, \\ 0, & \dfrac{1}{2n} < x. \end{cases}$$

Show that

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f(x)\delta_n(x)\,dx = f(0),$$

assuming that $f(x)$ is locally integrable and continuous at $x = 0$.

Solution.

Local integrability is assumed so that the integrals exist; continuity at the single point $x = 0$ does not guarantee it. The rectangle has width $1/n$ and height n, so its integral is 1, and $f(0) = n\int_{-1/(2n)}^{1/(2n)} f(0)\,dx$. Subtracting this under the integral, and bounding the integrand by its supremum over an interval of length $1/n$, gives the useful explicit bound

$$\begin{aligned} \left|\int_{\mathbb{R}} f(x)\delta_n(x)dx - f(0)\right| &= \left|n\int_{-1/(2n)}^{1/(2n)} [f(x) - f(0)]dx\right| \\ &\le n\cdot\frac{1}{n}\sup_{|x|\le 1/(2n)} |f(x) - f(0)| = \sup_{|x|\le 1/(2n)} |f(x) - f(0)|. \end{aligned}$$

Continuity at zero makes the supremum tend to zero: for each $\epsilon > 0$, take $\eta > 0$ with $|f(x) - f(0)| < \epsilon$ when $|x| < \eta$, then take $n > 1/(2\eta)$. This proves the desired limit. The unspecified values at the two endpoints do not affect an integral.

1.11.2 For

$$\delta_n(x) = \frac{n}{\pi}\,\frac{1}{1+n^2x^2},$$

show that

$$\int_{-\infty}^{\infty} \delta_n(x)\,dx = 1.$$

Solution.

For $n > 0$ use $u = nx$, so $dx = du/n$ and the limits stay $\mp\infty$. Then

$$\int_{-\infty}^{\infty} \delta_n(x)dx = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{du}{1+u^2} = \frac{1}{\pi}[\arctan u]_{-\infty}^{\infty} = \frac{1}{\pi}\left(\frac{\pi}{2}+\frac{\pi}{2}\right) = 1.$$

This proves normalization for each n, not merely in the limit. It also displays why the numerator must contain the factor n.

1.11.3 Fejér's method of summing series is associated with the function

$$\delta_n(t) = \frac{1}{2\pi n}\left[\frac{\sin(nt/2)}{\sin(t/2)}\right]^2.$$

Show that $\delta_n(t)$ is a delta distribution on one period, for continuous f, in the sense that

$$\lim_{n\to\infty} \frac{1}{2\pi n}\int_{-\pi}^{\pi} f(t)\left[\frac{\sin(nt/2)}{\sin(t/2)}\right]^2 dt = f(0).$$

Solution.

The printed whole-line assertion is false. On one period, $\int_{-\pi}^{\pi} f(t)\delta_n(t)dt \to f(0)$ for continuous f. On the whole real line the distributional limit is instead the comb $\sum_{m\in\mathbb{Z}} \delta(t-2\pi m)$. For integer $n \geq 1$, the geometric sum of Exercise 1.8.4 gives $\left|\sum_{j=0}^{n-1} e^{ijt}\right| = |\sin(nt/2)/\sin(t/2)|$, since the phase factor $e^{i(n-1)t/2}$ has modulus 1. Expanding the modulus square as a double sum, and noting that exactly $n-|k|$ pairs (j,l) with $0 \leq j,l \leq n-1$ have $j-l=k$,

$$\delta_n(t) = \frac{1}{2\pi n}\left|\sum_{j=0}^{n-1} e^{ijt}\right|^2 = \frac{1}{2\pi n}\sum_{j,l=0}^{n-1} e^{i(j-l)t} = \frac{1}{2\pi}\sum_{k=-(n-1)}^{n-1}\left(1-\frac{|k|}{n}\right)e^{ikt}.$$

The modulus-square expression proves $\delta_n \geq 0$; integrating the finite Fourier sum over $[-\pi,\pi]$, where $\int_{-\pi}^{\pi} e^{ikt}dt = 0$ for $k \neq 0$, leaves only its constant term, giving $\int_{-\pi}^{\pi}\delta_n(t)dt = 1$. For any $0 < \eta < \pi$, if $\eta \leq |t| \leq \pi$ then $\sin^2(t/2) \geq \sin^2(\eta/2)$ and $\sin^2(nt/2) \leq 1$, so, the region having length less than 2π,

$$0 \leq \delta_n(t) \leq \frac{1}{2\pi n\sin^2(\eta/2)}, \qquad \int_{\eta\leq|t|\leq\pi}\delta_n(t)dt \leq \frac{1}{n\sin^2(\eta/2)} \to 0.$$

By the normalization, $\int_{-\pi}^{\pi} f(t)\delta_n(t)dt - f(0) = \int_{-\pi}^{\pi}[f(t)-f(0)]\delta_n(t)dt$. Split this integral into $|t| < \eta$ and its complement. Since $\delta_n \geq 0$ has total integral 1, the first part is at most $\sup_{|t|<\eta}|f(t)-f(0)|$, which continuity at 0 makes smaller than any given $\epsilon > 0$ once η is small enough. A function continuous on $[-\pi,\pi]$ is bounded, say $|f| \leq M$, so for this η the second part is at most $2M/[n\sin^2(\eta/2)]$, which is below ϵ for all large n. This proves the one-period result. For a smooth compactly supported test function on the whole line, split $\mathbb{R}$ into the periods $[(2m-1)\pi,(2m+1)\pi]$ and

use the 2π-periodicity of δ_n; the one-period result, applied to each continuous function $s \mapsto f(s+2\pi m)$ on $[-\pi, \pi]$, gives

$$\int_{\mathbb{R}} f(t)\delta_n(t)dt = \sum_{m\in\mathbb{Z}} \int_{-\pi}^{\pi} f(s+2\pi m)\delta_n(s)\,ds \longrightarrow \sum_{m\in\mathbb{Z}} f(2\pi m) \quad (n\to\infty).$$

Only finitely many terms occur for such a test function. Choosing it supported near 2π, with $f(2\pi) = 1$ and $f(0) = 0$, directly contradicts the original claim and verifies the correction.

1.11.4 Prove that

$$\delta[a(x-x_1)] = \frac{1}{|a|}\,\delta(x-x_1), \qquad a \neq 0.$$

Note. With the absolute value the relation holds for every real $a \neq 0$; the factor $1/a$ is correct only for $a > 0$.

Solution.

For every real $a \neq 0$ the factor is $1/|a|$, as computed below, so the textbook's $1/a$ is wrong for $a < 0$; the evenness of δ, $\int \phi(x)\delta(-x)\,dx = \phi(0)$, is a property of the Dirac distribution, not an optional convention. Test the identity against a smooth compactly supported ϕ and make the change of variable $u = a(x - x_1)$, so $x = x_1 + u/a$ and $dx = du/a$. For $a > 0$ the limits $x = \mp\infty$ become $u = \mp\infty$:

$$\int_{-\infty}^{\infty} \phi(x)\delta(a(x-x_1))dx = \frac{1}{a}\int_{-\infty}^{\infty} \phi(x_1+u/a)\delta(u)du = \frac{\phi(x_1)}{a}.$$

For $a < 0$ they become $u = \pm\infty$, and restoring their order costs a sign:

$$\int_{-\infty}^{\infty} \phi(x)\delta(a(x-x_1))dx = \frac{1}{a}\int_{\infty}^{-\infty} \phi(x_1+u/a)\delta(u)du = -\frac{1}{a}\int_{-\infty}^{\infty} \phi(x_1+u/a)\delta(u)du = \frac{\phi(x_1)}{|a|}.$$

In both cases the result is $\phi(x_1)/|a|$, the action of $\delta(x-x_1)/|a|$, proving the distribution identity. For $a > 0$ it reduces to the printed factor $1/a$. The value $a = 0$ is excluded: then $a(x - x_1) = 0$ for every x, the substitution $u = a(x - x_1)$ cannot be inverted, and $\delta(a(x - x_1))$ would be the meaningless constant $\delta(0)$; correspondingly, $1/|a|$ is undefined.

1.11.5 For distinct real x_1, x_2, show that

$$\delta[(x-x_1)(x-x_2)] = [\delta(x-x_1) + \delta(x-x_2)]/|x_1 - x_2|.$$

Hint. Try using Exercise 1.11.4.

Solution.

The roots must be real, since $\delta(x - x_i)$ is defined only for real x_i (for complex conjugate roots $(x-x_1)(x-x_2) = |x-x_1|^2 > 0$ for real x, so the left side is zero), and distinct, since for $x_1 = x_2$ the right side has the zero denominator $|x_1-x_2|$ (see the end of the solution). Let $g(x) = (x-x_1)(x-x_2)$, so that

$$g'(x) = 2x - x_1 - x_2, \quad g'(x_1) = x_1 - x_2, \quad g'(x_2) = x_2 - x_1.$$

Since g' vanishes only at the midpoint $(x_1 + x_2)/2$, each root x_i has a neighborhood U_i on which g is strictly monotone, with an inverse $x = x_i(u)$ and $dx = du/g'(x_i(u))$. Let ϕ be a test function. On the rest of its support g has no zeros, so $|g|$ is bounded away from zero there and $\delta(g(x))$ contributes nothing. On U_i substitute $u = g(x)$: as in Exercise 1.11.4, the limits are reversed when $g'(x_i) < 0$,

and in either case $\int_{U_i} \phi(x)\delta(g(x))dx = \int \phi(x_i(u))\delta(u)\,du/|g'(x_i(u))| = \phi(x_i)/|g'(x_i)|$. Adding the two contributions,

$$\int_{\mathbb{R}} \phi(x)\delta(g(x))dx = \frac{\phi(x_1)}{|x_1 - x_2|} + \frac{\phi(x_2)}{|x_2 - x_1|}.$$

Exactly one of $g'(x_1)$, $g'(x_2)$ is negative, so the absolute values are essential. Recognizing the two evaluation functionals proves

$$\delta((x - x_1)(x - x_2)) = \frac{\delta(x - x_1) + \delta(x - x_2)}{|x_1 - x_2|}.$$

For coincident roots the Jacobian vanishes and this formula does not apply; its singular denominator cannot simply be assigned a value.

1.11.6 Using the Gauss error curve delta sequence $\delta_n = \dfrac{n}{\sqrt{\pi}} e^{-n^2x^2}$, show that

$$x\frac{d}{dx}\delta(x) = -\delta(x),$$

treating $\delta(x)$ and its derivative as in Eq. (1.157).

Solution.

With the delta sequence, the claim means that $\lim_{n\to\infty} \int x\delta_n'(x)\phi(x)dx = -\phi(0)$ for every smooth compactly supported test function ϕ. For the specified Gaussian sequence,

$$\delta_n'(x) = \frac{n}{\sqrt{\pi}}(-2n^2x)e^{-n^2x^2} = -\frac{2n^3x}{\sqrt{\pi}} e^{-n^2x^2}.$$

Substitute $y = nx$, so $x^2 = y^2/n^2$ and $dx = dy/n$; the powers of n cancel ($n^3 \cdot n^{-2} \cdot n^{-1} = 1$), giving

$$\int_{\mathbb{R}} x\delta_n'(x)\phi(x)dx = -\frac{2}{\sqrt{\pi}} \int_{\mathbb{R}} x^2 n^3 e^{-n^2x^2} \phi(x)dx = -\frac{2}{\sqrt{\pi}} \int_{\mathbb{R}} y^2 e^{-y^2} \phi(y/n)dy.$$

As $n \to \infty$, $\phi(y/n) \to \phi(0)$ for every y, and dominated convergence, with the integrable bound $My^2e^{-y^2}$, where $M = \max_x |\phi(x)|$, lets the limit pass inside the integral. Integration by parts with $u = y$, $dv = ye^{-y^2}dy$, $v = -e^{-y^2}/2$ gives

$$\int_{\mathbb{R}} y^2 e^{-y^2} dy = -\frac{1}{2}[ye^{-y^2}]_{-\infty}^{\infty} + \frac{1}{2}\int_{\mathbb{R}} e^{-y^2} dy = \frac{\sqrt{\pi}}{2}.$$

Hence the limit is $-(2/\sqrt{\pi})(\sqrt{\pi}/2)\phi(0) = -\phi(0)$, exactly the action of $-\delta$. This proves $x\delta' = -\delta$. As an independent check, $x\delta(x) = 0$, since its action on ϕ is $[x\phi(x)]_{x=0} = 0$; the product rule then gives $0 = (x\delta)' = \delta + x\delta'$.

1.11.7 Show that

$$\int_{-\infty}^{\infty} \delta'(x) f(x)\,dx = -f'(0).$$

Here we assume that $f'(x)$ is continuous at $x = 0$.

Solution.

A distributional derivative is defined by its action on test functions, $\langle T', f\rangle = -\langle T, f'\rangle$, where $\langle T, f\rangle$ stands for $\int_{\mathbb{R}} T(x)f(x)\,dx$, the action of T on f (not an average), so that $\langle \delta, f\rangle = f(0)$; for

a differentiable function T this definition is ordinary integration by parts. Taking $T = \delta$, so that $-\langle \delta, f' \rangle = -f'(0)$, gives

$$\boxed{\int_{\mathbb{R}} \delta'(x) f(x) dx = -f'(0)}.$$

To check this from regular functions, use a smooth delta sequence δ_n and a continuously differentiable f that vanishes outside a bounded interval. Ordinary integration by parts gives

$$\int_{\mathbb{R}} \delta_n'(x) f(x) dx = [\delta_n(x) f(x)]_{-\infty}^{\infty} - \int_{\mathbb{R}} \delta_n(x) f'(x) dx = -\int_{\mathbb{R}} \delta_n(x) f'(x) dx.$$

The boundary term vanishes because f vanishes outside a bounded interval. Since f' is continuous at zero (and bounded), the delta-sequence property makes the last integral tend to $-f'(0)$, reproducing the distributional definition and fixing the minus sign. Only the values near zero matter. If f is continuously differentiable near zero, multiply it by a smooth cutoff χ that equals 1 near zero and vanishes outside a small interval; then χf is of the kind just treated, and $(\chi f)'(0) = f'(0)$ because $\chi'(0) = 0$ and $\chi(0) = 1$. Since δ' is supported at zero, the cutoff leaves its action unchanged. The identity therefore needs only this regularity of f near $x = 0$ and imposes no behavior on f at infinity.

1.11.8 Prove that

$$\delta(f(x)) = \frac{1}{|f'(x_0)|} \delta(x - x_0),$$

where f is real and C^1, and x_0 is its unique simple real zero, so $f(x_0) = 0$ and $f'(x_0) \neq 0$.

Hint. Use the local variable $u = f(x)$ and the absolute Jacobian $1/|f'(x_0)|$ at each simple zero.

Solution.

For a unique simple real zero x_0, the factor is $1/|f'(x_0)|$, as computed below; a factor $1/f'(x_0)$ without the absolute value would have the wrong sign when $f'(x_0) < 0$. For several simple zeros, their contributions add. Let f be a real C^1 function on the domain under consideration, with $f'(x_i) \neq 0$ at every zero x_i. By continuity, f' keeps the sign of $f'(x_i)$ on a neighborhood U_i of x_i, so $u = f(x)$ is strictly monotone there, with a local inverse $x = x_i(u)$ (the inverse function theorem) and $dx = du/f'(x_i(u))$. Restricting a test function ϕ to this neighborhood and changing variables gives

$$\int_{U_i} \phi(x) \delta(f(x)) dx = \int \frac{\phi(x_i(u))}{|f'(x_i(u))|} \delta(u) du = \frac{\phi(x_i)}{|f'(x_i)|};$$

for $f'(x_i) < 0$ the substitution reverses the limits, and restoring their order turns $1/f'$ into $1/|f'|$, as in Exercise 1.11.4. Simple zeros are isolated, so the compact support of ϕ contains only finitely many of them; on the rest of the support f has no zeros, so $|f|$ is bounded away from zero and $\delta(f(x))$ contributes nothing. Adding the contributions of the zeros yields

$$\boxed{\delta(f(x)) = \sum_{i : f(x_i) = 0} \frac{\delta(x - x_i)}{|f'(x_i)|}}.$$

For the unique simple zero x_0 of the statement the sum has the single term $\delta(x - x_0)/|f'(x_0)|$, as claimed. Without the absolute value the formula holds only when $f'(x_0) > 0$; a negative derivative reverses the limits, which is why the absolute value is necessary. A repeated zero has $f'(x_i) = 0$ and lies outside this formula's hypotheses.

1.11.9

(a) If we define a sequence $\delta_n(x) = n/(2\cosh^2 nx)$, show that

$$\int_{-\infty}^{\infty} \delta_n(x)\,dx = 1, \quad \text{independent of } n.$$

(b) Continuing this analysis, show that[2]

$$\int_{-\infty}^{x} \delta_n(t)\,dt = \frac{1}{2}[1 + \tanh nx] \equiv u_n(x)$$

and

$$\lim_{n\to\infty} u_n(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases}$$

This is the Heaviside unit step function (Fig. 1.23).

Solution.

(a) For $n > 0$, substitute $u = nx$, $du = n\,dx$, with the limits unchanged; since $d(\tanh u)/du = \operatorname{sech}^2 u$ and $\tanh u \to \pm 1$ as $u \to \pm\infty$,

$$\int_{-\infty}^{\infty} \frac{n}{2\cosh^2(nx)} dx = \frac{1}{2}\int_{-\infty}^{\infty} \operatorname{sech}^2 u\,du = \frac{1}{2}[\tanh u]_{-\infty}^{\infty} = \frac{1}{2}[1 - (-1)] = 1.$$

(b) The statement uses the dummy variable t, because x is the upper limit of the integral and cannot also be the variable of integration. Since $d(\tanh nt)/dt = n\operatorname{sech}^2(nt)$,

$$u_n(x) = \int_{-\infty}^{x} \delta_n(t)dt = \frac{1}{2}[\tanh(nt)]_{t=-\infty}^{t=x} = \frac{1}{2}[1 + \tanh(nx)].$$

If $x > 0$, $nx \to +\infty$ and $u_n(x) \to 1$; if $x < 0$, $nx \to -\infty$ and $u_n(x) \to 0$. At $x = 0$ every $u_n(0) = 1/2$, so this symmetric sequence selects the convention $u(0) = 1/2$. The value at a single point does not affect its distributional derivative, which is δ.

[2] Many other symbols are used for this function. This is the AMS-55 notation (in Additional Readings, see Abramowitz and Stegun): u for unit.

2 Chapter 2. Determinants and Matrices

Problems

2.1 Determinants

2.1.1 Evaluate the following determinants:

(a) $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$,

(b) $\begin{vmatrix} 1 & 2 & 0 \\ 3 & 1 & 2 \\ 0 & 3 & 1 \end{vmatrix}$,

(c) $\dfrac{1}{\sqrt{2}} \begin{vmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{vmatrix}$.

Solution.

Expand along a row or column containing as many zeros as possible.

(a) Only the middle entry of row 2 contributes:

$$D_a = 1(-1)^{2+2} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = 1(1 \cdot 0 - 1 \cdot 1) = \boxed{-1}.$$

(b) Expansion along row 1 gives

$$D_b = 1 \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} - 2 \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} = 1(1-6) - 2(3-0) = \boxed{-11}.$$

(c) Write $s = \sqrt{3}$. The factor $1/\sqrt{2}$ in the question multiplies the determinant once; it is not a factor on each row. Expand along row 1, whose only nonzero entry is $a_{12} = s$ with sign $(-1)^{1+2} = -1$, and then along column 1 of the resulting minor, whose only nonzero entry is s in position $(1,1)$:

$$D_c = -\frac{s}{\sqrt{2}} \begin{vmatrix} s & 2 & 0 \\ 0 & 0 & s \\ 0 & s & 0 \end{vmatrix} = -\frac{s^2}{\sqrt{2}} \begin{vmatrix} 0 & s \\ s & 0 \end{vmatrix}$$

$$= -\frac{3}{\sqrt{2}}(0 - 3) = \boxed{\frac{9}{\sqrt{2}}}.$$

2.1.2 Test the set of linear homogeneous equations

$$x + 3y + 3z = 0, \quad x - y + z = 0, \quad 2x + y + 3z = 0$$

to see if it possesses a nontrivial solution. In any case, find a solution to this equation set.

Solution.

The coefficient determinant, expanded along row 1, is

$$\begin{aligned} D &= \begin{vmatrix} 1 & 3 & 3 \\ 1 & -1 & 1 \\ 2 & 1 & 3 \end{vmatrix} \\ &= 1(-3-1) - 3(3-2) + 3(1+2) = -4 - 3 + 9 = 2. \end{aligned}$$

Since $D \neq 0$, only the trivial solution is possible. Directly, the second equation gives $x = y - z$. The first and third then become

$$(y - z) + 3y + 3z = 4y + 2z = 0, \qquad 2(y - z) + y + 3z = 3y + z = 0.$$

Thus $z = -2y$ and $z = -3y$, so $y = 0$, $z = 0$, and $x = 0$: $\boxed{(x, y, z) = (0, 0, 0)}$. Substitution into each original equation gives zero.

2.1.3 Given the pair of equations

$$x + 2y = 3, \quad 2x + 4y = 6,$$

(a) Show that the determinant of the coefficients vanishes.

(b) Show that the numerator determinants, see Eq. (2.18), also vanish.

(c) Find at least two solutions.

Solution.

(a), (b) The coefficient determinant D and the numerator determinants D_x, D_y of Cramer's rule are

$$D = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0, \quad D_x = \begin{vmatrix} 3 & 2 \\ 6 & 4 \end{vmatrix} = 12 - 12 = 0, \quad D_y = \begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} = 6 - 6 = 0.$$

(c) The quotients $D_x/D, D_y/D$ are undefined and cannot determine a unique solution. The second equation is twice the first, so put $y = t$. Then $x = 3 - 2t$, and every real solution is

$$\boxed{(x, y) = (3 - 2t, t), \quad t \in \mathbb{R}}.$$

(Complex t is allowed if complex solutions are sought.) For example $t = 0$ gives $(3, 0)$, and $t = 1$ gives $(1, 1)$. Both satisfy $x + 2y = 3$ and hence $2x + 4y = 6$.

2.1.4 If C_{ij} is the cofactor of element a_{ij}, formed by striking out the ith row and jth column and including a sign $(-1)^{i+j}$, show that

(a) $\displaystyle\sum_i a_{ij}C_{ij} = \sum_i a_{ji}C_{ji} = |\mathsf{A}|$, where $|\mathsf{A}|$ is the determinant with the elements a_{ij},

(b) $\displaystyle\sum_i a_{ij}C_{ik} = \sum_i a_{ji}C_{ki} = 0$, $j \neq k$.

Solution.

Let $A_{\widehat{i,j}}$ denote A with row i and column j struck out, so that $C_{ij} = (-1)^{i+j} \det A_{\widehat{i,j}}$. For fixed j, Laplace expansion down column j is $\det A = \sum_i a_{ij}(-1)^{i+j} \det A_{\widehat{i,j}} = \sum_i a_{ij}C_{ij}$. Expansion across row j similarly gives $\det A = \sum_i a_{ji}C_{ji}$. This proves (a), without assuming invertibility.

For (b), replace column k of A by its column j, with $j \neq k$, and call the result B. Columns j and k of B are identical, so $\det B = 0$. Expanding down column k, each minor is unchanged from A because that column is deleted; its cofactor is C_{ik}. The entry being multiplied by that cofactor is now a_{ij}. Therefore

$$0 = \det B = \sum_i a_{ij}C_{ik}.$$

Replacing row k by row j and expanding across row k gives, by the same argument, $0 = \sum_i a_{ji}C_{ki}$.

2.1.5 A determinant with all elements of order unity may be surprisingly small. The Hilbert determinant $H_{ij} = (i + j - 1)^{-1}$, $i, j = 1, 2, \ldots, n$ is notorious for its small values.

(a) Calculate the value of the Hilbert determinants of order n for $n = 1$, 2, and 3.

(b) If an appropriate subroutine is available, find the Hilbert determinants of order n for $n = 4$, 5, and 6.

ANS.

n	$\mathrm{Det}(H_n)$
1	1.
2	8.33333×10^{-2}
3	4.62963×10^{-4}
4	1.65344×10^{-7}
5	3.74930×10^{-12}
6	5.36730×10^{-18}.

Solution.

(a) Use exact fractions so that small determinants are not lost by subtracting rounded numbers. The first three determinants (D_3 expanded along row 1) are

$$D_1 = 1, \qquad D_2 = 1 \cdot \frac{1}{3} - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{12},$$

$$\begin{aligned} D_3 &= \left(\frac{1}{15} - \frac{1}{16}\right) - \frac{1}{2}\left(\frac{1}{10} - \frac{1}{12}\right) + \frac{1}{3}\left(\frac{1}{8} - \frac{1}{9}\right) \\ &= \frac{1}{240} - \frac{1}{120} + \frac{1}{216} = \frac{9 - 18 + 10}{2160} = \frac{1}{2160}. \end{aligned}$$

(b) For the larger matrices, Gaussian elimination uses

$$R_i \longleftarrow R_i - \frac{u_{ik}}{u_{kk}} R_k \quad (i = k + 1, \ldots, n;\ k = 1, \ldots, n - 1),$$

where u_{ik} is the current entry in row i, column k; row k no longer changes after step $k-1$, so u_{kk} is the kth pivot. Each operation preserves the determinant. For example, the first step on H_6 changes an entry with $i, j \geq 2$ into

$$\frac{1}{i+j-1} - \frac{1}{i}\frac{1}{j} = \frac{(i-1)(j-1)}{ij(i+j-1)}.$$

Every later step has the same structure. Suppose that before step k the entries with $i, j \geq k$ have the form $\alpha_i\beta_j/(i+j-1)$; for H_6 itself $\alpha_i = \beta_j = 1$. The multiplier for row i is then

$$\frac{u_{ik}}{u_{kk}} = \frac{\alpha_i\beta_k/(i+k-1)}{\alpha_k\beta_k/(2k-1)} = \frac{\alpha_i}{\alpha_k}\,\frac{2k-1}{i+k-1}.$$

Subtracting it times the row-k entry $\alpha_k\beta_j/(k+j-1)$ and using the identity (expand both products)

$$(i+k-1)(j+k-1) - (2k-1)(i+j-1) = ij - k(i+j) + k^2 = (i-k)(j-k)$$

turns the entry in row i, column j into

$$\alpha_i\beta_j\left[\frac{1}{i+j-1} - \frac{2k-1}{(i+k-1)(j+k-1)}\right] = \frac{\alpha_i\beta_j\,(i-k)(j-k)}{(i+k-1)(j+k-1)(i+j-1)}.$$

The form is preserved, with α_i multiplied by $(i-k)/(i+k-1)$ and β_j by $(j-k)/(j+k-1)$; for $k=1$ this is the formula above. Hence at step k

$$\alpha_k = \beta_k = \prod_{m=1}^{k-1}\frac{k-m}{k+m-1} = \frac{[(k-1)!]^2}{(2k-2)!}, \qquad u_{kk} = \frac{\alpha_k\beta_k}{2k-1} = \frac{[(k-1)!]^4}{(2k-2)!\,(2k-1)!},$$

because $\prod_{m=1}^{k-1}(k-m) = (k-1)!$, $\prod_{m=1}^{k-1}(k+m-1) = k(k+1)\cdots(2k-2) = (2k-2)!/(k-1)!$, and $(2k-2)!\,(2k-1) = (2k-1)!$. Successive pivots therefore have the ratio

$$\frac{u_{k+1,k+1}}{u_{kk}} = \frac{k^4}{(2k-1)(2k)\cdot(2k)(2k+1)} = \frac{k^2}{4(4k^2-1)} = \frac{1}{12},\ \frac{1}{15},\ \frac{9}{140},\ \frac{4}{63},\ \frac{25}{396} \quad (k=1,\ldots,5),$$

which, starting from $u_{11} = 1$, gives the pivots 1, 1/12, 1/180, 1/2800, 1/44100, 1/698544 for $k = 1, \ldots, 6$. The other entries of row k of the final triangular matrix are $u_{kj} = \alpha_k\beta_j/(k+j-1)$ for $j > k$, with $\beta_j = \prod_{m=1}^{k-1}(j-m)/(j+m-1)$. For completeness, the multipliers u_{ik}/u_{kk} in the successive steps are listed below. At step k, subtract the indicated multiple of the current row k from each listed row i:

k	i	u_{ik}/u_{kk} in the same order
1	2, 3, 4, 5, 6	1/2, 1/3, 1/4, 1/5, 1/6
2	3, 4, 5, 6	1, 9/10, 4/5, 5/7
3	4, 5, 6	3/2, 12/7, 25/14
4	5, 6	2, 25/9
5	6	5/2

For example, the second pivot is 1/12 and the entry below it in row 4 is 3/40, giving multiplier $(3/40)/(1/12) = 9/10$. These operations produce

$$U_6 = \begin{pmatrix} 1 & 1/2 & 1/3 & 1/4 & 1/5 & 1/6 \\ 0 & 1/12 & 1/12 & 3/40 & 1/15 & 5/84 \\ 0 & 0 & 1/180 & 1/120 & 1/105 & 5/504 \\ 0 & 0 & 0 & 1/2800 & 1/1400 & 1/1008 \\ 0 & 0 & 0 & 0 & 1/44100 & 1/17640 \\ 0 & 0 & 0 & 0 & 0 & 1/698544 \end{pmatrix}.$$

Elimination on a leading $n \times n$ block uses the same leading pivots, so D_n is the product of the first n diagonal entries. In particular

$$D_4 = \frac{D_3}{2800}, \qquad D_5 = \frac{D_4}{44100}, \qquad D_6 = \frac{D_5}{698544}.$$

The resulting exact values and six-significant-figure approximations are

n	D_n	decimal value
1	1	1
2	1/12	8.33333×10^{-2}
3	1/2160	4.62963×10^{-4}
4	1/6048000	1.65344×10^{-7}
5	1/266716800000	3.74930×10^{-12}
6	1/186313420339200000	5.36730×10^{-18}

Equivalently, $D_n = \prod_{k=1}^{n} [(k-1)!]^4 / [(2k-2)!\,(2k-1)!]$. As an independent check, evaluating $\det H_n$ directly in exact rational arithmetic (for instance by cofactor expansion in a computer-algebra system) reproduces the same six fractions.

2.1.6 Prove that the determinant consisting of the coefficients from a set of linearly dependent forms has the value zero.

Solution.

Write the n forms as $L_i(x) = \sum_{j=1}^{n} a_{ij} x_j$. Linear dependence means that constants c_i, not all zero, satisfy $\sum_i c_i L_i(x) = 0$ for every x. Since $\sum_i c_i L_i(x) = \sum_j \big(\sum_i c_i a_{ij}\big) x_j$, the choice $x = e_j$ (the jth unit vector) gives $\sum_i c_i a_{ij} = 0$ for each j; that is, the rows $R_i = (a_{i1}, \ldots, a_{in})$ of the coefficient matrix A satisfy $\sum_i c_i R_i = 0$. Choose r with $c_r \neq 0$. Then the row vectors obey

$$R_r = -\sum_{i \neq r} \frac{c_i}{c_r} R_i.$$

Replace R_r by $R_r + \sum_{i \neq r} (c_i/c_r) R_i$. Adding multiples of other rows leaves the determinant unchanged, but the new row is zero. Expansion across it therefore gives $\boxed{\det A = 0}$. The same proof applies if the forms are recorded as columns.

2.1.7 Solve the following set of linear simultaneous equations. Give the results to five decimal places.

$$\begin{aligned}
1.0x_1 + 0.9x_2 + 0.8x_3 + 0.4x_4 + 0.1x_5 &= 1.0 \\
0.9x_1 + 1.0x_2 + 0.8x_3 + 0.5x_4 + 0.2x_5 + 0.1x_6 &= 0.9 \\
0.8x_1 + 0.8x_2 + 1.0x_3 + 0.7x_4 + 0.4x_5 + 0.2x_6 &= 0.8 \\
0.4x_1 + 0.5x_2 + 0.7x_3 + 1.0x_4 + 0.6x_5 + 0.3x_6 &= 0.7 \\
0.1x_1 + 0.2x_2 + 0.4x_3 + 0.6x_4 + 1.0x_5 + 0.5x_6 &= 0.6 \\
0.1x_2 + 0.2x_3 + 0.3x_4 + 0.5x_5 + 1.0x_6 &= 0.5.
\end{aligned}$$

Note. These equations may also be solved by matrix inversion, as discussed in Section 2.2.

Solution.

Multiply every equation by 10 and put them in the order $5, 6, 3, 4, 1, 2$:

$$\left[\begin{array}{cccccc|c} 1 & 2 & 4 & 6 & 10 & 5 & 6 \\ 0 & 1 & 2 & 3 & 5 & 10 & 5 \\ 8 & 8 & 10 & 7 & 4 & 2 & 8 \\ 4 & 5 & 7 & 10 & 6 & 3 & 7 \\ 10 & 9 & 8 & 4 & 1 & 0 & 10 \\ 9 & 10 & 8 & 5 & 2 & 1 & 9 \end{array}\right].$$

Here R_i refers to this new ordering. Apply $R_3 - 8R_1$, $R_4 - 4R_1$, $R_5 - 10R_1$, and $R_6 - 9R_1$ to rows $3, 4, 5, 6$. The last four rows, omitting the zero first column, become

$$\left[\begin{array}{ccccc|c} -8 & -22 & -41 & -76 & -38 & -40 \\ -3 & -9 & -14 & -34 & -17 & -17 \\ -11 & -32 & -56 & -99 & -50 & -50 \\ -8 & -28 & -49 & -88 & -44 & -45 \end{array}\right].$$

Next add $8R_2, 3R_2, 11R_2, 8R_2$ to rows $3, 4, 5, 6$, respectively, which clears the x_2 column. In columns x_3, x_4, x_5, x_6 this gives

$$\left[\begin{array}{cccc|c} -6 & -17 & -36 & 42 & 0 \\ -3 & -5 & -19 & 13 & -2 \\ -10 & -23 & -44 & 60 & 5 \\ -12 & -25 & -48 & 36 & -5 \end{array}\right].$$

To eliminate x_3, replace

$$R_4 \leftarrow -2R_4 + R_3, \quad R_5 \leftarrow -3R_5 + 5R_3, \quad R_6 \leftarrow -R_6 + 2R_3.$$

Rows $4, 5, 6$, in columns x_4, x_5, x_6, are now

$$\left[\begin{array}{ccc|c} -7 & 2 & 16 & 4 \\ -16 & -48 & 30 & -15 \\ -9 & -24 & 48 & 5 \end{array}\right].$$

Eliminate x_4 by $R_5 \leftarrow -7R_5 + 16R_4$ and $R_6 \leftarrow -7R_6 + 9R_4$. The final two equations are

$$368x_5 + 46x_6 = 169, \qquad 186x_5 - 192x_6 = 1.$$

Finally $R_6 \leftarrow (368R_6 - 186R_5)/14$ gives $-5658x_6 = -2219$, because $368(-192) - 186 \cdot 46 = -79212 = 14(-5658)$ and $368 \cdot 1 - 186 \cdot 169 = -31066 = 14(-2219)$. Hence $x_6 = 2219/5658$. To back-substitute without fractions, write $x_r = y_r/5658$ and use, in turn, $368x_5 + 46x_6 = 169$, row 4 $(-7x_4 + 2x_5 + 16x_6 = 4)$, row 3 $(-6x_3 - 17x_4 - 36x_5 + 42x_6 = 0)$, and rows 2 and 1 of the reordered matrix:

$$\begin{aligned} y_6 &= 2219, \\ y_5 &= \frac{169 \cdot 5658 - 46y_6}{368} = \frac{956202 - 102074}{368} = 2321, \\ y_4 &= \frac{2y_5 + 16y_6 - 4 \cdot 5658}{7} = \frac{4642 + 35504 - 22632}{7} = 2502, \\ y_3 &= \frac{-17y_4 - 36y_5 + 42y_6}{6} = \frac{-42534 - 83556 + 93198}{6} = -5482, \\ y_2 &= 5 \cdot 5658 - 2y_3 - 3y_4 - 5y_5 - 10y_6 \\ &= 28290 + 10964 - 7506 - 11605 - 22190 = -2047, \\ y_1 &= 6 \cdot 5658 - 2y_2 - 4y_3 - 6y_4 - 10y_5 - 5y_6 \\ &= 33948 + 4094 + 21928 - 15012 - 23210 - 11095 = 10653. \end{aligned}$$

Since $5658 = 2 \cdot 3 \cdot 23 \cdot 41$ and the common factors of 5658 with $y_1, \ldots, y_6$ are 3, 23, 2, 6, 1, 1, the fractions $y_r/5658$ reduce to

$$x = \Big(\frac{3551}{1886}, -\frac{89}{246}, -\frac{2741}{2829}, \frac{417}{943}, \frac{2321}{5658}, \frac{2219}{5658}\Big)^T.$$

Thus, to five decimal places,

$$\boxed{x = (1.88282, -0.36179, -0.96889, 0.44221, 0.41022, 0.39219)^T}.$$

For an exact check, let B be ten times the original coefficient matrix and multiply it into the vector of numerators y_r:

$$\begin{aligned} B(10653, -2047, -5482, 2502, 2321, 2219)^T &= (56580, 50922, 45264, 39606, 33948, 28290)^T \\ &= 5658(10, 9, 8, 7, 6, 5)^T, \end{aligned}$$

verifying all six equations without rounding.

2.1.8 Show that (in 3-D space)

(a) $\displaystyle\sum_i \delta_{ii} = 3,$

(b) $\displaystyle\sum_{ij} \delta_{ij}\varepsilon_{ijk} = 0,$

(c) $\displaystyle\sum_{pq} \varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij},$

(d) $\displaystyle\sum_{ijk} \varepsilon_{ijk}\varepsilon_{ijk} = 6.$

Note. The symbol δ_{ij} is the Kronecker delta, defined in Eq. (1.164), and ε_{ijk} is the Levi-Civita symbol, Eq. (2.8).

Solution.

(a) Since $\delta_{11} = \delta_{22} = \delta_{33} = 1$, $\sum_i \delta_{ii} = 1 + 1 + 1 = 3$.

(b) The factor δ_{ij} eliminates every term with $i \neq j$, so $\sum_{ij} \delta_{ij}\varepsilon_{ijk} = \sum_i \varepsilon_{iik} = 0$. Every remaining Levi-Civita symbol has a repeated index.

(c) If $i = j$, the only nonzero terms have (p, q) equal to the two orderings of the other two indices. Their squares are both 1, so the sum is 2. If $i \neq j$, a nonzero ε_{ipq} requires $\{i, p, q\} = \{1, 2, 3\}$, so $j \neq i$ equals p or q. For the same p, q, ε_{jpq} then has a repeated index and vanishes. Thus the sum is 2 for $i = j$ and 0 otherwise: $\sum_{pq} \varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij}$.

(d) There are $3! = 6$ permutations of $(1, 2, 3)$. For each, $\varepsilon_{ijk}^2 = 1$, and every other term vanishes. Thus $\sum_{ijk} \varepsilon_{ijk}\varepsilon_{ijk} = 6$. Equivalently, set $j = i$ in (c) and sum over i: $\sum_i 2\delta_{ii} = 6$ by (a).

2.1.9 Show that (in 3-D space)

$$\sum_k \varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}.$$

Note. See Exercise 2.1.8 for definitions of δ_{ij} and ε_{ijk}.

Solution.

The textbook identity is correct. In the Instructor's Manual the final line has $\delta_{ip}\delta_{jp}$ in place of $\delta_{ip}\delta_{jq}$. For $i = p = 1$, $j = q = 2$, the left side is 1, whereas that misprinted right side is 0. If $i = j$ or $p = q$, a Levi-Civita factor on the left vanishes for every k; the two delta products on the right are identical and cancel. Now take $i \neq j$ and $p \neq q$. A nonzero product requires a k different from both i, j and from both p, q. In three dimensions k is then the one index missing from $\{i, j\}$ and also the one missing from $\{p, q\}$, so the unordered pairs $\{i, j\}$ and $\{p, q\}$ must agree. If they differ, the left side is zero, and so is the right side, since each delta product on the right requires $(p, q) = (i, j)$ or $(p, q) = (j, i)$. If $(p, q) = (i, j)$, the single allowed k contributes $\varepsilon_{ijk}^2 = 1$, and the right side is $1 - 0$. If $(p, q) = (j, i)$, antisymmetry gives $\varepsilon_{ijk}\varepsilon_{jik} = -1$, and the right side is $0 - 1$. These cases exhaust all indices, proving

$$\boxed{\sum_k \varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}}.$$

2.2 Matrices

2.2.1 Show that matrix multiplication is associative, $(\mathsf{AB})\mathsf{C} = \mathsf{A}(\mathsf{BC})$.

Solution.

Let A, B, C have sizes $m \times n$, $n \times p$, and $p \times q$. For every allowed i, j,

$$\begin{aligned}((AB)C)_{ij} &= \sum_{k=1}^{p} (AB)_{ik}c_{kj} = \sum_{k=1}^{p}\sum_{\ell=1}^{n} a_{i\ell}b_{\ell k}c_{kj} \\ &= \sum_{\ell=1}^{n} a_{i\ell}\left(\sum_{k=1}^{p} b_{\ell k}c_{kj}\right) = \sum_{\ell=1}^{n} a_{i\ell}(BC)_{\ell j} = (A(BC))_{ij}.\end{aligned}$$

The sums are finite, so their order can be interchanged. Equality of every entry proves $(AB)C = A(BC)$.

2.2.2 Show that

$$(\mathsf{A} + \mathsf{B})(\mathsf{A} - \mathsf{B}) = \mathsf{A}^2 - \mathsf{B}^2$$

if and only if A and B commute,

$$[\mathsf{A}, \mathsf{B}] = 0.$$

Solution.

Distribute while preserving factor order:

$$(A + B)(A - B) = A^2 - AB + BA - B^2 = A^2 - B^2 - [A, B].$$

If $[A, B] = 0$, the required equality follows. Conversely, if that equality holds, subtract $A^2 - B^2$ from both sides to get $-[A, B] = 0$. This proves both directions.

2.2.3

(a) Complex numbers, $a+ib$, with a and b real, may be represented by (or are isomorphic with) 2×2 matrices:

$$a+ib \longleftrightarrow \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

Show that this matrix representation is valid for (i) addition and (ii) multiplication.

(b) Find the matrix corresponding to $(a+ib)^{-1}$, assuming $a+ib\neq 0$.

Solution.

(a) Define $F(a+ib)=\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. (i) For real a,b,c,d,

$$F(a+ib)+F(c+id)=\begin{pmatrix} a+c & b+d \\ -(b+d) & a+c \end{pmatrix}=F((a+c)+i(b+d)).$$

(ii) Multiplication gives

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}\begin{pmatrix} c & d \\ -d & c \end{pmatrix}=\begin{pmatrix} ac-bd & ad+bc \\ -(ad+bc) & ac-bd \end{pmatrix},$$

which is $F((a+ib)(c+id))$ because $(a+ib)(c+id)=(ac-bd)+i(ad+bc)$. The representation is one-to-one, since its first row recovers a,b, and $F(1)=I_2$.

(b) For $a+ib\neq 0$, multiplying numerator and denominator by $a-ib$ gives $(a+ib)^{-1}=x+iy$ with $x=a/(a^2+b^2)$, $y=-b/(a^2+b^2)$, so

$$\boxed{F((a+ib)^{-1})=\begin{pmatrix} x & y \\ -y & x \end{pmatrix}=\frac{1}{a^2+b^2}\begin{pmatrix} a & -b \\ b & a \end{pmatrix}}.$$

As a check,

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}\begin{pmatrix} a & -b \\ b & a \end{pmatrix}=\begin{pmatrix} a^2+b^2 & 0 \\ 0 & a^2+b^2 \end{pmatrix}=(a^2+b^2)I_2,$$

so, after division by $a^2+b^2\neq 0$, the boxed matrix is the inverse of $F(a+ib)$; if $a=b=0$, neither inverse exists.

2.2.4 If A is an $n\times n$ matrix, show that

$$\det(-\mathsf{A})=(-1)^n\det\mathsf{A}.$$

Solution.

Changing A into $-A$ multiplies each of its n rows by -1. Linearity of the determinant in each row gives n such factors:

$$\det(-A)=\underbrace{(-1)(-1)\cdots(-1)}_{n\text{ rows}}\det A=\boxed{(-1)^n\det A}.$$

No nonsingularity assumption is needed.

2.2.5

(a) The matrix equation $\mathsf{A}^2 = 0$ does not imply $\mathsf{A} = 0$. Show that the most general 2×2 matrix whose square is zero may be written as

$$\begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix},$$

where a and b are real or complex numbers. Allow complex parameters for this complete parametrization, including when A is real. A parametrization using only real parameters needs the additional sign derived below.

(b) If $\mathsf{C} = \mathsf{A} + \mathsf{B}$, in general

$$\det \mathsf{C} \neq \det \mathsf{A} + \det \mathsf{B}.$$

Construct a specific numerical example to illustrate this inequality.

Solution.

(a) Let $A = \begin{pmatrix} u & v \\ w & t \end{pmatrix}$. Then

$$A^2 = \begin{pmatrix} u^2 + vw & v(u+t) \\ w(u+t) & t^2 + vw \end{pmatrix}.$$

If $v = w = 0$, the equations $u^2 = t^2 = 0$ force $A = 0$, represented by $a = b = 0$. Otherwise one of v, w is nonzero, so the off-diagonal equations give $t = -u$. The remaining condition is $u^2 + vw = 0$. If $v \neq 0$, choose a complex square root b of v and set $a = u/b$. Then $ab = u$, $b^2 = v$, and $-a^2 = -u^2/v = w$. If $v = 0, w \neq 0$, the condition gives $u = t = 0$; choose $b = 0$ and $a^2 = -w$. These cases prove completeness over $\mathbb{C}$. Conversely,

$$\begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix}^2 = \begin{pmatrix} a^2b^2 - a^2b^2 & ab^3 - ab^3 \\ -a^3b + a^3b & -a^2b^2 + a^2b^2 \end{pmatrix} = 0.$$

For real matrices a complete real parametrization is $s \begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix}$, with $a, b \in \mathbb{R}$ and $s = \pm 1$. For $v \neq 0$, take $s = \operatorname{sgn} v$, $b = \sqrt{|v|}$, $a = u/(sb)$; then $sab = u$, $sb^2 = s|v| = v$, and $-sa^2 = -su^2/|v| = -u^2/v = w$ (using $s^2 = 1$ and $v = s|v|$). For $v = 0, w \neq 0$, take $s = -\operatorname{sgn} w$, $a = \sqrt{|w|}$, $b = 0$; then $-sa^2 = \operatorname{sgn}(w)|w| = w$ and the other entries vanish. For example $\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$ would require $b^2 = -1$ in the unmodified real formula.

(b) Take $A = B = I_2$. Then $\det A + \det B = 2$ but $\det(A + B) = \det(2I_2) = 4$. Thus the determinant is not additive.

2.2.6 Given

$$\mathsf{K} = \begin{pmatrix} 0 & 0 & i \\ -i & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix},$$

show that

$$\mathsf{K}^n = \mathsf{K}\mathsf{K}\mathsf{K}\cdots(n \text{ factors}) = \mathbf{1}$$

(with the proper choice of n, $n \neq 0$).

Solution.

Multiplication gives

$$K^2 = \begin{pmatrix} 0 & -i & 0 \\ 0 & 0 & 1 \\ i & 0 & 0 \end{pmatrix}, \qquad K^3 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = -I_3.$$

For example $(K^2)_{12} = K_{13}K_{32} = i(-1) = -i$; each row has only one nonzero product. Multiplying K^2 by K gives diagonal entries $(-i)(-i) = -1$, $1(-1) = -1$, and $i(i) = -1$. Therefore $\boxed{K^6 = (K^3)^2 = I_3}$. This is the smallest positive exponent: $K, K^2, K^4 = -K, K^5 = -K^2$ are off-diagonal, and $K^3 = -I_3 \neq I_3$. In general $K^n = I_3$ exactly when n is a multiple of 6.

2.2.7 Verify the Jacobi identity,

$$[\mathsf{A}, [\mathsf{B}, \mathsf{C}]] = [\mathsf{B}, [\mathsf{A}, \mathsf{C}]] - [\mathsf{C}, [\mathsf{A}, \mathsf{B}]].$$

Solution.

Expand without reordering factors:

$$\begin{aligned} [A, [B, C]] &= ABC - ACB - BCA + CBA, \\ [B, [A, C]] &= BAC - BCA - ACB + CAB, \\ [C, [A, B]] &= CAB - CBA - ABC + BAC. \end{aligned}$$

Subtracting the third line from the second cancels BAC and CAB:

$$[B, [A, C]] - [C, [A, B]] = -BCA - ACB + CBA + ABC,$$

which is the first line rearranged. This proves the identity for arbitrary square matrices of the same size.

2.2.8 Show that the matrices

$$\mathsf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathsf{B} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathsf{C} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

satisfy the commutation relations

$$[\mathsf{A}, \mathsf{B}] = \mathsf{C}, \quad [\mathsf{A}, \mathsf{C}] = 0, \quad \text{and } [\mathsf{B}, \mathsf{C}] = 0.$$

Solution.

Let E_{ij} have one 1 at position (i, j). Entrywise,

$$(E_{ij}E_{kl})_{rs} = \sum_t \delta_{ri}\delta_{tj}\delta_{tk}\delta_{sl} = \delta_{jk}\delta_{ri}\delta_{sl}, \quad E_{ij}E_{kl} = \delta_{jk}E_{il}.$$

Here $A = E_{12}$, $B = E_{23}$, $C = E_{13}$. Therefore

$$\begin{array}{lll} AB = E_{13} = C, & BA = 0, & [A, B] = C, \\ AC = 0, & CA = 0, & [A, C] = 0, \\ BC = 0, & CB = 0, & [B, C] = 0. \end{array}$$

For example $BA = E_{23}E_{12} = 0$ because $3 \neq 1$.

2.2.9 Let

$$\mathsf{i} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad \mathsf{j} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

and

$$\mathsf{k} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

Show that

(a) $\mathsf{i}^2 = \mathsf{j}^2 = \mathsf{k}^2 = -\mathbf{1}$, where $\mathbf{1}$ is the unit matrix.

(b) $\mathsf{ij} = -\mathsf{ji} = \mathsf{k}$,
$\mathsf{jk} = -\mathsf{kj} = \mathsf{i}$,
$\mathsf{ki} = -\mathsf{ik} = \mathsf{j}$.

These three matrices (i, j, and k) plus the unit matrix $\mathbf{1}$ form a basis for **quaternions**. An alternate basis is provided by the four 2×2 matrices, $i\sigma_1$, $i\sigma_2$, $-i\sigma_3$, and $\mathbf{1}$, where the σ_i are the Pauli spin matrices of Example 2.2.1.

Solution.

(a) Track the four standard basis columns: column r of a matrix is its image of e_r. Reading off the columns, the given matrices act as

	e_1	e_2	e_3	e_4
i	$-e_2$	e_1	$-e_4$	e_3
j	e_4	e_3	$-e_2$	$-e_1$
k	e_3	$-e_4$	$-e_1$	e_2

Applying each row twice sends every e_r to $-e_r$; hence $\mathsf{i}^2 = \mathsf{j}^2 = \mathsf{k}^2 = -I_4$. For example $\mathsf{i}^2 e_1 = \mathsf{i}(-e_2) = -e_1$ and $\mathsf{j}^2 e_1 = \mathsf{j}(e_4) = -e_1$.

(b) For the cyclic products apply the right-hand factor first, e.g. $\mathsf{ij}e_1 = \mathsf{i}(\mathsf{j}e_1) = \mathsf{i}e_4 = e_3$. The successive images are

	e_1	e_2	e_3	e_4
ij	e_3	$-e_4$	$-e_1$	e_2
jk	$-e_2$	e_1	$-e_4$	e_3
ki	e_4	e_3	$-e_2$	$-e_1$

Thus these products equal $\mathsf{k}, \mathsf{i}, \mathsf{j}$. For the reversed products, for example, $\mathsf{ji} = (\mathsf{ki})\mathsf{i} = \mathsf{k}(-I_4) = -\mathsf{k}$. Likewise $\mathsf{kj} = (\mathsf{ij})\mathsf{j} = -\mathsf{i}$ and $\mathsf{ik} = (\mathsf{jk})\mathsf{k} = -\mathsf{j}$. The complete multiplication table is

	i	j	k
i	$-I_4$	k	$-\mathsf{j}$
j	$-\mathsf{k}$	$-I_4$	i
k	j	$-\mathsf{i}$	$-I_4$

.

The matrices $I_4, \mathsf{i}, \mathsf{j}, \mathsf{k}$ are linearly independent: the first row of $a_0 I_4 + a_1\mathsf{i} + a_2\mathsf{j} + a_3\mathsf{k}$ is $(a_0, a_1, -a_3, -a_2)$, which vanishes only if every $a_r = 0$. For the alternative basis, put $q_1 = i\sigma_1$, $q_2 = i\sigma_2$, $q_3 = -i\sigma_3$. The Pauli products of Exercise 2.2.11 give $q_r^2 = -\sigma_r^2 = -I_2$ and

$$q_1 q_2 = -\sigma_1\sigma_2 = -i\sigma_3 = q_3, \quad q_2 q_3 = \sigma_2\sigma_3 = i\sigma_1 = q_1, \quad q_3 q_1 = \sigma_3\sigma_1 = i\sigma_2 = q_2.$$

The reversed products change sign. If $a_0 I_2 + a_1 q_1 + a_2 q_2 + a_3 q_3 = 0$, its entries are

$$\begin{pmatrix} a_0 - ia_3 & ia_1 + a_2 \\ ia_1 - a_2 & a_0 + ia_3 \end{pmatrix} = 0.$$

Adding and subtracting the diagonal equations gives $a_0 = a_3 = 0$; the off-diagonal equations give $a_1 = a_2 = 0$. Thus these four matrices are linearly independent and realize the same quaternion algebra with real coefficients.

2.2.10 A matrix with elements $a_{ij} = 0$ for $j < i$ may be called upper right triangular. The elements in the lower left (below and to the left of the main diagonal) vanish. Show that the product of two upper right triangular matrices is an upper right triangular matrix.

Solution.

For $i > j$ the (i, j) entry is $(AB)_{ij} = \sum_k a_{ik} b_{kj}$. If $k < i$, then $a_{ik} = 0$ by upper triangularity of A. If $k \geq i$, then $k > j$ and $b_{kj} = 0$ by upper triangularity of B. Thus every summand is zero, and $(AB)_{ij} = 0$ whenever $i > j$. This is precisely upper triangularity of AB.

2.2.11 The three Pauli spin matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{and} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Show that

(a) $(\sigma_i)^2 = \mathbf{1}_2$,

(b) $\sigma_i \sigma_j = i\sigma_k$, $(i, j, k) = (1, 2, 3)$ or a cyclic permutation thereof,

(c) $\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} \mathbf{1}_2$; $\mathbf{1}_2$ is the 2×2 unit matrix.

Solution.

(a) Direct multiplication gives

$$\sigma_1^2 = I_2, \qquad \sigma_2^2 = \begin{pmatrix} (-i)i & 0 \\ 0 & i(-i) \end{pmatrix} = I_2, \qquad \sigma_3^2 = \begin{pmatrix} 1 & 0 \\ 0 & (-1)^2 \end{pmatrix} = I_2.$$

(b) The six products with distinct indices are

$$\begin{aligned} \sigma_1\sigma_2 &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_3, \quad \sigma_2\sigma_1 = -i\sigma_3, \\ \sigma_2\sigma_3 &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\sigma_1, \qquad \sigma_3\sigma_2 = -i\sigma_1, \\ \sigma_3\sigma_1 &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2, \quad \sigma_1\sigma_3 = -i\sigma_2. \end{aligned}$$

The reversed products in the right column follow in the same way, or from $\sigma_j \sigma_i = \sigma_j^\dagger \sigma_i^\dagger = (\sigma_i \sigma_j)^\dagger = (i\sigma_k)^\dagger = -i\sigma_k$, since each σ_i is Hermitian and $(XY)^\dagger = Y^\dagger X^\dagger$ (Exercise 2.2.32); for example $\sigma_2\sigma_1 = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -i\sigma_3$.

(c) Hence the anticommutator is zero for distinct indices; for equal indices it is $2\sigma_i^2 = 2I_2$. Together,

$$\boxed{\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} I_2}, \qquad \sigma_i \sigma_j = \delta_{ij} I_2 + i \sum_k \varepsilon_{ijk} \sigma_k.$$

2.2.12 One description of spin-1 particles uses the matrices

$$\mathsf{M}_x = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathsf{M}_y = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix},$$

and

$$\mathsf{M}_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Show that

(a) $[\mathsf{M}_x, \mathsf{M}_y] = i\mathsf{M}_z$, and so on (cyclic permutation of indices). Using the Levi-Civita symbol, we may write

$$[\mathsf{M}_i, \mathsf{M}_j] = i\sum_k \varepsilon_{ijk}\mathsf{M}_k.$$

(b) $\mathsf{M}^2 \equiv \mathsf{M}_x^2 + \mathsf{M}_y^2 + \mathsf{M}_z^2 = 2\,\mathbf{1}_3$, where $\mathbf{1}_3$ is the 3×3 unit matrix.

(c) $[\mathsf{M}^2, \mathsf{M}_i] = 0$,
$[\mathsf{M}_z, \mathsf{L}^+] = \mathsf{L}^+$,
$[\mathsf{L}^+, \mathsf{L}^-] = 2\mathsf{M}_z$,
where $\mathsf{L}^+ \equiv \mathsf{M}_x + i\mathsf{M}_y$ and $\mathsf{L}^- \equiv \mathsf{M}_x - i\mathsf{M}_y$.

Solution.

(a) The two products involving x, y are

$$M_xM_y = \frac{i}{2}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \quad M_yM_x = \frac{i}{2}\begin{pmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Their difference is $i\,\mathrm{diag}(1, 0, -1) = iM_z$. For the other cyclic pairs,

$$M_yM_z = \frac{i}{\sqrt{2}}\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad M_zM_y = \frac{i}{\sqrt{2}}\begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix},$$

$$M_zM_x = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \quad M_xM_z = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Subtraction gives

$$[M_y, M_z] = \frac{i}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = iM_x, \qquad [M_z, M_x] = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = iM_y,$$

where the last matrix is iM_y because $i(-i) = 1$ and $i \cdot i = -1$. Reversed commutators change sign and $[M_i, M_i] = 0$, yielding the Levi-Civita expression for all indices.

(b) The squares are

$$M_x^2 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad M_y^2 = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad M_z^2 = \mathrm{diag}(1, 0, 1).$$

Adding cancels the off-diagonal entries $\pm\frac{1}{2}$; the diagonal entries add to $\frac{1}{2}+\frac{1}{2}+1=2$, $1+1+0=2$, and $\frac{1}{2}+\frac{1}{2}+1=2$. Hence $\boxed{M^2=2I_3}$.

(c) Since $M^2=2I_3$, $[M^2,M_i]=2M_i-2M_i=0$. Next, the matrix iM_y found in (a) agrees with M_x above the diagonal and is its negative below it. Adding and subtracting, the raising and lowering matrices are

$$L^+=\sqrt{2}\begin{pmatrix}0&1&0\\0&0&1\\0&0&0\end{pmatrix},\qquad L^-=\sqrt{2}\begin{pmatrix}0&0&0\\1&0&0\\0&1&0\end{pmatrix}.$$

For a diagonal $D=\mathrm{diag}(d_1,d_2,d_3)$, $[D,L]_{ij}=(d_i-d_j)L_{ij}$. On the nonzero entries of L^+ the difference for M_z is 1, so $[M_z,L^+]=L^+$. Finally

$$L^+L^-=\mathrm{diag}(2,2,0),\qquad L^-L^+=\mathrm{diag}(0,2,2),$$

whose difference is $\mathrm{diag}(2,0,-2)=2M_z$.

2.2.13 Repeat Exercise 2.2.12, using the matrices for a spin of 3/2,

$$\mathsf{M}_x=\frac{1}{2}\begin{pmatrix}0&\sqrt{3}&0&0\\\sqrt{3}&0&2&0\\0&2&0&\sqrt{3}\\0&0&\sqrt{3}&0\end{pmatrix},\quad \mathsf{M}_y=\frac{i}{2}\begin{pmatrix}0&-\sqrt{3}&0&0\\\sqrt{3}&0&-2&0\\0&2&0&-\sqrt{3}\\0&0&\sqrt{3}&0\end{pmatrix},$$

and

$$\mathsf{M}_z=\frac{1}{2}\begin{pmatrix}3&0&0&0\\0&1&0&0\\0&0&-1&0\\0&0&0&-3\end{pmatrix}.$$

In repeating part (b), determine the new value of M^2: for these matrices the claim is $M^2=\frac{15}{4}I_4$. The numerical spin-1 value is not retained.

Solution.

The commutator identities are unchanged, with 4×4 matrices. All three parts use the ladder matrices $L^\pm=M_x\pm iM_y$, so form them first. Since $i\cdot\frac{i}{2}=-\frac{1}{2}$, the matrix iM_y agrees with M_x above the diagonal and is its negative below it; hence

$$L^+=M_x+iM_y=\begin{pmatrix}0&\sqrt{3}&0&0\\0&0&2&0\\0&0&0&\sqrt{3}\\0&0&0&0\end{pmatrix},\qquad L^-=M_x-iM_y=(L^+)^T.$$

The diagonal entries of M_z are $3/2,1/2,-1/2,-3/2$. As in Exercise 2.2.12, $[M_z,L]_{ij}=((M_z)_{ii}-(M_z)_{jj})L_{ij}$ for any L. The nonzero entries of L^+ are at positions $(i,i+1)$, where this difference is 1, and those of L^- are at $(i+1,i)$, where it is -1. Thus

$$[M_z,L^+]=L^+,\qquad [M_z,L^-]=-L^-.$$

Because $L^-=(L^+)^T$ is real, $(L^+L^-)_{ii}$ is the sum of the squares of row i of L^+ and $(L^-L^+)_{ii}$ that of column i; the off-diagonal entries vanish because each row and each column of L^+ has at most one nonzero entry. Thus

$$L^+L^-=\mathrm{diag}(3,4,3,0),\qquad L^-L^+=\mathrm{diag}(0,3,4,3),$$

so $[L^+, L^-] = \mathrm{diag}(3, 1, -1, -3) = 2M_z$.

(a) Now $M_x = (L^+ + L^-)/2$ and $M_y = (L^+ - L^-)/(2i)$. Using $[L^\pm, L^\pm] = 0$, $[L^-, L^+] = -[L^+, L^-]$, and $[M_z, L^\pm] = \pm L^\pm$,

$$\begin{aligned}
[M_x, M_y] &= \frac{1}{4i}[L^+ + L^-, L^+ - L^-] = \frac{1}{4i}\big(-[L^+, L^-] + [L^-, L^+]\big) = -\frac{4M_z}{4i} = iM_z, \\
[M_z, M_x] &= \tfrac{1}{2}\big([M_z, L^+] + [M_z, L^-]\big) = \tfrac{1}{2}(L^+ - L^-) = iM_y, \\
[M_y, M_z] &= -\tfrac{1}{2i}\big([M_z, L^+] - [M_z, L^-]\big) = \tfrac{1}{2i}(-L^+ - L^-) = iM_x.
\end{aligned}$$

Reversed commutators change sign and $[M_i, M_i] = 0$, so, as in Exercise 2.2.12(a), $[M_i, M_j] = i\sum_k \varepsilon_{ijk} M_k$ for all i, j.

(b) Expand the squares:

$$\begin{aligned}
M_x^2 + M_y^2 &= \tfrac{1}{4}\big((L^+)^2 + L^+L^- + L^-L^+ + (L^-)^2\big) \\
&\quad - \tfrac{1}{4}\big((L^+)^2 - L^+L^- - L^-L^+ + (L^-)^2\big) \\
&= \tfrac{1}{2}(L^+L^- + L^-L^+) = \mathrm{diag}(3/2, 7/2, 7/2, 3/2).
\end{aligned}$$

Adding $M_z^2 = \mathrm{diag}(9/4, 1/4, 1/4, 9/4)$ gives, since $\frac{3}{2} + \frac{9}{4} = \frac{7}{2} + \frac{1}{4} = \frac{15}{4}$, $\boxed{M^2 = \frac{15}{4} I_4}$.

(c) Since $M^2 = \frac{15}{4} I_4$, $[M^2, M_i] = \frac{15}{4}(M_i - M_i) = 0$. The identities $[M_z, L^+] = L^+$ and $[L^+, L^-] = 2M_z$ were established at the start of this solution.

2.2.14 If A is a diagonal matrix, with all diagonal elements different, and A and B commute, show that B is diagonal.

Solution.

Write $A = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Then $(AB)_{ij} = \lambda_i b_{ij}$ and $(BA)_{ij} = b_{ij}\lambda_j$, so $AB = BA$ implies

$$(\lambda_i - \lambda_j) b_{ij} = 0.$$

For $i \neq j$ the hypothesis gives $\lambda_i - \lambda_j \neq 0$, so $b_{ij} = 0$. Diagonal entries b_{ii} are unrestricted. Therefore B is diagonal.

2.2.15 If A and B are diagonal, show that A and B commute.

Solution.

For $A = \mathrm{diag}(a_1, \ldots, a_n)$ and $B = \mathrm{diag}(b_1, \ldots, b_n)$,

$$(AB)_{ij} = \sum_k a_i \delta_{ik} b_k \delta_{kj} = a_i b_i \delta_{ij}.$$

Similarly $(BA)_{ij} = b_i a_i \delta_{ij}$. Scalar multiplication commutes, so the entries agree and $AB = BA$.

2.2.16 Show that $\mathrm{trace}(\mathsf{ABC}) = \mathrm{trace}(\mathsf{CBA})$ if any two of the three matrices commute.

Solution.

For square matrices of the same size,

$$\mathrm{tr}(XY) = \sum_{i,j} x_{ij} y_{ji} = \sum_{j,i} y_{ji} x_{ij} = \mathrm{tr}(YX).$$

Taking $X = AB, Y = C$, and then $X = A, Y = BC$, gives the cyclic rule $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$. There are three possible commuting pairs:

$$\begin{aligned} AB = BA: &\quad \mathrm{tr}(ABC) = \mathrm{tr}(BAC) = \mathrm{tr}(CBA), \\ BC = CB: &\quad \mathrm{tr}(ABC) = \mathrm{tr}(ACB) = \mathrm{tr}(CBA), \\ CA = AC: &\quad \mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(BAC) = \mathrm{tr}(CBA). \end{aligned}$$

Every possible hypothesis therefore gives the required equality.

2.2.17 Angular momentum matrices satisfy a commutation relation

$$[\mathsf{M}_j, \mathsf{M}_k] = i\mathsf{M}_l, \quad j, k, l \text{ cyclic}.$$

Show that the trace of each angular momentum matrix vanishes. Use finite matrices, so all products and traces below exist.

Solution.

For finite square matrices, cyclicity gives $\mathrm{tr}[A, B] = \mathrm{tr}(AB) - \mathrm{tr}(BA) = 0$. Apply this to each cyclic angular momentum relation:

$$\begin{aligned} i\,\mathrm{tr}\, M_z &= \mathrm{tr}[M_x, M_y] = 0, \\ i\,\mathrm{tr}\, M_x &= \mathrm{tr}[M_y, M_z] = 0, \\ i\,\mathrm{tr}\, M_y &= \mathrm{tr}[M_z, M_x] = 0. \end{aligned}$$

Division by i proves that every trace vanishes. This uses ordinary finite-dimensional traces, not undefined traces of infinite-dimensional operators.

2.2.18 A and B anticommute: $\mathsf{AB} = -\mathsf{BA}$. Also, $\mathsf{A}^2 = \mathbf{1}$, $\mathsf{B}^2 = \mathbf{1}$. Show that $\mathrm{trace}(\mathsf{A}) = \mathrm{trace}(\mathsf{B}) = 0$.

Note. The Pauli and Dirac matrices are specific examples.

Solution.

From $AB = -BA$ and $B^2 = I$, we obtain $BAB = -AB^2 = -A$. Taking traces and cycling the first B to the end,

$$-\,\mathrm{tr}\, A = \mathrm{tr}(BAB) = \mathrm{tr}(ABB) = \mathrm{tr}\, A.$$

Hence $2\,\mathrm{tr}\, A = 0$. Similarly $A^2 = I$ gives $ABA = -BA^2 = -B$, and cycling the first A to the end, $-\,\mathrm{tr}\, B = \mathrm{tr}(ABA) = \mathrm{tr}(BAA) = \mathrm{tr}\, B$, so $2\,\mathrm{tr}\, B = 0$. Thus $\boxed{\mathrm{tr}\, A = \mathrm{tr}\, B = 0}$.

2.2.19

(a) If two nonsingular matrices anticommute, show that the trace of each one is zero. (Nonsingular means that the determinant of the matrix is nonzero.)

(b) For the conditions of part (a) to hold, A and B must be $n \times n$ matrices with n **even**. Show that if n is **odd**, a contradiction results.

Solution.

The Instructor's Manual's left multiplication by B^{-1} proves the vanishing of $\mathrm{tr}\, A$, not of $\mathrm{tr}\, B$ as printed. Interchanging the roles of A and B supplies the second trace. **(a)** Multiplying $AB = -BA$

on the right by B^{-1} gives $A = -BAB^{-1}$, that is, $BAB^{-1} = -A$. Taking the trace and cycling B to the end,

$$-\operatorname{tr} A = \operatorname{tr}(BAB^{-1}) = \operatorname{tr}(AB^{-1}B) = \operatorname{tr} A,$$

so $\operatorname{tr} A = 0$. Likewise, multiplying $BA = -AB$ on the right by A^{-1} gives $ABA^{-1} = -B$, and $-\operatorname{tr} B = \operatorname{tr}(ABA^{-1}) = \operatorname{tr}(BA^{-1}A) = \operatorname{tr} B$. Hence $\operatorname{tr} B = 0$.

(b) Taking determinants and using the product rule and $\det(-X) = (-1)^n \det X$ (Exercise 2.2.4) gives

$$\det A \det B = \det(AB) = \det(-BA) = (-1)^n \det B \det A.$$

Both determinants are nonzero; division by their product yields $1 = (-1)^n$. Odd n would give $1 = -1$, a contradiction over the real or complex numbers. Therefore $\boxed{n \text{ is even}}$.

2.2.20 If A^{-1} has elements

$$(\mathsf{A}^{-1})_{ij} = a_{ij}^{(-1)} = \frac{C_{ji}}{|\mathsf{A}|},$$

where C_{ji} is the jith cofactor of $|\mathsf{A}|$, show that

$$\mathsf{A}^{-1}\mathsf{A} = \mathbf{1}.$$

Hence A^{-1} is the inverse of A (if $|\mathsf{A}| \neq 0$).

Solution.

Let $B_{ij} = C_{ji}/\det A$, with $\det A \neq 0$. Then

$$(BA)_{ij} = \sum_k \frac{C_{ki}}{\det A} a_{kj} = \frac{1}{\det A} \sum_k a_{kj} C_{ki}.$$

For $i = j$, the numerator is the expansion down column i and equals $\det A$ (Exercise 2.1.4(a)). For $i \neq j$, it is the expansion of the determinant with column i replaced by column j, so it is zero (Exercise 2.1.4(b)). Thus $(BA)_{ij} = \delta_{ij}$ and $BA = I$. The corresponding row identities of Exercise 2.1.4 give

$$(AB)_{ij} = \frac{1}{\det A} \sum_k a_{ik} C_{jk} = \delta_{ij}.$$

The formula therefore produces the two-sided inverse $B = A^{-1}$.

2.2.21 Find the matrices M_L such that the product $\mathsf{M}_L\mathsf{A}$ will be A but with:

(a) The ith row multiplied by a constant k ($a_{ij} \to k a_{ij}$, $j = 1, 2, 3, \ldots$);

(b) The ith row replaced by the original ith row minus a multiple of the mth row ($a_{ij} \to a_{ij} - K a_{mj}$, $j = 1, 2, 3, \ldots$);

(c) The ith and mth rows interchanged ($a_{ij} \to a_{mj}$, $a_{mj} \to a_{ij}$, $j = 1, 2, 3, \ldots$).

Solution.

In (b), the varying index in the parentheses must be j, not i: i labels the one row being changed. For (c), the Instructor's Manual's $M_{mi} - M_{im} = 1$ must read $M_{mi} = M_{im} = 1$ when $i \neq m$. Both inserted entries are positive. Let E_{ab} have a single 1 in position (a, b). Its action is

$$(E_{ab}A)_{rj} = \sum_s \delta_{ra}\delta_{sb} a_{sj} = \delta_{ra} a_{bj}.$$

Thus $E_{ab}A$ has row b of A in row a and all other rows zero. For distinct i, m, the requested matrices are

$$\boxed{\begin{array}{ll}(\text{a}) & M_L = I + (k-1)E_{ii},\\ (\text{b}) & M_L = I - KE_{im},\\ (\text{c}) & M_L = I - E_{ii} - E_{mm} + E_{im} + E_{mi}.\end{array}}$$

In (a), row i becomes $R_i + (k-1)R_i = kR_i$. In (b), it becomes $R_i - KR_m$. In (c), the two subtractions remove rows i, m; the last two terms insert R_m, R_i in their places. Every other row retains only the identity contribution. If $i = m$, (b) reduces to scaling by $1 - K$, and the formula in (c) reduces to I.

2.2.22 Find the matrices M_R such that the product AM_R will be A but with:

(a) The ith column multiplied by a constant k ($a_{ji} \to ka_{ji}$, $j = 1, 2, 3, \ldots$);

(b) The ith column replaced by the original ith column minus a multiple of the mth column ($a_{ji} \to a_{ji} - ka_{jm}$, $j = 1, 2, 3, \ldots$);

(c) The ith and mth columns interchanged ($a_{ji} \to a_{jm}$, $a_{jm} \to a_{ji}$, $j = 1, 2, 3, \ldots$).

Solution.

For right multiplication,

$$(AE_{ab})_{rj} = \sum_s a_{rs}\delta_{sa}\delta_{jb} = a_{ra}\delta_{jb}.$$

Thus AE_{ab} puts column a of A in column b, giving

$$\boxed{\begin{array}{ll}(\text{a}) & M_R = I + (k-1)E_{ii},\\ (\text{b}) & M_R = I - kE_{mi},\\ (\text{c}) & M_R = I - E_{ii} - E_{mm} + E_{im} + E_{mi}.\end{array}}$$

Column i in (a) is $C_i + (k-1)C_i = kC_i$; in (b), it is $C_i - kC_m$; in (c), the two columns are removed and reinserted in exchanged positions. The off-diagonal entry in (b) is (m, i), the transpose of the row-operation entry (i, m); the Instructor's Manual's statement that the answers are the same requires this transpose. For rectangular A, M_L has order equal to the number of rows of A, whereas M_R has order equal to the number of columns.

2.2.23 Find the inverse of

$$\mathsf{A} = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 4 \end{pmatrix}.$$

Solution.

Expansion along row 1 gives

$$\det A = 3(8-1) - 2(8-1) + (2-2) = 21 - 14 = 7.$$

The cofactors (signed 2×2 minors) are

$$\begin{array}{lll} C_{11} = 8 - 1 = 7, & C_{12} = -(8-1) = -7, & C_{13} = 2 - 2 = 0, \\ C_{21} = -(8-1) = -7, & C_{22} = 12 - 1 = 11, & C_{23} = -(3-2) = -1, \\ C_{31} = 2 - 2 = 0, & C_{32} = -(3-2) = -1, & C_{33} = 6 - 4 = 2. \end{array}$$

By Exercise 2.2.20, $(A^{-1})_{ij} = C_{ji}/\det A$, so A^{-1} is the transposed cofactor matrix divided by 7. The cofactor matrix is symmetric, so its transpose is unchanged:

$$\boxed{A^{-1} = \frac{1}{7}\begin{pmatrix} 7 & -7 & 0 \\ -7 & 11 & -1 \\ 0 & -1 & 2 \end{pmatrix}.}$$

An independent multiplication checks every entry:

$$A\begin{pmatrix} 7 & -7 & 0 \\ -7 & 11 & -1 \\ 0 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 21-14 & -21+22-1 & -2+2 \\ 14-14 & -14+22-1 & -2+2 \\ 7-7 & -7+11-4 & -1+8 \end{pmatrix} = 7I_3.$$

2.2.24 Matrices are far too useful to remain the exclusive property of physicists. They may appear wherever there are linear relations. For instance, in a study of population movement the initial fraction of a fixed population in each of n areas (or industries or religions, etc.) is represented by an n-component column vector P. The movement of people from one area to another in a given time is described by an $n \times n$ (stochastic) matrix T. Here T_{ij} is the fraction of the population in the jth area that moves to the ith area. (Those not moving are covered by $i = j$.) With P describing the initial population distribution, the final population distribution is given by the matrix equation $\mathsf{TP} = \mathsf{Q}$. From its definition, $\sum_{i=1}^{n} P_i = 1$.

(a) Show that conservation of people requires that

$$\sum_{i=1}^{n} T_{ij} = 1, \quad j = 1, 2, \ldots, n.$$

(b) Prove that

$$\sum_{i=1}^{n} Q_i = 1$$

continues the conservation of people.

Solution.

(a) A source area j containing N_j people contributes $T_{1j}N_j, \ldots, T_{nj}N_j$ to the destination areas. Conservation requires the sum to be N_j; for $N_j > 0$, division gives $\sum_i T_{ij} = 1$. This requirement holds for every source area, even one initially empty, because conservation must hold for every possible initial distribution. Equivalently, start with $P = e_j$; the final total $\sum_i T_{ij}$ must be 1.

(b) Since $Q_i = \sum_j T_{ij}P_j$,

$$\sum_i Q_i = \sum_i \sum_j T_{ij}P_j = \sum_j P_j \left(\sum_i T_{ij}\right) = \sum_j P_j = 1.$$

Nonnegative fractions and $P_j \geq 0$ also imply $Q_i \geq 0$, so Q is again a valid population distribution.

2.2.25 Given a 6×6 matrix A with elements $a_{ij} = 0.5^{|i-j|}$, $i, j = 0, 1, 2, \ldots, 5$, find A^{-1}.

$$ANS.\ \mathsf{A}^{-1} = \frac{1}{3}\begin{pmatrix} 4 & -2 & 0 & 0 & 0 & 0 \\ -2 & 5 & -2 & 0 & 0 & 0 \\ 0 & -2 & 5 & -2 & 0 & 0 \\ 0 & 0 & -2 & 5 & -2 & 0 \\ 0 & 0 & 0 & -2 & 5 & -2 \\ 0 & 0 & 0 & 0 & -2 & 4 \end{pmatrix}.$$

Solution.

First retain a parameter r in $A_{ij} = r^{|i-j|}$ for $i, j = 0, \ldots, 5$. Subtract r times original row $i-1$ from row i for each $i = 1, \ldots, 5$. For $j < i$, the new entry is $r^{i-j} - r\, r^{i-1-j} = 0$; for $j \geq i$, it is $r^{j-i} - r\, r^{j-i+1} = (1-r^2)r^{j-i}$. The result is EA, where E has 1 on the diagonal and $-r$ just below it. Now subtract r times original column $j-1$ of EA from column j, for each $j = 1, \ldots, 5$, which is right multiplication by E^T. Above the diagonal this gives $r^j - r\, r^{j-1} = 0$ in row 0 and $(1-r^2)(r^{j-i} - r\, r^{j-1-i}) = 0$ for $j > i \geq 1$; the diagonal and lower entries are unchanged, because the entry to the left of each is 0. Hence

$$EAE^T = \Delta \equiv \mathrm{diag}(1, 1-r^2, \ldots, 1-r^2).$$

Since E is lower triangular with unit diagonal, $\det E = 1$, so $\det A = \det \Delta = (1-r^2)^5$: A is invertible exactly when $r^2 \neq 1$ (for $r = \pm 1$ every row of A is $\pm$ row 0). For $r^2 \neq 1$, $A = E^{-1}\Delta(E^T)^{-1}$ gives $A^{-1} = E^T\Delta^{-1}E$. Write $\Delta^{-1} = \mathrm{diag}(\delta_0, \ldots, \delta_5)$, so $\delta_0 = 1$ and $\delta_1 = \cdots = \delta_5 = 1/(1-r^2)$, and set $\delta_6 = 0$. Column i of E has only the entries 1 (row i) and $-r$ (row $i+1$), so $(E^T\Delta^{-1}E)_{ij} = \sum_k E_{ki}\delta_k E_{kj}$ receives contributions only from $k \in \{i, i+1\} \cap \{j, j+1\}$. It is $\delta_i + r^2\delta_{i+1}$ on the diagonal, $-r\delta_{\max(i,j)}$ for $|i-j| = 1$, and zero elsewhere. Explicitly,

$$\delta_0 + r^2\delta_1 = 1 + \frac{r^2}{1-r^2} = \frac{1}{1-r^2}, \qquad \delta_5 + r^2\delta_6 = \frac{1}{1-r^2},$$
$$\delta_i + r^2\delta_{i+1} = \frac{1+r^2}{1-r^2} \quad (1 \leq i \leq 4), \quad -r\delta_{\max(i,j)} = -\frac{r}{1-r^2}.$$

This is $T/(1-r^2)$, where T is tridiagonal with end diagonal entries 1, interior diagonal entries $1+r^2$, and adjacent entries $-r$. The result can be confirmed by multiplying A and T directly. For an interior column $j = 1, \ldots, 4$,

$$(AT)_{ij} = -rA_{i,j-1} + (1+r^2)A_{ij} - rA_{i,j+1}.$$

If $i < j$, put $d = j - i \geq 1$ to obtain $-rr^{d-1} + (1+r^2)r^d - rr^{d+1} = 0$. If $i > j$, put $d = i - j \geq 1$ to obtain $-rr^{d+1} + (1+r^2)r^d - rr^{d-1} = 0$. For $i = j$, the result is $-r^2 + (1+r^2) - r^2 = 1 - r^2$. At the first column,

$$(AT)_{i0} = A_{i0} - rA_{i1} = \begin{cases} 1-r^2, & i = 0, \\ r^i - rr^{i-1} = 0, & i > 0. \end{cases}$$

At the last column, likewise,

$$(AT)_{i5} = -rA_{i4} + A_{i5} = \begin{cases} 1-r^2, & i = 5, \\ -r\, r^{4-i} + r^{5-i} = 0, & i < 5. \end{cases}$$

Thus $AT = (1-r^2)I_6$, confirming $A^{-1} = T/(1-r^2)$. At $r = 1/2$, $1 - r^2 = 3/4$, so $A^{-1} = \frac{4}{3}T$ has end diagonal entries $\frac{4}{3}$, interior diagonal entries $\frac{4}{3}\cdot\frac{5}{4} = \frac{5}{3}$, and adjacent entries $\frac{4}{3}\left(-\frac{1}{2}\right) = -\frac{2}{3}$:

$$\boxed{A^{-1} = \frac{T}{1-r^2} = \frac{1}{3}\begin{pmatrix} 4 & -2 & 0 & 0 & 0 & 0 \\ -2 & 5 & -2 & 0 & 0 & 0 \\ 0 & -2 & 5 & -2 & 0 & 0 \\ 0 & 0 & -2 & 5 & -2 & 0 \\ 0 & 0 & 0 & -2 & 5 & -2 \\ 0 & 0 & 0 & 0 & -2 & 4 \end{pmatrix}}.$$

2.2.26 Show that the product of two orthogonal matrices is orthogonal.

Solution.

Let $A^TA = I$ and $B^TB = I$. Since transposition reverses products,

$$(AB)^T(AB) = B^TA^TAB = B^T(A^TA)B = B^TIB = B^TB = I.$$

This is exactly the orthogonality condition for AB.

2.2.27 If A is orthogonal, show that its determinant $= \pm 1$.

Solution.

Taking determinants of $A^TA = I$ and using $\det(A^T) = \det A$ gives

$$1 = \det(A^TA) = \det(A^T)\det A = (\det A)^2.$$

The two roots of $d^2 = 1$ are $1, -1$, so $\boxed{\det A = \pm 1}$.

2.2.28 Show that the trace of the product of a symmetric and an antisymmetric matrix is zero.

Solution.

Let $S^T = S$ and $K^T = -K$. Transposition preserves the trace:

$$\operatorname{tr}(SK) = \operatorname{tr}((SK)^T) = \operatorname{tr}(K^TS^T) = -\operatorname{tr}(KS) = -\operatorname{tr}(SK).$$

Thus $2\operatorname{tr}(SK) = 0$ and $\boxed{\operatorname{tr}(SK) = 0}$. Entrywise, each pair with $i < j$ cancels: $s_{ij}k_{ji} + s_{ji}k_{ij} = s_{ij}(-k_{ij}) + s_{ij}k_{ij} = 0$, while all diagonal terms vanish because $k_{ii} = 0$.

2.2.29 A is 2×2 and orthogonal. Find the most general form of

$$\mathsf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Solution.

For real orthogonal A, $A^TA = I$ means $A^T = A^{-1}$, so also $AA^T = I$: the rows are orthonormal. The first row is a real unit vector, so write $(a, b) = (\cos\theta, \sin\theta)$. The second row is a unit vector perpendicular to the first. The vectors perpendicular to $(\cos\theta, \sin\theta)$ form the line spanned by $(-\sin\theta, \cos\theta)$, whose only unit vectors are $\pm(-\sin\theta, \cos\theta)$. Hence

$$\boxed{A = \begin{pmatrix} \cos\theta & \sin\theta \\ -s\sin\theta & s\cos\theta \end{pmatrix}, \quad s = \pm 1, \quad \theta \in \mathbb{R}}.$$

The two row norms are 1, their inner product is 0, and $\det A = s(\cos^2\theta + \sin^2\theta) = s$. For $s = 1$, A is the two-dimensional rotation matrix: it rotates the coordinate axes through the angle θ, $x' = x\cos\theta + y\sin\theta$, $y' = -x\sin\theta + y\cos\theta$. For $s = -1$, $A = \mathrm{diag}(1,-1)$ times that rotation, i.e. the rotation followed by the reflection $y' \to -y'$. Both signs are required. The Instructor's Manual assumes $\det A = 1$ and omits the $\det A = -1$ reflection family.

If orthogonality is extended to complex A with $A^TA = I$, the algebraic form is $A = \begin{pmatrix} a & b \\ -sb & sa \end{pmatrix}$ with $a^2 + b^2 = 1$, $a, b \in \mathbb{C}$, and $s = \pm 1$. Indeed, as in the real case $A^TA = I$ gives $AA^T = I$, so $a^2 + b^2 = 1$, $ac + bd = 0$, and $c^2 + d^2 = 1$. Since $(a, b) \neq (0, 0)$, the solutions (c, d) of the single linear equation $ac + bd = 0$ form a one-dimensional space containing $(-b, a) \neq 0$, so $(c, d) = \lambda(-b, a)$; then $c^2+d^2 = \lambda^2(a^2+b^2) = \lambda^2 = 1$ gives $\lambda = s = \pm 1$. Conversely, every such A has $A^TA = (a^2+b^2)I = I$.

2.2.30 Show that

$$\det(\mathsf{A}^*) = (\det \mathsf{A})^* = \det(\mathsf{A}^\dagger).$$

Solution.

The permutation formula is

$$\det A = \sum_{\pi \in S_n} \mathrm{sgn}(\pi) \prod_{r=1}^{n} a_{r,\pi(r)}.$$

Because each permutation sign is real, conjugating the finite sum gives

$$(\det A)^* = \sum_{\pi} \mathrm{sgn}(\pi) \prod_{r} a^*_{r,\pi(r)} = \det(A^*).$$

Finally $A^\dagger = (A^*)^T$, and determinants are invariant under transposition. Thus $\det(A^\dagger) = \det(A^*) = (\det A)^*$.

2.2.31 Three angular momentum matrices satisfy the basic commutation relation

$$[\mathsf{J}_x, \mathsf{J}_y] = i\mathsf{J}_z$$

(and cyclic permutation of indices). If two of the matrices have real elements, show that the elements of the third must be pure imaginary.

Solution.

Suppose the real matrices are J_x, J_y. Each entry

$$R_{ab} := ([J_x, J_y])_{ab} = \sum_c \big((J_x)_{ac}(J_y)_{cb} - (J_y)_{ac}(J_x)_{cb}\big)$$

is real. The commutation relation gives $(J_z)_{ab} = R_{ab}/i = -iR_{ab}$, which is pure imaginary (possibly zero). For either other real pair, use the corresponding cyclic commutation relation in the same calculation.

2.2.32 Show that $(\mathsf{AB})^\dagger = \mathsf{B}^\dagger\mathsf{A}^\dagger$.

Solution.

For conformable matrices,

$$((AB)^\dagger)_{ij} = (AB)^*_{ji} = \left(\sum_k a_{jk}b_{ki}\right)^* = \sum_k a^*_{jk}b^*_{ki}$$
$$= \sum_k (B^\dagger)_{ik}(A^\dagger)_{kj} = (B^\dagger A^\dagger)_{ij}.$$

The scalar entries commute; no commutation of the matrices was assumed. Equality of all entries proves the reversed product rule.

2.2.33 A matrix $\mathsf{C} = \mathsf{S}^\dagger\mathsf{S}$. Show that the trace is positive definite unless S is the null matrix, in which case $\text{trace}\,(\mathsf{C}) = 0$.

Solution.

If S has entries s_{ki}, the diagonal entries of $C = S^\dagger S$ are

$$C_{ii} = \sum_k (S^\dagger)_{ik}S_{ki} = \sum_k s^*_{ki}s_{ki} = \sum_k |s_{ki}|^2.$$

Therefore

$$\boxed{\text{tr}(S^\dagger S) = \sum_{i,k} |s_{ki}|^2}.$$

Every summand is nonnegative. The sum is zero exactly when every entry is zero; otherwise it is strictly positive. The trace is thus a positive-definite quadratic form in the real and imaginary parts of the entries of S. This does not imply that C itself is positive definite: $S = \text{diag}(1, 0)$ gives $C = \text{diag}(1, 0)$, which has a zero eigenvalue.

2.2.34 If A and B are Hermitian matrices, show that $(\mathsf{AB}+\mathsf{BA})$ and $i(\mathsf{AB}-\mathsf{BA})$ are also Hermitian.

Solution.

Using $A^\dagger = A$, $B^\dagger = B$, and reversing the product order (Exercise 2.2.32),

$$(AB + BA)^\dagger = B^\dagger A^\dagger + A^\dagger B^\dagger = BA + AB = AB + BA.$$

For the second expression also conjugate the scalar i:

$$\big(i(AB - BA)\big)^\dagger = -i(B^\dagger A^\dagger - A^\dagger B^\dagger) = -i(BA - AB) = i(AB - BA).$$

Both expressions equal their adjoints and are therefore Hermitian.

2.2.35 The matrix C is not Hermitian. Show that then $\mathsf{C} + \mathsf{C}^\dagger$ and $i(\mathsf{C} - \mathsf{C}^\dagger)$ are Hermitian. This means that a non-Hermitian matrix may be resolved into two Hermitian parts,

$$\mathsf{C} = \frac{1}{2}(\mathsf{C} + \mathsf{C}^\dagger) + \frac{1}{2i}\, i(\mathsf{C} - \mathsf{C}^\dagger).$$

This decomposition of a matrix into two Hermitian matrix parts parallels the decomposition of a complex number z into $x + iy$, where $x = (z + z^*)/2$ and $y = (z - z^*)/2i$.

Solution.

For any square C, whether or not it is Hermitian, set

$$H = \frac{C + C^\dagger}{2}, \qquad K = \frac{C - C^\dagger}{2i}.$$

Taking adjoints with $(C^\dagger)^\dagger = C$ and $(\lambda X)^\dagger = \lambda^* X^\dagger$, where $(1/2i)^* = 1/(-2i)$,

$$H^\dagger = \frac{C^\dagger + C}{2} = H, \qquad K^\dagger = \frac{C^\dagger - C}{-2i} = \frac{C - C^\dagger}{2i} = K.$$

Also $C + C^\dagger = 2H$ and $i(C - C^\dagger) = i \cdot 2iK = -2K$. Real multiples of Hermitian matrices are Hermitian, so the two matrices $C + C^\dagger$ and $i(C - C^\dagger)$ named in the statement are Hermitian. Recombining, with $iK = i(C - C^\dagger)/(2i) = (C - C^\dagger)/2$,

$$H + iK = \frac{C + C^\dagger}{2} + \frac{C - C^\dagger}{2} = C.$$

This is the displayed decomposition: its second summand is the Hermitian matrix $i(C - C^\dagger) = -2K$ times the scalar $1/(2i)$, that is, iK. The summand $iK = (C - C^\dagger)/2$ itself is anti-Hermitian, just as iy is imaginary; the two Hermitian parts are H and K, the matrix analogues of $x = \operatorname{Re} z$ and $y = \operatorname{Im} z$. The decomposition is unique: if $C = H_1 + iK_1$ with Hermitian H_1, K_1, then $C^\dagger = H_1 - iK_1$; adding and subtracting give $C + C^\dagger = 2H_1$ and $C - C^\dagger = 2iK_1$, so $H_1 = H$ and $K_1 = K$.

2.2.36 A and B are two noncommuting Hermitian matrices:

$$\mathsf{AB} - \mathsf{BA} = i\mathsf{C}.$$

Prove that C is Hermitian.

Solution.

Take adjoints of both sides of $AB - BA = iC$. By the reversed product rule $(XY)^\dagger = Y^\dagger X^\dagger$ (Exercise 2.2.32) and $A^\dagger = A$, $B^\dagger = B$, the left side gives

$$(AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = BA - AB = -(AB - BA) = -iC,$$

whereas the right side gives $(iC)^\dagger = i^* C^\dagger = -iC^\dagger$. Therefore $-iC^\dagger = -iC$, and division by $-i$ proves $\boxed{C^\dagger = C}$.

2.2.37 Two matrices A and B are each Hermitian. Find a necessary and sufficient condition for their product AB to be Hermitian.

ANS. $[\mathsf{A}, \mathsf{B}] = 0$.

Solution.

By the reversed product rule (Exercise 2.2.32) and Hermiticity of A, B, $(AB)^\dagger = B^\dagger A^\dagger = BA$. If AB is Hermitian, then $AB = (AB)^\dagger = BA$; conversely, if $AB = BA$, then $(AB)^\dagger = BA = AB$. Thus AB is Hermitian precisely when $AB = BA$, or $\boxed{[A, B] = 0}$. This proves necessity and sufficiency.

2.2.38 Show that the reciprocal (that is, inverse) of a unitary matrix is unitary.

Solution.

Unitary U satisfies $U^{-1} = U^\dagger$. Put $V = U^{-1}$. Then $V^\dagger = (U^\dagger)^\dagger = U$, so

$$V^\dagger V = UU^{-1} = I, \qquad VV^\dagger = U^{-1}U = I.$$

Hence V is unitary.

2.2.39 Prove that the direct product of two unitary matrices is unitary.

Solution.

"Direct product" here means the tensor product defined in the text. The Instructor's Manual instead displays the proof for the ordinary product UV; the double-index proof below supplies the requested result. Let unitary U, V have orders m, n. Their direct (Kronecker) product $W = U \otimes V$ has rows and columns labeled by index pairs, with entries $W_{(i,a),(j,b)} = U_{ij}V_{ab}$. Its adjoint has entries $(W^\dagger)_{(j,b),(i,a)} = W^*_{(i,a),(j,b)} = U^*_{ij}V^*_{ab}$, and a matrix product sums over the intermediate pair (i, a). Thus

$$\begin{aligned}(W^\dagger W)_{(j,b),(k,c)} &= \sum_{i=1}^{m}\sum_{a=1}^{n} U^*_{ij}V^*_{ab}U_{ik}V_{ac} = \left(\sum_i U^*_{ij}U_{ik}\right)\left(\sum_a V^*_{ab}V_{ac}\right) \\ &= (U^\dagger U)_{jk}(V^\dagger V)_{bc} = \delta_{jk}\delta_{bc}.\end{aligned}$$

Therefore $W^\dagger W = I_{mn}$. In the reverse order the sum runs over the pair (j, b):

$$(WW^\dagger)_{(i,a),(k,c)} = \sum_{j,b} U_{ij}V_{ab}U^*_{kj}V^*_{cb} = (UU^\dagger)_{ik}(VV^\dagger)_{ac} = \delta_{ik}\delta_{ac},$$

so $WW^\dagger = I_{mn}$ as well, proving unitarity of the direct product. In matrix notation, the entry formula for $W^\dagger$ says $(U \otimes V)^\dagger = U^\dagger \otimes V^\dagger$, and with the direct-product rule $(A \otimes B)(C \otimes D) = AC \otimes BD$ the same calculation reads

$$(U \otimes V)^\dagger(U \otimes V) = (U^\dagger U) \otimes (V^\dagger V) = I_m \otimes I_n = I_{mn}.$$

2.2.40 If $\boldsymbol{\sigma}$ is the vector with the σ_i as components given in Eq. (2.61), and $\mathbf{p}$ is an ordinary vector, show that

$$(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = \mathbf{p}^2\mathbf{1}_2,$$

where $\mathbf{1}_2$ is a 2×2 unit matrix.

Solution.

In the Instructor's Manual's expansion, the term multiplying $p_y p_z$ (that is, p_2p_3) must contain $\sigma_2\sigma_3 + \sigma_3\sigma_2$, not the repeated $\sigma_1\sigma_2 + \sigma_2\sigma_1$. Both vanish, so the printed final result is unaffected. The scalar components p_i commute with each other and with the matrices, so

$$(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = \Big(\sum_i p_i\sigma_i\Big)\Big(\sum_j p_j\sigma_j\Big) = \sum_{i,j} p_ip_j\,\sigma_i\sigma_j.$$

Collect the three terms with $i = j$ and, for each pair $i < j$, the two terms (i, j) and (j, i):

$$\begin{aligned}(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = {} & p_1^2\sigma_1^2 + p_2^2\sigma_2^2 + p_3^2\sigma_3^2 \\ & + p_1p_2(\sigma_1\sigma_2 + \sigma_2\sigma_1) + p_1p_3(\sigma_1\sigma_3 + \sigma_3\sigma_1) \\ & + p_2p_3(\sigma_2\sigma_3 + \sigma_3\sigma_2).\end{aligned}$$

By Exercise 2.2.11, each square is $\sigma_i^2 = I_2$ and each mixed anticommutator $\sigma_i\sigma_j + \sigma_j\sigma_i$ $(i \neq j)$ is zero:

$$\boxed{(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = (p_1^2 + p_2^2 + p_3^2)I_2}.$$

The same reasoning applies to the mutually commuting free-particle momentum operators $-i\hbar\partial_i$ on functions for which the mixed derivatives commute.

2.2.41 Use the equations for the properties of direct products, Eqs. (2.57) and (2.58), to show that the four matrices $\boldsymbol{\gamma}^\mu$, $\mu = 0, 1, 2, 3$, satisfy the conditions listed in Eqs. (2.74) and (2.75).

Solution.

Explicitly, the Dirac matrices of Eqs. (2.70) and (2.72) are

$$\gamma^0 = \sigma_3 \otimes I_2, \qquad \gamma^i = J \otimes \sigma_i \quad (i = 1, 2, 3), \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2.$$

The properties of direct products in Eqs. (2.57) and (2.58) are the direct-product rule and the distributive laws,

$$(A \otimes B)(C \otimes D) = AC \otimes BD,$$
$$(A + B) \otimes C = A \otimes C + B \otimes C, \qquad C \otimes (A + B) = C \otimes A + C \otimes B.$$

By the entry formula $(A \otimes B)_{(i,a),(j,b)} = A_{ij}B_{ab}$ of Exercise 2.2.39, scalar factors also move freely, $(\lambda A) \otimes B = A \otimes (\lambda B) = \lambda(A \otimes B)$, and $I_2 \otimes I_2 = I_4$. Since $J^2 = (i\sigma_2)^2 = -\sigma_2^2 = -I_2$ and $\sigma_i^2 = I_2$ (Exercise 2.2.11),

$$(\gamma^0)^2 = \sigma_3^2 \otimes I_2^2 = I_2 \otimes I_2 = I_4, \qquad (\gamma^i)^2 = J^2 \otimes \sigma_i^2 = -I_2 \otimes I_2 = -I_4.$$

For the mixed products the rule gives $\gamma^0\gamma^i = (\sigma_3 \otimes I_2)(J \otimes \sigma_i) = \sigma_3 J \otimes \sigma_i$ and $\gamma^i\gamma^0 = J\sigma_3 \otimes \sigma_i$, where

$$\sigma_3 J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_1, \qquad J\sigma_3 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = -\sigma_1.$$

Hence, by the distributive law,

$$\gamma^0\gamma^i + \gamma^i\gamma^0 = \sigma_3 J \otimes \sigma_i + J\sigma_3 \otimes \sigma_i = (\sigma_3 J + J\sigma_3) \otimes \sigma_i = (\sigma_1 - \sigma_1) \otimes \sigma_i = 0.$$

For spatial indices $\gamma^i\gamma^j = J^2 \otimes \sigma_i\sigma_j = -I_2 \otimes \sigma_i\sigma_j$, so by the distributive law and Exercise 2.2.11(c)

$$\gamma^i\gamma^j + \gamma^j\gamma^i = -I_2 \otimes (\sigma_i\sigma_j + \sigma_j\sigma_i) = -I_2 \otimes 2\delta_{ij}I_2 = -2\delta_{ij}I_4.$$

For $i = j$ this repeats $(\gamma^i)^2 = -I_4$; for $i \neq j$ it is the anticommutation of two spatial matrices. These are the conditions (2.74)–(2.75). With the anticommutator $\{A, B\} = AB + BA$ they combine into the single (Clifford) relation

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I_4, \quad g = \mathrm{diag}(1, -1, -1, -1)}.$$

2.2.42 Show that $\boldsymbol{\gamma}^5$, Eq. (2.76), anticommutes with all four $\boldsymbol{\gamma}^\mu$.

Solution.

By definition, $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. Use the tensor forms of Exercise 2.2.41 and the direct-product rule twice, with $J^3 = J^2 J = -J$ and $\sigma_1\sigma_2\sigma_3 = (i\sigma_3)\sigma_3 = iI_2$ (Exercise 2.2.11):

$$\gamma^1\gamma^2\gamma^3 = J^3 \otimes (\sigma_1\sigma_2\sigma_3) = (-J) \otimes (iI_2) = -iJ \otimes I_2.$$

Hence, since $i(-i) = 1$ and $\sigma_3 J = \sigma_1$ (Exercise 2.2.41),

$$\gamma^5 = i(\sigma_3 \otimes I_2)(-iJ \otimes I_2) = (\sigma_3 J) \otimes I_2 = \sigma_1 \otimes I_2 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}.$$

Now $\sigma_1\sigma_3 = -\sigma_3\sigma_1$ (Exercise 2.2.11) and, with $J = i\sigma_2$, $\sigma_1 J = i\sigma_1\sigma_2 = -i\sigma_2\sigma_1 = -J\sigma_1$. By the direct-product rule and the distributive law, therefore,

$$\{\gamma^5, \gamma^0\} = (\sigma_1\sigma_3 + \sigma_3\sigma_1) \otimes I_2 = 0,$$

$$\{\gamma^5, \gamma^i\} = (\sigma_1 J + J\sigma_1) \otimes \sigma_i = 0 \quad (i = 1, 2, 3).$$

Thus γ^5 anticommutes with all four γ^μ.

2.2.43 In this problem, the summations are over $\mu = 0, 1, 2, 3$. Define $g_{\mu\nu} = g^{\mu\nu}$ by the relations

$$g_{00} = 1; \quad g_{kk} = -1, \quad k = 1, 2, 3; \quad g_{\mu\nu} = 0, \quad \mu \neq \nu;$$

and define γ_μ as $\sum_{\nu=0}^{3} g_{\nu\mu}\gamma^\nu$. Using these definitions, show that

(a) $\sum \gamma_\mu \gamma^\alpha \gamma^\mu = -2\gamma^\alpha$,

(b) $\sum \gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\mu = 4g^{\alpha\beta}\mathbf{1}_4$,

(c) $\sum \gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\nu \gamma^\mu = -2\gamma^\nu\gamma^\beta\gamma^\alpha$.

Solution.

The textbook prints the lowering sum without a summation index and with γ^μ in the summand, although μ is the free index. The intended definition is the contraction over the dummy index ν shown above; for this diagonal metric it gives $\gamma_\mu = g_{\mu\mu}\gamma^\mu$ with no sum on μ. In (b), a scalar denotes that scalar times I_4, made explicit in the statement. The sums in (a)–(c) run over $\mu = 0, 1, 2, 3$. Lowering an index gives $\gamma_0 = \gamma^0$, $\gamma_i = -\gamma^i$. The Clifford relation $\{\gamma^\mu, \gamma^\alpha\} = 2g^{\mu\alpha}I_4$ of Exercise 2.2.41 implies

$$\{\gamma_\mu, \gamma^\alpha\} = g_{\mu\mu}\{\gamma^\mu, \gamma^\alpha\} = 2g_{\mu\mu}g^{\mu\alpha}I_4 = 2\delta^\alpha_\mu I_4, \qquad \sum_\mu \gamma_\mu\gamma^\mu = 4I_4,$$

because $g^{\mu\alpha} = 0$ for $\alpha \neq \mu$ and $g_{\mu\mu}g^{\mu\mu} = (\pm 1)^2 = 1$. For the second identity, the time term is $(\gamma^0)^2 = I_4$ and each spatial term is $-(\gamma^i)^2 = I_4$.

(a) Move γ_μ past the first factor with $\gamma_\mu\gamma^\alpha = 2\delta^\alpha_\mu I_4 - \gamma^\alpha\gamma_\mu$; in the first term the Kronecker delta keeps only $\mu = \alpha$:

$$\sum_\mu \gamma_\mu\gamma^\alpha\gamma^\mu = \sum_\mu (2\delta^\alpha_\mu I_4 - \gamma^\alpha\gamma_\mu)\gamma^\mu = 2\gamma^\alpha - \gamma^\alpha \sum_\mu \gamma_\mu\gamma^\mu$$

$$= 2\gamma^\alpha - \gamma^\alpha(4I_4) = \boxed{-2\gamma^\alpha}.$$

(b) Use the same step and then (a) with α replaced by β:

$$\begin{aligned}\sum_\mu \gamma_\mu\gamma^\alpha\gamma^\beta\gamma^\mu &= \sum_\mu (2\delta^\alpha_\mu I_4 - \gamma^\alpha\gamma_\mu)\gamma^\beta\gamma^\mu \\ &= 2\gamma^\beta\gamma^\alpha - \gamma^\alpha\sum_\mu \gamma_\mu\gamma^\beta\gamma^\mu \\ &= 2\gamma^\beta\gamma^\alpha + 2\gamma^\alpha\gamma^\beta = 2\{\gamma^\alpha,\gamma^\beta\} = \boxed{4g^{\alpha\beta}I_4}.\end{aligned}$$

(c) Use the same step, then (b) with (α,β) replaced by (β,ν), and finally the Clifford relation in the form $\gamma^\beta\gamma^\nu - 2g^{\beta\nu}I_4 = -\gamma^\nu\gamma^\beta$:

$$\begin{aligned}\sum_\mu \gamma_\mu\gamma^\alpha\gamma^\beta\gamma^\nu\gamma^\mu &= 2\gamma^\beta\gamma^\nu\gamma^\alpha - \gamma^\alpha\sum_\mu \gamma_\mu\gamma^\beta\gamma^\nu\gamma^\mu \\ &= 2\gamma^\beta\gamma^\nu\gamma^\alpha - 4g^{\beta\nu}\gamma^\alpha \\ &= 2(\gamma^\beta\gamma^\nu - 2g^{\beta\nu}I_4)\gamma^\alpha = \boxed{-2\gamma^\nu\gamma^\beta\gamma^\alpha}.\end{aligned}$$

These derivations include coincident free indices; no distinctness assumption was made.

2.2.44 If $\mathsf{M} = \frac{1}{2}(1+\boldsymbol{\gamma}^5)$, where $\boldsymbol{\gamma}^5$ is given in Eq. (2.76), show that

$$\mathsf{M}^2 = \mathsf{M}.$$

Note that this equation is still satisfied if $\boldsymbol{\gamma}^5$ is replaced by any other Dirac matrix listed in Eq. (2.76) whose square is $\mathbf{1}_4$.

Solution.

The printed note lacks the qualification "whose square is $\mathbf{1}_4$": as shown below, $M_\Gamma = (I_4+\Gamma)/2$ satisfies $M_\Gamma^2 - M_\Gamma = (\Gamma^2 - I_4)/4$, which is $-I_4/2$ for the seven Dirac matrices with $\Gamma^2 = -I_4$. Since $\gamma^5 = \sigma_1 \otimes I_2$ (Exercise 2.2.42), $(\gamma^5)^2 = \sigma_1^2 \otimes I_2 = I_4$. Expanding the square, in which I_4 commutes with γ^5,

$$M^2 = \tfrac{1}{4}(I_4 + 2\gamma^5 + (\gamma^5)^2) = \tfrac{1}{4}(2I_4 + 2\gamma^5) = \tfrac{1}{2}(I_4+\gamma^5) = M.$$

For the note in the statement, let Γ be any 4×4 matrix and $M_\Gamma = (I_4+\Gamma)/2$. The same expansion gives $M_\Gamma^2 = \frac{1}{4}(I_4 + 2\Gamma + \Gamma^2)$, so

$$M_\Gamma^2 - M_\Gamma = \frac{I_4 + 2\Gamma + \Gamma^2}{4} - \frac{2I_4 + 2\Gamma}{4} = \frac{\Gamma^2 - I_4}{4},$$

and $M_\Gamma^2 = M_\Gamma$ exactly when $\Gamma^2 = I_4$. Each of the 16 Dirac matrices squares to $\pm I_4$:

Γ	Γ^2
$I_4,\ \gamma^5,\ \gamma^0,\ \gamma^5\gamma^i,\ \sigma^{ij}\ (1\le i<j\le 3)$	$+I_4$
$\gamma^i,\ \gamma^5\gamma^0,\ \sigma^{0i}\ (i=1,2,3)$	$-I_4$

where $\sigma^{\mu\nu} = i\gamma^\mu\gamma^\nu$ for $\mu<\nu$. The entries follow from $(\gamma^0)^2 = (\gamma^5)^2 = I_4$, $(\gamma^i)^2 = -I_4$, and the anticommutation relations of Exercises 2.2.41 and 2.2.42: moving the second γ^5 to the left past γ^μ, and the second γ^μ to the left past γ^ν,

$$\begin{aligned}(\gamma^5\gamma^\mu)^2 &= \gamma^5\gamma^\mu\gamma^5\gamma^\mu = -(\gamma^5)^2(\gamma^\mu)^2 = -(\gamma^\mu)^2, \\ (i\gamma^\mu\gamma^\nu)^2 &= -\gamma^\mu\gamma^\nu\gamma^\mu\gamma^\nu = \gamma^\mu\gamma^\mu\gamma^\nu\gamma^\nu = (\gamma^\mu)^2(\gamma^\nu)^2 \quad (\mu\ne\nu).\end{aligned}$$

Hence $(\gamma^5\gamma^0)^2 = -I_4$, $(\gamma^5\gamma^i)^2 = +I_4$, $(\sigma^{0i})^2 = (\gamma^0)^2(\gamma^i)^2 = -I_4$, and $(\sigma^{ij})^2 = (\gamma^i)^2(\gamma^j)^2 = +I_4$. Thus the replacement works for the matrices in the first row of the table and fails for the seven in the second row, for which $M_\Gamma^2 - M_\Gamma = -I_4/2$. Multiplying such a Γ by i makes its square $(i\Gamma)^2 = -\Gamma^2 = +I_4$, so that $M_{i\Gamma} = (I_4 + i\Gamma)/2$ again satisfies $M_{i\Gamma}^2 = M_{i\Gamma}$.

2.2.45 Prove that the 16 Dirac matrices form a linearly independent set.

Solution.

Let $\sigma_0 = I_2$. All four σ_a are Hermitian, so $\sigma_a^\dagger \sigma_b = \sigma_a \sigma_b$, and the Pauli products of Exercise 2.2.11 give

$$\operatorname{tr}(\sigma_a^\dagger \sigma_b) = 2\delta_{ab}, \qquad a, b = 0, 1, 2, 3.$$

For $a = b$ the product is $\sigma_a^2 = I_2$, of trace 2; for distinct indices the product is a traceless Pauli matrix or $\pm i$ times one. Also, the diagonal entries of $X \otimes Y$ are the products $X_{rr} Y_{ss}$, so

$$\operatorname{tr}(X \otimes Y) = \sum_{r,s} X_{rr} Y_{ss} = \operatorname{tr} X \ \operatorname{tr} Y.$$

Therefore the 16 tensor products $T_{ab} = \sigma_a \otimes \sigma_b$ obey, by $(X \otimes Y)^\dagger = X^\dagger \otimes Y^\dagger$ (Exercise 2.2.39) and the direct-product rule,

$$\operatorname{tr}(T_{ab}^\dagger T_{cd}) = \operatorname{tr}(\sigma_a^\dagger \sigma_c \otimes \sigma_b^\dagger \sigma_d) = \operatorname{tr}(\sigma_a^\dagger \sigma_c) \ \operatorname{tr}(\sigma_b^\dagger \sigma_d) = 4\delta_{ac}\delta_{bd}.$$

The explicit correspondence with Eq. (2.76) is

Dirac matrix	tensor form
I_4	$\sigma_0 \otimes \sigma_0$
γ^5	$\sigma_1 \otimes \sigma_0$
γ^0	$\sigma_3 \otimes \sigma_0$
$\gamma^5\gamma^0$	$-i\sigma_2 \otimes \sigma_0$
γ^i $(i = 1, 2, 3)$	$i\sigma_2 \otimes \sigma_i$
$\gamma^5\gamma^i$ $(i = 1, 2, 3)$	$-\sigma_3 \otimes \sigma_i$
σ^{0i} $(i = 1, 2, 3)$	$i\sigma_1 \otimes \sigma_i$
$\sigma^{12},\ \sigma^{13},\ \sigma^{23}$	$\sigma_0 \otimes \sigma_3,\ -\sigma_0 \otimes \sigma_2,\ \sigma_0 \otimes \sigma_1$

The first three rows are I_4 and the forms of γ^5 (Exercise 2.2.42) and γ^0 (Exercise 2.2.41), and the row of γ^i is $\gamma^i = J \otimes \sigma_i$ with $J = i\sigma_2$. The others follow from the direct-product rule and the Pauli products of Exercise 2.2.11:

$$\begin{aligned}
\gamma^5\gamma^0 &= \sigma_1\sigma_3 \otimes \sigma_0 = -i\sigma_2 \otimes \sigma_0, \qquad \gamma^5\gamma^i = \sigma_1(i\sigma_2) \otimes \sigma_i = -\sigma_3 \otimes \sigma_i, \\
\sigma^{0i} &= i\gamma^0\gamma^i = i\,\sigma_3(i\sigma_2) \otimes \sigma_i = -\sigma_3\sigma_2 \otimes \sigma_i = i\sigma_1 \otimes \sigma_i, \\
\sigma^{ij} &= i\gamma^i\gamma^j = i(i\sigma_2)^2 \otimes \sigma_i\sigma_j = -iI_2 \otimes \sigma_i\sigma_j = \varepsilon_{ijk} I_2 \otimes \sigma_k,
\end{aligned}$$

where in the last line $i < j$, k is the remaining index, and $\sigma_i\sigma_j = i\varepsilon_{ijk}\sigma_k$. The table uses each pair (a, b) exactly once, so each Dirac matrix is $\Gamma_r = \eta_r T_{a_r b_r}$ with a different pair (a_r, b_r) and a coefficient $\eta_r \in \{\pm 1, \pm i\}$ of modulus 1. Hence

$$\operatorname{tr}(\Gamma_r^\dagger \Gamma_s) = \eta_r^* \eta_s \operatorname{tr}(T_{a_r b_r}^\dagger T_{a_s b_s}) = 4|\eta_r|^2 \delta_{rs} = 4\delta_{rs}.$$

If $\sum_s c_s \Gamma_s = 0$, multiply by $\Gamma_r^\dagger$ and take the trace: $4c_r = 0$ for every r. This proves linear independence over $\mathbb{C}$, and hence over $\mathbb{R}$. Other sign or phase conventions (for example $\gamma^0\gamma^5 = -\gamma^5\gamma^0$ in place of $\gamma^5\gamma^0$) only change some η_r and do not affect the argument. The complex 4×4 matrix space has dimension 16, so these matrices also span it.

2.2.46 If we assume that a given 4×4 matrix A (with constant elements) can be written as a linear combination of the 16 Dirac matrices (which we denote here as Γ_i)

$$\mathsf{A} = \sum_{i=1}^{16} c_i \Gamma_i,$$

show that

$$c_i \sim \text{trace}(\mathsf{A}\Gamma_i).$$

Solution.

To avoid confusion with the imaginary unit, label the Dirac matrices Γ_r, $r = 1, \ldots, 16$, and the coefficients c_r. Write $\Gamma_r^2 = s_r I_4$, where $s_r = \pm 1$ as in the table of Exercise 2.2.44. In Exercise 2.2.45 each $\Gamma_r = \eta_r T_r$, where T_r is one of the products $\sigma_a \otimes \sigma_b$ and $\eta_r \in \{\pm 1, \pm i\}$. Each T_r is Hermitian, since $(\sigma_a \otimes \sigma_b)^\dagger = \sigma_a^\dagger \otimes \sigma_b^\dagger = \sigma_a \otimes \sigma_b$ (Exercise 2.2.39), and, by the direct-product rule, $T_r^2 = \sigma_a^2 \otimes \sigma_b^2 = I_4$; hence $\Gamma_r^2 = \eta_r^2 I_4$, so $s_r = \eta_r^2$, and

$$\Gamma_r^\dagger = \eta_r^* T_r = \frac{\eta_r^*}{\eta_r}\Gamma_r = (\eta_r^*)^2 \Gamma_r = s_r \Gamma_r,$$

because $\eta_r^*/\eta_r = (\eta_r^*)^2/|\eta_r|^2$, $|\eta_r| = 1$, and $(\eta_r^*)^2 = (\eta_r^2)^* = s_r$ is real. Since $s_r^2 = 1$, also $\Gamma_r = s_r \Gamma_r^\dagger$, and the trace relation of Exercise 2.2.45 gives

$$\text{tr}(\Gamma_r \Gamma_j) = s_r \,\text{tr}(\Gamma_r^\dagger \Gamma_j) = 4 s_r \delta_{rj}.$$

Multiplying $A = \sum_j c_j \Gamma_j$ by Γ_r and taking the trace, which is linear and satisfies $\text{tr}(\Gamma_j \Gamma_r) = \text{tr}(\Gamma_r \Gamma_j)$,

$$\text{tr}(A\Gamma_r) = \sum_j c_j \,\text{tr}(\Gamma_j \Gamma_r) = 4 s_r c_r.$$

Thus the full proportionality, with its normalization and sign, is

$$\boxed{c_r = \frac{1}{4 s_r}\,\text{tr}(A\Gamma_r) = \frac{s_r}{4}\,\text{tr}(A\Gamma_r) = \frac{1}{4}\,\text{tr}(\Gamma_r^\dagger A)}.$$

The last form follows from $s_r \Gamma_r = \Gamma_r^\dagger$ and $\text{tr}(A\Gamma_r^\dagger) = \text{tr}(\Gamma_r^\dagger A)$. The sign s_r matters for the basis matrices squaring to $-I_4$.

2.2.47 The matrix $\mathsf{C} = i\boldsymbol{\gamma}^2\boldsymbol{\gamma}^0$ is sometimes called the charge conjugation matrix. Show that $\mathsf{C}\boldsymbol{\gamma}^\mu\mathsf{C}^{-1} = -(\boldsymbol{\gamma}^\mu)^T$.

Solution.

Use the Dirac representation of Eqs. (2.70) and (2.72), $\gamma^0 = \sigma_3 \otimes I_2$, $\gamma^i = i\sigma_2 \otimes \sigma_i$ (Exercise 2.2.41). By the direct-product rule and $\sigma_2\sigma_3 = i\sigma_1$,

$$C = i\gamma^2\gamma^0 = i\,(i\sigma_2\sigma_3 \otimes \sigma_2) = -\sigma_2\sigma_3 \otimes \sigma_2 = -i\sigma_1 \otimes \sigma_2,$$

so $C^2 = (-i)^2\sigma_1^2 \otimes \sigma_2^2 = -I_4$ and $C^{-1} = -C$. The transpose of a direct product is $(A \otimes B)^T = A^T \otimes B^T$, since in the notation of Exercise 2.2.39 $((A \otimes B)^T)_{(k,a),(l,b)} = A_{lk}B_{ba} = (A^T)_{kl}(B^T)_{ab}$. The Pauli transposes are $\sigma_1^T = \sigma_1$, $\sigma_2^T = -\sigma_2$, $\sigma_3^T = \sigma_3$, so $(\gamma^0)^T = \sigma_3 \otimes I_2 = \gamma^0$ and $(\gamma^i)^T = (i\sigma_2)^T \otimes \sigma_i^T = -i\sigma_2 \otimes \sigma_i^T$, giving

$$(\gamma^0)^T = \gamma^0, \quad (\gamma^2)^T = \gamma^2, \quad (\gamma^1)^T = -\gamma^1, \quad (\gamma^3)^T = -\gamma^3.$$

On the other side, the product $\gamma^2\gamma^0$ anticommutes with γ^0 and γ^2: by $(\gamma^0)^2 = I_4$, $(\gamma^2)^2 = -I_4$, and $\gamma^0\gamma^2 = -\gamma^2\gamma^0$ (Exercise 2.2.41),

$$C\gamma^0 = i\gamma^2(\gamma^0)^2 = i\gamma^2, \qquad \gamma^0 C = i\gamma^0\gamma^2\gamma^0 = -i\gamma^2(\gamma^0)^2 = -i\gamma^2,$$
$$C\gamma^2 = i\gamma^2\gamma^0\gamma^2 = -i(\gamma^2)^2\gamma^0 = i\gamma^0, \quad \gamma^2 C = i(\gamma^2)^2\gamma^0 = -i\gamma^0.$$

Hence $C\gamma^\mu = -\gamma^\mu C$ for $\mu = 0, 2$; for instance, $C\gamma^0 = i\gamma^2 = -\gamma^0 C$. On the other hand, C commutes with γ^1, γ^3, since moving either to the left past the two factors γ^0 and γ^2 gives two minus signs: $C\gamma^\mu = i\gamma^2\gamma^0\gamma^\mu = i\gamma^\mu\gamma^2\gamma^0 = \gamma^\mu C$ for $\mu = 1, 3$. Multiplying these two relations on the right by C^{-1} gives

$$C\gamma^\mu C^{-1} = \begin{cases} -\gamma^\mu, & \mu = 0, 2, \\ +\gamma^\mu, & \mu = 1, 3. \end{cases}$$

Comparison with the transposes proves $\boxed{C\gamma^\mu C^{-1} = -(\gamma^\mu)^T}$ for every μ. The formula for $C\gamma^\mu C^{-1}$ above used only the relations (2.74)–(2.75), which also give $C^2 = -\gamma^2\gamma^0\gamma^2\gamma^0 = (\gamma^2)^2(\gamma^0)^2 = -I_4$ in any representation. The representation therefore enters through the transposes alone: the identity holds exactly when γ^0, γ^2 are symmetric and γ^1, γ^3 antisymmetric, as also in the Weyl representation of Exercise 2.2.49. It fails, for example, for the matrices $S\gamma^\mu S^{-1}$ with $S = \mathrm{diag}(1, 2, 1, 1)$. These also obey (2.74)–(2.75), because $(S\gamma^\mu S^{-1})(S\gamma^\nu S^{-1}) = S\gamma^\mu\gamma^\nu S^{-1}$ turns each anticommutator into $S\{\gamma^\mu, \gamma^\nu\}S^{-1} = 2g^{\mu\nu} I_4$, but $S\gamma^1 S^{-1}$ is not antisymmetric: for diagonal S its (k, l) entry is $S_{kk}(\gamma^1)_{kl}/S_{ll}$, so the entries $(\gamma^1)_{23} = 1$ and $(\gamma^1)_{32} = -1$ become 2 and $-\frac{1}{2}$.

2.2.48

(a) Show that, by substitution of the definitions of the $\boldsymbol{\gamma}^\mu$ matrices from Eqs. (2.70) and (2.72), that the Dirac equation, Eq. (2.73), takes the following form when written as 2×2 blocks (with ψ_L and ψ_S column vectors of dimension 2). Here L and S stand, respectively, for "large" and "small" because of their relative size in the nonrelativistic limit:

$$\begin{pmatrix} mc^2 - E & c(\sigma_1 p_1 + \sigma_2 p_2 + \sigma_3 p_3) \\ +c(\sigma_1 p_1 + \sigma_2 p_2 + \sigma_3 p_3) & -mc^2 - E \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_S \end{pmatrix} = 0.$$

(b) Take $m > 0$ and the positive-energy branch. To reach the nonrelativistic limit, make the substitution $E = mc^2 + \varepsilon$ and approximate $-2mc^2 - \varepsilon$ by $-2mc^2$. Then write the matrix equation as two simultaneous two-component equations and show that they can be rearranged to yield

$$\frac{1}{2m}\left(p_1^2 + p_2^2 + p_3^2\right)\psi_L = \varepsilon\psi_L,$$

which is just the Schrödinger equation for a free particle.

(c) Explain why is it reasonable to call ψ_L and ψ_S "large" and "small."

Solution.

The lower-left block in the printed question (textbook p. 121) has an incorrect minus sign. Equation (2.73) requires the positive sign shown in the corrected question above. The printed sign would give negative free-particle kinetic energy upon elimination of the small component: its second row would imply $\psi_S \simeq -P\psi_L/(2mc)$, and its first would then give $\varepsilon\psi_L = -\mathbf{p}^2\psi_L/(2m)$, where $P = \boldsymbol{\sigma}\cdot\mathbf{p}$.

In the Instructor's Manual, the block Hamiltonian also omits c from its momentum terms. Every momentum term of Eq. (2.73), like those of the question above, carries the factor c; all factors of c are retained below.

(a) Put $P = \boldsymbol{\sigma}\cdot\mathbf{p} = \sigma_1 p_1 + \sigma_2 p_2 + \sigma_3 p_3$ and $\alpha_i = \gamma^0\gamma^i$. Since $(\gamma^0)^2 = I_4$ (Exercise 2.2.41), multiplying on the left by the invertible matrix γ^0 turns $\gamma^0 E - c\sum_i \gamma^i p_i$ into $E - c\sum_i \alpha_i p_i$ and mc^2 into $mc^2\gamma^0$. The Dirac equation written with the γ^μ is therefore equivalent to the form used below:

$$\Big(\gamma^0 E - c\sum_i \gamma^i p_i\Big)\Psi = mc^2\Psi \quad\Longleftrightarrow\quad \Big(mc^2\gamma^0 + c\sum_i \alpha_i p_i\Big)\Psi = E\Psi.$$

The definitions in Eqs. (2.70) and (2.72) give

$$\gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \quad \alpha_i = \gamma^0\gamma^i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix};$$

in the block product $\gamma^0\gamma^i$ the upper-right block is $I_2\sigma_i = \sigma_i$, the lower-left block is $(-I_2)(-\sigma_i) = \sigma_i$, and the diagonal blocks vanish. Hence

$$mc^2\gamma^0 + c\sum_i \alpha_i p_i - EI_4 = \begin{pmatrix} (mc^2 - E)I_2 & cP \\ cP & (-mc^2 - E)I_2 \end{pmatrix},$$

and with $\Psi = (\psi_L, \psi_S)^T$ the Dirac equation $(mc^2\gamma^0 + c\sum_i \alpha_i p_i)\Psi = E\Psi$ is

$$\boxed{\begin{pmatrix} (mc^2 - E)I_2 & cP \\ cP & (-mc^2 - E)I_2 \end{pmatrix}\begin{pmatrix} \psi_L \\ \psi_S \end{pmatrix} = 0}.$$

Both off-diagonal blocks are the same.

(b) The qualification added to (b) is needed because the approximation $-2mc^2 - \varepsilon \simeq -2mc^2$ requires $|\varepsilon| \ll mc^2$, that is, $E \simeq +mc^2$ with $m > 0$; on the negative-energy branch $E \le -mc^2$, so $\varepsilon \le -2mc^2$, and there ψ_L, not ψ_S, is the small component. On the positive-energy branch set $E = mc^2 + \varepsilon$ with $|\varepsilon| \ll mc^2$. Then $mc^2 - E = -\varepsilon$ and $-mc^2 - E = -(2mc^2 + \varepsilon)$, so the two block rows of (a) are the two-component equations

$$-\varepsilon\psi_L + cP\psi_S = 0, \qquad cP\psi_L - (2mc^2 + \varepsilon)\psi_S = 0.$$

The second gives $\psi_S = cP\psi_L/(2mc^2 + \varepsilon)$. Substitution into the first yields, since $P^2 = \mathbf{p}^2 I_2$ with $\mathbf{p}^2 = p_1^2 + p_2^2 + p_3^2$ (Exercise 2.2.40),

$$\varepsilon\psi_L = \frac{c^2P^2}{2mc^2 + \varepsilon}\psi_L = \frac{c^2\mathbf{p}^2}{2mc^2 + \varepsilon}\psi_L.$$

Approximating $-2mc^2 - \varepsilon$ by $-2mc^2$, as the problem directs, replaces $(2mc^2 + \varepsilon)^{-1}$ by $(2mc^2)^{-1}$; since $c^2/(2mc^2) = 1/(2m)$, this gives

$$\boxed{\frac{\mathbf{p}^2}{2m}\psi_L = \varepsilon\psi_L}.$$

With $p_i = -i\hbar\partial_i$, this is $-\hbar^2\nabla^2\psi_L/(2m) = \varepsilon\psi_L$. As an exact check before approximation, take a plane wave of momentum $\mathbf{p}$ (so that $\mathbf{p}$ is a numerical vector) with $\psi_L \neq 0$. Multiplying by $2mc^2 + \varepsilon$ gives $\varepsilon(2mc^2 + \varepsilon) = c^2\mathbf{p}^2$; since $\varepsilon(2mc^2 + \varepsilon) = (E - mc^2)(E + mc^2) = E^2 - m^2c^4$, this is $E^2 = m^2c^4 + c^2\mathbf{p}^2$.

(c) For such a state $2mc^2+\varepsilon = E+mc^2 > 0$, and the exact relation of (b) is $\psi_S = cP\psi_L/(E+mc^2)$. For real $\mathbf{p}$, P is Hermitian, so $P^\dagger P = P^2 = \mathbf{p}^2 I_2$. Measure a two-component column by its magnitude $\|\psi\| = (\psi^\dagger\psi)^{1/2}$. Since $E + mc^2$ is real, $\psi_S^\dagger = c\,\psi_L^\dagger P^\dagger/(E + mc^2)$, and

$$\|\psi_S\|^2 = \psi_S^\dagger\psi_S = \frac{c^2\,\psi_L^\dagger P^\dagger P\psi_L}{(E + mc^2)^2} = \frac{c^2\mathbf{p}^2}{(E + mc^2)^2}\,\|\psi_L\|^2.$$

Hence, with $E \simeq mc^2$ in the nonrelativistic regime,

$$\frac{\|\psi_S\|}{\|\psi_L\|} = \frac{c|\mathbf{p}|}{E + mc^2} \simeq \frac{|\mathbf{p}|}{2mc} \ll 1 \quad \text{when } |\mathbf{p}| \ll mc.$$

This explains the names on the positive-energy, nonrelativistic branch. At rest the small component vanishes; the ratio need not be small in the relativistic regime, and it tends to 1 as $|\mathbf{p}| \to \infty$, where $E \simeq c|\mathbf{p}|$.

2.2.49 Show that it is consistent with the requirements that they must satisfy to take the Dirac gamma matrices to be (in 2×2 block form)

$$\boldsymbol{\gamma}^0 = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}, \quad \boldsymbol{\gamma}^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \ (i = 1, 2, 3).$$

This choice for the gamma matrices is called the Weyl representation.

Solution.

Write the proposed Weyl matrices as

$$\gamma^0 = \sigma_1 \otimes I_2, \qquad \gamma^i = J \otimes \sigma_i, \quad J = i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix};$$

the γ^i are the same as in the Dirac representation of Exercise 2.2.41. Then $(\gamma^0)^2 = \sigma_1^2 \otimes I_2 = I_4$ and $(\gamma^i)^2 = J^2 \otimes \sigma_i^2 = -I_4$. By the direct-product rule $\gamma^0\gamma^i = \sigma_1 J \otimes \sigma_i$ and $\gamma^i\gamma^0 = J\sigma_1 \otimes \sigma_i$, where

$$\sigma_1 J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -\sigma_3, \qquad J\sigma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_3.$$

Hence

$$\{\gamma^0, \gamma^i\} = (\sigma_1 J + J\sigma_1) \otimes \sigma_i = 0.$$

For spatial indices, as in Exercise 2.2.41, $\{\gamma^i, \gamma^j\} = J^2 \otimes \{\sigma_i, \sigma_j\} = -2\delta_{ij} I_4$. All the required relations (2.74)–(2.75) hold. Also, by $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$ (Exercise 2.2.39), $(\gamma^0)^\dagger = \sigma_1^\dagger \otimes I_2 = \gamma^0$ and, since $J^\dagger = J^T = -J$ (J is real) and $\sigma_i^\dagger = \sigma_i$, $(\gamma^i)^\dagger = J^\dagger \otimes \sigma_i^\dagger = -\gamma^i$. With the anticommutation just proved,

$$\alpha_i^\dagger = (\gamma^0\gamma^i)^\dagger = (\gamma^i)^\dagger(\gamma^0)^\dagger = -\gamma^i\gamma^0 = \gamma^0\gamma^i = \alpha_i.$$

The free Dirac Hamiltonian $mc^2\gamma^0 + c\sum_i \alpha_i p_i$ therefore has the required Hermiticity as well.

2.2.50 Show that the Dirac equation separates into independent 2×2 blocks in the Weyl representation (see Exercise 2.2.49) in the limit that the mass m approaches zero. This observation is important in the ultra relativistic regime where the rest mass is inconsequential, or for particles of negligible mass (e.g., neutrinos).

Solution.

The Instructor's Manual's momentum blocks again lack their factors of c. With the units of Eq. (2.73), the decoupled equations contain $\pm c\boldsymbol{\sigma}\cdot\mathbf{p}$. In the Weyl representation, block multiplication gives

$$\alpha_i = \gamma^0\gamma^i = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} = \begin{pmatrix} -\sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}, \qquad mc^2\gamma^0 = \begin{pmatrix} 0 & mc^2 I_2 \\ mc^2 I_2 & 0 \end{pmatrix},$$

so $c\sum_i \alpha_i p_i$ is block diagonal with diagonal blocks $-c\boldsymbol{\sigma}\cdot\mathbf{p}$ and $c\boldsymbol{\sigma}\cdot\mathbf{p}$. With $\Psi = (\psi_1, \psi_2)^T$, the Dirac equation in the form $(mc^2\gamma^0 + c\sum_i \alpha_i p_i)\Psi = E\Psi$ of Exercise 2.2.48(a), whose derivation used only $(\gamma^0)^2 = I_4$, true here too (Exercise 2.2.49), is

$$\begin{pmatrix} -c\boldsymbol{\sigma}\cdot\mathbf{p} & mc^2 I_2 \\ mc^2 I_2 & c\boldsymbol{\sigma}\cdot\mathbf{p} \end{pmatrix}\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = E\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}.$$

Equivalently,

$$-c\boldsymbol{\sigma}\cdot\mathbf{p}\,\psi_1 + mc^2\psi_2 = E\psi_1, \qquad mc^2\psi_1 + c\boldsymbol{\sigma}\cdot\mathbf{p}\,\psi_2 = E\psi_2.$$

At $m = 0$ the coupling terms disappear:

$$\boxed{-c\boldsymbol{\sigma}\cdot\mathbf{p}\,\psi_1 = E\psi_1, \qquad c\boldsymbol{\sigma}\cdot\mathbf{p}\,\psi_2 = E\psi_2}.$$

Each two-component equation can be solved independently. Squaring either gives the massless relation: applying $-c\boldsymbol{\sigma}\cdot\mathbf{p}$ to both sides of the first and using $(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = \mathbf{p}^2 I_2$ (Exercise 2.2.40),

$$c^2\mathbf{p}^2\psi_1 = E(-c\boldsymbol{\sigma}\cdot\mathbf{p}\,\psi_1) = E^2\psi_1,$$

and likewise $c^2\mathbf{p}^2\psi_2 = E^2\psi_2$. For a plane wave with a nonzero component this is $E^2 = c^2\mathbf{p}^2$.

2.2.51

(a) Given $\mathbf{r}' = \mathsf{U}\mathbf{r}$, with U a unitary matrix and $\mathbf{r}$ a (column) vector with complex elements, show that the magnitude of $\mathbf{r}$ is invariant under this operation.

(b) The square matrix U transforms any column vector $\mathbf{r}$ with complex elements into $\mathbf{r}'$, leaving the magnitude invariant: $\mathbf{r}^\dagger\mathbf{r} = \mathbf{r}'^\dagger\mathbf{r}'$. Show that U is unitary.

Solution.

(a) For a complex vector, $\|\mathbf{r}\|^2 = \mathbf{r}^\dagger\mathbf{r}$. If $\mathbf{r}' = U\mathbf{r}$ with $U^\dagger U = I$, then, since $(U\mathbf{r})^\dagger = \mathbf{r}^\dagger U^\dagger$ (Exercise 2.2.32),

$$\|\mathbf{r}'\|^2 = (U\mathbf{r})^\dagger U\mathbf{r} = \mathbf{r}^\dagger U^\dagger U\mathbf{r} = \mathbf{r}^\dagger\mathbf{r}.$$

Taking nonnegative square roots proves invariance of the magnitude.

(b) In the Instructor's Manual, $\mathbf{r}'^\dagger\mathbf{r}$ must be $\mathbf{r}'^\dagger\mathbf{r}'$. Here U is square and acts on vectors of the same dimension. For rectangular U, norm preservation implies only $U^\dagger U = I$, not two-sided unitarity. Set $H = U^\dagger U - I$. Since $\mathbf{r}'^\dagger\mathbf{r}' = \mathbf{r}^\dagger U^\dagger U\mathbf{r}$ as in (a), the hypothesis is $\mathbf{r}^\dagger H\mathbf{r} = 0$ for every complex $\mathbf{r}$; also $H^\dagger = (U^\dagger U)^\dagger - I = H$, so $H_{ji} = H_{ij}^*$. Let e_i be the ith unit column vector, so that $e_i^\dagger H e_j = H_{ij}$. Choose $\mathbf{r} = e_i$ to get $H_{ii} = 0$. For $i \neq j$, choose $\mathbf{r} = e_i + e_j$:

$$0 = (e_i + e_j)^\dagger H(e_i + e_j) = H_{ii} + H_{ij} + H_{ji} + H_{jj} = H_{ij} + H_{ij}^* = 2\,\mathrm{Re}\,H_{ij}.$$

Next choose $\mathbf{r} = e_i + ie_j$, for which $\mathbf{r}^\dagger = e_i^\dagger - ie_j^\dagger$ and $(-i)(i) = 1$:

$$0 = H_{ii} + iH_{ij} - iH_{ji} + H_{jj} = i(H_{ij} - H_{ij}^*) = i \cdot 2i \operatorname{Im} H_{ij} = -2 \operatorname{Im} H_{ij}.$$

Every entry vanishes, so $H = 0$ and $U^\dagger U = I$. Since U is square, taking determinants of $U^\dagger U = I$, with the product rule $\det(AB) = \det A \det B$ and $\det U^\dagger = (\det U)^*$ (Exercise 2.2.30), yields $\det U^\dagger \det U = (\det U)^* \det U = |\det U|^2 = 1$. Hence $\det U \neq 0$ and U is invertible; multiplying $U^\dagger U = I$ on the right by U^{-1} gives $U^\dagger = U^{-1}$, and then $UU^\dagger = UU^{-1} = I$. Thus U is unitary.

3 Chapter 3. Vector Analysis

Problems

3.2 Vectors in 3-D Space

3.2.1 If $\mathbf{P} = \hat{\mathbf{e}}_x P_x + \hat{\mathbf{e}}_y P_y$ and $\mathbf{Q} = \hat{\mathbf{e}}_x Q_x + \hat{\mathbf{e}}_y Q_y$ are any two nonparallel (also nonantiparallel) vectors in the xy-plane, show that $\mathbf{P} \times \mathbf{Q}$ is in the z-direction.

Solution.

Distribute the cross product, using $\hat{\mathbf{e}}_x \times \hat{\mathbf{e}}_y = \hat{\mathbf{e}}_z$, $\hat{\mathbf{e}}_y \times \hat{\mathbf{e}}_x = -\hat{\mathbf{e}}_z$, and $\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_i = 0$:

$$\begin{aligned}\mathbf{P} \times \mathbf{Q} &= P_x Q_x(\hat{\mathbf{e}}_x \times \hat{\mathbf{e}}_x) + P_x Q_y(\hat{\mathbf{e}}_x \times \hat{\mathbf{e}}_y) + P_y Q_x(\hat{\mathbf{e}}_y \times \hat{\mathbf{e}}_x) + P_y Q_y(\hat{\mathbf{e}}_y \times \hat{\mathbf{e}}_y) \\ &= P_x Q_y \hat{\mathbf{e}}_z - P_y Q_x \hat{\mathbf{e}}_z = \boxed{(P_x Q_y - P_y Q_x)\hat{\mathbf{e}}_z}.\end{aligned}$$

The coefficient is nonzero: its magnitude is $|\mathbf{P} \times \mathbf{Q}| = PQ \sin\theta$, where θ is the angle between the vectors, and $\sin\theta \neq 0$ because they are neither parallel ($\theta = 0$) nor antiparallel ($\theta = \pi$). Its sign determines whether the direction is $+z$ or $-z$.

3.2.2 Prove that $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{A} \times \mathbf{B}) = (AB)^2 - (\mathbf{A} \cdot \mathbf{B})^2$.

Solution.

For nonzero $\mathbf{A}, \mathbf{B}$, let θ be their angle. Then $|\mathbf{A} \times \mathbf{B}| = AB \sin\theta$ and $\mathbf{A} \cdot \mathbf{B} = AB \cos\theta$, so

$$(\mathbf{A} \times \mathbf{B})^2 = A^2 B^2 \sin^2\theta = A^2 B^2 (1 - \cos^2\theta) = \boxed{A^2 B^2 - (\mathbf{A} \cdot \mathbf{B})^2}.$$

If either vector is zero, both sides are zero directly. An algebraic check expands both sides in components. The left side is $(A_y B_z - A_z B_y)^2 + (A_z B_x - A_x B_z)^2 + (A_x B_y - A_y B_x)^2$; its squared terms are $\sum_{i \neq j} A_i^2 B_j^2$ and its mixed terms are $-2\sum_{i<j} A_i A_j B_i B_j$. On the right,

$$A^2 B^2 = \sum_{i,j} A_i^2 B_j^2, \qquad (\mathbf{A} \cdot \mathbf{B})^2 = \sum_i A_i^2 B_i^2 + 2\sum_{i<j} A_i A_j B_i B_j.$$

In the difference the $i = j$ terms cancel, leaving $\sum_{i \neq j} A_i^2 B_j^2 - 2\sum_{i<j} A_i A_j B_i B_j$, the same as the left side.

3.2.3 Using the vectors

$$\begin{aligned}\mathbf{P} &= \hat{\mathbf{e}}_x \cos\theta + \hat{\mathbf{e}}_y \sin\theta, \\ \mathbf{Q} &= \hat{\mathbf{e}}_x \cos\varphi - \hat{\mathbf{e}}_y \sin\varphi, \\ \mathbf{R} &= \hat{\mathbf{e}}_x \cos\varphi + \hat{\mathbf{e}}_y \sin\varphi,\end{aligned}$$

prove the familiar trigonometric identities

$$\sin(\theta + \varphi) = \sin\theta\cos\varphi + \cos\theta\sin\varphi, \quad \cos(\theta + \varphi) = \cos\theta\cos\varphi - \sin\theta\sin\varphi.$$

Solution.

All three vectors have unit length. Their directed angles from $+x$ are $\theta, -\varphi, \varphi$, respectively; in particular the directed angle from $\mathbf{Q}$ to $\mathbf{P}$ is $\theta + \varphi$. Geometrically the dot product and signed z-component of the cross product are

$$\mathbf{Q} \cdot \mathbf{P} = \cos(\theta + \varphi), \qquad (\mathbf{Q} \times \mathbf{P})_z = \sin(\theta + \varphi).$$

Here $|\mathbf{Q}\times\mathbf{P}| = |\sin(\theta+\varphi)|$, the sine of the angle between the unit vectors, and by the right-hand rule $\mathbf{Q}\times\mathbf{P}$ points along $+\hat{\mathbf{e}}_z$ exactly when the shorter turn from $\mathbf{Q}$ to $\mathbf{P}$ is counterclockwise, i.e. when $\sin(\theta+\varphi) > 0$; this fixes the sign. Evaluation from their Cartesian components gives

$$\mathbf{Q}\cdot\mathbf{P} = \cos\varphi\cos\theta - \sin\varphi\sin\theta,$$

$$(\mathbf{Q}\times\mathbf{P})_z = Q_xP_y - Q_yP_x = \cos\varphi\sin\theta + \sin\varphi\cos\theta.$$

Equating the geometric and component expressions proves both identities, including their signs. Directed angles extend the argument to all real θ, φ. The vector $\mathbf{R}$ is not needed here. Since its directed angle is φ, the same steps give $\mathbf{R}\cdot\mathbf{P} = \cos\varphi\cos\theta + \sin\varphi\sin\theta = \cos(\theta-\varphi)$ and $(\mathbf{R}\times\mathbf{P})_z = \cos\varphi\sin\theta - \sin\varphi\cos\theta = \sin(\theta-\varphi)$, the corresponding difference formulas.

3.2.4

(a) Find a vector $\mathbf{A}$ that is perpendicular to

$$\mathbf{U} = 2\hat{\mathbf{e}}_x + \hat{\mathbf{e}}_y - \hat{\mathbf{e}}_z, \quad \mathbf{V} = \hat{\mathbf{e}}_x - \hat{\mathbf{e}}_y + \hat{\mathbf{e}}_z.$$

(b) What is $\mathbf{A}$ if, in addition to this requirement, we demand that it have unit magnitude?

Solution.

(a) Write $\mathbf{A} = (a, b, c)$. The two perpendicularity conditions are $2a + b - c = 0$ and $a - b + c = 0$. Adding gives $3a = 0$, so $a = 0$ and then $b = c$. Thus every common perpendicular is

$$\mathbf{A} = t(0, 1, 1).$$

A nonzero choice is $(0, 1, 1)$; the cross product

$$\mathbf{U}\times\mathbf{V} = \big(1\cdot 1 - (-1)(-1),\ (-1)(1) - 2\cdot 1,\ 2(-1) - 1\cdot 1\big) = (0, -3, -3)$$

gives the same direction. **(b)** Its squared norm is $2t^2$. Unit magnitude requires $t = \pm 1/\sqrt{2}$, giving

$$\boxed{\mathbf{A} = \pm(\hat{\mathbf{e}}_y + \hat{\mathbf{e}}_z)/\sqrt{2}}.$$

Both orientations meet the requirements.

3.2.5 If four vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$ all lie in the same plane, show that

$$(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d}) = 0.$$

Hint. Consider the directions of the cross-product vectors.

Solution.

Let $\hat{\mathbf{n}}$ be a unit normal to the common plane. Both $\mathbf{a}\times\mathbf{b}$ and $\mathbf{c}\times\mathbf{d}$ are normal to it, so they have the forms $\lambda\hat{\mathbf{n}}$ and $\mu\hat{\mathbf{n}}$ (including zero if a pair is dependent). Hence their cross product is $\lambda\mu(\hat{\mathbf{n}}\times\hat{\mathbf{n}}) = 0$.

3.2.6 Derive the law of sines (see Fig. 3.4):

$$\frac{\sin\alpha}{|A|} = \frac{\sin\beta}{|B|} = \frac{\sin\gamma}{|C|}.$$

Solution.

Take a nondegenerate triangle with side lengths A, B, C opposite α, β, γ. Its area Δ is half the magnitude of the cross product of the two side vectors that leave any one vertex. At the vertex with angle α these two sides have lengths B and C (A is the opposite side), so $2\Delta = BC\sin\alpha$; the other two vertices are treated the same way:

$$2\Delta = BC\sin\alpha = CA\sin\beta = AB\sin\gamma.$$

Divide each expression by the nonzero product ABC:

$$\boxed{\frac{\sin\alpha}{A} = \frac{\sin\beta}{B} = \frac{\sin\gamma}{C} = \frac{2\Delta}{ABC}}.$$

This also follows by dropping the altitude from the vertex of angle γ onto the line of side C: computed from either of the other two sides, its length is $h = A\sin\beta = B\sin\alpha$, so $\sin\alpha/A = \sin\beta/B$. (If α or β is obtuse, the foot lies on the extension of C, and the right triangle contains the supplementary angle, which has the same sine.) The altitude from the vertex of angle α gives $C\sin\beta = B\sin\gamma$, i.e. $\sin\beta/B = \sin\gamma/C$, in the same way.

3.2.7 The magnetic induction $\mathbf{B}$ is defined by the Lorentz force equation,

$$\mathbf{F} = q(\mathbf{v}\times\mathbf{B}).$$

Carrying out three experiments, we find that if

$$\mathbf{v} = \hat{\mathbf{e}}_x, \quad \frac{\mathbf{F}}{q} = 2\hat{\mathbf{e}}_z - 4\hat{\mathbf{e}}_y,$$

$$\mathbf{v} = \hat{\mathbf{e}}_y, \quad \frac{\mathbf{F}}{q} = 4\hat{\mathbf{e}}_x - \hat{\mathbf{e}}_z,$$

$$\mathbf{v} = \hat{\mathbf{e}}_z, \quad \frac{\mathbf{F}}{q} = \hat{\mathbf{e}}_y - 2\hat{\mathbf{e}}_x.$$

From the results of these three separate experiments calculate the magnetic induction $\mathbf{B}$.

Solution.

Put $\mathbf{B} = (B_x, B_y, B_z)$. The three unit-velocity cross products are

$$\hat{\mathbf{e}}_x\times\mathbf{B} = (0, -B_z, B_y), \quad \hat{\mathbf{e}}_y\times\mathbf{B} = (B_z, 0, -B_x), \quad \hat{\mathbf{e}}_z\times\mathbf{B} = (-B_y, B_x, 0).$$

Matching components with the measured $\mathbf{F}/q$: the first measurement, $(0, -B_z, B_y) = (0, -4, 2)$, gives $B_z = 4, B_y = 2$; the second, $(B_z, 0, -B_x) = (4, 0, -1)$, gives $B_z = 4, B_x = 1$; the third, $(-B_y, B_x, 0) = (-2, 1, 0)$, independently gives $B_y = 2, B_x = 1$. All are consistent, so $\boxed{\mathbf{B} = \hat{\mathbf{e}}_x + 2\hat{\mathbf{e}}_y + 4\hat{\mathbf{e}}_z}$.

3.2.8 You are given the three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$,

$$\mathbf{A} = \hat{\mathbf{e}}_x + \hat{\mathbf{e}}_y, \quad \mathbf{B} = \hat{\mathbf{e}}_y + \hat{\mathbf{e}}_z, \quad \mathbf{C} = \hat{\mathbf{e}}_x - \hat{\mathbf{e}}_z.$$

(a) Compute the scalar triple product, $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$. Noting that $\mathbf{A} = \mathbf{B} + \mathbf{C}$, give a geometric interpretation of your result for the scalar triple product.

(b) Compute $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

Solution.

First compute, expanding along the first row,

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \hat{\mathbf{e}}_x & \hat{\mathbf{e}}_y & \hat{\mathbf{e}}_z \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix} = \hat{\mathbf{e}}_x(-1-0) - \hat{\mathbf{e}}_y(0-1) + \hat{\mathbf{e}}_z(0-1) = (-1, 1, -1).$$

(a) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (1, 1, 0) \cdot (-1, 1, -1) = \boxed{0}$. This is forced by $\mathbf{A} = \mathbf{B} + \mathbf{C}$: since $\mathbf{B} \times \mathbf{C}$ is perpendicular to both $\mathbf{B}$ and $\mathbf{C}$, $(\mathbf{B} + \mathbf{C}) \cdot (\mathbf{B} \times \mathbf{C}) = 0$. Geometrically, the three vectors lie in one plane, so the parallelepiped they span has zero volume. **(b)** Direct multiplication gives

$$\begin{aligned} \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= (1, 1, 0) \times (-1, 1, -1) \\ &= \big(1(-1) - 0 \cdot 1,\ 0(-1) - 1(-1),\ 1 \cdot 1 - 1(-1)\big) = \boxed{(-1, 1, 2)}. \end{aligned}$$

The triple-product identity, with $\mathbf{A} \cdot \mathbf{C} = 1$ and $\mathbf{A} \cdot \mathbf{B} = 1$, also gives $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{B} - \mathbf{C} = (-1, 1, 2)$.

3.2.9 Prove Jacobi's identity for vector products:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = 0.$$

Solution.

Expand the three vector triple products:

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}), \\ \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) &= \mathbf{c}(\mathbf{b} \cdot \mathbf{a}) - \mathbf{a}(\mathbf{b} \cdot \mathbf{c}), \\ \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) &= \mathbf{a}(\mathbf{c} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{c} \cdot \mathbf{a}). \end{aligned}$$

Dot products commute, so the coefficient of each of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in the sum is zero. The sum vanishes.

3.2.10 A vector $\mathbf{A}$ is decomposed into a radial vector $\mathbf{A}_r$ and a tangential vector $\mathbf{A}_t$. If $\hat{\mathbf{r}}$ is a unit vector in the radial direction, show that

(a) $\mathbf{A}_r = \hat{\mathbf{r}}(\mathbf{A} \cdot \hat{\mathbf{r}})$ and

(b) $\mathbf{A}_t = -\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{A})$.

Solution.

(a) The radial component must be $\mathbf{A}_r = a\hat{\mathbf{r}}$. The remainder is tangential if $0 = \hat{\mathbf{r}} \cdot (\mathbf{A} - a\hat{\mathbf{r}}) = \hat{\mathbf{r}} \cdot \mathbf{A} - a$. Thus

$$\boxed{\mathbf{A}_r = \hat{\mathbf{r}}(\mathbf{A} \cdot \hat{\mathbf{r}})}.$$

(b) The triple-product identity and $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = 1$ give

$$-\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{A}) = -\big[\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{A}) - \mathbf{A}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}})\big] = \mathbf{A} - \mathbf{A}_r = \boxed{\mathbf{A}_t}.$$

It is perpendicular to $\hat{\mathbf{r}}$ and the two parts sum to $\mathbf{A}$, verifying the decomposition.

3.2.11 Prove that a necessary and sufficient condition for the three (nonvanishing) vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ to be coplanar is the vanishing of the scalar triple product

$$\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = 0.$$

Solution.

If $\mathbf{B}, \mathbf{C}$ are independent, their cross product is a nonzero normal to their plane. Consequently $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = 0$ says exactly that $\mathbf{A}$ lies in that plane. This proves necessity and sufficiency in this case. If $\mathbf{B}, \mathbf{C}$ are dependent, their cross product is zero, so the triple product vanishes; and all three vectors lie in a plane containing $\mathbf{A}$ and the common line of $\mathbf{B}, \mathbf{C}$, so they are coplanar. Both conditions hold, and again they are equivalent. Collinear triples are included in this second case.

3.2.12 Three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are given by

$$\mathbf{A} = 3\hat{\mathbf{e}}_x - 2\hat{\mathbf{e}}_y + 2\hat{\mathbf{e}}_z, \quad \mathbf{B} = 6\hat{\mathbf{e}}_x + 4\hat{\mathbf{e}}_y - 2\hat{\mathbf{e}}_z, \quad \mathbf{C} = -3\hat{\mathbf{e}}_x - 2\hat{\mathbf{e}}_y - 4\hat{\mathbf{e}}_z.$$

Compute the values of $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$ and $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$, $\mathbf{C} \times (\mathbf{A} \times \mathbf{B})$ and $\mathbf{B} \times (\mathbf{C} \times \mathbf{A})$.

Solution.

The needed cross product, from $\mathbf{B} \times \mathbf{C} = (B_yC_z - B_zC_y,\ B_zC_x - B_xC_z,\ B_xC_y - B_yC_x)$, and dot products are

$$\begin{aligned}
\mathbf{B} \times \mathbf{C} &= \big(4(-4) - (-2)(-2),\ (-2)(-3) - 6(-4),\ 6(-2) - 4(-3)\big) = (-20, 30, 0),\\
\mathbf{A} \cdot \mathbf{B} &= 18 - 8 - 4 = 6,\\
\mathbf{A} \cdot \mathbf{C} &= -9 + 4 - 8 = -13, \qquad \mathbf{B} \cdot \mathbf{C} = -18 - 8 + 8 = -18.
\end{aligned}$$

Therefore the scalar triple product is

$$\boxed{\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = 3(-20) + (-2)(30) + 2(0) = -120}.$$

Using $\mathbf{X} \times (\mathbf{Y} \times \mathbf{Z}) = \mathbf{Y}(\mathbf{X} \cdot \mathbf{Z}) - \mathbf{Z}(\mathbf{X} \cdot \mathbf{Y})$,

$$\begin{aligned}
\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = -13\mathbf{B} - 6\mathbf{C}\\
&= (-78 + 18,\ -52 + 12,\ 26 + 24) = \boxed{(-60, -40, 50)},\\
\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) &= \mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A}) = -18\mathbf{A} + 13\mathbf{B}\\
&= (-54 + 78,\ 36 + 52,\ -36 - 26) = \boxed{(24, 88, -62)},\\
\mathbf{B} \times (\mathbf{C} \times \mathbf{A}) &= \mathbf{C}(\mathbf{B} \cdot \mathbf{A}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C}) = 6\mathbf{C} + 18\mathbf{A}\\
&= (-18 + 54,\ -12 - 36,\ -24 + 36) = \boxed{(36, -48, 12)}.
\end{aligned}$$

Their sum is $(0, 0, 0)$, as required by Jacobi's identity (Exercise 3.2.9).

3.2.13 Show that

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}).$$

Solution.

Write components using ε_{ijk}, summing every repeated index from 1 to 3. After the cyclic shifts $\varepsilon_{ijk} = \varepsilon_{jki}$ and $\varepsilon_{i\ell m} = \varepsilon_{\ell mi}$, the contraction identity $\sum_k \varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}$ of Exercise 2.1.9,

with (i,j,k,p,q) renamed (j,k,i,ℓ,m), reads $\varepsilon_{ijk}\varepsilon_{i\ell m} = \delta_{j\ell}\delta_{km} - \delta_{jm}\delta_{k\ell}$. Hence

$$\begin{aligned}(\mathbf{A}\times\mathbf{B})\cdot(\mathbf{C}\times\mathbf{D}) &= \varepsilon_{ijk}\varepsilon_{i\ell m}A_jB_kC_\ell D_m \\ &= (\delta_{j\ell}\delta_{km} - \delta_{jm}\delta_{k\ell})A_jB_kC_\ell D_m \\ &= \sum_{j,k} A_jC_jB_kD_k - \sum_{j,k} A_jD_jB_kC_k \\ &= (\mathbf{A}\cdot\mathbf{C})(\mathbf{B}\cdot\mathbf{D}) - (\mathbf{A}\cdot\mathbf{D})(\mathbf{B}\cdot\mathbf{C}).\end{aligned}$$

No independence or nonzero-vector assumption is required.

3.2.14 Show that

$$(\mathbf{A}\times\mathbf{B})\times(\mathbf{C}\times\mathbf{D}) = (\mathbf{A}\cdot\mathbf{B}\times\mathbf{D})\mathbf{C} - (\mathbf{A}\cdot\mathbf{B}\times\mathbf{C})\mathbf{D}.$$

Solution.

Apply the triple-product identity with $\mathbf{X} = \mathbf{A}\times\mathbf{B}$:

$$\mathbf{X}\times(\mathbf{C}\times\mathbf{D}) = \mathbf{C}(\mathbf{X}\cdot\mathbf{D}) - \mathbf{D}(\mathbf{X}\cdot\mathbf{C}).$$

Cyclicity of the scalar triple product gives $\mathbf{X}\cdot\mathbf{D} = (\mathbf{A}\times\mathbf{B})\cdot\mathbf{D} = \mathbf{A}\cdot(\mathbf{B}\times\mathbf{D})$ and likewise $\mathbf{X}\cdot\mathbf{C} = \mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})$. Substituting these two scalars,

$$(\mathbf{A}\times\mathbf{B})\times(\mathbf{C}\times\mathbf{D}) = \boxed{(\mathbf{A}\cdot\mathbf{B}\times\mathbf{D})\mathbf{C} - (\mathbf{A}\cdot\mathbf{B}\times\mathbf{C})\mathbf{D}}.$$

3.2.15 An electric charge q_1 moving with velocity $\mathbf{v}_1$ produces a magnetic induction $\mathbf{B}$ given by

$$\mathbf{B} = \frac{\mu_0}{4\pi}q_1\frac{\mathbf{v}_1\times\hat{\mathbf{r}}}{r^2} \quad \text{(mks units)},$$

where $\hat{\mathbf{r}}$ is a unit vector that points from q_1 to the point at which $\mathbf{B}$ is measured (Biot and Savart law).

(a) Show that the magnetic force exerted by q_1 on a second charge q_2, velocity $\mathbf{v}_2$, is given by the vector triple product

$$\mathbf{F}_2 = \frac{\mu_0}{4\pi}\frac{q_1q_2}{r^2}\,\mathbf{v}_2\times(\mathbf{v}_1\times\hat{\mathbf{r}}).$$

(b) Write out the corresponding magnetic force $\mathbf{F}_1$ that q_2 exerts on q_1. Define your unit radial vector. How do $\mathbf{F}_1$ and $\mathbf{F}_2$ compare?

(c) Calculate $\mathbf{F}_1$ and $\mathbf{F}_2$ for the case of q_1 and q_2 moving along parallel trajectories side by side.

ANS. (b) $\mathbf{F}_1 = -\dfrac{\mu_0}{4\pi}\dfrac{q_1q_2}{r^2}\,\mathbf{v}_1\times(\mathbf{v}_2\times\hat{\mathbf{r}})$. In general, there is no simple relation between $\mathbf{F}_1$ and $\mathbf{F}_2$. Specifically, Newton's third law, $\mathbf{F}_1 = -\mathbf{F}_2$, does not hold.

(c) $\mathbf{F}_1 = \dfrac{\mu_0}{4\pi}\dfrac{q_1q_2}{r^2}(\mathbf{v}_1\cdot\mathbf{v}_2)\hat{\mathbf{r}} = -\mathbf{F}_2$. Mutual attraction if $q_1q_2(\mathbf{v}_1\cdot\mathbf{v}_2) > 0$; repulsion if it is negative.

Solution.

(a) Set $K = \mu_0q_1q_2/(4\pi r^2)$ and let $\hat{\mathbf{r}}$ point from charge 1 to charge 2. The field of q_1 at q_2 is $\mathbf{B}_1 = (\mu_0/4\pi)q_1\mathbf{v}_1\times\hat{\mathbf{r}}/r^2$, so the magnetic Lorentz force on q_2 is

$$\mathbf{F}_2 = q_2\mathbf{v}_2\times\mathbf{B}_1 = K\mathbf{v}_2\times(\mathbf{v}_1\times\hat{\mathbf{r}}) = K\{\mathbf{v}_1(\mathbf{v}_2\cdot\hat{\mathbf{r}}) - \hat{\mathbf{r}}(\mathbf{v}_1\cdot\mathbf{v}_2)\}.$$

(b) For the field of charge 2 evaluated at charge 1, the unit vector from the source to the field point is $-\hat{\mathbf{r}}$, so $\mathbf{B}_2 = -(\mu_0/4\pi)q_2\mathbf{v}_2 \times \hat{\mathbf{r}}/r^2$ and

$$\mathbf{F}_1 = q_1\mathbf{v}_1 \times \mathbf{B}_2 = -K\mathbf{v}_1 \times (\mathbf{v}_2 \times \hat{\mathbf{r}}) = K\{\hat{\mathbf{r}}(\mathbf{v}_1 \cdot \mathbf{v}_2) - \mathbf{v}_2(\mathbf{v}_1 \cdot \hat{\mathbf{r}})\}.$$

Thus $\mathbf{F}_1 + \mathbf{F}_2 = K[\mathbf{v}_1(\mathbf{v}_2 \cdot \hat{\mathbf{r}}) - \mathbf{v}_2(\mathbf{v}_1 \cdot \hat{\mathbf{r}})]$, generally nonzero. For example, if $\mathbf{v}_1 = v\hat{\mathbf{r}}$ with $v \neq 0$ and $\mathbf{v}_2 \neq 0$ is perpendicular to $\hat{\mathbf{r}}$, then $\mathbf{F}_2 = 0$ (because $\mathbf{v}_1 \times \hat{\mathbf{r}} = 0$), whereas $\mathbf{F}_1 = -Kv\,\mathbf{v}_2 \neq 0$; so $\mathbf{F}_1 \neq -\mathbf{F}_2$. This concerns the magnetic forces on the particles; it is not a claim that particle-plus-field momentum fails to be conserved.

(c) For side-by-side trajectories, both velocities are perpendicular to $\hat{\mathbf{r}}$: $\mathbf{v}_1 \cdot \hat{\mathbf{r}} = \mathbf{v}_2 \cdot \hat{\mathbf{r}} = 0$. The first term of $\mathbf{F}_2$ and the second term of $\mathbf{F}_1$ then vanish, leaving

$$\boxed{\mathbf{F}_1 = K(\mathbf{v}_1 \cdot \mathbf{v}_2)\hat{\mathbf{r}} = -\mathbf{F}_2}.$$

Because $\hat{\mathbf{r}}$ points from charge 1 toward charge 2, the charges attract when $K(\mathbf{v}_1 \cdot \mathbf{v}_2) > 0$. The printed v^2 assumes equal velocities. Attraction requires $q_1q_2(\mathbf{v}_1 \cdot \mathbf{v}_2) > 0$; opposite sign gives repulsion. For equal velocities and like charges this reduces to the printed attractive result.

3.3 Coordinate Transformations

3.3.1 A rotation $\varphi_1 + \varphi_2$ about the z-axis is carried out as two successive rotations φ_1 and φ_2, each about the z-axis. Use the matrix representation of the rotations to derive the trigonometric identities

$$\cos(\varphi_1 + \varphi_2) = \cos\varphi_1 \cos\varphi_2 - \sin\varphi_1 \sin\varphi_2, \quad \sin(\varphi_1 + \varphi_2) = \sin\varphi_1 \cos\varphi_2 + \cos\varphi_1 \sin\varphi_2.$$

Solution.

For a passive rotation of axes about z,

$$Z(\varphi) = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The two successive rotations give $Z(\varphi_2)Z(\varphi_1)$ (the first rotation acts first, so its matrix stands on the right). Multiplying rows of $Z(\varphi_2)$ into columns of $Z(\varphi_1)$, its upper left block is

$$\begin{pmatrix} c_2c_1 - s_2s_1 & c_2s_1 + s_2c_1 \\ -s_2c_1 - c_2s_1 & -s_2s_1 + c_2c_1 \end{pmatrix}, \qquad c_i = \cos\varphi_i,\ s_i = \sin\varphi_i;$$

the third row and column are those of the identity. Turning the axes through φ_1 and then through φ_2 about the same axis is a single rotation through $\varphi_1 + \varphi_2$, so the product equals $Z(\varphi_1 + \varphi_2)$, whose $(1,1)$ and $(1,2)$ entries are $\cos(\varphi_1 + \varphi_2)$ and $\sin(\varphi_1 + \varphi_2)$. Equating these entries gives the stated cosine and sine addition formulas.

3.3.2 A corner reflector is formed by three mutually perpendicular reflecting surfaces. Show that a ray of light incident upon the corner reflector (striking all three surfaces) is reflected back along a line parallel to the line of incidence.

Hint. Consider the effect of a reflection on the components of a vector describing the direction of the light ray.

Solution.

Choose x, y, z perpendicular to the three reflecting planes, and let $\mathbf{v} = (v_x, v_y, v_z)$ be the propagation direction of the incident ray. Reflection reverses the normal component of a ray's direction and preserves its two tangential components. The three reflection matrices are therefore

$$R_x = \mathrm{diag}(-1,1,1), \quad R_y = \mathrm{diag}(1,-1,1), \quad R_z = \mathrm{diag}(1,1,-1).$$

Being diagonal, they commute, and $R_x R_y R_z = \mathrm{diag}\big((-1)(1)(1),\ (1)(-1)(1),\ (1)(1)(-1)\big) = -I_3$. After one reflection from each surface, in any order, the propagation direction is $-\mathbf{v} = (-v_x, -v_y, -v_z)$. The outgoing line is thus parallel to, and traversed oppositely from, the incident line; it can be displaced from that line.

3.3.3 Let $\mathbf{x}$ and $\mathbf{y}$ be column vectors. Under an orthogonal transformation S, they become $\mathbf{x}' = \mathsf{S}\mathbf{x}$ and $\mathbf{y}' = \mathsf{S}\mathbf{y}$. Show that $(\mathbf{x}')^T\mathbf{y}' = \mathbf{x}^T\mathbf{y}$, a result equivalent to the invariance of the dot product under a rotational transformation.

Solution.

Orthogonality means $S^T S = I$. Consequently

$$(\mathbf{x}')^T\mathbf{y}' = (S\mathbf{x})^T(S\mathbf{y}) = \mathbf{x}^T S^T S\mathbf{y} = \boxed{\mathbf{x}^T\mathbf{y}}.$$

This argument works for both proper rotations and orthogonal transformations with determinant -1.

3.3.4 Given the orthogonal transformation matrix S and vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathsf{S} = \begin{pmatrix} 0.80 & 0.60 & 0.00 \\ -0.48 & 0.64 & 0.60 \\ 0.36 & -0.48 & 0.80 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix},$$

(a) Calculate $\det(\mathsf{S})$.

(b) Verify that $\mathbf{a}\cdot\mathbf{b}$ is invariant under application of S to $\mathbf{a}$ and $\mathbf{b}$.

(c) Determine what happens to $\mathbf{a}\times\mathbf{b}$ under application of S to $\mathbf{a}$ and $\mathbf{b}$. Is this what is expected?

Solution.

(a) Expansion along the first row ($S_{13} = 0$) gives

$$\begin{aligned} \det S &= 0.80(0.64\cdot 0.80 + 0.60\cdot 0.48) - 0.60(-0.48\cdot 0.80 - 0.60\cdot 0.36) \\ &= 0.80(0.80) - 0.60(-0.60) = 0.64 + 0.36 = \boxed{1}. \end{aligned}$$

(b) Matrix multiplication (each component is a row of S dotted into the column vector) gives

$$\begin{aligned} \mathbf{a}' &= S\mathbf{a} = (0.80 + 0.00,\ -0.48 + 0.60,\ 0.36 + 0.80) = (0.80, 0.12, 1.16), \\ \mathbf{b}' &= S\mathbf{b} = \big(2(0.60) - 0.00,\ 2(0.64) - 0.60,\ 2(-0.48) - 0.80\big) = (1.20, 0.68, -1.76). \end{aligned}$$

Then $\mathbf{a}\cdot\mathbf{b} = 0 + 0 - 1 = -1$, and

$$\mathbf{a}'\cdot\mathbf{b}' = 0.80(1.20) + 0.12(0.68) + 1.16(-1.76) = 0.96 + 0.0816 - 2.0416 = -1,$$

so the dot product is invariant. **(c)** The original cross product and its transform are

$$\mathbf{a}\times\mathbf{b} = (0(-1)-1\cdot 2,\ 1\cdot 0-1(-1),\ 1\cdot 2-0\cdot 0) = (-2,1,2),$$
$$S(\mathbf{a}\times\mathbf{b}) = (-1.60+0.60+0,\ 0.96+0.64+1.20,\ -0.72-0.48+1.60) = (-1.00,2.80,0.40).$$

The cross product after transformation is

$$\begin{aligned}\mathbf{a}'\times\mathbf{b}' &= (0.12(-1.76)-1.16(0.68),\ 1.16(1.20)-0.80(-1.76),\ 0.80(0.68)-0.12(1.20))\\ &= (-0.2112-0.7888,\ 1.392+1.408,\ 0.544-0.144) = (-1.00,2.80,0.40).\end{aligned}$$

This is expected: the cross product of two vectors is a pseudovector (Section 3.3), i.e. for orthogonal S, $(S\mathbf{a})\times(S\mathbf{b}) = (\det S)S(\mathbf{a}\times\mathbf{b})$, and here $\det S = 1$. To verify this law, note that $\sum_{ijk}\varepsilon_{ijk}S_{i\ell}S_{jm}S_{kn}$ is the determinant of the matrix whose columns are columns ℓ, m, n of S, which is $(\det S)\varepsilon_{\ell mn}$: it vanishes when two columns coincide and changes sign under each interchange of columns. Hence

$$\left(S^T[(S\mathbf{a})\times(S\mathbf{b})]\right)_\ell = \sum_{ijkmn}\varepsilon_{ijk}S_{i\ell}S_{jm}S_{kn}a_m b_n = (\det S)(\mathbf{a}\times\mathbf{b})_\ell,$$

and multiplying by S, with $SS^T = I$, gives the law.

3.3.5 Using $\mathbf{a}$ and $\mathbf{b}$ as defined in Exercise 3.3.4, but with

$$\mathsf{S} = \begin{pmatrix} 0.60 & 0.00 & 0.80\\ -0.64 & -0.60 & 0.48\\ -0.48 & 0.80 & 0.36\end{pmatrix} \quad\text{and}\quad \mathbf{c} = \begin{pmatrix}2\\1\\3\end{pmatrix},$$

(a) Calculate $\det(\mathsf{S})$.

Apply S to $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, and determine what happens to

(b) $\mathbf{a}\times\mathbf{b}$,

(c) $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$,

(d) $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})$.

(e) Classify the expressions in (b) through (d) as scalar, vector, pseudovector, or pseudoscalar.

Solution.

(a) Expansion along the first row, whose middle entry is zero, gives

$$\begin{aligned}\det S &= 0.60(-0.60\cdot 0.36-0.48\cdot 0.80)+0.80(-0.64\cdot 0.80-(-0.60)(-0.48))\\ &= 0.60(-0.60)+0.80(-0.80) = -0.36-0.64 = \boxed{-1}.\end{aligned}$$

Multiplying S into the three column vectors gives

$$\mathbf{a}' = (0.60+0.80,\ -0.64+0.48,\ -0.48+0.36) = (1.40,-0.16,-0.12),$$
$$\mathbf{b}' = (0-0.80,\ -1.20-0.48,\ 1.60-0.36) = (-0.80,-1.68,1.24),$$
$$\mathbf{c}' = (1.20+0+2.40,\ -1.28-0.60+1.44,\ -0.96+0.80+1.08) = (3.60,-0.44,0.92).$$

(b) With $\mathbf{a}\times\mathbf{b} = (-2,1,2)$ from Exercise 3.3.4,

$$S(\mathbf{a}\times\mathbf{b}) = (-1.20+0+1.60,\ 1.28-0.60+0.96,\ 0.96+0.80+0.72) = (0.40,1.64,2.48),$$

whereas

$$\begin{aligned}\mathbf{a}' \times \mathbf{b}' &= \big((-0.16)(1.24) - (-0.12)(-1.68),\ (-0.12)(-0.80) - 1.40(1.24),\\ &\qquad 1.40(-1.68) - (-0.16)(-0.80)\big)\\ &= (-0.1984 - 0.2016,\ 0.096 - 1.736,\ -2.352 - 0.128)\\ &= (-0.40, -1.64, -2.48) = -S(\mathbf{a} \times \mathbf{b}).\end{aligned}$$

(c) The original triple product is $(-2, 1, 2) \cdot (2, 1, 3) = -4 + 1 + 6 = 3$; the new one is

$$(\mathbf{a}' \times \mathbf{b}') \cdot \mathbf{c}' = (-0.40)(3.60) + (-1.64)(-0.44) + (-2.48)(0.92) = -1.44 + 0.7216 - 2.2816 = -3.$$

(d) Since $\mathbf{a} \cdot \mathbf{c} = 2 + 0 + 3 = 5$ and $\mathbf{a} \cdot \mathbf{b} = -1$,

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}) = 5\mathbf{b} + \mathbf{c} = (0, 10, -5) + (2, 1, 3) = (2, 11, -2).$$

Dot products are unchanged by any orthogonal S (Exercise 3.3.3), so the same identity gives for the transformed vectors

$$\begin{aligned}\mathbf{a}' \times (\mathbf{b}' \times \mathbf{c}') &= 5\mathbf{b}' + \mathbf{c}' = (-4.00 + 3.60,\ -8.40 - 0.44,\ 6.20 + 0.92)\\ &= (-0.40, -8.84, 7.12) = S(2, 11, -2),\end{aligned}$$

since $S(2, 11, -2) = (1.20 + 0 - 1.60,\ -1.28 - 6.60 - 0.96,\ -0.96 + 8.80 - 0.72)$. **(e)** Thus (b) is a pseudovector, (c) a pseudoscalar, and (d) an ordinary vector. A pseudovector acquires the extra factor $\det S$, and a pseudoscalar changes sign under this improper rotation.

The Instructor's Manual, p. 36, prints $S_{32} = -0.80$ in its matrices; the textbook has $+0.80$, as used here. With $+0.80$ the rows of S are orthonormal; with -0.80 the second and third rows would have scalar product $(-0.64)(-0.48) + (-0.60)(-0.80) + 0.48(0.36) = 0.96 \neq 0$, so S would not be orthogonal. The transformed vectors listed in the Instructor's Manual also require the positive sign. For example, $b'_z = 0.80(2) + 0.36(-1) = 1.24$, whereas the negative sign gives -1.96.

3.4 Rotations in $\mathbb{R}^3$

3.4.1 Another set of Euler rotations in common use is

(1) a rotation about the x_3-axis through an angle φ, counterclockwise,

(2) a rotation about the x'_1-axis through an angle θ, counterclockwise,

(3) a rotation about the x''_3-axis through an angle ψ, counterclockwise.

If

$$\begin{aligned}&\alpha = \varphi - \pi/2 && \varphi = \alpha + \pi/2\\ &\beta = \theta \qquad \text{or} && \theta = \beta\\ &\gamma = \psi + \pi/2 && \psi = \gamma - \pi/2,\end{aligned}$$

show that the final systems are identical.

Solution.

Use passive rotation matrices

$$Z(t) = \begin{pmatrix} c_t & s_t & 0 \\ -s_t & c_t & 0 \\ 0 & 0 & 1 \end{pmatrix},\ Y(t) = \begin{pmatrix} c_t & 0 & -s_t \\ 0 & 1 & 0 \\ s_t & 0 & c_t \end{pmatrix},\ X(t) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_t & s_t \\ 0 & -s_t & c_t \end{pmatrix},$$

where $c_t = \cos t, s_t = \sin t$. The textbook convention (rotations about x_3, x_2', x_3'', each later rotation multiplying from the left) is $S(\alpha,\beta,\gamma) = Z(\gamma)Y(\beta)Z(\alpha)$. The alternative set, whose second rotation is about x_1', is likewise $Z(\psi)X(\theta)Z(\varphi)$. For the quarter turns, $c_{\pm\pi/2} = 0$ and $s_{\pm\pi/2} = \pm 1$, so

$$Z(-\pi/2) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Z(\pi/2) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplying $Y(\theta)$ on the right by $Z(-\pi/2)$ replaces its columns $(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3)$ by $(\mathbf{k}_2, -\mathbf{k}_1, \mathbf{k}_3)$:

$$Y(\theta)Z(-\pi/2) = \begin{pmatrix} 0 & -c_\theta & -s_\theta \\ 1 & 0 & 0 \\ 0 & -s_\theta & c_\theta \end{pmatrix},$$

and multiplying on the left by $Z(\pi/2)$ replaces the rows $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3)$ of this matrix by $(\mathbf{r}_2, -\mathbf{r}_1, \mathbf{r}_3)$:

$$Z(\pi/2)Y(\theta)Z(-\pi/2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\theta & s_\theta \\ 0 & -s_\theta & c_\theta \end{pmatrix} = X(\theta).$$

Consequently, under the stated substitutions, and with $Z(a+b) = Z(a)Z(b) = Z(b)Z(a)$ (Exercise 3.3.1),

$$\begin{aligned} S &= Z(\psi + \pi/2)Y(\theta)Z(\varphi - \pi/2) \\ &= Z(\psi)\left[Z(\pi/2)Y(\theta)Z(-\pi/2)\right]Z(\varphi) = Z(\psi)X(\theta)Z(\varphi). \end{aligned}$$

The matrices, and therefore the final frames, coincide. For use below, the explicit product $Z(\gamma)Y(\beta)Z(\alpha)$, the textbook's Eq. (3.37), is

$$S = \begin{pmatrix} c_\gamma c_\beta c_\alpha - s_\gamma s_\alpha & c_\gamma c_\beta s_\alpha + s_\gamma c_\alpha & -c_\gamma s_\beta \\ -s_\gamma c_\beta c_\alpha - c_\gamma s_\alpha & -s_\gamma c_\beta s_\alpha + c_\gamma c_\alpha & s_\gamma s_\beta \\ s_\beta c_\alpha & s_\beta s_\alpha & c_\beta \end{pmatrix}. \tag{3.37}$$

3.4.2 Suppose the Earth is moved (rotated) so that the north pole goes to 30° north, 20° west (original latitude and longitude system) and the 10° west meridian points due south (also in the original system).

(a) What are the Euler angles describing this rotation?

(b) Find the corresponding direction cosines.

ANS. (b) $\mathsf{S} = \begin{pmatrix} 0.9551 & -0.2552 & -0.1504 \\ 0.0052 & 0.5221 & -0.8529 \\ 0.2962 & 0.8138 & 0.5000 \end{pmatrix}$. To reproduce the printed matrix, take $+y$ along the equator at Greenwich and $+x$ at 90° west; the longitude convention otherwise leaves the Cartesian axes unspecified.

Solution.

A longitude reference for the Cartesian axes is needed. The printed matrix is consistent with the $+y$-axis through Greenwich and the $+x$-axis through 90° west; use that convention here. East longitude λ then has azimuth $90° + \lambda$, measured from $+x$ toward $+y$.

(a) The rows of S are the new unit vectors expressed in the original axes. By (3.37) the third row, the new z-axis (the new north pole), is $(\sin\beta\cos\alpha, \sin\beta\sin\alpha, \cos\beta)$, so β is the colatitude and α the azimuth of the new pole. The new pole has $\lambda = -20^\circ$ and colatitude $\beta = 90^\circ - 30^\circ = 60^\circ$, so $\alpha = 90^\circ - 20^\circ = 70^\circ$.

After the first two rotations the axes are the rows of

$$Y(\beta)Z(\alpha) = \begin{pmatrix} c_\beta c_\alpha & c_\beta s_\alpha & -s_\beta \\ -s_\alpha & c_\alpha & 0 \\ s_\beta c_\alpha & s_\beta s_\alpha & c_\beta \end{pmatrix}.$$

The first row is the spherical unit vector $\mathbf{e}_\theta$ at the new pole (polar angle β, azimuth α), which points due south; the second row is $\mathbf{e}_\varphi$ there, which points east. Indeed, the first row is $\partial/\partial\beta$ of the third row $(s_\beta c_\alpha, s_\beta s_\alpha, c_\beta)$, the direction in which the colatitude increases, and the second row is $(1/s_\beta)\,\partial/\partial\alpha$ of it, the direction in which the longitude increases. The last rotation, through γ about the new z-axis, multiplies on the left by $Z(\gamma)$; its rows $(c_\gamma, s_\gamma, 0)$ and $(-s_\gamma, c_\gamma, 0)$ recombine the first two rows into the final equatorial axes

$$\mathbf{e}'_1 = \cos\gamma\,\mathbf{e}_\theta + \sin\gamma\,\mathbf{e}_\varphi, \qquad \mathbf{e}'_2 = -\sin\gamma\,\mathbf{e}_\theta + \cos\gamma\,\mathbf{e}_\varphi.$$

By the same rule (azimuth $90^\circ + \lambda$), applied to the rotated axes, the body's 10° west meridian has azimuth $90^\circ - 10^\circ = 80^\circ$ measured from its x-axis. A meridian of azimuth a leaves the pole along $\cos a\,\mathbf{e}'_1 + \sin a\,\mathbf{e}'_2$, so by the addition formulas this tangent is

$$\cos 80^\circ\,\mathbf{e}'_1 + \sin 80^\circ\,\mathbf{e}'_2 = \cos(\gamma + 80^\circ)\mathbf{e}_\theta + \sin(\gamma + 80^\circ)\mathbf{e}_\varphi.$$

Requiring this tangent to point due south, i.e. to equal $+\mathbf{e}_\theta$, requires $\cos(\gamma + 80^\circ) = 1$, so $\gamma = -80^\circ$ (equivalently 280°). Thus

$$\boxed{(\alpha, \beta, \gamma) = (70^\circ, 60^\circ, -80^\circ)}.$$

(b) The direction cosines are the elements of S. Substitute $c_\alpha = \cos 70^\circ = 0.342020$, $s_\alpha = 0.939693$, $c_\beta = 1/2$, $s_\beta = \sqrt{3}/2 = 0.866025$, $c_\gamma = \cos 80^\circ = 0.173648$, $s_\gamma = -\sin 80^\circ = -0.984808$ into (3.37); for example $S_{11} = c_\gamma c_\beta c_\alpha - s_\gamma s_\alpha = 0.029696 + 0.925417 = 0.955113$. In all, computing with unrounded sines and cosines,

$$\boxed{S = \begin{pmatrix} 0.95511217 & -0.25523613 & -0.15038373 \\ 0.00523613 & 0.52209946 & -0.85286853 \\ 0.29619813 & 0.81379768 & 0.50000000 \end{pmatrix}},$$

which rounds to the printed four-place matrix. The third row is the pole direction and its z-component is $\sin 30^\circ = 1/2$, providing a direct geometric check. If instead $+x$ is Greenwich and $+y$ is 90° east, the azimuth is λ itself: the pole gives $\alpha = -20^\circ$, $\beta = 60^\circ$, and the 10° west meridian has azimuth -10°, so the condition becomes $\cos(\gamma - 10^\circ) = 1$. The same physical orientation thus has Euler angles $(-20^\circ, 60^\circ, 10^\circ)$; its numerical matrix changes because the reference axes change.

3.4.3 Verify that the Euler angle rotation matrix, Eq. (3.37), is invariant under the transformation

$$\alpha \to \alpha + \pi, \quad \beta \to -\beta, \quad \gamma \to \gamma - \pi.$$

Solution.

In (3.37), the substitutions give

$$(c_\alpha, s_\alpha) \mapsto (-c_\alpha, -s_\alpha), \quad (c_\beta, s_\beta) \mapsto (c_\beta, -s_\beta), \quad (c_\gamma, s_\gamma) \mapsto (-c_\gamma, -s_\gamma).$$

Each term in the upper left 2×2 block has either one α factor and one γ factor, or those two factors times c_β; it acquires two minus signs. For example,

$$S_{11} = c_\gamma c_\beta c_\alpha - s_\gamma s_\alpha \mapsto (-c_\gamma)c_\beta(-c_\alpha) - (-s_\gamma)(-s_\alpha) = S_{11}.$$

The $(1,3),(2,3)$ entries contain one γ factor and s_β, again two signs. The $(3,1),(3,2)$ entries contain one α factor and s_β, also two signs. Finally $(3,3) = c_\beta$ is unchanged. Every matrix entry is therefore invariant.

3.4.4 Show that the Euler angle rotation matrix $\mathsf{S}(\alpha,\beta,\gamma)$ satisfies the following relations:

(a) $\mathsf{S}^{-1}(\alpha,\beta,\gamma) = \tilde{\mathsf{S}}(\alpha,\beta,\gamma)$,

(b) $\mathsf{S}^{-1}(\alpha,\beta,\gamma) = \mathsf{S}(-\gamma,-\beta,-\alpha)$.

Solution.

(a) Recall from Exercise 3.4.1 that $S = Z(\gamma)Y(\beta)Z(\alpha)$. For each elementary matrix, $Z(t)^T = Z(-t) = Z(t)^{-1}$ and $Y(t)^T = Y(-t) = Y(t)^{-1}$; multiplying the 2×2 blocks verifies these using $\cos^2 t + \sin^2 t = 1$. Therefore

$$S^T S = Z(\alpha)^T Y(\beta)^T Z(\gamma)^T Z(\gamma)Y(\beta)Z(\alpha) = I,$$

after canceling the three adjacent inverse pairs. Hence $S^{-1} = S^T$, the transpose, which the statement writes $\tilde{\mathsf{S}}$. **(b)** Inverting a product reverses the order of its factors:

$$S^{-1} = Z(\alpha)^{-1}Y(\beta)^{-1}Z(\gamma)^{-1} = Z(-\alpha)Y(-\beta)Z(-\gamma) = \boxed{S(-\gamma,-\beta,-\alpha)},$$

because $S(\alpha',\beta',\gamma') = Z(\gamma')Y(\beta')Z(\alpha')$ with $\alpha' = -\gamma$, $\beta' = -\beta$, $\gamma' = -\alpha$ is exactly this product.

3.4.5 The coordinate system (x,y,z) is rotated through an angle Φ counterclockwise about an axis defined by the unit vector $\hat{\mathbf{n}}$ into system (x',y',z'). In terms of the new coordinates the radius vector becomes

$$\mathbf{r}' = \mathbf{r}\cos\Phi + \mathbf{r}\times\hat{\mathbf{n}}\sin\Phi + \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\mathbf{r})(1-\cos\Phi).$$

(a) Derive this expression from geometric considerations.

(b) Show that it reduces as expected for $\hat{\mathbf{n}} = \hat{\mathbf{e}}_z$. The answer, in matrix form, appears in Eq. (3.35).

(c) Verify that $r'^2 = r^2$.

Solution.

(a) Split $\mathbf{r} = \mathbf{r}_\parallel + \mathbf{r}_\perp$, where $\mathbf{r}_\parallel = \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\mathbf{r})$ lies along the axis and $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$ is perpendicular to it. Turning the axes counterclockwise through Φ changes the components of a fixed vector exactly as turning the vector clockwise through Φ (through $-\Phi$) with the axes held fixed; senses are as seen from the tip of $\hat{\mathbf{n}}$. This preserves $\mathbf{r}_\parallel$. In the plane perpendicular to $\hat{\mathbf{n}}$, the vectors $\mathbf{r}_\perp$ and $\mathbf{r}_\perp\times\hat{\mathbf{n}}$ are orthogonal, both have length $|\mathbf{r}_\perp|$, and $\mathbf{r}_\perp\times\hat{\mathbf{n}}$ is $\mathbf{r}_\perp$ turned clockwise through 90° (for instance $\hat{\mathbf{e}}_x\times\hat{\mathbf{e}}_z = -\hat{\mathbf{e}}_y$). Turning $\mathbf{r}_\perp$ clockwise through Φ therefore gives $\mathbf{r}_\perp\cos\Phi + (\mathbf{r}_\perp\times\hat{\mathbf{n}})\sin\Phi$, and

$$\mathbf{r}' = \mathbf{r}_\parallel + \mathbf{r}_\perp\cos\Phi + (\mathbf{r}_\perp\times\hat{\mathbf{n}})\sin\Phi.$$

Here $\mathbf{r}_\perp \times \hat{\mathbf{n}} = \mathbf{r} \times \hat{\mathbf{n}}$, because $\mathbf{r}_\parallel \times \hat{\mathbf{n}} = 0$. Substituting $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$ gives the required formula:

$$\begin{aligned}\mathbf{r}' &= \mathbf{r}_\parallel + (\mathbf{r} - \mathbf{r}_\parallel)\cos\Phi + (\mathbf{r} \times \hat{\mathbf{n}})\sin\Phi \\ &= \mathbf{r}\cos\Phi + \mathbf{r} \times \hat{\mathbf{n}}\sin\Phi + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{r})(1 - \cos\Phi).\end{aligned}$$

(b) For $\hat{\mathbf{n}} = \hat{\mathbf{e}}_z$, $\mathbf{r} \times \hat{\mathbf{e}}_z = (y, -x, 0)$ and $\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{r}) = (0, 0, z)$, so

$$\begin{aligned}\mathbf{r}' &= \big(x\cos\Phi + y\sin\Phi,\ y\cos\Phi - x\sin\Phi,\ z\cos\Phi + z(1 - \cos\Phi)\big) \\ &= (x\cos\Phi + y\sin\Phi,\ -x\sin\Phi + y\cos\Phi,\ z),\end{aligned}$$

which is $Z(\Phi)\mathbf{r}$, the matrix form of Eq. (3.35). **(c)** The three terms $\mathbf{r}_\parallel$, $\mathbf{r}_\perp\cos\Phi$, $(\mathbf{r}_\perp \times \hat{\mathbf{n}})\sin\Phi$ of the first expression for $\mathbf{r}'$ are mutually orthogonal, and $|\mathbf{r}_\perp \times \hat{\mathbf{n}}| = |\mathbf{r}_\perp|$. By Pythagoras, first for $\mathbf{r}'$ and then for $\mathbf{r} = \mathbf{r}_\parallel + \mathbf{r}_\perp$,

$$|\mathbf{r}'|^2 = |\mathbf{r}_\parallel|^2 + |\mathbf{r}_\perp|^2(\cos^2\Phi + \sin^2\Phi) = |\mathbf{r}_\parallel|^2 + |\mathbf{r}_\perp|^2 = |\mathbf{r}|^2.$$

3.5 Differential Vector Operators

3.5.1 If $S(x, y, z) = \big(x^2 + y^2 + z^2\big)^{-3/2}$, find

(a) ∇S at the point $(1, 2, 3)$,

(b) the magnitude of the gradient of S, $|\nabla S|$ at $(1, 2, 3)$, and

(c) the direction cosines of ∇S at $(1, 2, 3)$.

Solution.

(a) For $r = (x^2 + y^2 + z^2)^{1/2} > 0$, differentiate by the chain rule:

$$\partial_x S = -\tfrac{3}{2}(x^2 + y^2 + z^2)^{-5/2}(2x) = -3xr^{-5},$$

and similarly for y, z. Hence $\nabla S = -3(x, y, z)/r^5$. At $(1, 2, 3)$, $r^2 = 1 + 4 + 9 = 14$ and $r^5 = 14^2\sqrt{14} = 196\sqrt{14}$, giving

$$\boxed{\nabla S = -\frac{3}{14^{5/2}}(1, 2, 3) = -\frac{3}{196\sqrt{14}}(1, 2, 3)}.$$

(b) Since $|(1, 2, 3)| = \sqrt{14}$,

$$\boxed{|\nabla S| = \frac{3\sqrt{14}}{196\sqrt{14}} = \frac{3}{196}} \approx 0.0153.$$

(c) Divide the vector by its magnitude to obtain the direction cosines $\boxed{(-1, -2, -3)/\sqrt{14}}$, whose squares sum to 1.

3.5.2

(a) Find a unit vector perpendicular to the surface

$$x^2 + y^2 + z^2 = 3$$

at the point $(1, 1, 1)$.

(b) Derive the equation of the plane tangent to the surface at $(1, 1, 1)$.

ANS. (a) $(\hat{\mathbf{e}}_x + \hat{\mathbf{e}}_y + \hat{\mathbf{e}}_z)/\sqrt{3}$, (b) $x + y + z = 3$.

Solution.

(a) Let $f = x^2 + y^2 + z^2 - 3$. A displacement tangent to $f = 0$ satisfies $df = \nabla f \cdot d\mathbf{r} = 0$, so ∇f is a normal. At $(1, 1, 1)$, $\nabla f = (2x, 2y, 2z) = (2, 2, 2)$ with length $\sqrt{4 + 4 + 4} = 2\sqrt{3}$. An outward unit normal is therefore $(1, 1, 1)/\sqrt{3}$; its negative is the inward unit normal. **(b)** The tangent plane through $\mathbf{r}_0 = (1, 1, 1)$ consists of the points $\mathbf{r}$ for which $\mathbf{r} - \mathbf{r}_0$ is perpendicular to the normal:

$$(2, 2, 2) \cdot (\mathbf{r} - \mathbf{r}_0) = 0,$$

or $2(x - 1) + 2(y - 1) + 2(z - 1) = 0$. Thus $\boxed{x + y + z = 3}$.

3.5.3 Given a vector $\mathbf{r}_{12} = \hat{\mathbf{e}}_x(x_1 - x_2) + \hat{\mathbf{e}}_y(y_1 - y_2) + \hat{\mathbf{e}}_z(z_1 - z_2)$, show that $\nabla_1 r_{12}$ (gradient with respect to x_1, y_1, and z_1 of the magnitude r_{12}) is a unit vector in the direction of $\mathbf{r}_{12}$. Assume $r_{12} > 0$.

Solution.

For separated points $r_{12} > 0$,

$$r_{12} = [(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2]^{1/2}.$$

Keeping point 2 fixed, the chain rule applied to $r_{12} = (r_{12}^2)^{1/2}$ gives

$$\frac{\partial r_{12}}{\partial x_1} = \frac{1}{2r_{12}} \frac{\partial (r_{12}^2)}{\partial x_1} = \frac{2(x_1 - x_2)}{2r_{12}} = \frac{x_1 - x_2}{r_{12}},$$

and analogously for y_1 and z_1. Therefore

$$\boxed{\nabla_1 r_{12} = \frac{(x_1 - x_2, y_1 - y_2, z_1 - z_2)}{r_{12}} = \hat{\mathbf{r}}_{12}}.$$

Its norm is $r_{12}/r_{12} = 1$. At coincidence $r_{12} = 0$, the distance is not differentiable and no radial unit direction is defined.

3.5.4 If a vector function $\mathbf{F}$ depends on both space coordinates (x, y, z) and time t, show that

$$d\mathbf{F} = (d\mathbf{r} \cdot \nabla)\mathbf{F} + \frac{\partial \mathbf{F}}{\partial t}\, dt.$$

Solution.

In a fixed Cartesian basis, apply the total differential to each component:

$$dF_i = (\partial_x F_i)dx + (\partial_y F_i)dy + (\partial_z F_i)dz + (\partial_t F_i)dt.$$

Because $d\mathbf{r} \cdot \nabla = dx\, \partial_x + dy\, \partial_y + dz\, \partial_z$, the first three terms are $[(d\mathbf{r} \cdot \nabla)\mathbf{F}]_i$. Collecting the three components proves the stated identity. Along a path $\mathbf{r}(t)$ it gives $d\mathbf{F}/dt = \partial_t \mathbf{F} + (\mathbf{v} \cdot \nabla)\mathbf{F}$.

3.5.5 Show that $\nabla(uv) = v\nabla u + u\nabla v$, where u and v are differentiable scalar functions of x, y, and z.

Solution.

The scalar product rule holds for each coordinate: $\partial_i(uv) = v\,\partial_i u + u\,\partial_i v$. Multiply by the constant Cartesian unit vector $\hat{\mathbf{e}}_i$ and sum:

$$\nabla(uv) = \sum_i \hat{\mathbf{e}}_i \partial_i(uv) = v\sum_i \hat{\mathbf{e}}_i\partial_i u + u\sum_i \hat{\mathbf{e}}_i\partial_i v = \boxed{v\nabla u + u\nabla v}.$$

3.5.6 For a particle moving in a circular orbit $\mathbf{r} = \hat{\mathbf{e}}_x r\cos\omega t + \hat{\mathbf{e}}_y r\sin\omega t$:

(a) Evaluate $\mathbf{r}\times\dot{\mathbf{r}}$, with $\dot{\mathbf{r}} = d\mathbf{r}/dt = \mathbf{v}$.

(b) Show that $\ddot{\mathbf{r}} + \omega^2\mathbf{r} = 0$ with $\ddot{\mathbf{r}} = d\mathbf{v}/dt$.

Hint. The radius r and the angular velocity ω are constant.

ANS. (a) $\hat{\mathbf{e}}_z\omega r^2$.

Solution.

(a) Differentiating, with r, ω constant,

$$\dot{\mathbf{r}} = \omega r(-\sin\omega t, \cos\omega t, 0).$$

Both vectors lie in the xy-plane, so only the z-component $x\dot{y} - y\dot{x}$ of their cross product survives:

$$\begin{aligned}\mathbf{r}\times\dot{\mathbf{r}} &= \hat{\mathbf{e}}_z\big[(r\cos\omega t)(\omega r\cos\omega t) - (r\sin\omega t)(-\omega r\sin\omega t)\big]\\ &= \hat{\mathbf{e}}_z\omega r^2(\cos^2\omega t + \sin^2\omega t) = \boxed{\hat{\mathbf{e}}_z\omega r^2}.\end{aligned}$$

(b) One further derivative gives $\ddot{\mathbf{r}} = -\omega^2 r(\cos\omega t, \sin\omega t, 0) = -\omega^2\mathbf{r}$, proving $\boxed{\ddot{\mathbf{r}} + \omega^2\mathbf{r} = 0}$.

3.5.7 Vector $\mathbf{A}$ satisfies the vector transformation law, Eq. (3.26). Show directly that its time derivative $d\mathbf{A}/dt$ also satisfies Eq. (3.26) and is therefore a vector. Use a fixed, time-independent orthogonal transformation.

Solution.

The transformation is a fixed, time-independent change of Cartesian axes. A frame rotating with time requires an additional term. For $A'_i = \sum_j S_{ij}A_j$ with constant S_{ij},

$$\frac{dA'_i}{dt} = \sum_j S_{ij}\frac{dA_j}{dt}.$$

This is precisely the same vector transformation law. If $S = S(t)$ instead, the product rule gives $\dot{\mathbf{A}}' = S\dot{\mathbf{A}} + \dot{S}\mathbf{A}$; it is not legitimate to omit the second term.

3.5.8 Show, by differentiating components, that

(a) $\dfrac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = \dfrac{d\mathbf{A}}{dt}\cdot\mathbf{B} + \mathbf{A}\cdot\dfrac{d\mathbf{B}}{dt}$,

(b) $\dfrac{d}{dt}(\mathbf{A}\times\mathbf{B}) = \dfrac{d\mathbf{A}}{dt}\times\mathbf{B} + \mathbf{A}\times\dfrac{d\mathbf{B}}{dt}$,

just like the derivative of the product of two algebraic functions.

Solution.

(a) Differentiate the component sum of the dot product, using the ordinary product rule on each term:

$$\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = \sum_i \frac{d}{dt}(A_iB_i) = \sum_i (\dot{A}_iB_i + A_i\dot{B}_i) = \dot{\mathbf{A}}\cdot\mathbf{B} + \mathbf{A}\cdot\dot{\mathbf{B}}.$$

(b) Each component $(\mathbf{A}\times\mathbf{B})_i = \sum_{jk}\varepsilon_{ijk}A_jB_k$ has constant coefficients ε_{ijk}, so

$$\frac{d}{dt}(\mathbf{A}\times\mathbf{B})_i = \sum_{jk}\varepsilon_{ijk}(\dot{A}_jB_k + A_j\dot{B}_k) = (\dot{\mathbf{A}}\times\mathbf{B} + \mathbf{A}\times\dot{\mathbf{B}})_i.$$

The order of the cross-product factors is preserved, as required by its antisymmetry.

3.5.9 Prove $\nabla\cdot(\mathbf{a}\times\mathbf{b}) = \mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b})$.
Hint. Treat as a scalar triple product.

Solution.

Apply the product rule with repeated indices summed:

$$\begin{aligned}\nabla\cdot(\mathbf{a}\times\mathbf{b}) &= \partial_i(\varepsilon_{ijk}a_jb_k)\\ &= \varepsilon_{ijk}(\partial_ia_j)b_k + \varepsilon_{ijk}a_j(\partial_ib_k)\\ &= b_k\varepsilon_{kij}\partial_ia_j - a_j\varepsilon_{jik}\partial_ib_k\\ &= \boxed{\mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b})}.\end{aligned}$$

The first sign follows from the cyclic permutation $\varepsilon_{ijk} = \varepsilon_{kij}$; the second from the swap $\varepsilon_{ijk} = -\varepsilon_{jik}$. The last line uses $(\nabla\times\mathbf{a})_k = \varepsilon_{kij}\partial_ia_j$ and $(\nabla\times\mathbf{b})_j = \varepsilon_{jik}\partial_ib_k$.

3.5.10 Classically, orbital angular momentum is given by $\mathbf{L} = \mathbf{r}\times\mathbf{p}$, where $\mathbf{p}$ is the linear momentum. To go from classical mechanics to quantum mechanics, $\mathbf{p}$ is replaced (in units with $\hbar = 1$) by the operator $-i\nabla$. Show that the quantum mechanical angular momentum operator has Cartesian components

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \quad L_y = -i\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \quad L_z = -i\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

Solution.

Apply the operator to an arbitrary differentiable scalar f:

$$\mathbf{L}f = -i\,\mathbf{r}\times\nabla f = -i\begin{vmatrix}\hat{\mathbf{e}}_x & \hat{\mathbf{e}}_y & \hat{\mathbf{e}}_z\\ x & y & z\\ \partial_x f & \partial_y f & \partial_z f\end{vmatrix}.$$

Expanding along the first row, with $f_x = \partial f/\partial x$ and so on, gives $L_xf = -i(yf_z - zf_y)$, $L_yf = -i(zf_x - xf_z)$, and $L_zf = -i(xf_y - yf_x)$. Because f is arbitrary, the stated three operator formulas follow. The coordinate factors multiply on the left of the derivatives.

3.5.11 Using the angular momentum operators previously given, show that they satisfy commutation relations of the form

$$[L_x, L_y] \equiv L_x L_y - L_y L_x = iL_z$$

and hence

$$\mathbf{L} \times \mathbf{L} = i\mathbf{L}.$$

These commutation relations will be taken later as the defining relations of an angular momentum operator.

Solution.

Let f have commuting mixed second derivatives, and write $f_x = \partial f/\partial x$, $f_{xz} = \partial^2 f/\partial x\,\partial z$, and so on. Since $(-i)^2 = -1$,

$$L_x L_y f = -(y\partial_z - z\partial_y)(zf_x - xf_z),$$
$$L_y L_x f = -(z\partial_x - x\partial_z)(yf_z - zf_y).$$

The product rule gives an extra term only where a derivative acts on a factor of its own coordinate (∂_z on z): $y\partial_z(zf_x) = yf_x + yzf_{xz}$ and $-x\partial_z(-zf_y) = xf_y + xzf_{yz}$; in every other term the coordinate factor is constant for that derivative, which acts on f alone. Thus

$$L_x L_y f = -\{yf_x + yzf_{xz} - xyf_{zz} - z^2 f_{xy} + xzf_{yz}\},$$
$$L_y L_x f = -\{zyf_{xz} - z^2 f_{xy} - xyf_{zz} + xf_y + xzf_{yz}\}.$$

All second-derivative terms cancel on subtraction, leaving

$$[L_x, L_y]f = -(yf_x - xf_y) = xf_y - yf_x = iL_z f,$$

since $iL_z f = i(-i)(xf_y - yf_x) = xf_y - yf_x$. Cyclic relabeling of x, y, z in this explicit computation gives $[L_y, L_z] = iL_x$ and $[L_z, L_x] = iL_y$. Thus

$$\mathbf{L} \times \mathbf{L} = \big([L_y, L_z], [L_z, L_x], [L_x, L_y]\big) = \boxed{i\mathbf{L}}.$$

Unlike ordinary numbers, the components here do not commute, so a vector operator need not have zero cross product with itself.

In the last line on p. 39 of the Instructor's Manual, the coefficient of ∂_x is misprinted as $-u$; it is $-y$, as the preceding subtraction shows.

3.5.12 With the aid of the results of Exercise 3.5.11, show that if two vectors $\mathbf{a}$ and $\mathbf{b}$ commute with each other and with $\mathbf{L}$, that is, $[\mathbf{a}, \mathbf{b}] = [\mathbf{a}, \mathbf{L}] = [\mathbf{b}, \mathbf{L}] = 0$, show that

$$[\mathbf{a} \cdot \mathbf{L}, \mathbf{b} \cdot \mathbf{L}] = i(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{L}.$$

Solution.

Exercise 3.5.11 gives $[L_x, L_y] = iL_z$ and its two cyclic permutations. Together with $[L_i, L_i] = 0$ and $[L_j, L_i] = -[L_i, L_j]$ these are summarized by $[L_i, L_j] = i\sum_k \varepsilon_{ijk} L_k$. Interpret the commuting hypotheses componentwise. They permit a_i, b_j to be moved through each other and through every L_k:

$$\begin{aligned}[\mathbf{a} \cdot \mathbf{L}, \mathbf{b} \cdot \mathbf{L}] &= \sum_{ij} (a_i L_i b_j L_j - b_j L_j a_i L_i) \\ &= \sum_{ij} a_i b_j (L_i L_j - L_j L_i) \\ &= i \sum_{ijk} a_i b_j \varepsilon_{ijk} L_k = \boxed{i(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{L}}.\end{aligned}$$

The last equality uses $\varepsilon_{ijk} = \varepsilon_{kij}$, so that $\sum_{ij} \varepsilon_{kij} a_i b_j = (\mathbf{a} \times \mathbf{b})_k$.

3.5.13 Prove that the stream lines of $\mathbf{b}$ in Example 3.5.7 with nonzero distance from the z-axis are counterclockwise circles; points on the z-axis are stationary.

Solution.

Example 3.5.7 defines $\mathbf{b} = (-y, x, 0)$. A streamline is a curve everywhere tangent to $\mathbf{b}$; parametrized in the flow direction it obeys $d\mathbf{r}/ds = \mathbf{b}$, that is, $dx/ds = -y$, $dy/ds = x$, $dz/ds = 0$. Therefore

$$\frac{d}{ds}(x^2 + y^2) = 2x\frac{dx}{ds} + 2y\frac{dy}{ds} = 2x(-y) + 2yx = 0, \qquad z = z_0.$$

The streamlines through points off the z-axis are therefore circles $x^2 + y^2 = R^2$, $R > 0$, in planes $z = z_0$. Explicitly $x = R\cos(s + s_0)$, $y = R\sin(s + s_0)$; increasing s is counterclockwise as viewed from $+z$. At $(R, 0)$ the velocity is $(0, R)$, confirming the sense. On the z-axis the field vanishes, so these are stationary points rather than nonzero-radius circular streamlines.

3.6 Differential Vector Operators: Further Properties

3.6.1 Show that $\mathbf{u} \times \mathbf{v}$ is solenoidal if $\mathbf{u}$ and $\mathbf{v}$ are each irrotational.

Solution.

Irrotationality means $\nabla \times \mathbf{u} = \nabla \times \mathbf{v} = 0$. By the product rule proved in Exercise 3.5.9, with $\mathbf{a} = \mathbf{u}$ and $\mathbf{b} = \mathbf{v}$,

$$\nabla \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot (\nabla \times \mathbf{u}) - \mathbf{u} \cdot (\nabla \times \mathbf{v}) = 0 - 0 = 0.$$

Thus $\mathbf{u} \times \mathbf{v}$ is solenoidal.

3.6.2 If $\mathbf{A}$ is irrotational, show that $\mathbf{A} \times \mathbf{r}$ is solenoidal.

Solution.

In Cartesian coordinates $\mathbf{r} = (x, y, z)$, and each component of $\nabla \times \mathbf{r}$ is a difference of zero cross derivatives:

$$\nabla \times \mathbf{r} = (\partial_y z - \partial_z y,\ \partial_z x - \partial_x z,\ \partial_x y - \partial_y x) = (0, 0, 0).$$

Therefore, by the product rule of Exercise 3.5.9 with $\mathbf{a} = \mathbf{A}$ and $\mathbf{b} = \mathbf{r}$,

$$\nabla \cdot (\mathbf{A} \times \mathbf{r}) = \mathbf{r} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{r}) = 0.$$

The first term vanishes by the hypothesis, the second by the calculation.

3.6.3 A rigid body is rotating with constant angular velocity $\boldsymbol{\omega}$. Show that the linear velocity $\mathbf{v}$ is solenoidal.

Solution.

The velocity is $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, or

$$\mathbf{v} = (\omega_y z - \omega_z y,\ \omega_z x - \omega_x z,\ \omega_x y - \omega_y x).$$

Each component is independent of its matching coordinate when $\boldsymbol{\omega}$ is spatially constant: v_x contains only y, z, v_y only z, x, and v_z only x, y. Hence $\nabla \cdot \mathbf{v} = \partial_x v_x + \partial_y v_y + \partial_z v_z = 0 + 0 + 0 = 0$. A spatially uniform translational velocity can be added without changing this result.

3.6.4 If a vector function $\mathbf{V}(x, y, z)$ is not irrotational, show that if there exists a nowhere-zero scalar function $g(x, y, z)$ such that $g\mathbf{V}$ is irrotational, then

$$\mathbf{V} \cdot \nabla \times \mathbf{V} = 0.$$

Solution.

The integrating factor must be nonzero on the region where the conclusion is asserted; allowing $g \equiv 0$ makes the statement false. The product rule, componentwise $\varepsilon_{ijk}\partial_j(gV_k) = \varepsilon_{ijk}(\partial_j g)V_k + g\,\varepsilon_{ijk}\partial_j V_k$ (repeated indices summed), gives

$$0 = \nabla \times (g\mathbf{V}) = \nabla g \times \mathbf{V} + g\,\nabla \times \mathbf{V}.$$

Dot with $\mathbf{V}$. The scalar triple product with two factors $\mathbf{V}$ is zero, $\mathbf{V} \cdot (\nabla g \times \mathbf{V}) = \nabla g \cdot (\mathbf{V} \times \mathbf{V}) = 0$, so $g\,\mathbf{V} \cdot (\nabla \times \mathbf{V}) = 0$. Where $g \neq 0$, division proves $\boxed{\mathbf{V} \cdot (\nabla \times \mathbf{V}) = 0}$. To check why the qualification is necessary, take $\mathbf{V} = (-y, x, 1)$. Then $\nabla \times \mathbf{V} = (0, 0, 2)$ and $\mathbf{V} \cdot \nabla \times \mathbf{V} = 2$, yet $g\mathbf{V} = 0$ is irrotational when $g = 0$.

3.6.5 Verify the vector identity

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} - \mathbf{B}(\nabla \cdot \mathbf{A}) + \mathbf{A}(\nabla \cdot \mathbf{B}).$$

Solution.

For component i, with repeated indices summed from 1 to 3, contract the Levi-Civita symbols: by the cyclic shift $\varepsilon_{k\ell m} = \varepsilon_{\ell m k}$ and the identity $\sum_k \varepsilon_{ijk}\varepsilon_{\ell mk} = \delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}$ of Exercise 2.1.9, $\varepsilon_{ijk}\varepsilon_{k\ell m} = \delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}$. Then apply the product rule to each ∂_j:

$$\begin{aligned}
[\nabla \times (\mathbf{A} \times \mathbf{B})]_i &= \varepsilon_{ijk}\partial_j(\varepsilon_{k\ell m}A_\ell B_m) \\
&= (\delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell})\partial_j(A_\ell B_m) \\
&= \partial_j(A_i B_j) - \partial_j(A_j B_i) \\
&= B_j\partial_j A_i + A_i\partial_j B_j - B_i\partial_j A_j - A_j\partial_j B_i.
\end{aligned}$$

The four terms are respectively the i-components of $(\mathbf{B}\cdot\nabla)\mathbf{A}$, $\mathbf{A}(\nabla\cdot\mathbf{B})$, $-\mathbf{B}(\nabla\cdot\mathbf{A})$, and $-(\mathbf{A}\cdot\nabla)\mathbf{B}$, proving the identity.

3.6.6 As an alternative to the vector identity of Example 3.6.4 show that

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \times \nabla) \times \mathbf{B} + (\mathbf{B} \times \nabla) \times \mathbf{A} + \mathbf{A}(\nabla \cdot \mathbf{B}) + \mathbf{B}(\nabla \cdot \mathbf{A}).$$

Solution.

The derivatives in $(\mathbf{A} \times \nabla) \times \mathbf{B}$ act on $\mathbf{B}$ to their right, not on the coefficient $\mathbf{A}$. Sum repeated indices and write $(\mathbf{A} \times \nabla)_j = \varepsilon_{j\ell m}A_\ell\partial_m$. The cyclic shifts $\varepsilon_{ijk} = \varepsilon_{kij}$, $\varepsilon_{j\ell m} = \varepsilon_{\ell mj}$ and the identity of Exercise 2.1.9, $\sum_j \varepsilon_{kij}\varepsilon_{\ell mj} = \delta_{k\ell}\delta_{im} - \delta_{km}\delta_{i\ell}$, give $\varepsilon_{ijk}\varepsilon_{j\ell m} = \delta_{k\ell}\delta_{im} - \delta_{km}\delta_{i\ell}$. Thus

$$\begin{aligned}
[(\mathbf{A} \times \nabla) \times \mathbf{B}]_i &= \varepsilon_{ijk}\varepsilon_{j\ell m}A_\ell\partial_m B_k \\
&= (\delta_{im}\delta_{k\ell} - \delta_{i\ell}\delta_{km})A_\ell\partial_m B_k \\
&= A_k\partial_i B_k - A_i\partial_k B_k.
\end{aligned}$$

Interchanging A, B gives $[(\mathbf{B} \times \nabla) \times \mathbf{A}]_i = B_k\partial_i A_k - B_i\partial_k A_k$. Adding the i-components $A_i\partial_k B_k + B_i\partial_k A_k$ of $\mathbf{A}(\nabla\cdot\mathbf{B}) + \mathbf{B}(\nabla\cdot\mathbf{A})$ cancels the two divergence terms:

$$(A_k\partial_i B_k - A_i\partial_k B_k) + (B_k\partial_i A_k - B_i\partial_k A_k) + A_i\partial_k B_k + B_i\partial_k A_k = A_k\partial_i B_k + B_k\partial_i A_k.$$

By the product rule this is $\partial_i(A_k B_k)$, the i-component of $\nabla(\mathbf{A}\cdot\mathbf{B})$.

3.6.7 Verify the identity

$$\mathbf{A} \times (\nabla \times \mathbf{A}) = \frac{1}{2}\nabla(A^2) - (\mathbf{A} \cdot \nabla)\mathbf{A}.$$

Solution.

For each component, with repeated indices summed and the contraction $\varepsilon_{ijk}\varepsilon_{k\ell m} = \delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}$ used in Exercise 3.6.5,

$$\begin{aligned}
[\mathbf{A} \times (\nabla \times \mathbf{A})]_i &= \varepsilon_{ijk}\varepsilon_{k\ell m} A_j \partial_\ell A_m \\
&= (\delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}) A_j \partial_\ell A_m \\
&= A_j \partial_i A_j - A_j \partial_j A_i \\
&= \tfrac{1}{2}\partial_i \left(\sum_j A_j^2 \right) - (\mathbf{A} \cdot \nabla) A_i.
\end{aligned}$$

Here $\partial_i(A_j^2) = 2A_j \partial_i A_j$. Collecting components gives the stated identity.

3.6.8 If $\mathbf{A}$ and $\mathbf{B}$ are constant vectors, show that

$$\nabla(\mathbf{A} \cdot \mathbf{B} \times \mathbf{r}) = \mathbf{A} \times \mathbf{B}.$$

Solution.

Cyclicity changes the scalar triple product to $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{r}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{r}$. Write the constant vector $\mathbf{A} \times \mathbf{B} = \mathbf{C}$. Then $\nabla(C_x x + C_y y + C_z z) = (C_x, C_y, C_z) = \mathbf{C}$, which proves the result. Constancy matters because otherwise derivatives of $\mathbf{C}$ would also appear.

3.6.9 Verify Eq. (3.70),

$$\nabla \times (\nabla \times \mathbf{V}) = \nabla(\nabla \cdot \mathbf{V}) - \nabla \cdot \nabla \mathbf{V},$$

by direct expansion in Cartesian coordinates.

Solution.

Assume the components have commuting mixed second derivatives. Directly, the x-component of the double curl is

$$\begin{aligned}
[\nabla \times (\nabla \times \mathbf{V})]_x &= \partial_y(\partial_x V_y - \partial_y V_x) - \partial_z(\partial_z V_x - \partial_x V_z) \\
&= \partial_x \partial_y V_y + \partial_x \partial_z V_z - \partial_y^2 V_x - \partial_z^2 V_x \\
&= \partial_x(\partial_x V_x + \partial_y V_y + \partial_z V_z) - (\partial_x^2 + \partial_y^2 + \partial_z^2) V_x.
\end{aligned}$$

The second line uses $\partial_y \partial_x = \partial_x \partial_y$ and $\partial_z \partial_x = \partial_x \partial_z$; the third adds and subtracts $\partial_x^2 V_x$. Cyclic replacement $x \to y \to z \to x$ gives $[\nabla \times (\nabla \times \mathbf{V})]_y = \partial_y(\nabla \cdot \mathbf{V}) - \nabla^2 V_y$ and $[\nabla \times (\nabla \times \mathbf{V})]_z = \partial_z(\nabla \cdot \mathbf{V}) - \nabla^2 V_z$. Together these are $\nabla(\nabla \cdot \mathbf{V}) - \nabla^2 \mathbf{V}$, where the Cartesian vector Laplacian acts componentwise.

3.6.10 Prove that $\nabla \times (\varphi \nabla \varphi) = 0$.

Solution.

The product rule gives $\nabla(\varphi^2) = 2\varphi\nabla\varphi$, i.e. $\varphi\nabla\varphi = \frac{1}{2}\nabla(\varphi^2)$. For a C^2 scalar f, the i-component of $\nabla \times \nabla f$ is $\varepsilon_{ijk}\partial_j\partial_k f = 0$, since the derivatives are symmetric and ε_{ijk} antisymmetric. Hence $\nabla \times (\varphi\nabla\varphi) = \frac{1}{2}\nabla \times \nabla(\varphi^2) = 0$.

3.6.11 You are given that the curl of $\mathbf{F}$ equals the curl of $\mathbf{G}$. Show that $\mathbf{F}$ and $\mathbf{G}$ may differ by (a) a constant and (b) a gradient of a scalar function.

Solution.

Put $\mathbf{H} = \mathbf{F} - \mathbf{G}$. The hypothesis is $\nabla \times \mathbf{H} = 0$. **(a)** A constant vector $\mathbf{c}$ has zero derivatives, so $\nabla \times \mathbf{c} = 0$ and $\mathbf{F} = \mathbf{G} + \mathbf{c}$ has the same curl as $\mathbf{G}$. **(b)** Likewise $\nabla \times \nabla f = 0$ for every C^2 scalar f (Exercise 3.6.10), so $\mathbf{F} = \mathbf{G} + \nabla f$ has the same curl. The constant case is included by $f = \mathbf{c} \cdot \mathbf{r}$, since $\nabla(\mathbf{c} \cdot \mathbf{r}) = \mathbf{c}$.

Conversely, every such difference $\mathbf{H}$ is the gradient of a single-valued function when the domain is simply connected, but not on every domain. On a simply connected domain set $f(\mathbf{r}) = \int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{H} \cdot d\boldsymbol{\ell}$. Two paths from $\mathbf{r}_0$ to $\mathbf{r}$ together form a closed loop that bounds a surface S in the domain, so by Stokes' theorem the two path integrals differ by $\int_S (\nabla \times \mathbf{H}) \cdot d\boldsymbol{\sigma} = 0$; thus f is well defined. Extending a path by a segment parallel to the x-axis gives $f(x+h, y, z) - f(x, y, z) = \int_x^{x+h} H_x(s, y, z)\, ds$, so $\partial_x f = H_x$; likewise for y and z, hence $\nabla f = \mathbf{H}$. With the z-axis removed, $\mathbf{H} = (-y, x, 0)/(x^2 + y^2)$ has zero curl: its x- and y-components vanish because $H_z = 0$ and $\mathbf{H}$ does not depend on z, and $\partial_x H_y = (y^2 - x^2)/(x^2 + y^2)^2 = \partial_y H_x$. On the unit circle $\mathbf{r} = (\cos\varphi, \sin\varphi, 0)$, however, $\mathbf{H} \cdot d\mathbf{r} = (\sin^2\varphi + \cos^2\varphi)\, d\varphi = d\varphi$, so its circulation is 2π, whereas the closed line integral of the gradient of a single-valued function vanishes. Hence this $\mathbf{H}$ is not such a gradient there.

3.6.12 The Navier-Stokes equation of hydrodynamics contains a nonlinear term of the form $(\mathbf{v}\cdot\nabla)\mathbf{v}$. Show that the curl of this term may be written as $-\nabla \times [\mathbf{v} \times (\nabla \times \mathbf{v})]$.

Solution.

Exercise 3.6.7 with $\mathbf{A} = \mathbf{v}$, solved for $(\mathbf{v} \cdot \nabla)\mathbf{v}$, gives

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \tfrac{1}{2}\nabla(\mathbf{v} \cdot \mathbf{v}) - \mathbf{v} \times (\nabla \times \mathbf{v}).$$

Take the curl. The gradient term has zero curl, as shown in Exercise 3.6.10, so

$$\boxed{\nabla \times [(\mathbf{v} \cdot \nabla)\mathbf{v}] = -\nabla \times [\mathbf{v} \times (\nabla \times \mathbf{v})]}.$$

No incompressibility assumption is needed.

The Instructor's Manual, p. 41, omits the factor 1/2 in front of $\nabla(\mathbf{v} \cdot \mathbf{v})$. For $\mathbf{v} = (x, 0, 0)$, where $\nabla \times \mathbf{v} = 0$, the identity requires $\frac{1}{2}\nabla(\mathbf{v} \cdot \mathbf{v}) - (\mathbf{v} \cdot \nabla)\mathbf{v} = 0$, and indeed $(x, 0, 0) - (x, 0, 0) = 0$; without the 1/2 this difference would be $(2x, 0, 0) - (x, 0, 0) = (x, 0, 0) \neq 0$. Taking a curl hides this error because either gradient has zero curl.

3.6.13 Prove that $(\nabla u) \times (\nabla v)$ is solenoidal, where u and v are C^2 scalar functions.

Solution.

For a classical pointwise calculation assume u, v are C^2, so the required mixed derivatives exist and commute. Using Exercise 3.5.9 with $\mathbf{a} = \nabla u$ and $\mathbf{b} = \nabla v$,

$$\nabla \cdot (\nabla u \times \nabla v) = \nabla v \cdot (\nabla \times \nabla u) - \nabla u \cdot (\nabla \times \nabla v) = 0.$$

Each curl of a gradient vanishes by equality of mixed partial derivatives, $(\nabla \times \nabla u)_i = \sum_{jk} \varepsilon_{ijk}\partial_j\partial_k u = 0$ (Exercise 3.6.10), so the cross product is solenoidal.

3.6.14 The function φ is a scalar satisfying Laplace's equation, $\nabla^2\varphi = 0$. Show that $\nabla\varphi$ is both solenoidal and irrotational.

Solution.

For $\mathbf{A} = \nabla\varphi$, $\nabla \cdot \mathbf{A} = \sum_i \partial_i\partial_i\varphi = \nabla^2\varphi = 0$ by the given equation. Also

$$(\nabla \times \mathbf{A})_i = \varepsilon_{ijk}\partial_j\partial_k\varphi = 0$$

by commuting mixed derivatives. Thus the gradient is both solenoidal and irrotational.

3.6.15 Show that any solution of the equation

$$\nabla \times (\nabla \times \mathbf{A}) - k^2\mathbf{A} = 0$$

automatically satisfies the vector Helmholtz equation

$$\nabla^2\mathbf{A} + k^2\mathbf{A} = 0$$

and the solenoidal condition

$$\nabla \cdot \mathbf{A} = 0.$$

Hint. Let $\nabla\cdot$ operate on the first equation. Assume that k is constant and $k \neq 0$.

Solution.

The conclusion requires a constant $k \neq 0$; the example at the end shows that it fails for $k = 0$. Taking the divergence of the first equation gives

$$\nabla \cdot [\nabla \times (\nabla \times \mathbf{A})] - k^2\nabla \cdot \mathbf{A} = 0 - k^2\nabla \cdot \mathbf{A} = 0.$$

The divergence of any curl vanishes, $\nabla \cdot (\nabla \times \mathbf{C}) = \sum_{ijk} \varepsilon_{ijk}\partial_i\partial_j C_k = 0$, because the mixed derivatives are symmetric in i, j and ε_{ijk} is antisymmetric; here $\mathbf{C} = \nabla \times \mathbf{A}$. The constant k^2 passes through the divergence. Division by $k^2 \neq 0$ gives $\nabla \cdot \mathbf{A} = 0$. Substitution into the double-curl identity (3.70) of Exercise 3.6.9, $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A}$, then yields

$$0 = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A} - k^2\mathbf{A} = -\nabla^2\mathbf{A} - k^2\mathbf{A}.$$

This is the required vector Helmholtz equation. For $k = 0$, take $\mathbf{A} = \nabla(x^3) = (3x^2, 0, 0)$. Its curl and double curl vanish, but $\nabla \cdot \mathbf{A} = 6x$ and $\nabla^2\mathbf{A} = (6, 0, 0)$; both proposed conclusions fail in general.

3.6.16 The theory of heat conduction leads to an equation

$$\nabla^2\Psi = k\left|\nabla\Phi\right|^2,$$

where Φ is a potential satisfying Laplace's equation: $\nabla^2\Phi = 0$. Show that a solution of this equation is $\Psi = k\Phi^2/2$. Take Φ real-valued and k constant.

Solution.

For the real potential Φ and constant k, differentiate $\Psi = k\Phi^2/2$:

$$\partial_i\Psi = k\Phi\,\partial_i\Phi, \qquad \partial_i^2\Psi = k(\partial_i\Phi)^2 + k\Phi\,\partial_i^2\Phi.$$

Summing over the three coordinates, with $\sum_i(\partial_i\Phi)^2 = |\nabla\Phi|^2$ for real Φ, gives

$$\nabla^2\Psi = k|\nabla\Phi|^2 + k\Phi\nabla^2\Phi = \boxed{k|\nabla\Phi|^2}.$$

Thus it is a solution. One can add any harmonic function to Ψ because its Laplacian contributes zero. Both qualifications are needed. The complex harmonic function $\Phi = x + iy$ has $\nabla\Phi = (1, i, 0)$ and $\sum_i(\partial_i\Phi)^2 = 1 + i^2 = 0$, so $\nabla^2(k\Phi^2/2) = 0$, whereas $k|\nabla\Phi|^2 = 2k$; and for nonconstant k the product rule adds $\frac{1}{2}\Phi^2\nabla^2 k + 2\Phi\,\nabla k \cdot \nabla\Phi$, nonzero in general, to $\nabla^2\Psi$.

3.6.17 Given the three matrices

$$\mathsf{M}_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \mathsf{M}_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix},$$

and

$$\mathsf{M}_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

show that the matrix-vector equation

$$\left(\mathsf{M} \cdot \nabla + \mathbf{1}_3 \frac{1}{c} \frac{\partial}{\partial t} \right) \boldsymbol{\psi} = 0$$

reproduces the two curl equations of Maxwell's vacuum system. The full system additionally requires $\nabla \cdot \mathbf{E} = \nabla \cdot \mathbf{B} = 0$ initially. Here $\boldsymbol{\psi}$ is a column vector with components $\psi_j = B_j - iE_j/c$, $j = x, y, z$. Note that $\epsilon_0 \mu_0 = 1/c^2$ and that $\mathbf{1}_3$ is the 3×3 unit matrix.

Solution.

The matrix differential operator $\mathsf{M} \cdot \nabla = \mathsf{M}_x \partial_x + \mathsf{M}_y \partial_y + \mathsf{M}_z \partial_z$ is

$$\mathsf{M} \cdot \nabla = \begin{pmatrix} 0 & -i\partial_z & i\partial_y \\ i\partial_z & 0 & -i\partial_x \\ -i\partial_y & i\partial_x & 0 \end{pmatrix}.$$

Its action on a column $\boldsymbol{\psi}$ is, row by row,

$$(\mathsf{M} \cdot \nabla)\boldsymbol{\psi} = \begin{pmatrix} i(\partial_y \psi_z - \partial_z \psi_y) \\ i(\partial_z \psi_x - \partial_x \psi_z) \\ i(\partial_x \psi_y - \partial_y \psi_x) \end{pmatrix} = i \nabla \times \boldsymbol{\psi}.$$

The given equation is therefore $i\nabla \times \boldsymbol{\psi} + c^{-1} \partial_t \boldsymbol{\psi} = 0$. Insert $\boldsymbol{\psi} = \mathbf{B} - i\mathbf{E}/c$: the two terms become $i\nabla \times \mathbf{B} + c^{-1} \nabla \times \mathbf{E}$ and $c^{-1} \partial_t \mathbf{B} - ic^{-2} \partial_t \mathbf{E}$. Separating real and imaginary parts,

$$\frac{1}{c}(\nabla \times \mathbf{E} + \partial_t \mathbf{B}) + i \left(\nabla \times \mathbf{B} - \frac{1}{c^2} \partial_t \mathbf{E} \right) = 0.$$

For real fields both brackets vanish, which with $1/c^2 = \epsilon_0 \mu_0$ gives

$$\boxed{\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \qquad \nabla \times \mathbf{B} = \frac{1}{c^2} \partial_t \mathbf{E} = \epsilon_0 \mu_0 \, \partial_t \mathbf{E}}.$$

The matrix equation supplies the two vacuum curl equations. The full vacuum Maxwell system also needs $\nabla \cdot \mathbf{E} = \nabla \cdot \mathbf{B} = 0$ as initial constraints; they are not consequences of the matrix equation alone. Taking the divergence of the matrix equation and using $\nabla \cdot (\nabla \times \boldsymbol{\psi}) = 0$ gives $c^{-1} \partial_t (\nabla \cdot \boldsymbol{\psi}) = 0$, that is, $\partial_t (\nabla \cdot \mathbf{B}) - ic^{-1} \partial_t (\nabla \cdot \mathbf{E}) = 0$. Its real and imaginary parts give $\partial_t (\nabla \cdot \mathbf{B}) = \partial_t (\nabla \cdot \mathbf{E}) = 0$, so initial zero divergences are preserved. As a counterexample without those initial conditions, the static fields $\mathbf{B} = \mathbf{r}$, $\mathbf{E} = 0$ solve the matrix equation (both curls vanish, $\nabla \times \mathbf{r} = 0$ as in Exercise 3.6.2, and so do both time derivatives) but have $\nabla \cdot \mathbf{B} = 3$.

The Instructor's Manual, p. 42, labels its full operator matrix $\mathsf{M} \cdot \nabla$ even though it includes $c^{-1} \partial_t I$. After multiplying that equation by c, the imaginary part is $c[\nabla \times \mathbf{B} - c^{-2} \partial_t \mathbf{E}]$, so its displayed imaginary bracket is also missing an overall factor c. These notation and normalization errors do not affect the conclusion that both brackets vanish.

3.6.18 Using the Pauli matrices σ_i of Eq. (2.28), show that

$$(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b}) = (\mathbf{a}\cdot\mathbf{b})\mathbf{1}_2 + i\boldsymbol{\sigma}\cdot(\mathbf{a}\times\mathbf{b}).$$

Here

$$\boldsymbol{\sigma} \equiv \hat{\mathbf{e}}_x\sigma_1 + \hat{\mathbf{e}}_y\sigma_2 + \hat{\mathbf{e}}_z\sigma_3,$$

$\mathbf{a}$ and $\mathbf{b}$ are ordinary vectors, and $\mathbf{1}_2$ is the 2×2 unit matrix.

Solution.

The matrices of Eq. (2.28) are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Direct multiplication gives $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \mathbf{1}_2$ and

$$\sigma_1\sigma_2 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_3, \quad \sigma_2\sigma_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\sigma_1, \quad \sigma_3\sigma_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2,$$

while the products in reverse order have the opposite sign (for example $\sigma_2\sigma_1 = -i\sigma_3$). These nine products are summarized by the Pauli multiplication law $\sigma_i\sigma_j = \delta_{ij}\mathbf{1}_2 + i\sum_k \varepsilon_{ijk}\sigma_k$. The ordinary vector components commute with the Pauli matrices, so we may expand and use this law, together with $\varepsilon_{ijk} = \varepsilon_{kij}$:

$$\begin{aligned}
(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b}) &= \sum_{ij} a_i b_j \sigma_i\sigma_j \\
&= \sum_{ij} a_i b_j \left(\delta_{ij}\mathbf{1}_2 + i\sum_k \varepsilon_{ijk}\sigma_k\right) \\
&= \left(\sum_i a_i b_i\right)\mathbf{1}_2 + i\sum_k \sigma_k \left(\sum_{ij}\varepsilon_{kij} a_i b_j\right) \\
&= \boxed{(\mathbf{a}\cdot\mathbf{b})\mathbf{1}_2 + i\boldsymbol{\sigma}\cdot(\mathbf{a}\times\mathbf{b})}.
\end{aligned}$$

The last step uses $\sum_{ij}\varepsilon_{kij}a_i b_j = (\mathbf{a}\times\mathbf{b})_k$.

3.7 Vector Integration

3.7.1 The origin and the three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ (all of which start at the origin) define a tetrahedron. Taking the outward direction as positive, calculate the total vector area of the four tetrahedral surfaces.

Solution.

Suppose first $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) > 0$. The face OAB is a triangle with edges $\mathbf{A}$, $\mathbf{B}$ at O, so its vector area is $\pm\frac{1}{2}\mathbf{A}\times\mathbf{B}$. The vertex C not on this face satisfies $\mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) > 0$ (cyclic triple product), so $\mathbf{A}\times\mathbf{B}$ points from the face toward C, into the tetrahedron. The same argument with the opposite vertices A and B applies to the faces OBC and OCA. Hence the outward vector areas of faces OAB, OBC, OCA are respectively $-\mathbf{A}\times\mathbf{B}/2$, $-\mathbf{B}\times\mathbf{C}/2$, $-\mathbf{C}\times\mathbf{A}/2$. For face ABC, with edges $\mathbf{B}-\mathbf{A}$ and $\mathbf{C}-\mathbf{A}$ at A, expanding and using $\mathbf{A}\times\mathbf{A} = 0$, $-\mathbf{B}\times\mathbf{A} = \mathbf{A}\times\mathbf{B}$, $-\mathbf{A}\times\mathbf{C} = \mathbf{C}\times\mathbf{A}$,

$$\frac{1}{2}(\mathbf{B}-\mathbf{A})\times(\mathbf{C}-\mathbf{A}) = \frac{1}{2}(\mathbf{B}\times\mathbf{C} + \mathbf{C}\times\mathbf{A} + \mathbf{A}\times\mathbf{B}).$$

This is the outward vector area: its scalar product with $\mathbf{A}$, the vector from O to a point of the face, is $\frac{1}{2}\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})>0$, so it points away from O. Adding all four cancels every term: $\boxed{\sum_{\text{faces}}\boldsymbol{\sigma}=0}$. For negative scalar triple product, every outward orientation reverses relative to these expressions, and the sum remains zero. This is the sum of directed areas; the sum of their positive magnitudes does not vanish.

The Instructor's Manual, p. 42, sums the positive scalar face areas instead of the requested vector areas. For the tetrahedron with vertices $0, \hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z$, its scalar total is $3\cdot\frac{1}{2}+\frac{\sqrt{3}}{4}(\sqrt{2})^2=(3+\sqrt{3})/2$ (three right triangles and an equilateral triangle of side $\sqrt{2}$), while the outward vector areas still sum to zero.

3.7.2 Find the work $-\int_C \mathbf{F}\cdot d\mathbf{r}$ done moving on a unit circle in the xy-plane, doing work against a force field given by

$$\mathbf{F}=\frac{-\hat{\mathbf{e}}_x y}{x^2+y^2}+\frac{\hat{\mathbf{e}}_y x}{x^2+y^2}\,:$$

(a) Counterclockwise from 0 to π,

(b) Clockwise from 0 to $-\pi$.

Note that the work done depends on the path.

Solution.

The textbook writes $\oint \mathbf{F}\cdot d\mathbf{r}$, but these paths are semicircles, not closed loops. Also, work against the specified field is $-\int_C \mathbf{F}\cdot d\mathbf{r}$; $\int_C \mathbf{F}\cdot d\mathbf{r}$ is the work done by the field. Parametrize the unit circle by $\mathbf{r}(\varphi)=(\cos\varphi,\sin\varphi,0)$. Then $\mathbf{F}=(-\sin\varphi,\cos\varphi,0)$ (because $x^2+y^2=1$) and $d\mathbf{r}=(-\sin\varphi,\cos\varphi,0)d\varphi$, so $\mathbf{F}\cdot d\mathbf{r}=(\sin^2\varphi+\cos^2\varphi)\,d\varphi=d\varphi$. Thus the work by the field is

$$W_F^{(a)}=\int_0^{\pi} d\varphi=\pi,\qquad W_F^{(b)}=\int_0^{-\pi} d\varphi=-\pi.$$

The requested work against it is consequently

$$\boxed{W^{(a)}_{\text{against}}=-\pi,\qquad W^{(b)}_{\text{against}}=+\pi}.$$

Both paths join $(1,0)$ to $(-1,0)$ but give different work.

3.7.3 Calculate the work you do in going from point $(1,1)$ to point $(3,3)$. The force you exert is given by

$$\mathbf{F}=\hat{\mathbf{e}}_x(x-y)+\hat{\mathbf{e}}_y(x+y).$$

Specify clearly the path you choose. Note that this force field is nonconservative.

Solution.

Choose the straight path $x=y=t$, $1\le t\le 3$. On this path, $\mathbf{F}=(0,2t)$ and $d\mathbf{r}=(1,1)dt$, giving

$$\boxed{W=\int_1^3 2t\,dt=[t^2]_1^3=8}.$$

To exhibit path dependence, instead go horizontally from $(1,1)$ to $(3,1)$, then vertically to $(3,3)$. On the first leg $y=1$, $dy=0$, and $\mathbf{F}\cdot d\mathbf{r} = (x-1)\,dx$; on the second $x=3$, $dx=0$, and $\mathbf{F}\cdot d\mathbf{r} = (3+y)\,dy$. Hence

$$W = \int_1^3 (x-1)\,dx + \int_1^3 (3+y)\,dy = \left[\tfrac{1}{2}(x-1)^2\right]_1^3 + \left[\tfrac{1}{2}(3+y)^2\right]_1^3 = 2+10 = 12.$$

The curl is $\hat{\mathbf{e}}_z[\partial_x(x+y) - \partial_y(x-y)] = 2\hat{\mathbf{e}}_z$, consistent with nonconservative behavior. Indeed, going out along the second path and back along the first is a counterclockwise circuit of the triangle with vertices $(1,1)$, $(3,1)$, $(3,3)$, of area 2; by Stokes' theorem (Section 3.8) its circulation is $2\cdot 2 = 4 = 12-8$.

3.7.4 Evaluate $\oint \mathbf{r}\cdot d\mathbf{r}$ for a closed path of your choosing.

Solution.

Since $d(\mathbf{r}\cdot\mathbf{r}) = 2\mathbf{r}\cdot d\mathbf{r}$,

$$\oint_C \mathbf{r}\cdot d\mathbf{r} = \frac{1}{2}\oint_C d(r^2) = \frac{1}{2}(r^2_{\text{final}} - r^2_{\text{initial}}) = \boxed{0}.$$

This holds for every piecewise smooth closed path. For a concrete choice, a circle $\mathbf{r} = (R\cos t, R\sin t, 0)$ has $\mathbf{r}\cdot d\mathbf{r} = R^2(-\cos t\sin t + \sin t\cos t)dt = 0$ pointwise.

3.7.5 Evaluate

$$\frac{1}{3}\int_s \mathbf{r}\cdot d\boldsymbol{\sigma}$$

over the unit cube defined by the point $(0,0,0)$ and the unit intercepts on the positive x-, y-, and z-axes. Note that $\mathbf{r}\cdot d\boldsymbol{\sigma}$ is zero for three of the surfaces and that each of the three remaining surfaces contributes the same amount to the integral.

Solution.

Use outward normals. On $x=0, y=0, z=0$, respectively, $\mathbf{r}\cdot\hat{\mathbf{n}} = -x, -y, -z = 0$. On each of $x=1, y=1, z=1$, it is 1, and the face area is 1. Thus

$$\boxed{\frac{1}{3}\oint_S \mathbf{r}\cdot d\boldsymbol{\sigma} = \frac{1}{3}(0+0+0+1+1+1) = 1}.$$

As a volume check (Gauss' theorem, Exercise 3.8.2), $\frac{1}{3}\int_V \nabla\cdot\mathbf{r}\,d\tau = \frac{1}{3}\cdot 3\cdot 1 = 1$, since $\nabla\cdot\mathbf{r} = 3$ and the cube has volume 1.

3.8 Integral Theorems

3.8.1 Using Gauss' theorem, prove that

$$\oint_S d\boldsymbol{\sigma} = 0$$

if $S = \partial V$ is a closed surface.

Solution.

For each constant Cartesian unit vector $\hat{\mathbf{e}}_i$, Gauss' theorem gives

$$\hat{\mathbf{e}}_i \cdot \oint_S d\boldsymbol{\sigma} = \oint_S \hat{\mathbf{e}}_i \cdot d\boldsymbol{\sigma} = \int_V \nabla \cdot \hat{\mathbf{e}}_i \, d\tau = 0.$$

All three components vanish, so $\boxed{\oint_S d\boldsymbol{\sigma} = 0}$. The surface is oriented outward.

3.8.2 Show that

$$\frac{1}{3} \oint_S \mathbf{r} \cdot d\boldsymbol{\sigma} = V,$$

where V is the volume enclosed by the closed surface $S = \partial V$.

Note. This is a generalization of Exercise 3.7.5.

Solution.

Apply Gauss' theorem to $\mathbf{r} = (x, y, z)$:

$$\frac{1}{3} \oint_S \mathbf{r} \cdot d\boldsymbol{\sigma} = \frac{1}{3} \int_V (\partial_x x + \partial_y y + \partial_z z) \, d\tau = \frac{1}{3} \int_V 3 \, d\tau = \boxed{V}.$$

The result is independent of the position of the origin: adding a constant vector $\mathbf{a}$ to $\mathbf{r}$ adds $\frac{1}{3}\mathbf{a} \cdot \oint_S d\boldsymbol{\sigma} = 0$ by Exercise 3.8.1.

3.8.3 If $\mathbf{B} = \nabla \times \mathbf{A}$, show that

$$\oint_S \mathbf{B} \cdot d\boldsymbol{\sigma} = 0$$

for any closed surface S. Assume S bounds a region on a neighborhood of which $\mathbf{A}$ is a single-valued C^2 field.

Solution.

Assume $\mathbf{A}$ is a single-valued C^2 field on a neighborhood of the enclosed region; singular potentials require separate treatment. For example, $\mathbf{B} = \hat{\mathbf{r}}/r^2$ is the curl of the potential of Exercise 3.9.5, which is singular on the z-axis, and its flux through any sphere centered at the origin is 4π. The divergence of its curl is $\sum_{ijk} \varepsilon_{ijk} \partial_i \partial_j A_k = 0$, since mixed derivatives are symmetric in i, j while ε_{ijk} is antisymmetric. Therefore Gauss' theorem, applied to the region V enclosed by S, gives

$$\boxed{\oint_S \mathbf{B} \cdot d\boldsymbol{\sigma} = \int_V \nabla \cdot (\nabla \times \mathbf{A}) \, d\tau = 0}.$$

3.8.4 From Eq. (3.72), with $\mathbf{V}$ the electric field $\mathbf{E}$ and f the electrostatic potential φ, show that, for integration over all space,

$$\int \rho\varphi \, d\tau = \epsilon_0 \int E^2 \, d\tau.$$

This corresponds to a 3-D integration by parts.

Hint. $\mathbf{E} = -\nabla\varphi$, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$. You may assume that φ vanishes at large r at least as fast as r^{-1}. Assume both volume integrals converge and $\lim_{R\to\infty} \oint_{r=R} \varphi\mathbf{E} \cdot d\boldsymbol{\sigma} = 0$. For a regular localized charge distribution, $\varphi = O(R^{-1})$ and $\mathbf{E} = O(R^{-2})$ suffice for this boundary condition.

Solution.

The product rule and the electrostatic equations $\nabla\varphi = -\mathbf{E}$, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ give

$$\nabla \cdot (\varphi\mathbf{E}) = \nabla\varphi \cdot \mathbf{E} + \varphi\nabla \cdot \mathbf{E} = -E^2 + \frac{\rho\varphi}{\epsilon_0}.$$

Integrate over a ball of radius R, apply Gauss' theorem to the left side, and multiply by ϵ_0:

$$\int_{r<R} \rho\varphi \, d\tau = \epsilon_0 \int_{r<R} E^2 \, d\tau + \epsilon_0 \oint_{r=R} \varphi\mathbf{E} \cdot d\boldsymbol{\sigma}.$$

The hint alone does not suffice: $\varphi = \sin(r^3)/r$ is $O(r^{-1})$, yet $E_r = -\varphi' = -3r\cos r^3 + r^{-2}\sin r^3$ oscillates with an amplitude growing like r, so the surface term $4\pi R^2 \varphi E_r = -6\pi R^2 \sin(2R^3) + 4\pi R^{-1}\sin^2 R^3$ has no limit and $\int E^2 \, d\tau$ diverges. For ideal point charges both volume integrals diverge. Under the conditions $\varphi = O(R^{-1})$, $\mathbf{E} = O(R^{-2})$ of the statement, the surface term is $O(R^{-1})$:

$$\left| \oint_{r=R} \varphi\mathbf{E} \cdot d\boldsymbol{\sigma} \right| \le 4\pi R^2 \max_{r=R} |\varphi| \, |\mathbf{E}| = 4\pi R^2 \, O(R^{-1}) \, O(R^{-2}) = O(R^{-1}).$$

Taking $R \to \infty$ gives $\boxed{\int \rho\varphi \, d\tau = \epsilon_0 \int E^2 \, d\tau}$.

3.8.5 A particular steady-state electric current distribution is localized in space. Choosing a bounding surface far enough out so that the current density $\mathbf{J}$ is zero everywhere on the surface, show that

$$\int \mathbf{J} \, d\tau = 0.$$

Hint. Take one component of $\mathbf{J}$ at a time. With $\nabla \cdot \mathbf{J} = 0$, show that $J_i = \nabla \cdot (x_i\mathbf{J})$ and apply Gauss' theorem.

Solution.

Steady current obeys $\nabla \cdot \mathbf{J} = 0$. For each component,

$$\nabla \cdot (x_i\mathbf{J}) = \sum_j [(\partial_j x_i) J_j + x_i \partial_j J_j] = J_i + x_i \nabla \cdot \mathbf{J} = J_i,$$

because $\partial_j x_i = \delta_{ij}$. Hence, by Gauss' theorem, $\int_V J_i \, d\tau = \oint_{\partial V} x_i \mathbf{J} \cdot d\boldsymbol{\sigma} = 0$ because $\mathbf{J} = 0$ on the chosen boundary. All three components are zero, giving $\boxed{\int_V \mathbf{J} \, d\tau = 0}$.

3.8.6 Given a vector $\mathbf{t} = -\hat{\mathbf{e}}_x y + \hat{\mathbf{e}}_y x$, show, with the help of Stokes' theorem, that the integral of $\mathbf{t}$ around a simple, piecewise smooth, counterclockwise oriented closed curve in the xy-plane satisfies

$$\frac{1}{2}\oint \mathbf{t}\cdot d\boldsymbol{\lambda} = \frac{1}{2}\oint (x\,dy - y\,dx) = A,$$

where A is the area enclosed by the curve.

Solution.

Take a simple, piecewise smooth loop oriented counterclockwise as viewed from $+z$. Reversing orientation changes the sign; a self-intersecting loop gives signed area with multiplicity. For $\mathbf{t} = (-y, x, 0)$,

$$\nabla\times\mathbf{t} = \big(\partial_y 0 - \partial_z x,\ \partial_z(-y) - \partial_x 0,\ \partial_x x - \partial_y(-y)\big) = (0,0,2) = 2\hat{\mathbf{e}}_z.$$

Stokes' theorem, applied to the plane region S enclosed by C with upward surface normal ($d\boldsymbol{\sigma} = \hat{\mathbf{e}}_z\,dA$), gives

$$\frac{1}{2}\oint_C \mathbf{t}\cdot d\boldsymbol{\lambda} = \frac{1}{2}\int_S 2\hat{\mathbf{e}}_z\cdot\hat{\mathbf{e}}_z\,dA = \boxed{A}.$$

Since $d\boldsymbol{\lambda} = (dx, dy, 0)$ along the curve, $\mathbf{t}\cdot d\boldsymbol{\lambda} = -y\,dx + x\,dy$, and this also proves the second displayed expression.

3.8.7 The calculation of the magnetic moment of a current loop leads to the line integral

$$\oint \mathbf{r}\times d\mathbf{r}.$$

(a) Integrate around the perimeter of a current loop (in the xy-plane) and show that the scalar magnitude of this line integral is twice the area of the enclosed surface.

(b) The perimeter of an ellipse is described by $\mathbf{r} = \hat{\mathbf{e}}_x a\cos\theta + \hat{\mathbf{e}}_y b\sin\theta$. From part (a) show that the area of the ellipse is πab.

In (a), take a simple, piecewise smooth loop; in (b), take $a, b > 0$.

Solution.

(a) In the xy-plane, $\mathbf{r} = (x, y, 0)$ and $d\mathbf{r} = (dx, dy, 0)$, so $\mathbf{r}\times d\mathbf{r} = (y\cdot 0 - 0\cdot dy,\ 0\cdot dx - x\cdot 0,\ x\,dy - y\,dx) = (x\,dy - y\,dx)\hat{\mathbf{e}}_z$. For a simple positively oriented loop, Exercise 3.8.6 gives

$$\oint_C \mathbf{r}\times d\mathbf{r} = 2A\hat{\mathbf{e}}_z,$$

so the magnitude is $2A$. Reverse traversal reverses the vector but not its magnitude. **(b)** For the ellipse with $a, b > 0$, $dx = -a\sin\theta\,d\theta$ and $dy = b\cos\theta\,d\theta$. Then

$$x\,dy - y\,dx = (a\cos\theta)(b\cos\theta\,d\theta) - (b\sin\theta)(-a\sin\theta\,d\theta) = ab(\cos^2\theta + \sin^2\theta)d\theta = ab\,d\theta.$$

As θ increases from 0 to 2π the ellipse is traversed once counterclockwise, so by (a) $2A = \oint(x\,dy - y\,dx) = \int_0^{2\pi} ab\,d\theta = 2\pi ab$, hence $\boxed{A = \pi ab}$. The qualifications are needed: a self-intersecting loop in (a) gives signed areas counted with multiplicity (Exercise 3.8.6), and for $ab < 0$ in (b) the same parametrization runs clockwise and gives $\oint(x\,dy - y\,dx) = 2\pi ab < 0$, while the area is $\pi|ab|$.

3.8.8 Evaluate $\oint \mathbf{r} \times d\mathbf{r}$ by using the alternate form of Stokes' theorem given by Eq. (3.95):

$$\int_S (d\boldsymbol{\sigma} \times \nabla) \times \mathbf{P} = \oint d\boldsymbol{\lambda} \times \mathbf{P}.$$

Take the loop to be entirely in the xy-plane.

Solution.

Take $\mathbf{P} = \mathbf{r}$ in the alternate Stokes formula. Treat the normal area element as the coefficient of the differential operator. The component contraction is that of Exercise 3.6.6 with $\mathbf{A} \to d\boldsymbol{\sigma}$ (proved for general $\mathbf{P}$ in Exercise 3.8.13); with repeated indices summed, $\partial_i r_j = \delta_{ij}$, and $\partial_j r_j = 3$, it gives

$$[(d\boldsymbol{\sigma} \times \nabla) \times \mathbf{r}]_i = d\sigma_j \, \partial_i r_j - d\sigma_i \, \partial_j r_j = d\sigma_i - 3 d\sigma_i = -2 d\sigma_i.$$

Thus $\oint d\mathbf{r} \times \mathbf{r} = -2 \int_S d\boldsymbol{\sigma}$. For the plane region S enclosed by the loop, with upward normal (counterclockwise traversal), $\int_S d\boldsymbol{\sigma} = \hat{\mathbf{e}}_z \int_S dA = A \hat{\mathbf{e}}_z$, where A is the enclosed area. Reversing the order of the cross product, $\mathbf{r} \times d\mathbf{r} = -d\mathbf{r} \times \mathbf{r}$, gives

$$\boxed{\oint \mathbf{r} \times d\mathbf{r} = 2A \hat{\mathbf{e}}_z}.$$

The sign is reversed for clockwise traversal.

3.8.9 Prove that

$$\oint u \nabla v \cdot d\boldsymbol{\lambda} = -\oint v \nabla u \cdot d\boldsymbol{\lambda}.$$

Assume u, v are single-valued C^1 functions along the closed, piecewise smooth path.

Solution.

By the product rule (Exercise 3.5.5), $u\nabla v + v\nabla u = \nabla(uv)$, and $\nabla(uv) \cdot d\boldsymbol{\lambda} = d(uv)$ is the change of uv along the line element. Its closed line integral is

$$\oint_C (u\nabla v + v\nabla u) \cdot d\boldsymbol{\lambda} = \oint_C d(uv) = 0$$

for single-valued differentiable u, v, since uv returns to its starting value. Move the second integral to the right to obtain the stated integration-by-parts identity.

3.8.10 Prove that

$$\oint u \nabla v \cdot d\boldsymbol{\lambda} = \int_S (\nabla u) \times (\nabla v) \cdot d\boldsymbol{\sigma}.$$

Here the line integral is over the positively oriented boundary ∂S of an oriented, piecewise smooth surface, and u, v are C^2 on a neighborhood of S.

Solution.

Apply Stokes' theorem to $u\nabla v$:

$$\oint_{\partial S} u \nabla v \cdot d\boldsymbol{\lambda} = \int_S \nabla \times (u \nabla v) \cdot d\boldsymbol{\sigma}.$$

The product rule $\nabla \times (g\mathbf{V}) = \nabla g \times \mathbf{V} + g \nabla \times \mathbf{V}$ (as in Exercise 3.6.4), with $g = u$ and $\mathbf{V} = \nabla v$, gives $\nabla \times (u \nabla v) = \nabla u \times \nabla v + u \, \nabla \times \nabla v = \nabla u \times \nabla v$, because a curl of a gradient vanishes (Exercise 3.6.10). Substitution proves the result with the compatible boundary orientation. The fields must be regular on the spanning surface.

The first integral in the Instructor's Manual's displayed derivation on p. 45 must be $2 \oint_{\partial S} u \nabla v \cdot d\boldsymbol{\lambda}$, not $2 \int_S u \nabla v \cdot d\boldsymbol{\sigma}$. Stokes' theorem relates that line integral to the surface integral of its curl.

3.8.11 Prove that

$$\oint_{\partial V} d\boldsymbol{\sigma} \times \mathbf{P} = \int_V \nabla \times \mathbf{P}\, d\tau.$$

Solution.

Take a constant test vector $\mathbf{a}$. Scalar triple-product cyclicity, $\mathbf{a} \cdot (d\boldsymbol{\sigma} \times \mathbf{P}) = d\boldsymbol{\sigma} \cdot (\mathbf{P} \times \mathbf{a})$, and Gauss' theorem give

$$\begin{aligned} \mathbf{a} \cdot \oint_{\partial V} d\boldsymbol{\sigma} \times \mathbf{P} &= \oint_{\partial V} (\mathbf{P} \times \mathbf{a}) \cdot d\boldsymbol{\sigma} \\ &= \int_V \nabla \cdot (\mathbf{P} \times \mathbf{a})\, d\tau \\ &= \int_V \mathbf{a} \cdot (\nabla \times \mathbf{P})\, d\tau. \end{aligned}$$

The last step is Exercise 3.5.9, $\nabla \cdot (\mathbf{P} \times \mathbf{a}) = \mathbf{a} \cdot (\nabla \times \mathbf{P}) - \mathbf{P} \cdot (\nabla \times \mathbf{a})$, with $\nabla \times \mathbf{a} = 0$. Taking $\mathbf{a}$ to be each Cartesian unit vector proves the vector identity.

3.8.12 Prove that

$$\int_S d\boldsymbol{\sigma} \times \nabla\varphi = \oint_{\partial S} \varphi\, d\mathbf{r}.$$

Solution.

For any constant $\mathbf{a}$, $\mathbf{a} \cdot \oint \varphi\, d\mathbf{r} = \oint (\varphi \mathbf{a}) \cdot d\mathbf{r}$, and Stokes' theorem applied to $\varphi\mathbf{a}$ gives

$$\begin{aligned} \mathbf{a} \cdot \oint_{\partial S} \varphi\, d\mathbf{r} &= \int_S [\nabla \times (\varphi \mathbf{a})] \cdot d\boldsymbol{\sigma} \\ &= \int_S (\nabla\varphi \times \mathbf{a}) \cdot d\boldsymbol{\sigma} \\ &= \mathbf{a} \cdot \int_S d\boldsymbol{\sigma} \times \nabla\varphi. \end{aligned}$$

The second equality is the product rule $\nabla \times (\varphi\mathbf{a}) = \nabla\varphi \times \mathbf{a} + \varphi \nabla \times \mathbf{a}$ with $\nabla \times \mathbf{a} = 0$; the last is the cyclic scalar triple product $(\nabla\varphi \times \mathbf{a}) \cdot d\boldsymbol{\sigma} = \mathbf{a} \cdot (d\boldsymbol{\sigma} \times \nabla\varphi)$. Equality for all $\mathbf{a}$ proves the claimed vector equation.

3.8.13 Prove that

$$\int_S (d\boldsymbol{\sigma} \times \nabla) \times \mathbf{P} = \oint_{\partial S} d\mathbf{r} \times \mathbf{P}.$$

Solution.

Let $\mathbf{a}$ be constant. By the cyclic triple product $\mathbf{a} \cdot (d\mathbf{r} \times \mathbf{P}) = (\mathbf{P} \times \mathbf{a}) \cdot d\mathbf{r}$ and Stokes' theorem,

$$\mathbf{a} \cdot \oint_{\partial S} d\mathbf{r} \times \mathbf{P} = \oint_{\partial S} (\mathbf{P} \times \mathbf{a}) \cdot d\mathbf{r} = \int_S [\nabla \times (\mathbf{P} \times \mathbf{a})] \cdot d\boldsymbol{\sigma}.$$

The curl-product rule of Exercise 3.6.5, with $\mathbf{A} \to \mathbf{P}$, $\mathbf{B} \to \mathbf{a}$, and all derivatives of $\mathbf{a}$ zero, gives $\nabla \times (\mathbf{P} \times \mathbf{a}) = (\mathbf{a} \cdot \nabla)\mathbf{P} - \mathbf{a}(\nabla \cdot \mathbf{P})$. With repeated indices summed, the integrand is therefore $[(\mathbf{a} \cdot \nabla)\mathbf{P} - \mathbf{a}(\nabla \cdot \mathbf{P})] \cdot d\boldsymbol{\sigma} = a_i [d\sigma_j \partial_i P_j - d\sigma_i \partial_j P_j]$. On the other hand, the Levi-Civita contraction of Exercise 3.6.6, with $\mathbf{A} \to d\boldsymbol{\sigma}$ (not differentiated) and $\mathbf{B} \to \mathbf{P}$, gives

$$[(d\boldsymbol{\sigma} \times \nabla) \times \mathbf{P}]_i = \varepsilon_{ijk} \varepsilon_{j\ell m}\, d\sigma_\ell\, \partial_m P_k = d\sigma_j \partial_i P_j - d\sigma_i \partial_j P_j,$$

so $\mathbf{a} \cdot \int_S (d\boldsymbol{\sigma} \times \nabla) \times \mathbf{P} = \int_S a_i [d\sigma_j \partial_i P_j - d\sigma_i \partial_j P_j]$ as well. Both sides have the same dot product with every constant $\mathbf{a}$, so the vector integrals agree.

3.9 Potential Theory

3.9.1 If a force $\mathbf{F}$ is given by

$$\mathbf{F} = \left(x^2 + y^2 + z^2\right)^n \left(\hat{\mathbf{e}}_x x + \hat{\mathbf{e}}_y y + \hat{\mathbf{e}}_z z\right),$$

find

(a) $\nabla \cdot \mathbf{F}$.

(b) $\nabla \times \mathbf{F}$.

(c) A scalar potential $\varphi(x, y, z)$ so that $\mathbf{F} = -\nabla\varphi$.

(d) For what value of the exponent n does the scalar potential diverge at both the origin and infinity?

Take $n \in \mathbb{R}$ and initially work at $r > 0$; any extension across the origin requires a separate regularity check.

ANS. (a) $(2n+3)r^{2n}$ (b) 0
(c) $-r^{2n+2}/(2n+2),\ n \neq -1$ (d) $n = -1,\ \varphi = -\ln r$.

Solution.

Write $\mathbf{F} = r^{2n}\mathbf{r}$, with $r > 0$ and $r^2 = x^2 + y^2 + z^2 = \sum_i x_i^2$. **(a)** Since $\partial_i r = x_i/r$, the chain rule gives $\partial_i(r^{2n}) = 2nr^{2n-1}x_i/r = 2nr^{2n-2}x_i$, so

$$\nabla \cdot \mathbf{F} = \sum_i \partial_i(r^{2n}x_i) = \sum_i [2nr^{2n-2}x_i^2 + r^{2n}] = 2nr^{2n} + 3r^{2n} = \boxed{(2n+3)r^{2n}}.$$

(b) By the product rule of Exercise 3.6.4, with $\nabla(r^{2n}) = 2nr^{2n-2}\mathbf{r}$ from (a) and $\nabla \times \mathbf{r} = 0$ (Exercise 3.6.2), $\nabla \times \mathbf{F} = \nabla(r^{2n}) \times \mathbf{r} + r^{2n}\nabla \times \mathbf{r} = 2nr^{2n-2}\mathbf{r} \times \mathbf{r} = \boxed{0}$. **(c)** Since $\mathbf{F}$ is radial, try $\varphi = \varphi(r)$. Then $\nabla\varphi = \varphi'(r)\nabla r = \varphi'(r)\mathbf{r}/r$, and $\mathbf{F} = -\nabla\varphi$ requires $r^{2n}\mathbf{r} = -\varphi'(r)\mathbf{r}/r$, that is, $-\varphi'(r) = r^{2n+1}$. Integration gives

$$\boxed{\varphi(r) = \begin{cases} -r^{2n+2}/(2n+2) + C, & n \neq -1, \\ -\ln(r/r_0) + C, & n = -1, \end{cases}}$$

where $r_0 > 0$ fixes the additive reference. With $C = 0$ (and $r_0 = 1$ in the logarithm) these are the textbook's answers. **(d)** For $n \neq -1$, $\varphi - C$ is a constant multiple of r^{2n+2}. If $2n + 2 > 0$ it tends to 0 as $r \to 0$ and diverges as $r \to \infty$; if $2n + 2 < 0$ the reverse holds. For $n = -1$, $-\ln(r/r_0) \to +\infty$ as $r \to 0$ and $-\ln(r/r_0) \to -\infty$ as $r \to \infty$. Hence only the logarithmic case $\boxed{n = -1}$ diverges in magnitude at both ends. These differential formulas hold away from singularities. For instance, at $n = -3/2$ the divergence is zero for $r > 0$, but its distributional extension is $4\pi\delta^{(3)}(\mathbf{r})$, as the flux through every sphere is 4π.

3.9.2 A sphere of radius a is uniformly charged (throughout its volume). Construct the electrostatic potential $\varphi(r)$ for $0 \leq r < \infty$.

Solution.

Let the total charge be Q, so the uniform density is $\rho_0 = 3Q/(4\pi a^3)$. Choose $\varphi(\infty) = 0$. By spherical symmetry $\mathbf{E} = E_r(r)\hat{\mathbf{r}}$, and Gauss' law for the sphere of radius r gives

$$4\pi r^2 E_r(r) = \frac{Q_{\text{enclosed}}(r)}{\epsilon_0}, \qquad Q_{\text{enclosed}}(r) = \begin{cases} \rho_0 \cdot \frac{4}{3}\pi r^3 = Qr^3/a^3, & r \leq a, \\ Q, & r \geq a. \end{cases}$$

Thus $E_r = Qr/(4\pi\epsilon_0 a^3)$ inside and $E_r = Q/(4\pi\epsilon_0 r^2)$ outside. Since $E_r = -d\varphi/dr$ and $\varphi(\infty) = 0$, integrating inward from infinity gives

$$\varphi(r) = \int_r^\infty E_r(s)\,ds = \Big[-\frac{Q}{4\pi\epsilon_0 s}\Big]_r^\infty = \frac{Q}{4\pi\epsilon_0 r} \quad (r \ge a).$$

Inside, using $\int_r^a s\,ds = (a^2 - r^2)/2$,

$$\begin{aligned}\varphi(r) &= \varphi(a) + \int_r^a \frac{Qs}{4\pi\epsilon_0 a^3}\,ds \\ &= \frac{Q}{4\pi\epsilon_0 a} + \frac{Q(a^2 - r^2)}{8\pi\epsilon_0 a^3} = \frac{Q}{8\pi\epsilon_0 a}\Big(2 + 1 - \frac{r^2}{a^2}\Big).\end{aligned}$$

Consequently

$$\boxed{\varphi(r) = \begin{cases} \dfrac{Q}{8\pi\epsilon_0 a}(3 - r^2/a^2), & 0 \le r \le a, \\ \dfrac{Q}{4\pi\epsilon_0 r}, & r \ge a. \end{cases}}$$

At $r = a$ the inner and outer formulas agree: both give $\varphi(a) = Q/(4\pi\epsilon_0 a)$ and $E_r(a) = -\varphi'(a) = Q/(4\pi\epsilon_0 a^2)$. Also $\varphi(0) = 3Q/(8\pi\epsilon_0 a)$ is finite.

3.9.3 The origin of the Cartesian coordinates is at the Earth's center. The moon is on the z-axis, a fixed distance R away (center-to-center distance). The tidal force exerted by the moon on a particle at the Earth's surface (point x, y, z) is given by

$$F_x = -GMm\frac{x}{R^3}, \quad F_y = -GMm\frac{y}{R^3}, \quad F_z = +2GMm\frac{z}{R^3}.$$

Find the potential that yields this tidal force.

$$\textit{ANS.}\ -\frac{GMm}{R^3}\left(z^2 - \frac{1}{2}x^2 - \frac{1}{2}y^2\right).$$

Solution.

Set $K = GMm/R^3$. From $\mathbf{F} = -\nabla\varphi$, $\partial_x\varphi = Kx$, so $\varphi = Kx^2/2 + f(y, z)$. The y equation gives $\partial_y f = Ky$, hence $f = Ky^2/2 + g(z)$. The z equation gives $g' = -2Kz$, so $g = -Kz^2 + C$. Thus

$$\boxed{\varphi = \frac{GMm}{R^3}\left(\frac{x^2 + y^2}{2} - z^2\right) + C}.$$

Its negative gradient is $(-Kx, -Ky, 2Kz)$, exactly the given force. With $C = 0$ this is the textbook's answer $-K(z^2 - \frac{1}{2}x^2 - \frac{1}{2}y^2)$.

The Instructor's Manual, p. 46, gives the opposite overall sign for the potential; applying $-\nabla$ to its expression reverses every required force component. Its intermediate expression also includes an extra factor r^2 before $(3z^2 - r^2)$. The consistent chain, with $r^2 = x^2 + y^2 + z^2$ and $P_2(\cos\theta) = \frac{1}{2}(3\cos^2\theta - 1) = (3z^2 - r^2)/(2r^2)$, is $V = -GmM(3z^2 - r^2)/(2R^3) = -GmMr^2P_2(\cos\theta)/R^3$, as in the textbook answer.

3.9.4 A long, straight wire carrying a current I produces a magnetic induction $\mathbf{B}$ with components

$$\mathbf{B} = \frac{\mu_0 I}{2\pi}\left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0\right).$$

Find a magnetic vector potential $\mathbf{A}$. Work off the ideal wire, where $x^2+y^2>0$.

ANS. $\mathbf{A} = -\hat{\mathbf{e}}_z(\mu_0 I/4\pi)\ln(x^2+y^2)$. (This solution is not unique.)

Solution.

Choose $\mathbf{A} = A_z(x,y)\hat{\mathbf{e}}_z$. Then $\nabla\times\mathbf{A} = (\partial_y A_z, -\partial_x A_z, 0)$. With $K = \mu_0 I/(2\pi)$, the required y-component implies $\partial_x A_z = -Kx/(x^2+y^2)$. Integration in x, using $\partial_x \ln(x^2+y^2) = 2x/(x^2+y^2)$, gives $A_z = -(K/2)\ln(x^2+y^2) + f(y)$. The required x-component, $\partial_y A_z = -Ky/(x^2+y^2) + f'(y) = -Ky/(x^2+y^2)$, sets $f'(y) = 0$, so f is a constant. Equivalently, with $\rho = \sqrt{x^2+y^2} > 0$ and reference length ρ_0,

$$\boxed{\mathbf{A} = K\ln(\rho_0/\rho)\hat{\mathbf{e}}_z}.$$

Indeed $\partial_y A_z = -Ky/\rho^2$ and $-\partial_x A_z = Kx/\rho^2$. The choice $\rho_0 = 1$ (in the units used) gives $-K\ln\rho = -\frac{K}{2}\ln(x^2+y^2) = -(\mu_0 I/4\pi)\ln(x^2+y^2)$, the textbook's answer. Adding any regular gradient leaves the curl unchanged, so the potential is not unique. The ideal-wire axis is excluded.

3.9.5 If

$$\mathbf{B} = \frac{\hat{\mathbf{r}}}{r^2} = \left(\frac{x}{r^3}, \frac{y}{r^3}, \frac{z}{r^3}\right),$$

find a local vector potential, for example off the z-axis, $\mathbf{A}$ such that $\nabla\times\mathbf{A} = \mathbf{B}$.

ANS. One possible solution is $\mathbf{A} = \hat{\mathbf{e}}_x \dfrac{yz}{r(x^2+y^2)} - \hat{\mathbf{e}}_y \dfrac{xz}{r(x^2+y^2)}$.

Solution.

The displayed potential is valid off the z-axis. No globally smooth, single-valued potential exists on all of $\mathbb{R}^3\setminus\{0\}$ for this field. Write $\rho = \sqrt{x^2+y^2}$ and $r = \sqrt{\rho^2+z^2}$. Then $\mathbf{r} = \rho\hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$ and $\mathbf{B} = \mathbf{r}/r^3 = (\rho/r^3)\hat{\mathbf{e}}_\rho + (z/r^3)\hat{\mathbf{e}}_z$. Try $\mathbf{A} = A_\varphi(\rho,z)\hat{\mathbf{e}}_\varphi$. By the curl in cylindrical coordinates (Section 3.10), its curl has components $B_\rho = -\partial_z A_\varphi$, $B_\varphi = 0$, and $B_z = \rho^{-1}\partial_\rho(\rho A_\varphi)$. The last condition, $\partial_\rho(\rho A_\varphi) = z\rho/(\rho^2+z^2)^{3/2}$, integrates to $\rho A_\varphi = -z/r + g(z)$; take $g = 0$. This choice, $A_\varphi = -z/(\rho r)$, with $\partial_z(1/r) = -z/r^3$, $\partial_\rho(1/r) = -\rho/r^3$, and $r^2 - z^2 = \rho^2$, gives

$$-\partial_z A_\varphi = \frac{1}{\rho}\left(\frac{1}{r} - \frac{z^2}{r^3}\right) = \frac{\rho}{r^3}, \qquad \frac{1}{\rho}\partial_\rho(\rho A_\varphi) = \frac{1}{\rho}\frac{z\rho}{r^3} = \frac{z}{r^3}.$$

Since $\hat{\mathbf{e}}_\varphi = (-y/\rho, x/\rho, 0)$,

$$\boxed{\mathbf{A} = \left(\frac{yz}{r\rho^2}, -\frac{xz}{r\rho^2}, 0\right)}.$$

A direct Cartesian check does not need cylindrical coordinates. Since $A_z = 0$ and $\partial_z(z/r) = 1/r - z^2/r^3 = \rho^2/r^3$,

$$(\nabla\times\mathbf{A})_x = -\partial_z A_y = \frac{x}{\rho^2}\,\partial_z\frac{z}{r} = \frac{x}{r^3}, \qquad (\nabla\times\mathbf{A})_y = \partial_z A_x = \frac{y}{\rho^2}\,\partial_z\frac{z}{r} = \frac{y}{r^3}.$$

For the third component put $f = 1/(r\rho^2)$, so that $A_x = yzf$ and $A_y = -xzf$. As f depends on x, y only through ρ, $x\partial_x f + y\partial_y f = \rho\,\partial_\rho f = \rho\big(-\rho^{-1}r^{-3} - 2\rho^{-3}r^{-1}\big) = -1/r^3 - 2f$; hence

$$(\nabla\times\mathbf{A})_z = \partial_x A_y - \partial_y A_x = -z\big(2f + x\partial_x f + y\partial_y f\big) = \frac{z}{r^3}.$$

For the global obstruction, the flux through a sphere is $\int r^{-2} dS = 4\pi$. If a regular single-valued potential existed on that sphere, applying Stokes' theorem to its upper and lower hemispheres would express the flux as two opposite equatorial line integrals, whose sum is zero. The contradiction proves the stated global limitation.

3.9.6 Show that the pair of equations

$$\mathbf{A} = \frac{1}{2}(\mathbf{B} \times \mathbf{r}), \quad \mathbf{B} = \nabla \times \mathbf{A},$$

is satisfied by any constant magnetic induction $\mathbf{B}$.

Solution.

Use the curl-product rule of Exercise 3.6.5, with $\mathbf{A} \to \mathbf{B}$ and $\mathbf{B} \to \mathbf{r}$, for constant $\mathbf{B}$:

$$\nabla \times (\mathbf{B} \times \mathbf{r}) = (\mathbf{r} \cdot \nabla)\mathbf{B} - (\mathbf{B} \cdot \nabla)\mathbf{r} - \mathbf{r}(\nabla \cdot \mathbf{B}) + \mathbf{B}(\nabla \cdot \mathbf{r}).$$

The four terms are $0, -\mathbf{B}, 0, 3\mathbf{B}$: the derivatives of the constant $\mathbf{B}$ vanish, $(\mathbf{B} \cdot \nabla)(x, y, z) = (B_x \partial_x + B_y \partial_y + B_z \partial_z)(x, y, z) = \mathbf{B}$, and $\nabla \cdot \mathbf{r} = 3$. Thus the curl is $2\mathbf{B}$, and $\boxed{\nabla \times [\frac{1}{2}(\mathbf{B} \times \mathbf{r})] = \mathbf{B}}$.

3.9.7 Vector $\mathbf{B}$ is formed by the product of two gradients

$$\mathbf{B} = (\nabla u) \times (\nabla v),$$

where u and v are C^2 scalar functions.

(a) Show that $\mathbf{B}$ is solenoidal.

(b) Show that

$$\mathbf{A} = \frac{1}{2}(u \nabla v - v \nabla u)$$

is a vector potential for $\mathbf{B}$, in that

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Solution.

(a) By Exercise 3.5.9 with $\mathbf{a} = \nabla u$, $\mathbf{b} = \nabla v$, and with commuting mixed derivatives ($\nabla \times \nabla u = \nabla \times \nabla v = 0$, Exercise 3.6.10),

$$\nabla \cdot \mathbf{B} = \nabla v \cdot (\nabla \times \nabla u) - \nabla u \cdot (\nabla \times \nabla v) = 0.$$

This is Exercise 3.6.13. **(b)** For the proposed potential, the product rule $\nabla \times (g\mathbf{V}) = \nabla g \times \mathbf{V} + g \nabla \times \mathbf{V}$ gives $\nabla \times (u \nabla v) = \nabla u \times \nabla v + u \nabla \times \nabla v = \nabla u \times \nabla v$, and likewise $\nabla \times (v \nabla u) = \nabla v \times \nabla u$. Hence

$$\begin{aligned} 2\nabla \times \mathbf{A} &= \nabla \times (u \nabla v) - \nabla \times (v \nabla u) \\ &= (\nabla u \times \nabla v) - (\nabla v \times \nabla u) \\ &= 2 \nabla u \times \nabla v = 2\mathbf{B}, \end{aligned}$$

by antisymmetry of the cross product. Thus $\mathbf{A}$ is a vector potential for $\mathbf{B}$.

3.9.8 The magnetic induction $\mathbf{B}$ is related to the magnetic vector potential $\mathbf{A}$ by $\mathbf{B} = \nabla \times \mathbf{A}$. By Stokes' theorem

$$\int \mathbf{B} \cdot d\boldsymbol{\sigma} = \oint \mathbf{A} \cdot d\mathbf{r}.$$

Show that each side of this equation is invariant under the gauge transformation, $\mathbf{A} \to \mathbf{A} + \nabla\varphi$.

Note. Take the function φ to be single-valued.

Solution.

Let $\mathbf{A}' = \mathbf{A} + \nabla\varphi$, with regular single-valued φ on the relevant surface. Then $\mathbf{B}' = \nabla \times \mathbf{A}' = \mathbf{B} + \nabla \times \nabla\varphi = \mathbf{B}$, so the flux does not change. On the closed boundary, since $\nabla\varphi \cdot d\mathbf{r} = d\varphi$,

$$\oint \mathbf{A}' \cdot d\mathbf{r} - \oint \mathbf{A} \cdot d\mathbf{r} = \oint \nabla\varphi \cdot d\mathbf{r} = \varphi(\text{end}) - \varphi(\text{start}) = 0.$$

Single-valuedness makes the final equality valid: the end and start points coincide, and so do the values of φ there.

3.9.9 Show that the value of the electrostatic potential φ at any point P is equal to the average of the potential over any spherical surface centered on P, provided that there are no electric charges on or within the sphere.

Hint. Use Green's theorem, Eq. (3.85), with $u = r^{-1}$, where r is the distance from P, and $v = \varphi$. Equation (3.120) will also be useful.

Solution.

The hint is reworded to make clear that the distance from P is r, not $u = r^{-1}$. Place P at the origin and let the sphere have radius a. With no charge on or within the sphere, $\nabla^2\varphi = -\rho/\epsilon_0 = 0$: the potential is harmonic throughout the ball. We use Green's theorem in the form

$$\int_V (u\nabla^2 v - v\nabla^2 u)\, d\tau = \oint_{\partial V} (u\nabla v - v\nabla u) \cdot d\boldsymbol{\sigma}.$$

To avoid applying it through the singularity of $u = 1/r$ at P, take for V the shell $\epsilon < r < a$. Both $\nabla^2 u$ and $\nabla^2\varphi$ vanish there; for u, note that $\nabla(1/r) = -\mathbf{r}/r^3$ is $-\mathbf{F}$ of Exercise 3.9.1 with $n = -3/2$, whose divergence $(2n+3)r^{2n}$ is zero. With ∂_n the derivative along the outward normal of the shell,

$$0 = \oint_{\partial\text{shell}} \left(\frac{1}{r}\partial_n\varphi - \varphi\partial_n\frac{1}{r} \right) dS.$$

On the outer sphere, $\partial_n = \partial_r$ and $\partial_r(1/r) = -1/r^2$, so its contribution is $a^{-1}\oint_{r=a} \partial_r\varphi\, dS + a^{-2}\oint_{r=a} \varphi\, dS$. The first integral is zero by Gauss' theorem and harmonicity in the ball: $\oint_{r=a} \partial_r\varphi\, dS = \oint_{r=a} \nabla\varphi \cdot d\boldsymbol{\sigma} = \int_{r<a} \nabla^2\varphi\, d\tau = 0$. On the inner sphere the outward shell normal is $-\hat{\mathbf{e}}_r$, so $\partial_n = -\partial_r$, giving

$$-\epsilon^{-1}\oint_{r=\epsilon} \partial_r\varphi\, dS - \epsilon^{-2}\oint_{r=\epsilon} \varphi\, dS.$$

The first term tends to zero: its magnitude is at most $\epsilon^{-1} \cdot 4\pi\epsilon^2 \max|\nabla\varphi| = O(\epsilon)$. The second is -4π times the mean value of φ over the small sphere (of area $4\pi\epsilon^2$), which tends to $-4\pi\varphi(P)$ by continuity. Letting $\epsilon \to 0$ gives $a^{-2}\oint_{r=a} \varphi\, dS - 4\pi\varphi(P) = 0$, or

$$\boxed{\varphi(P) = \frac{1}{4\pi a^2}\oint_{r=a} \varphi\, dS}.$$

Using instead $\nabla^2(1/r) = -4\pi\delta(\mathbf{r})$ on the whole ball gives the same result at once: the left side of Green's theorem is then $4\pi\varphi(P)$. The small sphere does the work of the delta function.

3.9.10 Using Maxwell's equations, show that for a system (steady current) the magnetic vector potential $\mathbf{A}$ satisfies a vector Poisson equation,

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J},$$

provided we require $\nabla \cdot \mathbf{A} = 0$. Use magnetostatics in a medium with spatially constant scalar permeability μ.

Solution.

The permeability must be spatially constant: by the product rule of Exercise 3.6.4, $\nabla \times (\mu \mathbf{H}) = \nabla\mu \times \mathbf{H} + \mu \nabla \times \mathbf{H}$, so for nonuniform μ Ampère's law below would contain the extra term $\nabla\mu \times \mathbf{H}$. For steady currents $\partial \mathbf{D}/\partial t = 0$, so the Maxwell equation $\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D}/\partial t$ with $\mathbf{B} = \mu \mathbf{H}$ and constant μ gives Ampère's law $\nabla \times \mathbf{B} = \mu \nabla \times \mathbf{H} = \mu \mathbf{J}$. Insert $\mathbf{B} = \nabla \times \mathbf{A}$ and expand the double curl by Eq. (3.70) (Exercise 3.6.9):

$$\mu \mathbf{J} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

The stated gauge condition $\nabla \cdot \mathbf{A} = 0$ removes the first term, leaving $\boxed{\nabla^2 \mathbf{A} = -\mu \mathbf{J}}$. Taking the divergence gives $-\mu \nabla \cdot \mathbf{J} = \nabla \cdot \nabla^2 \mathbf{A} = \nabla^2 (\nabla \cdot \mathbf{A}) = 0$, consistent with steady-current conservation.

3.9.11 Derive, assuming the Lorentz gauge, Eq. (3.109):

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}.$$

Hint. Eq. (3.70) will be helpful.

Solution.

Use $\mathbf{B} = \nabla \times \mathbf{A}$, $\mathbf{E} = -\nabla\varphi - \partial_t \mathbf{A}$, and the Maxwell equation $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + c^{-2} \partial_t \mathbf{E}$ (with $\epsilon_0 \mu_0 = 1/c^2$). By Eq. (3.70), $\nabla \times \mathbf{B} = \nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$, while $c^{-2} \partial_t \mathbf{E} = -c^{-2} \nabla \partial_t \varphi - c^{-2} \partial_t^2 \mathbf{A}$. Substitution yields

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J} - \frac{1}{c^2} \nabla \partial_t \varphi - \frac{1}{c^2} \partial_t^2 \mathbf{A}.$$

The Lorentz gauge condition (also called the Lorenz gauge) is $\nabla \cdot \mathbf{A} + c^{-2} \partial_t \varphi = 0$. Its gradient, $\nabla(\nabla \cdot \mathbf{A}) = -c^{-2} \nabla \partial_t \varphi$, makes the two gradient terms identical on the two sides, so they cancel. Rearrangement gives

$$\boxed{\frac{1}{c^2} \partial_t^2 \mathbf{A} - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}}.$$

3.9.12 Prove that an arbitrary solenoidal vector $\mathbf{B}$ can be described as $\mathbf{B} = \nabla \times \mathbf{A}$, with

$$\mathbf{A} = -\hat{\mathbf{e}}_x \int_{y_0}^{y} B_z(x, y, z)\, dy - \hat{\mathbf{e}}_z \left[\int_{x_0}^{x} B_y(x, y_0, z)\, dx - \int_{y_0}^{y} B_x(x, y, z)\, dy \right].$$

Take $\mathbf{B}$ to be C^1 on a coordinate box containing the indicated integration paths; the formula is a local construction on more general regions.

Solution.

The construction is valid on a coordinate box, or on a region containing all the indicated integration paths with a regular solenoidal field. Solenoidality alone does not ensure a global potential on an arbitrary punctured domain; see Exercise 3.9.5. Use distinct dummy variables:

$$A_x = -\int_{y_0}^{y} B_z(x, \eta, z)\, d\eta, \quad A_y = 0, \quad A_z = -\int_{x_0}^{x} B_y(\xi, y_0, z)\, d\xi + \int_{y_0}^{y} B_x(x, \eta, z)\, d\eta.$$

Since $A_y = 0$, and the first term of A_z does not depend on y, the fundamental theorem of calculus immediately gives $(\nabla \times \mathbf{A})_x = \partial_y A_z - \partial_z A_y = \partial_y A_z = B_x(x, y, z)$ and $(\nabla \times \mathbf{A})_z = \partial_x A_y - \partial_y A_x = -\partial_y A_x = B_z(x, y, z)$. For the remaining component, differentiate both integrals: in x and z, which are not integration variables, under the integral sign ($\mathbf{B}$ is C^1), and the upper limit of the ξ integral by the fundamental theorem of calculus:

$$\begin{aligned}(\nabla \times \mathbf{A})_y &= \partial_z A_x - \partial_x A_z \\ &= -\int_{y_0}^{y} \partial_z B_z \, d\eta + B_y(x, y_0, z) - \int_{y_0}^{y} \partial_x B_x \, d\eta \\ &= B_y(x, y_0, z) + \int_{y_0}^{y} \partial_\eta B_y(x, \eta, z) \, d\eta \\ &= B_y(x, y_0, z) + B_y(x, y, z) - B_y(x, y_0, z) = B_y(x, y, z).\end{aligned}$$

The third line uses $\partial_x B_x + \partial_\eta B_y + \partial_z B_z = 0$. All three curl components are correct.

3.10 Curvilinear Coordinates

The coordinate differential formulas below are classical formulas on regular charts: $\rho > 0$ in cylindrical coordinates and $r > 0$, $0 < \theta < \pi$ in spherical coordinates. At a coordinate axis or the origin, an extension must be checked from the underlying Cartesian field.

3.10.1 The u-, v-, z-coordinate system frequently used in electrostatics and in hydrodynamics is defined by

$$xy = u, \quad x^2 - y^2 = v, \quad z = z.$$

This u-, v-, z-system is orthogonal.

(a) In words, describe briefly the nature of each of the three families of coordinate surfaces.

(b) Sketch the system in the xy-plane showing the intersections of surfaces of constant u and surfaces of constant v with the xy-plane.

(c) Indicate the directions of the unit vectors $\hat{\mathbf{e}}_u$ and $\hat{\mathbf{e}}_v$ in all four quadrants.

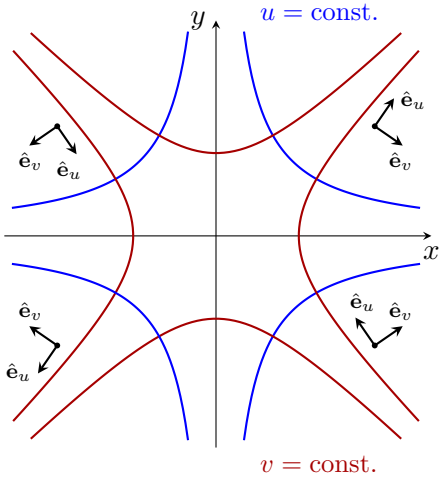
(d) Finally, is this u-, v-, z-system right-handed ($\hat{\mathbf{e}}_u \times \hat{\mathbf{e}}_v = +\hat{\mathbf{e}}_z$) or left-handed ($\hat{\mathbf{e}}_u \times \hat{\mathbf{e}}_v = -\hat{\mathbf{e}}_z$)?

Use a regular local branch with $x^2 + y^2 > 0$.

Solution.

(a) The surfaces $xy = u$ are rectangular hyperbolic cylinders parallel to z: for $u > 0$ their sections occupy quadrants I and III, and for $u < 0$ quadrants II and IV. Their asymptotes are the coordinate axes. The surfaces $x^2 - y^2 = v$ are another family of rectangular hyperbolic cylinders, with asymptotes $y = \pm x$; they open along x for $v > 0$ and along y for $v < 0$. Constant z gives horizontal planes. The zero levels are pairs of intersecting planes: $u = 0$ gives the planes $x = 0$ and $y = 0$, and $v = 0$ the planes $y = x$ and $y = -x$.

(b) Here blue curves have $u = \pm 1$ and red curves have $v = \pm 1.2$. Arrows show increasing coordinates:



(c) For $\rho = \sqrt{x^2+y^2} > 0$, $\nabla u = (y, x, 0)$ and $\nabla v = (2x, -2y, 0)$. They are orthogonal, since their dot product is $2xy - 2xy = 0$. For orthogonal coordinates the gradient formula of Section 3.10, $\nabla\psi = \sum_i \hat{\mathbf{e}}_i\, h_i^{-1}\, \partial\psi/\partial q_i$ with scale factors $h_i > 0$, applied to $\psi = u$ and $\psi = v$ gives $\nabla u = \hat{\mathbf{e}}_u/h_u$ and $\nabla v = \hat{\mathbf{e}}_v/h_v$. Hence the increasing-coordinate unit vectors are $\nabla u/|\nabla u|$ and $\nabla v/|\nabla v|$, with $|\nabla u| = \sqrt{y^2+x^2} = \rho$ and $|\nabla v| = 2\rho$:

$$\boxed{\hat{\mathbf{e}}_u = \frac{(y, x, 0)}{\rho}, \qquad \hat{\mathbf{e}}_v = \frac{(x, -y, 0)}{\rho}}.$$

Their (x, y) component signs in quadrants I, II, III, IV are respectively $(+,+), (+,-), (-,-), (-,+)$ for $\hat{\mathbf{e}}_u$ and $(+,-), (-,-), (-,+), (+,+)$ for $\hat{\mathbf{e}}_v$. **(d)** The cross product of the unit vectors is

$$\hat{\mathbf{e}}_u \times \hat{\mathbf{e}}_v = \rho^{-2}(y, x, 0) \times (x, -y, 0) = \rho^{-2}\big(0, 0, y(-y) - x \cdot x\big) = -\hat{\mathbf{e}}_z,$$

so the ordered system (u, v, z) is left-handed. The chart is singular on the z-axis and is only local unless a branch is chosen, because (x, y) and $(-x, -y)$ have the same (u, v).

One can also check handedness by the Jacobian. Differentiating the definitions gives

$$\begin{pmatrix} du \\ dv \end{pmatrix} = \begin{pmatrix} y & x \\ 2x & -2y \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}, \qquad \det \frac{\partial(u, v)}{\partial(x, y)} = -2y^2 - 2x^2 = -2(x^2 + y^2).$$

The rows of this matrix are ∇u and ∇v, and $\nabla u \times \nabla v = (0, 0, -2y^2 - 2x^2)$ is this determinant times $\hat{\mathbf{e}}_z$; so the negative determinant is the left-handedness found in (d). Inverting the 2×2 matrix with the rule

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}^{-1} = \frac{1}{\alpha\delta - \beta\gamma} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}$$

gives

$$\frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{-2\rho^2} \begin{pmatrix} -2y & -x \\ -2x & y \end{pmatrix} = \begin{pmatrix} y/\rho^2 & x/(2\rho^2) \\ x/\rho^2 & -y/(2\rho^2) \end{pmatrix},$$

whose determinant is the reciprocal of the one above,

$$\det \frac{\partial(x, y)}{\partial(u, v)} = -\frac{y^2}{2\rho^4} - \frac{x^2}{2\rho^4} = -\frac{1}{2\rho^2}.$$

The columns of the inverse matrix are the tangent vectors $\partial\mathbf{r}/\partial u = (y, x, 0)/\rho^2$ and $\partial\mathbf{r}/\partial v = (x, -y, 0)/(2\rho^2)$ of the coordinate lines; as a check on (c), they point along $\hat{\mathbf{e}}_u$ and $\hat{\mathbf{e}}_v$, with lengths

$h_u = 1/\rho$ and $h_v = 1/(2\rho)$. The Instructor's Manual, p. 49, omits the factor $1/2$ in this inverse determinant. Its earlier differential must likewise read $dv = 2x\,dx - 2y\,dy$, with a minus sign, since $\partial_y(x^2 - y^2) = -2y$.

3.10.2 The elliptic cylindrical coordinate system consists of three families of surfaces:

$$(1)\ \frac{x^2}{a^2\cosh^2 u} + \frac{y^2}{a^2\sinh^2 u} = 1; \quad (2)\ \frac{x^2}{a^2\cos^2 v} - \frac{y^2}{a^2\sin^2 v} = 1; \quad (3)\ z = z.$$

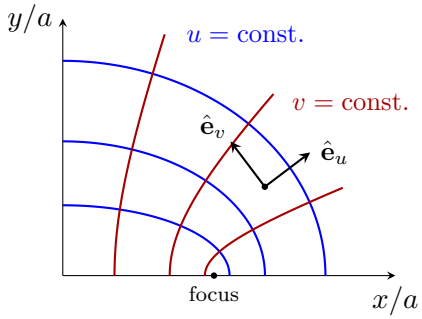
Sketch the coordinate surfaces $u =$ constant and $v =$ constant as they intersect the first quadrant of the xy-plane. Show the unit vectors $\hat{\mathbf{e}}_u$ and $\hat{\mathbf{e}}_v$. The range of u is $0 \le u < \infty$. The range of v is $0 \le v \le 2\pi$. Take $a > 0$. Interpret the zero-denominator limiting surfaces by the parametrization below; the unit basis is undefined at the foci, where $\sinh^2 u + \sin^2 v = 0$.

Solution.

Use the parametrization

$$x = a\cosh u\cos v, \qquad y = a\sinh u\sin v, \qquad z = z, \quad a > 0.$$

Dividing its squared coordinates by $a^2\cosh^2 u, a^2\sinh^2 u$ and adding gives 1; dividing by $a^2\cos^2 v, a^2\sin^2 v$ and subtracting also gives 1. Thus it generates both stated families. Constant $u > 0$ gives ellipses with semi-axes $a\cosh u > a\sinh u$, whose foci $(\pm c, 0)$ have $c^2 = a^2\cosh^2 u - a^2\sinh^2 u = a^2$; constant v (not a multiple of $\pi/2$) gives hyperbolas with semi-axes $a|\cos v|$ and $a|\sin v|$, whose foci $(\pm c, 0)$ have $c^2 = a^2\cos^2 v + a^2\sin^2 v = a^2$. Both families are therefore confocal, with foci $(\pm a, 0)$. In the first quadrant use $u > 0$, $0 < v < \pi/2$. The sketch shows the ellipses $u = 0.45, 0.8, 1.15$ (blue), the hyperbolas $v = 20°, 45°, 70°$ (red), and the unit vectors at the point $u = 1$, $v = 30°$:



Differentiate the position:

$$\frac{\partial \mathbf{r}}{\partial u} = a(\sinh u\cos v,\ \cosh u\sin v,\ 0), \qquad \frac{\partial \mathbf{r}}{\partial v} = a(-\cosh u\sin v,\ \sinh u\cos v,\ 0).$$

Both have squared length

$$\begin{aligned} a^2(\sinh^2 u\cos^2 v + \cosh^2 u\sin^2 v) &= a^2[\sinh^2 u(1 - \sin^2 v) + (1 + \sinh^2 u)\sin^2 v] \\ &= a^2(\sinh^2 u + \sin^2 v), \end{aligned}$$

and their dot product $a^2(-\sinh u\cosh u\sin v\cos v + \cosh u\sinh u\sin v\cos v)$ is zero. Normalizing,

$$\boxed{\begin{aligned} \hat{\mathbf{e}}_u &= \frac{(\sinh u\cos v,\ \cosh u\sin v,\ 0)}{\sqrt{\sinh^2 u + \sin^2 v}}, \\ \hat{\mathbf{e}}_v &= \frac{(-\cosh u\sin v,\ \sinh u\cos v,\ 0)}{\sqrt{\sinh^2 u + \sin^2 v}}. \end{aligned}}$$

The common scale factor is $h_u = h_v = a\sqrt{\sinh^2 u + \sin^2 v}$. The first vector points outward and northeast, the second along an ellipse toward increasing v (northwest) in this quadrant, since for $u > 0$, $0 < v < \pi/2$ both $\sinh u \cos v$ and $\cosh u \sin v$ are positive. At limiting coordinate surfaces where a displayed denominator of the implicit equations vanishes, use the parametrization and its limit: in the xy-plane, $u = 0$ gives the segment $|x| \le a$ of the x-axis, $v = 0$ and $v = \pi$ give the rays $x \ge a$ and $x \le -a$ of the x-axis, and $v = \pi/2$ and $v = 3\pi/2$ give the positive and negative y-axis.

3.10.3 Develop arguments to show that dot and cross products (not involving ∇) in orthogonal curvilinear coordinates in $\mathbb{R}^3$ proceed, as in Cartesian coordinates, **with no involvement of scale factors**. Use physical components along orthonormal unit vectors; the usual cross-product signs require a right-handed ordering.

Solution.

At a fixed regular point, expand in the local orthonormal basis: $\mathbf{A} = \sum_i A_i \hat{\mathbf{e}}_i$ and $\mathbf{B} = \sum_j B_j \hat{\mathbf{e}}_j$. Both products are bilinear, and the unit vectors of an orthogonal system satisfy $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \delta_{ij}$, so

$$\mathbf{A} \cdot \mathbf{B} = \sum_{ij} A_i B_j \, \hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \sum_{ij} A_i B_j \delta_{ij} = \sum_i A_i B_i.$$

For a right-handed basis, $\hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_2 = \hat{\mathbf{e}}_3$ and its cyclic permutations hold, the reversed products change sign, and $\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_i = 0$; together, $\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j = \sum_k \varepsilon_{ijk} \hat{\mathbf{e}}_k$, so

$$\mathbf{A} \times \mathbf{B} = \sum_{ij} A_i B_j \, \hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j = \sum_{ijk} \varepsilon_{ijk} A_i B_j \hat{\mathbf{e}}_k.$$

These are exactly the Cartesian component rules. Only the values of the unit vectors at the one point enter, never their derivatives; the scale factors appear only through $ds_i = h_i \, dq_i$ when ∇ differentiates along the coordinate lines, which these products do not do. The usual cross-product sign assumes a right-handed unit basis; a left-handed ordering, such as (u, v, z) in Exercise 3.10.1, contributes an overall minus sign. Also these formulas use physical components along unit vectors. Components along the unnormalized coordinate tangents $\partial \mathbf{r}/\partial q_i = h_i \hat{\mathbf{e}}_i$ would acquire the corresponding scale factors.

3.10.4 With $\hat{\mathbf{e}}_1$ a unit vector in the direction of increasing q_1, show that

(a) $\nabla \cdot \hat{\mathbf{e}}_1 = \dfrac{1}{h_1 h_2 h_3} \dfrac{\partial (h_2 h_3)}{\partial q_1}$

(b) $\nabla \times \hat{\mathbf{e}}_1 = \dfrac{1}{h_1} \left[\hat{\mathbf{e}}_2 \dfrac{1}{h_3} \dfrac{\partial h_1}{\partial q_3} - \hat{\mathbf{e}}_3 \dfrac{1}{h_2} \dfrac{\partial h_1}{\partial q_2} \right]$.

Note that even though $\hat{\mathbf{e}}_1$ is a unit vector, its divergence and curl **do not necessarily vanish**. Use a regular, right-handed orthogonal coordinate chart with positive scale factors h_i.

Solution.

(a) For physical components V_i in an orthogonal system, the flux through opposite q_i faces gives

$$\nabla \cdot \mathbf{V} = \frac{1}{h_1 h_2 h_3} \sum_i \partial_{q_i} (h_j h_k V_i), \qquad \{i, j, k\} = \{1, 2, 3\}.$$

For $\mathbf{V} = \hat{\mathbf{e}}_1$, $V_1 = 1, V_2 = V_3 = 0$, so only the $i = 1$ term survives, $\partial_{q_1}(h_2 h_3 \cdot 1)$, and $\boxed{\nabla \cdot \hat{\mathbf{e}}_1 = (h_1 h_2 h_3)^{-1} \partial_{q_1} (h_2 h_3)}$.

(b) For the curl, the gradient formula $\nabla\psi = \sum_j \hat{\mathbf{e}}_j h_j^{-1}\partial_{q_j}\psi$ with $\psi = q_1$ gives $\nabla q_1 = \hat{\mathbf{e}}_1/h_1$, i.e. $\hat{\mathbf{e}}_1 = h_1\nabla q_1$. By the product rule $\nabla\times(g\mathbf{V}) = \nabla g\times\mathbf{V} + g\nabla\times\mathbf{V}$ (Exercise 3.6.4) and $\nabla\times\nabla q_1 = 0$ (Exercise 3.6.10),

$$\nabla\times\hat{\mathbf{e}}_1 = \nabla h_1\times\nabla q_1 = \left(\sum_j \frac{\hat{\mathbf{e}}_j}{h_j}\partial_{q_j}h_1\right)\times\frac{\hat{\mathbf{e}}_1}{h_1}.$$

The $j=1$ term vanishes because $\hat{\mathbf{e}}_1\times\hat{\mathbf{e}}_1 = 0$. In a right-handed basis $\hat{\mathbf{e}}_2\times\hat{\mathbf{e}}_1 = -\hat{\mathbf{e}}_3$ and $\hat{\mathbf{e}}_3\times\hat{\mathbf{e}}_1 = \hat{\mathbf{e}}_2$, so the other two terms give

$$\boxed{\nabla\times\hat{\mathbf{e}}_1 = \frac{1}{h_1}\left[\frac{\hat{\mathbf{e}}_2}{h_3}\partial_{q_3}h_1 - \frac{\hat{\mathbf{e}}_3}{h_2}\partial_{q_2}h_1\right]}.$$

For a left-handed ordering both cross products, and hence the result of (b), change sign; this is why the qualification in the statement asks for a right-handed chart. The basis direction can vary even though its magnitude is fixed.

3.10.5 Show that a set of orthogonal unit vectors $\hat{\mathbf{e}}_i$ may be defined by

$$\hat{\mathbf{e}}_i = \frac{1}{h_i}\frac{\partial\mathbf{r}}{\partial q_i}.$$

In particular, show that $\hat{\mathbf{e}}_i\cdot\hat{\mathbf{e}}_i = 1$ leads to an expression for h_i in agreement with Eq. (3.131).

The above equation for $\hat{\mathbf{e}}_i$ may be taken as a starting point for deriving

$$\frac{\partial\hat{\mathbf{e}}_i}{\partial q_j} = \hat{\mathbf{e}}_j\frac{1}{h_i}\frac{\partial h_j}{\partial q_i}, \quad i\neq j$$

and

$$\frac{\partial\hat{\mathbf{e}}_i}{\partial q_i} = -\sum_{j\neq i}\hat{\mathbf{e}}_j\frac{1}{h_j}\frac{\partial h_i}{\partial q_j}.$$

Assume the coordinates form a regular orthogonal chart and $h_i > 0$; arbitrary curvilinear coordinates need not be orthogonal.

Solution.

On a regular orthogonal chart, the tangent to increasing q_i is $\mathbf{r}_i = \partial_{q_i}\mathbf{r}$, and ∂_i below means $\partial/\partial q_i$. Since $d\mathbf{r} = \sum_i \mathbf{r}_i\,dq_i$, the system is orthogonal exactly when $\mathbf{r}_i\cdot\mathbf{r}_j = 0$ for $i\neq j$ (no cross terms in $ds^2 = d\mathbf{r}\cdot d\mathbf{r}$), so the vectors $\hat{\mathbf{e}}_i = \mathbf{r}_i/h_i$ are mutually orthogonal. Requiring $\hat{\mathbf{e}}_i\cdot\hat{\mathbf{e}}_i = \mathbf{r}_i\cdot\mathbf{r}_i/h_i^2 = 1$ gives

$$h_i^2 = \mathbf{r}_i\cdot\mathbf{r}_i = \left(\frac{\partial x}{\partial q_i}\right)^2 + \left(\frac{\partial y}{\partial q_i}\right)^2 + \left(\frac{\partial z}{\partial q_i}\right)^2,$$

in agreement with Eq. (3.131); then $ds^2 = \sum_i h_i^2 dq_i^2$, so these h_i are the scale factors.

To derive the basis derivatives, write $\mathbf{r}_{ij} = \partial_j\mathbf{r}_i = \partial_i\partial_j\mathbf{r}$, which is symmetric in i,j because mixed partials commute. Differentiating $\mathbf{r}_i = h_i\hat{\mathbf{e}}_i$ gives $\mathbf{r}_{ij} = (\partial_j h_i)\hat{\mathbf{e}}_i + h_i\partial_j\hat{\mathbf{e}}_i$, so $\mathbf{r}_{ij} = \mathbf{r}_{ji}$ reads

$$(\partial_j h_i)\hat{\mathbf{e}}_i + h_i\partial_j\hat{\mathbf{e}}_i = (\partial_i h_j)\hat{\mathbf{e}}_j + h_j\partial_i\hat{\mathbf{e}}_j. \tag{*}$$

Differentiating $\hat{\mathbf{e}}_i\cdot\hat{\mathbf{e}}_i = 1$ with respect to q_j gives $\hat{\mathbf{e}}_i\cdot\partial_j\hat{\mathbf{e}}_i = 0$. For three distinct indices i,j,k, differentiate the orthogonality relations $\mathbf{r}_i\cdot\mathbf{r}_k = 0$, $\mathbf{r}_i\cdot\mathbf{r}_j = 0$, $\mathbf{r}_j\cdot\mathbf{r}_k = 0$ with respect to q_j, q_k, q_i,

respectively; for example the first gives $\mathbf{r}_{ij}\cdot\mathbf{r}_k+\mathbf{r}_i\cdot\mathbf{r}_{kj}=0$. With the symmetry of $\mathbf{r}_{ij}$ the three results give, in turn,

$$\mathbf{r}_k\cdot\mathbf{r}_{ij}=-\mathbf{r}_i\cdot\mathbf{r}_{jk}=\mathbf{r}_j\cdot\mathbf{r}_{ik}=-\mathbf{r}_k\cdot\mathbf{r}_{ij},$$

so $\mathbf{r}_k\cdot\mathbf{r}_{ij}=0$. Dotting the expansion of $\mathbf{r}_{ij}$ with $\mathbf{r}_k=h_k\hat{\mathbf{e}}_k$ and using $\hat{\mathbf{e}}_k\cdot\hat{\mathbf{e}}_i=0$ gives $\mathbf{r}_k\cdot\mathbf{r}_{ij}=h_kh_i\,\hat{\mathbf{e}}_k\cdot\partial_j\hat{\mathbf{e}}_i$; hence $\hat{\mathbf{e}}_k\cdot\partial_j\hat{\mathbf{e}}_i=0$. Dotting (*) with $\hat{\mathbf{e}}_j$ for $i\neq j$, and using $\hat{\mathbf{e}}_j\cdot\hat{\mathbf{e}}_i=0$ and $\hat{\mathbf{e}}_j\cdot\partial_i\hat{\mathbf{e}}_j=0$, gives the only remaining component: $h_i\hat{\mathbf{e}}_j\cdot\partial_j\hat{\mathbf{e}}_i=\partial_ih_j$. Therefore

$$\boxed{\partial_j\hat{\mathbf{e}}_i=\hat{\mathbf{e}}_j\,\frac{\partial_ih_j}{h_i}\quad(i\neq j)}.$$

Finally, $\partial_i\hat{\mathbf{e}}_i$ has no $\hat{\mathbf{e}}_i$ component. For $j\neq i$ differentiate $\hat{\mathbf{e}}_i\cdot\hat{\mathbf{e}}_j=0$ with respect to q_i and use the boxed formula with i and j interchanged, $\partial_i\hat{\mathbf{e}}_j=\hat{\mathbf{e}}_i(\partial_jh_i)/h_j$: $\hat{\mathbf{e}}_j\cdot\partial_i\hat{\mathbf{e}}_i=-\hat{\mathbf{e}}_i\cdot\partial_i\hat{\mathbf{e}}_j=-(\partial_jh_i)/h_j$. Expanding in the perpendicular basis gives $\boxed{\partial_i\hat{\mathbf{e}}_i=-\sum_{j\neq i}\hat{\mathbf{e}}_j(\partial_jh_i)/h_j}$.

3.10.6 Resolve the circular cylindrical unit vectors into their Cartesian components (see Fig. 3.23).

$$\begin{aligned}&\hat{\mathbf{e}}_\rho=\hat{\mathbf{e}}_x\cos\varphi+\hat{\mathbf{e}}_y\sin\varphi,\\ ANS.\quad&\hat{\mathbf{e}}_\varphi=-\hat{\mathbf{e}}_x\sin\varphi+\hat{\mathbf{e}}_y\cos\varphi,\\ &\hat{\mathbf{e}}_z=\hat{\mathbf{e}}_z.\end{aligned}$$

Solution.

Start with $\mathbf{r}=(\rho\cos\varphi,\rho\sin\varphi,z)$. Its coordinate derivatives are

$$\partial_\rho\mathbf{r}=(\cos\varphi,\sin\varphi,0),\quad\partial_\varphi\mathbf{r}=\rho(-\sin\varphi,\cos\varphi,0),\quad\partial_z\mathbf{r}=(0,0,1).$$

Their lengths are $(\cos^2\varphi+\sin^2\varphi)^{1/2}=1$, $\rho(\sin^2\varphi+\cos^2\varphi)^{1/2}=\rho$, and 1, for $\rho>0$. Dividing by these lengths gives

$$\boxed{\hat{\mathbf{e}}_\rho=(\cos\varphi,\sin\varphi,0),\quad\hat{\mathbf{e}}_\varphi=(-\sin\varphi,\cos\varphi,0),\quad\hat{\mathbf{e}}_z=(0,0,1)}.$$

Their dot products are δ_{ij} (for instance $\hat{\mathbf{e}}_\rho\cdot\hat{\mathbf{e}}_\varphi=-\cos\varphi\sin\varphi+\sin\varphi\cos\varphi=0$), and $\hat{\mathbf{e}}_\rho\times\hat{\mathbf{e}}_\varphi=(0,0,\cos^2\varphi+\sin^2\varphi)=\hat{\mathbf{e}}_z$, so the ordering (ρ,φ,z) is right-handed.

3.10.7 Resolve the Cartesian unit vectors into their circular cylindrical components (see Fig. 3.23).

$$\begin{aligned}&\hat{\mathbf{e}}_x=\hat{\mathbf{e}}_\rho\cos\varphi-\hat{\mathbf{e}}_\varphi\sin\varphi,\\ ANS.\quad&\hat{\mathbf{e}}_y=\hat{\mathbf{e}}_\rho\sin\varphi+\hat{\mathbf{e}}_\varphi\cos\varphi,\\ &\hat{\mathbf{e}}_z=\hat{\mathbf{e}}_z.\end{aligned}$$

Solution.

Exercise 3.10.6 gives $\hat{\mathbf{e}}_\rho,\hat{\mathbf{e}}_\varphi,\hat{\mathbf{e}}_z$ as the rows of an orthogonal matrix acting on $(\hat{\mathbf{e}}_x,\hat{\mathbf{e}}_y,\hat{\mathbf{e}}_z)$; its inverse is its transpose, whose first two rows are the first two columns $(\cos\varphi,-\sin\varphi,0)$ and $(\sin\varphi,\cos\varphi,0)$ of the original matrix. These rows give the coefficients of the combinations, which check directly:

$$\begin{aligned}\cos\varphi\,\hat{\mathbf{e}}_\rho-\sin\varphi\,\hat{\mathbf{e}}_\varphi&=(\cos^2\varphi+\sin^2\varphi,\ \cos\varphi\sin\varphi-\sin\varphi\cos\varphi,\ 0)=\hat{\mathbf{e}}_x,\\ \sin\varphi\,\hat{\mathbf{e}}_\rho+\cos\varphi\,\hat{\mathbf{e}}_\varphi&=(\sin\varphi\cos\varphi-\cos\varphi\sin\varphi,\ \sin^2\varphi+\cos^2\varphi,\ 0)=\hat{\mathbf{e}}_y.\end{aligned}$$

Together with unchanged $\hat{\mathbf{e}}_z$, these are the requested inverse formulas.

3.10.8 From the results of Exercise 3.10.6, show that

$$\frac{\partial \hat{\mathbf{e}}_\rho}{\partial \varphi} = \hat{\mathbf{e}}_\varphi, \quad \frac{\partial \hat{\mathbf{e}}_\varphi}{\partial \varphi} = -\hat{\mathbf{e}}_\rho$$

and that all other first derivatives of the circular cylindrical unit vectors with respect to the circular cylindrical coordinates vanish.

Solution.

Differentiating the Cartesian expressions gives

$$\partial_\varphi \hat{\mathbf{e}}_\rho = (-\sin\varphi, \cos\varphi, 0) = \hat{\mathbf{e}}_\varphi,$$

$$\partial_\varphi \hat{\mathbf{e}}_\varphi = (-\cos\varphi, -\sin\varphi, 0) = -\hat{\mathbf{e}}_\rho.$$

Neither depends on ρ or z, so those derivatives vanish. The vector $\hat{\mathbf{e}}_z = (0, 0, 1)$ is constant, so all three of its coordinate derivatives vanish.

3.10.9 Compare $\nabla \cdot \mathbf{V}$ as given for cylindrical coordinates in Eq. (3.148) with the result of its computation by applying to $\mathbf{V}$ the operator

$$\nabla = \hat{\mathbf{e}}_\rho \frac{\partial}{\partial \rho} + \hat{\mathbf{e}}_\varphi \frac{1}{\rho}\frac{\partial}{\partial \varphi} + \hat{\mathbf{e}}_z \frac{\partial}{\partial z}$$

Note that ∇ acts both on the unit vectors and on the components of $\mathbf{V}$.

Solution.

Write $\mathbf{V} = V_\rho \hat{\mathbf{e}}_\rho + V_\varphi \hat{\mathbf{e}}_\varphi + V_z \hat{\mathbf{e}}_z$. The basis has no ρ or z dependence, giving $\hat{\mathbf{e}}_\rho \cdot \partial_\rho \mathbf{V} = \partial_\rho V_\rho$ and $\hat{\mathbf{e}}_z \cdot \partial_z \mathbf{V} = \partial_z V_z$. For the azimuthal derivative, differentiate coefficients and basis, using $\partial_\varphi \hat{\mathbf{e}}_\rho = \hat{\mathbf{e}}_\varphi$ and $\partial_\varphi \hat{\mathbf{e}}_\varphi = -\hat{\mathbf{e}}_\rho$ (Exercise 3.10.8):

$$\partial_\varphi \mathbf{V} = (\partial_\varphi V_\rho - V_\varphi)\hat{\mathbf{e}}_\rho + (\partial_\varphi V_\varphi + V_\rho)\hat{\mathbf{e}}_\varphi + (\partial_\varphi V_z)\hat{\mathbf{e}}_z.$$

The operator gives $\nabla \cdot \mathbf{V} = \hat{\mathbf{e}}_\rho \cdot \partial_\rho \mathbf{V} + \rho^{-1}\hat{\mathbf{e}}_\varphi \cdot \partial_\varphi \mathbf{V} + \hat{\mathbf{e}}_z \cdot \partial_z \mathbf{V}$, and with $\hat{\mathbf{e}}_\varphi \cdot \partial_\varphi \mathbf{V} = \partial_\varphi V_\varphi + V_\rho$ from the last display, the three dot products yield

$$\boxed{\nabla \cdot \mathbf{V} = \partial_\rho V_\rho + \frac{V_\rho}{\rho} + \frac{1}{\rho}\partial_\varphi V_\varphi + \partial_z V_z = \frac{1}{\rho}\partial_\rho(\rho V_\rho) + \frac{1}{\rho}\partial_\varphi V_\varphi + \partial_z V_z},$$

where $\rho^{-1}\partial_\rho(\rho V_\rho) = \partial_\rho V_\rho + V_\rho/\rho$. This agrees with Eq. (3.148). The V_ρ/ρ term is exactly the contribution of the varying basis.

The Instructor's Manual, p. 51, calls the exercise's operator incorrect. The textbook explicitly says that it acts on the unit vectors as well as the components, so the operator as stated is correct: the basis derivative supplies exactly the V_ρ/ρ term computed above.

3.10.10

(a) Show that $\mathbf{r} = \hat{\mathbf{e}}_\rho \rho + \hat{\mathbf{e}}_z z$.

(b) Working entirely in circular cylindrical coordinates, show that

$$\nabla \cdot \mathbf{r} = 3 \quad \text{and} \quad \nabla \times \mathbf{r} = 0.$$

Solution.

(a) Since $x = \rho\cos\varphi, y = \rho\sin\varphi$, $x\hat{\mathbf{e}}_x + y\hat{\mathbf{e}}_y = \rho(\cos\varphi\,\hat{\mathbf{e}}_x + \sin\varphi\,\hat{\mathbf{e}}_y) = \rho\hat{\mathbf{e}}_\rho$ by Exercise 3.10.6, so $\mathbf{r} = \rho\hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$.

(b) The physical cylindrical components of $\mathbf{r}$ are $(V_\rho, V_\varphi, V_z) = (\rho, 0, z)$. Therefore Eq. (3.148) gives

$$\nabla\cdot\mathbf{r} = \frac{1}{\rho}\partial_\rho(\rho^2) + \frac{1}{\rho}\partial_\varphi 0 + \partial_z z = 2 + 0 + 1 = 3.$$

For the curl, expand the determinant form of the curl in orthogonal coordinates (Section 3.10),

$$\nabla\times\mathbf{V} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{\mathbf{e}}_1 & h_2\hat{\mathbf{e}}_2 & h_3\hat{\mathbf{e}}_3 \\ \partial_{q_1} & \partial_{q_2} & \partial_{q_3} \\ h_1V_1 & h_2V_2 & h_3V_3 \end{vmatrix},$$

with the right-handed ordering $(q_1, q_2, q_3) = (\rho, \varphi, z)$ (Exercise 3.10.6) and $h_\rho = 1$, $h_\varphi = \rho$, $h_z = 1$. For instance, the $\hat{\mathbf{e}}_\rho$ component is $\rho^{-1}[\partial_\varphi V_z - \partial_z(\rho V_\varphi)] = \rho^{-1}\partial_\varphi V_z - \partial_z V_\varphi$, since ρ does not depend on z. The three components are

$$(\nabla\times\mathbf{V})_\rho = \frac{1}{\rho}\partial_\varphi V_z - \partial_z V_\varphi, \quad (\nabla\times\mathbf{V})_\varphi = \partial_z V_\rho - \partial_\rho V_z, \quad (\nabla\times\mathbf{V})_z = \frac{1}{\rho}[\partial_\rho(\rho V_\varphi) - \partial_\varphi V_\rho].$$

For $\mathbf{V} = \mathbf{r}$,

$$(\nabla\times\mathbf{r})_\rho = \rho^{-1}\partial_\varphi z - \partial_z 0 = 0, \quad (\nabla\times\mathbf{r})_\varphi = \partial_z\rho - \partial_\rho z = 0,$$

and $(\nabla\times\mathbf{r})_z = \rho^{-1}[\partial_\rho(\rho\,0) - \partial_\varphi\rho] = 0$. Thus $\boxed{\nabla\times\mathbf{r} = 0}$.

3.10.11

(a) Show that the parity operation (reflection through the origin) on a point (ρ, φ, z) relative to fixed x-, y-, z-axes consists of the transformation

$$\rho\to\rho, \quad \varphi\to\varphi\pm\pi, \quad z\to -z.$$

(b) Show that $\hat{\mathbf{e}}_\rho$ and $\hat{\mathbf{e}}_\varphi$ have odd parity (reversal of direction) and that $\hat{\mathbf{e}}_z$ has even parity.

Note. The Cartesian unit vectors $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$ remain constant.

Solution.

(a) Under $(x, y, z) \mapsto (-x, -y, -z)$, $\rho = \sqrt{x^2 + y^2}$ is unchanged. Both $\cos\varphi = x/\rho$ and $\sin\varphi = y/\rho$ change sign. Since $\cos(\varphi\pm\pi) = -\cos\varphi$, $\sin(\varphi\pm\pi) = -\sin\varphi$, and the pair $(\cos\varphi, \sin\varphi)$ fixes the angle modulo 2π, $\varphi\mapsto\varphi\pm\pi$ (the two signs give the same point); and $z\mapsto -z$.

(b) Substituting this angle into the unit-vector functions of Exercise 3.10.6 gives

$$\begin{aligned}
\hat{\mathbf{e}}_\rho(\varphi+\pi) &= (-\cos\varphi, -\sin\varphi, 0) = -\hat{\mathbf{e}}_\rho(\varphi), \\
\hat{\mathbf{e}}_\varphi(\varphi+\pi) &= (\sin\varphi, -\cos\varphi, 0) = -\hat{\mathbf{e}}_\varphi(\varphi), \\
\hat{\mathbf{e}}_z(\varphi+\pi) &= (0, 0, 1) = \hat{\mathbf{e}}_z.
\end{aligned}$$

This is evaluation of the curvilinear basis at the reflected point relative to fixed Cartesian unit vectors, as specified in the note.

3.10.12 A rigid body is rotating about a fixed axis with a constant angular velocity $\boldsymbol{\omega}$. Take $\boldsymbol{\omega}$ to lie along the z-axis. Express the position vector $\mathbf{r}$ in circular cylindrical coordinates and using circular cylindrical coordinates,

(a) calculate $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$,

(b) calculate $\nabla \times \mathbf{v}$.

ANS. (a) $\mathbf{v} = \hat{\mathbf{e}}_\varphi \omega \rho$ (b) $\nabla \times \mathbf{v} = 2\boldsymbol{\omega}$.

Solution.

(a) With $\boldsymbol{\omega} = \omega \hat{\mathbf{e}}_z$ and $\mathbf{r} = \rho \hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$ (Exercise 3.10.10), and the right-handed products $\hat{\mathbf{e}}_z \times \hat{\mathbf{e}}_\rho = \hat{\mathbf{e}}_\varphi$, $\hat{\mathbf{e}}_z \times \hat{\mathbf{e}}_z = 0$,

$$\mathbf{v} = \omega \hat{\mathbf{e}}_z \times (\rho \hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z) = \omega \rho \, \hat{\mathbf{e}}_z \times \hat{\mathbf{e}}_\rho = \boxed{\omega \rho \hat{\mathbf{e}}_\varphi}.$$

(b) Its components are $(v_\rho, v_\varphi, v_z) = (0, \omega\rho, 0)$ and are independent of z, φ. Hence, with the cylindrical curl components of Exercise 3.10.10, $(\nabla \times \mathbf{v})_\rho = \rho^{-1}\partial_\varphi 0 - \partial_z(\omega\rho) = 0$ and $(\nabla \times \mathbf{v})_\varphi = \partial_z 0 - \partial_\rho 0 = 0$, while

$$(\nabla \times \mathbf{v})_z = \frac{1}{\rho}[\partial_\rho(\rho\,\omega\rho) - \partial_\varphi 0] = \frac{2\omega\rho}{\rho} = 2\omega.$$

Thus $\boxed{\nabla \times \mathbf{v} = 2\omega\hat{\mathbf{e}}_z = 2\boldsymbol{\omega}}$.

3.10.13 Find the circular cylindrical components of the velocity and acceleration of a moving particle,

$$\begin{aligned} v_\rho &= \dot{\rho}, & a_\rho &= \ddot{\rho} - \rho\dot{\varphi}^2, \\ v_\varphi &= \rho\dot{\varphi}, & a_\varphi &= \rho\ddot{\varphi} + 2\dot{\rho}\dot{\varphi}, \\ v_z &= \dot{z}, & a_z &= \ddot{z}. \end{aligned}$$

Hint. $\mathbf{r}(t) = \hat{\mathbf{e}}_\rho(t)\rho(t) + \hat{\mathbf{e}}_z z(t)$
$= [\hat{\mathbf{e}}_x \cos\varphi(t) + \hat{\mathbf{e}}_y \sin\varphi(t)]\rho(t) + \hat{\mathbf{e}}_z z(t)$.
Note. $\dot{\rho} = d\rho/dt$, $\ddot{\rho} = d^2\rho/dt^2$, and so on.

Solution.

Along the trajectory the unit vectors depend on t only through $\varphi(t)$, so by the chain rule and Exercise 3.10.8 the basis derivatives are $\dot{\hat{\mathbf{e}}}_\rho = \dot{\varphi}\,\partial_\varphi \hat{\mathbf{e}}_\rho = \dot{\varphi}\hat{\mathbf{e}}_\varphi$, $\dot{\hat{\mathbf{e}}}_\varphi = \dot{\varphi}\,\partial_\varphi \hat{\mathbf{e}}_\varphi = -\dot{\varphi}\hat{\mathbf{e}}_\rho$, and $\dot{\hat{\mathbf{e}}}_z = 0$. (Differentiating the hint's Cartesian form directly gives the same: $\frac{d}{dt}(\cos\varphi, \sin\varphi, 0) = \dot{\varphi}(-\sin\varphi, \cos\varphi, 0)$.) Differentiate $\mathbf{r} = \rho\hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$ by the product rule:

$$\mathbf{v} = \dot{\rho}\hat{\mathbf{e}}_\rho + \rho\dot{\varphi}\hat{\mathbf{e}}_\varphi + \dot{z}\hat{\mathbf{e}}_z.$$

Differentiate each term once more, again with $\dot{\hat{\mathbf{e}}}_\rho = \dot{\varphi}\hat{\mathbf{e}}_\varphi$ and $\dot{\hat{\mathbf{e}}}_\varphi = -\dot{\varphi}\hat{\mathbf{e}}_\rho$:

$$\mathbf{a} = \underbrace{\ddot{\rho}\hat{\mathbf{e}}_\rho + \dot{\rho}\dot{\varphi}\hat{\mathbf{e}}_\varphi}_{d(\dot{\rho}\hat{\mathbf{e}}_\rho)/dt} + \underbrace{(\dot{\rho}\dot{\varphi} + \rho\ddot{\varphi})\hat{\mathbf{e}}_\varphi - \rho\dot{\varphi}^2\hat{\mathbf{e}}_\rho}_{d(\rho\dot{\varphi}\hat{\mathbf{e}}_\varphi)/dt} + \ddot{z}\hat{\mathbf{e}}_z.$$

Collecting coefficients gives

$$\boxed{\begin{aligned} v_\rho &= \dot{\rho}, & v_\varphi &= \rho\dot{\varphi}, & v_z &= \dot{z}, \\ a_\rho &= \ddot{\rho} - \rho\dot{\varphi}^2, & a_\varphi &= \rho\ddot{\varphi} + 2\dot{\rho}\dot{\varphi}, & a_z &= \ddot{z}. \end{aligned}}$$

3.10.14 In right circular cylindrical coordinates, a particular vector function is given by

$$\mathbf{V}(\rho,\varphi)=\hat{\mathbf{e}}_\rho V_\rho(\rho,\varphi)+\hat{\mathbf{e}}_\varphi V_\varphi(\rho,\varphi).$$

Show that $\nabla\times\mathbf{V}$ has only a z-component. Note that this result will hold for any vector confined to a surface $q_3=$ constant as long as the products h_1V_1 and h_2V_2 are each independent of q_3.

Solution.

Use the cylindrical curl components of Exercise 3.10.10 with $V_z=0$ and the absence of z dependence:

$$(\nabla\times\mathbf{V})_\rho=\rho^{-1}\partial_\varphi V_z-\partial_z V_\varphi=0,\qquad(\nabla\times\mathbf{V})_\varphi=\partial_z V_\rho-\partial_\rho V_z=0.$$

The remaining component need not vanish:

$$\boxed{\nabla\times\mathbf{V}=\frac{\hat{\mathbf{e}}_z}{\rho}[\partial_\rho(\rho V_\varphi)-\partial_\varphi V_\rho]}.$$

More generally, expanding the determinant of Exercise 3.10.10 for general orthogonal coordinates gives the first two components

$$(\nabla\times\mathbf{V})_1=\frac{\partial_2(h_3V_3)-\partial_3(h_2V_2)}{h_2h_3},\qquad(\nabla\times\mathbf{V})_2=\frac{\partial_3(h_1V_1)-\partial_1(h_3V_3)}{h_3h_1},$$

with $\partial_i=\partial/\partial q_i$. For a vector confined to the surfaces $q_3=$ constant, $V_3=0$, and these reduce to

$$-(h_2h_3)^{-1}\partial_3(h_2V_2),\qquad(h_3h_1)^{-1}\partial_3(h_1V_1).$$

Both vanish when h_1V_1 and h_2V_2 are independent of q_3, as stated in the problem, leaving only component 3. (Cylindrical coordinates with $q_3=z$, $h_1V_1=V_\rho$, $h_2V_2=\rho V_\varphi$ are the case above.)

For this $V_z=0$ field, the transverse curl components are $-\partial_z V_\varphi$ and $\partial_z V_\rho$. The Instructor's Manual, p. 52, gives the first with the wrong sign and the second with an extra factor $1/\rho$. Both are zero here because the components are independent of z, so the conclusion survives those errors.

3.10.15 A conducting wire along the z-axis carries a current I. The resulting magnetic vector potential is given by

$$\mathbf{A}=\hat{\mathbf{e}}_z\frac{\mu I}{2\pi}\ln\frac{1}{\rho}.$$

Show that the magnetic induction $\mathbf{B}$ is given by

$$\mathbf{B}=\hat{\mathbf{e}}_\varphi\frac{\mu I}{2\pi\rho}.$$

Solution.

Let $K=\mu I/(2\pi)$, so $A_z=K\ln(1/\rho)=-K\ln\rho$, $A_\rho=A_\varphi=0$. With the cylindrical curl components of Exercise 3.10.10, $B_\rho=\rho^{-1}\partial_\varphi A_z-\partial_z A_\varphi=0$ because A_z does not depend on φ, and $B_z=\rho^{-1}[\partial_\rho(\rho A_\varphi)-\partial_\varphi A_\rho]=0$. The remaining one is

$$B_\varphi=\partial_z A_\rho-\partial_\rho A_z=0-(-K/\rho)=K/\rho.$$

Thus $\boxed{\mathbf{B}=\mu I\,\hat{\mathbf{e}}_\varphi/(2\pi\rho)}$, valid for $\rho>0$ outside the ideal wire. Replacing $\ln(1/\rho)$ by $\ln(\rho_0/\rho)$ makes the logarithm dimensionless and adds only a constant to A_z, leaving $\mathbf{B}$ unchanged.

3.10.16 A force is described by

$$\mathbf{F} = -\hat{\mathbf{e}}_x \frac{y}{x^2+y^2} + \hat{\mathbf{e}}_y \frac{x}{x^2+y^2}.$$

(a) Express $\mathbf{F}$ in circular cylindrical coordinates.

Operating entirely in circular cylindrical coordinates for (b) and (c),

(b) Calculate the curl of $\mathbf{F}$ and

(c) Calculate the work done by $\mathbf{F}$ in encircling the unit circle once counterclockwise.

(d) How do you reconcile the results of (b) and (c)?

Solution.

(a) Substitute $x = \rho\cos\varphi, y = \rho\sin\varphi$, so $x^2+y^2 = \rho^2$:

$$\mathbf{F} = \frac{(-\rho\sin\varphi, \rho\cos\varphi, 0)}{\rho^2} = \frac{1}{\rho}(-\sin\varphi, \cos\varphi, 0) = \boxed{\hat{\mathbf{e}}_\varphi/\rho}.$$

(b) For $\rho > 0$, $F_\rho = F_z = 0$ and $F_\varphi = 1/\rho$ depends on ρ only. With the cylindrical curl components of Exercise 3.10.10, the transverse components $\rho^{-1}\partial_\varphi F_z - \partial_z F_\varphi$ and $\partial_z F_\rho - \partial_\rho F_z$ vanish, and $(\nabla\times\mathbf{F})_z = \rho^{-1}[\partial_\rho(\rho\cdot\rho^{-1}) - \partial_\varphi F_\rho] = \rho^{-1}\partial_\rho 1 = 0$. So $\boxed{\nabla\times\mathbf{F} = 0}$ for $\rho > 0$. **(c)** In cylindrical coordinates $d\mathbf{r} = \hat{\mathbf{e}}_\rho d\rho + \hat{\mathbf{e}}_\varphi \rho\, d\varphi + \hat{\mathbf{e}}_z dz$; on the unit circle ($\rho = 1$, $z = 0$) $d\mathbf{r} = \hat{\mathbf{e}}_\varphi d\varphi$ and $\mathbf{F}\cdot d\mathbf{r} = d\varphi$, so $\boxed{W = \oint \mathbf{F}\cdot d\mathbf{r} = \int_0^{2\pi} d\varphi = 2\pi}$. **(d)** The field is undefined on the axis. A disk spanning the circle crosses that singularity, so Stokes' theorem cannot use the ordinary zero curl over that disk. On an annulus excluding the axis, the outer counterclockwise circulation 2π and the inner clockwise circulation -2π cancel, in full agreement with zero curl. (The calculation in (c) gives 2π on a circle of any radius a, since $\mathbf{F}\cdot d\mathbf{r} = a^{-1}\, a\, d\varphi$.) Locally $\mathbf{F} = \nabla\varphi$, since $\nabla\varphi = \hat{\mathbf{e}}_\varphi\rho^{-1}\partial_\varphi\varphi = \hat{\mathbf{e}}_\varphi/\rho$, but φ is not a global single-valued potential around the axis: it increases by 2π once around, which is the circulation found in (c).

3.10.17 A calculation of the magnetohydrodynamic pinch effect involves the evaluation of $(\mathbf{B}\cdot\nabla)\mathbf{B}$. If the magnetic induction $\mathbf{B}$ is taken to be $\mathbf{B} = \hat{\mathbf{e}}_\varphi B_\varphi(\rho)$, show that

$$(\mathbf{B}\cdot\nabla)\mathbf{B} = -\hat{\mathbf{e}}_\rho B_\varphi^2/\rho.$$

Solution.

For $\mathbf{B} = B_\varphi(\rho)\hat{\mathbf{e}}_\varphi$, dotting into $\nabla = \hat{\mathbf{e}}_\rho\partial_\rho + \hat{\mathbf{e}}_\varphi\rho^{-1}\partial_\varphi + \hat{\mathbf{e}}_z\partial_z$ keeps only the middle term, so the directional derivative operator is $\mathbf{B}\cdot\nabla = (B_\varphi/\rho)\partial_\varphi$. It must differentiate the basis as well as the scalar component:

$$(\mathbf{B}\cdot\nabla)\mathbf{B} = \frac{B_\varphi}{\rho}\left[(\partial_\varphi B_\varphi)\hat{\mathbf{e}}_\varphi + B_\varphi\partial_\varphi\hat{\mathbf{e}}_\varphi\right] = \boxed{-\frac{B_\varphi^2}{\rho}\hat{\mathbf{e}}_\rho}.$$

Here $\partial_\varphi B_\varphi = 0$ and $\partial_\varphi\hat{\mathbf{e}}_\varphi = -\hat{\mathbf{e}}_\rho$ (Exercise 3.10.8).

3.10.18 Express the spherical polar unit vectors in terms of Cartesian unit vectors.

$$\begin{aligned} \hat{\mathbf{e}}_r &= \hat{\mathbf{e}}_x \sin\theta\cos\varphi + \hat{\mathbf{e}}_y \sin\theta\sin\varphi + \hat{\mathbf{e}}_z \cos\theta, \\ \textit{ANS.}\quad \hat{\mathbf{e}}_\theta &= \hat{\mathbf{e}}_x \cos\theta\cos\varphi + \hat{\mathbf{e}}_y \cos\theta\sin\varphi - \hat{\mathbf{e}}_z \sin\theta, \\ \hat{\mathbf{e}}_\varphi &= -\hat{\mathbf{e}}_x \sin\varphi + \hat{\mathbf{e}}_y \cos\varphi. \end{aligned}$$

Solution.

The position map is $\mathbf{r} = r(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. Its r, θ, φ derivatives are

$$\begin{aligned} \partial_r \mathbf{r} &= (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \\ \partial_\theta \mathbf{r} &= r(\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta), \\ \partial_\varphi \mathbf{r} &= r\sin\theta(-\sin\varphi, \cos\varphi, 0), \end{aligned}$$

with lengths $1, r, r\sin\theta$ on a regular patch $r > 0$, $0 < \theta < \pi$ (for $\partial_r\mathbf{r}$ and $\partial_\theta\mathbf{r}/r$, the squared components sum to $\sin^2\theta + \cos^2\theta = 1$ after $\cos^2\varphi + \sin^2\varphi = 1$). Normalize each derivative to obtain

$$\boxed{\begin{aligned} \hat{\mathbf{e}}_r &= (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \\ \hat{\mathbf{e}}_\theta &= (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta), \\ \hat{\mathbf{e}}_\varphi &= (-\sin\varphi, \cos\varphi, 0). \end{aligned}}$$

Their norms are one, their pairwise dot products vanish (for instance $\hat{\mathbf{e}}_r \cdot \hat{\mathbf{e}}_\theta = \sin\theta\cos\theta(\cos^2\varphi + \sin^2\varphi) - \cos\theta\sin\theta = 0$), and

$$\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta = \big(-\sin\varphi(\sin^2\theta + \cos^2\theta),\ \cos\varphi(\cos^2\theta + \sin^2\theta),\ 0\big) = \hat{\mathbf{e}}_\varphi,$$

so (r, θ, φ) is right-handed. On the polar axis ($\sin\theta = 0$) the azimuth φ is undefined and $\partial_\varphi\mathbf{r}$ vanishes, so it cannot be normalized; moreover the boxed $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\varphi$ depend on φ, so the angular unit vectors have no unique value on the axis.

3.10.19 Resolve the Cartesian unit vectors into their spherical polar components:

$$\begin{aligned} \hat{\mathbf{e}}_x &= \hat{\mathbf{e}}_r \sin\theta\cos\varphi + \hat{\mathbf{e}}_\theta \cos\theta\cos\varphi - \hat{\mathbf{e}}_\varphi \sin\varphi, \\ \hat{\mathbf{e}}_y &= \hat{\mathbf{e}}_r \sin\theta\sin\varphi + \hat{\mathbf{e}}_\theta \cos\theta\sin\varphi + \hat{\mathbf{e}}_\varphi \cos\varphi, \\ \hat{\mathbf{e}}_z &= \hat{\mathbf{e}}_r \cos\theta - \hat{\mathbf{e}}_\theta \sin\theta. \end{aligned}$$

Solution.

Any vector is the sum of its projections on an orthonormal basis. For example, $\hat{\mathbf{e}}_x = \sum_{j=r,\theta,\varphi}(\hat{\mathbf{e}}_x \cdot \hat{\mathbf{e}}_j)\hat{\mathbf{e}}_j$. The projection coefficients $\hat{\mathbf{e}}_x \cdot \hat{\mathbf{e}}_r = \sin\theta\cos\varphi$, $\hat{\mathbf{e}}_x \cdot \hat{\mathbf{e}}_\theta = \cos\theta\cos\varphi$, $\hat{\mathbf{e}}_x \cdot \hat{\mathbf{e}}_\varphi = -\sin\varphi$ are precisely the x components of the vectors in Exercise 3.10.18. Repeating with their y and z components gives

$$\boxed{\begin{aligned} \hat{\mathbf{e}}_x &= \sin\theta\cos\varphi\,\hat{\mathbf{e}}_r + \cos\theta\cos\varphi\,\hat{\mathbf{e}}_\theta - \sin\varphi\,\hat{\mathbf{e}}_\varphi, \\ \hat{\mathbf{e}}_y &= \sin\theta\sin\varphi\,\hat{\mathbf{e}}_r + \cos\theta\sin\varphi\,\hat{\mathbf{e}}_\theta + \cos\varphi\,\hat{\mathbf{e}}_\varphi, \\ \hat{\mathbf{e}}_z &= \cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta. \end{aligned}}$$

Thus the inverse change-of-basis matrix is the transpose of the matrix whose rows are the vectors of Exercise 3.10.18 (the matrix T of Exercise 3.10.20).

3.10.20

(a) Explain why it is not possible to relate a column vector $\mathbf{r}$ (with components x, y, z) to another column vector $\mathbf{r}'$ (with components r, θ, φ), via a matrix equation of the form $\mathbf{r}' = \mathsf{B}\mathbf{r}$ with one constant matrix B valid at all positions.

(b) One can write a matrix equation relating the Cartesian components of a vector to its components in spherical polar coordinates. Find the transformation matrix and determine whether it is orthogonal.

Solution.

(a) The qualification "one constant matrix" is needed: at any one point $\mathbf{r} \neq 0$ the position-dependent matrix $\mathsf{B} = \mathbf{r}'\mathbf{r}^T/r^2$ does give $\mathsf{B}\mathbf{r} = \mathbf{r}'(\mathbf{r}^T\mathbf{r})/r^2 = \mathbf{r}'$. Write $\mathbf{x} = (x, y, z)^T$ for the column $\mathbf{r}$, to keep it apart from the radius r. The radius $r = \sqrt{x^2 + y^2 + z^2}$ is not a linear function: $r(-\mathbf{x}) = r(\mathbf{x})$, whereas the first component of a constant matrix map is linear, $(\mathsf{B}(-\mathbf{x}))_1 = -(\mathsf{B}\mathbf{x})_1$. If $(\mathsf{B}\mathbf{x})_1 = r(\mathbf{x})$ held for all $\mathbf{x}$, then $r(\mathbf{x}) = r(-\mathbf{x}) = (\mathsf{B}(-\mathbf{x}))_1 = -(\mathsf{B}\mathbf{x})_1 = -r(\mathbf{x})$, so $r(\mathbf{x}) = 0$, which is false for $\mathbf{x} \neq 0$. The angular coordinates are nonlinear as well; for example $\theta(2\mathbf{x}) = \theta(\mathbf{x})$, while a linear function doubles.

(b) Physical components of a vector at a fixed point are its projections on the unit vectors, $V_r = \hat{\mathbf{e}}_r \cdot \mathbf{V} = \sum_a (\hat{\mathbf{e}}_r)_a V_a$ and similarly for V_θ, V_φ, with the Cartesian components $(\hat{\mathbf{e}}_r)_a$ etc. of Exercise 3.10.18. Hence they obey

$$\boxed{\begin{pmatrix} V_r \\ V_\theta \\ V_\varphi \end{pmatrix} = T \begin{pmatrix} V_x \\ V_y \\ V_z \end{pmatrix}, \quad T = \begin{pmatrix} \sin\theta\cos\varphi & \sin\theta\sin\varphi & \cos\theta \\ \cos\theta\cos\varphi & \cos\theta\sin\varphi & -\sin\theta \\ -\sin\varphi & \cos\varphi & 0 \end{pmatrix}.}$$

The rows are the three unit vectors from Exercise 3.10.18, so the entries of TT^T are their inner products, $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \delta_{ij}$; thus $TT^T = I$, and T is orthogonal. Its determinant is the triple product $\hat{\mathbf{e}}_r \cdot (\hat{\mathbf{e}}_\theta \times \hat{\mathbf{e}}_\varphi) = \hat{\mathbf{e}}_r \cdot \hat{\mathbf{e}}_r = 1$ (a rotation). Thus $T^{-1} = T^T$, but T depends on the observation point. In particular $T(x, y, z)^T = (r, 0, 0)^T$, the components of the position vector, rather than the coordinate labels $(r, \theta, \varphi)^T$.

3.10.21 Find the transformation matrix that converts the components of a vector in spherical polar coordinates into its components in circular cylindrical coordinates. Then find the matrix of the inverse transformation.

Solution.

The common azimuthal vector is unchanged, while

$$\hat{\mathbf{e}}_\rho = \sin\theta\,\hat{\mathbf{e}}_r + \cos\theta\,\hat{\mathbf{e}}_\theta, \qquad \hat{\mathbf{e}}_z = \cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta.$$

These follow from the Cartesian forms in Exercises 3.10.6 and 3.10.18. For instance, the x component of $\sin\theta\,\hat{\mathbf{e}}_r + \cos\theta\,\hat{\mathbf{e}}_\theta$ is $\sin^2\theta\cos\varphi + \cos^2\theta\cos\varphi = \cos\varphi$, its y component is likewise $\sin\varphi$, and its z component is $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$; similarly $\cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta$ has components $(0, 0, \cos^2\theta + \sin^2\theta) = (0, 0, 1)$. Project $\mathbf{V} = V_r\hat{\mathbf{e}}_r + V_\theta\hat{\mathbf{e}}_\theta + V_\varphi\hat{\mathbf{e}}_\varphi$ onto these directions: $V_\rho = \hat{\mathbf{e}}_\rho \cdot \mathbf{V} = \sin\theta\,V_r + \cos\theta\,V_\theta$, $V_z = \hat{\mathbf{e}}_z \cdot \mathbf{V} = \cos\theta\,V_r - \sin\theta\,V_\theta$, and V_φ is shared. In the specified component orders,

$$\boxed{\begin{pmatrix} V_\rho \\ V_\varphi \\ V_z \end{pmatrix} = C \begin{pmatrix} V_r \\ V_\theta \\ V_\varphi \end{pmatrix}, \quad C = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{pmatrix}.}$$

The rows are orthonormal (e.g. the first and third give $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$), so C is orthogonal and

$$\boxed{C^{-1} = C^T = \begin{pmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{pmatrix}}.$$

Multiplication of these matrices gives I using $\sin^2\theta + \cos^2\theta = 1$.

3.10.22

(a) From the results of Exercise 3.10.18, calculate the partial derivatives of $\hat{\mathbf{e}}_r$, $\hat{\mathbf{e}}_\theta$, and $\hat{\mathbf{e}}_\varphi$ with respect to r, θ, and φ.

(b) With ∇ given by

$$\hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\varphi \frac{1}{r\sin\theta}\frac{\partial}{\partial \varphi}$$

(greatest space rate of change), use the results of part (a) to calculate $\nabla \cdot \nabla\psi$. This is an alternate derivation of the Laplacian.

Note. The derivatives of the left-hand ∇ operate on the unit vectors of the right-hand ∇ before the dot product is evaluated.

Solution.

(a) None of the unit vectors in Exercise 3.10.18 depends on r. Differentiating their Cartesian expressions with respect to the angles,

$$\partial_\theta \hat{\mathbf{e}}_r = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta), \quad \partial_\varphi \hat{\mathbf{e}}_r = \sin\theta(-\sin\varphi, \cos\varphi, 0),$$
$$\partial_\theta \hat{\mathbf{e}}_\theta = -(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \quad \partial_\varphi \hat{\mathbf{e}}_\theta = \cos\theta(-\sin\varphi, \cos\varphi, 0),$$

$\partial_\theta \hat{\mathbf{e}}_\varphi = 0$, and $\partial_\varphi \hat{\mathbf{e}}_\varphi = (-\cos\varphi, -\sin\varphi, 0)$. Comparing with the unit vectors themselves gives

	∂_r	∂_θ	∂_φ
$\hat{\mathbf{e}}_r$	0	$\hat{\mathbf{e}}_\theta$	$\sin\theta\,\hat{\mathbf{e}}_\varphi$
$\hat{\mathbf{e}}_\theta$	0	$-\hat{\mathbf{e}}_r$	$\cos\theta\,\hat{\mathbf{e}}_\varphi$
$\hat{\mathbf{e}}_\varphi$	0	0	$-\sin\theta\,\hat{\mathbf{e}}_r - \cos\theta\,\hat{\mathbf{e}}_\theta$

For the last entry, the projections of $(-\cos\varphi, -\sin\varphi, 0)$ on $\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\varphi$ are $-\sin\theta(\cos^2\varphi + \sin^2\varphi) = -\sin\theta$, $-\cos\theta(\cos^2\varphi + \sin^2\varphi) = -\cos\theta$, and $\sin\varphi\cos\varphi - \cos\varphi\sin\varphi = 0$, respectively.

(b) First apply the given ∇ to any vector $\mathbf{W} = W_r\hat{\mathbf{e}}_r + W_\theta\hat{\mathbf{e}}_\theta + W_\varphi\hat{\mathbf{e}}_\varphi$. By the product rule and the derivative table,

$$\partial_\theta \mathbf{W} = (\partial_\theta W_r - W_\theta)\hat{\mathbf{e}}_r + (W_r + \partial_\theta W_\theta)\hat{\mathbf{e}}_\theta + \partial_\theta W_\varphi\,\hat{\mathbf{e}}_\varphi,$$
$$\partial_\varphi \mathbf{W} = (\partial_\varphi W_r - \sin\theta W_\varphi)\hat{\mathbf{e}}_r + (\partial_\varphi W_\theta - \cos\theta W_\varphi)\hat{\mathbf{e}}_\theta + (\sin\theta W_r + \cos\theta W_\theta + \partial_\varphi W_\varphi)\hat{\mathbf{e}}_\varphi,$$

while $\partial_r \mathbf{W}$ has components $\partial_r W_r, \partial_r W_\theta, \partial_r W_\varphi$. Hence

$$\hat{\mathbf{e}}_r \cdot \partial_r \mathbf{W} = \partial_r W_r, \quad \hat{\mathbf{e}}_\theta \cdot \partial_\theta \mathbf{W} = W_r + \partial_\theta W_\theta,$$

$$\hat{\mathbf{e}}_\varphi \cdot \partial_\varphi \mathbf{W} = \sin\theta W_r + \cos\theta W_\theta + \partial_\varphi W_\varphi.$$

The given operator dotted into $\mathbf{W}$ is $\nabla\cdot\mathbf{W} = \hat{\mathbf{e}}_r\cdot\partial_r\mathbf{W} + r^{-1}\hat{\mathbf{e}}_\theta\cdot\partial_\theta\mathbf{W} + (r\sin\theta)^{-1}\hat{\mathbf{e}}_\varphi\cdot\partial_\varphi\mathbf{W}$; inserting the three dot products and collecting the W_r terms ($W_r/r + \sin\theta\, W_r/(r\sin\theta) = 2W_r/r$),

$$\nabla\cdot\mathbf{W} = \partial_r W_r + \frac{2W_r}{r} + \frac{\partial_\theta W_\theta + \cot\theta W_\theta}{r} + \frac{\partial_\varphi W_\varphi}{r\sin\theta}.$$

Now put $\mathbf{W} = \nabla\psi$, whose physical components are $W_r = \psi_r$, $W_\theta = \psi_\theta/r$, $W_\varphi = \psi_\varphi/(r\sin\theta)$ (subscripts on ψ denote partial derivatives). Then $\partial_\theta W_\theta = \psi_{\theta\theta}/r$ and $\partial_\varphi W_\varphi = \psi_{\varphi\varphi}/(r\sin\theta)$, and substitution yields

$$\begin{aligned}\nabla^2\psi &= \psi_{rr} + \frac{2}{r}\psi_r + \frac{1}{r^2}(\psi_{\theta\theta} + \cot\theta\,\psi_\theta) + \frac{\psi_{\varphi\varphi}}{r^2\sin^2\theta} \\ &= \boxed{\frac{1}{r^2}\partial_r(r^2\partial_r\psi) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta\psi) + \frac{1}{r^2\sin^2\theta}\partial_\varphi^2\psi}.\end{aligned}$$

The last step uses $r^{-2}\partial_r(r^2\psi_r) = \psi_{rr} + 2\psi_r/r$ and $(\sin\theta)^{-1}\partial_\theta(\sin\theta\,\psi_\theta) = \psi_{\theta\theta} + \cot\theta\,\psi_\theta$.

The Instructor's Manual, p. 54, prints $\tan\theta/r^2$ instead of $\cot\theta/r^2$ as the coefficient of ∂_θ. Its following divergence-form identity must have $1/(r^2\sin\theta)$, not $1/(r^2\sin^2\theta)$, before $\partial_\theta(\sin\theta\,\partial_\theta)$. For a quick independent check, $\psi = r\cos\theta = z$ has zero Cartesian Laplacian. The correct formula gives $2\cos\theta/r - \cos\theta/r - \cos\theta/r = 0$; the printed $\tan\theta$ version instead gives $\cos(2\theta)/(r\cos\theta)$.

3.10.23 A rigid body is rotating about a fixed axis with a constant angular velocity $\boldsymbol{\omega}$. Take $\boldsymbol{\omega}$ to be along the z-axis. Using spherical polar coordinates,

(a) calculate $\mathbf{v} = \boldsymbol{\omega}\times\mathbf{r}$.

(b) calculate $\nabla\times\mathbf{v}$.

ANS. (a) $\mathbf{v} = \hat{\mathbf{e}}_\varphi\omega r\sin\theta$. (b) $\nabla\times\mathbf{v} = 2\boldsymbol{\omega}$.

Solution.

(a) In spherical components, $\boldsymbol{\omega} = \omega\hat{\mathbf{e}}_z = \omega(\cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta)$ (Exercise 3.10.19) and $\mathbf{r} = r\hat{\mathbf{e}}_r$. With $\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_r = 0$ and $\hat{\mathbf{e}}_\theta\times\hat{\mathbf{e}}_r = -\hat{\mathbf{e}}_\varphi$, the cross product is

$$\mathbf{v} = \omega r(\cos\theta\,\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta\times\hat{\mathbf{e}}_r) = \boxed{\omega r\sin\theta\,\hat{\mathbf{e}}_\varphi}.$$

(b) Expanding the determinant of Exercise 3.10.10 with the right-handed ordering $(q_1,q_2,q_3) = (r,\theta,\varphi)$ (Exercise 3.10.18) and $h_r = 1$, $h_\theta = r$, $h_\varphi = r\sin\theta$ gives the spherical components below. For instance, in the expansion along the first row the $\hat{\mathbf{e}}_\theta$ term (cofactor sign -1) has the coefficient $(r^2\sin\theta)^{-1}\,r\,[\partial_\varphi V_r - \partial_r(r\sin\theta\,V_\varphi)]$, which is the θ formula because $\sin\theta$ does not depend on r:

$$\begin{aligned}(\nabla\times\mathbf{V})_r &= \frac{1}{r\sin\theta}[\partial_\theta(\sin\theta\,V_\varphi) - \partial_\varphi V_\theta], \qquad (\nabla\times\mathbf{V})_\varphi = \frac{1}{r}[\partial_r(rV_\theta) - \partial_\theta V_r], \\ (\nabla\times\mathbf{V})_\theta &= \frac{1}{r}\Big[\frac{1}{\sin\theta}\partial_\varphi V_r - \partial_r(rV_\varphi)\Big].\end{aligned}$$

For $v_r = v_\theta = 0$, $v_\varphi = \omega r\sin\theta$ these give

$$(\nabla\times\mathbf{v})_r = \frac{1}{r\sin\theta}\partial_\theta(\omega r\sin^2\theta) = \frac{2\omega r\sin\theta\cos\theta}{r\sin\theta} = 2\omega\cos\theta,$$

$$(\nabla\times\mathbf{v})_\theta = -\frac{1}{r}\partial_r(\omega r^2\sin\theta) = -\frac{2\omega r\sin\theta}{r} = -2\omega\sin\theta, \qquad (\nabla\times\mathbf{v})_\varphi = 0.$$

Thus the result is $2\omega(\cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta) = \boxed{2\boldsymbol{\omega}}$.

3.10.24 A certain vector $\mathbf{V}$ has no radial component. Its curl has no tangential components. What does this imply about the radial dependence of the tangential components of $\mathbf{V}$?

Solution.

Since $V_r = 0$, the tangential curl components of Exercise 3.10.23, $(\nabla\times\mathbf{V})_\theta = r^{-1}[(\sin\theta)^{-1}\partial_\varphi V_r - \partial_r(rV_\varphi)]$ and $(\nabla\times\mathbf{V})_\varphi = r^{-1}[\partial_r(rV_\theta) - \partial_\theta V_r]$, reduce to

$$(\nabla\times\mathbf{V})_\theta = -\frac{1}{r}\partial_r(rV_\varphi), \qquad (\nabla\times\mathbf{V})_\varphi = \frac{1}{r}\partial_r(rV_\theta).$$

Setting both to zero means $\partial_r(rV_\theta) = \partial_r(rV_\varphi) = 0$, so rV_θ and rV_φ are independent of r, say $rV_\theta = f(\theta,\varphi)$ and $rV_\varphi = g(\theta,\varphi)$. Hence, on $r > 0$,

$$\boxed{V_\theta = \frac{f(\theta,\varphi)}{r}, \qquad V_\varphi = \frac{g(\theta,\varphi)}{r}}.$$

If the region includes the origin and boundedness there is also required, then these two functions must vanish; that extra regularity was not part of the stated assumptions.

3.10.25 Modern physics lays great stress on the property of parity (whether a quantity remains invariant or changes sign under an inversion of the coordinate system). In Cartesian coordinates this means $x \to -x$, $y \to -y$, and $z \to -z$.

(a) Show that the inversion (reflection through the origin) of a point (r,θ,φ) relative to fixed x-, y-, z-axes consists of the transformation

$$r \to r, \quad \theta \to \pi - \theta, \quad \varphi \to \varphi \pm \pi.$$

(b) Show that $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\varphi$ have odd parity (reversal of direction) and that $\hat{\mathbf{e}}_\theta$ has even parity.

Solution.

(a) Inversion leaves $r = \sqrt{x^2+y^2+z^2}$ unchanged. Since $\cos\theta = z/r$ changes sign and $0 \le \theta \le \pi$, the new angle is the one in $[0,\pi]$ with cosine $-\cos\theta$, namely $\theta \mapsto \pi - \theta$; then $\sin\theta$ is unchanged. Both $x = r\sin\theta\cos\varphi$ and $y = r\sin\theta\sin\varphi$ change sign with $r\sin\theta$ fixed, so $\cos\varphi$ and $\sin\varphi$ both change sign, i.e. $\varphi \mapsto \varphi \pm \pi$ (modulo 2π, as in Exercise 3.10.11).

(b) Under these substitutions, $\sin\theta$ is unchanged, $\cos\theta$ changes sign, and $\sin\varphi, \cos\varphi$ both change sign. The component formulas in Exercise 3.10.18 then give

$$\begin{aligned}
\hat{\mathbf{e}}_r(\pi-\theta,\varphi+\pi) &= (-\sin\theta\cos\varphi, -\sin\theta\sin\varphi, -\cos\theta) = -\hat{\mathbf{e}}_r(\theta,\varphi),\\
\hat{\mathbf{e}}_\theta(\pi-\theta,\varphi+\pi) &= \big((-\cos\theta)(-\cos\varphi), (-\cos\theta)(-\sin\varphi), -\sin\theta\big) = \hat{\mathbf{e}}_\theta(\theta,\varphi),\\
\hat{\mathbf{e}}_\varphi(\varphi+\pi) &= (\sin\varphi, -\cos\varphi, 0) = -\hat{\mathbf{e}}_\varphi(\varphi).
\end{aligned}$$

Thus the stated basis-field parities follow relative to fixed Cartesian axes.

3.10.26 With $\mathbf{A}$ any vector,

$$\mathbf{A}\cdot\nabla\mathbf{r} = \mathbf{A}.$$

(a) Verify this result in Cartesian coordinates.

(b) Verify this result using spherical polar coordinates. Equation (3.156) provides ∇.

Solution.

(a) In Cartesian coordinates, $[(\mathbf{A}\cdot\nabla)\mathbf{r}]_j = \sum_i A_i\partial_i x_j = \sum_i A_i\delta_{ij} = A_j$. **(b)** In spherical coordinates $\mathbf{r} = r\hat{\mathbf{e}}_r$, so, with $\partial_\theta\hat{\mathbf{e}}_r = \hat{\mathbf{e}}_\theta$ and $\partial_\varphi\hat{\mathbf{e}}_r = \sin\theta\,\hat{\mathbf{e}}_\varphi$ (Exercise 3.10.22),

$$\partial_r\mathbf{r} = \hat{\mathbf{e}}_r, \quad \partial_\theta\mathbf{r} = r\hat{\mathbf{e}}_\theta, \quad \partial_\varphi\mathbf{r} = r\sin\theta\,\hat{\mathbf{e}}_\varphi.$$

Dotting $\mathbf{A} = A_r\hat{\mathbf{e}}_r + A_\theta\hat{\mathbf{e}}_\theta + A_\varphi\hat{\mathbf{e}}_\varphi$ into the gradient operator of Eq. (3.156) gives $\mathbf{A}\cdot\nabla = A_r\partial_r + (A_\theta/r)\partial_\theta + (A_\varphi/(r\sin\theta))\partial_\varphi$. Therefore

$$(\mathbf{A}\cdot\nabla)\mathbf{r} = A_r\partial_r\mathbf{r} + \frac{A_\theta}{r}\partial_\theta\mathbf{r} + \frac{A_\varphi}{r\sin\theta}\partial_\varphi\mathbf{r} = A_r\hat{\mathbf{e}}_r + A_\theta\hat{\mathbf{e}}_\theta + A_\varphi\hat{\mathbf{e}}_\varphi = \boxed{\mathbf{A}}.$$

The derivatives act on $\mathbf{r}$, not on the coefficients of $\mathbf{A}$.

3.10.27 Find the spherical coordinate components of the velocity and acceleration of a moving particle:

$$\begin{aligned} v_r &= \dot r, & a_r &= \ddot r - r\dot\theta^2 - r\sin^2\theta\,\dot\varphi^2,\\ v_\theta &= r\dot\theta, & a_\theta &= r\ddot\theta + 2\dot r\dot\theta - r\sin\theta\cos\theta\,\dot\varphi^2,\\ v_\varphi &= r\sin\theta\,\dot\varphi, & a_\varphi &= r\sin\theta\,\ddot\varphi + 2\dot r\sin\theta\,\dot\varphi + 2r\cos\theta\,\dot\theta\dot\varphi. \end{aligned}$$

Hint. $\mathbf{r}(t) = \hat{\mathbf{e}}_r(t)r(t)$
$= [\hat{\mathbf{e}}_x\sin\theta(t)\cos\varphi(t) + \hat{\mathbf{e}}_y\sin\theta(t)\sin\varphi(t) + \hat{\mathbf{e}}_z\cos\theta(t)]r(t)$.

Note. The dot in $\dot r$, $\dot\theta$, $\dot\varphi$ means time derivative: $\dot r = dr/dt$, $\dot\theta = d\theta/dt$, $\dot\varphi = d\varphi/dt$.

Solution.

The unit vectors depend on t through $\theta(t)$ and $\varphi(t)$, so by the chain rule $\dot{\hat{\mathbf{e}}} = \dot\theta\,\partial_\theta\hat{\mathbf{e}} + \dot\varphi\,\partial_\varphi\hat{\mathbf{e}}$ for each of them. With the basis derivatives in Exercise 3.10.22,

$$\begin{aligned} \dot{\hat{\mathbf{e}}}_r &= \dot\theta\hat{\mathbf{e}}_\theta + \sin\theta\dot\varphi\hat{\mathbf{e}}_\varphi,\\ \dot{\hat{\mathbf{e}}}_\theta &= -\dot\theta\hat{\mathbf{e}}_r + \cos\theta\dot\varphi\hat{\mathbf{e}}_\varphi,\\ \dot{\hat{\mathbf{e}}}_\varphi &= -\sin\theta\dot\varphi\hat{\mathbf{e}}_r - \cos\theta\dot\varphi\hat{\mathbf{e}}_\theta. \end{aligned}$$

Differentiate $\mathbf{r} = r\hat{\mathbf{e}}_r$ by the product rule:

$$\mathbf{v} = \dot r\hat{\mathbf{e}}_r + r\dot{\hat{\mathbf{e}}}_r = \dot r\hat{\mathbf{e}}_r + r\dot\theta\hat{\mathbf{e}}_\theta + r\sin\theta\dot\varphi\hat{\mathbf{e}}_\varphi,$$

so $\boxed{v_r = \dot r,\ v_\theta = r\dot\theta,\ v_\varphi = r\sin\theta\,\dot\varphi}$. Differentiating the three terms of $\mathbf{v}$ by the product rule (with $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$) and inserting the basis derivatives above gives

$$\begin{aligned} \frac{d}{dt}(\dot r\hat{\mathbf{e}}_r) &= \ddot r\hat{\mathbf{e}}_r + \dot r\dot\theta\hat{\mathbf{e}}_\theta + \dot r\sin\theta\dot\varphi\hat{\mathbf{e}}_\varphi,\\ \frac{d}{dt}(r\dot\theta\hat{\mathbf{e}}_\theta) &= (\dot r\dot\theta + r\ddot\theta)\hat{\mathbf{e}}_\theta - r\dot\theta^2\hat{\mathbf{e}}_r + r\cos\theta\dot\theta\dot\varphi\hat{\mathbf{e}}_\varphi,\\ \frac{d}{dt}(r\sin\theta\dot\varphi\hat{\mathbf{e}}_\varphi) &= (\dot r\sin\theta\dot\varphi + r\cos\theta\dot\theta\dot\varphi + r\sin\theta\ddot\varphi)\hat{\mathbf{e}}_\varphi\\ &\qquad - r\sin^2\theta\dot\varphi^2\hat{\mathbf{e}}_r - r\sin\theta\cos\theta\dot\varphi^2\hat{\mathbf{e}}_\theta. \end{aligned}$$

Collecting each basis vector (for example, the $\hat{\mathbf{e}}_\theta$ terms are $\dot r\dot\theta$, $\dot r\dot\theta + r\ddot\theta$, and $-r\sin\theta\cos\theta\dot\varphi^2$) yields

$$\boxed{\begin{aligned} a_r &= \ddot r - r\dot\theta^2 - r\sin^2\theta\dot\varphi^2,\\ a_\theta &= r\ddot\theta + 2\dot r\dot\theta - r\sin\theta\cos\theta\dot\varphi^2,\\ a_\varphi &= r\sin\theta\ddot\varphi + 2\dot r\sin\theta\dot\varphi + 2r\cos\theta\dot\theta\dot\varphi. \end{aligned}}$$

3.10.28 Express $\partial/\partial x$, $\partial/\partial y$, $\partial/\partial z$ in spherical polar coordinates.

$$\textit{ANS.}\quad \begin{aligned} \frac{\partial}{\partial x} &= \sin\theta\cos\varphi\frac{\partial}{\partial r} + \cos\theta\cos\varphi\frac{1}{r}\frac{\partial}{\partial\theta} - \frac{\sin\varphi}{r\sin\theta}\frac{\partial}{\partial\varphi},\\ \frac{\partial}{\partial y} &= \sin\theta\sin\varphi\frac{\partial}{\partial r} + \cos\theta\sin\varphi\frac{1}{r}\frac{\partial}{\partial\theta} + \frac{\cos\varphi}{r\sin\theta}\frac{\partial}{\partial\varphi},\\ \frac{\partial}{\partial z} &= \cos\theta\frac{\partial}{\partial r} - \sin\theta\frac{1}{r}\frac{\partial}{\partial\theta}. \end{aligned}$$

Hint. Equate ∇_{xyz} and $\nabla_{r\theta\varphi}$.

Solution.

The chain rule is $\partial_x = r_x\partial_r + \theta_x\partial_\theta + \varphi_x\partial_\varphi$, and similarly for y, z; subscripts on r, θ, φ denote partial derivatives with respect to x, y, z. From $r^2 = x^2+y^2+z^2$, $2rr_x = 2x$, so $r_x = x/r = \sin\theta\cos\varphi$; likewise $r_y = y/r = \sin\theta\sin\varphi$ and $r_z = z/r = \cos\theta$. Differentiating $\cos\theta = z/r$ with respect to x gives $-\sin\theta\,\theta_x = -zr_x/r^2 = -zx/r^3$, hence

$$\theta_x = \frac{zx}{r^3\sin\theta} = \frac{(r\cos\theta)(r\sin\theta\cos\varphi)}{r^3\sin\theta} = \frac{\cos\theta\cos\varphi}{r},$$

and in the same way $\theta_y = zy/(r^3\sin\theta) = \cos\theta\sin\varphi/r$. With respect to z, $-\sin\theta\,\theta_z = 1/r - z^2/r^3 = (x^2+y^2)/r^3 = \sin^2\theta/r$, so $\theta_z = -\sin\theta/r$. Finally, on a regular local branch of $\varphi = \operatorname{atan2}(y,x)$ with $x \neq 0$, differentiate $\tan\varphi = y/x$: $\sec^2\varphi\,\varphi_x = -y/x^2$ and $\sec^2\varphi\,\varphi_y = 1/x$, where $\sec^2\varphi = 1+\tan^2\varphi = (x^2+y^2)/x^2$; hence $\varphi_x = -y/(x^2+y^2)$ and $\varphi_y = x/(x^2+y^2)$ (where $x = 0$, differentiating $\cot\varphi = x/y$ gives the same), while $\varphi_z = 0$ because φ does not depend on z. With $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, and $x^2+y^2 = r^2\sin^2\theta$ these are $-\sin\varphi/(r\sin\theta)$, $\cos\varphi/(r\sin\theta)$, 0. The full Jacobian obtained this way is

	∂_x	∂_y	∂_z
r	$\sin\theta\cos\varphi$	$\sin\theta\sin\varphi$	$\cos\theta$
θ	$\cos\theta\cos\varphi/r$	$\cos\theta\sin\varphi/r$	$-\sin\theta/r$
φ	$-\sin\varphi/(r\sin\theta)$	$\cos\varphi/(r\sin\theta)$	0

Substitution of each column into the chain rule gives all three displayed operator formulas in the question. They hold on $r > 0, \sin\theta \neq 0$, where these coordinates are nonsingular. Following the hint gives the same result faster: dotting $\hat{\mathbf{e}}_x$ into $\hat{\mathbf{e}}_x\partial_x + \hat{\mathbf{e}}_y\partial_y + \hat{\mathbf{e}}_z\partial_z = \hat{\mathbf{e}}_r\partial_r + \hat{\mathbf{e}}_\theta r^{-1}\partial_\theta + \hat{\mathbf{e}}_\varphi(r\sin\theta)^{-1}\partial_\varphi$ and using the projections $\hat{\mathbf{e}}_x\cdot\hat{\mathbf{e}}_r = \sin\theta\cos\varphi$, $\hat{\mathbf{e}}_x\cdot\hat{\mathbf{e}}_\theta = \cos\theta\cos\varphi$, $\hat{\mathbf{e}}_x\cdot\hat{\mathbf{e}}_\varphi = -\sin\varphi$ of Exercise 3.10.19 reproduces the ∂_x formula; $\hat{\mathbf{e}}_y$ and $\hat{\mathbf{e}}_z$ give the other two.

3.10.29 Using results from Exercise 3.10.28, show that

$$-i\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) = -i\frac{\partial}{\partial\varphi}.$$

This is the quantum mechanical operator corresponding to the z-component of orbital angular momentum.

Solution.

Insert the derivatives from Exercise 3.10.28 and $x = r\sin\theta\cos\varphi, y = r\sin\theta\sin\varphi$. In $x\partial_y - y\partial_x$, the coefficient of ∂_r is $x\sin\theta\sin\varphi - y\sin\theta\cos\varphi = r\sin^2\theta(\cos\varphi\sin\varphi - \sin\varphi\cos\varphi) = 0$. The coefficient of ∂_θ is $(x\cos\theta\sin\varphi - y\cos\theta\cos\varphi)/r = \sin\theta\cos\theta(\cos\varphi\sin\varphi - \sin\varphi\cos\varphi) = 0$. The coefficient of ∂_φ is $x\cos\varphi/(r\sin\theta) + y\sin\varphi/(r\sin\theta) = \cos^2\varphi + \sin^2\varphi = 1$. Thus $x\partial_y - y\partial_x = \partial_\varphi$ and $\boxed{L_z = -i\partial_\varphi}$.

3.10.30 With the quantum mechanical orbital angular momentum operator defined as $\mathbf{L} = -i(\mathbf{r}\times\nabla)$, show that

(a) $L_x + iL_y = e^{i\varphi}\left(\dfrac{\partial}{\partial\theta} + i\cot\theta\dfrac{\partial}{\partial\varphi}\right)$,

(b) $L_x - iL_y = -e^{-i\varphi}\left(\dfrac{\partial}{\partial\theta} - i\cot\theta\dfrac{\partial}{\partial\varphi}\right)$.

Solution.

Using $\mathbf{r} = r\hat{\mathbf{e}}_r$ in the spherical gradient of Eq. (3.156), the radial term drops out ($\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_r = 0$), and $\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\varphi$, $\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_\varphi = -\hat{\mathbf{e}}_\theta$ give $\mathbf{r}\times\nabla = \hat{\mathbf{e}}_\varphi\partial_\theta - \hat{\mathbf{e}}_\theta\csc\theta\,\partial_\varphi$, so

$$\mathbf{L} = -i(\mathbf{r}\times\nabla) = i\left(\hat{\mathbf{e}}_\theta\csc\theta\,\partial_\varphi - \hat{\mathbf{e}}_\varphi\partial_\theta\right).$$

Projection on x, y, with the Cartesian components $\hat{\mathbf{e}}_\theta = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$ and $\hat{\mathbf{e}}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ of Exercise 3.10.18, gives $L_x = i(\sin\varphi\,\partial_\theta + \cot\theta\cos\varphi\,\partial_\varphi)$ and $L_y = i(-\cos\varphi\,\partial_\theta + \cot\theta\sin\varphi\,\partial_\varphi)$ (this is Exercise 3.10.32(a),(b)). Therefore, using $i\cdot i = -1$,

$$\begin{aligned} L_x + iL_y &= (\cos\varphi + i\sin\varphi)\partial_\theta + (i\cos\varphi - \sin\varphi)\cot\theta\,\partial_\varphi \\ &= \boxed{e^{i\varphi}(\partial_\theta + i\cot\theta\,\partial_\varphi)}, \\ L_x - iL_y &= (-\cos\varphi + i\sin\varphi)\partial_\theta + (i\cos\varphi + \sin\varphi)\cot\theta\,\partial_\varphi \\ &= \boxed{-e^{-i\varphi}(\partial_\theta - i\cot\theta\,\partial_\varphi)}, \end{aligned}$$

by Euler's formula: $\cos\varphi + i\sin\varphi = e^{i\varphi}$, $i\cos\varphi - \sin\varphi = i(\cos\varphi + i\sin\varphi) = ie^{i\varphi}$, $-\cos\varphi + i\sin\varphi = -e^{-i\varphi}$, and $i\cos\varphi + \sin\varphi = i(\cos\varphi - i\sin\varphi) = ie^{-i\varphi}$.

3.10.31 Verify that $\mathbf{L}\times\mathbf{L} = i\mathbf{L}$ in spherical polar coordinates. $\mathbf{L} = -i(\mathbf{r}\times\nabla)$, the quantum mechanical orbital angular momentum operator.

Written in component form, this relation is

$$L_yL_z - L_zL_y = iL_x, \quad L_zL_x - L_xL_z = iL_y, \quad L_xL_y - L_yL_x = iL_z.$$

Using the commutator notation, $[A, B] = AB - BA$, and the definition of the Levi-Civita symbol ϵ_{ijk}, the above can also be written

$$[L_i, L_j] = i\epsilon_{ijk}L_k,$$

where i, j, k are x, y, z in any order.

Hint. Use spherical polar coordinates for $\mathbf{L}$ but Cartesian components for the cross product.

Solution.

As printed on textbook p. 201, the second component relation has $-L_y$ on the right; it must be $[L_z, L_x] = iL_y$, the cyclic permutation $x\to z\to y\to x$ of $[L_x, L_y] = iL_z$, as the calculation below confirms. Use the Cartesian components expressed in spherical coordinates (Exercise 3.10.32(b)): $L_x = iX$, $L_y = iY$, $L_z = -i\partial_\varphi$, where

$$X = \sin\varphi\,\partial_\theta + \cot\theta\cos\varphi\,\partial_\varphi, \quad Y = -\cos\varphi\,\partial_\theta + \cot\theta\sin\varphi\,\partial_\varphi.$$

For first-order operators $X = a^\theta\partial_\theta + a^\varphi\partial_\varphi$ and $Y = b^\theta\partial_\theta + b^\varphi\partial_\varphi$, applying both orders to a smooth f and subtracting, the second-derivative terms $a^\mu b^\nu \partial_\mu\partial_\nu f$ cancel because mixed partials commute, leaving

$$[X,Y] = (Xb^\theta - Ya^\theta)\partial_\theta + (Xb^\varphi - Ya^\varphi)\partial_\varphi.$$

Put $t = \cot\theta$ and $q = \csc^2\theta$, so $\partial_\theta t = -q$. Here $a^\theta = \sin\varphi$, $a^\varphi = t\cos\varphi$, $b^\theta = -\cos\varphi$, $b^\varphi = t\sin\varphi$. The coefficient of ∂_θ in $[X,Y]$ is $Xb^\theta - Ya^\theta = t\cos\varphi\sin\varphi - t\sin\varphi\cos\varphi = 0$. The coefficient of ∂_φ is

$$Xb^\varphi - Ya^\varphi = [-q\sin^2\varphi + t^2\cos^2\varphi] - [q\cos^2\varphi - t^2\sin^2\varphi] = t^2 - q = -1,$$

since $\cot^2\theta - \csc^2\theta = -1$. Thus $[L_x, L_y] = i^2[X,Y] = -[X,Y] = \partial_\varphi = iL_z$. For the other two commutators, $(-i)(i) = 1$ and $(i)(-i) = 1$ give $[L_z, L_x] = [\partial_\varphi, X]$ and $[L_y, L_z] = [Y, \partial_\varphi]$. The commutator formula above holds for any two first-order operators; with ∂_φ as one of them, whose coefficients 0 and 1 are constant and hence annihilated by X and Y, it reduces to $[\partial_\varphi, X] = (\partial_\varphi a^\theta)\partial_\theta + (\partial_\varphi a^\varphi)\partial_\varphi$ and $[Y, \partial_\varphi] = -(\partial_\varphi b^\theta)\partial_\theta - (\partial_\varphi b^\varphi)\partial_\varphi$:

$$\begin{aligned}
[L_z, L_x] &= [\partial_\varphi, X] = \cos\varphi\,\partial_\theta - \cot\theta\sin\varphi\,\partial_\varphi = -Y = iL_y,\\
[L_y, L_z] &= [Y, \partial_\varphi] = -\sin\varphi\,\partial_\theta - \cot\theta\cos\varphi\,\partial_\varphi = -X = iL_x.
\end{aligned}$$

The components of the cross product of the operator vector with itself are $(\mathbf{L}\times\mathbf{L})_x = L_yL_z - L_zL_y$ and cyclically, so

$$\big([L_y, L_z], [L_z, L_x], [L_x, L_y]\big) = \boxed{i\mathbf{L}}.$$

3.10.32

(a) Using Eq. (3.156) show that

$$\mathbf{L} = -i(\mathbf{r}\times\nabla) = i\left(\hat{\mathbf{e}}_\theta \frac{1}{\sin\theta}\frac{\partial}{\partial\varphi} - \hat{\mathbf{e}}_\varphi\frac{\partial}{\partial\theta}\right).$$

(b) Resolving $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\varphi$ into Cartesian components, determine L_x, L_y, and L_z in terms of θ, φ, and their derivatives.

(c) From $L^2 = L_x^2 + L_y^2 + L_z^2$ show that

$$L^2 = -\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) - \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} = -r^2\nabla^2 + \frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right).$$

Solution.

(a) With $\mathbf{r} = r\hat{\mathbf{e}}_r$ and Eq. (3.156),

$$\mathbf{r}\times\nabla = r\hat{\mathbf{e}}_r \times \Big(\hat{\mathbf{e}}_r\partial_r + \hat{\mathbf{e}}_\theta\frac{1}{r}\partial_\theta + \hat{\mathbf{e}}_\varphi\frac{1}{r\sin\theta}\partial_\varphi\Big).$$

The radial derivative term vanishes because $\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_r = 0$. Since $\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\varphi$ and $\hat{\mathbf{e}}_r\times\hat{\mathbf{e}}_\varphi = -\hat{\mathbf{e}}_\theta$, $\mathbf{r}\times\nabla = \hat{\mathbf{e}}_\varphi\partial_\theta - \hat{\mathbf{e}}_\theta\csc\theta\,\partial_\varphi$, and multiplying by $-i$,

$$\boxed{\mathbf{L} = i(\hat{\mathbf{e}}_\theta\csc\theta\,\partial_\varphi - \hat{\mathbf{e}}_\varphi\partial_\theta)}.$$

The unit vectors stand to the left of the derivatives, so they are not differentiated. **(b)** Insert the Cartesian basis components of Exercise 3.10.18: $L_x = \hat{\mathbf{e}}_x\cdot\mathbf{L}$ uses $\hat{\mathbf{e}}_x\cdot\hat{\mathbf{e}}_\theta = \cos\theta\cos\varphi$ and

$\hat{\mathbf{e}}_x \cdot \hat{\mathbf{e}}_\varphi = -\sin\varphi$; L_y uses $\cos\theta\sin\varphi$ and $\cos\varphi$; L_z uses $-\sin\theta$ and 0. With $\cos\theta\csc\theta = \cot\theta$ and $\sin\theta\csc\theta = 1$,

$$\boxed{\begin{aligned} L_x &= i(\sin\varphi\,\partial_\theta + \cot\theta\cos\varphi\,\partial_\varphi), \\ L_y &= i(-\cos\varphi\,\partial_\theta + \cot\theta\sin\varphi\,\partial_\varphi), \\ L_z &= -i\partial_\varphi. \end{aligned}}$$

(c) Apply the squared operators to an arbitrary smooth f, including derivatives of their coefficients. Write $s = \sin\varphi$, $c = \cos\varphi$, $t = \cot\theta$, $q = \csc^2\theta$, so that $\partial_\varphi s = c$, $\partial_\varphi c = -s$, $\partial_\theta t = -q$, and subscripts on f denote partial derivatives. Since $L_x = i(s\partial_\theta + tc\partial_\varphi)$, $L_x^2 f = -(s\partial_\theta + tc\partial_\varphi)(sf_\theta + tcf_\varphi)$, where

$$\begin{aligned} (s\partial_\theta + tc\partial_\varphi)(sf_\theta + tcf_\varphi) &= s(sf_{\theta\theta} - qcf_\varphi + tcf_{\theta\varphi}) + tc(cf_\theta + sf_{\theta\varphi} - tsf_\varphi + tcf_{\varphi\varphi}) \\ &= s^2 f_{\theta\theta} + 2tscf_{\theta\varphi} + t^2c^2 f_{\varphi\varphi} + tc^2 f_\theta - sc(q + t^2)f_\varphi. \end{aligned}$$

$L_y = i(-c\partial_\theta + ts\partial_\varphi)$ is L_x with $s \to -c$, $c \to s$ (that is, $\varphi \to \varphi - \pi/2$, which preserves $\partial_\varphi s = c$, $\partial_\varphi c = -s$), so the same substitution gives $L_y^2 f$. Altogether

$$\begin{aligned} L_x^2 f &= -s^2 f_{\theta\theta} - 2tscf_{\theta\varphi} - t^2c^2 f_{\varphi\varphi} - tc^2 f_\theta + sc(q + t^2)f_\varphi, \\ L_y^2 f &= -c^2 f_{\theta\theta} + 2tscf_{\theta\varphi} - t^2s^2 f_{\varphi\varphi} - ts^2 f_\theta - sc(q + t^2)f_\varphi, \\ L_z^2 f &= (-i)^2 f_{\varphi\varphi} = -f_{\varphi\varphi}. \end{aligned}$$

Mixed terms and the terms proportional to f_φ cancel. Using $s^2 + c^2 = 1$ and $1 + t^2 = q$,

$$L^2 f = -f_{\theta\theta} - \cot\theta f_\theta - \csc^2\theta f_{\varphi\varphi} = -\frac{1}{\sin\theta}\partial_\theta(\sin\theta f_\theta) - \frac{1}{\sin^2\theta} f_{\varphi\varphi},$$

since $(\sin\theta)^{-1}\partial_\theta(\sin\theta f_\theta) = f_{\theta\theta} + \cot\theta f_\theta$. Multiplying the Laplacian of Exercise 3.10.22 by r^2 gives

$$r^2\nabla^2 f = \partial_r(r^2\partial_r f) + \frac{1}{\sin\theta}\partial_\theta(\sin\theta f_\theta) + \frac{1}{\sin^2\theta} f_{\varphi\varphi} = \partial_r(r^2\partial_r f) - L^2 f,$$

so

$$\boxed{L^2 = -r^2\nabla^2 + \partial_r(r^2\partial_r)}.$$

In the Instructor's Manual's part (a), p. 56, the coefficient of $\hat{\mathbf{e}}_\theta \csc\theta\,\partial_\varphi$ is printed as -1. It must be $+i$, as follows directly from $-i\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\varphi = +i\hat{\mathbf{e}}_\theta$.

3.10.33 With $\mathbf{L} = -i\mathbf{r} \times \nabla$, verify the operator identities

(a) $\nabla = \hat{\mathbf{e}}_r \dfrac{\partial}{\partial r} - i\dfrac{\mathbf{r} \times \mathbf{L}}{r^2}$,

(b) $\mathbf{r}\nabla^2 - \nabla\left(1 + r\dfrac{\partial}{\partial r}\right) = i\nabla \times \mathbf{L}$.

Solution.

In (b) the first factor is the position vector $\mathbf{r}$, not the scalar r: every other term of (b) is a vector operator. All identities below act on an arbitrary smooth scalar f. **(a)** The ordinary triple-product identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$ applies after ∇ has acted, with $\mathbf{a} = \mathbf{b} = \mathbf{r}$, $\mathbf{c} = \nabla f$:

$$\mathbf{r} \times \mathbf{L}f = -i\,\mathbf{r} \times (\mathbf{r} \times \nabla f) = -i[\mathbf{r}(\mathbf{r} \cdot \nabla f) - r^2\nabla f].$$

Since $\mathbf{r}\cdot\nabla f = r\hat{\mathbf{e}}_r\cdot\nabla f = r\partial_r f$ (the radial component of the gradient, Eq. (3.156)) and $(-i)(-i) = -1$,

$$-\frac{i}{r^2}\,\mathbf{r}\times\mathbf{L}f = -\frac{1}{r^2}[\mathbf{r}\,r\partial_r f - r^2\nabla f] = \nabla f - \hat{\mathbf{e}}_r\partial_r f.$$

Rearranging, $\nabla f = \hat{\mathbf{e}}_r\partial_r f - i\,\mathbf{r}\times\mathbf{L}f/r^2$ for every f, which is (a).

(b) Set $D = \mathbf{r}\cdot\nabla = r\partial_r$. Since $i\mathbf{L}f = i(-i)\mathbf{r}\times\nabla f = \mathbf{r}\times\nabla f$, the curl-product rule of Exercise 3.6.5, with $\mathbf{A} = \mathbf{r}$ and $\mathbf{B} = \nabla f$, gives

$$\begin{aligned} i\nabla\times(\mathbf{L}f) &= \nabla\times(\mathbf{r}\times\nabla f) \\ &= (\nabla f\cdot\nabla)\mathbf{r} - (\mathbf{r}\cdot\nabla)\nabla f - \nabla f\,(\nabla\cdot\mathbf{r}) + \mathbf{r}\,\nabla^2 f \\ &= \mathbf{r}\,\nabla^2 f - 2\nabla f - D(\nabla f), \end{aligned}$$

because $(\nabla f\cdot\nabla)\mathbf{r} = \nabla f$ (Exercise 3.10.26 with $\mathbf{A} = \nabla f$) and $\nabla\cdot\mathbf{r} = 3$. Componentwise, with the repeated index j summed, $\partial_i(Df) = \partial_i(x_j\partial_j f) = \delta_{ij}\partial_j f + x_j\partial_j\partial_i f = \partial_i f + x_j\partial_j\partial_i f$, so $\nabla(Df) = \nabla f + D(\nabla f)$, i.e. $D(\nabla f) = \nabla(Df) - \nabla f$. Substitution yields

$$i\nabla\times(\mathbf{L}f) = \mathbf{r}\nabla^2 f - 2\nabla f - \nabla(Df) + \nabla f = \mathbf{r}\nabla^2 f - \nabla(f + r\partial_r f),$$

which is precisely the operator identity (b), $\mathbf{r}\nabla^2 - \nabla(1 + r\partial_r) = i\nabla\times\mathbf{L}$.

3.10.34 Show that the following three forms (spherical coordinates) of $\nabla^2\psi(r)$ are equivalent:

$$\text{(a)}\ \frac{1}{r^2}\frac{d}{dr}\left[r^2\frac{d\psi(r)}{dr}\right];\quad \text{(b)}\ \frac{1}{r}\frac{d^2}{dr^2}[r\psi(r)];\quad \text{(c)}\ \frac{d^2\psi(r)}{dr^2} + \frac{2}{r}\frac{d\psi(r)}{dr}.$$

The second form is particularly convenient in establishing a correspondence between spherical polar and Cartesian descriptions of a problem.

Solution.

For $r > 0$, expand the first expression:

$$\frac{1}{r^2}\frac{d}{dr}(r^2\psi') = \frac{2r\psi' + r^2\psi''}{r^2} = \psi'' + \frac{2}{r}\psi'.$$

For the second, first differentiate $r\psi$ to get $\psi + r\psi'$, then differentiate again to get $2\psi' + r\psi''$. Division by r again gives $\psi'' + 2\psi'/r$. Thus all three expressions agree. At $r = 0$, values are obtained by regular limits when they exist; the displayed divisions by r do not themselves define a value at the origin.

3.10.35 A certain force field is given in spherical polar coordinates by

$$\mathbf{F} = \hat{\mathbf{e}}_r\frac{2P\cos\theta}{r^3} + \hat{\mathbf{e}}_\theta\frac{P}{r^3}\sin\theta,\quad r \geq P/2.$$

(a) Examine $\nabla\times\mathbf{F}$ to see if a potential exists.

(b) Calculate $\oint\mathbf{F}\cdot d\mathbf{r}$ for a unit circle in the plane $\theta = \pi/2$. What does this indicate about the force being conservative or nonconservative?

(c) If you believe that $\mathbf{F}$ may be described by $\mathbf{F} = -\nabla\psi$, find ψ. Otherwise simply state that no acceptable potential exists.

Take P constant and $r > 0$. In (b), the unit circle belongs to the stated region only if $P/2 \le 1$ in the chosen units.

Solution.

(a) Use the spherical curl components of Exercise 3.10.23 with $F_r = 2P\cos\theta/r^3$, $F_\theta = P\sin\theta/r^3$, $F_\varphi = 0$. For constant P, $\partial_r(rF_\theta) = \partial_r(P\sin\theta/r^2) = -2P\sin\theta/r^3$ and $\partial_\theta F_r = -2P\sin\theta/r^3$, so the only possibly nonzero curl component is

$$\begin{aligned}(\nabla\times\mathbf{F})_\varphi &= \frac{1}{r}[\partial_r(rF_\theta) - \partial_\theta F_r] \\ &= \frac{1}{r}\left[-\frac{2P\sin\theta}{r^3} - \left(-\frac{2P\sin\theta}{r^3}\right)\right] = 0.\end{aligned}$$

The other components, $(r\sin\theta)^{-1}[\partial_\theta(\sin\theta F_\varphi) - \partial_\varphi F_\theta]$ and $r^{-1}[(\sin\theta)^{-1}\partial_\varphi F_r - \partial_r(rF_\varphi)]$, vanish because $F_\varphi = 0$ and no component depends on φ. Thus $\nabla\times\mathbf{F} = 0$, so a potential may exist; (c) finds it.

(b) On the equatorial unit circle ($r = 1$, $\theta = \pi/2$), $d\mathbf{r} = \hat{\mathbf{e}}_\varphi\, r\sin\theta\, d\varphi = \hat{\mathbf{e}}_\varphi d\varphi$, so the integrand $F_\varphi\, d\varphi$ is zero and $\boxed{\oint \mathbf{F}\cdot d\mathbf{r} = 0}$. (The circle has $r = 1$, so it lies in the stated region $r \ge P/2$ when $P \le 2$; the field formulas themselves hold at every $r > 0$.) This is consistent with a conservative force, but a zero integral on one loop alone would not prove conservativeness; here an explicit potential settles it.

(c) With the gradient of Eq. (3.156), $\mathbf{F} = -\nabla\psi$ requires $-\partial_r\psi = F_r$, $-r^{-1}\partial_\theta\psi = F_\theta$, and $-(r\sin\theta)^{-1}\partial_\varphi\psi = F_\varphi = 0$. Integrate $-\partial_r\psi = 2P\cos\theta/r^3$ in r (using $\int r^{-3}dr = -r^{-2}/2$): $\psi = P\cos\theta/r^2 + f(\theta,\varphi)$. The equation $-r^{-1}\partial_\theta\psi = P\sin\theta/r^3$ becomes $P\sin\theta/r^3 - r^{-1}\partial_\theta f = P\sin\theta/r^3$, so $\partial_\theta f = 0$, and $F_\varphi = 0$ gives $\partial_\varphi f = 0$. Hence

$$\boxed{\psi = P\cos\theta/r^2 + C}.$$

It is single-valued and regular throughout the stated region away from the origin, so the force is conservative there.

3.10.36

(a) Show that $\mathbf{A} = -\hat{\mathbf{e}}_\varphi\cot\theta/r$ is a solution of $\nabla\times\mathbf{A} = \hat{\mathbf{e}}_r/r^2$.

(b) Show that this spherical polar coordinate solution agrees with the solution given for Exercise 3.9.5:

$$\mathbf{A} = \hat{\mathbf{e}}_x\frac{yz}{r(x^2+y^2)} - \hat{\mathbf{e}}_y\frac{xz}{r(x^2+y^2)}.$$

Note that the solution diverges for $\theta = 0, \pi$ corresponding to $x, y = 0$.

(c) Finally, show that $\mathbf{A} = -\hat{\mathbf{e}}_\theta\varphi\sin\theta/r$ is a solution. Note that although this solution does not diverge ($r \ne 0$), it is no longer single-valued for all possible azimuth angles.

Use a regular local spherical chart with a chosen azimuth branch; the potentials are not globally regular single-valued potentials on all of $r > 0$.

Solution.

(a) For $A_\varphi = -\cot\theta/r$, $A_r = A_\theta = 0$, the spherical curl components of Exercise 3.10.23 give

$$(\nabla\times\mathbf{A})_r = \frac{1}{r\sin\theta}\partial_\theta(\sin\theta A_\varphi) = \frac{1}{r\sin\theta}\partial_\theta(-\cos\theta/r) = \frac{1}{r\sin\theta}\cdot\frac{\sin\theta}{r} = \frac{1}{r^2}.$$

The other nontrivial candidate is $(\nabla\times\mathbf{A})_\theta = -r^{-1}\partial_r(rA_\varphi) = -r^{-1}\partial_r(-\cot\theta) = 0$, and the φ component $r^{-1}[\partial_r(rA_\theta) - \partial_\theta A_r]$ is zero. Hence $\nabla\times\mathbf{A} = \hat{\mathbf{e}}_r/r^2$. **(b)** With $\rho = r\sin\theta$ and $z = r\cos\theta$, use $\cot\theta = z/\rho$, $\rho^2 = x^2+y^2$, and $\hat{\mathbf{e}}_\varphi = (-\sin\varphi, \cos\varphi, 0) = (-y/\rho, x/\rho, 0)$:

$$\mathbf{A} = -\frac{z}{r\rho}(-y/\rho, x/\rho, 0) = \left(\frac{yz}{r(x^2+y^2)}, -\frac{xz}{r(x^2+y^2)}, 0\right).$$

This is the vector potential of Exercise 3.9.5. **(c)** For $A_\theta = -\varphi\sin\theta/r$, $A_r = A_\varphi = 0$,

$$(\nabla\times\mathbf{A})_r = -\frac{1}{r\sin\theta}\partial_\varphi A_\theta = \frac{1}{r\sin\theta}\cdot\frac{\sin\theta}{r} = \frac{1}{r^2}, \qquad (\nabla\times\mathbf{A})_\varphi = \frac{1}{r}\partial_r(rA_\theta) = \frac{1}{r}\partial_r(-\varphi\sin\theta) = 0.$$

The θ component $r^{-1}[(\sin\theta)^{-1}\partial_\varphi A_r - \partial_r(rA_\varphi)]$ is also zero. Both are local potentials: (a) excludes the polar axis, and (c) requires an azimuthal branch. Neither is a globally regular single-valued potential for the monopole field, and none can be: for such an $\mathbf{A}$ on $r>0$, Stokes' theorem on the upper and lower hemispheres of the unit sphere (both bounded by the equator, traversed in opposite senses) would make the total flux of $\nabla\times\mathbf{A}$ through the sphere zero, whereas the flux of $\hat{\mathbf{e}}_r/r^2$ is 4π. Indeed increasing φ by 2π changes the latter potential by $-2\pi\sin\theta\,\hat{\mathbf{e}}_\theta/r = \nabla(2\pi\cos\theta)$. The curl is unchanged within overlapping regular patches, while the potential itself is not periodic in φ.

3.10.37 An electric dipole of moment $\mathbf{p}$ is located at the origin. The dipole creates an electric potential at $\mathbf{r}$ given by

$$\psi(\mathbf{r}) = \frac{\mathbf{p}\cdot\mathbf{r}}{4\pi\epsilon_0 r^3}.$$

Find the electric field, $\mathbf{E} = -\nabla\psi$ at $\mathbf{r}$.

Solution.

The dipole moment is constant and $r>0$. Apply the product rule $\nabla(fg) = f\nabla g + g\nabla f$ to the scalars $f = \mathbf{p}\cdot\mathbf{r}$ and $g = r^{-3}$:

$$\nabla\psi = \frac{1}{4\pi\epsilon_0}\left[r^{-3}\nabla(\mathbf{p}\cdot\mathbf{r}) + (\mathbf{p}\cdot\mathbf{r})\nabla(r^{-3})\right].$$

Here $\nabla(\mathbf{p}\cdot\mathbf{r}) = \mathbf{p}$, since $\partial_i(p_j x_j) = p_i$ for constant p_j, and $\nabla r^{-3} = -3r^{-4}\nabla r = -3r^{-4}\hat{\mathbf{e}}_r = -3\mathbf{r}/r^5$. Thus

$$\mathbf{E} = -\nabla\psi = \frac{1}{4\pi\epsilon_0}\left[\frac{3(\mathbf{p}\cdot\mathbf{r})\mathbf{r}}{r^5} - \frac{\mathbf{p}}{r^3}\right],$$

and writing $\mathbf{r} = r\hat{\mathbf{e}}_r$,

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^3}\left[3(\mathbf{p}\cdot\hat{\mathbf{e}}_r)\hat{\mathbf{e}}_r - \mathbf{p}\right]}.$$

For $\mathbf{p} = p\hat{\mathbf{e}}_z = p(\cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta)$, $\mathbf{p}\cdot\hat{\mathbf{e}}_r = p\cos\theta$ and the bracket is $3p\cos\theta\,\hat{\mathbf{e}}_r - p\cos\theta\,\hat{\mathbf{e}}_r + p\sin\theta\,\hat{\mathbf{e}}_\theta$, so this becomes $\mathbf{E} = p(2\cos\theta\,\hat{\mathbf{e}}_r + \sin\theta\,\hat{\mathbf{e}}_\theta)/(4\pi\epsilon_0 r^3)$. This is the field of Exercise 3.10.35 with $P = p/(4\pi\epsilon_0)$, and correspondingly $\psi = p\cos\theta/(4\pi\epsilon_0 r^2)$ is the potential found there.

4 Chapter 4. Tensors and Differential Forms

Problems

4.1 Tensor Analysis

4.1.1 Show that if all the components of any tensor of any rank vanish in one particular coordinate system, they vanish in all coordinate systems.

Note. This point takes on special importance in the four-dimensional (4-D) curved space of general relativity. If a quantity, expressed as a tensor, exists in one coordinate system, it exists in all coordinate systems and is not just a consequence of a choice of a coordinate system (as are centrifugal and Coriolis forces in Newtonian mechanics).

Solution.

Let T have p contravariant indices and q covariant indices, so its rank is $p+q$. At a given point, the components in two coordinate systems related by a nonsingular change of coordinates obey the tensor transformation law

$$T'^{i_1\cdots i_p}{}_{j_1\cdots j_q} = \frac{\partial x'^{i_1}}{\partial x^{a_1}}\cdots\frac{\partial x'^{i_p}}{\partial x^{a_p}}\frac{\partial x^{b_1}}{\partial x'^{j_1}}\cdots\frac{\partial x^{b_q}}{\partial x'^{j_q}}T^{a_1\cdots a_p}{}_{b_1\cdots b_q}.$$

Every repeated index is summed. If every unprimed component on the right is zero, each term in every sum is zero. Thus every primed component is zero, regardless of the values of the finite Jacobian factors. For rank zero the transformation law is simply $T' = T$, with the same conclusion. This proves the assertion pointwise; if the tensor vanishes throughout a region, apply the argument at each point of that region.

4.1.2 The components of tensor A are equal to the corresponding components of tensor B in one particular coordinate system, denoted by the superscript 0; that is,

$$A^0_{ij} = B^0_{ij}.$$

Show that tensor A is equal to tensor B, $A_{ij} = B_{ij}$, in all coordinate systems.

Solution.

The difference $C_{ij} = A_{ij} - B_{ij}$ is a tensor because the transformation law is linear. Denote the coordinates of the specified system by x^a_0; the label 0 (a superscript on components, a subscript on coordinates) names that system and is not a tensor index. There $C^0_{ab} = A^0_{ab} - B^0_{ab} = 0$. In any other system,

$$C_{ij} = \frac{\partial x^a_0}{\partial x^i}\frac{\partial x^b_0}{\partial x^j}C^0_{ab} = 0.$$

Consequently $A_{ij} - B_{ij} = 0$ in every system.

4.1.3 The last three components of a 4-D vector vanish in each of two reference frames. If the second reference frame is not merely a rotation of the first about the x_0 axis, meaning that at least one of the coefficients $\partial(x')^i/\partial x^0$ $(i = 1, 2, 3)$ is nonzero, show that the zeroth component vanishes in all reference frames. Translated into relativistic mechanics, this means that if momentum is conserved in two Lorentz frames, then energy is conserved in all Lorentz frames.

Solution.

Write the vector in the first frame as $V^\mu = (V^0, 0, 0, 0)$. Its spatial components in the second frame are

$$V'^i = \frac{\partial x'^i}{\partial x^\mu} V^\mu = \frac{\partial x'^i}{\partial x^0} V^0, \qquad i = 1, 2, 3.$$

By hypothesis all three left sides vanish, and at least one coefficient $\partial x'^i/\partial x^0$ is nonzero. The corresponding equation therefore gives $V^0 = 0$. All four components vanish in the first frame, so Exercise 4.1.1 gives $V'^\mu = 0$ in every frame.

For the conservation statement, apply this argument to the difference between total final and initial four-momenta,

$$\Delta P^\mu = (\Delta E/c, \Delta p_x, \Delta p_y, \Delta p_z).$$

Conservation of three-momentum in the two frames supplies the two sets of three zero components. Two frames in relative motion satisfy the hypothesis on the coefficients. Indeed, $\partial x'^i/\partial x^0$, taken at fixed x^1, x^2, x^3, is the rate at which the primed spatial coordinates of a point at rest in the first frame change with x^0; if all three vanished, every point at rest in the first frame would also be at rest in the second. For example, for a boost with velocity βc along x^1, $x'^1 = \gamma(x^1 - \beta x^0)$ with $\gamma = (1 - \beta^2)^{-1/2}$, so $\partial x'^1/\partial x^0 = -\gamma\beta \neq 0$. The argument above then forces $\Delta E = 0$, and hence conservation of all four components in every Lorentz frame.

4.1.4 From an analysis of the behavior of a general second-rank tensor under 90° and 180° rotations about the coordinate axes, show that an isotropic second-rank tensor in 3-D space must be a multiple of δ^i_j.

Solution.

In orthonormal Cartesian coordinates, write the components as a 3×3 matrix T. A rotation Q gives $T'_{ij} = Q_{ik}Q_{jl}T_{kl}$, that is, $T' = QTQ^{\mathsf{T}}$; isotropy means $QTQ^{\mathsf{T}} = T$ for every proper rotation.

First use the 180° rotations

$$R_1 = \mathrm{diag}(1, -1, -1), \qquad R_2 = \mathrm{diag}(-1, 1, -1).$$

If $R = \mathrm{diag}(s_1, s_2, s_3)$, then $(RTR^{\mathsf{T}})_{ij} = s_i s_j T_{ij}$. For R_1, $s_1 s_2 = s_1 s_3 = -1$ and $s_2 s_3 = 1$, so R_1 changes the signs of $T_{12}, T_{13}, T_{21}, T_{31}$, and invariance forces each to equal its negative, hence to vanish. For R_2, $s_2 s_3 = -1$, so R_2 likewise forces $T_{23} = T_{32} = 0$. We have proved, without assuming symmetry in advance, that $T = \mathrm{diag}(a, b, c)$.

Now use 90° rotations about the third and first axes:

$$Q_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Q_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Multiplying first by Q on the left,

$$Q_3 T = \begin{pmatrix} 0 & -b & 0 \\ a & 0 & 0 \\ 0 & 0 & c \end{pmatrix}, \qquad Q_1 T = \begin{pmatrix} a & 0 & 0 \\ 0 & 0 & -c \\ 0 & b & 0 \end{pmatrix},$$

and then by Q^{T} on the right gives

$$Q_3 T Q_3^{\mathsf{T}} = \mathrm{diag}(b, a, c), \qquad Q_1 T Q_1^{\mathsf{T}} = \mathrm{diag}(a, c, b).$$

Equality with T requires $a = b$ and $b = c$. Therefore

$$\boxed{T^i{}_j = \lambda \delta^i_j}.$$

Conversely $Q(\lambda I)Q^{\mathsf{T}} = \lambda QQ^{\mathsf{T}} = \lambda I$, so every tensor of this form is isotropic.

4.1.5 The 4-D fourth-rank Riemann-Christoffel curvature tensor of general relativity, R_{iklm}, satisfies the symmetry relations

$$R_{iklm} = -R_{ikml} = -R_{kilm}.$$

With the indices running from 0 to 3, show that the number of independent components is reduced from 256 to 36 and that the condition

$$R_{iklm} = R_{lmik}$$

further reduces the number of independent components to 21. Finally, if the components satisfy an identity $R_{iklm} + R_{ilmk} + R_{imkl} = 0$, show that the number of independent components is reduced to 20.

Note. The final three-term identity furnishes new information only if all four indices are different.

Solution.

There are initially $4^4 = 256$ components. Antisymmetry in (i, k) makes every component with $i = k$ zero and determines a component with $i > k$ as the negative of one with $i < k$. The six possible ordered representatives of a pair are

$$01, \quad 02, \quad 03, \quad 12, \quad 13, \quad 23.$$

The same argument applies independently to (l, m). Thus the tensor is specified by a 6×6 array $C_{(ik),(lm)} = R_{iklm}$, containing $6^2 = 36$ potentially independent entries.

Pair exchange, $R_{iklm} = R_{lmik}$, says that C is symmetric. Its six diagonal entries and $6 \cdot 5/2 = 15$ entries above the diagonal determine all the others. The number is now $6 + 15 = 21$.

Consider the cyclic identity. Its left side is unchanged by the cyclic permutation $(k, l, m) \to (l, m, k)$, which merely permutes its three terms. So if two of k, l, m coincide, we may take them to be $k = l$. The left side then has the form

$$R_{ikkm} + R_{ikmk} + R_{imkk} = R_{ikkm} - R_{ikkm} + 0 = 0.$$

If i coincides with one of k, l, m, the same cyclic freedom lets us take $i = k$. Then

$$R_{iilm} + R_{ilmi} + R_{imil} = 0 - R_{ilim} + R_{ilim} = 0,$$

because $R_{ilmi} = -R_{ilim}$ by antisymmetry in the last pair, and $R_{imil} = R_{ilim}$ by pair exchange. Thus a new restriction is possible only when all four indices differ. For the four available indices it is

$$R_{0123} + R_{0231} + R_{0312} = 0, \quad \text{or} \quad \boxed{C_{01,23} - C_{02,13} + C_{03,12} = 0}.$$

All other orderings give this same equation, possibly with the opposite sign. To see why, set $B_{iklm} = R_{iklm} + R_{ilmk} + R_{imkl}$. Interchanging k, l changes all three terms' signs, by antisymmetry in the last pair:

$$B_{ilkm} = R_{ilkm} + R_{ikml} + R_{imlk} = -R_{ilmk} - R_{iklm} - R_{imkl} = -B_{iklm}.$$

Interchanging l, m likewise reverses all three signs:

$$B_{ikml} = R_{ikml} + R_{imlk} + R_{ilkm} = -R_{iklm} - R_{imkl} - R_{ilmk} = -B_{iklm}.$$

Interchanging i, k gives

$$B_{kilm} = R_{kilm} + R_{klmi} + R_{kmil} = -R_{iklm} - R_{imkl} - R_{ilmk} = -B_{iklm}.$$

Here $R_{kilm} = -R_{iklm}$, while pair exchange followed by antisymmetry gives $R_{klmi} = R_{mikl} = -R_{imkl}$ and $R_{kmil} = R_{ilkm} = -R_{ilmk}$. Every permutation of the four slots is a product of interchanges of neighboring slots, so B is completely antisymmetric. Its only independent entry in four dimensions is B_{0123}. The displayed relation is nontrivial: it links three different entries above the diagonal of the symmetric array C, which are independent among the 21 parameters. It therefore removes exactly one of them. The final number is

$$\boxed{21 - 1 = 20}.$$

4.1.6 T_{iklm} is antisymmetric with respect to all pairs of indices. How many independent components has it (in 3-D space)?

Solution.

Four indices chosen from the three values $1, 2, 3$ must include a repeated value. Interchanging the two slots containing that repeated value leaves a component unchanged. Complete antisymmetry says that the same interchange changes its sign. Therefore each component obeys $T_{iklm} = -T_{iklm}$ and is zero. The number of independent components is

$$\boxed{0}.$$

Equivalently, the space of completely antisymmetric rank-four tensors in three dimensions has dimension $\binom{3}{4} = 0$.

4.1.7 If $T_{\ldots i}$ is a tensor of rank n, show that $\partial T_{\ldots i}/\partial x^j$ is a tensor of rank $n + 1$ (Cartesian coordinates).

Note. In non-Cartesian coordinate systems the coefficients a_{ij} are, in general, functions of the coordinates, and the derivatives of the components of a tensor of rank n do not form a tensor except in the special case $n = 0$. In this case the derivative does yield a covariant vector (tensor of rank 1).

Solution.

A change between orthonormal Cartesian systems has the form $x'^i = Q_{ia}x^a + s^i$, where Q and the shift s^i are constant and $Q^{\mathsf{T}}Q = I$. Multiplying by Q_{ib}, summing over i, and using $Q_{ib}Q_{ia} = \delta_{ba}$ gives the inverse map $x^b = Q_{ib}(x'^i - s^i)$, so $\partial x^b/\partial x'^j = Q_{jb}$. The chain rule then gives

$$\partial'_j \equiv \frac{\partial}{\partial x'^j} = \frac{\partial x^b}{\partial x'^j}\frac{\partial}{\partial x^b} = Q_{jb}\partial_b.$$

For a rank-n tensor, its Cartesian component law is

$$T'_{i_1\cdots i_n} = Q_{i_1a_1}\cdots Q_{i_na_n}T_{a_1\cdots a_n}.$$

Differentiate this equation. The Q's are constant, so the product rule produces no derivatives of them:

$$\partial'_j T'_{i_1\cdots i_n} = Q_{i_1a_1}\cdots Q_{i_na_n}Q_{jb}\partial_b T_{a_1\cdots a_n}.$$

There is one transformation factor for each of the original n indices and one more for j. This is precisely a tensor law of rank $n+1$. For general coordinates the Jacobian factors depend on position, and differentiating them gives extra terms. For a covariant vector, for instance,

$$\partial'_j V'_i = \frac{\partial}{\partial x'^j}\left(\frac{\partial x^a}{\partial x'^i} V_a\right) = \frac{\partial x^a}{\partial x'^i}\frac{\partial x^b}{\partial x'^j}\partial_b V_a + \frac{\partial^2 x^a}{\partial x'^j \partial x'^i} V_a,$$

and the last term spoils the tensor law. A scalar has no such factors, so its ordinary first derivative remains a covector in general coordinates. The word "of" in the Note is supplied: the printed sentence lacks it ("the derivatives the components"), but it is the derivatives *of* the components that fail to form a tensor.

4.1.8 If $T_{ijk\ldots}$ is a tensor of rank n, show that $\sum_j \partial T_{ijk\ldots}/\partial x^j$ is a tensor of rank $n-1$ (Cartesian coordinates).

Solution.

Let $D_{ik\ldots} = \partial_j T_{ijk\ldots}$, with the repeated j summed. Use the same constant orthogonal matrix as in Exercise 4.1.7:

$$\begin{aligned} D'_{ik\ldots} &= \partial'_j T'_{ijk\ldots} \\ &= Q_{jd}\partial_d\big(Q_{ia}Q_{jb}Q_{kc}\cdots T_{abc\ldots}\big) \\ &= Q_{ia}Q_{kc}\cdots\left(\sum_j Q_{jd}Q_{jb}\right)\partial_d T_{abc\ldots} \\ &= Q_{ia}Q_{kc}\cdots\delta_{db}\partial_d T_{abc\ldots} \\ &= Q_{ia}Q_{kc}\cdots\partial_b T_{abc\ldots} = Q_{ia}Q_{kc}\cdots D_{ac\ldots}. \end{aligned}$$

The constant Q's were taken outside the derivative, and the fourth line uses orthogonality, $\sum_j Q_{jd}Q_{jb} = (Q^{\mathsf{T}}Q)_{db} = \delta_{db}$. Differentiation introduces one index and contraction removes two, leaving $n+1-2 = n-1$ free indices. The displayed law verifies that the result is a rank-$(n-1)$ tensor. In the rank-one case the same calculation gives the scalar divergence $\partial_j T_j$.

4.1.9 The operator

$$\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$$

may be written as

$$\sum_{i=1}^{4}\frac{\partial^2}{\partial x_i^2},$$

using $x_4 = ict$. This is the 4-D Laplacian, sometimes called the d'Alembertian and denoted by $\Box^2$. Show that it is a scalar operator, that is, invariant under Lorentz transformations, represented here by complex orthogonal transformations in the space of vectors (x_1, x_2, x_3, x_4).

Solution.

Since $x_4 = ict$, differentiation at fixed spatial coordinates gives

$$\frac{\partial}{\partial x_4} = \frac{1}{ic}\frac{\partial}{\partial t}, \qquad \frac{\partial^2}{\partial x_4^2} = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}.$$

Thus $\sum_{a=1}^{4} \partial_a^2$ is the stated operator. In these complex coordinates a Lorentz transformation $x' = Qx$ is represented by a constant complex orthogonal matrix Q: it preserves $x_1^2+x_2^2+x_3^2+x_4^2 = \mathbf{x}^2 - c^2t^2$, that is, $x^{\mathsf{T}}Mx = 0$ for every event x, where $M = Q^{\mathsf{T}}Q - I$ is symmetric. Sums of events are events, so polarization, as in Exercise 4.1.10, gives

$$u^{\mathsf{T}}Mv = \tfrac{1}{2}\{(u+v)^{\mathsf{T}}M(u+v) - u^{\mathsf{T}}Mu - v^{\mathsf{T}}Mv\} = 0$$

for any two events u, v. Events have real x_1, x_2, x_3 and imaginary x_4, so u, v may be taken among the unit columns e_1, e_2, e_3 and ie_4; this gives $M_{bd} = 0$, $iM_{b4} = 0$ $(b, d \le 3)$ and $-M_{44} = 0$. Hence $M = 0$, that is, $Q^{\mathsf{T}}Q = I$. For example, the boost $x_1' = \gamma(x_1 - \beta ct)$, $t' = \gamma(t - \beta x_1/c)$, with $\gamma = (1-\beta^2)^{-1/2}$, reads

$$x_1' = \gamma(x_1 + i\beta x_4), \quad x_4' = \gamma(x_4 - i\beta x_1), \qquad \begin{pmatrix} Q_{11} & Q_{14} \\ Q_{41} & Q_{44} \end{pmatrix} = \begin{pmatrix} \gamma & i\beta\gamma \\ -i\beta\gamma & \gamma \end{pmatrix},$$

and this block satisfies $Q^{\mathsf{T}}Q = I$: the diagonal entries of the product are $\gamma^2 + (i\beta\gamma)^2 = \gamma^2(1-\beta^2) = 1$, and the off-diagonal ones are $i\beta\gamma^2 - i\beta\gamma^2 = 0$. These are complex rotations, not arbitrary real Euclidean rotations mixing the physical time and space coordinates. Since $x = Q^{\mathsf{T}}x'$, that is, $x_b = Q_{ab}x_a'$, the chain rule gives $\partial_a' = (\partial x_b/\partial x_a')\partial_b = Q_{ab}\partial_b$. Hence, acting on a scalar function,

$$\begin{aligned} \sum_a \partial_a'^2 &= \sum_{a,b,d} Q_{ab}Q_{ad}\partial_b\partial_d \\ &= \sum_{b,d} (Q^{\mathsf{T}}Q)_{bd}\partial_b\partial_d = \sum_{b,d} \delta_{bd}\partial_b\partial_d = \sum_b \partial_b^2. \end{aligned}$$

This proves the operator's Lorentz invariance. No derivative of Q appears because inertial Lorentz transformations are linear.

4.1.10 The double summation $K_{ij}A_iB_j$ is invariant for any two vectors A_i and B_j. Prove that K_{ij} is a second-rank tensor.

Note. In the form $ds^2(\text{invariant}) = g_{ij}\,dx^i\,dx^j$, this result shows that the symmetric matrix g_{ij} is a tensor.

Solution.

Use orthonormal Cartesian coordinates, so vectors transform as $A_p' = Q_{pi}A_i$ and $B_q' = Q_{qj}B_j$. Invariance of the given scalar means

$$K_{ij}A_iB_j = K_{pq}'A_p'B_q' = K_{pq}'Q_{pi}Q_{qj}A_iB_j.$$

Choose A and B successively to be each pair of coordinate unit vectors, $A_i = \delta_{im}$ and $B_j = \delta_{jn}$; every such choice is permitted. The sums over i and j then collapse, leaving $K_{mn} = Q_{pm}Q_{qn}K_{pq}'$ for all m, n. Renaming m, n as i, j,

$$K_{ij} = Q_{pi}Q_{qj}K_{pq}'.$$

Multiply by $Q_{ri}Q_{sj}$ and sum i, j. Orthogonality gives

$$Q_{ri}Q_{sj}K_{ij} = (Q_{ri}Q_{pi})(Q_{sj}Q_{qj})K_{pq}' = \delta_{rp}\delta_{sq}K_{pq}' = K_{rs}'.$$

Thus $K_{rs}' = Q_{ri}Q_{sj}K_{ij}$, the second-rank tensor law.

The line-element application uses the symmetry of the metric. A quadratic expression alone cannot determine an antisymmetric part of g_{ij}, because that part contracts to zero with $dx^i dx^j$.

For a symmetric metric, polarization recovers the bilinear expression from the invariant quadratic form $q(U) = g_{ij}U^iU^j$:

$$g_{ij}U^iV^j = \frac{1}{2}\{q(U+V) - q(U) - q(V)\}.$$

To see the polarization step explicitly, expand

$$q(U+V) = q(U) + q(V) + g_{ij}U^iV^j + g_{ij}V^iU^j = q(U) + q(V) + 2g_{ij}U^iV^j,$$

where symmetry and relabeling i, j equate the cross terms. The left side of the polarization formula is consequently invariant for independent U, V. For a general nonsingular coordinate change, write $U^i = J^i{}_aU'^a$, $V^j = J^j{}_bV'^b$, where $J^i{}_a = \partial x^i/\partial x'^a$. Then

$$g'_{ab}U'^aV'^b = g_{ij}J^i{}_aJ^j{}_bU'^aV'^b.$$

Taking U', V' to be each pair of coordinate unit vectors yields $g'_{ab} = J^i{}_aJ^j{}_bg_{ij}$, the covariant metric transformation law.

4.1.11 The equation $K_{ij}A_{jk} = B_{ik}$ holds for all orientations of the coordinate system. If A and B are arbitrary second-rank tensors, show that K is a second-rank tensor also.

Solution.

Write the contraction as the matrix equation $KA = B$. Under a Cartesian rotation Q, the two known tensors transform as $A' = QAQ^{\mathsf{T}}$ and $B' = QBQ^{\mathsf{T}}$. Substituting $B = KA$ and inserting $Q^{\mathsf{T}}Q = I$ between K and A,

$$B' = QKAQ^{\mathsf{T}} = QK(Q^{\mathsf{T}}Q)AQ^{\mathsf{T}} = (QKQ^{\mathsf{T}})(QAQ^{\mathsf{T}}) = (QKQ^{\mathsf{T}})A'.$$

But the hypothesis states that the equation has the same form in the new coordinates: $K'A' = B'$. Subtracting gives

$$(K' - QKQ^{\mathsf{T}})A' = 0.$$

This must hold for every second-rank A'. In particular, set $A' = I$; then $K' - QKQ^{\mathsf{T}} = 0$. Hence

$$\boxed{K' = QKQ^{\mathsf{T}}},$$

which is the required tensor law. The word "arbitrary" is essential: if the equation were supplied only for one singular A, it would not in general determine K; for example $A = B = 0$ imposes no condition.

4.2 Pseudotensors, Dual Tensors

4.2.1 An antisymmetric square array is given by

$$\begin{pmatrix} 0 & C_3 & -C_2 \\ -C_3 & 0 & C_1 \\ C_2 & -C_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & C_{12} & C_{13} \\ -C_{12} & 0 & C_{23} \\ -C_{13} & -C_{23} & 0 \end{pmatrix},$$

where (C_1, C_2, C_3) form a pseudovector. Assuming that the relation

$$C_i = \frac{1}{2!}\epsilon_{ijk}C_{jk}$$

holds in all orthonormal Cartesian coordinate systems, prove that C_{jk} is a tensor. (This is another form of the quotient theorem.)

Solution.

The numerical Levi-Civita symbol in this question uses orthonormal Cartesian axes. General curvilinear coordinates require the metric-dependent volume tensor, rather than the same numerical symbol. First invert the stated relation: multiply it by $\epsilon_{\ell mi}$ and sum over i. The contraction identity

$$\epsilon_{\ell mi}\epsilon_{ijk} = \delta_{\ell j}\delta_{mk} - \delta_{\ell k}\delta_{mj}$$

gives

$$\epsilon_{\ell mi}C_i = \frac{1}{2}(\delta_{\ell j}\delta_{mk} - \delta_{\ell k}\delta_{mj})C_{jk} = \frac{1}{2}(C_{\ell m} - C_{m\ell}) = C_{\ell m}.$$

Here antisymmetry was essential in the final step. Thus $C_{jk} = \epsilon_{jki}C_i$, in every orthonormal Cartesian system.

For an orthogonal change of Cartesian axes Q, put $d = \det Q = \pm 1$. A pseudovector obeys $C'_i = dQ_{ia}C_a$. For the numerical Levi-Civita symbols, $Q_{jb}Q_{kc}Q_{i\ell}\epsilon_{bc\ell} = (\det Q)\epsilon_{jki}$. Indeed, by the expansion $\det A = \epsilon_{bc\ell}A_{1b}A_{2c}A_{3\ell}$, the left side is the determinant of the matrix whose rows are rows j, k, i of Q: for $(j, k, i) = (1, 2, 3)$ it is $\det Q$, an interchange of two rows reverses its sign, and a repeated row gives zero, exactly as for $(\det Q)\epsilon_{jki}$. Multiplying by d and using $d^2 = 1$ yields

$$\epsilon_{jki} = dQ_{jb}Q_{kc}Q_{i\ell}\epsilon_{bc\ell}.$$

Substituting both identities into the inverted relation gives

$$\begin{aligned} C'_{jk} &= \epsilon_{jki}C'_i \\ &= d^2Q_{jb}Q_{kc}Q_{i\ell}Q_{ia}\epsilon_{bc\ell}C_a \\ &= Q_{jb}Q_{kc}\delta_{\ell a}\epsilon_{bc\ell}C_a \\ &= Q_{jb}Q_{kc}C_{bc}. \end{aligned}$$

We used $d^2 = 1$, $\sum_i Q_{i\ell}Q_{ia} = \delta_{\ell a}$, and, in the last line, the inverted relation $\epsilon_{bca}C_a = C_{bc}$ in the original axes. This is the ordinary second-rank tensor law, including reflections.

4.2.2 Show that the vector product is unique to 3-D space, that is, only in three dimensions can we establish a linear one-to-one correspondence between the components of an antisymmetric tensor (second-rank) and the components of a vector.

Solution.

In n dimensions, an antisymmetric tensor has zero diagonal entries, and each entry below the diagonal is minus the corresponding entry above it. Its independent components are therefore indexed by unordered pairs of distinct indices, whose number is

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

A vector has n independent components. A linear one-to-one correspondence between the full component spaces requires equal dimensions:

$$\frac{n(n-1)}{2} = n \quad \Longrightarrow \quad n(n-3) = 0.$$

For a positive space dimension this forces $n = 3$. In three dimensions the correspondence actually exists: $C_i = \frac{1}{2}\epsilon_{ijk}C_{jk}$ and $C_{jk} = \epsilon_{jki}C_i$ are inverse maps. Exercise 4.2.1 showed that the second undoes the first. In the other order, start from C_j and form $C_{mi} = \epsilon_{mij}C_j = \epsilon_{ijm}C_j$. Setting $k = m$ in the contraction identity of Exercise 4.2.1 and summing gives $\epsilon_{\ell mi}\epsilon_{ijm} = 3\delta_{\ell j} - \delta_{\ell j} = 2\delta_{\ell j}$, since $\delta_{mm} = 3$; hence $\frac{1}{2}\epsilon_{\ell mi}C_{mi} = \frac{1}{2}\epsilon_{\ell mi}\epsilon_{ijm}C_j = C_\ell$. The associated vector is a pseudovector when orientation reversals are included.

The word "linear" is added to the statement because, as mere sets, $\mathbb{R}^{n(n-1)/2}$ and $\mathbb{R}^n$ can be put in one-to-one correspondence for every $n \geq 2$. The uniqueness of the vector product is meant in the sense of this linear correspondence with all antisymmetric second-rank tensors; the dimension count does not exclude other, differently defined products bearing the name cross product.

4.2.3 Write $\nabla \cdot \nabla \times \mathbf{A}$ and $\nabla \times \nabla\varphi$ in tensor (index) notation in $\mathbb{R}^3$ so that it becomes obvious that each expression vanishes. Assume that the fields have continuous second derivatives.

ANS.
$$\nabla \cdot \nabla \times \mathbf{A} = \epsilon_{ijk}\frac{\partial}{\partial x^i}\frac{\partial}{\partial x^j}A^k$$
$$(\nabla \times \nabla\varphi)_i = \epsilon_{ijk}\frac{\partial}{\partial x^j}\frac{\partial}{\partial x^k}\varphi.$$

Solution.

Assume the fields have continuous second derivatives so that mixed partial derivatives commute. With repeated indices summed,

$$\nabla \cdot (\nabla \times \mathbf{A}) = \partial_i(\epsilon_{ijk}\partial_j A^k) = \epsilon_{ijk}\partial_i\partial_j A^k.$$

Interchanging the dummy indices i, j changes ϵ_{ijk} to its negative but leaves $\partial_i\partial_j A^k$ unchanged. More explicitly, relabeling gives $\epsilon_{ijk}\partial_i\partial_j A^k = \epsilon_{jik}\partial_j\partial_i A^k = -\epsilon_{ijk}\partial_j\partial_i A^k$, and the average of these two forms is

$$\epsilon_{ijk}\partial_i\partial_j A^k = \frac{1}{2}\epsilon_{ijk}(\partial_i\partial_j - \partial_j\partial_i)A^k = 0.$$

Likewise,

$$(\nabla \times \nabla\varphi)_i = \epsilon_{ijk}\partial_j\partial_k\varphi = \frac{1}{2}\epsilon_{ijk}(\partial_j\partial_k - \partial_k\partial_j)\varphi = 0.$$

Both identities are contractions of an antisymmetric pair of indices with a symmetric pair.

4.2.4 Verify that each of the following fourth-rank tensors is isotropic, that is, that it has the same form independent of any rotation of the coordinate systems.

(a) $A^{ik}_{jl} = \delta^i_j\delta^k_l$,

(b) $B^{ij}_{kl} = \delta^i_k\delta^j_l + \delta^i_l\delta^j_k$,

(c) $C^{ij}_{kl} = \delta^i_k\delta^j_l - \delta^i_l\delta^j_k$.

Solution.

The mixed identity tensor is invariant. Indeed, under a rotation,

$$\delta'^i_{\ j} = Q_{ia}Q_{jb}\delta^a_b = \sum_a Q_{ia}Q_{ja} = (QQ^{\mathsf{T}})_{ij} = \delta^i_j.$$

For (a), apply the four-index transformation law and group the factors associated with each Kronecker delta:

$$\begin{aligned} A'^{ik}_{jl} &= Q_{ia}Q_{kc}Q_{jb}Q_{ld}\delta^a_b\delta^c_d \\ &= \left(\sum_a Q_{ia}Q_{ja}\right)\left(\sum_c Q_{kc}Q_{lc}\right) = \delta^i_j\delta^k_l = A^{ik}_{jl}. \end{aligned}$$

The same calculation with the indices arranged as in (b) gives

$$(\delta^i_k\delta^j_l)' = \delta^i_k\delta^j_l, \qquad (\delta^i_l\delta^j_k)' = \delta^i_l\delta^j_k.$$

The second pairs the indices crosswise; with upper indices i, j and lower indices k, l its transformation reads

$$Q_{ia}Q_{jc}Q_{kb}Q_{ld}\delta^a_d\delta^c_b = \Big(\sum_a Q_{ia}Q_{la}\Big)\Big(\sum_c Q_{jc}Q_{kc}\Big) = \delta^i_l\delta^j_k.$$

Since transformation is linear, adding the two invariance equations proves $B'^{ij}_{kl} = B^{ij}_{kl}$, and subtracting them proves $C'^{ij}_{kl} = C^{ij}_{kl}$. Thus all three tensors are isotropic. The argument in fact also allows orthogonal reflections.

4.2.5 Show that the two-index Levi-Civita symbol ϵ_{ij} is a second-rank pseudotensor (in two-dimensional [2-D] space). Does this contradict the uniqueness of δ^i_j (Exercise 4.1.4)?

Solution.

Use the orientation convention $\epsilon_{12} = 1$, so

$$E = (\epsilon_{ij}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For $Q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, direct multiplication gives

$$QEQ^{\mathsf{T}} = \begin{pmatrix} -b & a \\ -d & c \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} 0 & ad-bc \\ bc-ad & 0 \end{pmatrix} = (\det Q)E.$$

In particular, for orthogonal Q the pseudotensor law is

$$E' = (\det Q)QEQ^{\mathsf{T}} = (\det Q)^2E = E.$$

Thus the same numerical symbol in every oriented coordinate convention is a second-rank pseudotensor. A proper rotation has $\det Q = 1$; a reflection has $\det Q = -1$ and contributes the extra sign appropriate to a pseudotensor.

There is no contradiction with Exercise 4.1.4, which concerns *three* dimensions. In two dimensions E is a nonzero antisymmetric matrix invariant under all proper rotations; the rotations about three distinct axes used in that exercise are unavailable. Moreover, E is not invariant under the ordinary tensor transformation law for reflections, whereas the identity is.

4.2.6 Represent ϵ_{ij} by a 2×2 matrix, and using the 2×2 rotation matrix of Eq. (3.23), show that ϵ_{ij} is invariant under proper orthogonal similarity transformations.

Solution.

Here "orthogonal similarity transformations" must be restricted to proper rotations ($\det Q = 1$). For general orthogonal Q the correct relation is $QEQ^{-1} = (\det Q)E$.

With $c = \cos\phi$, $s = \sin\phi$, the rotation matrix in the question is

$$R = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}, \qquad R^{-1} = R^{\mathsf{T}} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}.$$

Since $E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, multiply explicitly:

$$RER^{-1} = \begin{pmatrix} -s & c \\ -c & -s \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} 0 & s^2+c^2 \\ -(c^2+s^2) & 0 \end{pmatrix} = E.$$

This proves the desired rotational invariance. To check the necessary qualification, take the reflection $Q = \mathrm{diag}(1,-1)$. Then

$$QEQ^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -E.$$

Under the *pseudotensor* law its additional factor $\det Q = -1$ restores E, exactly as in Exercise 4.2.5.

4.2.7 Given $A_k = \frac{1}{2}\epsilon_{ijk}B^{ij}$ with $B^{ij} = -B^{ji}$, antisymmetric, show that

$$B^{mn} = \epsilon^{mnk}A_k.$$

Solution.

Multiply the given equation by ϵ^{mnk} and sum k:

$$\begin{aligned} \epsilon^{mnk}A_k &= \frac{1}{2}\epsilon^{mnk}\epsilon_{ijk}B^{ij} \\ &= \frac{1}{2}(\delta_i^m\delta_j^n - \delta_j^m\delta_i^n)B^{ij} \\ &= \frac{1}{2}(B^{mn} - B^{nm}) \\ &= \frac{1}{2}(B^{mn} + B^{mn}) = B^{mn}. \end{aligned}$$

The second line is the contraction identity of Exercise 4.2.1, applied to $\epsilon^{mnk}\epsilon_{kij}$ after the cyclic rearrangement $\epsilon_{ijk} = \epsilon_{kij}$ (in Cartesian coordinates upper and lower indices are equivalent), and the last line uses $B^{nm} = -B^{mn}$. This is the stated inverse relation. It also checks the factor $1/2$: without it the two antisymmetric terms would produce $2B^{mn}$.

4.3 Tensors in General Coordinates

4.3.1 For the special case of 3-D space ($\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \boldsymbol{\varepsilon}_3$ defining a right-handed coordinate system, not necessarily orthogonal), show that

$$\boldsymbol{\varepsilon}^i = \frac{\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k}{\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k \cdot \boldsymbol{\varepsilon}_i}, \quad i, j, k = 1, 2, 3 \text{ and cyclic permutations.}$$

Note. These contravariant basis vectors $\boldsymbol{\varepsilon}^i$ define the reciprocal lattice space of Example 3.2.1.

Solution.

The reciprocal basis is characterized by $\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_\ell = \delta^i_\ell$: with $\boldsymbol{\varepsilon}^i = \nabla q^i$ and $\boldsymbol{\varepsilon}_\ell = \partial \mathbf{r}/\partial q^\ell$, the chain rule gives $\nabla q^i \cdot \partial \mathbf{r}/\partial q^\ell = \partial q^i/\partial q^\ell = \delta^i_\ell$. Let

$$D = (\boldsymbol{\varepsilon}_1 \times \boldsymbol{\varepsilon}_2) \cdot \boldsymbol{\varepsilon}_3.$$

Linear independence implies $D \neq 0$, and the specified right-handedness makes $D > 0$. Cyclic permutations preserve this scalar triple product, so for cyclic (i, j, k) the denominator in the formula is $(\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k) \cdot \boldsymbol{\varepsilon}_i = D$, and the right side of the formula is

$$\mathbf{b}^i = \frac{\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k}{D}.$$

The cross product is perpendicular to both its factors, so

$$\mathbf{b}^i \cdot \boldsymbol{\varepsilon}_j = 0, \qquad \mathbf{b}^i \cdot \boldsymbol{\varepsilon}_k = 0.$$

Its remaining scalar product is

$$\mathbf{b}^i \cdot \boldsymbol{\varepsilon}_i = \frac{(\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k) \cdot \boldsymbol{\varepsilon}_i}{D} = 1.$$

These are exactly the three reciprocal-basis conditions. They determine a vector uniquely: the difference of any two solutions would be orthogonal to three linearly independent vectors and hence zero. Therefore $\mathbf{b}^i = \boldsymbol{\varepsilon}^i$, proving the formula.

4.3.2 If the covariant vectors $\boldsymbol{\varepsilon}_i$ are orthogonal, show that

(a) g_{ij} is diagonal,

(b) $g^{ii} = 1/g_{ii}$ (no summation),

(c) $|\boldsymbol{\varepsilon}^i| = 1/|\boldsymbol{\varepsilon}_i|$.

Solution.

Put $h_i = |\boldsymbol{\varepsilon}_i| > 0$. Orthogonality means

$$g_{ij} = \boldsymbol{\varepsilon}_i \cdot \boldsymbol{\varepsilon}_j = \begin{cases} h_i^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

Thus (a) the covariant metric is diagonal. The contravariant metric $g^{ij} = \boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}^j$ is its inverse, $\sum_j g^{ij} g_{jk} = \delta^i_k$ (Exercise 4.3.3). For a diagonal g_{jk} only the term $j = k$ survives, so this reads $g^{ik} h_k^2 = \delta^i_k$ (no sum), and

$$g^{ik} = 0 \quad (i \neq k), \qquad \boxed{g^{ii} = h_i^{-2} = 1/g_{ii}} \quad \text{(no sum)}.$$

This proves (b). To obtain (c), observe that $\boldsymbol{\varepsilon}_i/h_i^2$ has scalar product 1 with $\boldsymbol{\varepsilon}_i$ and 0 with all the other basis vectors. By the uniqueness argument of Exercise 4.3.1, it is consequently the reciprocal vector:

$$\boldsymbol{\varepsilon}^i = \frac{\boldsymbol{\varepsilon}_i}{h_i^2}, \qquad \boxed{|\boldsymbol{\varepsilon}^i| = \frac{h_i}{h_i^2} = \frac{1}{|\boldsymbol{\varepsilon}_i|}}.$$

4.3.3 Prove that $(\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}^j)(\boldsymbol{\varepsilon}_j \cdot \boldsymbol{\varepsilon}_k) = \delta^i_k$.

Solution.

The dual-basis expansion of any vector $\mathbf{v}$ is

$$\mathbf{v} = \sum_j (\boldsymbol{\varepsilon}_j \cdot \mathbf{v})\boldsymbol{\varepsilon}^j.$$

To verify it, take the scalar product with each $\boldsymbol{\varepsilon}_k$; the right side gives $\sum_j (\boldsymbol{\varepsilon}_j \cdot \mathbf{v})\delta^j_k = \boldsymbol{\varepsilon}_k \cdot \mathbf{v}$. A vector is determined by these three scalar products. Apply the expansion to $\mathbf{v} = \boldsymbol{\varepsilon}_k$ and then dot with $\boldsymbol{\varepsilon}^i$:

$$\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_k = \sum_j (\boldsymbol{\varepsilon}_j \cdot \boldsymbol{\varepsilon}_k)(\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}^j) = \delta^i_k.$$

This is the desired identity, also written $g^{ij}g_{jk} = \delta^i_k$.

4.3.4 Show that $\Gamma^m_{jk} = \Gamma^m_{kj}$.

Solution.

For a smooth coordinate map $\mathbf{r}(q^1, q^2, q^3)$,

$$\boldsymbol{\varepsilon}_j = \frac{\partial \mathbf{r}}{\partial q^j}, \qquad \frac{\partial \boldsymbol{\varepsilon}_j}{\partial q^k} = \Gamma^m_{jk}\boldsymbol{\varepsilon}_m.$$

Commuting the mixed second derivatives of $\mathbf{r}$ gives

$$\Gamma^m_{jk}\boldsymbol{\varepsilon}_m = \frac{\partial^2 \mathbf{r}}{\partial q^k \partial q^j} = \frac{\partial^2 \mathbf{r}}{\partial q^j \partial q^k} = \Gamma^m_{kj}\boldsymbol{\varepsilon}_m.$$

Linear independence of the basis forces equality of each coefficient: $\boxed{\Gamma^m_{jk} = \Gamma^m_{kj}}$. Equivalently, take the scalar product of both sides with $\boldsymbol{\varepsilon}^n$ and use $\boldsymbol{\varepsilon}_m \cdot \boldsymbol{\varepsilon}^n = \delta^n_m$ to obtain $\Gamma^n_{jk} = \Gamma^n_{kj}$. The symmetry rests on the basis vectors $\boldsymbol{\varepsilon}_j = \partial \mathbf{r}/\partial q^j$ being derivatives of a single position vector $\mathbf{r}$. Coefficients defined in the same way for a basis not of this form need not be symmetric: for the cylindrical unit vectors, $\partial_\phi \hat{\mathbf{e}}_\rho = \hat{\mathbf{e}}_\phi$ but $\partial_\rho \hat{\mathbf{e}}_\phi = 0$, so the analogue of $\Gamma^\phi_{\rho\phi}$ is 1 while that of $\Gamma^\phi_{\phi\rho}$ is 0.

4.3.5 Derive the covariant and contravariant metric tensors for circular cylindrical coordinates.

Solution.

Choose $q^1 = \rho$, $q^2 = \phi$, $q^3 = z$, with

$$\mathbf{r} = (\rho\cos\phi, \rho\sin\phi, z).$$

Differentiating gives the coordinate basis

$$\boldsymbol{\varepsilon}_\rho = (\cos\phi, \sin\phi, 0), \quad \boldsymbol{\varepsilon}_\phi = (-\rho\sin\phi, \rho\cos\phi, 0), \quad \boldsymbol{\varepsilon}_z = (0, 0, 1).$$

Its squared lengths are $1, \rho^2, 1$. The only potentially nonzero off-diagonal scalar product is $\boldsymbol{\varepsilon}_\rho \cdot \boldsymbol{\varepsilon}_\phi = -\rho\cos\phi\sin\phi + \rho\sin\phi\cos\phi = 0$; the other off-diagonal scalar products also vanish, because $\boldsymbol{\varepsilon}_z$ has no x, y components while $\boldsymbol{\varepsilon}_\rho$ and $\boldsymbol{\varepsilon}_\phi$ have no z component. The basis is orthogonal, so by Exercise 4.3.2(b) the contravariant metric has the reciprocal diagonal entries. Therefore

$$\boxed{(g_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad (g^{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^{-2} & 0 \\ 0 & 0 & 1 \end{pmatrix}.}$$

As checks, their product is the identity and $ds^2 = g_{ij}\, dq^i dq^j = d\rho^2 + \rho^2 d\phi^2 + dz^2$. These coordinate formulas apply for $\rho > 0$; at the axis the angular coordinate is singular.

4.3.6 Show that the covariant derivative of a covariant vector is given by

$$V_{i;j} \equiv \frac{\partial V_i}{\partial q^j} - V_k \Gamma^k_{ij}.$$

Hint. Differentiate

$$\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_j = \delta^i_j.$$

Solution.

Differentiate the reciprocal-basis identity of the hint, written as $\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_k = \delta^i_k$, with respect to q^j, using $\partial_j \boldsymbol{\varepsilon}_k = \Gamma^m_{kj} \boldsymbol{\varepsilon}_m$:

$$0 = \partial_j(\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_k) = (\partial_j \boldsymbol{\varepsilon}^i) \cdot \boldsymbol{\varepsilon}_k + \boldsymbol{\varepsilon}^i \cdot (\Gamma^m_{kj} \boldsymbol{\varepsilon}_m).$$

The last term is Γ^i_{kj}, so $(\partial_j \boldsymbol{\varepsilon}^i) \cdot \boldsymbol{\varepsilon}_k = -\Gamma^i_{kj}$. Expanding this derivative in the reciprocal basis, $\mathbf{w} = (\mathbf{w} \cdot \boldsymbol{\varepsilon}_k)\boldsymbol{\varepsilon}^k$ (Exercise 4.3.3), therefore gives

$$\partial_j \boldsymbol{\varepsilon}^i = -\Gamma^i_{kj} \boldsymbol{\varepsilon}^k.$$

Now express $\mathbf{V}$ using covariant components, $\mathbf{V} = V_i \boldsymbol{\varepsilon}^i$. Differentiation gives

$$\begin{aligned} \partial_j \mathbf{V} &= (\partial_j V_i)\boldsymbol{\varepsilon}^i + V_i \partial_j \boldsymbol{\varepsilon}^i \\ &= (\partial_j V_i)\boldsymbol{\varepsilon}^i - V_i \Gamma^i_{kj} \boldsymbol{\varepsilon}^k \\ &= (\partial_j V_i - V_k \Gamma^k_{ij})\boldsymbol{\varepsilon}^i. \end{aligned}$$

In the last line the dummy indices in the second term were relabeled, $i \leftrightarrow k$. The coefficients of the reciprocal basis are the covariant components of the derivative, hence

$$\boxed{V_{i;j} = \partial_j V_i - V_k \Gamma^k_{ij}}.$$

4.3.7 Verify that $V_{i;j} = g_{ik} V^k_{;j}$ by showing that

$$\frac{\partial V_i}{\partial q^j} - V_k \Gamma^k_{ij} = g_{ik} \left[\frac{\partial V^k}{\partial q^j} + V^m \Gamma^k_{mj} \right].$$

Solution.

Start from $V_i = g_{ik}V^k$. We first need the derivative of the metric. Differentiate its basis-vector definition with the product rule and use $\partial_j \boldsymbol{\varepsilon}_i = \Gamma^m_{ij}\boldsymbol{\varepsilon}_m$:

$$\begin{aligned}\partial_j g_{ik} &= \partial_j(\boldsymbol{\varepsilon}_i \cdot \boldsymbol{\varepsilon}_k)\\ &= (\Gamma^m_{ij}\boldsymbol{\varepsilon}_m)\cdot\boldsymbol{\varepsilon}_k + \boldsymbol{\varepsilon}_i\cdot(\Gamma^m_{kj}\boldsymbol{\varepsilon}_m)\\ &= \Gamma^m_{ij}g_{mk} + \Gamma^m_{kj}g_{im}.\end{aligned}$$

Use this identity in the product rule:

$$\begin{aligned}V_{i;j} &= \partial_j(g_{ik}V^k) - V_\ell\Gamma^\ell_{ij}\\ &= g_{ik}\partial_j V^k + (\partial_j g_{ik})V^k - g_{\ell k}V^k\Gamma^\ell_{ij}\\ &= g_{ik}\partial_j V^k + \Gamma^m_{ij}g_{mk}V^k + \Gamma^m_{kj}g_{im}V^k - g_{mk}V^k\Gamma^m_{ij}\\ &= g_{ik}\partial_j V^k + g_{im}\Gamma^m_{kj}V^k\\ &= g_{ik}\big(\partial_j V^k + \Gamma^k_{mj}V^m\big) = \boxed{g_{ik}V^k{}_{;j}}.\end{aligned}$$

In the second line $V_\ell = g_{\ell k}V^k$; in the third the dummy ℓ of the last term was renamed m. The two terms containing $\Gamma^m_{ij}g_{mk}V^k$ then cancel exactly; only the dummy-index relabeling $m \leftrightarrow k$ is used in the last line.

4.3.8 From the circular cylindrical metric tensor g_{ij}, calculate the Γ^k_{ij} for circular cylindrical coordinates.

Note. There are only three nonvanishing Γ.

Solution.

For $q^i = (\rho, \phi, z)$, Exercise 4.3.5 gives $g_{ij} = \mathrm{diag}(1, \rho^2, 1)$ and $g^{ij} = \mathrm{diag}(1, \rho^{-2}, 1)$. Use

$$\Gamma^k_{ij} = \frac{1}{2}g^{k\ell}(\partial_i g_{\ell j} + \partial_j g_{\ell i} - \partial_\ell g_{ij}).$$

All metric derivatives vanish except $\partial_\rho g_{\phi\phi} = 2\rho$. For an upper ρ index the inverse metric selects $\ell = \rho$; only the last derivative can survive, and only when $i = j = \phi$:

$$\Gamma^\rho_{\phi\phi} = \frac{1}{2}g^{\rho\rho}(0 + 0 - \partial_\rho g_{\phi\phi}) = -\rho.$$

For an upper ϕ index the inverse metric selects $\ell = \phi$. The last derivative, $\partial_\phi g_{ij}$, is zero; the first survives only for $i = \rho$, $j = \phi$, and the second only for $i = \phi$, $j = \rho$:

$$\Gamma^\phi_{\rho\phi} = \frac{1}{2}g^{\phi\phi}(\partial_\rho g_{\phi\phi} + 0 - 0) = \frac{1}{2\rho^2}(2\rho) = \frac{1}{\rho}, \qquad \Gamma^\phi_{\phi\rho} = \frac{1}{\rho}.$$

With an upper z index, every term differentiates a constant metric entry or differentiates with respect to z, so every term is zero. There are no other nonzero possibilities. Thus the complete list is

$$\boxed{\Gamma^\rho_{\phi\phi} = -\rho, \qquad \Gamma^\phi_{\rho\phi} = \Gamma^\phi_{\phi\rho} = \rho^{-1}}.$$

As a check against the definition $\partial_j\boldsymbol{\varepsilon}_i = \Gamma^k_{ij}\boldsymbol{\varepsilon}_k$, differentiate the basis vectors of Exercise 4.3.5: $\partial_\phi\boldsymbol{\varepsilon}_\phi = (-\rho\cos\phi, -\rho\sin\phi, 0) = -\rho\,\boldsymbol{\varepsilon}_\rho$ and $\partial_\phi\boldsymbol{\varepsilon}_\rho = \partial_\rho\boldsymbol{\varepsilon}_\phi = (-\sin\phi, \cos\phi, 0) = \boldsymbol{\varepsilon}_\phi/\rho$, while all other derivatives vanish.

4.3.9 Using the Γ^k_{ij} from Exercise 4.3.8, write out the covariant derivatives $V^i_{;j}$ of a vector $\mathbf{V}$ in circular cylindrical coordinates.

Solution.

Use the contravariant derivative formula

$$V^i{}_{;j} = \partial_j V^i + \Gamma^i_{mj} V^m.$$

For $i = \rho$, the only nonzero connection term occurs at $m = j = \phi$ and equals $-\rho V^\phi$. For $i = \phi$, the terms are V^ϕ/ρ when $j = \rho$ and V^ρ/ρ when $j = \phi$. There are no connection terms for $i = z$. Thus all nine derivatives are

$$\boxed{\left(V^i{}_{;j}\right) = \begin{pmatrix} \partial_\rho V^\rho & \partial_\phi V^\rho - \rho V^\phi & \partial_z V^\rho \\ \partial_\rho V^\phi + V^\phi/\rho & \partial_\phi V^\phi + V^\rho/\rho & \partial_z V^\phi \\ \partial_\rho V^z & \partial_\phi V^z & \partial_z V^z \end{pmatrix}}.$$

Rows are labeled by $i = (\rho, \phi, z)$ and columns by $j = (\rho, \phi, z)$. These are components in the coordinate basis, whose angular vector has length ρ. If $\mathbf{V} = V_{\hat\rho}\hat{\mathbf{e}}_\rho + V_{\hat\phi}\hat{\mathbf{e}}_\phi + V_{\hat z}\hat{\mathbf{e}}_z$, then

$$V^\rho = V_{\hat\rho}, \qquad V^\phi = V_{\hat\phi}/\rho, \qquad V^z = V_{\hat z}.$$

For example, the first row's angular entry becomes $\partial_\phi V^\rho - \rho V^\phi = \partial_\phi V_{\hat\rho} - V_{\hat\phi}$. Since $\partial_j \mathbf{V} = V^i{}_{;j}\boldsymbol{\varepsilon}_i$ (the relation that defines $V^i{}_{;j}$) and $\boldsymbol{\varepsilon}_\rho = \hat{\mathbf{e}}_\rho$, this entry is the $\hat{\mathbf{e}}_\rho$ component of $\partial_\phi \mathbf{V}$. Direct differentiation of the unit-vector expansion, with $\partial_\phi \hat{\mathbf{e}}_\rho = \hat{\mathbf{e}}_\phi$, $\partial_\phi \hat{\mathbf{e}}_\phi = -\hat{\mathbf{e}}_\rho$ and $\partial_\phi \hat{\mathbf{e}}_z = 0$, agrees:

$$\partial_\phi \mathbf{V} = (\partial_\phi V_{\hat\rho} - V_{\hat\phi})\hat{\mathbf{e}}_\rho + (\partial_\phi V_{\hat\phi} + V_{\hat\rho})\hat{\mathbf{e}}_\phi + (\partial_\phi V_{\hat z})\hat{\mathbf{e}}_z.$$

Likewise, because $\boldsymbol{\varepsilon}_\phi = \rho\,\hat{\mathbf{e}}_\phi$, the $\hat{\mathbf{e}}_\phi$ component is $\rho V^\phi{}_{;\phi} = \rho\,\partial_\phi(V_{\hat\phi}/\rho) + V_{\hat\rho} = \partial_\phi V_{\hat\phi} + V_{\hat\rho}$.

4.3.10 Show that for the metric tensor $g_{ij;k} = g^{ij}_{;k} = 0$.

Solution.

The covariant derivative acts on each covariant index with a minus connection term:

$$g_{ij;k} = \partial_k g_{ij} - \Gamma^m_{ik} g_{mj} - \Gamma^m_{jk} g_{im}.$$

This rule follows as in Exercise 4.3.6. Write a second-rank tensor as $T_{ij}\boldsymbol{\varepsilon}^i\boldsymbol{\varepsilon}^j$, a sum of direct products of reciprocal basis vectors, and differentiate by the product rule, using $\partial_k \boldsymbol{\varepsilon}^i = -\Gamma^i_{mk}\boldsymbol{\varepsilon}^m$ for each basis vector:

$$\begin{aligned} \partial_k(T_{ij}\boldsymbol{\varepsilon}^i\boldsymbol{\varepsilon}^j) &= (\partial_k T_{ij})\boldsymbol{\varepsilon}^i\boldsymbol{\varepsilon}^j - T_{ij}\Gamma^i_{mk}\boldsymbol{\varepsilon}^m\boldsymbol{\varepsilon}^j - T_{ij}\Gamma^j_{mk}\boldsymbol{\varepsilon}^i\boldsymbol{\varepsilon}^m \\ &= (\partial_k T_{ij} - \Gamma^m_{ik}T_{mj} - \Gamma^m_{jk}T_{im})\boldsymbol{\varepsilon}^i\boldsymbol{\varepsilon}^j, \end{aligned}$$

after relabeling $i \leftrightarrow m$ in the second term and $j \leftrightarrow m$ in the third; the coefficient is $T_{ij;k}$. With contravariant components, $T^{ij}\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_j$, and $\partial_k \boldsymbol{\varepsilon}_i = \Gamma^m_{ik}\boldsymbol{\varepsilon}_m$ instead, each upper index gives a plus term.

Differentiate $g_{ij} = \boldsymbol{\varepsilon}_i \cdot \boldsymbol{\varepsilon}_j$ as in Exercise 4.3.7 to obtain

$$\partial_k g_{ij} = \Gamma^m_{ik} g_{mj} + \Gamma^m_{jk} g_{im}.$$

Substitution cancels both terms, so $\boxed{g_{ij;k} = 0}$.

For the inverse metric, differentiate $g^{ia}g_{ab} = \delta^i_b$:

$$(\partial_k g^{ia})g_{ab} + g^{ia}\partial_k g_{ab} = 0.$$

Multiply by g^{bj} and sum b. Since $g_{ab}g^{bj} = \delta^j_a$, the first term becomes $\partial_k g^{ij}$, which is thereby isolated:

$$\begin{aligned}\partial_k g^{ij} &= -g^{ia}(\partial_k g_{ab})g^{bj} \\ &= -g^{ia}(\Gamma^m_{ak}g_{mb} + \Gamma^m_{bk}g_{am})g^{bj} \\ &= -g^{ia}\Gamma^j_{ak} - \Gamma^i_{bk}g^{bj}.\end{aligned}$$

The last line uses $g_{mb}g^{bj} = \delta^j_m$ and $g^{ia}g_{am} = \delta^i_m$. As shown above, the contravariant derivative has a plus connection term for each upper index. Renaming the dummies a, b as m,

$$\begin{aligned}g^{ij}{}_{;k} &= \partial_k g^{ij} + \Gamma^i_{mk}g^{mj} + \Gamma^j_{mk}g^{im} \\ &= -g^{im}\Gamma^j_{mk} - \Gamma^i_{mk}g^{mj} + \Gamma^i_{mk}g^{mj} + \Gamma^j_{mk}g^{im} = 0.\end{aligned}$$

Both cancellations rely on the metric-compatible Christoffel connection used in this section.

4.3.11 Starting with the divergence in tensor notation, Eq. (4.70), develop the divergence of a vector in spherical polar coordinates, Eq. (3.157).

Solution.

The tensor divergence is

$$V^i{}_{;i} = \partial_i V^i + \Gamma^i_{mi}V^m = \frac{1}{\sqrt{g}}\partial_i(\sqrt{g}\,V^i), \qquad g = \det(g_{ij}).$$

For completeness, the contraction in the middle expression satisfies

$$\Gamma^i_{mi} = \frac{1}{2}g^{i\ell}(\partial_m g_{\ell i} + \partial_i g_{\ell m} - \partial_\ell g_{mi}) = \frac{1}{2}g^{i\ell}\partial_m g_{i\ell} = \partial_m \ln\sqrt{g}.$$

The last two derivatives in parentheses cancel after interchanging the dummy indices i, ℓ in one of them and using the symmetry of $g^{i\ell}$ and $g_{\ell m}$. The last equality is the determinant derivative $\partial_m g = g\,g^{i\ell}\partial_m g_{\ell i}$: by the chain rule, $\partial_m g$ is the sum over entries of $(\partial g/\partial g_{\ell i})\,\partial_m g_{\ell i}$, and $\partial g/\partial g_{\ell i}$ is the cofactor of $g_{\ell i}$, which equals $g\,g^{i\ell}$ by the cofactor formula for the inverse matrix. The product rule then gives the first displayed formula:

$$\frac{1}{\sqrt{g}}\partial_i(\sqrt{g}\,V^i) = \partial_i V^i + V^m\partial_m \ln\sqrt{g} = \partial_i V^i + \Gamma^i_{mi}V^m.$$

In spherical polar coordinates $q^i = (r, \theta, \phi)$, differentiating $\mathbf{r} = (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$ gives the coordinate basis

$$\begin{aligned}\boldsymbol{\varepsilon}_r &= (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \\ \boldsymbol{\varepsilon}_\theta &= r(\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \\ \boldsymbol{\varepsilon}_\phi &= r\sin\theta(-\sin\phi, \cos\phi, 0).\end{aligned}$$

Their pairwise scalar products vanish, and their squared lengths are 1, r^2, $r^2\sin^2\theta$. Hence

$$g_{ij} = \mathrm{diag}(1, r^2, r^2\sin^2\theta), \qquad g = r^4\sin^2\theta, \qquad \sqrt{g} = r^2\sin\theta$$

on the usual regular chart $r > 0$, $0 < \theta < \pi$, where $\sin\theta > 0$. Let V_r, V_θ, V_ϕ denote components along the *unit* vectors. The coordinate basis vectors have lengths $\sqrt{g_{ii}} = 1, r, r\sin\theta$, so comparing $\mathbf{V} = V^i \boldsymbol{\varepsilon}_i$ with the unit-vector expansion gives

$$V^r = V_r, \qquad V^\theta = \frac{V_\theta}{r}, \qquad V^\phi = \frac{V_\phi}{r\sin\theta}.$$

Thus $\sqrt{g}\,V^r = r^2\sin\theta\,V_r$, $\sqrt{g}\,V^\theta = r\sin\theta\,V_\theta$, and $\sqrt{g}\,V^\phi = rV_\phi$. Insert these quantities before differentiating:

$$\begin{aligned}\nabla\cdot\mathbf{V} &= \frac{1}{r^2\sin\theta}\left[\partial_r(r^2\sin\theta V_r) + \partial_\theta(r\sin\theta V_\theta) + \partial_\phi(rV_\phi)\right] \\ &= \boxed{\frac{1}{r^2}\frac{\partial(r^2V_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta V_\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial V_\phi}{\partial\phi}}.\end{aligned}$$

Expanding the products gives the equivalent form $\partial_r V_r + 2V_r/r + (\partial_\theta V_\theta)/r + (\cot\theta)V_\theta/r + (\partial_\phi V_\phi)/(r\sin\theta)$.

4.3.12 The covariant vector A_i is the gradient of a twice continuously differentiable scalar. Show that the difference of covariant derivatives $A_{i;j} - A_{j;i}$ vanishes.

Solution.

Write $A_i = \partial_i\Phi$ for the scalar potential. Then

$$A_{i;j} = \partial_j\partial_i\Phi - \Gamma^k_{ij}\partial_k\Phi, \qquad A_{j;i} = \partial_i\partial_j\Phi - \Gamma^k_{ji}\partial_k\Phi.$$

Subtract:

$$A_{i;j} - A_{j;i} = (\partial_j\partial_i - \partial_i\partial_j)\Phi - (\Gamma^k_{ij} - \Gamma^k_{ji})\partial_k\Phi = 0.$$

The first term vanishes for a twice continuously differentiable scalar; the second vanishes by Exercise 4.3.4. Thus the covariant Hessian of a scalar is symmetric for this connection.

4.4 Jacobians

4.4.1 Assuming the functions u and v to be continuously differentiable, and interpreting part (a) locally where at least one of $\nabla u, \nabla v$ is nonzero,

(a) Show that a necessary and sufficient condition that $u(x, y, z)$ and $v(x, y, z)$ are related by a regular C^1 relation $f(u, v) = 0$, with $(f_u, f_v) \neq (0, 0)$, is that $(\nabla u) \times (\nabla v) = 0$;

(b) If $u = u(x, y)$ and $v = v(x, y)$, show that the condition $(\nabla u) \times (\nabla v) = 0$ leads to the 2-D Jacobian

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix} = 0.$$

Solution.

The usual equivalence is local and needs regularity. A meaningful relation is a C^1 function f with $(f_u, f_v) \neq (0, 0)$ on the image near the point in question; otherwise $f \equiv 0$ would relate every pair u, v, and necessity would fail: $u = x$, $v = y$ satisfy $f(u, v) = 0$, yet $\nabla u \times \nabla v = \hat{\mathbf{e}}_z \neq 0$. For

the converse, work where at least one of $\nabla u, \nabla v$ is nonzero. Mere differentiability, without these qualifications, does not establish the printed global "necessary and sufficient" statement.

(a) First suppose $f(u(x,y,z), v(x,y,z)) = 0$ is a regular relation. It holds identically near the point, so its partial derivatives vanish; by the chain rule, $\partial_x f(u,v) = f_u u_x + f_v v_x = 0$, and likewise for y and z. Together,

$$f_u \nabla u + f_v \nabla v = 0.$$

If $f_v \neq 0$, then $\nabla v = -(f_u/f_v)\nabla u$, and their cross product is zero. If $f_v = 0$, regularity forces $f_u \neq 0$, whence $\nabla u = 0$ and the cross product again vanishes. This proves necessity.

Conversely suppose $\nabla u \times \nabla v = 0$ in a neighborhood and $\nabla u \neq 0$ at the point under consideration. Relabel x, y, z if necessary so that $u_x \neq 0$ there. Continuity lets us shrink the neighborhood so that $u_x \neq 0$ throughout. The coordinate change

$$(s, \eta, \zeta) = (u(x,y,z), y, z)$$

has Jacobian

$$\frac{\partial(s,\eta,\zeta)}{\partial(x,y,z)} = \begin{vmatrix} u_x & u_y & u_z \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = u_x \neq 0,$$

so the inverse function theorem makes it locally invertible. The vanishing cross product says that $\nabla v = \lambda \nabla u$, with $\lambda = v_x/u_x$: its z- and y-components, $u_x v_y - u_y v_x = 0$ and $u_z v_x - u_x v_z = 0$, give $v_y = \lambda u_y$ and $v_z = \lambda u_z$, while $v_x = \lambda u_x$ by the choice of λ. In differential notation this is $dv = \lambda\, du$. Let $\widetilde{v}(s,\eta,\zeta)$ denote v expressed in the new coordinates. Since $s = u$, we have $du = ds$, and the relation reads

$$d\widetilde{v} = \frac{\partial \widetilde{v}}{\partial s}\, ds + \frac{\partial \widetilde{v}}{\partial \eta}\, d\eta + \frac{\partial \widetilde{v}}{\partial \zeta}\, d\zeta = \lambda\, ds.$$

The differentials $ds, d\eta, d\zeta$ of a coordinate system are linearly independent, so the coefficients agree: $\widetilde{v}_s = \lambda$ and $\widetilde{v}_\eta = \widetilde{v}_\zeta = 0$. Let (s_0, η_0, ζ_0) be the image of the point under consideration. On a sufficiently small coordinate box $|s - s_0| < \delta$, $|\eta - \eta_0| < \delta$, $|\zeta - \zeta_0| < \delta$, the mean value theorem applied in η and then in ζ gives $\widetilde{v}(s,\eta,\zeta) = \widetilde{v}(s,\eta_0,\zeta_0) \equiv F(s)$, a C^1 function of s alone, because $\widetilde{v}$ is v composed with the inverse coordinate map, and both are C^1 (the inverse function theorem supplies a C^1 inverse). Therefore

$$v = F(u), \qquad f(u,v) = v - F(u) = 0,$$

and this relation is regular because $f_v = 1$. If instead only ∇v is known to be nonzero, interchange u and v in the argument.

The restriction at critical points is substantive. For $u = x^2$, $v = x^3$, the gradients $(2x, 0, 0)$ and $(3x^2, 0, 0)$ are parallel, so the cross product vanishes everywhere, and both gradients vanish at $x = 0$. Yet no regular C^1 relation describes the image near $(u,v) = (0,0)$. If $f_v(0,0) \neq 0$, the implicit function theorem would make the zero set of f near the origin a graph $v = F(u)$; this is contradicted by x and $-x$ producing the same u and opposite v. If $f_u(0,0) \neq 0$, it would give a C^1 function with $u = G(v)$, that is, $G(x^3) = x^2$, so $G(v) = |v|^{2/3}$, whose difference quotient $|v|^{2/3}/v$ at zero is unbounded. The singular relation $v^2 - u^3 = 0$, with $f_u = -3u^2$ and $f_v = 2v$ both zero at the origin, does exist. So at a critical point a vanishing cross product may yield only a singular relation, which is why the converse in part (a) is restricted to points where ∇u or ∇v is nonzero.

(b) If the functions are independent of z, write their gradients as $(u_x, u_y, 0)$ and $(v_x, v_y, 0)$. Expanding the cross product gives

$$\nabla u \times \nabla v = \big(0, 0, u_x v_y - u_y v_x\big) = \hat{\mathbf{e}}_z \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}.$$

Thus its vanishing is exactly $\boxed{\partial(u,v)/\partial(x,y)=0}$. This algebraic identity itself does not require the regular-point hypothesis in part (a).

4.4.2 A 2-D orthogonal system is described by the coordinates q_1 and q_2. Show that the Jacobian J satisfies the equation

$$J \equiv \frac{\partial(x,y)}{\partial(q_1,q_2)} \equiv \frac{\partial x}{\partial q_1}\frac{\partial y}{\partial q_2} - \frac{\partial x}{\partial q_2}\frac{\partial y}{\partial q_1} = \pm h_1 h_2.$$

Hint. It's easier to work with the square of each side of this equation.

Solution.

The orientation-independent statement is $|J| = h_1 h_2$. The textbook prints $J = h_1 h_2$, whose positive sign assumes an orientation-preserving coordinate chart; the orthogonal chart $x = q_1$, $y = -q_2$ checked below has $J = -h_1 h_2$.

At a given point set

$$a = x_{q_1}, \quad b = y_{q_1}, \quad c = x_{q_2}, \quad d = y_{q_2}.$$

The coordinate basis vectors are $\partial\mathbf{r}/\partial q_1 = (a,b)$ and $\partial\mathbf{r}/\partial q_2 = (c,d)$, and the scale factors are their lengths, $h_i = |\partial\mathbf{r}/\partial q_i|$ (from $ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2$). Hence

$$h_1^2 = a^2 + b^2, \qquad h_2^2 = c^2 + d^2, \qquad ac + bd = 0.$$

The last equality is precisely orthogonality, $(\partial\mathbf{r}/\partial q_1)\cdot(\partial\mathbf{r}/\partial q_2) = 0$. Now expand the squares:

$$\begin{aligned} h_1^2 h_2^2 - (ac+bd)^2 &= (a^2+b^2)(c^2+d^2) - (a^2c^2 + 2abcd + b^2d^2) \\ &= a^2d^2 + b^2c^2 - 2abcd \\ &= (ad-bc)^2 = J^2. \end{aligned}$$

Because $ac + bd = 0$, we obtain $J^2 = h_1^2 h_2^2$. Scale factors are positive on a regular chart, hence $|J| = h_1 h_2$. Positive orientation selects $J = +h_1 h_2$; negative orientation selects $J = -h_1 h_2$. For an explicit check, $x = q_1$, $y = -q_2$ is orthogonal with $h_1 = h_2 = 1$ but $J = -1$, so orthogonality alone cannot fix the sign.

4.4.3 For the transformation $u = x + y$, $v = x/y$, with $x \geq 0$ and $y > 0$, find the Jacobian $\dfrac{\partial(x,y)}{\partial(u,v)}$

(a) By direct computation,

(b) By first computing J^{-1}.

Solution.

The ratio $v = x/y$ requires $y > 0$ on the stated nonnegative quadrant; the boundary $y = 0$ is excluded. Then $u > 0$, $v \geq 0$, and the inverse transformation follows directly from $x = vy$:

$$u = x + y = (v+1)y, \qquad \boxed{y = \frac{u}{v+1}, \qquad x = \frac{uv}{v+1}}.$$

(a) Differentiate the inverse formulas; for x_v the quotient rule gives $x_v = u[(v+1) - v]/(v+1)^2$:

$$x_u = \frac{v}{v+1}, \qquad x_v = \frac{u}{(v+1)^2}, \qquad y_u = \frac{1}{v+1}, \qquad y_v = -\frac{u}{(v+1)^2}.$$

Their determinant is

$$J = x_u y_v - x_v y_u = \frac{v}{v+1}\left(-\frac{u}{(v+1)^2}\right) - \frac{u}{(v+1)^2}\cdot\frac{1}{v+1}$$
$$= -\frac{uv}{(v+1)^3} - \frac{u}{(v+1)^3} = -\frac{u(v+1)}{(v+1)^3} = \boxed{-\frac{u}{(v+1)^2}}.$$

(b) For the forward transformation,

$$u_x = 1, \quad u_y = 1, \quad v_x = \frac{1}{y}, \quad v_y = -\frac{x}{y^2}.$$

Thus

$$J^{-1} = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & 1 \\ 1/y & -x/y^2 \end{vmatrix} = -\frac{x}{y^2} - \frac{1}{y} = -\frac{x+y}{y^2} = -\frac{u}{y^2}.$$

It is nonzero on the admissible domain. The chain rule makes the Jacobian matrices of a map and its inverse mutually inverse, and the determinant of a product of matrices is the product of their determinants, so $\big[\partial(x,y)/\partial(u,v)\big]\big[\partial(u,v)/\partial(x,y)\big] = 1$, and with $y = u/(v+1)$,

$$J = -\frac{y^2}{u} = -\frac{1}{u}\left(\frac{u}{v+1}\right)^2 = -\frac{u}{(v+1)^2},$$

in agreement with (a). The negative sign records orientation reversal; an ordinary unoriented area integral uses $|J| = u/(v+1)^2$.

4.5 Differential Forms

4.5.1 Using the rules for the application of the Hodge star operator, verify the results given in Eq. (4.82) for its application to all linearly independent differential forms in Minkowski space. In the rule on p. 236, count the negative metric factors in the input form being starred, rather than in its complement.

Solution.

Use the ordered basis of 1-forms

$$e^0 = dt, \quad e^1 = dx_1, \quad e^2 = dx_2, \quad e^3 = dx_3, \qquad \eta = \mathrm{diag}(1,-1,-1,-1),$$

and write $e^{a_1\cdots a_p} = e^{a_1}\wedge\cdots\wedge e^{a_p}$, with $\mathrm{vol} = e^{0123}$. For an increasing index set I, let J be its increasing complement, m the number of spatial indices in I, and N the number of inversions in the concatenation (I,J), that is, of pairs of entries standing in decreasing order; for example, for $I = \{1,3\}$ the concatenation $(1,3,0,2)$ has the three inversions $(1,0)$, $(3,0)$, $(3,2)$, so $N = 3$. Then

$$e^I \wedge e^J = (-1)^N \mathrm{vol}, \qquad \langle e^I, e^I\rangle = (-1)^m:$$

the first because sorting (I,J) into the order 0123 takes N interchanges of adjacent factors, each of which changes the sign (each interchange of an adjacent pair standing in decreasing order removes exactly one inversion); the second because the metric induces on p-forms the inner product $\langle e^K, e^I\rangle = \det\big[\eta^{ab}\big]_{a\in K,\, b\in I}$, which for $K = I$ is the product of the diagonal metric entries η^{aa}, $a\in I$, m of which equal -1. For $K \neq I$ this determinant vanishes: since η is diagonal, the row of an index $a\in K$ not in I is zero. The Hodge star of a p-form β is fixed by the standard defining relation

$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \mathrm{vol}$ for every p-form α. Take $\beta = e^I$. For $\alpha = e^K$ with $K \neq I$ the right side is zero by this orthogonality, and e^K has a nonzero wedge product only with the basis form of its own complement; hence $*e^I$ has no component along the complement of any $K \neq I$, and $*e^I = s\, e^J$ for some number s. Then $\alpha = e^I$ gives $s(-1)^N = (-1)^m$, so

$$\boxed{*e^I = (-1)^{N+m} e^J}.$$

The sign combines the parity $(-1)^N$ of the permutation that orders $e^I \wedge e^J$ with a factor -1 for each negative metric entry of the form e^I being starred.

The rule in the first paragraph of textbook p. 236 says to count negative metric entries in the complementary form ω'. That is a typographical error: count them in the form ω being starred. The worked example and Eq. (4.82) use this corrected rule: for $\omega = 1$ the complement $\omega' = \mathrm{vol}$ has three negative metric factors, so the printed rule would give $*1 = -\mathrm{vol}$, whereas Eq. (4.82) has $*1 = +\mathrm{vol}$. Since ω and ω' together contain all three spatial differentials, the two counts always differ in parity, so the printed rule would reverse the sign of every result in Eq. (4.82). For instance, $*1 = \mathrm{vol}$ has $m = 0$, while $*\mathrm{vol} = -1$ has $m = 3$.

Here is the explicit calculation for every basis form. The complement column is written before its final sign is applied.

input	complement	N	m	Hodge star
1	e^{0123}	0	0	e^{0123}
e^0	e^{123}	0	0	e^{123}
e^1	e^{023}	1	1	e^{023}
e^2	e^{013}	2	1	$-e^{013}$
e^3	e^{012}	3	1	e^{012}
e^{01}	e^{23}	0	1	$-e^{23}$
e^{02}	e^{13}	1	1	e^{13}
e^{03}	e^{12}	2	1	$-e^{12}$
e^{12}	e^{03}	2	2	e^{03}
e^{13}	e^{02}	3	2	$-e^{02}$
e^{23}	e^{01}	4	2	e^{01}
e^{012}	e^3	0	2	e^3
e^{013}	e^2	1	2	$-e^2$
e^{023}	e^1	2	2	e^1
e^{123}	e^0	3	3	e^0
e^{0123}	1	0	3	-1

The table includes $1 + 4 + 6 + 4 + 1 = 16$ independent forms. Take (i, j, k) to be $(1, 2, 3)$, $(2, 3, 1)$, or $(3, 1, 2)$. Rewriting the table in cyclic order absorbs the displayed minus signs associated with reordering spatial indices. These occur in the rows e^2, e^{13}, e^{013}; using $dx_1 \wedge dx_3 = -dx_3 \wedge dx_1$ (also inside the starred forms),

$$*dx_2 = -dt \wedge dx_1 \wedge dx_3 = dt \wedge dx_3 \wedge dx_1, \qquad *(dx_3 \wedge dx_1) = -*(dx_1 \wedge dx_3) = dt \wedge dx_2,$$
$$*(dt \wedge dx_3 \wedge dx_1) = -*(dt \wedge dx_1 \wedge dx_3) = dx_2,$$

while the row e^{02} gives $*(dt \wedge dx_2) = dx_1 \wedge dx_3 = -dx_3 \wedge dx_1$. The table thus gives exactly

$$*1 = dt \wedge dx_1 \wedge dx_2 \wedge dx_3, \qquad *dt = dx_1 \wedge dx_2 \wedge dx_3,$$
$$*dx_i = dt \wedge dx_j \wedge dx_k,$$
$$*(dx_j \wedge dx_k) = dt \wedge dx_i, \qquad *(dt \wedge dx_i) = -dx_j \wedge dx_k,$$
$$*(dx_1 \wedge dx_2 \wedge dx_3) = dt, \qquad *(dt \wedge dx_i \wedge dx_j) = dx_k,$$
$$*(dt \wedge dx_1 \wedge dx_2 \wedge dx_3) = -1.$$

As an independent sign check, apply the boxed rule twice to a p-form: $*e^I = (-1)^{N_I + m_I} e^J$ and $*e^J = (-1)^{N_J + m_J} e^I$, where N_J counts the inversions of (J, I). Since I and J are both listed in increasing order, no two indices from the same set form an inversion, while each of the $p(4-p)$ pairs made of one index from I and one from J is an inversion in exactly one of the orders (I, J) and (J, I); so $N_I + N_J = p(4-p)$. Also $m_I + m_J = 3$, the total number of spatial indices. Hence

$$**\omega_p = (-1)^{p(4-p)+3} \omega_p,$$

which is negative for even p and positive for odd p, in agreement with every pair of rows in the table.

4.5.2 If the force field is constant and moving a particle from the origin to $(3, 0, 0)$ requires a units of work, from $(-1, -1, 0)$ to $(-1, 1, 0)$ takes b units of work, and from $(0, 0, 4)$ to $(0, 0, 5)$ c units of work, find the 1-form of the work.

Solution.

Let $\mathbf{F} = (F_x, F_y, F_z)$ be the constant force that does the quoted work (if the work is done against the field, $\mathbf{F}$ is minus the field's force). Its work 1-form is $\omega = F_x\,dx + F_y\,dy + F_z\,dz$, and its work between two points is $\mathbf{F} \cdot \Delta\mathbf{r}$, because each force component can be taken outside the line integral. The three given displacements are

$$(3, 0, 0), \qquad (0, 2, 0), \qquad (0, 0, 1).$$

They yield $3F_x = a$, $2F_y = b$, and $F_z = c$. Therefore

$$\boxed{\omega = \frac{a}{3}\,dx + \frac{b}{2}\,dy + c\,dz}.$$

For a direct check, integrating along the three paths multiplies these coefficients by $3, 2, 1$, respectively, returning a, b, c. The form is also exact: $\omega = d(ax/3 + by/2 + cz)$, so the result does not depend on the shapes of the paths.

4.6 Differentiating Forms

4.6.1 Given the two 1-forms $\omega_1 = x\,dy + y\,dx$ and $\omega_2 = x\,dy - y\,dx$, calculate

(a) $d\omega_1$,

(b) $d\omega_2$.

(c) For each of your answers to (a) or (b) that is nonzero, apply the operator d a second time and verify that $d(d\omega_i) = 0$.

Solution.

Use $d(f\alpha) = df \wedge \alpha + f\, d\alpha$ and $d(dx) = d(dy) = 0$.

(a) For the first form,

$$\begin{aligned} d\omega_1 &= d(x\,dy) + d(y\,dx) \\ &= dx \wedge dy + x\,d(dy) + dy \wedge dx + y\,d(dx) \\ &= dx \wedge dy - dx \wedge dy = \boxed{0}. \end{aligned}$$

Indeed $\omega_1 = d(xy)$.

(b) Changing the sign between the terms, and using the two derivatives $d(x\,dy) = dx \wedge dy$ and $d(y\,dx) = dy \wedge dx$ found in (a), gives

$$\begin{aligned} d\omega_2 &= d(x\,dy) - d(y\,dx) \\ &= dx \wedge dy - dy \wedge dx = \boxed{2\,dx \wedge dy}. \end{aligned}$$

(c) The nonzero result is a constant-coefficient 2-form. Applying d again, and using the product rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p\, \alpha \wedge d\beta$ for a p-form α (here $\alpha = dx$ with $p = 1$, hence the minus sign),

$$d(d\omega_2) = 2\,d(dx \wedge dy) = 2\{d(dx) \wedge dy - dx \wedge d(dy)\} = 0.$$

The already-zero result for ω_1 also has zero exterior derivative.

4.6.2 Apply the operator d twice to $\omega_3 = xy\,dz + xz\,dy - yz\,dx$. Verify that the second application of d yields a zero result.

Solution.

First differentiate each coefficient and keep every wedge term:

$$\begin{aligned} d\omega_3 &= d(xy) \wedge dz + d(xz) \wedge dy - d(yz) \wedge dx \\ &= (y\,dx + x\,dy) \wedge dz + (z\,dx + x\,dz) \wedge dy - (z\,dy + y\,dz) \wedge dx \\ &= y\,dx \wedge dz + x\,dy \wedge dz + z\,dx \wedge dy + x\,dz \wedge dy \\ &\quad - z\,dy \wedge dx - y\,dz \wedge dx. \end{aligned}$$

The two terms with coefficient x cancel. Reordering the last two terms reverses their signs, so the terms with coefficients y and z each add rather than cancel:

$$\boxed{d\omega_3 = 2y\,dx \wedge dz + 2z\,dx \wedge dy}.$$

Apply d again. Derivatives of the basis differentials vanish, leaving $d(2y) \wedge dx \wedge dz + d(2z) \wedge dx \wedge dy$ with $d(2y) = 2\,dy$ and $d(2z) = 2\,dz$:

$$\begin{aligned} d(d\omega_3) &= 2\,dy \wedge dx \wedge dz + 2\,dz \wedge dx \wedge dy \\ &= -2\,dx \wedge dy \wedge dz + 2\,dx \wedge dy \wedge dz = \boxed{0}. \end{aligned}$$

The first permutation needs one interchange; the second needs two, which explains their opposite signs.

4.6.3 For ω_2 and ω_3 the 1-forms with these names in Exercises 4.6.1 and 4.6.2, evaluate $d(\omega_2 \wedge \omega_3)$:

(a) By forming the exterior product and then differentiating, and

(b) Using the formula for differentiating a product of two forms.

Verify that both approaches give the same result.

Solution.

(a) Expand the product, keeping the order of differentials:

$$\begin{aligned}\omega_2 \wedge \omega_3 &= (x\,dy - y\,dx) \wedge (xy\,dz + xz\,dy - yz\,dx)\\ &= x^2 y\,dy \wedge dz + x^2 z\,dy \wedge dy - xyz\,dy \wedge dx\\ &\quad - xy^2\,dx \wedge dz - xyz\,dx \wedge dy + y^2 z\,dx \wedge dx\\ &= x^2 y\,dy \wedge dz - xy^2\,dx \wedge dz.\end{aligned}$$

The repeated-differential terms vanish, and the two xyz terms cancel because $dy \wedge dx = -dx \wedge dy$. Differentiating the surviving coefficients, $d(x^2y) = 2xy\,dx + x^2\,dy$ and $d(xy^2) = y^2\,dx + 2xy\,dy$, gives the following; in the third line the terms $x^2\,dy \wedge dy \wedge dz$ and $-y^2\,dx \wedge dx \wedge dz$ are dropped because they contain a repeated differential:

$$\begin{aligned}d(\omega_2 \wedge \omega_3) &= (2xy\,dx + x^2\,dy) \wedge dy \wedge dz\\ &\quad - (y^2\,dx + 2xy\,dy) \wedge dx \wedge dz\\ &= 2xy\,dx \wedge dy \wedge dz - 2xy\,dy \wedge dx \wedge dz\\ &= 2xy\,dx \wedge dy \wedge dz + 2xy\,dx \wedge dy \wedge dz = \boxed{4xy\,dx \wedge dy \wedge dz}.\end{aligned}$$

(b) Since ω_2 is a 1-form, the product rule quoted in Exercise 4.6.1(c) has a minus sign:

$$d(\omega_2 \wedge \omega_3) = d\omega_2 \wedge \omega_3 - \omega_2 \wedge d\omega_3.$$

Use the preceding exercises to compute the terms separately:

$$\begin{aligned}d\omega_2 \wedge \omega_3 &= 2\,dx \wedge dy \wedge (xy\,dz + xz\,dy - yz\,dx)\\ &= 2xy\,dx \wedge dy \wedge dz,\\ \omega_2 \wedge d\omega_3 &= (x\,dy - y\,dx) \wedge (2y\,dx \wedge dz + 2z\,dx \wedge dy)\\ &= 2xy\,dy \wedge dx \wedge dz = -2xy\,dx \wedge dy \wedge dz.\end{aligned}$$

All terms not displayed in the simplified lines contain a repeated dx or dy and are zero. Subtracting the second result from the first gives $(2xy - (-2xy))\,dx \wedge dy \wedge dz = 4xy\,dx \wedge dy \wedge dz$, exactly as in (a).

4.7 Integrating Forms

4.7.1 Use differential-forms relations to transform the integral $A(x,y,z)\,dx\wedge dy\wedge dz$ to the equivalent expression in $du\wedge dv\wedge dw$, where u,v,w is an invertible linear transformation of x,y,z, and thereby find the determinant that can be identified as the Jacobian of the transformation.

Solution.

To serve as a change of three independent coordinates, the linear transformation must be invertible. Write its inverse as

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} \equiv L \begin{pmatrix} u \\ v \\ w \end{pmatrix}.$$

The coefficients are constant, so

$$dx = a_1\,du + a_2\,dv + a_3\,dw, \quad dy = b_1\,du + b_2\,dv + b_3\,dw, \quad dz = c_1\,du + c_2\,dv + c_3\,dw.$$

Expand their exterior product. Any term containing two copies of a differential vanishes; precisely six terms remain, one for each ordering of du, dv, dw:

$$\begin{aligned} dx\wedge dy\wedge dz = {} & a_1b_2c_3\,du\wedge dv\wedge dw + a_1b_3c_2\,du\wedge dw\wedge dv \\ & + a_2b_1c_3\,dv\wedge du\wedge dw + a_2b_3c_1\,dv\wedge dw\wedge du \\ & + a_3b_1c_2\,dw\wedge du\wedge dv + a_3b_2c_1\,dw\wedge dv\wedge du. \end{aligned}$$

Each interchange of two factors changes the sign. The orderings $du\wedge dw\wedge dv$, $dv\wedge du\wedge dw$, $dw\wedge dv\wedge du$ are one interchange away from $du\wedge dv\wedge dw$ and acquire a factor -1; the cyclic shifts $dv\wedge dw\wedge du$ and $dw\wedge du\wedge dv$ are two interchanges away and keep their sign. Hence

$$dx\wedge dy\wedge dz = (a_1b_2c_3 - a_1b_3c_2 - a_2b_1c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1)\,du\wedge dv\wedge dw.$$

The coefficient is $a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$, the first-row expansion of the determinant of L. Since $x_u = a_1$, $x_v = a_2$, …, $z_w = c_3$, it is

$$J = \det L = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix} = \frac{\partial(x,y,z)}{\partial(u,v,w)}.$$

Therefore the transformed 3-form is

$$\boxed{A(x(u,v,w),y(u,v,w),z(u,v,w))\,J\,du\wedge dv\wedge dw}.$$

Its integral is taken over the corresponding parameter region with the induced orientation. Differential forms retain the sign of J; a scalar integral written using positive, unoriented volume elements instead uses $|J|$. If the forward transformation has matrix M, then $L = M^{-1}$ and $J = 1/\det M$.

4.7.2 Write Oersted's law,

$$\int_{\partial S} \mathbf{H}\cdot d\mathbf{r} = \int_S \nabla\times\mathbf{H}\cdot d\mathbf{a} \sim I,$$

in differential form notation.

Solution.

Associate the 1-form

$$h = H_x\,dx + H_y\,dy + H_z\,dz$$

with the magnetic field. Its exterior derivative is $dh = dH_x \wedge dx + dH_y \wedge dy + dH_z \wedge dz$. Expanding each $dH_i = \partial_x H_i\,dx + \partial_y H_i\,dy + \partial_z H_i\,dz$, dropping the terms with a repeated differential, and writing every pair in the order $dy \wedge dz$, $dz \wedge dx$, or $dx \wedge dy$,

$$\begin{aligned} dH_x \wedge dx &= \partial_y H_x\,dy \wedge dx + \partial_z H_x\,dz \wedge dx = -\partial_y H_x\,dx \wedge dy + \partial_z H_x\,dz \wedge dx, \\ dH_y \wedge dy &= \partial_x H_y\,dx \wedge dy + \partial_z H_y\,dz \wedge dy = \partial_x H_y\,dx \wedge dy - \partial_z H_y\,dy \wedge dz, \\ dH_z \wedge dz &= \partial_x H_z\,dx \wedge dz + \partial_y H_z\,dy \wedge dz = -\partial_x H_z\,dz \wedge dx + \partial_y H_z\,dy \wedge dz. \end{aligned}$$

Grouping terms with the same ordered differential pair,

$$\begin{aligned} dh = {} & (\partial_y H_z - \partial_z H_y)\,dy \wedge dz \\ & + (\partial_z H_x - \partial_x H_z)\,dz \wedge dx \\ & + (\partial_x H_y - \partial_y H_x)\,dx \wedge dy. \end{aligned}$$

These three coefficients are the components of $\nabla \times \mathbf{H}$. To integrate over S, parametrize it as $\mathbf{r}(s,t)$, with (s,t) ordered so that $\mathbf{r}_s \times \mathbf{r}_t$ points along the chosen normal; then $d\mathbf{a} = \mathbf{r}_s \times \mathbf{r}_t\,ds\,dt$. On S,

$$dy \wedge dz = (y_s\,ds + y_t\,dt) \wedge (z_s\,ds + z_t\,dt) = (y_s z_t - y_t z_s)\,ds \wedge dt,$$

and $y_s z_t - y_t z_s$ is the x-component of $\mathbf{r}_s \times \mathbf{r}_t$; cyclically ($x \to y \to z \to x$), $dz \wedge dx$ and $dx \wedge dy$ give its y- and z-components. Hence a 2-form $P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy$ integrates over S to the flux $\int_S (P,Q,R) \cdot d\mathbf{a}$; in particular $\int_S dh = \int_S \nabla \times \mathbf{H} \cdot d\mathbf{a}$. Along ∂S the 1-form h is $\mathbf{H} \cdot d\mathbf{r}$, so $\int_{\partial S} h = \int_{\partial S} \mathbf{H} \cdot d\mathbf{r}$. Stokes' theorem thus gives the requested notation:

$$\boxed{\int_{\partial S} h = \int_S dh \;\sim\; I}.$$

The boundary orientation is the one induced by S. More explicitly, if the current density is $\mathbf{J}$, its flux 2-form is

$$j = J_x\,dy \wedge dz + J_y\,dz \wedge dx + J_z\,dx \wedge dy, \qquad I = \int_S j.$$

With the proportionality constant κ appropriate to the chosen units ($\kappa = 1$ in SI units, where $\nabla \times \mathbf{H} = \mathbf{J}$; $\kappa = 4\pi/c$ in Gaussian units), the magnetostatic field equation is $dh = \kappa j$, and the integral relation reads $\int_{\partial S} h = \kappa \int_S j = \kappa I$. This preserves the unspecified proportionality convention in the question.

4.7.3 A 1-form $A\,dx + B\,dy$ is defined as closed if $\dfrac{\partial A}{\partial y} = \dfrac{\partial B}{\partial x}$. It is called exact if there is a function f such that $\dfrac{\partial f}{\partial x} = A$ and $\dfrac{\partial f}{\partial y} = B$. Determine which of the following 1-forms are closed, or exact, and find the corresponding functions f for those that are exact. Use their natural real domains for the first four forms, and state explicitly when a primitive is only local. For the fifth, take a C^1 complex-valued coefficient on an open domain in $\mathbb{C}$ and give the conditions for closedness and global exactness:

$$y\,dx + x\,dy, \qquad \frac{y\,dx + x\,dy}{x^2 + y^2}, \qquad [\ln(xy) + 1]\,dx + \frac{x}{y}\,dy,$$

$$-\frac{y\,dx}{x^2+y^2}+\frac{x\,dy}{x^2+y^2},\qquad f(z)\,dz \text{ with } z=x+iy.$$

Solution.

For a C^1 1-form $\omega = A\,dx + B\,dy$, the terms $A_x\,dx\wedge dx$ and $B_y\,dy\wedge dy$ vanish, so

$$d\omega = dA\wedge dx + dB\wedge dy = A_y\,dy\wedge dx + B_x\,dx\wedge dy = (B_x - A_y)\,dx\wedge dy.$$

Thus closedness means $B_x = A_y$. A globally exact form $\omega = d\Phi$ is closed, because $A_y = \Phi_{xy} = \Phi_{yx} = B_x$ (equality of mixed partial derivatives, Schwarz's theorem; Φ is C^2 since its first derivatives A, B are C^1), and it has zero integral around every closed curve Γ, because $\int_\Gamma d\Phi$ is the difference of the values of Φ at the endpoints of Γ, which coincide. We use Φ for a potential to distinguish it from the last question's f.

Exactness depends on the domain. In particular the angular form below is only locally exact on the punctured plane. For the last form, closedness is equivalent to holomorphy of f, and a closed $f\,dz$ is globally exact exactly when $\oint_\Gamma f\,dz = 0$ for every closed curve Γ in the domain, which holds on every simply connected domain but not in general (dz/z on $\mathbb{C}\setminus\{0\}$ is a counterexample).

First form. For $\omega = y\,dx + x\,dy$, $A_y = B_x = 1$, so the form is closed. Integrate $\Phi_x = y$ with respect to x: $\Phi = xy + h(y)$. The other equation gives $\Phi_y = x + h'(y) = x$, hence $h' = 0$. Therefore

$$\boxed{\omega = d(xy), \qquad \Phi = xy + C},$$

and the form is globally exact on $\mathbb{R}^2$.

Second form. Its domain excludes the origin. Set $A = y/(x^2+y^2)$ and $B = x/(x^2+y^2)$. The quotient rule gives

$$A_y = \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}, \qquad B_x = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

These derivatives differ except on the lines $y = \pm x$. Those lines contain no nonempty open subset of the plane, and a form that is exact on an open set is closed there. Consequently

$$\boxed{\frac{y\,dx + x\,dy}{x^2+y^2} \text{ is neither closed nor exact}}$$

on any nonempty open part of its natural domain.

Third form. For real logarithms the domain is $xy > 0$. The two coefficients obey

$$\partial_y[\ln(xy)+1] = \frac{x}{xy} = \frac{1}{y}, \qquad \partial_x(x/y) = \frac{1}{y},$$

so the form is closed. Integration in x gives $\Phi = x\ln(xy) + h(y)$, because, by the product rule and $\partial_x \ln(xy) = y/(xy) = 1/x$, $\partial_x[x\ln(xy)] = \ln(xy) + x\cdot(1/x) = \ln(xy)+1$. Then $\Phi_y = x/y + h'(y) = x/y$, so h is constant. Thus

$$\boxed{[\ln(xy)+1]\,dx + \frac{x}{y}\,dy = d[x\ln(xy)]}.$$

It is exact on each component of $xy > 0$, and also on their union with the displayed potential; arbitrary additive constants may be chosen independently on the two components.

Fourth form. Here $A = -y/(x^2+y^2)$ and $B = x/(x^2+y^2)$, so

$$A_y = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2} = B_x,$$

with B_x as computed for the second form. The form is closed everywhere on $\mathbb{R}^2 \setminus \{0\}$. On a region with a continuous angle coordinate θ, use $x = r\cos\theta$, $y = r\sin\theta$, so that $dx = \cos\theta\, dr - r\sin\theta\, d\theta$ and $dy = \sin\theta\, dr + r\cos\theta\, d\theta$. The dr terms cancel and the $d\theta$ terms combine through $\sin^2\theta + \cos^2\theta = 1$:

$$\begin{aligned} -y\,dx + x\,dy = &\ -r\sin\theta(\cos\theta\, dr - r\sin\theta\, d\theta) \\ &+ r\cos\theta(\sin\theta\, dr + r\cos\theta\, d\theta) = r^2\, d\theta. \end{aligned}$$

Dividing by $x^2 + y^2 = r^2$, $\omega = d\theta$ locally. For example, on $x > 0$ a potential is $\Phi = \arctan(y/x) + C$, since

$$\Phi_x = \frac{-y/x^2}{1 + y^2/x^2} = -\frac{y}{x^2 + y^2}, \qquad \Phi_y = \frac{1/x}{1 + y^2/x^2} = \frac{x}{x^2 + y^2}.$$

For a complete counterclockwise circle $x = R\cos t$, $y = R\sin t$, $0 \le t \le 2\pi$, where $dx = -R\sin t\, dt$ and $dy = R\cos t\, dt$,

$$\oint \omega = \int_0^{2\pi} \frac{R^2\sin^2 t + R^2\cos^2 t}{R^2}\, dt = 2\pi.$$

A single-valued potential would make this integral zero. Therefore

$$\boxed{\omega \text{ is closed but not globally exact on the punctured plane}}.$$

The Instructor's Manual, p. 65, prints $d\tan(y/x)$; the local primitive is $\arctan(y/x)$, and its assertion of exactness needs the domain qualification just proved.

Fifth form. Write the complex-valued coefficient as $f(x, y)$ and use $dz = dx + i\, dy$. Thus $A = f$ and $B = if$, so $B_x - A_y = if_x - f_y = 2i \cdot \frac{1}{2}(f_x + if_y)$, giving

$$d(f\, dz) = (if_x - f_y)\, dx \wedge dy = 2i\frac{\partial f}{\partial \bar{z}}\, dx \wedge dy, \qquad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}(f_x + if_y).$$

For C^1 coefficients the form is closed exactly when $f_y = if_x$. With $f = u + iv$ this reads $u_y + iv_y = iu_x - v_x$, that is, $u_x = v_y$ and $u_y = -v_x$: the Cauchy–Riemann conditions (Section 11.2) for f to be holomorphic. Under that condition it is locally exact. One can construct a primitive on a small rectangle with sides parallel to the axes, centred at a point (x_0, y_0) and contained in the domain, without presupposing a complex antiderivative. For (x, y) in the rectangle, the horizontal segment from (x_0, y_0) to (x, y_0) and the vertical segment from (x, y_0) to (x, y) stay inside it, so both integrals in

$$\Phi(x, y) = \int_{x_0}^{x} f(s, y_0)\, ds + i\int_{y_0}^{y} f(x, t)\, dt$$

are defined. Only the second integral depends on y, so the fundamental theorem of calculus gives Φ_y. For Φ_x, differentiate the second integral under the integral sign (allowed for C^1 f), then use $if_x = f_y$ and the fundamental theorem in t:

$$\Phi_y = if(x, y), \qquad \Phi_x = f(x, y_0) + i\int_{y_0}^{y} f_x(x, t)\, dt = f(x, y_0) + \int_{y_0}^{y} f_y(x, t)\, dt = f(x, y).$$

Hence $d\Phi = \Phi_x\, dx + \Phi_y\, dy = f\,(dx + i\, dy) = f\, dz$. On a simply connected complex domain, every closed curve Γ is contractible (it can be shrunk continuously to a point without leaving the domain). The shrinking sweeps out a surface S in the domain whose boundary is Γ, and $d(f\, dz) = 0$ on S, so Stokes' theorem gives $\oint_\Gamma f\, dz = \int_S d(f\, dz) = 0$ (this is Cauchy's integral theorem, Section 11.3).

Two paths with the same endpoints together form a closed curve, so they have equal integrals. The local primitives thus give a global primitive

$$\boxed{F(z) = \int_{z_0}^{z} f(\zeta)\,d\zeta + C, \qquad dF = f(z)\,dz},$$

where the integral is path independent. To see that $dF = f\,dz$, fix a point z_1 and let Φ be a local primitive, constructed as above, on a small rectangle about z_1 inside the domain. For z in that rectangle, path independence lets us integrate from z_0 to z_1 and then from z_1 to z inside the rectangle, where $f\,d\zeta = d\Phi$; hence

$$F(z) = F(z_1) + \int_{z_1}^{z} d\Phi = F(z_1) + \Phi(z) - \Phi(z_1).$$

So near z_1, F differs from Φ by a constant, and $dF = d\Phi = f\,dz$. More generally, this formula defines a global primitive precisely when $\oint_\Gamma f(z)\,dz = 0$ for every closed curve Γ in the domain. The domain condition cannot be omitted: $f(z) = 1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$, but on $z = Re^{it}$, $0 \le t \le 2\pi$, where $dz = iRe^{it}\,dt$,

$$\oint \frac{dz}{z} = \int_0^{2\pi} i\,dt = 2\pi i \neq 0,$$

so it is not globally exact there. If f is not holomorphic, even closedness can fail: $f = \bar{z} = x - iy$ has $f_x = 1$, $f_y = -i$, so $d(f\,dz) = (i + i)\,dx \wedge dy = 2i\,dx \wedge dy$. Accordingly, the Instructor's Manual's unqualified claim that $f(z)\,dz$ is exact must be read with both the analyticity and domain conditions above.

5 Chapter 5. Vector Spaces

Problems

5.1 Vectors in Function Spaces

5.1.1 A function $f(x)$ is expanded in a series of orthonormal functions, with convergence in the Hilbert-space norm,

$$f(x) = \sum_{n=0}^{\infty} a_n \varphi_n(x).$$

Show that the series expansion is unique for a given set of $\varphi_n(x)$. The functions $\varphi_n(x)$ are being taken here as the basis vectors in an infinite-dimensional Hilbert space.

Solution.

Write $S_N = \sum_{n=0}^{N} a_n \varphi_n$; the expansion means that $S_N \to f$ in the Hilbert-space norm. The textbook does not say in what sense the series converges, and uniqueness depends on it: the argument below uses norm convergence, whereas for convergence almost everywhere uniqueness fails (Menshov constructed a trigonometric series, with coefficients not all zero, that converges to zero almost everywhere). For a fixed k and $N \geq k$,

$$\langle \varphi_k | S_N \rangle = \sum_{n=0}^{N} a_n \langle \varphi_k | \varphi_n \rangle = \sum_{n=0}^{N} a_n \delta_{kn} = a_k.$$

The scalar product is continuous: by linearity in the second argument and Schwarz's inequality, for every $N \geq k$,

$$|\langle \varphi_k | f \rangle - a_k| = |\langle \varphi_k | f - S_N \rangle| \leq \|\varphi_k\| \, \|f - S_N\| = \|f - S_N\| \longrightarrow 0.$$

The left side does not depend on N, so it is zero. Each coefficient is therefore determined by f alone:

$$\boxed{a_k = \langle \varphi_k | f \rangle}.$$

Any other convergent expansion in the same orthonormal set has the same scalar products and hence the same coefficients. Equality of functions in this Hilbert space means equality almost everywhere; changes at isolated points do not produce a different Hilbert-space vector.

5.1.2 A function $f(x)$ is represented by a finite set of basis functions $\varphi_i(x)$,

$$f(x) = \sum_{i=1}^{N} c_i \varphi_i(x).$$

Show that the components c_i are unique, that no different set c_i' exists.

Note. Your basis functions are automatically linearly independent. They are not necessarily orthogonal.

Solution.

Suppose also $f = \sum_{i=1}^{N} c_i' \varphi_i$. Subtract the two finite sums:

$$0 = \sum_{i=1}^{N} (c_i - c_i') \varphi_i.$$

Linear independence means precisely that a zero linear combination of the φ_i has every coefficient zero. Consequently $c_i - c_i' = 0$ for $i = 1, \ldots, N$, proving uniqueness without any orthogonality assumption.

As a scalar-product check, let $G_{ij} = \langle \varphi_i | \varphi_j \rangle$. For any nonzero column d, linear independence makes $\sum_i d_i \varphi_i$ a nonzero function, so

$$d^\dagger G d = \sum_{i,j} d_i^* \langle \varphi_i | \varphi_j \rangle d_j = \left\| \sum_i d_i \varphi_i \right\|^2 > 0.$$

Hence $Gd \neq 0$ whenever $d \neq 0$, i.e., G is invertible. Taking the scalar product of the expansion of f with φ_i gives the equations $\sum_j G_{ij} c_j = \langle \varphi_i | f \rangle$, which therefore also have exactly one solution.

5.1.3 A square-integrable function $f(x)$ is approximated by a power series $\sum_{i=0}^{n-1} c_i x^i$ over the interval $[0, 1]$. Show that minimizing the mean square error leads to a set of linear equations

$$\mathsf{A}\mathbf{c} = \mathbf{b},$$

where

$$A_{ij} = \int_0^1 x^{i+j}\, dx = \frac{1}{i+j+1}, \quad i, j = 0, 1, 2, \ldots, n-1$$

and

$$b_i = \int_0^1 x^i f(x)\, dx, \quad i = 0, 1, 2, \ldots, n-1.$$

Note. The A_{ij} are the elements of the Hilbert matrix of order n. The determinant of this Hilbert matrix is a rapidly decreasing function of n. For $n = 5$, $\det \mathsf{A} = 3.7 \times 10^{-12}$ and the set of equations $\mathsf{A}\mathbf{c} = \mathbf{b}$ is becoming ill-conditioned and unstable.

Solution.

The added square integrability is needed: otherwise the mean square error is infinite for every choice of the c_i, because $|f|^2 \le 2|f-p|^2 + 2|p|^2$ and the polynomial $p = \sum_i c_i x^i$ is bounded on $[0,1]$. It also makes each b_i finite, by Schwarz's inequality $|b_i| \le (\int_0^1 x^{2i} dx)^{1/2} (\int_0^1 |f|^2 dx)^{1/2}$. For real f and real coefficients, put $p(x) = \sum_{j=0}^{n-1} c_j x^j$. Expanding the error before differentiating gives

$$\begin{aligned} E(c) &= \int_0^1 (f-p)^2\, dx \\ &= \int_0^1 f^2\, dx - 2\sum_j c_j \int_0^1 x^j f\, dx + \sum_{j,k} c_j c_k \int_0^1 x^{j+k}\, dx. \end{aligned}$$

In the notation of the statement, $E(c) = \int_0^1 f^2 dx - 2\sum_j b_j c_j + \sum_{j,k} A_{jk} c_j c_k$. The last matrix is symmetric, $A_{jk} = A_{kj}$, so $\partial(\sum_{j,k} A_{jk} c_j c_k)/\partial c_i = \sum_k A_{ik} c_k + \sum_j A_{ji} c_j = 2\sum_j A_{ij} c_j$, and

$$\frac{\partial E}{\partial c_i} = -2b_i + 2\sum_{j=0}^{n-1} A_{ij} c_j = 0, \qquad A_{ij} = \left[\frac{x^{i+j+1}}{i+j+1} \right]_0^1 = \frac{1}{i+j+1}.$$

Hence the normal equations are $\boxed{Ac = b}$. They determine the global minimum, not merely a stationary point: for any nonzero vector v,

$$v^T A v = \int_0^1 \left(\sum_{j=0}^{n-1} v_j x^j \right)^2 dx > 0.$$

A nonzero polynomial cannot vanish on an interval, so the Hessian $2A$ is positive definite. In particular $Av \neq 0$ for $v \neq 0$, so A is invertible and the normal equations have exactly one solution c_*. To complete the square, put $c = c_* + d$ in $E(c) = \int_0^1 f^2 dx - 2b^T c + c^T Ac$ and use $b = Ac_*$, $A^T = A$:

$$\begin{aligned} E(c_* + d) &= \int_0^1 f^2 dx - 2c_*^T A(c_* + d) + (c_* + d)^T A(c_* + d) \\ &= \int_0^1 f^2 dx - c_*^T Ac_* + d^T Ad = E(c_*) + d^T Ad. \end{aligned}$$

Since $d^T Ad > 0$ for $d \neq 0$, c_* is the unique minimizer.

For complex f and complex c, minimize $E(c) = \int_0^1 |f - p|^2 dx$. The x^j are real, so $\int_0^1 f^* p\, dx = b^\dagger c$ and $\int_0^1 f p^* dx = c^\dagger b$, with the same, now complex, b_i, and

$$E(c) = \int_0^1 |f|^2 dx - b^\dagger c - c^\dagger b + c^\dagger Ac.$$

If $Ac_* = b$, then $b^\dagger = c_*^\dagger A$ because A is real and symmetric, and the same completion of the square gives

$$E(c_* + d) = \int_0^1 |f|^2 dx - c_*^\dagger Ac_* + d^\dagger Ad = E(c_*) + d^\dagger Ad,$$

where $d^\dagger Ad = \int_0^1 |\sum_j d_j x^j|^2 dx > 0$ for $d \neq 0$. The same equations $Ac = b$, now with complex c and b, thus give the unique minimizer.

For $n = 5$, Gaussian elimination in exact fractions, without row interchanges, gives the pivots 1, 1/12, 1/180, 1/2800, 1/44100 (Exercise 2.1.5, where the indices start at 1), and the determinant is their product:

$$\det A = \frac{1}{12 \cdot 180 \cdot 2800 \cdot 44100} = \frac{1}{266716800000} = 3.7492951325\ldots \times 10^{-12}.$$

The Note infers ill-conditioning from the small determinant, but a small determinant alone is not a condition-number test (rescaling any well-conditioned matrix can make its determinant small). For this particular matrix the condition number $\kappa_2(A) = \sigma_{\max}/\sigma_{\min} \simeq 4.7661 \times 10^5$, the ratio of the extreme singular values, confirms sensitivity of the equations. The real symmetric matrix A has orthonormal eigenvectors $u_1, \ldots, u_5$, $Au_k = \lambda_k u_k$ (Chapter 6), and positive definiteness makes every $\lambda_k = u_k^T Au_k$ positive. Since $A^T Au_k = A^2 u_k = \lambda_k^2 u_k$, the singular values of A, the square roots of the eigenvalues of $A^T A$, are its eigenvalues λ_k. Computing them numerically (with any symmetric eigenvalue routine, e.g., the Jacobi method) gives $\lambda_{\max} \simeq 1.56705$ and $\lambda_{\min} \simeq 3.28793 \times 10^{-6}$, whose ratio is the quoted κ_2. To see what this ratio measures, let b change by δb, so that c changes by $\delta c = A^{-1}\delta b$. Write $\delta b = \sum_k \beta_k u_k$ and $c = \sum_k \gamma_k u_k$; then $\delta c = \sum_k (\beta_k/\lambda_k) u_k$ and $Ac = \sum_k \lambda_k \gamma_k u_k$, and Pythagoras gives, for Euclidean lengths,

$$\begin{aligned} |\delta c|^2 &= \sum_k \frac{|\beta_k|^2}{\lambda_k^2} \leq \frac{1}{\lambda_{\min}^2} \sum_k |\beta_k|^2 = \frac{|\delta b|^2}{\lambda_{\min}^2}, \\ |b|^2 = |Ac|^2 &= \sum_k \lambda_k^2 |\gamma_k|^2 \leq \lambda_{\max}^2 \sum_k |\gamma_k|^2 = \lambda_{\max}^2 |c|^2. \end{aligned}$$

Taking square roots and dividing,

$$\frac{|\delta c|}{|c|} \leq \frac{|\delta b|/\lambda_{\min}}{|b|/\lambda_{\max}} = \kappa_2(A)\, \frac{|\delta b|}{|b|},$$

with equality when c is an eigenvector for $\lambda_{\max}$ and δb one for $\lambda_{\min}$. A relative error in b can thus be magnified by a factor of up to about 5×10^5 in the solution c.

5.1.4 Let F and φ_n be real, square-integrable functions, with the φ_n orthonormal for a nonnegative weight w. In place of the expansion of $F(x)$ given by

$$F(x) = \sum_{n=0}^{\infty} a_n \varphi_n(x),$$

with

$$a_n = \int_a^b F(x)\varphi_n(x)w(x)\,dx,$$

take the finite series approximation

$$F(x) \approx \sum_{n=0}^{m} c_n \varphi_n(x).$$

Show that the mean square error

$$\int_a^b \left[F(x) - \sum_{n=0}^{m} c_n \varphi_n(x)\right]^2 w(x)\,dx$$

is minimized by taking $c_n = a_n$.

Note. The values of the coefficients are independent of the number of terms in the finite series. This independence is a consequence of orthogonality and would not hold for a least-squares fit using powers of x.

Solution.

The added hypotheses are implicit in the textbook: its coefficient formula (no complex conjugate) and its error (an ordinary square) are those for real functions, a nonnegative weight makes the error a weighted mean square, and square integrability of F keeps the error finite (otherwise, since $F^2 \le 2(F-p)^2 + 2p^2$ and $\int_a^b p^2 w\,dx = \sum_n c_n^2$ for $p = \sum_n c_n\varphi_n$, it would be infinite for every choice of the c_n). Orthonormality means $\int_a^b \varphi_j\varphi_k w\,dx = \delta_{jk}$. Expanding the finite error,

$$\begin{aligned} E(c) &= \int_a^b F^2 w\,dx - 2\sum_{n=0}^{m} c_n \int_a^b F\varphi_n w\,dx + \sum_{n,k=0}^{m} c_n c_k \delta_{nk} \\ &= \|F\|^2 - 2\sum_{n=0}^{m} c_n a_n + \sum_{n=0}^{m} c_n^2 \\ &= \|F\|^2 - \sum_{n=0}^{m} a_n^2 + \sum_{n=0}^{m} (c_n - a_n)^2. \end{aligned}$$

Here $\|F\|^2 = \int_a^b F^2 w\,dx$, and the last line uses $c_n^2 - 2c_n a_n = (c_n - a_n)^2 - a_n^2$ to complete the square term by term. Only the last sum depends on the trial coefficients. It is nonnegative and vanishes exactly when

$$\boxed{c_n = a_n \quad (0 \le n \le m)}, \qquad E_{\min} = \|F\|^2 - \sum_{n=0}^{m} a_n^2.$$

Adding another orthonormal function only adds its own square to this calculation; it does not alter the earlier coefficients. If the full expansion converges to F in norm (as it does when the φ_n form a complete set), Parseval's identity $\|F\|^2 = \sum_{n=0}^{\infty} a_n^2$ gives $E_{\min} = \sum_{n>m} a_n^2$. For complex functions the corresponding formula uses $a_n = \int \varphi_n^* F w\,dx$ and $E = \int |F - \sum c_n\varphi_n|^2 w\,dx$; the last sum is then $\sum |c_n - a_n|^2$, with the same conclusion.

5.1.5 From Example 5.1.6,

$$f(x) = \begin{cases} \dfrac{h}{2}, & 0 < x < \pi \\ -\dfrac{h}{2}, & -\pi < x < 0 \end{cases} = \frac{2h}{\pi} \sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}.$$

(a) Show that

$$\int_{-\pi}^{\pi} [f(x)]^2 \, dx = \frac{\pi}{2} h^2 = \frac{4h^2}{\pi} \sum_{n=0}^{\infty} (2n+1)^{-2}.$$

For a finite upper limit this would be Bessel's inequality. For the upper limit ∞, this is Parseval's identity.

(b) Verify that

$$\frac{\pi}{2} h^2 = \frac{4h^2}{\pi} \sum_{n=0}^{\infty} (2n+1)^{-2}$$

by evaluating the series.

Hint. The series can be expressed in terms of the Riemann zeta function $\zeta(2) = \pi^2/6$.

Solution.

(a) Except at the irrelevant endpoints and jump, $f^2 = h^2/4$. Thus

$$\int_{-\pi}^{\pi} f^2 \, dx = 2\pi \frac{h^2}{4} = \frac{\pi h^2}{2}.$$

For positive integers k, ℓ, the product-to-sum identity gives

$$\int_{-\pi}^{\pi} \sin kx \sin \ell x \, dx = \frac{1}{2} \int_{-\pi}^{\pi} [\cos(k-\ell)x - \cos(k+\ell)x] dx = \pi \delta_{k\ell}.$$

The norm squared of the partial sum f_N through $n = N$ is therefore

$$\begin{aligned} \int_{-\pi}^{\pi} f_N^2 \, dx &= \frac{4h^2}{\pi^2} \sum_{n,m=0}^{N} \frac{\pi \delta_{nm}}{(2n+1)(2m+1)} \\ &= \frac{4h^2}{\pi} \sum_{n=0}^{N} \frac{1}{(2n+1)^2}. \end{aligned}$$

The residual is orthogonal to this partial sum. Indeed, $f \sin(2n+1)x$ is even, so

$$\int_{-\pi}^{\pi} f \sin(2n+1)x \, dx = 2 \cdot \frac{h}{2} \int_0^{\pi} \sin(2n+1)x \, dx = h \left[-\frac{\cos(2n+1)x}{2n+1} \right]_0^{\pi} = \frac{2h}{2n+1},$$

and hence

$$\langle f_N | f \rangle = \frac{2h}{\pi} \sum_{n=0}^{N} \frac{1}{2n+1} \frac{2h}{2n+1} = \frac{4h^2}{\pi} \sum_{n=0}^{N} \frac{1}{(2n+1)^2} = \|f_N\|^2,$$

i.e., $\langle f_N | f - f_N \rangle = 0$. By Pythagoras,

$$\|f\|^2 = \|f_N\|^2 + 2\langle f_N | f - f_N \rangle + \|f - f_N\|^2 = \|f_N\|^2 + \|f - f_N\|^2 \geq \|f_N\|^2,$$

which is Bessel's inequality. The functions $1, \cos kx, \sin kx$ $(k \geq 1)$ form a complete orthogonal set on $[-\pi, \pi]$, so the Fourier series of every square-integrable function converges to it in the mean-square norm. Here all cosine coefficients vanish (f is odd), and so do the even-order sine coefficients, since $\int_0^\pi \sin 2mx\, dx = 0$. The odd-order sine coefficients are the integrals $2h/(2n+1)$ computed above divided by $\int_{-\pi}^{\pi} \sin^2(2n+1)x\, dx = \pi$, i.e., $2h/[\pi(2n+1)]$, exactly the coefficients of f_N. The partial sums of the Fourier series of f are therefore the f_N, and $\|f - f_N\| \to 0$. Letting $N \to \infty$ in the Pythagoras identity proves the stated Parseval equality.

(b) Split the positive integers into odd and even ones:

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \sum_{k=1}^{\infty} \frac{1}{k^2} - \sum_{k=1}^{\infty} \frac{1}{(2k)^2} = \left(1 - \frac{1}{4}\right) \zeta(2) = \frac{\pi^2}{8}.$$

Substitution yields $(4h^2/\pi)(\pi^2/8) = \pi h^2/2$, an independent check of the equality.

5.1.6 Derive the Schwarz inequality for real square-integrable functions from the identity

$$\left[\int_a^b f(x)g(x)\, dx\right]^2 = \int_a^b f^2(x)\, dx \int_a^b g^2(x)\, dx - \frac{1}{2}\int_a^b dx \int_a^b dy\, [f(x)g(y) - f(y)g(x)]^2.$$

Solution.

The qualification is needed for the derivation: square integrability makes every integral below finite, and for real functions the integrand of the double integral is a nonnegative square. (The identity is pure algebra and also holds for complex functions, but then that integrand need not be nonnegative.) To check the identity, expand its double integral. Fubini's theorem applies since $\int_a^b f^2 dx$ and $\int_a^b g^2 dx$ are finite and $\int_a^b |fg|\, dx < \infty$. The latter fact follows from $2|fg| \leq f^2 + g^2$, so it does not assume the Schwarz inequality we are proving:

$$\begin{aligned} D &= \int_a^b \int_a^b [f(x)g(y) - f(y)g(x)]^2\, dx\, dy \\ &= \int_a^b \int_a^b \{f(x)^2 g(y)^2 + f(y)^2 g(x)^2 - 2f(x)g(x)f(y)g(y)\}\, dx\, dy \\ &= 2\left(\int_a^b f^2 dx\right)\left(\int_a^b g^2 dx\right) - 2\left(\int_a^b fg\, dx\right)^2. \end{aligned}$$

In the last step each term, a function of x times a function of y, integrates to a product of single integrals, e.g., $\iint f(x)^2 g(y)^2\, dx\, dy = \int f^2 dx \int g^2 dy$; the second term gives the same product with x and y renamed, and the third gives $-2(\int fg\, dx)(\int fg\, dy)$. Dividing by 2 gives

$$\frac{1}{2}D = \int_a^b f^2 dx \int_a^b g^2 dx - \left(\int_a^b fg\, dx\right)^2,$$

which is the identity. Since the original integrand in D is a square, $D \geq 0$, and the identity gives $(\int fg\, dx)^2 \leq \int f^2 dx \int g^2 dx$. Taking nonnegative square roots proves

$$\boxed{\left|\int_a^b fg\, dx\right| \leq \left(\int_a^b f^2 dx\right)^{1/2} \left(\int_a^b g^2 dx\right)^{1/2}}.$$

If $\int g^2 dx > 0$, set $\lambda = (\int fg)/(\int g^2)$. Then

$$\int_a^b (f-\lambda g)^2 dx = \int f^2 dx - 2\lambda \int fg\, dx + \lambda^2 \int g^2 dx$$
$$= \int f^2 dx - \frac{(\int fg\, dx)^2}{\int g^2 dx}.$$

The right side vanishes exactly in the case of equality, and the left side exactly when $f = \lambda g$ almost everywhere. Thus equality holds exactly when $f = \lambda g$ almost everywhere; the zero-function cases also give equality. For complex functions the Schwarz inequality is $|\int f^* g|^2 \le \int |f|^2 \int |g|^2$; ordinary squares in the displayed real identity must be replaced by appropriate modulus squares.

5.1.7 Starting from

$$I = \left\langle f - \sum_i a_i \varphi_i \,\middle|\, f - \sum_j a_j \varphi_j \right\rangle \ge 0,$$

derive Bessel's inequality, $\langle f|f\rangle \ge \sum_n |a_n|^2$. Here the φ_n are orthonormal and $a_n = \langle \varphi_n | f\rangle$; begin with finite partial sums.

Solution.

The added hypotheses are needed: for $f = \varphi_0$, the arbitrary choice $a_0 = 2$ (all other $a_n = 0$) gives $\sum |a_n|^2 = 4 > \langle f|f\rangle = 1$, and $a_n = \langle \varphi_n|f\rangle$ with the nonorthogonal choice $\varphi_1 = \varphi_0$ gives $|a_0|^2 + |a_1|^2 = 2 > 1$. Finite sums come first because the convergence of $\sum_i a_i\varphi_i$, which the printed I presupposes, is not known in advance. Take a finite set of orthonormal functions first, and use the projection coefficients $a_j = \langle \varphi_j | f\rangle$. The scalar product is conjugate-linear in its first argument, so

$$I_N = \left\langle f - \sum_{i=0}^{N} a_i \varphi_i \,\middle|\, f - \sum_{j=0}^{N} a_j \varphi_j \right\rangle$$
$$= \langle f|f\rangle - \sum_j a_j \langle f|\varphi_j\rangle - \sum_i a_i^* \langle \varphi_i|f\rangle + \sum_{i,j} a_i^* a_j \delta_{ij}$$
$$= \|f\|^2 - \sum_j |a_j|^2 - \sum_i |a_i|^2 + \sum_i |a_i|^2 = \|f\|^2 - \sum_{i=0}^{N} |a_i|^2 \ge 0.$$

The passage to the last line uses $\langle \varphi_i|f\rangle = a_i$ and $\langle f|\varphi_j\rangle = \langle \varphi_j|f\rangle^* = a_j^*$. The partial sums of the nonnegative series $\sum |a_i|^2$ increase and are bounded by $\|f\|^2$. Taking their limit gives

$$\boxed{\sum_{i=0}^{\infty} |a_i|^2 \le \langle f|f\rangle}.$$

No completeness assumption on the orthonormal set is needed for this inequality. To check the equality case, let $S_N = \sum_{i=0}^{N} a_i\varphi_i$. For $M > N$, orthonormality gives $\|S_M - S_N\|^2 = \sum_{i=N+1}^{M} |a_i|^2$, which tends to 0 as $N \to \infty$ (a tail of the convergent series), so the S_N form a Cauchy sequence. A Hilbert space is complete in the sense that every Cauchy sequence converges in it (a property of the space, not of the orthonormal set), so $S_N \to S$ for some S. This S lies in the closed span of the φ_i,

the set of all norm limits of finite linear combinations of the φ_i. Letting $N \to \infty$ in the residual identity $\|f - S_N\|^2 = I_N = \|f\|^2 - \sum_{i=0}^{N} |a_i|^2$ (the norm is continuous) gives

$$\|f - S\|^2 = \|f\|^2 - \sum_{i=0}^{\infty} |a_i|^2.$$

Thus equality implies $f = S$ in that closed span. Conversely, if f is in the closed span, the residual $f - S$ belongs to it and is orthogonal to every φ_k, since $\langle \varphi_k | f - S \rangle = a_k - \lim_N \langle \varphi_k | S_N \rangle = a_k - a_k = 0$; by linearity and continuity it is orthogonal to the entire closed span. Its squared norm is therefore zero, proving equality.

5.1.8 Expand the function $\sin \pi x$ in a series of functions φ_i that are orthogonal (but not normalized) on the range $0 \le x \le 1$ when the scalar product has definition

$$\langle f | g \rangle = \int_0^1 f^*(x) g(x)\, dx.$$

Keep the first four terms of the expansion. The first four φ_i are:

$$\varphi_0 = 1, \quad \varphi_1 = 2x - 1, \quad \varphi_2 = 6x^2 - 6x + 1, \quad \varphi_3 = 20x^3 - 30x^2 + 12x - 1.$$

Note. The integrals that are needed are the subject of Example 1.10.5.

Solution.

For an orthogonal but unnormalized family, $c_n = \langle \varphi_n | f \rangle / \langle \varphi_n | \varphi_n \rangle$. Direct polynomial integration gives the four squared norms

$$\int_0^1 \varphi_n^2 dx = 1,\ \frac{1}{3},\ \frac{1}{5},\ \frac{1}{7} \qquad (n = 0, 1, 2, 3).$$

For the nonconstant squared norms, square the polynomials,

$$\varphi_1^2 = 4x^2 - 4x + 1, \qquad \varphi_2^2 = 36x^4 - 72x^3 + 48x^2 - 12x + 1,$$
$$\varphi_3^2 = 400x^6 - 1200x^5 + 1380x^4 - 760x^3 + 204x^2 - 24x + 1,$$

and integrate term by term with $\int_0^1 x^k\, dx = 1/(k+1)$:

$$\|\varphi_1\|^2 = \frac{4}{3} - \frac{4}{2} + 1 = \frac{1}{3},$$
$$\|\varphi_2\|^2 = \frac{36}{5} - \frac{72}{4} + \frac{48}{3} - \frac{12}{2} + 1 = \frac{1}{5},$$
$$\|\varphi_3\|^2 = \frac{400}{7} - \frac{1200}{6} + \frac{1380}{5} - \frac{760}{4} + \frac{204}{3} - \frac{24}{2} + 1 = \frac{1}{7}.$$

To compute all numerators, put $I_j = \int_0^1 x^j \sin \pi x\, dx$. The first two integrals are

$$I_0 = \frac{2}{\pi}, \qquad I_1 = \left[-\frac{x \cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} \right]_0^1 = \frac{1}{\pi}.$$

Two integrations by parts, for $j \geq 2$, give

$$\begin{aligned} I_j &= \left[-\frac{x^j \cos \pi x}{\pi}\right]_0^1 + \frac{j}{\pi}\int_0^1 x^{j-1}\cos \pi x\, dx \\ &= \frac{1}{\pi} + \frac{j}{\pi}\left(\left[\frac{x^{j-1}\sin \pi x}{\pi}\right]_0^1 - \frac{j-1}{\pi}\int_0^1 x^{j-2}\sin \pi x\, dx\right) \\ &= \frac{1}{\pi} - \frac{j(j-1)}{\pi^2} I_{j-2}, \end{aligned}$$

since $\cos \pi = -1$ and $\sin \pi = \sin 0 = 0$. Hence

$$I_2 = \frac{1}{\pi} - \frac{2}{\pi^2} I_0 = \frac{1}{\pi} - \frac{4}{\pi^3}, \qquad I_3 = \frac{1}{\pi} - \frac{6}{\pi^2} I_1 = \frac{1}{\pi} - \frac{6}{\pi^3}.$$

Substituting,

$$\begin{aligned} \langle \varphi_0 | f \rangle &= I_0 = 2/\pi, \\ \langle \varphi_1 | f \rangle &= 2I_1 - I_0 = 0, \\ \langle \varphi_2 | f \rangle &= 6I_2 - 6I_1 + I_0 = 2/\pi - 24/\pi^3, \\ \langle \varphi_3 | f \rangle &= 20I_3 - 30I_2 + 12I_1 - I_0 = \frac{20 - 30 + 12 - 2}{\pi} + \frac{-120 + 120}{\pi^3} = 0. \end{aligned}$$

Dividing by the squared norms $\|\varphi_0\|^2 = 1$ and $\|\varphi_2\|^2 = 1/5$ gives

$$c_0 = \frac{2}{\pi}, \qquad c_2 = 5\Big(\frac{2}{\pi} - \frac{24}{\pi^3}\Big) = \frac{10(\pi^2 - 12)}{\pi^3} \approx -0.6871,$$

while $c_1 = c_3 = 0$. The requested four-term projection is therefore

$$\boxed{\sin \pi x \;\approx\; \frac{2}{\pi}\varphi_0 + 0\varphi_1 + \frac{10(\pi^2 - 12)}{\pi^3}\varphi_2 + 0\varphi_3}.$$

The two zero coefficients can also be checked by reflection:

$$\sin \pi(1-x) = \sin \pi x, \qquad \text{whereas} \qquad \varphi_1(1-x) = -\varphi_1(x), \quad \varphi_3(1-x) = -\varphi_3(x).$$

The substitution $x \to 1 - x$, which maps $[0, 1]$ onto itself and so leaves the value of an integral over $[0, 1]$ unchanged, therefore turns each of $\int_0^1 \varphi_1 \sin \pi x\, dx$ and $\int_0^1 \varphi_3 \sin \pi x\, dx$ into its own negative; both integrals vanish. Because each c_n is a projection coefficient, the residual $\sin \pi x - \sum_{n=0}^{3} c_n \varphi_n$ has zero scalar product with each of the four retained functions.

5.1.9 Expand the function e^{-x} in Laguerre polynomials $L_n(x)$, which are orthonormal on the range $0 \leq x < \infty$ with scalar product

$$\langle f | g \rangle = \int_0^\infty f^*(x) g(x) e^{-x}\, dx.$$

Keep the first four terms of the expansion. The first four $L_n(x)$ are

$$L_0 = 1, \quad L_1 = 1 - x, \quad L_2 = \frac{2 - 4x + x^2}{2}, \quad L_3 = \frac{6 - 18x + 9x^2 - x^3}{6}.$$

Solution.

The weight contributes an additional factor e^{-x}, so

$$a_n = \langle L_n | e^{-x} \rangle = \int_0^\infty L_n(x) e^{-2x} dx.$$

Set $J_k = \int_0^\infty x^k e^{-2x} dx$. Then $J_0 = 1/2$, and for $k \geq 1$ integration by parts gives

$$J_k = \left[-\frac{x^k e^{-2x}}{2} \right]_0^\infty + \frac{k}{2} \int_0^\infty x^{k-1} e^{-2x} dx = \frac{k}{2} J_{k-1},$$

hence $J_k = k!/2^{k+1}$, i.e., $(J_0, J_1, J_2, J_3) = (1/2, 1/4, 1/4, 3/8)$. Inserting the powers of x in each L_n, the four coefficients are

$$\begin{aligned}
a_0 &= J_0 = \frac{1}{2}, \\
a_1 &= J_0 - J_1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}, \\
a_2 &= J_0 - 2J_1 + \frac{1}{2} J_2 = \frac{1}{2} - \frac{1}{2} + \frac{1}{8} = \frac{1}{8}, \\
a_3 &= J_0 - 3J_1 + \frac{3}{2} J_2 - \frac{1}{6} J_3 = \frac{1}{2} - \frac{3}{4} + \frac{3}{8} - \frac{1}{16} = \frac{1}{16}.
\end{aligned}$$

Consequently

$$\boxed{e^{-x} \approx \frac{1}{2} L_0 + \frac{1}{4} L_1 + \frac{1}{8} L_2 + \frac{1}{16} L_3 = \frac{15}{16} - \frac{11x}{16} + \frac{5x^2}{32} - \frac{x^3}{96}}.$$

Collecting powers of x: the constant is $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{15}{16}$, the coefficient of x is $-\frac{1}{4} - \frac{1}{4} - \frac{3}{16} = -\frac{11}{16}$, that of x^2 is $\frac{1}{16} + \frac{3}{32} = \frac{5}{32}$, and that of x^3 is $-\frac{1}{96}$. As a check of both the weight and the normalization, the residual identity of Exercise 5.1.7, with $\|e^{-x}\|^2 = \int_0^\infty e^{-2x} e^{-x} dx$, gives the squared error norm of the four-term sum f_3:

$$\begin{aligned}
\|e^{-x} - f_3\|^2 &= \int_0^\infty e^{-3x} dx - \sum_{n=0}^{3} |a_n|^2 \\
&= \frac{1}{3} - \left(\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} \right) = \frac{1}{3} - \frac{85}{256} = \frac{1}{768}.
\end{aligned}$$

Expanding the square gives the same value:

$$\begin{aligned}
\int_0^\infty (e^{-x} - f_3)^2 e^{-x} dx &= \int_0^\infty e^{-3x} dx - 2 \int_0^\infty f_3 e^{-2x} dx + \int_0^\infty f_3^2 e^{-x} dx \\
&= \frac{1}{3} - 2 \cdot \frac{85}{256} + \frac{85}{256} = \frac{1}{768},
\end{aligned}$$

where the polynomial form of f_3 and the J_k give $\int_0^\infty f_3 e^{-2x} dx = \frac{15}{32} - \frac{11}{64} + \frac{5}{128} - \frac{1}{256} = \frac{85}{256}$, and orthonormality gives $\int_0^\infty f_3^2 e^{-x} dx = \sum_{n=0}^{3} a_n^2 = \frac{85}{256}$.

5.1.10 The explicit form of a function f is not known, but the coefficients a_n of its expansion in the orthonormal set φ_n are available. Assuming that the φ_n and the members of another orthonormal set, χ_n, are available, use Dirac notation to obtain a formula for the coefficients for the expansion of f in the χ_n set.

Solution.

Write $|f\rangle = \sum_n a_n|\varphi_n\rangle$. Taking the scalar product with the mth new basis vector gives

$$\boxed{b_m = \langle\chi_m|f\rangle = \sum_n \langle\chi_m|\varphi_n\rangle a_n}.$$

Thus only overlaps between known basis functions are required. In matrix notation $b = Ua$, where $U_{mn} = \langle\chi_m|\varphi_n\rangle$. For an infinite expansion the interchange of $\langle\chi_m|$ and the sum follows by taking scalar products with its norm-convergent partial sums, as in Exercise 5.1.1.

If both sets are complete in the same space, then $(U^\dagger)_{jm} = U^*_{mj} = \langle\varphi_j|\chi_m\rangle$, and inserting the resolution of the identity $\sum_m |\chi_m\rangle\langle\chi_m| = 1$ gives

$$(U^\dagger U)_{jk} = \sum_m \langle\varphi_j|\chi_m\rangle\langle\chi_m|\varphi_k\rangle = \langle\varphi_j|\varphi_k\rangle = \delta_{jk}.$$

Similarly, $\sum_k |\varphi_k\rangle\langle\varphi_k| = 1$ gives

$$(UU^\dagger)_{mn} = \sum_k \langle\chi_m|\varphi_k\rangle\langle\varphi_k|\chi_n\rangle = \langle\chi_m|\chi_n\rangle = \delta_{mn},$$

i.e., $UU^\dagger = I$. If the χ_m do not span f, the same formula instead gives the coefficients of its orthogonal projection; it cannot reconstruct a component outside their closed span.

5.1.11 Using conventional vector notation, evaluate $\sum_j |\hat{\mathbf{e}}_j\rangle\langle\hat{\mathbf{e}}_j|\mathbf{a}\rangle$, where $\mathbf{a}$ is an arbitrary vector in the space spanned by the orthonormal $\hat{\mathbf{e}}_j$.

Solution.

For the usual orthonormal unit vectors, $\langle\hat{\mathbf{e}}_j|\mathbf{a}\rangle = \hat{\mathbf{e}}_j \cdot \mathbf{a} = a_j$. Therefore

$$\boxed{\sum_j |\hat{\mathbf{e}}_j\rangle\langle\hat{\mathbf{e}}_j|\mathbf{a}\rangle = \sum_j \hat{\mathbf{e}}_j(\hat{\mathbf{e}}_j \cdot \mathbf{a}) = \sum_j a_j\hat{\mathbf{e}}_j = \mathbf{a}}.$$

In matrix notation, with each $\hat{\mathbf{e}}_j$ a column, this is $(\sum_j \hat{\mathbf{e}}_j\hat{\mathbf{e}}_j^{\mathsf{T}})\mathbf{a} = I\mathbf{a}$. Orthonormality is needed here; unit length and spanning alone are insufficient. For example, the unit vectors $(1,0)$ and $(1,1)/\sqrt{2}$ span $\mathbb{R}^2$, but the displayed sum applied to $(1,0)$ is

$$(1,0)\,[(1,0)\cdot(1,0)] + \frac{(1,1)}{\sqrt{2}}\Big[\frac{(1,1)}{\sqrt{2}}\cdot(1,0)\Big] = (1,0) + \frac{1}{2}(1,1) = \Big(\frac{3}{2},\frac{1}{2}\Big),$$

not $(1,0)$.

5.1.12 Letting $\mathbf{a} = a_1\hat{\mathbf{e}}_1 + a_2\hat{\mathbf{e}}_2$ and $\mathbf{b} = b_1\hat{\mathbf{e}}_1 + b_2\hat{\mathbf{e}}_2$ be vectors in $\mathbb{R}^2$, for what values of k, if any, is

$$\langle \mathbf{a}|\mathbf{b}\rangle = a_1b_1 - a_1b_2 - a_2b_1 + ka_2b_2$$

a valid definition of a scalar product?

Solution.

The proposed form is bilinear and symmetric for every real k, because it equals $(a_1, a_2)\, G\, (b_1, b_2)^T$ with the matrix

$$G = \begin{pmatrix} 1 & -1 \\ -1 & k \end{pmatrix} = G^T.$$

It remains to test positive definiteness. Completing the square,

$$\langle \mathbf{a}|\mathbf{a}\rangle = a_1^2 - 2a_1a_2 + ka_2^2 = (a_1 - a_2)^2 + (k-1)a_2^2.$$

If $k > 1$, both terms are nonnegative and their sum can vanish only if $a_2 = 0$ and then $a_1 = 0$. If $k = 1$, the nonzero vector $(1, 1)$ has zero norm. If $k < 1$, this same vector has squared norm $k - 1 < 0$. Thus all scalar-product axioms hold exactly for

$$\boxed{k > 1}.$$

The determinant check $\det G = k - 1 > 0$, together with $G_{11} = 1 > 0$ (Sylvester's criterion for positive definiteness), agrees with the direct square-completion argument.

5.2 Gram-Schmidt Orthogonalization

For the Gram-Schmidt constructions in Exercises 5.2.1 through 5.2.6, use a scalar product of the form given in Eq. (5.7) with the specified interval and weight.

5.2.1 Following the Gram-Schmidt procedure, construct a set of polynomials $P_n^*(x)$ orthogonal (unit weighting factor) over the range $[0, 1]$ from the set $[1, x, x^2, \ldots]$. Scale so that $P_n^*(1) = 1$.

ANS.
$$\begin{aligned} P_0^*(x) &= 1, \\ P_1^*(x) &= 2x - 1, \\ P_2^*(x) &= 6x^2 - 6x + 1, \\ P_3^*(x) &= 20x^3 - 30x^2 + 12x - 1. \end{aligned}$$

These are the first four shifted Legendre polynomials.

Note. The "*" is the standard notation for "shifted": $[0, 1]$ instead of $[-1, 1]$. It does not mean complex conjugate.

Solution.

The first answer on textbook p. 273 is printed $P_n^*(x) = 1$; its subscript must be 0, because P_n^* has degree n, so the constant answer is P_0^*. It is convenient first to construct monic orthogonal polynomials q_n:

$$q_n = x^n - \sum_{j=0}^{n-1} \frac{\int_0^1 q_j x^n\, dx}{\int_0^1 q_j^2\, dx}\, q_j, \qquad P_n^* = \frac{q_n}{q_n(1)}.$$

All the required integrals reduce to $\int_0^1 x^k dx = 1/(k+1)$. Start with $q_0 = 1$, $\|q_0\|^2 = 1$. Then

$$q_1 = x - \frac{1}{2}, \qquad \|q_1\|^2 = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}.$$

For the next polynomial,

$$\langle q_0|x^2\rangle = \frac{1}{3}, \qquad \langle q_1|x^2\rangle = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

Dividing by $\|q_0\|^2 = 1$ and $\|q_1\|^2 = 1/12$, the coefficients are $1/3$ and 1, so

$$q_2 = x^2 - \frac{1}{3} - q_1 = x^2 - x + \frac{1}{6},$$

$$q_2^2 = x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36}, \qquad \|q_2\|^2 = \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}.$$

For x^3 the three projection numerators are

$$\langle q_0|x^3\rangle = \frac{1}{4}, \qquad \langle q_1|x^3\rangle = \frac{1}{5} - \frac{1}{8} = \frac{3}{40}, \qquad \langle q_2|x^3\rangle = \frac{1}{6} - \frac{1}{5} + \frac{1}{24} = \frac{1}{120}.$$

Dividing by $1, 1/12, 1/180$, respectively, gives the coefficients $1/4$, $9/10$, $3/2$ and

$$\begin{aligned} q_3 &= x^3 - \frac{1}{4} - \frac{9}{10}\left(x - \frac{1}{2}\right) - \frac{3}{2}\left(x^2 - x + \frac{1}{6}\right) \\ &= x^3 - \frac{3}{2}x^2 + \frac{3}{5}x - \frac{1}{20}, \end{aligned}$$

since the coefficient of x is $-\frac{9}{10} + \frac{3}{2} = \frac{3}{5}$ and the constant is $-\frac{1}{4} + \frac{9}{20} - \frac{1}{4} = -\frac{1}{20}$. The endpoint values are $q_0(1) = 1$, $q_1(1) = 1/2$, $q_2(1) = 1/6$, and $q_3(1) = (20 - 30 + 12 - 1)/20 = 1/20$. Hence

$$\boxed{P_0^* = 1, \quad P_1^* = 2x - 1, \quad P_2^* = 6x^2 - 6x + 1, \quad P_3^* = 20x^3 - 30x^2 + 12x - 1}.$$

The construction extends to the complete set. A formula with the same orthogonality and endpoint normalization is

$$\boxed{P_n^*(x) = \frac{1}{n!}\frac{d^n}{dx^n}[x^n(x-1)^n] = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}\binom{n+k}{k}x^k}.$$

The finite sum follows by first expanding and then differentiating:

$$x^n(x-1)^n = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}x^{n+k}, \qquad \frac{1}{n!}\frac{d^n x^{n+k}}{dx^n} = \frac{(n+k)!}{n!k!}x^k = \binom{n+k}{k}x^k.$$

To verify orthogonality, multiply the derivative formula by x^j, $j < n$, and integrate by parts n times. All boundary terms vanish because $x^n(x-1)^n$ and its first $n-1$ derivatives vanish at both endpoints; the remaining integral vanishes because $d^n x^j/dx^n = 0$. At $x = 1$, use the Leibniz rule

$$\frac{d^n}{dx^n}[x^n(x-1)^n] = \sum_{k=0}^{n}\binom{n}{k}\frac{d^k x^n}{dx^k}\,\frac{d^{n-k}(x-1)^n}{dx^{n-k}}.$$

For $k \geq 1$ the last factor is $(x-1)^k\, n!/k!$, which vanishes at $x = 1$, so only the term differentiating $(x-1)^n$ all n times ($k = 0$) survives; it equals $1^n\, n! = n!$, so $P_n^*(1) = 1$. Finally, these are the Gram–Schmidt polynomials: the polynomial given by the derivative formula and the Gram–Schmidt P_n^* (orthogonal to $P_0^*, \ldots, P_{n-1}^*$, which span the polynomials of degree below n) both have degree

n, are orthogonal to every polynomial of lower degree, and equal 1 at $x = 1$. If R and S are two such polynomials, $R - \lambda S$, with λ the ratio of their leading coefficients, has lower degree and is orthogonal to itself, hence zero; $x = 1$ then gives $\lambda = 1$, so $R = S$.

A useful check is the norm formula. The leading coefficient is $\binom{2n}{n}$ and lower powers are orthogonal to P_n^*, so

$$\int_0^1 (P_n^*)^2 dx = \binom{2n}{n} \int_0^1 x^n P_n^*(x) dx$$
$$= \binom{2n}{n} \int_0^1 x^n (1-x)^n dx = \frac{(2n)!}{(n!)^2} \frac{(n!)^2}{(2n+1)!} = \frac{1}{2n+1}.$$

The second equality again uses n integrations by parts with vanishing boundary terms, now moving all derivatives onto x^n:

$$\int_0^1 x^n P_n^* dx = \frac{(-1)^n}{n!} \int_0^1 \frac{d^n x^n}{dx^n} x^n (x-1)^n \, dx = (-1)^n \int_0^1 x^n (x-1)^n dx = \int_0^1 x^n (1-x)^n dx.$$

To evaluate the last integral without a table, set $J_{a,b} = \int_0^1 x^a (1-x)^b dx$. For integers $a \geq 0$, $b \geq 1$, integration by parts gives

$$J_{a,b} = \left[\frac{x^{a+1}(1-x)^b}{a+1} \right]_0^1 + \frac{b}{a+1} J_{a+1,b-1} = \frac{b}{a+1} J_{a+1,b-1}.$$

Repeating b times and using $J_{a+b,0} = 1/(a+b+1)$ yields

$$J_{a,b} = \frac{b}{a+1} \frac{b-1}{a+2} \cdots \frac{1}{a+b} \frac{1}{a+b+1} = \frac{a!\, b!}{(a+b+1)!};$$

take $a = b = n$.

5.2.2 Apply the Gram-Schmidt procedure to form the first three Laguerre polynomials:

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad 0 \leq x < \infty, \quad w(x) = e^{-x}.$$

The conventional normalization is

$$\int_0^\infty L_m(x) L_n(x) e^{-x} \, dx = \delta_{mn}.$$

ANS. $L_0 = 1, \quad L_1 = (1-x), \quad L_2 = \dfrac{2 - 4x + x^2}{2}.$

Solution.

All scalar products reduce to the moments $M_k = \int_0^\infty x^k e^{-x} dx$. For $k \geq 1$, integration by parts gives

$$M_k = [-x^k e^{-x}]_0^\infty + k M_{k-1} = k M_{k-1}, \qquad M_0 = 1.$$

The boundary term is zero, so repeated application gives

$$M_k = k!, \qquad M_0 = 1, \; M_1 = 1, \; M_2 = 2, \; M_3 = 6, \; M_4 = 24.$$

Begin with $q_0 = 1$ and subtract its projection from x:

$$q_1 = x - \frac{M_1}{M_0} = x - 1, \qquad \|q_1\|^2 = M_2 - 2M_1 + M_0 = 1.$$

For x^2, the projections onto q_0 and q_1 have coefficients

$$\frac{\langle q_0|x^2\rangle}{\|q_0\|^2} = \frac{M_2}{M_0} = 2, \qquad \frac{\langle q_1|x^2\rangle}{\|q_1\|^2} = M_3 - M_2 = 4.$$

Therefore

$$q_2 = x^2 - 2 - 4(x-1) = x^2 - 4x + 2.$$

Since $q_2^2 = x^4 - 8x^3 + 20x^2 - 16x + 4$,

$$\|q_2\|^2 = M_4 - 8M_3 + 20M_2 - 16M_1 + 4M_0 = 24 - 48 + 40 - 16 + 4 = 4.$$

Choose the conventional signs $L_n(0) > 0$. Dividing by the norms gives

$$\boxed{L_0 = 1, \qquad L_1 = -q_1 = 1 - x, \qquad L_2 = q_2/2 = 1 - 2x + \frac{x^2}{2}}.$$

Gram–Schmidt makes the cross integrals zero, and the displayed norm calculations make all three diagonal integrals one. Normalization alone leaves each polynomial's overall sign free.

5.2.3 You are given

(a) a set of functions $u_n(x) = x^n$, $n = 0, 1, 2, \ldots$,

(b) an interval $(0, \infty)$,

(c) a weighting function $w(x) = xe^{-x}$.

Use the Gram-Schmidt procedure to construct the first three orthonormal functions from the set $u_n(x)$ for this interval and this weighting function.

$$ANS.\ \varphi_0(x) = 1, \quad \varphi_1(x) = (x-2)/\sqrt{2}, \quad \varphi_2(x) = \frac{x^2 - 6x + 6}{2\sqrt{3}}.$$

Solution.

With the extra factor x in the weight, the moments are those of Exercise 5.2.2 with k raised by one:

$$M_k = \int_0^\infty x^k x e^{-x} dx = (k+1)!, \quad (M_0, M_1, M_2, M_3, M_4) = (1, 2, 6, 24, 120).$$

The first function is $q_0 = \varphi_0 = 1$. For the second,

$$q_1 = x - \frac{M_1}{M_0} = x - 2, \qquad \|q_1\|^2 = 6 - 4(2) + 4(1) = 2, \quad \varphi_1 = \frac{x-2}{\sqrt{2}}.$$

For the third polynomial,

$$\langle q_0|x^2\rangle = M_2 = 6, \qquad \frac{\langle q_1|x^2\rangle}{\|q_1\|^2} = \frac{M_3 - 2M_2}{2} = \frac{24 - 12}{2} = 6.$$

Consequently

$$q_2 = x^2 - 6 - 6(x-2) = x^2 - 6x + 6.$$

Its square is $x^4 - 12x^3 + 48x^2 - 72x + 36$, so

$$\|q_2\|^2 = 120 - 12(24) + 48(6) - 72(2) + 36 = 12.$$

Thus

$$\boxed{\varphi_0 = 1, \qquad \varphi_1 = \frac{x-2}{\sqrt{2}}, \qquad \varphi_2 = \frac{x^2-6x+6}{2\sqrt{3}}}.$$

The printed slash notation for φ_2 means division by the entire quantity $2\sqrt{3}$; writing a fraction removes the ambiguity. As a direct orthogonality check, $\langle 1|q_2\rangle = M_2 - 6M_1 + 6M_0 = 6 - 12 + 6 = 0$ and, since $(x-2)(x^2-6x+6) = x^3 - 8x^2 + 18x - 12$, $\langle q_1|q_2\rangle = 24 - 8(6) + 18(2) - 12 = 0$.

5.2.4 Using the Gram-Schmidt orthogonalization procedure, construct the lowest three Hermite polynomials:

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad -\infty < x < \infty, \quad w(x) = e^{-x^2}.$$

For this set of polynomials the usual normalization is

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) w(x)\, dx = \delta_{mn} 2^m m!\, \pi^{1/2}.$$

ANS. $H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2.$

Solution.

Let $M_k = \int_{-\infty}^{\infty} x^k e^{-x^2} dx$; the odd moments vanish by parity. For $j \geq 1$, write $x^{2j} = x^{2j-1} \cdot x$ and use $x e^{-x^2} dx = -\frac{1}{2} d(e^{-x^2})$ in integration by parts:

$$M_{2j} = -\frac{1}{2}[x^{2j-1} e^{-x^2}]_{-\infty}^{\infty} + \frac{2j-1}{2} \int_{-\infty}^{\infty} x^{2j-2} e^{-x^2} dx.$$

The boundary term vanishes. The even-moment recurrence is therefore

$$M_{2j} = \int_{-\infty}^{\infty} x^{2j} e^{-x^2} dx = \frac{2j-1}{2} M_{2j-2}, \qquad M_0 = \sqrt{\pi}.$$

Thus $M_2 = \sqrt{\pi}/2$ and $M_4 = 3\sqrt{\pi}/4$. Start with $q_0 = 1$. The projection of x onto 1 vanishes, so $q_1 = x$. For x^2 the projection onto x vanishes, and

$$q_2 = x^2 - \frac{M_2}{M_0} = x^2 - \frac{1}{2}.$$

Their squared norms, using $q_2^2 = x^4 - x^2 + \frac{1}{4}$, are

$$\|q_0\|^2 = \sqrt{\pi}, \qquad \|q_1\|^2 = M_2 = \frac{\sqrt{\pi}}{2}, \qquad \|q_2\|^2 = M_4 - M_2 + \frac{1}{4} M_0 = \left(\frac{3}{4} - \frac{1}{2} + \frac{1}{4}\right)\sqrt{\pi} = \frac{\sqrt{\pi}}{2}.$$

The required squared norms $2^n n!\, \pi^{1/2}$ for degrees $0, 1, 2$ are respectively $\sqrt{\pi}, 2\sqrt{\pi}, 8\sqrt{\pi}$. The needed positive scale factors s_n, with $s_n^2 \|q_n\|^2$ equal to these, are therefore 1, $\sqrt{2/(1/2)} = 2$, and $\sqrt{8/(1/2)} = 4$. Hence

$$\boxed{H_0 = 1, \qquad H_1 = 2x, \qquad H_2 = 4x^2 - 2}.$$

Parity verifies the odd cross integrals, while $\langle H_0|H_2\rangle = 4M_2 - 2M_0 = 0$ checks the remaining pair.

5.2.5 Use the Gram-Schmidt orthogonalization scheme to construct the first three Chebyshev polynomials (type I):

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad -1 \le x \le 1, \quad w(x) = (1 - x^2)^{-1/2}.$$

Take the normalization

$$\int_{-1}^{1} T_m(x) T_n(x) w(x)\, dx = \delta_{mn} \begin{cases} \pi, & m = n = 0, \\ \dfrac{\pi}{2}, & m = n \ge 1. \end{cases}$$

Hint. The needed integrals are given in Exercise 13.3.2.

ANS. $T_0 = 1, \quad T_1 = x, \quad T_2 = 2x^2 - 1, \quad (T_3 = 4x^3 - 3x).$

Solution.

Substitute $x = \cos t$. Since $dx = -\sin t\, dt$, $\sqrt{1 - x^2} = \sin t$ on $0 \le t \le \pi$, and $x = -1, 1$ correspond to $t = \pi, 0$ (the minus sign of dx is absorbed by interchanging these limits),

$$M_{2j} = \int_{-1}^{1} \frac{x^{2j}}{\sqrt{1 - x^2}} dx = \int_0^{\pi} \cos^{2j} t\, dt, \quad M_{2j} = \frac{2j - 1}{2j} M_{2j-2}.$$

To see every term of the reduction, integrate $\cos^{2j-1} t\, d(\sin t)$ by parts:

$$M_{2j} = [\cos^{2j-1} t \sin t]_0^{\pi} + (2j - 1) \int_0^{\pi} \sin^2 t \cos^{2j-2} t\, dt = (2j - 1)(M_{2j-2} - M_{2j}).$$

The boundary term vanishes because $\sin 0 = \sin \pi = 0$, and the last step uses $\sin^2 t = 1 - \cos^2 t$. Moving the last term to the left gives $2jM_{2j} = (2j - 1)M_{2j-2}$, the stated recurrence. It yields $M_0 = \pi$, $M_2 = \pi/2$, $M_4 = 3\pi/8$, and $M_6 = 5\pi/16$; all odd moments vanish. Gram–Schmidt therefore gives

$$q_0 = 1, \qquad q_1 = x, \qquad q_2 = x^2 - \frac{M_2}{M_0} = x^2 - \frac{1}{2}.$$

Their squared norms are

$$\|q_0\|^2 = \pi, \quad \|q_1\|^2 = M_2 = \frac{\pi}{2}, \quad \|q_2\|^2 = M_4 - M_2 + \frac{1}{4} M_0 = \frac{3\pi}{8} - \frac{\pi}{2} + \frac{\pi}{4} = \frac{\pi}{8}.$$

Thus q_0 and q_1 already have the requested normalization, and q_2 needs multiplication by 2:

$$\boxed{T_0 = 1, \qquad T_1 = x, \qquad T_2 = 2x^2 - 1}.$$

To check the additional answer T_3, only the projection of x^3 onto x survives parity; its coefficient is $\langle x | x^3 \rangle / \langle x | x \rangle = M_4 / M_2$:

$$q_3 = x^3 - \frac{M_4}{M_2} x = x^3 - \frac{3}{4} x,$$

$$\|q_3\|^2 = M_6 - \frac{3}{2} M_4 + \frac{9}{16} M_2 = \frac{5\pi}{16} - \frac{9\pi}{16} + \frac{9\pi}{32} = \frac{\pi}{32}.$$

Multiplying q_3 by 4 gives $T_3 = 4x^3 - 3x$ and norm squared $\pi/2$.

5.2.6 Use the Gram-Schmidt orthogonalization scheme to construct the first three Chebyshev polynomials (type II):

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad -1 \le x \le 1, \quad w(x) = (1 - x^2)^{+1/2}.$$

Take the normalization to be

$$\int_{-1}^{1} U_m(x) U_n(x) w(x)\, dx = \delta_{mn} \frac{\pi}{2}.$$

Hint.

$$\int_{-1}^{1} (1 - x^2)^{1/2} x^{2n}\, dx = \frac{\pi}{2} \times \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{4 \cdot 6 \cdot 8 \cdots (2n+2)}, \quad n = 1, 2, 3, \ldots$$
$$= \frac{\pi}{2}, \quad n = 0.$$

ANS. $U_0 = 1, \quad U_1 = 2x, \quad U_2 = 4x^2 - 1.$

Solution.

The hint gives the even moments

$$M_0 = \frac{\pi}{2}, \qquad M_2 = \frac{\pi}{2} \cdot \frac{1}{4} = \frac{\pi}{8}, \qquad M_4 = \frac{\pi}{2} \cdot \frac{1 \cdot 3}{4 \cdot 6} = \frac{\pi}{16},$$

and symmetry makes the odd moments zero. The monic Gram–Schmidt polynomials are

$$q_0 = 1, \qquad q_1 = x, \qquad q_2 = x^2 - \frac{\langle 1 | x^2 \rangle}{\langle 1 | 1 \rangle} = x^2 - \frac{1}{4}.$$

There is no projection onto x in the last line because $\langle x | x^2 \rangle = M_3 = 0$. The squared norms are, with $q_2^2 = x^4 - \frac{1}{2}x^2 + \frac{1}{16}$,

$$\|q_0\|^2 = \frac{\pi}{2}, \qquad \|q_1\|^2 = M_2 = \frac{\pi}{8},$$

$$\|q_2\|^2 = M_4 - \frac{1}{2} M_2 + \frac{1}{16} M_0 = \frac{\pi}{16} - \frac{\pi}{16} + \frac{\pi}{32} = \frac{\pi}{32}.$$

Scale these by 1, 2, and 4 so that every squared norm is $\pi/2$:

$$\boxed{U_0 = 1, \qquad U_1 = 2x, \qquad U_2 = 4x^2 - 1}.$$

The remaining even cross integral is $\langle U_0 | U_2 \rangle = 4M_2 - M_0 = 0$; the other cross integrals are odd.

5.2.7 As a modification of Exercise 5.2.4, apply the Gram-Schmidt orthogonalization procedure to the set $u_n(x) = x^n$, $n = 0, 1, 2, \ldots$, $0 \le x < \infty$. Take $w(x)$ to be $\exp(-x^2)$. Find the first two nonvanishing polynomials. Normalize so that the coefficient of the highest power of x is unity. In Exercise 5.2.4, the interval $(-\infty, \infty)$ led to the Hermite polynomials. The functions found here are certainly not the Hermite polynomials.

ANS. $\varphi_0 = 1, \quad \varphi_1 = x - \pi^{-1/2}.$

Solution.

The textbook (p. 275) twice refers to Exercise 5.2.5, which constructs the Chebyshev polynomials; both references should be to Exercise 5.2.4, which uses the Gaussian weight on $(-\infty, \infty)$ and gives the Hermite polynomials.

On the half-line the moments required for the first two polynomials are as follows; M_0 is half the full-line value $\sqrt{\pi}$ because the integrand is even:

$$M_0 = \int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}, \qquad M_1 = \int_0^\infty xe^{-x^2}dx = \left[-\frac{1}{2}e^{-x^2}\right]_0^\infty = \frac{1}{2}.$$

The monic constant is $\varphi_0 = 1$, and subtraction of its projection from x gives

$$\boxed{\varphi_1 = x - \frac{\langle 1|x\rangle}{\langle 1|1\rangle} = x - \frac{1}{\sqrt{\pi}}}.$$

Indeed, $\langle\varphi_0|\varphi_1\rangle = 1/2 - (1/\sqrt{\pi})(\sqrt{\pi}/2) = 0$. For an independent norm check, $M_2 = \sqrt{\pi}/4$ (half the full-line value of Exercise 5.2.4), so

$$\|\varphi_1\|^2 = M_2 - \frac{2}{\sqrt{\pi}}M_1 + \frac{1}{\pi}M_0 = \frac{\sqrt{\pi}}{4} - \frac{1}{\sqrt{\pi}} + \frac{1}{2\sqrt{\pi}} = \frac{\pi - 2}{4\sqrt{\pi}} > 0.$$

On the full line the first moment is zero; on the half-line it is $1/2$. This explicitly explains the constant term absent from the odd Hermite polynomial.

5.2.8 Form a set of three orthonormal vectors by the Gram-Schmidt process using these input vectors in the order given:

$$\mathbf{c}_1 = \begin{pmatrix}1\\1\\1\end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix}1\\1\\2\end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix}1\\0\\2\end{pmatrix}.$$

Solution.

Normalize the first vector:

$$\mathbf{e}_1 = \frac{\mathbf{c}_1}{\|\mathbf{c}_1\|} = \frac{1}{\sqrt{3}}(1,1,1)^T.$$

The projection coefficient of $\mathbf{c}_2$ onto it is $\mathbf{e}_1 \cdot \mathbf{c}_2 = 4/\sqrt{3}$. Subtract and normalize:

$$\mathbf{q}_2 = (1,1,2)^T - \frac{4}{3}(1,1,1)^T = \frac{1}{3}(-1,-1,2)^T, \qquad \|\mathbf{q}_2\|^2 = \frac{2}{3},$$

$$\mathbf{e}_2 = \frac{1}{\sqrt{6}}(-1,-1,2)^T.$$

For the third vector,

$$\mathbf{e}_1 \cdot \mathbf{c}_3 = \frac{3}{\sqrt{3}} = \sqrt{3}, \qquad \mathbf{e}_2 \cdot \mathbf{c}_3 = \frac{-1+4}{\sqrt{6}} = \frac{3}{\sqrt{6}}.$$

Since $\sqrt{3}\,\mathbf{e}_1 = (1,1,1)^T$ and $(3/\sqrt{6})\,\mathbf{e}_2 = \frac{1}{2}(-1,-1,2)^T$, subtracting both projections gives

$$\mathbf{q}_3 = (1,0,2)^T - (1,1,1)^T - \frac{1}{2}(-1,-1,2)^T$$

$$= (1/2,-1/2,0)^T, \qquad \|\mathbf{q}_3\|^2 = \frac{1}{2},$$

so the requested set is

$$\boxed{\mathbf{e}_1 = \frac{(1,1,1)^T}{\sqrt{3}}, \quad \mathbf{e}_2 = \frac{(-1,-1,2)^T}{\sqrt{6}}, \quad \mathbf{e}_3 = \frac{(1,-1,0)^T}{\sqrt{2}}}.$$

The three cross-dot-product numerators are $-1-1+2 = 0$, $1-1 = 0$, and $-1+1 = 0$; their squared lengths are respectively $3/3$, $6/6$, and $2/2$, independently verifying orthonormality.

5.3 Operators

5.3.1 Show (without introducing matrix representations) that the adjoint of the adjoint of a bounded, everywhere-defined operator restores the original operator, i.e., that $(\mathcal{A}^\dagger)^\dagger = \mathcal{A}$.

Solution.

For the identity as an equality of operators, assume the operators are bounded and everywhere defined, as they automatically are in finite dimension. For a densely defined unbounded operator, the double adjoint is its closure when the operator is closable; equality then additionally requires that it be closed.

Let f, g be arbitrary vectors. The adjoint definition and conjugate symmetry of the scalar product give, step by step,

$$\begin{aligned}\langle f|Ag\rangle &= \langle A^\dagger f|g\rangle = \langle g|A^\dagger f\rangle^* \\ &= \langle (A^\dagger)^\dagger g|f\rangle^* = \langle f|(A^\dagger)^\dagger g\rangle.\end{aligned}$$

The first and third equalities are the definition $\langle u|Bv\rangle = \langle B^\dagger u|v\rangle$, applied with $(B, u, v) = (A, f, g)$ and $(A^\dagger, g, f)$, respectively; the second and fourth are $\langle u|v\rangle = \langle v|u\rangle^*$. Hence $\langle f|[A - (A^\dagger)^\dagger]g\rangle = 0$ for every f. For fixed g, choose $f = [A - (A^\dagger)^\dagger]g$. Its squared norm is zero, so it is the zero vector. Since g was arbitrary,

$$\boxed{(A^\dagger)^\dagger = A}.$$

5.3.2 $\mathcal{U}$ and $\mathcal{V}$ are two arbitrary bounded, everywhere-defined operators. Without introducing matrix representations of these operators, show that

$$(\mathcal{U}\mathcal{V})^\dagger = \mathcal{V}^\dagger\mathcal{U}^\dagger.$$

Note the resemblance to adjoint matrices.

Solution.

The unrestricted equality is valid for bounded, everywhere-defined operators, including all finite matrices. For general unbounded operators their domains matter; the formal algebra alone does not prove equality of the adjoint domains. For arbitrary f, g in the bounded setting, apply the adjoint definition twice, first to U acting on the vector Vg, then to V with $U^\dagger f$ in the left slot:

$$\langle f|UVg\rangle = \langle U^\dagger f|Vg\rangle = \langle V^\dagger U^\dagger f|g\rangle.$$

By definition also $\langle f|UVg\rangle = \langle (UV)^\dagger f|g\rangle$. Subtracting, $\langle [(UV)^\dagger - V^\dagger U^\dagger]f|g\rangle = 0$ for every g; the choice $g = [(UV)^\dagger - V^\dagger U^\dagger]f$ shows, as in the previous exercise, that this vector is zero. Since f is arbitrary,

$$\boxed{(UV)^\dagger = V^\dagger U^\dagger}.$$

The order reverses because U, the operator applied last, is the first to be moved across the scalar product.

5.3.3 Consider a Hilbert space spanned by the three functions $\varphi_1 = x_1$, $\varphi_2 = x_2$, $\varphi_3 = x_3$, and a scalar product defined by $\langle x_\nu | x_\mu \rangle = \delta_{\nu\mu}$.

(a) Form the 3×3 matrix of each of the following operators:

$$\mathcal{A}_1 = \sum_{i=1}^{3} x_i \left(\frac{\partial}{\partial x_i} \right), \quad \mathcal{A}_2 = x_1 \left(\frac{\partial}{\partial x_2} \right) - x_2 \left(\frac{\partial}{\partial x_1} \right).$$

(b) Form the column vector representing $\psi = x_1 - 2x_2 + 3x_3$.

(c) Form the matrix equation corresponding to $\chi = (\mathcal{A}_1 - \mathcal{A}_2)\psi$ and verify that the matrix equation reproduces the result obtained by direct application of $\mathcal{A}_1 - \mathcal{A}_2$ to ψ.

Solution.

(a) Because $\langle x_\nu | x_\mu \rangle = \delta_{\nu\mu}$, the matrix element $A_{ij} = \langle x_i | A x_j \rangle$ is the coefficient of x_i in $A x_j$: the jth matrix column is the expansion of $A\varphi_j$ in the ordered basis (x_1, x_2, x_3). Since $\partial x_j / \partial x_i = \delta_{ij}$,

$$A_1 x_j = \sum_i x_i \delta_{ij} = x_j.$$

For the second operator,

$$A_2 x_1 = -x_2, \qquad A_2 x_2 = x_1, \qquad A_2 x_3 = 0.$$

Reading off the columns gives

$$\boxed{A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}.$$

(b) The coefficient column of ψ is $c = (1, -2, 3)^T$.

(c) Subtraction and multiplication give

$$(A_1 - A_2)c = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1+2 \\ 1-2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix}.$$

Independently, differentiating the function itself yields

$$A_1\psi = x_1 - 2x_2 + 3x_3, \qquad A_2\psi = x_1(-2) - x_2(1) = -2x_1 - x_2.$$

Thus $\chi = A_1\psi - A_2\psi = (x_1 - 2x_2 + 3x_3) - (-2x_1 - x_2) = 3x_1 - x_2 + 3x_3$, exactly the function represented by the matrix result $(3, -1, 3)^T$.

5.3.4

(a) Obtain the matrix representation of $\mathcal{A} = x(d/dx)$ in a basis of Legendre polynomials, keeping terms through P_3. Use the orthonormal forms of these polynomials as given in Example 5.2.1 and the scalar product defined there.

(b) Expand x^3 in the orthonormal Legendre polynomial basis.

(c) Verify that $\mathcal{A}x^3$ is given correctly by its matrix representation.

Solution.

The textbook says only "5.2.1." The intended reference is *Example* 5.2.1 (textbook pp. 270–271), which uses $[-1,1]$. Exercise 5.2.1 instead uses shifted polynomials on $[0,1]$. Write the orthonormal basis without confusing it with the conventionally normalized P_n:

$$p_0 = \frac{1}{\sqrt{2}}, \quad p_1 = \sqrt{\frac{3}{2}}x, \quad p_2 = \sqrt{\frac{5}{2}}\frac{3x^2-1}{2}, \quad p_3 = \sqrt{\frac{7}{2}}\frac{5x^3-3x}{2}.$$

(a) Differentiate each polynomial and then re-expand, using

$$3x^2 = (3x^2-1)+1, \qquad 15x^3 - 3x = 3(5x^3-3x) + 6x.$$

This gives

$$\begin{aligned}
Ap_0 &= 0, \qquad Ap_1 = \sqrt{\frac{3}{2}}x = p_1, \\
Ap_2 &= \sqrt{\frac{5}{2}}3x^2 = 2p_2 + \sqrt{\frac{5}{2}} = 2p_2 + \sqrt{5}\,p_0, \\
Ap_3 &= \sqrt{\frac{7}{2}}\frac{15x^3-3x}{2} \\
&= 3\sqrt{\frac{7}{2}}\frac{5x^3-3x}{2} + 3\sqrt{\frac{7}{2}}x = 3p_3 + \sqrt{21}\,p_1.
\end{aligned}$$

The leftover lower-degree terms are re-expressed with $1 = \sqrt{2}\,p_0$ and $x = \sqrt{2/3}\,p_1$: $\sqrt{5/2} = \sqrt{5}\,p_0$ and $3\sqrt{7/2}\,x = 3\sqrt{7/3}\,p_1 = \sqrt{21}\,p_1$. All outputs remain in this four-dimensional span. The matrix element $A_{ij} = \langle p_i | Ap_j \rangle$ is the coefficient of p_i in Ap_j, so the jth column lists the coefficients of Ap_j. With row and column order $0,1,2,3$, the matrix is

$$\boxed{A = \begin{pmatrix} 0 & 0 & \sqrt{5} & 0 \\ 0 & 1 & 0 & \sqrt{21} \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}}.$$

(b) The coefficients of the even polynomials vanish by parity. The two odd coefficients are

$$\begin{aligned}
\langle p_1 | x^3 \rangle &= \sqrt{\frac{3}{2}}\int_{-1}^{1} x^4 dx = \sqrt{\frac{3}{2}}\frac{2}{5} = \frac{\sqrt{6}}{5}, \\
\langle p_3 | x^3 \rangle &= \frac{1}{2}\sqrt{\frac{7}{2}}\int_{-1}^{1} (5x^6 - 3x^4)\,dx = \frac{1}{2}\sqrt{\frac{7}{2}}\left(5\frac{2}{7} - 3\frac{2}{5}\right) \\
&= \frac{1}{2}\sqrt{\frac{7}{2}}\frac{8}{35} = \frac{4}{35}\frac{\sqrt{14}}{2} = \frac{2\sqrt{14}}{35}.
\end{aligned}$$

Therefore

$$\boxed{x^3 = \frac{\sqrt{6}}{5}p_1 + \frac{2\sqrt{14}}{35}p_3}, \qquad c = \begin{pmatrix} 0 \\ \sqrt{6}/5 \\ 0 \\ 2\sqrt{14}/35 \end{pmatrix}.$$

Substitution of the polynomials checks the expansion directly: since

$$\frac{\sqrt{6}}{5}\sqrt{\frac{3}{2}} = \frac{\sqrt{9}}{5} = \frac{3}{5}, \qquad \frac{2\sqrt{14}}{35}\cdot\frac{1}{2}\sqrt{\frac{7}{2}} = \frac{\sqrt{49}}{35} = \frac{1}{5},$$

the sum is $(3/5)x + (1/5)(5x^3 - 3x) = x^3$.

(c) Matrix multiplication gives

$$Ac = \begin{pmatrix} 0 \\ \sqrt{6}/5 + \sqrt{21}(2\sqrt{14}/35) \\ 0 \\ 6\sqrt{14}/35 \end{pmatrix} = \begin{pmatrix} 0 \\ 3\sqrt{6}/5 \\ 0 \\ 6\sqrt{14}/35 \end{pmatrix} = 3c,$$

because $\sqrt{21}\sqrt{14} = \sqrt{294} = 7\sqrt{6}$ makes the second entry $\sqrt{6}/5 + 14\sqrt{6}/35 = \sqrt{6}/5 + 2\sqrt{6}/5 = 3\sqrt{6}/5$. This represents $3x^3$, agreeing with the direct derivative $x(d/dx)x^3 = 3x^3$.

5.4 Self-Adjoint Operators

5.4.1 Work with bounded, everywhere-defined operators.

(a) $\mathcal{A}$ is a non-Hermitian operator. Show that the operators $\mathcal{A} + \mathcal{A}^\dagger$ and $i(\mathcal{A} - \mathcal{A}^\dagger)$ are Hermitian.

(b) Using the preceding result, show that every non-Hermitian operator may be written as a linear combination of two Hermitian operators.

Solution.

Use the adjoint identities $(\lambda A)^\dagger = \lambda^* A^\dagger$, $(A + B)^\dagger = A^\dagger + B^\dagger$, and $(A^\dagger)^\dagger = A$ for bounded, everywhere-defined operators. General unbounded operators require additional domain assumptions, as noted in Exercises 5.3.1–5.3.2.

(a) If $B = A + A^\dagger$, then

$$B^\dagger = A^\dagger + (A^\dagger)^\dagger = A^\dagger + A = B.$$

If $C = i(A - A^\dagger)$, conjugating the scalar i is essential:

$$C^\dagger = -i\{A^\dagger - (A^\dagger)^\dagger\} = -i(A^\dagger - A) = i(A - A^\dagger) = C.$$

Thus both operators are Hermitian.

(b) Solve these two definitions for A: since $iC = i^2(A - A^\dagger) = -(A - A^\dagger)$,

$$B - iC = (A + A^\dagger) + (A - A^\dagger) = 2A, \qquad \text{so} \qquad \boxed{A = \frac{1}{2}B - \frac{i}{2}C.}$$

Equivalently, define the Hermitian real and imaginary parts

$$H = \frac{A + A^\dagger}{2}, \qquad K = \frac{A - A^\dagger}{2i}; \qquad \boxed{A = H + iK}.$$

Because $1/(2i) = -i/2$, these are $H = B/2$ and $K = -C/2$, real multiples of the operators of (a), hence Hermitian, and $H + iK = B/2 - iC/2$ is the first boxed formula. Directly, substitution verifies the reconstruction because the $A^\dagger$ terms cancel and the two $A/2$ terms add. This decomposition works also if A happens already to be Hermitian, in which case $K = 0$.

5.4.2 Prove that the product of two bounded, everywhere-defined Hermitian operators is Hermitian if and only if the two operators commute.

Solution.

For Hermitian A and B, the product-adjoint identity $(AB)^\dagger = B^\dagger A^\dagger$ (Exercise 5.3.2) gives

$$(AB)^\dagger = B^\dagger A^\dagger = BA.$$

If AB is Hermitian, then $AB = (AB)^\dagger = BA$, so the operators commute. Conversely, if $AB = BA$, then $(AB)^\dagger = BA = AB$, so the product is Hermitian. Thus

$$\boxed{AB \text{ is Hermitian} \iff [A, B] = 0.}$$

This statement has its usual finite-dimensional or bounded-operator meaning. For unbounded operators one must also establish the domains of the product and its adjoint; formal commutation on a subdomain alone is not a proof of self-adjointness.

5.4.3 $\mathcal{A}$ and $\mathcal{B}$ are noncommuting Hermitian, bounded, everywhere-defined quantum mechanical operators, and $\mathcal{C}$ is given by the formula

$$\mathcal{AB} - \mathcal{BA} = i\mathcal{C}.$$

Show that $\mathcal{C}$ is Hermitian. For unbounded operators the domains of $\mathcal{AB}$, $\mathcal{BA}$ and of their adjoints would also have to be checked; the solution treats the bounded case.

Solution.

The conclusion requires A and B themselves to be Hermitian, as when they represent quantum observables. Being merely "quantum mechanical operators" or merely noncommuting does not imply this. With that assumption and the indicated adjoint/domain conditions, the product-adjoint identity of Exercise 5.3.2 gives

$$\begin{aligned}(AB - BA)^\dagger &= B^\dagger A^\dagger - A^\dagger B^\dagger \\ &= BA - AB = -(AB - BA) = -iC.\end{aligned}$$

But the adjoint of the right side of $AB - BA = iC$ is $(iC)^\dagger = i^* C^\dagger = -iC^\dagger$. Thus $-iC^\dagger = -iC$, and multiplication by i gives

$$\boxed{C^\dagger = C.}$$

Equivalently $C = -i[A, B]$ is Hermitian because $[A, B]$ is anti-Hermitian. Noncommutation only ensures $C \neq 0$; it is not needed for Hermiticity. To check why the added hypothesis matters, take

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

where B is Hermitian but A is not. Multiplying out gives $AB = 0$ and $BA = A$, so $[A, B] = -A$, and then $iC = -A$ gives

$$C = -\frac{A}{i} = iA = \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix} \neq C^\dagger = \begin{pmatrix} 0 & 0 \\ -i & 0 \end{pmatrix}.$$

Thus C is not Hermitian, and this counterexample rules out the unqualified statement.

5.4.4 The operator $\mathcal{L}$ is Hermitian. Show that $\langle \mathcal{L}^2 \rangle \geq 0$, meaning that for all $\psi \in D(\mathcal{L}^2)$, $\langle \psi | \mathcal{L}^2 | \psi \rangle \geq 0$.

Solution.

For an unbounded L, the expression in the question requires $\psi \in D(L^2) = \{\psi \in D(L) : L\psi \in D(L)\}$, not merely $\psi \in D(L)$. Write $L^2\psi = L\chi$ with $\chi = L\psi$, move the left factor of L across the scalar product by the adjoint definition $\langle \psi | L\chi \rangle = \langle L^\dagger \psi | \chi \rangle$, and use $L^\dagger = L$:

$$\boxed{\langle \psi | L^2 | \psi \rangle = \langle L^\dagger \psi | L\psi \rangle = \langle L\psi | L\psi \rangle = \|L\psi\|^2 \geq 0.}$$

Equality occurs exactly when $L\psi = 0$. For a normalized state this is the stated expectation value. For an unnormalized nonzero state, dividing by $\langle \psi | \psi \rangle > 0$ preserves the inequality.

5.4.5 Consider a Hilbert space whose members are functions defined on the surface of the unit sphere, with a scalar product of the form

$$\langle f | g \rangle = \int d\Omega \, f^* g,$$

where $d\Omega$ is the element of solid angle. Note that the total solid angle of the sphere is 4π. We work here with the three functions $\varphi_1 = Cx/r$, $\varphi_2 = Cy/r$, $\varphi_3 = Cz/r$, with C assigned a value that makes the φ_i normalized.

(a) Find C, and show that the φ_i are also mutually orthogonal.

(b) Form the 3×3 matrices of the angular momentum operators

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \quad L_y = -i\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \quad L_z = -i\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

(c) Verify that the matrix representations of the components of $\mathbf{L}$ satisfy the angular momentum commutator $[L_x, L_y] = iL_z$.

Solution.

(a) Put $n_x = x/r = \sin\theta\cos\phi$, $n_y = \sin\theta\sin\phi$, $n_z = \cos\theta$, and $d\Omega = \sin\theta \, d\theta \, d\phi$. In the θ integrals substitute $u = \cos\theta$, $du = -\sin\theta \, d\theta$; the limits $\theta = 0, \pi$ become $u = 1, -1$, and reversing them absorbs the minus sign, while $\sin^2\theta = 1 - u^2$. With $\int_0^{2\pi} \cos^2\phi \, d\phi = \int_0^{2\pi} \sin^2\phi \, d\phi = \pi$ (their sum is 2π and their difference is $\int_0^{2\pi} \cos 2\phi \, d\phi = 0$),

$$\begin{aligned}
\int n_x^2 d\Omega &= \left(\int_0^{2\pi} \cos^2\phi \, d\phi\right)\left(\int_0^{\pi} \sin^3\theta \, d\theta\right) \\
&= \pi \int_{-1}^{1} (1 - u^2) du = \pi\left(2 - \frac{2}{3}\right) = \frac{4\pi}{3}, \\
\int n_y^2 d\Omega &= \frac{4\pi}{3} \quad (\cos^2\phi \to \sin^2\phi), \qquad \int n_z^2 d\Omega = 2\pi \int_{-1}^{1} u^2 du = \frac{4\pi}{3}.
\end{aligned}$$

The mixed xy integral contains $\int_0^{2\pi} \sin\phi\cos\phi \, d\phi = 0$; the xz and yz integrals contain respectively the zero integrals of $\cos\phi$ and $\sin\phi$. Normalization requires $\langle \varphi_i | \varphi_i \rangle = C^*C \, (4\pi/3) = |C|^2(4\pi/3) = 1$ for each φ_i, and choosing the common constant real and positive gives

$$\boxed{C = \sqrt{\frac{3}{4\pi}}}, \qquad \langle \varphi_i | \varphi_j \rangle = \delta_{ij}.$$

(b) Rotational derivatives annihilate the radial factor. For example,

$$(y\partial_z - z\partial_y)\,\frac{1}{r} = y\frac{-z}{r^3} - z\frac{-y}{r^3} = 0.$$

The same cancellation occurs for $z\partial_x - x\partial_z$ and $x\partial_y - y\partial_x$. The product rule therefore leaves only differentiation of x, y, z; for example, $L_x\varphi_2 = -iC(y\cdot 0 - z\cdot 1)/r = iCz/r = i\varphi_3$. The nine actions are

	φ_1	φ_2	φ_3
L_x	0	$i\varphi_3$	$-i\varphi_2$
L_y	$-i\varphi_3$	0	$i\varphi_1$
L_z	$i\varphi_2$	$-i\varphi_1$	0

Because the φ_i are orthonormal by (a), the matrix element $(L_x)_{ij} = \langle\varphi_i|L_x\varphi_j\rangle$ is the coefficient of φ_i in $L_x\varphi_j$. Thus column j of each matrix lists the coefficients of the table entry under φ_j in that operator's row; e.g. $L_x\varphi_2 = i\varphi_3$ gives the second column $(0,0,i)^T$ of L_x:

$$\boxed{L_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad L_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad L_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.}$$

Conjugate transposition checks that each is Hermitian.

(c) Each product has a single nonzero entry, $(L_xL_y)_{21} = (L_x)_{23}(L_y)_{31} = (-i)(-i) = -1$ and $(L_yL_x)_{12} = (L_y)_{13}(L_x)_{32} = i\cdot i = -1$; every other row–column pairing meets a zero. Hence

$$L_xL_y = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad L_yL_x = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Therefore

$$[L_x, L_y] = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = i\begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \boxed{iL_z}.$$

5.5 Unitary Operators

5.5.1 Show that the corrected matrix for Example 5.5.1,

$$\mathsf{U} = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 & i & 0 & 0 & 0 \\ 1 & i & 0 & 0 & 0 \\ 0 & 0 & -i & 1 & 0 \\ 0 & 0 & i & 1 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} \end{pmatrix},$$

correctly transforms the vector $f(\theta,\varphi) = 3\chi_1 + 2i\chi_2 - \chi_3 + \chi_5$ to the $\{\chi'_i\}$ basis by

(a) (1) Making a column vector $\mathbf{c}$ that represents $f(\theta,\varphi)$ in the $\{\chi_i\}$ basis, (2) forming $\mathbf{c}' = \mathsf{U}\mathbf{c}$, and (3) comparing the expansion $\sum_i c'_i\chi'_i(\theta,\varphi)$ with $f(\theta,\varphi)$;

(b) Verifying that U is unitary.

Solution.

The matrix displayed in Example 5.5.1 (p. 290) has four incorrect imaginary signs for the basis functions defined on p. 289. In fact its worked integral gives $u_{22} = +i/\sqrt{2}$, while the displayed matrix has $-i/\sqrt{2}$. The matrix must be corrected before part (a) is true. We derive the correction directly from those functions, without relying on the displayed matrix. Euler's identities give

$$\chi_1' = -\frac{\chi_1 + i\chi_2}{\sqrt{2}}, \qquad \chi_2' = \frac{\chi_1 - i\chi_2}{\sqrt{2}},$$
$$\chi_3' = \frac{\chi_4 + i\chi_3}{\sqrt{2}}, \qquad \chi_4' = \frac{\chi_4 - i\chi_3}{\sqrt{2}}, \qquad \chi_5' = \chi_5.$$

On p. 289, $\chi_1', \ldots, \chi_4'$ depend on φ through $e^{i\varphi}$, $e^{-i\varphi}$, $e^{2i\varphi}$, $e^{-2i\varphi}$, where the real $\chi_1, \ldots, \chi_4$ contain instead $\cos\varphi$, $\sin\varphi$, $\sin\varphi\cos\varphi$, $\cos 2\varphi$, and $\chi_5' = \chi_5$. Thus $e^{\pm i\varphi} = \cos\varphi \pm i\sin\varphi$ yields the first two identities, and $e^{\pm 2i\varphi} = \cos 2\varphi \pm i\sin 2\varphi$ together with $\sin 2\varphi = 2\sin\varphi\cos\varphi$ yields the next two, with exactly the normalizing constants in the example. Since $U_{ij} = \langle \chi_i' | \chi_j \rangle$, the coefficients in these identities must be conjugated to obtain the rows of U: if $\chi_i' = \sum_k a_{ik}\chi_k$, the orthonormality of the χ_k gives $U_{ij} = \sum_k a_{ik}^* \langle \chi_k | \chi_j \rangle = a_{ij}^*$. For instance $U_{12} = (-i/\sqrt{2})^* = i/\sqrt{2}$. Thus

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & i & 0 & 0 & 0 \\ 1 & i & 0 & 0 & 0 \\ 0 & 0 & -i & 1 & 0 \\ 0 & 0 & i & 1 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} \end{pmatrix}.$$

(a) The original and transformed coefficient columns are

$$c = \begin{pmatrix} 3 \\ 2i \\ -1 \\ 0 \\ 1 \end{pmatrix}, \qquad c' = Uc = \frac{1}{\sqrt{2}} \begin{pmatrix} -3 + i(2i) \\ 3 + i(2i) \\ (-i)(-1) \\ i(-1) \\ \sqrt{2} \end{pmatrix} = \boxed{\begin{pmatrix} -5/\sqrt{2} \\ 1/\sqrt{2} \\ i/\sqrt{2} \\ -i/\sqrt{2} \\ 1 \end{pmatrix}}.$$

Substituting the basis relations back into the expansion checks it:

$$\begin{aligned} \sum_j c_j' \chi_j' &= \frac{5}{2}(\chi_1 + i\chi_2) + \frac{1}{2}(\chi_1 - i\chi_2) \\ &\quad + \frac{i}{2}(\chi_4 + i\chi_3) - \frac{i}{2}(\chi_4 - i\chi_3) + \chi_5 \\ &= 3\chi_1 + 2i\chi_2 - \chi_3 + \chi_5 = f, \end{aligned}$$

since the coefficients of χ_1, χ_2, χ_3 collect to $\frac{5}{2} + \frac{1}{2}$, $\frac{5i - i}{2}$, and $\frac{i^2}{2} + \frac{i^2}{2}$, while the two χ_4 terms cancel. For comparison, the matrix as printed differs from U only in the signs of its four imaginary entries, so it is the complex conjugate U^*. It gives $\widetilde{c}' = U^* c = (-1, 5, -i, i, \sqrt{2})^T/\sqrt{2}$, and the same basis relations then reconstruct

$$\begin{aligned} \sum_j \widetilde{c}_j' \chi_j' &= \frac{1}{2}(\chi_1 + i\chi_2) + \frac{5}{2}(\chi_1 - i\chi_2) \\ &\quad - \frac{i}{2}(\chi_4 + i\chi_3) + \frac{i}{2}(\chi_4 - i\chi_3) + \chi_5 \\ &= 3\chi_1 - 2i\chi_2 + \chi_3 + \chi_5, \end{aligned}$$

which is not the requested f. This explicitly verifies the need for the correction.

(b) The corrected U has two 2×2 blocks and a final 1. Their products with their adjoints are

$$\frac{1}{2}\begin{pmatrix}-1 & i\\ 1 & i\end{pmatrix}\begin{pmatrix}-1 & 1\\ -i & -i\end{pmatrix}=\frac{1}{2}\begin{pmatrix}2 & -1+1\\ -1+1 & 2\end{pmatrix}=I_2,$$

$$\frac{1}{2}\begin{pmatrix}-i & 1\\ i & 1\end{pmatrix}\begin{pmatrix}i & -i\\ 1 & 1\end{pmatrix}=\frac{1}{2}\begin{pmatrix}2 & -1+1\\ -1+1 & 2\end{pmatrix}=I_2.$$

The blocks have disjoint support and $(\sqrt{2}/\sqrt{2})^2=1$ for the last entry, so $UU^\dagger=I_5$; for a square matrix a right inverse is also a left inverse, hence $U^\dagger U=I_5$ as well. The printed matrix U^* is also unitary, since $U^*(U^*)^\dagger=(UU^\dagger)^*=I_5$; that test by itself therefore cannot detect a matrix that represents the wrong basis transformation.

5.5.2

(a) Given (in $\mathbb{R}^3$) the basis $\varphi_1=x$, $\varphi_2=y$, $\varphi_3=z$, consider the basis transformation $x=z'$, $y=y'$, $z=-x'$ (old coordinate functions written in new coordinates). Find the 3×3 matrix U for this transformation.

(b) This transformation corresponds to a rotation of the coordinate axes. Identify the rotation and reconcile your transformation matrix with an appropriate matrix $\mathsf{S}(\alpha,\beta,\gamma)$ of the form given in Eq. (3.37).

(c) Form the column vector $\mathbf{c}$ representing (in the original basis) $f=2x-3y+z$, find the result of applying U to $\mathbf{c}$, and show that this is consistent with the basis transformation of part (a).

Note. You do not need to be able to form scalar products to handle this exercise; a knowledge of the linear relationship between the original and transformed functions is sufficient.

Solution.

The textbook writes the transformation as $x\to z$, $y\to y$, $z\to -x$. The arrows require a convention: here, consistent with the coefficient transformation in Section 5.5, they express the old coordinate functions in new coordinates, $x=z'$, $y=y'$, $z=-x'$. Equivalently the new functions are $x'=-z$, $y'=y$, $z'=x$.

(a) Each column of U contains the coefficients of an old basis function in the new basis (x',y',z'): for $f=\varphi_j$ the old column is $c=e_j$, and $c'=Uc$ is column j of U. The identities

$$x=0x'+0y'+1z',\qquad y=0x'+1y'+0z',\qquad z=-1x'+0y'+0z'$$

therefore give

$$\boxed{U=\begin{pmatrix}0 & 0 & -1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{pmatrix}.}$$

(b) Since $x'=-z$ and $z'=x$, the new coordinate axes are $\hat{e}'_x=-\hat{e}_z$, $\hat{e}'_y=\hat{e}_y$, $\hat{e}'_z=\hat{e}_x$. They result from a right-handed rotation of the axes by $\pi/2$ about $+y$: such a rotation by β carries $\hat{e}_x\to\cos\beta\,\hat{e}_x-\sin\beta\,\hat{e}_z$ and $\hat{e}_z\to\sin\beta\,\hat{e}_x+\cos\beta\,\hat{e}_z$, which for $\beta=\pi/2$ are $-\hat{e}_z$ and $\hat{e}_x$. For the Euler convention of Eq. (3.37), setting $\alpha=\gamma=0$ ($\cos\alpha=\cos\gamma=1$, $\sin\alpha=\sin\gamma=0$) leaves

$$S(0,\beta,0)=\begin{pmatrix}\cos\beta & 0 & -\sin\beta\\ 0 & 1 & 0\\ \sin\beta & 0 & \cos\beta\end{pmatrix},$$

whose rows are the components of the three rotated axes just described. Putting $\beta = \pi/2$ gives $S(0, \pi/2, 0) = U$, as required.

(c) The original coefficient column and its transform are

$$c = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, \qquad c' = Uc = \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix}.$$

The function itself satisfies

$$f = 2x - 3y + z = 2z' - 3y' - x' = -x' - 3y' + 2z',$$

which verifies the coefficient column without scalar products.

If instead the arrows are used to *define* the new ordered basis as $(z, y, -x)$ in the old coordinates, the coefficient transformation for the same fixed f is U^{-1} and gives $(1, -3, -2)^T$. Its new axes $\hat{e}'_x = \hat{e}_z$, $\hat{e}'_y = \hat{e}_y$, $\hat{e}'_z = -\hat{e}_x$ come from the formulas of (b) with $\beta = -\pi/2$, a rotation in the opposite sense. An active substitution $x \mapsto z$, $z \mapsto -x$ in f gives yet another reading: it produces a different function, $-x - 3y + 2z$, whose coefficients in the old basis are the entries of c'. Keeping the primes explicit distinguishes these readings.

5.5.3 Construct the matrix representing the inverse of the transformation in Exercise 5.5.2, and show that this matrix and the transformation matrix of that exercise are matrix inverses of each other.

Solution.

Using the convention established in Exercise 5.5.2, solve the transformation $x = z'$, $y = y'$, $z = -x'$ for the primed variables: $x' = -z$, $y' = y$, $z' = x$. To express a primed expansion in the old basis, read the coefficients of these three functions as columns (as in 5.5.2(a), with the roles of the two bases exchanged):

$$\boxed{V = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} = U^T = U^\dagger.}$$

Both orders of multiplication give the identity:

$$UV = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$VU = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = I_3.$$

(Each row of V has a single nonzero entry, so the rows of VU are row 3 of U, row 2 of U, and minus row 1 of U, namely $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.) As a vector check, $V(-1, -3, 2)^T = (2, -3, 1)^T$, recovering the original coefficients from 5.5.2.

5.5.4 Consider two successive changes of coefficient columns, first from a basis $\{\varphi_i\}$ to $\{\varphi'_i\}$ by U, then to $\{\chi_i\}$ by V. The bases need not be orthonormal. Take real θ with $\sin\theta\cos\theta(\cos^2\theta - \sin^2\theta) \neq 0$ so both maps are invertible. Their matrix representations are

$$\mathsf{U} = \begin{pmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos\theta & i\sin\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathsf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & i\sin\theta \\ 0 & \cos\theta & -i\sin\theta \end{pmatrix}.$$

Given the function $f(x) = 3\varphi_1(x) - \varphi_2(x) - 2\varphi_3(x)$,

(a) By applying U, form the vector representing $f(x)$ in the $\{\varphi'_i\}$ basis and then by applying V form the vector representing $f(x)$ in the $\{\chi_i\}$ basis. Use this result to write $f(x)$ as a linear combination of the χ_i.

(b) Form the matrix products UV and VU and then apply each to the vector representing $f(x)$ in the $\{\varphi_i\}$ basis. Verify that the results of these applications differ and that only one of them gives the result corresponding to part (a).

Solution.

Put $s = \sin\theta$, $c = \cos\theta$, and $a = (3, -1, -2)^T$. First retain the matrices exactly as printed and evaluate the requested products.

(a) Row-by-column multiplication gives

$$Ua = \begin{pmatrix} (is)\cdot 3 + c\cdot(-1) \\ (-c)\cdot 3 + (is)\cdot(-1) \\ -2 \end{pmatrix} = \begin{pmatrix} 3is - c \\ -3c - is \\ -2 \end{pmatrix}.$$

Applying V to this column gives

$$b = V(Ua) = \begin{pmatrix} 3is - c \\ c(-3c - is) + is(-2) \\ c(-3c - is) - is(-2) \end{pmatrix} = \boxed{\begin{pmatrix} -c + 3is \\ -3c^2 - is(c+2) \\ -3c^2 + is(2-c) \end{pmatrix}}.$$

Where these matrices are invertible and interpreted as successive coefficient changes, the formal expansion is

$$f = (-c + 3is)\chi_1 + [-3c^2 - is(c+2)]\chi_2 + [-3c^2 + is(2-c)]\chi_3.$$

These signs also correct the Instructor's Manual's multiplication: the second entry of the initial vector is -1, so the first two entries of Ua are $3is - c$ and $-3c - is$.

(b) Form both matrix products before acting on a; each entry contains at most one nonzero product, e.g. $(UV)_{12} = c\cdot c$, $(UV)_{23} = (is)(is) = -s^2$, $(VU)_{21} = c\,(-c)$, $(VU)_{32} = c\,(is)$:

$$UV = \begin{pmatrix} is & c^2 & ics \\ -c & ics & -s^2 \\ 0 & c & -is \end{pmatrix}, \qquad VU = \begin{pmatrix} is & c & 0 \\ -c^2 & ics & is \\ -c^2 & ics & -is \end{pmatrix}.$$

Consequently

$$(UV)a = \begin{pmatrix} -c^2 + is(3-2c) \\ -3c - ics + 2s^2 \\ -c + 2is \end{pmatrix}, \qquad (VU)a = \begin{pmatrix} -c + 3is \\ -3c^2 - is(c+2) \\ -3c^2 + is(2-c) \end{pmatrix} = b.$$

Only VU corresponds to applying U first and V second. In fact, the two resulting columns differ for every real θ: equality of their first entries, $-c^2+3is-2isc = -c+3is$, would require $c(c-1)+2isc = 0$; its real part $c(c-1)$ must vanish, so $c = 0$ or $c = 1$. For $c = 0$, their second entries are 2 and $-2is$ and differ; for $c = 1$, $s = 0$ and their third entries are -1 and -3 and differ.

The textbook's premise that both printed matrices are unitary is false; this is why the corrected question permits general bases and excludes singular angles. An explicit check, with

$$U^\dagger = \begin{pmatrix} -is & -c & 0 \\ c & -is & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad V^\dagger = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & c \\ 0 & -is & is \end{pmatrix},$$

gives (for instance $(U^\dagger U)_{12} = (-is)c + (-c)(is) = -2isc$ and $(V^\dagger V)_{33} = (-is)(is) + (is)(-is) = 2s^2$)

$$U^\dagger U = \begin{pmatrix} 1 & -2isc & 0 \\ 2isc & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad V^\dagger V = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2c^2 & 0 \\ 0 & 0 & 2s^2 \end{pmatrix}.$$

Thus U is unitary only when $sc = 0$, whereas V is unitary only when $c^2 = s^2 = 1/2$. These conditions never hold simultaneously, because $sc = 0$ forces $c^2 = 0$ or $c^2 = 1$. Moreover, expanding each determinant along the row that contains its isolated entry 1,

$$\det U = (is)(is) - c(-c) = c^2 - s^2, \qquad \det V = c(-is) - (is)c = -2isc.$$

At their zeros even an invertible change of basis is impossible. Away from those zeros the calculations above describe general changes of coefficients, but not successive changes between orthonormal bases.

One consistent repair, not uniquely determined by the printed problem, is to use the following unitary families instead:

$$\widehat{U} = \begin{pmatrix} is & c & 0 \\ c & is & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \widehat{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & is \\ 0 & is & c \end{pmatrix}.$$

Their nontrivial blocks are $isI + c\sigma_1$ and $cI + is\sigma_1$, with $\sigma_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\sigma_1^2 = I$; multiplication by their adjoints cancels the cross terms:

$$\begin{aligned} (-isI + c\sigma_1)(isI + c\sigma_1) &= s^2 I - isc\,\sigma_1 + isc\,\sigma_1 + c^2\sigma_1^2 = (s^2 + c^2)I = I, \\ (cI - is\sigma_1)(cI + is\sigma_1) &= c^2 I + isc\,\sigma_1 - isc\,\sigma_1 + s^2\sigma_1^2 = I. \end{aligned}$$

With these repaired data, the full calculation becomes

$$\widehat{U}a = \begin{pmatrix} -c + 3is \\ 3c - is \\ -2 \end{pmatrix}, \qquad \widehat{V}\widehat{U}a = \begin{pmatrix} -c + 3is \\ 3c^2 - is(c+2) \\ s^2 - 2c + 3ics \end{pmatrix},$$

$$\widehat{U}\widehat{V} = \begin{pmatrix} is & c^2 & ics \\ c & ics & -s^2 \\ 0 & is & c \end{pmatrix}, \qquad \widehat{V}\widehat{U} = \begin{pmatrix} is & c & 0 \\ c^2 & ics & is \\ ics & -s^2 & c \end{pmatrix},$$

$$\widehat{U}\widehat{V}a = \begin{pmatrix} -c^2 + is(3-2c) \\ 3c - ics + 2s^2 \\ -2c - is \end{pmatrix}.$$

The coefficients of the repaired expansion in its final basis are the entries of $\widehat{V}\widehat{U}a$. The repaired matrices commute only for $\theta \equiv 0 \pmod{2\pi}$, where $\widehat{V} = I$: up to sign, the entries of $\widehat{U}\widehat{V} - \widehat{V}\widehat{U}$ are $c(c-1)$, isc and $s(s+i)$, and for real θ the last vanishes only if $s = 0$, after which the first vanishes only if $c = 1$. At every other angle the two columns differ as well, by the argument used above for U and V: $\widehat{U}\widehat{V}a$ and $\widehat{V}\widehat{U}a$ have the same first entries as $(UV)a$ and $(VU)a$, and for $c = 0$ the same second entries 2 and $-2is$.

5.5.5 Three functions which are orthogonal with unit weight on the range $-1 \le x \le 1$ are $P_0 = 1$, $P_1 = x$, and $P_2 = \frac{3}{2}x^2 - \frac{1}{2}$. Another set of functions that are orthogonal and span the same space are $F_0 = x^2$, $F_1 = x$, $F_2 = 5x^2 - 3$. Although much of this exercise can be done by inspection, write down and evaluate all the integrals that lead to the results when they are obtained in terms of scalar products.

(a) Normalize each of the P_i and F_i.

(b) Find the unitary matrix U that transforms from the normalized P_i basis to the normalized F_i basis.

(c) Find the unitary matrix V that transforms from the normalized F_i basis to the normalized P_i basis.

(d) Show that U and V are unitary, and that $\mathsf{V} = \mathsf{U}^{-1}$.

(e) Expand $f(x) = 5x^2 - 3x + 1$ in terms of the normalized versions of both bases, and verify that the transformation matrix U converts the P-basis expansion of $f(x)$ into its F-basis expansion.

Solution.

Use $M_k = \int_{-1}^{1} x^k dx$, so $M_k = 0$ for odd k and $M_k = 2/(k+1)$ for even k: $M_0 = 2$, $M_2 = 2/3$, $M_4 = 2/5$.

(a) The normalization integrals are

$$\|P_0\|^2 = M_0 = 2, \qquad \|P_1\|^2 = M_2 = \frac{2}{3},$$
$$\|P_2\|^2 = \frac{1}{4}(9M_4 - 6M_2 + M_0) = \frac{1}{4}\left(\frac{18}{5} - 4 + 2\right) = \frac{2}{5},$$
$$\|F_0\|^2 = M_4 = \frac{2}{5}, \qquad \|F_1\|^2 = M_2 = \frac{2}{3},$$
$$\|F_2\|^2 = 25M_4 - 30M_2 + 9M_0 = 10 - 20 + 18 = 8.$$

The normalized functions are therefore

$$p_0 = \frac{1}{\sqrt{2}}, \quad p_1 = \sqrt{\frac{3}{2}}x, \quad p_2 = \sqrt{\frac{5}{2}}\frac{3x^2 - 1}{2},$$

$$g_0 = \sqrt{\frac{5}{2}}x^2, \quad g_1 = \sqrt{\frac{3}{2}}x, \quad g_2 = \frac{5x^2-3}{2\sqrt{2}}.$$

For completeness, the original off-diagonal integrals are

$$\langle P_0|P_1\rangle = M_1 = 0, \quad \langle P_0|P_2\rangle = (3M_2 - M_0)/2 = 0, \quad \langle P_1|P_2\rangle = (3M_3 - M_1)/2 = 0,$$

$$\langle F_0|F_1\rangle = M_3 = 0, \quad \langle F_0|F_2\rangle = 5M_4 - 3M_2 = 0, \quad \langle F_1|F_2\rangle = 5M_3 - 3M_1 = 0.$$

(b) With the convention $U_{ij} = \langle \varphi_i'|\varphi_j\rangle$ of Section 5.5, used in Exercise 5.5.1 (new functions g_i, old functions p_j), evaluate all nine overlaps $U_{ij} = \int_{-1}^{1} g_i p_j dx$:

$$U_{00} = \frac{\sqrt{5}}{2}M_2 = \frac{\sqrt{5}}{3}, \qquad U_{01} = \frac{\sqrt{15}}{2}M_3 = 0,$$
$$U_{02} = \frac{5}{4}(3M_4 - M_2) = \frac{2}{3}, \qquad U_{10} = \frac{\sqrt{3}}{2}M_1 = 0,$$
$$U_{11} = \frac{3}{2}M_2 = 1, \qquad U_{12} = \frac{\sqrt{15}}{4}(3M_3 - M_1) = 0,$$
$$U_{20} = \frac{1}{4}(5M_2 - 3M_0) = -\frac{2}{3}, \qquad U_{21} = \frac{\sqrt{3}}{4}(5M_3 - 3M_1) = 0,$$
$$U_{22} = \frac{\sqrt{5}}{8}(15M_4 - 14M_2 + 3M_0) = \frac{\sqrt{5}}{3}.$$

In U_{02}, U_{20}, U_{22} the moments evaluate as

$$\frac{5}{4}\Big(\frac{6}{5} - \frac{2}{3}\Big) = \frac{5}{4}\cdot\frac{8}{15} = \frac{2}{3}, \qquad \frac{1}{4}\Big(\frac{10}{3} - 6\Big) = -\frac{2}{3}, \qquad \frac{\sqrt{5}}{8}\Big(6 - \frac{28}{3} + 6\Big) = \frac{\sqrt{5}}{8}\cdot\frac{8}{3} = \frac{\sqrt{5}}{3};$$

for U_{22} the constants combine as $(2\sqrt{2})^{-1}\sqrt{5/2}\cdot\frac{1}{2} = \sqrt{5}/8$ and $(5x^2-3)(3x^2-1) = 15x^4 - 14x^2 + 3$. Thus

$$\boxed{U = \begin{pmatrix} \sqrt{5}/3 & 0 & 2/3 \\ 0 & 1 & 0 \\ -2/3 & 0 & \sqrt{5}/3 \end{pmatrix}.}$$

(c) Reversing the real integrands gives $V_{ij} = \int p_i g_j dx = U_{ji}$; these are the same nine evaluated integrals with their indices interchanged. Therefore

$$\boxed{V = U^T = \begin{pmatrix} \sqrt{5}/3 & 0 & -2/3 \\ 0 & 1 & 0 \\ 2/3 & 0 & \sqrt{5}/3 \end{pmatrix}.}$$

(d) Let $a = \sqrt{5}/3$, $b = 2/3$, so $a^2 + b^2 = 5/9 + 4/9 = 1$. The $0,2$ block of UV is

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} a^2+b^2 & -ab+ab \\ -ab+ab & a^2+b^2 \end{pmatrix} = I_2.$$

The middle entry is 1 and other entries vanish, so $UV = I_3$; the $0,2$ block of VU is the same computation with $b \to -b$, so also $VU = I_3$. Since $V = U^\dagger$, both matrices are unitary and $V = U^{-1}$.

(e) Evaluate the six coefficient integrals, retaining their moment expressions to check every term:

$$a_0 = \int p_0 f\,dx = \frac{5M_2 - 3M_1 + M_0}{\sqrt{2}} = \frac{8\sqrt{2}}{3},$$
$$a_1 = \int p_1 f\,dx = \sqrt{\frac{3}{2}}(5M_3 - 3M_2 + M_1) = -\sqrt{6},$$
$$a_2 = \int p_2 f\,dx = \frac{1}{2}\sqrt{\frac{5}{2}}(15M_4 - 9M_3 - 2M_2 + 3M_1 - M_0) = \frac{2\sqrt{10}}{3},$$
$$b_0 = \int g_0 f\,dx = \sqrt{\frac{5}{2}}(5M_4 - 3M_3 + M_2) = \frac{4\sqrt{10}}{3},$$
$$b_1 = \int g_1 f\,dx = a_1 = -\sqrt{6} \qquad (g_1 = p_1),$$
$$b_2 = \int g_2 f\,dx = \frac{25M_4 - 15M_3 - 10M_2 + 9M_1 - 3M_0}{2\sqrt{2}} = -\frac{2\sqrt{2}}{3}.$$

(All integrals in this part have limits $-1, 1$.) The moment combinations in a_0, a_1, a_2, b_0, b_2 are

$$\frac{10}{3} + 2 = \frac{16}{3}, \qquad -2, \qquad 6 - \frac{4}{3} - 2 = \frac{8}{3}, \qquad 2 + \frac{2}{3} = \frac{8}{3}, \qquad 10 - \frac{20}{3} - 6 = -\frac{8}{3},$$

where the integrands of a_2 and b_2 use

$$(3x^2 - 1)f = 15x^4 - 9x^3 - 2x^2 + 3x - 1, \qquad (5x^2 - 3)f = 25x^4 - 15x^3 - 10x^2 + 9x - 3.$$

With $\sqrt{3/2} = \sqrt{6}/2$ and $\sqrt{5/2} = \sqrt{10}/2$ these give the values shown, e.g.

$$a_0 = \frac{16}{3\sqrt{2}} = \frac{8\sqrt{2}}{3}, \qquad a_2 = \frac{1}{2} \cdot \frac{\sqrt{10}}{2} \cdot \frac{8}{3} = \frac{2\sqrt{10}}{3}, \qquad b_2 = -\frac{8}{3} \cdot \frac{1}{2\sqrt{2}} = -\frac{2\sqrt{2}}{3}.$$

Thus

$$\boxed{f = \frac{8\sqrt{2}}{3}p_0 - \sqrt{6}\,p_1 + \frac{2\sqrt{10}}{3}p_2 = \frac{4\sqrt{10}}{3}g_0 - \sqrt{6}\,g_1 - \frac{2\sqrt{2}}{3}g_2.}$$

Direct substitution yields, respectively, $8/3 - 3x + (5/3)(3x^2 - 1)$ and $(20/3)x^2 - 3x - (5x^2 - 3)/3$, both equal to f. Finally,

$$Ua = \begin{pmatrix} (\sqrt{5}/3)(8\sqrt{2}/3) + (2/3)(2\sqrt{10}/3) \\ -\sqrt{6} \\ (-2/3)(8\sqrt{2}/3) + (\sqrt{5}/3)(2\sqrt{10}/3) \end{pmatrix} = \begin{pmatrix} 4\sqrt{10}/3 \\ -\sqrt{6} \\ -2\sqrt{2}/3 \end{pmatrix} = b,$$

because $\sqrt{5}\sqrt{2} = \sqrt{10}$ and $\sqrt{5}\sqrt{10} = 5\sqrt{2}$ make the first entry $(8\sqrt{10} + 4\sqrt{10})/9 = 12\sqrt{10}/9$ and the third $(-16\sqrt{2} + 10\sqrt{2})/9 = -6\sqrt{2}/9$. This checks the coefficient transformation independently of reconstruction.

5.6 Transformations of Operators

5.6.1

(a) Using the two spin functions $\varphi_1 = \alpha$ and $\varphi_2 = \beta$ as an orthonormal basis (so $\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$, $\langle\alpha|\beta\rangle = 0$), and the relations

$$S_x\alpha = \frac{1}{2}\beta, \quad S_x\beta = \frac{1}{2}\alpha, \quad S_y\alpha = \frac{i}{2}\beta, \quad S_y\beta = -\frac{i}{2}\alpha, \quad S_z\alpha = \frac{1}{2}\alpha, \quad S_z\beta = -\frac{1}{2}\beta,$$

construct the 2×2 matrices of S_x, S_y, and S_z.

(b) Taking now the basis $\varphi_1' = C(\alpha + \beta)$, $\varphi_2' = C(\alpha - \beta)$:

(i) Verify that φ_1' and φ_2' are orthogonal,

(ii) Assign C a value that makes φ_1' and φ_2' normalized,

(iii) Find the unitary matrix for the transformation $\{\varphi_i\} \to \{\varphi_i'\}$.

(c) Find the matrices of S_x, S_y, and S_z in the $\{\varphi_i'\}$ basis.

Solution.

(a) In the order (α, β), an operator's first column gives its action on α and its second gives its action on β. The six given relations therefore give

$$\boxed{S_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.}$$

For example, $S_y\alpha = (i/2)\beta$ supplies the column $(0, i/2)^T$; this fixes the imaginary signs without an appeal to a memorized matrix.

(b) Expand the scalar products using orthonormality:

$$\langle\alpha + \beta|\alpha - \beta\rangle = 1 - 0 + 0 - 1 = 0,$$

$$\|\alpha \pm \beta\|^2 = 1 \pm 0 \pm 0 + 1 = 2.$$

Hence $\|\varphi_{1,2}'\|^2 = C^*C\,\|\alpha \pm \beta\|^2 = 2|C|^2 = 1$, and $C = 1/\sqrt{2}$ is a real positive normalizing choice. The overlap matrix $U_{ij} = \langle\varphi_i'|\varphi_j\rangle$ has, for example, $U_{21} = \langle(\alpha - \beta)/\sqrt{2}|\alpha\rangle = 1/\sqrt{2}$ and $U_{22} = \langle(\alpha - \beta)/\sqrt{2}|\beta\rangle = -1/\sqrt{2}$; altogether

$$\boxed{U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}, \qquad U^\dagger = U, \quad U^2 = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

Thus $U^\dagger U = U^2 = I$: U is unitary.

(c) For $U_{ij} = \langle\varphi_i'|\varphi_j\rangle$ an operator matrix transforms as $A' = UAU^\dagger$: expanding in the orthonormal φ_k, $\varphi_j' = \sum_k \langle\varphi_k|\varphi_j'\rangle\varphi_k = \sum_k U_{jk}^*\varphi_k = \sum_k (U^\dagger)_{kj}\varphi_k$, and so $A_{ij}' = \langle\varphi_i'|A\varphi_j'\rangle = \sum_{k,l} U_{ik}A_{kl}(U^\dagger)_{lj}$,

where conjugate linearity in the first slot turned $(U^\dagger)^*_{ki}$ into U_{ik}. Hence $S'_j = US_jU^\dagger$. Put $H = \sqrt{2}U$, so that $US_jU^\dagger = \frac{1}{2}HS_jH$ because $U^\dagger = U$. The three unscaled products are

$$H\begin{pmatrix}0&1\\1&0\end{pmatrix}H = \begin{pmatrix}1&1\\-1&1\end{pmatrix}\begin{pmatrix}1&1\\1&-1\end{pmatrix} = \begin{pmatrix}2&0\\0&-2\end{pmatrix},$$
$$H\begin{pmatrix}0&-i\\i&0\end{pmatrix}H = \begin{pmatrix}i&-i\\-i&-i\end{pmatrix}\begin{pmatrix}1&1\\1&-1\end{pmatrix} = \begin{pmatrix}0&2i\\-2i&0\end{pmatrix},$$
$$H\begin{pmatrix}1&0\\0&-1\end{pmatrix}H = \begin{pmatrix}1&-1\\1&1\end{pmatrix}\begin{pmatrix}1&1\\1&-1\end{pmatrix} = \begin{pmatrix}0&2\\2&0\end{pmatrix}.$$

Each must be multiplied by $1/4$ (two factors $1/\sqrt{2}$ and the operator's $1/2$), giving

$$\boxed{S'_x = \frac{1}{2}\begin{pmatrix}1&0\\0&-1\end{pmatrix}, \quad S'_y = \frac{1}{2}\begin{pmatrix}0&i\\-i&0\end{pmatrix}, \quad S'_z = \frac{1}{2}\begin{pmatrix}0&1\\1&0\end{pmatrix}.}$$

A direct action check is $S_x(\alpha \pm \beta) = \pm(\alpha \pm \beta)/2$, which confirms the diagonal S'_x and the order of its eigenvalues.

5.6.2 For the basis $\varphi_1 = Cxe^{-r^2}$, $\varphi_2 = Cye^{-r^2}$, $\varphi_3 = Cze^{-r^2}$, where $r^2 = x^2 + y^2 + z^2$, with the scalar product defined as an unweighted integral over $\mathbb{R}^3$ and with C chosen to make the φ_i normalized:

(a) Find the 3×3 matrix of $L_x = -i\left(y\dfrac{\partial}{\partial z} - z\dfrac{\partial}{\partial y}\right)$;

(b) Using the transformation matrix $\mathsf{U} = \begin{pmatrix}1&0&0\\0&1/\sqrt{2}&-i/\sqrt{2}\\0&1/\sqrt{2}&i/\sqrt{2}\end{pmatrix}$, find the transformed matrix of L_x;

(c) Find the new basis functions φ'_i defined by the transformation U, and write explicitly (in terms of x, y, and z) the functional forms of $L_x\varphi'_i$, $i = 1, 2, 3$.

Hint. Use $\int e^{-r^2}d^3r = \pi^{3/2}$, $\int x^2e^{-r^2}d^3r = \frac{1}{2}\pi^{3/2}$; the integrals are over $\mathbb{R}^3$.

Solution.

First determine the common normalization, choosing C real and positive. Scaling the Gaussian integral in the hint gives

$$\int_{\mathbb{R}^3} e^{-ar^2}d^3r = \pi^{3/2}a^{-3/2}, \qquad \int_{\mathbb{R}^3} x^2e^{-ar^2}d^3r = \frac{\pi^{3/2}}{2a^{5/2}}.$$

For the second formula substitute $(X, Y, Z) = \sqrt{a}(x, y, z)$ in the hint: $x^2d^3r = X^2d^3R/a^{5/2}$ and $e^{-ar^2} = e^{-R^2}$. The squared wavefunctions involve $a = 2$, not $a = 1$: $\|\varphi_1\|^2 = C^2\int x^2e^{-2r^2}d^3r = C^2\pi^{3/2}/(2 \cdot 2^{5/2})$, so

$$1 = C^2\frac{\pi^{3/2}}{2^{7/2}}, \qquad \boxed{C = \frac{2^{7/4}}{\pi^{3/4}}}.$$

By symmetry the same C normalizes φ_2 and φ_3. The mixed integrals vanish by odd parity (e.g. $\int xy\,e^{-2r^2}d^3r$ is odd in x), so the basis is orthonormal.

(a) Write $E = e^{-r^2}$. Its rotational derivative is zero:

$$(y\partial_z - z\partial_y)E = y(-2zE) - z(-2yE) = 0.$$

The product rule then gives

$$L_x(xE) = 0, \qquad L_x(yE) = -i(0-z)E = izE, \qquad L_x(zE) = -i(y-0)E = -iyE.$$

The common factor C does not change the coefficients, and in the orthonormal basis column j holds the coefficients of $L_x\varphi_j$, so

$$\boxed{L_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.}$$

(b) The nontrivial block of U has rows $(1,-i)/\sqrt{2}$ and $(1,i)/\sqrt{2}$, whose norms are one and whose mutual Hermitian scalar product is $(1+(-i)(-i))/2 = 0$. Thus $U^{-1} = U^\dagger$, and the transformed matrix is $L'_x = UL_xU^\dagger$ (Exercise 5.6.1(c)). Explicitly,

$$UL_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & -1/\sqrt{2} & -i/\sqrt{2} \end{pmatrix}, \qquad U^\dagger = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & i/\sqrt{2} & -i/\sqrt{2} \end{pmatrix}.$$

The first row of UL_x vanishes. Row 2 of UL_x times columns 2 and 3 of $U^\dagger$ gives $\frac{1}{2} + \frac{(-i)(i)}{2} = 1$ and $\frac{1}{2} + \frac{(-i)(-i)}{2} = 0$; row 3 gives $-\frac{1}{2} + \frac{(-i)(i)}{2} = 0$ and $-\frac{1}{2} + \frac{(-i)(-i)}{2} = -1$. Hence

$$\boxed{L'_x = UL_xU^\dagger = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.}$$

(c) Because $U_{ij} = \langle\varphi'_i|\varphi_j\rangle$, the new ket functions are $\varphi'_i = \sum_j U^*_{ij}\varphi_j$ (Exercise 5.6.1(c)), i.e. they use the *conjugates* of the rows of U:

$$\boxed{\varphi'_1 = CxE, \quad \varphi'_2 = \frac{C}{\sqrt{2}}(y+iz)E, \quad \varphi'_3 = \frac{C}{\sqrt{2}}(y-iz)E.}$$

Applying the differential operator, using the actions in (a), gives

$$\begin{aligned} L_x\varphi'_1 &= 0, \\ L_x\varphi'_2 &= \frac{C}{\sqrt{2}}[iz + i(-iy)]E = \frac{C}{\sqrt{2}}(y+iz)E = \varphi'_2, \\ L_x\varphi'_3 &= \frac{C}{\sqrt{2}}[iz - i(-iy)]E = -\frac{C}{\sqrt{2}}(y-iz)E = -\varphi'_3. \end{aligned}$$

These are exactly the three columns of the diagonal matrix in (b).

5.6.3 The Gram-Schmidt process for converting an arbitrary basis χ_ν into an orthonormal set φ_ν is described in Section 5.2 in a way that introduces coefficients of the form $-\langle\varphi_\mu|\chi_\nu\rangle$. For bases consisting of three functions, convert the formulation so that φ_ν is expressed entirely in terms of the χ_μ, thereby obtaining an expression for the upper-triangular matrix T appearing in Eq. (5.81).

Solution.

Use the complex scalar-product convention of the chapter and define

$$S_{ij} = \langle\chi_i|\chi_j\rangle, \quad S_{ji} = S_{ij}^*, \qquad D_1 = S_{11}, \quad D_2 = S_{11}S_{22} - S_{12}S_{21}, \quad D_3 = \det S.$$

Since the χ_i are linearly independent, $v^\dagger S v = \sum_{i,j} v_i^* S_{ij} v_j = \|\sum_i v_i\chi_i\|^2 > 0$ for every nonzero column v; so S, and likewise its leading 1×1 and 2×2 blocks (the Gram matrices of χ_1 and of χ_1, χ_2), are positive definite. Their determinants D_1, D_2, D_3 are products of positive eigenvalues, hence positive, and all square roots below are well defined and positive. Gram–Schmidt begins with

$$\varphi_1 = \frac{\chi_1}{\sqrt{D_1}}, \qquad \psi_2 = \chi_2 - \langle\varphi_1|\chi_2\rangle\varphi_1 = \chi_2 - \frac{S_{12}}{D_1}\chi_1.$$

Expanding the norm, including complex conjugation, gives with $a = S_{12}/D_1$, $a^* = S_{21}/D_1$,

$$\begin{aligned}\|\psi_2\|^2 &= S_{22} - aS_{21} - a^*S_{12} + a^*aS_{11} \\ &= S_{22} - \frac{S_{12}S_{21}}{D_1} - \frac{S_{21}S_{12}}{D_1} + \frac{S_{21}S_{12}S_{11}}{D_1^2} \\ &= S_{22} - \frac{S_{12}S_{21}}{D_1} = \frac{S_{11}S_{22} - S_{12}S_{21}}{D_1} = \frac{D_2}{D_1},\end{aligned}$$

because $S_{11} = D_1$ makes the last term cancel one of the two middle ones. Consequently

$$\varphi_2 = \frac{\psi_2}{\|\psi_2\|} = \sqrt{\frac{D_1}{D_2}}\Big(\chi_2 - \frac{S_{12}}{D_1}\chi_1\Big) = \frac{D_1\chi_2 - S_{12}\chi_1}{\sqrt{D_1D_2}}.$$

For the third step define $K = D_1S_{23} - S_{21}S_{13}$. The two projection coefficients, with the conjugation on the first argument accounted for (D_1 is real and $S_{12}^* = S_{21}$), are

$$\langle\varphi_1|\chi_3\rangle = \frac{S_{13}}{\sqrt{D_1}}, \qquad \langle\varphi_2|\chi_3\rangle = \frac{D_1S_{23} - S_{12}^*S_{13}}{\sqrt{D_1D_2}} = \frac{K}{\sqrt{D_1D_2}}.$$

Substitute the already-expanded φ_1, φ_2 into $\psi_3 = \chi_3 - \langle\varphi_1|\chi_3\rangle\varphi_1 - \langle\varphi_2|\chi_3\rangle\varphi_2$ and collect the coefficient of each χ_i:

$$\begin{aligned}\psi_3 &= \chi_3 - \frac{S_{13}}{D_1}\chi_1 - \frac{K}{D_1D_2}(D_1\chi_2 - S_{12}\chi_1) \\ &= \chi_3 - \frac{K}{D_2}\chi_2 + \frac{-D_2S_{13} + S_{12}K}{D_1D_2}\chi_1.\end{aligned}$$

The last numerator simplifies without any reality assumption, since the two terms $\pm S_{12}S_{21}S_{13}$ cancel:

$$\begin{aligned}-D_2S_{13} + S_{12}K &= -(D_1S_{22} - S_{12}S_{21})S_{13} + S_{12}(D_1S_{23} - S_{21}S_{13}) \\ &= D_1(S_{12}S_{23} - S_{22}S_{13}).\end{aligned}$$

Thus, with $-K = S_{21}S_{13} - D_1 S_{23}$ and the factor D_1 cancelled, the unnormalized third vector is

$$\psi_3 = \chi_3 + \frac{S_{21}S_{13} - D_1 S_{23}}{D_2}\chi_2 + \frac{S_{12}S_{23} - S_{22}S_{13}}{D_2}\chi_1.$$

Its norm can be evaluated by subtracting the squared orthogonal projection lengths: $\chi_3 = \psi_3 + \langle\varphi_1|\chi_3\rangle\varphi_1 + \langle\varphi_2|\chi_3\rangle\varphi_2$ is a sum of three mutually orthogonal vectors, so by Pythagoras $\|\chi_3\|^2 = \|\psi_3\|^2 + |\langle\varphi_1|\chi_3\rangle|^2 + |\langle\varphi_2|\chi_3\rangle|^2$, i.e.

$$\|\psi_3\|^2 = S_{33} - \frac{S_{31}S_{13}}{D_1} - \frac{K^*K}{D_1 D_2} = \frac{D_3}{D_2}.$$

To verify the determinant simplification explicitly, multiply by $D_1 D_2$ and use $K^* = D_1 S_{32} - S_{12}S_{31}$:

$$\begin{aligned} D_1 D_2\|\psi_3\|^2 &= D_1(D_1 S_{22} - S_{12}S_{21})S_{33} - (D_1 S_{22} - S_{12}S_{21})S_{31}S_{13} \\ &\quad - (D_1 S_{32} - S_{12}S_{31})(D_1 S_{23} - S_{21}S_{13}) \\ &= D_1\{D_1 S_{22}S_{33} - S_{12}S_{21}S_{33} - S_{22}S_{31}S_{13} \\ &\qquad\qquad - D_1 S_{32}S_{23} + S_{32}S_{21}S_{13} + S_{12}S_{31}S_{23}\} \\ &= D_1 \det S. \end{aligned}$$

In the second step the two products $S_{12}S_{21}S_{31}S_{13}$ (from the second and third terms) cancel and every remaining term contains D_1; with $D_1 = S_{11}$ the six terms in braces are exactly the six terms of the expansion $\det S = S_{11}S_{22}S_{33} + S_{12}S_{23}S_{31} + S_{13}S_{21}S_{32} - S_{13}S_{22}S_{31} - S_{12}S_{21}S_{33} - S_{11}S_{23}S_{32}$. Dividing ψ_3 by $\|\psi_3\| = \sqrt{D_3/D_2}$ now expresses all three φ_j in the original χ_i. Since the coefficients of $\varphi_j = \sum_i T_{ij}\chi_i$ form column j, the answer is

$$\boxed{T = \begin{pmatrix} \dfrac{1}{\sqrt{D_1}} & -\dfrac{S_{12}}{\sqrt{D_1 D_2}} & \dfrac{S_{12}S_{23} - S_{22}S_{13}}{\sqrt{D_2 D_3}} \\[2ex] 0 & \sqrt{\dfrac{D_1}{D_2}} & \dfrac{S_{21}S_{13} - D_1 S_{23}}{\sqrt{D_2 D_3}} \\[2ex] 0 & 0 & \sqrt{\dfrac{D_2}{D_3}} \end{pmatrix}}.$$

Every column has unit S-norm and is S-orthogonal to the others, because $(T^\dagger S T)_{\mu\nu} = \sum_{i,j} T^*_{i\mu} S_{ij} T_{j\nu} = \langle\varphi_\mu|\varphi_\nu\rangle = \delta_{\mu\nu}$; so $T^\dagger S T = I$. Finally, if $A_{ij} = \langle\chi_i|\mathcal{A}\chi_j\rangle$, expanding both arguments gives

$$\langle\varphi_\mu|\mathcal{A}\varphi_\nu\rangle = \sum_{i,j} T^*_{i\mu} A_{ij} T_{j\nu} = (T^\dagger A T)_{\mu\nu},$$

which also verifies the orientation of T in Eq. (5.81).

5.7 Invariants

5.7.1 Using the formal properties of unitary transformations, show that the operator commutator $[x, p] = iI$, on a common domain of the products, is invariant under unitary conjugation when the domain is carried along by the same unitary map.

Solution.

Let $x' = UxU^\dagger$, $p' = UpU^\dagger$, with $U^\dagger U = UU^\dagger = I$. Multiply before simplifying:

$$\begin{aligned}[x', p'] &= (UxU^\dagger)(UpU^\dagger) - (UpU^\dagger)(UxU^\dagger) \\ &= Ux(U^\dagger U)pU^\dagger - Up(U^\dagger U)xU^\dagger \\ &= U(xp - px)U^\dagger = U(iI)U^\dagger = iUU^\dagger = \boxed{iI}.\end{aligned}$$

The canonical relation $[x, p] = iI$ cannot hold for finite square matrices: $\mathrm{tr}(xp - px) = \mathrm{tr}(xp) - \mathrm{tr}(px) = 0$ by the cyclic property $\mathrm{tr}(AB) = \mathrm{tr}(BA)$, whereas $\mathrm{tr}(iI_N) = iN \neq 0$. It is an operator relation in an infinite-dimensional space. For unbounded position and momentum the calculation holds on a common domain of the products, carried to the new domain by U; it is not a claim that those products are defined on every Hilbert-space vector. For example, the usual relation holds for $x\psi = x\psi(x)$ and $p\psi = -i\psi'(x)$ on smooth rapidly decreasing functions, since

$$x(-i\psi') - [-i(x\psi)'] = -ix\psi' + i(\psi + x\psi') = i\psi.$$

This explicitly supplies a domain and realization for the formal relation.

5.7.2 The Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

have commutator $[\sigma_1, \sigma_2] = 2i\sigma_3$. Show that this relationship continues to be valid if these matrices are transformed by

$$\mathsf{U} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

Solution.

Let $c = \cos\theta$, $s = \sin\theta$. The columns of U are real orthonormal vectors, so $U^\dagger = U^T = U^{-1}$. Direct multiplication gives

$$\begin{aligned}\sigma_1' = U\sigma_1 U^T &= \begin{pmatrix} s & c \\ c & -s \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} 2sc & c^2 - s^2 \\ c^2 - s^2 & -2sc \end{pmatrix}, \\ \sigma_2' = U\sigma_2 U^T &= \begin{pmatrix} is & -ic \\ ic & is \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \\ \sigma_3' = U\sigma_3 U^T &= \begin{pmatrix} c & -s \\ -s & -c \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} c^2 - s^2 & -2sc \\ -2sc & -(c^2 - s^2) \end{pmatrix}.\end{aligned}$$

In σ_2' the diagonal entries are $isc - ics = 0$ and $-isc + isc = 0$, and $c^2 + s^2 = 1$ reduces the off-diagonal ones $-i(s^2 + c^2)$ and $i(c^2 + s^2)$ to $-i$ and i. Write $C = \cos 2\theta = c^2 - s^2$ and $S = \sin 2\theta = 2sc$. Then

$$\sigma_1'\sigma_2' = \begin{pmatrix} S & C \\ C & -S \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} iC & -iS \\ -iS & -iC \end{pmatrix} = i\sigma_3',$$

$$\sigma_2'\sigma_1' = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} S & C \\ C & -S \end{pmatrix} = \begin{pmatrix} -iC & iS \\ iS & iC \end{pmatrix} = -i\sigma_3'.$$

Subtracting proves

$$\boxed{[\sigma_1', \sigma_2'] = 2i\sigma_3'.}$$

This also follows from $[UAU^\dagger, UBU^\dagger] = U[A, B]U^\dagger$ (the computation of Exercise 5.7.1), but the displayed products check the signs and factors directly.

5.7.3

(a) The operator L_x is defined as

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right).$$

Verify that the basis $\varphi_1 = Cxe^{-r^2}$, $\varphi_2 = Cye^{-r^2}$, $\varphi_3 = Cze^{-r^2}$, where $r^2 = x^2+y^2+z^2$, forms a closed set under the operation of L_x, meaning that when L_x is applied to any member of this basis the result is a function within the basis space, and construct the 3×3 matrix of L_x in this basis from the result of the application of L_x to each basis function.

(b) Verify that $L_x(x+iy)e^{-r^2} = -ze^{-r^2}$, and note that this result, using the $\{\varphi_i\}$ basis, can be written $L_x(\varphi_1 + i\varphi_2) = -\varphi_3$.

(c) Express the equation of part (b) in matrix form, and write the matrix equation that results when each of the quantities is transformed using the transformation matrix

$$\mathsf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix}.$$

(d) Regarding the transformation U as producing a new basis $\{\varphi_i'\}$, find the explicit form (in x, y, z) of the φ_i'.

(e) Using the operator form of L_x and the explicit forms of the φ_i', verify the validity of the transformed equation found in part (c).

Hint. The results of Exercise 5.6.2 may be useful.

Solution.

Put $E = e^{-r^2}$ and use the same positive normalization $C = 2^{7/4}/\pi^{3/4}$ as in Exercise 5.6.2.

(a) Since $(y\partial_z - z\partial_y)E = 0$, the product rule gives, as in Exercise 5.6.2(a),

$$L_x\varphi_1 = 0, \qquad L_x\varphi_2 = i\varphi_3, \qquad L_x\varphi_3 = -i\varphi_2.$$

Every result belongs to the span, so the set is closed under L_x. Column j of the matrix holds the coefficients of $L_x\varphi_j$, so

$$L = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

(b) Applying the same actions without the factor C gives

$$L_x[(x+iy)E] = L_x(xE) + iL_x(yE) = 0 + i(izE) = -zE.$$

Multiplying by C produces $L_x(\varphi_1 + i\varphi_2) = -\varphi_3$.

(c) The input and output coefficient columns are $a = (1, i, 0)^T$ and $b = (0, 0, -1)^T$, so the original equation is

$$La = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}\begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ i^2 \end{pmatrix} = b.$$

Transform both vectors and the operator, not just the vectors:

$$a' = Ua = \begin{pmatrix} 1 \\ i/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix}, \quad b' = Ub = \begin{pmatrix} 0 \\ i/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}, \quad L' = ULU^\dagger = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

with L' taken from Exercise 5.6.2(b). For instance, the second entry of a' is $\frac{1}{\sqrt{2}} \cdot i + \left(-\frac{i}{\sqrt{2}}\right) \cdot 0$ and the third entry of b' is $\frac{i}{\sqrt{2}} \cdot (-1)$. The equation survives the transformation because $U^\dagger U = I$: $L'a' = ULU^\dagger Ua = U(La) = Ub = b'$. The transformed equation is therefore

$$\boxed{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ i/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 0 \\ i/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}.}$$

(d) Conjugating the rows of U to form the new ket functions, $\varphi'_i = \sum_j U^*_{ij}\varphi_j$ as in Exercise 5.6.2(c), yields

$$\varphi'_1 = CxE, \qquad \varphi'_2 = \frac{C}{\sqrt{2}}(y + iz)E, \qquad \varphi'_3 = \frac{C}{\sqrt{2}}(y - iz)E.$$

(e) Direct differentiation, including the cancellation of the radial terms as in (a), gives

$$L_x\varphi'_1 = 0, \qquad L_x\varphi'_2 = \frac{C}{\sqrt{2}}[iz + i(-iy)]E = \varphi'_2,$$

$$L_x\varphi'_3 = \frac{C}{\sqrt{2}}[iz - i(-iy)]E = -\varphi'_3.$$

The transformed matrix equation thus means exactly

$$L_x\left(\varphi'_1 + \frac{i}{\sqrt{2}}\varphi'_2 + \frac{i}{\sqrt{2}}\varphi'_3\right) = \frac{i}{\sqrt{2}}\varphi'_2 - \frac{i}{\sqrt{2}}\varphi'_3.$$

By the three actions just found, the left side is $0 + \frac{i}{\sqrt{2}}\varphi'_2 + \frac{i}{\sqrt{2}}(-\varphi'_3)$, which is the right side; this verifies the transformed equation. To see that it is the equation of (b), substitute the explicit forms, using $\frac{i}{\sqrt{2}} \cdot \frac{C}{\sqrt{2}} = \frac{iC}{2}$. The input is

$$CxE + \frac{iC}{2}[(y + iz) + (y - iz)]E = CxE + \frac{iC}{2}(2y)E = C(x + iy)E,$$

and the output is

$$\frac{iC}{2}[(y + iz) - (y - iz)]E = \frac{iC}{2}(2iz)E = -CzE.$$

This is precisely the functional equation from (b), verifying the invariance using both matrices and differential operators.

6 Chapter 6. Eigenvalue Problems

Problems

6.2 Matrix Eigenvalue Problems

Find the eigenvalues and corresponding normalized eigenvectors of the matrices in Exercises 6.2.1 through 6.2.14. Orthogonalize any degenerate eigenvectors.

In this section write $v = (x, y, z)^{\mathsf{T}}$ and $p(\lambda) = \det(A - \lambda I)$. A normalized vector satisfies $v^{\dagger} v = 1$; an overall sign or complex phase does not change its eigendirection.

6.2.1 $\mathsf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 1, 2$.

Solution.

Expanding the determinant along the second row gives

$$p(\lambda) = (1-\lambda)\big[(1-\lambda)^2 - 1\big] = (1-\lambda)\lambda(\lambda - 2).$$

Thus the eigenvalues are $0, 1, 2$. The component equations are

$$(1-\lambda)x + z = 0, \qquad (1-\lambda)y = 0, \qquad x + (1-\lambda)z = 0.$$

For $\lambda = 0$, they give $y = 0$, $z = -x$. For $\lambda = 1$, the first and third give $z = x = 0$, leaving y free. For $\lambda = 2$, they give $y = 0$, $z = x$. The resulting raw vectors have squared lengths $2, 1, 2$, so the normalized eigenpairs are

λ	u_λ
0	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
1	$(0, 1, 0)^{\mathsf{T}}$
2	$(1, 0, 1)^{\mathsf{T}}/\sqrt{2}$

The first and third vectors have scalar product $(1-1)/2 = 0$, and both are perpendicular to the middle vector. Thus this is an orthonormal basis, with no degeneracy to resolve.

6.2.2 $\mathsf{A} = \begin{pmatrix} 1 & \sqrt{2} & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

ANS. $\lambda = -1, 0, 2$.

Solution.

The third coordinate decouples. Expanding on that coordinate,

$$p(\lambda) = -\lambda\big[(1-\lambda)(-\lambda) - 2\big] = -\lambda(\lambda^2 - \lambda - 2) = -\lambda(\lambda - 2)(\lambda + 1).$$

The first two component equations are

$$(1-\lambda)x + \sqrt{2}\,y = 0, \qquad \sqrt{2}\,x - \lambda y = 0,$$

while the last is $-\lambda z = 0$. For $\lambda = -1$, $z = 0$ and $y = -\sqrt{2}\,x$. For $\lambda = 0$, the second equation forces $x = 0$, then the first forces $y = 0$; z is free. For $\lambda = 2$, $z = 0$ and $x = \sqrt{2}\,y$. The nontrivial two-coordinate vectors each have squared length 3:

λ	u_λ
-1	$(1, -\sqrt{2}, 0)^{\mathsf{T}}/\sqrt{3}$
0	$(0, 0, 1)^{\mathsf{T}}$
2	$(\sqrt{2}, 1, 0)^{\mathsf{T}}/\sqrt{3}$

The two vectors in the xy plane have scalar product $(\sqrt{2} - \sqrt{2})/3 = 0$; the third coordinate vector is perpendicular to both.

6.2.3 $\mathsf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = -1, 1, 2$.

Solution.

Expansion along the first row (whose third entry is zero) gives

$$\begin{aligned} p(\lambda) &= (1-\lambda)\begin{vmatrix} -\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda)\big[-\lambda(1-\lambda) - 1\big] - (1-\lambda) = (1-\lambda)\big[\lambda^2 - \lambda - 1 - 1\big] \\ &= (1-\lambda)(\lambda^2 - \lambda - 2) = (1-\lambda)(\lambda - 2)(\lambda + 1). \end{aligned}$$

The equations for an eigenvector are

$$(1-\lambda)x + y = 0, \quad x - \lambda y + z = 0, \quad y + (1-\lambda)z = 0.$$

For $\lambda = -1$, the first and third yield $y = -2x = -2z$; the middle then holds because $x + y + z = x - 2x + x = 0$. For $\lambda = 1$, $y = 0$ and the middle equation gives $z = -x$. For $\lambda = 2$, the first and third give $y = x = z$, which also satisfies the middle equation. Their squared lengths before normalization are $6, 2, 3$, respectively. Hence

λ	u_λ
-1	$(1, -2, 1)^{\mathsf{T}}/\sqrt{6}$
1	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
2	$(1, 1, 1)^{\mathsf{T}}/\sqrt{3}$

The raw pairwise scalar products are $1 - 1$, $1 - 2 + 1$, and $1 - 1$; all are zero.

6.2.4 $\mathsf{A} = \begin{pmatrix} 1 & \sqrt{8} & 0 \\ \sqrt{8} & 1 & \sqrt{8} \\ 0 & \sqrt{8} & 1 \end{pmatrix}$.

ANS. $\lambda = -3, 1, 5$.

Solution.

Set $q = 1 - \lambda$ and note that $\sqrt{8} = 2\sqrt{2}$. Expanding the tridiagonal determinant along the first row,

$$\begin{aligned} p(\lambda) &= q\begin{vmatrix} q & \sqrt{8} \\ \sqrt{8} & q \end{vmatrix} - \sqrt{8}\begin{vmatrix} \sqrt{8} & \sqrt{8} \\ 0 & q \end{vmatrix} = q(q^2 - 8) - \sqrt{8} \cdot \sqrt{8}\, q \\ &= q(q^2 - 16) = q(q-4)(q+4). \end{aligned}$$

Thus $q = 4, 0, -4$, giving $\lambda = -3, 1, 5$. The component equations are

$$qx + 2\sqrt{2}\,y = 0, \quad 2\sqrt{2}\,x + qy + 2\sqrt{2}\,z = 0, \quad 2\sqrt{2}\,y + qz = 0.$$

When $q = 4$, they give $y = -\sqrt{2}\,x$ and $z = x$; the middle equation then reads $2\sqrt{2}x - 4\sqrt{2}x + 2\sqrt{2}x = 0$. When $q = 0$, they give $y = 0$ and $z = -x$. When $q = -4$, they give $y = \sqrt{2}\,x$ and $z = x$, and the middle equation again vanishes. Normalization gives

λ	u_λ
-3	$(1, -\sqrt{2}, 1)^{\mathsf{T}}/2$
1	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
5	$(1, \sqrt{2}, 1)^{\mathsf{T}}/2$

The raw squared norms are $4, 2, 4$; the first and last scalar product is $1 - 2 + 1 = 0$, and both are perpendicular to the middle vector.

6.2.5 $\mathsf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 1, 2$.

Solution.

The first coordinate is separate from the lower 2×2 block:

$$p(\lambda) = (1 - \lambda)\big[(1 - \lambda)^2 - 1\big] = (1 - \lambda)\lambda(\lambda - 2).$$

The equations are

$$(1 - \lambda)x = 0, \qquad (1 - \lambda)y + z = 0, \qquad y + (1 - \lambda)z = 0.$$

At $\lambda = 0$, $x = 0$ and $z = -y$. At $\lambda = 1$, the last two equations give $y = z = 0$, while x is free. At $\lambda = 2$, $x = 0$ and $z = y$. Thus

λ	u_λ
0	$(0, 1, -1)^{\mathsf{T}}/\sqrt{2}$
1	$(1, 0, 0)^{\mathsf{T}}$
2	$(0, 1, 1)^{\mathsf{T}}/\sqrt{2}$

The vectors' squared norms are one; the only nontrivial scalar product is $(1 - 1)/2 = 0$.

6.2.6 $\mathsf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}$.

ANS. $\lambda = -1, 1, 2$.

Solution.

The block determinant is

$$\begin{aligned} p(\lambda) &= (1 - \lambda)\big[(1 - \lambda)(-\lambda) - 2\big] \\ &= (1 - \lambda)(\lambda^2 - \lambda - 2) = (1 - \lambda)(\lambda - 2)(\lambda + 1). \end{aligned}$$

The eigenvector equations are

$$(1-\lambda)x = 0, \qquad (1-\lambda)y + \sqrt{2}\,z = 0, \qquad \sqrt{2}\,y - \lambda z = 0.$$

For $\lambda = -1$, $x = 0$ and $z = -\sqrt{2}\,y$. For $\lambda = 1$, the last two equations force $z = y = 0$, leaving x free. For $\lambda = 2$, $x = 0$ and $y = \sqrt{2}\,z$. Therefore

λ	u_λ
-1	$(0, 1, -\sqrt{2})^{\mathsf{T}}/\sqrt{3}$
1	$(1, 0, 0)^{\mathsf{T}}$
2	$(0, \sqrt{2}, 1)^{\mathsf{T}}/\sqrt{3}$

Each two-coordinate raw vector has squared norm 3, and their scalar product is $\sqrt{2} - \sqrt{2} = 0$. The Instructor's Manual, p. 81, prints $(0, \sqrt{2}, 0)^{\mathsf{T}}/\sqrt{3}$ for the eigenvalue 2. Its last component must be $1/\sqrt{3}$, as above: the printed vector has squared norm $2/3$ and does not satisfy $Av = 2v$. Indeed, for the printed vector v_{p},

$$Av_{\mathrm{p}} - 2v_{\mathrm{p}} = \frac{1}{\sqrt{3}}(0, -\sqrt{2}, 2)^{\mathsf{T}} \neq 0.$$

6.2.7 $\mathsf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$.

ANS. $\lambda = -\sqrt{2}, 0, \sqrt{2}$.

Solution.

Expanding along the first row,

$$\begin{aligned} p(\lambda) &= -\lambda \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 1 \\ 0 & -\lambda \end{vmatrix} \\ &= (-\lambda)(\lambda^2 - 1) - (-\lambda) = -\lambda(\lambda^2 - 2), \end{aligned}$$

so $\lambda = 0, \pm\sqrt{2}$. The three equations are

$$y = \lambda x, \qquad x + z = \lambda y, \qquad y = \lambda z.$$

For $\lambda = 0$, $y = 0$ and $z = -x$. For $\lambda \neq 0$, the first and third imply $z = x$; substituting into the middle gives $2x = \lambda^2 x$. A nonzero vector then has $\lambda = \pm\sqrt{2}$ and $y = \pm\sqrt{2}\,x$. Thus

λ	u_λ
$-\sqrt{2}$	$(1, -\sqrt{2}, 1)^{\mathsf{T}}/2$
0	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
$\sqrt{2}$	$(1, \sqrt{2}, 1)^{\mathsf{T}}/2$

The nonzero-eigenvalue raw vectors have squared length $1 + 2 + 1 = 4$; the zero-eigenvalue vector has squared length 2. Pairwise orthogonality follows from $1 - 2 + 1 = 0$ and $1 - 1 = 0$.

6.2.8 $\mathsf{A} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 2, 2$.

Solution.

Expansion on the first coordinate gives

$$p(\lambda) = (2-\lambda)\big[(1-\lambda)^2 - 1\big] = (2-\lambda)\lambda(\lambda-2) = -\lambda(\lambda-2)^2.$$

For $\lambda = 0$, the equations give $x = 0$ and $y + z = 0$, hence $u_0 = (0, 1, -1)^{\mathsf{T}}/\sqrt{2}$. For the repeated eigenvalue $\lambda = 2$,

$$(A - 2I)v = 0 \quad \Longleftrightarrow \quad -y + z = 0,$$

with x free. The entire eigenspace is

$$v = x(1, 0, 0)^{\mathsf{T}} + y(0, 1, 1)^{\mathsf{T}}.$$

Its two displayed spanning vectors are already orthogonal, with lengths 1 and $\sqrt{2}$. Normalizing them gives the required orthonormal choice within the degenerate eigenspace:

λ	orthonormal eigenvectors
0	$(0, 1, -1)^{\mathsf{T}}/\sqrt{2}$
2	$(1, 0, 0)^{\mathsf{T}}$, $(0, 1, 1)^{\mathsf{T}}/\sqrt{2}$

Both eigenvectors for 2 have zero scalar product with u_0.

6.2.9 $\mathsf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

ANS. $\lambda = -1, -1, 2$.

Solution.

Expanding along the first row,

$$\begin{aligned} p(\lambda) &= -\lambda \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 1 & -\lambda \end{vmatrix} + \begin{vmatrix} 1 & -\lambda \\ 1 & 1 \end{vmatrix} \\ &= -\lambda(\lambda^2 - 1) + (\lambda + 1) + (\lambda + 1) = (\lambda + 1)\big[-\lambda(\lambda - 1) + 2\big] \\ &= (\lambda + 1)(2 + \lambda - \lambda^2) = (2 - \lambda)(\lambda + 1)^2, \end{aligned}$$

since $(2 - \lambda)(1 + \lambda) = 2 + \lambda - \lambda^2$. To obtain the eigenspaces without repeated elimination, set $s = x + y + z$. Then

$$Av = (s - x, s - y, s - z)^{\mathsf{T}}.$$

If $x = y = z$, this equals $2v$, giving $u_2 = (1, 1, 1)^{\mathsf{T}}/\sqrt{3}$. For $\lambda = -1$, the equation $Av = -v$ is exactly $s = 0$, a plane of dimension two. Choose the independent vectors

$$a = (1, -1, 0)^{\mathsf{T}}, \qquad b = (1, 0, -1)^{\mathsf{T}}.$$

They are not orthogonal: $a^{\mathsf{T}}b = 1$ and $a^{\mathsf{T}}a = 2$. Gram–Schmidt gives

$$b_\perp = b - \frac{a^{\mathsf{T}}b}{a^{\mathsf{T}}a}a = b - \frac{1}{2}a = (\tfrac{1}{2}, \tfrac{1}{2}, -1)^{\mathsf{T}}, \qquad \|b_\perp\|^2 = \frac{3}{2}.$$

Consequently a normalized orthogonal choice is

λ	orthonormal eigenvectors
-1	$(1,-1,0)^{\mathsf{T}}/\sqrt{2}$, $(1,1,-2)^{\mathsf{T}}/\sqrt{6}$
2	$(1,1,1)^{\mathsf{T}}/\sqrt{3}$

The coordinate sum of each eigenvector for $\lambda = -1$ is zero, so each is also perpendicular to u_2.

6.2.10 $\mathsf{A} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$.

ANS. $\lambda = -1, 2, 2$.

Solution.

Let J be the matrix whose nine entries are all 1. Here $A = 2I - J$, and $Jv = (x+y+z)(1,1,1)^{\mathsf{T}}$. On the line spanned by $w = (1,1,1)^{\mathsf{T}}$, $Jw = 3w$, so $Aw = (2-3)w = -w$. On the plane $x+y+z = 0$, $Jv = 0$, so $Av = 2v$. The line and plane have dimensions one and two and together span the whole space. In a basis made of w and two independent vectors of the plane, A becomes $\mathrm{diag}(-1, 2, 2)$. Since $\det(A - \lambda I)$ does not depend on the basis (Exercise 6.4.1),

$$p(\lambda) = (-1 - \lambda)(2 - \lambda)^2,$$

so $-1, 2, 2$ are all the eigenvalues. To orthogonalize the degenerate plane, take $a = (1,-1,0)^{\mathsf{T}}$ and $b = (1,0,-1)^{\mathsf{T}}$. Since $a \cdot b = 1$ and $a \cdot a = 2$,

$$b_\perp = b - \frac{1}{2}a = \frac{1}{2}(1,1,-2)^{\mathsf{T}}, \qquad \|a\| = \sqrt{2}, \quad \|b_\perp\| = \sqrt{3/2}.$$

Hence

λ	orthonormal eigenvectors
-1	$(1,1,1)^{\mathsf{T}}/\sqrt{3}$
2	$(1,-1,0)^{\mathsf{T}}/\sqrt{2}$, $(1,1,-2)^{\mathsf{T}}/\sqrt{6}$

The coordinate-sum condition also makes both eigenvectors for 2 perpendicular to the eigenvector for -1.

6.2.11 $\mathsf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 0, 3$.

Solution.

For this all-ones matrix,

$$Av = (x + y + z)(1,1,1)^{\mathsf{T}}.$$

The vector $w = (1,1,1)^{\mathsf{T}}$ satisfies $Aw = 3w$. Every vector in the plane $x + y + z = 0$ is annihilated, so 0 is an eigenvalue with a two-dimensional eigenspace. These dimensions add to three, so in a basis made of w and two independent vectors of the plane, A becomes $\mathrm{diag}(3, 0, 0)$. Since $\det(A - \lambda I)$ does not depend on the basis (Exercise 6.4.1),

$$p(\lambda) = (3 - \lambda)\lambda^2.$$

For an explicit orthogonalization of the zero eigenspace, choose $a = (1,-1,0)^{\mathsf{T}}$ and $b = (1,0,-1)^{\mathsf{T}}$. Subtract the projection of b on a:

$$b_\perp = b - \frac{1}{2}a = \tfrac{1}{2}(1,1,-2)^{\mathsf{T}}.$$

Their lengths are $\sqrt{2}$ and $\sqrt{3/2}$, respectively. Therefore

λ	orthonormal eigenvectors
0	$(1,-1,0)^{\mathsf{T}}/\sqrt{2}$, $(1,1,-2)^{\mathsf{T}}/\sqrt{6}$
3	$(1,1,1)^{\mathsf{T}}/\sqrt{3}$

The last vector is perpendicular to the zero eigenspace because its scalar product with v is $(x+y+z)/\sqrt{3}$.

6.2.12 $\mathsf{A} = \begin{pmatrix} 5 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{pmatrix}$.

ANS. $\lambda = 1, 1, 6$.

Solution.

The middle coordinate decouples, so

$$\begin{aligned} p(\lambda) &= (1-\lambda)\big[(5-\lambda)(2-\lambda) - 4\big] \\ &= (1-\lambda)(\lambda^2 - 7\lambda + 6) = (1-\lambda)(\lambda-1)(\lambda-6) = (1-\lambda)^2(6-\lambda). \end{aligned}$$

At $\lambda = 1$, the first and third equations are $4x + 2z = 0$ and $2x + z = 0$, which are equivalent to $z = -2x$; y is free. Thus

$$E_1 = \operatorname{span}\{(1,0,-2)^{\mathsf{T}}, (0,1,0)^{\mathsf{T}}\}.$$

The displayed vectors are already orthogonal, with squared norms 5 and 1. At $\lambda = 6$, the equations give $y = 0$, $x = 2z$, producing a raw vector of squared norm 5. The normalized answer is

λ	orthonormal eigenvectors
1	$(1,0,-2)^{\mathsf{T}}/\sqrt{5}$, $(0,1,0)^{\mathsf{T}}$
6	$(2,0,1)^{\mathsf{T}}/\sqrt{5}$

The remaining scalar product is $(2-2)/5 = 0$, completing the orthonormality check.

6.2.13 $\mathsf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

ANS. $\lambda = 0, 0, 2$.

Solution.

The third row gives the determinant factor $-\lambda$:

$$p(\lambda) = -\lambda\big[(1-\lambda)^2 - 1\big] = -\lambda \cdot \lambda(\lambda - 2) = (2-\lambda)\lambda^2.$$

For $\lambda = 0$, the first two equations both reduce to $x + y = 0$, and z is free. Two orthogonal spanning vectors are $(1, -1, 0)^{\mathsf{T}}$ and $(0, 0, 1)^{\mathsf{T}}$, with lengths $\sqrt{2}$ and 1. For $\lambda = 2$, the third equation forces $z = 0$, and the first gives $y = x$. Thus

λ	orthonormal eigenvectors
0	$(1, -1, 0)^{\mathsf{T}}/\sqrt{2}$, $(0, 0, 1)^{\mathsf{T}}$
2	$(1, 1, 0)^{\mathsf{T}}/\sqrt{2}$

The two vectors in the degenerate zero eigenspace are orthonormal as chosen, and each is perpendicular to the eigenvector for 2.

6.2.14 $\mathsf{A} = \begin{pmatrix} 5 & 0 & \sqrt{3} \\ 0 & 3 & 0 \\ \sqrt{3} & 0 & 3 \end{pmatrix}$.

ANS. $\lambda = 2, 3, 6$.

Solution.

Expansion on the middle coordinate gives

$$\begin{aligned} p(\lambda) &= (3 - \lambda)\big[(5 - \lambda)(3 - \lambda) - 3\big] \\ &= (3 - \lambda)(\lambda^2 - 8\lambda + 12) = (3 - \lambda)(\lambda - 2)(\lambda - 6). \end{aligned}$$

The nontrivial component equations are

$$(5 - \lambda)x + \sqrt{3}\, z = 0, \qquad (3 - \lambda)y = 0, \qquad \sqrt{3}\, x + (3 - \lambda)z = 0.$$

At $\lambda = 2$, $y = 0$ and $z = -\sqrt{3}\, x$. At $\lambda = 3$, the last equation forces $x = 0$, then the first forces $z = 0$, leaving y free. At $\lambda = 6$, $y = 0$ and $x = \sqrt{3}\, z$. The first and third raw vectors have squared norm 4, so

λ	u_λ
2	$(1, 0, -\sqrt{3})^{\mathsf{T}}/2$
3	$(0, 1, 0)^{\mathsf{T}}$
6	$(\sqrt{3}, 0, 1)^{\mathsf{T}}/2$

Their only nontrivial scalar product is $(\sqrt{3} - \sqrt{3})/4 = 0$.

6.2.15 Describe the geometric properties of the surface

$$x^2 + 2xy + 2y^2 + 2yz + z^2 = 1.$$

How is it oriented in 3-D space? Is it a conic section? If so, which kind? It is a quadric surface rather than a planar conic: classify it and its cross sections perpendicular to its axis.

Solution.

The quadratic form is

$$q = \begin{pmatrix} x & y & z \end{pmatrix} \underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}}_{B} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Notice first that $q = (x+y)^2 + (y+z)^2$. It is nonnegative, but it vanishes on the entire line $(x,y,z) = t(1,-1,1)$, not only at the origin. Thus the form is positive semidefinite rather than positive definite. The characteristic determinant, expanded along the first row, is

$$\begin{aligned}\det(B-\lambda I) &= (1-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} - 1\cdot\begin{vmatrix} 1 & 1 \\ 0 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda)\big[(2-\lambda)(1-\lambda)-1\big] - (1-\lambda) \\ &= (1-\lambda)(\lambda^2-3\lambda) = \lambda(1-\lambda)(\lambda-3).\end{aligned}$$

The equations $(B-\lambda I)v = 0$ are

$$(1-\lambda)x + y = 0, \qquad x + (2-\lambda)y + z = 0, \qquad y + (1-\lambda)z = 0.$$

For $\lambda = 0$, they are $x+y=0$, $y+z=0$, and their sum $x+2y+z=0$, giving $(1,-1,1)$. For $\lambda = 1$, the first and third give $y=0$ and then the middle gives $x+z=0$, giving $(1,0,-1)$. For $\lambda = 3$, the first and third give $y = 2x$ and $y = 2z$, so $z = x$ (the middle equation $x - y + z = 0$ then holds), giving $(1,2,1)$. Their pairwise scalar products, $1-1$, $1-2+1$, and $1-1$, vanish. Normalizing these mutually perpendicular vectors gives

$$e_1 = \frac{(1,0,-1)^{\mathsf T}}{\sqrt{2}}, \qquad e_3 = \frac{(1,2,1)^{\mathsf T}}{\sqrt{6}}, \qquad e_0 = \frac{(1,-1,1)^{\mathsf T}}{\sqrt{3}}.$$

Their subscripts label the eigenvalues. Define orthogonal coordinates

$$\xi = e_1^{\mathsf T} r = \frac{x-z}{\sqrt{2}}, \quad \eta = e_3^{\mathsf T} r = \frac{x+2y+z}{\sqrt{6}}, \quad \zeta = e_0^{\mathsf T} r = \frac{x-y+z}{\sqrt{3}}.$$

The inverse relation is $r = \xi e_1 + \eta e_3 + \zeta e_0$. Because $Be_1 = e_1$, $Be_3 = 3e_3$, $Be_0 = 0$, we have $Br = \xi e_1 + 3\eta e_3$, and orthonormality removes all mixed products:

$$q = r^{\mathsf T} B r = (\xi e_1 + \eta e_3 + \zeta e_0)^{\mathsf T}(\xi e_1 + 3\eta e_3) = \xi^2 + 3\eta^2.$$

Hence the surface is

$$\boxed{\xi^2 + 3\eta^2 = 1, \quad \zeta \text{ arbitrary.}}$$

This is an **elliptic cylinder**. Its axis is the line through the origin in direction $(1,-1,1)$; its transverse semiaxes have lengths 1 along e_1 and $1/\sqrt{3}$ along e_3. A parametrization displaying both the elliptical cross section and the unrestricted axis is

$$r(t,s) = e_1\cos t + \frac{e_3}{\sqrt{3}}\sin t + s e_0, \qquad 0 \le t < 2\pi, \quad s \in \mathbb{R}.$$

A surface in three-dimensional space is a quadric here, not itself a planar conic section. Its sections perpendicular to the axis are ellipses. The Instructor's Manual, p. 82, incorrectly calls the form positive definite and describes an ellipsoid or ellipse; the zero eigenvalue and unrestricted ζ prove that the full surface is an elliptic cylinder.

6.4 Hermitian Matrix Diagonalization

6.4.1 Show that the eigenvalues of a matrix are unaltered if the matrix is transformed by a similarity transformation—a transformation that need not be unitary, but of the form given in Eq. (5.78).

This property is not limited to symmetric or Hermitian matrices. It holds for any matrix satisfying an eigenvalue equation of the type $\mathsf{A}\mathbf{x} = \lambda\mathbf{x}$. If our matrix can be brought into diagonal form by a similarity transformation, then two immediate consequences are that:

1. The trace (sum of eigenvalues) is invariant under a similarity transformation.

2. The determinant (product of eigenvalues) is invariant under a similarity transformation.

Note. The invariance of the trace and determinant are often demonstrated by using the Cayley-Hamilton theorem, which states that a matrix satisfies its own characteristic (secular) equation.

Solution.

Let S be invertible and write the similar matrix as $B = S^{-1}AS$. If $Ax = \lambda x$ with $x \neq 0$, set $y = S^{-1}x$. Invertibility ensures $y \neq 0$, and

$$By = S^{-1}AS(S^{-1}x) = S^{-1}Ax = \lambda S^{-1}x = \lambda y.$$

Conversely, since $A = SBS^{-1}$, an eigenvector y of B maps to an eigenvector $x = Sy$ of A, with the same eigenvalue. Algebraic multiplicities are also preserved. Since $S^{-1}(\lambda I)S = \lambda I$ and the determinant of a product is the product of the determinants,

$$\begin{aligned}\det(B - \lambda I) &= \det\big[S^{-1}(A - \lambda I)S\big] \\ &= \det(S^{-1})\det(A - \lambda I)\det S = \det(A - \lambda I),\end{aligned}$$

because $\det(S^{-1})\det S = \det(S^{-1}S) = \det I = 1$. Thus the entire characteristic polynomial is unchanged.

The determinant follows directly by putting $\lambda = 0$: $\det B = (\det S)^{-1}\det A\det S = \det A$. For the trace, an index calculation avoids any diagonalizability assumption:

$$\operatorname{tr} B = \sum_{i,j,k}(S^{-1})_{ij}A_{jk}S_{ki} = \sum_{j,k}A_{jk}\underbrace{\sum_{i}S_{ki}(S^{-1})_{ij}}_{\delta_{kj}} = \sum_{j}A_{jj} = \operatorname{tr} A.$$

These invariances therefore hold even for defective matrices that cannot be diagonalized. When a diagonal form $S^{-1}AS = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ exists, its diagonal entries are the eigenvalues, so $\operatorname{tr} A = \sum_i \lambda_i$ and $\det A = \prod_i \lambda_i$: these are the sum and product interpretations stated in the question.

6.4.2 As a converse of the theorem that Hermitian matrices have real eigenvalues and that eigenvectors corresponding to distinct eigenvalues are orthogonal, show that if

(a) the eigenvalues of a matrix are real and

(b) a complete set of n eigenvectors for the $n \times n$ matrix satisfies $\mathbf{x}_i^\dagger\mathbf{x}_j = \delta_{ij}$,

then the matrix is Hermitian.

Solution.

The orthonormal eigenvectors must form a complete basis: there must be n of them for an $n \times n$ matrix. Let U have those eigenvectors as its columns and set $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. The (i,j) element of $U^\dagger U$ is $x_i^\dagger x_j = \delta_{ij}$, so $U^\dagger U = I$. Since U is square, $\det U^\dagger \det U = \det I = 1$ shows that U is invertible, and then $U^\dagger = (U^\dagger U)U^{-1} = U^{-1}$, so that also $UU^\dagger = I$. Column j of AU is $Ax_j = \lambda_j x_j$, which is column j of UD. Hence

$$U^\dagger U = UU^\dagger = I, \qquad AU = UD.$$

Multiply the second equation by $U^\dagger$ on the right: $A = UDU^\dagger$. Because every λ_i is real, $D^\dagger = D$. Taking the adjoint now gives

$$A^\dagger = (UDU^\dagger)^\dagger = UD^\dagger U^\dagger = UDU^\dagger = A.$$

Hence A is Hermitian.

Completeness is essential. The matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has the real eigenvalue 1 and a normalized eigenvector $(1,0)^\mathsf{T}$, but its eigenspace has only one dimension and the matrix is not Hermitian. A partial orthonormal set does not provide $UU^\dagger = I$.

6.4.3 Show that a real matrix that is not symmetric cannot be diagonalized by an orthogonal or unitary transformation. For unitary diagonalization, add the hypothesis that all eigenvalues are real.

Hint. Assume that the nonsymmetric real matrix can be diagonalized and develop a contradiction.

Solution.

The claim is correct for real orthogonal diagonalization. For unitary diagonalization it is false without the additional hypothesis that the eigenvalues are real.

First suppose a real orthogonal Q diagonalizes the real matrix A: $Q^\mathsf{T} A Q = D$. The matrix D is diagonal, and real because Q and A are real. Multiplying on the left by Q and on the right by Q^T and using $QQ^\mathsf{T} = I$ recovers A; then take its transpose, using $D^\mathsf{T} = D$:

$$A = QDQ^\mathsf{T}, \qquad A^\mathsf{T} = (QDQ^\mathsf{T})^\mathsf{T} = QD^\mathsf{T}Q^\mathsf{T} = QDQ^\mathsf{T} = A.$$

This contradicts the assumed nonsymmetry.

For a unitary transformation, however, the diagonal entries need not be real. A counterexample is

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}.$$

Here $A^\mathsf{T} = -A \neq A$, but the columns $u_+ = (1,-i)^\mathsf{T}/\sqrt{2}$ and $u_- = (1,i)^\mathsf{T}/\sqrt{2}$ of U satisfy

$$Au_\pm = \frac{1}{\sqrt{2}}(\pm i, 1)^\mathsf{T} = \pm i\, u_\pm,$$

and they are orthonormal: $u_\pm^\dagger u_\pm = (1+1)/2 = 1$ and $u_+^\dagger u_- = (1 \cdot 1 + i \cdot i)/2 = 0$. Thus $U^\dagger U = I$ and $AU = U\,\operatorname{diag}(i,-i)$. Consequently $U^\dagger A U = \operatorname{diag}(i,-i)$ is a valid unitary diagonalization of a real nonsymmetric matrix.

If one additionally assumes real eigenvalues, a unitary diagonalization would instead give $A = UDU^\dagger$ with $D^\dagger = D$, implying $A^\dagger = A$; because A is real, this is $A^\mathsf{T} = A$. That proves the corrected unitary version. The distinction between transpose and adjoint prevents the orthogonal proof from being applied to an arbitrary complex unitary matrix.

6.4.4 The matrices representing the angular momentum components L_x, L_y, and L_z are all Hermitian. Show that the eigenvalues of L^2, where $L^2 = L_x^2 + L_y^2 + L_z^2$, are real and nonnegative.

Solution.

For each component $L_\alpha^\dagger = L_\alpha$, so

$$(L_\alpha^2)^\dagger = L_\alpha^\dagger L_\alpha^\dagger = L_\alpha^2.$$

Therefore $L^2 = \sum_{\alpha=x,y,z} L_\alpha^2$ is Hermitian. More explicitly, let $L^2 v = \ell v$ with $v \neq 0$. Multiply by $v^\dagger$ and use the Hermiticity of each component:

$$\begin{aligned} \ell\, v^\dagger v = v^\dagger L^2 v &= \sum_\alpha v^\dagger L_\alpha^2 v \\ &= \sum_\alpha (L_\alpha v)^\dagger (L_\alpha v) = \sum_\alpha \|L_\alpha v\|^2. \end{aligned}$$

Each norm square is real and nonnegative, while $v^\dagger v > 0$. Thus

$$\boxed{\ell = \frac{\|L_x v\|^2 + \|L_y v\|^2 + \|L_z v\|^2}{\|v\|^2} \in [0, \infty).}$$

The value $\ell = 0$ occurs exactly when all three $L_\alpha v$ vanish.

6.4.5 A has eigenvalues λ_i and corresponding eigenvectors $|x_i\rangle$. Show that A^{-1} has the same eigenvectors but with eigenvalues λ_i^{-1}.

Solution.

The notation A^{-1} presupposes that A is invertible. An eigenvalue of an invertible matrix cannot be zero: $Ax_i = 0$ would imply $x_i = A^{-1}0 = 0$, contradicting the definition of eigenvector. Thus $\lambda_i \neq 0$. Multiply $Ax_i = \lambda_i x_i$ by A^{-1}:

$$x_i = \lambda_i A^{-1} x_i, \qquad \boxed{A^{-1} x_i = \lambda_i^{-1} x_i.}$$

Conversely, if $A^{-1} y = \mu y$ with $y \neq 0$, then $\mu \neq 0$ by the same argument applied to the invertible matrix A^{-1}, and multiplying by A gives $y = \mu A y$, that is, $Ay = \mu^{-1} y$. So A and A^{-1} have exactly the same eigenvectors, with reciprocal eigenvalues. No complete eigenbasis is required for this argument.

6.4.6 A square matrix with zero determinant is labeled singular.

(a) If A is singular, show that there is at least one nonzero column vector $\mathbf{v}$ such that

$$\mathsf{A}|\mathbf{v}\rangle = 0.$$

(b) If there is a nonzero vector $|\mathbf{v}\rangle$ such that

$$\mathsf{A}|\mathbf{v}\rangle = 0,$$

show that A is a singular matrix. This means that if a matrix (or operator) has zero as an eigenvalue, the matrix (or operator) has no inverse and its determinant is zero in the finite-dimensional matrix case; a general infinite-dimensional operator need not have a determinant.

Solution.

Write the columns of A as $a_1, \ldots, a_n$.

(a) A zero determinant means that these columns are linearly dependent. Indeed, if they were independent, the n columns would form a basis, so each unit vector could be written as $e_k = \sum_j b_{jk} a_j = A b_k$, with $b_k = (b_{1k}, \ldots, b_{nk})^{\mathsf{T}}$. The matrix B with columns b_k would satisfy $AB = I$, hence $\det A \det B = 1$, contradicting $\det A = 0$. Hence there exist coefficients $v_1, \ldots, v_n$, not all zero, such that

$$v_1 a_1 + \cdots + v_n a_n = 0.$$

For the nonzero column vector $v = (v_1, \ldots, v_n)^{\mathsf{T}}$, the left side is exactly Av. Thus $Av = 0$.

(b) Conversely, if $Av = 0$ with $v \neq 0$, expanding the matrix-vector product gives precisely such a nontrivial linear dependence $\sum_j v_j a_j = 0$ among the columns. Choose r with $v_r \neq 0$. Multiplying column r by v_r multiplies the determinant by v_r, and then adding v_j times column j to column r for every $j \neq r$ does not change it. Column r has become $\sum_j v_j a_j = 0$, so

$$v_r \det A = \det\Big(a_1, \ldots, \sum_j v_j a_j, \ldots, a_n\Big) = \det(a_1, \ldots, 0, \ldots, a_n) = 0.$$

Since $v_r \neq 0$, $\det A = 0$, and A is singular. An inverse is impossible: if A^{-1} existed, $Av = 0$ would give $v = A^{-1}Av = 0$, a contradiction.

The equation $Av = 0v$ is the eigenvalue equation for eigenvalue zero. We have proved the equivalence between a zero eigenvalue, a nontrivial kernel, and zero determinant for a finite square matrix, including matrices that are not diagonalizable. For a general infinite-dimensional operator a determinant need not be defined. A nonzero vector in its kernel still precludes a two-sided inverse; the determinant statement is a finite-dimensional one.

6.4.7 Two Hermitian matrices A and B have the same eigenvalues including multiplicities. Show that A and B are related by a unitary transformation.

Solution.

"The same eigenvalues" must include their algebraic multiplicities. The spectral theorem gives orthonormal eigenbases $a_1, \ldots, a_n$ for A and $b_1, \ldots, b_n$ for B, ordered so that

$$A a_i = \lambda_i a_i, \qquad B b_i = \lambda_i b_i.$$

Define

$$W = \sum_i |b_i\rangle\langle a_i|.$$

Its adjoint is $W^\dagger = \sum_i |a_i\rangle\langle b_i|$. Multiplying and using orthonormality gives

$$W^\dagger W = \sum_{i,j} |a_i\rangle \underbrace{\langle b_i | b_j\rangle}_{\delta_{ij}} \langle a_j| = \sum_i |a_i\rangle\langle a_i| = I,$$

the last step being the completeness relation of the orthonormal basis a_i. In the same way $WW^\dagger = \sum_{i,j} |b_i\rangle\langle a_i|a_j\rangle\langle b_j| = \sum_i |b_i\rangle\langle b_i| = I$. Thus W is unitary. Moreover $W a_i = \sum_j |b_j\rangle\langle a_j|a_i\rangle = b_i$, and likewise $W^\dagger b_i = a_i$. On each basis vector b_i,

$$W A W^\dagger b_i = W A a_i = \lambda_i W a_i = \lambda_i b_i = B b_i.$$

Because the b_i span the space, $\boxed{B = WAW^\dagger}$. Within a degenerate eigenspace any orthonormal basis may be used. If multiplicities were ignored, the statement would fail, for example for $\mathrm{diag}(1,1,2)$ and $\mathrm{diag}(1,2,2)$: both have the two distinct eigenvalues 1 and 2, but their traces, 4 and 5, differ, whereas a unitary transformation $WAW^\dagger = WAW^{-1}$ is a similarity transformation and leaves the trace unchanged (Exercise 6.4.1).

6.4.8 Find the eigenvalues and an orthonormal set of eigenvectors for each of the matrices of Exercise 2.2.12.

Solution.

The matrices in Exercise 2.2.12 (textbook p. 116) are

$$M_x = \frac{1}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}, \qquad M_y = \frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix}, \qquad M_z = \mathrm{diag}(1,0,-1).$$

For $M_x v = \lambda v$, the equations are

$$y/\sqrt{2} = \lambda x, \qquad (x+z)/\sqrt{2} = \lambda y, \qquad y/\sqrt{2} = \lambda z.$$

If $\lambda = 0$, then $y = 0$ and $z = -x$. If $\lambda \neq 0$, the first and third equations give $y = \sqrt{2}\lambda x = \sqrt{2}\lambda z$, so $z = x$ (and $x \neq 0$ for a nonzero vector). Substitution into the middle equation gives $\sqrt{2}\,x = \lambda y = \sqrt{2}\lambda^2 x$, so $\lambda^2 = 1$. The nonzero-eigenvalue vectors have the form $(1, \sqrt{2}\lambda, 1)^\mathsf{T}$, whose squared norm is 4. (Since $\sqrt{2}M_x$ is the matrix of Exercise 6.2.7, these agree with the eigenvectors found there.) Therefore

	$\lambda = 1$	$\lambda = 0$	$\lambda = -1$
M_x	$\dfrac{(1,\sqrt{2},1)^\mathsf{T}}{2}$	$\dfrac{(1,0,-1)^\mathsf{T}}{\sqrt{2}}$	$\dfrac{(1,-\sqrt{2},1)^\mathsf{T}}{2}$

Their pairwise inner products vanish, as follows from $1 - 2 + 1 = 0$ and $1 - 1 = 0$.

For M_y, use the diagonal unitary matrix $D = \mathrm{diag}(d_1,d_2,d_3) = \mathrm{diag}(1,i,-1)$. Since $(DM_xD^\dagger)_{jk} = d_j(M_x)_{jk}d_k^*$, the $(1,2)$ and $(2,3)$ entries of M_x are multiplied by $d_1d_2^* = d_2d_3^* = -i$ and the $(2,1)$ and $(3,2)$ entries by $d_2d_1^* = d_3d_2^* = i$:

$$DM_xD^\dagger = \frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix} = M_y.$$

If $M_xu = \lambda u$, then, because $M_yD = DM_xD^\dagger D = DM_x$, $M_y(Du) = DM_xu = \lambda(Du)$. Since $D^\dagger D = I$, multiplication by D preserves all norms and inner products. Multiplying the components of each eigenvector of M_x by $1, i, -1$ therefore gives an orthonormal answer for M_y:

	$\lambda = 1$	$\lambda = 0$	$\lambda = -1$
M_y	$\dfrac{(1,i\sqrt{2},-1)^\mathsf{T}}{2}$	$\dfrac{(1,0,1)^\mathsf{T}}{\sqrt{2}}$	$\dfrac{(1,-i\sqrt{2},-1)^\mathsf{T}}{2}$

Finally, M_z is already diagonal:

$$\boxed{\lambda = 1:\ (1,0,0)^\mathsf{T}, \qquad \lambda = 0:\ (0,1,0)^\mathsf{T}, \qquad \lambda = -1:\ (0,0,1)^\mathsf{T}.}$$

Every matrix has the spectrum $\{1,0,-1\}$ and the displayed orthonormal eigenbasis.

6.4.9 The unit process in the iterative matrix diagonalization procedure known as the Jacobi method is a unitary transformation that operates on rows/columns i and j of a real symmetric matrix A to make $a_{ij} = a_{ji} = 0$. If this transformation (from basis functions φ_i and φ_j to φ'_i and φ'_j) is written

$$\varphi'_i = \varphi_i \cos\theta - \varphi_j \sin\theta, \quad \varphi'_j = \varphi_i \sin\theta + \varphi_j \cos\theta,$$

(a) Show that a_{ij} is transformed to zero if $\tan 2\theta = \dfrac{2a_{ij}}{a_{jj} - a_{ii}}$ when $a_{jj} \neq a_{ii}$; also give a valid angle when $a_{jj} = a_{ii}$ or $a_{ij} = 0$,

(b) Show that $a_{\mu\nu}$ remains unchanged if neither μ nor ν is i or j,

(c) Find a'_{ii} and a'_{jj} and show that the trace of A is not changed by the transformation,

(d) Find $a'_{i\mu}$ and $a'_{j\mu}$ (where μ is neither i nor j) and show that the sum of the squares of the off-diagonal elements of A is reduced by the amount $2a_{ij}^2$.

Solution.

Put $c = \cos\theta$ and $s = \sin\theta$. Only the two basis vectors i, j are altered; the new matrix elements are $a'_{\mu\nu} = \langle \varphi'_\mu | A | \varphi'_\nu \rangle$. All coefficients are real, and $a_{ji} = a_{ij}$.

(a) Expanding the off-diagonal entry gives

$$\begin{aligned} a'_{ij} &= \langle c\varphi_i - s\varphi_j | A | s\varphi_i + c\varphi_j \rangle \\ &= cs\, a_{ii} + c^2 a_{ij} - s^2 a_{ji} - sc\, a_{jj} \\ &= cs\,(a_{ii} - a_{jj}) + (c^2 - s^2) a_{ij} \\ &= \frac{1}{2}(a_{ii} - a_{jj}) \sin 2\theta + a_{ij} \cos 2\theta, \end{aligned}$$

using $a_{ji} = a_{ij}$, $2cs = \sin 2\theta$, and $c^2 - s^2 = \cos 2\theta$. Setting this to zero gives $\frac{1}{2}(a_{jj} - a_{ii}) \sin 2\theta = a_{ij} \cos 2\theta$. If $a_{jj} \neq a_{ii}$, then $\cos 2\theta \neq 0$ (if $\cos 2\theta = 0$, the equation would force $\sin 2\theta = 0$ as well, which is impossible), and division gives

$$\boxed{\tan 2\theta = \frac{2a_{ij}}{a_{jj} - a_{ii}}}.$$

Conversely, every θ with this tangent satisfies the preceding sine–cosine equation and so makes $a'_{ij} = 0$. The tangent formula needs its limiting cases: if $a_{ii} = a_{jj}$ and $a_{ij} \neq 0$, choose $\theta = \pi/4$; if $a_{ij} = 0$, choose $\theta = 0$ and no elimination is needed. These choices avoid dividing by zero.

(b) If neither μ nor ν is i or j, neither basis vector changes. Thus $a'_{\mu\nu} = \langle \varphi_\mu | A | \varphi_\nu \rangle = a_{\mu\nu}$.

(c) Expand the two diagonal entries:

$$\begin{aligned} a'_{ii} &= \langle c\varphi_i - s\varphi_j | A | c\varphi_i - s\varphi_j \rangle = c^2 a_{ii} + s^2 a_{jj} - 2cs\, a_{ij}, \\ a'_{jj} &= \langle s\varphi_i + c\varphi_j | A | s\varphi_i + c\varphi_j \rangle = s^2 a_{ii} + c^2 a_{jj} + 2cs\, a_{ij}. \end{aligned}$$

Their sum is

$$a'_{ii} + a'_{jj} = (c^2 + s^2)(a_{ii} + a_{jj}) = a_{ii} + a_{jj}.$$

Every other diagonal element is unchanged by (b), so the total trace is unchanged.

(d) For $\mu \neq i, j$, the basis vector φ_μ is unchanged, so

$$a'_{i\mu} = \langle c\varphi_i - s\varphi_j | A | \varphi_\mu \rangle, \qquad a'_{j\mu} = \langle s\varphi_i + c\varphi_j | A | \varphi_\mu \rangle,$$

that is,

$$\boxed{a'_{i\mu} = c\,a_{i\mu} - s\,a_{j\mu}, \qquad a'_{j\mu} = s\,a_{i\mu} + c\,a_{j\mu}.}$$

Squaring and adding exhibits the cancellation of cross terms:

$$\begin{aligned}(a'_{i\mu})^2 + (a'_{j\mu})^2 &= c^2 a_{i\mu}^2 + s^2 a_{j\mu}^2 - 2cs\,a_{i\mu}a_{j\mu} \\ &\quad + s^2 a_{i\mu}^2 + c^2 a_{j\mu}^2 + 2cs\,a_{i\mu}a_{j\mu} \\ &= a_{i\mu}^2 + a_{j\mu}^2.\end{aligned}$$

Since A' is again symmetric, the transposed entries $a'_{\mu i}, a'_{\mu j}$ give the same result. Entries with neither index i, j are unchanged by (b), and with θ chosen as in (a) the pair itself becomes $a'_{ij} = a'_{ji} = 0$. Thus, in the sum $S = \sum_{\mu\neq\nu} a_{\mu\nu}^2$, the only net change is the removal of a_{ij} and a_{ji} themselves:

$$\boxed{S' = S - a_{ij}^2 - a_{ji}^2 = S - 2a_{ij}^2.}$$

Both members of each symmetric off-diagonal pair are included in S, which accounts for the factor 2.

6.5 Normal Matrices

6.5.1 Find the eigenvalues and corresponding eigenvectors for

$$\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}.$$

Note that the eigenvectors are not orthogonal.

ANS. $\lambda_1 = 0,\ \mathbf{c}_1 = (2,-1); \quad \lambda_2 = 4,\ \mathbf{c}_2 = (2,1).$

Solution.

The characteristic determinant is

$$\det(A - \lambda I) = (2-\lambda)^2 - 4 = \lambda^2 - 4\lambda = \lambda(\lambda - 4).$$

For $\lambda = 0$, the first row gives $2x + 4y = 0$, or $x = -2y$; $(2,-1)^{\mathsf{T}}$ is a nonzero solution. For $\lambda = 4$, the first row gives $-2x + 4y = 0$, or $x = 2y$; choose $(2,1)^{\mathsf{T}}$. In each case the second row is the same equation up to a constant factor. Normalizing the vectors, whose squared norms are 5, gives

$$\boxed{\lambda_0 = 0, \quad u_0 = \frac{(2,-1)^{\mathsf{T}}}{\sqrt{5}}; \qquad \lambda_4 = 4, \quad u_4 = \frac{(2,1)^{\mathsf{T}}}{\sqrt{5}}.}$$

Their inner product is $(4-1)/5 = 3/5 \neq 0$, so they are not orthogonal. This causes no contradiction: the matrix is not Hermitian, and in fact

$$AA^{\mathsf{T}} = \begin{pmatrix} 20 & 10 \\ 10 & 5 \end{pmatrix}, \qquad A^{\mathsf{T}}A = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}$$

show that it is not normal either.

6.5.2 If A is a 2×2 matrix, show that its eigenvalues λ satisfy the secular equation

$$\lambda^2 - \lambda \,\mathrm{trace}(\mathsf{A}) + \det(\mathsf{A}) = 0.$$

Solution.

Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. An eigenvalue makes $A - \lambda I$ singular, so

$$\begin{aligned} 0 = \det(A - \lambda I) &= (a - \lambda)(d - \lambda) - bc \\ &= ad - (a + d)\lambda + \lambda^2 - bc \\ &= \lambda^2 - \lambda \,\mathrm{tr}\, A + \det A. \end{aligned}$$

This proves the formula for any real or complex 2×2 matrix; no symmetry or diagonalizability is needed. It also shows that the two roots, counted with multiplicity, have sum $a + d$ and product $ad - bc$.

6.5.3 Assuming a unitary matrix U to satisfy an eigenvalue equation $\mathsf{U}\mathbf{c} = \lambda\mathbf{c}$, show that the eigenvalues of the unitary matrix have unit magnitude. This same result holds for real orthogonal matrices.

Solution.

Let $Uc = \lambda c$ with $c \neq 0$. Unitarity means $U^\dagger U = I$, so the squared norm is preserved:

$$c^\dagger c = c^\dagger U^\dagger U c = (Uc)^\dagger (Uc) = (\lambda c)^\dagger (\lambda c) = \lambda^* \lambda \, c^\dagger c = |\lambda|^2 c^\dagger c.$$

Divide by $c^\dagger c > 0$ to obtain

$$\boxed{|\lambda|^2 = 1, \qquad |\lambda| = 1.}$$

For a real orthogonal matrix Q, $Q^\dagger = Q^{\mathsf{T}}$ and $Q^{\mathsf{T}} Q = I$, so the identical argument applies, allowing complex eigenvectors and eigenvalues even though the entries of Q are real.

6.5.4 Since an orthogonal matrix describing a rotation in real 3-D space is a special case of a unitary matrix, such an orthogonal matrix can be diagonalized by a unitary transformation.

(a) Show that the sum of the three eigenvalues is $1 + 2\cos\varphi$, where φ is the net angle of rotation about a single fixed axis.

(b) Given that one eigenvalue is 1, show that the other two eigenvalues must be $e^{i\varphi}$ and $e^{-i\varphi}$.

Our orthogonal rotation matrix (real elements) has generally complex eigenvalues.

Solution.

(a) Let R be the rotation matrix, and choose an orthonormal basis whose third axis is the fixed rotation axis. In this basis the matrix is, up to the sign convention for the angle,

$$R_0 = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

A change of orthonormal basis is a similarity, so the trace is the same in the original coordinates (Exercise 6.4.1). The unitary diagonalization $U^\dagger R U = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$ is also a similarity, so the trace equals the sum of the eigenvalues. Thus

$$\boxed{\lambda_1 + \lambda_2 + \lambda_3 = \mathrm{tr}\, R_0 = 1 + 2\cos\varphi.}$$

(b) Take $\lambda_1 = 1$, corresponding to the rotation axis. A proper rotation has determinant 1, and by the same similarity argument the determinant is the product $\lambda_1\lambda_2\lambda_3$ of the eigenvalues, so $\lambda_2\lambda_3 = 1$; part (a) gives $\lambda_2 + \lambda_3 = 2\cos\varphi$. Therefore λ_2 and λ_3 are the roots of

$$(\lambda - \lambda_2)(\lambda - \lambda_3) = \lambda^2 - (\lambda_2 + \lambda_3)\lambda + \lambda_2\lambda_3 = \lambda^2 - 2\cos\varphi\,\lambda + 1 = 0.$$

Equivalently the planar block has characteristic determinant $(\cos\varphi - \lambda)^2 + \sin^2\varphi$. By the quadratic formula, since $\cos^2\varphi - 1 = -\sin^2\varphi$, the roots are $\lambda = \cos\varphi \pm \sqrt{\cos^2\varphi - 1} = \cos\varphi \pm i\sin\varphi$:

$$\boxed{\lambda_2 = \cos\varphi + i\sin\varphi = e^{i\varphi}, \qquad \lambda_3 = \cos\varphi - i\sin\varphi = e^{-i\varphi}.}$$

The unordered pair is unchanged if the passive-rotation convention reverses φ. These two eigenvalues are generally nonreal, but they are real when φ is an integer multiple of π: both are 1 for a full turn and both are -1 for a half-turn.

6.5.5 A is an nth-order Hermitian matrix with orthonormal eigenvectors $|x_i\rangle$ and real eigenvalues $\lambda_1 \le \lambda_2 \le \lambda_3 \le \cdots \le \lambda_n$. Show that for a unit magnitude vector $|y\rangle$,

$$\lambda_1 \le \langle y|\mathsf{A}|y\rangle \le \lambda_n.$$

Solution.

Expand the unit vector in the complete orthonormal eigenbasis:

$$|y\rangle = \sum_i c_i|x_i\rangle, \qquad c_i = \langle x_i|y\rangle, \qquad 1 = \langle y|y\rangle = \sum_i |c_i|^2.$$

Then

$$\begin{aligned}\langle y|A|y\rangle &= \sum_{i,j} c_i^* c_j \langle x_i|A|x_j\rangle \\ &= \sum_{i,j} c_i^* c_j \lambda_j \delta_{ij} = \sum_i \lambda_i |c_i|^2.\end{aligned}$$

Each weight $|c_i|^2$ is nonnegative. Since $\lambda_1 \le \lambda_i \le \lambda_n$, multiplication by these weights and summation give

$$\lambda_1 \sum_i |c_i|^2 \le \sum_i \lambda_i |c_i|^2 \le \lambda_n \sum_i |c_i|^2.$$

The two outer sums equal one, proving the stated bounds. Equality at an endpoint occurs precisely when y has components only in the eigenspace of that endpoint eigenvalue. Indeed, $\sum_i |c_i|^2 = 1$ gives

$$\langle y|A|y\rangle - \lambda_1 = \sum_i (\lambda_i - \lambda_1)|c_i|^2, \qquad \lambda_n - \langle y|A|y\rangle = \sum_i (\lambda_n - \lambda_i)|c_i|^2,$$

sums of nonnegative terms. The first vanishes exactly when $c_i = 0$ for every i with $\lambda_i > \lambda_1$, the second exactly when $c_i = 0$ for every i with $\lambda_i < \lambda_n$.

6.5.6 A particular matrix is both Hermitian and unitary. Show that its eigenvalues are all ± 1.

Note. The Pauli and Dirac matrices are specific examples.

Solution.

For a Hermitian matrix each eigenvalue is real. For a unitary matrix Exercise 6.5.3 gives $|\lambda| = 1$. Combining these facts yields $\lambda = 1$ or $\lambda = -1$. One may also use both matrix properties directly: $A^\dagger = A$ and $A^\dagger A = I$ imply $A^2 = A^\dagger A = I$. Thus for $Av = \lambda v$,

$$v = A^2 v = \lambda^2 v.$$

Since $v \neq 0$, $\lambda^2 = 1$, again giving $\boxed{\lambda \in \{-1, 1\}}$. Either sign may occur with any allowed multiplicity; a particular matrix need not contain both signs.

6.5.7 For his relativistic electron theory Dirac required a set of four anticommuting matrices. Assume that these matrices are to be Hermitian and unitary. If these are $n \times n$ matrices, show that n must be even. With 2×2 matrices inadequate (why?), this demonstrates that the smallest possible matrices forming a set of four anticommuting, Hermitian, unitary matrices are 4×4.

Solution.

Write the four matrices as Γ_a, with $\Gamma_a \Gamma_b = -\Gamma_b \Gamma_a$ when $a \neq b$. Unitarity makes every determinant nonzero: $|\det \Gamma_a|^2 = \det(\Gamma_a^\dagger \Gamma_a) = \det I = 1$. Taking determinants of an anticommutation relation, and using $\det(-M) = (-1)^n \det M$ (each of the n rows changes sign), gives

$$\det \Gamma_a \det \Gamma_b = \det(-\Gamma_b \Gamma_a) = (-1)^n \det \Gamma_b \det \Gamma_a.$$

Cancel the nonzero determinants. Then $(-1)^n = 1$, so n must be even.

We next exclude $n = 2$, not merely odd n. Hermiticity and unitarity give $\Gamma_a^2 = \Gamma_a^\dagger \Gamma_a = I$ and, by Exercise 6.5.6, eigenvalues ± 1. The first matrix cannot be I or $-I$, because anticommuting with either would give $\pm 2\Gamma_b = 0$ for every other matrix. Being Hermitian, Γ_1 is diagonalized by a unitary V: $V^\dagger \Gamma_1 V = \mathrm{diag}(\epsilon_1, \epsilon_2)$ with $\epsilon_1, \epsilon_2 \in \{1, -1\}$. If $\epsilon_1 = \epsilon_2 = \epsilon$, then $\Gamma_1 = V(\epsilon I)V^\dagger = \epsilon I$, which was just excluded. So Γ_1 has the eigenvalues 1 and -1, and, after ordering the columns of V, $V^\dagger \Gamma_1 V = \mathrm{diag}(1, -1)$. Replacing every Γ_a by $V^\dagger \Gamma_a V$ preserves Hermiticity, unitarity, and all anticommutation relations, so we may take

$$\Gamma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Write another Hermitian matrix as $B = \begin{pmatrix} a & z \\ z^* & d \end{pmatrix}$, with a, d real. Its anticommutator with Γ_1 is

$$\Gamma_1 B + B\Gamma_1 = \begin{pmatrix} a & z \\ -z^* & -d \end{pmatrix} + \begin{pmatrix} a & -z \\ z^* & -d \end{pmatrix} = \begin{pmatrix} 2a & 0 \\ 0 & -2d \end{pmatrix}.$$

Therefore every remaining matrix must have the form

$$B(z) = \begin{pmatrix} 0 & z \\ z^* & 0 \end{pmatrix}, \qquad B(z)^2 = \begin{pmatrix} zz^* & 0 \\ 0 & z^* z \end{pmatrix} = |z|^2 I,$$

and unitarity, $B(z)^\dagger B(z) = B(z)^2 = I$, forces $|z| = 1$. For two such matrices,

$$B(z)B(w) + B(w)B(z) = \begin{pmatrix} zw^* + wz^* & 0 \\ 0 & z^* w + w^* z \end{pmatrix} = 2\Re(zw^*) I.$$

Writing $z = x_1 + iy_1$ and $w = x_2 + iy_2$ gives $\Re(zw^*) = x_1x_2 + y_1y_2$, the scalar product of the real vectors (x_1, y_1) and (x_2, y_2). Thus the three remaining matrices would require three nonzero, mutually orthogonal real vectors (x, y) in a two-dimensional space. After choosing two independent orthogonal vectors, a vector orthogonal to both is zero; this contradicts $|z| = 1$. Hence four such 2×2 matrices cannot exist.

To prove that 4×4 matrices do exist, define the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each is Hermitian and has square I_2, and direct multiplication gives

$$\sigma_x\sigma_y = i\sigma_z = -\sigma_y\sigma_x, \quad \sigma_y\sigma_z = i\sigma_x = -\sigma_z\sigma_y, \quad \sigma_z\sigma_x = i\sigma_y = -\sigma_x\sigma_z,$$

so each anticommutes with either of the other two. Use the four block matrices

$$\Gamma_1 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad \Gamma_2 = \begin{pmatrix} 0 & -iI_2 \\ iI_2 & 0 \end{pmatrix}, \quad \Gamma_3 = \begin{pmatrix} \sigma_x & 0 \\ 0 & -\sigma_x \end{pmatrix}, \quad \Gamma_4 = \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix}.$$

They are Hermitian and their squares are I_4, so they are also unitary: $\Gamma_a^\dagger\Gamma_a = \Gamma_a^2 = I_4$. Equivalently, as direct products, $\Gamma_1 = \sigma_x \otimes I_2$, $\Gamma_2 = \sigma_y \otimes I_2$, $\Gamma_3 = \sigma_z \otimes \sigma_x$, and $\Gamma_4 = \sigma_z \otimes \sigma_y$. By the rule $(P \otimes Q)(P' \otimes Q') = PP' \otimes QQ'$,

$$\begin{aligned} \{\Gamma_1, \Gamma_2\} &= \{\sigma_x, \sigma_y\} \otimes I_2 = 0, \\ \{\sigma_\alpha \otimes I_2, \sigma_z \otimes \sigma_\beta\} &= \{\sigma_\alpha, \sigma_z\} \otimes \sigma_\beta = 0 \qquad (\alpha, \beta \in \{x, y\}), \\ \{\Gamma_3, \Gamma_4\} &= \sigma_z^2 \otimes \{\sigma_x, \sigma_y\} = 0. \end{aligned}$$

The middle line covers the four pairs formed by Γ_1 or Γ_2 with Γ_3 or Γ_4, so all six pairs anticommute. Therefore the smallest possible size is

$$\boxed{n = 4.}$$

6.5.8 A is a normal matrix with eigenvalues λ_n and orthonormal eigenvectors $|x_n\rangle$. Show that A may be written as

$$\mathsf{A} = \sum_n \lambda_n |x_n\rangle\langle x_n|.$$

Hint. Show that both this eigenvector form of A and the original A give the same result acting on an arbitrary vector $|y\rangle$.

Solution.

Let $B = \sum_n \lambda_n |x_n\rangle\langle x_n|$ be the proposed expression. Because A is normal, it is diagonalized by a unitary matrix (Section 6.5), whose columns are orthonormal eigenvectors, one for each eigenvalue counted with its multiplicity. So there are as many orthonormal $|x_n\rangle$ as the dimension of the space, and they form a complete basis. Expand an arbitrary vector as

$$|y\rangle = \sum_j c_j |x_j\rangle, \qquad c_j = \langle x_j | y\rangle.$$

The original matrix acts as

$$A|y\rangle = \sum_j c_j A|x_j\rangle = \sum_j c_j \lambda_j |x_j\rangle.$$

The proposed matrix acts as

$$B|y\rangle = \sum_{n,j} \lambda_n c_j |x_n\rangle \underbrace{\langle x_n|x_j\rangle}_{\delta_{nj}}$$
$$= \sum_j c_j \lambda_j |x_j\rangle = A|y\rangle.$$

Since this holds for every y, $A = B$, proving

$$\boxed{A = \sum_n \lambda_n |x_n\rangle\langle x_n|.}$$

Repeated eigenvalues cause no difficulty; any orthonormal basis in each degenerate eigenspace gives the same operator.

6.5.9 A has eigenvalues 1 and -1 and corresponding eigenvectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$. Construct A.

ANS. $\mathsf{A} = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$.

Solution.

Let $e_1 = (1,0)^{\mathsf{T}}$ and $e_2 = (0,1)^{\mathsf{T}}$. The product Ae_j is the jth column of A. Thus $Ae_1 = e_1$ fixes the first column as $(1,0)^{\mathsf{T}}$, while $Ae_2 = -e_2$ fixes the second as $(0,-1)^{\mathsf{T}}$. Therefore

$$\boxed{A = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.}$$

The spectral formula of Exercise 6.5.8 gives the same result explicitly:

$$A = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2| = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} - \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

6.5.10 A non-Hermitian matrix A has eigenvalues λ_i and corresponding eigenvectors $|u_i\rangle$. The adjoint matrix $\mathsf{A}^\dagger$ has the conjugate eigenvalues λ_i^*, with corresponding eigenvectors $|v_i\rangle$. Show that the eigenvectors form a biorthogonal set in the sense that

$$\langle v_i|u_j\rangle = 0 \quad \text{for } \lambda_i \neq \lambda_j.$$

Solution.

In general the adjoint has the conjugate eigenvalues, not the same eigenvalues. With the consistent pairing $A^\dagger v_i = \lambda_i^* v_i$, the condition for the displayed orthogonality is $\lambda_i \neq \lambda_j$. The conjugation of the spectrum follows from $A^\dagger - \mu I = (A - \overline{\mu} I)^\dagger$, since transposition does not change a determinant and conjugating every entry conjugates it:

$$\det(A^\dagger - \mu I) = \overline{\det(A - \overline{\mu} I)};$$

its zeros are the complex conjugates of the zeros of the original characteristic polynomial, including multiplicities.

Using the corrected pairing, take the adjoint of the second eigenvalue equation, with $(A^\dagger v_i)^\dagger = v_i^\dagger A$ and $(\lambda_i^* v_i)^\dagger = \lambda_i v_i^\dagger$:

$$Au_j = \lambda_j u_j, \qquad A^\dagger v_i = \lambda_i^* v_i \quad \Longrightarrow \quad \langle v_i|A = \lambda_i \langle v_i|.$$

Evaluate the same scalar in these two ways:

$$\langle v_i|A|u_j\rangle = \lambda_j \langle v_i|u_j\rangle = \lambda_i \langle v_i|u_j\rangle.$$

Subtracting gives

$$(\lambda_j - \lambda_i)\langle v_i|u_j\rangle = 0, \qquad \boxed{\langle v_i|u_j\rangle = 0 \quad \text{if } \lambda_i \neq \lambda_j.}$$

If one instead denotes the eigenvalue of $A^\dagger$ by μ_i, without pairing it in advance, the same calculation yields $(\lambda_j - \mu_i^*)\langle v_i|u_j\rangle = 0$. This is the textbook's condition $\lambda_i^* \neq \lambda_j$, read with λ_i denoting the eigenvalue of $A^\dagger$ that belongs to v_i. No orthogonality is forced when the two paired eigenvalues coincide, and the argument does not assume that a non-Hermitian matrix has a complete eigenbasis.

6.5.11 You are given a pair of equations:

$$\mathsf{A}|f_n\rangle = \lambda_n |g_n\rangle$$

$$\tilde{\mathsf{A}}|g_n\rangle = \lambda_n |f_n\rangle \quad \text{with } \mathsf{A} \text{ real.}$$

(a) Prove that $|f_n\rangle$ is an eigenvector of $(\tilde{\mathsf{A}}\mathsf{A})$ with eigenvalue λ_n^2.

(b) Prove that $|g_n\rangle$ is an eigenvector of $(\mathsf{A}\tilde{\mathsf{A}})$ with eigenvalue λ_n^2.

(c) State how you know that

(1) The $|f_n\rangle$ can be chosen to form an orthogonal set.

(2) The $|g_n\rangle$ can be chosen to form an orthogonal set.

(3) λ_n^2 is real.

Solution.

Here $\widetilde{A} = A^{\mathsf{T}} = A^\dagger$ because A is real. Take the indicated f_n, g_n to be nonzero.

(a) Apply A^{T} to the first equation:

$$A^{\mathsf{T}} A f_n = \lambda_n A^{\mathsf{T}} g_n = \lambda_n(\lambda_n f_n) = \boxed{\lambda_n^2 f_n}.$$

(b) Apply A to the second equation:

$$A A^{\mathsf{T}} g_n = \lambda_n A f_n = \lambda_n(\lambda_n g_n) = \boxed{\lambda_n^2 g_n}.$$

(c) Both products are real symmetric:

$$(A^{\mathsf{T}}A)^{\mathsf{T}} = A^{\mathsf{T}}A, \qquad (AA^{\mathsf{T}})^{\mathsf{T}} = AA^{\mathsf{T}}.$$

They are also positive semidefinite. In particular,

$$\lambda_n^2 = \frac{f_n^\dagger A^{\mathsf{T}} A f_n}{f_n^\dagger f_n} = \frac{\|Af_n\|^2}{\|f_n\|^2} \geq 0,$$

so the eigenvalue λ_n^2 is real and nonnegative. A number whose square is real and nonnegative is itself real, so λ_n is real; replacing g_n by $-g_n$ reverses its sign, and its nonnegative choice is the singular value.

For distinct squared eigenvalues, symmetry proves orthogonality:

$$\lambda_j^2 f_i^\dagger f_j = f_i^\dagger (A^\mathsf{T} A) f_j = ((A^\mathsf{T} A) f_i)^\dagger f_j = \lambda_i^2 f_i^\dagger f_j,$$

where the middle step uses $(A^\mathsf{T} A)^\dagger = A^\mathsf{T} A$ and the last uses the reality of λ_i^2. Thus $f_i^\dagger f_j = 0$ when $\lambda_i^2 \neq \lambda_j^2$. The identical argument with AA^T proves the assertion for g_i, g_j.

Eigenvectors in a repeated eigenspace are not automatically orthogonal; the correct assertion is that they can be chosen orthonormal. For example, if $A = I_2$, the unit vectors $f_1 = g_1 = (1,0)^\mathsf{T}$ and $f_2 = g_2 = (1,1)^\mathsf{T}/\sqrt{2}$ satisfy both paired equations with $\lambda_1 = \lambda_2 = 1$, but their inner product is $1/\sqrt{2}$.

To make a consistent orthonormal choice, first orthonormalize the f_n in each eigenspace of $A^\mathsf{T} A$. For a nonzero singular value $\sigma_n = |\lambda_n|$, set $g_n = Af_n/\sigma_n$. Then

$$g_i^\dagger g_j = \frac{f_i^\dagger A^\mathsf{T} A f_j}{\sigma_i \sigma_j} = \frac{\sigma_j^2}{\sigma_i \sigma_j} \delta_{ij} = \delta_{ij}, \qquad A^\mathsf{T} g_n = \frac{A^\mathsf{T} A f_n}{\sigma_n} = \frac{\sigma_n^2 f_n}{\sigma_n} = \sigma_n f_n,$$

so both pair equations hold with $\lambda_n = \sigma_n$. This also handles repeated positive singular values. For zero singular values, note that $\ker(A^\mathsf{T} A) = \ker A$, because $A^\mathsf{T} Af = 0$ implies $\|Af\|^2 = f^\dagger A^\mathsf{T} Af = 0$; likewise $\ker(AA^\mathsf{T}) = \ker A^\mathsf{T}$. Choose orthonormal vectors f_n in $\ker A$ and g_n in $\ker A^\mathsf{T}$; both pair equations then hold with $\lambda_n = 0$. For a square $N \times N$ matrix these kernels have the same dimension, $N - \operatorname{rank} A = N - \operatorname{rank} A^\mathsf{T}$, by the rank–nullity theorem ($\dim \ker M = N - \operatorname{rank} M$ for an $N \times N$ matrix M) and the equality of row and column rank ($\operatorname{rank} A^\mathsf{T} = \operatorname{rank} A$), so they can be paired. These f_n are orthogonal to the earlier ones, which belong to different eigenvalues of $A^\mathsf{T} A$; these g_n are orthogonal to the earlier ones $g_m = Af_m/\sigma_m$, since $h \in \ker A^\mathsf{T}$ gives $h^\dagger Af_m = (A^\mathsf{T} h)^\dagger f_m = 0$.

Both families are complete. Because $A^\mathsf{T} A$ is real symmetric, its eigenspaces together span the whole space, so there are N orthonormal f_n. Those with $\sigma_n = 0$ form a basis of $\ker(A^\mathsf{T} A) = \ker A$, so $N - \dim \ker A = \operatorname{rank} A$ of them have $\sigma_n > 0$ and give the $g_m = Af_m/\sigma_m$. Together with the $N - \operatorname{rank} A$ vectors g_n in $\ker A^\mathsf{T}$, there are also N orthonormal, hence linearly independent, g_n, which therefore form a basis. These are the complete orthonormal families used in the following exercises.

6.5.12 Prove that A of the preceding exercise may be written as

$$\mathsf{A} = \sum_n \lambda_n |g_n\rangle\langle f_n|,$$

with complete orthonormal families $|g_n\rangle$ and $|f_n\rangle$ chosen as in Exercise 6.5.11.

Hint. Expand an arbitrary vector as a linear combination of $|f_n\rangle$.

Solution.

The right vectors f_n must form a complete orthonormal basis, as chosen in Exercise 6.5.11; unit norms alone are insufficient. Expand an arbitrary vector using that basis:

$$|y\rangle = \sum_n c_n |f_n\rangle, \qquad c_n = \langle f_n | y\rangle;$$

the coefficient formula follows by applying $\langle f_n|$ to the expansion and using $\langle f_n | f_j\rangle = \delta_{nj}$. By linearity and the paired equation $A|f_n\rangle = \lambda_n |g_n\rangle$,

$$A|y\rangle = \sum_n c_n A|f_n\rangle = \sum_n c_n \lambda_n |g_n\rangle.$$

For the proposed expression, orthonormality gives

$$\left(\sum_n \lambda_n |g_n\rangle\langle f_n|\right)|y\rangle = \sum_{n,j} \lambda_n c_j |g_n\rangle \underbrace{\langle f_n|f_j\rangle}_{\delta_{nj}} = \sum_j c_j \lambda_j |g_j\rangle = A|y\rangle.$$

Agreement on every vector proves

$$\boxed{A = \sum_n \lambda_n |g_n\rangle\langle f_n|.}$$

Terms with $\lambda_n = 0$ vanish, but including the corresponding right vectors makes the expansion of an arbitrary y complete. For nonzero λ_n, which is real by Exercise 6.5.11(c), so that $|\lambda_n|^2 = \lambda_n^2$, a unit right vector has a unit left vector when the paired equations hold: $g_n = Af_n/\lambda_n$, $(Af_n)^\dagger = f_n^\dagger A^\mathsf{T}$ for real A, and $A^\mathsf{T}Af_n = \lambda_n^2 f_n$ by Exercise 6.5.11(a), so

$$\|g_n\|^2 = \frac{(Af_n)^\dagger Af_n}{\lambda_n^2} = \frac{f_n^\dagger A^\mathsf{T} Af_n}{\lambda_n^2} = \frac{\lambda_n^2 f_n^\dagger f_n}{\lambda_n^2} = \|f_n\|^2 = 1.$$

The nonnegative choices of λ_n give the usual singular-value decomposition; a negative real factor may instead be absorbed by reversing the sign of its g_n.

The unit-normalized but nonorthogonal example in Exercise 6.5.11 shows why the qualification matters. There $A = I_2$, $f_1 = g_1 = (1,0)^\mathsf{T}$, $f_2 = g_2 = (1,1)^\mathsf{T}/\sqrt{2}$ and $\lambda_1 = \lambda_2 = 1$, so the formula would give

$$|f_1\rangle\langle f_1| + |f_2\rangle\langle f_2| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \neq I_2,$$

even though both paired factors are 1 and the actual matrix is I_2.

6.5.13 Given

$$\mathsf{A} = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 2 \\ 1 & -4 \end{pmatrix},$$

(a) Construct the transpose $\tilde{\mathsf{A}}$ and the symmetric forms $\tilde{\mathsf{A}}\mathsf{A}$ and $\mathsf{A}\tilde{\mathsf{A}}$.

(b) From $\mathsf{A}\tilde{\mathsf{A}}|g_n\rangle = \lambda_n^2|g_n\rangle$, find λ_n and $|g_n\rangle$. Normalize the $|g_n\rangle$.

(c) From $\tilde{\mathsf{A}}\mathsf{A}|f_n\rangle = \lambda_n^2|f_n\rangle$, find λ_n [same as (b)] and $|f_n\rangle$. Normalize the $|f_n\rangle$.

(d) Verify that $\mathsf{A}|f_n\rangle = \lambda_n|g_n\rangle$ and $\tilde{\mathsf{A}}|g_n\rangle = \lambda_n|f_n\rangle$.

(e) Verify that $\mathsf{A} = \sum_n \lambda_n |g_n\rangle\langle f_n|$.

Solution.

The right side in part (c) must contain f_n, not g_n, because by Exercise 6.5.11(a) the f_n are the eigenvectors of $\tilde{\mathsf{A}}\mathsf{A}$. The correction is marked in the question.

(a) The matrix and its transpose are

$$A = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 2 \\ 1 & -4 \end{pmatrix}, \qquad A^\mathsf{T} = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 1 \\ 2 & -4 \end{pmatrix}.$$

Multiply in both orders, retaining the individual products:

$$A^{\mathsf{T}}A = \frac{1}{5}\begin{pmatrix} 2\cdot 2+1\cdot 1 & 2\cdot 2+1\cdot(-4) \\ 2\cdot 2+(-4)\cdot 1 & 2\cdot 2+(-4)(-4)\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & 4\end{pmatrix},$$
$$AA^{\mathsf{T}} = \frac{1}{5}\begin{pmatrix} 2\cdot 2+2\cdot 2 & 2\cdot 1+2\cdot(-4) \\ 1\cdot 2+(-4)\cdot 2 & 1\cdot 1+(-4)(-4)\end{pmatrix} = \frac{1}{5}\begin{pmatrix}8 & -6\\ -6 & 17\end{pmatrix}.$$

(b) Write $\mu = \lambda^2$. The characteristic determinant for AA^{T} is

$$\det(AA^{\mathsf{T}} - \mu I) = \frac{1}{25}\begin{vmatrix}8-5\mu & -6\\ -6 & 17-5\mu\end{vmatrix} = \frac{(8-5\mu)(17-5\mu)-36}{25}$$
$$= \frac{25\mu^2 - 125\mu + 100}{25} = \mu^2 - 5\mu + 4 = (\mu-1)(\mu-4).$$

Thus the nonnegative singular values are $\lambda_1 = 1$, $\lambda_2 = 2$. For $g = (g_x, g_y)^{\mathsf{T}}$, the first row of $5(AA^{\mathsf{T}} - \mu I)g = 0$ reads $(8-5\mu)g_x - 6g_y = 0$. For $\mu = 1$ it gives $3g_x - 6g_y = 0$, so $g_x = 2g_y$. For $\mu = 4$ it gives $-12g_x - 6g_y = 0$, so $g_y = -2g_x$. Taking $g_y = 1$ in the first case and $g_x = 1$ in the second, the raw vectors $(2,1)^{\mathsf{T}}$ and $(1,-2)^{\mathsf{T}}$ have squared length 5 and scalar product $2-2=0$, giving

$$g_1 = \frac{(2,1)^{\mathsf{T}}}{\sqrt{5}}, \qquad g_2 = \frac{(1,-2)^{\mathsf{T}}}{\sqrt{5}}.$$

(c) The matrix $A^{\mathsf{T}}A = \mathrm{diag}(1,4)$ is already diagonal. Its eigenvalues $\mu = 1, 4$ are those of (b), so again $\lambda_1 = 1$, $\lambda_2 = 2$, and the matching orthonormal right vectors are the coordinate vectors

$$f_1 = (1,0)^{\mathsf{T}}, \qquad f_2 = (0,1)^{\mathsf{T}}.$$

(d) The relative signs of left and right vectors are confirmed directly:

$$Af_1 = \frac{1}{\sqrt{5}}\begin{pmatrix}2\\1\end{pmatrix} = g_1, \qquad Af_2 = \frac{1}{\sqrt{5}}\begin{pmatrix}2\\-4\end{pmatrix} = 2g_2,$$
$$A^{\mathsf{T}}g_1 = \frac{1}{5}\begin{pmatrix}2\cdot 2+1\cdot 1\\ 2\cdot 2-4\cdot 1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} = f_1, \qquad A^{\mathsf{T}}g_2 = \frac{1}{5}\begin{pmatrix}2\cdot 1+1\cdot(-2)\\ 2\cdot 1-4\cdot(-2)\end{pmatrix} = \begin{pmatrix}0\\2\end{pmatrix} = 2f_2.$$

(e) Finally, form the two outer products. Since f_n is the nth coordinate vector, $g_n f_n^{\mathsf{T}}$ has the column g_n in column n and zeros elsewhere:

$$\lambda_1 g_1 f_1^{\mathsf{T}} + \lambda_2 g_2 f_2^{\mathsf{T}} = \frac{1}{\sqrt{5}}\begin{pmatrix}2 & 0\\ 1 & 0\end{pmatrix} + \frac{1}{\sqrt{5}}\begin{pmatrix}0 & 2\\ 0 & -4\end{pmatrix} = \boxed{\frac{1}{\sqrt{5}}\begin{pmatrix}2 & 2\\ 1 & -4\end{pmatrix} = A.}$$

6.5.14 Given the factors in the paired equations $\lambda_1 = 1$, $\lambda_2 = -1$ and the corresponding right and left vectors

$$|f_1\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |g_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \quad |f_2\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \quad \text{and} \quad |g_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix},$$

(a) construct A;

(b) verify that $\mathsf{A}|f_n\rangle = \lambda_n|g_n\rangle$;

(c) verify that $\tilde{\mathsf{A}}|g_n\rangle = \lambda_n |f_n\rangle$.

ANS. $\mathsf{A} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$.

Solution.

The numbers $1, -1$ are factors in the paired equations $Af_n = \lambda_n g_n$, not ordinary eigenvalues of A. The vectors f_n and g_n are right and left singular vectors with a sign convention. With this wording correction the exercise is well-defined, even though the Instructor's Manual, p. 86, says to disregard it.

(a) Both given families are orthonormal: the f_n are coordinate vectors, while $g_1^\dagger g_2 = (1-1)/2 = 0$ and each g_n has norm one. Use Exercise 6.5.12; as in Exercise 6.5.13(e), $g_n f_n^{\mathsf{T}}$ has the column g_n in column n and zeros elsewhere:

$$\begin{aligned} A &= g_1 f_1^{\mathsf{T}} - g_2 f_2^{\mathsf{T}} \\ &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} - \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} = \boxed{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}}. \end{aligned}$$

(b) Multiplication by the coordinate vectors selects the columns:

$$Af_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = g_1, \qquad Af_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = -g_2.$$

(c) The transpose is $A^{\mathsf{T}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, so

$$A^{\mathsf{T}} g_1 = \frac{1}{2}\begin{pmatrix} 1+1 \\ -1+1 \end{pmatrix} = f_1, \qquad A^{\mathsf{T}} g_2 = \frac{1}{2}\begin{pmatrix} 1-1 \\ -1-1 \end{pmatrix} = -f_2.$$

All paired equations hold with the given signs. To check explicitly why the textbook's terms "eigenvalues" and "eigenvectors" do not apply, compute the characteristic polynomial of A:

$$\det(A - \mu I) = \left(\frac{1}{\sqrt{2}} - \mu\right)^2 + \frac{1}{2} = \mu^2 - \sqrt{2}\mu + 1, \qquad \mu = \frac{\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{1 \pm i}{\sqrt{2}}.$$

These are the actual eigenvalues of A, and neither equals 1 or -1. Its ordinary singular values are $1, 1$, since

$$A^{\mathsf{T}} A = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I;$$

replacing g_2 by $-g_2$ converts the second paired factor to $+1$.

6.5.15 Two matrices U and H are related by

$$\mathsf{U} = e^{ia\mathsf{H}},$$

with a real. For the converses in (b) and (d), the stated property is required for every a in an interval about 0, with H fixed.

(a) If H is Hermitian, show that U is unitary.

(b) If U is unitary, show that H is Hermitian. (H is independent of a.)

(c) If $\operatorname{trace} \mathsf{H} = 0$, show that $\det \mathsf{U} = +1$.

(d) If $\det \mathsf{U} = +1$, show that $\operatorname{trace} \mathsf{H} = 0$.

Hint. When H is Hermitian, it can be diagonalized unitarily; then U is also diagonal in that basis. The corresponding eigenvalues are given by $u_j = \exp(iah_j)$.

Solution.

Parts (b) and (d) require the indicated property of $U(a) = e^{iaH}$ for every real a in an interval about 0, with the same fixed matrix H. Either converse can fail when the property is assumed for only one fixed value of a; see the counterexample at the end of this solution.

(a) The matrix exponential is defined by its convergent power series. Taking its adjoint term by term, with $\big((ia)^k H^k\big)^\dagger = (-ia)^k (H^\dagger)^k$ for real a, and using $H^\dagger = H$,

$$U(a)^\dagger = \sum_{k=0}^{\infty} \frac{(-ia)^k (H^\dagger)^k}{k!} = e^{-iaH} = U(a)^{-1}.$$

The last equality follows because iaH and $-iaH$ commute, so the product of their exponentials is $e^0 = I$. Thus $U(a)$ is unitary.

(b) Suppose now that $U(a)^\dagger U(a) = I$ throughout an interval about zero. The power series and its adjoint (as in (a), now without assuming $H^\dagger = H$) begin

$$U(a) = I + iaH + O(a^2), \qquad U(a)^\dagger = I - iaH^\dagger + O(a^2),$$

so

$$U(0) = I, \qquad U'(0) = iH, \qquad \left.\frac{dU^\dagger}{da}\right|_{a=0} = -iH^\dagger.$$

Differentiate the unitary identity by the product rule, $(U^\dagger U)' = (U^\dagger)' U + U^\dagger U'$, and evaluate at zero:

$$0 = \left.\frac{d}{da}(U^\dagger U)\right|_0 = (-iH^\dagger) I + I(iH) = i(H - H^\dagger).$$

Therefore $\boxed{H = H^\dagger}$.

For Hermitian H the hint settles (c) at once: in the eigenbasis of H, $\det U = \prod_j e^{iah_j} = e^{ia\sum_j h_j} = e^{ia \operatorname{tr} H}$. But in (c) and (d) H need not be Hermitian or even diagonalizable, which is why the hint is restricted: $H = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has only the eigenvalue 0, so a diagonal $S^{-1}HS$, whose diagonal entries are the eigenvalues of H (Exercise 6.4.1), would have to be 0, giving $H = S\,0\,S^{-1} = 0$, which is false. We therefore establish the same determinant identity for every finite square matrix H, whether or not it is diagonalizable. Let $F(a) = \det U(a)$. The exponential is invertible, with inverse e^{-iaH} (as in (a), the exponentials of the commuting matrices iaH and $-iaH$ multiply to $e^0 = I$), and termwise differentiation of its series gives

$$U'(a) = \sum_{k=1}^{\infty} \frac{k\, i^k a^{k-1} H^k}{k!} = iH \sum_{k=1}^{\infty} \frac{(iaH)^{k-1}}{(k-1)!} = iHU(a).$$

Let C_{ij} denote the cofactor of entry U_{ij}. The Laplace expansion along row i, $F = \sum_k U_{ik} C_{ik}$, contains U_{ij} only in the term $U_{ij}C_{ij}$, because no cofactor C_{ik} involves row i; hence $\partial F / \partial U_{ij} = C_{ij}$. The chain rule and the inverse-cofactor formula $U^{-1} = \operatorname{adj} U / F$, with $(\operatorname{adj} U)_{ji} = C_{ij}$, then give

$$F' = \sum_{i,j} C_{ij} U'_{ij}, \qquad C_{ij} = F(U^{-1})_{ji}.$$

Consequently, since $\sum_i (U^{-1})_{ji} U'_{ij} = (U^{-1}U')_{jj}$,

$$F' = F \sum_{i,j} (U^{-1})_{ji} U'_{ij} = F \operatorname{tr}(U^{-1}U'),$$

which is Jacobi's formula with all indices accounted for. Hence

$$F' = i \operatorname{tr}(U^{-1}HU)F = i(\operatorname{tr} H)F, \qquad F(0) = \det I = 1,$$

where trace invariance was proved in Exercise 6.4.1. Solving this scalar constant-coefficient equation with the integrating factor $e^{-ia \operatorname{tr} H}$,

$$\frac{d}{da}\left[e^{-ia \operatorname{tr} H} F(a)\right] = e^{-ia \operatorname{tr} H}\left[F' - i(\operatorname{tr} H)F\right] = 0, \qquad e^{-ia \operatorname{tr} H} F(a) = F(0) = 1,$$

yields

$$\boxed{\det U(a) = e^{ia \operatorname{tr} H}.}$$

(c) If $\operatorname{tr} H = 0$, this formula immediately gives $\det U(a) = 1$.

(d) If $\det U(a) = 1$ for every a near zero, differentiate at zero: $0 = F'(0) = i \operatorname{tr} H$. Therefore $\boxed{\operatorname{tr} H = 0}$ under the corrected hypothesis. For a single nonzero a_0, however,

$$e^{ia_0 \operatorname{tr} H} = 1 \quad \Longleftrightarrow \quad \operatorname{tr} H = \frac{2\pi k}{a_0}, \qquad k \in \mathbb{Z},$$

which need not vanish. At $a_0 = 0$ there is no restriction at all.

Here is a nonzero-parameter counterexample to the unqualified part (b), as well as to (d). Take

$$P = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \qquad P^2 = P, \qquad H = 2\pi P, \qquad a_0 = 1.$$

This H is not Hermitian ($P^\dagger = P^{\mathsf{T}} \neq P$) and has trace 2π. Nevertheless the idempotence, which gives $P^k = P$ for every $k \geq 1$ by induction, allows its exponential to be summed exactly:

$$e^{iH} = I + \sum_{k=1}^{\infty} \frac{(2\pi i)^k}{k!} P = I + (e^{2\pi i} - 1)P = I.$$

Thus $U(1)$ is unitary with determinant one, confirming why information about the whole parameter family is required for both converses.

6.5.16 An $n \times n$ matrix A has n eigenvalues A_i. If $\mathsf{B} = e^{\mathsf{A}}$, show that B has every eigenvector of A as an eigenvector, with the corresponding eigenvalues B_i given by $B_i = \exp(A_i)$.

Solution.

Let $Av = \alpha v$ with $v \neq 0$. Starting from $A^0 v = v$, induction gives

$$A^{k+1}v = A(A^k v) = A(\alpha^k v) = \alpha^k Av = \alpha^{k+1} v,$$

so $A^k v = \alpha^k v$ for every nonnegative integer k. Apply the convergent exponential series:

$$e^A v = \sum_{k=0}^{\infty} \frac{A^k v}{k!} = \sum_{k=0}^{\infty} \frac{\alpha^k}{k!} v = \boxed{e^\alpha v}.$$

Thus every eigenvector of A remains an eigenvector of $B = e^A$, with its eigenvalue exponentiated. In the question's notation, $\boxed{B_i = e^{A_i}}$. The proof works for each eigenvector even if A is defective and lacks a complete eigenbasis. When A does have n linearly independent eigenvectors v_i (for instance, when A is normal), let them be the columns of S. Then $BS = S\,\mathrm{diag}(e^{A_i})$, so $B = S\,\mathrm{diag}(e^{A_i})S^{-1}$, and by the similarity invariance of the characteristic polynomial (Exercise 6.4.1) the e^{A_i} are all n eigenvalues of B, counted with multiplicity. The same holds for defective A. By Schur's theorem, $A = QTQ^\dagger$ with Q unitary and T upper triangular with the eigenvalues A_i on its diagonal. A product of upper triangular matrices is upper triangular, with the products of the diagonal entries on its diagonal, so $A^k = QT^kQ^\dagger$ (since $Q^\dagger Q = I$) with T^k upper triangular with diagonal entries A_i^k. Summing the series, $B = Qe^TQ^\dagger$, where e^T is upper triangular with diagonal entries e^{A_i}. Hence, by Exercise 6.4.1 and because the determinant of a triangular matrix is the product of its diagonal entries, $\det(B - \lambda I) = \det(e^T - \lambda I) = \prod_i (e^{A_i} - \lambda)$.

The textbook's wording, that B has "the same eigenvectors" as A, should mean this forward implication; e^A can have additional eigendirections when different eigenvalues of A have the same exponential. For example, the powers of a diagonal matrix are diagonal with the powers of its entries, so

$$A = \mathrm{diag}(0, 2\pi i), \qquad e^A = \mathrm{diag}(e^0, e^{2\pi i}) = I_2.$$

The vector $(1,1)^\mathsf{T}$ is an eigenvector of e^A but not of A, since $A(1,1)^\mathsf{T} = (0, 2\pi i)^\mathsf{T}$ is not a multiple of $(1,1)^\mathsf{T}$. The distinct eigenvalues 0 and $2\pi i$ of A, each with a one-dimensional eigenspace, have the same exponential 1, whose eigenspace for e^A is the whole plane.

6.5.17 A matrix P is a projection operator satisfying the condition

$$\mathsf{P}^2 = \mathsf{P}.$$

Show that the corresponding eigenvalues $(\rho^2)_\lambda$ and ρ_λ satisfy the relation

$$(\rho^2)_\lambda = (\rho_\lambda)^2 = \rho_\lambda.$$

This means that the eigenvalues of P belong to $\{0, 1\}$.

Solution.

Let $Pv = \rho_\lambda v$ with $v \neq 0$. Applying P once more,

$$P^2 v = P(\rho_\lambda v) = \rho_\lambda P v = (\rho_\lambda)^2 v.$$

Thus the eigenvalue of P^2 on this same vector is $(\rho^2)_\lambda = (\rho_\lambda)^2$. But $P^2 = P$, so it is also ρ_λ: indeed $\big[(\rho_\lambda)^2 - \rho_\lambda\big]v = P^2 v - Pv = 0$ with $v \neq 0$. Hence

$$(\rho_\lambda)^2 = \rho_\lambda, \qquad \rho_\lambda(\rho_\lambda - 1) = 0, \qquad \boxed{\rho_\lambda \in \{0, 1\}.}$$

The interpretation is direct. If v lies in the range of P, say $v = Pw$, then $Pv = P^2 w = Pw = v$; conversely, if $Pv = v$, then $v = Pv$ lies in the range. If v lies in the kernel, $Pv = 0$. Every vector decomposes as $w = Pw + (I - P)w$, with the first part in the range and the second in the kernel because $P(I - P) = P - P^2 = 0$. So the range is the eigenspace for 1, the kernel is the eigenspace for 0, and together they span the whole space. The eigenvalues belong to $\{0, 1\}$; both need not occur. For example, $P = 0$ has only eigenvalue 0, and $P = I$ has only eigenvalue 1.

6.5.18 In the matrix eigenvector-eigenvalue equation

$$\mathsf{A}|x_i\rangle = \lambda_i|x_i\rangle,$$

A is an $n \times n$ Hermitian matrix. For simplicity assume that its n real eigenvalues are distinct, λ_1 being the largest. If $|x\rangle$ is an approximation to $|x_1\rangle$,

$$|x\rangle = |x_1\rangle + \sum_{i=2}^{n} \delta_i |x_i\rangle,$$

show that

$$\frac{\langle x|\mathsf{A}|x\rangle}{\langle x|x\rangle} \leq \lambda_1$$

and that the error in λ_1 is of the order $|\delta_i|^2$. Take $|\delta_i| \ll 1$.

Hint. The n vectors $|x_i\rangle$ form a complete orthogonal set spanning the n-dimensional (complex) space.

Solution.

First allow the mutually orthogonal eigenvectors, with $\langle x_i|x_j\rangle = 0$ for $i \neq j$, to have arbitrary nonzero squared norms $s_i = \langle x_i|x_i\rangle > 0$. Write the given expansion as $|x\rangle = \sum_{i=1}^{n} c_i|x_i\rangle$ with $c_1 = 1$ and $c_i = \delta_i$ for $i \geq 2$; then $A|x\rangle = \sum_j c_j\lambda_j|x_j\rangle$ and $\langle x| = \sum_i c_i^*\langle x_i|$, so

$$\langle x|x\rangle = \sum_{i,j} c_i^* c_j \langle x_i|x_j\rangle = \sum_i |c_i|^2 s_i = s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i,$$

$$\langle x|A|x\rangle = \sum_{i,j} c_i^* c_j \lambda_j \langle x_i|x_j\rangle = \lambda_1 s_1 + \sum_{i=2}^{n} \lambda_i |\delta_i|^2 s_i.$$

There are no linear terms in δ_i because every such term contains an inner product of two distinct eigenvectors. Subtract the Rayleigh quotient $R(x) = \langle x|A|x\rangle/\langle x|x\rangle$ from λ_1, combining the terms over a common denominator:

$$\begin{aligned}\lambda_1 - R(x) &= \frac{\lambda_1\big(s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i\big) - \big(\lambda_1 s_1 + \sum_{i=2}^{n} \lambda_i |\delta_i|^2 s_i\big)}{s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i} \\ &= \boxed{\frac{\sum_{i=2}^{n} (\lambda_1 - \lambda_i)|\delta_i|^2 s_i}{s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i}} \geq 0.\end{aligned}$$

Every gap $\lambda_1 - \lambda_i$ is positive and the denominator is positive. Thus $R(x) \leq \lambda_1$, with equality exactly when every $\delta_i = 0$ under the distinct-eigenvalue assumption.

For an orthonormal choice, all $s_i = 1$. Put $S = \sum_{i=2}^{n} |\delta_i|^2 = \|\delta\|^2$ and $N = \sum_{i=2}^{n} (\lambda_1 - \lambda_i)|\delta_i|^2$. The exact deficit is $N/(1 + S)$. Both are of second order: if $\lambda_{\min}$ is the smallest eigenvalue, $0 \leq N \leq (\lambda_1 - \lambda_{\min})S$, so $NS = O(\|\delta\|^4)$. With the geometric series $1/(1 + S) = 1 - S + O(S^2)$,

$$\lambda_1 - R(x) = N(1 - S + O(S^2)) = \sum_{i=2}^{n} (\lambda_1 - \lambda_i)|\delta_i|^2 + O(\|\delta\|^4).$$

This proves the claimed quadratic error and explicitly identifies its leading coefficient. The preceding weighted formula proves the same order for any fixed choice of eigenvector norms.

6.5.19 Two equal masses are connected to each other and to walls by springs as shown in Fig. 6.5. The masses are constrained to stay on a horizontal line.

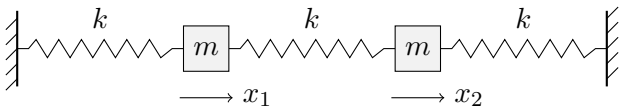


Fig. 6.5 (schematic): equal masses and three equal spring constants.

(a) Set up the Newtonian acceleration equation for each mass.

(b) Solve the secular equation for the eigenvalues.

(c) Determine the eigenvectors and thus the normal modes of motion.

Solution.

In Fig. 6.5 the three springs have the same constant k, and the two masses the same mass m. Take $m, k > 0$ and measure both displacements to the right from equilibrium.

(a) The left outer spring exerts $-kx_1$ on mass 1. The middle spring's extension is $x_2 - x_1$, so its forces are $+k(x_2 - x_1)$ on mass 1 and $-k(x_2 - x_1)$ on mass 2. The right outer spring exerts $-kx_2$ on mass 2. Newton's equations are therefore

$$\boxed{m\ddot{x}_1 = -2kx_1 + kx_2, \qquad m\ddot{x}_2 = kx_1 - 2kx_2.}$$

In matrix notation,

$$m\ddot{\boldsymbol{x}} + k\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\boldsymbol{x} = 0.$$

(b) Seek a normal mode $\boldsymbol{x}(t) = \boldsymbol{X}e^{i\omega t}$ and use $\ddot{\boldsymbol{x}} = -\omega^2\boldsymbol{x}$. After cancelling the nonzero common factor $e^{i\omega t}$, the matrix equation becomes

$$-m\omega^2\boldsymbol{X} + k\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\boldsymbol{X} = 0;$$

dividing by k and moving the first term to the right,

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\boldsymbol{X} = \lambda\boldsymbol{X}, \qquad \lambda = \frac{m\omega^2}{k}.$$

The secular equation is

$$0 = \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

Thus $\lambda = 1$ or 3, and $\omega = \sqrt{\lambda k/m}$ gives

$$\boxed{\omega_+ = \sqrt{k/m}, \qquad \omega_- = \sqrt{3k/m}.}$$

(The subscripts refer to the symmetric and antisymmetric modes found in (c).) Part (b) asks for eigenvectors in the textbook; the secular equation determines the eigenvalues. The eigenvectors are then found in part (c), as follows.

(c) For $\lambda = 1$, the first row, $(2-\lambda)X_1 - X_2 = 0$, is $X_1 - X_2 = 0$; choose the unit vector $u_+ = (1,1)^{\mathsf{T}}/\sqrt{2}$. The masses move in phase with equal displacement, so the middle spring has no changing extension. For $\lambda = 3$, the first row is $-X_1 - X_2 = 0$; choose $u_- = (1,-1)^{\mathsf{T}}/\sqrt{2}$. The masses move with equal amplitudes and opposite signs, stretching and compressing the middle spring. These vectors have norm one and inner product zero.

The corresponding normal coordinates are

$$Q_+ = \frac{x_1 + x_2}{\sqrt{2}}, \qquad Q_- = \frac{x_1 - x_2}{\sqrt{2}}.$$

Adding and subtracting the Newton equations gives

$$m(\ddot{x}_1 + \ddot{x}_2) = -k(x_1 + x_2), \qquad m(\ddot{x}_1 - \ddot{x}_2) = -3k(x_1 - x_2),$$

and dividing by $\sqrt{2}$,

$$m\ddot{Q}_+ + kQ_+ = 0, \qquad m\ddot{Q}_- + 3kQ_- = 0.$$

Each normal coordinate oscillates harmonically, $Q_\pm = C_\pm \cos(\omega_\pm t + \delta_\pm)$, and $\boldsymbol{x} = Q_+ u_+ + Q_- u_-$, because $u_\pm$ are orthonormal and $Q_\pm = u_\pm^{\mathsf{T}} \boldsymbol{x}$. Therefore the general real motion can be written

$$\boxed{\boldsymbol{x}(t) = C_+ u_+ \cos(\omega_+ t + \delta_+) + C_- u_- \cos(\omega_- t + \delta_-),}$$

where the amplitudes and phases are fixed by the initial positions and velocities.

6.5.20 Given a normal matrix A with eigenvalues λ_j, show that $\mathsf{A}^\dagger$ has eigenvalues λ_j^*, its real part $(\mathsf{A} + \mathsf{A}^\dagger)/2$ has eigenvalues $\Re(\lambda_j)$, and its imaginary part $(\mathsf{A} - \mathsf{A}^\dagger)/2i$ has eigenvalues $\Im(\lambda_j)$.

Solution.

First show that a right eigenvector of a normal matrix is also an eigenvector of its adjoint with the conjugate eigenvalue. If $Ax_j = \lambda_j x_j$, put $B = A - \lambda_j I$, so that $B^\dagger = A^\dagger - \lambda_j^* I$. It is normal because the terms $-\lambda_j^* A - \lambda_j A^\dagger + |\lambda_j|^2 I$ occur in both products and cancel:

$$BB^\dagger - B^\dagger B = AA^\dagger - A^\dagger A = 0.$$

Using $Bx_j = 0$, compute

$$\|B^\dagger x_j\|^2 = x_j^\dagger B B^\dagger x_j = x_j^\dagger B^\dagger B x_j = \|Bx_j\|^2 = 0.$$

A vector with zero norm is zero, so $B^\dagger x_j = 0$, or $\boxed{A^\dagger x_j = \lambda_j^* x_j}$. Now add and subtract the two eigenvalue equations:

$$\frac{A + A^\dagger}{2} x_j = \frac{\lambda_j + \lambda_j^*}{2} x_j = \boxed{\Re(\lambda_j) x_j},$$

$$\frac{A - A^\dagger}{2i} x_j = \frac{\lambda_j - \lambda_j^*}{2i} x_j = \boxed{\Im(\lambda_j) x_j}.$$

Thus every eigenvector of A is an eigenvector of the adjoint and of both Hermitian parts. A normal matrix has a complete orthonormal set of eigenvectors x_j (Section 6.5; see Exercise 6.5.8), so each of the three matrices is diagonal in that basis, with diagonal entries λ_j^*, $\Re(\lambda_j)$, and $\Im(\lambda_j)$, respectively. These are therefore all of their eigenvalues, counted with multiplicity, including degenerate eigenspaces.

6.5.21 Consider a rotation given by Euler angles $\alpha = \pi/4$, $\beta = \pi/2$, $\gamma = 5\pi/4$.

(a) Using the formula of Eq. (3.37), construct the matrix U representing this rotation.

(b) Find the eigenvalues and eigenvectors of U, and from them describe this rotation by specifying a single rotation axis and an angle of rotation about that axis.

Note. This technique provides a representation of rotations alternative to the Euler angles.

Solution.

(a) Equation (3.37), reproduced in Exercise 3.4.1, is the product $U = S_3(\gamma)S_2(\beta)S_1(\alpha)$ of rotations of the coordinate axes,

$$S_1(\alpha) = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2(\beta) = \begin{pmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{pmatrix},$$

and $S_3(\gamma)$ of the same form as S_1, with γ in place of α. Set $c = 1/\sqrt{2}$. The specified angles give

$$\cos\alpha = \sin\alpha = c, \quad \cos\beta = 0, \quad \sin\beta = 1, \quad \cos\gamma = \sin\gamma = -c.$$

The three matrices and their first product are

$$S_1 = \begin{pmatrix} c & c & 0 \\ -c & c & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad S_3 = \begin{pmatrix} -c & -c & 0 \\ c & -c & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$S_2S_1 = \begin{pmatrix} 0 & 0 & -1 \\ -c & c & 0 \\ c & c & 0 \end{pmatrix};$$

its rows are minus the third row, the second row, and the first row of S_1. Multiplying by S_3 on the left: if r_1, r_2, r_3 denote the rows of S_2S_1, the rows of U are

$$-c(r_1 + r_2) = (c^2, -c^2, c), \qquad c(r_1 - r_2) = (c^2, -c^2, -c), \qquad r_3 = (c, c, 0).$$

With $c^2 = 1/2$,

$$\boxed{U = \begin{pmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ 1/2 & -1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix}.}$$

Its columns have norm one and pairwise scalar products zero; its determinant is 1, as also follows from the three proper-rotation factors.

(b) To display the characteristic calculation without fractions, expand the determinant of $2(U - \lambda I)$, which equals $2^3 \det(U - \lambda I)$, along its first row:

$$\begin{aligned} 8\det(U - \lambda I) &= \begin{vmatrix} 1-2\lambda & -1 & \sqrt{2} \\ 1 & -1-2\lambda & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} & -2\lambda \end{vmatrix} \\ &= (1-2\lambda)\big[(1+2\lambda)2\lambda + 2\big] + \big[-2\lambda + 2\big] + \sqrt{2}\big[\sqrt{2} + (1+2\lambda)\sqrt{2}\big] \\ &= (1-2\lambda)(4\lambda^2 + 2\lambda + 2) + (2 - 2\lambda) + 4(1+\lambda) \\ &= (2 - 2\lambda - 8\lambda^3) + (6 + 2\lambda) = 8(1 - \lambda^3). \end{aligned}$$

Thus $\lambda^3 = 1$. Since $1 - \lambda^3 = (1-\lambda)(1+\lambda+\lambda^2)$, the roots are $\lambda = 1$ and the two roots of $\lambda^2 + \lambda + 1 = 0$, which by the quadratic formula are $\lambda = (-1 \pm \sqrt{1-4})/2$:

$$\boxed{\lambda_0 = 1, \quad \lambda_\pm = \frac{-1 \pm i\sqrt{3}}{2} = e^{\pm 2\pi i/3}.}$$

For an eigenvector $(x,y,z)^{\mathsf{T}}$, the rows of $2(U-\lambda I)(x,y,z)^{\mathsf{T}} = 0$ are

$$(1-2\lambda)x - y + \sqrt{2}z = 0, \qquad x - (1+2\lambda)y - \sqrt{2}z = 0, \qquad \sqrt{2}x + \sqrt{2}y - 2\lambda z = 0.$$

Adding the first two rows and dividing by 2, and dividing the third by $\sqrt{2}$, gives

$$(1-\lambda)x - (1+\lambda)y = 0, \qquad x + y = \sqrt{2}\lambda z.$$

Once these hold, the first row implies the second, because the sum of the two is satisfied. At $\lambda = 1$ the first equation gives $y = 0$ and the second gives $x = \sqrt{2}z$. The first row, $-x + \sqrt{2}z = 0$, also holds. Normalizing yields the fixed-axis vector

$$\boxed{n = \frac{(\sqrt{2},0,1)^{\mathsf{T}}}{\sqrt{3}}.}$$

For $\lambda = \lambda_\pm$, substitute $y = \sqrt{2}\lambda z - x$ into the first equation:

$$(1-\lambda)x + (1+\lambda)x - \sqrt{2}\lambda(1+\lambda)z = 2x - \sqrt{2}\lambda(1+\lambda)z = 0.$$

Since $\lambda^2 + \lambda + 1 = 0$, we have $\lambda(1+\lambda) = -1$, so $2x + \sqrt{2}z = 0$, i.e. $x = -z/\sqrt{2}$, and then, with $1 + 2\lambda_\pm = \pm i\sqrt{3}$,

$$y = \sqrt{2}\lambda z + \frac{z}{\sqrt{2}} = \frac{1+2\lambda}{\sqrt{2}}z = \pm i\sqrt{\frac{3}{2}}\,z.$$

The first row then holds as well: $(1-2\lambda)x - y + \sqrt{2}z = \big[-(1-2\lambda) - (1+2\lambda) + 2\big]z/\sqrt{2} = 0$, so $Uu_\pm = \lambda_\pm u_\pm$ for the normalized vectors below. For $z = 1$ the squared Hermitian norm is $|x|^2 + |y|^2 + |z|^2 = 1/2 + 3/2 + 1 = 3$, so

$$\boxed{u_\pm = \begin{pmatrix} -1/\sqrt{6} \\ \pm i/\sqrt{2} \\ 1/\sqrt{3} \end{pmatrix}.}$$

Also $n^\dagger u_\pm = -1/3 + 1/3 = 0$ and $u_+^\dagger u_- = 1/6 - 1/2 + 1/3 = 0$, so these three vectors are orthonormal.

The rotation axis is the line through the origin parallel to $(\sqrt{2},0,1)$, the direction left fixed by U. A rotation through an angle φ about this axis turns the plane perpendicular to it through φ, so by Exercise 6.5.4(b) its other two eigenvalues are $e^{\pm i\varphi}$; comparison with $\lambda_\pm = e^{\pm 2\pi i/3}$ gives the unsigned angle $2\pi/3 = 120°$. The sign can be fixed, rather than inferred only from a conjugate pair of eigenvalues. Define $[n]_\times v = n \times v$; for $n = (n_1,n_2,n_3)^{\mathsf{T}}$,

$$[n]_\times = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix}.$$

Direct subtraction, and this form with $\sqrt{3}\,n = (\sqrt{2}, 0, 1)^{\mathsf{T}}$, give

$$U - U^{\mathsf{T}} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -\sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix} = \sqrt{3}\,[n]_{\times}.$$

An active right-handed rotation through φ about the unit vector n has the Rodrigues form $R = \cos\varphi\, I + \sin\varphi\,[n]_{\times} + (1-\cos\varphi)\,nn^{\mathsf{T}}$. Indeed, $Rn = n$ because $[n]_{\times}n = n \times n = 0$ and $nn^{\mathsf{T}}n = n$; for $v \perp n$, $nn^{\mathsf{T}}v = 0$ gives $Rv = \cos\varphi\, v + \sin\varphi\, n \times v$. Here $n \times v$ is perpendicular to n and v, has the length of v, and satisfies $v \times (n \times v) = n(v \cdot v) - v(v \cdot n) = |v|^2 n$ by the BAC–CAB rule, so R turns the plane perpendicular to n through φ from v toward $n \times v$, counterclockwise as seen from the tip of n. In R the first and third terms are symmetric and $[n]_{\times}^{\mathsf{T}} = -[n]_{\times}$. Hence, using $\operatorname{tr}[n]_{\times} = 0$ and $\operatorname{tr}(nn^{\mathsf{T}}) = 1$,

$$R - R^{\mathsf{T}} = 2\sin\varphi\,[n]_{\times}, \qquad \operatorname{tr} R = 3\cos\varphi + (1-\cos\varphi) = 1 + 2\cos\varphi.$$

For U, $\operatorname{tr} U = \frac{1}{2} - \frac{1}{2} + 0 = 0$ gives $\cos\varphi = -\frac{1}{2}$, and $U - U^{\mathsf{T}} = \sqrt{3}\,[n]_{\times}$ gives $\sin\varphi = \sqrt{3}/2$. Thus active application of U to a column vector is a rotation by $\varphi = +120^{\circ}$ about n. Equation (3.37) is a passive coordinate transformation: the new coordinates are $x' = Ux$, so the new axes, the rows of U, are $e_i' = U^{\mathsf{T}}e_i$, and $U^{\mathsf{T}} = U^{-1}$, the inverse of the rotation by $+120^{\circ}$ about n, is the rotation by -120° about n (in the comparison above, $\operatorname{tr} U^{\mathsf{T}} = 0$ is unchanged, while $U^{\mathsf{T}} - (U^{\mathsf{T}})^{\mathsf{T}} = U^{\mathsf{T}} - U = -\sqrt{3}\,[n]_{\times}$ reverses the sign of $\sin\varphi$). The corresponding rotation of the coordinate axes is therefore -120° about n, equivalently $+120^{\circ}$ about $-n$. The directed angle must be paired with the chosen direction of the axis.

7 Chapter 7. Ordinary Differential Equations

Problems

7.2 First-Order Equations

7.2.1 From Kirchhoff's law the current I in an RC (resistance-capacitance) circuit (Fig. 7.1) obeys the equation

$$R\frac{dI}{dt} + \frac{1}{C}I = 0.$$

(a) Find $I(t)$.

(b) For a capacitance of 10,000 μF charged to 100 V and discharging through a resistance of 1 MΩ, find the current I for $t = 0$ and for $t = 100$ seconds.

Note. The initial voltage is I_0R or Q/C, where $Q = \int_0^\infty I(t)\,dt$.

Solution.

(a) For $R, C > 0$, separate the variables and integrate from 0 to t, with $I_0 = I(0) \neq 0$:

$$\frac{dI}{I} = -\frac{dt}{RC}, \qquad \ln\left|\frac{I(t)}{I_0}\right| = -\frac{t}{RC}, \qquad \boxed{I(t) = I_0 e^{-t/(RC)}}.$$

The absolute value drops because $I(t)/I_0 = \pm e^{-t/(RC)}$ is continuous, never zero, and equal to 1 at $t = 0$. The identically zero solution is included by $I_0 = 0$. Differentiation checks $RI' + I/C = -I/C + I/C = 0$.

(b) Convert the given units before finding the time constant:

$$C = 10^4 \times 10^{-6}\ \text{F} = 10^{-2}\ \text{F}, \quad R = 10^6\ \Omega, \quad RC = 10^4\ \text{s}.$$

By the Note, the initial voltage is $V_0 = I_0R = 100$ V, so $I_0 = V_0/R$; at $t = 100$ s the exponent is $t/(RC) = 100/10^4 = 0.01$. Thus

$$\boxed{I(0) = \frac{100}{10^6}\ \text{A} = 100\ \mu\text{A}}, \qquad \boxed{I(100\ \text{s}) = 100e^{-0.01}\ \mu\text{A} = 99.00498337\ldots\ \mu\text{A}}.$$

As a charge check, $\int_0^\infty I(t)dt = I_0RC\left[-e^{-t/(RC)}\right]_0^\infty = I_0RC = (10^{-4}\ \text{A})(10^4\ \text{s}) = 1$ C, which equals $CV_0 = (10^{-2}\ \text{F})(100\ \text{V})$, as the Note requires.

7.2.2 The Laplace transform of Bessel's equation ($n = 0$) leads to

$$(s^2 + 1)f'(s) + sf(s) = 0.$$

Solve for $f(s)$.

Solution.

For real s, $1 + s^2$ is positive. Separate and integrate, using $d\ln(1 + s^2)/ds = 2s/(1 + s^2)$:

$$\frac{f'}{f} = -\frac{s}{1 + s^2}, \qquad \ln|f| = -\frac{1}{2}\ln(1 + s^2) + K.$$

Exponentiating gives $|f| = e^K(1+s^2)^{-1/2}$; absorbing the sign into an arbitrary real constant gives

$$\boxed{f(s) = \frac{A}{\sqrt{1+s^2}}}.$$

This includes $f = 0$. Indeed, $f' = -As(1+s^2)^{-3/2}$, so $(1+s^2)f' = -As(1+s^2)^{-1/2} = -sf$. Conversely, every solution has this form, even one that vanishes somewhere (where the logarithm fails): by the product rule, $d\big[\sqrt{1+s^2}\,f\big]/ds = \big[(1+s^2)f' + sf\big]/\sqrt{1+s^2} = 0$, so $\sqrt{1+s^2}\,f$ is a constant A. The ODE alone does not determine A. If the intended transform is specifically that of J_0, the initial-value theorem of the Laplace transform, $\lim_{s\to+\infty} sf(s) = J_0(0) = 1$, together with $sf(s) = As/\sqrt{1+s^2} \to A$, gives $A = 1$. For complex s, the same formula holds with a complex constant on a simply connected domain avoiding $s = \pm i$, after choosing a square-root branch. On the Laplace half-plane $\operatorname{Re} s > 0$, choose the analytic branch positive for positive real s; it has $\sqrt{1+s^2} \sim s$ along the positive real axis.

7.2.3 The decay of a population by catastrophic two-body collisions is described by

$$\frac{dN}{dt} = -kN^2.$$

This is a first-order, nonlinear differential equation. Derive the solution

$$N(t) = N_0\left(1 + \frac{t}{\tau_0}\right)^{-1},$$

where $\tau_0 = (kN_0)^{-1}$, with $k, N_0 > 0$. The backward extrapolation has a pole as $t \downarrow -\tau_0$; $N(-\tau_0)$ itself is undefined.

Solution.

For $N_0 > 0$ and $k > 0$, integrate from the initial state:

$$\int_{N_0}^{N(t)} \frac{dN}{N^2} = -k\int_0^t ds, \qquad -\frac{1}{N(t)} + \frac{1}{N_0} = -kt.$$

Hence $1/N(t) = 1/N_0 + kt = (1 + kN_0t)/N_0$, and taking the reciprocal gives

$$\boxed{N(t) = \frac{N_0}{1+kN_0t} = N_0\left(1 + \frac{t}{\tau_0}\right)^{-1}}, \qquad \tau_0 = \frac{1}{kN_0}.$$

Direct differentiation yields $N' = -kN_0^2/(1+kN_0t)^2 = -kN^2$. The initial-value solution with positive population has mathematical interval $t > -\tau_0$ and a pole at its left endpoint; physical decay uses $t \geq 0$. The textbook's closing remark that this implies an infinite population at $t = -\tau_0$, qualified by the red note in the statement, should be read accordingly: the pole is a backward extrapolation of the model, not an attained infinite population at a finite time within its solution interval. If $N_0 = 0$, the separate solution is $N \equiv 0$; if $k = 0$, $N \equiv N_0$.

7.2.4 The rate of a particular chemical reaction A + B → C is proportional to the concentrations of the reactants A and B:

$$\frac{dC(t)}{dt} = \alpha[A(0) - C(t)][B(0) - C(t)].$$

(a) Find $C(t)$ for $A(0) \neq B(0)$.

(b) Find $C(t)$ for $A(0) = B(0)$.

The initial condition is that $C(0) = 0$.

Solution.

Let $A_0 = A(0)$ and $B_0 = B(0)$, and assume nonnegative initial concentrations and $\alpha \geq 0$.

(a) If $A_0 \neq B_0$ and both are positive, partial fractions give

$$\frac{1}{(A_0 - C)(B_0 - C)} = \frac{1}{B_0 - A_0}\left(\frac{1}{A_0 - C} - \frac{1}{B_0 - C}\right),$$

as recombining the right side over the common denominator shows. Separate the variables, $dC/[(A_0 - C)(B_0 - C)] = \alpha\, dt$, and integrate from 0 to t. The constant $C = \min(A_0, B_0)$ is an equilibrium solution. The right side $\alpha(A_0 - C)(B_0 - C)$ is a polynomial in C, so by the uniqueness theorem for $C' = f(C)$ with continuously differentiable f two solutions cannot take the same value at the same time; hence the solution starting at $C(0) = 0$ never reaches the equilibrium. Below it both factors are positive, so $C' \geq 0$ and C cannot fall below $C(0) = 0$. Thus $0 \leq C < \min(A_0, B_0)$, where

$$\int_0^C \frac{du}{A_0 - u} = \ln\frac{A_0}{A_0 - C}, \qquad \int_0^C \frac{du}{B_0 - u} = \ln\frac{B_0}{B_0 - C}.$$

The difference of these logarithms, divided by $B_0 - A_0$, yields

$$\frac{1}{B_0 - A_0}\ln\left[\frac{B_0 - C}{A_0 - C}\frac{A_0}{B_0}\right] = \alpha t.$$

Set $E = e^{\alpha(B_0 - A_0)t}$. Multiplying by $B_0 - A_0$, exponentiating, clearing the denominators, and collecting the terms in C give

$$A_0(B_0 - C) = B_0E(A_0 - C), \qquad C(B_0E - A_0) = A_0B_0(E - 1).$$

Thus $C = A_0B_0(E - 1)/(B_0E - A_0)$; dividing numerator and denominator by E gives

$$\boxed{C(t) = \frac{A_0B_0(1 - e^{-\alpha(B_0 - A_0)t})}{B_0 - A_0e^{-\alpha(B_0 - A_0)t}}}.$$

It starts at zero and approaches $\min(A_0, B_0)$ for $\alpha > 0$: if $B_0 > A_0$ the exponential tends to 0 and $C \to A_0B_0/B_0 = A_0$; if $B_0 < A_0$ it grows without bound, and $C \to A_0B_0/A_0 = B_0$. To check the result, put $\lambda = \alpha(B_0 - A_0)$ and $D = B_0 - A_0e^{-\lambda t}$, so $C = A_0B_0(1 - e^{-\lambda t})/D$. Then $A_0 - C = A_0(B_0 - A_0)e^{-\lambda t}/D$ and $B_0 - C = B_0(B_0 - A_0)/D$, so $\alpha(A_0 - C)(B_0 - C) = \alpha A_0B_0(B_0 - A_0)^2e^{-\lambda t}/D^2$. Since $D' = \lambda A_0e^{-\lambda t}$, the quotient rule gives

$$C' = \frac{A_0B_0\lambda e^{-\lambda t}\big[D - A_0(1 - e^{-\lambda t})\big]}{D^2} = \frac{A_0B_0\lambda e^{-\lambda t}(B_0 - A_0)}{D^2},$$

the same expression because $\lambda = \alpha(B_0 - A_0)$. If either concentration is zero, $C(t) = 0$ directly; the formula also reduces to that value when $A_0 \neq B_0$.

(b) With $A_0 = B_0 = a > 0$, separate instead as

$$\int_0^{C(t)} \frac{du}{(a-u)^2} = \alpha t, \qquad \frac{1}{a-C} - \frac{1}{a} = \alpha t.$$

Hence $1/(a-C) = (1+\alpha a t)/a$, i.e., $a - C = a/(1+\alpha a t)$, and

$$\boxed{C(t) = a - \frac{a}{1+\alpha a t} = \frac{\alpha a^2 t}{1+\alpha a t}}.$$

The formula includes $a = 0$ and $\alpha = 0$. It satisfies the equation: $C' = \alpha a^2/(1+\alpha a t)^2 = \alpha(a-C)^2$.

7.2.5 A boat, coasting through the water, experiences a resisting force proportional to v^n, v being the boat's instantaneous velocity. Newton's second law leads to

$$m\frac{dv}{dt} = -kv^n.$$

With $v(t=0) = v_0$, $x(t=0) = 0$, integrate to find v as a function of time and v as a function of distance. Take $m, k, v_0 > 0$, $n \in \mathbb{R}$, and use the real positive-speed branch until it reaches zero, if it does.

Solution.

Take $m, k, v_0 > 0$ and interpret v as speed while the boat moves forward. Put $\gamma = k/m$. For $n \neq 1$, separate and integrate from 0 to t:

$$v^{-n}dv = -\gamma dt, \qquad \frac{v^{1-n} - v_0^{1-n}}{1-n} = -\gamma t,$$

so $v^{1-n} = v_0^{1-n} - (1-n)\gamma t$, and raising to the power $1/(1-n)$,

$$\boxed{v(t) = \left[v_0^{1-n} + (n-1)\gamma t\right]^{1/(1-n)}} \quad (n \neq 1).$$

The real positive branch is used while the bracket is positive. For $n = 1$, the integration is logarithmic:

$$\ln(v/v_0) = -\gamma t, \qquad \boxed{v(t) = v_0 e^{-\gamma t}}.$$

To eliminate time use $dv/dt = (dv/dx)(dx/dt) = v\,dv/dx$ while $v > 0$. Then $mv\,dv/dx = -kv^n$, i.e., $v^{1-n}dv = -\gamma dx$; integrating from $x = 0$, for $n \neq 2$,

$$\frac{v^{2-n} - v_0^{2-n}}{2-n} = -\gamma x, \qquad \boxed{v(x) = \left[v_0^{2-n} + (n-2)\gamma x\right]^{1/(2-n)}}.$$

For the omitted case $n = 2$, $dv/v = -\gamma dx$, giving

$$\boxed{v(x) = v_0 e^{-\gamma x}}.$$

The general formulas also cover the value excluded in the other variable: $n = 1$ in $v(x)$ gives $v(x) = v_0 - \gamma x$, and $n = 2$ in $v(t)$ gives $v(t) = (v_0^{-1} + \gamma t)^{-1} = v_0/(1+\gamma v_0 t)$. Differentiating the two integrated identities, $d[v^{1-n}/(1-n)]/dt = v^{-n}v' = -\gamma$ and $d[v^{2-n}/(2-n)]/dx = v^{1-n}\,dv/dx = -\gamma$, checks both equations for every $n \neq 1$ and every $n \neq 2$, respectively; the exponentials satisfy $v' = -\gamma v$ $(n = 1)$ and $v\,dv/dx = -\gamma v^2$ $(n = 2)$ directly.

For $n < 1$ the bracket in $v(t)$ decreases to zero, so the positive-speed branch reaches zero at $t_* = v_0^{1-n}/[(1-n)\gamma]$; likewise, for $n < 2$ the bracket in $v(x)$ vanishes at the limiting travel distance $x_* = v_0^{2-n}/[(2-n)\gamma]$. For $n > 1$ the bracket in $v(t)$ grows without bound, so $v \to 0$ only as $t \to \infty$ (as for the exponential at $n = 1$): stopping takes infinite time. In the same way, for $n \geq 2$ the speed $v(x)$ is positive at every x, so the traveled distance is unbounded: were $x(t) \leq L$ for all t, then, since $dv/dx = -\gamma v^{n-1} < 0$, the speed would stay at least $v(L) > 0$, so $x(t) \geq v(L)\,t$ would exceed L for $t > L/v(L)$, a contradiction. The positive-speed drag model does not authorize continuing a fractional or negative power through $v = 0$. At or after stopping an appropriate force law must be specified; the formulas above state their actual domain.

7.2.6 In the first-order differential equation $dy/dx = f(x, y)$, the function $f(x, y)$ is a function of the ratio y/x:

$$\frac{dy}{dx} = g(y/x).$$

Show that the substitution of $u = y/x$ leads to a separable equation in u and x.

Solution.

Work on an interval with $x \neq 0$. Since $y = xu$,

$$\frac{dy}{dx} = u + x\frac{du}{dx} = g(u), \qquad x\frac{du}{dx} = g(u) - u.$$

Thus, whenever $g(u) - u \neq 0$,

$$\boxed{\frac{du}{g(u) - u} = \frac{dx}{x}}, \qquad \int^{u} \frac{ds}{g(s) - s} = \ln|x| + C.$$

Division by $g(u) - u$ omits its zeros. Each constant u_* satisfying $g(u_*) = u_*$ supplies the additional straight-line solution $y = u_* x$, which must be checked before declaring the solution set complete.

7.2.7 The differential equation

$$P(x, y)\,dx + Q(x, y)\,dy = 0$$

is exact. Assume P, Q have continuous first partial derivatives and that the rectangular integration path lies inside the domain. Show that its solution is of the form

$$\varphi(x, y) = \int_{x_0}^{x} P(\xi, y)\,d\xi + \int_{y_0}^{y} Q(x_0, \eta)\,d\eta = \text{constant}.$$

Solution.

Use distinct integration variables, so the proposed potential is

$$\varphi(x, y) = \int_{x_0}^{x} P(\xi, y)d\xi + \int_{y_0}^{y} Q(x_0, \eta)d\eta.$$

This is the line integral along the path first from (x_0, y_0) to (x_0, y), then to (x, y). Exactness means $P\,dx + Q\,dy = d\Phi$ for a potential Φ, i.e., $\Phi_x = P$ and $\Phi_y = Q$. Hence, by the fundamental theorem of calculus,

$$\int_{y_0}^{y} Q(x_0, \eta)d\eta = \Phi(x_0, y) - \Phi(x_0, y_0), \qquad \int_{x_0}^{x} P(\xi, y)d\xi = \Phi(x, y) - \Phi(x_0, y),$$

and the sum is $\varphi(x,y) = \Phi(x,y) - \Phi(x_0,y_0)$. It follows that $d\varphi = d\Phi = P\,dx + Q\,dy$ and the solution curves satisfy

$$\boxed{\varphi(x,y) = C}.$$

More explicitly, differentiation (under the integral sign in the first term, then the exactness condition $P_y = Q_x$ and the fundamental theorem of calculus in ξ; see 7.2.8) gives

$$\varphi_x = P(x,y), \quad \varphi_y = \int_{x_0}^{x} P_y(\xi,y)d\xi + Q(x_0,y) = Q(x,y) - Q(x_0,y) + Q(x_0,y) = Q(x,y).$$

Along any solution curve $(x(t), y(t))$, the chain rule gives $d\varphi/dt = P\dot{x} + Q\dot{y} = 0$. Conversely this same calculation shows that a differentiable level curve solves the differential form. The displayed rectangular path is understood to lie inside the domain; otherwise one uses an admissible path for the same exact differential.

7.2.8 The differential equation

$$P(x,y)\,dx + Q(x,y)\,dy = 0$$

is exact. Assume P, Q have continuous first partial derivatives and that the rectangular integration path lies inside the domain. If

$$\varphi(x,y) = \int_{x_0}^{x} P(\xi,y)\,d\xi + \int_{y_0}^{y} Q(x_0,\eta)\,d\eta,$$

show that

$$\frac{\partial\varphi}{\partial x} = P(x,y), \quad \frac{\partial\varphi}{\partial y} = Q(x,y).$$

Hence, $\varphi(x,y) =$ constant is a solution of the original differential equation.

Solution.

Write the formula with dummy variables:

$$\varphi(x,y) = \int_{x_0}^{x} P(\xi,y)d\xi + \int_{y_0}^{y} Q(x_0,\eta)d\eta.$$

The fundamental theorem of calculus, with y held fixed, gives $\boxed{\varphi_x = P(x,y)}$; the second term does not depend on x. Differentiating with respect to y requires both terms: the first is differentiated under the integral sign (allowed because P_y is continuous), the second by the fundamental theorem of calculus in η:

$$\begin{aligned}\varphi_y &= \int_{x_0}^{x} P_y(\xi,y)d\xi + Q(x_0,y)\\ &= \int_{x_0}^{x} Q_x(\xi,y)d\xi + Q(x_0,y)\\ &= Q(x,y) - Q(x_0,y) + Q(x_0,y) = \boxed{Q(x,y)}.\end{aligned}$$

The second line uses the exactness condition $P_y = Q_x$; the third integrates Q_x in ξ by the fundamental theorem of calculus. Therefore $d\varphi = \varphi_x dx + \varphi_y dy = Pdx + Qdy$. A level curve $\varphi = C$ has $d\varphi = 0$ and satisfies the original equation. As in 7.2.7, the chosen paths must lie within the domain.

7.2.9 Prove that Eq. (7.12) is exact in the sense of Eq. (7.9), provided that $\alpha(x)$ satisfies Eq. (7.14).

Solution.

The referenced equations are $\alpha y' + \alpha p y = \alpha q$ and $\alpha' = \alpha p$. In differential form (multiply the first by dx and move $\alpha q\, dx$ to the left) it is

$$M(x,y)dx + N(x,y)dy = 0, \qquad M = \alpha(x)[p(x)y - q(x)], \quad N = \alpha(x).$$

The exactness test is

$$M_y = \alpha p, \qquad N_x = \alpha',$$

so $M_y = N_x$ follows precisely from Eq. (7.14). A direct potential is

$$\boxed{\Phi(x,y) = \alpha(x)y - \int^x \alpha(s)q(s)ds},$$

because $\Phi_y = \alpha = N$ and, by Eq. (7.14), $\Phi_x = \alpha' y - \alpha q = \alpha(py - q) = M$. For equivalence with the original ODE choose a nonzero integrating factor, $\alpha = e^{\int p\, dx}$; the identically zero factor would give only $0 = 0$ and discard the equation.

7.2.10 A certain differential equation has the form

$$f(x)\, dx + g(x)h(y)\, dy = 0,$$

with none of the functions $f(x)$, $g(x)$, $h(y)$ identically zero. Show that a necessary and sufficient condition for this equation to be exact is that $g(x) =$ constant. Work on a product of connected intervals, with C^1 coefficients.

Solution.

On a product of connected intervals, with the stated functions continuously differentiable where needed, exactness requires

$$\frac{\partial f(x)}{\partial y} = 0 = \frac{\partial [g(x)h(y)]}{\partial x} = g'(x)h(y).$$

Since h is not identically zero, choose one y_* with $h(y_*) \neq 0$. The equality at (x, y_*) forces $g'(x) = 0$ for every x, so g is a constant. Conversely, if $g = g_0$, then

$$f(x)dx + g_0 h(y)dy = d\left[\int^x f(s)ds + g_0 \int^y h(t)dt\right]$$

is exact. Hence $\boxed{g(x) = g_0}$ is necessary and sufficient. The connected product-domain assumption specifies the usual meaning of this exercise. Without it, h could vanish throughout an entire component, imposing no condition on g there, or different x-components could carry different constants.

7.2.11 Show that

$$y(x) = \exp\left[-\int^x p(t)\, dt\right]\left\{\int^x \left[\exp\left(\int^s p(t)\, dt\right)\right] q(s)\, ds + C\right\}$$

is a solution of

$$\frac{dy}{dx} + p(x)y(x) = q(x)$$

by differentiating the expression for $y(x)$ and substituting into the differential equation.

Solution.

Choose a fixed base point and abbreviate

$$P_0(x) = \int_{x_0}^{x} p(t)dt, \qquad J(x) = \int_{x_0}^{x} e^{P_0(s)} q(s)ds + C.$$

Then the proposed solution is $y = e^{-P_0}J$, with $P_0' = p$ and $J' = e^{P_0}q$. The product and chain rules give

$$y' = -pe^{-P_0}J + e^{-P_0}e^{P_0}q = -py + q.$$

Therefore $\boxed{y' + py = q}$. Changing the fixed lower limits merely changes the arbitrary constant or rescales the integrating factor and does not change the resulting family of solutions.

7.2.12 The motion of a body falling in a resisting medium may be described by

$$m\frac{dv}{dt} = mg - bv$$

when the retarding force is proportional to the velocity, v. Find the velocity. Evaluate the constant of integration by demanding that $v(0) = 0$.

Solution.

For $b > 0$, divide by m and multiply by the integrating factor $e^{bt/m}$; by the product rule, $(e^{bt/m}v)' = e^{bt/m}(v' + bv/m)$, so

$$v' + \frac{b}{m}v = g, \qquad \frac{d}{dt}(e^{bt/m}v) = ge^{bt/m}.$$

Integrating from 0 to t, using $v(0) = 0$ and $\int_0^t ge^{bs/m}ds = (mg/b)(e^{bt/m} - 1)$, gives

$$e^{bt/m}v(t) = \frac{mg}{b}(e^{bt/m} - 1), \qquad \boxed{v(t) = \frac{mg}{b}(1 - e^{-bt/m})}.$$

The check is $mv' = mge^{-bt/m} = mg - bv$. The terminal velocity is mg/b; as $b \to 0$, the expansion $1 - e^{-bt/m} = bt/m + O(b^2)$ gives the drag-free solution $v = gt$.

7.2.13 Radioactive nuclei decay according to the law

$$\frac{dN}{dt} = -\lambda N,$$

N being the concentration of a given nuclide and λ, the particular decay constant. In a radioactive series of two different nuclides, with concentrations $N_1(t)$ and $N_2(t)$, we have

$$\frac{dN_1}{dt} = -\lambda_1 N_1, \quad \frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2.$$

Find $N_2(t)$ for the conditions $N_1(0) = N_0$ and $N_2(0) = 0$.

Solution.

The first equation integrates to $N_1 = N_0 e^{-\lambda_1 t}$. The second becomes a linear equation; multiplying it by the integrating factor $e^{\lambda_2 t}$ and using $(e^{\lambda_2 t}N_2)' = e^{\lambda_2 t}(N_2' + \lambda_2 N_2)$,

$$N_2' + \lambda_2 N_2 = \lambda_1 N_0 e^{-\lambda_1 t}, \qquad \frac{d}{dt}(e^{\lambda_2 t}N_2) = \lambda_1 N_0 e^{(\lambda_2 - \lambda_1)t}.$$

Integrate from 0 to t, with $N_2(0) = 0$. If $\lambda_1 \neq \lambda_2$,

$$e^{\lambda_2 t} N_2(t) = \lambda_1 N_0 \int_0^t e^{(\lambda_2 - \lambda_1)s} ds = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} \big(e^{(\lambda_2 - \lambda_1)t} - 1\big),$$

and multiplication by $e^{-\lambda_2 t}$ gives

$$\boxed{N_2(t) = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1}(e^{-\lambda_1 t} - e^{-\lambda_2 t})}.$$

The coincident-rate case must be handled separately rather than dividing by $\lambda_2 - \lambda_1$. If both rates equal λ,

$$e^{\lambda t} N_2 = \lambda N_0 \int_0^t ds = \lambda N_0 t, \qquad \boxed{N_2 = \lambda N_0 t e^{-\lambda t}}.$$

This is also the limit of the preceding expression: with $\lambda_1 = \lambda$ and $\lambda_2 \to \lambda$, $(e^{-\lambda t} - e^{-\lambda_2 t})/(\lambda_2 - \lambda)$ tends to the derivative of $-e^{-\lambda_2 t}$ with respect to λ_2 at $\lambda_2 = \lambda$, which is $te^{-\lambda t}$. Both forms satisfy the initial condition and the inhomogeneous equation by differentiation.

7.2.14 The rate of evaporation from a particular spherical drop of liquid (constant density) is proportional to its surface area. Assuming this to be the sole mechanism of mass loss, find the radius of the drop as a function of time.

Solution.

Let ρ be the fixed mass density and let $j > 0$ be the mass lost per unit area per unit time. Then

$$M = \frac{4\pi}{3}\rho r^3, \qquad \frac{dM}{dt} = 4\pi\rho r^2 r' = -j(4\pi r^2).$$

While $r > 0$, cancellation gives $r' = -j/\rho$. With $r(0) = r_0$,

$$\boxed{r(t) = r_0 - \frac{j}{\rho}t, \qquad 0 \le t \le t_* := \frac{\rho r_0}{j}}.$$

At t_* the drop is exhausted. The model's physical continuation has zero radius and mass, not a negative radius. If the proportionality constant is defined for volume loss instead of mass loss, put $\kappa = j/\rho$ and write $r = r_0 - \kappa t$.

7.2.15 In the linear homogeneous differential equation

$$\frac{dv}{dt} = -av$$

the variables are separable. When the variables are separated, the equation is exact. Solve this differential equation subject to $v(0) = v_0$ by the following three methods:

(a) Separating variables and integrating.

(b) Treating the separated variable equation as exact.

(c) Using the result for a linear homogeneous differential equation.

ANS. $v(t) = v_0 e^{-at}$.

Solution.

(a) For $v_0 \neq 0$, separation gives

$$\int_{v_0}^{v(t)} \frac{du}{u} = -a \int_0^t ds, \quad \ln|v(t)/v_0| = -at, \quad v(t) = v_0 e^{-at}.$$

The sign stays that of v_0; $v \equiv 0$ is separately included when $v_0 = 0$.

(b) The separated form $a\,dt + dv/v = 0$, i.e., $P\,dt + Q\,dv = 0$ with $P = a$ and $Q = 1/v$, is exact on either half-plane $v > 0$ or $v < 0$, since both cross partial derivatives vanish: $\partial P/\partial v = 0 = \partial Q/\partial t$. Its potential is $\Phi(t, v) = at + \ln|v|$, because $\Phi_t = a$ and $\Phi_v = 1/v$. The level condition $\Phi(t, v) = \Phi(0, v_0) = \ln|v_0|$ gives $\ln|v/v_0| = -at$, so $|v| = |v_0|e^{-at}$; since v stays in the half-plane of v_0, this is the same exponential $v = v_0 e^{-at}$. The zero solution was excluded only by division by v and must be restored.

(c) In the linear first-order formula of Exercise 7.2.11 take $p = a$, $q = 0$:

$$v = e^{-\int a\,dt}(C + 0) = Ce^{-at}.$$

The initial condition gives $C = v_0$. Thus all three methods give $\boxed{v = v_0 e^{-at}}$, and $v' = -av$ checks it directly.

7.2.16

(a) Solve Example 7.2.1, assuming that the parachute opens when the parachutist's velocity has reached $v_i = 60$ mi/h (regard this time as $t = 0$). Find $v(t)$.

(b) For a skydiver in free fall use the friction coefficient $b = 0.25$ kg/m and mass $m = 70$ kg. What is the limiting velocity in this case?

Solution.

In Example 7.2.1 the downward speed obeys $mv' = mg - bv^2$. Put $v_T = \sqrt{mg/b}$ and $T = \sqrt{m/(bg)}$, so $v_T/T = g$.

(a) Since $mg - bv^2 = b(v_T^2 - v^2)$, separation gives $dv/(v_T^2 - v^2) = (b/m)\,dt$. By partial fractions,

$$\frac{1}{v_T^2 - u^2} = \frac{1}{2v_T}\left(\frac{1}{v_T - u} + \frac{1}{v_T + u}\right), \qquad \int \frac{du}{v_T^2 - u^2} = \frac{1}{2v_T}\ln\left|\frac{v_T + u}{v_T - u}\right|.$$

Integrating from the nonzero initial speed v_i at $t = 0$ therefore gives

$$\int_{v_i}^{v} \frac{du}{v_T^2 - u^2} = \frac{b}{m}t, \qquad \frac{1}{2v_T}\ln\left|\frac{v_T + v}{v_T - v}\frac{v_T - v_i}{v_T + v_i}\right| = \frac{b}{m}t.$$

Here $2v_T b/m = 2\sqrt{bg/m} = 2/T$, so the logarithm equals $2t/T$. For $v_i \neq v_T$, put $q = (v_i - v_T)/(v_i + v_T)$. Exponentiating and inverting the ratio give

$$\left|\frac{v - v_T}{v + v_T}\right| = |q|\, e^{-2t/T}.$$

The right side never vanishes, so $v \neq v_T$ at every t; by continuity $v - v_T$ keeps the sign of $v_i - v_T$, which is the sign of q. (Since $v' = (b/m)(v_T^2 - v^2)$, the speed moves monotonically from $v_i \geq 0$ toward v_T, so $v + v_T > 0$.) Hence

$$\frac{v - v_T}{v + v_T} = qe^{-2t/T}, \qquad v(1 - qe^{-2t/T}) = v_T(1 + qe^{-2t/T}).$$

Divide by $1 - qe^{-2t/T}$ and multiply numerator and denominator by $v_i + v_T$. This gives

$$\boxed{v(t) = v_T \frac{(v_i + v_T) + (v_i - v_T)e^{-2t/T}}{(v_i + v_T) - (v_i - v_T)e^{-2t/T}}} = v_T \frac{v_i + v_T \tanh(t/T)}{v_T + v_i \tanh(t/T)}.$$

For the second form write the numerator as $v_i(1 + e^{-2t/T}) + v_T(1 - e^{-2t/T})$ and the denominator as $v_i(1 - e^{-2t/T}) + v_T(1 + e^{-2t/T})$, divide both by $1 + e^{-2t/T}$, and use $(1 - e^{-2u})/(1 + e^{-2u}) = \tanh u$. For $v_i = v_T$, separation divided by zero; direct substitution instead gives the equilibrium $v(t) = v_T$, which the boxed formula also includes. For every $v_i \geq 0$ and $t \geq 0$, $-1 \leq q < 1$, so its denominator is positive. It gives $v(0) = v_i$ and approaches v_T. For the example's $m = 70$ kg, $b = 700$ kg/m, $g = 9.8$ m/s^2, and with 1 mi/h $= 1609.344$ m/3600 s $= 0.44704$ m/s,

$$\begin{aligned} v_i &= 60(0.44704) = 26.8224 \text{ m/s}, \\ v_T &= \sqrt{0.98} = 0.9899494936\ldots \text{ m/s}, \\ T &= 1/\sqrt{98} = 0.1010152544\ldots \text{ s}. \end{aligned}$$

Here $v_i > v_T$. The Instructor's Manual's real expression involving $\operatorname{arctanh}(v_i/v_T)$ is therefore inapplicable, since $\operatorname{arctanh} w$ is real only for $|w| < 1$. The equivalent real hyperbolic form for this case is

$$v(t) = v_T \coth\left(t/T + \operatorname{arccoth}(v_i/v_T)\right), \quad \operatorname{arccoth} w = \tfrac{1}{2} \ln \frac{w+1}{w-1} \quad (w > 1).$$

Indeed, with $\phi = \operatorname{arccoth}(v_i/v_T)$ and $u = t/T$, the addition formula $\coth(u + \phi) = (\coth u \coth \phi + 1)/(\coth u + \coth \phi)$, multiplied above and below by $v_T \tanh u$, gives $v_T \coth(u + \phi) = v_T(v_i + v_T \tanh u)/(v_T + v_i \tanh u)$, the tanh form above. Differentiating either real form verifies $v' = g(1 - v^2/v_T^2)$, which is $mv' = mg - bv^2$ divided by m; e.g., $\operatorname{csch}^2 = \coth^2 - 1$ and $v_T/T = g$ give $v' = -(v_T/T)\operatorname{csch}^2(u + \phi) = g(1 - v^2/v_T^2)$. The speed initially decreases, as it must when the parachute opens above terminal speed. Numerically, $v_i/v_T = 27.0947\ldots$, so $\phi = \frac{1}{2}\ln[(v_i/v_T + 1)/(v_i/v_T - 1)] = 0.0369243\ldots$, and $1/T = \sqrt{98}$ s^{-1} $= 9.89949\ldots$ s^{-1}; with t in seconds,

$$v(t) \approx 0.989949 \coth(9.89949\, t + 0.0369243) \text{ m/s}.$$

For example, $v(0.1 \text{ s}) = 1.2812\ldots$ m/s, already within 0.3 m/s of v_T.

(b) With the free-fall drag coefficient $b = 0.25$ kg/m,

$$\boxed{v_T = \sqrt{\frac{70(9.8)}{0.25}} = \sqrt{2744} = 52.38320341\ldots \text{ m/s} = 188.5795\ldots \text{ km/h}},$$

where 1 m/s $= 3.6$ km/h.

7.2.17 Solve the ODE

$$(xy^2 - y)\,dx + x\,dy = 0.$$

Solution.

On $x \neq 0$, divide by x:

$$y' - \frac{y}{x} = -y^2.$$

The solution $y = 0$ is immediate. For $y \neq 0$, this Bernoulli equation becomes linear for $u = 1/y$: then $u' = -y'/y^2$, and multiplication by $-y^{-2}$ gives $-y'/y^2 + 1/(xy) = 1$, i.e.,

$$u' + \frac{u}{x} = 1, \qquad (xu)' = xu' + u = x.$$

Integrating yields $xu = x^2/2 + C$, hence $u = (x^2 + 2C)/(2x)$ and $y = 1/u$:

$$\boxed{y(x) = \frac{2x}{x^2 + K}}, \qquad \boxed{y \equiv 0},$$

where $K = 2C$. Substitution into the original, undivided equation, which is $xy' = y - xy^2$, checks it: by the quotient rule $y' = [2(x^2 + K) - 4x^2]/(x^2 + K)^2 = 2(K - x^2)/(x^2 + K)^2$, and

$$xy' = \frac{2x(K - x^2)}{(x^2 + K)^2} = \frac{2x(x^2 + K) - 4x^3}{(x^2 + K)^2} = y - xy^2.$$

The rational solution is valid wherever its denominator is nonzero, including its smooth extension through $x = 0$ when $K \neq 0$. The equation itself forces $y(0) = 0$ if a solution is defined there. For $K = 0$, $y = 2/x$ has a pole at zero and is used on either side separately.

7.2.18 Solve the ODE

$$(x^2 - y^2 e^{y/x})\,dx + (x^2 + xy)e^{y/x}\,dy = 0.$$

Hint. Note that the quantity y/x in the exponents is of combined degree zero and does not affect the determination of homogeneity.

Solution.

The original equation is defined on $x \neq 0$. Set $u = y/x$, so $y = xu$ and $dy = u\,dx + x\,du$. Substitution gives

$$\begin{aligned} 0 &= x^2(1 - u^2 e^u)dx + x^2(1 + u)e^u(u\,dx + x\,du) \\ &= x^2(1 + ue^u)dx + x^3(1 + u)e^u du, \end{aligned}$$

because the coefficient of dx is $x^2[1 - u^2 e^u + (u + u^2)e^u] = x^2(1 + ue^u)$. After division by x^2 this is already exact, since $d(ue^u) = (1 + u)e^u du$ and the product rule give

$$(1 + ue^u)dx + x(1 + u)e^u du = d[x(1 + ue^u)] = 0.$$

Therefore $x(1 + ue^u) = C$; with $xu = y$ this reads

$$\boxed{x + ye^{y/x} = C}.$$

A direct check avoids division by $1 + ue^u$ or $1 + u$. Using $d(y/x) = (x\,dy - y\,dx)/x^2$,

$$\begin{aligned} d(x + ye^{y/x}) &= dx + e^u dy + ye^u \frac{x\,dy - y\,dx}{x^2} \\ &= (1 - u^2 e^u)dx + (1 + u)e^u dy, \end{aligned}$$

which is exactly the original differential form divided by x^2. For real u, $ue^u \geq -1/e$ (its minimum, at $u = -1$, where $(ue^u)' = (1 + u)e^u$ vanishes), so $1 + ue^u > 0$. A constant-ratio solution $u = \text{const}$ would need $(1 + ue^u)dx = 0$ in the reduced form, so none was lost. At $u = -1$ the coefficient of dy vanishes, and the level curve can have a vertical tangent; the implicit differential-form solution remains valid there.

7.3 ODEs with Constant Coefficients

Find the general solutions to the following ODEs. Write the solutions in forms that are entirely real (i.e., that contain no complex quantities).

7.3.1 $y''' - 2y'' - y' + 2y = 0.$

Solution.

Substitute $y = e^{rx}$ and cancel the nonzero exponential. The characteristic polynomial factors by grouping:

$$r^3 - 2r^2 - r + 2 = r^2(r-2) - (r-2) = (r-2)(r^2-1) = (r-2)(r-1)(r+1).$$

Its three distinct real roots give

$$\boxed{y = C_1 e^{2x} + C_2 e^{x} + C_3 e^{-x}}.$$

Each exponential solves the ODE because its exponent is a root. In the Wronskian, the kth derivative of e^{rx} is $r^k e^{rx}$, so the factor e^{rx} comes out of each column:

$$W = e^{(2+1-1)x} \det \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 4 & 1 & 1 \end{pmatrix} = e^{2x}[(1+1) - (2+4) + (2-4)] = -6e^{2x} \neq 0,$$

expanding along the first row. So these three solutions are independent and span the three-dimensional solution space.

7.3.2 $y''' - 2y'' + y' - 2y = 0.$

Solution.

For $y = e^{rx}$ the characteristic equation is

$$r^3 - 2r^2 + r - 2 = r^2(r-2) + (r-2) = (r-2)(r^2+1) = 0.$$

The roots are $2, i, -i$. Euler's identity $e^{\pm ix} = \cos x \pm i \sin x$ converts the last pair into the real solutions $(e^{ix} + e^{-ix})/2 = \cos x$ and $(e^{ix} - e^{-ix})/(2i) = \sin x$, giving

$$\boxed{y = C_1 e^{2x} + C_2 \cos x + C_3 \sin x}.$$

Direct differentiation verifies all three terms. Their Wronskian at $x = 0$ (rows: the values, first and second derivatives of $e^{2x}, \cos x, \sin x$) is, expanding along the first row,

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 4 & -1 & 0 \end{pmatrix} = 1 \cdot (0+1) - 1 \cdot (0-4) = 5 \neq 0.$$

By Abel's formula a Wronskian of solutions is either identically zero or never zero, so the three solutions are independent and the three arbitrary real constants provide the general real solution.

7.3.3 $y''' - 3y' + 2y = 0.$

Solution.

The exponential trial gives the cubic $r^3 - 3r + 2$, which vanishes at $r = 1$; dividing by $r - 1$ and factoring the quadratic,

$$r^3 - 3r + 2 = (r - 1)(r^2 + r - 2) = (r - 1)^2(r + 2).$$

The repeated root $r = 1$ needs a second solution in addition to e^x. To see its form without assuming a multiplicity rule, set $y = e^x u$ and write $D = d/dx$. Then $(D - 1)(e^x u) = e^x u'$ and $(D + 2)(e^x u) = e^x(u' + 3u)$; applying $D - 1$ twice to $e^x(u' + 3u)$ differentiates the factor $u' + 3u$ twice, so

$$(D - 1)^2(D + 2)y = e^x(u^{(3)} + 3u'') = 0.$$

Taking $u = 1$ and $u = x$ supplies e^x and xe^x; the remaining solution $u = e^{-3x}$ gives e^{-2x}, from the root $r = -2$. Hence

$$\boxed{y = (C_1 + C_2 x)e^x + C_3 e^{-2x}}.$$

At $x = 0$ the values, first and second derivatives of e^x, xe^x, e^{-2x} are $(1, 0, 1)$, $(1, 1, -2)$, $(1, 2, 4)$, so the Wronskian at 0 is $\det\left(\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & -2 \\ 1 & 2 & 4 \end{smallmatrix}\right) = 1 \cdot (4 + 4) + 1 \cdot (2 - 1) = 9$. Since the ODE has no y'' term, Abel's formula makes the Wronskian constant, so it equals $9 \neq 0$ everywhere, which checks independence and completeness.

7.3.4 $y'' + 2y' + 2y = 0.$

Solution.

The characteristic equation is $r^2 + 2r + 2 = (r + 1)^2 + 1 = 0$, with roots $-1 \pm i$. Equivalently, substituting $y = e^{-x}u$ gives

$$y' = e^{-x}(u' - u), \quad y'' = e^{-x}(u'' - 2u' + u), \quad y'' + 2y' + 2y = e^{-x}(u'' + u).$$

Thus $u'' + u = 0$, so $u = C_1 \cos x + C_2 \sin x$, and

$$\boxed{y = e^{-x}(C_1 \cos x + C_2 \sin x)}.$$

The same real functions are e^{-x} times the real and imaginary parts of e^{ix} in $e^{(-1 \pm i)x} = e^{-x}(\cos x \pm i \sin x)$. For a common factor f, $W(fu_1, fu_2) = f^2 W(u_1, u_2)$ (the terms with f' cancel), so the Wronskian of the two displayed real functions is $e^{-2x}(\cos^2 x + \sin^2 x) = e^{-2x}$, which is nonzero for every real x.

7.4 Second-Order Linear ODEs

7.4.1 Show that Legendre's equation has regular singularities at $x = -1$ and 1, and at ∞ when $l(l + 1) \neq 0$. Classify ∞ also when $l = 0$ or $l = -1$.

Solution.

Write $\lambda = l(l + 1)$ and divide Legendre's equation by $1 - x^2$:

$$y'' + P(x)y' + Q(x)y = 0, \qquad P = -\frac{2x}{1 - x^2}, \quad Q = \frac{\lambda}{1 - x^2}.$$

At $x = 1$, $(x-1)P = 2x/(x+1)$ and $(x-1)^2Q = -\lambda(x-1)/(x+1)$ are analytic. At $x = -1$, $(x+1)P = 2x/(x-1)$ and $(x+1)^2Q = -\lambda(x+1)/(x-1)$ are analytic. The simple poles of P are nonremovable, so both are regular singular points for every λ.

At infinity put $z = 1/x$ and $Y(z) = y(1/z)$. Since $dz/dx = -1/x^2 = -z^2$, the chain rule gives

$$y' = -z^2 Y', \qquad y'' = -z^2 \frac{d}{dz}(-z^2 Y') = z^4 Y'' + 2z^3 Y'.$$

Substitution into the original equation, with $1 - x^2 = (z^2-1)/z^2$ and $-2xy' = 2zY'$, gives

$$(z^2-1)(z^2Y'' + 2zY') + 2zY' + \lambda Y = z^2(z^2-1)Y'' + 2z^3Y' + \lambda Y = 0,$$

and division by $z^2(z^2-1)$ gives

$$Y'' + \frac{2z}{z^2-1}Y' + \frac{\lambda}{z^2(z^2-1)}Y = 0.$$

Thus $z\widetilde{P} = 2z^2/(z^2-1)$ and $z^2\widetilde{Q} = \lambda/(z^2-1)$ are analytic at zero. For $\lambda \neq 0$ the double pole of $\widetilde{Q}$ makes infinity regular singular. The original unqualified question has an exception: if $l = 0$ or $l = -1$, then $\lambda = 0$ and both transformed coefficients are analytic at $z = 0$; infinity is then an ordinary point.

7.4.2 Show that Laguerre's equation, like the Bessel equation, has a regular singularity at $x = 0$ and an irregular singularity at $x = \infty$.

Solution.

For Laguerre's equation $xy'' + (1-x)y' + ny = 0$,

$$P = \frac{1}{x} - 1, \qquad Q = \frac{n}{x}.$$

At $x = 0$, $xP = 1 - x$ and $x^2Q = nx$ are analytic, whereas P has a simple pole; zero is regular singular. To analyze infinity, put $z = 1/x$, so $y' = -z^2Y'$ and $y'' = z^4Y'' + 2z^3Y'$ (as in 7.4.1). A normalized ODE $y'' + P(x)y' + Q(x)y = 0$ becomes $z^4Y'' + 2z^3Y' - z^2P(1/z)Y' + Q(1/z)Y = 0$, and dividing by z^4 gives

$$Y'' + \left[\frac{2}{z} - \frac{P(1/z)}{z^2}\right]Y' + \frac{Q(1/z)}{z^4}Y = 0.$$

Here $P(1/z) = z - 1$ and $Q(1/z) = nz$, so the transformed coefficients are

$$\widetilde{P} = \frac{2}{z} - \frac{z-1}{z^2} = \frac{1}{z} + \frac{1}{z^2}, \qquad \widetilde{Q} = \frac{nz}{z^4} = \frac{n}{z^3}.$$

Since $z\widetilde{P} = 1 + 1/z$ is singular, infinity is irregular even when $n = 0$.

For comparison, Bessel's equation has $P = 1/x$, $Q = 1 - \nu^2/x^2$. At zero, $xP = 1$ and $x^2Q = x^2 - \nu^2$ are analytic. At infinity, $P(1/z) = z$ and $Q(1/z) = 1 - \nu^2 z^2$, so

$$\widetilde{P} = \frac{2}{z} - \frac{z}{z^2} = \frac{1}{z}, \qquad \widetilde{Q} = \frac{1-\nu^2z^2}{z^4} = \frac{1}{z^4} - \frac{\nu^2}{z^2},$$

and $z^2\widetilde{Q} = 1/z^2 - \nu^2$ is singular. These establish the same classification for Bessel's equation.

7.4.3 Show that Chebyshev's equation, like the Legendre equation when $l(l+1) \neq 0$, has regular singularities at $x = -1$, 1, and ∞.

Solution.

Chebyshev's equation is $(1-x^2)y'' - xy' + n^2 y = 0$, so

$$P = -\frac{x}{1-x^2}, \qquad Q = \frac{n^2}{1-x^2}.$$

At 1, $(x-1)P = x/(x+1)$ and $(x-1)^2 Q = -n^2(x-1)/(x+1)$ are analytic; at -1 the corresponding expressions are $x/(x-1)$ and $-n^2(x+1)/(x-1)$. The poles of P ensure both points are regular singular.

With $z = 1/x$, $P(1/z) = -(1/z)/(1-z^{-2}) = z/(1-z^2)$ and $Q(1/z) = n^2/(1-z^{-2}) = n^2 z^2/(z^2-1)$, so the transformed coefficients of 7.4.2 are

$$\widetilde{P} = \frac{2}{z} - \frac{1}{z(1-z^2)} = \frac{1-2z^2}{z(1-z^2)}, \qquad \widetilde{Q} = \frac{n^2}{z^2(z^2-1)}.$$

Then $z\widetilde{P} = (1-2z^2)/(1-z^2)$ and $z^2\widetilde{Q} = n^2/(z^2-1)$ are analytic at zero. Since $z\widetilde{P} \to 1$ as $z \to 0$, $\widetilde{P}$ always has a simple pole, and infinity is regular singular even for $n = 0$. The Instructor's Manual's first displayed normalized Chebyshev equation has $+x/(1-x^2)$ multiplying y'. Division of the original equation gives $-x/(1-x^2)$ instead; its transformed P coefficient must therefore also be the one calculated above. The singularity classification is unchanged.

7.4.4 Show that Hermite's equation has no singularity other than an irregular singularity at $x = \infty$.

Solution.

In $y'' - 2xy' + 2\alpha y = 0$, the normalized coefficients $P = -2x$ and $Q = 2\alpha$ are polynomials. They have no singularities at any finite complex x. For $z = 1/x$, $P(1/z) = -2/z$ and $Q(1/z) = 2\alpha$, so the transformation of 7.4.2 gives $\widetilde{P} = 2/z + (2/z)/z^2$, $\widetilde{Q} = 2\alpha/z^4$, i.e., the transformed equation is

$$Y'' + \left(\frac{2}{z} + \frac{2}{z^3}\right) Y' + \frac{2\alpha}{z^4} Y = 0.$$

Multiplying the coefficient of Y' by z gives $2 + 2/z^2$, which is not analytic or finite at zero. Thus infinity is an irregular singular point for every α, including $\alpha = 0$.

7.4.5 Show that the substitution

$$x \to \frac{1-x}{2}, \quad a = -l, \quad b = l+1, \quad c = 1$$

converts the hypergeometric equation into Legendre's equation. Use $t(t-1)u'' + [(1+a+b)t - c]u' + ab\,u = 0$; Table 7.1 has the wrong sign on c.

Solution.

Distinguish the hypergeometric variable t from the new variable x:

$$t(t-1)u_{tt} + [(1+a+b)t - c]u_t + ab\,u = 0, \qquad t = \frac{1-x}{2}, \quad y(x) = u(t).$$

The coefficient is $(1+a+b)t - c$, as in Exercise 7.5.5. Table 7.1 of the textbook prints $+c$ in this sign convention; with it the coefficient of u_t would be $2t + 1 = 2 - x$, and the steps below would

give $(1-x^2)y'' + (4-2x)y' + l(l+1)y = 0$, which is not Legendre's equation. Since $x = 1-2t$, $dx/dt = -2$, and the chain rule gives $u_t = -2y'$ and $u_{tt} = 4y''$. Also $t - 1 = -(1+x)/2$, so $t(t-1) = -(1-x)(1+x)/4 = (x^2-1)/4$. With $a = -l$, $b = l+1$, $c = 1$,

$$1 + a + b = 2, \qquad 2t - 1 = -x, \qquad ab = -l(l+1).$$

Substitution therefore gives

$$(x^2-1)y'' + (-x)(-2y') - l(l+1)y = 0.$$

Multiplying by -1 yields exactly

$$\boxed{(1-x^2)y'' - 2xy' + l(l+1)y = 0}.$$

7.5 Series Solutions—Frobenius' Method

7.5.1 *Uniqueness theorem.* The function $y(x)$ satisfies a second-order, linear, homogeneous differential equation. At $x = x_0$, $y(x) = y_0$ and $dy/dx = y_0'$. Show that $y(x)$ is unique, in that no other solution of this differential equation passes through the points (x_0, y_0) with a slope of y_0'. Assume a connected interval on which the normalized coefficients are continuous. For the Taylor-series hint, assume they are analytic and x_0 is an ordinary point.

Hint. Assume a second solution satisfying these conditions and compare the Taylor series expansions.

Solution.

Uniqueness is guaranteed when the normalized coefficients are continuous near the initial point (in particular at an ordinary point); at a singular point it can fail. The hint using Taylor series additionally requires analytic coefficients. The unqualified claim, and the Instructor's Manual's extension to regular singular points, are false. For example, $xy'' - y' = 0$ (regular singular at 0) has the family $y = Cx^2$, since $x(2C) - 2Cx = 0$; all members have $y(0) = y'(0) = 0$, although they differ elsewhere.

For the Taylor proof, write $y'' + P(x)y' + Q(x)y = 0$, with P, Q analytic at x_0. If y and $\widetilde{y}$ share the initial data, their difference $u = y - \widetilde{y}$ solves the same equation (it is linear and homogeneous) and has $u(x_0) = u'(x_0) = 0$. The equation gives $u''(x_0) = -P(x_0)u'(x_0) - Q(x_0)u(x_0) = 0$. Differentiating k times and applying the product rule gives

$$u^{(k+2)} = -\sum_{j=0}^{k} \binom{k}{j} \{P^{(j)}u^{(k-j+1)} + Q^{(j)}u^{(k-j)}\}.$$

The right side involves derivatives of u only up to order $k+1$ (the term $j = 0$). If all derivatives through order $k+1$ vanish at x_0, the right side vanishes there, and so does $u^{(k+2)}(x_0)$. Induction on k therefore makes every Taylor coefficient of u zero. The hint presupposes that u, like the series solutions constructed in Section 7.5, is analytic at the ordinary point x_0, i.e., equal to its convergent Taylor series there. This is true, but Section 7.5 only constructs series solutions; that *every* solution is analytic is normally derived from the uniqueness theorem itself, so the integral estimate below is the self-contained proof. Granting the hint's assumption, $u = 0$ in a neighborhood of x_0. The same argument applies at every point of the connected ordinary interval: the set where $u = u' = 0$ is

closed (by continuity) and open (by this argument), and it contains x_0, so it is the whole interval. This proves uniqueness throughout it.

For merely continuous P, Q, one can replace the Taylor argument by an integral estimate. With zero initial data, integrating u' and $u'' = -Pu' - Qu$ from x_0 gives

$$u(x) = \int_{x_0}^{x} u'(t)dt, \qquad u'(x) = -\int_{x_0}^{x} [P(t)u'(t) + Q(t)u(t)]dt.$$

On a compact interval $[x_0, x_1]$ to the right, P and Q are continuous, hence bounded, so we can choose a constant K with $K \geq 1 + |P(t)| + |Q(t)|$ for all $t \in [x_0, x_1]$, and put $F(x) = \int_{x_0}^{x} (|u| + |u'|)dt$. The two integrals give $|u(x)| \leq F(x)$ and $|u'(x)| \leq \int_{x_0}^{x} (|P||u'| + |Q||u|)dt \leq (K-1)F(x)$, so $F' = |u| + |u'| \leq KF$, and $F(x_0) = 0$. Thus $(e^{-K(x-x_0)}F)' = e^{-K(x-x_0)}(F' - KF) \leq 0$, so $e^{-K(x-x_0)}F(x) \leq F(x_0) = 0$, while $F \geq 0$; this forces $F = 0$, hence $F' = |u| + |u'| = 0$ and $u = u' = 0$. Reverse the integration direction for points to the left. This establishes the continuous-coefficient version without assuming that a smooth function equals its Taylor series.

7.5.2 A series solution of Eq. (7.23) is attempted, expanding about the point $x = x_0$. If x_0 is an ordinary point, show that the indicial equation has roots $s = 0, 1$.

Solution.

Set $t = x - x_0$ and write the normalized equation as $y'' + P(t)y' + Q(t)y = 0$, where P, Q are analytic at $t = 0$. For a Frobenius trial

$$y = \sum_{j=0}^{\infty} a_j t^{s+j}, \quad a_0 \neq 0, \quad y' = \sum_{j} (s+j)a_j t^{s+j-1}, \quad y'' = \sum_{j} (s+j)(s+j-1)a_j t^{s+j-2},$$

the lowest possible power is t^{s-2}. Only y'' contributes to it: since $P(t) = P(0) + O(t)$ and $Q(t) = Q(0) + O(t)$ have no negative powers, Py' starts at t^{s-1} and Qy at t^s. Hence

$$a_0 s(s-1) = 0, \qquad \boxed{s = 0 \text{ or } s = 1}.$$

These represent a solution that can be nonzero at x_0 and one with a simple zero there. The coefficient of t^{s-1} receives $(s+1)sa_1$ from y'' (term $j = 1$) and $P(0)sa_0$ from Py', so it is $s(s+1)a_1 + sP(0)a_0 = 0$; for $s = 1$ this gives $a_1 = -P(0)a_0/2$, so ordinary-point exponents alone do not generally force a_1 to vanish.

7.5.3 In the development of a series solution of the simple harmonic oscillator (SHO) equation, the second series coefficient a_1 was neglected except to set it equal to zero. From the coefficient of the next-to-the-lowest power of x, x^{s-1}, develop a second-indicial type equation.

(a) (SHO equation with $s = 0$). Show that a_1 may be assigned any finite value (including zero).

(b) (SHO equation with $s = 1$). Show that a_1 must be set equal to zero.

Solution.

Use $y'' + \omega^2 y = 0$ and $y = \sum_{j \geq 0} a_j x^{s+j}$ with $a_0 \neq 0$. Substitution gives

$$\sum_{j \geq 0} (s+j)(s+j-1)a_j x^{s+j-2} + \omega^2 \sum_{j \geq 0} a_j x^{s+j} = 0.$$

The coefficients of x^{s-2} and x^{s-1} come from the first sum only ($j = 0$ and $j = 1$); they are respectively

$$s(s-1)a_0 = 0, \qquad \boxed{s(s+1)a_1 = 0}.$$

Replacing j by $j+2$ in the first sum, the coefficient of x^{s+j}, $j \geq 0$, gives

$$(s+j+2)(s+j+1)a_{j+2} + \omega^2 a_j = 0.$$

(a) For $s = 0$ the second equation is $0 = 0$, leaving a_1 arbitrary. The subsequent equations give

$$a_{j+2} = -\frac{\omega^2 a_j}{(j+2)(j+1)},$$

so the even coefficients depend on a_0 and the odd coefficients on a_1: iterating, $a_{2k} = (-\omega^2)^k a_0/(2k)!$ and $a_{2k+1} = (-\omega^2)^k a_1/(2k+1)!$. In particular, comparison with the Maclaurin series of $\cos\omega x$ and $\sin\omega x$ gives $y = a_0\cos\omega x + (a_1/\omega)\sin\omega x$ for $\omega \neq 0$.

(b) For $s = 1$ the second equation is $2a_1 = 0$, hence $a_1 = 0$. The recurrence becomes $a_{j+2} = -\omega^2 a_j/[(j+3)(j+2)]$, so $a_{2k} = (-\omega^2)^k a_0/(2k+1)!$ and the series $a_0\sum_k (-1)^k\omega^{2k}x^{2k+1}/(2k+1)!$ is $a_0\sin(\omega x)/\omega$. Thus setting $a_1 = 0$ in the $s = 0$ series is harmless only when the separate $s = 1$ solution is also retained. For $\omega = 0$ the two solutions reduce to $1, x$.

7.5.4 Analyze the series solutions of the following differential equations to see when a_1 may be set equal to zero without irrevocably losing anything and when a_1 must be set equal to zero.

(a) Legendre, (b) Chebyshev, (c) Bessel, (d) Hermite.

ANS. (a) Legendre, (b) Chebyshev, and (d) Hermite: For $s = 0$, a_1 may be set equal to zero; for $s = 1$, a_1 must be set equal to zero.
(c) Bessel: a_1 must be set equal to zero (except for $s = \pm n = -\frac{1}{2}$).

Solution.

Insert $y = \sum_{j\geq 0} a_j x^{s+j}$, $a_0 \neq 0$. For Legendre, Chebyshev, and Hermite the coefficient multiplying y' starts with x, so their lowest two powers come only from y'':

$$s(s-1)a_0 = 0, \qquad s(s+1)a_1 = 0.$$

The coefficient of x^{s+j} receives $(s+j+2)(s+j+1)a_{j+2}$ from y'' and multiples of a_j from the other terms: $-(s+j)(s+j-1)-2(s+j)+l(l+1) = -(s+j)(s+j+1)+l(l+1)$ from $-x^2y''-2xy'+l(l+1)y$ (Legendre), $-(s+j)(s+j-1)-(s+j)+n^2 = -(s+j)^2+n^2$ from $-x^2y''-xy'+n^2y$ (Chebyshev), and $-2(s+j)+2\alpha$ from $-2xy'+2\alpha y$ (Hermite). Setting the coefficient to zero gives the explicit recurrences, respectively,

$$\begin{aligned}
\text{Legendre:} \quad & a_{j+2} = \frac{(s+j)(s+j+1)-l(l+1)}{(s+j+2)(s+j+1)}a_j, \\
\text{Chebyshev:} \quad & a_{j+2} = \frac{(s+j)^2-n^2}{(s+j+2)(s+j+1)}a_j, \\
\text{Hermite:} \quad & a_{j+2} = \frac{2(s+j-\alpha)}{(s+j+2)(s+j+1)}a_j.
\end{aligned}$$

Thus (a), (b), and (d) have arbitrary a_1 for $s = 0$, but require $a_1 = 0$ for $s = 1$. Each recurrence links a_{j+2} to a_j only, so it preserves parity. Moreover, each depends on s and j only through the

power $s+j$; hence the odd coefficients $a_1, a_3, \ldots$ of the $s=0$ branch obey the same recurrence as $a_0, a_2, \ldots$ of the $s=1$ branch, and the $s=0$ series with $a_1 \neq 0$ is a_0 times the even solution plus a_1 times the $s=1$ solution. Choosing $a_1 = 0$ in the $s=0$ branch gives the even solution; the omitted odd branch is then supplied separately by $s=1$. It would be lost if only the even branch were retained.

(c) Bessel's equation is $x^2y'' + xy' + (x^2 - n^2)y = 0$. At x^{s+j} the first two terms contribute $[(s+j)(s+j-1) + (s+j)]a_j = (s+j)^2 a_j$ and $x^2 y$ contributes a_{j-2}. Its lowest two coefficients and subsequent recurrence are therefore

$$(s^2 - n^2)a_0 = 0, \quad [(s+1)^2 - n^2]a_1 = 0, \quad [(s+j)^2 - n^2]a_j + a_{j-2} = 0 \quad (j \geq 2).$$

At either indicial root $s^2 = n^2$, the second coefficient reduces to $(s+1)^2 - n^2 = s^2 + 2s + 1 - n^2 = 2s+1$, so $(2s+1)a_1 = 0$. Hence a_1 is forced to zero except when

$$\boxed{s = -\tfrac{1}{2}, \qquad n^2 = \tfrac{1}{4}}.$$

In that exceptional lower-root branch, a_1 is free and generates the other, $s = +1/2$, solution. This conclusion concerns a_1 only; for integer orders a later vanishing denominator in the lower-root recurrence can require a logarithmic second solution.

7.5.5 Obtain a series solution of the hypergeometric equation

$$x(x-1)y'' + [(1+a+b)x - c]y' + aby = 0.$$

Test your solution for convergence.

Solution.

Multiply the equation by -1 to use $x(1-x)y'' + [c - (a+b+1)x]y' - aby = 0$. Insert $y = \sum_{j \geq 0} A_j x^{s+j}$, $A_0 \neq 0$. The contributions from $xy'' + cy'$ at power x^{s+j-1} are $[(s+j)(s+j-1) + c(s+j)]A_j = (s+j)(s+j+c-1)A_j$; those from $-x^2y'' - (a+b+1)xy' - aby$ at power x^{s+j} are

$$-[(s+j)(s+j-1) + (a+b+1)(s+j) + ab]A_j = -(s+j+a)(s+j+b)A_j,$$

because the bracket equals $(s+j)^2 + (a+b)(s+j) + ab$. The lowest power x^{s-1} gives the indicial equation; replacing j by $j+1$ in the first group, the coefficient of x^{s+j} gives $(s+j+1)(s+j+c)A_{j+1} - (s+j+a)(s+j+b)A_j = 0$. Thus

$$s(s+c-1) = 0, \quad \boxed{A_{j+1} = \frac{(s+j+a)(s+j+b)}{(s+j+1)(s+j+c)} A_j}.$$

With $(q)_j = q(q+1)\cdots(q+j-1)$ and $(q)_0 = 1$, iterating from $A_0 = 1$ for $s=0$ gives $A_j = \prod_{i=0}^{j-1} \frac{(i+a)(i+b)}{(i+1)(i+c)} = \frac{(a)_j (b)_j}{(c)_j j!}$; for $s = 1-c$ the factors become $(i+a-c+1)(i+b-c+1)/[(i+1)(i+2-c)]$. The two generic Frobenius solutions are therefore

$$y_1 = \sum_{j=0}^{\infty} \frac{(a)_j (b)_j}{(c)_j j!} x^j = 1 + \frac{ab}{c}x + \frac{a(a+1)b(b+1)}{2c(c+1)}x^2 + \cdots,$$

$$y_2 = x^{1-c} \sum_{j=0}^{\infty} \frac{(a-c+1)_j (b-c+1)_j}{(2-c)_j j!} x^j.$$

For noninteger c no denominator $(j+1)(j+c)$ or $(j+1)(j+2-c)$ vanishes, and the two solutions are independent: their leading powers are 1 and x^{1-c}, and with $y_1 = 1+O(x)$, $y_2 = x^{1-c}[1+O(x)]$ their Wronskian $y_1y_2' - y_1'y_2$ starts as $(1-c)x^{-c} \neq 0$. A branch of the fractional power is fixed when complex x is used.

The displayed coefficient formulas require nonzero denominators. At integer c, one cannot automatically assert two independent solutions of these exact forms. A valid first series always comes from the root $s = 0$ if $c \geq 1$ and $s = 1-c$ if $c \leq 0$: its recurrence denominators do not vanish. When c is an integer, the roots 0 and $1-c$ coincide ($c = 1$) or differ by a nonzero integer; then the lower root either repeats the first series or its recurrence meets a zero denominator (at $j = c-2$ if $c \geq 2$, at $j = -c$ if $c \leq 0$), so the second solution may require a logarithm. To construct it, normalize the equation and split the first-derivative coefficient:

$$P(x) = \frac{c-(a+b+1)x}{x(1-x)} = \frac{c}{x} + \frac{c-a-b-1}{1-x}.$$

Consequently $e^{-\int P\,dx} = x^{-c}(1-x)^{c-a-b-1}$, and reduction of order (the second-solution formula, Eq. (7.67) of Section 7.6) gives

$$y_2 = y_1 \int^x \frac{t^{-c}(1-t)^{c-a-b-1}}{y_1(t)^2}\,dt.$$

Expand the analytic factors and integrate each power near zero; a t^{-1} term produces $\ln t$. On a branch avoiding $0, 1$ and zeros of y_1, its Wronskian is $W(y_1, y_2) = y_1^2\,(y_2/y_1)' = x^{-c}(1-x)^{c-a-b-1} \neq 0$. This supplies the missing independent solution in every integer-c case. If a numerator factor $s+j+a$ or $s+j+b$ vanishes at some j before a denominator factor does, then $A_{j+1} = 0$ and the series terminates; the resulting polynomial (times x^s) is a solution, because the undivided recurrence $(s+j+1)(s+j+c)A_{j+1} = (s+j+a)(s+j+b)A_j$ then holds for every j, reading $0 = 0$ beyond the termination, including at a zero denominator.

For a nonterminating series, its successive terms satisfy

$$\left|\frac{A_{j+1}x^{j+1}}{A_jx^j}\right| = |x|\left|\frac{(j+s+a)(j+s+b)}{(j+s+1)(j+s+c)}\right| \longrightarrow |x|.$$

By the ratio test it converges absolutely for $|x| < 1$ and diverges for $|x| > 1$. For boundary convergence put $\sigma = c-a-b$. Writing $(q)_j = \Gamma(j+q)/\Gamma(q)$, the coefficients of y_1 are

$$A_j = \frac{\Gamma(c)}{\Gamma(a)\Gamma(b)}\,\frac{\Gamma(j+a)}{\Gamma(j+1)}\,\frac{\Gamma(j+b)}{\Gamma(j+c)}.$$

For large j use Stirling's series (Section 13.4) with $\Gamma(z+1) = z\Gamma(z)$: $\ln\Gamma(z) = (z-\frac12)\ln z - z + \frac12\ln 2\pi + O(z^{-1})$. For fixed u, $\ln(j+u) = \ln j + u/j + O(j^{-2})$, so $(j+u-\frac12)\ln(j+u) = (j+u-\frac12)\ln j + u + O(j^{-1})$ and

$$\ln\Gamma(j+u) = (j+u-\tfrac12)\ln j - j + \tfrac12\ln 2\pi + O(j^{-1}).$$

Subtracting the same expansion with v leaves $(u-v)\ln j + O(j^{-1})$, and exponentiating gives $\Gamma(j+u)/\Gamma(j+v) = j^{u-v}[1+O(j^{-1})]$. With $(u,v) = (a,1)$ and (b,c) this gives, in a nonterminating generic case (a, b, c not zero or negative integers),

$$A_j = Kj^{(a-1)+(b-c)}[1+O(j^{-1})] = Kj^{-\sigma-1}[1+O(j^{-1})], \qquad K = \frac{\Gamma(c)}{\Gamma(a)\Gamma(b)} \neq 0.$$

Therefore at $x = 1$ convergence holds precisely when $\operatorname{Re}\sigma > 0$. Indeed, absolute convergence is then comparison with $j^{-1-\operatorname{Re}\sigma}$. If $-1 < \operatorname{Re}\sigma \le 0$, compare the leading powers $Kj^{-\sigma-1}$ with $\int_1^N t^{-\sigma-1}dt = (N^{-\sigma}-1)/(-\sigma)$, or $\ln N$ when $\sigma = 0$. Since $|d(t^{-\sigma-1})/dt| = |\sigma+1|\,t^{-\operatorname{Re}\sigma-2}$, for $j \le t \le j+1$ we have $|j^{-\sigma-1} - t^{-\sigma-1}| \le |\sigma+1|\,j^{-\operatorname{Re}\sigma-2}$, which is summable. Splitting the integral at the integers,

$$\sum_{j=1}^{N-1} j^{-\sigma-1} - \int_1^N t^{-\sigma-1}dt = \sum_{j=1}^{N-1}\int_j^{j+1}\bigl(j^{-\sigma-1} - t^{-\sigma-1}\bigr)dt$$

therefore tends to a finite limit. The partial sums thus contain a term proportional to $N^{-\sigma}$, or $\ln N$, with no limit: $|N^{-\sigma}| = N^{-\operatorname{Re}\sigma} \to \infty$ if $\operatorname{Re}\sigma < 0$, while $N^{-i\tau} = e^{-i\tau\ln N}$ keeps circling the unit circle if $\sigma = i\tau \neq 0$. The $O(j^{-\operatorname{Re}\sigma-2})$ correction is summable. For $\operatorname{Re}\sigma \le -1$ even the individual terms fail to tend to zero. At $x = -1$, put $p = \sigma + 1$. When $\operatorname{Re}p > 0$, $j^{-p} \to 0$ and $(j+1)^{-p} - j^{-p} = O(j^{-\operatorname{Re}p-1})$ is absolutely summable. Summation by parts with the bounded partial sums of $(-1)^j$ (Dirichlet's test) therefore proves convergence. The coefficient correction is also absolutely summable. Thus the condition there is $\operatorname{Re}\sigma > -1$, with absolute convergence only for $\operatorname{Re}\sigma > 0$. The second series has the same σ, since $(2-c) - (a-c+1) - (b-c+1) = c-a-b$. For any other point on $|x| = 1$ with $x \neq 1$, the partial sums of x^j are bounded by $2/|1-x|$, so the same summation-by-parts argument as at -1 gives convergence precisely when $\operatorname{Re}\sigma > -1$. A terminating power-series factor is a polynomial and converges for every finite x; the second solution still retains its possibly singular or multivalued prefactor x^{1-c}.

7.5.6 Obtain two series solutions of the confluent hypergeometric equation

$$xy'' + (c-x)y' - ay = 0.$$

Test your solutions for convergence. For integer c, allow a generalized Frobenius solution containing a logarithm; two independent pure power-series forms need not exist.

Solution.

Put $y = \sum_{j\ge0} A_j x^{s+j}$, $A_0 \neq 0$. Then $xy''+cy'$ contributes $[(s+j)(s+j-1)+c(s+j)]A_j x^{s+j-1} = (s+j)(s+j+c-1)A_j x^{s+j-1}$, whereas $-xy' - ay$ contributes $-(s+j+a)A_j x^{s+j}$. Matching powers (the lowest, x^{s-1}, then x^{s+j} after replacing j by $j+1$ in the first group),

$$s(s+c-1) = 0, \qquad \boxed{A_{j+1} = \frac{s+j+a}{(s+j+1)(s+j+c)}A_j}.$$

Iterating from $A_0 = 1$ gives $A_j = (a)_j/[(c)_j j!]$ for $s = 0$; for $s = 1-c$ the factors are $(i+a-c+1)/[(i+1)(i+2-c)]$, $i = 0,\dots,j-1$. For noninteger c no denominator vanishes, and the independent solutions, each with leading coefficient one, are

$$y_1 = \sum_{j=0}^{\infty}\frac{(a)_j}{(c)_j j!}x^j = 1 + \frac{a}{c}x + \frac{a(a+1)}{2c(c+1)}x^2 + \cdots,$$

$$y_2 = x^{1-c}\sum_{j=0}^{\infty}\frac{(a-c+1)_j}{(2-c)_j j!}x^j.$$

The first is conventionally $M(a,c,x)$; the second is $x^{1-c}M(a-c+1, 2-c, x)$. As in 7.5.5, $y_1 = 1+O(x)$ and $y_2 = x^{1-c}[1+O(x)]$, so their Wronskian has leading term $(1-c)x^{-c}$, verifying independence. For either coefficient sequence,

$$\left|\frac{A_{j+1}x^{j+1}}{A_j x^j}\right| = |x|\left|\frac{s+j+a}{(s+j+1)(s+j+c)}\right| \sim \frac{|x|}{j} \longrightarrow 0.$$

Thus both power-series factors converge for every finite x. The prefactor x^{1-c} may still be singular or multivalued at zero; convergence of its power-series factor does not remove that singularity. A terminating power-series factor is a polynomial; the second solution still includes its displayed power prefactor.

For integer c, two independent pure Frobenius series need not exist; a logarithmic solution may be necessary. The ordinary formulas above must not be used through a zero denominator.

A constructive series prescription for all integer c is as follows. Choose $s = \max(0, 1-c)$ and obtain $y_1 = x^s A(x)$ with $A(0) = 1$ by the displayed recurrence; its denominators are nonzero. Write

$$A(x)^2 = \sum_{j\ge 0} B_j x^j, \quad B_j = \sum_{r=0}^{j} A_r A_{j-r}, \quad \frac{e^x}{A(x)^2} = \sum_{j\ge 0} h_j x^j.$$

Since $B_0 = 1$, comparing the coefficients of x^k in $A(x)^2 \sum_j h_j x^j = e^x$, i.e., $\sum_{j=0}^{k} B_j h_{k-j} = 1/k!$, gives $h_0 = 1$ and $h_k = 1/k! - \sum_{j=1}^{k} B_j h_{k-j}$. The normalized equation has $P = c/x - 1$, so $e^{-\int P\,dx} = x^{-c}e^x$, and reduction of order gives

$$y_2 = y_1 \int^x \frac{t^{-c}e^t}{y_1(t)^2}\,dt = y_1 \int^x t^{-c-2s}\frac{e^t}{A(t)^2}\,dt = y_1 \int^x \sum_{k\ge 0} h_k t^{k-c-2s}\,dt.$$

Integrating term by term, t^{k-c-2s} gives $x^{k-c-2s+1}/(k-c-2s+1)$, except for $k = c+2s-1$, where it gives $\ln x$. This yields the explicit generalized series

$$y_2 = x^s A(x)\left[\sum_{\substack{k\ge 0\\ k\ne c+2s-1}} \frac{h_k x^{k-c-2s+1}}{k-c-2s+1} + h_{c+2s-1}\ln x\right].$$

Here $c+2s-1$ is a nonnegative integer ($c-1\ge 0$ if $c\ge 1$, and $1-c\ge 1$ if $c\le 0$), and the logarithm is omitted if its coefficient vanishes. This expansion converges locally, near zero where $A(x)\ne 0$; its Wronskian with y_1 is $y_1^2(y_2/y_1)' = e^x x^{-c} \ne 0$. For example, $a = c = 1$ gives $s = 0$, $A_j = (1)_j/[(1)_j j!] = 1/j!$, so $y_1 = e^x$; then $e^x/A^2 = e^{-x}$, $h_k = (-1)^k/k!$, $c+2s-1 = 0$, and $y_2 = e^x[\ln x + \sum_{k\ge 1}(-x)^k/(kk!)]$, displaying the necessary logarithm explicitly. Real solutions on the negative half-line use $\ln|x|$ and an appropriate real power convention.

7.5.7 A quantum mechanical analysis of the Stark effect (parabolic coordinates) leads to the differential equation

$$\frac{d}{d\xi}\left(\xi\frac{du}{d\xi}\right) + \left(\frac{1}{2}E\xi + \alpha - \frac{m^2}{4\xi} - \frac{1}{4}F\xi^2\right)u = 0.$$

Here α is a constant, E is the total energy, and F is a constant such that Fz is the potential energy added to the system by the introduction of an electric field.

Using the larger root of the indicial equation, develop a power-series solution about $\xi = 0$. Evaluate the first three coefficients in terms of a_0.

ANS. Indicial equation $s^2 - \dfrac{m^2}{4} = 0$,

$$M = |m|,\ u(\xi) = a_0\xi^{M/2}\left[1 - \frac{\alpha}{M+1}\xi + \left(\frac{\alpha^2}{2(M+1)(M+2)} - \frac{E}{4(M+2)}\right)\xi^2 + \cdots\right].$$

Note that the perturbation F does not appear until a_3 is included.

Solution.

The larger indicial root is $|m|/2$ for a real azimuthal quantum number. The printed answer implicitly assumes $m \geq 0$; use $M = |m|$ for either sign. Expand the derivative and multiply by ξ:

$$\xi^2 u'' + \xi u' - \frac{m^2}{4} u + \alpha \xi u + \frac{E}{2} \xi^2 u - \frac{F}{4} \xi^3 u = 0.$$

For $u = \sum_{j \geq 0} a_j \xi^{s+j}$, set negative-index coefficients to zero. At ξ^{s+j}, $\xi^2 u'' + \xi u'$ contributes $[(s+j)(s+j-1) + (s+j)]a_j = (s+j)^2 a_j$, while $\alpha \xi u$, $(E/2)\xi^2 u$ and $-(F/4)\xi^3 u$ contribute αa_{j-1}, $(E/2)a_{j-2}$ and $-(F/4)a_{j-3}$. Matching ξ^{s+j} gives

$$\left[(s+j)^2 - \frac{m^2}{4}\right] a_j + \alpha a_{j-1} + \frac{E}{2} a_{j-2} - \frac{F}{4} a_{j-3} = 0.$$

At $j = 0$ this yields $s^2 - m^2/4 = 0$, so $s = \pm M/2$; choose the larger root $s = M/2$. For $j \geq 1$ the coefficient of a_j is $(M/2 + j)^2 - M^2/4 = j(j + M) \neq 0$, so

$$a_j = \frac{-\alpha a_{j-1} - (E/2)a_{j-2} + (F/4)a_{j-3}}{j(j+M)}.$$

In particular, $j = 1$ and $j = 2$ give, with $-\alpha a_1 = \alpha^2 a_0/(M+1)$,

$$a_1 = -\frac{\alpha}{M+1} a_0, \quad a_2 = \frac{-\alpha a_1 - (E/2)a_0}{2(M+2)} = a_0 \left[\frac{\alpha^2}{2(M+1)(M+2)} - \frac{E}{4(M+2)}\right].$$

Therefore the requested result is

$$\boxed{u = a_0 \xi^{M/2} \left[1 - \frac{\alpha}{M+1}\xi + \left(\frac{\alpha^2}{2(M+1)(M+2)} - \frac{E}{4(M+2)}\right)\xi^2 + \cdots\right]}.$$

The next coefficient ($j = 3$, with $-(E/2)a_1 = E\alpha a_0/[2(M+1)]$) is

$$a_3 = \frac{1}{3(M+3)} \left[-\alpha a_2 + \frac{E\alpha}{2(M+1)} a_0 + \frac{F}{4} a_0\right],$$

so F first enters at a_3, as claimed.

7.5.8 For the special case of no azimuthal dependence, the quantum mechanical analysis of the hydrogen molecular ion leads to the equation

$$\frac{d}{d\eta}\left[(1-\eta^2)\frac{du}{d\eta}\right] + \alpha u + \beta \eta^2 u = 0.$$

Develop a power-series solution for $u(\eta)$. Evaluate the first three coefficients not forced to vanish by parity (a_0, a_2, a_4) in terms of a_0; special parameter values may make some of them zero.

ANS. Indicial equation $s(s-1) = 0$,

$$u_{k=1} = a_0 \eta \left[1 + \frac{2-\alpha}{6}\eta^2 + \left(\frac{(2-\alpha)(12-\alpha)}{120} - \frac{\beta}{20}\right)\eta^4 + \cdots\right].$$

Solution.

Expanding the differential operator gives $(1-\eta^2)u'' - 2\eta u' + \alpha u + \beta \eta^2 u = 0$. For $u = \sum_{j \geq 0} a_j \eta^{s+j}$, the terms $-\eta^2 u''$, $-2\eta u'$, αu start at η^s and $\beta \eta^2 u$ at η^{s+2}, so the two lowest powers η^{s-2}, η^{s-1} come from u'' alone; their coefficients are

$$s(s-1)a_0 = 0, \qquad s(s+1)a_1 = 0.$$

Thus $s = 0, 1$. For $s = 1$ the second equation, $2a_1 = 0$, forces $a_1 = 0$; for $s = 0$ it leaves a_1 free, and we take $a_1 = 0$ to separate the even and odd solutions (as in 7.5.4, the odd part of the $s = 0$ series is the $s = 1$ solution). At η^{s+j}, u'' contributes $(s+j+2)(s+j+1)a_{j+2}$, the terms $-\eta^2 u'' - 2\eta u' + \alpha u$ contribute $[-(s+j)(s+j-1) - 2(s+j) + \alpha]a_j = [\alpha - (s+j)(s+j+1)]a_j$, and $\beta\eta^2 u$ contributes βa_{j-2}. Equating the coefficients of η^{s+j} to zero then gives

$$\boxed{a_{j+2} = \frac{[(s+j)(s+j+1) - \alpha]a_j - \beta a_{j-2}}{(s+j+2)(s+j+1)}}, \qquad a_{-2} = a_{-1} = 0.$$

For $s = 1$, $j = 0$ has $(s+j)(s+j+1) = 2$ over the denominator $3 \cdot 2 = 6$, and $j = 2$ has $3 \cdot 4 = 12$ over $5 \cdot 4 = 20$:

$$a_2 = \frac{2-\alpha}{6}a_0, \qquad a_4 = \frac{(12-\alpha)a_2 - \beta a_0}{20} = a_0\left[\frac{(2-\alpha)(12-\alpha)}{120} - \frac{\beta}{20}\right].$$

This reproduces the displayed odd solution and fulfills the request for a series solution. For completeness, the even branch $s = 0$ ($j = 0$ has $0 \cdot 1 = 0$ over $2 \cdot 1 = 2$, and $j = 2$ has $2 \cdot 3 = 6$ over $4 \cdot 3 = 12$) has

$$a_2 = -\frac{\alpha}{2}a_0, \qquad a_4 = \frac{(6-\alpha)a_2 - \beta a_0}{12} = a_0\left[\frac{\alpha(\alpha-6)}{24} - \frac{\beta}{12}\right].$$

Thus a normalized pair is

$$u_{\text{even}} = 1 - \frac{\alpha}{2}\eta^2 + \left[\frac{\alpha(\alpha-6)}{24} - \frac{\beta}{12}\right]\eta^4 + \cdots,$$
$$u_{\text{odd}} = \eta\left[1 + \frac{2-\alpha}{6}\eta^2 + \left(\frac{(2-\alpha)(12-\alpha)}{120} - \frac{\beta}{20}\right)\eta^4 + \cdots\right].$$

The initial values $(u, u') = (1, 0)$ and $(0, 1)$ at zero establish independence. Their linear combination is the general local solution; both series converge at least for $|\eta| < 1$, because a power-series solution about an ordinary point converges in every disk about it in which the normalized coefficients are analytic, and here $P = -2\eta/(1-\eta^2)$ and $Q = (\alpha + \beta\eta^2)/(1-\eta^2)$ are analytic for $|\eta| < 1$ (P has poles at $\eta = \pm 1$). At special parameter values some displayed coefficients can vanish, so these are the first three coefficients not forced to vanish by parity, not necessarily three nonzero numbers.

7.5.9 To a good approximation, the interaction of two nucleons may be described by a mesonic potential

$$V = \frac{Ae^{-ax}}{x},$$

attractive for A negative. Show that the resultant Schrödinger wave equation

$$\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + (E - V)\psi = 0$$

has the following series solution through x^3, allowing a displayed coefficient to vanish at special parameter values:

$$\psi = a_0\left\{x + \frac{1}{2}A'x^2 + \frac{1}{6}\left[\frac{1}{2}A'^2 - E' - aA'\right]x^3 + \cdots\right\},$$

where the prime indicates multiplication by $2m/\hbar^2$.

Solution.

Put $A' = 2mA/\hbar^2$, $E' = 2mE/\hbar^2$. The equation becomes

$$\psi'' + \left(E' - \frac{A'e^{-ax}}{x}\right)\psi = 0.$$

Multiplied by x^2 it reads $x^2\psi'' + (E'x^2 - A'xe^{-ax})\psi = 0$; the coefficient of ψ vanishes at $x = 0$, so for $\psi \sim x^s$ only $x^2\psi''$ contributes to the lowest power. The indicial term is therefore $s(s-1)$, and the regular vanishing branch has $s = 1$. Write $\psi = \sum_{j\geq 0} a_j x^{j+1}$ and use $e^{-ax} = \sum_{k\geq 0}(-a)^k x^k/k!$. Then $\psi'' = \sum_{j\geq 1}(j+1)ja_j x^{j-1}$ has coefficient $(j+2)(j+1)a_{j+1}$ at x^j, $E'\psi$ has coefficient $E'a_{j-1}$, and the Cauchy product $(e^{-ax}/x)\psi = \sum_{k\geq 0}\frac{(-a)^k x^k}{k!}\sum_{i\geq 0} a_i x^i$ has coefficient $\sum_{k=0}^{j}(-a)^k a_{j-k}/k!$. At power x^j, the recurrence is therefore

$$(j+2)(j+1)a_{j+1} + E'a_{j-1} - A'\sum_{k=0}^{j}\frac{(-a)^k}{k!}a_{j-k} = 0, \qquad a_{-1} = 0.$$

For $j = 0$ this gives $2a_1 - A'a_0 = 0$, and for $j = 1$ it gives $6a_2 + E'a_0 - A'(a_1 - aa_0) = 0$, i.e., $6a_2 = A'a_1 - aA'a_0 - E'a_0 = (A'^2/2 - aA' - E')a_0$. Consequently

$$a_1 = \frac{A'}{2}a_0, \qquad a_2 = \frac{a_0}{6}\left(\frac{A'^2}{2} - E' - aA'\right),$$

and

$$\boxed{\psi = a_0\left[x + \frac{A'}{2}x^2 + \frac{1}{6}\left(\frac{A'^2}{2} - E' - aA'\right)x^3 + \cdots\right]}.$$

The same coefficients follow by substituting $e^{-ax}/x = x^{-1} - a + (a^2/2)x + \cdots$ and equating the constant and linear terms of the differential equation. The other branch may have a logarithm; the displayed series is the regular branch requested here. The qualification in the statement only makes the claim precise: the series is determined here through x^3, and its displayed coefficients need not be nonzero ($a_1 = A'a_0/2$ vanishes when $A' = 0$, and a_2 when $E' = A'^2/2 - aA'$).

7.5.10 If the parameter b^2 in Eq. (7.53) is equal to 2, Eq. (7.53) becomes

$$y'' + \frac{1}{x^2}y' - \frac{2}{x^2}y = 0.$$

From the indicial equation and the recurrence relation, derive a solution $y = 1 + 2x + 2x^2$. Verify that this is indeed a solution by substituting back into the differential equation.

Solution.

Multiplying by x^2 gives $x^2y'' + y' - 2y = 0$ for $x \neq 0$. For $y = \sum_{j\geq 0} a_j x^{s+j}$, the lowest power x^{s-1} comes only from y' (the terms x^2y'' and $-2y$ start at x^s) and has coefficient sa_0. With $a_0 \neq 0$, the indicial equation $sa_0 = 0$ gives $s = 0$. Then

$$x^2y'' = \sum_{j\geq 0} j(j-1)a_j x^j, \quad y' = \sum_{j\geq 0}(j+1)a_{j+1}x^j, \quad -2y = -\sum_{j\geq 0} 2a_j x^j,$$

the middle sum after replacing j by $j+1$. Matching the coefficients of x^j gives

$$[j(j-1) - 2]a_j + (j+1)a_{j+1} = 0, \qquad \boxed{a_{j+1} = \frac{2 - j(j-1)}{j+1}a_j}.$$

Thus $j = 0, 1, 2$ give $a_1 = 2a_0$, $a_2 = \frac{2}{2}a_1 = 2a_0$, and $a_3 = \frac{2-2}{3}a_2 = 0$, so every later coefficient is zero as well. Taking $a_0 = 1$ yields $\boxed{y = 1 + 2x + 2x^2}$. The direct check is

$$y' = 2 + 4x, \quad y'' = 4, \quad x^2y'' + y' - 2y = 4x^2 + (2 + 4x) - 2(1 + 2x + 2x^2) = 0.$$

The original normalized equation is undefined at zero even though this particular solution extends there analytically. Zero is irregular singular ($P = 1/x^2$, so $xP = 1/x$ is not analytic there), so finding this terminating series does not imply that every solution admits an ordinary Frobenius series there.

7.5.11 The modified Bessel function $I_0(x)$ satisfies the differential equation

$$x^2 \frac{d^2}{dx^2} I_0(x) + x \frac{d}{dx} I_0(x) - x^2 I_0(x) = 0.$$

Given that the leading term in an asymptotic expansion as $x \to +\infty$ is known to be

$$I_0(x) \sim \frac{e^x}{\sqrt{2\pi x}},$$

assume a series of the form

$$I_0(x) \sim \frac{e^x}{\sqrt{2\pi x}} \left\{1 + b_1 x^{-1} + b_2 x^{-2} + \cdots\right\}.$$

Determine the coefficients b_1 and b_2.

ANS. $b_1 = \dfrac{1}{8}, \quad b_2 = \dfrac{9}{128}.$

Solution.

Write $I_0 = Ef$, where $E = e^x/\sqrt{2\pi x}$. Since $\ln E = x - \frac{1}{2}\ln x - \frac{1}{2}\ln(2\pi)$, the derivatives of the prefactor satisfy

$$\frac{E'}{E} = 1 - \frac{1}{2x}, \qquad \frac{E''}{E} = \left(\frac{E'}{E}\right)' + \left(\frac{E'}{E}\right)^2 = 1 - \frac{1}{x} + \frac{3}{4x^2},$$

because $(E'/E)' = 1/(2x^2)$ and $(E'/E)^2 = 1 - 1/x + 1/(4x^2)$. Since $I_0' = E'f + Ef'$ and $I_0'' = E''f + 2E'f' + Ef''$, substitution into $x^2 I_0'' + xI_0' - x^2 I_0 = 0$ and division by E give

$$x^2 f'' + \left(2x^2 \frac{E'}{E} + x\right) f' + \left(x^2 \frac{E''}{E} + x \frac{E'}{E} - x^2\right) f = 0.$$

The coefficient of f' is $(2x^2 - x) + x = 2x^2$, and that of f is $(x^2 - x + \frac{3}{4}) + (x - \frac{1}{2}) - x^2 = \frac{1}{4}$: the terms in $x^2 f$ and xf cancel, leaving

$$\boxed{x^2 f'' + 2x^2 f' + \frac{1}{4} f = 0}.$$

Set $f \sim \sum_{j \geq 0} b_j x^{-j}$, $b_0 = 1$. Termwise,

$$x^2 f'' = \sum_{j \geq 0} j(j+1) b_j x^{-j}, \qquad 2x^2 f' = -\sum_{j \geq 1} 2j b_j x^{1-j} = -\sum_{j \geq 0} 2(j+1) b_{j+1} x^{-j},$$

the last sum after replacing j by $j+1$. Since $j(j+1)+\frac14=(j+\frac12)^2$, the coefficient of x^{-j} gives

$$[j(j+1)+\tfrac14]b_j-2(j+1)b_{j+1}=0,\quad b_{j+1}=\frac{(j+1/2)^2}{2(j+1)}b_j.$$

Therefore ($j=0$ and $j=1$)

$$\boxed{b_1=\frac{1/4}{2}=\frac18},\qquad \boxed{b_2=\frac{9/4}{4}\frac18=\frac{9}{128}}.$$

These are asymptotic coefficients for large positive x, not a convergent series at fixed x: the ratio of successive inverse-power terms, $b_{j+1}x^{-j-1}/(b_jx^{-j})=(j+\frac12)^2/[2(j+1)x]$, behaves as $j/(2x)$ for large j and grows beyond one. Truncation must be understood in the asymptotic sense. The qualification $x\to+\infty$ in the statement is needed because $I_0(x)=\sum_k(x/2)^{2k}/(k!)^2$ is an even function: as $x\to-\infty$, $I_0(x)=I_0(|x|)\sim e^{|x|}/\sqrt{2\pi|x|}$ grows, whereas $e^x/\sqrt{2\pi x}$ is imaginary there and tends to zero.

7.5.12 The even power-series solution of Legendre's equation is given by Exercise 8.3.1. Take $a_0=1$ and n not an even integer, say $n=0.5$. Calculate the partial sums of the series through x^{200}, x^{400}, x^{600}, …, x^{2000} for $x=0.95(0.01)1.00$. Also, write out the individual term corresponding to each of these powers.

Note. This calculation does not constitute proof of convergence at $x=0.99$ or divergence at $x=1.00$, but perhaps you can see the difference in the behavior of the sequence of partial sums for these two values of x.

Solution.

For $n=1/2$, $n(n+1)=\frac34$ and Legendre's equation is $(1-x^2)y''-2xy'+\frac34y=0$. Insert the even series $y=\sum_{j\ge0}a_{2j}x^{2j}$ with $a_0=1$. The coefficient of x^{2j} receives $(2j+2)(2j+1)a_{2j+2}$ from y'' (the term $a_{2j+2}x^{2j+2}$ differentiated twice), $-2j(2j-1)a_{2j}$ from $-x^2y''$, $-4ja_{2j}$ from $-2xy'$, and $\frac34a_{2j}$ from $\frac34y$. Since $2j(2j-1)+4j=2j(2j+1)$, it gives

$$(2j+2)(2j+1)a_{2j+2}+[\tfrac34-2j(2j+1)]a_{2j}=0,$$

so

$$\boxed{a_{2j+2}=\frac{2j(2j+1)-3/4}{(2j+2)(2j+1)}a_{2j}}.$$

For example $j=0$ gives $a_2=(-\frac34)/2=-3/8$, and $j=1$ gives $a_4=\frac{6-3/4}{12}a_2=\frac{7}{16}\left(-\frac38\right)=-21/128$. For each specified x, update the term and sum without repeatedly forming large powers:

$$T_0=1,\quad S_0=1,\quad T_{j+1}=T_jx^2\frac{2j(2j+1)-3/4}{(2j+2)(2j+1)},\quad S_{j+1}=S_j+T_{j+1}.$$

The first table gives $S_J=\sum_{j=0}^{J}a_{2j}x^{2j}$ through the power $N=2J$. The second gives the individual term $T_J=a_Nx^N$ at that same power. The entries are rounded to seven significant digits, a final zero not being printed (0.147621 stands for 0.1476210). The recursion is numerically benign: ordinary double-precision arithmetic reproduces every entry, and so does a high-precision sum of the closed-form coefficients a_{2j} given below.

Partial sums S_J

N	$x=0.95$	0.96	0.97	0.98	0.99	1.00
200	0.2154301	0.1476299	0.0605814	−0.06073655	−0.2554509	−0.6500133
400	0.2154292	0.1476211	0.06048901	−0.06176434	−0.2684287	−0.8540902
600	0.2154292	0.147621	0.0604889	−0.06177416	−0.2694032	−0.9735833
800	0.2154292	0.147621	0.0604889	−0.06177427	−0.2694942	−1.058397
1000	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695036	−1.124197
1200	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695046	−1.177967
1400	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.223433
1600	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.262819
1800	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.297563
2000	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.328643

Last terms T_J

N	$x=0.95$	0.96	0.97	0.98	0.99	1.00
200	-1.036298×10^{-7}	-8.414151×10^{-7}	-6.685141×10^{-6}	-5.199707×10^{-5}	−0.000396098	−0.002956403
400	-1.814542×10^{-12}	-1.19624×10^{-10}	-7.551248×10^{-9}	-4.568307×10^{-7}	-2.650958×10^{-5}	−0.001476811
600	-4.238976×10^{-17}	-2.269016×10^{-14}	-1.13799×10^{-11}	-5.354799×10^{-9}	-2.367089×10^{-6}	−0.0009842326
800	-1.114231×10^{-21}	-4.842588×10^{-18}	-1.92965×10^{-14}	-7.062389×10^{-11}	-2.378191×10^{-7}	−0.000738059
1000	-3.124249×10^{-26}	-1.102487×10^{-21}	-3.490395×10^{-17}	-9.936101×10^{-13}	-2.548795×10^{-8}	−0.0005903918
1200	-9.125533×10^{-31}	-2.614638×10^{-25}	-6.576775×10^{-20}	-1.456206×10^{-14}	-2.845545×10^{-9}	−0.0004919624
1400	-2.741657×10^{-35}	-6.378109×10^{-29}	-1.274658×10^{-22}	-2.195188×10^{-16}	-3.267669×10^{-10}	−0.0004216632
1600	-8.408677×10^{-40}	-1.588298×10^{-32}	-2.521936×10^{-25}	-3.378161×10^{-18}	-3.830633×10^{-11}	−0.0003689429
1800	-2.619901×10^{-44}	-4.018045×10^{-36}	-5.068939×10^{-28}	-5.281188×10^{-20}	-4.561899×10^{-12}	−0.0003279407
2000	-8.264934×10^{-49}	-1.029188×10^{-39}	-1.031567×10^{-30}	-8.359498×10^{-22}	-5.500701×10^{-13}	−0.0002951405

The tables show the difference the Note refers to. At $x=0.99$ every further 200 powers reduce the last term by a factor between about 8 and 15, and the partial sum is fixed to all seven digits, -0.2695047, from $N=1400$ on. At $x=1.00$ the last term only halves from $N=1000$ to $N=2000$, and the partial sums keep falling: by 0.204 from $N=200$ to 400, and still by 0.031 from $N=1800$ to 2000. Their total fall from $N=200$ to 2000, 0.679, matches $\sum_{j=101}^{1000} 0.295/j \approx 0.295\ln(1000/100) = 0.679$ (recall $N=2j$), as expected from the asymptotic form $a_{2j}\approx -0.295/j$ derived below.

The convergence distinction can also be proved rather than inferred from these tables. Since $2j(2j+1)-\frac{3}{4} = 4(j-\frac{1}{4})(j+\frac{3}{4})$ and $(2j+2)(2j+1) = 4(j+\frac{1}{2})(j+1)$, the recursion reads

$$\frac{a_{2j+2}}{a_{2j}} = \frac{(j-\frac{1}{4})(j+\frac{3}{4})}{(j+\frac{1}{2})(j+1)},$$

and because the Pochhammer symbol satisfies $(c)_{j+1} = (c)_j(c+j)$, $(c)_0=1$, the exact coefficients are

$$a_{2j} = \frac{(-1/4)_j(3/4)_j}{(1/2)_j j!}.$$

The ratio of consecutive terms, $a_{2j+2}x^{2j+2}/(a_{2j}x^{2j})$, tends to x^2, so the ratio test gives absolute convergence for $|x|<1$. For $x=1$ write $(c)_j = \Gamma(c+j)/\Gamma(c)$ and use $\Gamma(j+a)/\Gamma(j+b)\sim j^{a-b}$ as $j\to\infty$ (Stirling's formula):

$$a_{2j} = \frac{\Gamma(1/2)}{\Gamma(-1/4)\Gamma(3/4)}\frac{\Gamma(j-\frac{1}{4})}{\Gamma(j+\frac{1}{2})}\frac{\Gamma(j+\frac{3}{4})}{\Gamma(j+1)} \sim \frac{\Gamma(1/2)}{\Gamma(-1/4)\Gamma(3/4)}\, j^{-3/4}j^{-1/4} = -\frac{\sqrt{\pi}}{4\Gamma(3/4)^2}\frac{1}{j},$$

where $\Gamma(-\frac{1}{4}) = \Gamma(\frac{3}{4})/(-\frac{1}{4}) = -4\Gamma(\frac{3}{4})$ by $\Gamma(z+1)=z\Gamma(z)$; the constant is -0.29509. (The last-term table agrees: for $x=1$ and $N=2000$ it gives $1000\,T_{1000} = -0.2951$.) All coefficients after a_0 are negative, since $a_2=-3/8$ and every later ratio ($j\ge 1$ above) is positive. By limit comparison with the harmonic series $\sum 1/j$, the series therefore diverges to $-\infty$ at $x=1$. For every listed $x<1$, the additional factor x^{2j} instead makes the terms exponentially small. The finite numerical sequences alone would not establish either conclusion.

7.5.13

(a) The odd power-series solution of Hermite's equation is given by Exercise 8.3.3. Take $a_0 = 1$. Evaluate this series for $\alpha = 0$, $x = 1, 2, 3$. Cut off your calculation after the last term calculated has dropped below the maximum term by a factor of 10^6 or more. Set an upper bound to the error made in ignoring the remaining terms in the infinite series.

(b) As a check on the calculation of part (a), show that the Hermite series $y_{\text{odd}}(\alpha = 0)$ corresponds to $\int_0^x e^{t^2}\,dt$.

(c) Calculate this integral for $x = 1, 2, 3$.

Solution.

(a) For $\alpha = 0$, Hermite's equation is $y'' - 2xy' = 0$. The odd solution with leading coefficient $a_0 = 1$ has $y(0) = 0$, $y'(0) = 1$. Substitute $y = \sum_{k\ge0} c_k x^{2k+1}$, $c_0 = 1$; after the shift $k \to k+1$ in the second derivative,

$$y'' = \sum_{k\ge0}(2k+3)(2k+2)c_{k+1}x^{2k+1}, \qquad 2xy' = \sum_{k\ge0} 2(2k+1)c_k x^{2k+1},$$

and matching the coefficients of x^{2k+1} gives

$$(2k+3)(2k+2)c_{k+1} = 2(2k+1)c_k, \qquad c_0 = 1.$$

Since $2/(2k+2) = 1/(k+1)$, this says $c_{k+1}/c_k = (2k+1)/[(k+1)(2k+3)]$, which is also the ratio of consecutive values of $1/[k!(2k+1)]$; induction therefore gives $c_k = 1/[k!(2k+1)]$. Thus

$$\boxed{y(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{k!(2k+1)}}.$$

For a fixed positive x, define $t_k = x^{2k+1}/[k!(2k+1)]$. The iteration uses

$$\frac{t_{k+1}}{t_k} = \frac{x^2(2k+1)}{(k+1)(2k+3)}, \qquad S_K = \sum_{k=0}^{K} t_k.$$

The ratio decreases with k, since apart from the constant factor x^2

$$\frac{2k+1}{(k+1)(2k+3)} - \frac{2k+3}{(k+2)(2k+5)} = \frac{4k^2+8k+1}{(k+1)(k+2)(2k+3)(2k+5)} > 0,$$

so the terms rise while the ratio exceeds one and fall afterwards. For $x = 1$ the ratio is $\frac{1}{3}$ already at $k = 0$; for $x = 2$ it is $\frac{4}{3}, \frac{6}{5}, \frac{20}{21}$ at $k = 0, 1, 2$; for $x = 3$ it is $\frac{39}{35} > 1$ at $k = 6$ but $\frac{135}{136} < 1$ at $k = 7$. So the largest term is $t_0 = 1$ for $x = 1$, $t_2 = 3.2$ for $x = 2$, and $t_7 = 3^{15}/(7! \cdot 15) = 189.80$ for $x = 3$. Stop at the first K for which $t_K \le 10^{-6} \max_{j\le K} t_j$. The thresholds are therefore 10^{-6}, 3.2×10^{-6} and 1.898×10^{-4}; the terms $t_8 = 1.459 \times 10^{-6}$, $t_{16} = 1.244 \times 10^{-5}$ and $t_{27} = 2.913 \times 10^{-4}$ still exceed them, while t_9, t_{17} and t_{28} (tabulated below) do not, so $K = 9, 17, 28$. For every subsequent ratio, $t_{j+1}/t_j < x^2/(j+1) \le q := x^2/(K+2)$ when $j \ge K+1$, the first inequality because $(2j+1)/(2j+3) < 1$. Hence $t_{K+1+i} \le q^i t_{K+1}$, and the cutoffs below have $q = \frac{1}{11}, \frac{4}{19}, \frac{3}{10} < 1$, so summing the geometric series gives the rigorous bound

$$\boxed{0 < y - S_K \le B_K := t_{K+1}\sum_{i\ge0} q^i = \frac{t_{K+1}}{1 - x^2/(K+2)}}.$$

This bounds the entire remaining sum, rather than assuming that the last calculated term is itself an error bound. The tabulated bounds are rounded upward; the sums and last terms are rounded to nearest.

x	K	S_K	t_K	B_K
1	9	1.4626517316055	1.450385×10^{-7}	1.443479×10^{-8}
2	17	16.452627042558	2.760027×10^{-6}	7.349021×10^{-7}
3	28	1444.5450849204	9.034328×10^{-5}	3.869592×10^{-5}

(b) The textbook prints the integral as $\int_0^x \exp(x^2)\,dx$, reusing x as integration variable and upper limit; the dummy variable t makes its intended meaning explicit. Since the exponential series $e^{t^2} = \sum_{k\geq 0} t^{2k}/k!$ converges uniformly on every finite integration interval, termwise integration is valid:

$$\int_0^x e^{t^2}\,dt = \sum_{k=0}^{\infty} \frac{1}{k!} \int_0^x t^{2k} dt = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{k!(2k+1)} = y(x).$$

Alternatively $y' = e^{x^2}$ directly gives $y'' = 2xe^{x^2} = 2xy'$ and the correct initial data $y(0) = 0$, $y'(0) = 1$.

(c) Numerical quadrature evaluates the integral independently of the series: a 25-point Gauss–Legendre rule on $[0, x]$, carried out in 30-digit arithmetic, gives

$$\int_0^1 e^{t^2} dt = 1.462651745907181608\ldots,$$
$$\int_0^2 e^{t^2} dt = 16.45262776550723022\ldots,$$
$$\int_0^3 e^{t^2} dt = 1444.5451228927141547\ldots.$$

Every printed digit agrees with library values of $(\sqrt{\pi}/2)\,\mathrm{erfi}(x)$, where $\mathrm{erfi}(x) = -i\,\mathrm{erf}(ix) = (2/\sqrt{\pi})\int_0^x e^{t^2} dt$ is the imaginary error function (put $u = it$ in $\mathrm{erf}(z) = (2/\sqrt{\pi})\int_0^z e^{-u^2} du$); the relative error of the quadrature is below 10^{-26}. The differences from the finite sums in (a), 1.430×10^{-8}, 7.229×10^{-7} and 3.797×10^{-5}, are positive and smaller than the bounds B_K, as an independent numerical check.

7.6 Other Solutions

7.6.1 You know that the three unit vectors $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$ are mutually perpendicular (orthogonal). Show that $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$ are linearly independent. Specifically, show that no relation of the form of Eq. (7.54) exists for $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$.

Solution.

Suppose $a\hat{e}_x + b\hat{e}_y + c\hat{e}_z = 0$. Take the dot product successively with the three unit vectors. Their orthonormality gives

$$a(1) + b(0) + c(0) = 0, \qquad a(0) + b(1) + c(0) = 0, \qquad a(0) + b(0) + c(1) = 0.$$

Thus $a = b = c = 0$. No nontrivial constant linear relation exists, which is exactly linear independence. Equivalently their component matrix is the identity and has determinant one.

7.6.2 The criterion for the linear independence of three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ is that the equation

$$a\mathbf{A} + b\mathbf{B} + c\mathbf{C} = 0,$$

analogous to Eq. (7.54), has no solution other than the trivial $a = b = c = 0$. Using components $\mathbf{A} = (A_1, A_2, A_3)$, and so on, set up the determinant criterion for the existence or nonexistence of a nontrivial solution for the coefficients a, b, and c. Show that your criterion is equivalent to the scalar triple product $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} \neq 0$.

Solution.

The vector equation is the homogeneous component system

$$\begin{pmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0.$$

It has only the zero solution precisely when its square coefficient matrix is invertible, or equivalently when its determinant is nonzero. Expansion down the first column gives

$$\begin{aligned} \det[A\ B\ C] = {} & A_1(B_2C_3 - B_3C_2) - A_2(B_1C_3 - B_3C_1) \\ & + A_3(B_1C_2 - B_2C_1) = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}), \end{aligned}$$

because the three brackets, with the middle sign absorbed, are the components $(B_2C_3 - B_3C_2,\ B_3C_1 - B_1C_3,\ B_1C_2 - B_2C_1)$ of $\mathbf{B} \times \mathbf{C}$. Therefore

$$\boxed{\mathbf{A}, \mathbf{B}, \mathbf{C} \text{ independent } \iff \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) \neq 0}.$$

If the determinant is zero, the matrix has a nonzero null vector and there is a nontrivial relation; this proves the converse as well.

7.6.3 Using the Wronskian determinant, show that the set of functions

$$\left\{1, \frac{x^n}{n!}, (n = 1, 2, \ldots, N)\right\}$$

is linearly independent.

Solution.

The set is $\{1, x, x^2/2!, \ldots, x^N/N!\}$, that is, $x^n/n!$ for $n = 0, \ldots, N$. For columns indexed by $n = 0, \ldots, N$ and rows by derivative order $j = 0, \ldots, N$,

$$\frac{d^j}{dx^j}\frac{x^n}{n!} = \begin{cases} x^{n-j}/(n-j)!, & j \leq n, \\ 0, & j > n. \end{cases}$$

The Wronskian matrix is upper triangular with diagonal entries one:

$$W = \det \begin{pmatrix} 1 & x & x^2/2! & \cdots & x^N/N! \\ 0 & 1 & x & \cdots & x^{N-1}/(N-1)! \\ 0 & 0 & 1 & \cdots & x^{N-2}/(N-2)! \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} = \boxed{1}.$$

If a constant linear combination of the functions were zero, its first N derivatives would be zero too. Invertibility of this matrix would then force every coefficient to vanish.

Without the constant function, the set $\{x, x^2/2!, \dots, x^N/N!\}$ is still independent, but its Wronskian is no longer constant. With rows $j = 0, \dots, N-1$, the entry of column $n = 1, \dots, N$ is $n(n-1)\cdots(n-j+1)x^{n-j}/n!$. Taking $x^n/n!$ out of column n and x^{-j} out of row j leaves the matrix of falling factorials $n(n-1)\cdots(n-j+1)$. Each is n^j plus lower powers of n with coefficients independent of n, so subtracting multiples of earlier rows turns this matrix into the Vandermonde matrix $[n^j]$, of determinant $\prod_{1\le i<k\le N}(k-i) = \prod_{k=2}^{N}(k-1)!$. Hence

$$W\{x, x^2/2!, \dots, x^N/N!\} = \frac{\prod_{k=1}^{N-1} k!}{\prod_{n=1}^{N} n!}\, x^{\sum_{n=1}^{N} n - \sum_{j=0}^{N-1} j} = \boxed{\frac{x^N}{N!}},$$

since the two exponent sums are $N(N+1)/2$ and $N(N-1)/2$. (For $N = 2$: $x \cdot x - 1 \cdot x^2/2 = x^2/2$.) This Wronskian is nonzero at every $x \neq 0$, enough to prove independence; its isolated zero at zero does not imply dependence.

7.6.4 If the Wronskian of two functions y_1 and y_2 is identically zero, show by direct integration that

$$y_1 = cy_2,$$

that is, that y_1 and y_2 are linearly dependent. Assume the functions have continuous derivatives and that at least one of the functions does not vanish in the interval under consideration. Interchange the labels first, if necessary, so that y_2 is nowhere zero; without this convention $y_1 = cy_2$ can fail (for example when $y_2 \equiv 0$ and $y_1 \neq 0$), although y_1 and y_2 are still linearly dependent.

Solution.

Suppose first that y_2 is nowhere zero on the interval. The quotient rule gives

$$\left(\frac{y_1}{y_2}\right)' = \frac{y_1' y_2 - y_1 y_2'}{y_2^2} = -\frac{W}{y_2^2} = 0.$$

Integrating from any fixed x_* yields $y_1(x)/y_2(x) = y_1(x_*)/y_2(x_*) = c$, hence $\boxed{y_1 = cy_2}$. This argument also allows y_1 to have zeros.

If the nowhere-zero function is instead y_1, the directly valid statement is $y_2 = cy_1$. The literal requested orientation $y_1 = cy_2$ fails when $y_2 \equiv 0$ and $y_1 \neq 0$. In that case the same quotient calculation with roles reversed proves linear dependence. If its constant is nonzero one can invert it and recover the displayed orientation; if it is zero, dependence still holds because one function is identically zero. Thus the dependence conclusion requires no division by a function that may vanish.

7.6.5 The Wronskian of two functions is found to be zero at $x_0 - \epsilon \le x \le x_0 + \epsilon$ for arbitrarily small $\epsilon > 0$. Show that this Wronskian vanishes for all x and that the functions are linearly dependent. Assume both functions are analytic on one connected interval containing these points, or that they solve the same normalized second-order ODE with continuous coefficients there.

Solution.

The textbook's statement is false for general differentiable, or even infinitely differentiable, functions, as the counterexample below shows. It needs analyticity on a connected interval, or a common regular second-order ODE. A smooth counterexample on $\mathbb{R}$ is

$$y_1(x) = 1, \qquad y_2(x) = \begin{cases} 0, & x \le 1, \\ e^{-1/(x-1)^2}, & x > 1. \end{cases}$$

To check smoothness at 1, set $u = 1/(x-1)$ on the right. Since $du/dx = -u^2$, so that $d/dx = -u^2\, d/du$, induction shows that every derivative has the form $p(u)e^{-u^2}$ for a polynomial p (differentiating $p(u)e^{-u^2}$ gives $-u^2[p'(u) - 2up(u)]e^{-u^2}$). Each tends to zero as $u \to +\infty$, i.e. as $x \to 1^+$; by the mean value theorem the right-hand derivatives of every order at 1 are then zero, matching every derivative of the zero branch. Thus y_2 is smooth. Its Wronskian with 1 is

$$W = y_2' = \begin{cases} 0, & x \le 1, \\ \dfrac{2e^{-1/(x-1)^2}}{(x-1)^3}, & x > 1. \end{cases}$$

It vanishes in every sufficiently small interval around $x_0 = 0$, but not for $x > 1$. The functions are independent: a relation $a + by_2 = 0$ first forces $a = 0$ on $x \le 1$, then $b = 0$ on $x > 1$.

Under the added analytic hypothesis, $W = y_1y_2' - y_1'y_2$ is analytic. Its vanishing on a nonempty interval makes all its Taylor coefficients there zero; the analytic identity theorem propagates $W = 0$ through the connected domain. If, say, y_2 is not identically zero, its zeros are isolated (identity theorem again), so choose a small interval where it is nonzero. Exercise 7.6.4 gives $y_1 = cy_2$ there, and the analytic identity theorem applied to $y_1 - cy_2$ extends that relation throughout the domain. An identically zero member gives dependence immediately.

Alternatively, for two solutions of the same normalized ODE with continuous P, Q, Abel's identity gives $W(x) = W(x_*)e^{-\int_{x_*}^{x} P(t)dt}$. A single zero then makes W identically zero. At any point x_* the Wronskian matrix $\left(\begin{smallmatrix} y_1 & y_2 \\ y_1' & y_2' \end{smallmatrix}\right)$ is then singular, so it has a null vector $(a, b) \neq (0, 0)$, and $u = ay_1 + by_2$ solves the ODE with $u(x_*) = u'(x_*) = 0$; initial-value uniqueness gives $u \equiv 0$, which is linear dependence. The ODE or analytic assumption is the missing reason local information can be extended globally.

7.6.6 The three functions $\sin x$, e^x, and e^{-x} are linearly independent. No one function can be written as a linear combination of the other two. Show that the Wronskian of $\sin x$, e^x, and e^{-x} vanishes but only at isolated points.

ANS. $W = 4\sin x$, $W = 0$ for $x = \pm n\pi$, $n = 0, 1, 2, \ldots$.

Solution.

Form the determinant with functions as columns:

$$W = \det \begin{pmatrix} \sin x & e^x & e^{-x} \\ \cos x & e^x & -e^{-x} \\ -\sin x & e^x & e^{-x} \end{pmatrix}.$$

Factor e^x and e^{-x} from the last two columns; their product is one. Expanding the remaining determinant

$$\det \begin{pmatrix} \sin x & 1 & 1 \\ \cos x & 1 & -1 \\ -\sin x & 1 & 1 \end{pmatrix}$$

along its first row, with the 2×2 minors $1 - (-1) = 2$, $\cos x - \sin x$ and $\cos x + \sin x$, gives

$$W = 2\sin x - (\cos x - \sin x) + (\cos x + \sin x) = \boxed{4\sin x}.$$

It vanishes exactly at the isolated points $x = k\pi$, $k \in \mathbb{Z}$. Since it is nonzero, for example, at $x = \pi/2$, the functions are linearly independent. The special theorem that an ODE Wronskian is either everywhere zero or nowhere zero concerns solutions of one common regular ODE of the matching order; arbitrary independent functions can have isolated Wronskian zeros, as this example demonstrates.

7.6.7 Consider two functions $\varphi_1 = x$ and $\varphi_2 = |x|$. Since $\varphi_1' = 1$ and $\varphi_2' = x/|x|$, $W(\varphi_1, \varphi_2) = 0$ for any interval, including $[-1, +1]$. Does the vanishing of the Wronskian over $[-1, +1]$ prove that φ_1 and φ_2 are linearly dependent? Clearly, they are not. What is wrong?

Solution.

For $x > 0$, $|x| = x$ and $W = x(1) - 1(x) = 0$. For $x < 0$, $|x| = -x$ and $W = x(-1) - 1(-x) = 0$. But $|x|$ has left derivative -1 and right derivative 1 at zero, so

$$\boxed{W(x, |x|) \text{ is undefined at } x = 0, \text{ not zero there.}}$$

This is the flaw the question asks us to find; the printed argument is an intentional trap. The continuous-derivative hypothesis needed for the usual theorem fails. Furthermore, both functions vanish at $x = 0$, so the quotient argument of Exercise 7.6.4, which needs one function that is nowhere zero, cannot connect the constants on the two sides.

They are independent on that full interval: if $ax + b|x| = 0$, positive x gives $a + b = 0$ and negative x gives $a - b = 0$. Therefore $a = b = 0$. On $0 < x \leq 1$ alone they are dependent, since $|x| = x$ there, and on $-1 \leq x < 0$ alone, since $|x| = -x$ there.

7.6.8 Explain that linear independence does not mean the absence of any dependence. Illustrate your argument with $\cosh x$ and e^x.

Solution.

Linear dependence requires a relation with constant coefficients, $a \cosh x + be^x = 0$, valid throughout an interval. Since $\cosh x = (e^x + e^{-x})/2$, it would read $(\frac{a}{2} + b)e^x + \frac{a}{2}e^{-x} = 0$, and multiplication by e^x gives

$$\left(\frac{a}{2} + b\right) e^{2x} + \frac{a}{2} = 0.$$

Differentiation gives $2(a/2 + b)e^{2x} = 0$, hence $a/2 + b = 0$; the original relation then gives $a/2 = 0$ and $b = 0$. Thus they are linearly independent. Their Wronskian checks this directly:

$$W(\cosh x, e^x) = \cosh x\, e^x - \sinh x\, e^x = e^x(\cosh x - \sinh x) = e^x e^{-x} = 1.$$

They nevertheless satisfy the nonlinear relation

$$\boxed{2\cosh x - e^x - \frac{1}{e^x} = 0}.$$

Writing $Y_1 = \cosh x$, $Y_2 = e^x$, this is $2Y_1 - Y_2 - Y_2^{-1} = 0$. Linear independence excludes constant linear relations only, not all functional or nonlinear relations.

7.6.9 Legendre's differential equation

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$$

has a regular solution $P_n(x)$ and an irregular solution $Q_n(x)$. Show that the Wronskian of P_n and Q_n is given by

$$P_n(x)Q_n'(x) - P_n'(x)Q_n(x) = \frac{A_n}{1 - x^2},$$

with A_n independent of x.

Solution.

Let $u = P_n$, $v = Q_n$ and $W = uv' - u'v$ on an interval not crossing $x = \pm 1$. Differentiate so the cross terms cancel:

$$W' = u'v' + uv'' - u''v - u'v' = uv'' - u''v.$$

The normalized Legendre equation gives $y'' = 2xy'/(1-x^2) - n(n+1)y/(1-x^2)$. Substitution for u'' and v'' yields

$$W' = \frac{2x(uv' - u'v) - n(n+1)(uv - uv)}{1-x^2} = \frac{2x}{1-x^2}W.$$

Equivalently, $[(1-x^2)W]' = -2xW + (1-x^2)W' = 0$. Integrating gives $(1-x^2)W = A_n$, a constant, that is,

$$\boxed{W(P_n, Q_n) = \frac{A_n}{1-x^2}},$$

with A_n constant on the interval. Its value depends on the normalization of the chosen solutions; the differential equation alone determines the x dependence.

7.6.10 Show, by means of the Wronskian, that a linear, second-order, homogeneous ODE of the form

$$y''(x) + P(x)y'(x) + Q(x)y(x) = 0$$

cannot have three independent solutions. Assume P, Q are continuous on a connected interval.

Hint. Assume a third solution and show that the Wronskian vanishes for all x.

Solution.

Work on a connected interval where P, Q are continuous, so the normalized second-order initial-value problem is unique. Suppose y_1, y_2, y_3 solve the equation. In their 3×3 Wronskian,

$$W_3 = \det \begin{pmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{pmatrix},$$

the third row equals $-Q(x)$ times the first row minus $P(x)$ times the second. Hence $W_3(x) = 0$ for every x.

A vanishing Wronskian alone does not prove dependence of arbitrary differentiable functions: x^2 and $x|x|$, whose derivatives are $2x$ and $2|x|$, have $W = x^2 \cdot 2|x| - 2x \cdot x|x| = 0$ everywhere, yet they are independent on $[-1, 1]$ by the argument of Exercise 7.6.7. So complete the argument with the ODE: choose any point x_0 of the interval. The two homogeneous equations

$$\sum_{j=1}^{3} c_j y_j(x_0) = 0, \qquad \sum_{j=1}^{3} c_j y_j'(x_0) = 0$$

have a nonzero constant solution (c_1, c_2, c_3) because there are three unknowns and only two equations. The function $u = \sum c_j y_j$ solves the same ODE and has $u(x_0) = u'(x_0) = 0$. Uniqueness forces $u \equiv 0$. This supplies a nontrivial constant linear relation, proving that three independent solutions cannot exist.

7.6.11 Show the following when the linear second-order differential equation $py'' + qy' + ry = 0$ is expressed in self-adjoint form: Use the coefficients after conversion, so $q = p'$, and work on an interval with $p \neq 0$; part (b) initially requires $y_1 \neq 0$.

(a) The Wronskian is equal to a constant divided by p:

$$W(x) = \frac{C}{p(x)}.$$

(b) A second solution $y_2(x)$ is obtained from a first solution $y_1(x)$ as

$$y_2(x) = Cy_1(x) \int^x \frac{dt}{p(t)[y_1(t)]^2}.$$

Solution.

In self-adjoint form the equation is

$$(py')' + ry = 0, \qquad \text{so } py'' + p'y' + ry = 0.$$

Here p denotes the coefficient after any integrating factor has been applied; the self-adjoint condition is $q = p'$. Work where $p \neq 0$.

(a) For solutions y_1, y_2, subtract y_2 times the first equation from y_1 times the second. The terms involving r cancel:

$$p(y_1y_2'' - y_1''y_2) + p'(y_1y_2' - y_1'y_2) = 0.$$

Since $W = y_1y_2' - y_1'y_2$ has $W' = y_1y_2'' - y_1''y_2$ (the terms $y_1'y_2'$ cancel), this is $pW' + p'W = (pW)' = 0$, and

$$\boxed{W = C/p}.$$

(b) On a subinterval where $y_1 \neq 0$,

$$\left(\frac{y_2}{y_1}\right)' = \frac{y_1y_2' - y_1'y_2}{y_1^2} = \frac{C}{py_1^2}.$$

Integration gives

$$\boxed{y_2 = y_1\left[D + C\int^x \frac{dt}{p(t)y_1(t)^2}\right]}.$$

The D term only repeats the known solution; set $D = 0$ and choose $C \neq 0$ for an independent second solution. The formula is initially local around points with $y_1 \neq 0$ and may then be continued by the ODE.

7.6.12 Transform our linear, second-order ODE

$$y'' + P(x)y' + Q(x)y = 0$$

with P continuously differentiable and Q continuous on the interval, by the substitution

$$y = z \exp\left[-\frac{1}{2}\int^x P(t)\,dt\right]$$

and show that the resulting differential equation for z is

$$z'' + q(x)z = 0,$$

where

$$q(x) = Q(x) - \frac{1}{2}P'(x) - \frac{1}{4}P^2(x).$$

Note. This substitution can be derived by the technique of Exercise 7.6.25.

Solution.

Set $E(x) = \exp[-\frac{1}{2}\int^x P(t)dt]$, so $y = Ez$ and, differentiating $E' = -PE/2$ once more,

$$E' = -\frac{P}{2}E, \qquad E'' = -\frac{P'}{2}E - \frac{P}{2}E' = \left(-\frac{P'}{2} + \frac{P^2}{4}\right)E.$$

The product rule gives

$$y' = E\left(z' - \frac{P}{2}z\right), \qquad y'' = E\left[z'' - Pz' + \left(-\frac{P'}{2} + \frac{P^2}{4}\right)z\right].$$

Substitute all three terms of the ODE:

$$\begin{aligned} y'' + Py' + Qy &= E\left[z'' - Pz' + \left(-\frac{P'}{2} + \frac{P^2}{4}\right)z + Pz' - \frac{P^2}{2}z + Qz\right] \\ &= E\left[z'' + \left(Q - \frac{P'}{2} - \frac{P^2}{4}\right)z\right]. \end{aligned}$$

Since E never vanishes, division by it proves

$$\boxed{z'' + qz = 0, \qquad q = Q - \tfrac{1}{2}P' - \tfrac{1}{4}P^2}.$$

This calculation requires P differentiable where the substitution is used, in addition to the usual existence assumptions on the coefficients.

7.6.13 Use the result of Exercise 7.6.12 to show that the replacement of $\varphi(r)$ by $r\varphi(r)$ may be expected to eliminate the first derivative from the Laplacian in spherical polar coordinates. See also Exercise 3.10.34.

Solution.

The radial part of the spherical Laplacian is

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\varphi}{dr}\right) = \varphi'' + \frac{2}{r}\varphi', \qquad r > 0.$$

Thus $P(r) = 2/r$ and the factor in 7.6.12 is $\exp[-\frac{1}{2}\int(2/r)dr] = e^{-\ln r} = 1/r$, up to a constant. The new dependent variable is therefore $u = r\varphi$, or $\varphi = u/r$. Direct differentiation confirms the cancellation:

$$\varphi' = \frac{u'}{r} - \frac{u}{r^2}, \qquad \varphi'' = \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3},$$

so that

$$\varphi'' + \frac{2}{r}\varphi' = \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3} + \frac{2u'}{r^2} - \frac{2u}{r^3}, \qquad \boxed{\varphi'' + \frac{2}{r}\varphi' = \frac{u''}{r}}.$$

The potential correction from the substitution also vanishes: $-P'/2 - P^2/4 = 1/r^2 - 1/r^2 = 0$, so $q = Q$ in 7.6.12. If angular dependence is present, it is unaffected: writing $\nabla^2 = r^{-2}\partial_r(r^2\partial_r) + r^{-2}\Delta_\Omega$, where Δ_Ω is the angular operator in θ and the azimuth (it does not act on r), $\nabla^2(u/r) = u_{rr}/r + \Delta_\Omega u/r^3$. The eliminated derivative is the first *radial* derivative. At the origin a separate condition is needed: for $\varphi = u/r$ to stay finite there, $u = r\varphi$ must vanish at $r = 0$.

7.6.14 By direct differentiation and substitution show that

$$y_2(x) = y_1(x) \int^x \frac{\exp\left[-\int^s P(t)\,dt\right]}{[y_1(s)]^2}\,ds$$

satisfies, like $y_1(x)$, the ODE

$$y_2''(x) + P(x)y_2'(x) + Q(x)y_2(x) = 0.$$

Note. The Leibniz formula for the derivative of an integral is

$$\frac{d}{d\alpha}\int_{g(\alpha)}^{h(\alpha)} f(x,\alpha)\,dx = \int_{g(\alpha)}^{h(\alpha)} \frac{\partial f(x,\alpha)}{\partial \alpha}\,dx + f[h(\alpha),\alpha]\frac{dh(\alpha)}{d\alpha} - f[g(\alpha),\alpha]\frac{dg(\alpha)}{d\alpha}.$$

Apply the integral formula on an interval with continuous P, Q and a nowhere-zero first solution y_1; choose all integration limits within that interval.

Solution.

On an interval where $y_1 \neq 0$, abbreviate

$$E = e^{-\int^x P(t)dt}, \qquad h = E/y_1^2, \qquad J = \int^x h(s)ds,$$

so $E' = -PE$, $J' = h$ (the Leibniz formula of the note, with only the upper limit depending on x), and $y_2 = y_1 J$. Differentiate explicitly; $h' = E'/y_1^2 - 2Ey_1'/y_1^3$ and the product rule give

$$h' = -Ph - 2\frac{y_1'}{y_1}h, \quad y_2' = y_1'J + y_1 h, \quad y_2'' = y_1''J + 2y_1'h + y_1 h'.$$

Substitution groups the terms as

$$\begin{aligned} y_2'' + Py_2' + Qy_2 &= J(y_1'' + Py_1' + Qy_1) + (2y_1' + Py_1)h + y_1 h' \\ &= 0 + (2y_1' + Py_1)h + y_1\left(-P - 2\frac{y_1'}{y_1}\right)h = 0. \end{aligned}$$

Thus the displayed expression is a solution. Its Wronskian with y_1 is $y_1(y_1'J + y_1 h) - y_1'(y_1 J) = y_1^2 h = E \neq 0$, independently confirming that it is a second, independent solution.

7.6.15 In the equation

$$y_2(x) = y_1(x) \int^x \frac{\exp\left[-\int^s P(t)\,dt\right]}{[y_1(s)]^2}\,ds,$$

$y_1(x)$ satisfies

$$y_1'' + P(x)y_1' + Q(x)y_1 = 0.$$

The function $y_2(x)$ is a linearly independent second solution of the same equation. Show that the inclusion of lower limits on the two integrals leads to nothing new, that is, that it generates only an overall constant factor and a constant multiple of the known solution $y_1(x)$. Apply the integral formula on an interval with continuous P, Q and a nowhere-zero first solution y_1; choose all integration limits within that interval.

Solution.

Distinguish the inner and outer lower limits by defining

$$E_a(s) = \exp\left[-\int_a^s P(t)dt\right], \qquad Y_{a,c}(x) = y_1(x)\int_c^x \frac{E_a(s)}{y_1(s)^2}ds.$$

For another inner lower limit b,

$$\int_a^s P = \int_b^s P - \int_b^a P, \qquad E_a(s) = KE_b(s), \quad K = \exp\left(\int_b^a P(t)dt\right) \neq 0.$$

Changing the outer lower limit from c to d, with $\int_c^x = \int_d^x - \int_d^c$, then gives

$$\begin{aligned} Y_{a,c} &= Ky_1 \int_c^x \frac{E_b(s)}{y_1(s)^2} ds \\ &= Ky_1 \left[\int_d^x \frac{E_b(s)}{y_1(s)^2} ds - \int_d^c \frac{E_b(s)}{y_1(s)^2} ds\right] \\ &= \boxed{KY_{b,d} + Dy_1}, \qquad D = -K \int_d^c \frac{E_b(s)}{y_1(s)^2} ds. \end{aligned}$$

Both K and D are constants. Hence lower limits only rescale the second solution and add a multiple of the first. The limits must lie in an interval where the integrals and the nonzero-denominator formula are well defined.

7.6.16 Given that one solution of

$$R'' + \frac{1}{r}R' - \frac{m^2}{r^2}R = 0$$

is $R = r^m$, show that Eq. (7.67) predicts a second solution, $R = r^{-m}$ for $m \neq 0$, and $R = \ln r$ for $m = 0$, on $r > 0$.

Solution.

On $r > 0$, $P = 1/r$ and $e^{-\int P dr} = 1/r$, up to a constant. With $R_1 = r^m$, reduction of order gives

$$R_2 = r^m \int^r \frac{ds/s}{s^{2m}} = r^m \int^r s^{-2m-1} ds.$$

For $m \neq 0$, $\int^r s^{-2m-1} ds = -r^{-2m}/(2m)$, so

$$R_2 = -\frac{1}{2m} r^{-m},$$

and rescaling yields the requested independent solution $\boxed{R_2 = r^{-m}}$. With $R_2' = -mr^{-m-1}$ and $R_2'' = m(m+1)r^{-m-2}$, direct substitution gives $m(m+1)r^{-m-2} - mr^{-m-2} - m^2 r^{-m-2} = 0$. When $m = 0$, $r^m = r^{-m} = 1$ does not provide two independent solutions. The integral becomes $\int ds/s$, and the second solution is $\ln r$. Its derivatives $1/r, -1/r^2$ satisfy the $m = 0$ equation.

7.6.17 Using

$$y_1(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

as a solution of the linear oscillator equation, follow the analysis that proceeds through Eq. (7.81) and show that in that equation $c_n = 0$, so that in this case the second solution does not contain a logarithmic term.

Solution.

The given first solution is $y_1 = \sin x = xA(x^2)$ with $A(w) = \sum_{m\geq 0}(-1)^m w^m/(2m+1)!$, so $A(0) = 1$; the oscillator equation $y'' + y = 0$ has $P = 0$, so $e^{-\int P\,dx} = 1$. Its reduction-of-order integrand is

$$\frac{1}{y_1^2} = \frac{1}{x^2}\frac{1}{A(x^2)^2} = \frac{1}{x^2}\sum_{j\geq 0} b_j x^{2j} = \sum_{j\geq 0} b_j x^{2j-2}.$$

The reciprocal series exists near zero because $A(0) \neq 0$. A logarithm can arise on termwise integration only from a power x^{-1}. But $2j - 2 = -1$ would require $j = 1/2$, not an integer. Thus the logarithm coefficient denoted c_n in the referenced analysis is zero: the indicial roots of $y'' + y = 0$ at $x = 0$ are 1 and 0, so $n = 1$, the integrand is $x^{-2}(1 + c_1 x + c_2 x^2 + \cdots)$, and c_1, the coefficient of x in the even series $1/A(x^2)^2$, vanishes. Indeed,

$$y_2 = \sin x \int^x \frac{ds}{\sin^2 s} = \sin x(-\cot x) = -\cos x,$$

since $(\cot s)' = -1/\sin^2 s$; the integration constant adds a multiple of $\sin x$. The Taylor series $-1 + x^2/2! - x^4/4! + \cdots$ contains no logarithm and verifies the coefficient argument directly.

7.6.18 Show that when n is not an integer in Bessel's ODE, Eq. (7.40), the second solution of Bessel's equation, obtained from Eq. (7.67), does not contain a logarithmic term.

Solution.

For a real noninteger order set $\nu = |n|$; for a complex noninteger order the same proof works with $\nu = n$. Bessel's equation $x^2y'' + xy' + (x^2 - n^2)y = 0$ depends only on $n^2 = \nu^2$. Substituting $y = \sum_{j\geq 0} a_j x^{\nu+j}$ gives $[(\nu+j)^2 - \nu^2]a_j + a_{j-2} = 0$, that is, $j(2\nu+j)a_j = -a_{j-2}$, with $a_{-1} = a_{-2} = 0$. Here $2\nu + j \neq 0$ for $j \geq 1$, so $a_1 = 0$ and every odd coefficient vanishes: the Frobenius solution with exponent ν has the form

$$y_1 = x^\nu A(x^2), \qquad A(0) = a_0 \neq 0.$$

Since $P = 1/x$ and $e^{-\int P\,dx} = 1/x$,

$$\frac{e^{-\int P dx}}{y_1^2} = x^{-2\nu-1}\frac{1}{A(x^2)^2} = \sum_{k\geq 0} b_k x^{2k-2\nu-1}.$$

A logarithm could arise only if $2k - 2\nu - 1 = -1$, or $k = \nu$. There is no such nonnegative integer k when ν is noninteger. Termwise integration therefore gives

$$y_2 = x^\nu A(x^2)\sum_{k\geq 0}\frac{b_k}{2k-2\nu}x^{2k-2\nu} = x^{-\nu} \times \text{a power series in } x^2,$$

with no logarithm. The power-series integrations are local, and a consistent branch of x^ν is used for complex x. The leading coefficient is nonzero, so this is independent of $x^\nu A(x^2)$ and proportional, up to addition of that first solution, to $J_{-\nu}$. This also explains why half-integer orders, though their indicial roots differ by an integer, still produce no logarithm: parity prevents the resonant x^{-1} term.

7.6.19

(a) One solution of Hermite's differential equation

$$y'' - 2xy' + 2\alpha y = 0$$

for $\alpha = 0$ is $y_1(x) = 1$. Find a second solution, $y_2(x)$, using Eq. (7.67). Show that your second solution is equivalent to y_{odd} (Exercise 8.3.3).

(b) Find a second solution for $\alpha = 1$, where $y_1(x) = x$, using Eq. (7.67). Show that your second solution is equivalent to y_{even} (Exercise 8.3.3).

Solution.

For Hermite's equation $P = -2x$, so $e^{-\int P dx} = e^{x^2}$.

(a) When $\alpha = 0$ and $y_1 = 1$,

$$\boxed{y_2 = \int_0^x e^{t^2}\,dt = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{k!(2k+1)}}.$$

The exponential series converges uniformly on finite intervals, justifying its termwise integral. This is the normalized odd solution: $y_2(0) = 0$, $y_2'(0) = 1$, and $y_2'' - 2xy_2' = 0$ directly. To compare with Exercise 8.3.3, substitute $y = \sum_j a_j x^j$ into Hermite's equation: $(j+2)(j+1)a_{j+2} - 2ja_j + 2\alpha a_j = 0$, so

$$a_{j+2} = \frac{2(j-\alpha)}{(j+2)(j+1)} a_j.$$

For $\alpha = 0$ and $j = 2k+1$ this gives $a_{2k+3}/a_{2k+1} = (2k+1)/[(k+1)(2k+3)]$, the ratio of consecutive coefficients $1/[k!(2k+1)]$ above (as in Exercise 7.5.13); with the same leading coefficient the two odd series coincide: $x + x^3/3 + x^5/10 + \cdots$.

(b) With $\alpha = 1$ and $y_1 = x$, reduction of order gives $x\int^x e^{t^2}/t^2\,dt$. Differentiate e^{t^2}/t:

$$\frac{d}{dt}\left(\frac{e^{t^2}}{t}\right) = 2e^{t^2} - \frac{e^{t^2}}{t^2}, \quad \int^x \frac{e^{t^2}}{t^2}dt = -\frac{e^{x^2}}{x} + 2\int_0^x e^{t^2}\,dt.$$

Multiplying by x gives a solution with value -1 at zero. Reverse its sign to normalize the even solution to one:

$$\boxed{y_{\text{even}} = e^{x^2} - 2x\int_0^x e^{t^2}\,dt}.$$

Its series follows from $e^{x^2} = \sum_k x^{2k}/k!$ and the series of part (a), with $k \to k-1$ in the second sum:

$$\begin{aligned} y_{\text{even}} &= \sum_{k\geq 0} \frac{x^{2k}}{k!} - 2\sum_{k\geq 0} \frac{x^{2k+2}}{k!(2k+1)} \\ &= 1 + \sum_{k\geq 1} \left[\frac{1}{k!} - \frac{2}{(k-1)!(2k-1)}\right] x^{2k} \\ &= 1 - \sum_{k\geq 1} \frac{x^{2k}}{k!(2k-1)} = 1 - x^2 - \frac{x^4}{6} - \frac{x^6}{30} - \cdots, \end{aligned}$$

because $2/(k-1)! = 2k/k!$ and

$$\frac{1}{k!} - \frac{2k}{k!(2k-1)} = \frac{(2k-1)-2k}{k!(2k-1)} = -\frac{1}{k!(2k-1)}.$$

This is the even Hermite series at $\alpha = 1$: the recursion of part (a) with $\alpha = 1$, $j = 2k$, gives $a_2 = -a_0$ and, for $k \geq 1$, $a_{2k+2}/a_{2k} = (2k-1)/[(k+1)(2k+1)]$, which is exactly the ratio of consecutive coefficients $-1/[k!(2k-1)]$. Setting $J = \int_0^x e^{t^2} dt$, so $J' = e^{x^2}$, gives $y' = 2xe^{x^2} - 2J - 2xe^{x^2} = -2J$, $y'' = -2e^{x^2}$, and $y'' - 2xy' + 2y = -2e^{x^2} + 4xJ + 2(e^{x^2} - 2xJ) = 0$. The integral construction is initially used on $x \neq 0$; the displayed even function extends smoothly through zero.

7.6.20 One solution of Laguerre's differential equation

$$xy'' + (1-x)y' + ny = 0$$

for $n = 0$ is $y_1(x) = 1$. Using Eq. (7.67), develop a second, linearly independent solution. Exhibit the logarithmic term explicitly.

Solution.

Normalize the $n = 0$ equation to $y'' + (1/x - 1)y' = 0$ on $x \neq 0$. Since $\int (1/x - 1)dx = \ln|x| - x$,

$$e^{-\int (1/x-1)dx} = \frac{e^x}{|x|} = \pm\frac{e^x}{x}$$

on either real half-line; the constant sign is dropped. With $y_1 = 1$, $y_1^2 = 1$, Eq. (7.67) gives

$$y_2 = \int^x \frac{e^s}{s} ds.$$

Expand $e^s/s = s^{-1} + \sum_{k\geq 1} s^{k-1}/k!$ and integrate:

$$\boxed{y_2 = \ln|x| + \sum_{k=1}^{\infty} \frac{x^k}{k\,k!}}.$$

An additive constant repeats y_1 and is omitted. The series factor converges for all finite x, by its factorial denominator. Its derivative is e^x/x, so $W(1, y_2) = e^x/x \neq 0$, proving independence. The logarithmic singularity at zero is explicit.

7.6.21 For Laguerre's equation with $n = 0$,

$$y_2(x) = \int^x \frac{e^s}{s}\, ds.$$

(a) Write $y_2(x)$ as a logarithm plus a power series.

(b) Verify that the integral form of $y_2(x)$, previously given, is a solution of Laguerre's equation ($n = 0$) by direct differentiation of the integral and substitution into the differential equation.

(c) Verify that the series form of $y_2(x)$, part (a), is a solution by differentiating the series and substituting back into Laguerre's equation.

Solution.

(a) Separate the integrand's pole before integrating:

$$\frac{e^s}{s} = \frac{1}{s} + \sum_{k=1}^{\infty} \frac{s^{k-1}}{k!}, \qquad \boxed{y_2 = \ln|x| + \sum_{k=1}^{\infty} \frac{x^k}{k\,k!} + C}, \quad x \neq 0.$$

The factorial series and its derivatives converge locally uniformly, so these integrations and differentiations are justified away from zero.

(b) The fundamental theorem gives $y_2' = e^x/x$, and the quotient rule gives $y_2'' = e^x/x - e^x/x^2$. Consequently

$$xy_2'' + (1-x)y_2' = e^x - \frac{e^x}{x} + \frac{e^x}{x} - e^x = 0.$$

(c) Differentiate the series explicitly, using $[x^k/(k\,k!)]' = x^{k-1}/k!$:

$$y_2' = \frac{1}{x} + \sum_{k\geq 1} \frac{x^{k-1}}{k!}, \qquad y_2'' = -\frac{1}{x^2} + \sum_{k\geq 1} \frac{(k-1)x^{k-2}}{k!}.$$

Hence

$$xy_2'' = -\frac{1}{x} + \sum_{k\geq 1} \frac{(k-1)x^{k-1}}{k!}, \qquad (1-x)y_2' = \frac{1}{x} - 1 + \sum_{k\geq 1} \frac{x^{k-1}}{k!} - \sum_{k\geq 1} \frac{x^k}{k!}.$$

Adding, the $1/x$ terms cancel and $(k-1)+1 = k$, so

$$\begin{aligned} xy_2'' + (1-x)y_2' &= -1 + \sum_{k\geq 1} \frac{kx^{k-1}}{k!} - \sum_{k\geq 1} \frac{x^k}{k!} \\ &= -1 + \sum_{k\geq 1} \frac{x^{k-1}}{(k-1)!} - (e^x - 1) = -1 + e^x - (e^x - 1) = 0. \end{aligned}$$

This verifies the series independently of recognizing it as the integral.

7.6.22 One solution of the Chebyshev equation

$$(1-x^2)y'' - xy' + n^2 y = 0$$

for $n = 0$ is $y_1 = 1$.

(a) Using Eq. (7.67), develop a second, linearly independent solution.

(b) Find a second solution by direct integration of the Chebyshev equation.

Hint. Let $v = y'$ and integrate. Compare your result with the second solution given in Section 18.4.

ANS. (a) $y_2 = \sin^{-1} x$ on $-1 < x < 1$. (b) The second solution, $V_n(x)$, is not defined for $n = 0$ under the nonzero-n convention of Section 18.4.

Solution.

Work on $-1 < x < 1$. (a) The normalized coefficient is $P = -x/(1-x^2)$; with $w = 1-x^2$, $dw = -2x\,dx$,

$$\int P dx = \int \frac{dw}{2w} = \tfrac{1}{2}\ln(1-x^2), \qquad e^{-\int P dx} = (1-x^2)^{-1/2}.$$

Since $y_1 = 1$, reduction of order yields

$$\boxed{y_2 = \int_0^x \frac{ds}{\sqrt{1-s^2}} = \arcsin x}.$$

Its Wronskian with 1 is $1/\sqrt{1-x^2} \neq 0$.

(b) Put $v = y'$ in the $n = 0$ equation:

$$(1-x^2)v' - xv = 0, \quad \frac{dv}{v} = \frac{x\,dx}{1-x^2}, \quad \ln|v| = -\tfrac{1}{2}\ln(1-x^2) + C.$$

Therefore $v = A/\sqrt{1-x^2}$ and $\boxed{y = A\arcsin x + B}$, reproducing the same pair.

Section 18.4 explicitly introduces $V_n(x) = \sin(n \arccos x)$ for $n \neq 0$, so the printed answer follows its convention. The sine expression has a continuous extension to $n = 0$, but this gives the zero function, which cannot be an independent second solution. The expression $V_n = \sqrt{1-x^2}\,U_{n-1}$ also falls outside the usual nonnegative polynomial indexing when $n = 0$. A nonzero limiting second solution is $\lim_{n\to 0} V_n(x)/n = \arccos x = \pi/2 - \arcsin x$, differing only by a sign and a multiple of the first solution.

7.6.23 One solution of the Chebyshev equation

$$(1-x^2)y'' - xy' + n^2 y = 0$$

for $n = 1$ is $y_1(x) = x$. Set up the Wronskian double integral solution and derive a second solution, $y_2(x)$.

ANS. $y_2 = -(1-x^2)^{1/2}$.

Solution.

On $-1 < x < 1$, use $P = -x/(1-x^2)$ and $y_1 = x$. As in Exercise 7.6.22, the inner integral is $\int^s P(t)dt = \frac{1}{2}\ln(1-s^2)$, so $\exp[-\int^s P(t)dt] = (1-s^2)^{-1/2}$, and the Wronskian double integral gives, initially on either side of zero,

$$y_2 = x\int^x \frac{\exp[-\int^s(-t/(1-t^2))dt]}{s^2}ds = x\int^x \frac{ds}{s^2\sqrt{1-s^2}}.$$

The antiderivative follows directly from the product rule, since $(\sqrt{1-s^2})' = -s/\sqrt{1-s^2}$:

$$\frac{d}{ds}\left(-\frac{\sqrt{1-s^2}}{s}\right) = \frac{1}{\sqrt{1-s^2}} + \frac{\sqrt{1-s^2}}{s^2} = \frac{s^2 + (1-s^2)}{s^2\sqrt{1-s^2}} = \frac{1}{s^2\sqrt{1-s^2}}.$$

Thus $y_2 = x\cdot(-\sqrt{1-x^2}/x)$, that is,

$$\boxed{y_2 = -\sqrt{1-x^2}},$$

up to a multiple of x from the integration constant. Its derivatives are, by the chain rule and then the quotient rule,

$$y_2' = \frac{x}{\sqrt{1-x^2}}, \qquad y_2'' = \frac{\sqrt{1-x^2} + x^2/\sqrt{1-x^2}}{1-x^2} = (1-x^2)^{-3/2},$$

so substitution gives

$$(1-x^2)y_2'' - xy_2' + y_2 = \frac{1-x^2}{\sqrt{1-x^2}} - \sqrt{1-x^2} = 0.$$

Finally $W(x, y_2) = xy_2' - y_2 = x^2/\sqrt{1-x^2} + \sqrt{1-x^2} = 1/\sqrt{1-x^2} \neq 0$. Although the reduction integrand was singular at zero, the final solution and this Wronskian extend there regularly.

7.6.24 The radial Schrödinger wave equation for a spherically symmetric potential can be written in the form

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dr^2} + l(l+1)\frac{\hbar^2}{2mr^2} + V(r)\right] y(r) = Ey(r).$$

Take $l = 0, 1, 2, \ldots$ and $r > 0$. The potential energy $V(r)$ may be expanded about the origin as

$$V(r) = \frac{b_{-1}}{r} + b_0 + b_1 r + \cdots .$$

(a) Show that there is one (regular) solution $y_1(r)$ starting with r^{l+1}.

(b) From Eq. (7.69) show that the irregular solution $y_2(r)$ has leading behavior r^{-l}, diverging for $l \geq 1$ but approaching a nonzero constant for $l = 0$.

Solution.

Let $\kappa = 2m/\hbar^2$ and take the physical $l = 0, 1, 2, \ldots$. Multiplying by $-\kappa$ and collecting all terms on the left, the equation becomes

$$y'' - \frac{l(l+1)}{r^2}y + \kappa[E - V(r)]y = 0.$$

(a) Insert a Frobenius trial $y = \sum_{j\geq 0} a_j r^{s+j}$, $a_0 \neq 0$. The potential has at most a simple pole, so $\kappa(E - V)y$ begins at r^{s-1}, and the lowest power r^{s-2} comes only from $y'' - l(l+1)y/r^2$, with coefficient

$$[s(s-1) - l(l+1)]a_0 = (s - l - 1)(s + l)a_0 = 0.$$

The roots are $l + 1$ and $-l$. For $s = l + 1$, the coefficient of a_j in the recurrence is

$$(s + j)(s + j - 1) - l(l+1) = (l + j + 1)(l + j) - l(l+1) = j(2l + j + 1) > 0 \quad (j \geq 1).$$

In detail, the coefficient of r^{s+j-2} also receives $\kappa E a_{j-2}$ from $\kappa E y$ and, since $Vy = \sum_q \sum_i b_q a_i r^{s+i+q}$, the terms $-\kappa b_q a_i$ with $i + q = j - 2$ from $-\kappa V y$. With negative-index coefficients zero, this gives

$$j(2l + j + 1)a_j = \kappa \left[\sum_{q=-1}^{j-2} b_q a_{j-2-q} - E a_{j-2}\right].$$

Thus an arbitrary nonzero a_0 determines every following coefficient; for example $a_1 = \kappa b_{-1} a_0/[2(l + 1)]$. The regular solution is $\boxed{y_1 = a_0 r^{l+1}[1 + O(r)]}$.

(b) There is no first derivative ($P = 0$, $e^{-\int P} = 1$), so reduction of order is $y_2 = y_1 \int^r ds/y_1(s)^2$. Its leading terms are

$$\frac{1}{y_1(s)^2} = a_0^{-2} s^{-2l-2}[1 + O(s)], \quad \int^r \frac{ds}{y_1(s)^2} \sim -\frac{a_0^{-2}}{2l+1} r^{-2l-1} :$$

integrating termwise, s^{-2l-2} gives $r^{-2l-1}/(-2l - 1)$, while the higher terms give the higher powers $r^{-2l}, r^{-2l+1}, \ldots$ and possibly $\ln r$ (from an s^{-1} term), all smaller as $r \to 0$. Multiplying by $y_1 = a_0 r^{l+1}[1 + O(r)]$ gives

$$\boxed{y_2(r) \sim -\frac{1}{a_0(2l+1)} r^{-l}}.$$

The textbook states that y_2 diverges at the origin as r^{-l}. For $l \geq 1$ it does; for $l = 0$, however, y_2 tends to a nonzero constant and does not diverge. The physical radial function $R = y/r$ still behaves as $1/r$ and is irregular. For $l = 0$ the next terms may contain $r \ln r$, which still tend to zero and do not change the constant leading behavior.

7.6.25 Show that if a second solution, y_2, is assumed to be related to the first solution, y_1, according to $y_2(x) = y_1(x)f(x)$, substitution back into the original equation

$$y_2'' + P(x)y_2' + Q(x)y_2 = 0$$

leads to

$$f(x) = \int^x \frac{\exp\left[-\int^s P(t)\,dt\right]}{[y_1(s)]^2}\,ds,$$

in agreement with Eq. (7.67). Apply the integral formula on an interval with continuous P, Q and a nowhere-zero first solution y_1; choose all integration limits within that interval.

Solution.

Set $y_2 = y_1 f$. Two differentiations give

$$y_2' = y_1' f + y_1 f', \qquad y_2'' = y_1'' f + 2y_1' f' + y_1 f''.$$

Substitution and use of the equation for y_1 leave

$$0 = f(y_1'' + Py_1' + Qy_1) + (2y_1' + Py_1)f' + y_1 f'' = y_1 f'' + (2y_1' + Py_1)f'.$$

On an interval with $y_1 \neq 0$, let $g = f'$; then

$$g' + \left(2\frac{y_1'}{y_1} + P\right) g = 0, \qquad \frac{g'}{g} = -2\frac{y_1'}{y_1} - P.$$

Integrating gives $\ln|g| = -2\ln|y_1| - \int^x P\,dt + c$ on an interval where $g \neq 0$; exponentiating, with the constant sign of g absorbed, gives $g = Ce^{-\int P dx}/y_1^2$, and a second integration gives

$$\boxed{f = D + C\int^x \frac{e^{-\int^s P(t)dt}}{y_1(s)^2}ds}.$$

Here $C = 0$ recovers the constant-f branch excluded by division by g. For an independent second solution choose $C \neq 0$, and omit the Dy_1 contribution. Taking $C = 1, D = 0$ gives exactly the formula in the question; its Wronskian is $y_1(y_1' f + y_1 f') - y_1' y_1 f = y_1^2 f' = e^{-\int P dx} \neq 0$.

7.6.26 Use $x > 0$ and $a_0 \neq 0$; the primed labels in parts (b) and (c) name limiting solutions, not derivatives.

(a) Show that

$$y'' + \frac{1-\alpha^2}{4x^2}y = 0$$

has two solutions:

$$y_1(x) = a_0 x^{(1+\alpha)/2}, \quad y_2(x) = a_0 x^{(1-\alpha)/2}.$$

(b) For $\alpha = 0$ the two linearly independent solutions of part (a) reduce to the single solution $y_1' = a_0 x^{1/2}$. Using Eq. (7.68) derive a second solution,

$$y_2'(x) = a_0 x^{1/2}\ln x.$$

Verify that y_2' is indeed a solution.

(c) Show that the second solution from part (b) may be obtained as a limiting case from the two solutions of part (a):

$$y_2'(x) = \lim_{\alpha\to 0}\left(\frac{y_1 - y_2}{\alpha}\right).$$

Solution.

Take $x > 0$ and $a_0 \neq 0$. (a) For a trial $y = a_0 x^s$,

$$y'' + \frac{1-\alpha^2}{4x^2}y = a_0 x^{s-2}\left[s(s-1) + \frac{1-\alpha^2}{4}\right].$$

The quadratic is $s^2 - s + \frac{1}{4} - \frac{\alpha^2}{4} = (s - \frac{1}{2})^2 - (\frac{\alpha}{2})^2$, which factors as $[s - (1+\alpha)/2][s - (1-\alpha)/2]$, so both stated powers solve the equation. Their Wronskian is

$$W = a_0^2\left(\frac{1-\alpha}{2} - \frac{1+\alpha}{2}\right)x^{(1+\alpha)/2+(1-\alpha)/2-1} = -\alpha a_0^2.$$

They are independent precisely when $\alpha \neq 0$.

(b) At $\alpha = 0$ the common solution is $u_1 = a_0\sqrt{x}$; here u_1, u_2 denote the primed labels in the question, not derivatives. Since $P = 0$,

$$u_1 \int^x \frac{ds}{u_1(s)^2} = a_0\sqrt{x}\int^x \frac{ds}{a_0^2 s} = \frac{1}{a_0}\sqrt{x}\ln x.$$

An arbitrary nonzero rescaling (by a_0^2) gives $\boxed{u_2 = a_0\sqrt{x}\ln x}$. Explicitly, by the product rule,

$$u_2' = a_0\left(\frac{\ln x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}\right) = a_0 x^{-1/2}(1 + \tfrac{1}{2}\ln x),$$

$$u_2'' = a_0\left[-\tfrac{1}{2}x^{-3/2}(1+\tfrac{1}{2}\ln x) + x^{-1/2}\frac{1}{2x}\right] = -\frac{a_0}{4}x^{-3/2}\ln x,$$

so

$$u_2'' + \frac{u_2}{4x^2} = -\frac{a_0}{4}x^{-3/2}\ln x + \frac{a_0}{4}x^{-3/2}\ln x = 0.$$

Also $W(u_1, u_2) = u_1 u_2' - u_1' u_2 = a_0^2(1 + \frac{1}{2}\ln x) - \frac{1}{2}a_0^2\ln x = a_0^2 \neq 0$.

(c) Write the powers as exponentials, using $x^{(1\pm\alpha)/2} = \sqrt{x}\, e^{\pm(\alpha/2)\ln x}$:

$$\frac{y_1 - y_2}{\alpha} = a_0\sqrt{x}\frac{e^{(\alpha/2)\ln x} - e^{-(\alpha/2)\ln x}}{\alpha}.$$

Since $e^{hL} = 1 + hL + \frac{1}{2}h^2L^2 + O(h^3)$, applied with $L = \ln x$ and $h = \pm\alpha/2$, the constant and the equal quadratic terms cancel and the numerator is $\alpha\ln x + O(\alpha^3)$ (the odd terms double), and hence

$$\boxed{\lim_{\alpha\to 0}\frac{y_1 - y_2}{\alpha} = a_0\sqrt{x}\ln x = u_2}.$$

The Instructor's Manual drops a_0 and gives an extra factor $1/2$ in this limit. The explicit expansion above retains a_0 and shows that no extra $1/2$ is present.

7.7 Inhomogeneous Linear ODEs

7.7.1 If our linear, second-order ODE is inhomogeneous, that is, of the form of Eq. (7.94), the most general solution is

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x),$$

where y_1 and y_2 are independent solutions of the homogeneous equation, on a connected interval where the normalized coefficients and source are continuous. Show that

$$y_p(x) = y_2(x) \int^x \frac{y_1(s)F(s)\,ds}{W\{y_1(s), y_2(s)\}} - y_1(x) \int^x \frac{y_2(s)F(s)\,ds}{W\{y_1(s), y_2(s)\}},$$

with $W\{y_1(x), y_2(x)\}$ the Wronskian of $y_1(s)$ and $y_2(s)$.

Solution.

For fixed independent homogeneous solutions, the general solution is $y = C_1 y_1 + C_2 y_2 + y_p$; the two arbitrary coefficients must be explicit. Write the normalized equation as $L[y] = y'' + Py' + Qy = F$ and seek $y_p = u_1 y_1 + u_2 y_2$. Then $y_p' = u_1 y_1' + u_2 y_2' + (u_1' y_1 + u_2' y_2)$; impose the auxiliary condition $u_1' y_1 + u_2' y_2 = 0$, so that

$$y_p' = u_1 y_1' + u_2 y_2', \qquad y_p'' = u_1 y_1'' + u_2 y_2'' + u_1' y_1' + u_2' y_2'.$$

Collecting the terms multiplied by u_1 and by u_2,

$$L[y_p] = u_1 L[y_1] + u_2 L[y_2] + u_1' y_1' + u_2' y_2'.$$

Because $L[y_1] = L[y_2] = 0$, the remaining equation is $u_1' y_1' + u_2' y_2' = F$. Thus

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ F \end{pmatrix}.$$

Its determinant is $W = y_1 y_2' - y_1' y_2$, nonzero on the regular interval: if $W(x_0) = 0$, a null vector $(a, b) \neq (0, 0)$ of this matrix at x_0 would make $ay_1 + by_2$ a homogeneous solution with zero value and slope at x_0, hence identically zero by initial-value uniqueness (as in Exercise 7.6.5), contradicting independence. Cramer's rule gives

$$u_1' = \frac{1}{W} \det \begin{pmatrix} 0 & y_2 \\ F & y_2' \end{pmatrix} = -\frac{y_2 F}{W}, \qquad u_2' = \frac{1}{W} \det \begin{pmatrix} y_1 & 0 \\ y_1' & F \end{pmatrix} = \frac{y_1 F}{W}.$$

Integrate and substitute:

$$\boxed{y_p = y_2(x) \int^x \frac{y_1(s)F(s)}{W(s)} ds - y_1(x) \int^x \frac{y_2(s)F(s)}{W(s)} ds}.$$

Constants in the integrals contribute only multiples of y_1, y_2. As a direct check, these u_1', u_2' satisfy both conditions: $u_1' y_1 + u_2' y_2 = (-y_2 y_1 + y_1 y_2)F/W = 0$ and $L[y_p] = u_1' y_1' + u_2' y_2' = (y_1 y_2' - y_1' y_2)F/W = F$. Finally any other solution minus this particular solution satisfies the homogeneous equation, and every homogeneous solution y_h is a combination $C_1 y_1 + C_2 y_2$: by Exercise 7.6.10, y_h, y_1, y_2 satisfy a nontrivial relation $cy_h + ay_1 + by_2 = 0$, and $c \neq 0$ because y_1, y_2 are independent. This proves that $C_1 y_1 + C_2 y_2 + y_p$ is the general solution.

Find the general solutions to the following inhomogeneous ODEs:

7.7.2 $y'' + y = 1$.

Solution.

The homogeneous characteristic equation is $r^2 + 1 = 0$, with roots $r = \pm i$, giving $y_h = C_1 \cos x + C_2 \sin x$. For a constant particular solution $y_p = A$, the equation requires $0 + A = 1$, so $A = 1$. Therefore

$$\boxed{y = C_1 \cos x + C_2 \sin x + 1}.$$

For a variation-of-parameters check, take $y_1 = \cos x$, $y_2 = \sin x$, $W = \cos x \cos x - (-\sin x)\sin x = 1$, $F = 1$. Then, by Exercise 7.7.1, $u_1' = -y_2 F/W = -\sin x$, $u_2' = y_1 F/W = \cos x$, so one choice is $u_1 = \cos x$, $u_2 = \sin x$. Their combination is $y_p = u_1 y_1 + u_2 y_2 = \cos^2 x + \sin^2 x = 1$, agreeing with the constant trial. The Instructor's Manual labels both derivatives u_1' in its variation-of-parameters calculation; the second must be $u_2' = \cos x$, as Cramer's rule above shows.

7.7.3 $y'' + 4y = e^x$.

Solution.

The homogeneous roots satisfy $r^2 + 4 = 0$, so $r = \pm 2i$ and the real homogeneous solution is $C_1 \cos 2x + C_2 \sin 2x$. Try $y_p = Ae^x$; then

$$y_p'' + 4y_p = (A + 4A)e^x = 5Ae^x.$$

Matching the right side sets $A = 1/5$. Thus

$$\boxed{y = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{5}e^x}.$$

The homogeneous Wronskian is $W(\cos 2x, \sin 2x) = 2\cos^2 2x + 2\sin^2 2x = 2$, so the two constants are independent. Differentiating the displayed solution twice verifies that its trigonometric contributions cancel and its exponential contribution is exactly e^x.

Variation of parameters (Exercise 7.7.1) gives the same particular solution: with $y_1 = \cos 2x$, $y_2 = \sin 2x$, $W = 2$, $F = e^x$,

$$u_1' = -\frac{y_2 F}{W} = -\frac{e^x \sin 2x}{2}, \qquad u_2' = \frac{y_1 F}{W} = \frac{e^x \cos 2x}{2}.$$

The real and imaginary parts of $\int e^{(1+2i)x} dx = e^{(1+2i)x}/(1+2i) = e^x(\cos 2x + i \sin 2x)(1-2i)/5$ give $\int e^x \cos 2x\, dx = e^x(\cos 2x + 2\sin 2x)/5$ and $\int e^x \sin 2x\, dx = e^x(\sin 2x - 2\cos 2x)/5$, so

$$u_1 = \frac{e^x(2\cos 2x - \sin 2x)}{10}, \qquad u_2 = \frac{e^x(\cos 2x + 2\sin 2x)}{10},$$

and $y_p = u_1 \cos 2x + u_2 \sin 2x = e^x(2\cos^2 2x + 2\sin^2 2x)/10 = e^x/5$. In the Instructor's Manual's variation-of-parameters calculation, $u_2' = e^x \cos(2x)/2$, not $e^x \sin(2x)/2$: Cramer's rule gives $y_1 F/W = \cos(2x)e^x/2$. Its printed antiderivative and final answer agree with this corrected derivative.

7.7.4 $y'' - 3y' + 2y = \sin x$.

Solution.

The homogeneous characteristic polynomial is $r^2-3r+2 = (r-1)(r-2)$, giving $y_h = C_1e^x+C_2e^{2x}$. For $y_p = A\sin x + B\cos x$,

$$y_p' = A\cos x - B\sin x, \qquad y_p'' = -A\sin x - B\cos x.$$

Hence, collecting $\sin x$ terms $(-A + 3B + 2A)$ and $\cos x$ terms $(-B - 3A + 2B)$,

$$y_p'' - 3y_p' + 2y_p = (A + 3B)\sin x + (B - 3A)\cos x.$$

Matching coefficients gives $A + 3B = 1$, $B - 3A = 0$; hence $B = 3A$, $10A = 1$, $A = 1/10$, $B = 3/10$. The general solution is

$$\boxed{y = C_1e^x + C_2e^{2x} + \frac{\sin x + 3\cos x}{10}}.$$

The coefficient equations above are also a direct residual check of the particular solution; the homogeneous Wronskian is $e^{3x} \neq 0$.

7.7.5 $xy'' - (1 + x)y' + y = x^2$.

Solution.

First find homogeneous solutions. For a linear trial $y = ax + b$, $xy'' - (1 + x)y' + y = b - a$, so $y_1 = x + 1$ is a solution. The exponential $y_2 = e^x$ also works, since $xe^x - (1 + x)e^x + e^x = 0$. Their Wronskian is

$$W = (x + 1)e^x - e^x = xe^x,$$

nonzero on either interval $x > 0$ or $x < 0$. After division by x the equation reads $y'' - \frac{1+x}{x}y' + \frac{1}{x}y = x$, so the source term for variation of parameters is $F = x$, not x^2. Therefore

$$u_1' = -\frac{y_2F}{W} = -\frac{e^x x}{xe^x} = -1, \qquad u_2' = \frac{y_1F}{W} = \frac{(x + 1)x}{xe^x} = (x + 1)e^{-x}.$$

Integration by parts gives $\int(x + 1)e^{-x}dx = -(x + 1)e^{-x} + \int e^{-x}dx = -(x + 2)e^{-x}$, so one choice of antiderivatives is $u_1 = -x$, $u_2 = -(x + 2)e^{-x}$, giving

$$y_p = u_1y_1 + u_2y_2 = -x(x + 1) - (x + 2) = -x^2 - 2x - 2.$$

Since $-2(x + 1)$ is homogeneous, it can be absorbed into its arbitrary coefficient; take simply $y_p = -x^2$. Thus

$$\boxed{y = C_1(x + 1) + C_2e^x - x^2}.$$

The direct check for the particular solution, with $y_p' = -2x$ and $y_p'' = -2$, is $x(-2) - (1 + x)(-2x) - x^2 = -2x+2x+2x^2-x^2 = x^2$. The formula extends smoothly through zero, although the normalized ODE and the Wronskian argument are applied on intervals away from zero. It also describes every C^2 solution across zero: write the formula with constants $C_1^{\pm}, C_2^{\pm}$ on the two sides. Continuity of y'' gives $C_2^- = C_2^+$ because $y''(0) = C_2 - 2$; continuity of y then gives $C_1^- = C_1^+$. Thus no additional C^2 branch is lost at zero.

7.8 Nonlinear Differential Equations

7.8.1 Consider the Riccati equation $y' = y^2 - y - 2$. A particular solution to this equation is $y = 2$. Find a more general solution.

Solution.

Use the given solution by setting $y = 2 + u$. Then

$$u' = (2+u)^2 - (2+u) - 2 = u^2 + 3u.$$

The branch $u = 0$ gives the original solution $y = 2$. Since $u^2 + 3u$ is smooth, initial-value problems for u have unique solutions, so a solution with $u = 0$ at one point is this zero solution; any other solution has $u \neq 0$ throughout its interval. For such u put $v = 1/u$; then

$$v' = -\frac{u'}{u^2} = -\frac{u^2+3u}{u^2} = -1 - 3v, \qquad (e^{3x}v)' = e^{3x}(v'+3v) = -e^{3x},$$

the second form after multiplying by the integrating factor e^{3x}. Integrating gives $e^{3x}v = -\frac{1}{3}e^{3x} + C$, so $v = Ce^{-3x} - 1/3$ and $u = 1/v = 3/(3Ce^{-3x} - 1)$. Renaming $3C$ as K,

$$\boxed{y = 2 + \frac{3}{Ke^{-3x} - 1}}, \qquad \boxed{y \equiv 2}.$$

The constant solution $y = -1$ is already included at $K = 0$. Every nonconstant expression is used on a maximal interval where its denominator is nonzero. To check it, write $D = Ke^{-3x} - 1$, so $u = 3/D$ and $D' = -3Ke^{-3x}$: $u' = -3D'/D^2 = 9Ke^{-3x}/D^2$, while $u^2 + 3u = (9 + 9D)/D^2 = 9(1+D)/D^2 = 9Ke^{-3x}/D^2$. The integrating-constant formula for v must have the particular term $-1/3$; the Instructor's Manual's intermediate $+1/3$ does not satisfy $v' + 3v = -1$.

7.8.2 A particular solution to $y' = y^2/x^3 - y/x + 2x$ is $y = x^2$. Find a more general solution.

Solution.

Work on $x \neq 0$ and set $y = x^2 + u$. Expanding every term gives

$$2x + u' = \frac{x^4 + 2x^2u + u^2}{x^3} - \frac{x^2+u}{x} + 2x = 2x + \frac{u}{x} + \frac{u^2}{x^3}.$$

Thus $u' = u/x + u^2/x^3$. The branch $u = 0$ is $y = x^2$. Since the right side is smooth in u for $x \neq 0$, uniqueness of initial-value problems (as in Exercise 7.8.1) means that any other solution has $u \neq 0$ throughout its interval. For such u let $v = 1/u$; then $v' = -u'/u^2 = -1/(xu) - 1/x^3$, that is,

$$v' = -\frac{v}{x} - \frac{1}{x^3}, \qquad v' + \frac{v}{x} = -\frac{1}{x^3}, \qquad (xv)' = -\frac{1}{x^2},$$

the last form after multiplying by the integrating factor $e^{\int dx/x} = x$ (up to sign). Integrating gives $xv = 1/x + C$, so $v = (1+Cx)/x^2$ and $u = 1/v = x^2/(1+Cx)$. Therefore

$$\boxed{y = x^2 + \frac{x^2}{1+Cx}}, \qquad \boxed{y = x^2}.$$

The first family is valid where $x \neq 0$ and $1 + Cx \neq 0$; the second is the lost $u = 0$ branch. For a check, $u = x^2/(1+Cx)$ gives, by the quotient rule,

$$u' = \frac{2x(1+Cx) - Cx^2}{(1+Cx)^2} = \frac{2x + Cx^2}{(1+Cx)^2}, \qquad \frac{u}{x} + \frac{u^2}{x^3} = \frac{x}{1+Cx} + \frac{x}{(1+Cx)^2} = \frac{2x+Cx^2}{(1+Cx)^2}.$$

The Instructor's Manual's intermediate linear equation is missing the factor $1/x$ multiplying v; its displayed final solution agrees with the corrected equation derived here.

7.8.3 Solve the Bernoulli equation $y' + xy = xy^3$.

Solution.

The solution $y = 0$ must be recorded before dividing by a power of y. Since $xy^3 - xy$ is smooth in y, initial-value problems for this equation have unique solutions, so a solution that vanishes at one point is the zero solution. Every other solution is nowhere zero and, being continuous, keeps one sign on its interval. For such a solution set $u = y^{-2}$. Then

$$u' = -2y^{-3}y' = -2y^{-3}(xy^3 - xy) = -2x + 2xu.$$

Thus, multiplying the linear equation by the integrating factor e^{-x^2},

$$u' - 2xu = -2x, \qquad (e^{-x^2}u)' = -2xe^{-x^2} = (e^{-x^2})'.$$

After integration $e^{-x^2}u = e^{-x^2} + C$, so $u = 1 + Ce^{x^2}$. The real solutions are

$$\boxed{y = 0 \quad \text{or} \quad y = \pm\frac{1}{\sqrt{1 + Ce^{x^2}}}}.$$

Both signs and the zero solution are needed; taking only the positive square root loses real solutions. Choose a connected interval where $1 + Ce^{x^2} > 0$ and a fixed sign. In particular $C = 0$ gives $y = \pm 1$. Differentiation of $y = \pm(1 + Ce^{x^2})^{-1/2}$ gives the first line below, and the right side of the equation, written as $y' = xy^3 - xy$, gives the same expression:

$$y' = \mp\tfrac{1}{2}(1 + Ce^{x^2})^{-3/2}\, 2xCe^{x^2} = -\frac{xCe^{x^2}y}{1 + Ce^{x^2}},$$
$$xy^3 - xy = xy\left[\frac{1}{1 + Ce^{x^2}} - 1\right] = -\frac{xCe^{x^2}y}{1 + Ce^{x^2}}.$$

7.8.4 ODEs of the form $y = xy' + f(y')$ are known as Clairaut equations. The first step in solving an equation of this type is to differentiate it, assuming a twice differentiable solution, yielding

$$y' = y' + xy'' + f'(y')y'', \quad \text{or } y''\left[x + f'(y')\right] = 0.$$

Solutions may therefore be obtained both from $y'' = 0$ and from $f'(y') = -x$. The so-called general solution comes from $y'' = 0$.

For $f(y') = (y')^2$,

(a) Obtain the general solution (note that it contains a single constant).

(b) Obtain the so-called singular solution from $f'(y') = -x$. By substituting back into the original ODE show that this singular solution contains no adjustable constants.

Note. The singular solution is the envelope of the general solutions.

Solution.

For $f(p) = p^2$, differentiating the equation gives $y''(x + 2y') = 0$.

(a) On the branch $y'' = 0$, write $y = ax + b$ and $y' = a$. Substitute into the original, rather than only its derivative:

$$ax + b = xa + a^2 \quad \Longrightarrow \quad b = a^2.$$

Hence the one-parameter family is

$$\boxed{y = ax + a^2}, \qquad a \text{ arbitrary.}$$

(b) On the other branch $x + 2y' = 0$, so $y' = -x/2$ and $y = -x^2/4 + C$. The original equation requires

$$-\frac{x^2}{4} + C = x\left(-\frac{x}{2}\right) + \left(-\frac{x}{2}\right)^2 = -\frac{x^2}{4},$$

so $C = 0$ and

$$\boxed{y = -x^2/4}.$$

To verify the envelope statement, eliminate the parameter a from $y = ax + a^2$ and its parameter derivative $0 = x + 2a$. This gives $a = -x/2$ and $y = -x^2/4$. Each line touches the parabola at $x = -2a$: both values there are $-a^2$ and both slopes are a.

If only C^1 solutions of the original first-order equation are required, the two types can also be joined at a contact point. For example, $y = 0$ for $x \le 0$ and $y = -x^2/4$ for $x \ge 0$ has continuous value and derivative at zero, and each branch satisfies $y = xy' + (y')^2$. The second derivatives there are 0 and $-1/2$, so this joined solution lies outside the differentiation argument. The requested line family and singular curve are the standard smooth branches; they do not exhaust such C^1 joins.

8 Chapter 8. Sturm-Liouville Theory

Problems

8.2 Hermitian Operators

8.2.1 Show that Laguerre's ODE, Table 7.1, may be put into self-adjoint form by multiplying by e^{-x} and that $w(x) = e^{-x}$ is the weighting function.

Solution.

Write the equation as $xy'' + (1-x)y' + ny = 0$. Multiplication by $w = e^{-x}$ gives

$$xe^{-x}y'' + (1-x)e^{-x}y' + ne^{-x}y = 0.$$

Here $(xe^{-x})' = e^{-x} - xe^{-x} = (1-x)e^{-x}$: the coefficient of y' is the derivative of the coefficient of y'', which is the self-adjointness condition $p_0' = p_1$ for $p_0y'' + p_1y' + p_2y$. By the product rule, $(xe^{-x}y')' = xe^{-x}y'' + (1-x)e^{-x}y'$, so the first two terms combine without changing the eigenvalue term:

$$\boxed{(xe^{-x}y')' + ne^{-x}y = 0}, \qquad p(x) = xe^{-x}, \quad w(x) = e^{-x}.$$

Thus $Ay = -w^{-1}(py')' = -e^{x}(xe^{-x}y')' = -xy'' - (1-x)y'$, and the equation reads $Ay = ny$: the eigenvalue is n. For functions in a common domain, write $w\,Av = -(pv')'$ and integrate once by parts in each scalar product $\langle u, v\rangle_w = \int_a^b u^* v\, e^{-x}\, dx$ (p and w are real):

$$\langle u, Av\rangle_w = -\int_a^b u^*(pv')'\,dx = -[u^*pv']_a^b + \int_a^b p\,u'^*v'\,dx,$$
$$\langle Au, v\rangle_w = -\int_a^b (pu'^*)'v\,dx = -[pu'^*v]_a^b + \int_a^b p\,u'^*v'\,dx.$$

Subtracting cancels the identical integrals and gives

$$\langle u, Av\rangle_w - \langle Au, v\rangle_w = -\big[p(u^*v' - u'^*v)\big]_a^b.$$

This is the required weighted self-adjoint differential form; vanishing of the displayed boundary form makes the operator Hermitian on that domain. For polynomial solutions on $[0,\infty)$, both endpoint terms vanish, as checked in Exercise 8.2.4(d).

8.2.2 Show that the Hermite ODE, Table 7.1, may be put into self-adjoint form by multiplying by e^{-x^2} and that this gives $w(x) = e^{-x^2}$ as the appropriate weighting function.

Solution.

Hermite's equation is $y'' - 2xy' + 2ny = 0$. Since

$$\frac{d}{dx}e^{-x^2} = -2xe^{-x^2},$$

the product rule gives $(e^{-x^2}y')' = e^{-x^2}y'' - 2xe^{-x^2}y'$, and multiplication by e^{-x^2} gives

$$e^{-x^2}y'' - 2xe^{-x^2}y' + 2ne^{-x^2}y = (e^{-x^2}y')' + 2ne^{-x^2}y = 0.$$

Therefore

$$\boxed{p = w = e^{-x^2}, \qquad A = -w^{-1}\frac{d}{dx}\Big(p\frac{d}{dx}\Big) = -\frac{d^2}{dx^2} + 2x\frac{d}{dx}},$$

since $Ay = -e^{x^2}(e^{-x^2}y')' = -y'' + 2xy'$; the equation reads $Ay = 2ny$. In the scalar product $\langle u, v\rangle_w = \int u^* v e^{-x^2}\,dx$, the calculation of Exercise 8.2.1 with $p = e^{-x^2}$ (one integration by parts in each of $\langle u, Av\rangle_w$ and $\langle Au, v\rangle_w$) leaves only

$$\langle u, Av\rangle_w - \langle Au, v\rangle_w = -\big[e^{-x^2}(u^* v' - u'^* v)\big]_a^b.$$

It vanishes on $(-\infty, \infty)$ for polynomial solutions and their linear combinations (Exercise 8.2.4(c)), establishing the claimed weight and Hermiticity.

8.2.3 Show that the Chebyshev ODE, Table 7.1, may be put into self-adjoint form by multiplying by $(1 - x^2)^{-1/2}$ and that this gives $w(x) = (1 - x^2)^{-1/2}$ as the appropriate weighting function.

Solution.

On $-1 < x < 1$, multiply $(1-x^2)y'' - xy' + n^2 y = 0$ by $(1-x^2)^{-1/2}$. Since $(1-x^2)(1-x^2)^{-1/2} = \sqrt{1-x^2}$, this gives

$$\sqrt{1-x^2}\,y'' - \frac{x}{\sqrt{1-x^2}}\,y' + \frac{n^2}{\sqrt{1-x^2}}\,y = 0.$$

Because

$$\frac{d}{dx}\sqrt{1-x^2} = -\frac{x}{\sqrt{1-x^2}},$$

the product rule shows that the first two terms are $(\sqrt{1-x^2}\,y')'$, and the resulting equation is

$$\boxed{\Big(\sqrt{1-x^2}\,y'\Big)' + \frac{n^2}{\sqrt{1-x^2}}y = 0}.$$

Thus $p = \sqrt{1-x^2}$, $w = (1-x^2)^{-1/2}$ and $Ay = -w^{-1}(py')'$ has eigenvalue n^2; as in Exercise 8.2.1, $\langle u, Av\rangle_w - \langle Au, v\rangle_w = -[p(u^* v' - u'^* v)]_{-1}^{1}$. The scalar product is

$$\boxed{\langle u, v\rangle_w = \int_{-1}^{1} \frac{u^*(x)v(x)}{\sqrt{1-x^2}}\,dx}.$$

The singularities of this weight are integrable: near $x = 1$ put $t = 1 - x$, so that $1 - x^2 = t(2 - t)$ and, for $0 < t \le 1$,

$$w = t^{-1/2}(2-t)^{-1/2} \le t^{-1/2}, \qquad \int_0^1 t^{-1/2}\,dt = \big[2\sqrt{t}\,\big]_0^1 = 2;$$

the endpoint $x = -1$ is the mirror image ($t = 1 + x$). Polynomial solutions therefore have finite norm, and p times any product of their values and first derivatives vanishes at either endpoint, since $p = \sqrt{1-x^2} \to 0$ while the polynomials stay bounded.

8.2.4 The Legendre, Chebyshev, Hermite, and Laguerre equations, given in Table 7.1, have solutions that are polynomials. Show that ranges of integration that guarantee that the Hermitian operator boundary conditions will be satisfied are

(a) Legendre $[-1, 1]$, (b) Chebyshev $[-1, 1]$,

(c) Hermite $(-\infty, \infty)$, (d) Laguerre $[0, \infty)$.

Solution.

For $(pu')' + \lambda wu = 0$, the boundary condition needed in the orthogonality proof is (Exercise 8.2.1)

$$\big[p(u^*v' - u'^*v)\big]_a^b = 0.$$

If u, v are polynomials, then $u^*v' - u'^*v$ is another polynomial (on the real axis). We can therefore check each endpoint explicitly.

(a) Legendre: $p = 1-x^2$ and $w = 1$, since Legendre's equation is already self-adjoint: $((1-x^2)y')' = (1 - x^2)y'' - 2xy'$. The polynomial factor is bounded at $x = \pm 1$, while $1 - x^2 \to 0$. Each boundary term is zero, and the norm integral over $[-1, 1]$ is finite.

(b) Chebyshev: $p = \sqrt{1 - x^2} \to 0$ at ± 1. Bounded polynomial values and derivatives again make each boundary term zero. The norm exists because $w = (1 - x^2)^{-1/2}$ is integrable (Exercise 8.2.3).

(c) Hermite: $p = w = e^{-x^2}$. Every polynomial is bounded in magnitude by $C(1 + |x|^N)$ for suitable C, N, and $|x|^N e^{-x^2} \to 0$ as $x \to \pm\infty$. Thus both boundary terms vanish, and the weighted norms converge because, by symmetry and the substitution $t = x^2$, $dx = dt/(2\sqrt{t})$,

$$\int_{-\infty}^{\infty} |x|^k e^{-x^2}\, dx = 2\int_0^{\infty} x^k e^{-x^2}\, dx = \int_0^{\infty} t^{(k-1)/2} e^{-t}\, dt = \Gamma\Big(\frac{k+1}{2}\Big)$$

is finite for every $k \geq 0$.

(d) Laguerre: $p = xe^{-x}$ and $w = e^{-x}$. At 0, the factor x multiplies a bounded polynomial; at ∞, with C, N as in (c), $|p(u^*v' - u'^*v)| \leq C(x + x^{N+1})e^{-x} \to 0$. Both boundary terms vanish, and the weighted norms on $[0, \infty)$ converge because $\int_0^{\infty} x^k e^{-x}\, dx = k!$.

These conclusions use the polynomial hypothesis. The same intervals alone do not make arbitrary nonpolynomial solutions satisfy the boundary conditions, as Exercises 8.2.6 and 8.2.7 demonstrate: Q_0 is unbounded at $x = \pm 1$, while $V_1 = \sqrt{1 - x^2}$ is bounded but has an unbounded derivative there.

8.2.5 The functions $u_1(x)$ and $u_2(x)$ are eigenfunctions of the same Hermitian operator but for distinct eigenvalues λ_1 and λ_2. Prove that $u_1(x)$ and $u_2(x)$ are linearly independent.

Solution.

Suppose $c_1u_1 + c_2u_2 = 0$, where eigenfunctions are nonzero by definition. Apply A and then subtract λ_2 times the original relation:

$$c_1\lambda_1u_1 + c_2\lambda_2u_2 = 0, \qquad c_1(\lambda_1 - \lambda_2)u_1 = 0.$$

Since $\lambda_1 \neq \lambda_2$ and $u_1 \neq 0$, we obtain $c_1 = 0$. The original relation then reads $c_2u_2 = 0$, and $u_2 \neq 0$ implies $c_2 = 0$. Hence the functions are linearly independent. Hermiticity is sufficient but is not needed for this particular conclusion; only linearity and distinct eigenvalues were used.

8.2.6 Given that

$$P_1(x) = x \quad \text{and} \quad Q_0(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

are solutions of Legendre's differential equation (Table 7.1) corresponding to different eigenvalues:

(a) Evaluate their orthogonality integral

$$\int_{-1}^{1} \frac{x}{2}\ln\left(\frac{1+x}{1-x}\right)\,dx.$$

(b) Explain why these two functions are not orthogonal, that is, why the proof of orthogonality does not apply.

Solution.

(a) Put $Q = Q_0 = \frac{1}{2}\ln((1+x)/(1-x)) = \frac{1}{2}[\ln(1+x) - \ln(1-x)]$. Its derivative is

$$Q' = \frac{1}{2}\left(\frac{1}{1+x} + \frac{1}{1-x}\right) = \frac{1}{1-x^2}.$$

Integrate by parts on a closed subinterval of $(-1,1)$, differentiating Q and integrating x. Choosing the antiderivative $(x^2-1)/2$ of x makes the remaining integrand a constant:

$$\int xQ\,dx = \frac{x^2-1}{2}Q - \frac{1}{2}\int (x^2-1)\frac{dx}{1-x^2} = \frac{x^2-1}{2}Q + \frac{x}{2}.$$

The endpoint products vanish: near $x = 1$ put $t = 1-x$, so that $x^2 - 1 = -t(2-t)$, $1+x = 2-t$, and

$$\frac{x^2-1}{2}Q = -\frac{t(2-t)}{4}\big[\ln(2-t) - \ln t\big] \longrightarrow 0 \qquad (t \downarrow 0),$$

because $t\ln t \to 0$. Since Q is odd, $\frac{1}{2}(x^2-1)Q$ is odd, and its limit at $x = -1$ is also zero. Taking both improper limits therefore gives

$$\boxed{\int_{-1}^{1} P_1 Q_0\,dx = \Big[\frac{x^2-1}{2}Q + \frac{x}{2}\Big]_{-1}^{1} = \Big(0 + \frac{1}{2}\Big) - \Big(0 - \frac{1}{2}\Big) = 1}.$$

(b) Legendre's equation is $Ay = n(n+1)y$ with $Ay = -[(1-x^2)y']'$. The eigenvalues of $P_1 = x$ and Q are 2 and 0: $[(1-x^2)P_1']' = -2x = -2P_1$, while $(1-x^2)Q' = 1$ gives $[(1-x^2)Q']' = 0$. Their boundary form is not zero:

$$(1-x^2)(P_1Q' - P_1'Q) = x - (1-x^2)Q \longrightarrow \begin{cases} 1, & x \to 1, \\ -1, & x \to -1, \end{cases}$$

because $(1-x^2)Q = -2\cdot\frac{1}{2}(x^2-1)Q \to 0$ at both endpoints, by (a). Its endpoint difference is 2. Indeed, with $p = 1-x^2$, the product rule (the terms $pP_1'Q'$ cancel) and the two differential equations give

$$\big[p(P_1Q' - P_1'Q)\big]' = P_1(pQ')' - Q(pP_1')' = 0 + 2P_1Q,$$

and integration over $(-1,1)$ gives, in agreement with (a),

$$\big[(1-x^2)(P_1Q' - P_1'Q)\big]_{-1}^{1} = (2-0)\int_{-1}^{1} P_1Q\,dx = 2.$$

Although Q is square integrable (near $x = \pm 1$, Q^2 grows only like $\frac{1}{4}\ln^2(1 \mp x)$, and $t = e^{-u}$ gives $\int_0^1 \ln^2 t\,dt = \int_0^\infty u^2 e^{-u}\,du = 2$), it is singular at the endpoints and does not belong to the same boundary-condition domain as the Legendre polynomials. The Hermitian orthogonality theorem thus does not apply to this pair.

8.2.7 $T_0(x) = 1$ and $V_1(x) = (1 - x^2)^{1/2}$ are solutions of the Chebyshev differential equation corresponding to different eigenvalues. Explain, in terms of the boundary conditions, why these two functions are not orthogonal on the range $(-1, 1)$ with the weighting function found in Exercise 8.2.3.

Solution.

With $u = T_0 = 1$, $v = V_1 = \sqrt{1 - x^2}$, $p = \sqrt{1 - x^2}$ and $w = 1/p$, the weighted inner product is

$$\boxed{\langle u, v\rangle_w = \int_{-1}^{1} 1\, dx = 2 \neq 0}.$$

Here $u' = 0$, $v' = -x/p$, so

$$p(uv' - u'v) = -x, \qquad \big[p(uv' - u'v)\big]_{-1}^{1} = -2.$$

The two eigenvalues are 0 and 1: explicitly, $(pu')' = 0$ and $(pv')' = (-x)' = -1 = -wv$. Consequently the boundary identity of Exercise 8.2.6(b),

$$\big[p(uv' - u'v)\big]' = u(pv')' - v(pu')' = -1 = -wuv,$$

integrated over $(-1, 1)$, gives $-2 = (0 - 1)\langle u, v\rangle_w$, consistently with the integral. The divergence of v' cancels the zero of p; the boundary form does not vanish even though v itself vanishes at both endpoints. They are therefore not eigenfunctions in a common Hermitian domain containing the Chebyshev polynomials.

8.2.8 A set of functions $u_n(x)$ satisfies the Sturm-Liouville equation

$$\frac{d}{dx}\left[p(x)\frac{d}{dx}u_n(x)\right] + \lambda_n w(x) u_n(x) = 0.$$

The functions $u_m(x)$ and $u_n(x)$ satisfy boundary conditions that lead to orthogonality. The corresponding eigenvalues λ_m and λ_n are distinct. Prove that for appropriate boundary conditions, $u_m'(x)$ and $u_n'(x)$ are orthogonal with $p(x)$ as a weighting function.

Solution.

Multiply the equation for u_n by u_m^* and integrate:

$$0 = \int_a^b u_m^*(pu_n')'\, dx + \lambda_n \int_a^b w u_m^* u_n\, dx.$$

Integration by parts, with u_m^* differentiated, gives

$$\int_a^b u_m^*(pu_n')'\, dx = \big[pu_m^* u_n'\big]_a^b - \int_a^b p\, u_m'^* u_n'\, dx;$$

substituting this into the previous line gives the exact identity

$$\boxed{\int_a^b p\, u_m'^* u_n'\, dx = \big[pu_m^* u_n'\big]_a^b + \lambda_n \int_a^b w u_m^* u_n\, dx}.$$

For distinct eigenvalues the last integral is zero by the assumed orthogonality. Thus the additional sufficient boundary condition is

$$\big[pu_m^* u_n'\big]_a^b = 0,$$

which yields $\int_a^b p\, u_m'^* u_n'\, dx = 0$. This holds, for example, under common Dirichlet conditions, common Neumann conditions, or a common choice of Dirichlet or Neumann condition at each endpoint. It also holds under periodic conditions $u(a) = u(b)$, $p(a)u'(a) = p(b)u'(b)$, since the two endpoint products are then equal and cancel. An endpoint where $p \to 0$ while $u_m^* u_n'$ remains bounded contributes nothing.

The vanishing of the antisymmetrized boundary form alone need not suffice. Under real Robin conditions $u'(a) = h_a u(a)$ and $u'(b) = h_b u(b)$ that form vanishes, but the boundary term here is

$$p(b)h_b u_m^*(b)u_n(b) - p(a)h_a u_m^*(a)u_n(a),$$

which requires a separate check. For example, $u'' + \lambda u = 0$ on $[0,1]$ ($p = w = 1$) with $u'(0) = 0$ and $u'(1) = -u(1)$ ($h_a = 0$, $h_b = -1$) has the eigenfunctions $u = \cos kx$, $\lambda = k^2$, for which $u'(1) = -u(1)$ reads $k \sin k = \cos k$, i.e., $k \tan k = 1$. The two lowest, u_1 and u_2 with $k_1 \approx 0.8603$ and $k_2 \approx 3.4256$, are orthogonal, but by the boxed identity $\int_0^1 u_1' u_2'\, dx$ equals the displayed term, $-\cos k_1 \cos k_2 \approx 0.626 \neq 0$.

For a normalized eigenfunction with $[p\, u_n^* u_n']_a^b = 0$ (the condition above with $m = n$), the boxed identity with $m = n$ also gives $\int_a^b p|u_n'|^2\, dx = \lambda_n$.

8.2.9 Linear operator $\mathcal{A}$ has n distinct eigenvalues and n corresponding eigenfunctions: $\mathcal{A}\psi_i = \lambda_i \psi_i$. Show that the n eigenfunctions are linearly independent. Do not assume $\mathcal{A}$ to be Hermitian.

Hint. Assume linear dependence, i.e., that $\psi_n = \sum_{i=1}^{n-1} a_i \psi_i$. Use this relation and the operator-eigenfunction equation first in one order and then in the reverse order. Show that a contradiction results.

Solution.

We prove the statement by induction, so that comparison of coefficients will not presuppose the independence we are trying to establish. One nonzero eigenfunction is independent. Suppose any $n-1$ eigenfunctions with distinct eigenvalues are independent, and consider

$$\sum_{i=1}^{n} c_i \psi_i = 0.$$

Apply $\mathcal{A} - \lambda_n I$ to this equality. Linearity and the eigenfunction equations give $(\mathcal{A} - \lambda_n I)\psi_i = (\lambda_i - \lambda_n)\psi_i$, which is zero for $i = n$; hence

$$\sum_{i=1}^{n-1} c_i (\lambda_i - \lambda_n)\psi_i = 0.$$

By the induction hypothesis every coefficient in this shorter sum is zero. Since $\lambda_i - \lambda_n \neq 0$, $c_i = 0$ for $i < n$. The original equality becomes $c_n \psi_n = 0$, hence $c_n = 0$. This completes the induction without Hermiticity and without division by any individual eigenvalue (which may be zero).

8.2.10 The ultraspherical polynomials $C_n^{(\alpha)}(x)$ are solutions of the differential equation

$$\left\{(1-x^2)\frac{d^2}{dx^2} - (2\alpha+1)x\frac{d}{dx} + n(n+2\alpha)\right\} C_n^{(\alpha)}(x) = 0.$$

(a) Transform this differential equation into self-adjoint form.

(b) Find an interval of integration and weighting factor that make $C_n^{(\alpha)}(x)$ of the same α but different n orthogonal.

Note. Assume that your solutions are polynomials. Take real $\alpha > -1/2$. At $\alpha = 0$, use nonzero limiting normalizations for degrees $n \geq 1$.

Solution.

For the weight $(1-x^2)^{\alpha-1/2}$ to be positive and integrable on $(-1,1)$, so that polynomials of every degree have finite norm, assume real $\alpha > -\frac{1}{2}$ (see (b)). Eigenfunctions must be nonzero; at $\alpha = 0$ use the nonzero Chebyshev limits rather than the identically zero standard $C_n^{(0)}$ for $n \geq 1$.

(a) If w is the multiplier and $p = (1-x^2)w$, the coefficient of y' must be $p' = -(2\alpha+1)xw$ (the condition $p_0' = p_1$ of Exercise 8.2.1). Expanding this condition with the product rule gives

$$(1-x^2)w' - 2xw = -(2\alpha+1)xw, \qquad \frac{w'}{w} = \frac{(1-2\alpha)x}{1-x^2}.$$

Since $d\ln(1-x^2)/dx = -2x/(1-x^2)$, integration yields

$$\ln w = -\frac{1-2\alpha}{2}\ln(1-x^2) + \text{const} = \left(\alpha - \frac{1}{2}\right)\ln(1-x^2) + \text{const},$$

so that

$$w = (1-x^2)^{\alpha-1/2}, \qquad p = (1-x^2)^{\alpha+1/2},$$

where the irrelevant positive constant has been set to one. As a check, $p' = (\alpha+\frac{1}{2})(1-x^2)^{\alpha-1/2}(-2x) = -(2\alpha+1)xw$. Hence

$$\boxed{\left[(1-x^2)^{\alpha+1/2}y'\right]' + n(n+2\alpha)(1-x^2)^{\alpha-1/2}y = 0}.$$

(b) On $(-1,1)$ this weight is positive. Near $x = \pm 1$ put $\delta = 1 \mp x$; then $1-x^2 = \delta(2-\delta)$, and $w = \delta^{\alpha-1/2}(2-\delta)^{\alpha-1/2}$ is asymptotic to $2^{\alpha-1/2}\delta^{\alpha-1/2}$. The integral of $\delta^{\alpha-1/2}$ over $0 < \delta \leq 1$ is finite exactly when $\alpha - \frac{1}{2} > -1$, i.e., $\alpha > -1/2$. The same inequality makes $p = (1-x^2)^{\alpha+1/2} \to 0$ at ± 1, so the boundary form vanishes for polynomial solutions. For $m \neq n$ in $\{0,1,2,\ldots\}$, the eigenvalues $\lambda_n = n(n+2\alpha)$ satisfy

$$\lambda_n - \lambda_m = n^2 - m^2 + 2\alpha(n-m) = (n-m)(n+m+2\alpha) \neq 0,$$

because $n+m \geq 1$ and $2\alpha > -1$. Write C_k for $C_k^{(\alpha)}$. Multiplying the self-adjoint equation for C_n by C_m and subtracting the equation for C_m multiplied by C_n gives, as in Exercise 8.2.6(b),

$$\left[p(C_mC_n' - C_m'C_n)\right]' = C_m(pC_n')' - C_n(pC_m')' = (\lambda_m - \lambda_n)wC_mC_n.$$

Integration over $[-1,1]$ makes the left side zero; dividing by $\lambda_m - \lambda_n \neq 0$ gives

$$\boxed{\int_{-1}^{1}(1-x^2)^{\alpha-1/2}C_m^{(\alpha)}(x)C_n^{(\alpha)}(x)\,dx = 0 \quad (m \neq n)}.$$

To explain the qualification at zero, the generating function $(1-2xt+t^2)^{-\alpha} = \sum_{n\geq 0} C_n^{(\alpha)}(x)t^n$ is 1 when $\alpha = 0$, so $C_n^{(0)} = 0$ for $n \geq 1$. For $n \geq 1$ its nonzero limiting normalization is $\lim_{\alpha\to 0} C_n^{(\alpha)}(x)/\alpha = 2T_n(x)/n$. To see the coefficient explicitly, note that because $C_n^{(0)} = 0$, this limit is $\partial C_n^{(\alpha)}/\partial\alpha$ at $\alpha = 0$, i.e., the coefficient of t^n in $\partial(1-2xt+t^2)^{-\alpha}/\partial\alpha|_{\alpha=0} = -\ln(1-2xt+t^2)$. Put

$x = \cos\theta$ and factor $1-2xt+t^2 = (1-e^{i\theta}t)(1-e^{-i\theta}t)$. For $|t|<1$, the series $-\ln(1-z) = \sum_{n\geq 1} z^n/n$ with $z = e^{\pm i\theta}t$ gives

$$\begin{aligned} -\ln(1-2xt+t^2) &= \sum_{n\geq 1} \frac{e^{in\theta}+e^{-in\theta}}{n} t^n \\ &= 2\sum_{n\geq 1} \frac{\cos(n\theta)}{n} t^n = 2\sum_{n\geq 1} \frac{T_n(x)}{n} t^n. \end{aligned}$$

The resulting Chebyshev family has precisely the weight in Exercise 8.2.3.

8.3 ODE Eigenvalue Problems

8.3.1 Solve the Legendre equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

by direct series substitution.

(a) Verify that the indicial equation is

$$s(s-1) = 0.$$

(b) Using $s=0$ and setting the coefficient $a_1 = 0$, obtain a series of even powers of x:

$$y_{\text{even}} = a_0\left[1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 + \cdots\right],$$

where

$$a_{j+2} = \frac{j(j+1)-n(n+1)}{(j+1)(j+2)} a_j.$$

(c) Using $s=1$ and noting that the coefficient a_1 must be zero, develop a series of odd powers of x:

$$y_{\text{odd}} = a_0\left[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 + \cdots\right],$$

where

$$a_{j+2} = \frac{(j+1)(j+2)-n(n+1)}{(j+2)(j+3)} a_j.$$

(d) Show that both solutions, y_{even} and y_{odd}, diverge for $x = \pm 1$ if the series continue to infinity. (Compare with Exercise 1.2.5.)

(e) Finally, show that by an appropriate choice of n, one series at a time may be converted into a polynomial, thereby avoiding the divergence catastrophe. In quantum mechanics this restriction of n to integral values corresponds to quantization of angular momentum.

Solution.

(a) Substitute $y = \sum_{j\geq 0} a_j x^{j+s}$, with $a_0 \neq 0$. The term y'' gives $\sum_{j\geq 0}(j+s)(j+s-1)a_j x^{j+s-2}$, while in $-x^2y'' - 2xy' + n(n+1)y$ every power is x^{j+s}, with coefficient $n(n+1) - (j+s)(j+s-1) - 2(j+s) = n(n+1) - (j+s)(j+s+1)$. Before combining powers, the equation therefore reads

$$\sum_{j\geq 0}(j+s)(j+s-1)a_j x^{j+s-2} + \sum_{j\geq 0}\{n(n+1) - (j+s)(j+s+1)\}a_j x^{j+s} = 0.$$

The coefficient of x^{s-2} is $s(s-1)a_0$, and that of x^{s-1} is $s(s+1)a_1$ (the $j=0,1$ terms of the first sum). Thus $s=0$ or 1; a_1 is free for $s=0$ and is zero for $s=1$. For $j\ge 0$ the coefficient of x^{j+s} comes from the first sum at index $j+2$ and from the second at index j:

$$(j+s+2)(j+s+1)a_{j+2}+\{n(n+1)-(j+s)(j+s+1)\}a_j=0.$$

Since $(j+s)(j+s+1)-n(n+1)=(j+s-n)(j+s+n+1)$ (expand the right side),

$$\boxed{a_{j+2}=\frac{(j+s-n)(j+s+n+1)}{(j+s+1)(j+s+2)}a_j}.$$

In the unfactored form, $s=0$ and $s=1$ give the recurrences printed in (b) and (c). The point $x=0$ is ordinary; the two values of s simply select the even and odd Taylor solutions.

(b) For $s=0$, set $a_1=0$. The recurrence with $j=0$ and $j=2$ gives

$$a_2=-\frac{n(n+1)}{2}a_0,\qquad a_4=\frac{(2-n)(n+3)}{3\cdot 4}a_2=\frac{(n-2)n(n+1)(n+3)}{4!}a_0.$$

In general, since the denominators multiply to $\prod_{r=0}^{k-1}(2r+1)(2r+2)=(2k)!$, and with an empty product equal to one,

$$y_e=a_0\sum_{k=0}^{\infty}\frac{\prod_{r=0}^{k-1}(2r-n)(2r+n+1)}{(2k)!}x^{2k}.$$

This gives the stated even series and its recurrence.

(c) For $s=1$, $a_1=0$ is forced by $2a_1=0$. Now

$$a_2=-\frac{(n-1)(n+2)}{3!}a_0,\qquad a_4=\frac{(3-n)(n+4)}{4\cdot 5}a_2=\frac{(n-3)(n-1)(n+2)(n+4)}{5!}a_0.$$

The denominators now multiply to $\prod_{r=0}^{k-1}(2r+2)(2r+3)=(2k+1)!$. Thus the independent odd solution is

$$y_o=a_0\sum_{k=0}^{\infty}\frac{\prod_{r=0}^{k-1}(2r+1-n)(2r+n+2)}{(2k+1)!}x^{2k+1}.$$

The two occurrences of a_0 are independent arbitrary constants.

(d) Let c_k be the coefficient of x^{2k+s} in either nonterminating branch, with $s=0$ or 1; no c_k vanishes, since a zero coefficient would terminate the branch. The exact recurrence and $(2k+s+1)(2k+s+2)-(2k+s)(2k+s+1)=2(2k+s+1)$ imply

$$\frac{c_{k+1}}{c_k}=\frac{(2k+s)(2k+s+1)-n(n+1)}{(2k+s+1)(2k+s+2)}=1-\frac{2(2k+s+1)+n(n+1)}{(2k+s+1)(2k+s+2)}.$$

The subtracted numerator is $4k+O(1)$ and the denominator is $4k^2+O(k)$, so the ratio is $1-1/k+O(k^{-2})$. Consequently

$$\frac{(k+1)c_{k+1}}{kc_k}=\Big(1+\frac{1}{k}\Big)\Big(1-\frac{1}{k}+O(k^{-2})\Big)=1+O(k^{-2}).$$

Let $b_k=kc_k$. For real n the ratios b_{k+1}/b_k are real and tend to 1, so they are positive for $k\ge k_0$ (with $k_0\ge 2$), and

$$b_k=b_{k_0}\prod_{i=k_0}^{k-1}\frac{b_{i+1}}{b_i},\qquad \ell_i=\ln\frac{b_{i+1}}{b_i}=O(i^{-2})$$

by $\ln(1+z) = O(z)$. These logarithms have an absolutely convergent sum, so the product has a finite nonzero limit and $b_k \to C \neq 0$. Choose M with $|\ell_i| \le M/i^2$. Since $b_k = b_{k_0} \exp \sum_{i=k_0}^{k-1} \ell_i$ and $C = b_{k_0} \exp \sum_{i=k_0}^{\infty} \ell_i$, only the tail of the sum remains in their quotient:

$$\frac{b_k}{C} = \exp\Big(-\sum_{i\ge k} \ell_i\Big), \qquad \Big|\sum_{i\ge k} \ell_i\Big| \le M \sum_{i\ge k} \frac{1}{i^2} \le M \int_{k-1}^{\infty} \frac{dt}{t^2} = \frac{M}{k-1}.$$

Since $e^z = 1 + O(z)$ for bounded z, this gives $b_k = C[1 + O(k^{-1})]$, and

$$c_k = \frac{C}{k} + O(k^{-2}), \qquad C \neq 0.$$

At $x = 1$, the C/k terms have divergent partial sums, while the $O(k^{-2})$ remainder is absolutely summable. At $x = -1$, every term acquires the same factor $(-1)^{2k+s} = (-1)^s$, so it diverges there too. (Equivalently, $c_k/c_{k+1} = 1 + 1/k + O(k^{-2})$, and Gauss' test with $h = 1$ gives divergence, as in Exercise 1.2.5.) The ratio test gives convergence for $|x| < 1$, since the ratio of successive terms is $(c_{k+1}/c_k)x^2 \to x^2$. The Instructor's Manual's observation that $c_{k+1}/c_k \to 1$ is not by itself a divergence proof; the $1/k$ asymptotic above supplies the necessary endpoint argument.

(e) A branch terminates when an exponent $r = 2k + s \ge 0$ satisfies $(r - n)(r + n + 1) = 0$. With the conventional representative $n \ge 0$, this means $n = r$: even nonnegative n truncates the even branch, and odd positive n truncates the odd branch. The other branch remains nonterminating, so by (d) its series diverges at $x = \pm 1$. If negative parameters are allowed, $n = -r - 1$ produces the same eigenvalue $n(n+1) = r(r+1)$ and the same polynomial; restricting to $n \ge 0$ removes this duplicate labeling. Normalizing the surviving polynomial by $P_n(1) = 1$ gives the Legendre polynomial, for example $1, x, (3x^2 - 1)/2, (5x^3 - 3x)/2$ for $n = 0, 1, 2, 3$. For instance, $n = 2$ gives $y_e = a_0(1 - 3x^2)$, and $y_e(1) = -2a_0 = 1$ requires $a_0 = -\frac{1}{2}$; $n = 3$ gives $y_o = a_0(x - \frac{5}{3}x^3)$, and $y_o(1) = -\frac{2}{3}a_0 = 1$ requires $a_0 = -\frac{3}{2}$.

8.3.2 Show that with the weight factor $\exp(-x^2)$ and the interval $-\infty < x < \infty$ for the scalar product, the Hermite ODE eigenvalue problem is Hermitian. Use a common domain with vanishing boundary form at both infinities, such as the span of the polynomial eigenfunctions.

Solution.

A common domain satisfying the endpoint condition is required in addition to the weight and interval. The polynomial eigenfunctions satisfy it. Write the eigenvalue problem as $Ay = 2\alpha y$, where

$$A = -\frac{d^2}{dx^2} + 2x\frac{d}{dx} = -e^{x^2}\frac{d}{dx}\left(e^{-x^2}\frac{d}{dx}\right).$$

For finite endpoints a, b, use $e^{-x^2}Av = -(e^{-x^2}v')'$; integration by parts yields

$$\begin{aligned}
\langle u, Av\rangle_w &= -\int_a^b u^*(e^{-x^2}v')'\,dx \\
&= -[u^*e^{-x^2}v']_a^b + \int_a^b u'^*e^{-x^2}v'\,dx, \\
\langle Au, v\rangle_w &= -\int_a^b (e^{-x^2}u'^*)'v\,dx \\
&= -[u'^*e^{-x^2}v]_a^b + \int_a^b u'^*e^{-x^2}v'\,dx.
\end{aligned}$$

Their difference is

$$\langle u, Av\rangle_w - \langle Au, v\rangle_w = -\left[e^{-x^2}(u^*v' - u'^*v)\right]_a^b.$$

For polynomials u, v, the Gaussian defeats every power of $|x|$ (Exercise 8.2.4(c)), so both endpoint limits are zero as $a \to -\infty$, $b \to \infty$. All displayed integrals converge. The identity $\langle u, Av\rangle_w = \langle Au, v\rangle_w$ follows on the domain of finite linear combinations of these polynomials, which A maps into itself ($Ax^N = 2Nx^N - N(N-1)x^{N-2}$ does not raise the degree). More generally it holds on any common operator domain where the integrals exist and the same boundary form vanishes.

Weighted square integrability alone, as asserted in the Instructor's Manual, is insufficient to establish this boundary condition. For example $u = 1$ and $v = \sin(e^{x^2})$ are both in $L^2(\mathbb{R}, e^{-x^2}dx)$, since they are bounded, but $v' = 2xe^{x^2}\cos(e^{x^2})$, so

$$e^{-x^2}(uv' - u'v) = 2x\cos(e^{x^2})$$

has no zero limit: at $x_k = \sqrt{\ln(2\pi k)}$, where $e^{x_k^2} = 2\pi k$, it equals $2x_k \to \infty$. Such a function is excluded from the domain used in the Hermitian problem; square integrability does not replace an operator-domain condition.

8.3.3

(a) Develop series solutions for Hermite's differential equation

$$y'' - 2xy' + 2\alpha y = 0.$$

ANS. $s(s-1) = 0$, indicial equation.

For $s = 0$,

$$a_{j+2} = 2a_j\frac{j-\alpha}{(j+1)(j+2)} \quad (j \text{ even}),$$

$$y_{\text{even}} = a_0\left[1 + \frac{2(-\alpha)x^2}{2!} + \frac{2^2(-\alpha)(2-\alpha)x^4}{4!} + \cdots\right].$$

For $s = 1$,

$$a_{j+2} = 2a_j\frac{j+1-\alpha}{(j+2)(j+3)} \quad (j \text{ even}),$$

$$y_{\text{odd}} = a_1\left[x + \frac{2(1-\alpha)x^3}{3!} + \frac{2^2(1-\alpha)(3-\alpha)x^5}{5!} + \cdots\right].$$

(b) Show that both series solutions are convergent for all x, the ratio of successive coefficients behaving, for a large index, like the corresponding ratio in the expansion of $\exp(x^2)$.

(c) Show that by appropriate choice of α, the series solutions may be cut off and converted to finite polynomials. (These polynomials, properly normalized, become the Hermite polynomials in Section 18.1.)

Solution.

(a) Substitution of $y = \sum_{j\ge0} a_j x^{j+s}$ gives $y'' = \sum_{j\ge0}(j+s)(j+s-1)a_j x^{j+s-2}$ and $-2xy' + 2\alpha y = \sum_{j\ge0} 2(\alpha - j - s)a_j x^{j+s}$, so

$$\sum_{j\ge0}(j+s)(j+s-1)a_j x^{j+s-2} + \sum_{j\ge0} 2(\alpha - j - s)a_j x^{j+s} = 0.$$

The $j = 0, 1$ terms of the first sum are the only ones with x^{s-2} and x^{s-1}; for $j \geq 0$, the coefficient of x^{j+s} is $(j+s+2)(j+s+1)a_{j+2} + 2(\alpha - j - s)a_j$. Thus $s(s-1) = 0$, $s(s+1)a_1 = 0$, and

$$\boxed{a_{j+2} = \frac{2(j+s-\alpha)}{(j+s+1)(j+s+2)} a_j},$$

which for $s = 0$ and $s = 1$ is the printed recurrence. For the even solution take $s = 0$, $a_1 = 0$, $a_0 = A$. Successive substitutions ($j = 0$, then $j = 2$) give

$$a_2 = -\alpha A, \qquad a_4 = \frac{2(2-\alpha)}{3 \cdot 4}(-\alpha A) = \frac{2^2(-\alpha)(2-\alpha)}{4!} A,$$

and hence, since each step contributes a factor 2 and the denominators multiply to $(2k)!$,

$$y_e = A \sum_{k \geq 0} \frac{2^k \prod_{r=0}^{k-1}(2r - \alpha)}{(2k)!} x^{2k} = A \left[1 - \alpha x^2 + \frac{\alpha(\alpha-2)}{6} x^4 + \cdots \right].$$

For the odd solution take $s = 1$. Its Frobenius coefficient a_1 is forced to zero; call its free a_0 coefficient B, which is the coefficient of x (denoted a_1 in the printed answer). Then

$$a_2 = \frac{2(1-\alpha)}{2 \cdot 3} B, \qquad a_4 = \frac{2(3-\alpha)}{4 \cdot 5} a_2,$$

and the denominators now multiply to $\prod_{r=0}^{k-1}(2r+2)(2r+3) = (2k+1)!$, so

$$y_o = B \sum_{k \geq 0} \frac{2^k \prod_{r=0}^{k-1}(2r + 1 - \alpha)}{(2k+1)!} x^{2k+1}.$$

Its first terms are exactly $B[x + 2(1-\alpha)x^3/3! + 2^2(1-\alpha)(3-\alpha)x^5/5! + \cdots]$.

(b) For a fixed x, the ratio of successive nonzero terms of either parity has magnitude

$$|x|^2 \left| \frac{2(j+s-\alpha)}{(j+s+1)(j+s+2)} \right| \sim \frac{2|x|^2}{j} \longrightarrow 0.$$

Both series therefore converge absolutely for every finite real or complex x. With $j = 2k$, the ratio is asymptotic to x^2/k, which is the large-k behavior of the ratio $x^2/(k+1)$ for $e^{x^2} = \sum_{k \geq 0} x^{2k}/k!$.

(c) Termination requires $j + s = \alpha$ at a nonnegative integer exponent. Thus $\alpha = N \in \{0, 1, 2, \ldots\}$ gives one polynomial: the even branch for even N and the odd branch for odd N. The other branch does not terminate, and its coefficient must be set to zero if a polynomial is required. Choosing leading coefficient 2^N yields H_N, beginning with

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x.$$

For example, $\alpha = 2$ gives $y_e = A(1 - 2x^2)$, and $A = -2$ makes the leading coefficient 4; $\alpha = 3$ gives $y_o = B(x - \frac{2}{3}x^3)$, and $B = -12$ makes it 8.

8.3.4 Laguerre's ODE is

$$xL_n''(x) + (1-x)L_n'(x) + nL_n(x) = 0.$$

Develop a series solution and select the parameter n to make your series a polynomial.

Solution.

Use $y = \sum_{j\ge 0} a_j x^{j+s}$. The $xy'' + y'$ terms combine as

$$xy'' + y' = \sum_{j\ge 0}(j+s)\big[(j+s-1)+1\big]a_j x^{j+s-1} = \sum_{j\ge 0}(j+s)^2 a_j x^{j+s-1},$$

whereas $-xy' + ny = \sum_{j\ge 0}(n-j-s)a_j x^{j+s}$. The lowest coefficient is $s^2 a_0$, so the repeated indicial root is $s = 0$. Matching x^{j+s} (index $j+1$ in the first sum, j in the second) gives

$$(j+s+1)^2 a_{j+1} + (n-j-s)a_j = 0, \qquad \boxed{a_{j+1} = \frac{j-n}{(j+1)^2}a_j \quad (s=0)}.$$

The recurrence advances by one index; the Instructor's Manual's a_{j+2} on the left is a typographical error. Iteration yields

$$a_1 = -na_0, \quad a_2 = \frac{n(n-1)}{(2!)^2}a_0, \quad a_3 = -\frac{n(n-1)(n-2)}{(3!)^2}a_0,$$

and, in general, since each step multiplies by $-(n-j)/(j+1)^2$,

$$y = a_0 \sum_{j=0}^{\infty} \frac{(-1)^j n(n-1)\cdots(n-j+1)}{(j!)^2}x^j.$$

For a nonterminating solution the term ratio is asymptotic to x/j, so this regular solution is entire. It becomes a polynomial precisely when $n = N$ is a nonnegative integer; then $a_{N+1} = 0$ and all later coefficients vanish. With $a_0 = 1$, and $N(N-1)\cdots(N-j+1)/j! = \binom{N}{j}$,

$$\boxed{L_N(x) = \sum_{j=0}^{N}(-1)^j \binom{N}{j}\frac{x^j}{j!}}.$$

For example $L_0 = 1$, $L_1 = 1 - x$, and $L_2 = 1 - 2x + x^2/2$. There is only one analytic solution near zero up to scale, since a_0 determines every coefficient. Reduction of order (Exercise 7.6.11(b), with the self-adjoint coefficient $p = xe^{-x}$ of Exercise 8.2.1) supplies a second local solution

$$y_2 = y_1 \int \frac{e^x}{x\, y_1(x)^2}\, dx,$$

where $y_1(0) = 1$. Since $e^x/y_1^2 = 1 + O(x)$, its integrand is $1/x + O(1)$, so the integral is $\ln x + O(x)$ plus a constant; y_2 contains $y_1 \ln x$ and cannot be a regular polynomial at zero.

8.3.5 Solve the Chebyshev equation

$$(1-x^2)T_n'' - xT_n' + n^2 T_n = 0,$$

by series substitution. What restrictions are imposed on n if you demand that the series solution converge for $x = \pm 1$?

ANS. The infinite series does converge for $x = \pm 1$ and no restriction on n exists (compare with Exercise 1.2.6).

Solution.

Substitute $y = \sum_{j\ge 0} a_j x^{j+s}$. As in Exercise 8.3.1, y'' supplies the powers x^{j+s-2}, while in $-x^2y'' - xy' + n^2 y$ the coefficient of x^{j+s} is $n^2 - (j+s)(j+s-1) - (j+s) = n^2 - (j+s)^2$. The coefficients of x^{s-2} and x^{s-1} give $s(s-1) = 0$ and $s(s+1)a_1 = 0$. The other coefficients give

$$(j+s+1)(j+s+2)a_{j+2} + [n^2 - (j+s)^2]a_j = 0,$$

so

$$\boxed{a_{j+2} = \frac{(j+s)^2 - n^2}{(j+s+1)(j+s+2)} a_j}.$$

The independent even and odd solutions, normalized to $y_e(0) = 1$ and $y_o'(0) = 1$, are therefore (the denominators multiply to $(2k)!$ and $(2k+1)!$, as in Exercise 8.3.1)

$$y_e = \sum_{k\ge 0} \frac{\prod_{r=0}^{k-1}[(2r)^2 - n^2]}{(2k)!} x^{2k} = 1 - \frac{n^2x^2}{2!} + \frac{n^2(n^2-4)x^4}{4!} + \cdots,$$

$$y_o = \sum_{k\ge 0} \frac{\prod_{r=0}^{k-1}[(2r+1)^2 - n^2]}{(2k+1)!} x^{2k+1} = x + \frac{(1-n^2)x^3}{3!} + \frac{(1-n^2)(9-n^2)x^5}{5!} + \cdots.$$

For a nonterminating branch put $c_k = a_{2k}$ in its own Frobenius notation. Since $(2k+s+1)(2k+s+2) - (2k+s)^2 = 3(2k+s) + 2$, its ratio is

$$\frac{c_{k+1}}{c_k} = \frac{(2k+s)^2 - n^2}{(2k+s+1)(2k+s+2)} = 1 - \frac{3(2k+s) + 2 + n^2}{(2k+s+1)(2k+s+2)}.$$

The fraction subtracted from one is $(6k+O(1))/(4k^2+O(k)) = 3/(2k) + O(k^{-2})$. Since $(1+1/k)^{3/2} = 1 + 3/(2k) + O(k^{-2})$, multiplying c_k by $k^{3/2}$ makes the new successive ratio

$$\frac{(k+1)^{3/2}c_{k+1}}{k^{3/2}c_k} = \Big(1 + \frac{3}{2k} + O(k^{-2})\Big)\Big(1 - \frac{3}{2k} + O(k^{-2})\Big) = 1 + O(k^{-2}),$$

and, as in Exercise 8.3.1(d), the product of these ratios converges to a nonzero constant. Thus $c_k \sim Ck^{-3/2}$. Comparison with $\sum k^{-3/2}$ proves absolute convergence at both $x = 1$ and $x = -1$; this is Gauss' test with $h = 3/2 > 1$, as in Exercise 1.2.6. Terminating branches also converge there. Hence

$$\boxed{\text{endpoint convergence imposes no restriction on } n.}$$

As a consistency check, which also identifies the sums, set $x = \sin t$ with $|t| < \pi/2$. Writing $Y(t) = y(\sin t)$, the chain rule gives

$$Y' = y'\cos t, \qquad Y'' = y''\cos^2 t - y'\sin t = (1-x^2)y'' - xy'.$$

Thus $Y''+n^2Y=0$. Since $Y(0)=y(0)$ and $Y'(0)=y'(0)$, the initial conditions $y_e(0)=1$, $y_e'(0)=0$ and $y_o(0)=0$, $y_o'(0)=1$ select $Y=\cos nt$ and $Y=\sin(nt)/n$ (or $Y=t$ if $n=0$), with $t=\arcsin x$:

$$y_e=\cos(n\arcsin x),\qquad y_o=\begin{cases}\sin(n\arcsin x)/n, & n\neq 0,\\ \arcsin x, & n=0.\end{cases}$$

These closed forms hold for $|x|<1$. Since the series converge at $x=\pm1$, Abel's theorem (a power series is continuous up to an endpoint at which it converges) makes their sums there equal to the limits of the closed forms as $x\to\pm1$, where $\arcsin x\to\pm\pi/2$: $y_e(\pm1)=\cos(n\pi/2)$ and $y_o(\pm1)=\pm\sin(n\pi/2)/n$ ($\pm\pi/2$ if $n=0$). Finite limits alone would not prove convergence: $1/(1+x)=\sum_{j\ge0}(-x)^j\to\frac12$ as $x\to1^-$, but the series diverges at $x=1$.

Polynomial solutions instead require $n^2=N^2$, N a nonnegative integer, and selection of the branch of parity N. Endpoint convergence and polynomial termination are therefore different requirements.

8.3.6 Solve

$$(1-x^2)U_n''(x)-3xU_n'(x)+n(n+2)U_n(x)=0,$$

choosing the root of the indicial equation to obtain a series of odd powers of x. Since the nonterminating series will diverge for $x=1$, choose n to convert it into a polynomial.

Solution.

Substitution of $y=\sum_{j\ge0}a_jx^{j+s}$ gives, as in Exercise 8.3.5, $s(s-1)=0$ and $s(s+1)a_1=0$; now the coefficient of x^{j+s} in $-x^2y''-3xy'+n(n+2)y$ is $n(n+2)-(j+s)(j+s-1)-3(j+s)=n(n+2)-(j+s)(j+s+2)$, so

$$a_{j+2}=\frac{(j+s)(j+s+2)-n(n+2)}{(j+s+1)(j+s+2)}a_j.$$

For the odd branch choose $s=1$, which forces $a_1=0$. Factor the numerator, $(j+1)(j+3)-n(n+2)=(j+1-n)(j+n+3)$, to obtain

$$\boxed{a_{j+2}=\frac{(j+1-n)(j+n+3)}{(j+2)(j+3)}a_j\quad(j=0,2,4,\dots)}.$$

In particular, $j=0$ and $j=2$ give

$$a_2=-\frac{(n-1)(n+3)}{3!}a_0,\qquad a_4=\frac{(n-3)(n-1)(n+3)(n+5)}{5!}a_0,$$

so

$$y=a_0\left[x-\frac{(n-1)(n+3)}{3!}x^3+\frac{(n-3)(n-1)(n+3)(n+5)}{5!}x^5+\cdots\right].$$

For the nonterminating odd series, $c_k=a_{2k}$ satisfies, since $(2k+2)(2k+3)-(2k+1)(2k+3)=2k+3$,

$$\frac{c_{k+1}}{c_k}=\frac{(2k+1)(2k+3)-n(n+2)}{(2k+2)(2k+3)}=1-\frac{2k+3+n(n+2)}{(2k+2)(2k+3)}.$$

The subtracted fraction is $(2k+O(1))/(4k^2+O(k))=1/(2k)+O(k^{-2})$. Hence

$$\frac{(k+1)^{1/2}c_{k+1}}{k^{1/2}c_k}=\Big(1+\frac{1}{2k}+O(k^{-2})\Big)\Big(1-\frac{1}{2k}+O(k^{-2})\Big)=1+O(k^{-2}),$$

and, as in Exercise 8.3.1(d), a convergent product shows $c_k \sim Ck^{-1/2}$, $C \neq 0$; the terms have a fixed eventual sign for real n, so by comparison with $\sum k^{-1/2}$ the series diverges at $x = 1$ (and at -1, where every term is multiplied by -1). The statement's "nonterminating" is a necessary qualification: a terminating series is a polynomial, finite at $x = 1$. Termination occurs at an odd exponent $N = 2k + 1$ when

$$(N - n)(N + n + 2) = 0.$$

With the usual convention $n \geq 0$, choose $n = N = 1, 3, 5, \ldots$; then the degree is N. Allowing negative labels also gives $n = -N - 2 = -3, -5, \ldots$, which produce the same eigenvalue and polynomial. The standard second-kind Chebyshev normalization is $a_0 = (-1)^{(N-1)/2}(N + 1)$. The first three polynomials are

$$U_1 = 2x, \qquad U_3 = 8x^3 - 4x, \qquad U_5 = 32x^5 - 32x^3 + 6x.$$

Indeed, for $n = 3$ the series is $a_0(x - 2x^3)$, and $a_0 = -4$ gives U_3; for $n = 5$ it is $a_0(x - \frac{16}{3}x^3 + \frac{16}{3}x^5)$, and $a_0 = 6$ gives U_5. This normalization can be checked from $U_N(\cos\theta) = \sin((N + 1)\theta)/\sin\theta$. Since $d/dx = -(\sin\theta)^{-1}d/d\theta$, differentiation at $x = 0$ ($\theta = \pi/2$) gives, by the quotient rule with $\sin\frac{\pi}{2} = 1$ and $\cos\frac{\pi}{2} = 0$,

$$\begin{aligned} a_0 = U_N'(0) &= -\frac{(N+1)\cos\frac{(N+1)\pi}{2}\sin\frac{\pi}{2} - \sin\frac{(N+1)\pi}{2}\cos\frac{\pi}{2}}{\sin^3\frac{\pi}{2}} \\ &= -(N+1)\cos\frac{(N+1)\pi}{2} = (-1)^{(N-1)/2}(N+1), \end{aligned}$$

where the last equality uses odd $N = 2m + 1$: $\cos((m + 1)\pi) = (-1)^{m+1}$.

8.4 Variation Method

8.4.1 A function that is normalized on the interval $0 \leq x < \infty$ with an unweighted scalar product is

$$\psi = 2\alpha^{3/2}xe^{-\alpha x}, \qquad \alpha > 0 \text{ real}.$$

(a) Verify the normalization.

(b) Verify that for this ψ, $\langle x^{-1}\rangle = \alpha$.

(c) Verify that for this ψ, $\langle d^2/dx^2\rangle = -\alpha^2$.

(d) Use the variation method to find the value of α that minimizes

$$\left\langle \psi \left| -\frac{1}{2}\frac{d^2}{dx^2} - \frac{1}{x} \right| \psi \right\rangle,$$

and find the minimum value of this expectation value.

Solution.

Take $\alpha > 0$ real; this is necessary for the stated normalized trial function: $\alpha = 0$ gives $\psi = 0$; for $\operatorname{Re}\alpha \leq 0$, $\alpha \neq 0$, the normalization integral diverges; and for nonreal α with $\operatorname{Re}\alpha > 0$ it equals

$|\alpha|^3/(\operatorname{Re}\alpha)^3 > 1$. For $\beta > 0$, define $I_k(\beta) = \int_0^\infty x^k e^{-\beta x}\,dx$. Direct integration gives $I_0 = 1/\beta$, and integration by parts gives

$$I_k = \left[-\frac{x^k e^{-\beta x}}{\beta}\right]_0^\infty + \frac{k}{\beta} I_{k-1} = \frac{k}{\beta} I_{k-1} \quad (k \geq 1).$$

Therefore $I_k = k!/\beta^{k+1}$.

(a) The normalization integral is

$$\int_0^\infty |\psi|^2\,dx = 4\alpha^3 I_2(2\alpha) = 4\alpha^3 \frac{2!}{(2\alpha)^3} = 1.$$

(b) Multiplication by x^{-1} removes one power of x:

$$\boxed{\langle x^{-1}\rangle = 4\alpha^3 I_1(2\alpha) = 4\alpha^3 \frac{1!}{(2\alpha)^2} = \alpha}.$$

The apparent singularity at 0 is harmless because the integrand is proportional to $xe^{-2\alpha x}$.

(c) Differentiate with the exponential retained:

$$\psi' = 2\alpha^{3/2}(1-\alpha x)e^{-\alpha x}, \qquad \psi'' = 2\alpha^{3/2}\big[-\alpha - \alpha(1-\alpha x)\big]e^{-\alpha x} = 2\alpha^{3/2}(\alpha^2 x - 2\alpha)e^{-\alpha x}.$$

The factor $e^{-\alpha x}$ is missing from the Instructor's Manual's displayed second derivative, although its following integral is correct. Consequently, with $\psi\psi'' = 4\alpha^3(\alpha^2 x^2 - 2\alpha x)e^{-2\alpha x}$ and $I_2(2\alpha)$, $I_1(2\alpha)$ from above,

$$\begin{aligned}\langle d^2/dx^2\rangle &= 4\alpha^3 \int_0^\infty (\alpha^2 x^2 - 2\alpha x)e^{-2\alpha x}\,dx \\ &= 4\alpha^3\left[\alpha^2 \frac{2!}{(2\alpha)^3} - 2\alpha \frac{1!}{(2\alpha)^2}\right] = \boxed{-\alpha^2}.\end{aligned}$$

The sign also follows by integration by parts: $\int \psi\psi'' = [\psi\psi']_0^\infty - \int(\psi')^2 = -\int(\psi')^2$, since $\psi(0) = 0$ and the exponential removes the boundary term at infinity. Indeed,

$$\begin{aligned}\int_0^\infty (\psi')^2\,dx &= 4\alpha^3\big[I_0(2\alpha) - 2\alpha I_1(2\alpha) + \alpha^2 I_2(2\alpha)\big] \\ &= 4\alpha^3\left[\frac{1}{2\alpha} - \frac{2\alpha}{4\alpha^2} + \frac{2\alpha^2}{8\alpha^3}\right] = \alpha^2.\end{aligned}$$

(d) By linearity, the expectation value is $-\frac{1}{2}\langle d^2/dx^2\rangle - \langle x^{-1}\rangle$; by (b) and (c),

$$E(\alpha) = -\tfrac{1}{2}(-\alpha^2) - \alpha = \frac{\alpha^2}{2} - \alpha = \frac{(\alpha-1)^2}{2} - \frac{1}{2}.$$

Thus the unique minimum for $\alpha > 0$ is

$$\boxed{\alpha_{\text{opt}} = 1, \qquad E_{\min} = -\tfrac{1}{2}}.$$

Equivalently $E' = \alpha - 1 = 0$ and $E'' = 1 > 0$. As a direct check, at $\alpha = 1$ we have $\psi = 2xe^{-x}$ and $\psi'' = 2(x-2)e^{-x}$, so

$$\left(-\frac{1}{2}\frac{d^2}{dx^2} - \frac{1}{x}\right)\psi = (2-x)e^{-x} - 2e^{-x} = -xe^{-x} = -\frac{1}{2}\psi;$$

the optimized trial function is an exact eigenfunction in this case.

9 Chapter 9. Partial Differential Equations

Problems

9.2 First-Order Equations

Find the general solutions of the PDEs in Exercises 9.2.1 to 9.2.4.

9.2.1 $\dfrac{\partial\psi}{\partial x} + 2\dfrac{\partial\psi}{\partial y} + (2x - y)\psi = 0.$

Solution.

Let $s = x$ and $u = y - 2x$, so $y = u + 2s$. The chain rule gives

$$\partial_x = \partial_s - 2\partial_u, \qquad \partial_y = \partial_u, \qquad 2x - y = 2s - (u + 2s) = -u.$$

Hence $\partial_x + 2\partial_y = \partial_s - 2\partial_u + 2\partial_u = \partial_s$, and the PDE reduces, at fixed u, to $\psi_s - u\psi = 0$. Multiplying by e^{-us} gives $\partial_s(e^{-us}\psi) = e^{-us}(\psi_s - u\psi) = 0$, so $e^{-us}\psi = F(u)$ does not depend on s. Returning to $s = x$, $u = y - 2x$,

$$\boxed{\psi(x, y) = e^{x(y-2x)}F(y - 2x)},$$

where F is an arbitrary differentiable function; this includes the zero solution and does not require taking $\ln\psi$. Indeed $D = \partial_x + 2\partial_y$ annihilates u and satisfies $D(xu) = u$, hence $D\psi = u\psi = -(2x - y)\psi$. The coordinate map has determinant one, so it loses no local solutions.

9.2.2 $\dfrac{\partial\psi}{\partial x} - 2\dfrac{\partial\psi}{\partial y} + x + y = 0.$

Solution.

Put $s = x$, $u = y + 2x$. The chain rule gives $\partial_x = \partial_s + 2\partial_u$ and $\partial_y = \partial_u$, so $\partial_x - 2\partial_y = \partial_s$; also $x + y = s + (u - 2s) = u - s$. This reduces the PDE to

$$\psi_s = s - u.$$

Integration with u held fixed gives

$$\psi = \tfrac{1}{2}s^2 - us + F(u) = \tfrac{1}{2}x^2 - (y + 2x)x + F(y + 2x) = -\tfrac{3}{2}x^2 - xy + F(y + 2x).$$

Since $\tfrac{1}{2}(x + y)^2 - (-\tfrac{3}{2}x^2 - xy) = 2x^2 + 2xy + \tfrac{1}{2}y^2 = \tfrac{1}{2}(y + 2x)^2$, the difference is $\tfrac{1}{2}u^2$, a function of u alone, and can be absorbed into the arbitrary function F. An equivalent convenient form is

$$\boxed{\psi(x, y) = \tfrac{1}{2}(x + y)^2 + F(y + 2x)}.$$

Direct differentiation gives $\psi_x = (x + y) + 2F'$ and $\psi_y = (x + y) + F'$, so $\psi_x - 2\psi_y + x + y = 0$ exactly.

9.2.3 $\dfrac{\partial\psi}{\partial x} + \dfrac{\partial\psi}{\partial y} = \dfrac{\partial\psi}{\partial z}.$

Solution.

Move the z derivative to the left and put $s = x$, $u = y - x$, $v = z + x$. The inverse transformation is $x = s$, $y = u + s$, $z = v - s$, and

$$\partial_x = \partial_s - \partial_u + \partial_v, \quad \partial_y = \partial_u, \quad \partial_z = \partial_v.$$

Thus $(\partial_x + \partial_y - \partial_z)\psi = (\partial_s - \partial_u + \partial_v + \partial_u - \partial_v)\psi = \psi_s = 0$, so ψ does not depend on s at fixed (u, v), giving

$$\boxed{\psi(x, y, z) = F(y - x, z + x)}.$$

Here F is any differentiable function of two arguments. Its derivatives are $\psi_x = -F_u + F_v$, $\psi_y = F_u$, $\psi_z = F_v$, which directly verify the original equation. The characteristic lines have tangent $(1, 1, -1)$ and keep both arguments of F constant.

9.2.4 $\dfrac{\partial \psi}{\partial x} + \dfrac{\partial \psi}{\partial y} + \dfrac{\partial \psi}{\partial z} = x - y.$

Solution.

Choose $s = z$, $u = x - z$, $v = y - z$, whose inverse is $x = u + s$, $y = v + s$, $z = s$. The chain rule gives $\partial_x = \partial_u$, $\partial_y = \partial_v$ and $\partial_z = \partial_s - \partial_u - \partial_v$, so the sum of the three original derivatives is ∂_s; also $x - y = u - v$. Therefore

$$\psi_s = u - v, \qquad \psi = s(u - v) + F(u, v),$$

or

$$\boxed{\psi(x, y, z) = z(x - y) + F(x - z, y - z)}.$$

The derivative sum annihilates $x - z$, $y - z$ and $x - y$, while it sends z to one. Applied to the boxed expression it gives $x - y$, as required. Every characteristic is a straight line in the direction $(1, 1, 1)$.

9.2.5

(a) Show that the PDE

$$y\frac{\partial \psi}{\partial x} + x\frac{\partial \psi}{\partial y} = 0$$

can be transformed into a readily soluble form by writing it in the new variables $u = xy$, $v = x^2 - y^2$, and find its general local solution away from the origin; distinguish connected characteristic branches when describing global solutions.

(b) Discuss this result in terms of characteristics.

Solution.

(a) For $u = xy$, $v = x^2 - y^2$, the chain rule gives

$$\psi_x = y\psi_u + 2x\psi_v, \qquad \psi_y = x\psi_u - 2y\psi_v.$$

Thus

$$y\psi_x + x\psi_y = y(y\psi_u + 2x\psi_v) + x(x\psi_u - 2y\psi_v) = (x^2 + y^2)\psi_u.$$

Since $(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2 = v^2 + 4u^2$, in the new variables alone the PDE reads $\sqrt{v^2 + 4u^2}\,\psi_u = 0$. Away from $(0, 0)$ the coefficient is nonzero, so $\psi_u = 0$ and

$$\boxed{\psi = F(v) = F(x^2 - y^2) \quad \text{locally}}.$$

The Jacobian is $\partial(u, v)/\partial(x, y) = u_x v_y - u_y v_x = y(-2y) - x(2x) = -2(x^2 + y^2)$, confirming local invertibility at every such point.

(b) A characteristic parametrized by s satisfies $x' = y$, $y' = x$, $\psi' = 0$: along such a curve $\psi' = x'\psi_x + y'\psi_y = y\psi_x + x\psi_y$, which the PDE sets to zero. Adding and subtracting the first two relations gives $(x+y)' = x+y$ and $(x-y)' = -(x-y)$, so

$$x + y = Ae^s, \qquad x - y = Be^{-s}, \qquad x^2 - y^2 = (x+y)(x-y) = AB.$$

The nonzero level sets are hyperbolas, and the zero level consists of the lines $y = \pm x$, with each ray a separate characteristic; the origin is a stationary point. The solution is constant along every connected characteristic.

Away from the origin, a single function $F(x^2 - y^2)$ is the general local form. Globally, distinct connected branches of the same level set can carry different functions, subject to smooth matching across the separating lines and at the origin. More explicitly, on the whole plane a C^1 solution can use four functions F_R, F_L for $v > 0$ (right and left sectors) and F_U, F_D for $v < 0$ (upper and lower sectors). These functions must extend C^1 to $v = 0$ with a common value C and a common first derivative. Set $\psi = C$ on both lines $y = \pm x$. The common value follows because each characteristic ray approaches the origin; the common derivative follows by crossing each ray at a nonzero point, where $\nabla v \neq 0$. Conversely these matching conditions make the piecewise formula C^1: off the lines, $\nabla\psi = F'(v)(2x, -2y)$; across a nonzero ray both limits agree, and at the origin the gradient tends to zero. For a smooth solution all one-sided derivatives of the four F's must likewise agree at zero. For example, write $v = x^2 - y^2$ and set

$$\psi = \begin{cases} e^{-1/v^2}, & v > 0,\ x > 0, \\ -e^{-1/v^2}, & v > 0,\ x < 0, \\ 0, & v \leq 0. \end{cases}$$

Each derivative of e^{-1/v^2} is that exponential times a polynomial in $1/v$: differentiating preserves this form because $(e^{-1/v^2})' = 2v^{-3}e^{-1/v^2}$. Every resulting term tends to zero: with $w = 1/v$, $w^m e^{-w^2} \to 0$ for every fixed m. Thus all four branch functions have zero derivatives of every order at zero; equivalently, $G(v) = e^{-1/v^2}$ for $v > 0$, $G(v) = 0$ for $v \leq 0$, is smooth on $\mathbb{R}$ with $G^{(j)}(0) = 0$ for all j. Now $\psi = G(v)$ on the half-plane $x > 0$, $\psi = -G(v)$ on $x < 0$, and $\psi = 0$ on the open set $v < 0$, which contains the y-axis minus the origin. These three open sets cover the plane minus the origin, and on each of them ψ is a smooth function of v alone, so there it is smooth and satisfies the PDE. Off the origin each partial derivative of ψ of order j is a sum of terms $\pm G^{(i)}(v)$, $i \leq j$, times polynomials in x, y, so it tends to zero as $(x, y) \to (0, 0)$: then $v \to 0$ and $G^{(i)}(v) \to 0$, while the polynomials stay bounded. A function continuous at the origin whose gradient tends to zero there is differentiable at the origin with zero gradient, by the mean value theorem on the segment from the origin. Applying this to ψ and then to each of its partial derivatives in turn shows that ψ is smooth at the origin too, with all derivatives zero. Its values at (x, y) and $(-x, -y)$ with $v > 0$ differ, proving that it cannot be represented by one global $F(v)$. At the origin the PDE itself is $0 = 0$; continuity and differentiability govern the matching there.

9.2.6 Find the general local solution away from the origin to the PDE

$$x\frac{\partial\psi}{\partial x} - y\frac{\partial\psi}{\partial y} = 0,$$

and account for characteristic branches globally. *Hint.* The solution to Exercise 9.2.5 may provide a suggestion as to how to proceed.

Solution.

Use the same coordinates $u = xy$, $v = x^2 - y^2$ as in Exercise 9.2.5. Now

$$x\psi_x - y\psi_y = x(y\psi_u + 2x\psi_v) - y(x\psi_u - 2y\psi_v) = 2(x^2 + y^2)\psi_v.$$

Off the origin the coefficient $2(x^2 + y^2)$ is nonzero, so $\psi_v = 0$ and

$$\boxed{\psi(x, y) = F(xy) \quad \text{locally}}.$$

Along a curve with $x' = x$, $y' = -y$ the PDE gives $\psi' = x'\psi_x + y'\psi_y = x\psi_x - y\psi_y = 0$. The characteristic equations and their integrals are therefore

$$x' = x, \quad y' = -y, \quad \psi' = 0, \qquad x = Ae^s, \quad y = Be^{-s}, \quad xy = AB.$$

Consequently the characteristics are connected branches of $xy =$ constant, together with the four open coordinate-axis rays and the stationary origin.

As in Exercise 9.2.5, different branches can have different functions of the invariant; smoothness must be imposed where they meet. A single $F(xy)$ need not describe every global solution. On the whole plane assign functions $F_{\mathrm{I}}, F_{\mathrm{III}}$ on $u > 0$ to the first and third quadrants, and $F_{\mathrm{II}}, F_{\mathrm{IV}}$ on $u < 0$ to the other quadrants. For a C^1 solution they extend to $u = 0$ with a common value C and first derivative; set $\psi = C$ on the axes. Necessity follows by approaching the origin along each ray and by crossing each nonzero ray transversely. Sufficiency follows from $\nabla\psi = F'(u)(y, x)$, whose limiting values match on each ray and tend to zero at the origin. Smooth solutions require matching derivatives of every order, exactly as in Exercise 9.2.5. For a concrete example let $u = xy$ and take $\psi = \operatorname{sgn}(x)e^{-1/u^2}$ when $u > 0$, and $\psi = 0$ when $u \le 0$. It is smooth: with the flat function G of Exercise 9.2.5, $\psi = G(u)$ on each of the half-planes $x > 0$ and $y > 0$, and $\psi = -G(u)$ on each of $x < 0$ and $y < 0$, because $G(xy) \neq 0$ only where x and y have the same sign. These four open half-planes cover the plane minus the origin, and at the origin the argument of Exercise 9.2.5 applies with u in place of v. Both u and the sign of x are constant along each of its nonzero characteristics, so it solves the PDE, but its values in the first and third quadrants differ at the same positive u.

9.3 Second-Order Equations

9.3.1 Let a, b, c be real constants with $c > 0$, so the displayed coordinates are real and defined. Show that by making a change of variables to $\xi = c^{1/2}x - c^{-1/2}by$, $\eta = c^{-1/2}y$, the operator L of Eq. (9.18) can be brought to the form

$$L = (ac - b^2)\frac{\partial^2}{\partial\xi^2} + \frac{\partial^2}{\partial\eta^2}.$$

Solution.

The coefficient $ac - b^2$ multiplies only ∂_ξ^2, as in the textbook's Eq. (9.21). The displayed change of coordinates requires $c \neq 0$; take $c > 0$ for real square roots. The coefficients are constants. Expanding Eq. (9.18) gives $L = a\partial_x^2 + 2b\partial_x\partial_y + c\partial_y^2$. For the given ξ, η we have $\xi_x = \sqrt{c}$, $\eta_x = 0$, $\xi_y = -b/\sqrt{c}$, $\eta_y = 1/\sqrt{c}$, so the chain rule $\partial_x = \xi_x\partial_\xi + \eta_x\partial_\eta$ (and likewise for ∂_y) gives

$$\partial_x = \sqrt{c}\,\partial_\xi, \qquad \partial_y = -\frac{b}{\sqrt{c}}\partial_\xi + \frac{1}{\sqrt{c}}\partial_\eta.$$

As these coefficients are constant, a second differentiation gives

$$\begin{aligned}\partial_x^2 &= c\partial_\xi^2,\\ \partial_x\partial_y &= -b\partial_\xi^2 + \partial_\xi\partial_\eta,\\ \partial_y^2 &= \frac{b^2}{c}\partial_\xi^2 - \frac{2b}{c}\partial_\xi\partial_\eta + \frac{1}{c}\partial_\eta^2.\end{aligned}$$

Substituting and collecting coefficients yields

$$L = (ac - 2b^2 + b^2)\partial_\xi^2 + (2b - 2b)\partial_\xi\partial_\eta + \partial_\eta^2 = \boxed{(ac - b^2)\partial_\xi^2 + \partial_\eta^2}.$$

If $\Delta = ac - b^2 > 0$, a further rescaling $\widetilde{\eta} = \sqrt{\Delta}\,\eta$ does give $L = \Delta(\partial_\xi^2 + \partial_{\widetilde{\eta}}^2)$. That is a different coordinate change from the one stated.

9.4 Separation of Variables

9.4.1 By letting the operator $\nabla^2 + k^2$ act on the general form $a_1\psi_1(x,y,z) + a_2\psi_2(x,y,z)$, show that it is linear, i.e., that $(\nabla^2 + k^2)(a_1\psi_1 + a_2\psi_2) = a_1(\nabla^2 + k^2)\psi_1 + a_2(\nabla^2 + k^2)\psi_2$.

Solution.

For constants a_1, a_2, each second derivative obeys

$$\partial_j^2(a_1\psi_1 + a_2\psi_2) = a_1\partial_j^2\psi_1 + a_2\partial_j^2\psi_2 \quad (j = x, y, z).$$

Adding these three identities and distributing the multiplication by k^2 gives

$$\begin{aligned}(\nabla^2 + k^2)(a_1\psi_1 + a_2\psi_2) &= a_1\nabla^2\psi_1 + a_2\nabla^2\psi_2 + a_1k^2\psi_1 + a_2k^2\psi_2\\ &= a_1(\nabla^2 + k^2)\psi_1 + a_2(\nabla^2 + k^2)\psi_2.\end{aligned}$$

This proves linearity for arbitrary twice-differentiable functions. In particular, a constant linear combination of solutions of the homogeneous equation is another solution. The statement remains true if k^2 is a prescribed position-dependent coefficient, since multiplication by a fixed function is also linear.

9.4.2 Show that the Helmholtz equation,

$$\nabla^2\psi + k^2\psi = 0,$$

is still separable in circular cylindrical coordinates if k^2 is generalized to $k^2 + f(\rho) + (1/\rho^2)g(\varphi) + h(z)$.

Solution.

Set $\psi = R(\rho)\Phi(\varphi)Z(z)$. Substitution of

$$\nabla^2 = \frac{1}{\rho}\partial_\rho(\rho\partial_\rho) + \frac{1}{\rho^2}\partial_\varphi^2 + \partial_z^2$$

and division by the product, where it is nonzero, gives

$$\frac{(\rho R')'}{\rho R} + k^2 + f(\rho) + \frac{1}{\rho^2}\left(\frac{\Phi''}{\Phi} + g(\varphi)\right) + \frac{Z''}{Z} + h(z) = 0.$$

Only the last two terms depend on z, so set $Z''/Z + h = -\kappa^2$. The remaining equation, multiplied by ρ^2, is

$$\frac{\rho(\rho R')'}{R} + \rho^2[k^2 - \kappa^2 + f(\rho)] + \frac{\Phi''}{\Phi} + g(\varphi) = 0.$$

Set the angular part equal to $-\mu$. The three ordinary differential equations are therefore

$$\boxed{\begin{aligned} Z'' + [h(z) + \kappa^2]Z &= 0, \\ \Phi'' + [g(\varphi) + \mu]\Phi &= 0, \\ \rho^2 R'' + \rho R' + \{\rho^2[k^2 - \kappa^2 + f(\rho)] - \mu\}R &= 0. \end{aligned}}$$

Their products satisfy the original PDE by direct substitution, including at zeros of individual factors by continuity. The notation κ^2 does not presume positivity; boundary conditions determine the allowed constants. When $g = 0$ and the angular function is single-valued and periodic, $\mu = m^2$ with integer m.

9.4.3 Separate variables in the Helmholtz equation in spherical polar coordinates, splitting off the radial dependence first. Show that your separated equations have the same form as Eqs. (9.74), (9.77), and (9.78).

Solution.

Separate the radius first by writing $\psi = R(r)Y(\theta, \varphi)$. Define the angular differential expression

$$\Delta_\Omega Y = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta Y) + \frac{1}{\sin^2\theta}\partial_\varphi^2 Y.$$

In spherical polar coordinates $\nabla^2 = r^{-2}\partial_r(r^2\partial_r) + r^{-2}\Delta_\Omega$, so the Helmholtz equation becomes

$$\frac{Y}{r^2}(r^2R')' + \frac{R}{r^2}\Delta_\Omega Y + k^2 RY = 0.$$

Multiplication by $r^2/(RY)$ isolates the independent variables:

$$\frac{(r^2R')'}{R} + k^2r^2 = -\frac{\Delta_\Omega Y}{Y} = \lambda.$$

Hence

$$(r^2R')' + (k^2r^2 - \lambda)R = 0, \qquad \Delta_\Omega Y + \lambda Y = 0.$$

Now set $Y = \Theta(\theta)\Phi(\varphi)$ and multiply its equation by $\sin^2\theta/(\Theta\Phi)$. The result is

$$\frac{\sin\theta}{\Theta}(\sin\theta\,\Theta')' + \lambda\sin^2\theta = -\frac{\Phi''}{\Phi} = m^2.$$

Thus all three separated equations are

$$\boxed{\begin{aligned} \Phi'' + m^2\Phi &= 0, \\ \frac{1}{\sin\theta}(\sin\theta\,\Theta')' + \left(\lambda - \frac{m^2}{\sin^2\theta}\right)\Theta &= 0, \\ \frac{1}{r^2}(r^2R')' + \left(k^2 - \frac{\lambda}{r^2}\right)R &= 0. \end{aligned}}$$

These are Eqs. (9.74), (9.77), and (9.78), respectively. Periodicity of a nonzero Φ on the full azimuth makes m integral; regularity on the sphere selects $\lambda = l(l+1)$ with $l \geq |m|$, as in the associated Legendre problem. The separation itself does not require these boundary restrictions.

9.4.4 Verify that

$$\nabla^2\psi(r,\theta,\varphi)+\left[k^2+f(r)+\frac{1}{r^2}g(\theta)+\frac{1}{r^2\sin^2\theta}h(\varphi)\right]\psi(r,\theta,\varphi)=0$$

is separable (in spherical polar coordinates). The functions f, g, and h are functions only of the variables indicated; k^2 is a constant.

Solution.

Substitute $\psi = R(r)\Theta(\theta)\Phi(\varphi)$ in the PDE, write the Laplacian as in Exercise 9.4.3 with $\Delta_\Omega = (\sin\theta)^{-1}\partial_\theta(\sin\theta\,\partial_\theta) + (\sin\theta)^{-2}\partial_\varphi^2$, and multiply by r^2/ψ. We obtain

$$\frac{(r^2R')'}{R}+r^2[k^2+f(r)]+\frac{(\sin\theta\,\Theta')'}{\Theta\sin\theta}+g(\theta)+\frac{1}{\sin^2\theta}\left(\frac{\Phi''}{\Phi}+h(\varphi)\right)=0.$$

The radial part depends only on r; set it equal to λ. Multiply the remaining angular equation by $\sin^2\theta$ to find

$$\frac{\sin\theta}{\Theta}(\sin\theta\,\Theta')'+[g(\theta)+\lambda]\sin^2\theta=-\frac{\Phi''}{\Phi}-h(\varphi)=\mu.$$

The desired ODEs are

$$\boxed{\begin{aligned}(r^2R')'+\{r^2[k^2+f(r)]-\lambda\}R&=0,\\ \frac{1}{\sin\theta}(\sin\theta\,\Theta')'+\left[g(\theta)+\lambda-\frac{\mu}{\sin^2\theta}\right]\Theta&=0,\\ \Phi''+[h(\varphi)+\mu]\Phi&=0.\end{aligned}}$$

Every coefficient in each equation depends on its own independent variable alone. Conversely, multiplying the three equations by $\Theta\Phi/r^2$, $R\Phi/r^2$ and $R\Theta/(r^2\sin^2\theta)$, respectively, and adding gives the PDE: the separation constants enter as $-\lambda\psi/r^2+\lambda\psi/r^2$ and $-\mu\psi/(r^2\sin^2\theta)+\mu\psi/(r^2\sin^2\theta)$, which cancel. For general g, h, the separation constants need not be the unperturbed values $l(l+1)$ and m^2.

9.4.5 An atomic (quantum mechanical) particle is confined inside a rectangular box of sides a, b, and c. The particle is described by a wave function ψ that satisfies the Schrödinger wave equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi=E\psi.$$

The wave function is required to vanish at each surface of the box (but not to be identically zero). This condition imposes constraints on the separation constants and therefore on the energy E. What is the smallest value of E for which such a solution can be obtained?

$$\textit{ANS. } E=\frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right).$$

Solution.

Place the box at $0<x<a$, $0<y<b$, $0<z<c$, with $a,b,c,m>0$. For a product $\psi = XYZ$, division of the Schrödinger equation by XYZ gives

$$\frac{X''}{X}+\frac{Y''}{Y}+\frac{Z''}{Z}=-\frac{2mE}{\hbar^2}.$$

Each ratio depends on a different variable, so each must be a constant, written $-k_x^2, -k_y^2, -k_z^2$; then $E = \hbar^2(k_x^2 + k_y^2 + k_z^2)/(2m)$. For $X'' = -k_x^2 X$ with $X(0) = X(a) = 0$, multiply by X^* and integrate by parts; the boundary term vanishes because $X(0) = X(a) = 0$:

$$-k_x^2 \int_0^a |X|^2 dx = \int_0^a X^* X'' dx = \left[X^* X'\right]_0^a - \int_0^a |X'|^2 dx = -\int_0^a |X'|^2 dx.$$

Hence $k_x^2 \int_0^a |X|^2 dx = \int_0^a |X'|^2 dx > 0$ for a nonzero X (if $X' \equiv 0$, then $X \equiv X(0) = 0$), so $k_x^2 > 0$; take $k_x > 0$. Thus $X = A \sin k_x x + B \cos k_x x$; $X(0) = 0$ makes $B = 0$, and $X(a) = A \sin k_x a = 0$ with $A \neq 0$ requires $k_x = n_x \pi/a$, $n_x = 1, 2, \ldots$. The other two directions give $k_y = n_y \pi/b$, $k_z = n_z \pi/c$. Consequently

$$E_{n_x n_y n_z} = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right).$$

All three contributions are minimized at $n_x = n_y = n_z = 1$, yielding

$$\boxed{E_{\min} = \frac{\pi^2 \hbar^2}{2m} \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)}.$$

A normalized state attaining it is

$$\psi_{111} = \sqrt{\frac{8}{abc}} \sin \frac{\pi x}{a} \sin \frac{\pi y}{b} \sin \frac{\pi z}{c},$$

since each sine-squared integral is half the corresponding side length. To see that a nonseparable state cannot have lower energy, put $s(x) = \sin(\pi x/a)$ and let X now denote any continuously differentiable, possibly complex, function on $[0, a]$ with $X(0) = X(a) = 0$ (not necessarily a solution of $X'' = -k_x^2 X$); write $X = sV$ on $0 < x < a$. Since $s'' = -(\pi/a)^2 s$,

$$|X'|^2 = (s')^2 |V|^2 + s^2 |V'|^2 + s s' \big(V'^* V + V^* V' \big),$$
$$\frac{d}{dx}(s s' |V|^2) = \left[(s')^2 - \frac{\pi^2}{a^2} s^2 \right] |V|^2 + s s' \big(V'^* V + V^* V' \big).$$

Subtracting the second identity from the first and using $s^2 |V|^2 = |X|^2$ gives $|X'|^2 - (\pi/a)^2 |X|^2 = s^2 |V'|^2 + (s s' |V|^2)'$; integrating,

$$\int_0^a \left(|X'|^2 - \frac{\pi^2}{a^2} |X|^2 \right) dx = \int_0^a \sin^2(\pi x/a) |V'|^2 dx \geq 0.$$

The boundary term $[s s' |V|^2]_0^a = [s' |X|^2 / s]_0^a$ vanishes: since X is continuously differentiable with $X(0) = 0$, $|X|^2 = O(x^2)$ while $s \sim \pi x/a$ as $x \to 0$, and similarly at $x = a$. Apply this inequality to each slice $x \mapsto \psi(x, y, z)$ at fixed (y, z) and integrate over y and z: $\int |\partial_x \psi|^2 dV \geq (\pi/a)^2 \int |\psi|^2 dV$, and likewise with $(\pi/b)^2$ and $(\pi/c)^2$ for $\partial_y \psi$ and $\partial_z \psi$. Multiplying the Schrödinger equation by ψ^* and integrating by parts, with $\psi = 0$ on the walls, gives $E \int |\psi|^2 dV = (\hbar^2/2m) \int |\nabla \psi|^2 dV$. Adding the three inequalities yields $E \geq E_{\min}$, with equality for the state above.

9.4.6 The quantum mechanical angular momentum operator is given by $\mathbf{L} = -i(\mathbf{r} \times \nabla)$. Show that

$$\mathbf{L} \cdot \mathbf{L}\psi = l(l+1)\psi$$

leads to the associated Legendre equation.

Hint. Section 8.3 and Exercise 8.3.1 may be helpful.

Solution.

In the given units $L_i = -i\epsilon_{ijk} x_j \partial_k$, so $L_i L_i \psi = (-i)^2 \epsilon_{ijk}\epsilon_{ilm} x_j \partial_k (x_l \partial_m \psi)$ (summed over repeated indices). The derivative in the first operator must also act on the coordinate in the second. Contracting the two epsilon symbols with $\epsilon_{ijk}\epsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}$, and then using $\partial_k x_j = \delta_{jk}$ and $\partial_k x_k = 3$, gives

$$\begin{aligned} L_i L_i \psi &= -(\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}) x_j \partial_k (x_l \partial_m \psi) \\ &= -\sum_{j,k} x_j \partial_k (x_j \partial_k \psi) + \sum_{j,k} x_j \partial_k (x_k \partial_j \psi) \\ &= -\sum_{j} x_j \partial_j \psi - r^2 \nabla^2 \psi + 3 \sum_{j} x_j \partial_j \psi + \sum_{j,k} x_j x_k \partial_k \partial_j \psi \\ &= 2\mathbf{r} \cdot \nabla \psi - r^2 \nabla^2 \psi + \sum_{j,k} x_j x_k \partial_j \partial_k \psi. \end{aligned}$$

If $D = \mathbf{r} \cdot \nabla = r\partial_r$, then $D^2\psi = x_j \partial_j (x_k \partial_k \psi) = D\psi + \sum_{j,k} x_j x_k \partial_j \partial_k \psi$. Substituting $\sum_{j,k} x_j x_k \partial_j \partial_k \psi = D^2\psi - D\psi$ above, and using $D^2 + D = r\partial_r(r\partial_r) + r\partial_r = r^2\partial_r^2 + 2r\partial_r$, we obtain

$$\mathbf{L}^2 = -r^2\nabla^2 + D^2 + D = -r^2\nabla^2 + r^2\partial_r^2 + 2r\partial_r = -\Delta_\Omega,$$

because the spherical Laplacian gives $r^2\nabla^2 = \partial_r(r^2\partial_r) + \Delta_\Omega = r^2\partial_r^2 + 2r\partial_r + \Delta_\Omega$, with Δ_Ω the angular operator of Exercise 9.4.3. The Instructor's Manual, p. 114, prints $L^2 = +\Delta_\Omega$; the sign must be negative for the stated $L_i = -i(\mathbf{r} \times \nabla)_i$. For example, $\Delta_\Omega \cos\theta = -2\cos\theta$, whereas $L^2 \cos\theta = 2\cos\theta$. The eigenvalue equation is consequently

$$\frac{1}{\sin\theta}\partial_\theta(\sin\theta\, \partial_\theta \psi) + \frac{1}{\sin^2\theta}\partial_\varphi^2 \psi + l(l+1)\psi = 0.$$

Separate an angular mode as $\Theta(\theta)\Phi(\varphi)$ and multiply by $\sin^2\theta/(\Theta\Phi)$, as in Exercise 9.4.3 with $\lambda = l(l+1)$. Setting $\Phi''/\Phi = -m^2$ and multiplying by $\Theta/\sin^2\theta$ leaves

$$\frac{1}{\sin\theta}(\sin\theta\, \Theta')' + \left[l(l+1) - \frac{m^2}{\sin^2\theta} \right] \Theta = 0.$$

Put $u = \cos\theta$ and $\Theta(\theta) = P(u)$. Then $d/d\theta = -\sin\theta\, d/du$, so $\Theta' = -\sin\theta P'$, $\sin\theta\,\Theta' = -(1-u^2)P'$, and

$$\frac{1}{\sin\theta}(\sin\theta\,\Theta')' = \frac{1}{\sin\theta}\frac{d}{d\theta}[-(1-u^2)P'] = \frac{d}{du}[(1-u^2)P'] = (1-u^2)P'' - 2uP'.$$

Thus

$$\boxed{(1-u^2)P'' - 2uP' + \left[l(l+1) - \frac{m^2}{1-u^2} \right] P = 0},$$

the associated Legendre equation. Radial factors are unrestricted by this angular eigenvalue equation alone.

9.4.7 The 1-D Schrödinger wave equation for a particle in a potential field $V = \frac{1}{2}kx^2$ is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2\psi = E\psi(x).$$

(a) Defining

$$a = \left(\frac{mk}{\hbar^2}\right)^{1/4}, \quad \lambda = \frac{2E}{\hbar}\left(\frac{m}{k}\right)^{1/2},$$

and setting $\xi = ax$, show that

$$\frac{d^2\psi(\xi)}{d\xi^2} + (\lambda - \xi^2)\psi(\xi) = 0.$$

(b) Substituting

$$\psi(\xi) = y(\xi)e^{-\xi^2/2},$$

show that $y(\xi)$ satisfies the Hermite differential equation.

Solution.

Assume the harmonic-oscillator parameters $m, k, \hbar$ are positive. **(a)** Since $\xi = ax$, $d/dx = a\, d/d\xi$ and $x^2 = \xi^2/a^2$. Substitution gives

$$-\frac{\hbar^2 a^2}{2m}\psi_{\xi\xi} + \frac{k\xi^2}{2a^2}\psi = E\psi.$$

Multiply by $-2m/(\hbar^2 a^2)$ and move all terms to the left:

$$\psi_{\xi\xi} + \left[\frac{2mE}{\hbar^2 a^2} - \frac{mk}{\hbar^2 a^4}\xi^2\right]\psi = 0.$$

The definition $a^4 = mk/\hbar^2$ makes the second coefficient one, while $a^2 = \sqrt{mk}/\hbar$ gives

$$\frac{2mE}{\hbar^2 a^2} = \frac{2E}{\hbar}\sqrt{\frac{m}{k}} = \lambda.$$

Hence $\boxed{\psi_{\xi\xi} + (\lambda - \xi^2)\psi = 0}$.

(b) Differentiating $\psi = ye^{-\xi^2/2}$ by the product rule, with $(e^{-\xi^2/2})' = -\xi e^{-\xi^2/2}$, gives $\psi' = e^{-\xi^2/2}(y' - \xi y)$ and

$$\psi'' = e^{-\xi^2/2}(y'' - y - \xi y') - \xi e^{-\xi^2/2}(y' - \xi y) = e^{-\xi^2/2}[y'' - 2\xi y' + (\xi^2 - 1)y].$$

Substitution in part (a) gives $e^{-\xi^2/2}[y'' - 2\xi y' + (\xi^2 - 1)y + (\lambda - \xi^2)y] = 0$. The terms $+\xi^2 y$ and $-\xi^2 y$ cancel, and dividing by the nonzero exponential leaves

$$\boxed{y'' - 2\xi y' + (\lambda - 1)y = 0}.$$

This is Hermite's equation with parameter $2\alpha = \lambda - 1$. When $\alpha = n$ is a nonnegative integer, the polynomial solutions in Exercise 8.3.3 give $\psi = H_n(\xi)e^{-\xi^2/2}$ and $E = \hbar\sqrt{k/m}(n + \frac{1}{2})$.

9.5 Laplace and Poisson Equations

9.5.1 Verify that the following are solutions of Laplace's equation:

(a) $\psi_1 = 1/r$, $r \neq 0$, (b) $\psi_2 = \dfrac{1}{2r} \ln \dfrac{r+z}{r-z}$, $x^2 + y^2 > 0$ (exclude the entire z-axis).

Solution.

(a) For a radial function in three dimensions, $\nabla^2 f = r^{-2}(r^2 f')'$. At $r > 0$, with $f = 1/r$,

$$f' = -r^{-2}, \qquad (r^2 f')' = (-1)' = 0, \qquad \boxed{\nabla^2(1/r) = 0 \quad (r > 0)}.$$

This is a local statement away from the singularity. Its nonzero flux through a sphere, $\int \partial_r(1/r)\, dS = -4\pi$, is why it cannot be extended as a source-free solution through the origin.

(b) The domain must exclude the whole z-axis, where the logarithm is singular; $r \neq 0$ alone would not suffice. With $z = r\cos\theta$ and $0 < \theta < \pi$,

$$\psi_2 = \frac{g(\theta)}{r}, \qquad g(\theta) = \frac{1}{2} \ln \frac{1+\cos\theta}{1-\cos\theta} = \ln \cot \frac{\theta}{2}.$$

Here $(1+\cos\theta)/(1-\cos\theta) = 2\cos^2(\theta/2)/[2\sin^2(\theta/2)] = \cot^2(\theta/2)$, and $\cot(\theta/2) > 0$. In the spherical Laplacian

$$\nabla^2 = \frac{1}{r^2}\partial_r(r^2\partial_r) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{r^2\sin^2\theta}\partial_\varphi^2$$

the radial and azimuthal contributions vanish:

$$\partial_r(r^2\partial_r\psi_2) = \partial_r[-g(\theta)] = 0, \qquad \partial_\varphi^2\psi_2 = 0.$$

For the remaining angular term, direct differentiation gives

$$g' = -\frac{\sin\theta}{2}\left(\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta}\right) = -\frac{\sin\theta}{2}\,\frac{2}{1-\cos^2\theta} = -\frac{1}{\sin\theta},$$

so $(\sin\theta\, g')' = (-1)' = 0$. Consequently

$$\boxed{\nabla^2\psi_2 = \frac{1}{r^3\sin\theta}(\sin\theta\, g')' = 0 \quad \text{where } x^2 + y^2 > 0}.$$

No claim of harmonicity across the excluded axis follows from this calculation.

9.5.2 If Ψ is a solution of Laplace's equation, $\nabla^2\Psi = 0$, show that $\partial\Psi/\partial z$ is also a solution.

Solution.

Where the third derivatives exist and are continuous, interchange of mixed derivatives gives

$$\begin{aligned}\nabla^2(\partial_z\Psi) &= \partial_x^2\partial_z\Psi + \partial_y^2\partial_z\Psi + \partial_z^3\Psi \\ &= \partial_z(\partial_x^2\Psi + \partial_y^2\Psi + \partial_z^2\Psi) = \partial_z(\nabla^2\Psi) = 0.\end{aligned}$$

Thus $\partial_z\Psi$ solves Laplace's equation on the same open region. The conclusion also holds in the distributional sense, where partial derivatives commute without classical differentiability assumptions. Boundary values generally change on differentiation; the assertion concerns the PDE, not preservation of prescribed boundary data.

9.5.3 Show that an argument based on Eq. (9.88) can be used to prove that the Laplace and Poisson equations with Dirichlet boundary conditions have unique solutions on a bounded region with a piecewise smooth boundary, for solutions regular enough for Green's identity and continuous up to the boundary.

Solution.

Use the bounded region and regularity assumptions of the text's uniqueness argument. An unbounded region requires an appropriate condition at infinity as well. Let ψ_1, ψ_2 satisfy the same Poisson equation $\nabla^2 \psi_i = f$ on a bounded region Ω, with the same Dirichlet values on its entire boundary. Their difference $u = \psi_1 - \psi_2$ satisfies

$$\nabla^2 u = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

The product rule followed by the divergence theorem gives Green's identity, including complex conjugation if the functions are complex:

$$\int_{\partial\Omega} u^* \partial_n u \, dS = \int_\Omega \nabla \cdot (u^* \nabla u) \, dV = \int_\Omega \left(|\nabla u|^2 + u^* \nabla^2 u \right) dV.$$

The boundary integral is zero because $u = 0$ there, and the Laplacian term is zero in the interior. It follows that

$$\int_\Omega |\nabla u|^2 \, dV = 0.$$

The integrand is nonnegative, so $\nabla u = 0$ (everywhere for a classical smooth solution). Thus u is constant on each connected component, and its zero boundary values make each constant zero. Hence $\psi_1 = \psi_2$. Taking $f = 0$ proves the Laplace case as well. This proves uniqueness when a solution exists; it does not assert existence for arbitrary boundary data. For the need for a condition at infinity, $u = z$ and $u = 0$ are distinct harmonic functions in the half-space $z > 0$ with the same zero values on $z = 0$.

9.6 Wave Equation

Solve the wave equation, Eq. (9.89), subject to the indicated conditions.

9.6.1 Determine $\psi(x, t)$ given that at $t = 0$ $\psi_0(x) = \sin x$ and $\partial\psi(x)/\partial t = \cos x$.

Solution.

Take $c > 0$, the speed in $\psi_{tt} = c^2 \psi_{xx}$. To incorporate both initial data, write $\psi = F(x - ct) + G(x + ct)$. For initial displacement f and velocity v,

$$F + G = f, \qquad -cF' + cG' = v.$$

Differentiate the first equation and solve these two linear equations:

$$F' = \tfrac{1}{2}(f' - v/c), \qquad G' = \tfrac{1}{2}(f' + v/c).$$

Integrating, $F(\zeta) = \frac{1}{2} f(\zeta) - \frac{1}{2c} \int_0^\zeta v(s)\, ds + K$ and $G(\zeta) = \frac{1}{2} f(\zeta) + \frac{1}{2c} \int_0^\zeta v(s)\, ds - K$, where the two constants of integration add to zero because $F + G = f$. Evaluating F at $x - ct$ and G at $x + ct$, and combining the two integrals into one, yields d'Alembert's formula

$$\psi(x, t) = \frac{f(x - ct) + f(x + ct)}{2} + \frac{1}{2c} \int_{x - ct}^{x + ct} v(s)\, ds.$$

For $f(x) = \sin x$, $v(x) = \cos x$, the integral is $\sin(x+ct) - \sin(x-ct) = 2\cos x \sin ct$ by the addition formula $\sin(x \pm ct) = \sin x \cos ct \pm \cos x \sin ct$. The same formula makes the first term $[\sin(x-ct) + \sin(x+ct)]/2 = \sin x \cos ct$, so

$$\boxed{\psi(x,t) = \sin x \cos ct + \frac{\cos x}{c} \sin ct}.$$

At $t = 0$ this is $\sin x$, and $\psi_t = -c \sin x \sin ct + \cos x \cos ct$ becomes $\cos x$. Each term obeys $\psi_{tt} = c^2 \psi_{xx}$, completing the check.

9.6.2 Determine $\psi(x,t)$ given that at $t = 0$ $\psi_0(x) = \delta(x)$ (Dirac delta function) and the initial time derivative of ψ is zero.

Solution.

With initial velocity zero, d'Alembert's formula, interpreted for distributions, gives

$$\boxed{\psi(x,t) = \tfrac{1}{2}\delta(x-ct) + \tfrac{1}{2}\delta(x+ct)}.$$

The Instructor's Manual prints $\delta(x-ct)$ twice. The second pulse must have argument $x+ct$ to obtain zero initial velocity. For any smooth compactly supported test function φ,

$$\langle \psi(\cdot,t), \varphi \rangle = \tfrac{1}{2}[\varphi(ct) + \varphi(-ct)].$$

At $t = 0$ this is $\varphi(0)$, while its time derivative is $\frac{c}{2}[\varphi'(ct) - \varphi'(-ct)]$, which is zero at 0. The second time derivative is $\frac{c^2}{2}[\varphi''(ct) + \varphi''(-ct)]$, exactly the action of $c^2 \partial_x^2 \psi$ on the same test function, which by definition of the distributional derivative is $c^2 \langle \psi(\cdot,t), \varphi'' \rangle$. Thus both the PDE and the initial data hold distributionally. The unit impulse splits into equal masses moving with speeds $+c$ and $-c$.

9.6.3 Determine $\psi(x,t)$ given that at $t = 0$ $\psi_0(x)$ is a single square-wave pulse as defined below, and the initial time derivative of ψ is zero.

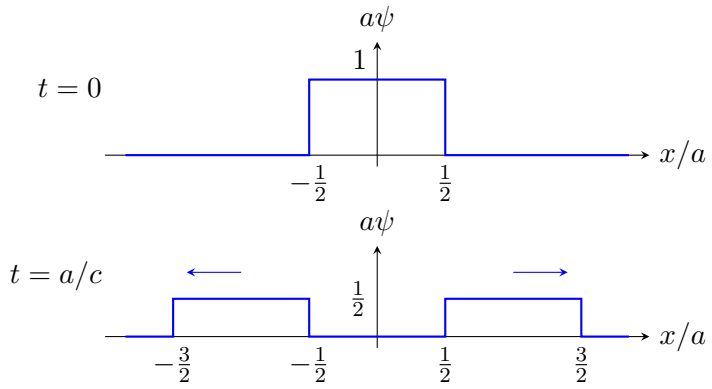
$$\psi_0(x) = 0, \ |x| > a/2, \quad \psi_0(x) = 1/a, \ |x| < a/2.$$

Solution.

Take $a, c > 0$ and write $f(x) = a^{-1}\mathbf{1}_{\{|x|<a/2\}}$, where $\mathbf{1}$ denotes an indicator function. Since the initial velocity is zero, d'Alembert's formula (Exercise 9.6.1) reduces to $\psi = \frac{1}{2}[f(x-ct) + f(x+ct)]$:

$$\boxed{\psi(x,t) = \frac{1}{2a}\left[\mathbf{1}_{\{|x-ct|<a/2\}} + \mathbf{1}_{\{|x+ct|<a/2\}}\right]}.$$

The right-moving pulse occupies $(ct - a/2, ct + a/2)$ and the left-moving pulse occupies $(-ct - a/2, -ct + a/2)$; each has height $1/(2a)$ and integral $1/2$. At $t = 0$ they coincide to reproduce f. The two intervals overlap in $(ct - a/2,\ a/2 - ct)$, which is nonempty exactly when $2ct < a$. Thus for $0 < t < a/(2c)$ their overlap has height $1/a$; for $t \geq a/(2c)$ the pulses are disjoint. Endpoint values may be assigned the half-jump convention; they do not affect the associated distribution. As for any traveling profile, each term solves the wave equation weakly, and their time derivatives cancel initially.



9.6.4 Determine $\psi(x,t)$ given that at $t=0$ $\psi_0 = 0$ for all x, but $\partial\psi/\partial t = \sin(x)$.

Solution.

Here $f = 0$ and $v(x) = \sin x$. The initial-displacement terms in d'Alembert's formula vanish, leaving

$$\begin{aligned}\psi(x,t) &= \frac{1}{2c}\int_{x-ct}^{x+ct} \sin s\, ds \\ &= \frac{1}{2c}[\cos(x-ct) - \cos(x+ct)] = \boxed{\frac{\sin x \sin ct}{c}},\end{aligned}$$

since $\cos(x \mp ct) = \cos x \cos ct \pm \sin x \sin ct$ makes the bracket $2\sin x \sin ct$. Direct checks give $\psi(x,0) = 0$, $\psi_t(x,0) = \sin x$, and $\psi_{tt} = -c\sin x \sin ct = c^2\psi_{xx}$.

9.7 Heat-Flow, or Diffusion PDE

9.7.1 For a homogeneous spherical solid with constant thermal diffusivity, K, and no heat sources, the equation of heat conduction becomes

$$\frac{\partial T(r,t)}{\partial t} = K\nabla^2 T(r,t).$$

Assume a solution of the form

$$T = R(r)\mathcal{T}(t)$$

and separate variables. Show that the radial equation may take on the standard form

$$r^2\frac{d^2R}{dr^2} + 2r\frac{dR}{dr} + \alpha^2 r^2 R = 0,$$

and that $\sin\alpha r/r$ and $\cos\alpha r/r$ are its solutions. For a linearly independent pair take $\alpha \neq 0$; also give the independent pair at $\alpha = 0$.

Solution.

Spherical symmetry removes the angular derivatives, so

$$R(r)\mathcal{T}'(t) = K\mathcal{T}(t)\frac{1}{r^2}(r^2R')'.$$

Divide by $KR\mathcal{T}$ and equate the independent-variable expressions to a separation constant, conventionally $-\alpha^2$:

$$\frac{\mathcal{T}'}{K\mathcal{T}} = \frac{(r^2R')'}{r^2R} = -\alpha^2.$$

Integrating $\mathcal{T}' = -K\alpha^2\mathcal{T}$, and writing $(r^2R')' = r^2R'' + 2rR'$ in $(r^2R')' + \alpha^2r^2R = 0$, gives the decaying time factor and radial equation

$$\mathcal{T} = Ce^{-K\alpha^2t}, \qquad \boxed{r^2R'' + 2rR' + \alpha^2r^2R = 0}.$$

To solve the latter, let $u = rR$. Then

$$R' = \frac{u'}{r} - \frac{u}{r^2}, \qquad R'' = \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3},$$

so $r^2R'' + 2rR' + \alpha^2r^2R = r(u'' + \alpha^2u)$. The radial equation is therefore $u'' + \alpha^2u = 0$ for $r > 0$. For $\alpha \neq 0$ its solutions are $u = A\sin\alpha r + B\cos\alpha r$, so

$$\boxed{R = A\frac{\sin\alpha r}{r} + B\frac{\cos\alpha r}{r}}.$$

These two functions form a basis only for $\alpha \neq 0$. At $\alpha = 0$, the independent radial solutions are 1 and $1/r$. Indeed, at $\alpha = 0$ the sine solution vanishes identically, while $u'' = 0$ gives $u = A + Br$, that is, $R = A/r + B$. If the solid contains its center and the temperature is finite there, the cosine term (or $1/r$ term) must be absent. The sine term is regular because $\sin\alpha r/r = \alpha - \alpha^3r^2/6 + \cdots$. The choice $-\alpha^2$ describes the usual decaying modes for $K > 0$; allowed values depend on the surface boundary condition, which was not specified.

9.7.2 Separate variables in the thermal diffusion equation of Exercise 9.7.1 in circular cylindrical coordinates. Assume that you can neglect end effects and take $T = T(\rho, t)$.

Solution.

With no dependence on z or φ, the cylindrical Laplacian is $\rho^{-1}\partial_\rho(\rho\partial_\rho)$. Set $T(\rho,t) = R(\rho)\mathcal{T}(t)$. The PDE becomes

$$R\mathcal{T}' = K\mathcal{T}\frac{1}{\rho}(\rho R')', \qquad \frac{\mathcal{T}'}{K\mathcal{T}} = \frac{(\rho R')'}{\rho R} = -\alpha^2.$$

The time equation $\mathcal{T}' = -K\alpha^2\mathcal{T}$ integrates to an exponential, and the radial equation $(\rho R')' + \alpha^2\rho R = 0$, multiplied by ρ with $\rho(\rho R')' = \rho^2R'' + \rho R'$, becomes

$$\boxed{\mathcal{T} = Ce^{-K\alpha^2t}, \qquad \rho^2R'' + \rho R' + \alpha^2\rho^2R = 0}.$$

For nonzero α put $s = \alpha\rho$, so that $R_\rho = \alpha R_s$ and $R_{\rho\rho} = \alpha^2R_{ss}$; then $\rho^2R_{\rho\rho} = s^2R_{ss}$, $\rho R_\rho = sR_s$, and $s^2R_{ss} + sR_s + s^2R = 0$, the order-zero Bessel equation. Its solution is $R = AJ_0(\alpha\rho) + BY_0(\alpha\rho)$. If the axis is included, finiteness there selects J_0 since Y_0 has a logarithmic singularity. For $\alpha = 0$, integrate $(\rho R')' = 0$ directly: $\rho R' = B$ and $R = A + B\ln\rho$, again leaving only a constant on the axis. A radial boundary condition would determine the allowed α; none is supplied here.

9.7.3 Solve the PDE

$$\frac{\partial \psi}{\partial t} = a^2 \frac{\partial^2 \psi}{\partial x^2},$$

to obtain $\psi(x,t)$ for a rod of infinite extent (in both the $+x$ and $-x$ directions), with a heat pulse at time $t = 0$ that corresponds to $\psi_0(x) = A\delta(x)$.

Solution.

Let $D = a^2 > 0$. Use the Fourier transform convention

$$\widehat{\psi}(k,t) = \int_{\mathbb{R}} \psi(x,t) e^{-ikx}\, dx, \qquad \psi(x,t) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\psi}(k,t) e^{ikx}\, dk.$$

Integrating by parts twice, with boundary terms that vanish for ψ decaying as $|x| \to \infty$ (or by distributional differentiation), the transform of ψ_{xx} is $(ik)^2 \widehat{\psi} = -k^2 \widehat{\psi}$. Hence $\partial_t \widehat{\psi} = -Dk^2 \widehat{\psi}$, and $\widehat{\psi}(k,0) = \int_{\mathbb{R}} A\delta(x) e^{-ikx}\, dx = A$. Solving this scalar ODE yields $\widehat{\psi} = Ae^{-Dk^2 t}$, so $\psi = (A/2\pi) I(x)$, where for $t > 0$ we put $I(x) = \int_{\mathbb{R}} e^{-Dtk^2} e^{ikx}\, dk$. Differentiate under the integral sign, use $ke^{-Dtk^2} = -(2Dt)^{-1} \partial_k e^{-Dtk^2}$, and integrate by parts in k:

$$I'(x) = -\frac{i}{2Dt} \int_{\mathbb{R}} \partial_k (e^{-Dtk^2}) e^{ikx}\, dk = \frac{i}{2Dt} \int_{\mathbb{R}} e^{-Dtk^2} (ix) e^{ikx}\, dk = -\frac{x}{2Dt} I(x).$$

The boundary term vanishes because of the Gaussian factor. This linear ODE gives $[e^{x^2/(4Dt)} I(x)]' = 0$, and $I(0) = \int_{\mathbb{R}} e^{-Dtk^2} dk = \sqrt{\pi/(Dt)}$ from the Gaussian integral. Thus $I(x) = \sqrt{\pi/(Dt)} e^{-x^2/(4Dt)}$ and, since $(2\pi)^{-1} \sqrt{\pi/(Dt)} = 1/\sqrt{4\pi Dt}$,

$$\boxed{\psi(x,t) = \frac{A}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)} \quad (t > 0)}.$$

In terms of a, the denominator is $2|a|\sqrt{\pi t}$. The form $\psi = A\, e^{-x^2/(4a^2 t)}/(2a\sqrt{\pi t})$ presumes $a > 0$: the PDE contains only the diffusivity a^2, and for $a < 0$ that expression would make the pulse negative. Direct differentiation, with $\psi_x = -x\psi/(2Dt)$ and hence $\psi_{xx} = [-1/(2Dt) + x^2/(4D^2 t^2)]\psi$, gives

$$\psi_t = \left(-\frac{1}{2t} + \frac{x^2}{4Dt^2} \right) \psi, \qquad D\psi_{xx} = \left(-\frac{1}{2t} + \frac{x^2}{4Dt^2} \right) \psi.$$

The total heat is $\int \psi\, dx = A$: substitute $x = 2\sqrt{Dt}\, u$ and use $\int e^{-u^2} du = \sqrt{\pi}$. To check the initial impulse, substitute $x = 2\sqrt{Dt}\, u$ in its action on a test function:

$$\int \psi(x,t) \varphi(x)\, dx = \frac{A}{\sqrt{\pi}} \int e^{-u^2} \varphi(2\sqrt{Dt}\, u)\, du \longrightarrow A\varphi(0).$$

Dominated convergence applies because φ is bounded and the Gaussian is integrable. This verifies the initial condition as a distribution, rather than assigning a finite value to $\delta(0)$. If $a = 0$, the PDE is $\psi_t = 0$ and the initial distribution $A\delta(x)$ remains unchanged.

9.7.4 Solve the same PDE as in Exercise 9.7.3 for a rod of length L, with positive diffusivity $a^2 > 0$ and $L > 0$, with position on the rod given by the variable x, with the two ends of the rod at $x = 0$ and $x = L$ kept (at all times t) at the respective temperatures $T = 1$ and $T = 0$, and with the rod initially at $T(x) = 0$, for $0 < x \le L$.

Solution.

Take $D = a^2 > 0$ and $L > 0$. Here $L > 0$ is the rod's length, and $a^2 > 0$ excludes the degenerate case $a = 0$, in which the PDE is $T_t = 0$: the interior then keeps its initial temperature 0 for all t and never relaxes to the steady profile S found below. The steady solution satisfying the two endpoint temperatures is $S(x) = 1 - x/L$, since $S'' = 0$, $S(0) = 1$, $S(L) = 0$. Write $T = S + u$. Then

$$u_t = Du_{xx}, \quad u(0,t) = u(L,t) = 0, \quad u(x,0) = \frac{x}{L} - 1 \quad (0 < x < L).$$

Separate its zero-boundary modes: a product $X(x)\mathcal{T}(t)$ gives $\mathcal{T}'/(D\mathcal{T}) = X''/X = -\kappa^2$. As in Exercise 9.4.5, $X'' = -\kappa^2 X$ with $X(0) = X(L) = 0$ forces $\kappa^2 > 0$, $X \propto \sin \kappa x$ and $\sin \kappa L = 0$; this gives $X_n = \sin(n\pi x/L)$, $\kappa_n = n\pi/L$, and the time factors $e^{-D\kappa_n^2 t}$. Hence

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-D(n\pi/L)^2 t}.$$

Setting $t = 0$ and using the orthogonality relation $\int_0^L \sin(n\pi x/L) \sin(m\pi x/L)\, dx = (L/2)\delta_{nm}$ gives

$$\begin{aligned} b_n &= \frac{2}{L} \int_0^L \left(\frac{x}{L} - 1\right) \sin \frac{n\pi x}{L}\, dx \\ &= \frac{2}{L} \left[-\frac{(x/L - 1)\cos(n\pi x/L)}{n\pi/L} + \frac{\sin(n\pi x/L)}{L(n\pi/L)^2} \right]_0^L = -\frac{2}{n\pi}. \end{aligned}$$

The antiderivative comes from one integration by parts, differentiating $x/L - 1$ and integrating the sine. At $x = L$ both terms vanish ($x/L - 1 = 0$ and $\sin n\pi = 0$), while at $x = 0$ the bracket equals $1/(n\pi/L) = L/(n\pi)$; hence $b_n = (2/L)[0 - L/(n\pi)] = -2/(n\pi)$. Therefore

$$\boxed{T(x,t) = 1 - \frac{x}{L} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{L} e^{-D(n\pi/L)^2 t}}.$$

Fix $t_0 > 0$ and put $\eta_0 = D\pi^2 t_0/L^2$. For $t \geq t_0$ the nth terms of the series and of its termwise derivatives ∂_t, ∂_x and ∂_x^2 are bounded in modulus by constant multiples (depending only on D and L) of $n\, e^{-\eta_0 n^2}$, and $\sum_n n\, e^{-\eta_0 n^2} < \infty$. By the Weierstrass M-test these series converge uniformly for $0 \leq x \leq L$, $t \geq t_0$, so termwise differentiation is valid; each term satisfies $u_t = Du_{xx}$, hence so does the sum, and $T = S + u$ solves the PDE because $S'' = 0$. At $x = 0, L$ all sine terms vanish, giving the required endpoint values. For $0 < x < L$, its initial limit is zero because

$$\sum_{n=1}^{\infty} \frac{\sin(n\theta)}{n} = \frac{\pi - \theta}{2} \quad (0 < \theta < 2\pi), \qquad \theta = \frac{\pi x}{L},$$

so at $t = 0$ the series term is $(2/\pi)(\pi - \pi x/L)/2 = 1 - x/L$, which cancels $S(x)$. This identity follows by taking the imaginary part of $\sum_{n \geq 1} r^n e^{in\theta}/n = -\ln(1 - re^{i\theta})$ ($0 \leq r < 1$, principal logarithm) and letting $r \uparrow 1$. By Abel's theorem the limit of the left side equals the series at $r = 1$, which converges by Dirichlet's test because the partial sums of $\sin(n\theta)$ are bounded (see below). The right side tends to $-\ln(1 - e^{i\theta})$, because $1 - re^{i\theta}$ has positive real part $1 - r\cos\theta$ for $0 \leq r \leq 1$ and the principal logarithm is continuous off the negative real axis. Since $1 - e^{i\theta} = -2i \sin(\theta/2) e^{i\theta/2} = 2\sin(\theta/2) e^{i(\theta - \pi)/2}$ with $\sin(\theta/2) > 0$, we have $\arg(1 - e^{i\theta}) = (\theta - \pi)/2$, and the imaginary part of $-\ln(1 - e^{i\theta})$ is $(\pi - \theta)/2$.

The initial limit is justified without assuming uniform convergence at $x = 0$. Write $\eta = D\pi^2 t/L^2$, so the nth term carries $e^{-\eta n^2}$, and fix $0 < \theta < \pi$. Summing $2\sin(\theta/2)\sin(n\theta) = \cos((n - \frac{1}{2})\theta) - \cos((n + \frac{1}{2})\theta)$ telescopically gives, for $m \geq N \geq 0$,

$$B_m = \sum_{n=N+1}^{m} \sin(n\theta) = \frac{\cos((N + \frac{1}{2})\theta) - \cos((m + \frac{1}{2})\theta)}{2\sin(\theta/2)}, \qquad |B_m| \leq \frac{1}{\sin(\theta/2)},$$

with $B_N = 0$ (empty sum), so that $\sin(n\theta) = B_n - B_{n-1}$ for $n > N$. Let $M > N$ and $c_n = e^{-\eta n^2}/n$, which decreases in n. Substituting $\sin(n\theta) = B_n - B_{n-1}$ and shifting the index in the sum containing B_{n-1} (summation by parts) gives

$$\begin{aligned}\sum_{n=N+1}^{M} c_n \sin(n\theta) &= \sum_{n=N+1}^{M} c_n B_n - \sum_{n=N}^{M-1} c_{n+1} B_n \\ &= c_M B_M + \sum_{n=N+1}^{M-1} (c_n - c_{n+1}) B_n,\end{aligned}$$

since the $n = N$ term of the second sum is $c_{N+1}B_N = 0$. As $c_n - c_{n+1} \geq 0$ and these differences add up to $c_{N+1} - c_M$, the modulus is at most $[c_M + (c_{N+1} - c_M)]/\sin(\theta/2) = c_{N+1}/\sin(\theta/2)$. Letting $M \to \infty$ bounds the tail starting at $N + 1$ by $1/[(N + 1)\sin(\theta/2)]$, uniformly for $\eta \geq 0$. Hence the series at time t and at $t = 0$ differ by at most $\sum_{n \leq N}(1 - e^{-\eta n^2})/n + 2/[(N + 1)\sin(\theta/2)]$. Take $t \downarrow 0$ in the finite initial portion and then let $N \to \infty$. The long-time limit is $S(x)$, since the modulus of the series is at most $\sum_{n \geq 1} e^{-\eta n^2} \leq \sum_{n \geq 1} e^{-\eta n} = 1/(e^{\eta} - 1)$, which tends to zero as $t \to \infty$.

The initial interior temperature tends to zero as $x \downarrow 0$, whereas the imposed endpoint is one. There is consequently a corner discontinuity at $(0, 0)$; the initial limit is taken at fixed interior x, as the question specifies, not jointly with $x \to 0$. For numerical evaluation at $t > 0$, so that $\eta = D\pi^2 t/L^2 > 0$, truncation after $N \geq 1$ terms has the uniform error bound

$$|R_N| \leq \frac{2}{\pi} \sum_{n > N} \frac{e^{-\eta n^2}}{n} \leq \frac{2}{\pi} \int_N^{\infty} \frac{e^{-\eta s^2}}{s}\, ds \leq \frac{e^{-\eta N^2}}{\pi \eta N^2}.$$

The first inequality uses $|\sin| \leq 1$; the second holds because $e^{-\eta s^2}/s$ decreases, so each term with $n > N$ is at most the integral of that function over $[n - 1, n]$. The last inequality uses $1/s \leq s/N^2$ for $s \geq N$ and integrates $se^{-\eta s^2}$ exactly: $\int_N^{\infty} se^{-\eta s^2} ds = e^{-\eta N^2}/(2\eta)$.

10 Chapter 10. Green's Functions

Problems

10.1 One-Dimensional Problems

10.1.1 Show that

$$G(x,t) = \begin{cases} x, & 0 \le x < t, \\ t, & t < x \le 1, \end{cases}$$

is the Green's function for the operator $L = -d^2/dx^2$ and the boundary conditions $y(0) = 0$, $y'(1) = 0$.

Solution.

For fixed $0 < t < 1$, write the proposed kernel as $G(x,t) = \min(x,t)$. It is continuous at $x = t$, and its two derivatives there are

$$G_x(t^-,t) = 1, \qquad G_x(t^+,t) = 0,$$

because the branch x has slope 1, while the branch t does not depend on x. Thus $[G_x]_{t^-}^{t^+} = -1$. A continuous, piecewise smooth function has a delta function in its second derivative with coefficient equal to this derivative jump. Consequently

$$-G_{xx} = \delta(x - t).$$

Away from t, both branches are linear, so there is no ordinary part of $-G_{xx}$. The endpoint conditions also hold: $G(0,t) = 0$ and $G_x(1,t) = 0$.

To check the resulting solution directly, let f be continuous and split the integral at $t = x$ ($G = t$ for $t < x$ and $G = x$ for $t > x$):

$$y(x) = \int_0^1 G(x,t) f(t)\, dt = \int_0^x t f(t)\, dt + x \int_x^1 f(t)\, dt.$$

The product rule and the fundamental theorem of calculus give

$$\begin{aligned} y'(x) &= x f(x) + \int_x^1 f(t)\, dt - x f(x) = \int_x^1 f(t)\, dt, \\ y''(x) &= -f(x). \end{aligned}$$

Hence $-y'' = f$. At $x = 0$ the first integral has zero length and the second term has the factor $x = 0$, so $y(0) = 0$; also $y'(1) = \int_1^1 f(t)\, dt = 0$. Finally, the difference of two solutions satisfies $-y'' = 0$, so it is $ax + b$; the condition at $x = 0$ forces $b = 0$, and the one at $x = 1$ forces $a = 0$. Thus this kernel gives the unique solution.

10.1.2 Find the Green's function for

(a) $Ly(x) = \dfrac{d^2y(x)}{dx^2} + y(x), \quad y(0) = 0, \quad y'(1) = 0.$

(b) $Ly(x) = \dfrac{d^2y(x)}{dx^2} - y(x), \quad y(x)$ finite for $-\infty < x < \infty$, meaning bounded as $x \to \pm\infty$.

Solution.

(a) For $y'' + y = 0$, choose a left solution and a right solution

$$u(x) = \sin x, \qquad v(x) = \cos(1 - x).$$

They satisfy $u(0) = 0$ and $v'(1) = 0$, respectively. Continuity at $x = t$ is built into the form

$$G(x,t) = C \begin{cases} u(x)v(t), & x < t, \\ u(t)v(x), & x > t. \end{cases}$$

For $x < t$, $G_x = Cu'(x)v(t)$, and for $x > t$, $G_x = Cu(t)v'(x)$; hence the derivative jump at $x = t$ is

$$G_x(t^+, t) - G_x(t^-, t) = C\big[u(t)v'(t) - u'(t)v(t)\big].$$

Integrating $G_{xx} + G = \delta(x - t)$ over $(t - \varepsilon, t + \varepsilon)$ and letting $\varepsilon \to 0$ (the bounded term G contributes nothing) shows that this jump must be $+1$, the reciprocal of the coefficient 1 of y''. Since $v'(x) = \sin(1 - x)$, the needed Wronskian is

$$\begin{aligned} u(t)v'(t) - u'(t)v(t) &= \sin t \sin(1 - t) - \cos t \cos(1 - t) \\ &= -\cos\big(t + (1 - t)\big) = -\cos 1. \end{aligned}$$

Therefore $C(-\cos 1) = 1$, and

$$\boxed{G(x,t) = -\frac{1}{\cos 1} \begin{cases} \sin x \cos(1 - t), & 0 \le x \le t, \\ \sin t \cos(1 - x), & t \le x \le 1. \end{cases}}$$

Both branches solve $G_{xx} + G = 0$ away from the source, are continuous at the source, have derivative jump 1, and satisfy the specified endpoint conditions. These facts prove $(\partial_x^2 + 1)G = \delta(x - t)$. There is no zero mode, that is, no nonzero solution h of $h'' + h = 0$ satisfying both endpoint conditions: $h(0) = 0$ gives $h = A \sin x$, and $h'(1) = A \cos 1 = 0$ gives $A = 0$ since $\cos 1 \neq 0$. Hence the Green's function is unique: the difference of two of them would be such an h.

(b) "Finite for $-\infty < x < \infty$" must here mean bounded as $x \to \pm\infty$. Finiteness only at each finite point would allow both e^x and e^{-x} and would not select a unique Green's function. The bounded homogeneous solution on the left is proportional to e^x; on the right it is proportional to e^{-x}. Impose continuity by writing

$$G(x,t) = C \begin{cases} e^{x-t}, & x < t, \\ e^{t-x}, & x > t. \end{cases}$$

Its x-derivative tends to C as $x \uparrow t$ (from Ce^{x-t}) and to $-C$ as $x \downarrow t$ (from Ce^{t-x}), so the derivative jump is $-C - C = -2C$. As in (a), it must equal 1, and $-2C = 1$ gives

$$\boxed{G(x,t) = -\frac{1}{2} e^{-|x-t|}.}$$

It decays at both infinities, solves $G_{xx} - G = 0$ for $x \neq t$, and has derivative jump 1. Thus $y(x) = \int_{\mathbb{R}} G(x,t) f(t)\, dt$ solves $y'' - y = f$ whenever the integral is defined, for example for bounded continuous f. In that example, with $\|f\|_\infty = \sup_t |f(t)|$ and since $\int_{\mathbb{R}} e^{-|s|}\, ds = 2$, $|y(x)| \le \|f\|_\infty \int_{\mathbb{R}} e^{-|x-t|}/2\, dt = \|f\|_\infty$. No nonzero combination $Ae^x + Be^{-x}$ is bounded at both infinities, which proves uniqueness in the bounded class.

10.1.3 Show that the function $y(x)$ defined by Eq. (10.26) satisfies the initial-value problem defined by Eq. (10.24) and its initial conditions $y(0) = y'(0) = 0$.

Solution.

The referenced formulas, Eqs. (10.24) and (10.26) on textbook pp. 453–454, are

$$y'' + y = f(x), \qquad y(x) = \int_0^x \sin(x-t) f(t)\, dt.$$

For continuous f, the Leibniz rule

$$\frac{d}{dx}\int_0^x g(x,t)\, dt = g(x,x) + \int_0^x \partial_x g(x,t)\, dt$$

includes both an endpoint term and a derivative of the integrand. Applying it once, with $g = \sin(x-t)f(t)$, gives

$$\begin{aligned} y'(x) &= \sin(x-x)f(x) + \int_0^x \cos(x-t) f(t)\, dt \\ &= \int_0^x \cos(x-t) f(t)\, dt. \end{aligned}$$

Apply the rule a second time, now with $g = \cos(x-t)f(t)$:

$$\begin{aligned} y''(x) &= \cos(x-x)f(x) - \int_0^x \sin(x-t) f(t)\, dt \\ &= f(x) - y(x). \end{aligned}$$

This proves $y'' + y = f$. At $x = 0$, both displayed integrals have zero length, so $y(0) = y'(0) = 0$.

For forward evolution $x \geq 0$ with source times $t \geq 0$, the equivalent causal kernel is $G(x,t) = \Theta(x-t)\sin(x-t)$, where Θ is the Heaviside unit step function ($\Theta(s) = 0$ for $s < 0$, $\Theta(s) = 1$ for $s > 0$; it is the $u(s)$ of Exercise 1.11.9). The displayed Volterra integral itself also solves the initial-value problem for negative x when its integral is given the usual orientation, since the Leibniz rule above holds for $x < 0$ as well. The kernel vanishes for $x < t$, is continuous with value zero at $x = t$, and its first derivative jumps from 0 to $\cos 0 = 1$. Thus $(\partial_x^2 + 1)G = \delta(x-t)$. The difference of any two solutions is $A\sin x + B\cos x$; its zero initial position and velocity force $A = B = 0$, so the verified solution is unique.

10.1.4 Find the Green's function for the equation

$$-\frac{d^2y}{dx^2} - \frac{y}{4} = f(x),$$

with boundary conditions $y(0) = y(\pi) = 0$.

$$\textit{ANS. } G(x,t) = \begin{cases} 2\sin(x/2)\cos(t/2), & 0 \leq x < t, \\ 2\cos(x/2)\sin(t/2), & t < x \leq \pi. \end{cases}$$

Solution.

For $y'' + y/4 = 0$, take

$$u(x) = \sin(x/2), \qquad v(x) = \cos(x/2).$$

Here $u(0)=0$ and $v(\pi)=0$. A continuous kernel with the required endpoint behavior is $Cu(x)v(t)$ for $x<t$ and $Cu(t)v(x)$ for $x>t$. With $u'=\frac{1}{2}\cos(x/2)$ and $v'=-\frac{1}{2}\sin(x/2)$, the jump formula of Exercise 10.1.2(a) gives

$$\begin{aligned} G_x(t^+,t)-G_x(t^-,t) &= C\big[u(t)v'(t)-u'(t)v(t)\big] \\ &= C\left[-\frac{1}{2}\sin^2(t/2)-\frac{1}{2}\cos^2(t/2)\right]=-\frac{C}{2}. \end{aligned}$$

Because the operator is $-\partial_x^2-1/4$, integrating $(-\partial_x^2-\frac{1}{4})G=\delta(x-t)$ across $x=t$ gives $-[G_x]_{t^-}^{t^+}=1$, so this jump must be -1. Therefore $C=2$ and

$$\boxed{G(x,t)=\begin{cases} 2\sin(x/2)\cos(t/2), & 0\le x\le t, \\ 2\cos(x/2)\sin(t/2), & t\le x\le \pi. \end{cases}}$$

The branches solve the homogeneous equation off the diagonal; continuity and the jump prove $(-\partial_x^2-1/4)G=\delta(x-t)$.

For a direct check of the source sign, put

$$I(x)=\int_0^x \sin(t/2)f(t)\,dt, \qquad J(x)=\int_x^\pi \cos(t/2)f(t)\,dt.$$

Splitting $y=\int_0^\pi G(x,t)f(t)\,dt$ at $t=x$ gives $y=2\cos(x/2)I+2\sin(x/2)J$. Since $I'=\sin(x/2)f$ and $J'=-\cos(x/2)f$, the terms of the first derivative that contain I' and J', $2\cos(x/2)\sin(x/2)f-2\sin(x/2)\cos(x/2)f$, cancel:

$$y'=-\sin(x/2)I+\cos(x/2)J.$$

Differentiating again, with the same I' and J', gives

$$y''=-\tfrac{1}{2}\cos(x/2)I-\tfrac{1}{2}\sin(x/2)J-\big[\sin^2(x/2)+\cos^2(x/2)\big]f=-\frac{y}{4}-f,$$

because $y/4=\frac{1}{2}\cos(x/2)I+\frac{1}{2}\sin(x/2)J$; thus $-y''-y/4=f$. Also $I(0)=J(\pi)=0$ gives the two Dirichlet conditions: $y(0)=2I(0)=0$ and $y(\pi)=2J(\pi)=0$. A homogeneous solution satisfying $y(0)=0$ is $A\sin(x/2)$, and its value at π is A, so the homogeneous boundary problem has only the zero solution.

10.1.5 Construct the Green's function for

$$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(k^2x^2-1)y=0,$$

subject to the boundary conditions $y(0)=0$, $y(1)=0$. Use the operator exactly as printed with ordinary dt source integration. Construct its inverse for nonresonant real k, treat $k=0$ separately, and explain the obstruction at a positive zero of $J_1(k)$.

Solution.

The normalization of the operator matters: dividing it by x multiplies its Green's function by t. The Instructor's Manual, p. 119, first divides the printed ODE by x. Its symmetric kernel is the Green's function of that divided operator. For the operator exactly as printed, with an ordinary dt source integral, the kernel has an additional factor $1/t$. An ordinary inverse also requires a nonresonant parameter. Both conventions are specified below. Write

$$D=x^2\frac{d^2}{dx^2}+x\frac{d}{dx}+k^2x^2-1, \qquad L=\frac{1}{x}D=\frac{d}{dx}\left(x\frac{d}{dx}\right)+k^2x-\frac{1}{x}.$$

First take $k > 0$ with $J_1(k) \neq 0$, and let $0 < t < 1$. On either side of the source, $Dy = 0$ is Bessel's equation of order one in the variable $z = kx$, $z^2 y_{zz} + z y_z + (z^2 - 1)y = 0$. Its independent solutions are $J_1(kx)$ and $Y_1(kx)$. Their small-x behavior is

$$J_1(kx) \sim \frac{kx}{2}, \qquad Y_1(kx) \sim -\frac{2}{\pi kx}.$$

Thus the left homogeneous solution obeying $y(0) = 0$ is $u(x) = J_1(kx)$. Choose the right solution

$$v(x) = Y_1(kx) - \frac{Y_1(k)}{J_1(k)} J_1(kx), \qquad v(1) = 0.$$

To determine the normalization, define $W = uv' - u'v$, with derivatives with respect to x. Dividing the homogeneous ODE by x^2 gives $y'' = -y'/x - (k^2 - x^{-2})y$ for both $y = u$ and $y = v$. In $uv'' - u''v$ the terms $(k^2 - x^{-2})uv$ cancel, and

$$W' = uv'' - u''v = -\frac{1}{x}(uv' - u'v) = -\frac{W}{x}.$$

Hence $(xW)' = W + xW' = 0$: $xW = w_0$ is constant (Abel's formula), and $W = w_0/x$. The multiple of J_1 in v does not change W, because the Wronskian of $J_1(kx)$ with itself vanishes. The constant w_0 is the limit of xW as $x \downarrow 0$, which follows from the displayed small-x forms of J_1, Y_1 and their x-derivatives $k/2$ and $2/(\pi k x^2)$:

$$W \sim \frac{kx}{2}\frac{2}{\pi k x^2} - \frac{k}{2}\left(-\frac{2}{\pi kx}\right) = \frac{2}{\pi x}.$$

Therefore $w_0 = 2/\pi$, and the exact Wronskian is $W(x) = 2/(\pi x)$.

For $LG_L = \delta(x - t)$, integrate $(xG_{L,x})_x + (k^2 x - 1/x)G_L = \delta(x - t)$ across $x = t$: only the flux $xG_{L,x}$ can jump, so the jump condition is $t[G_{L,x}]_{t^-}^{t^+} = 1$. Set $G_L = Cu(x_<)v(x_>)$, where $x_< = \min(x, t)$ and $x_> = \max(x, t)$. By the jump formula of Exercise 10.1.2(a), its jump is $CW(t)$, giving $t\,C\,W(t) = tC \cdot 2/(\pi t) = 2C/\pi = 1$, or $C = \pi/2$. Consequently

$$\boxed{G_L(x,t) = \frac{\pi}{2} J_1(kx_<)\left[Y_1(kx_>) - \frac{Y_1(k)}{J_1(k)} J_1(kx_>)\right].}$$

This is the symmetric kernel of the Instructor's Manual. Since $D = xL$,

$$D_x G_L(x,t) = x\delta(x - t) = t\delta(x - t).$$

Thus the answer for the printed operator is

$$\boxed{G_D(x,t) = \frac{G_L(x,t)}{t}, \qquad D_x G_D(x,t) = \delta(x - t).}$$

In particular $[G_{D,x}] = 1/t^2$, exactly as required by the leading coefficient x^2. The two equivalent source formulas are

$$Dy = F \quad \Longrightarrow \quad y(x) = \int_0^1 G_D(x,t)F(t)\,dt = \int_0^1 G_L(x,t)\frac{F(t)}{t}\,dt.$$

For each interior t, both kernels are continuous at $x = t$, vanish at $x = 1$, and tend to zero as $x \downarrow 0$.

The singular endpoint also restricts allowable sources if the integrated solution is required to vanish at zero. Sources supported inside $(0,1)$ are sufficient; more generally $F(x) = O(x^\alpha)$ for $\alpha > 0$ is sufficient. To see this, split $y(x) = \int_0^1 G_D(x,t)F(t)\,dt$ at $t = x$, and let c denote constants. For $t < x$ the kernel is $\frac{\pi}{2}[J_1(kt)/t]\,v(x)$, where $J_1(kt)/t$ is bounded and $|v(x)| \le c/x$; for $t > x$ it is $\frac{\pi}{2}J_1(kx)\,v(t)/t$, where $|J_1(kx)| \le cx$ and $|v(t)/t| \le c/t^2$. The two parts of y are therefore bounded by constants times $x^{-1}\int_0^x |F(t)|dt$ and $x\int_x^1 |F(t)|t^{-2}dt$. For $|F(t)| \le ct^\alpha$, up to constant multipliers, these bounds are respectively

$$\frac{1}{x}\int_0^x t^\alpha\,dt = \frac{x^\alpha}{\alpha+1}, \qquad x\int_x^1 t^{\alpha-2}\,dt = \begin{cases} \dfrac{x-x^\alpha}{\alpha-1}, & \alpha \ne 1, \\ -x\ln x, & \alpha = 1. \end{cases}$$

For every $\alpha > 0$ both tend to zero, including the logarithmic case. An arbitrary nonzero constant source at zero is not covered by this boundary-value inverse. For example, $Dy = 1$ has the power-series solution $-1 + \frac{k^2}{3}x^2 + \cdots$ (the coefficient of x^n gives $(n^2-1)a_n + k^2 a_{n-2} = \delta_{n0}$), and the other solutions bounded at 0 differ from it by multiples of $J_1(kx)$. All of them tend to -1 as $x \downarrow 0$, so $y(0) = 0$ cannot hold.

At a positive root $J_1(k) = 0$, the nonzero function $u = J_1(kx)$ satisfies both endpoint conditions. The denominator vanishes for a reason: the homogeneous problem has a zero mode. Let y solve $Dy = F$ with $y(0) = y(1) = 0$, for a source F supported inside $(0,1)$. Then $Ly = F/x$ and $Lu = 0$, and Lagrange's identity

$$uLy - yLu = u(xy')' - y(xu')' = \big[x(uy' - u'y)\big]'$$

gives

$$\int_0^1 u(x)\frac{F(x)}{x}\,dx = \int_0^1 (uLy - yLu)\,dx = \big[x(uy' - u'y)\big]_0^1 = 0.$$

The boundary term vanishes at $x = 1$ because $u(1) = y(1) = 0$. Near $x = 0$ the source vanishes, so there y solves $Dy = 0$ and, being regular at 0, equals cu for a constant c; hence $uy' - u'y = 0$ near 0. For the delta source $F = \delta(x-t)$ the left side is $J_1(kt)/t$, which tends to $k/2 \ne 0$ as $t \downarrow 0$. Thus a general source does not satisfy this compatibility condition, and a compatible solution is not unique because one can add any multiple of u. Therefore no ordinary inverse Green's function exists at these resonances.

The case $k = 0$ must be treated separately, rather than counted as a positive Bessel zero. Now $Dy = x^2y'' + xy' - y = 0$ has the solutions x and $1/x$ (for $y = x^m$ it gives $m(m-1) + m - 1 = m^2 - 1 = 0$). Choose $u = x$ and $v = x - 1/x$, so that $u(0) = 0$ and $v(1) = 0$; then $W = x(1 + x^{-2}) - (x - x^{-1}) = 2/x$. The jump condition $tCW(t) = 2C = 1$ gives $C = 1/2$, and the kernels are

$$\boxed{G_{L,0}(x,t) = \frac{1}{2}x_<\left(x_> - \frac{1}{x_>}\right), \qquad G_{D,0}(x,t) = \frac{1}{2t}x_<\left(x_> - \frac{1}{x_>}\right).}$$

These are also the $k \downarrow 0$ limits of the previous expressions. Indeed, as $k \to 0$, $J_1(kx) \approx kx/2$, $Y_1(kx) \approx -2/(\pi kx)$ and $Y_1(k)/J_1(k) \approx -4/(\pi k^2)$; the neglected terms are smaller by factors $O(k^2\ln k)$, since as $z \to 0$, $J_1(z) = (z/2)[1 + O(z^2)]$ and, from the series of Y_1 with its logarithmic term (Eq. (14.61) with $n = 1$), $Y_1(z) = -2/(\pi z) + (z/\pi)\ln(z/2) + O(z)$. Hence

$$G_L(x,t) \approx \frac{\pi}{2}\,\frac{kx_<}{2}\left[-\frac{2}{\pi kx_>} + \frac{4}{\pi k^2}\,\frac{kx_>}{2}\right] = \frac{1}{2}x_<\left(x_> - \frac{1}{x_>}\right).$$

No nonzero multiple of x vanishes at 1, so this case is nonresonant. For real negative k, replace it by $|k|$ throughout, since the printed operator depends only on k^2.

10.1.6 Given that

$$L = (1 - x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx}$$

and that $G(\pm 1, t)$ remains finite, show that no Green's function can be constructed by the techniques of this section.

Note. The solutions to $L = 0$ needed for the regions $x < t$ and $x > t$ are linearly dependent.

Solution.

Put the operator in divergence form:

$$Ly = \frac{d}{dx}\big[(1 - x^2)y'(x)\big].$$

For a homogeneous solution, one integration gives $(1 - x^2)y' = C$. By partial fractions, $y' = C/(1 - x^2) = \frac{C}{2}\big[1/(1 + x) + 1/(1 - x)\big]$, and a second integration gives

$$y(x) = D + \frac{C}{2}\big[\ln(1 + x) - \ln(1 - x)\big] = D + \frac{C}{2}\ln\frac{1 + x}{1 - x}.$$

The logarithm diverges at either endpoint. Thus a homogeneous solution finite at -1 is a constant, and a homogeneous solution finite at $+1$ is also a constant. These two solutions are therefore linearly dependent, as the Note says: their Wronskian vanishes, so no continuous combination of them can have the jump that the source requires, as shown next.

For an interior source $-1 < t < 1$, the proposed Green's function would therefore have to be

$$G(x, t) = \begin{cases} A(t), & x < t, \\ B(t), & x > t. \end{cases}$$

It must be continuous at t: a jump in G would put a delta function into G_x, and hence its derivative $\delta'(x - t)$ into LG, which is absent from the required source. So $A(t) = B(t)$. Both ordinary derivatives are zero, and hence

$$\big[(1 - x^2)G_x(x, t)\big]_{t^-}^{t^+} = 0.$$

But integrating $LG = \delta(x - t)$ across t requires this flux jump to be 1. This contradiction rules out an ordinary Green's function under the specified finite-endpoint conditions.

The obstruction is the constant zero mode $L1 = 0$. For a regular source f, endpoint finiteness of a solution to $Ly = f$ forces the endpoint fluxes $(1 - x^2)y'$ to vanish: a nonzero limiting flux would integrate to the logarithmic divergence displayed above. Indeed, if $(1 - x^2)y' \to \Phi \neq 0$ as $x \uparrow 1$, then $y' \approx \Phi/[2(1 - x)]$ near $x = 1$, and $y \approx -\frac{\Phi}{2}\ln(1 - x)$ is unbounded; the same holds at $x = -1$. Consequently, integrating $Ly = [(1 - x^2)y']' = f$ over $(-1, 1)$ gives

$$\int_{-1}^{1} f(x)\,dx = \big[(1 - x^2)y'(x)\big]_{-1}^{1} = 0.$$

A delta source has integral 1 and violates that condition. Even when a source is compatible, an arbitrary additive constant remains. A Green's function with the constant mode projected out would solve a modified source equation, not $LG = \delta$ requested here.

10.1.7 Find the Green's function for

$$\frac{d^2\psi}{dt^2} + k\frac{d\psi}{dt} = f(t),$$

subject to the initial conditions $\psi(0) = \psi'(0) = 0$, and solve this ODE for $t > 0$ given $f(t) = \exp(-t)$.

Solution.

Use u for the source time and t for the observation time. For $t < u$, the zero initial data require the causal kernel to vanish. For $t > u$, the homogeneous equation is $G_{tt} + kG_t = 0$, a first-order equation for G_t with solution proportional to e^{-kt}. When $k \neq 0$, a further integration gives $A(u)+B(u)e^{-kt}$. Continuity at $t = u$ requires $A+Be^{-ku} = 0$, so it can be written $C(u)(1-e^{-k(t-u)})$. Its first derivative at u^+ is $kC(u)$, while that at u^- is zero. The coefficient of ψ'' is one; therefore the jump must be one, and $C(u) = 1/k$. We obtain

$$\boxed{G(t,u) = \Theta(t-u)\frac{1-e^{-k(t-u)}}{k} \quad (k \neq 0).}$$

For $k = 0$, integrate $G_{tt} = 0$ on each side and use the same matching conditions, or take the limit:

$$\boxed{G(t,u) = \Theta(t-u)(t-u) \quad (k = 0).}$$

For either expression G is continuous with value zero at the source and $[G_t] = 1$, proving $(\partial_t^2+k\partial_t)G = \delta(t-u)$.

The Instructor's Manual's kernel on pp. 119–120 is normalized for the multiplied equation $(e^{kt}\psi')' = e^{kt}f$. It is $e^{-ku}G(t,u)$ and must be integrated against $e^{ku}f(u)$. The kernel above acts directly on the original source $f(u)$. Also, the apparent singularities at $k = 0$ and $k = 1$ in the final algebraic formula are removable and need the cases given below.

For $f(u) = e^{-u}$ and $k \neq 0, 1$, the convolution is computed with $\int_0^t e^{-u}\,du = 1 - e^{-t}$ and $e^{-k(t-u)}e^{-u} = e^{-kt}e^{(k-1)u}$:

$$\begin{aligned}
\psi(t) &= \frac{1}{k}\int_0^t \left(1 - e^{-k(t-u)}\right) e^{-u}\,du \\
&= \frac{1}{k}\left[1 - e^{-t} - e^{-kt}\int_0^t e^{(k-1)u}\,du\right] \\
&= \frac{1}{k}\left[1 - e^{-t} - e^{-kt}\,\frac{e^{(k-1)t}-1}{k-1}\right] = \frac{1}{k}\left[1 - e^{-t} - \frac{e^{-t}-e^{-kt}}{k-1}\right] \\
&= \frac{(k-1)(1-e^{-t}) - e^{-t} + e^{-kt}}{k(k-1)} = \boxed{\frac{(k-1) - ke^{-t} + e^{-kt}}{k(k-1)}}.
\end{aligned}$$

Its derivatives are

$$\psi'(t) = \frac{e^{-t} - e^{-kt}}{k-1}, \qquad \psi''(t) = \frac{-e^{-t} + ke^{-kt}}{k-1},$$

so $\psi'' + k\psi' = e^{-t}$ and $\psi(0) = \psi'(0) = 0$. For the exceptional values, evaluate the integrals directly; for $k = 0$ integrate by parts:

$$\begin{aligned}
k = 1: \quad \psi(t) &= \int_0^t (e^{-u} - e^{-t})\,du = (1 - e^{-t}) - te^{-t} = \boxed{1 - (1+t)e^{-t}}, \\
k = 0: \quad \psi(t) &= \int_0^t (t-u)e^{-u}\,du = \Big[-(t-u)e^{-u}\Big]_0^t - \int_0^t e^{-u}\,du \\
&= t - (1 - e^{-t}) = \boxed{t - 1 + e^{-t}}.
\end{aligned}$$

For $k = 1$, the derivatives are te^{-t} and $(1-t)e^{-t}$; for $k = 0$, they are $1 - e^{-t}$ and e^{-t}. These verify the ODE and zero initial data in both cases. There is no boundary condition at infinite time, so a negative real k is also permitted in the causal formula despite the possible growth of the response.

10.1.8 Take $k > 0$ and time dependence $e^{-i\omega\tau}$, with $\omega > 0$. Verify that the Green's function

$$G(x, x') = -\frac{i}{2k} \exp\left(ik|x - x'|\right)$$

yields an outgoing wave solution to the ODE

$$\left(\frac{d^2}{dx^2} + k^2\right) \psi(x) = g(x).$$

Note. Compare with Example 10.1.3.

Solution.

Take $k > 0$ and the time convention $e^{-i\omega\tau}$ used in Example 10.1.3. Its Eq. (10.28), textbook p. 454, omits k inside the exponential; the expression in this exercise and Table 10.1 has the correct factor $ik|x - x'|$. Let $s = x - x'$. Away from $s = 0$, each branch is proportional to e^{iks} or e^{-iks}, and hence solves $G_{xx} + k^2 G = 0$. Both branches have the common value $-i/(2k)$ at $s = 0$. Their first derivatives are

$$G_x(x, x') = \begin{cases} -\frac{1}{2}e^{ik(x'-x)}, & x < x', \\ \frac{1}{2}e^{ik(x-x')}, & x > x'. \end{cases}$$

Thus $G_x(x'^+, x') - G_x(x'^-, x') = 1$. Continuity rules out a delta derivative, and this jump gives

$$(\partial_x^2 + k^2)G(x, x') = \delta(x - x').$$

For a continuous compactly supported source g, define

$$F(x) = \int_{-\infty}^{x} e^{-ikt} g(t)\, dt, \qquad H(x) = \int_{x}^{\infty} e^{ikt} g(t)\, dt.$$

Since $e^{ik|x-t|}$ equals $e^{ikx}e^{-ikt}$ for $t < x$ and $e^{-ikx}e^{ikt}$ for $t > x$, the convolution $\psi(x) = \int_{-\infty}^{\infty} G(x, t)g(t)\, dt$ has the explicit form

$$\psi(x) = -\frac{i}{2k}\left[e^{ikx}F(x) + e^{-ikx}H(x)\right].$$

Because $F' = e^{-ikx}g$ and $H' = -e^{ikx}g$, the terms of ψ' that contain F' and H' add up to $-\frac{i}{2k}(g - g) = 0$, and only the derivatives of the exponentials remain:

$$\psi'(x) = -\frac{i}{2k}\left[ike^{ikx}F - ike^{-ikx}H\right] = \frac{1}{2}\left[e^{ikx}F(x) - e^{-ikx}H(x)\right].$$

In the second derivative the terms with F' and H' give $\frac{1}{2}(g + g) = g$:

$$\psi''(x) = \frac{ik}{2}\left[e^{ikx}F(x) + e^{-ikx}H(x)\right] + g(x).$$

Since $k^2\psi = -ik[e^{ikx}F + e^{-ikx}H]/2$, the sum $\psi'' + k^2\psi$ is g.

If the source is supported in $[a, b]$, then for $x > b$ we have $H(x) = 0$ and $F(x) = F(b)$, so only the first term remains: $\psi = C_+e^{ikx}$ with $C_+ = -iF(b)/(2k)$. For $x < a$, $F(x) = 0$ and $H(x) = H(a)$, so only the second remains: $\psi = C_-e^{-ikx}$ with $C_- = -iH(a)/(2k)$. With the time factor $e^{-i\omega\tau}$, constant phase on the right obeys $kx - \omega\tau =$ constant and moves to increasing x; on the left it obeys $-kx - \omega\tau =$ constant and moves to decreasing x. Both waves travel away from the source, which verifies the outgoing condition.

10.1.9 Construct the 1-D Green's function for the modified Helmholtz equation, with $k > 0$ in the displayed answer,

$$\left(\frac{d^2}{dx^2} - k^2\right)\psi(x) = f(x).$$

The boundary conditions are that the Green's function must vanish for $x \to \infty$ and $x \to -\infty$.

ANS. $G(x_1, x_2) = -\dfrac{1}{2k}\exp\left(-k|x_1 - x_2|\right)$.

Solution.

The printed formula assumes $k > 0$. For a real nonzero parameter of either sign, set $\kappa = |k|$; the operator contains $k^2 = \kappa^2$, while decay requires the positive root. For $x < t$, decay as $x \to -\infty$ selects $e^{\kappa x}$; for $x > t$, decay as $x \to +\infty$ selects $e^{-\kappa x}$. Continuity can be imposed by a single coefficient:

$$G(x,t) = C\begin{cases} e^{\kappa(x-t)}, & x < t, \\ e^{-\kappa(x-t)}, & x > t. \end{cases}$$

Its derivative tends to κC as $x \uparrow t$ and to $-\kappa C$ as $x \downarrow t$, so the jump of its derivative is $-\kappa C - \kappa C = -2\kappa C$. The coefficient of ψ'' is one, so this must equal one. Hence

$$\boxed{G(x,t) = -\frac{1}{2\kappa}e^{-\kappa|x-t|}.}$$

It is continuous at $x = t$, solves $G_{xx} - \kappa^2 G = 0$ off the diagonal, and decays at both infinities. Its derivative limits are $-1/2$ and $+1/2$, so $(\partial_x^2 - k^2)G = \delta(x-t)$. Therefore, splitting $\int_{-\infty}^{\infty} G(x,t)f(t)\,dt$ at $t = x$, where $e^{-\kappa|x-t|}$ equals $e^{-\kappa x}e^{\kappa t}$ for $t < x$ and $e^{\kappa x}e^{-\kappa t}$ for $t > x$,

$$\psi(x) = -\frac{1}{2\kappa}\left[e^{-\kappa x}\int_{-\infty}^{x} e^{\kappa t}f(t)\,dt + e^{\kappa x}\int_{x}^{\infty} e^{-\kappa t}f(t)\,dt\right]$$

is the required solution for, for example, a continuous compactly supported source. A homogeneous solution $Ae^{\kappa x} + Be^{-\kappa x}$ can decay at both ends only if $A = B = 0$, proving uniqueness.

For $k = 0$, no Green's function for d^2/dx^2 can vanish at both infinities. Each branch would be affine; decay at its corresponding infinity would make that branch zero, leaving no derivative jump. The divergent constant in the $\kappa \downarrow 0$ expansion, obtained from $e^{-\kappa|x-t|} = 1 - \kappa|x-t| + O(\kappa^2)$,

$$G(x,t) = -\frac{1}{2\kappa} + \frac{1}{2}|x-t| + O(\kappa)$$

reflects that obstruction. Subtracting the constant would yield the Laplace fundamental solution, but it would not satisfy decay at infinity.

10.1.10 From the eigenfunction expansion of the Green's function show that

(a) $$\frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin n\pi x\,\sin n\pi t}{n^2} = \begin{cases} x(1-t), & 0 \le x < t, \\ t(1-x), & t < x \le 1. \end{cases}$$

(b) $$\frac{2}{\pi^2}\sum_{n=0}^{\infty}\frac{\sin(n+\frac{1}{2})\pi x\,\sin(n+\frac{1}{2})\pi t}{(n+\frac{1}{2})^2} = \begin{cases} x, & 0 \le x < t, \\ t, & t < x \le 1. \end{cases}$$

Solution.

In both parts use $L = -d^2/dx^2$ on $[0,1]$. For a normalized eigenfunction ϕ_n, $L\phi_n = \lambda_n\phi_n$, the inverse has coefficients $1/\lambda_n$:

$$G(x,t) = \sum_n \frac{\phi_n(x)\phi_n(t)}{\lambda_n}.$$

There is no complex conjugation to display here because the chosen eigenfunctions are real.

(a) With $\phi(0) = \phi(1) = 0$, the equation $-\phi'' = \lambda\phi$ has positive eigenvalues. Indeed, multiplying it by ϕ^* and integrating by parts gives

$$\lambda\int_0^1 |\phi|^2 dx = -\int_0^1 \phi^*\phi''\,dx = -\big[\phi^*\phi'\big]_0^1 + \int_0^1 |\phi'|^2 dx = \int_0^1 |\phi'|^2 dx,$$

since the boundary term vanishes when $\phi(0) = \phi(1) = 0$. Hence $\lambda \geq 0$, and $\lambda = 0$ would give $\phi' = 0$, making ϕ a constant vanishing at the endpoints, i.e., $\phi = 0$. Write $\lambda = q^2$ with $q > 0$. In $\phi = A\sin(qx) + B\cos(qx)$ the condition at zero gives $B = 0$, and $\phi(1) = A\sin q = 0$ gives $q = n\pi$, $n \geq 1$. Since

$$\int_0^1 \sin^2(n\pi x)\,dx = \left[\frac{x}{2} - \frac{\sin(2n\pi x)}{4n\pi}\right]_0^1 = \frac{1}{2},$$

we have $\phi_n(x) = \sqrt{2}\sin(n\pi x)$ and $\lambda_n = n^2\pi^2$. The eigenfunction formula is therefore

$$G_D(x,t) = \frac{2}{\pi^2}\sum_{n=1}^{\infty} \frac{\sin(n\pi x)\sin(n\pi t)}{n^2}.$$

Construct the same kernel directly. The left and right homogeneous solutions are x and $1-x$, and continuity suggests $Cx_<(1-x_>)$. Its derivative is $C(1-t)$ for $x < t$ and $-Ct$ for $x > t$, so its jump is $-Ct - C(1-t) = -C$. Requiring $-G_{xx} = \delta$, i.e., a jump of -1 as in Exercise 10.1.1, gives $C = 1$. Since $\lambda = 0$ is not an eigenvalue (shown above), the homogeneous problem has only the zero solution, so the Green's function is unique, and the eigenfunction series and the closed form are the same kernel (the final paragraph of this solution also verifies this directly through the Fourier coefficients). Thus

$$\boxed{\frac{2}{\pi^2}\sum_{n=1}^{\infty} \frac{\sin(n\pi x)\sin(n\pi t)}{n^2} = \begin{cases} x(1-t), & x \leq t, \\ t(1-x), & t \leq x. \end{cases}}$$

(b) Now impose $\phi(0) = 0$, $\phi'(1) = 0$. The same energy identity shows that the eigenvalues are positive: its boundary term $-\phi^*(1)\phi'(1) + \phi^*(0)\phi'(0)$ vanishes also for these mixed conditions. Again $\phi = A\sin(qx)$, while the second endpoint condition gives $\phi'(1) = Aq\cos q = 0$, i.e., $\cos q = 0$. Thus

$$q_n = (n + \tfrac{1}{2})\pi, \qquad \lambda_n = (n+\tfrac{1}{2})^2\pi^2, \qquad n = 0, 1, 2, \ldots.$$

The normalization is unchanged because $\sin(2q_n) = 0$: $\int_0^1 \sin^2(q_n x)dx = 1/2$. Hence

$$G_M(x,t) = \frac{2}{\pi^2}\sum_{n=0}^{\infty} \frac{\sin((n+\frac{1}{2})\pi x)\sin((n+\frac{1}{2})\pi t)}{(n+\frac{1}{2})^2}.$$

Exercise 10.1.1 constructed the unique kernel for precisely these conditions, $G_M(x,t) = \min(x,t)$. It follows that

$$\boxed{\frac{2}{\pi^2}\sum_{n=0}^{\infty} \frac{\sin((n+\frac{1}{2})\pi x)\sin((n+\frac{1}{2})\pi t)}{(n+\frac{1}{2})^2} = \begin{cases} x, & x \leq t, \\ t, & t \leq x. \end{cases}}$$

The identification can also be checked without differentiating either series twice term by term. For each fixed interior t, integrate by parts on $(0,t)$ and $(t,1)$, using $-G_{xx} = \delta(x-t)$ and the matching endpoint conditions. Because each closed kernel is affine on either side of t ($G_{xx} = 0$ there, so no integral remains), integration by parts twice gives

$$\begin{aligned}\int_0^1 G(-\phi_n'')\,dx &= \big[-G\phi_n' + G_x\phi_n\big]_0^{t^-} + \big[-G\phi_n' + G_x\phi_n\big]_{t^+}^1 \\ &= \big[G_x(t^-,t) - G_x(t^+,t)\big]\phi_n(t) = \phi_n(t).\end{aligned}$$

Here the outer-endpoint terms vanish: at $x = 0$ both G and ϕ_n are zero, and at $x = 1$ we have $G = \phi_n = 0$ in (a) and $G_x = \phi_n' = 0$ in (b). The terms with $\phi_n'(t)$ cancel because G is continuous, and the derivative jump is -1. Since $-\phi_n'' = \lambda_n\phi_n$, this gives

$$\lambda_n \int_0^1 G(x,t)\phi_n(x)\,dx = \int_0^1 G(x,t)(-\phi_n''(x))\,dx = \phi_n(t).$$

Thus the closed kernels have exactly the stated Fourier coefficients. By the completeness of the eigenfunctions, two square-integrable functions with the same expansion coefficients are equal in the mean-square sense. Both series converge absolutely and uniformly by the Weierstrass M test, because their terms are bounded by $2/(\pi^2 n^2)$ and $2/[\pi^2(n+\frac{1}{2})^2]$, so their sums are continuous. Equality therefore holds everywhere, including $x = t$ and the endpoints by continuity.

10.1.11 Derive an integral equation corresponding to

$$y''(x) - y(x) = 0, \quad y(1) = 1, \quad y(-1) = 1,$$

(a) by integrating twice.

(b) by forming the Green's function.

$$\textit{ANS. } y(x) = 1 - \int_{-1}^1 K(x,t)\,y(t)\,dt,$$
$$K(x,t) = \begin{cases} \frac{1}{2}(1-x)(t+1), & x > t, \\ \frac{1}{2}(1-t)(x+1), & x < t. \end{cases}$$

Solution.

(a) Rewrite the ODE as $y'' = y$, and set $c = y'(-1)$. Integrating first to s and then to x gives

$$y'(s) = c + \int_{-1}^s y(t)\,dt, \qquad y(x) = 1 + c(x+1) + \int_{-1}^x \int_{-1}^s y(t)\,dt\,ds.$$

Interchanging the order on the triangle $-1 \le t \le s \le x$ turns the double integral into a single one:

$$\int_{-1}^x \int_{-1}^s y(t)\,dt\,ds = \int_{-1}^x y(t)\left(\int_t^x ds\right)dt = \int_{-1}^x (x-t)y(t)\,dt.$$

At $x = 1$, the condition $y(1) = 1$ gives

$$1 = 1 + 2c + \int_{-1}^1 (1-t)y(t)\,dt, \qquad c = -\frac{1}{2}\int_{-1}^1 (1-t)y(t)\,dt.$$

Substitution yields

$$y(x) = 1 + \int_{-1}^{x} (x-t)y(t)\,dt - \frac{x+1}{2}\int_{-1}^{1} (1-t)y(t)\,dt.$$

Split the last integral at $t = x$. On $t < x$, the combined coefficient of $y(t)$ is

$$(x-t) - \frac{x+1}{2}(1-t) = \frac{2x - 2t - (x+1-xt-t)}{2} = \frac{(x-1)+t(x-1)}{2} = -\frac{(1-x)(t+1)}{2}.$$

On $t > x$, only the last integral contributes, with coefficient $-(x+1)(1-t)/2$. We obtain

$$\boxed{y(x) = 1 - \int_{-1}^{1} K(x,t)y(t)\,dt, \quad K(x,t) = \begin{cases} \frac{1}{2}(1-x)(t+1), & t \le x, \\ \frac{1}{2}(1-t)(x+1), & x \le t. \end{cases}}$$

(b) Homogenize the boundary conditions by setting $u = y - 1$. Then $u(-1) = u(1) = 0$, and since $u'' = y'' = y$, the original equation becomes $-u'' = -y$. Construct the Green's function of $-d^2/dx^2$ on $[-1, 1]$: the left and right homogeneous solutions are $x + 1$ and $1 - x$. The continuous kernel $C(x_< + 1)(1 - x_>)$ has x-derivative $C(1 - t)$ for $x < t$ and $-C(t + 1)$ for $x > t$, hence the derivative jump

$$-C(t+1) - C(1-t) = -2C.$$

For $-G_{xx} = \delta(x - t)$ the jump must be -1 (as in Exercise 10.1.1), so $C = 1/2$, which is exactly K above. Thus

$$u(x) = \int_{-1}^{1} K(x,t)(-y(t))\,dt,$$

and adding 1 recovers the same integral equation.

Conversely, $K(\pm 1, t) = 0$ makes any solution of that integral equation satisfy $y(\pm 1) = 1$. Applying $-d^2/dx^2$ and using $-\partial_x^2 K(x,t) = \delta(x-t)$ gives $-y'' = -y$, or $y'' - y = 0$. As an explicit check, write $y = A\cosh x + B\sinh x$. The two endpoint values give $A\cosh 1 + B\sinh 1 = 1$ and $A\cosh 1 - B\sinh 1 = 1$, so $B = 0$ and $A = 1/\cosh 1$:

$$\boxed{y(x) = \frac{\cosh x}{\cosh 1}.}$$

To verify the integral equation with this solution, integrate the two branches of K against $\cosh t$, using the antiderivatives (found by integrating by parts)

$$\int (t+1)\cosh t\,dt = (t+1)\sinh t - \cosh t, \qquad \int (1-t)\cosh t\,dt = (1-t)\sinh t + \cosh t.$$

They give

$$\begin{aligned} \int_{-1}^{1} K(x,t)\cosh t\,dt &= \tfrac{1}{2}(1-x)\big[(x+1)\sinh x - \cosh x + \cosh 1\big] \\ &\quad + \tfrac{1}{2}(x+1)\big[\cosh 1 - (1-x)\sinh x - \cosh x\big] \\ &= \cosh 1 - \cosh x, \end{aligned}$$

because the $\sinh x$ terms cancel and $\frac{1}{2}(1-x) + \frac{1}{2}(x+1) = 1$. Hence $1 - \int_{-1}^{1} K(x,t)\,\frac{\cosh t}{\cosh 1}\,dt = 1 - \frac{\cosh 1 - \cosh x}{\cosh 1} = \frac{\cosh x}{\cosh 1}$, as required.

10.1.12 The general second-order linear ODE with constant coefficients is

$$y''(x) + a_1 y'(x) + a_2 y(x) = 0.$$

Given the boundary conditions $y(0) = y(1) = 0$, integrate twice and develop the integral equation

$$y(x) = \int_0^1 K(x,t)\, y(t)\, dt,$$

with

$$K(x,t) = \begin{cases} a_2 t(1-x) + a_1(x-1), & t < x, \\ a_2 x(1-t) + a_1 x, & x < t. \end{cases}$$

Note that $K(x,t)$ is symmetric and continuous if $a_1 = 0$. How is this related to self-adjointness of the ODE? For this self-adjointness discussion take a_1, a_2 real, and distinguish the ordinary unit-weight inner product from a weighted one.

Solution.

Let $c = y'(0)$ and use $y(0) = 0$. One integration of $y'' = -a_1 y' - a_2 y$ over $(0,x)$, with $\int_0^x y'(t)\, dt = y(x) - y(0) = y(x)$, gives

$$y'(x) = c - a_1 y(x) - a_2 \int_0^x y(t)\, dt.$$

Integrate once more over $(0,x)$. Reversing the order in the double integral, as in Exercise 10.1.11, $\int_0^x \int_0^s y(t)\, dt\, ds = \int_0^x (x-t) y(t)\, dt$, so

$$y(x) = cx - a_1 \int_0^x y(t)\, dt - a_2 \int_0^x (x-t) y(t)\, dt.$$

The second boundary condition, $y(1) = 0$, supplies

$$0 = c - a_1 \int_0^1 y(t)\, dt - a_2 \int_0^1 (1-t) y(t)\, dt.$$

Solve this for c and substitute into the preceding expression:

$$\begin{aligned} y(x) = {} & a_1 x \int_0^1 y(t)\, dt - a_1 \int_0^x y(t)\, dt \\ & + a_2 x \int_0^1 (1-t) y(t)\, dt - a_2 \int_0^x (x-t) y(t)\, dt. \end{aligned}$$

For $t < x$, the coefficient of $y(t)$ is

$$a_1(x-1) + a_2\big[x(1-t) - (x-t)\big] = a_1(x-1) + a_2 t(1-x).$$

For $t > x$, it is $a_1 x + a_2 x(1-t)$. Hence

$$\boxed{y(x) = \int_0^1 K(x,t) y(t)\, dt, \qquad K(x,t) = \begin{cases} a_2 t(1-x) + a_1(x-1), & t < x, \\ a_2 x(1-t) + a_1 x, & t > x. \end{cases}}$$

The value of K at the single point $t = x$ does not affect the integral.

The reverse implication can also be checked without assuming the ODE. Let the integral on the right be $F(x)$ and split it at $t = x$. The first branch at the diagonal minus the second branch there is $[a_2x(1-x) + a_1(x-1)] - [a_2x(1-x) + a_1x] = -a_1$, and the x-derivatives of the two branches are $-a_2t + a_1$ and $a_2(1-t) + a_1$, so the Leibniz rule gives

$$F'(x) = -a_1y(x) + \int_0^x (-a_2t + a_1)y(t)\,dt + \int_x^1 \big[a_2(1-t) + a_1\big]y(t)\,dt.$$

A second differentiation (the integrands no longer depend on x, so only the endpoint terms appear) gives

$$F'' = -a_1y' + \big[(-a_2x + a_1) - (a_2(1-x) + a_1)\big]y = -a_1y' - a_2y.$$

Thus $y = F$ implies the original ODE. Also $K(0,t) = K(1,t) = 0$, so the integral equation incorporates both boundary conditions.

The self-adjointness discussion assumes real a_1, a_2 and the unit-weight inner product. A nonzero real a_1 does not make the ODE intrinsically incapable of a self-adjoint formulation; it requires an integrating factor, or equivalently a weighted inner product. For real coefficients and zero Dirichlet data, integrate by parts twice in the ψ'' term and once in the ψ' term:

$$\int_0^1 \phi^*(\psi'' + a_1\psi' + a_2\psi)\,dx = \int_0^1 (\phi'' - a_1\phi' + a_2\phi)^*\psi\,dx + \big[\phi^*\psi' - (\phi^*)'\psi + a_1\phi^*\psi\big]_0^1.$$

The boundary term vanishes when ϕ and ψ vanish at both ends. Hence

$$\left(\frac{d^2}{dx^2} + a_1\frac{d}{dx} + a_2\right)^\dagger = \frac{d^2}{dx^2} - a_1\frac{d}{dx} + a_2.$$

So the displayed operator is formally self-adjoint in unit weight precisely when $a_1 = 0$. In this case

$$K(x,t) = a_2x_<(1 - x_>),$$

which is continuous and symmetric. When $a_1 \neq 0$, the kernel jumps by $a_1x - a_1(x-1) = a_1$ as t passes from below to above x, and it is not symmetric: for $t < x$,

$$K(x,t) - K(t,x) = \big[a_2t(1-x) + a_1(x-1)\big] - \big[a_2t(1-x) + a_1t\big] = -a_1(1 + t - x),$$

which is not identically zero. Since $(py')' = p(y'' + a_1y')$ for $p(x) = e^{a_1x}$, multiplication of the ODE by p instead gives

$$(py')' + a_2py = 0.$$

This differential expression is self-adjoint with the Dirichlet domain; the original expression is symmetric in the weighted scalar product $\int_0^1 p(x)\phi(x)^*\psi(x)dx$. Indeed, with $L = d^2/dx^2 + a_1d/dx + a_2$ we have $pL\psi = (p\psi')' + a_2p\psi$, and one integration by parts (its boundary term $[p\phi^*\psi']_0^1$ vanishes because $\phi = 0$ at both ends) gives

$$\int_0^1 p\,\phi^*L\psi\,dx = -\int_0^1 p\,(\phi^*)'\psi'\,dx + \int_0^1 a_2p\,\phi^*\psi\,dx,$$

which is unchanged when ϕ and ψ are interchanged and complex conjugated.

The homogeneous integral equation need not have a nonzero solution for arbitrary coefficients. To identify its exceptional parameters, put $y = e^{-a_1 x/2} v$. Then

$$y' = e^{-a_1 x/2}(v' - a_1 v/2), \qquad y'' = e^{-a_1 x/2}(v'' - a_1 v' + a_1^2 v/4),$$

and substitution reduces the ODE to

$$v'' + (a_2 - a_1^2/4)v = 0, \qquad v(0) = v(1) = 0,$$

because the v' terms cancel ($-a_1 v' + a_1 v' = 0$) and the coefficient of v is $a_1^2/4 - a_1^2/2 + a_2 = a_2 - a_1^2/4$. For real coefficients, nonzero solutions exist exactly when $a_2 - a_1^2/4 = n^2\pi^2$, $n = 1, 2, \ldots$ (the Dirichlet eigenvalues found in Exercise 10.1.10(a), with $v = \sin(n\pi x)$); they are $y = Ce^{-a_1 x/2}\sin(n\pi x)$. Otherwise the only solution of either formulation is $y = 0$.

10.1.13 Transform the ODE

$$\frac{d^2 y(r)}{dr^2} - k^2 y(r) + V_0 \frac{e^{-r}}{r} y(r) = 0$$

and the boundary conditions $y(0) = y(\infty) = 0$ into an integral equation of the form

$$y(r) = -V_0 \int_0^\infty G(r,t)\, \frac{e^{-t}}{t}\, y(t)\, dt.$$

The quantities V_0 and k^2 are constants. Take $k > 0$ (or, more generally, $\Re k > 0$ for the displayed decaying kernel), and the regular branch $y(r) = O(r)$ at the origin. The ODE is derived from the Schrödinger wave equation with a mesonic potential:

$$G(r,t) = \begin{cases} -\dfrac{1}{k} e^{-kt} \sinh kr, & 0 \le r < t, \\ -\dfrac{1}{k} e^{-kr} \sinh kt, & t < r < \infty. \end{cases}$$

Solution.

The decaying kernel in the question assumes k is the positive square root of a positive k^2 (more generally $\Re k > 0$). The singular endpoint is interpreted using the regular solution $y(r) = O(r)$ as $r \downarrow 0$. The Instructor's Manual's final sentence on p. 120, claiming that the question's G is minus the Green's function, is incorrect for the chapter's convention $LG = +\delta$. Isolate the free operator and source:

$$L_0 = \frac{d^2}{dr^2} - k^2, \qquad L_0 y(r) = -V_0 \frac{e^{-r}}{r} y(r).$$

For $L_0 u = 0$, choose $u(r) = \sinh(kr)$, which vanishes at zero, and $v(r) = e^{-kr}$, which decays at infinity. Their Wronskian is

$$\begin{aligned} uv' - u'v &= -k\sinh(kr)e^{-kr} - k\cosh(kr)e^{-kr} \\ &= -ke^{-kr}\big[\sinh(kr) + \cosh(kr)\big] = -k. \end{aligned}$$

By the jump formula of Exercise 10.1.2(a), the continuous kernel $Cu(r_<)v(r_>)$ has derivative jump $C(uv' - u'v) = -kC$. For $L_0 G = \delta(r - t)$ the jump must be 1, so $C = -1/k$:

$$\boxed{G(r,t) = -\frac{1}{k}\sinh(kr_<)e^{-kr_>}, \qquad r_< = \min(r,t), \quad r_> = \max(r,t).}$$

It satisfies $G(0,t) = 0$ and $G(\infty,t) = 0$. The matching calculation shows directly that it is the Green's function with a positive unit delta source. Equivalently, the method-of-images form is

$$G(r,t) = -\frac{1}{2k}\left[e^{-k|r-t|} - e^{-k(r+t)}\right].$$

Indeed, for $r < t$ the bracket is $e^{-kt}(e^{kr} - e^{-kr}) = 2e^{-kt}\sinh kr$, and for $r > t$ it is $2e^{-kr}\sinh kt$. The second term cancels the first at $r = 0$ and is homogeneous as a function of positive r.

Apply this inverse to the source $-V_0 e^{-t}y(t)/t$:

$$\boxed{y(r) = -V_0\int_0^\infty G(r,t)\frac{e^{-t}}{t}\,y(t)\,dt.}$$

Thus the negative sign outside the integral and the negative sign in G are both required. In split form, the same equation reads

$$y(r) = \frac{V_0}{k}\left[e^{-kr}\int_0^r \sinh(kt)\frac{e^{-t}}{t}y(t)\,dt + \sinh(kr)\int_r^\infty e^{-kt}\frac{e^{-t}}{t}y(t)\,dt\right].$$

On the regular branch $y(t) = O(t)$, the source factor $y(t)/t$ is bounded near zero. For example, substitute $y = At + Bt^2 + \cdots$ into the ODE: $y'' = 2B + O(t)$, $k^2 y = O(t)$, and $V_0 e^{-t}y/t = V_0 A + O(t)$, so the t^0 terms give $2B + V_0 A = 0$ and $y(t) = At - (V_0 A/2)t^2 + \cdots$. At infinity the exponential factors ensure convergence for a decaying y.

Conversely, apply L_0 to the integral equation. The delta identity gives $L_0 y = -V_0 e^{-r}y/r$, and the kernel enforces the endpoint conditions. There is no omitted homogeneous term: $Ae^{kr} + Be^{-kr}$ can decay at infinity only if $A = 0$, and its value at zero then forces $B = 0$. The integral equation is therefore equivalent to the original boundary problem within this regular, decaying class; it is a homogeneous eigenvalue equation for the allowed parameters, not an assertion that all V_0, k admit a nonzero solution.

The limit $k \downarrow 0$ gives $G(r,t) \to -\min(r,t)$ (since $\sinh(kr_<)/k \to r_<$ and $e^{-kr_>} \to 1$), which approaches $-t$, not zero, as $r \to \infty$. Accordingly there is no inverse of d^2/dr^2 satisfying these same two zero boundary values for an arbitrary source: its right homogeneous branch would have to be identically zero. This explains why the assumption $k > 0$ (more generally $\Re k > 0$), which makes the right branch e^{-kr} decay, is needed.

10.2 Problems in Two and Three Dimensions

10.2.1 Show that the fundamental Green's function for the 1-D Laplace equation, $|x_1 - x_2|/2$, is consistent with the form found in Example 10.1.1.

Solution.

Write $x = x_1$, $t = x_2$. The fundamental solution $G_0(x,t) = |x - t|/2$ has derivative $-1/2$ for $x < t$ and $+1/2$ for $x > t$. It is continuous, linear on each side, and its derivative jump is 1, so (as in Exercise 10.1.1)

$$\partial_x^2 G_0(x,t) = \delta(x - t).$$

Example 10.1.1 solves $-y'' = f$ with $y(0) = y(1) = 0$. To impose these zero Dirichlet values at $x = 0, 1$, add a homogeneous solution in x, $h(x,t) = a(t)x + b(t)$. The endpoint equations are

$$0 = G_0(0,t) + b(t) = t/2 + b(t),$$

$$0 = G_0(1,t) + a(t) + b(t) = (1-t)/2 + a(t) - t/2.$$

Hence $b = -t/2$, and then the second equation gives $a = t/2 - (1-t)/2 = t - 1/2$, so that $h = (t - \frac{1}{2})x - \frac{t}{2}$ and

$$G_D(x,t) = \frac{1}{2}|x-t| + xt - \frac{x+t}{2}.$$

This added affine function has zero second derivative and cannot change the source normalization. Evaluating the two branches gives

$$x < t: \quad G_D = \frac{t-x}{2} + xt - \frac{x+t}{2} = -x(1-t),$$
$$x > t: \quad G_D = \frac{x-t}{2} + xt - \frac{x+t}{2} = -t(1-x).$$

Thus

$$\boxed{G_D(x,t) = -\begin{cases} x(1-t), & x \le t, \\ t(1-x), & t \le x. \end{cases}}$$

The operator in Example 10.1.1 is $-d^2/dx^2$, whereas the Laplace operator here is $+d^2/dx^2$. Their Green's functions therefore differ by a minus sign. Consistency requires retaining that operator sign, as also shown in Example 10.2.1, textbook p. 464. Indeed $\partial_x^2 G_D = \delta$, while $(-\partial_x^2)(-G_D) = \delta$; the latter positive kernel is exactly the one in Example 10.1.1.

10.2.2 Show that if

$$L\psi(\mathbf{r}) \equiv \nabla \cdot [p(\mathbf{r})\nabla\psi(\mathbf{r})] + q(\mathbf{r})\psi(\mathbf{r}),$$

then L is Hermitian for $p(\mathbf{r})$ and $q(\mathbf{r})$ real, assuming Dirichlet boundary conditions on the boundary of a region (homogeneous and common to both functions, with sufficient regularity for Green's identity and, if the region is unbounded, a vanishing boundary form at infinity) and that the scalar product is an integral over that region with unit weight.

Solution.

For an operator domain, the Dirichlet conditions used here are homogeneous: both test functions vanish at the boundary. Nonzero prescribed boundary data must first be subtracted, using a fixed function that takes those boundary values. On an unbounded region the corresponding boundary form at infinity must also vanish. Let Ω have a smooth boundary, with sufficiently regular p, q, ϕ, ψ for the integrations below, and take p, q real. The unit-weight scalar product is

$$\langle \phi, \psi \rangle = \int_\Omega \phi^* \psi \, dV.$$

The product rule gives

$$\phi^* \nabla \cdot (p \nabla \psi) = \nabla \cdot (p \phi^* \nabla \psi) - p \nabla \phi^* \cdot \nabla \psi.$$

Integrate and use the divergence theorem:

$$\begin{aligned} \langle \phi, L\psi \rangle &= \int_{\partial\Omega} p \phi^* \partial_n \psi \, dS - \int_\Omega p \nabla \phi^* \cdot \nabla \psi \, dV + \int_\Omega q \phi^* \psi \, dV \\ &= -\int_\Omega p \nabla \phi^* \cdot \nabla \psi \, dV + \int_\Omega q \phi^* \psi \, dV, \end{aligned}$$

because $\phi = 0$ on $\partial\Omega$. Reality of the coefficients gives $(L\phi)^* = \nabla\cdot(p\nabla\phi^*) + q\phi^*$. Repeating the integration by parts with ψ outside the divergence yields

$$\begin{aligned}\langle L\phi, \psi\rangle &= \int_{\partial\Omega} p\psi\partial_n\phi^*\, dS - \int_\Omega p\nabla\psi\cdot\nabla\phi^*\, dV + \int_\Omega q\phi^*\psi\, dV \\ &= -\int_\Omega p\nabla\psi\cdot\nabla\phi^*\, dV + \int_\Omega q\phi^*\psi\, dV.\end{aligned}$$

because $\psi = 0$ on $\partial\Omega$. The remaining volume integrals agree term by term, so

$$\boxed{\langle \phi, L\psi\rangle = \langle L\phi, \psi\rangle.}$$

Equivalently, their difference is the boundary form

$$\int_{\partial\Omega} p\,(\phi^*\partial_n\psi - \psi\partial_n\phi^*)\, dS = 0.$$

This proves the Hermitian (symmetric) property on the stated common domain. If "self-adjoint" is used in its stronger operator-theoretic sense, a domain realization must also be specified; the usual Dirichlet realization on a smooth bounded region, with smooth real coefficients and $p \geq p_0 > 0$ (uniform ellipticity), is one such setting.

10.2.3 Show that the terms $+k^2$ in the Helmholtz operator and $-k^2$ in the modified Helmholtz operator do not affect the behavior of $G(\mathbf{r}_1, \mathbf{r}_2)$ in the immediate vicinity of the singular point $\mathbf{r}_1 = \mathbf{r}_2$. Specifically, show that

$$\lim_{\varepsilon\downarrow 0} \int_{|\mathbf{r}_2 - \mathbf{r}_1| < \varepsilon} k^2 G(\mathbf{r}_1, \mathbf{r}_2)\, d^3 r_2 = 0.$$

Solution.

The printed limit, including its value -1, is incorrect for a shrinking neighborhood of the source (textbook p. 466). The integration region must be specified as a ball of radius ε, and the local limit is 0. The Instructor's Manual, p. 121, computes a different, whole-space Fourier integral, formally giving $\int G\, d^3 r_2 = +1/k^2$, the value at $p = 0$ of the Helmholtz transform $1/(k^2 - p^2)$, so that $\int k^2 G\, d^3 r_2 = +1$. It does not establish the printed local claim. Fix $\mathbf{r}_1$ and put $s = |\mathbf{r}_2 - \mathbf{r}_1|$. The three-dimensional outgoing and modified kernels are respectively

$$G_H(s) = -\frac{e^{iks}}{4\pi s}, \qquad G_M(s) = -\frac{e^{-ks}}{4\pi s}, \qquad k > 0.$$

Since $e^{iks} = 1 + iks + O(s^2)$ and $e^{-ks} = 1 - ks + O(s^2)$, their expansions at $s = 0$ start with

$$G_H(s) = -\frac{1}{4\pi s} - \frac{ik}{4\pi} + O(s), \qquad G_M(s) = -\frac{1}{4\pi s} + \frac{k}{4\pi} + O(s).$$

Both therefore have the same leading singularity as the Laplace kernel. Since $d^3 r_2 = 4\pi s^2 ds$ for a radial integral,

$$\begin{aligned}\int_{s<\varepsilon} k^2 G(s)\, d^3 r_2 &= 4\pi k^2 \int_0^\varepsilon \left[-\frac{1}{4\pi s} + O(1)\right] s^2\, ds \\ &= -\frac{k^2\varepsilon^2}{2} + O(\varepsilon^3).\end{aligned}$$

Consequently the corrected assertion is

$$\boxed{\lim_{\varepsilon\downarrow 0}\int_{|\mathbf{r}_2-\mathbf{r}_1|<\varepsilon} k^2 G(\mathbf{r}_1,\mathbf{r}_2)\,d^3r_2 = 0.}$$

The same conclusion holds for $-k^2G$.

The exact radial integrals $I_H(\varepsilon)$ and $I_M(\varepsilon)$ of k^2G_H and k^2G_M over the ball independently show the limit. For Helmholtz, use the antiderivative $\int se^{iks}\,ds = e^{iks}\big(\frac{s}{ik}+\frac{1}{k^2}\big)$ (differentiate to check) and $-k^2/(ik) = ik$:

$$\begin{aligned} I_H(\varepsilon) &= 4\pi k^2\int_0^\varepsilon \left(-\frac{e^{iks}}{4\pi s}\right) s^2\,ds = -k^2\int_0^\varepsilon se^{iks}\,ds \\ &= -k^2\left[e^{iks}\left(\frac{s}{ik}+\frac{1}{k^2}\right)\right]_0^\varepsilon = ik\varepsilon e^{ik\varepsilon} - e^{ik\varepsilon} + 1 \\ &= e^{ik\varepsilon}(ik\varepsilon - 1) + 1 \longrightarrow 0. \end{aligned}$$

For modified Helmholtz, $\frac{d}{ds}\big[(1+ks)e^{-ks}\big] = ke^{-ks} - k(1+ks)e^{-ks} = -k^2se^{-ks}$, so

$$\begin{aligned} I_M(\varepsilon) &= -k^2\int_0^\varepsilon se^{-ks}\,ds \\ &= \Big[(1+ks)e^{-ks}\Big]_0^\varepsilon = (1+k\varepsilon)e^{-k\varepsilon} - 1 \longrightarrow 0. \end{aligned}$$

Integrating the defining equation $(\nabla^2 \pm k^2)G = \delta^{(3)}$ (upper sign for Helmholtz) over the ball, with the divergence theorem for $\nabla^2 G = \nabla\cdot\nabla G$, now gives

$$\int_{s=\varepsilon} \nabla G\cdot d\mathbf{S} \pm \int_{s<\varepsilon} k^2 G\,dV = 1.$$

The second integral tends to zero, while the flux tends to one: for these radial kernels the flux is $4\pi\varepsilon^2 G'(\varepsilon)$, which equals $(1-ik\varepsilon)e^{ik\varepsilon}$ for G_H and $(1+k\varepsilon)e^{-k\varepsilon}$ for G_M. (With the values of I_H and I_M above, both identities in fact hold exactly for every $\varepsilon > 0$.) Therefore it is the Laplacian term that supplies the delta singularity; the zeroth-order $\pm k^2$ term cannot change its coefficient. For comparison, taking the entire space for the modified kernel means $\varepsilon\to\infty$, and gives $I_M \to -1$. That global value must not be substituted for the shrinking-ball limit.

10.2.4 Show that

$$-\frac{\exp(ik|\mathbf{r}_1-\mathbf{r}_2|)}{4\pi|\mathbf{r}_1-\mathbf{r}_2|}$$

satisfies the appropriate criteria and therefore is a Green's function for the Helmholtz equation. Use $k>0$ and the outgoing convention with time factor $e^{-i\omega\tau}$, $\omega>0$.

Solution.

Let $s = |\mathbf{r}_1-\mathbf{r}_2|$ and take $k>0$, with time dependence $e^{-i\omega\tau}$. For a radial function $G(s)$ at $s>0$,

$$\nabla^2 G = G'' + \frac{2}{s}G'.$$

Write $G = F(s)/s$. Then

$$G' = \frac{F'}{s} - \frac{F}{s^2}, \qquad G'' = \frac{F''}{s} - \frac{2F'}{s^2} + \frac{2F}{s^3},$$

so in $G'' + \frac{2}{s}G'$ the terms in F' and F cancel, leaving F''/s, and

$$(\nabla^2 + k^2)G = \frac{F'' + k^2 F}{s}.$$

The homogeneous radial solutions are consequently given by $F'' + k^2 F = 0$, $F = Ae^{iks} + Be^{-iks}$, i.e., $G = (Ae^{iks} + Be^{-iks})/s$. The $e^{iks - i\omega\tau}$ phase moves outward, whereas $e^{-iks - i\omega\tau}$ moves inward. Thus the outgoing condition selects $B = 0$. (This is why the convention in the statement is needed: with the time factor $e^{+i\omega\tau}$ the roles of $e^{\pm iks}$ would be exchanged, and the displayed kernel would be incoming.)

For $G = Ae^{iks}/s$, $G' = Ae^{iks}(ik/s - 1/s^2)$, and the flux through a sphere of radius ε centered on the source is

$$4\pi\varepsilon^2 G'(\varepsilon) = 4\pi A e^{ik\varepsilon}(ik\varepsilon - 1) \longrightarrow -4\pi A.$$

Exercise 10.2.3 shows that the integral of $k^2 G$ over that small ball vanishes. A positive unit delta source therefore requires $-4\pi A = 1$, giving

$$\boxed{G(\mathbf{r}_1, \mathbf{r}_2) = -\frac{e^{ik|\mathbf{r}_1 - \mathbf{r}_2|}}{4\pi|\mathbf{r}_1 - \mathbf{r}_2|}.}$$

It solves the homogeneous Helmholtz equation away from the source, is smooth there, and has a locally integrable $1/s$ singularity. It also satisfies the outgoing (Sommerfeld) radiation condition $s(\partial_s - ik)G \to 0$ as $s \to \infty$: with $A = -1/(4\pi)$, $\partial_s G - ikG = Ae^{iks}(ik/s - 1/s^2) - ikAe^{iks}/s = -Ae^{iks}/s^2$, so

$$s(\partial_s - ik)G = \frac{e^{iks}}{4\pi s} \longrightarrow 0 \qquad (s \to \infty).$$

For the incoming kernel $-e^{-iks}/(4\pi s)$ the same combination is $(2iks + 1)e^{-iks}/(4\pi s)$, which does not tend to zero.

To verify the delta identity against an arbitrary smooth compactly supported test function η, apply Green's second identity,

$$\int_V \big(G\nabla^2\eta - \eta\nabla^2 G\big)\, dV = \oint_{\partial V} \big(G\,\partial_n\eta - \eta\,\partial_n G\big)\, dS,$$

to the region V between a sphere of radius ε around the source and a large sphere enclosing the support of η. The outer boundary contributes zero, while the outward normal of V at the inner sphere points in the direction $-\widehat{\mathbf{s}}$, so $\partial_n = -\partial_s$ there. Since $(\nabla^2 + k^2)G = 0$ in V, the integrand on the left equals $G(\nabla^2 + k^2)\eta - \eta(\nabla^2 + k^2)G = G(\nabla^2 + k^2)\eta$, and the identity reads

$$\int_{s>\varepsilon} G(\nabla^2 + k^2)\eta\, dV = \int_{s=\varepsilon} [-G\,\partial_s\eta + \eta\,\partial_s G]\, dS.$$

As $\varepsilon \to 0$, the left side tends to $\int G(\nabla^2 + k^2)\eta\, dV$, because G is locally integrable. The first surface term is $O(\varepsilon)$, because $G = O(1/\varepsilon)$ and the sphere has area $O(\varepsilon^2)$. In the second term, $\eta = \eta(\mathbf{r}_2) + O(\varepsilon)$ and the flux $\int_{s=\varepsilon} \partial_s G\, dS = 4\pi\varepsilon^2 G'(\varepsilon)$ tends to one, so its limit is $\eta(\mathbf{r}_2)$. Hence

$$\int G(\mathbf{r}_1, \mathbf{r}_2)\, (\nabla_1^2 + k^2)\eta(\mathbf{r}_1)\, dV_1 = \eta(\mathbf{r}_2)$$

for every such η. For the singular function G this is what the delta identity means: the operator is moved from G onto the test function, as in the integration by parts that defines δ'. This proves, with its sign and normalization,

$$(\nabla_1^2 + k^2)G(\mathbf{r}_1, \mathbf{r}_2) = \delta^{(3)}(\mathbf{r}_1 - \mathbf{r}_2).$$

Convolution with a compactly supported source therefore produces the outgoing solution of the inhomogeneous Helmholtz equation.

10.2.5 Find the Green's function for the 3-D Helmholtz equation, Exercise 10.2.4, when the wave is a standing wave. Find the customary equal average of outgoing and incoming kernels, and state the remaining freedom to add a regular standing homogeneous solution.

Solution.

With $s = |\mathbf{r}_1 - \mathbf{r}_2|$, the outgoing and incoming Green's functions are

$$G_{\text{out}} = -\frac{e^{iks}}{4\pi s}, \qquad G_{\text{in}} = -\frac{e^{-iks}}{4\pi s}.$$

For real k, they are complex conjugates. Each solves $(\nabla^2 + k^2)G = \delta^{(3)}$; its small-sphere flux tends to 1. For G_{in}, which is G_{out} with k replaced by $-k$, the calculation of Exercise 10.2.4 applies unchanged: G_{in} solves the homogeneous equation off the source, is $O(1/s)$, and has flux $(1 + ik\varepsilon)e^{-ik\varepsilon} \to 1$, which is all the test-function argument there uses. The equal average therefore still has a unit, rather than doubled, delta source:

$$\begin{aligned} G_{\text{st}} &= \frac{1}{2}(G_{\text{out}} + G_{\text{in}}) \\ &= -\frac{e^{iks} + e^{-iks}}{8\pi s} = \boxed{-\frac{\cos(ks)}{4\pi s}}. \end{aligned}$$

Multiplying by $e^{-i\omega\tau}$ and taking the real part gives $-\cos(ks)\cos(\omega\tau)/(4\pi s)$, a standing radial wave. Equivalently, the outgoing and incoming amplitudes are equal, so their opposite radial fluxes cancel. Directly, for a wave $u = \text{Re}(\psi e^{-i\omega\tau})$ the radial energy flux is proportional to $-\partial_\tau u\, \partial_s u$, whose time average is $\frac{\omega}{2}\,\text{Im}(\psi^* \partial_s \psi)$ (write $u = \frac{1}{2}(\psi e^{-i\omega\tau} + \psi^* e^{i\omega\tau})$; the terms with $e^{\mp 2i\omega\tau}$ average to zero). With G'_{out} from Exercise 10.2.4, $G^*_{\text{out}} G'_{\text{out}} = (ik/s - 1/s^2)/(16\pi^2 s)$, so $\text{Im}(G^*_{\text{out}} G'_{\text{out}}) = +k/(16\pi^2 s^2)$, an outward flux; for the conjugate G_{in} it is $-k/(16\pi^2 s^2)$. But $\text{Im}(G^*_{\text{st}} G'_{\text{st}}) = 0$ because G_{st} is real.

Its derivative and source flux are

$$G'_{\text{st}} = \frac{k\sin(ks)}{4\pi s} + \frac{\cos(ks)}{4\pi s^2}, \qquad 4\pi s^2 G'_{\text{st}} = ks\sin(ks) + \cos(ks) \longrightarrow 1.$$

Also $F(s) = sG_{\text{st}} = -\cos(ks)/(4\pi)$ satisfies $F'' + k^2 F = 0$. These checks establish the same differential operator and source as in Exercise 10.2.4.

A standing-wave condition alone does not uniquely specify the regular homogeneous part. The answer above uses the customary equal average of the outgoing and incoming Green's functions. For any real constant C, adding $C\sin(ks)/s$ gives another real radial standing-wave kernel with the same delta source; an additional phase convention or boundary condition is needed to exclude it. Indeed $\sin(ks)/s = k - k^3 s^2/6 + \cdots$ is smooth at the origin, its small-sphere flux $4\pi s^2\,[\sin(ks)/s]' = 4\pi\big[ks\cos(ks) - \sin(ks)\big]$ tends to zero, and it satisfies the homogeneous Helmholtz equation everywhere ($F = \sin(ks)$ satisfies $F'' + k^2 F = 0$). Thus it does not alter the singular normalization.

10.2.6 Verify that the formula given for the 3-D Green's function of the modified Helmholtz equation in Table 10.1 is correct when the boundary conditions of the problem are that G vanish at infinity. Use the positive decay root $k > 0$ in the table and also identify its $k \downarrow 0$ limit.

Solution.

The three-dimensional entry in Table 10.1, textbook p. 463, is

$$\boxed{G(\mathbf{r}_1, \mathbf{r}_2) = -\frac{e^{-ks}}{4\pi s}, \qquad s = |\mathbf{r}_1 - \mathbf{r}_2|.}$$

Here $k > 0$ is the decay constant. For a real nonzero parameter of either sign, replace k by $|k|$. The limit $k = 0$ gives the decaying three-dimensional Laplace kernel $-1/(4\pi s)$. For $s > 0$, put $F = sG = -e^{-ks}/(4\pi)$. The radial calculation of Exercise 10.2.4 gives

$$(\nabla^2 - k^2)G = \frac{F'' - k^2 F}{s} = 0.$$

The kernel decays at infinity, and its leading singularity is $-1/(4\pi s)$.

Check the entire source normalization on a sphere of arbitrary radius $R > 0$, not just in a limiting expansion. The derivative is

$$G'(s) = \frac{e^{-ks}}{4\pi}\left(\frac{k}{s} + \frac{1}{s^2}\right),$$

so its flux is

$$\Phi(R) = 4\pi R^2 G'(R) = (1 + kR)e^{-kR}.$$

The integral of G over the ball follows from $\frac{d}{ds}\left[e^{-ks}\left(\frac{s}{k} + \frac{1}{k^2}\right)\right] = -se^{-ks}$:

$$\begin{aligned} I(R) &= 4\pi \int_0^R G(s)s^2\, ds = -\int_0^R se^{-ks}\, ds \\ &= \left[e^{-ks}\left(\frac{s}{k} + \frac{1}{k^2}\right)\right]_0^R = \frac{(1 + kR)e^{-kR} - 1}{k^2}. \end{aligned}$$

Therefore

$$\int_{s<R} (\nabla^2 - k^2)G\, dV = \Phi(R) - k^2 I(R) = (1 + kR)e^{-kR} - \left[(1 + kR)e^{-kR} - 1\right] = 1.$$

In particular $\Phi(R) \to 1$ and $k^2 I(R) \to 0$ as $R \downarrow 0$. The test-function argument from Exercise 10.2.4 applies without change to the singularity, proving

$$(\nabla_1^2 - k^2)G = \delta^{(3)}(\mathbf{r}_1 - \mathbf{r}_2).$$

For uniqueness in the class of kernels with that singularity and decay, the difference h of two such solutions is regular everywhere, vanishes at infinity, and obeys $\nabla^2 h = k^2 h$. Apply this to its real part: a positive maximum would be attained at a finite point and would have $\nabla^2 h \le 0$, contradicting $k^2 h > 0$; a negative minimum gives the opposite contradiction. Hence the real part is zero, and the same argument applies to the imaginary part. This establishes uniqueness for $k > 0$ under these conditions.

10.2.7 An electrostatic potential (mks units) is

$$\varphi(\mathbf{r}) = Z\,\frac{e^{-ar}}{4\pi\epsilon_0 r}, \qquad a > 0.$$

Reconstruct the electrical charge distribution that will produce this potential. Note that $\varphi(\mathbf{r})$ vanishes exponentially for large r, showing that the net charge is zero.

$$ANS.\ \ \rho(\mathbf{r}) = Z\,\delta(\mathbf{r}) - \frac{Za^2e^{-ar}}{4\pi r}.$$

Solution.

The exponential-decay and zero-net-charge conclusion assumes $a > 0$. At $a = 0$ the potential is Coulombic and its net charge is Z, not zero. Use Poisson's equation in SI units,

$$\nabla^2\varphi = -\frac{\rho}{\epsilon_0}, \qquad \rho = -\epsilon_0\nabla^2\varphi.$$

First compute away from the origin. For $h(r) = e^{-ar}/r$,

$$h' = -e^{-ar}\left(\frac{a}{r} + \frac{1}{r^2}\right), \qquad r^2h' = -(ar+1)e^{-ar},$$

and therefore

$$\frac{d}{dr}(r^2h') = -ae^{-ar} + a(ar+1)e^{-ar} = a^2re^{-ar}.$$

Thus $\nabla^2h = r^{-2}(r^2h')' = a^2e^{-ar}/r$ for $r > 0$. This ordinary calculation omits the point singularity. Its coefficient is fixed by the flux through a small sphere:

$$\lim_{R\downarrow 0} 4\pi R^2h'(R) = -4\pi\lim_{R\downarrow 0}(aR+1)e^{-aR} = -4\pi.$$

The integral of a^2h over that ball, $4\pi a^2\int_0^R re^{-ar}\,dr \le 2\pi a^2R^2$, tends to zero. As in Exercise 10.2.6 (with k replaced by a), the full distributional identity is therefore

$$\nabla^2\frac{e^{-ar}}{r} = a^2\frac{e^{-ar}}{r} - 4\pi\delta^{(3)}(\mathbf{r}).$$

Insert $\varphi = Zh/(4\pi\epsilon_0)$ into Poisson's equation:

$$\rho = -\epsilon_0\nabla^2\varphi = -\frac{Z}{4\pi}\nabla^2h = -\frac{Z}{4\pi}\left[a^2\frac{e^{-ar}}{r} - 4\pi\delta^{(3)}(\mathbf{r})\right],$$

that is,

$$\boxed{\rho(\mathbf{r}) = Z\delta^{(3)}(\mathbf{r}) - \frac{Za^2e^{-ar}}{4\pi r}.}$$

The first term is a point charge Z at the origin; the second is a spherically symmetric cloud of opposite sign. Its $1/r$ density singularity is integrable because the radial volume element contains r^2dr.

Compute its total charge explicitly. Integration by parts gives

$$\int_0^\infty re^{-ar}\,dr = \left[-\frac{r}{a}e^{-ar}\right]_0^\infty + \frac{1}{a}\int_0^\infty e^{-ar}\,dr = \frac{1}{a^2}.$$

Consequently

$$Q_{\text{cloud}} = 4\pi \int_0^\infty r^2 \left(-\frac{Za^2 e^{-ar}}{4\pi r} \right) dr = -Za^2 \int_0^\infty r e^{-ar}\, dr = -Z, \qquad Q_{\text{total}} = Z - Z = 0.$$

As an independent check, the enclosed charge at a finite radius follows from the same integration by parts on $(0, R)$, $\int_0^R r e^{-ar}\, dr = [1 - (1 + aR)e^{-aR}]/a^2$:

$$Q(R) = Z - Za^2 \int_0^R r e^{-ar}\, dr = Z(1 + aR)e^{-aR}.$$

The electric field obtained by differentiating the given potential is

$$E_r = -\varphi'(r) = \frac{Ze^{-ar}}{4\pi\epsilon_0} \left(\frac{a}{r} + \frac{1}{r^2} \right),$$

and Gauss' law is satisfied exactly: $4\pi\epsilon_0 R^2 E_r(R) = Z(aR+1)e^{-aR} = Q(R)$. It tends to Z as $R \downarrow 0$ and to zero as $R \to \infty$, confirming both the point charge and the screening cloud.

11 Chapter 11. Complex Variable Theory

Problems

11.2 Cauchy-Riemann Conditions

11.2.1 Show whether or not the function $f(z) = \Re(z) = x$ is analytic.

Solution.

Here $u = x$ and $v = 0$, so $u_x = 1$ whereas $v_y = 0$. The necessary Cauchy–Riemann equation $u_x = v_y$ fails at every point. Equivalently, for real h, $[f(z+h) - f(z)]/h = 1$, whereas $[f(z+ih) - f(z)]/(ih) = 0$. These direction-dependent limits disagree. Thus f is nowhere complex differentiable and analytic on no open set.

11.2.2 Having shown that the real part $u(x, y)$ and the imaginary part $v(x, y)$ of an analytic function $w(z)$ each satisfy Laplace's equation, show that neither $u(x, y)$ nor $v(x, y)$ can have either a maximum or a minimum in the interior of any region in which $w(z)$ is analytic and nonconstant, with the region connected. Any stationary points of the components are saddles.

Solution.

Exclude the constant case: a constant harmonic function has both non-strict maxima and minima everywhere. Let w be nonconstant and z_0 an interior point. Not all Taylor coefficients a_m ($m \geq 1$) of w at z_0 vanish; otherwise w would be constant near z_0, hence constant on its connected domain by the identity theorem. With m the smallest index of a nonzero one,

$$w(z_0 + h) = w(z_0) + a_m h^m + O(h^{m+1}), \qquad a_m \neq 0, \quad m \geq 1.$$

Writing $a_m = |a_m| e^{i\alpha}$ and $h = re^{i\theta}$, we have $a_m h^m = |a_m| r^m e^{i(\alpha + m\theta)}$, whose real part gives

$$u(z_0 + h) - u(z_0) = |a_m| r^m \cos(\alpha + m\theta) + O(r^{m+1}).$$

Choose $\theta = -\alpha/m$ and $(\pi - \alpha)/m$, where the cosine is $+1$ and -1. The remainder is bounded by Kr^{m+1} for $0 < r < r_0$, with some constants $K, r_0 > 0$, so for $0 < r < \min(r_0, |a_m|/K)$ the leading term dominates it and the two differences have opposite signs. Thus nearby values of u lie both above and below $u(z_0)$, which is neither a maximum nor a minimum. For v the imaginary part gives $v(z_0 + h) - v(z_0) = |a_m| r^m \sin(\alpha + m\theta) + O(r^{m+1})$, and $\alpha + m\theta = \pi/2, 3\pi/2$ give the two signs. A stationary point of u or v is a zero of $w' = u_x - iu_y = v_y + iv_x$, i.e., a point with $m \geq 2$. There $\cos(\alpha + m\theta)$ and $\sin(\alpha + m\theta)$ change sign $2m$ times as θ runs around z_0: a saddle. At other points the gradients are nonzero.

11.2.3 Find the analytic function

$$w(z) = u(x, y) + iv(x, y)$$

(a) if $u(x, y) = x^3 - 3xy^2$, (b) if $v(x, y) = e^{-y} \sin x$.

Solution.

(a) Cauchy–Riemann gives $v_y = u_x = 3x^2 - 3y^2$. Integrating in y gives $v = 3x^2 y - y^3 + g(x)$. Then $v_x = 6xy + g'(x)$ must equal $-u_y = 6xy$, which forces $g' = 0$, i.e., $g = C$, a real constant. Since $(x + iy)^3 = x^3 - 3xy^2 + i(3x^2 y - y^3)$,

$$\boxed{w(z) = z^3 + iC, \qquad C \in \mathbb{R}}.$$

(b) Here $u_x = v_y = -e^{-y}\sin x$, so $u = e^{-y}\cos x + g(y)$. Then $u_y = -e^{-y}\cos x + g'(y)$ must equal $-v_x = -e^{-y}\cos x$, so $g' = 0$ and $g = C$. Therefore

$$\boxed{w(z) = e^{iz} + C, \qquad C \in \mathbb{R}},$$

using $e^{iz} = e^{-y}(\cos x + i\sin x)$. The constants preserve the specified real or imaginary component.

11.2.4 If there is a common connected region in which $w_1 = u(x,y) + iv(x,y)$ and $w_2 = w_1^* = u(x,y) - iv(x,y)$ are both analytic, prove that $u(x,y)$ and $v(x,y)$ are constants.

Solution.

For w_1, $u_x = v_y$ and $u_y = -v_x$. For $w_2 = u - iv$, $u_x = -v_y$ and $u_y = v_x$. Adding and subtracting $u_x = v_y$ and $u_x = -v_y$ gives $u_x = 0$ and $v_y = 0$; likewise $u_y = -v_x$ and $u_y = v_x$ give $u_y = 0$ and $v_x = 0$. Hence $u_x = u_y = v_x = v_y = 0$. Along every path in a connected component, $du = u_x dx + u_y dy = 0$ and $dv = v_x dx + v_y dy = 0$, so both functions are constant there. Connectedness is needed: on the union of two disjoint disks, $w_1 = 0$ on one and $w_1 = 1$ on the other makes w_1 and $w_1^* = w_1$ analytic, yet u is not constant. The two resulting complex constants are conjugates. The Instructor's Manual further writes $w_1 = w_2$, but equality requires $v = 0$. For example, $w_1 = 1 + i$ and $w_2 = 1 - i$ are both analytic and satisfy the hypotheses without being equal.

11.2.5 Starting from $f(z) = 1/(x+iy)$, show that $1/z$ is analytic in the entire finite z plane except at the point $z = 0$. This extends our discussion of the analyticity of z^n to negative integer powers n.

Solution.

Rationalizing the denominator gives $u = x/(x^2+y^2)$ and $v = -y/(x^2+y^2)$. Set $D = x^2+y^2 > 0$. Then

$$u_x = \frac{y^2 - x^2}{D^2} = v_y, \qquad u_y = -\frac{2xy}{D^2} = -v_x.$$

These partial derivatives are continuous off the origin, proving analyticity there. The derivative is

$$f' = u_x + iv_x = \frac{y^2 - x^2 + 2ixy}{D^2} = -\frac{(x-iy)^2}{D^2} = -\frac{(z^*)^2}{(zz^*)^2} = \boxed{-z^{-2}},$$

using $x - iy = z^*$ and $D = zz^*$. At zero there is a pole. Products of $1/z$ consequently give analytic negative integer powers on $\mathbb{C} \setminus \{0\}$.

11.2.6 Show that, assuming real differentiability of u, v at the point, the Cauchy-Riemann equations imply that the derivative $f'(z)$ has the same value for $dz = a\,dx + ib\,dy$ (with neither a nor b zero) as it has for $dz = dx$.

Solution.

Assume real differentiability of u, v at the point; continuous first partial derivatives nearby are sufficient. The increment $h = a\,dx + ib\,dy$ has $\Delta x = a\,dx$ and $\Delta y = b\,dy$, so the real differential of $f = u + iv$ is the first line below. Cauchy–Riemann, $u_y = -v_x$ and $v_y = u_x$, gives the second line, and $-v_x + iu_x = i(u_x + iv_x)$ gives the third:

$$\begin{aligned} df &= (u_x + iv_x)a\,dx + (u_y + iv_y)b\,dy \\ &= (u_x + iv_x)a\,dx + (-v_x + iu_x)b\,dy \\ &= (u_x + iv_x)(a\,dx + ib\,dy). \end{aligned}$$

The remainder in this real differential is $o(|h|)$. Dividing by $h \neq 0$ and taking the limit yields $\boxed{f' = u_x + iv_x}$ independently of direction. Cauchy–Riemann equalities at an isolated point, without real differentiability, are not alone sufficient for this conclusion. For a concrete reason for the hypothesis, set

$$f(x+iy) = \begin{cases} xy/\sqrt{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Here $v = 0$, and both partial derivatives of u at zero are zero because u vanishes on each axis, so Cauchy–Riemann holds there. Along the real axis the difference quotient is zero. Along $h = t(1+i)$ with $t > 0$, however,

$$\frac{f(h)}{h} = \frac{t/\sqrt{2}}{t(1+i)} = \frac{1-i}{2\sqrt{2}} \neq 0.$$

Thus real differentiability cannot be omitted.

11.2.7 Using $f(re^{i\theta}) = R(r,\theta)e^{i\Theta(r,\theta)}$, in which $R(r,\theta)$ and $\Theta(r,\theta)$ are differentiable real functions of r and θ, show that the Cauchy-Riemann conditions in polar coordinates become on a local patch with $r > 0$ and $f \neq 0$

(a) $\dfrac{\partial R}{\partial r} = \dfrac{R}{r}\dfrac{\partial \Theta}{\partial \theta}$, (b) $\dfrac{1}{r}\dfrac{\partial R}{\partial \theta} = -R\dfrac{\partial \Theta}{\partial r}$.

Hint. Set up the derivative first with δz radial and then with δz tangential.

Solution.

On a regular polar patch $r > 0$, use a differentiable phase for $f = Re^{i\Theta}$. The product rule gives $\partial_r f = (R_r + iR\Theta_r)e^{i\Theta}$ and $\partial_\theta f = (R_\theta + iR\Theta_\theta)e^{i\Theta}$. A radial displacement is $dz = e^{i\theta}dr$, giving

$$f'_r = e^{-i\theta}\partial_r f = e^{i(\Theta-\theta)}(R_r + iR\Theta_r).$$

A tangential displacement is $dz = ire^{i\theta}d\theta$, giving, with $1/i = -i$,

$$f'_\theta = \frac{e^{-i\theta}}{ir}\partial_\theta f = e^{i(\Theta-\theta)}\left(\frac{R\Theta_\theta}{r} - i\frac{R_\theta}{r}\right).$$

Analyticity requires $f'_r = f'_\theta$. Cancel the common phase and equate real and imaginary parts:

$$\boxed{R_r = \frac{R}{r}\Theta_\theta, \qquad \frac{1}{r}R_\theta = -R\Theta_r}.$$

A smooth argument is a local choice away from zeros of f; it is not generally defined at a zero or at $r = 0$.

11.2.8 As an extension of Exercise 11.2.7 show that on a nonzero analytic patch with $r > 0$, $\Theta(r,\theta)$ satisfies the 2-D Laplace equation in polar coordinates,

$$\frac{\partial^2\Theta}{\partial r^2} + \frac{1}{r}\frac{\partial\Theta}{\partial r} + \frac{1}{r^2}\frac{\partial^2\Theta}{\partial\theta^2} = 0.$$

Solution.

Use a local nonzero analytic branch, so $R > 0$ and a smooth single-valued phase Θ is available. Put $L = \ln R$, so $L_r = R_r/R$ and $L_\theta = R_\theta/R$. Dividing the equations of 11.2.7 by R, and multiplying the second by r, gives $L_r = \Theta_\theta/r$ and $L_\theta = -r\Theta_r$. Differentiating the first in θ and the second in r,

$$L_{r\theta} = \Theta_{\theta\theta}/r, \qquad L_{\theta r} = -\Theta_r - r\Theta_{rr}.$$

L and Θ are the real and imaginary parts of the analytic function $\ln f = L + i\Theta$, so they have continuous second derivatives and the mixed derivatives agree: $\Theta_{\theta\theta}/r = -\Theta_r - r\Theta_{rr}$. Moving all terms to one side and dividing by r gives

$$\boxed{\Theta_{rr} + \frac{1}{r}\Theta_r + \frac{1}{r^2}\Theta_{\theta\theta} = 0}.$$

11.2.9 For each of the following functions $f(z)$, find $f'(z)$ and identify the maximal region within which $f(z)$ is analytic.

(a) $f(z) = \dfrac{\sin z}{z}$, (d) $f(z) = e^{-1/z}$,

(b) $f(z) = \dfrac{1}{z^2+1}$, (e) $f(z) = z^2 - 3z + 2$,

(c) $f(z) = \dfrac{1}{z(z+1)}$, (f) $f(z) = \tan(z)$, (g) $f(z) = \tanh(z)$.

Solution.

Quotient and chain rules give the following derivatives. Domains listed are in the finite complex plane:

	$f'(z)$	domain
(a)	$(z\cos z - \sin z)/z^2$	$\mathbb{C}$ after setting $f(0) = 1$
(b)	$-2z/(z^2+1)^2$	$\mathbb{C} \setminus \{i, -i\}$
(c)	$-(2z+1)/[z^2(z+1)^2]$	$\mathbb{C} \setminus \{0, -1\}$
(d)	$e^{-1/z}/z^2$	$\mathbb{C} \setminus \{0\}$
(e)	$2z - 3$	$\mathbb{C}$
(f)	$1/\cos^2 z$	$\mathbb{C} \setminus \{\pi/2 + n\pi : n \in \mathbb{Z}\}$
(g)	$1/\cosh^2 z$	$\mathbb{C} \setminus \{i(\pi/2 + n\pi) : n \in \mathbb{Z}\}$

For (a), $\sin z/z = 1 - z^2/3! + z^4/5! - \cdots$ gives $f'(0) = 0$. The literal quotient is undefined at zero; the entire function is its removable extension. In (b),(c) the numerator 1 cannot cancel the denominator zeros; (d) has an essential singularity at zero. For (f), $\cos z = (e^{iz} + e^{-iz})/2 = 0$ is equivalent to $e^{2iz} = -1 = e^{i(2n+1)\pi}$, i.e., $2iz = (2n+1)\pi i$, giving the listed real sequence. For (g), $\cosh z = (e^z + e^{-z})/2 = 0$ gives $e^{2z} = -1$, $2z = (2n+1)\pi i$, the listed imaginary sequence.

11.2.10 For what complex values do each of the following functions $f(z)$ have a derivative after choosing single-valued branches (and specifying any extension at a branch point)?

(a) $f(z) = z^{3/2}$, (b) $f(z) = z^{-3/2}$, (c) $f(z) = \tan^{-1}(z)$, (d) $f(z) = \tanh^{-1}(z)$.

Solution.

These expressions require branches. Derivatives below are local analytic derivatives away from branch points; a chosen cut is a further boundary of that branch. For the powers choose $-\pi < \arg z < \pi$, i.e., $z^c = \exp(c \operatorname{Log} z)$, where $\operatorname{Log} z = \ln|z| + i \arg z$ with $-\pi < \arg z \le \pi$ is the principal logarithm, analytic off $(-\infty, 0]$. On the slit plane the chain rule and $(\operatorname{Log} z)' = 1/z$ give $(z^c)' = cz^c/z = cz^{c-1}$, so

$$(z^{3/2})' = \tfrac{3}{2} z^{1/2}, \qquad (z^{-3/2})' = -\tfrac{3}{2} z^{-5/2}.$$

Other cuts allow local analytic branches through every nonzero point. The origin is a branch point of both powers, and the second diverges there. For (a), the pointwise derivative at zero does exist if a single-valued selection is defined with $f(0) = 0$: its value is zero. The Instructor's Manual's exclusion of that derivative is incorrect; this does not give analyticity on a neighborhood of the branch point. Indeed $|f(h)/h| = |h|^{1/2} \to 0$, independently of the square-root sign.

Solving $z = \tan w = (e^{2iw} - 1)/[i(e^{2iw} + 1)]$ for e^{2iw} gives $e^{2iw} = (1 + iz)/(1 - iz)$; similarly $z = \tanh w = (e^{2w} - 1)/(e^{2w} + 1)$ gives $e^{2w} = (1 + z)/(1 - z)$. Near zero we therefore choose the inverse-function branches

$$\arctan z = \frac{\operatorname{Log}(1 + iz) - \operatorname{Log}(1 - iz)}{2i}, \qquad \operatorname{arctanh} z = \frac{\operatorname{Log}(1 + z) - \operatorname{Log}(1 - z)}{2}.$$

Differentiating the logarithms, $[\operatorname{Log}(1 \pm iz)]' = \pm i/(1 \pm iz)$ and $[\operatorname{Log}(1 \pm z)]' = \pm 1/(1 \pm z)$, gives

$$(\arctan z)' = \frac{1}{2i}\left(\frac{i}{1 + iz} + \frac{i}{1 - iz}\right) = \frac{1}{2}\,\frac{(1 - iz) + (1 + iz)}{(1 + iz)(1 - iz)} = \boxed{\frac{1}{1 + z^2}},$$

$$(\operatorname{arctanh} z)' = \frac{1}{2}\left(\frac{1}{1 + z} + \frac{1}{1 - z}\right) = \frac{1}{2}\,\frac{(1 - z) + (1 + z)}{(1 + z)(1 - z)} = \boxed{\frac{1}{1 - z^2}}.$$

Local branches have derivatives away from $\pm i$ and ± 1, respectively. The standard cuts are where an argument $1 \pm iz$ or $1 \pm z$ of a logarithm is real and ≤ 0: they run outward along the imaginary axis from $\pm i$ for (c), and along the real axis from ± 1 for (d).

11.2.11 Two-dimensional irrotational fluid flow is conveniently described by a complex potential $f(z) = u(x, y) + iv(x, y)$. We label the real part, $u(x, y)$, the *velocity potential*, and the imaginary part, $v(x, y)$, the *stream function*. The fluid velocity $\mathbf{V}$ is given by $\mathbf{V} = \nabla u$. If $f(z)$ is analytic:

(a) Show that $df/dz = V_x - iV_y$.

(b) Show that $\nabla \cdot \mathbf{V} = 0$ (no sources or sinks).

(c) Show that $\nabla \times \mathbf{V} = 0$ (irrotational, nonturbulent flow).

Solution.

Write $\mathbf{V} = (u_x, u_y, 0)$. **(a)** A real increment gives $f' = u_x + iv_x$. Since $v_x = -u_y$, $\boxed{f' = V_x - iV_y}$. **(b)** Differentiating $u_x = v_y$ in x and $u_y = -v_x$ in y gives $u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$; the mixed partials are equal because the components of an analytic function have continuous partial derivatives of all orders. Thus $\nabla \cdot \mathbf{V} = 0$. **(c)** The only possibly nonzero curl component is $\partial_x u_y - \partial_y u_x = 0$ by equality of mixed derivatives. The field is incompressible and irrotational on its analytic region. The first occurrence of $u(x, v)$ in the printed problem is a typographical error; the potential depends on the coordinates x, y.

11.2.12 The function $f(z)$ is analytic. Show that the ordinary complex derivative with respect to the conjugate variable exists at z_0 exactly when $f'(z_0) = 0$. Deduce that it exists throughout a connected region exactly when f is constant. Distinguish this from the Wirtinger partial derivative.

Hint. For the Wirtinger interpretation, use the chain rule and take $x = (z + z^*)/2$, $y = (z - z^*)/2i$.

Note. This result emphasizes that our analytic function $f(z)$ is not just a complex function of two real variables x and y. It is a function of the complex variable $x + iy$.

Solution.

Distinguish the ordinary complex derivative in the conjugate variable from the Wirtinger partial derivative. The latter exists and is zero for every analytic f. The former can exist at isolated critical points, but exists throughout a connected region only for a constant function. For the ordinary interpretation, divide $f(z_0+h)-f(z_0)=f'(z_0)h+o(|h|)$ by h^*, which has $|h^*|=|h|$:

$$\frac{f(z_0+h)-f(z_0)}{h^*}=f'(z_0)\frac{h}{h^*}+o(1).$$

For real increments $h/h^*=1$, and for purely imaginary ones $h/h^*=-1$; the quotient then tends to $f'(z_0)$ and $-f'(z_0)$, respectively. The limit therefore exists only if $f'(z_0)=0$. This condition is also sufficient because the remaining term tends to zero. For example $f(z)=z^2$ has this derivative zero at $z_0=0$. If the condition holds throughout a connected region, $f'=0$ there and f is constant.

The chain rule in the hint instead defines the Wirtinger derivative. Treating z and z^* as independent variables, $x=(z+z^*)/2$ and $y=(z-z^*)/2i$ give $\partial x/\partial z^*=\frac{1}{2}$ and $\partial y/\partial z^*=-1/(2i)=i/2$, so

$$\partial_{z^*}f=\tfrac{1}{2}(\partial_x+i\partial_y)(u+iv)=\tfrac{1}{2}[(u_x-v_y)+i(v_x+u_y)]=\boxed{0}.$$

This is exactly the Cauchy–Riemann condition, not a condition that f be constant.

11.3 Cauchy's Integral Theorem

11.3.1 Show that $\displaystyle\int_{z_1}^{z_2}f(z)\,dz=-\int_{z_2}^{z_1}f(z)\,dz$, with the same path traversed in reverse.

Solution.

Parametrize the specified path C by $\gamma(t)$, $0\le t\le 1$, from z_1 to z_2. Its reverse $\widetilde{C}$ is $\widetilde{\gamma}(t)=\gamma(1-t)$, with $\widetilde{\gamma}'(t)=-\gamma'(1-t)$ by the chain rule:

$$\int_{\widetilde{C}}f\,dz=-\int_0^1 f(\gamma(1-t))\gamma'(1-t)\,dt=-\int_0^1 f(\gamma(s))\gamma'(s)\,ds.$$

In the second step $s=1-t$, $ds=-dt$, maps $t=0,1$ to $s=1,0$; the sign of ds and the reversal of the limits cancel. The last member equals $-\int_C f\,dz$, proving the identity. The reverse integral follows the same geometric path in the opposite direction; two different paths need not give this relation.

11.3.2 Prove that

$$\left|\int_C f(z)\,dz\right|\le |f|_{\max}\cdot L,$$

where f is continuous on the piecewise smooth finite-length contour C, $|f|_{\max}$ is the maximum value of $|f(z)|$ along the contour C and L is the length of the contour.

Solution.

For a piecewise smooth parametrization $\gamma:[a,b]\to C$, the modulus of an integral is at most the integral of the modulus:

$$\left|\int_C f\,dz\right|=\left|\int_a^b f(\gamma(t))\gamma'(t)\,dt\right|\le\int_a^b|f(\gamma(t))|\,|\gamma'(t)|\,dt.$$

To prove this inequality for the complex integrand $g(t) = f(\gamma(t))\gamma'(t)$, write $\int_a^b g\,dt = \left|\int_a^b g\,dt\right| e^{i\varphi}$. Then

$$\left|\int_a^b g\,dt\right| = \int_a^b e^{-i\varphi} g\,dt = \int_a^b \Re\big(e^{-i\varphi} g\big)\,dt \le \int_a^b |g|\,dt,$$

since the left side is real (so the imaginary part of the middle integral vanishes) and $\Re w \le |w|$. Bound $|f(\gamma(t))|$ by $M = |f|_{\max}$ and use $L = \int_a^b |\gamma'(t)|dt$: $\boxed{\left|\int_C f\,dz\right| \le ML}$. Continuity on the compact contour guarantees a maximum; a finite supremum suffices for the same argument.

11.3.3 Show that the integral

$$\int_{3+4i}^{4-3i} (4z^2 - 3iz)\,dz$$

has the same value on the two paths: (a) the straight line connecting the integration limits, and (b) an arc on the circle $|z| = 5$.

Solution.

The polynomial has the entire primitive $P(z) = 4z^3/3 - 3iz^2/2$, $P'(z) = 4z^2 - 3iz$. Along a smooth path γ the chain rule gives $dP(\gamma(t))/dt = (4\gamma(t)^2 - 3i\gamma(t))\gamma'(t)$, so by the fundamental theorem of calculus (applied to real and imaginary parts) the integral equals P at the final point minus P at the initial point. Let $a = 3 + 4i$, $b = 4 - 3i$. Since $|a| = |b| = 5$, both endpoints lie on the circle $|z| = 5$, at the angles $\arctan\frac{4}{3}$ and $-\arctan\frac{3}{4}$. Thus the straight path $\gamma(t) = a + t(b - a)$, $0 \le t \le 1$, and either circular arc $\gamma(\theta) = 5e^{i\theta}$ between these angles both give $P(b) - P(a)$. The endpoint powers are

$$a^2 = -7 + 24i, \quad b^2 = 7 - 24i, \quad a^3 = -117 + 44i, \quad b^3 = -44 - 117i,$$

so $b^3 - a^3 = 73 - 161i$ and $b^2 - a^2 = 14 - 48i$. With $\frac{3i}{2}(14 - 48i) = 72 + 21i$,

$$\begin{aligned} P(b) - P(a) &= \tfrac{4}{3}(b^3 - a^3) - \tfrac{3i}{2}(b^2 - a^2) = \tfrac{4}{3}(73 - 161i) - \tfrac{3i}{2}(14 - 48i) \\ &= \frac{292 - 644i}{3} - (72 + 21i) = \frac{292 - 216}{3} - \frac{644 + 63}{3}\,i = \boxed{\frac{76 - 707i}{3}}. \end{aligned}$$

11.3.4 Let $F(z) = \displaystyle\int_{\pi(1+i)}^{z} \cos 2\zeta\,d\zeta$. Show that $F(z)$ is independent of the path connecting the limits of integration, and evaluate $F(\pi i)$.

Solution.

The integrand has the entire primitive $\frac{1}{2}\sin 2\zeta$. Thus every path gives

$$F(z) = \tfrac{1}{2}[\sin 2z - \sin(2\pi + 2\pi i)].$$

At $z = \pi i$ the bracket is $\sin 2\pi i - \sin(2\pi i + 2\pi)$, and periodicity $\sin(w + 2\pi) = \sin w$ makes the two terms cancel. Therefore $\boxed{F(\pi i) = 0}$, and the primitive proves path independence throughout the plane.

11.3.5 Evaluate $\oint_C (x^2 - iy^2)\,dz$, where the integration is (a) clockwise around the unit circle, (b) on a square with vertices at $\pm 1 \pm i$. Explain why the results of parts (a) and (b) are or are not identical.

Solution.

Expand in real differentials:

$$(x^2 - iy^2)(dx + i\,dy) = (x^2 dx + y^2 dy) + i(x^2 dy - y^2 dx).$$

The real part is $d[(x^3 + y^3)/3]$, whose closed integral is zero. **(a)** For clockwise traversal set $x = \cos t, y = -\sin t$, $0 \le t \le 2\pi$, so $dx = -\sin t\,dt$ and $dy = -\cos t\,dt$. The imaginary part becomes $x^2 dy - y^2 dx = (-\cos^3 t + \sin^3 t)dt$. Each odd power changes sign under $t \to t + \pi$ and so has zero integral over a full period; the answer is 0. **(b)** Traverse the square counterclockwise from $-1 - i$. On a horizontal edge $dz = dx$; on a vertical edge $dz = i\,dy$. The four contributions are

$$\begin{aligned} I_{\text{bottom}} &= \int_{-1}^{1} (x^2 - i)\,dx = \frac{2}{3} - 2i, \\ I_{\text{right}} &= \int_{-1}^{1} (1 - iy^2) i\,dy = \frac{2}{3} + 2i, \\ I_{\text{top}} &= \int_{1}^{-1} (x^2 - i)\,dx = -\frac{2}{3} + 2i, \\ I_{\text{left}} &= \int_{1}^{-1} (1 - iy^2) i\,dy = -\frac{2}{3} - 2i. \end{aligned}$$

Opposite edges cancel pairwise, giving $\boxed{0}$. Reversing the orientation reverses all four signs and still gives zero.

The results are identical, but not by Cauchy's theorem: $u = x^2$, $v = -y^2$ give $u_x = 2x$, $v_y = -2y$, so Cauchy–Riemann holds only on the line $y = -x$ and fails on every open set. The reason is symmetry. The real part is exact, and by Green's theorem the imaginary part is, for counterclockwise traversal (clockwise reverses the sign),

$$\oint (x^2 dy - y^2 dx) = \iint \left[\frac{\partial(x^2)}{\partial x} - \frac{\partial(-y^2)}{\partial y} \right] dx\,dy = \iint (2x + 2y)\,dx\,dy = 2A(\bar{x} + \bar{y}),$$

where A is the enclosed area and $(\bar{x}, \bar{y})$ its centroid. Both the disk and the square are centered at the origin, so both integrals vanish; a region whose centroid has $\bar{x} + \bar{y} \neq 0$ would give a nonzero value.

11.3.6 Verify that

$$\int_0^{1+i} z^*\,dz$$

depends on the path by evaluating the integral for the two paths shown in Fig. 11.7. Recall that $f(z) = z^*$ is not an analytic function of z and that Cauchy's integral theorem therefore does not apply.

Solution.

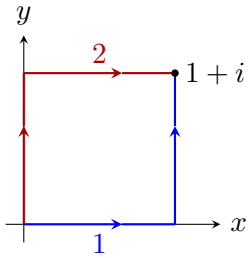
In Fig. 11.7, path 1 is $0 \to 1 \to 1 + i$ and path 2 is $0 \to i \to 1 + i$. Along path 1, $z = x$, $z^* = x$, $dz = dx$ on the first segment and $z = 1 + iy$, $z^* = 1 - iy$, $dz = i\,dy$ on the second:

$$I_1 = \int_0^1 x\,dx + \int_0^1 (1 - iy) i\,dy = \frac{1}{2} + i + \frac{1}{2} = \boxed{1 + i}.$$

Along the first segment of path 2, $z = iy$, $z^* = -iy$, $dz = i\,dy$; on the second, $z = x + i$, $z^* = x - i$, $dz = dx$:

$$I_2 = \int_0^1 (-iy)i\,dy + \int_0^1 (x - i)\,dx = \frac{1}{2} + \frac{1}{2} - i = \boxed{1 - i}.$$

The Instructor's Manual's second-path calculation omits the factor i in $dz = i\,dy$ and prints an incorrect second value. The difference $2i$ establishes path dependence.



11.3.7 Show that

$$\oint_C \frac{dz}{z^2 + z} = 0,$$

in which the contour C is a counterclockwise circle defined by $|z| = R > 1$.

Hint. Direct use of the Cauchy integral theorem is illegal. The integral may be evaluated by expanding into partial fractions and then treating the two terms individually. This yields 0 for $R > 1$ and $2\pi i$ for $0 < R < 1$.

Solution.

Partial fractions give $1/[z(z+1)] = 1/z - 1/(z+1)$, since $(z+1) - z = 1$. For a counterclockwise circle, $\oint dz/(z - a) = 2\pi i$ if a lies inside: the circle may be shrunk to $|z - a| = \rho$ without crossing the singularity, and $z - a = \rho e^{i\theta}$ gives $\int_0^{2\pi} i\,d\theta$. If a lies outside, the integrand is analytic on and inside the circle and Cauchy's theorem gives zero. For $R > 1$, both poles lie inside. The two terms therefore contribute $2\pi i$ and $-2\pi i$, giving $\boxed{0}$. For $0 < R < 1$ only the origin lies inside; the term $1/(z + 1)$ gives zero, so the integral is $2\pi i$. At $R = 1$ the path hits -1, so the ordinary integral is undefined. The orientation matters for the hint's value $2\pi i$ at $0 < R < 1$, which a clockwise circle changes to $-2\pi i$; the bound $R > 0$ excludes the degenerate circle $|z| = 0$. The zero-integral theorem cannot be applied directly over the singularities, which is why the decomposition is useful.

11.4 Cauchy's Integral Formula

Unless explicitly stated otherwise, closed contours occurring in these exercises are to be understood as traversed in the mathematically positive (counterclockwise) direction.

11.4.1 Show that

$$\frac{1}{2\pi i} \oint z^{m-n-1}\,dz, \quad m \text{ and } n \text{ integers}$$

(with the contour encircling the origin once), is a representation of the Kronecker δ_{mn}.

Solution.

On the circle $z = Re^{i\theta}$, $0 \le \theta \le 2\pi$, we have $dz = iRe^{i\theta}d\theta$, so $z^{m-n-1}dz = iR^{m-n}e^{i(m-n)\theta}d\theta$ and

$$\frac{1}{2\pi i} \oint z^{m-n-1}\,dz = \frac{R^{m-n}}{2\pi} \int_0^{2\pi} e^{i(m-n)\theta}\,d\theta.$$

For $m = n$ this is 1. For the nonzero integer $k = m - n$, the integral is $[e^{ik\theta}/(ik)]_0^{2\pi} = 0$. Thus the result is $\boxed{\delta_{mn}}$. Contour deformation gives the same value for any contour avoiding zero with positive winding number one.

11.4.2 Evaluate

$$\oint_C \frac{dz}{z^2 - 1},$$

where C is the circle $|z - 1| = 1$.

Solution.

The poles at $1, -1$ have distances $0, 2$ from the center 1. Only 1 is inside the specified unit-radius circle. Using

$$\frac{1}{z^2 - 1} = \frac{1}{2}\left(\frac{1}{z - 1} - \frac{1}{z + 1}\right),$$

the first term contributes $\frac{1}{2} \cdot 2\pi i = \pi i$, since the circle encloses 1 (as in 11.3.7), and the second, analytic on and inside the circle, integrates to zero by Cauchy's theorem. Therefore $\boxed{\oint_C dz/(z^2 - 1) = \pi i}$.

The Instructor's Manual's answer 0 is incorrect for this contour: the textbook (p. 491) prints the circle $|z - 1| = 1$, which encloses only the pole at $z = 1$.

11.4.3 Assuming that $f(z)$ is analytic on and within a closed contour C and that the point z_0 is within C, show that

$$\oint_C \frac{f'(z)}{z - z_0}\, dz = \oint_C \frac{f(z)}{(z - z_0)^2}\, dz.$$

Solution.

Along the contour, the single-valued function $f(z)/(z - z_0)$ obeys

$$d\left(\frac{f(z)}{z - z_0}\right) = \left[\frac{f'(z)}{z - z_0} - \frac{f(z)}{(z - z_0)^2}\right] dz.$$

Its closed integral is zero: the integral of the derivative of the single-valued function $f(z)/(z - z_0)$, analytic on a neighborhood of C, is the change of that function around the closed path. Moving one term to the opposite side proves the stated equality. Both integrals also equal $2\pi i f'(z_0)$ by Cauchy's formulas.

11.4.4 You know that $f(z)$ is analytic on and within a closed contour C, with z_0 inside C and n a nonnegative integer. You suspect that the nth derivative $f^{(n)}(z_0)$ is given by

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz.$$

Using mathematical induction (Section 1.4), prove that this expression is correct.

Solution.

The base case $n = 0$ is Cauchy's integral formula. Assume the formula holds for n at every point inside C. The point z_0 has a positive distance δ from C, so the disk $|\zeta - z_0| < \delta$, which contains z_0

and does not meet C, lies inside C. Let $0 < |\Delta| < \delta/2$. Then $z_0 + \Delta$ is inside C; subtracting the formula at z_0 from the formula at $z_0 + \Delta$ and dividing by Δ gives

$$\frac{f^{(n)}(z_0+\Delta) - f^{(n)}(z_0)}{\Delta} = \frac{n!}{2\pi i}\oint_C f(z)\,\frac{q(\Delta)-q(0)}{\Delta}\,dz, \qquad q(s) = (z - z_0 - s)^{-n-1}.$$

Fix z on C. By the chain rule,

$$q'(s) = -(n+1)(z-z_0-s)^{-n-2}(-1) = (n+1)(z-z_0-s)^{-n-2},$$
$$q''(s) = (n+1)(n+2)(z-z_0-s)^{-n-3}.$$

On the segment from 0 to Δ, $|z-z_0-s| \geq \delta - |s| > \delta/2$, so $|q''(s)| \leq K$ with $K = (n+1)(n+2)(2/\delta)^{n+3}$. Integrating q'' from 0 to s along the segment gives $|q'(s) - q'(0)| \leq K|s| \leq K|\Delta|$ (Exercise 11.3.2), and integrating q' gives

$$\left|\frac{q(\Delta)-q(0)}{\Delta} - q'(0)\right| = \left|\frac{1}{\Delta}\int_0^{\Delta}[q'(s)-q'(0)]\,ds\right| \leq \frac{K|\Delta|\cdot|\Delta|}{|\Delta|} = K|\Delta|,$$

a bound independent of z. Hence, by Exercise 11.3.2 with L the length of C,

$$\left|\frac{f^{(n)}(z_0+\Delta)-f^{(n)}(z_0)}{\Delta} - \frac{n!}{2\pi i}\oint_C f(z)\,q'(0)\,dz\right| \leq \frac{n!}{2\pi}\Big(\max_C |f|\Big)\,L\,K|\Delta| \to 0.$$

The difference quotient therefore converges as $\Delta \to 0$, and with $q'(0) = (n+1)(z-z_0)^{-n-2}$ its limit is

$$f^{(n+1)}(z_0) = \frac{n!}{2\pi i}\oint_C \frac{(n+1)f(z)}{(z-z_0)^{n+2}}\,dz = \frac{(n+1)!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+2}}\,dz.$$

This is the induction step, proving the result for every $n \geq 0$.

11.4.5

(a) A function $f(z)$ is analytic within a closed contour C (and continuous on C). If $f(z) \neq 0$ within C and $|f(z)| \geq m > 0$ on C, show that

$$|f(z)| \geq m$$

for all points within C.

Hint. Consider $w(z) = 1/f(z)$.

(b) If $f(z) = 0$ within the contour C, show that the foregoing result does not hold and that it is possible to have $|f(z)| = 0$ at one or more points in the interior with $|f(z)| > 0$ over the entire bounding contour. Cite a specific example of an analytic function that behaves this way.

Solution.

As printed on p. 491, (a) reads: if $f(z) \neq 0$ within C and $|f(z)| \leq M$ on C, then $|f(z)| \leq M$ for all points within C. This maximum-modulus statement does not require $f \neq 0$, so (b)'s claim that it fails when f has zeros is false. The hint $w = 1/f$ points to the intended minimum-modulus statement given above. We prove both. For the **original upper-bound version**, continuity gives a maximum of $|f|$ on the closed bounded region (the inside of C together with C). Suppose it exceeded M. Then it would occur at an interior point z_0, with $f(z_0) \neq 0$ because $|f(z_0)| > M \geq 0$. A constant f

is impossible, since its modulus would equal its boundary value, at most M. For nonconstant f some Taylor coefficient at z_0 with $k \geq 1$ is nonzero (identity theorem); with the smallest such k, $f(z_0+h) = f(z_0)+a_k h^k+O(h^{k+1})$, $a_k \neq 0$. Write $f(z_0) = |f(z_0)|e^{i\beta}$, $a_k = |a_k|e^{i\alpha}$, and take $h = re^{i\theta}$ with $\theta = (\beta-\alpha)/k$. Then $a_k h^k = |a_k|r^k e^{i\beta}$ has the phase of $f(z_0)$, so $|f(z_0)+a_k h^k| = |f(z_0)|+|a_k|r^k$. The remainder is bounded by Kr^{k+1} for $0 < r < r_0$, with some constants $K, r_0 > 0$, so

$$|f(z_0+h)| \geq |f(z_0)| + |a_k|r^k - Kr^{k+1} > |f(z_0)| \qquad \text{for } 0 < r < \min(r_0, |a_k|/K),$$

contradicting maximality. Hence $\boxed{|f| \leq M}$ throughout, whether or not zeros occur.

For the **corrected lower bound in (a)**, assume f has no zero on the closed region and $|f| \geq m > 0$ on C. Then $1/f$ is analytic and obeys $|1/f| \leq 1/m$ on the boundary. The upper-bound result gives $|1/f| \leq 1/m$ inside, hence $|f| \geq m$. For **(b)**, $f(z) = z$ on the unit disk has boundary modulus 1 and $f(0) = 0$. This disproves a positive interior *lower* bound without nonvanishing, while still satisfying the printed upper bound.

11.4.6 Evaluate

$$\oint_C \frac{e^{iz}}{z^3}\,dz,$$

for the contour a square with sides of length $a > 1$, centered at $z = 0$.

Solution.

Let $g = e^{iz}$, so $g'' = i^2 e^{iz} = -e^{iz}$. The square surrounds zero once, so Cauchy's formula for the second derivative gives

$$\oint_C \frac{e^{iz}}{z^3}\,dz = \frac{2\pi i}{2!}g''(0) = \pi i[-e^{iz}]_{z=0} = \boxed{-\pi i}.$$

Any positive side length suffices; $a > 1$ is stronger than needed.

11.4.7 Evaluate

$$\oint_C \frac{\sin^2 z - z^2}{(z-a)^3}\,dz,$$

where the contour encircles the point $z = a$.

Solution.

Set $g(z) = \sin^2 z - z^2$. Then, with $2\sin z\cos z = \sin 2z$, $g' = \sin 2z - 2z$ and $g'' = 2\cos 2z - 2$. Cauchy's derivative formula gives

$$\oint_C \frac{g(z)}{(z-a)^3}\,dz = \frac{2\pi i}{2!}g''(a) = \boxed{2\pi i(\cos 2a - 1)}.$$

The formula still applies when a numerator zero lowers or removes the apparent pole.

11.4.8 Evaluate

$$\oint_C \frac{dz}{z(2z+1)},$$

for the contour the unit circle.

Solution.

Decompose

$$\frac{1}{z(2z+1)} = \frac{1}{z} - \frac{2}{2z+1} = \frac{1}{z} - \frac{1}{z+1/2}.$$

Both poles are inside the unit circle. Their contributions, $2\pi i$ and $-2\pi i$, cancel, giving $\boxed{0}$.

11.4.9 Evaluate

$$\oint_C \frac{f(z)}{z(2z+1)^2}\,dz,$$

for the contour the unit circle, assuming f is analytic on a neighborhood of the closed disk.

Hint. Make a partial fraction expansion.

Solution.

Assume f is analytic on and within the unit circle; additional singularities would supply additional contributions. Write $1/[z(2z+1)^2] = A/z + B/(2z+1) + D/(2z+1)^2$. Clearing denominators yields $1 = A(2z+1)^2 + Bz(2z+1) + Dz = (4A+2B)z^2 + (4A+B+D)z + A$. Constant, quadratic, and linear coefficients give, respectively, $A = 1$, $4A+2B = 0$, $4A+B+D = 0$; hence $B = -2$ and $D = -4 - B = -2$. Since $2/(2z+1) = 1/(z+1/2)$ and $2/(2z+1)^2 = \frac{1}{2}/(z+1/2)^2$,

$$\frac{f(z)}{z(2z+1)^2} = \frac{f(z)}{z} - \frac{f(z)}{z+1/2} - \frac{f(z)/2}{(z+1/2)^2}.$$

Both points 0 and $-1/2$ lie inside the unit circle. Cauchy's formula $\oint f(z)\,dz/(z-z_0) = 2\pi i f(z_0)$ and its derivative form $\oint f(z)\,dz/(z-z_0)^2 = 2\pi i f'(z_0)$ now give

$$2\pi i f(0) - 2\pi i f(-1/2) - \tfrac{1}{2}\cdot 2\pi i f'(-1/2) = \boxed{2\pi i f(0) - 2\pi i f(-1/2) - \pi i f'(-1/2)}.$$

11.5 Laurent Expansion

11.5.1 Develop the Taylor expansion of $\ln(1+z)$.

ANS. $\displaystyle\sum_{n=1}^{\infty}(-1)^{n-1}\frac{z^n}{n}$.

Solution.

Choose the logarithm analytic near $z = 0$ with value zero there. For $|z| < 1$, the geometric series gives $1/(1+z) = \sum_{n=0}^{\infty}(-1)^n z^n$. It converges uniformly on every smaller closed disk, allowing termwise integration from zero to z along the segment. Since $[\ln(1+z)]' = 1/(1+z)$ and $\int_0^z \zeta^n\,d\zeta = z^{n+1}/(n+1)$, and with the index shift $n+1 \to n$ in the second form,

$$\boxed{\ln(1+z) = \sum_{n=0}^{\infty}\frac{(-1)^n z^{n+1}}{n+1} = \sum_{n=1}^{\infty}(-1)^{n-1}\frac{z^n}{n}}.$$

Both sides vanish at zero, fixing the integration constant. The nearest singularity is $z = -1$, so the Taylor radius is one.

11.5.2 Derive the binomial expansion

$$(1+z)^m = 1 + mz + \frac{m(m-1)}{1\cdot 2}z^2 + \cdots = \sum_{n=0}^{\infty}\binom{m}{n}z^n$$

for m, any real number. The expansion is convergent for $|z| < 1$. Why?

Solution.

Choose $(1+z)^m = \exp[m \operatorname{Log}(1+z)]$ with value 1 at zero. For any real exponent s the chain rule gives $\frac{d}{dz}(1+z)^s = \exp[s \operatorname{Log}(1+z)] \frac{s}{1+z} = s(1+z)^{s-1}$, so repeated differentiation gives

$$\frac{d^n}{dz^n}(1+z)^m = m(m-1)\cdots(m-n+1)(1+z)^{m-n}.$$

Set $z = 0$ and divide by $n!$ to obtain

$$\boxed{(1+z)^m = \sum_{n=0}^{\infty} \frac{m(m-1)\cdots(m-n+1)}{n!} z^n}.$$

The empty product for $n = 0$ is one. For a nonterminating series the coefficients c_n have $c_{n+1}/c_n = (m-n)/(n+1)$, so the absolute successive-term ratio is $|z|\,|m-n|/(n+1) \to |z|$; by the ratio test the series converges for $|z| < 1$ and diverges for $|z| > 1$. The reason is the singularity at $z = -1$, where $1 + z = 0$: this branch is analytic on the open disk $|z| < 1$, which reaches up to -1, so by Taylor's theorem the series converges to $(1+z)^m$ there, and no larger disk about 0 avoids -1. For nonnegative integer m the series terminates and is valid everywhere; for other real m the Taylor radius is one.

11.5.3 A function $f(z)$ is analytic on and within the unit circle. Also, $|f(z)| < 1$ for $|z| \le 1$ and $f(0) = 0$. Show that $|f(z)| < |z|$ for $0 < |z| \le 1$.

Hint. One approach is to show that $f(z)/z$ is analytic and then to express $[f(z_0)/z_0]^n$ by the Cauchy integral formula. Finally, consider absolute magnitudes and take the nth root. This exercise is sometimes called Schwarz's theorem.

Solution.

The strict inequality excludes $z = 0$, where both sides are zero. It holds for $0 < |z| \le 1$. The series $f(z) = a_1 z + a_2 z^2 + \cdots$ shows that $g(z) = f(z)/z$ extends analytically to zero by $g(0) = a_1$. Compactness of the unit circle and the strict hypothesis give $|g(\zeta)| = |f(\zeta)| \le M < 1$ there. For $|z_0| < 1$, Cauchy's formula applied to the analytic function g^n gives the first relation below. On the circle $|g(\zeta)^n| \le M^n$ and $|\zeta - z_0| \ge 1 - |z_0|$, so the bound of Exercise 11.3.2, with length 2π, gives the second:

$$g(z_0)^n = \frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{g(\zeta)^n}{\zeta - z_0}\, d\zeta, \qquad |g(z_0)|^n \le \frac{1}{2\pi} \cdot 2\pi \cdot \frac{M^n}{1 - |z_0|} = \frac{M^n}{1 - |z_0|}.$$

Taking nth roots, $|g(z_0)| \le M(1 - |z_0|)^{-1/n}$; letting $n \to \infty$ gives $|g(z_0)| \le M$. This also holds on the boundary. Thus $|f(z)| \le M|z| < |z|$ for nonzero z, with equality $0 = 0$ at zero.

11.5.4 If $f(z)$ is a real function of the complex variable $z = x + iy$, that is, $f(x) = f^*(x)$ for real x near zero, and the Laurent expansion about the origin, $f(z) = \sum a_n z^n$ on $0 < |z| < R$, has $a_n = 0$ for $n < -N$, show that all of the coefficients a_n are real.

Hint. Show that $z^N f(z)$ is analytic (via Morera's theorem, Section 11.4).

Solution.

Here *real* means real-valued for real arguments near the Laurent center, not throughout an open complex region. The finite principal part makes

$$g(z) = z^N f(z) = \sum_{j=0}^{\infty} a_{j-N} z^j$$

analytic after its removable extension to zero. For nonzero real x near zero, $g(x) = x^N f(x)$ is real, and continuity gives a real value at zero. Each analytic derivative at zero may be computed along the real line; derivatives of this real-valued restriction are real. Hence $a_{j-N} = g^{(j)}(0)/j! \in \mathbb{R}$ for every $j \geq 0$. The remaining coefficients vanish and are real as well.

11.5.5 Prove that the Laurent expansion of a given function about a given point is unique; that is, if

$$f(z) = \sum_{n=-N}^{\infty} a_n (z - z_0)^n = \sum_{n=-N}^{\infty} b_n (z - z_0)^n,$$

with both series representing f on a common annulus, show that $a_n = b_n$ for all n.

Hint. Use the Cauchy integral formula.

Solution.

Choose a circle $|\zeta - z_0| = r$ in the common convergence annulus. Uniform convergence there permits termwise integration:

$$\begin{aligned} \frac{1}{2\pi i} \oint f(\zeta)(\zeta - z_0)^{-k-1}\, d\zeta &= \sum_n a_n \frac{1}{2\pi i} \oint (\zeta - z_0)^{n-k-1}\, d\zeta \\ &= \sum_n a_n \delta_{nk} = a_k. \end{aligned}$$

The circle integrals equal δ_{nk} by Exercise 11.4.1, applied in the variable $\zeta - z_0$. The second series gives b_k for the identical integral. Thus $\boxed{a_k = b_k}$ for every integer k. The proof also works for an infinite principal part on a common annulus. The common annulus is needed because the proof integrates both series over one circle on which both converge to f; about one point a function can have different Laurent series on different annuli, e.g., the two expansions of $1/[z(z-1)]$ about $z = 1$ in Exercise 11.6.10. For the displayed series, whose principal parts are finite, the condition holds once both represent f near z_0: each then does so on a punctured disk $0 < |z - z_0| < \varepsilon$, and the smaller of the two disks is a common annulus.

11.5.6 Obtain the Laurent expansion of e^z/z^2 about $z = 0$.

Solution.

Divide the everywhere-convergent exponential series by z^2:

$$\boxed{\frac{e^z}{z^2} = \sum_{n=0}^{\infty} \frac{z^{n-2}}{n!} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2} + \frac{z}{6} + \frac{z^2}{24} + \cdots}.$$

The annulus is $0 < |z| < \infty$. The pole has order two and residue one.

11.5.7 Obtain the Laurent expansion of $ze^z/(z-1)$ about $z = 1$.

Solution.

Put $w = z - 1$, so $z = 1 + w$ and $e^z = e\, e^w$. Then

$$\frac{ze^z}{z-1} = \frac{(1+w)\, e\, e^w}{w} = e\left(\frac{1}{w} + 1\right) \sum_{n=0}^{\infty} \frac{w^n}{n!}.$$

The w^{-1} coefficient is e, from $n = 0$ in the product with $1/w$. For $m \geq 0$ the w^m coefficient comes from $n = m+1$ in the product with $1/w$ and from $n = m$ in the product with 1: $e[1/(m+1)!+1/m!] = e[1 + (m + 1)]/(m + 1)! = e(m + 2)/(m + 1)!$. Consequently

$$\boxed{\frac{ze^z}{z-1} = \frac{e}{z-1} + e\sum_{m=0}^{\infty} \frac{m+2}{(m+1)!}(z-1)^m}, \qquad 0 < |z-1| < \infty.$$

The first regular terms are $2e + \frac{3}{2}e(z-1) + \frac{2}{3}e(z-1)^2$.

11.5.8 Obtain the Laurent expansion of $(z-1)e^{1/z}$ about $z = 0$.

Solution.

Multiplication of the exponential series gives

$$(z-1)e^{1/z} = \sum_{n=0}^{\infty} \frac{z^{1-n}}{n!} - \sum_{n=0}^{\infty} \frac{z^{-n}}{n!}.$$

The leading term is z ($n = 0$ in the first sum), and the constants ($n = 1$ in the first sum, $n = 0$ in the second) cancel. For $k \geq 1$ the coefficient of z^{-k} comes from $n = k + 1$ in the first sum and $n = k$ in the second: $1/(k + 1)! - 1/k! = [1 - (k + 1)]/(k + 1)! = -k/(k + 1)!$. Thus

$$\boxed{(z-1)e^{1/z} = z - \sum_{k=1}^{\infty} \frac{k}{(k+1)!} z^{-k} = z - \frac{1}{2z} - \frac{1}{3z^2} - \frac{1}{8z^3} - \cdots}.$$

It converges for $0 < |z| < \infty$; its infinite principal part makes zero an essential singularity.

11.6 Singularities

11.6.1 As an example of an essential singularity consider $e^{1/z}$ as z approaches zero. For any complex number z_0, $z_0 \neq 0$, show that

$$e^{1/z} = z_0$$

has an infinite number of solutions.

Solution.

Write $z_0 = \rho e^{i\alpha}$ with $\rho > 0$, choosing one argument. Since $e^w = e^{w'}$ exactly when $w - w' \in 2\pi i\mathbb{Z}$, all solutions of $e^w = z_0 = e^{\ln\rho + i\alpha}$ are $w = \ln\rho + i(\alpha + 2\pi n)$, $n \in \mathbb{Z}$. Therefore

$$\boxed{z_n = \frac{1}{\ln\rho + i(\alpha + 2\pi n)}}$$

whenever the denominator is nonzero. Substitution gives $e^{1/z_n} = \rho e^{i\alpha} = z_0$. A zero denominator needs $\ln\rho = 0$ and $\alpha + 2\pi n = 0$, so at most one integer has one. All remaining denominators are distinct, since their imaginary parts differ by multiples of 2π, so their reciprocals are distinct. Thus infinitely many distinct solutions remain, with $|z_n| \leq 1/|\alpha + 2\pi n| \to 0$ as $|n| \to \infty$. For $z_0 = 1$ with the choice $\alpha = 0$, simply omit $n = 0$. The Instructor's Manual argues from roots of truncated exponential polynomials. Passing to the limit does not by itself establish distinct roots of the original equation. The explicit logarithmic preimages above supply that missing argument.

11.6.2 Show that the function

$$w(z) = (z^2 - 1)^{1/2}$$

is single-valued if we make branch cuts on the real axis for $x \geq 1$ and for $x \leq -1$.

Solution.

Remove both rays including their endpoints: $D = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$. For $z = x + iy$, $1 - z^2 = (1 - x^2 + y^2) - 2ixy$ is nonpositive real only if $xy = 0$ and $1 - x^2 + y^2 \leq 0$. The choice $x = 0$ gives $1 + y^2 > 0$; otherwise $y = 0$, $|x| \geq 1$, exactly the removed rays. The principal square root is analytic off the ray $(-\infty, 0]$, so $\sqrt{1 - z^2}$ is analytic on D, and the explicit branch

$$\boxed{w(z) = i\sqrt{1 - z^2}}$$

is single-valued and satisfies $w^2 = z^2 - 1$. Its negative is the other branch; one reference value selects its sign. The branch points ± 1 themselves are excluded from the analytic domain, not merely the open rays. The Instructor's Manual instead discusses a cut joining -1 to 1. That is the different branch domain in Example 11.6.4, not the two outward rays specified here.

11.6.3 A function $f(z)$ can be represented by

$$f(z) = \frac{f_1(z)}{f_2(z)},$$

in which $f_1(z)$ and $f_2(z)$ are analytic. The denominator, $f_2(z)$, vanishes at $z = z_0$, showing that $f(z)$ has a pole at $z = z_0$. However, $f_1(z_0) \neq 0$, $f_2'(z_0) \neq 0$. Show that a_{-1}, the coefficient of $(z - z_0)^{-1}$ in a Laurent expansion of $f(z)$ at $z = z_0$, is given by

$$a_{-1} = \frac{f_1(z_0)}{f_2'(z_0)}.$$

Solution.

With $h = z - z_0$, Taylor expansion gives

$$f_1(z) = f_1(z_0) + O(h), \quad f_2(z) = f_2'(z_0)h[1 + O(h)].$$

Each $O(h)$ here is a convergent power series in h without constant term. The last factor has an analytic reciprocal $1 + O(h)$ near zero because its constant term is one. Hence

$$f(z) = \frac{f_1(z_0) + O(h)}{f_2'(z_0)h}[1 + O(h)] = \frac{f_1(z_0)}{f_2'(z_0)}h^{-1} + O(1).$$

Therefore $\boxed{a_{-1} = f_1(z_0)/f_2'(z_0)}$. The assumptions make this coefficient nonzero and the pole simple.

11.6.4 Determine a unique branch for the function of Exercise 11.6.2 that will cause the value it yields for $f(i)$ to be the same as that found for $f(i)$ in Example 11.6.4. Although Exercise 11.6.2 and Example 11.6.4 describe the same multivalued function, the specific values assigned for various z will not agree everywhere, due to the difference in the location of the branch cuts. Identify the portions of the complex plane where both these descriptions do and do not agree, and characterize the differences.

Solution.

Write $z-1=\rho_1 e^{i\varphi_1}$ and $z+1=\rho_2 e^{i\varphi_2}$. The segment-cut branch G of Example 11.6.4 is $G(z)=(\rho_1\rho_2)^{1/2}e^{i(\varphi_1+\varphi_2)/2}$, positive on the real axis $z>1$, where $\varphi_1=\varphi_2=0$. At $z=i$, the arguments of $i-1$ and $i+1$ are $3\pi/4$ and $\pi/4$ and their moduli are both $\sqrt{2}$. Its value is

$$(\sqrt{2})^{1/2}(\sqrt{2})^{1/2}e^{i(3\pi/4+\pi/4)/2}=\boxed{i\sqrt{2}}.$$

The outward-cut branch with this value is $F(z)=i\sqrt{1-z^2}$ from 11.6.2, since $F(i)=i\sqrt{1-i^2}=i\sqrt{2}$. Since its domain is connected, the reference value fixes its sign; the other branch, $-F$, has the value $-i\sqrt{2}$ there.

Where both are defined, $F^2=G^2=z^2-1$ and neither vanishes; thus F/G is continuous and takes only ±1, so it is constant on each connected piece of the common domain. The two systems of cuts together cover the real axis, so the common domain is the plane minus the real axis, whose components are the upper and lower half-planes. Equality at i implies $F=G$ in the upper half-plane. At $-i$, continuing G from $z>1$ through the lower half-plane, the arguments of $-i-1$ and $-i+1$ are $-3\pi/4$ and $-\pi/4$, so $G(-i)=\sqrt{2}\,e^{-i\pi/2}=-i\sqrt{2}$, while $F(-i)=i\sqrt{1-(-i)^2}=i\sqrt{2}$. Hence $F=-G$ in the lower half-plane. On the real axis at most one of them is defined, since G is cut along $[-1,1]$ and F along $|x|\ge1$. Each is continuous across the other's cut, so by the half-plane relations its value there is the limit of the other from above, written with argument $x+i0$, and minus the limit from below, $x-i0$. For $|x|>1$, where $\varphi_1=\varphi_2$ is 0 ($x>1$) or π ($x<-1$),

$$G(x)=\operatorname{sgn}(x)\sqrt{x^2-1}=F(x+i0)=-F(x-i0),$$

and for $|x|<1$, $F(x)=i\sqrt{1-x^2}=G(x+i0)=-G(x-i0)$. These statements specify precisely their agreement and sign difference. The Instructor's Manual gives $f(i)=2i$. Each factor contributes modulus $2^{1/4}$, not $\sqrt{2}$, so their product has modulus $\sqrt{2}$. Also $(i\sqrt{2})^2=-2=i^2-1$, whereas $(2i)^2=-4$.

11.6.5 Find all singularities of

$$z^{-1/3}+\frac{z^{-1/4}}{(z-3)^3}+(z-2)^{1/2},$$

and identify their types (e.g., second-order branch point, fifth-order pole, ...). Include any singularities at the point at infinity.

Note. A branch point is of nth order if it requires n, but no fewer, circuits around the point to restore the original value.

Solution.

At zero, since $(z-3)^3=(-3)^3(1-z/3)^3$,

$$\frac{z^{-1/4}}{(z-3)^3}=-\frac{1}{27}z^{-1/4}(1-z/3)^{-3},$$

where $(1-z/3)^{-3}$ and the third term $(z-2)^{1/2}$ are analytic near zero. After N circuits about zero the first two terms therefore acquire multipliers $e^{-2\pi iN/3}$ and $e^{-2\pi iN/4}$, and the third is unchanged, so the function changes by

$$c_1z^{-1/3}+c_2z^{-1/4}(1-z/3)^{-3},\qquad c_1=e^{-2\pi iN/3}-1,\quad c_2=-\tfrac{1}{27}\big(e^{-2\pi iN/4}-1\big).$$

These changes cannot cancel. If $c_1 \neq 0$, the ratio of the second term to the first, $(c_2/c_1)z^{1/12}(1 - z/3)^{-3}$, tends to zero as $z \to 0$ (because $-1/4 - (-1/3) = 1/12 > 0$), so the change is nonzero near zero; if $c_1 = 0$, the change $c_2 z^{-1/4}(1 - z/3)^{-3}$ vanishes only if $c_2 = 0$. Hence the function returns to its initial value iff both multipliers equal 1, i.e., iff N is divisible by 3 and 4; the least such positive integer is 12. Zero is a branch point of order 12 in the question's convention. At 3, the chosen local numerator is analytic and nonzero and the other two terms are analytic, giving a third-order pole. At 2, only $(z-2)^{1/2}$ branches, with order 2. For infinity put $w = 1/z$; with $(1/w - 3)^3 = (1 - 3w)^3/w^3$ and $1/w - 2 = (1 - 2w)/w$,

$$f(1/w) = w^{1/3} + \frac{w^{13/4}}{(1-3w)^3} + w^{-1/2}(1-2w)^{1/2}.$$

The analytic factors $(1-3w)^{-3}$ and $(1-2w)^{1/2}$ have nonzero constant terms, so N circuits about $w = 0$ multiply the three terms by $e^{2\pi i N/3}$, $e^{2\pi i(13N/4)} = e^{2\pi i N/4}$ (since $e^{2\pi i \cdot 3N} = 1$) and $e^{-\pi i N}$. As at zero, the changes cannot cancel: among the terms whose multiplier is not 1, the one with the smallest exponent of w (the exponents are $-1/2 < 1/3 < 13/4$) dominates the others as $w \to 0$, so the change is nonzero near $w = 0$. So the function returns to its initial value iff all three multipliers equal 1, i.e., iff N is divisible by 3, 4 and 2; their least common multiple is 12. Infinity is an order-12 branch point, not an isolated meromorphic pole. There are no other finite singularities.

11.6.6 The function $F(z) = \ln(z^2 + 1)$ is made single-valued by straight-line branch cuts from $(x, y) = (0, -1)$ to $(-\infty, -1)$ and from $(0, +1)$ to $(0, +\infty)$. See Fig. 11.17. If $F(0) = -2\pi i$, find the value of $F(i - 2)$.

Solution.

Factor $z^2 + 1 = (z - i)(z + i)$, so every branch has the form $\ln|z - i| + \ln|z + i| + i[\arg(z - i) + \arg(z + i)] + 2\pi i k$. The upward cut from i consists of the points with $\arg(z - i) = \pi/2$; choosing $-3\pi/2 < \arg(z - i) < \pi/2$ makes this argument continuous off that cut. The leftward cut from $-i$ consists of the points with $\arg(z + i) = \pi$; choosing $-\pi < \arg(z + i) < \pi$ makes this argument continuous off it. At zero, $z - i = -i$ and $z + i = i$ have arguments $-\pi/2, \pi/2$ and both moduli are one, so $F(0) = 2\pi i k = -2\pi i$ requires $k = -1$. The required branch is therefore

$$F(z) = \ln|z - i| + \ln|z + i| + i[\arg(z - i) + \arg(z + i) - 2\pi].$$

At $z = -2 + i$, the factors are -2 and $-2 + 2i$. Their arguments in these ranges are $-\pi$ and $3\pi/4$; their moduli are 2 and $2\sqrt{2}$. Thus

$$F(i-2) = \ln 2 + \ln(2\sqrt{2}) + i\left(-\pi + \frac{3\pi}{4} - 2\pi\right) = \boxed{\ln(4\sqrt{2}) - 9\pi i/4}.$$

Exponentiation verifies the result: $e^F = 4\sqrt{2}\, e^{-9\pi i/4} = 4\sqrt{2}\, e^{-i\pi/4} = 4 - 4i = (-2 + i)^2 + 1$.

11.6.7 Show that negative numbers have logarithms in the complex plane. In particular, find $\ln(-1)$.

ANS. $\mathrm{Log}(-1) = i\pi$ (principal value).

Solution.

For $r > 0$, $-r = re^{i(\pi + 2\pi n)}$, $n \in \mathbb{Z}$. All logarithms are consequently

$$\boxed{\ln(-r) = \ln r + i(2n+1)\pi}.$$

In particular $\ln(-1) = i(2n+1)\pi$. The printed $i\pi$ is the principal value, or one chosen branch value, rather than the complete multivalued answer.

11.6.8 For noninteger m, show that the binomial expansion of Exercise 11.5.2 holds only for a suitably defined branch of the function $(1+z)^m$. Show how the z-plane is cut. Explain why $|z|<1$ may be taken as the circle of convergence for the expansion of this branch, in light of the cut you have chosen.

Solution.

Choose $(1+z)^m = \exp[m\,\mathrm{Log}(1+z)]$ with $-\pi < \arg(1+z) < \pi$. The cut is $(-\infty,-1]$ and the value at zero is 1. This analytic branch has the Taylor coefficients derived in 11.5.2, and the open unit disk avoids the cut. For noninteger real m, a circuit about -1 increases $\arg(1+z)$ by 2π and so multiplies the value by $e^{2\pi i m} \neq 1$. Thus -1 is a genuine branch point, distance one from the Taylor center. The radius is exactly one: if the series converged on a disk $|z|<\rho$ with $\rho>1$, its sum S would be analytic there and equal to the branch on $|z|<1$, hence (by analytic continuation) on the disk minus the cut. Continuing along a small circle about -1 inside the disk would then return S to its starting value, contradicting the factor $e^{2\pi i m}$.

11.6.9 The Taylor expansion of Exercises 11.5.2 and 11.6.8 is not suitable for branches other than the one suitably defined branch of the function $(1+z)^m$ for noninteger m. (Note that other branches cannot have the same Taylor expansion since they must be distinguishable.) Using the same branch cut of the earlier exercises for all other branches, find the corresponding Taylor expansions, detailing the phase assignments and Taylor coefficients.

Solution.

For the same cut, let

$$\mathrm{Log}_k(1+z) = \ln|1+z| + i\arg(1+z) + 2\pi i k, \quad k\in\mathbb{Z},$$

where the displayed argument lies in $(-\pi,\pi)$; the branch k assigns $1+z$ the phase $\arg(1+z)+2\pi k$. Then $(1+z)^m_k = \exp[m\,\mathrm{Log}_k(1+z)] = e^{2\pi i m k}\exp[m\,\mathrm{Log}_0(1+z)] = e^{2\pi i m k}(1+z)^m_0$, where $k=0$ is the branch of 11.6.8, so

$$\boxed{(1+z)^m_k = e^{2\pi i m k}\sum_{n=0}^{\infty}\binom{m}{n}z^n, \qquad |z|<1}.$$

Every coefficient is multiplied by the same phase; the constant coefficient already distinguishes the branches. For $m=p/q$ in lowest terms, k modulo q gives q distinct branches. For irrational m, all integer k give distinct ones.

11.6.10

(a) Develop a Laurent expansion of $f(z)=[z(z-1)]^{-1}$ about the point $z=1$ valid for small values of $|z-1|$. Specify the exact range over which your expansion holds. This is an analytic continuation of the infinite series in Eq. (11.49).

(b) Determine the Laurent expansion of $f(z)$ about $z=1$ but for $|z-1|$ large.

Hint. Make a partial fraction decomposition of this function and use the geometric series.

Solution.

Put $w=z-1$, so $f=1/[w(1+w)]=1/w-1/(1+w)$, since $(1+w)-w=1$. **(a)** For $|w|<1$, the geometric series $1/(1+w)=\sum_{n=0}^{\infty}(-1)^n w^n$ gives

$$\boxed{f=\frac{1}{w}-1+w-w^2+\cdots=\sum_{n=-1}^{\infty}(-1)^{n+1}w^n, \qquad 0<|w|<1}.$$

The pole at $w = 0$ excludes the center; the pole at $w = -1$ sets the outer radius. **(b)** For $|w| > 1$ write $w(1 + w) = w^2(1 + w^{-1})$ and expand $(1 + w^{-1})^{-1} = \sum_{n=0}^{\infty}(-1)^n w^{-n}$, a geometric series in w^{-1} with $|w^{-1}| < 1$:

$$\boxed{f = w^{-2}(1 + w^{-1})^{-1} = \sum_{n=0}^{\infty}(-1)^n w^{-n-2}, \qquad |w| > 1}.$$

The annuli differ, so uniqueness of Laurent coefficients on a fixed annulus is not contradicted. The Instructor's Manual writes the outer series as $\sum_{n=-2}^{\infty}(-1)^n(z-1)^{-n}$. Its lower index should be 2; otherwise it begins with growing powers and cannot equal a function that decays as $(z-1)^{-2}$ at infinity.

11.6.11

(a) Given $f_1(z) = \displaystyle\int_0^{\infty} e^{-zt}\,dt$ (with t real), show that the domain in which $f_1(z)$ exists (and is analytic) is $\Re(z) > 0$.

(b) Show that $f_2(z) = 1/z$ equals $f_1(z)$ over $\Re(z) > 0$ and is therefore an analytic continuation of $f_1(z)$ over the entire z-plane except for $z = 0$.

(c) Expand $1/z$ about the point $z = -i$. You will have

$$f_3(z) = \sum_{n=0}^{\infty} a_n(z + i)^n.$$

What is the domain of this formula for $f_3(z)$?

$$\textit{ANS.}\ \ \frac{1}{z} = i\sum_{n=0}^{\infty} i^{-n}(z + i)^n, \quad |z + i| < 1.$$

Solution.

(a) For $z \neq 0$,

$$\int_0^T e^{-zt}\,dt = (1 - e^{-zT})/z.$$

Here $|e^{-zT}| = e^{-T\Re z}$. If $\Re z > 0$, the exponential tends to zero as $T \to \infty$, so $f_1(z) = 1/z$, which is analytic there. If $\Re z < 0$, $|e^{-zT}|$ grows without bound. If $\Re z = 0$, $z = iy \neq 0$, then e^{-iyT} has modulus one and keeps rotating, without a limit. For $z = 0$, the integral equals T. Thus ordinary convergence occurs exactly on $\Re z > 0$. **(b)** By (a), $f_2 = 1/z$ equals f_1 on $\Re z > 0$, and $1/z$ is analytic on $\mathbb{C} \setminus \{0\}$. By the identity theorem it is the only analytic function on this connected domain that agrees with f_1 on $\Re z > 0$; it is the analytic continuation of f_1. The continuation has no negative-integer singularities; the Instructor's Manual's mention of those points is extraneous to this function. **(c)** Set $w = z + i$. Then $z = w - i = -i(1 + iw)$ and $1/(-i) = i$; the geometric series in $-iw$ converges for $|w| < 1$, and $(-i)^n = i^{-n}$ because $-i = 1/i$, so

$$\boxed{\frac{1}{z} = \frac{i}{1 + iw} = i\sum_{n=0}^{\infty}(-i)^n w^n = i\sum_{n=0}^{\infty} i^{-n}(z + i)^n, \quad |z + i| < 1}.$$

The pole at zero is distance one from the center $-i$, fixing the exact convergence radius. The Instructor's Manual expands about $+i$ in part (c), while the question requests $-i$. The expansion here uses $z + i$ and has constant coefficient $1/(-i) = i$.

11.7 Calculus of Residues

11.7.1 Determine the nature of the singularities of each of the following functions and evaluate the residues ($a > 0$).

(a) $\dfrac{1}{z^2+a^2}$. (b) $\dfrac{1}{(z^2+a^2)^2}$. (c) $\dfrac{z^2}{(z^2+a^2)^2}$.

(d) $\dfrac{\sin 1/z}{z^2+a^2}$. (e) $\dfrac{ze^{iz}}{z^2+a^2}$. (f) $\dfrac{ze^{iz}}{z^2-a^2}$.

(g) $\dfrac{e^{iz}}{z^2-a^2}$. (h) $\dfrac{z^{-k}}{z+1}$, $0<k<1$.

Hint. For the point at infinity, use the transformation $w = 1/z$ for $|z| \to \infty$. For the residue, transform $f(z)\,dz$ into $g(w)\,dw$ and look at the behavior of $g(w)$.

Solution.

Infinity corresponds to $w = 1/z \to 0$, hence to $|z| \to \infty$, not to $|z| \to 0$ as the printed hint says. The nature of the singularity at infinity is the nature of $f(1/w)$ at zero. For the residue, $z = 1/w$ gives $dz = -dw/w^2$, so $f(z)\,dz = g(w)\,dw$ with $g(w) = -f(1/w)/w^2$, and, as the hint says, the residue at infinity is that of g at $w = 0$:

$$\operatorname{Res}_\infty f = \operatorname{Res}_{w=0}\left[-\frac{f(1/w)}{w^2}\right].$$

These are different transformations: the second includes the differential.

(a)–(c) Factor $z^2+a^2 = (z-ia)(z+ia)$. For (a), both poles are simple, and evaluating $(z \mp ia)f(z) = 1/(z \pm ia)$ at $z = \pm ia$ gives

$$\operatorname{Res}_{ia}\frac{1}{z^2+a^2} = \frac{1}{2ia}, \qquad \operatorname{Res}_{-ia}\frac{1}{z^2+a^2} = -\frac{1}{2ia}.$$

For (b), the poles are double. The double-pole formula $\operatorname{Res}_{z_0} f = \frac{d}{dz}[(z-z_0)^2 f(z)]_{z=z_0}$, with $(z-ia)^2 f(z) = (z+ia)^{-2}$ and $(2ia)^3 = 8i^3a^3 = -8ia^3$, gives

$$\operatorname{Res}_{ia}\frac{1}{(z^2+a^2)^2} = \frac{d}{dz}(z+ia)^{-2}\bigg|_{ia} = -\frac{2}{(2ia)^3} = \frac{1}{4ia^3}.$$

Each of the functions (a)–(c) depends on a only through a^2, so replacing a by $-a$ interchanges the poles $\pm ia$: the residue at $-ia$ is the residue at ia with $a \to -a$. For (b) it is $-1/(4ia^3)$. In (c) the numerator z^2 equals $-a^2 \neq 0$ at $\pm ia$, so the poles are again double, and the double-pole formula, now with $(z-ia)^2 f(z) = z^2/(z+ia)^2$, gives

$$\frac{d}{dz}\frac{z^2}{(z+ia)^2}\bigg|_{ia} = \frac{2z(z+ia)-2z^2}{(z+ia)^3}\bigg|_{ia} = \frac{-2a^2}{-8ia^3} = \frac{1}{4ia}.$$

The residue at $-ia$ is $-1/(4ia)$. At infinity the three functions $f(1/w)$ become, after multiplying numerator and denominator by w^2 or w^4,

$$\frac{w^2}{1+a^2w^2}, \quad \frac{w^4}{(1+a^2w^2)^2}, \quad \frac{w^2}{(1+a^2w^2)^2}.$$

Thus infinity is removable, with a zero of order $2, 4, 2$, respectively. The transformed differentials $g(w) = -f(1/w)/w^2$,

$$-\frac{1}{1+a^2w^2}, \quad -\frac{w^2}{(1+a^2w^2)^2}, \quad -\frac{1}{(1+a^2w^2)^2},$$

are analytic at $w = 0$, so all three residues at infinity are zero. Consistently, in each case the two finite residues add to zero.

(d) Zero is an essential singularity: if f had at most a pole there, so would $\sin(1/z) = (z^2+a^2)f(z)$, whose Laurent series $\sum_{k\ge0}(-1)^k z^{-2k-1}/(2k+1)!$ has infinitely many negative powers. The points $\pm ia$ are simple poles; as in (a), the residue there is $\sin(1/z)/(z \pm ia)$ evaluated at $z = \pm ia$, where $1/(\pm ia) = \mp i/a$ and $\sin(\mp i/a) = \mp i \sinh(1/a)$ (from $\sin iy = i \sinh y$):

$$\operatorname{Res}_{ia} f = \frac{\sin(-i/a)}{2ia} = -\frac{\sinh(1/a)}{2a}, \qquad \operatorname{Res}_{-ia} f = \frac{\sin(i/a)}{-2ia} = -\frac{\sinh(1/a)}{2a}.$$

To obtain the residue at zero directly, multiply two series that converge absolutely on $0 < |z| < a$, the Laurent series of $\sin(1/z)$ above and the geometric series $1/(z^2+a^2) = a^{-2}(1+z^2/a^2)^{-1} = \sum_{j\ge0}(-1)^j z^{2j}/a^{2j+2}$:

$$\frac{\sin(1/z)}{z^2+a^2} = \sum_{j,k\ge0} \frac{(-1)^{j+k} z^{2j-2k-1}}{a^{2j+2}(2k+1)!}.$$

A term has exponent $2j - 2k - 1 = -1$ exactly when $j = k$, and then $(-1)^{j+k} = 1$. Consequently

$$\boxed{\operatorname{Res}_0 f = \sum_{k=0}^{\infty} \frac{1}{a^{2k+2}(2k+1)!} = \frac{1}{a}\sum_{k=0}^{\infty} \frac{(1/a)^{2k+1}}{(2k+1)!} = \frac{\sinh(1/a)}{a}}.$$

At infinity $f(1/w) = w^2 \sin w/(1+a^2w^2)$ has a zero of order three, since $\sin w = w + O(w^3)$, and $g(w) = -\sin w/(1+a^2w^2)$ is analytic at $w = 0$, so the residue at infinity is zero. Consistently, the three finite residues add to zero.

(e)–(g) If D has a simple zero p and N is analytic at p, the residue of N/D is $N(p)/D'(p)$, here with $D'(p) = 2p$. Applying this rule gives

	finite poles	residues in the same order	$\operatorname{Res}_\infty f$
(e)	$ia,\ -ia$	$e^{-a}/2,\ e^{a}/2$	$-\cosh a$
(f)	$a,\ -a$	$e^{ia}/2,\ e^{-ia}/2$	$-\cos a$
(g)	$a,\ -a$	$e^{ia}/(2a),\ -e^{-ia}/(2a)$	$-i \sin a/a$

For example, the first two rows follow from $pe^{ip}/(2p) = e^{ip}/2$ (for $p = \pm ia$ this is $e^{\mp a}/2$), and the third from $e^{ip}/(2p)$ with $p = \pm a$. All these finite poles are simple, since the numerators do not vanish there. Each function has an essential singularity at infinity: $f(1/w)$ is $e^{i/w}$ times a rational function of w that is not identically zero, so if $f(1/w)$ had at most a pole at $w = 0$, so would $e^{i/w} = \sum_{m\ge0} i^m w^{-m}/m!$, which has infinitely many negative powers. Nevertheless its residue there is defined. Under $z = 1/w$ a small counterclockwise circle about $w = 0$ becomes a large clockwise circle $|z| = R$, so $\operatorname{Res}_\infty f = -\frac{1}{2\pi i}\oint_{|z|=R} f\,dz$ with the counterclockwise orientation. For $R > a$ the residue theorem gives $\operatorname{Res}_\infty f = -\sum_{\text{finite } p} \operatorname{Res}_p f$. The sums are $(e^{-a}+e^{a})/2 = \cosh a$, $(e^{ia}+e^{-ia})/2 = \cos a$, and $(e^{ia}-e^{-ia})/(2a) = i \sin a/a$, which yield the displayed answers. In (d), the Instructor's Manual gives positive residues at $\pm ia$; both must be $-\sinh(1/a)/(2a)$, as the displayed substitution shows. It reverses the paired exponents in (e), giving $e^{\pm a}/2$ at $\pm ia$. Its (f) incorrectly uses poles $\pm ia$ and residues $-ie^{\mp a}/2$, although the denominator is $z^2 - a^2$. In (g) it

reverses the exponential signs as well. Substitution into $N(p)/D'(p)$ gives the table above, with each residue paired to its pole.

(h) The nonintegral power produces branch points at zero and infinity: continuation once around zero multiplies z^{-k} by $e^{-2\pi ik} \neq 1$, while $f(1/w) = (1/w)^{-k}/(1/w+1) = w^k \cdot w/(1+w) = w^{k+1}/(1+w)$ is also multivalued at $w = 0$. A rational $k = r/q$ in lowest terms gives q sheets; irrational k gives infinitely many. On any local branch at -1 the numerator is analytic and nonzero, so there is a simple pole whose residue is the value of z^{-k} at -1 on that branch, $e^{-ik\arg(-1)}$ with $\arg(-1) = (2m+1)\pi$:

$$\boxed{\operatorname{Res}_{-1} f = (-1)^{-k} = e^{-ik(2m+1)\pi}}.$$

For the branch $0 < \arg z < 2\pi$, $\arg(-1) = \pi$: take $m = 0$, so the residue is $e^{-ik\pi}$. The residue depends on the branch. Ordinary Laurent residues at the branch points zero and infinity are not defined.

11.7.2 Evaluate the residues at $z = 0$ and $z = -1$ of $\pi \cot \pi z / z(z+1)$.

Solution.

From $\cot x = \cos x / \sin x = x^{-1}(1 - x^2/2 + \cdots)(1 - x^2/6 + \cdots)^{-1} = x^{-1} - x/3 + O(x^3)$ with $x = \pi z$, $\pi \cot \pi z = z^{-1} - (\pi^2/3)z + O(z^3)$. At zero, divide this by z and expand $1/(1+z)$ in a geometric series:

$$\frac{\pi \cot \pi z}{z(z+1)} = \left(\frac{1}{z^2} - \frac{\pi^2}{3} + O(z^2)\right)(1 - z + z^2 - \cdots) = \frac{1}{z^2} - \frac{1}{z} + O(1).$$

At $z = -1$, put $w = z + 1$. The period one of $\cot \pi z$ gives $\pi \cot \pi z = \pi \cot \pi w = w^{-1} + O(w)$, while $z(z+1) = (w-1)w$ and $1/(w-1) = -(1 + w + w^2 + \cdots)$:

$$\frac{\pi \cot \pi z}{z(z+1)} = \frac{w^{-1} + O(w)}{w(w-1)} = -\frac{1}{w^2}(1 + w + w^2 + \cdots) + O(1) = -\frac{1}{w^2} - \frac{1}{w} + O(1).$$

Thus both points are double poles and

$$\boxed{\operatorname{Res}_0 f = \operatorname{Res}_{-1} f = -1}.$$

11.7.3 The classical definition of the exponential integral $\operatorname{Ei}(x)$ for $x > 0$ is the Cauchy principal value integral

$$\operatorname{Ei}(x) = \int_{-\infty}^{x} \frac{e^t}{t}\, dt,$$

where the integration range is cut at $t = 0$. Show that this definition yields a convergent result for positive x.

Solution.

The printed problem says the range is cut at $x = 0$; the singular point of the integrand is $t = 0$, while $x > 0$ is the upper limit. Fix $x > 0$ and choose $0 < \delta < x$. Split the negative side at $-x$ and change $t = -u$, $dt = -du$, in the interval $[-x, -\delta]$:

$$\int_{-x}^{-\delta} \frac{e^t}{t}\, dt = \int_{x}^{\delta} \frac{e^{-u}}{-u}(-du) = -\int_{\delta}^{x} \frac{e^{-u}}{u}\, du.$$

Inserting this after the split,

$$\int_{-\infty}^{-\delta}\frac{e^t}{t}\,dt+\int_{\delta}^{x}\frac{e^t}{t}\,dt=\int_{-\infty}^{-x}\frac{e^t}{t}\,dt-\int_{\delta}^{x}\frac{e^{-u}}{u}\,du+\int_{\delta}^{x}\frac{e^u}{u}\,du$$
$$=\int_{-\infty}^{-x}\frac{e^t}{t}\,dt+\int_{\delta}^{x}\frac{e^u-e^{-u}}{u}\,du.$$

The first integral converges absolutely, since $|e^t/t|\le e^t/x$ for $t\le -x$ and $\int_{-\infty}^{-x}e^t\,dt=e^{-x}$. In the second, $e^u-e^{-u}=2u+O(u^3)$, so the integrand has the finite limit 2 at zero; it extends continuously to $[0,x]$, and the integral over $[\delta,x]$ tends to the integral over $[0,x]$ as $\delta\to 0$. Therefore

$$\boxed{\mathrm{Ei}(x)=\int_{-\infty}^{-x}\frac{e^t}{t}\,dt+\int_{0}^{x}\frac{2\sinh u}{u}\,du}$$

is finite. The cancellation uses equal distances δ on the two sides of the pole, precisely the principal-value prescription.

11.7.4 Writing a Cauchy principal value integral to deal with the singularity at $x=1$, show that, if $0<p<1$,

$$\int_0^\infty\frac{x^{-p}}{x-1}\,dx=\pi\cot p\pi.$$

Solution.

For the printed denominator $x-1$, the answer is $+\pi\cot(\pi p)$, not $-\pi\cot(\pi p)$. The Instructor's Manual derives its negative answer for the different denominator $1-x$. Write the intended integral as

$$I=\lim_{\delta\downarrow 0}\left(\int_0^{1-\delta}+\int_{1+\delta}^{\infty}\right)\frac{x^{-p}}{x-1}\,dx.$$

The endpoint estimates are $O(x^{-p})$ at zero and $O(x^{-p-1})$ at infinity, so $0<p<1$ makes both endpoints integrable. Only $x=1$ requires a principal value.

Use $F(z)=z^{-p}/(z-1)$ with $z^{-p}=|z|^{-p}e^{-ip\arg z}$, $0<\arg z<2\pi$, and a keyhole contour about the positive real axis, with an origin circle $|z|=\epsilon<1$ and an outer circle $|z|=R>1$. On the upper bank ($\arg z=0$) $z^{-p}=x^{-p}$; on the lower bank ($\arg z=2\pi$) it equals qx^{-p}, where $q=e^{-2\pi ip}$. The upper bank, directed outward, gives I; the lower bank, directed inward, gives $\int_\infty^0 qx^{-p}(x-1)^{-1}\,dx=-qI$; together they give $(1-q)I$. Exclude the boundary pole at 1 by semicircles of radius δ into the domain. Near $z=1$, $z^{-p}\to 1$ on the upper bank and $z^{-p}\to q$ on the lower one, so there $F(z)=1/(z-1)+O(1)$ and $F(z)=q/(z-1)+O(1)$, respectively. With $z-1=\delta e^{i\theta}$, $\int dz/(z-1)=i\int d\theta$, while the bounded remainder contributes $O(\delta)$. The upper semicircle runs from $\theta=\pi$ to $\theta=0$ (clockwise) and contributes $-i\pi$; the lower one runs from $\theta=0$ to $\theta=-\pi$ (also clockwise) and contributes $-i\pi q$. On the origin circle $|F|\le\epsilon^{-p}/(1-\epsilon)$ and on the outer circle $|F|\le R^{-p}/(R-1)$; times the lengths $2\pi\epsilon$ and $2\pi R$, these give $O(\epsilon^{1-p})$ and $O(R^{-p})$, so both vanish. No pole remains inside the indented contour, hence

$$(1-q)I-i\pi(1+q)=0.$$

Finally, factoring out $e^{-\pi ip}$,

$$1+q=e^{-\pi ip}(e^{\pi ip}+e^{-\pi ip})=2e^{-\pi ip}\cos\pi p,\qquad 1-q=2ie^{-\pi ip}\sin\pi p,$$

and therefore

$$\boxed{⨍_0^\infty \frac{x^{-p}}{x-1}\,dx = i\pi\frac{1+q}{1-q} = i\pi\frac{\cos\pi p}{i\sin\pi p} = \pi\cot\pi p}.$$

An independent check of the sign: $x = 1/u$ gives $dx = -du/u^2$, $x^{-p} = u^p$ and $x-1 = (1-u)/u$, so $x^{-p}\,dx/(x-1) = -u^{p-1}\,du/(1-u)$, while $x = 1+\delta, \infty$ become $u = 1/(1+\delta), 0$. Thus the part beyond $1+\delta$ becomes $\int_0^{1/(1+\delta)} u^{p-1}(1-u)^{-1}\,du$. Writing the part below $1-\delta$ as $-\int_0^{1-\delta} x^{-p}(1-x)^{-1}\,dx$, splitting the u-integral at $1-\delta$, and renaming u as x in its first piece, the two integrals in the definition of I add up to

$$\int_0^{1-\delta} \frac{x^{p-1}-x^{-p}}{1-x}\,dx + \int_{1-\delta}^{1/(1+\delta)} \frac{u^{p-1}}{1-u}\,du.$$

The last integral runs over an interval of length $\delta^2/(1+\delta)$, with integrand at most $(1-\delta)^{p-1}(1+\delta)/\delta$, so it vanishes as $\delta \to 0$. Hence

$$⨍_0^\infty \frac{x^{-p}}{x-1}\,dx = \int_0^1 \frac{x^{p-1}-x^{-p}}{1-x}\,dx,$$

an ordinary integral, since the numerator vanishes at $x = 1$. For $0 < p < \frac{1}{2}$ we have $p-1 < -p$, so the integrand is positive on $(0,1)$; the integral is positive, like $\pi\cot\pi p$, whereas $-\pi\cot\pi p < 0$.

11.7.5 Explain why Eq. (11.88) is not directly applicable to the product expansion of $\sin z$. Show how the expansion, Eq. (11.89), can be obtained by expanding instead $\sin z/z$.

Solution.

The product formula quoted in the text contains the factors $f(0)$ and $\exp[zf'(0)/f(0)]$. Since $\sin 0 = 0$, direct substitution would both annihilate the product and divide by zero. Instead let

$$g(z) = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The Taylor expansion $g(z) = 1 - z^2/3! + z^4/5! - \cdots$ gives $g(0) = 1$ and $g'(0) = 0$. Its zeros are the nonzero points $\pm n\pi$, each simple. To justify the product for this specific function, differentiate $\ln g = \ln\sin z - \ln z$ and use the cotangent pole expansion (11.81), together with $\frac{d}{dz}\operatorname{Log}[1 - z^2/(n^2\pi^2)] = [-2z/(n^2\pi^2)]/[1 - z^2/(n^2\pi^2)]$: on $|z| < \pi$,

$$\frac{g'(z)}{g(z)} = \cot z - \frac{1}{z} = \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2\pi^2} = \sum_{n=1}^{\infty} \frac{d}{dz}\operatorname{Log}\left(1 - \frac{z^2}{n^2\pi^2}\right).$$

The removable value at zero is understood. On every smaller closed disk the summands are bounded by a constant times n^{-2}, so the series converges uniformly and may be integrated from zero to z. Choose the analytic logarithms that vanish at zero; since g has no zero in this disk and $g(0) = 1$, integration gives

$$\operatorname{Log} g(z) = \sum_{n=1}^{\infty} \operatorname{Log}\left(1 - \frac{z^2}{n^2\pi^2}\right).$$

Exponentiating yields the paired product. Equivalently, the canonical factors of (11.88) for g, with $g(0) = 1$, $g'(0) = 0$ (so the exponential prefactor is 1) and the opposite zeros $\pm n\pi$, give

$$g(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n\pi}\right) e^{z/(n\pi)} \left(1 + \frac{z}{n\pi}\right) e^{-z/(n\pi)}.$$

Pairing the positive and negative zeros cancels the exponentials and multiplies the two linear factors:

$$\boxed{\sin z = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2\pi^2}\right)}.$$

The paired product converges uniformly on compact sets because $\sum_{n\geq 1} |z|^2/(n^2\pi^2)$ does. It defines an entire function, so its equality to g on $|z| < \pi$ extends to the whole plane by the identity theorem. Multiplication by z proves the displayed formula, including its simple zero and derivative one at zero. The statement preceding (11.88) needs additional restrictions: zeros and the two values $f(0), f'(0)$ do not determine an arbitrary entire function. For example, e^{z^2} has no zeros, $f(0) = 1$, and $f'(0) = 0$, while that formula would give 1. The logarithmic-derivative argument above establishes the sine product without this unjustified generalization.

11.7.6 Starting from the observations

1. $f(z) = a_n z^n$ has n zeros counted with multiplicity, where $a_n \neq 0$ and $n \geq 1$, and
2. for sufficiently large $|R|$, $\left|\sum_{m=0}^{n-1} a_m R^m\right| < |a_n R^n|$,

use Rouché's theorem to prove the fundamental theorem of algebra (namely that every polynomial of degree n has n roots).

Solution.

The monomial $a_n z^n$ has the single zero $z = 0$, so it has n zeros only when they are counted with multiplicity; and it has degree $n \geq 1$ only if $a_n \neq 0$. Let

$$P(z) = a_n z^n + \sum_{m=0}^{n-1} a_m z^m, \qquad a_n \neq 0, \quad S = \sum_{m=0}^{n-1} |a_m|.$$

Choose $R > \max(1, S/|a_n|)$. At every point of $|z| = R$, the triangle inequality, $R^m \leq R^{n-1}$ for $m \leq n-1$ (as $R > 1$), and $S < |a_n|R$ give

$$\left|\sum_{m=0}^{n-1} a_m z^m\right| \leq \sum_{m=0}^{n-1} |a_m| R^m \leq S R^{n-1} < |a_n| R^n = |a_n z^n|.$$

Rouché's theorem (if $|g| < |f|$ on a closed contour, then f and $f+g$ have the same number of zeros inside it), with $f = a_n z^n$ and $g = P - a_n z^n$, says that P and $a_n z^n$ have the same number of zeros inside this circle. The monomial has its sole zero at zero, of multiplicity n, so P has exactly n zeros there. The same inequality holds for every larger radius. Any additional finite zero would lie within some such larger circle and increase the count, which is impossible. Thus P has exactly n complex roots with multiplicity.

11.7.7 Using Rouché's theorem, show that all the zeros of $F(z) = z^6 - 4z^3 + 10$ lie between the circles $|z| = 1$ and $|z| = 2$.

Solution.

Rouché's theorem: if $|g| < |f|$ on a closed contour, then f and $f+g$ have the same number of zeros inside it. On $|z| = 1$ compare F with the constant $f = 10$, taking $g = z^6 - 4z^3$:

$$|z^6 - 4z^3| \leq |z|^6 + 4|z|^3 = 1 + 4 = 5 < 10.$$

The constant has no zeros, so Rouché's theorem gives no zeros in $|z| < 1$; on the circle itself $|F| \geq 10 - 5 > 0$, so there is no zero there either. On $|z| = 2$, compare with $f = z^6$, taking $g = -4z^3 + 10$:

$$|-4z^3 + 10| \leq 4(2^3) + 10 = 42 < 2^6 = 64.$$

Since z^6 has six zeros (at the origin, with multiplicity), there are six zeros of F in $|z| < 2$, counted with multiplicity, and none on its boundary, where $|F| \geq 64 - 42 > 0$. A degree-six polynomial has exactly six zeros, so all lie in $1 < |z| < 2$. As an independent check, setting $w = z^3$ yields $w^2 - 4w + 10 = 0$, hence $w = (4 \pm \sqrt{16 - 40})/2 = 2 \pm i\sqrt{6}$. Both values have modulus $\sqrt{4+6} = \sqrt{10}$, so every root satisfies $|z| = |w|^{1/3} = 10^{1/6} \approx 1.468$, strictly between one and two. The Instructor's Manual claims $|z^6 + 10| > |-4z^3|$ for all points inside $|z| = 2$. It fails when $z^3 = i\sqrt{10}$: the left side is zero and the right side is positive. The comparisons above apply on the two boundary circles, as required by Rouché's theorem.

11.7.8 Derive the pole expansions of $\sec z$ and $\csc z$ given in Eqs. (11.82) and (11.83).

Solution.

A simple pole of $1/h$ at a simple zero a of h has residue $1/h'(a)$. Thus, with $a_n = (n + \frac{1}{2})\pi$, $(\cos z)' = -\sin z$, $\sin(\pm a_n) = \pm(-1)^n$, and $(\sin z)' = \cos z$, $\cos n\pi = (-1)^n$,

$$\operatorname{Res}_{a_n} \sec z = (-1)^{n+1}, \quad \operatorname{Res}_{-a_n} \sec z = (-1)^n, \qquad \operatorname{Res}_{n\pi} \csc z = (-1)^n.$$

To justify a pole expansion, integrate $F(\zeta)/(\zeta - z)$ on expanding rectangles, with z fixed inside. For $\sec\zeta$, use vertical sides $\Re\zeta = \pm N\pi$ and horizontal sides $\Im\zeta = \pm N\pi$. On a vertical side $\cos(\pm N\pi + iy) = \cos N\pi \cosh y \mp i \sin N\pi \sinh y = (-1)^N \cosh y$, so $|\sec\zeta| = \operatorname{sech} y$, and $|\zeta - z| \geq N\pi - |z|$; its integral is at most $(N\pi - |z|)^{-1} \int_{\mathbb{R}} \operatorname{sech} y\, dy = \pi/(N\pi - |z|)$, which is $O(N^{-1})$; here $u = e^y$ gives $\operatorname{sech} y = 2u/(u^2+1)$ and $dy = du/u$, so $\int_{\mathbb{R}} \operatorname{sech} y\, dy = 2\int_0^\infty du/(1+u^2) = \pi$. On a horizontal side $|\cos(x \pm iN\pi)|^2 = \cos^2 x + \sinh^2 N\pi \geq \sinh^2 N\pi$, so its integral is at most $2N\pi/[(N\pi - |z|)\sinh N\pi]$ and tends to zero exponentially. For $\csc\zeta$, choose vertical sides $\Re\zeta = \pm(N + \frac{1}{2})\pi$ and the same horizontal heights. There $\sin(\pm(N+\frac{1}{2})\pi + iy) = \pm(-1)^N \cosh y$, so $|\csc\zeta| = \operatorname{sech} y$, while $|\sin(x \pm iN\pi)|^2 = \sin^2 x + \sinh^2 N\pi$; the estimates are the same.

Inside the rectangle, $F(\zeta)/(\zeta - z)$ has residue $F(z)$ at $\zeta = z$ and residue $\operatorname{Res}_a F/(a - z)$ at each pole a of F. Since the contour integral tends to zero, the residue theorem gives, in the limit,

$$0 = F(z) + \sum_a \frac{\operatorname{Res}_a F}{a - z}, \qquad F(z) = \sum_a \frac{\operatorname{Res}_a F}{z - a}.$$

The Nth rectangle contains the poles $\pm a_n$, $n < N$, for sec, and $0, \pm\pi, \ldots, \pm N\pi$ for csc, so the sums are limits of symmetric partial sums. For $\sec z$ pair the poles a_n and $-a_n$:

$$\frac{(-1)^{n+1}}{z - a_n} + \frac{(-1)^n}{z + a_n} = \frac{2(-1)^n a_n}{a_n^2 - z^2}.$$

For $\csc z$ separate the pole at zero and pair $\pm n\pi$:

$$\frac{(-1)^n}{z - n\pi} + \frac{(-1)^n}{z + n\pi} = \frac{2(-1)^n z}{z^2 - n^2\pi^2}.$$

Consequently, using $2a_n = (2n+1)\pi$ and $a_n^2 = (n + \frac{1}{2})^2\pi^2$, off their poles,

$$\boxed{\sec z = \pi \sum_{n=0}^{\infty} \frac{(-1)^n(2n+1)}{(n+\frac{1}{2})^2\pi^2 - z^2}}, \qquad \boxed{\csc z = \frac{1}{z} + 2z \sum_{n=1}^{\infty} \frac{(-1)^n}{z^2 - n^2\pi^2}}.$$

The second paired series is absolutely locally uniformly convergent. In the first, the leading term is the convergent alternating series $2(-1)^n/a_n$, and subtracting that leading term leaves $2(-1)^n z^2/[a_n(a_n^2 - z^2)]$, a locally uniform $O(n^{-3})$ series. This specifies the pairing and convergence needed for the pole expansion.

11.7.9 Given that $f(z) = (z^2 - 3z + 2)/z$, apply a partial fraction decomposition to f'/f and show directly that $\oint_C f'(z)/f(z)\,dz = 2\pi i(N_f - P_f)$, where N_f and P_f are, respectively, the numbers of zeros and poles encircled by C (including their multiplicities), for a simple positively oriented contour avoiding $0, 1, 2$.

Solution.

Here $f(z) = z - 3 + 2/z = (z-1)(z-2)/z$ and $f'(z) = 1 - 2/z^2 = (z^2-2)/z^2$, so

$$\frac{f'}{f} = \frac{z^2 - 2}{z(z-1)(z-2)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z-2}.$$

Multiplying by each linear factor and setting z equal to its root gives $A = (-2)/[(-1)(-2)] = -1$, $B = (-1)/[1 \cdot (-1)] = 1$, and $C = 2/(2 \cdot 1) = 1$:

$$\frac{f'}{f} = \frac{1}{z-1} + \frac{1}{z-2} - \frac{1}{z}.$$

For a positively oriented simple closed contour through none of $0, 1, 2$, $\oint_C dz/(z-c)$ equals $2\pi i$ if c is inside (Cauchy's integral formula) and zero if c is outside (Cauchy's theorem). So each of the first two fractions integrates to $2\pi i$ if its pole is inside and to zero otherwise. The last contributes $-2\pi i$ if zero is inside. The zeros of f are $1, 2$ and its only pole is 0, all simple. Hence, writing $\mathbf{1}_{\{c \text{ inside}\}}$ for 1 if c is inside C and 0 otherwise,

$$\boxed{\oint_C \frac{f'}{f}\,dz = 2\pi i\big(\mathbf{1}_{\{1 \text{ inside}\}} + \mathbf{1}_{\{2 \text{ inside}\}} - \mathbf{1}_{\{0 \text{ inside}\}}\big) = 2\pi i(N_f - P_f)}.$$

For a general closed curve the same formula uses winding-number weighted counts; the unweighted interpretation assumes a simple positively oriented contour avoiding all zeros and poles.

11.7.10 The statement that the integral halfway around a singular point is equal to one-half the integral all the way around was limited to the case of a simple pole (and, even then, holds only in the limit of a shrinking semicircle traversed in the same sense as the circle). Show, by a specific example, that

$$\int_{\text{Semicircle}} f(z)\,dz = \frac{1}{2}\oint_{\text{Circle}} f(z)\,dz$$

does not necessarily hold if the integral encircles a pole of higher order.

Hint. Try $f(z) = z^{-2}$.

Solution.

Even for a simple pole the equality holds only in the limit: if $f = a_{-1}/(z - z_0) + g$ with g analytic, the semicircle $z - z_0 = Re^{i\theta}$, $0 \le \theta \le \pi$, gives $i\pi a_{-1} + O(R)$, which tends to $\frac{1}{2}(2\pi i a_{-1})$, half the full-circle integral, as $R \to 0$. Take $f(z) = z^{-2}$ and the upper semicircle $z = Re^{i\theta}$, $0 \le \theta \le \pi$. Since $dz = iRe^{i\theta}\,d\theta$, the integrand is $iRe^{i\theta}/(R^2e^{2i\theta})\,d\theta = (i/R)e^{-i\theta}\,d\theta$, and with the antiderivative $e^{-i\theta}/(-i)$ and $e^{-i\pi} = -1$,

$$\int_{\text{upper arc}} \frac{dz}{z^2} = \frac{i}{R}\int_0^{\pi} e^{-i\theta}\,d\theta = -\frac{1}{R}[e^{-i\theta}]_0^{\pi} = -\frac{1}{R}(-1 - 1) = \frac{2}{R}.$$

On the full circle the same computation gives $-R^{-1}[e^{-i\theta}]_0^{2\pi} = 0$, since $e^{-2\pi i} = 1$; equivalently, the residue of z^{-2} at zero is zero. Thus $2/R \neq \frac{1}{2}(0)$ for every $R > 0$; the semicircle integral even diverges as the radius shrinks. The cancellation on the other half of the circle is essential for a pole of order two.

11.7.11 A function $f(z)$ is analytic along the real axis except for a third-order pole at $z = x_0$. The Laurent expansion about $z = x_0$ has the form

$$f(z) = \frac{a_{-3}}{(z - x_0)^3} + \frac{a_{-1}}{z - x_0} + g(z),$$

with $g(z)$ analytic at $z = x_0$. Assume the integrals converge separately at both infinite endpoints. Show that the Cauchy principal value technique is applicable, in the sense that

(a) $\lim_{\delta \to 0} \left\{ \int_{-\infty}^{x_0 - \delta} f(x)\, dx + \int_{x_0 + \delta}^{\infty} f(x)\, dx \right\}$ is finite.

(b) $\lim_{\delta \downarrow 0} \int_{C_{x_0}} f(z)\, dz = \pm i\pi a_{-1}$,

where C_{x_0} denotes a semicircle of radius δ about $z = x_0$.

Solution.

The hypothesis controls the singularity at x_0, but does not ensure convergence at infinity. Part (a) needs convergent tails, or a separately specified principal value there. For instance $f(z) = (z - x_0)^{-3} + 1$ satisfies the local hypothesis and has divergent tails. Also, the equality in (b) is a limit as the semicircle shrinks. Assume the tails in (a) converge, and choose $h > 0$ within the Laurent expansion's disk. Put $t = x - x_0$. The singular terms are odd, so for $j = 1, 3$ and every $0 < \delta < h$,

$$\int_{-h}^{-\delta} t^{-j}\, dt + \int_{\delta}^{h} t^{-j}\, dt = 0;$$

indeed, the substitution $t = -u$, $dt = -du$, gives $\int_{-h}^{-\delta} t^{-j}\, dt = \int_h^{\delta} (-u)^{-j}(-du) = -\int_{\delta}^{h} u^{-j}\, du$ for odd j. Meanwhile continuity of g gives

$$\lim_{\delta \downarrow 0} \left(\int_{x_0 - h}^{x_0 - \delta} g(x)\, dx + \int_{x_0 + \delta}^{x_0 + h} g(x)\, dx \right) = \int_{x_0 - h}^{x_0 + h} g(x)\, dx.$$

Adding the two nonsingular convergent tails proves local principal-value existence and thus the qualified statement (a).

On a semicircle set $z - x_0 = \delta e^{i\theta}$, so $dz = i\delta e^{i\theta}\, d\theta$ and $dz/(z - x_0)^3 = i\delta^{-2} e^{-2i\theta}\, d\theta$. Then

$$\int \frac{a_{-3}\, dz}{(z - x_0)^3} = i a_{-3} \delta^{-2} \int_{\theta_1}^{\theta_2} e^{-2i\theta}\, d\theta = i a_{-3} \delta^{-2} \left[\frac{e^{-2i\theta}}{-2i} \right]_{\theta_1}^{\theta_2} = 0, \qquad \theta_2 - \theta_1 = \pm\pi,$$

because the endpoint values of $e^{-2i\theta}$ are equal ($e^{\mp 2\pi i} = 1$). The simple pole contributes $a_{-1} \int i\, d\theta = i a_{-1}(\theta_2 - \theta_1) = \pm i\pi a_{-1}$. Finally, by the ML inequality, the analytic part is bounded by the length $\pi\delta$ times $\max_{|z - x_0| = \delta} |g(z)|$, which tends to zero. Therefore

$$\boxed{\lim_{\delta \downarrow 0} \int_{C_{x_0}} f(z)\, dz = \pm i\pi a_{-1}},$$

with plus for counterclockwise traversal and minus for clockwise traversal.

11.7.12 The unit step function is defined as (compare Exercise 1.11.9)

$$u(s-a)=\begin{cases}0, & s<a\\ 1, & s>a.\end{cases}$$

Show that $u(s)$ has the integral representations

(a) $u(s)=\lim_{\epsilon\to 0^+}\dfrac{1}{2\pi i}\displaystyle\int_{-\infty}^{\infty}\frac{e^{ixs}}{x-i\epsilon}\,dx.$

(b) $u(s)=\dfrac{1}{2}+\dfrac{1}{2\pi i}\,⨍_{-\infty}^{\infty}\dfrac{e^{ixs}}{x}\,dx.$

Note. The parameter s is real and nonzero; the symmetric convention at $s=0$ is discussed below.

Solution.

The published erratum removes the principal-value symbol in (a): for fixed $\epsilon>0$ the pole is off the real axis. For $s\neq 0$ the ordinary oscillatory improper integral exists. At $s=0$ its two tails do not converge separately; a symmetric limit assigns $u(0)=1/2$, while the problem's definition leaves $u(0)$ unspecified. The printed cross-reference, Exercise 1.15.13, does not exist in this edition, whose Chapter 1 ends with Section 1.11; the step function is introduced in Exercise 1.11.9 as the limit of $u_n(x)=\frac{1}{2}(1+\tanh nx)$. For $s>0$, close the contour in the upper half-plane. The exponential has modulus $|e^{izs}|=e^{-s\,\Im z}\le 1$ there, and $1/(z-i\epsilon)=O(R^{-1})$ on the arc $|z|=R$, so Jordan's estimate makes the large semicircle contribution tend to zero. The sole pole is $z=i\epsilon$ and has residue $e^{i(i\epsilon)s}=e^{-\epsilon s}$. Thus

$$\frac{1}{2\pi i}\int_{\mathbb{R}}\frac{e^{ixs}}{x-i\epsilon}\,dx=\frac{2\pi i\,e^{-\epsilon s}}{2\pi i}=e^{-\epsilon s}\longrightarrow 1.$$

For $s<0$, close below, where $|e^{izs}|=e^{|s|\Im z}\le 1$ and Jordan's estimate applies again. There are no poles below the real axis, so the clockwise contour gives zero. These are exactly the two prescribed step values.

For (b), evaluate $J(s)=⨍_{\mathbb{R}}e^{ixs}/x\,dx$ directly. The integrand e^{izs}/z has a simple pole at zero with residue 1. For $s>0$ close above and indent the origin into the upper half-plane. The small clockwise arc has limit $-i\pi\cdot 1$; the indented domain contains no pole. Hence $J(s)-i\pi=0$. For $s<0$, close below; the lower indentation is counterclockwise and contributes $+i\pi$, so $J(s)+i\pi=0$. Consequently $J(s)=i\pi\operatorname{sgn}s$ for $s\neq 0$, that is, $J(s)/(2\pi i)=\frac{1}{2}\operatorname{sgn}s$, and

$$\boxed{\frac{1}{2}+\frac{J(s)}{2\pi i}=\frac{1+\operatorname{sgn}s}{2}=u(s)\quad(s\neq 0)}.$$

For completeness the connection between the prescriptions is visible in

$$\frac{1}{x-i\epsilon}=\frac{x}{x^2+\epsilon^2}+i\frac{\epsilon}{x^2+\epsilon^2}.$$

The second term has integral $i\pi$ in the limit: put $x=\epsilon t$, then use dominated convergence with $(1+t^2)^{-1}$. The first tends to the principal-value term: the difference $x/(x^2+\epsilon^2)-1/x=-\epsilon^2/[x(x^2+\epsilon^2)]$ is odd, so pairing x with $-x$ cancels its product with $\cos xs$ and doubles that with $i\sin xs$; then $x=\epsilon t$ gives

$$\begin{aligned}\int_{\mathbb{R}}\frac{x\,e^{ixs}}{x^2+\epsilon^2}\,dx-J(s)&=-2i\int_0^{\infty}\frac{\epsilon^2\sin xs}{x(x^2+\epsilon^2)}\,dx\\&=-2i\int_0^{\infty}\frac{\sin(\epsilon ts)}{t(1+t^2)}\,dt,\end{aligned}$$

whose modulus is at most $2\epsilon|s|\int_0^\infty dt/(1+t^2) = \pi\epsilon|s|$, by $|\sin y| \le |y|$. At $s=0$, the symmetric integral of the first term is zero and that of the second is $i\pi$, giving $u(0) = 1/2$, the value of the symmetric convention, in both representations.

11.8 Evaluation of Definite Integrals

11.8.1 Generalizing Example 11.8.1, for real a, b, show that

$$\int_0^{2\pi} \frac{d\theta}{a \pm b\cos\theta} = \int_0^{2\pi} \frac{d\theta}{a \pm b\sin\theta} = \frac{2\pi}{(a^2-b^2)^{1/2}}, \quad \text{for } a > |b|.$$

What happens if $|b| > |a|$?

Solution.

Take a, b real (the condition $a > |b|$ and the final question presuppose this) with $a > |b|$, and first consider $a + b\cos\theta$. If $b = 0$, the integral is $2\pi/a$. Otherwise set $z = e^{i\theta}$, $d\theta = dz/(iz)$, $\cos\theta = (z + z^{-1})/2$, and $D = \sqrt{a^2 - b^2} > 0$. Then $a + b\cos\theta = (bz^2 + 2az + b)/(2z)$, so

$$I = \oint_{|z|=1} \frac{2z}{bz^2 + 2az + b}\,\frac{dz}{iz} = \frac{2}{i}\oint_{|z|=1} \frac{dz}{bz^2 + 2az + b}.$$

By the quadratic formula the roots of the denominator are $(-a \pm D)/b$; multiplying numerator and denominator by $a + D$ and using $D^2 - a^2 = -b^2$,

$$r_+ = \frac{-a + D}{b} = -\frac{b}{a + D}, \qquad r_- = \frac{-a - D}{b}, \qquad r_+ r_- = \frac{b}{b} = 1.$$

Since $|r_+| = |b|/(a + D) < 1$ and $|r_-| = 1/|r_+| > 1$, only r_+ is inside. The residue of the rational integrand there is the reciprocal of the derivative $2bz + 2a$ of its denominator at r_+:

$$\frac{1}{2br_+ + 2a} = \frac{1}{2(-a + D) + 2a} = \frac{1}{2D}.$$

The residue theorem gives

$$\boxed{I = \frac{2}{i}(2\pi i)\frac{1}{2D} = \frac{2\pi}{\sqrt{a^2 - b^2}}}.$$

The derivation holds for either sign of b, so changing b to $-b$ leaves the result unchanged. Since $\sin\theta = \cos(\theta - \pi/2)$, the substitution $\varphi = \theta - \pi/2$ turns the sine integrals into cosine integrals over $[-\pi/2, 3\pi/2]$, and a full-period integral of a 2π-periodic function does not depend on where the period starts, proving the other two formulas.

If $|b| > |a|$, the denominator vanishes at the angles θ_0 with $\cos\theta_0 = -a/b$ (or the corresponding sine values), which exist because $|a/b| < 1$. Its derivative $-b\sin\theta_0$ there is nonzero because $|\cos\theta_0| < 1$. Locally the reciprocal therefore has the form $c/(\theta - \theta_0) + O(1)$, $c \neq 0$, which is not integrable. The ordinary improper integrals then do not exist. An analytic continuation of the displayed square root is not their value; any principal-value prescription must be specified separately. If principal values are taken at both zeros, the roots $r_\pm = (-a \pm i\sqrt{b^2 - a^2})/b$ lie on $|z| = 1$, and each contributes πi times its residue (half the usual $2\pi i$) to the contour integral, namely $\pi i/(2br_\pm + 2a) = \pi i/(\pm 2i\sqrt{b^2 - a^2})$. These cancel, so the principal value is zero.

11.8.2 Show that

$$\int_0^{\pi} \frac{d\theta}{(a+\cos\theta)^2} = \frac{\pi a}{(a^2-1)^{3/2}}, \quad a > 1.$$

Solution.

For $a > 1$, the substitution $\theta \mapsto 2\pi - \theta$, which leaves $\cos\theta$ unchanged, maps $[\pi, 2\pi]$ onto $[0, \pi]$; so the integral over $[0, \pi]$ is half the full-period integral of Exercise 11.8.1 with $b = 1$:

$$J(a) = \int_0^{\pi} \frac{d\theta}{a+\cos\theta} = \frac{1}{2} \cdot \frac{2\pi}{\sqrt{a^2-1}} = \frac{\pi}{\sqrt{a^2-1}}.$$

On any closed interval of parameters lying in $(1, \infty)$, the denominator is bounded away from zero, and the a-derivative $-(a+\cos\theta)^{-2}$ of the integrand is continuous. We may therefore differentiate under the finite integral, and by the chain rule $\frac{d}{da}(a^2-1)^{-1/2} = -\frac{1}{2}(a^2-1)^{-3/2} \cdot 2a$:

$$J'(a) = -\int_0^{\pi} \frac{d\theta}{(a+\cos\theta)^2} = -\pi a (a^2-1)^{-3/2}.$$

Multiplying by -1 proves

$$\boxed{\int_0^{\pi} \frac{d\theta}{(a+\cos\theta)^2} = \frac{\pi a}{(a^2-1)^{3/2}}}.$$

11.8.3 Show that

$$\int_0^{2\pi} \frac{d\theta}{1-2t\cos\theta+t^2} = \frac{2\pi}{1-t^2}, \quad \text{for } |t| < 1.$$

What happens if $|t| > 1$? What happens if $|t| = 1$?

Solution.

For $t \neq 0$ write, with $z = e^{i\theta}$ and $z + 1/z = 2\cos\theta$,

$$(1-tz)(1-t/z) = 1 - t(z+1/z) + t^2 = 1 - 2t\cos\theta + t^2.$$

With $d\theta = dz/(iz)$ and $z(1-t/z) = z - t$,

$$I(t) = \oint_{|z|=1} \frac{dz}{iz(1-tz)(1-t/z)} = \frac{1}{i} \oint_{|z|=1} \frac{dz}{(1-tz)(z-t)}.$$

The two poles are $z = t$ and $z = 1/t$. If $|t| < 1$, only t is inside and its residue is $1/(1-t\cdot t) = 1/(1-t^2)$, so $I = (1/i)(2\pi i)/(1-t^2)$. If $|t| > 1$, only $1/t$ is inside; writing $1 - tz = -t(z - 1/t)$, its residue is $1/[-t(1/t - t)] = 1/(t^2-1)$. Therefore

$$\boxed{I(t) = \begin{cases} 2\pi/(1-t^2), & |t| < 1, \\ 2\pi/(t^2-1), & |t| > 1. \end{cases}}$$

The case $t = 0$ directly gives 2π, consistent with the first branch. These calculations also hold for complex t off the unit circle.

For real t with $|t| = 1$, the denominator is $2(1-\cos\theta) = 4\sin^2(\theta/2)$ at $t = 1$ and $2(1+\cos\theta) = 4\cos^2(\theta/2)$ at $t = -1$. Each has a quadratic zero (at $\theta = 0, 2\pi$ and at $\theta = \pi$, respectively), near which the positive integrand behaves like $(\theta - \theta_0)^{-2}$, and the integral diverges to $+\infty$. If complex t is admitted, the other points $|t| = 1$ put simple poles on the integration path instead. Those ordinary improper integrals are also undefined; they need not diverge to positive infinity.

11.8.4 Evaluate

$$\int_0^{2\pi} \frac{\cos 3\theta \, d\theta}{5 - 4\cos\theta}.$$

ANS. $\pi/12$.

Solution.

Because $\sin 3\theta/(5 - 4\cos\theta)$ changes sign under $\theta \mapsto 2\pi - \theta$, which maps $[0, 2\pi]$ onto itself, its integral equals its own negative and vanishes. We may thus replace $\cos 3\theta$ by $e^{3i\theta} = \cos 3\theta + i \sin 3\theta$ without changing the desired integral. With $z = e^{i\theta}$, $e^{3i\theta} = z^3$, $d\theta = dz/(iz)$, and $5 - 4\cos\theta = 5 - 2(z + z^{-1}) = (-2z^2 + 5z - 2)/z$, so

$$I = \oint_{|z|=1} \frac{z^3 \cdot z}{-2z^2 + 5z - 2} \frac{dz}{iz} = \frac{1}{i} \oint_{|z|=1} \frac{z^3 \, dz}{-2z^2 + 5z - 2}.$$

The denominator is $-(2z - 1)(z - 2) = -2(z - \frac{1}{2})(z - 2)$, so only $z = 1/2$ lies inside. Its residue is

$$\left. \frac{z^3}{-2(z-2)} \right|_{z=1/2} = \frac{1/8}{(-2)(-3/2)} = \frac{1/8}{3} = \frac{1}{24}.$$

It follows that

$$\boxed{I = \frac{1}{i}(2\pi i)\frac{1}{24} = \frac{\pi}{12}}.$$

11.8.5 With the calculus of residues, show that

$$\int_0^{\pi} \cos^{2n}\theta \, d\theta = \pi \frac{(2n)!}{2^{2n}(n!)^2} = \pi \frac{(2n-1)!!}{(2n)!!}, \quad n = 0, 1, 2, \ldots.$$

The double factorial notation is defined in Eq. (1.76).

Hint. $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \frac{1}{2}(z + z^{-1}), \quad |z| = 1.$

Solution.

The even power has period π, since $\cos^{2n}(\theta + \pi) = (-\cos\theta)^{2n} = \cos^{2n}\theta$, so twice the required integral is its integral over $[0, 2\pi]$. Set $z = e^{i\theta}$, so that $\cos^{2n}\theta = 2^{-2n}(z + z^{-1})^{2n}$, $d\theta = dz/(iz)$, and $(z + z^{-1})^{2n} = z^{-2n}(z^2 + 1)^{2n}$:

$$2I_n = \frac{1}{i\,2^{2n}} \oint_{|z|=1} \frac{(z + z^{-1})^{2n}}{z} \, dz = \frac{1}{i\,2^{2n}} \oint_{|z|=1} \frac{(z^2+1)^{2n}}{z^{2n+1}} \, dz.$$

Expand the numerator, $(z^2 + 1)^{2n} = \sum_{j=0}^{2n} \binom{2n}{j} z^{2j}$. The z^{-1} term occurs when $2j - (2n + 1) = -1$, or $j = n$, and has coefficient $\binom{2n}{n}$, which is therefore the residue. Hence, by the residue theorem,

$$2I_n = \frac{2\pi i}{i\,2^{2n}} \binom{2n}{n} = \frac{2\pi}{2^{2n}} \binom{2n}{n}, \qquad \boxed{I_n = \pi \frac{(2n)!}{2^{2n}(n!)^2}}.$$

Finally $(2n)!! = 2 \cdot 4 \cdots (2n) = 2^n n!$ and $(2n-1)!! = (2n)!/(2n)!! = (2n)!/(2^n n!)$, so $(2n-1)!!/(2n)!! = (2n)!/(2^{2n}(n!)^2)$, the factorial ratio above. For $n = 0$ the empty-product convention $(-1)!! = 0!! = 1$ gives $I_0 = \pi$ as required.

11.8.6 Verify that simplification of the expression in Eq. (11.112) yields the result given in Eq. (11.113).

Solution.

In the textbook example that contains Eqs. (11.112) and (11.113), $I = \int_0^\infty \ln x/(1+x^3)\,dx$. Put $q = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. Equation (11.112), as printed in the textbook, reads

$$(1-q)I - \frac{2\pi i}{3} q\,\frac{2\pi}{3\sqrt{3}} = (2\pi i)\frac{\pi i}{9} q^{-1}.$$

Move the second term to the right and use $i^2 = -1$:

$$(1-q)I = -\frac{2\pi^2}{9} q^{-1} + \frac{4\pi^2 i}{9\sqrt{3}} q.$$

Since $q^{-1} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ and $iq = -\frac{\sqrt{3}}{2} - \frac{i}{2}$, the two terms are $-\frac{2\pi^2}{9}q^{-1}$ and $\frac{4\pi^2}{9\sqrt{3}}(iq)$, that is,

$$\frac{\pi^2}{9} + \frac{i\pi^2\sqrt{3}}{9}, \qquad -\frac{2\pi^2}{9} - \frac{2i\pi^2}{9\sqrt{3}}.$$

Since $\sqrt{3}/9 - 2/(9\sqrt{3}) = (3-2)/(9\sqrt{3})$, their sum is $-\pi^2/9 + i\pi^2/(9\sqrt{3}) = -(2\pi^2/27)(\frac{3}{2} - i\frac{\sqrt{3}}{2})$, where $\sqrt{3}/27 = 1/(9\sqrt{3})$ was used. But $1 - q = \frac{3}{2} - i\frac{\sqrt{3}}{2} \neq 0$. Canceling this factor gives

$$\boxed{I = -\frac{2\pi^2}{27}},$$

which is Eq. (11.113).

11.8.7 Complete the details of Example 11.8.8 by verifying that there is no contribution to the contour integral from either the small or the large circles of the contour, and that Eq. (11.115) simplifies to the result given as (11.116).

Solution.

Use $F(z) = z^p/(1+z^2)$ with $0 < \arg z < 2\pi$, and let the keyhole run outward along the upper bank and inward along the lower bank. This orientation reverses the incorrectly drawn lower arrows in the textbook's Fig. 11.26, as specified in the published errata. On the small circle $|z| = \epsilon < 1$, $|z^p| = \epsilon^p$ and $|1+z^2| \geq 1 - \epsilon^2$; with the length $2\pi\epsilon$,

$$\left| \int_{\text{small}} F(z)\,dz \right| \leq \frac{2\pi\epsilon^{p+1}}{1-\epsilon^2} \longrightarrow 0.$$

On the large circle $|z| = R > 1$, $|z^p| = R^p$ and $|1+z^2| \geq R^2 - 1$:

$$\left| \int_{\text{large}} F(z)\,dz \right| \leq \frac{2\pi R^{p+1}}{R^2-1} = O(R^{p-1}) \longrightarrow 0.$$

Thus the two arc contributions vanish for $-1 < p < 1$, in particular for the stated $0 < p < 1$.

The upper bank ($\arg z = 0$, $z^p = x^p$) contributes I; on the lower bank $z^p = x^p e^{2\pi i p}$ and the direction is inward, so it contributes $-e^{2\pi i p} I$. The residues at i and $-i$ are $z^p/(z+i)$ at i and $z^p/(z-i)$ at $-i$; with arguments $\pi/2$ and $3\pi/2$ on this branch they are

$$\frac{e^{ip\pi/2}}{2i}, \qquad -\frac{e^{3ip\pi/2}}{2i}.$$

Multiplying their sum by $2\pi i$ and factoring out $e^{ip\pi}$,

$$(1-e^{2\pi ip})I=\pi(e^{ip\pi/2}-e^{3ip\pi/2})=\pi e^{ip\pi}(e^{-ip\pi/2}-e^{ip\pi/2})=-2\pi i e^{ip\pi}\sin(p\pi/2).$$

Also $1-e^{2\pi ip}=e^{ip\pi}(e^{-ip\pi}-e^{ip\pi})=-2ie^{ip\pi}\sin p\pi$. For $0<|p|<1$ these factors are nonzero, and so is $\sin(p\pi/2)$. Dividing, $I=\pi\sin(p\pi/2)/\sin p\pi$, and $\sin p\pi=2\sin(p\pi/2)\cos(p\pi/2)$ gives

$$\boxed{I=\int_0^\infty \frac{x^p}{1+x^2}\,dx=\frac{\pi}{2\cos(p\pi/2)}}.$$

Since the arc estimates hold for $-1<p<1$, this proves the result for $0<|p|<1$; $p=0$ follows by continuity or directly from $\int_0^\infty dx/(1+x^2)=\pi/2$.

11.8.8 Evaluate

$$\int_{-\infty}^{\infty}\frac{\cos bx-\cos ax}{x^2}\,dx,\quad a>b>0.$$

ANS. $\pi(a-b)$.

Solution.

Consider $F(z)=(e^{ibz}-e^{iaz})/z^2$. From $e^{icz}=1+icz-c^2z^2/2+O(z^3)$, $e^{ibz}-e^{iaz}=i(b-a)z+(a^2-b^2)z^2/2+O(z^3)$, so

$$F(z)=\frac{e^{ibz}-e^{iaz}}{z^2}=\frac{i(b-a)}{z}+\frac{a^2-b^2}{2}+O(z).$$

Close the real line above and indent the origin by a small clockwise upper semicircle of radius ϵ. On the large arc $|z|=R$, $|e^{iaz}|,|e^{ibz}|\le 1$ (since $\Im z\ge 0$), so $|F|\le 2/R^2$ and, with the length πR, the arc integral is bounded by $2\pi/R$ and vanishes. On the small arc, $z=\epsilon e^{i\theta}$ with θ from π to 0, the term $i(b-a)/z$ contributes $i(b-a)\int_\pi^0 i\,d\theta=-i\pi[i(b-a)]=\pi(b-a)$, while the bounded remainder contributes $O(\epsilon)$. There are no poles inside the indented contour, hence

$$⨍_{\mathbb{R}}F(x)\,dx+\pi(b-a)=0,\qquad ⨍_{\mathbb{R}}F(x)\,dx=\pi(a-b).$$

Taking real parts removes the simple pole: the real integrand $(\cos bx-\cos ax)/x^2$ tends to $(a^2-b^2)/2$ at zero and is absolutely integrable at infinity. Thus it needs no principal value, and

$$\boxed{\int_{\mathbb{R}}\frac{\cos bx-\cos ax}{x^2}\,dx=\pi(a-b)}.$$

The same proof allows $b=0$, since the large-arc estimate still holds.

11.8.9 Prove that

$$\int_{-\infty}^{\infty}\frac{\sin^2 x}{x^2}\,dx=\pi.$$

Hint. $\sin^2 x=\frac{1}{2}(1-\cos 2x)$.

Solution.

Use $\sin^2 x=(1-\cos 2x)/2$ and the case $b=0$, $a=2$ of Exercise 11.8.8, which gives $\int_{\mathbb{R}}(1-\cos 2x)x^{-2}\,dx=\pi(2-0)=2\pi$:

$$\int_{\mathbb{R}}\frac{\sin^2 x}{x^2}\,dx=\frac{1}{2}\int_{\mathbb{R}}\frac{1-\cos 2x}{x^2}\,dx=\frac{1}{2}(2\pi)=\boxed{\pi}.$$

The origin is removable with limiting value one, and the tails are bounded by x^{-2}, so this is an ordinary absolutely convergent integral. The textbook prints the limits $-\infty$ to ∞ but gives $\pi/2$, and the errata list of the Instructor's Manual (p. 3) also states $\pi/2$; that is the half-line value $\int_0^\infty (\sin x/x)^2\,dx$. The Instructor's Manual's worked solution (p. 138) correctly gives π for the full line and $\pi/2$ for the half-line. Evenness and the contour calculation above prove that the full-line value is π.

11.8.10 Show that

$$\int_0^\infty \frac{x\sin x}{x^2+1}\,dx = \frac{\pi}{2e}.$$

Solution.

Integrate $F(z) = ze^{iz}/(z^2+1)$ over an upper semicircle. Its rational factor is bounded by $R/(R^2-1) = O(R^{-1})$ on the large arc, and $|e^{iz}| = e^{-\Im z}$ decays in the upper half-plane, so Jordan's estimate ($\int_{\text{arc}} |e^{iz}|\,|dz| < \pi$) bounds that arc by $\pi R/(R^2-1)$, which vanishes. The only enclosed pole is i:

$$\operatorname{Res}_i F = \left.\frac{ze^{iz}}{z+i}\right|_{z=i} = \frac{ie^{-1}}{2i} = \frac{e^{-1}}{2}.$$

Therefore

$$\int_{\mathbb{R}} \frac{xe^{ix}}{x^2+1}\,dx = 2\pi i\,\frac{e^{-1}}{2} = \frac{\pi i}{e}.$$

The real part $x\cos x/(x^2+1)$ is odd and integrates to zero over $[-R,R]$. The imaginary part $x\sin x/(x^2+1)$ is even, so its full-line integral, π/e, is twice the required one:

$$\boxed{\int_0^\infty \frac{x\sin x}{x^2+1}\,dx = \frac{\pi}{2e}}.$$

The tails converge by Dirichlet's test (the amplitude $x/(x^2+1)$ decreases to zero for $x > 1$), although they are not absolutely convergent.

11.8.11 A quantum mechanical calculation of a transition probability leads to the function $f(t,\omega) = 2(1-\cos\omega t)/\omega^2$. Show that

$$\int_{-\infty}^{\infty} f(t,\omega)\,d\omega = 2\pi|t|, \qquad t \in \mathbb{R}.$$

Solution.

For a general real time parameter the answer is $2\pi|t|$. The printed $2\pi t$ assumes $t \geq 0$. If $t = 0$ the integrand is zero. If $t \neq 0$, put $y = |t|\omega$, which maps $\mathbb{R}$ onto $\mathbb{R}$ preserving direction; then $d\omega = dy/|t|$, $1/\omega^2 = t^2/y^2$, and $\cos\omega t = \cos(\pm y) = \cos y$. Thus

$$\begin{aligned}\int_{\mathbb{R}} \frac{2(1-\cos\omega t)}{\omega^2}\,d\omega &= 2|t|\int_{\mathbb{R}} \frac{1-\cos y}{y^2}\,dy \\ &= 2|t|\,\pi = \boxed{2\pi|t|},\end{aligned}$$

where Exercise 11.8.8 with $a = 1, b = 0$ evaluates the last integral. At $\omega = 0$ the integrand extends continuously with value t^2; the tails are $O(\omega^{-2})$, so the substitution is justified in an absolutely convergent integral.

11.8.12 Show that ($a > 0$, $k \in \mathbb{R}$):

(a) $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{x^2+a^2}\,dx = \frac{\pi}{a}e^{-a}.$

How is the right side modified if $\cos x$ is replaced by $\cos kx$?

(b) $\displaystyle\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+a^2}\,dx = \pi e^{-a}.$

How is the right side modified if $\sin x$ is replaced by $\sin kx$?

Solution.

The statement leaves k unspecified; it must be real, since for $\Im k \neq 0$ the factors $\cos kx$ and $\sin kx$ grow like $\frac{1}{2}e^{|\Im k|\,|x|}$ and both integrals diverge. Let $k > 0$. Closing above for $F(z) = e^{ikz}/(z^2+a^2)$ encloses only ia, whose residue is $e^{ik(ia)}/(2ia) = e^{-ka}/(2ia)$. On the large arc $|e^{ikz}| = e^{-k\Im z} \le 1$ and $|z^2+a^2| \ge R^2 - a^2$, so the arc integral is at most $\pi R/(R^2-a^2)$ and vanishes. The result $2\pi i\, e^{-ka}/(2ia)$ is real, and the imaginary part $\int \sin kx/(x^2+a^2)\,dx$ vanishes by oddness; taking real parts gives

$$\int_{\mathbb{R}} \frac{\cos kx}{x^2+a^2}\,dx = 2\pi i \frac{e^{-ka}}{2ia} = \frac{\pi}{a}e^{-ka}.$$

Cosine is even, so $\cos kx = \cos|k|x$ and $k < 0$ gives $(\pi/a)e^{-|k|a}$; and $k = 0$ gives $\int_{\mathbb{R}} dx/(x^2+a^2) = a^{-1}[\arctan(x/a)]_{-\infty}^{\infty} = \pi/a$. Hence

$$\boxed{\text{(a)}\quad \int_{\mathbb{R}} \frac{\cos kx}{x^2+a^2}\,dx = \frac{\pi}{a}e^{-a|k|} \qquad (k \in \mathbb{R}).}$$

For (b), integrate $G(z) = ze^{ikz}/(z^2+a^2)$. For $k > 0$ the rational factor is only $O(R^{-1})$, but Jordan's estimate ($\int_{\text{arc}} |e^{ikz}|\,|dz| < \pi/k$) again removes the upper arc, and $\operatorname{Res}_{ia} G = ia\,e^{-ka}/(2ia) = e^{-ka}/2$. Thus $\int_{\mathbb{R}} xe^{ikx}/(x^2+a^2)\,dx = 2\pi i\, e^{-ka}/2 = \pi i e^{-ka}$, and taking imaginary parts yields $\int_{\mathbb{R}} x\sin kx/(x^2+a^2)\,dx = \pi e^{-ka}$. For $k < 0$, $\sin kx = -\sin|k|x$, so the integral changes sign; for $k = 0$ the integrand is zero. Hence

$$\boxed{\text{(b)}\quad \int_{\mathbb{R}} \frac{x\sin kx}{x^2+a^2}\,dx = \begin{cases} \pi\ \operatorname{sgn}(k) e^{-a|k|}, & k \neq 0, \\ 0, & k = 0. \end{cases}}$$

Setting $k = 1$ recovers both stated formulas. The second formula has a jump in its limits as $k \to 0$; the integral at $k = 0$ is exactly zero. Differentiation of the first formula across $k = 0$ would therefore be invalid. For nonzero k, convergence of the second integral's tails follows from Dirichlet's test, not absolute convergence.

11.8.13 Use the contour shown (Fig. 11.28) with $R \to \infty$ to prove that

$$\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx = \pi.$$

Solution.

Use exactly the rectangular contour of Fig. 11.28: its vertical sides are $\Re z = \pm R$, its top is $\Im z = R$, and its bottom follows the real axis except for a clockwise semicircular indentation of

radius ϵ at zero. Integrate $F(z) = e^{iz}/z$ counterclockwise around this indented domain. The pole at zero has been excluded, so the contour integral is zero.

On either vertical side, $z = \pm R + iy$, $0 \le y \le R$, we have $|e^{iz}| = e^{-y}$ and $|z| \ge R$, so

$$\left| \int_{\text{side}} F(z)\, dz \right| \le \frac{1}{R} \int_0^R e^{-y}\, dy \le \frac{1}{R}.$$

On the top, $|z| \ge R$, $|e^{iz}| = e^{-R}$ and the length is $2R$, so the bound is $2R \cdot e^{-R}/R = 2e^{-R}$. All three contributions vanish. On the small arc, $e^{iz} = 1 + iz + O(z^2)$ gives $e^{iz}/z = 1/z + i + O(z)$. With $z = \epsilon e^{i\theta}$, the term $1/z$ gives $\int_\pi^0 i\, d\theta = -i\pi$, since the argument decreases from π to zero, while the bounded remainder contributes $O(\epsilon)$. Consequently

$$⨍_{\mathbb{R}} \frac{e^{ix}}{x}\, dx - i\pi = 0.$$

Since $e^{ix}/x = \cos x/x + i \sin x/x$, taking imaginary parts gives

$$\boxed{\int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx = \pi}.$$

Here $\sin x/x \to 1$ at zero and each infinite tail converges by Dirichlet's test; the real integral itself does not need a principal-value qualification.

11.8.14 In the quantum theory of atomic collisions, we encounter the integral

$$I = \int_{-\infty}^{\infty} \frac{\sin t}{t} e^{ipt}\, dt,$$

in which p is real. Show that

$$I = 0, \quad |p| > 1$$
$$I = \pi, \quad |p| < 1.$$

What happens if $p = \pm 1$?

Solution.

Use $\sin t = (e^{it} - e^{-it})/(2i)$ to write

$$I(p) = \frac{1}{2i} \left\{ ⨍_{\mathbb{R}} \frac{e^{i(p+1)t}}{t}\, dt - ⨍_{\mathbb{R}} \frac{e^{i(p-1)t}}{t}\, dt \right\}.$$

The pole terms at zero cancel in their difference, because $\sin t/t$ is continuous there. For any real nonzero q, the upper or lower indented contour gives $⨍_{\mathbb{R}} e^{iqt}/t\, dt = i\pi \operatorname{sgn} q$ (this is $J(q)$ of Exercise 11.7.12(b)). Thus, for $p \ne \pm 1$, $I(p) = (2i)^{-1}[i\pi \operatorname{sgn}(p+1) - i\pi \operatorname{sgn}(p-1)]$. For $|p| < 1$ the two signs are $+1$ and -1; for $p > 1$ both are $+1$, and for $p < -1$ both are -1. Hence

$$\boxed{I(p) = \frac{\pi}{2}\{\operatorname{sgn}(p+1) - \operatorname{sgn}(p-1)\} = \begin{cases} \pi, & |p| < 1, \\ 0, & |p| > 1. \end{cases}}$$

Both frequencies are then nonzero, so each tail of the original integral converges by Dirichlet's test.

At $p = \pm 1$ the ordinary improper complex integral does not exist: its two imaginary tails diverge logarithmically. A symmetric principal value at infinity, or Abel regularization, gives $\pi/2$. Indeed for $p = 1$, using $\sin t \cos t = \frac{1}{2} \sin 2t$,

$$\frac{\sin t}{t} e^{it} = \frac{\sin 2t}{2t} + i \frac{\sin^2 t}{t}.$$

The real part integrates to $\pi/2$: with $u = 2t$, $\int_{\mathbb{R}} \sin 2t/(2t)\, dt = \frac{1}{2} \int_{\mathbb{R}} \sin u/u\, du$, and Exercise 11.8.13 gives π for the last integral. The imaginary part is odd, so it cancels on symmetric intervals, but, by $\sin^2 t = (1 - \cos 2t)/2$,

$$\int_1^R \frac{\sin^2 t}{t}\, dt = \frac{1}{2} \ln R - \frac{1}{2} \int_1^R \frac{\cos 2t}{t}\, dt$$

diverges as $\frac{1}{2} \ln R$, since the second term has a finite limit as $R \to \infty$ (Dirichlet's test). For $p = -1$ its sign is reversed. This proves both the failure of ordinary convergence and the symmetric value at the endpoints.

11.8.15 Show that

$$\int_0^\infty \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{4a^3}, \quad a > 0.$$

Solution.

The function $(z^2 + a^2)^{-2}$ is bounded by $(R^2 - a^2)^{-2} = O(R^{-4})$ on a large upper semicircle, so that arc, of length πR, contributes $O(R^{-3})$. Its sole enclosed pole is the double pole at ia, with residue (as in Exercise 11.7.1(b))

$$\frac{d}{dz} (z + ia)^{-2} \Big|_{ia} = -\frac{2}{(2ia)^3} = \frac{1}{4ia^3}.$$

Thus

$$\int_{\mathbb{R}} \frac{dx}{(x^2 + a^2)^2} = 2\pi i \frac{1}{4ia^3} = \frac{\pi}{2a^3}.$$

The integrand is even, giving

$$\boxed{\int_0^\infty \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{4a^3}}.$$

11.8.16 Evaluate

$$\int_{-\infty}^\infty \frac{x^2}{1 + x^4}\, dx.$$

ANS. $\pi/\sqrt{2}$.

Solution.

The roots of $1 + z^4$, i.e., of $z^4 = -1 = e^{i\pi(2m+1)}$, are $e^{i\pi/4}$, $e^{3i\pi/4}$, $e^{5i\pi/4}$, $e^{7i\pi/4}$. Only the first two lie above the real axis. At a root p the residue of $z^2/(1 + z^4)$ is, by the rule $N(p)/D'(p)$ for a simple zero of the denominator, $p^2/(4p^3) = 1/(4p)$; and $1/p = e^{-i\pi/4}$, $e^{-3i\pi/4}$ for the two upper roots. Their sum is

$$\frac{1}{4} \left(e^{-i\pi/4} + e^{-3i\pi/4} \right) = \frac{1}{4} \left(\frac{1 - i}{\sqrt{2}} + \frac{-1 - i}{\sqrt{2}} \right) = -\frac{i}{2\sqrt{2}}.$$

On $|z| = R$ the integrand is bounded by $R^2/(R^4 - 1) = O(R^{-2})$, so its large semicircle integral, of length πR, is $O(R^{-1})$ and vanishes. The residue theorem therefore yields

$$\boxed{\int_{\mathbb{R}} \frac{x^2}{1+x^4}\,dx = 2\pi i\left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}}.$$

11.8.17 Evaluate

$$\int_0^\infty \frac{x^p \ln x}{x^2+1}\,dx, \quad 0 < p < 1.$$

ANS. $\dfrac{\pi^2 \sin(\pi p/2)}{4\cos^2(\pi p/2)}$.

Solution.

From Exercise 11.8.7,

$$J(p) = \int_0^\infty \frac{x^p}{1+x^2}\,dx = \frac{\pi}{2\cos(\pi p/2)}, \qquad -1 < p < 1.$$

Fix p in this interval and choose $\eta > 0$ so that $-1 < p - \eta < p + \eta < 1$. The p-derivative of the integrand is $x^p \ln x/(1 + x^2)$, since $\partial_p x^p = x^p \ln x$. For every p' with $|p' - p| \le \eta$, this derivative is bounded on $0 < x < 1$ by $x^{p-\eta}|\ln x|$ (there $x^{p'} \le x^{p-\eta}$), and on $x > 1$ by $x^{p+\eta-2}\ln x$ (there $x^{p'} \le x^{p+\eta}$ and $1 + x^2 > x^2$). Both bounds are integrable, because $p - \eta > -1$ and $p + \eta - 2 < -1$. Differentiation under the integral is therefore valid, and $\frac{d}{dp}[\cos(\pi p/2)]^{-1} = (\pi/2)\sin(\pi p/2)/\cos^2(\pi p/2)$:

$$\begin{aligned}\int_0^\infty \frac{x^p \ln x}{1+x^2}\,dx = J'(p) &= \frac{\pi}{2}\frac{(\pi/2)\sin(\pi p/2)}{\cos^2(\pi p/2)} \\ &= \boxed{\frac{\pi^2 \sin(\pi p/2)}{4\cos^2(\pi p/2)}}.\end{aligned}$$

This includes the stated range $0 < p < 1$.

11.8.18 Evaluate

$$\int_0^\infty \frac{(\ln x)^2}{1+x^2}\,dx,$$

(a) by appropriate series expansion of the integrand to obtain

$$4\sum_{n=0}^{\infty}(-1)^n(2n+1)^{-3},$$

(b) and by contour integration to obtain

$$\frac{\pi^3}{8}.$$

Hint. $x \to z = e^t$. Try the contour shown in Fig. 11.29, letting $R \to \infty$.

Solution.

Let I denote the required integral. **(a)** In its part from one to infinity set $x = 1/u$. Then $dx = -du/u^2$, $(\ln x)^2 = (\ln u)^2$, and $1 + x^2 = (1 + u^2)/u^2$; the limits $x = 1, \infty$ become $u = 1, 0$, so

$$\int_1^\infty \frac{(\ln x)^2}{1+x^2}\,dx = \int_1^0 \frac{(\ln u)^2 u^2}{1+u^2}\left(-\frac{du}{u^2}\right) = \int_0^1 \frac{(\ln u)^2}{1+u^2}\,du,$$

the part from zero to one. Expanding $1/(1 + x^2) = \sum_{n\ge0}(-1)^n x^{2n}$ for $0 \le x < 1$,

$$I = 2\int_0^1 \frac{(\ln x)^2}{1+x^2}\,dx = 2\sum_{n=0}^{\infty}(-1)^n \int_0^1 x^{2n}(\ln x)^2\,dx.$$

To evaluate the coefficient put $x = e^{-u}$, so $dx = -e^{-u}\,du$, $(\ln x)^2 = u^2$, $x^{2n} = e^{-2nu}$, and $x = 0, 1$ correspond to $u = \infty, 0$:

$$\int_0^1 x^{2n}(\ln x)^2\,dx = \int_0^\infty u^2 e^{-(2n+1)u}\,du = \frac{2}{(2n+1)^3}.$$

For $\alpha = 2n + 1 > 0$, the two integrations by parts are

$$\begin{aligned}\int_0^\infty u^2 e^{-\alpha u}\,du &= \left[-\frac{u^2 e^{-\alpha u}}{\alpha}\right]_0^\infty + \frac{2}{\alpha}\int_0^\infty u e^{-\alpha u}\,du \\ &= \frac{2}{\alpha}\left(\left[-\frac{u e^{-\alpha u}}{\alpha}\right]_0^\infty + \frac{1}{\alpha}\int_0^\infty e^{-\alpha u}\,du\right) = \frac{2}{\alpha^3}.\end{aligned}$$

Each boundary term vanishes because exponential decay dominates the polynomial at infinity, and the polynomial is zero at the origin. Since $\sum 2/(2n+1)^3$ converges, the integrals of the absolute values of the series terms are summable, justifying the interchange even at the endpoint. Therefore

$$\boxed{I = 4\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^3}}.$$

(b) Set $x = e^t$, so $dx = e^t\,dt$ and $(\ln x)^2 = t^2$, while $x = 0, \infty$ correspond to $t = -\infty, \infty$. Because $e^t/(1 + e^{2t}) = 1/(e^{-t} + e^t) = 1/(2\cosh t)$, we have

$$I = \int_{-\infty}^{\infty}\frac{t^2}{2\cosh t}\,dt.$$

Integrate $F(z) = z^2/(2\cosh z)$ around the rectangle with vertices $-R, R, R + i\pi, -R + i\pi$, as in Fig. 11.29. On the vertical sides $|\cosh(\pm R + iy)|^2 = \sinh^2 R + \cos^2 y \ge \sinh^2 R$ and $|z|^2 \le R^2 + \pi^2$; with the length π, their integrals are at most $\pi(R^2 + \pi^2)/(2\sinh R) = O(R^2 e^{-R})$. On the top, $z = t + i\pi$ and $\cosh(t + i\pi) = \cosh t\cos\pi + i\sinh t\sin\pi = -\cosh t$, and its direction is from R to $-R$, so

$$\int_{\text{top}} F\,dz = \int_R^{-R}\frac{(t+i\pi)^2}{-2\cosh t}\,dt = \int_{-R}^{R}\frac{(t+i\pi)^2}{2\cosh t}\,dt.$$

After adding bottom and top, expanding $t^2 + (t + i\pi)^2 = 2t^2 - \pi^2 + 2i\pi t$, and passing to the limit, the contour integral is

$$\int_{\mathbb{R}}\frac{t^2 + (t+i\pi)^2}{2\cosh t}\,dt = 2I - \frac{\pi^2}{2}\int_{\mathbb{R}}\operatorname{sech} t\,dt.$$

The term $2i\pi t$ integrates to zero by oddness. Also, with $u = e^t$, $\operatorname{sech} t = 2u/(u^2+1)$ and $dt = du/u$, so

$$\int_{\mathbb{R}} \operatorname{sech} t \, dt = 2\int_0^\infty \frac{du}{1+u^2} = \pi.$$

The zeros of $\cosh z$ are $z = i(m+\frac{1}{2})\pi$, $m \in \mathbb{Z}$, so the only pole in the rectangle is $z = i\pi/2$. By the rule N/D', with $(2\cosh z)' = 2\sinh z$ and $\sinh(i\pi/2) = i\sin(\pi/2) = i$, its residue is

$$\frac{(i\pi/2)^2}{2\sinh(i\pi/2)} = \frac{-\pi^2/4}{2i} = \frac{i\pi^2}{8}.$$

Hence $2I - \pi^3/2 = 2\pi i(i\pi^2/8) = -\pi^3/4$, so $2I = \pi^3/2 - \pi^3/4 = \pi^3/4$. Solving, and comparing with (a), which gives the series as $I/4$,

$$\boxed{I = \frac{\pi^3}{8}}, \qquad \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3} = \frac{\pi^3}{32}.$$

11.8.19 Prove that

$$\int_0^\infty \frac{\ln(1+x^2)}{1+x^2}\, dx = \pi \ln 2.$$

Solution.

Use the principal logarithm and consider separately

$$F_+(z) = \frac{\operatorname{Log}(z+i)}{z^2+1}, \qquad F_-(z) = \frac{\operatorname{Log}(z-i)}{z^2+1}.$$

The logarithm in F_+ has its cut where $z+i \le 0$, i.e., on the ray $\{x - i : x \le 0\}$, below the real axis. Close its contour above; the pole at i has residue $\operatorname{Log}(z+i)/(z+i)$ at $z = i$, that is, $\operatorname{Log}(2i)/(2i)$. The logarithm in F_- has its cut on $\{x + i : x \le 0\}$, above the real axis; close below, clockwise, enclosing the pole at $-i$ with residue $\operatorname{Log}(-2i)/(-2i)$. On the large arcs $|\operatorname{Log}(z \pm i)| \le \ln(R+1) + \pi$ and $|z^2+1| \ge R^2 - 1$; with the length πR, both arc integrals are $O((\ln R)/R)$ and vanish. Therefore, with the factors $2\pi i$ and $-2\pi i$ (clockwise),

$$\int_{\mathbb{R}} F_+(x)\, dx = 2\pi i \frac{\operatorname{Log}(2i)}{2i} = \pi \operatorname{Log}(2i), \qquad \int_{\mathbb{R}} F_-(x)\, dx = -2\pi i \frac{\operatorname{Log}(-2i)}{-2i} = \pi \operatorname{Log}(-2i).$$

For real x, $x-i$ is the conjugate of $x+i$ and neither lies on the cut, so the two logarithms are conjugates and their sum is $2\ln|x+i| = \ln(x^2+1)$. Also $\operatorname{Log}(2i) = \ln 2 + i\pi/2$ and $\operatorname{Log}(-2i) = \ln 2 - i\pi/2$. Adding the two equations yields

$$\int_{\mathbb{R}} \frac{\ln(1+x^2)}{1+x^2}\, dx = 2\pi \ln 2.$$

Evenness then gives

$$\boxed{\int_0^\infty \frac{\ln(1+x^2)}{1+x^2}\, dx = \pi \ln 2}.$$

11.8.20 Show that

$$\int_0^\infty \frac{x^a}{(x+1)^2}\,dx = \frac{\pi a}{\sin \pi a},$$

where $-1 < a < 1$, with the quotient interpreted as its limit 1 at $a = 0$.

Hint. Use the contour shown in Fig. 11.26, noting that $z = 0$ is a branch point and the positive x-axis can be chosen to be a cut line.

Solution.

For $a \neq 0$ use $F(z) = z^a/(1+z)^2$ on $0 < \arg z < 2\pi$ and a keyhole contour about the positive real axis. Since $|z^a| = |z|^a$, $|1+z| \geq 1-\epsilon$ on $|z| = \epsilon$, and $|1+z| \geq R-1$ on $|z| = R$, the small circle is bounded by $2\pi\epsilon^{a+1}/(1-\epsilon)^2$, and the large circle by $2\pi R^{a+1}/(R-1)^2 = O(R^{a-1})$. Both vanish exactly when $-1 < a < 1$. The upper bank, where $z^a = x^a$, contributes I; the lower bank, where $z^a = x^a e^{2\pi i a}$, is traversed inward and contributes $-e^{2\pi i a} I$; together they give $(1 - e^{2\pi i a})I$.

The enclosed pole at -1 is double, so its residue is $\frac{d}{dz}[(z+1)^2 F(z)]$ at $z = -1$. Since $\arg(-1) = \pi$, $(-1)^{a-1} = e^{i\pi(a-1)}$, and

$$\operatorname{Res}_{-1} F = \frac{d}{dz} z^a \bigg|_{-1} = a(-1)^{a-1} = a e^{i\pi(a-1)} = -a e^{i\pi a}.$$

It follows from the residue theorem that

$$(1 - e^{2\pi i a})I = 2\pi i(-a e^{i\pi a}) = -2\pi i a e^{i\pi a}.$$

Using $1 - e^{2\pi i a} = e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) = -2i e^{i\pi a} \sin \pi a$ and dividing, $I = -2\pi i a e^{i\pi a}/(-2i e^{i\pi a} \sin \pi a)$, which gives

$$\boxed{I = \frac{\pi a}{\sin \pi a} \qquad (-1 < a < 1,\ a \neq 0)}.$$

At $a = 0$ the displayed quotient means its removable limit, equal to one. Directly, $\int_0^\infty (1+x)^{-2}\,dx = [-(1+x)^{-1}]_0^\infty = 1$. This also verifies the continuous value at zero.

11.8.21 Show that

$$\int_{-\infty}^\infty \frac{x^2\,dx}{x^4 - 2x^2\cos 2\theta + 1} = \frac{\pi}{2|\sin\theta|} = \frac{\pi}{2^{1/2}(1-\cos 2\theta)^{1/2}}.$$

Assume θ is real and $\sin\theta \neq 0$. Exercise 11.8.16 is a special case of this result.

Solution.

The formula with $\sin\theta$ in the denominator requires $\sin\theta > 0$, for example $0 < \theta < \pi$. For any real $\theta \notin \pi\mathbb{Z}$, the general answer is $\pi/(2|\sin\theta|)$. At $\theta \in \pi\mathbb{Z}$ the integral diverges. First take $0 < \theta < \pi$, $\theta \neq \pi/2$, and factor

$$D(z) = (z^2 - e^{2i\theta})(z^2 - e^{-2i\theta}).$$

Its zeros are $\pm e^{i\theta}$ and $\pm e^{-i\theta}$. Since $\Im e^{i\theta} = \Im(-e^{-i\theta}) = \sin\theta > 0$, the upper poles are $u = e^{i\theta}$ and $v = -e^{-i\theta}$; the other two are their negatives. Since $\theta \neq \pi/2$, $\sin 2\theta = 2\sin\theta\cos\theta \neq 0$, so $e^{2i\theta} \neq e^{-2i\theta}$ and the four zeros are distinct, hence simple. Since $D'(z) = 4z^3 - 4z\cos 2\theta = 4z(z^2 - \cos 2\theta)$, the residue of z^2/D at a simple zero p is $p^2/D'(p) = p/[4(p^2 - \cos 2\theta)]$, and $u^2 - \cos 2\theta = e^{2i\theta} - \cos 2\theta = i\sin 2\theta$, $v^2 - \cos 2\theta = e^{-2i\theta} - \cos 2\theta = -i\sin 2\theta$. Hence

$$\operatorname{Res}_u \frac{z^2}{D(z)} = \frac{u}{4i\sin 2\theta}, \qquad \operatorname{Res}_v \frac{z^2}{D(z)} = -\frac{v}{4i\sin 2\theta} = \frac{e^{-i\theta}}{4i\sin 2\theta}.$$

With $e^{i\theta}+e^{-i\theta}=2\cos\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$, their sum simplifies to

$$\frac{2\cos\theta}{4i(2\sin\theta\cos\theta)} = \frac{1}{4i\sin\theta}.$$

The large upper arc vanishes because the rational function is $O(R^{-2})$ and the arc has length πR. Hence

$$I = 2\pi i\,\frac{1}{4i\sin\theta} = \frac{\pi}{2\sin\theta}.$$

At $\theta=\pi/2$ the poles coalesce, since $\cos 2\theta=-1$ and $D(x)=(x^2+1)^2$, but the integral can be evaluated directly, with $\int_{\mathbb{R}} dx/(x^2+1)=\pi$ and $\int_{\mathbb{R}} dx/(x^2+1)^2=\pi/2$ (Exercise 11.8.15 with $a=1$):

$$\frac{x^2}{(x^2+1)^2} = \frac{1}{x^2+1} - \frac{1}{(x^2+1)^2}, \qquad I = \pi - \frac{\pi}{2} = \frac{\pi}{2}.$$

This agrees with the formula. In general the denominator depends on θ only through $\sin^2\theta$, because $\cos 2\theta = 1-2\sin^2\theta$ gives $D(x)=(x^2-1)^2+4x^2\sin^2\theta$. For real $\theta\notin\pi\mathbb{Z}$ choose $\theta'\in(0,\pi)$ with $\sin\theta' = |\sin\theta|$; then $I(\theta)=I(\theta')=\pi/(2\sin\theta')=\pi/(2|\sin\theta|)$. Also $1-\cos 2\theta = 2\sin^2\theta$, so $\sqrt{2}\sqrt{1-\cos 2\theta}=2|\sin\theta|$. Finally

$$\boxed{I = \frac{\pi}{2|\sin\theta|} = \frac{\pi}{\sqrt{2}\,\sqrt{1-\cos 2\theta}}}.$$

If $\sin\theta=0$, then $D(x)=(x^2-1)^2=(x-1)^2(x+1)^2$, and the real poles at ± 1 behave like $\frac{1}{4}(x\mp 1)^{-2}$ and cause divergence. Choosing $\theta=\pi/4$, where $\cos 2\theta=0$ and $\sin\theta=1/\sqrt{2}$, recovers Exercise 11.8.16: $\pi/(2/\sqrt{2})=\pi/\sqrt{2}$.

11.8.22 For real $n>1$, show that

$$\int_0^\infty \frac{dx}{1+x^n} = \frac{\pi/n}{\sin(\pi/n)}.$$

Hint. Try the contour shown in Fig. 11.30, with $\theta = 2\pi/n$.

Solution.

The convergence condition is $n>1$ (for real n); it is necessary and was omitted from the statement. Choose a branch of z^n with argument in the sector $0<\arg z<\alpha=2\pi/n$, and integrate $1/(1+z^n)$ over the sector boundary, with small and large circular arcs of radii ϵ and R. On the small arc $|1+z^n|\ge 1-\epsilon^n$ and the length is $\alpha\epsilon$, so it is $O(\epsilon)$; on the large one $|1+z^n|\ge R^n-1$ and the length is αR, so it is $O(R^{1-n})$. Both vanish for $n>1$. On the outgoing real ray the integral is I, while on the incoming ray $z=re^{i\alpha}$, $dz=e^{i\alpha}\,dr$, traversed from $r=R$ to $r=\epsilon$, it is $\int_\infty^0 e^{i\alpha}\,dr/(1+r^n) = -e^{i\alpha}I$, since $z^n = r^n e^{in\alpha} = r^n e^{2\pi i} = r^n$.

A pole $z=re^{i\varphi}$ in the sector satisfies $r^n e^{in\varphi}=-1$ with $0<n\varphi<2\pi$, so $r=1$ and $n\varphi=\pi$: there is one pole, $z_0=e^{i\pi/n}$, where $z_0^n=e^{i\pi}=-1$. By the rule N/D' with $(1+z^n)'=nz^{n-1}$,

$$\operatorname{Res}_{z_0}\frac{1}{1+z^n} = \frac{1}{nz_0^{n-1}} = \frac{z_0}{nz_0^n} = -\frac{e^{i\pi/n}}{n}.$$

Thus, by the residue theorem, with $e^{i\alpha}=e^{2\pi i/n}$,

$$(1-e^{2\pi i/n})I = -\frac{2\pi i}{n}e^{i\pi/n}.$$

Since $1 - e^{2i\beta} = e^{i\beta}(e^{-i\beta} - e^{i\beta}) = -2ie^{i\beta}\sin\beta$, with $\beta = \pi/n$ we obtain

$$I = \frac{-(2\pi i/n)e^{i\pi/n}}{-2ie^{i\pi/n}\sin(\pi/n)}, \qquad \boxed{\int_0^\infty \frac{dx}{1+x^n} = \frac{\pi}{n\sin(\pi/n)} \qquad (n>1)}.$$

The branch argument also covers nonintegral real $n > 1$; the small circle excludes its branch point at zero. For $0 < n \le 1$ the tail is comparable to x^{-n} and diverges; for $n \le 0$ the integrand does not even tend to zero (it tends to 1 for $n < 0$ and equals $\frac{1}{2}$ for $n = 0$). The Instructor's Manual loses a minus sign when replacing z_0^{1-n}/n by an exponential. Since $z_0^n = -1$, the residue is $-z_0/n = -e^{i\pi/n}/n$, not $+e^{i\pi/n}/n$. Its positive final integral agrees with the correctly signed derivation above.

11.8.23

(a) Show that

$$f(z) = z^4 - 2z^2\cos 2\theta + 1$$

has zeros at $e^{i\theta}$, $e^{-i\theta}$, $-e^{i\theta}$, and $-e^{-i\theta}$.

(b) For real θ with $\sin\theta \neq 0$, show that

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 - 2x^2\cos 2\theta + 1} = \frac{\pi}{2|\sin\theta|} = \frac{\pi}{2^{1/2}(1-\cos 2\theta)^{1/2}}.$$

Exercise 11.8.22 ($n = 4$) is a special case of this result.

Solution.

(a) Since $2\cos 2\theta = e^{2i\theta} + e^{-2i\theta}$ and $e^{2i\theta}e^{-2i\theta} = 1$,

$$\begin{aligned} z^4 - 2z^2\cos 2\theta + 1 &= z^4 - (e^{2i\theta} + e^{-2i\theta})z^2 + e^{2i\theta}e^{-2i\theta} = (z^2 - e^{2i\theta})(z^2 - e^{-2i\theta}) \\ &= (z - e^{i\theta})(z + e^{i\theta})(z - e^{-i\theta})(z + e^{-i\theta}), \end{aligned}$$

the last step by $z^2 - c^2 = (z-c)(z+c)$ with $c = e^{\pm i\theta}$. The four listed numbers are exactly the zeros, with repetitions counted when they coincide.

(b) On the positive half-line put $x = 1/u$, so that $x = 0, \infty$ correspond to $u = \infty, 0$. Then

$$dx = -\frac{du}{u^2}, \qquad x^4 - 2x^2\cos 2\theta + 1 = \frac{1 - 2u^2\cos 2\theta + u^4}{u^4},$$

so

$$\begin{aligned} \int_0^\infty \frac{dx}{x^4 - 2x^2\cos 2\theta + 1} &= \int_\infty^0 \frac{u^4}{u^4 - 2u^2\cos 2\theta + 1}\left(-\frac{du}{u^2}\right) \\ &= \int_0^\infty \frac{u^2\,du}{u^4 - 2u^2\cos 2\theta + 1}. \end{aligned}$$

Both integrands are even, so each integral over $\mathbb{R}$ is twice the corresponding half-line integral, and Exercise 11.8.21 gives $\int_{\mathbb{R}} u^2\,du/(u^4 - 2u^2\cos 2\theta + 1) = \pi/(2|\sin\theta|)$. Consequently

$$\int_{\mathbb{R}} \frac{dx}{x^4 - 2x^2\cos 2\theta + 1} = 2\int_0^\infty \frac{u^2\,du}{u^4 - 2u^2\cos 2\theta + 1} = \int_{\mathbb{R}} \frac{u^2\,du}{u^4 - 2u^2\cos 2\theta + 1},$$

and, with $1 - \cos 2\theta = 2\sin^2\theta$, so that $\sqrt{2}\sqrt{1-\cos 2\theta} = 2|\sin\theta|$,

$$\boxed{\int_{\mathbb{R}} \frac{dx}{x^4 - 2x^2\cos 2\theta + 1} = \frac{\pi}{2|\sin\theta|} = \frac{\pi}{\sqrt{2}\sqrt{1-\cos 2\theta}}}.$$

As in 11.8.21, the printed expression without the absolute value requires $\sin\theta > 0$, and $\sin\theta = 0$ is divergent. At $\theta = \pi/4$ we have $\cos 2\theta = 0$ and $\sin\theta = 1/\sqrt{2}$, so the integrand is $1/(x^4+1)$, the integral over $\mathbb{R}$ is $\pi/(2/\sqrt{2}) = \pi/\sqrt{2}$, and the half-line value is $\pi/(2\sqrt{2})$. This is the $n = 4$ case of Exercise 11.8.22, $(\pi/4)/\sin(\pi/4) = \pi/(2\sqrt{2})$.

11.8.24 Show that

$$\int_0^\infty \frac{x^{-a}}{x+1}\,dx = \frac{\pi}{\sin a\pi},$$

where $0 < a < 1$.

Hint. You have a branch point and you will need a cut line. Try the contour shown in Fig. 11.26.

Solution.

Use $F(z) = z^{-a}/(1+z)$ with $z^{-a} = |z|^{-a}e^{-ia\arg z}$, $0 < \arg z < 2\pi$, so that the cut lies along the positive real axis. The keyhole contour of Fig. 11.26 consists of the upper bank from ϵ to R, the circle $|z| = R$ counterclockwise, the lower bank from R to ϵ, and the circle $|z| = \epsilon$ clockwise $(0 < \epsilon < 1 < R)$. On the small circle $|z^{-a}| = \epsilon^{-a}$, $|1+z| \geq 1-\epsilon$, and the length is $2\pi\epsilon$, so its integral is bounded by $2\pi\epsilon^{1-a}/(1-\epsilon)$; on the large circle $|z^{-a}| = R^{-a}$ and $|1+z| \geq R-1$, giving the bound $2\pi R^{1-a}/(R-1)$. The first tends to zero if $a < 1$ and the second if $a > 0$. On the upper bank $(\arg z \to 0)$ $z^{-a} = x^{-a}$, which gives the desired integral $I = \int_0^\infty x^{-a}\,dx/(1+x)$; on the lower bank $(\arg z \to 2\pi)$ $z^{-a} = x^{-a}e^{-2\pi ia}$, and the inward direction gives $\int_\infty^0 e^{-2\pi ia}x^{-a}\,dx/(1+x) = -e^{-2\pi ia}I$. The two straight banks therefore give $(1-e^{-2\pi ia})I$. The only pole, $z = -1 = e^{i\pi}$, is simple, and its residue is the numerator there:

$$\operatorname{Res}_{-1} F = (e^{i\pi})^{-a} = e^{-i\pi a}.$$

Therefore, by the residue theorem in the limit $\epsilon \to 0$, $R \to \infty$,

$$(1-e^{-2\pi ia})I = 2\pi i e^{-i\pi a}.$$

Using $1 - e^{-2\pi ia} = e^{-i\pi a}(e^{i\pi a} - e^{-i\pi a}) = 2ie^{-i\pi a}\sin\pi a$ yields $I = 2\pi i e^{-i\pi a}/(2ie^{-i\pi a}\sin\pi a)$, that is,

$$\boxed{\int_0^\infty \frac{x^{-a}}{1+x}\,dx = \frac{\pi}{\sin\pi a} \qquad (0 < a < 1)}.$$

The real integral converges at both endpoints: the integrand is at most x^{-a}, integrable at zero because $a < 1$, and at most x^{-1-a}, integrable at infinity because $a > 0$.

11.8.25 Show that

$$\int_0^\infty \frac{\cosh bx}{\cosh x}\,dx = \frac{\pi}{2\cos(\pi b/2)}, \quad |b| < 1.$$

Hint. Choose a contour that encloses one zero of $\cosh z$, hence one pole of $1/\cosh z$.

Solution.

First let b be real, $|b| < 1$, and integrate $F(z) = e^{bz}/\cosh z$ around the rectangle $-R, R, R+i\pi, -R+i\pi$. From $\cosh(x+iy) = \cosh x\cos y + i\sinh x\sin y$,

$$|\cosh(x+iy)|^2 = \cosh^2 x\cos^2 y + \sinh^2 x\sin^2 y = \sinh^2 x + \cos^2 y \geq \sinh^2 x.$$

On either vertical side $|e^{bz}| = e^{\pm bR} \leq e^{|b|R}$ and $|\cosh(\pm R + iy)| \geq \sinh R$, so with the length π each side contributes at most $\pi e^{|b|R}/\sinh R = O(e^{-(1-|b|)R})$, which tends to zero. Let

$$J = \int_{\mathbb{R}} \frac{e^{bx}}{\cosh x}\,dx,$$

which converges absolutely because $e^{bx}/\cosh x \le 2e^{|b||x|-|x|}$. On the top, $z = x + i\pi$, where $\cosh(x+i\pi) = -\cosh x$ and $e^{b(x+i\pi)} = e^{i\pi b}e^{bx}$, so $F(x+i\pi) = -e^{i\pi b}F(x)$. The top is traversed from $R + i\pi$ to $-R + i\pi$, so it gives $\int_R^{-R}(-e^{i\pi b})F(x)\,dx = e^{i\pi b}\int_{-R}^{R} F(x)\,dx \to e^{i\pi b}J$, while the bottom gives J. The zeros of $\cosh z$ are $z = i(m + \frac{1}{2})\pi$, $m \in \mathbb{Z}$, so the one pole inside is $i\pi/2$. It is simple, and the rule N/D', with $(\cosh z)' = \sinh z$ and $\sinh(i\pi/2) = i\sin(\pi/2) = i$, gives

$$\operatorname{Res}_{i\pi/2} F = \frac{e^{i\pi b/2}}{\sinh(i\pi/2)} = \frac{e^{i\pi b/2}}{i}.$$

The residue theorem, in the limit $R \to \infty$, gives $(1 + e^{i\pi b})J = 2\pi i\, e^{i\pi b/2}/i$, that is,

$$(1 + e^{i\pi b})J = 2\pi e^{i\pi b/2}.$$

Now $1 + e^{i\pi b} = e^{i\pi b/2}(e^{-i\pi b/2} + e^{i\pi b/2}) = 2e^{i\pi b/2}\cos(\pi b/2)$, so $J = \pi/\cos(\pi b/2)$. Pairing x with $-x$ in J (substitute $x \to -x$ on $(-\infty, 0)$ and use $\cosh(-x) = \cosh x$),

$$J = \int_0^\infty \frac{e^{bx} + e^{-bx}}{\cosh x}\,dx = 2\int_0^\infty \frac{\cosh bx}{\cosh x}\,dx.$$

Thus

$$\boxed{\int_0^\infty \frac{\cosh bx}{\cosh x}\,dx = \frac{\pi}{2\cos(\pi b/2)}}.$$

The same argument holds for complex b with $|b| < 1$, as printed, because then $|\Re b| \le |b| < 1$. On the vertical sides, where $z = \pm R + iy$ with $0 \le y \le \pi$, $|e^{bz}| = e^{\Re(bz)} = e^{\pm R\Re b - y\Im b} \le e^{|\Re b|R + \pi|\Im b|}$, so each side contributes $O(e^{-(1-|\Re b|)R})$; J converges absolutely because $|e^{bx}| = e^{x\Re b}$; and $\cos(\pi b/2) \ne 0$, because it vanishes only when b is an odd integer. Every other step is unchanged. The printed hint calls this a pole of $\cosh z$. That function is entire; its simple zero at $i\pi/2$ produces the pole of the reciprocal used here.

11.8.26 Show that

$$\int_0^\infty \cos(t^2)\,dt = \int_0^\infty \sin(t^2)\,dt = \frac{\sqrt{\pi}}{2\sqrt{2}}.$$

Hint. Try the contour shown in Fig. 11.30, with $\theta = \pi/4$.

Note. These are the Fresnel integrals for the special case of infinity as the upper limit. For the general case of a varying upper limit, asymptotic expansions of the Fresnel integrals are the topic of Exercise 12.6.1.

Solution.

Integrate the entire function e^{iz^2} around the sector $0 \le \arg z \le \pi/4$, $|z| \le R$ (Fig. 11.30 with opening angle $\theta = \pi/4$). The real segment contributes $I_R = \int_0^R e^{it^2}\,dt$. On the returning ray put $z = e^{i\pi/4}t$; then $dz = e^{i\pi/4}\,dt$, $iz^2 = i \cdot it^2 = -t^2$, and t runs from R to 0, so its contribution is $-e^{i\pi/4}\int_0^R e^{-t^2}\,dt$. On the circular arc $z = Re^{i\varphi}$, $0 \le \varphi \le \pi/4$, $iz^2 = iR^2(\cos 2\varphi + i\sin 2\varphi)$ has real part $-R^2\sin 2\varphi$, so

$$|e^{iz^2}| = e^{-R^2\sin 2\varphi}.$$

Sine is concave on $[0, \pi/2]$, so it lies above its chord there: $\sin\psi \ge 2\psi/\pi$. With $\psi = 2\varphi$ this gives $\sin 2\varphi \ge 4\varphi/\pi$ for $0 \le \varphi \le \pi/4$, and with $|dz| = R\,d\varphi$,

$$\left|\int_{\text{arc}} e^{iz^2}\,dz\right| \le R\int_0^{\pi/4} e^{-4R^2\varphi/\pi}\,d\varphi = \frac{\pi}{4R}\bigl(1 - e^{-R^2}\bigr) \le \frac{\pi}{4R} \longrightarrow 0.$$

Since e^{iz^2} is entire, Cauchy's theorem gives $I_R + \int_{\text{arc}} e^{iz^2}\,dz - e^{i\pi/4}\int_0^R e^{-t^2}\,dt = 0$, and letting $R \to \infty$,

$$\lim_{R\to\infty} I_R = e^{i\pi/4}\int_0^\infty e^{-t^2}\,dt.$$

The Gaussian value follows by squaring its integral and using polar coordinates $x = r\cos\varphi$, $y = r\sin\varphi$, $dx\,dy = r\,dr\,d\varphi$, in the first quadrant:

$$\left(\int_0^\infty e^{-t^2}\,dt\right)^2 = \int_0^\infty\int_0^\infty e^{-(x^2+y^2)}\,dx\,dy = \int_0^{\pi/2}\int_0^\infty e^{-r^2} r\,dr\,d\varphi = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4}.$$

The integral is positive, so it is $\sqrt{\pi}/2$. Since $e^{i\pi/4} = (1+i)/\sqrt{2}$, the limit is $(1+i)\sqrt{\pi}/(2\sqrt{2})$, and taking real and imaginary parts gives

$$\boxed{\int_0^\infty \cos(t^2)\,dt = \int_0^\infty \sin(t^2)\,dt = \frac{\sqrt{\pi}}{2\sqrt{2}}}.$$

The vanishing arc estimate proves convergence of these improper oscillatory integrals as well as their values.

11.8.27 Show that

$$\int_0^1 \frac{1}{(x^2-x^3)^{1/3}}\,dx = 2\pi/\sqrt{3}.$$

Hint. Try the contour shown in Fig. 11.31.

Solution.

The real integrand is $x^{-2/3}(1-x)^{-1/3}$, with its positive real cube root. Both endpoint exponents exceed -1, so it is integrable. To use the dogbone contour of Fig. 11.31, define on $\mathbb{C}\setminus[0,1]$

$$F(z) = \frac{e^{i\pi/3}}{z}\left(1-\frac{1}{z}\right)^{-1/3},$$

where the power in parentheses uses its principal logarithm. The map $1-1/z$ meets the nonpositive real axis exactly on the removed segment: the equation $1-1/z = -s$ with $s \ge 0$ has the single solution $z = 1/(1+s)$, and these points fill $(0,1]$ as s runs over $[0,\infty)$. So this branch is analytic on $\mathbb{C}\setminus[0,1]$. It satisfies $F(z)^3 = e^{i\pi}z^{-3}\big((z-1)/z\big)^{-1} = -1/[z^2(z-1)] = 1/[z^2(1-z)]$, and hence $|F(z)| = |z|^{-2/3}|z-1|^{-1/3}$.

For $0 < x < 1$, $1-1/x = -(1-x)/x < 0$. Approaching from above, $z = x+i\eta$ with $\eta \downarrow 0$, the number $1-1/z = 1-(x-i\eta)/(x^2+\eta^2)$ has positive imaginary part, so $\arg(1-1/z) \to \pi$, whereas approaching from below makes it tend to $-\pi$. With $|1-1/x| = (1-x)/x$, the two boundary values are

$$F_\pm(x) = \frac{e^{i\pi/3}}{x}\left(\frac{1-x}{x}\right)^{-1/3} e^{\mp i\pi/3},$$

that is,

$$F_+(x) = x^{-2/3}(1-x)^{-1/3}, \qquad F_-(x) = e^{2\pi i/3}F_+(x).$$

The dogbone consists of the two banks of the cut from ϵ to $1-\epsilon$, joined by the circles $|z| = \epsilon$ and $|z-1| = \epsilon$. Enclose it in a circle $|z| = R$ with $R > 1$; F is analytic in the region between the two curves. Orient this outer circle counterclockwise; the dogbone, as the inner boundary of that

region, then goes from zero to one above the cut and from one to zero below it. On $|z| = \epsilon$ we have $|z-1| \geq 1-\epsilon$, and on $|z-1| = \epsilon$ we have $|z| \geq 1-\epsilon$, so by the modulus formula

$$|F| \leq \epsilon^{-2/3}(1-\epsilon)^{-1/3} \text{ on } |z| = \epsilon, \qquad |F| \leq (1-\epsilon)^{-2/3}\epsilon^{-1/3} \text{ on } |z-1| = \epsilon.$$

With lengths at most $2\pi\epsilon$, the endpoint circles contribute $O(\epsilon^{1/3})$ and $O(\epsilon^{2/3})$, hence vanish. If I is the desired integral, the upper bank gives $\int_0^1 F_+\,dx = I$ and the lower bank $\int_1^0 F_-\,dx = -e^{2\pi i/3}I$, together $(1-e^{2\pi i/3})I$. At infinity, the binomial series $(1-1/z)^{-1/3} = 1 + 1/(3z) + O(z^{-2})$ gives $F(z) = e^{i\pi/3}/z + O(z^{-2})$; the remainder contributes $O(R^{-1})$ on the outer circle, of length $2\pi R$, so the outer circle tends to $e^{i\pi/3}\oint dz/z = 2\pi i e^{i\pi/3}$. Since F is analytic between the two curves, Cauchy's theorem gives

$$(1-e^{2\pi i/3})I + 2\pi i e^{i\pi/3} = 0.$$

Since $1 - e^{2\pi i/3} = e^{i\pi/3}(e^{-i\pi/3} - e^{i\pi/3}) = -2i\sin(\pi/3)\,e^{i\pi/3} = -i\sqrt{3}\,e^{i\pi/3}$, we get $I = -2\pi i e^{i\pi/3}/(-i\sqrt{3}\,e^{i\pi/3})$:

$$\boxed{I = \frac{2\pi}{\sqrt{3}}}.$$

Independently, $x = t/(1+t)$ maps $0 < t < \infty$ onto $0 < x < 1$, with $dx = dt/(1+t)^2$, $x^{-2/3} = t^{-2/3}(1+t)^{2/3}$, and $(1-x)^{-1/3} = (1+t)^{1/3}$. It transforms I into $\int_0^\infty t^{-2/3}(1+t)^{2/3+1/3-2}\,dt = \int_0^\infty t^{-2/3}/(1+t)\,dt$, the $a = 2/3$ case of 11.8.24, $\pi/\sin(2\pi/3) = 2\pi/\sqrt{3}$.

11.8.28 Evaluate

$$\int_{-\infty}^{\infty} \frac{\tan^{-1} ax\,dx}{x(x^2+b^2)},$$

for a and b positive, with $ab < 1$. Explain why the integrand does not have a singularity at $x = 0$.

Hint. Try the contour shown in Fig. 11.32, and use Eq. (1.137) to represent $\tan^{-1} az$. After cancellation, the integrals on segments B and B' combine to give an elementary integral.

Solution.

Let

$$A(z) = \arctan(az) = \frac{1}{2i}\{\mathrm{Log}(1+iaz) - \mathrm{Log}(1-iaz)\}, \qquad F(z) = \frac{A(z)}{z(z^2+b^2)},$$

using the principal logarithms, so that $A(0) = 0$. For real x the numbers $1 \pm iax$ are conjugate with positive real part, so $A(x) = \frac{1}{2i}\cdot 2i\,\mathrm{Arg}(1+iax) = \tan^{-1} ax$, the real integrand. The Taylor expansion $A(z) = az + O(z^3)$ gives $A(z)/z = a + O(z^2)$ and $F(z) = a/b^2 + O(z^2)$ at zero, so the apparent singularity is removable: the integrand tends to a/b^2 at $x = 0$.

The principal $\mathrm{Log}(1+iaz)$ is cut where $1 + iaz \leq 0$, that is, on $z = iy$, $y \geq 1/a$; similarly $\mathrm{Log}(1-iaz)$ is cut on $z = iy$, $y \leq -1/a$, outside the upper half-plane. Use the upper contour of Fig. 11.32, with the slit from i/a to $i\infty$ excluded. The assumption $ab < 1$ places the pole ib below the branch point i/a. On the small circle of radius ρ around i/a, $1 + iaz = ia(z - i/a)$ has modulus $a\rho$, so $|\mathrm{Log}(1+iaz)| \leq |\ln(a\rho)| + \pi$, while $1 - iaz = 2 - ia(z - i/a)$ stays near 2, so $\mathrm{Log}(1-iaz)$ is bounded; and $1/[z(z^2+b^2)]$ is bounded there because $z^2 + b^2 = b^2 - 1/a^2 \neq 0$ at $z = i/a$. Hence $|F| = O(|\ln\rho|)$ on this circle, and with its length at most $2\pi\rho$ its integral is $O(\rho|\ln\rho|) \to 0$. On the large arc $|z| = R$, $|1+iaz|/|1-iaz| = |z - i/a|/|z + i/a| \to 1$ and the arguments are bounded by π, so $A(z)$ stays bounded, while $|z(z^2+b^2)| \geq R(R^2 - b^2)$; with the length at most πR, the arc

integral is $O(R^{-2})$ and vanishes. The enclosed pole ib is simple, and $(z-ib)F(z) = A(z)/[z(z+ib)]$ at $z = ib$ gives

$$\operatorname{Res}_{ib} F = \frac{A(ib)}{(ib)(2ib)} = -\frac{A(ib)}{2b^2} = -\frac{i}{4b^2}\ln\frac{1+ab}{1-ab},$$

where, since $1 + ia(ib) = 1 - ab$, $1 - ia(ib) = 1 + ab$, and $1/(2i) = -i/2$,

$$A(ib) = \frac{1}{2i}\left[\ln(1-ab) - \ln(1+ab)\right] = \frac{i}{2}\ln\frac{1+ab}{1-ab}.$$

For $z = iy$, $y > 1/a$, we have $1 + iaz = 1 - ay < 0$ and $1 - iaz = 1 + ay > 0$. Just right of the slit, $z = iy + \delta$ with small $\delta > 0$, so $1 + iaz = 1 - ay + ia\delta$ lies in the upper half-plane and its argument tends to $+\pi$; just left of it the argument tends to $-\pi$. The second logarithm has no jump. Hence $A_{\text{right}} - A_{\text{left}} = \frac{1}{2i}(i\pi + i\pi) = \pi$. The right bank is traversed downward and the left bank upward; with $z = iy$, $dz = i\,dy$, and $z(z^2 + b^2) = iy(b^2 - y^2)$, their combined contribution is

$$\begin{aligned}\int_{\infty}^{1/a} \frac{A_{\text{right}}\, i\,dy}{iy(b^2-y^2)} + \int_{1/a}^{\infty} \frac{A_{\text{left}}\, i\,dy}{iy(b^2-y^2)} &= -i\int_{1/a}^{\infty} \frac{(A_{\text{right}} - A_{\text{left}})\,dy}{iy(b^2-y^2)} \\ &= -i\int_{1/a}^{\infty} \frac{\pi\,dy}{iy(b^2-y^2)} = \pi\int_{1/a}^{\infty} \frac{dy}{y(y^2-b^2)}.\end{aligned}$$

The elementary antiderivative is

$$\int \frac{dy}{y(y^2-b^2)} = \frac{1}{2b^2}\ln\left(1 - \frac{b^2}{y^2}\right),$$

as direct differentiation verifies:

$$\frac{d}{dy}\frac{1}{2b^2}\ln\left(1-\frac{b^2}{y^2}\right) = \frac{1}{2b^2}\cdot\frac{2b^2/y^3}{1-b^2/y^2} = \frac{1}{y(y^2-b^2)}.$$

The antiderivative tends to $\ln 1 = 0$ as $y \to \infty$ and equals $\ln(1-a^2b^2)/(2b^2)$ at $y = 1/a$, so its definite value is $-\ln(1-a^2b^2)/(2b^2)$. Calling the real-axis integral I, the residue theorem, with $2\pi i \cdot (-i) = 2\pi$ in $2\pi i \operatorname{Res}_{ib} F$, gives

$$I - \frac{\pi}{2b^2}\ln(1-a^2b^2) = \frac{\pi}{2b^2}\ln\frac{1+ab}{1-ab}.$$

Combine the logarithms, whose arguments are positive under $ab < 1$, and use $1 - a^2b^2 = (1-ab)(1+ab)$:

$$\boxed{I = \frac{\pi}{2b^2}\ln\left[\frac{1+ab}{1-ab}(1-a^2b^2)\right] = \frac{\pi}{b^2}\ln(1+ab)}.$$

The tails are $O(|x|^{-3})$ and the origin is removable, so this real integral is absolutely convergent.

11.9 Evaluation of Sums

11.9.1 Show that if $f(z)$ is analytic at $z = z_0$ and $g(z)$ has a simple pole at $z = z_0$ with residue b_0, then $f(z)g(z)$ has at most a simple pole at $z = z_0$, with residue $f(z_0)b_0$.

Solution.

Write the Taylor and Laurent expansions, respectively:

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + O((z - z_0)^2), \qquad g(z) = \frac{b_0}{z - z_0} + b_1 + O(z - z_0).$$

Multiplication gives, since the cross term $f'(z_0)(z - z_0) \cdot b_0/(z - z_0) = f'(z_0)b_0$ is a constant,

$$f(z)g(z) = \frac{f(z_0)b_0}{z - z_0} + \{f(z_0)b_1 + f'(z_0)b_0\} + O(z - z_0).$$

No power below $(z-z_0)^{-1}$ occurs, so fg has at most a simple pole at z_0. The coefficient of $(z-z_0)^{-1}$ is therefore

$$\boxed{\operatorname{Res}_{z_0}(fg) = f(z_0)b_0}.$$

The claim that the product also has a simple pole requires $f(z_0) \neq 0$. If $f(z_0) = 0$, the pole is canceled and the product has a removable singularity, with residue zero. For example, $f(z) = z - z_0$ and $g(z) = 1/(z - z_0)$ give the constant one.

11.9.2 Show that $\cot z$ has magnitude $O(1)$ away from fixed neighborhoods of its poles. Deduce that $I_N = \oint_{C_N} f(z)\pi \cot(\pi z)\, dz \to 0$ on $C_N : |z| = N + \frac{1}{2}$ if $(N + \frac{1}{2}) \max_{C_N} |f(z)| \to 0$.

Solution.

For $z = x + iy$, the addition formulas with $\cos iy = \cosh y$ and $\sin iy = i \sinh y$ give $\sin z = \sin x \cosh y + i \cos x \sinh y$ and $\cos z = \cos x \cosh y - i \sin x \sinh y$; with $\cosh^2 y = 1 + \sinh^2 y$, the identities

$$|\sin z|^2 = \sin^2 x + \sinh^2 y, \qquad |\cos z|^2 = \cos^2 x + \sinh^2 y$$

follow. Hence

$$|\cot z|^2 = \frac{\cos^2 x + \sinh^2 y}{\sin^2 x + \sinh^2 y}.$$

If $|y| \geq \delta > 0$, then

$$|\cot z|^2 \leq \frac{1 + \sinh^2 y}{\sinh^2 y} = \coth^2 |y| \leq \coth^2 \delta.$$

In the remaining strip $|y| \leq \delta$, impose a fixed positive distance $d \leq \pi/2$ from the poles $\pi\mathbb{Z}$. The period π of $\cot z$ reduces x to $[-\pi/2, \pi/2]$, where the only pole is $z = 0$; the resulting set $\{|x| \leq \pi/2,\ |y| \leq \delta,\ |z| \geq d\}$ is compact and contains no zero of $\sin z$. The continuous quotient $\cos z / \sin z$ is therefore bounded there too. Thus $\cot z$ is uniformly bounded away from fixed neighborhoods of its poles. The textbook's *order one*, written $O(1)$ in the statement, means boundedness, not a nonzero lower bound: $\cot z$ has zeros at $(n + \frac{1}{2})\pi$ even for arbitrarily large $|z|$. The exclusion of the poles must have a fixed positive width.

For the text's circles C_N, with radius $R_N = N + \frac{1}{2}$,

$$|z - n| \geq \big||z| - |n|\big| = \big|N + \tfrac{1}{2} - |n|\big| \geq \tfrac{1}{2} \qquad (z \in C_N,\ n \in \mathbb{Z}),$$

because $N+\frac{1}{2}-|n|$ is an odd multiple of $\frac{1}{2}$. Thus $w=\pi z$ stays at distance at least $\pi/2$ from the poles $\pi\mathbb{Z}$ of $\cot w$, and the bound just proved, with $d=\pi/2$, gives $|\pi\cot\pi z|\le M$ on every C_N, with one constant M independent of N. If $R_N\max_{C_N}|f(z)|\to 0$, the estimation lemma gives

$$|I_N|=\left|\oint_{C_N} f(z)\pi\cot\pi z\,dz\right|\le 2\pi R_N M\max_{C_N}|f(z)|\longrightarrow 0.$$

Since $|z|=R_N$ on C_N, the hypothesis of the statement is $\max_{C_N}|zf(z)|\to 0$: the decay of $zf(z)$ used here is uniform on the contours. Directional or pointwise decay alone does not supply this bound.

11.9.3 Evaluate $\dfrac{1}{1^3}-\dfrac{1}{3^3}+\dfrac{1}{5^3}-\cdots$.

Solution.

Exercise 11.8.18 established both

$$\int_0^\infty\frac{(\ln x)^2}{1+x^2}\,dx=4\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^3}\quad\text{and}\quad\int_0^\infty\frac{(\ln x)^2}{1+x^2}\,dx=\frac{\pi^3}{8}.$$

Dividing the second value by the factor four in the first identity gives

$$\boxed{1-\frac{1}{3^3}+\frac{1}{5^3}-\cdots=\frac{\pi^3}{32}}.$$

That earlier derivation justified the series interchange by absolute integrability and evaluated the integral with the rectangular contour of Fig. 11.29.

11.9.4 Evaluate $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n(n+2)}$.

Solution.

Resolve the summand into partial fractions:

$$\frac{1}{n(n+2)}=\frac{1}{2}\left(\frac{1}{n}-\frac{1}{n+2}\right).$$

For a finite upper limit $N\ge 2$, shift the index of the second sum to $m=n+2$; every term from $1/3$ through $1/N$ then occurs in both sums and cancels:

$$S_N=\frac{1}{2}\left(\sum_{n=1}^{N}\frac{1}{n}-\sum_{m=3}^{N+2}\frac{1}{m}\right)=\frac{1}{2}\left(1+\frac{1}{2}-\frac{1}{N+1}-\frac{1}{N+2}\right).$$

Taking $N\to\infty$ gives

$$\boxed{\sum_{n=1}^{\infty}\frac{1}{n(n+2)}=\frac{3}{4}}.$$

The summands are $O(n^{-2})$, which also confirms convergence before performing the cancellation.

11.9.5 Evaluate $\displaystyle\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2}$, where a is real and not an integer.

Solution.

Integrate

$$F(z) = \frac{\pi \csc \pi z}{(z+a)^2}$$

on the expanding rectangles with vertical sides $\Re z = \pm(N+\frac{1}{2})$ and horizontal sides $\Im z = \pm N$, where $N > |a|$ is an integer. With the identities of Exercise 11.9.2: on a vertical side, $\sin(\pi(m+\frac{1}{2})+i\pi y) = (-1)^m \cosh \pi y$ with $m = N$ or $m = -N-1$, so $|\csc \pi z| = \operatorname{sech}(\pi \Im z) \le 1$; on the horizontal sides $|\sin \pi z| \ge \sinh \pi N$, so $|\csc \pi z| \le 1/\sinh \pi N$ decays exponentially. Also $|z+a| \ge N - |a|$, so $(z+a)^{-2} = O(N^{-2})$ on the boundary, whose length is $8N+2$. Thus the boundary integral is $O(N^{-1})$ and tends to zero. The poles of $\pi \csc \pi z = \pi/\sin \pi z$ are the simple zeros n of $\sin \pi z$; the residue of a quotient p/q at a simple zero of q is p/q' there, which gives $\pi/(\pi \cos \pi n) = (-1)^n$. Since $(z+a)^{-2}$ is analytic at n, Exercise 11.9.1 gives the residue $(-1)^n/(n+a)^2$ of F there.

Since a is not an integer, the remaining pole at $-a$ is double and distinct from all the integer poles. For a double pole of $g(z)/(z+a)^2$ with g analytic at $-a$, the residue is $g'(-a)$; here $g(z) = \pi \csc \pi z$, with $(\csc w)' = -\csc w \cot w$, and csc, cot are odd, so

$$\left.\frac{d}{dz}(\pi \csc \pi z)\right|_{z=-a} = -\pi^2 \csc(-\pi a)\cot(-\pi a) = -\pi^2 \csc \pi a \cot \pi a.$$

For $N > |a|$ the residue theorem gives

$$\oint F\,dz = 2\pi i \left[\sum_{n=-N}^{N} \frac{(-1)^n}{(n+a)^2} - \pi^2 \csc \pi a \cot \pi a\right],$$

and since the boundary integral vanishes as $N \to \infty$,

$$0 = \sum_{n\in\mathbb{Z}} \frac{(-1)^n}{(n+a)^2} - \pi^2 \csc \pi a \cot \pi a.$$

Therefore

$$\boxed{\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2} = \pi^2 \csc \pi a \cot \pi a}.$$

The series is absolutely convergent, so the limiting rectangular ordering does not restrict the meaning of the sum.

11.9.6

(a) Using a method based on contour integration, evaluate $\displaystyle\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$.

(b) Check your work by relating your answer to an appropriate expression involving zeta functions.

Solution.

(a) Integrate $F(z) = \pi \tan(\pi z)/z^2$ over the square with sides $\Re z = \pm N$, $\Im z = \pm N$, where N is a positive integer. On the vertical sides the period one of $\tan \pi z$ gives $\tan(\pi(\pm N + iy)) = \tan(i\pi y) = i \tanh \pi y$, so its modulus is at most one. On the horizontal sides, by the identities of Exercise 11.9.2,

$$|\tan(x+iy)|^2 = \frac{\sin^2 x + \sinh^2 y}{\cos^2 x + \sinh^2 y} \le \frac{1+\sinh^2 y}{\sinh^2 y} = \coth^2 y,$$

so with $y = \pm\pi N$ its modulus is bounded by $\coth \pi N \le \coth \pi$. Since $|z| \ge N$ on the square, $|F| \le \pi \coth \pi / N^2$ there; with the perimeter $8N$, the contour integral is $O(N^{-1})$ and vanishes.

The expansion $\tan \pi z = \pi z + O(z^3)$ gives $F(z) = \pi^2/z + O(z)$, hence the residue π^2 at zero. The other poles are the zeros $p = n + \frac{1}{2}$, $n \in \mathbb{Z}$, of $\cos \pi z$. They are simple, so the residue of $\pi \tan \pi z = \pi \sin \pi z / \cos \pi z$ is the numerator divided by the derivative of the denominator:

$$\operatorname{Res}_p(\pi \tan \pi z) = \frac{\pi \sin \pi p}{-\pi \sin \pi p} = -1,$$

and since $1/z^2$ is analytic at p, the residue of F is $-1/p^2$ (Exercise 11.9.1). The square contains zero and the $2N$ poles $n + \frac{1}{2}$ with $-N \le n \le N - 1$, so the residue theorem gives

$$\oint F\, dz = 2\pi i \left[\pi^2 - \sum_{n=-N}^{N-1} \frac{1}{(n + \frac{1}{2})^2} \right],$$

and letting $N \to \infty$ yields

$$\sum_{n \in \mathbb{Z}} \frac{1}{(n + \frac{1}{2})^2} = \pi^2.$$

The substitution $n \to -1 - n$ maps $n + \frac{1}{2}$ to $-(n + \frac{1}{2})$, so the terms with $n \le -1$ repeat those with $n \ge 0$; with $1/(n + \frac{1}{2})^2 = 4/(2n + 1)^2$ this gives

$$\sum_{n \in \mathbb{Z}} \frac{1}{(n + \frac{1}{2})^2} = 2 \sum_{n=0}^{\infty} \frac{4}{(2n+1)^2} = 8 \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}.$$

Thus

$$\boxed{\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}}.$$

The discussion on textbook p. 545 assigns residue $+1$ to $\pi \tan(\pi z)$. The quotient calculation above shows that each residue is -1; this sign is required in the contour sum.

(b) Split the defining zeta series into its odd and even parts:

$$\zeta(2) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} + \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} + \frac{1}{4}\zeta(2).$$

The odd sum is $(3/4)\zeta(2)$. Using $\zeta(2) = \pi^2/6$ gives $(3/4)(\pi^2/6) = \pi^2/8$, in agreement with the contour result; conversely that result supplies the same value of $\zeta(2)$.

11.9.7 Show that $\dfrac{1}{\cosh(\pi/2)} - \dfrac{1}{3\cosh(3\pi/2)} + \dfrac{1}{5\cosh(5\pi/2)} - \cdots = \dfrac{\pi}{8}$.

Solution.

Let

$$S = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)\cosh((2n+1)\pi/2)}, \qquad F(z) = \frac{1}{z \cos z \cosh z}.$$

Integrate F over the square with sides $\Re z = \pm R$, $\Im z = \pm R$, choosing $R = N\pi$ with a positive integer N. On a vertical side, $\cos(\pm N\pi + iy) = \cos(N\pi)\cos iy = (-1)^N \cosh y$ (the term with

$\sin(\pm N\pi)$ vanishes), so $|\cos z| = \cosh y$; also $|\cosh(\pm R + iy)| \geq \sinh R$ (Exercise 11.8.25), while $|z| \geq R$. Since $\int_{\mathbb{R}} \operatorname{sech} y\, dy = \pi$ (Exercise 11.8.18), the integral over the side is bounded by

$$\frac{1}{R\sinh R}\int_{-R}^{R} \operatorname{sech} y\, dy \leq \frac{\pi}{R\sinh R}.$$

On a horizontal side $z = x \pm iR$ the roles of cosine and hyperbolic cosine interchange: $|\cos z|^2 = \cos^2 x + \sinh^2 R \geq \sinh^2 R$ (Exercise 11.9.2) and $\cosh(x \pm iN\pi) = (-1)^N \cosh x$, giving the same bound. Thus the integral over the square is at most $4\pi/(R\sinh R)$ and tends to zero.

Since $\cos z \cosh z = 1 + O(z^2)$, $F(z) = z^{-1} + O(z)$ and the residue at zero is one. Put $a_n = (n+\frac{1}{2})\pi$, $n \geq 0$. The other poles are the zeros $\pm a_n$ of $\cos z$ and $\pm i a_n$ of $\cosh z$ (recall $\cosh iy = \cos y$); they are simple and distinct. At each, the residue is $1/D'$ for $D = z\cos z\cosh z$, and D' reduces to the product with the vanishing factor differentiated. At the positive real poles, with $(\cos z)' = -\sin z$ and $\sin a_n = (-1)^n$,

$$\operatorname{Res}_{a_n} F = \frac{1}{-a_n \sin a_n \cosh a_n} = \frac{(-1)^{n+1}}{a_n \cosh a_n}.$$

At the positive imaginary poles, with $(\cosh z)' = \sinh z$, $\cos(ia_n) = \cosh a_n$, and $\sinh(ia_n) = i\sin a_n = i(-1)^n$,

$$\operatorname{Res}_{ia_n} F = \frac{1}{(ia_n)\cos(ia_n)\sinh(ia_n)} = \frac{1}{(ia_n)\cosh a_n\, i(-1)^n} = \frac{(-1)^{n+1}}{a_n\cosh a_n}.$$

Because F is odd, its residue at $-p$ equals its residue c at p: near $-p$, $F(z) = -F(-z) = -c/(-z-p) + \cdots = c/(z+p) + \cdots$. So the poles $-a_n$ and $-ia_n$ have the same residue, and, with $1/a_n = 2/[(2n+1)\pi]$, the four poles for each n contribute

$$4\frac{(-1)^{n+1}}{a_n\cosh a_n} = -\frac{8}{\pi}\,\frac{(-1)^n}{(2n+1)\cosh((2n+1)\pi/2)}.$$

The square contains zero and the four poles with $a_n < N\pi$, that is, $n \leq N-1$, so the residue theorem gives

$$\oint F\, dz = 2\pi i\left[1 - \frac{8}{\pi}\sum_{n=0}^{N-1}\frac{(-1)^n}{(2n+1)\cosh((2n+1)\pi/2)}\right].$$

The series converges absolutely by exponential decay. Letting $N \to \infty$ gives $0 = 2\pi i(1 - 8S/\pi)$, and hence

$$\boxed{S = \frac{\pi}{8}}.$$

11.9.8 For $-\pi \leq \varphi \leq +\pi$, show that $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{\sin n\varphi}{n^3} = \frac{\varphi}{12}(\varphi^2 - \pi^2)$.

Solution.

The original series converges absolutely and uniformly in real φ, since $|\sin n\varphi|/n^3 \leq 1/n^3$. To justify the differentiations, first insert $0 < r < 1$:

$$S_r(\varphi) = \sum_{n=1}^{\infty}\frac{(-1)^n r^n \sin n\varphi}{n^3}.$$

The once and twice differentiated series have terms bounded by r^n/n^2 and r^n/n, so they converge uniformly (Weierstrass M-test) and may be differentiated term by term. Since $(\sin n\varphi)'' = -n^2 \sin n\varphi$, differentiating twice gives the first series below; the second equality is the imaginary part of the logarithmic series $\operatorname{Log}(1+w) = \sum_{n\ge1}(-1)^{n+1}w^n/n$, $|w|<1$, at $w = re^{i\varphi}$, where $\Im w^n = r^n \sin n\varphi$:

$$S_r''(\varphi) = -\sum_{n=1}^{\infty} \frac{(-1)^n r^n \sin n\varphi}{n} = \Im \operatorname{Log}(1+re^{i\varphi}).$$

The real part $1 + r\cos\varphi \ge 1 - r$ of $1+re^{i\varphi}$ is positive, so this imaginary part, its principal argument, is bounded in absolute value by $\pi/2$. For $-\pi < \varphi < \pi$,

$$1 + e^{i\varphi} = e^{i\varphi/2}(e^{-i\varphi/2} + e^{i\varphi/2}) = 2\cos(\varphi/2)e^{i\varphi/2},$$

with $\cos(\varphi/2) > 0$, so its principal argument is $\varphi/2$; by continuity of Log off the negative real axis,

$$\lim_{r\uparrow1} S_r''(\varphi) = \frac{\varphi}{2}.$$

For $0 \le \varphi \le \pi$, Taylor's formula with integral remainder, $S_r(\varphi) = S_r(0) + S_r'(0)\varphi + \int_0^\varphi (\varphi - t)S_r''(t)\,dt$, with $S_r(0) = 0$ and $C_r = S_r'(0)$, gives

$$S_r(\varphi) = C_r\varphi + \int_0^\varphi (\varphi - t)\Im \operatorname{Log}(1+re^{it})\,dt.$$

Also $S_r(\pi) = 0$, term by term, since $\sin n\pi = 0$; setting $\varphi = \pi$ and solving for C_r,

$$C_r = -\frac{1}{\pi}\int_0^\pi (\pi - t)\Im \operatorname{Log}(1+re^{it})\,dt \longrightarrow -\frac{1}{\pi}\int_0^\pi \frac{(\pi-t)t}{2}\,dt = -\frac{\pi^2}{12},$$

because $\int_0^\pi (\pi - t)t\,dt = [\pi t^2/2 - t^3/3]_0^\pi = \pi^3/6$. The bounded argument justifies dominated convergence in both integrals (the single point $t = \pi$, where the limit is not $t/2$, does not matter). Since $0 \le 1 - r^n = (1-r)(1 + r + \cdots + r^{n-1}) \le n(1-r)$,

$$|S(\varphi) - S_r(\varphi)| \le \sum_{n=1}^{\infty} \frac{1-r^n}{n^3} \le (1-r)\sum_{n=1}^{\infty}\frac{1}{n^2} = (1-r)\frac{\pi^2}{6} \to 0,$$

so $S_r \to S$ uniformly. Letting $r \uparrow 1$ in the formula for $S_r(\varphi)$, with $\int_0^\varphi (\varphi - t)t\,dt = \varphi^3/2 - \varphi^3/3 = \varphi^3/6$,

$$S(\varphi) = -\frac{\pi^2\varphi}{12} + \int_0^\varphi \frac{(\varphi-t)t}{2}\,dt = -\frac{\pi^2\varphi}{12} + \frac{\varphi^3}{12}.$$

Both sides are odd, so the formula extends to negative φ. Continuity includes the endpoints, where both sides vanish:

$$\boxed{\sum_{n=1}^{\infty} \frac{(-1)^n \sin n\varphi}{n^3} = \frac{\varphi}{12}(\varphi^2 - \pi^2), \quad -\pi \le \varphi \le \pi}.$$

11.10 Miscellaneous Topics

11.10.1 A function $f(z) = u(x, y) + iv(x, y)$ satisfies the conditions for the Schwarz reflection principle. Show that

(a) u is an even function of y. (b) v is an odd function of y.

Solution.

Under those conditions (f analytic in a connected domain symmetric about the real axis that contains a real interval on which f is real), the Schwarz reflection principle gives $f(z^*) = f^*(z)$ throughout that domain, where $f^*(z)$ denotes the complex conjugate $[f(z)]^*$. Writing $z = x + iy$, so that $z^* = x - iy$, $f(z^*) = u(x, -y) + iv(x, -y)$ and $f^*(z) = u(x, y) - iv(x, y)$; hence

$$u(x, -y) + iv(x, -y) = u(x, y) - iv(x, y),$$

and equating real and imaginary parts gives

$$\boxed{u(x, -y) = u(x, y)}, \qquad \boxed{v(x, -y) = -v(x, y)}.$$

The real part is even in y and the imaginary part is odd in y, at all pairs of points related by the reflection.

11.10.2 A function $f(z)$ can be expanded in a Laurent series about the origin with the coefficients a_n real. Show that the complex conjugate of this function of z is the same function of the complex conjugate of z; that is,

$$f^*(z) = f(z^*).$$

Verify this explicitly for

(a) $f(z) = z^n$, n an integer. (b) $f(z) = \sin z$.

If $f(z) = iz$ ($a_1 = i$), show that the foregoing statement does not hold.

Solution.

On the Laurent series' annulus of convergence,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n, \qquad a_n \in \mathbb{R}.$$

The annulus is invariant under complex conjugation. Conjugating its convergent partial sums and then taking the limit gives, with $f^*(z) = [f(z)]^*$,

$$f^*(z) = \sum_{n=-\infty}^{\infty} a_n^* \, (z^*)^n = \sum_{n=-\infty}^{\infty} a_n (z^*)^n = f(z^*).$$

(a) For $f(z) = z^n$ this follows directly from conjugation of products if $n \geq 0$ and of reciprocals if $n < 0$ (the latter require $z \neq 0$). Explicitly, with $z = \rho e^{i\phi}$, $z^* = \rho e^{-i\phi}$,

$$(z^n)^* = (\rho^n e^{in\phi})^* = \rho^n e^{-in\phi} = (\rho e^{-i\phi})^n = (z^*)^n.$$

(b) For the sine function, the addition formula with $\cos iy = \cosh y$ and $\sin iy = i\sinh y$ gives

$$\sin(x+iy) = \sin x\cosh y + i\cos x\sinh y.$$

Replacing y by $-y$ changes only the sign of its imaginary part, since cosh is even and sinh odd, so $\sin(z^*) = \sin x\cosh y - i\cos x\sinh y = (\sin z)^*$ explicitly. The Instructor's Manual's sine series omits both alternating signs and the odd powers. The correct series is $\sin z = \sum_{n=0}^{\infty}(-1)^n z^{2n+1}/(2n+1)!$, whose real coefficients agree with the direct Cartesian verification above.

In contrast, for $f(z) = iz$,

$$f^*(z) = (iz)^* = -iz^*, \qquad f(z^*) = iz^*.$$

They disagree for every $z \neq 0$, showing why the reality of the coefficients is essential.

11.10.3 The function $f(z)$ is analytic in a domain that is connected, invariant under complex conjugation, and includes a real interval. When z is real ($z = x$), $f(x)$ is pure imaginary.

(a) Show that

$$f(z^*) = -[f(z)]^*.$$

(b) For the specific case $f(z) = iz$, develop the Cartesian forms of $f(z)$, $f(z^*)$, and $f^*(z)$. Do not quote the general result of part (a).

Solution.

(a) Define $g(z) = -if(z)$, analytic in the same domain. If $f(x) = i\beta(x)$ with $\beta(x)$ real on the real interval, then $g(x) = \beta(x)$ is real there. Schwarz reflection gives $g(z^*) = [g(z)]^*$; since $g(z^*) = -if(z^*)$ and $[g(z)]^* = (-i)^*[f(z)]^* = i[f(z)]^*$,

$$-if(z^*) = i[f(z)]^*.$$

Dividing by $-i$ proves

$$\boxed{f(z^*) = -[f(z)]^*}.$$

The domain conditions of the statement are needed because $f(z^*)$ must be defined wherever the identity is asserted, and the identity theorem behind Schwarz reflection needs a connected domain containing the real interval. For a domain D not symmetric about the real axis, the identity is guaranteed only on the connected component of $D \cap \{z^* : z \in D\}$ that contains the interval.

(b) Compute the three expressions separately from $z = x + iy$:

$$f(z) = i(x+iy) = -y + ix, \qquad f(z^*) = i(x-iy) = y + ix, \qquad f^*(z) = -y - ix.$$

The second is the negative of the third, directly confirming (a) without invoking it.

11.10.4 How do circles centered on the origin in the z-plane transform for

(a) $w_1(z) = z + \dfrac{1}{z}$, (b) $w_2(z) = z - \dfrac{1}{z}$, for $z \neq 0$?

What happens when $|z| \to 1$?

Solution.

Parameterize a circle by $z = re^{i\theta}$, $r > 0$. Then $z^{-1} = r^{-1}(\cos\theta - i\sin\theta)$. **(a)** For $w_1 = u + iv = z + z^{-1}$,

$$u = (r + r^{-1})\cos\theta, \qquad v = (r - r^{-1})\sin\theta.$$

If $r \neq 1$, then $\cos\theta = u/(r + r^{-1})$ and $\sin\theta = v/(r - r^{-1})$, and $\cos^2\theta + \sin^2\theta = 1$ eliminates θ, giving the ellipse

$$\boxed{\frac{u^2}{(r + r^{-1})^2} + \frac{v^2}{(r - r^{-1})^2} = 1}.$$

Its horizontal semiaxis is $r + r^{-1}$ and its vertical semiaxis is $|r - r^{-1}|$. For $r < 1$ the sign of the sine coefficient reverses the direction of traversal, but not the geometric ellipse.

(b) For $w_2 = z - z^{-1}$ the coordinates are

$$u = (r - r^{-1})\cos\theta, \qquad v = (r + r^{-1})\sin\theta,$$

giving the ellipse

$$\boxed{\frac{u^2}{(r - r^{-1})^2} + \frac{v^2}{(r + r^{-1})^2} = 1 \qquad (r \neq 1)}.$$

Here the major axis is vertical. Since $(r + r^{-1})^2 - (r - r^{-1})^2 = 4$, the ellipses of (a) all have foci ± 2 and those of (b) foci $\pm 2i$. As $r \to 1$, one semiaxis in each case tends to zero, and the ellipses collapse onto the segment joining the foci. At $r = 1$ the parameterizations, rather than the divided equations, give $w_1 = 2\cos\theta$ and $w_2 = 2i\sin\theta$. The images are the segments $[-2, 2]$ and $[-2i, 2i]$, respectively, each traced twice during a full circuit of the unit circle.

11.10.5 What part of the z-plane corresponds to the interior of the unit circle in the w-plane if

(a) $w = \dfrac{z - 1}{z + 1}$? (b) $w = \dfrac{z - i}{z + i}$?

Solution.

(a) For $z \neq -1$, $|w| = |z - 1|/|z + 1|$, so the condition $|w| < 1$ is $|z - 1| < |z + 1|$. Squaring (both sides are nonnegative) and substituting $z = x + iy$ gives

$$(x - 1)^2 + y^2 < (x + 1)^2 + y^2 \quad \Longleftrightarrow \quad -2x < 2x \quad \Longleftrightarrow \quad x > 0.$$

Thus the disk corresponds exactly to the open right half-plane. The pole $z = -1$ is outside it; the imaginary axis maps to its boundary.

(b) Similarly,

$$|z - i| < |z + i| \quad \Longleftrightarrow \quad x^2 + (y - 1)^2 < x^2 + (y + 1)^2 \quad \Longleftrightarrow \quad -2y < 2y \quad \Longleftrightarrow \quad y > 0.$$

The disk corresponds to the open upper half-plane. The pole $z = -i$ lies outside it, and the real axis maps to the boundary. The inverse formulas $z = (1 + w)/(1 - w)$ and $z = i(1 + w)/(1 - w)$, obtained by solving $w(z + 1) = z - 1$ and $w(z + i) = z - i$ for z, show that every point of $|w| < 1$ is attained, since $w = 1$ is not in the open disk.

11.10.6

(a) Writing $z = x + iy$, $w = u + iv$, show that if $w = 1/z$, the circle in the xy plane defined by $(x - a)^2 + (y - b)^2 = r^2$ transforms into $(u - A)^2 + (v - B)^2 = R^2$ when $r > 0$ and $a^2 + b^2 \neq r^2$. Determine the image line when the circle passes through zero, omitting zero from the original circle.

(b) Does the center of the circle in the z plane transform into the center of the corresponding circle in the w plane?

Solution.

Inversion maps a circle through the origin to a straight line. The claimed circle-to-circle form requires $a^2 + b^2 - r^2 \neq 0$; the origin itself is excluded from the map.

(a) Because $z = 1/w = w^*/|w|^2 = (u - iv)/(u^2 + v^2)$,

$$x = \frac{u}{u^2 + v^2}, \qquad y = -\frac{v}{u^2 + v^2}, \qquad x^2 + y^2 = \frac{1}{u^2 + v^2}.$$

Expand the original circle as $x^2 + y^2 - 2ax - 2by + a^2 + b^2 - r^2 = 0$ and multiply by $u^2 + v^2$; the terms $x^2 + y^2$, $-2ax$, $-2by$ become 1, $-2au$, $+2bv$. With $D = a^2 + b^2 - r^2$, the result is

$$1 - 2au + 2bv + D(u^2 + v^2) = 0.$$

If $D \neq 0$, divide by D, $u^2 - 2(a/D)u + v^2 + 2(b/D)v + 1/D = 0$, and complete both squares:

$$\left(u - \frac{a}{D}\right)^2 + \left(v + \frac{b}{D}\right)^2 = \frac{a^2 + b^2}{D^2} - \frac{1}{D} = \frac{a^2 + b^2 - D}{D^2} = \frac{r^2}{D^2}.$$

Therefore

$$\boxed{A = \frac{a}{D}, \qquad B = -\frac{b}{D}, \qquad R = \frac{r}{|D|}}.$$

The Instructor's Manual writes the signed quantity $r/(r^2 - a^2 - b^2)$ for the radius. Its square gives the correct circle equation, but the geometric radius must be nonnegative: $R = r/|r^2 - a^2 - b^2|$. Every step is reversible for $w \neq 0$, and $w = 0$ is not on this circle, because $A^2 + B^2 - R^2 = (a^2 + b^2 - r^2)/D^2 = 1/D \neq 0$; so the image is the whole circle. If $D = 0$, the equation is instead the line $\boxed{2au - 2bv = 1}$, which does not contain $w = 0$; each of its points is the image of a point of the circle other than zero.

(b) When $a + ib \neq 0$, the image of the original center is

$$\frac{1}{a + ib} = \frac{a - ib}{a^2 + b^2},$$

whereas the center of the image circle is $A + iB = (a - ib)/(a^2 + b^2 - r^2)$. For a genuine circle $r > 0$ these are different whenever both are defined, since the numerators agree and are nonzero while the denominators differ by r^2. If the original center is zero, its image is undefined even though the image circle has center zero. Thus a circle's center does not in general map to its image's center.

11.10.7 Assume that a curve in the xy plane passes through point z_0 in the direction $dz = e^{i\theta} ds$, where s indicates arc length on the curve. Then, if $w = f(z)$, with $f(z)$ analytic at $z = z_0$, we have $dw = (dw/dz)\, dz = f'(z) e^{i\theta} ds$, where dw is in the direction the mapping of the xy curve passes through $w_0 = f(z_0)$ in the w plane. Use this observation to prove that if $f'(z_0) \neq 0$, the angle at which two curves intersect in the z plane is the same (both in magnitude and direction) as the angle of intersection of their mappings in the w plane.

Solution.

Since $f'(z_0) \neq 0$, write it uniquely in modulus-argument form $f'(z_0) = \rho e^{i\alpha}$ with $\rho > 0$ and α defined modulo 2π. Let the two oriented tangents be

$$dz_1 = e^{i\theta_1} ds_1, \qquad dz_2 = e^{i\theta_2} ds_2, \qquad ds_1, ds_2 > 0.$$

Their images satisfy

$$dw_j = f'(z_0)dz_j = \rho e^{i(\alpha+\theta_j)}ds_j, \qquad j = 1, 2.$$

Thus each tangent rotates by the same angle α and each infinitesimal length is multiplied by the same positive factor ρ. The oriented angle between the image tangents is consequently

$$(\alpha + \theta_2) - (\alpha + \theta_1) = \theta_2 - \theta_1 \pmod{2\pi}.$$

It equals the original oriented angle, preserving both magnitude and sense. The condition $f'(z_0) \neq 0$ ensures that neither image tangent collapses to zero, so its direction is defined.

12 Chapter 12. Further Topics in Analysis

Problems

12.1 Orthogonal Polynomials

12.1.1 Starting from the Rodrigues formula in Table 12.1 for the Hermite polynomials H_n, derive the generating function for the H_n given in that table.

Solution.

The convention in Table 12.1 is

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}.$$

Taylor's theorem applied to the entire function $f(x) = e^{-x^2}$ gives

$$f(x-t) = \sum_{n=0}^{\infty} \frac{(-t)^n}{n!} f^{(n)}(x).$$

Multiply by e^{x^2}. By the Rodrigues formula each term becomes

$$e^{x^2} \frac{(-t)^n}{n!} f^{(n)}(x) = (-1)^n e^{x^2} f^{(n)}(x) \frac{t^n}{n!} = H_n(x) \frac{t^n}{n!},$$

so

$$\sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!} = e^{x^2} e^{-(x-t)^2} = e^{x^2 - (x^2 - 2xt + t^2)} = \boxed{e^{2xt - t^2}}.$$

The Taylor series converges for every finite complex t, so the generating function is an identity, not only a formal expansion. As a normalization check, $e^{2xt-t^2} = 1 + (2xt - t^2) + \frac{1}{2}(2xt)^2 + O(t^3)$, so its first three coefficients are $1, 2x, 2x^2 - 1$, corresponding to $H_0 = 1$, $H_1 = 2x$, $H_2 = 4x^2 - 2$ after multiplying by $n!$.

12.1.2

(a) Starting from the Laguerre ODE,

$$xy'' + (1-x)y' + \lambda y = 0,$$

obtain the Rodrigues formula for its polynomial solutions $L_n(x)$.

(b) From the Rodrigues formula, scaled as in Table 12.1, derive the generating function for the $L_n(x)$ given in that table.

Solution.

(a) If y is a polynomial of degree n with leading coefficient $a_n \neq 0$, the coefficient of x^n in the ODE is $(\lambda - n)a_n$: the term xy'' has degree at most $n-1$, y' has degree $n-1$, and only $-xy'$ (contributing $-na_n x^n$) and λy (contributing $\lambda a_n x^n$) reach degree n. Thus $\lambda = n$. The weight needed to put the equation in self-adjoint form satisfies $(pw)' = qw$ with $p = x$, $q = 1 - x$:

$$(xw)' = (1-x)w, \quad xw' + w = w - xw, \quad w' = -w,$$

so choose $w = e^{-x}$. The Rodrigues construction $y_n = w^{-1}D^n(wp^n)$ with $p = x$, multiplied by the factor $1/n!$ used in Table 12.1, gives

$$\boxed{L_n(x) = \frac{e^x}{n!}\frac{d^n}{dx^n}(x^n e^{-x})}.$$

We can verify this construction directly. Put $F = x^n e^{-x}$, so $xF' = x(nx^{n-1} - x^n)e^{-x} = (n - x)F$. Differentiate $n + 1$ times. Since both x and $n - x$ are linear, the Leibniz product rule leaves

$$xF^{(n+2)} + (n + 1)F^{(n+1)} = (n - x)F^{(n+1)} - (n + 1)F^{(n)}.$$

With $G = F^{(n)}$ this becomes $xG'' + (x + 1)G' + (n + 1)G = 0$. Now $G = n!e^{-x}L_n$, so $G' = n!e^{-x}(L'_n - L_n)$ and $G'' = n!e^{-x}(L''_n - 2L'_n + L_n)$. Substitution and division by the common factor $n!e^{-x}$ give

$$x(L''_n - 2L'_n + L_n) + (x + 1)(L'_n - L_n) + (n + 1)L_n = xL''_n + (1 - x)L'_n + nL_n = 0,$$

exactly the required ODE.

Leibniz's rule also shows that the result is a polynomial:

$$\frac{d^n}{dx^n}(x^n e^{-x}) = \sum_{k=0}^{n}\binom{n}{k}\frac{d^{n-k}x^n}{dx^{n-k}}\,\frac{d^k e^{-x}}{dx^k}, \qquad L_n(x) = \sum_{k=0}^{n}\binom{n}{k}\frac{(-x)^k}{k!}.$$

Indeed, differentiating x^n $n - k$ times gives $n!x^k/k!$, while the remaining k derivatives of e^{-x} give $(-1)^k e^{-x}$; the factors $n!$ and e^{-x} cancel against the prefactor $e^x/n!$. This fixes the scale $L_n(0) = 1$ and gives leading coefficient $(-1)^n/n!$.

(b) Sum that explicit polynomial formula, exchanging the order of the double sum over $0 \le k \le n$:

$$\begin{aligned}\sum_{n=0}^{\infty} L_n(x)t^n &= \sum_{k=0}^{\infty}\frac{(-x)^k}{k!}\sum_{n=k}^{\infty}\binom{n}{k}t^n = \sum_{k=0}^{\infty}\frac{(-x)^k}{k!}\frac{t^k}{(1-t)^{k+1}}\\ &= \frac{1}{1-t}\sum_{k=0}^{\infty}\frac{1}{k!}\left(-\frac{xt}{1-t}\right)^k = \boxed{\frac{1}{1-t}\exp\left(-\frac{xt}{1-t}\right)}, \qquad |t| < 1.\end{aligned}$$

The inner sum follows by differentiating the geometric series k times,

$$\frac{k!}{(1-t)^{k+1}} = \frac{d^k}{dt^k}\sum_{n=0}^{\infty}t^n = \sum_{n=k}^{\infty}\frac{n!}{(n-k)!}t^{n-k},$$

and multiplying by $t^k/k!$. Interchanging the sums is justified because the corresponding absolute sum is at most $(1 - |t|)^{-1}\exp[|x||t|/(1 - |t|)] < \infty$.

12.1.3 Carry out in detail the steps needed to confirm that the $(n + 1)$-fold differentiation of Eq. (12.10) leads to Eq. (12.12).

Solution.

Set $F = wp^n$ and $A = (n-1)p' + q$. Equation (12.10) is $pF' = AF$ (indeed $pF' = pw'p^n + np'wp^n$ and $pw' = (q - p')w$). Because p has degree at most two, $p^{(j)} = 0$ for $j \ge 3$; because A is at most linear, $A^{(j)} = 0$ for $j \ge 2$. Leibniz's formula therefore gives, with every surviving term displayed,

$$\begin{aligned}D^{n+1}(pF') &= pF^{(n+2)} + (n + 1)p'F^{(n+1)} + \frac{n(n+1)}{2}p''F^{(n)},\\ D^{n+1}(AF) &= AF^{(n+1)} + (n + 1)A'F^{(n)}.\end{aligned}$$

The last term in the first line has coefficient zero if $n = 0$. Bring the right side to the left and divide by nonzero w. The coefficient of $F^{(n+1)}/w$ is

$$(n+1)p' - [(n-1)p' + q] = 2p' - q.$$

The coefficient of $F^{(n)}/w = y_n$ is, with $A' = (n-1)p'' + q'$,

$$\begin{aligned}\frac{n(n+1)}{2}p'' - (n+1)[(n-1)p'' + q'] &= (n+1)\left[\frac{n}{2} - (n-1)\right]p'' - (n+1)q' \\ &= \frac{(n+1)(2-n)}{2}p'' - (n+1)q' \\ &= -\left[\frac{n^2-n-2}{2}p'' + (n+1)q'\right],\end{aligned}$$

since $(n+1)(2-n) = -(n^2-n-2)$. Hence

$$\boxed{\frac{p}{w}F^{(n+2)} + \frac{2p'-q}{w}F^{(n+1)} - \left[\frac{n^2-n-2}{2}p'' + (n+1)q'\right]y_n = 0},$$

which is Eq. (12.12).

12.1.4 Confirm the algebraic steps that convert Eq. (12.12) into Eq. (12.16).

Solution.

Write $G = D^n(wp^n) = wy_n$ and

$$S = \frac{w'}{w} = \frac{q-p'}{p},$$

where the second form follows from the weight condition $(pw)' = qw$, i.e. $p'w + pw' = qw$. Since $w' = Sw$ and hence $w'' = (Sw)' = (S' + S^2)w$, two product differentiations of $G = wy_n$ give

$$\frac{G'}{w} = y_n' + Sy_n, \qquad \frac{G''}{w} = y_n'' + 2Sy_n' + (S' + S^2)y_n.$$

Put $A = (n^2-n-2)p''/2 + (n+1)q'$, so that Eq. (12.12) reads $(p/w)G'' + [(2p'-q)/w]G' - Ay_n = 0$. Substitution gives

$$py_n'' + [2pS + 2p' - q]y_n' + [pS' + pS^2 + (2p'-q)S - A]y_n = 0.$$

The first-derivative coefficient simplifies, with $pS = q - p'$, to $2(q-p') + 2p' - q = q$. In the last coefficient,

$$pS^2 + (2p'-q)S = S[(q-p') + 2p' - q] = p'S,$$

so

$$\begin{aligned}pS' + p'S - A = (pS)' - A &= q' - p'' - \frac{n^2-n-2}{2}p'' - (n+1)q' \\ &= -\frac{n(n-1)}{2}p'' - nq'.\end{aligned}$$

Thus

$$\boxed{py_n'' + qy_n' - \left[\frac{n(n-1)}{2}p'' + nq'\right]y_n = 0},$$

exactly Eq. (12.16), with eigenvalue $\lambda_n = -n(n-1)\alpha - n\mu$ when $p'' = 2\alpha$, $q' = \mu$.

The textbook's intermediate Eqs. (12.13)–(12.15), on p. 553, contain algebraic errors. Equation (12.16) is correct. The consistent intermediate identities are the following:

$$\begin{aligned}
\frac{p}{w}G'' &= py_n'' + \frac{2(q-p')}{w}G' - \left[p'' - q' + \frac{q(q-p')}{p}\right]y_n, \\
0 &= py_n'' + \frac{q}{w}G' - \left[\frac{n(n-1)}{2}p'' + nq' + \frac{q(q-p')}{p}\right]y_n, \\
\frac{q}{w}G' &= qy_n' + \frac{q(q-p')}{p}y_n.
\end{aligned}$$

For the first identity, multiply G''/w by p and eliminate y_n' from the term $2pSy_n'$ using $y_n' = G'/w - Sy_n$ and $pS = q - p'$:

$$\frac{p}{w}G'' = py_n'' + \frac{2(q-p')}{w}G' + p(S' - S^2)y_n.$$

Here $pS' = q' - p'' - p'(q-p')/p$ and $pS^2 = (q-p')^2/p$, so $p(S'-S^2) = q' - p'' - (q-p')(p'+q-p')/p = q' - p'' - q(q-p')/p$. Inserting the first identity into Eq. (12.12) gives the second, because $2(q-p') + 2p' - q = q$ and $p'' - q' + A = n(n-1)p''/2 + nq'$. The third is q times $G'/w = y_n' + Sy_n$; substituting it into the second gives Eq. (12.16). The opposite signs printed in (12.14) and (12.15) would cancel each other, but neither intermediate identity is valid with those signs.

12.1.5 Given the following integral representations, in which $n \in \mathbb{Z}$ and the contours encircle the origin once counterclockwise but no other singular points, derive the corresponding generating functions:

(a) Bessel functions:

$$J_n(x) = \frac{1}{2\pi i}\oint e^{(x/2)(t-1/t)}t^{-n-1}\,dt.$$

(b) Modified Bessel functions:

$$I_n(x) = \frac{1}{2\pi i}\oint e^{(x/2)(t+1/t)}t^{-n-1}\,dt.$$

Solution.

Both functions $f(t) = e^{(x/2)(t\mp 1/t)}$ are analytic in the punctured t-plane $0 < |t| < \infty$, so each has a Laurent expansion $f(t) = \sum_{n=-\infty}^{\infty} c_n t^n$ there, with

$$c_n = \frac{1}{2\pi i}\oint f(t)\,t^{-n-1}\,dt$$

for any contour encircling the origin once counterclockwise. These are exactly the given integrals, so $c_n = J_n(x)$, respectively $I_n(x)$. This is why the statement requires $n \in \mathbb{Z}$ and a single counterclockwise circuit: for non-integer n the factor t^{-n-1} is not single-valued around the origin, so the closed-contour integral is not defined, and a contour winding k times about the origin ($k < 0$ for clockwise winding) would give $k\,c_n$ instead of c_n. Uniqueness of the Laurent expansion therefore gives

$$\boxed{e^{(x/2)(t-1/t)} = \sum_{n=-\infty}^{\infty} J_n(x)t^n}, \qquad \boxed{e^{(x/2)(t+1/t)} = \sum_{n=-\infty}^{\infty} I_n(x)t^n}.$$

For $x \neq 0$, the essential singularity at zero requires the full integer range rather than a Taylor series. At $x = 0$ both generating functions are one, so only the $n = 0$ coefficient survives. To exhibit the coefficients directly, multiply the two exponential series (upper sign for J_n, lower sign for I_n):

$$e^{xt/2}e^{\mp x/(2t)} = \sum_{j,k\geq 0} \frac{(x/2)^{j+k}(\mp 1)^k}{j!k!} t^{j-k}.$$

For $n \geq 0$, the terms with $j - k = n$ have $j = k + n$ and give

$$J_n(x) = \sum_{k\geq 0} \frac{(-1)^k (x/2)^{n+2k}}{k!(n+k)!}, \quad I_n(x) = \sum_{k\geq 0} \frac{(x/2)^{n+2k}}{k!(n+k)!}.$$

For $n = -m < 0$, the terms have $k = j + m$, so the coefficient of t^{-m} is

$$\sum_{j\geq 0} \frac{(\mp 1)^{j+m}(x/2)^{m+2j}}{j!(m+j)!} = (\mp 1)^m \times (\text{coefficient of } t^m),$$

that is, $J_{-m} = (-1)^m J_m$ and $I_{-m} = I_m$. On any compact annulus in t, the product of exponential series converges absolutely and uniformly, justifying collection of coefficients and contour integration term by term.

12.1.6 Expand the generating function for the Legendre polynomials, $(1 - 2tz + t^2)^{-1/2}$, in powers of t. Assume that t is small. Collect the coefficients of t^0, t^1, and t^2.

ANS. $a_0 = P_0(z) = 1, \quad a_1 = P_1(z) = z, \quad a_2 = P_2(z) = \frac{1}{2}(3z^2 - 1).$

Solution.

Use the binomial series

$$(1+u)^{-1/2} = 1 - \frac{1}{2}u + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}u^2 + O(u^3) = 1 - \frac{u}{2} + \frac{3u^2}{8} + O(u^3)$$

with $u = -2tz + t^2 = O(t)$. To retain only powers through t^2,

$$-\frac{u}{2} = zt - \frac{t^2}{2}, \qquad \frac{3u^2}{8} = \frac{3}{8}(4z^2t^2 + O(t^3)) = \frac{3}{2}z^2t^2 + O(t^3).$$

Consequently

$$(1 - 2tz + t^2)^{-1/2} = 1 + zt - \frac{t^2}{2} + \frac{3}{2}z^2t^2 + O(t^3) = 1 + zt + \frac{3z^2 - 1}{2}t^2 + O(t^3).$$

Thus $\boxed{P_0 = 1,\ P_1 = z,\ P_2 = (3z^2 - 1)/2}$. Here z is fixed and the branch is chosen to equal one at $t = 0$; the expansion is valid in a sufficiently small disk about zero.

12.1.7 The set of Chebyshev polynomials usually denoted $U_n(x)$ has the generating-function formula

$$\frac{1}{1 - 2xt + t^2} = \sum_{n=0}^{\infty} U_n(x)t^n.$$

Derive a recurrence formula (for integer $n \geq 0$) connecting three U_n of consecutive n, using $U_{-1}(x) = 0$ at $n = 0$.

Solution.

Multiply the generating-function identity by its denominator:

$$(1-2xt+t^2)\sum_{n=0}^{\infty}U_n(x)t^n=\sum_{n=0}^{\infty}U_nt^n-2x\sum_{n=0}^{\infty}U_nt^{n+1}+\sum_{n=0}^{\infty}U_nt^{n+2}=1.$$

The constant term gives $U_0=1$, and the coefficient of t gives $U_1-2xU_0=0$, so $U_1=2x$. At power t^{n+1} for $n\geq 1$, the three sums contribute U_{n+1}, $-2xU_n$, and U_{n-1}, and the right side contributes nothing:

$$U_{n+1}-2xU_n+U_{n-1}=0.$$

Hence

$$\boxed{U_{n+1}=2xU_n-U_{n-1}\quad(n\geq 1)}.$$

To write this recurrence at $n=0$ as requested, define $U_{-1}=0$; otherwise the three-term formula starts at $n=1$ and the $n=0$ case is the separate initial equation $U_1=2xU_0$. The recurrence gives $U_2=2x(2x)-1=4x^2-1$ and $U_3=2x(4x^2-1)-2x=8x^3-4x$, which also follow by direct expansion of the rational generating function.

12.2 Bernoulli Numbers

12.2.1 Verify the identities, Eqs. (12.32) and (12.46).

Solution.

For Eq. (12.32), multiply the numerator and denominator involving e^{-t} by e^t, then write $e^t=(e^t-1)+1$:

$$\frac{-t}{e^{-t}-1}=\frac{-te^t}{1-e^t}=\frac{te^t}{e^t-1}=\frac{t(e^t-1)+t}{e^t-1}=t+\frac{t}{e^t-1}.$$

Subtracting t gives

$$\boxed{\frac{t}{e^t-1}=\frac{-t}{e^{-t}-1}-t}.$$

The first two equalities already prove Eq. (12.46):

$$\boxed{\frac{te^t}{e^t-1}=\frac{-t}{e^{-t}-1}}.$$

These identities hold away from the denominator's zeros and extend through the removable point $t=0$ by continuity; both quotients there have limiting value one.

12.2.2 Show that the first Bernoulli polynomials are

$$\begin{aligned}B_0(s)&=1\\B_1(s)&=s-\tfrac{1}{2}\\B_2(s)&=s^2-s+\tfrac{1}{6}.\end{aligned}$$

Note that $B_n(0)=B_n$, the Bernoulli number.

Solution.

The defining generating function is $te^{st}/(e^t-1) = \sum_{n\ge 0} B_n(s)t^n/n!$. First invert the series $(e^t-1)/t = \sum_{k\ge 0} t^k/(k+1)! = 1 + t/2 + t^2/6 + O(t^3)$. If its inverse is $1 + at + bt^2 + O(t^3)$, then

$$\left(1 + \frac{t}{2} + \frac{t^2}{6}\right)(1 + at + bt^2) = 1 + \left(a + \frac{1}{2}\right)t + \left(b + \frac{a}{2} + \frac{1}{6}\right)t^2 + O(t^3)$$

must equal one, which requires $a + 1/2 = 0$ and $b + a/2 + 1/6 = 0$. Thus $a = -1/2$, $b = 1/4 - 1/6 = 1/12$. Multiply $t/(e^t-1) = 1 - t/2 + t^2/12 + O(t^3)$ by $e^{st} = 1 + st + s^2t^2/2 + O(t^3)$:

$$\frac{te^{st}}{e^t-1} = 1 + \left(s - \frac{1}{2}\right)t + \left(\frac{s^2}{2} - \frac{s}{2} + \frac{1}{12}\right)t^2 + O(t^3).$$

Comparing with $B_0(s) + B_1(s)t + B_2(s)t^2/2! + \cdots$, so that $B_2(s)$ is twice the coefficient of t^2, yields

$$\boxed{B_0(s) = 1, \quad B_1(s) = s - \frac{1}{2}, \quad B_2(s) = s^2 - s + \frac{1}{6}}.$$

At $s = 0$ the generating function is the Bernoulli-number generating function itself, proving $B_n(0) = B_n$ for every n.

12.2.3 Show that

$$\tan x = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}2^{2n}(2^{2n}-1)B_{2n}}{(2n)!}x^{2n-1}, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

Hint. $\tan x = \cot x - 2\cot 2x$.

Solution.

First verify the hinted identity, using $\cos 2x = \cos^2 x - \sin^2 x$ and $\sin 2x = 2\sin x \cos x$:

$$\cot x - 2\cot 2x = \frac{\cos x}{\sin x} - \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} = \frac{\cos^2 x - (\cos^2 x - \sin^2 x)}{\sin x \cos x} = \frac{\sin x}{\cos x} = \tan x.$$

The identity extends through zero by continuity after the poles cancel. Next express the cotangent through exponentials:

$$\cot x = \frac{i(e^{ix} + e^{-ix})}{e^{ix} - e^{-ix}} = \frac{i(e^{2ix}+1)}{e^{2ix}-1} = i + \frac{2i}{e^{2ix}-1} = i + \frac{1}{x}\sum_{k=0}^{\infty} B_k \frac{(2ix)^k}{k!},$$

where the Bernoulli generating function $t/(e^t-1) = \sum_k B_k t^k/k!$ was used with $t = 2ix$; it converges for $|t| < 2\pi$, that is, $|x| < \pi$, because the singularities of $t/(e^t-1)$ nearest the origin are the poles $t = \pm 2\pi i$ (the zeros of $e^t - 1$ are $t = 2\pi i k$, $k \in \mathbb{Z}$, and the one at $t = 0$ is removable). The $k = 0$ term is $1/x$; the $k = 1$ term, $2iB_1 = -i$, cancels the initial i; all odd Bernoulli numbers beyond B_1 vanish (Eq. (12.32), verified in Exercise 12.2.1, shows that $t/(e^t-1) + t/2$ is even); and $(2ix)^{2n} = (-1)^n 2^{2n} x^{2n}$. Hence

$$\cot x = \frac{1}{x} + \sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n} B_{2n}}{(2n)!}x^{2n-1}, \quad |x| < \pi.$$

Substitute this series at x and at $2x$, noting $2(2x)^{2n-1} = 2^{2n}x^{2n-1}$:

$$\cot x - 2\cot 2x = \left(\frac{1}{x} - \frac{2}{2x}\right) + \sum_{n\geq 1} \frac{(-1)^n 2^{2n} B_{2n}}{(2n)!}[1 - 2^{2n}]x^{2n-1}$$

$$= \boxed{\sum_{n=1}^{\infty} \frac{(-1)^{n-1} 2^{2n}(2^{2n}-1)B_{2n}}{(2n)!} x^{2n-1}}.$$

In the last step $(-1)^n(1-2^{2n}) = (-1)^{n-1}(2^{2n}-1)$. Both cotangent expansions converge for $|x| < \pi/2$ (the one at $2x$ needs $|2x| < \pi$). The nearest poles of $\tan x$ are at $\pm\pi/2$, so this is also its exact Taylor radius. With $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$ the first terms are $x + x^3/3 + 2x^5/15 + \cdots$.

12.3 Euler-Maclaurin Integration Formula

12.3.1 The Euler-Maclaurin integration formula may be used for the evaluation of finite series:

$$\sum_{m=1}^{n} f(m) = \int_1^n f(x)\,dx + \frac{1}{2}f(1) + \frac{1}{2}f(n) + \frac{B_2}{2!}\left[f'(n) - f'(1)\right] + \cdots.$$

Show that

(a) $\displaystyle\sum_{m=1}^{n} m = \tfrac{1}{2}n(n+1)$.

(b) $\displaystyle\sum_{m=1}^{n} m^2 = \tfrac{1}{6}n(n+1)(2n+1)$.

(c) $\displaystyle\sum_{m=1}^{n} m^3 = \tfrac{1}{4}n^2(n+1)^2$.

(d) $\displaystyle\sum_{m=1}^{n} m^4 = \tfrac{1}{30}n(n+1)(2n+1)(3n^2+3n-1)$.

Solution.

For each polynomial, the Euler–Maclaurin correction sequence terminates because sufficiently high derivatives vanish; the remainder, an integral containing such a derivative, is then zero, so the formula is exact. The term following the one displayed in the statement is $(B_4/4!)[f'''(n) - f'''(1)]$. Use $B_2/2! = 1/12$ and $B_4/4! = -1/720$ (from $B_2 = 1/6$, $B_4 = -1/30$).

(a) For $f(x) = x$, $\int_1^n x\,dx = (n^2-1)/2$ and the first-derivative difference is $1 - 1 = 0$, so

$$\sum_{m=1}^{n} m = \frac{n^2-1}{2} + \frac{1+n}{2} = \frac{n^2+n}{2} = \boxed{\frac{n(n+1)}{2}}.$$

(b) For $f(x) = x^2$, $f' = 2x$ and higher odd derivatives vanish. Hence, over the common denominator 6,

$$\sum_{m=1}^{n} m^2 = \frac{n^3-1}{3} + \frac{1+n^2}{2} + \frac{1}{12}(2n-2) = \frac{2(n^3-1) + 3(1+n^2) + (n-1)}{6}$$

$$= \frac{2n^3+3n^2+n}{6} = \boxed{\frac{n(n+1)(2n+1)}{6}},$$

since $n(n+1)(2n+1) = n(2n^2+3n+1)$.

(c) For $f(x) = x^3$, $f' = 3x^2$ and $f^{(3)} = 6$, whose endpoint difference is zero. Thus, over the common denominator 4,

$$\sum_{m=1}^{n} m^3 = \frac{n^4-1}{4} + \frac{1+n^3}{2} + \frac{1}{12}(3n^2-3) = \frac{(n^4-1)+2(1+n^3)+(n^2-1)}{4}$$
$$= \frac{n^4+2n^3+n^2}{4} = \boxed{\frac{n^2(n+1)^2}{4}}.$$

(d) For $f(x) = x^4$, $f' = 4x^3$, $f^{(3)} = 24x$, and $f^{(5)} = 0$. Retain both nonzero endpoint corrections, and note $(4n^3-4)/12 = 10(n^3-1)/30$ and $(24n-24)/720 = (n-1)/30$:

$$\sum_{m=1}^{n} m^4 = \frac{n^5-1}{5} + \frac{1+n^4}{2} + \frac{1}{12}(4n^3-4) - \frac{1}{720}(24n-24)$$
$$= \frac{6(n^5-1)+15(1+n^4)+10(n^3-1)-(n-1)}{30}$$
$$= \frac{6n^5+15n^4+10n^3-n}{30}$$
$$= \boxed{\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}}.$$

The constants cancel in the second line, $-6+15-10+1=0$. The last factorization follows from $(n+1)(2n+1)(3n^2+3n-1) = 6n^4+15n^3+10n^2-1$.

12.3.2 The Euler-Maclaurin integration formula provides a way of calculating the Euler-Mascheroni constant γ to high accuracy. Using $f(x) = 1/x$ in Eq. (12.57) (with interval $[1, n]$) and the definition of γ, Eq. (1.13), we obtain

$$\gamma = \sum_{s=1}^{n} s^{-1} - \ln n - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{(2k)n^{2k}} + R_{n,N}.$$

Using double-precision arithmetic, calculate the approximation for $N = 1, 2, \ldots$ at $n = 1000$, as in the printed answer below.

Note. See D. E. Knuth, Euler's constant to 1271 places. *Math. Comput.* **16**: 275 (1962).

ANS. For $n = 1000$, $N = 2$ $\quad \gamma \approx 0.5772\,1566\,4901$
(truncated, rather than rounded, to twelve decimal places).

Solution.

The textbook prints this finite-N formula as an equality without the remainder $R_{n,N}$; for finite N it holds only with $R_{n,N}$ included, as derived below. Put $H_n = \sum_{s=1}^{n} 1/s$ and define the computable approximation

$$A_{n,N} = H_n - \ln n - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k\, n^{2k}}.$$

To derive the formula with its signs and remainder, apply Euler–Maclaurin to $f(x) = 1/x$ on the interval $[n, M]$:

$$\sum_{s=n}^{M} \frac{1}{s} = \int_n^M \frac{dx}{x} + \frac{f(n)+f(M)}{2} + \sum_{k=1}^{N} \frac{B_{2k}}{(2k)!}\left[f^{(2k-1)}(M) - f^{(2k-1)}(n)\right] + \text{remainder}.$$

The left side is $H_M - H_n + 1/n$, and $f^{(2k-1)}(x) = -(2k-1)!/x^{2k}$, so $B_{2k}[f^{(2k-1)}(M) - f^{(2k-1)}(n)]/(2k)! = B_{2k}(n^{-2k} - M^{-2k})/(2k)$. Moving $1/n$ to the right turns $\frac{1}{2n} + \frac{1}{2M}$ into $\frac{1}{2M} - \frac{1}{2n}$:

$$H_M - H_n = \ln(M/n) + \frac{1}{2M} - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k}(n^{-2k} - M^{-2k}) + \text{remainder}.$$

Move $\ln(M/n) = \ln M - \ln n$ to the left, which becomes $(H_M - \ln M) - (H_n - \ln n)$, let $M \to \infty$, and use $H_M - \ln M \to \gamma$ (Eq. (1.13)); the terms containing M vanish. This yields $\gamma = A_{n,N} + R_{n,N}$, where $R_{n,N}$ is the limit of the remainder.

The statement's interval $[1, n]$ leads to the same result. On it, $\int_1^n dx/x = \ln n$, $\frac{1}{2}[f(1) + f(n)] = \frac{1}{2} + \frac{1}{2n}$, and $B_{2k}[f^{(2k-1)}(n) - f^{(2k-1)}(1)]/(2k)! = B_{2k}(1 - n^{-2k})/(2k)$, so Euler–Maclaurin for H_n, with the terms containing n moved to the left, reads

$$A_{n,N} = \frac{1}{2} + \sum_{k=1}^{N} \frac{B_{2k}}{2k} + (\text{remainder on } [1, n]).$$

As $n \to \infty$ the left side tends to γ by Eq. (1.13) and the remainder to the remainder on $[1, \infty)$. Subtracting the identity from this limit leaves $\gamma - A_{n,N}$ equal to the remainder on $[n, \infty)$, because the remainder is an integral over the interval. A rigorous signed remainder bound, derived in Exercise 12.6.6, is

$$0 < (-1)^N R_{n,N} < \frac{|B_{2N+2}|}{(2N+2)n^{2N+2}}.$$

The computation in IEEE double precision runs as follows. The exact value is $H_{1000} = 7.4854708605503449126\ldots$; summing the terms $1/s$, $s = 1, \ldots, 1000$, with compensated (Kahan) summation, in either order, gives its correctly rounded double $H_{1000} = 7.485470860550345$. Plain summation is less accurate here: left to right it gives 7.485470860550343 and right to left 7.485470860550341, which would lower every entry of the table below by 1.8×10^{-15}, respectively 4.4×10^{-15}, without changing the twelve-decimal result. From the correctly rounded H_{1000} subtract $\ln 1000 = 6.907755278982137$ and then $1/2000$; finally add the corrections $B_{2k}/(2k \cdot 1000^{2k})$, $k = 1, \ldots, N$. This gives

N	$A_{1000,N}$	last correction
1	0.5772156649015416	$8.3333333333 \times 10^{-8}$
2	0.5772156649015333	$-8.3333333333 \times 10^{-15}$
3	0.5772156649015333	$3.9682539683 \times 10^{-21}$
4	0.5772156649015333	$-4.1666666667 \times 10^{-27}$
5	0.5772156649015333	$7.5757575758 \times 10^{-33}$
6	0.5772156649015333	$-2.1092796093 \times 10^{-38}$

The "last correction" column lists the $k = N$ term $B_{2N}/(2N \cdot 1000^{2N})$; for instance $B_4/(4 \cdot 1000^4) = -1/(120 \cdot 10^{12})$. The Bernoulli numbers needed are $B_2, B_4, \ldots, B_{12} = \frac{1}{6}, -\frac{1}{30}, \frac{1}{42}, -\frac{1}{30}, \frac{5}{66}, -\frac{691}{2730}$. The exact truncation error at $N = 2$ is below $|B_6|/(6 \cdot 1000^6) = 1/(252 \cdot 1000^6) = 3.9682539683 \times 10^{-21}$. The remaining difference of about 4.5×10^{-16} from $\gamma = 0.5772156649015328606\ldots$ is double-precision rounding: H_{1000} and $\ln 1000$ lie near 7, where adjacent doubles are 8.9×10^{-16} apart, so each is rounded by up to 4.4×10^{-16}. Here H_{1000} is rounded up by 1.4×10^{-16} and $\ln 1000$ down by 2.4×10^{-16}, both of which raise $H_{1000} - \ln 1000$, and the roundings after subtracting $1/2000$ and

after adding the corrections contribute a further 0.7×10^{-16}. More corrections cannot recover those already lost digits. To twelve decimal places the result is $\boxed{\gamma = 0.577215664902}$, consistent with the textbook's truncated display 0.5772 1566 4901.

12.4 Dirichlet Series

12.4.1 From $B_{2n} = (-1)^{n-1}\dfrac{2(2n)!}{(2\pi)^{2n}}\zeta(2n)$, show that

(a) $\zeta(2) = \dfrac{\pi^2}{6}$, (d) $\zeta(8) = \dfrac{\pi^8}{9450}$,

(b) $\zeta(4) = \dfrac{\pi^4}{90}$, (e) $\zeta(10) = \dfrac{\pi^{10}}{93{,}555}$,

(c) $\zeta(6) = \dfrac{\pi^6}{945}$.

Solution.

Rearrange the stated identity before inserting numbers:

$$\zeta(2n) = \frac{(-1)^{n-1}B_{2n}(2\pi)^{2n}}{2(2n)!}.$$

With $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, $B_8 = -1/30$, $B_{10} = 5/66$, $(2\pi)^{2n} = 2^{2n}\pi^{2n}$, and $(2n)! = 2$, 24, 720, 40320, 3628800, the five evaluations are

$$\begin{aligned}
\zeta(2) &= \frac{(1/6)4\pi^2}{2\cdot 2} = \frac{4\pi^2}{24} = \boxed{\pi^2/6},\\
\zeta(4) &= \frac{(1/30)16\pi^4}{2\cdot 24} = \frac{16\pi^4}{1440} = \boxed{\pi^4/90},\\
\zeta(6) &= \frac{(1/42)64\pi^6}{2\cdot 720} = \frac{64\pi^6}{60480} = \boxed{\pi^6/945},\\
\zeta(8) &= \frac{(1/30)256\pi^8}{2\cdot 40320} = \frac{256\pi^8}{2419200} = \boxed{\pi^8/9450},\\
\zeta(10) &= \frac{(5/66)1024\pi^{10}}{2\cdot 3628800} = \frac{5120\pi^{10}}{479001600} = \boxed{\pi^{10}/93555}.
\end{aligned}$$

Each alternating sign cancels the sign of the corresponding Bernoulli number, leaving the positive values required by the defining zeta series.

12.4.2 The integral

$$\int_0^1 \frac{dx}{x}\,[\ln(1-x)]^2$$

appears in the fourth-order correction to the magnetic moment of the electron. Show that it equals $2\zeta(3)$.

Hint. Let $1 - x = e^{-t}$.

Solution.

Let $1-x=e^{-t}$, so $x=1-e^{-t}$, $dx=e^{-t}dt$, and $\ln(1-x)=-t$. The endpoints $x=0,1$ become $t=0,\infty$, and $[\ln(1-x)]^2/x=t^2/(1-e^{-t})$, giving

$$I=\int_0^\infty \frac{t^2e^{-t}}{1-e^{-t}}dt.$$

For $t>0$, the geometric series gives $e^{-t}/(1-e^{-t})=\sum_{m=1}^\infty e^{-mt}$. All summands are nonnegative, so Tonelli's theorem permits interchange:

$$I=\sum_{m=1}^\infty \int_0^\infty t^2e^{-mt}dt.$$

Put $u=mt$, $dt=du/m$, in each integral: $\int_0^\infty t^2e^{-mt}dt=m^{-3}\int_0^\infty u^2e^{-u}du$. Two integrations by parts, whose boundary terms $[-u^2e^{-u}]_0^\infty$ and $[-ue^{-u}]_0^\infty$ vanish, give

$$\int_0^\infty u^2e^{-u}du=2\int_0^\infty ue^{-u}du=2\int_0^\infty e^{-u}du=2.$$

Consequently

$$\boxed{I=2\sum_{m=1}^\infty \frac{1}{m^3}=2\zeta(3)}.$$

The finite value also proves convergence of the original improper integral, including its logarithmic endpoint singularity.

12.4.3

(a) Show that

$$\int_0^\infty \frac{(\ln z)^2}{1+z^2}\,dz=4\left(1-\frac{1}{3^3}+\frac{1}{5^3}-\frac{1}{7^3}+\cdots\right).$$

(b) By contour integration show that the integral, equivalently four times the parenthesized series, evaluates to $\pi^3/8$.

Solution.

(a) Split $I=\int_0^\infty(\ln z)^2dz/(1+z^2)$ at $z=1$. In the part from 1 to ∞, set $z=1/t$. Since $dz=-dt/t^2$ and $1+z^2=(1+t^2)/t^2$, $dz/(1+z^2)=-dt/(1+t^2)$; also $(\ln z)^2=(\ln t)^2$, and $z=1\to\infty$ becomes $t=1\to 0$. The minus sign reverses the limits, so this part equals the part from 0 to 1:

$$I=2\int_0^1 \frac{(\ln t)^2}{1+t^2}dt.$$

Expand the denominator geometrically, $1/(1+t^2)=\sum_{k\ge0}(-1)^kt^{2k}$ for $0\le t<1$. The absolute sum of the integrals is finite, because

$$\int_0^1 t^{2k}(\ln t)^2dt=\int_0^\infty u^2e^{-(2k+1)u}du=\frac{2}{(2k+1)^3}.$$

Here $t=e^{-u}$, $dt=-e^{-u}du$, $(\ln t)^2=u^2$ was used, with $t=0\to1$ becoming $u=\infty\to0$; the last integral is the one of Exercise 12.4.2 with $m=2k+1$. By the Fubini–Tonelli theorem

for sums and integrals, this finite absolute sum permits termwise integration. Directly: by the finite geometric sum, $1/(1+t^2)$ minus its partial sum through t^{2K} is $(-1)^{K+1}t^{2K+2}/(1+t^2)$, of modulus at most t^{2K+2}, so the Kth partial sum of the termwise integrals differs from $I/2$ by at most $\int_0^1 t^{2K+2}(\ln t)^2 dt = 2/(2K+3)^3 \to 0$. Thus termwise integration gives

$$\boxed{I = 2\sum_{k=0}^{\infty}(-1)^k \frac{2}{(2k+1)^3} = 4\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)^3} = 4\beta(3)}.$$

(b) Integrate $F(z) = (\log z)^2/(1+z^2)$ round an upper semicircle of radius R, indenting the origin by a small semicircle of radius ϵ. Here $\log z = \ln|z| + i\arg z$ with $-\pi/2 < \arg z < 3\pi/2$; its branch cut along the negative imaginary axis avoids the contour, so F is analytic on and inside the contour except at $z = i$, and $\arg z = 0$ on the positive, $\arg z = \pi$ on the negative real axis. On the large arc $|\log z| \le \ln R + \pi$ and $|1+z^2| \ge R^2 - 1$; on the small arc $|\log z| \le |\ln\epsilon| + \pi$ and $|1+z^2| \ge 1-\epsilon^2$. Multiplying by the arc lengths πR and $\pi\epsilon$ shows that the large arc is $O((\ln R)^2/R)$ and the small arc is $O(\epsilon|\ln\epsilon|^2)$; both vanish. The sole enclosed pole is $z = i$, a simple zero of $1+z^2$ with $(1+z^2)' = 2i$ there and $\log i = i\pi/2$, so

$$\operatorname{Res}_{z=i} F = \frac{(i\pi/2)^2}{2i} = \frac{i\pi^2}{8}, \qquad \oint F dz = 2\pi i\,\frac{i\pi^2}{8} = -\frac{\pi^3}{4}.$$

On the positive real axis F gives I in the limit. On the negative real edge, put $z = -t$, $dz = -dt$, so that $z = -R \to -\epsilon$ becomes $t = R \to \epsilon$ and $\log z = \ln t + i\pi$:

$$\int_{-R}^{-\epsilon} F\,dz = \int_{\epsilon}^{R} \frac{(\ln t + i\pi)^2}{1+t^2}dt, \qquad (\ln t + i\pi)^2 = (\ln t)^2 + 2i\pi\ln t - \pi^2.$$

The two real-axis contributions therefore add to

$$2I + 2i\pi J - \pi^2 K, \qquad J = \int_0^{\infty}\frac{\ln t}{1+t^2}dt, \quad K = \int_0^{\infty}\frac{dt}{1+t^2} = \frac{\pi}{2}.$$

The substitution $t = 1/u$, $dt = -du/u^2$, gives

$$J = \int_{\infty}^{0}\frac{-\ln u}{1+u^{-2}}\left(-\frac{du}{u^2}\right) = \int_{\infty}^{0}\frac{\ln u}{u^2+1}du = -J,$$

so $J = 0$. Equating to the residue result yields $2I - \pi^3/2 = -\pi^3/4$, hence

$$\boxed{I = \pi^3/8}, \qquad \boxed{\beta(3) = \pi^3/32}.$$

The value $\pi^3/8$ refers to the integral, or to four times the series in parentheses in (a). The unmultiplied alternating odd-cube series has value $\pi^3/32$.

12.4.4 Show that Catalan's constant, $\beta(2)$, may be written as

$$\beta(2) = 2\sum_{k=1}^{\infty}(4k-3)^{-2} - \frac{\pi^2}{8}.$$

Hint. $\pi^2 = 6\zeta(2)$.

Solution.

Separate the positive odd integers into their two residue classes:

$$A = \sum_{k\geq 1}(4k-3)^{-2}, \qquad B = \sum_{k\geq 1}(4k-1)^{-2}.$$

Absolute convergence permits grouping. By definition $\beta(2) = 1 - 3^{-2} + 5^{-2} - \cdots = A - B$, while the odd integers make up all positive integers except the even ones $2j$, so

$$A + B = \sum_{j\geq 0}(2j+1)^{-2} = \zeta(2) - \sum_{j\geq 1}(2j)^{-2} = \zeta(2) - \frac{1}{4}\zeta(2) = \frac{3}{4}\zeta(2) = \frac{\pi^2}{8},$$

using the hint $\zeta(2) = \pi^2/6$ in the last step. Eliminate $B = \pi^2/8 - A$ to obtain $\beta(2) = A - (\pi^2/8 - A)$, that is,

$$\boxed{\beta(2) = 2\sum_{k=1}^{\infty}(4k-3)^{-2} - \frac{\pi^2}{8}}.$$

12.4.5 Show that

(a) $\displaystyle\int_0^1 \frac{\ln(1+x)}{x}\,dx = \frac{1}{2}\zeta(2)$, (b) $\displaystyle\lim_{a\to 1^-}\int_0^a \frac{\ln(1-x)}{x}\,dx = -\zeta(2)$.

Note that the integrand in part (b) diverges for $a = 1$ but that the integral is convergent.

Solution.

(a) For $0 \leq x < 1$, expand $\ln(1+x)/x = \sum_{k\geq 1}(-1)^{k+1}x^{k-1}/k$; each term integrates to $\int_0^1 x^{k-1}dx/k = 1/k^2$ in absolute value. The sum of the integrals of the absolute values is thus $\sum_{k\geq 1} 1/k^2 < \infty$, which justifies termwise integration up to the endpoint (the Fubini–Tonelli theorem for sums and integrals). Directly: for $0 \leq x \leq 1$ the terms x^{k-1}/k decrease in k, so by the alternating-series bound the remainder after K terms has modulus at most $x^K/(K+1)$, whose integral over $[0,1]$ is $1/(K+1)^2 \to 0$. Therefore, subtracting the even terms twice,

$$\int_0^1 \frac{\ln(1+x)}{x}dx = \sum_{k\geq 1}\frac{(-1)^{k+1}}{k^2} = \zeta(2) - 2\sum_{j\geq 1}\frac{1}{(2j)^2} = \zeta(2) - \frac{2}{4}\zeta(2) = \boxed{\frac{1}{2}\zeta(2)}.$$

(b) For $0 < a < 1$, the logarithm series $\ln(1-x)/x = -\sum_{k\geq 1} x^{k-1}/k$ is uniformly convergent on $[0,a]$ by the Weierstrass M-test, since its terms are bounded by a^{k-1}/k and $\sum_k a^{k-1}/k < \infty$; at $x = 0$ the quotient is given its limiting value -1, which is also the sum of the series there. Since $\int_0^a x^{k-1}dx/k = a^k/k^2$, termwise integration gives

$$\int_0^a \frac{\ln(1-x)}{x}dx = -\sum_{k=1}^{\infty}\frac{a^k}{k^2}.$$

Dominated convergence, with dominating summable sequence $1/k^2$, permits $a \uparrow 1$, so

$$\boxed{\lim_{a\to 1^-}\int_0^a \frac{\ln(1-x)}{x}dx = -\zeta(2)}.$$

The printed right-hand side in (b) has the wrong sign. The integral is negative since $\ln(1-x)/x < 0$ for $0 < x < 1$. The integrand diverges logarithmically at one, but its absolute integral is $\zeta(2)$ and is finite.

12.4.6

(a) Show that the equation $\ln 2 = \sum_{s=1}^{\infty}(-1)^{s+1}s^{-1}$, Eq. (1.53), may be rewritten as

$$\ln 2 = \sum_{s=2}^{n} 2^{-s}\zeta(s) + \sum_{p=1}^{\infty}(2p)^{-n-1}\left[1 - \frac{1}{2p}\right]^{-1}.$$

Hint. Take the terms in pairs.

(b) Calculate $\ln 2$ to six significant figures.

Solution.

(a) Pair neighboring terms of the convergent alternating harmonic series (grouping terms of a convergent series does not change its sum):

$$\ln 2 = \sum_{p=1}^{\infty}\left(\frac{1}{2p-1} - \frac{1}{2p}\right) = \sum_{p=1}^{\infty}\frac{1}{(2p-1)2p} = \sum_{p=1}^{\infty}\frac{(2p)^{-2}}{1-(2p)^{-1}},$$

where the last form divides numerator and denominator by $(2p)^2$. For $r = (2p)^{-1}$ the finite geometric identity is

$$\frac{r^2}{1-r} = \sum_{s=2}^{n} r^s + \frac{r^{n+1}}{1-r},$$

since $\sum_{s=2}^{n} r^s = (r^2 - r^{n+1})/(1-r)$. Sum it over p. The first sum has finitely many s, and all terms are nonnegative; with $\sum_{p\geq 1}(2p)^{-s} = 2^{-s}\zeta(s)$ this gives exactly

$$\boxed{\ln 2 = \sum_{s=2}^{n} 2^{-s}\zeta(s) + \sum_{p=1}^{\infty}\frac{(2p)^{-n-1}}{1-(2p)^{-1}}}.$$

The usual nonempty first sum takes $n \geq 2$; the $n = 1$ case also holds if that sum is interpreted as empty.

(b) Call the two sums on the right S_n and R_n. The remainder R_n is positive and, since $1 - (2p)^{-1} \geq 1/2$, obeys

$$0 < R_n \leq 2\sum_{p\geq 1}(2p)^{-n-1} = 2\cdot 2^{-n-1}\zeta(n+1) = 2^{-n}\zeta(n+1).$$

Take $n = 24$, for which this bound is below 10^{-7}. Direct summation of the 23 terms $2^{-s}\zeta(s)$, $s = 2, \ldots, 24$, gives

$$S_{24} = 0.6931471209552993497\ldots, \qquad R_{24} \leq 2^{-24}\zeta(25) = 5.9604646552\times 10^{-8}.$$

Both ends of the interval $S_{24} < \ln 2 \leq S_{24} + 2^{-24}\zeta(25) = 0.6931471805599\ldots$ round to the same six-significant-figure value:

$$\boxed{\ln 2 = 0.693147 \quad \text{(six significant figures)}}.$$

The error bound comes from the derived remainder, independently of any precomputed logarithm value.

12.4.7

(a) Show that the equation $\pi/4 = \sum_{s=1}^{\infty}(-1)^{s+1}(2s-1)^{-1}$, Eq. (12.72), may be rewritten as

$$\frac{\pi}{4} = 1 - 2\sum_{s=1}^{n} 4^{-2s}\zeta(2s) - 2\sum_{p=1}^{\infty}(4p)^{-2n-2}\left[1 - \frac{1}{(4p)^2}\right]^{-1}.$$

(b) Calculate $\pi/4$ to six significant figures.

Solution.

(a) Separate the first term of the arctangent series, then pair the rest ($-1/3 + 1/5$, $-1/7 + 1/9$, ...); as in Exercise 12.4.6, grouping terms of a convergent series does not change its sum:

$$\frac{\pi}{4} = 1 + \sum_{p=1}^{\infty}\left(-\frac{1}{4p-1} + \frac{1}{4p+1}\right) = 1 - 2\sum_{p=1}^{\infty}\frac{1}{(4p)^2-1},$$

because $-(4p+1) + (4p-1) = -2$ and $(4p-1)(4p+1) = (4p)^2 - 1$. With $r = (4p)^{-2}$ (divide numerator and denominator by $(4p)^2$) and the finite geometric sum $\sum_{s=1}^{n} r^s = (r - r^{n+1})/(1-r)$,

$$\frac{1}{(4p)^2-1} = \frac{r}{1-r} = \sum_{s=1}^{n} r^s + \frac{r^{n+1}}{1-r}.$$

Summing over p, with $\sum_{p\geq 1}(4p)^{-2s} = 4^{-2s}\zeta(2s)$, yields

$$\boxed{\frac{\pi}{4} = 1 - 2\sum_{s=1}^{n} 4^{-2s}\zeta(2s) - 2\sum_{p=1}^{\infty}\frac{(4p)^{-2n-2}}{1-(4p)^{-2}}}.$$

(b) Write the result as $\pi/4 = S_n - R_n$, with $S_n = 1 - 2\sum_{s=1}^{n} 4^{-2s}\zeta(2s)$; the omitted correction $-R_n$ has negative sign. Its magnitude is bounded using $1 - (4p)^{-2} \geq 15/16$:

$$0 < R_n := 2\sum_{p\geq 1}\frac{(4p)^{-2n-2}}{1-(4p)^{-2}} \leq \frac{32}{15}\sum_{p\geq 1}(4p)^{-2n-2} = \frac{32}{15}\,4^{-2n-2}\zeta(2n+2).$$

For $n = 6$ the bound is below 10^{-8}. Summing the six terms $4^{-2s}\zeta(2s)$, $s = 1, \ldots, 6$, gives the finite part, and the bound is evaluated with $\zeta(14)$:

$$S_6 = 0.7853981713451978430\ldots, \qquad R_6 \leq \tfrac{32}{15}4^{-14}\zeta(14) = 7.9477727265 \times 10^{-9}.$$

Thus $0.7853981633974\ldots = S_6 - \frac{32}{15}4^{-14}\zeta(14) \leq \pi/4 < S_6$, whose endpoints both round to

$$\boxed{\pi/4 = 0.785398 \quad \text{(six significant figures)}}.$$

12.5 Infinite Products

12.5.1 Assume $a_n \geq 0$, no factor vanishes, and the factors are eventually positive. Product convergence means a finite nonzero limit. Using

$$\ln \prod_{n=1}^{\infty}(1 \pm a_n) = \sum_{n=1}^{\infty} \ln(1 \pm a_n)$$

and the Maclaurin expansion of $\ln(1 \pm a_n)$, show that the infinite product $\prod_{n=1}^{\infty}(1 \pm a_n)$ converges or diverges with the infinite series $\sum_{n=1}^{\infty} a_n$.

Solution.

The unqualified statement is false for signed or complex a_n. It is valid in its usual comparison-test form when $a_n \geq 0$ and the factors are eventually positive. A convergent infinite product here means a finite, nonzero limit, with no zero factor. Discarding finitely many initial nonzero factors only multiplies all later partial products by a fixed nonzero constant. We therefore apply real logarithms to the eventually positive tail. For that tail, with partial products P_N, convergence of P_N to some $L > 0$ is equivalent to convergence of $\ln P_N = \sum_{n \leq N} \ln(1 \pm a_n)$, because ln and exp are continuous and $P_N = e^{\ln P_N}$. If $a_n \not\to 0$, a nonzero product limit is already impossible, since the ratio $P_n/P_{n-1} = 1 \pm a_n$ of two successive partial products would not tend to one; and $\sum a_n$ diverges as well. Otherwise, eventually $0 \leq a_n \leq 1/2$. For $0 \leq t \leq 1/2$ one has $\frac{1}{2} \leq 1/(1+t) \leq 1$ and $1 \leq 1/(1-t) \leq 2$, so integration of $1/(1+t)$ and $1/(1-t)$ on $[0, a_n]$ gives

$$\frac{a_n}{2} \leq \ln(1+a_n) \leq a_n, \qquad a_n \leq -\ln(1-a_n) \leq 2a_n.$$

The comparison test, applied to the nonnegative series $\sum \ln(1+a_n)$ and $\sum[-\ln(1-a_n)]$, now proves the stated equivalence for both choices of sign. If $\sum a_n$ diverges, then $\sum \ln(1+a_n) = +\infty$ and $\sum \ln(1-a_n) = -\infty$: the plus-product grows without bound and the minus-product tends to zero.

To see why the hypotheses matter, take $a_n = (-1)^n/\sqrt{n+1}$. Its series converges by the alternating test, and every $1 + a_n$ is positive. But the logarithm expansion gives

$$\ln(1+a_n) = a_n - \frac{1}{2}a_n^2 + O(|a_n|^3).$$

Here $\sum a_n^2 = \sum 1/(n+1) = \infty$, while $\sum |a_n|^3 = \sum (n+1)^{-3/2} < \infty$. Consequently the logarithmic partial sums, a convergent series minus $\frac{1}{2}\sum a_n^2$ plus an absolutely convergent one, tend to $-\infty$, and $\prod(1+a_n)$ tends to zero, not a nonzero limit.

A useful corrected version for signed or complex terms is: if $a_n \to 0$, $\sum |a_n|^2 < \infty$, and no factor vanishes, then $\prod(1+a_n)$ converges to a nonzero limit exactly when $\sum a_n$ converges. Indeed $\log(1+a_n) - a_n = O(|a_n|^2)$, using the logarithm branch near one; its error series is absolutely convergent. To justify necessity as well as sufficiency, suppose the partial products $P_n \to L \neq 0$. Choose a disk about L avoiding zero and one analytic logarithm ℓ there. For large n, both $\ell(P_n) - \ell(P_{n-1})$ and $\log(1+a_n)$ are logarithms of $P_n/P_{n-1} = 1 + a_n$, so they differ by an integer multiple of $2\pi i$; both tend to zero, so the multiple is zero and $\ell(P_n) - \ell(P_{n-1}) = \log(1+a_n)$. The logarithmic tail $\sum_{n=N+1}^{M} \log(1+a_n) = \ell(P_M) - \ell(P_N)$ therefore telescopes to the convergent limit $\ell(L) - \ell(P_N)$. Conversely, a convergent logarithmic series exponentiates to a nonzero product. Subtracting the absolutely convergent error proves the claim about $\sum a_n$. The minus case follows by replacing a_n by $-a_n$.

12.5.2 An infinite product appears in the form

$$\prod_{n=1}^{\infty}\left(\frac{1+a/n}{1+b/n}\right),$$

where a and b are constants not equal to negative integers, so all factors are defined and nonzero. Show that this infinite product converges to a finite nonzero limit only if $a=b$.

Solution.

Use the standard finite, nonzero meaning of product convergence, and assume every denominator and numerator factor is nonzero. If a zero limit were counted as convergence, the asserted "only if" would fail. For fixed real or complex a, b, the tail has a logarithm branch near one. From $\log(1+u) = u + O(u^2)$, its expansion is

$$\log\frac{1+a/n}{1+b/n} = \log\left(1+\frac{a}{n}\right) - \log\left(1+\frac{b}{n}\right) = \frac{a-b}{n} + O(n^{-2}).$$

The error series converges absolutely. Hence, summing the logarithms over the tail $n_0 \le n \le N$ and collecting all constants in C, the partial product T_N of the tail satisfies, with $H_N = \sum_{n\le N} 1/n = \ln N + \gamma + o(1)$ by Eq. (1.13),

$$\log T_N = (a-b)H_N + C + o(1) = (a-b)\log N + C_1 + o(1), \qquad C_1 = C + (a-b)\gamma.$$

The finitely many initial factors only alter the nonzero multiplicative constant. Thus, exponentiating, the full partial product is $P_N = KN^{a-b}(1+o(1))$, $K \neq 0$, where $|N^{a-b}| = N^{\Re(a-b)}$. If $\Re(a-b) > 0$, its modulus tends to infinity; if $\Re(a-b) < 0$, it tends to zero. If $a - b = i\beta$ with real $\beta \neq 0$, then $N^{i\beta} = e^{i\beta \ln N}$ has modulus one, but its phase $\beta \ln N$ has no limit modulo 2π: for the integers N_k nearest $e^{2\pi k/|\beta|}$, $\ln N_k = 2\pi k/|\beta| + o(1)$ and so $N_k^{i\beta} \to 1$, while the integers nearest $e^{(2\pi k+\pi)/|\beta|}$ give the limit -1. Then P_N has the two limit points $K \neq -K$. Hence no finite nonzero limit exists unless $a = b$. When $a = b$, every defined factor equals one, so

$$\boxed{\text{the nonzero product converges exactly when } a = b.}$$

For a concrete zero-limit exception, $a = 0, b = 1$ gives $P_N = \prod_{n=1}^{N} n/(n+1) = 1/(N+1) \to 0$. If a is a negative integer a numerator factor is zero; if b is a negative integer the written product is undefined at that factor.

12.5.3 Show that the infinite product representations of $\sin x$ and $\cos x$ are consistent with the identity $2\sin x\cos x = \sin 2x$.

Solution.

The product formulas are

$$\sin x = x\prod_{n=1}^{\infty}\left(1-\frac{x^2}{n^2\pi^2}\right), \quad \cos x = \prod_{n=1}^{\infty}\left(1-\frac{4x^2}{(2n-1)^2\pi^2}\right).$$

In the sine product rewrite $x^2/(n^2\pi^2) = 4x^2/[(2n)^2\pi^2]$. The two products then supply the even values $m = 2n$ and the odd values $m = 2n-1$ of the positive integers m, and $4x^2 = (2x)^2$:

$$\begin{aligned} 2\sin x\cos x &= 2x\prod_{n\ge1}\left(1-\frac{4x^2}{(2n)^2\pi^2}\right)\prod_{n\ge1}\left(1-\frac{4x^2}{(2n-1)^2\pi^2}\right) \\ &= 2x\prod_{m=1}^{\infty}\left(1-\frac{(2x)^2}{m^2\pi^2}\right) = \boxed{\sin 2x}, \end{aligned}$$

the last step being the sine product with x replaced by $2x$. The deviations of the factors from one are bounded on compact x sets by a constant times $1/n^2$. Their absolute summability justifies interleaving the factors. At zeros of a factor the equality extends by continuity, so it holds for all complex x.

12.5.4 Determine the limit to which $\displaystyle\prod_{n=2}^{\infty}\left(1+\frac{(-1)^n}{n}\right)$ converges.

Solution.

Pair the factor with even index $2k$ with the following odd index:

$$\left(1+\frac{1}{2k}\right)\left(1-\frac{1}{2k+1}\right)=\frac{2k+1}{2k}\frac{2k}{2k+1}=1.$$

To ensure that pairing has not hidden a subsequence issue, let $P_N=\prod_{n=2}^{N}[1+(-1)^n/n]$. The factors $n=2,\dots,2M+1$ form the M pairs $k=1,\dots,M$, and $P_{2M}=P_{2M-1}(1+1/(2M))$. Then exactly

$$P_{2M+1}=1,\qquad P_{2M}=\frac{2M+1}{2M}.$$

Both subsequences tend to one. Therefore the complete sequence of partial products converges and

$$\boxed{\prod_{n=2}^{\infty}\left(1+\frac{(-1)^n}{n}\right)=1}.$$

12.5.5 Show that $\displaystyle\prod_{n=2}^{\infty}\left[1-\frac{2}{n(n+1)}\right]=\frac{1}{3}$.

Solution.

Factor each term before taking any limit:

$$1-\frac{2}{n(n+1)}=\frac{n^2+n-2}{n(n+1)}=\frac{(n-1)(n+2)}{n(n+1)}.$$

The finite product telescopes in two separate pieces:

$$\begin{aligned}P_N&=\left(\prod_{n=2}^{N}\frac{n-1}{n}\right)\left(\prod_{n=2}^{N}\frac{n+2}{n+1}\right)\\&=\left(\frac{1}{2}\frac{2}{3}\cdots\frac{N-1}{N}\right)\left(\frac{4}{3}\frac{5}{4}\cdots\frac{N+2}{N+1}\right)\\&=\frac{1}{N}\frac{N+2}{3}=\frac{N+2}{3N}=\frac{1}{3}+\frac{2}{3N}.\end{aligned}$$

Taking $N\to\infty$ proves $\boxed{P_\infty=1/3}$.

12.5.6 Prove that $\displaystyle\prod_{n=2}^{\infty}\left(1-\frac{1}{n^2}\right)=\frac{1}{2}$.

Solution.

Use $1-1/n^2=(n^2-1)/n^2=(n-1)(n+1)/n^2$. For a finite upper limit,

$$\begin{aligned}P_N&=\left(\prod_{n=2}^{N}\frac{n-1}{n}\right)\left(\prod_{n=2}^{N}\frac{n+1}{n}\right)=\left(\frac{1}{2}\frac{2}{3}\cdots\frac{N-1}{N}\right)\left(\frac{3}{2}\frac{4}{3}\cdots\frac{N+1}{N}\right)\\&=\frac{1}{N}\frac{N+1}{2}=\frac{N+1}{2N}.\end{aligned}$$

Both telescoping factors are exact for every $N\geq 2$. Thus $\boxed{\lim_{N\to\infty}P_N=1/2}$.

12.5.7 Verify the Euler identity $\displaystyle\prod_{p=1}^{\infty}(1+z^p)=\prod_{q=1}^{\infty}(1-z^{2q-1})^{-1}$, $|z|<1$.

Solution.

For $|z|<1$, all factors $1\pm z^n$ are nonzero and $\sum_{n\geq 1}|z|^n<\infty$. The corresponding products and their logarithmic tails converge absolutely, permitting the following factorization and regrouping, based on $1+z^p=(1-z^{2p})/(1-z^p)$:

$$\prod_{p=1}^{\infty}(1+z^p)=\frac{\prod_{p=1}^{\infty}(1-z^{2p})}{\prod_{p=1}^{\infty}(1-z^p)}.$$

Split the denominator into even and odd indices:

$$\prod_{p\geq 1}(1-z^p)=\prod_{q\geq 1}(1-z^{2q})\prod_{q\geq 1}(1-z^{2q-1}).$$

The even-index product is nonzero, so it cancels legitimately, leaving

$$\boxed{\prod_{p=1}^{\infty}(1+z^p)=\prod_{q=1}^{\infty}(1-z^{2q-1})^{-1}}.$$

The same argument is uniform on $|z|\leq r<1$. No boundary claim at $|z|=1$ follows from it.

12.5.8 Show that $\prod_{r=1}^{\infty}(1+x/r)e^{-x/r}$ converges for all finite x (except for the zeros of $1+x/r$).

Hint. Write the nth factor as $1+a_n$. Here convergence means a nonzero product; at a negative integer x the partial products instead have the zero limit.

Solution.

Write the rth factor as $1+a_r$ with $u=x/r$:

$$\begin{aligned}(1+u)e^{-u}&=(1+u)\left(1-u+\frac{u^2}{2}-\frac{u^3}{6}+\cdots\right)\\&=1-u+\frac{u^2}{2}-\frac{u^3}{6}+u-u^2+\frac{u^3}{2}+O(u^4)\\&=1-\frac{u^2}{2}+\frac{u^3}{3}+O(u^4).\end{aligned}$$

More precisely, for $k \ge 1$ the coefficient of u^k in $(1+u)e^{-u}$ is $(-1)^k/k! + (-1)^{k-1}/(k-1)! = (-1)^{k-1}(k-1)/k!$, which vanishes for $k = 1$, so

$$a_r = \sum_{k\ge 2} \frac{(-1)^{k-1}(k-1)}{k!} u^k, \qquad |a_r| \le |u|^2 \sum_{k\ge 2} \frac{k-1}{k!} = |u|^2 \quad (|u| \le 1),$$

because $\sum_{k\ge 2}(k-1)/k! = \sum_{k\ge 2} 1/(k-1)! - \sum_{k\ge 2} 1/k! = (e-1) - (e-2) = 1$. Hence $a_r = -x^2/(2r^2) + O(r^{-3})$ for fixed finite complex x, and $|a_r| \le |x|^2/r^2$ for $r \ge |x|$, so $\sum |a_r| < \infty$ by comparison with $\sum r^{-2}$. Once $|a_r| < 1/2$, which holds for $r > \sqrt{2}\,|x|$, the convergent logarithm series has the bound

$$|\log(1+a_r)| \le \sum_{k\ge 1} \frac{|a_r|^k}{k} \le |a_r| \sum_{k\ge 0} |a_r|^k = \frac{|a_r|}{1-|a_r|} \le 2|a_r|.$$

The logarithmic tail therefore converges absolutely, proving a finite nonzero product whenever no factor vanishes. The exceptional points are exactly $x = -1, -2, \ldots$, since exponentials never vanish.

The estimate is uniform for $|x| \le R$: $|a_r| \le R^2/r^2$ for $r \ge R$, and hence $|\log(1+a_r)| \le 2R^2/r^2$ for $r > \sqrt{2}\,R$. By the Weierstrass M-test the logarithmic tail converges uniformly on $|x| \le R$, so the product even defines an entire function. At the excluded points it still converges as a sequence of partial products, with value zero; they are excluded only from the convention of a nonzero convergent infinite product.

12.5.9 Derive the formula, for $0 < x < \pi$ with real logarithms,

$$\ln \sin x = \ln x + \sum a_n x^n,$$

giving the explicit form for the coefficients a_n.

Hint. $d(\ln \sin x)/dx = \cot x$.

Solution.

It is convenient to remove the known logarithmic singularity first:

$$\frac{d}{dx} \ln \frac{\sin x}{x} = \cot x - \frac{1}{x}.$$

The Bernoulli expansion of the right side, derived in the solution of Exercise 12.2.3, is

$$\cot x - \frac{1}{x} = \sum_{k=1}^{\infty} \frac{(-1)^k 2^{2k} B_{2k}}{(2k)!} x^{2k-1}, \qquad |x| < \pi.$$

Integrate term by term from zero, using $\int_0^x t^{2k-1}dt = x^{2k}/(2k)$ and $\sin x/x \to 1$ to fix the integration constant to zero:

$$\boxed{\ln \frac{\sin x}{x} = \sum_{k=1}^{\infty} \frac{(-1)^k 2^{2k} B_{2k}}{2k(2k)!} x^{2k}}.$$

Inserting $B_{2k} = (-1)^{k-1} 2(2k)!\,\zeta(2k)/(2\pi)^{2k}$, the formula used in Exercise 12.4.1, gives

$$\frac{(-1)^k 2^{2k} B_{2k}}{2k(2k)!} = \frac{(-1)^{2k-1}\, 2^{2k} \cdot 2(2k)!\,\zeta(2k)}{2k(2k)!\, 2^{2k}\pi^{2k}} = -\frac{\zeta(2k)}{k\pi^{2k}}.$$

Therefore

$$\boxed{a_{2k} = \frac{(-1)^k 2^{2k} B_{2k}}{2k(2k)!} = -\frac{\zeta(2k)}{k\pi^{2k}}, \qquad a_{2k-1} = 0 \quad (k \geq 1).}$$

The constant coefficient is $a_0 = 0$ by the normalization at zero. In particular, with $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$, $\zeta(6) = \pi^6/945$, $\ln(\sin x/x) = -x^2/6 - x^4/180 - x^6/2835 - \cdots$. For real logarithms the question's form $\ln \sin x = \ln x + \cdots$ uses $0 < x < \pi$. For $-\pi < x < 0$, write $\ln|\sin x| = \ln|x| + \cdots$. In complex form, the quotient has an analytic logarithm on $|x| < \pi$ normalized to zero at the origin; any separate logarithms must use consistent branches.

12.5.10 Using the infinite product representations of $\sin z$, show that for $|z| < \pi$

$$z \cot z = 1 - 2 \sum_{m,n=1}^{\infty} \left(\frac{z}{n\pi}\right)^{2m},$$

and hence that the Bernoulli numbers are given by the formula

$$B_{2n} = (-1)^{n-1} \frac{2(2n)!}{(2\pi)^{2n}} \zeta(2n).$$

This is an alternate route to Eq. (12.38).

Hint. The result of Exercise 12.5.9 will be helpful.

Solution.

Take the logarithmic derivative of the sine product $\sin z = z \prod_{n\geq 1}(1 - z^2/(n^2\pi^2))$ on a compact subset avoiding its zeros; this is the derivative of the logarithm used in Exercise 12.5.9. Each factor contributes

$$\frac{d}{dz} \ln\left(1 - \frac{z^2}{n^2\pi^2}\right) = \frac{-2z/(n^2\pi^2)}{1 - z^2/(n^2\pi^2)} = \frac{-2z}{n^2\pi^2 - z^2}.$$

Local uniform convergence of the derivative series follows from its $O(n^{-2})$ terms. Thus

$$\cot z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{-2z}{n^2\pi^2 - z^2}, \qquad z \cot z = 1 - 2\sum_{n=1}^{\infty} \frac{z^2}{n^2\pi^2 - z^2}.$$

The requested double power series is restricted to $|z| < \pi$; it is not a representation for arbitrary z away from the poles. In that disk $w = (z/(n\pi))^2$ has $|w| < 1$ for every $n \geq 1$, so each term can be expanded geometrically:

$$\frac{z^2}{n^2\pi^2 - z^2} = \frac{w}{1 - w} = \sum_{m=1}^{\infty} \left(\frac{z}{n\pi}\right)^{2m}.$$

For $|z| \leq r < \pi$ the double absolute sum is bounded by

$$\sum_{n\geq 1} \frac{r^2/(n\pi)^2}{1 - r^2/(n\pi)^2} \leq \frac{r^2\zeta(2)}{\pi^2[1 - r^2/\pi^2]} < \infty,$$

so the two summations may be interchanged. Summing first over n, $\sum_{n\geq 1}(z/(n\pi))^{2m} = \zeta(2m) z^{2m}/\pi^{2m}$. Hence

$$\boxed{z \cot z = 1 - 2 \sum_{m,n\geq 1} (z/(n\pi))^{2m}} = 1 - 2\sum_{m\geq 1} \frac{\zeta(2m)}{\pi^{2m}} z^{2m}.$$

Independently, the Bernoulli generating function gives, by multiplying the cotangent expansion of Exercise 12.2.3 by z,

$$z\cot z = 1 + \sum_{m\ge 1} \frac{(-1)^m 2^{2m} B_{2m}}{(2m)!} z^{2m}.$$

Both are power series of the same function on $|z| < \pi$, so by uniqueness their coefficients of z^{2m} agree:

$$\frac{(-1)^m 2^{2m} B_{2m}}{(2m)!} = -\frac{2\zeta(2m)}{\pi^{2m}}.$$

Multiply by $(-1)^m (2m)!/2^{2m}$ and use $-(-1)^m = (-1)^{m-1}$ to solve for B_{2m}:

$$\boxed{B_{2m} = (-1)^{m-1} \frac{2(2m)!}{(2\pi)^{2m}} \zeta(2m)}.$$

12.6 Asymptotic Series

12.6.1 Integrating by parts, develop asymptotic expansions as $x \to +\infty$ of the Fresnel integrals

$$\text{(a) } C(x) = \int_0^x \cos\frac{\pi u^2}{2}\,du, \quad \text{(b) } s(x) = \int_0^x \sin\frac{\pi u^2}{2}\,du.$$

These integrals appear in the analysis of a knife-edge diffraction pattern.

Solution.

Take $x \to +\infty$ and write $S(x) = s(x)$ for the sine integral in the question. Put $\theta = \pi x^2/2$. Rotating the Gaussian integration ray through $\pi/4$, i.e. substituting $u = e^{i\pi/4} t$, so that $i\pi u^2/2 = -\pi t^2/2$, gives

$$\int_0^\infty e^{i\pi u^2/2} du = e^{i\pi/4} \int_0^\infty e^{-\pi t^2/2} dt = \frac{1+i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1+i}{2},$$

by $\int_0^\infty e^{-\alpha t^2} dt = \tfrac{1}{2}\sqrt{\pi/\alpha}$ with $\alpha = \pi/2$. The rotation is justified by Cauchy's theorem: the integrand is entire, so its integrals over $[0, R]$, over the arc $u = Re^{i\phi}$, $0 \le \phi \le \pi/4$, and back along the ray $\arg u = \pi/4$ add to zero, and the arc contribution tends to zero. Indeed, on the arc $|e^{i\pi u^2/2}| = e^{-\pi R^2 \sin(2\phi)/2}$ and $|du| = R\,d\phi$, and Jordan's inequality $\sin 2\phi \ge 4\phi/\pi$ $(0 \le 2\phi \le \pi/2)$ gives

$$R\int_0^{\pi/4} e^{-\pi R^2 \sin(2\phi)/2} d\phi \le R\int_0^\infty e^{-2R^2\phi} d\phi = \frac{1}{2R}.$$

Therefore, since $\int_0^x = \int_0^\infty - \int_x^\infty$,

$$C(x) + iS(x) = \frac{1+i}{2} - T_0(x), \qquad T_m(x) = \int_x^\infty u^{-2m} e^{i\pi u^2/2} du.$$

Since $d(e^{i\pi u^2/2}) = i\pi u e^{i\pi u^2/2} du$, integration by parts gives

$$\begin{aligned} T_m &= \left[\frac{e^{i\pi u^2/2}}{i\pi u^{2m+1}}\right]_x^\infty + \frac{2m+1}{i\pi}\int_x^\infty u^{-2m-2} e^{i\pi u^2/2} du \\ &= \frac{i e^{i\theta}}{\pi x^{2m+1}} - \frac{i(2m+1)}{\pi} T_{m+1}. \end{aligned}$$

Substitute the formula for T_1 into that for T_0, then the one for T_2, and so on. After k steps the coefficient of T_k is $\prod_{m=0}^{k-1}[-i(2m+1)/\pi] = (-i)^k(2k-1)!!/\pi^k$, and it multiplies the boundary term $ie^{i\theta}/(\pi x^{2k+1})$ of T_k. This gives the finite identity

$$T_0 = \frac{ie^{i\theta}}{\pi x}\sum_{k=0}^{N-1}\frac{(-i)^k(2k-1)!!}{(\pi x^2)^k} + \frac{(-i)^N(2N-1)!!}{\pi^N}T_N,$$

where $(-1)!! = 1$. The same integration-by-parts identity with $m = N$, together with the absolute bound $|T_{N+1}| \le \int_x^\infty u^{-2N-2}du = x^{-2N-1}/(2N+1)$, gives

$$|T_N| \le \frac{1}{\pi x^{2N+1}} + \frac{2N+1}{\pi}\int_x^\infty u^{-2N-2}du = \frac{2}{\pi x^{2N+1}}.$$

Thus the remainder $(-i)^N(2N-1)!!\,T_N/\pi^N$ has modulus at most $2(2N-1)!!/(\pi^{N+1}x^{2N+1})$, the order of the first omitted term ($k = N$), for every fixed N; this proves asymptotic validity.

In the sum, the factor $i(-i)^k$ equals $i(-1)^j$ for $k = 2j$ and $(-1)^j$ for $k = 2j+1$. Separate the even and odd k contributions accordingly by defining

$$f(x) \sim \frac{1}{\pi x}\left[1 - \frac{3}{\pi^2x^4} + \frac{105}{\pi^4x^8} - \cdots\right] = \sum_{j\ge0}\frac{(-1)^j(4j-1)!!}{\pi^{2j+1}x^{4j+1}},$$
$$g(x) \sim \frac{1}{\pi^2x^3}\left[1 - \frac{15}{\pi^2x^4} + \frac{945}{\pi^4x^8} - \cdots\right] = \sum_{j\ge0}\frac{(-1)^j(4j+1)!!}{\pi^{2j+2}x^{4j+3}}.$$

Then $T_0 \sim e^{i\theta}(g + if)$, and

$$\frac{1+i}{2} - (\cos\theta + i\sin\theta)(g+if) = \Big(\frac12 - g\cos\theta + f\sin\theta\Big) + i\Big(\frac12 - f\cos\theta - g\sin\theta\Big),$$

so the requested real expansions are

$$\text{(a)}\ \boxed{C(x) \sim \frac12 + f(x)\sin\theta - g(x)\cos\theta}, \qquad \text{(b)}\ \boxed{S(x) \sim \frac12 - f(x)\cos\theta - g(x)\sin\theta}.$$

For example $C = 1/2 + \sin\theta/(\pi x) - \cos\theta/(\pi^2x^3) + O(x^{-5})$ and $S = 1/2 - \cos\theta/(\pi x) - \sin\theta/(\pi^2x^3) + O(x^{-5})$. For large negative x, use the oddness of both integrals (their integrands are even), $C(-x) = -C(x)$, $S(-x) = -S(x)$, rather than the positive-infinity constants $1/2$.

12.6.2 Rederive the asymptotic expansions of $\mathrm{Ci}(x)$ and $\mathrm{si}(x)$ as $x \to +\infty$ by repeated integration by parts.

Hint. $\mathrm{Ci}(x) + i\,\mathrm{si}(x) = -\displaystyle\int_x^\infty \frac{e^{it}}{t}\,dt.$

Solution.

Use the definitions in the hint, $\mathrm{Ci}(x) = -\int_x^\infty \cos t\,dt/t$ and $\mathrm{si}(x) = -\int_x^\infty \sin t\,dt/t = \mathrm{Si}(x) - \pi/2$ (because $\int_0^\infty \sin t\,dt/t = \pi/2$), and take $x \to +\infty$; for $x < 0$ the hint's integral would run through the singularity of e^{it}/t at $t = 0$. Set

$$J_m(x) = \int_x^\infty \frac{e^{it}}{t^{m+1}}dt.$$

Integration by parts with $dv = e^{it}dt$ gives

$$J_m = \left[\frac{e^{it}}{it^{m+1}}\right]_x^\infty + \frac{m+1}{i}J_{m+1} = \frac{ie^{ix}}{x^{m+1}} - i(m+1)J_{m+1}.$$

Substituting repeatedly, after k steps the coefficient of J_k is $\prod_{m=0}^{k-1}[-i(m+1)] = (-i)^k k!$, and it multiplies the boundary term ie^{ix}/x^{k+1} of J_k. Consequently

$$\mathrm{Ci}(x) + i\,\mathrm{si}(x) = -J_0 = -\frac{ie^{ix}}{x}\sum_{k=0}^{N-1}\frac{(-i)^k k!}{x^k} - (-i)^N N! J_N.$$

The recurrence with $m = N$ and the bound $|J_{N+1}| \le \int_x^\infty t^{-N-2}dt = x^{-N-1}/(N+1)$ bound the remainder integral by

$$|J_N| \le x^{-N-1} + (N+1)\int_x^\infty t^{-N-2}dt = 2x^{-N-1}.$$

The residual $-(-i)^N N! J_N$ after N displayed terms therefore has modulus at most $2N!/x^{N+1}$, of the order of the first omitted term. In the sum, $-i(-i)^k$ equals $-i(-1)^j$ for $k = 2j$ and $-(-1)^j$ for $k = 2j+1$. Collect the real even/odd subsequences:

$$A(x) \sim \sum_{j\ge0}\frac{(-1)^j(2j)!}{x^{2j}} = 1 - \frac{2!}{x^2} + \frac{4!}{x^4} - \cdots,$$
$$B(x) \sim \sum_{j\ge0}\frac{(-1)^j(2j+1)!}{x^{2j}} = 1 - \frac{3!}{x^2} + \frac{5!}{x^4} - \cdots.$$

Since $-J_0 \sim e^{ix}[-iA/x - B/x^2]$, multiplication by $\cos x + i\sin x$ gives the real part $A\sin x/x - B\cos x/x^2$ and the imaginary part $-A\cos x/x - B\sin x/x^2$, which yields

$$\boxed{\mathrm{Ci}(x) \sim \frac{\sin x}{x}A(x) - \frac{\cos x}{x^2}B(x)},$$

$$\boxed{\mathrm{si}(x) \sim -\frac{\cos x}{x}A(x) - \frac{\sin x}{x^2}B(x)}.$$

These are fixed-order asymptotic truncations; the factorial coefficients do not define convergent infinite series at a fixed finite x.

12.6.3 Derive the asymptotic expansion of the Gauss error function as $x \to +\infty$

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt$$
$$\approx 1 - \frac{e^{-x^2}}{\sqrt{\pi}\,x}\left(1 - \frac{1}{2x^2} + \frac{1\cdot3}{2^2x^4} - \frac{1\cdot3\cdot5}{2^3x^6} + \cdots + (-1)^n\frac{(2n-1)!!}{2^n x^{2n}}\right).$$

Hint. $\mathrm{erf}(x) = 1 - \mathrm{erfc}(x) = 1 - \dfrac{2}{\sqrt{\pi}}\displaystyle\int_x^\infty e^{-t^2}\,dt.$

Normalized so that $\mathrm{erf}(\infty) = 1$, this function plays an important role in probability theory. It may be expressed in terms of the Fresnel integrals (Exercise 12.6.1), the incomplete gamma functions (Section 13.6), or the confluent hypergeometric functions (Section 18.5).

Solution.

For $x > 0$, let

$$I_m(x) = \int_x^\infty t^{-2m} e^{-t^2} dt, \qquad \operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} I_0(x).$$

Use $d(e^{-t^2}) = -2te^{-t^2}dt$ in integration by parts:

$$I_m = \left[-\frac{e^{-t^2}}{2t^{2m+1}} \right]_x^\infty - \frac{2m+1}{2} \int_x^\infty t^{-2m-2} e^{-t^2} dt = \frac{e^{-x^2}}{2x^{2m+1}} - \frac{2m+1}{2} I_{m+1}.$$

After k repetitions the coefficient of I_k is $\prod_{m=0}^{k-1}[-(2m+1)/2] = (-1)^k(2k-1)!!/2^k$, and it multiplies the boundary term $e^{-x^2}/(2x^{2k+1})$ of I_k. After N repetitions, therefore,

$$I_0 = \frac{e^{-x^2}}{2x} \sum_{k=0}^{N-1} \frac{(-1)^k (2k-1)!!}{2^k x^{2k}} + \frac{(-1)^N (2N-1)!!}{2^N} I_N.$$

All I_m are positive, so the recurrence with $m = N$, $I_N = e^{-x^2}/(2x^{2N+1}) - \frac{2N+1}{2} I_{N+1}$, implies $0 < I_N < e^{-x^2}/(2x^{2N+1})$. Multiplying the finite identity by $2/\sqrt{\pi}$,

$$\operatorname{erfc}(x) = \frac{e^{-x^2}}{\sqrt{\pi} x} \sum_{k=0}^{N-1} \frac{(-1)^k (2k-1)!!}{2^k x^{2k}} + R_N, \qquad R_N = \frac{2}{\sqrt{\pi}} \frac{(-1)^N (2N-1)!!}{2^N} I_N.$$

So the remainder R_N in erfc has sign $(-1)^N$ and magnitude less than its first omitted term:

$$|R_N| < \frac{e^{-x^2}}{\sqrt{\pi} x} \frac{(2N-1)!!}{2^N x^{2N}}.$$

Finally erf $= 1 -$ erfc gives

$$\boxed{\operatorname{erf}(x) \sim 1 - \frac{e^{-x^2}}{\sqrt{\pi} x} \left[1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \frac{15}{8x^6} + \cdots \right]}.$$

The term with index $k = n$ is $(-1)^n(2n-1)!!/(2^n x^{2n})$, as requested. The remainder bound proves the expansion for $x \to +\infty$. For $x \to -\infty$ use $\operatorname{erf}(-x) = -\operatorname{erf}(x)$.

12.6.4 The asymptotic expressions for the various Bessel functions, Section 14.6, contain the series

$$P_\nu(z) \sim 1 + \sum_{n=1}^\infty (-1)^n \frac{\prod_{s=1}^{2n} [4\nu^2 - (2s-1)^2]}{(2n)!(8z)^{2n}},$$
$$Q_\nu(z) \sim \sum_{n=1}^\infty (-1)^{n+1} \frac{\prod_{s=1}^{2n-1} [4\nu^2 - (2s-1)^2]}{(2n-1)!(8z)^{2n-1}}.$$

Show that these are asymptotic expansions for fixed ν as $z = x \to +\infty$; distinguish terminating half-integer orders from divergent cases. Then specify a complex sector of validity.

Solution.

Divergence at fixed z does not by itself prove that a series is asymptotic to a particular function. Also, for half-integer ν the products terminate, so the blanket divergence argument in the Instructor's

Manual has exceptions. We first prove the asymptotic statement for fixed real ν and $z = x \to +\infty$, then give its complex extension on a specified sector. Put

$$\mu = \nu^2 - \frac{1}{4}, \qquad A_k = \frac{\prod_{s=1}^{k}[4\nu^2 - (2s-1)^2]}{k!8^k}, \quad A_0 = 1.$$

The outgoing Bessel solution in Section 14.6 is written

$$H_\nu^{(1)}(x) = \sqrt{\frac{2}{\pi x}} e^{i(x - \nu\pi/2 - \pi/4)} w(x), \qquad w = P_\nu + iQ_\nu.$$

Substituting $H = x^{-1/2}U$ into $x^2H'' + xH' + (x^2 - \nu^2)H = 0$, the first-derivative terms cancel and the equation becomes $x^{3/2}[U'' + (1 - \mu/x^2)U] = 0$. Then $U = e^{ix}w$ (up to a constant factor) has $U'' = e^{ix}(w'' + 2iw' - w)$, so

$$w'' + 2iw' - \frac{\mu}{x^2} w = 0, \qquad w(x) \to 1.$$

Only the leading outgoing normalization $w \to 1$ from Section 14.6 is used here; the higher coefficients and their remainder will be proved. For $w \sim \sum_{k\ge 0} c_k x^{-k}$, termwise $w'' = \sum_{k\ge 0} k(k+1)c_k x^{-k-2}$ and, shifting the index (w' has no $k = 0$ term), $2iw' = -\sum_{k\ge 0} 2i(k+1)c_{k+1}x^{-k-2}$. So the coefficient of x^{-k-2} is

$$[k(k+1) - \mu]c_k - 2i(k+1)c_{k+1} = 0.$$

Since $k(k+1) - \mu = (k + \frac{1}{2})^2 - \nu^2 = [(2k+1)^2 - 4\nu^2]/4$ and $1/(2i) = -i/2$, this gives, with $c_0 = 1$,

$$c_{k+1} = \frac{i[4\nu^2 - (2k+1)^2]}{8(k+1)} c_k, \qquad c_k = i^k A_k.$$

Because $i^{2n} = (-1)^n$, $i^{2n-1} = i(-1)^{n+1}$ and $A_k z^{-k} = \prod_{s=1}^{k}[4\nu^2 - (2s-1)^2]/(k!(8z)^k)$, the real even powers and the imaginary odd powers (divided by i) are exactly the two series in the question, including their signs.

Here is a remainder argument. For an integrable forcing f, define

$$(Kf)(x) = \frac{1}{2i} \int_x^\infty [e^{2i(t-x)} - 1] f(t)\, dt.$$

Differentiating with Leibniz's rule (the boundary term of the first derivative vanishes because the kernel is zero at $t = x$; that of the second derivative is $+f(x)$) gives

$$(Kf)' = -\int_x^\infty e^{2i(t-x)} f(t)\, dt, \qquad (Kf)'' = f(x) + 2i \int_x^\infty e^{2i(t-x)} f(t)\, dt.$$

Thus $(Kf)'' + 2i(Kf)' = f$, and $Kf \to 0$ at infinity, since $|Kf(x)| \le \int_x^\infty |f|\, dt$. Apply this to $f = \mu w/x^2$, which is integrable because $w \to 1$. Since $w'' + 2iw' = f$ as well, the difference $d = w - 1 - Kf$ solves $d'' + 2id' = 0$, so $d = \alpha + \beta e^{-2ix}$. Since d has period π and tends to 0 on the positive axis, it vanishes identically, so $\alpha = \beta = 0$. This proves the Volterra equation

$$w(x) = 1 + \frac{\mu}{2i} \int_x^\infty [e^{2i(t-x)} - 1] \frac{w(t)}{t^2} dt \equiv 1 + T[w](x).$$

Its kernel has absolute value at most 1 after division by $2i$, so $|T[w](x)| \le |\mu| \sup|w| \int_x^\infty t^{-2} dt = |\mu| \sup|w|/x$. On $x \ge R > |\mu|$, the operator T therefore has sup-norm at most $|\mu|/R < 1$, so by

the contraction-mapping principle the bounded outgoing solution is unique and is obtained by successive substitution. Indeed, two bounded solutions $w, \tilde{w}$ satisfy $\sup|w-\tilde{w}| = \sup|T[w-\tilde{w}]| \le (|\mu|/R)\sup|w-\tilde{w}|$, which forces $\tilde{w} = w$; and the iterates $w^{(0)} = 1$, $w^{(m+1)} = 1 + T[w^{(m)}]$ satisfy $\sup|w^{(m+1)} - w^{(m)}| \le (|\mu|/R)^m \sup|T[1]|$, so they converge uniformly. Let $S_N = \sum_{k=0}^{N-1} c_k x^{-k}$, $N \ge 1$. In $S_N'' + 2iS_N' - \mu S_N/x^2$, the coefficient of x^{-k-2} is the left side of the recurrence for $k \le N-2$, hence zero; for $k = N-1$ the term c_N is absent. This leaves precisely

$$r_N := S_N'' + 2iS_N' - \mu S_N/x^2 = [N(N-1) - \mu]c_{N-1}x^{-N-1}.$$

Then $E_N = w - S_N$ satisfies $E_N'' + 2iE_N' = \mu E_N/x^2 - r_N$ and $E_N \to 0$, so the argument that gave the Volterra equation gives

$$E_N = T[E_N] - \frac{1}{2i}\int_x^\infty [e^{2i(t-x)} - 1]r_N(t)dt.$$

The explicit last integral is at most $\int_x^\infty |r_N|\,dt = |[N(N-1)-\mu]c_{N-1}|\,x^{-N}/N$, i.e. $O(x^{-N})$. In the weighted norm $\|E\|_N = \sup_{x\ge R} x^N|E(x)|$,

$$x^N|T[E](x)| \le |\mu|\|E\|_N x^N \int_x^\infty t^{-N-2}dt \le \frac{|\mu|}{(N+1)R}\|E\|_N.$$

So the map $E \mapsto T[E]$ − (last integral) is a contraction both in the sup norm and in $\|\cdot\|_N$. Its fixed point with $\|E\|_N < \infty$ is bounded, so by uniqueness of the bounded fixed point it equals $E_N = w - S_N$; hence $E_N = O(x^{-N})$. This is the required asymptotic remainder estimate for every fixed truncation. Taking real and imaginary parts proves the P_ν, Q_ν expansions (for real ν, $H_\nu^{(2)}(x) = \overline{H_\nu^{(1)}(x)}$ makes P_ν, Q_ν real on the axis). For real ν the $c_k = i^k A_k$ are real for even k and imaginary for odd k, so $\Re S_{2n+1} = \Re S_{2n+2}$ and $\Im S_{2n} = \Im S_{2n+1}$. Hence the P_ν series stopped after its x^{-2n} term, which is $\Re S_{2n+1}$, has error $\Re E_{2n+2} = O(x^{-2n-2})$, and the Q_ν series stopped after its $x^{-(2n-1)}$ term, which is $\Im S_{2n}$, has error $\Im E_{2n+1} = O(x^{-2n-1})$; each error is of the order of the first omitted term.

For nonterminating cases the ratio of two consecutive P terms is

$$\left|\frac{[4\nu^2 - (4n+1)^2][4\nu^2 - (4n+3)^2]}{(2n+1)(2n+2)(8x)^2}\right| \sim \frac{(16n^2)^2}{4n^2\cdot 64x^2} = \frac{n^2}{x^2},$$

and the Q ratio has the same growth. Thus they diverge for fixed x (ratio test), while their finite truncations have the proved large-x meaning. If $\nu = \pm(j+1/2)$, $j = 0, 1, \ldots$, a numerator factor vanishes at $s = j+1$ and all later coefficients are zero; the resulting finite expansions remain valid, and in fact solve the transformed equation exactly: for $N = j+1$, $N(N-1)-\mu = j(j+1)-[(j+\frac{1}{2})^2-\frac{1}{4}] = 0$, so $r_N = 0$. These terminating cases satisfy the usual fixed-order remainder definition. They are excluded only if one imposes the textbook's extra divergence convention, Eq. (12.98); see Exercise 12.6.5.

The proof also supplies a complex sector, rather than assuming that an expansion on the real axis automatically extends to all directions. For $\Re z \ge 0$, $|z| \ge R$, integrate along $t = z + u$, $u \ge 0$, so that $e^{2i(t-z)} = e^{2iu}$ and the Volterra equation reads

$$w(z) = 1 + \frac{\mu}{2i}\int_0^\infty \left[e^{2iu} - 1\right]\frac{w(z+u)}{(z+u)^2}\,du.$$

Then

$$|e^{2iu} - 1| \le 2, \qquad |z+u| \ge (|z|+u)/\sqrt{2},$$

the latter because $|z+u|^2 = |z|^2 + 2u\Re z + u^2 \geq |z|^2 + u^2 \geq (|z|+u)^2/2$. Also $|z+u| \geq |z|$, so the path $z+u$ stays in the region $\Re z \geq 0$, $|z| \geq R$, and $\int_0^\infty 2\,du/(|z|+u)^2 = 2/|z|$; hence the unweighted operator norm on that region is at most $2|\mu|/R$. In the weighted norm, using $|z+u|^{-N-2} \leq 2^{(N+2)/2}(|z|+u)^{-N-2}$, its bound is

$$\frac{|\mu|\,2^{(N+2)/2}}{(N+1)R},$$

and the residual integral is bounded by $|[N(N-1)-\mu]c_{N-1}|\,2^{(N+1)/2}|z|^{-N}/N$. Taking R sufficiently large gives the same contraction and $O(|z|^{-N})$ remainder, uniformly in this closed right half-plane. The incoming solution obeys $w_-'' - 2iw_-' - \mu w_-/z^2 = 0$; its kernel uses $e^{-2iu}-1$, with the same bound, and its coefficients are $(-i)^k A_k$ (replace i by $-i$ throughout, since $H_\nu^{(2)}$ carries the factor e^{-iz}). Therefore $P = (w+w_-)/2$ selects the even terms and $Q = (w-w_-)/(2i)$ selects the odd terms, because $i^k + (-i)^k$ vanishes for odd k and $i^k - (-i)^k$ for even k. The successive-substitution iterates (starting from 1) are analytic for $\Re z > 0$, $|z| > R$: each is an integral, along paths $z+u$ that stay in that region, of a function analytic in z (differentiate under the integral sign). They converge uniformly, so by Weierstrass's theorem on uniform limits the fixed points w and w_- are analytic there. For real $z = x > R$ the paths stay on the axis, so by the uniqueness proved above (with $i \to -i$ for w_-) these fixed points coincide there with the Hankel normalizations, which are analytic in the same region; by the identity theorem they coincide throughout the sector. This argument also works for each fixed complex ν, since only $|\mu|$ enters the bounds. Wider continuation sectors require the corresponding Hankel branches; no expansion across an unspecified branch or Stokes boundary is being asserted.

12.6.5 For $x > 1$,

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n \frac{1}{x^{n+1}}.$$

Test this series to see if it is an asymptotic series.

Solution.

The answer depends on the stated convention. With the textbook's partial sum through x^{-m}, $m \geq 1$, finite geometric summation gives

$$s_m(x) = \sum_{k=1}^{m} (-1)^{k-1} x^{-k} = \frac{1}{x}\,\frac{1-(-1/x)^m}{1+1/x} = \frac{1-(-1/x)^m}{1+x},$$

so the remainder is

$$R_m(x) = \frac{1}{1+x} - s_m(x) = \frac{(-1/x)^m}{1+x} = \frac{(-1)^m}{x^m(1+x)}.$$

Consequently

$$x^m R_m(x) = \frac{(-1)^m}{1+x}.$$

For each fixed m this tends to zero as $x \to +\infty$, satisfying Eq. (12.97). For each fixed $x > 1$, however, its absolute value is the constant $1/(1+x)$; it does not diverge as $m \to \infty$. Thus the extra requirement in Eq. (12.98) is not met:

not an asymptotic series under the textbook's convention.

This agrees with the Instructor's Manual. Its convergence follows independently from the geometric ratio $|-1/x| < 1$.

Terminology qualification: the usual Poincaré definition uses the fixed-order condition (12.97) without the additional divergence requirement (12.98). The textbook itself acknowledges this alternative in footnote 5 on p. 583. Under that definition the same series is both convergent and asymptotic; the Instructor's Manual's answer is correct under the textbook's explicitly stricter convention. In that usual notation,

$$\frac{1}{1+x} \sim \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3} - \cdots, \qquad x \to +\infty.$$

The exact remainder above proves the claim under either convention without inferring asymptotic behavior merely from convergence or divergence.

12.6.6 Derive the following Bernoulli-number asymptotic series for the Euler-Mascheroni constant, defined in Eq. (1.13):

$$\gamma \sim \sum_{s=1}^{n} s^{-1} - \ln n - \frac{1}{2n} + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)n^{2k}}.$$

Here n plays the role of x.

Hint. Apply the Euler-Maclaurin integration formula to $f(x) = x^{-1}$ over the interval $[1, n]$ for $N = 1, 2, \ldots$.

Solution.

Write $H_n = \sum_{s=1}^{n} 1/s$. Apply the Euler–Maclaurin formula to $f(x) = 1/x$ on $[n, M]$ (equivalently, apply it on $[1, n]$ and on $[1, M]$, as in the hint, and subtract):

$$\sum_{s=n}^{M} f(s) = \int_n^M f\,dx + \frac{f(n) + f(M)}{2} + \sum_{k=1}^{N} \frac{B_{2k}}{(2k)!}\left[f^{(2k-1)}(M) - f^{(2k-1)}(n)\right] + \text{remainder}.$$

Here the left side is $H_M - H_n + 1/n$, the integral is $\ln(M/n)$, and $f^{(2k-1)}(x) = -(2k-1)!x^{-2k}$ turns the kth correction into $(B_{2k}/2k)(n^{-2k} - M^{-2k})$. Hence

$$H_M - H_n = \ln(M/n) + \frac{1}{2M} - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k}(n^{-2k} - M^{-2k}) + \text{remainder}.$$

Let $M \to \infty$ and use the definition of γ, $H_M - \ln M \to \gamma$. This produces

$$\gamma = H_n - \ln n - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k\,n^{2k}} + R_{n,N},$$

which defines the remainder $R_{n,N}$. To prove the asymptotic nature, rather than merely write formal corrections, we now derive a signed bound for $R_{n,N}$. Define

$$K(n) = \int_0^{\infty} e^{-nt}\left[\frac{1}{t} - \frac{1}{e^t - 1}\right] dt.$$

This integral is finite: the bracket tends to $1/2$ at zero (since $e^t - 1 = t + t^2/2 + O(t^3)$) and is $O(1/t)$ at infinity. Its consecutive difference is

$$\begin{aligned} K(n) - K(n+1) &= \int_0^\infty \frac{e^{-nt} - e^{-(n+1)t}}{t} dt - \int_0^\infty e^{-(n+1)t} dt \\ &= \ln(1 + 1/n) - \frac{1}{n+1}. \end{aligned}$$

The first equality used $e^{-nt}(1 - e^{-t})/(e^t - 1) = e^{-(n+1)t}$, because $(1 - e^{-t})/(e^t - 1) = e^{-t}$. For the logarithmic integral write $(e^{-nt} - e^{-(n+1)t})/t = \int_n^{n+1} e^{-at} da$ and integrate first in t (all terms are positive): $\int_n^{n+1} da/a = \ln(1 + 1/n)$. This is also the consecutive difference of $H_n - \ln n - \gamma$, because $(H_n - \ln n) - (H_{n+1} - \ln(n+1)) = \ln(1 + 1/n) - 1/(n+1)$. The bracket in $K(n)$ is bounded (it is continuous for $t > 0$ with finite limits at both ends), say by C, so $|K(n)| \le C \int_0^\infty e^{-nt} dt = C/n \to 0$, while $H_n - \ln n - \gamma \to 0$ by the definition of γ. The two sequences have equal consecutive differences, so they differ by a constant, and that constant tends to zero; hence $K(n) = H_n - \ln n - \gamma$. The bracket in $K(n)$ equals $\frac{1}{2} - q(t)$, with $q(t) = 1/(e^t - 1) - 1/t + 1/2$, and $\int_0^\infty e^{-nt} dt/2 = 1/(2n)$; thus

$$\gamma = H_n - \ln n - \frac{1}{2n} + \int_0^\infty e^{-nt} q(t) dt.$$

Now $\frac{1}{e^t - 1} + \frac{1}{2} = \frac{e^t + 1}{2(e^t - 1)} = \frac{1}{2} \coth \frac{t}{2}$. The cotangent partial fractions of Exercise 12.5.10, evaluated at iz, give $\coth z = i \cot(iz) = \frac{1}{z} + \sum_{j \ge 1} \frac{2z}{z^2 + j^2 \pi^2}$; with $z = t/2$,

$$q(t) = \frac{1}{2} \coth(t/2) - \frac{1}{t} = 2t \sum_{j=1}^\infty \frac{1}{(2\pi j)^2 + t^2}.$$

Expand each denominator by the *finite* geometric identity

$$\frac{1}{a^2 + t^2} = \sum_{k=1}^N \frac{(-1)^{k-1} t^{2k-2}}{a^{2k}} + \frac{(-1)^N t^{2N}}{a^{2N}(a^2 + t^2)}, \qquad a = a_j = 2\pi j.$$

Since $2 \sum_{j \ge 1} (-1)^{k-1} a_j^{-2k} = 2(-1)^{k-1} \zeta(2k)/(2\pi)^{2k} = B_{2k}/(2k)!$ by Eq. (12.38) (Exercise 12.5.10), this yields

$$q(t) = \sum_{k=1}^N \frac{B_{2k}}{(2k)!} t^{2k-1} + (-1)^N 2t^{2N+1} \sum_{j=1}^\infty \frac{1}{a_j^{2N}(a_j^2 + t^2)}.$$

The last sum is positive. Multiply by e^{-nt} and integrate. Since $\int_0^\infty e^{-nt} t^m dt = m!/n^{m+1}$, the kth term gives $\frac{B_{2k}}{(2k)!} \cdot \frac{(2k-1)!}{n^{2k}} = \frac{B_{2k}}{2k\, n^{2k}}$, the stated correction, and therefore (the j sum and the integral may be interchanged because all terms are positive)

$$R_{n,N} = (-1)^N 2 \sum_{j=1}^\infty a_j^{-2N} \int_0^\infty \frac{e^{-nt} t^{2N+1}}{a_j^2 + t^2} dt.$$

This gives the rigorous bound

$$\boxed{0 < (-1)^N R_{n,N} < \frac{|B_{2N+2}|}{(2N+2) n^{2N+2}}}.$$

Indeed, replacing $(a_j^2+t^2)^{-1}$ by the larger a_j^{-2} gives

$$(-1)^N R_{n,N} < \frac{2(2N+1)!}{n^{2N+2}} \sum_{j\geq 1} a_j^{-2N-2} = \frac{2(2N+1)!\,\zeta(2N+2)}{(2\pi)^{2N+2} n^{2N+2}} = \frac{|B_{2N+2}|}{(2N+2)n^{2N+2}}$$

by Eq. (12.38). Since B_{2N+2} has the sign $(-1)^N$, the remainder has the sign of the first omitted term and is smaller in magnitude. For each fixed N this remainder is smaller by two powers of n than the last correction, proving the requested asymptotic series. The coefficients grow factorially, so taking infinitely many terms at fixed n is not justified.

12.6.7 Develop an asymptotic series for

$$\int_0^\infty \frac{e^{-xv}}{1+v^2}\,dv.$$

Take x to be real and positive.

$$ANS.\ \ \frac{1}{x} - \frac{2!}{x^3} + \frac{4!}{x^5} - \cdots + \frac{(-1)^n(2n)!}{x^{2n+1}}.$$

Solution.

The published errata (Instructor's Manual, 6 April 2012, textbook p. 585) replace the original denominator $(1+v^2)^2$ by $1+v^2$. The corrected integral in the question now matches its printed answer. We give both versions below to make the change explicit. For the corrected integral $I(x)$, use the exact finite geometric identity $\sum_{k=0}^{N}(-w)^k = [1-(-w)^{N+1}]/(1+w)$ with $w=v^2$:

$$\frac{1}{1+v^2} = \sum_{k=0}^{N}(-1)^k v^{2k} + \frac{(-1)^{N+1}v^{2N+2}}{1+v^2}.$$

Multiply by e^{-xv} and integrate for $x>0$. With $u=xv$,

$$\int_0^\infty e^{-xv}v^{2k}dv = x^{-2k-1}\int_0^\infty e^{-u}u^{2k}du = \frac{(2k)!}{x^{2k+1}},$$

where repeated integration by parts evaluates the last integral. Consequently

$$\boxed{I(x) \sim \sum_{k=0}^{\infty} \frac{(-1)^k(2k)!}{x^{2k+1}} = \frac{1}{x} - \frac{2!}{x^3} + \frac{4!}{x^5} - \cdots}.$$

The actual remainder after $k=N$ is

$$R_N = (-1)^{N+1}\int_0^\infty \frac{e^{-xv}v^{2N+2}}{1+v^2}dv, \qquad 0 < (-1)^{N+1}R_N < \frac{(2N+2)!}{x^{2N+3}},$$

because $0 < v^{2N+2}/(1+v^2) < v^{2N+2}$ for $v>0$ and $\int_0^\infty e^{-xv}v^{2N+2}dv = (2N+2)!/x^{2N+3}$. This proves the asymptotic statement without interchanging an infinite geometric series outside its disk of convergence.

For comparison, the *original* squared-denominator integral has

$$\int_0^\infty \frac{e^{-xv}}{(1+v^2)^2}dv \sim \boxed{\sum_{k=0}^{\infty} \frac{(-1)^k(k+1)(2k)!}{x^{2k+1}}} = \frac{1}{x} - \frac{4}{x^3} + \frac{72}{x^5} - \cdots.$$

Its coefficients follow from $(1+w)^{-2} = \sum(-1)^k(k+1)w^k$ with $w = v^2$, the kth term integrating to $(-1)^k(k+1)(2k)!/x^{2k+1}$ ($2 \cdot 2! = 4$, $3 \cdot 4! = 72$). More precisely, differentiate the finite geometric identity $1/(1+w) = \sum_{k=0}^{N+1}(-w)^k + (-w)^{N+2}/(1+w)$ with respect to w. The derivative of the sum is $\sum_{k=1}^{N+1}(-1)^k k w^{k-1}$, which after the shift $k \to k+1$ is $-\sum_{k=0}^{N}(-1)^k(k+1)w^k$, and $(-1)^{N+2} = (-1)^N$; so

$$-\frac{1}{(1+w)^2} = -\sum_{k=0}^{N}(-1)^k(k+1)w^k + (-1)^N \frac{d}{dw}\frac{w^{N+2}}{1+w}.$$

Changing sign gives

$$\frac{1}{(1+w)^2} = \sum_{k=0}^{N}(-1)^k(k+1)w^k + (-1)^{N+1}\frac{w^{N+1}[(N+2)+(N+1)w]}{(1+w)^2},$$

since $\frac{d}{dw}\frac{w^{N+2}}{1+w} = \frac{w^{N+1}[(N+2)+(N+1)w]}{(1+w)^2}$. For $w = v^2 \geq 0$, $(N+2)+(N+1)w \leq (N+2)(1+w)$, so the magnitude of the finite remainder is at most $(N+2)v^{2N+2}$, and integration bounds the original integral's remainder by $(N+2)(2N+2)!/x^{2N+3}$. The factor $k+1$ shows exactly why the original problem and printed answer were inconsistent.

12.7 Method of Steepest Descents

We present here a rather small number of exercises on the method of steepest descents. Several additional exercises appear elsewhere in this book, in particular in Section 14.6, where the technique is applied to the contour integral representations of Bessel functions.

12.7.1 Let F be nonconstant and analytic on a connected open region. Derive the correct circle mean of $|F|^2$ about z_0 from the Taylor coefficients of F. Prove that $|F|$ has no interior local maximum and no positive interior local minimum. In particular, if F never vanishes, it has no interior modulus extremum.

Solution.

Both the asserted mean-value identity for $|F|^2$ (that its mean over a circle about z_0 equals $|F(z_0)|^2$) and the unqualified absence of minima are false. For $F(z) = z$ at $z_0 = 0$, the circle average is r^2, whereas $|F(0)|^2 = 0$ is a minimum. The nonconstant, nonzero case used in the textbook's discussion of saddle points needs a corrected argument. Expand in a disk about z_0 lying in the analytic region:

$$F(z_0 + re^{i\theta}) = \sum_{n=0}^{\infty} a_n r^n e^{in\theta}, \qquad a_0 = F(z_0).$$

Then $|F|^2 = F\overline{F} = \sum_{n,m\geq 0} a_n \overline{a_m} r^{n+m} e^{i(n-m)\theta}$; uniform convergence on a smaller circle permits this multiplication and integration term by term. Since $(2\pi)^{-1}\int_0^{2\pi} e^{i(n-m)\theta}d\theta = \delta_{nm}$,

$$\frac{1}{2\pi}\int_0^{2\pi} |F(z_0+re^{i\theta})|^2 d\theta = \sum_{n,m\geq 0} a_n\overline{a_m} r^{n+m}\delta_{nm}$$

$$= \boxed{|F(z_0)|^2 + \sum_{n=1}^{\infty} |a_n|^2 r^{2n}}.$$

At least one a_n, $n \geq 1$, is nonzero: otherwise the Taylor series would make F constant near z_0, and the identity theorem would make it constant throughout the connected region. Thus

$\sum_{n\ge1}|a_n|^2r^{2n}>0$ for $r>0$, and the mean strictly exceeds the center value on every sufficiently small circle. A local maximum would make every value on that circle no larger than the center, a contradiction. Therefore $|F|$ has no interior local maximum unless F is constant.

If $F(z_0)\neq0$, continuity gives a smaller disk where F never vanishes. Apply the same maximum argument to the analytic function $1/F$. A local minimum of $|F|$ would be a local maximum of $|1/F|$, again impossible for nonconstant F. Thus a nonconstant analytic function has no positive interior local minimum; zeros are allowed minima. A constant function is the usual exception to both statements.

For comparison, writing $\langle g\rangle=\frac{1}{2\pi}\int_0^{2\pi}g(z_0+re^{i\theta})\,d\theta$ for the mean over the circle, the true mean-value identities are $\langle F\rangle=F(z_0)$ (Cauchy's integral formula on the circle $z=z_0+re^{i\theta}$, $dz=ire^{i\theta}d\theta$), and, on a zero-free disk, $\langle\log|F|\rangle=\log|F(z_0)|$. The latter follows by taking the real part of the mean-value formula for a local analytic logarithm of F. It gives a second proof of the no-extremum assertion in the zero-free setting relevant to the saddle-point discussion: if $\log|F|\le\log|F(z_0)|$ (or $\ge$) near z_0, then on each small circle the continuous function $\log|F|$ lies on one side of its mean $\log|F(z_0)|$ and hence equals it, so $|F|=|F(z_0)|$ near z_0; then the boxed identity gives $\sum_{n\ge1}|a_n|^2r^{2n}=0$, and F is constant.

12.7.2 Find the steepest path and leading asymptotic expansion as $s\to+\infty$ for the Fresnel integrals $\displaystyle\int_0^s\cos x^2\,dx,\quad\int_0^s\sin x^2\,dx.$

Hint. Use $\displaystyle\int_0^1 e^{itz^2}\,dz.$

Solution.

Put $t=s^2>0$ and scale $x=sz$ (so $dx=s\,dz$ and $x^2=tz^2$):

$$\int_0^s e^{ix^2}dx=sJ(t),\qquad J(t)=\int_0^1 e^{itz^2}dz.$$

For $f(z)=iz^2$, the saddle is $z=0$, with $f''(0)=2i$. Writing $z=re^{i\theta}$ gives $\Re(itz^2)=-tr^2\sin2\theta$ and $\Im(itz^2)=tr^2\cos2\theta$. The descending ray from zero is therefore $\theta=\pi/4$; on it the exponential becomes e^{-tr^2}. The endpoint $z=1$ has a descending constant-phase path

$$z=\sqrt{1+iv},\quad 0\le v<\infty,\qquad itz^2=it-tv,\quad dz=\frac{i\,dv}{2\sqrt{1+iv}}.$$

Both paths approach the same direction at infinity, since $\sqrt{1+iv}\sim e^{i\pi/4}\sqrt{v}$ as $v\to\infty$. Deforming the finite interval to their difference is legitimate: the integrand is entire, so by Cauchy's theorem the integral over $[0,1]$ equals the integral along the ray $z=re^{i\pi/4}$ ($dz=e^{i\pi/4}dr$, $itz^2=-tr^2$) for $0\le r\le\sqrt{V}$, minus the integral along the endpoint path for $0\le v\le V$, minus the integral over the segment $z^2=\sigma+iV$, $0\le\sigma\le1$, joining their far ends. On that segment $|e^{itz^2}|=e^{-tV}$ and $|dz|=d\sigma/(2|z|)\le d\sigma/(2\sqrt{V})$, so its integral vanishes as $V\to\infty$. Hence the exact contour decomposition is

$$J(t)=e^{i\pi/4}\int_0^\infty e^{-tr^2}dr-\frac{i}{2}e^{it}\int_0^\infty\frac{e^{-tv}}{\sqrt{1+iv}}dv.$$

The first integral is the endpoint half-Gaussian $\frac12\sqrt{\pi/t}=\sqrt{\pi}/(2\sqrt{t})$. For the second, Taylor's theorem with integral remainder gives, for all $v\ge0$,

$$(1+iv)^{-1/2}=1-\frac{iv}{2}+\rho(v),\qquad|\rho(v)|\le\frac{3}{8}v^2,$$

because the second derivative $-\frac{3}{4}(1+iv)^{-5/2}$ has modulus at most $3/4$ ($|1+iv| \geq 1$). With $\int_0^\infty e^{-tv}dv = 1/t$, $\int_0^\infty ve^{-tv}dv = 1/t^2$ and $\int_0^\infty v^2 e^{-tv}dv = 2/t^3$,

$$-\frac{i}{2}e^{it}\int_0^\infty \frac{e^{-tv}}{\sqrt{1+iv}}dv = -\frac{i}{2}e^{it}\left[\frac{1}{t} - \frac{i}{2t^2}\right] + \eta, \qquad |\eta| \leq \frac{1}{2}\cdot\frac{3}{8}\cdot\frac{2}{t^3} = \frac{3}{8t^3}.$$

Therefore

$$J(t) = \frac{e^{i\pi/4}\sqrt{\pi}}{2\sqrt{t}} - \frac{ie^{it}}{2t} - \frac{e^{it}}{4t^2} + O(t^{-3}).$$

Multiply by s, with $t = s^2$: the terms become $e^{i\pi/4}\sqrt{\pi}/2 = (1+i)\sqrt{\pi/8}$, $-ie^{is^2}/(2s)$, $-e^{is^2}/(4s^3)$ and $s\,O(s^{-6}) = O(s^{-5})$. Since $\Re(-ie^{i\alpha}) = \sin\alpha$ and $\Im(-ie^{i\alpha}) = -\cos\alpha$, taking real and imaginary parts yields

$$\boxed{\begin{aligned}\int_0^s \cos x^2 dx &= \sqrt{\frac{\pi}{8}} + \frac{\sin s^2}{2s} - \frac{\cos s^2}{4s^3} + O(s^{-5}),\\ \int_0^s \sin x^2 dx &= \sqrt{\frac{\pi}{8}} - \frac{\cos s^2}{2s} - \frac{\sin s^2}{4s^3} + O(s^{-5}).\end{aligned}}$$

Both integrals are odd in s, so for $s \to -\infty$ the constants become $-\sqrt{\pi/8}$. The saddle is an endpoint, so it contributes half of a full Gaussian integral. The Instructor's Manual's limiting value is twice the correct $\sqrt{\pi/8}$.

12.7.3 Show that the formula

$$\Gamma(1+s) \approx \sqrt{2\pi s}\, s^s e^{-s}$$

holds for complex values of s (with $\Re(s)$ large and positive).

Hint. This involves assigning a phase to s and then demanding that $\Im[sf(z)]$ be constant in the vicinity of the saddle point.

Solution.

Use the principal $\operatorname{Log} s$ for $\Re s > 0$, and write $s = \rho e^{i\phi}$, $|\phi| < \pi/2$. Starting with $\Gamma(1+s) = \int_0^\infty e^{-u}u^s du$, set $u = sz$. As u runs over $(0,\infty)$, z runs along the ray $\arg z = -\phi$; this ray can be rotated to the positive z axis, i.e. the u path to the ray $\arg u = \phi$. By Cauchy's theorem this only requires the arcs $|u| = r$ between the two rays to contribute nothing as $r \to \infty$ and as $r \to 0$. On such an arc $u = re^{i\psi}$ with ψ between 0 and ϕ, so $|e^{-u}| = e^{-r\cos\psi} \leq e^{-r\cos\phi}$ and $|u^s| = |e^{s(\ln r + i\psi)}| = r^{\Re s}e^{-\psi\Im s} \leq r^{\Re s}e^{|\phi\,\Im s|}$; the length is $r|\phi|$, and the product tends to zero in both limits because $\cos\phi > 0$ and $\Re s > 0$. On the rotated path $z > 0$, so $\operatorname{Log} u = \operatorname{Log} s + \ln z$ and $u^s = s^s z^s$ with principal branches; also $du = s\,dz$. Thus

$$\Gamma(1+s) = s^{s+1}\int_0^\infty e^{sf(z)}dz, \qquad f(z) = \operatorname{Log} z - z.$$

The saddle is $z_0 = 1$, because $f' = 1/z - 1$; there $f(1) = -1$, $f''(1) = -1/z^2|_{z=1} = -1$ and $f'''(1) = 2/z^3|_{z=1} = 2$. Near the saddle, Taylor's formula gives

$$sf(z) = -s - \frac{s}{2}(z-1)^2 + \frac{s}{3}(z-1)^3 + \cdots.$$

To make the quadratic decrement real and negative, choose

$$z - 1 = e^{-i\phi/2}\frac{v}{\sqrt{\rho}} = \frac{v}{\sqrt{s}}, \qquad v \in \mathbb{R}.$$

Then $\frac{s}{2}(z-1)^2 = \frac{v^2}{2}$, so $sf(z) = -s - \frac{v^2}{2} + \frac{v^3}{3\sqrt{s}} + \cdots$, whose imaginary part is $-\Im s$ up to the cubic term. This is the local steepest path and keeps $\Im[sf(z)]$ constant to quadratic order. Near $z = 1$ the real z path may be turned onto this line by Cauchy's theorem, since $e^{sf(z)}$ is analytic for $\Re z > 0$. Since $dz = dv/\sqrt{s}$, the leading Gaussian contribution of the line is

$$\int e^{sf(z)}dz \sim \frac{e^{-s}}{\sqrt{s}} \int_{-\infty}^{\infty} e^{-v^2/2}dv = \frac{\sqrt{2\pi}e^{-s}}{\sqrt{s}}.$$

Equivalently, staying on the real axis with $z = 1 + y$, $\int_{-\infty}^{\infty} e^{-sy^2/2}dy = \sqrt{2\pi/s}$ for $\Re s > 0$ gives the same value. Therefore, with the same branch for all powers,

$$\boxed{\Gamma(1+s) \sim \sqrt{2\pi s}\, s^s e^{-s}}.$$

The factor $e^{-i\phi/2}$ from the path is essential; replacing the Gaussian width by $1/\sqrt{|s|}$ would lose that phase.

For completeness, the approximation remains valid throughout the stated limit $\Re s \to +\infty$, even if the phase changes with s. A useful uniform remainder follows from Euler–Maclaurin. Euler's limit (Section 13.1), $\Gamma(s) = \lim_{M\to\infty} M!\, M^s/[s(s+1)\cdots(s+M)]$, gives a logarithm of $\Gamma(s)$ (the limit below exists, as shown next):

$$\log\Gamma(s) = \lim_{M\to\infty} \left[\log M! + s\log M - \sum_{k=0}^{M} \operatorname{Log}(s+k) \right].$$

Apply the Euler–Maclaurin formula with one Bernoulli correction ($B_2 = 1/6$) to $h(t) = \operatorname{Log}(s+t)$:

$$\sum_{k=0}^{M} h(k) = \int_0^M h(t)dt + \frac{h(0)+h(M)}{2} + \frac{h'(M)-h'(0)}{12} - \frac{1}{2}\int_0^M \widetilde{B}_2(t)h''(t)dt,$$

where $\widetilde{B}_2(t) = B_2(t - \lfloor t \rfloor)$ is the periodic extension of the Bernoulli polynomial $B_2(x) = x^2 - x + \frac{1}{6}$ from $[0,1)$. Using $h' = 1/(s+t)$, $h'' = -1/(s+t)^2$, and $\int h = (s+t)\operatorname{Log}(s+t) - (s+t)$, the finite sum is

$$\begin{aligned}\sum_{k=0}^{M} \operatorname{Log}(s+k) = {} & (s+M+\tfrac{1}{2})\operatorname{Log}(s+M) - M \\ & - (s-\tfrac{1}{2})\operatorname{Log} s + \frac{1}{12(s+M)} - \frac{1}{12s} \\ & + \frac{1}{2}\int_0^M \frac{\widetilde{B}_2(t)}{(s+t)^2}dt.\end{aligned}$$

Stirling's formula for real arguments (the case of real s above; Section 13.4) gives $\log M! = (M+\frac{1}{2})\log M - M + \frac{1}{2}\log(2\pi) + o(1)$. After inserting both formulas in the Euler limit, the terms $-M$ and $+M$ cancel, and the remaining M-dependent logarithms are

$$(M+s+\tfrac{1}{2})\big[\log M - \operatorname{Log}(M+s)\big] = -(M+s+\tfrac{1}{2})\operatorname{Log}(1+s/M) \longrightarrow -s$$

for each fixed s with $\Re s > 0$, because $\operatorname{Log}(1+s/M) = s/M + O(M^{-2})$ makes the product $-s + O(1/M)$; also $1/(M+s) \to 0$, and the last integral converges absolutely as $M \to \infty$. Collecting

this limit $-s$, the constant $\frac{1}{2}\ln 2\pi$ from Stirling's formula, and the remaining terms of the finite sum with their signs reversed by the minus sign in front of the sum,

$$\log\Gamma(s) = (s-\tfrac{1}{2})\operatorname{Log} s - s + \tfrac{1}{2}\ln 2\pi + \frac{1}{12s} - \frac{1}{2}\int_0^\infty \frac{\widetilde{B}_2(t)}{(s+t)^2}dt.$$

Adding $\operatorname{Log} s$ to pass from $\Gamma(s)$ to $\Gamma(1+s) = s\Gamma(s)$ therefore yields

$$\log\Gamma(1+s) = (s+\tfrac{1}{2})\operatorname{Log} s - s + \tfrac{1}{2}\ln 2\pi + R(s), \qquad R(s) = \frac{1}{12s} - \frac{1}{2}\int_0^\infty \frac{\widetilde{B}_2(t)}{(s+t)^2}dt.$$

Since $B_2(x) = x^2 - x + \frac{1}{6}$ takes values in $[-\frac{1}{12}, \frac{1}{6}]$ for $0 \le x \le 1$, $|\widetilde{B}_2| \le 1/6$; with $|s+t| \ge \Re s + t$, $\int_0^\infty dt/(\Re s + t)^2 = 1/\Re s$ and $|s| \ge \Re s$,

$$|R(s)| \le \frac{1}{12|s|} + \frac{1}{12\Re s} \le \frac{1}{6\Re s} \longrightarrow 0.$$

Exponentiation proves that the ratio to the boxed expression tends to one for the entire requested right-half-plane limit.

12.8 Dispersion Relations

12.8.1 Assume that the function $f(z)$ satisfies the conditions for the dispersion relations. In addition, assume that $f(z) = f^*(z^*)$, i.e., that it meets the conditions of the Schwarz reflection principle, Eq. (11.127). Show that $f(z)$ is identically zero.

Solution.

The dispersion assumptions are analyticity in the upper half-plane and across the real axis, with sufficient decay for the large arc to vanish. On the real axis ($z = z^* = x$) the additional reflection condition gives $f(x) = f(x)^*$, so $v(x) = \Im f(x) = 0$. The first Hilbert-transform equation, the first of Eqs. (12.118), is

$$u(x_0) = \frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{v(x)}{x - x_0}dx.$$

Its integrand vanishes identically, hence $u(x_0) = 0$ for every real x_0 too. Thus $f = 0$ on the real axis. Because that axis lies in a region through which f is analytic, the identity theorem implies

$$\boxed{f(z) \equiv 0}$$

on the connected analytic upper-half-plane domain. Equivalently, Schwarz reflection would extend f to an entire function with decay in both half-planes, and Liouville's theorem gives the same result.

12.8.2 For $f(z)$ such that we may replace the closed contour of the Cauchy integral formula by an integral over the real axis we have

$$f(x_0) = \frac{1}{2\pi i}\left\{\int_{-\infty}^{x_0-\delta} \frac{f(x)}{x - x_0}\,dx + \int_{x_0+\delta}^{\infty} \frac{f(x)}{x - x_0}\,dx\right\} + \frac{1}{2\pi i}\int_C \frac{f(x)}{x - x_0}\,dx.$$

Here we take C to be a small semicircle about x_0 in the lower half-plane. Show that as $\delta \downarrow 0$ the formula for $f(x_0)$ reduces to

$$f(x_0) = \frac{1}{\pi i}\,⨍_{-\infty}^{\infty} \frac{f(x)}{x - x_0}\,dx,$$

which is Eq. (12.117).

Solution.

Parameterize the lower semicircle from its left endpoint to its right endpoint as

$$z = x_0 + \delta e^{i\theta}, \quad -\pi \le \theta \le 0, \qquad dz = i\delta e^{i\theta} d\theta.$$

Its orientation is counterclockwise around the lower half of the small circle, so

$$\int_C \frac{f(z)}{z - x_0} dz = i \int_{-\pi}^{0} f(x_0 + \delta e^{i\theta}) d\theta \longrightarrow i\pi f(x_0),$$

because f is continuous at x_0: the difference from $i\pi f(x_0)$ is at most $\pi \max_{|z-x_0|=\delta} |f(z) - f(x_0)| \to 0$. The two real integrals together tend, by definition, to the Cauchy principal value; the limit exists because every other term of the formula has a limit. Taking $\delta \to 0$ in the formula in the question therefore gives

$$f(x_0) = \frac{1}{2\pi i} \,\mathrm{PV} \int_{-\infty}^{\infty} \frac{f(x)}{x - x_0} dx + \frac{1}{2} f(x_0).$$

Subtract the half contribution and multiply by two:

$$\boxed{f(x_0) = \frac{1}{\pi i} \,\mathrm{PV} \int_{-\infty}^{\infty} \frac{f(x)}{x - x_0} dx}.$$

The cut integral sign $⨍$ in the requested formula denotes this Cauchy principal value, written $\mathrm{PV}\int$ here; an ordinary improper integral at the pole need not exist.

12.8.3

(a) The function $f(z) = e^{iz}$ does not vanish at the endpoints of the range of $\arg z$, 0 and π. Show, with the help of Jordan's lemma, Eq. (11.102), that Eq. (12.116) still holds.

(b) For $f(z) = e^{iz}$ verify by direct integration the dispersion relations, Eq. (12.117) or Eqs. (12.118).

Solution.

The textbook prints the first endpoint as the letter a; it is 0, since on the upper semicircle the range of $\arg z$ is $0 \le \arg z \le \pi$. (a) For z_0 fixed in the upper half-plane, put $z = Re^{i\theta}$ on the large semicircle, $0 \le \theta \le \pi$, with $R > |z_0|$. Since $|e^{iz}| = e^{-R\sin\theta}$, $|z - z_0| \ge R - |z_0|$ and $|dz| = R\,d\theta$,

$$\left| \int_{\mathrm{arc}} \frac{e^{iz}}{z - z_0} dz \right| \le \frac{R}{R - |z_0|} \int_0^{\pi} e^{-R\sin\theta} d\theta.$$

On $0 \le \theta \le \pi/2$, $\sin\theta \ge 2\theta/\pi$; the symmetry $\sin(\pi - \theta) = \sin\theta$ therefore bounds the last integral by $2\int_0^{\pi/2} e^{-2R\theta/\pi} d\theta = (\pi/R)(1 - e^{-R}) \le \pi/R$ (this is Jordan's lemma, Eq. (11.102)). The arc contribution is at most $\pi/(R - |z_0|) \to 0$, despite the nonvanishing endpoint values. Cauchy's formula for the closed half-disk consequently gives, as $R \to \infty$,

$$\boxed{e^{iz_0} = \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{ix}}{x - z_0} dx}, \qquad \Im z_0 > 0.$$

(b) Set $t = x - x_0$. The symmetric principal value is

$$\mathrm{PV}\int_{-\infty}^{\infty}\frac{e^{ix}}{x-x_0}dx = e^{ix_0}\left[\mathrm{PV}\int_{-\infty}^{\infty}\frac{\cos t}{t}dt + i\int_{-\infty}^{\infty}\frac{\sin t}{t}dt\right].$$

The cosine integral is zero by oddness: the contributions of $[\delta, L]$ and $[-L, -\delta]$ cancel. To evaluate the sine integral, first insert $e^{-\epsilon t}$ on $t > 0$. If $A(b) = \int_0^\infty e^{-\epsilon t}\sin(bt)dt/t$, then differentiation under the integral (allowed for $\epsilon > 0$) gives $A'(b) = \int_0^\infty e^{-\epsilon t}\cos(bt)\,dt = \Re\frac{1}{\epsilon - ib} = \epsilon/(\epsilon^2 + b^2)$, and $A(0) = 0$, so $A(1) = \int_0^1 \epsilon\,db/(\epsilon^2 + b^2) = \arctan(1/\epsilon) \to \pi/2$ as $\epsilon \to 0$. The damping can be removed uniformly in the tail: with $g(t) = e^{-\epsilon t}/t$, positive and decreasing, integration by parts against the bounded primitive $-\cos t$ of $\sin t$ gives

$$\int_L^\infty g(t)\sin t\,dt = g(L)\cos L + \int_L^\infty g'(t)\cos t\,dt, \qquad \left|\int_L^\infty g(t)\sin t\,dt\right| \le 2g(L) \le \frac{2}{L}$$

for every $\epsilon \ge 0$, since $\int_L^\infty |g'|\,dt = g(L)$. On $[0, L]$, dominated convergence (the integrand is bounded by 1) gives the limit $\epsilon \to 0$. Splitting $\int_0^\infty$ at L and using the tail bound both for $\epsilon = 0$ and for $\epsilon > 0$ therefore gives

$$\limsup_{\epsilon\to 0}\left|\int_0^\infty \frac{\sin t}{t}\,dt - A(1)\right| \le \frac{2}{L} + \frac{2}{L} = \frac{4}{L}.$$

Since L is arbitrary, $\int_0^\infty \sin t\,dt/t = \lim_{\epsilon\to 0} A(1) = \pi/2$, and by evenness the whole-line sine integral is π. Hence

$$\mathrm{PV}\int_{-\infty}^{\infty}\frac{e^{ix}}{x-x_0}dx = i\pi e^{ix_0} = -\pi\sin x_0 + i\pi\cos x_0.$$

Dividing by $i\pi$ verifies Eq. (12.117). Separating the imaginary and real parts also gives

$$\frac{1}{\pi}\,\mathrm{PV}\int\frac{\sin x}{x-x_0}dx = \cos x_0, \qquad -\frac{1}{\pi}\,\mathrm{PV}\int\frac{\cos x}{x-x_0}dx = \sin x_0,$$

with both integrals over the whole real line, verifying both equations in (12.118).

12.8.4 With $f(x) = u(x) + iv(x)$ and $f(x) = f^*(-x)$, assume the dispersion relations. For (a), require $v \in C^1$ and $v^{(j)}(x) = O(x^{-2-\epsilon-j})$; for (b), require $u \in C^1$ and $u^{(j)}(x) = O(x^{-1-\epsilon-j})$, $j = 0, 1$, with $0 < \epsilon < 1$. Show that as $x_0 \to +\infty$,

(a) $u(x_0) = -\dfrac{2}{\pi x_0^2}\displaystyle\int_0^\infty xv(x)\,dx + o(x_0^{-2})$,

(b) $v(x_0) = \dfrac{2}{\pi x_0}\displaystyle\int_0^\infty u(x)\,dx + o(x_0^{-1})$.

In quantum mechanics relations of this form are often called *sum rules*.

Solution.

The crossing condition $u(x) + iv(x) = f(x) = f^*(-x) = u(-x) - iv(-x)$ implies $u(-x) = u(x)$ and $v(-x) = -v(x)$. Split the Hilbert integrals, Eqs. (12.118), into positive and negative halves, substitute $x \mapsto -x$ on the latter, and combine denominators. For example,

$$\int_{-\infty}^0 \frac{v(x)\,dx}{x-a} = \int_0^\infty \frac{v(-x)\,dx}{-x-a} = \int_0^\infty \frac{v(x)\,dx}{x+a}, \qquad \frac{1}{x-a} + \frac{1}{x+a} = \frac{2x}{x^2-a^2};$$

for v, the evenness of u leads instead to $1/(x-a)-1/(x+a)=2a/(x^2-a^2)$. With $a=x_0>0$,

$$u(a)=\frac{2}{\pi}\,\mathrm{PV}\int_0^\infty \frac{xv(x)}{x^2-a^2}dx, \qquad v(a)=-\frac{2a}{\pi}\,\mathrm{PV}\int_0^\infty \frac{u(x)}{x^2-a^2}dx.$$

The displayed moments and the passage to large a require additional decay/tail control. Crossing and the ordinary dispersion assumptions alone do not suffice. Also, a zero leading moment gives a little-o conclusion, not an equivalence to zero. For example $f(z)=1/(1-iz)$ (pole at $z=-i$) is analytic above the real axis, decays, and satisfies crossing, $f^*(-x)=1/(1-ix)=f(x)$, but on the axis $f(x)=(1+ix)/(1+x^2)$, so $v(x)=x/(1+x^2)$ makes $\int_0^\infty xv(x)dx=\int_0^\infty x^2dx/(1+x^2)$ divergent.

Here are sufficient assumptions and a justified expansion. For (a), assume v, v' are continuous and, for some $0<\epsilon<1$, $v^{(j)}(x)=O(x^{-2-\epsilon-j})$, $j=0,1$. For (b), assume instead $u^{(j)}(x)=O(x^{-1-\epsilon-j})$, $j=0,1$. Each condition is imposed on the function in its corresponding integral; it makes the moment $\int_0^\infty xv\,dx$, respectively $\int_0^\infty u\,dx$, absolutely convergent. The exact kernel decompositions

$$\frac{x}{x^2-a^2}=-\frac{x}{a^2}+\frac{x^3}{a^2(x^2-a^2)}, \qquad \frac{1}{x^2-a^2}=-\frac{1}{a^2}+\frac{x^2}{a^2(x^2-a^2)}$$

give

$$u(a)=-\frac{2}{\pi a^2}\int_0^\infty xv(x)dx+\frac{2}{\pi a^2}\,\mathrm{PV}\int_0^\infty \frac{x^3v(x)}{x^2-a^2}dx,$$
$$v(a)=\frac{2}{\pi a}\int_0^\infty u(x)dx-\frac{2}{\pi a}\,\mathrm{PV}\int_0^\infty \frac{x^2u(x)}{x^2-a^2}dx.$$

Both final integrals tend to zero. To see the control near the moving pole, put $h=x^3v$ in (a) or $h=x^2u$ in (b). In both cases h is C^1 and the assumptions give $h(x)=O(x^{1-\epsilon})$ and $h'(x)=O(x^{-\epsilon})$ at large x (e.g. $h'=3x^2v+x^3v'$ in (a), $h'=2xu+x^2u'$ in (b)). Split the range at $a/2, 3a/2$. On the middle interval write $g(x)=h(x)/(x+a)$, so the principal value is

$$\mathrm{PV}\int_{a/2}^{3a/2}\frac{g(x)}{x-a}dx=\int_{a/2}^{3a/2}\frac{g(x)-g(a)}{x-a}dx.$$

The constant term cancels because $\mathrm{PV}\int_{a/2}^{3a/2}dx/(x-a)=0$ by symmetry about a. There $g'=h'/(x+a)-h/(x+a)^2=O(a^{-1-\epsilon})$, so by the mean value theorem the last integrand is $O(a^{-1-\epsilon})$, and the integral over an interval of length a is $O(a^{-\epsilon})$. On the lower interval, $|x^2-a^2|\ge 3a^2/4$ and, since h is bounded on $[0,1]$ and $O(x^{1-\epsilon})$ beyond, $\int_0^{a/2}|h|\,dx\le C_0+C_1a^{2-\epsilon}$ for constants C_0, C_1; so its contribution is $O(a^{-2})+O(a^{-\epsilon})$. On the upper interval $x^2-a^2\ge x^2-\frac{4}{9}x^2=\frac{5}{9}x^2$, giving $O(\int_{3a/2}^\infty x^{-1-\epsilon}dx)=O(a^{-\epsilon})$. After multiplication by $2/(\pi a^2)$, respectively $2/(\pi a)$, the second terms are $O(a^{-2-\epsilon})=o(a^{-2})$ and $O(a^{-1-\epsilon})=o(a^{-1})$. It follows that

$$\boxed{u(a)=-\frac{2}{\pi a^2}\int_0^\infty xv(x)dx+o(a^{-2})},$$

$$\boxed{v(a)=\frac{2}{\pi a}\int_0^\infty u(x)dx+o(a^{-1})}.$$

When the respective moment is nonzero, these imply exactly the two originally printed $\sim$ relations. When it vanishes, a further moment is needed to find the first nonzero term.

12.8.5

(a) Given the integral equation (valid for all real x_0)

$$\frac{1}{1+x_0^2} = \frac{1}{\pi} \fint_{-\infty}^{\infty} \frac{u(x)}{x - x_0}\, dx,$$

use Hilbert transforms to determine $u(x_0)$.

(b) Verify that the $u(x_0)$ found as your answer to part (a) actually satisfies the integral equation.

(c) From $f(z)|_{y=0} = u(x) + iv(x)$, replace x by z and determine $f(z)$. Verify that the conditions for the Hilbert transforms are satisfied.

(d) Are the crossing conditions satisfied?

ANS. (a) $u(x_0) = \dfrac{x_0}{1+x_0^2}$, (c) $f(z) = (z+i)^{-1}$.

Solution.

The cut integral sign denotes a Cauchy principal value, written PV$\int$ below; it is necessary at $x = x_0$. All integrals below have that meaning. Write a for x_0 and use the convention of Eqs. (12.118),

$$(Hg)(a) = \frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{g(x)}{x-a}dx, \qquad u = Hv, \quad v = -Hu.$$

(a) The given equation is $Hu = 1/(1+a^2)$, so $v(a) = -1/(1+a^2)$ and $u = Hv$. To evaluate the latter transform, use the partial fractions (check by recombining: $(1+x^2) - (x+a)(x-a) = 1+a^2$)

$$\frac{1}{(1+x^2)(x-a)} = \frac{1}{1+a^2}\left[\frac{1}{x-a} - \frac{x+a}{1+x^2}\right].$$

Integrate each piece over $[-L, L]$ and let $L \to \infty$; this symmetric cutoff is legitimate because the combined integrand is $O(x^{-2})$. The principal value of $1/(x-a)$ is $\ln[(L-a)/(L+a)] \to 0$; $x/(1+x^2)$ contributes zero by symmetric odd integration; and $\int dx/(1+x^2) = \pi$. Therefore

$$u(a) = -\frac{1}{\pi}\,\mathrm{PV}\int \frac{dx}{(1+x^2)(x-a)} = -\frac{1}{\pi}\cdot\frac{-a\pi}{1+a^2} = \boxed{\frac{a}{1+a^2}}.$$

Within the decaying Hilbert-transform class, inversion $H^2 = -1$ fixes this solution uniquely.

(b) Check the original transform directly with another decomposition:

$$\frac{x}{(1+x^2)(x-a)} = \frac{1}{1+a^2}\left[\frac{a}{x-a} + \frac{1-ax}{1+x^2}\right].$$

By the same three evaluations its integral is $[a\cdot 0 + \pi - a\cdot 0]/(1+a^2) = \pi/(1+a^2)$. Division by π reproduces the given equation, including its positive sign.

(c) On the real axis,

$$f(x) = u + iv = \frac{x-i}{x^2+1} = \frac{1}{x+i}.$$

Thus $\boxed{f(z) = 1/(z+i)}$. Its only pole is at $-i$, below the real axis, so it is analytic in and above the real axis. Also $f(z) = O(1/z)$ there. For the Cauchy kernel $f(z)/(z-z_0)$ the large arc is $O(1/R)$, verifying the needed dispersion condition.

(d) The standard crossing conditions are not satisfied: u is odd, whereas v is even. In fact

$$f(-x) = \frac{1}{-x+i} = -\frac{1}{x-i} = -f(x)^*,$$

with a minus sign instead of the requested $f(-x) = f(x)^*$.

12.8.6 Assume the full-frequency dispersion hypotheses for $f(\omega) = n^2(\omega) - 1$, including analyticity above the real axis and high-frequency decay. In (a), constancy means on the entire real frequency axis.

(a) If the real part of the complex index of refraction (squared) is constant (no optical dispersion), show that the imaginary part is zero (no absorption).

(b) Conversely, if there is absorption, show that there must be dispersion. In other words, if the imaginary part of $n^2 - 1$ is not zero, show that the real part of $n^2 - 1$ is not constant.

Solution.

The conclusion uses the dispersion hypotheses on $f(\omega) = n^2(\omega) - 1$, including its decay at high frequency, and constancy over the entire frequency axis. Constancy over only a finite frequency interval is not enough for this argument. Write $f = u + iv$. (a) If the real part of n^2 is constant, then $u = \Re(n^2 - 1) = K$ is constant. Since $f(\omega) \to 0$ at high frequency, $K = 0$. The second Hilbert-transform equation then gives

$$v(\omega_0) = -\frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{u(\omega)}{\omega - \omega_0} d\omega = 0.$$

Thus $\boxed{\Im n^2 = 0}$, meaning no absorption under the stated model. The constant real value of n^2 is in fact one because of the high-frequency normalization.

(b) This is the contrapositive of (a). If v is not identically zero and u were constant, the preceding argument would force $v = 0$, a contradiction. Hence

$$\boxed{\Im(n^2 - 1) \not\equiv 0 \quad \Longrightarrow \quad \Re(n^2 - 1) \text{ is not constant}}.$$

This asserts the existence of dispersion somewhere in the full spectrum; it does not demand that its size be appreciable at every frequency.

12.8.7 Given $u(x) = x/(x^2+1)$ and $v(x) = -1/(x^2+1)$, show by direct evaluation of each integral that

$$\int_{-\infty}^{\infty} |u(x)|^2\,dx = \int_{-\infty}^{\infty} |v(x)|^2\,dx.$$

$$\textit{ANS.} \quad \int_{-\infty}^{\infty} |u(x)|^2\,dx = \int_{-\infty}^{\infty} |v(x)|^2\,dx = \frac{\pi}{2}.$$

Solution.

Use $x = \tan\theta$, $dx = \sec^2\theta\,d\theta$, with $-\pi/2 < \theta < \pi/2$, so that $x^2 + 1 = \sec^2\theta$. For the real functions in the question,

$$\int_{-\infty}^{\infty} |u|^2 dx = \int_{-\pi/2}^{\pi/2} \frac{\tan^2\theta}{\sec^4\theta} \sec^2\theta\,d\theta = \int_{-\pi/2}^{\pi/2} \sin^2\theta\,d\theta,$$

$$\int_{-\infty}^{\infty} |v|^2 dx = \int_{-\pi/2}^{\pi/2} \frac{1}{\sec^4\theta} \sec^2\theta\,d\theta = \int_{-\pi/2}^{\pi/2} \cos^2\theta\,d\theta.$$

With $\sin^2\theta = (1 - \cos 2\theta)/2$ and $\cos^2\theta = (1 + \cos 2\theta)/2$, the antiderivatives are $\theta/2 \mp \sin 2\theta/4$, and the oscillatory endpoint term $\sin 2\theta/4$ vanishes at both limits $\theta = \pm\pi/2$. Each integral is therefore $[\theta/2]_{-\pi/2}^{\pi/2}$, i.e.

$$\boxed{\pi/2}.$$

This directly checks the Parseval equality without assuming it.

12.8.8 Take $u(x) = \delta(x)$, a delta function, and assume that the Hilbert transform equations hold. Interpret both displayed kernel integrals as distributions: replace each reciprocal $1/q$ by $q/(q^2+\epsilon^2)$, use the same $\epsilon > 0$ for both factors, integrate, and then take $\epsilon \downarrow 0$ against smooth compactly supported test functions.

(a) Show that

$$\delta(w) = \frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{dy}{y(y-w)}.$$

(b) With changes of variables $w = s - t$ and $x = s - y$, transform the δ representation of part (a) into

$$\delta(s-t) = \frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{dx}{(x-s)(x-t)}.$$

Note. The δ function is discussed in Section 1.11.

Solution.

These formulas are distributional kernel identities. They are not ordinary improper integrals, nor does the pointwise principal value at coincident poles exist. A consistent regularization is essential. (a) With $u(x) = \delta(x)$, the second Hilbert relation gives, since the delta function evaluates $1/(x-y)$ at $x = 0$,

$$v(y) = -\frac{1}{\pi}\,\mathrm{PV}\int \frac{\delta(x)}{x-y}dx = \frac{1}{\pi}\,\mathrm{PV}\,\frac{1}{y}.$$

The first relation $u = Hv$ therefore formally gives the expression in (a):

$$\delta(w) = u(w) = \frac{1}{\pi}\int \frac{v(y)\,dy}{y-w} = \frac{1}{\pi^2}\int \frac{dy}{y(y-w)}.$$

To specify and prove its meaning, regularize both kernels by the same positive ϵ and define

$$D_\epsilon(w) = \frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{y(y-w)}{(y^2+\epsilon^2)((y-w)^2+\epsilon^2)}dy.$$

For real $w \neq 0$, close the rational integrand in the upper half-plane. It is $O(y^{-2})$, so the arc vanishes as $O(1/R)$. Its poles are $y = \pm i\epsilon$ and $y = w \pm i\epsilon$; the upper ones are $i\epsilon$ and $w + i\epsilon$. Using $y^2+\epsilon^2 = (y-i\epsilon)(y+i\epsilon)$, $(i\epsilon - w)^2 + \epsilon^2 = w(w-2i\epsilon)$ and $(w+i\epsilon)^2+\epsilon^2 = w(w+2i\epsilon)$, the two residues before division by π^2 are

$$\frac{i\epsilon(i\epsilon - w)}{2i\epsilon\, w(w-2i\epsilon)} = \frac{-w+i\epsilon}{2w(w-2i\epsilon)}, \qquad \frac{(w+i\epsilon)\,i\epsilon}{w(w+2i\epsilon)\,2i\epsilon} = \frac{w+i\epsilon}{2w(w+2i\epsilon)}.$$

Over the common denominator $2w(w^2+4\epsilon^2)$ the numerator of their sum is $(-w+i\epsilon)(w+2i\epsilon) + (w+i\epsilon)(w-2i\epsilon) = -2i\epsilon w$, so the sum is $-i\epsilon/(w^2+4\epsilon^2)$. Multiplying by $2\pi i/\pi^2$,

$$\boxed{D_\epsilon(w) = \frac{2\epsilon}{\pi(w^2+4\epsilon^2)}}.$$

The formula extends to $w = 0$ by continuity; directly, with $y = \epsilon\tan\phi$,

$$D_\epsilon(0) = \frac{1}{\pi^2}\int_{-\infty}^{\infty}\frac{y^2\,dy}{(y^2+\epsilon^2)^2} = \frac{1}{\pi^2\epsilon}\int_{-\pi/2}^{\pi/2}\sin^2\phi\,d\phi = \frac{1}{2\pi\epsilon},$$

the same value. For every smooth compactly supported test function φ, substitute $w = 2\epsilon t$, so that $dw = 2\epsilon\, dt$ and $D_\epsilon(2\epsilon t) = 1/[2\pi\epsilon(1+t^2)]$:

$$\int_{-\infty}^{\infty} D_\epsilon(w)\varphi(w)dw = \int_{-\infty}^{\infty} \frac{\varphi(2\epsilon t)}{\pi(1+t^2)}dt \longrightarrow \varphi(0).$$

Dominated convergence applies because φ is bounded and the Cauchy density has integral one. Therefore $D_\epsilon \to \delta$ in the distributional sense, establishing the precise meaning of

$$\boxed{\delta(w) = \frac{1}{\pi^2}\int \frac{dy}{y(y-w)}}.$$

For $w \neq 0$ the separately interpreted principal values would give zero, because $1/(y(y-w)) = [1/(y-w) - 1/y]/w$ and each principal value vanishes, while at $w = 0$ the integral of $1/y^2$ diverges; this is why a pointwise calculation cannot replace the distributional limit.

(b) Set $w = s - t$ and $x = s - y$. Then $y = s - x$, $y - w = t - x$, and $dy = -dx$. Reversing the limits removes the minus sign, while

$$y(y-w) = (s-x)(t-x) = (x-s)(x-t).$$

Thus

$$\boxed{\delta(s-t) = \frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{dx}{(x-s)(x-t)}}$$

with the same regularization: its finite-ϵ kernel is $2\epsilon/[\pi((s-t)^2 + 4\epsilon^2)]$. This is also the kernel identity used in the textbook's derivation of the Parseval relation $\int |u|^2 dx = \int |v|^2 dx$ in Section 12.8 (checked directly in Exercise 12.8.7).

13 Chapter 13. Gamma Function

Problems

13.1 Definitions, Properties

13.1.1 Derive the recurrence relations

$$\Gamma(z+1) = z\Gamma(z)$$

from the Euler integral, Eq. (13.5),

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1}\, dt.$$

The Euler integral is used for $\Re z > 0$; the recurrence then extends meromorphically.

Solution.

Initially let $\Re z > 0$, so both Euler integrals and the boundary terms below converge. Integrate $\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt$ by parts with $u = t^z$ and $dv = e^{-t}dt$, so that $du = zt^{z-1}dt$ and $v = -e^{-t}$:

$$\Gamma(z+1) = \left[-t^z e^{-t}\right]_0^\infty + z\int_0^\infty t^{z-1} e^{-t} dt.$$

At zero, $|t^z e^{-t}| \le |t^z| = t^{\Re z} \to 0$; at infinity $|t^z e^{-t}| = t^{\Re z} e^{-t} \to 0$, since the exponential outweighs the power. The boundary term therefore vanishes, and the remaining integral is $\Gamma(z)$. Hence

$$\boxed{\Gamma(z+1) = z\Gamma(z)}.$$

Applying this N times, $\Gamma(z+N) = (z+N-1)(z+N-2)\cdots(z+1)z\,\Gamma(z)$, that is, $\Gamma(z) = \Gamma(z+N)/[z(z+1)\cdots(z+N-1)]$ with $\Re(z+N) > 0$. This extends the identity to the meromorphic gamma function; at poles it is interpreted by meromorphic continuation, not as multiplication of finite values. In particular $\Gamma(1) = \int_0^\infty e^{-t}dt = 1$, and $\Gamma(N+1) = N(N-1)\cdots 1\cdot\Gamma(1) = N!$ for every nonnegative integer N.

13.1.2 In a power-series solution for the Legendre functions of the second kind we encounter the expression

$$\frac{(n+1)(n+2)(n+3)\cdots(n+2s-1)(n+2s)}{2\cdot 4\cdot 6\cdot 8\cdots(2s-2)(2s)\cdot(2n+3)(2n+5)(2n+7)\cdots(2n+2s+1)},$$

in which s is a positive integer.

(a) Rewrite this expression in terms of factorials.

(b) Rewrite this expression using Pochhammer symbols; see Eq. (1.72).

Solution.

(a) For the factorials of integers take $n \ge 0$ integral. The numerator is $(n+2s)!/n!$, and the even product is $2\cdot 4\cdots(2s) = 2^s(1\cdot 2\cdots s) = 2^s s!$. The odd product is a quotient of odd double factorials.

Since $(2m+1)! = (2m+1)!!\,(2m)!!$ and $(2m)!! = 2^m m!$, each of them is $(2m+1)!! = (2m+1)!/(2^m m!)$; with $m = n+s$ and $m = n$,

$$\frac{(2n+2s+1)!!}{(2n+1)!!} = \frac{(2n+2s+1)!}{2^{n+s}(n+s)!}\frac{2^n n!}{(2n+1)!}.$$

In the product of the two denominator factors the powers of two cancel:

$$2^s s!\,\frac{(2n+2s+1)!\,2^n n!}{2^{n+s}(n+s)!\,(2n+1)!} = \frac{s!\,n!\,(2n+2s+1)!}{(n+s)!\,(2n+1)!}.$$

Dividing the numerator $(n+2s)!/n!$ by this product gives

$$\boxed{\frac{(n+2s)!(n+s)!(2n+1)!}{s!(n!)^2(2n+2s+1)!}}.$$

For nonintegral n, factorial notation means $q! = \Gamma(q+1)$, and the equality holds wherever the original denominator is nonzero, with any removable gamma quotients interpreted by limits.

(b) By definition $(q)_s = q(q+1)\cdots(q+s-1)$. The numerator is $(n+1)_{2s}$. Taking a factor 2 out of each factor of the two denominator products gives

$$2\cdot 4\cdots(2s) = 2^s(1)_s = 2^s s!,$$
$$(2n+3)(2n+5)\cdots(2n+2s+1) = 2^s\left(n+\tfrac{3}{2}\right)\left(n+\tfrac{5}{2}\right)\cdots\left(n+s+\tfrac{1}{2}\right) = 2^s\left(n+\tfrac{3}{2}\right)_s.$$

Thus

$$\boxed{\frac{(n+1)_{2s}}{4^s s!(n+\frac{3}{2})_s} = \frac{(\frac{n+1}{2})_s(\frac{n+2}{2})_s}{s!(n+\frac{3}{2})_s}}.$$

For the last equality, separate the $2s$ consecutive numerator factors into those in odd and even positions and take a factor 2 out of each:

$$\begin{aligned}(n+1)_{2s} &= \big[(n+1)(n+3)\cdots(n+2s-1)\big]\big[(n+2)(n+4)\cdots(n+2s)\big]\\ &= 2^s\left(\tfrac{n+1}{2}\right)_s\cdot 2^s\left(\tfrac{n+2}{2}\right)_s,\end{aligned}$$

so the factor 4^s cancels. These finite-product forms also cover complex n away from zeros of the original denominator.

13.1.3 Show that $\Gamma(z)$ may be written

$$\Gamma(z) = 2\int_0^\infty e^{-t^2}t^{2z-1}\,dt, \quad \Re(z) > 0,$$
$$\Gamma(z) = \int_0^1\left[\ln\left(\frac{1}{t}\right)\right]^{z-1}dt, \quad \Re(z) > 0.$$

Solution.

Start from $\Gamma(z) = \int_0^\infty e^{-u}u^{z-1}du$, with $\Re z > 0$. Put $u = t^2$. Then $du = 2t\,dt$ and $u^{z-1} = t^{2z-2}$, so $e^{-u}u^{z-1}du = 2e^{-t^2}t^{2z-1}dt$, and the limits remain zero and infinity:

$$\boxed{\Gamma(z) = 2\int_0^\infty e^{-t^2}t^{2z-1}\,dt}.$$

For the second representation put $u = -\ln t$. Then $t = e^{-u}$, $du = -dt/t$, and $e^{-u} = t$; the endpoints $u = 0, \infty$ correspond to $t = 1, 0$. Therefore

$$\Gamma(z) = \int_1^0 t(-\ln t)^{z-1}\frac{-dt}{t} = \boxed{\int_0^1 [\ln(1/t)]^{z-1}\,dt}.$$

All powers use the real logarithm of their positive bases. The condition $\Re z > 0$ controls the original lower endpoint and, after substitution, the corresponding endpoint of each transformed integral.

13.1.4 In a Maxwellian distribution the fraction of particles of mass m with speed between v and $v + dv$ is

$$\frac{dN}{N} = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2}\exp\left(-\frac{mv^2}{2kT}\right)v^2\,dv,$$

where N is the total number of particles, k is Boltzmann's constant, and T is the absolute temperature. The average or expectation value of v^n is defined as $\langle v^n\rangle = N^{-1}\int v^n\,dN$. Show that

$$\langle v^n\rangle = \left(\frac{2kT}{m}\right)^{n/2}\frac{\Gamma(\frac{n+3}{2})}{\Gamma(\frac{3}{2})}.$$

This is an extension of Example 13.1.1, in which the distribution was in kinetic energy $E = mv^2/2$, with $dE = mv\,dv$.

Assume $m, k, T > 0$ and $\Re n > -3$.

Solution.

Assume $m, k, T > 0$. The moment requires $\Re n > -3$ for convergence at zero. Set $b = m/(2kT) > 0$. Then $m/(2\pi kT) = b/\pi$, the prefactor is $4\pi(b/\pi)^{3/2} = 4b^{3/2}/\sqrt{\pi}$, and the normalized speed density is $(4/\sqrt{\pi})b^{3/2}v^2e^{-bv^2}$. Hence

$$\langle v^n\rangle = \frac{4b^{3/2}}{\sqrt{\pi}}\int_0^\infty v^{n+2}e^{-bv^2}\,dv.$$

Put $u = bv^2$; the limits $0, \infty$ are unchanged, $v = (u/b)^{1/2}$, $v^{n+2} = b^{-(n+2)/2}u^{(n+2)/2}$ and $dv = \frac{1}{2}b^{-1/2}u^{-1/2}du$. The powers combine to $b^{-(n+3)/2}u^{(n+1)/2}$, and since $b^{3/2}b^{-(n+3)/2} = b^{-n/2}$,

$$\begin{aligned}\langle v^n\rangle &= \frac{4b^{3/2}}{\sqrt{\pi}}\frac{1}{2}b^{-(n+3)/2}\int_0^\infty e^{-u}u^{(n+1)/2}\,du\\ &= \frac{2}{\sqrt{\pi}}b^{-n/2}\Gamma\left(\frac{n+3}{2}\right).\end{aligned}$$

Finally $b^{-n/2} = (2kT/m)^{n/2}$, and $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2) = \sqrt{\pi}/2$, i.e., $2/\sqrt{\pi} = 1/\Gamma(3/2)$. We obtain

$$\boxed{\langle v^n\rangle = \left(\frac{2kT}{m}\right)^{n/2}\frac{\Gamma((n+3)/2)}{\Gamma(3/2)}}.$$

At $n = 0$ this equals one, checking normalization; at $n = 2$, $\Gamma(5/2) = \frac{3}{2}\Gamma(3/2)$ gives $\langle v^2\rangle = 3kT/m$. The convergence restriction comes from the lower limit: $|v^{n+2}| = v^{\Re n+2}$ is integrable at $v = 0$ exactly when $\Re n + 2 > -1$.

13.1.5 By transforming the integral into a gamma function, show that

$$-\int_0^1 x^k \ln x \, dx = \frac{1}{(k+1)^2}, \quad k > -1.$$

Solution.

Put $u = -(k+1)\ln x$, which is positive for $0 < x < 1$ since $k > -1$. Then

$$x = e^{-u/(k+1)}, \quad dx = -\frac{1}{k+1} e^{-u/(k+1)} du, \quad -\ln x = \frac{u}{k+1},$$

and $x = 1$ and $x \to 0^+$ correspond to $u = 0$ and $u \to \infty$. The factor $x^k = e^{-ku/(k+1)}$ and the exponential in dx combine to e^{-u}. Reversing the transformed limits absorbs the minus sign of dx:

$$\begin{aligned} -\int_0^1 x^k \ln x \, dx &= \int_\infty^0 \frac{u}{k+1} \left(-\frac{e^{-u}}{k+1} \right) du = \frac{1}{(k+1)^2} \int_0^\infty u e^{-u} \, du \\ &= \frac{\Gamma(2)}{(k+1)^2} = \boxed{\frac{1}{(k+1)^2}}, \end{aligned}$$

since $\Gamma(2) = 1! = 1$. The restriction $k > -1$ ensures convergence at zero and makes the change of variable valid with these limits.

13.1.6 Show that

$$\int_0^\infty e^{-x^4} \, dx = \Gamma\left(\frac{5}{4}\right).$$

Solution.

With $t = x^4$, i.e., $x = t^{1/4}$, we have $dx = \frac{1}{4} t^{-3/4} dt$, and the limits $0, \infty$ are unchanged. Thus

$$\int_0^\infty e^{-x^4} dx = \frac{1}{4} \int_0^\infty e^{-t} t^{1/4 - 1} dt = \frac{1}{4} \Gamma\left(\frac{1}{4}\right) = \boxed{\Gamma\left(\frac{5}{4}\right)},$$

where the last equality is the recurrence from Exercise 13.1.1 with $z = \frac{1}{4}$. The transformed power $t^{-3/4}$ is integrable at zero.

13.1.7 Show that

$$\lim_{x \to 0} \frac{\Gamma(ax)}{\Gamma(x)} = \frac{1}{a}.$$

Take a fixed $a \neq 0$.

Solution.

The parameter a is fixed and nonzero. Use the recurrence in the form $\Gamma(w) = \Gamma(1+w)/w$, with $w = ax$ and $w = x$, to remove the two poles at $x = 0$:

$$\frac{\Gamma(ax)}{\Gamma(x)} = \frac{\Gamma(1+ax)/(ax)}{\Gamma(1+x)/x} = \frac{1}{a} \frac{\Gamma(1+ax)}{\Gamma(1+x)}.$$

As $x \to 0$, both $1 + ax$ and $1 + x$ tend to 1. The gamma function is continuous and nonzero near 1, with $\Gamma(1) = 1$, so the last quotient tends to 1. Consequently

$$\boxed{\lim_{x \to 0} \frac{\Gamma(ax)}{\Gamma(x)} = \frac{1}{a}}.$$

This argument also works for complex $a \neq 0$ and a complex limit $x \to 0$. For $a = 0$ the numerator would be $\Gamma(0)$ throughout, so the original ratio would not be defined.

13.1.8 Locate the poles of $\Gamma(z)$. Show that they are simple poles and determine the residues.

Solution.

The Euler integral is analytic for $\Re z > 0$. On a compact subset of that half-plane, where $0 < \sigma_1 \le \Re z \le \sigma_2$, say, the jth z-derivative $e^{-t}t^{z-1}(\ln t)^j$ of the integrand is bounded by the integrable function $e^{-t}(t^{\sigma_1-1} + t^{\sigma_2-1})|\ln t|^j$, because $|t^{z-1}| = t^{\Re z-1}$ is at most t^{σ_1-1} for $t \le 1$ and at most t^{σ_2-1} for $t \ge 1$; hence the integral may be differentiated under the integral sign. For any point outside that half-plane, shift far enough (the recurrence of Exercise 13.1.1, applied $N+1$ times) to use

$$\Gamma(z) = \frac{\Gamma(z+N+1)}{z(z+1)\cdots(z+N)}.$$

In the half-plane $\Re z > -N-1$ the numerator is analytic, because $\Re(z+N+1) > 0$; the only possible singularities there are the zeros $z = 0, -1, \ldots, -N$ of the denominator. Since N is arbitrary, the only possible singularities of Γ are $z = 0, -1, -2, \ldots$. At $z = -n$ choose $N = n$. The factor $z+n$ cancels the last denominator factor, and the residue is

$$\lim_{z\to -n}(z+n)\Gamma(z) = \lim_{z\to -n}\frac{\Gamma(z+n+1)}{z(z+1)\cdots(z+n-1)} = \frac{\Gamma(1)}{(-n)(-n+1)\cdots(-1)}$$
$$= \frac{1}{(-1)^n n!} = \boxed{\frac{(-1)^n}{n!}} \qquad (n = 0, 1, 2, \ldots).$$

For $n = 0$ the denominator product is empty and equals one. The denominator has exactly one vanishing factor, and the displayed residue is nonzero. Hence each listed singularity is a simple pole, and the shifted formula proves that there are no others.

13.1.9 Show that the equation $\Gamma(x) = k$, $k \neq 0$, has an infinite number of real roots.

The level k is real.

Solution.

Here k must be a nonzero real number. A nonreal k cannot equal $\Gamma(x)$ at a real nonsingular x. On the interval $(-N, -N+1)$, with $N \ge 1$, repeated recurrence gives

$$\Gamma(x) = \frac{\Gamma(x+N)}{x(x+1)\cdots(x+N-1)}.$$

Here $0 < x+N < 1$, so $\Gamma(x+N) > 0$, while all N denominator factors are negative. Hence $\Gamma(x)$ has the fixed sign $(-1)^N$ there. By the residues of Exercise 13.1.8, $\Gamma(x) \approx (-1)^N/[N!\,(x+N)]$ near $x = -N$, where $x+N > 0$, and $\Gamma(x) \approx (-1)^{N-1}/[(N-1)!\,(x+N-1)]$ near $x = -N+1$, where $x+N-1 < 0$. Therefore

$$\Gamma(x) \longrightarrow (-1)^N\infty \quad \text{as } x \downarrow -N \text{ or } x \uparrow -N+1.$$

At the midpoint, recurrence from $\Gamma(1/2) = \sqrt{\pi}$ gives (see Exercise 13.1.14)

$$\Gamma\left(\frac{1}{2} - N\right) = \frac{(-1)^N 2^N\sqrt{\pi}}{(2N-1)!!}.$$

Its magnitude tends to zero, since by $\Gamma(w) = \Gamma(w+1)/w$ with $w = -\frac{1}{2} - N$, $|\Gamma(\frac{1}{2} - N - 1)| = |\Gamma(\frac{1}{2} - N)|/(N + \frac{1}{2})$: the ratio of consecutive magnitudes is $1/(N+1/2) \to 0$. Choose arbitrarily large N of the parity making $(-1)^N = \operatorname{sgn} k$ and with midpoint magnitude less than $|k|$. The function is continuous between its poles, and its magnitude tends to infinity at either endpoint. The intermediate value theorem therefore gives a root of $\Gamma(x) = k$ on each side of the midpoint. Infinitely many disjoint intervals have the required parity, proving infinitely many real roots.

13.1.10 Show that, for integer s,

(a) $\displaystyle\int_0^\infty x^{2s+1}\exp(-ax^2)\,dx = \frac{s!}{2a^{s+1}}.$

(b) $\displaystyle\int_0^\infty x^{2s}\exp(-ax^2)\,dx = \frac{\Gamma(s+\frac{1}{2})}{2a^{s+1/2}} = \frac{(2s-1)!!}{2^{s+1}a^s}\sqrt{\frac{\pi}{a}}.$

These Gaussian integrals are of major importance in statistical mechanics.

Take $a > 0$ and $s = 0, 1, 2, \ldots$.

Solution.

For these convergent real Gaussian integrals, take $a > 0$ and integer $s \geq 0$. In both cases put $u = ax^2$, so $x = (u/a)^{1/2}$, $dx = \frac{1}{2}a^{-1/2}u^{-1/2}du$, and the limits $0, \infty$ are unchanged. **(a)** Since $x^{2s+1} = a^{-s-1/2}u^{s+1/2}$, the integrand becomes $a^{-s-1/2}u^{s+1/2}e^{-u}\cdot\frac{1}{2}a^{-1/2}u^{-1/2}du = \frac{1}{2}a^{-s-1}u^s e^{-u}du$, and

$$\int_0^\infty x^{2s+1}e^{-ax^2}dx = \frac{1}{2a^{s+1}}\int_0^\infty u^s e^{-u}du = \frac{\Gamma(s+1)}{2a^{s+1}} = \boxed{\frac{s!}{2a^{s+1}}}.$$

(b) With $x^{2s} = a^{-s}u^s$ instead, the integrand becomes $\frac{1}{2}a^{-s-1/2}u^{s-1/2}e^{-u}du$, and

$$\int_0^\infty x^{2s}e^{-ax^2}dx = \frac{1}{2a^{s+1/2}}\int_0^\infty u^{s-1/2}e^{-u}du = \frac{\Gamma(s+1/2)}{2a^{s+1/2}}.$$

Repeated recurrence gives

$$\Gamma\left(s+\frac{1}{2}\right) = \frac{2s-1}{2}\frac{2s-3}{2}\cdots\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{(2s-1)!!}{2^s}\sqrt{\pi}.$$

Thus, writing $2a^{s+1/2} = 2a^s\sqrt{a}$,

$$\boxed{\int_0^\infty x^{2s}e^{-ax^2}dx = \frac{(2s-1)!!}{2^{s+1}a^s}\sqrt{\frac{\pi}{a}}}.$$

The convention $(-1)!! = 1$ includes $s = 0$. Negative integer s would make the lower endpoint divergent in both parts.

13.1.11 Express the coefficient of the nth term of the expansion of $(1+x)^{1/2}$ in powers of x

(a) in terms of factorials of integers,

(b) in terms of the double factorial (!!) functions.

$$ANS.\ a_n = (-1)^{n+1}\frac{(2n-3)!}{2^{2n-2}n!(n-2)!} = (-1)^{n+1}\frac{(2n-3)!!}{(2n)!!}, \quad n = 2, 3, \ldots.$$

Solution.

The coefficient of x^n is the binomial coefficient

$$a_n = \binom{1/2}{n} = \frac{(1/2)(1/2-1)\cdots(1/2-n+1)}{n!}.$$

The first two coefficients are $a_0 = 1$ and $a_1 = 1/2$. For $n \geq 2$, pull a factor $1/2$ from each of the n numerator terms; the last $n-1$ of them are negative:

$$\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2n-3}{2}\right) = \frac{(-1)^{n-1}\,1\cdot 3\cdots(2n-3)}{2^n} = \frac{(-1)^{n-1}(2n-3)!!}{2^n}.$$

(b) Dividing by $n!$ and using $(2n)!! = 2^n n!$,

$$a_n = \frac{(-1)^{n-1}(2n-3)!!}{2^n n!} = \boxed{(-1)^{n+1}\frac{(2n-3)!!}{(2n)!!}}.$$

(a) To express the odd double factorial using ordinary factorials, separate the even factors of $(2n-3)!$, whose product is $2\cdot 4\cdots(2n-4) = 2^{n-2}(n-2)!$:

$$(2n-3)! = (2n-3)!!\,2^{n-2}(n-2)!.$$

Substituting $(2n-3)!! = (2n-3)!/[2^{n-2}(n-2)!]$ and $2^n 2^{n-2} = 2^{2n-2}$ gives

$$\boxed{a_n = (-1)^{n+1}\frac{(2n-3)!}{2^{2n-2}n!(n-2)!} \quad (n \geq 2)}.$$

For example $a_2 = -1/8$ and $a_3 = 1/16$, in agreement with direct multiplication of the first few binomial factors.

13.1.12 Express the coefficient of the nth term of the expansion of $(1+x)^{-1/2}$ in powers of x

(a) in terms of the factorials of integers,

(b) in terms of the double factorial (!!) functions.

$$ANS.\ a_n = (-1)^n\frac{(2n)!}{2^{2n}(n!)^2} = (-1)^n\frac{(2n-1)!!}{(2n)!!}, \quad n = 1, 2, 3, \ldots.$$

Solution.

For $n \geq 1$, all n numerator factors $-(2j-1)/2$, $j = 1, \ldots, n$, are negative, so

$$a_n = \binom{-1/2}{n} = \frac{(-1/2)(-3/2)\cdots(-(2n-1)/2)}{n!} = \frac{(-1)^n(2n-1)!!}{2^n n!}.$$

(b) Since $(2n)!! = 2^n n!$, the double-factorial result is

$$\boxed{a_n = (-1)^n\frac{(2n-1)!!}{(2n)!!}}.$$

(a) Splitting $(2n)!$ into its even and odd factors gives $(2n)! = (2n)!!(2n-1)!! = 2^n n!(2n-1)!!$. Substituting $(2n-1)!! = (2n)!/(2^n n!)$ in the first expression for a_n gives

$$\boxed{a_n = (-1)^n\frac{(2n)!}{2^{2n}(n!)^2}}.$$

The coefficient $a_0 = 1$ is also covered if $(-1)!! = 0!! = 1$. The first terms are $1 - x/2 + 3x^2/8 - 5x^3/16 + \cdots$.

13.1.13 The Legendre polynomial P_n may be written as

$$P_n(\cos\theta) = 2\frac{(2n-1)!!}{(2n)!!}\Bigg\{\cos n\theta + \frac{1}{1}\cdot\frac{n}{2n-1}\cos(n-2)\theta + \frac{1\cdot 3}{1\cdot 2}\frac{n(n-1)}{(2n-1)(2n-3)}\cos(n-4)\theta + \frac{1\cdot 3\cdot 5}{1\cdot 2\cdot 3}\frac{n(n-1)(n-2)}{(2n-1)(2n-3)(2n-5)}\cos(n-6)\theta + \cdots\Bigg\}.$$

Let $n = 2s+1$. Then the above can be written

$$P_n(\cos\theta) = P_{2s+1}(\cos\theta) = \sum_{m=0}^{s} a_m \cos(2m+1)\theta.$$

Find a_m in terms of factorials and double factorials.

Solution.

Let k count downward steps of two from the highest harmonic n. The term in the braces that contains $\cos(n-2k)\theta$ is

$$\frac{1\cdot 3\cdots(2k-1)}{1\cdot 2\cdots k}\,\frac{n(n-1)\cdots(n-k+1)}{(2n-1)(2n-3)\cdots(2n-2k+1)}\cos(n-2k)\theta,$$

and its three products are $(2k-1)!!/k!$, $n!/(n-k)!$, and $(2n-1)!!/(2n-2k-1)!!$. Including the prefactor, the coefficient of $\cos(n-2k)\theta$ is therefore

$$c_k = 2\frac{(2n-1)!!}{(2n)!!}\frac{(2k-1)!!}{k!}\frac{n!}{(n-k)!}\frac{(2n-2k-1)!!}{(2n-1)!!}.$$

Cancel $(2n-1)!!$ and use $(2n)!! = 2^n n!$; the factor $n!$ cancels too, leaving

$$c_k = 2^{1-n}\frac{(2k-1)!!(2n-2k-1)!!}{k!(n-k)!}.$$

For odd $n = 2s+1$ the series in braces ends with its $\cos\theta$ term, $k = s$. The printed "$+\cdots$" must stop there. Its general term vanishes only from $k = n+1$ on (for $n = 3$ the printed third term, $k = 2$, is $\frac{3}{5}\cos(-\theta)$), but since $c_{n-k} = c_k$ and $\cos(n-2k)\theta = \cos(2k-n)\theta$, the terms with $s < k \le n$ only repeat the harmonics already present and would double the sum to $2P_n$; the factor 2 in front already accounts for them. Thus $k = 0, 1, \ldots, s$ gives the harmonics $\cos(2m+1)\theta$ with $m = s, s-1, \ldots, 0$, and $n-2k = 2m+1$ implies $k = s-m$ and $n-k = s+m+1$; also $2n-2k-1 = 2s+2m+1$ and $2^{1-n} = 2^{-2s}$. Hence

$$\boxed{a_m = \frac{(2s-2m-1)!!(2s+2m+1)!!}{2^{2s}(s-m)!(s+m+1)!} \quad (0 \le m \le s)}.$$

The convention $(-1)!! = 1$ includes the highest harmonic $m = s$. Finally apply $(2q-1)!! = (2q)!/(2^q q!)$ with $q = s-m$ and $q = s+m+1$:

$$(2s-2m-1)!! = \frac{(2s-2m)!}{2^{s-m}(s-m)!}, \qquad (2s+2m+1)!! = \frac{(2s+2m+2)!}{2^{s+m+1}(s+m+1)!}.$$

Their powers of two multiply to 2^{2s+1}, which with the factor 2^{2s} already present makes 2^{4s+1}:

$$\boxed{a_m = \frac{(2s-2m)!(2s+2m+2)!}{2^{4s+1}[(s-m)!]^2[(s+m+1)!]^2}}.$$

For $s=1$ the double-factorial form gives $a_0 = 1!!\,3!!/(2^2\,1!\,2!) = 3/8$ and $a_1 = (-1)!!\,5!!/(2^2\,0!\,3!) = 5/8$, so $P_3(\cos\theta) = (3\cos\theta + 5\cos 3\theta)/8$. This also follows by inserting $\cos^3\theta = (\cos 3\theta + 3\cos\theta)/4$ in $P_3(x) = (5x^3 - 3x)/2$.

13.1.14

(a) Show that $\Gamma(\frac{1}{2}-n)\Gamma(\frac{1}{2}+n) = (-1)^n\pi$, where n is an integer.

(b) Express $\Gamma(\frac{1}{2}+n)$ and $\Gamma(\frac{1}{2}-n)$ separately in terms of $\pi^{1/2}$ and a double factorial function.

ANS. $\Gamma(\frac{1}{2}+n) = \dfrac{(2n-1)!!}{2^n}\pi^{1/2}$.

The displayed double-factorial answer uses $n \geq 0$ and $(-1)!! = 1$. For negative integer n, interchange the two half-integer formulas derived below.

Solution.

(b) First take $n = N \geq 0$. Recurrence upward from 1/2, $\Gamma(\frac{1}{2}+N) = (N-\frac{1}{2})(N-\frac{3}{2})\cdots\frac{1}{2}\,\Gamma(\frac{1}{2})$, yields

$$\Gamma\left(\frac{1}{2}+N\right) = \Gamma\left(\frac{1}{2}\right)\prod_{j=1}^{N}\left(j-\frac{1}{2}\right) = \frac{1\cdot 3\cdots(2N-1)}{2^N}\sqrt{\pi} = \boxed{\frac{(2N-1)!!}{2^N}\sqrt{\pi}}.$$

Recurrence downward, $\Gamma(\frac{1}{2}) = (-\frac{1}{2})\Gamma(-\frac{1}{2}) = (-\frac{1}{2})(-\frac{3}{2})\Gamma(-\frac{3}{2}) = \cdots = \prod_{j=1}^{N}(\frac{1}{2}-j)\,\Gamma(\frac{1}{2}-N)$, gives, since each of the N factors $\frac{1}{2}-j = -(2j-1)/2$ is negative,

$$\Gamma\left(\frac{1}{2}-N\right) = \frac{\Gamma(1/2)}{\prod_{j=1}^{N}(1/2-j)} = \frac{2^N\sqrt{\pi}}{(-1)^N(2N-1)!!} = \boxed{\frac{(-1)^N 2^N}{(2N-1)!!}\sqrt{\pi}}.$$

The products are empty when $N=0$, consistently with $(-1)!! = 1$.

(a) Multiplying the two results cancels the double factorial and the power of two:

$$\boxed{\Gamma(1/2-N)\Gamma(1/2+N) = (-1)^N\pi}.$$

For a negative integer $n = -N$, interchange the two boxed individual formulas. The product is unchanged and $(-1)^{-N} = (-1)^N$, so the product identity holds for every integer n without introducing undefined negative odd double factorials. The reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$ with $z = \frac{1}{2}+n$ confirms this directly: $\pi/\sin(\frac{\pi}{2}+n\pi) = \pi/\cos n\pi = (-1)^n\pi$.

13.1.15 Show that if $\Gamma(x+iy) = u+iv$, then $\Gamma(x-iy) = u-iv$.

This is a special case of the Schwarz reflection principle, Section 11.10.

Take real x, y and exclude poles of the gamma functions.

Solution.

For real $x > 0$ and real y, the Euler integral is absolutely convergent and $t^{x+iy-1} = \exp[(x+iy-1)\ln t]$ with real $\ln t$. Complex conjugation may therefore pass under the integral:

$$\overline{\Gamma(x+iy)} = \int_0^\infty e^{-t}\overline{t^{x+iy-1}}\,dt = \int_0^\infty e^{-t}t^{x-iy-1}\,dt = \Gamma(x-iy).$$

For any other nonsingular $z = x + iy$, choose an integer N with $\Re(z + N) > 0$ and use

$$\Gamma(z) = \frac{\Gamma(z + N)}{\prod_{j=0}^{N-1}(z + j)}.$$

Conjugating numerator and denominator and applying the proven half-plane result to $\Gamma(z + N)$ (note that $\Re(\bar{z} + N) = \Re(z + N) > 0$) gives

$$\overline{\Gamma(z)} = \frac{\overline{\Gamma(z + N)}}{\prod_{j=0}^{N-1}(\bar{z} + j)} = \frac{\Gamma(\bar{z} + N)}{\prod_{j=0}^{N-1}(\bar{z} + j)} = \Gamma(\bar{z})$$

there as well. Thus if $\Gamma(x + iy) = u + iv$,

$$\boxed{\Gamma(x - iy) = u - iv}.$$

At a pole neither side is a finite complex value; the equality is understood as the corresponding meromorphic reflection relation.

13.1.16 Prove that $|\Gamma(\alpha + i\beta)| = |\Gamma(\alpha)| \prod_{n=0}^{\infty} \left[1 + \frac{\beta^2}{(\alpha + n)^2}\right]^{-1/2}$.

This equation has been useful in calculations of beta decay theory.

Take real α, β with $\alpha \notin \{0, -1, -2, \ldots\}$.

Solution.

Assume real α, β and $\alpha \notin \{0, -1, -2, \ldots\}$, so every displayed factor and $\Gamma(\alpha)$ are defined. The Euler product-limit representation is

$$\Gamma(z) = \lim_{N\to\infty} \frac{N!\, N^z}{z(z + 1)\cdots(z + N)}.$$

Since $|N^{\alpha+i\beta}| = |e^{(\alpha+i\beta)\ln N}| = N^\alpha$, the factors $N!\,|N^z|$ cancel when the absolute value at $z = \alpha + i\beta$ is divided by that at $z = \alpha$ (a finite nonzero limit, $|\Gamma(\alpha)|$). This gives

$$\begin{aligned}\frac{|\Gamma(\alpha + i\beta)|}{|\Gamma(\alpha)|} &= \lim_{N\to\infty} \prod_{j=0}^{N} \frac{|\alpha + j|}{|\alpha + j + i\beta|} \\ &= \prod_{j=0}^{\infty} \left(1 + \frac{\beta^2}{(\alpha + j)^2}\right)^{-1/2},\end{aligned}$$

because each factor is $|\alpha + j|/\sqrt{(\alpha + j)^2 + \beta^2} = [1 + \beta^2/(\alpha + j)^2]^{-1/2}$. This product converges to a positive value: by $0 \le \ln(1 + w) \le w$ for $w \ge 0$, the logarithm of the jth factor has magnitude at most $\beta^2/[2(\alpha + j)^2]$, and these bounds form a convergent series. Multiplying by $|\Gamma(\alpha)|$ gives the requested identity. The argument applies to negative nonintegral α as well as positive α, because the product-limit formula holds for every $z \notin \{0, -1, -2, \ldots\}$. At a nonpositive integer α the written expression would contain an undefined pole times a zero factor; it must not be used literally there.

13.1.17 Show that for n, a positive integer,

$$|\Gamma(n+ib+1)| = \left(\frac{\pi b}{\sinh \pi b}\right)^{1/2} \prod_{s=1}^{n} (s^2+b^2)^{1/2}.$$

Take real b; at $b=0$, interpret $\pi b/\sinh(\pi b)$ by its limit, one.

Solution.

Take real b. For $b \neq 0$, reflection and recurrence give

$$\Gamma(ib)\Gamma(1-ib) = \frac{\pi}{\sin(\pi ib)}, \qquad \Gamma(1+ib) = ib\Gamma(ib).$$

Multiply the first relation by ib and use the second. Since $\sin(i\pi b) = (e^{-\pi b} - e^{\pi b})/(2i) = i\sinh(\pi b)$, and the two factors $\Gamma(1 \pm ib)$ are conjugates (Exercise 13.1.15), this yields

$$|\Gamma(1+ib)|^2 = \Gamma(1+ib)\Gamma(1-ib) = \frac{ib\pi}{i\sinh \pi b} = \frac{\pi b}{\sinh \pi b}.$$

Apply recurrence n more times, $\Gamma(n+1+ib) = (n+ib)\Gamma(n+ib) = \cdots$:

$$\Gamma(n+1+ib) = \left[\prod_{s=1}^{n}(s+ib)\right]\Gamma(1+ib).$$

Taking the absolute value and using $|s+ib| = \sqrt{s^2+b^2}$ gives

$$\boxed{|\Gamma(n+1+ib)| = \left(\frac{\pi b}{\sinh \pi b}\right)^{1/2} \prod_{s=1}^{n} \sqrt{s^2+b^2}}.$$

At $b=0$, the quotient $\pi b/\sinh \pi b$ is interpreted by its limit 1. The formula then gives $n! = \Gamma(n+1)$, as it should. Its square root is the nonnegative real one; the quotient is positive for either sign of nonzero real b.

13.1.18 Show that for all real values of x and y, $|\Gamma(x)| \geq |\Gamma(x+iy)|$.

Exclude $x = 0, -1, -2, \ldots$, where the displayed $\Gamma(x)$ is undefined.

Solution.

Exclude $x = 0, -1, -2, \ldots$, where $\Gamma(x)$ is not a finite defined value. For other real x and real y, Exercise 13.1.16 gives

$$\frac{|\Gamma(x+iy)|}{|\Gamma(x)|} = \prod_{n=0}^{\infty}\left(1+\frac{y^2}{(x+n)^2}\right)^{-1/2}.$$

Each factor is positive and at most one, and the product converges. Every partial product is therefore at most one, as is its limit. Consequently

$$\boxed{|\Gamma(x+iy)| \leq |\Gamma(x)|}.$$

Equality holds at $y=0$. If $y \neq 0$, already the $n=0$ factor is strictly smaller than one and all later factors are at most one, so the inequality is strict. This proof includes negative nonintegral x, where the original Euler integral cannot be used directly.

13.1.19 Show that $\left|\Gamma(\frac{1}{2}+iy)\right|^2 = \dfrac{\pi}{\cosh \pi y}$.

Take real y.

Solution.

Put $z = 1/2 + iy$ in the reflection formula. Then $1 - z = 1/2 - iy = \bar{z}$, so Exercise 13.1.15 gives

$$\left|\Gamma\left(\frac{1}{2}+iy\right)\right|^2 = \Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}.$$

The denominator is

$$\sin\left(\frac{\pi}{2}+i\pi y\right) = \cos(i\pi y) = \frac{e^{\pi y}+e^{-\pi y}}{2} = \cosh \pi y.$$

Hence

$$\boxed{\left|\Gamma\left(\frac{1}{2}+iy\right)\right|^2 = \frac{\pi}{\cosh \pi y}}.$$

For real y the denominator is positive, and at $y = 0$ the formula reduces to $\Gamma(1/2)^2 = \pi$.

13.1.20 The probability density associated with the normal distribution of statistics is given by

$$f(x) = \frac{1}{\sigma(2\pi)^{1/2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right],$$

with $(-\infty, \infty)$ for the range of x. Show that

(a) $\langle x \rangle$, the mean value of x, is equal to μ,

(b) the standard deviation $(\langle x^2 \rangle - \langle x \rangle^2)^{1/2}$ is given by σ.

Assume real μ and $\sigma > 0$.

Solution.

Take real μ and $\sigma > 0$, as required for the stated probability density. Set $u = (x-\mu)/\sigma$, so $x = \mu + \sigma u$, $dx = \sigma\, du$, the limits remain $\mp\infty$, and $f(x)dx = (2\pi)^{-1/2} e^{-u^2/2} du$. The Gaussian integral (Exercise 13.1.10(b) with $s = 0$ and $a = \frac{1}{2}$ gives $\int_0^\infty e^{-u^2/2} du = \sqrt{\pi/2}$, doubled for the even integrand) and oddness of the second integrand give

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2} du = \frac{2\sqrt{\pi/2}}{\sqrt{2\pi}} = 1, \qquad \int_{-\infty}^{\infty} u e^{-u^2/2} du = 0.$$

Writing $u e^{-u^2/2} du = d(-e^{-u^2/2})$ and integrating by parts also gives, since the boundary term vanishes at both ends,

$$\int_{-\infty}^{\infty} u^2 e^{-u^2/2} du = \left[-u e^{-u^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-u^2/2} du = \sqrt{2\pi}.$$

(a) Expanding the first moment now yields

$$\langle x \rangle = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma u) e^{-u^2/2} du = \mu \cdot 1 + \sigma \cdot 0 = \boxed{\mu}.$$

(b) Expanding $(\mu + \sigma u)^2 = \mu^2 + 2\mu\sigma u + \sigma^2 u^2$ and using the three integrals yields

$$\langle x^2 \rangle = \mu^2 + 2\mu\sigma \cdot 0 + \sigma^2, \qquad \boxed{\sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\sigma^2} = \sigma}.$$

The final equality uses the positive scale convention.

13.1.21 For the gamma distribution

$$f(x) = \begin{cases} \dfrac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & x \le 0, \end{cases}$$

show that

(a) $\langle x \rangle$, the mean value of x, is equal to $\alpha\beta$,

(b) σ^2, its variance, defined as $\langle x^2 \rangle - \langle x \rangle^2$, has the value $\alpha\beta^2$.

Assume real $\alpha, \beta > 0$.

Solution.

A gamma probability distribution requires $\alpha > 0$ and $\beta > 0$. For any moment exponent q with $\Re q > -\alpha$, substitute $x = \beta u$, $dx = \beta du$, so that $x^{\alpha+q-1}dx = \beta^{\alpha+q}u^{\alpha+q-1}du$:

$$\begin{aligned} \langle x^q \rangle &= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty x^{\alpha+q-1} e^{-x/\beta} dx \\ &= \frac{\beta^q}{\Gamma(\alpha)} \int_0^\infty u^{\alpha+q-1} e^{-u} du = \frac{\beta^q \Gamma(\alpha+q)}{\Gamma(\alpha)}. \end{aligned}$$

At $q = 0$ this is one, checking normalization. **(a)** With $q = 1$ and the gamma recurrence $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$,

$$\boxed{\langle x \rangle = \frac{\beta\Gamma(\alpha+1)}{\Gamma(\alpha)} = \alpha\beta}.$$

(b) With $q = 2$ and $\Gamma(\alpha+2) = (\alpha+1)\Gamma(\alpha+1) = (\alpha+1)\alpha\Gamma(\alpha)$, $\langle x^2 \rangle = \beta^2\alpha(\alpha+1)$, so

$$\boxed{\sigma^2 = \alpha(\alpha+1)\beta^2 - (\alpha\beta)^2 = \alpha\beta^2}.$$

13.1.22 The wave function of a particle scattered by a Coulomb potential is $\psi(r, \theta)$. Given that at the origin the wave function becomes

$$\psi(0) = e^{-\pi\gamma/2}\Gamma(1 + i\gamma),$$

where $\gamma > 0$ is a dimensionless parameter, show that

$$|\psi(0)|^2 = \frac{2\pi\gamma}{e^{2\pi\gamma} - 1}.$$

Solution.

The real exponential factor has squared magnitude $e^{-\pi\gamma}$. Using Exercise 13.1.17 with the empty product ($n = 0$), that is, its first step $|\Gamma(1+ib)|^2 = \pi b/\sinh \pi b$ with $b = \gamma$,

$$|\psi(0)|^2 = e^{-\pi\gamma}|\Gamma(1+i\gamma)|^2 = e^{-\pi\gamma}\frac{\pi\gamma}{\sinh \pi\gamma}.$$

Write $\sinh \pi\gamma = (e^{\pi\gamma} - e^{-\pi\gamma})/2$ and multiply numerator and denominator by $e^{\pi\gamma}$:

$$|\psi(0)|^2 = \frac{2\pi\gamma\, e^{-\pi\gamma}}{e^{\pi\gamma} - e^{-\pi\gamma}} = \boxed{\frac{2\pi\gamma}{e^{2\pi\gamma} - 1}}.$$

Since $e^{2\pi\gamma} - 1 = 2\pi\gamma + O(\gamma^2)$, the limit as $\gamma \downarrow 0$ is one, agreeing with $\Gamma(1) = 1$ in the original wave function.

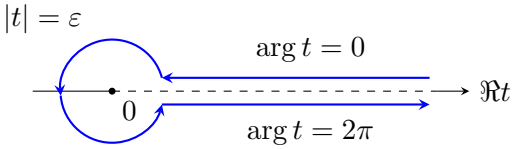
13.1.23 Derive the contour integral representation of Eq. (13.31),

$$2i\Gamma(\nu+1)\sin\nu\pi = \int_C e^{-t}e^{-\pi i\nu}t^\nu\,dt.$$

Use $0 < \arg t < 2\pi$: C runs inward on the upper lip of the positive real cut, counterclockwise around a fixed circle about zero, and outward on the lower lip. At negative integer ν, interpret the gamma–sine product by continuation.

Solution.

Use the branch $t^\nu = \exp[\nu(\ln|t| + i\arg t)]$ with $0 < \arg t < 2\pi$, cut along the positive real axis. The contour starts at positive infinity above the cut, runs inward to a small circle of radius ε, traverses that circle counterclockwise, then returns below the cut to positive infinity. The published errata (Instructor's Manual, 6 April 2012) replace the printed $(-t)^\nu$ by $e^{-\pi i\nu}t^\nu$ with this continuous branch convention. The two forms agree: for $0 < \arg t < 2\pi$ the principal argument of $-t$ is $\arg t - \pi \in (-\pi, \pi)$, so the principal power is $(-t)^\nu = \exp\{\nu[\ln|t| + i(\arg t - \pi)]\} = e^{-\pi i\nu}t^\nu$; the explicit form only states the branch.



The two lips of the cut are displaced in the sketch for visibility. Initially assume $\Re\nu > -1$. On the upper lip $t = x$ with $\arg t = 0$, so $t^\nu = x^\nu$, but the direction is inward (x runs from ∞ to ε). Its contribution is

$$I_{\text{upper}} = -e^{-\pi i\nu}\int_\varepsilon^\infty e^{-x}x^\nu dx.$$

On the lower lip the continuous argument is 2π, so $t^\nu = e^{2\pi i\nu}x^\nu$; since $e^{-\pi i\nu}e^{2\pi i\nu} = e^{\pi i\nu}$, the outward contribution is

$$I_{\text{lower}} = e^{\pi i\nu}\int_\varepsilon^\infty e^{-x}x^\nu dx.$$

On the small circle put $t = \varepsilon e^{i\theta}$, $0 \le \theta \le 2\pi$, so $dt = i\varepsilon e^{i\theta}d\theta$ and $t^\nu = \varepsilon^\nu e^{i\nu\theta}$. There $|e^{-t}| \le e^\varepsilon$ and $|e^{-\pi i\nu}t^\nu| = \varepsilon^{\Re\nu}e^{(\pi-\theta)\Im\nu} \le \varepsilon^{\Re\nu}e^{\pi|\Im\nu|}$, and the circle has length $2\pi\varepsilon$. The circle integral is therefore at most $2\pi e^{\pi|\Im\nu|}\varepsilon^{\Re\nu+1}e^\varepsilon$ in magnitude, which tends to zero because $\Re\nu + 1 > 0$. The integrand is analytic in the plane cut along the positive real axis, so by Cauchy's theorem $\int_C$ does not depend on ε: for radii $\varepsilon_1 < \varepsilon_2$ the two contours differ by the boundary of the cut annulus $\varepsilon_1 < |t| < \varepsilon_2$ (the two circles and the lip segments between them), which contains no singularity. Hence $\int_C$ is the limit as $\varepsilon \downarrow 0$ of $I_{\text{upper}} + I_{\text{lower}}$ plus the circle integral. In this limit the circle integral vanishes and the two ray integrals tend to $\int_0^\infty e^{-x}x^\nu dx = \Gamma(\nu+1)$; with $e^{\pi i\nu} - e^{-\pi i\nu} = 2i\sin\pi\nu$,

$$\begin{aligned}\int_C e^{-t}e^{-\pi i\nu}t^\nu dt &= (e^{\pi i\nu} - e^{-\pi i\nu})\int_0^\infty e^{-x}x^\nu dx \\ &= \boxed{2i\sin(\pi\nu)\Gamma(\nu+1)}.\end{aligned}$$

For $\Re\nu \le -1$, keep a nonzero circle radius; the individual ray integrals cannot be extended to zero separately. For fixed ε the complete contour integral is entire in ν: the circle is compact and the ray tails are dominated uniformly on compact parameter sets by an exponential times a fixed power. By

the Cauchy argument above, which did not use $\Re\nu > -1$, it is independent of ε. Therefore the identity continues analytically. By the reflection formula, $\Gamma(\nu+1)\Gamma(-\nu) = \pi/\sin\pi(\nu+1) = -\pi/\sin\pi\nu$, so its left side is the entire function

$$2i\Gamma(\nu+1)\sin\pi\nu = -\frac{2\pi i}{\Gamma(-\nu)}.$$

Only nonnegative integer ν makes the integrand entire in t and gives zero. At negative integers the integral is nonzero; the text's unqualified statement about integer ν needs this restriction. Indeed, for $\nu = -m$, $m = 1, 2, \ldots$, the integrand is $(-1)^m e^{-t}t^{-m}$. Since t^{-m} is single valued, the two lips carry the same integrand in opposite directions and cancel, so only the circle contributes. The coefficient of t^{-1} is $(-1)^m(-1)^{m-1}/(m-1)! = -1/(m-1)!$, so by the residue theorem the counterclockwise circle gives $2\pi i \cdot [-1/(m-1)!] = -2\pi i/(m-1)!$, which is the continued formula $-2\pi i/\Gamma(-\nu) = -2\pi i/\Gamma(m)$.

13.2 Digamma and Polygamma Functions

13.2.1 For "small" values of x,

$$\ln\Gamma(x+1) = -\gamma x + \sum_{n=2}^{\infty}(-1)^n\frac{\zeta(n)}{n}x^n,$$

where γ is the Euler-Mascheroni constant and $\zeta(n)$ the Riemann zeta function. For what values of x does this series converge?

ANS. $-1 < x \le 1$.

Note that if $x = 1$, we obtain

$$\gamma = \sum_{n=2}^{\infty}(-1)^n\frac{\zeta(n)}{n},$$

a series for the Euler-Mascheroni constant. The convergence of this series is exceedingly slow. For actual computation of γ, other, indirect, approaches are far superior (see Exercise 12.3.2).

Solution.

Write $\zeta(n) = 1 + d_n$. Since $k^{-n} \le \int_{k-1}^{k} t^{-n}dt$ for $k \ge 3$,

$$0 < d_n = \sum_{k=2}^{\infty} k^{-n} \le 2^{-n} + \int_2^{\infty} t^{-n}dt = 2^{-n} + \frac{2^{1-n}}{n-1} = 2^{-n}\left(1 + \frac{2}{n-1}\right).$$

Thus, apart from the term $-\gamma x$, the series is the sum of

$$\sum_{n=2}^{\infty}\frac{(-x)^n}{n} \qquad \text{and} \qquad \sum_{n=2}^{\infty}\frac{(-1)^n d_n}{n}x^n.$$

The first has radius one. The second, the correction, converges absolutely for $|x| < 2$, because $d_n/n \le 3\cdot 2^{-n}$ bounds its terms by $3(|x|/2)^n$. The sum converges absolutely for $|x| < 1$. At $x = 1$ the first series is the convergent alternating harmonic series, while at $x = -1$ it is the divergent harmonic tail $\sum_{n\ge2} 1/n$; the correction converges at both endpoints, so the whole series converges at $x = 1$ and diverges at $x = -1$. For $|x| > 1$ the terms $\zeta(n)x^n/n$ do not tend to zero, since $\zeta(n) > 1$ and $|x|^n/n \to \infty$. Hence the real convergence set is exactly

$$\boxed{-1 < x \le 1}.$$

To identify the endpoint sum, let $x \uparrow 1$ in the interior identity. By Abel's theorem the power series, which converges at $x = 1$, tends to its value there, while $\ln\Gamma(1+x) \to \ln\Gamma(2) = \ln 1 = 0$. Hence $0 = -\gamma + \sum_{n\geq 2}(-1)^n\zeta(n)/n$, that is,

$$\gamma = \sum_{n=2}^{\infty}(-1)^n\frac{\zeta(n)}{n}.$$

The leading $1/n$ behavior explains the slow convergence. For complex x, the same decomposition gives conditional convergence on $|x| = 1$ except at $x = -1$.

13.2.2 Show that the series expansion of $\ln\Gamma(x+1)$ (Exercise 13.2.1) may be written as

(a) $$\ln\Gamma(x+1) = \frac{1}{2}\ln\left(\frac{\pi x}{\sin\pi x}\right) - \gamma x - \sum_{n=1}^{\infty}\frac{\zeta(2n+1)}{2n+1}x^{2n+1},$$

(b) $$\ln\Gamma(x+1) = \frac{1}{2}\ln\left(\frac{\pi x}{\sin\pi x}\right) - \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) + (1-\gamma)x - \sum_{n=1}^{\infty}\left[\zeta(2n+1)-1\right]\frac{x^{2n+1}}{2n+1}.$$

Determine the range of convergence of each of these expressions.

Solution.

Use the real logarithm initially on $-1 < x < 1$ (with removable quotients at $x = 0$). Split the original series into even and odd powers:

$$\ln\Gamma(1+x) = -\gamma x + \sum_{n=1}^{\infty}\frac{\zeta(2n)x^{2n}}{2n} - \sum_{n=1}^{\infty}\frac{\zeta(2n+1)x^{2n+1}}{2n+1}.$$

Reflection and recurrence give $\Gamma(1+x)\Gamma(1-x) = x\Gamma(x)\Gamma(1-x) = \pi x/\sin\pi x$. Adding the series at x and $-x$ cancels $-\gamma x$ and the odd terms and doubles the even ones:

$$\ln\frac{\pi x}{\sin\pi x} = \ln\Gamma(1+x) + \ln\Gamma(1-x) = 2\sum_{n=1}^{\infty}\frac{\zeta(2n)x^{2n}}{2n}.$$

(a) Substituting half of this for the even series gives exactly the first requested form. Its remaining odd series has radius one; at $x = 1$ its positive terms behave as $1/(2n+1)$, and at $x = -1$ their negatives do. It therefore converges for $\boxed{-1 < x < 1}$ only. In particular, the finite limit of the complete expression at $x = 1$ requires cancellation of divergent pieces; direct endpoint substitution in the separated terms is invalid.

(b) Split $\zeta(2n+1) = 1 + [\zeta(2n+1) - 1]$ and use

$$\sum_{n=1}^{\infty}\frac{x^{2n+1}}{2n+1} = \frac{1}{2}\ln\frac{1+x}{1-x} - x.$$

This follows by integrating the geometric series $(1-t^2)^{-1} = \sum_{n\geq 0}t^{2n}$ from zero to x, which gives $\sum_{n\geq 0}x^{2n+1}/(2n+1) = \int_0^x dt/(1-t^2) = \frac{1}{2}\ln[(1+x)/(1-x)]$, and removing the linear term. Hence

$$-\sum_{n=1}^{\infty}\frac{\zeta(2n+1)x^{2n+1}}{2n+1} = -\frac{1}{2}\ln\frac{1+x}{1-x} + x - \sum_{n=1}^{\infty}\left[\zeta(2n+1)-1\right]\frac{x^{2n+1}}{2n+1},$$

and inserting this in (a) produces the second displayed form, with $-\gamma x + x = (1-\gamma)x$. By the bounds $2^{-m} < \zeta(m) - 1 \le 2^{-m}[1 + 2/(m-1)]$ (the lower one is the first term of d_m, the upper one is from Exercise 13.2.1), $2^{2n+1}[\zeta(2n+1) - 1] \to 1$, so the remaining series has radius two. At $x = \pm 2$ its terms behave as $\pm 1/(2n+1)$, a signed harmonic tail, and it diverges. Thus this series alone converges on $\boxed{-2 < x < 2}$, but the expression as written with separate real logarithms has the domain $-1 < x < 1$. If those logarithms are first combined, their contribution is

$$\frac{1}{2} \ln \frac{\pi x(1-x)}{(1+x)\sin \pi x}.$$

This has a removable limit $\frac{1}{2}\ln(1/2)$ at $x = 1$, because $\sin \pi x = \sin \pi(1-x) \approx \pi(1-x)$ there, so the quotient tends to $\pi/(2\pi) = 1/2$. It is real on $-1 < x < 2$, since for $1 < x < 2$ both $1 - x$ and $\sin \pi x$ are negative and the quotient stays positive. Both sides of the combined identity are analytic on $-1 < x < 2$ and agree on $-1 < x < 1$, so by the identity theorem (uniqueness of analytic continuation) the combined form extends the identity across $x = 1$ to that larger interval. This continuation must be distinguished from the domain of the separate logarithms.

13.2.3 Verify that for n, a positive integer, the following two forms of the digamma function are equal to each other:

$$\psi(n+1) = \sum_{j=1}^{n} \frac{1}{j} - \gamma \quad \text{and} \quad \psi(n+1) = \sum_{j=1}^{\infty} \frac{n}{j(n+j)} - \gamma.$$

Solution.

For a fixed positive integer n, partial fractions give

$$\frac{n}{j(n+j)} = \frac{1}{j} - \frac{1}{j+n}.$$

Sum only to a finite N before canceling. With $H_N = \sum_{j=1}^{N} 1/j$ and $\sum_{j=1}^{N} 1/(j+n) = \sum_{i=n+1}^{N+n} 1/i = H_{N+n} - H_n$,

$$\begin{aligned} \sum_{j=1}^{N} \frac{n}{j(n+j)} &= H_N - (H_{N+n} - H_n) \\ &= H_n - \sum_{j=N+1}^{N+n} \frac{1}{j}. \end{aligned}$$

The final sum lies between zero and $n/(N+1)$, hence tends to zero as $N \to \infty$. Therefore

$$\boxed{\sum_{j=1}^{\infty} \frac{n}{j(n+j)} = H_n}, \qquad \psi(n+1) = H_n - \gamma,$$

and the two representations agree. This argument avoids subtracting two divergent harmonic series separately.

13.2.4 Show that $\psi(z+1)$ has the series expansion

$$\psi(z+1) = -\gamma + \sum_{n=2}^{\infty} (-1)^n \zeta(n) z^{n-1}.$$

The expansion holds for $|z| < 1$.

Solution.

On every closed disk $|z| \le r < 1$, the Maclaurin series for $\ln\Gamma(1+z)$, Eq. (13.44), may be differentiated term by term. Indeed, the derivative terms $(d/dz)[(-1)^n\zeta(n)z^n/n] = (-1)^n\zeta(n)z^{n-1}$ have absolute values at most $\zeta(2)r^{n-1}$ (since $\zeta(n) \le \zeta(2)$), whose sum converges; by the Weierstrass M-test the differentiated series converges uniformly on the disk, which justifies the term-by-term differentiation. Since $\psi(1+z) = d\ln\Gamma(1+z)/dz$ and $(d/dz)(-\gamma z) = -\gamma$, differentiation gives

$$\boxed{\psi(1+z) = -\gamma + \sum_{n=2}^{\infty} (-1)^n \zeta(n) z^{n-1}}, \qquad |z| < 1.$$

The domain $|z| < 1$ is needed: on $|z| = 1$ the summands have magnitudes $\zeta(n) \to 1$, so none of that boundary belongs to the convergence set. The closest pole of the expanded function is at $z = -1$, consistent with radius one.

13.2.5 For a power-series expansion of $\ln\Gamma(z+1)$, AMS-55 (see Additional Readings for the reference) lists

$$\ln\Gamma(z+1) = -\ln(1+z) + z(1-\gamma) + \sum_{n=2}^{\infty} (-1)^n \left[\zeta(n) - 1\right] \frac{z^n}{n}.$$

(a) Show that this agrees with Eq. (13.44) for $|z| < 1$.

(b) What is the range of convergence of this new expression?

Solution.

(a) For $|z| < 1$,

$$\ln(1+z) = z - \sum_{n=2}^{\infty} (-1)^n \frac{z^n}{n}.$$

Add this to the series for $\ln\Gamma(1+z)$. The linear coefficients combine to $1-\gamma$, and for each $n \ge 2$ the coefficients $(-1)^n\zeta(n)/n$ and $-(-1)^n/n$ combine to $(-1)^n[\zeta(n)-1]/n$, giving

$$\ln\Gamma(1+z) + \ln(1+z) = (1-\gamma)z + \sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)-1}{n} z^n.$$

Moving $\ln(1+z)$ to the right proves the claimed agreement.

(b) Set $e_n = \zeta(n) - 1 - 2^{-n} = \sum_{k\ge 3} k^{-n}$. The integral comparison of Exercise 13.2.1, now starting at $k = 3$, gives the estimate

$$0 < e_n \le 3^{-n} + \int_3^{\infty} t^{-n} dt = 3^{-n}\left(1 + \frac{3}{n-1}\right).$$

Since $(-1)^n[\zeta(n)-1]z^n/n = (-z/2)^n/n + (-1)^n e_n z^n/n$, the new series is $\sum_{n=2}^{\infty}(-z/2)^n/n$ plus a correction absolutely convergent for $|z|<3$ (its terms are at most $4(|z|/3)^n$). Thus its radius is two; on $|z|=2$ it converges except at $z=-2$, by the Dirichlet test for the leading series $\sum w^n/n$, $|w|=1$, $w\neq 1$. Its real convergence set is $\boxed{-2<z\leq 2}$.

Convergence of the series is distinct from the domain of the complete logarithmic identity: with ordinary real logarithms that identity requires $z>-1$, and hence holds for $-1<z\leq 2$. It is undefined at $z=-1$. More precisely, since $\Gamma(2+z)=(1+z)\Gamma(1+z)$, part (a) shows that the analytic series equals $\ln\Gamma(2+z)-(1-\gamma)z$ for $|z|<1$. The function $\Gamma(2+z)$ has no zeros and its nearest pole is at $z=-2$, so $\ln\Gamma(2+z)$, continued from zero, is analytic on $|z|<2$, as is the series (radius two). Agreeing on $|z|<1$, the two agree on the whole disk $|z|<2$ by the identity theorem (uniqueness of analytic continuation). Subtracting $\ln(1+z)$ gives the requested identity on a slit disk, for example $|z|<2$ with $z\notin(-2,-1]$, using compatible analytic branches. The endpoint $z=2$ follows by Abel's theorem and continuity.

13.2.6 Show that

$$\frac{1}{2}\ln\left(\frac{\pi z}{\sin\pi z}\right)=\sum_{n=1}^{\infty}\frac{\zeta(2n)}{2n}z^{2n},\qquad |z|<1.$$

Hint. Use Eqs. (13.23) and (13.44).

Solution.

Write the expansion in Eq. (13.44) at z and $-z$ and add:

$$\begin{aligned}\ln\Gamma(1+z)+\ln\Gamma(1-z)&=\sum_{k=2}^{\infty}\frac{(-1)^k\zeta(k)}{k}[z^k+(-z)^k]\\&=2\sum_{n=1}^{\infty}\frac{\zeta(2n)}{2n}z^{2n}.\end{aligned}$$

The linear terms and all odd powers cancel. Reflection, $\Gamma(z)\Gamma(1-z)=\pi/\sin\pi z$, followed by $z\Gamma(z)=\Gamma(1+z)$, gives

$$\Gamma(1+z)\Gamma(1-z)=\frac{\pi z}{\sin\pi z}.$$

In $|z|<1$ the quotient has no zero or pole, with value one at zero. Choose its analytic logarithm to vanish there. The sum of the two gamma logarithms has that same value and exponential, so there is no additional $2\pi i$ constant. Dividing by two gives

$$\boxed{\frac{1}{2}\ln\frac{\pi z}{\sin\pi z}=\sum_{n=1}^{\infty}\frac{\zeta(2n)}{2n}z^{2n}},\qquad |z|<1.$$

The printed hint cites Eq. (13.35); the published errata (Instructor's Manual, 6 April 2012) replace it by Eq. (13.44), the Maclaurin series of $\ln\Gamma(1+z)$ used above.

13.2.7 Write out a Weierstrass infinite-product definition of $\ln\Gamma(z+1)$. Without differentiating, show that this leads directly to the Maclaurin expansion of $\ln\Gamma(z+1)$, Eq. (13.44).

Solution.

Weierstrass' product, after using $\Gamma(1+z)=z\Gamma(z)$, is

$$\frac{1}{\Gamma(1+z)}=e^{\gamma z}\prod_{k=1}^{\infty}\left(1+\frac{z}{k}\right)e^{-z/k}.$$

Taking the analytic logarithm that is zero at $z = 0$ gives

$$\ln\Gamma(1+z) = -\gamma z + \sum_{k=1}^{\infty}\left[\frac{z}{k} - \ln\left(1+\frac{z}{k}\right)\right].$$

For $|z| < 1$, expand each logarithm by $\ln(1+w) = w - w^2/2 + w^3/3 - \cdots$ with $w = z/k$, $|w| < 1$. Its linear term cancels:

$$\frac{z}{k} - \ln\left(1+\frac{z}{k}\right) = \sum_{n=2}^{\infty}\frac{(-1)^n z^n}{nk^n}.$$

The double series is absolutely convergent, since summing over k first gives $\zeta(n)|z|^n/n \le \zeta(2)|z|^n$ and

$$\sum_{n=2}^{\infty}\sum_{k=1}^{\infty}\frac{|z|^n}{nk^n} \le \zeta(2)\sum_{n=2}^{\infty}|z|^n < \infty.$$

Interchange the sums and identify $\sum_{k=1}^{\infty} k^{-n}$:

$$\boxed{\ln\Gamma(1+z) = -\gamma z + \sum_{n=2}^{\infty}(-1)^n\frac{\zeta(n)}{n}z^n}.$$

No differentiation of the product was used.

13.2.8 Derive the difference relation for the polygamma function,

$$\psi^{(m)}(z+2) = \psi^{(m)}(z+1) + (-1)^m\frac{m!}{(z+1)^{m+1}}, \quad m = 0, 1, 2, \ldots.$$

Solution.

Differentiate $\Gamma(z+2) = (z+1)\Gamma(z+1)$ logarithmically at regular arguments, i.e., differentiate $\ln\Gamma(z+2) = \ln(z+1) + \ln\Gamma(z+1)$. The result is

$$\psi(z+2) = \frac{1}{z+1} + \psi(z+1).$$

This already proves the case $m = 0$. For $m \ge 1$, differentiate this relation m times, using $\psi^{(m)} = d^m\psi/dz^m$; the mth derivative of the first term is

$$\frac{d^m}{dz^m}(z+1)^{-1} = (-1)(-2)\cdots(-m)(z+1)^{-m-1} = \frac{(-1)^m m!}{(z+1)^{m+1}}.$$

Hence

$$\boxed{\psi^{(m)}(z+2) = \psi^{(m)}(z+1) + \frac{(-1)^m m!}{(z+1)^{m+1}}}, \qquad m = 0, 1, 2, \ldots.$$

These are identities of meromorphic functions; a displayed value at a pole is not a finite number. For integer positive arguments, the formulas recursively reduce to the values at one.

13.2.9 The Pochhammer symbol $(a)_n$ is defined (for integral n) as

$$(a)_n = a(a+1)\cdots(a+n-1), \quad (a)_0 = 1.$$

(a) Express $(a)_n$ in terms of factorials.

(b) Find $(d/da)(a)_n$ in terms of $(a)_n$ and digamma functions.

$$\textit{ANS.}\ \frac{d}{da}(a)_n = (a)_n\,[\psi(a+n) - \psi(a)].$$

(c) Show that

$$(a)_{n+k} = (a+n)_k \cdot (a)_n.$$

The finite-product indices n, k are nonnegative integers. Products involving singular gamma or digamma factors are evaluated by their polynomial continuation.

Solution.

The finite-product definition here uses nonnegative integers n and, in part (c), k. **(a)** Repeated gamma recurrence gives $\Gamma(a+n) = (a+n-1)(a+n-2)\cdots a\,\Gamma(a) = (a)_n\Gamma(a)$, so

$$\boxed{(a)_n = \frac{\Gamma(a+n)}{\Gamma(a)} = \frac{(a+n-1)!}{(a-1)!}}.$$

The factorials have their generalized meaning. The finite product is a polynomial in a and defines any removable value of this quotient at gamma poles.

(b) At points where all factors are nonzero, logarithmic differentiation of the finite product gives

$$\frac{d}{da}(a)_n = (a)_n \sum_{j=0}^{n-1} \frac{1}{a+j}.$$

Iterating $\psi(a+1) - \psi(a) = 1/a$ (Exercise 13.2.8 with $m = 0$) identifies the finite sum by telescoping, $\psi(a+n) - \psi(a) = \sum_{j=0}^{n-1}[\psi(a+j+1) - \psi(a+j)] = \sum_{j=0}^{n-1} 1/(a+j)$:

$$\boxed{\frac{d}{da}(a)_n = (a)_n[\psi(a+n) - \psi(a)]}.$$

The same result follows from (a): $(d/da)[\Gamma(a+n)/\Gamma(a)] = [\Gamma(a+n)/\Gamma(a)][\psi(a+n) - \psi(a)]$. At exceptional arguments use the equivalent polynomial formula

$$\frac{d}{da}(a)_n = \sum_{j=0}^{n-1} \prod_{\substack{0 \le \ell < n \\ \ell \ne j}} (a+\ell).$$

For example, at $a = -j$ with $0 \le j < n$ only the term that omits the vanishing factor $a+j$ survives, and its value is

$$\prod_{\ell=0}^{j-1}(\ell - j) \prod_{\ell=j+1}^{n-1} (\ell - j) = (-1)^j j!\,(n-1-j)!,$$

even though the separated digamma factors have poles. For $n = 0$ the empty product is one and its derivative is zero.

(c) Split the product after its first n factors:

$$(a)_{n+k} = \left[\prod_{j=0}^{n-1}(a+j)\right]\left[\prod_{j=n}^{n+k-1}(a+j)\right]$$
$$= \boxed{(a)_n(a+n)_k}.$$

This polynomial proof also includes zeros and empty products.

13.2.10 Verify the following special values of the ψ form of the digamma and polygamma functions:

$$\psi(1) = -\gamma, \quad \psi^{(1)}(1) = \zeta(2), \quad \psi^{(2)}(1) = -2\zeta(3).$$

Solution.

Differentiating the Weierstrass form $\ln\Gamma(1+z) = -\gamma z + \sum_k [z/k - \ln(1+z/k)]$ of Exercise 13.2.7 term by term (each term has derivative $1/k - 1/(k+z) = z/[k(k+z)]$, and these converge locally uniformly away from the poles) gives the convergent digamma representation

$$\psi(1+z) = -\gamma + \sum_{k=1}^{\infty}\left(\frac{1}{k} - \frac{1}{k+z}\right).$$

Putting $z = 0$ makes every term of the sum vanish, so $\boxed{\psi(1) = -\gamma}$. Differentiation on compact sets away from the poles, with $(d/dz)(k+z)^{-1} = -(k+z)^{-2}$ and $(d/dz)(k+z)^{-2} = -2(k+z)^{-3}$, gives

$$\psi^{(1)}(1+z) = \sum_{k=1}^{\infty}\frac{1}{(k+z)^2}, \qquad \psi^{(2)}(1+z) = -2\sum_{k=1}^{\infty}\frac{1}{(k+z)^3}.$$

Both differentiated sums converge absolutely and locally uniformly. Therefore at $z = 0$,

$$\boxed{\psi^{(1)}(1) = \sum_{k=1}^{\infty}k^{-2} = \zeta(2)}, \qquad \boxed{\psi^{(2)}(1) = -2\sum_{k=1}^{\infty}k^{-3} = -2\zeta(3)}.$$

13.2.11 Verify:

(a) $\displaystyle\int_0^{\infty} e^{-r}\ln r\,dr = -\gamma.$

(b) $\displaystyle\int_0^{\infty} re^{-r}\ln r\,dr = 1-\gamma.$

(c) $\displaystyle\int_0^{\infty} r^n e^{-r}\ln r\,dr = (n-1)! + n\int_0^{\infty} r^{n-1}e^{-r}\ln r\,dr, \quad n = 1,2,3,\ldots.$

Hint. These may be verified by integration by parts, or by differentiating the Euler integral formula for $\Gamma(n+1)$ with respect to n.

Solution.

For s in a compact subset of $\Re s > 0$, on which $\Re s \geq \varepsilon$ for some $\varepsilon > 0$, differentiation of Euler's integral is justified by the integrable bounds $r^{\varepsilon-1}|\ln r|$ near zero and a polynomial times $e^{-r}|\ln r|$ at infinity. Hence

$$\Gamma'(s) = \int_0^{\infty} r^{s-1}e^{-r}\ln r\,dr = \Gamma(s)\psi(s).$$

(a) At $s = 1$, $\Gamma(1) = 1$ and $\psi(1) = -\gamma$, so the integral is $\boxed{-\gamma}$.

(c) Write $I_n = \int_0^\infty r^n e^{-r} \ln r \, dr$. For integers $n \geq 1$, integrate by parts with $u = r^n \ln r$, $dv = e^{-r} dr$, so that $du = (nr^{n-1} \ln r + r^{n-1}) dr$ and $v = -e^{-r}$. Since $r^n \ln r \to 0$ at zero and the exponential removes the upper boundary term, and since $\int_0^\infty r^{n-1} e^{-r} dr = \Gamma(n) = (n-1)!$,

$$\begin{aligned} I_n &= \left[-r^n e^{-r} \ln r \right]_0^\infty + \int_0^\infty e^{-r} (nr^{n-1} \ln r + r^{n-1}) dr \\ &= \boxed{nI_{n-1} + (n-1)!}. \end{aligned}$$

(b) In particular, $I_1 = I_0 + 0! = \boxed{1 - \gamma}$. The general closed form is $I_n = n!(H_n - \gamma)$, with the harmonic numbers $H_n = \sum_{j=1}^n 1/j$ of Exercise 13.2.3 and $H_0 = 0$, as is checked by this recurrence and $I_0 = -\gamma$: if $I_{n-1} = (n-1)!(H_{n-1} - \gamma)$, then $I_n = n!(H_{n-1} - \gamma) + n!/n = n!(H_n - \gamma)$. It agrees with $I_n = \Gamma'(n+1) = \Gamma(n+1)\psi(n+1)$ and Exercise 13.2.3. Integration by parts at $n = 0$ would have a divergent boundary term, which is why part (a) was evaluated before using this recurrence.

13.2.12 Dirac relativistic wave functions for hydrogen involve factors such as $\Gamma[2(1 - \alpha^2 Z^2)^{1/2} + 1]$ where α, the fine structure constant, is approximated here by $1/137$ and Z is the atomic number. Expand $\Gamma[2(1 - \alpha^2 Z^2)^{1/2} + 1]$ in a series of powers of $\alpha^2 Z^2$.

Solution.

Let $q = \alpha^2 Z^2$ and expand about $q = 0$, using the square-root branch that equals one there. The binomial series $\sqrt{1-q} = 1 - \frac{q}{2} - \frac{q^2}{8} - \frac{q^3}{16} + O(q^4)$ gives

$$2\sqrt{1-q} + 1 = 3 + \delta, \qquad \delta = -q - \frac{q^2}{4} - \frac{q^3}{8} + O(q^4).$$

Set $p = \psi(3)$, $r = \psi^{(1)}(3)$ and $s = \psi^{(2)}(3)$. The recurrence of Exercise 13.2.8, applied with $z = 0$ and $z = 1$, gives $\psi^{(m)}(3) = \psi^{(m)}(1) + (-1)^m m! \, (1 + 2^{-m-1})$; with the values at one from Exercise 13.2.10 this yields

$$p = \frac{3}{2} - \gamma, \qquad r = \zeta(2) - 1 - \frac{1}{4} = \zeta(2) - \frac{5}{4}, \qquad s = -2\zeta(3) + 2 + \frac{1}{4} = \frac{9}{4} - 2\zeta(3).$$

Differentiating $\Gamma' = \Gamma\psi$ and using $\Gamma' = \Gamma\psi$ again at each step gives $\Gamma'' = \Gamma'\psi + \Gamma\psi' = \Gamma(\psi^2 + \psi')$ and $\Gamma^{(3)} = \Gamma\psi(\psi^2 + \psi') + \Gamma(2\psi\psi' + \psi'')$, that is,

$$\Gamma'' = \Gamma(\psi^2 + \psi'), \qquad \Gamma^{(3)} = \Gamma(\psi^3 + 3\psi\psi' + \psi'').$$

Since $\Gamma(3) = 2$, Taylor expansion in δ is

$$\Gamma(3 + \delta) = 2\left[1 + p\delta + \frac{p^2 + r}{2}\delta^2 + \frac{p^3 + 3pr + s}{6}\delta^3 + O(\delta^4) \right].$$

Now $\delta^2 = q^2 + q^3/2 + O(q^4)$ and $\delta^3 = -q^3 + O(q^4)$. Inside the brackets the coefficient of q is $-p$; that of q^2 is $-p/4 + (p^2 + r)/2$; and that of q^3 is $-p/8 + (p^2 + r)/4 - (p^3 + 3pr + s)/6$. Multiplying by two,

$$\begin{aligned} \Gamma(2\sqrt{1-q} + 1) = 2 - 2pq + \left(p^2 + r - \frac{p}{2} \right) q^2 \\ + \left[-\frac{p}{4} + \frac{p^2 + r}{2} - \frac{p^3 + 3pr + s}{3} \right] q^3 + O(q^4). \end{aligned}$$

Here $-2p = 2\gamma - 3$ and

$$p^2 + r - \frac{p}{2} = \left(\frac{9}{4} - 3\gamma + \gamma^2\right) + \zeta(2) - \frac{5}{4} - \frac{3}{4} + \frac{\gamma}{2} = \gamma^2 - \frac{5}{2}\gamma + \zeta(2) + \frac{1}{4}.$$

In particular, the first three terms in the original variables are

$$\boxed{2 + (2\gamma - 3)\alpha^2 Z^2 + \left[\gamma^2 - \frac{5}{2}\gamma + \zeta(2) + \frac{1}{4}\right]\alpha^4 Z^4 + O(\alpha^6 Z^6)}.$$

For the q^3 coefficient, multiplying out the products and adding $s = \frac{9}{4} - 2\zeta(3)$ gives

$$p^3 = \frac{27}{8} - \frac{27}{4}\gamma + \frac{9}{2}\gamma^2 - \gamma^3, \qquad 3pr = \frac{9}{2}\zeta(2) - \frac{45}{8} + \frac{15}{4}\gamma - 3\gamma\zeta(2),$$

$$p^3 + 3pr + s = -\gamma^3 + \frac{9}{2}\gamma^2 - 3\gamma - 3\gamma\zeta(2) + \frac{9}{2}\zeta(2) - 2\zeta(3),$$

$$-\frac{p}{4} + \frac{p^2 + r}{2} = \frac{\gamma^2}{2} - \frac{5}{4}\gamma + \frac{\zeta(2)}{2} + \frac{1}{8},$$

and subtracting one third of the second line from the third gives the q^3 coefficient $\gamma^3/3 - \gamma^2 - \gamma/4 + (\gamma - 1)\zeta(2) + \frac{2}{3}\zeta(3) + \frac{1}{8}$. Numerically,

$$\Gamma(2\sqrt{1-q} + 1) = 2 - 1.845569\,q + 0.785073\,q^2 - 0.182458\,q^3 + O(q^4),$$

which can be checked by evaluating the gamma function directly: at $q = 0.1$, $\Gamma(2\sqrt{0.9} + 1) = \Gamma(2.8973666) = 1.8231159$, while the cubic gives 1.8231114. The difference, about 4.5×10^{-6}, is the size of the omitted q^4 term: its coefficient, obtained in the same way from $\Gamma^{(4)}$ and $\psi^{(3)}(3)$, is 0.04416, so the term is 4.4×10^{-6} at $q = 0.1$. Higher terms follow from the same two Taylor expansions. The expansion is local at $q = 0$ and converges for $|q| < 1$; there the chosen square root has positive real part, so its gamma argument encounters no gamma pole. Physically, a short truncation requires $\alpha^2 Z^2$ small. The value of α does not enter the coefficients; the statement's $1/137$ approximates the measured $\alpha \simeq 1/137.036$.

13.2.13 The quantum mechanical description of a particle in a Coulomb field requires a knowledge of the argument of $\Gamma(z)$ when z is complex. Determine the argument of $\Gamma(1 + ib)$ for small, real b.

Solution.

Choose the argument continuously from $b = 0$, where $\Gamma(1) = 1$ and the argument is zero; it is then the imaginary part of the logarithm of Eq. (13.44), which vanishes at $z = 0$. For $|b| < 1$, substitute $z = ib$ in

$$\ln\Gamma(1 + z) = -\gamma z + \sum_{n=2}^{\infty}(-1)^n \frac{\zeta(n)}{n} z^n.$$

The linear term $-\gamma ib$ contributes $-\gamma b$ to the imaginary part. Even powers $(ib)^{2m} = (-1)^m b^{2m}$ are real. For the odd powers $n = 2m + 1$, $(-1)^{2m+1} i^{2m+1} = -i\,(i^2)^m = (-1)^{m+1} i$. Therefore

$$\arg\Gamma(1 + ib) = \Im \ln\Gamma(1 + ib) = -\gamma b + \sum_{m=1}^{\infty}(-1)^{m+1} \frac{\zeta(2m+1)}{2m+1} b^{2m+1},$$

that is,

$$\boxed{\arg\Gamma(1 + ib) = -\gamma b + \frac{\zeta(3)}{3}b^3 - \frac{\zeta(5)}{5}b^5 + \frac{\zeta(7)}{7}b^7 - \cdots}.$$

The leading approximation is $-\gamma b$, with an error of order b^3. This is the continuous local phase (and agrees with the principal phase for sufficiently small b); arbitrary arguments differ by 2π multiples. Its oddness in b also follows from $\Gamma(1 - ib) = \overline{\Gamma(1 + ib)}$.

13.2.14 Using digamma and polygamma functions, sum the series

$$\text{(a)}\ \sum_{n=1}^{\infty} \frac{1}{n(n+1)}, \quad \text{(b)}\ \sum_{n=2}^{\infty} \frac{1}{n^2-1}.$$

Note. You can use Exercise 13.2.8 to calculate the needed digamma functions.

Solution.

Use the digamma identity

$$\psi(1+z)+\gamma = \sum_{j=1}^{\infty} \frac{z}{j(j+z)},$$

which is the representation of Exercise 13.2.10 with $1/j - 1/(j+z) = z/[j(j+z)]$. **(a)** At $z=1$ the requested sum is $\psi(2)+\gamma$. The recurrence of Exercise 13.2.8 ($m=0$, $z=0$), $\psi(2) = \psi(1)+1 = -\gamma+1$, gives

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1}.$$

(b) Put $j = n-1$; then $n^2-1 = (n-1)(n+1) = j(j+2)$. At $z=2$ the identity gives $\sum_j 2/[j(j+2)] = \psi(3)+\gamma$, and the recurrence ($m=0$, $z=1$) gives $\psi(3) = \psi(2) + \frac{1}{2} = \frac{3}{2} - \gamma$. Hence

$$\begin{aligned}\sum_{n=2}^{\infty} \frac{1}{n^2-1} &= \sum_{j=1}^{\infty} \frac{1}{j(j+2)} = \frac{1}{2}[\psi(3)+\gamma] \\ &= \frac{1}{2}\left(1+\frac{1}{2}\right) = \boxed{\frac{3}{4}}.\end{aligned}$$

As a separate check, the partial fractions $1/[n(n+1)] = 1/n - 1/(n+1)$ and $1/(n^2-1) = \frac{1}{2}[1/(n-1) - 1/(n+1)]$ make the finite partial sums telescope to $1 - 1/(N+1)$ in (a) and $\frac{1}{2}[1 + 1/2 - 1/N - 1/(N+1)]$ in (b), respectively; their limits give the same answers.

13.2.15 Show that

$$\sum_{n=1}^{\infty} \frac{1}{(n+a)(n+b)} = \frac{1}{(b-a)}\left[\psi(1+b) - \psi(1+a)\right],$$

where $a \neq b$, and neither a nor b is a negative integer. It is of some interest to compare this summation with the corresponding integral,

$$\int_1^{\infty} \frac{dx}{(x+a)(x+b)} = \frac{1}{b-a}\left[\ln(1+b) - \ln(1+a)\right].$$

The relation between $\psi(x)$ and $\ln x$ is made explicit in the analysis leading to Stirling's formula.

For the displayed ordinary real improper integral and real logarithms, require real $a, b > -1$ as well as $a \neq b$. The series has the wider domain stated above.

Solution.

For the stated admissible a, b, every denominator is nonzero and the summands are $O(n^{-2})$, so the series converges absolutely. Write the digamma representation of Exercise 13.2.10, $\psi(1+z) =$

$-\gamma+\sum_{n=1}^{\infty}[1/n-1/(n+z)]$, at $z=b$ and at $z=a$, and subtract the two convergent series term by term; the terms $-\gamma$ and $1/n$ cancel. Since $1/(n+a)-1/(n+b)=[(n+b)-(n+a)]/[(n+a)(n+b)]$,

$$\begin{aligned}\psi(1+b)-\psi(1+a)&=\sum_{n=1}^{\infty}\left(\frac{1}{n+a}-\frac{1}{n+b}\right)\\&=(b-a)\sum_{n=1}^{\infty}\frac{1}{(n+a)(n+b)}.\end{aligned}$$

Division by $b-a$ proves

$$\boxed{\sum_{n=1}^{\infty}\frac{1}{(n+a)(n+b)}=\frac{\psi(1+b)-\psi(1+a)}{b-a}}.$$

For $b=a$ the summand is $(n+a)^{-2}$, and the right side has the removable value $\lim_{b\to a}[\psi(1+b)-\psi(1+a)]/(b-a)=\psi^{(1)}(1+a)$; this agrees with $\sum_{n=1}^{\infty}(n+a)^{-2}=\psi^{(1)}(1+a)$ from Exercise 13.2.10.

The accompanying real improper integral requires stronger conditions, for example real $a,b>-1$. Merely excluding negative integers does not prevent an interior pole: $a=-3/2$, $b=0$ gives one at $x=3/2$. Under $a,b>-1$ both factors $x+a$ and $x+b$ are positive on $[1,\infty)$, and the partial fractions

$$\frac{1}{(x+a)(x+b)}=\frac{1}{b-a}\left(\frac{1}{x+a}-\frac{1}{x+b}\right)$$

give

$$\begin{aligned}\int_1^R\frac{dx}{(x+a)(x+b)}&=\frac{1}{b-a}\big[\ln(x+a)-\ln(x+b)\big]_1^R\\&=\frac{1}{b-a}\left[\ln\frac{R+a}{R+b}+\ln(1+b)-\ln(1+a)\right].\end{aligned}$$

As $R\to\infty$, $(R+a)/(R+b)\to 1$, so the first logarithm tends to zero, proving the quoted integral. When $b\to a>-1$, its value tends to $1/(1+a)$, the derivative of $\ln(1+a)$, whereas the series tends to the trigamma value above. The sum always exceeds the integral: the summand $f(x)=1/[(x+a)(x+b)]$ decreases on $[1,\infty)$, so $f(n)>\int_n^{n+1}f\,dx$ for every $n\ge 1$. For example, $a=0$, $b=1$ gives the sum $\psi(2)-\psi(1)=1$ of Exercise 13.2.14(a) but the integral $\ln 2=0.693$.

13.3 The Beta Function

13.3.1 Verify the following beta function identities:

(a) $B(a,b)=B(a+1,b)+B(a,b+1)$,

(b) $B(a,b)=\dfrac{a+b}{b}B(a,b+1)$,

(c) $B(a,b)=\dfrac{b-1}{a}B(a+1,b-1)$,

(d) $B(a,b)B(a+b,c)=B(b,c)B(a,b+c)$.

Solution.

Start where the beta integrals converge; in part (c) take $\Re b > 1$ as well as $\Re a > 0$. The results then extend as meromorphic identities wherever the displayed values are defined. Parts (b)–(d) use the gamma form $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$; parts (b) and (c) also use the recurrence $\Gamma(s+1) = s\Gamma(s)$.

(a) Insert $1 = t + (1-t)$ into the defining integral and split it:

$$\begin{aligned} B(a,b) &= \int_0^1 t^{a-1}(1-t)^{b-1}[t + (1-t)]\,dt \\ &= \int_0^1 t^{a}(1-t)^{b-1}\,dt + \int_0^1 t^{a-1}(1-t)^{b}\,dt \\ &= \boxed{B(a+1,b) + B(a,b+1)}. \end{aligned}$$

(b) The recurrence, applied to $\Gamma(b+1)$ and to $\Gamma(a+b+1)$, gives

$$B(a,b+1) = \frac{\Gamma(a)b\Gamma(b)}{(a+b)\Gamma(a+b)} = \frac{b}{a+b}B(a,b),$$

so $\boxed{B(a,b) = \dfrac{a+b}{b}B(a,b+1)}$.

(c) Use $\Gamma(a+1) = a\Gamma(a)$ and $\Gamma(b-1) = \Gamma(b)/(b-1)$:

$$B(a+1,b-1) = \frac{a\Gamma(a)\Gamma(b)}{(b-1)\Gamma(a+b)} = \frac{a}{b-1}B(a,b).$$

Multiplying by $(b-1)/a$ gives $\boxed{B(a,b) = \dfrac{b-1}{a}B(a+1,b-1)}$. Where a factor on the right has a pole, the identity holds as a limit, not as a product of undefined values. For example, at $b = 1$ the factor $B(a+1,b-1)$ has a pole while $(b-1)/a$ vanishes; since $(b-1)\Gamma(b-1) = \Gamma(b)$,

$$\frac{b-1}{a}B(a+1,b-1) = \frac{\Gamma(a+1)\Gamma(b)}{a\,\Gamma(a+b)} \longrightarrow \frac{\Gamma(a+1)}{a\,\Gamma(a+1)} = \frac{1}{a} = B(a,1) \qquad (b \to 1).$$

(d) Multiplication cancels the intermediate gamma factor:

$$\begin{aligned} B(a,b)B(a+b,c) &= \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}\frac{\Gamma(a+b)\Gamma(c)}{\Gamma(a+b+c)} \\ &= \frac{\Gamma(a)\Gamma(b)\Gamma(c)}{\Gamma(a+b+c)} \\ &= \frac{\Gamma(b)\Gamma(c)}{\Gamma(b+c)}\frac{\Gamma(a)\Gamma(b+c)}{\Gamma(a+b+c)} \\ &= \boxed{B(b,c)B(a,b+c)}. \end{aligned}$$

13.3.2

(a) Show that

$$\int_{-1}^{1}(1-x^2)^{1/2}x^{2n}\,dx=\begin{cases}\pi/2, & n=0,\\ \pi\dfrac{(2n-1)!!}{(2n+2)!!}, & n=1,2,3,\ldots.\end{cases}$$

(b) Show that

$$\int_{-1}^{1}(1-x^2)^{-1/2}x^{2n}\,dx=\begin{cases}\pi, & n=0,\\ \pi\dfrac{(2n-1)!!}{(2n)!!}, & n=1,2,3,\ldots.\end{cases}$$

Solution.

Both integrands are even. In twice the integral from zero to one, put $u=x^2$, so $x^{2n}=u^n$, $1-x^2=1-u$ and $2\,dx=u^{-1/2}du$; the limits remain zero and one. Repeated use of $\Gamma(s+1)=s\Gamma(s)$ gives the half-integer values

$$\Gamma\left(n+\frac{1}{2}\right)=\left(n-\frac{1}{2}\right)\left(n-\frac{3}{2}\right)\cdots\frac{1}{2}\,\Gamma\left(\frac{1}{2}\right)=\frac{(2n-1)!!\sqrt{\pi}}{2^n},$$

because each of the n factors is an odd integer divided by two and $\Gamma(1/2)=\sqrt{\pi}$. **(a)** This gives

$$\begin{aligned}\int_{-1}^{1}(1-x^2)^{1/2}x^{2n}dx&=2\int_{0}^{1}(1-x^2)^{1/2}x^{2n}dx=\int_{0}^{1}u^{n-1/2}(1-u)^{1/2}du\\&=B\left(n+\frac{1}{2},\frac{3}{2}\right)=\frac{\Gamma(n+1/2)\Gamma(3/2)}{\Gamma(n+2)}\\&=\frac{[(2n-1)!!\sqrt{\pi}/2^n](\sqrt{\pi}/2)}{(n+1)!}=\frac{\pi(2n-1)!!}{2^{n+1}(n+1)!}\\&=\boxed{\pi\frac{(2n-1)!!}{(2n+2)!!}}.\end{aligned}$$

Here $\Gamma(3/2)=\frac{1}{2}\Gamma(\frac{1}{2})=\sqrt{\pi}/2$ and $(2n+2)!!=2^{n+1}(n+1)!$. With $(-1)!!=1$ this includes $n=0$, giving $\pi/2$.

(b) The same substitution changes the second integral to

$$\begin{aligned}\int_{0}^{1}u^{n-1/2}(1-u)^{-1/2}du&=B\left(n+\frac{1}{2},\frac{1}{2}\right)=\frac{\Gamma(n+1/2)\Gamma(1/2)}{\Gamma(n+1)}\\&=\frac{(2n-1)!!\pi}{2^n n!}=\boxed{\pi\frac{(2n-1)!!}{(2n)!!}},\end{aligned}$$

since $(2n)!!=2^n n!$. This also includes $n=0$, because $0!!=1$. The endpoint powers $\pm1/2$ are greater than -1, so both improper integrals exist.

13.3.3 Show that

$$\int_{-1}^{1}(1-x^2)^n\,dx=\begin{cases}2^{2n+1}\dfrac{n!\,n!}{(2n+1)!}, & n>-1,\\[2mm] 2\dfrac{(2n)!!}{(2n+1)!!}, & n=0,1,2,\ldots.\end{cases}$$

Solution.

Evenness followed by $u=x^2$ yields

$$\int_{-1}^{1}(1-x^2)^n dx=2\int_{0}^{1}(1-x^2)^n dx=\int_{0}^{1}u^{-1/2}(1-u)^n du=B\left(\frac{1}{2},n+1\right).$$

This converges exactly for $n>-1$, the condition for $(1-u)^n$ to be integrable at $u=1$. For such n, $B(\frac{1}{2},n+1)=\Gamma(\frac{1}{2})\Gamma(n+1)/\Gamma(n+\frac{3}{2})$, and the Legendre duplication formula $\Gamma(z)\Gamma(z+\frac{1}{2})=2^{1-2z}\sqrt{\pi}\,\Gamma(2z)$ at $z=n+1$ gives $\Gamma(n+\frac{3}{2})=2^{-2n-1}\sqrt{\pi}\,\Gamma(2n+2)/\Gamma(n+1)$; hence

$$B\left(\frac{1}{2},n+1\right)=\frac{\sqrt{\pi}\,[\Gamma(n+1)]^2}{2^{-2n-1}\sqrt{\pi}\,\Gamma(2n+2)}=\boxed{2^{2n+1}\frac{n!\,n!}{(2n+1)!}},$$

where $n!=\Gamma(n+1)$ and $(2n+1)!=\Gamma(2n+2)$ for nonintegral n. For $n=0,1,2,\ldots$ the double-factorial form follows directly: $\Gamma(1/2)=\sqrt{\pi}$, $\Gamma(n+1)=n!$, and the half-integer formula of Exercise 13.3.2 with $n+1$ in place of n gives $\Gamma(n+3/2)=(2n+1)!!\sqrt{\pi}/2^{n+1}$, so

$$B\left(\frac{1}{2},n+1\right)=\frac{\sqrt{\pi}\,n!}{(2n+1)!!\sqrt{\pi}/2^{n+1}}=\frac{2\cdot 2^n n!}{(2n+1)!!}=\boxed{\frac{2(2n)!!}{(2n+1)!!}},$$

by $(2n)!!=2^n n!$. The two boxed forms agree for integer n: splitting $(2n+1)!=(2n)!!\,(2n+1)!!$ into its even and odd factors and using $2^n n!=(2n)!!$ turns the first into $2[(2n)!!]^2/[(2n)!!\,(2n+1)!!]$. At $n=0$ the value is two, the length of the interval, checking the empty-product convention.

13.3.4 Evaluate $\displaystyle\int_{-1}^{1}(1+x)^a(1-x)^b\,dx$ in terms of the beta function.

ANS. $2^{a+b+1}B(a+1,b+1)$.

Require $\Re a,\Re b>-1$ for this improper integral.

Solution.

The improper integral requires $\Re a>-1$ and $\Re b>-1$: near $x=-1$ the integrand behaves like $2^b(1+x)^a$, and near $x=1$ like $2^a(1-x)^b$. The bases inside the interval are positive. Put $u=(1+x)/2$, so $x=2u-1$, $dx=2du$, $1+x=2u$ and $1-x=2(1-u)$. The limits $x=-1,1$ become $u=0,1$. Therefore

$$\begin{aligned}\int_{-1}^{1}(1+x)^a(1-x)^b dx&=\int_{0}^{1}(2u)^a[2(1-u)]^b\,2\,du\\&=2^{a+b+1}\int_{0}^{1}u^a(1-u)^b du\\&=\boxed{2^{a+b+1}B(a+1,b+1)}\\&=2^{a+b+1}\frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)}.\end{aligned}$$

The factor 2^{a+b+1} includes the Jacobian two. Outside the stated convergence domain the gamma expression may continue meromorphically, but that continuation is not the original convergent improper integral.

13.3.5 Show, by means of the beta function, that

$$\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^{\alpha}} = \frac{\pi}{\sin \pi\alpha}, \quad 0 < \alpha < 1.$$

Take real $t < z$.

Solution.

Take real $t < z$ for the displayed real integral; for $t > z$ both bases $z - x$ and $x - t$ are negative between the limits. Set $u = (x - t)/(z - t)$, which runs from 0 to 1 as x runs from t to z. Then

$$x - t = (z - t)u, \quad z - x = (z - t)(1 - u), \quad dx = (z - t)du.$$

The denominator has total power $(1 - \alpha) + \alpha = 1$, so $(z - x)^{1-\alpha}(x - t)^{\alpha} = (z - t)(1 - u)^{1-\alpha}u^{\alpha}$, and its factor $z - t$ cancels the Jacobian. Hence

$$\begin{aligned}\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^{\alpha}} &= \int_0^1 u^{-\alpha}(1-u)^{\alpha-1}du \\ &= B(1-\alpha,\alpha) = \frac{\Gamma(1-\alpha)\Gamma(\alpha)}{\Gamma(1)} \\ &= \boxed{\frac{\pi}{\sin \pi\alpha}}.\end{aligned}$$

The conditions $1 - \alpha > 0$ and $\alpha > 0$ are exactly the two endpoint convergence requirements. The last equality uses $\Gamma(1) = 1$ and the reflection formula $\Gamma(\alpha)\Gamma(1 - \alpha) = \pi/\sin \pi\alpha$; the interval length has canceled completely.

13.3.6 Show that the Dirichlet integral

$$\iint x^p y^q \, dx \, dy = \frac{p!\, q!}{(p + q + 2)!} = \frac{B(p + 1, q + 1)}{p + q + 2},$$

where the range of integration is the triangle bounded by the positive x- and y-axes and the line $x + y = 1$.

The integral requires $\Re p, \Re q > -1$; factorials outside nonnegative integers mean gamma functions.

Solution.

The integral converges for $\Re p, \Re q > -1$: the inner integral below needs $\Re q > -1$ at $y = 0$, and the outer one $\Re p > -1$ at $x = 0$. Ordinary integer factorials correspond to $p, q \geq 0$. Describe the triangle by $0 \leq x \leq 1$ and $0 \leq y \leq 1-x$. Integrate with respect to y first, using $\int_0^{1-x} y^q \, dy = (1-x)^{q+1}/(q+1)$:

$$\begin{aligned}\int_0^1 \int_0^{1-x} x^p y^q \, dy \, dx &= \frac{1}{q+1}\int_0^1 x^p(1-x)^{q+1}dx \\ &= \frac{B(p+1,q+2)}{q+1} \\ &= \frac{\Gamma(p+1)\Gamma(q+2)}{(q+1)\Gamma(p+q+3)} \\ &= \boxed{\frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+q+3)}}.\end{aligned}$$

Since $\Gamma(q+2) = (q+1)\Gamma(q+1)$, the inner-integral factor has canceled. The recurrence $\Gamma(p+q+3) = (p+q+2)\Gamma(p+q+2)$ in the remaining denominator gives

$$\frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+q+3)} = \frac{\Gamma(p+1)\Gamma(q+1)}{(p+q+2)\Gamma(p+q+2)} = \boxed{\frac{B(p+1,q+1)}{p+q+2}} = \frac{p!\,q!}{(p+q+2)!}$$

for nonnegative integers. At $p = q = 0$ the value is $1/2$, the area of the triangle.

13.3.7 Show that

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2+2xy\cos\theta)}\,dx\,dy = \frac{\theta}{2\sin\theta}.$$

What are the limits on θ?

Hint. Consider oblique xy-coordinates.

ANS. $-\pi < \theta < \pi$ for the formula $\theta/(2\sin\theta)$ as written, with value $1/2$ at $\theta = 0$ by continuity. The integral itself converges for every real θ except odd multiples of π.

Solution.

Denote the double integral by $I(\theta)$, and first take $0 < \theta < \pi$. Introduce Cartesian variables

$$X = x + y\cos\theta, \qquad Y = y\sin\theta.$$

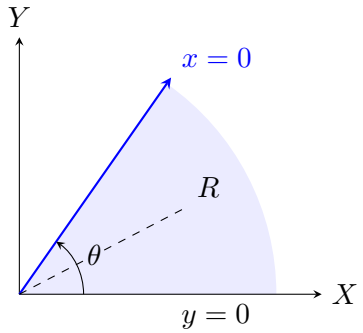
Then

$$X^2 + Y^2 = x^2 + 2xy\cos\theta + y^2(\cos^2\theta + \sin^2\theta) = x^2 + y^2 + 2xy\cos\theta$$

and

$$\frac{\partial(X,Y)}{\partial(x,y)} = \begin{vmatrix} 1 & \cos\theta \\ 0 & \sin\theta \end{vmatrix} = \sin\theta > 0, \qquad dx\,dy = \frac{dX\,dY}{\sin\theta}.$$

The linear map sends the positive quadrant, the set of combinations $x(1,0) + y(0,1)$ with $x, y \geq 0$, onto the combinations $x(1,0) + y(\cos\theta, \sin\theta)$ with $x, y \geq 0$: the sector between the positive X-axis ($y = 0$) and the ray at angle θ ($x = 0$).



In polar coordinates $X = R\cos\phi$, $Y = R\sin\phi$, the sector is $0 \leq \phi \leq \theta$, $R \geq 0$, and $dX\,dY = R\,dR\,d\phi$, so

$$\begin{aligned} I(\theta) &= \frac{1}{\sin\theta}\int_0^\theta d\phi \int_0^\infty e^{-R^2} R\,dR \\ &= \frac{\theta}{\sin\theta}\left[-\frac{1}{2}e^{-R^2}\right]_0^\infty = \boxed{\frac{\theta}{2\sin\theta}}. \end{aligned}$$

The original integral depends on θ only through $\cos\theta$, and $\theta/\sin\theta$ is even, so the result extends to $-\pi < \theta < 0$. At zero the exponent is $-(x+y)^2$; use $u = x+y$, $v = x$ (Jacobian of absolute value one), which maps the quadrant onto $0 \le v \le u$:

$$I(0) = \int_0^\infty e^{-u^2} \int_0^u dv\, du = \int_0^\infty u e^{-u^2} du = \frac{1}{2},$$

the removable limit of the formula, since $\theta/\sin\theta \to 1$.

The stated interval $-\pi < \theta < \pi$ is the range for the unreduced formula $\theta/(2\sin\theta)$, including its removable value at zero. For unrestricted real θ, the integral is periodic and converges except at odd multiples of π. Indeed, if $c = \cos\theta > -1$, the quadratic form on $x, y \ge 0$ is at least $\min(1, 1+c)(x^2+y^2)$: for $c \ge 0$ drop $2cxy \ge 0$, and for $-1 < c < 0$ write the form as $(1+c)(x^2+y^2) - c(x-y)^2$. Hence the integrand is at most $e^{-\min(1,1+c)(x^2+y^2)}$, an integrable Gaussian. If $c = -1$, the integrand is $e^{-(x-y)^2}$; on $x \ge 1$, $x \le y \le x+1$ it is at least e^{-1}, giving infinite integral. In general reduce to $\vartheta = \arccos(\cos\theta) \in [0, \pi]$ and use $\vartheta/(2\sin\vartheta)$, with the endpoint conventions just established.

13.3.8 Evaluate (using the beta function)

(a) $$\int_0^{\pi/2} \cos^{1/2}\theta\, d\theta = \frac{(2\pi)^{3/2}}{16[\Gamma(5/4)]^2},$$

(b) $$\int_0^{\pi/2} \cos^n\theta\, d\theta = \int_0^{\pi/2} \sin^n\theta\, d\theta = \frac{\sqrt{\pi}\,[(n-1)/2]!}{2(n/2)!}$$

$$= \begin{cases} \dfrac{(n-1)!!}{n!!} & \text{for } n \text{ odd}, \\ \dfrac{\pi}{2} \cdot \dfrac{(n-1)!!}{n!!} & \text{for } n \text{ even}. \end{cases}$$

The general integral requires $\Re n > -1$; its odd/even forms use nonnegative integer n and $(-1)!! = 0!! = 1$.

Solution.

For $u = \sin^2\theta$, the limits become zero and one and

$$du = 2\sin\theta\cos\theta\, d\theta, \qquad d\theta = \frac{du}{2\sqrt{u(1-u)}}, \qquad \cos^n\theta = (1-u)^{n/2},$$

because $\sin\theta = \sqrt{u}$ and $\cos\theta = \sqrt{1-u}$ are nonnegative on $[0, \pi/2]$. Since $(1-u)^{n/2}(1-u)^{-1/2} = (1-u)^{(n-1)/2}$, the common form is

$$I_n = \int_0^{\pi/2} \cos^n\theta\, d\theta = \frac{1}{2}\int_0^1 u^{-1/2}(1-u)^{(n-1)/2} du = \frac{1}{2} B\left(\frac{1}{2}, \frac{n+1}{2}\right)$$
$$= \frac{\sqrt{\pi}\,\Gamma((n+1)/2)}{2\Gamma((n+2)/2)},$$

with $\Gamma(1/2) = \sqrt{\pi}$ and $\frac{1}{2} + \frac{n+1}{2} = \frac{n+2}{2}$. This integral formula requires $\Re n > -1$, since $\cos^n\theta$ behaves like $(\pi/2 - \theta)^n$ near $\theta = \pi/2$; the odd/even factorial forms below use nonnegative integer n. Replacing θ by $\pi/2 - \theta$ proves the same value for $\int_0^{\pi/2} \sin^n\theta\, d\theta$.

(a) At $n = 1/2$, $I_{1/2} = \sqrt{\pi}\,\Gamma(3/4)/[2\Gamma(5/4)]$. Reflection, with $\sin(\pi/4) = 1/\sqrt{2}$, and recurrence give

$$\Gamma(1/4)\Gamma(3/4) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}, \qquad \Gamma(1/4) = 4\Gamma(5/4).$$

Hence $\Gamma(3/4) = \pi\sqrt{2}/\Gamma(1/4) = \pi\sqrt{2}/[4\Gamma(5/4)]$, and

$$\boxed{I_{1/2} = \frac{\sqrt{\pi}\,\pi\sqrt{2}}{8[\Gamma(5/4)]^2} = \frac{\pi\sqrt{2\pi}}{8[\Gamma(5/4)]^2} = \frac{(2\pi)^{3/2}}{16[\Gamma(5/4)]^2}},$$

where the last step uses $(2\pi)^{3/2} = 2\pi\sqrt{2\pi}$.

(b) Since $\Gamma((n+1)/2) = [(n-1)/2]!$ and $\Gamma((n+2)/2) = (n/2)!$, the gamma expression is precisely $\sqrt{\pi}\,[(n-1)/2]!/[2(n/2)!]$. For $n = 2m+1$, use $\Gamma(m+3/2) = (2m+1)!!\sqrt{\pi}/2^{m+1}$ (Exercise 13.3.2 with $m+1$ in place of n):

$$I_{2m+1} = \frac{\sqrt{\pi}\,m!}{2\Gamma(m+3/2)} = \frac{\sqrt{\pi}\,m!\,2^{m+1}}{2(2m+1)!!\sqrt{\pi}} = \frac{2^m m!}{(2m+1)!!} = \boxed{\frac{(2m)!!}{(2m+1)!!}}.$$

For $n = 2m$, use $\Gamma(m+1/2) = (2m-1)!!\sqrt{\pi}/2^m$:

$$I_{2m} = \frac{\sqrt{\pi}\,\Gamma(m+1/2)}{2\,m!} = \frac{\pi}{2}\frac{(2m-1)!!}{2^m m!} = \boxed{\frac{\pi}{2}\frac{(2m-1)!!}{(2m)!!}}.$$

Both boxed forms use $(2m)!! = 2^m m!$. The case $m = 0$ in the even formula gives $I_0 = \pi/2$.

13.3.9 Evaluate $\displaystyle\int_0^1 (1-x^4)^{-1/2}\,dx$ as a beta function.

ANS. $\dfrac{[\Gamma(5/4)]^2 \cdot 4}{(2\pi)^{1/2}} = 1.311028777.$

Solution.

Put $u = x^4$, so $x = u^{1/4}$, $dx = \frac{1}{4}u^{-3/4}du$ and $1 - x^4 = 1 - u$; the limits remain zero and one. The endpoint exponents $-3/4$ and $-1/2$ are both integrable, and

$$I = \frac{1}{4}\int_0^1 u^{-3/4}(1-u)^{-1/2}du = \boxed{\frac{1}{4}B\left(\frac{1}{4},\frac{1}{2}\right)} = \frac{\Gamma(1/4)\sqrt{\pi}}{4\Gamma(3/4)}.$$

Use $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ (reflection, as in Exercise 13.3.8(a)) to replace $\Gamma(3/4)$ by $\pi\sqrt{2}/\Gamma(1/4)$, then $\Gamma(1/4) = 4\Gamma(5/4)$:

$$I = \frac{[\Gamma(1/4)]^2\sqrt{\pi}}{4\pi\sqrt{2}} = \frac{[\Gamma(1/4)]^2}{4\sqrt{2\pi}} = \boxed{\frac{4[\Gamma(5/4)]^2}{\sqrt{2\pi}}} = 1.311028777146\ldots,$$

with $\Gamma(5/4) = 0.9064024770554\ldots$. For an independent numerical quadrature one can also put $x = \sin\phi$: then $1 - x^4 = (1-x^2)(1+x^2) = \cos^2\phi\,(1+\sin^2\phi)$ and $dx = \cos\phi\,d\phi$, and cancellation of $\cos\phi$ gives the nonsingular integral $\int_0^{\pi/2}(1+\sin^2\phi)^{-1/2}d\phi$. Its integrand is smooth, has period π and is even about both endpoints, so the trapezoidal rule converges very fast: in the Euler–Maclaurin formula (Section 12.3) every correction term contains an odd-order derivative of the integrand at $\phi = 0$ and $\phi = \pi/2$, and all of these vanish for a function even about both points, so the error falls faster than any power of the step. Eight equal subintervals already give 1.311028777146 (error 2×10^{-13}).

13.3.10 Using beta functions, show that the integral representation

$$J_\nu(z) = \frac{2}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_0^{\pi/2} \sin^{2\nu}\theta\cos(z\cos\theta)\,d\theta, \quad \Re(\nu) > -\tfrac{1}{2},$$

reduces to the Bessel series

$$J_\nu(z) = \sum_{s=0}^{\infty}(-1)^s \frac{1}{s!\,\Gamma(s+\nu+1)}\left(\frac{z}{2}\right)^{2s+\nu},$$

thereby confirming its validity.

Solution.

Expand the cosine, keeping the same branch of $(z/2)^\nu$ throughout:

$$\cos(z\cos\theta) = \sum_{s=0}^{\infty} \frac{(-1)^s z^{2s}\cos^{2s}\theta}{(2s)!}.$$

For fixed z, since $|\cos\theta| \le 1$, the sum of the absolute values of the terms is at most $\sum_s |z|^{2s}/(2s)! = \cosh|z| \le e^{|z|}$. The remaining weight has magnitude $\sin^{2\Re\nu}\theta$, which is integrable at zero exactly when $\Re\nu > -1/2$. Dominated convergence therefore permits integration term by term. For each term put $u = \sin^2\theta$ as in Exercise 13.3.8, so that $\sin^{2\nu}\theta\cos^{2s}\theta\,d\theta = \frac{1}{2}u^{\nu-1/2}(1-u)^{s-1/2}du$; the beta integral is

$$\int_0^{\pi/2}\sin^{2\nu}\theta\cos^{2s}\theta\,d\theta = \frac{1}{2}B\left(\nu+\frac{1}{2}, s+\frac{1}{2}\right) = \frac{\Gamma(\nu+1/2)\Gamma(s+1/2)}{2\Gamma(s+\nu+1)}.$$

Insert this in the proposed representation and cancel its 2 and $\Gamma(\nu+1/2)$ factors:

$$J_\nu(z) = \left(\frac{z}{2}\right)^\nu \sum_{s=0}^{\infty}\frac{(-1)^s z^{2s}}{\sqrt{\pi}(2s)!}\frac{\Gamma(s+1/2)}{\Gamma(s+\nu+1)}.$$

Finally, removing the even factors $(2s)!! = 2^s s!$ from $(2s)!$ gives $(2s-1)!! = (2s)!/(2^s s!)$, so the half-integer formula of Exercise 13.3.2 becomes

$$\Gamma\left(s+\frac{1}{2}\right) = \frac{(2s-1)!!\sqrt{\pi}}{2^s} = \frac{(2s)!\sqrt{\pi}}{4^s s!}, \qquad \frac{\Gamma(s+1/2)}{\sqrt{\pi}(2s)!} = \frac{1}{4^s s!}.$$

Thus, with $z^{2s}/4^s = (z/2)^{2s}$ and $(z/2)^\nu(z/2)^{2s} = (z/2)^{2s+\nu}$,

$$\boxed{J_\nu(z) = \sum_{s=0}^{\infty}\frac{(-1)^s}{s!\Gamma(s+\nu+1)}\left(\frac{z}{2}\right)^{2s+\nu}},$$

the defining Bessel series. For noninteger ν both representations use a chosen branch of z^ν; a value at zero is taken only when the corresponding limit exists.

13.3.11 Given the associated Legendre function, defined in Chapter 15,

$$P_m^m(x) = (2m-1)!!\,(1-x^2)^{m/2},$$

show that

(a) $\displaystyle\int_{-1}^{1} [P_m^m(x)]^2\,dx = \frac{2}{2m+1}(2m)!, \quad m = 0, 1, 2, \ldots,$

(b) $\displaystyle\int_{-1}^{1} [P_m^m(x)]^2 \frac{dx}{1-x^2} = 2\cdot(2m-1)!, \quad m = 1, 2, 3, \ldots.$

Solution.

Square the stated expression: $[P_m^m(x)]^2 = [(2m-1)!!]^2(1-x^2)^m$. **(a)** Exercise 13.3.3 with $n = m$ gives

$$\int_{-1}^{1} [P_m^m(x)]^2 dx = [(2m-1)!!]^2 \frac{2(2m)!!}{(2m+1)!!}$$
$$= \frac{2(2m-1)!!(2m)!!}{2m+1} = \boxed{\frac{2(2m)!}{2m+1}}.$$

We used $(2m+1)!! = (2m+1)(2m-1)!!$ and $(2m-1)!!(2m)!! = (2m)!$, the product of all odd and all even factors up to $2m$. The case $m = 0$ gives two.

(b) Division by $1-x^2$ reduces the exponent to $m-1$. For $m \geq 1$, Exercise 13.3.3 with $n = m-1$ gives $\int_{-1}^{1}(1-x^2)^{m-1}dx = 2(2m-2)!!/(2m-1)!!$, so

$$\int_{-1}^{1} [P_m^m(x)]^2 \frac{dx}{1-x^2} = [(2m-1)!!]^2 \frac{2(2m-2)!!}{(2m-1)!!}$$
$$= 2(2m-1)!!(2m-2)!! = \boxed{2(2m-1)!},$$

where now the odd and even factors up to $2m-1$ combine. At $m = 0$ the second integral diverges at both endpoints, explaining its different lower limit on m. An alternative associated-Legendre convention introduces $(-1)^m$, which has no effect on either squared integral.

13.3.12 Show that, for integers p and q,

(a) $\displaystyle\int_0^1 x^{2p+1}(1-x^2)^{-1/2}\,dx = \frac{(2p)!!}{(2p+1)!!},$

(b) $\displaystyle\int_0^1 x^{2p}(1-x^2)^q\,dx = \frac{(2p-1)!!\,(2q)!!}{(2p+2q+1)!!}.$

Take $p \geq 0$ in (a), and $p, q \geq 0$ in (b).

Solution.

For the stated integer parameters, require $p \geq 0$ in (a), and $p, q \geq 0$ in (b). Negative integer values produce nonintegrable endpoint powers. Use $u = x^2$, $dx = du/(2\sqrt{u})$; the limits remain zero

and one, and the half-integer gamma values come from the formula $\Gamma(k+1/2) = (2k-1)!!\sqrt{\pi}/2^k$ of Exercise 13.3.2. **(a)** Since $x^{2p+1}dx = u^p du/2$,

$$\begin{aligned}\int_0^1 x^{2p+1}(1-x^2)^{-1/2}dx &= \frac{1}{2}\int_0^1 u^p(1-u)^{-1/2}du = \frac{1}{2}B\left(p+1,\frac{1}{2}\right)\\ &= \frac{p!\sqrt{\pi}}{2\Gamma(p+3/2)} = \frac{p!\sqrt{\pi}\,2^{p+1}}{2(2p+1)!!\sqrt{\pi}}\\ &= \frac{2^p p!}{(2p+1)!!} = \boxed{\frac{(2p)!!}{(2p+1)!!}}.\end{aligned}$$

(b) Here $x^{2p}dx = u^{p-1/2}du/2$, so

$$\begin{aligned}\int_0^1 x^{2p}(1-x^2)^q dx &= \frac{1}{2}\int_0^1 u^{p-1/2}(1-u)^q du = \frac{1}{2}B\left(p+\frac{1}{2},q+1\right)\\ &= \frac{\Gamma(p+1/2)q!}{2\Gamma(p+q+3/2)} = \frac{[(2p-1)!!\sqrt{\pi}/2^p]q!}{2[(2p+2q+1)!!\sqrt{\pi}/2^{p+q+1}]}\\ &= \frac{(2p-1)!!\,2^q q!}{(2p+2q+1)!!} = \boxed{\frac{(2p-1)!!(2q)!!}{(2p+2q+1)!!}},\end{aligned}$$

since $2^{p+q+1}/(2\cdot 2^p) = 2^q$ and $2^q q! = (2q)!!$. These formulas include $p=0$ or $q=0$ with the usual empty products $(-1)!! = 0!! = 1$.

13.3.13 A particle of mass m moving in a symmetric potential that is well described by $V(x) = A|x|^n$ has a total energy $\frac{1}{2}m(dx/dt)^2 + V(x) = E$. Solving for dx/dt and integrating we find that the period of motion is

$$\tau = 2\sqrt{2m}\int_0^{x_{\max}} \frac{dx}{(E-Ax^n)^{1/2}},$$

where $x_{\max}$ is a classical turning point given by $Ax_{\max}^n = E$. Show that

$$\tau = \frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{1/n}\frac{\Gamma(1/n)}{\Gamma(1/n+\frac{1}{2})}.$$

Assume $m, A, E, n > 0$.

Solution.

For the bounded oscillation considered here, take $m, A, E > 0$ and $n > 0$. The positive turning point is $x_{\max} = (E/A)^{1/n}$. On the outward quarter-cycle, energy conservation gives

$$\frac{dx}{dt} = \sqrt{\frac{2}{m}}\sqrt{E-Ax^n}, \qquad \frac{\tau}{4} = \sqrt{\frac{m}{2}}\int_0^{x_{\max}}\frac{dx}{\sqrt{E-Ax^n}}.$$

Because $V(-x) = V(x)$, the legs $0 \to x_{\max} \to 0 \to -x_{\max} \to 0$ of one period take equal times, and $4\sqrt{m/2} = 2\sqrt{2m}$; this explains the factor in the question. Set $u = Ax^n/E = (x/x_{\max})^n$. Then

$$x = x_{\max}u^{1/n}, \quad dx = \frac{x_{\max}}{n}u^{1/n-1}du, \quad (E-Ax^n)^{-1/2} = E^{-1/2}(1-u)^{-1/2}.$$

Both endpoints become zero and one, respectively. The constant factors combine as

$$2\sqrt{2m}\,\frac{x_{\max}}{n}\,E^{-1/2} = \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{1/n}.$$

Consequently

$$\begin{aligned}\tau &= \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{1/n}\int_0^1 u^{1/n-1}(1-u)^{-1/2}du \\ &= \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{1/n}B\left(\frac{1}{n},\frac{1}{2}\right) = \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{1/n}\frac{\Gamma(1/n)\sqrt{\pi}}{\Gamma(1/n+1/2)} \\ &= \boxed{\frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{1/n}\frac{\Gamma(1/n)}{\Gamma(1/n+1/2)}}.\end{aligned}$$

As a check, $n=2$ gives $\tau = \sqrt{2\pi m/E}\,\sqrt{E/A}\,\Gamma(1/2)/\Gamma(1) = \pi\sqrt{2m/A}$; with $A = m\omega^2/2$, $\sqrt{2m/A} = 2/\omega$ and $\tau = 2\pi/\omega$, the harmonic-oscillator period, independent of E.

13.3.14 Referring to Exercise 13.3.13,

(a) Determine the limit as $n\to\infty$ of

$$\frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{1/n}\frac{\Gamma(1/n)}{\Gamma(1/n+\frac{1}{2})}.$$

(b) Find $\lim_{n\to\infty}\tau$ from the behavior of the integrand $(E-Ax^n)^{-1/2}$.

(c) Investigate the behavior of the physical system (potential well) as $n\to\infty$. Obtain the period from inspection of this limiting physical system.

For a dimensionful family, write $V_n(x) = A(|x|/L)^n$ with fixed $A, L > 0$; the displayed formula uses length units $L = 1$.

Solution.

(a) Use the recurrence $(1/n)\Gamma(1/n) = \Gamma(1+1/n)$ before taking the limit; by continuity of Γ at 1 and 1/2,

$$\frac{\Gamma(1/n)}{n\Gamma(1/n+1/2)} = \frac{\Gamma(1+1/n)}{\Gamma(1/n+1/2)} \longrightarrow \frac{\Gamma(1)}{\Gamma(1/2)} = \frac{1}{\sqrt{\pi}}.$$

Also $(E/A)^{1/n}\to 1$ for fixed positive numerical E/A. Hence the period written in the question, $2\sqrt{2\pi m/E}\,(E/A)^{1/n}\,\Gamma(1/n)/[n\Gamma(1/n+1/2)]$, tends to

$$\tau_\infty = 2\sqrt{\frac{2\pi m}{E}}\cdot\frac{1}{\sqrt{\pi}} = \boxed{2\sqrt{\frac{2m}{E}}}.$$

When comparing dimensionful potentials at different n, fix a length L and write $V_n(x) = A(|x|/L)^n$ with fixed energy scale A. The question uses length units with $L = 1$; the dimensionful limiting period is $2L\sqrt{2m/E}$.

(b) To justify passing to the limit despite the moving turning point, set $x = x_{\max}u$, where $x_{\max} = L(E/A)^{1/n}$ solves $V_n(x_{\max}) = E$. Then $E - V_n(x) = E - Ex_{\max}^{-n}x^n = E(1-u^n)$ and $dx = x_{\max}du$, so

$$\tau_n = 2\sqrt{2m}\int_0^{x_{\max}}\frac{dx}{\sqrt{E-V_n(x)}} = 2\sqrt{\frac{2m}{E}}\,x_{\max}\int_0^1(1-u^n)^{-1/2}du.$$

For $0 \le u < 1$, this integrand tends to one. For $n \ge 1$, $u^n \le u$, so

$$0 < (1-u^n)^{-1/2} \le (1-u)^{-1/2}, \qquad \int_0^1 (1-u)^{-1/2} du = 2.$$

Dominated convergence gives an integral limit of one, and $x_{\max} = L(E/A)^{1/n} \to L$, so

$$\boxed{\lim_{n\to\infty} \tau_n = 2L\sqrt{\frac{2m}{E}}},$$

the limit found in (a) (where $L = 1$), now obtained without ignoring the endpoint singularity.

(c) For $|x| < L$, $V_n(x) \to 0$; for $|x| > L$, $V_n(x) \to \infty$. The limiting motion is free flight between reflecting walls at $\pm L$, with speed $v = \sqrt{2E/m}$. A round trip covers distance $4L$, so

$$\boxed{\tau_\infty = \frac{4L}{v} = 4L\sqrt{\frac{m}{2E}} = 2L\sqrt{\frac{2m}{E}}}.$$

The isolated value $V_n(\pm L) = A$ does not affect the limiting travel time derived in (b).

13.3.15 Show that

$$\int_0^\infty \frac{\sinh^\alpha x}{\cosh^\beta x}\, dx = \frac{1}{2} B\left(\frac{\alpha+1}{2}, \frac{\beta-\alpha}{2}\right), \quad -1 < \alpha < \beta.$$

Hint. Let $\sinh^2 x = u$.

Solution.

The endpoint behavior explains the given restrictions: near zero $\sinh x \sim x$ and $\cosh x \to 1$, so $\sinh^\alpha x / \cosh^\beta x \sim x^\alpha$, whereas at infinity $\sinh x \sim \cosh x \sim e^x/2$, so the integrand is asymptotic to $(e^x/2)^{\alpha-\beta} = 2^{\beta-\alpha} e^{(\alpha-\beta)x}$. Thus for real parameters convergence requires $\alpha > -1$ and $\beta > \alpha$.

Set $u = \sinh^2 x$, which runs from 0 to ∞. Since $\cosh^2 x = 1 + u$ and $\sinh x, \cosh x \ge 0$ for $x \ge 0$,

$$du = 2\sinh x \cosh x\, dx = 2\sqrt{u(1+u)}\, dx, \qquad \sinh^\alpha x = u^{\alpha/2}, \qquad \cosh^\beta x = (1+u)^{\beta/2}.$$

Since $dx = \frac{1}{2} u^{-1/2}(1+u)^{-1/2} du$ lowers both exponents by one half, it follows that

$$\int_0^\infty \frac{\sinh^\alpha x}{\cosh^\beta x} dx = \frac{1}{2}\int_0^\infty u^{(\alpha-1)/2}(1+u)^{-(\beta+1)/2} du.$$

To recognize the beta integral, put $v = u/(1+u)$, which runs from 0 to 1; then $u = v/(1-v)$, $1 + u = 1/(1-v)$ and $du = dv/(1-v)^2$, so

$$u^{p-1}(1+u)^{-p-q} du = v^{p-1}(1-v)^{1-p}(1-v)^{p+q}(1-v)^{-2} dv = v^{p-1}(1-v)^{q-1} dv.$$

In general this shows

$$\int_0^\infty u^{p-1}(1+u)^{-p-q} du = \int_0^1 v^{p-1}(1-v)^{q-1} dv = B(p,q).$$

Matching $p - 1 = (\alpha-1)/2$ and $p + q = (\beta+1)/2$ gives $p = (\alpha+1)/2$, $q = (\beta-\alpha)/2$, both positive. Therefore

$$\boxed{\int_0^\infty \frac{\sinh^\alpha x}{\cosh^\beta x} dx = \frac{1}{2} B\left(\frac{\alpha+1}{2}, \frac{\beta-\alpha}{2}\right)}.$$

For complex parameters the corresponding absolute-convergence conditions are $\Re\alpha > -1$ and $\Re(\beta - \alpha) > 0$.

13.3.16 The beta distribution of probability theory has a probability density

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1},$$

with x restricted to the interval $(0,1)$. Show that

(a) $\langle x\rangle$, the mean value, is $\dfrac{\alpha}{\alpha+\beta}$.

(b) σ^2, its variance, is $\langle x^2\rangle - \langle x\rangle^2 = \dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

Assume real $\alpha, \beta > 0$.

Solution.

A beta probability density requires real $\alpha, \beta > 0$: a density must be real and nonnegative, and $x^{\alpha-1}(1-x)^{\beta-1}$ is integrable at $x = 0$ only for $\alpha > 0$ and at $x = 1$ only for $\beta > 0$. Its prefactor is $\Gamma(\alpha+\beta)/[\Gamma(\alpha)\Gamma(\beta)] = 1/B(\alpha,\beta)$. Consequently, for $\Re q > -\alpha$,

$$\begin{aligned}\langle x^q\rangle &= \frac{1}{B(\alpha,\beta)}\int_0^1 x^{\alpha+q-1}(1-x)^{\beta-1}dx = \frac{B(\alpha+q,\beta)}{B(\alpha,\beta)} \\ &= \frac{\Gamma(\alpha+q)\Gamma(\beta)}{\Gamma(\alpha+\beta+q)}\cdot\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{\Gamma(\alpha+q)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+q)}.\end{aligned}$$

At $q = 0$ this equals one and verifies normalization. **(a)** For $q = 1$, apply gamma recurrence to numerator and denominator:

$$\langle x\rangle = \frac{[\alpha\Gamma(\alpha)]\Gamma(\alpha+\beta)}{\Gamma(\alpha)[(\alpha+\beta)\Gamma(\alpha+\beta)]} = \boxed{\frac{\alpha}{\alpha+\beta}}.$$

(b) For $q = 2$, the recurrence applied twice gives $\Gamma(\alpha+2) = (\alpha+1)\alpha\Gamma(\alpha)$ and $\Gamma(\alpha+\beta+2) = (\alpha+\beta+1)(\alpha+\beta)\Gamma(\alpha+\beta)$, and the same cancellation gives

$$\langle x^2\rangle = \frac{\alpha(\alpha+1)\Gamma(\alpha)\Gamma(\alpha+\beta)}{\Gamma(\alpha)(\alpha+\beta)(\alpha+\beta+1)\Gamma(\alpha+\beta)} = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}.$$

Subtract the squared mean using a common denominator:

$$\begin{aligned}\sigma^2 &= \frac{\alpha(\alpha+1)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)} \\ &= \frac{\alpha[(\alpha^2+\alpha\beta+\alpha+\beta) - (\alpha^2+\alpha\beta+\alpha)]}{(\alpha+\beta)^2(\alpha+\beta+1)} \\ &= \boxed{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}.\end{aligned}$$

13.3.17 From

$$\lim_{n\to\infty} \frac{\displaystyle\int_0^{\pi/2} \sin^{2n}\theta\, d\theta}{\displaystyle\int_0^{\pi/2} \sin^{2n+1}\theta\, d\theta} = 1,$$

derive the Wallis formula for π:

$$\frac{\pi}{2} = \frac{2\cdot 2}{1\cdot 3}\cdot\frac{4\cdot 4}{3\cdot 5}\cdot\frac{6\cdot 6}{5\cdot 7}\cdots.$$

Solution.

Write $I_k = \int_0^{\pi/2} \sin^k\theta\, d\theta$. Exercise 13.3.8(b), where the sine and cosine integrals are equal, gives

$$I_{2n} = \frac{\pi}{2}\frac{(2n-1)!!}{(2n)!!}, \qquad I_{2n+1} = \frac{(2n)!!}{(2n+1)!!}.$$

Therefore

$$\frac{I_{2n}}{I_{2n+1}} = \frac{\pi}{2}\frac{(2n-1)!!(2n+1)!!}{[(2n)!!]^2}.$$

Define the finite product

$$P_n = \prod_{j=1}^{n} \frac{(2j)^2}{(2j-1)(2j+1)} = \frac{[(2n)!!]^2}{(2n-1)!!(2n+1)!!};$$

its numerators multiply to $[2\cdot 4\cdots(2n)]^2 = [(2n)!!]^2$, and its denominators to $[1\cdot 3\cdots(2n-1)][3\cdot 5\cdots(2n+1)] = (2n-1)!!(2n+1)!!$. The previous ratio is $(\pi/2)/P_n$, so $P_n = (\pi/2)\, I_{2n+1}/I_{2n}$. Taking the stated limit $I_{2n}/I_{2n+1}\to 1$ gives

$$\boxed{\frac{\pi}{2} = \lim_{n\to\infty} P_n = \frac{2\cdot 2}{1\cdot 3}\frac{4\cdot 4}{3\cdot 5}\frac{6\cdot 6}{5\cdot 7}\cdots}.$$

For completeness, the limit used here also follows without a gamma asymptotic formula: since $0\le \sin\theta\le 1$, $\sin^{k+1}\theta \le \sin^k\theta$ and so $I_{2n+1}\le I_{2n}\le I_{2n-1}$. The already evaluated odd integrals satisfy

$$\frac{I_{2n-1}}{I_{2n+1}} = \frac{(2n-2)!!}{(2n-1)!!}\cdot\frac{(2n+1)!!}{(2n)!!} = \frac{2n+1}{2n},$$

so dividing the inequalities by I_{2n+1} gives

$$1 \le \frac{I_{2n}}{I_{2n+1}} \le 1 + \frac{1}{2n} \longrightarrow 1.$$

This proves convergence of the infinite product as well as its value.

13.4 Stirling's Series

13.4.1 Rewrite Stirling's series to give $\Gamma(z+1)$ instead of $\ln\Gamma(z+1)$.

$$\textit{ANS. } \Gamma(z+1) \sim \sqrt{2\pi}\, z^{z+1/2} e^{-z} \left(1 + \frac{1}{12z} + \frac{1}{288z^2} - \frac{139}{51{,}840z^3} + \cdots\right).$$

Interpret this as an asymptotic expansion as $|z| \to \infty$ in a fixed sector $|\arg z| \le \pi - \delta$, $\delta > 0$, with consistent logarithm branches.

Solution.

Use Stirling's logarithmic expansion on the positive real axis, or on a fixed closed sector $|\arg z| \le \pi - \delta$ with a consistent logarithm branch; its correction terms $B_{2n}/[2n(2n-1)z^{2n-1}]$ with $B_2 = 1/6$, $B_4 = -1/30$ give

$$\ln\Gamma(z+1) = \frac{1}{2}\ln(2\pi) + \left(z + \frac{1}{2}\right)\ln z - z + \frac{1}{12z} - \frac{1}{360z^3} + O(z^{-5}).$$

The first three terms exponentiate to $\sqrt{2\pi}\, z^{z+1/2}e^{-z}$. Let $u = 1/(12z) - 1/(360z^3) + O(z^{-5})$ and exponentiate the rest. Since $u = O(z^{-1})$,

$$u^2 = \frac{1}{144z^2} + O(z^{-4}), \qquad u^3 = \frac{1}{1728z^3} + O(z^{-5}), \qquad u^k = O(z^{-4}) \quad (k \ge 4),$$

so through order z^{-3},

$$\begin{aligned} e^u &= 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + O(z^{-4}) \\ &= 1 + \frac{1}{12z} + \frac{1}{2\cdot 12^2 z^2} + \left(-\frac{1}{360} + \frac{1}{6\cdot 12^3}\right) z^{-3} + O(z^{-4}). \end{aligned}$$

Here $2\cdot 12^2 = 288$ and $6\cdot 12^3 = 10368$; with $51840 = 144\cdot 360 = 5\cdot 10368$ the cubic coefficient is $-1/360 + 1/10368 = (-144+5)/51840 = -139/51840$. Thus

$$\boxed{\Gamma(z+1) \sim \sqrt{2\pi}\, z^{z+1/2} e^{-z} \left(1 + \frac{1}{12z} + \frac{1}{288z^2} - \frac{139}{51840z^3} + \cdots\right).}$$

This is an asymptotic expansion, not a convergent infinite power series in $1/z$. A finite truncation and its remainder order are meaningful; the ellipsis does not license summing infinitely many terms at a fixed z.

13.4.2 Use Stirling's formula to estimate 52!, the number of possible rearrangements of cards in a standard deck of playing cards.

Solution.

The leading Stirling estimate is

$$52! \approx \sqrt{2\pi\cdot 52}\left(\frac{52}{e}\right)^{52}.$$

Compute its logarithm first; the powers of 52 contribute $\frac{1}{2}\ln 52 + 52\ln 52 = \frac{105}{2}\ln 52$, so

$$L_0 = \frac{1}{2}\ln(2\pi) + \frac{105}{2}\ln 52 - 52 = 156.3592337587296\ldots.$$

Exponentiation gives

$$\boxed{52! \approx e^{L_0} = 8.05290203839 \times 10^{67} \approx 8.1 \times 10^{67}}.$$

The first logarithmic correction gives

$$e^{L_0+1/(12\cdot 52)} = 8.06581767642 \times 10^{67}.$$

For comparison, direct integer multiplication $1 \cdot 2 \cdots 52$ gives $52! = 8.065817517094\ldots \times 10^{67}$. The leading estimate has relative error $(e^{L_0} - 52!)/52! = -0.00160126096\ldots$, about -0.1601%; including $1/(12 \cdot 52)$ reduces the relative error to 1.97534×10^{-8}. The size and sign are consistent with the omitted next logarithmic term $-1/(360 \cdot 52^3) = -1.97555 \times 10^{-8}$: since $e^\delta - 1 \approx \delta$ for small δ, leaving out this term makes the estimate too large by about 1.97555×10^{-8} of its value. Including it leaves a relative error of -2.1×10^{-12}, the size of the following term $1/(1260 \cdot 52^5)$.

13.4.3 Show that the constants C_1 and C_2 in Stirling's formula have the respective values zero and $\frac{1}{2}\ln 2\pi$ by using the logarithm of the Legendre duplication formula and that of the recurrence formula.

Solution.

Let $F(z) = \ln\Gamma(z+1)$ for large positive z. Before the constants are determined, the integrated expansion is

$$F(z) = C_2 + \left(z + \frac{1}{2}\right)\ln z + (C_1 - 1)z + O(z^{-1});$$

it is the integral of the digamma expansion $\psi(z+1) = F'(z) = C_1 + \ln z + 1/(2z) + O(z^{-2})$, since $\int \ln z\, dz = z\ln z - z$. The recurrence $\Gamma(z+1) = z\Gamma(z)$ gives $F(z) - F(z-1) = \ln z$. Substitute the expansion at z and at $z-1$ (whose remainder $O((z-1)^{-1})$ is again $O(z^{-1})$): the constants C_2 cancel and the linear terms leave $(C_1 - 1)[z - (z-1)] = C_1 - 1$, so

$$\left(z + \frac{1}{2}\right)\ln z - \left(z - \frac{1}{2}\right)\ln(z-1) + C_1 - 1 = \ln z + O(z^{-1}).$$

Moving $\ln z$ to the left yields

$$0 = \left(z - \frac{1}{2}\right)\ln\frac{z}{z-1} + C_1 - 1 + O(z^{-1}).$$

Since $\ln[z/(z-1)] = -\ln(1 - 1/z) = 1/z + O(z^{-2})$, the first term is $1 + O(z^{-1})$. Letting $z \to \infty$ gives $\boxed{C_1 = 0}$.

Now use duplication. Multiply the Legendre duplication formula $\Gamma(z)\Gamma(z+\frac{1}{2}) = 2^{1-2z}\sqrt{\pi}\,\Gamma(2z)$ by z; since $z\Gamma(z) = \Gamma(z+1)$ and $z\Gamma(2z) = \frac{1}{2}\Gamma(2z+1)$,

$$\Gamma(z+1)\Gamma\left(z + \frac{1}{2}\right) = 2^{-2z}\sqrt{\pi}\,\Gamma(2z+1).$$

Its logarithm, with $\ln\Gamma(z+\frac{1}{2}) = F(z-\frac{1}{2})$, is

$$F(z) + F\left(z - \frac{1}{2}\right) = \frac{1}{2}\ln\pi - 2z\ln 2 + F(2z).$$

The needed shifted expansion (with $C_1 = 0$, and $(z - \frac{1}{2}) + \frac{1}{2} = z$) follows from $\ln(z - \frac{1}{2}) = \ln z + \ln(1 - \frac{1}{2z})$:

$$\begin{aligned} F\left(z - \frac{1}{2}\right) &= C_2 + z\ln\left(z - \frac{1}{2}\right) - z + \frac{1}{2} + O(z^{-1}) \\ &= C_2 + z\ln z + z\left[-\frac{1}{2z} + O(z^{-2})\right] - z + \frac{1}{2} + O(z^{-1}) \\ &= C_2 + z\ln z - z + O(z^{-1}). \end{aligned}$$

Consequently the left side of duplication is $2C_2 + (2z + 1/2)\ln z - 2z + O(z^{-1})$. On the right side, $F(2z) = C_2 + (2z + \frac{1}{2})\ln(2z) - 2z + O(z^{-1})$ with $(2z + \frac{1}{2})\ln(2z) = (2z + \frac{1}{2})\ln z + 2z\ln 2 + \frac{1}{2}\ln 2$; the terms $\pm 2z\ln 2$ cancel, so the right side is

$$C_2 + \left(2z + \frac{1}{2}\right)\ln z - 2z + \frac{1}{2}\ln\pi + \frac{1}{2}\ln 2 + O(z^{-1}).$$

Canceling the common terms and letting $z \to \infty$ gives $2C_2 = C_2 + \frac{1}{2}\ln\pi + \frac{1}{2}\ln 2$, that is,

$$\boxed{C_2 = \frac{1}{2}\ln(2\pi)}.$$

The recurrence added to the instruction is needed. With C_1 left free, the linear term of $F(z - \frac{1}{2})$ is $(C_1 - 1)(z - \frac{1}{2})$ instead of $-(z - \frac{1}{2})$, which adds $C_1 z - \frac{1}{2}C_1$ to the shifted expansion above:

$$F\left(z - \frac{1}{2}\right) = C_2 + z\ln z + (C_1 - 1)z - \frac{C_1}{2} + O(z^{-1}).$$

Since $F(z)$ and $F(2z)$ likewise gain $C_1 z$ and $2C_1 z$, the left and right sides of the logarithmic duplication formula become

$$\begin{aligned} \text{left:}\quad & 2C_2 + \left(2z + \frac{1}{2}\right)\ln z + 2(C_1 - 1)z - \frac{C_1}{2} + O(z^{-1}), \\ \text{right:}\quad & C_2 + \left(2z + \frac{1}{2}\right)\ln z + 2(C_1 - 1)z + \frac{1}{2}\ln(2\pi) + O(z^{-1}). \end{aligned}$$

All terms that grow with z agree for every value of C_1, so the comparison gives only the single relation $2C_2 - \frac{1}{2}C_1 = C_2 + \frac{1}{2}\ln(2\pi)$, that is, $C_2 = \frac{1}{2}\ln(2\pi) + \frac{1}{2}C_1$.

13.4.4 Without using Stirling's series show that

(a) $\ln(n!) < \displaystyle\int_1^{n+1} \ln x\,dx$, (b) $\ln(n!) > \displaystyle\int_1^{n} \ln x\,dx$; n is an integer ≥ 2.

Note that the arithmetic mean of these two integrals gives a good approximation for Stirling's series.

Solution.

The logarithm is strictly increasing. **(a)** On each interval $[j, j+1]$, $\ln x > \ln j$ except at its left endpoint. Integrating this continuous inequality over the unit interval gives $\int_j^{j+1} \ln x\,dx > \ln j$, and summing $j = 1, \ldots, n$ gives

$$\boxed{\ln(n!) = \sum_{j=1}^{n} \ln j < \sum_{j=1}^{n} \int_j^{j+1} \ln x\,dx = \int_1^{n+1} \ln x\,dx}.$$

(b) On $[j-1, j]$ for $j = 2, \ldots, n$, $\ln x < \ln j$ except at the right endpoint, so $\int_{j-1}^{j} \ln x\, dx < \ln j$. Summing, and using $\ln 1 = 0$ in $\ln(n!)$, gives

$$\boxed{\int_1^n \ln x\, dx = \sum_{j=2}^{n} \int_{j-1}^{j} \ln x\, dx < \sum_{j=2}^{n} \ln j = \ln(n!)}.$$

Evaluating the elementary antiderivative $x \ln x - x$, $\int_1^n \ln x\, dx = n \ln n - n + 1$ and $\int_1^{n+1} \ln x\, dx = (n+1)\ln(n+1) - (n+1) + 1$, gives the explicit two-sided bound

$$n \ln n - n + 1 < \ln(n!) < (n+1)\ln(n+1) - n.$$

Since $(n+1)\ln(n+1) = (n+1)\ln n + (n+1)\ln(1+1/n) = (n+1)\ln n + 1 + O(n^{-1})$, the arithmetic mean of these bounds is

$$\frac{n \ln n + (n+1)\ln(n+1) - 2n + 1}{2} = \left(n + \frac{1}{2}\right)\ln n - n + 1 + O(n^{-1}),$$

which reproduces Stirling's growing terms without assuming that formula. It does not determine Stirling's exact constant term: Stirling's series has $\frac{1}{2}\ln(2\pi) = 0.9189\ldots$ in place of 1, so the mean exceeds $\ln(n!)$ by $1 - \frac{1}{2}\ln(2\pi) + O(n^{-1}) = 0.0811 + O(n^{-1})$ (for example 0.0969 at $n = 10$), a bounded error that is small compared with $\ln(n!)$.

13.4.5 Test for convergence

$$\sum_{p=0}^{\infty} \left[\frac{\Gamma(p+\frac{1}{2})}{p!}\right]^2 \frac{2p+1}{2p+2} = \pi \sum_{p=0}^{\infty} \frac{(2p-1)!!\,(2p+1)!!}{(2p)!!\,(2p+2)!!}.$$

This series arises in an attempt to describe the magnetic field created by and enclosed by a current loop.

Solution.

Let

$$a_p = \left[\frac{\Gamma(p+1/2)}{\Gamma(p+1)}\right]^2 \frac{2p+1}{2p+2} > 0.$$

Stirling's formula, in the form of Exercise 13.4.6 with $x = p$, $a = -1/2$ and $b = 0$, gives the gamma quotient: there $x^{b-a}\Gamma(x+a+1)/\Gamma(x+b+1) = p^{1/2}\Gamma(p+1/2)/\Gamma(p+1)$, and its logarithm is $(a-b)(a+b+1)/(2x) + O(x^{-2}) = -1/(8p) + O(p^{-2})$, so, exponentiating,

$$\frac{\Gamma(p+1/2)}{\Gamma(p+1)} = p^{-1/2}\left[1 - \frac{1}{8p} + O(p^{-2})\right].$$

Since also $(2p+1)/(2p+2) \to 1$, it follows that

$$p\, a_p = \left[p^{1/2}\frac{\Gamma(p+1/2)}{\Gamma(p+1)}\right]^2 \frac{2p+1}{2p+2} \longrightarrow 1.$$

By limit comparison with $\sum_{p=1}^{\infty} 1/p$ (for all large p, $a_p > 1/(2p)$, and the harmonic series diverges),

$$\boxed{\sum_{p=0}^{\infty} a_p \text{ diverges to } +\infty}.$$

The finite $p = 0$ term is $[\Gamma(1/2)]^2/2 = \pi/2$ and does not alter convergence. For the second form, $\Gamma(p + 1/2) = (2p - 1)!!\sqrt{\pi}/2^p$ (Exercise 13.3.2) and $p! = (2p)!!/2^p$ give $\Gamma(p + 1/2)/p! = \sqrt{\pi}(2p - 1)!!/(2p)!!$; then absorb the factors $2p + 1$ and $2p + 2$ into the double factorials:

$$\left[\frac{\Gamma(p + 1/2)}{p!}\right]^2 \frac{2p + 1}{2p + 2} = \pi \, \frac{(2p - 1)!!}{(2p)!!} \cdot \frac{(2p - 1)!!\,(2p + 1)}{(2p)!!\,(2p + 2)}$$
$$= \pi \, \frac{(2p - 1)!!\,(2p + 1)!!}{(2p)!!\,(2p + 2)!!}.$$

This proves that it is the same positive series. A ratio tending to one alone would not establish its divergence; the harmonic comparison above does.

13.4.6 Show that $\lim_{x\to\infty} x^{b-a}\dfrac{\Gamma(x + a + 1)}{\Gamma(x + b + 1)} = 1$.

Solution.

Take fixed a, b and $x \to +\infty$. Complex a, b are also allowed with logarithms continued in the large right half-plane. For either fixed parameter c, Stirling's series (as in Exercise 13.4.1, with $z = x + c$) gives

$$\ln\Gamma(x + c + 1) = \left(x + c + \frac{1}{2}\right)\ln(x + c) - (x + c) + \frac{1}{2}\ln(2\pi) + \frac{1}{12(x + c)} + O(x^{-3}).$$

Expand $\ln(x + c) = \ln x + \ln(1 + c/x) = \ln x + c/x - c^2/(2x^2) + O(x^{-3})$. Multiplication by $x + c + \frac{1}{2}$ gives $(x + c + \frac{1}{2})\ln x$ plus

$$\left(x + c + \frac{1}{2}\right)\left(\frac{c}{x} - \frac{c^2}{2x^2}\right) + O(x^{-2}) = c + \frac{c(c + \frac{1}{2}) - \frac{1}{2}c^2}{x} + O(x^{-2}) = c + \frac{c(c + 1)}{2x} + O(x^{-2}).$$

With $1/[12(x + c)] = 1/(12x) + O(x^{-2})$ and cancellation of the constant c against $-(x + c)$, this gives

$$\ln\Gamma(x + c + 1) = \left(x + \frac{1}{2}\right)\ln x - x + \frac{1}{2}\ln(2\pi) + c\ln x$$
$$+ \frac{c(c + 1)/2 + 1/12}{x} + O(x^{-2}).$$

Subtract the $c = b$ expression from the $c = a$ expression: the terms without c cancel, $a(a + 1) - b(b + 1) = (a - b)(a + b + 1)$, and adding $(b - a)\ln x$ removes $(a - b)\ln x$:

$$\ln\left[x^{b-a}\frac{\Gamma(x + a + 1)}{\Gamma(x + b + 1)}\right] = \frac{(a - b)(a + b + 1)}{2x} + O(x^{-2}) \longrightarrow 0.$$

Exponentiating (the exponential is continuous at zero) proves

$$\boxed{\lim_{x\to\infty} x^{b-a}\frac{\Gamma(x + a + 1)}{\Gamma(x + b + 1)} = 1}.$$

The calculation also gives the first relative correction $(a - b)(a + b + 1)/(2x)$, useful for checking the rate of approach.

13.4.7 Show that $\lim_{n\to\infty} \dfrac{(2n-1)!!}{(2n)!!} n^{1/2} = \pi^{-1/2}$.

Solution.

Express the double factorials in gamma form, using $\Gamma(n+1/2) = (2n-1)!!\sqrt{\pi}/2^n$ (Exercise 13.3.2) and $(2n)!! = 2\cdot 4\cdots(2n) = 2^n n!$:

$$(2n-1)!! = \frac{2^n \Gamma(n+1/2)}{\sqrt{\pi}}, \qquad (2n)!! = 2^n \Gamma(n+1).$$

Their normalized ratio is consequently

$$\frac{(2n-1)!!}{(2n)!!}\sqrt{n} = \frac{1}{\sqrt{\pi}}\, n^{1/2} \frac{\Gamma(n+1/2)}{\Gamma(n+1)}.$$

Exercise 13.4.6 with $x = n$, $a = -1/2$ and $b = 0$ (so that $x^{b-a} = n^{1/2}$) shows that the final factor tends to one, so

$$\boxed{\lim_{n\to\infty} \frac{(2n-1)!!}{(2n)!!}\sqrt{n} = \pi^{-1/2}}.$$

More precisely, the logarithm of the final factor is $(a-b)(a+b+1)/(2n) + O(n^{-2}) = -1/(8n) + O(n^{-2})$ by the same exercise, so the ratio equals $\pi^{-1/2}[1 - 1/(8n) + O(n^{-2})]$, approaching the limit from below for sufficiently large n.

13.4.8 A set of N distinguishable particles is assigned to states ψ_i, $i = 1, 2, \ldots, M$. If the numbers of particles in the various states are $n_1, n_2, \ldots, n_M$ (with $M \ll N$), the number of ways this can be done is

$$W = \frac{N!}{n_1!\, n_2! \cdots n_M!}.$$

The entropy associated with this assignment is $S = k \ln W$, where k is Boltzmann's constant. In the limit $N \to \infty$, with $n_i = p_i N$ (so p_i is the fraction of the particles in state i), find S as a function of N and the p_i.

(a) In the limit of large N, find the entropy associated with an arbitrary set of n_i. Is the entropy an extensive function of the system size (i.e., is it proportional to N)?

(b) Find the set of p_i that maximize S.

Hint. Remember that $\sum_i p_i = 1$ and that this is a constrained maximization (see Section 22.3).

Note. These formulas correspond to classical, or Boltzmann, statistics.

Take M fixed and the fractions p_i fixed as $N \to \infty$. Extensivity refers to the leading term; $M \ll N$ alone is insufficient if M grows with N.

Solution.

Use $0! = 1$ and $0 \ln 0 = 0$ for unoccupied states. **(a)** With fixed M and fixed fractions p_i, $\sum_i n_i = N$ and $n_i = N p_i$. The leading Stirling terms, $\ln n! = n \ln n - n + O(\ln n)$ for each occupied

$n_i \to \infty$ (an unoccupied state contributes $\ln 0! = 0 = 0 \ln 0 - 0$), give

$$\begin{aligned}\frac{S}{k} &= \ln N! - \sum_i \ln n_i! \\ &= N \ln N - N - \sum_i (n_i \ln n_i - n_i) + O(\ln N) \\ &= N \ln N - \sum_i N p_i (\ln N + \ln p_i) + O(\ln N) \\ &= -N \sum_i p_i \ln p_i + O(\ln N).\end{aligned}$$

From the second to the third line, $-N + \sum_i n_i = 0$ and $\ln n_i = \ln N + \ln p_i$; in the last line, $\sum_i p_i = 1$ cancels $N \ln N$. The error terms, one from $\ln N!$ and one from each of the fixed number M of states, are each $O(\ln N)$. Thus

$$\boxed{S \sim -Nk \sum_{i=1}^{M} p_i \ln p_i}$$

when the entropy per particle is nonzero; more generally the additive estimate in the preceding line includes the zero-entropy case. This leading term is extensive.

If q fractions are positive, retaining the half-logarithm terms of $\ln n! = n \ln n - n + \frac{1}{2} \ln(2\pi n) + O(n^{-1})$ for N and for the q occupied $n_i = N p_i$ adds

$$\frac{1}{2} \ln(2\pi N) - \frac{1}{2} \sum_{p_i > 0} \ln(2\pi N p_i) = -\frac{q-1}{2} \ln(2\pi N) - \frac{1}{2} \sum_{p_i > 0} \ln p_i,$$

which makes the finite-size correction explicit:

$$\frac{S}{k} = -N \sum_{p_i > 0} p_i \ln p_i - \frac{q-1}{2} \ln(2\pi N) - \frac{1}{2} \sum_{p_i > 0} \ln p_i + O(N^{-1}).$$

Here every occupied n_i tends to infinity; zero fractions were omitted before applying Stirling's formula. Extensivity is the leading large-N statement for fixed state fractions and a fixed number of states. The exact finite-N entropy differs from this extensive leading term by the $O(\ln N)$ corrections displayed above. The condition $M \ll N$ alone does not ensure extensivity if M also grows: for example $M = \sqrt{N}$, $p_i = 1/M$ gives a leading $(k/2) N \ln N$.

(b) Maximize $H = -\sum_i p_i \ln p_i$ subject to $\sum_i p_i = 1$. In the interior of the probability simplex, introduce $\mathcal{L} = H - \lambda(\sum_i p_i - 1)$. Then

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \lambda = 0, \qquad p_i = e^{-1-\lambda}.$$

All probabilities are equal, and normalization $M e^{-1-\lambda} = 1$ gives $p_i = 1/M$. Then $H = -M \cdot \frac{1}{M} \ln \frac{1}{M} = \ln M$, so

$$\boxed{p_i = \frac{1}{M} \quad (i = 1, \ldots, M)}, \qquad \boxed{S_{\max} = Nk \ln M + O(\ln N)}.$$

The Hessian of H is $\partial^2 H / \partial p_i \partial p_j = -\delta_{ij}/p_i$, so for every nonzero tangent vector $v = (v_1, \ldots, v_M)$ $(\sum_i v_i = 0)$ the quadratic form $\sum_{i,j} v_i v_j \, \partial^2 H / \partial p_i \partial p_j = -\sum_i v_i^2 / p_i$ is strictly negative. Thus H is

strictly concave on the simplex, and this critical point is the unique maximum. A boundary with only $q < M$ occupied states has, by the same argument, maximum $\ln q < \ln M$ and cannot improve it.

For a finite integer assignment, if $n_i \geq n_j + 2$, moving one particle from i to j changes the multiplicity by

$$\frac{W_{\text{new}}}{W_{\text{old}}} = \frac{n_i!\, n_j!}{(n_i - 1)!\, (n_j + 1)!} = \frac{n_i}{n_j + 1} > 1.$$

Thus the exact maximizing counts differ by at most one; they are exactly uniform when M divides N, and tend to the probabilities above otherwise.

13.5 Riemann Zeta Function

13.5.1 Show that the symmetrical functional relation

$$\Gamma\left(\frac{z}{2}\right) \pi^{-z/2} \zeta(z) = \Gamma\left(\frac{1-z}{2}\right) \pi^{-(1-z)/2} \zeta(1-z)$$

follows from the equation

$$\left(e^{2\pi i z} - 1\right) \zeta(z) = \zeta(1-z) \frac{(2\pi)^z}{\Gamma(z)} \left(e^{3\pi i z/2} - e^{\pi i z/2}\right).$$

Solution.

Initially avoid integer z and poles of the gamma factors so that the following divisions are valid. Factor the exponentials, taking $e^{\pi i z}$ out of each difference:

$$e^{2\pi i z} - 1 = e^{\pi i z} \left(e^{\pi i z} - e^{-\pi i z}\right) = 2i e^{\pi i z} \sin \pi z,$$
$$e^{3\pi i z/2} - e^{\pi i z/2} = e^{\pi i z} \left(e^{\pi i z/2} - e^{-\pi i z/2}\right) = 2i e^{\pi i z} \sin \frac{\pi z}{2}.$$

Cancel the nonvanishing factor $2ie^{\pi i z}$ in the given relation to obtain

$$\sin \pi z\, \zeta(z) = \frac{(2\pi)^z}{\Gamma(z)} \sin \frac{\pi z}{2}\, \zeta(1-z).$$

The reflection formula $\Gamma(w)\Gamma(1-w) = \pi / \sin \pi w$, at $w = z$ and at $w = z/2$, gives

$$\sin \pi z = \frac{\pi}{\Gamma(z)\Gamma(1-z)}, \qquad \sin \frac{\pi z}{2} = \frac{\pi}{\Gamma(z/2)\Gamma(1-z/2)}.$$

Substituting both,

$$\frac{\pi\, \zeta(z)}{\Gamma(z)\Gamma(1-z)} = \frac{(2\pi)^z}{\Gamma(z)} \frac{\pi}{\Gamma(z/2)\Gamma(1-z/2)} \zeta(1-z),$$

and multiplication by $\Gamma(z)\Gamma(1-z)/\pi$ gives

$$\zeta(z) = \frac{(2\pi)^z \Gamma(1-z)}{\Gamma(z/2)\Gamma(1-z/2)} \zeta(1-z).$$

The Legendre duplication formula $\Gamma(w)\Gamma(w + \frac{1}{2}) = 2^{1-2w}\sqrt{\pi}\, \Gamma(2w)$ with $w = (1-z)/2$, for which $w + \frac{1}{2} = 1 - \frac{z}{2}$, $2w = 1 - z$ and $1 - 2w = z$, reads

$$\Gamma\left(\frac{1-z}{2}\right) \Gamma\left(1 - \frac{z}{2}\right) = 2^z \sqrt{\pi}\, \Gamma(1-z).$$

Substitute $\Gamma(1-z)=\Gamma(\frac{1-z}{2})\Gamma(1-\frac{z}{2})/(2^z\sqrt{\pi})$, cancel $\Gamma(1-z/2)$, and use $(2\pi)^z/(2^z\sqrt{\pi})=\pi^{z-1/2}$:

$$\zeta(z)=\pi^{z-1/2}\frac{\Gamma((1-z)/2)}{\Gamma(z/2)}\zeta(1-z).$$

Multiplication by $\Gamma(z/2)\pi^{-z/2}$, with $\pi^{-z/2}\pi^{z-1/2}=\pi^{-(1-z)/2}$, yields

$$\boxed{\Gamma\left(\frac{z}{2}\right)\pi^{-z/2}\zeta(z)=\Gamma\left(\frac{1-z}{2}\right)\pi^{-(1-z)/2}\zeta(1-z)}.$$

Both sides are meromorphic in z and agree off the discrete set of excluded points (the integers), so by the identity theorem they are the same meromorphic function. At an excluded point the two sides therefore have the same Laurent expansion, an identity between the full products, not between the canceled zero and pole factors taken separately. Being one function, both sides are finite at the integers other than 0 and 1: the left side visibly is for $z\geq 2$, and the right side for $z\leq -1$, where $1-z\geq 2$. At $z=0$ and $z=1$ both have simple poles with equal residues: using $\Gamma(w/2)=\Gamma(1+w/2)/(w/2)=2/w+O(1)$ as $w\to 0$, $\Gamma(\frac{1}{2})\pi^{-1/2}=1$, $\zeta(0)=-\frac{1}{2}$ and the residue 1 of ζ at 1 (Exercise 13.5.10), each side is $-1/z+O(1)$ near $z=0$ and $1/(z-1)+O(1)$ near $z=1$.

13.5.2 Prove that

$$\int_0^\infty \frac{x^n e^x\,dx}{(e^x-1)^2}=n!\,\zeta(n).$$

Assuming n to be real, show that each side of the equation diverges if $n=1$. Hence the preceding equation carries the condition $n>1$. Integrals such as this appear in the quantum theory of transport effects: thermal and electrical conductivity.

Solution.

Since $d[(e^x-1)^{-1}]/dx=-e^x/(e^x-1)^2$, integration by parts gives

$$\begin{aligned} I_n &= \int_0^\infty \frac{x^n e^x}{(e^x-1)^2}dx \\ &= \left[-\frac{x^n}{e^x-1}\right]_0^\infty + n\int_0^\infty \frac{x^{n-1}}{e^x-1}dx. \end{aligned}$$

At infinity the boundary term vanishes exponentially. At zero $x^n/(e^x-1)\sim x^{n-1}$, so it vanishes for real $n>1$. Expanding $(e^x-1)^{-1}=e^{-x}/(1-e^{-x})=\sum_{k=1}^\infty e^{-kx}$ and integrating the nonnegative terms one by one (monotone convergence), with $t=kx$ in the kth term, then gives

$$I_n=n\sum_{k=1}^\infty\int_0^\infty x^{n-1}e^{-kx}dx=n\sum_{k=1}^\infty\frac{1}{k^n}\int_0^\infty t^{n-1}e^{-t}dt=n\sum_{k=1}^\infty\frac{\Gamma(n)}{k^n},$$

that is,

$$I_n=\boxed{\Gamma(n+1)\zeta(n)=n!\zeta(n)}.$$

For a noninteger n, the notation $n!$ means $\Gamma(n+1)$. Near zero, since $(e^x-1)^2/e^x=(e^{x/2}-e^{-x/2})^2=4\sinh^2(x/2)$ and $\sinh(x/2)=(x/2)[1+O(x^2)]$, $e^x/(e^x-1)^2=1/[4\sinh^2(x/2)]=x^{-2}[1+O(x^2)]$, so the original integrand behaves as x^{n-2}; it is integrable at zero exactly when $n-2>-1$. At $n=1$ the left side is therefore the logarithmically divergent $\int_0 dx/x$, and the right side is $1!\,\zeta(1)=\sum_{k\geq 1}1/k$, the divergent harmonic series. Hence the equation carries the condition $n>1$; for real n the integral converges exactly then. For complex n the same argument, with absolute convergence, holds for $\Re n>1$.

13.5.3 The Bloch-Grüneisen approximation for the resistance in a monovalent metal at absolute temperature T is

$$\rho = C\frac{T^5}{\Theta^6}\int_0^{\Theta/T}\frac{x^5\,dx}{(e^x-1)(1-e^{-x})},$$

where Θ is the Debye temperature characteristic of the metal.

(a) For $T \to \infty$, show that

$$\rho \approx \frac{C}{4}\cdot\frac{T}{\Theta^2}.$$

(b) For $T \to 0$, show that

$$\rho \approx 5!\,\zeta(5)\,C\frac{T^5}{\Theta^6}.$$

Solution.

Take $T, \Theta > 0$ and set $y = \Theta/T$. First rewrite, using $1 - e^{-x} = e^{-x}(e^x - 1)$,

$$\frac{1}{(e^x-1)(1-e^{-x})} = \frac{e^x}{(e^x-1)^2}.$$

(a) For high temperature, $y \to 0$. The Bernoulli generating function $x/(e^x-1) = 1 - x/2 + B_2x^2/2! + B_4x^4/4! + \cdots$, with $B_2 = 1/6$ and $B_4 = -1/30$, divided by x gives the local expansion $(e^x-1)^{-1} = x^{-1} - 1/2 + x/12 - x^3/720 + \cdots$. Since $d(e^x-1)^{-1}/dx = -e^x/(e^x-1)^2$, differentiating it and changing sign gives

$$\frac{e^x}{(e^x-1)^2} = \frac{1}{x^2} - \frac{1}{12} + \frac{x^2}{240} - \cdots = \frac{1}{x^2} - \frac{1}{12} + O(x^2).$$

Multiply by x^5, giving $x^3 - x^5/12 + O(x^7)$, and integrate from 0 to y:

$$\int_0^y \frac{x^5e^x}{(e^x-1)^2}dx = \frac{y^4}{4} - \frac{y^6}{72} + O(y^8).$$

Hence, factoring out $\frac{1}{4}(\Theta/T)^4$ and using $C(T^5/\Theta^6)(\Theta/T)^4 = CT/\Theta^2$ and $\frac{1}{72}/\frac{1}{4} = \frac{1}{18}$,

$$\begin{aligned}\rho &= C\frac{T^5}{\Theta^6}\left[\frac{1}{4}\left(\frac{\Theta}{T}\right)^4 - \frac{1}{72}\left(\frac{\Theta}{T}\right)^6 + O\left(\frac{\Theta^8}{T^8}\right)\right]\\ &= \boxed{\frac{C}{4}\frac{T}{\Theta^2}}\left[1 - \frac{1}{18}\left(\frac{\Theta}{T}\right)^2 + O\left(\frac{\Theta^4}{T^4}\right)\right].\end{aligned}$$

(b) For low temperature, $y \to \infty$. The positive integral tends to the complete one from Exercise 13.5.2 at $n = 5$:

$$\int_0^\infty \frac{x^5e^x}{(e^x-1)^2}dx = 5!\zeta(5).$$

Thus

$$\boxed{\rho \sim 5!\zeta(5)C\frac{T^5}{\Theta^6}}.$$

The omitted tail is exponentially small. Explicitly, for $x \ge y > 0$, dividing numerator and denominator by e^{2x} and using $1 - e^{-x} \ge 1 - e^{-y}$,

$$\frac{e^x}{(e^x-1)^2} = \frac{e^{-x}}{(1-e^{-x})^2} \le \frac{e^{-x}}{(1-e^{-y})^2},$$

so the tail $\int_y^\infty x^5 e^x (e^x-1)^{-2} dx$ is at most

$$\frac{e^{-y}}{(1-e^{-y})^2}\left(y^5 + 5y^4 + 20y^3 + 60y^2 + 120y + 120\right).$$

The polynomial is $e^y \int_y^\infty x^5 e^{-x} dx$: the integration by parts $\int_y^\infty x^j e^{-x} dx = y^j e^{-y} + j \int_y^\infty x^{j-1} e^{-x} dx$, applied for $j = 5, 4, 3, 2, 1$, leaves the boundary terms $(5!/j!)\, y^j e^{-y}$, and the remaining integral $5! \int_y^\infty e^{-x} dx = 5!\, e^{-y}$ is the term $j = 0$; in all, $\int_y^\infty x^5 e^{-x} dx = e^{-y} \sum_{j=0}^5 (5!/j!)\, y^j$. Relative to $5!\zeta(5)$ the bound is $O(y^5 e^{-y}) \to 0$, which proves the limiting replacement.

13.5.4 Derive the following expansion of the Debye function for $n \ge 1$:

$$\int_0^x \frac{t^n\, dt}{e^t - 1} = x^n \left[\frac{1}{n} - \frac{x}{2(n+1)} + \sum_{k=1}^\infty \frac{B_{2k} x^{2k}}{(2k+n)(2k)!}\right], \quad |x| < 2\pi.$$

The complete integral $(0, \infty)$ equals $n!\, \zeta(n+1)$ (Exercise 13.5.6).

For complex x, take integer $n \ge 1$. For noninteger real $n \ge 1$, the real-power version uses $0 < x < 2\pi$; complex continuation needs a specified power branch.

Solution.

The Bernoulli generating function is analytic for $|t| < 2\pi$:

$$\frac{t}{e^t - 1} = 1 - \frac{t}{2} + \sum_{k=1}^\infty \frac{B_{2k}}{(2k)!} t^{2k},$$

where $B_1 = -\frac{1}{2}$ and the odd B_{2k+1}, $k \ge 1$, vanish. The denominator vanishes at $t = 2\pi i k$, $k \in \mathbb{Z}$; the zero at $t = 0$ is canceled by the factor t, so the nearest poles are at $t = \pm 2\pi i$ and the series converges for $|t| < 2\pi$. Multiply by t^{n-1}:

$$\frac{t^n}{e^t - 1} = t^{n-1} - \frac{t^n}{2} + \sum_{k=1}^\infty \frac{B_{2k}}{(2k)!} t^{2k+n-1}.$$

Integrate from zero to x, using $\int_0^x t^{m-1} dt = x^m/m$ for $m = n,\ n+1,\ n+2k$. For integer $n \ge 1$ uniform convergence on every smaller disk justifies termwise integration, giving

$$\int_0^x \frac{t^n}{e^t - 1} dt = \frac{x^n}{n} - \frac{x^{n+1}}{2(n+1)} + \sum_{k=1}^\infty \frac{B_{2k} x^{n+2k}}{(n+2k)(2k)!}$$

$$= \boxed{x^n \left[\frac{1}{n} - \frac{x}{2(n+1)} + \sum_{k=1}^\infty \frac{B_{2k} x^{2k}}{(2k+n)(2k)!}\right]}.$$

Every denominator $n + 2k$ comes from integrating a monomial; the lower-limit terms vanish because every exponent m is positive. The same calculation is valid for real $n > 0$ and $0 < x < 2\pi$, with the integrable factor t^{n-1}; for complex x and noninteger n the powers t^{n-1} and x^n are multivalued, so a consistent power branch is needed. The full integral to infinity is instead $\Gamma(n+1)\zeta(n+1)$, as derived in Exercise 13.5.6.

13.5.5 The equilibrium radiation energy per unit volume of a blackbody is given by

$$u = \frac{8\pi k^4 T^4}{c^3 h^3} \int_0^\infty \frac{x^3}{e^x - 1}\, dx.$$

Show that the integral in this expression is equal to $3!\,\zeta(4)$. The final result is the Stefan-Boltzmann law.

Solution.

For $x > 0$ the geometric series gives $(e^x - 1)^{-1} = \sum_{m=1}^{\infty} e^{-mx}$. The terms are nonnegative, so integrate them by monotone convergence. In each integral put $t = mx$, so $x^3 dx = t^3 dt/m^4$ with unchanged limits:

$$\begin{aligned}\int_0^\infty \frac{x^3}{e^x - 1} dx &= \sum_{m=1}^{\infty} \int_0^\infty x^3 e^{-mx} dx \\ &= \sum_{m=1}^{\infty} \frac{1}{m^4} \int_0^\infty t^3 e^{-t} dt = \boxed{3!\zeta(4)} = \frac{\pi^4}{15}.\end{aligned}$$

Here $\int_0^\infty t^3 e^{-t} dt = \Gamma(4) = 3! = 6$ and $\zeta(4) = (2\pi)^4 |B_4|/(2 \cdot 4!) = \pi^4/90$ (Exercise 12.4.1(b), with $B_4 = -1/30$), so $3!\zeta(4) = 6\pi^4/90 = \pi^4/15$. Substitution gives

$$u = \frac{8\pi k^4 T^4}{c^3 h^3} \cdot \frac{\pi^4}{15}, \qquad \boxed{u = \frac{8\pi^5 k^4}{15 c^3 h^3} T^4}.$$

The printed formula describes radiation energy per unit volume, not total radiated energy or power per unit area: $(kT)^4/(hc)^3$ has the dimensions of energy per volume. Isotropic radiation of energy density u moving with speed c carries the energy $cu/(4\pi)$ per unit time, area and solid angle across a surface normal to its direction; a surface whose normal makes the angle θ with that direction receives the projected fraction $\cos\theta$. Integrating over the outgoing hemisphere gives the outgoing flux through a surface,

$$j^\star = \frac{cu}{4\pi} \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos\theta \sin\theta\, d\theta = \frac{cu}{4\pi} \cdot 2\pi \cdot \frac{1}{2} = \frac{cu}{4}.$$

Thus, with $\sigma = (c/4) \cdot 8\pi^5 k^4/(15 c^3 h^3)$, the Stefan–Boltzmann flux law is

$$\boxed{j^\star = \sigma T^4, \qquad \sigma = \frac{2\pi^5 k^4}{15 c^2 h^3}}.$$

The factor $c/4$ relating $j^\star$ to u is why σ contains c^{-2} while u contains c^{-3}.

13.5.6 As a generalization of the result in Exercise 13.5.5, show that

$$\int_0^\infty \frac{x^s\, dx}{e^x - 1} = s!\,\zeta(s+1), \quad \Re(s) > 0.$$

Solution.

Let $\sigma = \Re s > 0$ and use $x^s = e^{s \ln x}$ for $x > 0$, so $|x^s| = x^\sigma$. Since $e^x - 1 \geq x$, the integrand is at most $x^{\sigma-1}$ in absolute value near zero, and at infinity it has exponential decay. Moreover, by the substitution $t = nx$ used below,

$$\sum_{n=1}^{\infty} \int_0^\infty |x^s e^{-nx}| dx = \sum_{n=1}^{\infty} \int_0^\infty x^\sigma e^{-nx} dx = \Gamma(\sigma+1) \sum_{n=1}^{\infty} n^{-\sigma-1} < \infty.$$

Thus the geometric expansion $(e^x-1)^{-1}=\sum_{n\ge1}e^{-nx}$ may be integrated term by term (Fubini's theorem):

$$\begin{aligned}\int_0^\infty \frac{x^s}{e^x-1}dx &= \sum_{n=1}^\infty \int_0^\infty x^s e^{-nx}dx\\ &= \sum_{n=1}^\infty n^{-s-1}\int_0^\infty t^s e^{-t}dt\\ &= \boxed{\Gamma(s+1)\zeta(s+1)=s!\zeta(s+1)}.\end{aligned}$$

The substitution in the second line is $t=nx$, including its Jacobian $dx=dt/n$. The condition $\Re s>0$ is exactly what makes both the endpoint and the absolutely summed integrals converge.

13.5.7 Prove that

$$\int_0^\infty \frac{x^s\,dx}{e^x+1}=s!\,(1-2^{-s})\,\zeta(s+1),\quad \Re(s)>0.$$

Exercises 13.5.6 and 13.5.7 give the Mellin integral transform of $1/(e^x\pm1)$; this transform is defined in Eq. (20.9).

Solution.

The algebraic identity

$$\frac{1}{e^x+1}=\frac{1}{e^x-1}-\frac{2}{e^{2x}-1}$$

follows by taking the common denominator $e^{2x}-1=(e^x-1)(e^x+1)$: the right side is $[(e^x+1)-2]/[(e^x-1)(e^x+1)]=1/(e^x+1)$. For $\Re s>0$ both integrals on the right converge separately, and Exercise 13.5.6 gives

$$\begin{aligned}\int_0^\infty \frac{x^s}{e^x+1}dx &= \Gamma(s+1)\zeta(s+1)-2\int_0^\infty \frac{x^s}{e^{2x}-1}dx\\ &= \Gamma(s+1)\zeta(s+1)-2\cdot 2^{-s-1}\int_0^\infty \frac{t^s}{e^t-1}dt\\ &= \boxed{\Gamma(s+1)(1-2^{-s})\zeta(s+1)}.\end{aligned}$$

In the second integral $t=2x$, so $x^s=2^{-s}t^s$ and $dx=dt/2$ give the factor 2^{-s-1}, the limits being unchanged; the last integral is $\Gamma(s+1)\zeta(s+1)$ again, and $2\cdot2^{-s-1}=2^{-s}$.

The original Fermi integral actually converges for $\Re s>-1$, because its denominator tends to two at zero. Both sides are analytic in that half-plane: the integral converges uniformly on compact subsets; on the right $\Gamma(s+1)$ is analytic for $\Re s>-1$ (its poles are at $s=-1,-2,\ldots$), and the only pole of $\zeta(s+1)$, at $s=0$ (Exercise 13.5.10), is canceled by the zero of $1-2^{-s}$. They agree for $\Re s>0$, so the identity theorem extends the answer to $\Re s>-1$. At $s=0$, where $1-2^{-s}=1-e^{-s\ln2}=s\ln2+O(s^2)$ and $\zeta(1+s)=1/s+O(1)$, the product is removable and $\Gamma(1)=1$; directly, $1/(e^x+1)=e^{-x}/(1+e^{-x})$ has the primitive $-\ln(1+e^{-x})$:

$$(1-2^{-s})\zeta(s+1)\longrightarrow\ln2,\qquad \int_0^\infty\frac{dx}{e^x+1}=\big[-\ln(1+e^{-x})\big]_0^\infty=\ln2.$$

The extension does not permit separating the two divergent Bose integrals when $-1<\Re s\le0$.

13.5.8 The neutrino energy density (Fermi distribution) in the early history of the universe is given by

$$\rho_\nu = \frac{4\pi}{h^3}\int_0^\infty \frac{x^3}{\exp(x/kT)+1}\,dx.$$

Show that

$$\rho_\nu = \frac{7\pi^5}{30h^3}(kT)^4.$$

Use $kT > 0$ and the convention $c = 1$. With x an energy in conventional units, insert c^{-3} in both formulas.

Solution.

For positive kT, put $t = x/(kT)$, so $x = kTt$, $dx = kT\,dt$, $x^3dx = (kT)^4t^3dt$, and the limits $0, \infty$ are unchanged. Then

$$\rho_\nu = \frac{4\pi(kT)^4}{h^3}\int_0^\infty \frac{t^3}{e^t+1}dt.$$

At $s = 3$ the preceding exercise, with $\zeta(4) = \pi^4/90$ (Exercise 13.5.5), gives

$$\int_0^\infty \frac{t^3}{e^t+1}dt = 3!(1-2^{-3})\zeta(4) = 6\cdot\frac{7}{8}\cdot\frac{\pi^4}{90} = \frac{7\pi^4}{120}.$$

Multiplying by the prefactor, $4\pi\cdot 7\pi^4/120 = 7\pi^5/30$, yields

$$\boxed{\rho_\nu = \frac{7\pi^5}{30h^3}(kT)^4}.$$

As an energy density with x an energy, the printed formula uses $c = 1$. Restoring conventional length/time units requires a factor c^{-3} on both the stated integral and its answer. Indeed for a massless particle the energy x and the momentum p satisfy $x = pc$, so the radial phase-space factor p^2dp becomes x^2dx/c^3; one additional factor x supplies the energy.

13.5.9 Prove that

$$\psi^{(n)}(z) = (-1)^{n+1}\int_0^\infty \frac{t^n e^{-zt}}{1-e^{-t}}\,dt, \quad \Re(z) > 0.$$

The derivative order is an integer $n \geq 1$.

Solution.

Require an integer $n \geq 1$ as well as $\Re z > 0$. For $n = 0$ the written integral diverges as $\int_0 dt/t$ and cannot equal the finite digamma function. For $t > 0$, expand $(1-e^{-t})^{-1} = \sum_{p=0}^\infty e^{-pt}$. Absolute convergence justifies termwise integration, because

$$\sum_{p=0}^\infty \int_0^\infty t^n e^{-(\Re z+p)t}dt = n!\sum_{p=0}^\infty (\Re z+p)^{-n-1} < \infty.$$

Therefore

$$\begin{aligned}(-1)^{n+1}\int_0^\infty \frac{t^n e^{-zt}}{1-e^{-t}}dt &= (-1)^{n+1}\sum_{p=0}^\infty \int_0^\infty t^n e^{-(z+p)t}dt \\ &= (-1)^{n+1}n!\sum_{p=0}^\infty \frac{1}{(z+p)^{n+1}} \\ &= \boxed{\psi^{(n)}(z)}.\end{aligned}$$

The second equality is the scaled gamma integral $\int_0^\infty t^n e^{-wt}dt = n!/w^{n+1}$ for $\Re w > 0$, here with $w = z+p$. The last is the differentiated digamma series: from $\psi(z) = -\gamma+\sum_{p=0}^\infty[(p+1)^{-1}-(z+p)^{-1}]$ (the series $\psi(1+z) = -\gamma + \sum_{k\ge1}[k^{-1} - (k+z)^{-1}]$ of Exercise 13.2.10 with $z-1$ in place of z and $k = p+1$), which converges uniformly on compact subsets of $\Re z > 0$ and may be differentiated term by term, and $d^n[-(z+p)^{-1}]/dz^n = (-1)^{n+1}n!\,(z+p)^{-n-1}$,

$$\psi^{(n)}(z) = (-1)^{n+1}n!\sum_{p=0}^\infty \frac{1}{(z+p)^{n+1}}, \qquad n \ge 1.$$

Near $t = 0$, $1 - e^{-t} = t + O(t^2)$, so the integrand behaves as t^{n-1}, which for integer n is integrable at zero exactly when $n \ge 1$; this is why the condition $n \ge 1$ is required.

13.5.10 Show that $\zeta(s)$ is analytic in the entire finite complex plane except at $s = 1$, where it has a simple pole with a residue of $+1$.

Hint. The contour integral representation will be useful.

Solution.

Use the positive-real-axis cut and continuous argument $0 < \arg t < 2\pi$ from Exercise 13.1.23. Let C_ε run inward along the upper lip, counterclockwise around the circle $|t| = \varepsilon$, then outward along the lower lip, where $0 < \varepsilon < 2\pi$. Define

$$H(s) = \int_{C_\varepsilon} \frac{t^{s-1}}{e^t - 1}dt.$$

The integrand has poles where $e^t = 1$, at $t = 2\pi ik$, $k \in \mathbb{Z}$: $t = 0$ lies inside the circle, and $t = \pm2\pi i, \pm4\pi i, \ldots$ lie outside it, so none is on the path. Keeping the circle radius fixed makes H entire in s: the integrand is entire in s on the compact circle, and the two ray tails decay exponentially and uniformly for s in a compact set. Moreover H does not depend on $\varepsilon \in (0, 2\pi)$: as in Exercise 13.1.23, two such contours differ by the boundary of the cut annulus $\varepsilon_1 < |t| < \varepsilon_2$, which contains no pole, so Cauchy's theorem gives equal integrals.

For $\Re s > 1$ the circle can therefore shrink to zero. On it $|t^{s-1}| = \varepsilon^{\Re s-1}e^{-\theta\Im s} \le \varepsilon^{\Re s-1}e^{2\pi|\Im s|}$ with $\theta = \arg t$, and $|e^t - 1|^{-1} \le M/\varepsilon$ for $\varepsilon \le 1$, where M is the maximum of $|t/(e^t-1)|$ on $|t| \le 1$, finite because $t/(e^t-1)$ (equal to 1 at $t = 0$) is analytic there, its nearest poles being $\pm2\pi i$. Its length being $2\pi\varepsilon$, the circle contributes $O(\varepsilon^{\Re s-1}) \to 0$. On the upper lip $t = x$ and $t^{s-1} = x^{s-1}$; on the lower lip $\arg t = 2\pi$ and $t^{s-1} = e^{2\pi i(s-1)}x^{s-1} = e^{2\pi is}x^{s-1}$. By Exercise 13.5.6 with $s-1$ in place of s, $\int_0^\infty x^{s-1}(e^x-1)^{-1}dx = \Gamma(s)\zeta(s)$. Hence the inward upper ray contributes $-\Gamma(s)\zeta(s)$ and the outward lower ray contributes $e^{2\pi is}\Gamma(s)\zeta(s)$. Thus

$$H(s) = (e^{2\pi is} - 1)\Gamma(s)\zeta(s).$$

Now $e^{2\pi is} - 1 = 2ie^{\pi is}\sin\pi s$, and the reflection formula gives $\Gamma(s)\sin\pi s = \pi/\Gamma(1-s)$, so $(e^{2\pi is} - 1)\Gamma(s) = 2\pi i\, e^{\pi is}/\Gamma(1-s)$. Solving for ζ,

$$\boxed{\zeta(s) = \frac{e^{-\pi is}\Gamma(1-s)}{2\pi i}\,H(s)}.$$

The right side is meromorphic on the whole finite plane and agrees with the original zeta function on a nonempty open set. It therefore supplies its meromorphic continuation. Since H is entire, its only possible poles are the poles of $\Gamma(1-s)$, namely $s = 1, 2, 3, \ldots$.

At an integer $s = m$ the power t^{m-1} is single valued ($e^{2\pi i(m-1)} = 1$), so the two lips carry the same integrand in opposite directions and their contributions cancel; only the circle remains. For $m \geq 2$ the integrand is analytic at zero, because

$$\frac{t^{m-1}}{e^t - 1} = t^{m-2}\left(1 - \frac{t}{2} + O(t^2)\right).$$

Hence, by Cauchy's theorem, its circle integral is zero and $H(m) = 0$. Since H is entire, $H(s) = (s-m)K(s)$ with K entire, and $(s-m)\Gamma(1-s)$ is regular at m: this zero cancels the simple pole of $\Gamma(1-s)$, proving regularity at every $m \geq 2$.

At $s = 1$ the circle integrand is $(e^t - 1)^{-1} = t^{-1} - 1/2 + O(t)$, with residue 1, so the counterclockwise circle gives $H(1) = 2\pi i$. Writing $s = 1+\eta$ and using $\Gamma(-\eta) = \Gamma(1-\eta)/(-\eta)$ with $\Gamma(1-\eta) = 1 + O(\eta)$,

$$\Gamma(1-s) = \Gamma(-\eta) = -\frac{1}{\eta} + O(1), \qquad e^{-\pi i s} = -e^{-\pi i \eta} = -1 + O(\eta), \qquad H(s) = 2\pi i + O(\eta).$$

Substitution yields

$$\zeta(s) = \frac{[-1 + O(\eta)][-\eta^{-1} + O(1)][2\pi i + O(\eta)]}{2\pi i}, \qquad \boxed{\zeta(s) = \frac{1}{s-1} + O(1)},$$

so the pole is simple and its residue is $+1$. At nonpositive integers $\Gamma(1-s)$ is regular, giving no additional poles. For example, at $s = 0$ the lips again cancel and

$$\frac{t^{-1}}{e^t - 1} = \frac{1}{t^2}\left(1 - \frac{t}{2} + \frac{t^2}{12} + \cdots\right) = \frac{1}{t^2} - \frac{1}{2t} + \frac{1}{12} + \cdots,$$

so the residue $-1/2$ gives $H(0) = -\pi i$ and $\zeta(0) = e^0\Gamma(1)(-\pi i)/(2\pi i) = -1/2$.

An alternative argument continues ζ to $\Re s > 0$ through the quotient

$$\zeta(s) = \frac{1}{1 - 2^{1-s}} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s},$$

which holds for $\Re s > 1$ because the alternating sum equals $\zeta(s) - 2\sum_{n\geq 1}(2n)^{-s} = (1 - 2^{1-s})\zeta(s)$, and whose alternating series converges for $\Re s > 0$. That argument does not by itself prove regularity throughout $\Re s > 0$ apart from $s = 1$: the denominator $1 - 2^{1-s}$ vanishes also at $s = 1 + 2\pi i k/\ln 2$ for nonzero integers k, where $2^{1-s} = e^{-2\pi i k} = 1$. Cancellation at those points must be justified. The contour argument above, in which H is entire, establishes that all such points are regular and leaves only the pole at $s = 1$.

13.6 Other Related Functions

13.6.1 Show that $\gamma(a, x) = e^{-x} \sum_{n=0}^{\infty} \frac{(a-1)!}{(a+n)!} x^{a+n}$

(a) by repeatedly integrating by parts,

(b) by transforming it into Eq. (13.76).

Take $x > 0$ and $\Re a > 0$ for the defining lower-gamma integral; factorials mean gamma functions.

Solution.

For the defining lower incomplete-gamma integral, take $x > 0$ and $\Re a > 0$, because $e^{-t}t^{a-1}$ is integrable at $t = 0$ only for $\Re a > 0$; factorials here mean gamma functions and do not require integral a. **(a)** Integrate $e^{-t}t^{a-1}$ by parts, integrating the power and differentiating the exponential ($u = e^{-t}$, $dv = t^{a-1}dt$, $v = t^a/a$); the boundary term at $t = 0$ vanishes because $|t^a| = t^{\Re a} \to 0$:

$$\gamma(a,x) = \left[\frac{t^a e^{-t}}{a}\right]_0^x + \frac{1}{a}\int_0^x t^a e^{-t}dt = \frac{x^a e^{-x}}{a} + \frac{\gamma(a+1,x)}{a}.$$

Repeat this step N times. With $(a)_N = a(a+1)\cdots(a+N-1)$,

$$\gamma(a,x) = e^{-x}\sum_{n=0}^{N-1}\frac{x^{a+n}}{(a)_{n+1}} + \frac{\gamma(a+N,x)}{(a)_N}.$$

Indeed, $N = 1$ is the step above, and the same step with $a+N$ in place of a, divided by $(a)_N$, gives $\gamma(a+N,x)/(a)_N = x^{a+N}e^{-x}/(a)_{N+1} + \gamma(a+N+1,x)/(a)_{N+1}$, which advances N by one. Let $\sigma = \Re a > 0$. Since $|a+k| \geq \Re(a+k) = \sigma + k$, $|(a)_N| \geq (\sigma)_N$; and $|\gamma(a+N,x)| \leq \int_0^x t^{\sigma+N-1}dt = x^{\sigma+N}/(\sigma+N)$ because $e^{-t} \leq 1$. Using $(\sigma+N)(\sigma)_N = (\sigma)_{N+1} = \Gamma(\sigma+N+1)/\Gamma(\sigma)$, the final remainder has the bound

$$\left|\frac{\gamma(a+N,x)}{(a)_N}\right| \leq \frac{x^{\sigma+N}}{(\sigma+N)(\sigma)_N} = \frac{\Gamma(\sigma)x^{\sigma+N}}{\Gamma(\sigma+N+1)} \longrightarrow 0.$$

The ratio of successive bounds is $x/(\sigma+N+1) \to 0$. Since $1/(a)_{n+1} = \Gamma(a)/\Gamma(a+n+1) = (a-1)!/(a+n)!$, therefore

$$\boxed{\gamma(a,x) = e^{-x}\sum_{n=0}^{\infty}\frac{\Gamma(a)}{\Gamma(a+n+1)}x^{a+n}}.$$

(b) To transform this into Eq. (13.76), use the beta integral $B(a,n+1) = \Gamma(a)\Gamma(n+1)/\Gamma(a+n+1)$ (Section 13.3), divided by $\Gamma(n+1) = n!$:

$$\frac{\Gamma(a)}{\Gamma(a+n+1)} = \frac{1}{n!}\int_0^1 u^{a-1}(1-u)^n du.$$

The exponential series converges uniformly on $0 \leq u \leq 1$, and $u^{\Re a-1}$ is integrable. Thus, summing the exponential series $\sum_n [x(1-u)]^n/n! = e^{x(1-u)}$ in the second line and expanding e^{-xu} in the third,

$$\begin{aligned}
e^{-x}\sum_{n=0}^{\infty}\frac{\Gamma(a)x^{a+n}}{\Gamma(a+n+1)} &= x^a\int_0^1 u^{a-1}e^{-x}\sum_{n=0}^{\infty}\frac{[x(1-u)]^n}{n!}du \\
&= x^a\int_0^1 u^{a-1}e^{-xu}du \\
&= x^a\sum_{m=0}^{\infty}\frac{(-x)^m}{m!}\int_0^1 u^{a+m-1}du \\
&= \boxed{x^a\sum_{m=0}^{\infty}\frac{(-1)^m x^m}{m!(a+m)}}.
\end{aligned}$$

This is exactly the lower incomplete-gamma series in Eq. (13.76). (The second line is $\gamma(a,x)$ itself by the substitution $t = xu$, and $\int_0^1 u^{a+m-1}du = 1/(a+m)$ gives the last line.) Both series converge for every finite positive x in the stated parameter domain, although they need not be equally convenient numerically.

13.6.2 Show that

(a) $\dfrac{d^m}{dx^m}\left[x^{-a}\gamma(a,x)\right] = (-1)^m x^{-a-m}\gamma(a+m,x),$

(b) $\dfrac{d^m}{dx^m}\left[e^x\gamma(a,x)\right] = e^x\dfrac{\Gamma(a)}{\Gamma(a-m)}\gamma(a-m,x).$

Take $x > 0$ and integer $m \geq 0$. The direct integral proofs use $\Re a > 0$ in (a) and $\Re a > m$ in (b); beyond these domains continue the whole products meromorphically.

Solution.

Let m be a nonnegative integer and $x > 0$. **(a)** For $\Re a > 0$, put $t = xu$ in the defining integral, so $dt = x\,du$, $t^{a-1} = x^{a-1}u^{a-1}$, and the limits become $u = 0, 1$:

$$\gamma(a,x) = x^a\int_0^1 u^{a-1}e^{-xu}du, \qquad x^{-a}\gamma(a,x) = \int_0^1 u^{a-1}e^{-xu}du.$$

Differentiation under this finite integral is justified by the integrable weight $u^{\Re a-1}$, which bounds every x-derivative of the integrand since $0 \leq u \leq 1$. Each derivative supplies $-u$, and the same substitution with $a+m$ in place of a identifies the result:

$$\begin{aligned}\frac{d^m}{dx^m}[x^{-a}\gamma(a,x)] &= (-1)^m\int_0^1 u^{a+m-1}e^{-xu}du \\ &= \boxed{(-1)^m x^{-a-m}\gamma(a+m,x)}.\end{aligned}$$

(b) Initially take $\Re a > m$ so every lower gamma integral appearing below is defined. By the fundamental theorem of calculus $d\gamma(a,x)/dx = x^{a-1}e^{-x}$, so the product rule gives the first derivative

$$\frac{d}{dx}[e^x\gamma(a,x)] = e^x\gamma(a,x) + e^x\cdot x^{a-1}e^{-x} = e^x\gamma(a,x) + x^{a-1}.$$

The Instructor's Manual keeps a factor e^{-x} in the second term of this intermediate derivative; that factor, from $d\gamma(a,x)/dx$, cancels against e^x. The lower recurrence of Exercise 13.6.3(a), with $a-1$ in place of a, gives $\gamma(a,x) = (a-1)\gamma(a-1,x) - x^{a-1}e^{-x}$, so $e^x\gamma(a,x) = (a-1)e^x\gamma(a-1,x) - x^{a-1}$. Substitution cancels the power term:

$$\frac{d}{dx}[e^x\gamma(a,x)] = (a-1)e^x\gamma(a-1,x).$$

Apply the same formula with a replaced successively by $a-1,\ldots,a-m+1$ (each needs $\Re(a-j) > 1$, $j \leq m-1$, which holds since $\Re a > m$):

$$\frac{d^m}{dx^m}[e^x\gamma(a,x)] = (a-1)\frac{d^{m-1}}{dx^{m-1}}[e^x\gamma(a-1,x)] = \cdots,$$

and with $\Gamma(a) = (a-1)(a-2)\cdots(a-m)\Gamma(a-m)$ from $\Gamma(b+1) = b\Gamma(b)$,

$$\boxed{\frac{d^m}{dx^m}[e^x\gamma(a,x)] = e^x(a-1)(a-2)\cdots(a-m)\gamma(a-m,x) = e^x\frac{\Gamma(a)}{\Gamma(a-m)}\gamma(a-m,x)}.$$

Outside the initial integral domain, the whole product $(a-1)\cdots(a-m)\gamma(a-m,x)$ on the right must be interpreted by meromorphic continuation. A zero coefficient cannot be multiplied literally by a pole of $\gamma(a-m,x)$. For example, when $a=m=1$ the left side is $d(e^x-1)/dx = e^x$, not zero. The continued product has precisely this finite value: by the series of Eq. (13.76),

$$(a-1)\gamma(a-1,x) = x^{a-1}\left[1+(a-1)\sum_{n=1}^{\infty}\frac{(-x)^n}{n!\,(a-1+n)}\right] \longrightarrow 1 \qquad (a\to 1),$$

so the right side tends to e^x. At $m=0$ both identities are trivial: each side of (a) is $x^{-a}\gamma(a,x)$, and each side of (b) is $e^x\gamma(a,x)$ because $\Gamma(a)/\Gamma(a)=1$.

13.6.3 Show that $\gamma(a,x)$ and $\Gamma(a,x)$ satisfy the recurrence relations

(a) $\gamma(a+1,x) = a\,\gamma(a,x) - x^a e^{-x}$,

(b) $\Gamma(a+1,x) = a\,\Gamma(a,x) + x^a e^{-x}$.

Take $x>0$. The lower-integral proof starts at $\Re a>0$ and then continues meromorphically; the upper-integral identity holds for every complex a.

Solution.

(a) For $x>0$, $\Re a>0$, integrate $\gamma(a+1,x)=\int_0^x t^a e^{-t}dt$ by parts with $u=t^a$, $dv=e^{-t}dt$, so $du=at^{a-1}dt$ and $v=-e^{-t}$:

$$\begin{aligned}\gamma(a+1,x) &= \left[-t^a e^{-t}\right]_0^x + a\int_0^x t^{a-1}e^{-t}dt \\ &= \boxed{a\gamma(a,x) - x^a e^{-x}}.\end{aligned}$$

The lower boundary term vanishes because $|t^a| = t^{\Re a}\to 0$ for $\Re a>0$.

(b) The same integration by parts over $[x,\infty)$ gives, since $t^a e^{-t}\to 0$ as $t\to\infty$ for every a,

$$\begin{aligned}\Gamma(a+1,x) &= \left[-t^a e^{-t}\right]_x^\infty + a\int_x^\infty t^{a-1}e^{-t}dt \\ &= \boxed{a\Gamma(a,x) + x^a e^{-x}}.\end{aligned}$$

For fixed $x>0$ this upper-integral proof works for every complex a, because zero is outside the interval and the exponential controls infinity. Adding the two relations cancels the endpoint terms and, by $\gamma(a,x)+\Gamma(a,x)=\Gamma(a)$, recovers $\Gamma(a+1)=a\Gamma(a)$. The lower relation also continues meromorphically in a.

13.6.4 Show that the asymptotic expansion (for large x) of the incomplete gamma function $\Gamma(a,x)$ has the form

$$\Gamma(a,x) \sim x^{a-1}e^{-x}\sum_{n=0}^{\infty}\frac{\Gamma(a)}{\Gamma(a-n)}\cdot\frac{1}{x^n},$$

and that the above expression is equivalent to

$$\Gamma(a,x) \sim x^{a-1}e^{-x}\sum_{n=0}^{\infty}(a-n)_n\frac{1}{x^n}.$$

Take fixed a and $x \to +\infty$. Interpret $\Gamma(a)/\Gamma(a-n)$ as the polynomial $(a-1)\cdots(a-n)$ at gamma poles.

Solution.

For fixed a and $x \to +\infty$, integrate the upper gamma integral by parts in the direction of decreasing parameter ($u = t^{a-1}$, $dv = e^{-t}dt$; this is Exercise 13.6.3(b) with $a-1$ in place of a):

$$\Gamma(a,x) = \left[-t^{a-1}e^{-t}\right]_x^{\infty} + (a-1)\int_x^{\infty} t^{a-2}e^{-t}dt = x^{a-1}e^{-x} + (a-1)\Gamma(a-1,x).$$

After N steps this is the exact identity

$$\Gamma(a,x) = x^{a-1}e^{-x}\sum_{n=0}^{N-1}\frac{c_n(a)}{x^n} + c_N(a)\Gamma(a-N,x), \qquad c_n(a) = \prod_{j=1}^{n}(a-j), \quad c_0 = 1.$$

Indeed, the first step with $a-N$ in place of a, multiplied by $c_N(a)$, gives $c_N(a)\Gamma(a-N,x) = x^{a-1}e^{-x}c_N(a)/x^N + c_{N+1}(a)\Gamma(a-N-1,x)$, since $x^{a-N-1} = x^{a-1}/x^N$ and $c_N(a)\,(a-N-1) = c_{N+1}(a)$; this advances N by one. The remainder has the required order: putting $t = x+u$, so that $t^{a-N-1} = x^{a-N-1}(1+u/x)^{a-N-1}$, $e^{-t} = e^{-x}e^{-u}$ and the limits become $u = 0, \infty$, gives

$$\Gamma(a-N,x) = e^{-x}x^{a-N-1}\int_0^{\infty} e^{-u}(1+u/x)^{a-N-1}du.$$

For $x \geq 1$, $|(1+u/x)^{a-N-1}| = (1+u/x)^{\Re a-N-1}$ is at most 1 if the exponent is not positive, and at most $(1+u)^{\Re a-N-1}$ otherwise, because $1 \leq 1+u/x \leq 1+u$. So the integral is bounded in absolute value by $\int_0^{\infty} e^{-u}(1+u)^{\max(\Re a-N-1,0)}du < \infty$. Thus the remainder is $O(e^{-x}x^{\Re a-N-1})$, the order of the first omitted term $c_N x^{a-1-N}e^{-x}$; relative to the prefactor $x^{a-1}e^{-x}$ it is $O(x^{-N})$ for every N. This establishes the asymptotic expansion

$$\boxed{\Gamma(a,x) \sim x^{a-1}e^{-x}\sum_{n=0}^{\infty}\frac{c_n(a)}{x^n}}.$$

The proof avoids integrating term by term the binomial series of $(1+u/x)^{a-1}$ in the case $N = 0$ above. Unless a is a positive integer, that would not be justified: the series diverges for $u > x$, while the integral runs over all $u > 0$. The gamma recurrence applied n times, $\Gamma(a) = (a-1)\Gamma(a-1) = \cdots = (a-1)(a-2)\cdots(a-n)\Gamma(a-n)$, and reversal of the finite product give

$$c_n(a) = \frac{\Gamma(a)}{\Gamma(a-n)} = (a-n)(a-n+1)\cdots(a-1) = (a-n)_n,$$

which proves both requested forms. At gamma poles the coefficient is the finite polynomial $c_n(a)$, not an undefined quotient of gamma values. For example $a = 0$ gives $c_n(0) = \prod_{j=1}^{n}(-j) = (-1)^n n!$. If a is a positive integer, $c_n(a)$ contains the factor $a-a = 0$ for $n \geq a$, so the remainder $c_N\Gamma(a-N,x)$ vanishes for $N \geq a$ and the expansion terminates exactly. Otherwise the ratio of successive term magnitudes, $|c_{n+1}(a)/x^{n+1}|/|c_n(a)/x^n| = |a-n-1|/x$, behaves as n/x for fixed x, so the infinite series diverges even though every fixed truncation is asymptotic.

13.6.5 A series expansion of the incomplete beta function yields

$$B_x(p,q) = x^p\left\{\frac{1}{p} + \frac{1-q}{p+1}x + \frac{(1-q)(2-q)}{2!\,(p+2)}x^2 + \cdots \right.$$
$$\left. + \frac{(1-q)(2-q)\cdots(n-q)}{n!\,(p+n)}x^n + \cdots\right\}.$$

Given that $0 \le x \le 1$, $p > 0$, and $q > 0$, test this series for convergence. What happens at $x = 1$?

Solution.

Let the coefficient after removing x^p be

$$a_n = \frac{(1-q)_n}{n!(p+n)}, \qquad n = 0, 1, \ldots.$$

Here $(1-q)_n = (1-q)(2-q)\cdots(n-q)$ is the product in the numerators of the given series, with $(1-q)_0 = 1$. If q is a positive integer, $(1-q)_n$ contains the factor $q-q=0$ for $n \ge q$, so $(1-q)_n = 0$ there, the series is a finite polynomial times x^p, and convergence is immediate. Otherwise no such termination occurs, and, with $(1-q)_{n+1}/(1-q)_n = n+1-q$ and $n!/(n+1)! = 1/(n+1)$,

$$\frac{a_{n+1}x^{n+1}}{a_n x^n} = x\,\frac{n+1-q}{n+1}\,\frac{p+n}{p+n+1} \longrightarrow x.$$

By the ratio test the series converges absolutely for $0 \le x < 1$. In the Instructor's Manual's reverse ratio (term $n-1$)/(term n), the factor is $1/x$, not x, because the power of x drops by one; its stated large-n limit must be $1/x$.

At $x = 1$ the ratio test is inconclusive (the limit is 1), so use the gamma quotient for the coefficients. For q not a positive integer, $(1-q)_n = \Gamma(n+1-q)/\Gamma(1-q)$, and Exercise 13.4.6 (with $x = n$, $a = -q$, $b = 0$) gives $\Gamma(n+1-q)/\Gamma(n+1) \sim n^{-q}$; with $p+n \sim n$,

$$a_n = \frac{\Gamma(n+1-q)}{\Gamma(1-q)\Gamma(n+1)(p+n)} \sim \frac{n^{-q-1}}{\Gamma(1-q)}.$$

Since $q > 0$, comparison with the convergent positive series $\sum n^{-q-1}$ proves *absolute* convergence at the endpoint, regardless of the finite initial pattern of signs. In fact $\sum |a_n| < \infty$, and $|a_n x^n| \le |a_n|$ on $0 \le x \le 1$, so by the Weierstrass M-test the series is uniformly convergent there; its sum, times x^p, is continuous on $0 \le x \le 1$. The integral $B_x(p,q) = \int_0^x t^{p-1}(1-t)^{q-1}dt$ is also continuous as $x \uparrow 1$ for $p, q > 0$, and the two agree for $0 \le x < 1$. Indeed, the binomial series $(1-t)^{q-1} = \sum_{n\ge 0}\binom{q-1}{n}(-t)^n$, with $(-1)^n\binom{q-1}{n} = (1-q)_n/n!$, converges uniformly on $0 \le t \le x < 1$; multiplied by t^{p-1} and integrated term by term with $\int_0^x t^{p+n-1}dt = x^{p+n}/(p+n)$, it gives exactly the given series. Hence

$$\boxed{B_1(p,q) = \sum_{n=0}^{\infty}\frac{(1-q)_n}{n!(p+n)} = B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}}.$$

Thus the full specified interval, including $x = 1$, belongs to the convergence set. At $x = 0$ its value is zero because $p > 0$.

13.6.6 Using the definitions of the various functions, show that

(a) $\operatorname{si}(x) = \dfrac{1}{2i}\,[E_1(ix) - E_1(-ix)]$,

(b) $\operatorname{Ci}(x) = -\dfrac{1}{2}\,[E_1(ix) + E_1(-ix)]$,

(c) $E_1(ix) = -\operatorname{Ci}(x) + i\operatorname{si}(x)$.

Take $x > 0$, principal E_1, and $\operatorname{si}(x) = \operatorname{Si}(x) - \pi/2$.

Solution.

Take $x > 0$ and the principal branch of E_1. The real sine integral here is $\operatorname{si}(x) = \operatorname{Si}(x) - \pi/2$, where $\operatorname{Si}(x) = \int_0^x \sin t\,dt/t$, because $\operatorname{si}(x) = -\int_x^\infty \sin t\,dt/t$ and $\int_0^\infty \sin t\,dt/t = \pi/2$ (Exercise 13.6.13). For $\Re z > 0$ the substitution $t = zu$ turns the defining integral into $E_1(z) = \int_1^\infty e^{-zu}du/u$ (for $z > 0$ directly, then by analytic continuation). For $\varepsilon > 0$, with $t = xu$,

$$E_1((\varepsilon \pm i)x) = \int_1^\infty \frac{e^{-(\varepsilon\pm i)xu}}{u}\,du = \int_x^\infty \frac{e^{-(\varepsilon\pm i)t}}{t}\,dt.$$

The principal E_1 is analytic off the negative real axis, so $E_1(\pm ix) = \lim_{\varepsilon\downarrow 0} E_1((\varepsilon \pm i)x)$. Let $\varepsilon \downarrow 0$. The oscillatory sine and cosine tails converge by the Dirichlet test. To justify equality with the Abel limits, set $g_\varepsilon(t) = e^{-\varepsilon t}/t$, which decreases to zero. An integration by parts,

$$\int_R^\infty g_\varepsilon(t)e^{\pm it}dt = \left[\frac{g_\varepsilon(t)e^{\pm it}}{\pm i}\right]_R^\infty - \frac{1}{\pm i}\int_R^\infty g'_\varepsilon(t)e^{\pm it}dt,$$

bounds its magnitude by

$$g_\varepsilon(R) + \int_R^\infty |g'_\varepsilon(t)|dt = 2g_\varepsilon(R) \le \frac{2}{R},$$

since $\int_R^\infty |g'_\varepsilon|dt = g_\varepsilon(R)$ for a decreasing function with limit zero. This bound is uniform for $\varepsilon \ge 0$; on $[x, R]$ ordinary dominated convergence applies. First let $\varepsilon \downarrow 0$, then $R \to \infty$, to obtain the ordinary improper integrals

$$E_1(\pm ix) = \int_x^\infty \frac{e^{\mp it}}{t}\,dt = \int_x^\infty \frac{\cos t}{t}dt \mp i\int_x^\infty \frac{\sin t}{t}dt.$$

Using the definitions

$$\operatorname{si}(x) = -\int_x^\infty \frac{\sin t}{t}dt, \qquad \operatorname{Ci}(x) = -\int_x^\infty \frac{\cos t}{t}dt,$$

we obtain

$$\boxed{E_1(ix) = -\operatorname{Ci}(x) + i\operatorname{si}(x)}, \qquad E_1(-ix) = -\operatorname{Ci}(x) - i\operatorname{si}(x).$$

This proves (c). Subtracting the two identities gives $E_1(ix) - E_1(-ix) = 2i\operatorname{si}(x)$, and dividing by $2i$ proves (a); adding gives $E_1(ix) + E_1(-ix) = -2\operatorname{Ci}(x)$, and multiplying by $-1/2$ proves (b). The integrals for $E_1(\pm ix)$ converge only conditionally, and it is this limit from $\Re z > 0$ that identifies them with the principal E_1; their path to infinity (the real t-axis beyond x) may not be moved without checking the contribution of the arcs at infinity.

13.6.7 The potential produced by a $1s$ hydrogen electron is given by

$$V(r) = \frac{q}{4\pi\epsilon_0 a_0}\left[\frac{1}{2r}\,\gamma(3,2r) + \Gamma(2,2r)\right].$$

(a) For $r \ll 1$, show that

$$V(r) = \frac{q}{4\pi\epsilon_0 a_0}\left[1 - \frac{2}{3}r^2 + \cdots\right].$$

(b) For $r \gg 1$, show that

$$V(r) \sim \frac{q}{4\pi\epsilon_0 a_0}\cdot\frac{1}{r}.$$

Here r is expressed in units of a_0, the Bohr radius.

Note. $V(r)$ is illustrated in Fig. 13.8.

Solution.

Write $V(r) = [q/(4\pi\epsilon_0 a_0)]F(r)$. For positive integer parameters n the incomplete gamma functions are elementary,

$$\gamma(n,y) = (n-1)!\Big[1 - e^{-y}\sum_{s=0}^{n-1}\frac{y^s}{s!}\Big], \qquad \Gamma(n,y) = (n-1)!\,e^{-y}\sum_{s=0}^{n-1}\frac{y^s}{s!}.$$

The second follows by induction on n from $\Gamma(1,y) = \int_y^\infty e^{-t}dt = e^{-y}$ and the recurrence of Exercise 13.6.3(b):

$$\Gamma(n+1,y) = n\Gamma(n,y) + y^n e^{-y} = n!\,e^{-y}\Big[\sum_{s=0}^{n-1}\frac{y^s}{s!} + \frac{y^n}{n!}\Big];$$

the first then follows from $\gamma(n,y) = \Gamma(n) - \Gamma(n,y)$ with $\Gamma(n) = (n-1)!$. With $n = 3$, $n = 2$ and $y = 2r$ they give

$$\gamma(3,2r) = 2\left[1 - e^{-2r}(1 + 2r + 2r^2)\right], \qquad \Gamma(2,2r) = e^{-2r}(1+2r).$$

Insert these expressions before taking either limit, and use $(1 + 2r + 2r^2)/r - (1 + 2r) = 1/r + 1$:

$$F(r) = \frac{1 - e^{-2r}(1+2r+2r^2)}{r} + e^{-2r}(1+2r) = \frac{1}{r} - e^{-2r}\left(\frac{1}{r} + 1\right)$$
$$= \boxed{\frac{1 - (1+r)e^{-2r}}{r}}.$$

(a) For small positive r, multiply the exponential Taylor series $e^{-2r} = 1 - 2r + 2r^2 - \frac{4}{3}r^3 + \frac{2}{3}r^4 - \frac{4}{15}r^5 + O(r^6)$ by $1 + r$; collecting powers gives

$$(1+r)e^{-2r} = 1 - r + \frac{2}{3}r^3 - \frac{2}{3}r^4 + \frac{2}{5}r^5 + O(r^6).$$

Thus $1 - (1+r)e^{-2r} = r - \frac{2}{3}r^3 + \frac{2}{3}r^4 - \frac{2}{5}r^5 + O(r^6)$, and the apparent zero-over-zero singularity of F at $r = 0$ is removable; dividing by r,

$$F(r) = 1 - \frac{2}{3}r^2 + \frac{2}{3}r^3 - \frac{2}{5}r^4 + O(r^5), \qquad \boxed{V(r) = \frac{q}{4\pi\epsilon_0 a_0}\left[1 - \frac{2}{3}r^2 + O(r^3)\right]}.$$

In particular $F(0) = 1$ and $F'(0) = 0$.

(b) The exact expression also gives

$$F(r) = \frac{1}{r} - \left(1 + \frac{1}{r}\right) e^{-2r} = \frac{1}{r}[1 - (1 + r)e^{-2r}].$$

Therefore

$$\boxed{V(r) \sim \frac{q}{4\pi\epsilon_0 a_0 r} \quad (r \to \infty)}.$$

This is the leading Coulomb potential; its relative correction is the explicitly displayed exponentially small term $-(1 + r)e^{-2r}$. Because $F(r)$ differs from $1/r$ by $-(1 + 1/r)e^{-2r} \neq 0$, the printed equality in (b) holds only asymptotically, hence the $\sim$ in the statement.

13.6.8 The potential produced by a $2p$ hydrogen electron can be shown to be

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{24a_0} \left[\frac{1}{r}\gamma(5, r) + \Gamma(4, r)\right] - \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{120a_0} \left[\frac{1}{r^3}\gamma(7, r) + r^2\Gamma(2, r)\right] P_2(\cos\theta).$$

Here r is expressed in units of a_0, the Bohr radius. $P_2(\cos\theta)$ is a Legendre polynomial (Section 15.1).

(a) For $r \ll 1$, show that

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a_0} \left[\frac{1}{4} - \frac{1}{120}r^2 P_2(\cos\theta) + \cdots\right].$$

(b) For $r \gg 1$, show that

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a_0 r} \left[1 - \frac{6}{r^2} P_2(\cos\theta) + \cdots\right].$$

Solution.

Factor out $C = q/(4\pi\epsilon_0 a_0)$ and write $V = C[A_0(r) - A_2(r)P_2(\cos\theta)]$, where

$$A_0 = \frac{\gamma(5, r)/r + \Gamma(4, r)}{24}, \qquad A_2 = \frac{\gamma(7, r)/r^3 + r^2\Gamma(2, r)}{120}.$$

(a) Integrating the exponential power series, $\gamma(j, r) = \int_0^r t^{j-1}(1 - t + O(t^2))dt$, gives

$$\gamma(j, r) = \frac{r^j}{j} - \frac{r^{j+1}}{j + 1} + O(r^{j+2}), \qquad \Gamma(j, r) = (j - 1)! - \gamma(j, r).$$

Thus $\gamma(5, r)/r = r^4/5 + O(r^5)$, $\Gamma(4, r) = 6 - r^4/4 + O(r^5)$, $\gamma(7, r)/r^3 = r^4/7 + O(r^5)$ and $\Gamma(2, r) = 1 - r^2/2 + O(r^3)$. Consequently

$$24A_0 = \frac{r^4}{5} + O(r^5) + 6 - \frac{r^4}{4} + O(r^5) = 6 - \frac{r^4}{20} + O(r^5),$$
$$120A_2 = \frac{r^4}{7} + O(r^5) + r^2\left[1 - \frac{r^2}{2} + O(r^3)\right] = r^2 - \frac{5}{14}r^4 + O(r^5).$$

Divide by the normalizing constants 24 and 120, using $\frac{1}{20}/24 = \frac{1}{480}$ and $\frac{5}{14}/120 = \frac{1}{336}$:

$$A_0 = \frac{1}{4} - \frac{r^4}{480} + O(r^5), \qquad A_2 = \frac{r^2}{120} - \frac{r^4}{336} + O(r^5).$$

Substituting in $V = C[A_0 - A_2 P_2(\cos\theta)]$ gives

$$\boxed{V = C\left[\frac{1}{4} - \frac{r^2}{120}P_2(\cos\theta) + r^4\left(-\frac{1}{480} + \frac{P_2(\cos\theta)}{336}\right) + O(r^5)\right].}$$

This includes the requested leading terms and identifies the first omitted orders.

(b) At integer j, as in Exercise 13.6.7, $\Gamma(j,r) = (j-1)!e^{-r}\sum_{k=0}^{j-1} r^k/k!$. Write $S_j(r) = \sum_{k=0}^{j} r^k/k!$, so $\Gamma(j,r) = (j-1)!e^{-r}S_{j-1}(r)$. Substituting these finite upper-gamma formulas and $\gamma(j,r) = (j-1)! - \Gamma(j,r)$ gives

$$24A_0 = \frac{24 - 24e^{-r}S_4}{r} + 6e^{-r}S_3, \qquad 120A_2 = \frac{720 - 720e^{-r}S_6}{r^3} + r^2e^{-r}(1+r),$$

that is,

$$A_0 = \frac{1}{r} - e^{-r}\left[\frac{S_4(r)}{r} - \frac{S_3(r)}{4}\right], \qquad A_2 = \frac{6}{r^3} - e^{-r}\left[\frac{6S_6(r)}{r^3} - \frac{r^2(1+r)}{120}\right].$$

For the first bracket, $S_4/r = 1/r + 1 + r/2 + r^2/6 + r^3/24$ and $S_3/4 = 1/4 + r/4 + r^2/8 + r^3/24$. Their difference is $1/r + 3/4 + r/4 + r^2/24$. For the second bracket,

$$\frac{6S_6}{r^3} = \frac{6}{r^3} + \frac{6}{r^2} + \frac{3}{r} + 1 + \frac{r}{4} + \frac{r^2}{20} + \frac{r^3}{120};$$

subtracting $r^2/120 + r^3/120$ leaves the r^2 coefficient $1/20 - 1/120 = 1/24$ and cancels the r^3 term. Thus the exact expressions are

$$A_0 = \frac{1}{r} - e^{-r}\left(\frac{1}{r} + \frac{3}{4} + \frac{r}{4} + \frac{r^2}{24}\right),$$
$$A_2 = \frac{6}{r^3} - e^{-r}\left(\frac{6}{r^3} + \frac{6}{r^2} + \frac{3}{r} + 1 + \frac{r}{4} + \frac{r^2}{24}\right).$$

Hence $V = C[A_0 - A_2P_2] = C[1/r - (6/r^3)P_2]$ plus exponential terms of size $O(r^2e^{-r})$ (the largest power in the parentheses is r^2), which become $O(r^3e^{-r})$ when the factor $1/r$ is taken out. Thus

$$\boxed{V = \frac{C}{r}\left[1 - \frac{6}{r^2}P_2(\cos\theta) + O(r^3e^{-r})\right].}$$

The monopole and quadrupole coefficients are fixed by $\Gamma(5) = 24$ and $\Gamma(7) = 720$, respectively; $720/120 = 6$ explains the second coefficient.

13.6.9 Prove that the exponential integral has the expansion

$$\int_x^\infty \frac{e^{-t}}{t}\,dt = -\gamma - \ln x - \sum_{n=1}^{\infty}\frac{(-1)^n x^n}{n\cdot n!},$$

where γ is the Euler-Mascheroni constant.

Take $x > 0$ for this real improper integral and logarithm.

Solution.

Take $x > 0$. For $a > 0$, $\Gamma(a,x) = \Gamma(a) - \gamma(a,x)$; insert the lower incomplete-gamma series $\gamma(a,x) = x^a \sum_{n\geq 0}(-1)^n x^n/[n!(a+n)]$ of Eq. (13.76) and separate its $n=0$ term x^a/a:

$$\Gamma(a,x) = \left[\Gamma(a) - \frac{x^a}{a}\right] - x^a \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n!\,(a+n)}.$$

Let $a \downarrow 0$. On the left, $\Gamma(a,x) \to \Gamma(0,x) = E_1(x)$ by dominated convergence, since for $0 < a \leq 1$ the integrand $t^{a-1}e^{-t}$ is at most $\max(1,1/x)e^{-t}$ on $[x,\infty)$. The series $\sum_{n\geq 1}(-1)^n x^n/[n!(a+n)]$ is uniformly convergent for $|a| \leq \frac{1}{2}$, because there $|a+n| \geq n/2$ and its terms are bounded by $2x^n/(n\,n!)$; so it may be evaluated termwise at $a = 0$, while its prefactor $x^a \to 1$. Hence

$$E_1(x) = \lim_{a\to 0}\left[\Gamma(a) - \frac{x^a}{a}\right] - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n\,n!}.$$

In the singular pair, use $a\Gamma(a) = \Gamma(1+a)$: $\Gamma(a) - x^a/a = [\Gamma(1+a) - x^a]/a$. The numerator vanishes at $a = 0$, so the limit is its derivative there; with $dx^a/da = x^a \ln x$ and $\Gamma'(1) = \Gamma(1)\psi(1) = -\gamma$,

$$\lim_{a\to 0} \frac{\Gamma(1+a) - x^a}{a} = \Gamma'(1) - \ln x = \Gamma(1)\psi(1) - \ln x = -\gamma - \ln x.$$

The evaluation subscript in the textbook's Eq. (13.82) must be $a = 0$, not $a = 1$; the limit being evaluated is $a \to 0$. Therefore

$$\boxed{E_1(x) = \int_x^{\infty} \frac{e^{-t}}{t}\,dt = -\gamma - \ln x - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n\,n!}}.$$

The ratio of successive absolute series terms is $xn/(n+1)^2 \to 0$, so the series converges for every finite x. As an independent differential check,

$$\frac{d}{dx}\left[-\gamma - \ln x - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n\,n!}\right] = -\frac{1}{x} - \frac{e^{-x}-1}{x} = -\frac{e^{-x}}{x},$$

because $\sum_{n\geq 1}(-1)^n x^{n-1}/n! = (e^{-x}-1)/x$. This is the derivative of the defining integral, so the check confirms the right side up to an additive constant, the $-\gamma$ found above. For complex x, use the principal $\operatorname{Log} x$ on the plane cut along the negative real axis; zero remains a logarithmic singularity.

13.6.10 Show that $E_1(z)$ may be written as

$$E_1(z) = e^{-z} \int_0^{\infty} \frac{e^{-zt}}{1+t}\,dt.$$

Show also that we must impose the condition $|\arg z| \leq \pi/2$ and $z \neq 0$; convergence on the imaginary axis is conditional.

Solution.

Initially let $\Re z > 0$. The scaled exponential integral is

$$E_1(z) = \int_1^{\infty} \frac{e^{-zu}}{u}\,du.$$

For positive real z this follows directly from $v = zu$ in $\int_z^\infty e^{-v}dv/v$; both sides are analytic in the right half-plane, so the equality holds there by analytic continuation. Set $u = 1 + t$, with $du = dt$ and new limits $t = 0, \infty$, so that $e^{-zu} = e^{-z}e^{-zt}$, to obtain

$$\boxed{E_1(z) = e^{-z}\int_0^\infty \frac{e^{-zt}}{1+t}dt}.$$

If $\Re z > 0$, the integral is absolutely convergent, since $|e^{-zt}|/(1+t) \le e^{-t\Re z}$. If $z = ib$, $b \neq 0$, the primitive $e^{-ibt}/(-ib)$ of e^{-ibt} is bounded and $(1+t)^{-1}$ decreases to zero, so the Dirichlet test gives conditional convergence. Taking the limit from $\Re z > 0$ identifies this value with the principal E_1 on the imaginary axis: with $z = \varepsilon + ib$, $\varepsilon \downarrow 0$, the integration by parts of Exercise 13.6.6, applied to the decreasing function $e^{-\varepsilon t}/(1+t)$, bounds the tail beyond R by $2/[|b|(1+R)]$ uniformly in $\varepsilon \ge 0$, so the limit may be taken under the integral, and E_1 is continuous at ib. At $z = 0$ the integral is $\int_0^\infty dt/(1+t) = \lim_{R\to\infty}\ln(1+R) = \infty$, matching the logarithmic singularity of E_1 there.

If $\Re z < 0$, even conditional convergence fails. Fix $L > 0$; then $|e^{-zL}| = e^{L|\Re z|} > 1$, so $1 - e^{-zL} \neq 0$. For t in $[R, R+L]$, $|1/(1+t) - 1/R| = |R - 1 - t|/[R(1+t)] \le (L+1)/R^2$ and $|e^{-zt}| = e^{t|\Re z|} \le e^{L|\Re z|}|e^{-zR}|$, so replacing $1/(1+t)$ by $1/R$ changes the integral over $[R, R+L]$ by at most $L(L+1)e^{L|\Re z|}|e^{-zR}|/R^2$, that is, by $|e^{-zR}/R|$ times $O(R^{-1})$. With $\int_R^{R+L} e^{-zt}dt = e^{-zR}(1 - e^{-zL})/z$, the integral over $[R, R+L]$ therefore satisfies

$$\int_R^{R+L}\frac{e^{-zt}}{1+t}dt = \frac{e^{-zR}}{R}\left[\frac{1-e^{-zL}}{z} + O(R^{-1})\right].$$

Since $|e^{-zR}| = e^{R|\Re z|}$ and the bracket tends to $(1 - e^{-zL})/z \neq 0$, its magnitude grows exponentially, contradicting the Cauchy criterion for convergence of an improper integral. Therefore

$$\boxed{|\arg z| \le \frac{\pi}{2}, \qquad z \neq 0}$$

is the convergence domain of this particular representation. The exclusion $z = 0$ is essential, and the boundary $|\arg z| = \pi/2$ is conditional rather than absolute convergence. The function itself has analytic continuation beyond this integral's convergence sector.

13.6.11 Related to the exponential integral by a simple change of variable is the function

$$E_n(x) = \int_1^\infty \frac{e^{-xt}}{t^n}\,dt.$$

Show that $E_n(x)$ satisfies the recurrence relation

$$E_{n+1}(x) = \frac{1}{n}e^{-x} - \frac{x}{n}E_n(x), \quad n = 1, 2, 3, \cdots.$$

Take $x > 0$ for the displayed recurrence. At $x = 0$, use the limiting interpretation when $n = 1$.

Solution.

For $x > 0$, integrate $E_{n+1}(x)$ by parts with $u = e^{-xt}$ and $dv = t^{-n-1}dt$, so $du = -xe^{-xt}dt$ and $v = -t^{-n}/n$:

$$\begin{aligned} E_{n+1}(x) &= \left[-\frac{e^{-xt}}{nt^n}\right]_1^\infty - \frac{x}{n}\int_1^\infty \frac{e^{-xt}}{t^n}dt \\ &= \boxed{\frac{e^{-x}}{n} - \frac{x}{n}E_n(x)}, \qquad n = 1, 2, \ldots. \end{aligned}$$

The upper boundary term vanishes because $e^{-xt} \to 0$ for $x > 0$, and the lower one is $+e^{-x}/n$. When $n > 1$, the identity also holds at $x = 0$ using the finite values of both integrals. For $n = 1$, $E_1(0)$ is divergent, so the extension must instead use the limit $xE_1(x) = -\gamma x - x \ln x + O(x^2) \to 0$, from the series of Exercise 13.6.9 and $x \ln x \to 0$. It then gives $E_2(0) = \lim_{x \downarrow 0}[e^{-x} - xE_1(x)] = 1$ without assigning a value to $0 \cdot E_1(0)$.

13.6.12 With $E_n(x)$ as defined in Exercise 13.6.11, show that for $n > 1$,

$$E_n(0) = 1/(n-1).$$

Solution.

At $x = 0$ the exponential is one. For $n > 1$ evaluate the improper integral through a finite upper limit:

$$E_n(0) = \lim_{R \to \infty} \int_1^R t^{-n} dt = \lim_{R \to \infty} \frac{R^{1-n} - 1}{1-n} = \boxed{\frac{1}{n-1}}.$$

The exponent $1 - n$ is negative, making $R^{1-n} \to 0$. At $n = 1$ the integral is $\ln R \to \infty$, while for $n < 1$ it diverges as a positive power of R. Thus the stated real condition $n > 1$ is necessary as well as sufficient. Dominated convergence also shows $E_n(x) \to E_n(0)$ as $x \downarrow 0$ for these n, since $0 \le e^{-xt} t^{-n} \le t^{-n}$.

13.6.13 Develop the following power-series expansions:

(a) $\operatorname{si}(x) = -\dfrac{\pi}{2} + \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!}$,

(b) $\operatorname{Ci}(x) = \gamma + \ln x + \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2n(2n)!}$.

Use $x > 0$ for the displayed real logarithm; the sine-integral series also has its finite limit at zero.

Solution.

(a) For $x > 0$, split $\operatorname{si}(x) = -\int_x^\infty \sin t\, dt/t = -\int_0^\infty \sin t\, dt/t + \int_0^x \sin t\, dt/t$, and use the convergent Dirichlet integral $\int_0^\infty \sin t\, dt/t = \pi/2$ to write

$$\operatorname{si}(x) = -\frac{\pi}{2} + \int_0^x \frac{\sin t}{t} dt.$$

The constant $\pi/2$ follows by Abel regularization: $J(\varepsilon) = \int_0^\infty e^{-\varepsilon t} \sin t\, dt/t$, $\varepsilon > 0$, may be differentiated under the integral (the derivative of the integrand is bounded by $e^{-\varepsilon_0 t}$ for $\varepsilon \ge \varepsilon_0 > 0$), so

$$J'(\varepsilon) = -\int_0^\infty e^{-\varepsilon t} \sin t\, dt = -\Im \int_0^\infty e^{-(\varepsilon - i)t} dt = -\Im \frac{1}{\varepsilon - i} = -\frac{1}{1+\varepsilon^2}.$$

Since $|\sin t/t| \le 1$, $|J(\varepsilon)| \le 1/\varepsilon$, so $J(\infty) = 0$; hence $J(\varepsilon) = \int_\varepsilon^\infty d\eta/(1+\eta^2) = \arctan(1/\varepsilon)$ and $J(0^+) = \pi/2$. The tail bound of Exercise 13.6.6 (uniform in $\varepsilon \ge 0$) shows that $J(0^+)$ is the improper integral $\int_0^\infty \sin t\, dt/t$. Now the sine series divided by t, $\sin t/t = \sum_{n \ge 0} (-1)^n t^{2n}/(2n+1)!$, is uniformly convergent on every finite interval. Integrating term by term, with $\int_0^x t^{2n} dt = x^{2n+1}/(2n+1)$, gives

$$\boxed{\operatorname{si}(x) = -\frac{\pi}{2} + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!}}.$$

At $x = 0$ the series gives $\mathrm{si}(0) = -\pi/2 = -\int_0^\infty \sin t\, dt/t$, so (a) holds there too.

(b) Exercise 13.6.6 gives $\mathrm{Ci}(x) = -[E_1(ix) + E_1(-ix)]/2$. Insert the series from Exercise 13.6.9, $E_1(z) = -\gamma - \mathrm{Log}\, z - \sum_{n\ge1}(-1)^n z^n/(n\, n!)$, which holds at $z = \pm ix$ by analytic continuation, both sides being analytic in the plane cut along the negative real axis. The logarithms satisfy

$$\mathrm{Log}(ix) = \ln x + i\pi/2, \qquad \mathrm{Log}(-ix) = \ln x - i\pi/2,$$

so their imaginary constants cancel and their sum is $2\ln x$. In the power series, $(ix)^n + (-ix)^n = i^n x^n[1 + (-1)^n]$ vanishes for odd n, while for $n = 2m$ it is $2i^{2m}x^{2m} = 2(-1)^m x^{2m}$ and the sign $(-1)^n = (-1)^{2m} = 1$. Hence

$$E_1(ix) + E_1(-ix) = -2\gamma - 2\ln x - 2\sum_{m=1}^{\infty} \frac{(-1)^m x^{2m}}{2m\,(2m)!}.$$

Multiplying by $-1/2$ and renaming m as n,

$$\boxed{\mathrm{Ci}(x) = \gamma + \ln x + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2n(2n)!}}.$$

Both displayed power series have infinite radius of convergence; Ci itself retains the explicitly separated logarithmic singularity. The formulas for real logarithms here use $x > 0$; complex continuation requires the same chosen logarithm branch.

13.6.14 An analysis of a center-fed linear antenna leads to the expression

$$\int_0^x \frac{1 - \cos t}{t}\, dt.$$

Show that this is equal to $\gamma + \ln x - \mathrm{Ci}(x)$.

Take $x > 0$ in the displayed real-logarithm identity; extend continuously at zero.

Solution.

For $x > 0$, Exercise 13.6.13(b) gives $\mathrm{Ci}(x) = \gamma + \ln x + \sum_{n\ge1}(-1)^n x^{2n}/[2n(2n)!]$. Since $\int_0^x t^{2n-1}dt = x^{2n}/(2n)$, the series is the termwise integral of $(\cos t - 1)/t = \sum_{n\ge1}(-1)^n t^{2n-1}/(2n)!$; termwise integration is allowed because this power series converges uniformly on $[0, x]$. Hence

$$\mathrm{Ci}(x) = \gamma + \ln x + \int_0^x \frac{\cos t - 1}{t} dt.$$

The integral is finite at its lower endpoint because $(1 - \cos t)/t = t/2 + O(t^3)$. Moving the integral to the other side proves

$$\boxed{\int_0^x \frac{1 - \cos t}{t} dt = \gamma + \ln x - \mathrm{Ci}(x)}.$$

One can also check the identity by differentiation: $\mathrm{Ci}'(x) = \cos x/x$, so the derivative of the right side is exactly $(1 - \cos x)/x$, the derivative of the left side, and both sides tend to zero as $x \downarrow 0$, the right side because $\mathrm{Ci}(x) - \gamma - \ln x \to 0$. For negative real x, the substitution $t = -s$ gives $\int_0^x (1 - \cos t)dt/t = \int_0^{|x|}(1 - \cos s)ds/s$, so the original integral is even in x, and its real form is $\gamma + \ln|x| - \mathrm{Ci}(|x|)$. At zero the continuous value is zero.

13.6.15 Using the relation

$$\Gamma(a) = \gamma(a,x) + \Gamma(a,x),$$

show that if $\gamma(a,x)$ satisfies the relations of Exercise 13.6.2, then $\Gamma(a,x)$ must satisfy the same relations.

Take $x > 0$ and integer $m \geq 0$; interpret the gamma quotient as the finite polynomial $(a-1)\cdots(a-m)$.

Solution.

Initially take $x > 0$, $m \geq 0$ integral and $\Re a > m$, so all complete and lower gamma factors used separately below are regular. Write $\Gamma(a,x) = \Gamma(a) - \gamma(a,x)$.

(a) Direct differentiation of the complete-gamma term, with $d^m x^{-a}/dx^m = (-a)(-a-1)\cdots(-a-m+1)x^{-a-m} = (-1)^m (a)_m x^{-a-m}$ and $(a)_m \Gamma(a) = \Gamma(a+m)$, gives

$$\frac{d^m}{dx^m}[x^{-a}\Gamma(a)] = (-1)^m (a)_m x^{-a-m}\Gamma(a) = (-1)^m x^{-a-m}\Gamma(a+m).$$

Subtract the corresponding lower-gamma derivative from Exercise 13.6.2(a):

$$\begin{aligned}\frac{d^m}{dx^m}[x^{-a}\Gamma(a,x)] &= (-1)^m x^{-a-m}[\Gamma(a+m) - \gamma(a+m,x)] \\ &= \boxed{(-1)^m x^{-a-m}\Gamma(a+m,x)}.\end{aligned}$$

(b) Every derivative of $e^x\Gamma(a)$ equals $e^x\Gamma(a)$. Put $P_m(a) = (a-1)\cdots(a-m)$, so $\Gamma(a) = P_m(a)\Gamma(a-m)$. Subtraction of Exercise 13.6.2(b) then gives

$$\begin{aligned}\frac{d^m}{dx^m}[e^x\Gamma(a,x)] &= e^x\Gamma(a) - e^x P_m(a)\gamma(a-m,x) \\ &= e^x P_m(a)[\Gamma(a-m) - \gamma(a-m,x)] \\ &= \boxed{e^x \frac{\Gamma(a)}{\Gamma(a-m)}\Gamma(a-m,x)}.\end{aligned}$$

For fixed $x > 0$, upper incomplete gamma functions are entire in their parameter, because the integral over $[x,\infty)$, which avoids $t = 0$, converges uniformly for a in compact sets. Thus both identities extend to every complex a, with the quotient in (b) interpreted as the polynomial $P_m(a)$ at exceptional gamma arguments. For instance $a = 1$, $m = 1$ gives zero on both sides of (b), since $e^x\Gamma(1,x) = 1$.

13.6.16 For $x > 0$, show that

$$\int_x^\infty \frac{t^n\,dt}{e^t - 1} = \sum_{k=1}^\infty e^{-kx}\left[\frac{x^n}{k} + \frac{nx^{n-1}}{k^2} + \frac{n(n-1)x^{n-2}}{k^3} + \cdots + \frac{n!}{k^{n+1}}\right].$$

Take $n = 0, 1, 2, \ldots$ so the bracket terminates.

Solution.

The terminating bracket requires a nonnegative integer n; for other n the binomial expansion of $(x+u)^n$ used below does not terminate. For $t \geq x > 0$, expand the positive geometric series:

$$\frac{1}{e^t - 1} = \sum_{k=1}^\infty e^{-kt}.$$

Tonelli's theorem permits interchange of sum and integral. In each term set $t = x + u$, so $e^{-kt} = e^{-kx}e^{-ku}$ and the limits become $u = 0, \infty$, and expand the finite binomial:

$$\begin{aligned}\int_x^\infty \frac{t^n}{e^t - 1}dt &= \sum_{k=1}^{\infty} e^{-kx} \int_0^\infty (x+u)^n e^{-ku}du \\ &= \sum_{k=1}^{\infty} e^{-kx} \sum_{j=0}^{n} \binom{n}{j} x^{n-j} \int_0^\infty u^j e^{-ku}du.\end{aligned}$$

Substitution $v = ku$ gives $\int_0^\infty u^j e^{-ku}du = k^{-j-1}\int_0^\infty v^j e^{-v}dv = j!/k^{j+1}$. Cancel $j!$ against the binomial coefficient, $\binom{n}{j}j! = n!/(n-j)!$, to obtain

$$\boxed{\int_x^\infty \frac{t^n}{e^t - 1}dt = \sum_{k=1}^{\infty} e^{-kx} \sum_{j=0}^{n} \frac{n!}{(n-j)!}\frac{x^{n-j}}{k^{j+1}}}.$$

Writing the finite inner sum out, its terms $j = 0, 1, 2, \ldots, n$ are x^n/k, nx^{n-1}/k^2, $n(n-1)x^{n-2}/k^3$, $\ldots$, $n!/k^{n+1}$, exactly the bracket in the question. For fixed $x > 0$ its outer sum converges absolutely by its exponential factor. In the boundary case $n = 0$ it is $\sum_{k=1}^{\infty} e^{-kx}/k = -\ln(1 - e^{-x})$ (the series $-\ln(1-w) = \sum_k w^k/k$ at $w = e^{-x} < 1$), also obtained by direct integration: $(e^t-1)^{-1} = e^{-t}/(1-e^{-t})$ has the primitive $\ln(1 - e^{-t})$, which tends to zero as $t \to \infty$.

14 Chapter 14. Bessel Functions

Problems

14.1 Bessel Functions of the First Kind, $J_\nu(x)$

14.1.1 From the product of the generating functions $g(x,t)g(x,-t)$, show that

$$1 = [J_0(x)]^2 + 2[J_1(x)]^2 + 2[J_2(x)]^2 + \cdots$$

and therefore that $|J_0(x)| \le 1$ and $|J_n(x)| \le 1/\sqrt{2}$, $n = 1, 2, 3, \ldots$.

Hint. Use uniqueness of power series, (Section 1.2).

Take x real for the two inequalities.

Solution.

The inequalities require real x. The identity of squares also holds for complex x, but those squares are then not nonnegative. First obtain the negative-order relation from the generating function $g(x,t) = e^{(x/2)(t-1/t)} = \sum_m J_m(x)t^m$. Both $g(x,1/t)$ and $g(x,-t)$ equal $e^{(x/2)(1/t-t)}$; expanding the first in powers of $1/t$ (and renaming $m \to -m$) and the second in powers of $-t$ gives

$$g(x,1/t) = g(x,-t), \qquad \sum_{m=-\infty}^{\infty} J_{-m}(x)t^m = \sum_{m=-\infty}^{\infty} (-1)^m J_m(x)t^m.$$

Uniqueness of Laurent coefficients gives $J_{-m} = (-1)^m J_m$. Next, the exponents $\frac{x}{2}(t - t^{-1})$ of $g(x,t)$ and $\frac{x}{2}(-t + t^{-1})$ of $g(x,-t)$ cancel, so $g(x,t)g(x,-t) = 1$. Multiply the two absolutely convergent Laurent expansions:

$$1 = g(x,t)g(x,-t) = \sum_{m,n\in\mathbb{Z}} J_m(x)J_n(x)(-1)^n t^{m+n}.$$

The left side is the constant 1, so by uniqueness of Laurent coefficients the coefficient of t^0 on the right is 1. It comes from $n = -m$, and $(-1)^{-m}J_{-m} = (-1)^{-m}(-1)^m J_m = J_m$. Thus

$$\begin{aligned} 1 &= \sum_{m\in\mathbb{Z}} (-1)^{-m} J_m J_{-m} = \sum_{m\in\mathbb{Z}} J_m^2 \\ &= J_0^2 + \sum_{m=1}^{\infty} (J_m^2 + J_{-m}^2) = \boxed{J_0^2 + 2\sum_{m=1}^{\infty} J_m^2}. \end{aligned}$$

For real x, every summand is nonnegative. Dropping all terms but J_0^2 gives $J_0^2 \le 1$; retaining only $2J_n^2$ gives $2J_n^2 \le 1$. Taking nonnegative square roots proves the two bounds.

14.1.2 Using a generating function $g(x,t) = g(u+v,t) = g(u,t)g(v,t)$, show that

(a) $J_n(u+v) = \sum_{s=-\infty}^{\infty} J_s(u)J_{n-s}(v)$,

(b) $J_0(u+v) = J_0(u)J_0(v) + 2\sum_{s=1}^{\infty} J_s(u)J_{-s}(v)$.

These are addition theorems for the Bessel functions.

Solution.

The exponent is linear in its first argument, so

$$e^{(u+v)(t-t^{-1})/2} = e^{u(t-t^{-1})/2} e^{v(t-t^{-1})/2}.$$

Insert the Laurent series on both sides:

$$\sum_{n\in\mathbb{Z}} J_n(u+v)t^n = \sum_{s,r\in\mathbb{Z}} J_s(u)J_r(v)t^{s+r}.$$

(a) To obtain t^n on the right, set $r = n - s$. Coefficient comparison gives

$$\boxed{J_n(u+v) = \sum_{s=-\infty}^{\infty} J_s(u)J_{n-s}(v).}$$

(b) Set $n = 0$ and separate $s = 0$ from the pairs $s, -s$:

$$J_0(u+v) = J_0(u)J_0(v) + \sum_{s=1}^{\infty} \big[J_s(u)J_{-s}(v) + J_{-s}(u)J_s(v)\big].$$

Because $J_{-s}(z) = (-1)^s J_s(z)$ (Exercise 14.1.1), the two terms inside each bracket are equal, both being $(-1)^s J_s(u)J_s(v)$. Hence

$$\boxed{J_0(u+v) = J_0(u)J_0(v) + 2\sum_{s=1}^{\infty} J_s(u)J_{-s}(v).}$$

Absolute convergence on any circle $0 < |t| < \infty$ justifies the product and coefficient collection; alternatively they follow by multiplying the two convergent exponential series defining the generating function.

14.1.3 Using only the generating function

$$e^{(x/2)(t-1/t)} = \sum_{n=-\infty}^{\infty} J_n(x)t^n$$

and not the explicit series form of $J_n(x)$, show that $J_n(x)$ has odd or even parity according to whether n is odd or even, that is,

$$J_n(x) = (-1)^n J_n(-x).$$

Solution.

Changing both signs leaves the exponent unchanged:

$$g(-x,-t) = \exp\left[\frac{-x}{2}\left(-t + \frac{1}{t}\right)\right] = g(x,t).$$

Expand these equal functions in powers of t:

$$\sum_{n\in\mathbb{Z}} J_n(x)t^n = \sum_{n\in\mathbb{Z}} J_n(-x)(-t)^n = \sum_{n\in\mathbb{Z}} (-1)^n J_n(-x)t^n.$$

For every integer n, equality of Laurent coefficients gives

$$\boxed{J_n(x) = (-1)^n J_n(-x).}$$

Thus even integer orders give even functions of x, and odd integer orders give odd functions. No explicit coefficient series for J_n was used.

14.1.4 Use the basic recurrence formulas, Eqs. (14.7) and (14.8), to prove the following formulas:

(a) $\dfrac{d}{dx}\big[x^n J_n(x)\big] = x^n J_{n-1}(x)$,

(b) $\dfrac{d}{dx}\big[x^{-n} J_n(x)\big] = -x^{-n} J_{n+1}(x)$,

(c) $J_n(x) = J'_{n+1} + \dfrac{n+1}{x} J_{n+1}(x)$.

Solution.

The two given recurrences are

$$J_{n-1} + J_{n+1} = \frac{2n}{x} J_n, \qquad J_{n-1} - J_{n+1} = 2J'_n.$$

Adding them gives $2J_{n-1} = 2nJ_n/x + 2J'_n$; subtracting the second from the first gives $2J_{n+1} = 2nJ_n/x - 2J'_n$. After division by 2,

$$J'_n + \frac{n}{x} J_n = J_{n-1}, \qquad J'_n - \frac{n}{x} J_n = -J_{n+1}.$$

(a) The product rule and the first identity yield

$$\frac{d}{dx}(x^n J_n) = nx^{n-1} J_n + x^n J'_n = x^n \left(J'_n + \frac{n}{x} J_n\right) = \boxed{x^n J_{n-1}}.$$

(b) Using the second identity instead,

$$\frac{d}{dx}(x^{-n} J_n) = -nx^{-n-1} J_n + x^{-n} J'_n = x^{-n} \left(J'_n - \frac{n}{x} J_n\right) = \boxed{-x^{-n} J_{n+1}}.$$

(c) Replace n by $n+1$ in the first identity:

$$\boxed{J_n = J'_{n+1} + \frac{n+1}{x} J_{n+1}.}$$

The manipulations hold at $x \neq 0$; any expression with a removable singularity at zero is extended there by its series limit.

14.1.5 Derive the Jacobi-Anger expansion

$$e^{i\rho\cos\varphi} = \sum_{m=-\infty}^{\infty} i^m J_m(\rho) e^{im\varphi}.$$

This is an expansion of a plane wave in a series of cylindrical waves.

Solution.

In the generating function, put $x = \rho$ and $t = ie^{i\varphi}$. Since $t^{-1} = -ie^{-i\varphi}$ (because $1/i = -i$),

$$t - t^{-1} = i(e^{i\varphi} + e^{-i\varphi}) = 2i\cos\varphi,$$

so the exponent of the generating function is $(\rho/2)(t - t^{-1}) = i\rho\cos\varphi$. Therefore

$$e^{i\rho\cos\varphi} = \sum_{m=-\infty}^{\infty} J_m(\rho)(ie^{i\varphi})^m$$

$$= \boxed{\sum_{m=-\infty}^{\infty} i^m J_m(\rho) e^{im\varphi}}.$$

The factors $e^{im\varphi}$ are the angular cylindrical modes; their radial coefficients are $J_m(\rho)$.

14.1.6 Show that

(a) $\cos x = J_0(x) + 2\sum_{n=1}^{\infty}(-1)^n J_{2n}(x)$,

(b) $\sin x = 2\sum_{n=0}^{\infty}(-1)^n J_{2n+1}(x)$.

Solution.

Set $\varphi = 0$, $\rho = x$ in Exercise 14.1.5:

$$e^{ix} = \sum_{m\in\mathbb{Z}} i^m J_m(x).$$

Pair the terms of orders m and $-m$. Since $i^{-m}J_{-m} = i^{-m}(-1)^m J_m = (-i)^m(-1)^m J_m = i^m J_m$, the terms of orders $\pm m$ are equal. Then split the sum into even orders $m = 2n$, where $i^{2n} = (-1)^n$, and odd orders $m = 2n+1$, where $i^{2n+1} = i(-1)^n$:

$$e^{ix} = J_0 + 2\sum_{m=1}^{\infty} i^m J_m = J_0 + 2\sum_{n=1}^{\infty}(-1)^n J_{2n} + 2i\sum_{n=0}^{\infty}(-1)^n J_{2n+1}.$$

For real x every $J_m(x)$ is real, so comparison of the real and imaginary parts with $e^{ix} = \cos x + i\sin x$ gives

$$\boxed{\cos x = J_0(x) + 2\sum_{n=1}^{\infty}(-1)^n J_{2n}(x)}, \qquad \boxed{\sin x = 2\sum_{n=0}^{\infty}(-1)^n J_{2n+1}(x)}.$$

Both sides are entire functions, so the identities also extend to complex x.

14.1.7 To help remove the generating function from the realm of magic, show that it can be derived from the recurrence relation, Eq. (14.7).

Hint.

(a) Assume a generating function of the form

$$g(x,t) = \sum_{m=-\infty}^{\infty} J_m(x)t^m.$$

(b) Multiply Eq. (14.7) by t^n and sum over n.

(c) Rewrite the preceding result as

$$\left(t + \frac{1}{t}\right) g(x,t) = \frac{2t}{x}\frac{\partial g(x,t)}{\partial t}.$$

(d) Integrate and adjust the "constant" of integration (a function of x) so that the coefficient of the zeroth power, t^0, is $J_0(x)$ as given by Eq. (14.6).

Solution.

Start with the proposed Laurent series $g = \sum_m J_m(x)t^m$. Multiply $J_{n-1} + J_{n+1} = 2nJ_n/x$ by t^n and sum. Shifting the indices on the left ($m = n-1$ and $m = n+1$, respectively) gives

$$\sum_n J_{n-1}t^n = t\sum_m J_m t^m = tg, \qquad \sum_n J_{n+1}t^n = t^{-1}\sum_m J_m t^m = t^{-1}g,$$

while on the right $\sum_n (2n/x)J_n t^n = (2t/x)\sum_n nJ_n t^{n-1} = (2t/x)\partial_t g$. This is the form of hint (c):

$$\left(t + \frac{1}{t}\right) g = \frac{2t}{x}\frac{\partial g}{\partial t}.$$

Thus, multiplying by $x/(2t)$ for fixed $x \neq 0$,

$$\partial_t g = \frac{x}{2}(1 + t^{-2})g.$$

Since $\partial_t[\frac{x}{2}(t - t^{-1})] = \frac{x}{2}(1 + t^{-2})$, this says $\partial_t\{g\, e^{-(x/2)(t-t^{-1})}\} = 0$, so the bracket is independent of t:

$$g(x,t) = C(x)\exp\left[\frac{x}{2}(t - t^{-1})\right].$$

To find C, multiply the two elementary exponential series $e^{xt/2} = \sum_r (x/2)^r t^r/r!$ and $e^{-x/(2t)} = \sum_s (-1)^s (x/2)^s t^{-s}/s!$:

$$e^{xt/2}e^{-x/(2t)} = \sum_{r,s\geq 0} \frac{(-1)^s (x/2)^{r+s}}{r!s!} t^{r-s}.$$

Only $r = s$ contributes to t^0. Its coefficient is the series for $J_0(x)$ given by Eq. (14.6):

$$\sum_{s=0}^{\infty} \frac{(-1)^s (x/2)^{2s}}{(s!)^2} = J_0(x).$$

The coefficient of t^0 in g is consequently $C(x)J_0(x)$, which must equal $J_0(x)$. This yields $C(x) = 1$ wherever $J_0(x) \neq 0$. At an isolated zero of J_0, matching that one coefficient alone does not determine C. Continuity in x of the Bessel family fixes the same value there; the recurrence by itself leaves an overall normalization undetermined. The limit at $x = 0$ likewise gives $g(0,t) = 1$. Hence

$$\boxed{g(x,t) = e^{(x/2)(t-t^{-1})}.}$$

14.1.8 Show, by direct differentiation, that

$$J_\nu(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!\Gamma(s+\nu+1)}\left(\frac{x}{2}\right)^{\nu+2s}$$

satisfies the two recurrence relations

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x}J_\nu(x),$$
$$J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x),$$

and Bessel's differential equation

$$x^2 J''_\nu(x) + xJ'_\nu(x) + (x^2 - \nu^2)J_\nu(x) = 0.$$

Solution.

Use a fixed branch of x^ν (in particular $x>0$ for real ν), and write

$$J_\nu(x)=\sum_{s=0}^{\infty}a_s x^{\nu+2s},\qquad a_s=\frac{(-1)^s}{2^{\nu+2s}s!\Gamma(s+\nu+1)}.$$

The convergent series may be differentiated term by term on compact sets away from a branch point, so $J_\nu'=\sum_s(\nu+2s)a_s x^{\nu+2s-1}$ and $(\nu/x)J_\nu=\sum_s \nu a_s x^{\nu+2s-1}$. First, since $(\nu+2s)+\nu=2(\nu+s)$,

$$\begin{aligned}J_\nu'+\frac{\nu}{x}J_\nu&=\sum_{s=0}^{\infty}2(\nu+s)a_s x^{\nu+2s-1}\\&=\sum_{s=0}^{\infty}\frac{(-1)^s x^{\nu-1+2s}}{2^{\nu-1+2s}s!\Gamma(s+\nu)}=J_{\nu-1},\end{aligned}$$

where $(s+\nu)/\Gamma(s+\nu+1)=1/\Gamma(s+\nu)$ (from $\Gamma(z+1)=z\Gamma(z)$) and $2/2^{\nu+2s}=1/2^{\nu-1+2s}$ were used. On subtracting rather than adding the ν/x term, the coefficient becomes $(\nu+2s)-\nu=2s$, so the $s=0$ term vanishes. Put $s=r+1$ in the remaining sum and use $2(r+1)/(r+1)!=2/r!$, $2/2^{\nu+2r+2}=1/2^{\nu+1+2r}$, and $(-1)^{r+1}=-(-1)^r$:

$$\begin{aligned}J_\nu'-\frac{\nu}{x}J_\nu&=\sum_{s=1}^{\infty}2sa_s x^{\nu+2s-1}\\&=-\sum_{r=0}^{\infty}\frac{(-1)^r x^{\nu+1+2r}}{2^{\nu+1+2r}r!\Gamma(r+\nu+2)}=-J_{\nu+1}.\end{aligned}$$

Adding the last two identities gives $J_{\nu-1}-J_{\nu+1}=2J_\nu'$; subtracting gives $J_{\nu-1}+J_{\nu+1}=2\nu J_\nu/x$.

For the differential equation, the coefficient contributed by $x^2J_\nu''+xJ_\nu'-\nu^2J_\nu$ at $x^{\nu+2s}$ is

$$\big[(\nu+2s)(\nu+2s-1)+(\nu+2s)-\nu^2\big]a_s=4s(s+\nu)a_s.$$

The term $x^2J_\nu=\sum_s a_s x^{\nu+2s+2}$ contributes a_{s-1} at $x^{\nu+2s}$ for $s\ge1$. Using $s/s!=1/(s-1)!$, $(s+\nu)/\Gamma(s+\nu+1)=1/\Gamma(s+\nu)$, and $4/2^{\nu+2s}=1/2^{\nu+2s-2}$,

$$4s(s+\nu)a_s=\frac{(-1)^s}{2^{\nu+2s-2}(s-1)!\,\Gamma(s+\nu)}=-a_{s-1},$$

so $4s(s+\nu)a_s+a_{s-1}=0$. The initial coefficient ($s=0$) already vanishes. Every power therefore cancels, proving

$$\boxed{x^2J_\nu''+xJ_\nu'+(x^2-\nu^2)J_\nu=0.}$$

The identity $z/\Gamma(z+1)=1/\Gamma(z)$ used above holds for every complex z, both sides being zero at the poles $z=0,-1,-2,\dots$ of Γ. Hence no order is exceptional: for a negative integer ν the vanishing leading coefficients simply drop out.

14.1.9 Prove that

$$\frac{\sin x}{x}=\int_0^{\pi/2}J_0(x\cos\theta)\cos\theta\,d\theta,\qquad\frac{1-\cos x}{x}=\int_0^{\pi/2}J_1(x\cos\theta)\,d\theta.$$

Hint. The definite integral

$$\int_0^{\pi/2} \cos^{2s+1}\theta\, d\theta = \frac{2\cdot 4\cdot 6\cdots(2s)}{1\cdot 3\cdot 5\cdots(2s+1)}$$

may be useful.

Solution.

The cosine integral of the hint can be written in factorial form. Since $(2s)!! = 2\cdot 4\cdots(2s) = 2^s s!$ and $(2s+1)!! = (2s+1)!/(2s)!! = (2s+1)!/(2^s s!)$,

$$I_s = \int_0^{\pi/2} \cos^{2s+1}\theta\, d\theta = \frac{(2s)!!}{(2s+1)!!} = \frac{2^{2s}(s!)^2}{(2s+1)!}.$$

Expand $J_0(x\cos\theta) = \sum_s (-1)^s (x\cos\theta)^{2s}/[2^{2s}(s!)^2]$, multiply by $\cos\theta$, and integrate each term:

$$\begin{aligned}\int_0^{\pi/2} J_0(x\cos\theta)\cos\theta\, d\theta &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s}}{2^{2s}(s!)^2} I_s \\ &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s}}{(2s+1)!} = \boxed{\frac{\sin x}{x}},\end{aligned}$$

the last step being the Maclaurin series $\sin x = \sum_s (-1)^s x^{2s+1}/(2s+1)!$ divided by x. For $J_1(x\cos\theta) = \sum_s (-1)^s (x\cos\theta)^{2s+1}/[2^{2s+1}s!(s+1)!]$, the power of $\cos\theta$ is again $2s+1$. The factor $I_s/[2^{2s+1}s!(s+1)!] = s!/[2(s+1)!(2s+1)!] = 1/[2(s+1)(2s+1)!]$, and $2(s+1)(2s+1)! = (2s+2)!$, so

$$\begin{aligned}\int_0^{\pi/2} J_1(x\cos\theta)\, d\theta &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{2^{2s+1}s!(s+1)!} I_s \\ &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{2(s+1)(2s+1)!} \\ &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{(2s+2)!} = \boxed{\frac{1-\cos x}{x}}.\end{aligned}$$

For the last step, put $k = s+1$ in $1-\cos x = -\sum_{k\ge 1}(-1)^k x^{2k}/(2k)! = \sum_{s\ge 0}(-1)^s x^{2s+2}/(2s+2)!$ and divide by x. For fixed x, the Bessel series converge uniformly on this finite integration interval, justifying the termwise integrations. At $x = 0$, the two apparent quotients have their continuous values 1 and 0, which also follow immediately from the integrands.

14.1.10 Derive

$$J_n(x) = (-1)^n x^n \left(\frac{1}{x}\frac{d}{dx}\right)^n J_0(x).$$

Hint. Try mathematical induction (Section 1.4).
Here n is a nonnegative integer; at $x = 0$ use the continuous limit.

Solution.

Let $D = x^{-1}d/dx$ and take $n = 0, 1, 2, \ldots$. For $n = 0$, the claimed identity is simply $J_0 = J_0$. Assume that $J_n = (-1)^n x^n D^n J_0$. Then

$$x^{-n} J_n = (-1)^n D^n J_0.$$

Differentiate this equation and use Exercise 14.1.4(b) on the left and $d/dx = xD$ on the right:

$$-x^{-n}J_{n+1} = (-1)^n \frac{d}{dx}(D^n J_0) = (-1)^n xD(D^n J_0) = (-1)^n xD^{n+1}J_0.$$

Multiplying by $-x^n$ gives

$$J_{n+1} = (-1)^{n+1}x^{n+1}D^{n+1}J_0.$$

This is the required induction step, so

$$\boxed{J_n(x) = (-1)^n x^n \left(\frac{1}{x}\frac{d}{dx}\right)^n J_0(x)}.$$

The proof is for $x \neq 0$. The even power series of J_0 shows that the apparent singularities of the final expression are removable at zero.

14.1.11 Show that between any two consecutive zeros of $J_n(x)$ there is one and only one zero of $J_{n+1}(x)$.

Hint. Equations (14.10) and (14.11) may be useful.

Use consecutive strictly positive zeros, excluding the origin.

Solution.

Use consecutive positive zeros, excluding the origin. For integer $n \geq 1$, the origin is also a zero of J_n; the assertion would be false on the interval between zero and its first positive zero. Indeed, let j be the first positive zero of J_n. If $J_{n+1}(c) = 0$ with $0 < c < j$, then $x^{n+1}J_{n+1}$ vanishes at 0 and at c, so by Rolle's theorem its derivative $x^{n+1}J_n$ (Exercise 14.1.4(a)) vanishes at some point of $(0, c)$, contradicting the choice of j. Hence J_{n+1} has no zero in $(0, j)$.

Let $a < b$ be consecutive positive zeros of J_n. The function $f = x^{-n}J_n$ also vanishes at a, b. Rolle's theorem gives at least one $c \in (a, b)$ for which $f'(c) = 0$. By Exercise 14.1.4(b),

$$f' = -x^{-n}J_{n+1},$$

so $J_{n+1}(c) = 0$.

Suppose there were two such zeros $c < d$ inside (a, b). Apply Rolle's theorem to $g = x^{n+1}J_{n+1}$ on $[c, d]$. Exercise 14.1.4(a) gives

$$g' = x^{n+1}J_n,$$

so g' would vanish at some point of (c, d). That would be a zero of J_n between its consecutive zeros a, b, a contradiction. There is therefore exactly one zero of J_{n+1} in (a, b).

For strict interlacing there are no shared positive zeros. If both $J_n(a)$ and $J_{n+1}(a)$ vanished, the recurrence $J_n' = nJ_n/x - J_{n+1}$ would give $J_n'(a) = 0$. At a positive point, Bessel's ODE is regular; zero value and zero derivative force the identically zero solution, contrary to the nonzero defining series. The same argument shows that each positive zero is simple.

14.1.12 An analysis of antenna radiation patterns for a system with a circular aperture involves the equation

$$g(u) = \int_0^1 f(r)J_0(ur)r\,dr.$$

If $f(r) = 1 - r^2$, show that

$$g(u) = \frac{2}{u^2}J_2(u).$$

Solution.

For $u \neq 0$, put $x = ur$, so $r\,dr = x\,dx/u^2$:

$$g(u) = \frac{1}{u^2}\int_0^u \left(1 - \frac{x^2}{u^2}\right) x J_0(x)\,dx.$$

The derivative identity $(xJ_1)' = xJ_0$ (Exercise 14.1.4(a) with $n = 1$) allows integration by parts, with $d(1 - x^2/u^2)/dx = -2x/u^2$:

$$\begin{aligned} g(u) &= \frac{1}{u^2}\left[\left(1 - \frac{x^2}{u^2}\right) x J_1(x)\right]_0^u + \frac{2}{u^4}\int_0^u x^2 J_1(x)\,dx \\ &= \frac{2}{u^4}\left[x^2 J_2(x)\right]_0^u = \boxed{\frac{2J_2(u)}{u^2}}. \end{aligned}$$

The first endpoint expression vanishes at $x = u$ because of its prefactor and at $x = 0$ because $xJ_1(x) = O(x^2)$. The second line uses $(x^2J_2)' = x^2J_1$ (Exercise 14.1.4(a) with $n = 2$) and $x^2J_2(x) = O(x^4)$. At $u = 0$, direct integration gives $g(0) = \int_0^1 (r - r^3)dr = 1/4$; the answer has the same continuous limit because $J_2(u) = u^2/8 + O(u^4)$.

14.1.13 The differential cross section in a nuclear scattering experiment is given by $d\sigma/d\Omega = |f(\theta)|^2$. An approximate treatment leads to

$$f(\theta) = \frac{-ik}{2\pi}\int_0^{2\pi}\int_0^R \exp[ik\rho\sin\theta\sin\varphi]\rho\,d\rho\,d\varphi.$$

Here θ is an angle through which the scattered particle is scattered. R is the nuclear radius. Show that

$$\frac{d\sigma}{d\Omega} = (\pi R^2)\frac{1}{\pi}\left[\frac{J_1(kR\sin\theta)}{\sin\theta}\right]^2.$$

Solution.

Put $b = k\sin\theta$. The order-zero angular representation is

$$\frac{1}{2\pi}\int_0^{2\pi} e^{ib\rho\sin\varphi}\,d\varphi = J_0(b\rho).$$

It follows from the generating function with $t = e^{i\varphi}$, for which $t - t^{-1} = 2i\sin\varphi$: $e^{iz\sin\varphi} = \sum_m J_m(z)e^{im\varphi}$, and integrating over $0 \le \varphi \le 2\pi$ leaves only the $m = 0$ term, $2\pi J_0(z)$. The integrand is continuous on the rectangle of integration, so the φ integral may be performed first. This gives

$$f(\theta) = -ik\int_0^R \rho J_0(b\rho)\,d\rho.$$

For $b \neq 0$, change variable to $x = b\rho$ (so $\rho\,d\rho = x\,dx/b^2$, with new upper limit bR) and integrate $xJ_0(x) = (xJ_1(x))'$:

$$\int_0^R \rho J_0(b\rho)\,d\rho = \frac{1}{b^2}\left[xJ_1(x)\right]_0^{bR} = \frac{R}{b}J_1(bR).$$

Consequently, since $k/b = 1/\sin\theta$,

$$f(\theta) = -ik\frac{R}{b}\,J_1(bR) = -iR\frac{J_1(kR\sin\theta)}{\sin\theta}.$$

For real k, R, θ, the quotient is real, and $|-i|^2 = 1$. Thus

$$\boxed{\frac{d\sigma}{d\Omega} = R^2 \left[\frac{J_1(kR\sin\theta)}{\sin\theta}\right]^2 = (\pi R^2)\frac{1}{\pi}\left[\frac{J_1(kR\sin\theta)}{\sin\theta}\right]^2.}$$

The forward and backward axial directions are understood by continuity: $J_1(z) = z/2 + O(z^3)$ gives $f \to -ikR^2/2$ and $d\sigma/d\Omega \to k^2R^4/4$. The apparent division by $\sin\theta = 0$ is therefore removable.

14.1.14 A set of functions $C_n(x)$ satisfies the recurrence relations

$$C_{n-1}(x) - C_{n+1}(x) = \frac{2n}{x}C_n(x),$$
$$C_{n-1}(x) + C_{n+1}(x) = 2C_n'(x).$$

(a) What linear second-order ODE does the $C_n(x)$ satisfy?

(b) By a change of variable transform your ODE into Bessel's equation. This suggests that $C_n(x)$ may be expressed in terms of Bessel functions of transformed argument.

Solution.

(a) Adding the given recurrences gives $2C_{n-1} = 2nC_n/x + 2C_n'$; subtracting the first from the second gives $2C_{n+1} = 2C_n' - 2nC_n/x$. After division by 2,

$$C_{n-1} = C_n' + \frac{n}{x}C_n, \qquad C_{n+1} = C_n' - \frac{n}{x}C_n.$$

Differentiate the second identity:

$$C_{n+1}' = C_n'' - \frac{n}{x}C_n' + \frac{n}{x^2}C_n.$$

The first identity with n replaced by $n+1$ also says

$$C_{n+1}' = C_n - \frac{n+1}{x}C_{n+1} = C_n - \frac{n+1}{x}\left(C_n' - \frac{n}{x}C_n\right).$$

Equate these two expressions, move everything to the left, and collect coefficients: the C_n' terms give $(-n+n+1)/x = 1/x$, and the C_n terms give $n/x^2 - 1 - n(n+1)/x^2 = -1 - n^2/x^2$. Hence

$$C_n'' + \frac{1}{x}C_n' - \left(1 + \frac{n^2}{x^2}\right)C_n = 0.$$

Multiplication by x^2 gives the modified Bessel equation

$$\boxed{x^2C_n'' + xC_n' - (x^2 + n^2)C_n = 0.}$$

The ODE displayed in the Instructor's Manual, p. 197, has x^2C_n'; the correct first-derivative term is xC_n', as the derivation above shows.

(b) Put $z = ix$ and $C_n(x) = Y(z)$. Then $C_n' = iY'$, $C_n'' = -Y''$, $x = -iz$, and $x^2 = -z^2$. Substituting these into the boxed ODE gives

$$(-z^2)(-Y'') + (-iz)(iY') - (-z^2 + n^2)Y = 0, \quad \text{i.e.,} \quad \boxed{z^2Y'' + zY' + (z^2 - n^2)Y = 0}.$$

This is Bessel's equation of order n. Its solutions can consequently be expressed using Bessel functions with imaginary argument. For example $I_n(x) = i^{-n}J_n(ix)$ is the regular modified Bessel function; the phase depending on n makes it satisfy the given recurrences. Indeed, with $C_n = i^{-n}J_n(ix)$ and $i^{-(n+1)} = -i^{1-n}$, the recurrences (14.7) and (14.8) at the argument ix give

$$C_{n-1} - C_{n+1} = i^{1-n}[J_{n-1}(ix) + J_{n+1}(ix)] = i^{1-n}\frac{2n}{ix}J_n(ix) = \frac{2n}{x}C_n,$$
$$C_{n-1} + C_{n+1} = i^{1-n}[J_{n-1}(ix) - J_{n+1}(ix)] = 2i^{1-n}J_n'(ix) = 2C_n',$$

since $C_n'(x) = i^{-n}\cdot iJ_n'(ix)$.

14.1.15

(a) Show by direct differentiation and substitution that

$$J_\nu(x) = \frac{1}{2\pi i}\int_C e^{(x/2)(t-1/t)}t^{-\nu-1}\,dt$$

(this is the Schlaefli integral representation of J_ν), and that the equivalent equation,

$$J_\nu(x) = \frac{1}{2\pi i}\left(\frac{x}{2}\right)^\nu \int_C e^{s-x^2/4s}s^{-\nu-1}\,ds,$$

both satisfy Bessel's equation. C is the contour shown in Fig. 14.5. The negative real axis is the cut line.

Hint. This exercise is aimed at providing details of the discussion that starts at Eq. (14.38).

(b) Show that the first integral (with n an integer) may be transformed into

$$J_n(x) = \frac{1}{2\pi}\int_0^{2\pi} e^{i(x\sin\theta - n\theta)}\,d\theta = \frac{i^{-n}}{2\pi}\int_0^{2\pi} e^{i(x\cos\theta + n\theta)}\,d\theta.$$

Take $x > 0$ initially. The open contour starts below the negative-real cut at $-\infty$, circles zero counterclockwise, and returns above the cut; use $-\pi < \arg t < \pi$.

Solution.

For noninteger ν, the contour in Fig. 14.5 is an open Hankel contour: it starts at $-\infty$ below the cut, circles zero counterclockwise, and returns to $-\infty$ above the cut. Take $x > 0$ and $-\pi < \arg t < \pi$ initially. The two unbounded ends then decay exponentially: on either bank $t = -r$ with $r \to \infty$, and the exponent of the integrand has real part $-\frac{x}{2}(r - r^{-1}) \to -\infty$, while $|t^{-\nu-1}|$ grows at most like a power of r. Write

$$f(t,x) = e^{x(t-t^{-1})/2}t^{-\nu-1}, \quad S = t - t^{-1}, \quad T = t + t^{-1}.$$

(a) Differentiation with respect to x gives $f_x = Sf/2$ and $f_{xx} = S^2f/4$. Consequently the Bessel operator applied under the integral has integrand

$$x^2f_{xx} + xf_x + (x^2 - \nu^2)f = f\left(\frac{x^2(S^2+4)}{4} + \frac{xS}{2} - \nu^2\right)$$
$$= f\left(\frac{x^2T^2}{4} + \frac{xS}{2} - \nu^2\right),$$

since $S^2+4=t^2-2+t^{-2}+4=T^2$. This is a total derivative. Indeed, $tS'=t(1+t^{-2})=T$ and $tT'=t(1-t^{-2})=S$, so the three factors below have derivatives $[xT/(2t)]\,e^{xS/2}$, $-\nu t^{-\nu}/t$, and $xS/(2t)$, and the product rule gives

$$\begin{aligned}\frac{d}{dt}\left[e^{xS/2}t^{-\nu}\left(\nu+\frac{xT}{2}\right)\right] &= f\left[\left(\frac{xT}{2}-\nu\right)\left(\nu+\frac{xT}{2}\right)+\frac{xS}{2}\right]\\ &= f\left(\frac{x^2T^2}{4}+\frac{xS}{2}-\nu^2\right).\end{aligned}$$

Its integral is zero because the bracketed primitive tends to zero at both ends of the contour (it contains the same decaying exponential times a power of r). Therefore $F(x)=(2\pi i)^{-1}\int_C f\,dt$, differentiated under the integral sign, satisfies $x^2F''+xF'+(x^2-\nu^2)F=0$.

In the same integral put $s=xt/2$, i.e., $t=2s/x$ and $dt=(2/x)\,ds$. Then

$$\frac{x}{2}(t-t^{-1})=s-\frac{x^2}{4s},\qquad t^{-\nu-1}dt=\left(\frac{2s}{x}\right)^{-\nu-1}\frac{2}{x}\,ds=(x/2)^{\nu}s^{-\nu-1}ds,$$

which gives the second representation. Positive dilation changes neither the cut nor the contour's winding around zero; it may be deformed back to a fixed contour without crossing a singularity. This proves its equivalence and hence its differential equation.

For completeness the normalization is also fixed, rather than merely identifying an unspecified solution of the ODE. Expand the second exponential, $e^{-x^2/(4s)}=\sum_r(-x^2/4)^r s^{-r}/r!$, and use the Hankel formula for the reciprocal gamma function, $(2\pi i)^{-1}\int_C e^s s^{-z}ds=1/\Gamma(z)$, with $z=\nu+r+1$ and $(-x^2/4)^r=(-1)^r(x/2)^{2r}$:

$$\begin{aligned}F(x) &= (x/2)^{\nu}\sum_{r=0}^{\infty}\frac{(-x^2/4)^r}{r!}\frac{1}{2\pi i}\int_C e^s s^{-\nu-r-1}\,ds\\ &= \sum_{r=0}^{\infty}\frac{(-1)^r(x/2)^{\nu+2r}}{r!\Gamma(\nu+r+1)}=J_\nu(x).\end{aligned}$$

The contour can have a fixed positive distance from zero, so the series is uniformly dominated there by $e^{x^2/(4|s|)}$, and the negative-real tails decay exponentially. These facts justify the termwise integration and differentiation. Other admissible x follow by analytic continuation on a fixed branch.

(b) For integer n the factor t^{-n-1} is single valued, so the integrand takes the same values on the two banks of the cut, which are traversed in opposite directions; their contributions cancel, leaving a counterclockwise circle, which may be taken as $|t|=1$. Set $t=e^{i\theta}$, so $t-t^{-1}=2i\sin\theta$ and $t^{-n-1}dt=e^{-i(n+1)\theta}\,ie^{i\theta}d\theta=ie^{-in\theta}d\theta$; the factor i cancels that of $2\pi i$:

$$\boxed{J_n(x)=\frac{1}{2\pi}\int_0^{2\pi}e^{i(x\sin\theta-n\theta)}\,d\theta}.$$

Now set $\theta=\pi/2-\phi$. Then $\sin\theta=\cos\phi$, and the phase becomes $x\cos\phi+n\phi-n\pi/2$, with $e^{-in\pi/2}=i^{-n}$. Also $d\theta=-d\phi$, and $\theta:0\to2\pi$ becomes $\phi:\pi/2\to-3\pi/2$. Reversing the integration limits removes the minus sign, and the 2π periodicity of the integrand in ϕ (for integer n) allows the interval $[-3\pi/2,\pi/2]$ to be replaced by $[0,2\pi]$. This yields

$$\boxed{J_n(x)=\frac{i^{-n}}{2\pi}\int_0^{2\pi}e^{i(x\cos\phi+n\phi)}\,d\phi}.$$

14.1.16 The contour C in Exercise 14.1.15 is deformed to the path $-\infty$ to -1, unit circle $e^{-i\pi}$ to $e^{i\pi}$, and finally -1 to $-\infty$. Show that

$$J_\nu(x) = \frac{1}{\pi}\int_0^\pi \cos(\nu\theta - x\sin\theta)\,d\theta - \frac{\sin\nu\pi}{\pi}\int_0^\infty e^{-\nu\theta - x\sinh\theta}\,d\theta.$$

This is Bessel's integral.

Hint. The negative values of the variable of integration u must be represented in a manner consistent with the presence of the branch cut, for example, by writing $u = te^{\pm i\pi}$.

Take $x > 0$ initially and use the cut convention of Exercise 14.1.15.

Solution.

The hint's $e^{\pm ix}$ must be $e^{\pm i\pi}$: these are the arguments of the two banks of the negative real axis. Use the open contour and branch specified in Exercise 14.1.15, with $x > 0$ initially. Split its integral into a lower ray, the unit circle, and an upper ray. (The hint's integration variable u is called t here, as in Exercise 14.1.15, and the hint's positive t is written e^u, so its $u = te^{\pm i\pi}$ becomes $t = e^{u\pm i\pi}$.) On the lower ray put $t = e^{u-i\pi}$ with $u : \infty \to 0$. On the upper ray use $t = e^{u+i\pi}$ with $u : 0 \to \infty$. In both cases $t = -e^u$, so $t - t^{-1} = -e^u + e^{-u} = -2\sinh u$ and the exponential becomes $e^{-x\sinh u}$, whereas $t^{-\nu-1}dt = t^{-\nu}(dt/t) = t^{-\nu}du$ gives

$$t^{-\nu-1}dt = e^{-\nu u}e^{+i\pi\nu}du \quad \text{(lower)}, \qquad t^{-\nu-1}dt = e^{-\nu u}e^{-i\pi\nu}du \quad \text{(upper)}.$$

Reversing the limits of the lower ray changes its sign, and $-e^{i\pi\nu} + e^{-i\pi\nu} = -2i\sin\pi\nu$. Thus their combined contribution is

$$\frac{1}{2\pi i}\left[-e^{i\pi\nu} + e^{-i\pi\nu}\right]\int_0^\infty e^{-\nu u - x\sinh u}\,du = -\frac{\sin\pi\nu}{\pi}\int_0^\infty e^{-\nu u - x\sinh u}\,du.$$

On the circle $t = e^{i\theta}$, $-\pi \le \theta \le \pi$, so $t - t^{-1} = 2i\sin\theta$ and $t^{-\nu-1}dt = ie^{-i\nu\theta}d\theta$; the contribution is

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(x\sin\theta - \nu\theta)}d\theta = \frac{1}{\pi}\int_0^\pi \cos(\nu\theta - x\sin\theta)d\theta.$$

Here the phase $x\sin\theta - \nu\theta$ is an odd function of θ, so the sine of the phase is odd and integrates to zero over $[-\pi, \pi]$, while its cosine is even and integrates to twice its integral over $[0, \pi]$. Adding the three parts gives exactly the stated formula. For integer $\nu = n$, the ray contribution vanishes because $\sin\pi n = 0$.

14.1.17

(a) Show that

$$J_\nu(x) = \frac{2}{\pi^{1/2}\Gamma(\nu + \frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_0^{\pi/2} \cos(x\sin\theta)\cos^{2\nu}\theta\,d\theta,$$

where $\nu > -\frac{1}{2}$.

Hint. Here is a chance to use series expansion and term-by-term integration. The formulas of Section 13.3 will prove useful.

(b) Transform the integral in part (a) into

$$\begin{aligned} J_\nu(x) &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_0^\pi \cos(x\cos\theta)\sin^{2\nu}\theta\, d\theta \\ &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_0^\pi e^{\pm ix\cos\theta}\sin^{2\nu}\theta\, d\theta \\ &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_{-1}^1 e^{\pm ipx}(1-p^2)^{\nu-1/2}\, dp. \end{aligned}$$

These are alternate integral representations of $J_\nu(x)$.

Solution.

(a) Expand the cosine, $\cos(x\sin\theta)=\sum_r (-1)^r x^{2r}\sin^{2r}\theta/(2r)!$, and integrate each term. For real $\nu > -1/2$, the endpoint weight is integrable; uniform convergence of the cosine series on the finite interval justifies the operation. The beta integral of Section 13.3, $\int_0^{\pi/2}\sin^{2a-1}\theta\cos^{2b-1}\theta\, d\theta = \frac{1}{2}B(a,b) = \Gamma(a)\Gamma(b)/[2\Gamma(a+b)]$, with $a = r+\frac{1}{2}$, $b=\nu+\frac{1}{2}$, gives

$$\begin{aligned} \int_0^{\pi/2}\sin^{2r}\theta\cos^{2\nu}\theta\, d\theta &= \frac{1}{2}B(r+\tfrac{1}{2},\nu+\tfrac{1}{2}) \\ &= \frac{\Gamma(r+\frac{1}{2})\Gamma(\nu+\frac{1}{2})}{2\Gamma(r+\nu+1)}. \end{aligned}$$

Since $\Gamma(r+1/2) = \frac{(2r-1)!!}{2^r}\sqrt{\pi}$ and $(2r-1)!! = (2r)!/(2^r r!)$, we have $\Gamma(r+1/2) = \sqrt{\pi}(2r)!/(4^r r!)$, and the proposed right-hand side becomes

$$\begin{aligned} &\frac{2(x/2)^\nu}{\sqrt{\pi}\Gamma(\nu+1/2)}\sum_{r=0}^\infty \frac{(-1)^r x^{2r}}{(2r)!}\frac{\Gamma(r+1/2)\Gamma(\nu+1/2)}{2\Gamma(r+\nu+1)} \\ &\quad = (x/2)^\nu \sum_{r=0}^\infty \frac{(-1)^r x^{2r}}{4^r r!\,\Gamma(r+\nu+1)} = \sum_{r=0}^\infty \frac{(-1)^r (x/2)^{\nu+2r}}{r!\Gamma(r+\nu+1)} = J_\nu(x), \end{aligned}$$

where $\Gamma(\nu+1/2)$, $\sqrt{\pi}$, 2 and $(2r)!$ cancel, and $x^{2r}/4^r = (x/2)^{2r}$. The factor x^ν is taken on the same branch as J_ν.

(b) Replacing θ by $\pi/2-\theta$ first changes the integrand to $\cos(x\cos\theta)\sin^{2\nu}\theta$; the limits $0,\pi/2$ are interchanged and $d\theta\to -d\theta$, so the interval is again $[0,\pi/2]$. Reflection about $\pi/2$ ($\theta\to\pi-\theta$, with $\cos(\pi-\theta) = -\cos\theta$, $\sin(\pi-\theta)=\sin\theta$, and the cosine even) leaves this integrand unchanged, so

$$2\int_0^{\pi/2}\cos(x\cos\theta)\sin^{2\nu}\theta\, d\theta = \int_0^\pi \cos(x\cos\theta)\sin^{2\nu}\theta\, d\theta.$$

Inserting this into (a), the factor 2 of the prefactor there and the factor $\frac{1}{2}$ from this relation combine to 1, which gives the first requested form. The integral of $\sin(x\cos\theta)\sin^{2\nu}\theta$ over $[0,\pi]$ is zero: the substitution $\theta\mapsto\pi-\theta$ reverses its sign. Since $e^{\pm ix\cos\theta} = \cos(x\cos\theta)\pm i\sin(x\cos\theta)$, we may consequently replace the cosine by either $e^{ix\cos\theta}$ or $e^{-ix\cos\theta}$. Finally $p=\cos\theta$ gives, with $\sin\theta = (1-p^2)^{1/2}\ge 0$ on $[0,\pi]$,

$$dp = -\sin\theta\, d\theta, \qquad \sin^{2\nu}\theta\, d\theta = \sin^{2\nu-1}\theta\,\sin\theta\, d\theta = -(1-p^2)^{\nu-1/2}dp.$$

The limits $\theta = 0,\pi$ become $p=1,-1$; reversing them absorbs the minus sign and yields the final integral in the question. The same proof works for complex ν with $\Re\nu > -1/2$.

14.1.18 Given that C is the contour in Fig. 14.5,

(a) From

$$J_\nu(x) = \frac{1}{2\pi i}\left(\frac{x}{2}\right)^\nu \int_C t^{-\nu-1} e^{t-x^2/4t}\, dt$$

derive the recurrence relation

$$J'_\nu(x) = \frac{\nu}{x} J_\nu(x) - J_{\nu+1}(x).$$

(b) From

$$J_\nu(x) = \frac{1}{2\pi i}\int_C t^{-\nu-1} e^{(x/2)(t-1/t)}\, dt$$

derive the recurrence relation

$$J'_\nu(x) = \frac{1}{2}\big[J_{\nu-1}(x) - J_{\nu+1}(x)\big].$$

The integrals are over the open Hankel contour of Exercise 14.1.15, with its fixed branch and $x > 0$ initially.

Solution.

Use a fixed open Hankel contour, the branch convention of Exercise 14.1.15, and initially $x > 0$. Its exponentially decaying tails justify differentiating under the integral.

(a) Both the prefactor and the exponential depend on x: $d(x/2)^\nu/dx = (\nu/x)(x/2)^\nu$ and $\partial_x[-x^2/(4t)] = -x/(2t)$. In the second line $(x/2)^\nu(x/2) = (x/2)^{\nu+1}$ and $t^{-\nu-1}t^{-1} = t^{-(\nu+1)-1}$, so the remaining term is the same representation with ν replaced by $\nu+1$:

$$\begin{aligned} J'_\nu(x) &= \frac{\nu}{x}J_\nu(x) + \frac{(x/2)^\nu}{2\pi i}\int_C t^{-\nu-1}\left(-\frac{x}{2t}\right) e^{t-x^2/(4t)} dt \\ &= \frac{\nu}{x}J_\nu(x) - \frac{(x/2)^{\nu+1}}{2\pi i}\int_C t^{-(\nu+1)-1} e^{t-x^2/(4t)} dt \\ &= \boxed{\frac{\nu}{x}J_\nu(x) - J_{\nu+1}(x)}. \end{aligned}$$

(b) Differentiating the other representation brings down $\partial_x[\frac{x}{2}(t-t^{-1})] = \frac{1}{2}(t-t^{-1})$, and $t^{-\nu-1}(t-t^{-1}) = t^{-\nu} - t^{-\nu-2}$. Writing $t^{-\nu} = t^{-(\nu-1)-1}$ and $t^{-\nu-2} = t^{-(\nu+1)-1}$ identifies the two integrals as the representations of orders $\nu \mp 1$:

$$\begin{aligned} J'_\nu(x) &= \frac{1}{4\pi i}\int_C \left[t^{-\nu} - t^{-\nu-2}\right] e^{x(t-t^{-1})/2} dt \\ &= \frac{1}{2}\left[\frac{1}{2\pi i}\int_C t^{-(\nu-1)-1} e^{x(t-t^{-1})/2} dt - \frac{1}{2\pi i}\int_C t^{-(\nu+1)-1} e^{x(t-t^{-1})/2} dt\right] \\ &= \boxed{\tfrac{1}{2}[J_{\nu-1}(x) - J_{\nu+1}(x)]}. \end{aligned}$$

The identities then continue to the domains of the Bessel functions.

14.1.19 Show that the recurrence relation

$$J_n'(x) = \frac{1}{2}\big[J_{n-1}(x) - J_{n+1}(x)\big]$$

follows directly from differentiation of

$$J_n(x) = \frac{1}{\pi}\int_0^\pi \cos(n\theta - x\sin\theta)\,d\theta.$$

The order n in this trigonometric integral is an integer.

Solution.

For integer n, differentiating the finite-interval integral under the integral sign (the integrand is smooth in x and θ), with $\partial_x \cos(n\theta - x\sin\theta) = \sin(n\theta - x\sin\theta)\sin\theta$, gives

$$J_n'(x) = \frac{1}{\pi}\int_0^\pi \sin(n\theta - x\sin\theta)\sin\theta\,d\theta.$$

Set $A = n\theta - x\sin\theta$. The elementary subtraction identity $\cos(A-\theta) - \cos(A+\theta) = 2\sin A\sin\theta$ therefore gives

$$\begin{aligned} 2J_n'(x) &= \frac{1}{\pi}\int_0^\pi \{\cos[(n-1)\theta - x\sin\theta] - \cos[(n+1)\theta - x\sin\theta]\}\,d\theta \\ &= J_{n-1}(x) - J_{n+1}(x). \end{aligned}$$

Dividing by two proves the recurrence. The restriction to integer order matters: for noninteger order the extra ray integral from Exercise 14.1.16 must also be differentiated.

14.1.20 Evaluate

$$\int_0^\infty e^{-ax}J_0(bx)\,dx, \qquad a, b > 0.$$

Actually the results hold for $a \geq 0$, $-\infty < b < \infty$. This is a Laplace transform of J_0.

Hint. Either an integral representation of J_0 or a series expansion will be helpful.

For real $a \geq 0$ and b, exclude $(a,b) = (0,0)$; the $a = 0$, $b \neq 0$ integral is conditionally convergent.

Solution.

For $a > 0$ and real b, use the integer-order integral representation $J_0(z) = \pi^{-1}\int_0^\pi \cos(z\sin\theta)\,d\theta$ (Bessel's integral, Exercise 14.1.16 with $\nu = 0$) at $z = bx$:

$$\begin{aligned} L(a,b) &= \frac{1}{\pi}\int_0^\pi \int_0^\infty e^{-ax}\cos(bx\sin\theta)\,dx\,d\theta \\ &= \frac{1}{\pi}\int_0^\pi \frac{a}{a^2 + b^2\sin^2\theta}\,d\theta. \end{aligned}$$

The interchange is justified by the integrable bound e^{-ax}. The inner Laplace integral is the real part of

$$\int_0^\infty e^{-(a - ib\sin\theta)x}dx = \frac{1}{a - ib\sin\theta} = \frac{a + ib\sin\theta}{a^2 + b^2\sin^2\theta}.$$

The integrand $a/(a^2+b^2\sin^2\theta)$ is symmetric about $\pi/2$ ($\sin(\pi-\theta)=\sin\theta$), so the integral over $[0,\pi]$ is twice that over $[0,\pi/2]$. There put $u=\tan\theta$, with $d\theta = du/(1+u^2)$, $\sin^2\theta = u^2/(1+u^2)$, and limits $u: 0\to\infty$; then $d\theta/(a^2+b^2\sin^2\theta) = du/[a^2(1+u^2)+b^2u^2]$, and

$$L(a,b) = \frac{2a}{\pi}\int_0^{\pi/2}\frac{d\theta}{a^2+b^2\sin^2\theta} = \frac{2a}{\pi}\int_0^{\infty}\frac{du}{a^2+(a^2+b^2)u^2}$$
$$= \frac{2}{\pi\sqrt{a^2+b^2}}\left[\arctan\frac{u\sqrt{a^2+b^2}}{a}\right]_0^{\infty} = \boxed{\frac{1}{\sqrt{a^2+b^2}}}.$$

Here, with $c=\sqrt{a^2+b^2}$, an antiderivative of $1/(a^2+c^2u^2)$ is $\arctan(cu/a)/(ac)$, and the arctangent increases by $\pi/2$ from $u=0$ to $u=\infty$. The claimed extension to $a\ge 0$ excludes $(a,b)=(0,0)$, where the integrand is 1 and the integral diverges. At $a=0$, $b\neq 0$, the integral converges conditionally and equals $1/|b|$. To justify that boundary value, $J_0(bx) = J_0(|b|x)$ and its large-x expansion (Section 14.6) is an oscillating cosine times $x^{-1/2}$ plus an $O(x^{-3/2})$ remainder. The first term converges by integration by parts and the remainder absolutely. Abel's theorem for this convergent improper integral then gives its value as $\lim_{a\downarrow 0} L(a,b) = 1/|b|$.

14.1.21 Using the symmetries of the trigonometric functions, confirm that for integer n,

$$\frac{1}{2\pi}\int_0^{2\pi}\cos(x\sin\theta-n\theta)\,d\theta = \frac{1}{\pi}\int_0^{\pi}\cos(x\sin\theta-n\theta)\,d\theta.$$

Solution.

Let $F(\theta)=\cos(x\sin\theta-n\theta)$. For integer n, using $\sin(2\pi-\theta)=-\sin\theta$, then the 2π periodicity of the cosine (to drop $-2\pi n$), and finally its evenness,

$$F(2\pi-\theta) = \cos[-x\sin\theta-2\pi n+n\theta]$$
$$= \cos[-(x\sin\theta-n\theta)] = F(\theta).$$

In the integral from π to 2π, substitute $u=2\pi-\theta$. Then

$$\int_{\pi}^{2\pi}F(\theta)d\theta = \int_0^{\pi}F(2\pi-u)du = \int_0^{\pi}F(u)du.$$

Hence the integral over $[0,2\pi]$ is twice that over $[0,\pi]$. Division by 2π proves the stated equality.

14.1.22

(a) Plot the intensity, Φ^2 of Eq. (14.25), as a function of $(\sin\alpha/\lambda)$ along a diameter of the circular diffraction pattern. Locate the first two minima.

(b) Estimate the fraction of the total light intensity that falls within the central maximum.

Hint. $[J_1(x)]^2/x$ may be written as a derivative and the area integral of the intensity integrated by inspection.

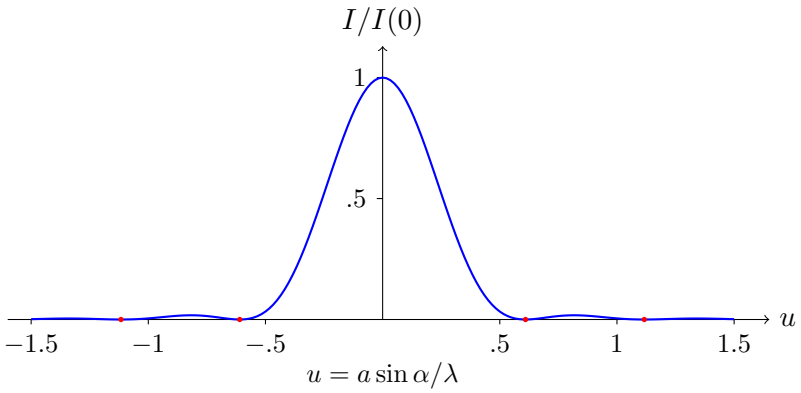
For the total-intensity estimate use the paraxial diffraction plane and an aperture large compared with the wavelength.

Solution.

(a) Write $q = \sin\alpha/\lambda$ and $u = aq$. Equation (14.25) gives $\Phi \propto (\lambda a/\sin\alpha)J_1(2\pi a \sin\alpha/\lambda) = (a/q)J_1(2\pi aq)$. Since $J_1(z) = z/2 + O(z^3)$, $(a/q)J_1(2\pi aq) \to \pi a^2$ as $q \to 0$, so the intensity normalized by its central value is

$$\boxed{\frac{I(q)}{I(0)} = \left[\frac{2J_1(2\pi aq)}{2\pi aq}\right]^2}.$$

The function is even, so the diameter profile is symmetric:



The noncentral minima occur at the nonzero zeros of J_1, where the intensity vanishes, and only there. Indeed, with $x = 2\pi aq$, Exercise 14.1.4(b) with $n = 1$ gives $[J_1(x)/x]' = -J_2(x)/x$, so $I = I(0)[2J_1(x)/x]^2$ has $dI/dx = -8I(0)J_1(x)J_2(x)/x^2$. Up to the first positive zero j_{11} of J_1, neither J_1 nor J_2 vanishes (for J_2 see Exercise 14.1.11), so both keep the positive signs of their leading terms $x/2$ and $x^2/8$, and I decreases to 0. Between consecutive positive zeros of J_1, J_2 has exactly one zero (Exercise 14.1.11 with $n = 1$), so dI/dx vanishes at a single interior point. Since $I > 0$ inside and $I = 0$ at both ends, I has an interior maximum, necessarily at that point (a bright ring), and no interior minimum. The first two positive zeros, found from $J_1(j) = 0$ (e.g., by Newton's method from the starting values 3.8 and 7.0, with $J_1' = J_0 - J_1/x$), are

$$j_{11} = 3.8317059702, \qquad j_{12} = 7.0155866698.$$

Setting $2\pi aq = j_{1k}$, i.e., $q = j_{1k}/(2\pi a)$, gives their positions on either side of the center:

$$\boxed{\frac{\sin\alpha}{\lambda} = \pm\frac{0.6098349456}{a}, \qquad \frac{\sin\alpha}{\lambda} = \pm\frac{1.1165652972}{a}}.$$

The point $q = 0$ is a maximum, not a zero: the apparent $0/0$ there has limit one. Physical angles require $|\sin\alpha| \le 1$, so the kth minimum exists only if $\lambda j_{1k}/(2\pi a) \le 1$, i.e., $a \ge 0.6098\lambda$ for the first and $a \ge 1.1166\lambda$ for the second.

(b) In the small-angle diffraction plane the annular area element is $2\pi q\,dq$. Put $x = 2\pi aq$; then $q\,dq \propto x\,dx$ and $I \propto [J_1(x)/x]^2$, so the energy within $x \le X$ is proportional to $\int_0^X [J_1(x)/x]^2 x\,dx = \int_0^X J_1(x)^2\,dx/x$. Using $J_0' = -J_1$ and $J_1' = J_0 - J_1/x$ (Exercise 14.1.4(b) with $n = 0$, and the identity $J_n' = J_{n-1} - nJ_n/x$ derived there, with $n = 1$),

$$\frac{d}{dx}(J_0^2 + J_1^2) = 2J_0(-J_1) + 2J_1(J_0 - J_1/x) = -\frac{2J_1^2}{x},$$
$$\int_0^X \frac{J_1(x)^2}{x}dx = \frac{1}{2}[1 - J_0(X)^2 - J_1(X)^2],$$

the second line by integrating the first from 0 to X with $J_0(0) = 1$, $J_1(0) = 0$. Both squared Bessel functions tend to zero as $X \to \infty$, so the total radial integral is $1/2$. At the first dark ring, $X = j_{11}$ and $J_1(X) = 0$. The central fraction is the ratio $\frac{1}{2}[1 - J_0(j_{11})^2]/\frac{1}{2}$; with $J_0(j_{11}) = -0.4027593957$ it is consequently

$$\boxed{1 - J_0(j_{11})^2 = 0.8377848691\ldots \simeq 83.8\%}.$$

This is the usual paraxial, large-aperture estimate ($a/\lambda \gg 1$). The Instructor's Manual, p. 201, says a/λ is small when extending the pattern to infinity; the regime is reversed. Likewise the textbook, p. 650, calls $2\pi a/\lambda$ small while giving its value as 57120, which is large. For a subwavelength aperture, the first dark ring may not exist at any physical angle, and the 83.8% plane-area estimate is not applicable.

14.1.23 The fraction of light incident on a circular aperture (normal incidence) that is transmitted is given by

$$T = 2\int_0^{2ka} J_2(x)\frac{dx}{x} - \frac{1}{2ka}\int_0^{2ka} J_2(x)\,dx.$$

Here a is the radius of the aperture and k is the wave number, $2\pi/\lambda$. Show that

(a) $T = 1 - \dfrac{1}{ka}\displaystyle\sum_{n=0}^{\infty} J_{2n+1}(2ka)$,

(b) $T = 1 - \dfrac{1}{2ka}\displaystyle\int_0^{2ka} J_0(x)\,dx$.

Solution.

Put $A = 2ka > 0$. The recurrence from Exercise 14.1.4(b) with $n = 1$ gives $[J_1(x)/x]' = -J_2(x)/x$, while $J_1(x)/x \to 1/2$ at zero. Thus

$$2\int_0^A \frac{J_2(x)}{x}\,dx = -2\left[\frac{J_1(x)}{x}\right]_0^A = -2\left(\frac{J_1(A)}{A} - \frac{1}{2}\right) = 1 - \frac{2J_1(A)}{A}.$$

Also $2J_1' = J_0 - J_2$ (Eq. (14.8) with $n = 1$), so $J_2 = J_0 - 2J_1'$ and

$$\int_0^A J_2(x)dx = \int_0^A J_0(x)dx - 2J_1(A),$$

because $J_1(0) = 0$. Substitution into the defining formula for T cancels the two terms containing $J_1(A)$ and proves part (b):

$$T = 1 - \frac{2J_1(A)}{A} - \frac{1}{A}\left[\int_0^A J_0(x)dx - 2J_1(A)\right] = \boxed{1 - \frac{1}{A}\int_0^A J_0(x)dx}.$$

For part (a), successively use $J_{2r} = 2J_{2r+1}' + J_{2r+2}$, which is Eq. (14.8) with $n = 2r+1$. Integrating it from 0 to A gives $\int_0^A J_{2r}dx = 2J_{2r+1}(A) + \int_0^A J_{2r+2}dx$, because $J_{2r+1}(0) = 0$. Applying this for $r = 0, 1, \ldots, N-1$ in turn,

$$\int_0^A J_0(x)dx = 2\sum_{r=0}^{N-1} J_{2r+1}(A) + \int_0^A J_{2N}(x)dx.$$

The series definition bounds $|J_m(x)|$ for $0 \le x \le A$: since $\Gamma(m+s+1) = (m+s)! \ge m!$,

$$|J_m(x)| \le \sum_{s=0}^{\infty} \frac{(x/2)^{m+2s}}{s!\,(m+s)!} \le \frac{(A/2)^m}{m!} \sum_{s=0}^{\infty} \frac{(A/2)^{2s}}{s!} = \frac{(A/2)^m e^{A^2/4}}{m!}.$$

Thus $|\int_0^A J_{2N}\,dx| \le A(A/2)^{2N} e^{A^2/4}/(2N)! \to 0$ as $N \to \infty$. With $2/A = 1/(ka)$, part (b) then gives

$$\boxed{T = 1 - \frac{1}{ka} \sum_{r=0}^{\infty} J_{2r+1}(2ka)}.$$

At $ka = 0$ the expressions with denominators have their continuous limit $T = 0$, as follows from $A^{-1}\int_0^A J_0 \to 1$.

14.1.24 The amplitude $U(\rho, \varphi, t)$ of a vibrating circular membrane of radius a satisfies the wave equation

$$\nabla^2 U \equiv \frac{\partial^2 U}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial U}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 U}{\partial \varphi^2} = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2}.$$

Here v is the phase velocity of the wave, determined by the properties of the membrane.

(a) Show that a physically relevant solution is

$$U(\rho, \varphi, t) = J_m(k\rho)\big(c_1 e^{im\varphi} + c_2 e^{-im\varphi}\big)\big(b_1 e^{i\omega t} + b_2 e^{-i\omega t}\big).$$

(b) From the Dirichlet boundary condition $J_m(ka) = 0$, find the allowable values of k.

The separated wave satisfies $\omega = vk$; the two angular factors coincide for $m = 0$.

Solution.

(a) Write $U = P(\rho)\Phi(\varphi)T(t)$ and divide the wave equation by $P\Phi T$. The result is

$$\frac{P''}{P} + \frac{P'}{\rho P} + \frac{\Phi''}{\rho^2 \Phi} = \frac{T''}{v^2 T} = -k^2.$$

Here the left side depends only on ρ, φ and $T''/(v^2T)$ only on t, so both equal a constant, written $-k^2$. Therefore $T'' + v^2k^2T = 0$. Defining $\omega = vk$ gives $T = b_1 e^{i\omega t} + b_2 e^{-i\omega t}$. Multiplication of the remaining equation by ρ^2 gives $\rho^2 P''/P + \rho P'/P + k^2\rho^2 = -\Phi''/\Phi$; the left side depends only on ρ and the right only on φ, so both equal a constant m^2:

$$\frac{\Phi''}{\Phi} = -m^2, \qquad \rho^2 P'' + \rho P' + (k^2\rho^2 - m^2)P = 0.$$

Single-valuedness under $\varphi \mapsto \varphi + 2\pi$ requires integer m, so $\Phi = c_1 e^{im\varphi} + c_2 e^{-im\varphi}$. With $z = k\rho$ and $P(\rho) = y(z)$, so that $\rho P' = zy'$ and $\rho^2 P'' = z^2 y''$, the radial equation becomes Bessel's equation $z^2y'' + zy' + (z^2 - m^2)y = 0$. Finiteness at the center excludes the second solution $Y_m(k\rho)$ (the Neumann function of Section 14.3, which is unbounded as $\rho \to 0$), leaving $P = J_m(k\rho)$ up to a constant. Multiplying the three factors proves the displayed solution. Its parameters must satisfy $\omega = vk$; they are not independent. For $m = 0$ the two angular exponentials are the same constant and represent only one angular mode.

(b) Let α_{mn} be the nth positive zero of J_m, $m = 0, 1, \ldots$, $n = 1, 2, \ldots$. The fixed-edge condition gives

$$ka = \alpha_{mn}, \qquad \boxed{k_{mn} = \frac{\alpha_{mn}}{a}, \qquad \omega_{mn} = \frac{v\alpha_{mn}}{a}}.$$

The negative-m radial solutions duplicate these since $J_{-m} = (-1)^m J_m$. For $m > 0$ the two independent real angular factors are $\cos m\varphi$ and $\sin m\varphi$. A zero value of k gives a regular harmonic radial function proportional to ρ^m (a constant for $m = 0$); the edge condition then forces its coefficient to vanish, so it supplies no additional normal mode.

14.1.25 Example 14.1.2 describes the TM modes of electromagnetic cavity oscillation. To obtain the transverse electric (TE) modes, we set $E_z = 0$ and work from the z component of the magnetic induction B:

$$\nabla^2 B_z + \alpha^2 B_z = 0$$

with boundary conditions

$$B_z(0) = B_z(l) = 0 \quad \text{and} \quad \left.\frac{\partial B_z}{\partial \rho}\right|_{\rho=a} = 0.$$

Show that the TE resonant frequencies are given by

$$\omega_{mnp} = c\sqrt{\frac{\beta_{mn}^2}{a^2} + \frac{p^2\pi^2}{l^2}}, \quad p = 1, 2, 3, \ldots,$$

and identify the quantities β_{mn}.

(Here $\alpha^2 = \omega^2/c^2$, with ω the angular frequency; the height of the cavity is l.)

The TE fields must also obey $\nabla \cdot B = 0$ and the perfectly conducting side condition $B_\rho = 0$; these exclude a spurious constant transverse scalar mode.

Solution.

Write $B_z = P(\rho)\Phi(\varphi)Z(z)$ with harmonic time factor $e^{-i\omega t}$. Dividing $\nabla^2 B_z + (\omega^2/c^2)B_z = 0$ by $P\Phi Z$, the z dependence separates as $Z''/Z = -g^2$, and multiplying the rest by ρ^2 separates $\Phi''/\Phi = -m^2$ exactly as in Exercise 14.1.24. This gives

$$\Phi'' + m^2\Phi = 0, \quad Z'' + g^2 Z = 0, \quad \rho^2 P'' + \rho P' + (q^2\rho^2 - m^2)P = 0, \qquad q^2 = \frac{\omega^2}{c^2} - g^2.$$

Periodicity makes m an integer. The end conditions $Z(0) = Z(l) = 0$ give $Z = \sin(p\pi z/l)$ and $g = p\pi/l$, $p = 1, 2, \ldots$. Regularity at the axis gives $P = J_m(q\rho)$ for $q > 0$. The side condition $\partial B_z/\partial\rho = P'(\rho)\Phi Z = 0$ at $\rho = a$ is

$$P'(a) = qJ'_m(qa) = 0.$$

Thus, with β_{mn} the nth *positive zero of* J'_m, $qa = \beta_{mn}$, and $\omega^2/c^2 = q^2 + g^2$ gives

$$q = \frac{\beta_{mn}}{a}, \qquad \boxed{\omega_{mnp} = c\sqrt{\frac{\beta_{mn}^2}{a^2} + \frac{p^2\pi^2}{l^2}}}, \quad m \geq 0,\ n \geq 1,\ p \geq 1.$$

The Instructor's Manual, p. 202, mistakenly identifies β_{mn} as a zero of J_m; it must be a zero of J'_m. Its separated sine also needs the factor z, and its oscillatory time factor needs i: $\sin(p\pi z/l)$ and $e^{\pm i\omega t}$.

The scalar Helmholtz equation and the displayed scalar boundary conditions alone also admit a constant transverse profile at $q = 0$. Excluding it for a TE electromagnetic mode requires Maxwell's equations, not merely the scalar equation. For a Neumann transverse eigenfunction $\psi = P\Phi$, which satisfies $\nabla_\perp^2 \psi = -q^2\psi$ on the disk D, integration by parts (Green's first identity, whose boundary term contains $\partial_\rho \psi = 0$ at $\rho = a$) gives

$$q^2 \int_D |P\Phi|^2 \, dA = \int_D |\nabla_\perp (P\Phi)|^2 \, dA \geq 0.$$

If $q = 0$, the profile is constant over the disk. For a perfectly conducting side wall $B_\rho = 0$; integrating $\nabla \cdot B = 0$ over a thin disk-shaped slab therefore gives $d(\int_D B_z \, dA)/dz = 0$. The end condition $B_z = 0$ makes this flux zero, which rules out a nonzero constant transverse profile. Hence only the positive derivative zeros represent TE modes.

14.1.26 A conducting cylinder can accommodate traveling electromagnetic waves; when used for this purpose it is called a wave guide. The equations describing traveling waves are the same as those of Example 14.1.2, but there is no boundary condition on E_z at $z = 0$ or $z = h$ other than that its z dependence be oscillatory. For each TM mode (values of m and j of Example 14.1.2), there is a minimum frequency that can be transmitted through a wave guide of radius a. Explain why this is so, and give a formula for the cutoff frequencies.

Solution.

As in Example 14.1.2, E_z satisfies $\nabla^2 E_z + (\omega^2/c^2)E_z = 0$, is regular on the axis, and, being tangential to the conducting wall, vanishes at $\rho = a$. The radial factor is therefore $J_m(\alpha_{mj}\rho/a)$ with α_{mj} a positive zero of J_m, exactly as in the cavity; only the z dependence changes. Take a separated traveling mode with axial wave number k_z:

$$E_z = J_m(\alpha_{mj}\rho/a)e^{im\varphi}e^{ik_z z - i\omega t}, \qquad J_m(\alpha_{mj}) = 0, \quad \alpha_{mj} > 0.$$

By Bessel's equation for $J_m(k\rho)$ with $k = \alpha_{mj}/a$, the radial and angular factors satisfy $(\partial_\rho^2 + \rho^{-1}\partial_\rho + \rho^{-2}\partial_\varphi^2)[J_m(k\rho)e^{im\varphi}] = -k^2 J_m(k\rho)e^{im\varphi}$, i.e., they have transverse Laplacian eigenvalue $-\alpha_{mj}^2/a^2$, and the axial derivative supplies $-k_z^2$. Thus the Helmholtz equation becomes

$$-\frac{\alpha_{mj}^2}{a^2} - k_z^2 + \frac{\omega^2}{c^2} = 0, \qquad k_z^2 = \frac{\omega^2}{c^2} - \frac{\alpha_{mj}^2}{a^2}.$$

Below $c\alpha_{mj}/a$, k_z is imaginary and the axial field is evanescent. Above that value k_z is real and nonzero, so traveling waves exist. The cutoff is therefore

$$\boxed{\omega_{c,mj} = \frac{c\alpha_{mj}}{a}, \qquad f_{c,mj} = \frac{c\alpha_{mj}}{2\pi a}}.$$

This is why each TM mode has a minimum frequency: the wall fixes the transverse wave number at α_{mj}/a, and only the excess of ω^2/c^2 over α_{mj}^2/a^2 is available for propagation along the guide. At equality $k_z = 0$; this is the onset of propagation, with axial group velocity $d\omega/dk_z = c^2 k_z/\omega = 0$ (differentiate $\omega^2 = c^2k_z^2 + c^2\alpha_{mj}^2/a^2$). Strictly propagating modes require $\omega > \omega_c$. The ordinary-frequency formula in the Instructor's Manual, p. 203, omits the factor c. The dispersion relation above also shows the dimensionally necessary factor explicitly.

14.1.27 Plot the three lowest TM and the three lowest TE angular resonant frequencies, ω_{mnp}, as a function of the ratio radius/length (a/l) for $0 \le a/l \le 1.5$.

Hint. Try plotting ω^2 (in units of c^2/a^2) vs. $(a/l)^2$. Why this choice?

Count the $\pm m$ degeneracy as one mode family. The endpoint $a/l = 0$ is the infinite-length limit.

Solution.

Use the dimensionless coordinates suggested in the hint:

$$r = a/l, \qquad X = r^2, \qquad Y = (a\omega/c)^2.$$

Multiplying the TM frequencies $\omega_{mnp}^2 = c^2(\alpha_{mn}^2/a^2 + p^2\pi^2/l^2)$ of Example 14.1.2 and the TE frequencies of Exercise 14.1.25 by a^2/c^2 shows that each individual mode is a straight line,

$$Y_{mnp}^{\mathrm{TM}} = \alpha_{mn}^2 + p^2\pi^2 X, \quad p \ge 0; \qquad Y_{mnp}^{\mathrm{TE}} = \beta_{mn}^2 + p^2\pi^2 X, \quad p \ge 1.$$

This is the reason for the hint's choice: in the (X, Y) plane every mode is a line with intercept α_{mn}^2 or β_{mn}^2 and slope $p^2\pi^2$, so the curves are trivial to draw and their crossings follow from linear equations. The requested angular frequencies follow from $\omega = (c/a)\sqrt{Y}$ and $r = \sqrt{X}$; $0 \le r \le 1.5$ means $0 \le X \le 2.25$. The three lowest mode families change as the aspect ratio changes. The two angular polarizations for $m > 0$ have the same frequency and are counted as one family here; at crossings the plotted families coincide. The point $r = 0$ is an infinite-length limit, not a finite cavity, and the axial indices then become degenerate.

For TM modes the lowest transverse roots are the first positive zeros $\alpha_{01} = 2.4048255577$, $\alpha_{11} = 3.8317059702$, $\alpha_{21} = 5.1356223018$ of J_0, J_1, J_2 (Newton's method as in Exercise 14.1.22, started at 2.4, 3.8, 5.1, with $J_0' = -J_1$ and $J_m' = J_{m-1} - mJ_m/x$); their squares are

$$A = \alpha_{01}^2 = 5.783185963, \quad B = \alpha_{11}^2 = 14.68197064, \quad C = \alpha_{21}^2 = 26.37461643.$$

To find the first three modes, it suffices to sort A, $A + \pi^2 X$, $A + 4\pi^2 X$, B, and C: a mode lying above three other modes cannot be among the lowest three. For $X > 0$, $A + p^2\pi^2 X$ with $p \ge 3$ lies above A, $A + \pi^2 X$, $A + 4\pi^2 X$; $B + p^2\pi^2 X$ with $p \ge 1$ lies above A, $A + \pi^2 X$, B; and $C + p^2\pi^2 X$ with $p \ge 1$ lies above A, $A + \pi^2 X$, C. Higher transverse eigenvalues cannot improve the first three (see the end of this solution). The crossover equations and their solutions are

$$A + 4\pi^2 r_1^2 = B, \quad A + \pi^2 r_2^2 = B, \quad A + \pi^2 r_3^2 = C,$$

$$r_1 = \frac{\sqrt{B - A}}{2\pi}, \quad r_2 = \frac{\sqrt{B - A}}{\pi}, \quad r_3 = \frac{\sqrt{C - A}}{\pi},$$

$$(r_1, r_2, r_3) = (0.4747724183,\ 0.9495448366,\ 1.444419644).$$

Thus the ordering, with indices written as (m, n, p), is

r interval	lowest	second	third
$0 < r < r_1$	TM_{010}	TM_{011}	TM_{012}
$r_1 < r < r_2$	TM_{010}	TM_{011}	TM_{110}
$r_2 < r < r_3$	TM_{010}	TM_{110}	TM_{011}
$r_3 < r \le 1.5$	TM_{010}	TM_{110}	TM_{210}

For TE modes the lowest transverse roots are the first positive zeros $\beta_{11} = 1.8411837813$, $\beta_{21} = 3.0542369282$ of J_1', J_2' (Newton's method started at 1.8 and 3.0, with $J_m'' = -J_m'/x - (1-m^2/x^2)J_m$ from Bessel's equation) and $\beta_{01} = \alpha_{11}$, because $J_0' = -J_1$ (Exercise 14.1.4(b) with $n = 0$); their squares are

$$D = \beta_{11}^2 = 3.389957717, \quad E = \beta_{21}^2 = 9.328363214, \quad F = \beta_{01}^2 = B = 14.68197064.$$

Here the candidates are $D + \pi^2 X$, $D + 4\pi^2 X$, $D + 9\pi^2 X$, $E + \pi^2 X$, and $F + \pi^2 X$. As for TM, for $X > 0$, $D + p^2\pi^2 X$ with $p \geq 4$ lies above the first three members of its branch, while $E + p^2\pi^2 X$ and $F + p^2\pi^2 X$ with $p \geq 2$ lie above $D + \pi^2 X$, $D + 4\pi^2 X$, and their own $p = 1$ member. The crossovers satisfy

$$D + 9\pi^2 s_1^2 = E + \pi^2 s_1^2, \quad D + 4\pi^2 s_2^2 = E + \pi^2 s_2^2, \quad D + 4\pi^2 s_3^2 = F + \pi^2 s_3^2,$$

so that

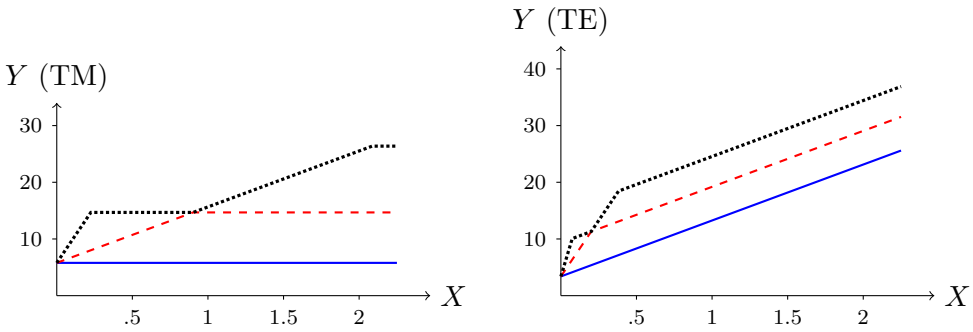
$$s_1 = \sqrt{\frac{E-D}{8\pi^2}}, \quad s_2 = \sqrt{\frac{E-D}{3\pi^2}}, \quad s_3 = \sqrt{\frac{F-D}{3\pi^2}},$$

$$(s_1, s_2, s_3) = (0.2742458463,\ 0.4478415918,\ 0.6175543472).$$

This gives

r interval	lowest	second	third
$0 < r < s_1$	TE_{111}	TE_{112}	TE_{113}
$s_1 < r < s_2$	TE_{111}	TE_{112}	TE_{211}
$s_2 < r < s_3$	TE_{111}	TE_{211}	TE_{112}
$s_3 < r \leq 1.5$	TE_{111}	TE_{211}	TE_{011}

These orderings give the following plots of the three lowest frequencies in squared coordinates. Blue solid is the lowest, red dashed the second, and black dotted the third:



The transverse ordering used above can also be checked without surveying indefinitely many roots. The radial eigenvalues for given m are the min–max values of the Rayleigh quotient $\int_0^a (|P'|^2 + m^2|P|^2/\rho^2)\rho\,d\rho / \int_0^a |P|^2\rho\,d\rho$. Increasing m^2 increases the nonnegative term $m^2 \int |P|^2 d\rho/\rho$, so each radial eigenvalue increases with m; each fixed-m sequence increases with its radial index. For TM, the third-lowest value is at most C, since A, B, C are all modes. Every other branch has either $n \geq 2$, so $\alpha_{mn} \geq \alpha_{02} = 5.520078$, or $n = 1$ and $m \geq 3$, so $\alpha_{m1} \geq \alpha_{31} = 6.380162$. Both exceed $\alpha_{21} = 5.135622$, so these branches lie above C. For TE, the zero Neumann eigenvalue is excluded as in Exercise 14.1.25, and the third-lowest value is at most $F + \pi^2 X$. Every other branch has $n = 1$ and $m \geq 3$, so $\beta_{m1} \geq \beta_{31} = 4.201189$, or $n \geq 2$, so $\beta_{0n} \geq \beta_{02} > \beta_{01}$ and, for $m \geq 1$, $\beta_{mn} \geq \beta_{12} = 5.331443$. All exceed β_{01}, so with $p \geq 1$ these branches lie above $F + \pi^2 X$. In all these labels $n = 1$ means the first *positive* root; in particular $\beta_{01} = 3.831706$, not zero.

14.1.28 Show that the integral

$$\int_0^a x^m J_n(x)\,dx, \quad m \geq n \geq 0,$$

(a) is integrable for $m+n$ odd in terms of Bessel functions and powers of x, i.e., is expressible as linear combinations of $a^p J_q(a)$;

(b) may be reduced for $m+n$ even to integrated terms plus $\displaystyle\int_0^a J_0(x)\,dx$.

Here m, n are nonnegative integers with $m \geq n$.

Solution.

The parity language presumes nonnegative integer m, n. Define $I_{m,n}(a) = \int_0^a x^m J_n(x)dx$. Since $[x^{n+1}J_{n+1}(x)]' = x^{n+1}J_n(x)$, integration by parts gives

$$\begin{aligned} I_{m,n} &= \int_0^a x^{m-n-1}\, d[x^{n+1}J_{n+1}(x)] \\ &= a^m J_{n+1}(a) - (m-n-1)I_{m-1,n+1}(a). \end{aligned} \tag{*}$$

Explicitly, for $x > 0$ the product rule gives

$$\frac{d}{dx}\big[x^m J_{n+1}\big] = \frac{d}{dx}\big[x^{m-n-1}\cdot x^{n+1}J_{n+1}\big] = (m-n-1)x^{m-1}J_{n+1} + x^m J_n,$$

and integrating from 0 to a gives (*) for every $m \geq 1$, $n \geq 0$, including $m-n-1 < 0$, which occurs in (b). The lower boundary term is zero because $x^m J_{n+1}(x) = O(x^{m+n+1})$. Every repetition decreases the power by one, increases the order by one, and leaves their sum $N = m+n$ unchanged.

(a) If N is odd, then $m-n$ is odd. After r repetitions the indices are $(m-r, n+r)$, and the next coefficient in (*) is $(m-r)-(n+r)-1 = m-n-1-2r$. After $r = (m-n-1)/2$ repetitions the remaining indices therefore satisfy $m-r = (n+r)+1$, and at that stage the coefficient of the remaining integral in (*) is zero. Thus the procedure ends with the antiderivative $I_{j+1,j} = a^{j+1}J_{j+1}(a)$, $j = n+r$, and all earlier terms are constants times $a^{m-r}J_{n+r+1}(a)$. For example, $m-n-1 = 2$ for $I_{n+3,n}$, so

$$I_{n+3,n} = a^{n+3}J_{n+1}(a) - 2I_{n+2,n+1} = a^{n+3}J_{n+1}(a) - 2a^{n+2}J_{n+2}(a).$$

This supplies a finite constructive expression of the required type.

(b) If N is even, each coefficient $m-n-1-2r$ in (*) is odd and cannot vanish. Continue until the power becomes zero, i.e., m repetitions, after which the order is $n+m = N$. All the terms already integrated have the desired form, and only a constant multiple of $I_{0,N}$ remains. For even $N = 2h$, the recurrence $J_{2j} = J_{2j-2} - 2J'_{2j-1}$ (Eq. (14.8) with $n = 2j-1$), integrated from 0 to a with $J_{2j-1}(0) = 0$, gives $I_{0,2j} = I_{0,2j-2} - 2J_{2j-1}(a)$; applying this for $j = h, h-1, \ldots, 1$,

$$\begin{aligned} I_{0,2h}(a) &= I_{0,2h-2}(a) - 2J_{2h-1}(a) \\ &= \boxed{\int_0^a J_0(x)dx - 2\sum_{j=1}^{h} J_{2j-1}(a)}. \end{aligned}$$

All odd-order Bessel functions vanish at zero, so no omitted lower boundary constants occur. Inserting this into the repeated formula (*) proves the requested reduction.

14.1.29 Show that

$$\int_0^{\alpha_{0n}} \left(1 - \frac{y}{\alpha_{0n}}\right) J_0(y) y \, dy = \frac{1}{\alpha_{0n}} \int_0^{\alpha_{0n}} J_0(y) \, dy.$$

Here α_{0n} is the nth zero of $J_0(y)$. This relation is useful (see Exercise 14.2.9): The expression on the right is easier and quicker to evaluate, and is much more accurate. Taking the difference of two terms in the expression on the left leads to a large relative error.

Solution.

Set $A = \alpha_{0n} > 0$, so $J_0(A) = 0$. By Exercise 14.1.4(a) with $n = 1$, $[yJ_1(y)]' = yJ_0(y)$. Integrate by parts with $u = 1 - y/A$, $du = -dy/A$, and $dv = yJ_0(y)dy$, $v = yJ_1(y)$:

$$\begin{aligned}\int_0^A (1 - y/A) y J_0(y) dy &= [(1 - y/A) y J_1(y)]_0^A + \frac{1}{A} \int_0^A y J_1(y) dy \\ &= \frac{1}{A} \int_0^A y J_1(y) dy.\end{aligned}$$

The upper boundary vanishes because $1 - A/A = 0$, and the lower boundary because $yJ_1(y) = O(y^2)$. Now $J_1 = -J_0'$ (Exercise 14.1.4(b) with $n = 0$), and a second integration by parts, with $u = y$ and $v = J_0(y)$, gives

$$\begin{aligned}\frac{1}{A} \int_0^A y J_1(y) dy &= -\frac{1}{A} [y J_0(y)]_0^A + \frac{1}{A} \int_0^A J_0(y) dy \\ &= \boxed{\frac{1}{A} \int_0^A J_0(y) dy}.\end{aligned}$$

The last boundary term is zero by the defining root condition. The identity avoids separately evaluating the two oscillatory moments $\int yJ_0$ and $A^{-1} \int y^2 J_0$ and then subtracting them; the smaller final quantity is computed directly.

14.2 Orthogonality

14.2.1 Show that

$$\int_0^a J_\nu(kx) J_\nu(k'x) x \, dx = \frac{a[k' J_\nu(ka) J_\nu'(k'a) - k J_\nu'(ka) J_\nu(k'a)]}{(k^2 - k'^2)},$$

where $J_\nu'(ka) = \dfrac{d}{d(kx)} J_\nu(kx)\Big|_{x=a}$, and that

$$\int_0^a [J_\nu(kx)]^2 x \, dx = \frac{a^2}{2} \left\{ [J_\nu'(ka)]^2 + \left(1 - \frac{\nu^2}{k^2 a^2}\right) [J_\nu(ka)]^2 \right\}, \qquad \nu > -1.$$

These two integrals are usually called the first and second Lommel integrals.

Take real $\nu > -1$, $a > 0$, and $k, k' > 0$. The first displayed quotient requires $k^2 \neq k'^2$; its coincident limit is the second integral.

Solution.

Let $y(x) = J_\nu(kx)$ and $z(x) = J_\nu(k'x)$, initially with $k, k' > 0$, $k^2 \neq k'^2$. Bessel's equation for $J_\nu(kx)$, $x^2y'' + xy' + (k^2x^2 - \nu^2)y = 0$, divided by x, gives the self-adjoint forms

$$(xy')' + (k^2 x - \nu^2/x) y = 0, \qquad (xz')' + (k'^2 x - \nu^2/x) z = 0.$$

Multiply the second by y, the first by z, and subtract. Since $\frac{d}{dx}\{x(yz'-y'z)\} = y(xz')' - z(xy')'$ (the terms $xy'z'$ cancel),

$$\frac{d}{dx}\{x(yz'-y'z)\} = (k^2-k'^2)xyz.$$

Integrate first from $\epsilon > 0$ to a:

$$\big[x(yz'-y'z)\big]_\epsilon^a = (k^2-k'^2)\int_\epsilon^a xyz\,dx.$$

As $\epsilon \to 0$ the integral converges, since $xyz = O(x^{2\nu+1})$ with $2\nu+1 > -1$. The bracket needs care: for $-1 < \nu < 0$ the separate products xyz' and $xy'z$ each behave like $x^{2\nu}$ and diverge as $x \to 0$, so its value at the origin must be found as a limit. For this, write the first two terms of the series as

$$y = Ax^\nu\left[1 - px^2 + O(x^4)\right], \quad z = Bx^\nu\left[1 - qx^2 + O(x^4)\right],$$

with $A = (k/2)^\nu/\Gamma(\nu+1)$, $B = (k'/2)^\nu/\Gamma(\nu+1)$, $p = k^2/[4(\nu+1)]$, and $q = k'^2/[4(\nu+1)]$. Then

$$\begin{aligned} yz' - y'z &= ABx^{2\nu-1}\big[(1-px^2)(\nu-(\nu+2)qx^2) \\ &\qquad -(\nu-(\nu+2)px^2)(1-qx^2) + O(x^4)\big] \\ &= ABx^{2\nu-1}[2(p-q)x^2 + O(x^4)]\,: \end{aligned}$$

the leading powers cancel in the bracket, leaving

$$x(yz'-y'z) = \frac{AB(k^2-k'^2)}{2(\nu+1)}x^{2\nu+2} + O(x^{2\nu+4}) \longrightarrow 0 \qquad (\nu > -1).$$

At a, the chain rule gives $y' = kJ_\nu'(ka)$ and $z' = k'J_\nu'(k'a)$. Hence

$$\boxed{\int_0^a xyz\,dx = \frac{a[k'J_\nu(ka)J_\nu'(k'a) - kJ_\nu'(ka)J_\nu(k'a)]}{k^2-k'^2}}.$$

For the second integral let $k' \to k$. The left side is continuous in k': for k' in a closed interval about k inside $(0,\infty)$, the integrand is bounded by a constant times $x^{2\nu+1}$, since $J_\nu(w)w^{-\nu}$ is bounded for bounded $w > 0$; and $x^{2\nu+1}$ is integrable for $\nu > -1$, so dominated convergence applies. On the right, numerator and denominator both vanish at $k' = k$; their k' derivatives are

$$a[J_\nu(ka)J_\nu'(k'a) + k'aJ_\nu(ka)J_\nu''(k'a) - kaJ_\nu'(ka)J_\nu'(k'a)] \quad \text{and} \quad -2k'.$$

By l'Hôpital's rule, writing $J = J_\nu(ka)$ and using primes for argument derivatives,

$$\int_0^a xJ_\nu(kx)^2dx = \frac{a[J(J'+kaJ'') - ka(J')^2]}{-2k}.$$

Bessel's equation at ka says $kaJ'' = -J' - (ka - \nu^2/(ka))J$. Therefore

$$J(J'+kaJ'') - ka(J')^2 = -ka\left[(J')^2 + \left(1 - \frac{\nu^2}{k^2a^2}\right)J^2\right],$$

which gives

$$\boxed{\int_0^a xJ_\nu(kx)^2dx = \frac{a^2}{2}\left[J_\nu'(ka)^2 + \left(1 - \frac{\nu^2}{k^2a^2}\right)J_\nu(ka)^2\right]}.$$

14.2.2

(a) If $\beta_{\nu m}$ is the mth positive zero of J'_ν, so that $(d/d\rho)J_\nu(\beta_{\nu m}\rho/a)|_{\rho=a} = 0$, show that the Bessel functions are orthogonal over the interval $[0, a]$ with an orthogonality integral

$$\int_0^a J_\nu\left(\beta_{\nu m}\frac{\rho}{a}\right) J_\nu\left(\beta_{\nu n}\frac{\rho}{a}\right)\rho\, d\rho = 0, \qquad m \neq n, \quad \nu > -1.$$

(b) Derive the corresponding normalization integral ($m = n$).

ANS. (b) $\dfrac{a^2}{2}\left(1 - \dfrac{\nu^2}{\beta_{\nu m}^2}\right)[J_\nu(\beta_{\nu m})]^2$, $\nu > -1$.

These formulas refer to positive derivative roots. For $\nu = 0$ the additional constant mode has norm $a^2/2$. For $-1 < \nu < 0$, J'_ν also has the imaginary zeros $\pm i\kappa$, where $\kappa > 0$ and $I'_\nu(\kappa) = 0$ (differentiate $J_\nu(\pm ix) = (\pm i)^\nu I_\nu(x)$); the corresponding mode $I_\nu(\kappa\rho/a)$ is treated in Exercise 14.2.5.

Solution.

Interpret $\beta_{\nu m}$ as a positive argument satisfying $J'_\nu(\beta_{\nu m}) = 0$, so the derivative with respect to ρ vanishes at $\rho = a$. Put $k = \beta_{\nu m}/a$ and $k' = \beta_{\nu n}/a$ in the first Lommel integral. For $m \neq n$ the denominator is nonzero, while both terms in its numerator contain a vanishing derivative. Thus

$$\boxed{\int_0^a \rho J_\nu(\beta_{\nu m}\rho/a)J_\nu(\beta_{\nu n}\rho/a)d\rho = 0 \quad (m \neq n)}.$$

For $m = n$, use the second Lommel integral instead. The squared derivative term is zero, and $k^2a^2 = \beta_{\nu m}^2$, giving

$$\boxed{N_{\nu m} = \frac{a^2}{2}\left(1 - \frac{\nu^2}{\beta_{\nu m}^2}\right) J_\nu(\beta_{\nu m})^2}.$$

The origin terms vanish for $\nu > -1$ by the calculation in Exercise 14.2.1. The integral defining $N_{\nu m}$ is positive, because its integrand $\rho J_\nu(\beta_{\nu m}\rho/a)^2$ is nonnegative and not identically zero. Hence $(1 - \nu^2/\beta_{\nu m}^2)J_\nu(\beta_{\nu m})^2 > 0$, which forces $\beta_{\nu m}^2 > \nu^2$: $\beta_{\nu m} > |\nu|$ for every positive derivative zero under consideration. For $\nu = 0$ the Neumann problem has the additional constant mode $J_0(0\rho/a) = 1$, of norm $a^2/2$. It is not one of the positive roots, and the displayed expression containing ν^2/β^2 must not be used as $0/0$. Its orthogonality to the positive modes follows from $\int_0^a \rho J_0(\beta\rho/a)d\rho = (a/\beta)^2 \int_0^\beta u J_0(u)du = (a/\beta)^2[uJ_1(u)]_0^\beta = a^2 J_1(\beta)/\beta = 0$ (substitute $u = \beta\rho/a$ and use $[uJ_1(u)]' = uJ_0(u)$, Exercise 14.1.4(a)); the value is zero because $J_1 = -J'_0$ vanishes at the derivative root β.

14.2.3 Verify that the orthogonality equation, Eq. (14.44), and the normalization equation, Eq. (14.46), hold for $\nu > -1$.

Hint. Using power-series expansions, examine the behavior of Eq. (14.43) as $\rho \to 0$.

Solution.

The two equations to be checked are the Dirichlet orthogonality and normalization formulas

$$\int_0^a \rho J_\nu(\alpha_{\nu m}\rho/a)J_\nu(\alpha_{\nu n}\rho/a)d\rho = 0 \quad (m \neq n),$$

$$\int_0^a \rho J_\nu(\alpha_{\nu n}\rho/a)^2 d\rho = \frac{a^2}{2}J_{\nu+1}(\alpha_{\nu n})^2.$$

The only possible issue when $-1 < \nu < 0$ is the singular-looking origin. Put $k = \alpha_{\nu m}/a$, $k' = \alpha_{\nu n}/a$. The series begin

$$J_\nu(k\rho) = \frac{(k/2)^\nu}{\Gamma(\nu+1)}\rho^\nu\left[1 - \frac{k^2\rho^2}{4(\nu+1)} + O(\rho^4)\right].$$

This is the form used in Exercise 14.2.1, with $J_\nu(k\rho) = A\rho^\nu[\ldots]$, $J_\nu(k'\rho) = B\rho^\nu[\ldots]$, $A = (k/2)^\nu/\Gamma(\nu+1)$, and $B = (k'/2)^\nu/\Gamma(\nu+1)$. In the boundary expression $\rho[J_\nu(k\rho)\partial_\rho J_\nu(k'\rho) - \partial_\rho J_\nu(k\rho)J_\nu(k'\rho)]$ (the bracket of Eq. (14.43), up to sign), both products begin with $\nu AB\rho^{2\nu}$, so the terms of order $\rho^{2\nu}$ cancel exactly. The next term, computed in Exercise 14.2.1, is $AB(k^2 - k'^2)\rho^{2\nu+2}/[2(\nu+1)]$, that is,

$$\frac{(kk'/4)^\nu(k^2-k'^2)}{2(\nu+1)\Gamma(\nu+1)^2}\rho^{2\nu+2},$$

which tends to zero as $\rho \to 0$ for precisely the stated range $\nu > -1$. At $\rho = a$ both function values vanish for distinct Dirichlet roots, so the first Lommel formula gives orthogonality. For $m = n$, the second Lommel integral of Exercise 14.2.1 with $k = \alpha_{\nu n}/a$ and $J_\nu(\alpha_{\nu n}) = 0$ reduces to $a^2 J'_\nu(\alpha_{\nu n})^2/2$. The recurrence $J'_\nu = \nu J_\nu/x - J_{\nu+1}$ (from Eqs. (14.7) and (14.8), valid for any order) yields $J'_\nu(\alpha_{\nu n}) = -J_{\nu+1}(\alpha_{\nu n})$, proving normalization. All relevant integrals converge because their origin integrands behave as $\rho^{2\nu+1}$.

14.2.4 From Eq. (14.44), develop a proof that $J_\nu(z)$, $\nu > -1$ has no complex roots (with a nonzero imaginary part).

Hint.

(a) Use the series form of $J_\nu(z)$ to exclude pure imaginary roots.

(b) Assume $\alpha_{\nu m}$ to be complex and take $\alpha_{\nu n}$ to be $\alpha^*_{\nu m}$.

Solution.

Take real $\nu > -1$ and consider nonzero roots on a fixed branch. The factor z^ν can be removed by defining the entire even function

$$F_\nu(z) = \Gamma(\nu+1)(2/z)^\nu J_\nu(z) = \sum_{r=0}^{\infty}\frac{(-1)^r\Gamma(\nu+1)(z^2/4)^r}{r!\Gamma(\nu+r+1)}.$$

(a) If $z = iy$ with real $y \neq 0$, every term of $F_\nu(iy)$ is positive, because $(-1)^r(iy)^{2r} = y^{2r}$ and all gamma arguments $\nu + r + 1$ are positive. Thus $F_\nu(iy) > 0$, excluding nonzero pure imaginary roots.

(b) For $z \neq 0$, $J_\nu(z) = 0$ exactly when $F_\nu(z) = 0$, on any branch of z^ν. The series coefficients of F_ν are real, so $F_\nu(z^*) = F_\nu(z)^*$, and if z_0 is a root then z_0^* is also a root. Suppose both its real and imaginary parts are nonzero. Then $z_0^2 - (z_0^*)^2 = 4i\,\Re z_0\,\Im z_0 \neq 0$. For complex k and $0 < x \leq 1$ put $u_k(x) = x^\nu F_\nu(kx)$, with the real power x^ν. For $k > 0$ this is a constant multiple of $J_\nu(kx)$. Its series contains only k^2: $u_k = \sum_{r\geq 0} c_r k^{2r} x^{\nu+2r}$ with $c_r = (-1)^r\Gamma(\nu+1)/[4^r r!\,\Gamma(\nu+r+1)]$, so that $4r(\nu+r)c_r = -c_{r-1}$ for $r \geq 1$. Since $(\nu+2r)^2 - \nu^2 = 4r(\nu+r)$, termwise differentiation gives, for every complex k,

$$\begin{aligned}(xu'_k)' - \frac{\nu^2}{x}u_k &= \sum_{r\geq 1} 4r(\nu+r)c_r k^{2r}x^{\nu+2r-1}\\ &= -k^2 x\sum_{r\geq 1} c_{r-1}k^{2r-2}x^{\nu+2r-2} = -k^2 x\,u_k,\end{aligned}$$

that is, $(xu_k')' + (k^2x - \nu^2/x)u_k = 0$. Hence the derivation of the first Lommel identity in Exercise 14.2.1 (the derivation of Eqs. (14.43)–(14.44)) applies to $y = u_{z_0}$ and $w = u_{z_0^*}$:

$$\frac{d}{dx}\{x(yw' - y'w)\} = (z_0^2 - z_0^{*2})\, xyw.$$

Integrate from 0 to 1. At $x = 1$ the bracket vanishes, because $y(1) = F_\nu(z_0) = 0$ and $w(1) = F_\nu(z_0^*) = 0$. As $x \to 0$ it is $O(x^{2\nu+2})$ by the expansion in Exercise 14.2.1 (with $A = B = 1$; the algebra is unchanged for complex k, k'). For real x, $w(x) = y(x)^*$, so

$$0 = (z_0^2 - z_0^{*2}) \int_0^1 xyw\, dx = (z_0^2 - z_0^{*2}) \int_0^1 x^{2\nu+1} |F_\nu(z_0 x)|^2 dx.$$

The first factor is nonzero. But the integrand is nonnegative, continuous on $(0, 1]$, and positive near $x = 0$, where $F_\nu(z_0 x) \to F_\nu(0) = 1$; so the integral is strictly positive. This contradiction excludes every remaining nonreal possibility. Therefore all nonzero zeros are real. The printed statement cites Eq. (11.49), an unrelated equation of Chapter 11; the intended reference is the orthogonality relation, Eq. (14.44), whose derivation is used above.

14.2.5

(a) In the series expansion

$$f(\rho) = \sum_{m=1}^{\infty} c_{\nu m} J_\nu\left(\alpha_{\nu m} \frac{\rho}{a}\right), \quad 0 \le \rho \le a, \quad \nu > -1,$$

with $J_\nu(\alpha_{\nu m}) = 0$, show that the coefficients are given by

$$c_{\nu m} = \frac{2}{a^2 [J_{\nu+1}(\alpha_{\nu m})]^2} \int_0^a f(\rho) J_\nu\left(\alpha_{\nu m} \frac{\rho}{a}\right) \rho\, d\rho.$$

(b) In the series expansion

$$f(\rho) = f_{\text{extra}}(\rho) + \sum_{m=1}^{\infty} d_{\nu m} J_\nu\left(\beta_{\nu m} \frac{\rho}{a}\right), \quad 0 \le \rho \le a, \quad \nu > -1,$$

with $(d/d\rho) J_\nu(\beta_{\nu m}\rho/a)\big|_{\rho=a} = 0$, show that the coefficients are given by

$$d_{\nu m} = \frac{2}{a^2 (1 - \nu^2/\beta_{\nu m}^2)[J_\nu(\beta_{\nu m})]^2} \int_0^a f(\rho) J_\nu\left(\beta_{\nu m} \frac{\rho}{a}\right) \rho\, d\rho.$$

Use real $\nu > -1$ and convergence in $L^2((0, a), \rho\, d\rho)$; the roots are positive. In (b), $f_{\text{extra}} = 0$ for $\nu > 0$; for $\nu = 0$ it is a constant mode, and for $-1 < \nu < 0$ a multiple of $I_\nu(\kappa\rho/a)$ with $I_\nu'(\kappa) = 0$, $\kappa > 0$. Its coefficient is determined below.

Solution.

Use the weighted inner product $\langle g, h\rangle = \int_0^a g(\rho)^* h(\rho) \rho\, d\rho$. The coefficient formulas hold whenever the stated expansions converge in this weighted L^2 space: then the inner product of the series with a fixed function ϕ may be taken term by term, because the partial sums S_N satisfy $|\langle \phi, f - S_N\rangle| \le \|\phi\|\, \|f - S_N\| \to 0$ (Cauchy–Schwarz).

(a) Multiply the Dirichlet expansion by $J_\nu(\alpha_{\nu n}\rho/a)\rho$ and integrate. Orthogonality, Eq. (14.44), removes every term except $m = n$, whose integral is given by the normalization, Eq. (14.46) (both valid for $\nu > -1$ by Exercise 14.2.3):

$$\int_0^a f(\rho)J_\nu(\alpha_{\nu n}\rho/a)\rho d\rho = c_{\nu n}\frac{a^2}{2}J_{\nu+1}(\alpha_{\nu n})^2.$$

Division gives the asserted $c_{\nu n}$.

(b) Repeat with $J_\nu(\beta_{\nu n}\rho/a)\rho$ instead. The term f_{extra} contributes nothing, because it is orthogonal to every $J_\nu(\beta_{\nu n}\rho/a)$ (see the erratum of Exercise 14.2.2 for $\nu = 0$, and the check below for $-1 < \nu < 0$). The Neumann orthogonality and norm from Exercise 14.2.2 then give

$$\int_0^a f(\rho)J_\nu(\beta_{\nu n}\rho/a)\rho d\rho = d_{\nu n}\frac{a^2}{2}\left(1 - \frac{\nu^2}{\beta_{\nu n}^2}\right)J_\nu(\beta_{\nu n})^2.$$

Solving for $d_{\nu n}$ proves the second displayed coefficient formula for each positive root.

Positive derivative roots alone do not give an expansion of an arbitrary function throughout the stated range $\nu > -1$. For $\nu = 0$ one must add the constant mode with coefficient $d_{00} = 2a^{-2}\int_0^a f(\rho)\rho\,d\rho$. For $-1 < \nu < 0$ there is no constant mode; one must instead include a negative-eigenvalue mode $I_\nu(\kappa\rho/a)$, where $I'_\nu(\kappa) = 0$, $\kappa > 0$. These omissions do not change the coefficient formulas for the positive-root terms. Here is a direct check of the latter qualification. For $-1 < \nu < 0$, $I_\nu(x) > 0$ by its series, its derivative is negative near zero ($I_\nu \sim (x/2)^\nu/\Gamma(\nu+1)$), and positive for large x (where I_ν grows like $e^x/\sqrt{2\pi x}$). Thus it has a positive derivative zero. Put $w = xI'_\nu/I_\nu$, which has the sign of I'_ν. Differentiating, $xw' = w + x^2I''_\nu/I_\nu - w^2$, and the modified Bessel equation $x^2I''_\nu = (x^2+\nu^2)I_\nu - xI'_\nu$ turns this into $xw' = x^2 + \nu^2 - w^2$. At every zero of w its derivative is positive, so it crosses zero only once. The corresponding radial function has eigenvalue $-\kappa^2/a^2$ and is orthogonal to all positive-eigenvalue modes by the boundary-bracket argument of Exercise 14.2.1. Indeed, $y = I_\nu(\kappa\rho/a)$ satisfies $(\rho y')' - (\kappa^2\rho/a^2 + \nu^2/\rho)y = 0$, the equation of Exercise 14.2.1 with $k^2 = -\kappa^2/a^2$, so with $z = J_\nu(\beta_{\nu n}\rho/a)$ the identity derived there becomes

$$\frac{d}{d\rho}\{\rho(yz' - y'z)\} = -\frac{\kappa^2 + \beta_{\nu n}^2}{a^2}\,\rho yz.$$

The bracket vanishes at $\rho = a$, where $y' = z' = 0$, and it is $O(\rho^{2\nu+2})$ as $\rho \to 0$ by the expansion of Exercise 14.2.1, now with $p = k^2/[4(\nu+1)] = -\kappa^2/[4(\nu+1)a^2]$. Integrating from 0 to a therefore gives $\int_0^a \rho yz\,d\rho = 0$, since $\kappa^2 + \beta_{\nu n}^2 > 0$. For clarity, derive its norm rather than assuming a modified Lommel formula. Put $y(x) = I_\nu(x)$ and

$$F(x) = \frac{1}{2}\{x^2[y(x)^2 - y'(x)^2] + \nu^2 y(x)^2\}.$$

Differentiation gives

$$F' = x(y^2 - y'^2) + (x^2+\nu^2)yy' - x^2y'y''.$$

The modified equation says $x^2y'' = (x^2+\nu^2)y - xy'$. Substitution cancels the two yy' terms and the two xy'^2 terms, leaving $F' = xy^2$. Near zero, the series of I_ν gives $y = Cx^\nu[1 + O(x^2)]$ and $y' = Cx^{\nu-1}[\nu + O(x^2)]$, with $C = 2^{-\nu}/\Gamma(\nu+1)$. So in $2F = x^2y^2 - x^2y'^2 + \nu^2y^2$ the last two terms are $C^2x^{2\nu}[-\nu^2 + O(x^2)]$ and $C^2x^{2\nu}[\nu^2 + O(x^2)]$, whose leading parts cancel, while $x^2y^2 = O(x^{2\nu+2})$. Hence $F = O(x^{2\nu+2}) \to 0$ for $\nu > -1$. Now set $x = \kappa\rho/a$ and use $I'_\nu(\kappa) = 0$:

$$N_- = \int_0^a I_\nu(\kappa\rho/a)^2\rho\,d\rho = \frac{a^2}{\kappa^2}\int_0^\kappa xI_\nu(x)^2\,dx = \frac{a^2}{\kappa^2}[F(\kappa) - F(0)].$$

Here $F(0)$ means the limit 0 found above, and $y'(\kappa) = 0$ gives $F(\kappa) = \frac{1}{2}(\kappa^2 + \nu^2)I_\nu(\kappa)^2$. Thus its norm is

$$N_- = \frac{a^2}{2}\left(1 + \frac{\nu^2}{\kappa^2}\right) I_\nu(\kappa)^2,$$

so its coefficient is $N_-^{-1}\int_0^a f(\rho)I_\nu(\kappa\rho/a)\rho d\rho$. At the Dirichlet endpoint $\rho = a$, a series of zero-valued basis functions is zero; nonzero prescribed endpoint values are understood as interior limits or in L^2, rather than equality at that single point.

14.2.6 A right circular cylinder has an electrostatic potential of $\psi(\rho, \varphi)$ on both ends. The potential on the curved cylindrical surface is zero. Find the potential at all interior points.

Hint. Choose your coordinate system and adjust your z dependence to exploit the symmetry of your potential.

Assume bounded, piecewise smooth end data, and seek the bounded harmonic interior solution. At an incompatible rim, impose the face limits away from the junction.

Solution.

Let the cylinder have radius a and length l, with its ends at $z = \pm l/2$. Denote the sought interior potential by V and the identical prescribed end potential by $\psi(\rho, \varphi)$. Substituting $V = R(\rho)e^{im\varphi}Z(z)$, $m \in \mathbb{Z}$, in $\nabla^2 V = 0$ and separating with the constant k^2 gives

$$Z'' - k^2 Z = 0, \qquad \rho^2 R'' + \rho R' + (k^2\rho^2 - m^2)R = 0,$$

Bessel's equation of order $|m|$ in $k\rho$. Regularity on the axis selects $R = J_{|m|}(k\rho)$, since $Y_{|m|}$ is unbounded at $\rho = 0$. The side condition forces $k = \alpha_{|m|n}/a$, where $J_{|m|}(\alpha_{|m|n}) = 0$. (A negative separation constant $-\kappa^2$ would give the regular radial factor $I_{|m|}(\kappa\rho)$, which is positive for $\rho > 0$ and cannot vanish at $\rho = a$; a zero constant gives $\rho^{|m|}$, which cannot either.) Identical end data are even in z; hence use $\cosh(kz)$ rather than the odd $\sinh(kz)$. Normalize the axial factor to one on both ends:

$$\boxed{V(\rho, \varphi, z) = \sum_{m=-\infty}^{\infty}\sum_{n=1}^{\infty} A_{mn}J_{|m|}(\alpha_{|m|n}\rho/a)e^{im\varphi}\frac{\cosh(\alpha_{|m|n}z/a)}{\cosh(\alpha_{|m|n}l/(2a))}}.$$

At either end this is the double Fourier–Bessel expansion of ψ. Multiply it by $e^{-im\varphi}J_{|m|}(\alpha_{|m|n}\rho/a)\rho$ and integrate over the disk. Write the summation indices of the series as m', n'. Since $\int_0^{2\pi} e^{i(m'-m)\varphi}d\varphi = 2\pi\delta_{mm'}$, only terms with $m' = m$ survive, and among these the radial orthogonality, Eq. (14.44) of order $|m|$, keeps only $n' = n$. The angular norm is 2π and the radial norm, Eq. (14.46), is $a^2 J_{|m|+1}(\alpha_{|m|n})^2/2$. Therefore

$$\boxed{A_{mn} = \frac{\displaystyle\int_0^{2\pi}\int_0^a \psi(\rho, \varphi)e^{-im\varphi}J_{|m|}(\alpha_{|m|n}\rho/a)\rho\, d\rho\, d\varphi}{\pi a^2 J_{|m|+1}(\alpha_{|m|n})^2}}.$$

Every term satisfies Laplace's equation, the grounded side, and reflection symmetry; the coefficient projection supplies both end conditions. For real data $A_{-m,n} = A_{mn}^*$, making the potential real. For compatible boundary data and finite-energy solutions, the difference $U = V_1 - V_2$ has zero boundary values. Green's identity gives $\int |\nabla U|^2 d^3x = 0$, hence $U = 0$.

A discontinuous rim need not give a finite-energy field, so that energy argument should not be applied to each such field without qualification. Uniqueness still holds among bounded harmonic

solutions with the specified face limits. To see this, let Γ be the two rim circles and introduce the positive harmonic barrier

$$B(\mathbf{x}) = \int_\Gamma \frac{d\ell_{\mathbf{y}}}{|\mathbf{x}-\mathbf{y}|}.$$

It is finite in the interior, harmonic there, and diverges to $+\infty$ at the rim. Indeed, let d be the distance from $\mathbf{x}$ to Γ, attained at $\mathbf{y}_0 \in \Gamma$, and let s be arc length along that rim circle, measured from $\mathbf{y}_0$. On the arc $|s| \le \delta$ (a fixed δ, $0 < \delta < \pi a$) a chord is no longer than its arc, so $|\mathbf{x}-\mathbf{y}| \le d + |\mathbf{y}-\mathbf{y}_0| \le d + |s| \le \sqrt{2}\,(d^2+s^2)^{1/2}$ (since $2d|s| \le d^2+s^2$). The rest of the integrand is positive; hence

$$B(\mathbf{x}) \ge \frac{1}{\sqrt{2}} \int_{-\delta}^{\delta} \frac{ds}{\sqrt{d^2+s^2}} = \sqrt{2}\,\operatorname{arsinh}\frac{\delta}{d} \longrightarrow \infty \qquad (d \to 0).$$

For any $\epsilon > 0$, both $U - \epsilon B$ and $-U - \epsilon B$ have nonpositive boundary upper limits, including the rim because U is bounded. The maximum principle gives $|U(\mathbf{x})| \le \epsilon B(\mathbf{x})$ in the interior. Letting $\epsilon \downarrow 0$ proves $U = 0$. The series therefore supplies the bounded interior solution and, at incompatible rims, the face limits away from the junction.

14.2.7 A function $f(x)$ is expressed as a Bessel series:

$$f(x) = \sum_{n=1}^{\infty} a_n J_m(\alpha_{mn} x),$$

with α_{mn} the nth root of J_m. Prove the Parseval relation,

$$\int_0^1 [f(x)]^2 x\,dx = \frac{1}{2}\sum_{n=1}^{\infty} a_n^2 [J_{m+1}(\alpha_{mn})]^2.$$

Let $m > -1$ be real, let α_{mn} enumerate the positive zeros, and assume convergence to f in $L^2((0,1), x\,dx)$.

Solution.

First take a finite partial sum $S_N(x) = \sum_{n=1}^{N} a_n J_m(\alpha_{mn}x)$. Expanding its square, then using orthogonality and the Dirichlet norm, gives

$$\begin{aligned}\int_0^1 S_N(x)^2 x\,dx &= \sum_{n=1}^{N}\sum_{j=1}^{N} a_n a_j \int_0^1 J_m(\alpha_{mn}x) J_m(\alpha_{mj}x) x\,dx \\ &= \frac{1}{2}\sum_{n=1}^{N} a_n^2 J_{m+1}(\alpha_{mn})^2.\end{aligned}$$

To pass to an infinite series, the expansion must converge to f in $L^2((0,1), x\,dx)$; merely writing a formal series does not justify Parseval's equality. Under that assumption, write $f^2 - S_N^2 = (f - S_N)(f + S_N)$; the Cauchy–Schwarz inequality gives

$$\left|\int_0^1 (f^2 - S_N^2) x\,dx\right| \le \|f - S_N\|\,\|f + S_N\| \longrightarrow 0,$$

because $\|f - S_N\| \to 0$ while $\|f + S_N\| \le 2\|f\| + \|S_N - f\|$ stays bounded. The finite identity therefore tends to the formula in the question. For real f and coefficients this is the usual Parseval identity. For complex functions, the norm version is

$$\int_0^1 |f(x)|^2 x\,dx = \frac{1}{2}\sum_{n=1}^{\infty} |a_n|^2 J_{m+1}(\alpha_{mn})^2;$$

the bilinear square identity also follows by the same limiting argument. The assumption holds for every f in $L^2((0,1), x\,dx)$ when the a_n are its Fourier–Bessel coefficients (those of Exercise 14.2.5(a) with $a = 1$), because the Dirichlet Bessel functions are complete; without completeness the partial sums would converge only to the orthogonal projection of f onto their closed span.

14.2.8 Prove that

$$\sum_{n=1}^{\infty} (\alpha_{mn})^{-2} = \frac{1}{4(m+1)}.$$

Hint. Expand x^m in a Bessel series and apply the Parseval relation.
Here $m > -1$ is real and α_{mn} is the nth positive zero of J_m.

Solution.

Expand $f(x) = x^m$ on $0 < x < 1$ in the Dirichlet Bessel basis. For $m > -1$, $\int_0^1 |x^m|^2 x\,dx = 1/[2(m+1)]$ is finite, so x^m lies in $L^2((0,1), x\,dx)$. By the completeness of the Dirichlet Bessel functions (end of Exercise 14.2.7), its Fourier–Bessel series then converges to x^m in this space, which is the hypothesis of the Parseval relation. The coefficient formula of Exercise 14.2.5(a), with $a = 1$ and $\nu = m$, gives

$$a_n = \frac{2}{J_{m+1}(\alpha_{mn})^2} \int_0^1 x^{m+1} J_m(\alpha_{mn} x) dx.$$

Set $\alpha = \alpha_{mn}$. The recurrence $d[u^{m+1} J_{m+1}(u)]/du = u^{m+1} J_m(u)$ (Exercise 14.1.4(a) with $n = m+1$, valid for any order), with $u = \alpha x$ and $x^{m+1} J_{m+1}(\alpha x) = \alpha^{-m-1} u^{m+1} J_{m+1}(u)$, implies

$$\frac{d}{dx}[x^{m+1} J_{m+1}(\alpha x)] = \alpha x^{m+1} J_m(\alpha x).$$

Its lower boundary vanishes as x^{2m+2}; hence

$$\int_0^1 x^{m+1} J_m(\alpha x) dx = \frac{J_{m+1}(\alpha)}{\alpha}, \qquad a_n = \frac{2}{\alpha_{mn} J_{m+1}(\alpha_{mn})}.$$

Insert these coefficients into Parseval's relation:

$$\frac{1}{2(m+1)} = \frac{1}{2}\sum_{n=1}^{\infty} \frac{4 J_{m+1}(\alpha_{mn})^2}{\alpha_{mn}^2 J_{m+1}(\alpha_{mn})^2} = 2\sum_{n=1}^{\infty} \alpha_{mn}^{-2}.$$

Thus

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{\alpha_{mn}^2} = \frac{1}{4(m+1)}}.$$

The expansion is used in the weighted L^2 sense. At $x = 1$ every basis function is zero whereas $x^m = 1$; that single endpoint has zero measure and does not affect the integral or Parseval's identity.

14.2.9 A right circular cylinder of length l and radius a has on its end caps a potential

$$\psi\Big|_{z=\pm\frac{l}{2}} = 100\left(1-\frac{\rho}{a}\right).$$

The potential on the curved surface (the side) is zero. Using the Bessel series from Exercise 14.2.6, calculate the electrostatic potential for $\rho/a = 0.0(0.2)1.0$ and $z/l = 0.0(0.1)0.5$. Take $a/l = 0.5$.

Hint. From Exercise 14.1.29 you have

$$\int_0^{\alpha_{0n}} \left(1-\frac{y}{\alpha_{0n}}\right) J_0(y)y\,dy.$$

Show that this equals

$$\frac{1}{\alpha_{0n}}\int_0^{\alpha_{0n}} J_0(y)\,dy.$$

Numerical evaluation of this latter form rather than the former is both faster and more accurate.

Note. For $\rho/a = 0.0$ and $z/l = 0.5$ the convergence is slow, 20 terms giving only 98.4 rather than 100.

Check value. For $\rho/a = 0.4$ and $z/l = 0.3$, $\psi = 24.559$.

Solution.

The end data are independent of φ, so $\int_0^{2\pi} e^{-im\varphi}d\varphi = 0$ for $m \neq 0$ and only $m = 0$ remains in Exercise 14.2.6. Put $r = \rho/a$, $s = z/l$, and $\alpha_n = \alpha_{0n}$. Then $z/a = (z/l)(l/a) = 2s$ and $l/(2a) = 1$, so the axial factor is $\cosh(2\alpha_n s)/\cosh\alpha_n$. With $\rho = ar$, the coefficient formula of Exercise 14.2.6 becomes

$$A_n = \frac{2\pi\cdot 100\,a^2\int_0^1(1-r)J_0(\alpha_n r)r\,dr}{\pi a^2 J_1(\alpha_n)^2}.$$

Hence

$$\boxed{V(r,s) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r)\frac{\cosh(2\alpha_n s)}{\cosh\alpha_n}}, \qquad A_n = \frac{200}{J_1(\alpha_n)^2}\int_0^1 (1-r)J_0(\alpha_n r)r\,dr.$$

With $y = \alpha_n r$, the last integral is

$$\frac{1}{\alpha_n^2}\int_0^{\alpha_n}(1-y/\alpha_n)yJ_0(y)dy = \frac{1}{\alpha_n^3}\int_0^{\alpha_n} J_0(y)dy,$$

where the final equality is proved by the two integrations by parts in Exercise 14.1.29. Consequently the stable coefficients are

$$\boxed{A_n = \frac{200}{\alpha_n^3 J_1(\alpha_n)^2}\int_0^{\alpha_n} J_0(y)dy}.$$

The resulting potentials are:

$\rho/a \backslash z/l$	0.0	0.1	0.2	0.3	0.4	0.5
0.0	14.108	15.805	21.353	32.341	52.640	100.000
0.2	13.290	14.881	20.061	30.213	48.293	80.000
0.4	10.989	12.287	16.485	24.559	38.270	60.000
0.6	7.617	8.504	11.359	16.786	25.844	40.000
0.8	3.747	4.179	5.568	8.204	12.648	20.000
1.0	0.000	0.000	0.000	0.000	0.000	0.000

The last column is the prescribed boundary value $100(1-r)$; the last row is the grounded side. For $s < 1/2$ the series converges exponentially: $|J_0| \le 1$ and $\cosh(2\alpha_n s)/\cosh\alpha_n \le e^{2\alpha_n s}/(e^{\alpha_n}/2) = 2e^{-\alpha_n(1-2s)}$, with $\alpha_n \approx (n-\frac{1}{4})\pi$; for $s \le 0.4$ this factor is at most $2e^{-0.2\alpha_n}$. At $s = 1/2$ it equals one and only the algebraic decay of the A_n remains, which is the slow convergence mentioned in the Note. Quantitatively, the large-x forms $J_0(x) \approx \sqrt{2/(\pi x)}\cos(x-\pi/4)$ and $J_1(x) \approx \sqrt{2/(\pi x)}\cos(x-3\pi/4)$ (Section 14.6) give $\alpha_n \approx (n-\frac{1}{4})\pi$ and $J_1(\alpha_n)^2 \approx 2/(\pi\alpha_n)$, while $\int_0^{\alpha_n} J_0\,dy \to \int_0^{\infty} J_0\,dy = 1$ (Exercise 14.1.20 with $a = 0$, $b = 1$). Hence

$$A_n \approx \frac{100\pi}{\alpha_n^2} \approx \frac{100}{\pi(n-\frac{1}{4})^2}.$$

The main relative correction, $-\int_{\alpha_n}^{\infty} J_0\,dy \approx (-1)^{n+1}\sqrt{2/(\pi\alpha_n)}$ (integrate the cosine form by parts), alternates in sign and so largely cancels in sums. At the top center, where $J_0(0) = 1$ and the axial factor is one, the modes after the twentieth therefore add about

$$\sum_{n\ge 21} \frac{100}{\pi(n-\frac{1}{4})^2} \approx \frac{100}{\pi}\int_{20.5}^{\infty} \frac{dt}{(t-\frac{1}{4})^2} = \frac{100}{20.25\,\pi} \approx 1.57,$$

which is the shortfall $100 - 98.42$ quoted in the Note. For the interior entries, the integrals $\int_0^{\alpha_n} J_0(y)dy$ were accumulated as sums of quadratures over the intervals between consecutive zeros of J_0, and 120 modes were summed. Comparing 80 and 120 modes changed every interior entry by less than 10^{-20}, comfortably below the 0.001 displayed precision. This convergence comparison is a numerical check, not a universal truncation-error theorem. The printed check value 24.558 is not correctly rounded to three decimals: the converged value is $V(0.4, 0.3) = 24.5585639709\ldots$, hence 24.559. At the top center, a 20-mode sum is $98.424595978\ldots$; prescribing the exact boundary value avoids reporting its slowly convergent truncation as the boundary potential. Symmetry gives $V(r,-s) = V(r,s)$ for the lower half of the cylinder.

14.3 Neumann Functions, Bessel Functions of the Second Kind

14.3.1 Prove that the Neumann functions Y_n (with n an integer) satisfy the recurrence relations

$$Y_{n-1}(x) + Y_{n+1}(x) = \frac{2n}{x}Y_n(x),$$
$$Y_{n-1}(x) - Y_{n+1}(x) = 2Y_n'(x).$$

Hint. These relations may be proved by differentiating the recurrence relations for J_ν or by using the limit form of Y_ν but not dividing everything by zero.

Solution.

Begin with noninteger ν, where the defining denominator is nonzero. Abbreviate $C = \cos\pi\nu$ and $S = \sin\pi\nu$. Since $\cos\pi(\nu \pm 1) = -C$ and $\sin\pi(\nu \pm 1) = -S$,

$$Y_{\nu-1} = \frac{CJ_{\nu-1} + J_{-\nu+1}}{S}, \qquad Y_{\nu+1} = \frac{CJ_{\nu+1} + J_{-\nu-1}}{S}.$$

The recurrences (14.7) and (14.8) hold for every order; at the orders ν and $-\nu$ they read

$$J_{\nu-1} + J_{\nu+1} = \frac{2\nu}{x}J_\nu, \qquad J_{\nu-1} - J_{\nu+1} = 2J'_\nu,$$
$$J_{-\nu-1} + J_{-\nu+1} = -\frac{2\nu}{x}J_{-\nu}, \qquad J_{-\nu-1} - J_{-\nu+1} = 2J'_{-\nu}.$$

Add the two expressions for $Y_{\nu\mp1}$ and use the first column:

$$Y_{\nu-1} + Y_{\nu+1} = \frac{C(2\nu J_\nu/x) - 2\nu J_{-\nu}/x}{S} = \frac{2\nu}{x}\frac{CJ_\nu - J_{-\nu}}{S} = \frac{2\nu}{x}Y_\nu.$$

Subtracting instead and using the second column, together with $Y'_\nu = (CJ'_\nu - J'_{-\nu})/S$, gives

$$\begin{aligned} Y_{\nu-1} - Y_{\nu+1} &= \frac{C(J_{\nu-1} - J_{\nu+1}) + (J_{-\nu+1} - J_{-\nu-1})}{S} \\ &= \frac{2CJ'_\nu - 2J'_{-\nu}}{S} = 2Y'_\nu. \end{aligned}$$

Now let $\nu \to n$. The defining removable limits give $Y_\nu \to Y_n$ and $Y_{\nu\pm1} \to Y_{n\pm1}$. Because $Y_\nu(x)$, with these limits inserted at the integers, depends analytically on ν and on $x > 0$ jointly, the limit may also be taken after the x differentiation: $Y'_\nu \to Y'_n$. Both recurrences therefore hold for integer n, with no division by $\sin\pi n = 0$.

14.3.2 Show that for integer n

$$Y_{-n}(x) = (-1)^n Y_n(x).$$

Solution.

For noninteger ν, solve the definition for the opposite-order first-kind function:

$$J_{-\nu} = \cos\pi\nu\, J_\nu - \sin\pi\nu\, Y_\nu.$$

Apply the definition once more with order $-\nu$, using $\cos(-\pi\nu) = \cos\pi\nu$, $\sin(-\pi\nu) = -\sin\pi\nu$, and then $\cos^2\pi\nu - 1 = -\sin^2\pi\nu$:

$$\begin{aligned} Y_{-\nu} &= \frac{\cos\pi\nu\, J_{-\nu} - J_\nu}{-\sin\pi\nu} \\ &= \frac{(\cos^2\pi\nu - 1)J_\nu - \cos\pi\nu\sin\pi\nu\, Y_\nu}{-\sin\pi\nu} \\ &= \frac{-\sin^2\pi\nu\, J_\nu - \cos\pi\nu\sin\pi\nu\, Y_\nu}{-\sin\pi\nu} \\ &= \sin\pi\nu\, J_\nu + \cos\pi\nu\, Y_\nu. \end{aligned}$$

As $\nu \to n \in \mathbb{Z}$, $\sin\pi\nu \to 0$ while $J_\nu \to J_n$ stays finite, $\cos\pi\nu \to (-1)^n$, and $Y_{\pm\nu} \to Y_{\pm n}$ by the defining removable limits. This gives

$$\boxed{Y_{-n}(x) = (-1)^n Y_n(x)}.$$

This also includes $n = 0$, where the equality is immediate.

14.3.3 Show that

$$Y_0'(x) = -Y_1(x).$$

Solution.

Set $n = 0$ in the derivative recurrence of Exercise 14.3.1:

$$2Y_0' = Y_{-1} - Y_1.$$

Exercise 14.3.2 gives $Y_{-1} = -Y_1$, and hence

$$2Y_0' = -2Y_1, \qquad \boxed{Y_0' = -Y_1}.$$

The Instructor's Manual's last equation in this exercise, p. 209, prints J_1, J_0' where Y_1, Y_0' are required. The latter are the functions used in the recurrence above.

14.3.4 If X and Z are any two solutions of Bessel's equation, show that

$$X_\nu(x)Z_\nu'(x) - X_\nu'(x)Z_\nu(x) = \frac{A_\nu}{x},$$

in which A_ν may depend on ν but is independent of x. This is a special case of Exercise 7.6.11.

Solution.

Define $W = X_\nu Z_\nu' - X_\nu' Z_\nu$ on an interval not containing $x = 0$. Differentiation cancels the mixed terms:

$$W' = X_\nu Z_\nu'' - X_\nu'' Z_\nu.$$

Both functions satisfy Bessel's equation $x^2y'' + xy' + (x^2 - \nu^2)y = 0$, that is (dividing by x^2), $y'' = -y'/x - (1 - \nu^2/x^2)y$. Substitution therefore gives

$$\begin{aligned} W' &= X_\nu \left[-\frac{Z_\nu'}{x} - \left(1 - \frac{\nu^2}{x^2}\right) Z_\nu \right] - \left[-\frac{X_\nu'}{x} - \left(1 - \frac{\nu^2}{x^2}\right) X_\nu \right] Z_\nu \\ &= -\frac{1}{x}(X_\nu Z_\nu' - X_\nu' Z_\nu) = -\frac{W}{x}. \end{aligned}$$

Thus $(xW)' = W + xW' = 0$, proving

$$\boxed{W(x) = \frac{A_\nu}{x}}.$$

The constant depends on the chosen solutions and their order, but not on x. It is zero for linearly dependent solutions.

14.3.5 Verify the Wronskian formulas

$$J_\nu(x)J_{-\nu+1}(x) + J_{-\nu}(x)J_{\nu-1}(x) = \frac{2\sin\nu\pi}{\pi x},$$

$$J_\nu(x)Y_\nu'(x) - J_\nu'(x)Y_\nu(x) = \frac{2}{\pi x}.$$

Solution.

For noninteger ν, use the derivative recurrences $J_{\mu-1} = J'_\mu + (\mu/x)J_\mu$ and $J_{\mu+1} = -J'_\mu + (\mu/x)J_\mu$ (half the sum and half the difference of Eqs. (14.7) and (14.8), valid for any order; see Exercise 14.1.4). The second with $\mu = -\nu$ and the first with $\mu = \nu$ give

$$J_{1-\nu} = -J'_{-\nu} - \frac{\nu}{x}J_{-\nu}, \qquad J_{\nu-1} = J'_\nu + \frac{\nu}{x}J_\nu.$$

Multiply the first by J_ν and the second by $J_{-\nu}$ and add; the terms proportional to ν/x cancel:

$$J_\nu J_{1-\nu} + J_{-\nu}J_{\nu-1} = -J_\nu J'_{-\nu} + J'_\nu J_{-\nu} = -W[J_\nu, J_{-\nu}].$$

The Instructor's Manual's corresponding line on p. 209 has a minus sign before the second product; it must be plus, as the cancellation above shows. By Exercise 14.3.4, $W[J_\nu, J_{-\nu}] = J_\nu J'_{-\nu} - J'_\nu J_{-\nu} = A/x$ with a constant A. The leading series terms are

$$J_\nu = \frac{(x/2)^\nu}{\Gamma(\nu+1)}[1 + O(x^2)], \qquad J_{-\nu} = \frac{(x/2)^{-\nu}}{\Gamma(1-\nu)}[1 + O(x^2)].$$

Differentiating, $J'_{\pm\nu} = \pm(\nu/x)J_{\pm\nu}[1 + O(x^2)]$, and the powers $(x/2)^{\pm\nu}$ cancel in each product:

$$J_\nu J'_{-\nu} = -\frac{\nu}{x}\,\frac{1 + O(x^2)}{\Gamma(\nu+1)\Gamma(1-\nu)}, \qquad J'_\nu J_{-\nu} = \frac{\nu}{x}\,\frac{1 + O(x^2)}{\Gamma(\nu+1)\Gamma(1-\nu)}.$$

Since $A = xW = x(J_\nu J'_{-\nu} - J'_\nu J_{-\nu})$ is constant, let $x \to 0$; then use $\Gamma(\nu+1) = \nu\Gamma(\nu)$ and the reflection formula $\Gamma(\nu)\Gamma(1-\nu) = \pi/\sin\pi\nu$:

$$A = -\frac{2\nu}{\Gamma(\nu+1)\Gamma(1-\nu)} = -\frac{2}{\Gamma(\nu)\Gamma(1-\nu)} = -\frac{2\sin\pi\nu}{\pi}.$$

Therefore, since the left side of the first formula equals $-W[J_\nu, J_{-\nu}] = -A/x$,

$$\boxed{J_\nu J_{1-\nu} + J_{-\nu}J_{\nu-1} = \frac{2\sin\pi\nu}{\pi x}}.$$

Finally $Y_\nu = (\cos\pi\nu\, J_\nu - J_{-\nu})/\sin\pi\nu$, so bilinearity of the Wronskian and $W[J_\nu, J_\nu] = 0$ give

$$W[J_\nu, Y_\nu] = \frac{\cos\pi\nu\, W[J_\nu, J_\nu] - W[J_\nu, J_{-\nu}]}{\sin\pi\nu} = \frac{2\sin\pi\nu/(\pi x)}{\sin\pi\nu} = \boxed{\frac{2}{\pi x}}.$$

Both formulas extend to integer orders by their removable order limits. In the first, both sides then vanish: $\sin n\pi = 0$, and $J_{-n} = (-1)^n J_n$, $J_{1-n} = (-1)^{n-1}J_{n-1}$ turn the left side into $[(-1)^{n-1} + (-1)^n]J_n J_{n-1} = 0$. The second remains nonzero.

14.3.6 As an alternative to letting x approach zero in the evaluation of the Wronskian constant, we may invoke the uniqueness of power-series expansions. The coefficient of x^{-1} in the series expansion of $u_\nu(x)v'_\nu(x) - u'_\nu(x)v_\nu(x)$ is then A_ν. Show by series expansion that the coefficients of x^0 and x^1 of $J_\nu(x)J'_{-\nu}(x) - J'_\nu(x)J_{-\nu}(x)$ are each zero.

Solution.

For noninteger ν, write the first two series terms as

$$J_\nu = c\,x^\nu(1 - ax^2 + O(x^4)), \quad J_{-\nu} = d\,x^{-\nu}(1 - bx^2 + O(x^4)),$$

where

$$c = \frac{2^{-\nu}}{\Gamma(\nu+1)}, \quad d = \frac{2^{\nu}}{\Gamma(1-\nu)}, \quad a = \frac{1}{4(\nu+1)}, \quad b = \frac{1}{4(1-\nu)}$$

(the letters c, d avoid a clash with the Wronskian constant A_ν). Here J_ν is x^ν times a power series in x^2 and $J'_{-\nu}$ is $x^{-\nu-1}$ times a power series in x^2, so all powers in $J_\nu J'_{-\nu}$, and likewise in $J'_\nu J_{-\nu}$, are x^{2r-1} with integer $r \geq 0$. Consequently an x^0 term cannot occur. For x^1, differentiate the two displayed series,

$$J'_\nu = c\,[\nu x^{\nu-1} - a(\nu+2)x^{\nu+1} + O(x^{\nu+3})],$$
$$J'_{-\nu} = d\,[-\nu x^{-\nu-1} - b(2-\nu)x^{1-\nu} + O(x^{3-\nu})],$$

and multiply:

$$J_\nu J'_{-\nu} = cd\left[-\frac{\nu}{x} + \{a\nu - b(2-\nu)\}x + O(x^3)\right],$$
$$J'_\nu J_{-\nu} = cd\left[\frac{\nu}{x} + \{-b\nu - a(\nu+2)\}x + O(x^3)\right].$$

Subtracting, the x coefficient is

$$\begin{aligned} cd[a\nu - b(2-\nu) + b\nu + a(\nu+2)] &= 2cd[(\nu+1)a + (\nu-1)b] \\ &= 2cd\left[\frac{1}{4} - \frac{1}{4}\right] = 0. \end{aligned}$$

Thus both requested coefficients vanish, leaving $-2\nu cd/x + O(x^3)$ at this order; that is, $A_\nu = -2\nu cd = -2/[\Gamma(\nu)\Gamma(1-\nu)] = -2\sin(\pi\nu)/\pi$, as in Exercise 14.3.5. At integer $\nu = n$ the negative-order relation makes the entire Wronskian identically zero, so the same coefficient conclusion holds there without using intermediate gamma factors at their poles.

14.3.7

(a) By differentiating and substituting into Bessel's ODE for $\nu = 0$, show that $\displaystyle\int_0^\infty \cos(x\cosh t)\,dt$ is a solution.

Hint. Rearrange the final integral to $\displaystyle\int_0^\infty \frac{d}{dt}\left[x\sin(x\cosh t)\sinh t\right]dt$.

(b) Show that $\displaystyle Y_0(x) = -\frac{2}{\pi}\int_0^\infty \cos(x\cosh t)\,dt$ is linearly independent of $J_0(x)$.

Take $x > 0$. Justify differentiation through an upper-half-plane convergence factor before taking its real-axis boundary limit.

Solution.

For real $x > 0$, the integral is conditionally convergent. Its formal x derivatives are not convergent ordinary improper integrals, and the proposed boundary term $x\sin(x\cosh t)\sinh t$ does not vanish as $t \to \infty$. The Instructor's Manual's unregularized integration by parts on p. 211 therefore needs the limiting argument below.

(a) Begin in the upper half-plane, with

$$H(z) = \int_0^\infty e^{iz\cosh t}dt, \qquad \Im z > 0.$$

The factor $|e^{iz\cosh t}| = e^{-\Im z\cosh t}$ makes the integral and all local z derivatives absolutely convergent, so $H' = \int_0^\infty i\cosh t\, e^{iz\cosh t}dt$ and $H'' = -\int_0^\infty \cosh^2 t\, e^{iz\cosh t}dt$. With $\cosh^2 t - 1 = \sinh^2 t$,

$$\begin{aligned} z^2H'' + zH' + z^2H &= \int_0^\infty [-z^2\sinh^2 t + iz\cosh t]e^{iz\cosh t}dt \\ &= \int_0^\infty \frac{d}{dt}\left[iz\sinh t\, e^{iz\cosh t}\right]dt = 0, \end{aligned}$$

since the product rule gives $\frac{d}{dt}[iz\sinh t\, e^{iz\cosh t}] = (iz\cosh t - z^2\sinh^2 t)e^{iz\cosh t}$. At $t = 0$ the bracket is zero, and at infinity exponential damping overcomes $\sinh t$. Thus H solves the order-zero Bessel equation in the upper half-plane. There J_0 and Y_0 are analytic solutions with Wronskian $2/(\pi z) \neq 0$ (Exercise 14.3.5, continued analytically), so $H = c_1J_0 + c_2Y_0$ with constants c_1, c_2. Both functions are continuous up to each $x > 0$ and real there; hence

$$\lim_{\epsilon\downarrow 0} \Re H(x + i\epsilon) = (\Re c_1)J_0(x) + (\Re c_2)Y_0(x).$$

Now put $u = \cosh t$, $du = \sinh t\, dt = \sqrt{u^2-1}\, dt$. Then

$$\Re H(x+i\epsilon) = \int_1^\infty \frac{\cos(xu)\, e^{-\epsilon u}}{\sqrt{u^2-1}}du, \qquad F(x) = \int_1^\infty \frac{\cos(xu)}{\sqrt{u^2-1}}du = \int_0^\infty \cos(x\cosh t)dt.$$

The singularity at $u = 1$ is integrable, and the tail of F converges by Dirichlet's test: $1/\sqrt{u^2-1}$ decreases to zero and $|\int_1^U \cos(xu)du| \le 2/x$. Abel's theorem for integrals then gives $\lim_{\epsilon\downarrow 0}\Re H(x + i\epsilon) = F(x)$. Indeed, $G(u) = \int_u^\infty \cos(xv)(v^2-1)^{-1/2}dv$ is continuous and bounded on $[1,\infty)$ with $G(1) = F(x)$ and $G(\infty) = 0$, and integration by parts gives

$$\Re H(x+i\epsilon) = G(1)e^{-\epsilon} - \epsilon\int_1^\infty G(u)e^{-\epsilon u}du \longrightarrow G(1),$$

because for every $R > 1$ the last term is at most $\epsilon\int_1^R |G|\, du + \sup_{u\ge R}|G(u)|$ in absolute value. Therefore $F = (\Re c_1)J_0 + (\Re c_2)Y_0$ is a solution:

$$\boxed{x^2F'' + xF' + x^2F = 0 \quad (x > 0)}.$$

(b) To compare it with J_0, find the behavior of F as $x \downarrow 0$. Split off an elementary logarithmic integral:

$$F(x) = \int_1^\infty \frac{\cos(xu)}{u}\, du + \int_1^\infty \cos(xu)\left(\frac{1}{\sqrt{u^2-1}} - \frac{1}{u}\right)du.$$

In the first, put $t = xu$: it equals $\int_x^\infty t^{-1}\cos t\, dt = -\operatorname{Ci}(x)$, and, by Exercise 13.6.14, $\operatorname{Ci}(x) = \gamma + \ln x + \int_0^x t^{-1}(\cos t - 1)dt = \gamma + \ln x + O(x^2)$, because $|\cos t - 1| \le t^2/2$ bounds the last integral by $x^2/4$. The difference in parentheses is positive, $O((u-1)^{-1/2})$ as $u \to 1$ and $O(u^{-3})$ as $u \to \infty$, hence integrable; since $|\cos(xu)| \le 1$, dominated convergence gives for the second integral the limit

$$\int_1^\infty \left(\frac{1}{\sqrt{u^2-1}} - \frac{1}{u}\right)du = \lim_{R\to\infty}[\operatorname{arcosh} R - \ln R] = \lim_{R\to\infty}\ln\frac{R + \sqrt{R^2-1}}{R} = \ln 2.$$

Consequently $F(x) = -\ln x - \gamma + \ln 2 + o(1)$, that is,

$$-\frac{2}{\pi}F(x) = \frac{2}{\pi}[\ln(x/2) + \gamma] + o(1).$$

This diverges logarithmically as $x \downarrow 0$, whereas $J_0(x)$ tends to one. If $k_1 J_0 + k_2 F \equiv 0$, then $k_2 = 0$ (otherwise F would be bounded near zero) and then $k_1 = 0$: they are linearly independent. The normalization also agrees with Y_0: Eq. (14.61) with $n = 0$ and $\psi(1) = -\gamma$ gives

$$Y_0(x) = \frac{2}{\pi} J_0(x) \ln(x/2) + \frac{2\gamma}{\pi} + O(x^2) = \frac{2}{\pi}[\ln(x/2) + \gamma] + o(1).$$

Hence $D = -2F/\pi - Y_0$ is a solution tending to zero as $x \downarrow 0$. Writing $D = d_1 J_0 + d_2 Y_0$, the logarithm forces $d_2 = 0$, and then $d_1 = \lim_{x\to 0} D = 0$. Hence $Y_0(x) = -2F(x)/\pi$ with the stated normalization.

14.3.8 Verify the expansion formula for $Y_n(x)$ given in Eq. (14.61).

Hint. Start from Eq. (14.60) and perform the indicated differentiations on the power-series expansions of J_ν and $J_{-\nu}$. The digamma functions ψ arise from the differentiation of the gamma function. You will need the identity (not derived in this book) $\lim_{z\to -n} \psi(z)/\Gamma(z) = (-1)^{n-1} n!$, where n is a positive integer.

Take a nonnegative integer n and $x > 0$ initially; a complex continuation uses a consistent logarithm branch.

Solution.

Let $L = \ln(x/2)$, take integer $n \geq 0$, and use a fixed branch of the logarithm. Equation (14.60) reads

$$Y_n = \frac{1}{\pi}\left[\partial_\nu J_\nu|_n - (-1)^n\, \partial_\nu J_{-\nu}|_n\right].$$

Differentiate the series $J_\nu = \sum_{r\geq 0} (-1)^r (x/2)^{\nu+2r}/[r!\,\Gamma(\nu + r + 1)]$ term by term: $\partial_\nu (x/2)^{\nu+2r} = L\,(x/2)^{\nu+2r}$ and $\partial_\nu[1/\Gamma(\nu+r+1)] = -\psi(\nu+r+1)/\Gamma(\nu+r+1)$. At $\nu = n$, where $\Gamma(n+r+1) = (n+r)!$,

$$\partial_\nu J_\nu|_n = J_n L - \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{n+2r} \psi(n + r + 1)}{r!(n + r)!}. \tag{1}$$

For the second, use $J_{-\nu} = \sum_{r\geq 0} (-1)^r (x/2)^{2r-\nu}/[r!\,\Gamma(r - \nu + 1)]$. The power $(x/2)^{-\nu}$ contributes $-L$, while by the chain rule ($d(r - \nu + 1)/d\nu = -1$) the derivative of $1/\Gamma(r - \nu + 1)$ contributes $+\psi(r - \nu + 1)/\Gamma(r - \nu + 1)$. Thus

$$\partial_\nu J_{-\nu}|_n = -J_{-n} L + \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{2r-n}}{r!} \left.\frac{\psi(z)}{\Gamma(z)}\right|_{z=r-n+1}. \tag{2}$$

The values at gamma poles are limits. To derive them, set $z = -q + \epsilon$, $q = 0, 1, \ldots$. Gamma has a simple pole at $-q$ with residue $(-1)^q/q!$, so $\Gamma(-q + \epsilon) = (-1)^q/(q!\,\epsilon) + O(1)$. Hence the entire function $1/\Gamma$ has a simple zero there; with $\psi/\Gamma = \Gamma'/\Gamma^2 = -(1/\Gamma)'$,

$$\frac{1}{\Gamma(-q + \epsilon)} = (-1)^q q!\epsilon + O(\epsilon^2), \quad \frac{\psi}{\Gamma}(-q + \epsilon) = -\left(\frac{1}{\Gamma}\right)'(-q + \epsilon) \longrightarrow (-1)^{q+1} q!.$$

For $q \geq 1$ this is the identity quoted in the hint. Split (2) at $r = n$. For $r < n$, $q = n - r - 1$, and its summands become $(-1)^n (n - r - 1)!(x/2)^{2r-n}/r!$. For $r \geq n$, put $r = n + k$ and use $\Gamma(k + 1) = k!$. Since $J_{-n} = (-1)^n J_n$, this yields

$$\partial_\nu J_{-\nu}|_n = (-1)^{n+1} J_n L + (-1)^n \sum_{r=0}^{n-1} \frac{(n - r - 1)!}{r!} (x/2)^{2r-n} + (-1)^n \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{n+2k} \psi(k + 1)}{k!(n + k)!}.$$

Substitute this and (1), with its summation index renamed k, into the order-limit formula. In the bracket, J_nL from (1) and $-(-1)^n(-1)^{n+1}J_nL = J_nL$ add to $2J_nL$; the finite sum enters with a minus sign; and the two infinite sums, both with a minus sign, combine into one with $\psi(k+1)+\psi(n+k+1)$. Dividing by π,

$$\boxed{Y_n(x) = \frac{2}{\pi}J_n(x)\ln(x/2) - \frac{1}{\pi}\sum_{r=0}^{n-1}\frac{(n-r-1)!}{r!}(x/2)^{2r-n} - \frac{1}{\pi}\sum_{k=0}^{\infty}\frac{(-1)^k[\psi(k+1)+\psi(n+k+1)]}{k!(n+k)!}(x/2)^{n+2k}.}$$

This is Eq. (14.61). When $n = 0$ the finite sum is empty; negative integer orders follow from $Y_{-n} = (-1)^nY_n$. The gamma-pole limit above also supplies $q = 0$, needed by the last finite-sum term.

14.3.9 If Bessel's ODE (with solution J_ν) is differentiated with respect to ν, one obtains

$$x^2\frac{d^2}{dx^2}\left(\frac{\partial J_\nu}{\partial\nu}\right) + x\frac{d}{dx}\left(\frac{\partial J_\nu}{\partial\nu}\right) + (x^2-\nu^2)\frac{\partial J_\nu}{\partial\nu} = 2\nu J_\nu.$$

Use the above equation to show that $Y_n(x)$ is a solution to Bessel's ODE.

Hint. Equation (14.60) will be useful.

Solution.

Denote the Bessel operator by $L_\nu = x^2\partial_x^2 + x\partial_x + x^2 - \nu^2$. Differentiate $L_\nu J_\nu = 0$ with respect to the order. The ν-derivative commutes with the x-derivatives in L_ν ($J_\nu(x)$ is analytic in ν and in $x > 0$), and $\partial_\nu L_\nu = -2\nu$, so the product rule gives

$$L_\nu(\partial_\nu J_\nu) - 2\nu J_\nu = 0,$$

which is the equation of the statement. Because $J_{-\nu}$ satisfies the same equation with the same ν^2, differentiating $L_\nu J_{-\nu} = 0$ likewise gives

$$L_\nu(\partial_\nu J_{-\nu}) = 2\nu J_{-\nu}.$$

At $\nu = n$, apply L_n to Eq. (14.60):

$$\begin{aligned} L_nY_n &= \frac{1}{\pi}L_n\left[\partial_\nu J_\nu|_n - (-1)^n\,\partial_\nu J_{-\nu}|_n\right] \\ &= \frac{2n}{\pi}[J_n - (-1)^nJ_{-n}] \\ &= \frac{2n}{\pi}[J_n - (-1)^{2n}J_n] = 0, \end{aligned}$$

where the last step uses $J_{-n} = (-1)^nJ_n$. Thus $\boxed{x^2Y_n'' + xY_n' + (x^2-n^2)Y_n = 0}$, including $n = 0$, for which both inhomogeneous terms already vanish.

14.3.10 For the case $m = 0$, $a = 1$, and $b = 2$, the coaxial wave-guide TM boundary conditions become $f(\lambda) = 0$, with

$$f(x) = \frac{J_0(2x)}{Y_0(2x)} - \frac{J_0(x)}{Y_0(x)}.$$

This function is plotted in Fig. 14.9.

(a) Calculate $f(x)$ for $x = 0.0(0.1)10.0$ and plot $f(x)$ vs. x to find the approximate location of the roots.

(b) Call a root-finding program to determine the first three roots to higher precision.

ANS. (b) 3.1230, 6.2734, 9.4182.

Note. The higher roots can be expected to appear at intervals whose length approaches π. Why? AMS-55 (see Additional Readings) gives an approximate formula for the roots. The function $g(x) = J_0(x)Y_0(2x) - J_0(2x)Y_0(x)$ is much better behaved than the $f(x)$ previously discussed.

Tabulate $x = 0$ as undefined and use the values $0.1, 0.2, \ldots, 10.0$ elsewhere. The limiting zero at the origin is not an annular eigenvalue.

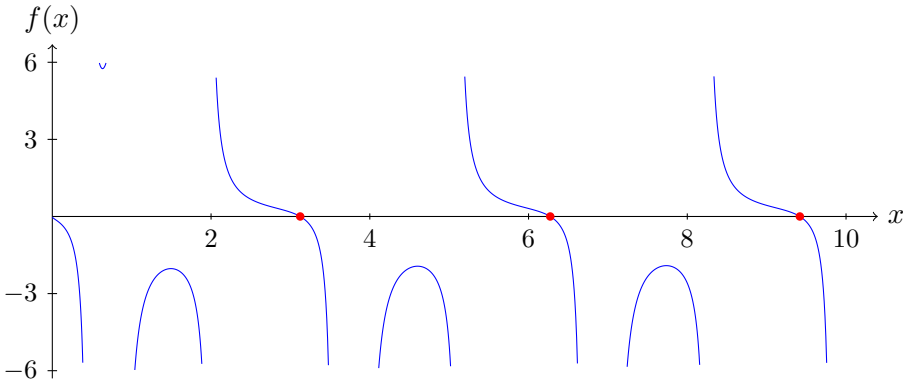
Solution.

At $x = 0$ the quoted ratios are undefined. Their difference has limiting value zero, but this is not an eigenvalue of the annular Dirichlet problem. Poles of either denominator must also be distinguished from roots; a sign change across a pole is not a zero.

(a) The requested grid values, rounded to four decimals, are below. The origin entry is undefined; its right-hand limit is zero.

x	$f(x)$	x	$f(x)$	x	$f(x)$	x	$f(x)$
0.0	undef.	2.6	0.5341	5.2	5.2300	7.8	-1.9419
0.1	-0.2656	2.7	0.4305	5.3	2.2930	7.9	-2.1253
0.2	-0.6690	2.8	0.3441	5.4	1.4109	8.0	-2.5934
0.3	-1.7451	2.9	0.2613	5.5	0.9937	8.1	-3.8402
0.4	-8.1648	3.0	0.1673	5.6	0.7530	8.2	-10.3590
0.5	10.7813	3.1	0.0388	5.7	0.5959	8.3	9.6855
0.6	5.8986	3.2	-0.1741	5.8	0.4827	8.4	2.9820
0.7	6.2993	3.3	-0.6075	5.9	0.3925	8.5	1.6783
0.8	10.8328	3.4	-1.8044	6.0	0.3110	8.6	1.1349
0.9	-142.7633	3.5	-9.5528	6.1	0.2248	8.7	0.8408
1.0	-8.2314	3.6	11.3692	6.2	0.1151	8.8	0.6571
1.1	-4.2257	3.7	6.8360	6.3	-0.0542	8.9	0.5298
1.2	-2.9376	3.8	8.0074	6.4	-0.3728	9.0	0.4326
1.3	-2.3652	3.9	18.3403	6.5	-1.1445	9.1	0.3496
1.4	-2.1021	4.0	-22.6755	6.6	-4.2622	9.2	0.2677
1.5	-2.0284	4.1	-6.4402	6.7	28.1279	9.3	0.1711
1.6	-2.1260	4.2	-3.7568	6.8	7.4939	9.4	0.0333
1.7	-2.4670	4.3	-2.7318	6.9	7.4278	9.5	-0.2061
1.8	-3.3644	4.4	-2.2430	7.0	12.9089	9.6	-0.7246
1.9	-6.8081	4.5	-2.0077	7.1	-70.6154	9.7	-2.3274
2.0	23.0048	4.6	-1.9344	7.2	-8.1293	9.8	-21.9636
2.1	3.6951	4.7	-2.0067	7.3	-4.2476	9.9	9.2367
2.2	1.8835	4.8	-2.2745	7.4	-2.9405	10.0	7.0840
2.3	1.2180	4.9	-2.9403	7.5	-2.3396		
2.4	0.8780	5.0	-4.9933	7.6	-2.0424		
2.5	0.6728	5.1	-45.1412	7.7	-1.9218		

The plot draws the branch of f on each interval between consecutive poles (listed below) as a separate curve, clipped to $|f| \le 6$. The branches on $(3.5430, 3.9577)$, $(6.6805, 7.0861)$ and $(9.8207, 10]$ stay above 6 (their minima are about 6.79, 6.98 and 7.06) and therefore do not appear. Red points mark the first three roots.



The poles of f are the zeros of $Y_0(x)$, at $x = 0.8936$, 3.9577, 7.0861, and of $Y_0(2x)$, at $x = 0.4468$, 1.9788, 3.5430, 5.1112, 6.6805, 8.2505, 9.8207 (half the zeros of Y_0). Every sign change of the table in an interval containing one of these is a pole, not a root. The remaining sign changes bracket the roots in $(3.1, 3.2)$, $(6.2, 6.3)$, and $(9.4, 9.5)$, but root finding should use the continuous determinant

$$g(x) = J_0(x)Y_0(2x) - J_0(2x)Y_0(x), \qquad f(x) = -\frac{g(x)}{Y_0(x)Y_0(2x)}.$$

No pole lies in these three intervals, so g changes sign in each of them.

(b) Solving $g(x) = 0$ inside those intervals, for example by bisection (34 halvings of an interval of length 0.1 reduce it below 10^{-11}), gives

$$\boxed{\lambda_1 = 3.1230309196, \qquad \lambda_2 = 6.2734357140, \qquad \lambda_3 = 9.4182075423}.$$

At these roots both denominators in f are nonzero, and direct substitution verifies both $g(\lambda_j) = 0$ and $f(\lambda_j) = 0$. The corresponding radial eigenfunction

$$R_j(\rho) = J_0(\lambda_j\rho)Y_0(\lambda_j) - Y_0(\lambda_j\rho)J_0(\lambda_j)$$

vanishes at $\rho = 1$ identically and at $\rho = 2$ by $g = 0$. At zero parameter the radial equation has solution $A + B\ln\rho$; both boundary values zero force $A = B = 0$. Consistently, $g(0^+) = 2\ln 2/\pi \neq 0$: since $J_0(x)$ and $J_0(2x)$ are $1 + O(x^2)$, $g(x) = Y_0(2x) - Y_0(x) + O(x^2\ln x)$, and the small-$x$ form $Y_0(x) = \frac{2}{\pi}[\ln(x/2) + \gamma] + o(1)$ of Exercise 14.3.7(b) gives $Y_0(2x) - Y_0(x) = \frac{2}{\pi}[\ln x - \ln(x/2)] + o(1) \to \frac{2}{\pi}\ln 2$. On the other hand $f = -g/[Y_0(x)Y_0(2x)] \to 0$, because each Y_0 grows logarithmically.

For large x, substitute the leading terms of the asymptotic forms of Section 14.6, $J_0(x) = \sqrt{2/(\pi x)}[\cos(x - \pi/4) + O(x^{-1})]$ and $Y_0(x) = \sqrt{2/(\pi x)}[\sin(x - \pi/4) + O(x^{-1})]$. The products carry the factor $\sqrt{2/(\pi x)}\sqrt{2/(2\pi x)} = \sqrt{2}/(\pi x)$, and the sine subtraction identity $\sin A\cos B - \cos A\sin B = \sin(A - B)$, with $A = 2x - \pi/4$ and $B = x - \pi/4$, gives

$$g(x) = \frac{\sqrt{2}}{\pi x}\{\sin[(2x - \pi/4) - (x - \pi/4)] + O(x^{-1})\} = \frac{\sqrt{2}}{\pi x}\{\sin x + O(x^{-1})\}.$$

To justify the ordering of the roots as well as the spacing, set $u(\rho) = \sqrt{\rho}\,R(\rho)$. Differentiating $R = \rho^{-1/2}u$ gives $R' = \rho^{-1/2}u' - \frac{1}{2}\rho^{-3/2}u$ and $R'' = \rho^{-1/2}u'' - \rho^{-3/2}u' + \frac{3}{4}\rho^{-5/2}u$, so $R'' + \rho^{-1}R' =$

$\rho^{-1/2}[u'' + u/(4\rho^2)]$, and $R'' + \rho^{-1}R' + \lambda^2 R = 0$ becomes

$$-u'' - \frac{u}{4\rho^2} = \lambda^2 u, \qquad u(1) = u(2) = 0.$$

On $[1,2]$, $-1/4 \le -1/(4\rho^2) \le -1/16$. Hence for every $u \ne 0$ with $u(1) = u(2) = 0$ the Rayleigh quotient $Q[u] = \int_1^2 [u'^2 - u^2/(4\rho^2)]d\rho / \int_1^2 u^2 d\rho$ satisfies $Q_0[u] - \frac{1}{4} \le Q[u] \le Q_0[u] - \frac{1}{16}$, where $Q_0[u] = \int_1^2 u'^2 d\rho / \int_1^2 u^2 d\rho$ is the quotient of $-u''$, whose Dirichlet eigenvalues on an interval of length one are $(j\pi)^2$. The minimum–maximum characterization of the jth eigenvalue therefore gives, for every $j \ge 1$,

$$\sqrt{(j\pi)^2 - \tfrac{1}{4}} \le \lambda_j \le \sqrt{(j\pi)^2 - \tfrac{1}{16}}.$$

These disjoint intervals contain the three computed roots for $j = 1, 2, 3$. Every positive root λ of g is one of the λ_k, because R as above, with λ in place of λ_j, is an eigenfunction (it is not identically zero, since $J_0(\lambda)$ and $Y_0(\lambda)$ cannot both vanish: their Wronskian is $2/(\pi\lambda)$, Exercise 14.3.5); and λ_k lies in the kth interval. Since the intervals are disjoint, the root found in the jth interval is λ_j, so the computed roots are indeed the first three. The same bounds give $\lambda_j = j\pi + O(j^{-1})$, hence $\lambda_{j+1} - \lambda_j \to \pi$.

14.4 Hankel Functions

14.4.1 Verify the Wronskian formulas

(a) $J_\nu(x)H_\nu^{(1)\prime}(x) - J_\nu'(x)H_\nu^{(1)}(x) = \dfrac{2i}{\pi x}$,

(b) $J_\nu(x)H_\nu^{(2)\prime}(x) - J_\nu'(x)H_\nu^{(2)}(x) = -\dfrac{2i}{\pi x}$,

(c) $Y_\nu(x)H_\nu^{(1)\prime}(x) - Y_\nu'(x)H_\nu^{(1)}(x) = -\dfrac{2}{\pi x}$,

(d) $Y_\nu(x)H_\nu^{(2)\prime}(x) - Y_\nu'(x)H_\nu^{(2)}(x) = -\dfrac{2}{\pi x}$,

(e) $H_\nu^{(1)}(x)H_\nu^{(2)\prime}(x) - H_\nu^{(1)\prime}(x)H_\nu^{(2)}(x) = -\dfrac{4i}{\pi x}$,

(f) $H_\nu^{(2)}(x)H_{\nu+1}^{(1)}(x) - H_\nu^{(1)}(x)H_{\nu+1}^{(2)}(x) = \dfrac{4}{i\pi x}$,

(g) $J_{\nu-1}(x)H_\nu^{(1)}(x) - J_\nu(x)H_{\nu-1}^{(1)}(x) = \dfrac{2}{i\pi x}$.

Solution.

Write $J = J_\nu$, $Y = Y_\nu$, $H^{(1)} = J + iY$, $H^{(2)} = J - iY$, $W[f,g] = fg' - f'g$, and $W[J,Y] = JY' - J'Y = 2/(\pi x)$ (Exercise 14.3.5). Expand each requested expression; the products JJ' and

YY' always cancel:

$$\begin{aligned}
&\text{(a)}\quad W[J,H^{(1)}] = J(J'+iY') - J'(J+iY) = iW[J,Y] = \frac{2i}{\pi x},\\
&\text{(b)}\quad W[J,H^{(2)}] = J(J'-iY') - J'(J-iY) = -iW[J,Y] = -\frac{2i}{\pi x},\\
&\text{(c)}\quad W[Y,H^{(1)}] = YJ' - Y'J + i(YY' - Y'Y) = -W[J,Y] = -\frac{2}{\pi x},\\
&\text{(d)}\quad W[Y,H^{(2)}] = YJ' - Y'J - i(YY' - Y'Y) = -W[J,Y] = -\frac{2}{\pi x},\\
&\text{(e)}\quad W[H^{(1)},H^{(2)}] = (J+iY)(J'-iY') - (J'+iY')(J-iY)\\
&\qquad\qquad = (JJ' + YY' - iJY' + iJ'Y) - (JJ' + YY' + iJY' - iJ'Y)\\
&\qquad\qquad = -2i(JY' - J'Y) = -\frac{4i}{\pi x}.
\end{aligned}$$

For (f) and (g) note that J_ν and Y_ν satisfy the recurrences (14.7) and (14.8) for every order (Exercise 14.3.1), and so do $H_\nu^{(1,2)} = J_\nu \pm iY_\nu$, whose coefficients do not depend on ν. Half the sum and half the difference of these recurrences give, for each such function Z_ν,

$$Z_{\nu-1} = Z'_\nu + \frac{\nu}{x}Z_\nu, \qquad Z_{\nu+1} = \frac{\nu}{x}Z_\nu - Z'_\nu.$$

For (f), substitute the second form in the left side; its two ν/x products cancel:

$$\begin{aligned}
H_\nu^{(2)}H_{\nu+1}^{(1)} - H_\nu^{(1)}H_{\nu+1}^{(2)} &= H_\nu^{(2)}\Big(\frac{\nu}{x}H_\nu^{(1)} - H_\nu^{(1)\prime}\Big) - H_\nu^{(1)}\Big(\frac{\nu}{x}H_\nu^{(2)} - H_\nu^{(2)\prime}\Big)\\
&= H_\nu^{(1)}H_\nu^{(2)\prime} - H_\nu^{(1)\prime}H_\nu^{(2)} = W[H^{(1)},H^{(2)}] = -\frac{4i}{\pi x} = \frac{4}{i\pi x},
\end{aligned}$$

by (e) and $-i = 1/i$. For (g), use the first form for $J_{\nu-1} = J'_\nu + \nu J_\nu/x$ and $H_{\nu-1}^{(1)} = H_\nu^{(1)\prime} + \nu H_\nu^{(1)}/x$; again the ν/x products cancel, and (a) gives

$$J_{\nu-1}H_\nu^{(1)} - J_\nu H_{\nu-1}^{(1)} = J'_\nu H_\nu^{(1)} - J_\nu H_\nu^{(1)\prime} = -W[J,H^{(1)}] = -\frac{2i}{\pi x} = \frac{2}{i\pi x}.$$

This verifies all seven signs with the same Wronskian convention.

14.4.2 Show that the integral forms

$$\text{(a)}\quad \frac{1}{i\pi}\int_{0C_1}^{\infty e^{i\pi}} e^{(x/2)(t-1/t)}\frac{dt}{t^{\nu+1}} = H_\nu^{(1)}(x),$$

$$\text{(b)}\quad \frac{1}{i\pi}\int_{\infty e^{-i\pi}C_2}^{0} e^{(x/2)(t-1/t)}\frac{dt}{t^{\nu+1}} = H_\nu^{(2)}(x)$$

satisfy Bessel's ODE. The contours C_1 and C_2 are shown in Fig. 14.10.

Take $x > 0$ and $-\pi < \arg t < \pi$. Approach zero from positive t and the negative-real endpoints on their indicated banks.

Solution.

Take $x > 0$ and the branch $-\pi < \arg t < \pi$, with limiting values on either bank of the negative axis. Put $f = e^{x(t-t^{-1})/2}t^{-\nu-1}$. The computation in Exercise 14.1.15(a) (there $f_x = \frac{1}{2}(t-t^{-1})f$, $f_{xx} = \frac{1}{4}(t-t^{-1})^2 f$, and the product rule is applied to Q below) gives the exact identity

$$(x^2\partial_x^2 + x\partial_x + x^2 - \nu^2)f = \frac{\partial}{\partial t}Q(t,x),$$

$$Q(t,x) = e^{x(t-t^{-1})/2} t^{-\nu} \left[\nu + \frac{x}{2}(t+t^{-1})\right].$$

We now check the different endpoints of these contours. On the positive axis as $t \downarrow 0$, the exponential is $e^{-x/(2t)+O(t)}$; it dominates every power of t^{-1} and makes $Q \to 0$. On either bank $t = Re^{\pm i\pi} = -R$ as $R \to \infty$, $t - t^{-1} = -R + R^{-1}$, so the exponential is $e^{-xR/2+O(R^{-1})}$; it dominates the powers of R in Q, and again $Q \to 0$. The same estimates bound f, f_x and f_{xx} uniformly for x in compact positive intervals, which justifies differentiation under the integrals. They also explain the qualification in the statement: for $x > 0$ the factor $e^{-x/(2t)}$ decays as $t \to 0$ only with $\Re t > 0$, and $e^{xt/2}$ decays as $|t| \to \infty$ only with $\Re t < 0$, so the endpoints must be approached as stated.

(a) Let $F(x) = (\pi i)^{-1} \int_{C_1} f\, dt$. Differentiating under the integral and integrating the identity from 0^+ along C_1 to the upper negative-real infinity gives

$$x^2F'' + xF' + (x^2 - \nu^2)F = \frac{1}{\pi i}\int_{C_1} \frac{\partial Q}{\partial t} dt = \frac{1}{\pi i}[Q]_{0^+}^{\infty e^{i\pi}} = 0,$$

so the first integral satisfies Bessel's equation.

(b) Along C_2 the orientation is from the lower negative-real infinity to 0^+. Both endpoint values still vanish, $[Q]_{\infty e^{-i\pi}}^{0^+} = 0$, so the second integral satisfies the same equation. These endpoint estimates establish the ODE, while Exercises 14.4.3–14.4.4 establish that their normalizations are precisely $J_\nu \pm iY_\nu$.

14.4.3 Show that the substitution $t = e^{i\pi}/s$ into Eq. (14.90) for $H_\nu^{(1)}(x)$ not only produces the integrand for the similar integral representation of $H_{-\nu}^{(1)}(x)$ but that the contour in s is the same geometric contour up to deformation, but with reversed orientation.

Solution.

Choose a convenient representative of C_1: the positive segment $0 < t \le 1$, the upper unit semicircle, and the upper bank of the negative ray from -1 to $-\infty$. This can be deformed to the contour in Fig. 14.10 without crossing the cut or the origin. Under $s = e^{i\pi}/t = -1/t$,

$$t : 0^+ \longmapsto s : \infty e^{i\pi}, \qquad t : i \longmapsto s : i, \qquad t : \infty e^{i\pi} \longmapsto s : 0^+.$$

The two rays are exchanged. On the semicircle, $t = e^{i\theta}$ maps to $s = e^{i(\pi-\theta)}$, so that arc is also unchanged as a set but traversed backwards. Thus the entire transformed contour is C_1 with reversed orientation.

On this branch, $\log t = i\pi - \log s$: on C_1, $0 \le \arg t \le \pi$, and $\arg s = \pi - \arg t$ lies in the same range, so $\log s = -\ln|t| + i(\pi - \arg t) = i\pi - \log t$. Since $t = -1/s$, we have $t^{-1} = -s$ and $dt = s^{-2}ds$, while $t^{-\nu-1} = e^{-(\nu+1)(i\pi - \log s)} = -e^{-i\pi\nu}s^{\nu+1}$. Hence

$$t - t^{-1} = s - s^{-1}, \qquad t^{-\nu-1}dt = -e^{-i\pi\nu}s^{\nu-1}ds.$$

Write $-C_1$ for C_1 traversed backwards. The minus sign from the differential cancels the reversed orientation:

$$\begin{aligned} H_\nu^{(1)}(x) &= \frac{1}{\pi i}\int_{-C_1} e^{x(s-s^{-1})/2}\left(-e^{-i\pi\nu}\right)s^{\nu-1}ds \\ &= \frac{e^{-i\pi\nu}}{\pi i}\int_{C_1} e^{x(s-s^{-1})/2}s^{\nu-1}ds = \boxed{e^{-i\pi\nu}H_{-\nu}^{(1)}(x)}, \end{aligned}$$

because $s^{\nu-1} = s^{-(-\nu)-1}$: the last integral is Eq. (14.90) with ν replaced by $-\nu$. Equivalently, $H_{-\nu}^{(1)}(x) = e^{i\pi\nu}H_\nu^{(1)}(x)$. For any other admissible representative, "the same contour" means the same deformable contour class with the same branch and endpoints; the representative just chosen makes the reversal explicit.

14.4.4 Using the integrals and contours given in Exercise 14.4.2, show that

$$\frac{1}{2i}\left[H_\nu^{(1)}(x) - H_\nu^{(2)}(x)\right] = Y_\nu(x).$$

Solution.

Temporarily denote the two contour integrals of Exercise 14.4.2 by A_ν (along C_1) and B_ν (along C_2), without using their algebraic Hankel definitions. For C_1 take the representative of Exercise 14.4.3: the segment $0 < t \le 1$, the upper unit semicircle, and the upper bank from -1 to $-\infty$. For C_2 take its mirror image, traversed from $-\infty$ along the lower bank to -1, along the lower unit semicircle to 1, and then to 0^+. In $A_\nu + B_\nu$ the segment $(0, 1]$ is traversed once in each direction, so its two contributions cancel. What remains runs from $-\infty$ below the cut to -1, once counterclockwise around the unit circle, and back to $-\infty$ above the cut. This is the contour C of Fig. 14.5 in the Schlaefli integral $J_\nu = (2\pi i)^{-1}\int_C e^{x(t-t^{-1})/2}t^{-\nu-1}dt$ of Exercise 14.1.15(a). Since $1/(\pi i) = 2/(2\pi i)$,

$$2J_\nu = A_\nu + B_\nu. \tag{1}$$

Exercise 14.4.3 gives $A_\nu = e^{-i\pi\nu}A_{-\nu}$, that is, $A_{-\nu} = e^{i\pi\nu}A_\nu$. The same inversion on the lower contour, now with $t = e^{-i\pi}/s$ and $\log t = -i\pi - \log s$ (on C_2, $-\pi \le \arg t \le 0$ and $\arg s = -\pi - \arg t$), gives $t^{-\nu-1}dt = -e^{i\pi\nu}s^{\nu-1}ds$ and exchanges the endpoints $\infty e^{-i\pi}$ and 0^+. As there, the minus sign cancels the reversed orientation, so $B_\nu = e^{i\pi\nu}B_{-\nu}$, that is, $B_{-\nu} = e^{-i\pi\nu}B_\nu$. Applying (1) at order $-\nu$ therefore yields

$$2J_{-\nu} = e^{i\pi\nu}A_\nu + e^{-i\pi\nu}B_\nu. \tag{2}$$

For noninteger ν, insert (1) and (2) into the definition $Y_\nu = (\cos\pi\nu\, J_\nu - J_{-\nu})/\sin\pi\nu$ and use $\cos\pi\nu - e^{\pm i\pi\nu} = \mp i\sin\pi\nu$:

$$\begin{aligned} Y_\nu &= \frac{\cos\pi\nu\,(A_\nu + B_\nu) - e^{i\pi\nu}A_\nu - e^{-i\pi\nu}B_\nu}{2\sin\pi\nu} \\ &= \frac{-i\sin\pi\nu\,A_\nu + i\sin\pi\nu\,B_\nu}{2\sin\pi\nu} = \frac{-i(A_\nu - B_\nu)}{2} = \boxed{\frac{A_\nu - B_\nu}{2i}}. \end{aligned}$$

For integer $\nu = n$, Y_n is the limit of Y_ν as $\nu \to n$, and A_ν, B_ν are continuous in ν (on a fixed representative the integrals converge uniformly for ν in bounded sets), so the result persists in the limit. Combining this result with (1) gives $A_\nu = J_\nu + iY_\nu$ and $B_\nu = J_\nu - iY_\nu$, verifying both identifications from the contours themselves.

14.4.5 Show that the integrals in Exercise 14.4.2 may be transformed to yield

(a) $H_\nu^{(1)}(x) = \dfrac{1}{\pi i}\displaystyle\int_{C_3} e^{x\sinh\gamma - \nu\gamma}\,d\gamma,$

(b) $H_\nu^{(2)}(x) = \dfrac{1}{\pi i}\displaystyle\int_{C_4} e^{x\sinh\gamma - \nu\gamma}\,d\gamma,$

where C_3 and C_4 are the contours in Fig. 14.11.

Take $x > 0$ initially and retain the upper/lower logarithm branches and the orientations of C_1, C_2.

Solution.

Put $t = e^\gamma$ using a continuous logarithm on each contour. Then

$$\frac{dt}{t} = d\gamma, \qquad t - t^{-1} = 2\sinh\gamma, \qquad t^{-\nu} = e^{-\nu\gamma}.$$

The transformed integrand is consequently

$$e^{x(t-t^{-1})/2}t^{-\nu-1}dt = e^{x\sinh\gamma-\nu\gamma}d\gamma.$$

Here $\gamma = \ln|t| + i\arg t$ with $-\pi \le \arg t \le \pi$, the branch of Exercise 14.4.2; as the statement's qualification requires, this branch fixes the heights $\pm\pi$ of the images of the two banks, and the orientations of C_1, C_2 fix the directions of C_3, C_4.

(a) Use the representative of C_1 from Exercise 14.4.3. The segment $0 < t \le 1$ maps to the real axis $-\infty < \gamma \le 0$, so positive $t \downarrow 0$ corresponds to $\gamma \to -\infty$; the upper semicircle $t = e^{i\theta}$, $0 \le \theta \le \pi$, maps to the segment $\gamma = i\theta$ from 0 to $i\pi$; and the upper bank $t = Re^{i\pi}$, $R \ge 1$, maps to $\gamma = \ln R + i\pi$, which tends to $+\infty + i\pi$. Thus the image of C_1 is

$$C_3: \quad -\infty \longrightarrow 0 \longrightarrow i\pi \longrightarrow +\infty + i\pi,$$

and $H_\nu^{(1)} = (\pi i)^{-1}\int_{C_3} e^{x\sinh\gamma-\nu\gamma}d\gamma$.

(b) Likewise the lower bank $t = Re^{-i\pi}$ of C_2, traversed from $R = \infty$ to $R = 1$, maps to $\gamma = \ln R - i\pi$, from $+\infty - i\pi$ to $-i\pi$; the lower semicircle $t = e^{i\theta}$, $\theta: -\pi \to 0$, maps to $\gamma = i\theta$; and the final segment $t: 1 \to 0^+$ maps to $\gamma: 0 \to -\infty$. With this orientation,

$$C_4: \quad +\infty - i\pi \longrightarrow -i\pi \longrightarrow 0 \longrightarrow -\infty.$$

This gives the second requested integral with its stated sign. The integrand is entire in γ, so these piecewise paths may be smoothed to those in Fig. 14.11. For $x > 0$, it decays at $\Re\gamma \to -\infty$ on height zero, where $\sinh\gamma \to -\infty$, and at $\Re\gamma = u \to +\infty$ on heights $\pm\pi$, where $\sinh(u \pm i\pi) = -\sinh u \to -\infty$; in both cases the factor $e^{x\sinh\gamma}$ overwhelms $e^{-\nu\gamma}$, validating the endpoint deformations.

14.4.6

(a) Transform $H_0^{(1)}(x)$, Eq. (14.90), into

$$H_0^{(1)}(x) = \frac{1}{i\pi}\int_C e^{ix\cosh s}\,ds,$$

where the contour C runs from $-\infty - i\pi/2$ through the origin of the s-plane to $\infty + i\pi/2$.

(b) Justify rewriting $H_0^{(1)}(x)$ as

$$H_0^{(1)}(x) = \frac{2}{i\pi}\int_0^{\infty+i\pi/2} e^{ix\cosh s}\,ds.$$

(c) Verify that this integral representation actually satisfies Bessel's differential equation. (The $i\pi/2$ in the upper limit is not essential. It serves as a convergence factor. We can replace it by $ia\pi/2$ and take the limit $a \to 0$.)

Take $x > 0$ and keep a positive imaginary contour height, or a positive imaginary part of x, during differentiation. Take the damping limit only afterward.

Solution.

(a) In the order-zero formula, Eq. (14.90) with $\nu = 0$,

$$H_0^{(1)}(x) = \frac{1}{\pi i}\int_{C_1} e^{x(t-t^{-1})/2}\frac{dt}{t},$$

set $t = ie^s = e^{s+i\pi/2}$, that is, $s = \log t - i\pi/2$. Since $1/i = -i$,

$$t - t^{-1} = ie^s - (-i)e^{-s} = i(e^s + e^{-s}) = 2i\cosh s, \qquad dt/t = ds,$$

so the exponent becomes $(x/2)(2i\cosh s) = ix\cosh s$. The initial positive-origin endpoint ($\arg t = 0$, $|t| \to 0$) maps to $s = -\infty - i\pi/2$, the point $t = i$ maps to $s = 0$, and the final upper negative infinity ($\arg t = \pi$, $|t| \to \infty$) maps to $s = +\infty + i\pi/2$. Therefore

$$\boxed{H_0^{(1)}(x) = \frac{1}{\pi i}\int_C e^{ix\cosh s}ds}.$$

(b) The integrand is entire, so only the two ends of C matter. Choose the contour to consist of a path C_+ from 0 to $\infty + i\pi/2$ and its reflected path $s = -u$, $u \in C_+$, from $-\infty - i\pi/2$ to 0. On the reflected half, $\cosh s = \cosh(-u) = \cosh u$ and $ds = -du$, and reversing the limits removes the minus sign:

$$\int_{-\infty-i\pi/2}^{0} e^{ix\cosh s}ds = \int_{\infty+i\pi/2}^{0} e^{ix\cosh u}(-du) = \int_0^{\infty+i\pi/2} e^{ix\cosh u}du.$$

So the reflected half contributes as much as C_+. Hence

$$\boxed{H_0^{(1)}(x) = \frac{2}{\pi i}\int_0^{\infty+i\pi/2} e^{ix\cosh s}ds}.$$

(c) For $x > 0$, choose a contour from 0 whose final ray $s = R + i\eta$, $R \to \infty$, has fixed height η, $0 < \eta \le \pi/2$. Since $\cosh(R + i\tau) = \cosh R\cos\tau + i\sinh R\sin\tau$,

$$|e^{ix\cosh(R+i\tau)}| = e^{-x\sinh R\sin\tau}.$$

For τ between η and $\pi/2$ this is at most $e^{-x\sinh R\sin\eta} \to 0$, so the vertical segment joining the two heights at $\Re s = R$ contributes nothing as $R \to \infty$: the end of the contour may be moved from height $\pi/2$ to η. On the ray this bound decays faster than the factors $\cosh s$ and $\cosh^2 s$ produced by two x derivatives grow, uniformly for x in compact positive intervals, so differentiation under the integral is legitimate. With $\partial_x e^{ix\cosh s} = i\cosh s\, e^{ix\cosh s}$, $\partial_x^2 e^{ix\cosh s} = -\cosh^2 s\, e^{ix\cosh s}$, and $\cosh^2 s - 1 = \sinh^2 s$,

$$\begin{aligned}(x^2\partial_x^2 + x\partial_x + x^2)e^{ix\cosh s} &= [-x^2\sinh^2 s + ix\cosh s]e^{ix\cosh s}\\ &= \frac{d}{ds}[ix\sinh s\, e^{ix\cosh s}],\end{aligned}$$

by the product rule, since $\frac{d}{ds}e^{ix\cosh s} = ix\sinh s\, e^{ix\cosh s}$. Integrating along the contour, the lower boundary term is zero since $\sinh 0 = 0$; the upper one is at most $x|\sinh(R + i\eta)|e^{-x\sinh R\sin\eta} \to 0$. Thus the integral solves the Bessel ODE of order zero. The real-axis form is a boundary limit, not a license to discard an undamped boundary term after differentiating. Take $\eta \downarrow 0$ only after the convergent-contour calculation, or use $x + i\epsilon$ and Abel convergence as in Exercise 14.3.7.

14.4.7 From

$$H_0^{(1)}(x) = \frac{2}{i\pi}\int_0^\infty e^{ix\cosh s}\,ds$$

show that

(a) $J_0(x) = \dfrac{2}{\pi}\displaystyle\int_0^\infty \sin(x\cosh s)\,ds,$

(b) $J_0(x) = \dfrac{2}{\pi}\displaystyle\int_1^\infty \frac{\sin(xt)}{\sqrt{t^2-1}}\,dt.$

This last result is a Fourier sine transform.

These ordinary oscillatory integrals hold for $x > 0$; the value at zero is obtained after integration by a limit.

Solution.

These real integrals are for $x > 0$. The sine-transform formula cannot be extended to $x < 0$ unchanged, since its right side is odd in x whereas J_0 is even. At $x = 0$ the limit must be taken after integration. First, for $x > 0$ the given real-axis formula holds with an ordinary improper integral (Exercise 14.4.6 used contours whose end has positive height). For $R > 0$ and $0 < \eta \le \pi/2$, the path from 0 along the real axis to R, up to $R + i\eta$, and then along the ray of height η is admissible in Exercise 14.4.6(c), so the integral of $e^{ix\cosh s}$ along it is $(i\pi/2)H_0^{(1)}(x)$. On the vertical segment $|e^{ix\cosh(R+i\tau)}| = e^{-x\sinh R\sin\tau}$, as in Exercise 14.4.6(c), and $\sin\tau \ge 2\tau/\pi$ for $0 \le \tau \le \pi/2$ (sine is concave there), so this segment contributes at most

$$\int_0^\eta e^{-x\sinh R\sin\tau}d\tau \le \int_0^\infty e^{-2x\tau\sinh R/\pi}d\tau = \frac{\pi}{2x\sinh R} \longrightarrow 0 \quad (R\to\infty).$$

The ray beyond $R + i\eta$ contributes the tail of an absolutely convergent integral, which also tends to zero. Hence $\int_0^R e^{ix\cosh s}ds \to (i\pi/2)H_0^{(1)}(x)$ as $R \to \infty$: the formula holds, and the real and imaginary parts used below converge as ordinary improper integrals. Now write the prefactor as $2/(i\pi) = -2i/\pi$ and expand the exponential, $e^{ix\cosh s} = \cos(x\cosh s) + i\sin(x\cosh s)$. Since $-(2i/\pi)\,i = 2/\pi$,

$$\begin{aligned} H_0^{(1)}(x) &= -\frac{2i}{\pi}\int_0^\infty [\cos(x\cosh s) + i\sin(x\cosh s)]ds \\ &= \frac{2}{\pi}\int_0^\infty \sin(x\cosh s)ds - \frac{2i}{\pi}\int_0^\infty \cos(x\cosh s)ds. \end{aligned}$$

(a) Here $H_0^{(1)} = J_0 + iY_0$, where J_0 and Y_0 are real on the positive axis (their series, Eq. (14.61) for Y_0, have real coefficients), and both integrals are real. Comparing real parts,

$$\boxed{J_0(x) = \frac{2}{\pi}\int_0^\infty \sin(x\cosh s)ds}.$$

(b) Put $t = \cosh s$. The limits become $1, \infty$, and, since $\sinh s = +\sqrt{\cosh^2 s - 1}$ for $s \ge 0$, $dt = \sinh s\,ds = \sqrt{t^2-1}\,ds$. Thus

$$\boxed{J_0(x) = \frac{2}{\pi}\int_1^\infty \frac{\sin(xt)}{\sqrt{t^2-1}}dt}.$$

The convergence of this t integral, which follows from that of the s integral, can also be checked directly. The singularity at $t = 1$ is integrable, since $1/\sqrt{t^2-1} \sim 1/\sqrt{2(t-1)}$ there. The tail converges by Dirichlet's test: $|\int_2^T \sin(xt)dt| \le 2/x$ for every T, and $1/\sqrt{t^2-1}$ decreases monotonically to zero.

14.4.8 From $H_0^{(1)}(x) = \dfrac{2}{i\pi}\displaystyle\int_0^\infty e^{ix\cosh s}\,ds$ (see Exercises 14.4.5 and 14.4.6), show that

(a) $Y_0(x) = -\dfrac{2}{\pi}\displaystyle\int_0^\infty \cos(x\cosh s)\,ds$,

(b) $Y_0(x) = -\dfrac{2}{\pi}\displaystyle\int_1^\infty \frac{\cos(xt)}{\sqrt{t^2-1}}\,dt$.

These are the integral representations in Eq. (14.63). This last result is a Fourier cosine transform.

Take $x > 0$ for both ordinary oscillatory integrals.

Solution.

For $x > 0$ the real-axis formula holds with an ordinary improper integral whose real and imaginary parts converge separately (Exercise 14.4.7). Multiplication by $2/(i\pi) = -2i/\pi$ gives

$$H_0^{(1)}(x) = \frac{2}{\pi}\int_0^\infty \sin(x\cosh s)ds - \frac{2i}{\pi}\int_0^\infty \cos(x\cosh s)ds.$$

(a) Both integrals are real, and so are J_0 and Y_0 on the positive axis (Exercise 14.4.7). Equating imaginary parts with $J_0 + iY_0$ therefore yields

$$\boxed{Y_0(x) = -\frac{2}{\pi}\int_0^\infty \cos(x\cosh s)ds}.$$

(b) With $t = \cosh s$, use $ds = dt/\sqrt{t^2-1}$ (as in Exercise 14.4.7) and the limits $t = 1, \infty$:

$$\boxed{Y_0(x) = -\frac{2}{\pi}\int_1^\infty \frac{\cos(xt)}{\sqrt{t^2-1}}dt}.$$

As in Exercise 14.4.7, the finite endpoint is integrable and the tail converges conditionally for $x > 0$; at $x = 0$ the integral $\int_1^\infty dt/\sqrt{t^2-1}$ diverges, in accordance with the logarithmic singularity of Y_0, which is why the qualification requires $x > 0$. The minus sign in (a) and (b) comes from the factor $1/i = -i$ of the prefactor.

14.5 Modified Bessel Functions, $I_\nu(x)$ and $K_\nu(x)$

14.5.1 Show that $e^{(x/2)(t+1/t)} = \displaystyle\sum_{n=-\infty}^{\infty} I_n(x)t^n$, thus generating modified Bessel functions, $I_n(x)$.

Solution.

Start from the Bessel generating function $\exp[X(T - T^{-1})/2] = \sum_{n\in\mathbb{Z}} J_n(X)T^n$, valid for all complex X and $T \neq 0$, and put $X = ix$, $T = -it$. Since $T^{-1} = 1/(-it) = i/t$, we have $T - T^{-1} = -it - i/t$, and since $(-i)^n = i^{-n}$ for every integer n, the definition $I_n(x) = i^{-n}J_n(ix)$ gives

$$\frac{ix}{2}(-it - i/t) = \frac{x}{2}(t + t^{-1}), \qquad J_n(ix)(-it)^n = i^{-n}J_n(ix)t^n = I_n(x)t^n.$$

Thus, for $t \neq 0$,

$$\boxed{e^{x(t+t^{-1})/2} = \sum_{n=-\infty}^{\infty} I_n(x)t^n}.$$

Absolute convergence on compact annuli follows either from the original generating series or from the product of the two exponential series $e^{xt/2}e^{x/(2t)}$.

14.5.2 Verify the following identities

(a) $1 = I_0(x) + 2\sum_{n=1}^{\infty}(-1)^n I_{2n}(x)$,

(b) $e^x = I_0(x) + 2\sum_{n=1}^{\infty} I_n(x)$,

(c) $e^{-x} = I_0(x) + 2\sum_{n=1}^{\infty}(-1)^n I_n(x)$,

(d) $\cosh x = I_0(x) + 2\sum_{n=1}^{\infty} I_{2n}(x)$,

(e) $\sinh x = 2\sum_{n=1}^{\infty} I_{2n-1}(x)$.

Solution.

The generating function of Exercise 14.5.1 is invariant under $t \mapsto 1/t$, so $\sum_n I_n t^n = \sum_n I_n t^{-n} = \sum_n I_{-n} t^n$; uniqueness of Laurent coefficients gives $I_{-n} = I_n$. Pair positive and negative orders:

$$e^{x(t+t^{-1})/2} = I_0 + \sum_{n=1}^{\infty} I_n(t^n + t^{-n}).$$

(a) Put $t = i$. Since $i + i^{-1} = i - i = 0$, the left side is $e^0 = 1$. On the right, $i^n + i^{-n} = i^n + (-i)^n$, so odd-$n$ pairs cancel, while for $n = 2j$, $i^{2j} + i^{-2j} = 2(-1)^j$. Hence

$$\boxed{1 = I_0 + 2\sum_{j=1}^{\infty}(-1)^j I_{2j}}.$$

(b) Put $t = 1$; all pairs equal two, so

$$\boxed{e^x = I_0 + 2\sum_{n\geq 1} I_n}.$$

(c) Put $t = -1$; each pair is $2(-1)^n$, giving

$$\boxed{e^{-x} = I_0 + 2\sum_{n\geq 1}(-1)^n I_n}.$$

(d) Add (b) and (c) and divide by two. Odd orders cancel and even orders double, so

$$\boxed{\cosh x = I_0 + 2\sum_{n\geq 1} I_{2n}}.$$

(e) Subtract (c) from (b) and divide by two. The constant and even orders cancel, giving

$$\boxed{\sinh x = 2\sum_{n\geq 1} I_{2n-1}}.$$

All these substitutions are on the unit circle, where the generating series converges absolutely.

14.5.3

(a) From the generating function of Exercise 14.5.1 show that

$$I_n(x) = \frac{1}{2\pi i}\oint e^{(x/2)(t+1/t)}\frac{dt}{t^{n+1}}.$$

(b) For $n = \nu$, not an integer, show that the preceding integral representation may be generalized to

$$I_\nu(x) = \frac{1}{2\pi i}\int_C e^{(x/2)(t+1/t)}\frac{dt}{t^{\nu+1}}.$$

The contour C is the same as that for $J_\nu(x)$ (Fig. 14.5).

In (b) use $x > 0$ initially and the open Hankel contour with $-\pi < \arg t < \pi$.

Solution.

(a) Divide the Laurent expansion from Exercise 14.5.1 by t^{n+1} and integrate counterclockwise around any circle centered at zero; the series converges uniformly on the circle, so it may be integrated term by term. The integral of t^{m-n-1} is $2\pi i$ if $m = n$ and zero otherwise. Thus coefficient extraction gives

$$\boxed{I_n(x) = \frac{1}{2\pi i}\oint e^{x(t+t^{-1})/2}t^{-n-1}dt}.$$

The exponential has an essential singularity at zero, but its Laurent coefficient is extracted exactly as for any isolated singularity.

(b) For noninteger order the contour is the open Hankel contour of Fig. 14.5, not an ordinary closed circle, because $t^{-\nu-1}$ is then not single valued around zero. Use the two banks of the negative-real cut and a fixed circle about zero, with $-\pi < \arg t < \pi$; take $x > 0$ initially. Put $s = xt/2$, i.e., $t = 2s/x$ and $dt = (2/x)ds$. Then

$$\frac{x}{2}(t + t^{-1}) = s + \frac{x^2}{4s}, \qquad t^{-\nu-1}dt = \left(\frac{2s}{x}\right)^{-\nu-1}\frac{2}{x}\,ds = \left(\frac{x}{2}\right)^{\nu} s^{-\nu-1}ds.$$

A positive dilation changes neither the cut nor the winding about zero, so the image is again a Hankel contour C (as in Exercise 14.1.15(a)), and the proposed integral becomes

$$\frac{(x/2)^\nu}{2\pi i}\int_C e^{s+x^2/(4s)}s^{-\nu-1}ds.$$

Expand only the second exponential, $e^{x^2/(4s)} = \sum_r (x^2/4)^r s^{-r}/r!$, and use the reciprocal-gamma Hankel integral $(2\pi i)^{-1}\int_C e^s s^{-z}ds = 1/\Gamma(z)$ (quoted in Exercise 14.1.15(a)) with $z = \nu + r + 1$; also $(x/2)^\nu(x^2/4)^r = (x/2)^{\nu+2r}$:

$$\frac{(x/2)^\nu}{2\pi i}\sum_{r=0}^{\infty}\frac{(x^2/4)^r}{r!}\int_C e^s s^{-\nu-r-1}ds = \sum_{r=0}^{\infty}\frac{(x/2)^{\nu+2r}}{r!\Gamma(\nu+r+1)}$$
$$= \boxed{I_\nu(x)}.$$

The last sum is the series of I_ν: by $I_\nu(x) = i^{-\nu}J_\nu(ix)$ and $(ix/2)^{\nu+2r} = i^\nu(-1)^r(x/2)^{\nu+2r}$, the factors $(-1)^r$ of the J_ν series are cancelled. A fixed-radius contour stays away from zero, so the expanded factor is uniformly bounded; its tails have the decaying factor e^s. This justifies termwise integration. The equality then continues to compatible complex x branches.

14.5.4 For $\nu > -\frac{1}{2}$ show that $I_\nu(z)$ may be represented by

$$\begin{aligned} I_\nu(z) &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_0^\pi e^{\pm z\cos\theta}\sin^{2\nu}\theta\, d\theta \\ &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_{-1}^1 e^{\pm zp}(1-p^2)^{\nu-1/2}\, dp \\ &= \frac{2}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_0^{\pi/2} \cosh(z\cos\theta)\sin^{2\nu}\theta\, d\theta. \end{aligned}$$

Use one consistent branch of z^ν and of $I_\nu(z)$; at zero use an existing limit.

Solution.

For $\nu > -1/2$ the endpoint weight $\sin^{2\nu}\theta$ is integrable, since $2\nu > -1$. Expand $e^{z\cos\theta} = \sum_k z^k\cos^k\theta/k!$; the sum of the absolute values of its terms is at most $e^{|z|}$, so dominated convergence permits termwise integration. Under the reflection $\theta \mapsto \pi - \theta$, $\cos\theta \mapsto -\cos\theta$ and $\sin\theta \mapsto \sin\theta$. Hence the integral over $[0,\pi]$ of an odd power of $\cos\theta$ times $\sin^{2\nu}\theta$ vanishes, and that of an even power is twice the integral over $[0,\pi/2]$. This leaves

$$\int_0^\pi e^{z\cos\theta}\sin^{2\nu}\theta d\theta = \sum_{r=0}^\infty \frac{z^{2r}}{(2r)!}\int_0^\pi \cos^{2r}\theta\sin^{2\nu}\theta d\theta.$$

With $u = \sin^2\theta$ on $[0,\pi/2]$, as in Exercise 13.3.10, the remaining integral is a beta function:

$$\begin{aligned} \int_0^\pi \cos^{2r}\theta\sin^{2\nu}\theta\, d\theta = 2\int_0^{\pi/2}\cos^{2r}\theta\sin^{2\nu}\theta\, d\theta &= B\left(r+\tfrac{1}{2},\nu+\tfrac{1}{2}\right) \\ &= \frac{\Gamma(r+1/2)\Gamma(\nu+1/2)}{\Gamma(r+\nu+1)}. \end{aligned}$$

Multiplication by the prescribed prefactor cancels $\Gamma(\nu+1/2)$. With $\Gamma(r+1/2)/[\sqrt{\pi}(2r)!] = 1/(4^r r!)$ (Exercise 13.3.10) and $(z/2)^\nu z^{2r}/4^r = (z/2)^{\nu+2r}$ it gives

$$\begin{aligned} \frac{(z/2)^\nu}{\sqrt{\pi}\Gamma(\nu+1/2)}\sum_{r=0}^\infty \frac{z^{2r}}{(2r)!}\frac{\Gamma(r+1/2)\Gamma(\nu+1/2)}{\Gamma(r+\nu+1)} &= \sum_{r=0}^\infty \frac{(z/2)^{\nu+2r}}{r!\Gamma(r+\nu+1)} \\ &= I_\nu(z), \end{aligned}$$

the series of I_ν found in Exercise 14.5.3. Changing z to $-z$ inside the exponential does not change any surviving even term, proving both signs in the first integral.

With $p = \cos\theta$, $dp = -\sin\theta d\theta$, and $\sin\theta = (1-p^2)^{1/2} \geq 0$ on $[0,\pi]$; hence

$$\sin^{2\nu}\theta d\theta = \sin^{2\nu-1}\theta\,\sin\theta d\theta = -(1-p^2)^{\nu-1/2}dp.$$

As θ runs from 0 to π, p runs from 1 to -1; reversing the limits removes the minus sign and gives the second representation. Finally, averaging the two signs of the first representation replaces $e^{\pm z\cos\theta}$ by $\cosh(z\cos\theta)$. Since $\cosh(z\cos(\pi-\theta)) = \cosh(z\cos\theta)$, its integral over $[0,\pi]$ is twice its integral over $[0,\pi/2]$. This proves the third form. All factors z^ν use the same branch as $I_\nu(z)$, since the identity was proved term by term with the common factor $(z/2)^\nu$. At $z=0$, a branch point of z^ν for noninteger ν, both sides are understood as limits; by the $r=0$ term of the series both behave like $(z/2)^\nu/\Gamma(\nu+1)$, so the limit is 0 for $\nu > 0$ and 1 for $\nu = 0$, while for $-1/2 < \nu < 0$ no finite limit exists.

14.5.5 The cylindrical cavity depicted in Fig. 14.4 has radius a and height h. For this exercise, the end caps $z = 0$ and h are at zero potential, while the cylindrical wall $\rho = a$ has a potential of functional form $V = V(\varphi, z)$.

(a) Show that the electrostatic potential $\Phi(\rho, \varphi, z)$ has the functional form

$$\Phi(\rho, \varphi, z) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} I_m(k_n \rho) \, (a_{mn} \sin m\varphi + b_{mn} \cos m\varphi) \sin k_n z,$$

where $k_n = n\pi/h$.

(b) Show that the coefficients a_{mn} and b_{mn} are given by

$$\left\{ \begin{array}{c} a_{mn} \\ b_{mn} \end{array} \right\} = \frac{2 - \delta_{m0}}{\pi h \, I_m(k_n a)} \int_0^{2\pi} \int_0^h V(\varphi, z) \left\{ \begin{array}{c} \sin m\varphi \\ \cos m\varphi \end{array} \right\} \sin k_n z \, dz \, d\varphi.$$

Hint. Expand $V(\varphi, z)$ as a double series and use the orthogonality of the trigonometric functions. Set $a_{0n} = 0$ by convention, since the zero angular sine contributes nothing.

Solution.

(a) Set $\Phi = P(\rho)F(\varphi)Z(z)$ in Laplace's equation $\rho^{-1}(\rho\Phi_\rho)_\rho + \rho^{-2}\Phi_{\varphi\varphi} + \Phi_{zz} = 0$ and divide by PFZ:

$$\frac{(\rho P')'}{\rho P} + \frac{F''}{\rho^2 F} + \frac{Z''}{Z} = 0.$$

Only the last term depends on z, so Z''/Z is a constant λ. With $Z(0) = Z(h) = 0$, $\lambda = 0$ ($Z = A + Bz$) and $\lambda > 0$ ($Z = A \sinh \sqrt{\lambda} z + B \cosh \sqrt{\lambda} z$) allow only $Z \equiv 0$. Hence $\lambda = -k^2 < 0$, $Z = \sin kz$ with $\sin kh = 0$:

$$Z'' + k_n^2 Z = 0, \quad Z(0) = Z(h) = 0, \quad Z = \sin k_n z, \quad k_n = n\pi/h, \quad n \geq 1.$$

Multiplying the remaining equation by ρ^2 isolates F''/F, which must therefore be a constant; single-valuedness in φ (period 2π) makes it $-m^2$ with m an integer, and $m \geq 0$ suffices:

$$F'' + m^2 F = 0, \quad F = a \sin m\varphi + b \cos m\varphi, \quad m = 0, 1, \ldots.$$

The radial equation is then $\rho(\rho P')' - (k_n^2 \rho^2 + m^2)P = 0$, that is,

$$\rho^2 P'' + \rho P' - (k_n^2 \rho^2 + m^2)P = 0,$$

the modified Bessel equation of order m in the variable $k_n \rho$. Its regular solution is $I_m(k_n \rho)$; K_m is singular at the axis. Adding all separated solutions gives precisely the double series in part (a).

(b) At $\rho = a$ the series must reproduce the wall potential:

$$V(\varphi, z) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} I_m(k_n a)(a_{mn} \sin m\varphi + b_{mn} \cos m\varphi) \sin k_n z.$$

Multiply by $\sin m'\varphi$ or $\cos m'\varphi$ and by $\sin k_{n'} z$, and integrate over the wall, $0 \leq \varphi \leq 2\pi$, $0 \leq z \leq h$. The orthogonality relations ($m, m' \geq 0$)

$$\int_0^{2\pi} \sin m\varphi \cos m'\varphi \, d\varphi = 0, \qquad \int_0^{2\pi} \sin m\varphi \sin m'\varphi \, d\varphi = \pi \delta_{mm'} \quad (m \geq 1),$$

$$\int_0^{2\pi} \cos m\varphi \cos m'\varphi \, d\varphi = \pi(1 + \delta_{m0})\delta_{mm'}, \qquad \int_0^h \sin k_n z \sin k_{n'} z \, dz = \frac{h}{2} \delta_{nn'}$$

remove every term except $m = m'$, $n = n'$. For $m > 0$ the surviving product of norms is $\pi \cdot h/2$; for $m = 0$ (cosine only) it is $2\pi \cdot h/2 = \pi h$. In both cases its reciprocal is $(2 - \delta_{m0})/(\pi h)$. Dividing also by $I_m(k_n a)$, which is positive because every term of its series is positive, gives

$$\boxed{\begin{pmatrix} a_{mn} \\ b_{mn} \end{pmatrix} = \frac{2 - \delta_{m0}}{\pi h I_m(k_n a)} \int_0^{2\pi} \int_0^h V(\varphi, z) \begin{pmatrix} \sin m\varphi \\ \cos m\varphi \end{pmatrix} \sin k_n z \, dz \, d\varphi}.$$

The height is h throughout, as in the published erratum; the original l in the coefficient formula must be replaced by h. Also $\sin 0\varphi = 0$, so a_{0n} is not determined by orthogonality and has no effect on the solution. The displayed formula makes the convenient choice $a_{0n} = 0$.

14.5.6 Verify that $K_\nu(x)$ as defined in Eq. (14.106) is equivalent to

$$K_\nu(x) = \frac{\pi}{2} \frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu\pi}$$

and from this show that

$$K_\nu(x) = K_{-\nu}(x).$$

Take $x > 0$ initially; at integer order interpret the quotient by its removable order limit.

Solution.

For noninteger ν, put $S = \sin \pi\nu$ and $C = \cos \pi\nu$. The definitions $H_\nu^{(1)} = J_\nu + iY_\nu$ and $Y_\nu = (CJ_\nu - J_{-\nu})/S$ imply

$$\begin{aligned} H_\nu^{(1)}(ix) &= J_\nu(ix) + i\frac{CJ_\nu(ix) - J_{-\nu}(ix)}{S} = \frac{(S + iC)J_\nu(ix) - iJ_{-\nu}(ix)}{S} \\ &= \frac{i}{S}[e^{-i\pi\nu} J_\nu(ix) - J_{-\nu}(ix)], \end{aligned}$$

because $S + iC = i(C - iS) = ie^{-i\pi\nu}$. Using the branch-compatible relations $J_\nu(ix) = i^\nu I_\nu(x)$ and $J_{-\nu}(ix) = i^{-\nu} I_{-\nu}(x)$, which restate $I_{\pm\nu}(x) = i^{\mp\nu} J_{\pm\nu}(ix)$ with $i^\nu = e^{i\pi\nu/2}$, we get $i^{\nu+1} \cdot i \cdot i^\nu = i^{2\nu+2}$ and $i^{\nu+1} \cdot i \cdot i^{-\nu} = i^2$, so

$$\begin{aligned} K_\nu(x) &= \frac{\pi}{2} i^{\nu+1} H_\nu^{(1)}(ix) = \frac{\pi}{2S}\left[i^{2\nu+2} e^{-i\pi\nu} I_\nu - i^2 I_{-\nu}\right] \\ &= \frac{\pi}{2S}\left[-I_\nu + I_{-\nu}\right] = \boxed{\frac{\pi}{2}\frac{I_{-\nu} - I_\nu}{\sin \pi\nu}}, \end{aligned}$$

since $i^{2\nu+2} = e^{i\pi(\nu+1)} = -e^{i\pi\nu}$ and $i^2 = -1$. Changing ν to $-\nu$ reverses the signs of both numerator and denominator, so

$$\boxed{K_{-\nu} = K_\nu}.$$

At integer order n, where $I_{-n} = I_n$ and $\sin n\pi = 0$ make the quotient $0/0$, both quotients are understood through their removable limits in ν. The definition (14.106) is continuous in ν, because $Y_\nu \to Y_n$, so $K_n = \lim_{\nu\to n} K_\nu$ is the limit of the quotient, and $K_{-n} = \lim_{\nu\to n} K_{-\nu} = \lim_{\nu\to n} K_\nu = K_n$.

14.5.7 Show that $K_\nu(x)$ satisfies the following recurrence relations:

$$K_{\nu-1}(x) - K_{\nu+1}(x) = -\frac{2\nu}{x}K_\nu(x),$$
$$K_{\nu-1}(x) + K_{\nu+1}(x) = -2K'_\nu(x).$$

Note. These differ from the recurrence relations for I_ν.

Solution.

Let $c = \pi/(2\sin\pi\nu)$ and first take noninteger ν, so that $K_\nu = c(I_{-\nu} - I_\nu)$ by Exercise 14.5.6. Since $\sin\pi(\nu \pm 1) = -\sin\pi\nu$, the same formula at orders $\nu \mp 1$ gives

$$K_{\nu-1} = -c(I_{-\nu+1} - I_{\nu-1}), \qquad K_{\nu+1} = -c(I_{-\nu-1} - I_{\nu+1}).$$

The modified functions satisfy

$$I_{\mu-1} - I_{\mu+1} = \frac{2\mu}{x}I_\mu, \qquad I_{\mu-1} + I_{\mu+1} = 2I'_\mu$$

for every order μ: the derivation in Exercise 14.1.14 uses only Eqs. (14.7), (14.8) at argument ix and $i^{-(\mu+1)} = -i^{1-\mu}$, which hold for any μ. At $\mu = -\nu$ the first reads $I_{-\nu-1} - I_{-\nu+1} = -(2\nu/x)I_{-\nu}$, that is, $I_{-\nu+1} - I_{-\nu-1} = (2\nu/x)I_{-\nu}$. Hence, for the difference,

$$\begin{aligned} K_{\nu-1} - K_{\nu+1} &= -c[(I_{-\nu+1} - I_{-\nu-1}) - (I_{\nu-1} - I_{\nu+1})] \\ &= -c\left[\frac{2\nu}{x}I_{-\nu} - \frac{2\nu}{x}I_\nu\right] = \boxed{-\frac{2\nu}{x}K_\nu}. \end{aligned}$$

For the sum, use the second recurrence at $\mu = -\nu$ and $\mu = \nu$:

$$\begin{aligned} K_{\nu-1} + K_{\nu+1} &= -c[(I_{-\nu+1} + I_{-\nu-1}) - (I_{\nu-1} + I_{\nu+1})] \\ &= -2c(I'_{-\nu} - I'_\nu) = \boxed{-2K'_\nu}. \end{aligned}$$

Taking order limits gives the same relations for integer ν: $K_\nu(x) = \frac{\pi}{2}i^{\nu+1}[J_\nu(ix) + iY_\nu(ix)]$, like Y_ν (Exercise 14.3.1), depends analytically on ν and x jointly, so both K_ν and K'_ν are continuous in the order (Exercise 14.5.6 shows this for K_ν itself). The opposite signs relative to the I recurrences arise from the sign reversal of $\sin\pi\nu$, not from a change in the ODE.

14.5.8 If $\mathcal{K}_\nu = e^{\nu\pi i}K_\nu$, show that $\mathcal{K}_\nu$ satisfies the same recurrence relations as I_ν.

Solution.

The neighboring phase factors satisfy $e^{i\pi(\nu-1)} = e^{i\pi(\nu+1)} = -e^{i\pi\nu}$. Multiplying the K recurrences of Exercise 14.5.7 by $-e^{i\pi\nu}$, which is this common phase, therefore gives

$$\mathcal{K}_{\nu-1} - \mathcal{K}_{\nu+1} = -e^{i\pi\nu}(K_{\nu-1} - K_{\nu+1}) = \frac{2\nu}{x}e^{i\pi\nu}K_\nu = \boxed{\frac{2\nu}{x}\mathcal{K}_\nu},$$
$$\mathcal{K}_{\nu-1} + \mathcal{K}_{\nu+1} = -e^{i\pi\nu}(K_{\nu-1} + K_{\nu+1}) = 2e^{i\pi\nu}K'_\nu = \boxed{2\mathcal{K}'_\nu}.$$

The order-dependent phase is constant with respect to x, so it passes through the derivative. These are exactly the recurrences for I_ν.

14.5.9 Show that when K_0 is evaluated from its series expansion about $x = 0$, the formula given as Eq. (14.110) only follows if a specific branch of its logarithmic term is chosen.

Solution.

By Eq. (14.61) with $n = 0$ and $\psi(1) = -\gamma$ (as in Exercise 14.3.7), $Y_0(z) = (2/\pi)J_0(z)\ln(z/2) + 2\gamma/\pi + O(z^2)$, where ln is whichever branch of the logarithm is used in the series, and $J_0(z) = 1 + O(z^2)$. Hence the small-argument Hankel expansion is

$$H_0^{(1)}(z) = J_0(z) + iY_0(z) = 1 + \frac{2i}{\pi}[\ln(z/2) + \gamma] + O(z^2 \ln z).$$

Insert $z = ix$, $x > 0$, in $K_0(x) = (\pi i/2)H_0^{(1)}(ix)$ (Eq. (14.106) with $\nu = 0$). Since $(\pi i/2)(2i/\pi) = -1$,

$$\begin{aligned} K_0(x) &= \frac{\pi i}{2} - \ln(ix/2) - \gamma + O(x^2 \ln x) \\ &= -\ln(x/2) - \gamma - 2\pi ik + O(x^2 \ln x), \end{aligned}$$

where $\ln(ix/2) = \ln(x/2) + i\pi/2 + 2\pi ik$ with an integer k labeling the branch. Thus the real expression in Eq. (14.110) requires

$$\boxed{\arg(ix) = \pi/2, \qquad k = 0}.$$

On this branch the $+\pi i/2$ prefactor contribution cancels the imaginary part of the logarithm, leaving $K_0 = -\ln x - \gamma + \ln 2 + \cdots$. In the full series the only logarithm is $(2/\pi)J_0(z)\ln(z/2)$ in Y_0; at $z = ix$ it contributes $(\pi i/2)\, i\, (2/\pi)J_0(ix)\ln(ix/2) = -I_0(x)\ln(ix/2)$ to K_0, because $J_0(ix) = I_0(x)$. So the logarithm has coefficient $-I_0(x)$, and selecting another branch changes the full function by $-2\pi ikI_0(x)$, not merely its leading constant. This is why the branch is part of the definition of the real positive-axis function K_0.

14.5.10 For $\nu > -\frac{1}{2}$ show that $K_\nu(z)$ may be represented by

$$\begin{aligned} K_\nu(z) &= \frac{\pi^{1/2}}{\Gamma(\nu + \frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_0^\infty e^{-z\cosh t} \sinh^{2\nu} t\, dt, \qquad -\frac{\pi}{2} < \arg z < \frac{\pi}{2} \\ &= \frac{\pi^{1/2}}{\Gamma(\nu + \frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_1^\infty e^{-zp}(p^2 - 1)^{\nu - 1/2}\, dp. \end{aligned}$$

Solution.

Let $\nu > -1/2$ and initially $z > 0$. Denote the second proposed expression by $F(z)$, and put $c = \sqrt{\pi}/[2^\nu \Gamma(\nu + 1/2)]$, so that $cz^\nu = \sqrt{\pi}(z/2)^\nu/\Gamma(\nu + 1/2)$. Thus

$$F(z) = cz^\nu G(z), \qquad G(z) = \int_1^\infty e^{-zp}(p^2 - 1)^{\nu - 1/2} dp.$$

Both endpoints are integrable: at $p = 1$, where $p^2 - 1 \approx 2(p - 1)$, the power is $\nu - 1/2 > -1$, and at infinity the exponential dominates. The same holds after multiplication by powers of p, so G may be differentiated under the integral: $G' = -\int_1^\infty pe^{-zp}(p^2 - 1)^{\nu - 1/2} dp$ and $G'' = \int_1^\infty p^2 e^{-zp}(p^2 - 1)^{\nu - 1/2} dp$. From $F' = c(\nu z^{\nu - 1}G + z^\nu G')$ and $F'' = c[\nu(\nu - 1)z^{\nu - 2}G + 2\nu z^{\nu - 1}G' + z^\nu G'']$, collect the terms in G, G', G''; those in $\nu^2 G$ and νG cancel:

$$\begin{aligned} z^2F'' + zF' - (z^2 + \nu^2)F &= c\big[\nu(\nu - 1) + \nu - \nu^2\big]z^\nu G + c(2\nu + 1)z^{\nu + 1}G' + cz^{\nu + 2}(G'' - G) \\ &= cz^\nu[z^2 G'' + (2\nu + 1)zG' - z^2 G]. \end{aligned}$$

Inserting the integrals for G, G', G'',

$$z^2F'' + zF' - (z^2+\nu^2)F = cz^\nu \int_1^\infty e^{-zp}(p^2-1)^{\nu-1/2}[z^2(p^2-1) - (2\nu+1)zp]dp$$
$$= -cz^{\nu+1}\int_1^\infty \frac{d}{dp}[e^{-zp}(p^2-1)^{\nu+1/2}]dp = 0.$$

The second line uses the product rule,

$$\frac{d}{dp}[e^{-zp}(p^2-1)^{\nu+1/2}] = e^{-zp}(p^2-1)^{\nu-1/2}[-z(p^2-1)+(2\nu+1)p].$$

The boundary term vanishes at infinity because of e^{-zp}, and at $p=1$ because $\nu+1/2>0$. Thus F solves the modified Bessel equation.

Its scale is fixed by the large-positive-z limit. Put $p = 1+u/z$, $dp = du/z$. Then $e^{-zp} = e^{-z}e^{-u}$ and $p^2-1 = (p-1)(p+1) = (2u/z)[1+u/(2z)]$, so

$$F(z) = cz^\nu e^{-z}\left(\frac{2}{z}\right)^{\nu-1/2}\frac{1}{z}\int_0^\infty e^{-u}u^{\nu-1/2}\left(1+\frac{u}{2z}\right)^{\nu-1/2}du.$$

The powers of z combine to $z^{-1/2}$, and $c\,2^{\nu-1/2} = \sqrt{\pi/2}/\Gamma(\nu+1/2)$; hence

$$F(z) = \sqrt{\frac{\pi}{2z}}\frac{e^{-z}}{\Gamma(\nu+1/2)}\int_0^\infty e^{-u}u^{\nu-1/2}\left(1+\frac{u}{2z}\right)^{\nu-1/2}du.$$

As $z\to\infty$ the last factor tends to one. It is at most one if $\nu\le 1/2$, and at most $(1+u)^{\nu-1/2}$ for $z\ge 1/2$ if $\nu>1/2$. Dominated convergence therefore makes the integral tend to $\int_0^\infty e^{-u}u^{\nu-1/2}du = \Gamma(\nu+1/2)$, so $F(z)\sim\sqrt{\pi/(2z)}e^{-z}$. This is the normalization of K_ν from its Hankel definition. Insert the leading form $H_\nu^{(1)}(w)\sim\sqrt{2/(\pi w)}\,e^{i(w-\pi\nu/2-\pi/4)}$ at $w=iz$ in Eq. (14.106), with $i^{\nu+1} = ie^{i\pi\nu/2}$ and $(iz)^{-1/2} = e^{-i\pi/4}z^{-1/2}$. The factors $e^{\pm i\pi\nu/2}$ cancel, and so do the remaining phases, since $i\,e^{-i\pi/4}e^{-i\pi/4} = ie^{-i\pi/2} = 1$:

$$\frac{\pi}{2}i^{\nu+1}\sqrt{\frac{2}{\pi iz}}e^{i(iz-\pi\nu/2-\pi/4)} = \frac{\pi}{2}\sqrt{\frac{2}{\pi z}}e^{-z} = \sqrt{\frac{\pi}{2z}}e^{-z}.$$

Here the leading Hankel asymptotic form, established independently in Section 14.6, is used only for the behavior of the K functions as $z\to\infty$. Two decaying modified-Bessel solutions are proportional: by Abel's formula the Wronskian of two solutions of $y''+y'/z-(1+\nu^2/z^2)y=0$ is A/z with a constant A. For F and K_ν the functions and their derivatives are $O(z^{-1/2}e^{-z})$ (for F' apply the substitution above to the integral for G'; for K_ν' use $K_\nu' = -\frac{1}{2}(K_{\nu-1}+K_{\nu+1})$ from Exercise 14.5.7 and the leading form $K_\mu\sim\sqrt{\pi/(2z)}\,e^{-z}$ just derived, which holds for every order μ), so $A = zW\to 0$ as $z\to\infty$. Hence $A=0$, $W\equiv 0$, and $F=\lambda K_\nu$; the common leading coefficient then gives $\lambda=1$, $F=K_\nu$.

Finally, substitute $p=\cosh t$, $dp=\sinh t\,dt$, with limits $t=0,\infty$; since $p^2-1=\sinh^2 t$,

$$(p^2-1)^{\nu-1/2}dp = \sinh^{2\nu-1}t\,\sinh t\,dt = \sinh^{2\nu}t\,dt,$$

which gives the first representation. Absolute convergence and analyticity extend both formulas to $\Re z>0$, equivalently $-\pi/2<\arg z<\pi/2$, with a consistent branch of z^ν.

14.5.11 Show that $I_\nu(x)$ and $K_\nu(x)$ satisfy the Wronskian relation

$$I_\nu(x)K'_\nu(x) - I'_\nu(x)K_\nu(x) = -\frac{1}{x}.$$

Solution.

Use the definitions at argument $w = ix$:

$$I_\nu(x) = i^{-\nu}J_\nu(w), \qquad K_\nu(x) = \frac{\pi}{2}i^{\nu+1}H_\nu^{(1)}(w).$$

Each x derivative supplies an additional factor i from $dw/dx = i$. The constant factors combine to $i^{-\nu}i^{\nu+1}i = i^2 = -1$, and the bracket is the Wronskian of Exercise 14.4.1(a), $2i/(\pi w)$; that identity between analytic functions continues from $w > 0$ to complex w off the cut. Thus

$$\begin{aligned} I_\nu K'_\nu - I'_\nu K_\nu &= \frac{\pi}{2}i^{-\nu}i^{\nu+1}i[J_\nu(w)H_\nu^{(1)\prime}(w) - J'_\nu(w)H_\nu^{(1)}(w)] \\ &= -\frac{\pi}{2}\frac{2i}{\pi w} = -\frac{i}{ix} = \boxed{-\frac{1}{x}}. \end{aligned}$$

The derivative primes inside the brackets refer to the argument w, whereas those on I, K refer to x; keeping that distinction is essential for the sign and normalization.

14.5.12 Verify that the coefficient in the axial Green's function of Eq. (14.117) is -1.

Use the cylindrical unit-source convention, whose radial equation has source $\delta(\rho_1 - \rho_2)/\rho_1$. The inconsistent source factors in Eqs. (14.114) and (14.116) are corrected below.

Solution.

The coefficient -1 is correct for the cylindrical point-source normalization. On p. 684, Eq. (14.114) must have ρ_1^{-1}, not ρ_1^{-2}, multiplying the three coordinate delta functions. Equation (14.116) must have $\delta(\rho_1 - \rho_2)/\rho_1$ on its right, not the unweighted delta function printed there. Indeed the cylindrical volume element is $\rho_1 d\rho_1 d\varphi_1 dz_1$, so a unit point source (one whose integral over this volume element is one) is $\delta(\rho_1 - \rho_2)\delta(\varphi_1 - \varphi_2)\delta(z_1 - z_2)/\rho_1$. Expand G and $\delta(\varphi_1 - \varphi_2)\delta(z_1 - z_2)$ in the same functions $e^{im(\varphi_1-\varphi_2)}\cos k(z_1 - z_2)$, with the same constant factors; acting on $e^{im\varphi_1}\cos kz_1$, the Laplacian contributes $-m^2/\rho_1^2 - k^2$. Put $r = \rho_1$, $b = \rho_2 > 0$. For $\nabla_1^2 G = \delta(\mathbf{r}_1 - \mathbf{r}_2)$ the correct radial equation is therefore

$$\left[\frac{d^2}{dr^2} + \frac{1}{r}\frac{d}{dr} - \frac{m^2}{r^2} - k^2\right] g(r,b) = \frac{\delta(r-b)}{r}, \qquad k > 0.$$

Away from $r = b$ this is the modified Bessel equation of order m in kr. Write a continuous candidate $g = AI_m(kr_<)K_m(kr_>)$, regular at the axis and decaying at infinity. For $r > b$, $g = AI_m(kb)K_m(kr)$; for $r < b$, $g = AI_m(kr)K_m(kb)$. With the Wronskian of Exercise 14.5.11 at $x = kb$, the derivative jump is

$$\begin{aligned} g'(b^+) - g'(b^-) &= Ak[I_m(kb)K'_m(kb) - I'_m(kb)K_m(kb)] \\ &= Ak\left(-\frac{1}{kb}\right) = -\frac{A}{b}. \end{aligned}$$

Multiplied by r, the ODE reads $(rg')' - (m^2/r + k^2 r)g = \delta(r-b)$. Integrating it from $b-\epsilon$ to $b+\epsilon$ and letting $\epsilon \to 0$, the bounded g terms drop out, so $[rg']_{b^-}^{b^+} = 1$, i.e., $b(-A/b) = -A = 1$. Hence

$$\boxed{g(r,b) = -I_m(kr_<)K_m(kr_>)}.$$

If one instead uses the literally printed unweighted radial delta, the right side after multiplication by r is $r\delta(r-b) = b\delta(r-b)$, so the jump $g'(b^+) - g'(b^-)$ must be one and the coefficient would be $A = -b$, not -1. The printed ρ_1^{-2} source, $\delta(r-b)/r^2$, would give $[rg']_{b^-}^{b^+} = 1/b$ and require $A = -1/b$.

The proof uses $k > 0$. For integer $|m| > 0$ (the radial equation contains only m^2, so m may be replaced by $|m|$), the $k \downarrow 0$ limit follows from the leading small-argument forms. By its series, $I_m(w) = (w/2)^m/m!\,[1+O(w^2)]$. In Eq. (14.61) the dominant term of $Y_m(w)$ is $-(m-1)!(w/2)^{-m}/\pi$; inserted in $K_m(w) = (\pi/2)i^{m+1}[J_m(iw) + iY_m(iw)]$ it gives $K_m(w) \simeq -\frac{1}{2}i^{m+2}(m-1)!\,i^{-m}(2/w)^m = \frac{1}{2}(m-1)!(2/w)^m$. Hence

$$I_m(kr_<)K_m(kr_>) \longrightarrow \frac{r_<^m}{2^m m!}\,\frac{(m-1)!\,2^m}{2r_>^m} = \frac{1}{2m}\left(\frac{r_<}{r_>}\right)^m,$$

and $g \to -(r_</r_>)^{|m|}/(2|m|)$. For $m = 0$, since $I_0(0) = 1$ and $K_0(w) = -\ln(w/2) - \gamma + o(1)$ (Exercise 14.5.9), $g = \ln(kr_>/2) + \gamma + o(1)$: the limit has an additive logarithmic divergence. Indeed the zero-wave-number radial equation with $m = 0$ has the solutions 1 and $\ln r$, none both regular at the axis and decaying at infinity. This component must be handled as a limiting Fourier component, not by substituting $I_0(0)K_0(0)$ into the formula.

14.5.13 If $r = (x^2+y^2)^{1/2}$, prove that

$$\frac{1}{r} = \frac{2}{\pi}\int_0^\infty \cos(xt)K_0(yt)\,dt.$$

This is a Fourier cosine transform of K_0.

Take real x and $y > 0$. For negative y, use $K_0(|y|t)$; $y = 0$ is only a limit.

Solution.

For the ordinary real integral take $y > 0$ and real x, since K_0 is complex on the negative axis (its cut) and infinite at zero. If negative y is to be allowed, replace $K_0(yt)$ by $K_0(|y|t)$. The case $y = 0$ is a limit of the transform, not a pointwise-defined integral of $K_0(0)$. Insert the convergent representation $K_0(yt) = \int_0^\infty e^{-yt\cosh u}du$ (Exercise 14.5.15, valid for $yt > 0$):

$$\int_0^\infty \cos(xt)K_0(yt)dt = \int_0^\infty\int_0^\infty \cos(xt)e^{-yt\cosh u}\,dt\,du.$$

The absolute double integral is bounded by

$$\int_0^\infty\int_0^\infty e^{-yt\cosh u}\,dt\,du = \int_0^\infty \frac{du}{y\cosh u} = \frac{1}{y}\Big[2\arctan e^u\Big]_0^\infty = \frac{\pi}{2y},$$

because $\frac{d}{du}2\arctan e^u = 2e^u/(1+e^{2u}) = 1/\cosh u$; so changing the order is justified (Fubini). With $a = y\cosh u > 0$, the inner elementary Laplace integral is

$$\int_0^\infty e^{-at}\cos(xt)dt = \Re\int_0^\infty e^{-(a-ix)t}dt = \Re\frac{1}{a-ix} = \frac{a}{a^2+x^2} = \frac{y\cosh u}{y^2\cosh^2 u + x^2}.$$

Now put $v = \sinh u$, $dv = \cosh u\,du$, $\cosh^2 u = 1+v^2$, and use $r = \sqrt{x^2+y^2}$; since $\frac{d}{dv}\arctan(yv/r) = yr/(y^2v^2+r^2)$,

$$\begin{aligned}\int_0^\infty \cos(xt)K_0(yt)dt &= y\int_0^\infty \frac{dv}{y^2(1+v^2)+x^2} = y\int_0^\infty \frac{dv}{y^2v^2+r^2}\\ &= \frac{1}{r}\left[\arctan\frac{yv}{r}\right]_0^\infty = \frac{\pi}{2r}.\end{aligned}$$

Multiplying by $2/\pi$ proves the formula. For $x \neq 0$, the limit $y \downarrow 0$ of this evaluated transform is $1/|x|$.

14.5.14 Derive the integral representation

$$I_n(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}\cos(n\theta)\,d\theta,$$

where n is an integer.

Hint. Start with the corresponding integral representation of $J_n(x)$. Equation (14.112) is a special case of this representation.

Solution.

For integer n, take the coefficient-extraction contour of Exercise 14.5.3(a) to be the unit circle $t = e^{i\theta}$, $-\pi \le \theta \le \pi$. Then $(x/2)(t+t^{-1}) = x\cos\theta$ and $t^{-n-1}dt = t^{-n}\,dt/t = ie^{-in\theta}d\theta$, whose factor i cancels that of $2\pi i$:

$$\begin{aligned} I_n(x) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{x\cos\theta}e^{-in\theta}d\theta \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{x\cos\theta}[\cos n\theta - i\sin n\theta]d\theta. \end{aligned}$$

Since $e^{x\cos\theta}$ is even in θ, the sine term is odd and integrates to zero over $[-\pi,\pi]$, while the cosine term is even and gives twice its integral over $[0,\pi]$. Therefore

$$\boxed{I_n(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}\cos(n\theta)d\theta}.$$

This holds for negative integers too, because both $I_{-n} = I_n$ and $\cos(-n\theta) = \cos(n\theta)$. For $n = 0$ it gives $I_0(x) = \pi^{-1}\int_0^\pi e^{x\cos\theta}d\theta = \pi^{-1}\int_0^\pi \cosh(x\cos\theta)d\theta$ (average the integrand over θ and $\pi-\theta$), the special case cited in the hint. The published qualification that n is an integer is necessary: noninteger order has an additional branch-cut contribution, so an ordinary closed-circle coefficient extraction is unavailable. Split the open contour of Exercise 14.5.3(b), as in Exercise 14.1.16, into the lower bank $t = e^{u-i\pi}$ $(u:\infty\to 0)$, the unit circle, and the upper bank $t = e^{u+i\pi}$ $(u:0\to\infty)$. On both banks $t = -e^u$, so $t + t^{-1} = -2\cosh u$, and $t^{-\nu-1}dt = t^{-\nu}\,dt/t = e^{-\nu u}e^{\pm i\pi\nu}du$, with the upper sign on the lower bank. Reversing the limits on the lower bank, the banks contribute

$$\frac{e^{-i\pi\nu} - e^{i\pi\nu}}{2\pi i}\int_0^\infty e^{-x\cosh u - \nu u}du = -\frac{\sin\nu\pi}{\pi}\int_0^\infty e^{-x\cosh u-\nu u}du.$$

On the circle, $-\pi \le \theta \le \pi$ lies on the branch, so the computation above with n replaced by ν (including the parity argument) gives $\pi^{-1}\int_0^\pi e^{x\cos\theta}\cos\nu\theta\,d\theta$. Together,

$$I_\nu(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}\cos\nu\theta\,d\theta - \frac{\sin\nu\pi}{\pi}\int_0^\infty e^{-x\cosh u-\nu u}du.$$

For noninteger ν and $x > 0$ the last term is nonzero, since $\sin\nu\pi \ne 0$ and its integrand is positive.

14.5.15 Show that

$$K_0(z) = \int_0^\infty e^{-z\cosh t}\,dt$$

satisfies the modified Bessel equation. How can you establish that this form is linearly independent of $I_0(z)$?

Take $\Re z > 0$ for the integral and its differentiated forms.

Solution.

Take $\Re z > 0$, which ensures absolute convergence and allows two derivatives under the integral, since $|e^{-z\cosh t}| = e^{-\Re z \cosh t}$ dominates the factors $\cosh t$ and $\cosh^2 t$ they produce. With $F(z) = \int_0^\infty e^{-z\cosh t}dt$, we have $F' = -\int_0^\infty \cosh t\, e^{-z\cosh t}dt$ and $F'' = \int_0^\infty \cosh^2 t\, e^{-z\cosh t}dt$; using $z^2\cosh^2 t - z^2 = z^2\sinh^2 t$,

$$\begin{aligned} z^2F'' + zF' - z^2F &= \int_0^\infty [z^2\sinh^2 t - z\cosh t]e^{-z\cosh t}dt \\ &= -\int_0^\infty \frac{d}{dt}[z\sinh t\, e^{-z\cosh t}]dt = 0, \end{aligned}$$

because, by the product rule, $\frac{d}{dt}[z\sinh t\, e^{-z\cosh t}] = [z\cosh t - z^2\sinh^2 t]e^{-z\cosh t}$. At zero the bracket vanishes because $\sinh 0 = 0$; at infinity the exponential dominates the hyperbolic factor. Hence F solves the order-zero modified Bessel equation. Exercise 14.5.10 at $\nu = 0$ has prefactor $\sqrt{\pi}(z/2)^0/\Gamma(1/2) = 1$ and identifies its normalization as $F = K_0$.

To prove independence from I_0, take positive real z and fix any $T > 0$. Positivity gives

$$F(z) \geq \int_0^T e^{-z\cosh t}dt \longrightarrow T \quad (z \downarrow 0).$$

The limit holds because the integrand tends to one uniformly on $[0, T]$. Since T is arbitrary, $F(z) \to \infty$, whereas $I_0(z) = 1 + O(z^2)$. If $\alpha F + \beta I_0 \equiv 0$ with $\alpha \neq 0$, then $F = -(\beta/\alpha)I_0$ would stay bounded as $z \downarrow 0$; so $\alpha = 0$, and then $\beta = 0$ because $I_0 \not\equiv 0$. The two solutions are therefore linearly independent. Equivalently, their Wronskian $I_0K_0' - I_0'K_0 = -1/z$ (Exercise 14.5.11) never vanishes.

14.5.16 The cylindrical cavity of Exercise 14.5.5 has along the cylinder walls the potential

$$V(z) = \begin{cases} 100\,\dfrac{z}{h}, & 0 \leq \dfrac{z}{h} \leq 1/2, \\ 100\left(1 - \dfrac{z}{h}\right), & 1/2 \leq \dfrac{z}{h} \leq 1. \end{cases}$$

With the radius-height ratio $a/h = 0.5$, calculate the potential for $z/h = 0.1(0.1)0.5$ and $\rho/a = 0.0(0.2)1.0$.

Check value. For $z/h = 0.3$ and $\rho/a = 0.8$, $V = 26.396$.

Solution.

The boundary data are independent of φ, so only the $m = 0$ cosine modes of Exercise 14.5.5 survive: for $m \geq 1$, $\int_0^{2\pi}\sin m\varphi\, d\varphi = \int_0^{2\pi}\cos m\varphi\, d\varphi = 0$ makes $a_{mn} = b_{mn} = 0$. Put $r = \rho/a$, $s = z/h$. For $m = 0$ the formula of Exercise 14.5.5(b), with the φ integral equal to 2π and $z = hs$, gives

$$b_{0n}I_0(k_na) = \frac{2\pi}{\pi h}\int_0^h V\sin k_nz\, dz = 2\int_0^1 V(s)\sin n\pi s\, ds \equiv B_n,$$

the Fourier sine coefficient of the wall data, $V(s) = \sum_{n\geq 1} B_n \sin n\pi s$. Reflection $s \mapsto 1 - s$ leaves V unchanged and multiplies the sine by $(-1)^{n+1}$, because $\sin(n\pi - n\pi s) = -\cos n\pi\, \sin n\pi s$. Therefore even-$n$ coefficients vanish. For odd n the two halves add, so $B_n = 2\cdot 2\int_0^{1/2} 100s\sin n\pi s\, ds$. Integrating by parts with $u = s$, $dv = \sin n\pi s\, ds$, $v = -\cos(n\pi s)/(n\pi)$, and using $\cos(n\pi/2) = 0$ for

odd n,

$$B_n = 400\int_0^{1/2} s\sin n\pi s\,ds$$
$$= 400\left[-\frac{s\cos n\pi s}{n\pi}+\frac{\sin n\pi s}{n^2\pi^2}\right]_0^{1/2} = \frac{400\sin(n\pi/2)}{n^2\pi^2}.$$

The final expression also gives zero for even n. Since $a/h = 1/2$, $k_n\rho = n\pi(\rho/a)(a/h) = n\pi r/2$ and $k_n a = n\pi/2$, so $b_{0n}I_0(k_n\rho) = B_n I_0(n\pi r/2)/I_0(n\pi/2)$ and the potential is

$$\boxed{\Phi(r,s) = \frac{400}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin(n\pi/2)\sin(n\pi s)}{n^2}\frac{I_0(n\pi r/2)}{I_0(n\pi/2)}}.$$

It vanishes on $s = 0, 1$, stays regular at $r = 0$, and at $r = 1$ reproduces the triangular wall potential, whose maximum is 50. The numerical results are

$\rho/a\backslash z/h$	0.1	0.2	0.3	0.4	0.5
0.0	7.118	13.657	19.006	22.551	23.798
0.2	7.262	13.954	19.461	23.138	24.437
0.4	7.682	14.836	20.843	24.958	26.441
0.6	8.345	16.261	23.180	28.212	30.130
0.8	9.165	18.088	26.396	33.252	36.364
1.0	10.000	20.000	30.000	40.000	50.000

The wall row uses the prescribed boundary values. The interior entries sum the modes through $n = 160$; comparison with $n = 100$ changes them by less than 10^{-14}. Each term is evaluated through the ratio $I_0(n\pi r/2)/I_0(n\pi/2) \le 1$ (I_0 increases on the positive axis), which decays roughly like $e^{-n\pi(1-r)/2}$, rather than as the product of the tiny $b_{0n} = B_n/I_0(n\pi/2)$ and the huge $I_0(n\pi r/2)$: for $n = 159$, the last nonzero mode, $I_0(159\pi/2) \approx 7.4\times 10^{106}$. The requested check is $\Phi(0.8, 0.3) = 26.3955684295\ldots \simeq 26.396$, and the symmetry $\Phi(r, 1-s) = \Phi(r, s)$ determines the other half of the cylinder.

14.6 Asymptotic Expansions

14.6.1 Determine the asymptotic dependence of the modified Bessel functions $I_\nu(x)$, given

$$I_\nu(x) = \frac{1}{2\pi i}\int_C e^{(x/2)(t+1/t)}\frac{dt}{t^{\nu+1}}.$$

The contour starts and ends at $t = -\infty$, encircling the origin in a positive sense. There are two saddle points. Only the one at $z = +1$ contributes significantly to the asymptotic form.

Find the leading form as $x \to +\infty$ with ν fixed; use the open contour and branch of Exercise 14.5.3.

Solution.

Take $x \to +\infty$ with order ν fixed. The rapidly varying exponent is $w(t) = x(t + t^{-1})/2$, with

$$w'(t) = \frac{x}{2}(1 - t^{-2}), \qquad w''(t) = xt^{-3}.$$

Thus the saddles are $t = \pm 1$, not $\pm x$. The Instructor's Manual, p. 219, prints $t = \pm x$ in identifying the saddles; the differentiated exponent gives $t = \pm 1$. Their exponential factors are e^x and e^{-x}, so the positive saddle supplies the dominant positive-real-axis asymptotic form. To see this, deform C into the lower bank of the negative axis from $-\infty$ to -1, the unit circle $t = e^{i\theta}$, $-\pi < \theta < \pi$, and the upper bank from -1 back to $-\infty$. On the two banks $t = -s$ with $s \geq 1$, so $w = -x(s+s^{-1})/2 \leq -x$, and these two pieces are $O(e^{-x})$. On the circle $w(e^{i\theta}) = x\cos\theta$ is real, so the circle is a path of constant $\Im w$ through $t = 1$. Along it $|e^w|$ is largest, e^x, at $\theta = 0$ and falls to e^{-x} at the other saddle $t = -1$ ($\theta = \pm\pi$). At $\theta = 0$ the circle runs upward: $dt = ie^{i\theta}d\theta = i\,d\theta$ there. Near $t = 1$, put $t = 1 + iu$ along this upward local descent direction. Since $1/(1+iu) = 1 - iu - u^2 + O(u^3)$, we have $t + t^{-1} = 2 - u^2 + O(u^3)$ and

$$w(1+iu) = x - \frac{xu^2}{2} + O(xu^3), \quad t^{-\nu-1} = 1 + O(u), \quad dt = i\,du.$$

The contributing width is $u = O(x^{-1/2})$. Keeping the leading terms and extending the local Gaussian integral to the real line,

$$\begin{aligned} I_\nu(x) &\sim \frac{1}{2\pi i} e^x i \int_{-\infty}^{\infty} e^{-xu^2/2} du \\ &= \frac{e^x}{2\pi}\sqrt{\frac{2\pi}{x}} = \boxed{\frac{e^x}{\sqrt{2\pi x}}}. \end{aligned}$$

The next relative terms are powers of $1/x$ for fixed order; this leading approximation is not uniform when ν grows with x. The branch-cut contour is understood as in Exercise 14.5.3: C begins and ends at $-\infty$ and is not closed, so the integral sign is the open $\int_C$ marked in the statement; the saddle called $z = +1$ there is $t = +1$.

14.6.2 Determine the asymptotic dependence of the modified Bessel function of the second kind, $K_\nu(x)$, by using

$$K_\nu(x) = \frac{1}{2}\int_0^\infty e^{(-x/2)(s+1/s)} \frac{ds}{s^{1-\nu}}.$$

Find the leading form as $x \to +\infty$ with ν fixed.

Solution.

For $x \to +\infty$ and fixed ν, write the given integral as

$$K_\nu(x) = \frac{1}{2}\int_0^\infty e^{w(s)} s^{\nu-1} ds, \qquad w(s) = -\frac{x}{2}(s + s^{-1}).$$

Since $w(s) + x = -x(s-1)^2/(2s) \leq 0$ for positive s, with equality only at $s = 1$, the unique real maximum of w is at $s = 1$. In detail,

$$w'(s) = -\frac{x}{2}(1 - s^{-2}), \quad w''(s) = -xs^{-3}, \quad w''(1) = -x, \quad w(1) = -x.$$

Put $s = 1+u$. Then $(s-1)^2/s = u^2(1-u+\cdots)$ gives $w = -x - xu^2/2 + O(xu^3)$, and $s^{\nu-1} = 1 + O(u)$. Outside a fixed interval $|u| < \delta$, $w + x$ stays below a negative multiple of x, so those parts are exponentially smaller than the peak. Only a neighborhood of width $x^{-1/2}$ contributes at leading

order, so the endpoints can be extended to the full Gaussian line in the local variable:

$$K_\nu(x) \sim \frac{1}{2}e^{-x}\int_{-\infty}^{\infty} e^{-xu^2/2}du = \frac{1}{2}e^{-x}\sqrt{\frac{2\pi}{x}}$$
$$= \boxed{\sqrt{\frac{\pi}{2x}}e^{-x}}.$$

Both original endpoints are exponentially suppressed; no contour deformation is needed for positive real x.

14.6.3 Verify that the integral representations

$$I_n(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}\cos(nt)\,dt,$$
$$K_\nu(z) = \int_0^\infty e^{-z\cosh t}\cosh(\nu t)\,dt, \qquad \Re(z) > 0,$$

satisfy the modified Bessel equation by direct substitution into that equation (the index n is an integer). How can you check the normalization?

Solution.

For the I integral put $f(t) = e^{z\cos t}$ and $c(t) = \cos nt$. Direct differentiation gives

$$f'' = (-z\cos t + z^2\sin^2 t)f, \qquad c'' = -n^2 c.$$

Here primes on f, c denote t derivatives. The z derivatives are $\partial_z f = \cos t\, f$ and $\partial_z^2 f = \cos^2 t\, f$, so

$$(z^2\partial_z^2 + z\partial_z - z^2)f = (z^2\cos^2 t + z\cos t - z^2)f = (z\cos t - z^2\sin^2 t)f = -f'',$$

while $-n^2 c = c''$. Applying the modified Bessel operator with respect to z under the integral therefore gives

$$(z^2\partial_z^2 + z\partial_z - z^2 - n^2)I_n = \frac{1}{\pi}\int_0^\pi [-f''c + fc'']dt$$
$$= \frac{1}{\pi}[fc' - f'c]_0^\pi = 0.$$

At both endpoints $f' = -z\sin t\, f = 0$, and $c' = -n\sin nt = 0$ because n is an integer. This explains the published integer qualification. The integral converges for every real n; the qualification is needed only for the boundary terms, since for noninteger n the term at π survives: $\pi^{-1}f(\pi)c'(\pi) = -(n/\pi)e^{-z}\sin n\pi \neq 0$.

For the K integral use $f(t) = e^{-z\cosh t}$, $c(t) = \cosh \nu t$. Now $f'' = (-z\cosh t + z^2\sinh^2 t)f$ and $c'' = \nu^2 c$, while $\partial_z f = -\cosh t\, f$ and $\partial_z^2 f = \cosh^2 t\, f$ give

$$(z^2\partial_z^2 + z\partial_z - z^2)f = (z^2\cosh^2 t - z\cosh t - z^2)f = (z^2\sinh^2 t - z\cosh t)f = f''.$$

Since $-\nu^2 c = -c''$,

$$(z^2\partial_z^2 + z\partial_z - z^2 - \nu^2)K_\nu = \int_0^\infty [f''c - fc'']dt$$
$$= [f'c - fc']_0^\infty = 0.$$

At zero both first derivatives vanish. At infinity, $\Re z > 0$ gives double-exponential damping, stronger than the fixed-order factors $\cosh \nu t$ and $\sinh \nu t$.

To check normalization, note first that each integral is a solution, so only its multiple of a standard solution is in question. The I integral is an entire function of z, while the second solution K_n has a logarithmic singularity at $z = 0$; hence the integral is a constant multiple of I_n, and comparing the lowest power of z fixes that constant. Expand $e^{z\cos t} = \sum_k z^k \cos^k t/k!$. For $k < n$, $\cos^k t$ is a combination of $\cos jt$ with $j \le k < n$, each orthogonal to $\cos nt$ on $[0, \pi]$; so for integer $n \ge 1$ the first nonzero term of the I integral comes from $z^n \cos^n t/n!$. Writing $\cos^n t = 2^{-n}(e^{it} + e^{-it})^n$, the terms $e^{\pm int}$ show that the coefficient of $\cos nt$ in $\cos^n t$ is 2^{1-n}, and $\int_0^\pi \cos^2 nt\, dt = \pi/2$. Thus the leading term is $\pi^{-1}(z^n/n!)\, 2^{1-n}(\pi/2) = z^n/(2^n n!)$, exactly the defining I_n coefficient. For $n = 0$ its value at zero is one; negative orders agree by $I_{-n} = I_n$. For K, the integral decays as $z \to +\infty$, so it contains no multiple of I_ν, which grows like $e^z/\sqrt{2\pi z}$. The dominant large-positive-z region is $t = 0$, where $\cosh t = 1 + t^2/2 + \cdots$ and $\cosh \nu t = 1 + \cdots$. Hence

$$K_\nu(z) \sim e^{-z} \int_0^\infty e^{-zt^2/2} dt = \sqrt{\frac{\pi}{2z}} e^{-z},$$

which is the leading term of Eq. (14.133) with coefficient one and so fixes the standard decaying normalization.

14.6.4

(a) Show that when K_ν is defined by Eq. (14.128),

$$\frac{dK_0(z)}{dz} = -K_1(z).$$

(b) Show that the indefinite integral of $-K_1(x)$ as defined by Eq. (14.128) has in the limit of small z the value $-\ln z + C$, and therefore, by comparison with Eq. (14.110), that K_0 as defined by Eq. (14.128) has the correct normalization.

To identify the full K_0 solution, also determine the constant C or require decay at infinity; the logarithmic coefficient alone permits addition of I_0.

Solution.

(a) Equation (14.128) reads $K_\nu(z) = \sqrt{\pi}(z/2)^\nu \Gamma(\nu + \frac{1}{2})^{-1} \int_1^\infty e^{-zp}(p^2 - 1)^{\nu - 1/2} dp$. Its prefactor is $\sqrt{\pi}/\Gamma(\frac{1}{2}) = 1$ for $\nu = 0$ and $\sqrt{\pi}(z/2)/\Gamma(\frac{3}{2}) = z$ for $\nu = 1$, so the two instances are

$$K_0(z) = \int_1^\infty \frac{e^{-zp}}{\sqrt{p^2 - 1}} dp, \qquad K_1(z) = z \int_1^\infty e^{-zp} \sqrt{p^2 - 1}\, dp.$$

For $\Re z > 0$, differentiate the first and use $d\sqrt{p^2 - 1}/dp = p/\sqrt{p^2 - 1}$:

$$\begin{aligned} K_0'(z) &= -\int_1^\infty e^{-zp}\, d\sqrt{p^2 - 1} \\ &= -[e^{-zp}\sqrt{p^2 - 1}]_1^\infty - z \int_1^\infty e^{-zp} \sqrt{p^2 - 1} dp = \boxed{-K_1(z)}. \end{aligned}$$

The boundary bracket vanishes at both ends.

(b) Set $u = zp$ in the K_1 integral for positive z:

$$K_1(z) = \frac{1}{z} \int_z^\infty e^{-u} \sqrt{u^2 - z^2}\, du.$$

As $z \downarrow 0$, the integrand (taken as zero on $0 < u < z$) is bounded by ue^{-u} and tends to it; by dominated convergence its integral tends to $\int_0^\infty ue^{-u}du = \Gamma(2) = 1$. Thus $K_1(z) \sim 1/z$ and an antiderivative of $-K_1$ has leading logarithmic term $-\ln z$. By (a), K_0 itself is such an antiderivative. That leading coefficient fixes the scale of the singular solution but does not by itself exclude adding a multiple of I_0. To identify the full normalization, fix the constant or the decaying behavior as well. The constant can be found directly from the K_0 integral:

$$K_0(z) = \int_1^\infty \frac{e^{-zp}}{p}dp + \int_1^\infty e^{-zp}\left(\frac{1}{\sqrt{p^2-1}} - \frac{1}{p}\right)dp.$$

With $u = zp$ the first integral is the exponential integral $E_1(z) = \int_z^\infty e^{-u}du/u$, whose series $E_1(z) = -\gamma - \ln z - \sum_{k\geq 1}(-z)^k/(k\,k!)$ gives $E_1(z) = -\ln z - \gamma + o(1)$. In the second, the factor multiplying e^{-zp} is positive and integrable: it behaves like $[2(p-1)]^{-1/2}$ near $p = 1$ and like $1/(2p^3)$ at infinity. By dominated convergence the second integral therefore tends, using $\operatorname{arcosh} R = \ln(R + \sqrt{R^2-1})$, to

$$\int_1^\infty \left(\frac{1}{\sqrt{p^2-1}} - \frac{1}{p}\right)dp = \lim_{R\to\infty}(\operatorname{arcosh} R - \ln R) = \lim_{R\to\infty} \ln\frac{R+\sqrt{R^2-1}}{R} = \ln 2.$$

Therefore

$$\boxed{K_0(z) = -\ln z + \ln 2 - \gamma + o(1)},$$

which reproduces both the logarithmic scale and the constant in Eq. (14.110).

14.6.5 Verify that Eq. (14.132) can be rearranged to the form given as Eq. (14.133).

Solution.

The coefficient of $(2z)^{-r}/r!$ in Eq. (14.132) is

$$\frac{\Gamma(\nu + r + 1/2)}{\Gamma(\nu - r + 1/2)}.$$

In the derivation of Eq. (14.132) it arises as $r!$ times the binomial coefficient $\Gamma(\nu+\frac{1}{2})/[r!\,\Gamma(\nu-r+\frac{1}{2})]$ of $(t/2z)^r$ in $(1+t/2z)^{\nu-1/2}$, times the termwise integral $\int_0^\infty e^{-t}t^{\nu+r-1/2}dt = \Gamma(\nu+r+\frac{1}{2})$, divided by the $\Gamma(\nu+\frac{1}{2})$ of the prefactor, which cancels the numerator of the binomial coefficient. For $r \geq 1$, apply the gamma recurrence $\Gamma(a+1) = a\Gamma(a)$ a total of $2r$ times, starting from $a = \nu - r + \frac{1}{2}$:

$$\Gamma(\nu + r + \tfrac{1}{2}) = (\nu + r - \tfrac{1}{2})(\nu + r - \tfrac{3}{2})\cdots(\nu - r + \tfrac{1}{2})\,\Gamma(\nu - r + \tfrac{1}{2}).$$

The $2r$ factors are $\nu + k - \frac{1}{2}$, $k = 1-r, \ldots, r$. Pair $k = j$ with $k = 1-j$ ($j = 1, \ldots, r$), i.e., factors equidistant from ν, and use $(\nu + j - \frac{1}{2})(\nu - j + \frac{1}{2}) = \nu^2 - (j-\frac{1}{2})^2 = \frac{1}{4}[4\nu^2 - (2j-1)^2]$:

$$\begin{aligned}\frac{\Gamma(\nu + r + 1/2)}{\Gamma(\nu - r + 1/2)} &= \prod_{j=1}^{r}\left(\nu + j - \tfrac{1}{2}\right)\left(\nu - j + \tfrac{1}{2}\right) \\ &= 4^{-r}\prod_{j=1}^{r}[4\nu^2 - (2j-1)^2].\end{aligned}$$

Since $4^r(2z)^r = (8z)^r$, the expansion is

$$\boxed{K_\nu(z) \sim \sqrt{\frac{\pi}{2z}}e^{-z}\sum_{r=0}^{\infty}\frac{\prod_{j=1}^{r}[4\nu^2 - (2j-1)^2]}{r!(8z)^r}}.$$

The empty product at $r = 0$ is one. Its first three terms are

$$1 + \frac{4\nu^2 - 1}{8z} + \frac{(4\nu^2 - 1)(4\nu^2 - 9)}{2!(8z)^2} + \cdots,$$

as in Eq. (14.133). At exceptional half-integer orders the product form also makes the terminating factors explicit. In general this is an asymptotic series, not a convergent infinite sum at a fixed z.

14.6.6

(a) Show that

$$y(z) = z^\nu \int e^{-zt}(t^2 - 1)^{\nu - 1/2}\, dt$$

satisfies the modified Bessel equation, provided the contour is chosen so that

$$e^{-zt}(t^2 - 1)^{\nu + 1/2}$$

has the same value at the initial and final points of the contour.

(b) Verify that the contours shown in Fig. 14.14 are suitable for this problem.

Require convergence of the integral and its differentiated forms. For the two endpoint contours in (b), take $\Re\nu > -1/2$; the unbounded contour additionally needs $\Re z > 0$.

Solution.

(a) Put $F(z) = \int_C e^{-zt}(t^2 - 1)^{\nu - 1/2} dt$, so $y = z^\nu F$. The product rule gives

$$y' = \nu z^{\nu - 1} F + z^\nu F', \qquad y'' = \nu(\nu - 1) z^{\nu - 2} F + 2\nu z^{\nu - 1} F' + z^\nu F''.$$

In the substitution the terms proportional to $z^\nu F$ cancel, their coefficient being $\nu(\nu - 1) + \nu - \nu^2 = 0$. Differentiating under the integral, $F' = -\int_C t\, e^{-zt}(t^2 - 1)^{\nu - 1/2} dt$ and $F'' = \int_C t^2 e^{-zt}(t^2 - 1)^{\nu - 1/2} dt$, so $F'' - F$ carries the factor $t^2 - 1$. Hence

$$\begin{aligned} z^2 y'' + z y' - (z^2 + \nu^2) y &= z^{\nu + 2}(F'' - F) + (2\nu + 1) z^{\nu + 1} F' \\ &= z^{\nu + 1} \int_C e^{-zt}(t^2 - 1)^{\nu - 1/2} [z(t^2 - 1) - (2\nu + 1)t] dt \\ &= -z^{\nu + 1} \int_C \frac{d}{dt} [e^{-zt}(t^2 - 1)^{\nu + 1/2}] dt, \end{aligned}$$

because

$$\frac{d}{dt}[e^{-zt}(t^2 - 1)^{\nu + 1/2}] = e^{-zt}(t^2 - 1)^{\nu - 1/2}[(2\nu + 1)t - z(t^2 - 1)].$$

Thus matching the two endpoint values of the bracket makes the ODE residual zero. Differentiation under the integral requires a fixed contour with convergent derivatives, as in the examples below; a formal endpoint condition alone does not ensure the integral exists.

(b) In Fig. 14.14, contour (1) joins $t = 1$ and positive-real infinity, and contour (2) joins $t = -1$ and $t = 1$. Their orientations change the solution's sign but not the ODE test. For these endpoint contours take $\Re\nu > -1/2$. The unbounded contour (1) also requires $\Re z > 0$ as drawn; without these conditions the stated suitability is not automatic. Under these assumptions, at $t = \pm 1$ the bracket vanishes like $(t \mp 1)^{\nu + 1/2}$, and the integrand is locally integrable like $(t \mp 1)^{\nu - 1/2}$. On contour (1), exponential decay makes the bracket and the differentiated integrands vanish at infinity. On contour (2), both endpoint bracket values are zero, with one branch of $(t^2 - 1)^{\nu - 1/2}$ continued continuously along the whole contour. Hence both contours satisfy the required conditions.

14.6.7 Use the asymptotic expansions to verify the following Wronskian formulas:

(a) $J_\nu(x)J_{-\nu-1}(x) + J_{-\nu}(x)J_{\nu+1}(x) = -2\sin\nu\pi/\pi x$,

(b) $J_\nu(x)Y_{\nu+1}(x) - J_{\nu+1}(x)Y_\nu(x) = -2/\pi x$,

(c) $J_\nu(x)H^{(2)}_{\nu-1}(x) - J_{\nu-1}(x)H^{(2)}_\nu(x) = 2/i\pi x$,

(d) $I_\nu(x)K'_\nu(x) - I'_\nu(x)K_\nu(x) = -1/x$,

(e) $I_\nu(x)K_{\nu+1}(x) + I_{\nu+1}(x)K_\nu(x) = 1/x$.

Solution.

For fixed real order and $x \to +\infty$, put $A = \sqrt{2/(\pi x)}$, $\chi = x - \nu\pi/2 - \pi/4$. Then

$$J_\nu \sim A\cos\chi, \quad Y_\nu \sim A\sin\chi, \quad H^{(2)}_\nu \sim Ae^{-i\chi},$$

$$I_\nu \sim \frac{e^x}{\sqrt{2\pi x}}, \qquad K_\nu \sim \sqrt{\frac{\pi}{2x}}e^{-x}, \qquad I'_\nu \sim I_\nu, \quad K'_\nu \sim -K_\nu.$$

To exhibit the Wronskians explicitly, use $J_{-\nu-1} = J'_{-\nu} - \nu J_{-\nu}/x$, $J_{\nu+1} = \nu J_\nu/x - J'_\nu$, $Y_{\nu+1} = \nu Y_\nu/x - Y'_\nu$, and $H^{(2)}_{\nu-1} = H^{(2)\prime}_\nu + \nu H^{(2)}_\nu/x$. These are the recurrences $Z_{\mu-1} = Z'_\mu + \mu Z_\mu/x$ and $Z_{\mu+1} = \mu Z_\mu/x - Z'_\mu$, valid for every cylinder function Z_μ (with $\mu = -\nu$ for $J_{-\nu-1}$). The ν/x products cancel in each pair; for example, in (a)

$$J_\nu\Big(J'_{-\nu} - \frac{\nu}{x}J_{-\nu}\Big) + J_{-\nu}\Big(\frac{\nu}{x}J_\nu - J'_\nu\Big) = J_\nu J'_{-\nu} - J'_\nu J_{-\nu}.$$

This leaves

$$\text{(a) } W[J_\nu, J_{-\nu}], \qquad \text{(b) } -W[J_\nu, Y_\nu], \qquad \text{(c) } W[J_\nu, H^{(2)}_\nu].$$

Part (d) is already a Wronskian and (e) is its negative. For two solutions of $y'' + y'/x + (\pm 1 - \nu^2/x^2)y = 0$ (Bessel's equation or the modified one), Abel's identity gives $W' = -W/x$, so each Wronskian is exactly a constant divided by x. Thus computing its leading coefficient at infinity determines the exact identity, not just an approximation.

(a) The phases for orders $-\nu$, $-\nu-1$, and $\nu+1$ are $\chi + \nu\pi$, $\chi + \nu\pi + \pi/2$, and $\chi - \pi/2$; thus $J_{-\nu-1} \sim -A\sin(\chi + \nu\pi)$ and $J_{\nu+1} \sim A\sin\chi$. Consequently, by $\sin a\cos b - \cos a\sin b = \sin(a-b)$ with $a = \chi$, $b = \chi + \nu\pi$,

$$\begin{aligned} J_\nu J_{-\nu-1} + J_{-\nu}J_{\nu+1} &\sim A^2[-\cos\chi\sin(\chi+\nu\pi) + \cos(\chi+\nu\pi)\sin\chi] \\ &= -A^2\sin\nu\pi = \boxed{-\frac{2\sin\nu\pi}{\pi x}}. \end{aligned}$$

(b) Since $Y_{\nu+1} \sim -A\cos\chi$ and $J_{\nu+1} \sim A\sin\chi$,

$$J_\nu Y_{\nu+1} - J_{\nu+1}Y_\nu \sim -A^2(\cos^2\chi + \sin^2\chi) = \boxed{-\frac{2}{\pi x}}.$$

The Y notation here is the published correction to N.

(c) The order-$\nu-1$ phase is $\chi+\pi/2$, so $H^{(2)}_{\nu-1}\sim Ae^{-i\chi}e^{-i\pi/2}=-iAe^{-i\chi}$ and $J_{\nu-1}\sim A\cos(\chi+\pi/2)=-A\sin\chi$. Since $-i\cos\chi+\sin\chi=-i(\cos\chi+i\sin\chi)=-ie^{i\chi}$,

$$\begin{aligned}J_\nu H^{(2)}_{\nu-1}-J_{\nu-1}H^{(2)}_\nu &\sim A^2e^{-i\chi}[-i\cos\chi+\sin\chi]\\ &=-iA^2=\boxed{\frac{2}{i\pi x}}.\end{aligned}$$

(d) The product of the two leading modified functions is $I_\nu K_\nu\sim(e^x/\sqrt{2\pi x})\sqrt{\pi/(2x)}\,e^{-x}=1/(2x)$. Therefore

$$I_\nu K'_\nu-I'_\nu K_\nu\sim-2I_\nu K_\nu=\boxed{-\frac{1}{x}}.$$

(e) Changing the fixed order by one does not alter these leading factors, giving $I_\nu K_{\nu+1}+I_{\nu+1}K_\nu\sim 1/x$. To confirm its exact Wronskian status, substitute $K_{\nu+1}=-K'_\nu+\nu K_\nu/x$ and $I_{\nu+1}=I'_\nu-\nu I_\nu/x$; these follow, after halving, by subtracting $I_{\nu-1}-I_{\nu+1}=(2\nu/x)I_\nu$ from $I_{\nu-1}+I_{\nu+1}=2I'_\nu$ and $K_{\nu-1}-K_{\nu+1}=-(2\nu/x)K_\nu$ from $K_{\nu-1}+K_{\nu+1}=-2K'_\nu$ (the modified-Bessel recurrences). The order terms cancel,

$$I_\nu\Big(-K'_\nu+\frac{\nu}{x}K_\nu\Big)+\Big(I'_\nu-\frac{\nu}{x}I_\nu\Big)K_\nu=-(I_\nu K'_\nu-I'_\nu K_\nu),$$

leaving $-W[I_\nu,K_\nu]$. Thus, by (d), it is exactly $\boxed{1/x}$.

14.6.8 Verify that the Green's function for the 2-D Helmholtz equation (operator ∇^2+k^2) with outgoing-wave boundary conditions is

$$G(\boldsymbol{\rho}_1,\boldsymbol{\rho}_2)=-\frac{i}{4}H^{(1)}_0(k|\boldsymbol{\rho}_1-\boldsymbol{\rho}_2|).$$

Hint. $H^{(1)}_0(k\rho)$ is known to be an outgoing-wave solution to the homogeneous Helmholtz equation. Use $k>0$, time dependence $e^{-i\omega t}$, and $(\nabla^2+k^2)G=+\delta^{(2)}$. The displayed minus sign uses this positive-source convention.

Solution.

The textbook prints $+iH^{(1)}_0/4$, which has source $-\delta^{(2)}$. With the book's Chapter 10 convention $LG=+\delta$, the coefficient must instead be $-i/4$; the Instructor's Manual, p. 223, derives this minus sign. Both conventions are common, so the source sign must be stated explicitly. Let $r=|\boldsymbol{\rho}_1-\boldsymbol{\rho}_2|$, $k>0$, and $G=CH^{(1)}_0(kr)$. For $r>0$, G depends on r alone, so $\nabla^2G=G''+G'/r$, and $(\nabla^2+k^2)G=0$ is Bessel's equation of order zero in kr, which $H^{(1)}_0(kr)$ satisfies. Near the source, $J_0(kr)=1+O(r^2)$ and $Y_0(kr)=(2/\pi)[\ln(kr/2)+\gamma]+O(r^2\ln r)$ give

$$H^{(1)}_0(kr)=1+\frac{2i}{\pi}[\ln(kr/2)+\gamma]+O(r^2\ln r),\quad G'(r)=\frac{2iC}{\pi r}+O(r\ln r).$$

Integrate the operator over a disk of radius ϵ and use the divergence theorem; the flux of ∇G through the circle $r=\epsilon$ is $2\pi\epsilon\,G'(\epsilon)$:

$$\begin{aligned}\int_{r<\epsilon}(\nabla^2+k^2)G\,d^2r &=2\pi\epsilon G'(\epsilon)+2\pi k^2\int_0^\epsilon G(r)r\,dr\\ &\longrightarrow 4iC.\end{aligned}$$

Here $2\pi\epsilon G'(\epsilon) = 2\pi\epsilon \cdot 2iC/(\pi\epsilon) + O(\epsilon^2|\ln\epsilon|) = 4iC + O(\epsilon^2|\ln\epsilon|)$, and, since $|G| = O(|\ln r|)$, the last integral tends to zero as $O(\epsilon^2|\ln\epsilon|)$. A unit positive delta source therefore requires $4iC = 1$:

$$\boxed{G_+(\boldsymbol{\rho}_1, \boldsymbol{\rho}_2) = -\frac{i}{4}H_0^{(1)}(kr), \qquad (\nabla^2 + k^2)G_+ = \delta^{(2)}(\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2)}.$$

Changing the sign gives the textbook's $+iH_0^{(1)}/4$, whose source is the negative unit source $-\delta^{(2)}$. Finally the large-r form $H_0^{(1)}(kr) \sim \sqrt{2/(\pi kr)}e^{i(kr-\pi/4)}$ satisfies $(\partial_r - ik)G = O(r^{-3/2})$, the outgoing condition for time dependence $e^{-i\omega t}$.

14.6.9 From the asymptotic form of $K_\nu(z)$, Eq. (14.134), derive the asymptotic form of $H_\nu^{(1)}(z)$, Eq. (14.138). Note particularly the phase, $(\nu + \frac{1}{2})\pi/2$.

Solution.

The definition $K_\nu(x) = (\pi/2)i^{\nu+1}H_\nu^{(1)}(ix)$, taken at $x = -iz$ (so that $ix = z$), gives $H_\nu^{(1)}(z) = (2/\pi)i^{-\nu-1}K_\nu(-iz)$. Take $z > 0$ first and rotate clockwise to $-iz = ze^{-i\pi/2}$, fixing the square-root and power branches. Equation (14.134) then gives

$$K_\nu(-iz) \sim \sqrt{\frac{\pi}{2(-iz)}}e^{iz}[P_\nu(z) + iQ_\nu(z)],$$

because $i(-iz) = z$. On this branch $(-iz)^{-1/2} = z^{-1/2}e^{i\pi/4}$. Combining the prefactors, with $(2/\pi)\sqrt{\pi/2} = \sqrt{2/\pi}$ and $i^{-\nu-1} = e^{-i(\nu+1)\pi/2}$,

$$\frac{2}{\pi}i^{-\nu-1}\sqrt{\frac{\pi}{2z}}e^{i\pi/4} = \sqrt{\frac{2}{\pi z}}e^{-i(\nu+1)\pi/2+i\pi/4} = \sqrt{\frac{2}{\pi z}}e^{-i(\nu+1/2)\pi/2}.$$

Hence

$$\boxed{H_\nu^{(1)}(z) \sim \sqrt{\frac{2}{\pi z}}e^{i[z-(\nu+1/2)\pi/2]}[P_\nu(z) + iQ_\nu(z)]}.$$

The additional $\pi/4$ is the phase of the rotated square root; omitting it would give an incorrect outgoing-wave phase. Extension to complex z uses this same continuation: the sector $-3\pi/2 < \arg(-iz) < 3\pi/2$, in which the K_ν expansion of Eq. (14.134) holds, becomes $-\pi < \arg z < 2\pi$. The asymptotic estimates hold on closed subsectors away from their boundary rays, with order fixed.

14.6.10 Apply Stokes' method for obtaining an asymptotic expansion for the Hankel function $H_\nu^{(1)}$ as follows:

(a) Replace the Bessel function in Bessel's equation by $x^{-1/2}y(x)$ and show that $y(x)$ satisfies

$$y''(x) + \left(1 - \frac{\nu^2 - \frac{1}{4}}{x^2}\right)y(x) = 0.$$

(b) Develop a power-series solution with negative powers of x starting with the assumed form

$$y(x) \sim e^{ix}\sum_{n=0}^{\infty} a_n x^{-n}.$$

Obtain the recurrence relation giving a_{n+1} in terms of a_n. Check your result against the asymptotic series, Eq. (14.138).

(c) From Eq. (14.125), determine the initial coefficient, a_0.

The inverse-power ansatz in (b) is an asymptotic expansion at positive infinity for fixed ν, rather than a generally convergent series.

Solution.

(a) Put $F = x^{-1/2}y$. Its derivatives are

$$F' = x^{-1/2}y' - \tfrac{1}{2}x^{-3/2}y, \qquad F'' = x^{-1/2}y'' - x^{-3/2}y' + \tfrac{3}{4}x^{-5/2}y.$$

Hence

$$\begin{aligned} x^2F'' &= x^{3/2}y'' - x^{1/2}y' + \tfrac{3}{4}x^{-1/2}y, \\ xF' &= x^{1/2}y' - \tfrac{1}{2}x^{-1/2}y, \\ (x^2-\nu^2)F &= x^{3/2}y - \nu^2 x^{-1/2}y. \end{aligned}$$

Adding these in $x^2F'' + xF' + (x^2-\nu^2)F = 0$ cancels the y' terms, and $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$, so

$$x^{3/2}y'' + x^{3/2}y + (1/4 - \nu^2)x^{-1/2}y = 0.$$

After division by $x^{3/2}$, $\boxed{y'' + [1 - (\nu^2 - 1/4)/x^2]y = 0}$.

(b) Write $y = e^{ix}h$, $h \sim \sum_{n\ge 0} a_n x^{-n}$, and put $\mu = \nu^2 - 1/4$. Since $y' = e^{ix}(h' + ih)$ and $y'' = e^{ix}(h'' + 2ih' - h)$, we have $y'' + y = e^{ix}(h'' + 2ih')$, and the equation becomes

$$h'' + 2ih' - \mu x^{-2}h = 0.$$

Termwise, $h'' = \sum_{n\ge 0} n(n+1)a_n x^{-n-2}$ and, since the $n = 0$ term of h' vanishes, $h' = -\sum_{n\ge 1} n a_n x^{-n-1} = -\sum_{n\ge 0}(n+1)a_{n+1}x^{-n-2}$. So the coefficient of x^{-n-2} is

$$[n(n+1) - \mu]a_n - 2i(n+1)a_{n+1} = 0.$$

With $\mu - n(n+1) = \nu^2 - (n+\frac{1}{2})^2 = \frac{1}{4}[4\nu^2 - (2n+1)^2]$, solving gives $a_{n+1} = i[\mu - n(n+1)]a_n/[2(n+1)]$, that is,

$$\boxed{a_{n+1} = \frac{i[4\nu^2 - (2n+1)^2]}{8(n+1)}a_n}, \qquad a_n = a_0 \frac{i^n}{8^n n!}\prod_{j=1}^{n}[4\nu^2 - (2j-1)^2].$$

The second formula follows by iterating the first from a_0. Even $n = 2r$ have factor $i^{2r} = (-1)^r$ and build P_ν; odd $n = 2r+1$ have $i^{2r+1} = i(-1)^r$ and build iQ_ν. Hence $y \sim a_0 e^{ix}[P_\nu(x) + iQ_\nu(x)]$, and $x^{-1/2}y$ reproduces Eq. (14.138) term by term once a_0 is fixed in (c).

(c) The leading Hankel form, Eq. (14.125), is $H_\nu^{(1)}(x) \sim \sqrt{2/(\pi x)}\,e^{i(x - \nu\pi/2 - \pi/4)}$. Since $H = x^{-1/2}y$, the leading term $a_0 e^{ix}$ of y must equal $x^{1/2}\sqrt{2/(\pi x)}\,e^{ix}e^{-i(\nu\pi/2 + \pi/4)}$, so

$$\boxed{a_0 = \sqrt{\frac{2}{\pi}}e^{-i(\nu+1/2)\pi/2}}.$$

The formal inverse-power series is generally asymptotic; for half-integer order a factor in the recurrence can terminate it.

14.6.11 Using the method of steepest descents, evaluate the second Hankel function given by

$$H_\nu^{(2)}(t) = \frac{1}{\pi i}\int_{C_2} e^{(t/2)(z-1/z)}\frac{dz}{z^{\nu+1}},$$

with contour C_2 as shown in Fig. 14.10.

ANS. $H_\nu^{(2)}(t) \approx \sqrt{\dfrac{2}{\pi t}}\, e^{-i(t-\pi/4-\nu\pi/2)}$.

Take $t \to +\infty$ with order ν fixed.

Solution.

Take positive $t \to \infty$, with ν fixed. Here $w(z) = t(z - z^{-1})/2$, $g(z) = z^{-\nu-1}$, and

$$w'(z) = \frac{t}{2}(1 + z^{-2}), \qquad w''(z) = -tz^{-3}.$$

The saddles solve $z^2 = -1$, so $z = \pm i$. The lower contour goes through $z_0 = -i$, where $w(z_0) = (t/2)(-i - 1/(-i)) = (t/2)(-2i) = -it$ and, since $(-i)^3 = i$, $w''(z_0) = -t/i = it$. The contour C_2 runs from $-\infty$ on the lower bank ($\arg z = -\pi$) to the origin, which it approaches from the right half-plane, where $e^{-t/(2z)}$ decays; so it crosses the lower half-plane from left to right. Its oriented descent direction is therefore $e^{i\pi/4}$ rather than $-e^{i\pi/4}$: writing $z = -i + e^{i\pi/4}u$ gives $\frac{1}{2}w''(z_0)(z - z_0)^2 = \frac{1}{2}(it)(iu^2) = -tu^2/2$, so

$$w(z) = -it - \frac{tu^2}{2} + O(tu^3), \qquad dz = e^{i\pi/4}du.$$

On the lower branch $\arg(-i) = -\pi/2$, so $g(-i) = e^{i(\nu+1)\pi/2}$. With $\int_{-\infty}^{\infty} e^{-tu^2/2}du = \sqrt{2\pi/t}$, the Gaussian approximation is

$$H_\nu^{(2)}(t) \sim \frac{1}{\pi i}e^{i(\nu+1)\pi/2}e^{-it}e^{i\pi/4}\sqrt{\frac{2\pi}{t}}$$
$$= \boxed{\sqrt{\frac{2}{\pi t}}e^{-i(t-\nu\pi/2-\pi/4)}}.$$

In the last step $\frac{1}{\pi i}\sqrt{2\pi/t} = e^{-i\pi/2}\sqrt{2/(\pi t)}$, and the phases add to $-\frac{\pi}{2} + \frac{(\nu+1)\pi}{2} + \frac{\pi}{4} - t = -(t - \frac{\nu\pi}{2} - \frac{\pi}{4})$. This is the incoming-wave counterpart of the upper-contour asymptotic expression.

14.6.12

(a) In applying the method of steepest descents to the Hankel function $H_\nu^{(1)}(t)$, show that $w(z,t)$, which appears in Eq. (14.121), satisfies

$$\Re[w(z,t)] < \Re[w(z_0,t)] = 0$$

for z on the contour C_1 (Fig. 14.10) but away from the point $z = z_0 = i$.

(b) For general values of $z = re^{i\theta}$, show that

$$\Re[w(z,t)] > 0 \quad \text{for } 0 < r < 1, \quad \begin{cases} \pi/2 < \theta \le \pi \\ -\pi \le \theta < -\pi/2 \end{cases}$$

and

$$\Re[w(z,t)] > 0 \quad \text{for } r > 1, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Your demonstration verifies that the distribution of the sign of w is as shown schematically in Fig. 14.15.

(c) Explain why the contour C_1 (Fig. 14.10) cannot be deformed to go through both saddle points, and why it may not go through the saddle point at $-i$ if it is to end at $z = -\infty$ with argument $+\pi$.

Take $t > 0$. In (a) and (c) use an admissible descending representative of C_1; the claimed restriction concerns steepest descent, not arbitrary analytic detours through regions of exponential growth.

Solution.

For $t > 0$ and $z = re^{i\theta}$, direct separation into real and imaginary parts gives

$$\boxed{\Re w = \frac{t}{2}(r - r^{-1})\cos\theta, \qquad \Im w = \frac{t}{2}(r + r^{-1})\sin\theta}.$$

(a) At the saddle, $w(i) = (t/2)(i - 1/i) = it$. A steepest path through it keeps $\Im w = t$, that is, $(r + r^{-1})\sin\theta = 2$, or $r^2\sin\theta - 2r + \sin\theta = 0$. Its roots are $r = (1 \mp \cos\theta)/\sin\theta$, namely $\tan(\theta/2)$ and $\cot(\theta/2)$; the second is the path of steepest ascent ($\Re w = t\cos^2\theta/\sin\theta \geq 0$ on it). The first gives an explicit upper descent contour,

$$r = \tan(\theta/2), \qquad 0 < \theta < \pi.$$

It starts at 0^+, passes through i, and ends at infinity on the upper bank of the negative axis. Since, by the half-angle formulas, $r + r^{-1} = 2/\sin\theta$ and $r - r^{-1} = -2\cot\theta$,

$$\Im w = t, \qquad \Re w = -\frac{t\cos^2\theta}{\sin\theta} \leq 0.$$

Equality occurs only at $\theta = \pi/2$, namely $z = i$. Thus the saddle is the unique maximum of the exponential modulus on this admissible representative of C_1.

(b) If $r < 1$, the factor $r - r^{-1}$ is negative. Therefore $\Re w > 0$ precisely in the left half-plane, $\pi/2 < \theta \leq \pi$ or $-\pi \leq \theta < -\pi/2$. If $r > 1$, that factor is positive, so $\Re w > 0$ in the right half-plane $-\pi/2 < \theta < \pi/2$. The textbook prints $\Re w < 0$ in the second inequality; it is corrected to $\Re w > 0$ in the statement. The endpoint $-\pi/2$ of the lower range in the first case, as marked in the statement, is likewise a corrected sign. Both corrected signs agree with Fig. 14.15. The remaining open regions have $\Re w < 0$; the unit circle and imaginary axis have $\Re w = 0$.

(c) A descending path leaving 0^+ lies in the right half of the unit disk, where $\Re w < 0$. Its passage through $+i$ connects to the upper part of the exterior left half-plane, which reaches the upper negative-real endpoint. Passage through $-i$ instead connects to the lower exterior left half-plane and the lower bank of the cut. To move from that lower branch to the upper endpoint one must cross the cut, or leave the decaying regions through a region with $\Re w > 0$. A single steepest path also has constant $\Im w$, whereas the two saddles have $\Im w = +t$ and $-t$. The claimed impossibility is about an admissible descent contour for this saddle approximation. An arbitrary analytic contour can be deformed to detour through both saddles while entering growth regions; Cauchy's theorem does not forbid that. Such a detour cannot be evaluated by retaining only independent decaying Gaussian neighborhoods of the two saddles.

14.6.13 Calculate the first 15 partial sums of $P_0(x)$ and $Q_0(x)$, Eqs. (14.135) and (14.136). Let x vary from 4 to 10 in unit steps. Determine the number of terms to be retained for maximum accuracy and the accuracy achieved as a function of x. Specifically, how small may x be without raising the error above 3×10^{-6}?

ANS. $x_{\min} = 6$.

Measure absolute error separately in P_0 and Q_0 on the stated integer grid; the quoted minimum is a grid minimum.

Solution.

For order zero define the exact coefficients recursively by

$$a_0 = 1, \qquad a_{j+1} = -\frac{(2j+1)^2}{8(j+1)} a_j .$$

These are the values at $\nu = 0$ of the coefficients $\prod_{l=1}^{j}[4\nu^2 - (2l-1)^2]/(j!\,8^j)$ of Eqs. (14.135) and (14.136) (compare Exercise 14.7.1), since each factor becomes $-(2l-1)^2$; those equations give P_0 the even-index terms $(-1)^r a_{2r} x^{-2r}$ and Q_0 the odd-index terms $(-1)^r a_{2r+1} x^{-2r-1}$. The N-term partial sums requested are

$$P_N(x) = \sum_{r=0}^{N-1} \frac{(-1)^r a_{2r}}{x^{2r}}, \qquad Q_N(x) = \sum_{r=0}^{N-1} \frac{(-1)^r a_{2r+1}}{x^{2r+1}} .$$

Here the subscript N counts the retained terms; the order is zero throughout, so P_N and Q_N are not the amplitude functions of order N. For example $P_2 = 1 - 9/(128x^2)$ and $Q_2 = -1/(8x) + 225/(3072x^3)$. The first fifteen P_N are:

$N\backslash x$	4	5	6	7	8	9	10
1	1.0	1.0	1.0	1.0	1.0	1.0	1.0
2	0.99560546875	0.9971875	0.998046875	0.99856505102	0.99890136719	0.99913194444	0.999296875
3	0.99604356289	0.99736694336	0.99813341211	0.9986117616	0.99892874807	0.9991490382	0.99930809021
4	0.99590379203	0.99733030327	0.99812114142	0.99860689542	0.99892656415	0.99914796093	0.99930751771
5	0.99599647456	0.99734585282	0.99812475775	0.99860794906	0.99892692619	0.99914810204	0.99930757845
6	0.99589155404	0.99733458706	0.99812293827	0.99860755959	0.99892682373	0.99914807048	0.99930756745
7	0.99607263834	0.99734703108	0.99812433395	0.99860777908	0.99892686794	0.99914808124	0.99930757049
8	0.99562993071	0.99732756059	0.99812281746	0.99860760386	0.99892684092	0.99914807605	0.9993075693
9	0.99708580943	0.99736853993	0.99812503394	0.99860779201	0.99892686314	0.99914807942	0.99930756992
10	0.99088757806	0.99725688252	0.99812083998	0.99860753045	0.99892683949	0.99914807658	0.9993075695
11	1.0240553954	0.99763928142	0.99813081449	0.99860798749	0.99892687112	0.99914807958	0.99930756986
12	0.8061441789	0.99603138045	0.99810168909	0.998607007	0.99892681917	0.99914807569	0.99930756948
13	2.5306366164	1.0041750657	0.99820412944	0.99860954066	0.99892692196	0.99914808177	0.99930756996
14	-13.649436996	0.95527379385	0.99777695255	0.99860177835	0.99892668085	0.9991480705	0.99930756923
15	163.94765864	1.2987965369	0.99986087045	0.9986295992	0.99892734245	0.99914809495	0.99930757051

The first fifteen Q_N are:

$N\backslash x$	4	5	6	7	8	9	10
1	-0.03125	-0.025	-0.020833333333	-0.017857142857	-0.015625	-0.013888888889	-0.0125
2	-0.03010559082	-0.0244140625	-0.020494249132	-0.017643608783	-0.015481948853	-0.013788419496	-0.012426757812
3	-0.030327375978	-0.024486737061	-0.020523455408	-0.017657121486	-0.015488879639	-0.01379226559	-0.012429028893
4	-0.03022192386	-0.024464622149	-0.020517283542	-0.017655023566	-0.015488055794	-0.013791904365	-0.01242885612
5	-0.0303149282	-0.02447710498	-0.020519702798	-0.017655627738	-0.015488237443	-0.013791967295	-0.0124288805
6	-0.030183479479	-0.024465813621	-0.020518183117	-0.017655348909	-0.015488173259	-0.013791949726	-0.012428874987
7	-0.030455541237	-0.024480770374	-0.020519581033	-0.017655537349	-0.01548820647	-0.013791956909	-0.012428876813
8	-0.029679880565	-0.02445347924	-0.020517809689	-0.01765536192	-0.015488182799	-0.013791952864	-0.01242887598
9	-0.032594314255	-0.024519106453	-0.02052076772	-0.017655577152	-0.015488205034	-0.013791955866	-0.012428876481
10	-0.018638099211	-0.024317976197	-0.02051447217	-0.017655240606	-0.015488178415	-0.013791953026	-0.012428876097
11	-0.10160699937	-0.025083229231	-0.020531106254	-0.017655893912	-0.015488217977	-0.013791956361	-0.012428876462
12	0.49794492096	-0.021544099364	-0.020477683307	-0.017654352383	-0.015488146505	-0.013791951601	-0.01242887604
13	-4.6776880071	-0.041097087738	-0.020682649385	-0.017658697607	-0.015488300751	-0.013791959718	-0.012428876623
14	47.926278172	0.086091498118	-0.019756769828	-0.017644276747	-0.015487908821	-0.013791943422	-0.012428875675
15	-573.85493274	-0.87606832271	-0.024620742001	-0.017699935591	-0.015489066979	-0.013791981471	-0.012428877467

To measure accuracy, recover the exact amplitude functions from the two Bessel functions, with $\chi = x - \pi/4$:

$$P(x) = \sqrt{\frac{\pi x}{2}}[J_0(x)\cos\chi + Y_0(x)\sin\chi],$$

$$Q(x) = \sqrt{\frac{\pi x}{2}}[Y_0(x)\cos\chi - J_0(x)\sin\chi].$$

These formulas invert the J_0, Y_0 amplitude-phase formulas $J_0 = \sqrt{2/(\pi x)}(P\cos\chi - Q\sin\chi)$, $Y_0 = \sqrt{2/(\pi x)}(P\sin\chi + Q\cos\chi)$ and provide references independent of truncating the asymptotic series. With $J_0(x)$ and $Y_0(x)$ evaluated to 20 significant digits (for example from their ascending series in extended-precision arithmetic), minimizing the absolute error separately among the fifteen sums gives:

x	best N_P	$\lvert P_{N_P} - P\rvert$	best N_Q	$\lvert Q_{N_Q} - Q\rvert$	relative Q error
4	5	4.40308e-05	4	4.36604e-05	1.44258e-03
5	6	5.39862e-06	5	5.36775e-06	2.19345e-04
6	7	6.72951e-07	6	6.70164e-07	3.26609e-05
7	8	8.48751e-08	7	8.46089e-08	4.79223e-06
8	9	1.07992e-08	8	1.07726e-08	6.95539e-07
9	10	1.38345e-09	9	1.38072e-09	1.00110e-07
10	11	1.78200e-10	10	1.77910e-10	1.43143e-08

The optimum is $N_P = x + 1$, $N_Q = x$ for these seven integer arguments. Extra terms eventually make the errors larger: successive terms $a_j x^{-j}$ have the modulus ratio

$$\frac{|a_{j+1}|x^{-j-1}}{|a_j|x^{-j}} = \frac{(2j+1)^2}{8(j+1)x},$$

which grows like $j/(2x)$. The convex quadratic $(2j+1)^2 - 8(j+1)x$ equals $1 - 8x < 0$ at both $j = 0$ and $j = 2x - 1$, and $+1$ at $j = 2x$; so for integer x the ratio is below one for $j \le 2x - 1$ and above one for $j \ge 2x$. For each x of the table the smallest term therefore has index $j = 2x$: the optimal P_{x+1} stops at that term ($j = 2r \le 2x$), and Q_x stops just before it ($j = 2r + 1 \le 2x - 1$).

The textbook's answer $x_{\min} = 6$ is reproduced by absolute errors in both P and Q on the integer grid specified in the question. If "error" means relative error in both amplitude functions, the corresponding minimum on this grid is 8, because Q is small. No threshold for all real x between grid

points is established by the seven sampled arguments alone. Under the absolute-error interpretation, at $x = 5$ both optimal errors exceed 5.3×10^{-6}, whereas at $x = 6$ both are below 6.8×10^{-7}. Thus the requested grid threshold is $\boxed{x_{\min} = 6}$. The error table uses the full precision of the computed sums, rather than the rounded values in the two partial-sum tables.

14.7 Spherical Bessel Functions

Unless a problem explicitly states otherwise, the spherical index n (and m) is a nonnegative integer. Singular formulas are initially for nonzero argument; a value at zero requires its regular limit.

14.7.1 Show how one can obtain Eq. (14.162) starting from Eq. (14.161).

Solution.

Put $\nu = n + \frac{1}{2}$ in Eq. (14.161). The coefficient of i^s/x^s in $P_\nu(x) + iQ_\nu(x)$ is

$$a_s = \frac{\prod_{j=1}^{s}[4\nu^2 - (2j-1)^2]}{s!8^s},$$

because P_ν contains the even orders $s = 2r$ with sign $(-1)^r = i^{2r}$ and Q_ν the odd orders $s = 2r+1$ with sign $(-1)^r$, which iQ_ν turns into i^{2r+1}. For this half-integer order each factor is

$$(2n+1)^2 - (2j-1)^2 = 4(n+j)(n-j+1).$$

For $0 \le s \le n$ the two products are

$$\prod_{j=1}^{s}(n+j) = \frac{(n+s)!}{n!}, \qquad \prod_{j=1}^{s}(n-j+1) = \frac{n!}{(n-s)!},$$

and the factor at $j = n+1$ is zero. Hence $a_s = 4^s(n+s)!/[(n-s)!\,s!\,8^s] = (n+s)!/[s!\,2^s(n-s)!]$ for $s \le n$ and $a_s = 0$ for $s > n$, so the expansion terminates:

$$P_{n+1/2}(x) + iQ_{n+1/2}(x) = \sum_{s=0}^{n} \frac{i^s(n+s)!}{s!(2x)^s(n-s)!}.$$

For $\nu = n + \frac{1}{2}$ the Hankel phase $(\nu + \frac{1}{2})\pi/2$ is $(n+1)\pi/2$, and the prefactor of $h_n^{(1)}$ is $\sqrt{\pi/(2x)}\sqrt{2/(\pi x)} = 1/x$. Since $e^{-i(n+1)\pi/2} = (-i)^{n+1}$, Eq. (14.161) becomes

$$\boxed{h_n^{(1)}(x) = (-i)^{n+1}\frac{e^{ix}}{x}\sum_{s=0}^{n}\frac{i^s(n+s)!}{s!(2x)^s(n-s)!}.}$$

Finally, $(2r)!! = 2^r r!$ gives

$$\frac{(2n+2s)!!}{(2n-2s)!!} = 2^{2s}\frac{(n+s)!}{(n-s)!}.$$

Consequently the same sum is $\sum_{s=0}^{n} i^s(2n+2s)!!/[s!(8x)^s(2n-2s)!!]$, the other form in Eq. (14.162). The finite expression is exact for $x \neq 0$, not merely a large-x approximation. Indeed, for $\nu = n + \frac{1}{2}$ the constant $\nu^2 - \frac{1}{4}$ of Exercise 14.6.10 equals $n(n+1)$, so the recurrence there gives a zero coefficient at order $n+1$ and hence at every higher order: e^{ix} times the terminating sum solves the equation of Exercise 14.6.10(a) exactly. Dividing such a solution y by x gives $x^{-1/2}(x^{-1/2}y)$, a solution of the spherical Bessel equation; so the finite sum S (the boxed right side) solves it exactly, and for real x so does $\overline{S} \sim i^{n+1}e^{-ix}/x$, which is independent of S. Thus $h_n^{(1)} - S = AS + B\overline{S}$ for constants A, B. By Eq. (14.161) this difference is $o(1/x)$, whereas $x(AS + B\overline{S}) = A(-i)^{n+1}e^{ix} + Bi^{n+1}e^{-ix} + O(1/x)$ tends to zero only if $A = B = 0$.

14.7.2 Show that if

$$y_n(x) = \sqrt{\frac{\pi}{2x}}\, Y_{n+1/2}(x),$$

it automatically equals

$$(-1)^{n+1}\sqrt{\frac{\pi}{2x}}\, J_{-n-1/2}(x).$$

Solution.

For noninteger ν, the defining relation for Y_ν is

$$Y_\nu(x) = \frac{\cos(\pi\nu)J_\nu(x) - J_{-\nu}(x)}{\sin(\pi\nu)}.$$

At $\nu = n + \frac{1}{2}$, with integer n, $\cos(\pi\nu) = 0$ and $\sin(\pi\nu) = (-1)^n$. Hence

$$Y_{n+1/2}(x) = -(-1)^n J_{-n-1/2}(x) = (-1)^{n+1} J_{-n-1/2}(x).$$

Multiplication by $\sqrt{\pi/(2x)}$ gives the asserted formula. For complex x, use the same branch of $x^{1/2}$ in both expressions; their resulting spherical functions agree by analytic continuation.

14.7.3 Derive the trigonometric-polynomial forms of $j_n(z)$ and $y_n(z)$[3]:

$$j_n(z) = \frac{1}{z}\sin\left(z - \frac{n\pi}{2}\right)\sum_{s=0}^{[n/2]} \frac{(-1)^s(n+2s)!}{(2s)!(2z)^{2s}(n-2s)!} + \frac{1}{z}\cos\left(z - \frac{n\pi}{2}\right)\sum_{s=0}^{[(n-1)/2]} \frac{(-1)^s(n+2s+1)!}{(2s+1)!(2z)^{2s+1}(n-2s-1)!},$$

$$y_n(z) = \frac{(-1)^{n+1}}{z}\cos\left(z + \frac{n\pi}{2}\right)\sum_{s=0}^{[n/2]} \frac{(-1)^s(n+2s)!}{(2s)!(2z)^{2s}(n-2s)!} - \frac{(-1)^{n+1}}{z}\sin\left(z + \frac{n\pi}{2}\right)\sum_{s=0}^{[(n-1)/2]} \frac{(-1)^s(n+2s+1)!}{(2s+1)!(2z)^{2s+1}(n-2s-1)!}.$$

Solution.

Separate the even and odd indices in the finite sum of Exercise 14.7.1: the terms $s = 2k$ carry $i^{2k} = (-1)^k$ and the terms $s = 2k+1$ carry $i^{2k+1} = i(-1)^k$, so the sum is $P + iQ$ with (renaming k as s)

$$P = \sum_{s=0}^{\lfloor n/2\rfloor} \frac{(-1)^s(n+2s)!}{(2s)!(2z)^{2s}(n-2s)!},$$

$$Q = \sum_{s=0}^{\lfloor (n-1)/2\rfloor} \frac{(-1)^s(n+2s+1)!}{(2s+1)!(2z)^{2s+1}(n-2s-1)!}.$$

[3]The upper summation limit $[n/2]$ means the largest integer that does not exceed $n/2$.

An empty sum is zero, so $Q = 0$ when $n = 0$. With $\chi = z - n\pi/2$ the prefactor is $(-i)^{n+1}e^{iz} = e^{i[z-(n+1)\pi/2]} = e^{i(\chi-\pi/2)} = -ie^{i\chi} = \sin\chi - i\cos\chi$. Multiplying out $(\sin\chi - i\cos\chi)(P + iQ)$ gives

$$h_n^{(1)}(z) = \frac{1}{z}\{P\sin\chi + Q\cos\chi + i[-P\cos\chi + Q\sin\chi]\}.$$

For positive real z, P, Q are real and $h_n^{(1)} = j_n + iy_n$; separating its two parts gives

$$\boxed{j_n = \frac{P\sin\chi + Q\cos\chi}{z}, \qquad y_n = \frac{-P\cos\chi + Q\sin\chi}{z}.}$$

These finite analytic identities then extend to complex $z \neq 0$. To put the second one in the question's phase convention, shift the angle by $n\pi$, $\chi = (z + n\pi/2) - n\pi$:

$$\cos\chi = (-1)^n\cos(z + n\pi/2), \qquad \sin\chi = (-1)^n\sin(z + n\pi/2).$$

Thus

$$\boxed{y_n(z) = \frac{(-1)^{n+1}}{z}\{P\cos(z + n\pi/2) - Q\sin(z + n\pi/2)\}.}$$

There are two corrections to the printed formulas. The power in the odd sum for j_n is $2s + 1$, as in the published errata. In the displayed phase convention for y_n, the sign before its second sum must be minus; the textbook, p. 710, prints plus. For $n = 1$, $P = 1$, $Q = 1/z$; the correct answer is $y_1 = -\sin z/z - \cos z/z^2$. The printed plus would change the last sign and fail the spherical Bessel equation. The same substitution gives $j_1 = \sin z/z^2 - \cos z/z$, which also checks the corrected odd power.

14.7.4 Use the integral representation of $J_\nu(x)$,

$$J_\nu(x) = \frac{1}{\pi^{1/2}\Gamma(\nu + \frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_{-1}^{1} e^{\pm ixp}(1 - p^2)^{\nu-1/2}\,dp,$$

to show that the spherical Bessel functions $j_n(x)$ are expressible in terms of trigonometric functions; that is, for example,

$$j_0(x) = \frac{\sin x}{x}, \qquad j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}.$$

Solution.

Insert $\nu = n + \frac{1}{2}$ and multiply by $\sqrt{\pi/(2x)}$. Since $\Gamma(n+1) = n!$ and $\sqrt{\pi/(2x)}\,(x/2)^{n+1/2}/\sqrt{\pi} = x^n/2^{n+1}$,

$$j_n(x) = \frac{x^n}{2^{n+1}n!}\int_{-1}^{1} e^{ixp}(1 - p^2)^n\,dp.$$

Expand $(1 - p^2)^n = \sum_{r=0}^{n}\binom{n}{r}(-1)^r p^{2r}$ and introduce

$$S(x) = \int_{-1}^{1} e^{ixp}dp = \frac{e^{ix} - e^{-ix}}{ix} = \frac{2\sin x}{x}.$$

Differentiation under this finite-interval integral gives

$$S^{(2r)}(x) = \int_{-1}^{1}(ip)^{2r}e^{ixp}dp = (-1)^r\int_{-1}^{1} p^{2r}e^{ixp}dp.$$

The two factors $(-1)^r$ cancel in the binomial expansion, so

$$\boxed{j_n(x) = \frac{x^n}{2^{n+1}n!}\sum_{r=0}^{n}\binom{n}{r}S^{(2r)}(x).}$$

Every derivative of $\sin x/x$ is a finite combination of $\sin x, \cos x$ multiplied by integer negative powers of x. This proves the requested statement for every nonnegative integer n. In particular,

$$j_0 = S/2 = \frac{\sin x}{x}.$$

For $n = 1$, differentiating $S' = 2\cos x/x - 2\sin x/x^2$ once more,

$$S'' = -\frac{2\sin x}{x} - \frac{4\cos x}{x^2} + \frac{4\sin x}{x^3},$$

so that $S + S'' = 4\sin x/x^3 - 4\cos x/x^2$, whence

$$j_1 = \frac{x}{4}(S + S'') = \frac{\sin x}{x^2} - \frac{\cos x}{x}.$$

The apparent singularities at $x = 0$ cancel, as is already clear from the original integral multiplied by x^n.

14.7.5

(a) Derive the recurrence relations

$$f_{n-1}(x) + f_{n+1}(x) = \frac{2n+1}{x}f_n(x),$$
$$nf_{n-1}(x) - (n+1)f_{n+1}(x) = (2n+1)f_n'(x)$$

satisfied by the spherical Bessel functions $j_n(x)$, $y_n(x)$, $h_n^{(1)}(x)$, and $h_n^{(2)}(x)$.

(b) Show, from these two recurrence relations, that the spherical Bessel function $f_n(x)$ satisfies the differential equation

$$x^2 f_n''(x) + 2xf_n'(x) + \big[x^2 - n(n+1)\big]f_n(x) = 0.$$

Solution.

(a) For any cylindrical solution F_ν, set $f_n = \sqrt{\pi/(2x)}F_{n+1/2}$. Its two cylindrical recurrences, $F_{\nu-1} + F_{\nu+1} = (2\nu/x)F_\nu$ and $F_{\nu-1} - F_{\nu+1} = 2F_\nu'$ at $\nu = n + \frac{1}{2}$, multiplied by $\sqrt{\pi/(2x)}$, give, respectively,

$$f_{n-1} + f_{n+1} = \frac{2n+1}{x}f_n, \qquad f_n' = \frac{1}{2}(f_{n-1} - f_{n+1}) - \frac{f_n}{2x}.$$

The last term arises from differentiating $x^{-1/2}$: $f_n' = \sqrt{\pi/(2x)}F_{n+1/2}' - f_n/(2x)$. Replace f_n/x by $(f_{n-1} + f_{n+1})/(2n+1)$:

$$\begin{aligned}(2n+1)f_n' &= \frac{2n+1}{2}(f_{n-1} - f_{n+1}) - \frac{1}{2}(f_{n-1} + f_{n+1})\\ &= nf_{n-1} - (n+1)f_{n+1}.\end{aligned}$$

Eliminating f_{n+1}, respectively f_{n-1}, from the right side with the first recurrence gives

$$(2n+1)f_n' = (2n+1)f_{n-1} - \frac{(n+1)(2n+1)}{x}f_n = \frac{n(2n+1)}{x}f_n - (2n+1)f_{n+1},$$

that is, the useful equivalent pair

$$f_n' = f_{n-1} - \frac{n+1}{x}f_n = \frac{n}{x}f_n - f_{n+1}.$$

All four functions in the question inherit these relations by linearity from their cylindrical counterparts.

(b) Differentiate $f_{n+1} = nf_n/x - f_n'$:

$$f_{n+1}' = \frac{n}{x}f_n' - \frac{n}{x^2}f_n - f_n''.$$

Use the first equivalent derivative identity at order $n+1$, and then substitute $f_{n+1} = nf_n/x - f_n'$ once more:

$$f_{n+1}' = f_n - \frac{n+2}{x}f_{n+1} = f_n - \frac{n(n+2)}{x^2}f_n + \frac{n+2}{x}f_n'.$$

Equate the right sides and move all terms to one side. The f_n' terms combine to $[(n+2)-n]f_n'/x = 2f_n'/x$, and the f_n terms to $[1 - n(n+2)/x^2 + n/x^2]f_n = [1 - n(n+1)/x^2]f_n$; so

$$f_n'' + \frac{2}{x}f_n' + \left(1 - \frac{n(n+1)}{x^2}\right)f_n = 0.$$

Multiplication by x^2 proves the required equation.

14.7.6 Prove by mathematical induction (Section 1.4) that

$$j_n(x) = (-1)^n x^n \left(\frac{1}{x}\frac{d}{dx}\right)^n \left(\frac{\sin x}{x}\right)$$

for n, an arbitrary nonnegative integer.

Solution.

Let $D = x^{-1}d/dx$. The assertion is true for $n = 0$ because $j_0 = \sin x/x$. Assume it is true at n. The derivative recurrence in Exercise 14.7.5 and the product rule give

$$j_{n+1} = \frac{n}{x}j_n - j_n' = -x^n \frac{d}{dx}(x^{-n}j_n).$$

Insert the induction hypothesis, $x^{-n}j_n = (-1)^n D^n j_0$, before differentiating, and write $x^n\, d/dx = x^{n+1}D$:

$$\begin{aligned} j_{n+1} &= -x^n \frac{d}{dx}\{(-1)^n D^n j_0\} \\ &= (-1)^{n+1}x^{n+1}D(D^n j_0) = (-1)^{n+1}x^{n+1}D^{n+1}j_0. \end{aligned}$$

This is the assertion for $n+1$, completing the induction. The displayed differential expression is initially interpreted at $x \neq 0$; its continuous limit at the origin is supplied by the power series of j_n.

14.7.7 From the discussion of orthogonality of the spherical Bessel functions, show that a Wronskian relation for $j_n(x)$ and $y_n(x)$ is

$$j_n(x)y_n'(x) - j_n'(x)y_n(x) = \frac{1}{x^2}.$$

Solution.

The spherical equation is self-adjoint in the form

$$(x^2 f_n')' + [x^2 - n(n+1)]f_n = 0.$$

Multiply the equation for y_n by j_n, the equation for j_n by y_n, and subtract. The undifferentiated terms cancel:

$$j_n(x^2 y_n')' - y_n(x^2 j_n')' = \frac{d}{dx}\{x^2(j_n y_n' - j_n' y_n)\} = 0.$$

Thus $x^2 W$ is a constant. To determine it, use the leading series at the origin,

$$j_n = \frac{x^n + O(x^{n+2})}{(2n+1)!!}, \qquad y_n = -(2n-1)!!\,[x^{-n-1} + O(x^{-n+1})],$$

where $(-1)!! = 1$. Differentiating termwise, $j_n' = [nx^{n-1} + O(x^{n+1})]/(2n+1)!!$ and $y_n' = (2n-1)!!\,[(n+1)x^{-n-2} + O(x^{-n})]$. Hence $j_n y_n'$ and $-j_n' y_n$ equal the two terms below up to errors $O(1)$, so

$$\begin{aligned} W &\sim \frac{(n+1)(2n-1)!!}{(2n+1)!!}\frac{1}{x^2} + \frac{n(2n-1)!!}{(2n+1)!!}\frac{1}{x^2} \\ &= \frac{(2n+1)(2n-1)!!}{(2n+1)!!}\frac{1}{x^2} = \frac{1}{x^2}. \end{aligned}$$

Thus $x^2 W = 1 + O(x^2)$; because $x^2 W$ is exactly constant, the small-x computation fixes the exact result $\boxed{W[j_n, y_n] = x^{-2}}$. The second function is y_n, correcting the printed n_n in accordance with the published errata.

14.7.8 Verify

$$h_n^{(1)}(x)h_n^{(2)\prime}(x) - h_n^{(1)\prime}(x)h_n^{(2)}(x) = -\frac{2i}{x^2}.$$

Solution.

Write $j = j_n$, $y = y_n$. Bilinearity and antisymmetry of the Wronskian give

$$\begin{aligned} W[h_n^{(1)}, h_n^{(2)}] &= W[j + iy, j - iy] \\ &= W[j,j] - iW[j,y] + iW[y,j] + W[y,y] \\ &= -iW[j,y] - iW[j,y] = -2iW[j,y]. \end{aligned}$$

The two same-function Wronskians vanish. Substitution of Exercise 14.7.7 yields $\boxed{-2i/x^2}$, including its sign.

14.7.9 Verify Poisson's integral representation of the spherical Bessel function,

$$j_n(z) = \frac{z^n}{2^{n+1}n!}\int_0^\pi \cos(z\cos\theta)\sin^{2n+1}\theta\,d\theta.$$

Solution.

Expand the cosine, $\cos(z\cos\theta) = \sum_r (-1)^r z^{2r}\cos^{2r}\theta/(2r)!$; this series is uniformly convergent on the finite θ interval for every fixed complex z. Its rth term requires

$$\begin{aligned}\int_0^\pi \cos^{2r}\theta\sin^{2n+1}\theta\,d\theta &= \int_{-1}^1 p^{2r}(1-p^2)^n dp \\ &= \int_0^1 u^{r-1/2}(1-u)^n du \\ &= \frac{\Gamma(r+1/2)n!}{\Gamma(n+r+3/2)}.\end{aligned}$$

In the first line $p=\cos\theta$, $dp = -\sin\theta\,d\theta$, and $\sin^{2n}\theta = (1-p^2)^n$. In the second, the even integrand was doubled on $0<p<1$ and $u=p^2$, $dp = du/(2\sqrt{u})$, was used. The third is the beta integral $B(r+\frac12, n+1) = \Gamma(r+\frac12)\Gamma(n+1)/\Gamma(n+r+\frac32)$. Since $\Gamma(r+1/2) = \sqrt{\pi}(2r)!/(4^r r!)$, the proposed right side becomes

$$\begin{aligned}\frac{z^n}{2^{n+1}n!}\sum_{r=0}^\infty \frac{(-1)^r z^{2r}}{(2r)!}\frac{\Gamma(r+1/2)n!}{\Gamma(n+r+3/2)} &= \frac{\sqrt{\pi}z^n}{2^{n+1}}\sum_{r=0}^\infty \frac{(-1)^r z^{2r}}{4^r r!\Gamma(n+r+3/2)} \\ &= \sqrt{\frac{\pi}{2z}}J_{n+1/2}(z) = j_n(z).\end{aligned}$$

The last step is the ascending series $J_{n+1/2}(z) = \sum_r (-1)^r (z/2)^{n+2r+1/2}/[r!\,\Gamma(n+r+\frac32)]$ multiplied by $\sqrt{\pi/(2z)}$, because $\sqrt{\pi/(2z)}\,(z/2)^{n+1/2} = \sqrt{\pi}z^n/2^{n+1}$ and $(z/2)^{2r} = z^{2r}/4^r$. This proves the formula. At $z=0$ it has its regular limiting value, 1 for $n=0$ and 0 for $n>0$.

14.7.10 A well-known integral representation for $K_\nu(x)$ has the form

$$K_\nu(x) = \frac{2^\nu\Gamma(\nu+\frac12)}{\sqrt{\pi}\,x^\nu}\int_0^\infty \frac{\cos xt}{(t^2+1)^{\nu+1/2}}\,dt.$$

Starting from this formula, show that

$$k_n(x) = \frac{2^{n+2}(n+1)!}{\pi x^{n+1}}\int_0^\infty \frac{k^2 j_0(kx)}{(k^2+1)^{n+2}}\,dk.$$

Take $x>0$ and nonnegative integer n. The starting cosine integral converges for real $\nu > -1/2$ (absolutely when $\nu>0$), and the required order is $\nu = n+1/2$.

Solution.

The book uses

$$k_n(x) = \sqrt{\frac{2}{\pi x}}K_{n+1/2}(x),$$

so its normalization differs from that of j_n. Insert $\nu = n+\frac{1}{2}$ in the given integral: then $\Gamma(\nu+\frac{1}{2}) = n!$, $(t^2+1)^{\nu+1/2} = (1+t^2)^{n+1}$, and the prefactor times $\sqrt{2/(\pi x)}$ is $\sqrt{2/(\pi x)}\,2^{n+1/2}n!/(\sqrt{\pi}x^{n+1/2}) = 2^{n+1}n!/(\pi x^{n+1})$. Hence

$$k_n(x) = \frac{2^{n+1}n!}{\pi x^{n+1}}\int_0^\infty \frac{\cos(xt)}{(1+t^2)^{n+1}}dt.$$

Integrate this last integral by parts, taking the cosine as the derivative of $\sin(xt)/x$ and using $d(1+t^2)^{-n-1}/dt = -2(n+1)t(1+t^2)^{-n-2}$; in the last step $t\sin(xt)/x = t^2\sin(xt)/(xt) = t^2 j_0(xt)$:

$$\begin{aligned}\int_0^\infty \frac{\cos(xt)}{(1+t^2)^{n+1}}dt &= \left[\frac{\sin(xt)}{x(1+t^2)^{n+1}}\right]_0^\infty + \frac{2(n+1)}{x}\int_0^\infty \frac{t\sin(xt)}{(1+t^2)^{n+2}}dt \\ &= 2(n+1)\int_0^\infty \frac{t^2 j_0(xt)}{(1+t^2)^{n+2}}dt.\end{aligned}$$

The boundary term is zero at both ends for $x>0$, $n\ge 0$: it contains $\sin 0 = 0$ at $t=0$ and decays like t^{-2n-2} at infinity. Multiplying the prefactor by $2(n+1)$, with $2^{n+1}n!\cdot 2(n+1) = 2^{n+2}(n+1)!$, and renaming t as k gives exactly

$$\boxed{k_n(x) = \frac{2^{n+2}(n+1)!}{\pi x^{n+1}}\int_0^\infty \frac{k^2 j_0(kx)}{(1+k^2)^{n+2}}dk.}$$

Both integrals are absolutely convergent for these parameters.

14.7.11 Show that

$$\int_0^\infty J_\mu(x)J_\nu(x)\frac{dx}{x} = \frac{2}{\pi}\frac{\sin[(\mu-\nu)\pi/2]}{\mu^2-\nu^2}, \qquad \mu+\nu>0.$$

Take real orders. At $\mu=\nu>0$, interpret the quotient by continuity as $1/(2\nu)$.

Solution.

Take real orders with $\mu+\nu>0$. The two Bessel equations in self-adjoint form are

$$(xJ_\alpha')' + \left(x - \frac{\alpha^2}{x}\right)J_\alpha = 0.$$

Multiply the μ equation by J_ν, the ν equation by J_μ, and subtract. The terms $xJ_\mu J_\nu$ cancel, and $J_\nu(xJ_\mu')' - J_\mu(xJ_\nu')' = \frac{d}{dx}\{x(J_\nu J_\mu' - J_\nu' J_\mu)\}$ because the two products $xJ_\nu' J_\mu'$ cancel. The result is

$$\frac{d}{dx}\{x(J_\nu J_\mu' - J_\nu' J_\mu)\} = \frac{\mu^2-\nu^2}{x}J_\mu J_\nu.$$

Integrate on $[\epsilon, R]$:

$$(\mu^2-\nu^2)\int_\epsilon^R J_\mu J_\nu\frac{dx}{x} = \Big[x(J_\nu J_\mu' - J_\nu' J_\mu)\Big]_\epsilon^R.$$

At the lower endpoint, $J_\alpha = (x/2)^\alpha[1+O(x^2)]/\Gamma(\alpha+1)$ makes the bracket $(\mu-\nu)(\epsilon/2)^{\mu+\nu}[1+O(\epsilon^2)]/[\Gamma(\mu+1)\Gamma(\nu+1)] = O(\epsilon^{\mu+\nu})$, which tends to zero; when a leading coefficient vanishes at a negative integer order, the actual power only improves this estimate. At the upper endpoint put $\chi_\alpha = x - \alpha\pi/2 - \pi/4$ and use

$$J_\alpha = \sqrt{\frac{2}{\pi x}}[\cos\chi_\alpha + O(x^{-1})], \quad J_\alpha' = \sqrt{\frac{2}{\pi x}}[-\sin\chi_\alpha + O(x^{-1})].$$

It follows that

$$x(J_\nu J'_\mu - J'_\nu J_\mu) = \frac{2}{\pi}[-\cos\chi_\nu \sin\chi_\mu + \sin\chi_\nu \cos\chi_\mu] + O(x^{-1})$$
$$\longrightarrow \frac{2}{\pi}\sin\frac{(\mu-\nu)\pi}{2},$$

since the bracket is $\sin(\chi_\nu - \chi_\mu)$ and $\chi_\nu - \chi_\mu = (\mu-\nu)\pi/2$. The integrand is integrable at zero and $O(x^{-2})$ at infinity, so both limiting operations are justified. For $\mu^2 \neq \nu^2$,

$$\boxed{\int_0^\infty J_\mu J_\nu \frac{dx}{x} = \frac{2}{\pi}\frac{\sin[(\mu-\nu)\pi/2]}{\mu^2-\nu^2}.}$$

At equal orders the printed quotient is 0/0 and must be interpreted by continuity. Here $\mu = \nu > 0$ and its limit is $1/(2\nu)$. The other vanishing denominator, $\mu = -\nu$, is excluded by $\mu + \nu > 0$. Indeed, dividing numerator and denominator by $\mu - \nu$ gives the limit $(2/\pi)(\pi/2)/(2\nu) = 1/(2\nu)$. Dominated convergence is valid in a sufficiently small neighborhood of any fixed positive ν, using the same origin and infinity bounds.

14.7.12 Derive Eq. (14.190):

$$\int_{-\infty}^{\infty} j_m(x) j_n(x)\, dx = 0, \qquad \begin{cases} m \neq n, \\ m, n \geq 0. \end{cases}$$

Solution.

The power series of j_n (Exercise 14.7.9), x^n times a series in x^2, gives $j_n(-x) = (-1)^n j_n(x)$. The product is integrable at the origin and decays as $O(x^{-2})$ at infinity. If $m+n$ is odd, its integral over the whole real line therefore vanishes by oddness.

If $m+n$ is even, the product is even. On the positive half-line, $j_m j_n = [\pi/(2x)] J_{m+1/2} J_{n+1/2}$; use Exercise 14.7.11 with $\mu = m + \frac{1}{2}$, $\nu = n + \frac{1}{2}$ (so $\mu + \nu = m + n + 1 > 0$) and $(m+\frac{1}{2})^2 - (n+\frac{1}{2})^2 = (m-n)(m+n+1)$:

$$\begin{aligned} \int_0^\infty j_m(x) j_n(x) dx &= \frac{\pi}{2}\int_0^\infty J_{m+1/2}(x) J_{n+1/2}(x)\frac{dx}{x} \\ &= \frac{\sin[(m-n)\pi/2]}{(m+1/2)^2 - (n+1/2)^2} \\ &= \frac{\sin[(m-n)\pi/2]}{(m-n)(m+n+1)}. \end{aligned}$$

For distinct nonnegative integers of the same parity, $m-n$ is a nonzero even integer: the numerator is zero and the denominator is nonzero. Doubling this zero proves the whole-line result. The denominator in this half-line intermediate formula is $(m-n)(m+n+1)$, not $m^2 - n^2$ as in the Instructor's Manual, p. 228. The final orthogonality conclusion is unaffected.

14.7.13 Derive Eq. (14.191):

$$\int_{-\infty}^{\infty} \big[j_n(x)\big]^2 dx = \frac{\pi}{2n+1}.$$

Solution.

Use the equal-order limit established in Exercise 14.7.11, now with $\nu = n + \frac{1}{2} > 0$:

$$\int_0^\infty \frac{J_{n+1/2}(x)^2}{x}\,dx = \frac{1}{2(n+1/2)} = \frac{1}{2n+1}.$$

Since $j_n^2 = \pi J_{n+1/2}^2/(2x)$ on $x > 0$,

$$\int_0^\infty j_n(x)^2 dx = \frac{\pi}{2(2n+1)}.$$

The square is even, regardless of the parity of n. Its integral over the whole line is twice the last expression, hence $\boxed{\pi/(2n+1)}$.

14.7.14 The Fresnel integrals (Fig. 14.19 and Exercise 12.7.2) occurring in diffraction theory are given by

$$x(t) = \sqrt{\frac{\pi}{2}}\,C\left(\sqrt{\frac{2}{\pi}}\,t\right) = \int_0^t \cos(v^2)\,dv,$$
$$y(t) = \sqrt{\frac{\pi}{2}}\,S\left(\sqrt{\frac{2}{\pi}}\,t\right) = \int_0^t \sin(v^2)\,dv.$$

Show that these integrals may be expanded in series of spherical Bessel functions as follows:

$$x(\sqrt{s}) = \frac{1}{2}\int_0^s j_{-1}(u)u^{1/2}\,du = s^{1/2}\sum_{n=0}^\infty j_{2n}(s),$$

$$y(\sqrt{s}) = \frac{1}{2}\int_0^s j_0(u)u^{1/2}\,du = s^{1/2}\sum_{n=0}^\infty j_{2n+1}(s).$$

Hint. To establish the equality of the integral and the sum, you may wish to work with their derivatives. The spherical Bessel analogs of Eqs. (14.8) and (14.12) may be helpful.

Use the standard $C(q) = \int_0^q \cos(\pi u^2/2)\,du$ and $S(q) = \int_0^q \sin(\pi u^2/2)\,du$. In the subsequent formulas set $s = t^2 \geq 0$.

Solution.

Use capital X, Y temporarily to distinguish the two functions from their arguments. With the standard Fresnel convention $C(q) = \int_0^q \cos(\pi u^2/2)du$ and $S(q) = \int_0^q \sin(\pi u^2/2)du$, the substitution $u = \sqrt{2/\pi}\,v$ gives

$$X(t) = \sqrt{\frac{\pi}{2}}C\left(\sqrt{\frac{2}{\pi}}\,t\right), \qquad Y(t) = \sqrt{\frac{\pi}{2}}S\left(\sqrt{\frac{2}{\pi}}\,t\right).$$

Next put $s = t^2$ for $t \geq 0$, and define $F(s) = X(\sqrt{s})$, $H(s) = Y(\sqrt{s})$. In the defining integrals set $u = v^2$, so $dv = du/(2\sqrt{u})$:

$$F(s) = \frac{1}{2}\int_0^s \frac{\cos u}{\sqrt{u}}du = \frac{1}{2}\int_0^s \sqrt{u}\,j_{-1}(u)du,$$
$$H(s) = \frac{1}{2}\int_0^s \frac{\sin u}{\sqrt{u}}du = \frac{1}{2}\int_0^s \sqrt{u}\,j_0(u)du,$$

where $j_{-1}(u) = \cos u/u$ and $j_0(u) = \sin u/u$. The Fresnel arguments printed on p. 711 are $\sqrt{\pi/2}\,t$; the correct arguments are $\sqrt{2/\pi}\,t$. Also the functions in the subsequent s-integrals are $X(\sqrt{s})$ and $Y(\sqrt{s})$, not the same functions evaluated at s. The printed second integral has a stray x after $u^{1/2}$, which must be deleted. In the definition of $y(t)$ the Fresnel function is the sine integral S, as marked, because the right side is $\int_0^t \sin(v^2)\,dv$.

To prove the series, form finite sums

$$E_N(s) = \sqrt{s}\sum_{n=0}^{N} j_{2n}(s) = \sqrt{\frac{\pi}{2}}\sum_{n=0}^{N} J_{2n+1/2}(s),$$

$$O_N(s) = \sqrt{s}\sum_{n=0}^{N} j_{2n+1}(s) = \sqrt{\frac{\pi}{2}}\sum_{n=0}^{N} J_{2n+3/2}(s).$$

Use $2J'_\alpha = J_{\alpha-1} - J_{\alpha+1}$. For example,

$$E'_N(s) = \frac{1}{2}\sqrt{\frac{\pi}{2}}\sum_{n=0}^{N}[J_{2n-1/2}(s) - J_{2n+3/2}(s)],$$

and since $2(n+1) - \frac{1}{2} = 2n + \frac{3}{2}$ the second member of each term cancels the first member of the next. Consecutive terms cancel in this way, leaving the explicit terminal terms

$$E'_N(s) = \frac{1}{2}\sqrt{\frac{\pi}{2}}[J_{-1/2}(s) - J_{2N+3/2}(s)],$$

$$O'_N(s) = \frac{1}{2}\sqrt{\frac{\pi}{2}}[J_{1/2}(s) - J_{2N+5/2}(s)].$$

With $J_{-1/2}(s) = \sqrt{2/(\pi s)}\cos s$ and $J_{1/2}(s) = \sqrt{2/(\pi s)}\sin s$, the first terms are $\frac{1}{2}\sqrt{\pi/2}\,J_{-1/2}(s) = \cos s/(2\sqrt{s}) = F'(s)$ and $\frac{1}{2}\sqrt{\pi/2}\,J_{1/2}(s) = \sin s/(2\sqrt{s}) = H'(s)$. For $\alpha \geq 0$ we have $\Gamma(k+\alpha+1) = (\alpha+k)(\alpha+k-1)\cdots(\alpha+1)\,\Gamma(\alpha+1) \geq \Gamma(\alpha+1)$, since each factor is at least one, so the series $J_\alpha(s) = \sum_k (-1)^k (s/2)^{2k+\alpha}/[k!\,\Gamma(k+\alpha+1)]$ bounds the terminal term on $0 \leq s \leq R$ by $(R/2)^\alpha e^{R^2/4}/\Gamma(\alpha+1)$, which tends to zero as $\alpha = 2N + \frac{3}{2}$ or $2N + \frac{5}{2}$ grows. Integrate these finite identities from zero to s; both sums vanish at zero, as do F and H:

$$E_N(s) = F(s) - \frac{1}{2}\sqrt{\frac{\pi}{2}}\int_0^s J_{2N+3/2}(u)\,du,$$

$$O_N(s) = H(s) - \frac{1}{2}\sqrt{\frac{\pi}{2}}\int_0^s J_{2N+5/2}(u)\,du.$$

The remaining integrals are at most s times the bound above, so the limits of E_N and O_N are $F(s)$ and $H(s)$, respectively. This establishes

$$\boxed{F(s) = \sqrt{s}\sum_{n=0}^{\infty} j_{2n}(s), \qquad H(s) = \sqrt{s}\sum_{n=0}^{\infty} j_{2n+1}(s).}$$

Equivalently, in the original parameter,

$$\boxed{X(t) = t\sum_{n=0}^{\infty} j_{2n}(t^2), \qquad Y(t) = t\sum_{n=0}^{\infty} j_{2n+1}(t^2).}$$

These last formulas also extend to negative real t by oddness.

14.7.15 A hollow sphere of radius a (Helmholtz resonator) contains standing sound waves. Find the minimum positive frequency of oscillation in terms of the radius a and the velocity of sound v. The sound waves satisfy the wave equation

$$\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

and the boundary condition $\dfrac{\partial\psi}{\partial r} = 0$, $r = a$.

The spatial part of this PDE is the same as the PDE discussed in Example 14.7.1, but here we have a Neumann boundary condition, in contrast to the Dirichlet boundary condition of that example.

ANS. $\nu_{\min} = 0.3313\, v/a$, $\quad \lambda_{\max} = 3.018\, a$.

Solution.

Separate a time-harmonic mode as $\psi = R_l(r)Y_{lm}(\theta,\phi)e^{-i\omega t}$ and set $k = \omega/v$. The radial equation is

$$(r^2R_l')' + [k^2r^2 - l(l+1)]R_l = 0.$$

Regularity at the center selects $R_l(r) = j_l(kr)$ for $k > 0$. The rigid-wall condition is $j_l'(ka) = 0$. A spatially constant Neumann mode has $k = \omega = 0$. The quoted minimum means the lowest *positive* oscillation frequency; the zero mode must be excluded explicitly.

For $l = 1$ differentiation of $j_1(x) = \sin x/x^2 - \cos x/x$ gives

$$j_1'(x) = \frac{F(x)}{x^3}, \qquad F(x) = (x^2-2)\sin x + 2x\cos x.$$

Differentiate the numerator completely:

$$F' = 2x\sin x + (x^2-2)\cos x + 2\cos x - 2x\sin x = x^2\cos x.$$

Thus F increases from $F(0) = 0$ to $x = \pi/2$, then decreases strictly to $F(\pi) = -2\pi < 0$. It has exactly one zero in $(\pi/2, \pi)$ and none earlier. Newton's iteration $x \mapsto x - F(x)/F'(x) = x - F(x)/(x^2\cos x)$, started at $x = 2$, converges to it in five steps (bisection of F on $(\pi/2,\pi)$ works as well):

$$x_1 = ka = 2.0815759778181006\ldots.$$

To establish that this is the global first positive eigenvalue, first consider $l = 0$. The derivative numerator is $x\cos x - \sin x$, with derivative $-x\sin x < 0$ on $(0,\pi)$. It starts at zero, hence j_0' has no positive root below π. For every $l \geq 2$, multiply the radial equation by $\overline{R}_l$ and integrate over $0 < r < a$. By parts, $\int_0^a \overline{R}_l(r^2R_l')'dr = [r^2\overline{R}_lR_l']_0^a - \int_0^a r^2|R_l'|^2dr$, and the boundary term vanishes: $r^2 \to 0$ at the regular origin and $R_l'(a) = 0$ at the wall. Solving for k^2 and then using $r^2 \leq a^2$ in the denominator,

$$k^2 = \frac{\int_0^a [r^2|R_l'|^2 + l(l+1)|R_l|^2]dr}{\int_0^a r^2|R_l|^2dr} \geq \frac{l(l+1)}{a^2} \geq \frac{6}{a^2}.$$

Since $x_1 < \sqrt{6}$, no higher angular order can lower this frequency. The $l = 1$ root is consequently the required one. Converting angular frequency to cycles per unit time,

$$\boxed{\nu_{\min} = \frac{x_1}{2\pi}\frac{v}{a} = 0.3312931062\ldots\frac{v}{a}, \qquad \lambda_{\max} = \frac{2\pi a}{x_1} = 3.0184751237\ldots a.}$$

At $k = 0$ the regular radial solutions are r^l; their Neumann condition permits only $l = 0$, confirming the constant zero mode.

14.7.16

(a) Show that the parity of $i_n(x)$ (the behavior under $x \to -x$) is $(-1)^n$.

(b) Show that $k_n(x)$ has no definite parity.

Solution.

(a) From the series of $I_{n+1/2}$,

$$i_n(x) = \sqrt{\frac{\pi}{2x}} I_{n+1/2}(x) = \frac{\sqrt{\pi} x^n}{2^{n+1}} \sum_{s=0}^{\infty} \frac{x^{2s}}{4^s s! \Gamma(n+s+3/2)}.$$

Here $I_{n+1/2}(x) = \sum_s (x/2)^{n+2s+1/2}/[s!\,\Gamma(n+s+\frac{3}{2})]$ and $\sqrt{\pi/(2x)}\,(x/2)^{n+1/2} = \sqrt{\pi} x^n/2^{n+1}$. The series after x^n contains only even powers and is entire. Hence $i_n(-x) = (-1)^n i_n(x)$. This argument also handles the square-root branches: their product has the displayed integer power series, which supplies the continuation through the origin.

(b) Use $K_\nu(x) = (\pi/2) i^{\nu+1} H_\nu^{(1)}(ix)$ and $H^{(1)}_{n+1/2}(ix) = \sqrt{2ix/\pi}\, h_n^{(1)}(ix)$, with $(ix)^{1/2} = x^{1/2} e^{i\pi/4}$. For $x > 0$ the book's normalization gives

$$k_n(x) = \sqrt{\frac{2}{\pi x}}\, \frac{\pi}{2}\, i^{n+3/2} \sqrt{\frac{2x}{\pi}}\, e^{i\pi/4} h_n^{(1)}(ix) = i^{n+3/2} e^{i\pi/4} h_n^{(1)}(ix) = -i^n h_n^{(1)}(ix),$$

since $i^{3/2} e^{i\pi/4} = e^{3i\pi/4} e^{i\pi/4} = -1$. In the finite sum of Exercise 14.7.1 at argument ix (valid there: both sides are analytic in the plane cut along the negative real axis and agree for positive argument, as noted in Exercise 14.7.3), $e^{i(ix)} = e^{-x}$, $i^s/(2ix)^s = 1/(2x)^s$, and the constant factors combine to $-i^n(-i)^{n+1}/i = 1$. Hence the finite half-integer K formula, with the book's normalization, is

$$k_n(x) = \frac{e^{-x}}{x} \sum_{s=0}^{n} \frac{(n+s)!}{s!(2x)^s (n-s)!}.$$

Its continuation to $-x$ is

$$k_n(-x) = -\frac{e^{x}}{x} \sum_{s=0}^{n} \frac{(-1)^s (n+s)!}{s!(2x)^s (n-s)!}.$$

For positive $x \to \infty$ both finite sums tend to 1, so $k_n(-x)/k_n(x) \sim -e^{2x}$. This ratio is neither identically $+1$ nor identically -1. Therefore k_n is neither even nor odd for any nonnegative integer n. Already $k_0(x) = e^{-x}/x$ exhibits the failure.

14.7.17 Show that the Wronskian of the spherical modified Bessel functions is given by

$$i_n(x) k_n'(x) - i_n'(x) k_n(x) = -\frac{1}{x^2}.$$

Solution.

Keep the two normalization factors distinct:

$$i_n = a(x) I_\nu(x), \quad k_n = b(x) K_\nu(x), \quad \nu = n + \tfrac{1}{2}, \quad a = \sqrt{\frac{\pi}{2x}}, \quad b = \sqrt{\frac{2}{\pi x}}.$$

Differentiate each product and subtract:

$$\begin{aligned} i_n k_n' - i_n' k_n &= aI_\nu(b'K_\nu + bK_\nu') - (a'I_\nu + aI_\nu')bK_\nu \\ &= ab(I_\nu K_\nu' - I_\nu' K_\nu) + (ab' - a'b)I_\nu K_\nu. \end{aligned}$$

Both a and b are constants times $x^{-1/2}$, so $a'/a = b'/b = -1/(2x)$ and the last coefficient, $ab' - a'b = ab(b'/b - a'/a)$, is zero; also $ab = \sqrt{\pi/(2x)}\sqrt{2/(\pi x)} = 1/x$. Exercise 14.5.11 supplies $W[I_\nu, K_\nu] = -1/x$. Consequently

$$\boxed{W[i_n, k_n] = \frac{1}{x}\left(-\frac{1}{x}\right) = -\frac{1}{x^2}.}$$

Using $\sqrt{\pi/(2x)}$ for both functions would introduce an extra factor $\pi/2$; the result here uses precisely the book's Eq. (14.194) definition of k_n.

15 Chapter 15. Legendre Functions

Problems

15.1 Legendre Polynomials

15.1.1 Derive the Legendre ODE by manipulation of the Legendre polynomial recurrence relations. Suggested starting point: Eqs. (15.24) and (15.25).

Solution.

The two needed recurrences, Eqs. (15.24) and (15.25), are

$$P'_{n-1} = -nP_n + xP'_n, \qquad (1-x^2)P'_n = nP_{n-1} - nxP_n.$$

Differentiate the second, including both product derivatives:

$$(1-x^2)P''_n - 2xP'_n = nP'_{n-1} - nP_n - nxP'_n.$$

Insert n times the first recurrence for nP'_{n-1} on the right; the terms $\pm nxP'_n$ cancel:

$$n(-nP_n + xP'_n) - nP_n - nxP'_n = -n^2P_n - nP_n = -n(n+1)P_n.$$

Moving this term to the left gives

$$\boxed{(1-x^2)P''_n - 2xP'_n + n(n+1)P_n = 0}.$$

For $n = 0$, $P_0 = 1$ verifies the equation directly, so no undefined P_{-1} is needed at that endpoint.

15.1.2 Derive the following closed formula for the Legendre polynomials $P_n(x)$,

$$P_n(x) = \sum_{k=0}^{[n/2]} (-1)^k \frac{(2n-2k)!}{2^n k!(n-k)!(n-2k)!}\, x^{n-2k},$$

where $[n/2]$ stands for the integer part of $n/2$.

Hint. Further expand Eq. (15.12) and rearrange the resulting double sum.

Solution.

Start in a sufficiently small disk about $t = 0$, where the binomial series converges:

$$(1-2xt+t^2)^{-1/2} = \sum_{j=0}^{\infty} \binom{-1/2}{j} (-2xt+t^2)^j.$$

Expand each power by the binomial theorem, $(t^2-2xt)^j = \sum_{k=0}^{j} \binom{j}{k} (t^2)^k (-2xt)^{j-k}$: choosing k copies of t^2 from the j factors, the other $j-k$ copies contribute $(-2xt)^{j-k}$, so the resulting term is

$$\binom{-1/2}{j}\binom{j}{k}(-2x)^{j-k}t^{j+k}.$$

For $2|xt| + |t|^2 < 1$ this double series converges absolutely (with every term replaced by its absolute value it sums to $(1 - 2|xt| - |t|^2)^{-1/2}$), so its terms may be collected by powers of t. To obtain t^n, set $j = n - k$. The condition $0 \le k \le j$ becomes $2k \le n$, i.e., $0 \le k \le \lfloor n/2 \rfloor$. Therefore

$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{-1/2}{n-k} \binom{n-k}{k} (-2x)^{n-2k}.$$

Use

$$\binom{-1/2}{j} = \frac{(-\frac{1}{2})(-\frac{3}{2}) \cdots (-\frac{2j-1}{2})}{j!} = (-1)^j \frac{(2j-1)!!}{2^j j!} = (-1)^j \frac{(2j)!}{2^{2j}(j!)^2}$$

(since $(2j-1)!! = (2j)!/(2^j j!)$) with $j = n - k$, together with $\binom{n-k}{k} = (n-k)!/[k!(n-2k)!]$. The sign becomes $(-1)^{n-k+n-2k} = (-1)^k$ and the power of two is $2^{n-2k-2(n-k)} = 2^{-n}$. The remaining factorials combine as

$$\frac{(2n-2k)!}{[(n-k)!]^2} \frac{(n-k)!}{k!(n-2k)!} = \frac{(2n-2k)!}{k!(n-k)!(n-2k)!},$$

so that

$$\boxed{P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{2^n k!(n-k)!(n-2k)!} x^{n-2k}}.$$

Although the generating series was used locally in t, the coefficient is a polynomial identity and holds for every finite complex x.

15.1.3 By differentiation and direct substitution of the series form given in Exercise 15.1.2, show that $P_n(x)$ satisfies the Legendre ODE. Note that there is no restriction on x. We may have any x, $-\infty < x < \infty$, and indeed any z in the entire finite complex plane.

Solution.

Write $P_n = \sum_{k=0}^{\lfloor n/2 \rfloor} A_k x^{n-2k}$, with $A_k = (-1)^k (2n-2k)!/[2^n k!(n-k)!(n-2k)!]$ as in Exercise 15.1.2. With $D = d/dx$, the Legendre operator $L_n = (1-x^2)D^2 - 2xD + n(n+1)$ acts on a monomial as

$$L_n[x^j] = (1-x^2) j(j-1) x^{j-2} - 2jx^j + n(n+1)x^j = j(j-1)x^{j-2} + [n(n+1) - j(j+1)]x^j,$$

since $-j(j-1) - 2j = -j(j+1)$. For $1 \le k \le \lfloor n/2 \rfloor$, the power x^{n-2k} in $L_n[P_n]$ receives the second term of $L_n[A_k x^{n-2k}]$ and the first term of $L_n[A_{k-1} x^{n-2k+2}]$ (with $j = n - 2k + 2$). Its total coefficient is

$$(n-2k+2)(n-2k+1)A_{k-1} + 2k(2n-2k+1)A_k,$$

because $n(n+1) - (n-2k)(n-2k+1) = 4kn - 4k^2 + 2k = 2k(2n-2k+1)$. The ratio of consecutive coefficients is

$$\begin{aligned} \frac{A_k}{A_{k-1}} &= -\frac{(2n-2k)!}{(2n-2k+2)!} \frac{(k-1)!}{k!} \frac{(n-k+1)!}{(n-k)!} \frac{(n-2k+2)!}{(n-2k)!} \\ &= -\frac{(n-k+1)(n-2k+2)(n-2k+1)}{k(2n-2k+2)(2n-2k+1)} = -\frac{(n-2k+2)(n-2k+1)}{2k(2n-2k+1)}, \end{aligned}$$

where the last step uses $2n - 2k + 2 = 2(n-k+1)$. Thus every displayed coefficient is zero. The leading coefficient ($k = 0$) is multiplied by $n(n+1) - n(n+1) = 0$. At the bottom, the last monomial is either 1 or x, whose second derivative is zero, so no lower power remains. Hence $L_n[P_n] = 0$ identically. All manipulations concern finite polynomials, which proves the claim on the entire finite complex plane.

15.1.4 The shifted Legendre polynomials, designated by the symbol $P_n^*(x)$ (where the asterisk does not mean complex conjugate) are orthogonal with unit weight on $[0,1]$, with normalization integral $\langle P_n^*|P_n^*\rangle = 1/(2n+1)$. The P_n^* through $n=6$ are shown in Table 15.2.

Table 15.2 Shifted Legendre Polynomials

$P_0^*(x) = 1$
$P_1^*(x) = 2x - 1$
$P_2^*(x) = 6x^2 - 6x + 1$
$P_3^*(x) = 20x^3 - 30x^2 + 12x - 1$
$P_4^*(x) = 70x^4 - 140x^3 + 90x^2 - 20x + 1$
$P_5^*(x) = 252x^5 - 630x^4 + 560x^3 - 210x^2 + 30x - 1$
$P_6^*(x) = 924x^6 - 2772x^5 + 3150x^4 - 1680x^3 + 420x^2 - 42x + 1$

(a) Find the recurrence relation satisfied by the P_n^*.

(b) Show that all the coefficients of the P_n^* are integers.

Hint. Look at the closed formula in Exercise 15.1.2.

Solution.

The listed normalization and signs are obtained by $P_n^*(x) = P_n(2x-1)$. Indeed, $y = 2x-1$ (so $dx = dy/2$, and $x = 0, 1$ become $y = -1, 1$) gives

$$\int_0^1 P_n^* P_m^* dx = \frac{1}{2}\int_{-1}^1 P_n(y)P_m(y)dy = \frac{1}{2}\,\frac{2\delta_{nm}}{2n+1} = \frac{\delta_{nm}}{2n+1}.$$

The Instructor's Manual's last expression on p. 232 retains an extra factor 1/2. The change of variable contributes 1/2, but the unshifted norm contributes $2/(2n+1)$, leaving $1/(2n+1)$ as stated correctly in the textbook. (a) Replace the argument in $(n+1)P_{n+1}(y) = (2n+1)yP_n(y) - nP_{n-1}(y)$ by $y = 2x-1$:

$$\boxed{(n+1)P_{n+1}^*(x) = (2n+1)(2x-1)P_n^*(x) - nP_{n-1}^*(x)}.$$

The initial polynomials are $P_0^* = 1$, $P_1^* = 2x-1$. For example, $n=1$ gives $2P_2^* = 3(2x-1)^2 - 1 = 12x^2 - 12x + 2$, i.e., $P_2^* = 6x^2 - 6x + 1$, as in the table.

(b) The recurrence alone involves division by $n+1$, so an explicit coefficient formula is more useful for proving integrality. Transform Rodrigues's formula $P_n(y) = D_y^n(y^2-1)^n/(2^n n!)$, where $D_y = d/dy$, using $D_y = \frac{1}{2}D_x$ (with $D_x = d/dx$, since $dy = 2\,dx$) and $y^2 - 1 = 4x^2 - 4x = 4x(x-1)$:

$$P_n^*(x) = \frac{1}{2^n n!}\,\frac{1}{2^n}D_x^n[4^n x^n(x-1)^n] = \frac{1}{n!}D_x^n[x^n(x-1)^n].$$

Expand $(x-1)^n = \sum_{k=0}^n (-1)^{n-k}\binom{n}{k}x^k$, so that $x^n(x-1)^n = \sum_{k=0}^n (-1)^{n-k}\binom{n}{k}x^{n+k}$, and differentiate with $D_x^n x^{n+k} = \frac{(n+k)!}{k!}x^k$:

$$P_n^*(x) = \sum_{k=0}^n (-1)^{n-k}\binom{n}{k}\frac{(n+k)!}{n!k!}x^k = \boxed{\sum_{k=0}^n (-1)^{n-k}\binom{n}{k}\binom{n+k}{k}x^k}.$$

Every coefficient is a signed product of integers.

15.1.5 Given the series

$$\alpha_0 + \alpha_2 \cos^2\theta + \alpha_4 \cos^4\theta + \alpha_6 \cos^6\theta = a_0P_0 + a_2P_2 + a_4P_4 + a_6P_6,$$

where the arguments of the P_n are $\cos\theta$, express the coefficients α_i as a column vector $\boldsymbol{\alpha}$ and the coefficients a_i as a column vector $\mathbf{a}$ and determine the matrices A and B such that

$$\mathsf{A}\boldsymbol{\alpha} = \mathbf{a} \quad \text{and} \quad \mathsf{B}\mathbf{a} = \boldsymbol{\alpha}.$$

Check your computation by showing that $\mathsf{AB} = \mathbf{1}$ (unit matrix). Repeat for the odd case

$$\alpha_1 \cos\theta + \alpha_3 \cos^3\theta + \alpha_5 \cos^5\theta + \alpha_7 \cos^7\theta = a_1P_1 + a_3P_3 + a_5P_5 + a_7P_7.$$

Note. $P_n(\cos\theta)$ and $\cos^n\theta$ are tabulated in terms of each other in AMS-55 (see Additional Readings for the complete reference).

Solution.

Put $x = \cos\theta$ and order the even columns as $\boldsymbol{\alpha} = (\alpha_0, \alpha_2, \alpha_4, \alpha_6)^T$, $\mathbf{a} = (a_0, a_2, a_4, a_6)^T$. The coefficients of P_0, P_2, P_4, P_6 in ascending powers of x give the columns of B_e:

$$\boxed{\mathsf{B}_e = \begin{pmatrix} 1 & -1/2 & 3/8 & -5/16 \\ 0 & 3/2 & -15/4 & 105/16 \\ 0 & 0 & 35/8 & -315/16 \\ 0 & 0 & 0 & 231/16 \end{pmatrix}}.$$

The rows of $\boldsymbol{\alpha} = \mathsf{B}_e\mathbf{a}$ read

$$\begin{aligned} &\alpha_6 = \tfrac{231}{16}a_6, \qquad \alpha_4 = \tfrac{35}{8}a_4 - \tfrac{315}{16}a_6, \\ &\alpha_2 = \tfrac{3}{2}a_2 - \tfrac{15}{4}a_4 + \tfrac{105}{16}a_6, \qquad \alpha_0 = a_0 - \tfrac{1}{2}a_2 + \tfrac{3}{8}a_4 - \tfrac{5}{16}a_6. \end{aligned}$$

Solve them from the last one upward, inserting each result in the next:

$$\begin{aligned} a_6 &= \tfrac{16}{231}\alpha_6, \\ a_4 &= \tfrac{8}{35}\big(\alpha_4 + \tfrac{315}{16}a_6\big) = \tfrac{8}{35}\alpha_4 + \tfrac{24}{77}\alpha_6, \\ a_2 &= \tfrac{2}{3}\big(\alpha_2 + \tfrac{15}{4}a_4 - \tfrac{105}{16}a_6\big) = \tfrac{2}{3}\big(\alpha_2 + \tfrac{6}{7}\alpha_4 + \tfrac{5}{7}\alpha_6\big) = \tfrac{2}{3}\alpha_2 + \tfrac{4}{7}\alpha_4 + \tfrac{10}{21}\alpha_6, \\ a_0 &= \alpha_0 + \tfrac{1}{2}a_2 - \tfrac{3}{8}a_4 + \tfrac{5}{16}a_6 = \alpha_0 + \tfrac{1}{3}\alpha_2 + \tfrac{1}{5}\alpha_4 + \tfrac{1}{7}\alpha_6. \end{aligned}$$

Inside the parentheses of a_2, the coefficient of α_6 is $\frac{15}{4}\cdot\frac{24}{77} - \frac{105}{16}\cdot\frac{16}{231} = \frac{90}{77} - \frac{35}{77} = \frac{5}{7}$; in a_0, those of α_4 and α_6 are $\frac{2}{7} - \frac{3}{35} = \frac{1}{5}$ and $\frac{5}{21} - \frac{9}{77} + \frac{5}{231} = \frac{33}{231} = \frac{1}{7}$. Thus

$$\boxed{\mathsf{A}_e = \begin{pmatrix} 1 & 1/3 & 1/5 & 1/7 \\ 0 & 2/3 & 4/7 & 10/21 \\ 0 & 0 & 8/35 & 24/77 \\ 0 & 0 & 0 & 16/231 \end{pmatrix}}.$$

Read by columns, A_e gives the Legendre expansions $x^2 = \frac{1}{3}P_0 + \frac{2}{3}P_2$, $x^4 = \frac{1}{5}P_0 + \frac{4}{7}P_2 + \frac{8}{35}P_4$, and $x^6 = \frac{1}{7}P_0 + \frac{10}{21}P_2 + \frac{24}{77}P_4 + \frac{16}{231}P_6$. The diagonal entries of $\mathsf{A}_e\mathsf{B}_e$ are products of diagonal entries, $1\cdot 1$, $\frac{2}{3}\cdot\frac{3}{2}$, $\frac{8}{35}\cdot\frac{35}{8}$, $\frac{16}{231}\cdot\frac{231}{16}$, all equal to one. The six upper off-diagonal entries (row i of A_e times column j of B_e, $i<j$, in the order $(1,2),(1,3),(1,4)$ and $(2,3),(2,4),(3,4)$) are, explicitly,

$$-\tfrac{1}{2} + \tfrac{1}{2}, \quad \tfrac{3}{8} - \tfrac{5}{4} + \tfrac{7}{8}, \quad -\tfrac{5}{16} + \tfrac{35}{16} - \tfrac{63}{16} + \tfrac{33}{16},$$

$$-\tfrac{5}{2}+\tfrac{5}{2},\quad \tfrac{35}{8}-\tfrac{45}{4}+\tfrac{55}{8},\quad -\tfrac{9}{2}+\tfrac{9}{2}.$$

All vanish, and entries below the diagonal vanish by triangularity. Hence $\mathsf{A}_e\mathsf{B}_e=\mathsf{1}$.

For the odd case use the order $1,3,5,7$. The coefficients of P_1,P_3,P_5,P_7 give the columns of B_o below, and the same back substitution, from $\alpha_7=\frac{429}{16}a_7$, $\alpha_5=\frac{63}{8}a_5-\frac{693}{16}a_7$, $\alpha_3=\frac{5}{2}a_3-\frac{35}{4}a_5+\frac{315}{16}a_7$, $\alpha_1=a_1-\frac{3}{2}a_3+\frac{15}{8}a_5-\frac{35}{16}a_7$, gives

$$\begin{aligned}
a_7&=\tfrac{16}{429}\alpha_7, \qquad a_5=\tfrac{8}{63}\big(\alpha_5+\tfrac{693}{16}a_7\big)=\tfrac{8}{63}\alpha_5+\tfrac{8}{39}\alpha_7,\\
a_3&=\tfrac{2}{5}\big(\alpha_3+\tfrac{35}{4}a_5-\tfrac{315}{16}a_7\big)=\tfrac{2}{5}\big(\alpha_3+\tfrac{10}{9}\alpha_5+\tfrac{35}{33}\alpha_7\big)=\tfrac{2}{5}\alpha_3+\tfrac{4}{9}\alpha_5+\tfrac{14}{33}\alpha_7,\\
a_1&=\alpha_1+\tfrac{3}{2}a_3-\tfrac{15}{8}a_5+\tfrac{35}{16}a_7=\alpha_1+\tfrac{3}{5}\alpha_3+\tfrac{3}{7}\alpha_5+\tfrac{1}{3}\alpha_7.
\end{aligned}$$

Inside the parentheses of a_3, the coefficient of α_7 is $\frac{35}{4}\cdot\frac{8}{39}-\frac{315}{16}\cdot\frac{16}{429}=\frac{770-315}{429}=\frac{35}{33}$, and in a_1 the coefficients of α_5 and α_7 are $\frac{2}{3}-\frac{5}{21}=\frac{3}{7}$ and $\frac{7}{11}-\frac{5}{13}+\frac{35}{429}=\frac{143}{429}=\frac{1}{3}$. Therefore

$$\boxed{\mathsf{B}_o=\begin{pmatrix}1&-3/2&15/8&-35/16\\0&5/2&-35/4&315/16\\0&0&63/8&-693/16\\0&0&0&429/16\end{pmatrix}},\qquad \boxed{\mathsf{A}_o=\begin{pmatrix}1&3/5&3/7&1/3\\0&2/5&4/9&14/33\\0&0&8/63&8/39\\0&0&0&16/429\end{pmatrix}}.$$

In the same order as for the even case, the six upper off-diagonal entries of $\mathsf{A}_o\mathsf{B}_o$ are

$$-\tfrac{3}{2}+\tfrac{3}{2},\quad \tfrac{15}{8}-\tfrac{21}{4}+\tfrac{27}{8},\quad -\tfrac{35}{16}+\tfrac{189}{16}-\tfrac{297}{16}+\tfrac{143}{16},$$
$$-\tfrac{7}{2}+\tfrac{7}{2},\quad \tfrac{63}{8}-\tfrac{77}{4}+\tfrac{91}{8},\quad -\tfrac{11}{2}+\tfrac{11}{2},$$

again zero, with diagonal entries $1\cdot1$, $\frac{2}{5}\cdot\frac{5}{2}$, $\frac{8}{63}\cdot\frac{63}{8}$, $\frac{16}{429}\cdot\frac{429}{16}$, all one. Therefore $\mathsf{A}_o\mathsf{B}_o=\mathsf{1}$. In both cases $\mathbf{a}=\mathsf{A}\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}=\mathsf{B}\mathbf{a}$.

15.1.6 By differentiating the generating function $g(x,t)$ with respect to t, multiplying by $2t$, and then adding $g(x,t)$, show that inside the generating function's Taylor disk (in particular, for $-1\le x\le 1$ and $|t|<1$)

$$\frac{1-t^2}{(1-2tx+t^2)^{3/2}}=\sum_{n=0}^{\infty}(2n+1)P_n(x)t^n.$$

This result is useful in calculating the charge induced on a grounded metal sphere by a nearby point charge.

Solution.

Write $Q=1-2xt+t^2$ and $g=Q^{-1/2}$. Then

$$g_t=-\frac{1}{2}Q^{-3/2}(-2x+2t)=(x-t)Q^{-3/2}.$$

Consequently, since $2t(x-t)+Q=2xt-2t^2+1-2xt+t^2=1-t^2$,

$$2tg_t+g=\frac{2t(x-t)+Q}{Q^{3/2}}=\frac{1-t^2}{(1-2xt+t^2)^{3/2}}.$$

Termwise differentiation of the generating series $g=\sum_{n\ge0}P_n(x)t^n$ in its convergence disk gives $2tg_t=\sum_{n\ge0}2nP_n(x)t^n$, hence $2tg_t+g=\sum_{n\ge0}(2n+1)P_n(x)t^n$. Equating the two expressions proves the identity. In particular, for real $-1\le x\le 1$ the zeros $t=x\pm i\sqrt{1-x^2}$ of Q have $|t|^2=x^2+(1-x^2)=1$, so g is analytic in $|t|<1$, its Taylor series converges there, and the identity holds for $|t|<1$.

15.1.7

(a) Derive Eq. (15.26),

$$(1-x^2)P_n'(x) = (n+1)xP_n(x) - (n+1)P_{n+1}(x).$$

(b) Write out the relation of Eq. (15.26) to preceding equations in symbolic form analogous to the symbolic forms for Eqs. (15.22) to (15.25).

Solution.

(a) Begin with Eq. (15.25), $(1-x^2)P_n' = nP_{n-1} - nxP_n$, and eliminate nP_{n-1} using the three-term recurrence $(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}$:

$$nP_{n-1} = (2n+1)xP_n - (n+1)P_{n+1}.$$

It follows that

$$(1-x^2)P_n' = [(2n+1)-n]xP_n - (n+1)P_{n+1} = \boxed{(n+1)(xP_n - P_{n+1})}.$$

This elimination uses $n \geq 1$. For $n = 0$, $P_0 = 1$ and $P_1 = x$ give $0 = x - x$ directly, so no P_{-1} is required. The Instructor's Manual drops the factor x from $(n+1)xP_n$ in its last line for part (a), p. 233. The textbook equation and the derivation above retain it. (b) To express the result in the same symbolic style as the textbook, use $(15.24)_{n\mapsto n+1}$ and $x(15.23)$:

$$P_n' = -(n+1)P_{n+1} + xP_{n+1}', \qquad xP_{n+1}' = (n+1)xP_n + x^2P_n'.$$

Substitute the second equation into the first,

$$P_n' = -(n+1)P_{n+1} + (n+1)xP_n + x^2P_n',$$

and move x^2P_n' to the left; this is Eq. (15.26). Thus the indicated combination is

$$\boxed{(15.24)_{n\mapsto n+1} + x(15.23) \implies (15.26)},$$

where the notation means eliminating their common xP_{n+1}' term.

15.1.8 Prove that

$$P_n'(1) = \frac{d}{dx}P_n(x)\Big|_{x=1} = \frac{1}{2}n(n+1).$$

Solution.

Since P_n is a polynomial, substitute $x = 1$ directly into its ODE:

$$(1-1)P_n''(1) - 2P_n'(1) + n(n+1)P_n(1) = 0.$$

The first term vanishes, so $2P_n'(1) = n(n+1)P_n(1)$. The generating function at $x = 1$ is $(1-2t+t^2)^{-1/2} = (1-t)^{-1} = \sum_{n\geq 0} t^n$ for $|t| < 1$, so $P_n(1) = 1$. Therefore

$$\boxed{P_n'(1) = \frac{n(n+1)}{2}}.$$

There is no singular limiting derivative here: the factor $1-x^2$ multiplies a finite polynomial derivative. The formula also gives $P_0'(1) = 0$.

15.1.9 Show that $P_n(\cos\theta) = (-1)^n P_n(-\cos\theta)$ by use of the recurrence relation relating P_n, P_{n+1}, and P_{n-1} and your knowledge of P_0 and P_1.

Solution.

The base cases are $P_0(-x) = 1 = P_0(x)$ and $P_1(-x) = -x = -P_1(x)$. Assume $P_n(-x) = (-1)^n P_n(x)$ and $P_{n-1}(-x) = (-1)^{n-1} P_{n-1}(x)$. Evaluate the three-term recurrence $(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$ at $-x$ and insert the induction hypothesis; then use $-(-1)^n = (-1)^{n+1}$ and $(-1)^{n-1} = (-1)^{n+1}$:

$$\begin{aligned}(n+1)P_{n+1}(-x) &= (2n+1)(-x)P_n(-x) - nP_{n-1}(-x)\\ &= -(2n+1)x(-1)^n P_n(x) - n(-1)^{n-1}P_{n-1}(x)\\ &= (-1)^{n+1}[(2n+1)xP_n(x) - nP_{n-1}(x)]\\ &= (-1)^{n+1}(n+1)P_{n+1}(x).\end{aligned}$$

Divide by $n+1$ and apply induction. Thus $P_n(-x) = (-1)^n P_n(x)$ for every $n \geq 0$. Putting $x = \cos\theta$ gives the requested form.

15.1.10 From Eq. (15.27) write out the coefficient of t^2 in terms of $\cos n\theta$, $n \leq 2$. This coefficient is $P_2(\cos\theta)$.

Solution.

Since $1 - 2t\cos\theta + t^2 = (1 - te^{i\theta})(1 - te^{-i\theta})$, Eq. (15.27) factors the generating function as

$$(1 - te^{i\theta})^{-1/2}(1 - te^{-i\theta})^{-1/2}.$$

Each factor has the binomial series $(1-u)^{-1/2} = 1 + \frac{1}{2}u + \frac{3}{8}u^2 + \cdots$ with $u = te^{\pm i\theta}$. The coefficient of t^2 in the product therefore receives exactly three contributions: the u^2 term of either factor times the constant 1 of the other, and the product of the two linear terms:

$$\frac{3}{8}e^{2i\theta} + \left(\frac{1}{2}e^{i\theta}\right)\left(\frac{1}{2}e^{-i\theta}\right) + \frac{3}{8}e^{-2i\theta}.$$

Since $e^{2i\theta} + e^{-2i\theta} = 2\cos 2\theta$ and $\cos 2\theta = 2\cos^2\theta - 1$,

$$\boxed{P_2(\cos\theta) = \frac{1}{4} + \frac{3}{4}\cos 2\theta = \frac{1}{2}(3\cos^2\theta - 1)}.$$

The only angular frequencies are 0 and 2.

15.1.11 Derive, for integers $n \geq 1$, the recurrence relation

$$(1 - x^2)P_n'(x) = nP_{n-1}(x) - nxP_n(x)$$

from the Legendre polynomial generating function.

Solution.

Let $g = (1 - 2xt + t^2)^{-1/2} = \sum P_n(x)t^n$ and $Q = 1 - 2xt + t^2$. Termwise differentiation gives $tg_t = \sum_{n\geq 0} nP_n t^n$. Put $j = n - 1$ in the first part of the proposed right-hand side and use $\sum_{j\geq 0}(j+1)P_j t^j = tg_t + g$:

$$\begin{aligned}\sum_{n\geq 1}(nP_{n-1} - nxP_n)t^n &= t\sum_{j\geq 0}(j+1)P_j t^j - xtg_t\\ &= t(tg_t + g) - xtg_t = tg + t(t-x)g_t.\end{aligned}$$

On the other hand, differentiating $Q^{-1/2}$ gives $g_x = -\frac{1}{2}Q^{-3/2}(-2t) = tQ^{-3/2}$ and $g_t = -\frac{1}{2}Q^{-3/2}(2t-2x) = (x-t)Q^{-3/2}$. With $Q-(x-t)^2 = 1-2xt+t^2-x^2+2xt-t^2 = 1-x^2$,

$$tg + t(t-x)g_t = \frac{t[Q-(x-t)^2]}{Q^{3/2}} = \frac{t(1-x^2)}{Q^{3/2}} = (1-x^2)g_x.$$

Since $(1-x^2)g_x = \sum_{n\ge 0}(1-x^2)P_n'(x)t^n$, compare coefficients of t^n:

$$\boxed{(1-x^2)P_n' = nP_{n-1} - nxP_n}, \qquad n \ge 1.$$

The $n=0$ coefficient is the separate identity $P_0' = 0$.

15.1.12 Evaluate $\displaystyle\int_0^1 P_n(x)\,dx$.

ANS. $n = 2s$, 1 for $s = 0$, 0 for $s > 0$;
$n = 2s+1$, $P_{2s}(0)/(2s+2) = (-1)^s(2s-1)!!/(2s+2)!!$.

Hint. Use a recurrence relation to replace $P_n(x)$ by derivatives and then integrate by inspection. Alternatively, you can integrate the generating function.

Solution.

For $n \ge 1$, use $P_{n+1}' - P_{n-1}' = (2n+1)P_n$ and integrate:

$$I_n := \int_0^1 P_n(x)dx = \frac{[P_{n+1}-P_{n-1}]_0^1}{2n+1} = \frac{P_{n-1}(0)-P_{n+1}(0)}{2n+1},$$

because both endpoint values at one equal one. If $n = 2s > 0$, both polynomials in the numerator are odd and vanish at zero, so $I_{2s} = 0$. Separately, $I_0 = 1$. If $n = 2s+1$, the recurrence $(m+1)P_{m+1} = (2m+1)xP_m - mP_{m-1}$ at $x = 0$ with $m = 2s+1$ gives

$$P_{2s+2}(0) = -\frac{2s+1}{2s+2}P_{2s}(0).$$

Thus

$$I_{2s+1} = \frac{P_{2s}(0)-P_{2s+2}(0)}{4s+3} = \frac{P_{2s}(0)}{4s+3}\left(1+\frac{2s+1}{2s+2}\right) = \frac{P_{2s}(0)}{2s+2}.$$

Starting with $P_0(0) = 1$ and repeating the same recurrence yields

$$P_{2s}(0) = \prod_{j=1}^{s}\left(-\frac{2j-1}{2j}\right) = (-1)^s\frac{1\cdot 3\cdots(2s-1)}{2\cdot 4\cdots(2s)} = (-1)^s\frac{(2s-1)!!}{(2s)!!},$$

whence, with $(2s)!!\,(2s+2) = (2s+2)!!$,

$$\boxed{I_{2s+1} = (-1)^s\frac{(2s-1)!!}{(2s+2)!!}}.$$

Use $(-1)!! = 0!! = 1$; at $s = 0$ the formula gives $I_1 = 1/2$. The denominator shown in dark red in the answer applies the published correction for textbook p. 723 in the Instructor's Manual errata dated 6 April 2012.

15.1.13 Show that each term in the summation

$$\sum_{r=[n/2]+1}^{n} \left(\frac{d}{dx}\right)^n \frac{(-1)^r n!}{r!\,(n-r)!}\, x^{2n-2r}$$

vanishes (r and n integral). Here $[n/2]$ is the largest integer $\le n/2$.

Solution.

For every integer $r \ge \lfloor n/2 \rfloor + 1$, one has $2r > n$, because $\lfloor n/2 \rfloor > n/2 - 1$; so the exponent $d = 2n - 2r$ satisfies $0 \le d < n$ when $r \le n$. The derivative of a monomial of degree d is zero after $d+1$ derivatives, and hence, with $D = d/dx$,

$$D^n x^{2n-2r} = 0.$$

The scalar factorial coefficient cannot change this result. Therefore each individual summand vanishes, not only their sum. For $n = 0$ the indicated index set is empty, with sum zero.

15.1.14 Show that $\displaystyle\int_{-1}^{1} x^m P_n(x)\,dx = 0$ when m, n are integers with $0 \le m < n$.

Hint. Use Rodrigues formula or expand x^m in Legendre polynomials.

Solution.

Take nonnegative integers $m < n$. Rodrigues's formula, $P_n = D^n(x^2-1)^n/(2^n n!)$ with $D = d/dx$, gives

$$I = \frac{1}{2^n n!}\int_{-1}^{1} x^m D^n (x^2-1)^n\,dx.$$

The polynomial $F = (x^2-1)^n = (x-1)^n(x+1)^n$ and all its derivatives of orders $0, \dots, n-1$ vanish at both endpoints. Each integration by parts has the form

$$\int_{-1}^{1} u\,D^k F\,dx = \Big[u\,D^{k-1}F\Big]_{-1}^{1} - \int_{-1}^{1} u'\,D^{k-1}F\,dx, \qquad 1 \le k \le n,$$

with u a derivative of x^m, so every boundary term from the n integrations by parts is zero. Moving all derivatives onto x^m in this way leaves

$$I = \frac{(-1)^n}{2^n n!}\int_{-1}^{1} (D^n x^m)(x^2-1)^n\,dx = 0,$$

because $D^n x^m = 0$. This proves the required orthogonality.

15.1.15 Show that

$$\int_{-1}^{1} x^n P_n(x)\,dx = \frac{2n!}{(2n+1)!!}.$$

Note. You are expected to use the Rodrigues formula and integrate by parts, but also see if you can get the result from Eq. (15.14) by inspection.

Solution.

Here $2n!$ means $2 \cdot n!$, not $(2n)!$: for example, at $n = 2$ the integral is $\int_{-1}^{1} x^2 \cdot \frac{1}{2}(3x^2-1)\,dx = \frac{3}{5} - \frac{1}{3} = \frac{4}{15} = 2 \cdot 2!/5!!$, whereas $(2n)!/(2n+1)!!$ would give $\frac{24}{15}$. Move the n Rodrigues derivatives onto x^n as in Exercise 15.1.14, using $D^n x^n = n!$ and $(-1)^n(x^2-1)^n = (1-x^2)^n$:

$$I_n = \frac{1}{2^n n!}\int_{-1}^{1} x^n D^n (x^2-1)^n\,dx = \frac{(-1)^n}{2^n n!}\int_{-1}^{1} n!(x^2-1)^n\,dx = \frac{1}{2^n}\int_{-1}^{1}(1-x^2)^n\,dx.$$

The integrand is even. Put $u = x^2$ on the positive half, so $x = \sqrt{u}$ and $dx = du/(2\sqrt{u})$:

$$\int_{-1}^{1}(1-x^2)^n dx = 2\int_{0}^{1}(1-x^2)^n dx = \int_0^1 u^{-1/2}(1-u)^n\,du = B(\tfrac{1}{2}, n+1) = \frac{\Gamma(1/2)n!}{\Gamma(n+3/2)}.$$

Using $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(n+3/2) = (2n+1)!!\sqrt{\pi}/2^{n+1}$, and $(2n+1)! = (2n+1)!!\,2^n n!$,

$$\int_{-1}^{1}(1-x^2)^n dx = \frac{2^{n+1}n!}{(2n+1)!!} = \frac{2^{2n+1}(n!)^2}{(2n+1)!}.$$

Dividing by 2^n yields

$$\boxed{I_n = \frac{2n!}{(2n+1)!!} = \frac{2^{n+1}(n!)^2}{(2n+1)!}}.$$

For the alternative inspection argument, the leading coefficient of P_n (the $k=0$ term of Exercise 15.1.2) is $(2n)!/[2^n(n!)^2] = (2n-1)!!/n!$, so $x^n = n!P_n/(2n-1)!! +$ a polynomial of degree below n. The lower-degree polynomial is orthogonal to P_n (Exercise 15.1.14), and $\int P_n^2 dx = 2/(2n+1)$ gives the same result:

$$\int_{-1}^{1} x^n P_n\,dx = \frac{n!}{(2n-1)!!}\,\frac{2}{2n+1} = \frac{2n!}{(2n+1)!!}.$$

15.1.16 Show that

$$\int_{-1}^{1} x^{2r} P_{2n}(x)\,dx = \frac{2^{2n+1}(2r)!\,(r+n)!}{(2r+2n+1)!\,(r-n)!}, \qquad r, n \in \mathbb{Z}, \quad r \geq n \geq 0.$$

Solution.

For integers $r \geq n \geq 0$, apply Rodrigues, $P_{2n} = D^{2n}(x^2-1)^{2n}/[2^{2n}(2n)!]$, and integrate by parts $2n$ times as in Exercise 15.1.14. The endpoint terms vanish because $(x^2-1)^{2n}$ has zeros of order $2n$ at both ends. With $D^{2n}x^{2r} = \frac{(2r)!}{(2r-2n)!}x^{2r-2n}$ and $(-1)^{2n}(x^2-1)^{2n} = (1-x^2)^{2n}$,

$$I = \frac{(2r)!}{2^{2n}(2n)!(2r-2n)!}\int_{-1}^{1} x^{2r-2n}(1-x^2)^{2n}dx.$$

The integrand is even, so the integral is twice that over $[0,1]$; with $u = x^2$, $dx = du/(2\sqrt{u})$, it becomes

$$\int_0^1 u^{r-n-1/2}(1-u)^{2n}du = B(r-n+\tfrac{1}{2}, 2n+1) = \frac{\Gamma(r-n+1/2)(2n)!}{\Gamma(r+n+3/2)}.$$

For a nonnegative integer j, $\Gamma(j+1/2) = (2j)!\sqrt{\pi}/(4^j j!)$; use it with $j = r-n$ and $j = r+n+1$. Substitution, after cancelling $(2n)!$, gives

$$I = \frac{(2r)!}{2^{2n}(2r-2n)!}\frac{(2r-2n)!\sqrt{\pi}}{4^{r-n}(r-n)!}\frac{4^{r+n+1}(r+n+1)!}{(2r+2n+2)!\sqrt{\pi}} = \frac{2^{2n+2}(2r)!(r+n+1)!}{(r-n)!(2r+2n+2)!}$$
$$= \boxed{\frac{2^{2n+1}(2r)!(r+n)!}{(2r+2n+1)!(r-n)!}}.$$

The powers of two combine as $4^{r+n+1}/(2^{2n}4^{r-n}) = 2^{2n+2}$. The last step uses $(2r+2n+2)! = 2(r+n+1)(2r+2n+1)!$ and $(r+n+1)! = (r+n+1)(r+n)!$, so the factor $2(r+n+1)$ cancels.

15.1.17 As a generalization of Exercises 15.1.15 and 15.1.16, show that, for nonnegative integers r, the Legendre expansions of x^s are

(a) $x^{2r} = \displaystyle\sum_{n=0}^{r} \frac{2^{2n}(4n+1)(2r)!\,(r+n)!}{(2r+2n+1)!\,(r-n)!}\,P_{2n}(x), \qquad s = 2r,$

(b) $x^{2r+1} = \displaystyle\sum_{n=0}^{r} \frac{2^{2n+1}(4n+3)(2r+1)!\,(r+n+1)!}{(2r+2n+3)!\,(r-n)!}\,P_{2n+1}(x), \qquad s = 2r+1.$

Solution.

For a polynomial of degree s, the expansion terminates at P_s, because the nonzero leading coefficients of $P_0, \ldots, P_s$ make them a basis for polynomials of degree at most s. Orthogonality gives

$$a_j = \frac{2j+1}{2}\int_{-1}^{1} x^s P_j(x)dx.$$

Parity eliminates all j of parity opposite to s.

(a) With $s = 2r$, $j = 2n$, use Exercise 15.1.16 and multiply by $(4n+1)/2$:

$$\boxed{x^{2r} = \sum_{n=0}^{r} \frac{2^{2n}(4n+1)(2r)!(r+n)!}{(2r+2n+1)!(r-n)!}P_{2n}(x)}.$$

(b) For clarity, evaluate the odd moment separately. Moving $2n+1$ Rodrigues derivatives onto x^{2r+1}, with $D^{2n+1}x^{2r+1} = \frac{(2r+1)!}{(2r-2n)!}x^{2r-2n}$, and substituting $u = x^2$ as in Exercise 15.1.16 gives

$$\int_{-1}^{1} x^{2r+1}P_{2n+1}(x)dx = \frac{(2r+1)!}{2^{2n+1}(2n+1)!(2r-2n)!}B(r-n+\tfrac{1}{2}, 2n+2).$$

The two signs, $(-1)^{2n+1}$ from integration by parts and $(x^2-1)^{2n+1} = -(1-x^2)^{2n+1}$, cancel. Now $B(r-n+\frac{1}{2}, 2n+2) = \Gamma(r-n+\frac{1}{2})(2n+1)!/\Gamma(r+n+\frac{5}{2})$. Evaluating both gammas by the same half-integer formula as before (with $j = r-n$ and $j = r+n+2$) and cancelling $(2n+1)!$ gives

$$\begin{aligned}\int_{-1}^{1} x^{2r+1}P_{2n+1}(x)dx &= \frac{(2r+1)!}{2^{2n+1}(2r-2n)!}\frac{(2r-2n)!\sqrt{\pi}}{4^{r-n}(r-n)!}\frac{4^{r+n+2}(r+n+2)!}{(2r+2n+4)!\sqrt{\pi}} \\ &= \frac{2^{2n+3}(2r+1)!(r+n+2)!}{(r-n)!(2r+2n+4)!} = \frac{2^{2n+2}(2r+1)!(r+n+1)!}{(2r+2n+3)!(r-n)!},\end{aligned}$$

where $4^{r+n+2}/(2^{2n+1}4^{r-n}) = 2^{2n+3}$ and the last step cancels $2(r+n+2)$ between $(2r+2n+4)!$ and $(r+n+2)!$. Multiplication by $(4n+3)/2$ therefore yields

$$\boxed{x^{2r+1} = \sum_{n=0}^{r} \frac{2^{2n+1}(4n+3)(2r+1)!(r+n+1)!}{(2r+2n+3)!(r-n)!}P_{2n+1}(x)}.$$

15.1.18 In numerical work (for e.g., the Gauss-Legendre quadrature), it is useful to establish that $P_n(x)$ has n real zeros in the interior of $[-1, 1]$. Show that this is so.

Hint. Rolle's theorem shows that the first derivative of $(x^2 - 1)^n$ has one zero in the interior of $[-1, 1]$. Extend this argument to the second, third, and ultimately the nth derivative.

Solution.

To connect the proposed Rolle argument with Rodrigues's formula, start with $(x^2 - 1)^n$, not the printed $(x^2 - 1)^{2n}$. The nth derivative of the latter is not proportional to P_n. Let $F(x) = (x^2 - 1)^n$, $n \geq 1$. Its endpoints are zeros of multiplicity n. We prove inductively that $F^{(k)}$ has at least k distinct zeros strictly between -1 and 1, for $0 \leq k \leq n$. The assertion for $k = 0$ is empty. If it holds for $k - 1 < n$, then $F^{(k-1)}$ still vanishes at both endpoints, because their multiplicities are $n - k + 1 \geq 1$. Its $k - 1$ interior zeros, together with the two endpoints, determine k disjoint open intervals. Rolle's theorem supplies a zero of $F^{(k)}$ in each interval. At $k = n$, Rodrigues gives $F^{(n)} = 2^n n! P_n$, so P_n has at least n distinct interior zeros. Its degree is exactly n, so these are all its zeros and each is simple. For $n = 0$ the statement means no zeros, consistent with $P_0 = 1$.

15.2 Orthogonality

15.2.1 Using a Rodrigues formula, show that the $P_n(x)$ are orthogonal and that

$$\int_{-1}^{1} [P_n(x)]^2 \, dx = \frac{2}{2n + 1}.$$

Hint. Integrate by parts.

Solution.

Suppose $m < n$. Rodrigues's formula $P_n = D^n(x^2 - 1)^n/(2^n n!)$, with $D = d/dx$, and n integrations by parts, as in Exercise 15.1.14, give

$$\int_{-1}^{1} P_m P_n \, dx = \frac{(-1)^n}{2^n n!} \int_{-1}^{1} P_m^{(n)}(x)(x^2 - 1)^n \, dx = 0.$$

Every boundary term vanishes because $(x^2 - 1)^n$ and its first $n - 1$ derivatives vanish at ± 1; the remaining derivative of P_m is zero, since P_m has degree $m < n$. Symmetry covers $m > n$.

For $m = n$, the leading coefficient $c_n = (2n)!/[2^n (n!)^2]$ of P_n (the $k = 0$ term of Exercise 15.1.2) gives $P_n^{(n)} = n! c_n$. The same integrations by parts yield

$$\int_{-1}^{1} P_n^2 dx = \frac{(-1)^n}{2^n n!} \int_{-1}^{1} n! c_n (x^2 - 1)^n \, dx = \frac{c_n}{2^n} \int_{-1}^{1} (1 - x^2)^n dx.$$

The beta integral in Exercise 15.1.15 is $\int_{-1}^{1} (1 - x^2)^n dx = 2^{2n+1}(n!)^2/(2n + 1)!$. Therefore

$$\int_{-1}^{1} P_n^2 dx = \frac{(2n)!}{2^{2n}(n!)^2} \frac{2^{2n+1}(n!)^2}{(2n + 1)!} = \frac{2\,(2n)!}{(2n + 1)!} = \boxed{\frac{2}{2n + 1}}.$$

Together these establish $\int P_m P_n = 2\delta_{mn}/(2n + 1)$.

15.2.2 You have constructed a set of orthogonal functions by the Gram-Schmidt process (Section 5.2), taking $u_n(x) = x^n$, $n = 0, 1, 2, \ldots$, in increasing order with $w(x) = 1$ and an interval $-1 \le x \le 1$. Prove that the nth such function constructed in this way is proportional to $P_n(x)$.

Hint. Use mathematical induction (Section 1.4).

Solution.

Without normalization, let the Gram–Schmidt output be

$$g_n = x^n - \sum_{j=0}^{n-1} \frac{\langle g_j, x^n\rangle}{\langle g_j, g_j\rangle} g_j.$$

Its leading coefficient is one, since every subtracted polynomial has degree below n. For $n = 0$, $g_0 = 1 = P_0$. Assume the earlier g_j are proportional to P_j. Both g_n and P_n are orthogonal to all polynomials of degree less than n. For g_n this follows from the construction and the induction hypothesis: $g_0, \ldots, g_{n-1}$ are proportional to $P_0, \ldots, P_{n-1}$ and hence mutually orthogonal (Exercise 15.2.1), so for $k < n$ only the term $j = k$ of the sum survives in

$$\langle g_k, g_n\rangle = \langle g_k, x^n\rangle - \sum_{j=0}^{n-1} \frac{\langle g_j, x^n\rangle}{\langle g_j, g_j\rangle}\langle g_k, g_j\rangle = \langle g_k, x^n\rangle - \langle g_k, x^n\rangle = 0.$$

Thus g_n is orthogonal to $g_0, \ldots, g_{n-1}$, which are monic of degrees $0, \ldots, n-1$ and hence span the polynomials of degree less than n. For P_n it follows from Rodrigues's formula (Exercise 15.1.14). Put $c_n = (2n)!/[2^n(n!)^2]$, the leading coefficient of P_n (the $k = 0$ term of Exercise 15.1.2). Then $h = g_n - P_n/c_n$ has degree below n and is orthogonal to all such polynomials, including itself. Thus $\int_{-1}^{1} h(x)^2 dx = 0$, so the continuous polynomial h is zero. Consequently

$$\boxed{g_n = \frac{2^n(n!)^2}{(2n)!} P_n}.$$

Any optional nonzero normalization in Gram–Schmidt changes only this scalar factor. This completes the induction.

15.2.3 Expand the Dirac delta function $\delta(x)$ in a series of Legendre polynomials using the interval $-1 \le x \le 1$, with convergence understood in the distributional sense.

Solution.

The coefficient of P_n is obtained by pairing the delta distribution with the polynomial:

$$a_n = \frac{2n+1}{2}\langle \delta, P_n\rangle = \frac{2n+1}{2} P_n(0).$$

Odd n vanish. For $n = 2s$, use $P_{2s}(0) = (-1)^s(2s-1)!!/(2s)!!$:

$$\boxed{\delta(x) = \sum_{s=0}^{\infty} (-1)^s \frac{4s+1}{2} \frac{(2s-1)!!}{(2s)!!} P_{2s}(x)}.$$

This is a distributional expansion, not a pointwise function series for the delta distribution. To verify it, let ϕ be smooth on $[-1, 1]$, with Legendre coefficients $c_n = (2n+1)\int \phi P_n/2$, i.e., $\int_{-1}^{1} \phi P_n dx = 2c_n/(2n+1)$. Pairing the proposed series with ϕ term by term gives

$$\sum_{n=0}^{\infty} \frac{2n+1}{2} P_n(0) \int_{-1}^{1} \phi P_n \, dx = \sum_{n=0}^{\infty} c_n P_n(0) = \phi(0),$$

where the last equality is the Legendre expansion of ϕ evaluated at $x = 0$. Here is the convergence justification for that pointwise evaluation. Set $L = -D[(1 - x^2)D]$, with $D = d/dx$, so $LP_n = n(n + 1)P_n$ by the Legendre ODE. For $n \geq 1$, transfer L from P_n to ϕ by two integrations by parts:

$$\int_{-1}^{1} \phi\, LP_n\, dx = \Big[(1 - x^2)(\phi' P_n - \phi P_n')\Big]_{-1}^{1} + \int_{-1}^{1} (L\phi)P_n\, dx = \int_{-1}^{1} (L\phi)P_n\, dx,$$

because both boundary terms contain $1 - x^2$ and vanish. Hence $c_n = \frac{2n+1}{2n(n+1)} \int_{-1}^{1} (L\phi)P_n\, dx$, and repeating this k times gives

$$c_n = \frac{2n + 1}{2[n(n + 1)]^k} \int_{-1}^{1} (L^k\phi)(x)P_n(x)\, dx.$$

Since $|P_n(x)| \leq 1$ on this interval, with $\|L^k\phi\|_\infty = \max_{-1\leq x\leq 1} |L^k\phi(x)|$,

$$\left|\int_{-1}^{1} (L^k\phi)P_n\, dx\right| \leq 2\|L^k\phi\|_\infty, \qquad \text{hence} \qquad |c_n| \leq \frac{(2n + 1)\|L^k\phi\|_\infty}{[n(n + 1)]^k} = O(n^{1-2k}).$$

Taking $k = 2$ makes $\sum |c_n|$ converge, so $\sum c_n P_n$ converges uniformly. Its L^2 limit is ϕ by polynomial completeness, and continuity then gives equality at every point. Evaluation at zero therefore proves the required delta action.

15.2.4 Verify the Dirac delta function expansions as distributions on $[-1, 1]$

$$\delta(1 - x) = \sum_{n=0}^{\infty} \frac{2n + 1}{2} P_n(x),$$
$$\delta(1 + x) = \sum_{n=0}^{\infty} (-1)^n \frac{2n + 1}{2} P_n(x).$$

These expressions appear in a resolution of the Rayleigh plane wave expansion (Exercise 15.2.24) into incoming and outgoing spherical waves.

Note. Assume that the entire Dirac delta function is covered when integrating over $[-1, 1]$.

Solution.

Use the full endpoint-mass convention specified in the question. Orthogonal projection gives

$$a_n^{(+)} = \frac{2n + 1}{2} P_n(1) = \frac{2n + 1}{2}, \qquad a_n^{(-)} = \frac{2n + 1}{2} P_n(-1) = (-1)^n \frac{2n + 1}{2}.$$

Therefore the two proposed expansions follow. For a direct distributional verification, expand a smooth test function as $\phi = \sum c_n P_n$, with $c_n = \frac{2n+1}{2} \int_{-1}^{1} \phi P_n\, dx$ as in Exercise 15.2.3. Pairing the first series with ϕ gives $\sum c_n = \sum c_n P_n(1) = \phi(1)$, while pairing the second gives $\sum (-1)^n c_n = \sum c_n P_n(-1) = \phi(-1)$, since $P_n(1) = 1$ and $P_n(-1) = (-1)^n$. These are precisely the actions of $\delta(1 - x)$ and $\delta(1 + x)$, respectively. The uniform convergence proved in Exercise 15.2.3 justifies both endpoint evaluations. The endpoint convention matters: inserting a half-weight convention would change both coefficients by a factor of two. These are distributional identities, with no assertion of pointwise convergence.

15.2.5 Neutrons (mass 1) are being scattered by a nucleus of mass A ($A > 1$). In the center-of-mass system the scattering is isotropic. Then, in the laboratory system the average of the cosine of the angle of deflection of the neutron is

$$\langle \cos\psi \rangle = \frac{1}{2}\int_0^\pi \frac{A\cos\theta + 1}{(A^2 + 2A\cos\theta + 1)^{1/2}} \sin\theta\, d\theta.$$

Show, by expansion of the denominator, that $\langle \cos\psi \rangle = 2/(3A)$.

Solution.

Set $x = \cos\theta$, so $\sin\theta\, d\theta = -dx$; the limits $\theta = 0, \pi$ become $x = 1, -1$, and the minus sign reverses them. Then divide numerator and denominator by $A > 1$, i.e., factor A out of $Ax + 1$ and A out of the square root:

$$\langle \cos\psi \rangle = \frac{1}{2}\int_{-1}^1 \frac{Ax + 1}{(A^2 + 2Ax + 1)^{1/2}}\, dx = \frac{1}{2}\int_{-1}^1 (x + A^{-1})(1 + 2x/A + A^{-2})^{-1/2} dx.$$

The generating function at $t = -1/A$ gives

$$(1 + 2x/A + A^{-2})^{-1/2} = \sum_{n=0}^{\infty} (-A^{-1})^n P_n(x).$$

Since $|P_n(x)| \le 1$ and $\sum A^{-n}$ converges, the series converges uniformly on $[-1, 1]$ (Weierstrass M-test), so it may be integrated termwise. As $1 = P_0$ and $x = P_1$, orthogonality kills every term except $n = 0$ in the constant part and $n = 1$ in the x part:

$$\langle \cos\psi \rangle = \frac{1}{2}\left[\frac{1}{A}\int_{-1}^1 P_0^2 dx - \frac{1}{A}\int_{-1}^1 P_1^2 dx\right] = \frac{1}{2}\left(\frac{2}{A} - \frac{2}{3A}\right) = \boxed{\frac{2}{3A}}.$$

15.2.6 A particular function $f \in L^2[-1, 1]$ is expanded in a Legendre series over this same interval, converging in L^2. Show that the expansion is unique.

Solution.

In the usual $L^2[-1, 1]$ interpretation of a Legendre expansion, suppose $f = \sum a_n P_n = \sum b_n P_n$, both sums converging in L^2. Taking the inner product with a fixed P_m is continuous in that norm, by the Cauchy–Schwarz inequality: for the partial sums $S_N = \sum_{n \le N} a_n P_n$ with $N \ge m$, orthogonality gives $\int_{-1}^1 S_N P_m\, dx = 2a_m/(2m + 1)$, and $|\int_{-1}^1 (f - S_N) P_m\, dx| \le \|f - S_N\|_2 \|P_m\|_2 \to 0$, where $\|g\|_2 = (\int_{-1}^1 |g|^2 dx)^{1/2}$. The same holds for the b_n. Hence

$$\int_{-1}^1 f P_m dx = \frac{2a_m}{2m + 1} = \frac{2b_m}{2m + 1},$$

so $a_m = b_m$ for every m. The same proof applies to any stronger convergence that permits these integrations. Thus the unique coefficients are

$$\boxed{a_m = \frac{2m + 1}{2}\int_{-1}^1 f(x) P_m(x) dx}.$$

A convergence class is needed to justify extracting a coefficient from an infinite series; the argument above uses the standard square-integrable Fourier–Legendre setting.

15.2.7 A real function $f \in L^2[-1,1]$ is expanded in its L^2-convergent Legendre series $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$. Show that

$$\int_{-1}^{1} [f(x)]^2\,dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}.$$

This is a statement that the Legendre polynomials form a complete set.

Solution.

For a finite partial sum $S_N = \sum_{n=0}^{N} a_n P_n$, orthogonality gives, with the cross terms written explicitly,

$$\int_{-1}^{1} S_N^2 dx = \sum_{n,m=0}^{N} a_n a_m \int_{-1}^{1} P_n P_m dx = \sum_{n=0}^{N} \frac{2a_n^2}{2n+1}.$$

For a real square-integrable f, its Legendre projections converge in L^2. One way to establish completeness is that polynomials approximate continuous functions uniformly on the compact interval, continuous functions are dense in L^2, and the P_n span all polynomials. Density gives convergence of the S_N because S_N is the best approximation to f among the polynomials p of degree at most N. By the coefficient formula of Exercise 15.2.6, $\int_{-1}^{1}(f - S_N)P_m\,dx = \int_{-1}^{1} fP_m\,dx - 2a_m/(2m+1) = 0$ for $m \le N$, so $f - S_N$ is orthogonal to $P_0, \ldots, P_N$ and hence to $S_N - p$, and

$$\|f - p\|_2^2 = \|f - S_N\|_2^2 + \|S_N - p\|_2^2 \ge \|f - S_N\|_2^2.$$

Given $\varepsilon > 0$, choose a polynomial p with $\|f - p\|_2 < \varepsilon$; then $\|f - S_N\|_2 < \varepsilon$ for every $N \ge \deg p$. Moreover, writing $f^2 - S_N^2 = (f - S_N)(f + S_N)$ and using the Cauchy–Schwarz and triangle inequalities,

$$\left|\int_{-1}^{1} (f^2 - S_N^2)\,dx\right| \le \|f - S_N\|_2(\|f\|_2 + \|S_N\|_2) \longrightarrow 0,$$

since $\|S_N\|_2 \le \|f\|_2 + \|f - S_N\|_2$ stays bounded. Taking the limit proves

$$\boxed{\int_{-1}^{1} f^2 dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}}.$$

The norm identity for complex f uses $|f|^2$ and $|a_n|^2$. The printed bilinear identity with squares also holds for complex L^2 functions, but is then not the nonnegative norm formula.

15.2.8

(a) For

$$f(x) = \begin{cases} +1, & 0 < x < 1, \\ -1, & -1 < x < 0, \end{cases}$$

show that

$$\int_{-1}^{1} [f(x)]^2\,dx = 2\sum_{n=0}^{\infty} (4n+3)\left[\frac{(2n-1)!!}{(2n+2)!!}\right]^2.$$

(b) By testing the series, prove that it is convergent.

(c) The value of the integral in part (a) is 2. Check the rate at which the series converges by summing its first 10 terms.

Solution.

(a) The function is odd, so only odd Legendre coefficients occur. For $j = 2n+1$,

$$a_{2n+1} = \frac{4n+3}{2}\int_{-1}^{1} \operatorname{sgn}(x) P_{2n+1}(x)\,dx = (4n+3)\int_{0}^{1} P_{2n+1}(x)\,dx,$$

because $\operatorname{sgn}(x)P_{2n+1}(x)$ is even. Exercise 15.1.12 gives $a_{2n+1} = (-1)^n(4n+3)(2n-1)!!/(2n+2)!!$. Insert this in Parseval (Exercise 15.2.7), whose $j = 2n+1$ term is $2a_{2n+1}^2/(4n+3) = 2(4n+3)[(2n-1)!!/(2n+2)!!]^2$:

$$\int_{-1}^{1} f^2\,dx = 2\sum_{n=0}^{\infty}(4n+3)\left[\frac{(2n-1)!!}{(2n+2)!!}\right]^2.$$

(b) Since $(2n-1)!!/(2n)!! = (2n)!/[(2n)!!]^2 = (2n)!/[4^n(n!)^2]$ and $(2n+2)!! = 2(n+1)(2n)!!$, write

$$\frac{(2n-1)!!}{(2n+2)!!} = \frac{1}{2(n+1)}\frac{\binom{2n}{n}}{4^n}.$$

Stirling's formula gives $\binom{2n}{n}/4^n \sim (\pi n)^{-1/2}$. Thus the entire summand, including its leading factor 2, is

$$\frac{2(4n+3)}{4(n+1)^2}\left[\frac{\binom{2n}{n}}{4^n}\right]^2 \sim \frac{8n}{4n^2}\frac{1}{\pi n} = \frac{2}{\pi n^2}.$$

Comparison with $\sum n^{-2}$ proves convergence independently of invoking Parseval. It also predicts a tail $\sum_{n\ge N} 2/(\pi n^2) \approx 2/(\pi N)$, of order $1/N$.

(c) The terms are $\frac{3}{2}$, $\frac{7}{32}$, $\frac{11}{128}$, $\frac{375}{8192}$, …, and the partial sums S_N of the first N terms are, for $N = 1, \dots, 9$,

$$1.5,\ 1.71875,\ 1.80469,\ 1.85046,\ 1.87888,\ 1.89822,\ 1.91224,\ 1.92287,\ 1.93120.$$

The first ten terms ($n = 0, \dots, 9$), evaluated by exact rational arithmetic, give

$$\boxed{S_{10} = \frac{66586053015}{34359738368} = 1.9379091977316420525\dots}.$$

The missing amount is $2 - S_{10} = 0.0620908022683579475\dots$, or about 3.105% of the integral, close to the estimate $2/(10\pi) \approx 0.064$ of part (b). The slow $1/N$ tail is visible even though all terms are positive. The Instructor's Manual, p. 237, prints 1.943… for ten terms. That is instead the eleven-term value $S_{11} = 267102370903/137438953472 = 1.943425529338128\dots$. The requested ten terms run from $n = 0$ through $n = 9$.

15.2.9 Prove that

$$\int_{-1}^{1} x(1-x^2)P_n' P_m'\,dx = \frac{2n(n^2-1)}{4n^2-1}\,\delta_{m,n-1} + \frac{2n(n+2)(n+1)}{(2n+1)(2n+3)}\,\delta_{m,n+1}.$$

Solution.

Let I_{nm} denote the integral. Integrate by parts with $u = x(1-x^2)P_n'$ and $dv = P_m' dx$, $v = P_m$. The boundary term vanishes because $x(1-x^2)P_n'P_m$ is zero at $x = \pm 1$, and $u' = (1-x^2)P_n' + x\{(1-x^2)P_n'\}'$, so

$$I_{nm} = -\int_{-1}^{1} P_m \left[(1-x^2)P_n' + x\{(1-x^2)P_n'\}'\right] dx.$$

The ODE says $\{(1-x^2)P_n'\}' = -n(n+1)P_n$. Also $(1-x^2)P_n' = nP_{n-1} - nxP_n$. The bracket is therefore $nP_{n-1} - nxP_n - n(n+1)xP_n = nP_{n-1} - n(n+2)xP_n$. Consequently

$$I_{nm} = \int_{-1}^{1} P_m[n(n+2)xP_n - nP_{n-1}]dx.$$

Replace xP_n by $[(n+1)P_{n+1} + nP_{n-1}]/(2n+1)$ (the three-term recurrence). The coefficient of P_{n-1} becomes $n[n(n+2)-(2n+1)]/(2n+1) = n(n^2-1)/(2n+1)$, and that of P_{n+1} is $n(n+1)(n+2)/(2n+1)$. Orthogonality, $\int_{-1}^{1} P_m P_{n-1}dx = 2\delta_{m,n-1}/(2n-1)$ and $\int_{-1}^{1} P_m P_{n+1}dx = 2\delta_{m,n+1}/(2n+3)$, now yields

$$\boxed{I_{nm} = \frac{2n(n^2-1)}{(2n-1)(2n+1)}\delta_{m,n-1} + \frac{2n(n+1)(n+2)}{(2n+1)(2n+3)}\delta_{m,n+1}}.$$

For $n = 0$ the original derivative is zero; for $n = 1$ the first coefficient is zero. Thus the boundary index cases are included.

15.2.10 The coincidence counting rate, $W(\theta)$, in a gamma-gamma angular correlation experiment has the form

$$W(\theta) = \sum_{n=0}^{\infty} a_{2n} P_{2n}(\cos\theta).$$

Show that data in the range $\pi/2 \le \theta \le \pi$ can, in principle, define the function $W(\theta)$ (and permit a determination of the coefficients a_{2n}). This means that although data in the range $0 \le \theta < \pi/2$ may be useful as a check, they are not essential.

Solution.

Every P_{2n} is even, so with $x = \cos\theta$ the stated expansion implies $W(-x) = W(x)$, equivalently $W(\pi - \theta) = W(\theta)$. Data for $\pi/2 \le \theta \le \pi$ give $W(x)$ on $[-1, 0]$; reflection supplies its values on $[0, 1]$. The coefficient formula makes this explicit:

$$a_{2n} = \frac{4n+1}{2}\int_{-1}^{1} W(x)P_{2n}(x)dx = (4n+1)\int_{-1}^{0} W(x)P_{2n}(x)dx.$$

The second form holds because WP_{2n} is even. Changing back to θ ($x = \cos\theta$ maps $[\pi/2, \pi]$ onto $[0, -1]$, and $dx = -\sin\theta\, d\theta$) gives

$$\boxed{a_{2n} = (4n+1)\int_{\pi/2}^{\pi} W(\theta)P_{2n}(\cos\theta)\sin\theta\, d\theta}.$$

Thus ideal knowledge on the stated half-range fixes every coefficient and the full even function. The other half-range tests the assumed even symmetry and improves experimental precision.

15.2.11 A conducting sphere of radius r_0 is placed in an initially uniform electric field, E_0. Show the following:

(a) The induced surface charge density is $\sigma = 3\varepsilon_0 E_0 \cos\theta$,

(b) The induced electric dipole moment is $P = 4\pi r_0^3 \varepsilon_0 E_0$.

Note. The induced electric dipole moment can be calculated either from the surface charge [part (a)] or by noting that the final electric field E is the result of superimposing a dipole field on the original uniform field.

Solution.

Choose the z axis along $\mathbf{E}_0$ and take the initially uncharged conductor. The imposed potential is $-E_0 z = -E_0 r P_1(\cos\theta)$. Outside the sphere the potential is a sum of terms $(a_l r^l + b_l r^{-l-1})P_l(\cos\theta)$. Far away V must approach the imposed potential, $V + E_0 z \to 0$ (this also fixes the additive constant), so $a_1 = -E_0$ and all other a_l vanish, including the constant a_0; and on the sphere the potential is constant. Matching each P_l separately therefore leaves only the imposed term and the decaying dipole harmonic $A\cos\theta/r^2$, with $A = b_1$ (an $l = 0$ term b_0/r would carry a net charge, absent for the neutral sphere). So outside the sphere

$$V = -E_0 r\cos\theta + \frac{A}{r^2}\cos\theta.$$

A conducting surface is equipotential. At $r = r_0$ the coefficient of $\cos\theta$, $-E_0 r_0 + A/r_0^2$, must vanish, giving $A = E_0 r_0^3$. Therefore

$$E_r = -\partial_r V = E_0\cos\theta + \frac{2A}{r^3}\cos\theta = E_0\left(1 + \frac{2r_0^3}{r^3}\right)\cos\theta.$$

The field inside the conductor is zero, so the normal jump condition gives, with $E_r(r_0) = 3E_0\cos\theta$,

$$\boxed{\sigma = \varepsilon_0 E_r(r_0) = 3\varepsilon_0 E_0\cos\theta}.$$

It integrates to zero, since $\int_0^\pi \cos\theta\sin\theta\,d\theta = 0$, as required for induced charge on a neutral sphere. The dipole moment has only a z component by axial symmetry; with $dA = r_0^2\sin\theta\,d\theta\,d\varphi$ and the φ integral giving 2π,

$$\begin{aligned} p_z &= \int_{\text{surface}} (r_0\cos\theta)\sigma\,dA \\ &= 6\pi\varepsilon_0 E_0 r_0^3 \int_0^\pi \cos^2\theta\sin\theta\,d\theta \\ &= 6\pi\varepsilon_0 E_0 r_0^3\left(\frac{2}{3}\right) = \boxed{4\pi\varepsilon_0 E_0 r_0^3}. \end{aligned}$$

The coefficient of the induced potential independently gives the same answer: a dipole has $V_{\text{dip}} = p_z\cos\theta/(4\pi\varepsilon_0 r^2)$, and comparison with $A\cos\theta/r^2$ gives $p_z = 4\pi\varepsilon_0 A = 4\pi\varepsilon_0 E_0 r_0^3$.

15.2.12 Obtain as a Legendre expansion the electrostatic potential of the circular ring of Example 15.2.3, for points (r,θ) with $r<a$.

Solution.

Let $C=q/(4\pi\varepsilon_0)$ and a be the ring radius, as in Example 15.2.3 (the ring lies in the plane $\theta=\pi/2$, centered on the polar axis). Inside the sphere $r<a$ there is no charge, and the axisymmetric potential is regular at the origin. Hence it has the form $\psi=\sum_{n\ge0}b_n r^n P_n(\cos\theta)$. On the positive axis ($\theta=0$, $r=z$) every ring element is the same distance $\sqrt{a^2+z^2}$ away. The binomial series $(1+u)^{-1/2}=\sum_s \binom{-1/2}{s}u^s=\sum_s(-1)^s\frac{(2s-1)!!}{(2s)!!}u^s$ ($|u|<1$, here $u=z^2/a^2$) gives, for $0\le z<a$,

$$\psi(z,0)=\frac{C}{\sqrt{a^2+z^2}}=\frac{C}{a}\sum_{s=0}^{\infty}(-1)^s\frac{(2s-1)!!}{(2s)!!}\left(\frac{z}{a}\right)^{2s}.$$

On the other hand, $\psi(z,0)=\sum_n b_n z^n$, since $P_n(1)=1$. Equality of these power-series coefficients gives $b_{2s}=C(-1)^s(2s-1)!!/[a^{2s+1}(2s)!!]=CP_{2s}(0)/a^{2s+1}$ and $b_{2s+1}=0$. Therefore

$$\boxed{\psi(r,\theta)=\frac{q}{4\pi\varepsilon_0 a}\sum_{s=0}^{\infty}P_{2s}(0)\left(\frac{r}{a}\right)^{2s}P_{2s}(\cos\theta)},\qquad r<a.$$

The series converges uniformly on every $r\le\rho<a$, and gives $\psi(0)=C/a$. Its axis values and regular solid harmonics determine all coefficients uniquely.

15.2.13 Calculate the electric field produced by the charged conducting ring of Example 15.2.3 for

(a) $r>a$, (b) $r<a$.

Solution.

Put $C=q/(4\pi\varepsilon_0)$, $c_s=P_{2s}(0)$, and $x=\cos\theta$. There is no azimuthal dependence, so $E_\varphi=0$. For any potential $\psi(r,x)$, $E_r=-\psi_r$ and, since $\partial_\theta=-\sin\theta\,\partial_x$, $E_\theta=-r^{-1}\partial_\theta\psi=(\sin\theta/r)\psi_x$.

(a) The exterior ring potential of Example 15.2.3 is $\psi=C\sum c_s a^{2s}r^{-2s-1}P_{2s}(x)$. Termwise differentiation, with $-\partial_r r^{-2s-1}=(2s+1)r^{-2s-2}$, gives

$$\boxed{E_r=\frac{C}{r^2}\sum_{s=0}^{\infty}(2s+1)c_s\left(\frac{a}{r}\right)^{2s}P_{2s}(x)},$$

$$\boxed{E_\theta=\frac{C\sin\theta}{r^2}\sum_{s=0}^{\infty}c_s\left(\frac{a}{r}\right)^{2s}P'_{2s}(x)},\qquad r>a.$$

(b) Differentiate the interior result of Exercise 15.2.12, $\psi=(C/a)\sum c_s(r/a)^{2s}P_{2s}(x)$, using $\partial_r(r/a)^{2s}=(2s/a)(r/a)^{2s-1}$ and $(1/r)(r/a)^{2s}=(1/a)(r/a)^{2s-1}$:

$$\boxed{E_r=-\frac{C}{a^2}\sum_{s=1}^{\infty}2s\,c_s\left(\frac{r}{a}\right)^{2s-1}P_{2s}(x)},$$

$$\boxed{E_\theta=\frac{C\sin\theta}{a^2}\sum_{s=1}^{\infty}c_s\left(\frac{r}{a}\right)^{2s-1}P'_{2s}(x)},\qquad r<a.$$

The $s = 0$ term is a constant and contributes no field. Both interior series begin at $s = 1$ with terms proportional to r/a, and the later terms carry higher powers of r/a, so E_r and E_θ tend to zero as $r \to 0$ from every direction: the field at the center of the ring is $\mathbf{E}(0) = 0$, as symmetry requires. Uniform convergence on compact subregions justifies the derivatives; the singular charged ring itself is excluded.

15.2.14 As an extension of Example 15.2.3, find the potential $\psi(r, \theta)$ produced by a charged conducting disk, Fig. 15.4, for $r > a$, where a is the radius of the disk. The charge density σ (on each side of the disk) is

$$\sigma(\rho) = \frac{q}{4\pi a(a^2 - \rho^2)^{1/2}}, \qquad \rho^2 = x^2 + y^2.$$

Hint. The definite integral you get can be evaluated as a beta function, Section 13.3. For more details see section 5.03 of Smythe in Additional Readings.

$$\textit{ANS. } \psi(r, \theta) = \frac{q}{4\pi\varepsilon_0 r} \sum_{l=0}^{\infty} (-1)^l \frac{1}{2l+1} \left(\frac{a}{r}\right)^{2l} P_{2l}(\cos\theta).$$

Solution.

Both faces contribute. A circular annulus of radius ρ and width $d\rho$ carries

$$dq = 2\sigma(\rho)\, 2\pi\rho\, d\rho = \frac{q}{a} \frac{\rho\, d\rho}{\sqrt{a^2 - \rho^2}}.$$

Its integral from zero to a is $(q/a)\big[-\sqrt{a^2 - \rho^2}\,\big]_0^a = q$, confirming the normalization. By Example 15.2.3 with the radius a replaced by ρ, the annulus produces, for $r > \rho$, the potential $(dq/4\pi\varepsilon_0 r) \sum_l P_{2l}(0)(\rho/r)^{2l} P_{2l}(\cos\theta)$. Adding the annuli gives

$$\psi = \frac{1}{4\pi\varepsilon_0 r} \sum_{l=0}^{\infty} \frac{P_{2l}(0) P_{2l}(\cos\theta)}{r^{2l}} \int_0^a \rho^{2l}\, dq.$$

For $r > a$ the series is dominated by $\sum (a/r)^{2l}$ (as $|P_{2l}| \le 1$), so it can be integrated termwise, even with the integrable edge singularity in σ. Using $u = \rho^2/a^2$, so that $\rho^{2l+1} d\rho = \frac{1}{2} a^{2l+2} u^l\, du$ and $\sqrt{a^2 - \rho^2} = a\sqrt{1-u}$,

$$\begin{aligned} \int_0^a \rho^{2l}\, dq &= \frac{q}{a} \int_0^a \frac{\rho^{2l+1}}{\sqrt{a^2 - \rho^2}} d\rho = \frac{q a^{2l}}{2} \int_0^1 u^l (1-u)^{-1/2} du \\ &= \frac{q a^{2l}}{2} B(l+1, \tfrac{1}{2}) = q a^{2l} \frac{2^{2l} (l!)^2}{(2l+1)!}. \end{aligned}$$

The last step uses $B(l+1, \frac{1}{2}) = \Gamma(l+1)\Gamma(\frac{1}{2})/\Gamma(l+\frac{3}{2}) = l!\, 2^{l+1}/(2l+1)!! = 2^{2l+1}(l!)^2/(2l+1)!$, by $\Gamma(l+\frac{3}{2}) = (2l+1)!!\sqrt{\pi}/2^{l+1}$ and $(2l+1)!! = (2l+1)!/(2^l l!)$. Multiply by $P_{2l}(0) = (-1)^l (2l)!/[2^{2l}(l!)^2]$; all factors cancel except $(-1)^l (2l)!/(2l+1)! = (-1)^l/(2l+1)$. Hence

$$\boxed{\psi(r, \theta) = \frac{q}{4\pi\varepsilon_0 r} \sum_{l=0}^{\infty} \frac{(-1)^l}{2l+1} \left(\frac{a}{r}\right)^{2l} P_{2l}(\cos\theta)}.$$

15.2.15 The hemisphere defined by $r = a$, $0 \le \theta < \pi/2$, has an electrostatic potential $+V_0$. The hemisphere $r = a$, $\pi/2 < \theta \le \pi$ has an electrostatic potential $-V_0$. Show that the potential at interior points is

$$V = V_0 \sum_{n=0}^{\infty} \frac{4n+3}{2n+2} \left(\frac{r}{a}\right)^{2n+1} P_{2n}(0) P_{2n+1}(\cos\theta)$$
$$= V_0 \sum_{n=0}^{\infty} (-1)^n \frac{(4n+3)(2n-1)!!}{(2n+2)!!} \left(\frac{r}{a}\right)^{2n+1} P_{2n+1}(\cos\theta).$$

Hint. You need Exercise 15.1.12.

Solution.

Let $x = \cos\theta$. The boundary potential is $V(a, x) = V_0 \operatorname{sgn} x$, except for the unspecified equatorial jump. Its Legendre coefficients $b_l = \frac{2l+1}{2} \int_{-1}^{1} V(a,x) P_l(x)\, dx$ are V_0 times those of Exercise 15.2.8: for even l the integrand is odd, while for $l = 2n+1$ it is even, and Exercise 15.1.12 gives the integral over $[0, 1]$:

$$b_{2n} = 0, \qquad b_{2n+1} = V_0(4n+3) \int_0^1 P_{2n+1}(x) dx = V_0 \frac{4n+3}{2n+2} P_{2n}(0).$$

Regularity at the origin selects r^l, while the other radial solution r^{-l-1} is excluded. To match each boundary coefficient at $r = a$, multiply it by $(r/a)^l$. Thus

$$\boxed{V(r,\theta) = V_0 \sum_{n=0}^{\infty} \frac{4n+3}{2n+2} P_{2n}(0) \left(\frac{r}{a}\right)^{2n+1} P_{2n+1}(\cos\theta)}.$$

Substituting the double-factorial value $P_{2n}(0) = (-1)^n (2n-1)!!/(2n)!!$ and using $(2n)!!\,(2n+2) = (2n+2)!!$ gives the second series in the question. For $r < a$ it converges locally uniformly and is harmonic. Its boundary limit is $\pm V_0$ away from the equator and the symmetric value zero at the jump. This also shows $V(0) = 0$ and the reflection antisymmetry.

15.2.16 A conducting sphere of radius a is divided into two electrically separate hemispheres by a thin insulating barrier at its equator. The top hemisphere is maintained at a potential V_0, and the bottom hemisphere at $-V_0$. Take $V \to 0$ at infinity and the ideal limit of zero barrier width.

(a) Show that the electrostatic potential exterior to the two hemispheres is

$$V(r,\theta) = V_0 \sum_{s=0}^{\infty} (-1)^s (4s+3) \frac{(2s-1)!!}{(2s+2)!!} \left(\frac{a}{r}\right)^{2s+2} P_{2s+1}(\cos\theta).$$

(b) Calculate the electric charge density σ on the outside surface. Interpret the boundary derivative by its radial Abel limit. Show that the pole densities are finite although the unregularized series diverges there; identify the equatorial singularity that produces the infinite ideal capacitance.

ANS. (b) With $\rho = a/r$, away from the equator,

$$\sigma = \varepsilon_0 \lim_{r \downarrow a} E_r = \frac{\varepsilon_0 V_0}{a} \lim_{\rho \uparrow 1} \sum_{s=0}^{\infty} (-1)^s (4s+3) \frac{(2s-1)!!}{(2s)!!} \rho^{2s+3} P_{2s+1}(\cos\theta).$$

Solution.

(a) The boundary data are those of Exercise 15.2.15, so the boundary coefficients are $b_{2s+1} = V_0(-1)^s(4s+3)(2s-1)!!/(2s+2)!!$ and the even ones are zero. The exterior solution that vanishes at infinity uses r^{-l-1}: the term $b_l(a/r)^{l+1}P_l(\cos\theta)$ reduces to $b_lP_l(\cos\theta)$ at $r=a$, and $(a/r)^{l+1} = (a/r)^{2s+2}$ for $l = 2s+1$. So

$$\boxed{V = V_0\sum_{s=0}^{\infty}(-1)^s(4s+3)\frac{(2s-1)!!}{(2s+2)!!}\left(\frac{a}{r}\right)^{2s+2}P_{2s+1}(\cos\theta)}, \quad r > a.$$

(b) Differentiate first at $r > a$, where this is legitimate: $-\partial_r(a/r)^{2s+2} = \frac{2s+2}{a}(a/r)^{2s+3}$ and $(2s+2)/(2s+2)!! = 1/(2s)!!$, so

$$E_r = \frac{V_0}{a}\sum_{s=0}^{\infty}(-1)^s(4s+3)\frac{(2s-1)!!}{(2s)!!}\left(\frac{a}{r}\right)^{2s+3}P_{2s+1}(\cos\theta).$$

Consequently, with $\sigma = \varepsilon_0 E_r$ just outside the surface and $\rho = a/r$,

$$\boxed{\sigma(\theta) = \frac{\varepsilon_0 V_0}{a}\lim_{\rho\uparrow 1}\sum_{s=0}^{\infty}(-1)^s(4s+3)\frac{(2s-1)!!}{(2s)!!}\rho^{2s+3}P_{2s+1}(\cos\theta)}.$$

This is the radial, or Abel, boundary limit. The printed answer is missing $1/a$. Moreover, ordinary series divergence at the poles does not mean the physical charge density is infinite there. The physical singularity and infinite capacitance occur at the equatorial potential jump.

For an explicit pole check, sum the series on the axis in closed form. For axisymmetric boundary data $f(u)$, $u = \cos\theta$, the exterior solution is $V = \sum_l b_l\rho^{l+1}P_l(\cos\theta)$ with $b_l = \frac{2l+1}{2}\int_{-1}^{1} f(u)P_l(u)\,du$. On the positive axis $P_l(1) = 1$; for $\rho < 1$ the sum may be taken under the integral ($|P_l| \le 1$), and Exercise 15.1.6 sums it:

$$V(r,0) = \frac{\rho}{2}\int_{-1}^{1} f(u)\sum_{l=0}^{\infty}(2l+1)\rho^l P_l(u)\,du = \frac{\rho(1-\rho^2)}{2}\int_{-1}^{1}\frac{f(u)\,du}{(1-2\rho u+\rho^2)^{3/2}}.$$

For $f = V_0\operatorname{sgn}u$ this spherical Poisson kernel gives

$$\frac{V(r,0)}{V_0} = \frac{\rho(1-\rho^2)}{2}\int_{-1}^{1}\frac{\operatorname{sgn}u}{(1-2\rho u+\rho^2)^{3/2}}du = 1 - \frac{1-\rho^2}{\sqrt{1+\rho^2}}, \quad \rho = \frac{a}{r}.$$

The antiderivative of the integrand without the sign is $[\rho\sqrt{1-2\rho u+\rho^2}]^{-1}$; for $0 < \rho < 1$ it gives

$$\int_0^1 - \int_{-1}^0 = \frac{1}{\rho}\left[\frac{1}{1-\rho} - \frac{2}{\sqrt{1+\rho^2}} + \frac{1}{1+\rho}\right] = \frac{2}{\rho}\left[\frac{1}{1-\rho^2} - \frac{1}{\sqrt{1+\rho^2}}\right],$$

and multiplication by $\rho(1-\rho^2)/2$ gives the last equality. Since $\rho = a/r$, $-\partial_r = (\rho^2/a)\partial_\rho$, and

$$\frac{d}{d\rho}\left[1 - \frac{1-\rho^2}{\sqrt{1+\rho^2}}\right] = \frac{2\rho}{\sqrt{1+\rho^2}} + \frac{\rho(1-\rho^2)}{(1+\rho^2)^{3/2}} = \frac{\rho(\rho^2+3)}{(1+\rho^2)^{3/2}},$$

which equals $\sqrt{2}$ at $\rho = 1$. With $\sigma = \varepsilon_0 E_r$ and the antisymmetry $V(r,\pi-\theta) = -V(r,\theta)$ (only odd P_l occur), this yields

$$\boxed{\sigma(0) = \frac{\sqrt{2}\varepsilon_0 V_0}{a}, \qquad \sigma(\pi) = -\frac{\sqrt{2}\varepsilon_0 V_0}{a}}.$$

The unsmoothed series has terms of magnitude proportional to $\sqrt{s}$ at those poles: there $|P_{2s+1}| = 1$, and $(2s-1)!!/(2s)!! \sim (\pi s)^{-1/2}$ (Exercise 15.2.8(b)) makes the coefficient grow like $4\sqrt{s/\pi}$. So an ordinary termwise boundary sum cannot give these finite values.

The poles are not special in this respect. For fixed $0 < \theta < \pi$, the standard large-n asymptotic form

$$P_n(\cos\theta) \approx \Big(\frac{2}{\pi n \sin\theta}\Big)^{1/2} \cos\Big[\Big(n+\frac{1}{2}\Big)\theta - \frac{\pi}{4}\Big],$$

with $n = 2s+1$, gives terms of magnitude about $4(\pi\sqrt{\sin\theta})^{-1}|\cos\beta_s|$, where $\beta_s = (2s+\frac{3}{2})\theta - \frac{\pi}{4}$, since $4\sqrt{s/\pi}\,[2/(\pi(2s+1)\sin\theta)]^{1/2} \to 4/(\pi\sqrt{\sin\theta})$. If $\cos\beta_s$ tended to zero, the β_s would approach odd multiples of $\pi/2$, so their constant step $\beta_{s+1} - \beta_s = 2\theta$ would be a multiple of π; for $0 < \theta < \pi$ this happens only on the equator, $\theta = \pi/2$, where indeed every $P_{2s+1}(0) = 0$. Hence the termwise boundary series diverges at every angle off the equator, and the radial limit in the boxed σ is needed everywhere, not only at the poles.

To locate the actual singularity of σ, let Ω be a unit vector on the north hemisphere. The axis formula extends to arbitrary boundary data $f(\Omega')$ with the polar axis taken along Ω: average V over the rotations about that axis. The average is again harmonic outside the sphere and vanishes at infinity; its boundary data are the averages of f over the circles $\Omega\cdot\Omega' = u$; and on the axis it equals V, because these rotations fix the points $r\Omega$. In spherical coordinates (u, φ') about Ω, with $u = \Omega\cdot\Omega'$ the cosine of the polar angle, the averaged data are $\bar f(u) = \frac{1}{2\pi}\int_0^{2\pi} f\,d\varphi'$. Apply the axis formula to $\bar f$; its kernel depends on u only, so it may be taken inside the φ' integral:

$$\begin{aligned} V(r\Omega) &= \frac{\rho(1-\rho^2)}{2}\int_{-1}^{1}\frac{\bar f(u)\,du}{(1-2\rho u+\rho^2)^{3/2}} \\ &= \frac{\rho(1-\rho^2)}{4\pi}\int_{-1}^{1}\int_0^{2\pi}\frac{f\,d\varphi'\,du}{(1-2\rho u+\rho^2)^{3/2}}. \end{aligned}$$

Since $d\varphi'\,du$ is the solid-angle element $d\Omega'$ and $1 - 2\rho\,\Omega\cdot\Omega' + \rho^2 = |\rho\Omega - \Omega'|^2$, this is the exterior Poisson integral

$$V(r\Omega) = \frac{\rho(1-\rho^2)}{4\pi}\int\frac{f(\Omega')\,d\Omega'}{|\rho\Omega-\Omega'|^3}.$$

Split the boundary data into V_0 on the whole sphere minus $2V_0$ on the south. The constant data V_0 give $V = V_0 a/r = V_0\rho$, so, with $J(\rho) = \frac{1}{2\pi}\int_{\text{south}} d\Omega'/|\rho\Omega-\Omega'|^3$,

$$\frac{V}{V_0} = \rho - \frac{\rho(1-\rho^2)}{2\pi}\int_{\text{south}}\frac{d\Omega'}{|\rho\Omega-\Omega'|^3} = \rho - \rho(1-\rho^2)J(\rho).$$

At a point strictly north of the equator, $|\rho\Omega - \Omega'|$ stays away from zero for Ω' in the south, so J and J' remain finite as $\rho \uparrow 1$. By the product rule, $\frac{d}{d\rho}[\rho(1-\rho^2)J] = (1-3\rho^2)J + \rho(1-\rho^2)J'$, which equals $-2J(1)$ at $\rho = 1$. Using $-\partial_r = (\rho^2/a)\partial_\rho$ therefore gives $\sigma = (\varepsilon_0 V_0/a)[1 + 2J(1)]$, i.e.,

$$\sigma(\Omega) = \frac{\varepsilon_0 V_0}{a}\left[1 + \frac{1}{\pi}\int_{\text{south}}\frac{d\Omega'}{|\Omega-\Omega'|^3}\right].$$

At the north pole, $|\hat{\mathbf{e}}_z - \Omega'|^2 = 2 - 2u$ with $u = \cos\theta'$, and $\int_{\text{south}} d\Omega'/|\hat{\mathbf{e}}_z - \Omega'|^3 = 2\pi\int_{-1}^{0}(2-2u)^{-3/2}du = \pi(\sqrt{2}-1)$, which reproduces $\sigma(0) = \sqrt{2}\varepsilon_0 V_0/a$. Now let Ω lie at a small angular distance $h > 0$ north of the equator. Near the equatorial point closest to Ω, describe Ω' by its longitude X measured from that point and its angular distance $Y \ge 0$ south of the equator. For small X, Y, h the sphere is locally flat: $|\Omega - \Omega'|^2 \approx X^2 + (Y+h)^2$ and $d\Omega' = \cos Y\,dX\,dY \approx dX\,dY$.

The south hemisphere becomes the half-plane $Y > 0$, and the leading singular part of the integral is (using $\int_{-\infty}^{\infty} dX/(X^2+b^2)^{3/2} = 2/b^2$)

$$\int_0^\infty \int_{-\infty}^\infty \frac{dX\,dY}{[X^2+(Y+h)^2]^{3/2}} = \int_0^\infty \frac{2\,dY}{(Y+h)^2} = \frac{2}{h}.$$

To justify this local approximation, fix a small angular patch of radius δ about the closest equatorial point. The spherical area element and distance in that patch differ from their tangent-plane values by multiplicative factors tending to one as $\delta \to 0$. The integral outside the patch stays bounded for fixed δ. Inside, substitute $X = h\xi$, $Y = h\eta$: the area contributes h^2 and the denominator h^3. After multiplying by h, the patch expands to the whole half-plane and its integral tends to 2. The multiplicative bounds tend to one as $\delta \to 0$, proving

$$\sigma \sim \frac{2\varepsilon_0 V_0}{\pi a h} \qquad (h \downarrow 0).$$

An equatorial strip has area $2\pi a^2\,dh$ to leading order, so its charge grows as $4\varepsilon_0 V_0 a \int dh/h$. The hemisphere charge, and hence its ratio to the voltage difference $2V_0$, diverges logarithmically. This establishes the infinite ideal capacitance at the equator.

15.2.17 By writing $\varphi_s(x) = \sqrt{(2s+1)/2}\,P_s(x)$, a Legendre polynomial is renormalized to unity. Explain how $|\varphi_s\rangle\langle\varphi_s|$ acts as a projection operator. In particular, show that if $|f\rangle = \sum_n a'_n|\varphi_n\rangle$, then

$$|\varphi_s\rangle\langle\varphi_s|f\rangle = a'_s|\varphi_s\rangle.$$

Solution.

The normalization gives

$$\langle\varphi_s|\varphi_n\rangle = \frac{\sqrt{(2s+1)(2n+1)}}{2}\int_{-1}^1 P_s P_n\,dx = \frac{\sqrt{(2s+1)(2n+1)}}{2}\,\frac{2\delta_{sn}}{2n+1} = \delta_{sn}.$$

For $\Pi_s = |\varphi_s\rangle\langle\varphi_s|$, the scalar product $\langle\varphi_s|$ may be taken term by term in the convergent expansion of $|f\rangle$ (it is continuous, by the Schwarz inequality):

$$\Pi_s|f\rangle = |\varphi_s\rangle\sum_n a'_n\langle\varphi_s|\varphi_n\rangle = \boxed{a'_s|\varphi_s\rangle}.$$

It is a projection because

$$\Pi_s^2 = |\varphi_s\rangle\langle\varphi_s|\varphi_s\rangle\langle\varphi_s| = \Pi_s, \qquad \Pi_s^\dagger = \Pi_s.$$

Its image is the one-dimensional span of φ_s, and the discarded remainder is orthogonal to that span: $\langle\varphi_s, (1-\Pi_s)f\rangle = 0$. In the x representation its kernel is $\Pi_s(x,y) = \varphi_s(x)\varphi_s(y)^*$.

15.2.18 Expand x^8 as a Legendre series. Determine the Legendre coefficients from Eq. (15.40),

$$a_m = \frac{2m+1}{2}\int_{-1}^1 x^8 P_m(x)\,dx.$$

Check your values against AMS-55, Table 22.9. (For the complete reference, see Additional Readings.) This illustrates the expansion of a simple function $f(x)$.

Hint. Gaussian quadrature can be used to evaluate the integral.

Solution.

Parity makes every odd coefficient zero. Since x^8 has degree eight, coefficients above a_8 also vanish by Exercise 15.1.14. For $m = 2n$, $0 \le n \le 4$, the exact moment formula of Exercise 15.1.16 with $r = 4$ gives (as in Exercise 15.1.17(a))

$$a_{2n} = \frac{4n+1}{2}\int_{-1}^{1} x^8 P_{2n}(x)\,dx = \frac{2^{2n}(4n+1)8!(4+n)!}{(2n+9)!(4-n)!}.$$

Substitute $n = 0, 1, 2, 3, 4$ and cancel the factorials, using $8!/(2n+9)! = 1/[9\cdot 10\cdots(2n+9)]$:

$$a_0 = \frac{8!\,4!}{9!\,4!} = \frac{1}{9}, \qquad a_2 = \frac{4\cdot 5\cdot 8!\,5!}{11!\,3!} = \frac{20\cdot 20}{9\cdot 10\cdot 11} = \frac{40}{99},$$
$$a_4 = \frac{16\cdot 9\cdot 8!\,6!}{13!\,2!} = \frac{144\cdot 360}{9\cdot 10\cdot 11\cdot 12\cdot 13} = \frac{48}{143}, \qquad a_6 = \frac{64\cdot 13\cdot 8!\,7!}{15!} = \frac{832\cdot 5040}{9\cdot 10\cdots 15} = \frac{64}{495},$$
$$a_8 = \frac{256\cdot 17\cdot 8!\,8!}{17!} = \frac{4352\cdot 40320}{9\cdot 10\cdots 17} = \frac{128}{6435}.$$

In summary,

m	0	2	4	6	8
a_m	1/9	40/99	48/143	64/495	128/6435

Over the common denominator $6435 = 9\cdot 715 = 99\cdot 65 = 143\cdot 45 = 495\cdot 13$ this is

$$\boxed{x^8 = \frac{715P_0 + 2600P_2 + 2160P_4 + 832P_6 + 128P_8}{6435}}.$$

Expansion of the five polynomials cancels all powers below eight and leaves coefficient one on x^8 (the leading coefficient of P_8 is $6435/128$), providing a separate exact check. Following the hint, Gauss–Legendre quadrature with nine or more nodes reproduces these coefficients exactly (apart from rounding), since $x^8 P_m$ has degree at most $16 \le 2\cdot 9 - 1$. The x^8 column of AMS-55, Table 22.9, p. 798 lists denominator 6435 and numerators $715, 2600, 2160, 832, 128$ in its P_0, P_2, P_4, P_6, P_8 rows, respectively. These agree with all five coefficients derived above.

15.2.19 Calculate and tabulate the electrostatic potential created by a ring of charge, Example 15.2.3, for $r/a = 1.5(0.5)5.0$ and $\theta = 0°(15°)90°$. Carry terms through $P_{22}(\cos\theta)$.

Note. The convergence of your series will be slow for $r/a = 1.5$. Truncating the series at P_{22} limits you to about four-significant-figure accuracy.

Check value. For $r/a = 2.5$ and $\theta = 60°$, $\psi = 0.40272(q/4\pi\varepsilon_0 a)$.

Solution.

Set $R = r/a$ and report the dimensionless potential $F = 4\pi\varepsilon_0 a\,\psi/q$. The exterior expansion of Example 15.2.3, $\psi = (q/4\pi\varepsilon_0 r)\sum_s P_{2s}(0)(a/r)^{2s}P_{2s}(\cos\theta)$, with $P_{2s}(0) = (-1)^s(2s-1)!!/(2s)!! = (-1)^s\binom{2s}{s}/4^s$, gives the required truncation

$$F_{22}(R,\theta) = \sum_{s=0}^{11}(-1)^s\frac{\binom{2s}{s}}{4^s}\frac{P_{2s}(\cos\theta)}{R^{2s+1}}.$$

For each angle start $P_0 = 1$, $P_1 = \cos\theta$ and use $P_{n+1} = [(2n+1)\cos\theta\,P_n - nP_{n-1}]/(n+1)$ up to P_{22}. Summing the displayed twelve terms (ordinary double precision suffices) gives the following values, rounded to six decimals; the six decimals describe the truncated sum F_{22}, not a uniform six-decimal accuracy claim for the exact potential.

R	0°	15°	30°	45°	60°	75°	90°
1.5	0.554696	0.560829	0.579888	0.613868	0.665263	0.729928	0.768045
2.0	0.447214	0.450895	0.461951	0.480116	0.503626	0.526419	0.536591
2.5	0.371391	0.373649	0.380258	0.390536	0.402722	0.413298	0.417623
3.0	0.316228	0.317677	0.321844	0.328102	0.335155	0.340944	0.343220
3.5	0.274721	0.275694	0.278459	0.282517	0.286950	0.290473	0.291829
4.0	0.242536	0.243216	0.245133	0.247901	0.250864	0.253173	0.254050
4.5	0.216930	0.217422	0.218801	0.220769	0.222847	0.224443	0.225045
5.0	0.196116	0.196483	0.197504	0.198952	0.200464	0.201615	0.202046

For example, $F_{22}(2.5, 60°) = 0.402721525118444\ldots$. The printed check value uses the wrong length in its prefactor: 0.40272 multiplies $q/(4\pi\varepsilon_0 a)$, not $q/(4\pi\varepsilon_0 r)$. With the latter prefactor the coefficient is $RF = 1.00680381279611\ldots$. A simple rigorous truncation bound follows from $|P_n| \le 1$ and $\binom{2s}{s}/4^s \le 1$, by summing the geometric series:

$$|F - F_{22}| \le \sum_{s=12}^{\infty} R^{-2s-1} = \frac{R^{-25}}{1 - R^{-2}}.$$

It is 7.1×10^{-5} at $R = 1.5$ and 4.0×10^{-8} at $R = 2$, which explains the four-figure warning of the note for $R = 1.5$. As an independent check, the exact potential follows from Coulomb's law: the ring element $q\,d\varphi'/2\pi$ at $(a\cos\varphi', a\sin\varphi', 0)$ is at distance $a(R^2 + 1 - 2R\sin\theta\cos\varphi')^{1/2}$ from the field point $(r\sin\theta, 0, r\cos\theta)$, so

$$F(R,\theta) = \frac{1}{2\pi}\int_0^{2\pi} \frac{d\varphi'}{(R^2 + 1 - 2R\sin\theta\cos\varphi')^{1/2}}.$$

For this smooth periodic integrand the trapezoidal rule with 64 equally spaced φ' is accurate to about 10^{-12} on the whole grid; on the axis, $F(R,0) = (1+R^2)^{-1/2}$ exactly. The largest difference between F and F_{22} on the grid is at $R = 1.5$, $\theta = 0°$, where $F = 1/\sqrt{3.25} = 0.554700$ exceeds F_{22} by 4.5×10^{-6}, within the bound; for $R \ge 2$ the differences are below 4×10^{-9}.

15.2.20 Calculate and tabulate the electrostatic potential created by a charged disk (Exercise 15.2.14), for $r/a = 1.5(0.5)5.0$ and $\theta = 0°(15°)90°$. Carry terms through $P_{22}(\cos\theta)$.

Check value. For $r/a = 2.0$ and $\theta = 15°$, $\psi = 0.46638(q/4\pi\varepsilon_0 a)$.

Solution.

Use the same normalization as in Exercise 15.2.19, $F = 4\pi\varepsilon_0 a\,\psi/q$ with $R = r/a$. Multiplying the answer of Exercise 15.2.14 by $4\pi\varepsilon_0 a/q$ turns its factor $q(a/r)^{2l}/(4\pi\varepsilon_0 r)$ into $(a/r)^{2l+1} = R^{-2l-1}$; keeping $l = 0, \ldots, 11$ (through P_{22}) and writing s for l gives the twelve-term approximation

$$F_{22}(R,\theta) = \sum_{s=0}^{11} \frac{(-1)^s}{2s+1}\frac{P_{2s}(\cos\theta)}{R^{2s+1}}.$$

The same three-term polynomial recurrence used in Exercise 15.2.19 evaluates all the needed polynomials; only the multipole weights change, from $(-1)^s\binom{2s}{s}/4^s$ to $(-1)^s/(2s+1)$. Summing the twelve terms (ordinary double precision suffices) and rounding to six decimals gives the requested grid:

R	0°	15°	30°	45°	60°	75°	90°
1.5	0.588001	0.592890	0.607783	0.633185	0.668769	0.708689	0.729727
2.0	0.463648	0.466381	0.474451	0.487268	0.503036	0.517445	0.523599
2.5	0.380506	0.382123	0.386792	0.393878	0.401998	0.408801	0.411517
3.0	0.321751	0.322766	0.325659	0.329924	0.334617	0.338380	0.339837
3.5	0.278300	0.278973	0.280872	0.283620	0.286569	0.288874	0.289752
4.0	0.244979	0.245445	0.246752	0.248620	0.250592	0.252108	0.252680
4.5	0.218669	0.219005	0.219940	0.221264	0.222647	0.223699	0.224093
5.0	0.197396	0.197644	0.198335	0.199308	0.200314	0.201074	0.201358

At the check point, $F_{22}(2, 15°) = 0.466380536818770\ldots$, which rounds to the stated 0.46638. Again the printed numerical coefficient multiplies $q/(4\pi\varepsilon_0 a)$. If the prefactor is instead $q/(4\pi\varepsilon_0 r)$, its coefficient is $RF = 0.932761073637540\ldots$. For all the tabulated points, $|P_{2s}(\cos\theta)| \le 1$ and $1/(2s+1) \le 1/25$ for $s \ge 12$, so summing the geometric series gives

$$|F - F_{22}| \le \sum_{s=12}^{\infty} \frac{R^{-2s-1}}{2s+1} \le \frac{1}{25} \sum_{s=12}^{\infty} R^{-2s-1} = \frac{R^{-25}}{25(1-R^{-2})}.$$

The bound is 2.9×10^{-6} at $R = 1.5$ and 1.6×10^{-9} at $R = 2$. Thus for $R \ge 2$ the truncation error is far below the six-decimal rounding, while at $R = 1.5$ it can change the last displayed digit; as in Exercise 15.2.19, the table describes the truncated sum F_{22}.

For an independent check, two columns of the table, the axis ($\theta = 0°$) and the plane of the disk ($\theta = 90°$), can be summed in closed form for the exact potential. On the axis $P_{2s}(1) = 1$, and in the plane of the disk $(-1)^s P_{2s}(0) = \binom{2s}{s}/4^s$, so the Maclaurin series of arctan and arcsin give

$$F(R, 0°) = \sum_{s=0}^{\infty} \frac{(-1)^s}{(2s+1)R^{2s+1}} = \arctan\frac{1}{R}, \qquad F(R, 90°) = \sum_{s=0}^{\infty} \binom{2s}{s} \frac{R^{-2s-1}}{4^s(2s+1)} = \arcsin\frac{1}{R}.$$

For the second, $\arcsin y = \int_0^y (1-t^2)^{-1/2} dt$; the binomial series with $\binom{-1/2}{s} = (-1)^s\binom{2s}{s}/4^s$ (Exercise 15.1.2) gives $(1-t^2)^{-1/2} = \sum_s \binom{2s}{s} t^{2s}/4^s$ for $|t| < 1$, and integrating term by term gives $\arcsin y = \sum_s \binom{2s}{s} y^{2s+1}/[4^s(2s+1)]$ for $|y| < 1$; put $y = 1/R$. At other angles, add the potentials of the rings that make up the disk. With $\rho = a\sin t$, the annulus charge of Exercise 15.2.14, $dq = (q/a)\rho\, d\rho/\sqrt{a^2-\rho^2}$, becomes $dq = q \sin t\, dt$. As in the Coulomb integral of Exercise 15.2.19, the element of this ring at azimuth φ' is at distance $a(R^2 + \sin^2 t - 2R\sin t \sin\theta\cos\varphi')^{1/2}$ from the field point, so

$$F(R, \theta) = \int_0^{\pi/2} \frac{\sin t}{2\pi} \int_0^{2\pi} \frac{d\varphi'\, dt}{(R^2 + \sin^2 t - 2R\sin t\sin\theta\cos\varphi')^{1/2}}.$$

The substitution has removed the edge singularity of σ, so the integrand is smooth. A 20-point Gauss–Legendre rule in t and the 64-point trapezoidal rule in φ' agree with the two closed forms to about 10^{-13}, and doubling both numbers of nodes changes no value by more than that. The largest difference between F and F_{22} on the grid is at $R = 1.5$, $\theta = 0°$, where $F = \arctan(2/3) = 0.588003$ exceeds F_{22} by 1.1×10^{-6}, within the bound; for $R \ge 2$ the differences are below 10^{-9}.

15.2.21 Calculate the first five (nonvanishing) coefficients in the Legendre series expansion of $f(x) = 1 - |x|$, evaluating the coefficients in the series by numerical integration. Actually these coefficients can be obtained in closed form. Compare your coefficients with those listed in Exercise 18.4.26.

ANS. $a_0 = 0.5000$, $a_2 = -0.6250$, $a_4 = 0.1875$, $a_6 = -0.1016$, $a_8 = 0.0664$.

Solution.

The coefficients are $a_k = \frac{2k+1}{2}\int_{-1}^{1} f(x)P_k(x)\,dx$. Since f is even, the integrand is odd for odd k, so $a_{2n+1} = 0$; for $k = 2n$ it is even, and the integral is twice that over $[0,1]$, where $f = 1 - x$:

$$a_{2n} = (4n+1)\int_0^1 (1-x)P_{2n}(x)\,dx.$$

For the numerical integration note that $(1-x)P_{2n}(x)$ is a polynomial of degree $2n+1 \le 9$ when $2n \le 8$. The 9-point Gauss–Legendre rule, with the standard nodes ξ_i and weights w_i of $[-1,1]$ mapped to $[0,1]$ (nodes $(1+\xi_i)/2$, weights $w_i/2$), is exact through degree 17 (an N-point rule is exact through degree $2N-1$, so five nodes would already suffice), so it gives these integrals up to rounding error:

$2n$	a_{2n} (9-point rule)	exact a_{2n}
0	0.5000000000	$1/2$
2	−0.6250000000	$-5/8$
4	0.1875000000	$3/16$
6	−0.1015625000	$-13/128$
8	0.0664062500	$17/256$

The numerical values agree with the exact fractions derived next.

For the closed form, $a_0 = \int_0^1 (1-x)dx = 1/2$. For $n \ge 1$, $\int_0^1 P_{2n}dx = 0$ (Exercise 15.1.12), so $a_{2n} = -(4n+1)\int_0^1 xP_{2n}dx$. Write $I_k = \int_0^1 P_k\,dx$. The three-term recurrence gives $xP_{2n} = [(2n+1)P_{2n+1} + 2nP_{2n-1}]/(4n+1)$, and Exercise 15.1.12 gives $I_{2n+1} = P_{2n}(0)/(2n+2)$ and $I_{2n-1} = P_{2n-2}(0)/(2n)$. The recurrence $P_{2n}(0) = -\frac{2n-1}{2n}P_{2n-2}(0)$ used there turns the latter into $I_{2n-1} = -P_{2n}(0)/(2n-1)$. Hence

$$\begin{aligned}\int_0^1 xP_{2n}dx &= \frac{(2n+1)I_{2n+1} + 2nI_{2n-1}}{4n+1} = \frac{P_{2n}(0)}{4n+1}\left[\frac{2n+1}{2n+2} - \frac{2n}{2n-1}\right] \\ &= \frac{P_{2n}(0)}{4n+1}\,\frac{(4n^2-1)-(4n^2+4n)}{(2n+2)(2n-1)} = -\frac{P_{2n}(0)}{(2n-1)(2n+2)}.\end{aligned}$$

Therefore, with $P_{2n}(0) = (-1)^n(2n-1)!!/(2n)!!$,

$$a_{2n} = \frac{(4n+1)P_{2n}(0)}{(2n-1)(2n+2)} = (-1)^n(4n+1)\frac{(2n-1)!!}{(2n-1)\,(2n)!!\,(2n+2)},$$

and $(2n-1)!!/(2n-1) = (2n-3)!!$, $(2n)!!\,(2n+2) = (2n+2)!!$ give

$$\boxed{a_{2n} = (-1)^n(4n+1)\frac{(2n-3)!!}{(2n+2)!!}}, \quad n \ge 1,$$

with $(-1)!! = 1$. For $n = 1,\ldots,4$ this gives $-5/8$, $3/16$, $-13/128$, $17/256$, the exact column of the table. The expansion of $|x|$ in Exercise 18.4.26(a) has constant $1/2$ and the opposite nonconstant coefficients, exactly as $1 - |x|$ requires.

15.2.22 Calculate and tabulate the exterior electrostatic potential created by the two charged hemispheres of Exercise 15.2.16, for $r/a = 1.5(0.5)5.0$ and $\theta = 0°(15°)90°$. Carry terms through $P_{23}(\cos\theta)$.

Check value. For $r/a = 2.0$ and $\theta = 45°$, $V = 0.27066V_0$.

Solution.

Write $R = r/a$ and $G = V/V_0$. In the exterior solution of Exercise 15.2.16(a), $(2s-1)!!/(2s+2)!! = [(2s-1)!!/(2s)!!]/(2s+2) = \binom{2s}{s}/[4^s(2s+2)]$ (as in Exercise 15.2.19, $(2s-1)!!/(2s)!! = \binom{2s}{s}/4^s$), and P_{23} is the term $s = 11$. Keeping $s = 0, \ldots, 11$ gives

$$G_{23}(R,\theta) = \sum_{s=0}^{11} (-1)^s \frac{(4s+3)\binom{2s}{s}}{4^s(2s+2)} \frac{P_{2s+1}(\cos\theta)}{R^{2s+2}}.$$

The polynomial recurrence of Exercise 15.2.19 is evaluated through P_{23}, and the twelve contributions are summed (ordinary double precision suffices). Rounded to six decimals:

R	$0°$	$15°$	$30°$	$45°$	$60°$	$75°$	$90°$
1.5	0.537744	0.532551	0.515181	0.478972	0.406992	0.260331	0.000000
2.0	0.329180	0.323560	0.305432	0.270657	0.211743	0.120305	0.000000
2.5	0.220080	0.215255	0.200089	0.172575	0.129952	0.070921	0.000000
3.0	0.156726	0.152795	0.140654	0.119398	0.088122	0.047211	0.000000
3.5	0.116968	0.113786	0.104077	0.087472	0.063803	0.033839	0.000000
4.0	0.090491	0.087897	0.080046	0.066831	0.048383	0.025503	0.000000
4.5	0.072020	0.069879	0.063440	0.052724	0.037977	0.019937	0.000000
5.0	0.058643	0.056853	0.051498	0.042657	0.030616	0.016027	0.000000

The equatorial values are exactly zero because all retained polynomials are odd, so $P_{2s+1}(0) = 0$. The check point gives

$$G_{23}(2, 45°) = 0.270656664912169\ldots,$$

which rounds to the stated 0.27066. Since $\binom{2s}{s}/4^s \le 1$, $(4s+3)/(2s+2) < 2$ and $|P_{2s+1}| \le 1$, summing the geometric tail gives

$$|G - G_{23}| \le 2\sum_{s=12}^{\infty} R^{-2s-2} = \frac{2R^{-26}}{1-R^{-2}}.$$

This is 9.5×10^{-5} at $R = 1.5$, 4.0×10^{-8} at $R = 2$ and 1.1×10^{-10} at $R = 2.5$. The table therefore shows the requested truncation G_{23}, which for $R = 1.5$ need not agree with G to six decimals. For an independent check, the axis values are known in closed form from Exercise 15.2.16: $G(R, 0°) = 1 - (1 - R^{-2})/\sqrt{1 + R^{-2}}$. At $R = 1.5$ this is 0.537750, which exceeds $G_{23}(1.5, 0°)$ by 5.8×10^{-6}, within the bound; at $R = 2$ the difference is 3.8×10^{-9}.

15.2.23

(a) Given $f(x) = 2.0$, $|x| < 0.5$ and $f(x) = 0$, $0.5 < |x| < 1.0$, expand $f(x)$ in a Legendre series and calculate the coefficients a_n through a_{80} (analytically).

(b) Evaluate $\displaystyle\sum_{n=0}^{80} a_n P_n(x)$ for $x = 0.400(0.005)0.600$. Plot your results.

Note. This illustrates the Gibbs phenomenon of Section 19.3 and the danger of trying to calculate with a series expansion in the vicinity of a discontinuity.

Solution.

(a) With $a_n = \frac{2n+1}{2}\int_{-1}^{1} f(x)P_n(x)\,dx$, and $f = 2$ only on $|x| < 1/2$,

$$a_n = (2n+1)\int_{-1/2}^{1/2} P_n(x)dx.$$

Every odd coefficient is zero because P_n is then odd, and $a_0 = \int_{-1/2}^{1/2} dx = 1$. For even $n \geq 2$, integrate the recurrence $(2n+1)P_n = P'_{n+1} - P'_{n-1}$. Since $n \pm 1$ are odd, $P_{n\pm1}(-1/2) = -P_{n\pm1}(1/2)$, and

$$a_n = \Big[P_{n+1}(x) - P_{n-1}(x)\Big]_{-1/2}^{1/2} = \boxed{2[P_{n+1}(1/2) - P_{n-1}(1/2)] \quad (n = 2, 4, \ldots, 80)}.$$

This is an exact analytic formula for all the requested coefficients. For an expression involving only integers and powers of two, put $x = 1/2$ in the closed formula of Exercise 15.1.2, where $2^{-j}(1/2)^{j-2k} = 2^{-(2j-2k)}$:

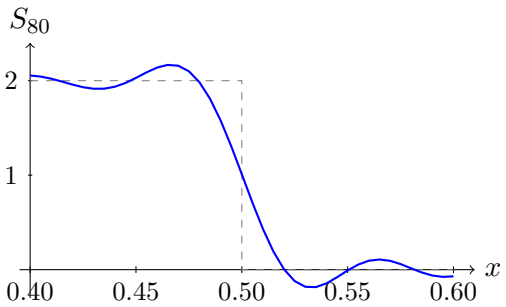
$$P_j(1/2) = \sum_{k=0}^{\lfloor j/2 \rfloor} \frac{(-1)^k (2j-2k)!}{2^{2j-2k} k!(j-k)!(j-2k)!}.$$

For example, $P_1(1/2) = 1/2$ and $P_3(1/2) = (5/8 - 3/2)/2 = -7/16$, so $a_2 = 2(-7/16 - 1/2) = -15/8$. All 41 nonzero coefficients ($n = 0, 2, \ldots, 80$) were calculated as exact rational numbers from these polynomial values. Their decimal displays are:

n	a_n	n	a_n
0	1.000000000	42	0.115695743
2	-1.875000000	44	-0.429575813
4	1.054687500	46	0.309464967
6	0.266601563	48	0.108557716
8	-0.982086182	50	-0.403296282
10	0.663539886	52	0.291080519
12	0.203865767	54	0.102595591
14	-0.751645535	56	-0.381317431
16	0.524470681	58	0.275626635
18	0.171100741	60	0.097518496
20	-0.632362136	62	-0.362580937
22	0.447317591	64	0.262399140
24	0.150272358	66	0.093127210
26	-0.556355477	68	-0.346360592
28	0.396552617	70	0.250910070
30	0.135556642	72	0.089280100
32	-0.502500382	74	-0.332139273
34	0.359902602	76	0.240809175
36	0.124454578	78	0.085873358
38	-0.461771439	80	-0.319537401
40	0.331841571		

(b) For each specified rational x, compute $P_0 = 1$, $P_1 = x$, then $P_{n+1} = [(2n+1)xP_n - nP_{n-1}]/(n+1)$ up to P_{80}, and sum $S_{80} = \sum_{n=0}^{80} a_n P_n$. Exact rational arithmetic avoids cancellation error; only the final display is rounded.

x	$S_{80}(x)$	x	$S_{80}(x)$
0.400	2.055021750	0.505	0.710618291
0.405	2.043444528	0.510	0.430206382
0.410	2.020966606	0.515	0.189030075
0.415	1.990922807	0.520	0.002436961
0.420	1.958725538	0.525	-0.121105466
0.425	1.931073999	0.530	-0.181043732
0.430	1.914789974	0.535	-0.184550845
0.435	1.915445056	0.540	-0.145134072
0.440	1.936002637	0.545	-0.080266216
0.445	1.975722662	0.550	-0.008445837
0.450	2.029559995	0.555	0.053799979
0.455	2.088226666	0.560	0.094441298
0.460	2.138990947	0.565	0.107554625
0.465	2.167165489	0.570	0.093742176
0.470	2.158112203	0.575	0.059315097
0.475	2.099484848	0.580	0.014435104
0.480	1.983362297	0.585	-0.029385877
0.485	1.807911778	0.590	-0.061902155
0.490	1.578269431	0.595	-0.076321894
0.495	1.306431978	0.600	-0.070616739
0.500	1.010104145		



The dashed line is the original step function; the solid line connects all 41 requested computed values. On this grid the largest value is $S_{80}(0.465) = 2.1671654888\ldots$ and the smallest is $S_{80}(0.535) = -0.1845508446\ldots$. At the jump the partial sums tend to the midpoint value 1: the same computation gives $S_N(1/2) = 1.0386$, 0.9893, 1.0101, 0.9972, 1.0026 for $N = 20$, 40, 80, 160, 320. The overshoot near the jump, however, does not die out. On the grid $x = 0.40000(0.00001)0.50000$ (the same recurrence, in double precision) the maximum of S_{80} is 2.1690, near $x = 0.4665$, about 8.4% of the jump above $f = 2$; for larger N the maximum moves toward $x = 1/2$, while the overshoot approaches the Gibbs fraction of about 9% of the jump found for Fourier series in Section 19.3 (S_{640} peaks at 2.1797 on the same grid). This is the Gibbs phenomenon; it is not a failure of the exact coefficient calculation.

15.2.24 A plane wave may be expanded in a series of spherical waves by the Rayleigh equation,

$$e^{ikr\cos\gamma} = \sum_{n=0}^{\infty} a_n j_n(kr) P_n(\cos\gamma).$$

Show that $a_n = i^n(2n+1)$.

Hint.

1. Use the orthogonality of the P_n to solve for $a_n j_n(kr)$.

2. Differentiate n times with respect to (kr) and set $r = 0$ to eliminate the r-dependence.

3. Evaluate the remaining integral by Exercise 15.1.15.

Note. This problem may also be treated by noting that both sides of the equation satisfy the Helmholtz equation. The equality can be established by showing that the solutions have the same behavior at the origin and also behave alike at large distances.

Solution.

Put $z = kr$ and $x = \cos\gamma$. Multiply the proposed expansion by $P_n(x)$ and integrate over $-1 \le x \le 1$; by orthogonality, $\int_{-1}^{1} P_m P_n\, dx = 2\delta_{mn}/(2n+1)$, only the term $m = n$ survives:

$$a_n j_n(z) = \frac{2n+1}{2} \int_{-1}^{1} e^{izx} P_n(x) dx.$$

Expand $e^{izx} = \sum_{j\ge 0} (iz)^j x^j / j!$; the series converges uniformly for $-1 \le x \le 1$, so it may be integrated term by term. All powers x^j with $j < n$ have zero integral against P_n by Exercise 15.1.14, and for $j = n+1$ the integrand $x^{n+1} P_n$ is odd. With the moment $\int_{-1}^{1} x^n P_n\, dx = 2 \cdot n!/(2n+1)!!$ of Exercise 15.1.15, the right-hand side is therefore

$$\frac{2n+1}{2} \frac{(iz)^n}{n!} \int_{-1}^{1} x^n P_n(x) dx + O(z^{n+2}) = \frac{(2n+1) i^n z^n}{(2n+1)!!} + O(z^{n+2}).$$

The power series of j_n (Section 14.7), $j_n(z) = \sum_{k\ge 0} (-1)^k z^{n+2k} / [2^k k!\,(2n+2k+1)!!]$, gives $j_n(z) = z^n/(2n+1)!! + O(z^{n+2})$, so the left-hand side is $a_n z^n/(2n+1)!! + O(z^{n+2})$. Comparing the coefficients of z^n gives

$$\boxed{a_n = i^n (2n+1)}.$$

This is the procedure of the hint: differentiating both sides n times and setting $z = 0$ removes all other terms and leaves $a_n\, n!/(2n+1)!! = (2n+1) i^n\, n!/(2n+1)!!$; the common factor $n!/(2n+1)!!$ cancels. Comparing power-series coefficients avoids any division by $j_n(z)$, which has zeros. That the series with these coefficients does sum to e^{izx} is proved by the recurrence argument in Exercise 15.2.25; Exercise 15.2.27 checks the coefficients independently, by evaluating $\frac{2n+1}{2} \int_{-1}^{1} e^{izx} P_n\, dx = i^n (2n+1) j_n(z)$ directly.

15.2.25 Verify the Rayleigh equation of Exercise 15.2.24 by starting with the following steps:

(a) Differentiate with respect to (kr) to establish

$$\sum_n a_n j_n'(kr) P_n(\cos\gamma) = i \sum_n a_n j_n(kr) \cos\gamma\, P_n(\cos\gamma).$$

(b) Use a recurrence relation to replace $\cos\gamma\, P_n(\cos\gamma)$ by a linear combination of P_{n-1} and P_{n+1}.

(c) Use a recurrence relation to replace j_n' by a linear combination of j_{n-1} and j_{n+1}.

Solution.

Again put $z = kr$, $x = \cos\gamma$, and $F = \sum a_n j_n(z) P_n(x)$. (a) If $F = e^{izx}$, then $F_z = ixF$. Differentiating the series term by term (assumed in this derivation of the a_n; for the coefficients

found, it is justified in the sufficiency argument below) gives $F_z = \sum_n a_n j_n'(z) P_n(x)$, while $ixF = i\sum_n a_n j_n(z)\, xP_n(x)$; this is the displayed equation.

(b) Use the three-term recurrence $xP_n = [(n+1)P_{n+1} + nP_{n-1}]/(2n+1)$:

$$i\sum_n a_n j_n\, xP_n = i\sum_n a_n j_n \left[\frac{n+1}{2n+1}P_{n+1} + \frac{n}{2n+1}P_{n-1}\right].$$

A given P_m comes from $n = m-1$ in the first term, with factor $(n+1)/(2n+1) = m/(2m-1)$, and from $n = m+1$ in the second, with factor $n/(2n+1) = (m+1)/(2m+3)$. So the coefficient of P_m on the right is

$$i\left[\frac{m\, a_{m-1}}{2m-1} j_{m-1} + \frac{(m+1)a_{m+1}}{2m+3} j_{m+1}\right].$$

The first term is absent for $m = 0$.

(c) The spherical Bessel recurrence $mj_{m-1} - (m+1)j_{m+1} = (2m+1)j_m'$ of Exercise 14.7.5 gives

$$j_m' = \frac{m}{2m+1} j_{m-1} - \frac{m+1}{2m+1} j_{m+1}$$

for $m \geq 1$; at $m = 0$, where the coefficient of j_{m-1} vanishes, it reads $j_0' = -j_1$. Multiplying both sides of (a) by P_m and integrating over $-1 \leq x \leq 1$ (again term by term), orthogonality shows that the coefficient $a_m j_m'$ of P_m on the left must equal the expression in (b). Inserting the recurrence and collecting terms,

$$\left[\frac{m\, a_m}{2m+1} - \frac{i\, m\, a_{m-1}}{2m-1}\right] j_{m-1} = \left[\frac{(m+1)a_m}{2m+1} + \frac{i(m+1)a_{m+1}}{2m+3}\right] j_{m+1}.$$

Near $z = 0$, j_{m-1} and j_{m+1} begin with the different powers z^{m-1} and z^{m+1}, so they are linearly independent and both brackets vanish (for $m = 0$ only the right one occurs). Dividing by m and by $m+1$:

$$\frac{a_m}{2m+1} = \frac{i\, a_{m-1}}{2m-1}, \qquad -\frac{a_m}{2m+1} = \frac{i\, a_{m+1}}{2m+3}.$$

The first with m replaced by $m+1$ reads $a_{m+1}/(2m+3) = ia_m/(2m+1)$; the second, multiplied by $-i$ (and $-i \cdot i = 1$), reads the same. So both give the single recurrence $a_{m+1} = i(2m+3)a_m/(2m+1)$. Setting $z = 0$ in the expansion, where $j_n(0) = \delta_{n0}$ and $P_0 = 1$, gives $a_0 = 1$; and if $a_m = i^m(2m+1)$, then $a_{m+1} = i(2m+3)i^m = i^{m+1}(2m+3)$. By induction $a_n = i^n(2n+1)$.

For sufficiency, insert these coefficients in F. The spherical Bessel power series supplies a concrete uniform bound: for $|z| \leq M$, using $(2n+2k+1)!! \geq (2n+1)!!$,

$$|j_n(z)| \leq \sum_{k=0}^{\infty} \frac{M^{n+2k}}{2^k k!(2n+2k+1)!!} \leq \frac{M^n}{(2n+1)!!} \sum_{k=0}^{\infty} \frac{(M^2/2)^k}{k!} = \frac{M^n}{(2n+1)!!} e^{M^2/2}.$$

After multiplication by $2n+1$, the ratio of consecutive bounds is $M(2n-1)!!/(2n+1)!! = M/(2n+1) \to 0$. Thus, by the ratio test and the Weierstrass M test, $\sum(2n+1)|j_n(z)|$ converges uniformly on that disk. The derivative recurrence above gives $|j_n'| \leq |j_{n-1}| + |j_{n+1}|$, hence the same conclusion for $\sum(2n+1)|j_n'(z)|$, with $j_0' = -j_1$ handled separately. Together with $|P_n(x)| \leq 1$, these bounds justify differentiating F term by term for $-1 \leq x \leq 1$. The calculation above, read backwards, proves $F_z = ixF$. At zero, $j_0(0) = 1$ and $j_n(0) = 0$ for $n > 0$, so $F(0,x) = 1$. For fixed x, $\partial_z[e^{-izx}F] = e^{-izx}(F_z - ixF) = 0$, so $F(z,x) = F(0,x)e^{izx} = e^{izx}$, verifying the Rayleigh identity for all finite z.

15.2.26 From Exercise 15.2.24 show that

$$j_n(kr) = \frac{1}{2i^n}\int_{-1}^{1} e^{ikr\mu}P_n(\mu)\,d\mu.$$

This means that (apart from a constant factor) the spherical Bessel function $j_n(kr)$ is an integral transform of the Legendre polynomial $P_n(\mu)$.

Solution.

Write the Rayleigh expansion of Exercise 15.2.24 with $z = kr$, $\mu = \cos\gamma$ and $a_n = i^n(2n+1)$, multiply it by $P_m(\mu)$ and integrate:

$$\int_{-1}^{1} e^{iz\mu}P_m(\mu)d\mu = \sum_{n\geq 0} i^n(2n+1)j_n(z)\frac{2\delta_{nm}}{2n+1} = 2i^m j_m(z).$$

Only $n = m$ survives, by the orthogonality relation $\int_{-1}^{1} P_nP_m\,d\mu = 2\delta_{nm}/(2n+1)$. Uniform convergence for $\mu \in [-1,1]$ and fixed z (Exercise 15.2.25: $\sum(2n+1)|j_n(z)|$ converges and $|P_n| \leq 1$) justifies the interchange of sum and integral. Divide by $2i^m$ and rename the index:

$$\boxed{j_n(z) = \frac{1}{2i^n}\int_{-1}^{1} e^{iz\mu}P_n(\mu)d\mu}.$$

The integral is entire in z, so the formula also holds for complex z and at zero, with the usual removable values of j_n.

15.2.27 Rewriting the formula of Exercise 15.2.26 as

$$j_n(z) = \frac{1}{2}(-i)^n\int_0^{\pi} e^{iz\cos\theta}P_n(\cos\theta)\sin\theta\,d\theta, \qquad n = 0,1,2,\ldots,$$

verify it by transforming the right-hand side into

$$\frac{z^n}{2^{n+1}n!}\int_0^{\pi}\cos(z\cos\theta)\sin^{2n+1}\theta\,d\theta$$

and using Exercise 14.7.9.

Solution.

Call the right-hand side I. Put $x = \cos\theta$: then $dx = -\sin\theta\,d\theta$, and $\theta = 0, \pi$ become $x = 1, -1$, so reversing the limits absorbs the minus sign. Then insert Rodrigues' formula, $P_n = D^n(x^2-1)^n/(2^n n!)$ with $D = d/dx$:

$$I = \frac{(-i)^n}{2}\int_{-1}^{1} e^{izx}P_n(x)dx = \frac{(-i)^n}{2^{n+1}n!}\int_{-1}^{1} e^{izx}D^n(x^2-1)^n dx.$$

Integrate by parts n times, moving one D at a time onto the exponential. As in Exercise 15.1.14, $D^k(x^2-1)^n$ vanishes at $x = \pm 1$ for $k < n$, so every boundary term vanishes, and since $De^{izx} = ize^{izx}$ each step contributes a factor $-iz$:

$$\int_{-1}^{1} e^{izx}D^n(x^2-1)^n dx = (-iz)^n\int_{-1}^{1} e^{izx}(x^2-1)^n dx.$$

Since $(-i)^n(-i)^n = (-1)^n$ and $(x^2-1)^n = (-1)^n(1-x^2)^n$, the two signs cancel:

$$I = \frac{z^n}{2^{n+1}n!}\int_{-1}^{1} e^{izx}(1-x^2)^n dx.$$

Write $e^{izx} = \cos zx + i\sin zx$. The sine part is odd in x and integrates to zero, even for complex z. Returning to θ in the cosine part ($1-x^2 = \sin^2\theta$, $dx \to \sin\theta\, d\theta$ as before) gives exactly

$$I = \frac{z^n}{2^{n+1}n!}\int_0^{\pi} \cos(z\cos\theta)\sin^{2n+1}\theta\, d\theta,$$

the Poisson integral that Exercise 14.7.9 identifies with $j_n(z)$. To verify the spherical Bessel value directly, expand $\cos zx = \sum_k (-1)^k z^{2k}x^{2k}/(2k)!$ (uniformly convergent on $[-1,1]$) and integrate each term; with $u = x^2$ on the even integrand,

$$\int_{-1}^{1} x^{2k}(1-x^2)^n dx = \int_0^1 u^{k-1/2}(1-u)^n\, du = B(k+\tfrac{1}{2}, n+1) = \frac{\Gamma(k+1/2)n!}{\Gamma(k+n+3/2)}.$$

With $\Gamma(k+\frac{1}{2}) = (2k)!\sqrt{\pi}/(4^k k!)$ and $\Gamma(k+n+\frac{3}{2}) = (2n+2k+1)!!\sqrt{\pi}/2^{k+n+1}$, the kth term of I is

$$\frac{z^n}{2^{n+1}n!}\,\frac{(-1)^k z^{2k}}{(2k)!}\,\frac{(2k)!\,n!}{4^k k!}\,\frac{2^{k+n+1}}{(2n+2k+1)!!} = \frac{(-1)^k z^{n+2k}}{2^k k!\,(2n+2k+1)!!}.$$

Therefore

$$I = \sum_{k=0}^{\infty} \frac{(-1)^k z^{n+2k}}{2^k k!(2n+2k+1)!!} = j_n(z),$$

the power series of j_n (Section 14.7). This reproduces the Poisson representation of Exercise 14.7.9 without assuming it.

15.3 Physical Interpretation of Generating Function

15.3.1 Develop the electrostatic potential for the array of charges shown in Fig. 15.7. This is a linear electric quadrupole.

Solution.

In Fig. 15.7 the charges are $q, -2q, q$ at $z = -a, 0, a$. Coulomb's law therefore gives, at every point away from these charges,

$$V = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{r^2+a^2+2ar\cos\theta}} - \frac{2}{r} + \frac{1}{\sqrt{r^2+a^2-2ar\cos\theta}}\right].$$

For $r > a$, expand the two displaced charges with the generating function, $t = \pm a/r$:

$$\frac{1}{\sqrt{r^2+a^2\mp 2ar\cos\theta}} = \frac{1}{r}\Big[1 \mp 2\frac{a}{r}\cos\theta + \frac{a^2}{r^2}\Big]^{-1/2} = \sum_{n=0}^{\infty}\frac{(\pm a)^n}{r^{n+1}}P_n(\cos\theta).$$

Their nth coefficients add to $1+(-1)^n$, which is 0 for odd n and 2 for even n. The $n=0$ term, $2/r$, cancels the central charge's $-2/r$, and $n = 2s$ leaves $2a^{2s}/r^{2s+1}$:

$$\boxed{V = \frac{2q}{4\pi\varepsilon_0 r}\sum_{s=1}^{\infty}\left(\frac{a}{r}\right)^{2s} P_{2s}(\cos\theta)}.$$

In particular,

$$V = \frac{2qa^2}{4\pi\varepsilon_0 r^3} P_2(\cos\theta) + O(a^4/r^5) = \frac{qa^2(3\cos^2\theta - 1)}{4\pi\varepsilon_0 r^3} + O(a^4/r^5).$$

The total charge and dipole moment are zero, while $\sum q_i z_i^2 = 2qa^2 \neq 0$, identifying the leading quadrupole.

15.3.2 Calculate the electrostatic potential of the array of charges shown in Fig. 15.8. Here is an example of two equal but oppositely directed quadrupoles. The quadrupole contributions cancel. The octopole terms do not cancel.

Solution.

Figure 15.8 has charges $-q, 2q, -2q, q$ at $-2a, -a, a, 2a$. Thus the exact potential is

$$V = \frac{q}{4\pi\varepsilon_0}\left[-\frac{1}{|\mathbf{r} + 2a\hat{\mathbf{z}}|} + \frac{2}{|\mathbf{r} + a\hat{\mathbf{z}}|} - \frac{2}{|\mathbf{r} - a\hat{\mathbf{z}}|} + \frac{1}{|\mathbf{r} - 2a\hat{\mathbf{z}}|}\right].$$

For $r > 2a$, expand each term with the generating function, $1/|\mathbf{r} - c\hat{\mathbf{z}}| = \sum_n c^n P_n(\cos\theta)/r^{n+1}$ for $|c| < r$, with $c = -2a, -a, a, 2a$. The coefficient of $(q/4\pi\varepsilon_0)a^n P_n(\cos\theta)/r^{n+1}$ is

$$-(-2)^n + 2(-1)^n - 2 + 2^n = [1 - (-1)^n](2^n - 2).$$

For even n the four terms cancel in pairs; for odd n they give $2^{n+1} - 4 = 2(2^n - 2)$. So the coefficient is zero for every even n and also for $n = 1$. Putting $n = 2s + 1$ leaves

$$\boxed{V = \frac{2q}{4\pi\varepsilon_0 r}\sum_{s=1}^{\infty}(2^{2s+1} - 2)\left(\frac{a}{r}\right)^{2s+1} P_{2s+1}(\cos\theta)}.$$

The leading term ($s = 1$, with $2^3 - 2 = 6$) is

$$\boxed{V_{\text{oct}} = \frac{12qa^3}{4\pi\varepsilon_0 r^4} P_3(\cos\theta)}.$$

This also follows from the axial moments $\mu_n = \sum_i q_i z_i^n$, in terms of which the same expansion reads $V = \sum_n \mu_n P_n(\cos\theta)/(4\pi\varepsilon_0 r^{n+1})$: $\mu_0 = \mu_1 = \mu_2 = 0$ and $\mu_3 = q[-(-2)^3 + 2(-1)^3 - 2 + 2^3]a^3 = q[8 - 2 - 2 + 8]a^3 = 12qa^3$.

15.3.3 Show that the electrostatic potential produced by a charge q at $z = a$ for $r < a$ is

$$\varphi(r) = \frac{q}{4\pi\varepsilon_0 a}\sum_{n=0}^{\infty}\left(\frac{r}{a}\right)^n P_n(\cos\theta).$$

Solution.

The source-point distance is $|\mathbf{r} - a\hat{\mathbf{z}}| = \sqrt{a^2 + r^2 - 2ar\cos\theta}$. For $0 \leq r < a$, factor out a and use the generating function with $t = r/a$:

$$\varphi = \frac{q}{4\pi\varepsilon_0 a}\left[1 - 2\frac{r}{a}\cos\theta + \left(\frac{r}{a}\right)^2\right]^{-1/2} = \boxed{\frac{q}{4\pi\varepsilon_0 a}\sum_{n=0}^{\infty}\left(\frac{r}{a}\right)^n P_n(\cos\theta)}.$$

For each fixed $r < a$ the series converges uniformly in θ, since $|P_n(\cos\theta)| \leq 1$ bounds its terms by those of the geometric series $\sum_n (r/a)^n$. At $r = 0$ the value is $q/(4\pi\varepsilon_0 a)$, with all higher terms vanishing.

15.3.4 Using $\mathbf{E} = -\nabla\varphi$, determine the components of the electric field corresponding to the (pure) electric dipole potential,

$$\varphi(r) = \frac{2aq\,P_1(\cos\theta)}{4\pi\varepsilon_0 r^2}.$$

Here it is assumed that $r \gg a$.

$$\textit{ANS.}\ E_r = +\frac{4aq\cos\theta}{4\pi\varepsilon_0 r^3}, \quad E_\theta = +\frac{2aq\sin\theta}{4\pi\varepsilon_0 r^3}, \quad E_\varphi = 0.$$

Solution.

Since $P_1(\cos\theta) = \cos\theta$, put $C = 2aq/(4\pi\varepsilon_0)$ and write $\varphi = Cr^{-2}\cos\theta$. The spherical gradient gives

$$E_r = -\partial_r\varphi = 2Cr^{-3}\cos\theta, \qquad E_\theta = -\frac{1}{r}\partial_\theta\varphi = Cr^{-3}\sin\theta, \qquad E_\varphi = -\frac{1}{r\sin\theta}\partial_\varphi\varphi = 0.$$

Substituting C yields all three stated components. For a pair of charges separated by $2a$ these are its leading $r \gg a$ terms; for an ideal point dipole of fixed moment $2aq$ they are exact at $r > 0$.

15.3.5 Operating in spherical polar coordinates, show that for $r > 0$

$$\frac{\partial}{\partial z}\left[\frac{P_l(\cos\theta)}{r^{l+1}}\right] = -(l+1)\frac{P_{l+1}(\cos\theta)}{r^{l+2}}.$$

This is the key step in the mathematical argument that the derivative of one multipole leads to the next higher multipole.

Hint. Compare with Exercise 3.10.28.

Solution.

At fixed Cartesian x, y, write $\rho = \sqrt{x^2+y^2}$, $r = \sqrt{\rho^2+z^2}$ and $u = z/r = \cos\theta$. Then

$$\frac{\partial r}{\partial z} = \frac{z}{r} = u, \qquad \frac{\partial u}{\partial z} = \frac{1}{r} - \frac{z}{r^2}\frac{\partial r}{\partial z} = \frac{1}{r} - \frac{z^2}{r^3} = \frac{1-u^2}{r}.$$

Hence, by the product and chain rules,

$$\begin{aligned}\partial_z\left[r^{-l-1}P_l(u)\right] &= -(l+1)r^{-l-2}u\,P_l(u) + r^{-l-1}P_l'(u)\frac{1-u^2}{r} \\ &= r^{-l-2}\left[-(l+1)uP_l(u) + (1-u^2)P_l'(u)\right].\end{aligned}$$

Exercise 15.1.7 gives $(1-u^2)P_l' = (l+1)(uP_l - P_{l+1})$. The uP_l terms cancel, leaving

$$\boxed{\partial_z\left[\frac{P_l(\cos\theta)}{r^{l+1}}\right] = -(l+1)\frac{P_{l+1}(\cos\theta)}{r^{l+2}}}, \qquad r > 0.$$

Both sides are singular at the origin, where θ is undefined; hence the restriction $r > 0$ added to the statement. This chain-rule calculation is equivalent to $\partial_z = \cos\theta\,\partial_r - (\sin\theta/r)\partial_\theta$.

15.3.6 A point electric dipole of strength $p^{(1)}$ is placed at $z = a$; a second point electric dipole of equal but opposite strength is at the origin. Keeping the product $p^{(1)}a$ constant, let $a \to 0$. Show that this results in a point electric quadrupole.

Hint. Exercise 15.3.5 (when proved) will be helpful.

Solution.

Define $F_1(\mathbf{r}) = P_1(\cos\theta)/r^2$. The positive z dipole at $a\hat{\mathbf{z}}$ and its negative at zero have combined potential

$$V_a(\mathbf{r}) = \frac{p^{(1)}}{4\pi\varepsilon_0}[F_1(\mathbf{r} - a\hat{\mathbf{z}}) - F_1(\mathbf{r})].$$

At each fixed point away from the origin, Taylor expansion in a gives $F_1(\mathbf{r} - a\hat{\mathbf{z}}) - F_1(\mathbf{r}) = -a\,\partial_z F_1 + O(a^2)$. Exercise 15.3.5 with $l = 1$ evaluates $\partial_z F_1 = -2P_2/r^3$, so

$$V_a = \frac{p^{(1)}}{4\pi\varepsilon_0}\left[\frac{2aP_2(\cos\theta)}{r^3} + O(a^2)\right] = \frac{2p^{(1)}a\,P_2(\cos\theta)}{4\pi\varepsilon_0 r^3} + p^{(1)}O(a^2).$$

Holding $K = p^{(1)}a$ fixed, the error is $p^{(1)}O(a^2) = O(Ka) \to 0$, so

$$\boxed{V_{\text{quad}} = \frac{2K}{4\pi\varepsilon_0 r^3}P_2(\cos\theta)}.$$

In the convention $V_{\text{quad}} = p^{(2)}P_2/(4\pi\varepsilon_0 r^3)$, the resulting quadrupole strength is $p^{(2)} = 2ap^{(1)}$. The dipole terms cancel, while a nonzero $r^{-3}P_2$ term remains.

15.3.7 A point electric octupole may be constructed by placing a point electric quadrupole (pole strength $p^{(2)}$ in the z-direction) at $z = a$ and an equal but opposite point electric quadrupole at $z = 0$ and then letting $a \to 0$, subject to $p^{(2)}a =$ constant. Find the electrostatic potential corresponding to a point electric octupole. Show from the construction of the point electric octupole that the corresponding potential may be obtained by differentiating the point quadrupole potential.

Solution.

Use the same strength convention: $V^{(2)} = p^{(2)}P_2(\cos\theta)/(4\pi\varepsilon_0 r^3)$. For a positive quadrupole displaced to $z = a$ and its negative at zero,

$$V_a^{(3)} = \frac{p^{(2)}}{4\pi\varepsilon_0}\left[F_2(\mathbf{r} - a\hat{\mathbf{z}}) - F_2(\mathbf{r})\right], \qquad F_2 = \frac{P_2(\cos\theta)}{r^3}.$$

The Taylor difference is $-a\partial_z F_2 + O(a^2)$, and Exercise 15.3.5 with $l = 2$ gives $\partial_z F_2 = -3P_3/r^4$, so $V_a^{(3)} = 3ap^{(2)}P_3/(4\pi\varepsilon_0 r^4) + p^{(2)}O(a^2)$. Therefore, keeping $K_2 = ap^{(2)}$ fixed and using $P_3 = \frac{1}{2}(5\cos^3\theta - 3\cos\theta)$,

$$\boxed{V^{(3)} = \frac{3K_2}{4\pi\varepsilon_0 r^4}P_3(\cos\theta) = \frac{3K_2}{8\pi\varepsilon_0 r^4}(5\cos^3\theta - 3\cos\theta)}.$$

Thus $p^{(3)} = 3ap^{(2)}$ in this convention, and the construction is precisely $V^{(3)} = -a\,\partial_z V^{(2)}$ in the limiting sense. The sign is set by putting the positive source at $+a$. The Instructor's Manual, p. 245, instead writes $48a^3q$ for the numerator of the octupole moment, although this exercise supplies $p^{(2)}$, not a charge q. In the stated strength convention the moment is $3ap^{(2)}$. That form would agree only after the additional, unstated assignment $p^{(2)} = 16a^2q$.

15.3.8 A point charge q is in the interior of a hollow conducting sphere of radius r_0. The charge q is displaced a distance $0 < a < r_0$ from the center of the sphere. If the conducting sphere is grounded, show that the potential in the interior produced by q and the distributed induced charge is the same as that produced by q and its image charge q'. The image charge is at a distance $a' = r_0^2/a$ from the center, collinear with q and the origin (Fig. 15.9).

Hint. Calculate the electrostatic potential for $a < r_0 < a'$. Show that the potential vanishes for $r = r_0$ if we take $q' = -qr_0/a$.

Solution.

For $0 < a < r_0$, place $q' = -qr_0/a$ at $a' = r_0^2/a > r_0$. The restriction added to the statement is needed: for $a = 0$ there is no image point a' (that case is treated as a limit at the end), and $a < r_0$ puts q inside the sphere and its image outside. Consider

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{|\mathbf{r} - a\hat{\mathbf{z}}|} - \frac{qr_0/a}{|\mathbf{r} - a'\hat{\mathbf{z}}|}\right].$$

On $r = r_0$, $|\mathbf{r} - a\hat{\mathbf{z}}|^2 = r_0^2 + a^2 - 2ar_0\cos\theta$, and the two distances satisfy

$$|\mathbf{r} - a'\hat{\mathbf{z}}|^2 = r_0^2 + \frac{r_0^4}{a^2} - \frac{2r_0^3}{a}\cos\theta = \frac{r_0^2}{a^2}|\mathbf{r} - a\hat{\mathbf{z}}|^2.$$

Their positive square roots therefore satisfy $|\mathbf{r} - a'\hat{\mathbf{z}}| = (r_0/a)|\mathbf{r} - a\hat{\mathbf{z}}|$, and the two potential terms cancel exactly.

The Legendre version of the same cancellation uses expansions valid for $a < r < a'$, a range that contains the sphere $r = r_0$. There $r > a$ and $r < a'$, so the generating function gives $1/|\mathbf{r} - a\hat{\mathbf{z}}| = \sum_n a^n P_n/r^{n+1}$ and $1/|\mathbf{r} - a'\hat{\mathbf{z}}| = \sum_n r^n P_n/(a')^{n+1}$ (as in Exercise 15.3.3), hence

$$V = \frac{1}{4\pi\varepsilon_0}\sum_{n=0}^{\infty}\left[\frac{qa^n}{r^{n+1}} + \frac{q'r^n}{(a')^{n+1}}\right]P_n(\cos\theta).$$

At $r = r_0$, inserting $q' = -qr_0/a$ and $(a')^{n+1} = r_0^{2n+2}/a^{n+1}$ makes each bracket zero:

$$\frac{q'r_0^n}{(a')^{n+1}} = -\frac{qr_0}{a}\,\frac{r_0^n\,a^{n+1}}{r_0^{2n+2}} = -\frac{qa^n}{r_0^{n+1}}.$$

The image term is harmonic everywhere inside the sphere, because its source is outside; the real charge has exactly the required Coulomb singularity. The difference from any other grounded solution is harmonic, including across the canceled source singularity, and zero on the boundary. Uniqueness for the Dirichlet problem proves the result throughout the interior, including $r < a$, where the first expansion diverges. For $a = 0$, take the limit directly: as $a \to 0$, $a' \to \infty$ and the image term tends to $-(qr_0/a)/a' = -q/r_0$, so $V = q(1/r - 1/r_0)/(4\pi\varepsilon_0)$.

15.4 Associated Legendre Equation

15.4.1 Apply the Frobenius method to Eq. (15.72) to obtain Eq. (15.73) and verify that the numerator of that equation becomes zero if $\lambda = l(l+1)$ and $j = l - m$.

Solution.

Write U for the factor multiplying $(1 - x^2)^{m/2}$ in the associated Legendre function; Eq. (15.72) is

$$(1 - x^2)U'' - 2(m+1)xU' + [\lambda - m(m+1)]U = 0.$$

For $U = \sum_{j\ge 0} a_j x^{j+k}$, the lowest power x^{k-2} has coefficient $k(k-1)a_0$, so $k = 0$ or 1. The next power, x^{k-1}, has coefficient $k(k+1)a_1$. Thus $k = 1$ forces $a_1 = 0$, whereas $k = 0$ leaves a_0, a_1 independent; these are the even and odd branches at the ordinary point $x = 0$. In general, x^{j+k} receives $(j+k+2)(j+k+1)a_{j+2} - (j+k)(j+k-1)a_j$ from $(1-x^2)U''$ (the first part from the term $a_{j+2}x^{j+k+2}$), $-2(m+1)(j+k)a_j$ from $-2(m+1)xU'$, and $[\lambda - m(m+1)]a_j$ from the last term. So the coefficient of x^{j+k} is

$$(j+k+2)(j+k+1)a_{j+2} + \{\lambda - m(m+1) - (j+k)(j+k-1) - 2(m+1)(j+k)\}a_j.$$

Set it to zero, using $(j+k)(j+k-1) + 2(m+1)(j+k) = (j+k)^2 + (2m+1)(j+k)$:

$$a_{j+2} = \frac{(j+k)^2 + (2m+1)(j+k) - \lambda + m(m+1)}{(j+k+1)(j+k+2)} a_j.$$

For $k = 0$ the numerator is $j^2 + (2m+1)j + m(m+1) - \lambda = (j+m)(j+m+1) - \lambda$, and this is exactly Eq. (15.73). If $\lambda = l(l+1)$, put $p = j + m$; the numerator factors as

$$p(p+1) - l(l+1) = (p-l)(p+l) + (p-l) = (j+m-l)(j+m+l+1),$$

which vanishes at $j = l - m$. The parity of $l - m$ selects the even or odd power series. For nonnegative integers $m \le l$, that branch terminates in a polynomial of degree $l - m$.

15.4.2 Starting from the entries for P_2^2 and P_2^1 in Table 15.3, apply a recurrence formula to obtain P_2^0 (which is P_2), P_2^{-1}, and P_2^{-2}. Compare your results with the value of P_2 from Table 15.1 and with values of P_2^{-1} and P_2^{-2} obtained by applying Eq. (15.81) to entries from Table 15.3.

Solution.

Put $s = \sqrt{1-x^2}$ on $-1 < x < 1$. Table 15.3 gives $P_2^2 = 3(1-x^2) = 3s^2$ and $P_2^1 = -3xs$. Use the fixed-degree recurrence of Exercise 15.4.8 with $l = 2$, so that $l(l+1) = 6$:

$$P_2^{m+1} + \frac{2mx}{s} P_2^m + [6 - m(m-1)]P_2^{m-1} = 0.$$

At $m = 1$, this gives $3s^2 + \frac{2x}{s}(-3xs) + 6P_2^0 = 3s^2 - 6x^2 + 6P_2^0 = 0$, so $P_2^0 = x^2 - \frac{1}{2}(1-x^2)$, i.e.

$$\boxed{P_2^0 = \frac{3x^2 - 1}{2}},$$

which is P_2 of Table 15.1. At $m = 0$ the middle term is absent and $6 - 0 = 6$: $P_2^1 + 6P_2^{-1} = 0$, so

$$\boxed{P_2^{-1} = -\frac{1}{6}(-3xs) = \frac{1}{2}xs}.$$

At $m = -1$, $6 - (-1)(-2) = 4$ and $P_2^0 - (2x/s)P_2^{-1} + 4P_2^{-2} = 0$. Substituting the two results, $(2x/s)(xs/2) = x^2$ and $4P_2^{-2} = x^2 - (3x^2-1)/2 = (1-x^2)/2 = s^2/2$, so

$$\boxed{P_2^{-2} = \frac{1}{8}(1-x^2)}.$$

For the comparison the exercise asks for, Eq. (15.81), $P_l^{-m} = (-1)^m[(l-m)!/(l+m)!]P_l^m$, applied to the Table 15.3 entries predicts $P_2^{-1} = -(1!/3!)P_2^1 = xs/2$ and $P_2^{-2} = (0!/4!)P_2^2 = 3s^2/24 = s^2/8$, in agreement. (For P_2^{-1} and P_2^{-2} this is a consistency check rather than an independent derivation, since Exercise 15.4.8 proves the recurrence for $m \le 0$ from Eq. (15.81), in the form of Exercise 15.4.3.) The endpoint values follow by continuity.

15.4.3 Prove that, for integers $0 \le m \le l$ and $-1 < x < 1$,

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x),$$

where $P_l^m(x)$ is defined by

$$P_l^m(x) = \frac{(-1)^m}{2^l l!}(1-x^2)^{m/2}\frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l.$$

Hint. One approach is to apply Leibniz's formula to $(x+1)^l(x-1)^l$.

Solution.

Let $F = (x-1)^l(x+1)^l$, $D = d/dx$, and take integers $0 \le m \le l$. Leibniz's rule, $D^N(fg) = \sum_i \binom{N}{i} D^i f\, D^{N-i} g$, with $D^i(x \mp 1)^l = \frac{l!}{(l-i)!}(x \mp 1)^{l-i}$ for $i \le l$ (and zero for $i > l$), gives for $N = l - m$, with $i = j$,

$$D^{l-m}F = \sum_{j=0}^{l-m} \frac{(l-m)!(l!)^2(x-1)^{l-j}(x+1)^{m+j}}{j!(l-m-j)!(l-j)!(m+j)!}.$$

For $D^{l+m}F$, the number i of derivatives on the first factor must lie between m and l, because neither factor survives more than l derivatives. Write $i = m + j$, so the second factor receives $l - j$ derivatives:

$$D^{l+m}F = \sum_{j=0}^{l-m} \frac{(l+m)!(l!)^2(x-1)^{l-m-j}(x+1)^{j}}{(m+j)!(l-j)!(l-m-j)!j!}.$$

Multiplication of the second sum by $(x^2-1)^m = (x-1)^m(x+1)^m$ makes its powers $(x-1)^{l-j}(x+1)^{m+j}$, exactly those of the first sum. The remaining factors $(l!)^2/[j!(l-m-j)!(l-j)!(m+j)!]$ are the same in both sums, so the coefficients differ only by the ratio $(l-m)!/(l+m)!$. Thus

$$D^{l-m}F = \frac{(l-m)!}{(l+m)!}(x^2-1)^m D^{l+m}F.$$

Insert this identity into the Rodrigues definition with order $-m$, where $(-1)^{-m} = (-1)^m$ and $(x^2-1)^m = (-1)^m(1-x^2)^m$:

$$\begin{aligned} P_l^{-m} &= \frac{(-1)^m}{2^l l!}(1-x^2)^{-m/2}\frac{(l-m)!}{(l+m)!}(-1)^m(1-x^2)^m D^{l+m}F \\ &= (-1)^m \frac{(l-m)!}{(l+m)!}\,\frac{(-1)^m}{2^l l!}(1-x^2)^{m/2}D^{l+m}F = \boxed{(-1)^m\frac{(l-m)!}{(l+m)!}P_l^m}. \end{aligned}$$

On $(-1,1)$ choose the positive square root of $1-x^2$. The restrictions added to the statement are needed: for $m > l$ the definition of P_l^{-m} would require a derivative of negative order $l-m$, and $(l-m)!$ is undefined; at $x = \pm 1$ its factor $(1-x^2)^{-m/2}$ is undefined for $m \ge 1$, although both sides extend continuously there. (Orders $-l \le m < 0$ give the same identity solved for $P_l^{|m|}$.) The identity also extends with any consistently chosen complex branch.

15.4.4 Show that

$$P_{2l}^1(0) = 0,$$
$$P_{2l+1}^1(0) = (-1)^{l+1}\frac{(2l+1)!!}{(2l)!!},$$

by each of these three methods:

(a) Use of recurrence relations,

(b) Expansion of the generating function,

(c) Rodrigues formula.

Solution.

Write the index of the statement as s, so the claims are $P_{2s}^1(0) = 0$ and $P_{2s+1}^1(0) = (-1)^{s+1}(2s+1)!!/(2s)!!$.

(a) The fixed-order recurrence $(2l+1)xP_l^m = (l+m)P_{l-1}^m + (l-m+1)P_{l+1}^m$ (also used in Exercise 15.4.12) at $x = 0$, with $m = 1$, gives

$$lP_{l+1}^1(0) = -(l+1)P_{l-1}^1(0), \qquad l \geq 1.$$

The starting values are $P_0^1(0) = 0$ and $P_1^1(0) = -1$, from $P_0^1 = 0$ and $P_1^1 = -\sqrt{1-x^2}$. Odd l link the even indices, so every even-index value is zero. For the odd subsequence, $l = 2k$ gives $P_{2k+1}^1(0) = -\frac{2k+1}{2k}P_{2k-1}^1(0)$; using $k = 1, 2, \ldots, s$ in turn,

$$P_{2s+1}^1(0) = (-1)^s\frac{3\cdot 5\cdots(2s+1)}{2\cdot 4\cdots 2s}P_1^1(0) = \boxed{(-1)^{s+1}\frac{(2s+1)!!}{(2s)!!}}.$$

(b) Differentiate the generating function $\sum_l P_l(x)t^l = (1-2xt+t^2)^{-1/2}$ with respect to x: $\sum_l P_l'(x)t^l = t(1-2xt+t^2)^{-3/2}$. At $x = 0$, $P_l^1(0) = -\sqrt{1-0}\,P_l'(0) = -P_l'(0)$, so the binomial series gives

$$\sum_{l\geq 0} P_l^1(0)t^l = -t(1+t^2)^{-3/2} = -\sum_{s\geq 0}(-1)^s\frac{(2s+1)!!}{2^s s!}t^{2s+1},$$

because

$$\binom{-3/2}{s} = \frac{(-\frac{3}{2})(-\frac{5}{2})\cdots(-\frac{2s+1}{2})}{s!} = (-1)^s\frac{(2s+1)!!}{2^s s!}.$$

There are no even powers, and $2^s s! = (2s)!!$, so the coefficient of t^{2s+1} is $(-1)^{s+1}(2s+1)!!/(2s)!!$: the same values.

(c) Since $P_n^1(0) = -P_n'(0)$, which is minus the coefficient of x in P_n, Rodrigues' formula gives

$$P_n^1(0) = -\frac{1}{2^n n!}D^{n+1}(x^2-1)^n\Big|_{x=0}, \qquad (x^2-1)^n = \sum_{k=0}^{n}\binom{n}{k}(-1)^{n-k}x^{2k}.$$

At $x = 0$, $D^{n+1}x^{2k}$ is nonzero only for $2k = n+1$, when it equals $(n+1)!$. For $n = 2s$ no such k exists, so $P_{2s}^1(0) = 0$. For $n = 2s+1$ the term is $k = s+1$, with $(-1)^{n-k} = (-1)^s$:

$$P_{2s+1}^1(0) = -\frac{(-1)^s}{2^{2s+1}(2s+1)!}\binom{2s+1}{s+1}(2s+2)! = (-1)^{s+1}\frac{(2s+2)!}{2^{2s+1}s!\,(s+1)!}.$$

With $(2s+2)! = (2s+1)!!\,(2s+2)!! = (2s+1)!!\,2^{s+1}(s+1)!$ this is $(-1)^{s+1}(2s+1)!!/(2^s s!) = (-1)^{s+1}(2s+1)!!/(2s)!!$. This proves both formulas by the third requested method.

15.4.5 Evaluate $P_l^m(0)$ for integers $-l \le m \le l$.

$$ANS.\ P_l^m(0) = \begin{cases} (-1)^{(l+m)/2}\dfrac{(l+m-1)!!}{(l-m)!!}, & l+m \text{ even}, \\ 0, & l+m \text{ odd}. \end{cases}$$

Solution.

For integers $0 \le m \le l$, the definition $P_l^m = (-1)^m(1-x^2)^{m/2}D^m P_l$ $(D = d/dx)$ gives $P_l^m(0) = (-1)^m P_l^{(m)}(0)$, and by Taylor's formula $P_l^{(m)}(0) = m!\,c_m$, where c_m is the coefficient of x^m in P_l. The powers in P_l are $l, l-2, \ldots$ (Exercise 15.1.2), so $c_m = 0$ when $l+m$ is odd. If $l+m$ is even, the power $x^{l-2k} = x^m$ of Exercise 15.1.2 has $k = (l-m)/2$, so that $2l-2k = l+m$, $l-k = (l+m)/2$ and $l-2k = m$:

$$c_m = (-1)^{(l-m)/2}\frac{(l+m)!}{2^l[(l-m)/2]!\,[(l+m)/2]!\,m!}.$$

Multiplying by $(-1)^m m!$ and using $(-1)^m(-1)^{(l-m)/2} = (-1)^{(l+m)/2}$ gives

$$P_l^m(0) = (-1)^{(l+m)/2}\frac{(l+m)!}{2^l[(l-m)/2]![(l+m)/2]!}.$$

Write $2^l = 2^{(l+m)/2}\,2^{(l-m)/2}$ and split the factorials into even and odd factors, $(l+m)! = (l+m)!!\,(l+m-1)!!$ with $(l+m)!! = 2^{(l+m)/2}[(l+m)/2]!$:

$$\frac{(l+m)!}{2^{(l+m)/2}[(l+m)/2]!} = (l+m-1)!!, \qquad 2^{(l-m)/2}[(l-m)/2]! = (l-m)!!.$$

Hence the stated nonzero value is

$$\boxed{P_l^m(0) = (-1)^{(l+m)/2}\frac{(l+m-1)!!}{(l-m)!!}}.$$

For $m = 0$ this is $P_l(0) = (-1)^{l/2}(l-1)!!/l!!$ for even l, as in Exercise 15.1.12, and $m = 1$ reproduces Exercise 15.4.4.

Negative orders need no separate formula. For $m = -\mu$ with $1 \le \mu \le l$, Exercise 15.4.3 gives $P_l^{-\mu}(0) = (-1)^\mu \frac{(l-\mu)!}{(l+\mu)!}P_l^\mu(0)$, which vanishes when $l+\mu$, and hence $l+m = l-\mu$, is odd. Otherwise, by $(l-\mu)!/(l-\mu)!! = (l-\mu-1)!!$, $(l+\mu)!/(l+\mu-1)!! = (l+\mu)!!$ and $(-1)^{\mu+(l+\mu)/2} = (-1)^{(l-\mu)/2+2\mu} = (-1)^{(l-\mu)/2}$,

$$P_l^{-\mu}(0) = (-1)^\mu\frac{(l-\mu)!}{(l+\mu)!}(-1)^{(l+\mu)/2}\frac{(l+\mu-1)!!}{(l-\mu)!!} = (-1)^{(l-\mu)/2}\frac{(l-\mu-1)!!}{(l+\mu)!!},$$

which is the boxed formula with $m = -\mu$. The restriction $|m| \le l$ in the statement is needed: for integer $m > l$ the defining derivative makes $P_l^m = 0$, while for even $l+m$ the factorial expression would contain $(l-m)!!$ with $l-m < 0$; for $m < -l$ the Rodrigues definition would require a derivative of negative order $l+m$.

15.4.6 Starting from the potential of a finite dipole, Eq. (15.65), verify the formulas for the electric field components given as Eqs. (15.130) and (15.131).

Solution.

For charges $\pm q$ at $z = \pm a$, $V = (q/4\pi\varepsilon_0)[|\mathbf{r} - a\hat{\mathbf{z}}|^{-1} - |\mathbf{r} + a\hat{\mathbf{z}}|^{-1}]$; the generating-function expansions (as in Exercise 15.3.1) have nth coefficients $[1-(-1)^n]a^n/r^{n+1}$, so only odd $n = 2s+1$ survive, with a factor 2. This is Eq. (15.65), the convergent finite-dipole expansion

$$V = \frac{2q}{4\pi\varepsilon_0}\sum_{s=0}^{\infty}\frac{a^{2s+1}}{r^{2s+2}}P_{2s+1}(\cos\theta), \qquad r > a.$$

Radial differentiation, $-\partial_r r^{-2s-2} = (2s+2)r^{-2s-3}$, with $2q(2s+2)/(4\pi\varepsilon_0) = q(s+1)/(\pi\varepsilon_0)$, gives

$$E_r = -\partial_r V = \frac{q}{\pi\varepsilon_0}\sum_{s\geq 0}(s+1)\frac{a^{2s+1}}{r^{2s+3}}P_{2s+1} = \boxed{\frac{qa}{\pi\varepsilon_0 r^3}\left[P_1 + 2(a/r)^2 P_3 + 3(a/r)^4 P_5 + \cdots\right]}.$$

For the angular derivative, $dP_l(\cos\theta)/d\theta = -\sin\theta\, P_l'(\cos\theta) = P_l^1(\cos\theta)$, by $P_l^1(x) = -\sqrt{1-x^2}\, P_l'(x)$ with $\sqrt{1-x^2} = \sin\theta$. Since $(1/r)\,2q/(4\pi\varepsilon_0) = q/(2\pi\varepsilon_0 r)$, therefore

$$E_\theta = -\frac{1}{r}\partial_\theta V = -\frac{q}{2\pi\varepsilon_0}\sum_{s\geq 0}\frac{a^{2s+1}}{r^{2s+3}}P_{2s+1}^1 = \boxed{-\frac{qa}{2\pi\varepsilon_0 r^3}\left[P_1^1 + (a/r)^2 P_3^1 + (a/r)^4 P_5^1 + \cdots\right]}.$$

These are Eqs. (15.130) and (15.131), with $E_\varphi = 0$ since V does not depend on φ. Because $P_1^1(\cos\theta) = -\sin\theta$, the leading terms are $E_r = qa\cos\theta/(\pi\varepsilon_0 r^3)$ and $E_\theta = qa\sin\theta/(2\pi\varepsilon_0 r^3)$: the components found in Exercise 15.3.4 for the dipole moment $2aq$, since $4aq/(4\pi\varepsilon_0) = qa/(\pi\varepsilon_0)$ and $2aq/(4\pi\varepsilon_0) = qa/(2\pi\varepsilon_0)$. For positive qa the leading E_θ points in the positive θ direction, since $\sin\theta \geq 0$, while the leading E_r has the sign of $\cos\theta$: outward for $\theta < \pi/2$ and inward for $\theta > \pi/2$.

15.4.7 Show that

$$P_l^l(\cos\theta) = (-1)^l(2l-1)!!\,\sin^l\theta, \qquad l = 0, 1, 2, \ldots, \quad 0 \leq \theta \leq \pi.$$

Solution.

The Rodrigues definition at $m = l$ involves $D^{2l}(x^2-1)^l$ ($D = d/dx$). Only the leading monomial x^{2l} of this polynomial of degree $2l$ survives the $2l$ derivatives, with value $(2l)!$. Thus, using $2^l l! = (2l)!!$ and $(2l)! = (2l)!!\,(2l-1)!!$,

$$P_l^l(x) = \frac{(-1)^l(2l)!}{2^l l!}(1-x^2)^{l/2} = (-1)^l(2l-1)!!(1-x^2)^{l/2}.$$

For $x = \cos\theta$ and $0 \leq \theta \leq \pi$, $\sqrt{1-x^2} = \sin\theta \geq 0$, so

$$\boxed{P_l^l(\cos\theta) = (-1)^l(2l-1)!!\,\sin^l\theta}.$$

The convention $(-1)!! = 1$ includes $l = 0$.

15.4.8 Derive the associated Legendre recurrence relation, for $-1 < x < 1$ and integer orders $-l \le m \le l$, $l \ge 1$, with $P_l^{l+1} = 0$ and, at $m = -l$, the zero-coefficient term of order $-l-1$ omitted,

$$P_l^{m+1}(x) + \frac{2mx}{(1-x^2)^{1/2}} P_l^m(x) + \big[l(l+1) - m(m-1)\big] P_l^{m-1}(x) = 0.$$

Solution.

For $m \ge 1$, differentiate the ordinary Legendre equation $(1-x^2)P_l'' - 2xP_l' + l(l+1)P_l = 0$ a total of $m-1$ times, and put $u = D^{m-1}P_l$ ($D = d/dx$). Leibniz's rule, with the derivatives $-2x$, -2 of $1-x^2$ and 2 of $2x$, gives

$$D^{m-1}[(1-x^2)P_l''] = (1-x^2)u'' - 2(m-1)xu' - (m-1)(m-2)u,$$
$$D^{m-1}[2xP_l'] = 2xu' + 2(m-1)u,$$

where $-(m-1)(m-2)$ is $\binom{m-1}{2}(-2)$. Subtracting and adding $l(l+1)u$,

$$(1-x^2)u'' - 2mxu' + [l(l+1) - m(m-1)]u = 0,$$

because the u' coefficient is $-2(m-1)x - 2x = -2mx$ and the u subtraction is $(m-1)(m-2) + 2(m-1) = m(m-1)$. Multiply this equation by $(-1)^{m+1}(1-x^2)^{(m-1)/2}$ and recall $P_l^k = (-1)^k(1-x^2)^{k/2}D^kP_l$. Since $u'' = D^{m+1}P_l$, $u' = D^mP_l$, $(-1)^{m+1} = -(-1)^m$ and $(1-x^2)^{(m-1)/2} = (1-x^2)^{m/2}/\sqrt{1-x^2}$, its three terms become, respectively,

$$P_l^{m+1}, \qquad \frac{2mx}{\sqrt{1-x^2}} P_l^m, \qquad [l(l+1) - m(m-1)]P_l^{m-1}.$$

Their sum is zero, proving the recurrence. For the extension to negative orders, denote the displayed left-hand side by R_m and write $c_k = (-1)^k(l-k)!/(l+k)!$, so $P_l^{-k} = c_kP_l^k$ (Exercise 15.4.3). For $1 \le k < l$, the adjacent ratios are

$$c_{k-1} = -(l+k)(l-k+1)c_k, \qquad (l-k)(l+k+1)c_{k+1} = -c_k,$$

since $(l-k+1)!/(l+k-1)! = (l+k)(l-k+1)\,(l-k)!/(l+k)!$ and $(l-k-1)!/(l+k+1)! = (l-k)!/[(l+k)!\,(l-k)(l+k+1)]$. In R_{-k} the last coefficient is $l(l+1)-(-k)(-k-1) = (l-k)(l+k+1)$, so

$$R_{-k} = c_{k-1}P_l^{k-1} - \frac{2kx}{\sqrt{1-x^2}} c_kP_l^k + (l-k)(l+k+1)c_{k+1}P_l^{k+1}.$$

Substituting the ratios and using $(l+k)(l-k+1) = l(l+1) - k(k-1)$ gives

$$R_{-k} = -c_k\left\{P_l^{k+1} + \frac{2kx}{\sqrt{1-x^2}} P_l^k + [l(l+1) - k(k-1)]P_l^{k-1}\right\} = -c_kR_k = 0.$$

At $m = 0$ the recurrence is $P_l^1 + l(l+1)P_l^{-1} = 0$, which is exactly the negative-order formula, since $c_1 = -1/[l(l+1)]$. At $m = l$, Rodrigues's formula gives $P_l^{l+1} = 0$, so $R_l = (2lx/\sqrt{1-x^2})P_l^l + 2lP_l^{l-1} = 0$. At $m = -l$ the coefficient $l(l+1) - l(l+1)$ of the out-of-range order $-l-1$ is zero and that term is omitted; with $c_{l-1} = -2l\,c_l$,

$$R_{-l} = c_{l-1}P_l^{l-1} - \frac{2lx}{\sqrt{1-x^2}} c_lP_l^l = -c_l\left[2lP_l^{l-1} + \frac{2lx}{\sqrt{1-x^2}} P_l^l\right] = -c_lR_l = 0.$$

The $l = 0$ case is simply $P_0^1 = 0$. At $x = \pm 1$ the factor $2mx/\sqrt{1-x^2}$ is infinite, but for $m \ne 0$ its product with P_l^m, which is $(1-x^2)^{|m|/2}$ times a polynomial (for $m < 0$ by Exercise 15.4.3), tends to a finite limit; every term extends continuously to $x = \pm 1$, and the recurrence holds there in that limiting sense.

15.4.9 Develop a recurrence relation that will yield $P_l^1(x)$, for $l \geq 1$ and $-1 < x < 1$, as

$$P_l^1(x) = f_1(x,l)P_l(x) + f_2(x,l)P_{l-1}(x).$$

Follow either of the procedures (a) or (b):

(a) Derive a recurrence relation of the preceding form. Give $f_1(x,l)$ and $f_2(x,l)$ explicitly.

(b) Find the appropriate recurrence relation in print.

(1) Give the source.

(2) Verify the recurrence relation.

ANS. (a) $P_l^1(x) = \dfrac{lx}{(1-x^2)^{1/2}}P_l - \dfrac{l}{(1-x^2)^{1/2}}P_{l-1}.$

Solution.

Use the derivation option (a). The defining relation is $P_l^1 = -\sqrt{1-x^2}P_l' = -(1-x^2)P_l'/\sqrt{1-x^2}$, while the ordinary recurrence, Eq. (15.25), gives $(1-x^2)P_l' = lP_{l-1} - lxP_l$. Divide the latter by $\sqrt{1-x^2}$ and reverse the sign:

$$\boxed{P_l^1(x) = \frac{lx}{\sqrt{1-x^2}}P_l(x) - \frac{l}{\sqrt{1-x^2}}P_{l-1}(x)}, \qquad -1 < x < 1, \quad l \geq 1.$$

Thus $f_1 = lx/\sqrt{1-x^2}$ and $f_2 = -l/\sqrt{1-x^2}$. The numerator $l[xP_l - P_{l-1}]$ vanishes at $x = \pm 1$, because $P_l(\pm 1) = (\pm 1)^l$; there the associated function, which is zero, is recovered as a limit. For $l = 0$, $P_0^1 = 0$ directly.

15.4.10 Show that, for $0 \leq \theta \leq \pi$, $\sin\theta \dfrac{d}{d\cos\theta}P_n(\cos\theta) = -P_n^1(\cos\theta)$.

Solution.

For $0 \leq \theta \leq \pi$, set $x = \cos\theta$ in $P_n^1(x) = -\sqrt{1-x^2}\,dP_n/dx$. Because the square root is $\sin\theta$, this gives

$$\boxed{\sin\theta \frac{d}{d\cos\theta}P_n(\cos\theta) = -P_n^1(\cos\theta)}.$$

The minus sign is the published erratum for textbook p. 754, listed in the Instructor's Manual dated 6 April 2012. It is required by the book's Condon–Shortley definition. For instance $n = 1$ gives $\sin\theta = -(-\sin\theta)$.

15.4.11 Show that for nonnegative integer degrees and admissible orders $|m| \leq \min(l, l')$ in part (a)

(a) $$\int_0^\pi \left(\frac{dP_l^m}{d\theta}\frac{dP_{l'}^m}{d\theta} + \frac{m^2 P_l^m P_{l'}^m}{\sin^2\theta}\right)\sin\theta\,d\theta = \frac{2l(l+1)}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'},$$

(b) $$\int_0^\pi \left(\frac{P_l^1}{\sin\theta}\frac{dP_{l'}^1}{d\theta} + \frac{P_{l'}^1}{\sin\theta}\frac{dP_l^1}{d\theta}\right)\sin\theta\,d\theta = 0.$$

These integrals occur in the theory of scattering of electromagnetic waves by spheres.

Solution.

(a) Put $u = P_l^m(\cos\theta)$, $v = P_{l'}^m(\cos\theta)$. Primes on u and v denote $d/d\theta$, while P_l' denotes dP_l/dx. The self-adjoint angular equation for u is

$$(\sin\theta\, u')' + \left[l(l+1)\sin\theta - \frac{m^2}{\sin\theta}\right] u = 0.$$

This is the associated Legendre equation in the self-adjoint form $[(1-x^2)P']' + [l(l+1) - m^2/(1-x^2)]P = 0$ (Exercise 15.4.14) written in θ: for $x = \cos\theta$, $d/dx = -(\sin\theta)^{-1}d/d\theta$ and $1 - x^2 = \sin^2\theta$, so $(1-x^2)\,dP/dx = -\sin\theta\, u'$ and $[(1-x^2)P']' = (\sin\theta)^{-1}(\sin\theta\, u')'$; multiply by $\sin\theta$. Multiply by v and integrate over $0 \le \theta \le \pi$. Integration by parts, $\int_0^\pi (\sin\theta\, u')' v\, d\theta = [\sin\theta\, u' v]_0^\pi - \int_0^\pi \sin\theta\, u' v'\, d\theta$, then yields

$$\int_0^\pi \left(u'v' + \frac{m^2 uv}{\sin^2\theta}\right)\sin\theta\, d\theta = l(l+1)\int_0^\pi uv\sin\theta\, d\theta + [\sin\theta\, u' v]_0^\pi.$$

For admissible integer orders, the boundary term is zero. For $m = 0$, $u' = -\sin\theta\, P_l'(\cos\theta) = O(\sin\theta)$. For $|m| > 0$, both functions have the form $\sin^{|m|}\theta$ times a polynomial in $\cos\theta$ (for negative orders by Exercise 15.4.3), so $u' = O(\sin^{|m|-1}\theta)$ and $\sin\theta\, u' v = O(\sin^{2|m|}\theta)$. The fixed-$m$ orthogonality and normalization, $\int_0^\pi uv\sin\theta\, d\theta = \frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'}$, give

$$\boxed{\int_0^\pi \left(u'v' + \frac{m^2 uv}{\sin^2\theta}\right)\sin\theta\, d\theta = \frac{2l(l+1)}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'}}.$$

(b) With $u = P_l^1(\cos\theta)$, $v = P_{l'}^1(\cos\theta)$, the denominators $\sin\theta$ cancel the integration weight $\sin\theta$. The integrand becomes $uv' + vu' = (uv)'$:

$$\int_0^\pi (uv' + vu')d\theta = [uv]_0^\pi = 0,$$

because both order-one functions contain the factor $\sin\theta$ and vanish at the two poles.

15.4.12 As a repeat of Exercise 15.2.9, show, using associated Legendre functions, that for integers $n \ge 2$ and $m \ge 0$

$$\int_{-1}^1 x(1-x^2)P_n'(x)P_m'(x)\,dx = \frac{n+1}{2n+1}\frac{2}{2n-1}\frac{n!}{(n-2)!}\,\delta_{m,n-1} + \frac{n}{2n+1}\frac{2}{2n+3}\frac{(n+2)!}{n!}\,\delta_{m,n+1}.$$

Solution.

Since $P_n^1 = -\sqrt{1-x^2}P_n'$, we have $(1-x^2)P_n'P_m' = P_n^1 P_m^1$, so the required integral is $I_{nm} = \int_{-1}^1 xP_n^1 P_m^1\,dx$. The fixed-order recurrence $(2l+1)xP_l^k = (l+k)P_{l-1}^k + (l-k+1)P_{l+1}^k$ with $k = 1$, $l = n$ gives

$$(2n+1)xP_n^1 = (n+1)P_{n-1}^1 + nP_{n+1}^1.$$

Multiply by P_m^1 and integrate. The order-one functions are orthogonal for different degrees, with $\int_{-1}^1 (P_j^1)^2 dx = \frac{2}{2j+1}\frac{(j+1)!}{(j-1)!} = \frac{2j(j+1)}{2j+1}$, so only two norms remain:

$$I_{nm} = \frac{n+1}{2n+1}\frac{2(n-1)n}{2n-1}\delta_{m,n-1} + \frac{n}{2n+1}\frac{2(n+1)(n+2)}{2n+3}\delta_{m,n+1}.$$

Here the norm was used with $j = n-1$ and $j = n+1$, after dividing the recurrence by $2n+1$. Writing $n(n-1) = n!/(n-2)!$ and $(n+1)(n+2) = (n+2)!/n!$ gives exactly the displayed formula. The factorial form of its first term assumes $n \geq 2$. The polynomial form above remains meaningful at $n = 0, 1$ and gives the correct zero term, without a negative factorial.

15.4.13 Evaluate $\displaystyle\int_0^\pi \sin^2\theta\, P_n^1(\cos\theta)\, d\theta$.

Solution.

Insert $P_n^1(\cos\theta) = -\sin\theta\, P_n'(\cos\theta)$ and then put $x = \cos\theta$:

$$I_n = -\int_0^\pi \sin^3\theta\, P_n'(\cos\theta) d\theta = -\int_{-1}^1 (1-x^2) P_n'(x) dx.$$

Integration by parts has a zero endpoint term, since $1 - x^2 = 0$ at $x = \pm 1$:

$$\int_{-1}^1 (1-x^2) P_n'\, dx = \Big[(1-x^2)P_n\Big]_{-1}^1 + \int_{-1}^1 2xP_n\, dx, \qquad I_n = -2\int_{-1}^1 xP_n(x) dx.$$

Because $x = P_1$ and $\int_{-1}^1 P_1 P_n\, dx = \frac{2}{3}\delta_{n1}$,

$$\boxed{I_n = -\frac{4}{3}\,\delta_{n1}}, \qquad n = 0, 1, 2, \ldots.$$

In particular, $P_1^1 = -\sin\theta$ directly gives $-\int_0^\pi \sin^3\theta\, d\theta = -4/3$. The Instructor's Manual's answer on p. 251 has the opposite sign, $+4\delta_{n1}/3$. The negative sign follows from the book's $P_1^1 = -\sin\theta$ convention and the direct $n = 1$ integral.

15.4.14 The associated Legendre function $P_l^m(x)$ satisfies the self-adjoint ODE

$$(1-x^2)\frac{d^2P_l^m(x)}{dx^2} - 2x\frac{dP_l^m(x)}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2}\right] P_l^m(x) = 0.$$

From the differential equations for $P_l^m(x)$ and $P_l^k(x)$ show that for integer orders $0 \leq m, k \leq l$ with $k \neq m$,

$$\int_{-1}^1 P_l^m(x) P_l^k(x) \frac{dx}{1-x^2} = 0.$$

Solution.

The eigenvalue that is compared is m^2, so the required condition is $k^2 \neq m^2$, or distinct nonnegative orders. If $k = -m \neq 0$, the two functions are proportional and the stated zero integral is false. Let $u = P_l^m$, $v = P_l^k$, with nonnegative integer orders $m, k \leq l$ and $m \neq k$. In self-adjoint form their equations are

$$[(1-x^2)u']' + \left[l(l+1) - \frac{m^2}{1-x^2}\right] u = 0$$

and the analogous equation for v, with k in place of m. Multiply the first by v, the second by u, and subtract. The $l(l+1)uv$ terms cancel, and $v[(1-x^2)u']' - u[(1-x^2)v']' = \frac{d}{dx}[(1-x^2)(vu' - uv')]$ because the terms $(1-x^2)u'v'$ cancel in the derivative on the right. So

$$\frac{d}{dx}[(1-x^2)(vu' - uv')] + \frac{k^2 - m^2}{1-x^2} uv = 0.$$

At least one order is positive. The endpoint products in the bracket vanish: an order-m function is $(1-x^2)^{m/2}$ times a polynomial, so $u = O((1-x^2)^{m/2})$ and, for $m \geq 1$, $u' = O((1-x^2)^{m/2-1})$ (for $m = 0$, u' is bounded); likewise for v with k. Hence the weighted Wronskian $(1-x^2)(vu' - uv')$ is $O((1-x^2)^{(m+k)/2})$ and tends to zero at $x = \pm 1$. Also $uv/(1-x^2) = O((1-x^2)^{(m+k)/2-1})$, which is integrable because $(m+k)/2 - 1 > -1$, i.e. $m + k > 0$. Integration over $-1 < x < 1$ therefore gives

$$(k^2 - m^2) \int_{-1}^{1} \frac{P_l^m P_l^k}{1-x^2} dx = 0.$$

Division by the nonzero eigenvalue difference proves the result. As a concrete excluded case, $P_1^1 = -\sqrt{1-x^2}$ and $P_1^{-1} = \sqrt{1-x^2}/2$ give integral -1, not zero.

15.4.15 Determine the vector potential and the magnetic induction field of a magnetic quadrupole by differentiating the magnetic dipole potential.

$$ANS.\ \mathbf{A}_{MQ} = -\frac{\mu_0}{2}(Ia^2)(dz)\frac{P_2^1(\cos\theta)}{r^3}\,\hat{\mathbf{e}}_\varphi + \text{higher-order terms},$$

$$\mathbf{B}_{MQ} = \mu_0(Ia^2)(dz)\left[\frac{3P_2(\cos\theta)}{r^4}\hat{\mathbf{e}}_r - \frac{P_2^1(\cos\theta)}{r^4}\hat{\mathbf{e}}_\theta + \cdots\right].$$

This corresponds to placing a current loop of radius a at $z \to dz$ and an oppositely directed current loop at $z \to -dz$. The vector potential and magnetic induction field of a point quadrupole are given by the leading terms in these expansions if we take the limit $dz \to 0$, $a \to 0$, and $I \to \infty$ subject to $Ia^2 dz =$ constant.

Solution.

Take a positive current loop at $+h$ and the opposite loop at $-h$, where $h = dz$ in the question. A small loop at the origin has, to leading order in a/r (the first term of the series of Example 15.4.2, i.e. the dipole potential of Exercise 15.4.17 with $\mathbf{m} = I\pi a^2\hat{\mathbf{z}}$, since $\hat{\mathbf{z}} \times \mathbf{r} = r\sin\theta\,\hat{\mathbf{e}}_\varphi$),

$$A_\varphi^{(D)} = \frac{\mu_0 I a^2}{4}\frac{\sin\theta}{r^2} = -\frac{\mu_0 I a^2}{4}\frac{P_1^1(\cos\theta)}{r^2}.$$

Translation along the z axis does not rotate $\hat{\mathbf{e}}_\varphi$. In cylindrical coordinates (ρ, φ, z) the two translated potentials are $A_\varphi^{(D)}(\rho, z-h)$ and $-A_\varphi^{(D)}(\rho, z+h)$; Taylor expansion in h cancels the even powers, so their sum is $-2h\,\partial_z A_\varphi^{(D)} + O(h^3)$. For fixed cylindrical radius ρ, $\sin\theta/r^2 = \rho/r^3 = \rho(\rho^2+z^2)^{-3/2}$, so, with $P_2^1(\cos\theta) = -3\cos\theta\sin\theta$,

$$\partial_z(\sin\theta/r^2) = -\frac{3\rho z}{r^5} = -\frac{3\sin\theta\cos\theta}{r^3} = \frac{P_2^1(\cos\theta)}{r^3}.$$

Thus $A_\varphi = -2h(\mu_0 I a^2/4)P_2^1/r^3$; writing $C = \mu_0 I a^2 h/2$,

$$\boxed{\mathbf{A}_{MQ} = -C\frac{P_2^1(\cos\theta)}{r^3}\hat{\mathbf{e}}_\varphi}.$$

The axisymmetric curl has components $B_r = (r\sin\theta)^{-1}\partial_\theta(\sin\theta A_\varphi)$ and $B_\theta = -r^{-1}\partial_r(rA_\varphi)$. With $x = \cos\theta$, $\sin\theta\,P_l^1(\cos\theta) = -(1-x^2)P_l'(x)$ and $d/d\theta = -\sin\theta\,d/dx$; the Legendre ODE $[(1-x^2)P_l']' = -l(l+1)P_l$ then gives $\partial_\theta(\sin\theta P_l^1) = -l(l+1)\sin\theta P_l$, and for $l = 2$, $\partial_\theta(\sin\theta P_2^1) = -6\sin\theta P_2$. Consequently

$$B_r = \frac{-C}{r^4\sin\theta}(-6\sin\theta P_2) = \frac{6C}{r^4}P_2, \qquad B_\theta = \frac{CP_2^1}{r}\,\partial_r(r^{-2}) = -\frac{2C}{r^4}P_2^1, \qquad B_\varphi = 0.$$

These are exactly the stated coefficients after substituting C. For fixed observation points away from the origin, the next translation term is $O(Ia^2h^3) = O((Ia^2h)h^2)$, and the next finite-radius loop term is $O(Ia^4h) = O((Ia^2h)a^2)$. Both vanish in the simultaneous point limit at fixed Ia^2h. The final sentence of the problem calls this a point dipole. This construction and its r^{-4} field describe a point quadrupole.

15.4.16 A single circular wire loop of radius a carries a constant current I. Model the wire as a line of zero cross section; choose $I > 0$ counterclockwise when seen from $+z$, with the flux normal along $+z$.

(a) Find the magnetic induction $\mathbf{B}$ for $r < a$, $\theta = \pi/2$.

(b) Calculate the integral of the magnetic flux $(\mathbf{B} \cdot d\boldsymbol{\sigma})$ over the area of the current loop, that is,

$$\int_0^a r\,dr \int_0^{2\pi} d\varphi\, B_z\left(r, \theta = \frac{\pi}{2}\right).$$

ANS. ∞.

The Earth is within such a ring current, in which I approximates millions of amperes arising from the drift of charged particles in the Van Allen belt.

Solution.

Let $\rho = r/a < 1$ in the equatorial plane. To obtain an interior expansion, first use the exact (Biot–Savart) field on the axis and the binomial series, with $\binom{-3/2}{n} = (-1)^n(2n+1)!!/(2^n n!) = (-1)^n(2n+1)!!/(2n)!!$ as in Exercise 15.4.4(b):

$$B_z(z) = \frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}} = \frac{\mu_0 I}{2a}(1+z^2/a^2)^{-3/2} = \frac{\mu_0 I}{2a}\sum_{n=0}^{\infty}(-1)^n\frac{(2n+1)!!}{(2n)!!}(z/a)^{2n}, \quad |z| < a.$$

In the source-free ball $r < a$, write $\mathbf{B} = -\nabla\Psi$ with Ψ harmonic, axially symmetric and regular, so $\Psi = \sum_l A_l r^l P_l(\cos\theta)$, and on the positive axis ($P_l(1) = 1$) $\Psi = \sum_l A_l z^l$. Choosing $\Psi(0) = 0$ and integrating $-d\Psi/dz = B_z$ term by term, with $(2n+1)!!/(2n+1) = (2n-1)!!$,

$$\Psi(z) = -\frac{\mu_0 I}{2}\sum_{n=0}^{\infty}(-1)^n\frac{(2n-1)!!}{(2n)!!}\left(\frac{z}{a}\right)^{2n+1}.$$

These axis values fix the A_l, so replacing z^{2n+1} by $r^{2n+1}P_{2n+1}(\cos\theta)$ gives

$$\Psi = -\frac{\mu_0 I}{2}\sum_{n=0}^{\infty}(-1)^n\frac{(2n-1)!!}{(2n)!!}\left(\frac{r}{a}\right)^{2n+1}P_{2n+1}(\cos\theta).$$

(a) At the equator $P_{2n+1}(0) = 0$, so $B_r = -\partial_r\Psi = 0$. For $B_\theta = -r^{-1}\partial_\theta\Psi$, use $\partial_\theta P_l(\cos\theta) = P_l^1(\cos\theta)$ (as in Exercise 15.4.6) and, from Exercise 15.4.4, $P_{2n+1}^1(0) = (-1)^{n+1}(2n+1)!!/(2n)!!$:

$$B_\theta(r, \tfrac{\pi}{2}) = \frac{\mu_0 I}{2r}\sum_{n=0}^{\infty}(-1)^n\frac{(2n-1)!!}{(2n)!!}\rho^{2n+1}(-1)^{n+1}\frac{(2n+1)!!}{(2n)!!} = -\frac{\mu_0 I}{2a}\sum_{n=0}^{\infty}(2n+1)\left[\frac{(2n-1)!!}{(2n)!!}\right]^2\rho^{2n}.$$

With $(2n-1)!!/(2n)!! = \binom{2n}{n}/4^n$ and $\hat{\mathbf{z}} = -\hat{\mathbf{e}}_\theta$ on the equator, this gives

$$\boxed{\mathbf{B}(r, \pi/2) = \frac{\mu_0 I}{2a}\sum_{n=0}^{\infty}(2n+1)\left[\frac{\binom{2n}{n}}{4^n}\right]^2\rho^{2n}\,\hat{\mathbf{z}}}.$$

The first terms are $\mu_0 I[1 + \frac{3}{4}\rho^2 + \frac{45}{64}\rho^4 + \cdots]/(2a)$.

(b) All coefficients $c_n = (2n+1)[\binom{2n}{n}/4^n]^2$ are positive, so for $I > 0$ every term of the series for B_z is positive. Tonelli's theorem (monotone convergence) permits integrating term by term up to the edge, as an extended nonnegative integral; with $\int_0^a r(r/a)^{2n}dr = a^2/(2n+2)$,

$$\Phi = 2\pi \int_0^a rB_z\, dr = 2\pi \frac{\mu_0 I}{2a} \sum_{n=0}^{\infty} c_n \frac{a^2}{2n+2} = \pi\mu_0 I a \sum_{n=0}^{\infty} \frac{c_n}{2n+2}.$$

Stirling's formula gives $\binom{2n}{n}/4^n \sim 1/\sqrt{\pi n}$, so $c_n \sim (2n+1)/(\pi n) \to 2/\pi$, and the summand is asymptotic to $1/(\pi n)$. Comparison with the harmonic series proves

$$\boxed{\Phi = +\infty \quad (I > 0)}$$

for an ideal zero-thickness wire, with the sign reversed if the current or surface orientation is reversed. For $r < a$ the field itself is finite; the divergence occurs only when the flux integral reaches the wire.

15.4.17 The vector potential $\mathbf{A}$ of a magnetic dipole, dipole moment $\mathbf{m}$, is given by $\mathbf{A}(\mathbf{r}) = (\mu_0/4\pi)(\mathbf{m} \times \mathbf{r}/r^3)$. Show by direct computation that the magnetic induction $\mathbf{B} = \nabla \times \mathbf{A}$ is given, for $r > 0$, by

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m}}{r^3}.$$

Solution.

The formula describes points $r > 0$. Put $\mathbf{F} = \mathbf{r}/r^3$. For constant $\mathbf{m}$, the vector identity $\nabla \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\nabla \cdot \mathbf{b}) - \mathbf{b}(\nabla \cdot \mathbf{a}) + (\mathbf{b} \cdot \nabla)\mathbf{a} - (\mathbf{a} \cdot \nabla)\mathbf{b}$ loses the two terms that differentiate $\mathbf{m}$:

$$\nabla \times (\mathbf{m} \times \mathbf{F}) = \mathbf{m}(\nabla \cdot \mathbf{F}) - (\mathbf{m} \cdot \nabla)\mathbf{F}.$$

The required derivatives can be computed componentwise, with $\partial_j r_i = \delta_{ij}$ and $\partial_j r = r_j/r$, so $\partial_j r^{-3} = -3r_j/r^5$:

$$\partial_j F_i = \frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5}, \qquad \nabla \cdot \mathbf{F} = \frac{3}{r^3} - \frac{3r^2}{r^5} = 0.$$

Therefore, summing over j,

$$[(\mathbf{m} \cdot \nabla)\mathbf{F}]_i = m_j \partial_j F_i = \frac{m_i}{r^3} - \frac{3r_i(\mathbf{m} \cdot \mathbf{r})}{r^5}.$$

Multiplying its negative by $\mu_0/(4\pi)$ and writing $\mathbf{r} = r\hat{\mathbf{r}}$ yields

$$\boxed{\mathbf{B} = \frac{\mu_0}{4\pi r^3}[3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m}]}.$$

The ordinary-function field excludes $r = 0$. Its distributional curl contains an additional contact term, not a value assigned to r^{-3} at the origin. To find that term explicitly, apply $\partial_j F_i$ to a smooth test function ϕ: $\langle \partial_j F_i, \phi \rangle = -\int F_i \partial_j \phi\, d^3r$. Integrate $-F_i \partial_j \phi$ outside a sphere of radius ϵ and integrate by parts. This leaves $\int_{r>\epsilon} \phi\, \partial_j F_i\, d^3r$ plus a surface term on $r = \epsilon$, where the outward normal of the region is $-\hat{\mathbf{r}}$, $F_i = \hat{r}_i/\epsilon^2$ and $dS = \epsilon^2 d\Omega$; so the surface term is $\int \hat{r}_i \hat{r}_j \phi\, d\Omega$, which tends to

$$\phi(0) \int_{S^2} \hat{r}_i \hat{r}_j\, d\Omega = \frac{4\pi}{3} \delta_{ij} \phi(0).$$

The off-diagonal integrals vanish by reflection; the three diagonal integrals are equal and sum to $\int d\Omega = 4\pi$. Therefore

$$\partial_j F_i = \mathrm{PV}\left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5}\right) + \frac{4\pi}{3}\delta_{ij}\delta(\mathbf{r}),$$

where the principal value uses spherical exclusions. Its trace is $\nabla \cdot \mathbf{F} = 4\pi\delta(\mathbf{r})$. In the curl formula, the two source terms combine as $\mu_0(4\pi - 4\pi/3)\mathbf{m}\delta/(4\pi) = (2\mu_0/3)\mathbf{m}\delta(\mathbf{r})$, in addition to the principal value of the displayed field.

15.4.18

(a) Show that in the point dipole limit the magnetic induction field of the current loop becomes

$$B_r(r,\theta) = \frac{\mu_0 m}{2\pi r^3} P_1(\cos\theta),$$
$$B_\theta(r,\theta) = -\frac{\mu_0 m}{4\pi r^3} P_1^1(\cos\theta),$$

with $m = I\pi a^2$ and $r > 0$.

(b) Compare these results with the magnetic induction of the point magnetic dipole of Exercise 15.4.17. Take $\mathbf{m} = \hat{\mathbf{z}}m$.

Solution.

(a) The leading term of the loop's exterior vector potential (the first term of the series of Example 15.4.2) is $A_\varphi = \mu_0 I a^2 \sin\theta/(4r^2) = \mu_0 m \sin\theta/(4\pi r^2)$, where $m = I\pi a^2$. The following terms carry additional factors $(a/r)^2$, so they vanish in the point-dipole limit $a \to 0$ at fixed m and fixed $r > 0$. The curl of the leading term gives

$$B_r = \frac{1}{r\sin\theta}\partial_\theta(\sin\theta A_\varphi) = \frac{\mu_0 m}{4\pi r^3 \sin\theta} 2\sin\theta\cos\theta = \frac{\mu_0 m}{2\pi r^3}\cos\theta,$$

$$B_\theta = -\frac{1}{r}\partial_r(rA_\varphi) = -\frac{\mu_0 m \sin\theta}{4\pi r}\,\partial_r\frac{1}{r} = \frac{\mu_0 m}{4\pi r^3}\sin\theta.$$

Thus, with $P_1(\cos\theta) = \cos\theta$ and $P_1^1(\cos\theta) = -\sin\theta$,

$$\boxed{B_r = \frac{\mu_0 m}{2\pi r^3} P_1(\cos\theta), \qquad B_\theta = -\frac{\mu_0 m}{4\pi r^3} P_1^1(\cos\theta)}.$$

The printed B_θ denominator 2π must be 4π. Its value as written is too large by a factor of two.

(b) Insert $\mathbf{m} = m\hat{\mathbf{z}}$ into Exercise 15.4.17 and use $\hat{\mathbf{r}} \cdot \mathbf{m} = m\cos\theta$ and $\hat{\mathbf{z}} = \cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta$. Then

$$3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m} = 3m\cos\theta\,\hat{\mathbf{e}}_r - m\hat{\mathbf{z}} = 2m\cos\theta\,\hat{\mathbf{e}}_r + m\sin\theta\,\hat{\mathbf{e}}_\theta,$$

so $B_r = \mu_0 m\cos\theta/(2\pi r^3)$ and $B_\theta = \mu_0 m\sin\theta/(4\pi r^3)$, which agrees with both components of (a), including the corrected B_θ: the point-dipole limit of the loop is the point magnetic dipole.

15.4.19 A uniformly charged spherical shell is rotating with constant angular velocity.

(a) Calculate the magnetic induction $\mathbf{B}$ along the axis of rotation outside the sphere.

(b) Using the vector potential series of Example 15.4.2, find $\mathbf{A}$ and then $\mathbf{B}$ for all points outside the sphere.

Solution.

Let the shell radius be a, its uniform surface charge density be σ, and its angular velocity be $\omega\hat{\mathbf{z}}$. The strip between the polar angles α and $\alpha + d\alpha$ has circumference $2\pi a \sin\alpha$ and width $a\,d\alpha$, hence charge $dq = 2\pi\sigma a^2 \sin\alpha\, d\alpha$. This charge passes a fixed point once per period $2\pi/\omega$, so the strip carries the current $dI = (\omega/2\pi)dq = \sigma\omega a^2 \sin\alpha\, d\alpha$. It is a circular loop of radius $a\sin\alpha$ centered on the axis at $z = a\cos\alpha$.

(a) By the Biot–Savart law, a circular loop of radius R carrying current I produces on its axis, at distance d from its center, the field $B_z = \mu_0 I R^2/[2(R^2+d^2)^{3/2}]$. For the strip, $R = a\sin\alpha$, $d = z - a\cos\alpha$, and $R^2 + d^2 = z^2 + a^2 - 2az\cos\alpha$. Adding the strips and putting $u = \cos\alpha$, $du = -\sin\alpha\, d\alpha$, $\sin^2\alpha = 1 - u^2$ (the limits $\alpha = 0, \pi$ become $u = 1, -1$),

$$B_z(z) = \int_0^\pi \frac{\mu_0\sigma\omega a^4 \sin^3\alpha\, d\alpha}{2(z^2+a^2-2az\cos\alpha)^{3/2}} = \frac{\mu_0\sigma\omega a^4}{2}\int_{-1}^{1} \frac{1-u^2}{(z^2+a^2-2azu)^{3/2}}du, \qquad z > a.$$

Set $t = z^2 + a^2 - 2azu$. Then $du = -dt/(2az)$, the limits $u = -1$ and $u = 1$ become $t = (z+a)^2$ and $t = (z-a)^2$, and $1 - u^2 = [4a^2z^2 - (z^2+a^2-t)^2]/(4a^2z^2)$, so the integral becomes

$$\frac{1}{8a^3z^3}\int_{(z-a)^2}^{(z+a)^2} [4a^2z^2 - (z^2+a^2-t)^2]t^{-3/2}dt.$$

Since $(z^2+a^2)^2 - 4a^2z^2 = (z^2-a^2)^2$, the integrand is $-(z^2-a^2)^2t^{-3/2} + 2(z^2+a^2)t^{-1/2} - t^{1/2}$, with the antiderivative

$$2(z^2-a^2)^2t^{-1/2} + 4(z^2+a^2)t^{1/2} - \frac{2}{3}t^{3/2}.$$

Since $z > a$, the endpoint square roots are $z+a$ and $z-a$. With $1/(z+a) - 1/(z-a) = -2a/(z^2-a^2)$ and $(z+a)^3 - (z-a)^3 = 6az^2 + 2a^3$, the difference of the antiderivative values is

$$\begin{aligned} &2(z^2-a^2)^2\left(\frac{1}{z+a} - \frac{1}{z-a}\right) + 8a(z^2+a^2) - \frac{2}{3}[(z+a)^3 - (z-a)^3] \\ &\quad = -4a(z^2-a^2) + 8a(z^2+a^2) - 4az^2 - \frac{4}{3}a^3 = \frac{32a^3}{3}. \end{aligned}$$

Multiplication by $1/(8a^3z^3)$ leaves $4/(3z^3)$. Therefore

$$\boxed{\mathbf{B}(z\hat{\mathbf{z}}) = \frac{2\mu_0\sigma\omega a^4}{3|z|^3}\hat{\mathbf{z}}}, \qquad |z| > a.$$

For $z < -a$ the integral representation of B_z still holds, and replacing z by $-z$ and u by $-u$ leaves it unchanged; so B_z depends only on $|z|$.

(b) Outside the shell, $r > a$, there is no current, so use the same decaying vector-potential expansion as for the current loop of Example 15.4.2:

$$A_\varphi = \sum_{l\ge1} c_l r^{-l-1} P_l^1(\cos\theta).$$

With $x = \cos\theta$, $P_l^1(x) = -(1-x^2)^{1/2}P_l'(x)$ and $\partial_\theta = -\sin\theta\, d/dx$, we have $\sin\theta\, P_l^1 = -(1-x^2)P_l'$ and, by Legendre's equation,

$$\partial_\theta[\sin\theta\, P_l^1(\cos\theta)] = \sin\theta\,\frac{d}{dx}[(1-x^2)P_l'(x)] = -l(l+1)\sin\theta\, P_l(x).$$

Hence the radial component of the curl is

$$B_r = \frac{1}{r\sin\theta}\partial_\theta(\sin\theta A_\varphi) = -\sum_{l\geq 1} l(l+1)c_l r^{-l-2}P_l(\cos\theta).$$

On the positive axis ($\theta = 0$, $r = z$), $B_r = B_z$ and $P_l(1) = 1$, so part (a) requires $-\sum_l l(l+1)c_l z^{-l-2} = 2\mu_0\sigma\omega a^4/(3z^3)$ for all $z > a$. Uniqueness of the power series in $1/z$ therefore gives $-2c_1 = 2\mu_0\sigma\omega a^4/3$ and $l(l+1)c_l = 0$ for $l > 1$, that is, $c_1 = -\mu_0\sigma\omega a^4/3$ and $c_l = 0$ for $l > 1$. Consequently, for $r > a$ and with $P_1^1(\cos\theta) = -\sin\theta$,

$$\boxed{A_\varphi = -\frac{\mu_0\sigma\omega a^4}{3r^2}P_1^1 = \frac{\mu_0\sigma\omega a^4}{3r^2}\sin\theta}.$$

The radial component is $B_r = -2c_1 r^{-3}\cos\theta$, and

$$B_\theta = -\frac{1}{r}\partial_r(rA_\varphi) = -\frac{\mu_0\sigma\omega a^4\sin\theta}{3r}\,\partial_r\frac{1}{r} = \frac{\mu_0\sigma\omega a^4}{3r^3}\sin\theta,$$

while $B_\varphi = r^{-1}[\partial_r(rA_\theta) - \partial_\theta A_r] = 0$ because $A_r = A_\theta = 0$. Thus, for $r > a$,

$$\boxed{B_r = \frac{2\mu_0\sigma\omega a^4}{3r^3}\cos\theta, \qquad B_\theta = \frac{\mu_0\sigma\omega a^4}{3r^3}\sin\theta, \qquad B_\varphi = 0}.$$

Comparison with the point-dipole field of Exercise 15.4.18, $B_r = \mu_0 m\cos\theta/(2\pi r^3)$, $B_\theta = \mu_0 m\sin\theta/(4\pi r^3)$, shows that this is exactly a dipole field with $m = 4\pi\sigma\omega a^4/3 = Q\omega a^2/3$, $Q = 4\pi a^2\sigma$. In the Instructor's Manual, p. 252, the solution of part (a) has an extra factor 3 in its denominator: its displayed coefficient is 2/9 instead of 2/3. This also conflicts with its own radial field on the positive axis. In its expressions in terms of P_1^1, both A_φ and B_θ require a minus sign. The printed positive signs conflict with $P_1^1 = -\sin\theta$ and with the printed positive radial field.

15.4.20 In the liquid-drop model of the nucleus, a spherical nucleus is subjected to small deformations. Consider a sphere of radius r_0 that is deformed so that its new surface is given by

$$r = r_0\big[1 + \alpha_2 P_2(\cos\theta)\big].$$

Find the area of the deformed sphere through terms of order α_2^2.

Hint.

$$dA = \left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{1/2} r\sin\theta\,d\theta\,d\varphi.$$

ANS. $A = 4\pi r_0^2\left[1 + \frac{4}{5}\alpha_2^2 + O\big(\alpha_2^3\big)\right]$.

Note. The area element dA follows from noting that the line element ds for fixed φ is given by

$$ds = (r^2 d\theta^2 + dr^2)^{1/2} = \left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{1/2} d\theta.$$

Solution.

Put $\alpha = \alpha_2$, $P = P_2(\cos\theta)$ and $\dot{P} = dP_2(\cos\theta)/d\theta$. Then $r = r_0(1+\alpha P)$ and $r_\theta \equiv dr/d\theta = r_0\alpha\dot{P}$. For small α the radius is positive and $r_\theta/r = O(\alpha)$. The binomial series $\sqrt{1+\varepsilon} = 1 + \varepsilon/2 + O(\varepsilon^2)$

with $\varepsilon = (r_\theta/r)^2 = O(\alpha^2)$, together with $r^2 = r_0^2(1 + 2\alpha P + \alpha^2 P^2)$ and $r_\theta^2 = r_0^2\alpha^2\dot{P}^2$, gives the area integrand

$$\begin{aligned} r\sqrt{r^2 + r_\theta^2} &= r^2\sqrt{1 + (r_\theta/r)^2} \\ &= r^2 + \frac{1}{2}r_\theta^2 + O(\alpha^3) \\ &= r_0^2[1 + 2\alpha P + \alpha^2(P^2 + \tfrac{1}{2}\dot{P}^2)] + O(\alpha^3). \end{aligned}$$

With $x = \cos\theta$, $\sin\theta\, d\theta = -dx$, each $\int_0^\pi (\cdots)\sin\theta\, d\theta$ becomes $\int_{-1}^1 (\cdots)dx$. The zeroth-order term gives $\int_{-1}^1 dx = 2$. The first-order integral vanishes because $\int_{-1}^1 P_2 dx = 0$ (orthogonality to $P_0 = 1$). The norm $2/(2l+1)$ gives $\int_{-1}^1 P_2^2 dx = 2/5$. Also, from $P = (3\cos^2\theta - 1)/2$, $\dot{P} = -3\cos\theta\sin\theta$, so $\dot{P}^2 = 9x^2(1 - x^2)$, whence

$$\int_0^\pi \dot{P}^2 \sin\theta\, d\theta = 9\int_{-1}^1 (x^2 - x^4)dx = 9\left(\frac{2}{3} - \frac{2}{5}\right) = \frac{12}{5}.$$

Integrating over φ (a factor 2π) now gives, with $\frac{1}{2}\cdot\frac{12}{5} = \frac{6}{5}$,

$$A = 2\pi r_0^2\left[2 + \alpha^2\left(\frac{2}{5} + \frac{6}{5}\right) + O(\alpha^3)\right] = \boxed{4\pi r_0^2[1 + \tfrac{4}{5}\alpha_2^2 + O(\alpha_2^3)]}.$$

Here r_0 is the fixed constant of the given surface equation; no volume-conserving readjustment of r_0 is made.

15.5 Spherical Harmonics

15.5.1 Show that the parity of $Y_l^m(\theta, \varphi)$ is $(-1)^l$. Note the disappearance of any m dependence.

Hint. For the parity operation in spherical polar coordinates, see Exercise 3.10.25.

Solution.

Spatial parity sends $\mathbf{r}$ to $-\mathbf{r}$, so, by Exercise 3.10.25, $(\theta, \varphi) \mapsto (\pi - \theta, \varphi + \pi)$ and $x = \cos\theta \mapsto \cos(\pi - \theta) = -x$. In Rodrigues's formula (as in Exercise 15.4.3), valid for integers $-l \le m \le l$,

$$P_l^m(x) = \frac{(-1)^m}{2^l l!}(1 - x^2)^{m/2}\frac{d^{l+m}}{dx^{l+m}}(x^2 - 1)^l,$$

the factor $(1 - x^2)^{m/2}$ is even, while $(x^2 - 1)^l$ is even, so its $(l + m)$th derivative has parity $(-1)^{l+m}$ (each derivative reverses the parity). Hence $P_l^m(-x) = (-1)^{l+m}P_l^m(x)$. The azimuthal factor changes by $e^{im(\varphi+\pi)} = e^{im\pi}e^{im\varphi} = (-1)^m e^{im\varphi}$. The normalization constant is unchanged. Multiplying the two signs,

$$\boxed{Y_l^m(\pi - \theta, \varphi + \pi) = (-1)^{l+2m}Y_l^m(\theta, \varphi) = (-1)^l Y_l^m(\theta, \varphi)}.$$

Thus it is spatial parity, rather than θ reflection alone, that removes the m dependence.

15.5.2 Prove that $Y_l^m(0,\varphi) = \left(\dfrac{2l+1}{4\pi}\right)^{1/2} \delta_{m0}$.

Solution.

At $\theta = 0$, $x = \cos\theta = 1$, so the definition, Eq. (15.137) (displayed in Exercise 15.5.3), gives

$$Y_l^m(0,\varphi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\, P_l^m(1) e^{im\varphi}.$$

For $m > 0$, $P_l^m(x) = (-1)^m(1-x^2)^{m/2} P_l^{(m)}(x)$, where $P_l^{(m)} = d^m P_l/dx^m$. The derivative is a finite polynomial, so at $x = 1$ the factor $(1-x^2)^{m/2}$ forces $P_l^m(1) = 0$. For $m < 0$, Exercise 15.4.3 makes P_l^m a constant multiple of $P_l^{|m|}$, which gives the same zero. For $m = 0$, $P_l(1) = 1$, the factorial normalization ratio $(l-m)!/(l+m)!$ is one, and the azimuthal exponential is one. Therefore

$$\boxed{Y_l^m(0,\varphi) = \sqrt{\frac{2l+1}{4\pi}}\delta_{m0}}.$$

The result is independent of the otherwise undefined azimuthal coordinate at the pole.

15.5.3 In the theory of Coulomb excitation of nuclei we encounter $Y_l^m(\pi/2, 0)$. Show that

$$\begin{aligned} Y_l^m\left(\frac{\pi}{2},0\right) &= \left(\frac{2l+1}{4\pi}\right)^{1/2} \frac{[(l-m)!\,(l+m)!]^{1/2}}{(l-m)!!\,(l+m)!!}\,(-1)^{(l+m)/2}, \quad l+m \text{ even}, \\ &= 0, \qquad l+m \text{ odd}. \end{aligned}$$

Solution.

The book's definition, Eq. (15.137), is

$$Y_l^m(\theta,\varphi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\, P_l^m(\cos\theta) e^{im\varphi},$$

with the Condon–Shortley sign already in P_l^m. At the point in question, $x = \cos(\pi/2) = 0$ and the exponential equals one, so only $P_l^m(0)$ is needed; Exercise 15.4.5 gives it for all integers $-l \le m \le l$. If $l+m$ is odd, Exercise 15.4.5 gives zero. If $l+m$ is even, substitution of $P_l^m(0) = (-1)^{(l+m)/2}(l+m-1)!!/(l-m)!!$ gives

$$Y_l^m(\tfrac{\pi}{2},0) = (-1)^{(l+m)/2}\sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\,\frac{(l+m-1)!!}{(l-m)!!}.$$

Use $(l+m)! = (l+m)!!(l+m-1)!!$, that is, $(l+m-1)!! = (l+m)!/(l+m)!!$, to rewrite the magnitude:

$$\sqrt{\frac{(l-m)!}{(l+m)!}}\,\frac{(l+m)!}{(l+m)!!\,(l-m)!!} = \frac{\sqrt{(l-m)!(l+m)!}}{(l-m)!!(l+m)!!},$$

so that

$$\boxed{Y_l^m(\tfrac{\pi}{2},0) = (-1)^{(l+m)/2}\sqrt{\frac{2l+1}{4\pi}}\,\frac{\sqrt{(l-m)!(l+m)!}}{(l-m)!!(l+m)!!}}.$$

Since Exercise 15.4.5 covers negative orders, the result holds for $-l \le m < 0$ as well. The printed phase $(-1)^{(l-m)/2}$ differs by $(-1)^m$ from the phase required by the book's own definition. For $l = m = 1$, $Y_1^1(\pi/2,0) = -\sqrt{3/(8\pi)}$, which directly disproves the printed positive sign. The Instructor's Manual obtains the correct $(l+m)/2$ phase before claiming agreement with the printed result.

15.5.4 The orthogonal azimuthal functions yield a useful representation of the Dirac delta function. Show that as a distribution on the circle

$$\delta(\varphi_1 - \varphi_2) = \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} e^{im(\varphi_1 - \varphi_2)}.$$

Note. This formula assumes that φ_1 and φ_2 are restricted to $0 \le \varphi < 2\pi$. Without this restriction there will be additional delta-function contributions at intervals of 2π in $\varphi_1 - \varphi_2$.

Solution.

On the circle, the normalized functions are $\phi_m(\varphi) = e^{im\varphi}/\sqrt{2\pi}$, and the proposed kernel is the sum of $\phi_m(\varphi_1)\phi_m(\varphi_2)^*$ over all m. For a smooth periodic f, its Fourier coefficients are $c_m = (2\pi)^{-1}\int_0^{2\pi} f(\varphi_2)e^{-im\varphi_2}d\varphi_2$. Pair the proposed kernel with f, taking the sum as the limit of the symmetric partial sums $|m| \le N$ and integrating each of these term by term:

$$\int_0^{2\pi} \frac{1}{2\pi} \sum_{|m|\le N} e^{im(\varphi_1-\varphi_2)} f(\varphi_2)d\varphi_2 = \sum_{|m|\le N} c_m e^{im\varphi_1} \longrightarrow f(\varphi_1) \quad (N \to \infty).$$

For the limit, let f be twice continuously differentiable. Two integrations by parts, whose boundary terms cancel by periodicity, give $c_m = -(2\pi m^2)^{-1}\int_0^{2\pi} f''(\varphi_2)e^{-im\varphi_2}d\varphi_2$ for $m \ne 0$, so $|c_m| \le \max|f''|/m^2$. The Fourier series of f therefore converges absolutely and uniformly, and by Dirichlet's theorem for Fourier series (Chapter 19) its sum is f. This proves the periodic delta identity. The series itself converges at no point, since its terms all have modulus $1/(2\pi)$; it has a meaning only through this action on test functions, as the qualification in the statement records.

One explicit regularization is to insert $\rho^{|m|}$ with $0 < \rho < 1$. Summing the geometric series over $m \ge 0$ and over $m \le -1$ separately ($|\rho e^{\pm iu}| = \rho < 1$) gives

$$\frac{1}{1-\rho e^{iu}} + \frac{\rho e^{-iu}}{1-\rho e^{-iu}} = \frac{1 - \rho e^{-iu} + \rho e^{-iu} - \rho^2}{(1-\rho e^{iu})(1-\rho e^{-iu})},$$

whose numerator is $1-\rho^2$ and whose denominator is $1 - \rho(e^{iu}+e^{-iu}) + \rho^2 = 1 - 2\rho\cos u + \rho^2$, so that

$$K_\rho(u) \equiv \frac{1}{2\pi}\sum_m \rho^{|m|}e^{imu} = \frac{1-\rho^2}{2\pi(1-2\rho\cos u + \rho^2)}, \quad 0 < \rho < 1.$$

It is nonnegative, since $1 - 2\rho\cos u + \rho^2 \ge (1-\rho)^2 > 0$. Its integral over a period is one, since the series converges uniformly in u and integrating term by term leaves only $m = 0$. It tends uniformly to zero away from $u = 0$ modulo 2π: for fixed $0 < \eta < \pi$ and $\eta \le |u| \le \pi$, $1 - 2\rho\cos u + \rho^2 = (1-\rho)^2 + 2\rho(1-\cos u) \ge 2\rho(1-\cos\eta)$, so $K_\rho(u) \le (1-\rho^2)/[4\pi\rho(1-\cos\eta)] \to 0$. Consequently, for continuous periodic f, substitute $u = \varphi_1 - \varphi_2$, which turns $\int_0^{2\pi} d\varphi_2$ into $\int_{\varphi_1-2\pi}^{\varphi_1} du$; the integrand is 2π-periodic in u, so this interval may be shifted to $[-\pi, \pi]$. Writing $f(\varphi_1) = \int_{-\pi}^{\pi} K_\rho(u) f(\varphi_1)\, du$ (unit integral) then gives

$$\int_0^{2\pi} K_\rho(\varphi_1 - \varphi_2) f(\varphi_2) d\varphi_2 - f(\varphi_1) = \int_{-\pi}^{\pi} K_\rho(u)[f(\varphi_1 - u) - f(\varphi_1)]du.$$

Since $K_\rho \ge 0$ has unit integral, the part $|u| < \eta$ is at most $\max_{|u|<\eta}|f(\varphi_1 - u) - f(\varphi_1)|$, which is small for small η by continuity, and the part $|u| \ge \eta$ is at most $4\pi\max|f|$ times the bound on K_ρ, which tends to zero. Choosing first η and then ρ close to 1 makes both parts arbitrarily small; hence K_ρ tends to the periodic delta as $\rho \uparrow 1$. On the real line the same distribution is $\delta_{2\pi}(u) \equiv \sum_{k\in\mathbb{Z}} \delta(u - 2\pi k)$, one delta in each period, in agreement with the note.

15.5.5 Derive the spherical harmonic closure relation as an identity of distributions for spherical area measure, using $0 < \theta_1 < \pi$ in its coordinate form and the periodic delta in azimuth:

$$\sum_{l=0}^{\infty} \sum_{m=-l}^{+l} Y_l^m(\theta_1, \varphi_1) \left[Y_l^m(\theta_2, \varphi_2) \right]^* = \frac{1}{\sin\theta_1} \, \delta(\theta_1 - \theta_2) \, \delta(\varphi_1 - \varphi_2)$$
$$= \delta(\cos\theta_1 - \cos\theta_2) \, \delta(\varphi_1 - \varphi_2).$$

Solution.

Let $d\Omega = \sin\theta \, d\theta \, d\varphi$ and define

$$K(\Omega_1, \Omega_2) = \sum_{l,m} Y_l^m(\Omega_1) Y_l^m(\Omega_2)^*.$$

For a smooth function $f = \sum f_{lm} Y_l^m$, with $f_{lm} = \int Y_l^m(\Omega)^* f(\Omega) d\Omega$, integration term by term gives

$$\int K(\Omega_1, \Omega_2) f(\Omega_2) d\Omega_2 = \sum_{l,m} Y_l^m(\Omega_1) \int Y_l^m(\Omega_2)^* f(\Omega_2) d\Omega_2 = \sum_{l,m} f_{lm} Y_l^m(\Omega_1) = f(\Omega_1).$$

Thus K is the delta distribution for spherical area measure.

For completeness of the basis used here, first expand in Fourier modes in φ. At fixed m, the polar factors P_l^m, $l = |m|, |m|+1, \ldots$, are $(1-x^2)^{|m|/2}$ times polynomials of successive degrees $0, 1, 2, \ldots$ in $x = \cos\theta$, and $d\Omega = dx \, d\varphi$. If an L^2 function of x is orthogonal on $[-1, 1]$ to every such factor, its product with $(1-x^2)^{|m|/2}$ is orthogonal to every polynomial. Polynomial density (Weierstrass) makes that product zero almost everywhere, and since $(1-x^2)^{|m|/2} > 0$ for $|x| < 1$, the original function is zero almost everywhere. This proves completeness of each polar family. Now let F be orthogonal to every Y_l^m. Then, for each m, its Fourier coefficient $F_m(x) = \int_0^{2\pi} F e^{-im\varphi} d\varphi$ is orthogonal to every P_l^m with $l \geq |m|$, hence zero almost everywhere; by completeness of the $e^{im\varphi}$, $F = 0$ almost everywhere. So the spherical harmonics are complete.

The distributional passage to the infinite kernel can be made without assuming pointwise convergence of an arbitrary L^2 series. Let K_N retain degrees $l \leq N$, so integration against f gives its orthogonal projection $S_N f$. For every smooth test function h, the Cauchy–Schwarz inequality and completeness (Parseval: $\|S_N f - f\|_2^2 = \sum_{l>N} \sum_m |f_{lm}|^2 \to 0$) give

$$\left| \int h^*(\Omega_1) \left[\int K_N(\Omega_1, \Omega_2) f(\Omega_2) d\Omega_2 - f(\Omega_1) \right] d\Omega_1 \right| \leq \|h\|_2 \|S_N f - f\|_2 \longrightarrow 0.$$

Thus the limiting kernel acts as the identity when tested in both variables, which is precisely the delta kernel. For smooth f this agrees with the evaluation formula used above.

The coordinate kernel that reproduces the same test-function action is, with the periodic delta $\delta_{2\pi}$ of Exercise 15.5.4,

$$\boxed{K = \frac{\delta(\theta_1 - \theta_2)\delta_{2\pi}(\varphi_1 - \varphi_2)}{\sin\theta_1} = \delta(\cos\theta_1 - \cos\theta_2)\delta_{2\pi}(\varphi_1 - \varphi_2)}.$$

Indeed, for $0 < \theta_1 < \pi$,

$$\int_0^{2\pi} \int_0^{\pi} \frac{\delta(\theta_1 - \theta_2)\delta_{2\pi}(\varphi_1 - \varphi_2)}{\sin\theta_1} f(\theta_2, \varphi_2) \sin\theta_2 \, d\theta_2 \, d\varphi_2 = f(\theta_1, \varphi_1),$$

and, since $g(\theta_2) = \cos\theta_1 - \cos\theta_2$ has the single zero $\theta_2 = \theta_1$ in $(0, \pi)$ with $g'(\theta_1) = \sin\theta_1 > 0$, $\delta(\cos\theta_1 - \cos\theta_2) = \delta(\theta_1 - \theta_2)/\sin\theta_1$: the Jacobian of $d(\cos\theta) = -\sin\theta \, d\theta$ cancels. The closure formula is a distributional identity. Its polar coordinate form is used for $0 < \theta_1 < \pi$; at a pole the invariant spherical delta, rather than division by a zero sine, specifies its meaning.

15.5.6 In some circumstances it is desirable to replace the imaginary exponential of our spherical harmonic by sine or cosine. Morse and Feshbach (see Additional Readings) define

$$\overset{e}{Y}_{ml} = P_l^m(\cos\theta)\cos m\varphi, \qquad m \geq 0,$$
$$\overset{o}{Y}_{ml} = P_l^m(\cos\theta)\sin m\varphi, \qquad m > 0,$$

and their normalization integrals are

$$\int_0^{2\pi}\int_0^{\pi} \left[\overset{e \text{ or } o}{Y}_{mn}(\theta,\varphi)\right]^2 \sin\theta\, d\theta\, d\varphi = \frac{4\pi}{2(2n+1)}\frac{(n+m)!}{(n-m)!}, \qquad 1 \leq m \leq n,$$
$$= \frac{4\pi}{2n+1}, \qquad m = 0,\ n \geq 1 \quad \text{(cosine family)},$$
$$= 4\pi, \qquad n = 0.$$

These spherical harmonics are often named according to the patterns of their positive and negative regions on the surface of a sphere: zonal harmonics for $m = 0$, sectoral harmonics for $m = n$, and tesseral harmonics for $0 < m < n$. For $\overset{e}{Y}_{mn}$, $n = 4$, $m = 0, 2, 4$, indicate on a diagram of a hemisphere (one diagram for each spherical harmonic) the regions in which the spherical harmonic is positive.

Solution.

In the two definitions the degree is written l; it is the n of the normalization integrals and of the rest of the statement, and we write n throughout. We also write Y^e_{mn} and Y^o_{mn} for $\overset{e}{Y}_{mn}$ and $\overset{o}{Y}_{mn}$, and $\|Y\|^2 = \int_0^{2\pi}\int_0^{\pi} Y^2 \sin\theta\, d\theta\, d\varphi$ for a normalization integral. Since each Y is a function of θ times a function of φ, this integral is the product of a polar and an azimuthal integral. With $x = \cos\theta$, the polar one is the associated Legendre normalization $\int_{-1}^{1}[P_n^m(x)]^2 dx = 2(n+m)!/[(2n+1)(n-m)!]$. For $m > 0$, both $\int_0^{2\pi}\cos^2 m\varphi\, d\varphi$ and its sine counterpart equal π (write $\cos^2 m\varphi = (1+\cos 2m\varphi)/2$ and $\sin^2 m\varphi = (1-\cos 2m\varphi)/2$; the $\cos 2m\varphi$ term integrates to zero), giving the stated value

$$\pi\,\frac{2(n+m)!}{(2n+1)(n-m)!} = \frac{4\pi}{2(2n+1)}\frac{(n+m)!}{(n-m)!}, \qquad 1 \leq m \leq n.$$

For $m = 0$, the cosine factor is identically one, its square integrates to 2π, and the polar integral is $2/(2n+1)$, so

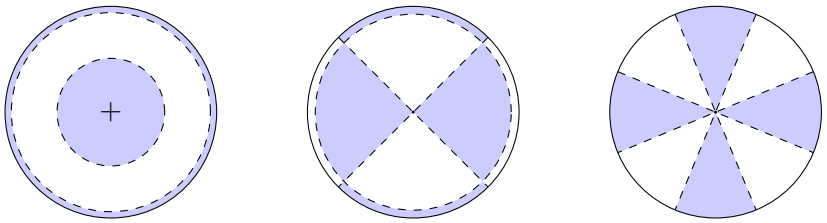
$$\boxed{\|Y^e_{0n}\|^2 = \frac{4\pi}{2n+1}}.$$

The printed normalization for $n \geq 1$ is valid for $m > 0$. For the included $m = 0$ harmonics it is smaller than the correct norm by a factor of two. The separate $n = 0$ value 4π is correct.

For $n = 4$, use $P_4^m = (-1)^m(1-x^2)^{m/2}d^mP_4/dx^m$ with $P_4 = (35x^4 - 30x^2 + 3)/8$ and $x = \cos\theta$; the sign $(-1)^m$ is $+1$ for even m. Since $d^2P_4/dx^2 = (420x^2 - 60)/8 = \frac{15}{2}(7x^2 - 1)$ and $d^4P_4/dx^4 = 35 \cdot 24/8 = 105$, the three real functions are

$$Y^e_{04} = \frac{1}{8}(35x^4 - 30x^2 + 3), \quad Y^e_{24} = \frac{15}{2}(1-x^2)(7x^2-1)\cos 2\varphi, \quad Y^e_{44} = 105(1-x^2)^2\cos 4\varphi.$$

The first polynomial in the Instructor's Manual, p. 253, prints $35\cos^2\theta$ in the leading term. It must be $35\cos^4\theta$, as in P_4 above; the squared power would incorrectly reduce this degree-four polynomial to degree two. On the north hemisphere, viewed from above, put $\rho = \sin\theta$, so its orthographic projection is the unit disk. In the diagrams blue regions are positive and white regions negative; the dashed curves are the zeros, the solid circle is the equator, and the meridian $\varphi = 0$ points to the right.



$m = 0$ (zonal) $m = 2$ (tesseral) $m = 4$ (sectoral)

For $m = 0$, the latitude zeros solve $35y^2 - 30y + 3 = 0$ with $y = x^2$, so $x^2 = (30 \pm \sqrt{900 - 420})/70 = (15 \pm 2\sqrt{30})/35$, that is, $x \approx 0.861$ and 0.340 ($\theta \approx 30.6°$ and $70.1°$, $\rho \approx 0.508$ and 0.940). Since $P_4(1) = 1 > 0$ and $P_4(0) = 3/8 > 0$, the polar cap and the equatorial band are positive, and the band between them is negative. For $m = 2$, the factor $7x^2 - 1$ gives the latitude zero $x = 1/\sqrt{7}$ ($\rho = \sqrt{6/7}$), and the zeros of $\cos 2\varphi$ give the meridians $\varphi = \pi/4 + j\pi/2$. Inside that latitude circle ($x > 1/\sqrt{7}$, where $7x^2 - 1 > 0$) the sign is that of $\cos 2\varphi$; outside it the sign reverses. For $m = 4$ there are no interior latitude zeros, and positivity is $\cos 4\varphi > 0$, namely sectors centered at $\varphi = j\pi/2$ of half-width $\pi/8$. The pole is also a zero for $m = 2, 4$. Since $n + m$ is even in all three cases, each function is even in x ($P_n^m(-x) = (-1)^{n+m} P_n^m(x)$, Exercise 15.5.1), so the southern hemisphere has the same pattern.

15.5.7 Suppose $f(r, \theta, \varphi)$ has the regular Laplace series below, converging uniformly on the closed ball under consideration:

$$f(r, \theta, \varphi) = \sum_{l,m} a_{lm} r^l Y_l^m(\theta, \varphi).$$

Letting $\langle \cdots \rangle_{\text{sphere}}$ denote the average over a sphere centered on the origin, show that

$$\big\langle f(r, \theta, \varphi) \big\rangle_{\text{sphere}} = f(0, 0, 0).$$

Solution.

Use $Y_0^0 = 1/\sqrt{4\pi}$ and orthogonality:

$$\int Y_l^m \, d\Omega = \sqrt{4\pi} \int Y_l^m (Y_0^0)^* d\Omega = \sqrt{4\pi}\, \delta_{l0} \delta_{m0}.$$

The average over the sphere of radius r is $\langle f \rangle_{\text{sphere}} = (4\pi)^{-1} \int f(r, \theta, \varphi) \, d\Omega$. Integrating the regular solid-harmonic expansion term by term, which the assumed uniform convergence permits, therefore leaves only its constant term:

$$\langle f \rangle_{\text{sphere}} = \frac{1}{4\pi} \sum_{l,m} a_{lm} r^l \int Y_l^m \, d\Omega = \frac{a_{00}}{\sqrt{4\pi}}.$$

At the origin every $r^l Y_l^m$ with $l > 0$ vanishes, whereas the $l = 0$ term has the same value $a_{00} Y_0^0 = a_{00}/\sqrt{4\pi}$. Thus

$$\boxed{\langle f \rangle_{\text{sphere}} = f(0, 0, 0)}.$$

This is the mean-value property for the assumed regular harmonic expansion. It does not hold for an arbitrary function; for example, $f = r^2$ has sphere average r^2 but $f(0) = 0$.

15.6 Legendre Functions of the Second Kind

15.6.1 Use the real branches specified in the solution, with $x \in \mathbb{R} \setminus \{-1, 1\}$ and integer $l \geq 0$. Show that if l is even, $Q_l(-x) = -Q_l(x)$, and that if l is odd, $Q_l(-x) = Q_l(x)$.

Solution.

Use the real Legendre functions with

$$Q_0(x) = \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|, \qquad Q_1(x) = xQ_0(x) - 1, \qquad x \neq \pm 1.$$

Replacing x by $-x$ replaces the argument $|(1+x)/(1-x)|$ of the logarithm by its reciprocal: $Q_0(-x) = \frac{1}{2}\ln|(1-x)/(1+x)| = -Q_0(x)$, so Q_0 is odd, and then $Q_1(-x) = (-x)(-Q_0(x)) - 1 = Q_1(x)$, so Q_1 is even. Thus $Q_l(-x) = (-1)^{l+1}Q_l(x)$ holds for $l = 0, 1$. The Q_l obey the same three-term recurrence as the P_l, $(n+1)Q_{n+1}(x) = (2n+1)xQ_n(x) - nQ_{n-1}(x)$. Assume $Q_n(-x) = (-1)^{n+1}Q_n(x)$ and $Q_{n-1}(-x) = (-1)^n Q_{n-1}(x)$ for some $n \geq 1$. Then the recurrence at $-x$ and $-(-1)^{n+1} = (-1)^{n+2} = (-1)^n$ give

$$\begin{aligned}(n+1)Q_{n+1}(-x) &= -(2n+1)x(-1)^{n+1}Q_n(x) - n(-1)^n Q_{n-1}(x)\\ &= (-1)^{n+2}[(2n+1)xQ_n(x) - nQ_{n-1}(x)]\\ &= (-1)^{n+2}(n+1)Q_{n+1}(x).\end{aligned}$$

Induction proves

$$\boxed{Q_l(-x) = (-1)^{l+1}Q_l(x)}.$$

In particular even degree gives an odd function and odd degree an even function. The assertion refers to these real functions (the usual real Q_l both for $|x| < 1$ and for $|x| > 1$); another choice of branch of the logarithm would add a multiple of P_l, whose parity $(-1)^l$ is the opposite one.

15.6.2 Show that, for nonnegative integers p and the real analytic Q_l branch at $x = 0$,

(a) $$Q_{2p}(x) = (-1)^p 2^{2p}\sum_{s=0}^{p}(-1)^s\frac{(p+s)!\,(p-s)!}{(2s+1)!\,(2p-2s)!}\,x^{2s+1} + 2^{2p}\sum_{s=p+1}^{\infty}\frac{(p+s)!\,(2s-2p)!}{(2s+1)!\,(s-p)!}\,x^{2s+1}, \qquad |x| < 1,$$

(b) $$Q_{2p+1}(x) = (-1)^{p+1}2^{2p}\sum_{s=0}^{p}(-1)^s\frac{(p+s)!\,(p-s)!}{(2s)!\,(2p-2s+1)!}\,x^{2s} + 2^{2p+1}\sum_{s=p+1}^{\infty}\frac{(p+s)!\,(2s-2p-2)!}{(2s)!\,(s-p-1)!}\,x^{2s}, \qquad |x| < 1.$$

Solution.

Use the real analytic branch at $x = 0$. As in Exercise 15.6.1, another branch of the logarithm would add a multiple of P_l (for example, the exterior Q_l continued onto $-1 < x < 1$ from above or below the real axis equals this real function $\mp\frac{i\pi}{2}P_l(x)$), and P_l has only powers of the parity that is absent from the stated series. Substituting $y = \sum_j a_j x^j$ into $(1-x^2)y'' - 2xy' + l(l+1)y = 0$, the

coefficient of x^j is $(j+2)(j+1)a_{j+2} - j(j-1)a_j - 2ja_j + l(l+1)a_j$, so the Legendre equation gives the coefficient recurrence for any Taylor solution:

$$(j+2)(j+1)a_{j+2} = [j(j+1) - l(l+1)]a_j.$$

By Exercise 15.6.1, Q_{2p} is odd and Q_{2p+1} is even, so their Taylor series contain only odd and only even powers, respectively, and the recurrence determines them from $Q'_{2p}(0)$ and $Q_{2p+1}(0)$. These initial scales follow from $Q_0 = x + \cdots$, $Q_1 = -1 + \cdots$ and the degree recurrences:

$$Q'_{2p}(0) = (-1)^p \frac{(2p)!!}{(2p-1)!!}, \qquad Q_{2p+1}(0) = (-1)^{p+1} \frac{(2p)!!}{(2p+1)!!}.$$

For the odd-degree values, the three-term recurrence $(n+1)Q_{n+1} = (2n+1)xQ_n - nQ_{n-1}$ at $x = 0$ with $n = 2p$ gives $(2p+1)Q_{2p+1}(0) = -2pQ_{2p-1}(0)$; iterating down to $Q_1(0) = -1$,

$$Q_{2p+1}(0) = \prod_{k=1}^{p} \left(-\frac{2k}{2k+1} \right) Q_1(0) = (-1)^{p+1} \frac{(2p)!!}{(2p+1)!!}.$$

For the even-degree derivative, the identity $(1-x^2)Q'_n = n(Q_{n-1} - xQ_n)$, which the Q_n share with the P_n, gives with $n = 2p$ at $x = 0$, and the odd-degree value for $2p - 1 = 2(p-1) + 1$,

$$Q'_{2p}(0) = 2pQ_{2p-1}(0) = (-1)^p \frac{2p(2p-2)!!}{(2p-1)!!} = (-1)^p \frac{(2p)!!}{(2p-1)!!} \quad (p \geq 1).$$

The case $p = 0$ comes from Q_0. With $(2p)!! = 2^p p!$, $(2p-1)!! = (2p)!/(2^p p!)$ and $(2p+1)!! = (2p+1)!/(2^p p!)$, these are the values C_0 and D_0 below.

(a) Write $Q_{2p} = \sum_{s \geq 0} C_s x^{2s+1}$. For $j = 2s+1$ and $l = 2p$, $(2s+1)(2s+2) - 2p(2p+1) = 2(s+p+1)(2s-2p+1)$, so the Taylor recurrence is

$$\frac{C_{s+1}}{C_s} = \frac{(s+p+1)(2s-2p+1)}{(s+1)(2s+3)}, \qquad C_0 = (-1)^p 2^{2p} \frac{(p!)^2}{(2p)!}.$$

For $0 \leq s \leq p$, its product gives

$$C_s = (-1)^{p+s} 2^{2p} \frac{(p+s)!(p-s)!}{(2s+1)!(2p-2s)!},$$

which equals C_0 at $s = 0$. To check the product explicitly, the factorial ratio between adjacent terms of this formula ($0 \leq s \leq p-1$) is, with $(2p-2s)/(p-s) = 2$,

$$-\frac{p+s+1}{p-s} \frac{(2p-2s)(2p-2s-1)}{(2s+3)(2s+2)} = \frac{(p+s+1)(2s-2p+1)}{(s+1)(2s+3)}.$$

At the transition, the formula gives $C_p = 2^{2p}/(2p+1)$, and the recurrence with $s = p$ gives $C_{p+1} = C_p(2p+1)/[(p+1)(2p+3)] = 2^{2p}/[(p+1)(2p+3)]$. For $s \geq p+1$, the all-positive continuation is

$$C_s = 2^{2p} \frac{(p+s)!(2s-2p)!}{(2s+1)!(s-p)!},$$

which reproduces C_{p+1} at $s = p+1$. Its adjacent ratio,

$$\frac{(p+s+1)(2s-2p+2)(2s-2p+1)}{(2s+3)(2s+2)(s-p+1)},$$

reduces to the same recurrence by canceling $2s-2p+2=2(s-p+1)$ against $s-p+1$. These formulas give exactly the two sums in part (a).

(b) Put $Q_{2p+1}=\sum_{s\ge 0} D_s x^{2s}$. This time $j=2s$, $l=2p+1$, and $2s(2s+1)-(2p+1)(2p+2)=2(s+p+1)(2s-2p-1)$, so

$$\frac{D_{s+1}}{D_s}=\frac{(s+p+1)(2s-2p-1)}{(s+1)(2s+1)}, \qquad D_0=(-1)^{p+1}2^{2p}\frac{(p!)^2}{(2p+1)!}.$$

For $0\le s\le p$, repeated multiplication gives

$$D_s=(-1)^{p+1+s}2^{2p}\frac{(p+s)!(p-s)!}{(2s)!(2p-2s+1)!},$$

which equals D_0 at $s=0$. Its ratio ($0\le s\le p-1$) is $-(p+s+1)(2p-2s+1)(2p-2s)/[(p-s)(2s+2)(2s+1)]$, equal to the required ratio after canceling $(2p-2s)/(p-s)=2$ against $2s+2=2(s+1)$ and writing $-(2p-2s+1)=2s-2p-1$. At the transition, the formula gives $D_p=-2^{2p}$, and the recurrence with $s=p$ gives $D_{p+1}=D_p(2p+1)(-1)/[(p+1)(2p+1)]=2^{2p}/(p+1)$. For the positive tail,

$$D_s=2^{2p+1}\frac{(p+s)!(2s-2p-2)!}{(2s)!(s-p-1)!}, \qquad s\ge p+1,$$

which reproduces $D_{p+1}=2^{2p+1}/(2p+2)$ at $s=p+1$. Its adjacent ratio is $(p+s+1)(2s-2p)(2s-2p-1)/[(2s+2)(2s+1)(s-p)]$, and canceling $2s-2p=2(s-p)$ against $s-p$ again gives the same recurrence. The initial values, the ratios within each range and the transition values therefore establish both requested sums for every $s\ge 0$. Their term ratios tend to x^2, proving convergence for $|x|<1$; uniqueness of the Taylor solution identifies them with Q_l.

15.6.3

(a) Use integer $l\ge 0$ and the exterior Q_l branch decaying at infinity, on $|x|>1$. Starting with the assumed form

$$Q_l(x)=\sum_{j=0}^{\infty} b_{lj}x^{k-j},$$

show that

$$Q_l(x)=b_{l0}x^{-l-1}\sum_{s=0}^{\infty}\frac{(l+s)!\,(l+2s)!\,(2l+1)!}{s!\,(l!)^2(2l+2s+1)!}\,x^{-2s}.$$

(b) The standard choice of b_{l0} is

$$b_{l0}=\frac{2^l(l!)^2}{(2l+1)!},$$

leading to the final result

$$Q_l(x)=x^{-l-1}\sum_{s=0}^{\infty}\frac{(l+2s)!}{(2s)!!\,(2l+2s+1)!!}\,x^{-2s}.$$

Show that this choice of b_{l0} brings this negative power-series form of $Q_n(x)$ into agreement with the closed-form solutions.

Solution.

(a) For a monomial x^α, the Legendre operator $L_l = (1-x^2)\,d^2/dx^2 - 2x\,d/dx + l(l+1)$ gives $(1-x^2)\alpha(\alpha-1)x^{\alpha-2} - 2\alpha x^\alpha + l(l+1)x^\alpha$, that is,

$$L_l[x^\alpha] = \alpha(\alpha-1)x^{\alpha-2} + [l(l+1) - \alpha(\alpha+1)]x^\alpha.$$

The leading term $b_{l0}x^k$ ($b_{l0} \neq 0$) of the proposed series gives the highest power x^k, with coefficient $[l(l+1) - k(k+1)]b_{l0}$; the equation therefore requires $k(k+1) = l(l+1)$, giving $k = l$ or $k = -l-1$. The solution that decays at infinity uses the latter root. The power $x^{k-j} = x^{-l-1-j}$ then receives $[l(l+1) - (l+j)(l+j+1)]b_{lj} = -j(2l+j+1)b_{lj}$ from the term j, and, through $\alpha(\alpha-1)x^{\alpha-2}$ with $\alpha = -l-j+1$, $(l+j-1)(l+j)b_{l,j-2}$ from the term $j-2$. Hence

$$j(2l+j+1)b_{lj} = (l+j)(l+j-1)b_{l,j-2}, \qquad b_{l,-1} = 0.$$

At $j = 1$, the power x^{-l-2}, this says $2(l+1)b_{l1} = 0$. It follows that $b_{l1} = 0$, and the recurrence then makes every odd j vanish. Set $c_s = b_{l,2s}$. At power $x^{-l-1-2s}$ ($j = 2s$) the equation gives

$$2s(2l+2s+1)c_s = (l+2s)(l+2s-1)c_{s-1}.$$

Hence

$$\frac{c_s}{c_0} = \prod_{q=1}^{s} \frac{(l+2q)(l+2q-1)}{2q(2l+2q+1)} = \frac{(l+s)!(l+2s)!(2l+1)!}{s!(l!)^2(2l+2s+1)!}.$$

For the last equality, the numerator product is $(l+1)(l+2)\cdots(l+2s) = (l+2s)!/l!$, while the denominator product is $2^s s!\,(2l+3)(2l+5)\cdots(2l+2s+1) = 2^s s!\,(2l+2s+1)!!/(2l+1)!!$; with $(2N+1)!! = (2N+1)!/(2^N N!)$ for $N = l+s$ and $N = l$, this is $s!\,l!(2l+2s+1)!/[(l+s)!(2l+1)!]$. This proves the requested series. Since $c_s/c_{s-1} = (l+2s)(l+2s-1)/[2s(2l+2s+1)] \to 1$, the ratio of successive terms tends to x^{-2}, and the series converges for $|x| > 1$.

(b) The normalization can be checked against the closed logarithmic solutions for every degree by the integral

$$R_l(z) = \frac{1}{2}\int_{-1}^{1} \frac{P_l(t)}{z-t}dt, \qquad |z| > 1.$$

Directly, since $-\operatorname{Log}(z-t)$ is an antiderivative of $1/(z-t)$, $R_0 = \frac{1}{2}[\operatorname{Log}(z+1) - \operatorname{Log}(z-1)] = \frac{1}{2}\operatorname{Log}[(z+1)/(z-1)]$ with the branch vanishing at infinity (for real $z > 1$ this is $\frac{1}{2}\ln[(z+1)/(z-1)] = Q_0(z)$), and, with $t/(z-t) = z/(z-t) - 1$, $R_1 = \frac{1}{2}\int_{-1}^{1}[z/(z-t) - 1]dt = zR_0 - 1 = Q_1(z)$. For $l \geq 1$, multiply the polynomial degree recurrence $(l+1)P_{l+1}(t) = (2l+1)tP_l(t) - lP_{l-1}(t)$ by $1/[2(z-t)]$, integrate over $[-1,1]$, and use $t/(z-t) = z/(z-t) - 1$ and $\int_{-1}^{1} P_l\,dt = 0$; the middle term becomes $(2l+1)[zR_l - \frac{1}{2}\int_{-1}^{1} P_l\,dt] = (2l+1)zR_l$. This gives

$$(l+1)R_{l+1} = (2l+1)zR_l - lR_{l-1}.$$

The closed forms Q_l obey the same recurrence and have the same first two members, so $R_l = Q_l$ for every l by induction. For $|t| \leq 1 < |z|$ the geometric series $1/(z-t) = \sum_{j\geq 0} t^j z^{-j-1}$ converges uniformly in t, so it may be integrated term by term under the finite integral: $R_l(z) = \sum_{j\geq 0} z^{-j-1}\,\frac{1}{2}\int_{-1}^{1} t^j P_l(t)dt$. All moments with $j < l$ vanish by orthogonality (t^j is a combination of $P_0, \ldots, P_j$), so the series starts with z^{-l-1}, as in part (a), and its leading coefficient is the moment with $j = l$. Since P_l has leading coefficient $(2l)!/[2^l(l!)^2]$, t^l equals $2^l(l!)^2P_l(t)/(2l)!$ plus a polynomial of lower degree; with $\int_{-1}^{1} P_l^2 dt = 2/(2l+1)$,

$$b_{l0} = \frac{1}{2}\int_{-1}^{1} t^l P_l(t)dt = \frac{2^l(l!)^2}{(2l)!}\,\frac{1}{2l+1} = \boxed{\frac{2^l(l!)^2}{(2l+1)!}}.$$

Because $R_l = Q_l$ satisfies Legendre's equation and, for $|z| > 1$, equals this convergent series in descending powers beginning with z^{-l-1} (the form assumed in part (a), with $k = -l-1$), the recurrence of part (a) applies to its coefficients; they are therefore b_{l0} times the coefficients found there:

$$b_{l0}\frac{(l+s)!(l+2s)!(2l+1)!}{s!(l!)^2(2l+2s+1)!} = \frac{2^l(l+s)!(l+2s)!}{s!(2l+2s+1)!},$$

and splitting even and odd factorials, $(2l+2s+1)! = 2^{l+s}(l+s)!\,(2l+2s+1)!!$ and $2^s s! = (2s)!!$, gives

$$\boxed{Q_l(z) = z^{-l-1}\sum_{s=0}^{\infty}\frac{(l+2s)!}{(2s)!!(2l+2s+1)!!}z^{-2s}}, \qquad |z| > 1.$$

For example, for $l = 0$ the coefficients are $(2s)!/[(2s)!!(2s+1)!!] = (2s)!/(2s+1)! = 1/(2s+1)$, and $\sum_s z^{-2s-1}/(2s+1) = \frac{1}{2}\ln[(z+1)/(z-1)]$ is the familiar expansion of Q_0. For complex z with $|z| > 1$ the integral R_l and the series define the same branch of Q_l, the one that is analytic outside $[-1, 1]$ and vanishes at infinity.

15.6.4

(a) Using the recurrence relations, prove (independent of the Wronskian relation) that, for integers $n \geq 1$ on a common consistent branch away from $x = \pm 1$,

$$n\big[P_n(x)Q_{n-1}(x) - P_{n-1}(x)Q_n(x)\big] = P_1(x)Q_0(x) - P_0(x)Q_1(x).$$

(b) By direct substitution show that the right-hand side of this equation equals 1.

Solution.

(a) Let $D_n = P_nQ_{n-1} - P_{n-1}Q_n$. For $n \geq 2$, both families obey the three-term recurrence (with n shifted to $n-1$)

$$nF_n = (2n-1)xF_{n-1} - (n-1)F_{n-2}.$$

Substitute this for nP_n and nQ_n in $nD_n = (nP_n)Q_{n-1} - P_{n-1}(nQ_n)$; the terms $\pm(2n-1)xP_{n-1}Q_{n-1}$ cancel:

$$\begin{aligned} nD_n &= [(2n-1)xP_{n-1} - (n-1)P_{n-2}]Q_{n-1} \\ &\quad - P_{n-1}[(2n-1)xQ_{n-1} - (n-1)Q_{n-2}] \\ &= (n-1)[P_{n-1}Q_{n-2} - P_{n-2}Q_{n-1}] \\ &= (n-1)D_{n-1}. \end{aligned}$$

Repeating gives $nD_n = (n-1)D_{n-1} = (n-2)D_{n-2} = \cdots = 1 \cdot D_1$, which is the stated identity, since $D_1 = P_1Q_0 - P_0Q_1$; the case $n = 1$ is trivially included.

(b) Finally $P_0 = 1$, $P_1 = x$, and $Q_1 = xQ_0 - 1$, so

$$D_1 = xQ_0 - (xQ_0 - 1) = 1.$$

Therefore

$$\boxed{n[P_nQ_{n-1} - P_{n-1}Q_n] = 1}.$$

The argument uses only degree recurrences and starting values, as requested, and is valid on any common consistent branch away from the singular endpoints.

16 Chapter 16. Angular Momentum

Problems

Throughout this chapter use $\hbar = 1$, as in the textbook. Scalar spherical harmonics are normalized with the Condon–Shortley phase; their indices satisfy $L \in \{0, 1, \ldots\}$ and $-L \le M \le L$. Terms beyond a ladder endpoint mean zero, rather than an undefined harmonic. Abstract spin multiplets may also have half-integral L; in that case $L \pm M$ and $2L$ are nonnegative integers.

16.1 Angular Momentum Operators

16.1.1 The quantum mechanical angular momentum operators $L_x \pm iL_y$ in 3-D physical space are given by

$$L_x + iL_y = e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\,\frac{\partial}{\partial\varphi}\right),$$
$$L_x - iL_y = -e^{-i\varphi}\left(\frac{\partial}{\partial\theta} - i\cot\theta\,\frac{\partial}{\partial\varphi}\right).$$

Show that

(a) $(L_x + iL_y)Y_L^M(\theta,\varphi) = \sqrt{(L-M)(L+M+1)}\,Y_L^{M+1}(\theta,\varphi)$,

(b) $(L_x - iL_y)Y_L^M(\theta,\varphi) = \sqrt{(L+M)(L-M+1)}\,Y_L^{M-1}(\theta,\varphi)$.

Solution.

Use $\hbar = 1$ and the Condon–Shortley normalization

$$Y_L^M = N_{LM}P_L^M(\cos\theta)e^{iM\varphi}, \qquad N_{LM} = \sqrt{\frac{2L+1}{4\pi}\frac{(L-M)!}{(L+M)!}}.$$

Put $P = P_L^M(\cos\theta)$, $A = P_L^{M+1}(\cos\theta)$ and $B = (L+M)(L-M+1)P_L^{M-1}(\cos\theta)$. In the variable $x = \cos\theta$ the associated-Legendre recurrences (15.87) and (15.91) read

$$A + \frac{2Mx}{\sqrt{1-x^2}}\,P + B = 0, \qquad \sqrt{1-x^2}\,\frac{dP}{dx} = \frac{B-A}{2}.$$

Since $\sqrt{1-x^2} = \sin\theta$, $x/\sqrt{1-x^2} = \cot\theta$ and $d/d\theta = -\sin\theta\,d/dx$, they become

$$-M\cot\theta\,P = \frac{A+B}{2}, \qquad \frac{dP}{d\theta} = \frac{A-B}{2}.$$

Adding them, or subtracting the second from the first, gives

$$(\partial_\theta - M\cot\theta)P = A, \qquad (-\partial_\theta - M\cot\theta)P = B.$$

The first identity also follows directly from the form $P_L^M = c\sin^M\theta\,f(x)$ of Exercise 15.4.3, valid for $-L \le M \le L$, where $c = (-1)^M/(2^L L!)$ and $f = d^{L+M}(x^2-1)^L/dx^{L+M}$:

$$\partial_\theta P = M\cot\theta\,P - c\sin^{M+1}\theta\,f'(x) = M\cot\theta\,P + P_L^{M+1},$$

because the same form of P_L^{M+1} has the constant $(-1)^{M+1}/(2^L L!) = -c$ and the function f'.

Since $\partial_\varphi e^{iM\varphi} = iMe^{iM\varphi}$, the azimuthal term of L_+ acts on Y_L^M as $i\cot\theta \cdot iM = -M\cot\theta$, so

$$L_+ Y_L^M = N_{LM} e^{i(M+1)\varphi}(\partial_\theta - M\cot\theta)P = N_{LM} P_L^{M+1} e^{i(M+1)\varphi} = \frac{N_{LM}}{N_{L,M+1}} Y_L^{M+1}.$$

For $M < L$ the normalization ratio is

$$\frac{N_{LM}^2}{N_{L,M+1}^2} = \frac{(L-M)!(L+M+1)!}{(L+M)!(L-M-1)!} = (L-M)(L+M+1),$$

proving (a). Similarly, the azimuthal term of L_- acts as $-i\cot\theta \cdot iM = M\cot\theta$, so

$$\begin{aligned} L_- Y_L^M &= N_{LM} e^{i(M-1)\varphi}(-\partial_\theta - M\cot\theta)P = N_{LM} B\, e^{i(M-1)\varphi} \\ &= (L+M)(L-M+1)\frac{N_{LM}}{N_{L,M-1}} Y_L^{M-1}, \end{aligned}$$

and, for $M > -L$,

$$\frac{N_{LM}^2}{N_{L,M-1}^2} = \frac{(L-M)!(L+M-1)!}{(L+M)!(L-M+1)!} = \frac{1}{(L+M)(L-M+1)}.$$

The coefficient of Y_L^{M-1} is therefore

$$\frac{(L+M)(L-M+1)}{\sqrt{(L+M)(L-M+1)}} = \sqrt{(L+M)(L-M+1)}.$$

Taking positive square roots establishes

$$\boxed{L_\pm Y_L^M = \sqrt{(L \mp M)(L \pm M + 1)}\, Y_L^{M\pm 1}}.$$

These formulas hold for integer $-L \le M \le L$ with the endpoint understood as annihilation: $L_+ Y_L^L = L_- Y_L^{-L} = 0$. Indeed $Y_L^{\pm L}$ is proportional to $\sin^L\theta\, e^{\pm iL\varphi}$, and direct substitution gives zero:

$$L_\pm(\sin^L\theta\, e^{\pm iL\varphi}) = \pm e^{\pm i(L+1)\varphi}(L\sin^{L-1}\theta\cos\theta - L\cot\theta\,\sin^L\theta) = 0.$$

No harmonic outside the permitted M range is needed.

16.1.2 With $L_\pm$ given by

$$L_\pm = L_x \pm iL_y = \pm e^{\pm i\varphi}\left(\frac{\partial}{\partial\theta} \pm i\cot\theta\,\frac{\partial}{\partial\varphi}\right),$$

show that

(a) $Y_l^m = \sqrt{\dfrac{(l+m)!}{(2l)!\,(l-m)!}}\,(L_-)^{l-m} Y_l^l$,

(b) $Y_l^m = \sqrt{\dfrac{(l-m)!}{(2l)!\,(l+m)!}}\,(L_+)^{l+m} Y_l^{-l}$.

Solution.

Each application of L_- changes m to $m-1$ with the coefficient proved in 16.1.1. Starting at $m = l$, step j ($j = 0, 1, \ldots$) acts on magnetic index $m = l - j$, so its coefficient is $\sqrt{(l+m)(l-m+1)} = \sqrt{(2l-j)(j+1)}$. After k steps we therefore have

$$(L_-)^k Y_l^l = \left[\prod_{j=0}^{k-1}(2l-j)(j+1)\right]^{1/2} Y_l^{l-k}.$$

The two products are $\prod_{j=0}^{k-1}(2l-j) = 2l(2l-1)\cdots(2l-k+1) = (2l)!/(2l-k)!$ and $\prod_{j=0}^{k-1}(j+1) = k!$. Set $k = l - m$, so that $2l - k = l + m$:

$$(L_-)^{l-m} Y_l^l = \sqrt{\frac{(2l)!(l-m)!}{(l+m)!}}\, Y_l^m.$$

Dividing by this positive coefficient proves

$$\boxed{Y_l^m = \sqrt{\frac{(l+m)!}{(2l)!(l-m)!}}\,(L_-)^{l-m} Y_l^l}.$$

For raising from $m = -l$, the successive squared coefficients are again $(2l-j)(j+1)$, because at step j the magnetic index is $m = -l + j$ and $(l-m)(l+m+1) = (2l-j)(j+1)$. Thus

$$(L_+)^k Y_l^{-l} = \sqrt{\frac{(2l)!\,k!}{(2l-k)!}}\, Y_l^{-l+k}.$$

Putting $k = l + m$, so that $2l - k = l - m$, gives

$$(L_+)^{l+m} Y_l^{-l} = \sqrt{\frac{(2l)!(l+m)!}{(l-m)!}}\, Y_l^m,$$

and solving for the final harmonic gives

$$\boxed{Y_l^m = \sqrt{\frac{(l-m)!}{(2l)!(l+m)!}}\,(L_+)^{l+m} Y_l^{-l}}.$$

For $k = 0$ the product is empty and equals one, so both extreme states are included.

16.1.3 Using the known forms of L_+ and L_- (Exercise 16.1.2), show that

$$\int \left[Y_L^M\right]^* L_-\left(L_+ Y_L^M\right) d\Omega = \int \left(L_+ Y_L^M\right)^* \left(L_+ Y_L^M\right) d\Omega.$$

Here $d\Omega$ is the element of solid angle ($\sin\theta\, d\theta\, d\varphi$), and the integration is over the entire angular space.

Solution.

We prove the needed adjoint identity by integration by parts rather than assuming it. Let u, v be smooth single-valued functions on the sphere, with φ-periodic coordinate expressions. Since

$$L_- v = -e^{-i\varphi} v_\theta + i e^{-i\varphi} \cot\theta\, v_\varphi,$$

we have, with $d\Omega = \sin\theta\, d\theta\, d\varphi$ and $\langle u, w\rangle = \int u^* w\, d\Omega$,

$$\langle u, L_- v\rangle = \int_0^{2\pi}\int_0^{\pi} \left[-u^* e^{-i\varphi}\sin\theta\, v_\theta + i u^* e^{-i\varphi}\cos\theta\, v_\varphi\right] d\theta\, d\varphi.$$

Integrate the first term by parts in θ and the second in φ:

$$-\int_0^{\pi} u^* e^{-i\varphi}\sin\theta\, v_\theta\, d\theta = \int_0^{\pi} v\, e^{-i\varphi}(\sin\theta\, u_\theta^* + \cos\theta\, u^*)\, d\theta,$$
$$\int_0^{2\pi} i u^* e^{-i\varphi}\cos\theta\, v_\varphi\, d\varphi = -\int_0^{2\pi} v\, e^{-i\varphi}\cos\theta\, (i u_\varphi^* + u^*)\, d\varphi,$$

where $\partial_\varphi(iu^* e^{-i\varphi}) = e^{-i\varphi}(iu_\varphi^* + u^*)$. The θ boundary term $[-u^* e^{-i\varphi}\sin\theta\, v]_0^{\pi}$ vanishes because $\sin\theta = 0$ at the poles; the φ boundary term vanishes by periodicity. The remaining integrand is

$$v\left[e^{-i\varphi}\sin\theta\, u_\theta^* + e^{-i\varphi}\cos\theta\, u^* - e^{-i\varphi}\cos\theta\, u^* - i e^{-i\varphi}\cos\theta\, u_\varphi^*\right].$$

The middle two terms cancel. Dividing the rest by $\sin\theta$ to restore $d\Omega$, and using $(L_+ u)^* = e^{-i\varphi}(u_\theta^* - i\cot\theta\, u_\varphi^*)$, we are left with

$$\langle u, L_- v\rangle = \int v\, e^{-i\varphi}(u_\theta^* - i\cot\theta\, u_\varphi^*)\, d\Omega = \langle L_+ u, v\rangle.$$

Take $u = Y_L^M$ and $v = L_+ Y_L^M$, both regular spherical harmonics or zero. This is exactly the requested equality. Moreover, since $L_+ Y_L^M = \sqrt{(L-M)(L+M+1)}\, Y_L^{M+1}$ by 16.1.1 and $\int |Y_L^{M+1}|^2 d\Omega = 1$, both sides equal $\boxed{(L-M)(L+M+1)}$, including zero at the upper endpoint.

16.1.4

(a) Show that $\mathbf{J}^2 = \frac{1}{2}\left[J_+ J_- + J_- J_+\right] + J_z^2$.

(b) Use the result from part (a) and the explicit formulas for L_+ and L_- from Exercise 16.1.2 to verify that all the spherical harmonics with $l = 2$ are eigenfunctions of $\mathbf{L}^2$ with eigenvalue $l(l+1) = 6$.

Solution.

(a) Preserve operator order while expanding:

$$J_+ J_- = J_x^2 + J_y^2 + i J_y J_x - i J_x J_y,$$
$$J_- J_+ = J_x^2 + J_y^2 + i J_x J_y - i J_y J_x.$$

Their sum is $2J_x^2 + 2J_y^2$ because the mixed terms cancel. Consequently

$$\boxed{\mathbf{J}^2 = \tfrac{1}{2}(J_+ J_- + J_- J_+) + J_z^2}.$$

(b) For an arbitrary smooth test function f, direct product differentiation in the given ladder operators yields

$$L_+ L_- f = -f_{\theta\theta} - \cot\theta f_\theta - \cot^2\theta f_{\varphi\varphi} - i f_\varphi,$$
$$L_- L_+ f = -f_{\theta\theta} - \cot\theta f_\theta - \cot^2\theta f_{\varphi\varphi} + i f_\varphi.$$

For the first, write $c = \cot\theta$, so $c' = -\csc^2\theta$, and $L_-f = -e^{-i\varphi}(f_\theta - icf_\varphi)$. Then $L_+L_-f = e^{i\varphi}\partial_\theta(L_-f) + e^{i\varphi}ic\,\partial_\varphi(L_-f)$, and since $\partial_\varphi(-e^{-i\varphi}) = ie^{-i\varphi}$ the two parts are

$$\begin{aligned} e^{i\varphi}\partial_\theta(L_-f) &= -f_{\theta\theta} + ic'f_\varphi + icf_{\theta\varphi}, \\ e^{i\varphi}ic\,\partial_\varphi(L_-f) &= ic\,[\,i(f_\theta - icf_\varphi) - (f_{\theta\varphi} - icf_{\varphi\varphi})\,] \\ &= -cf_\theta - icf_{\theta\varphi} + c^2 if_\varphi - c^2 f_{\varphi\varphi}. \end{aligned}$$

The mixed derivatives cancel and $i(c' + c^2) = -i$, because $\csc^2\theta - \cot^2\theta = 1$. For the second, replacing i by $-i$ in the coefficients turns L_+ into $-L_-$ and L_- into $-L_+$, hence turns L_+L_- into $(-L_-)(-L_+) = L_-L_+$. The computation above uses only $i^2 = -1$ and $\partial_\varphi e^{\pm i\varphi} = \pm ie^{\pm i\varphi}$, which remain true with $-i$ in place of i, so the same replacement in its result (with f unchanged) gives L_-L_+f. This changes only the last first-order term, from $-if_\varphi$ to $+if_\varphi$. The Instructor's Manual (printed p. 257) writes the difference $(L_+L_- - L_-L_+)/2$ when extracting the common second-order terms. It must use the sum. The half-difference is $L_z = -i\partial_\varphi$, not $\mathbf{L}^2 - L_z^2$. In the half-sum the terms $\mp if_\varphi$ cancel: $\frac{1}{2}(L_+L_- + L_-L_+)f = -f_{\theta\theta} - \cot\theta f_\theta - \cot^2\theta f_{\varphi\varphi}$. Since $L_z^2 f = -f_{\varphi\varphi}$ and $\cot^2\theta + 1 = \csc^2\theta$, (a) gives

$$\mathbf{L}^2 f = -f_{\theta\theta} - \cot\theta f_\theta - \csc^2\theta f_{\varphi\varphi}.$$

From here on write $s = \sin\theta$ and $c = \cos\theta$ (so c no longer denotes $\cot\theta$). The five $l = 2$ harmonics are nonzero constant multiples of $s^2e^{\pm 2i\varphi}$, $sc\,e^{\pm i\varphi}$, and $3c^2 - 1$. The operator acts on $F(\theta)e^{im\varphi}$ as $[-F'' - \cot\theta F' + m^2F/s^2]e^{im\varphi}$. Explicitly,

F	m^2	F', F''	$-F'' - (c/s)F' + m^2F/s^2$
s^2	4	$2sc,\ 2(c^2 - s^2)$	$-2c^2 + 2s^2 - 2c^2 + 4 = 6s^2$
sc	1	$c^2 - s^2,\ -4sc$	$4sc - c^3/s + cs + c/s = 6sc$
$3c^2 - 1$	0	$-6sc,\ -6(c^2 - s^2)$	$12c^2 - 6s^2 = 6(3c^2 - 1)$

The last column uses $s^2 + c^2 = 1$: $4 = 4s^2 + 4c^2$ in the first row, $c/s - c^3/s = c(1 - c^2)/s = cs$ in the second, and $-6s^2 = 6c^2 - 6$ in the third. Both signs of m have the same m^2, so the first two rows each verify two harmonics. Multiplying by their normalization constants does not change the eigenvalue: $\boxed{\mathbf{L}^2 Y_2^m = 6Y_2^m}$ for all five m.

16.1.5 Derive the following relations for normalized eigenstates of the Hermitian angular-momentum algebra in one irreducible multiplet, with relative phases chosen so the ladder coefficients are positive. Do not assume coordinate formulas for ψ_L^M:

(a) $\psi_L^M(\theta,\varphi) = \sqrt{\dfrac{(L+M)!}{(2L)!\,(L-M)!}}\,(L_-)^{L-M}\psi_L^L(\theta,\varphi)$,

(b) $\psi_L^M(\theta,\varphi) = \sqrt{\dfrac{(L-M)!}{(2L)!\,(L+M)!}}\,(L_+)^{L+M}\psi_L^{-L}(\theta,\varphi)$.

Solution.

The numerical coefficients require normalized states in a single angular-momentum multiplet and a consistent ladder phase convention. Eigenvalue equations alone leave arbitrary magnitudes, phases, and possible degeneracy labels. Assume the Hermitian angular-momentum algebra, and choose states with $\mathbf{L}^2\psi_L^M = L(L+1)\psi_L^M$ and $L_z\psi_L^M = M\psi_L^M$. The commutators $[L_z, L_\pm] = \pm L_\pm$

and $[\mathbf{L}^2, L_\pm] = 0$ show that $L_\pm \psi_L^M$ has the same L and magnetic index $M \pm 1$: $L_z L_\pm \psi_L^M = L_\pm (L_z \pm 1)\psi_L^M = (M \pm 1) L_\pm \psi_L^M$. Within a fixed irreducible multiplet it is therefore proportional to $\psi_L^{M\pm1}$.

From $[L_x, L_y] = iL_z$, $[L_+, L_-] = -i[L_x, L_y] + i[L_y, L_x] = 2L_z$, while the identity of 16.1.4(a) gives $L_+L_- + L_-L_+ = 2(\mathbf{L}^2 - L_z^2)$. Adding and subtracting these two equations and halving gives

$$L_-L_+ = \mathbf{L}^2 - L_z^2 - L_z, \qquad L_+L_- = \mathbf{L}^2 - L_z^2 + L_z.$$

Since $L_\pm^\dagger = L_\mp$, normalization determines the squared proportionality factors:

$$\|L_+\psi_L^M\|^2 = \langle \psi_L^M, L_-L_+\psi_L^M\rangle = L(L+1) - M^2 - M = (L-M)(L+M+1),$$

$$\|L_-\psi_L^M\|^2 = \langle \psi_L^M, L_+L_-\psi_L^M\rangle = L(L+1) - M^2 + M = (L+M)(L-M+1).$$

Choose the relative phases so these factors have positive square roots. This works for raising and lowering at once: if $L_-\psi_L^{M+1} = c\,\psi_L^M$ with $c > 0$, then $L_+\psi_L^M$, being proportional to ψ_L^{M+1}, has the coefficient $\langle \psi_L^{M+1}, L_+\psi_L^M\rangle = \langle L_-\psi_L^{M+1}, \psi_L^M\rangle = c$. Thus fixing the phases of $\psi_L^{L-1}, \dots, \psi_L^{-L}$ by positive lowering coefficients makes every raising coefficient positive too. As in 16.1.2, lowering step j acts on $M = L - j$, where $(L+M)(L-M+1) = (2L-j)(j+1)$. Iteration now gives, without any coordinate formula,

$$(L_-)^k \psi_L^L = \sqrt{\prod_{j=0}^{k-1}(2L-j)(j+1)}\,\psi_L^{L-k} = \sqrt{\frac{(2L)!k!}{(2L-k)!}}\,\psi_L^{L-k}.$$

Setting $k = L - M$ and inverting the coefficient yields (a). Starting at $-L$ and raising gives the identical product; setting $k = L + M$ yields (b):

$$\boxed{\psi_L^M = \sqrt{\frac{(L+M)!}{(2L)!(L-M)!}}(L_-)^{L-M}\psi_L^L},$$

$$\boxed{\psi_L^M = \sqrt{\frac{(L-M)!}{(2L)!(L+M)!}}(L_+)^{L+M}\psi_L^{-L}}.$$

The argument also applies to abstract half-integral multiplets: $2L, L \pm M$ are still nonnegative integers. Ordinary scalar spherical harmonics are the integral-L realization.

16.1.6 Derive the operator equations for integers $n \geq 0$, on $0 < \theta < \pi$, with the derivatives in $\cos\theta$ taken at fixed φ:

$$(L_+)^n Y_L^M(\theta,\varphi) = (-1)^n e^{in\varphi}\sin^{n+M}\theta\,\frac{d^n \sin^{-M}\theta\, Y_L^M(\theta,\varphi)}{(d\cos\theta)^n},$$

$$(L_-)^n Y_L^M(\theta,\varphi) = e^{-in\varphi}\sin^{n-M}\theta\,\frac{d^n \sin^{M}\theta\, Y_L^M(\theta,\varphi)}{(d\cos\theta)^n}.$$

Hint. Try mathematical induction (Section 1.4).

Solution.

All derivatives with respect to $x = \cos\theta$ below hold φ fixed, so $\partial_\theta = -\sin\theta\,\partial_x$; the interior range $0 < \theta < \pi$ keeps $\cot\theta$ and the powers $\sin^{\pm M}\theta$ finite. Since $\partial_\theta \sin^{\mp m}\theta = \mp m\cot\theta\,\sin^{\mp m}\theta$, a

function $g_m = F(\theta)e^{im\varphi}$, for which $\partial_\varphi g_m = im g_m$, satisfies

$$\begin{aligned} L_+ g_m &= e^{i\varphi}(\partial_\theta - m\cot\theta)g_m = e^{i\varphi}\sin^m\theta\,\partial_\theta(\sin^{-m}\theta\, g_m) \\ &= -e^{i\varphi}\sin^{m+1}\theta\,\partial_x(\sin^{-m}\theta\, g_m), \\ L_- g_m &= -e^{-i\varphi}(\partial_\theta + m\cot\theta)g_m = -e^{-i\varphi}\sin^{-m}\theta\,\partial_\theta(\sin^m\theta\, g_m) \\ &= e^{-i\varphi}\sin^{1-m}\theta\,\partial_x(\sin^m\theta\, g_m). \end{aligned}$$

These establish both claims for $n = 1$; $n = 0$ is the identity.

Assume the raising formula at n: $g_{M+n} = (L_+)^n Y_L^M = (-1)^n e^{in\varphi}\sin^{M+n}\theta\,\partial_x^n(\sin^{-M}\theta\, Y_L^M)$. It has magnetic index $M+n$, because $Y_L^M \propto e^{iM\varphi}$ and ∂_x at fixed φ keeps that factor. Apply the first identity with $m = M+n$. The two opposite powers of sine cancel inside the derivative, $\sin^{-(M+n)}\theta\, g_{M+n} = (-1)^n e^{in\varphi}\partial_x^n(\sin^{-M}\theta\, Y_L^M)$, and $e^{in\varphi}$ is constant in x, so

$$L_+ g_{M+n} = (-1)^{n+1}e^{i(n+1)\varphi}\sin^{M+n+1}\theta\,\partial_x^{n+1}(\sin^{-M}\theta\, Y_L^M).$$

This is the $n+1$ formula.

For lowering, the induction hypothesis is $g_{M-n} = (L_-)^n Y_L^M = e^{-in\varphi}\sin^{n-M}\theta\,\partial_x^n(\sin^M\theta\, Y_L^M)$. Use the second identity with $m = M-n$; its factor $\sin^m\theta$ cancels $\sin^{n-M}\theta$: $\sin^{M-n}\theta\, g_{M-n} = e^{-in\varphi}\partial_x^n(\sin^M\theta\, Y_L^M)$. Since $\sin^{1-m}\theta = \sin^{n+1-M}\theta$, this leaves

$$L_- g_{M-n} = e^{-i(n+1)\varphi}\sin^{n+1-M}\theta\,\partial_x^{n+1}(\sin^M\theta\, Y_L^M).$$

Thus both stated equations hold for every integer $n \geq 0$. They are coordinate expressions on $0 < \theta < \pi$; the resulting harmonics extend regularly to the poles.

16.1.7 Show, using $(L_-)^n$, that

$$Y_L^{-M}(\theta,\varphi) = (-1)^M\big[Y_L^M(\theta,\varphi)\big]^*.$$

Solution.

First let $M \geq 0$. Iterating the raising coefficient from zero to M gives

$$Y_L^M = c_M(L_+)^M Y_L^0, \qquad c_M = \sqrt{\frac{(L-M)!}{(L+M)!}}.$$

Indeed step j acts on magnetic index j, with squared coefficient $(L-j)(L+j+1)$ by 16.1.1, and the squared product is

$$\prod_{j=0}^{M-1}(L-j)\prod_{j=0}^{M-1}(L+j+1) = \frac{L!}{(L-M)!}\cdot\frac{(L+M)!}{L!} = \frac{(L+M)!}{(L-M)!}.$$

The differential forms show, for any differentiable f,

$$(L_+ f)^* = e^{-i\varphi}(\partial_\theta - i\cot\theta\,\partial_\varphi)f^* = -L_- f^*.$$

Applying this M times gives $[(L_+)^M f]^* = (-1)^M(L_-)^M f^*$. Because Y_L^0 is real, therefore

$$(Y_L^M)^* = (-1)^M c_M(L_-)^M Y_L^0.$$

Lowering from zero to $-M$ has the same squared coefficient product $(L+M)!/(L-M)!$, since step j acts on $m=-j$ and $(L+m)(L-m+1)=(L-j)(L+j+1)$. Hence $c_M(L_-)^M Y_L^0 = Y_L^{-M}$, so $(Y_L^M)^* = (-1)^M Y_L^{-M}$. Multiplying by $(-1)^M$,

$$\boxed{Y_L^{-M} = (-1)^M (Y_L^M)^*}.$$

Replacing M by $-M$ and conjugating proves the identical statement for negative M, since $(-1)^{-M} = (-1)^M$ for integral M. The Instructor's Manual (printed p. 258) says to start with Y_0^0. For a fixed degree L the starting function must be Y_L^0. Indeed both ladders annihilate Y_0^0, so they cannot generate any nonconstant harmonic from it.

16.1.8 Verify by explicit calculation that

(a) $L_+ Y_1^0(\theta,\varphi) = -\sqrt{\dfrac{3}{4\pi}}\sin\theta\, e^{i\varphi} = \sqrt{2}\, Y_1^1(\theta,\varphi)$,

(b) $L_- Y_1^0(\theta,\varphi) = +\sqrt{\dfrac{3}{4\pi}}\sin\theta\, e^{-i\varphi} = \sqrt{2}\, Y_1^{-1}(\theta,\varphi)$.

The signs have the indicated values because the spherical harmonics were defined to be consistent with the results obtained using the ladder operators L_+ and L_- (Condon–Shortley phase).

Solution.

The relevant normalized harmonics are

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta, \qquad Y_1^1 = -\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\varphi}, \qquad Y_1^{-1} = \sqrt{\frac{3}{8\pi}}\sin\theta\, e^{-i\varphi}.$$

Since Y_1^0 is independent of φ, the azimuthal derivatives in $L_\pm$ vanish. Direct differentiation, with $\partial_\theta \cos\theta = -\sin\theta$ and $\sqrt{3/(4\pi)} = \sqrt{2}\sqrt{3/(8\pi)}$, gives

$$L_+ Y_1^0 = e^{i\varphi}\sqrt{\frac{3}{4\pi}}(-\sin\theta) = -\sqrt{\frac{3}{4\pi}}\sin\theta\, e^{i\varphi} = \boxed{\sqrt{2}\, Y_1^1},$$

$$L_- Y_1^0 = -e^{-i\varphi}\sqrt{\frac{3}{4\pi}}(-\sin\theta) = \sqrt{\frac{3}{4\pi}}\sin\theta\, e^{-i\varphi} = \boxed{\sqrt{2}\, Y_1^{-1}}.$$

The minus sign in Y_1^1 is precisely the phase needed to make both ladder coefficients positive.

16.2 Angular Momentum Coupling

16.2.1 Derive recursion relations for Clebsch–Gordan coefficients. Use them to calculate $C(11J\,|\,m_1 m_2 M)$ for $J=0,1,2$.

Hint. Use the known matrix elements of $J_+ = J_{1+} + J_{2+}$, J_{i+}, and $\mathbf{J}^2 = (\mathbf{J}_1+\mathbf{J}_2)^2$, etc.

Solution.

Write $|m_1,m_2\rangle = |j_1 m_1\rangle|j_2 m_2\rangle$ and

$$|JM\rangle = \sum_{m_1+m_2=M} C_M(m_1,m_2)|m_1,m_2\rangle.$$

Define $a_i^{\pm}(m) = \sqrt{(j_i \mp m)(j_i \pm m + 1)}$, so that $J_{i\pm}|j_i m\rangle = a_i^{\pm}(m)|j_i, m\pm 1\rangle$. Apply $J_+ = J_{1+} + J_{2+}$ to this expansion and collect the coefficient of the *target* state $|m_1, m_2\rangle$ ($m_1 + m_2 = M + 1$); it comes from $|m_1 - 1, m_2\rangle$ through J_{1+} and from $|m_1, m_2 - 1\rangle$ through J_{2+}, while the left side is $\sqrt{(J-M)(J+M+1)}\,|J, M+1\rangle$:

$$\sqrt{(J-M)(J+M+1)}\,C_{M+1}(m_1,m_2) = a_1^+(m_1-1)C_M(m_1-1,m_2) + a_2^+(m_2-1)C_M(m_1,m_2-1).$$

Similarly, applying J_- gives

$$\sqrt{(J+M)(J-M+1)}\,C_{M-1}(m_1,m_2) = a_1^-(m_1+1)C_M(m_1+1,m_2) + a_2^-(m_2+1)C_M(m_1,m_2+1).$$

Coefficients outside the allowed magnetic ranges are zero. A recursion within a fixed M follows from

$$\mathbf{J}^2 = \mathbf{J}_1^2 + \mathbf{J}_2^2 + 2J_{1z}J_{2z} + J_{1+}J_{2-} + J_{1-}J_{2+},$$

since $J_{1+}J_{2-} + J_{1-}J_{2+} = 2(J_{1x}J_{2x} + J_{1y}J_{2y})$, so the last three terms are $2\mathbf{J}_1\cdot\mathbf{J}_2$. The first three terms multiply $|m_1, m_2\rangle$ by $j_1(j_1+1) + j_2(j_2+1) + 2m_1m_2$; $J_{1+}J_{2-}$ reaches $|m_1, m_2\rangle$ from $|m_1 - 1, m_2 + 1\rangle$ and $J_{1-}J_{2+}$ from $|m_1 + 1, m_2 - 1\rangle$. Equating coefficients of $|m_1, m_2\rangle$ in $\mathbf{J}^2|JM\rangle = J(J+1)|JM\rangle$ gives

$$\begin{aligned}[J(J+1) - j_1(j_1+1) - j_2(j_2+1) - 2m_1m_2]C_M(m_1,m_2) \\ = a_1^+(m_1-1)a_2^-(m_2+1)C_M(m_1-1,m_2+1) \\ + a_1^-(m_1+1)a_2^+(m_2-1)C_M(m_1+1,m_2-1).\end{aligned}$$

Together with normalization and phase, these determine the coefficients.

For $j_1 = j_2 = 1$, choose positive lowering coefficients and a positive coefficient of the largest m_1 in each highest-M state, consistent with the text's Condon–Shortley convention. For $j_i = 1$, $J_{i-}|1\rangle = \sqrt{2}|0\rangle$, $J_{i-}|0\rangle = \sqrt{2}|-1\rangle$ and $J_{i-}|-1\rangle = 0$; the total ladder factors are $J_-|J,M\rangle = \sqrt{(J+M)(J-M+1)}\,|J, M-1\rangle$. Each lowering below applies the J_- recursion to all coefficients at once: the coefficient of $|m_1, m_2\rangle$ in $J_-|JM\rangle$ is the right side of that recursion, and dividing by $\sqrt{(J+M)(J-M+1)}$ gives $C_{M-1}(m_1, m_2)$. Start with $|2,2\rangle = |1,1\rangle$. Since $J_-|1,1\rangle = \sqrt{2}|0,1\rangle + \sqrt{2}|1,0\rangle$ and $J_-|2,2\rangle = 2|2,1\rangle$, the first lowering gives

$$2|2,1\rangle = \sqrt{2}(|1,0\rangle + |0,1\rangle).$$

Next, $J_-(|1,0\rangle + |0,1\rangle) = \sqrt{2}(|0,0\rangle + |1,-1\rangle) + \sqrt{2}(|-1,1\rangle + |0,0\rangle)$ and $J_-|2,1\rangle = \sqrt{6}|2,0\rangle$, so, after dividing by the $\sqrt{2}$ in $|2,1\rangle$,

$$\sqrt{6}|2,0\rangle = |1,-1\rangle + 2|0,0\rangle + |-1,1\rangle.$$

Lowering this numerator gives $\sqrt{2}|0,-1\rangle + 2\sqrt{2}(|-1,0\rangle + |0,-1\rangle) + \sqrt{2}|-1,0\rangle = 3\sqrt{2}(|0,-1\rangle + |-1,0\rangle)$; after dividing by its existing $\sqrt{6}$ and the next ladder factor $\sqrt{6}$, $|2,-1\rangle = 3\sqrt{2}(|0,-1\rangle + |-1,0\rangle)/6 = (|0,-1\rangle + |-1,0\rangle)/\sqrt{2}$. The final lowering gives $2|2,-2\rangle = (\sqrt{2}+\sqrt{2})|-1,-1\rangle/\sqrt{2} = 2|-1,-1\rangle$. Thus the complete quintet is

M	$\lvert 2, M\rangle$
2	$\lvert 1,1\rangle$
1	$(\lvert 1,0\rangle + \lvert 0,1\rangle)/\sqrt{2}$
0	$(\lvert 1,-1\rangle + 2\lvert 0,0\rangle + \lvert -1,1\rangle)/\sqrt{6}$
-1	$(\lvert 0,-1\rangle + \lvert -1,0\rangle)/\sqrt{2}$
-2	$\lvert -1,-1\rangle$

The normalized vector at $M=1$ orthogonal to $|2,1\rangle$, with a positive coefficient of $|1,0\rangle$, is $|1,1\rangle_{\text{coupled}} = (|1,0\rangle - |0,1\rangle)/\sqrt{2}$. Its two lowerings, with ladder factors $\sqrt{(1+M)(2-M)} = \sqrt{2}$ for $M = 1, 0$, are

$$\begin{aligned} J_-|1,1\rangle_{\text{coupled}} &= \tfrac{1}{\sqrt{2}}\big[\sqrt{2}(|0,0\rangle + |1,-1\rangle) - \sqrt{2}(|-1,1\rangle + |0,0\rangle)\big] \\ &= |1,-1\rangle - |-1,1\rangle = \sqrt{2}|1,0\rangle_{\text{coupled}}, \end{aligned}$$

$$J_-|1,0\rangle_{\text{coupled}} = \tfrac{1}{\sqrt{2}}\big(\sqrt{2}|0,-1\rangle - \sqrt{2}|-1,0\rangle\big) = |0,-1\rangle - |-1,0\rangle = \sqrt{2}|1,-1\rangle_{\text{coupled}}.$$

Hence the triplet is

M	$\lvert 1,M\rangle_{\text{coupled}}$
1	$(\lvert 1,0\rangle - \lvert 0,1\rangle)/\sqrt{2}$
0	$(\lvert 1,-1\rangle - \lvert -1,1\rangle)/\sqrt{2}$
-1	$(\lvert 0,-1\rangle - \lvert -1,0\rangle)/\sqrt{2}$

Finally write the singlet as $u|1,-1\rangle + v|0,0\rangle + w|-1,1\rangle$. Orthogonality to $|2,0\rangle$ and $|1,0\rangle_{\text{coupled}}$ gives $u+2v+w=0$ and $u-w=0$, so $w=u, v=-u$. Normalization gives $3u^2=1$, and the chosen sign gives $u>0$:

$$\boxed{|0,0\rangle_{\text{coupled}} = \frac{|1,-1\rangle - |0,0\rangle + |-1,1\rangle}{\sqrt{3}}}.$$

The fixed-M recursion checks all three $M=0$ states at once. For $m_1=1$, $m_2=-1$ its bracket is $J(J+1)-2-2+2$, and only the term with $a_1^+(0)a_2^-(0)=2$ survives on the right (the other needs $m_1=2$), so $C_0(0,0) = \frac{1}{2}[J(J+1)-2]\,C_0(1,-1)$. This gives $2C_0(1,-1)$, 0 and $-C_0(1,-1)$ for $J=2,1,0$, as in the tables.

Every Clebsch–Gordan coefficient requested is the coefficient of its product ket in these nine normalized states; all unlisted coefficients are zero. Their dimensions $5+3+1=9$ exhaust the product basis.

16.2.2 Defining $\big(Y_l\chi\big)^M_J$ by the formula

$$\big(Y_l\chi\big)^M_J = \sum C\big(l\tfrac{1}{2}J\,\big|\,m_l m_s M\big)Y_l^{m_l}\chi_{m_s},$$

where $\chi_{\pm1/2}$ are the spin up and down eigenfunctions of $\sigma_3=\sigma_z$, show that $\big(Y_l\chi\big)^M_J$ is a J,M eigenfunction.

Solution.

Let $\chi_+ = \chi_{1/2}$ and $\chi_- = \chi_{-1/2}$, so that $S_z\chi_\pm = \pm\chi_\pm/2$ with $\mathbf{S}=\boldsymbol{\sigma}/2$. Every nonzero Clebsch–Gordan term has $m_l+m_s=M$; therefore $J_z = L_z + S_z$ acting on the sum gives M times that sum.

To check $\mathbf{J}^2$ explicitly, consider the fixed-M product basis

$$u = Y_l^{M-1/2}\chi_+, \qquad v = Y_l^{M+1/2}\chi_-,$$

omitting a vector if its orbital magnetic index is outside $[-l,l]$. In this basis,

$$\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2L_zS_z + L_+S_- + L_-S_+ = \big[l(l+1)+\tfrac{1}{4}\big]\,I + \begin{pmatrix} M & q \\ q & -M \end{pmatrix},$$

where $K = l + \tfrac12$ and $q = \sqrt{K^2 - M^2}$. Indeed, by 16.1.1 and $S_-\chi_+ = \chi_-$, $S_+\chi_- = \chi_+$,

$$L_+S_-u = \sqrt{(l - M + \tfrac12)(l + M + \tfrac12)}\,v = \sqrt{(K - M)(K + M)}\,v = qv,$$

and likewise $L_-S_+v = qu$, while $S_-\chi_- = S_+\chi_+ = 0$ make $L_+S_-v = L_-S_+u = 0$. With $\mathbf{S}^2 = \tfrac34$ and $2L_zS_z = 2(M \mp \tfrac12)(\pm\tfrac12)$ on u and v, respectively, the diagonal terms are $l(l+1) + \tfrac34 \pm (M \mp \tfrac12) = l(l+1) + \tfrac14 \pm M$. The square of the last matrix is $(M^2 + q^2)I = K^2 I$, so its eigenvalues are $\pm K$; its trace is zero, so one is $+K$ and the other $-K$.

The corresponding normalized eigenvectors are

$$\Psi_+ = \sqrt{\frac{K+M}{2K}}\,u + \sqrt{\frac{K-M}{2K}}\,v, \qquad \Psi_- = -\sqrt{\frac{K-M}{2K}}\,u + \sqrt{\frac{K+M}{2K}}\,v.$$

Indeed $q\sqrt{K-M} = (K-M)\sqrt{K+M}$ and $q\sqrt{K+M} = (K+M)\sqrt{K-M}$. The two components of the matrix times Ψ_+ are consequently $[M+K-M]\sqrt{K+M}$ and $[K+M-M]\sqrt{K-M}$, divided by $\sqrt{2K}$; these are the components of $K\Psi_+$. For Ψ_- the components are $[-M+K+M]\sqrt{K-M}$ and $[-K+M-M]\sqrt{K+M}$, divided by $\sqrt{2K}$, which equal $-K\Psi_-$. The complete eigenvalues are thus

$$l(l+1) + \tfrac14 + K = (l+\tfrac12)(l+\tfrac32), \qquad l(l+1) + \tfrac14 - K = (l-\tfrac12)(l+\tfrac12).$$

These are $J(J+1)$ for $J = l + 1/2$ and $J = l - 1/2$.

The eigenvectors also carry the Condon–Shortley phases. Write $u_M, v_M, \Psi_\pm(M)$ for the vectors above at magnetic index M. Since $J_- = L_- + S_-$, 16.1.1 and $S_-\chi_+ = \chi_-$, $S_-\chi_- = 0$ give

$$J_-u_M = \sqrt{(K+M-1)(K-M+1)}\,u_{M-1} + v_{M-1}, \qquad J_-v_M = q\,v_{M-1}.$$

Hence, combining the v_{M-1} terms with $q\sqrt{K \mp M} = (K \mp M)\sqrt{K \pm M}$,

$$\begin{aligned}
J_-\Psi_+(M) &= \sqrt{\tfrac{K+M}{2K}}\big[\sqrt{(K+M-1)(K-M+1)}\,u_{M-1} + (K-M+1)\,v_{M-1}\big] \\
&= \sqrt{(K+M)(K-M+1)}\,\Psi_+(M-1), \\
J_-\Psi_-(M) &= \sqrt{\tfrac{K-M}{2K}}\big[-\sqrt{(K+M-1)(K-M+1)}\,u_{M-1} + (K+M-1)\,v_{M-1}\big] \\
&= \sqrt{(K+M-1)(K-M)}\,\Psi_-(M-1).
\end{aligned}$$

These are the positive ladder factors $\sqrt{(J+M)(J-M+1)}$ for $J = K$ and $J = K-1$. The top state of each multiplet has a positive coefficient of Y_l^l: at $M = K$, $\Psi_+ = u = Y_l^l\chi_+$, and at $M = K-1$ the coefficient of $v = Y_l^l\chi_-$ in Ψ_- is $\sqrt{(2K-1)/2K} > 0$. Hence $\Psi_\pm$ are the Condon–Shortley states, and since $K \pm M = l \pm M + \tfrac12$ and $2K = 2l+1$, their components are the Clebsch–Gordan coefficients for $j_2 = \tfrac12$:

$$C(l\tfrac12 J \,|\, M \mp \tfrac12, \pm\tfrac12, M) = \begin{cases} \sqrt{\dfrac{l \pm M + \frac12}{2l+1}}, & J = l + \tfrac12, \\[2ex] \mp\sqrt{\dfrac{l \mp M + \frac12}{2l+1}}, & J = l - \tfrac12. \end{cases}$$

For $l = 1$ these coefficients reproduce the states obtained by explicit lowering in 16.2.3. At the extreme $M = \pm(l + 1/2)$ only the permitted basis vector survives. The lower multiplet exists only for $l \geq 1$. Consequently

$$\boxed{\mathbf{J}^2(Y_l\chi)_J^M = J(J+1)(Y_l\chi)_J^M, \qquad J_z(Y_l\chi)_J^M = M(Y_l\chi)_J^M}.$$

16.2.3 Find the (j, m) states of a p electron ($l = 1$), in which the orbital angular momentum of the electron is coupled to its spin angular momentum ($s = 1/2$) to form states whose conventional labelings are ${}^2p_{1/2}$ and ${}^2p_{3/2}$. The notation is of the general form ${}^{2s+1}(\text{symbol})_j$, where "symbol" is that indicating the l value (i.e., s, p, ...).

Solution.

Write $p_m = Y_1^m$ and $\alpha = \chi_{1/2}$, $\beta = \chi_{-1/2}$. The tensor products are normalized and mutually orthogonal. Since $1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$, there are four ${}^2p_{3/2}$ states and two ${}^2p_{1/2}$ states.

The highest state is $|\frac{3}{2}, \frac{3}{2}\rangle = p_1\alpha$. Use $L_-p_1 = \sqrt{2}p_0$, $L_-p_0 = \sqrt{2}p_{-1}$, $S_-\alpha = \beta$, $S_-\beta = 0$:

$$J_-(p_1\alpha) = \sqrt{2}p_0\alpha + p_1\beta.$$

Divide by $\sqrt{3}$, the $J = 3/2$ ladder coefficient $\sqrt{(J+M)(J-M+1)}$ at $M = 3/2$, to get $|\frac{3}{2}, \frac{1}{2}\rangle$ below. Lower its numerator once more:

$$J_-(\sqrt{2}p_0\alpha + p_1\beta) = \sqrt{2}(\sqrt{2}p_{-1}\alpha + p_0\beta) + \sqrt{2}p_0\beta = 2p_{-1}\alpha + 2\sqrt{2}p_0\beta.$$

Dividing by the previous $\sqrt{3}$ and the next ladder coefficient $\sqrt{2 \cdot 2} = 2$ gives $|\frac{3}{2}, -\frac{1}{2}\rangle = (p_{-1}\alpha + \sqrt{2}p_0\beta)/\sqrt{3}$. The last lowering is $J_-(p_{-1}\alpha + \sqrt{2}p_0\beta)/\sqrt{3} = (p_{-1}\beta + 2p_{-1}\beta)/\sqrt{3} = \sqrt{3}p_{-1}\beta$, and the ladder coefficient at $M = -1/2$ is $\sqrt{1 \cdot 3} = \sqrt{3}$. Therefore

$$\boxed{\begin{aligned} |\tfrac{3}{2}, \tfrac{3}{2}\rangle &= p_1\alpha, \\ |\tfrac{3}{2}, \tfrac{1}{2}\rangle &= \sqrt{\tfrac{2}{3}}p_0\alpha + \sqrt{\tfrac{1}{3}}p_1\beta, \\ |\tfrac{3}{2}, -\tfrac{1}{2}\rangle &= \sqrt{\tfrac{1}{3}}p_{-1}\alpha + \sqrt{\tfrac{2}{3}}p_0\beta, \\ |\tfrac{3}{2}, -\tfrac{3}{2}\rangle &= p_{-1}\beta. \end{aligned}}$$

For the remaining $M = 1/2$ state, orthogonality to $|\frac{3}{2}, \frac{1}{2}\rangle$ requires $\sqrt{2}u + v = 0$ in $up_0\alpha + vp_1\beta$. Normalization gives $|u|^2 + |v|^2 = 3u^2 = 1$; choosing the coefficient of $p_1\beta$ ($m_l = l$) positive, as in 16.2.1, fixes $u = -1/\sqrt{3}$, $v = \sqrt{2/3}$. Its lowering coefficient for $J = 1/2$ is $\sqrt{1 \cdot 1} = 1$:

$$\begin{aligned} J_-\left(-\frac{p_0\alpha}{\sqrt{3}} + \sqrt{\frac{2}{3}}p_1\beta\right) &= -\sqrt{\frac{2}{3}}p_{-1}\alpha - \frac{p_0\beta}{\sqrt{3}} + \frac{2}{\sqrt{3}}p_0\beta \\ &= -\sqrt{\frac{2}{3}}p_{-1}\alpha + \frac{p_0\beta}{\sqrt{3}}. \end{aligned}$$

Thus

$$\boxed{\begin{aligned} |\tfrac{1}{2}, \tfrac{1}{2}\rangle &= -\sqrt{\tfrac{1}{3}}p_0\alpha + \sqrt{\tfrac{2}{3}}p_1\beta, \\ |\tfrac{1}{2}, -\tfrac{1}{2}\rangle &= -\sqrt{\tfrac{2}{3}}p_{-1}\alpha + \sqrt{\tfrac{1}{3}}p_0\beta. \end{aligned}}$$

The Instructor's Manual's displayed doublet changes the relative phase between its two M states and does not obey $J_-|\frac{1}{2}, \frac{1}{2}\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$. The explicit lowering above fixes that phase. A common overall sign change of both states would be harmless. The superscript 2 is $2s + 1$, and the letter p records $l = 1$.

16.2.4 Repeat Exercise 16.2.3 for $l = 1$, $s = 3/2$. Apply the conventional labels to the j, m states.

Solution.

Here $1 \otimes \frac{3}{2} = \frac{5}{2} \oplus \frac{3}{2} \oplus \frac{1}{2}$. Because $2s + 1 = 4$ and $l = 1$, the labels are ${}^4p_{5/2}$, ${}^4p_{3/2}$, and ${}^4p_{1/2}$, with $6, 4, 2$ states, respectively. Write $p_+ = Y_1^1$, $p_0 = Y_1^0$, $p_- = Y_1^{-1}$ and let χ_μ be the normalized spin-3/2 state with $S_z\chi_\mu = \mu\chi_\mu$. The elementary lowering actions are

$$L_-p_+ = \sqrt{2}p_0, \quad L_-p_0 = \sqrt{2}p_-, \quad L_-p_- = 0,$$

$$S_-\chi_{3/2} = \sqrt{3}\chi_{1/2}, \quad S_-\chi_{1/2} = 2\chi_{-1/2}, \quad S_-\chi_{-1/2} = \sqrt{3}\chi_{-3/2}, \quad S_-\chi_{-3/2} = 0.$$

These specify every matrix element used below.

For a compact complete table, express a state of given M by its coefficient column in the ordered product basis

$$B_M = (p_+\chi_{M-1},\ p_0\chi_M,\ p_-\chi_{M+1}), \qquad |JM\rangle = B_M c_M^{(J)}.$$

A nonexistent spin state is omitted, and its coefficient is zero. Let $\sigma(\mu) = \sqrt{(3/2 + \mu)(5/2 - \mu)}$ for allowed μ; set $\sigma = 0$ outside the spin range. Thus $\sigma(\frac{3}{2}) = \sqrt{3}$, $\sigma(\frac{1}{2}) = 2$, $\sigma(-\frac{1}{2}) = \sqrt{3}$, $\sigma(-\frac{3}{2}) = 0$, the S_- coefficients above. Applying $J_- = L_- + S_-$ and collecting the three target coefficients gives the explicit rule

$$\sqrt{(J + M)(J - M + 1)}\, c_{M-1}^{(J)} = \begin{pmatrix} \sigma(M-1) & 0 & 0 \\ \sqrt{2} & \sigma(M) & 0 \\ 0 & \sqrt{2} & \sigma(M+1) \end{pmatrix} c_M^{(J)}. \qquad (*)$$

Its columns are the images of the three basis states: $J_-(p_+\chi_{M-1}) = \sigma(M-1)p_+\chi_{M-2} + \sqrt{2}p_0\chi_{M-1}$, $J_-(p_0\chi_M) = \sigma(M)p_0\chi_{M-1} + \sqrt{2}p_-\chi_M$, and $J_-(p_-\chi_{M+1}) = \sigma(M+1)p_-\chi_M$. Each row of the tables below lists the complete coefficient column $c_M^{(J)}$ of $|JM\rangle$ in the basis B_M.

For $J = 5/2$, start with $c_{5/2} = (1, 0, 0)^T$. The first lowering gives $(\sigma(\frac{3}{2}), \sqrt{2}, 0)^T = (\sqrt{3}, \sqrt{2}, 0)^T$, whose norm $\sqrt{5}$ equals the ladder factor $\sqrt{5 \cdot 1}$. The next, with $\sigma(\frac{1}{2}) = 2$, $\sigma(\frac{3}{2}) = \sqrt{3}$, gives

$$\frac{1}{\sqrt{5}}(2\sqrt{3},\ \sqrt{2}\sqrt{3} + \sqrt{3}\sqrt{2},\ \sqrt{2}\sqrt{2})^T = \frac{1}{\sqrt{5}}(2\sqrt{3},\ 2\sqrt{6},\ 2)^T;$$

dividing by the next factor $\sqrt{4 \cdot 2} = 2\sqrt{2}$ gives the $M = 1/2$ row. For the remaining steps the factors in (*) are $\sqrt{3 \cdot 3} = 3$, $\sqrt{2 \cdot 4} = 2\sqrt{2}$, $\sqrt{1 \cdot 5} = \sqrt{5}$, and the numerators are ($\sigma(-\frac{3}{2}) = 0$ in the last two)

$$M = \tfrac{1}{2}: \quad \big(\sqrt{3}\sqrt{\tfrac{3}{10}},\ \sqrt{2}\sqrt{\tfrac{3}{10}} + 2\sqrt{\tfrac{3}{5}},\ \sqrt{2}\sqrt{\tfrac{3}{5}} + \sqrt{3}\sqrt{\tfrac{1}{10}}\big)^T = \big(\tfrac{3}{\sqrt{10}},\ 3\sqrt{\tfrac{3}{5}},\ 3\sqrt{\tfrac{3}{10}}\big)^T,$$

$$M = -\tfrac{1}{2}: \quad \big(0,\ \sqrt{2}\sqrt{\tfrac{1}{10}} + \sqrt{3}\sqrt{\tfrac{3}{5}},\ \sqrt{2}\sqrt{\tfrac{3}{5}} + 2\sqrt{\tfrac{3}{10}}\big)^T = \big(0,\ \tfrac{4}{\sqrt{5}},\ 2\sqrt{\tfrac{6}{5}}\big)^T,$$

$$M = -\tfrac{3}{2}: \quad \big(0,\ 0,\ \sqrt{2}\sqrt{\tfrac{2}{5}} + \sqrt{3}\sqrt{\tfrac{3}{5}}\big)^T = (0,\ 0,\ \sqrt{5})^T,$$

where $\sqrt{2}\sqrt{3/10} = \sqrt{3/5}$, $\sqrt{2}\sqrt{3/5} = \sqrt{6/5} = 2\sqrt{3/10}$, $\sqrt{3}\sqrt{1/10} = \sqrt{3/10}$ and $\sqrt{2}\sqrt{1/10} = 1/\sqrt{5}$. Dividing by the factors 3, $2\sqrt{2}$, $\sqrt{5}$ gives the last three rows of the complete sextet:

M	c_+	c_0	c_-
$5/2$	1	0	0
$3/2$	$\sqrt{3/5}$	$\sqrt{2/5}$	0
$1/2$	$\sqrt{3/10}$	$\sqrt{3/5}$	$\sqrt{1/10}$
$-1/2$	$\sqrt{1/10}$	$\sqrt{3/5}$	$\sqrt{3/10}$
$-3/2$	0	$\sqrt{2/5}$	$\sqrt{3/5}$
$-5/2$	0	0	1

$({}^4p_{5/2})$.

At $M=3/2$, the highest state of the $J=3/2$ multiplet must be orthogonal to $(\sqrt{3/5},\sqrt{2/5},0)^T$. A positive first coefficient ($m_l=l$, as in 16.2.1) fixes it to $(\sqrt{2/5},-\sqrt{3/5},0)^T$. Its three lowerings, with factors $\sqrt{3\cdot1}=\sqrt{3}$, $\sqrt{2\cdot2}=2$ and $\sqrt{1\cdot3}=\sqrt{3}$, have numerators

$$\begin{aligned}
M=\tfrac{3}{2}:&\quad \big(2\sqrt{\tfrac{2}{5}},\ \sqrt{2}\sqrt{\tfrac{2}{5}}-\sqrt{3}\sqrt{\tfrac{3}{5}},\ -\sqrt{2}\sqrt{\tfrac{3}{5}}\big)^T=\big(2\sqrt{\tfrac{2}{5}},\ -\tfrac{1}{\sqrt{5}},\ -\sqrt{\tfrac{6}{5}}\big)^T,\\
M=\tfrac{1}{2}:&\quad \big(\sqrt{3}\sqrt{\tfrac{8}{15}},\ \sqrt{2}\sqrt{\tfrac{8}{15}}-\tfrac{2}{\sqrt{15}},\ -\tfrac{\sqrt{2}}{\sqrt{15}}-\sqrt{3}\sqrt{\tfrac{2}{5}}\big)^T=\big(2\sqrt{\tfrac{2}{5}},\ \tfrac{2}{\sqrt{15}},\ -2\sqrt{\tfrac{8}{15}}\big)^T,\\
M=-\tfrac{1}{2}:&\quad \big(0,\ \sqrt{2}\sqrt{\tfrac{2}{5}}+\tfrac{\sqrt{3}}{\sqrt{15}},\ \tfrac{\sqrt{2}}{\sqrt{15}}-2\sqrt{\tfrac{8}{15}}\big)^T=\big(0,\ \tfrac{3}{\sqrt{5}},\ -\sqrt{\tfrac{6}{5}}\big)^T,
\end{aligned}$$

where $\sqrt{2}\sqrt{2/5}=2/\sqrt{5}$, $\sqrt{3}\sqrt{3/5}=3/\sqrt{5}$, $\sqrt{2}\sqrt{8/15}=4/\sqrt{15}$, $\sqrt{3}\sqrt{2/5}=3\sqrt{2/15}$ and $2\sqrt{8/15}=4\sqrt{2/15}$; for example the last component is $\sqrt{2/15}-4\sqrt{2/15}=-3\sqrt{2/15}=-\sqrt{6/5}$. Dividing by the factors gives

M	c_+	c_0	c_-
$3/2$	$\sqrt{2/5}$	$-\sqrt{3/5}$	0
$1/2$	$\sqrt{8/15}$	$-1/\sqrt{15}$	$-\sqrt{2/5}$
$-1/2$	$\sqrt{2/5}$	$1/\sqrt{15}$	$-\sqrt{8/15}$
$-3/2$	0	$\sqrt{3/5}$	$-\sqrt{2/5}$

$({}^4p_{3/2})$.

For the remaining highest state at $M=1/2$, write $c=(A,B,C)^T$. Orthogonality to the two existing $M=1/2$ rows (multiplied by $\sqrt{10}$ and $\sqrt{15}$, respectively) gives

$$\sqrt{3}A+\sqrt{6}B+C=0,\qquad 2\sqrt{2}A-B-\sqrt{6}C=0.$$

Substituting $C=-\sqrt{3}A-\sqrt{6}B$ in the second equation yields $2\sqrt{2}A-B+3\sqrt{2}A+6B=5\sqrt{2}A+5B=0$, so $B=-\sqrt{2}A$ and $C=-\sqrt{3}A+2\sqrt{3}A=\sqrt{3}A$. Normalization gives $(1+2+3)A^2=6A^2=1$; choose $A=1/\sqrt{6}>0$. Then $c_{1/2}=(1/\sqrt{6},-1/\sqrt{3},1/\sqrt{2})^T$. Applying (*) with $\sigma(-\frac{1}{2})=\sqrt{3}$, $\sigma(\frac{1}{2})=2$, $\sigma(\frac{3}{2})=\sqrt{3}$ and its factor $\sqrt{1\cdot1}=1$ produces

$$\left(\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{3}}-\frac{2}{\sqrt{3}},\ -\sqrt{\frac{2}{3}}+\sqrt{\frac{3}{2}}\right)^T=\left(\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{6}}\right)^T.$$

Thus the doublet is

M	c_+	c_0	c_-
$1/2$	$1/\sqrt{6}$	$-1/\sqrt{3}$	$1/\sqrt{2}$
$-1/2$	$1/\sqrt{2}$	$-1/\sqrt{3}$	$1/\sqrt{6}$

$({}^4p_{1/2})$.

Each row has norm one; rows of the same M but different J are orthogonal, being eigenvectors of the Hermitian operator $\mathbf{J}^2$ with different eigenvalues. For instance, at $M=1/2$ the $J=5/2$ and $J=3/2$ rows give $\sqrt{3/10}\sqrt{8/15}-\sqrt{3/5}/\sqrt{15}-\sqrt{1/10}\sqrt{2/5}=(2-1-1)/5=0$. The dimension $6+4+2=12=3\cdot4$ confirms that every product-basis state is included. A common sign change within a complete multiplet changes no physics; the tables use the same ladder phase convention as 16.2.1. The Instructor's Manual's $J=3/2, M=-1/2$ row contains an invalid spin label on its p_+ term. That label must be $-3/2$, since $1+(-3/2)=-1/2$. The corrected table above obeys $m_l+m_s=M$ in every entry. Its $J=3/2$ multiplet differs from the Instructor's Manual by one common, harmless overall sign. Its intermediate lowering work on printed p. 261 also has two p_0 labels that must be p_-: the term with spin $-1/2$ in $J=5/2, M=-3/2$, and the term with spin $+1/2$ in $J=1/2, M=-1/2$. The sums m_l+m_s fix both labels.

16.2.5 A deuterium atom consists of a proton, a neutron, and an electron. Each of these particles has spin 1/2. The coupling of these three spins can produce J values of 3/2 and 1/2. We consider here only states with no orbital angular momentum.

(a) Show that these J, M states consist of one quartet ($J = 3/2$) and two linearly independent doublets ($J = 1/2$).

Hint. Make a vector-model diagram.

(b) One way to analyze this problem is to couple the spins of the proton and neutron to form a nuclear triplet or singlet, and then to couple the resultant nuclear spin to the electron spin. Find the states that are obtained in this way (designate the single-particle states p_α, p_β, n_α, n_β, e_α, e_β).

(c) Another way to analyze this problem is to couple the spins of the proton and electron to form an atomic triplet or singlet, and then to couple that resultant to the neutron spin. Find the states that result from this coupling scheme.

(d) Show that the coupling schemes of parts (b) and (c) span the same Hilbert space.

Note. The actual interaction energies among these angular momenta cause the scheme of part (b) to be the better way of treating this problem (the triplet nuclear state is substantially the more stable), and the system actually looks like a spin-1 deuterium nucleus plus an electron.

Solution.

Use the fixed tensor order proton–neutron–electron. The particles are distinct, so reordering their tensor factors while writing a coupling scheme introduces no fermionic exchange sign.

(a) The product space has dimension $2^3 = 8$. Its numbers of states at $M = 3/2, 1/2, -1/2, -3/2$ are respectively $1, 3, 3, 1$, the numbers $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}$ of ways to choose which of the three spins point down. The unique highest state generates one $J = 3/2$ quartet, removing one state from each row. The remainder has two states at each of $M = \pm 1/2$, hence two independent $J = 1/2$ doublets. Equivalently, the coupling tree is

$$(\tfrac{1}{2} \otimes \tfrac{1}{2}) \otimes \tfrac{1}{2} = (1 \oplus 0) \otimes \tfrac{1}{2} = (\tfrac{3}{2} \oplus \tfrac{1}{2}) \oplus \tfrac{1}{2}.$$

(b) Couple p and n first. Their triplet and singlet are

$$t_+ = p_\alpha n_\alpha, \qquad t_0 = \frac{p_\alpha n_\beta + p_\beta n_\alpha}{\sqrt{2}}, \qquad t_- = p_\beta n_\beta, \qquad s = \frac{p_\alpha n_\beta - p_\beta n_\alpha}{\sqrt{2}}.$$

With $J_- = J_{p-} + J_{n-}$ on the nuclear states, lowering each spin in turn gives $J_- t_+ = p_\beta n_\alpha + p_\alpha n_\beta = \sqrt{2}\, t_0$ and $J_- t_0 = 2 p_\beta n_\beta / \sqrt{2} = \sqrt{2}\, t_-$, while $J_\pm s = 0$; so the t_m have the ladder coefficients of $p_m = Y_1^m$ in 16.2.3. Applying the $1 \otimes 1/2$ formulas of 16.2.3 with $p_m \to t_m$, $\alpha \to e_\alpha$, $\beta \to e_\beta$ gives the quartet

$$\begin{aligned} Q_{3/2} &= t_+ e_\alpha = p_\alpha n_\alpha e_\alpha, \\ Q_{1/2} &= \sqrt{\tfrac{2}{3}} t_0 e_\alpha + \sqrt{\tfrac{1}{3}} t_+ e_\beta, \\ Q_{-1/2} &= \sqrt{\tfrac{1}{3}} t_- e_\alpha + \sqrt{\tfrac{2}{3}} t_0 e_\beta, \\ Q_{-3/2} &= t_- e_\beta = p_\beta n_\beta e_\beta, \end{aligned}$$

and two doublets, denoted T_M for nuclear triplet and S_M for nuclear singlet:

$$T_{1/2} = \sqrt{\tfrac{2}{3}}t_+e_\beta - \sqrt{\tfrac{1}{3}}t_0e_\alpha, \qquad S_{1/2} = se_\alpha,$$
$$T_{-1/2} = \sqrt{\tfrac{1}{3}}t_0e_\beta - \sqrt{\tfrac{2}{3}}t_-e_\alpha, \quad S_{-1/2} = se_\beta.$$

To display all product coefficients and compare the next scheme, define

$$A = p_\alpha n_\alpha e_\beta, \quad B = p_\alpha n_\beta e_\alpha, \quad C = p_\beta n_\alpha e_\alpha,$$
$$D = p_\alpha n_\beta e_\beta, \quad E = p_\beta n_\alpha e_\beta, \quad F = p_\beta n_\beta e_\alpha.$$

Substitution of t_0, s then gives

M	Q_M	T_M	S_M
$1/2$	$(A+B+C)/\sqrt{3}$	$(2A-B-C)/\sqrt{6}$	$(B-C)/\sqrt{2}$
$-1/2$	$(D+E+F)/\sqrt{3}$	$(D+E-2F)/\sqrt{6}$	$(D-E)/\sqrt{2}$

together with the two extreme quartet states already written. For example $t_+e_\beta = A$ and $t_0e_\alpha = (B+C)/\sqrt{2}$, so $T_{1/2} = \sqrt{2/3}\,A - \sqrt{1/3}\,(B+C)/\sqrt{2} = (2A-B-C)/\sqrt{6}$; the other entries follow in the same way. Lowering each spin in turn gives $J_-A = D+E$, $J_-B = D+F$ and $J_-C = E+F$ (for example, in $A = p_\alpha n_\alpha e_\beta$ lowering p gives E, lowering n gives D, and e_β is annihilated). Consequently $J_-(2A-B-C) = D+E-2F$ and $J_-(B-C) = D-E$. Since the $J = 1/2$ ladder coefficient at $M = 1/2$ is $\sqrt{1\cdot 1} = 1$, this explicitly checks the doublet ladder phases.

(c) Replace the first pair by pe:

$$t'_+ = p_\alpha e_\alpha, \quad t'_0 = (p_\alpha e_\beta + p_\beta e_\alpha)/\sqrt{2}, \quad t'_- = p_\beta e_\beta, \quad s' = (p_\alpha e_\beta - p_\beta e_\alpha)/\sqrt{2}.$$

Couple these to n with the same coefficients:

$$T'_{1/2} = \sqrt{\tfrac{2}{3}}t'_+n_\beta - \sqrt{\tfrac{1}{3}}t'_0n_\alpha, \qquad S'_{1/2} = s'n_\alpha,$$
$$T'_{-1/2} = \sqrt{\tfrac{1}{3}}t'_0n_\beta - \sqrt{\tfrac{2}{3}}t'_-n_\alpha, \quad S'_{-1/2} = s'n_\beta.$$

Reordering the displayed factors back to pne (for example $t'_+n_\beta = p_\alpha n_\beta e_\alpha = B$ and $t'_0n_\alpha = (A+C)/\sqrt{2}$, so $T'_{1/2} = \sqrt{2/3}\,B - \sqrt{1/3}\,(A+C)/\sqrt{2}$) gives

M	T'_M	S'_M
$1/2$	$(2B-A-C)/\sqrt{6}$	$(A-C)/\sqrt{2}$
$-1/2$	$(D-2E+F)/\sqrt{6}$	$(D-F)/\sqrt{2}$

and the same four Q_M: the quartet is totally symmetric under permuting its spin labels.

(d) At each $M = \pm 1/2$ the product space is three-dimensional, with orthonormal basis Q_M, T_M, S_M. The states T'_M, S'_M have $J = 1/2$, so they are orthogonal to the $J = 3/2$ state Q_M (eigenvectors of the Hermitian $\mathbf{J}^2$ with different eigenvalues), and therefore $T'_M = \langle T_M, T'_M\rangle T_M + \langle S_M, T'_M\rangle S_M$, and similarly for S'_M. For $M = 1/2$ the coefficient vectors in the basis (A, B, C) are $T_M = (2,-1,-1)/\sqrt{6}$, $S_M = (0,1,-1)/\sqrt{2}$, $T'_M = (-1,2,-1)/\sqrt{6}$, $S'_M = (1,0,-1)/\sqrt{2}$, and their dot products give

$$\langle T_M, T'_M\rangle = (-2-2+1)/6 = -\tfrac{1}{2}, \qquad \langle S_M, T'_M\rangle = (2+1)/\sqrt{12} = \tfrac{\sqrt{3}}{2}.$$

The corresponding products with S'_M are $\langle T_M, S'_M\rangle = (2+1)/\sqrt{12} = \sqrt{3}/2$ and $\langle S_M, S'_M\rangle = 1/2$. Using the second rows of the tables, in the basis (D, E, F), gives exactly the same four numbers at $M = -1/2$. Thus, for both magnetic values,

$$\boxed{\begin{pmatrix} T'_M \\ S'_M \end{pmatrix} = \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix} \begin{pmatrix} T_M \\ S_M \end{pmatrix}}.$$

The real matrix squares to the identity and has orthonormal rows. It therefore changes basis in the same two-dimensional doublet space at each M. Together with the unchanged quartet, it is a unitary change of basis in the full eight-dimensional Hilbert space. The Instructor's Manual repeats the relative doublet-phase mismatch from 16.2.3 in its triplet–electron coupling. The expansions here use $J_- T_{1/2} = T_{-1/2}$ and $J_- T'_{1/2} = T'_{-1/2}$, so the same recoupling matrix applies to both M values.

16.3 Spherical Tensors

16.3.1 For $l = 1$, Eq. (16.52) becomes

$$Y_1^m(\theta', \varphi') = \sum_{m'=-1}^{1} D^1_{m'm}(\alpha, \beta, \gamma)\, Y_1^{m'}(\theta, \varphi).$$

Rewrite these spherical harmonics in Cartesian form. Show that the resulting Cartesian coordinate equations are equivalent to the Euler rotation matrix $\mathsf{A}(\alpha, \beta, \gamma)$, Eq. (3.37).

Solution.

Use the passive rotations of Eq. (3.37): first α about z, then β about the intermediate y, then γ about the new z. Define

$$Z(t) = \begin{pmatrix} \cos t & \sin t & 0 \\ -\sin t & \cos t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B(\beta) = \begin{pmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{pmatrix}.$$

Thus the Cartesian column transforms by $\mathbf{r}' = A\mathbf{r}$, where $A = Z(\gamma)B(\beta)Z(\alpha)$.

The $l = 1$ harmonics are linear combinations of the Cartesian components (x, y, z) of $\hat{\mathbf{r}}$: since $\sin\theta\, e^{\pm i\varphi} = x \pm iy$ and $\cos\theta = z$, $Y_1^{\pm 1} = \mp\sqrt{3/(8\pi)}\,(x \pm iy)$ and $Y_1^0 = \sqrt{3/(4\pi)}\, z$, that is,

$$\mathcal{Y} = \begin{pmatrix} Y_1^{-1} \\ Y_1^0 \\ Y_1^1 \end{pmatrix} = \sqrt{\frac{3}{4\pi}}\, Q\hat{\mathbf{r}}, \qquad Q = \begin{pmatrix} 1/\sqrt{2} & -i/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & -i/\sqrt{2} & 0 \end{pmatrix}.$$

Direct row products give $QQ^\dagger = I$; for a square matrix this also gives $Q^\dagger Q = I$, so $\hat{\mathbf{r}} = \sqrt{4\pi/3}\, Q^\dagger \mathcal{Y}$. In the index convention of Eq. (16.52), $Y_1^m(\Omega') = \sum_{m'} D^1_{m'm} Y_1^{m'}(\Omega)$, so the mth component of $\mathcal{Y}'$ is $\sum_{m'} (D^1)^T_{mm'} \mathcal{Y}_{m'}$: the matrix acting on this column is $(D^1)^T$. On the other hand,

$$\mathcal{Y}' = \sqrt{\frac{3}{4\pi}}\, Q\hat{\mathbf{r}}' = \sqrt{\frac{3}{4\pi}}\, QA\hat{\mathbf{r}} = QAQ^\dagger \mathcal{Y}.$$

The three functions Y_1^m are linearly independent, so the two matrices agree:

$$\boxed{(D^1)^T = QAQ^\dagger, \quad A = Q^\dagger (D^1)^T Q}.$$

The transpose here follows from the displayed summation indices.

To compare with the explicit rotation matrix D^1, compute the elementary spherical-basis matrices

$$QZ(t)Q^\dagger = E(t) = \mathrm{diag}(e^{it}, 1, e^{-it}),$$

$$QB(\beta)Q^\dagger = H(\beta) = \begin{pmatrix} (1+c_\beta)/2 & -s_\beta/\sqrt{2} & (1-c_\beta)/2 \\ s_\beta/\sqrt{2} & c_\beta & -s_\beta/\sqrt{2} \\ (1-c_\beta)/2 & s_\beta/\sqrt{2} & (1+c_\beta)/2 \end{pmatrix},$$

where $c_t = \cos t$, $s_t = \sin t$. To obtain them, write $(u_{-1}, u_0, u_1)^T = Q\hat{\mathbf{r}}$, i.e. $u_{\mp 1} = \pm(x \mp iy)/\sqrt{2}$, $u_0 = z$; inversely $x = (u_{-1} - u_1)/\sqrt{2}$, $iy = -(u_{-1} + u_1)/\sqrt{2}$, $z = u_0$. Under $Z(t)$, $x' \mp iy' = (x \mp iy)(\cos t \pm i \sin t) = e^{\pm it}(x \mp iy)$ and $z' = z$, which gives $E(t)$. Under $B(\beta)$, $x' = c_\beta x - s_\beta z$, $y' = y$, $z' = s_\beta x + c_\beta z$, so for example

$$u'_{-1} = \frac{x' - iy'}{\sqrt{2}} = \frac{c_\beta(u_{-1} - u_1) + (u_{-1} + u_1)}{2} - \frac{s_\beta u_0}{\sqrt{2}},$$

which is the first row of H; the rows for $u'_0 = z'$ and $u'_1 = -(x' + iy')/\sqrt{2}$ follow in the same way. Inserting $Q^\dagger Q = I$ between the factors of $QZ(\gamma)B(\beta)Z(\alpha)Q^\dagger$ gives $(D^1)^T = E(\gamma)H(\beta)E(\alpha)$. Since the diagonal entries of $E(t)$ are e^{-imt} ($m = -1, 0, 1$), this is, entry by entry, the standard Euler-angle form of the $l = 1$ rotation matrix:

$$D^1_{m'm}(\alpha, \beta, \gamma) = e^{-im'\alpha}\, d^1_{m'm}(\beta)\, e^{-im\gamma}, \qquad d^1_{m'm}(\beta) = H_{mm'}(\beta),$$

that is, $d^1_{11} = d^1_{-1,-1} = (1 + c_\beta)/2$, $d^1_{1,-1} = d^1_{-1,1} = (1 - c_\beta)/2$, $d^1_{00} = c_\beta$, $d^1_{10} = d^1_{0,-1} = -s_\beta/\sqrt{2}$ and $d^1_{01} = d^1_{-1,0} = s_\beta/\sqrt{2}$. Conversely, inserting this explicit D^1 in $\mathcal{Y}' = (D^1)^T\mathcal{Y}$ and multiplying by $\sqrt{4\pi/3}\, Q^\dagger$ gives the Cartesian equations

$$\hat{\mathbf{r}}' = Q^\dagger (D^1)^T Q\, \hat{\mathbf{r}} = Q^\dagger E(\gamma) Q\, Q^\dagger H(\beta) Q\, Q^\dagger E(\alpha) Q\, \hat{\mathbf{r}} = Z(\gamma)B(\beta)Z(\alpha)\, \hat{\mathbf{r}},$$

because $Q^\dagger E(t) Q = Z(t)$ and $Q^\dagger H(\beta) Q = B(\beta)$ by the unitarity of Q. Multiplying out the two right-hand factors gives

$$B(\beta)Z(\alpha) = \begin{pmatrix} c_\beta c_\alpha & c_\beta s_\alpha & -s_\beta \\ -s_\alpha & c_\alpha & 0 \\ s_\beta c_\alpha & s_\beta s_\alpha & c_\beta \end{pmatrix}.$$

Multiplying by $Z(\gamma)$ on the left replaces the first row by c_γ times the first row plus s_γ times the second, and the second row by $-s_\gamma$ times the first plus c_γ times the second; the third row is unchanged. This yields

$$\boxed{A = \begin{pmatrix} c_\gamma c_\beta c_\alpha - s_\gamma s_\alpha & c_\gamma c_\beta s_\alpha + s_\gamma c_\alpha & -c_\gamma s_\beta \\ -s_\gamma c_\beta c_\alpha - c_\gamma s_\alpha & -s_\gamma c_\beta s_\alpha + c_\gamma c_\alpha & s_\gamma s_\beta \\ s_\beta c_\alpha & s_\beta s_\alpha & c_\beta \end{pmatrix}}.$$

This is exactly Eq. (3.37), with the same passive-angle convention.

16.3.2 In proving the addition theorem, we assumed that $Y^k_l(\theta_1, \varphi_1)$ could be expanded in a series of $Y^m_l(\theta_2, \varphi_2)$, in which m varied from $-l$ to $+l$ but l was held fixed. What arguments can you develop to justify summing only over the upper index, m, and not over the lower index, l?

Hints. Examine the homogeneous harmonic polynomials $r^l Y^m_l$. Their restrictions to the sphere have terms of the form $\cos^{l-p}\theta \sin^p\theta e^{im\varphi}$, or $x^{l-p-s}y^p z^s/r^l$. Alternatively examine the rotation

invariance of the angular Laplacian. The textbook's angular hint omits the azimuthal factor; homogeneity alone also needs the harmonicity condition to exclude lower-degree traces.

Solution.

An expansion of $Y_l^k(\theta_1,\varphi_1)$ in functions of (θ_2,φ_2) requires (θ_1,φ_1) to be determined by (θ_2,φ_2): the two pairs are the angles of one direction referred to two sets of axes related by a fixed rotation. Let R be the rotation matrix with $\hat{\mathbf{r}}_1 = R\hat{\mathbf{r}}_2$ for the component columns, and let $U_R f(\mathbf{r}) = f(R\mathbf{r})$ denote its action on functions; then $Y_l^k(\theta_1,\varphi_1)$, as a function of (θ_2,φ_2), is $U_R Y_l^k$. A rotation preserves r, dot products, and the Cartesian Laplacian. The identity

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\mathbf{L}^2}{r^2}$$

can be solved as $\mathbf{L}^2 = \partial_r(r^2\partial_r) - r^2\nabla^2$. Each term on the right commutes with U_R: ∇^2 is rotation invariant, and r and ∂_r are unchanged because $R(r\hat{\mathbf{r}}) = r\,R\hat{\mathbf{r}}$. Therefore $U_R\mathbf{L}^2 = \mathbf{L}^2 U_R$. Expand the rotated harmonic in the complete orthonormal spherical-harmonic basis:

$$U_R Y_l^k = \sum_{n=0}^{\infty}\sum_{m=-n}^{n} a_{nm} Y_n^m.$$

Applying $\mathbf{L}^2$ in the two orders, $\mathbf{L}^2 U_R Y_l^k = U_R \mathbf{L}^2 Y_l^k = l(l+1)U_R Y_l^k$, gives

$$l(l+1)\sum_{nm} a_{nm} Y_n^m = \sum_{nm} n(n+1) a_{nm} Y_n^m.$$

Multiply by $Y_{n'}^{m'*}$ and integrate, using orthonormality:

$$\big[n'(n'+1) - l(l+1)\big]\, a_{n'm'} = 0.$$

(Equivalently, $n'(n'+1)a_{n'm'} = \langle \mathbf{L}^2 Y_{n'}^{m'}, U_R Y_l^k\rangle = \langle Y_{n'}^{m'}, \mathbf{L}^2 U_R Y_l^k\rangle = l(l+1)a_{n'm'}$ by the Hermiticity of $\mathbf{L}^2$, without differentiating the series.) Since $n(n+1)$ increases strictly for $n \geq 0$, the factor is zero only for $n' = l$. All other coefficients vanish, proving

$$\boxed{Y_l^k(\theta_1,\varphi_1) = \sum_{m=-l}^{l} a_{lm} Y_l^m(\theta_2,\varphi_2)}.$$

The polynomial hint gives the same conclusion when its harmonic condition is retained. The solid harmonic $r^l Y_l^k$ is a homogeneous harmonic polynomial of degree l. Linear substitution by a rotation preserves its degree, and rotation invariance of ∇^2 preserves its being harmonic. It therefore remains in the space spanned by $r^l Y_l^{-l}, \ldots, r^l Y_l^l$: writing a homogeneous polynomial of degree l as $r^l f(\hat{\mathbf{r}})$, the displayed form of ∇^2 gives $\nabla^2(r^l f) = r^{l-2}[l(l+1)f - \mathbf{L}^2 f]$, so harmonicity means $\mathbf{L}^2 f = l(l+1)f$, and the coefficient argument above leaves only the Y_l^m. Homogeneity alone would not suffice: a general homogeneous polynomial may contain traces such as r^2 times a lower-degree harmonic. Harmonicity excludes those traces.

16.3.3 An atomic electron with angular momentum l and magnetic quantum number m has a wave function

$$\psi(r,\theta,\varphi) = f(r)Y_l^m(\theta,\varphi).$$

Show that the sum of the electron densities in a given complete shell is spherically symmetric; that is, $\sum_{m=-l}^{l}\psi^*(r,\theta,\varphi)\psi(r,\theta,\varphi)$ is independent of θ and φ.

Solution.

All members of the specified shell have the same radial factor. Consequently

$$\sum_{m=-l}^{l}\psi_m^*\psi_m = |f(r)|^2\sum_{m=-l}^{l}|Y_l^m(\theta,\varphi)|^2.$$

Apply the addition theorem to two identical directions. Their separation angle is zero, so $\cos\gamma = 1$, and $P_l(1) = 1$:

$$1 = P_l(1) = \frac{4\pi}{2l+1}\sum_{m=-l}^{l}Y_l^{m*}(\theta,\varphi)Y_l^m(\theta,\varphi),$$

that is, $\sum_m |Y_l^m(\theta,\varphi)|^2 = (2l+1)/(4\pi)$ for every direction. Thus

$$\boxed{\sum_{m=-l}^{l}\psi_m^*\psi_m = \frac{2l+1}{4\pi}|f(r)|^2}.$$

This depends only on r. Equal occupancy of all magnetic states is the complete-shell condition used by the stated sum; a common spin-occupancy factor merely multiplies this radial density.

16.3.4 The potential energy of an electron at point $\mathbf{r}_e$ in the field of Z protons at points $\mathbf{r}_p$ is

$$\Phi = -\frac{e^2}{4\pi\varepsilon_0}\sum_{p=1}^{Z}\frac{1}{|\mathbf{r}_e-\mathbf{r}_p|}.$$

Show that for r_e larger than all r_p, this may be written as

$$\Phi = -\frac{e^2}{4\pi\varepsilon_0 r_e}\sum_{p=1}^{Z}\sum_{L,M}\left(\frac{r_p}{r_e}\right)^L\frac{4\pi}{2L+1}Y_L^M(\theta_p,\varphi_p)^*Y_L^M(\theta_e,\varphi_e).$$

How should Φ be written for $r_e < r_p$?

Solution.

The displayed expression contains the charge product $(-e)(+e) = -e^2$, so it is an interaction energy. The electric potential of the protons alone has coefficient $+e/(4\pi\varepsilon_0)$. For each proton define γ_p as the angle between $\mathbf{r}_e$ and $\mathbf{r}_p$, so that $|\mathbf{r}_e-\mathbf{r}_p|^2 = r_e^2 - 2r_e r_p\cos\gamma_p + r_p^2$. With $t = r_p/r_e < 1$, the Legendre generating function gives

$$\frac{1}{|\mathbf{r}_e-\mathbf{r}_p|} = \frac{1}{r_e}(1-2t\cos\gamma_p+t^2)^{-1/2} = \frac{1}{r_e}\sum_{L=0}^{\infty}t^L P_L(\cos\gamma_p).$$

The addition theorem is

$$P_L(\cos\gamma_p) = \frac{4\pi}{2L+1}\sum_{M=-L}^{L} Y_L^{M*}(\Omega_p)Y_L^M(\Omega_e).$$

Substitution into each term of the proton sum, with the factor $-e^2/(4\pi\varepsilon_0)$, gives exactly the first requested expansion.

For $r_e < r_p$, factor out r_p instead: $|\mathbf{r}_e - \mathbf{r}_p|^{-1} = r_p^{-1}(1 - 2t'\cos\gamma_p + t'^2)^{-1/2}$ with $t' = r_e/r_p < 1$. The same angular factor then gives

$$\boxed{\Phi = -\frac{e^2}{4\pi\varepsilon_0}\sum_{p=1}^{Z}\frac{1}{r_p}\sum_{L=0}^{\infty}\left(\frac{r_e}{r_p}\right)^L\frac{4\pi}{2L+1}\sum_{M=-L}^{L}Y_L^{M*}(\Omega_p)Y_L^M(\Omega_e)}$$

when the electron lies inside every proton radius. More generally, the expansion can be used term by term, even when some protons are inside r_e and others outside:

$$\boxed{\Phi = -\frac{e^2}{4\pi\varepsilon_0}\sum_{p=1}^{Z}\sum_{L=0}^{\infty}\frac{r_{<p}^L}{r_{>p}^{L+1}}\frac{4\pi}{2L+1}\sum_{M=-L}^{L}Y_L^{M*}(\Omega_p)Y_L^M(\Omega_e)},$$

where $r_{<p} = \min(r_e, r_p)$ and $r_{>p} = \max(r_e, r_p)$. The strictly unequal radii assumed above make each generating series convergent. Reversing which radius is larger changes only the radial factor, not the angular addition theorem.

16.3.5 Two protons are uniformly distributed within the same spherical volume. If the coordinates of one element of charge are $(r_1, \theta_1, \varphi_1)$ and the coordinates of the other are $(r_2, \theta_2, \varphi_2)$ and r_{12} is the distance between them, the element of repulsion energy will be given by

$$d\psi = \rho^2\frac{d\tau_1\, d\tau_2}{r_{12}} = \rho^2\frac{r_1^2 dr_1\sin\theta_1\, d\theta_1\, d\varphi_1\, r_2^2 dr_2\sin\theta_2\, d\theta_2\, d\varphi_2}{r_{12}},$$

where

$$\rho = \frac{\text{charge}}{\text{volume}} = \frac{3e}{4\pi R^3} \qquad \text{and} \qquad r_{12}^2 = r_1^2 + r_2^2 - 2r_1r_2\cos\gamma.$$

Here ρ is the charge density and γ is the angle between $\mathbf{r}_1$ and $\mathbf{r}_2$. Calculate the total electrostatic energy (of repulsion) of the two protons. This calculation is used in accounting for the mass difference in "mirror" nuclei, such as O^{15} and N^{15}.

ANS. $\dfrac{6}{5}\dfrac{e^2}{R}$.

Solution.

The angular average of the Coulomb expansion retains only $L = 0$. As in 16.3.4, for $r_1 \neq r_2$,

$$\frac{1}{r_{12}} = \sum_{L=0}^{\infty}\frac{r_<^L}{r_>^{L+1}}P_L(\cos\gamma), \qquad r_< = \min(r_1, r_2),\ r_> = \max(r_1, r_2).$$

By the addition theorem and $\int Y_L^M\, d\Omega = \sqrt{4\pi}\int Y_0^{0*}Y_L^M\, d\Omega = \sqrt{4\pi}\,\delta_{L0}\delta_{M0}$,

$$\int P_L(\hat{\mathbf{r}}_1\cdot\hat{\mathbf{r}}_2)\, d\Omega_2 = \frac{4\pi}{2L+1}\sum_M Y_L^{M*}(\Omega_1)\sqrt{4\pi}\,\delta_{L0}\delta_{M0} = 4\pi\delta_{L0},$$

since $Y_0^0 = 1/\sqrt{4\pi}$. The remaining Ω_1 integral gives another factor 4π, so

$$\int d\Omega_1 \int d\Omega_2 \, \frac{1}{r_{12}} = \frac{16\pi^2}{\max(r_1, r_2)}.$$

After these angular integrations, $U = \int d\psi$ is the double radial integral in the first line below. The diagonal $r_1 = r_2$ has zero measure in it. The integrand is symmetric in r_1, r_2, and on the triangle $r_2 < r_1$ we have $\max(r_1, r_2) = r_1$. Split its square into the two equal triangles:

$$\begin{aligned} U &= 16\pi^2\rho^2 \int_0^R \int_0^R \frac{r_1^2 r_2^2}{\max(r_1, r_2)} \, dr_2 \, dr_1 \\ &= 32\pi^2\rho^2 \int_0^R r_1 \, dr_1 \int_0^{r_1} r_2^2 \, dr_2 \\ &= 32\pi^2\rho^2 \int_0^R r_1 \, \frac{r_1^3}{3} \, dr_1 = \frac{32\pi^2\rho^2}{3} \int_0^R r_1^4 \, dr_1 = \frac{32\pi^2\rho^2 R^5}{15}. \end{aligned}$$

Substitute $\rho^2 = 9e^2/(16\pi^2 R^6)$:

$$\boxed{U = \frac{32 \cdot 9}{15 \cdot 16} \frac{e^2}{R} = \frac{288}{240} \frac{e^2}{R} = \frac{6}{5} \frac{e^2}{R}}.$$

There is no factor 1/2: this is the mutual energy of two distinct charge distributions, and each pair with one element from each distribution is counted once. The problem uses a Coulomb constant of one; in SI units the result is multiplied by $1/(4\pi\varepsilon_0)$.

16.3.6 Each of the two $1s$ electrons in helium may be described by a hydrogenic wave function

$$\psi(r) = \left(\frac{Z^3}{\pi a_0^3} \right)^{1/2} e^{-Zr/a_0}$$

in the absence of the other electron. Here Z, the atomic number, is 2. The symbol a_0 is the Bohr radius, $\hbar^2/me^2$. Find the mutual potential energy of the two electrons, given by

$$\int \psi^*(\mathbf{r}_1)\psi^*(\mathbf{r}_2) \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} \, \psi(\mathbf{r}_1)\psi(\mathbf{r}_2) \, d^3r_1 \, d^3r_2.$$

ANS. $\dfrac{5e^2 Z}{8a_0}$.

Solution.

Put $\lambda = 2Z/a_0$ and $N = Z^3/(\pi a_0^3)$, so $|\psi(r)|^2 = Ne^{-\lambda r}$. Its normalization is

$$4\pi N \int_0^\infty r^2 e^{-\lambda r} \, dr = 4\pi N \, \frac{2}{\lambda^3} = \frac{8\pi Z^3}{\pi a_0^3} \, \frac{a_0^3}{8Z^3} = 1.$$

The angular integration is the same as in 16.3.5: it replaces $1/|\mathbf{r}_1 - \mathbf{r}_2|$ by $16\pi^2/\max(r_1, r_2)$, so

$$U = 16\pi^2 e^2 N^2 \int_0^\infty \int_0^\infty \frac{r_1^2 r_2^2 e^{-\lambda(r_1 + r_2)}}{\max(r_1, r_2)} \, dr_2 \, dr_1.$$

The integrand is symmetric, so splitting the radial quadrant into the equal halves $r_2 < r_1$ and $r_1 < r_2$ gives

$$U = 32\pi^2 e^2 N^2 \int_0^\infty r_1 e^{-\lambda r_1}\, dr_1 \int_0^{r_1} r_2^2 e^{-\lambda r_2}\, dr_2.$$

Set $x = \lambda r_2$, $y = \lambda r_1$, so that $r_1\, dr_1\, r_2^2\, dr_2 = \lambda^{-5} y\, dy\, x^2\, dx$ and the region $0 < r_2 < r_1$ becomes $0 < x < y$, and reverse the order:

$$U = \frac{32\pi^2 e^2 N^2}{\lambda^5} \int_0^\infty x^2 e^{-x} \left(\int_x^\infty y e^{-y}\, dy \right) dx.$$

Integration by parts gives

$$\int_x^\infty y e^{-y} dy = [-y e^{-y}]_x^\infty + \int_x^\infty e^{-y} dy = (x+1)e^{-x}.$$

The x integrand is therefore $x^2(x+1)e^{-2x} = (x^3 + x^2)e^{-2x}$. For $I_k = \int_0^\infty x^k e^{-2x} dx$ and $k \geq 1$,

$$I_k = \left[-\frac{x^k e^{-2x}}{2} \right]_0^\infty + \frac{k}{2} \int_0^\infty x^{k-1} e^{-2x} dx = \frac{k}{2} I_{k-1}.$$

Since $I_0 = 1/2$, this gives $I_1 = 1/4$, $I_2 = 1/4$, and $I_3 = 3/8$, or generally $I_k = k!/2^{k+1}$. Hence the dimensionless integral is

$$\int_0^\infty (x^3 + x^2) e^{-2x}\, dx = \frac{3!}{2^4} + \frac{2!}{2^3} = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}.$$

The prefactor simplifies without an approximation, using $N^2 = Z^6/(\pi^2 a_0^6)$ and $(2Z/a_0)^5 = 32Z^5/a_0^5$:

$$\frac{32\pi^2 e^2}{(2Z/a_0)^5} \frac{Z^6}{\pi^2 a_0^6} = \frac{32 e^2 a_0^5}{32 Z^5} \frac{Z^6}{a_0^6} = \frac{e^2 Z}{a_0}.$$

Thus

$$\boxed{U = \frac{5e^2 Z}{8a_0} = \frac{5e^2}{4a_0} \quad (Z = 2)}.$$

This evaluates the stated independent hydrogenic-orbital integral; it uses the same Coulomb-unit convention as the problem.

16.3.7 The probability of finding a $1s$ hydrogen electron in a volume element $r^2 dr \sin\theta\, d\theta\, d\varphi$ is

$$\frac{1}{\pi a_0^3}\, e^{-2r/a_0}\, r^2 dr \sin\theta\, d\theta\, d\varphi,$$

where r is the distance of the electron from the nucleus. Find the electrostatic potential of this charge distribution at points $\mathbf{r}_1$, where you may not assume that $\mathbf{r}_1$ is on the polar axis of your coordinate system. Calculate the potential from

$$V(\mathbf{r}_1) = -\frac{e}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r}_2)}{r_{12}}\, d^3 r_2,$$

where $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$ and $\rho(r) = e^{-2r/a_0}/(\pi a_0^3)$ is the probability density. Expand $1/r_{12}$. Apply the Legendre polynomial addition theorem and show that the angular dependence of $V(\mathbf{r}_1)$ drops out.

ANS. $V(r_1) = -\dfrac{e}{4\pi\varepsilon_0} \left[\dfrac{1}{2r_1} \gamma\left(3, \dfrac{2r_1}{a_0}\right) + \dfrac{1}{a_0} \Gamma\left(2, \dfrac{2r_1}{a_0}\right) \right]$, where γ and Γ are incomplete gamma functions, Eq. (13.73).

Solution.

The textbook's instruction "Expand r_{12}" must instead say "Expand $1/r_{12}$", since $1/r_{12}$ is the Coulomb kernel in the integral. Write $R = r_1$ and $r = r_2$. Expand the inverse separation:

$$\frac{1}{r_{12}} = \sum_{l=0}^{\infty} \frac{r_<^l}{r_>^{l+1}} \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_l^m(\Omega_1) Y_l^{m*}(\Omega_2).$$

The density is independent of Ω_2. Because $\int Y_l^{m*}\, d\Omega_2 = \sqrt{4\pi}\delta_{l0}\delta_{m0}$ and $Y_0^0(\Omega_1) = 1/\sqrt{4\pi}$, its angular integral is $4\pi \cdot Y_0^0(\Omega_1)\sqrt{4\pi}/r_> = 4\pi/r_>$. Thus no choice of polar axis is needed. With $d^3r_2 = r^2 dr\, d\Omega_2$, the factor $4\pi/(\pi a_0^3) = 4/a_0^3$, and $r_> = R$ for $r < R$, $r_> = r$ for $r > R$,

$$V(R) = -\frac{e}{4\pi\varepsilon_0}\frac{4}{a_0^3}\left[\frac{1}{R}\int_0^R r^2 e^{-2r/a_0}\, dr + \int_R^{\infty} r e^{-2r/a_0}\, dr\right].$$

Put $q = 2R/a_0$ and $r = a_0 t/2$, so $dr = (a_0/2)\, dt$. By the definitions $\gamma(s,q) = \int_0^q t^{s-1}e^{-t}dt$ and $\Gamma(s,q) = \int_q^{\infty} t^{s-1}e^{-t}dt$,

$$\int_0^R r^2 e^{-2r/a_0}\, dr = \frac{a_0^3}{8}\,\gamma(3,q), \qquad \int_R^{\infty} r e^{-2r/a_0}\, dr = \frac{a_0^2}{4}\,\Gamma(2,q),$$

and multiplying by $4/a_0^3$ gives

$$\boxed{V(R) = -\frac{e}{4\pi\varepsilon_0}\left[\frac{\gamma(3,q)}{2R} + \frac{\Gamma(2,q)}{a_0}\right]}.$$

For clarity, the required integrations by parts are

$$\begin{aligned}
\int_0^q te^{-t}dt &= [-te^{-t}]_0^q + \int_0^q e^{-t}dt = 1 - (q+1)e^{-q},\\
\gamma(3,q) = \int_0^q t^2e^{-t}dt &= [-t^2e^{-t}]_0^q + 2\int_0^q te^{-t}dt\\
&= 2 - e^{-q}(q^2+2q+2),\\
\Gamma(2,q) = \int_q^{\infty} te^{-t}dt &= [-te^{-t}]_q^{\infty} + \int_q^{\infty} e^{-t}dt = e^{-q}(q+1).
\end{aligned}$$

Since $q/(2R) = 1/a_0$, so that $q^2/(2R) = q/a_0$ and $2q/(2R) = 2/a_0$, substitution gives

$$\frac{\gamma(3,q)}{2R} + \frac{\Gamma(2,q)}{a_0} = \frac{1}{R} - e^{-q}\left(\frac{q}{a_0} + \frac{2}{a_0} + \frac{1}{R}\right) + e^{-q}\left(\frac{q}{a_0} + \frac{1}{a_0}\right),$$

in which the q/a_0 terms cancel and $2/a_0 - 1/a_0 = 1/a_0$, leaving

$$\boxed{V(R) = -\frac{e}{4\pi\varepsilon_0}\left[\frac{1}{R} - e^{-2R/a_0}\left(\frac{1}{R} + \frac{1}{a_0}\right)\right]}.$$

For example, writing the bracket as $(1-e^{-2R/a_0})/R - e^{-2R/a_0}/a_0$ and using $(1-e^{-2R/a_0})/R \to 2/a_0$ as $R \to 0$ (since $1 - e^{-x} = x + O(x^2)$) gives the limit $2/a_0 - 1/a_0$, so $V(0) = -e/(4\pi\varepsilon_0 a_0)$. At large R the exponential disappears and the potential is that of charge $-e$, as required.

16.3.8 A hydrogen electron in a $2p$ orbital has a charge distribution

$$\rho = -\frac{e}{64\pi a_0^5}\, r^2 e^{-r/a_0} \sin^2\theta,$$

where $a_0 = \hbar^2/me^2$ is the Bohr radius, and r is the distance between the electron and the nucleus. Find the electrostatic electron–nucleus interaction energy, treating the nucleus as a point charge $+e$ at the origin. Also find the electric potential of the electron distribution at an arbitrary field point.

Solution.

For the hydrogen atom, the stated electrostatic energy is interpreted as the electron–nucleus interaction energy; the classical self-energy $\frac{1}{2}\int \rho V\, d^3r$ of the electron cloud alone would be a different quantity. The Instructor's Manual instead displays a position-dependent electric potential; both quantities are obtained here.

Let the field point be (R, Θ, Φ) and write $a = a_0$. Since $P_2(\cos\theta) = \frac{1}{2}(3\cos^2\theta - 1)$, we have $\cos^2\theta = \frac{1}{3}[1 + 2P_2(\cos\theta)]$, hence the identity $\sin^2\theta = 1 - \cos^2\theta = \frac{2}{3}[1 - P_2(\cos\theta)]$; with $\frac{2}{3}\cdot\frac{1}{64} = \frac{1}{96}$ it gives

$$\rho(r,\theta) = -\frac{e}{96\pi a^5} r^2 e^{-r/a}[1 - P_2(\cos\theta)].$$

Its total charge is

$$-\frac{e}{96\pi a^5}\,(4\pi)\int_0^\infty r^4 e^{-r/a} dr = -\frac{e}{24a^5}(4!\,a^5) = -e,$$

because $\int P_2\, d\Omega = 0$ (orthogonality to P_0), so the angular integral of $1 - P_2$ is 4π, and $\int_0^\infty r^n e^{-r/a} dr = n!\, a^{n+1}$.

The addition theorem expands the Coulomb kernel, with $r_< = \min(r, R)$ and $r_> = \max(r, R)$, as

$$\frac{1}{|\mathbf{R} - \mathbf{r}|} = \sum_{k=0}^{\infty} \frac{4\pi}{2k+1} \frac{r_<^k}{r_>^{k+1}} \sum_{m=-k}^{k} Y_k^{m*}(\theta,\varphi) Y_k^m(\Theta,\Phi).$$

Write $P_l(\cos\theta) = \sqrt{4\pi/(2l+1)}\, Y_l^0(\theta,\varphi)$ and integrate over the source angles: orthonormality, $\int Y_l^0 Y_k^{m*}\, d\Omega = \delta_{kl}\delta_{m0}$, keeps only the term $k = l$, $m = 0$, so

$$\int \frac{P_l(\cos\theta)}{|\mathbf{R} - \mathbf{r}|}\, d\Omega = \frac{4\pi}{2l+1}\frac{r_<^l}{r_>^{l+1}} \sqrt{\frac{4\pi}{2l+1}}\, Y_l^0(\Theta,\Phi) = \frac{4\pi}{2l+1}\frac{r_<^l}{r_>^{l+1}} P_l(\cos\Theta).$$

The angular factor $P_0 - P_2$ of ρ therefore leaves only the monopole ($l = 0$, factor 4π) and the quadrupole ($l = 2$, factor $4\pi/5$). With $d^3r = r^2 dr\, d\Omega$ and $4\pi/(96\pi a^5) = 1/(24a^5)$,

$$V(R,\Theta) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\mathbf{r})}{|\mathbf{R} - \mathbf{r}|}\, d^3r = -\frac{e}{4\pi\varepsilon_0}\frac{1}{24a^5}\left[I_0(R) - \frac{P_2(\cos\Theta)}{5} I_2(R)\right],$$

where $I_l(R) = \int_0^\infty r^4 e^{-r/a}\, r_<^l / r_>^{l+1}\, dr$. Splitting each radial integral at $r = R$ gives

$$I_0(R) = \frac{1}{R}\int_0^R r^4 e^{-r/a} dr + \int_R^\infty r^3 e^{-r/a} dr,$$
$$I_2(R) = \frac{1}{R^3}\int_0^R r^6 e^{-r/a} dr + R^2 \int_R^\infty r e^{-r/a} dr.$$

Use the incomplete gamma functions $\gamma(n,q)=\int_0^q t^{n-1}e^{-t}dt$ and $\Gamma(n,q)=\int_q^\infty t^{n-1}e^{-t}dt$, with $q=R/a$. The substitution $r=at$ (new limits $0,q$ or q,∞) gives

$$\int_0^R r^n e^{-r/a}dr = a^{n+1}\gamma(n+1,q), \qquad \int_R^\infty r^n e^{-r/a}dr = a^{n+1}\Gamma(n+1,q),$$

so $I_0=a^5\gamma(5,q)/R+a^4\Gamma(4,q)$ and $I_2=a^7\gamma(7,q)/R^3+R^2a^2\Gamma(2,q)$. Dividing by $24a^5$, and the quadrupole term by a further 5, gives

$$\boxed{V(R,\Theta)=-\frac{e}{4\pi\varepsilon_0}\left\{\frac{1}{24}\left[\frac{\gamma(5,q)}{R}+\frac{\Gamma(4,q)}{a}\right] - \frac{P_2(\cos\Theta)}{120}\left[\frac{a^2\gamma(7,q)}{R^3}+\frac{R^2\Gamma(2,q)}{a^3}\right]\right\}.}$$

The Instructor's Manual's potential (printed p. 265) lacks the overall factor $-e$ required by the given electron charge density; the boxed result includes the charge with its negative sign. For integer n, integrating by parts with $u=t^{n-1}$ and $dv=e^{-t}dt$ (the boundary term is $q^{n-1}e^{-q}$ from $t=q$ and zero at $t=\infty$) gives $\Gamma(n,q)=q^{n-1}e^{-q}+(n-1)\Gamma(n-1,q)$. Repeating this down to $\Gamma(1,q)=e^{-q}$ gives the elementary form $\Gamma(n,q)=(n-1)!\,e^{-q}\sum_{k=0}^{n-1}q^k/k!$, and $\gamma(n,q)+\Gamma(n,q)=\Gamma(n)=(n-1)!$ gives $\gamma(n,q)=(n-1)!-\Gamma(n,q)$. As a check, for $R\gg a$ the Γ terms decay exponentially while $\gamma(5,q)\to 4!$ and $\gamma(7,q)\to 6!$, so

$$V\to -\frac{e}{4\pi\varepsilon_0}\left[\frac{1}{R}-\frac{6a^2}{R^3}P_2(\cos\Theta)\right],$$

the potential of the total charge $-e$ plus a quadrupole term. A test charge q_{test} has energy $q_{\text{test}}V(R,\Theta)$.

For the hydrogen nucleus, $q_{\text{test}}=+e$ and $R=0$. Since $e^{-r/a}=1+O(r)$ near the origin, the interior contributions are

$$\frac{1}{R}\int_0^R r^4e^{-r/a}dr=\frac{R^4}{5}+O(R^5), \qquad \frac{1}{R^3}\int_0^R r^6e^{-r/a}dr=\frac{R^4}{7}+O(R^5).$$

Both vanish; the exterior quadrupole contribution is $R^2\int_R^\infty re^{-r/a}dr=O(R^2)$ and also vanishes. The surviving contribution in the braces is $\Gamma(4,0)/(24a)=3!/(24a)=1/(4a)$. Therefore

$$\boxed{U_{\text{electron–nucleus}}=eV(0)=-\frac{e^2}{4\pi\varepsilon_0}\frac{1}{4a_0}.}$$

Equivalently one may integrate the nuclear potential directly:

$$U=\int\rho\,\frac{e}{4\pi\varepsilon_0 r}\,d^3r=-\frac{e^2}{4\pi\varepsilon_0}\frac{1}{24a^5}\int_0^\infty r^3e^{-r/a}dr=-\frac{e^2}{4\pi\varepsilon_0}\frac{1}{4a},$$

since the angular integral of $1-P_2$ is again 4π, the factor $r^2/r=r$ of d^3r/r turns $r^2e^{-r/a}$ into $r^3e^{-r/a}$, and $\int_0^\infty r^3e^{-r/a}dr=3!\,a^4=6a^4$. In the Gaussian units used in the question's $a_0=\hbar^2/(me^2)$, set $1/(4\pi\varepsilon_0)=1$: the energy is $-e^2/(4a_0)$. This is the electrostatic interaction energy; the kinetic energy is not part of the stated electrostatic calculation.

16.3.9 Interpret the delta and both sums as distributions with respect to $d\Omega = \sin\theta\, d\theta\, d\varphi$.

(a) As a Laplace series and as an example of Eq. (5.27), show that

$$\delta(\Omega_1 - \Omega_2) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} Y_l^m(\theta_2, \varphi_2)^* Y_l^m(\theta_1, \varphi_1).$$

(b) Show also that this same representation of the Dirac delta function may be written as

$$\delta(\Omega_1 - \Omega_2) = \sum_{l=0}^{\infty} \frac{2l+1}{4\pi} P_l(\cos\gamma),$$

and identify γ. Now, if you can justify equating the summations over l term by term, you have an alternate derivation of the spherical harmonic addition theorem.

Solution.

These infinite sums represent a distribution on the sphere, with measure $d\Omega = \sin\theta\, d\theta\, d\varphi$; they are not ordinary pointwise convergent formulas for a function.

(a) For a smooth test function f, let $f_{lm} = \int Y_l^{m*}(\Omega_2) f(\Omega_2)\, d\Omega_2$. Completeness gives the Laplace series $f(\Omega_1) = \sum_{lm} f_{lm} Y_l^m(\Omega_1)$. Therefore, integrating term by term (the l, m term gives $Y_l^m(\Omega_1) f_{lm}$),

$$\int \left[\sum_{lm} Y_l^{m*}(\Omega_2) Y_l^m(\Omega_1) \right] f(\Omega_2)\, d\Omega_2 = \sum_{lm} f_{lm} Y_l^m(\Omega_1) = f(\Omega_1).$$

This is precisely the defining action of $\delta(\Omega_1 - \Omega_2)$, proving (a) in the distributional sense. It is the closure relation (5.27), $\delta(x-t) = \sum_j \psi_j(x)\psi_j^*(t)$, for the orthonormal set $\psi_j = Y_l^m$ on the sphere (j running over the pairs l, m), with $x = \Omega_1$ and $t = \Omega_2$. The Instructor's Manual (printed p. 266) says that reconstructing the arbitrary test function yields $Y_l^m(\Omega_1)$. Its preceding sum instead yields $f(\Omega_1)$, as the delta's defining action requires. A single harmonic is recovered only when f is that harmonic.

(b) Define the finite-dimensional projection kernel

$$K_l(\Omega_1, \Omega_2) = \sum_{m=-l}^{l} Y_l^m(\Omega_1) Y_l^{m*}(\Omega_2).$$

By Eq. (16.52), a rotation of the axes transforms the harmonics of degree l among themselves, $Y_l^m(\Omega') = \sum_\mu D^l_{\mu m} Y_l^\mu(\Omega)$, with a unitary matrix D^l (rotations preserve $d\Omega$ and hence the harmonic inner product). Applying the same rotation to both directions,

$$K_l(\Omega_1', \Omega_2') = \sum_{\mu,\nu} \Big(\sum_m D^l_{\mu m} D^{l*}_{\nu m} \Big) Y_l^\mu(\Omega_1) Y_l^{\nu*}(\Omega_2) = K_l(\Omega_1, \Omega_2),$$

because $\sum_m D^l_{\mu m} D^{l*}_{\nu m} = (D^l D^{l\dagger})_{\mu\nu} = \delta_{\mu\nu}$. Thus K_l is unchanged by a common rotation. Choose the rotation so that Ω_2 lies on the new polar axis. The other direction then has polar angle γ, their angular separation. At the pole the factor $(1-x^2)^{|m|/2}$ of P_l^m vanishes for $m \neq 0$, and $P_l(1) = 1$, so

$$Y_l^m(0, \varphi) = 0 \quad (m \neq 0), \qquad Y_l^0(0, \varphi) = \sqrt{\frac{2l+1}{4\pi}}.$$

It follows directly, without assuming the addition theorem, that

$$K_l(\Omega_1,\Omega_2) = Y_l^0(\gamma,0)\sqrt{\frac{2l+1}{4\pi}} = \boxed{\frac{2l+1}{4\pi}P_l(\cos\gamma)},$$

where, from the scalar product of the unit vectors $(\sin\theta_i\cos\varphi_i, \sin\theta_i\sin\varphi_i, \cos\theta_i)$,

$$\cos\gamma = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2\cos(\varphi_1-\varphi_2).$$

Summing these kernels over l and using (a) gives the stated Legendre representation of the delta distribution.

The Legendre representation also follows without the boxed identity, which is what makes the term-by-term identification below an alternate derivation. The proof of (a) used only the orthonormality and completeness of the harmonics, so (a) holds equally with the harmonics of rotated axes; and the delta distribution does not depend on the axes, because $d\Omega$ is rotation invariant. With Ω_2 on the new polar axis, the pole values above leave only the $m=0$ terms:

$$\delta(\Omega_1-\Omega_2) = \sum_{l,m} Y_l^m(\Omega_2')^*\, Y_l^m(\Omega_1') = \sum_{l=0}^{\infty} Y_l^0(0,\varphi)^*\, Y_l^0(\gamma,0) = \sum_{l=0}^{\infty}\frac{2l+1}{4\pi}P_l(\cos\gamma).$$

The boxed identity is the addition theorem. The term-by-term identification of the two delta representations is justified by orthogonal projection, rather than by subtracting two infinite sums. As functions of Ω_1, both K_k and $P_k(\cos\gamma)$ are combinations of the degree-k harmonics $Y_k^q(\Omega_1)$: in axes with Ω_2 on the polar axis, $P_k(\cos\gamma) = \sqrt{4\pi/(2k+1)}\,Y_k^0(\Omega_1')$, and Eq. (16.52) re-expresses this within degree k in the original axes. Apply the degree-l projector in Ω_1, $f(\Omega_1)\mapsto \sum_m Y_l^m(\Omega_1)\int Y_l^{m*}(\Omega)f(\Omega)\,d\Omega$. It annihilates every degree-k function with $k\neq l$, since $\int Y_l^{m*}Y_k^q\,d\Omega = \delta_{lk}\delta_{mq}$, and leaves degree-$l$ functions unchanged. Applied to the equality $\sum_k K_k = \sum_k \frac{2k+1}{4\pi}P_k(\cos\gamma)$ of the two independently obtained representations, it gives $K_l = \frac{2l+1}{4\pi}P_l(\cos\gamma)$ for each l separately: each degree block is uniquely determined. This is the alternate derivation of the addition theorem referred to in the problem.

16.3.10 Verify the following, with $L\geq 0$ and $|M|\leq L$; in (d) also require $L\geq 1$ and $|M+1|\leq L-1$:

(a) $$\int Y_L^M(\theta,\varphi)Y_0^0(\theta,\varphi)Y_L^{M*}(\theta,\varphi)\,d\Omega = \frac{1}{\sqrt{4\pi}},$$

(b) $$\int Y_L^M Y_1^0 Y_{L+1}^{M*}\,d\Omega = \sqrt{\frac{3}{4\pi}\frac{(L+M+1)(L-M+1)}{(2L+1)(2L+3)}},$$

(c) $$\int Y_L^M Y_1^1 Y_{L+1}^{M+1*}\,d\Omega = \sqrt{\frac{3}{8\pi}\frac{(L+M+1)(L+M+2)}{(2L+1)(2L+3)}},$$

(d) $$\int Y_L^M Y_1^1 Y_{L-1}^{M+1*}\,d\Omega = -\sqrt{\frac{3}{8\pi}\frac{(L-M)(L-M-1)}{(2L-1)(2L+1)}}.$$

These integrals were used in an investigation of the angular correlation of internal conversion electrons.

Solution.

Let

$$N_{LM} = \sqrt{\frac{2L+1}{4\pi}\frac{(L-M)!}{(L+M)!}}, \qquad Y_L^M = N_{LM}P_L^M(\cos\theta)e^{iM\varphi}.$$

All integrals use normalized harmonics and $d\Omega$. Each harmonic appearing in an integral must have nonnegative degree and $|m| \le l$. In particular, (d) is an integral of the displayed harmonics only when $L \ge 1$ and $|M+1| \le L-1$; absent terms in recurrence formulas are omitted.

(a) Since $Y_0^0 = 1/\sqrt{4\pi}$,

$$\int Y_L^M Y_0^0 Y_L^{M*} d\Omega = \frac{1}{\sqrt{4\pi}} \int |Y_L^M|^2 d\Omega = \boxed{\frac{1}{\sqrt{4\pi}}}.$$

(b) Put $x = \cos\theta$. Equation (15.88) gives

$$x P_L^M = \frac{L-M+1}{2L+1} P_{L+1}^M + \frac{L+M}{2L+1} P_{L-1}^M.$$

Multiply by $N_{LM} e^{iM\varphi}$ and write $P_{L\pm1}^M e^{iM\varphi} = Y_{L\pm1}^M / N_{L\pm1,M}$. For the upper term, $(L-M)!/(L-M+1)! = 1/(L-M+1)$ and $(L+M+1)!/(L+M)! = L+M+1$ give the normalization ratio

$$\frac{N_{LM}}{N_{L+1,M}} = \sqrt{\frac{2L+1}{2L+3}\frac{L+M+1}{L-M+1}}.$$

Multiplying by $(L-M+1)/(2L+1)$ therefore gives

$$A_+ = \sqrt{\frac{(L+M+1)(L-M+1)}{(2L+1)(2L+3)}} = \sqrt{\frac{(L+1)^2 - M^2}{(2L+1)(2L+3)}}.$$

For the lower term with $|M| \le L-1$, $(L-M)!/(L-M-1)! = L-M$ and $(L+M-1)!/(L+M)! = 1/(L+M)$ give

$$\frac{N_{LM}}{N_{L-1,M}} = \sqrt{\frac{2L+1}{2L-1}\frac{L-M}{L+M}}, \qquad A_- = \frac{L+M}{2L+1}\frac{N_{LM}}{N_{L-1,M}} = \sqrt{\frac{L^2-M^2}{(2L-1)(2L+1)}}.$$

At $M = \pm L$ the lower term is absent ($P_{L-1}^L \equiv 0$, and $L+M=0$ for $M=-L$), consistent with $A_- = 0$ there. Thus $\cos\theta\, Y_L^M = A_+ Y_{L+1}^M + A_- Y_{L-1}^M$. Multiply by Y_{L+1}^{M*}, integrate, and use $Y_1^0 = \sqrt{3/(4\pi)}\cos\theta$; orthonormality removes the Y_{L-1}^M term and leaves $\sqrt{3/(4\pi)}\,A_+$:

$$\boxed{\int Y_L^M Y_1^0 Y_{L+1}^{M*} d\Omega = \sqrt{\frac{3}{4\pi}\frac{(L+M+1)(L-M+1)}{(2L+1)(2L+3)}}}.$$

(c),(d) Equation (15.89) is

$$\sqrt{1-x^2}\, P_L^M = \frac{P_{L-1}^{M+1} - P_{L+1}^{M+1}}{2L+1}.$$

Multiply by $N_{LM} e^{i(M+1)\varphi}$, so that the left side becomes $\sin\theta\, e^{i\varphi} Y_L^M$, and write $P_{L\pm1}^{M+1} e^{i(M+1)\varphi} = Y_{L\pm1}^{M+1} / N_{L\pm1,M+1}$. The required ratios are

$$\frac{N_{LM}}{(2L+1)N_{L+1,M+1}} = \sqrt{\frac{(L+M+1)(L+M+2)}{(2L+1)(2L+3)}} = B_+,$$

$$\frac{N_{LM}}{(2L+1)N_{L-1,M+1}} = \sqrt{\frac{(L-M)(L-M-1)}{(2L-1)(2L+1)}} = B_-.$$

Indeed, with

$$N^2_{L+1,M+1} = \frac{2L+3}{4\pi}\frac{(L-M)!}{(L+M+2)!}, \qquad N^2_{L-1,M+1} = \frac{2L-1}{4\pi}\frac{(L-M-2)!}{(L+M)!},$$

the squares of the two ratios are

$$B_+^2 = \frac{(L+M+2)!/(L+M)!}{(2L+1)(2L+3)}, \qquad B_-^2 = \frac{(L-M)!/(L-M-2)!}{(2L-1)(2L+1)},$$

and $(L+M+2)!/(L+M)! = (L+M+2)(L+M+1)$, $(L-M)!/(L-M-2)! = (L-M)(L-M-1)$. For $M \geq L-1$ the lower term is absent ($P_{L-1}^{M+1} \equiv 0$), consistent with $B_- = 0$ there. Hence

$$\sin\theta\, e^{i\varphi} Y_L^M = -B_+ Y_{L+1}^{M+1} + B_- Y_{L-1}^{M+1}.$$

Because $Y_1^1 = -\sqrt{3/(8\pi)}\sin\theta\, e^{i\varphi}$, the integrands of (c) and (d) are $-\sqrt{3/(8\pi)}\,(\sin\theta\, e^{i\varphi} Y_L^M)\, Y_{L\pm1}^{M+1*}$, and orthonormality selects $-\sqrt{3/(8\pi)}\,(-B_+)$ and $-\sqrt{3/(8\pi)}\,B_-$:

$$\boxed{\begin{aligned} \int Y_L^M Y_1^1 Y_{L+1}^{M+1*}\, d\Omega &= \sqrt{\frac{3}{8\pi}\frac{(L+M+1)(L+M+2)}{(2L+1)(2L+3)}}, \\ \int Y_L^M Y_1^1 Y_{L-1}^{M+1*}\, d\Omega &= -\sqrt{\frac{3}{8\pi}\frac{(L-M)(L-M-1)}{(2L-1)(2L+1)}}. \end{aligned}}$$

16.3.11 For nonnegative integers L, N, show that

$$\text{(a)}\ \int_{-1}^{1} xP_L(x)P_N(x)\,dx = \begin{cases} \dfrac{2(L+1)}{(2L+1)(2L+3)}, & N = L+1, \\ \dfrac{2L}{(2L-1)(2L+1)}, & N = L-1, \\ 0, & \text{all other } N \geq 0. \end{cases}$$

$$\text{(b)}\ \int_{-1}^{1} x^2P_L(x)P_N(x)\,dx = \begin{cases} \dfrac{2(L+1)(L+2)}{(2L+1)(2L+3)(2L+5)}, & N = L+2, \\ \dfrac{2(2L^2+2L-1)}{(2L-1)(2L+1)(2L+3)}, & N = L, \\ \dfrac{2L(L-1)}{(2L-3)(2L-1)(2L+1)}, & N = L-2, \\ 0, & \text{all other } N \geq 0. \end{cases}$$

Solution.

Take L, N to be nonnegative integers. Write the three-term recurrence as

$$xP_L = a_L P_{L+1} + b_L P_{L-1}, \qquad a_L = \frac{L+1}{2L+1}, \quad b_L = \frac{L}{2L+1}.$$

At $L = 0$ this is $xP_0 = P_1$; the negative-degree term is omitted. Orthogonality is $\int_{-1}^{1} P_j P_N\,dx = 2\delta_{jN}/(2N+1)$.

(a) Multiplying the recurrence by P_N and integrating gives

$$\int_{-1}^{1} xP_L P_N\,dx = a_L\frac{2\delta_{N,L+1}}{2L+3} + b_L\frac{2\delta_{N,L-1}}{2L-1}.$$

For $N = L+1$ and $N = L-1$ this gives respectively $2(L+1)/[(2L+1)(2L+3)]$ and $2L/[(2L-1)(2L+1)]$; for every other N both Kronecker deltas vanish and the integral is 0.

(b) Apply the recurrence a second time, to P_{L+1} and to P_{L-1}:

$$\begin{aligned} x^2P_L &= a_L(a_{L+1}P_{L+2} + b_{L+1}P_L) + b_L(a_{L-1}P_L + b_{L-1}P_{L-2}) \\ &= A_LP_{L+2} + B_LP_L + C_LP_{L-2}, \end{aligned}$$

where $A_L = a_La_{L+1}$, $B_L = a_Lb_{L+1} + b_La_{L-1}$, $C_L = b_Lb_{L-1}$. With $a_{L+1} = (L+2)/(2L+3)$, $b_{L+1} = (L+1)/(2L+3)$, $a_{L-1} = L/(2L-1)$ and $b_{L-1} = (L-1)/(2L-1)$, the coefficients are

$$A_L = \frac{(L+1)(L+2)}{(2L+1)(2L+3)}, \qquad C_L = \frac{L(L-1)}{(2L+1)(2L-1)},$$

$$\begin{aligned} B_L &= \frac{(L+1)^2}{(2L+1)(2L+3)} + \frac{L^2}{(2L+1)(2L-1)} \\ &= \frac{(L+1)^2(2L-1) + L^2(2L+3)}{(2L+1)(2L-1)(2L+3)} \\ &= \frac{(2L+1)(2L^2+2L-1)}{(2L+1)(2L-1)(2L+3)} = \frac{2L^2+2L-1}{(2L-1)(2L+3)}, \end{aligned}$$

because $(L+1)^2(2L-1) = 2L^3+3L^2-1$ and $L^2(2L+3) = 2L^3+3L^2$ add to $4L^3+6L^2-1 = (2L+1)(2L^2+2L-1)$. For $L = 0, 1$, direct application of $xP_0 = P_1$ removes every negative-degree term; the surviving formulas still give these same A_L, B_L: $x^2P_0 = \frac{2}{3}P_2 + \frac{1}{3}P_0$ and $x^2P_1 = \frac{2}{5}P_3 + \frac{3}{5}P_1$. Integration against P_N multiplies the three coefficients by $2/(2N+1)$, that is, by $2/(2L+5)$, $2/(2L+1)$, and $2/(2L-3)$, respectively, and gives 0 for every other N. These are exactly the three stated answers. Both displayed case lists need a final case 0 for every other nonnegative integer N. A case whose index $N = L-1$ or $N = L-2$ would be negative does not occur.

16.3.12 Since $xP_n(x)$ is a polynomial (of degree $n+1$), it may be represented by the Legendre series

$$xP_n(x) = \sum_{s=0}^{\infty} a_sP_s(x).$$

(a) Show that $a_s = 0$ for $s < n-1$ and $s > n+1$.

(b) For $n \geq 1$, calculate a_{n-1}, a_n, and a_{n+1} and reproduce the recurrence relation, Eq. (15.18). Handle $n = 0$ separately, since a_{-1} is not an expansion coefficient.

Note. This argument may be put in a general form to demonstrate the existence of a three-term recurrence relation for any of our complete sets of orthogonal polynomials:

$$x\varphi_n = a_{n+1}\varphi_{n+1} + a_n\varphi_n + a_{n-1}\varphi_{n-1}.$$

Solution.

(a) Since the Legendre polynomials of degrees $0, \ldots, n+1$ form a basis of the polynomials of degree at most $n+1$, the expansion of xP_n is finite and $a_s = 0$ for $s > n+1$. Multiplying the

expansion by P_s and integrating, with $\int_{-1}^{1} P_t P_s\,dx = 2\delta_{ts}/(2s+1)$, gives each coefficient as an inner product; for $s < n-1$,

$$a_s = \frac{2s+1}{2}\int_{-1}^{1} xP_n(x)P_s(x)\,dx = \frac{2s+1}{2}\int_{-1}^{1} P_n(x)[xP_s(x)]\,dx.$$

The bracket has degree $s+1 < n$. It is a combination of $P_0, \ldots, P_{n-1}$, each orthogonal to P_n; hence $a_s = 0$. Only degrees $n-1, n, n+1$ can occur.

(b) The middle coefficient vanishes because $x[P_n(x)]^2$ is odd:

$$a_n = \frac{2n+1}{2}\int_{-1}^{1} x[P_n(x)]^2 dx = 0.$$

To find the upper coefficient without assuming the desired recurrence, use Rodrigues' formula, $P_n = (2^n n!)^{-1}\,d^n(x^2-1)^n/dx^n$. The highest term in $(x^2-1)^n$ is x^{2n}, whose nth derivative is $(2n)!x^n/n!$. Thus the leading coefficient of P_n is

$$\kappa_n = \frac{(2n)!}{2^n(n!)^2}.$$

Matching the coefficient of x^{n+1} in $xP_n = a_{n+1}P_{n+1} + a_{n-1}P_{n-1}$ (the left side has κ_n, the right side $a_{n+1}\kappa_{n+1}$, since P_{n-1} has lower degree) gives

$$a_{n+1} = \frac{\kappa_n}{\kappa_{n+1}} = \frac{(2n)!}{2^n(n!)^2}\cdot\frac{2^{n+1}[(n+1)!]^2}{(2n+2)!} = \frac{2(n+1)^2}{(2n+2)(2n+1)} = \frac{n+1}{2n+1}.$$

For $n \geq 1$, evaluate at $x = 1$, where every $P_j(1) = 1$:

$$1 = a_{n+1} + a_{n-1}, \qquad a_{n-1} = 1 - \frac{n+1}{2n+1} = \frac{n}{2n+1}.$$

Therefore

$$\boxed{(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1} \quad (n \geq 1)}.$$

For $n = 0$ the separate identity is $xP_0 = P_1$.

For the general form in the Note, the same degree and inner-product argument works for orthogonal polynomials with any positive weight on their interval: multiplication by x is symmetric in that inner product, $\langle x\varphi_n, \varphi_s\rangle = \langle \varphi_n, x\varphi_s\rangle = 0$ for $s+1 < n$, so only the neighboring degrees and the original degree can appear. The middle coefficient need not vanish unless the weight and interval supply the parity symmetry used here.

16.4 Vector Spherical Harmonics

16.4.1 Construct the vector spherical harmonics with total $J = 0, M = 0$ and total $J = 1, M = 0$, for every allowed orbital degree L. The notation is $\mathbf{Y}_{JLM}$: the first index is total angular momentum, and the second is orbital angular momentum.

ANS. $\mathbf{Y}_{010} = -\hat{\mathbf{r}}(4\pi)^{-1/2}$
$\mathbf{Y}_{000} = \mathbf{0}$ (formal zero only)
$\mathbf{Y}_{120} = -\hat{\mathbf{r}}(2\pi)^{-1/2}\cos\theta - \hat{\boldsymbol{\theta}}(8\pi)^{-1/2}\sin\theta$
$\mathbf{Y}_{110} = \hat{\boldsymbol{\varphi}}\, i\,(3/8\pi)^{1/2}\sin\theta$
$\mathbf{Y}_{100} = \hat{\mathbf{r}}(4\pi)^{-1/2}\cos\theta - \hat{\boldsymbol{\theta}}(4\pi)^{-1/2}\sin\theta.$

Solution.

Use the spherical Cartesian basis

$$\mathbf{e}_+ = -\frac{\mathbf{e}_x + i\mathbf{e}_y}{\sqrt{2}}, \qquad \mathbf{e}_0 = \mathbf{e}_z, \qquad \mathbf{e}_- = \frac{\mathbf{e}_x - i\mathbf{e}_y}{\sqrt{2}}.$$

The first index of $\mathbf{Y}_{JLM}$ is the total angular momentum; the second is the orbital degree. Its definition, Eq. (16.88), is

$$\mathbf{Y}_{JLM} = \sum_{m,q} C(L\,1\,J \mid m\,q\,M) Y_L^m \mathbf{e}_q.$$

The triangle condition $|L-1| \le J \le L+1$ excludes $J = L = 0$. The printed $\mathbf{Y}_{000} = 0$ is a useful formal zero, not a normalized vector harmonic. There is also no negative-degree orbital harmonic.

For $J = M = 0$, only $L = 1$ is allowed. The singlet coefficients from 16.2.1, $C(1\,1\,0 \mid m, -m, 0) = 1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}$ for $m = 1, 0, -1$, give

$$\mathbf{Y}_{010} = \frac{1}{\sqrt{3}}(Y_1^1\mathbf{e}_- - Y_1^0\mathbf{e}_0 + Y_1^{-1}\mathbf{e}_+).$$

Insert $Y_1^{\pm1} = \mp\sqrt{3/(8\pi)}\sin\theta\, e^{\pm i\varphi}$ and $Y_1^0 = \sqrt{3/(4\pi)}\cos\theta$. With $\mathbf{e}_\rho = \cos\varphi\,\mathbf{e}_x + \sin\varphi\,\mathbf{e}_y$,

$$e^{i\varphi}(\mathbf{e}_x - i\mathbf{e}_y) + e^{-i\varphi}(\mathbf{e}_x + i\mathbf{e}_y) = 2\cos\varphi\,\mathbf{e}_x + 2\sin\varphi\,\mathbf{e}_y = 2\mathbf{e}_\rho,$$

so the $m = \pm1$ terms combine to $Y_1^1\mathbf{e}_- + Y_1^{-1}\mathbf{e}_+ = -\sqrt{3/(8\pi)}\,(\sin\theta/\sqrt{2})\,2\mathbf{e}_\rho = -\sqrt{3/(4\pi)}\sin\theta\,\mathbf{e}_\rho$. The $m = 0$ term is $-Y_1^0\mathbf{e}_0 = -\sqrt{3/(4\pi)}\cos\theta\,\mathbf{e}_z$. Since $\hat{\mathbf{r}} = \sin\theta\,\mathbf{e}_\rho + \cos\theta\,\mathbf{e}_z$ and $\frac{1}{\sqrt{3}}\sqrt{\frac{3}{4\pi}} = \frac{1}{\sqrt{4\pi}}$,

$$\boxed{\mathbf{Y}_{010} = -\frac{\hat{\mathbf{r}}}{\sqrt{4\pi}}}.$$

For $J = 1, M = 0, L = 0$, coupling a scalar to spin one is immediate, $C(0\,1\,1 \mid 0\,0\,0) = 1$, and $\mathbf{e}_z = \cos\theta\,\hat{\mathbf{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}$:

$$\boxed{\mathbf{Y}_{100} = Y_0^0\mathbf{e}_0 = \frac{\cos\theta\,\hat{\mathbf{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}}{\sqrt{4\pi}}}.$$

For $L = 1$, the triplet coefficients of 16.2.1, $C(1\,1\,1 \mid m, -m, 0) = 1/\sqrt{2}, 0, -1/\sqrt{2}$ for $m = 1, 0, -1$, give

$$\mathbf{Y}_{110} = \frac{1}{\sqrt{2}}(Y_1^1\mathbf{e}_- - Y_1^{-1}\mathbf{e}_+).$$

Insert $Y_1^{\pm1} = \mp\sqrt{3/(8\pi)}\sin\theta\, e^{\pm i\varphi}$: the bracket equals $\sqrt{3/(8\pi)}\,(\sin\theta/\sqrt{2})$ times the Cartesian combination

$$-e^{i\varphi}(\mathbf{e}_x - i\mathbf{e}_y) + e^{-i\varphi}(\mathbf{e}_x + i\mathbf{e}_y) = 2i(-\sin\varphi\,\mathbf{e}_x + \cos\varphi\,\mathbf{e}_y) = 2i\hat{\boldsymbol{\varphi}},$$

so, with $\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\cdot 2 = 1$,

$$\boxed{\mathbf{Y}_{110} = i\sqrt{\frac{3}{8\pi}}\sin\theta\,\hat{\boldsymbol{\varphi}}}.$$

Finally consider $L = 2$. To obtain the $J = 1$ coefficients, first write its highest $M = 1$ state in the product basis $(m, q) = (2,-1),(1,0),(0,1)$ with coefficients (a, b, c). Its raising by $J_+ = L_+ + S_+$ must vanish. The ladder factors $\sqrt{(j-m)(j+m+1)}$ give $L_+Y_2^1 = 2Y_2^2$, $L_+Y_2^0 = \sqrt{6}\,Y_2^1$, $S_+\mathbf{e}_- = \sqrt{2}\,\mathbf{e}_0$ and $S_+\mathbf{e}_0 = \sqrt{2}\,\mathbf{e}_+$, so the coefficients of the two $M = 2$ products $(2,0)$ and $(1,1)$ yield

$$\sqrt{2}a + 2b = 0, \qquad \sqrt{2}b + \sqrt{6}c = 0.$$

Thus $b = -a/\sqrt{2}$, $c = -\sqrt{2}\,b/\sqrt{6} = a/\sqrt{6}$, and $a^2 + b^2 + c^2 = a^2(1 + \frac{1}{2} + \frac{1}{6}) = 5a^2/3 = 1$ fixes, with $a > 0$ (the Condon–Shortley choice of a positive coefficient for the largest orbital m), $(a, b, c) = (\sqrt{3/5}, -\sqrt{3/10}, 1/\sqrt{10})$. Lower once with $J_- = L_- + S_-$: by $L_-Y_2^2 = 2Y_2^1$, $L_-Y_2^1 = \sqrt{6}\,Y_2^0$, $L_-Y_2^0 = \sqrt{6}\,Y_2^{-1}$, $S_-\mathbf{e}_0 = \sqrt{2}\,\mathbf{e}_-$ and $S_-\mathbf{e}_+ = \sqrt{2}\,\mathbf{e}_0$, the products $(1,-1)$, $(0,0)$, $(-1,1)$ receive $2a + \sqrt{2}b$, $\sqrt{6}b + \sqrt{2}c$ and $\sqrt{6}c$. Dividing by the total ladder factor $\sqrt{(J+M)(J-M+1)} = \sqrt{2}$ (for $J = M = 1$) gives

$$\begin{aligned} C(2\,1\,1 \mid 1,-1,0) &= (2a + \sqrt{2}b)/\sqrt{2} = \sqrt{3/10}, \\ C(2\,1\,1 \mid 0,0,0) &= (\sqrt{6}b + \sqrt{2}c)/\sqrt{2} = -\sqrt{2/5}, \\ C(2\,1\,1 \mid -1,1,0) &= \sqrt{6}c/\sqrt{2} = \sqrt{3/10}. \end{aligned}$$

Consequently

$$\mathbf{Y}_{120} = \sqrt{\frac{3}{10}}(Y_2^1\mathbf{e}_- + Y_2^{-1}\mathbf{e}_+) - \sqrt{\frac{2}{5}}Y_2^0\mathbf{e}_z.$$

Use $Y_2^{\pm1} = \mp\sqrt{15/(8\pi)}\sin\theta\cos\theta e^{\pm i\varphi}$ and $Y_2^0 = \sqrt{5/(16\pi)}(3\cos^2\theta - 1)$. The $m = \pm1$ terms contain the same Cartesian combination $2\mathbf{e}_\rho$ as in $\mathbf{Y}_{010}$, so $Y_2^1\mathbf{e}_- + Y_2^{-1}\mathbf{e}_+ = -\sqrt{15/(4\pi)}\sin\theta\cos\theta\,\mathbf{e}_\rho$; with $\sqrt{3/10}\sqrt{15/(4\pi)} = 3/\sqrt{8\pi}$ and $\sqrt{2/5}\sqrt{5/(16\pi)} = 1/\sqrt{8\pi}$,

$$\mathbf{Y}_{120} = -\frac{1}{\sqrt{8\pi}}\left[3\sin\theta\cos\theta\,\mathbf{e}_\rho + (3\cos^2\theta - 1)\mathbf{e}_z\right].$$

The substitutions $\mathbf{e}_\rho = \sin\theta\,\hat{\mathbf{r}} + \cos\theta\,\hat{\boldsymbol{\theta}}$ and $\mathbf{e}_z = \cos\theta\,\hat{\mathbf{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}$ give radial coefficient $3\sin^2\theta\cos\theta + (3\cos^2\theta - 1)\cos\theta = 3\cos\theta - \cos\theta = 2\cos\theta$ and polar coefficient $3\sin\theta\cos^2\theta - (3\cos^2\theta - 1)\sin\theta = \sin\theta$. Therefore, with $2/\sqrt{8\pi} = 1/\sqrt{2\pi}$,

$$\boxed{\mathbf{Y}_{120} = -\frac{\cos\theta}{\sqrt{2\pi}}\hat{\mathbf{r}} - \frac{\sin\theta}{\sqrt{8\pi}}\hat{\boldsymbol{\theta}}}.$$

Together with the formal $\mathbf{Y}_{000} = 0$, these reproduce all five displayed expressions.

16.4.2 Use argument inversion $\hat{\mathbf{r}} \mapsto -\hat{\mathbf{r}}$ with a fixed Cartesian basis, and include only allowed index triples. Verify that the parity of $\mathbf{Y}_{L\,L+1\,M}$ is $(-1)^{L+1}$, that of $\mathbf{Y}_{L\,L\,M}$ is $(-1)^{L}$, and that of $\mathbf{Y}_{L\,L-1\,M}$ is $(-1)^{L+1}$. What happened to the M-dependence of the parity?

Hint. $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\varphi}}$ have odd parity; $\hat{\boldsymbol{\theta}}$ has even parity (compare with Exercise 3.10.25).

Solution.

Under inversion of the direction, $(\theta, \varphi) \mapsto (\pi - \theta, \varphi + \pi)$: indeed $\sin(\pi - \theta) = \sin\theta$, $\cos(\pi - \theta) = -\cos\theta$ and $\cos(\varphi + \pi) = -\cos\varphi$, $\sin(\varphi + \pi) = -\sin\varphi$ reverse all three Cartesian components of $\hat{\mathbf{r}}$. For integer m, the associated Legendre and azimuthal factors give

$$P_l^m(-\cos\theta) = (-1)^{l+m} P_l^m(\cos\theta), \qquad e^{im(\varphi+\pi)} = (-1)^m e^{im\varphi}.$$

The first follows from the Rodrigues form $P_l^m \propto (1-x^2)^{m/2}\, d^{l+m}(x^2-1)^l/dx^{l+m}$: the factors $(1-x^2)^{m/2}$ and $(x^2-1)^l$ are even, and each of the $l+m$ derivatives reverses the parity. The product of the two factors is $(-1)^{l+2m} = (-1)^l$ times the original harmonic. Thus $Y_l^m(-\hat{\mathbf{r}}) = (-1)^l Y_l^m(\hat{\mathbf{r}})$, independently of m.

In the fixed Cartesian spherical basis, $\mathbf{e}_q$ is independent of the field direction. Every term in

$$\mathbf{Y}_{JLM}(\hat{\mathbf{r}}) = \sum_{m,q} C(L\,1\,J \mid m\,q\,M) Y_L^m(\hat{\mathbf{r}}) \mathbf{e}_q$$

has the same orbital degree L. Therefore

$$\mathbf{Y}_{JLM}(-\hat{\mathbf{r}}) = (-1)^L \mathbf{Y}_{JLM}(\hat{\mathbf{r}}).$$

Applying this to the three orbital degrees $L+1, L, L-1$ gives

$$\boxed{\begin{aligned} \mathbf{Y}_{L,L+1,M}(-\hat{\mathbf{r}}) &= (-1)^{L+1} \mathbf{Y}_{L,L+1,M}, \\ \mathbf{Y}_{L,L,M}(-\hat{\mathbf{r}}) &= (-1)^{L} \mathbf{Y}_{L,L,M}, \\ \mathbf{Y}_{L,L-1,M}(-\hat{\mathbf{r}}) &= (-1)^{L+1} \mathbf{Y}_{L,L-1,M}. \end{aligned}}$$

The last two cases are present only when $L \geq 1$: for $L = 0$ the triangle condition excludes $\mathbf{Y}_{000}$ (see 16.4.1), and there is no orbital degree -1. The M dependence disappears because the two scalar parity factors $(-1)^{l+m}$ and $(-1)^m$ multiply to $(-1)^{l+2m} = (-1)^l$, with $2m$ even, while the Clebsch–Gordan coefficients and the fixed $\mathbf{e}_q$ are unaffected by the inversion.

This is the argument-parity convention in the question: the Cartesian component functions are evaluated at $-\hat{\mathbf{r}}$. If one instead defines parity on a physical polar-vector field by $(\mathcal{P}\mathbf{F})(\mathbf{r}) = -\mathbf{F}(-\mathbf{r})$, that operation includes an additional intrinsic minus sign. In the moving spherical basis the same calculation distributes the argument parity among the components and the basis vectors: $\hat{\mathbf{r}}, \hat{\boldsymbol{\varphi}}$ are odd and $\hat{\boldsymbol{\theta}}$ is even, as stated in the hint.

16.4.3 Verify the orthonormality of the vector spherical harmonics $\mathbf{Y}_{JLM_J}$ under the Hermitian inner product $\int \mathbf{Y}^* \cdot \mathbf{Y}'\, d\Omega$, for allowed indices.

Solution.

The textbook's Eq. (16.89) omits the complex conjugate on one vector harmonic. Orthonormality uses the Hermitian product $\int \mathbf{Y}^* \cdot \mathbf{Y}'\, d\Omega$. An unconjugated product cannot express it: already $\mathbf{e}_+ \cdot \mathbf{e}_+ = 0$, whereas $\mathbf{e}_+^* \cdot \mathbf{e}_+ = 1$. The spherical Cartesian basis obeys $\mathbf{e}_q^* \cdot \mathbf{e}_{q'} = \delta_{qq'}$; for example $\mathbf{e}_+^* \cdot \mathbf{e}_+ = \tfrac{1}{2}(\mathbf{e}_x - i\mathbf{e}_y)\cdot(\mathbf{e}_x + i\mathbf{e}_y) = \tfrac{1}{2}(1+1) = 1$ and $\mathbf{e}_+^* \cdot \mathbf{e}_- = -\tfrac{1}{2}(\mathbf{e}_x - i\mathbf{e}_y)\cdot(\mathbf{e}_x - i\mathbf{e}_y) = -\tfrac{1}{2}(1-1) = 0$.

Write $C^{JM}_{mq} = C(L\,1\,J \mid m\,q\,M)$. Expanding both vector harmonics before integrating, the factors $\mathbf{e}^*_q \cdot \mathbf{e}_{q'} = \delta_{qq'}$ and $\int Y_L^{m*} Y_{L'}^{m'}\, d\Omega = \delta_{LL'}\delta_{mm'}$ collapse the primed sums:

$$\begin{aligned}
&\int \mathbf{Y}^*_{JLM} \cdot \mathbf{Y}_{J'L'M'}\, d\Omega \\
&\quad = \sum_{m,q} \sum_{m',q'} (C^{JM}_{mq})^* C(L'\,1\,J' \mid m'\,q'\,M')(\mathbf{e}^*_q \cdot \mathbf{e}_{q'}) \int Y_L^{m*} Y_{L'}^{m'}\, d\Omega \\
&\quad = \delta_{LL'} \sum_{m,q} C(L\,1\,J \mid m\,q\,M)^* C(L\,1\,J' \mid m\,q\,M').
\end{aligned}$$

The remaining sum is the inner product of two normalized coupled states $|JM\rangle$, $|J'M'\rangle$ of the same L and spin 1, expressed in the orthonormal product basis $|m\rangle|q\rangle$. The Clebsch–Gordan change of basis is unitary (the coefficients are real, so the conjugation is immaterial), so the sum is $\delta_{JJ'}\delta_{MM'}$; this is the orthogonality relation of the Clebsch–Gordan coefficients. Therefore, for all allowed indices,

$$\boxed{\int \mathbf{Y}^*_{JLM} \cdot \mathbf{Y}_{J'L'M'}\, d\Omega = \delta_{JJ'}\delta_{LL'}\delta_{MM'}}.$$

This does not assign norm one to the excluded formal zero $\mathbf{Y}_{000}$.

16.4.4 Jackson's *Classical Electrodynamics* (see Additional Readings) defines $\mathbf{Y}_{LLM}$ by the equation

$$\mathbf{Y}_{LLM}(\theta,\varphi) = \frac{1}{\sqrt{L(L+1)}}\,\mathbf{L}\,Y_L^M(\theta,\varphi),$$

in which the angular momentum operator $\mathbf{L}$ is given by

$$\mathbf{L} = -i(\mathbf{r} \times \nabla).$$

Show that this definition agrees with Eq. (16.88) for $L \geq 1$ and $|M| \leq L$.

Solution.

The displayed definition requires $L \geq 1$. At $L = 0$ its denominator vanishes, and the formal $\mathbf{Y}_{000} = 0$ is not a normalized member of this family. In the spherical Cartesian basis of 16.4.1, express the vector operator in terms of $L_\pm = L_x \pm iL_y$. Since $L_x = \frac{1}{2}(L_+ + L_-)$ and $L_y = \frac{1}{2i}(L_+ - L_-)$, $\mathbf{e}_x L_x + \mathbf{e}_y L_y = \frac{1}{2}(\mathbf{e}_x - i\mathbf{e}_y)L_+ + \frac{1}{2}(\mathbf{e}_x + i\mathbf{e}_y)L_-$, and $\frac{1}{2}(\mathbf{e}_x \mp i\mathbf{e}_y)$ equals $\mathbf{e}_-/\sqrt{2}$ and $-\mathbf{e}_+/\sqrt{2}$, respectively. Hence

$$\mathbf{L} = \mathbf{e}_x L_x + \mathbf{e}_y L_y + \mathbf{e}_0 L_z = \frac{\mathbf{e}_- L_+ - \mathbf{e}_+ L_-}{\sqrt{2}} + \mathbf{e}_0 L_z.$$

Applying the scalar ladder formulas of 16.1.1, $L_\pm Y_L^M = \sqrt{(L \mp M)(L \pm M + 1)}\, Y_L^{M\pm1}$, and $L_z Y_L^M = MY_L^M$ gives

$$\begin{aligned}
\frac{\mathbf{L}Y_L^M}{\sqrt{L(L+1)}} &= \sqrt{\frac{(L-M)(L+M+1)}{2L(L+1)}}\, Y_L^{M+1}\mathbf{e}_- \\
&\quad + \frac{M}{\sqrt{L(L+1)}} Y_L^M \mathbf{e}_0 \\
&\quad - \sqrt{\frac{(L+M)(L-M+1)}{2L(L+1)}}\, Y_L^{M-1}\mathbf{e}_+.
\end{aligned}$$

These are the three Clebsch–Gordan coefficients for coupling orbital L and vector spin 1 to total $J = L$.

One may establish their phase and total J directly. At $M = L$ the coefficient of Y_L^{L+1} contains $L - M = 0$, and the two nonzero coefficients are $-\sqrt{2L/[2L(L+1)]} = -1/\sqrt{L+1}$ on $(m, q) = (L-1, 1)$ and $L/\sqrt{L(L+1)} = \sqrt{L/(L+1)}$ on $(L, 0)$. Their norm squared is $1/(L+1)+L/(L+1) = 1$. Apply $J_+ = L_+ + S_+$: $L_+ Y_L^{L-1} = \sqrt{2L}\, Y_L^L$, $S_+ \mathbf{e}_0 = \sqrt{2}\, \mathbf{e}_+$, while $L_+ Y_L^L = 0$ and $S_+ \mathbf{e}_+ = 0$. Raising therefore gives the coefficient

$$-\frac{\sqrt{2L}}{\sqrt{L+1}} + \sqrt{\frac{L}{L+1}}\sqrt{2} = 0$$

on $(L, 1)$, the only product reached. A state with $J_z = L$ that J_+ annihilates has $\mathbf{J}^2 = J_- J_+ + J_z^2 + J_z = L(L+1)$; thus this is the highest state of total $J = L$, with positive coefficient at the largest orbital m, as the Condon–Shortley convention requires.

The subsequent lowering uses the vector transformation law. On any scalar f, the total lowering generator acting on $\mathbf{L}f$ satisfies

$$J_-(\mathbf{L}f) = \mathbf{L}(L_- f).$$

To exhibit the cancellation, write $J_i = L_i + S_i$, where L_i acts on each Cartesian component and the Cartesian spin-one matrices $(S_i)_{jk} = -i\epsilon_{ijk}$ act on the vector index, whereas $[L_i, L_j] = i\sum_k \epsilon_{ijk} L_k$. Therefore, for each Cartesian i, the jth component of the difference is

$$\begin{aligned}[J_i(\mathbf{L}f) - \mathbf{L}(L_i f)]_j &= L_i L_j f + \sum_k (S_i)_{jk} L_k f - L_j L_i f \\ &= [L_i, L_j] f - i \sum_k \epsilon_{ijk} L_k f = 0.\end{aligned}$$

Taking the $x - iy$ combination proves the stated lowering identity. These S_i are the spin-one operators used above: $(S_i \mathbf{v})_j = -i\sum_k \epsilon_{ijk} v_k$ says $S_i \mathbf{v} = i\, \mathbf{e}_i \times \mathbf{v}$, so, for example, $S_z \mathbf{e}_\pm = i\, \mathbf{e}_z \times \mathbf{e}_\pm = \pm \mathbf{e}_\pm$ and $S_+ \mathbf{e}_0 = i(\mathbf{e}_x + i\mathbf{e}_y) \times \mathbf{e}_z = -\mathbf{e}_x - i\mathbf{e}_y = \sqrt{2}\, \mathbf{e}_+$. The same cross products give $S_z \mathbf{e}_q = q\, \mathbf{e}_q$ and $S_\pm \mathbf{e}_q = \sqrt{(1 \mp q)(2 \pm q)}\, \mathbf{e}_{q\pm 1}$ for every q, the standard spin-one ladder actions; hence this J_- is the lowering operator that generates the coefficients of Eq. (16.88). With $f = Y_L^M$ and $L_- Y_L^M = \sqrt{(L+M)(L-M+1)}\, Y_L^{M-1}$, it follows that lowering the displayed M expression gives $\sqrt{(L+M)(L-M+1)}$ times the expression at $M - 1$. The normalized highest state and these positive lowering factors uniquely generate the $J = L$ multiplet. Thus its coefficients are exactly those of Eq. (16.88), and

$$\boxed{\mathbf{Y}_{LLM} = \frac{\mathbf{L} Y_L^M}{\sqrt{L(L+1)}}}.$$

16.4.5 For $L \geq 1$, show that

$$\sum_{M=-L}^{L} \mathbf{Y}^*_{LLM}(\theta, \varphi) \cdot \mathbf{Y}_{LLM}(\theta, \varphi) = \frac{2L+1}{4\pi}.$$

Hint. One way is to use Exercise 16.4.4 with $\mathbf{L}$ expanded in Cartesian coordinates and to apply raising and lowering operators.

Solution.

This identity holds for $L \geq 1$. The textbook's formal $\mathbf{Y}_{000} = 0$ would give zero on the left for $L = 0$, whereas the stated right side would be $1/(4\pi)$. Use 16.4.4. Since Cartesian basis vectors are

orthonormal,

$$\sum_M \mathbf{Y}^*_{LLM}\cdot\mathbf{Y}_{LLM} = \frac{1}{L(L+1)}\sum_M \left(|L_xY_L^M|^2+|L_yY_L^M|^2+|L_zY_L^M|^2\right)$$
$$= \frac{1}{L(L+1)}\sum_M\left(\frac{|L_+Y_L^M|^2+|L_-Y_L^M|^2}{2}+M^2|Y_L^M|^2\right).$$

The second equality follows by expanding $L_x=(L_++L_-)/2$ and $L_y=(L_+-L_-)/(2i)$: with $u=L_+Y_L^M$ and $v=L_-Y_L^M$,

$$|L_xY_L^M|^2+|L_yY_L^M|^2=\tfrac14|u+v|^2+\tfrac14|u-v|^2=\tfrac12\left(|u|^2+|v|^2\right),$$

since the cross terms $\pm\frac14(uv^*+u^*v)$ cancel. By 16.1.1, $|L_+Y_L^M|^2=(L-M)(L+M+1)|Y_L^{M+1}|^2$ and $|L_-Y_L^M|^2=(L+M)(L-M+1)|Y_L^{M-1}|^2$, while $|L_zY_L^M|^2=M^2|Y_L^M|^2$. Now collect the coefficient of a fixed $|Y_L^k|^2$. The raising term comes from $M=k-1$, with factor $\frac12(L-k+1)(L+k)$, and the lowering term from $M=k+1$, with factor $\frac12(L+k+1)(L-k)$. Including the L_z term ($M=k$), the coefficient is

$$\frac12(L-k+1)(L+k)+\frac12(L+k+1)(L-k)+k^2$$
$$=\frac12[L(L+1)-k^2+k]+\frac12[L(L+1)-k^2-k]+k^2$$
$$=L(L+1).$$

At $k=\pm L$, the out-of-range source term has coefficient zero, so the same collection is valid at both endpoints. Hence

$$\sum_M \mathbf{Y}^*_{LLM}\cdot\mathbf{Y}_{LLM}=\sum_{k=-L}^{L}|Y_L^k|^2=\boxed{\frac{2L+1}{4\pi}},$$

where the last step is the scalar addition theorem at coincident directions ($\gamma=0$, $P_L(1)=1$).

16.4.6 For $L\geq 1$ and $|M|\leq L$, show that

$$\int \mathbf{Y}^*_{LLM}\cdot\left(\hat{\mathbf{r}}\times\mathbf{Y}_{LLM}\right)d\Omega=0.$$

The integrand represents an interference term in electromagnetic radiation that contributes to angular distributions but not to total intensity.

Solution.

The textbook's statement has no complex conjugate in this exercise. Its following statement about a nonzero angular interference term therefore cannot describe the integrand as printed, which vanishes identically (shown first). The physically relevant Hermitian version, with the first $\mathbf{Y}_{LLM}$ conjugated, can be nonzero pointwise; its angular integral also vanishes, as shown next.

The textbook's unconjugated version has a stronger but different property: its integrand is identically zero. For any vector $\mathbf{A}$, including one with complex components, the bilinear scalar triple product is

$$\mathbf{A}\cdot(\hat{\mathbf{r}}\times\mathbf{A})=\sum_{ijk}\epsilon_{ijk}A_i\hat{r}_jA_k=0.$$

Interchanging i and k changes ϵ_{ijk} to its negative but leaves the commuting product A_iA_k unchanged. The sum therefore equals its own negative and is zero at every point. Taking $\mathbf{A}=\mathbf{Y}_{LLM}$ proves the unconjugated integral is zero.

For the Hermitian version put $N=\sqrt{L(L+1)}$, $Y=Y_L^M$, and $\mathbf{A}=\mathbf{Y}_{LLM}=\mathbf{L}Y/N$ (16.4.4). With $\nabla=\hat{\mathbf{r}}\,\partial_r+\hat{\boldsymbol{\theta}}\,r^{-1}\partial_\theta+\hat{\boldsymbol{\varphi}}\,(r\sin\theta)^{-1}\partial_\varphi$, $\hat{\mathbf{r}}\times\hat{\boldsymbol{\theta}}=\hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{r}}\times\hat{\boldsymbol{\varphi}}=-\hat{\boldsymbol{\theta}}$, we have $\mathbf{r}\times\nabla=\hat{\boldsymbol{\varphi}}\,\partial_\theta-\hat{\boldsymbol{\theta}}(\sin\theta)^{-1}\partial_\varphi$. Using $\mathbf{L}=-i\hat{\boldsymbol{\varphi}}\,\partial_\theta+i\hat{\boldsymbol{\theta}}(\sin\theta)^{-1}\partial_\varphi$ and $\partial_\varphi Y=iMY$ gives

$$A_\theta=-\frac{M}{N\sin\theta}Y,\qquad A_\varphi=-\frac{i}{N}\partial_\theta Y,\qquad A_r=0.$$

Hence $\hat{\mathbf{r}}\times\mathbf{A}=A_\theta\hat{\boldsymbol{\varphi}}-A_\varphi\hat{\boldsymbol{\theta}}$. Its dot product with $\mathbf{A}^*$, where $A_\theta^*=-MY^*/(N\sin\theta)$ and $A_\varphi^*=(i/N)\partial_\theta Y^*$, is

$$\begin{aligned}\mathbf{A}^*\cdot(\hat{\mathbf{r}}\times\mathbf{A})&=-A_\theta^*A_\varphi+A_\varphi^*A_\theta\\&=-\frac{iM}{L(L+1)\sin\theta}\left[Y^*\partial_\theta Y+(\partial_\theta Y^*)Y\right]\\&=-\frac{iM}{L(L+1)\sin\theta}\partial_\theta|Y|^2.\end{aligned}$$

The modulus is independent of φ. In $d\Omega=\sin\theta\,d\theta\,d\varphi$ the factor $\sin\theta$ cancels the $1/\sin\theta$, the φ integration gives 2π, and the θ integral of a derivative leaves its end values:

$$\int\mathbf{A}^*\cdot(\hat{\mathbf{r}}\times\mathbf{A})\,d\Omega=-\frac{2\pi iM}{L(L+1)}\left[|Y_L^M(\theta,0)|^2\right]_{\theta=0}^{\theta=\pi}=0.$$

For $M=0$ the prefactor is zero; for $M\neq0$ the spherical harmonic vanishes at both poles, since P_L^M contains the factor $(1-x^2)^{|M|/2}=\sin^{|M|}\theta$. For example $L=M=1$ gives $|Y_1^1|^2=3\sin^2\theta/(8\pi)$ and hence

$$-\frac{i}{2\sin\theta}\frac{d}{d\theta}\left(\frac{3\sin^2\theta}{8\pi}\right)=-\frac{3i}{8\pi}\cos\theta,$$

which is nonzero away from the equator but integrates to zero. Thus the corrected Hermitian interpretation has the angular behavior described in the prose.

17 Chapter 17. Group Theory

Problems

17.1 Introduction to Group Theory

17.1.1 The Vierergruppe (German: four-membered group) is a group different from the C_4 group introduced in Example 17.1.4. The Vierergruppe has the multiplication table shown in Table 17.2. Determine whether this group is cyclic and whether it is abelian.

Solution.

The diagonal of Table 17.2 gives

$$I^2 = A^2 = B^2 = C^2 = I.$$

Thus the powers of a nonidentity element g alternate $I, g, I, g, \ldots$, generating only two elements. The powers of I generate only I. No element generates all four, so the group is $\boxed{\text{not cyclic}}$.

The table is symmetric across its diagonal. More explicitly,

$$AB = BA = C, \qquad BC = CB = A, \qquad CA = AC = B,$$

and products with I commute automatically. These exhaust all pairs of distinct elements; an element commutes with itself. Hence the Vierergruppe is $\boxed{\text{abelian}}$. It is isomorphic to $C_2 \times C_2$, for example under $I \mapsto (0,0)$, $A \mapsto (1,0)$, $B \mapsto (0,1)$, $C \mapsto (1,1)$ with componentwise addition modulo two.

17.1.2

(a) Show that the permutations of n distinct objects satisfy the group postulates.

(b) Construct the multiplication table for the permutations of three objects, giving each permutation a name of some sort. (Suggestion: Use I for the permutation that leaves the order unchanged.)

(c) Show that this permutation group (named S_3) is isomorphic with D_3 and identify corresponding operations. Is your identification unique?

Solution.

(a) A permutation is a bijection of $\{1, \ldots, n\}$ to itself; multiplication means composition, with the right-hand permutation acting first. The composition of two bijections is a bijection, proving closure. For every label j, $[(pq)r](j) = p(q(r(j))) = [p(qr)](j)$, proving associativity. The identity is $I(j) = j$. For each p, every target k has a unique preimage; define $p^{-1}(k)$ to be that preimage. Then $pp^{-1} = p^{-1}p = I$ and p^{-1} is another permutation. All four postulates hold.

(b) For three labels use cycle notation

$$r = (123), \qquad s = (23), \qquad r^2 = (132), \quad sr = (13), \quad sr^2 = (12).$$

Here (123) moves label 1 to 2, 2 to 3, and 3 to 1. For instance sr (r acting first) sends $1 \to 2 \to 3$, $2 \to 3 \to 2$, $3 \to 1 \to 1$, so $sr = (13)$; likewise sr^2 sends $1 \to 3 \to 2$, $2 \to 1 \to 1$, $3 \to 2 \to 3$, so $sr^2 = (12)$. The elementary relations are $r^3 = s^2 = I$ and $srs = r^{-1}$; the last holds because

srs sends $1 \to 1 \to 2 \to 3$, $2 \to 3 \to 1 \to 1$, $3 \to 2 \to 3 \to 2$, which is $r^{-1} = (132)$. Multiplying $srs = r^{-1}$ on the left by s gives $rs = sr^{-1}$, and repeating, $r^i s = sr^{-i}$. Hence

$$(s^\epsilon r^i)(s^\eta r^j) = s^{\epsilon+\eta} r^{(-1)^\eta i + j},$$

with the exponents of s and r reduced modulo two and three. This reduces any product to one of the six listed elements. For example, $r(sr) = (rs)r = sr^{-1}r = s$. The complete table (row times column) is

	I	r	r^2	s	sr	sr^2
I	I	r	r^2	s	sr	sr^2
r	r	r^2	I	sr^2	s	sr
r^2	r^2	I	r	sr	sr^2	s
s	s	sr	sr^2	I	r	r^2
sr	sr	sr^2	s	r^2	I	r
sr^2	sr^2	s	sr	r	r^2	I

.

(c) Number the triangle vertices as in Fig. 17.1. A counterclockwise 120° rotation has vertex action (123); the 180° rotation about the axis through vertex 1 fixes that vertex and interchanges 2 and 3. Hence an isomorphism is

$$\boxed{I \leftrightarrow I, \quad C_3 \leftrightarrow r, \quad C_3^2 \leftrightarrow r^2, \quad C_2 \leftrightarrow s, \quad C_2' \leftrightarrow sr, \quad C_2'' \leftrightarrow sr^2}.$$

The table just constructed is Table 17.1 with these substitutions, so every product is preserved. Equivalently the geometric action on the three vertices is injective, and both groups have six elements. The identification is not unique: C_3 can be assigned either 3-cycle and C_2 any of the three transpositions; the relations then determine the other images. These $2 \cdot 3 = 6$ choices correspond to the possible vertex relabelings.

17.1.3 *Rearrangement theorem*: Given a group of distinct elements $(I, a, b, \ldots, n)$, show that the set of products $(aI, a^2, ab, ac, \ldots, an)$ reproduces all the group elements in a new order.

Solution.

Fix $a \in \mathcal{G}$ and consider $L_a(g) = ag$. Closure ensures its values lie in $\mathcal{G}$. If $L_a(b) = L_a(c)$, then

$$ab = ac \quad \Longrightarrow \quad a^{-1}(ab) = a^{-1}(ac) \quad \Longrightarrow \quad b = c.$$

Thus no two distinct elements yield the same product. Furthermore, for every $h \in \mathcal{G}$, the element $g = a^{-1}h$ lies in $\mathcal{G}$ and satisfies $L_a(g) = a(a^{-1}h) = h$. Therefore L_a is also onto. It is a bijection with inverse $L_{a^{-1}}$. The list $(aI, a^2, ab, \ldots)$ consequently contains each group element exactly once. This proof works for infinite groups as well as for the finite list in the question.

17.1.4 A group $\mathcal{G}$ has a subgroup $\mathcal{H}$ with elements h_i. Let x be a fixed element of the original group $\mathcal{G}$ and not a member of $\mathcal{H}$. The transform

$$xh_i x^{-1}, \qquad i = 1, 2, \ldots$$

generates a conjugate subgroup $x\mathcal{H}x^{-1}$. Show that this conjugate subgroup satisfies each of the four group postulates and therefore is a group.

Solution.

Let $\mathcal{K} = x\mathcal{H}x^{-1}$. For $h, k \in \mathcal{H}$,

$$(xhx^{-1})(xkx^{-1}) = xh(x^{-1}x)kx^{-1} = x(hk)x^{-1} \in \mathcal{K},$$

because $hk \in \mathcal{H}$. This proves closure. Associativity is inherited from multiplication in $\mathcal{G}$; explicitly, for $h, k, \ell \in \mathcal{H}$, both associations of $(xhx^{-1})(xkx^{-1})(x\ell x^{-1})$ reduce to $x(hk\ell)x^{-1}$.

The identity is $xIx^{-1} = I$, which belongs to $\mathcal{K}$. The inverse of xhx^{-1} is $xh^{-1}x^{-1} \in \mathcal{K}$:

$$(xhx^{-1})(xh^{-1}x^{-1}) = xhh^{-1}x^{-1} = I,$$

and the reverse product is also I. Thus $\mathcal{K}$ satisfies all four group postulates.

In fact $h \mapsto xhx^{-1}$ is a group isomorphism from $\mathcal{H}$ to $\mathcal{K}$: the product formula proves the homomorphism property, and conjugation by x^{-1} is its inverse. The condition $x \notin \mathcal{H}$ is unnecessary for this conclusion; it only makes a potentially different conjugate subgroup of interest.

17.1.5

(a) A particular group is abelian. A second group is created by replacing g_i by g_i^{-1} for each element in the original group. Show that the two groups are isomorphic.

Note. This means showing that if $ab = c$, then $a^{-1}b^{-1} = c^{-1}$.

(b) Continuing part (a), show that the second group is also abelian.

Solution.

(a) First recall the inverse-of-product identity:

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = I, \qquad (b^{-1}a^{-1})(ab) = I.$$

Hence $(ab)^{-1} = b^{-1}a^{-1}$. In an abelian group a^{-1} and b^{-1} commute, so

$$f(ab) = (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1} = f(a)f(b), \qquad f(g) = g^{-1}.$$

Also $f(f(g)) = g$, making f bijective. It is therefore an isomorphism.

(b) Every pair in the set of inverses has the form a^{-1}, b^{-1}, and

$$a^{-1}b^{-1} = (ba)^{-1} = (ab)^{-1} = b^{-1}a^{-1},$$

by the inverse-of-product identity of part (a) in the first and last steps and by $ab = ba$ in the middle one. Thus the second group is abelian. In fact it is the same set $\mathcal{G}$, because inversion permutes a group's elements. The noteworthy conclusion is that inversion is an automorphism when the group is abelian; in a general group it reverses the multiplication order instead.

17.1.6 Consider a cubic crystal consisting of identical atoms at $\mathbf{r} = (la, ma, na)$, with l, m, and n taking on all integral values.

(a) Show that each Cartesian axis is a fourfold symmetry axis.

(b) The cubic point group will consist of all operations (rotations, reflections, inversion) that leave the simple cubic crystal invariant and that do not move the atom at $l = m = n = 0$. From a consideration of the permutation of the positive and negative coordinate axes, predict how many elements this cubic group will contain.

Solution.

(a) A 90° counterclockwise rotation about z takes

$$(la, ma, na) \longmapsto (-ma, la, na).$$

The new coordinates again have integer coefficients. Repeating the rotation gives $(-la, -ma, na)$, then $(ma, -la, na)$, then the original point. The same argument applies cyclically to x and y. These axes have exactly fourfold rotational symmetry: the four nearest lattice points in their perpendicular coordinate plane are at angular separations 90°, so a smaller positive rotation cannot permute them.

(b) A point symmetry fixing the origin preserves distances, so it permutes the six nearest lattice points $\{\pm a\mathbf{e}_x, \pm a\mathbf{e}_y, \pm a\mathbf{e}_z\}$. The image of $\mathbf{e}_x$ has six choices. Orthogonality leaves four choices for the image of $\mathbf{e}_y$, and then two for $\mathbf{e}_z$. Thus there are

$$\boxed{6 \cdot 4 \cdot 2 = 48}$$

possible signed permutation matrices. Every one really is a symmetry: it merely permutes the integers (l, m, n) and changes some signs. No other point symmetry is possible because the images of three orthonormal coordinate vectors determine the entire orthogonal transformation. Exactly 24 have determinant $+1$ and are rotations (the determinant is the sign of the permutation times the product of the three signs, and changing one sign switches it); multiplying by the inversion $-I_3$, of determinant -1, pairs these with the 24 operations of determinant -1. The full point group is commonly denoted O_h.

17.1.7 A plane is covered with regular hexagons, as shown in Fig. 17.3.

(a) Determine the rotational symmetry of an axis perpendicular to the plane through the common vertex of three hexagons (A). That is, if the axis has n-fold symmetry, show (with careful explanation) what n is.

(b) Repeat part (a) for an axis perpendicular to the plane through the geometric center of one hexagon (B).

(c) Find all the different kinds of axes within the plane of hexagons about which a 180° rotation is a symmetry element (this corresponds to turning the plane over by rotation about that axis).

Solution.

(a) At a vertex A, three edges leave at angular separations 120°. A rotation preserving the tiling must permute these three rays, so its angle is a multiple of 120°. Rotation through 120° about A permutes the three hexagons that meet at A, and it maps the whole tiling onto itself. Indeed, each neighbor of a tile is the mirror image of that tile in their common edge, so an isometry that maps a tile onto a tile also maps each neighbor of it onto a neighbor of the image tile; continuing from tile to tile across edges reaches every tile. Thus the axis through A is $\boxed{\text{threefold}}$.

(b) At a center B, the six vertices are equally spaced. A 60° rotation permutes the hexagon and its six neighboring tiles, and hence the entire tiling by the same edge-continuation argument. Any symmetry fixing B must permute its six nearest vertices, excluding a smaller positive angle. The axis through B is $\boxed{\text{sixfold}}$.

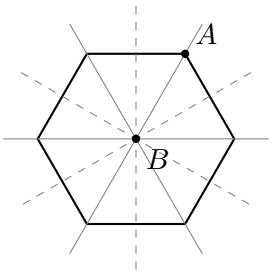
(c) A 180° rotation about a line in the plane acts, on that plane, as reflection in the line. The two kinds are

$$\boxed{\begin{array}{l}\text{lines through opposite vertices of a hexagon,}\\ \text{lines through midpoints of opposite edges of a hexagon.}\end{array}}$$

There are three directions of each kind through a given center. Their lattice translates give all the corresponding axes throughout the plane; an axis of the first kind also contains some tile edges.

To check completeness, take side length one and a hexagon centered at the origin with vertices $(\pm 1, 0)$, $(\pm 1/2, \pm\sqrt{3}/2)$. Its neighbors lie across its edges, at twice the apothem, $\sqrt{3}$, in the directions $30^\circ + 60^\circ k$ of the edge midpoints. Hence the centers form the lattice L of points $p\,\mathbf{g}_1 + q\,\mathbf{g}_2$ (p, q integers), with $\mathbf{g}_1 = (3/2, \sqrt{3}/2)$ and $\mathbf{g}_2 = (0, \sqrt{3})$. A symmetry maps tile centers to tile centers, so a reflection symmetry must permute the six shortest center-to-center vectors, which point at $30^\circ + 60^\circ k$. Reflection in a line at angle θ sends the direction φ to $2\theta - \varphi$, so $2\theta - 30^\circ$ must again be of the form $30^\circ + 60^\circ k$; that is, θ is a multiple of 30°, giving six directions.

For a horizontal axis $y = d$ the reflection is $(x, y) \mapsto (x, 2d - y)$. Its linear part $(x, y) \mapsto (x, -y)$ maps L onto itself, because it sends $\mathbf{g}_1$ to $(3/2, -\sqrt{3}/2) = \mathbf{g}_1 - \mathbf{g}_2$ and $\mathbf{g}_2$ to $-\mathbf{g}_2$. Hence the reflection maps L onto itself exactly when the image $(0, 2d)$ of the origin lies in L; each tile is symmetric under reflection in the horizontal and vertical lines through its center, so the tiling is then preserved as well. A lattice point with $x = 3p/2 = 0$ has $p = 0$, so $2d = q\sqrt{3}$ and $d = q\sqrt{3}/2$. This line contains the center $q\mathbf{g}_1 = (3q/2, q\sqrt{3}/2)$ and, being horizontal, passes through two opposite vertices of that hexagon. For a vertical axis $x = d$ the linear part $(x, y) \mapsto (-x, y)$ sends $\mathbf{g}_1$ to $(-3/2, \sqrt{3}/2) = \mathbf{g}_2 - \mathbf{g}_1$ and fixes $\mathbf{g}_2$, so by the same reasoning the condition is $(2d, 0) \in L$. The y component $p\sqrt{3}/2 + q\sqrt{3}$ vanishes only for $p = -2q$, so $2d = 3p/2 = -3q$ and $d = 3k/2$ with $k = -q$. The line $x = 3k/2$ contains the center $k\mathbf{g}_1$ and, being vertical, passes through the midpoints of two opposite edges of that hexagon. The 60° and 120° rotations about the origin are symmetries by part (b); they carry these two families onto the axes in the remaining four directions ($0^\circ \to 60^\circ, 120^\circ$ and $90^\circ \to 150^\circ, 30^\circ$), preserving their kind. Conversely, let σ be a reflection symmetry in a line ℓ of one of these four directions, and let ρ be the rotation about the origin through 60° or 120° for which $\rho^{-1}\ell$ is horizontal or vertical. Then $\rho^{-1}\sigma\rho$ is a symmetry that fixes every point of $\rho^{-1}\ell$, namely the reflection in $\rho^{-1}\ell$, so $\rho^{-1}\ell$ is one of the axes found above and ℓ is its image under ρ. No additional offsets or directions are possible.



Solid lines join opposite vertices; dashed lines join opposite edge midpoints. Both sets are reflection axes.

17.2 Representation of Groups

17.2.1 For any representation K of a group, and for any group element a, show that

$$\left[U^K(a)\right]^{-1} = U^K\left(a^{-1}\right).$$

Solution.

A matrix representation maps the identity to the identity matrix and preserves products. Hence

$$U^K(a)U^K(a^{-1}) = U^K(aa^{-1}) = U^K(I) = I,$$

and also

$$U^K(a^{-1})U^K(a) = U^K(a^{-1}a) = I.$$

Thus $U^K(a^{-1})$ is both a right and left inverse of $U^K(a)$. By uniqueness of the matrix inverse,

$$\boxed{[U^K(a)]^{-1} = U^K(a^{-1})}.$$

No assumption of a faithful representation or of unitarity is needed.

17.2.2 Show that these four matrices form a representation of the Vierergruppe, whose multiplication table is in Table 17.2.

$$\mathsf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathsf{A} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathsf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathsf{C} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Solution.

The given matrices obey $A = -I$ and $C = -B$. Direct multiplication gives

$$B^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = I,$$

so $A^2 = (-I)^2 = I$ and $C^2 = (-B)^2 = I$ as well. The products of distinct nonidentity matrices are

$$AB = BA = -B = C, \qquad AC = CA = B, \qquad BC = CB = -B^2 = -I = A.$$

Multiplication by I leaves every matrix unchanged. These statements give all 16 products:

	I	A	B	C
I	I	A	B	C
A	A	I	C	B
B	B	C	I	A
C	C	B	A	I

.

The table is exactly Table 17.2; therefore the four matrices form a representation of the Vierergruppe. Since all four are different, this particular representation is faithful.

17.2.3 Show that the matrices I, A, B, and C of Exercise 17.2.2 are reducible. Reduce them.

Note. This means transforming B and C to diagonal form (by the same unitary transformation).

Solution.

The eigenvectors of B are

$$v_+ = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \quad Bv_+ = v_+, \qquad v_- = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}, \quad Bv_- = -v_-.$$

Let $Q = (v_+\ v_-) = 2^{-1/2}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}$. Its columns are orthonormal, so $Q^\dagger Q = I$; that is, $Q^\dagger v_+ = (1,0)^T$ and $Q^\dagger v_- = (0,1)^T$. Since $BQ = (v_+\ \ -v_-)$, the columns of $Q^\dagger BQ$ are $(1,0)^T$ and $(0,-1)^T$. Because $C = -B$, the same eigenvectors diagonalize C:

$$Q^\dagger BQ = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}, \qquad Q^\dagger CQ = \begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix}.$$

The scalar matrices $I, A = -I$ are unchanged. Thus the transformed representation is a direct sum of two one-dimensional representations, with characters in the order (I, A, B, C)

$$\boxed{(1,-1,1,-1) \quad \text{and} \quad (1,-1,-1,1)}.$$

Each line span$\{v_\pm\}$ is invariant under every group element, proving reducibility and supplying the reduction.

17.2.4

(a) Once you have a matrix representation of any group, a 1-D representation can be obtained by taking the determinants of the matrices. Show that the multiplicative relations are preserved in this determinant representation.

(b) Use determinants to obtain a 1-D representation of D_3 from the 2×2 representation in Eq. (17.2).

Solution.

(a) If $U(ab) = U(a)U(b)$, the determinant product theorem gives

$$\chi_{\det}(ab) = \det U(ab) = \det U(a)\det U(b) = \chi_{\det}(a)\chi_{\det}(b).$$

Also $\chi_{\det}(I) = 1$ and $\chi_{\det}(a^{-1}) = 1/\chi_{\det}(a)$. Every $U(a)$ is invertible, so no determinant is zero. The determinants therefore define a one-dimensional representation.

(b) In Eq. (17.2), the matrices of C_3 and C_3^2 have determinant $(-1/2)^2 + (\sqrt{3}/2)^2 = 1$. The C_2 matrix has determinant -1; either other twofold matrix has determinant $(-1/2)(1/2) - (\sqrt{3}/2)^2 = -1$. Thus

g	I	C_3	C_3^2	C_2	C_2'	C_2''
$\chi_{\det}(g)$	1	1	1	-1	-1	-1

This is the A_2 representation of D_3. Products of two flips have determinant $+1$, products of a flip with a rotation have determinant -1, and products of rotations have determinant $+1$, as required by the group table.

17.2.5 Show that the cyclic group of n objects, C_n, may be represented by r^m, $m = 0, 1, 2, \ldots, n-1$. Here r is the image of an abstract generator, given by

$$r = \exp(2\pi i s/n).$$

The parameter s takes on the values $s = 1, 2, 3, \ldots, n$, with each value of s yielding a different 1-D (irreducible) representation of C_n.

Take $n \geq 1$ and complex representations; the scalar image need not have order n.

Solution.

Let g be the abstract generator of C_n, so $g^n = I$. For each integer $s \in \{1, \ldots, n\}$ set

$$r_s = e^{2\pi i s/n}, \qquad U_s(g^m) = r_s^m.$$

Because $r_s^n = e^{2\pi i s} = 1$, reducing m modulo n does not change this value. For arbitrary integers j, k,

$$U_s(g^j g^k) = r_s^{j+k} = r_s^j r_s^k = U_s(g^j)U_s(g^k).$$

Thus this is a representation; it is irreducible since its space is one-dimensional.

The representations are distinct: equivalence in one dimension cannot change the scalar $U_s(g)$, and the n numbers $e^{2\pi i s/n}$ are distinct for these s. They are all the irreducible representations. Indeed any complex representation matrix of g satisfies $U(g)^n = U(g^n) = I$, so its eigenvalues are nth roots of unity; each eigenline is invariant under every power of g. An irreducible representation must therefore have dimension one. A one-dimensional representation is fixed by the number $U(g)$, which satisfies $U(g)^n = 1$; hence $U(g) = e^{2\pi i s/n}$ for exactly one $s \in \{1, \ldots, n\}$, and the representation is U_s.

The scalar r_s is the image of a generator, not necessarily an element of order n. Its order is $n/\gcd(s, n)$. The representation is faithful only when $\gcd(s, n) = 1$; $s = n$ gives the trivial representation. To check the order, $r_s^k = 1$ is equivalent to n dividing sk. Write $d = \gcd(s, n)$, $s = ds'$, $n = dn'$; then $n \mid sk$ means $n' \mid s'k$, and since $\gcd(n', s') = 1$ this means $n' \mid k$. The smallest positive such k is therefore $n' = n/\gcd(s, n)$.

17.2.6 Develop the irreducible 2×2 matrix representation of the group of rotations (including those that turn it over) that transform a square into itself. Give the group multiplication table.

Note. This group has the name D_4 (see Fig. 17.5).

Solution.

Take r to be a counterclockwise 90° rotation and s a 180° turn about the x axis in the plane of the square. Their action on in-plane coordinates is

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $R^2 = -I$, $R^4 = S^2 = I$, and

$$SRS = S(RS) = S\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = R^{-1}.$$

The eight matrices, with $U(r^i s) = R^i S$, are

g	I	r	r^2	r^3
$U(g)$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

g	s	rs	r^2s	r^3s
$U(g)$	$\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$	$\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$	$\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}$	$\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}$

The first four preserve the displayed side of the square; the last four turn it over. All are real orthogonal.

Every element is $r^i s^\epsilon$, $0 \le i < 4$, $\epsilon = 0, 1$. The relation $srs = r^{-1}$ gives $sr = r^{-1}s$, and applying it j times, $sr^j = r^{-j}s$. Hence the product rule is

$$(r^i s^\epsilon)(r^j s^\eta) = r^{\,i+(-1)^\epsilon j} s^{\,\epsilon+\eta},$$

with exponents reduced modulo four and two. This gives the full table (row times column):

	I	r	r^2	r^3	s	rs	r^2s	r^3s
I	I	r	r^2	r^3	s	rs	r^2s	r^3s
r	r	r^2	r^3	I	rs	r^2s	r^3s	s
r^2	r^2	r^3	I	r	r^2s	r^3s	s	rs
r^3	r^3	I	r	r^2	r^3s	s	rs	r^2s
s	s	r^3s	r^2s	rs	I	r^3	r^2	r
rs	rs	s	r^3s	r^2s	r	I	r^3	r^2
r^2s	r^2s	rs	s	r^3s	r^2	r	I	r^3
r^3s	r^3s	r^2s	rs	s	r^3	r^2	r	I

To prove irreducibility even over $\mathbb{C}$, a proper nonzero invariant subspace would be a line. It would have to be an eigenline of R, whose distinct eigenvalues are $i, -i$ with eigenvectors $(1, -i)^T$ and $(1, i)^T$. But S interchanges these lines. Neither is invariant under the whole group, so the two-dimensional representation is irreducible.

17.3 Symmetry and Physics

17.3.1 Consider a quantum mechanics problem with D_3 symmetry, with the threefold symmetry axis taken as the z direction, and with orbitals $\varphi(\mathbf{r} - \mathbf{R}_j)$ located at the vertices of an equilateral triangle. This is the same system geometry as in Example 17.3.2, but in the present problem φ will no longer be chosen to have spherical symmetry.

Given that $\varphi(\mathbf{r}) = (z/r)f(r)$ (so φ has the symmetry of a p orbital oriented along the symmetry axis), construct linear combinations of the φ that are bases for irreducible representations of D_3, for each basis indicating its representation.

Solution.

Write $\varphi_j(\mathbf{r}) = \varphi(\mathbf{r} - \mathbf{R}_j)$, with the vertices numbered cyclically. Use active operations: $r = C_3$ sends $\varphi_1 \to \varphi_2 \to \varphi_3 \to \varphi_1$. A twofold rotation $s = C_2$ about the in-plane axis through vertex 1 also reverses z. Since each local orbital is proportional to z, its transformation is

$$s\varphi_1 = -\varphi_1, \qquad s\varphi_2 = -\varphi_3, \qquad s\varphi_3 = -\varphi_2.$$

Thus the sum

$$\boxed{\psi_{A_2} = \frac{\varphi_1 + \varphi_2 + \varphi_3}{\sqrt{3}}}$$

is unchanged by r and changes sign under s. Its characters on $(I, 2C_3, 3C_2)$ are $(1, 1, -1)$, identifying A_2.

Choose two independent zero-sum combinations

$$u = \frac{2\varphi_1 - \varphi_2 - \varphi_3}{\sqrt{6}}, \qquad v = \frac{\varphi_2 - \varphi_3}{\sqrt{2}}.$$

Substituting $\varphi_1 \to \varphi_2 \to \varphi_3 \to \varphi_1$ gives

$$ru = \frac{-\varphi_1 + 2\varphi_2 - \varphi_3}{\sqrt{6}} = -\frac{1}{2}u + \frac{\sqrt{3}}{2}v, \qquad rv = \frac{\varphi_3 - \varphi_1}{\sqrt{2}} = -\frac{\sqrt{3}}{2}u - \frac{1}{2}v.$$

Both images again have zero coefficient sum (a permutation of the φ_j does not change the sum), so they lie in the plane of zero-sum combinations spanned by u and v. The coefficient vectors $(2,-1,-1)/\sqrt{6}$ and $(0,1,-1)/\sqrt{2}$ of u and v are orthonormal, so each coefficient here is an inner product of coefficient vectors; for example, the coefficient of v in ru is $[0\cdot(-1)+1\cdot 2+(-1)(-1)]/\sqrt{12} = \sqrt{3}/2$. The substitution for s gives

$$su = \frac{-2\varphi_1 + \varphi_3 + \varphi_2}{\sqrt{6}} = -u, \qquad sv = \frac{-\varphi_3 + \varphi_2}{\sqrt{2}} = v.$$

Hence the coefficient matrices in the ordered basis (u, v) are

$$D_E(r) = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad D_E(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Their traces are -1 and 0, so the characters are $(2,-1,0)$. An invariant line would have to be an eigenline of $D_E(r)$, whose distinct eigenvalues $e^{\pm 2\pi i/3}$ have eigenvectors $(1, \mp i)^T$. But $D_E(s)(1,\mp i)^T = -(1,\pm i)^T$, so the flip interchanges these two lines, and the two-dimensional space is irreducible. As a check, the representation on $\mathrm{span}\{\varphi_1,\varphi_2,\varphi_3\}$ has characters $(3,0,-1)$: r moves every orbital to another site, while s contributes only the diagonal entry -1 from $s\varphi_1 = -\varphi_1$. Indeed $(3,0,-1) = (1,1,-1) + (2,-1,0)$. Therefore

$$\boxed{\mathrm{span}\{\varphi_1,\varphi_2,\varphi_3\} = A_2 \oplus E, \qquad E \text{ has basis } (u,v)}.$$

The Instructor's Manual gives $-I_2$ for a twofold operation in this E space; that matrix has trace -2, whereas every such operation has eigenvalues $+1, -1$ and trace zero, the E character, as the direct transformations show. The square-root factors normalize the coefficient vectors. If spatially overlapping orbitals are used, their actual Hilbert-space norms must be obtained from their overlap integrals; the symmetry assignments do not depend on those common scale choices.

17.4 Discrete Groups

17.4.1 The Vierergruppe has the multiplication table shown in Table 17.2.

(a) Divide its elements into classes.

(b) Using the class information, determine for the Vierergruppe its number of inequivalent irreducible representations and their dimensions.

(c) Construct a character table for the Vierergruppe.

Solution.

(a) The group is abelian, so $gag^{-1} = agg^{-1} = a$ for every a, g. Its conjugacy classes are therefore

$$\boxed{\{I\}, \quad \{A\}, \quad \{B\}, \quad \{C\}}.$$

(b) The number of inequivalent complex irreducible representations equals the number of classes, hence is four. Their positive integer dimensions satisfy $d_1^2 + d_2^2 + d_3^2 + d_4^2 = |\mathcal{G}| = 4$. Every term is at least one, so each $d_j = 1$.

(c) In a one-dimensional representation set $\chi(A) = a$, $\chi(B) = b$. Since $A^2 = B^2 = I$, both a, b are ± 1. The relation $C = AB$ fixes $\chi(C) = ab$. The four independent choices of a, b give

	I	A	B	C
A_1	1	1	1	1
A_2	1	1	-1	-1
A_3	1	-1	1	-1
A_4	1	-1	-1	1

.

Each row obeys the multiplication table. Its squared norm is four, and two distinct rows have two equal and two opposite entries, giving inner product zero. Thus the table also satisfies character orthogonality and contains all four irreducibles.

17.4.2 The group D_3 may be discussed as a permutation group of three objects. Operation C_3, for instance, moves vertex 1 to the position formerly occupied by vertex 3; likewise vertex 2 moves to the original position of vertex 1 and vertex 3 moves to the original position of vertex 2. So this shuffling could be described as the permutation of $(1, 2, 3)$ to $(2, 3, 1)$. Using now letters a, b, c to avoid notational confusion, this permutation $(abc) \to (bca)$ corresponds to the matrix equation

$$C_3 \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} b \\ c \\ a \end{pmatrix},$$

thereby identifying a 3×3 representation of the operation C_3.

(a) Develop analogous 3×3 representations for the other elements of D_3.

(b) Reduce your 3×3 representation to the direct sum of a 1×1 and a 2×2 representation. Note: This 3×3 representation must be reducible or Eq. (17.10) would be violated.

The printed statement has vertex 1 moving to the position of vertex 2, vertex 2 to that of vertex 3, and vertex 3 to that of vertex 1, which does not agree with the displayed matrix (see the solution); the vertex motion is reversed here so that the name C_3 follows the displayed matrix.

Solution.

The printed vertex motion is the active map $1 \to 2 \to 3 \to 1$. Afterwards position 1 holds the label c of vertex 3, position 2 holds a, and position 3 holds b, so it sends the column of labels at fixed positions from (a, b, c) to (c, a, b). The displayed matrix instead gives (b, c, a), the inverse reordering. We retain the displayed matrix for this exercise; using the printed motion would interchange the labels C_3 and C_3^2 and, through $C_2C_3 = C_2'$, also the labels C_2' and C_2''. Alternatively, the column can be read as listing the positions now occupied by vertices 1, 2, 3; read so, the printed motion does give (b, c, a). But then the matrix of a motion σ replaces entry j by entry $\sigma(j)$, and such matrices

multiply in the reverse order, $U(g)U(h) = U(hg)$; for instance, SR of part (a) would be the matrix of C_3C_2 rather than of C_2C_3. Only the reading by labels at fixed positions gives a representation. The printed motion, vertex action (123), is the counterclockwise C_3 of Exercise 17.1.2(c); hence, with the vertices numbered as there, the C_3, C_2', C_2'' of this exercise are the C_3^2, C_2'', C_2' of 17.1.2(c). For example, $U(C_2') = SR$ below interchanges the first two entries, so it is the turn about the axis through vertex 3.

(a) Put

$$R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Here S interchanges the second and third entries; as a right factor it interchanges the second and third columns. Direct multiplication gives $R^2 = U(C_3^2)$ below, $R^3 = S^2 = I$, and $SRS = (SR)S = R^{-1}$: interchanging the last two columns of SR (displayed below) gives $R^2 = R^T = R^{-1}$. Hence R, S generate a representation of D_3. The six matrices, named consistently with the multiplication table $C_2C_3 = C_2'$, are

$$U(I) = I_3, \quad U(C_3) = R, \qquad\qquad U(C_3^2) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

$$U(C_2) = S, \quad U(C_2') = SR = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad U(C_2'') = SR^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

(b) All permutations preserve the sum of the three components. Consequently

$$e_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

is fixed. The permutation matrices are orthogonal, so its orthogonal complement, the vectors with zero component sum, is invariant. Choose an orthonormal basis there:

$$e_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \qquad e_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \qquad Q = (e_0\ e_1\ e_2).$$

Since $R(a,b,c)^T = (b,c,a)^T$ and $S(a,b,c)^T = (a,c,b)^T$,

$$Re_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} = -\tfrac{1}{2}e_1 - \tfrac{\sqrt{3}}{2}e_2, \qquad Re_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \tfrac{\sqrt{3}}{2}e_1 - \tfrac{1}{2}e_2,$$

as one checks by adding the right-hand sides componentwise (note $\frac{\sqrt{3}}{2}e_2 = \frac{3}{2\sqrt{6}}(0,1,-1)^T$), and $Se_1 = e_1$, $Se_2 = -e_2$. These coefficients form the columns of the 2×2 blocks, while $Re_0 = Se_0 = e_0$ gives the entry 1:

$$Q^\dagger RQ = 1 \oplus \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad Q^\dagger SQ = 1 \oplus \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Every other group matrix is a product of R, S, so the same $1+2$ block structure holds for all six matrices. The scalar block is A_1; the two-dimensional block has characters $(2,-1,0)$ and is E. Hence

$$\boxed{U = A_1 \oplus E}.$$

As a check, the traces of I_3, R, S give the characters $(3,0,1)$ of U on $(I, 2C_3, 3C_2)$, and $(3,0,1) = (1,1,1) + (2,-1,0)$.

17.4.3 The group named D_4 has a fourfold axis of symmetry, and twofold axes in four directions perpendicular to the fourfold axis. See Fig. 17.5. D_4 has the following classes (the numbers preceding the class descriptors indicate the number of elements in the class): I, $2C_4$, C_2, $2C_2'$, $2C_2''$. The twofold axes marked with primes are in the plane of fourfold symmetry.

(a) Find the number and dimensions of the irreducible representations.

(b) Given that all the characters of the representations of dimension 1 are ± 1 and that $C_2 = C_4^2$, use the orthogonality conditions to construct a complete character table for D_4.

Solution.

(a) There are five classes, so there are five irreducible representations. Their dimensions have squares summing to eight. No dimension can exceed two, since $3^2 = 9 > 8$. If k dimensions are two and the other $5-k$ are one, then

$$4k + (5-k) = 8 \quad \Longrightarrow \quad k = 1.$$

Thus there are four one-dimensional representations and one two-dimensional representation.

(b) Write the group as $r^4 = s^2 = I$, $srs = r^{-1}$, as in 17.2.6. Its classes are

$$\{I\}, \quad \{r, r^3\}, \quad \{r^2\}, \quad \{s, r^2 s\}, \quad \{rs, r^3 s\}.$$

Indeed, with $sr^j = r^{-j}s$ from 17.2.6, conjugation by r gives $rsr^{-1} = r^2 s$ and $r(rs)r^{-1} = r^3 s$, conjugation by s gives $srs^{-1} = r^{-1} = r^3$, and r^2 commutes with every element. No element of one set is conjugate to an element of another: the conjugates of r^j are $r^{\pm j}$, while $r^i(r^j s)r^{-i} = r^{j+2i}s$ and $(r^i s)(r^j s)(r^i s)^{-1} = r^{2i-j}s$ keep the parity of j. By 17.2.6, s and $r^2 s$ (matrices $\mathrm{diag}(1,-1)$ and $\mathrm{diag}(-1,1)$) are the turns about the x and y axes, and rs, $r^3 s$ (matrices $\pm\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$) are those about the diagonals $y = \pm x$. With $C_2' = (x \to -x,\ y \to y)$ and $C_2'' = (x \to y,\ y \to x)$ as in Exercise 17.4.4, the five sets are therefore the classes $I, 2C_4, C_2, 2C_2', 2C_2''$, in that order. In one dimension, $a = U(r)$ and $b = U(s)$ are scalars. The relation $bab = a^{-1}$ and $b^2 = 1$ imply $a = a^{-1}$, so $a, b = \pm 1$. The class characters are therefore $(1, a, 1, b, ab)$, since $\chi(r^2) = a^2 = 1$ and $\chi(rs) = ab$. Their four choices give the first four rows below. The route suggested in the statement gives the same rows: with all one-dimensional characters ± 1 and $\chi(C_2) = \chi(C_4)^2 = 1$, a row other than A_1 has the form $(1, a, 1, b, c)$, and its orthogonality to A_1 with class weights $(1,2,1,2,2)$ requires $2 + 2(a+b+c) = 0$, i.e. $a+b+c = -1$; so exactly two of a, b, c equal -1, which gives three rows.

For the remaining representation, $\chi_E(I) = 2$. Column orthogonality between a nonidentity column g and the identity column requires

$$\sum_K \chi_K(I)^* \chi_K(g) = 0.$$

For r, r^3, the first four entries sum to zero, so $2\chi_E(r) = 0$. The same reasoning gives zero on both flip classes. For r^2, the four scalar entries sum to four, so $4 + 2\chi_E(r^2) = 0$, giving $\chi_E(r^2) = -2$. Thus the complete table is

D_4	I	$2C_4$	C_2	$2C_2'$	$2C_2''$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	-1	1	1	-1
B_2	1	-1	1	-1	1
E	2	0	-2	0	0

.

Using class weights $(1, 2, 1, 2, 2)$, each row has squared norm eight and distinct rows have inner product zero. For example $\langle E, E\rangle = 4 + 4 = 8$, while $\langle E, A_1\rangle = 2 - 2 = 0$. This verifies the required orthogonality; the E matrices in 17.2.6 realize its last row.

17.4.4 The eight functions $\pm x^3$, $\pm x^2y$, $\pm xy^2$, $\pm y^3$ span a reducible function space for D_4, with C_4 a 90° counterclockwise rotation in the xy plane, $C_2 = C_4^2$, $C_2' = (x \to -x,\ y \to y)$, $C_2'' = (x \to y,\ y \to x)$, and the remaining members of D_4 are additional members of the classes containing the above operations. Find the characters of the reducible representation on this span, and find the direct sum of irreducible representations of which it consists.

Also distinguish the formal permutation representation on eight independent signed labels, which is the interpretation used by the Instructor's Manual.

Solution.

As actual functions, f and $-f$ are linearly dependent. The eight listed signed functions are not an eight-dimensional basis: their function space has dimension four. The Instructor's Manual's eight-dimensional answer describes a different, formal permutation space whose eight signed labels are treated as independent. We give both interpretations explicitly.

For the space $\mathcal{P}_3$ of actual cubic functions (homogeneous cubic polynomials in x, y) take the independent basis

$$f_1 = x^3, \quad f_2 = x^2y, \quad f_3 = xy^2, \quad f_4 = y^3.$$

Let the quarter-turn act on functions by the substitution $r : (x, y) \mapsto (y, -x)$, that is, $(rf)(\mathbf{r}) = f(R^{-1}\mathbf{r})$ for the counterclockwise 90° rotation R. The opposite convention, $f(R\mathbf{r})$, assigns to each operation the matrix of its inverse; it gives the same characters, because every element of D_4 is conjugate to its inverse ($r^{-1} = r^3$ lies in the class $\{r, r^3\}$ found in Exercise 17.4.3, and every other element is its own inverse). The transformed basis is

$$r(f_1, f_2, f_3, f_4) = \big(y^3,\ y^2(-x),\ y(-x)^2,\ (-x)^3\big) = (f_4, -f_3, f_2, -f_1).$$

Thus no diagonal coefficient occurs and $\chi(r) = 0$. The half-turn sends every cubic to its negative, so $\chi(r^2) = -4$. For $s : (x, y) \mapsto (-x, y)$,

$$s(f_1, f_2, f_3, f_4) = (-f_1, f_2, -f_3, f_4),$$

whose trace is zero. The diagonal reflection $(x, y) \mapsto (y, x)$ exchanges $f_1 \leftrightarrow f_4$ and $f_2 \leftrightarrow f_3$, again with trace zero. Therefore

$$\chi_{\text{cubic}} = (4, 0, -4, 0, 0) \quad \text{on } (I, 2C_4, C_2, 2C_2', 2C_2'').$$

Only I and C_2 have nonzero characters, and every one-dimensional representation has $\chi(C_2) = 1$, so the multiplicity formula $n_K = 8^{-1} \sum_{\text{classes}} |C| \chi_K^* \chi$, where $|C|$ is the number of elements in the class C, gives

$$n_{A_1} = n_{A_2} = n_{B_1} = n_{B_2} = \frac{4-4}{8} = 0, \qquad n_E = \frac{4 \cdot 2 + (-4)(-2)}{8} = 2.$$

Hence the function-space answer is

$$\boxed{\mathcal{P}_3 = E \oplus E}.$$

Explicit invariant pairs are (x^3, y^3) and (x^2y, xy^2), as the transformation lists above show. In each pair a quarter-turn has eigenvalues $\pm i$: $r(x^3 \mp iy^3) = \pm i(x^3 \mp iy^3)$ and $r(x^2y \pm ixy^2) = \pm i(x^2y \pm ixy^2)$. The reflection s interchanges the eigenlines, since $s(x^3 - iy^3) = -(x^3 + iy^3)$ and $s(x^2y + ixy^2) = x^2y - ixy^2$. Hence neither pair has an invariant line, and each copy is irreducible.

For the formal eight-label interpretation, introduce independent symbols $e_{j,+}, e_{j,-}$ representing the labels $+f_j, -f_j$; they span an eight-dimensional space $\mathcal{V}_{\text{formal}}$. A group operation permutes these eight symbols, so its character counts fixed labels. The identity fixes eight. The quarter-turn and diagonal reflection fix none. The half-turn exchanges each + label with its − partner and fixes none. The reflection $x \mapsto -x$ fixes the four labels $\pm x^2y, \pm y^3$. Thus

$$\chi_{\text{formal}} = (8, 0, 0, 4, 0),$$

and

$$\boxed{\mathcal{V}_{\text{formal}} = 2A_1 \oplus 2B_1 \oplus 2E}.$$

Indeed its multiplicities are $(8 + 2 \cdot 4)/8 = 2$ for A_1, B_1, $(8 - 2 \cdot 4)/8 = 0$ for A_2, B_2, and $(8 \cdot 2)/8 = 2$ for E. The map $e_{j,\pm} \mapsto \pm f_j$ commutes with the group action and maps onto the cubic space; its kernel is spanned by the four vectors $e_{j,+} + e_{j,-}$. Each operation permutes these kernel vectors without signs (an operation taking f_j to $\pm f_k$ sends $e_{j,+} + e_{j,-}$ to $e_{k,+} + e_{k,-}$), so on the kernel the character of an operation is the number of j with $f_j \mapsto \pm f_j$. By the transformation lists above this is $(4, 0, 4, 4, 0)$, which equals $\chi_{\text{formal}} - \chi_{\text{cubic}}$, as it must, since the cubic space is the quotient by the kernel. The same multiplicity formula gives $(4 + 4 + 2 \cdot 4)/8 = 2$ for A_1, B_1 and zero for the others, i.e. the kernel is $2A_1 \oplus 2B_1$, and $\mathcal{V}_{\text{formal}}$ is the direct sum of this kernel and a copy of $\mathcal{P}_3 = E \oplus E$. This explains why the two legitimate spaces have different characters and dimensions.

17.4.5 The group C_{4v} has a fourfold symmetry axis in the z direction, reflection symmetries (σ_v) about the xz and yz planes, and additional reflection symmetries (σ_d, d = dihedral) about planes that contain the z axis but are 45° from the x and y axes. See Fig. 17.6. The character table for C_{4v} follows.

	I	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$
A_1	1	1	1	1	1
A_2	1	1	1	−1	−1
B_1	1	−1	1	1	−1
B_2	1	−1	1	−1	1
E	2	0	−2	0	0

(a) Construct the matrix representing one member of each class of C_{4v} using as a basis a p_z orbital at each of the points $(x, y) = (a, 0)$, $(0, a)$, $(-a, 0)$, $(0, -a)$, and therefrom extract the characters of the reducible representation for which these p_z orbitals form a basis. A p_z orbital has functional form $(z/r)f(r)$.

(b) Determine the irreducible representations contained in our reducible p_z representation.

(c) Form those linear combinations of our p_z functions that are bases for each of the irreducible representations found in part (b).

The printed reference in (c) says (a); the irreducible summands are determined in (b).

Solution.

Let f_1, f_2, f_3, f_4 denote the p_z orbitals centered, in order, at $+x, +y, -x, -y$. A C_{4v} operation leaves z unchanged, so the local p_z shape incurs no sign change; only the centers are permuted. We use matrices whose jth column gives the image of f_j.

(a) Choose $r = C_4$ counterclockwise, $s = \sigma_v$ reflection in the xz plane, and $d = \sigma_d$ reflection in $x = y$. On the centers, r acts as $+x \to +y \to -x \to -y \to +x$, r^2 exchanges opposite centers, s fixes $\pm x$ and exchanges $+y \leftrightarrow -y$, and d exchanges $+x \leftrightarrow +y$ and $-x \leftrightarrow -y$. Representatives of the five classes are therefore

$$P_I = I_4, \qquad P_r = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad P_{r^2} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

$$P_s = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad P_d = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The traces are, on $(I, 2C_4, C_2, 2\sigma_v, 2\sigma_d)$,

$$\boxed{\chi = (4, 0, 0, 2, 0)}.$$

For example s fixes the two orbitals on the x axis, while r, r^2, d fix none.

(b) With class sizes $(1, 2, 1, 2, 2)$, $n_K = 8^{-1} \sum_{\text{classes}} |C| \chi_K^* \chi$, in which only the classes I and $2\sigma_v$ contribute, gives

$$n_{A_1} = n_{B_1} = \frac{4 + 2(2)}{8} = 1, \qquad n_{A_2} = n_{B_2} = \frac{4 - 2(2)}{8} = 0, \qquad n_E = \frac{2(4)}{8} = 1.$$

Thus, writing Γ for the p_z representation, $\boxed{\Gamma = A_1 \oplus B_1 \oplus E}$.

(c) Suitable combinations are

$$\boxed{\begin{array}{ll} A_1: & a = \tfrac{1}{2}(f_1 + f_2 + f_3 + f_4), \\ B_1: & b = \tfrac{1}{2}(f_1 - f_2 + f_3 - f_4), \\ E: & e_x = (f_1 - f_3)/\sqrt{2}, \qquad e_y = (f_2 - f_4)/\sqrt{2}. \end{array}}$$

Substituting the center permutations gives

$$ra = a, \quad sa = a, \qquad rb = -b, \quad sb = b,$$

$$re_x = e_y, \quad re_y = -e_x, \qquad se_x = e_x, \quad se_y = -e_y, \qquad de_x = e_y, \quad de_y = e_x.$$

For instance, $rb = \frac{1}{2}(f_2 - f_3 + f_4 - f_1) = -b$, and $sb = \frac{1}{2}(f_1 - f_4 + f_3 - f_2) = b$ because s fixes f_1, f_3 and exchanges f_2, f_4; also $db = -b$. These generator actions identify the stated representations.

The function a is invariant, as A_1 requires. The function b has $\chi(r) = -1$, $\chi(\sigma_v) = 1$, $\chi(\sigma_d) = -1$, which is the B_1 row. On (e_x, e_y) the matrices of r and s are $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, the E matrices of Exercise 17.2.6, with characters $(2, 0, -2, 0, 0)$. The pair e_x, e_y is odd across the square and transforms like (x, y). The four coefficient vectors are mutually orthonormal and hence independent; their actual function norms, if orbitals overlap, are handled as explained in 17.3.1.

17.4.6 Using the notation and geometry of Exercise 17.4.5, repeat that exercise for the eight-member basis consisting of a p_x and a p_y orbital at each of the points $(x, y) = (a, 0)$, $(0, a)$, $(-a, 0)$, $(0, -a)$.

Solution.

Write $x_j = p_x(\mathbf{r} - \mathbf{R}_j)$ and $y_j = p_y(\mathbf{r} - \mathbf{R}_j)$, with the four sites ordered as in 17.4.5. All x_j have positive lobe along positive x, and all y_j along positive y. The ordered basis is

$$(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4).$$

Unlike p_z, these local orbitals rotate and reflect in their two-component orientation space.

(a) Let P_g be the site matrix from 17.4.5 and O_g the matrix acting on (p_x, p_y). The operation g carries the orbital of orientation p_α ($\alpha = x, y$) at site j to the site k with $(P_g)_{kj} = 1$, with orientation $\sum_\beta (O_g)_{\beta\alpha}\, p_\beta$. Hence

$$D(g)_{(k,\beta),(j,\alpha)} = (P_g)_{kj}(O_g)_{\beta\alpha}, \qquad \boxed{D(g) = P_g \otimes O_g}.$$

This formula determines every entry of the requested 8×8 matrices. Explicitly, put

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = I_2.$$

Because p_x and p_y transform like x and y, the orientation matrices are those of the point operations on (x, y): $O_r = R$, $O_{r^2} = R^2 = -H$, $O_s = S$, $O_d = T$. In the following block matrices each entry is 2×2, and block (k, j) is $(P_g)_{kj} O_g$:

$$D(I) = I_8, \qquad D(r) = \begin{pmatrix} 0 & 0 & 0 & R \\ R & 0 & 0 & 0 \\ 0 & R & 0 & 0 \\ 0 & 0 & R & 0 \end{pmatrix}, \qquad D(r^2) = \begin{pmatrix} 0 & 0 & -H & 0 \\ 0 & 0 & 0 & -H \\ -H & 0 & 0 & 0 \\ 0 & -H & 0 & 0 \end{pmatrix},$$

$$D(s) = \begin{pmatrix} S & 0 & 0 & 0 \\ 0 & 0 & 0 & S \\ 0 & 0 & S & 0 \\ 0 & S & 0 & 0 \end{pmatrix}, \qquad D(d) = \begin{pmatrix} 0 & T & 0 & 0 \\ T & 0 & 0 & 0 \\ 0 & 0 & 0 & T \\ 0 & 0 & T & 0 \end{pmatrix}.$$

For example $rx_1 = y_2$, $ry_1 = -x_2$, $sx_1 = x_1$, and $sy_1 = -y_1$, which explain the orientation signs. The trace of a tensor product is the product of traces, $\operatorname{tr}(P \otimes O) = \operatorname{tr} P \, \operatorname{tr} O$, so, class by class,

$$\chi_D = \chi_{\text{site}}\chi_{\text{orientation}} = (4, 0, 0, 2, 0)(2, 0, -2, 0, 0) = \boxed{(8, 0, 0, 0, 0)}.$$

(b) Only the identity contributes to the multiplicity sum: $n_K = 8^{-1}(8)\chi_K(I) = \dim K$. Therefore, writing Γ for the representation with matrices $D(g)$,

$$\boxed{\Gamma = A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus E \oplus E},$$

with total dimension $4 \cdot 1 + 2 \cdot 2 = 8$, as required.

(c) The following four combinations give the scalar representations:

$$\boxed{\begin{aligned} A_1: &\quad a_1 = (x_1 + y_2 - x_3 - y_4)/2, \\ A_2: &\quad a_2 = (y_1 - x_2 - y_3 + x_4)/2, \\ B_1: &\quad b_1 = (x_1 - y_2 - x_3 + y_4)/2, \\ B_2: &\quad b_2 = (y_1 + x_2 - y_3 - x_4)/2. \end{aligned}}$$

They arise from the site pair $(f_1 - f_3, f_2 - f_4)$ of 17.4.5 and the orientation pair (p_x, p_y), both of which transform like (x, y). Written as

$$\begin{aligned} a_1 &= \tfrac{1}{2}[(x_1 - x_3) + (y_2 - y_4)], \quad a_2 = \tfrac{1}{2}[(y_1 - y_3) - (x_2 - x_4)], \\ b_1 &= \tfrac{1}{2}[(x_1 - x_3) - (y_2 - y_4)], \quad b_2 = \tfrac{1}{2}[(y_1 - y_3) + (x_2 - x_4)], \end{aligned}$$

they are the scalar product (like $x^2 + y^2$), the antisymmetric product (like the z component $xy' - yx'$ of a cross product), the difference of diagonal products (like $x^2 - y^2$), and the symmetric off-diagonal product (like xy) of the two pairs. To check them directly, note that with site indices modulo four

$$r: \; x_j \mapsto y_{j+1}, \quad y_j \mapsto -x_{j+1}; \qquad s: \; x_j \mapsto x_{\bar{\jmath}}, \quad y_j \mapsto -y_{\bar{\jmath}},$$

where $\bar{1} = 1$, $\bar{2} = 4$, $\bar{3} = 3$, $\bar{4} = 2$. For example, $ra_1 = (y_2 - x_3 - y_4 + x_1)/2 = a_1$ and $sa_1 = (x_1 - y_4 - x_3 + y_2)/2 = a_1$. In the same way one finds the eigenvalues under the two generators:

	a_1	a_2	b_1	b_2
r	1	1	-1	-1
s	1	-1	1	-1

.

Since $d = rs$ (as matrices on (x, y), $RS = T$), its eigenvalues are the corresponding products, so these are exactly the four character-table rows.

The two independent E pairs are

$$\boxed{\begin{aligned} u_x &= (x_1 + x_2 + x_3 + x_4)/2, \quad u_y = (y_1 + y_2 + y_3 + y_4)/2, \\ v_x &= (x_1 - x_2 + x_3 - x_4)/2, \quad v_y = (y_1 - y_2 + y_3 - y_4)/2. \end{aligned}}$$

The first pair is the site A_1 combination times the orientation pair; the second uses the site B_1 combination. By the rules above, $ru_x = u_y$, $ru_y = -u_x$, while $rv_x = (y_2 - y_3 + y_4 - y_1)/2 = -v_y$ and $rv_y = (-x_2 + x_3 - x_4 + x_1)/2 = v_x$; also $su_x = u_x$, $su_y = -u_y$, $sv_x = v_x$, $sv_y = -v_y$. So their generator matrices are respectively (R, S) and $(-R, S)$. Both have characters $(2, 0, -2, 0, 0)$ and have no common invariant line: $-R$ has the same eigenlines $(1, \mp i)^T$ as R, and S interchanges them, as in 17.2.6. The eight coefficient columns displayed above have norm one and pairwise zero inner products, so they form a complete basis. As in 17.3.1, coefficient normalization is distinct from spatial normalization when local orbitals overlap.

17.5 Direct Products

17.5.1 The group C_{4v} has eight elements, corresponding to the rotational and reflection symmetries of a square that cannot be turned over. See Fig. 17.6. Symmetry rotations about the z axis are denoted C_4, C_2, C_4'. Reflections relative to the xz and yz planes are named σ_v and σ_v'; those at 45° relative to the xz and yz planes are called σ_d and σ_d' (d indicates "dihedral"). The character table for C_{4v} is in Table 17.4.

(a) Find the direct sum of irreducible representations of C_{4v} corresponding to the direct product $E \otimes E$.

(b) A basis for E (in the context of Fig. 17.5) consists of the two functions $\varphi_1 = x$, $\varphi_2 = y$. Apply a few of the group operations to this basis and verify the entries for E in the character table.

(c) Assume now that we have two sets of variables, x_1, y_1 and x_2, y_2, and we form the direct-product basis x_1x_2, x_1y_2, y_1x_2, y_1y_2. Determine how the direct-product basis functions can be combined to form bases for each of the irreducible representations in the direct sum corresponding to $E \otimes E$.

Solution.

(a) The matrices of $E \otimes E$ are $D(g) \otimes D(g)$, where $D(g)$ are those of E. For matrices A, B, $\mathrm{tr}(A \otimes B) = (\mathrm{tr}\, A)(\mathrm{tr}\, B)$, since the diagonal entries of $A \otimes B$ are the products $A_{ii}B_{jj}$. On the classes $(I, 2C_4, C_2, 2\sigma_v, 2\sigma_d)$ the product character is consequently

$$\chi_{E\otimes E} = (2, 0, -2, 0, 0)^2 = (4, 0, 4, 0, 0).$$

The multiplicity of an irreducible K is $n_K = \frac{1}{8}\sum_C |C|\, \chi_K(C)^* \chi_{E\otimes E}(C)$, summed over the classes C with sizes $|C| = (1, 2, 1, 2, 2)$, as in 17.4.5(b); only I and C_2 contribute. For each of the four one-dimensional irreducibles, $\chi(I) = \chi(C_2) = 1$, so its multiplicity is $(4 + 4)/8 = 1$. For E itself it is $[2(4) + (-2)(4)]/8 = 0$. Thus

$$\boxed{E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2}.$$

(b) Let $r = C_4$ (the counterclockwise quarter-turn), $s = \sigma_v$ (reflection in the xz plane), and $d = \sigma_d$ (reflection in the plane $x = y$). With functions transformed by $(gf)(\mathbf{r}) = f(g^{-1}\mathbf{r})$ as in 17.4.4, for instance $(rx)(x, y) = y$ because $r^{-1}(x, y) = (y, -x)$, they act as $rx = y$, $ry = -x$, $sx = x$, $sy = -y$, and $dx = y$, $dy = x$. In the ordered basis (x, y) their coefficient matrices, whose jth column gives the image of the jth basis function as in 17.4.5, are

$$D(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad D(r^2) = -I_2, \quad D(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D(d) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Their traces, together with $\mathrm{tr}\, I_2 = 2$, are $(2, 0, -2, 0, 0)$ as in the E row. One element of each class suffices, because conjugate elements have similar matrices and hence equal traces. For instance, the other members r^3, $\sigma_v' = r^2 s$, $\sigma_d' = r^3 s$ of the classes have the traceless matrices $D(r)^T$, $-D(s)$, $-D(d)$.

(c) The product basis has independent variables for particles 1 and 2. Its four convenient combinations are

$$\boxed{\begin{aligned} A_1: &\quad a_1 = (x_1x_2 + y_1y_2)/\sqrt{2}, \\ A_2: &\quad a_2 = (x_1y_2 - y_1x_2)/\sqrt{2}, \\ B_1: &\quad b_1 = (x_1x_2 - y_1y_2)/\sqrt{2}, \\ B_2: &\quad b_2 = (x_1y_2 + y_1x_2)/\sqrt{2}. \end{aligned}}$$

Apply r to each factor: $x_1x_2 \to y_1y_2$, $y_1y_2 \to x_1x_2$, $x_1y_2 \to -y_1x_2$, and $y_1x_2 \to -x_1y_2$. Since r interchanges x_1x_2 with y_1y_2, and x_1y_2 with y_1x_2 up to sign, the sums and differences of these pairs, which are the four combinations above, are its eigenvectors:

$$ra_1 = (y_1y_2 + x_1x_2)/\sqrt{2} = a_1, \qquad ra_2 = (-y_1x_2 + x_1y_2)/\sqrt{2} = a_2,$$
$$rb_1 = (y_1y_2 - x_1x_2)/\sqrt{2} = -b_1, \quad rb_2 = (-y_1x_2 - x_1y_2)/\sqrt{2} = -b_2.$$

Thus the four eigenvalues of r are $(1, 1, -1, -1)$. Under s a product acquires one minus sign for each y, so the eigenvalues are $(1, -1, 1, -1)$. Under d the variables x_i, y_i are exchanged, so $x_1x_2 \leftrightarrow y_1y_2$ and $x_1y_2 \leftrightarrow y_1x_2$, giving $(1, -1, -1, 1)$. Every element of C_{4v} is a product $r^j s^k$ (and $d = rs$: s followed by r sends $(x, y) \to (x, -y) \to (y, x)$, which is the reflection d), so each combination spans an invariant line, and r^2 has eigenvalue $(\pm1)^2 = 1$ on it. The eigenvalues on the classes $2C_4$, $2\sigma_v$, $2\sigma_d$ are those of r, s, d, and they reproduce the rows A_1, A_2, B_1, B_2 of the table. The coefficient vectors $(1, 0, 0, 1)$, $(0, 1, -1, 0)$, $(1, 0, 0, -1)$, $(0, 1, 1, 0)$, each divided by $\sqrt{2}$, in the basis $(x_1x_2, x_1y_2, y_1x_2, y_1y_2)$ are orthonormal, so the four combinations span the full four-dimensional tensor-product space.

17.6 Symmetric Group

17.6.1

(a) The objects $(abcd)$ are permuted to $(dacb)$. Write out a 4×4 matrix representation of this one permutation.

Hint. Compare with Exercise 17.4.2.

(b) Is the permutation $(abdc) \to (dacb)$ odd or even?

(c) Is either of the permutations in (a) and (b) a possible member of the D_4 group, which was the subject of Exercise 17.4.3? Why or why not?

As printed, (b) starts from $(abdc)$ rather than the $(abcd)$ of (a), so the two parts concern different permutations; (c), printed for a single permutation, is therefore answered for both, and for the permutation of (b) the answer depends on how the four slots are placed around the square.

Solution.

(a) The target column selects entries $4, 1, 3, 2$ of the original column:

$$\boxed{P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}}, \qquad P \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} d \\ a \\ c \\ b \end{pmatrix}.$$

As a map from old positions to new positions (a moves from 1 to 2, b from 2 to 4, d from 4 to 1), this is $(1\,2\,4)$ with position 3 fixed. It is a 3-cycle, has order three, and is even: a 3-cycle is a product of two transpositions, here $(1\,2\,4) = (1\,4)(1\,2)$ with the right factor applied first. Correspondingly $\det P = +1$.

(b) The printed input changes from $(abcd)$ in (a) to $(abdc)$ in (b). These are different permutations. The Instructor's Manual's answer *even* applies to (a), not to (b) as printed. For the literal

input (a,b,d,c), the target (d,a,c,b) selects entries $3,1,4,2$. Its matrix is

$$Q=\begin{pmatrix}0&0&1&0\\1&0&0&0\\0&0&0&1\\0&1&0&0\end{pmatrix}.$$

Its position cycle (a moves from 1 to 2, b from 2 to 4, c from 4 to 3, d from 3 to 1) is $(1\,2\,4\,3)$, which is a product of three transpositions, $(1\,2\,4\,3)=(1\,3)(1\,4)(1\,2)$. Thus the answer to (b) as printed is

$$\boxed{\text{odd},\qquad \det Q=-1}.$$

Equivalently the row-selection list $(3,1,4,2)$ has three inversions, also giving $(-1)^3=-1$.

(c) For the permutation in (a), the answer is unambiguously $\boxed{\text{no}}$: its order is three, while D_4 has only elements of orders one, two, and four (its classes I, $2C_4$, C_2, $2C_2'$, $2C_2''$ of 17.4.3 consist of the identity, two quarter-turns, and five half-turns). Alternatively, by Lagrange's theorem the order of every element divides $|D_4|=8$, and $3\nmid 8$.

For the literal permutation of (b), the answer depends on the placement of the slots around the square, which the printed statement does not specify. With slots $1,2,3,4$ placed in that cyclic order around the square, Q is not a symmetry: it sends the adjacent pair $(1,2)$ to $(2,4)$, a diagonal pair, whereas every symmetry of the square maps adjacent vertices to adjacent vertices. If instead the slots are placed in the cyclic order $1,2,4,3$, then $Q=(1\,2\,4\,3)$ is a quarter-turn. In the arrangement $(abdc)$ the slots $1,2,4,3$ hold a,b,c,d, so this is the placement with the objects in alphabetical order around the square, and Q moves a to the former place of b, b to that of c, c to that of d, and d to that of a. Thus the permutation of (b) is a member of D_4 (a quarter-turn) if a,b,c,d sit in this order around the square, but not if the slots $1,2,3,4$ do.

17.6.2

(a) The permutation group of four objects, S_4, has $4!=24$ elements. Treating the four elements of the cyclic group, C_4, as permutations, set up a 4×4 matrix representation of C_4. Note that C_4 is a subgroup of S_4.

(b) How do you know that this 4×4 matrix representation of C_4 must be reducible?

Solution.

(a) Let R cyclically shift a column of four entries to the left, $R(a,b,c,d)^T=(b,c,d,a)^T$. The four matrices are

$$U(I)=I_4,\qquad U(C_4)=R=\begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{pmatrix},$$

$$U(C_4^2)=R^2=\begin{pmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{pmatrix},\qquad U(C_4^3)=R^3=\begin{pmatrix}0&0&0&1\\1&0&0&0\\0&1&0&0\\0&0&1&0\end{pmatrix}.$$

Four shifts restore the original column, so $R^4=I_4$. The products satisfy $R^jR^k=R^{j+k\pmod 4}$, exactly the cyclic-group law. All four are permutation matrices, closed under products and containing

I_4 and the inverses $R^{-j} = R^{4-j}$, so they form a subgroup of the full permutation group S_4 (the printed note writes P_4; this denotes the permutation group S_4 just introduced).

(b) The representation must be reducible because every irreducible representation of the cyclic group C_4 is one-dimensional (Exercise 17.2.5). Alternatively, the squared dimensions of the inequivalent irreducible representations sum to the group order 4, so none has dimension four ($4^2 = 16 > 4$). Explicitly, the vector $(1,1,1,1)^T$ is fixed by all four matrices, giving a proper invariant line. More fully, let $\omega = i$ and define

$$v_k = \frac{1}{2}\begin{pmatrix} 1 \\ \omega^k \\ \omega^{2k} \\ \omega^{3k} \end{pmatrix}, \qquad k = 0, 1, 2, 3.$$

Cyclic shifting gives $Rv_k = \omega^k v_k$: component m of Rv_k ($m = 0, \ldots, 3$) is $\omega^{(m+1)k}/2 = \omega^k\,\omega^{mk}/2$; for $m = 3$ the shifted-in entry is the first one, $1/2 = \omega^{4k}/2$. Since $\omega^* = \omega^{-1}$, their inner products are

$$v_j^\dagger v_k = \frac{1}{4}\sum_{m=0}^{3} \omega^{(k-j)m} = \begin{cases} 1, & j = k, \\ 0, & j \neq k, \end{cases}$$

by the finite geometric-series formula: for $j \neq k$, $q = \omega^{k-j} \neq 1$ while $q^4 = 1$, so $\sum_{m=0}^{3} q^m = (1-q^4)/(1-q) = 0$. Thus $V = (v_0\, v_1\, v_2\, v_3)$ is unitary. Since $R^m v_k = \omega^{km} v_k$, $R^m V = V\,\mathrm{diag}(\omega^{km})$, and

$$\boxed{V^\dagger R^m V = \mathrm{diag}(1, i^m, (-1)^m, (-i)^m)}.$$

The representation reduces to the four one-dimensional irreducible representations of C_4.

17.6.3 The permutation group of four objects, S_4, has five classes.

(a) Determine the number of elements in each class of S_4 and identify one element of each class as a product of cycles.

(b) Two of the irreducible representations of S_4 are of dimension 1 (and are usually denoted A_1 and A_2). Noting that permutations can be classified as even or odd, find the characters of A_1 and A_2.

Hint. Set up a character table and fill in the A_1 and A_2 lines.

(c) One irreducible representation of S_4 (usually denoted E) is of dimension 2. Determine the dimensions of all the irreducible representations of S_4 other than A_1, A_2, and E.

(d) Complete the character table of S_4.

Hint. Only the even permutations have nonzero characters in the E representation.

Solution.

(a) Conjugation relabels a permutation's cycles and preserves their lengths. Conversely two permutations with the same cycle lengths are conjugate by a suitable relabeling. The five possibilities and their sizes are

cycle type	example	class size
$1+1+1+1$	I	1
$2+1+1$	(12)	$\binom{4}{2} = 6$
$2+2$	(12)(34)	$\frac{1}{2}\binom{4}{2} = 3$
$3+1$	(123)	$\binom{4}{3}(3-1)! = 8$
4	(1234)	$(4-1)! = 6$

.

For two transpositions, dividing by two removes the ordering of the two pairs. For a cycle of length k, cyclically shifting its notation describes the same cycle, leaving $k!/k = (k-1)!$ choices. The sizes sum to 24.

(b) The trivial character is $(1,1,1,1,1)$. The sign $\operatorname{sgn}\sigma = \pm 1$ of an even or odd permutation is the determinant of its permutation matrix, hence multiplicative, $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \operatorname{sgn}\tau$, so it is a one-dimensional representation. A cycle of length k is a product of $k-1$ transpositions, so the signs of the five classes are $(1,-1,1,1,-1)$. These give the A_1, A_2 rows below.

(c) There are as many irreducibles as classes, namely five, and the squares of their dimensions sum to the group order 24. The known dimensions $1,1,2$ account for $1+1+4=6$ of this sum. The other dimensions d, e obey $d^2 + e^2 = 24 - 6 = 18$. Trying $d = 1,2,3,4$ gives $e^2 = 17, 14, 9, 2$, of which only 9 is a square, so the only positive integer solution is $d = e = 3$.

(d) A useful independent construction supplies the remaining characters. The ordinary four-dimensional permutation representation has character equal to its number of fixed letters:

$$(4,2,0,1,0).$$

The all-ones vector is fixed by every permutation and spans an A_1 line; its orthogonal complement, the vectors with zero component sum, is invariant because permutation matrices are orthogonal, as in 17.4.2(b). Traces add over this direct sum, so the complement is a three-dimensional representation with character

$$\chi_{T_2} = (4,2,0,1,0) - (1,1,1,1,1) = (3,1,-1,0,-1).$$

Its squared character norm is $9 + 6(1) + 3(1) + 8(0) + 6(1) = 24 = |S_4|$. A character with $\sum_C |C|\,|\chi(C)|^2$ equal to the group order is irreducible, so this proves irreducibility. Multiplying by the sign representation gives a second irreducible (its squared norm is again 24, since the signs have modulus one),

$$\chi_{T_1} = (3,-1,-1,0,1).$$

This fixes our names T_1, T_2; interchanging their names is an equally valid convention.

To construct E, act on the three pair partitions

$$12|34, \qquad 13|24, \qquad 14|23.$$

The respective numbers of fixed partitions for the five classes are $(3,1,3,0,1)$. For example (12) fixes $12|34$ and exchanges the other two; $(12)(34)$ fixes all three; a 3-cycle cycles all three. A 4-cycle fixes $13|24$ and exchanges the other two. Remove the all-ones invariant line from this three-dimensional permutation space, exactly as above, to obtain $\chi_E = (3,1,3,0,1) - (1,1,1,1,1) = (2,0,2,-1,0)$. Its squared norm is $4 + 3(4) + 8(1) = 24$, again proving irreducibility.

The complete table is therefore

S_4	I	$6(12)$	$3(12)(34)$	$8(123)$	$6(1234)$
A_1	1	1	1	1	1
A_2	1	-1	1	1	-1
E	2	0	2	-1	0
T_1	3	-1	-1	0	1
T_2	3	1	-1	0	-1

.

Using the displayed class sizes, distinct rows are orthogonal; for instance $\langle T_1, T_2\rangle = 9 - 6 + 3 - 6 = 0$ and $\langle E, A_1\rangle = 2 + 6 - 8 = 0$. All five rows have norm squared 24, and their squared dimensions sum to 24, as required.

17.7 Continuous Groups

17.7.1 Determine three $SU(2)$ subgroups of $SU(3)$.

Solution.

Every $SU(2)$ matrix can be written

$$W = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

To see that this describes every such matrix, let its first row be (a, b), of norm one. Every unit row orthogonal to it has the form $e^{i\delta}(-b^*, a^*)$. The determinant is then $e^{i\delta}(|a|^2 + |b|^2) = e^{i\delta}$; requiring determinant one gives $e^{i\delta} = 1$. Conversely the displayed matrix has orthonormal rows and determinant $|a|^2 + |b|^2 = 1$. Embed this block into each of the three coordinate pairs:

$$\boxed{\begin{aligned} H_{12} &= \left\{ \begin{pmatrix} a & b & 0 \\ -b^* & a^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\}, \\ H_{13} &= \left\{ \begin{pmatrix} a & 0 & b \\ 0 & 1 & 0 \\ -b^* & 0 & a^* \end{pmatrix} \right\}, \\ H_{23} &= \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & a & b \\ 0 & -b^* & a^* \end{pmatrix} \right\}. \end{aligned}}$$

Each set has the stated constraint on a, b. Each such 3×3 matrix M is unitary, $M^\dagger M = I_3$, because the block W is unitary and the remaining diagonal entry is 1; expanding along the row of that isolated 1 gives $\det M = \det W = 1$. Hence each set lies in $SU(3)$. Products and adjoints act only in the chosen 2×2 block, so each set is a subgroup. Each embedding is injective and preserves multiplication, making these subgroups isomorphic to $SU(2)$. They fix, respectively, the third, second, and first basis vector.

For explicit generators, let E_{ij} have a single one in row i, column j. In the (i, j) block take

$$T_x = \frac{E_{ij} + E_{ji}}{2}, \qquad T_y = \frac{E_{ij} - E_{ji}}{2i}, \qquad T_z = \frac{E_{ii} - E_{jj}}{2}.$$

Using $E_{ab}E_{cd} = \delta_{bc}E_{ad}$, put $A = E_{ij}$, $B = E_{ji}$, and $D = E_{ii} - E_{jj}$. Then

$$[A, B] = D, \qquad [D, A] = 2A, \qquad [D, B] = -2B.$$

Substitution, with bilinearity and antisymmetry of the commutator giving $[A+B, A-B] = -2[A, B]$, $[A - B, D] = -[D, A] + [D, B]$, and $[D, A + B] = [D, A] + [D, B]$, displays all three commutators:

$$\begin{aligned} [T_x, T_y] &= \frac{[A + B, A - B]}{4i} = \frac{-2D}{4i} = i\,\frac{D}{2} = iT_z, \\ [T_y, T_z] &= \frac{[A - B, D]}{4i} = \frac{-2(A + B)}{4i} = i\,\frac{A + B}{2} = iT_x, \\ [T_z, T_x] &= \frac{[D, A + B]}{4} = \frac{A - B}{2} = i\,\frac{A - B}{2i} = iT_y. \end{aligned}$$

The Gell-Mann matrices $\lambda_1, \lambda_4, \lambda_6$ are $E_{ij} + E_{ji} = 2T_x$ and $\lambda_2, \lambda_5, \lambda_7$ are $-iE_{ij} + iE_{ji} = 2T_y$ for $(i,j) = (1,2), (1,3), (2,3)$, while $\lambda_3 = \mathrm{diag}(1,-1,0)$ and $\lambda_8 = \mathrm{diag}(1,1,-2)/\sqrt{3}$. In their terms the three triples are

$$\left(\frac{\lambda_1}{2}, \frac{\lambda_2}{2}, \frac{\lambda_3}{2}\right), \quad \left(\frac{\lambda_4}{2}, \frac{\lambda_5}{2}, \frac{\lambda_3 + \sqrt{3}\lambda_8}{4}\right), \quad \left(\frac{\lambda_6}{2}, \frac{\lambda_7}{2}, \frac{-\lambda_3 + \sqrt{3}\lambda_8}{4}\right).$$

Equations (17.62) and (17.63) omit a factor $1/2$ in their linear combinations. Direct addition gives $\sqrt{3}\lambda_8 - \lambda_3 = \mathrm{diag}(0,2,-2)$ and $\sqrt{3}\lambda_8 + \lambda_3 = \mathrm{diag}(2,0,-2)$. Their displayed diagonal matrices therefore equal half those combinations. The generator triples above include the needed factors.

17.7.2 Prove that the matrices $U(n)$ (unitary matrices of order n) form a group, and that $SU(n)$ (those with determinant unity) form a subgroup of $U(n)$.

Solution.

Let U, V be $n \times n$ unitary matrices: $U^\dagger U = UU^\dagger = I_n$ and likewise for V. Their product is unitary because

$$(UV)^\dagger (UV) = V^\dagger U^\dagger UV = V^\dagger V = I_n,$$

and similarly $(UV)(UV)^\dagger = I_n$. Thus $U(n)$ is closed. Matrix multiplication is associative, I_n is a unitary identity, and the inverse of U is $U^\dagger$, itself unitary: $(U^\dagger)^\dagger U^\dagger = UU^\dagger = I_n$. This proves all group axioms for $U(n)$.

For $SU(n)$, impose $\det U = 1$ in addition. If $U, V \in SU(n)$, then

$$\det(UV) = \det U \, \det V = 1,$$

so the product is in $SU(n)$. The identity has determinant one, and $\det U^{-1} = (\det U)^{-1} = 1$, so the inverse is in $SU(n)$. Associativity is inherited from $U(n)$. Hence $\boxed{SU(n) \text{ is a subgroup of } U(n)}$. For comparison, unitarity alone gives $|\det U|^2 = \det(U^\dagger U) = 1$; restricting this unit-modulus determinant to the value one is precisely the special-unitary condition.

17.7.3 Using Eq. (17.56) for the matrix elements of $SU(2)$ corresponding to rotations about the coordinate axes, find the matrix corresponding to a rotation defined by Euler angles (α, β, γ). The Euler angles are defined in Section 3.4.

Solution.

With the Euler convention of Section 3.4, the successive passive coordinate rotations are about z, then the intermediate y, then the final z. Equation (17.56) therefore prescribes

$$U(\alpha, \beta, \gamma) = e^{i\gamma\sigma_3/2} e^{i\beta\sigma_2/2} e^{i\alpha\sigma_3/2}.$$

Since $\sigma_j^2 = I$, separating even and odd powers of the exponential gives $e^{it\sigma_j/2} = I\cos(t/2) + i\sigma_j \sin(t/2)$. In particular, with $\sigma_3 = \mathrm{diag}(1,-1)$ and $i\sigma_2 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$,

$$Z(t) = \begin{pmatrix} e^{it/2} & 0 \\ 0 & e^{-it/2} \end{pmatrix}, \qquad Y(\beta) = \begin{pmatrix} \cos(\beta/2) & \sin(\beta/2) \\ -\sin(\beta/2) & \cos(\beta/2) \end{pmatrix}.$$

First multiply on the right: the diagonal $Z(\alpha)$ multiplies the first column of $Y(\beta)$ by $e^{i\alpha/2}$ and the second by $e^{-i\alpha/2}$, so

$$Y(\beta)Z(\alpha) = \begin{pmatrix} c\, e^{i\alpha/2} & s\, e^{-i\alpha/2} \\ -s\, e^{i\alpha/2} & c\, e^{-i\alpha/2} \end{pmatrix}, \qquad c = \cos(\beta/2), \ s = \sin(\beta/2).$$

Left multiplication by $Z(\gamma)$ multiplies its first row by $e^{i\gamma/2}$ and its second by $e^{-i\gamma/2}$. Hence

$$\boxed{U(\alpha,\beta,\gamma)=\begin{pmatrix} e^{i(\alpha+\gamma)/2}\cos(\beta/2) & e^{i(\gamma-\alpha)/2}\sin(\beta/2) \\ -e^{i(\alpha-\gamma)/2}\sin(\beta/2) & e^{-i(\alpha+\gamma)/2}\cos(\beta/2)\end{pmatrix}}.$$

Its determinant is $c^2+s^2=1$ and its columns are orthonormal, so it is in $SU(2)$. The rotation determines its $SU(2)$ matrix only up to sign, and the stated exponential product fixes that sign. Changing a single Euler angle by 2π, which leaves the rotation unchanged, changes the sign of U; the two matrices $U,-U$ correspond to the same spatial rotation, consistent with the double cover of $SO(3)$ by $SU(2)$.

17.7.4 For a product of three quarks, the member of $SU(3)$ representation **10** with $(I_3,Y)=\left(\frac{3}{2},1\right)$ is uuu.

(a) Apply operators in the root diagram for $SU(3)$, Fig. 17.8, to obtain all the remaining members of the decuplet comprising the representation **10**.

(b) The two representations **8** can be chosen to have for $I_3=\frac{1}{2}$, $Y=1$ the respective members $\psi_1\left(\frac{1}{2},1\right)=(ud-du)u$ and $\psi_2\left(\frac{1}{2},1\right)=2uud-udu-duu$. Briefly explain why this choice is possible.

(c) Using the operators in the root diagram and the above $\psi_1\left(\frac{1}{2},1\right)$, find expressions for $\psi_1\left(-\frac{1}{2},1\right)$, $\psi_1(-1,0)$, $\psi_1(1,0)$, $\psi_1\left(-\frac{1}{2},-1\right)$, and $\psi_1\left(\frac{1}{2},-1\right)$.

(d) Taking each of the six ψ_1 functions you now have, apply an operator that will convert it into $\psi_1(0,0)$. Show that you obtain exactly two linearly independent $\psi_1(0,0)$, thereby justifying the claim that the ψ_1 are an octet at the points shown in Fig. 17.10.

(e) Show that the octet built starting from $\psi_2\left(\frac{1}{2},1\right)$ is linearly independent from that built from ψ_1.

(f) Find the wave function $\psi(0,0)$ that is linearly independent of all the $\psi(0,0)$ functions found in parts (a)–(e). It is the sole member of the representation **1**.

Use the ordered flavor tensor space with u,d,s orthonormal, and the root convention $I_-u=d$, $U_-d=s$, $V_-u=s$ from Fig. 17.8. The conflicting U,V matrix labels in Eq. (17.67) are corrected below.

Solution.

The matrices labeled $U_\pm$ and $V_\pm$ in Eq. (17.67) are interchanged relative to Eqs. (17.64)–(17.66) and Fig. 17.8. We use the consistent root-diagram convention below.

Use an ordered tensor product, so $uds=u\otimes d\otimes s$; the 27 ordered words are orthonormal. The single-quark weights and root actions are

$$u:(1/2,1/3),\qquad d:(-1/2,1/3),\qquad s:(0,-2/3),$$

$$I_-u=d,\quad I_+d=u,\qquad U_-d=s,\quad U_+s=d,\qquad V_-u=s,\quad V_+s=u,$$

with all unlisted single-quark actions zero. On a word the operator acts on all three slots:

$$T(abc)=(Ta)bc+a(Tb)c+ab(Tc).$$

Thus the total generators are sums of the three single-slot generators, not products of them. A word with occupation numbers (n_u, n_d, n_s) has

$$I_3 = (n_u - n_d)/2, \qquad Y = (n_u + n_d - 2n_s)/3.$$

(a) The decuplet. Let $\mathcal{S}(aab) = aab + aba + baa$ for $a \neq b$, and

$$\mathcal{S}(uds) = uds + usd + dus + dsu + sud + sdu.$$

The ten normalized states are

I_3	Y	state
$3/2$	1	uuu
$1/2$	1	$\mathcal{S}(uud)/\sqrt{3}$
$-1/2$	1	$\mathcal{S}(ddu)/\sqrt{3}$
$-3/2$	1	ddd
1	0	$\mathcal{S}(uus)/\sqrt{3}$
0	0	$\mathcal{S}(uds)/\sqrt{6}$
-1	0	$\mathcal{S}(dds)/\sqrt{3}$
$1/2$	-1	$\mathcal{S}(ssu)/\sqrt{3}$
$-1/2$	-1	$\mathcal{S}(ssd)/\sqrt{3}$
0	-2	sss

To derive every coefficient, let $\mathcal{S}$ be the sum of the $N = 3!/(n_u!\, n_d!\, n_s!)$ distinct words with occupations (n_u, n_d, n_s), as in $\mathcal{S}(uud)$ above, and $D_{n_u,n_d,n_s} = \mathcal{S}/\sqrt{N}$ the normalized sum. It is also labeled by any word with these occupations, e.g. $D_{uud} = D_{2,1,0}$ and $D_{uss} = D_{ssu}$. A root operator $T_{b\leftarrow a}$ changing a into b produces each target word $n_b + 1$ times, once from each of its $n_b + 1$ letters b (changing that letter back into a recovers a source word). Thus $T_{b\leftarrow a}\mathcal{S} = (n_b + 1)\mathcal{S}'$, where $\mathcal{S}'$ is the sum of the N' words with occupations $n_a - 1$ and $n_b + 1$ in place of n_a and n_b, and the formula for N gives $N'/N = n_a!\, n_b!/[(n_a - 1)!\,(n_b + 1)!] = n_a/(n_b + 1)$. With $D' = \mathcal{S}'/\sqrt{N'}$, the action on normalized sums is therefore

$$T_{b\leftarrow a} D_{n_u,n_d,n_s} = \frac{(n_b + 1)\,\mathcal{S}'}{\sqrt{N}} = (n_b + 1)\sqrt{\frac{N'}{N}}\, D' = \sqrt{n_a(n_b + 1)}\, D'.$$

For example $I_- uuu = duu + udu + uud = \sqrt{3} D_{uud}$, and

$$I_- \mathcal{S}(uud) = (dud + udd) + (ddu + udd) + (ddu + dud) = 2\,\mathcal{S}(ddu),$$

each target word arising once from each of its two d's; since $D_{uud} = \mathcal{S}(uud)/\sqrt{3}$ and $D_{udd} = \mathcal{S}(ddu)/\sqrt{3}$, this is $I_- D_{uud} = 2 D_{udd}$, in agreement with $\sqrt{n_u(n_d + 1)} = \sqrt{2 \cdot 2}$. Successive I_- operations down the top row have coefficients $\sqrt{3}, 2, \sqrt{3}$. The remaining rows follow from

$$V_- D_{uuu} = \sqrt{3} D_{uus}, \quad I_- D_{uus} = \sqrt{2} D_{uds}, \quad I_- D_{uds} = \sqrt{2} D_{dds},$$
$$V_- D_{uus} = 2 D_{uss}, \qquad I_- D_{uss} = D_{dss}, \qquad V_- D_{uss} = \sqrt{3} D_{sss}.$$

These paths reach all ten states listed. The root actions preserve complete permutation symmetry; the symmetric three-quark space has $\binom{3+3-1}{3} = 10$ occupation states, so none is omitted.

(b) Choosing the two octet highest states. At weight $(1/2, 1)$ the ordered basis is (uud, udu, duu). The decuplet coefficient vector is $(1, 1, 1)/\sqrt{3}$. The two proposed vectors have coefficients

$$a = (0, 1, -1), \qquad b = (2, -1, -1).$$

Their sums are zero, so both are orthogonal to the decuplet. Also $a \cdot b = -1 + 1 = 0$ and their squared norms are $2, 6$. They span the two-dimensional remaining weight space. Since I_+ sends each of uud, udu, duu to uuu, it maps a combination of them to its coefficient sum times uuu; both sums vanish, so I_+ annihilates both vectors. U_+ and V_+ annihilate them because they contain no strange quark. They can therefore be taken as the two highest states. The constructions below exhibit all eight states generated from each, thereby identifying the two octets directly.

(c) The six nonzero weights of the first octet. Use unnormalized states to keep every replacement visible. Starting with $A = udu - duu$, the following path gives all requested weights:

state	(I_3, Y)	word expression and operation
A	$(1/2, 1)$	$udu - duu$
B	$(-1/2, 1)$	$I_- A = udd - dud$
C	$(-1, 0)$	$V_- B = sdd - dsd$
D	$(-1/2, -1)$	$U_- C = sds - dss$
E	$(1/2, -1)$	$I_+ D = sus - uss$
F	$(1, 0)$	$V_+ E = suu - usu$

.

For clarity the first and third steps are

$$I_-(udu - duu) = (ddu + udd) - (ddu + dud) = udd - dud,$$

$$U_-(sdd - dsd) = (ssd + sds) - (ssd + dss) = sds - dss.$$

The remaining steps are equally explicit:

$$\begin{aligned} V_-(udd - dud) &= sdd - dsd, \\ I_+(sds - dss) &= sus - uss, \\ V_+(sus - uss) &= (uus + suu) - (uus + usu) = suu - usu. \end{aligned}$$

Every listed vector has squared norm two. Dividing them by $\sqrt{2}$ produces normalized nonzero-weight states.

(d) The two zero-weight states. Approach weight $(0, 0)$ from each of the six preceding weights. Define

$$Z_1 = sdu + uds - dsu - dus, \qquad Z_2 = usd + uds - sud - dus.$$

The complete list of results is

operation	result	in the (Z_1, Z_2) basis
$V_- A$	$sdu + uds - dsu - dus$	Z_1
$U_- B$	$usd + uds - sud - dus$	Z_2
$I_+ C$	$sud + sdu - usd - dsu$	$Z_1 - Z_2$
$V_+ D$	$uds + sdu - dus - dsu$	Z_1
$U_+ E$	$dus + sud - uds - usd$	$-Z_2$
$I_- F$	$sdu + sud - dsu - usd$	$Z_1 - Z_2$

.

For example $V_- A$ changes either u in each term of $udu - duu$, giving $(sdu + uds) - (dsu + dus) = Z_1$; all six rows are obtained by the same slot-by-slot rule. In the third and last rows, $Z_1 - Z_2 = sdu + sud - dsu - usd$ because the terms uds and dus cancel. In the order $(uds, usd, dus, dsu, sud, sdu)$ of the

six zero-weight words, the coefficient vectors are $Z_1 = (1, 0, -1, -1, 0, 1)$ and $Z_2 = (1, 1, -1, 0, -1, 0)$, so the inner products are

$$\langle Z_1, Z_1\rangle = \langle Z_2, Z_2\rangle = 4, \qquad \langle Z_1, Z_2\rangle = 2.$$

Their Gram determinant is $16 - 4 = 12 > 0$, so they are independent, while the table proves that all six incoming results are in their span. One orthonormal zero-weight pair is

$$\boxed{z_1 = Z_1/2, \qquad z_2 = (2Z_2 - Z_1)/\sqrt{12}}.$$

Indeed the second numerator is orthogonal to Z_1 and has norm squared $4(4) + 4 - 4(2) = 12$. The six states at the six distinct nonzero weights and these two independent zero-weight states make exactly eight states.

(e) Independence of the second octet. Interchange slots 1 and 2, calling this unitary operation P_{12}. The first highest state is odd under it: $P_{12}(udu - duu) = -(udu - duu)$. The second is even: $P_{12}(2uud - udu - duu) = 2uud - udu - duu$. Every total root operator commutes with P_{12}, because it sums identical actions on all three slots. Hence all states generated from the first are odd and all from the second are even. These eigenspaces are orthogonal: if $P_{12}v = -v$ and $P_{12}w = w$, then $\langle v, w\rangle = \langle P_{12}v, P_{12}w\rangle = -\langle v, w\rangle = 0$.

For an explicit check that the second space also has eight states, use the same path I_-, V_-, U_-, I_+, V_+ as in (c):

(I_3, Y)	ψ_2
$(1/2, 1)$	$2uud - udu - duu$
$(-1/2, 1)$	$dud + udd - 2ddu$
$(-1, 0)$	$dsd + sdd - 2dds$
$(-1/2, -1)$	$2ssd - sds - dss$
$(1/2, -1)$	$2ssu - sus - uss$
$(1, 0)$	$usu + suu - 2uus$

.

For instance lowering the first row gives $2(dud+udd)-(ddu+udd)-(ddu+dud) = dud+udd-2ddu$. The second and fourth steps change the single u or d of each word; the third and fifth are

$$\begin{aligned} U_-(dsd + sdd - 2dds) &= (ssd + dss) + (ssd + sds) - 2(sds + dss) \\ &= 2ssd - sds - dss, \\ V_+(2ssu - sus - uss) &= 2(usu + suu) - (uus + suu) - (uus + usu) \\ &= usu + suu - 2uus. \end{aligned}$$

Applying V_- to its first row and U_- to its second yields

$$\begin{aligned} T_1 &= 2(sud + usd) - (sdu + uds) - (dsu + dus) \\ &= 2sud + 2usd - sdu - uds - dsu - dus, \\ T_2 &= (sud + dus) + (usd + uds) - 2(sdu + dsu) \\ &= sud + dus + usd + uds - 2sdu - 2dsu. \end{aligned}$$

In the word order used in (d) their coefficient vectors are $T_1 = (-1, 2, -1, -1, 2, -1)$ and $T_2 = (1, 1, 1, -2, 1, -2)$, so $|T_1|^2 = |T_2|^2 = 12$ and $\langle T_1, T_2\rangle = -1 + 2 - 1 + 2 + 2 + 2 = 6$. Their Gram matrix is $\begin{pmatrix} 12 & 6 \\ 6 & 12 \end{pmatrix}$ with determinant $108 > 0$, giving two more independent states at zero weight.

The six outer states have norm squared $4+1+1=6$; a normalized central pair is $T_1/\sqrt{12}$ and $(2T_2-T_1)/6$, since $\langle T_1, 2T_2-T_1\rangle = 12-12=0$ and $|2T_2-T_1|^2 = 4(12)+12-4(6)=36$.

For completeness, the slot-symmetric subspace under P_{12} has dimension $6\cdot 3=18$: the first two slots carry one of the six symmetric combinations uu, dd, ss, $ud+du$, $us+su$, $ds+sd$, and the third slot any of three quarks. (Likewise the three antisymmetric pairs $ud-du$, $us-su$, $ds-sd$ give the $3\cdot 3=9$-dimensional slot-antisymmetric space.) The eighteen-dimensional slot-symmetric subspace contains the completely symmetric decuplet, which takes ten dimensions, leaving eight. This orthogonal complement is invariant under $SU(3)$, because both the slot exchange and full symmetrization commute with $U\otimes U\otimes U$. The second construction supplies eight independent vectors there and hence exhausts that invariant complement. The first octet lies in the nine-dimensional slot-antisymmetric space; the remaining one-dimensional component is found next.

(f) The singlet. There are six ordered words at zero weight, in the order

$$(uds, usd, dus, dsu, sud, sdu).$$

The decuplet supplies the all-ones vector; the two octets supply Z_1, Z_2, T_1, T_2. These five vectors are independent: Z_1, Z_2 have odd P_{12} parity, while the other three have even parity; T_1, T_2 are independent and have coefficient sum zero, whereas the all-ones vector has coefficient sum six, so it is not in their span. The three even vectors therefore span the three-dimensional P_{12}-even part of the zero-weight space (spanned by $uds+dus$, $usd+sud$, $dsu+sdu$), and a vector orthogonal to them is P_{12}-odd:

$$\psi = \alpha(uds-dus)+\beta(usd-sud)+\gamma(dsu-sdu).$$

Since $Z_1 = (uds-dus)-(dsu-sdu)$ and $Z_2=(uds-dus)+(usd-sud)$, and each bracket has squared norm two, orthogonality to Z_1, Z_2 requires $2\alpha-2\gamma=0$ and $2\alpha+2\beta=0$. Taking $\alpha=\gamma=-\beta=1/\sqrt{6}$, the normalized vector orthogonal to all five is

$$\boxed{\psi_{\mathbf{1}} = \frac{uds-usd-dus+dsu+sud-sdu}{\sqrt{6}}}.$$

As a check, its coefficients $(1,-1,-1,1,1,-1)/\sqrt{6}$ have zero sum and zero dot products with the vectors Z_1, Z_2, T_1, T_2 listed above, and squared norm one.

Finally this is an invariant state, not merely an unused vector: it is the fully antisymmetric tensor $\psi_{\mathbf{1}} = \frac{1}{\sqrt{6}}\sum_{a,b,c}\varepsilon_{abc}\,q_aq_bq_c$, where $(q_1,q_2,q_3)=(u,d,s)$. A matrix U acts by $Uq_a=\sum_i U_{ia}q_i$, so

$$(U\otimes U\otimes U)\psi_{\mathbf{1}} = \frac{1}{\sqrt{6}}\sum_{i,j,k}\Big(\sum_{a,b,c}\varepsilon_{abc}U_{ia}U_{jb}U_{kc}\Big)q_iq_jq_k = \det U\,\psi_{\mathbf{1}}$$

by the Levi-Civita expansion of the determinant, $\sum_{a,b,c}\varepsilon_{abc}U_{ia}U_{jb}U_{kc}=\varepsilon_{ijk}\det U$; for $U\in SU(3)$ this is $\psi_{\mathbf{1}}$ itself. Equivalently each root operator produces pairs of opposite terms and annihilates it. Thus it spans the trivial representation. The first octet starts orthogonal to this invariant singlet and stays orthogonal under the unitary group action. It therefore fills the eight-dimensional complement of the singlet within the nine-dimensional slot-antisymmetric space.

One can also verify irreducibility directly from the displayed states. In either octet every state at a nonzero weight other than the starting state has a nonzero raising image: use I_+ on B, C, V_+ on D, U_+ on E, F. In the first octet $I_+B=(uud+udu)-(uud+duu)=A$ and $U_+F=duu-udu=-A$, while $I_+C=Z_1-Z_2$, $V_+D=Z_1$, and $U_+E=-Z_2$ are rows of the table in (d). In the second octet write $A_2=2uud-udu-duu$ for its highest state and $F_2=usu+suu-2uus$ for its state at $(1,0)$; the same five operators applied to its rows at

$(-\frac{1}{2},1)$, $(-1,0)$, $(-\frac{1}{2},-1)$, $(\frac{1}{2},-1)$, $(1,0)$ give A_2, T_1-T_2, T_1, $-T_2$, and $-A_2$. At zero weight the replacement rules give $I_+Z_1 = suu+uus-usu-uus = F$, $I_+Z_2 = usu+uus-suu-uus = -F$, $V_+Z_1 = (udu+udu)-(duu+duu) = 2A$, and $V_+Z_2 = uud+udu-uud-duu = A$; hence

$$I_+(aZ_1+bZ_2) = (a-b)F, \qquad V_+(aZ_1+bZ_2) = (2a+b)A.$$

Both vanish only if $a=b$ and $2a+b=0$, that is, $a=b=0$. The same equations hold with Z_j replaced by T_j and A,F by A_2,F_2, since $I_+T_1 = -I_+T_2 = F_2$ and $V_+T_1 = 2V_+T_2 = 2A_2$. Thus the common kernel of I_+,U_+,V_+ in each octet is exactly its one-dimensional highest-state line. In the decuplet the occupation formula shows that any d can be raised by I_+ and any s by V_+; only uuu is killed by all raising operators. Every nonzero invariant summand of a finite-dimensional unitary representation has a highest state. The commuting Hermitian operators Y, I_3 preserve that summand and admit a simultaneous eigenbasis there. Choose a weight vector of largest Y, then largest I_3 among that Y eigenspace. Both U_+,V_+ would increase Y, while I_+ would increase I_3 at the same Y; therefore all three annihilate the chosen vector. Two nonzero orthogonal summands would give two independent common-kernel vectors, which the preceding calculations exclude. All three constructed multiplets are therefore irreducible.

The dimensions and the explicit independent states give

$$\boxed{\mathbf{3}\otimes\mathbf{3}\otimes\mathbf{3} = \mathbf{10}\oplus\mathbf{8}\oplus\mathbf{8}\oplus\mathbf{1}}, \qquad 10+8+8+1=27.$$

17.8 Lorentz Group

17.8.1 Show that in $3+1$ dimensions (this means three spatial dimensions plus time), a boost in the xy plane at an angle θ from the x direction has, in coordinates (x^0,x^1,x^2,x^3), the generator

$$\mathsf{S} = i\begin{pmatrix} 0 & \cos\theta & \sin\theta & 0 \\ \cos\theta & 0 & 0 & 0 \\ \sin\theta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Use $x^0 = ct$, metric $\mathrm{diag}(1,-1,-1,-1)$, and $U(\rho) = e^{i\rho\mathsf{S}}$ for the passive boost. The exercise does not restate these conventions, and the form of S depends on them; for example, the active boost, or $U = e^{-i\rho\mathsf{S}}$, would give $-\mathsf{S}$.

Solution.

Let $x^0 = ct$ and $\mathbf{n} = (\cos\theta,\sin\theta,0)$. For an infinitesimal passive boost of rapidity $\delta\rho$, the frame velocity is $c\tanh(\delta\rho)\,\mathbf{n} = c\,\delta\rho\,\mathbf{n} + O(\delta\rho^3)$. To first order $\gamma = \cosh\delta\rho = 1$, and the spatial coordinates change as $\mathbf{r}' = \mathbf{r} - \mathbf{v}t$ with $t = x^0/c$. Thus

$$\delta x = -\delta\rho\cos\theta\,x^0, \qquad \delta y = -\delta\rho\sin\theta\,x^0, \qquad \delta z = 0.$$

Write the most general linear first-order time change as $\delta x^0 = -\delta\rho(ax+by+dz+ex^0)$. Preservation of $(x^0)^2 - x^2 - y^2 - z^2$ to first order gives

$$2x^0\delta x^0 - 2x\delta x - 2y\delta y = 2x^0\delta\rho\big[(\cos\theta-a)x + (\sin\theta-b)y - dz - ex^0\big] = 0.$$

Because this polynomial vanishes for every x, y, z, x^0, each coefficient vanishes: $a = \cos\theta$, $b = \sin\theta$, $d = e = 0$. Hence

$$x' = (I_4 - \delta\rho\, B)x + O(\delta\rho^2), \qquad B = \begin{pmatrix} 0 & \cos\theta & \sin\theta & 0 \\ \cos\theta & 0 & 0 & 0 \\ \sin\theta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The book defines a generator by $U(\delta\rho) = I_4 + i\delta\rho\, S + O(\delta\rho^2)$. Comparing first-order coefficients, $iS = -B$, so

$$\boxed{S = iB},$$

which is the requested generator. With $\eta = \mathrm{diag}(1, -1, -1, -1)$ one also checks $B^T\eta + \eta B = 0$, the infinitesimal Lorentz condition: it is the first-order part of $U^T\eta U = \eta$ for $U = I_4 - \delta\rho\, B$, since $(I_4 - \delta\rho\, B^T)\eta(I_4 - \delta\rho\, B) = \eta - \delta\rho\,(B^T\eta + \eta B) + O(\delta\rho^2)$.

17.8.2

(a) Show that the generator in Exercise 17.8.1 produces a Lorentz transformation matrix for rapidity ρ given by

$$U(\rho;\theta) = \begin{pmatrix} \cosh\rho & -\cos\theta\sinh\rho & -\sin\theta\sinh\rho & 0 \\ -\cos\theta\sinh\rho & \sin^2\theta + \cos^2\theta\cosh\rho & \cos\theta\sin\theta(\cosh\rho - 1) & 0 \\ -\sin\theta\sinh\rho & \cos\theta\sin\theta(\cosh\rho - 1) & \cos^2\theta + \sin^2\theta\cosh\rho & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Note. This transformation matrix is symmetric. All single boosts (in any spatial direction) have symmetric transformation matrices.

(b) Verify that the transformation matrix of part (a) is consistent with (1) rotating the spatial coordinates to align the boost direction with a coordinate axis, (2) performing a boost in the direction of that axis using Eq. (17.70), and (3) rotating back to the original coordinate system.

Solution.

(a) Put $c_\theta = \cos\theta$, $s_\theta = \sin\theta$, and let B be the real matrix in 17.8.1, so $S = iB$. Direct multiplication gives

$$B^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & c_\theta^2 & c_\theta s_\theta & 0 \\ 0 & c_\theta s_\theta & s_\theta^2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad B^3 = B,$$

using $c_\theta^2 + s_\theta^2 = 1$. Thus every positive odd power is B and every positive even power is B^2. Separating the exponential series,

$$\begin{aligned} U(\rho;\theta) = e^{i\rho S} = e^{-\rho B} &= I_4 - B\left(\rho + \frac{\rho^3}{3!} + \cdots\right) + B^2\left(\frac{\rho^2}{2!} + \frac{\rho^4}{4!} + \cdots\right) \\ &= I_4 - B\sinh\rho + B^2(\cosh\rho - 1). \end{aligned}$$

Writing $C=\cosh\rho$, $H=\sinh\rho$ and inserting the entries yields

$$\boxed{U(\rho;\theta)=\begin{pmatrix} C & -c_\theta H & -s_\theta H & 0\\ -c_\theta H & s_\theta^2+c_\theta^2 C & c_\theta s_\theta(C-1) & 0\\ -s_\theta H & c_\theta s_\theta(C-1) & c_\theta^2+s_\theta^2 C & 0\\ 0 & 0 & 0 & 1\end{pmatrix}}.$$

For example the xx entry is $1+c_\theta^2(C-1)=s_\theta^2+c_\theta^2 C$. The matrix is symmetric. Let $\mathbf{e}_0=(1,0,0,0)^T$, $\mathbf{e}_3=(0,0,0,1)^T$, $N=(0,c_\theta,s_\theta,0)^T$, and $N_\perp=(0,-s_\theta,c_\theta,0)^T$. Direct multiplication gives $B\mathbf{e}_0=N$, $BN=\mathbf{e}_0$, and $BN_\perp=B\mathbf{e}_3=0$, so the expression $U=I_4-BH+B^2(C-1)$ yields $U\mathbf{e}_0=C\mathbf{e}_0-HN$, $UN=-H\mathbf{e}_0+CN$, $UN_\perp=N_\perp$, and $U\mathbf{e}_3=\mathbf{e}_3$. Thus along the time and $\mathbf{n}$ directions U is the 2×2 block $M=\begin{pmatrix} C & -H\\ -H & C\end{pmatrix}$, and on the two perpendicular directions it is the identity. The basis $(\mathbf{e}_0,N,N_\perp,\mathbf{e}_3)$ is obtained from the standard one by a spatial rotation, so η keeps the form $\mathrm{diag}(1,-1,-1,-1)$ in it, and $U^T\eta U=\eta$ reduces to

$$M^T\begin{pmatrix}1&0\\0&-1\end{pmatrix}M=\begin{pmatrix}C^2-H^2&0\\0&H^2-C^2\end{pmatrix}=\begin{pmatrix}1&0\\0&-1\end{pmatrix},$$

while $\det U=\det M=C^2-H^2=1$.

(b) Rotate the spatial coordinates so that $\mathbf{n}$ becomes the x direction:

$$R=\begin{pmatrix}1&0&0&0\\0&c_\theta&s_\theta&0\\0&-s_\theta&c_\theta&0\\0&0&0&1\end{pmatrix},\qquad U_x=\begin{pmatrix}C&-H&0&0\\-H&C&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}.$$

First applying R, then U_x, then $R^{-1}=R^T$ gives R^TU_xR. The first multiplication is

$$U_xR=\begin{pmatrix}C&-Hc_\theta&-Hs_\theta&0\\-H&Cc_\theta&Cs_\theta&0\\0&-s_\theta&c_\theta&0\\0&0&0&1\end{pmatrix}.$$

Left multiplication by R^T forms c_θ times the x row (the second row) of U_xR minus s_θ times its y row (the third row) for the new x row, and s_θ times the x row plus c_θ times the y row for the new y row:

$$c_\theta(-H,Cc_\theta,Cs_\theta,0)-s_\theta(0,-s_\theta,c_\theta,0)=(-c_\theta H,\ s_\theta^2+c_\theta^2C,\ c_\theta s_\theta(C-1),\ 0),$$
$$s_\theta(-H,Cc_\theta,Cs_\theta,0)+c_\theta(0,-s_\theta,c_\theta,0)=(-s_\theta H,\ c_\theta s_\theta(C-1),\ c_\theta^2+s_\theta^2C,\ 0).$$

These are exactly the two middle rows of the boxed matrix. The other rows are unchanged. This verifies the rotation–boost–inverse-rotation construction.

17.8.3 Obtain the Lorentz transformation matrix for a boost of finite amount ρ' in the x direction followed by a finite boost ρ'' in the y direction, with both rapidities real and nonzero. Show that there are no values of ρ and θ that can bring this transformation to the form given in Exercise 17.8.2.

Solution.

Write $C'=\cosh\rho'$, $H'=\sinh\rho'$ and similarly C'',H'' for ρ''. Since the x boost acts first, the

product is $U_y(\rho'')U_x(\rho')$:

$$U_yU_x = \begin{pmatrix} C'' & 0 & -H'' & 0 \\ 0 & 1 & 0 & 0 \\ -H'' & 0 & C'' & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} C' & -H' & 0 & 0 \\ -H' & C' & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
$$= \boxed{\begin{pmatrix} C''C' & -C''H' & -H'' & 0 \\ -H' & C' & 0 & 0 \\ -H''C' & H''H' & C'' & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}.$$

The spatial xy entry is zero, whereas the yx entry is $H''H'$. For real nonzero ρ', ρ'', this product is nonzero. The matrix is therefore not symmetric. Every single boost in 17.8.2 is symmetric, so no choice of ρ, θ can produce this two-boost matrix.

The impossibility statement requires both rapidities to be nonzero. If $\rho'' = 0$, the result is the single x boost, with $\theta = 0$ and $\rho = \rho'$. If $\rho' = 0$, it is the single y boost, with $\theta = \pi/2$ and $\rho = \rho''$. If both vanish, the result is the identity.

17.9 Lorentz Covariance of Maxwell's Equations

17.9.1 Apply the Lorentz transformation of Eq. (17.80) to $F^{\mu\lambda}$ as given in Eq. (17.79). Verify that the result is a matrix F' whose elements confirm the results given in Eqs. (17.82) and (17.83).

Solution.

Factor out the common ε_0 in Eq. (17.79) and use $\beta = v/c$, $\gamma = (1-\beta^2)^{-1/2} > 0$, so $\gamma^2(1-\beta^2) = 1$. In matrix form, with $x^0 = ct$, the field tensor of Eq. (17.79) and the boost of Eq. (17.80), along $x^3 = z$, are

$$\frac{F}{\varepsilon_0} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -cB_z & cB_y \\ E_y & cB_z & 0 & -cB_x \\ E_z & -cB_y & cB_x & 0 \end{pmatrix}, \qquad U = \begin{pmatrix} \gamma & 0 & 0 & -\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta\gamma & 0 & 0 & \gamma \end{pmatrix}.$$

The transformation is $F' = UFU^T$; the particular boost matrix is symmetric, so this is UFU, as in Eq. (17.81). Left multiplication by U replaces row 0 by $\gamma(\text{row } 0) - \beta\gamma(\text{row } 3)$ and row 3 by $-\beta\gamma(\text{row } 0) + \gamma(\text{row } 3)$, leaving rows 1 and 2 unchanged. The first multiplication therefore gives

$$\frac{UF}{\varepsilon_0} = \begin{pmatrix} -\beta\gamma E_z & -\gamma E_x + \beta\gamma cB_y & -\gamma E_y - \beta\gamma cB_x & -\gamma E_z \\ E_x & 0 & -cB_z & cB_y \\ E_y & cB_z & 0 & -cB_x \\ \gamma E_z & \beta\gamma E_x - \gamma cB_y & \beta\gamma E_y + \gamma cB_x & \beta\gamma E_z \end{pmatrix}.$$

Right multiplication by U mixes only columns 0 and 3 in the same way: column 0 becomes $\gamma(\text{column } 0) - \beta\gamma(\text{column } 3)$ and column 3 becomes $-\beta\gamma(\text{column } 0) + \gamma(\text{column } 3)$. The six independent entries are therefore

$$F'^{01}/\varepsilon_0 = -\gamma(E_x - \beta cB_y), \qquad F'^{02}/\varepsilon_0 = -\gamma(E_y + \beta cB_x),$$
$$F'^{03}/\varepsilon_0 = (\beta^2\gamma^2 - \gamma^2)E_z = -E_z, \qquad F'^{12}/\varepsilon_0 = -cB_z,$$
$$F'^{13}/\varepsilon_0 = \gamma(cB_y - \beta E_x), \qquad F'^{23}/\varepsilon_0 = -\gamma(cB_x + \beta E_y).$$

For instance F'^{13}/ε_0 is row 1 of UF/ε_0 dotted with column 3 of U, namely $E_x(-\beta\gamma)+(cB_y)\gamma$, and F'^{03}/ε_0 is $-\beta\gamma(-\beta\gamma E_z)+\gamma(-\gamma E_z)$. The remaining off-diagonal entries are their negatives because $(UFU^T)^T = UF^TU^T = -UFU^T$; every diagonal entry is zero.

Define

$$X=\gamma(E_x-\beta cB_y),\quad Y=\gamma(E_y+\beta cB_x),\quad P=\gamma(cB_x+\beta E_y),\quad Q=\gamma(cB_y-\beta E_x).$$

The entire transformed matrix is

$$F'=\varepsilon_0\begin{pmatrix}0 & -X & -Y & -E_z\\ X & 0 & -cB_z & Q\\ Y & cB_z & 0 & -P\\ E_z & -Q & P & 0\end{pmatrix}.$$

Reading it in the same pattern as the original field tensor, $E'_x=X$, $E'_y=Y$, $cB'_x=P$, $cB'_y=Q$, and with $\beta c=v$,

$$\boxed{\begin{array}{lll}E'_x=\gamma(E_x-vB_y), & E'_y=\gamma(E_y+vB_x), & E'_z=E_z,\\ B'_x=\gamma(B_x+vE_y/c^2), & B'_y=\gamma(B_y-vE_x/c^2), & B'_z=B_z.\end{array}}$$

These are precisely Eqs. (17.82) and (17.83), including their signs.

17.9.2 Confirm that the generalization of Eqs. (17.82) and (17.83) to a boost corresponding to an arbitrary velocity $\mathbf{v}$ is properly given by Eq. (17.84).

At $\mathbf{v}=0$, interpret the direction-dependent projections through the continuous zero-velocity limit.

Solution.

For $0<|\mathbf{v}|<c$, put $\mathbf{n}=\mathbf{v}/|\mathbf{v}|$ and $\gamma=(1-|\mathbf{v}|^2/c^2)^{-1/2}$. Split the fields into parallel and perpendicular parts:

$$\mathbf{E}_\parallel=(\mathbf{E}\cdot\mathbf{n})\mathbf{n},\quad \mathbf{E}_\perp=\mathbf{E}-\mathbf{E}_\parallel,\qquad \mathbf{B}_\parallel=(\mathbf{B}\cdot\mathbf{n})\mathbf{n},\quad \mathbf{B}_\perp=\mathbf{B}-\mathbf{B}_\parallel.$$

Choose a proper spatial rotation taking $\mathbf{n}$ to the z axis. In those coordinates 17.9.1 says that the longitudinal components do not change, while the transverse electric components are $\gamma(E_x-vB_y, E_y+vB_x)$. Since $\mathbf{v}=v\hat{\mathbf{z}}$ there, with $v=|\mathbf{v}|$, $\mathbf{v}\times\mathbf{B}=v(-B_y,B_x,0)$ and $\mathbf{v}\times\mathbf{E}=v(-E_y,E_x,0)$, so these are exactly $\gamma(\mathbf{E}_\perp+\mathbf{v}\times\mathbf{B})$. Likewise the transverse magnetic components $\gamma(B_x+vE_y/c^2,\ B_y-vE_x/c^2)$ are $\gamma(\mathbf{B}_\perp-\mathbf{v}\times\mathbf{E}/c^2)$. Proper rotations preserve dot and cross products, so these vector statements remain valid after rotating back.

Adding the unchanged longitudinal parts gives

$$\begin{aligned}\mathbf{E}' &= \mathbf{E}_\parallel+\gamma(\mathbf{E}-\mathbf{E}_\parallel+\mathbf{v}\times\mathbf{B})\\ &= \boxed{\gamma(\mathbf{E}+\mathbf{v}\times\mathbf{B})+(1-\gamma)\mathbf{E}_\parallel},\\ \mathbf{B}' &= \mathbf{B}_\parallel+\gamma(\mathbf{B}-\mathbf{B}_\parallel-\mathbf{v}\times\mathbf{E}/c^2)\\ &= \boxed{\gamma(\mathbf{B}-\mathbf{v}\times\mathbf{E}/c^2)+(1-\gamma)\mathbf{B}_\parallel}.\end{aligned}$$

These are Eq. (17.84), with $\mathbf{E}_\parallel=\mathbf{E}_v$, $\mathbf{B}_\parallel=\mathbf{B}_v$.

At $\mathbf{v} = 0$ the direction $\hat{\mathbf{v}}$ is undefined; the equations mean their continuous limit, $\mathbf{E}' = \mathbf{E}$, $\mathbf{B}' = \mathbf{B}$. An equivalent form makes this endpoint explicit. Since $\gamma^2 - 1 = \gamma^2 v^2/c^2$,

$$\frac{\gamma - 1}{v^2} = \frac{\gamma^2}{(\gamma + 1)c^2}$$

(divide $(\gamma - 1)(\gamma + 1) = \gamma^2 v^2/c^2$ by $(\gamma + 1)v^2$). With $\mathbf{E}_\parallel = \mathbf{v}(\mathbf{v} \cdot \mathbf{E})/v^2$,

$$(1 - \gamma)\mathbf{E}_\parallel = -\frac{\gamma - 1}{v^2}\,\mathbf{v}(\mathbf{v} \cdot \mathbf{E}) = -\frac{\gamma^2}{\gamma + 1}\frac{\mathbf{v}(\mathbf{v} \cdot \mathbf{E})}{c^2},$$

and likewise for $\mathbf{B}_\parallel$. Thus, for all $|\mathbf{v}| < c$, including zero,

$$\boxed{\begin{aligned} \mathbf{E}' &= \gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1}\frac{\mathbf{v}(\mathbf{v} \cdot \mathbf{E})}{c^2}, \\ \mathbf{B}' &= \gamma(\mathbf{B} - \mathbf{v} \times \mathbf{E}/c^2) - \frac{\gamma^2}{\gamma + 1}\frac{\mathbf{v}(\mathbf{v} \cdot \mathbf{B})}{c^2}. \end{aligned}}$$

Both the longitudinal invariance and the z-boost component formulas follow from this expression. Since $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$ and $\gamma^2 v^2/c^2 = \gamma^2 - 1$,

$$\mathbf{v} \cdot \mathbf{E}' = \gamma\,\mathbf{v} \cdot \mathbf{E} - \frac{\gamma^2 - 1}{\gamma + 1}\,\mathbf{v} \cdot \mathbf{E} = [\gamma - (\gamma - 1)]\,\mathbf{v} \cdot \mathbf{E} = \mathbf{v} \cdot \mathbf{E},$$

and likewise $\mathbf{v} \cdot \mathbf{B}' = \mathbf{v} \cdot \mathbf{B}$. For $\mathbf{v} = v\hat{\mathbf{z}}$ the last terms change only the z components, so the x, y components are those of 17.9.1.

18 Chapter 18. More Special Functions

Problems

Unless an exercise says otherwise, polynomial degrees and derivative orders in this chapter are nonnegative integers. The associated Laguerre index k is a fixed nonnegative integer. Real and complex extensions are stated separately where needed.

18.1 Hermite Functions

18.1.1 Assume that the Hermite polynomials are known to be solutions of the Hermite ODE, Eq. (18.1). Assume further that the recurrence relation, Eq. (18.3), and the values of $H_n(0)$ are also known. Given the existence of a generating function

$$g(x,t) = \sum_{n=0}^{\infty} H_n(x)\,\frac{t^n}{n!},$$

(a) Differentiate $g(x,t)$ with respect to x and using the recurrence relation develop a first-order PDE for $g(x,t)$.

(b) Integrate with respect to x, holding t fixed.

(c) Evaluate $g(0,t)$ using the known values of $H_n(0)$.

(d) Finally, show that $g(x,t) = \exp(-t^2 + 2tx)$.

Solution.

The Instructor's Manual prints $g(x,0) = f(t)$ in part (c); the evaluation is $g(0,t) = f(t)$, because $g = e^{2tx} f(t)$ from part (b) reduces to $f(t)$ at $x = 0$.

First derive the derivative identity from precisely the stated data. Write $D = d/dx$ and $L_n = D^2 - 2xD + 2n$, so that the ODE reads $L_m H_m = 0$. Since $L_n = L_m + 2(n-m)$, the ODE gives

$$L_n H_{n+1} = -2H_{n+1}, \qquad L_n H_{n-1} = 2H_{n-1},$$

while the product rule, $D(xH_n) = xH_n' + H_n$ and $D^2(xH_n) = xH_n'' + 2H_n'$, gives

$$L_n(xH_n) = xL_nH_n + 2H_n' - 2xH_n = 2H_n' - 2xH_n.$$

Apply L_n to the known identity $H_{n+1} - 2xH_n + 2nH_{n-1} = 0$, and then use the same identity to replace H_{n+1}:

$$\begin{aligned} 0 &= -2H_{n+1} - 4H_n' + 4xH_n + 4nH_{n-1} \\ &= -2(2xH_n - 2nH_{n-1}) - 4H_n' + 4xH_n + 4nH_{n-1} \\ &= 8nH_{n-1} - 4H_n'. \end{aligned}$$

Thus $H_n' = 2nH_{n-1}$ for $n \geq 1$; $H_0' = 0$ because H_0, the polynomial solution of degree zero, is the constant $H_0(0) = 1$. We have not assumed the derivative formula separately.

(a) Differentiate the proposed generating series, initially as a formal power series. With $2n/n! = 2/(n-1)!$ and $m = n-1$,

$$\frac{\partial g}{\partial x} = \sum_{n=1}^{\infty} 2nH_{n-1}(x)\frac{t^n}{n!} = 2t\sum_{m=0}^{\infty} H_m(x)\frac{t^m}{m!} = \boxed{2tg}.$$

(b) Holding t fixed, $\partial_x(e^{-2tx}g) = e^{-2tx}(\partial_x g - 2tg) = 0$, so $e^{-2tx}g$ depends on t alone, say $f(t)$, and $g(x,t) = e^{2tx}f(t)$.

(c) At $x = 0$, the known values are $H_{2m}(0) = (-1)^m(2m)!/m!$ and $H_{2m+1}(0) = 0$. Therefore

$$f(t) = g(0,t) = \sum_{m=0}^{\infty} \frac{(-1)^m(2m)!}{m!}\frac{t^{2m}}{(2m)!} = \sum_{m=0}^{\infty}\frac{(-t^2)^m}{m!} = e^{-t^2}.$$

(d) Substitution gives

$$\boxed{g(x,t) = e^{2tx-t^2}}.$$

The resulting exponential is entire in t for each x, so its coefficient identity also establishes convergence of the initially formal generating series everywhere.

18.1.2 In developing the properties of the Hermite polynomials, start at a number of different points, such as:

1. Hermite's ODE, Eq. (18.1),
2. Rodrigues's formula, Eq. (18.8),
3. Integral representation, Eq. (18.10),
4. Generating function, Eq. (18.2),
5. Gram–Schmidt construction of a complete set of orthogonal polynomials over $(-\infty,\infty)$ with a weighting factor of $\exp(-x^2)$ (Section 5.2).

Outline how you can go from any one of these starting points to all the other points. In the ODE and Gram–Schmidt constructions, select the degree-n polynomial with leading coefficient 2^n.

Solution.

To identify a unique Hermite polynomial, supplement the ODE or Gram–Schmidt construction by degree n and leading coefficient 2^n. The ODE alone also has nonpolynomial solutions, and orthogonality alone does not fix the scale. Here is a closed chain, with both directions to the fifth starting point.

ODE $\longrightarrow$ Rodrigues. Let $w = e^{-x^2}$ and $D = d/dx$. Differentiating $w' + 2xw = 0$ exactly $n+1$ times, with Leibniz's rule $D^{n+1}(xw) = xw^{(n+1)} + (n+1)w^{(n)}$, gives

$$w^{(n+2)} + 2xw^{(n+1)} + 2(n+1)w^{(n)} = 0.$$

Put $w^{(n)} = (-1)^n wR_n$. Since $w' = -2xw$, the product rule gives

$$\begin{aligned} w^{(n+1)} &= (-1)^n w(R_n' - 2xR_n), \\ w^{(n+2)} &= (-1)^n w\big[R_n'' - 2xR_n' - 2R_n - 2x(R_n' - 2xR_n)\big] \\ &= (-1)^n w\big[R_n'' - 4xR_n' + (4x^2-2)R_n\big]. \end{aligned}$$

Substitution into the differentiated equation and division by $(-1)^n w$ give

$$\begin{aligned} 0 &= R_n'' - 4xR_n' + (4x^2-2)R_n + 2x(R_n' - 2xR_n) + 2(n+1)R_n \\ &= R_n'' - 2xR_n' + 2nR_n. \end{aligned}$$

Differentiation of w repeatedly produces w times a polynomial, since $D(wp) = w(p' - 2xp)$; each step multiplies the highest term by $-2x$, so the highest term of $w^{(n)}/w$ is $(-2x)^n$. Thus R_n has degree n and leading coefficient $(-1)^n(-2)^n = 2^n$. Two such polynomial solutions differ by a polynomial d of degree below n with $L_n d = 0$, where $L_n = D^2 - 2xD + 2n$. If $d \neq 0$, let cx^m, with $c \neq 0$ and $m < n$, be its leading term. The x^m term of $L_n d$ is then $(-2m + 2n)cx^m \neq 0$, because D^2 lowers the degree, contradicting $L_n d = 0$. Hence $d = 0$ and

$$H_n = R_n = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}.$$

Rodrigues $\longrightarrow$ generating function. Taylor's theorem for the entire function w gives

$$\sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!} = e^{x^2} \sum_{n=0}^{\infty} \frac{(-t)^n}{n!} w^{(n)}(x) = e^{x^2} w(x-t) = e^{x^2-(x-t)^2} = e^{-t^2+2xt}.$$

Generating function $\longleftrightarrow$ contour integral. Cauchy's coefficient formula on a positively oriented circle about zero is

$$H_n(x) = \frac{n!}{2\pi i} \oint_{|z|=R} \frac{e^{-z^2+2xz}}{z^{n+1}}\, dz.$$

Conversely, multiply by $t^n/n!$ and sum for $|t| < R$. The geometric series $\sum_n t^n/z^{n+1} = 1/(z-t)$ converges uniformly on that circle, so termwise integration and Cauchy's integral formula give

$$\sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!} = \frac{1}{2\pi i} \oint \frac{e^{-z^2+2xz}}{z-t}\, dz = e^{-t^2+2xt}.$$

Any fixed t is covered by choosing $R > |t|$.

Generating function $\longrightarrow$ ODE. For $g = e^{-t^2+2xt}$, direct differentiation gives $g_x = 2tg$, $g_{xx} = 4t^2 g$ and $g_t = (2x - 2t)g$, so

$$g_{xx} - 2xg_x + 2tg_t = [4t^2 - 4xt + 2t(2x-2t)]g = 0.$$

Since $t\,\partial_t(t^n/n!) = nt^n/n!$, the $t^n/n!$ coefficient of this equation is $H_n'' - 2xH_n' + 2nH_n = 0$. These coefficients are polynomials of degree n with leading coefficient 2^n, as the uniqueness argument in *ODE $\longrightarrow$ Rodrigues* requires: multiplying $e^{2xt} = \sum_j (2x)^j t^j/j!$ by $e^{-t^2} = \sum_k (-1)^k t^{2k}/k!$ and collecting the terms with $j + 2k = n$ give the coefficient of $t^n/n!$,

$$H_n(x) = n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2x)^{n-2k}}{k!\,(n-2k)!},$$

whose highest term, from $k = 0$, is $(2x)^n$. This closes the first four starting points.

Rodrigues $\longleftrightarrow$ Gram–Schmidt. For every polynomial q of degree less than n, use $wH_n = (-1)^n w^{(n)}$ and integrate by parts n times; each integration moves one derivative onto q with a factor -1:

$$\int_{-\infty}^{\infty} wH_n q\, dx = (-1)^n \int_{-\infty}^{\infty} w^{(n)} q\, dx = \int_{-\infty}^{\infty} w q^{(n)}\, dx = 0.$$

Every boundary term is a polynomial times e^{-x^2} and vanishes. For $q = H_n$, whose leading term is $2^n x^n$, the nth derivative is $2^n n!$, so the same calculation and $\int w\,dx = \sqrt{\pi}$ give

$$\int_{-\infty}^{\infty} w H_n^2\,dx = 2^n n!\sqrt{\pi}.$$

Therefore Rodrigues produces the orthogonal polynomials generated by Gram–Schmidt from $1, x, x^2, \ldots$, with the prescribed leading coefficients. Conversely, two degree-n polynomials with that leading coefficient and orthogonality to all lower degrees have difference d of degree below n. Orthogonality gives $\int w d^2 = 0$, hence $d = 0$. Thus Gram–Schmidt recovers Rodrigues and enters the preceding chain.

Completeness in $L^2(\mathbb{R}, w\,dx)$ can also be justified. If f is orthogonal to every polynomial, set $F(z) = \int_{\mathbb{R}} f(x)w(x)e^{zx}\,dx$. Cauchy–Schwarz bounds this integral and all its differentiated integrals uniformly on compact z sets by Gaussian moments, so F is entire. All $F^{(n)}(0) = \int_{\mathbb{R}} f(x)w(x)x^n\,dx$ vanish, so the Taylor series of F about zero vanishes, implying $F = 0$. In particular the Fourier transform of the integrable function fw is zero; Fourier uniqueness gives $fw = 0$ almost everywhere, hence $f = 0$.

18.1.3 Prove that $\big|H_n(x)\big| \le \big|H_n(ix)\big|$. Take x real.

Solution.

The assertion is for real x. It is false for arbitrary complex x: for $n = 2$, $x = i$, the two moduli are 6 and 2. The finite series is

$$H_n(x) = n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2x)^{n-2k}}{k!(n-2k)!}.$$

For real x, the triangle inequality gives

$$|H_n(x)| \le n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(2|x|)^{n-2k}}{k!(n-2k)!}.$$

On the other hand $(-1)^k i^{n-2k} = i^n (-1)^k (i^2)^{-k} = i^n$, so

$$H_n(ix) = i^n n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(2x)^{n-2k}}{k!(n-2k)!}.$$

The exponents $n - 2k$ all have the parity of n. For $x > 0$ all terms are nonnegative; for $x < 0$ they all have the sign $(-1)^n$. No cancellation occurs, so

$$|H_n(ix)| = n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(2|x|)^{n-2k}}{k!(n-2k)!},$$

which is exactly the bound just obtained for $|H_n(x)|$. At $x = 0$ the conclusion follows directly from the constant terms. Thus $\boxed{|H_n(x)| \le |H_n(ix)|}$ for every real x.

18.1.4 Rewrite the series form of $H_n(x)$, Eq. (18.9), as an ascending power series.

$$\text{ANS. } H_{2n}(x) = (-1)^n \sum_{s=0}^{n} (-1)^s (2x)^{2s} \frac{(2n)!}{(2s)!\,(n-s)!},$$

$$H_{2n+1}(x) = (-1)^n \sum_{s=0}^{n} (-1)^s (2x)^{2s+1} \frac{(2n+1)!}{(2s+1)!\,(n-s)!}.$$

Solution.

Start with the descending series $H_N(x) = N! \sum_{k=0}^{\lfloor N/2 \rfloor} (-1)^k (2x)^{N-2k}/[k!(N-2k)!]$. For $N = 2n$, put $s = n - k$, so $k = n - s$ and $N - 2k = 2s$. As k goes from 0 to n, s goes from n to 0, and reversing the finite sum and using $(-1)^{n-s} = (-1)^n (-1)^s$ give

$$H_{2n}(x) = (2n)! \sum_{s=0}^{n} \frac{(-1)^{n-s}(2x)^{2s}}{(n-s)!(2s)!}$$

$$= \boxed{(-1)^n \sum_{s=0}^{n} \frac{(-1)^s (2n)!(2x)^{2s}}{(2s)!(n-s)!}}.$$

For $N = 2n + 1$ we have $\lfloor N/2 \rfloor = n$, and the same change $s = n - k$ gives $N - 2k = 2s + 1$, hence

$$H_{2n+1}(x) = (2n+1)! \sum_{s=0}^{n} \frac{(-1)^{n-s}(2x)^{2s+1}}{(n-s)!(2s+1)!}$$

$$= \boxed{(-1)^n \sum_{s=0}^{n} \frac{(-1)^s (2n+1)!(2x)^{2s+1}}{(2s+1)!(n-s)!}}.$$

The even formula needs $(-1)^s$, not the textbook's $(-1)^{2s}$. For example the corrected $n = 1$ formula gives $H_2(x) = 4x^2 - 2$, directly checking the relative sign.

18.1.5

(a) Expand x^{2r} in a series of even-order Hermite polynomials.

(b) Expand x^{2r+1} in a series of odd-order Hermite polynomials.

$$\text{ANS. (a) } x^{2r} = \frac{(2r)!}{2^{2r}} \sum_{n=0}^{r} \frac{H_{2n}(x)}{(2n)!\,(r-n)!}$$

$$\text{(b) } x^{2r+1} = \frac{(2r+1)!}{2^{2r+1}} \sum_{n=0}^{r} \frac{H_{2n+1}(x)}{(2n+1)!\,(r-n)!}, \qquad r = 0, 1, 2, \ldots.$$

Hint. Use a Rodrigues representation and integrate by parts.

Solution.

Because H_n has degree n and leading coefficient 2^n, the polynomials $H_0, \ldots, H_m$ form a basis for polynomials of degree at most m. Write $x^m = \sum_{n=0}^{m} c_n H_n(x)$. Orthogonality gives

$$c_n = \frac{1}{2^n n! \sqrt{\pi}} \int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\, dx.$$

By Rodrigues's formula $e^{-x^2}H_n = (-1)^n D^n e^{-x^2}$, with $D = d/dx$. Each of n integrations by parts moves one derivative onto x^m with a factor -1, and every boundary term is a polynomial times a derivative of e^{-x^2}, which vanishes at $\pm\infty$. Since $D^n x^m = m!\,x^{m-n}/(m-n)!$ for $n \le m$,

$$\int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\,dx = (-1)^n \int_{-\infty}^{\infty} x^m D^n e^{-x^2}\,dx = \int_{-\infty}^{\infty} e^{-x^2} D^n x^m\,dx$$
$$= \frac{m!}{(m-n)!}\int_{-\infty}^{\infty} x^{m-n} e^{-x^2}\,dx.$$

Odd powers integrate to zero, the integrand being odd. For $m - n = 2j$, the integrand is even, and the substitution $u = x^2$, $dx = du/(2\sqrt{u})$, gives

$$\int_{-\infty}^{\infty} x^{2j} e^{-x^2}\,dx = 2\int_0^{\infty} x^{2j} e^{-x^2}\,dx = \int_0^{\infty} u^{j-1/2} e^{-u}\,du = \Gamma(j+\tfrac12) = \frac{(2j)!\sqrt{\pi}}{4^j j!}.$$

The last value follows from $\Gamma(j+\frac12) = (j-\frac12)(j-\frac32)\cdots\frac12\,\Gamma(\frac12) = 2^{-j}(2j-1)!!\sqrt{\pi}$ and $(2j-1)!! = (2j)!/(2^j j!)$. Consequently, with $(m-n)! = (2j)!$ and $2^n 4^j = 2^{n+2j} = 2^m$,

$$c_n = \frac{1}{2^n n!\sqrt{\pi}}\,\frac{m!}{(2j)!}\,\frac{(2j)!\sqrt{\pi}}{4^j j!} = \frac{m!}{2^n 4^j n! j!} = \frac{m!}{2^m n! j!}, \qquad n = m - 2j,$$

and $c_n = 0$ when $m - n$ is odd.

(a) Take $m = 2r$; only even $n = 2k$ contribute, with $j = r - k$:

$$\boxed{x^{2r} = \frac{(2r)!}{2^{2r}}\sum_{k=0}^{r}\frac{H_{2k}(x)}{(2k)!(r-k)!}}.$$

(b) Take $m = 2r+1$; only odd $n = 2k+1$ contribute, again with $j = r - k$:

$$\boxed{x^{2r+1} = \frac{(2r+1)!}{2^{2r+1}}\sum_{k=0}^{r}\frac{H_{2k+1}(x)}{(2k+1)!(r-k)!}}.$$

These finite identities include $r = 0$, giving $1 = H_0$ and $x = H_1/2$.

18.1.6 Show that

(a) $$\int_{-\infty}^{\infty} H_n(x)\exp\left[-\frac{x^2}{2}\right]dx = \begin{cases}\sqrt{2\pi}\,n!/(n/2)!, & n \text{ even},\\ 0, & n \text{ odd}.\end{cases}$$

(b) $$\int_{-\infty}^{\infty} xH_n(x)\exp\left[-\frac{x^2}{2}\right]dx = \begin{cases}0, & n \text{ even},\\ \sqrt{2\pi}\,\dfrac{(n+1)!}{((n+1)/2)!}, & n \text{ odd}.\end{cases}$$

Solution.

(a) Multiply the generating function by $e^{-x^2/2}$ and integrate. For real t, completion of the

square, $-\frac{1}{2}x^2+2tx-t^2=-\frac{1}{2}(x-2t)^2+t^2$, and the shift $x\to x+2t$ give

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{t^n}{n!}\int_{-\infty}^{\infty}e^{-x^2/2}H_n(x)\,dx &= \int_{-\infty}^{\infty}e^{-x^2/2+2tx-t^2}\,dx\\ &= e^{t^2}\int_{-\infty}^{\infty}e^{-(x-2t)^2/2}\,dx\\ &= \sqrt{2\pi}\,e^{t^2}=\sqrt{2\pi}\sum_{k=0}^{\infty}\frac{t^{2k}}{k!}.\end{aligned}$$

The right side contains only even powers, and $t^{2k}/k!=[(2k)!/k!]\,t^{2k}/(2k)!$. Comparing coefficients of $t^n/n!$ (with $n=2k$ for even n) gives

$$\boxed{\int_{-\infty}^{\infty}e^{-x^2/2}H_n(x)\,dx=\begin{cases}\sqrt{2\pi}\,n!/(n/2)!, & n \text{ even},\\ 0, & n \text{ odd}.\end{cases}}$$

The interchange of sum and integral is justified for $|t|\le R$ by dominated convergence. By Exercise 18.1.3, $|H_n(x)|\le|H_n(ix)|=n!\sum_k(2|x|)^{n-2k}/[k!(n-2k)!]$, so, with $R^n=R^{n-2k}R^{2k}$,

$$\sum_{n=0}^{\infty}|H_n(x)|\frac{R^n}{n!}\le\sum_{n=0}^{\infty}\sum_{k=0}^{\lfloor n/2\rfloor}\frac{(2R|x|)^{n-2k}R^{2k}}{k!(n-2k)!}=e^{2R|x|}e^{R^2},$$

the double sum being the Cauchy product of the two exponential series. Every partial sum is therefore bounded by the integrable function $e^{R^2}e^{-x^2/2+2R|x|}$.

(b) Inserting x, completing the same square and putting $y=x-2t$ give

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{t^n}{n!}\int_{-\infty}^{\infty}xe^{-x^2/2}H_n(x)\,dx &= e^{t^2}\int_{-\infty}^{\infty}(y+2t)e^{-y^2/2}\,dy\\ &= 2t\sqrt{2\pi}\,e^{t^2}=\sqrt{2\pi}\sum_{k=0}^{\infty}\frac{2t^{2k+1}}{k!}.\end{aligned}$$

The interchange of sum and integral is justified as in (a): for $|t|\le R$ the partial sums are now bounded by $|x|e^{R^2}e^{-x^2/2+2R|x|}$, which is still integrable. The odd Gaussian moment $\int ye^{-y^2/2}dy$ is zero, and $2t\int e^{-y^2/2}dy=2t\sqrt{2\pi}$. The right side has only odd powers, and

$$\frac{2t^{2k+1}}{k!}=\frac{2(2k+1)!}{k!}\,\frac{t^{2k+1}}{(2k+1)!}.$$

Thus even n give zero, and for $n=2k+1$ the answer is $\sqrt{2\pi}\,2(2k+1)!/k!$. Since $2(2k+1)!/k!=(2k+2)(2k+1)!/[(k+1)k!]=(2k+2)!/(k+1)!$, this is

$$\boxed{\int_{-\infty}^{\infty}xe^{-x^2/2}H_n(x)\,dx=\begin{cases}0, & n \text{ even},\\ \sqrt{2\pi}\,(n+1)!/((n+1)/2)!, & n \text{ odd}.\end{cases}}$$

Both square roots implement the published correction of 2π to $\sqrt{2\pi}$. The case $n=0$ in (a) is the usual Gaussian integral and already detects the original normalization error.

18.1.7

(a) Using the Cauchy integral formula, develop an integral representation of $H_n(x)$ based on Eq. (18.2) with the contour enclosing the point $z = -x$.

$$\textit{ANS. } H_n(x) = \frac{n!}{2\pi i}\, e^{x^2} \oint \frac{e^{-z^2}}{(z+x)^{n+1}}\, dz.$$

(b) Show by direct substitution that this result satisfies the Hermite equation.

Solution.

(a) Cauchy's coefficient formula applied to the generating function gives

$$H_n(x) = \frac{n!}{2\pi i} \oint_{|w|=R} \frac{e^{-w^2+2xw}}{w^{n+1}}\, dw.$$

Set $z = w - x$. Then $w = z + x$, $dw = dz$, and $-w^2 + 2xw = -(z+x)^2 + 2x(z+x) = x^2 - z^2$. The positively oriented circle now encloses $z = -x$, and hence

$$\boxed{H_n(x) = \frac{n!}{2\pi i} e^{x^2} \oint \frac{e^{-z^2}}{(z+x)^{n+1}}\, dz}.$$

Any positively oriented simple contour enclosing $-x$ gives the same result, since this is the only finite singularity of the integrand.

(b) Use the original w contour, which stays fixed as x varies; its integrand is entire in x, so differentiation under the finite contour integral is valid. Put $E = e^{-w^2+2xw}$, so that $\partial_x E = 2wE$. Then

$$H_n' = \frac{n!}{2\pi i} \oint \frac{2wE}{w^{n+1}}\, dw, \qquad H_n'' = \frac{n!}{2\pi i} \oint \frac{4w^2 E}{w^{n+1}}\, dw.$$

Combining all three terms of the ODE gives

$$\begin{aligned} H_n'' - 2xH_n' + 2nH_n &= \frac{n!}{2\pi i} \oint \frac{(4w^2 - 4xw + 2n)E}{w^{n+1}}\, dw \\ &= -2\,\frac{n!}{2\pi i} \oint \frac{d}{dw}\left(\frac{E}{w^n}\right) dw = 0. \end{aligned}$$

Indeed, since $\partial_w E = (-2w + 2x)E$,

$$\frac{d}{dw}\left(\frac{E}{w^n}\right) = \frac{(-2w+2x)E}{w^n} - \frac{nE}{w^{n+1}} = \frac{(-2w^2 + 2xw - n)E}{w^{n+1}},$$

and multiplication by -2 gives precisely the numerator $4w^2 - 4xw + 2n$. The integral of this derivative is zero because its primitive is single-valued on the closed contour, even though it can have a pole inside. This proves the equation directly, including $n = 0$.

18.2 Applications of Hermite Functions

18.2.1 Prove that $\left(2x - \dfrac{d}{dx}\right)^n 1 = H_n(x)$.

Hint. Check out the cases $n = 0$ and $n = 1$ and then use mathematical induction (Section 1.4).

Solution.

Let $A = 2x - D$, where $D = d/dx$ acts on everything to its right. The derivative formula $H_n' = 2nH_{n-1}$ and the recurrence (18.3), $H_{n+1} = 2xH_n - 2nH_{n-1}$, give

$$AH_n = 2xH_n - H_n' = 2xH_n - 2nH_{n-1} = H_{n+1}.$$

Now $A^0 1 = 1 = H_0$ and $A1 = 2x = H_1$. If $A^n 1 = H_n$, then

$$A^{n+1}1 = A(A^n 1) = AH_n = H_{n+1}.$$

Mathematical induction therefore proves $\boxed{(2x - D)^n 1 = H_n(x)}$ for every nonnegative integer n. The operator power means repeated composition; x and D do not commute, since $D(xf) = f + xDf$.

18.2.2 Show that $\displaystyle\int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\,dx = 0$ for m an integer, $0 \le m \le n - 1$.

Solution.

The Instructor's Manual's polynomial expansion must start at H_0, not H_1; its final integral is zero. For example, the constant polynomial cannot be expanded using only $H_1, \ldots, H_m$.

For integers $0 \le m < n$, Rodrigues gives

$$\begin{aligned}\int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\,dx &= (-1)^n \int_{-\infty}^{\infty} x^m \frac{d^n}{dx^n} e^{-x^2}\,dx \\ &= \int_{-\infty}^{\infty} e^{-x^2} \frac{d^n}{dx^n} x^m\,dx = \boxed{0}.\end{aligned}$$

There are n integrations by parts: their $(-1)^n$ cancels the Rodrigues sign. The $(i+1)$st of them, $0 \le i < n$, reads

$$\begin{aligned}\int_{-\infty}^{\infty} (D^i x^m)\, D^{n-i} e^{-x^2}\,dx &= \Big[(D^i x^m)\, D^{n-i-1} e^{-x^2}\Big]_{-\infty}^{\infty} \\ &\quad - \int_{-\infty}^{\infty} (D^{i+1} x^m)\, D^{n-i-1} e^{-x^2}\,dx,\end{aligned}$$

with $D = d/dx$. Every boundary term is a polynomial times a derivative of the Gaussian, hence a polynomial times e^{-x^2}, and vanishes at both infinities. Finally $D^n x^m = 0$ because $n > m$. For $n = 0$ no integer m meets the stated range, so there is no exceptional missing case.

18.2.3 The transition probability between two oscillator states m and n depends on

$$\int_{-\infty}^{\infty} x e^{-x^2} H_n(x) H_m(x)\,dx.$$

Show that this integral equals $\pi^{1/2} 2^{n-1} n!\, \delta_{m,n-1} + \pi^{1/2} 2^n (n+1)!\, \delta_{m,n+1}$. This result shows that such transitions can occur only between states of adjacent energy levels, $m = n \pm 1$.

Hint. Multiply the generating function, Eq. (18.2), by itself using two different sets of variables (x, s) and (x, t). Alternatively, the factor x may be eliminated by the recurrence relation, Eq. (18.3).

Solution.

The recurrence relation can be solved for multiplication by x:

$$xH_n = \tfrac{1}{2}H_{n+1} + nH_{n-1}.$$

Let I_{nm} denote the integral, and use the orthogonality integral $\int_{\mathbb{R}} e^{-x^2} H_j H_m \, dx = \sqrt{\pi}\, 2^j j!\, \delta_{jm}$. Multiplying the recurrence by $e^{-x^2} H_m$ and integrating yields

$$\begin{aligned} I_{nm} &= \tfrac{1}{2} \int_{\mathbb{R}} e^{-x^2} H_{n+1} H_m \, dx + n \int_{\mathbb{R}} e^{-x^2} H_{n-1} H_m \, dx \\ &= \tfrac{1}{2}\sqrt{\pi}\, 2^{n+1}(n+1)!\, \delta_{m,n+1} + n\sqrt{\pi}\, 2^{n-1}(n-1)!\, \delta_{m,n-1} \\ &= \boxed{\sqrt{\pi}\, 2^{n-1} n!\, \delta_{m,n-1} + \sqrt{\pi}\, 2^n (n+1)!\, \delta_{m,n+1}}, \end{aligned}$$

since $\frac{1}{2}\, 2^{n+1} = 2^n$ and $n\,(n-1)! = n!$. For $n = 0$, omit the lowering term before writing its factorial: $xH_0 = H_1/2$. The final formula still applies because $\delta_{m,-1} = 0$ for nonnegative m. Thus the position matrix element vanishes unless $m = n \pm 1$. Normalization of the wave functions changes its nonzero magnitude but not these zeros, giving the stated adjacent-level selection rule.

18.2.4 Show that $\displaystyle\int_{-\infty}^{\infty} x^2 e^{-x^2} H_n(x) H_n(x)\, dx = \pi^{1/2} 2^n n! \left(n + \frac{1}{2}\right)$. This integral occurs in the calculation of the mean-square displacement of our quantum oscillator.

Hint. Use the recurrence relation, Eq. (18.3), and the orthogonality integral.

Solution.

Square $xH_n = H_{n+1}/2 + nH_{n-1}$,

$$x^2 H_n^2 = \tfrac{1}{4} H_{n+1}^2 + nH_{n+1}H_{n-1} + n^2 H_{n-1}^2,$$

and integrate with Gaussian weight, using the orthogonality integral:

$$\begin{aligned} \int_{\mathbb{R}} x^2 e^{-x^2} H_n^2 \, dx &= \frac{1}{4} \int_{\mathbb{R}} e^{-x^2} H_{n+1}^2 \, dx + n \int_{\mathbb{R}} e^{-x^2} H_{n+1} H_{n-1} \, dx + n^2 \int_{\mathbb{R}} e^{-x^2} H_{n-1}^2 \, dx \\ &= \frac{1}{4}\sqrt{\pi}\, 2^{n+1}(n+1)! + n^2 \sqrt{\pi}\, 2^{n-1}(n-1)! \\ &= \sqrt{\pi}\, 2^{n-1} n! [(n+1) + n] \\ &= \boxed{\sqrt{\pi}\, 2^n n! \left(n + \tfrac{1}{2}\right)}. \end{aligned}$$

The mixed term is zero by orthogonality, and the third line uses $\frac{1}{4}\, 2^{n+1}(n+1)! = 2^{n-1} n!\,(n+1)$ and $n^2\, 2^{n-1}(n-1)! = 2^{n-1} n!\, n$. At $n = 0$ the terms with H_{-1} are omitted at the first line, leaving $\sqrt{\pi}/2$; this agrees with the final formula. Dividing by the norm $\sqrt{\pi} 2^n n!$ gives $\langle x^2 \rangle_n = n + 1/2$ in the dimensionless oscillator coordinate.

18.2.5 Evaluate

$$\int_{-\infty}^{\infty} x^2 e^{-x^2} H_n(x) H_m(x)\, dx$$

in terms of n and m and appropriate Kronecker delta functions.

ANS. $2^{n-1}\pi^{1/2}(2n+1)n!\,\delta_{nm} + 2^n\pi^{1/2}(n+2)!\,\delta_{n+2,m} + 2^{n-2}\pi^{1/2}n!\,\delta_{n-2,m}$.

Solution.

Apply the multiplication recurrence $xH_j = \frac{1}{2}H_{j+1} + jH_{j-1}$ twice, first with $j = n$ and then with $j = n \pm 1$, keeping each coefficient:

$$\begin{aligned} x^2 H_n &= \tfrac{1}{2} x H_{n+1} + n x H_{n-1} \\ &= \tfrac{1}{2}\left(\tfrac{1}{2}H_{n+2} + (n+1)H_n\right) + n\left(\tfrac{1}{2}H_n + (n-1)H_{n-2}\right) \\ &= \tfrac{1}{4}H_{n+2} + \tfrac{2n+1}{2}H_n + n(n-1)H_{n-2}. \end{aligned}$$

Multiplication by $e^{-x^2}H_m$ and integration therefore gives

$$\begin{aligned} \int_{\mathbb{R}} x^2 e^{-x^2} H_n H_m\, dx &= \tfrac{1}{4}\sqrt{\pi}2^{n+2}(n+2)!\,\delta_{m,n+2} \\ &\quad + \tfrac{2n+1}{2}\sqrt{\pi}2^n n!\,\delta_{mn} \\ &\quad + n(n-1)\sqrt{\pi}2^{n-2}(n-2)!\,\delta_{m,n-2} \\ &= \boxed{\begin{array}{l} \sqrt{\pi}2^{n-1}(2n+1)n!\,\delta_{mn} \\ \quad + \sqrt{\pi}2^n(n+2)!\,\delta_{m,n+2} + \sqrt{\pi}2^{n-2}n!\,\delta_{m,n-2} \end{array}}. \end{aligned}$$

The orthogonality integral gives each line, and the box uses $\frac{1}{4}\,2^{n+2} = 2^n$, $\frac{2n+1}{2}\,2^n = 2^{n-1}(2n+1)$ and $n(n-1)\,(n-2)! = n!$. For $n = 0, 1$ the last Hermite term is absent; the last delta in the boxed formula is zero for all allowed $m \geq 0$, so that form covers these cases without a negative factorial. Setting $m = n$ recovers Exercise 18.2.4.

18.2.6 Show that $\displaystyle\int_{-\infty}^{\infty} x^r e^{-x^2} H_n(x) H_{n+p}(x)\, dx = \begin{cases} 0, & p > r \\ 2^n\pi^{1/2}(n+r)!, & p = r, \end{cases}$ with n, p, and r nonnegative integers.

Hint. Apply the multiplication recurrence, Eq. (18.3), r times, to $x^r H_n$.

Solution.

The recurrence removes one factor x per application, so $x^r H_n$ needs exactly r applications, whatever the value of p; this is the count in the corrected hint. Multiplication by x raises a Hermite index by one with coefficient $1/2$, or lowers it by one with coefficient equal to the current index: $xH_j = H_{j+1}/2 + jH_{j-1}$. Iterating this relation r times gives

$$x^r H_n = 2^{-r} H_{n+r} + \text{a polynomial of degree at most } n + r - 2.$$

To see why the highest coefficient is exactly 2^{-r}, note that the only way to reach index $n + r$ in r steps is to raise the index at every step; each contributes $1/2$. Any lowering step decreases the final index by two relative to that highest index. This also follows by matching leading terms: $x^r H_n$ has leading term $2^n x^{n+r}$, whereas H_{n+r} has leading term $2^{n+r}x^{n+r}$.

If $p > r$, the polynomial $x^r H_n$ has degree below $n + p$, so it is a combination of $H_0, \ldots, H_{n+r}$ and is orthogonal to H_{n+p}. If $p = r$, the lower terms, of degree at most $n + r - 2$, are orthogonal to

H_{n+r}, and only the displayed highest Hermite term survives; it contributes $2^{-r}\int e^{-x^2}H_{n+r}^2\,dx = 2^{-r}\sqrt{\pi}\,2^{n+r}(n+r)!$. Consequently

$$\boxed{\int_{-\infty}^{\infty} x^r e^{-x^2} H_n H_{n+p}\,dx = \begin{cases} 0, & p > r, \\ 2^{-r}\sqrt{\pi}2^{n+r}(n+r)! = 2^n\sqrt{\pi}(n+r)!, & p = r. \end{cases}}$$

The case $r = 0, p = 0$ is exactly the normalization integral. All indices are nonnegative integers as required by the question.

18.2.7 With $\psi_n(x) = \dfrac{e^{-x^2/2}H_n(x)}{(2^n n!\,\pi^{1/2})^{1/2}}$, verify that

$$a\,\psi_n(x) = \frac{x+ip}{\sqrt{2}}\,\psi_n(x) = \frac{1}{\sqrt{2}}\left(x + \frac{d}{dx}\right)\psi_n(x) = n^{1/2}\psi_{n-1}(x),$$
$$a^\dagger\psi_n(x) = \frac{x-ip}{\sqrt{2}}\,\psi_n(x) = \frac{1}{\sqrt{2}}\left(x - \frac{d}{dx}\right)\psi_n(x) = (n+1)^{1/2}\psi_{n+1}(x).$$

Note. The usual quantum mechanical operator approach establishes these raising and lowering properties before the form of $\psi_n(x)$ is known. Use dimensionless $p = -i\,d/dx$; at $n = 0$ the lowering formula means $a\psi_0 = 0$.

Solution.

The Instructor's Manual's two reduced ladder expressions also need $e^{-x^2/2}$, the factor carried by ψ_n, in place of e^{-x^2}, and its recurrence lacks the index $n-1$; it must read $2xH_n - 2nH_{n-1} = H_{n+1}$.

In the stated dimensionless variables $p = -iD$, with $D = d/dx$, so $ip = D$. Thus $a = (x+D)/\sqrt{2}$ and $a^\dagger = (x-D)/\sqrt{2}$. These are the signs of the published correction: since $ip = d/dx$, they make $(x \pm ip)/\sqrt{2}$ agree with the printed $(x \pm d/dx)/\sqrt{2}$. Set $N_n = (2^n n!\sqrt{\pi})^{-1/2}$, so $\psi_n = N_n e^{-x^2/2}H_n$. The product rule gives

$$D\psi_n = N_n e^{-x^2/2}(H_n' - xH_n).$$

The normalization ratios are (the first for $n \geq 1$)

$$\frac{N_n}{N_{n-1}} = \left[\frac{2^{n-1}(n-1)!}{2^n n!}\right]^{1/2} = \frac{1}{\sqrt{2n}}, \qquad \frac{N_n}{N_{n+1}} = \left[\frac{2^{n+1}(n+1)!}{2^n n!}\right]^{1/2} = \sqrt{2(n+1)}.$$

For lowering, $(x+D)\psi_n = N_n e^{-x^2/2}(xH_n + H_n' - xH_n)$: the xH_n terms cancel, and $H_n' = 2nH_{n-1}$ gives

$$\begin{aligned} a\psi_n &= \frac{N_n}{\sqrt{2}}e^{-x^2/2}H_n' = \sqrt{2}\,nN_n e^{-x^2/2}H_{n-1} \\ &= \sqrt{2}\,n\frac{N_n}{N_{n-1}}\psi_{n-1} = \sqrt{2}\,n\frac{1}{\sqrt{2n}}\psi_{n-1} = \boxed{\sqrt{n}\,\psi_{n-1}} \qquad (n \geq 1). \end{aligned}$$

For $n = 0$, $H_0' = 0$, so $a\psi_0 = 0$ directly; no function ψ_{-1} is needed. For raising, $(x-D)\psi_n = N_n e^{-x^2/2}(2xH_n - H_n')$, and $2xH_n - H_n' = 2xH_n - 2nH_{n-1} = H_{n+1}$ by the recurrence (18.3), so

$$\begin{aligned} a^\dagger\psi_n &= \frac{N_n}{\sqrt{2}}e^{-x^2/2}(2xH_n - H_n') \\ &= \frac{N_n}{\sqrt{2}}e^{-x^2/2}H_{n+1} = \frac{N_n}{\sqrt{2}N_{n+1}}\psi_{n+1} \\ &= \frac{\sqrt{2(n+1)}}{\sqrt{2}}\psi_{n+1} = \boxed{\sqrt{n+1}\,\psi_{n+1}}. \end{aligned}$$

The normalization ratios, not only the polynomial recurrences, produce the square-root coefficients.

18.2.8

(a) Verify the operator identity

$$x - ip = x - \frac{d}{dx} = -\exp\left(\frac{x^2}{2}\right)\frac{d}{dx}\exp\left(-\frac{x^2}{2}\right).$$

(b) The normalized simple harmonic oscillator wave function is

$$\psi_n(x) = (\pi^{1/2}2^n n!)^{-1/2}\exp\left(-\frac{x^2}{2}\right)H_n(x).$$

Show that this may be written as

$$\psi_n(x) = (\pi^{1/2}2^n n!)^{-1/2}\left(x - \frac{d}{dx}\right)^n \exp\left(-\frac{x^2}{2}\right).$$

Note. This corresponds to an n-fold application of the raising operator of Exercise 18.2.7. Use the same dimensionless momentum $p = -i\,d/dx$.

Solution.

(a) An operator identity must act on a function. For any differentiable f, with $D = d/dx$, the product rule gives

$$\begin{aligned} -e^{x^2/2}D(e^{-x^2/2}f) &= -e^{x^2/2}\left[-xe^{-x^2/2}f + e^{-x^2/2}f'\right] \\ &= xf - f' = (x - D)f. \end{aligned}$$

Since $p = -iD$, $x - ip = x - D$, proving the identity as written with the published correction $x - ip$.

(b) Let M denote multiplication by $e^{x^2/2}$, and M^{-1} multiplication by $e^{-x^2/2}$. Part (a) says $x - D = M(-D)M^{-1}$. Consecutive inverse multiplication operators cancel under composition, $M^{-1}M = 1$, so if $(x - D)^n = M(-D)^n M^{-1}$, then $(x - D)^{n+1} = M(-D)M^{-1}\,M(-D)^n M^{-1} = M(-D)^{n+1}M^{-1}$. Induction therefore gives

$$(x - D)^n = M(-D)^n M^{-1}.$$

Apply this identity to $e^{-x^2/2}$, for which $M^{-1}e^{-x^2/2} = e^{-x^2}$:

$$\begin{aligned} (x - D)^n e^{-x^2/2} &= e^{x^2/2}(-1)^n D^n e^{-x^2} \\ &= e^{x^2/2}e^{-x^2}H_n(x) = e^{-x^2/2}H_n(x), \end{aligned}$$

where the second line is Rodrigues's formula. Therefore

$$\boxed{\psi_n(x) = (\sqrt{\pi}2^n n!)^{-1/2}(x - D)^n e^{-x^2/2}}.$$

Equivalently, $(a^\dagger)^n\psi_0 = \sqrt{n!}\,\psi_n$ follows from Exercise 18.2.7 by multiplying the successive factors $\sqrt{1}, \sqrt{2}, \ldots, \sqrt{n}$. Since $x - D = \sqrt{2}\,a^\dagger$ and $e^{-x^2/2} = \pi^{1/4}\psi_0$, this gives $(x - D)^n e^{-x^2/2} = 2^{n/2}\pi^{1/4}\sqrt{n!}\,\psi_n$, the same result. Both derivations include $n = 0$ as the identity operator.

18.3 Laguerre Functions

18.3.1 Show with the aid of the Leibniz formula that the series expansion of $L_n(x)$, Eq. (18.53), follows from the Rodrigues representation, Eq. (18.72).

Solution.

Use Eq. (18.72) with $k = 0$ and $D = d/dx$:

$$L_n(x) = \frac{e^x}{n!} D^n(x^n e^{-x}).$$

Leibniz's formula assigns j derivatives to x^n and $n - j$ to the exponential. Since

$$D^j x^n = \frac{n!}{(n-j)!} x^{n-j}, \qquad D^{n-j} e^{-x} = (-1)^{n-j} e^{-x}, \qquad \frac{1}{n!}\binom{n}{j}\frac{n!}{(n-j)!} = \frac{n!}{j!\,[(n-j)!]^2},$$

it gives

$$\begin{aligned} L_n(x) &= \frac{e^x}{n!} \sum_{j=0}^{n} \binom{n}{j} \frac{n!}{(n-j)!} x^{n-j} (-1)^{n-j} e^{-x} \\ &= \sum_{j=0}^{n} \frac{(-1)^{n-j} n!}{j!\,[(n-j)!]^2} x^{n-j}. \end{aligned}$$

Putting $m = n - j$ reverses the finite sum and yields

$$\boxed{L_n(x) = \sum_{m=0}^{n} \frac{(-1)^m n!}{(n-m)!(m!)^2} x^m}.$$

In descending order, the coefficient of x^{n-i} ($j = i$ above) is

$$\frac{(-1)^{n-i} n!}{i!\,[(n-i)!]^2} = \frac{(-1)^n}{n!}\,\frac{(-1)^i}{i!}\left[\frac{n!}{(n-i)!}\right]^2,$$

which for $i = 0, 1, 2$ and n gives the bracketed coefficients 1, $-n^2$, $n^2(n-1)^2/2!$ and $(-1)^n n!$. Thus the first descending terms are

$$\frac{(-1)^n}{n!}\left[x^n - n^2 x^{n-1} + \frac{n^2(n-1)^2}{2!} x^{n-2} - \cdots + (-1)^n n!\right],$$

which is Eq. (18.53). The empty differentiations and single-term sum at $n = 0$ both give $L_0 = 1$.

18.3.2

(a) Using the explicit series form, Eq. (18.53), show that

$$L_n'(0) = -n, \qquad L_n''(0) = \tfrac{1}{2} n(n-1).$$

(b) Repeat without using the explicit series form of $L_n(x)$.

Solution.

(a) In the ascending series, the coefficients of x and x^2 are

$$-\frac{n!}{(n-1)!} = -n, \qquad \frac{n!}{(n-2)!(2!)^2} = \frac{n(n-1)}{4}.$$

A first derivative at zero selects the first coefficient; a second derivative selects twice the second. Thus

$$\boxed{L_n'(0) = -n, \qquad L_n''(0) = \frac{n(n-1)}{2}}.$$

For $n = 0, 1$ the absent higher terms give zero exactly where the formulas predict it.

(b) Use the ODE $xL_n'' + (1-x)L_n' + nL_n = 0$ and the value $L_n(0) = 1$, which follows without the series from the generating function at $x = 0$: $\sum_n L_n(0)t^n = 1/(1-t) = \sum_n t^n$. Setting $x = 0$ in the ODE (the polynomial L_n'' is finite there) gives $L_n'(0) + nL_n(0) = L_n'(0) + n = 0$. Differentiate the ODE once:

$$xL_n''' + L_n'' + (1-x)L_n'' - L_n' + nL_n' = 0,$$

or

$$xL_n''' + (2-x)L_n'' + (n-1)L_n' = 0.$$

At zero, $2L_n''(0) + (n-1)L_n'(0) = 2L_n''(0) + (n-1)(-n) = 0$, giving the second formula $L_n''(0) = n(n-1)/2$. No explicit polynomial series has been used in this second proof.

18.3.3 Derive the normalization relation, Eq. (18.71) for the associated Laguerre polynomials, thereby also confirming Eq. (18.55) for the L_n.

Solution.

For fixed nonnegative integer k and $D = d/dx$, Rodrigues's formula is

$$e^{-x}x^k L_n^k(x) = \frac{1}{n!}D^n(e^{-x}x^{n+k}).$$

Assume first $m \le n$. Integrating by parts n times gives

$$\begin{aligned} I_{mn} &= \int_0^\infty e^{-x}x^k L_n^k L_m^k\,dx = \frac{1}{n!}\int_0^\infty D^n(e^{-x}x^{n+k})\,L_m^k\,dx \\ &= \frac{(-1)^n}{n!}\int_0^\infty e^{-x}x^{n+k}D^n L_m^k\,dx. \end{aligned}$$

The $(j+1)$st integration by parts, $0 \le j < n$, is

$$\begin{aligned} \int_0^\infty D^{n-j}(e^{-x}x^{n+k})\,D^j L_m^k\,dx = &\Big[D^{n-j-1}(e^{-x}x^{n+k})\,D^j L_m^k\Big]_0^\infty \\ &- \int_0^\infty D^{n-j-1}(e^{-x}x^{n+k})\,D^{j+1}L_m^k\,dx. \end{aligned}$$

At infinity the boundary terms decay exponentially. At zero, by Leibniz's rule every term of $D^{n-1-j}(e^{-x}x^{n+k})$ contains a power x^{n+k-i} with $i \le n-1-j$, so this derivative is $O(x^{k+j+1})$ and each boundary term also vanishes. If $m < n$, then $D^n L_m^k = 0$. If $m = n$, the leading term of L_n^k is $(-1)^n x^n/n!$: in Rodrigues's formula the highest power comes from letting all n derivatives act on e^{-x}, giving $D^n(e^{-x}x^{n+k}) = e^{-x}[(-1)^n x^{n+k} + \cdots]$. Hence $D^n L_n^k = (-1)^n$ and

$$I_{nn} = \frac{1}{n!}\int_0^\infty e^{-x}x^{n+k}\,dx = \frac{\Gamma(n+k+1)}{n!}.$$

Symmetry $I_{mn} = I_{nm}$ supplies the case $m > n$. Therefore

$$\boxed{\int_0^\infty e^{-x} x^k L_n^k(x) L_m^k(x)\, dx = \frac{(n+k)!}{n!}\delta_{mn}}.$$

For $k = 0$ the norm is one, proving Eq. (18.55). The same proof works for real $k > -1$, replacing factorials by gamma functions: the displayed zero-endpoint powers still tend to zero.

18.3.4 Expand x^r in a series of associated Laguerre polynomials $L_n^k(x)$, with k fixed and n ranging from 0 to r when r is a nonnegative integer. For other real $r > -k-1$, use an infinite expansion with the convergence interpretation below.

Hint. The Rodrigues form of $L_n^k(x)$ will be useful.

ANS. $x^r = (r+k)!\, r! \sum_{n=0}^{r} \dfrac{(-1)^n L_n^k(x)}{(n+k)!\,(r-n)!}$, $\quad 0 \le x < \infty$. The displayed finite answer applies to $r = 0, 1, 2, \ldots$. For $r > 0$ the infinite version converges locally uniformly on $[0, \infty)$; for $-k-1 < r < 0$ use its Abel sum on $x > 0$. Weighted L^2 convergence holds precisely when $r > -(k+1)/2$.

Solution.

Take the book's fixed nonnegative integer k. Write $h_n = \Gamma(n+k+1)/n!$ for the norm from Exercise 18.3.3. For a nonnegative integer r, x^r is a finite combination of $L_0^k, \ldots, L_r^k$, since their degrees are distinct and their leading coefficients are nonzero. Its coefficient is

$$c_n = \frac{1}{h_n} \int_0^\infty x^{r+k} e^{-x} L_n^k(x)\, dx.$$

Rodrigues's formula, $x^k e^{-x} L_n^k = D^n(e^{-x} x^{n+k})/n!$, cancels the $n!$ in $1/h_n$. Then n integrations by parts, as in Exercise 18.3.3 but moving the derivatives onto x^r, together with $D^n x^r = r!\, x^{r-n}/(r-n)!$ and $\int_0^\infty e^{-x} x^{r+k} dx = (r+k)!$, give for $n \le r$

$$\begin{aligned} c_n &= \frac{1}{\Gamma(n+k+1)} \int_0^\infty x^r D^n(e^{-x} x^{n+k})\, dx \\ &= \frac{(-1)^n}{\Gamma(n+k+1)} \frac{r!}{(r-n)!} \int_0^\infty e^{-x} x^{r+k}\, dx \\ &= \frac{(-1)^n (r+k)! r!}{(n+k)!(r-n)!}. \end{aligned}$$

For $n > r$ the derivative of x^r is zero, so $c_n = 0$. Every endpoint term is zero: at infinity by the exponential and at zero by a positive power at least $k+1$. Hence

$$\boxed{x^r = (r+k)! r! \sum_{n=0}^{r} \frac{(-1)^n L_n^k(x)}{(n+k)!(r-n)!}}, \qquad r = 0, 1, 2, \ldots.$$

The noninteger extension needs parameter and convergence conditions; a finite upper limit r and ordinary factorials no longer specify it. For real $r > -k-1$ the same integration is valid, even when intermediate derivatives are singular: every zero-endpoint boundary product has order x^{r+k+1}, and the final integral has power $r+k > -1$. Write the falling derivative as a polynomial in r,

$$D^n x^r = r(r-1)\cdots(r-n+1)\, x^{r-n} = (-1)^n (-r)_n x^{r-n},$$

with $(-r)_n = (-r)(-r+1)\cdots(-r+n-1)$. With $\int_0^\infty e^{-x}x^{r+k}dx = \Gamma(r+k+1)$, the two signs $(-1)^n$ cancel and

$$\boxed{c_n = \frac{\Gamma(r+k+1)(-r)_n}{\Gamma(n+k+1)}}.$$

This form also handles nonnegative integer r without undefined gamma quotients and terminates automatically.

For $r > 0$ the resulting infinite series converges absolutely and uniformly on each finite interval $0 \le x \le A$, and equals x^r. Here are explicit convergence justifications. The useful bound is

$$|L_n^k(x)| \le e^{x/2}\binom{n+k}{n}, \qquad x \ge 0.$$

For $k = 0$ it follows from the Gaussian identity

$$e^{-x/2}L_n(x) = \frac{1}{2^n n!\sqrt{\pi}} \int_{\mathbb{R}} e^{-y^2} H_n(y)^2 e^{i\sqrt{2x}\,y}\, dy.$$

Indeed $|e^{i\sqrt{2x}\,y}| = 1$, and without the phase the integral of the nonnegative function $e^{-y^2}H_n^2$ is the Hermite norm $2^n n!\sqrt{\pi}$, so the right side has modulus at most one. To prove this identity, multiply two Hermite generators and integrate with e^{-y^2+icy}, c real. Completion of the square, $-y^2 + (2u+2v+ic)y = -(y-w)^2 + w^2$ with $w = u+v+ic/2$, gives

$$\int_{\mathbb{R}} e^{-y^2+icy} e^{-u^2+2uy} e^{-v^2+2vy}\, dy = \sqrt{\pi}\, e^{w^2-u^2-v^2} = \sqrt{\pi}\, e^{2uv+ic(u+v)-c^2/4}.$$

Here $\int_{\mathbb{R}} e^{-(y-w)^2} dy = \sqrt{\pi}$ also for complex w, since its value at $w = 0$ is $\sqrt{\pi}$ and its derivative with respect to w vanishes:

$$\int_{\mathbb{R}} 2(y-w)e^{-(y-w)^2} dy = -\Big[e^{-(y-w)^2}\Big]_{-\infty}^{\infty} = 0.$$

On the left, expand both generators and integrate termwise. This is justified for $|u|, |v| \le R$: the double series of absolute values is at most $\big(\sum_n |H_n(y)|R^n/n!\big)^2 \le e^{4R|y|+2R^2}$ by the bound of Exercise 18.1.6, and $e^{-y^2+4R|y|}$ is integrable. The coefficient of $u^n v^n$ on the left is therefore $(n!)^{-2}\int e^{-y^2+icy}H_n^2\, dy$. On the right the terms $(2uv)^\ell/\ell!$ of e^{2uv} and $(icu)^{n-\ell}(icv)^{n-\ell}/[(n-\ell)!]^2$ of $e^{icu}e^{icv}$ give the coefficient

$$\sqrt{\pi}\, e^{-c^2/4} \sum_{\ell=0}^{n} \frac{2^\ell (-c^2)^{n-\ell}}{\ell!\,[(n-\ell)!]^2}.$$

Multiplying both by $(n!)^2/(2^n n!\sqrt{\pi})$, i.e., dividing the integral by the Hermite norm, and putting $j = n - \ell$ give

$$\frac{\int e^{-y^2+icy} H_n^2\, dy}{2^n n!\sqrt{\pi}} = e^{-c^2/4} \sum_{j=0}^{n} \frac{n!(-c^2/2)^j}{(n-j)!(j!)^2} = e^{-c^2/4} L_n(c^2/2),$$

by the series of Exercise 18.3.1. Set $c = \sqrt{2x}$, so that $c^2/2 = x$ and $c^2/4 = x/2$. For integer $k \ge 1$, the generating function $\sum_n L_n^k(x)t^n = (1-t)^{-k-1}e^{-xt/(1-t)}$ (derived in Exercise 18.3.5(b)) is $(1-t)^{-k}\sum_j L_j(x)t^j$. With $(1-t)^{-k} = \sum_i \binom{i+k-1}{k-1}t^i$, the coefficient of t^n gives

$$L_n^k(x) = \sum_{j=0}^{n} \binom{n-j+k-1}{k-1} L_j(x).$$

The coefficients are positive, and their sum

$$\sum_{i=0}^{n}\binom{i+k-1}{k-1}=\binom{n+k}{n}$$

is the coefficient of t^n in $(1-t)^{-k}(1-t)^{-1}=(1-t)^{-k-1}$. Since each $|L_j(x)|\le e^{x/2}$, this proves the bound. Consequently, for $0\le x\le A$, using $\binom{n+k}{n}/\Gamma(n+k+1)=1/(n!\,k!)$,

$$|c_nL_n^k(x)|\le\frac{e^{A/2}\Gamma(r+k+1)}{k!}\frac{|(-r)_n|}{n!}.$$

For noninteger $r>0$, the last ratio is $|\Gamma(n-r)/[\Gamma(-r)\Gamma(n+1)]|=O(n^{-r-1})$, since $\Gamma(n-r)/\Gamma(n+1)\sim n^{-r-1}$ (Exercise 13.4.6 with $a=-r-1$, $b=0$); this is summable, so the Weierstrass M-test gives absolute and uniform convergence on $0\le x\le A$. For integer r it terminates.

To identify the sum, the Laguerre polynomials are complete in $L^2((0,\infty),x^ke^{-x}dx)$. A brief proof is to take a function f orthogonal to all monomials and form $F(z)=\int_0^\infty f(x)x^ke^{-x}e^{zx}dx$. Cauchy–Schwarz makes F analytic for $\Re z<1/2$, all its derivatives $F^{(m)}(0)=\int_0^\infty f(x)x^{m+k}e^{-x}dx$ at zero vanish, and the identity theorem makes it zero throughout that half-plane. Fourier uniqueness on $z=i\xi$ then gives $f=0$ almost everywhere. Thus the computed orthogonal expansion converges to x^r in this norm whenever x^r lies in the space, i.e. whenever $\int_0^\infty x^{2r+k}e^{-x}dx<\infty$, or $r>-(k+1)/2$. Conversely, for $-k-1<r\le-(k+1)/2$ it does not converge in this norm: by orthogonality the Nth partial sum has squared norm $\sum_{n\le N}|c_n|^2h_n$, and, with $(-r)_n=\Gamma(n-r)/\Gamma(-r)$,

$$\begin{aligned}|c_n|^2h_n&=\frac{\Gamma(r+k+1)^2[(-r)_n]^2}{n!\,\Gamma(n+k+1)}=\left[\frac{\Gamma(r+k+1)}{\Gamma(-r)}\right]^2\frac{\Gamma(n-r)}{\Gamma(n+1)}\,\frac{\Gamma(n-r)}{\Gamma(n+k+1)}\\&\sim\left[\frac{\Gamma(r+k+1)}{\Gamma(-r)}\right]^2n^{-2r-k-2},\end{aligned}$$

because, by Exercise 13.4.6, the two gamma quotients behave as n^{-r-1} and n^{-r-k-1}. The exponent $-2r-k-2$ is at least -1, so these norms are unbounded.

For $r>0$ the partial sums S_N converge uniformly on each $[0,A]$ to a continuous sum S, so $\int_0^A|S_N-S|^2x^ke^{-x}dx\le k!\max_{[0,A]}|S_N-S|^2\to0$. They also converge to x^r in the weighted norm, hence in the norm over $(0,A)$. Limits in this norm are unique, so $S=x^r$ almost everywhere on $[0,A]$, and, both functions being continuous, $S(x)=x^r$ at every $x\ge0$. The case $r=0$ is the single term $L_0^k=1$.

For completeness, negative powers have a precise extension. Put $r=-\beta$ with $0<\beta<k+1$ and $x>0$. The same coefficients give the Abel sum

$$x^{-\beta}=\lim_{q\uparrow1}\sum_{n=0}^{\infty}c_nq^nL_n^k(x).$$

To prove it rather than assume ordinary pointwise convergence, write $c_n=\Gamma(k+1-\beta)(\beta)_n/\Gamma(n+k+1)$ with $(\beta)_n=\Gamma(\beta+n)/\Gamma(\beta)$. The beta integral

$$\int_0^1u^{\beta+n-1}(1-u)^{k-\beta}du=\frac{\Gamma(\beta+n)\Gamma(k+1-\beta)}{\Gamma(n+k+1)}$$

gives $c_n=\Gamma(\beta)^{-1}\int_0^1u^{\beta+n-1}(1-u)^{k-\beta}du$. Inserting this and summing the generating function $\sum_n(qu)^nL_n^k(x)=(1-qu)^{-k-1}e^{-xqu/(1-qu)}$ under the integral give, for $0\le q<1$,

$$\sum_{n=0}^{\infty}c_nq^nL_n^k(x)=\frac{1}{\Gamma(\beta)}\int_0^1\frac{u^{\beta-1}(1-u)^{k-\beta}}{(1-qu)^{k+1}}\exp\!\left(-\frac{xqu}{1-qu}\right)du.$$

The series interchange is uniform because $|qu| \le q < 1$: by the bound above, $\sum_n |L_n^k(x)|(qu)^n \le e^{x/2}(1-q)^{-k-1}$. For $q \ge 1/2$ and $u \ge 1/2$, put $\sigma = qu/(1-qu) \ge 1/3$; then

$$(1-qu)^{-k-1}e^{-x\sigma} = (1+\sigma)^{k+1}e^{-x\sigma} \le \max_{\sigma\ge 0}(1+\sigma)^{k+1}e^{-x\sigma} < \infty,$$

so the exponential bounds the inverse power uniformly; on $u \le 1/2$ the denominator satisfies $1-qu \ge 1/2$ directly. Thus an integrable constant multiple of the beta weight $u^{\beta-1}(1-u)^{k-\beta}$ dominates the integrand. Dominated convergence and the substitution $v = u/(1-u)$, for which $1-u = 1/(1+v)$, $du = dv/(1+v)^2$ and $u^{\beta-1}(1-u)^{-\beta-1}du = v^{\beta-1}dv$, yield

$$\lim_{q\uparrow 1}(\cdots) = \frac{1}{\Gamma(\beta)}\int_0^1 \frac{u^{\beta-1}}{(1-u)^{\beta+1}}e^{-xu/(1-u)}du = \frac{1}{\Gamma(\beta)}\int_0^\infty v^{\beta-1}e^{-xv}dv = x^{-\beta}.$$

If also $\beta < (k+1)/2$, the series has the weighted L^2 interpretation above. For negative powers the endpoint $x = 0$ is not a finite value. If $r \le -k-1$, even the coefficient integrals diverge at zero, so the unrestricted orthogonal-expansion prescription in the question does not apply there.

18.3.5 Expand e^{-ax} in a series of associated Laguerre polynomials $L_n^k(x)$, with k fixed and n ranging from 0 to ∞.

(a) Evaluate directly the coefficients in your assumed expansion.

(b) Develop the desired expansion from the generating function.

ANS. $e^{-ax} = \dfrac{1}{(1+a)^{1+k}}\displaystyle\sum_{n=0}^{\infty}\left(\frac{a}{1+a}\right)^n L_n^k(x), \qquad 0 \le x < \infty$. Assume $\Re a > -1/2$ for an expansion valid at every $x \ge 0$.

Solution.

For an expansion valid throughout $0 \le x < \infty$, take $\Re a > -1/2$. This is both the weighted L^2 condition and the condition $|a/(1+a)| < 1$ for the generating series.

(a) By orthogonality the coefficient of L_n^k is

$$c_n = \frac{n!}{\Gamma(n+k+1)}\int_0^\infty x^k e^{-(1+a)x}L_n^k(x)\,dx.$$

Rodrigues's formula, $x^k e^{-x}L_n^k = D^n(e^{-x}x^{n+k})/n!$, followed by n integrations by parts, as in Exercise 18.3.3, with $D^n e^{-ax} = (-a)^n e^{-ax}$ and, for $\Re(1+a) > 0$,

$$\int_0^\infty e^{-(1+a)x}x^{n+k}dx = \frac{\Gamma(n+k+1)}{(1+a)^{n+k+1}},$$

gives

$$\begin{aligned} c_n &= \frac{1}{\Gamma(n+k+1)}\int_0^\infty e^{-ax}D^n(e^{-x}x^{n+k})\,dx \\ &= \frac{(-1)^n(-a)^n}{\Gamma(n+k+1)}\int_0^\infty e^{-(1+a)x}x^{n+k}\,dx \\ &= \frac{a^n}{\Gamma(n+k+1)}\frac{\Gamma(n+k+1)}{(1+a)^{n+k+1}} \\ &= \frac{1}{(1+a)^{k+1}}\left(\frac{a}{1+a}\right)^n. \end{aligned}$$

The boundary products vanish at zero as positive powers and at infinity because $\Re(1+a)>0$. This weaker condition suffices to compute coefficients, but by itself does not ensure the requested series converges.

(b) Leibniz's rule in Rodrigues's formula, as in Exercise 18.3.1, with $n-j$ derivatives on x^{n+k} and j on e^{-x}, gives the explicit polynomial

$$\begin{aligned} L_n^k(x) &= \frac{e^x x^{-k}}{n!}\sum_{j=0}^{n}\binom{n}{j}\frac{(n+k)!}{(k+j)!}x^{k+j}(-1)^j e^{-x} \\ &= \sum_{j=0}^{n}\binom{n+k}{n-j}\frac{(-x)^j}{j!}, \end{aligned}$$

since

$$\frac{1}{n!}\binom{n}{j}\frac{(n+k)!}{(k+j)!} = \frac{(n+k)!}{j!\,(n-j)!\,(k+j)!} = \binom{n+k}{n-j}\frac{1}{j!}.$$

Put $n=j+m$ in the generating sum, so that $\binom{n+k}{n-j}=\binom{m+j+k}{m}$. The binomial series $\sum_m \binom{m+N}{m}t^m = (1-t)^{-N-1}$, with $N=j+k$, then yields

$$\begin{aligned} \sum_{n=0}^{\infty} L_n^k(x)t^n &= \sum_{j=0}^{\infty}\frac{(-xt)^j}{j!}\sum_{m=0}^{\infty}\binom{m+j+k}{m}t^m \\ &= \sum_{j=0}^{\infty}\frac{(-xt)^j}{j!}(1-t)^{-j-k-1} \\ &= (1-t)^{-k-1}\exp\!\left(-\frac{xt}{1-t}\right), \qquad |t|<1. \end{aligned}$$

For $|t|<1$ the sum of absolute values is bounded by the same calculation with $|x|,|t|$, so changing the order is justified. Set $t=a/(1+a)$, so $1-t=1/(1+a)$ and $t/(1-t)=a$; the generating function becomes $(1+a)^{k+1}e^{-ax}$. Division by $(1+a)^{k+1}$ gives

$$\boxed{e^{-ax} = \frac{1}{(1+a)^{k+1}}\sum_{n=0}^{\infty}\left(\frac{a}{1+a}\right)^n L_n^k(x)}.$$

The convergence condition $|t|<1$, i.e. $|a|<|1+a|$, follows from $|1+a|^2-|a|^2 = 1+2\Re a>0$. The Laguerre bound proved in Exercise 18.3.4 also gives absolute locally uniform convergence. At $a=0$ only the $n=0$ term remains. At $\Re a=-1/2$ the terms at $x=0$ have magnitudes proportional to $\binom{n+k}{n}$ and fail to tend to zero; this prevents including the boundary in a formula asserted for all $x\ge 0$.

18.3.6 Show that $\displaystyle\int_0^\infty e^{-x}x^{k+1}L_n^k(x)L_n^k(x)\,dx = \frac{(n+k)!}{n!}\,(2n+k+1)$.

Hint. Note that $xL_n^k = (2n+k+1)L_n^k - (n+k)L_{n-1}^k - (n+1)L_{n+1}^k$.

Solution.

Let $h_n = (n+k)!/n!$. Multiply the recurrence of the hint by $x^k e^{-x} L_n^k$ and integrate:

$$\int_0^\infty x^{k+1} e^{-x} (L_n^k)^2\, dx = (2n+k+1)\int_0^\infty x^k e^{-x}(L_n^k)^2\, dx - (n+k)\int_0^\infty x^k e^{-x} L_{n-1}^k L_n^k\, dx - (n+1)\int_0^\infty x^k e^{-x} L_{n+1}^k L_n^k\, dx.$$

Both mixed integrals vanish by orthogonality, and the first integral on the right is the norm h_n of Eq. (18.71), leaving

$$\boxed{\int_0^\infty x^{k+1} e^{-x} (L_n^k)^2\, dx = (2n+k+1)\frac{(n+k)!}{n!}}.$$

At $n=0$ the L_{-1}^k term is absent (equivalently define it as zero); directly the integral is $\Gamma(k+2) = (k+1)k!$, consistent with the formula. The Instructor's Manual's last denominator $k!$ must be $n!$, as in the norm (18.71); the question's displayed normalization is correct.

18.3.7 Assume that a particular problem in quantum mechanics has led to the ODE

$$\frac{d^2y}{dx^2} - \left[\frac{k^2-1}{4x^2} - \frac{2n+k+1}{2x} + \frac{1}{4}\right] y = 0$$

for nonnegative integers n, k. Write $y(x)$ as $y(x) = A(x)B(x)C(x)$, with the requirement that

(a) $A(x)$ be a **negative** exponential giving the required asymptotic behavior of $y(x)$, and

(b) $B(x)$ be a **positive** power of x giving the behavior of $y(x)$ for $0 \le x \ll 1$.

Determine $A(x)$ and $B(x)$. Find the relation between $C(x)$ and the associated Laguerre polynomial.

ANS. $A(x) = e^{-x/2}$, $B(x) = x^{(k+1)/2}$, $C(x) = L_n^k(x)$. Select the solution regular at $x=0$ (excluding the logarithmic branch when $k=0$).

Solution.

At large positive x, the constant coefficient gives $y'' - y/4 \simeq 0$, with exponential factors $e^{\pm x/2}$. The decaying factor is therefore $A = e^{-x/2}$; powers of x can still multiply it. At zero insert $y \sim x^p$ into the terms of order x^{p-2}, namely y'' and $-(k^2-1)y/(4x^2)$:

$$p(p-1) - \frac{k^2-1}{4} = 0, \qquad p = \frac{1 \pm \sqrt{1+k^2-1}}{2} = \frac{1 \pm k}{2}.$$

The regular positive-power choice is $p = (k+1)/2$, giving $B = x^{(k+1)/2}$. For $k=0$ the indicial root is repeated: selecting the regular Frobenius branch also excludes its logarithmic companion. The conditions fix a solution only up to overall normalization.

Set $p = (k+1)/2$ and $g = AB = e^{-x/2}x^p$. Then, using $g''/g = (g'/g)' + (g'/g)^2$,

$$\frac{g'}{g} = \frac{p}{x} - \frac{1}{2}, \qquad \frac{g''}{g} = -\frac{p}{x^2} + \left(\frac{p}{x} - \frac{1}{2}\right)^2 = \frac{p(p-1)}{x^2} - \frac{p}{x} + \frac{1}{4}.$$

Insert $y = gC$, so that $y'' = g''C + 2g'C' + gC''$, and divide by g:

$$C'' + 2\frac{g'}{g}C' + \left[\frac{g''}{g} - \frac{k^2-1}{4x^2} + \frac{2n+k+1}{2x} - \frac{1}{4}\right] C = 0.$$

The indicial equation $p(p-1) = (k^2-1)/4$ cancels the x^{-2} terms, and the constants $\pm\frac{1}{4}$ cancel. What remains is

$$C'' + \left(\frac{2p}{x} - 1\right) C' + \frac{(2n+k+1)/2 - p}{x} C = 0,$$

or, since $2p = k+1$ and $(2n+k+1)/2 - p = n$, after multiplication by x,

$$xC'' + (k+1-x)C' + nC = 0.$$

This is the associated Laguerre equation. More explicitly, insertion of $C = \sum_{j\ge 0} c_j x^j$ and collection of the coefficient of x^j (xC'' and $(k+1)C'$ contribute $(j+1)j\,c_{j+1} + (k+1)(j+1)c_{j+1}$, while $-xC' + nC$ contributes $(n-j)c_j$) give

$$(j+1)(j+k+1)c_{j+1} + (n-j)c_j = 0, \quad c_{j+1} = \frac{j-n}{(j+1)(j+k+1)} c_j.$$

For nonnegative integer n it terminates at degree n, because $c_{n+1} = 0$. Iterating,

$$c_j = c_0 \prod_{i=0}^{j-1} \frac{i-n}{(i+1)(i+k+1)} = c_0 \frac{(-1)^j n!\, k!}{(n-j)!\, j!\, (k+j)!}.$$

Choosing $c_0 = (n+k)!/(n!k!)$ gives $c_j = (-1)^j (n+k)!/[(n-j)!\, j!\, (k+j)!] = \binom{n+k}{n-j}(-1)^j/j!$, the coefficients of L_n^k (Exercise 18.3.5(b)), so exactly $C = L_n^k$; an arbitrary c_0 gives a constant multiple. Thus

$$\boxed{y(x) = \text{constant } e^{-x/2} x^{(k+1)/2} L_n^k(x)}.$$

Its full large-x leading behavior is proportional to $e^{-x/2} x^{n+(k+1)/2}$, since $L_n^k \sim (-1)^n x^n/n!$.

18.3.8 From Eq. (18.84) the normalized radial part of the hydrogenic wave function is

$$R_{nL}(r) = \left[\alpha^3 \frac{(n-L-1)!}{2n(n+L)!}\right]^{1/2} e^{-\alpha r/2} (\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r),$$

in which $\alpha = 2Z/na_0 = 2Zme^2/(4\pi\epsilon_0\hbar^2 n)$. Evaluate

(a) $\langle r \rangle = \displaystyle\int_0^\infty r R_{nL}(r) R_{nL}(r) r^2\, dr,$

(b) $\langle r^{-1} \rangle = \displaystyle\int_0^\infty r^{-1} R_{nL}(r) R_{nL}(r) r^2\, dr.$

The quantity $\langle r \rangle$ is the average displacement of the electron from the nucleus, whereas $\langle r^{-1} \rangle$ is the average of the reciprocal displacement.

ANS. $\langle r \rangle = \dfrac{a_0}{2Z}\big[3n^2 - L(L+1)\big]$, $\quad \langle r^{-1} \rangle = \dfrac{Z}{n^2 a_0}$. Use $Z > 0$, $a_0 > 0$, $n \ge 1$ and $0 \le L \le n-1$, with integer n, L.

Solution.

Equation (18.84) has $e^{-\alpha r/2}$, not the $e^{-\alpha r}$ printed in this exercise. The second expression for α also needs $1/n$. Once R_{nL} is defined as a function of r, the moments use $R_{nL}(r)$, not $R_{nL}(\alpha r)$. The printed answers further assume $Z = 1$; the corrected formulas below retain general Z. Take $Z > 0$, $n = 1, 2, \ldots$, and $L = 0, \ldots, n-1$. Put $q = n-L-1$, $k = 2L+1$, $x = \alpha r$ and $h_q = (q+k)!/q!$. Then

$(n-L-1)! = q!$, $(n+L)! = (q+k)!$, and $r^2dr = x^2dx/\alpha^3$ cancels the factor α^3, while $x^{2L}x^2 = x^{k+1}$. The corrected radial probability measure is therefore

$$R_{nL}(r)^2 r^2 dr = \frac{1}{2nh_q} e^{-x} x^{k+1} [L_q^k(x)]^2 dx, \qquad \alpha = \frac{2Z}{na_0}.$$

Since $2q + k + 1 = 2n$, Exercise 18.3.6 shows that its integral is 1, explicitly checking normalization.

(b) Multiplication by $r^{-1} = \alpha/x$ removes the additional power of x. The normalization integral (18.71), whose value is h_q, then gives

$$\langle r^{-1} \rangle = \frac{\alpha}{2nh_q} \int_0^\infty e^{-x} x^k [L_q^k(x)]^2 dx = \frac{\alpha}{2n} = \boxed{\frac{Z}{n^2 a_0}}.$$

(a) Multiplication by $r = x/\alpha$ instead gives $\langle r \rangle = J_2/(2n\alpha h_q)$, where $J_2 = \int_0^\infty e^{-x} x^k [x L_q^k(x)]^2 dx$. Use the recurrence of Exercise 18.3.6,

$$x L_q^k = (2q + k + 1) L_q^k - (q + k) L_{q-1}^k - (q + 1) L_{q+1}^k.$$

Squaring and integrating with the weight $e^{-x}x^k$ removes every mixed product by orthogonality, and each square integrates to its norm h_q, h_{q-1} or h_{q+1}. Hence

$$\begin{aligned} \frac{J_2}{h_q} &= (2q + k + 1)^2 + (q + k)^2 \frac{h_{q-1}}{h_q} + (q + 1)^2 \frac{h_{q+1}}{h_q} \\ &= (2q + k + 1)^2 + q(q + k) + (q + 1)(q + k + 1). \end{aligned}$$

Here $h_{q-1}/h_q = q/(q + k)$ and $h_{q+1}/h_q = (q + k + 1)/(q + 1)$. For $q = 0$ omit the first neighbor term before taking these ratios; its final contribution $q(q + k)$ is zero. Substituting $q = n - L - 1$ and $k = 2L + 1$ gives

$$\begin{aligned} q(q + k) &= (n - L - 1)(n + L) = n^2 - n - L(L + 1), \\ (q + 1)(q + k + 1) &= (n - L)(n + L + 1) = n^2 + n - L(L + 1), \\ J_2/h_q &= 4n^2 + 2n^2 - 2L(L + 1). \end{aligned}$$

Consequently, with $n\alpha = 2Z/a_0$,

$$\langle r \rangle = \frac{J_2}{2n\alpha h_q} = \frac{6n^2 - 2L(L + 1)}{2n\alpha} = \frac{3n^2 - L(L + 1)}{n\alpha} = \boxed{\frac{a_0}{2Z}[3n^2 - L(L + 1)]}.$$

At $n = 1$, $L = 0$ these become $3a_0/(2Z)$ and Z/a_0. Setting $Z = 1$ in the two boxed results gives the printed answers $\frac{a_0}{2}[3n^2 - L(L + 1)]$ and $1/(n^2 a_0)$.

18.3.9 Derive a recurrence formula for the hydrogen wave function expectation values:

$$\frac{s + 2}{n^2} \langle r^{s+1} \rangle - (2s + 3) a_0 \langle r^s \rangle + \frac{s + 1}{4} \Big[(2L + 1)^2 - (s + 1)^2 \Big] a_0^2 \langle r^{s-1} \rangle = 0,$$

with $s \geq -2L - 1$.

Hint. Transform Eq. (18.80) into a form analogous to Eq. (18.73). Multiply by $\rho^{s+2}u' - c\rho^{s+1}u$, with $u = \rho\Phi$. Adjust c to cancel terms that do not yield expectation values. Use hydrogen, $Z = 1$, with $n \geq 1$, $0 \leq L \leq n - 1$ and real s in the stated range.

Solution.

Use hydrogen, $Z = 1$, and $\rho = \alpha r$ with $\alpha = 2/(na_0)$. In terms of ρ, the radial equation (18.80) for $\Phi(\rho) = R_{nL}(\rho/\alpha)$, with its eigenvalue parameter equal to n, reads

$$\frac{1}{\rho^2} \frac{d}{d\rho} \left(\rho^2 \frac{d\Phi}{d\rho} \right) + \left(\frac{n}{\rho} - \frac{1}{4} - \frac{L(L + 1)}{\rho^2} \right) \Phi = 0.$$

Let $u(\rho) = \rho\Phi(\rho)$; its constant scale does not affect moment ratios. Since $u'' = \rho\Phi'' + 2\Phi' = \rho^{-1}(\rho^2\Phi')'$, multiplication of the radial equation by ρ gives the form analogous to Eq. (18.73),

$$u'' = Q(\rho)u, \qquad Q(\rho) = \frac{1}{4} - \frac{n}{\rho} + \frac{L(L+1)}{\rho^2}.$$

Because $u^2 d\rho = \rho^2 R_{nL}^2\, d\rho = \alpha^3 r^2 R_{nL}^2\, dr$, the normalized moments are

$$M_j = \frac{\int_0^\infty \rho^j u^2 d\rho}{\int_0^\infty u^2 d\rho} = \alpha^j \langle r^j \rangle.$$

The radial function may be chosen real. By Eq. (18.84), u is a constant times $\rho^{L+1} e^{-\rho/2} L_{n-L-1}^{2L+1}(\rho)$, so near zero $u = O(\rho^{L+1})$ and $u' = O(\rho^L)$, while at infinity u and u' are $e^{-\rho/2}$ times polynomials.

Put $f = \rho^{s+2}$ and $g = -c\rho^{s+1}$. Multiply $u'' - Qu = 0$ by $fu' + gu = fu' - c\rho^{s+1}u$ and integrate first on a finite interval:

$$\int fu'u'' + \int guu'' - \int fQuu' - \int gQu^2 = 0.$$

Since $u'u'' = \frac{1}{2}(u'^2)'$, the first term produces

$$\int fu'u'' = [fu'^2/2] - \frac{1}{2}\int f'u'^2.$$

For the second, integrate by parts and then use $uu' = \frac{1}{2}(u^2)'$ in the term $\int g'uu'$:

$$\begin{aligned}\int guu'' &= [guu'] - \int g'uu' - \int gu'^2 \\ &= [guu' - g'u^2/2] - \int gu'^2 + \frac{1}{2}\int g''u^2.\end{aligned}$$

Thus the unwanted integral in u'^2 cancels when $g = -f'/2$; since $f' = (s+2)\rho^{s+1}$, this means $\boxed{c = (s+2)/2}$. Then $-\int gQu^2 = \frac{1}{2}\int f'Qu^2$, and the same rule $uu' = \frac{1}{2}(u^2)'$ shows that the potential terms satisfy

$$-\int fQuu' + \frac{1}{2}\int f'Qu^2 = [-fQu^2/2] + \frac{1}{2}\int (fQ' + 2f'Q)u^2.$$

With $g = -f'/2$, the second identity reads

$$\int guu'' = \Big[-\frac{1}{2}f'uu' + \frac{1}{4}f''u^2\Big] + \frac{1}{2}\int f'u'^2 - \frac{1}{4}\int f'''u^2.$$

Adding these three identities and multiplying by two yields

$$0 = \left[fu'^2 - f'uu' + \frac{f''}{2}u^2 - fQu^2\right]_0^\infty + \int_0^\infty \left(2f'Q + fQ' - \frac{f'''}{2}\right)u^2 d\rho.$$

Since $f' = (s+2)\rho^{s+1}$, $f'' = (s+1)(s+2)\rho^s$ and, near zero, $Q = O(\rho^{-2})$, each of fu'^2, $f'uu'$, $f''u^2$ and fQu^2 is $O(\rho^{s+2L+2})$ there. Hence the zero-endpoint boundary bracket is $O(\rho^{s+2L+2})$; it vanishes under the given condition $s \geq -2L - 1$. The lowest moment integrand has order ρ^{s+2L+1} and is integrable there. The upper endpoint vanishes exponentially. These estimates justify taking the limits after the finite-interval integrations by parts.

Now compute the remaining coefficient explicitly. With $f''' = s(s+1)(s+2)\rho^{s-1}$ and $Q' = n/\rho^2 - 2L(L+1)/\rho^3$,

$$2f'Q = \frac{s+2}{2}\rho^{s+1} - 2n(s+2)\rho^s + 2(s+2)L(L+1)\rho^{s-1}, \qquad fQ' = n\rho^s - 2L(L+1)\rho^{s-1}.$$

Collecting powers, and then using $4L(L+1) = (2L+1)^2 - 1$ and $s(s+2) = (s+1)^2 - 1$ in the last bracket, gives

$$\begin{aligned} 2f'Q + fQ' - \tfrac{1}{2}f''' &= \frac{s+2}{2}\rho^{s+1} - n(2s+3)\rho^s \\ &\quad + \left[2(s+1)L(L+1) - \frac{s(s+1)(s+2)}{2}\right]\rho^{s-1} \\ &= \frac{s+2}{2}\rho^{s+1} - n(2s+3)\rho^s \\ &\quad + \frac{s+1}{2}\big[(2L+1)^2 - (s+1)^2\big]\rho^{s-1}. \end{aligned}$$

Dividing the integral identity by $\int u^2 d\rho$ gives

$$\frac{s+2}{2}M_{s+1} - n(2s+3)M_s + \frac{s+1}{2}\big[(2L+1)^2 - (s+1)^2\big]M_{s-1} = 0.$$

Substitute $M_j = \alpha^j\langle r^j\rangle$ and multiply by $a_0/(n\alpha^s)$. Since $\alpha = 2/(na_0)$, we have $\alpha a_0/(2n) = 1/n^2$ and $a_0/(2n\alpha) = a_0^2/4$, so

$$\boxed{\begin{aligned} &\frac{s+2}{n^2}\langle r^{s+1}\rangle - (2s+3)a_0\langle r^s\rangle \\ &\quad + \frac{s+1}{4}\big[(2L+1)^2 - (s+1)^2\big]a_0^2\langle r^{s-1}\rangle = 0 \end{aligned}}.$$

For $s = -1$ the last term vanishes and $\langle r^0\rangle = 1$, so $1/n^2 - a_0\langle r^{-1}\rangle = 0$, i.e. $\langle r^{-1}\rangle = 1/(n^2 a_0)$. For $s = 0$, since $\frac{1}{4}[(2L+1)^2 - 1] = L(L+1)$, the relation reads $2\langle r\rangle/n^2 - 3a_0 + L(L+1)a_0^2\langle r^{-1}\rangle = 0$; inserting $\langle r^{-1}\rangle$ gives $\langle r\rangle = a_0[3n^2 - L(L+1)]/2$, checking both moments from Exercise 18.3.8. For a hydrogenic ion replace a_0 by a_0/Z throughout. The proof also permits real $s > -2L - 2$, a slightly larger range than the one requested.

18.3.10 Show that $\displaystyle\int_{-\infty}^{\infty} x^n e^{-x^2} H_n(xy)\,dx = \sqrt{\pi}\,n!\,P_n(y)$, where P_n is a Legendre polynomial.

Solution.

The Instructor's Manual drops $1/[2(n+1)y]$ in its integration by parts and e^{-x^2} in the last term of its next recurrence. The subsequent integrals are with respect to x, not y. The finite calculation below also covers $y = 0$ without division by y.

The Hermite sum is finite, so termwise integration has no convergence interchange issue. Insert

$$H_n(xy) = n!\sum_{j=0}^{\lfloor n/2\rfloor}\frac{(-1)^j(2xy)^{n-2j}}{j!(n-2j)!}.$$

After multiplication by x^n, the Gaussian power is $x^n x^{n-2j} = x^{2n-2j}$. By the even moment of Exercise 18.1.5, with $n-j$ in place of j, its integral is

$$\int_{-\infty}^{\infty} x^{2(n-j)}e^{-x^2}dx = \frac{(2n-2j)!\sqrt{\pi}}{4^{n-j}(n-j)!}.$$

Thus the jth term contributes $n!\,(-1)^j 2^{n-2j} y^{n-2j}(2n-2j)!\sqrt{\pi}/[j!\,(n-2j)!\,4^{n-j}(n-j)!]$, and, using $2^{n-2j}/4^{n-j} = 2^{-n}$,

$$\int_{-\infty}^{\infty} x^n e^{-x^2} H_n(xy)\,dx = \frac{\sqrt{\pi}\,n!}{2^n} \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (2n-2j)!}{j!(n-j)!(n-2j)!} y^{n-2j}.$$

To identify the polynomial, expand the Legendre Rodrigues formula by the binomial theorem, $(y^2-1)^n = \sum_j \binom{n}{j}(-1)^j y^{2n-2j}$, and use, with $D_y = d/dy$, $D_y^n y^{2n-2j} = \frac{(2n-2j)!}{(n-2j)!} y^{n-2j}$ for $2j \le n$:

$$\begin{aligned} P_n(y) &= \frac{1}{2^n n!} D_y^n (y^2-1)^n \\ &= \frac{1}{2^n n!} \sum_{j=0}^{n} (-1)^j \binom{n}{j} D_y^n y^{2n-2j} \\ &= \frac{1}{2^n} \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (2n-2j)!}{j!(n-j)!(n-2j)!} y^{n-2j}. \end{aligned}$$

Terms with $2n-2j < n$, i.e. $j > n/2$, vanish upon differentiation, and

$$\frac{1}{2^n n!}\binom{n}{j}\frac{(2n-2j)!}{(n-2j)!} = \frac{1}{2^n}\frac{(2n-2j)!}{j!(n-j)!(n-2j)!}.$$

Comparing the two finite sums proves

$$\boxed{\int_{-\infty}^{\infty} x^n e^{-x^2} H_n(xy)\,dx = \sqrt{\pi}\, n! P_n(y)}.$$

Both sides are polynomials in y, so the identity also holds for complex y and includes $y = 0$ without division by that parameter.

18.4 Chebyshev Polynomials

18.4.1 By evaluating the generating function for special values of x, verify the special values

$$T_n(1) = 1, \qquad T_n(-1) = (-1)^n, \qquad T_{2n}(0) = (-1)^n, \qquad T_{2n+1}(0) = 0.$$

Solution.

Use the book's type-I generator, whose nonconstant terms carry a factor two:

$$G_T(x,t) = \frac{1-t^2}{1-2xt+t^2} = T_0(x) + 2\sum_{n=1}^{\infty} T_n(x) t^n, \qquad |t| < 1.$$

At $x = \pm 1$ the denominator is $(1 \mp t)^2$, and its factor $1 \mp t$ cancels against $1 - t^2 = (1-t)(1+t)$. Writing each reduced quotient as $2/(\text{its denominator}) - 1$ and expanding geometrically, with $2\sum_{n\ge 0} t^n - 1 =$

$1 + 2\sum_{n\ge1} t^n$, gives at the three requested arguments

$$G_T(1,t) = \frac{1+t}{1-t} = \frac{2}{1-t} - 1 = 1 + 2\sum_{n=1}^{\infty} t^n,$$
$$G_T(-1,t) = \frac{1-t}{1+t} = \frac{2}{1+t} - 1 = 1 + 2\sum_{n=1}^{\infty} (-1)^n t^n,$$
$$G_T(0,t) = \frac{1-t^2}{1+t^2} = \frac{2}{1+t^2} - 1 = 1 + 2\sum_{j=1}^{\infty} (-1)^j t^{2j}.$$

Equating coefficients with $T_0(x) + 2\sum_{n\ge1} T_n(x)t^n$, including the constant coefficient $T_0 = 1$, gives

$$\boxed{T_n(1) = 1, \quad T_n(-1) = (-1)^n, \quad T_{2j}(0) = (-1)^j, \quad T_{2j+1}(0) = 0}.$$

All expansions used are ordinary geometric series with $|t| < 1$.

18.4.2 By evaluating the generating function for special values of x, verify the special values

$$U_n(1) = n + 1, \qquad U_n(-1) = (-1)^n(n+1), \qquad U_{2n}(0) = (-1)^n, \qquad U_{2n+1}(0) = 0.$$

Solution.

The type-II generator has no extra factor two:

$$G_U(x,t) = \frac{1}{1-2xt+t^2} = \sum_{n=0}^{\infty} U_n(x)t^n.$$

For $|t| < 1$,

$$G_U(1,t) = (1-t)^{-2} = \sum_{n=0}^{\infty} (n+1)t^n,$$
$$G_U(-1,t) = (1+t)^{-2} = \sum_{n=0}^{\infty} (-1)^n(n+1)t^n,$$
$$G_U(0,t) = (1+t^2)^{-1} = \sum_{j=0}^{\infty} (-1)^j t^{2j}.$$

The first series is the derivative of $(1-t)^{-1} = \sum_{n\ge0} t^n$, namely $\sum_{n\ge1} nt^{n-1} = \sum_{n\ge0}(n+1)t^n$; the second is the first with t replaced by $-t$, and the third is the geometric series in $-t^2$. Comparison gives

$$\boxed{U_n(1) = n + 1, \quad U_n(-1) = (-1)^n(n+1), \quad U_{2j}(0) = (-1)^j, \quad U_{2j+1}(0) = 0}.$$

18.4.3 Another Chebyshev generating function is

$$\frac{1-xt}{1-2xt+t^2} = \sum_{n=0}^{\infty} X_n(x)t^n, \qquad |t| < 1.$$

How is $X_n(x)$ related to $T_n(x)$ and $U_n(x)$? Take $-1 \le x \le 1$ for the stated radius $|t| < 1$.

Solution.

Write the proposed generator in two ways. Since $(1 - 2xt + t^2) + (1 - t^2) = 2(1 - xt)$, the type-I generator, with $T_0 = 1$, gives

$$\frac{1 - xt}{1 - 2xt + t^2} = \frac{1}{2}\left[1 + \frac{1 - t^2}{1 - 2xt + t^2}\right] = \frac{1}{2}\left[1 + T_0(x) + 2\sum_{n=1}^{\infty} T_n(x)t^n\right]$$
$$= 1 + \sum_{n=1}^{\infty} T_n(x)t^n.$$

The type-II generator, with the index shift $xt\sum_{n\ge0} U_n t^n = \sum_{n\ge1} xU_{n-1}t^n$, gives

$$\frac{1 - xt}{1 - 2xt + t^2} = (1 - xt)\sum_{n=0}^{\infty} U_n(x)t^n$$
$$= U_0(x) + \sum_{n=1}^{\infty}[U_n(x) - xU_{n-1}(x)]t^n.$$

Therefore

$$\boxed{X_n = T_n = U_n - xU_{n-1} \quad (n \ge 1), \qquad X_0 = T_0 = U_0 = 1}.$$

Defining $U_{-1} = 0$ includes $n = 0$ in the first expression. The radius $|t| < 1$ of the statement needs $-1 \le x \le 1$ (the erratum there): for any other x the expansions hold only for $|t|$ less than the distance from 0 to the nearest pole of the generator, and this distance is less than 1. Indeed, $1 - 2xt + t^2$ vanishes at $t_\pm = x \pm \sqrt{x^2 - 1}$, with $t_+t_- = 1$. For $x = \cos\theta \in [-1, 1]$ these are $e^{\pm i\theta}$, of modulus 1. Conversely, if $|t_+| = |t_-| = 1$ then $t_- = 1/t_+ = \overline{t_+}$ and $x = (t_+ + t_-)/2 = \Re t_+ \in [-1, 1]$; so for every other x one root lies inside the unit circle. It is a pole of the generator, because the numerator $1 - xt$ vanishes only at $t = 1/x$ (never if $x = 0$), where $1 - 2xt + t^2 = (1 - x^2)/x^2 \ne 0$ for $x \ne \pm1$. The coefficient identities themselves are polynomial identities, so hold for every complex x.

18.4.4 Given

$$(1 - x^2)U_n''(x) - 3xU_n'(x) + n(n + 2)U_n(x) = 0,$$

show that $V_n(x)$, Eq. (18.106), satisfies

$$(1 - x^2)V_n''(x) - xV_n'(x) + n^2V_n(x) = 0,$$

which is Chebyshev's equation. Use $n \ge 1$ (so that U_{n-1} exists) and $-1 < x < 1$ (where $\sqrt{1 - x^2}$ is differentiable).

Solution.

On $-1 < x < 1$ put $h = \sqrt{1 - x^2}$, $u = U_{n-1}$ and $V_n = hu$, for $n \ge 1$. Since $h' = -x/h$,

$$h'' = -\frac{1}{h} + \frac{xh'}{h^2} = -\frac{h^2 + x^2}{h^3} = -\frac{1}{h^3},$$

and the product rule gives

$$V_n' = hu' + h'u = hu' - \frac{x}{h}u, \qquad V_n'' = hu'' + 2h'u' + h''u = hu'' - \frac{2x}{h}u' - \frac{1}{h^3}u.$$

Substitute these derivatives into the claimed equation, using $1-x^2=h^2$ and $-u/h+x^2u/h=-(1-x^2)u/h=-hu$:

$$\begin{aligned}(1-x^2)V_n''-xV_n'+n^2V_n &= h^3u''-2xhu'-\frac{u}{h}-xhu'+\frac{x^2u}{h}+n^2hu\\ &= h^3u''-3xhu'-hu+n^2hu\\ &= h\big[(1-x^2)u''-3xu'+(n^2-1)u\big].\end{aligned}$$

Because $(n-1)((n-1)+2)=n^2-1$, the bracket is exactly the given type-II equation at index $n-1$, and is zero. This proves the result. The formulas with h specify the positive root on this real interval; analytic continuation uses a consistent branch.

18.4.5 Show that the Wronskian of $T_n(x)$ and $V_n(x)$ is given by

$$T_n(x)V_n'(x)-T_n'(x)V_n(x)=-\frac{n}{(1-x^2)^{1/2}}.$$

This verifies that T_n and V_n ($n\neq 0$) are independent solutions of Eq. (18.97). Conversely, for $n=0$, we do not have linear independence. What happens at $n=0$? Where is the "second" solution? Work on $-1<x<1$, where $\sqrt{1-x^2}$ is differentiable and the right side is finite.

Solution.

Put $x=\cos\theta$, $0<\theta<\pi$, so $\sqrt{1-x^2}=\sin\theta$ and $d/dx=-(\sin\theta)^{-1}\,d/d\theta$. For $n\geq 1$, $T_n=\cos n\theta$ and, since $U_{n-1}(\cos\theta)=\sin n\theta/\sin\theta$, $V_n=\sqrt{1-x^2}\,U_{n-1}=\sin n\theta$. Hence

$$T_n'=\frac{n\sin n\theta}{\sin\theta},\qquad V_n'=-\frac{n\cos n\theta}{\sin\theta}.$$

Thus

$$T_nV_n'-T_n'V_n=-\frac{n(\cos^2 n\theta+\sin^2 n\theta)}{\sin\theta}=\boxed{-\frac{n}{\sqrt{1-x^2}}}.$$

It is nonzero throughout $(-1,1)$, proving independence there. At $n=0$ this particular family gives $T_0=1$ and $\sin(0\theta)=0$, so its second member has collapsed to zero. To find the second solution, transform Eq. (18.97) to θ: with $y'=-y_\theta/\sin\theta$,

$$y''=\frac{y_{\theta\theta}}{\sin^2\theta}-\frac{\cos\theta\,y_\theta}{\sin^3\theta},\qquad (1-x^2)y''-xy'+n^2y=y_{\theta\theta}+n^2y,$$

the two $\cot\theta\,y_\theta$ terms cancelling. At $n=0$ the equation is $d^2y/d\theta^2=0$, whose general solution is $A+B\theta$. The missing independent solution is therefore $\arccos x$, equivalently $\arcsin x$ up to a constant and a sign. It also follows from $\lim_{n\to 0}\sin(n\arccos x)/n=\arccos x$, treating the order as continuous in this limit.

18.4.6 Show that $W_n(x)=(1-x^2)^{-1/2}T_{n+1}(x)$ is a solution of

$$(1-x^2)W_n''(x)-3xW_n'(x)+n(n+2)W_n(x)=0.$$

Work on $-1<x<1$, since W_n is singular at $x=\pm 1$.

Solution.

Let $h=\sqrt{1-x^2}$ and $T=T_{n+1}$ on $-1<x<1$. Since $h'=-x/h$,

$$\left(\frac{1}{h}\right)'=\frac{x}{h^3},\qquad \left(\frac{x}{h^3}\right)'=\frac{1}{h^3}+\frac{3x^2}{h^5}=\frac{h^2+3x^2}{h^5}=\frac{1+2x^2}{h^5}.$$

Hence for $W = T/h$ the product rule gives

$$W' = \frac{T'}{h} + \frac{xT}{h^3}, \qquad W'' = \frac{T''}{h} + \frac{2xT'}{h^3} + \frac{(1+2x^2)T}{h^5}.$$

Insert these expressions and collect coefficients, using $(1-x^2)/h^3 = 1/h$ and $1+n(n+2) = (n+1)^2$:

$$\begin{aligned} h^2W'' - 3xW' + n(n+2)W &= hT'' - \frac{xT'}{h} + \frac{(1+2x^2-3x^2)T}{h^3} + \frac{n(n+2)T}{h} \\ &= hT'' - \frac{xT'}{h} + \frac{[1+n(n+2)]T}{h} \\ &= \frac{1}{h}\big[(1-x^2)T'' - xT' + (n+1)^2T\big] = 0. \end{aligned}$$

The last bracket is the type-I equation for T_{n+1}. Thus $\boxed{W_n = (1-x^2)^{-1/2}T_{n+1}}$ solves the type-II equation, including $n = 0$. The differential identity is on the open interval; W_n itself is singular at its endpoints.

18.4.7 Evaluate the Wronskian of $U_n(x)$ and $W_n(x) = (1-x^2)^{-1/2}T_{n+1}(x)$. Work on $-1 < x < 1$, since W_n is singular at $x = \pm 1$.

Solution.

Set $a = n+1$, $x = \cos\theta$ with $0 < \theta < \pi$, and $s = \sin\theta = \sqrt{1-x^2}$. Then $U_n = \sin(a\theta)/s$ and, since $T_{n+1}(\cos\theta) = \cos(a\theta)$, $W_n = T_{n+1}/s = \cos(a\theta)/s$. Their θ derivatives are

$$\frac{dU_n}{d\theta} = \frac{a\cos(a\theta)}{s} - \frac{\sin(a\theta)\cos\theta}{s^2}, \qquad \frac{dW_n}{d\theta} = -\frac{a\sin(a\theta)}{s} - \frac{\cos(a\theta)\cos\theta}{s^2}.$$

In the θ Wronskian the two terms with $\cos\theta$ cancel:

$$\begin{aligned} U_n\frac{dW_n}{d\theta} - \frac{dU_n}{d\theta}W_n &= -\frac{a\sin^2(a\theta)}{s^2} - \frac{\sin(a\theta)\cos(a\theta)\cos\theta}{s^3} \\ &\quad - \frac{a\cos^2(a\theta)}{s^2} + \frac{\sin(a\theta)\cos(a\theta)\cos\theta}{s^3} = -\frac{a}{s^2}. \end{aligned}$$

Each x derivative is $d\theta/dx = -1/s$ times the θ derivative, so multiplying by $-1/s$ gives

$$\boxed{U_nW_n' - U_n'W_n = \frac{n+1}{(1-x^2)^{3/2}}}.$$

It is nonzero for every $n \geq 0$ on $(-1,1)$, so here the second solution remains independent even at the lowest polynomial index.

18.4.8 $V_n(x) = (1-x^2)^{1/2}U_{n-1}(x)$ is not defined for $n = 0$. Show that a second and independent solution of the Chebyshev differential equation for $T_n(x)$ $(n = 0)$ is $V_0(x) = \arccos x$ (or $\arcsin x$).

Solution.

At $n = 0$, the equation is $(1-x^2)y'' - xy' = 0$. Writing $v = y'$ gives the linear first-order equation $(1-x^2)v' = xv$. By uniqueness for this equation, a solution that vanishes at one point of $(-1,1)$ vanishes identically; so either $v \equiv 0$, which gives the constant solution, or v has no zero and

$$\frac{v'}{v} = \frac{x}{1-x^2}, \qquad \ln|v| = -\frac{1}{2}\ln(1-x^2) + \text{constant},$$

so $v = C/\sqrt{1-x^2}$ on $(-1,1)$ and, since $\arcsin x = \pi/2 - \arccos x$,

$$y = A + C\arcsin x = \widetilde{A} + \widetilde{C}\arccos x, \qquad \widetilde{A} = A + \frac{\pi C}{2}, \quad \widetilde{C} = -C.$$

The constant solution is $T_0 = 1$. The derivative of $\arccos x$ is $-1/\sqrt{1-x^2}$, so its Wronskian with 1 is $1 \cdot (\arccos x)' - 0 = -1/\sqrt{1-x^2} \neq 0$, proving independence. Directly,

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}, \qquad (\arccos x)'' = -\frac{x}{(1-x^2)^{3/2}},$$

and the two terms of the ODE cancel:

$$(1-x^2)\Big(-\frac{x}{(1-x^2)^{3/2}}\Big) - x\Big(-\frac{1}{\sqrt{1-x^2}}\Big) = -\frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} = 0.$$

Calling this solution V_0 is a separate zero-order convention. It must not replace the sine-family value $V_0 = 0$ in recurrences or orthogonality formulas for $V_n = \sin(n \arccos x)$.

18.4.9 Show that $V_n(x)$ satisfies the same three-term recurrence relation as $T_n(x)$, Eq. (18.92). The recurrence holds for $n \geq 2$ with the book's $V_0 = \arccos x$. It also holds at $n = 1$ if instead the sine family is extended by $V_0 = 0$.

Solution.

With $x = \cos\theta$, $0 \leq \theta \leq \pi$, the sine family is $V_n = \sqrt{1-x^2}\,U_{n-1}(x) = \sin\theta \cdot \sin(n\theta)/\sin\theta = \sin(n\theta)$ (by continuity also at $\theta = 0, \pi$). Adding the two angle formulas $\sin(n\theta \pm \theta) = \sin n\theta \cos\theta \pm \cos n\theta \sin\theta$ gives

$$\begin{aligned} V_{n+1} + V_{n-1} &= \sin(n\theta + \theta) + \sin(n\theta - \theta) \\ &= 2\sin(n\theta)\cos\theta = 2xV_n. \end{aligned}$$

Hence $\boxed{V_{n+1} - 2xV_n + V_{n-1} = 0}$. Equivalently, for $n \geq 2$ multiply the type-II recurrence with index $n-1$, $U_n - 2xU_{n-1} + U_{n-2} = 0$, by $\sqrt{1-x^2}$. At $n = 1$ the recurrence reads $V_2 - 2xV_1 + V_0 = 0$. Since $V_2 - 2xV_1 = \sin 2\theta - 2\cos\theta\sin\theta = 0$, it holds there exactly for the sine-family value $V_0 = \sin 0 = 0$ and fails for the solution $V_0 = \arccos x$ of Exercise 18.4.8, as the qualification in the statement says; for $n \geq 2$ the value of V_0 does not enter.

18.4.10 Verify the series solutions for $T_n(x)$ and $U_n(x)$, Eqs. (18.114) and (18.115).

Solution.

The Instructor's Manual substitutes the series for T_n into Chebyshev's equation, a route not followed below. In its first ODE line, x^2T_n must be n^2T_n, as in Chebyshev's equation. After shifting the derivative sum, its factorial is $(n-2m+2)!$, not $(n-2m)!$. Its evaluation at zero also omits the prefactor $n/2$.

Start with the type-II generator near $t = 0$ and expand geometrically, then binomially:

$$\begin{aligned} \frac{1}{1-2xt+t^2} &= \sum_{j=0}^{\infty}(2xt - t^2)^j = \sum_{j=0}^{\infty}\sum_{m=0}^{j}\binom{j}{m}(2xt)^{j-m}(-t^2)^m \\ &= \sum_{j=0}^{\infty}\sum_{m=0}^{j}(-1)^m\binom{j}{m}(2x)^{j-m}t^{j+m}. \end{aligned}$$

For $2|x||t| + |t|^2 < 1$ the double series converges absolutely, so its terms may be collected by powers of t. For a fixed power t^n, $j = n - m$, and the condition $m \le j = n - m$ gives $0 \le m \le \lfloor n/2 \rfloor$. Since $\binom{n-m}{m} = (n-m)!/[m!(n-2m)!]$, the coefficient is

$$\boxed{U_n(x) = \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m (n-m)!}{m!(n-2m)!} (2x)^{n-2m}}.$$

For T_n, use $T_n = U_n - xU_{n-1}$ from Exercise 18.4.3, $n \ge 1$. Writing $x(2x)^{n-1-2m} = \frac{1}{2}(2x)^{n-2m}$,

$$xU_{n-1}(x) = \frac{1}{2} \sum_{m=0}^{\lfloor (n-1)/2 \rfloor} \frac{(-1)^m (n-m-1)!}{m!(n-2m-1)!} (2x)^{n-2m}.$$

With $(n-m)! = (n-m)(n-m-1)!$ and $1/(n-2m-1)! = (n-2m)/(n-2m)!$, the coefficient of $(-1)^m (2x)^{n-2m}$ in the difference is

$$\begin{aligned} \frac{(n-m)!}{m!(n-2m)!} - \frac{(n-m-1)!}{2\,m!\,(n-2m-1)!} &= \frac{(n-m-1)!}{m!(n-2m)!} \left[n - m - \frac{n-2m}{2} \right] \\ &= \frac{n}{2} \frac{(n-m-1)!}{m!(n-2m)!}. \end{aligned}$$

When $n = 2m$, the sum for xU_{n-1} stops at $\lfloor (n-1)/2 \rfloor = m - 1$, so the second term is absent; its rewritten form $(n-m-1)!\,(n-2m)/[2\,m!\,(n-2m)!]$ is zero there, and the bracket is still $n - m = n/2$. (Directly: $(n-m)!/[m!\,0!] = 1 = \frac{n}{2}(n-m-1)!/[m!\,0!]$.) Thus for $n \ge 1$,

$$\boxed{T_n(x) = \frac{n}{2} \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m (n-m-1)!}{m!(n-2m)!} (2x)^{n-2m}}.$$

The case $T_0 = 1$ is separate; it must not be evaluated through $0 \cdot (-1)!$. These coefficient extractions prove the two stated series including their normalization.

18.4.11 Transform the series form of $T_n(x)$, Eq. (18.114), into an ascending power series.

$$\textit{ANS. } T_{2n}(x) = (-1)^n n \sum_{m=0}^{n} (-1)^m \frac{(n+m-1)!}{(n-m)!\,(2m)!} (2x)^{2m},\ n \ge 1,$$

$$T_{2n+1}(x) = \frac{2n+1}{2} \sum_{m=0}^{n} \frac{(-1)^{m+n} (n+m)!}{(n-m)!\,(2m+1)!} (2x)^{2m+1}.$$

Solution.

In the finite series (18.114),

$$T_N(x) = \frac{N}{2} \sum_{j=0}^{\lfloor N/2 \rfloor} \frac{(-1)^j (N-j-1)!}{j!\,(N-2j)!} (2x)^{N-2j}, \qquad N \ge 1,$$

change the descending-power index j to $n - m$; as j runs from 0 to $n = \lfloor N/2 \rfloor$, m runs from n down to 0. For $N = 2n$, with $n \ge 1$, the prefactor is $N/2 = n$, the power is $N - 2j = 2m$,

$N-j-1=n+m-1$, $j!=(n-m)!$, $(N-2j)!=(2m)!$, and $(-1)^j=(-1)^{n-m}=(-1)^n(-1)^m$. Thus reversing the finite sum yields

$$\boxed{T_{2n}(x)=(-1)^n n\sum_{m=0}^{n}\frac{(-1)^m(n+m-1)!}{(n-m)!(2m)!}(2x)^{2m}}.$$

For $N=2n+1$, the corresponding values are $N/2=(2n+1)/2$, $N-2j=2m+1$, $N-j-1=n+m$, $j!=(n-m)!$, $(N-2j)!=(2m+1)!$ and $(-1)^j=(-1)^{n+m}$, giving

$$\boxed{T_{2n+1}(x)=\frac{2n+1}{2}\sum_{m=0}^{n}\frac{(-1)^{n+m}(n+m)!}{(n-m)!(2m+1)!}(2x)^{2m+1}}.$$

The even formula starts at $n=1$; $T_0=1$ is retained separately. The odd formula includes $n=0$, where it gives $T_1=x$.

18.4.12 Rewrite the series form of $U_n(x)$, Eq. (18.115), as an ascending power series.

$$\textit{ANS. } U_{2n}(x)=(-1)^n\sum_{m=0}^{n}(-1)^m\frac{(n+m)!}{(n-m)!\,(2m)!}\,(2x)^{2m},$$

$$U_{2n+1}(x)=(-1)^n\sum_{m=0}^{n}(-1)^m\frac{(n+m+1)!}{(n-m)!\,(2m+1)!}\,(2x)^{2m+1}.$$

Solution.

Use $U_N=\sum_{j=0}^{\lfloor N/2\rfloor}(-1)^j(N-j)!(2x)^{N-2j}/[j!(N-2j)!]$. For $N=2n$ put $j=n-m$ (m from n down to 0), so the numerator factorial $(N-j)!$ becomes $(n+m)!$, the denominator $j!(N-2j)!$ becomes $(n-m)!(2m)!$, the sign is $(-1)^{n-m}$, and the power becomes $2m$. Hence

$$\boxed{U_{2n}(x)=(-1)^n\sum_{m=0}^{n}\frac{(-1)^m(n+m)!}{(n-m)!(2m)!}(2x)^{2m}}.$$

For $N=2n+1$ the same substitution gives numerator $(n+m+1)!$, denominator $(n-m)!(2m+1)!$ and power $2m+1$, so

$$\boxed{U_{2n+1}(x)=(-1)^n\sum_{m=0}^{n}\frac{(-1)^m(n+m+1)!}{(n-m)!(2m+1)!}(2x)^{2m+1}}.$$

Both include $n=0$, giving $U_0=1$ and $U_1=2x$. The sign $(-1)^{n-m}$ was replaced by $(-1)^{n+m}$ because their exponents differ by the even integer $2m$.

18.4.13

(a) From the differential equation for T_n (in self-adjoint form) show that

$$\int_{-1}^{1}\frac{dT_m(x)}{dx}\frac{dT_n(x)}{dx}\,(1-x^2)^{1/2}\,dx=0,\qquad m\neq n.$$

(b) Confirm the preceding result by showing that

$$\frac{dT_n(x)}{dx}=nU_{n-1}(x).$$

Solution.

(a) Set $p=\sqrt{1-x^2}$ and $w=1/p$. Since $(pT_n')'=pT_n''-xT_n'/p$, dividing Chebyshev's equation $(1-x^2)T_n''-xT_n'+n^2T_n=0$ by p gives the self-adjoint equations

$$(pT_n')'+n^2wT_n=0, \qquad (pT_m')'+m^2wT_m=0.$$

Multiply the first by T_m and the second by T_n and subtract. Since $T_m(pT_n')'-T_n(pT_m')'=[p(T_mT_n'-T_nT_m')]'$ (the terms $pT_m'T_n'$ cancel), integration gives

$$[p(T_mT_n'-T_nT_m')]_{-1}^{1}+(n^2-m^2)\int_{-1}^{1}wT_mT_n\,dx=0.$$

The boundary term is zero because polynomials and their derivatives are finite there while $p\to 0$. For distinct nonnegative m,n, $n^2-m^2\neq 0$, so the integral is zero. Next multiply the first self-adjoint equation by T_m alone and integrate, using $\int_{-1}^{1}T_m(pT_n')'dx=[pT_mT_n']_{-1}^{1}-\int_{-1}^{1}pT_m'T_n'\,dx$:

$$\int_{-1}^{1}pT_m'T_n'\,dx=[pT_mT_n']_{-1}^{1}+n^2\int_{-1}^{1}wT_mT_n\,dx=\boxed{0}, \qquad m\neq n,$$

because the boundary term again contains $p(\pm1)=0$ and the last integral was just shown to vanish. All endpoint integrals converge since w has only square-root singularities.

(b) With $x=\cos\theta$, $dT_n/dx=(dT_n/d\theta)/(dx/d\theta)$ and $U_{n-1}(\cos\theta)=\sin n\theta/\sin\theta$, so

$$T_n'(x)=\frac{-n\sin n\theta}{-\sin\theta}=\boxed{nU_{n-1}(x)} \qquad (n\geq 1).$$

Both sides are polynomials, so the identity holds on all of $[-1,1]$. Therefore, by the type-II orthogonality of Exercise 18.4.19(c),

$$\int_{-1}^{1}T_m'T_n'\sqrt{1-x^2}\,dx=mn\int_{-1}^{1}U_{m-1}U_{n-1}\sqrt{1-x^2}\,dx=mn\,\frac{\pi}{2}\,\delta_{mn}, \qquad m,n\geq 1,$$

which vanishes for $m\neq n$. If either index is zero, its derivative is zero directly. As a normalization check, the same calculation gives $\int_{-1}^{1}p(T_n')^2dx=\pi n^2/2$ for $n\geq 1$.

18.4.14 The substitution $x=2x'-1$ converts $T_n(x)$ into the shifted Chebyshev polynomials $T_n^*(x')$. Verify that this produces the shifted polynomials shown in Table 18.5 and that they satisfy the orthogonality and normalization relation

$$\int_0^1 T_n^*(x')T_m^*(x')\left[x'(1-x')\right]^{-1/2}dx'=\frac{\delta_{mn}\pi}{2-\delta_{n0}}.$$

Table 18.5 Shifted Type I Chebyshev Polynomials

$T_0^*=1$
$T_1^*=2x-1$
$T_2^*=8x^2-8x+1$
$T_3^*=32x^3-48x^2+18x-1$
$T_4^*=128x^4-256x^3+160x^2-32x+1$
$T_5^*=512x^5-1280x^4+1120x^3-400x^2+50x-1$
$T_6^*=2048x^6-6144x^5+6912x^4-3584x^3+840x^2-72x+1$

Solution.

Put $y = x'$ and $S_n(y) = T_n(2y-1)$. The recurrence (18.92), $T_{n+1} = 2xT_n - T_{n-1}$, with $2x = 4y - 2$ becomes

$$S_{n+1} = (4y-2)S_n - S_{n-1}, \qquad S_0 = 1, \quad S_1 = 2y - 1.$$

Successive multiplication and subtraction give

$$\begin{aligned}
S_2 &= (4y-2)(2y-1) - 1 = 8y^2 - 8y + 1,\\
S_3 &= (4y-2)(8y^2-8y+1) - (2y-1)\\
&= (32y^3 - 48y^2 + 20y - 2) - (2y-1) = 32y^3 - 48y^2 + 18y - 1,\\
S_4 &= (4y-2)(32y^3 - 48y^2 + 18y - 1) - (8y^2 - 8y + 1)\\
&= (128y^4 - 256y^3 + 168y^2 - 40y + 2) - (8y^2 - 8y + 1)\\
&= 128y^4 - 256y^3 + 160y^2 - 32y + 1,\\
S_5 &= (4y-2)S_4 - S_3\\
&= (512y^5 - 1280y^4 + 1152y^3 - 448y^2 + 68y - 2) - (32y^3 - 48y^2 + 18y - 1)\\
&= 512y^5 - 1280y^4 + 1120y^3 - 400y^2 + 50y - 1,\\
S_6 &= (4y-2)S_5 - S_4\\
&= (2048y^6 - 6144y^5 + 7040y^4 - 3840y^3 + 1000y^2 - 104y + 2) - S_4\\
&= 2048y^6 - 6144y^5 + 6912y^4 - 3584y^3 + 840y^2 - 72y + 1.
\end{aligned}$$

For example the y^3 coefficient of S_5 is $4(160) - 2(-256) - 32 = 1120$. The y^3 coefficient of T_5^* is therefore 1120, not the 120 printed in Table 18.5. The integration formula needs a star on both polynomials and the same primed variable throughout. The relation is orthogonality with the stated norms, not unit normalization.

In the integral set $x = 2y-1$, so $dx = 2dy$ and $1 - x^2 = (1-x)(1+x) = (2-2y)(2y) = 4y(1-y)$; thus $dx/\sqrt{1-x^2} = dy/\sqrt{y(1-y)}$, and $y = 0, 1$ correspond to $x = -1, 1$. Then $x = \cos\theta$, with $dx/\sqrt{1-x^2} = -d\theta$ and reversed limits, gives

$$\begin{aligned}
\int_0^1 \frac{S_n(y)S_m(y)}{\sqrt{y(1-y)}}\,dy &= \int_{-1}^1 \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}}\,dx\\
&= \int_0^\pi \cos(n\theta)\cos(m\theta)\,d\theta\\
&= \boxed{\frac{\pi\delta_{mn}}{2-\delta_{n0}}}.
\end{aligned}$$

For the last line write $\cos n\theta \cos m\theta = \frac{1}{2}[\cos((n-m)\theta) + \cos((n+m)\theta)]$ and use $\int_0^\pi \cos k\theta\,d\theta = \pi$ for $k = 0$ and 0 for every other integer k: the result is π for $m = n = 0$, $\pi/2$ for $m = n \geq 1$, and zero for unequal nonnegative indices.

18.4.15 The expansion of a power of x in a Chebyshev series leads to the integral

$$I_{mn} = \int_{-1}^1 x^m T_n(x)\,\frac{dx}{\sqrt{1-x^2}}.$$

(a) Show that this integral vanishes for $m < n$.

(b) Show that this integral vanishes for $m+n$ odd.

Solution.

Assume the power and polynomial indices m, n are nonnegative integers. (a) The polynomials $T_0, \ldots, T_m$ form a basis for all polynomials of degree at most m: $T_0 = 1$ and T_j has nonzero leading coefficient 2^{j-1} for $j \geq 1$. Therefore $x^m = \sum_{j=0}^{m} c_j T_j$. If $m < n$, orthogonality gives

$$I_{mn} = \sum_{j=0}^{m} c_j \int_{-1}^{1} \frac{T_j T_n}{\sqrt{1-x^2}} dx = \boxed{0},$$

since no index j equals n.

(b) Every power x^{n-2k} in the finite series (18.114) has the parity of n, so $T_n(-x) = (-1)^n T_n(x)$. Changing x to $-u$ in the integral, which leaves the interval and the weight unchanged, thus gives

$$I_{mn} = \int_{-1}^{1} (-u)^m T_n(-u) \frac{du}{\sqrt{1-u^2}} = (-1)^{m+n} I_{mn}.$$

For odd $m+n$ this says $I_{mn} = -I_{mn}$, so again $\boxed{I_{mn} = 0}$. The weight is integrable at both endpoints, so this symmetry argument applies to ordinary convergent integrals.

18.4.16 Evaluate the integral

$$I_{mn} = \int_{-1}^{1} x^m T_n(x) \frac{dx}{\sqrt{1-x^2}}$$

for $m \geq n$ and $m+n$ even by each of two methods:

(a) Replacing $T_n(x)$ by its Rodrigues representation.

(b) Using $x = \cos\theta$ to transform the integral to a form with θ as the variable.

$$\textit{ANS. } I_{mn} = \pi \frac{m!\,(m-n-1)!!}{(m-n)!\,(m+n)!!}, \qquad m \geq n,\ m+n \text{ even}.$$

Solution.

The Instructor's Manual uses the incorrect Rodrigues power $n+1/2$ twice; it must be $n-1/2$, as in the type-I Rodrigues formula below. Its final gamma quotient must have $((m+n)/2)!$ in the denominator, not $((m-n)/2)!$, as the quotient $\Gamma(j+\frac{1}{2})/\Gamma(n+j+1)$ in (a) shows, with $n+j = (m+n)/2$.

Let $m-n = 2j \geq 0$, and write $D = d/dx$. (a) The Rodrigues representation $T_n(x) = (-1)^n \sqrt{\pi}\,(1-x^2)^{1/2} D^n (1-x^2)^{n-1/2} / [2^n \Gamma(n+1/2)]$ gives

$$\frac{T_n(x)}{\sqrt{1-x^2}} = \frac{(-1)^n \sqrt{\pi}}{2^n \Gamma(n+1/2)} D^n (1-x^2)^{n-1/2}.$$

Integrate by parts n times:

$$\int_{-1}^{1} x^m D^n f\, dx = \Big[\sum_{k=0}^{n-1} (-1)^k D^k(x^m)\, D^{n-1-k} f \Big]_{-1}^{1} + (-1)^n \int_{-1}^{1} D^n(x^m)\, f\, dx,$$

with $f = (1-x^2)^{n-1/2}$. By induction, $D^r f = (1-x^2)^{n-1/2-r} q_r(x)$ with a polynomial q_r, and $n - 1/2 - r \geq 1/2$ for $r \leq n-1$; so every boundary term vanishes at both endpoints. The factor

$(-1)^n$ cancels the sign of the Rodrigues formula, and $D^n x^m = m!\,x^{m-n}/(m-n)!$. For the remaining integral, evenness and $u = x^2$, $dx = du/(2\sqrt{u})$, give

$$\int_{-1}^{1} x^{2j}(1-x^2)^{n-1/2}dx = 2\int_0^1 x^{2j}(1-x^2)^{n-1/2}dx = \int_0^1 u^{j-1/2}(1-u)^{n-1/2}du = B(j+\tfrac12, n+\tfrac12).$$

Collecting these results, with $(m-n)! = (2j)!$ in the second line,

$$B(j+\tfrac12, n+\tfrac12) = \frac{\Gamma(j+\frac12)\Gamma(n+\frac12)}{\Gamma(n+j+1)}$$

in the third, and $\Gamma(j+\frac12) = \sqrt{\pi}\,(2j)!/(4^j j!)$, $2^n 4^j = 2^m$ in the last,

$$\begin{aligned} I_{mn} &= \frac{\sqrt{\pi}}{2^n\Gamma(n+1/2)}\frac{m!}{(m-n)!}\int_{-1}^{1} x^{m-n}(1-x^2)^{n-1/2}dx \\ &= \frac{\sqrt{\pi}m!}{2^n(2j)!\Gamma(n+1/2)}B(j+\tfrac12, n+\tfrac12) \\ &= \frac{\sqrt{\pi}m!\Gamma(j+1/2)}{2^n(2j)!\Gamma(n+j+1)} \\ &= \boxed{\frac{\pi m!}{2^m j!(n+j)!}}. \end{aligned}$$

For $n = 0$ no integration by parts is needed and the same beta calculation applies. Since $(2j-1)!! = (2j)!/(2^j j!)$ and $(m+n)!! = 2^{n+j}(n+j)!$, the answer is equivalently

$$\boxed{I_{mn} = \pi\frac{m!(m-n-1)!!}{(m-n)!(m+n)!!}},$$

with $(-1)!! = 1$ when $m = n$.

(b) Substitute $x = \cos\theta$: $I_{mn} = \int_0^\pi \cos^m\theta\cos n\theta\,d\theta$. The binomial expansion is

$$\cos^m\theta = 2^{-m}\sum_{\ell=0}^{m}\binom{m}{\ell}e^{i(m-2\ell)\theta}.$$

Pair the terms ℓ and $m-\ell$, which have equal binomial coefficients and opposite frequencies: each pair with $\ell < m/2$ gives $2^{1-m}\binom{m}{\ell}\cos((m-2\ell)\theta)$. Since $\int_0^\pi \cos k\theta\cos n\theta\,d\theta = 0$ for nonnegative integers $k \neq n$, only the frequency $m-2\ell = n$, i.e. $\ell = j$, survives. For $n > 0$ the coefficient of $\cos n\theta$ is thus $2^{1-m}\binom{m}{j}$, and multiplication by $\cos n\theta$ and integration multiplies it by $\int_0^\pi \cos^2 n\theta\,d\theta = \pi/2$. For $n = 0$ the unpaired constant coefficient is $2^{-m}\binom{m}{m/2}$ and its integral is π times that coefficient. Both cases give, since $m - j = n + j$, $\pi 2^{-m}\binom{m}{j} = \pi m!/[2^m j!(n+j)!]$, agreeing with (a).

18.4.17 Establish the following bounds, $-1 \le x \le 1$:

(a) $\big|U_n(x)\big| \le n+1$,

(b) $\left|\dfrac{d}{dx}T_n(x)\right| \le n^2$.

Solution.

(a) For $x = \cos\theta$ with $0 < \theta < \pi$, a finite geometric sum gives

$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta} = \sum_{j=0}^{n} e^{i(n-2j)\theta}.$$

Indeed, summing the geometric series with ratio $e^{-2i\theta} \neq 1$ and multiplying numerator and denominator by $e^{i\theta}$,

$$e^{in\theta}\sum_{j=0}^{n} e^{-2ij\theta} = e^{in\theta}\frac{1 - e^{-2i(n+1)\theta}}{1 - e^{-2i\theta}} = \frac{e^{i(n+1)\theta} - e^{-i(n+1)\theta}}{e^{i\theta} - e^{-i\theta}} = \frac{\sin((n+1)\theta)}{\sin\theta}.$$

There are $n+1$ unit-modulus terms, so the triangle inequality gives $\boxed{|U_n(x)| \leq n+1}$. By polynomial continuity the bound also holds at $x = \pm 1$, where the values are $n+1$ and $(-1)^n(n+1)$; thus it is sharp.

(b) Exercise 18.4.13 gives $T_n' = nU_{n-1}$ for $n \geq 1$. Applying (a) with index $n-1$ gives

$$\boxed{|T_n'(x)| \leq n \cdot n = n^2}, \qquad -1 \leq x \leq 1.$$

For $n = 0$ the derivative is zero, again consistent with the bound. For positive n its magnitude at either endpoint is n^2.

18.4.18

(a) Show that for $-1 \leq x \leq 1$, $\left|V_n(x)\right| \leq 1$.

(b) Show that $W_n(x)$ is unbounded in $-1 \leq x \leq 1$.

Use $n \geq 1$ for V_n and $n \geq 0$ for W_n; unboundedness refers to approach to the endpoints from inside $(-1, 1)$.

Solution.

(a) Use $x = \cos\theta$, $0 \leq \theta \leq \pi$. Then $V_n(x) = \sin(n\theta)$, so $\boxed{|V_n(x)| \leq 1}$. This refers to the sine family ($n \geq 1$, or its zero member $V_0 = 0$), not the separate solution $\arccos x$ at order zero, whose maximum is π.

(b) As $x \to 1^-$, $T_{n+1}(x) \to T_{n+1}(1) = 1$ (Exercise 18.4.1) and $\sqrt{1-x^2} = \sqrt{(1-x)(1+x)} \sim \sqrt{2(1-x)}$. Thus

$$W_n(x) \sim \frac{1}{\sqrt{2(1-x)}} \longrightarrow +\infty.$$

Similarly, as $x \to -1^+$, $T_{n+1}(x) \to (-1)^{n+1}$ and $\sqrt{1-x^2} \sim \sqrt{2(1+x)}$, so

$$W_n(x) \sim \frac{(-1)^{n+1}}{\sqrt{2(1+x)}}.$$

Therefore W_n is unbounded for every nonnegative integer n; it has no finite values at the two endpoints. Possible zeros in the interior cannot remove either endpoint singularity.

18.4.19 Verify the orthogonality-normalization integrals for

(a) $T_m(x)$, $T_n(x)$,

(b) $V_m(x)$, $V_n(x)$,

(c) $U_m(x)$, $U_n(x)$,

(d) $W_m(x)$, $W_n(x)$.

Hint. All these can be converted to trigonometric integrals. For V_m, V_n, take $m, n \geq 1$. The zero-norm entry in Eq. (18.117) uses the optional sine-family extension $V_0 = 0$, not the independent solution $\arccos x$.

Solution.

In part (d) of the Instructor's Manual the second cosine should be $\cos((n+1)\theta)$, not a repeated $\cos((m+1)\theta)$, since the integrand is $W_m W_n$.

Put $x = \cos\theta$, $0 < \theta < \pi$, so $dx = -\sin\theta\, d\theta$; reversing the integration limits gives $dx/\sqrt{1-x^2} = d\theta$ and $\sqrt{1-x^2}\, dx = \sin^2\theta\, d\theta$ in the integrals. With

$$T_m = \cos m\theta, \quad V_m = \sin m\theta, \quad U_m = \frac{\sin((m+1)\theta)}{\sin\theta}, \quad W_m = \frac{\cos((m+1)\theta)}{\sin\theta},$$

the factor $\sin^2\theta$ cancels the two denominators in the last two cases. The four transformations are

$$\int_{-1}^{1} \frac{T_m T_n}{\sqrt{1-x^2}} dx = \int_0^{\pi} \cos(m\theta)\cos(n\theta) d\theta,$$

$$\int_{-1}^{1} \frac{V_m V_n}{\sqrt{1-x^2}} dx = \int_0^{\pi} \sin(m\theta)\sin(n\theta) d\theta,$$

$$\int_{-1}^{1} \sqrt{1-x^2}\, U_m U_n\, dx = \int_0^{\pi} \sin((m+1)\theta)\sin((n+1)\theta) d\theta,$$

$$\int_{-1}^{1} \sqrt{1-x^2}\, W_m W_n\, dx = \int_0^{\pi} \cos((m+1)\theta)\cos((n+1)\theta) d\theta.$$

For the cosine product use $2\cos a\theta \cos b\theta = \cos((a-b)\theta) + \cos((a+b)\theta)$; for the sine product use $2\sin a\theta \sin b\theta = \cos((a-b)\theta) - \cos((a+b)\theta)$. Every nonzero integer-frequency cosine integrates to zero on $[0, \pi]$, whereas the zero-frequency cosine integrates to π. For nonnegative integers a, b, the difference $a - b$ is zero only for $a = b$ and the sum $a + b$ only for $a = b = 0$, so

$$\int_0^{\pi} \cos a\theta \cos b\theta\, d\theta = \frac{\pi}{2}(\delta_{ab} + \delta_{a0}\delta_{b0}), \qquad \int_0^{\pi} \sin a\theta \sin b\theta\, d\theta = \frac{\pi}{2}(\delta_{ab} - \delta_{a0}\delta_{b0}).$$

In (a) and (b) take $a = m$, $b = n$; in (c) and (d) take $a = m + 1 \geq 1$, $b = n + 1$, so the term $\delta_{a0}\delta_{b0}$ is absent. Consequently the four answers are, respectively,

$$\boxed{\frac{\pi\delta_{mn}}{2 - \delta_{n0}}}, \qquad \boxed{\frac{\pi}{2}\delta_{mn}(1 - \delta_{n0})}, \qquad \boxed{\frac{\pi}{2}\delta_{mn}}, \qquad \boxed{\frac{\pi}{2}\delta_{mn}}.$$

The factor $1 - \delta_{n0}$ in the second result comes from $V_0 = \sin 0 = 0$. The solution $\arccos x$ of Exercise 18.4.8 would not satisfy relation (b): with $x = \cos\theta$, its weighted integral with V_n is $\int_0^{\pi} \theta \sin n\theta\, d\theta = (-1)^{n+1}\pi/n \neq 0$ for $n \geq 1$. Although W_n is singular at the endpoints, its weight reduces the displayed integrals to bounded trigonometric integrals, so all four improper integrals converge.

18.4.20 Show whether

(a) $T_m(x)$ and $V_n(x)$ are or are not orthogonal over the interval $[-1,1]$ with respect to the weighting factor $(1-x^2)^{-1/2}$.

(b) $U_m(x)$ and $W_n(x)$ are or are not orthogonal over the interval $[-1,1]$ with respect to the weighting factor $(1-x^2)^{1/2}$.

Use $n \ge 1$ in part (a), with $V_n = \sin(n \arccos x)$: for the order-zero solution $\arccos x$ of Exercise 18.4.8 the integral is $\int_0^\pi \theta \cos m\theta \, d\theta$, which is $\pi^2/2$ for $m = 0$, whereas the formula below gives 0 at $n = 0$.

Solution.

For an integer j, define

$$S(j) = \int_0^\pi \sin(j\theta) d\theta = \Big[-\frac{\cos j\theta}{j}\Big]_0^\pi = \begin{cases} [1-(-1)^j]/j, & j \neq 0, \\ 0, & j = 0. \end{cases}$$

This formula includes negative integers; in particular $S(-j) = -S(j)$. Thus $S(j) = 0$ for even j and $S(j) = 2/j$ for odd j.

(a) Use the sine-family convention for V_n, the transformation of Exercise 18.4.19, and $2\cos(m\theta)\sin(n\theta) = \sin((n+m)\theta) + \sin((n-m)\theta)$:

$$\begin{aligned} A_{mn} &= \int_{-1}^1 \frac{T_m V_n}{\sqrt{1-x^2}} dx = \int_0^\pi \cos(m\theta)\sin(n\theta) d\theta \\ &= \tfrac{1}{2}[S(n+m) + S(n-m)] \\ &= \boxed{\begin{cases} 0, & m+n \text{ even}, \\ \dfrac{2n}{n^2-m^2}, & m+n \text{ odd}. \end{cases}} \end{aligned}$$

For $m+n$ even, $n \pm m$ are both even; for $m+n$ odd, both are odd, and $\frac{1}{2}[2/(n+m) + 2/(n-m)] = [(n-m)+(n+m)]/(n^2-m^2) = 2n/(n^2-m^2)$. The odd-parity case cannot have $m = n$, so its denominator is nonzero. For example $A_{0,1} = \int_0^\pi \sin\theta \, d\theta = 2$. Thus the two families are not mutually orthogonal, although all pairs of equal parity are.

(b) The weight cancels the two sine denominators in $U_m W_n$, as in Exercise 18.4.19, and $2\sin(a\theta)\cos(b\theta) = \sin((a+b)\theta) + \sin((a-b)\theta)$ with $a = m+1$, $b = n+1$:

$$\begin{aligned} B_{mn} &= \int_{-1}^1 \sqrt{1-x^2}\, U_m W_n \, dx \\ &= \int_0^\pi \sin((m+1)\theta)\cos((n+1)\theta) d\theta \\ &= \tfrac{1}{2}[S(m+n+2) + S(m-n)] \\ &= \boxed{\begin{cases} 0, & m+n \text{ even}, \\ \dfrac{2(m+1)}{(m+1)^2-(n+1)^2}, & m+n \text{ odd}. \end{cases}} \end{aligned}$$

Here $m+n+2$ and $m-n$ have the parity of $m+n$; for $m+n$ odd,

$$\frac{1}{m+n+2} + \frac{1}{m-n} = \frac{2(m+1)}{(m+n+2)(m-n)} = \frac{2(m+1)}{(m+1)^2-(n+1)^2}.$$

For example $B_{0,1} = \int_0^\pi \sin\theta \cos 2\theta \, d\theta = \frac{1}{2}[S(3) + S(-1)] = \frac{1}{3} - 1 = -\frac{2}{3}$, so again there is no mutual orthogonality across the full two families.

18.4.21 Derive

(a) $T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x)$,

(b) $T_{m+n}(x) + T_{m-n}(x) = 2T_m(x)T_n(x)$,

from the "corresponding" cosine identities.

Solution.

For $x = \cos\theta$, use $\cos(A + B) + \cos(A - B) = 2\cos A \cos B$. (a) Set $A = n\theta$, $B = \theta$:

$$T_{n+1}(x) + T_{n-1}(x) = \cos((n+1)\theta) + \cos((n-1)\theta) = \boxed{2xT_n(x)}.$$

(b) Set $A = m\theta$, $B = n\theta$:

$$T_{m+n}(x) + T_{m-n}(x) = \cos((m+n)\theta) + \cos((m-n)\theta) = \boxed{2T_m(x)T_n(x)}.$$

If $m < n$, interpret $T_{m-n} = T_{n-m}$, because cosine is even in its order; one may equivalently write $T_{|m-n|}$ throughout. The identities first hold on $[-1, 1]$, and then for all complex x because their two sides are polynomials agreeing on an interval.

18.4.22 A number of equations relate the two types of Chebyshev polynomials. As examples show that

$$T_n(x) = U_n(x) - xU_{n-1}(x)$$

and

$$(1 - x^2)U_n(x) = xT_{n+1}(x) - T_{n+2}(x).$$

The first identity uses $n \geq 1$, since it contains U_{n-1}; the second uses $n \geq 0$.

Solution.

Put $x = \cos\theta$ with $0 < \theta < \pi$, so that $U_n = \sin((n+1)\theta)/\sin\theta$ and $T_n = \cos n\theta$. For $n \geq 1$,

$$\begin{aligned} U_n - xU_{n-1} &= \frac{\sin((n+1)\theta) - \cos\theta \sin(n\theta)}{\sin\theta} \\ &= \frac{\cos(n\theta)\sin\theta}{\sin\theta} = \boxed{T_n}. \end{aligned}$$

The sine addition formula $\sin((n+1)\theta) = \sin n\theta \cos\theta + \cos n\theta \sin\theta$ gives the second line. For the other identity, expand the cosine of a sum, $\cos((n+2)\theta) = \cos((n+1)\theta)\cos\theta - \sin((n+1)\theta)\sin\theta$:

$$\begin{aligned} xT_{n+1} - T_{n+2} &= \cos\theta \cos((n+1)\theta) - \cos((n+2)\theta) \\ &= \sin\theta \sin((n+1)\theta) = \sin^2\theta \, \frac{\sin((n+1)\theta)}{\sin\theta} = \boxed{(1 - x^2)U_n}. \end{aligned}$$

Continuity supplies the endpoints, and polynomial identity extends both formulas to arbitrary complex x. The second holds for $n \geq 0$; the first also holds for $n = 0$ if $U_{-1} = 0$, in which case it reads $T_0 = U_0 = 1$.

18.4.23 Show that

$$\frac{d}{dx}\,V_n(x) = -\frac{nT_n(x)}{\sqrt{1-x^2}}$$

(a) using the trigonometric forms of V_n and T_n,

(b) using the Rodrigues representation.

Use $n \geq 1$ and $-1 < x < 1$: V_0 is not given by $\sqrt{1-x^2}\,U_{n-1}$, and the right side is singular at $x = \pm 1$.

Solution.

(a) For $n \geq 1$, $V_n = \sin(n\theta)$ and $x = \cos\theta$, so $d\theta/dx = -1/\sin\theta = -1/\sqrt{1-x^2}$ for $0 < \theta < \pi$ and

$$V_n'(x) = n\cos(n\theta)\frac{d\theta}{dx} = \boxed{-\frac{nT_n(x)}{\sqrt{1-x^2}}}, \qquad -1 < x < 1.$$

(b) The type-II Rodrigues formula is

$$U_n(x) = \frac{(-1)^n(n+1)\sqrt{\pi}}{2^{n+1}\Gamma(n+3/2)\sqrt{1-x^2}}D^n(1-x^2)^{n+1/2}, \qquad D = \frac{d}{dx}.$$

Replace n by $n-1$, so that $(n+1) \to n$, $2^{n+1} \to 2^n$, $\Gamma(n+3/2) \to \Gamma(n+1/2)$ and $n+1/2 \to n-1/2$, and multiply by $\sqrt{1-x^2}$:

$$V_n(x) = \frac{(-1)^{n-1}n\sqrt{\pi}}{2^n\Gamma(n+1/2)}D^{n-1}(1-x^2)^{n-1/2}.$$

Differentiate once, then compare with the type-I Rodrigues formula of Exercise 18.4.16,

$$\frac{T_n}{\sqrt{1-x^2}} = \frac{(-1)^n\sqrt{\pi}}{2^n\Gamma(n+1/2)}D^n(1-x^2)^{n-1/2},$$

using $(-1)^{n-1} = -(-1)^n$:

$$\begin{aligned} V_n' &= \frac{(-1)^{n-1}n\sqrt{\pi}}{2^n\Gamma(n+1/2)}D^n(1-x^2)^{n-1/2} \\ &= -n\left[\frac{(-1)^n\sqrt{\pi}}{2^n\Gamma(n+1/2)}D^n(1-x^2)^{n-1/2}\right] = -\frac{nT_n}{\sqrt{1-x^2}}. \end{aligned}$$

The signs and normalization factors agree exactly. At $n = 0$ the sine family has zero derivative, while the separately chosen independent solution $\arccos x$ is not covered by this formula.

18.4.24 Starting with $x = \cos\theta$ and $T_n(\cos\theta) = \cos n\theta$, expand

$$x^k = \left(\frac{e^{i\theta} + e^{-i\theta}}{2}\right)^k$$

and show that

$$x^k = \frac{1}{2^{k-1}}\left[T_k(x) + \binom{k}{1}T_{k-2}(x) + \binom{k}{2}T_{k-4} + \cdots\right],$$

the series in brackets terminating after the term containing T_1 or T_0. (If the series ends with a T_0 term, that term carries an additional factor of 1/2.)

Solution.

For a nonnegative integer k, expand binomially, with $(e^{i\theta})^{k-j}(e^{-i\theta})^{j} = e^{i(k-2j)\theta}$:

$$x^k = 2^{-k}\sum_{j=0}^{k}\binom{k}{j}e^{i(k-2j)\theta}.$$

The terms j and $k-j$ have equal binomial coefficients and opposite frequencies. Each such pair, $j < k/2$, sums by $e^{i\phi}+e^{-i\phi} = 2\cos\phi$ to $2\binom{k}{j}\cos((k-2j)\theta) = 2\binom{k}{j}T_{k-2j}(x)$. If k is odd all terms pair. If k is even, $j = k/2$ is an unpaired zero-frequency term. Thus the precise formula is

$$\boxed{x^k = 2^{1-k}\sum_{j=0}^{\lfloor (k-1)/2\rfloor}\binom{k}{j}T_{k-2j}(x) + \begin{cases} 2^{-k}\binom{k}{k/2}T_0(x), & k \text{ even},\\ 0, & k \text{ odd}.\end{cases}}$$

The first sum is empty at $k = 0$ and the last term is then 1. Factoring out 2^{1-k} leaves the bracketed terms $\binom{k}{j}T_{k-2j}$ of the statement and turns the T_0 term into $2^{1-k}\cdot\frac{1}{2}\binom{k}{k/2}T_0$; so the last T_0 coefficient, when present, has the additional factor 1/2 specified by the published erratum. For example $x^2 = (T_2+T_0)/2$ and $x^3 = (T_3+3T_1)/4$.

18.4.25 Develop the following Chebyshev expansions (for $[-1,1]$):

(a) $(1-x^2)^{1/2} = \dfrac{2}{\pi}\left[1 - 2\sum_{s=1}^{\infty}(4s^2-1)^{-1}T_{2s}(x)\right]$,

(b) $\begin{cases}+1, & 0 < x \le 1\\ -1, & -1 \le x < 0\end{cases} = \dfrac{4}{\pi}\sum_{s=0}^{\infty}(-1)^s(2s+1)^{-1}T_{2s+1}(x).$

Solution.

For $F(\theta) = f(\cos\theta)$, use the cosine-series convention

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}a_nT_n(x), \qquad a_n = \frac{2}{\pi}\int_0^{\pi}F(\theta)\cos(n\theta)d\theta.$$

(a) Here $F(\theta) = \sin\theta$ on $[0,\pi]$. The constant is $a_0 = (2/\pi)\int_0^{\pi}\sin\theta\,d\theta = 4/\pi$. The reflection $\theta \mapsto \pi-\theta$ leaves $\sin\theta$ unchanged and multiplies $\cos n\theta$ by $\cos(n\pi) = (-1)^n$, so every odd coefficient equals its own negative and is zero. For $s \ge 1$, the product-to-sum formula $2\sin A\cos B = \sin(A+B)+\sin(A-B)$ and $\int_0^{\pi}\sin(k\theta)d\theta = 2/k$ for odd k (Exercise 18.4.20) give

$$\begin{aligned}\int_0^{\pi}\sin\theta\cos(2s\theta)d\theta &= \frac{1}{2}\int_0^{\pi}[\sin((2s+1)\theta)+\sin((1-2s)\theta)]d\theta\\ &= \frac{1}{2s+1}+\frac{1}{1-2s} = \frac{2}{(2s+1)(1-2s)} = -\frac{2}{4s^2-1}.\end{aligned}$$

Thus $a_{2s} = -4/[\pi(4s^2-1)]$, and with $a_0/2 = 2/\pi$,

$$\boxed{\sqrt{1-x^2} = \frac{2}{\pi}\left[1 - 2\sum_{s=1}^{\infty}\frac{T_{2s}(x)}{4s^2-1}\right]}.$$

The coefficients are absolutely summable and $|T_n(x)| \le 1$ on $[-1,1]$, so the series converges uniformly, including both endpoints. Its sum is continuous with the same cosine coefficients as $\sin\theta$ (integrate term by term), and a continuous function on $[0,\pi]$ is determined by its cosine coefficients, so the sum is $\sqrt{1-x^2}$ there.

(b) Now $F = 1$ on $0 \le \theta < \pi/2$ and $F = -1$ on $\pi/2 < \theta \le \pi$. Its average is zero, so $a_0 = 0$. For $n \ge 1$,

$$\begin{aligned} a_n &= \frac{2}{\pi}\left[\int_0^{\pi/2} \cos(n\theta)d\theta - \int_{\pi/2}^{\pi} \cos(n\theta)d\theta\right] \\ &= \frac{2}{\pi n}\left[\sin\frac{n\pi}{2} - \left(\sin n\pi - \sin\frac{n\pi}{2}\right)\right] = \frac{4}{\pi n}\sin\frac{n\pi}{2}, \end{aligned}$$

since $\sin n\pi = 0$. This vanishes for even n, and, as $\sin((2s+1)\pi/2) = (-1)^s$, equals $4(-1)^s/[\pi(2s+1)]$ for $n = 2s+1$. Therefore

$$\boxed{\operatorname{sgn} x = \frac{4}{\pi}\sum_{s=0}^{\infty} \frac{(-1)^s}{2s+1} T_{2s+1}(x)}, \qquad -1 \le x \le 1,\ x \ne 0.$$

The cosine series is the Fourier series of the even 2π-periodic extension of F, which is 1 for $|\theta| < \pi/2$ and -1 for $\pi/2 < |\theta| \le \pi$. This extension is piecewise constant with jumps only at $\theta = \pm\pi/2$ in $[-\pi,\pi]$, so by Dirichlet's theorem the series converges to $F(\theta)$ at every other θ, including $\theta = 0$ and $\theta = \pi$, that is, $x = \pm 1$. At $x = 1$, for example, $T_{2s+1}(1) = 1$ and the series is $\frac{4}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \cdots\right) = \frac{4}{\pi}\cdot\frac{\pi}{4} = 1$. At the jump $x = 0$ the series equals the midpoint value 0. In fact every odd $T_{2s+1}(0)$ is zero, so all partial sums already have that value; the series is not uniformly convergent across the jump.

18.4.26

(a) For the interval $[-1,1]$ show that

$$\begin{aligned} |x| &= \frac{1}{2} + \sum_{s=1}^{\infty} (-1)^{s+1} \frac{(2s-3)!!}{(2s+2)!!}(4s+1)P_{2s}(x) \\ &= \frac{2}{\pi} + \frac{4}{\pi}\sum_{s=1}^{\infty} (-1)^{s+1} \frac{1}{4s^2-1} T_{2s}(x). \end{aligned}$$

(b) Show that the ratio of the coefficient of $T_{2s}(x)$ to that of $P_{2s}(x)$ approaches $(\pi s)^{-1/2}$ as $s \to \infty$. This illustrates the relatively rapid convergence of the Chebyshev series.

Hint. With the Legendre recurrence relations, rewrite $xP_n(x)$ as a linear combination of derivatives. The trigonometric substitution $x = \cos\theta$, $T_n(x) = \cos n\theta$ is most helpful for the Chebyshev part.

Solution.

(a) For the Legendre expansion the coefficient of P_N is $\frac{2N+1}{2}\int_{-1}^{1} |x| P_N(x)dx$. Parity removes all odd degrees, and the constant coefficient is $\frac{1}{2}\int_{-1}^{1}|x|dx = 1/2$. At even degree $N = 2s \ge 2$ the integrand is even, so the coefficient is $(2N+1)J_N$, where $J_N = \int_0^1 xP_N(x)dx$. Multiply the self-adjoint Legendre equation $[(1-x^2)P_N']' + N(N+1)P_N = 0$ by x and integrate:

$$0 = [x(1-x^2)P_N']_0^1 - \int_0^1 (1-x^2)P_N' dx + N(N+1)J_N.$$

The boundary term is zero (the factor x vanishes at 0 and $1-x^2$ at 1), while another integration by parts gives

$$\int_0^1 (1-x^2)P_N' dx = [(1-x^2)P_N]_0^1 + \int_0^1 2xP_N\,dx = -P_N(0) + 2J_N.$$

Consequently $0 = P_N(0) - 2J_N + N(N+1)J_N$, and since $N(N+1)-2 = (N-1)(N+2)$,

$$J_N = -\frac{P_N(0)}{(N-1)(N+2)}.$$

Use $P_{2s}(0) = (-1)^s(2s-1)!!/(2s)!!$, $(2s-1)!! = (2s-1)(2s-3)!!$ and $(2s)!!\,(2s+2) = (2s+2)!!$:

$$J_{2s} = -\frac{(-1)^s(2s-1)!!}{(2s)!!\,(2s-1)(2s+2)} = (-1)^{s+1}\frac{(2s-3)!!}{(2s+2)!!},$$

with $(-1)!! = 1$ at $s=1$. It follows that

$$\boxed{|x| = \frac{1}{2} + \sum_{s=1}^{\infty}(-1)^{s+1}(4s+1)\frac{(2s-3)!!}{(2s+2)!!}P_{2s}(x)}.$$

For the Chebyshev expansion, with the cosine-series convention of Exercise 18.4.25, $F(\theta) = |\cos\theta|$ has $a_0 = (2/\pi)\int_0^\pi |\cos\theta| d\theta = 4/\pi$. Under $\theta \mapsto \pi - \theta$, $|\cos\theta|$ is unchanged and $\cos n\theta$ acquires the factor $(-1)^n$, so the odd coefficients vanish and each even one is twice the integral over $[0, \pi/2]$. With $2\cos\theta\cos(2s\theta) = \cos((2s-1)\theta) + \cos((2s+1)\theta)$ and $\sin((2s\mp1)\pi/2) = \mp(-1)^s$, for $s \geq 1$,

$$\begin{aligned} a_{2s} &= \frac{4}{\pi}\int_0^{\pi/2} \cos\theta\cos(2s\theta)d\theta \\ &= \frac{2}{\pi}\left[\frac{\sin((2s-1)\pi/2)}{2s-1} + \frac{\sin((2s+1)\pi/2)}{2s+1}\right] \\ &= \frac{2}{\pi}(-1)^{s+1}\left[\frac{1}{2s-1} - \frac{1}{2s+1}\right] = \frac{4}{\pi}\frac{(-1)^{s+1}}{4s^2-1}. \end{aligned}$$

Hence, with $a_0/2 = 2/\pi$,

$$\boxed{|x| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{s=1}^{\infty}\frac{(-1)^{s+1}}{4s^2-1}T_{2s}(x)}.$$

Both expansions are valid on the whole closed interval. In the Legendre case the coefficients are $O(s^{-3/2})$ as shown below, and $|P_n(x)| \leq 1$ on this interval; in the Chebyshev case they are $O(s^{-2})$ and $|T_n(x)| \leq 1$. Thus both are absolutely uniformly convergent, and orthogonal completeness identifies their continuous sums with $|x|$.

(b) Let $A_s = (-1)^{s+1}(4s+1)(2s-3)!!/(2s+2)!!$ and $B_s = 4(-1)^{s+1}/[\pi(4s^2-1)]$ be the coefficients of P_{2s} and T_{2s}. Convert the double-factorial ratio to gamma functions with $(2k-1)!! = 2^k\Gamma(k+\frac{1}{2})/\sqrt{\pi}$ at $k = s-1$ and $(2k)!! = 2^k k!$ at $k = s+1$, and use $\Gamma(s+a)/\Gamma(s+b) \sim s^{a-b}$ as $s \to \infty$:

$$\frac{(2s-3)!!}{(2s+2)!!} = \frac{2^{s-1}\Gamma(s-1/2)}{\sqrt{\pi}\,2^{s+1}\Gamma(s+2)} = \frac{\Gamma(s-1/2)}{4\sqrt{\pi}\,\Gamma(s+2)} \sim \frac{1}{4\sqrt{\pi}}\,s^{-5/2}.$$

Therefore, with $4s+1 \sim 4s$ and $4s^2-1 \sim 4s^2$, the Legendre and Chebyshev coefficients satisfy

$$A_s \sim \frac{(-1)^{s+1}}{\sqrt{\pi}\,s^{3/2}}, \qquad B_s \sim \frac{(-1)^{s+1}}{\pi s^2},$$

and their ratio is

$$\boxed{\frac{B_s}{A_s} \sim \frac{1}{\sqrt{\pi s}}}.$$

This is the published exponent correction from -1 to $-1/2$. It compares the large-order coefficients; the two expansions use different orthogonality weights.

18.4.27 Show that

$$\frac{\pi^2}{8} = 1 + 2\sum_{s=1}^{\infty} (4s^2 - 1)^{-2}.$$

Hint. Apply Parseval's identity (or the completeness relation) to the results of Exercise 18.4.26.

Solution.

Use the uniformly convergent Chebyshev expansion of $|x|$ from Exercise 18.4.26, with $a_0/2 = 2/\pi$ and $a_{2s} = 4(-1)^{s+1}/[\pi(4s^2 - 1)]$. Orthogonality eliminates the mixed terms upon squaring and integrating with weight $(1-x^2)^{-1/2}$, and the norms of Exercise 18.4.19(a), $\int_{-1}^{1} T_0^2 (1-x^2)^{-1/2} dx = \pi$ and $\int_{-1}^{1} T_n^2 (1-x^2)^{-1/2} dx = \pi/2$ $(n \geq 1)$, give Parseval's identity

$$\int_{-1}^{1} \frac{|x|^2}{\sqrt{1-x^2}} dx = \left(\frac{a_0}{2}\right)^2 \pi + \sum_{s=1}^{\infty} a_{2s}^2 \frac{\pi}{2}.$$

Thus

$$\begin{aligned} \int_{-1}^{1} \frac{x^2}{\sqrt{1-x^2}} dx &= \left(\frac{2}{\pi}\right)^2 \pi + \sum_{s=1}^{\infty} \left(\frac{4}{\pi(4s^2-1)}\right)^2 \frac{\pi}{2} \\ &= \frac{4}{\pi} + \frac{8}{\pi} \sum_{s=1}^{\infty} (4s^2 - 1)^{-2}. \end{aligned}$$

The integral on the left is, with $x = \cos\theta$, $\int_0^{\pi} \cos^2\theta \, d\theta = \pi/2$. So $\pi/2 = 4/\pi + (8/\pi)\sum_{s\geq 1}(4s^2-1)^{-2}$, and multiplication by $\pi/4$ gives

$$\boxed{\frac{\pi^2}{8} = 1 + 2\sum_{s=1}^{\infty} (4s^2 - 1)^{-2}}.$$

For each partial sum S_N of the expansion, orthogonality gives this identity with $|x|$ replaced by S_N and the sum stopped at N. Passing to the limit $N \to \infty$ is justified because $S_N \to |x|$ uniformly on $[-1, 1]$, so $S_N^2 \to x^2$ uniformly, and $\int_{-1}^{1}(1-x^2)^{-1/2}dx = \pi$ is finite; alternatively, use the weighted L^2 Parseval identity.

18.4.28 Show that

(a) $\cos^{-1} x = \dfrac{\pi}{2} - \dfrac{4}{\pi} \displaystyle\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} T_{2n+1}(x),$

(b) $\sin^{-1} x = \dfrac{4}{\pi} \displaystyle\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} T_{2n+1}(x).$

Use principal real inverse trigonometric functions on $-1 \le x \le 1$.

Solution.

The intermediate constant coefficient in the Instructor's Manual should be $\pi/2$, not $\pi^2/2$, since $\frac{1}{\pi}\int_0^\pi \theta\, d\theta = \pi/2$. Its final expansion has the correct constant.

(a) Put $x = \cos\theta$, $0 \le \theta \le \pi$, so $\arccos x = \theta$. In the cosine-series convention of Exercise 18.4.25, the coefficients of this linear function are

$$a_0 = \frac{2}{\pi}\int_0^\pi \theta\, d\theta = \frac{2}{\pi}\cdot\frac{\pi^2}{2} = \pi,$$

and, for $n \ge 1$, integration by parts with $u = \theta$, $dv = \cos(n\theta)d\theta$, $v = \sin(n\theta)/n$, i.e. $\int \theta\cos(n\theta)d\theta = \theta\sin(n\theta)/n - \int \sin(n\theta)d\theta/n$, gives

$$\begin{aligned} a_n &= \frac{2}{\pi}\int_0^\pi \theta\cos(n\theta)d\theta \\ &= \frac{2}{\pi}\left[\frac{\theta\sin(n\theta)}{n} + \frac{\cos(n\theta)}{n^2}\right]_0^\pi = \frac{2}{\pi}\frac{(-1)^n - 1}{n^2}, \end{aligned}$$

since $\sin n\pi = \sin 0 = 0$, $\cos n\pi = (-1)^n$ and $\cos 0 = 1$. Even coefficients vanish, and for $n = 2j+1$ the result is $-4/[\pi(2j+1)^2]$. With $a_0/2 = \pi/2$, therefore

$$\boxed{\arccos x = \frac{\pi}{2} - \frac{4}{\pi}\sum_{j=0}^{\infty}\frac{T_{2j+1}(x)}{(2j+1)^2}}.$$

The coefficients are absolutely summable and $|T_n(x)| \le 1$, so the series converges uniformly on $[-1,1]$; as in Exercise 18.4.25(a), its continuous sum has the cosine coefficients of θ (integrate term by term) and therefore equals $\theta = \arccos x$, including at both endpoints.

(b) The principal real inverse functions satisfy $\arcsin x + \arccos x = \pi/2$ for $-1 \le x \le 1$. Subtracting the first expansion from $\pi/2$ yields

$$\boxed{\arcsin x = \frac{4}{\pi}\sum_{j=0}^{\infty}\frac{T_{2j+1}(x)}{(2j+1)^2}}.$$

At $x = 0$ all terms are zero; at $x = \pm 1$, where $T_{2j+1}(\pm 1) = \pm 1$, the sums give $\pm\pi/2$, using $\sum_{j\ge 0}(2j+1)^{-2} = \zeta(2) - \sum_{k\ge 1}(2k)^{-2} = \frac{3}{4}\zeta(2) = \pi^2/8$ (from $\zeta(2) = \pi^2/6$).

18.5 Hypergeometric Functions

18.5.1

(a) For c, an integer, and a and b nonintegral, show that

$$ {}_2F_1(a,b;c;x) \quad \text{and} \quad x^{1-c}\,{}_2F_1(a+1-c, b+1-c; 2-c; x)$$

provide only one usable series formula for a solution to the hypergeometric equation (counting coincident formulas once).

(b) What happens if a is an integer, say, $a = -1$, and $c = -2$?

Solution.

(a) Write $F(a,b;c;x) = {}_2F_1(a,b;c;x)$ and substitute $y = \sum_{j\ge 0} d_j x^j$ into $x(1-x)y'' + [c-(a+b+1)x]y' - aby = 0$. In the coefficient of x^j, the terms xy'' and cy' contribute $(j+1)jd_{j+1} + c(j+1)d_{j+1}$, and the other three contribute $-[j(j-1) + (a+b+1)j + ab]d_j = -(j+a)(j+b)d_j$. Hence

$$(j+1)(j+c)d_{j+1} = (j+a)(j+b)d_j.$$

For $y = x^s(1+O(x))$ the lowest power x^{s-1} gives the indicial equation $s(s-1)+cs = s(s+c-1) = 0$, so the two Frobenius exponents at zero are 0 and $1-c$. For nonintegral a, b, neither numerator series terminates. If $c = 1$, the two displayed expressions coincide term for term. If $c = 0, -1, -2, \ldots$, the first series encounters $(c)_j = c(c+1)\cdots(c+j-1) = 0$ for $j \ge 1-c$, with nonzero numerator $(a)_j(b)_j$, while the second has the positive integer denominator parameter $2-c$ and is well defined near zero. If $c = 2, 3, \ldots$, the first series is well defined and the second encounters the nonpositive integer parameter $2-c$, again with nonzero numerator $(a+1-c)_j(b+1-c)_j$ because a, b are nonintegral. Thus precisely one of these two unmodified formulas is usable (or they are identical when $c = 1$).

This does not mean that the second-order ODE has only one independent solution. Nor does a zero denominator parameter make the hypergeometric series zero: it makes the generic series undefined, since a vanishing factor of $(c)_j$ then stands in a denominator. On a subinterval avoiding $0, 1$ and zeros of a first solution y_1, reduction of order constructs a second solution:

$$y_2 = y_1 \int^x \frac{s^{-c}(1-s)^{c-a-b-1}}{y_1(s)^2}\, ds.$$

Indeed, written as $y'' + py' + qy = 0$, the ODE has, by partial fractions,

$$p = \frac{c-(a+b+1)x}{x(1-x)} = \frac{c}{x} + \frac{c-a-b-1}{1-x},$$

so $\exp(-\int p\,dx) = x^{-c}(1-x)^{c-a-b-1}$ is the numerator in this integral. The Wronskian $y_1y_2' - y_1'y_2 = y_1^2(y_2/y_1)'$ equals that nonzero numerator. Initial-value uniqueness then extends the independent pair across any zeros of y_1 to the whole interval avoiding $0, 1$. The resonant Frobenius construction generally introduces a logarithm; the loss is in the displayed pair of formulas, not the dimension of the solution space.

(b) For $a = -1, c = -2$, termination occurs before the zero denominator: $(a)_j = 0$ for $j \ge 2$, while $(c)_1 = -2$, $(c)_2 = 2$ and $(c)_j = 0$ only for $j \ge 3$. With $2-c = 4$, $a+1-c = 2$, $b+1-c = b+3$,

$$y_1 = 1 + \frac{(-1)b}{-2}x = 1 + \frac{b}{2}x, \qquad y_2 = x^3F(2, b+3; 4; x).$$

The equation becomes

$$x(1-x)y'' + (-2-bx)y' + by = 0.$$

Direct substitution of $y_1' = b/2$, $y_1'' = 0$ gives $(-2-bx)b/2 + b(1+bx/2) = 0$. For the other solution, set $y = x^3u$ and differentiate: $y' = 3x^2u + x^3u'$, $y'' = 6xu + 6x^2u' + x^3u''$. After substitution the u'' terms are $x^4(1-x)u''$, the u' terms $[6x^3(1-x) - 2x^3 - bx^4]u' = x^3[4-(b+6)x]u'$, and the u terms $[6x^2(1-x) - 6x^2 - 3bx^3 + bx^3]u = -2(b+3)x^3u$. Division by x^3 gives

$$x(1-x)u'' + [4-(b+6)x]u' - 2(b+3)u = 0,$$

which is the hypergeometric equation with parameters $2, b+3; 4$, since $2 + (b+3) + 1 = b+6$ and $2\cdot(b+3) = 2(b+3)$; its analytic solution with value one at zero is $F(2, b+3; 4; x)$. Since

$y_1 = 1 + O(x)$ and $y_2 = x^3 + O(x^4)$, they are independent: here, unlike part (a), both formulas give solutions, and no logarithm is needed. The coefficient recurrence also shows exactly the extra freedom: $d_1 = bd_0/2$, $d_2 = 0$, and at $j = 2$ both sides vanish, leaving d_3 free. Here $F(-1, b; -2; x)$ means the explicitly terminating polynomial; a generic parameter limit need not select the same solution without a specified limiting prescription.

18.5.2 Find the Legendre, Chebyshev I, and Chebyshev II recurrence relations corresponding to the hypergeometric contiguous function relation given as Eq. (18.126).

Solution.

To avoid confusing the hypergeometric argument with the polynomial variable, put $u = (1-z)/2$ and write the contiguous relation as $AF(a, b; c; u) = BF(a-1, b+1; c; u) + CF(a+1, b-1; c; u)$. In this notation Eq. (18.126) has

$$A = (a-b)\{c(a+b-1) + 1 - a^2 - b^2 + [(a-b)^2 - 1](1-u)\},$$
$$B = (c-a)(a-b+1)b, \qquad C = (c-b)(a-b-1)a.$$

We use the hypergeometric representations $P_n(z) = F(-n, n+1; 1; u)$, $T_n(z) = F(-n, n; \frac{1}{2}; u)$ and $U_n(z) = (n+1)F(-n, n+2; \frac{3}{2}; u)$. Their parameter sets give the following coefficients:

(a, b, c)	A	B	C
$(-n, n+1, 1)$	$-2n(n+1)(2n+1)z$	$-2n(n+1)^2$	$-2n^2(n+1)$
$(-n, n, \frac{1}{2})$	$-n(4n^2-1)z$	$-n(4n^2-1)/2$	$-n(4n^2-1)/2$
$(-n, n+2, \frac{3}{2})$	$-(n+1)(2n+1)(2n+3)z$	$-(n+2)(2n+1)(2n+3)/2$	$-n(2n+1)(2n+3)/2$

For example, in the Legendre row $a - b = -2n - 1$, $c(a+b-1) = 0$ and $(a-b)^2 - 1 = 4n(n+1)$, so the expression in braces on the left is

$$1 - n^2 - (n+1)^2 + 4n(n+1)(1-u) = -2n(n+1) + 4n(n+1)(1-u)$$
$$= 2n(n+1)(1-2u) = 2n(n+1)z.$$

This gives the first entry and illustrates the substitution used for each row. In the two Chebyshev rows, $c(a+b-1)+1-a^2-b^2$ equals $\frac{1}{2} - 2n^2 = -\frac{1}{2}(4n^2-1)$ and $-2n^2 - 4n - \frac{3}{2} = -\frac{1}{2}(2n+1)(2n+3)$, while $(a-b)^2 - 1$ equals $4n^2 - 1$ and $(2n+1)(2n+3)$. Since $(1-u) - \frac{1}{2} = z/2$, the braces are $(4n^2-1)z/2$ and $(2n+1)(2n+3)z/2$; multiplication by $a - b = -2n$ and $-2(n+1)$ gives the other two entries of A. The entries B, C are direct products, e.g. $B = (1+n)(-2n)(n+1)$ and $C = (-n)(-2n-2)(-n)$ in the Legendre row.

In each row, $F(a-1, b+1; c; u)$ and $F(a+1, b-1; c; u)$ are the hypergeometric functions of that row with n replaced by $n+1$ and $n-1$, since $a \mp 1 = -(n \pm 1)$ and b moves with it. For Legendre, the three functions are therefore P_n, P_{n+1}, P_{n-1}. Divide the first row by $-2n(n+1)$ for $n \geq 1$:

$$\boxed{(n+1)P_{n+1}(z) - (2n+1)zP_n(z) + nP_{n-1}(z) = 0}.$$

At $n = 0$ the original contiguous equation degenerates to $0 = 0$; the recurrence separately follows from $P_0 = 1, P_1 = z$.

For Chebyshev I the three functions are T_n, T_{n+1}, T_{n-1}. Divide the second row by $-n(4n^2-1)/2$, for $n \geq 1$, to get

$$\boxed{T_{n+1}(z) - 2zT_n(z) + T_{n-1}(z) = 0}.$$

The initial polynomials are $T_0 = 1, T_1 = z$. If the cosine extension $T_{-1} = T_1$ is adopted, the recurrence also holds at zero.

For Chebyshev II, include the normalization factors:

$$F(-n,n+2;\tfrac{3}{2};u)=\frac{U_n(z)}{n+1},\quad F(-n-1,n+3;\tfrac{3}{2};u)=\frac{U_{n+1}(z)}{n+2}.$$

For $n\ge 1$ the last function is U_{n-1}/n. In the third row the factors $n+1,n+2,n$ therefore cancel individually, leaving

$$-z(2n+1)(2n+3)U_n=-\frac{(2n+1)(2n+3)}{2}(U_{n+1}+U_{n-1}),$$

so

$$\boxed{U_{n+1}(z)-2zU_n(z)+U_{n-1}(z)=0}.$$

At $n=0$ the last contiguous coefficient is zero; with $U_{-1}=0$ it gives $U_1=2zU_0$, consistent with $U_0=1$.

18.5.3 Transform the following polynomials into hypergeometric functions of argument x^2:

(a) $T_{2n}(x)$;

(b) $x^{-1}T_{2n+1}(x)$;

(c) $U_{2n}(x)$;

(d) $x^{-1}U_{2n+1}(x)$.

ANS. (a) $T_{2n}(x)=(-1)^n\,{}_2F_1\big(-n,n;\tfrac{1}{2};x^2\big)$.
(b) $x^{-1}T_{2n+1}(x)=(-1)^n(2n+1)\,{}_2F_1\big(-n,n+1;\tfrac{3}{2};x^2\big)$.
(c) $U_{2n}(x)=(-1)^n\,{}_2F_1\big(-n,n+1;\tfrac{1}{2};x^2\big)$.
(d) $x^{-1}U_{2n+1}(x)=(-1)^n(2n+2)\,{}_2F_1\big(-n,n+2;\tfrac{3}{2};x^2\big)$.

Interpret the quotients at $x=0$ by their removable limits, since x^{-1} is undefined there.

Solution.

Take $n=0,1,2,\dots$ and $t=x^2$. The polynomials T_{2n} and U_{2n} are even and T_{2n+1}, U_{2n+1} are odd, so each has the form $f(t)$ or $xf(t)$ with f a polynomial. For an even function $Y(x)=f(t)$, the chain rule $d/dx=2x\,d/dt$ gives

$$Y'=2xf',\qquad Y''=2f'+4tf''.$$

For an odd function $Y(x)=xf(t)$,

$$Y'=f+2tf',\qquad Y''=2xf'+4xf'+4xtf''=x(6f'+4tf'').$$

Substitution into the type-I equation $(1-x^2)Y''-xY'+N^2Y=0$ turns its left side into $4t(1-t)f''+(2-4t)f'+N^2f$ for even Y and into $x[4t(1-t)f''+(6-8t)f'+(N^2-1)f]$ for odd Y. Division by 4 (and by x) gives, respectively,

$$t(1-t)f''+(\tfrac{1}{2}-t)f'+\frac{N^2}{4}f=0,$$

$$t(1-t)f''+(\tfrac{3}{2}-2t)f'+\frac{N^2-1}{4}f=0.$$

With $N = 2n$ in the first and $N = 2n+1$ in the second, comparison with $t(1-t)f''+[c-(a+b+1)t]f'-abf = 0$ gives $c = \frac{1}{2}$, $a+b = 0$, $ab = -n^2$ and $c = \frac{3}{2}$, $a+b = 1$, $ab = -[(2n+1)^2-1]/4 = -n(n+1)$, that is, $(a,b,c) = (-n, n, \frac{1}{2})$ and $(-n, n+1, \frac{3}{2})$.

For type II, use $(1-x^2)Y'' - 3xY' + N(N+2)Y = 0$ instead. Compared with type I, the extra term $-2xY'$ adds $-4tf'$ in the even case and $-x(2f+4tf')$ in the odd case, and N^2 becomes $N(N+2)$. Dividing as before, the corresponding equations are

$$t(1-t)f'' + (\tfrac{1}{2} - 2t)f' + \frac{N(N+2)}{4}f = 0,$$

$$t(1-t)f'' + (\tfrac{3}{2} - 3t)f' + \frac{N(N+2)-3}{4}f = 0.$$

With $N = 2n$, $a+b = 1$ and $ab = -n(n+1)$; with $N = 2n+1$, $a+b = 2$ and $ab = -[(2n+1)(2n+3)-3]/4 = -n(n+2)$. The parameters are therefore $(-n, n+1, \frac{1}{2})$ and $(-n, n+2, \frac{3}{2})$. All four analytic solutions at $t = 0$ are uniquely determined by their constant term, because c is positive and the coefficient recurrence $(j+1)(j+c)d_{j+1} = (j+a)(j+b)d_j$ of Exercise 18.5.1 has no zero factor $j+c$. The needed constants are the values $f(0)$; they equal $Y(0)$ for even Y and, by $Y' = f + 2tf'$ at $t = 0$, $Y'(0)$ for odd Y:

$$T_{2n}(0) = U_{2n}(0) = (-1)^n, \quad T'_{2n+1}(0) = (-1)^n(2n+1), \quad U'_{2n+1}(0) = (-1)^n(2n+2).$$

To obtain them put $x = \cos\theta$, so that $x = 0$ at $\theta = \pi/2$ and $d/dx = -\csc\theta\, d/d\theta$. From $T_N = \cos N\theta$, $T_N(0) = \cos(N\pi/2)$ and $T'_N = N\sin N\theta/\sin\theta$, so $T'_N(0) = N\sin(N\pi/2)$. From $U_N = \sin((N+1)\theta)/\sin\theta$, $U_N(0) = \sin((N+1)\pi/2)$ and, since $\cos\theta = 0$ at $\theta = \pi/2$, $U'_N(0) = -(N+1)\cos((N+1)\pi/2)$. With $N = 2n$ or $N = 2n+1$ these are the displayed values. Since $F(a,b;c;0) = 1$, the four answers are

$$\begin{aligned}
T_{2n}(x) &= \boxed{(-1)^n F(-n, n; \tfrac{1}{2}; x^2)}, \\
\frac{T_{2n+1}(x)}{x} &= \boxed{(-1)^n(2n+1)F(-n, n+1; \tfrac{3}{2}; x^2)}, \\
U_{2n}(x) &= \boxed{(-1)^n F(-n, n+1; \tfrac{1}{2}; x^2)}, \\
\frac{U_{2n+1}(x)}{x} &= \boxed{(-1)^n(2n+2)F(-n, n+2; \tfrac{3}{2}; x^2)}.
\end{aligned}$$

The parameter $-n$ terminates every series. The quotients at $x = 0$ are removable and take the derivative values just displayed; these are polynomial identities, so hold also for complex x.

18.5.4 Derive or verify the leading factor in the hypergeometric representations of the Chebyshev functions.

Solution.

In Eqs. (18.133)–(18.135), the hypergeometric argument is $u = (1-x)/2$, which is zero at $x = 1$. Each representation is a constant multiple of a hypergeometric series: with $x = 1-2u$, $d/dx = -\frac{1}{2}\,d/du$ and $1-x^2 = 4u(1-u)$, the type-I and type-II equations $(1-x^2)y'' - xy' + n^2y = 0$ and $(1-x^2)y'' - 3xy' + n(n+2)y = 0$ become

$$\begin{aligned}
u(1-u)y_{uu} + (\tfrac{1}{2} - u)y_u + n^2 y = 0, \\
u(1-u)y_{uu} + (\tfrac{3}{2} - 3u)y_u + n(n+2)y = 0,
\end{aligned}$$

the hypergeometric equations with $(a,b,c)=(-n,n,\frac12)$ and $(-n,n+2,\frac32)$. The polynomials T_n, U_n and $V_n/\sqrt{1-x^2}=U_{n-1}$ are analytic at $u=0$, where the other exponent $1-c=\pm\frac12$ is not an integer, so each is a constant times the corresponding $F(a,b;c;u)$ (for U_{n-1} replace n by $n-1$ in the type-II parameters, giving $F(-n+1,n+1;\frac32;u)$). Since that series starts with one, the constants follow from endpoint normalization, $\theta=\arccos x\to 0$:

$$T_n(1)=\cos 0=1,\qquad U_n(1)=\lim_{\theta\to0}\frac{\sin((n+1)\theta)}{\sin\theta}=n+1,$$

$$\lim_{x\to1^-}\frac{V_n(x)}{\sqrt{1-x^2}}=U_{n-1}(1)=n.$$

Thus the leading factors of T_n, U_n and V_n are, respectively,

$$\boxed{1,\qquad n+1,\qquad n\sqrt{1-x^2}}.$$

The last expression describes the sine family $V_n=\sqrt{1-x^2}U_{n-1}$ for $n\ge1$; at zero order its continuation is $V_0=0$, not the separate solution $\arccos x$.

For the four x^2 representations in Exercise 18.5.3, evaluate at $x=0$ instead. The even values are $(-1)^n$; for each odd quotient use $\lim_{x\to0}Y(x)/x=Y'(0)$. Differentiating the trigonometric representations (as in Exercise 18.5.3) gives

$$T_N'(0)=N\sin(N\pi/2),\qquad U_N'(0)=-(N+1)\cos((N+1)\pi/2).$$

At $N=2n+1$ these are $(-1)^n(2n+1)$ and $(-1)^n(2n+2)$. Hence the four leading factors, in order, are

$$\boxed{(-1)^n,\quad (-1)^n(2n+1),\quad (-1)^n,\quad (-1)^n(2n+2)}.$$

Both sets of representations are therefore scaled at a point where the hypergeometric series equals one.

18.5.5 Verify that the Legendre function of the second kind, $Q_\nu(z)$, is given by

$$Q_\nu(z)=\frac{\pi^{1/2}\nu!}{\Gamma\left(\nu+\frac32\right)(2z)^{\nu+1}}\,{}_2F_1\left(\frac{\nu}{2}+\frac12,\frac{\nu}{2}+1;\nu+\frac32;z^{-2}\right),$$

where $|z|>1$, $|\arg z|<\pi$, and $\nu\ne-1,-2,-3,\cdots$. When $\nu+3/2$ is a nonpositive integer, interpret ${}_2F_1/\Gamma(\nu+3/2)$ as the regularized hypergeometric function, as defined in the solution, since the series ${}_2F_1$ alone is then undefined; use $\nu!=\Gamma(\nu+1)$.

Solution.

Put $r=\nu+1$, $t=z^{-2}$ and $Q=z^{-r}f(t)$, so that $dt/dz=-2z^{-3}=-2t/z$. The product and chain rules give

$$Q'=-rz^{-r-1}f+z^{-r}f'\frac{dt}{dz}=z^{-r-1}[-rf-2tf'].$$

The bracket has derivative $-[(r+2)f'+2tf'']\,dt/dz=(2t/z)[(r+2)f'+2tf'']$, so

$$\begin{aligned}Q''&=-(r+1)z^{-r-2}[-rf-2tf']+2tz^{-r-2}[(r+2)f'+2tf'']\\&=z^{-r-2}[r(r+1)f+(4r+6)tf'+4t^2f''].\end{aligned}$$

Substitute into $(1-z^2)Q''-2zQ'+\nu(\nu+1)Q=0$, using $(1-z^2)z^{-2}=t-1$ and $\nu(\nu+1)=r(r-1)$. The left side becomes z^{-r} times

$$4t^2(t-1)f''+[(4r+6)t(t-1)+4t]f'+[r(r+1)(t-1)+2r+r(r-1)]f,$$

and the last bracket is $r(r+1)t$. Division by $-4tz^{-r}$ and $r=\nu+1$ give

$$t(1-t)f''+[\nu+\tfrac{3}{2}-(\nu+\tfrac{5}{2})t]f'-\frac{(\nu+1)(\nu+2)}{4}f=0.$$

Thus $c=\nu+\frac{3}{2}$, $a+b=\nu+\frac{3}{2}$ and $ab=(\nu+1)(\nu+2)/4$, so

$$a=\frac{\nu+1}{2},\quad b=\frac{\nu+2}{2},\quad \boxed{c=\nu+\tfrac{3}{2}},$$

confirming the published correction of the denominator parameter. For nonsingular c, the analytic solution at $t=0$ is $f=AF(a,b;c;t)$; its recurrence is

$$\frac{d_{j+1}}{d_j}=\frac{(j+(\nu+1)/2)(j+(\nu+2)/2)}{(j+1)(j+\nu+3/2)}.$$

To fix A, including for noninteger degree, start with real $\nu>0, z>1$ and the conventional exterior second-kind normalization

$$Q_\nu(z)=\int_0^\infty[z+\sqrt{z^2-1}\cosh u]^{-\nu-1}\,du.$$

This integral representation, with the conversion from Olver's normalization, is recorded in DLMF 14.12.6 and 14.3.10. Multiply by $z^{\nu+1}$ and let $z\to+\infty$. The integrand becomes $[1+\sqrt{1-z^{-2}}\cosh u]^{-\nu-1}$, which for $z\ge 2$ is at most $(\sqrt{3}/2)^{-\nu-1}(1+\cosh u)^{-\nu-1}$, an integrable function. Dominated convergence gives

$$\begin{aligned}A&=\int_0^\infty(1+\cosh u)^{-\nu-1}du\\&=2^{-\nu}\int_0^\infty\operatorname{sech}^{2\nu+2}v\,dv\qquad(u=2v)\\&=2^{-\nu-1}\int_0^1 s^{-1/2}(1-s)^\nu ds\qquad(s=\tanh^2 v)\\&=\frac{\sqrt{\pi}\,\Gamma(\nu+1)}{2^{\nu+1}\Gamma(\nu+3/2)}.\end{aligned}$$

Here $1+\cosh 2v=2\cosh^2 v$ and $du=2\,dv$; next $ds=2\tanh v\operatorname{sech}^2 v\,dv$ and $\operatorname{sech}^2 v=1-s$ give $\operatorname{sech}^{2\nu+2}v\,dv=\frac{1}{2}s^{-1/2}(1-s)^\nu ds$; the last integral is $B(\frac{1}{2},\nu+1)=\Gamma(\frac{1}{2})\Gamma(\nu+1)/\Gamma(\nu+\frac{3}{2})$. At $t=0$ the other Frobenius exponent is $1-c=-\nu-\frac{1}{2}<0$, so every solution of the f equation independent of F is unbounded as $t\to0$ (in Q it grows like $z^{-r}t^{1-c}=z^\nu$). Since Q_ν satisfies Legendre's equation and $f=z^{\nu+1}Q_\nu\to A$ stays bounded, $f=AF(a,b;c;t)$. Therefore

$$\boxed{Q_\nu(z)=\frac{\sqrt{\pi}\,\Gamma(\nu+1)}{\Gamma(\nu+3/2)(2z)^{\nu+1}}F\left(\frac{\nu+1}{2},\frac{\nu+2}{2};\nu+\tfrac{3}{2};z^{-2}\right)}.$$

Analytic continuation in degree and in z gives the requested exterior branch, with $(2z)^{\nu+1}=\exp((\nu+1)\operatorname{Log}(2z))$ and $|\arg z|<\pi$. The condition $|z|>1$ makes $|t|<1$. At nonnegative integer

$\nu = n$, the leading coefficient also reduces to $n!/(2n+1)!!$, since $\Gamma(n+\frac{3}{2}) = (2n+1)!!\sqrt{\pi}/2^{n+1}$, agreeing with the ordinary Q_n normalization.

At $\nu = -3/2, -5/2, \ldots$ (the case of the red note in the statement) the displayed product must be read as a regularized limit, since its separate hypergeometric factor is undefined. Using $\Gamma(c)(c)_j = \Gamma(c+j)$, a formula that includes these values directly is

$$Q_\nu(z) = \frac{\sqrt{\pi}\,\Gamma(\nu+1)}{(2z)^{\nu+1}} \sum_{j=0}^{\infty} \frac{((\nu+1)/2)_j((\nu+2)/2)_j}{\Gamma(\nu+3/2+j)j!}\, z^{-2j}.$$

Here $1/\Gamma$ is entire and vanishes at its nonpositive integer arguments. This is precisely $F/\Gamma(c)$, not an instruction to multiply an undefined infinity by zero. The regularized convention is independently consistent with DLMF 14.3.7. The excluded negative integer degrees are the poles specified in the question; no value at those poles is asserted.

18.5.6 The incomplete beta function was defined in Eq. (13.78) as

$$B_x(p,q) = \int_0^x t^{p-1}(1-t)^{q-1}\,dt.$$

Show that

$$B_x(p,q) = p^{-1}x^p\,{}_2F_1(p, 1-q; p+1; x).$$

Initially take $0 \le x < 1$ and $\Re p > 0$, so that the integral converges at $t = 0$; at $x = 1$ also require $\Re q > 0$, for convergence at $t = 1$.

Solution.

First take $0 < x < 1$, $\Re p > 0$ and arbitrary complex q, using real logarithms for positive t and $1-t$. The binomial series $(1-t)^{-\alpha} = \sum_j (\alpha)_j t^j/j!$ with $\alpha = 1-q$ is uniformly convergent for $0 \le t \le x$:

$$(1-t)^{q-1} = \sum_{j=0}^{\infty} \frac{(1-q)_j}{j!} t^j.$$

Its product with t^{p-1} is integrable at zero, and termwise integration, with $\int_0^x t^{p+j-1}dt = x^{p+j}/(p+j)$, therefore gives

$$B_x(p,q) = \sum_{j=0}^{\infty} \frac{(1-q)_j}{j!} \frac{x^{p+j}}{p+j}.$$

The finite-product cancellation

$$\frac{(p)_j}{(p+1)_j} = \frac{p(p+1)\cdots(p+j-1)}{(p+1)\cdots(p+j)} = \frac{p}{p+j}$$

holds also at $j = 0$. Insert it into the last series:

$$\boxed{B_x(p,q) = \frac{x^p}{p} \sum_{j=0}^{\infty} \frac{(p)_j(1-q)_j}{(p+1)_j j!} x^j = \frac{x^p}{p} F(p, 1-q; p+1; x)}.$$

At $x = 0$ both sides tend to zero under $\Re p > 0$. At $x = 1$, convergence of the defining integral additionally requires $\Re q > 0$, yielding the complete beta function. The series also converges there ($c - a - b = q$), both sides are continuous as $x \to 1^-$, and Exercise 18.5.8 confirms the value: $p^{-1}F(p, 1-q; p+1; 1) = \Gamma(p+1)\Gamma(q)/[p\,\Gamma(p+q)] = B(p,q)$. For complex x, continue along a path avoiding the branch points $0, 1$, with the same logarithms used in the integral and x^p. Parameter continuation beyond $\Re p > 0$ is a continuation of the function, not an ordinary convergent lower-endpoint integral.

18.5.7 Verify the integral representation

$$ {}_2F_1(a,b;c;z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-tz)^{-a}\,dt. $$

What restrictions must be placed on the parameters b and c?

Note. Although the power series used to establish this integral representation is only valid for $|z| < 1$, the principal representation is valid on $\mathbb{C} \setminus [1,\infty)$. Elsewhere specify matching analytic continuation or boundary limits. For nonintegral a, $[1,\infty)$ is the conventional cut.

Hint. The integral is suspiciously like a beta function and can be expanded into a series of beta functions.

ANS. $c > b > 0$ for real parameters; in general $\Re c > \Re b > 0$.

Solution.

For $|z| < 1$, expand the last factor:

$$ (1-tz)^{-a} = \sum_{j=0}^{\infty} \frac{(a)_j}{j!} t^j z^j. $$

Assume $\Re b > 0$ and $\Re(c-b) > 0$. The beta weight is absolutely integrable and, since $|tz| \le |z| < 1$, the binomial series is uniformly absolutely convergent for $t \in [0,1]$ (Weierstrass M-test), so integration and summation may be interchanged. The proposed right side becomes

$$ \begin{aligned} \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \sum_{j=0}^{\infty} \frac{(a)_j z^j}{j!} \int_0^1 t^{b+j-1}(1-t)^{c-b-1}dt &= \sum_{j=0}^{\infty} \frac{(a)_j z^j}{j!} \frac{\Gamma(c)\Gamma(b+j)}{\Gamma(b)\Gamma(c+j)} \\ &= \sum_{j=0}^{\infty} \frac{(a)_j (b)_j}{(c)_j j!} z^j = F(a,b;c;z). \end{aligned} $$

Here the integral is the beta function $B(b+j, c-b) = \Gamma(b+j)\Gamma(c-b)/\Gamma(c+j)$, and $\Gamma(b+j) = \Gamma(b)(b)_j$, $\Gamma(c+j) = \Gamma(c)(c)_j$. Near $t = 0$ and $t = 1$, for $z \neq 1$ off the cut, the additional factor is finite and nonzero. Ordinary endpoint convergence thus requires

$$ \boxed{\Re b > 0, \qquad \Re(c-b) > 0}; \qquad \boxed{c > b > 0 \quad \text{for real } b, c}. $$

There is no restriction on a for this initial domain. Choose the branch of $(1-tz)^{-a}$ continued from $z = 0$. For $t \in [0,1]$ the number $1 - tz$ is real and ≤ 0 only when $tz \ge 1$, that is, only for real $z \ge 1/t \ge 1$; hence for $z \notin [1,\infty)$ this branch is $(1-tz)^{-a} = \exp[-a\,\mathrm{Log}(1-tz)]$, continuous in (t,z) and analytic in z. For $z \in \mathbb{C} \setminus [1,\infty)$ the integral is analytic: on every compact subset the additional factor and its derivatives are uniformly bounded against the integrable endpoint weight. The identity theorem therefore extends the representation from the unit disk to this cut plane. On the cut itself a boundary value or another explicit prescription is needed; the textbook's "general z" does not guarantee an ordinary integral through $t = 1/z$. At $z = 1$, the endpoint powers combine and require $\Re(c-a-b) > 0$ instead at $t = 1$, as used next.

18.5.8 Prove that

$$ {}_2F_1(a,b;c;1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}, \qquad c \neq 0, -1, -2, \ldots, \quad c > a+b. $$

For complex parameters read $c > a + b$ as $\Re(c - a - b) > 0$, the condition for convergence of the series at 1.

Hint. Here is a chance to use the integral representation in Exercise 18.5.7.

Solution.

Initially impose $\Re b > 0$, $\Re(c-b) > 0$ and $\Re(c-a-b) > 0$, so Euler's integral remains convergent at $z = 1$. First, the series converges at $z = 1$. If it does not terminate ($a, b \notin \{0, -1, -2, \ldots\}$), its jth term is

$$\frac{(a)_j (b)_j}{(c)_j j!} = \frac{\Gamma(c)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(a+j)}{\Gamma(c+j)} \frac{\Gamma(b+j)}{\Gamma(1+j)},$$

and $\Gamma(j+\alpha)/\Gamma(j+\beta) \sim j^{\alpha-\beta}$ as $j \to \infty$ (a consequence of Stirling's formula) makes it $O(j^{\Re(a+b-c)-1})$ in magnitude. Thus the series converges absolutely when $\Re(c-a-b) > 0$, locally uniformly away from $c = 0, -1, -2, \ldots$; a terminating series is a finite sum. By Abel's theorem, $F(a, b; c; z) \to F(a, b; c; 1)$ as $z \to 1^-$. In Euler's integral, $0 \le z < 1$ gives $1 - tz \ge 1 - t$, so $|(1 - tz)^{-a}| = (1 - tz)^{-\Re a} \le \max\{1, (1 - t)^{-\Re a}\}$. The integrand is therefore dominated by an integrable function (at $t = 0$ because $\Re b > 0$, at $t = 1$ because $\Re(c - b) > 0$ and $\Re(c - a - b) > 0$), and dominated convergence lets $z \to 1^-$ under the integral. Combining its two factors involving $1 - t$ and using the beta integral $B(b, c - a - b)$ gives

$$\begin{aligned} F(a, b; c; 1) &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-a-b-1} dt \\ &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \frac{\Gamma(b)\Gamma(c-a-b)}{\Gamma(c-a)} \\ &= \boxed{\frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}}. \end{aligned}$$

Both sides are analytic in (a, b, c) on the connected set below: the series by its locally uniform convergence, the right side because $\Gamma(c)$ and $\Gamma(c - a - b)$ are analytic there and $1/\Gamma$ is entire. Analytic continuation in the parameters therefore removes the temporary Euler-integral conditions on b and $c - b$ and proves the stated formula throughout

$$c \notin \{0, -1, -2, \ldots\}, \qquad \Re(c - a - b) > 0.$$

For real parameters this is the question's $c > a + b$. A gamma function in a denominator is interpreted through its entire reciprocal, so a zero of that reciprocal is included when the other factors are finite. Terminating polynomials can obey a larger parameter range; Exercise 18.5.12 proves their finite identity without imposing this nonterminating convergence condition.

18.5.9 Prove that

$${}_2F_1(a, b; c; x) = (1 - x)^{-a} \, {}_2F_1\left(a, c - b; c; \frac{-x}{1 - x}\right).$$

Hint. Try an integral representation.

Note. This relation is useful in developing a Rodrigues representation of $T_n(x)$ (see Exercise 18.5.10). Take $c \notin \{0, -1, -2, \ldots\}$ and principal branches on the cut plane $x \notin [1, \infty)$; further continuation must follow matching paths, since both sides are multivalued around $x = 1$.

Solution.

Assume first $\Re b > 0$, $\Re(c - b) > 0$ and take x close to zero. In Euler's integral for $F(a, b; c; x)$

put $u = 1 - t$. Then $dt = -du$, the limits reverse, and

$$1 - tx = 1 - (1-u)x = (1-x)\left(1 + \frac{xu}{1-x}\right).$$

Consequently

$$\begin{aligned} F(a,b;c;x) &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 (1-u)^{b-1} u^{c-b-1} [1-x+xu]^{-a} du \\ &= (1-x)^{-a} \frac{\Gamma(c)}{\Gamma(c-b)\Gamma(b)} \int_0^1 u^{c-b-1}(1-u)^{b-1} \left(1 - u\frac{-x}{1-x}\right)^{-a} du \\ &= \boxed{(1-x)^{-a} F\left(a, c-b; c; \frac{-x}{1-x}\right)}. \end{aligned}$$

The last integral is Euler's integral (Exercise 18.5.7) for $F(a,b';c;X)$ with $b' = c - b$ and $X = -x/(1-x)$: its weight is $u^{b'-1}(1-u)^{c-b'-1}$, its conditions $\Re b' > 0$, $\Re(c-b') > 0$ are the ones assumed, and the prefactor $\Gamma(c)/[\Gamma(b')\Gamma(c-b')]$ is unchanged when b and $c-b$ exchange places. For x near zero, $|X| < 1$. This proves the identity in a neighborhood of $x = 0$. Both sides are analytic on the principal cut plane, with $(1-x)^{-a} = \exp[-a\,\mathrm{Log}(1-x)]$: if $x \notin [1,\infty)$ then $1 - x \notin (-\infty, 0]$, hence $1/(1-x) \notin (-\infty, 0]$ and $X = 1 - 1/(1-x) \notin [1,\infty)$. The identity theorem extends the identity to this whole cut plane. It also holds on matching analytic continuations; independent branch choices need not agree. Parameter continuation removes the temporary beta-integral restrictions whenever the functions are defined. The singular point $x = 1$ requires a separate limiting analysis.

18.5.10 Derive the Rodrigues representation of $T_n(x)$,

$$T_n(x) = \frac{(-1)^n \pi^{1/2} (1-x^2)^{1/2}}{2^n\, \Gamma\left(n + \frac{1}{2}\right)} \frac{d^n}{dx^n}\left[(1-x^2)^{n-1/2}\right].$$

Hint. One possibility is to use the hypergeometric function relation

$${}_2F_1(a,b;c;z) = (1-z)^{-a}\, {}_2F_1\left(a, c-b; c; \frac{-z}{1-z}\right),$$

with $z = (1-x)/2$. An alternate approach is to develop a first-order differential equation for $y = (1-x^2)^{n-1/2}$. Repeated differentiation of this equation leads to the Chebyshev equation. Derive the identity on $-1 < x < 1$; at $x = \pm 1$, where the nth derivative is singular, use the polynomial's continuous limits.

Solution.

Take $n = 0, 1, 2, \ldots$ and first $-1 < x < 1$. Set $g = (1-x^2)^{n-1/2}$; direct differentiation, $g' = -(2n-1)x(1-x^2)^{n-3/2}$, gives

$$(1-x^2)g' + (2n-1)xg = 0.$$

Differentiate $n+1$ times. The first factor has only two nonzero derivatives and x only one, so Leibniz's formula gives

$$\begin{aligned} 0 &= (1-x^2)g^{(n+2)} - 2(n+1)xg^{(n+1)} - n(n+1)g^{(n)} \\ &\quad + (2n-1)xg^{(n+1)} + (2n-1)(n+1)g^{(n)} \\ &= (1-x^2)g^{(n+2)} - 3xg^{(n+1)} + (n^2-1)g^{(n)}. \end{aligned}$$

Thus $u = g^{(n)}$ satisfies $(1-x^2)u'' - 3xu' + (n^2-1)u = 0$. With $h = \sqrt{1-x^2}$ we have $h' = -x/h$ and $h'' = -1/h - x^2/h^3 = -1/h^3$ (as $h^2 + x^2 = 1$), so $R = hu$ has $R' = hu' - xu/h$ and $R'' = hu'' - 2xu'/h - u/h^3$. Hence

$$\begin{aligned}(1-x^2)R'' - xR' + n^2R &= h^3u'' - 3xhu' + \Big(n^2h - \frac{1-x^2}{h}\Big)u \\ &= h[(1-x^2)u'' - 3xu' + (n^2-1)u] = 0.\end{aligned}$$

Moreover R is a polynomial. Indeed induction gives

$$g^{(j)} = (1-x^2)^{n-1/2-j}p_j(x), \quad p_0 = 1,$$

$$p_{j+1} = (1-x^2)p_j' - 2(n-1/2-j)xp_j, \qquad \deg p_j \le j,$$

since differentiating $(1-x^2)^\alpha p_j$ gives $(1-x^2)^{\alpha-1}[(1-x^2)p_j' - 2\alpha x p_j]$. At $j = n$, multiplication by h leaves $R = p_n$. The polynomial solution of this degree-n Chebyshev equation is a scalar multiple of T_n: substitution of $\sum_k a_k x^k$ gives $(k+2)(k+1)a_{k+2} + (n^2-k^2)a_k = 0$. If $a_k \neq 0$ for some k of parity opposite to n, then $k^2 \neq n^2$ makes $a_{k+2}, a_{k+4}, \dots$ all nonzero, impossible for a polynomial; for the parity of n, $a_k = (k+2)(k+1)a_{k+2}/(k^2-n^2)$ $(k < n)$ fixes every coefficient from a_n. As T_n is such a polynomial solution with $T_n(1) = 1$, $R = R(1)\,T_n$.

It remains to determine the scalar, rather than assume its sign. Write $g = (1-x)^{n-1/2}(1+x)^{n-1/2}$ and use Leibniz's formula

$$g^{(n)} = \sum_{j=0}^{n} \binom{n}{j} \frac{d^j(1-x)^{n-1/2}}{dx^j}\,\frac{d^{n-j}(1+x)^{n-1/2}}{dx^{n-j}}$$

as $x \to 1^-$. The jth term contains $(1-x)^{n-1/2-j}$; after multiplication by $h = (1-x)^{1/2}(1+x)^{1/2}$, every term with $j < n$ vanishes as $(1-x)^{n-j}$. The term $j = n$ is $(-1)^n(n-\frac12)(n-\frac32)\cdots\frac12\,(1-x)^{-1/2}(1+x)^{n-1/2}$, where $(n-\frac12)(n-\frac32)\cdots\frac12 = \Gamma(n+\frac12)/\Gamma(\frac12)$; multiplied by h, its last two factors become $(1+x)^{n-1/2}(1+x)^{1/2} \to 2^{n-1/2}\sqrt{2}$. Hence

$$\begin{aligned}R(1) &= (-1)^n\frac{\Gamma(n+1/2)}{\Gamma(1/2)}2^{n-1/2}\sqrt{2} \\ &= (-1)^n 2^n \frac{\Gamma(n+1/2)}{\sqrt{\pi}}.\end{aligned}$$

Since $R = R(1)\,T_n$, division by this value proves

$$\boxed{T_n(x) = \frac{(-1)^n\sqrt{\pi}}{2^n\Gamma(n+1/2)}\sqrt{1-x^2}\,\frac{d^n}{dx^n}(1-x^2)^{n-1/2}}.$$

The published notation correction is the gamma function $\Gamma(n+1/2)$ in the denominator. At $n = 0$ the two square-root factors cancel and give $T_0 = 1$. Although the derivative representation separately has endpoint singularities, its product is the polynomial just proved, so it extends continuously to $x = \pm 1$ and algebraically to complex x.

18.5.11 Show that the summation in Eq. (18.43),

$$\sum_{\nu=0}^{\min(m,n)} \frac{(-m)_\nu(-n)_\nu}{\nu!\left(\dfrac{1-m-n}{2}\right)_\nu}\left(\frac{a^2}{2(a^2-1)}\right)^\nu,$$

can be written as a hypergeometric function. Take integers $m, n \geq 0$, as in Eq. (18.43), and $a^2 \neq 1$, since the argument is undefined at $a^2 = 1$. Interpret a singular lower hypergeometric parameter by the terminating finite series, not an undefined infinite-series quotient.

Solution.

Let m, n be nonnegative integers and set

$$c = \frac{1-m-n}{2}, \qquad w = \frac{a^2}{2(a^2-1)}.$$

Compare each coefficient with the defining hypergeometric series:

$$F(A, B; C; w) = \sum_{j\geq 0} \frac{(A)_j(B)_j}{(C)_j j!} w^j.$$

The numerator parameters are $A = -m$, $B = -n$, and the denominator parameter is $C = c$. The Pochhammer symbols $(-m)_j = (-m)(-m+1)\cdots(-m+j-1)$ and $(-n)_j$ vanish for $j > m$ and $j > n$, respectively, so the series terminates at $j = \min(m, n)$ and the finite sum is

$$\boxed{F\left(-m, -n; \frac{1-m-n}{2}; \frac{a^2}{2(a^2-1)}\right)}.$$

Here one must specify what "terminates" means when c is itself a nonpositive integer. If $m + n = 2\ell + 1$, then $c = -\ell$ and $(c)_j = (-\ell)(-\ell+1)\cdots(-\ell+j-1)$ first vanishes at $j = \ell + 1$. But $2\min(m, n) \leq m + n = 2\ell + 1$ gives $\min(m, n) \leq \ell$, so every term actually retained has a nonzero denominator. If $m + n$ is even, c is a half-integer and no denominator factor vanishes. Thus the original finite sum is well defined in every degree case. At the singular integer denominator parameters the boxed notation therefore means this terminating polynomial (the convention of the red note in the statement), not a generic infinite series with later $0/0$ coefficients. For $a^2 \neq 1$ the argument is finite. At $a = \pm 1$ it is undefined as written; if one index is zero the polynomial is identically one and has that removable value. With both indices positive its nonzero highest coefficient gives divergence as $w \to \infty$. Any cancellation with other factors in the full Eq. (18.43) must be taken in that full expression, not in this isolated sum.

18.5.12 Verify that

$${}_2F_1(-n, b; c; 1) = \frac{(c-b)_n}{(c)_n},$$

where n is a nonnegative integer, so that the series terminates, and $(c)_n \neq 0$, so that the right side is defined. Use the terminating series if the lower parameter would make a later, unused coefficient singular.

Hint. Here is a chance to use the contiguous function relation Eq. (18.127) and mathematical induction (Section 1.4). Alternatively, use the integral representation and the beta function.

Solution.

For the finite identity take $n = 0, 1, 2, \ldots$ and initially $\Re b > 0$, $\Re(c-b) > 0$. In Euler's integral (Exercise 18.5.7) with first parameter $-n$ the factor is $(1-tz)^n$, so both sides are polynomials in z; the representation, proved for $|z| < 1$, therefore also holds at $z = 1$. With the beta integral $B(b, c-b+n)$ it gives

$$\begin{aligned} F(-n, b; c; 1) &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-t)^n dt \\ &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \frac{\Gamma(b)\Gamma(c-b+n)}{\Gamma(c+n)} \\ &= \frac{\Gamma(c-b+n)}{\Gamma(c-b)} \frac{\Gamma(c)}{\Gamma(c+n)} = \boxed{\frac{(c-b)_n}{(c)_n}}. \end{aligned}$$

Since $(-n)_j/j! = (-1)^j n!/[(n-j)!\, j!]$, the left side is the finite sum

$$\sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{(b)_j}{(c)_j};$$

hence both sides are rational functions of b, c. Multiply by $(c)_n$: for every $j \le n$, the quotient $(c)_n/(c)_j = (c+j)_{n-j}$ is a polynomial. The resulting polynomial identity, proved on an open parameter set, holds identically. Division back gives the formula whenever $(c)_n \neq 0$, without requiring the beta-integral restrictions. This includes singular denominator parameters whose first zero occurs strictly after termination, with the terminating convention of Exercise 18.5.11. At $n = 0$ both empty products and the single-term sum equal one.

The terminating version uses a nonnegative integer n; the published erratum's word "integer" should be understood in that polynomial setting. Negative integer n gives a nonterminating series, for which the convergence conditions of Gauss's formula in Exercise 18.5.8 must instead be imposed. At zeros of $(c)_n$ the literal formula is not defined unless a particular removable limit has been specified.

18.6 Confluent Hypergeometric Functions

18.6.1 Verify the confluent hypergeometric representation of the error function

$$\operatorname{erf}(x) = \frac{2x}{\pi^{1/2}} M\left(\frac{1}{2}, \frac{3}{2}, -x^2\right).$$

Solution.

By definition $M(a, c, z) = \sum_{j \ge 0} (a)_j z^j/[(c)_j j!]$. The finite products $(1/2)_j = \frac{1}{2} \cdot \frac{3}{2} \cdots (j - \frac{1}{2})$ and $(3/2)_j = \frac{3}{2} \cdots (j + \frac{1}{2})$ share all factors except the first and the last, so

$$\frac{(1/2)_j}{(3/2)_j} = \frac{1/2}{j + 1/2} = \frac{1}{2j+1},$$

including $j = 0$. Hence

$$\frac{2x}{\sqrt{\pi}} M(\tfrac{1}{2}, \tfrac{3}{2}, -x^2) = \frac{2}{\sqrt{\pi}} \sum_{j=0}^{\infty} \frac{(-1)^j x^{2j+1}}{(2j+1)j!}.$$

On every bounded complex path, the exponential series $e^{-u^2} = \sum_j (-1)^j u^{2j}/j!$ is uniformly absolutely convergent, so it may be integrated term by term:

$$\operatorname{erf} x = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du = \frac{2}{\sqrt{\pi}} \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \int_0^x u^{2j} du.$$

Evaluating the last monomial integral, $\int_0^x u^{2j} du = x^{2j+1}/(2j+1)$, gives the preceding series, proving the identity for every complex x, including its zero value at the origin. The entire integrand makes the integration path irrelevant.

18.6.2 Show that the Fresnel integrals $C(x)$ and $s(x)$ of Exercise 12.6.1 may be expressed in terms of the confluent hypergeometric function as

$$C(x) + i\, s(x) = xM\left(\frac{1}{2}, \frac{3}{2}, \frac{i\pi x^2}{2}\right).$$

Solution.

The definitions give

$$C(x) + is(x) = \int_0^x [\cos(\pi u^2/2) + i \sin(\pi u^2/2)] du = \int_0^x e^{i\pi u^2/2} du.$$

Expand the exponential and integrate term by term on the finite path (uniform absolute convergence justifies this):

$$\begin{aligned} C(x) + is(x) &= \sum_{j=0}^{\infty} \frac{(i\pi/2)^j}{j!} \frac{x^{2j+1}}{2j+1} \\ &= x \sum_{j=0}^{\infty} \frac{(1/2)_j}{(3/2)_j j!} \left(\frac{i\pi x^2}{2}\right)^j \\ &= \boxed{xM\left(\tfrac{1}{2}, \tfrac{3}{2}, \frac{i\pi x^2}{2}\right)}. \end{aligned}$$

The factor $1/(2j+1)$ was converted by the exact product identity in Exercise 18.6.1. Both sides are entire, so the proof includes negative real and complex x, with the same analytic definitions of the Fresnel integrals.

18.6.3 By direct differentiation and substitution verify that

$$y = ax^{-a} \int_0^x e^{-t} t^{a-1}\, dt = ax^{-a}\gamma(a, x)$$

satisfies

$$xy'' + (a + 1 + x)y' + ay = 0.$$

For the defining integral, take $x > 0$ and $\Re a > 0$; for $\Re a \le 0$ the integral diverges at $t = 0$.

Solution.

Take $x > 0$ and $\Re a > 0$, the domain in which the defining lower-endpoint integral converges. Write $G(x) = \gamma(a, x)$, so $G' = e^{-x}x^{a-1}$, and let $y = ax^{-a}G$. The product rule gives

$$y' = -a^2 x^{-a-1} G + ax^{-a} e^{-x} x^{a-1} = -\frac{a}{x} y + \frac{a}{x} e^{-x}.$$

Differentiate this equality once more and substitute for y':

$$\begin{aligned} y'' &= \frac{a}{x^2}y - \frac{a}{x}y' - \frac{a}{x^2}e^{-x} - \frac{a}{x}e^{-x} \\ &= \frac{a(a+1)}{x^2}y - \frac{a(a+1)}{x^2}e^{-x} - \frac{a}{x}e^{-x}. \end{aligned}$$

Thus the four contributions are

$$\begin{aligned} xy'' &= \frac{a(a+1)}{x}y - \frac{a(a+1)}{x}e^{-x} - ae^{-x}, \\ (a+1)y' &= -\frac{a(a+1)}{x}y + \frac{a(a+1)}{x}e^{-x}, \\ xy' &= -ay + ae^{-x}, \qquad ay = ay. \end{aligned}$$

Every term cancels upon addition, proving

$$\boxed{xy'' + (a+1+x)y' + ay = 0}.$$

Integrating $e^{-t} = \sum_j (-t)^j/j!$ termwise gives $\gamma(a,x) = \sum_j (-1)^j x^{a+j}/[j!\,(a+j)]$, so, with $a/(a+j) = (a)_j/(a+1)_j$ as in Exercise 18.5.6,

$$y = \sum_{j=0}^{\infty} \frac{a}{a+j}\frac{(-x)^j}{j!} = \sum_{j=0}^{\infty} \frac{(a)_j}{(a+1)_j}\frac{(-x)^j}{j!} = M(a, a+1, -x),$$

and $y(0) = 1$. This supplies analytic continuation with matching branches. For $\Re a \le 0$ the displayed integral is not an ordinary convergent integral (the red note in the statement); any extension uses that analytic continuation. For example its removable continuation at $a = 0$ is 1, not a direct evaluation of 0 times a divergent integral.

18.6.4 Show that the modified Bessel function of the second kind, $K_\nu(x)$, is given by

$$K_\nu(x) = \pi^{1/2} e^{-x} (2x)^\nu \, U\big(\nu + \tfrac{1}{2}, 2\nu + 1, 2x\big).$$

Take $x > 0$; first derive the identity for $\Re\nu > -1/2$, where the integral representations used converge, then continue in ν.

Solution.

Start with the integral representation of K_ν, Eq. (14.128) of Section 14.6, proved in Exercise 14.5.10 for real $\nu > -1/2$; since both sides are analytic in ν, it holds for $x > 0$ and $\Re\nu > -1/2$:

$$K_\nu(x) = \frac{\sqrt{\pi}(x/2)^\nu}{\Gamma(\nu + 1/2)} \int_1^\infty e^{-xs}(s^2-1)^{\nu-1/2} ds.$$

Set $s = 1 + 2t$, so $ds = 2dt$, the limits become 0 and ∞, $s^2 - 1 = (s-1)(s+1) = 4t(1+t)$ and $e^{-xs} = e^{-x}e^{-2xt}$. Then

$$\begin{aligned} K_\nu(x) &= \frac{\sqrt{\pi}(x/2)^\nu e^{-x}\, 2\, 4^{\nu-1/2}}{\Gamma(\nu+1/2)} \int_0^\infty e^{-2xt} t^{\nu-1/2}(1+t)^{\nu-1/2} dt \\ &= \sqrt{\pi} e^{-x} (2x)^\nu \frac{1}{\Gamma(\nu+1/2)} \int_0^\infty e^{-2xt} t^{\nu-1/2}(1+t)^{\nu-1/2} dt, \end{aligned}$$

since $2 \cdot 4^{\nu-1/2} = 2^{2\nu}$ and $(x/2)^\nu 2^{2\nu} = (2x)^\nu$. In the integral representation $U(a,c,z) = \Gamma(a)^{-1}\int_0^\infty e^{-zt}t^{a-1}(1+t)^{c-a-1}dt$ of Exercise 18.6.10(b), choose

$$a = \nu + \tfrac{1}{2}, \quad c = 2\nu + 1, \quad z = 2x,$$

so that $a - 1 = c - a - 1 = \nu - 1/2$ and $\Gamma(a) = \Gamma(\nu + \frac{1}{2})$; the last integral divided by $\Gamma(\nu + \frac{1}{2})$ is $U(\nu + \frac{1}{2}, 2\nu + 1, 2x)$. Therefore

$$\boxed{K_\nu(x) = \sqrt{\pi}e^{-x}(2x)^\nu U(\nu + \tfrac{1}{2}, 2\nu + 1, 2x)}.$$

The proof also works for $\Re x > 0$ with consistent powers; continuation in ν gives the remaining orders, and continuation in x uses the conventional cut and the same logarithm in $(2x)^\nu$. The integral restrictions belong to this derivation, not to an extra restriction on the continued identity.

The Instructor's Manual (with z for x) reaches the same integral from the form of Eq. (14.128) used for the asymptotic expansion in Section 14.6, whose variable t is $2z$ times ours,

$$K_\nu(z) = \sqrt{\frac{\pi}{2z}}\,\frac{e^{-z}}{\Gamma(\nu + \frac{1}{2})}\int_0^\infty e^{-t}t^{\nu-1/2}\Big(1 + \frac{t}{2z}\Big)^{\nu-1/2}dt,$$

by putting $t = 2zy$: then $dt = 2z\,dy$, $t^{\nu-1/2} = (2z)^{\nu-1/2}y^{\nu-1/2}$ and $1 + t/(2z) = 1 + y$, so the powers $(2z)^{-1/2}(2z)^{\nu-1/2}(2z)$ combine to $(2z)^\nu$ and y is our t. The exponential inside the integral is then e^{-2zy}, not e^{z-2zy} as the Instructor's Manual has it, since the factor e^{-z} already stands outside.

18.6.5 Show that the cosine and sine integrals of Section 13.6 may be expressed in terms of confluent hypergeometric functions as

$$\mathrm{Ci}(x) + i\,\mathrm{si}(x) = -e^{ix}U(1,1,-ix).$$

This relation is useful in numerical computation of $\mathrm{Ci}(x)$ and $\mathrm{si}(x)$ for large values of x. Take $x > 0$, since for $x \le 0$ the integral defining $\mathrm{Ci}(x)$ meets the singularity at $t = 0$; here $\mathrm{si}(x) = -\int_x^\infty (\sin t)/t\,dt$.

Solution.

For real $x > 0$, the convention in Section 13.6 is

$$\mathrm{Ci}\,x = -\int_x^\infty \frac{\cos t}{t}dt, \qquad \mathrm{si}\,x = -\int_x^\infty \frac{\sin t}{t}dt = \mathrm{Si}\,x - \frac{\pi}{2}.$$

These oscillatory improper integrals converge by integration by parts. For $\varepsilon > 0$, the Laplace representation of Exercise 18.6.10(b) with $a = c = 1$ (weight $u^{a-1}(1+u)^{c-a-1} = (1+u)^{-1}$ and $\Gamma(a) = 1$), valid because $\Re(\varepsilon - ix) = \varepsilon > 0$, gives

$$U(1,1,\varepsilon - ix) = \int_0^\infty \frac{e^{-(\varepsilon - ix)u}}{1+u}du.$$

Multiplying by $e^{-\varepsilon + ix}$ and setting $v = 1 + u$ gives

$$e^{-\varepsilon+ix}U(1,1,\varepsilon - ix) = \int_1^\infty \frac{e^{-(\varepsilon - ix)v}}{v}\,dv.$$

Let $\varepsilon \downarrow 0$. The left side tends to $e^{ix}U(1,1,-ix)$, since $U(1,1,z)$ is continuous at $z = -ix$, which is off its cut. On the right, with $w = \varepsilon - ix \neq 0$, integration by parts gives

$$\int_1^\infty \frac{e^{-wv}}{v}\,dv = \frac{e^{-w}}{w} - \frac{1}{w}\int_1^\infty \frac{e^{-wv}}{v^2}\,dv,$$

and this holds also at $\varepsilon = 0$ (as an improper integral), because the integrated term $e^{-wv}/(wv)$ still vanishes as $v \to \infty$. The last integrand is bounded by v^{-2} for all $\varepsilon \geq 0$, so by dominated convergence the right side tends to its value at $\varepsilon = 0$, the ordinary oscillatory integral. With $t = xv$ it becomes

$$e^{ix}U(1,1,-ix) = \int_1^\infty \frac{e^{ixv}}{v}\,dv = \int_x^\infty \frac{e^{it}}{t}\,dt.$$

By the definitions above, the last integral is

$$\int_x^\infty \frac{\cos t}{t}\,dt + i\int_x^\infty \frac{\sin t}{t}\,dt = -\operatorname{Ci} x - i \operatorname{si} x;$$

taking its negative gives

$$\boxed{\operatorname{Ci} x + i \operatorname{si} x = -e^{ix}U(1,1,-ix)}.$$

The value of U is continued from the right half-plane, fixing the branch at $-ix$. Equivalently, since $E_1(z) = e^{-z}U(1,1,z)$ on the principal cut plane (Exercise 18.6.12), the choice $z = -ix$, for which $e^{-z} = e^{ix}$, shows that the answer is $-E_1(-ix)$. The lowercase si is essential: replacing it by Si would require adding $i\pi/2$.

18.6.6 Verify the confluent hypergeometric form of the Hermite polynomial $H_{2n+1}(x)$, Eq. (18.149), by showing that

(a) $H_{2n+1}(x)/x$ satisfies the confluent hypergeometric equation with $a = -n$, $c = 3/2$ and argument x^2,

(b) $\displaystyle\lim_{x\to 0} \frac{H_{2n+1}(x)}{x} = (-1)^n \frac{2(2n+1)!}{n!}$.

Solution.

(a) Let $t = x^2$ and $H_{2n+1}(x) = xf(t)$. Then, as in Exercise 18.5.3,

$$H' = f + 2tf', \qquad H'' = x(6f' + 4tf'').$$

Insert these into $H'' - 2xH' + 2(2n+1)H = 0$:

$$x[6f' + 4tf'' - 2f - 4tf' + (4n+2)f] = 0.$$

For $x \neq 0$, division by $4x$ gives

$$\boxed{tf'' + (\tfrac{3}{2} - t)f' + nf = 0},$$

which is Kummer's equation with $a = -n, c = 3/2$. Since the odd polynomial is divisible by x, f is an actual polynomial in t and the equation extends to $x = 0$ as well.

(b) In the finite Hermite series

$$H_{2n+1}(x) = (2n+1)!\sum_{j=0}^{n} \frac{(-1)^j (2x)^{2n+1-2j}}{j!(2n+1-2j)!},$$

the coefficient of x comes only from $j = n$. It is $(-1)^n 2(2n+1)!/n!$. Therefore

$$\lim_{x\to 0} \frac{H_{2n+1}(x)}{x} = \boxed{(-1)^n \frac{2(2n+1)!}{n!}}.$$

Because $M(-n, 3/2, 0) = 1$ and the analytic solution is uniquely fixed by its constant term (Kummer's recurrence $(j+1)(j+c)d_{j+1} = (j+a)d_j$ has $j + c \neq 0$ for $c = \frac{3}{2}$), parts (a) and (b) together prove

$$H_{2n+1}(x) = (-1)^n \frac{2(2n+1)!}{n!} x M(-n, \tfrac{3}{2}, x^2).$$

For $n = 0$ this reduces to $H_1 = 2x$ without a limiting exception.

18.6.7 Show that the contiguous confluent hypergeometric function equation

$$(c-a)M(a-1,c,x)+(2a-c+x)M(a,c,x)-aM(a+1,c,x)=0$$

leads to the associated Laguerre function recurrence relation, Eq. (18.66).

Solution.

Use $a=-n$, $c=k+1$ and

$$M(-n,k+1,x)=A_nL_n^k(x),\qquad A_n=\frac{n!\,k!}{(n+k)!}.$$

This follows on comparing the two finite series

$$M(-n,k+1,x)=\sum_{j=0}^{n}\frac{(-n)_j}{(k+1)_j\,j!}x^j,\qquad L_n^k(x)=\sum_{j=0}^{n}\frac{(-1)^j(n+k)!}{(n-j)!\,(k+j)!\,j!}x^j,$$

with $(-n)_j=(-1)^jn!/(n-j)!$ and $(k+1)_j=(k+j)!/k!$. For $n\ge1$, with $c-a=n+k+1$, $2a-c+x=x-(2n+k+1)$, $-a=n$ and $a\mp1=-(n\pm1)$, the contiguous equation becomes

$$(n+k+1)A_{n+1}L_{n+1}^k+[x-(2n+k+1)]A_nL_n^k+nA_{n-1}L_{n-1}^k=0.$$

The two ratios are

$$\frac{A_{n+1}}{A_n}=\frac{(n+1)!\,(n+k)!}{n!\,(n+k+1)!}=\frac{n+1}{n+k+1},\qquad \frac{A_{n-1}}{A_n}=\frac{n+k}{n}.$$

Divide by A_n; the first and last coefficients become $(n+k+1)(n+1)/(n+k+1)=n+1$ and $n(n+k)/n=n+k$:

$$\boxed{(n+1)L_{n+1}^k-(2n+k+1-x)L_n^k+(n+k)L_{n-1}^k=0}.$$

This is Eq. (18.66). At $n=0$ the original coefficient $-a$ is zero, so its last term is absent. The result is $L_1^k=(k+1-x)L_0^k$, consistent with $L_0^k=1$ and the convention $L_{-1}^k=0$. No division by $n=0$ is used. Gamma functions extend this proof from integer $k\ge0$ to nonsingular parameter values, in particular real $k>-1$.

18.6.8 Verify the Kummer transformations:

(a) $M(a,c,x)=e^xM(c-a,c,-x)$,

(b) $U(a,c,x)=x^{1-c}U(a-c+1,2-c,x)$.

For M, take $c\notin\{0,-1,-2,\ldots\}$, where its series is defined; for U and powers of x, which are multivalued, use matching branches, initially $|\arg x|<\pi$.

Solution.

(a) Put $g(x)=M(c-a,c,-x)$. The function $w(z)=M(c-a,c,z)$ satisfies Kummer's equation $zw''+(c-z)w'-(c-a)w=0$; since $g(x)=w(-x)$ has $g'=-w'(-x)$ and $g''=w''(-x)$, setting $z=-x$ and multiplying by -1 gives

$$xg''+(c+x)g'+(c-a)g=0.$$

For $y = e^x g$, $y' = e^x(g + g')$ and $y'' = e^x(g + 2g' + g'')$. Hence

$$\begin{aligned} xy'' + (c - x)y' - ay &= e^x\{x(g + 2g' + g'') + (c - x)(g + g') - ag\} \\ &= e^x\{xg'' + (c + x)g' + (c - a)g\} = 0. \end{aligned}$$

Both y and $M(a, c, x)$ are analytic at zero with value one. For $c \notin \{0, -1, -2, \ldots\}$, their Taylor recurrence $(j + 1)(j + c)d_{j+1} = (j + a)d_j$ uniquely fixes every coefficient. Therefore

$$\boxed{M(a, c, x) = e^x M(c - a, c, -x)}.$$

Exercise 18.6.11 supplies a separate integral proof.

(b) Use the corrected connection formula, initially $c \notin \mathbb{Z}$:

$$U(a, c, x) = \frac{\Gamma(1 - c)}{\Gamma(a - c + 1)} M(a, c, x) + \frac{\Gamma(c - 1)}{\Gamma(a)} x^{1-c} M(a - c + 1, 2 - c, x).$$

It follows from Eq. (18.142) by the reflection identity, with its published replacement $\Gamma(-c) \to \Gamma(2-c)$; so corrected, Eq. (18.142) reads

$$U(a, c, x) = \frac{\pi}{\sin \pi c} \left[\frac{M(a, c, x)}{\Gamma(a - c + 1)\Gamma(c)} - \frac{x^{1-c} M(a - c + 1, 2 - c, x)}{\Gamma(a)\Gamma(2 - c)} \right].$$

Explicitly, $\Gamma(z)\Gamma(1 - z) = \pi / \sin \pi z$ with $z = c$ and with $z = c - 1$ (where $\sin \pi(c - 1) = -\sin \pi c$) gives $\pi/[\sin \pi c\, \Gamma(c)] = \Gamma(1-c)$ and $-\pi/[\sin \pi c\, \Gamma(2-c)] = \Gamma(c-1)$; substituting these into the bracket gives the connection formula above. Now replace (a, c) by $(a - c + 1, 2 - c)$; since $1 - (2 - c) = c - 1$, $(a - c + 1) - (2 - c) + 1 = a$ and $2 - (2 - c) = c$,

$$\begin{aligned} U(a - c + 1, 2 - c, x) = {} & \frac{\Gamma(c - 1)}{\Gamma(a)} M(a - c + 1, 2 - c, x) \\ & + \frac{\Gamma(1 - c)}{\Gamma(a - c + 1)} x^{c-1} M(a, c, x). \end{aligned}$$

Multiplication by x^{1-c} simply exchanges the two original terms. Thus

$$\boxed{U(a, c, x) = x^{1-c} U(a - c + 1, 2 - c, x)}.$$

For integer c, take the limit of the entire combination; the separate gamma terms may have poles which cancel. Parameter continuation includes other removable cases. The powers on both sides use the same logarithm, with the usual principal branch initially $|\arg x| < \pi$.

18.6.9 Prove that

(a) $\dfrac{d^n}{dx^n} M(a, c, x) = \dfrac{(a)_n}{(c)_n} M(a + n, c + n, x)$,

(b) $\dfrac{d^n}{dx^n} U(a, c, x) = (-1)^n (a)_n U(a + n, c + n, x)$.

Take $n \geq 0$; for M, exclude $c = 0, -1, -2, \ldots$, where its series is undefined. For U, differentiate on a fixed branch away from its branch point $x = 0$.

Solution.

Let n be a nonnegative integer. (a) Differentiate the entire defining series n times, using $d^n(x^j/j!)/dx^n = x^{j-n}/(j-n)!$ for $j \geq n$ and 0 for $j < n$, and then put $j = n + r$:

$$\begin{aligned}\frac{d^n}{dx^n} M(a,c,x) &= \sum_{j=n}^{\infty} \frac{(a)_j}{(c)_j} \frac{x^{j-n}}{(j-n)!} \\ &= \sum_{r=0}^{\infty} \frac{(a)_{n+r}}{(c)_{n+r}} \frac{x^r}{r!} \\ &= \frac{(a)_n}{(c)_n} \sum_{r=0}^{\infty} \frac{(a+n)_r}{(c+n)_r} \frac{x^r}{r!}.\end{aligned}$$

The last step uses $(a)_{n+r} = (a)_n (a+n)_r$ (both sides are the product of $a, a+1, \ldots, a+n+r-1$) and its c analogue. Therefore

$$\boxed{\frac{d^n}{dx^n} M(a,c,x) = \frac{(a)_n}{(c)_n} M(a+n, c+n, x)}.$$

The two denominator-parameter labels are c, not b, as in the published erratum. Initially exclude nonpositive integer c; terminating cases with n beyond the polynomial degree have zero derivative, exactly as $(a)_n = 0$ predicts.

(b) For $\Re a > 0$, $\Re x > 0$, differentiate the Laplace integral of Exercise 18.6.10(b); each derivative brings down a factor $-t$, and exponential decay justifies every finite number of derivatives under the integral:

$$\begin{aligned}\frac{d^n}{dx^n} U(a,c,x) &= \frac{(-1)^n}{\Gamma(a)} \int_0^{\infty} e^{-xt} t^{a+n-1} (1+t)^{c-a-1} dt \\ &= (-1)^n \frac{\Gamma(a+n)}{\Gamma(a)} U(a+n, c+n, x) \\ &= \boxed{(-1)^n (a)_n U(a+n, c+n, x)}.\end{aligned}$$

The exponent $c-a-1$ stays unchanged because both parameters increase by n: $(c+n)-(a+n)-1 = c-a-1$, so the integral is $\Gamma(a+n)\, U(a+n, c+n, x)$. Analytic continuation supplies the usual cut plane and other parameter values. For $a = -N$, the resulting polynomial has zero derivatives for $n > N$, consistently with the vanishing Pochhammer factor.

18.6.10 Verify the following integral representations:

(a) $M(a,c,x) = \dfrac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)} \displaystyle\int_0^1 e^{xt} t^{a-1} (1-t)^{c-a-1}\, dt, \quad c > a > 0,$

(b) $U(a,c,x) = \dfrac{1}{\Gamma(a)} \displaystyle\int_0^{\infty} e^{-xt} t^{a-1} (1+t)^{c-a-1}\, dt, \quad \Re x > 0,\ a > 0.$

Under what conditions can you accept $\Re x = 0$ in part (b)?

Solution.

(a) For complex parameters the endpoint conditions are $\Re a > 0$, $\Re(c-a) > 0$. Expand e^{xt} and integrate its uniformly convergent series against the beta weight:

$$\begin{aligned}\frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)}\int_0^1 e^{xt}t^{a-1}(1-t)^{c-a-1}dt &= \frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)}\sum_{j=0}^{\infty}\frac{x^j}{j!}B(a+j,c-a)\\ &= \sum_{j=0}^{\infty}\frac{\Gamma(c)\Gamma(a+j)}{\Gamma(a)\Gamma(c+j)}\frac{x^j}{j!}\\ &= \boxed{M(a,c,x)}.\end{aligned}$$

Here $B(a+j,c-a) = \Gamma(a+j)\Gamma(c-a)/\Gamma(c+j)$, and $\Gamma(a+j)/\Gamma(a) = (a)_j$, $\Gamma(c+j)/\Gamma(c) = (c)_j$. This holds for every finite complex x; for real parameters its restrictions are precisely $c > a > 0$.

(b) Denote the proposed integral by $I(a,c,x)$. It converges absolutely for $\Re a > 0$, $\Re x > 0$, with no restriction on c. Differentiation under the integral brings down $-t$ for I' and t^2 for I'', and $xt^2 - (c-x)t - a = xt(t+1) - ct - a$; so

$$\begin{aligned}xI'' + (c-x)I' - aI &= \frac{1}{\Gamma(a)}\int_0^{\infty} e^{-xt}t^{a-1}(1+t)^{c-a-1}[xt(t+1) - ct - a]dt\\ &= -\frac{1}{\Gamma(a)}\left[e^{-xt}t^a(1+t)^{c-a}\right]_0^{\infty} = 0.\end{aligned}$$

The integrand is an exact derivative: by the product rule,

$$\frac{d}{dt}\left[e^{-xt}t^a(1+t)^{c-a}\right] = e^{-xt}t^{a-1}(1+t)^{c-a-1}[-xt(1+t) + a(1+t) + (c-a)t],$$

and $a(1+t) + (c-a)t = a + ct$. The endpoint terms vanish by $\Re a > 0$ at zero and exponential decay at infinity. This verifies the differential equation, but we must also identify the particular solution and its scale.

For noninteger c, its local basis is $M(a,c,x)$ and $x^{1-c}M(a-c+1,2-c,x)$. Let their coefficients in I be A, B. Initially take real $a > 0, c < 1$. As $x \downarrow 0$, $M(a,c,x) \to 1$ and $x^{1-c} \to 0$, so $I \to A$; on the other hand dominated convergence (the integrand is bounded by its value at $x = 0$) gives

$$A = I(a,c,0) = \frac{1}{\Gamma(a)}\int_0^{\infty} t^{a-1}(1+t)^{c-a-1}dt = \frac{\Gamma(1-c)}{\Gamma(a-c+1)}.$$

Here the substitution $v = t/(1+t)$, i.e. $t = v/(1-v)$, $1 + t = 1/(1-v)$, $dt = dv/(1-v)^2$, turns the integrand into $v^{a-1}(1-v)^{-c}$ on $[0,1]$, so the integral is $B(a,1-c) = \Gamma(a)\Gamma(1-c)/\Gamma(a-c+1)$. To compute the coefficient B, take instead $1 < c < a+1$ and put $u = xt$, for which $t^{a-1}(1+t)^{c-a-1}dt = x^{1-c}u^{a-1}(x+u)^{c-a-1}du$:

$$x^{c-1}I = \frac{1}{\Gamma(a)}\int_0^{\infty} e^{-u}u^{a-1}(x+u)^{c-a-1}du \longrightarrow \frac{\Gamma(c-1)}{\Gamma(a)}.$$

The exponent $c-a-1$ is negative in this initial strip, so $(x+u)^{c-a-1} \le u^{c-a-1}$ and the integrand is bounded by $e^{-u}u^{c-2}$, an integrable function since $c > 1$; the limit is $\Gamma(a)^{-1}\int_0^{\infty} e^{-u}u^{c-2}du$. The analytic first solution vanishes after multiplication by x^{c-1}, while $x^{c-1}\cdot x^{1-c}M(a-c+1,2-c,x) \to 1$; hence this limit is B. Write $y_1 = M(a,c,x)$ and $y_2 = x^{1-c}M(a-c+1,2-c,x)$. Abel's formula for $y'' + py' + qy = 0$, here with $p = (c-x)/x = c/x - 1$, gives $W(y_1,y_2) = Ce^{-\int p\,dx} = Cx^{-c}e^x$.

As $x \to 0$, $y_1 = 1 + O(x)$, $y_1' = O(1)$, $y_2 = x^{1-c}[1 + O(x)]$ and $y_2' = (1-c)x^{-c}[1 + O(x)]$, so $y_1 y_2' - y_1' y_2 = (1-c)x^{-c} + O(x^{1-c})$, whence $C = 1 - c$ and $W(y_1, y_2) = (1-c)x^{-c}e^x \neq 0$. Hence the coefficients $A = W(I, y_2)/W(y_1, y_2)$ and $B = W(y_1, I)/W(y_1, y_2)$ are analytic in a, c for $\Re a > 0$, $c \notin \mathbb{Z}$. The values found on the two real parameter sets therefore hold throughout, and

$$I = \frac{\Gamma(1-c)}{\Gamma(a-c+1)} M(a, c, x) + \frac{\Gamma(c-1)}{\Gamma(a)} x^{1-c} M(a-c+1, 2-c, x).$$

This is the corrected defining connection formula for $\boxed{U(a, c, x)}$. Limits include integer c; continuation from positive real x gives the whole right half-plane. As a further scale check, $t = u/x$ in the original integral gives $x^a I \to \Gamma(a)/\Gamma(a) = 1$ as $x \to +\infty$.

The boundary $\Re x = 0$ must distinguish $x = 0$ from a nonzero purely imaginary x. At large t, the amplitude is

$$h(t) = t^{a-1}(1+t)^{c-a-1} = t^{c-2}[1 + O(t^{-1})], \quad h'(t) = O(t^{\Re c - 3}) \text{ in magnitude.}$$

At $x = 0$, the ordinary integral converges exactly when $\Re c < 1$ (still $\Re a > 0$ at zero); it is the beta value above. For $x = iy$, $y \neq 0$, integration by parts on $[T, R]$ gives

$$\int_T^R e^{-iyt} h(t)\, dt = \left[\frac{e^{-iyt} h(t)}{-iy}\right]_T^R + \frac{1}{iy} \int_T^R e^{-iyt} h'(t)\, dt.$$

If $\Re c < 2$, the endpoint at infinity vanishes and the last integral is absolutely convergent. The original integral is absolutely convergent for $\Re c < 1$ and conditionally convergent for $1 \leq \Re c < 2$. For $\Re c \geq 2$, the nonvanishing leading oscillatory boundary term prevents an ordinary improper limit; repeated integration by parts separates it from lower powers. Thus $\Re x = 0$ is acceptable, with $\Re a > 0$, at $x = 0$ when $\Re c < 1$ and at $x = iy \neq 0$ when $\Re c < 2$; there the convergent integral equals the limit of U as $\Re x \downarrow 0$ (Abel's theorem for Laplace integrals). A value assigned by $\Re x \downarrow 0$ outside these ranges is an Abel continuation, not that ordinary integral.

18.6.11 From the integral representation of $M(a, c, x)$, Exercise 18.6.10(a), show that

$$M(a, c, x) = e^x M(c-a, c, -x).$$

Hint. Replace the variable of integration t by $1 - s$ to release a factor e^x from the integral. For the integral derivation, take $\Re a > 0$ and $\Re(c-a) > 0$, where the integral converges at both endpoints; then continue in the nonsingular parameters.

Solution.

For $\Re a > 0$, $\Re(c-a) > 0$, put $s = 1 - t$ in the Euler integral. Reversing the limits cancels the sign from $dt = -ds$:

$$\begin{aligned} M(a, c, x) &= \frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)} \int_0^1 e^{x(1-s)} (1-s)^{a-1} s^{c-a-1}\, ds \\ &= e^x \frac{\Gamma(c)}{\Gamma(c-a)\Gamma(a)} \int_0^1 e^{-xs} s^{(c-a)-1} (1-s)^{c-(c-a)-1}\, ds \\ &= \boxed{e^x M(c-a, c, -x)}. \end{aligned}$$

Both the powers and the two gamma factors have exchanged roles, so no further normalization is needed. The finite-interval integral allows arbitrary complex x. Analytic continuation in a, c then gives the identity wherever M is defined, including its terminating polynomial cases. This is independent of the ODE uniqueness proof in Exercise 18.6.8(a).

18.6.12 From the integral representation of $U(a,c,x)$ in Exercise 18.6.10(b), show that the exponential integral is given by

$$E_1(x) = e^{-x}U(1,1,x).$$

Hint. Replace the variable of integration t in $E_1(x)$ by $x(1+s)$. Initially take $x>0$, so that the integral defining $E_1(x)$ runs along the positive axis; both sides then have matching continuation to the principal cut plane.

Solution.

For real $x>0$ start with $E_1(x)=\int_x^\infty e^{-t}dt/t$. Make exactly the suggested substitution $t=x(1+s)$: then $dt=x\,ds$, $t=x\Rightarrow s=0$, $t\to\infty\Rightarrow s\to\infty$, $e^{-t}=e^{-x}e^{-xs}$ and $dt/t=ds/(1+s)$, so

$$E_1(x)=e^{-x}\int_0^\infty \frac{e^{-xs}}{1+s}ds.$$

The Laplace integral for U with $a=c=1$ has $s^{a-1}=1$, $(1+s)^{c-a-1}=(1+s)^{-1}$ and $\Gamma(a)=1$. Thus

$$\boxed{E_1(x)=e^{-x}U(1,1,x)}.$$

Both sides are analytic on the right half-plane and continue to the principal cut plane $|\arg x|<\pi$ with the same branch. They have a logarithmic singularity at $x=0$; that point is not included as a finite value.

18.6.13 From the integral representations of $M(a,c,x)$ and $U(a,c,x)$ in Exercise 18.6.10, develop asymptotic expansions of (a) $M(a,c,x)$, (b) $U(a,c,x)$.

Hint. You can use the technique that was employed with $K_\nu(z)$ in Section 14.6.

ANS. (a) $\dfrac{\Gamma(c)}{\Gamma(a)}\dfrac{e^x}{x^{c-a}}\left[1+\dfrac{(1-a)(c-a)}{1!\,x}+\dfrac{(1-a)(2-a)(c-a)(c-a+1)}{2!\,x^2}+\cdots\right]$,

(b) $\dfrac{1}{x^a}\left[1+\dfrac{a(1+a-c)}{1!\,(-x)}+\dfrac{a(a+1)(1+a-c)(2+a-c)}{2!\,(-x)^2}+\cdots\right]$.

Use fixed parameters and $x\to+\infty$, since an asymptotic expansion depends on the direction of the limit. Initially assume $\Re c>\Re a>0$ in (a), $\Re a>0$ in (b), where the integrals of Exercise 18.6.10 converge; discuss terminating $a=0,-1,-2,\ldots$ separately, since then $1/\Gamma(a)=0$.

Solution.

We derive Poincaré expansions for $x\to+\infty$, with fixed parameters. A displayed infinite asymptotic series means that each fixed truncation has the stated remainder, not that the whole series converges.

(a) Initially take real $0<a<c$. In the Euler integral set $u=1-t$, so $e^{xt}=e^xe^{-xu}$, obtaining

$$M(a,c,x)=\frac{\Gamma(c)e^x}{\Gamma(a)\Gamma(c-a)}\int_0^1 e^{-xu}u^{c-a-1}(1-u)^{a-1}du.$$

Near $u=0$ the Taylor (binomial) expansion is

$$(1-u)^{a-1}=\sum_{j=0}^{N-1}\frac{(1-a)_j}{j!}u^j+R_N(u).$$

Choose a fixed $0<\delta<1$. On $[0,\delta]$, $|R_N(u)|\le K_\delta u^N$ for a constant K_δ, so the remainder integral is at most $K_\delta\int_0^\infty e^{-xu}u^{c-a+N-1}du=K_\delta\Gamma(c-a+N)x^{-(c-a+N)}$ (put $v=xu$). On $[\delta,1]$ the integrable endpoint weight is multiplied by at most $e^{-\delta x}$, making its contribution exponentially small.

Replacing each Taylor-term integral's upper limit by infinity also changes it by an exponentially small amount. Since

$$\int_0^\infty e^{-xu}u^{c-a+j-1}du = \frac{\Gamma(c-a+j)}{x^{c-a+j}},$$

multiplication by $\Gamma(c)e^x/[\Gamma(a)\Gamma(c-a)]$ and $\Gamma(c-a+j) = \Gamma(c-a)(c-a)_j$ give

$$\boxed{M(a,c,x) = \frac{\Gamma(c)}{\Gamma(a)}e^x x^{a-c}\left[\sum_{j=0}^{N-1}\frac{(1-a)_j(c-a)_j}{j!x^j} + O(x^{-N})\right]}.$$

The first three coefficients are 1, $(1-a)(c-a)$ and $(1-a)(2-a)(c-a)(c-a+1)/2$, as printed. The derivation extends to $\Re a > 0$, $\Re(c-a) > 0$ with complex parameters and the same fixed positive-real limit.

(b) For $\Re a > 0$, expand the Laplace-integral amplitude at zero by the binomial series, whose coefficients are $\binom{c-a-1}{j} = (c-a-1)(c-a-2)\cdots(c-a-j)/j! = (-1)^j(1+a-c)_j/j!$:

$$(1+t)^{c-a-1} = \sum_{j=0}^{N-1}\frac{(-1)^j(1+a-c)_j}{j!}t^j + O(t^N).$$

The same split at a fixed $0 < \delta < 1$ works; the tail grows only as a fixed power and is exponentially suppressed by e^{-xt}. Termwise Laplace integration, with $\int_0^\infty e^{-xt}t^{a+j-1}dt = \Gamma(a+j)x^{-a-j}$ and $\Gamma(a+j)/\Gamma(a) = (a)_j$, therefore gives

$$U(a,c,x) = \frac{1}{\Gamma(a)}\sum_{j=0}^{N-1}\frac{(-1)^j(1+a-c)_j}{j!}\frac{\Gamma(a+j)}{x^{a+j}} + O(x^{-\Re a-N})$$

$$= \boxed{x^{-a}\left[\sum_{j=0}^{N-1}\frac{(a)_j(1+a-c)_j}{j!(-x)^j} + O(x^{-N})\right]}.$$

In particular the first correction is $-a(1+a-c)/x$, fixing its sign.

As the red note in the statement says, these displayed forms need a limit direction and parameter qualifications; a positive-axis exponential expansion for M cannot be applied unchanged at a terminating a. If $a = -m$, $m = 0, 1, \ldots$, use the exact polynomial $M(-m,c,x)$, whose leading term is $(-1)^m x^m/(c)_m$ when $(c)_m \neq 0$, since $(-m)_m = (-1)^m m!$. The apparent exponential prefactor $1/\Gamma(a)$ then vanishes. The U expansion instead terminates at $j = m$, because $(-m)_j = 0$ for $j > m$, and gives its exact polynomial, with leading term x^m. To justify this without applying the integral outside its domain, use the connection formula from Exercise 18.6.8: its second term vanishes because $1/\Gamma(-m) = 0$, while

$$\frac{\Gamma(1-c)}{\Gamma(1-c-m)} = (1-c-m)(2-c-m)\cdots(-c) = (-1)^m(c)_m.$$

Thus $U(-m,c,x) = (-1)^m(c)_m M(-m,c,x)$, first for noninteger c and then by continuation. Reversing the finite powers of this polynomial gives the displayed terminating U expansion: with $(-m)_{m-j} = (-1)^{m-j}m!/j!$ and $(c)_m/(c)_{m-j} = (c+m-j)_j$, the coefficient of x^{m-j} in $(-1)^m(c)_m M(-m,c,x)$ is $(-1)^j\binom{m}{j}(c+m-j)_j$; in the U expansion with $a = -m$ it is $(-m)_j(1-m-c)_j/[j!\,(-1)^j] = \binom{m}{j}(1-m-c)_j$, the same number because $(1-m-c)_j = (-1)^j(c+m-j)_j$. On other complex rays one must choose the relevant asymptotic sector; the estimates above specifically establish the positive-real expansions requested by the given integrals.

18.6.14 Show that the Wronskian of the two confluent hypergeometric functions $M(a,c,x)$ and $U(a,c,x)$ is given by

$$MU' - M'U = -\frac{(c-1)!\,e^x}{(a-1)!\,x^c}.$$

What happens if a is 0 or a negative integer? Take $c \notin \{0,-1,-2,\ldots\}$, where M is defined, and $x \neq 0$ on a fixed branch of the multivalued x^{-c}. Interpret factorials by gamma functions, including reciprocal-gamma zeros.

Solution.

Assume initially $c \notin \{0,-1,-2,\ldots\}$ so that M is defined. Write $W = MU' - M'U$. Since both functions satisfy $y'' = (1-c/x)y' + (a/x)y$, differentiation gives

$$\begin{aligned} W' = MU'' - M''U &= M[(1-\tfrac{c}{x})U' + \tfrac{a}{x}U] - [(1-\tfrac{c}{x})M' + \tfrac{a}{x}M]U \\ &= (1-c/x)(MU' - M'U) = (1-c/x)W. \end{aligned}$$

Integration ($\ln W = x - c\ln x + \text{const}$) yields $W = C(a,c)e^x x^{-c}$ on a fixed branch. For real $0 < a < c$, Exercise 18.6.13 applies to M and U and, through Exercise 18.6.9, to $M' = (a/c)M(a+1,c+1,x)$ and $U' = -aU(a+1,c+1,x)$; since $(a/c)\Gamma(c+1)/\Gamma(a+1) = \Gamma(c)/\Gamma(a)$, it gives

$$M \sim \frac{\Gamma(c)}{\Gamma(a)} e^x x^{a-c}, \quad M' \sim M, \quad U \sim x^{-a}, \quad U' \sim -ax^{-a-1}.$$

Thus $e^{-x}x^c MU' \sim -a\Gamma(c)x^{-1}/\Gamma(a) \to 0$, while $e^{-x}x^c M'U \to \Gamma(c)/\Gamma(a)$, and

$$\lim_{x\to+\infty} e^{-x}x^c W = -\frac{\Gamma(c)}{\Gamma(a)}.$$

Because this quotient is the constant $C(a,c)$, the result is exact, and parameter continuation gives

$$\boxed{MU' - M'U = -\frac{\Gamma(c)}{\Gamma(a)} e^x x^{-c}}.$$

The factorials in the question are these gamma functions.

If $a = -m$ is zero or a negative integer, $1/\Gamma(a) = 0$ and $W = 0$. The Instructor's Manual's explanation that M does not exist is incorrect: it exists and terminates. The two solutions have become proportional. Their leading polynomial coefficients are $(-1)^m/(c)_m$ for M and 1 for U (Exercise 18.6.13), so, with $L_m^{c-1}(x) = (c)_m M(-m,c,x)/m!$ from Exercise 18.6.7 ($k = c-1$),

$$\boxed{U(-m,c,x) = (-1)^m (c)_m M(-m,c,x) = (-1)^m m! L_m^{c-1}(x)}.$$

The degree recurrence in the ODE uniquely fixes the polynomial from that leading coefficient: Kummer's recurrence $(j+1)(j+c)d_{j+1} = (j-m)d_j$ gives $d_{m-1},\ldots,d_0$ from d_m. At $m = 0$ both functions equal one. The ODE still has a second independent solution, but it is not supplied by this now-degenerate pair.

18.6.15 The Coulomb wave equation (radial part of the Schrödinger equation with Coulomb potential) is

$$\frac{d^2y}{dr^2} + \left[1 - \frac{2\eta}{r} - \frac{L(L+1)}{r^2}\right] y = 0.$$

Show that a regular solution $y = F_L(\eta, r)$ is given by

$$F_L(\eta, r) = C_L(\eta) r^{L+1} e^{-ir} M(L + 1 - i\eta, 2L + 2, 2ir).$$

For the physical radial problem, take $L \geq 0$ integral, η real and $r > 0$, so that the denominator parameter $2L + 2$ is a positive integer and r^{L+1} is the regular branch at $r = 0$.

Solution.

Set $z = 2ir$, $g(r) = r^{L+1}e^{-ir}$ and $y = Cg(r)f(z)$. The logarithmic derivatives of g are, using $g''/g = (g'/g)' + (g'/g)^2$,

$$\frac{g'}{g} = \frac{L+1}{r} - i, \qquad \frac{g''}{g} = \frac{L(L+1)}{r^2} - \frac{2i(L+1)}{r} - 1.$$

Since $dz/dr = 2i$, $y' = C(g'f + 2igf')$ and $y'' = C(g''f + 4ig'f' - 4gf'')$, so

$$\frac{y''}{Cg} = -4f'' + 4i\left(\frac{L+1}{r} - i\right) f' + \left[\frac{L(L+1)}{r^2} - \frac{2i(L+1)}{r} - 1\right] f.$$

Adding the potential term cancels $L(L+1)/r^2$ and 1. Use $1/r = 2i/z$ in the remainder: the f' coefficient is $4i(L+1)/r + 4 = 4 - 8(L+1)/z = -(4/z)(2L + 2 - z)$, and the f coefficient is $-(2/r)[i(L+1) + \eta] = -(4i/z)[i(L+1) + \eta] = (4/z)(L + 1 - i\eta)$. Hence

$$\frac{1}{Cg}\left\{y'' + \left[1 - \frac{2\eta}{r} - \frac{L(L+1)}{r^2}\right] y\right\} = -\frac{4}{z}\{zf'' + (2L + 2 - z)f' - (L + 1 - i\eta)f\}.$$

The braces vanish for $f = M(L + 1 - i\eta, 2L + 2, z)$, proving

$$\boxed{F_L(\eta, r) = C_L(\eta) r^{L+1} e^{-ir} M(L + 1 - i\eta, 2L + 2, 2ir)}.$$

For physical $L = 0, 1, 2, \ldots$, the denominator parameter is positive and $M = 1 + O(r)$ at zero. Thus $F_L \sim C_L r^{L+1}$, the regular Frobenius branch; the other exponent is $-L$. The constant C_L is set by the chosen normalization at large radius and is not fixed by this homogeneous ODE alone.

18.6.16

(a) Show that the radial part of the hydrogen-atom wave function, Eq. (18.81), may be written as

$$e^{-\alpha r/2}(\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r) = \frac{(n+L)!}{(n-L-1)!\,(2L+1)!}\, e^{-\alpha r/2}(\alpha r)^L M(L + 1 - n, 2L + 2, \alpha r).$$

(b) It was assumed previously that the total (kinetic + potential) energy E of the electron was negative. Rewrite the (unnormalized) radial wave function for an unbound hydrogenic electron, $E > 0$.

ANS. $e^{i\alpha r/2}(\alpha r)^L M(L + 1 - in, 2L + 2, -i\alpha r)$, outgoing wave (printed label; this function is the regular continuum wave, see the solution). This representation provides a powerful alternative technique for the calculation of photoionization and recombination coefficients.

In (a), $n \geq L+1$ with integer $n, L \geq 0$ and $\alpha > 0$. In (b), take $Z > 0$ and real $E > 0$; the continuum Coulomb parameter is not a discrete principal quantum number.

Solution.

(a) Exercise 18.6.7 gives, for integers $N, q \geq 0$,

$$L_N^q(x) = \frac{(N+q)!}{N!\,q!} M(-N, q+1, x).$$

Put $N = n - L - 1 \geq 0$ and $q = 2L+1$. Then $N + q = n + L$, $-N = L + 1 - n$ and $q + 1 = 2L + 2$, so at $x = \alpha r$

$$L_{n-L-1}^{2L+1}(\alpha r) = \frac{(n+L)!}{(n-L-1)!\,(2L+1)!} M(L+1-n, 2L+2, \alpha r).$$

Multiplication by $e^{-\alpha r/2}(\alpha r)^L$ yields exactly

$$\boxed{e^{-\alpha r/2}(\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r) = \frac{(n+L)!}{(n-L-1)!(2L+1)!} e^{-\alpha r/2}(\alpha r)^L M(L+1-n, 2L+2, \alpha r)}.$$

All factorials have nonnegative integer arguments in the bound-state range $n \geq L+1$.

(b) Derive the positive-energy parameters directly to fix their signs. For $\psi = R(r)Y_L^M$ in the Schrödinger equation with potential $-Ze^2/(4\pi\varepsilon_0 r)$, the function $u = rR$ satisfies, with the Bohr radius $a_0 = 4\pi\varepsilon_0\hbar^2/(m_e e^2)$,

$$u'' + \left[\frac{2m_e E}{\hbar^2} + \frac{2Z}{a_0 r} - \frac{L(L+1)}{r^2}\right] u = 0.$$

For $E > 0$, define

$$k = \frac{\sqrt{2m_e E}}{\hbar} > 0, \quad \alpha = 2k, \quad \rho = kr, \quad n = \frac{Z}{k a_0} > 0.$$

Here n is a continuous Coulomb parameter, not a bound-state principal quantum number. Since $d^2u/dr^2 = k^2\, d^2u/d\rho^2$, $2Z/(a_0 r) = k^2 \cdot 2n/\rho$ and $L(L+1)/r^2 = k^2 L(L+1)/\rho^2$, dividing the equation by k^2 gives

$$\frac{d^2u}{d\rho^2} + \left[1 + \frac{2n}{\rho} - \frac{L(L+1)}{\rho^2}\right] u = 0.$$

This is the Coulomb wave equation of Exercise 18.6.15 in the variable ρ, with $\eta = -n$. Its regular solution is

$$u = F_L(-n, \rho) = C_L\, \rho^{L+1} e^{-i\rho} M(L+1+in, 2L+2, 2i\rho).$$

With $\rho = kr = \alpha r/2$ we have $2i\rho = i\alpha r$ and $\rho^{L+1}/r = k\, 2^{-L}(\alpha r)^L$. Dividing by r and absorbing the constants into the arbitrary scale,

$$R_{\text{reg}}(r) \;\propto\; e^{-i\alpha r/2}(\alpha r)^L M(L+1+in, 2L+2, i\alpha r).$$

Kummer's transformation $M(a, c, z) = e^z M(c-a, c, -z)$ (Exercise 18.6.8(a)) with $a = L+1+in$, $c = 2L+2$, $z = i\alpha r$ has $c - a = L + 1 - in$; the factor $e^z = e^{i\alpha r}$ turns $e^{-i\alpha r/2}$ into $e^{i\alpha r/2}$. This gives the form printed in the question:

$$\boxed{R_{\text{reg}}(r) \;\propto\; e^{i\alpha r/2}(\alpha r)^L M(L+1-in, 2L+2, -i\alpha r)}.$$

It is regular, $R_{\text{reg}} = O(r^L)$, at the origin, because $M(a, c, 0) = 1$.

The printed label "outgoing wave" is incorrect for this M solution. It is the regular continuum wave, containing incoming and outgoing components. For real n, r, complex conjugation of the boxed expression produces the preceding (unboxed) one, and Kummer's transformation proves they are equal. Thus the boxed function is real, and its radial current, proportional to $\mathrm{Im}(R^* R')$, vanishes. A purely outgoing wave $R \simeq Ce^{i\phi(r)}/r$ with $\phi'(r) \to k > 0$ instead has $R^* R' = |C|^2(i\phi'/r^2 - 1/r^3)$, so $\mathrm{Im}(R^* R') = |C|^2\phi'/r^2 > 0$: it carries positive outward current.

For comparison, a purely outgoing radial solution, with time factor $e^{-iEt/\hbar}$, is

$$R_{\text{out}}(r) = e^{i\alpha r/2}(\alpha r)^L U(L + 1 - in, 2L + 2, -i\alpha r).$$

It solves the radial equation. The computation of Exercise 18.6.15 holds for every twice-differentiable f and uses i only through $i^2 = -1$, while the Coulomb equation contains no i; since $(-i)^2 = -1$ as well, it remains valid with i replaced by $-i$ throughout. With $\eta = -n$ and the variable ρ, it then shows that $u = \rho^{L+1}e^{i\rho}w(-2i\rho)$ is a solution whenever $w(z)$ satisfies Kummer's equation $zw'' + (c - z)w' - aw = 0$ with $a = L + 1 + i\eta = L + 1 - in$ and $c = 2L + 2$. Since $e^{i\rho} = e^{i\alpha r/2}$, $-2i\rho = -i\alpha r$ and $\rho^{L+1}/r \propto (\alpha r)^L$, the solution $w = M$ gives the boxed R_{reg} again, and the second solution $w = U$ of the same Kummer equation gives R_{out}.

The asymptotic $U(a, c, z) \sim z^{-a}$ holds for $|\arg z| \le 3\pi/2 - \delta$, including this ray $\arg z = -\pi/2$; see DLMF 13.7.3. Here $z = -i\alpha r = -2ikr$ and $a = L + 1 - in$, so

$$z^{-a} = (-i)^{-a}(2kr)^{-L-1}e^{in\ln(2kr)}, \qquad R_{\text{out}} \sim \frac{(-i)^{-a}}{2k}\,\frac{\exp\{i[kr + n\ln(2kr)]\}}{r},$$

using $e^{i\alpha r/2} = e^{ikr}$ and $(\alpha r)^L = (2kr)^L$. The constant $(-i)^{-a} = e^{i\pi a/2}$ is nonzero, and the phase $kr + n\ln(2kr)$ is the outward Coulomb phase.

This U solution is irregular at the origin. For $c = 2L + 2 \ge 2$, $U(a, c, z) \simeq \Gamma(c - 1)z^{1-c}/\Gamma(a)$ as $z \to 0$ (DLMF Section 13.2(iii); for noninteger c this is the dominant term of the connection formula in Exercise 18.6.8(b)), and $1/\Gamma(a) \neq 0$ because $\mathrm{Re}\, a = L + 1 > 0$. Hence, near $r = 0$,

$$R_{\text{out}} \simeq \frac{\Gamma(2L + 1)}{\Gamma(a)}\,(\alpha r)^L(-i\alpha r)^{-2L-1} \;\propto\; r^{-L-1}.$$

Regularity and a purely outgoing condition are different boundary conditions.

18.6.17 Evaluate

(a) $\displaystyle\int_0^\infty \big[M_{k\mu}(x)\big]^2 dx,$

(b) $\displaystyle\int_0^\infty \big[M_{k\mu}(x)\big]^2 \frac{dx}{x},$

(c) $\displaystyle\int_0^\infty \big[M_{k\mu}(x)\big]^2 \frac{dx}{x^{1-a}},$

where $2\mu = 0, 1, 2, \ldots$, $\quad k - \mu - \frac{1}{2} = 0, 1, 2, \ldots$, $\quad a > -2\mu - 1$.

ANS. (a) $(2\mu)!\,2k$, (b) $(2\mu)!$, (c) $(2\mu)!\,(2k)^a$.

With the definition in Eq. (18.152), answers (a) and (b) hold only for $\mu = 0$ or $k = \mu + \frac{1}{2}$, and (c) even then only at $a = 0, 1$; the corrected evaluations are shown below.

Solution.

Use the definition in Eq. (18.152), without changing its scale: $M_{k\mu}(x) = e^{-x/2}x^{\mu+1/2}M(\mu - k + \frac{1}{2}, 2\mu + 1, x)$. Put

$$q = 2\mu \in \{0, 1, 2, \ldots\}, \qquad N = k - \mu - \tfrac{1}{2} \in \{0, 1, 2, \ldots\}, \qquad 2k = 2N + q + 1.$$

Then $\mu - k + \frac{1}{2} = -N$, $2\mu + 1 = q + 1$ and $\mu + \frac{1}{2} = (q+1)/2$. The terminating Kummer–Laguerre relation of Exercise 18.6.7, $M(-N, q+1, x) = A\, L_N^q(x)$, therefore gives

$$M_{k\mu}(x) = Ae^{-x/2}x^{(q+1)/2}L_N^q(x), \qquad A = \frac{N!q!}{(N+q)!}.$$

Define the common norm factor

$$B = A^2\frac{(N+q)!}{N!} = \boxed{\frac{N!(q!)^2}{(N+q)!}}.$$

(b) Here $M_{k\mu}(x)^2/x = A^2e^{-x}x^q[L_N^q(x)]^2$, and the Laguerre norm (18.71), $\int_0^\infty e^{-x}x^q[L_N^q(x)]^2dx = (N+q)!/N!$, yields

$$\int_0^\infty M_{k\mu}(x)^2\frac{dx}{x} = A^2\int_0^\infty e^{-x}x^q[L_N^q(x)]^2dx = \boxed{B}.$$

(a) Now $M_{k\mu}(x)^2 = A^2e^{-x}x^{q+1}[L_N^q(x)]^2$, and the first-moment integral of Exercise 18.3.6 (its n, k replaced by N, q) gives

$$\int_0^\infty M_{k\mu}(x)^2dx = A^2(2N+q+1)\frac{(N+q)!}{N!} = \boxed{2kB}.$$

The printed answers omit the factor $N!q!/(N+q)! = 1/\binom{N+q}{N}$ in both (a) and (b); this factor equals one only for $N = 0$ or $q = 0$. For example, $N = q = 1$ gives $k = 2$, $\mu = 1/2$, $M(-1, 2, x) = 1 - x/2$ and $M_{2,1/2} = xe^{-x/2}(1 - x/2)$. With $\int_0^\infty x^je^{-x}dx = j!$, its two integrals are

$$\int_0^\infty \Big(x^2 - x^3 + \frac{x^4}{4}\Big)e^{-x}dx = 2 - 6 + 6 = 2, \qquad \int_0^\infty \Big(x - x^2 + \frac{x^3}{4}\Big)e^{-x}dx = 1 - 2 + \frac{3}{2} = \frac{1}{2},$$

rather than the printed $(2\mu)!\,2k = 4$ and $(2\mu)! = 1$.

(c) For real $a > -q - 1$, the integral is

$$I_a = A^2\int_0^\infty e^{-x}x^{q+a}[L_N^q(x)]^2dx.$$

The given condition $a > -2\mu - 1 = -q - 1$ is exactly the condition for convergence at zero: the Kummer polynomial has constant term one, so the original integrand behaves like x^{q+a} there. Infinity is harmless because of e^{-x}. A useful exact single sum follows by changing the Laguerre parameter. The generating functions of L_N^q and of L_j^{q+a} satisfy

$$\begin{aligned}\sum_{N=0}^\infty L_N^q(x)t^N &= (1-t)^{-q-1}e^{-xt/(1-t)}\\ &= (1-t)^a\sum_{j=0}^\infty L_j^{q+a}(x)t^j\\ &= \left[\sum_{r=0}^\infty \frac{(-a)_r}{r!}t^r\right]\left[\sum_{j=0}^\infty L_j^{q+a}(x)t^j\right].\end{aligned}$$

The second line uses the same generating function with parameter $q + a$, $\sum_j L_j^{q+a}(x)t^j = (1-t)^{-q-a-1}e^{-xt/(1-t)}$; the third is the binomial series $(1-t)^a = \sum_r \binom{a}{r}(-t)^r$ with $(-1)^r\binom{a}{r} = (-a)_r/r!$.

For noninteger $\nu = q + a$, L_j^ν is the polynomial of Exercise 18.3.5(b) with its factorials written as gamma functions,

$$L_j^\nu(x) = \sum_{s=0}^{j} \frac{\Gamma(j+\nu+1)}{(j-s)!\,\Gamma(\nu+s+1)} \frac{(-x)^s}{s!},$$

and the derivation of the generating function there holds unchanged for every real $\nu > -1$. The coefficient of t^N on the right collects the products with $r + j = N$, giving the finite connection formula

$$L_N^q(x) = \sum_{j=0}^{N} \frac{(-a)_{N-j}}{(N-j)!} L_j^{q+a}(x).$$

Square it and integrate with weight $e^{-x}x^{q+a}$. Since $q+a > -1$, orthogonality applies (Exercise 18.3.3, whose closing remark extends it to every real parameter above -1) and all unequal-index terms vanish; each diagonal term contributes the norm $\int_0^\infty e^{-x}x^{q+a}[L_j^{q+a}(x)]^2 dx = \Gamma(j+q+a+1)/j!$ (Eq. (18.71), with $(j+q+a)!$ written as $\Gamma(j+q+a+1)$). Thus the complete answer is

$$\boxed{I_a = \left[\frac{N!q!}{(N+q)!}\right]^2 \sum_{j=0}^{N} \frac{[(-a)_{N-j}]^2}{[(N-j)!]^2} \frac{\Gamma(j+q+a+1)}{j!}}.$$

Every term is finite and nonnegative for the stated real parameters. At $a = 0$, $(0)_r = 0$ for $r \geq 1$, so only $j = N$ survives, giving $I_0 = B$. At $a = 1$, $(-1)_0 = 1$, $(-1)_1 = -1$ and $(-1)_r = 0$ for $r \geq 2$, so only $j = N, N-1$ survive, giving

$$\begin{aligned} I_1 &= A^2 \left[\frac{\Gamma(N+q+2)}{N!} + \frac{\Gamma(N+q+1)}{(N-1)!}\right] \\ &= A^2 \frac{(N+q)!}{N!} [(N+q+1) + N] = (2N+q+1)B, \end{aligned}$$

with the second term omitted for $N = 0$. This recovers (a).

The printed general formula $(2\mu)!\,(2k)^a$ is also incorrect even when the normalization factor happens to be one. At $N = q = 0$, $k = 1/2$ and $M_{1/2,0} = x^{1/2}e^{-x/2}$, so $I_a = \int_0^\infty x^a e^{-x} dx = \Gamma(a+1)$; at $a = 2$ this is 2, while the printed answer is 1. More generally, for $a = 2$ the connection coefficients $(-2)_r/r!$ are $1, -2, 1$ for $r = 0, 1, 2$ and zero beyond, so the next integer moment is

$$\begin{aligned} I_2 &= A^2 \left[\frac{\Gamma(N+q+3)}{N!} + \frac{4\,\Gamma(N+q+2)}{(N-1)!} + \frac{\Gamma(N+q+1)}{(N-2)!}\right] \\ &= B\,[(N+q+2)(N+q+1) + 4N(N+q+1) + N(N-1)] \\ &= B\,[6N(N+q+1) + (q+1)(q+2)], \end{aligned}$$

with the terms having a negative factorial argument omitted. Since $(2k)^2 = (2N+q+1)^2 = 4N(N+q+1) + (q+1)^2$, it exceeds $(2k)^2 B$ by $B[2N(N+q+1) + q + 1] > 0$, so even restoring the missing normalization cannot turn all moments into powers of the first moment. In fact, when $N = 0$ or $q = 0$, so that the factor is one, the printed (c) holds only at $a = 0, 1$. For $a \neq b$ the Cauchy–Schwarz inequality, applied to $M_{k\mu}x^{(a-1)/2}$ and $M_{k\mu}x^{(b-1)/2}$, which are not proportional, gives $I_{(a+b)/2}^2 < I_a I_b$. Hence the continuous function $\ln I_a$ is strictly convex in a, and it meets the line $\ln[(2\mu)!] + a\ln(2k)$, which passes through its values at $a = 0$ and $a = 1$, at no other point.

18.7 Dilogarithm

18.7.1 Prove that the expansion of $\mathrm{Li}_2(z)$, Eq. (18.156), converges everywhere on the circle $|z| = 1$.

Solution.

For every $|z| \le 1$,

$$\left|\frac{z^n}{n^2}\right| \le \frac{1}{n^2}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty.$$

The Weierstrass comparison test therefore proves absolute and uniform convergence on the whole closed unit disk, in particular at every point of its boundary circle. More explicitly,

$$\left|\sum_{n=N+1}^{\infty} \frac{z^n}{n^2}\right| \le \sum_{n=N+1}^{\infty} \frac{1}{n^2} \le \int_N^{\infty} \frac{dt}{t^2} = \frac{1}{N},$$

since $1/n^2 \le \int_{n-1}^{n} dt/t^2$ for each $n \ge 2$. This controls every boundary point with the same remainder bound. As a uniform limit of continuous partial sums, the sum is continuous there. Convergence at $z = 1$ does not make it analytic across that point: its derivative, the termwise derivative $\sum_{n\ge1} z^{n-1}/n = -\log(1-z)/z$ for $|z| < 1$, has a logarithmic branch point at $z = 1$.

18.7.2 Use the functional relations, Eqs. (18.161) to (18.163), to find the principal value of $\mathrm{Li}_2(1/2)$.

Solution.

On the principal real branch for $0 < z < 1$, the reflection identity, Eq. (18.161), is

$$\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \frac{\pi^2}{6} - \ln z \ln(1-z).$$

At $z = 1/2$ the two dilogarithms are equal and both logarithms are $-\ln 2$. Thus

$$2\,\mathrm{Li}_2(1/2) = \frac{\pi^2}{6} - (\ln 2)^2, \qquad \boxed{\mathrm{Li}_2(1/2) = \frac{\pi^2}{12} - \frac{1}{2}(\ln 2)^2}.$$

As a check using a second functional relation, set $z = 1/2$ in Eq. (18.163), $\mathrm{Li}_2(z) + \mathrm{Li}_2(z/(z-1)) = -\frac{1}{2}\ln^2(1-z)$. Its transformed argument is $z/(z-1) = -1$, so $\mathrm{Li}_2(1/2) + \mathrm{Li}_2(-1) = -\frac{1}{2}\ln^2(1/2) = -\frac{1}{2}\ln^2 2$. Splitting the series into even and odd terms,

$$\mathrm{Li}_2(-1) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\zeta(2) + 2\sum_{n\ \text{even}} \frac{1}{n^2} = -\zeta(2) + \frac{\zeta(2)}{2} = -\frac{\pi^2}{12},$$

so the same answer follows.

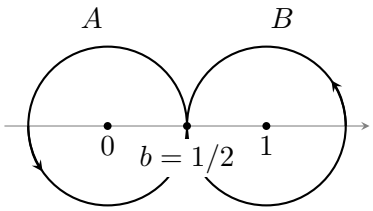
18.7.3 Find all the multiple values of $\mathrm{Li}_2(1/2)$.

Solution.

Let $b = 1/2$ and $F_0 = \pi^2/12 - \frac{1}{2}\ln^2 2$ denote the principal value (Exercise 18.7.2). Termwise differentiation of the series (18.156) gives $F'(z) = \sum_{n\ge1} z^{n-1}/n = -\log(1-z)/z$ for $|z| < 1$. To find every value, continue the differential

$$dF = -L_1(z)\frac{dz}{z}, \qquad L_1(z) = \log(1-z),$$

around loops based at b in $\mathbb{C}\setminus\{0,1\}$. Let A encircle zero once counterclockwise and B encircle one once counterclockwise. They generate all closed paths up to homotopy: every closed path at b in $\mathbb{C}\setminus\{0,1\}$ can be deformed there, without passing through 0 or 1, into a succession of the loops $A^{\pm1}$, $B^{\pm1}$, and by the monodromy theorem such a deformation does not change the value reached by analytic continuation. The two circles below are valid representatives.



On a sheet where $L_1 = \log(1-z) + 2\pi i n$ (principal logarithm plus $2\pi i n$), traversal of A leaves L_1 unchanged, because $1-z$ does not wind around zero, and changes F by

$$\Delta_A F = -\oint_A \frac{\log(1-z)}{z}\,dz - 2\pi i n \oint_A \frac{dz}{z} = 0 - 2\pi i n \cdot 2\pi i = 4\pi^2 n.$$

The first integral vanishes by Cauchy's theorem: $-\log(1-z)/z$ is analytic inside and on the zero circle (its singularity at $z=0$ is removable). For B, the additive constant $2\pi i n$ contributes $-2\pi i n \oint_B dz/z = 0$ because B has zero winding about zero. To find the remaining increment, let L_0 be the logarithm of z analytic on the closed disk bounded by B (which excludes $z=0$), with $L_0(1)=0$; then $dL_0 = dz/z$ and $L_0(b) = \ln b$. Along B, $1-z$ winds once counterclockwise about zero, so L_1 gains $2\pi i$ while L_0 returns to $\ln b$. Integration by parts, with $dL_1 = dz/(z-1)$ and Cauchy's integral formula for the last integral, gives

$$\begin{aligned}\Delta_B F = -\oint_B L_1\,dL_0 &= -[L_1 L_0]_{\text{start}}^{\text{end}} + \oint_B L_0\,dL_1 \\ &= -2\pi i \ln b + \oint_B \frac{L_0(z)}{z-1}\,dz = -2\pi i \ln b + 2\pi i L_0(1) = 2\pi i \ln 2.\end{aligned}$$

Because L_1 gains $2\pi i$, B also changes the integer sheet label from n to $n+1$. Inverse loops give the inverse increments: on sheet n, A^{-1} subtracts $4\pi^2 n$, while B^{-1} subtracts $2\pi i \ln 2$ and lowers the label to $n-1$. Hence $A^{\pm1}$ changes $F - 2\pi i n \ln 2$ by an integer multiple of $4\pi^2$, and $B^{\pm1}$ leaves it unchanged. Starting from F_0 with $n=0$, induction on the length of the word shows that every word in A, B ends at a value of the form

$$\boxed{F_0 + 2\pi i n \ln 2 + 4\pi^2 m, \qquad m, n \in \mathbb{Z}}.$$

Conversely, follow B, then A, then B^{-1}, then A^{-1}. The two B increments cancel, while A was traversed with sheet label one and A^{-1} with sheet label zero; this path adds $4\pi^2$ and returns the label to zero. Its repetitions and inverse give every m, after which B^n gives every n. Thus the boxed set is both necessary and sufficient, with no branch values omitted.

The Instructor's Manual's proposed correction $2n^2\pi^2 - 2\pi i n \ln 2$ to the principal value does not give the full branch set and is incorrect for odd n, whose real shift $2n^2\pi^2$ is not a multiple of $4\pi^2$. Substituting a multivalued logarithm into the principal reflection formula while equating its two dilogarithms assigns incompatible branches. For example one loop B gives $F_0 + 2\pi i \ln 2$ with no real shift, directly contradicting that proposed enumeration, in which the imaginary part $2\pi \ln 2$ requires $n=-1$ and hence the real shift $2\pi^2$.

18.7.4 Explain why Eq. (18.161) gives the expected result for $z = 0$ when on the principal branch of the dilogarithm.

Solution.

One cannot substitute $z = 0$ separately into $\ln z$. Take the principal-branch limit instead. For $z \to 0$ off its logarithmic cut,

$$\ln(1 - z) = -z + O(z^2), \qquad \ln z \ln(1 - z) = -z \ln z + O(z^2 \ln z).$$

Writing $z = re^{i\theta}$ with $|\theta| \le \pi$ gives $|z \ln z| \le r(|\ln r| + \pi) \to 0$; the remainder also vanishes. The dilogarithm series gives $\mathrm{Li}_2(0) = 0$, while its continuous boundary value at one is $\mathrm{Li}_2(1) = \sum n^{-2} = \pi^2/6$ (Exercise 18.7.1). The limit $\mathrm{Li}_2(1-z) \to \pi^2/6$ holds from every direction off the cut, also where $|1 - z| > 1$: since $\mathrm{Li}_2'(t) = -\log(1 - t)/t$, $\mathrm{Li}_2(w) - \mathrm{Li}_2(1) = -\int_1^w \log(1 - t)\, dt/t$ along the segment, which tends to zero as $w \to 1$ because the logarithmic singularity is integrable. Consequently the two sides of Eq. (18.161) both tend to

$$\boxed{\pi^2/6}.$$

This is a limit of the product, not an assignment of a finite value to $\ln 0$. The principal branch condition also matters: on other sheets the extra logarithmic terms must be continued consistently.

18.7.5 Show that

$$\mathrm{Li}_2\left(\frac{1 + z^{-1}}{2}\right) = -\mathrm{Li}_2\left(\frac{1 + z}{1 - z}\right) - \frac{1}{2}\ln^2\left(\frac{1 - z^{-1}}{2}\right).$$

Exclude $z = 0, 1$. With principal branches the identity holds for every z off the real segment $[0, 1]$; for $0 < z < 1$ both dilogarithm arguments lie on their cuts, so matching boundary values must be used (see the solution).

Solution.

Set $w = (1 + z^{-1})/2 = (z + 1)/(2z)$, with $z \neq 0, 1$. Two elementary reductions are

$$\frac{w}{w - 1} = \frac{(z + 1)/(2z)}{(1 - z)/(2z)} = \frac{1 + z}{1 - z}, \qquad 1 - w = \frac{z - 1}{2z} = \frac{1 - z^{-1}}{2}.$$

Insert them into Eq. (18.163), $\mathrm{Li}_2(w) + \mathrm{Li}_2(w/(w - 1)) = -\frac{1}{2}\log^2(1 - w)$, and move its second term to the right:

$$\boxed{\mathrm{Li}_2\left(\frac{1 + z^{-1}}{2}\right) = -\mathrm{Li}_2\left(\frac{1 + z}{1 - z}\right) - \frac{1}{2}\log^2\left(\frac{1 - z^{-1}}{2}\right)}.$$

For real $z > 1$ or $z < 0$, both dilogarithm arguments are less than one and the logarithm's argument is positive, so this is an equality of ordinary real principal values.

More generally, Eq. (18.163) holds with principal branches for every $w \notin [1, \infty)$. There w, $w/(w - 1)$ and $1 - w$ avoid the cuts $[1, \infty)$, $[1, \infty)$ and $(-\infty, 0]$ ($w/(w - 1) = 1$ is impossible, and if $w/(w - 1) = t > 1$, then $w = t/(t - 1) > 1$), so both sides are analytic in w on this cut plane; they agree for real $w < 1$, hence everywhere on it by analytic continuation. Since $w = (z + 1)/(2z) = t$ gives $z = 1/(2t - 1)$, which lies in $(0, 1]$ for $t \ge 1$, the boxed identity holds with principal values for every z off the real segment $[0, 1]$.

The formula is not an unrestricted identity between independently selected boundary values on the cut. For $0 < z < 1$, both dilogarithm arguments exceed one, but a small imaginary part of z puts them

on opposite sides of their cuts: for $z = x+i\varepsilon$, $z^{-1} = \bar{z}/|z|^2$ gives $\mathrm{Im}[(1+z^{-1})/2] = -\varepsilon/[2(x^2+\varepsilon^2)] < 0$, while $(1+z)(1-\bar{z}) = 1-|z|^2+2i\varepsilon$ gives

$$\mathrm{Im}\,\frac{1+z}{1-z} = \frac{\mathrm{Im}[(1+z)(1-\bar{z})]}{|1-z|^2} = \frac{2\varepsilon}{|1-z|^2} > 0.$$

For example, at $z = 1/3$ the arguments both equal 2, and the logarithm's argument is $(1-3)/2 = -1$, with $\log^2(-1 \pm i0) = -\pi^2$; the identity then demands that the two dilogarithm values add up to $\pi^2/2$. The reflection formula (18.161) holds with principal branches on the plane cut along $(-\infty, 0]$ and $[1, \infty)$, by the argument just given for (18.163): both sides are analytic there and agree for $0 < z < 1$. Applied at the point $2 \pm i0$, where $\log(1-(2 \pm i0)) = \log(-1 \mp i0) = \mp i\pi$, together with $\mathrm{Li}_2(-1) = -\pi^2/12$ (Exercise 18.7.2), it gives the boundary values $\mathrm{Li}_2(2 \pm i0) = \pi^2/6 + \pi^2/12 \pm i\pi \ln 2 = \pi^2/4 \pm i\pi \ln 2$. Selecting the same boundary value for both would incorrectly assert $2\,\mathrm{Li}_2(2 \pm i0) = \pi^2/2$, whereas the left side is $\pi^2/2 \pm 2\pi i \ln 2$. Opposite boundary values, $\mathrm{Li}_2(2+i0) + \mathrm{Li}_2(2-i0) = \pi^2/2$, with the matching logarithm, satisfy the continued identity.

18.7.6 The following integral arises in the computation of the electronic energy of the Li atom using a correlated wave function (one that explicitly includes the electron-electron distances as well as the distances of electrons from the nucleus):

$$I = \iiint d^3r_1\, d^3r_2\, d^3r_3\, \frac{e^{-\alpha_1 r_1 - \alpha_2 r_2 - \alpha_3 r_3}}{r_1 r_2 r_3 r_{12} r_{13} r_{23}},$$

where $r_i = |\mathbf{r}_i|$, $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$, and the integrations are over the entire three-dimensional space of each $\mathbf{r}_i$. For convergence of I, we require all $\alpha_j > 0$, but there are no restrictions on their relative magnitudes.

In terms of the auxiliary quantities

$$\zeta_1 = \frac{\alpha_1}{\alpha_2+\alpha_3}, \qquad \zeta_2 = \frac{\alpha_2}{\alpha_1+\alpha_3}, \qquad \zeta_3 = \frac{\alpha_3}{\alpha_1+\alpha_2},$$

this integral has the value

$$I = \frac{32\pi^3}{\alpha_1\alpha_2\alpha_3}\left(-\frac{\pi^2}{2} + \sum_{j=1}^{3}\left[\mathrm{Li}_2(\zeta_j) - \mathrm{Li}_2(-\zeta_j) + \ln\zeta_j \ln\left(\frac{1-\zeta_j}{1+\zeta_j}\right)\right]\right).$$

Rearrange I to a form (first found by Remiddi[4]), in which all terms in the final expression are guaranteed to evaluate to real quantities and can be evaluated as principal values. In the given complex expression use continuation from $0 < \zeta_j < 1$, with the logarithm boundary values matching those of $\mathrm{Li}_2(\zeta_j)$. At $\zeta_j = 1$ use the finite limit.

Solution.

For a positive variable ζ, denote one summand by

$$S(\zeta) = \mathrm{Li}_2(\zeta) - \mathrm{Li}_2(-\zeta) + \ln\zeta \log\frac{1-\zeta}{1+\zeta}.$$

Here ln denotes the real logarithm of a positive number, and log the principal logarithm of an argument, $1-\zeta$ or $(1-\zeta)/(1+\zeta)$, that becomes negative for $\zeta > 1$ and must then be continued.

[4] E. Remiddi, "Analytic value of the atomic three-electron correlation integral with Slater wave functions," *Physical Review A* **44**, 5492 (1991).

Start with $0 < \zeta < 1$, where all terms are real. Reflection, Eq. (18.161), and the logarithm of a positive quotient give

$$\mathrm{Li}_2(\zeta) = \frac{\pi^2}{6} - \mathrm{Li}_2(1-\zeta) - \ln\zeta \log(1-\zeta),$$

$$\log\frac{1-\zeta}{1+\zeta} = \log(1-\zeta) - \ln(1+\zeta).$$

Substituting both into $S(\zeta)$,

$$\begin{aligned} S(\zeta) = \frac{\pi^2}{6} &- \mathrm{Li}_2(1-\zeta) - \ln\zeta\log(1-\zeta) - \mathrm{Li}_2(-\zeta) \\ &+ \ln\zeta\log(1-\zeta) - \ln\zeta\ln(1+\zeta). \end{aligned}$$

The terms containing $\ln\zeta\log(1-\zeta)$ cancel, leaving

$$S(\zeta) = \frac{\pi^2}{6} - \mathrm{Li}_2(1-\zeta) - \mathrm{Li}_2(-\zeta) - \ln\zeta\ln(1+\zeta).$$

For $\zeta > 1$, continue consistently from above, $\zeta + i0$ (from below, change the sign of every $i\pi$). Then $\log(1-\zeta) = \ln(\zeta-1) - i\pi$, and Eq. (18.161), continued off the real axis as in Exercise 18.7.5, gives the first display with this value, so that $\mathrm{Im}\,\mathrm{Li}_2(\zeta+i0) = \pi\ln\zeta$. The quotient $(1-\zeta)/(1+\zeta) < 0$ is approached from below (at $\zeta + i\varepsilon$ its imaginary part is $-2\varepsilon/|1+\zeta+i\varepsilon|^2$), so its logarithm is $\ln[(\zeta-1)/(\zeta+1)] - i\pi$, and the second display holds with the same value of $\log(1-\zeta)$. The cancellation therefore goes through unchanged: the imaginary parts $+\pi\ln\zeta$ of $\mathrm{Li}_2(\zeta+i0)$ and $-\pi\ln\zeta$ of the logarithmic product cancel, and the right side, whose arguments $1-\zeta$, $-\zeta$ lie below one and ζ, $1+\zeta$ are positive, is real for every $\zeta > 0$. At $\zeta = 1$ it has the finite value $\pi^2/6 - \mathrm{Li}_2(-1) = \pi^2/4$, so no $0\log 0$ product needs to be evaluated separately.

Sum over $j = 1, 2, 3$. The three constants $\pi^2/6$ cancel the original $-\pi^2/2$ exactly:

$$-\frac{\pi^2}{2} + \sum_{j=1}^{3} S(\zeta_j) = -\sum_{j=1}^{3}\left[\mathrm{Li}_2(-\zeta_j) + \mathrm{Li}_2(1-\zeta_j) + \ln\zeta_j\ln(1+\zeta_j)\right].$$

Thus the desired form is

$$\boxed{I = -\frac{32\pi^3}{\alpha_1\alpha_2\alpha_3}\sum_{j=1}^{3}\left[\mathrm{Li}_2(-\zeta_j) + \mathrm{Li}_2(1-\zeta_j) + \ln\zeta_j\ln(1+\zeta_j)\right]}.$$

The assumptions $\alpha_j > 0$ imply $\zeta_j > 0$. Consequently $-\zeta_j < 0$, $1-\zeta_j < 1$, and both logarithm arguments are strictly positive. Every term in the final formula therefore has a real principal value, for arbitrary relative magnitudes of the three positive decay parameters. In particular, $\zeta_j = 1$ is harmless because $\mathrm{Li}_2(0) = 0$ and $\ln 1 = 0$.

18.8 Elliptic Integrals

18.8.1 The ellipse $x^2/a^2 + y^2/b^2 = 1$ may be represented parametrically by $x = a\sin\theta$, $y = b\cos\theta$. Show that the length of arc within the first quadrant is

$$a\int_0^{\pi/2}(1 - m\sin^2\theta)^{1/2}\,d\theta = aE(m).$$

Take $a \geq b > 0$, so $0 \leq m = (a^2 - b^2)/a^2 < 1$. The value $m = 1$ is the degenerate limit $b \downarrow 0$.

Solution.

Take $a \geq b > 0$, so the given parameter lies in $0 \leq m < 1$. Differentiating the parametrization gives $dx/d\theta = a\cos\theta$ and $dy/d\theta = -b\sin\theta$. The arc-length element is therefore

$$\begin{aligned} ds &= \sqrt{(dx)^2 + (dy)^2} = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\,d\theta \\ &= \sqrt{a^2 - (a^2 - b^2)\sin^2\theta}\,d\theta = a\sqrt{1 - \frac{a^2 - b^2}{a^2}\sin^2\theta}\,d\theta, \end{aligned}$$

using $\cos^2\theta = 1 - \sin^2\theta$. As θ increases from 0 to $\pi/2$, the point moves from $(0, b)$ to $(a, 0)$ along precisely the first-quadrant arc. Thus its length is

$$\boxed{\ell = a\int_0^{\pi/2}\sqrt{1 - m\sin^2\theta}\,d\theta = aE(m)}, \qquad m = 1 - b^2/a^2.$$

For $a = b$, $m = 0$ and $\ell = \pi a/2$, the quarter-circle length. The limit $b \downarrow 0$ gives $m \uparrow 1$ and $\ell \to a\int_0^{\pi/2}\cos\theta\,d\theta = a$; this is a degenerate limiting ellipse. If $b > a$, interchange the axis labels (or use $\theta \mapsto \pi/2 - \theta$, which turns $a^2\cos^2\theta + b^2\sin^2\theta$ into $b^2 - (b^2 - a^2)\sin^2\theta$) to obtain $bE(1 - a^2/b^2)$ in the conventional nonnegative-parameter form.

18.8.2 Derive the series expansion

$$E(m) = \frac{\pi}{2}\left[1 - \left(\frac{1}{2}\right)^2\frac{m}{1} - \left(\frac{1\cdot 3}{2\cdot 4}\right)^2\frac{m^2}{3} - \cdots\right].$$

The series is valid for $|m| \leq 1$, with the square root continued from $m = 0$.

Solution.

For $|m| < 1$, expand the integrand uniformly in θ:

$$\sqrt{1 - m\sin^2\theta} = 1 + \sum_{n=1}^{\infty}(-1)^n\binom{1/2}{n}m^n\sin^{2n}\theta.$$

For $n \geq 1$, direct multiplication of the binomial factors gives

$$\binom{1/2}{n} = \frac{\frac{1}{2}(\frac{1}{2} - 1)\cdots(\frac{1}{2} - n + 1)}{n!} = \frac{(-1)^{n-1}\,1\cdot 3\cdots(2n-3)}{2^n n!}, \qquad (-1)^n\binom{1/2}{n} = -\frac{(2n-3)!!}{(2n)!!},$$

where $(-1)!! = 1$ and $2^n n! = (2n)!!$. The sine moment is the beta integral

$$\int_0^{\pi/2}\sin^{2n}\theta\,d\theta = \frac{1}{2}B(n + \tfrac{1}{2}, \tfrac{1}{2}) = \frac{\Gamma(n + \frac{1}{2})\,\Gamma(\frac{1}{2})}{2\,\Gamma(n+1)} = \frac{\pi}{2}\frac{(2n-1)!!}{(2n)!!},$$

using $B(p,q)=\Gamma(p)\Gamma(q)/\Gamma(p+q)$, $\Gamma(\tfrac12)=\sqrt{\pi}$, $\Gamma(n+1)=n!$ and $\Gamma(n+\tfrac12)=(2n-1)!!\sqrt{\pi}/2^n$, so that the gamma quotient equals $\pi(2n-1)!!/(2^{n+1}n!)$. Integrating termwise, the m^n term is the product of these two factors, $-\frac{\pi}{2}\frac{(2n-3)!!\,(2n-1)!!}{[(2n)!!]^2}m^n$; with $(2n-3)!!=(2n-1)!!/(2n-1)$ this gives

$$\boxed{E(m)=\frac{\pi}{2}\left[1-\sum_{n=1}^{\infty}\frac{1}{2n-1}\left(\frac{(2n-1)!!}{(2n)!!}\right)^2 m^n\right].}$$

The first two nonconstant terms are $-(1/2)^2 m$ and $-(1\cdot 3/(2\cdot 4))^2 m^2/3$, which reproduce the requested series. In ordinary powers,

$$E(m)=\frac{\pi}{2}\left(1-\frac{m}{4}-\frac{3m^2}{64}-\frac{5m^3}{256}-\cdots\right).$$

Since $(2n-1)!!/(2n)!!=(2n)!/(2^n n!)^2\sim 1/\sqrt{\pi n}$ by Stirling's formula, the absolute coefficients are $O(n^{-2})$. Thus the series is absolutely uniformly convergent for $|m|\le 1$ and extends to the boundary by continuity. In particular $m=1$ gives $E(1)=1$, consistent with its defining integral $\int_0^{\pi/2}\cos\theta\,d\theta=1$.

18.8.3 Show that

$$\lim_{m\to 0}\frac{K(m)-E(m)}{m}=\frac{\pi}{4}.$$

Solution.

The statement incorporates the published erratum, which makes both elliptic arguments explicitly m. Subtract the defining integrands before taking the limit:

$$\begin{aligned}K(m)-E(m)&=\int_0^{\pi/2}\frac{1-(1-m\sin^2\theta)}{\sqrt{1-m\sin^2\theta}}d\theta\\&=m\int_0^{\pi/2}\frac{\sin^2\theta}{\sqrt{1-m\sin^2\theta}}d\theta.\end{aligned}$$

For $|m|\le\frac12$ we have $1-m\sin^2\theta\ge\frac12$, so the integrand is dominated by $\sqrt{2}\sin^2\theta$, and it tends to $\sin^2\theta$ as $m\to 0$. Dominated convergence gives

$$\boxed{\lim_{m\to 0}\frac{K(m)-E(m)}{m}=\int_0^{\pi/2}\sin^2\theta\,d\theta=\frac{\pi}{4}.}$$

Equivalently the two series start with $K=(\pi/2)(1+m/4+O(m^2))$ and $E=(\pi/2)(1-m/4+O(m^2))$ (Exercise 18.8.2; for K the m term is $\frac{m}{2}\int_0^{\pi/2}\sin^2\theta\,d\theta=\pi m/8$), so $(K-E)/m=(\pi/2)(\frac12)+O(m)$.

18.8.4 A circular loop of wire in the xy-plane, centered at the origin with radius $a>0$, carries a current I. Given that the vector potential is

$$A_\varphi(\rho,\varphi,z)=\frac{a\mu_0 I}{2\pi}\int_0^{\pi}\frac{\cos\alpha\,d\alpha}{(a^2+\rho^2+z^2-2a\rho\cos\alpha)^{1/2}},$$

show that

$$A_\varphi(\rho,\varphi,z)=\frac{\mu_0 I}{\pi k}\left(\frac{a}{\rho}\right)^{1/2}\left[\left(1-\frac{k^2}{2}\right)K(k^2)-E(k^2)\right],$$

where

$$k^2 = \frac{4a\rho}{(a+\rho)^2 + z^2}.$$

Note. For extension of this exercise to **B**, see Smythe.[5] Take $\rho > 0$ and real z, away from the wire $(\rho, z) = (a, 0)$; recover the axis by a limit.

Solution.

Assume $a > 0$, $\rho > 0$, away from the ideal wire $(\rho, z) = (a, 0)$. Set

$$D = (a+\rho)^2 + z^2, \quad m = k^2 = \frac{4a\rho}{D}, \quad 0 < m < 1,$$

where $m < 1$ because $D - 4a\rho = (a-\rho)^2 + z^2 > 0$ off the wire. Using $\cos\alpha = 2\cos^2(\alpha/2) - 1$, the denominator squared becomes

$$a^2 + \rho^2 + z^2 - 2a\rho\cos\alpha = D - 4a\rho\cos^2(\alpha/2) = D[1 - m\cos^2(\alpha/2)].$$

Put $u = \alpha/2$, so $d\alpha = 2\,du$, $0 \le u \le \pi/2$ and $\cos\alpha = 2\cos^2 u - 1$; then put $\theta = \pi/2 - u$, so $\cos u = \sin\theta$ and $d\theta = -du$ (which restores the order of the limits). The factor 2 from $d\alpha$ turns $a\mu_0 I/(2\pi)$ into $a\mu_0 I/\pi$, and $\sqrt{D}$ comes out of the root:

$$A_\varphi = \frac{a\mu_0 I}{\pi\sqrt{D}} \int_0^{\pi/2} \frac{2\sin^2\theta - 1}{\sqrt{1 - m\sin^2\theta}} d\theta.$$

The difference of the elliptic integrands gives, as in Exercise 18.8.3,

$$K(m) - E(m) = m \int_0^{\pi/2} \frac{\sin^2\theta}{\sqrt{1 - m\sin^2\theta}} d\theta.$$

Therefore, splitting the numerator $2\sin^2\theta - 1$, the remaining angular integral is

$$\frac{2}{m}[K(m) - E(m)] - K(m) = \frac{2}{m}\left[(1 - \tfrac{m}{2})K(m) - E(m)\right].$$

Finally

$$\frac{2a}{m\sqrt{D}} = \frac{2aD}{4a\rho\sqrt{D}} = \frac{\sqrt{D}}{2\rho} = \frac{\sqrt{D}}{2\sqrt{a\rho}}\sqrt{\frac{a}{\rho}} = \frac{1}{k}\sqrt{\frac{a}{\rho}},$$

because $k = 2\sqrt{a\rho}/\sqrt{D}$. Substitution of $A_\varphi = (\mu_0 I/\pi)\,[2a/(m\sqrt{D})]\,[(1 - \frac{m}{2})K - E]$ yields

$$\boxed{A_\varphi(\rho, \varphi, z) = \frac{\mu_0 I}{\pi k}\sqrt{\frac{a}{\rho}}\left[(1 - \tfrac{k^2}{2})K(k^2) - E(k^2)\right]}.$$

Rotational symmetry explains the absence of φ. At $\rho = 0$ the apparent singular factors are interpreted by a limit: the original integral is $\int_0^\pi \cos\alpha\, d\alpha/\sqrt{a^2 + z^2} = 0$, so $A_\varphi = 0$ on the axis. The formula reproduces this as $\rho \to 0$, where $k \to 0$: by the next exercise the bracket is $(m/2)f(m) = \pi m^2/32 + O(m^3)$, so, with $m^2 = k^4$ and $k^3 = 8(a\rho)^{3/2}/D^{3/2}$,

$$A_\varphi \simeq \frac{\mu_0 I}{\pi k}\sqrt{\frac{a}{\rho}}\,\frac{\pi k^4}{32} = \frac{\mu_0 I k^3}{32}\sqrt{\frac{a}{\rho}} = \frac{\mu_0 I a^2 \rho}{4D^{3/2}} \to 0.$$

At the ideal wire $k^2 = 1$, where K diverges logarithmically while $E(1) = 1$, the singularity is real and is not a removable limit.

[5] W. R. Smythe, *Static and Dynamic Electricity*, 3rd ed. (McGraw-Hill, New York, 1969), p. 270.

18.8.5 An analysis of the magnetic vector potential of a circular current loop leads to the expression

$$f(k^2) = k^{-2}\big[(2-k^2)K(k^2) - 2E(k^2)\big],$$

where $K(k^2)$ and $E(k^2)$ are the complete elliptic integrals of the first and second kinds. Show that for $k^2 \ll 1$ ($r \gg$ radius of loop)

$$f(k^2) \approx \frac{\pi k^2}{16}.$$

Here $r = \sqrt{\rho^2 + z^2}$ and the loop radius is a. The far-field condition $r \gg a$ is sufficient for $k^2 \ll 1$.

Solution.

Write $m = k^2$. Since the definition divides by m, retain the elliptic series through m^2 (one extra order beyond the desired answer):

$$K(m) = \frac{\pi}{2}\left(1 + \frac{m}{4} + \frac{9m^2}{64} + O(m^3)\right), \quad E(m) = \frac{\pi}{2}\left(1 - \frac{m}{4} - \frac{3m^2}{64} + O(m^3)\right).$$

The series for E is Exercise 18.8.2; the same binomial and sine-moment computation applied to $(1 - m\sin^2\theta)^{-1/2}$, whose binomial coefficients are $(2n-1)!!/(2n)!!$, gives

$$K(m) = \frac{\pi}{2}\sum_{n=0}^{\infty}\left[\frac{(2n-1)!!}{(2n)!!}\right]^2 m^n.$$

Then, since

$$(2-m)\Big(1 + \frac{m}{4} + \frac{9m^2}{64}\Big) = 2 + \frac{m}{2} + \frac{9m^2}{32} - m - \frac{m^2}{4} + O(m^3),$$

$$\begin{aligned}(2-m)K(m) - 2E(m) &= \frac{\pi}{2}\left[2 - \frac{m}{2} + \frac{m^2}{32} - 2 + \frac{m}{2} + \frac{3m^2}{32} + O(m^3)\right] \\ &= \frac{\pi}{2}\left[\frac{m^2}{8} + O(m^3)\right].\end{aligned}$$

The constant and linear terms cancel. Dividing by m gives

$$\boxed{f(k^2) = \frac{\pi k^2}{16} + O(k^4)}.$$

Keeping the next terms $25m^3/256$ in $2K/\pi$ and $-5m^3/256$ in $2E/\pi$ adds $(2\cdot\frac{25}{256} - \frac{9}{64} + 2\cdot\frac{5}{256})m^3 = \frac{3}{32}m^3$ to the bracket, and so gives $f(m) = \pi m/16 + 3\pi m^2/64 + O(m^3)$. Thus f has the removable value $f(0) = 0$.

For the distant loop field, $D = (a+\rho)^2 + z^2 = r^2 + 2a\rho + a^2 > r^2$, so $m = 4a\rho/D < 4a\rho/r^2 \le 4a/r$, using $\rho \le r$; hence $r \gg a$ makes $m \ll 1$, with $D \sim r^2$ and $m \sim 4a\rho/r^2$. The preceding exercise has $A_\varphi = (a\mu_0 I/\pi\sqrt{D})(2/m)[(1-\frac{m}{2})K - E] = a\mu_0 I f(m)/(\pi\sqrt{D})$. Consequently, with $\rho = r\sin\vartheta$ (ϑ the polar angle of the field point), its leading term is

$$A_\varphi \simeq \frac{a\mu_0 I}{\pi r}\cdot\frac{\pi}{16}\cdot\frac{4a\rho}{r^2} = \frac{\mu_0 I a^2 \rho}{4r^3} = \frac{\mu_0}{4\pi}\frac{(\pi a^2 I)\sin\vartheta}{r^2},$$

the vector potential $\mathbf{A} = (\mu_0/4\pi)(\pi a^2 I)\,\hat{\mathbf{z}}\times\hat{\mathbf{r}}/r^2$ of a magnetic dipole of moment $\pi a^2 I$ along the z-axis, since $\hat{\mathbf{z}}\times\hat{\mathbf{r}} = \sin\vartheta\,\hat{\boldsymbol{\varphi}}$. Large r/a is a sufficient way to make m small; it is not the only one, since $m \to 0$ also on approaching the symmetry axis.

18.8.6 Show that

(a) $\dfrac{d\,E(k^2)}{dk} = \dfrac{1}{k}\big(E(k^2) - K(k^2)\big)$,

(b) $\dfrac{d\,K(k^2)}{dk} = \dfrac{E(k^2)}{k(1-k^2)} - \dfrac{K(k^2)}{k}$.

Hint. For part (b) show that

$$E(k^2) = (1-k^2)\int_0^{\pi/2} (1 - k^2\sin^2\theta)^{-3/2}\,d\theta$$

by comparing series expansions. Initially take $0 < k < 1$; both derivatives have removable limits at $k = 0$.

Solution.

Take initially $0 < k < 1$, put $m = k^2$, and use differentiation under the integral on compact subintervals of this range. (a) The chain rule gives $\partial_k \sqrt{1 - k^2\sin^2\theta} = -k\sin^2\theta/\sqrt{1-k^2\sin^2\theta}$, so

$$\frac{dE(k^2)}{dk} = -k\int_0^{\pi/2} \frac{\sin^2\theta}{\sqrt{1-k^2\sin^2\theta}}\,d\theta.$$

Since $-k\sin^2\theta = [(1-k^2\sin^2\theta) - 1]/k$, the integrand equals $\frac{1}{k}\big[\sqrt{1-k^2\sin^2\theta} - 1/\sqrt{1-k^2\sin^2\theta}\,\big]$, whose integral is $(E-K)/k$:

$$\boxed{\frac{dE(k^2)}{dk} = \frac{E(k^2) - K(k^2)}{k}}.$$

(b) First verify the corrected hint by the requested series method. Define

$$a_n = \left[\frac{(1/2)_n}{n!}\right]^2, \qquad J(m) = \int_0^{\pi/2} (1 - m\sin^2\theta)^{-3/2} d\theta.$$

For $|m| < 1$, the binomial series $(1-x)^{-3/2} = \sum_n (3/2)_n x^n/n!$ and the sine moment $\int_0^{\pi/2}\sin^{2n}\theta\,d\theta = \frac{\pi}{2}(1/2)_n/n!$ of Exercise 18.8.2 (there written $\frac{\pi}{2}(2n-1)!!/(2n)!!$), integrated termwise, give

$$J(m) = \frac{\pi}{2}\sum_{n=0}^{\infty} \frac{(3/2)_n (1/2)_n}{(n!)^2} m^n = \frac{\pi}{2}\sum_{n=0}^{\infty} (2n+1)a_n m^n,$$

since $(3/2)_n = \frac{3}{2}\cdot\frac{5}{2}\cdots\frac{2n+1}{2} = (2n+1)(1/2)_n$. For $n \ge 1$, $(1/2)_n = (n-\frac{1}{2})(1/2)_{n-1}$ gives $a_n/a_{n-1} = (2n-1)^2/(4n^2)$. Write $2(1-m)J/\pi = \sum_n (2n+1)a_n m^n - \sum_{n\ge 1}(2n-1)a_{n-1}m^n$, shifting the index by one in the second sum. Hence, for $n \ge 1$, the m^n coefficient of $2(1-m)J/\pi$ is

$$(2n+1)a_n - (2n-1)a_{n-1} = a_n\left[(2n+1) - \frac{4n^2}{2n-1}\right] = -\frac{a_n}{2n-1},$$

because $(2n+1)(2n-1) - 4n^2 = -1$. The constant is one and $a_n = [(2n-1)!!/(2n)!!]^2$, so Exercise 18.8.2 identifies the entire series with $2E(m)/\pi$. Thus

$$\boxed{E(m) = (1-m)J(m)}, \qquad |m| < 1.$$

This uses the corrected integrand $1-k^2\sin^2\theta$ of the hint. With the printed $1-k\sin^2\theta$ the right side would be $(1-k^2)\frac{\pi}{2}(1+\frac{3}{4}k+\cdots)$, with a linear term $\frac{3\pi}{8}k$ that $E(k^2)=\frac{\pi}{2}-\frac{\pi}{8}k^2+\cdots$ lacks.

Now differentiate K, using $\partial_k(1-k^2\sin^2\theta)^{-1/2}=k\sin^2\theta\,(1-k^2\sin^2\theta)^{-3/2}$, and rewrite the numerator with $k\sin^2\theta=[1-(1-k^2\sin^2\theta)]/k$; the hint, $J(k^2)=E(k^2)/(1-k^2)$, gives the last step:

$$\begin{aligned}\frac{dK(k^2)}{dk} &= k\int_0^{\pi/2}\frac{\sin^2\theta}{(1-k^2\sin^2\theta)^{3/2}}d\theta \\ &= \frac{1}{k}\int_0^{\pi/2}\left[\frac{1}{(1-k^2\sin^2\theta)^{3/2}}-\frac{1}{\sqrt{1-k^2\sin^2\theta}}\right]d\theta \\ &= \frac{J(k^2)-K(k^2)}{k} = \boxed{\frac{E(k^2)}{k(1-k^2)}-\frac{K(k^2)}{k}}.\end{aligned}$$

All elliptic arguments in these formulas are the parameter k^2, as stated in the published erratum. At $k=0$ both derivatives have the removable value zero: $E(k^2)=\pi/2-\pi k^2/8+O(k^4)$ and $K(k^2)=\pi/2+\pi k^2/8+O(k^4)$, so $(E-K)/k=-\pi k/4+O(k^3)$, while $E/[k(1-k^2)]-K/k=[E-(1-k^2)K]/[k(1-k^2)]=\pi k/4+O(k^3)$. The singular endpoint $k=1$ is outside the finite derivative formulas and is approached only as a limit.

19 Chapter 19. Fourier Series

Problems

19.1 General Properties

19.1.1 A real-valued function $f(x)$ (quadratically integrable) is to be represented by a finite Fourier series. A convenient measure of the accuracy of the series is given by the integrated square of the deviation,

$$\Delta_p = \int_0^{2\pi} \left[f(x) - \frac{a_0}{2} - \sum_{n=1}^{p} (a_n \cos nx + b_n \sin nx) \right]^2 dx.$$

Show that the requirement that Δ_p be minimized, that is,

$$\frac{\partial \Delta_p}{\partial a_n} = 0, \qquad \frac{\partial \Delta_p}{\partial b_n} = 0,$$

for all n, leads to choosing a_n and b_n as given in Eqs. (19.2) and (19.3).

Note. Your coefficients a_n and b_n are independent of p. This independence is a consequence of orthogonality and would not hold if we expanded $f(x)$ in a power series. Take the coefficients real as well. For complex data, replace the squared residual by its modulus squared.

Solution.

Write $s_p = a_0/2 + \sum_{n=1}^{p}(a_n \cos nx + b_n \sin nx)$. The restriction to real values is needed because for complex f or coefficients $[f - s_p]^2$ is complex and cannot be minimized; the real measure of the error is then $|f - s_p|^2$. All integrals below exist: Cauchy–Schwarz on the finite interval gives $\int |f| \le \sqrt{2\pi}\,(\int |f|^2)^{1/2}$.

The orthogonality integrals on $(0, 2\pi)$ are, for $m, n \ge 1$,

$$\int_0^{2\pi} \cos mx\, dx = \int_0^{2\pi} \sin mx\, dx = \int_0^{2\pi} \sin nx \cos mx\, dx = 0,$$
$$\int_0^{2\pi} \cos nx \cos mx\, dx = \int_0^{2\pi} \sin nx \sin mx\, dx = \pi \delta_{nm}.$$

Hence, for $1 \le m \le p$,

$$\int_0^{2\pi} s_p \cos mx\, dx = \pi a_m, \qquad \int_0^{2\pi} s_p \sin mx\, dx = \pi b_m, \qquad \int_0^{2\pi} s_p\, dx = \frac{a_0}{2} 2\pi = \pi a_0.$$

Δ_p is a quadratic polynomial in the finitely many coefficients, with $\partial s_p/\partial a_m = \cos mx$ and $\partial s_p/\partial b_m = \sin mx$. Differentiating under the integral sign, for $1 \le m \le p$,

$$\frac{\partial \Delta_p}{\partial a_m} = -2 \int_0^{2\pi} (f - s_p) \cos mx\, dx = -2 \int_0^{2\pi} f \cos mx\, dx + 2\pi a_m,$$
$$\frac{\partial \Delta_p}{\partial b_m} = -2 \int_0^{2\pi} (f - s_p) \sin mx\, dx = -2 \int_0^{2\pi} f \sin mx\, dx + 2\pi b_m.$$

For the constant coefficient, $\partial s_p/\partial a_0 = 1/2$ gives instead

$$\frac{\partial \Delta_p}{\partial a_0} = -\int_0^{2\pi} (f - s_p)\, dx = -\int_0^{2\pi} f\, dx + \pi a_0.$$

Setting these derivatives to zero yields

$$\boxed{a_m = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos mx\,dx \quad (m \geq 0), \qquad b_m = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin mx\,dx \quad (m \geq 1).}$$

These stationary values are the unique minimum, not merely a stationary point. Let s_p^* have the displayed coefficients, with error Δ_p^*, and change the coefficients by $\alpha_0, \alpha_m, \beta_m$, so that $s_p = s_p^* + u$ with $u = \alpha_0/2 + \sum_{m=1}^{p}(\alpha_m \cos mx + \beta_m \sin mx)$. Then

$$\Delta_p = \int_0^{2\pi}(f - s_p^* - u)^2 dx = \Delta_p^* - 2\int_0^{2\pi}(f - s_p^*)u\,dx + \int_0^{2\pi} u^2 dx.$$

The middle integral vanishes, since the vanishing derivatives above say that $f - s_p^*$ is orthogonal to 1, $\cos mx$, $\sin mx$ $(m \leq p)$. The orthogonality integrals give $\int_0^{2\pi} u^2 dx = 2\pi(\alpha_0/2)^2 + \pi\sum_{m=1}^{p}(\alpha_m^2 + \beta_m^2)$, so

$$\Delta_p = \Delta_p^* + \frac{\pi}{2}\alpha_0^2 + \pi\sum_{m=1}^{p}(\alpha_m^2 + \beta_m^2).$$

Every added term is nonnegative and vanishes only when all changes vanish. No coefficient formula contains p: adding an orthogonal mode cannot change the projections onto modes already present.

19.1.2 In the analysis of a complex waveform (ocean tides, earthquakes, musical tones, etc.), it might be more convenient to have the Fourier series written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\alpha_n \cos(nx - \theta_n).$$

Show that this is equivalent to Eq. (19.1) with

$$a_n = \alpha_n \cos\theta_n, \qquad \alpha_n^2 = a_n^2 + b_n^2,$$
$$b_n = \alpha_n \sin\theta_n, \qquad \tan\theta_n = b_n/a_n.$$

Note. The coefficients α_n^2 as a function of n define what is called the *power spectrum*. The importance of α_n^2 lies in their invariance under a shift in the phase θ_n. The waveform and its sine/cosine coefficients are real. Take $\alpha_n \geq 0$; the tangent formula applies only when $a_n \neq 0$. Specify the phase quadrant and the zero-amplitude case.

Solution.

The cosine subtraction formula gives, for each harmonic,

$$\alpha_n \cos(nx - \theta_n) = \alpha_n \cos\theta_n \cos nx + \alpha_n \sin\theta_n \sin nx.$$

The Fourier coefficients of a function are unique (they are given by Eqs. (19.2) and (19.3)), so matching the coefficients of $\cos nx$ and $\sin nx$ with Eq. (19.1) gives $a_n = \alpha_n \cos\theta_n$, $b_n = \alpha_n \sin\theta_n$. Squaring and adding gives $a_n^2 + b_n^2 = \alpha_n^2(\cos^2\theta_n + \sin^2\theta_n) = \alpha_n^2$, and dividing b_n by a_n when $a_n \neq 0$ gives $\tan\theta_n$:

$$\boxed{\alpha_n = \sqrt{a_n^2 + b_n^2}, \qquad \tan\theta_n = b_n/a_n.}$$

We choose the nonnegative amplitude: a negative α_n gives the same term as $|\alpha_n|$ with θ_n increased by π, since $-\cos u = \cos(u - \pi)$. The tangent alone does not fix the quadrant and is undefined when

$a_n = 0$. For nonzero amplitude take θ_n to be the angle with $\cos\theta_n = a_n/\alpha_n$ and $\sin\theta_n = b_n/\alpha_n$ (the two-argument arctangent $\operatorname{atan2}(b_n, a_n)$), unique modulo 2π; for $a_n = b_n = 0$ the phase is arbitrary. Reality is needed because for complex data the amplitude–phase form can fail: $e^{ix} = \cos x + i\sin x$ has $a_1 = 1$, $b_1 = i$, so $\alpha_1^2 = a_1^2 + b_1^2 = 0$ although $a_1 \neq 0$. In particular, when $a_n = 0$ and $b_n \neq 0$, the phase is $\operatorname{sgn}(b_n)\pi/2$ modulo 2π.

Under a shift $x \mapsto x + x_0$, $\alpha_n \cos(n(x + x_0) - \theta_n) = \alpha_n \cos(nx - (\theta_n - nx_0))$: the phase becomes $\theta_n - nx_0$, while the amplitude stays unchanged. Equivalently, expanding $a_n' = \alpha_n \cos(\theta_n - nx_0)$ and $b_n' = \alpha_n \sin(\theta_n - nx_0)$ by the addition formulas, the coefficient pair is rotated:

$$\begin{pmatrix} a_n' \\ b_n' \end{pmatrix} = \begin{pmatrix} \cos nx_0 & \sin nx_0 \\ -\sin nx_0 & \cos nx_0 \end{pmatrix} \begin{pmatrix} a_n \\ b_n \end{pmatrix}, \qquad (a_n')^2 + (b_n')^2 = a_n^2 + b_n^2.$$

This is the asserted phase invariance of the power spectrum.

19.1.3 A function $f(x)$ is expanded in an exponential Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

If $f(x)$ is real, $f(x) = f^*(x)$, what restriction is imposed on the coefficients c_n?

Solution.

The coefficient formula, with f real so that only the exponential is conjugated, gives

$$c_n^* = \left[\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,dx\right]^* = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{inx}\,dx = c_{-n}.$$

Thus, with $n = 0$ giving $c_0^* = c_0$,

$$\boxed{c_{-n} = c_n^* \quad \text{for every integer } n; \qquad c_0 \in \mathbb{R}.}$$

Conversely, this relation makes every symmetric partial sum real because

$$c_n e^{inx} + c_{-n}e^{-inx} = c_n e^{inx} + (c_n e^{inx})^* = 2\,\mathrm{Re}(c_n e^{inx}).$$

Consequently its pointwise limit, wherever it exists, is real; its L^2 limit is real almost everywhere as well.

19.1.4 Assuming that f is real and $\displaystyle\int_{-\pi}^{\pi} [f(x)]^2\,dx$ is finite, show that

$$\lim_{m\to\infty} a_m = 0, \qquad \lim_{m\to\infty} b_m = 0.$$

Hint. Integrate $[f(x) - s_n(x)]^2$, where $s_n(x)$ is the nth partial sum, and use Bessel's inequality (Section 5.1). For our finite interval the assumption that $f(x)$ is square integrable ($\displaystyle\int_{-\pi}^{\pi} |f(x)|^2\,dx$ is finite) implies that $\displaystyle\int_{-\pi}^{\pi} |f(x)|\,dx$ is also finite. The converse does not hold.

Solution.

Let $s_N = a_0/2 + \sum_{n=1}^{N}(a_n \cos nx + b_n \sin nx)$ be the finite Fourier projection; all integrals are over $(-\pi, \pi)$. Expanding the square gives

$$0 \le \int_{-\pi}^{\pi} (f - s_N)^2\,dx = \int f^2\,dx - 2\int f s_N\,dx + \int s_N^2\,dx.$$

The coefficient formulas $\int f\,dx = \pi a_0$, $\int f\cos nx\,dx = \pi a_n$, $\int f\sin nx\,dx = \pi b_n$ give

$$\int f s_N\,dx = \frac{a_0}{2}\pi a_0 + \sum_{n=1}^{N}(a_n\pi a_n + b_n\pi b_n) = \frac{\pi}{2}a_0^2 + \pi\sum_{n=1}^{N}(a_n^2+b_n^2),$$

and orthogonality ($\int 1\,dx = 2\pi$, $\int \cos^2 nx\,dx = \int \sin^2 nx\,dx = \pi$, all cross terms zero) gives the same value for $\int s_N^2\,dx$. Therefore $0 \le \int f^2\,dx - \frac{\pi}{2}a_0^2 - \pi\sum_{n=1}^{N}(a_n^2+b_n^2)$, that is,

$$\frac{a_0^2}{2} + \sum_{n=1}^{N}(a_n^2+b_n^2) \le \frac{1}{\pi}\int_{-\pi}^{\pi} f^2\,dx < \infty.$$

This is Bessel's inequality. The partial sums of the nonnegative series $\sum(a_n^2+b_n^2)$ are nondecreasing and bounded, so the series converges, and the terms of a convergent series tend to zero: $a_n^2+b_n^2 \to 0$. Since each square is no larger than the sum,

$$\boxed{a_n \to 0, \qquad b_n \to 0.}$$

No pointwise convergence of the original Fourier series was used. The statement assumes f real because the argument uses $(f-s_N)^2 \ge 0$; for complex f the same argument with modulus squares, as in 19.1.1, gives $|a_n|^2 + |b_n|^2 \to 0$.

The integrability assertion in the hint follows from the Cauchy–Schwarz inequality applied to $|f|$ and 1:

$$\int_{-\pi}^{\pi} |f|\cdot 1\,dx \le \left(\int |f|^2\,dx\right)^{1/2}\sqrt{2\pi}.$$

For a counterexample to the converse, $f(x) = |x|^{-2/3}$ on $0<|x|<1$, and zero elsewhere, has $\int |f| = 2\int_0^1 x^{-2/3}\,dx = 2\big[3x^{1/3}\big]_0^1 = 6$ but $\int |f|^2 = 2\int_0^1 x^{-4/3}\,dx = 2\lim_{\epsilon\downarrow 0}\big[-3x^{-1/3}\big]_\epsilon^1 = \infty$.

19.1.5 Apply the summation technique of this section to show that

$$\sum_{n=1}^{\infty}\frac{\sin nx}{n} = \begin{cases} \dfrac{1}{2}(\pi - x), & 0 < x \le \pi, \\ -\dfrac{1}{2}(\pi + x), & -\pi \le x < 0. \end{cases}$$

This is the reverse sawtooth wave shown in Fig. 19.5.

Solution.

First insert an Abel factor $0<r<1$. With $z = re^{ix}$, $|z|<1$, the series $-\log(1-z) = \sum_{n\ge 1} z^n/n$ converges absolutely, and $\operatorname{Im} z^n = r^n \sin nx$, so

$$S_r(x) = \sum_{n=1}^{\infty}\frac{r^n\sin nx}{n} = \operatorname{Im}\{-\log(1-re^{ix})\}.$$

Use the principal logarithm; it is continuous on the right half-plane, which contains $1-re^{ix}$ (real part $1-r\cos x > 0$) and, for $0<x<2\pi$, also $1-e^{ix}$ (real part $1-\cos x > 0$). For $0<x<2\pi$,

$$1 - e^{ix} = e^{ix/2}\big(e^{-ix/2} - e^{ix/2}\big) = -2ie^{ix/2}\sin(x/2) = 2\sin(x/2)e^{i(x/2-\pi/2)},$$

with modulus $2\sin(x/2) > 0$ and argument $x/2 - \pi/2 \in (-\pi/2, \pi/2)$. Hence

$$\lim_{r\uparrow 1} S_r(x) = -\operatorname{Arg}(1 - e^{ix}) = -\Big(\frac{x}{2} - \frac{\pi}{2}\Big) = \frac{\pi - x}{2}, \qquad 0 < x < 2\pi.$$

The original series converges for $x \notin 2\pi\mathbb{Z}$ by Dirichlet's test: $1/n$ decreases to zero, and the partial sums of e^{inx} are bounded,

$$\Big|\sum_{n=1}^{N} e^{inx}\Big| = \Big|\frac{e^{ix}(1 - e^{iNx})}{1 - e^{ix}}\Big| \le \frac{2}{|1 - e^{ix}|}.$$

Abel's theorem (if $\sum c_n$ converges, then $\lim_{r\uparrow 1} \sum c_n r^n = \sum c_n$) then identifies its ordinary sum with this limit. Oddness supplies the negative interval: for $-\pi \le x < 0$ the sum is $-(\pi - |x|)/2 = -(\pi + x)/2$. Thus

$$\boxed{\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \begin{cases} (\pi - x)/2, & 0 < x \le \pi, \\ -(\pi + x)/2, & -\pi \le x < 0. \end{cases}}$$

At $x = 0$ every summand is zero; the sum is zero, the mean of the two one-sided limits. At $\pm\pi$ it is also zero. The complete function is 2π-periodic.

19.1.6 Sum the series $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n}$ and show that it equals $x/2$ for $-\pi < x < \pi$.

Solution.

For $0 < r < 1$, the alternating logarithmic series $\log(1 + z) = \sum_{n\ge 1} (-1)^{n+1} z^n/n$ with $z = re^{ix}$ gives

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{r^n \sin nx}{n} = \operatorname{Im} \log(1 + re^{ix}).$$

When $-\pi < x < \pi$,

$$1 + e^{ix} = e^{ix/2}\big(e^{-ix/2} + e^{ix/2}\big) = 2\cos(x/2) e^{ix/2}, \qquad \cos(x/2) > 0,$$

so its principal argument is $x/2$; as in 19.1.5 the logarithm is continuous as $r \uparrow 1$, since $1 + re^{ix}$ and $1 + e^{ix}$ have positive real parts. The series converges by Dirichlet's test, because the partial sums of $(-1)^{n+1} e^{inx} = -(-e^{ix})^n$ are bounded by $2/|1 + e^{ix}|$ when $e^{ix} \ne -1$. Abel's theorem, as in 19.1.5, therefore gives

$$\boxed{\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n} = \frac{x}{2}, \qquad -\pi < x < \pi.}$$

The interval is essential: the left side is periodic, whereas $x/2$ is not. At $x = (2k + 1)\pi$ its value is zero, not $x/2$. By 2π-periodicity, on $((2k - 1)\pi, (2k + 1)\pi)$ its value is $(x - 2k\pi)/2$.

19.1.7 Sum the trigonometric series $\displaystyle\sum_{n=0}^{\infty} \frac{\sin(2n + 1)x}{2n + 1}$ and show that it equals

$$\begin{cases} \pi/4, & 0 < x < \pi, \\ -\pi/4, & -\pi < x < 0. \end{cases}$$

Solution.

Subtract the even harmonics from the sum of all harmonics. Let $S(x) = \sum_{n\ge1} \sin nx/n$ be the function of 19.1.5, where the principal-logarithm computation gave $S(y) = (\pi - y)/2$ for every $0 < y < 2\pi$. Split a partial sum of S into odd and even n, writing $n = 2m$ in the even terms:

$$\sum_{n=1}^{2N} \frac{\sin nx}{n} = \sum_{n=0}^{N-1} \frac{\sin(2n+1)x}{2n+1} + \frac{1}{2}\sum_{m=1}^{N} \frac{\sin(2mx)}{m}.$$

For $0 < x < \pi$, as $N \to \infty$ the left side tends to $S(x)$ and the last sum to $S(2x)$, since x and $2x$ lie in $(0, 2\pi)$. Hence the odd series converges, and

$$\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1} = S(x) - \frac{1}{2}S(2x) = \frac{\pi - x}{2} - \frac{\pi - 2x}{4} = \frac{\pi}{4}.$$

The series is odd in x, giving $-\pi/4$ on $(-\pi, 0)$. Thus

$$\boxed{\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1} = \begin{cases} \pi/4, & 0 < x < \pi, \\ -\pi/4, & -\pi < x < 0. \end{cases}}$$

At every integer multiple of π, each sine vanishes and the sum is zero, the midpoint value at the jump.

19.1.8 Let $f(z) = \ln(1+z) = \sum_{n=1}^{\infty} \dfrac{(-1)^{n+1} z^n}{n}$. This series converges to $\ln(1+z)$ for $|z| \le 1$, except at the point $z = -1$.

(a) From the real parts show that

$$\ln\left(2\cos\frac{\theta}{2}\right) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cos n\theta}{n}, \qquad -\pi < \theta < \pi.$$

(b) Using a change of variable, transform part (a) into

$$-\ln\left(2\sin\frac{\theta}{2}\right) = \sum_{n=1}^{\infty} \frac{\cos n\theta}{n}, \qquad 0 < \theta < 2\pi.$$

Solution.

(a) Put $z = e^{i\theta}$. For $-\pi < \theta < \pi$ the identity

$$1 + e^{i\theta} = 2\cos(\theta/2) e^{i\theta/2}$$

has a positive real amplitude $2\cos(\theta/2)$ and argument $\theta/2 \in (-\pi/2, \pi/2)$, so the principal logarithm is

$$\log(1 + e^{i\theta}) = \ln(2\cos(\theta/2)) + i\theta/2.$$

By the statement the series converges to this value at $z = e^{i\theta}$, $\theta \ne \pi$. Taking real parts, with $\operatorname{Re} e^{in\theta} = \cos n\theta$, gives

$$\boxed{\ln(2\cos(\theta/2)) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cos n\theta}{n}, \qquad -\pi < \theta < \pi.}$$

At the included point $\theta = 0$ this is the convergent alternating harmonic series. The excluded endpoints have the logarithmic singularity.

(b) Replace the angle in (a) by $\theta - \pi$, which belongs to $(-\pi, \pi)$ when $0 < \theta < 2\pi$. Then

$$\cos\frac{\theta-\pi}{2} = \sin\frac{\theta}{2}, \qquad \cos(n\theta - n\pi) = (-1)^n \cos n\theta.$$

Consequently (a) becomes

$$\ln\Big(2\sin\frac{\theta}{2}\Big) = \sum_{n=1}^{\infty}(-1)^{n+1}(-1)^n\frac{\cos n\theta}{n} = -\sum_{n=1}^{\infty}\frac{\cos n\theta}{n},$$

since the two alternating factors multiply to $(-1)^{2n+1} = -1$. Hence

$$\boxed{-\ln(2\sin(\theta/2)) = \sum_{n=1}^{\infty}\frac{\cos n\theta}{n}, \qquad 0 < \theta < 2\pi.}$$

The Instructor's Manual's intermediate exponential identity in (b) has the wrong sign. The correct identity is $2\sin(\theta/2) = ie^{-i\theta/2}(1 - e^{i\theta})$. Its final real-part result agrees with the formula above.

19.1.9

(a) Expand $f(x) = x$ in the interval $(0, 2L)$. Sketch the series you have found (right-hand side of ANS.) over $(-2L, 2L)$.

$$\textit{ANS. } x = L - \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi x}{L}.$$

(b) Expand $f(x) = x$ as a sine series in the half interval $(0, L)$. Sketch the series you have found (right-hand side of ANS.) over $(-2L, 2L)$.

$$\textit{ANS. } x = \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{L}.$$

Solution.

(a) For period $2L$, put $k_n = n\pi/L$. The coefficient normalization on $(0, 2L)$ is $1/L$, so

$$a_0 = \frac{1}{L}\int_0^{2L} x\,dx = 2L.$$

Two elementary antiderivatives are

$$\int x\cos kx\,dx = \frac{x\sin kx}{k} + \frac{\cos kx}{k^2}, \qquad \int x\sin kx\,dx = -\frac{x\cos kx}{k} + \frac{\sin kx}{k^2}.$$

At $k = k_n$ and $x = 0, 2L$ the sines vanish and the cosines are one ($k_n 2L = 2n\pi$). Consequently

$$a_n = \frac{1}{L}\left[\frac{x\sin k_n x}{k_n} + \frac{\cos k_n x}{k_n^2}\right]_0^{2L} = \frac{1}{L}\,\frac{1-1}{k_n^2} = 0,$$

$$b_n = \frac{1}{L}\left[-\frac{x\cos k_n x}{k_n} + \frac{\sin k_n x}{k_n^2}\right]_0^{2L} = \frac{1}{L}\left(-\frac{2L}{k_n}\right) = -\frac{2L}{n\pi}.$$

With $a_0/2 = L$, this gives

$$\boxed{x = L - \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi x/L)}{n}, \qquad 0 < x < 2L.}$$

The series has period $2L$, so on $(-2L, 0)$ it equals $x + 2L$; at $x = 0$ its value is the midpoint L of the jump from $2L$ to 0 (every sine vanishes there).

(b) A sine series on $(0, L)$ uses the odd extension to $(-L, L)$, period $2L$, and the normalization $2/L$. The same antiderivative, with $\sin k_n L = \sin n\pi = 0$ and $\cos k_n L = (-1)^n$, gives

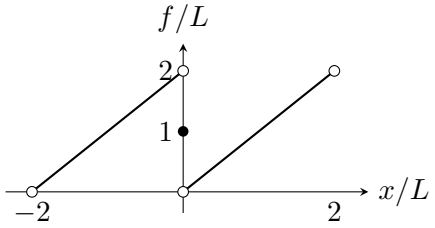
$$b_n = \frac{2}{L}\int_0^L x\sin k_n x\,dx = \frac{2}{L}\left[-\frac{x\cos k_n x}{k_n} + \frac{\sin k_n x}{k_n^2}\right]_0^L = \frac{2}{L}\,\frac{-L(-1)^n}{k_n} = \frac{2L(-1)^{n+1}}{n\pi}.$$

Therefore the corrected answer is

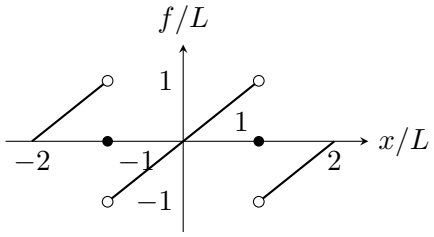
$$\boxed{x = \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{L}, \qquad 0 < x < L.}$$

The textbook's printed answer in (b), with coefficient $4L/\pi$ and only odd harmonics, is incorrect for $f(x) = x$: by 19.1.7, with $\pi x/L$ in place of x, that printed series equals the constant L throughout $(0, L)$. The ANS. line above shows the corrected series. The corrected series equals x on $(-L, L)$, $x + 2L$ on $(-2L, -L)$, and $x - 2L$ on $(L, 2L)$; its values at $\pm L$ are zero.

These are the requested sketches, with both axes scaled by L:



(a) Full-interval expansion



(b) Half-interval sine expansion

Open circles are one-sided limits, and filled circles give the series values at the indicated jumps.

19.1.10 In some problems it is convenient to approximate $\sin\pi x$ over the interval $[0, 1]$ by a parabola $ax(1 - x)$, where a is a constant. To get a feeling for the accuracy of this approximation, expand $4x(1 - x)$ in a Fourier sine series $(-1 \le x \le 1)$:

$$f(x) = \begin{cases} 4x(1-x), & 0 \le x \le 1 \\ 4x(1+x), & -1 \le x \le 0 \end{cases} = \sum_{n=1}^{\infty} b_n \sin n\pi x.$$

ANS. $b_n = \dfrac{32}{\pi^3 n^3}$, n odd, $\qquad b_n = 0$, n even.

This approximation is shown in Fig. 19.6.

Solution.

The stated extension is odd, since $f(-x) = 4(-x)(1 - x) = -f(x)$ for $0 \le x \le 1$, so all cosine coefficients vanish. The period is 2 $(L = 1)$, and the integrand $f(x)\sin n\pi x$, a product of two odd functions, is even. Put $k = n\pi$; the sine coefficient is

$$b_n = \int_{-1}^{1} f(x)\sin kx\,dx = 2\int_0^1 4x(1-x)\sin kx\,dx = 8I_n, \qquad I_n = \int_0^1 x(1-x)\sin kx\,dx.$$

Integrate by parts twice. The boundary terms vanish because $x(1-x)=0$ at $x=0,1$ and $\sin k=\sin 0=0$:

$$\begin{aligned} I_n &= \left[-\frac{x(1-x)\cos kx}{k}\right]_0^1 + \frac{1}{k}\int_0^1 (1-2x)\cos kx\,dx \\ &= \frac{1}{k}\left[\frac{(1-2x)\sin kx}{k}\right]_0^1 + \frac{2}{k^2}\int_0^1 \sin kx\,dx = \frac{2}{k^2}\,\frac{1-\cos k}{k} = \frac{2(1-\cos k)}{k^3}. \end{aligned}$$

With $\cos k=\cos n\pi=(-1)^n$ and $k^3=\pi^3 n^3$, it follows that

$$\boxed{b_n=\frac{16[1-(-1)^n]}{\pi^3 n^3}=\begin{cases}32/(\pi^3 n^3), & n \text{ odd},\\ 0, & n \text{ even}.\end{cases}}$$

Thus the fundamental sine has coefficient $32/\pi^3\approx 1.032$, and the next nonzero coefficient, $b_3=32/(27\pi^3)\approx 0.038$, is smaller by a factor 27. The series converges absolutely and uniformly, since its terms are bounded by $32/(\pi^3 n^3)$ and $\sum n^{-3}$ converges (Weierstrass M-test). It equals the continuous odd periodic extension, including its zero values at $-1,0,1$.

19.1.11 Verify that $\delta(\varphi_1-\varphi_2)=\dfrac{1}{2\pi}\displaystyle\sum_{m=-\infty}^{\infty} e^{im(\varphi_1-\varphi_2)}$ is a Dirac delta function by showing that it satisfies the definition,

$$\int_{-\pi}^{\pi} f(\varphi_1)\,\frac{1}{2\pi}\sum_{m=-\infty}^{\infty} e^{im(\varphi_1-\varphi_2)}\,d\varphi_1 = f(\varphi_2).$$

Hint. Represent $f(\varphi_1)$ by an exponential Fourier series. Use the periodic delta distribution on the circle, with a smooth 2π-periodic test function f; interpret the series as the limit $N\to\infty$ of its symmetric partial sums $\sum_{m=-N}^{N}$.

Solution.

This is the periodic delta distribution on the circle, not a pointwise convergent series of functions: the terms e^{imu} do not tend to zero, so the series diverges at every u. On the real line it is the comb $\delta_{2\pi}(u)=\sum_{k\in\mathbb{Z}}\delta(u-2\pi k)$. The interval endpoints are identified. Take a smooth 2π-periodic test function $f(\varphi)=\sum_{j\in\mathbb{Z}} c_j e^{ij\varphi}$, with $c_j=(2\pi)^{-1}\int_{-\pi}^{\pi} f(\varphi)e^{-ij\varphi}\,d\varphi$. For $j\neq 0$, integrating by parts twice, with boundary terms that cancel by periodicity, gives

$$c_j=\frac{1}{2\pi ij}\int_{-\pi}^{\pi} f'(\varphi)e^{-ij\varphi}\,d\varphi=-\frac{1}{2\pi j^2}\int_{-\pi}^{\pi} f''(\varphi)e^{-ij\varphi}\,d\varphi,$$

so $|c_j|\le j^{-2}(2\pi)^{-1}\int|f''|$ and the coefficient series is absolutely summable. For a finite symmetric kernel, interchange the finite sum with the integral and use $(2\pi)^{-1}\int_{-\pi}^{\pi} f(\varphi_1)e^{im\varphi_1}\,d\varphi_1=c_{-m}$:

$$\begin{aligned}\int_{-\pi}^{\pi} f(\varphi_1)\frac{1}{2\pi}\sum_{m=-N}^{N} e^{im(\varphi_1-\varphi_2)}\,d\varphi_1 &= \sum_{m=-N}^{N} c_{-m}e^{-im\varphi_2}\\ &= \sum_{j=-N}^{N} c_j e^{ij\varphi_2}\qquad (j=-m).\end{aligned}$$

The last expression is the symmetric partial sum of the Fourier series of f, which converges to $f(\varphi_2)$ as $N \to \infty$ because f is smooth (Section 19.1); the absolutely summable coefficients make the convergence uniform. This is exactly the defining distributional action, hence

$$\boxed{\delta_{2\pi}(\varphi_1 - \varphi_2) = \frac{1}{2\pi} \sum_{m \in \mathbb{Z}} e^{im(\varphi_1 - \varphi_2)}.}$$

For φ_2 strictly inside the integration interval, only one member of the delta comb is present. For a piecewise regular function with a jump at φ_2, symmetric Fourier sums give its midpoint value; one must not assert recovery of an arbitrary assigned value at the jump.

19.1.12 Show that integration of the Fourier expansion of $f(x) = x$, $-\pi < x < \pi$, leads to

$$\frac{\pi^2}{12} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \cdots .$$

Note. The series for $f(x) = x$ was the subject of Example 19.1.1. Confirm that the change in the defined range from $[0, 2\pi]$ to $[-\pi, \pi]$ has no effect on the expansion.

Solution.

From 19.1.6, or from the Fourier coefficients computed in 19.2.7,

$$x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx, \qquad -\pi < x < \pi.$$

Integrating from 0 to x term by term, with $\int_0^x \sin nt\, dt = (1 - \cos nx)/n$, gives

$$\frac{x^2}{2} = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} (1 - \cos nx).$$

The integrated series is absolutely and uniformly convergent (its terms are bounded by $4/n^2$). Termwise integration is justified as follows. The partial sums s_N of the series for $f(x) = x$ are its Fourier partial sums, and since the trigonometric functions form a complete set, the Fourier series of a square-integrable function converges to it in the mean: $\|f - s_N\|_2 \to 0$, where $\|g\|_2 = \left(\int_{-\pi}^{\pi} |g|^2\, dx\right)^{1/2}$. By Cauchy–Schwarz on the interval between 0 and x,

$$\left| \int_0^x (f - s_N)\, dt \right| \le \sqrt{|x|}\, \|f - s_N\|_2 \le \sqrt{\pi}\, \|f - s_N\|_2 \longrightarrow 0$$

uniformly in x, so the integrated partial sums converge to $\int_0^x t\, dt = x^2/2$. Average the integrated identity over $(-\pi, \pi)$; uniform convergence allows averaging term by term. Every cosine has zero average, so the right side averages to $2 \sum_{n \ge 1} (-1)^{n+1}/n^2$, while

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{x^2}{2}\, dx = \frac{1}{2\pi} \left[\frac{x^3}{6} \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \frac{2\pi^3}{6} = \frac{\pi^2}{6}.$$

Consequently $2 \sum_{n \ge 1} (-1)^{n+1}/n^2 = \pi^2/6$, that is,

$$\boxed{\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.}$$

The change of range in the note does not change the underlying periodic function: Example 19.1.1 defines it as x on $(0, \pi)$ and $x - 2\pi$ on $(\pi, 2\pi)$. Moving that latter interval left by 2π gives x on $(-\pi, 0)$. Since the integral of a 2π-periodic function is the same over every interval of length 2π, integrals over $[0, 2\pi]$ and over $[-\pi, \pi]$ give exactly these same Fourier coefficients.

19.1.13

(a) Assuming that the Fourier expansion of a real-valued $f(x)$ is uniformly convergent, show that

$$\frac{1}{\pi}\int_{-\pi}^{\pi}\big[f(x)\big]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}\big(a_n^2 + b_n^2\big).$$

This is Parseval's identity. Note that it is a completeness relation for the Fourier expansion.

(b) Given $x^2 = \dfrac{\pi^2}{3} + 4\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n\cos nx}{n^2}$, $-\pi \le x \le \pi$, apply Parseval's identity to obtain $\zeta(4)$ in closed form.

(c) The condition of uniform convergence is not necessary. Show this by applying the Parseval identity to the square wave

$$f(x) = \begin{cases} -1, & -\pi < x < 0 \\ 1, & 0 < x < \pi \end{cases} = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{2n-1}.$$

Solution.

(a) Square the partial sum $s_N = a_0/2 + \sum_{n=1}^{N}(a_n\cos nx + b_n\sin nx)$ and integrate over $(-\pi,\pi)$. All cross terms integrate to zero, while $\int(a_0/2)^2dx = \pi a_0^2/2$ and $\int\cos^2 nx\,dx = \int\sin^2 nx\,dx = \pi$. Hence for every finite partial sum

$$\frac{1}{\pi}\int_{-\pi}^{\pi} s_N^2\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{N}(a_n^2 + b_n^2).$$

If $s_N \to f$ uniformly, that is, $\|s_N - f\|_\infty \to 0$ with $\|g\|_\infty = \sup_{-\pi<x<\pi}|g(x)|$, then f is bounded, the s_N are uniformly bounded for large N ($\|s_N\|_\infty \le \|f\|_\infty + 1$ once $\|s_N - f\|_\infty \le 1$), and, since $s_N^2 - f^2 = (s_N - f)(s_N + f)$,

$$\left|\int(s_N^2 - f^2)\right| \le 2\pi\|s_N - f\|_\infty(\|s_N\|_\infty + \|f\|_\infty) \longrightarrow 0.$$

Letting $N \to \infty$ on both sides of the finite identity proves the stated identity. The displayed identity is for real functions; the complex version uses modulus squares. Also the Instructor's Manual's first display in 19.1.13(a) is missing the factor $1/\pi$ on its integral: for $f = 1$ its unnormalized left side is 2π, whereas its right side is 2. More generally L^2 convergence suffices: with $\|\cdot\|_2$ as in 19.1.12, the triangle inequality gives $|\|s_N\|_2 - \|f\|_2| \le \|s_N - f\|_2 \to 0$, so $\pi^{-1}\int_{-\pi}^{\pi} s_N^2\,dx = \|s_N\|_2^2/\pi \to \|f\|_2^2/\pi$.

(b) Reading off the given series for $f = x^2$: $a_0/2 = \pi^2/3$, so $a_0 = 2\pi^2/3$; $a_n = 4(-1)^n/n^2$, so $a_n^2 = 16/n^4$; and $b_n = 0$. The identity becomes

$$\frac{1}{\pi}\int_{-\pi}^{\pi} x^4\,dx = \frac{1}{\pi}\,\frac{2\pi^5}{5} = \frac{2\pi^4}{5} = \frac{1}{2}\left(\frac{2\pi^2}{3}\right)^2 + 16\sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{2\pi^4}{9} + 16\,\zeta(4).$$

Solving, with $2/5 - 2/9 = 8/45$,

$$\boxed{\zeta(4) = \frac{1}{16}\left(\frac{2}{5} - \frac{2}{9}\right)\pi^4 = \frac{1}{16}\,\frac{8\pi^4}{45} = \frac{\pi^4}{90}.}$$

The given x^2 series is uniformly convergent by comparison with $\sum n^{-2}$, so part (a) applies directly.

(c) The square wave has $f^2 = 1$, so $\pi^{-1}\int_{-\pi}^{\pi} f^2\,dx = 2$. From the given series it has no constant or cosine terms, $b_{2j-1} = 4/[\pi(2j-1)]$, and the even-n sine coefficients vanish. Hence Parseval reads

$$2 = \frac{16}{\pi^2}\sum_{j=1}^{\infty}\frac{1}{(2j-1)^2}.$$

To check this independently, the L^2 convergence of the square-wave series may not be used: by the expansion in 19.1.4, $\int(f-s_N)^2\,dx = \int f^2\,dx - \pi\sum_{n\le N} b_n^2$, so for this f it is equivalent to the identity being checked. Instead use the primitive $\int_0^x f\,dt = |x|$, which is continuous and piecewise smooth; its cosine series (19.2.9) converges to it at every point, and at $x = 0$ it gives

$$0 = \frac{\pi}{2} - \frac{4}{\pi}\sum_{j=1}^{\infty}\frac{1}{(2j-1)^2}.$$

Thus $\sum_{j\ge1}(2j-1)^{-2} = \pi^2/8$ (as also found in 19.2.13(b)), and the right side of Parseval is indeed $(16/\pi^2)(\pi^2/8) = 2$.

For any fixed partial sum s_N, which is continuous with $s_N(0) = 0$, $s_N(x) \to 0$ as $x \downarrow 0$, whereas the square wave equals one there. Thus $\sup_{0<x<\pi}|s_N(x) - 1| \ge 1$, ruling out uniform convergence. This supplies the requested counterexample to necessity of uniform convergence, while L^2 completeness still justifies Parseval.

19.1.14 Given

$$\varphi_1(x) \equiv \sum_{n=1}^{\infty}\frac{\sin nx}{n} = \begin{cases} -\dfrac{1}{2}(\pi + x), & -\pi \le x < 0, \\ \dfrac{1}{2}(\pi - x), & 0 < x \le \pi, \end{cases}$$

show by integrating that

$$\varphi_2(x) \equiv \sum_{n=1}^{\infty}\frac{\cos nx}{n^2} = \begin{cases} \dfrac{1}{4}(\pi + x)^2 - \dfrac{\pi^2}{12}, & -\pi \le x \le 0, \\ \dfrac{1}{4}(\pi - x)^2 - \dfrac{\pi^2}{12}, & 0 \le x \le \pi. \end{cases}$$

Solution.

The cosine series for φ_2 is absolutely and uniformly convergent by comparison with $\sum n^{-2}$. Integrating the sine series for φ_1 term by term from 0 to x, $0 < x \le \pi$, is valid because the series is the Fourier series of φ_1 (its coefficients $b_n = 1/n$ are computed in 19.2.8). Its partial sums s_N therefore converge to φ_1 in the mean, $\|\varphi_1 - s_N\|_2 \to 0$, as in 19.1.12, and by Cauchy–Schwarz on $(0, x)$,

$$\left|\int_0^x(\varphi_1 - s_N)\,dt\right| \le \sqrt{x}\,\|\varphi_1 - s_N\|_2 \le \sqrt{\pi}\,\|\varphi_1 - s_N\|_2 \longrightarrow 0.$$

Since $n^{-1}\int_0^x \sin nt\,dt = (1 - \cos nx)/n^2$, the term-by-term integral is $\sum_{n\ge1}(1-\cos nx)/n^2 = \varphi_2(0) - \varphi_2(x)$, and integrating the closed form instead gives

$$\varphi_2(0) - \varphi_2(x) = \int_0^x \frac{\pi - t}{2}\,dt = \frac{\pi x}{2} - \frac{x^2}{4}.$$

Write $C = \varphi_2(0)$, so $\varphi_2(x) = C - \pi x/2 + x^2/4$ on $[0, \pi]$. The cosine series is even in x, which gives, on $[-\pi, \pi]$, $\varphi_2(x) = C - \pi|x|/2 + x^2/4$. Every term of the uniformly convergent cosine series has zero integral over $[0, \pi]$, because $\int_0^\pi \cos nx\, dx = \sin(n\pi)/n = 0$, so termwise integration gives

$$0 = \int_0^\pi \varphi_2(x)\, dx = C\pi - \frac{\pi}{2}\frac{\pi^2}{2} + \frac{1}{4}\frac{\pi^3}{3} = C\pi - \frac{\pi^3}{4} + \frac{\pi^3}{12} = C\pi - \frac{\pi^3}{6}.$$

Therefore $C = \pi^2/6$. Completing the square, $\pi^2/6 - \pi|x|/2 + x^2/4 = (\pi - |x|)^2/4 - \pi^2/4 + \pi^2/6$, yields

$$\boxed{\varphi_2(x) = \frac{(\pi - |x|)^2}{4} - \frac{\pi^2}{12}, \qquad -\pi \le x \le \pi,}$$

which gives exactly the two requested branches ($|x| = -x$ for $x \le 0$ and $|x| = x$ for $x \ge 0$). The derivation covered every $0 \le x \le \pi$, and evenness every $-\pi \le x \le 0$, so the endpoints $x = 0, \pm\pi$ are included; in particular setting $x = 0$ proves $\sum n^{-2} = \varphi_2(0) = \pi^2/6$.

19.1.15 Given

$$\psi_{2s}(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n^{2s}}, \qquad \psi_{2s+1}(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{n^{2s+1}},$$

develop the following recurrence relations:

(a) $\psi_{2s}(x) = \displaystyle\int_0^x \psi_{2s-1}(t)\, dt,$

(b) $\psi_{2s+1}(x) = \zeta(2s+1) - \displaystyle\int_0^x \psi_{2s}(t)\, dt.$

Note. The functions $\psi_s(x)$ and $\varphi_s(x)$ of this and the preceding exercise are known as *Clausen functions*. In theory they may be used to improve the rate of convergence of a Fourier series. As is often the case, there is the question of how much analytical work we do and how much arithmetic work we demand that a computer do. As computers become steadily more powerful, the balance progressively shifts so that we are doing less and demanding that computers do more. Here $s = 1, 2, \ldots$. When $s = 1$, the first integral is improper at multiples of 2π crossed by the interval.

Solution.

Let s be a positive integer, as the statement requires: for $s = 0$, (b) would contain the divergent $\zeta(1) = \sum_n 1/n$, and (a) the divergent series $\psi_{-1} = \sum_n n \cos nx$. The statement writes the integration variable as t, since x is already the upper limit. For $s \ge 2$, the terms of the series for ψ_{2s-1} are bounded by $n^{-(2s-1)} \le n^{-3}$, so it converges absolutely and uniformly (Weierstrass M-test) and may be integrated term by term, with $\int_0^x \cos nt\, dt = \sin nx/n$. Thus

$$\int_0^x \psi_{2s-1}(t)\, dt = \sum_{n=1}^{\infty} \frac{1}{n^{2s-1}} \int_0^x \cos nt\, dt = \sum_{n=1}^{\infty} \frac{\sin nx}{n^{2s}} = \boxed{\psi_{2s}(x)}.$$

For every $s \ge 1$, the even-index series has terms bounded by $n^{-2s} \le n^{-2}$, so it too converges absolutely and uniformly. Termwise integration with $\int_0^x \sin nt\, dt = (1 - \cos nx)/n$, and splitting into two absolutely convergent series ($2s + 1 \ge 3$), gives

$$\begin{aligned}\int_0^x \psi_{2s}(t)\, dt &= \sum_{n=1}^{\infty} \frac{1 - \cos nx}{n^{2s+1}} = \sum_{n=1}^{\infty} \frac{1}{n^{2s+1}} - \sum_{n=1}^{\infty} \frac{\cos nx}{n^{2s+1}} \\ &= \zeta(2s+1) - \psi_{2s+1}(x).\end{aligned}$$

Solving for $\psi_{2s+1}(x)$ proves the second recurrence.

The first recurrence at $s = 1$ has an integrable logarithmic singularity at its lower endpoint: $\psi_1(t) = -\ln(2|\sin(t/2)|)$. The integral must be read as improper at every multiple of 2π it crosses; ψ_1 itself is not finite at those points. To justify this base case, insert r^n with $0 < r < 1$. Then $\sum_{n\geq 1} r^n \cos nt/n = \operatorname{Re}\{-\log(1 - re^{it})\} = -\ln|1 - re^{it}|$, and this series is uniformly convergent (its terms are bounded by r^n), so termwise integration gives

$$\int_0^x -\ln|1 - re^{it}|\,dt = \sum_{n=1}^{\infty} \frac{r^n \sin nx}{n^2}.$$

The right side tends uniformly to $\psi_2(x)$ as $r \uparrow 1$, because the difference is bounded by $\sum_n (1-r^n)/n^2$, whose terms tend to zero and are dominated by n^{-2}. On the left, for $t \notin 2\pi\mathbb{Z}$ the integrand tends to $-\ln|1 - e^{it}| = -\ln(2|\sin(t/2)|) = \psi_1(t)$. To dominate it, use

$$|1 - re^{it}|^2 = 1 - 2r\cos t + r^2 = (1-r)^2 + 4r\sin^2(t/2).$$

For $1/2 \leq r < 1$ this gives $2\sin^2(t/2) \leq |1 - re^{it}|^2 < 4$ (for the lower bound drop $(1-r)^2 \geq 0$ and use $4r \geq 2$; for the upper bound use $|1 - re^{it}| \leq 1 + r < 2$). Taking half the logarithm,

$$\tfrac{1}{2}\ln 2 + \ln|\sin(t/2)| \leq \ln|1 - re^{it}| < \ln 2,$$

and since $\ln|\sin(t/2)| \leq 0$ this gives $|\ln|1 - re^{it}|| \leq \ln 2 + |\ln|\sin(t/2)||$, independently of r. Let $d = \operatorname{dist}(t, 2\pi\mathbb{Z}) \in [0, \pi]$, so that $|\sin(t/2)| = \sin(d/2)$. By Jordan's inequality ($\sin u \geq 2u/\pi$ on $[0, \pi/2]$), $\sin(d/2) \geq d/\pi$, hence $|\ln|\sin(t/2)|| \leq \ln\pi - \ln d \leq \ln\pi + |\ln d|$. So on any bounded interval the integrand is bounded by $\ln(2\pi) + |\ln d|$, which is integrable near the finitely many singularities because $\int_0^\epsilon |\ln t|\,dt = \epsilon(1 - \ln\epsilon)$ for $0 < \epsilon < 1$. Dominated convergence therefore lets $r \uparrow 1$ under the integral, which proves the first recurrence also for $s = 1$.

19.1.16 Show that $f(x) = \displaystyle\sum_{n=1}^{\infty} \frac{\cos nx}{n+1}$ may be written as

$$f(x) = \psi_1(x) - \varphi_2(x) + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2(n+1)},$$

where $\psi_1(x)$ and $\varphi_2(x)$ are the Clausen functions defined in Exercises 19.1.14 and 19.1.15. For the finite-valued identity take $x \notin 2\pi\mathbb{Z}$.

Solution.

The elementary coefficient identity, over the common denominator $n^2(n+1)$, is

$$\frac{1}{n} - \frac{1}{n^2} + \frac{1}{n^2(n+1)} = \frac{n(n+1) - (n+1) + 1}{n^2(n+1)} = \frac{n^2}{n^2(n+1)} = \frac{1}{n+1}.$$

Multiply by $\cos nx$, sum first to a finite N, and then pass to the limit; for $x \notin 2\pi\mathbb{Z}$ each of the three series converges, so the limit of the sum is the sum of the limits. The first series is $\psi_1 = \sum \cos nx/n$ (the case $s = 0$ of ψ_{2s+1} in 19.1.15), the second is $\varphi_2 = \sum \cos nx/n^2$ of 19.1.14, and the last is absolutely convergent. Therefore

$$\boxed{f(x) = \psi_1(x) - \varphi_2(x) + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2(n+1)}.}$$

This equality is an ordinary finite-valued identity only for $x \notin 2\pi\mathbb{Z}$. At a multiple of 2π both f and ψ_1 diverge. Away from those points, Dirichlet's test proves convergence of the original series and of ψ_1, as in 19.1.5. The correction series has a simple uniform truncation bound: $|\cos nx|/(n^2(n+1)) < n^{-3}$, and $n^{-3} \le \int_{n-1}^{n} t^{-3}\,dt$, so

$$\left|\sum_{n>N} \frac{\cos nx}{n^2(n+1)}\right| \le \sum_{n>N} \frac{1}{n^3} \le \int_N^\infty \frac{dt}{t^3} = \frac{1}{2N^2}.$$

Thus the known closed forms of the first two terms, for $0 < x < 2\pi$,

$$\psi_1(x) = -\ln\Big(2\sin\frac{x}{2}\Big), \qquad \varphi_2(x) = \frac{(\pi - x)^2}{4} - \frac{\pi^2}{12},$$

leave a rapidly convergent remainder. The first is 19.1.8(b); the second is 19.1.14 on $(0, \pi]$, extended to $[\pi, 2\pi)$ by $\varphi_2(x) = \varphi_2(x - 2\pi) = \varphi_2(2\pi - x)$ (periodicity and evenness), which reproduces the same formula.

19.2 Applications of Fourier Series

19.2.1 Transform the Fourier expansion of a square wave, Eq. (19.26), into a power series (formally: expand each sine in powers of x and collect like powers; height $h \neq 0$). Show that the coefficients of x^1 form a divergent series. Repeat for the coefficients of x^3.

Note. A power series cannot handle a discontinuity. These infinite coefficients are the result of attempting to beat this basic limitation on power series.

Solution.

Equation (19.26) is

$$f(x) = \frac{h}{2} + \frac{2h}{\pi}\sum_{j=0}^{\infty} \frac{\sin((2j+1)x)}{2j+1}.$$

For each individual term its entire Taylor expansion is

$$\frac{\sin((2j+1)x)}{2j+1} = x - \frac{(2j+1)^2 x^3}{3!} + \frac{(2j+1)^4 x^5}{5!} - \cdots.$$

Retain only the first N harmonics before collecting powers. Each harmonic contributes $2h/\pi$ to the coefficient of x, so the linear coefficient is $2hN/\pi$. For the cubic coefficient expand $(2j+1)^2 = 4j^2 + 4j + 1$ and use $\sum_{j=0}^{N-1} j^2 = (N-1)N(2N-1)/6$, $\sum_{j=0}^{N-1} j = N(N-1)/2$:

$$\begin{aligned} -\frac{2h}{6\pi}\sum_{j=0}^{N-1}(2j+1)^2 &= -\frac{h}{3\pi}\left[4\frac{(N-1)N(2N-1)}{6} + 4\frac{N(N-1)}{2} + N\right] \\ &= -\frac{h}{3\pi}\,\frac{N(4N^2-1)}{3} = -\frac{hN(4N^2-1)}{9\pi}, \end{aligned}$$

since the bracket is $[(4N^3 - 6N^2 + 2N) + (6N^2 - 6N) + 3N]/3 = (4N^3 - N)/3$. For nonzero h neither coefficient has a finite limit as $N \to \infty$; their magnitudes grow as N and N^3, respectively. Formally, the coefficient of x^{2k+1} would be

$$\frac{2h}{\pi}\frac{(-1)^k}{(2k+1)!}\sum_{j=0}^{\infty}(2j+1)^{2k},$$

and it diverges for every $k \geq 0$, because every term of the sum is at least 1. The textbook's request to transform the Fourier series into a power series is therefore only formal, not a valid convergent power-series rearrangement. Interchanging the harmonic sum and Taylor sum is precisely the invalid step. A convergent power series on a neighborhood of zero is continuous there, so it cannot reproduce the jump between 0 and h. Away from jumps the square wave is locally constant and has a perfectly valid local constant Taylor series; the obstruction concerns a single series spanning the discontinuity.

19.2.2 Derive the Fourier series expansion of the Dirac delta function $\delta(x)$ in the interval $-\pi < x < \pi$.

(a) What significance can be attached to the constant term?

(b) In what region is this representation valid?

(c) With the identity

$$\sum_{n=1}^{N} \cos nx = \frac{\sin(Nx/2)}{\sin(x/2)} \cos\left[(N+1)\frac{x}{2}\right],$$

show that your Fourier representation of $\delta(x)$ is consistent with Eq. (5.27).

Interpret the expansion distributionally. In the finite-sum identity use the continuous limiting value where $\sin(x/2) = 0$.

Solution.

Apply the coefficient formulas to the distribution on $(-\pi, \pi)$, using the sampling rule $\int_{-\pi}^{\pi} \delta(x) g(x)\, dx = g(0)$:

$$a_0 = \frac{1}{\pi}\int \delta(x)\, dx = \frac{1}{\pi}, \quad a_n = \frac{1}{\pi}\int \delta(x) \cos nx\, dx = \frac{\cos 0}{\pi} = \frac{1}{\pi}, \quad b_n = \frac{\sin 0}{\pi} = 0.$$

Thus, with $a_0/2 = 1/(2\pi)$,

$$\boxed{\delta_{2\pi}(x) = \frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty} \cos nx.}$$

(a) The constant $1/(2\pi)$ is the mean density over one period. All other terms integrate to zero over $(-\pi, \pi)$; this term alone ensures unit total mass, $\int_{-\pi}^{\pi} dx/(2\pi) = 1$.

(b) The equality is distributional, and its global extension is periodic: $\delta_{2\pi}(x) = \sum_{k\in\mathbb{Z}} \delta(x - 2\pi k)$. It represents the single $\delta(x)$ within $(-\pi, \pi)$, not a single isolated delta on the whole real line. The series cannot be interpreted as an ordinary pointwise function.

(c) The textbook's finite-sum identity has an incorrect angle: its cosine must be $\cos((N+1)x/2)$, not $\cos((N+1/2)x/2)$, as the geometric sum below shows. Indeed, factoring $e^{iNx/2}$ from the numerator and $e^{ix/2}$ from the denominator, and using $e^{-iu} - e^{iu} = -2i \sin u$, the geometric sum is

$$\sum_{n=1}^{N} e^{inx} = e^{ix}\frac{1 - e^{iNx}}{1 - e^{ix}} = e^{ix}\,\frac{e^{iNx/2}(e^{-iNx/2} - e^{iNx/2})}{e^{ix/2}(e^{-ix/2} - e^{ix/2})}$$

$$= e^{i(N+1)x/2}\frac{\sin(Nx/2)}{\sin(x/2)}.$$

Taking real parts proves the corrected identity. Add the constant term and use $2\sin A\cos B = \sin(A+B)+\sin(A-B)$ with $A = Nx/2$, $B = (N+1)x/2$, so that $A+B = (N+1/2)x$ and $A-B = -x/2$. This yields

$$1+2\sum_{n=1}^{N}\cos nx = \frac{\sin(x/2)+\sin((N+1/2)x)-\sin(x/2)}{\sin(x/2)}$$
$$= \frac{\sin((N+1/2)x)}{\sin(x/2)} = D_N(x).$$

The quotient has the removable value $D_N(0) = 2N+1$ (the left side at $x=0$, or the limit $(N+1/2)/(1/2)$ of the quotient). Its normalized integral is one, $(2\pi)^{-1}\int_{-\pi}^{\pi} D_N\,dx = 1$, since each $\cos nx$ integrates to zero. Moreover, since $D_N(x) = \sum_{n=-N}^{N} e^{inx}$, for a periodic f with exponential Fourier coefficients c_n (as in 19.1.11)

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)D_N(x)\,dx = \sum_{n=-N}^{N} c_{-n} = \sum_{n=-N}^{N} c_n,$$

the symmetric partial sum of the Fourier series of f at $x=0$; in particular $D_N/(2\pi)$ reproduces the value at 0 of every Fourier polynomial of degree at most N. For smooth periodic f these partial sums tend to $f(0)$ as $N\to\infty$, as shown in 19.1.11. This is the delta limit.

Specifically Eq. (5.27) is the closure relation $\delta(x-t) = \sum_j \phi_j(x)\phi_j^*(t)$. For the orthonormal functions $1/\sqrt{2\pi}$, $\cos nx/\sqrt{\pi}$, $\sin nx/\sqrt{\pi}$, setting $t=0$ gives $1/(2\pi)+\pi^{-1}\sum_{n\ge1}(\cos nx\cos 0+\sin nx\sin 0)$, exactly the boxed formula. Thus the finite kernel and the general bilinear closure relation agree.

19.2.3 Expand $\delta(x-t)$ in a Fourier series. Compare your result with the bilinear form of Eq. (5.27).

ANS. $\delta(x-t) = \dfrac{1}{2\pi}+\dfrac{1}{\pi}\sum_{n=1}^{\infty}(\cos nx\cos nt+\sin nx\sin nt)$

$= \dfrac{1}{2\pi}+\dfrac{1}{\pi}\sum_{n=1}^{\infty}\cos n(x-t)$. Use the periodic distribution on $(-\pi,\pi)$, with t in that interval, or equivalently work on the circle.

Solution.

For t strictly inside $(-\pi,\pi)$, the delta's sampling rule $\int_{-\pi}^{\pi}\delta(x-t)g(x)\,dx = g(t)$, applied to $g = 1$, $\cos nx$, $\sin nx$ in the coefficient formulas, gives

$$a_0(t) = \frac{1}{\pi}, \qquad a_n(t) = \frac{\cos nt}{\pi}, \qquad b_n(t) = \frac{\sin nt}{\pi}.$$

Substitution into $a_0/2+\sum_{n\ge1}(a_n\cos nx+b_n\sin nx)$ gives

$$\delta_{2\pi}(x-t) = \frac{1}{2\pi}+\frac{1}{\pi}\sum_{n=1}^{\infty}(\cos nx\cos nt+\sin nx\sin nt)$$
$$= \boxed{\frac{1}{2\pi}+\frac{1}{\pi}\sum_{n=1}^{\infty}\cos n(x-t)}.$$

The last equality is the cosine subtraction identity $\cos nx\cos nt+\sin nx\sin nt = \cos n(x-t)$, applied first to finite sums. With the real orthonormal basis $1/\sqrt{2\pi}$, $\cos nx/\sqrt{\pi}$, $\sin nx/\sqrt{\pi}$ from 19.2.2,

the first line is precisely $\sum_j \phi_j(x)\phi_j^*(t)$ in Eq. (5.27), because $(1/\sqrt{2\pi})(1/\sqrt{2\pi}) = 1/(2\pi)$ and $(\cos nx/\sqrt{\pi})(\cos nt/\sqrt{\pi}) = \pi^{-1}\cos nx\cos nt$, and similarly for the sines. For example, integrating against $\phi_k(x)$ leaves, by orthonormality, only the kth term and gives $\phi_k(t)$; by the convergent Fourier expansion this extends to smooth test functions. The periodic/distributional and endpoint qualifications of 19.2.2 apply equally to the translated kernel.

19.2.4 Show that integrating the Fourier expansion of the Dirac delta function (Exercise 19.2.2) leads to the Fourier representation of the square wave, Eq. (19.26), with $h = 1$.

Note. Integrating the constant term $(1/2\pi)$ leads to a term $x/2\pi$. What are you going to do with this?

Solution.

Let $H(x)$ be zero for $-\pi < x < 0$ and one for $0 < x < \pi$. Within this open interval, $H' = \delta(x)$ and the symmetric value at zero is $H(0) = 1/2$. Integrate the delta Fourier series from zero term by term,

$$\int_0^x \Big[\frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\cos nt\Big]\,dt = \frac{x}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\sin nx}{n},$$

and add $H(0) = 1/2$. This gives

$$H(x) = \frac{1}{2} + \frac{x}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\sin nx}{n}, \qquad -\pi < x < \pi.$$

As a check, by 19.1.5 the sine series is $(\pi - x)/2$ on $(0, \pi)$ and $-(\pi + x)/2$ on $(-\pi, 0)$, so the right side is $1/2 + x/(2\pi) + (\pi - x)/(2\pi) = 1$ and $1/2 + x/(2\pi) - (\pi + x)/(2\pi) = 0$, respectively. The term $x/(2\pi)$ of the note must be retained; the answer to the note's question is to replace it by its own Fourier series, the sawtooth series $x = 2\sum_{n\ge1}(-1)^{n+1}\sin nx/n$ of 19.2.7, that is, $x/(2\pi) = \pi^{-1}\sum_{n\ge1}(-1)^{n+1}\sin nx/n$:

$$\begin{aligned} H(x) &= \frac{1}{2} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{1 + (-1)^{n+1}}{n}\sin nx \\ &= \boxed{\frac{1}{2} + \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{\sin((2j+1)x)}{2j+1}}. \end{aligned}$$

Since $1 + (-1)^{n+1}$ is 0 for even n and 2 for odd $n = 2j + 1$, the even terms cancel and the odd terms double, which is Eq. (19.26) with $h = 1$.

There is also a direct way to keep the periodic boundary bookkeeping. Integrate the series of 19.2.3 over $0 < t < \pi$. The constant term gives $\pi/(2\pi) = 1/2$, and with $u = x - t$, $\int_0^{\pi}\cos n(x-t)\,dt = \int_{x-\pi}^{x}\cos nu\,du = [\sin nx - \sin(n(x-\pi))]/n$, and $\sin(n(x-\pi)) = (-1)^n\sin nx$:

$$\begin{aligned} H(x) &= \int_0^{\pi}\delta_{2\pi}(x - t)\,dt \\ &= \frac{1}{2} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\sin nx - \sin(n(x-\pi))}{n} \\ &= \frac{1}{2} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{1 - (-1)^n}{n}\sin nx. \end{aligned}$$

For $0 < x < \pi$ the delta at $t = x$ lies in $(0, \pi)$ and the integral is 1; for $-\pi < x < 0$ the comb members at $t = x$ and $t = x + 2\pi$ both lie outside $(0, \pi)$ and it is 0. The coefficient $1 - (-1)^n$ again keeps only odd n, with value 2, reproducing the boxed series. The Instructor's Manual's final negative-x branch is printed as 1; it must be 0. Also the derivative of the periodic step is $\delta_{2\pi}(x) - \delta_{2\pi}(x - \pi)$, including the downward boundary jump. Integrating a single periodic delta produces the linear term above, not by itself a periodic primitive. At all jump points the Fourier series takes the midpoint 1/2.

19.2.5 Starting from the Fourier series given as lines 4 and 5 of Table 19.1, show that:

$$\sum_{n=0}^{\infty} \frac{\cos(2n+1)x}{2n+1} = \frac{1}{2} \ln \left| \cot \frac{x}{2} \right|.$$

Exclude $x \in \pi\mathbb{Z}$.

Solution.

On $0 < |x| < \pi$, the two logarithmic series in Table 19.1 read as follows; they are 19.1.8(b), extended to $-\pi < x < 0$ because both sides are even, and 19.1.8(a) multiplied by -1:

$$A(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{n} = -\ln(2|\sin(x/2)|), \qquad B(x) = \sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n} = -\ln(2\cos(x/2)).$$

In the Instructor's Manual's intermediate first sum, the denominator printed as x must be n, and that sum starts at $n = 1$. Subtracting the two convergent series term by term eliminates every even harmonic, since $1 - (-1)^n$ is 0 for even n and 2 for odd n:

$$A - B = \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \cos nx = 2 \sum_{j=0}^{\infty} \frac{\cos((2j+1)x)}{2j+1}.$$

The corresponding logarithmic difference is, since $\cos(x/2) > 0$ for $|x| < \pi$,

$$\ln(2\cos(x/2)) - \ln(2|\sin(x/2)|) = \ln \frac{\cos(x/2)}{|\sin(x/2)|} = \ln|\cot(x/2)|.$$

Dividing $A - B$ by 2 gives

$$\boxed{\sum_{j=0}^{\infty} \frac{\cos((2j+1)x)}{2j+1} = \frac{1}{2} \ln|\cot(x/2)|.}$$

Exclude $x \in \pi\mathbb{Z}$. At even multiples of π the series diverges to $+\infty$; at odd multiples it diverges to $-\infty$. Elsewhere its ordinary convergence follows from Dirichlet's test for $\sum e^{i(2j+1)x}$, whose geometric ratio is $e^{2ix} \neq 1$. Both sides have period 2π, so periodicity extends the displayed identity to every other real $x \notin \pi\mathbb{Z}$.

19.2.6 Develop the Fourier series representation of

$$f(t) = \begin{cases} 0, & -\pi \le \omega t \le 0, \\ \sin \omega t, & 0 \le \omega t \le \pi. \end{cases}$$

This is the output of a simple half-wave rectifier. It is also an approximation of the solar thermal effect that produces "tides" in the atmosphere.

$$\textit{ANS. } f(t) = \frac{1}{\pi} + \frac{1}{2}\sin\omega t - \frac{2}{\pi}\sum_{n=2,4,6,\ldots} \frac{\cos n\omega t}{n^2-1}.$$

Solution.

Put $x = \omega t$ and use the 2π-periodic extension. Only the positive half-period contributes, so

$$a_0 = \frac{1}{\pi}\int_0^\pi \sin x\,dx = \frac{2}{\pi}.$$

Orthogonality of sines on $(0,\pi)$ gives

$$b_n = \frac{1}{\pi}\int_0^\pi \sin x \sin nx\,dx = \frac{1}{2\pi}\int_0^\pi [\cos((n-1)x) - \cos((n+1)x)]\,dx = \frac{1}{2}\delta_{n1},$$

because for $n \geq 2$ both cosines integrate to zero, while for $n = 1$ the integrand is $1 - \cos 2x$, with integral π. For $n \neq 1$, the product-to-sum identity $\sin x \cos nx = \frac{1}{2}[\sin((n+1)x) + \sin((1-n)x)]$ and $\int_0^\pi \sin mx\,dx = [1-(-1)^m]/m$ $(m \neq 0)$ give

$$\begin{aligned} a_n &= \frac{1}{2\pi}\int_0^\pi [\sin((n+1)x) + \sin((1-n)x)]\,dx \\ &= \frac{1}{2\pi}\left[\frac{1-(-1)^{n+1}}{n+1} + \frac{1-(-1)^{1-n}}{1-n}\right]. \end{aligned}$$

For odd $n > 1$ both $n+1$ and $1-n$ are even, so both numerators vanish. For even n both are odd and both numerators equal 2, so

$$a_n = \frac{1}{\pi}\left(\frac{1}{n+1} + \frac{1}{1-n}\right) = \frac{1}{\pi}\,\frac{(1-n)+(n+1)}{(n+1)(1-n)} = -\frac{2}{\pi(n^2-1)}.$$

The exceptional coefficient is also zero: $a_1 = (2\pi)^{-1}\int_0^\pi \sin 2x\,dx = 0$. Consequently, with $a_0/2 = 1/\pi$ and $n = 2j$, $n^2 - 1 = 4j^2 - 1$,

$$\boxed{f(t) = \frac{1}{\pi} + \frac{1}{2}\sin\omega t - \frac{2}{\pi}\sum_{j=1}^{\infty}\frac{\cos(2j\omega t)}{4j^2-1}.}$$

The coefficients of the infinite remainder are $O(j^{-2})$, so the series converges absolutely and uniformly, including the joining points where the rectified waveform is continuous and zero.

19.2.7 A sawtooth wave is given by

$$f(x) = x, \qquad -\pi < x < \pi.$$

Show that

$$f(x) = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx.$$

Solution.

Oddness gives $a_0 = a_n = 0$, since x and $x\cos nx$ are odd. The integrand $x\sin nx$ is even, so with the antiderivative of 19.1.9, $\sin n\pi = 0$ and $\cos n\pi = (-1)^n$, the sine coefficient is

$$\begin{aligned} b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin nx\,dx = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx \\ &= \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} = \frac{2}{\pi}\left(-\frac{\pi(-1)^n}{n}\right) = \frac{2(-1)^{n+1}}{n}. \end{aligned}$$

Thus

$$\boxed{x = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx, \qquad -\pi < x < \pi.}$$

The periodic extension is piecewise smooth; its Fourier sum agrees with x at every point of this open interval. At $x = \pm\pi$ the periodic extension jumps from π to $-\pi$, and every sine term vanishes, giving the midpoint zero.

19.2.8 A different sawtooth wave is described by

$$f(x) = \begin{cases} -\dfrac{1}{2}(\pi + x), & -\pi \le x < 0, \\ +\dfrac{1}{2}(\pi - x), & 0 < x \le \pi. \end{cases}$$

Show that $f(x) = \displaystyle\sum_{n=1}^{\infty}(\sin nx/n)$.

Solution.

The function is odd, so only sine terms remain. For $n \ge 1$, using $\int_0^{\pi}\sin nx\,dx = [1-(-1)^n]/n$ and $\int_0^{\pi} x\sin nx\,dx = -\pi(-1)^n/n$ from 19.2.7,

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi}\frac{\pi - x}{2}\sin nx\,dx \\ &= \int_0^{\pi}\sin nx\,dx - \frac{1}{\pi}\int_0^{\pi} x\sin nx\,dx \\ &= \frac{1-(-1)^n}{n} - \frac{1}{\pi}\left(-\frac{\pi(-1)^n}{n}\right) = \frac{1-(-1)^n+(-1)^n}{n} = \frac{1}{n}. \end{aligned}$$

The Instructor's Manual's last integral in its coefficient calculation prints $x\sin x$; it must be $x\sin nx$. The n-dependence is retained above. Therefore

$$\boxed{f(x) = \sum_{n=1}^{\infty}\frac{\sin nx}{n}}$$

throughout the stated intervals. At zero the two limiting values are $\pm\pi/2$, and the Fourier value is zero. At $\pm\pi$ the function and the series are both zero. This coefficient calculation independently confirms the logarithmic summation in 19.1.5.

19.2.9 A triangular wave (Fig. 19.4) is represented by

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0. \end{cases}$$

Represent $f(x)$ by a Fourier series.

$$\textit{ANS. } f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\ldots} \frac{\cos nx}{n^2}.$$

Solution.

Here $f(x) = |x|$ is even, so $b_n = 0$ and the cosine integrals may be taken over $(0, \pi)$ and doubled. Its constant coefficient is

$$a_0 = \frac{2}{\pi} \int_0^\pi x \, dx = \frac{2}{\pi} \frac{\pi^2}{2} = \pi.$$

For $n \geq 1$, integration by parts (the antiderivative of 19.1.9, with $\sin n\pi = 0$, $\cos n\pi = (-1)^n$) gives

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi x \cos nx \, dx \\ &= \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^\pi = \frac{2[(-1)^n - 1]}{\pi n^2}. \end{aligned}$$

Thus the even coefficients vanish, while each odd one is $-4/(\pi n^2)$; with $a_0/2 = \pi/2$,

$$\boxed{|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{j=0}^{\infty} \frac{\cos((2j+1)x)}{(2j+1)^2}, \qquad -\pi \leq x \leq \pi.}$$

The periodic triangular wave is continuous, including at the period endpoints, and piecewise smooth, so by Dirichlet's theorem its Fourier series converges to it at every point; the corners at $x = 0$ and $x = \pm\pi$, where the slope jumps between -1 and $+1$, do not obstruct equality. The series converges absolutely and uniformly by comparison with $\sum n^{-2}$.

19.2.10 Expand

$$f(x) = \begin{cases} 1, & x^2 < x_0^2 \\ 0, & x^2 > x_0^2 \end{cases}$$

in the interval $[-\pi, \pi]$.

Note. This variable-width square wave is of some importance in electronic music. Take x_0 real and include zero width and full width; at a jump, state the Fourier midpoint value.

Solution.

Because the definition involves x_0^2, write $\alpha = |x_0|$. First assume $0 < \alpha < \pi$. The waveform depends only on x^2, so it is even, giving $b_n = 0$; on $(0, \pi)$ it is 1 for $x < \alpha$ and 0 for $x > \alpha$. Hence

$$a_0 = \frac{2}{\pi} \int_0^\alpha 1 \, dx = \frac{2\alpha}{\pi}, \qquad a_n = \frac{2}{\pi} \int_0^\alpha \cos nx \, dx = \frac{2}{\pi} \left[\frac{\sin nx}{n} \right]_0^\alpha = \frac{2 \sin n\alpha}{n\pi},$$

and, with $a_0/2 = \alpha/\pi$,

$$\boxed{f(x) = \frac{\alpha}{\pi} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\alpha}{n} \cos nx.}$$

The constant term is the duty fraction: the pulse occupies length 2α of a period 2π. The series is one or zero in the open constant regions. At the two jumps $x = \pm\alpha$ its value is the midpoint $1/2$: with $2\sin n\alpha\cos n\alpha = \sin 2n\alpha$ and 19.1.5 for $0 < 2\alpha < 2\pi$,

$$\frac{\alpha}{\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\sin 2n\alpha}{n} = \frac{\alpha}{\pi} + \frac{1}{\pi}\,\frac{\pi - 2\alpha}{2} = \frac{1}{2}.$$

The usual variable-width formula presumes $0 < |x_0| < \pi$. For $|x_0| \geq \pi$ the function is one almost everywhere in the interval; for $x_0 = 0$ it is zero almost everywhere. The equality points $x^2 = x_0^2$ in the piecewise definition can be assigned arbitrarily without changing any coefficient. One formula for all real x_0 is obtained by replacing α by $\min(|x_0|, \pi)$ above: it reduces to the appropriate constant series in the two limiting cases, since $\alpha = \pi$ gives $\alpha/\pi = 1$ and $\sin n\pi = 0$, and $\alpha = 0$ makes every term zero.

19.2.11 A metal cylindrical tube of radius a is split lengthwise into two nontouching halves. The top half is maintained at a potential $+V$, the bottom half at a potential $-V$. See Fig. 19.9. Separate the variables in Laplace's equation and solve for the electrostatic potential for $r \leq a$. Observe the resemblance between your solution for $r = a$ and the Fourier series for a square wave. Assume an infinitely long tube, negligible slit width and no axial variation. Impose the boundary values away from the two ideal slit lines.

Solution.

Use the ideal infinitely long tube, with translational symmetry along its axis and negligible slit width. Finite end effects or a finite open gap require additional boundary data. At the ideal slit endpoints the two assigned boundary limits disagree, so a single boundary value cannot be imposed there. Take $0 < \varphi < \pi$ to be the top half. With no dependence on the axial coordinate, Laplace's equation in the interior, in polar coordinates, is

$$\Phi_{rr} + \frac{1}{r}\Phi_r + \frac{1}{r^2}\Phi_{\varphi\varphi} = 0.$$

With $\Phi = R(r)\Theta(\varphi)$, multiplication by $r^2/(R\Theta)$ separates this into a function of r equal to a function of φ, hence to a constant, written n^2:

$$\frac{r^2R'' + rR'}{R} = -\frac{\Theta''}{\Theta} = n^2.$$

The angular equation $\Theta'' + n^2\Theta = 0$ has 2π-periodic (single-valued) solutions only for integer $n \geq 0$, namely $\Theta = \cos n\varphi, \sin n\varphi$ (for $n = 0$ the general solution $A + B\varphi$ changes by $2\pi B$ around the axis, so periodicity forces $B = 0$ and leaves the constant); a negative constant would give nonperiodic exponentials. The radial equation $r^2R'' + rR' - n^2R = 0$ is of Euler type: $R = r^k$ gives $k^2 = n^2$, so $R = r^n, r^{-n}$ for $n \geq 1$, while for $n = 0$ it reads $(rR')' = 0$, with solutions $R = 1, \ln r$. Boundedness at the axis excludes r^{-n} and $\ln r$, leaving

$$\Phi(r, \varphi) = C_0 + \sum_{n=1}^{\infty}(r/a)^n(C_n\cos n\varphi + D_n\sin n\varphi).$$

At $r = a$ this must be the Fourier series of the boundary function, V on $(0, \pi)$ and $-V$ on $(-\pi, 0)$. That function is odd and has zero mean, so $C_0 = C_n = 0$. Its sine coefficients, with the even integrand reduced to $(0, \pi)$ and $\int_0^\pi \sin n\varphi\, d\varphi = [1 - (-1)^n]/n$, are

$$D_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\Phi(a, \varphi)\sin n\varphi\, d\varphi = \frac{2}{\pi}\int_0^\pi V\sin n\varphi\, d\varphi = \frac{2V}{n\pi}[1 - (-1)^n],$$

that is, $4V/(n\pi)$ for odd n and 0 for even n. Therefore

$$\boxed{\Phi(r,\varphi) = \frac{4V}{\pi}\sum_{j=0}^{\infty}\frac{(r/a)^{2j+1}}{2j+1}\sin((2j+1)\varphi), \qquad r<a.}$$

The series and all differentiated series converge uniformly on compact subdisks, so it is harmonic. As $r\uparrow a$, it becomes the $\pm V$ square-wave series and attains the prescribed values away from the slits. Its radial limit on a slit is the midpoint zero.

For an additional closed-form check put $q=r/a<1$ and $z=qe^{i\varphi}$, so that the boxed series is $(4V/\pi)\operatorname{Im}\sum_{j\ge0} z^{2j+1}/(2j+1)$. Integrate the geometric series $\sum_{j\ge0} z^{2j} = 1/(1-z^2)$ from 0 to z, using the partial fractions $1/(1-z^2) = \frac{1}{2}[1/(1+z)+1/(1-z)]$:

$$\sum_{j=0}^{\infty}\frac{z^{2j+1}}{2j+1} = \frac{1}{2}\big[\log(1+z)-\log(1-z)\big] = \frac{1}{2}\log\frac{1+z}{1-z},$$

with principal logarithms; $1\pm z$ lie in the right half-plane, so their arguments lie in $(-\pi/2,\pi/2)$ and subtract without a 2π correction. Multiplying numerator and denominator by $1-\bar z$, and using $z-\bar z = 2iq\sin\varphi$,

$$\frac{1+z}{1-z} = \frac{(1+z)(1-\bar z)}{|1-z|^2} = \frac{1-q^2+2iq\sin\varphi}{|1-z|^2},$$

whose real part is positive. Thus

$$\Phi = \frac{2V}{\pi}\operatorname{Arg}\frac{1+z}{1-z} = \frac{2V}{\pi}\arctan\frac{2q\sin\varphi}{1-q^2},$$

and multiplying numerator and denominator by a^2 gives

$$\boxed{\Phi(r,\varphi) = \frac{2V}{\pi}\arctan\frac{2ar\sin\varphi}{a^2-r^2}, \qquad r<a.}$$

As $r\uparrow a$ the argument of the arctangent tends to $+\infty$ where $\sin\varphi>0$ and to $-\infty$ where $\sin\varphi<0$, so the limit is $+V$ above and $-V$ below. The bounded harmonic Dirichlet solution is unique: the difference of two such solutions has zero boundary data almost everywhere and its Poisson representation is zero.

19.2.12 A metal cylinder is placed in a (previously) uniform electric field, $\mathbf{E}_0$, with the axis of the cylinder perpendicular to that of the original field.

(a) Find the perturbed electrostatic potential.

(b) Find the induced surface charge on the cylinder as a function of angular position.

Assume an infinitely long neutral cylinder in vacuum. Take its radius to be a and set its constant potential to zero.

Solution.

The usual solution assumes an infinitely long, neutral cylinder in vacuum. The textbook's problem does not specify a net charge; allowing one adds a logarithmic monopole potential. Finite cylinders require end data. Let the radius be a and take the original field to be $E_0\hat{\mathbf{x}}$. Choose the conductor potential as zero, so the far-field potential is $-E_0 r\cos\varphi$.

(a) The separated exterior harmonic functions are those of 19.2.11, $1, \ln r, r^{\pm n}\cos n\varphi, r^{\pm n}\sin n\varphi$, now keeping r^{-n} and $\ln r$ because the axis is excluded. Thus

$$\Phi_{\text{out}} = A_0 + B_0 \ln r + \sum_{n=1}^{\infty}\big[(A_n r^n + B_n r^{-n})\cos n\varphi + (A'_n r^n + B'_n r^{-n})\sin n\varphi\big].$$

The applied field has only the $n = 1$ cosine component. Since the perturbation $\Phi_{\text{out}} + E_0 r\cos\varphi$ decays at infinity, $A_1 = -E_0$, every other A_n and every A'_n vanish, and $A_0 = B_0 = 0$ ($\ln r$ grows; it is the potential of a net line charge). The conductor condition $\Phi_{\text{out}}(a,\varphi) = 0$ for all φ makes every Fourier coefficient of the boundary values vanish: $B_n a^{-n} = 0$ for $n \geq 2$, $B'_n a^{-n} = 0$ for $n \geq 1$, and $-E_0 a + B_1/a = 0$, or $B_1 = E_0 a^2$. Hence

$$\boxed{\Phi_{\text{out}}(r,\varphi) = -E_0\left(r - \frac{a^2}{r}\right)\cos\varphi \quad (r \geq a), \qquad \Phi_{\text{in}} = 0.}$$

Each radial term solves the separated Laplace equation, and $\Phi_{\text{out}} + E_0 r\cos\varphi = E_0 a^2\cos\varphi/r \to 0$.

(b) Since $\partial_r(r - a^2/r) = 1 + a^2/r^2$, the outward normal field just outside the surface is

$$E_r = -\partial_r\Phi_{\text{out}} = E_0\left(1 + \frac{a^2}{r^2}\right)\cos\varphi, \qquad E_r(a^+) = 2E_0\cos\varphi.$$

Since the field inside the metal is zero, the boundary condition (Gauss's law for a thin pillbox across the surface) gives

$$\boxed{\sigma(\varphi) = \epsilon_0[E_r(a^+) - E_r(a^-)] = 2\epsilon_0 E_0\cos\varphi.}$$

Also $E_\varphi = -r^{-1}\partial_\varphi\Phi_{\text{out}} = -E_0(1 - a^2/r^2)\sin\varphi$ vanishes at $r = a$, as it must on a conductor, and the total induced charge per unit length is $a\int_0^{2\pi}\sigma\, d\varphi = 2\epsilon_0 E_0 a\int_0^{2\pi}\cos\varphi\, d\varphi = 0$, verifying neutrality. If a net line charge λ is specified instead, the more general solution contains

$$\Phi_{\text{mono}} = -\frac{\lambda}{2\pi\epsilon_0}\ln(r/a), \qquad \sigma_{\text{mono}} = \frac{\lambda}{2\pi a},$$

plus an arbitrary common potential constant. The decaying-perturbation condition used above excludes this monopole.

19.2.13

(a) Find the Fourier series representation of

$$f(x) = \begin{cases} 0, & -\pi < x \leq 0 \\ x, & 0 \leq x < \pi. \end{cases}$$

(b) From the Fourier expansion show that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.$$

Solution.

(a) Since $f = 0$ on $(-\pi, 0)$, all coefficient integrals run only from zero to π:

$$a_0 = \frac{1}{\pi}\int_0^{\pi} x\, dx = \frac{1}{\pi}\,\frac{\pi^2}{2} = \frac{\pi}{2}.$$

For $n \geq 1$, the antiderivatives of $x\cos nx$ and $x\sin nx$ from 19.1.9, with $\sin n\pi = 0$ and $\cos n\pi = (-1)^n$, give

$$a_n = \frac{1}{\pi}\int_0^{\pi} x\cos nx\,dx = \frac{1}{\pi}\left[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right]_0^{\pi} = \frac{(-1)^n - 1}{\pi n^2},$$
$$b_n = \frac{1}{\pi}\int_0^{\pi} x\sin nx\,dx = \frac{1}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} = \frac{(-1)^{n+1}}{n}.$$

Thus $a_0/2 = \pi/4$, and $a_n = -2/(\pi n^2)$ for odd n and 0 for even n:

$$\boxed{f(x) = \frac{\pi}{4} - \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{\cos((2j+1)x)}{(2j+1)^2} + \sum_{n=1}^{\infty}\frac{(-1)^{n+1}\sin nx}{n}.}$$

This also follows by splitting $f(x) = (|x| + x)/2$ and taking half the sum of the triangular expansion of 19.2.9 and the sawtooth expansion of 19.2.7. The function is piecewise smooth, so by Dirichlet's theorem the equality holds on $(-\pi, \pi)$, including zero, where f is continuous. At the period endpoints the series equals the midpoint $\pi/2$ of the jump between $f(\pi^-) = \pi$ and $f(-\pi^+) = 0$.

(b) Set $x = 0$, where $f(0) = 0$. All sine terms vanish and every cosine is one, so

$$0 = \frac{\pi}{4} - \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2}.$$

Solving, $\sum_j (2j+1)^{-2} = (\pi/2)(\pi/4)$, gives

$$\boxed{\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2} = \frac{\pi^2}{8}.}$$

19.2.14 Integrate the Fourier expansion of the unit step function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1, & 0 < x < \pi. \end{cases}$$

Show that your integrated series agrees with Exercise 19.2.13.

Solution.

The unit step has the series (Eq. (19.26) with $h = 1$; see 19.2.4)

$$H(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{\sin((2j+1)t)}{2j+1}.$$

Its integral $F(x) = \int_0^x H(t)\,dt$ is 0 for $-\pi < x < 0$, where $H = 0$ between x and 0, and x for $0 < x < \pi$; this is the function f of 19.2.13. Integrating term by term, with $\int_0^x \sin((2j+1)t)\,dt = [1 - \cos((2j+1)x)]/(2j+1)$, gives

$$F(x) = \frac{x}{2} + \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{1 - \cos((2j+1)x)}{(2j+1)^2}.$$

This step is justified in L^2: integration against the indicator of $(0, x)$ is a bounded linear functional, and the Fourier partial sums converge in L^2. The resulting series converges absolutely and uniformly, since its terms are bounded by $(4/\pi)(2j+1)^{-2}$. Splitting it into two absolutely convergent series and using $\sum_{j\ge0}(2j+1)^{-2} = \pi^2/8$ (19.2.13(b)), so that the constant part is $(2/\pi)(\pi^2/8) = \pi/4$, yields

$$F(x) = \frac{\pi}{4} + \frac{x}{2} - \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{\cos((2j+1)x)}{(2j+1)^2}.$$

Finally, on $(-\pi, \pi)$, half of the sawtooth series of 19.2.7 gives $x/2 = \sum_{n\ge1}(-1)^{n+1}\sin nx/n$. Substituting it produces exactly the constant, cosine and sine terms in 19.2.13. The integral F itself is not periodic, because of its linear term $x/2$; replacing that term by the sawtooth series supplies the required periodic extension, with midpoint values at the endpoints.

19.2.15 In the interval $(-\pi, \pi)$,

$$\delta_n(x) = \begin{cases} n, & |x| < 1/2n, \\ 0, & |x| > 1/2n. \end{cases}$$

This wave form is the pulse shown in Fig. 19.10.

(a) Expand $\delta_n(x)$ as a Fourier cosine series.

(b) Show that your Fourier series agrees with a Fourier expansion of $\delta(x)$ in the distributional limit as $n \to \infty$.

Solution.

Use a different letter m for the Fourier harmonic. The pulse has half-width $1/(2n) \le 1/2 < \pi$, so it lies inside the interval, and total area $n(1/n) = 1$. It is even, so $b_m = 0$. Since δ_n vanishes for $|x| > 1/(2n)$ and $\cos mx$ is even, its coefficients are

$$a_0 = \frac{1}{\pi}\int_{-1/(2n)}^{1/(2n)} n\,dx = \frac{1}{\pi},$$

$$a_m = \frac{1}{\pi}\int_{-1/(2n)}^{1/(2n)} n\cos mx\,dx = \frac{2n}{\pi}\int_0^{1/(2n)}\cos mx\,dx = \frac{2n}{\pi}\left[\frac{\sin mx}{m}\right]_0^{1/(2n)} = \frac{2n}{\pi m}\sin\frac{m}{2n}.$$

(a) Therefore, with $a_0/2 = 1/(2\pi)$,

$$\boxed{\delta_n(x) = \frac{1}{2\pi} + \frac{2n}{\pi}\sum_{m=1}^{\infty}\frac{\sin(m/(2n))}{m}\cos mx.}$$

By Dirichlet's theorem (the pulse is piecewise constant), for each fixed n the series converges to the pulse away from its two edges, and to the mean $(n+0)/2 = n/2$ at the edges $x = \pm 1/(2n)$.

(b) At every fixed harmonic m, the argument $u = m/(2n)$ tends to zero and $\sin u/u \to 1$, so

$$\frac{2n}{m}\sin\frac{m}{2n} = \frac{\sin(m/(2n))}{m/(2n)} \longrightarrow 1, \qquad a_m = \frac{1}{\pi}\frac{\sin(m/(2n))}{m/(2n)} \longrightarrow \frac{1}{\pi}.$$

The limiting coefficients therefore give $1/(2\pi) + (1/\pi)\sum_{m\ge1}\cos mx$, the delta expansion of 19.2.2. This is a distributional limit; taking the coefficient limit is not justification for a pointwise limit of

the infinite series. A direct proof uses any test function f continuous at zero. Since $n\int_{-1/(2n)}^{1/(2n)} dx = 1$, write $f(0) = n\int_{-1/(2n)}^{1/(2n)} f(0)\,dx$; the remaining integral runs over an interval of length $1/n$, so

$$\left|\int f(x)\delta_n(x)\,dx - f(0)\right| = \left|n\int_{-1/(2n)}^{1/(2n)} [f(x)-f(0)]\,dx\right| \le \sup_{|x|\le 1/(2n)} |f(x)-f(0)| \longrightarrow 0.$$

This proves $\delta_n \to \delta$ directly, without passing to the limit in the coefficients.

19.2.16 Confirm the delta function nature of your Fourier series of Exercise 19.2.15 by showing that for any $f(x)$ that is bounded and measurable in $[-\pi,\pi]$ and continuous at $x = 0$,

$$\lim_{n\to\infty}\lim_{M\to\infty}\int_{-\pi}^{\pi} f(x)S_{n,M}(x)\,dx = f(0).$$

Here $S_{n,M}$ is the partial sum of the Fourier series of the finite-width pulse δ_n through the harmonic $\cos Mx$.

Solution.

The displayed integrals require an integrable test function; the textbook's hypothesis that f be finite in $[-\pi,\pi]$ (pointwise finiteness) does not by itself ensure integrability. A bounded measurable f continuous at zero suffices. The Fourier expansion must first represent the finite-width pulse, followed by the pulse-width limit; no unrestricted interchange of two infinite limits is needed. By 19.2.15(a), for fixed n the partial sums are

$$S_{n,M}(x) = \frac{1}{2\pi} + \sum_{m=1}^{M} a_m \cos mx, \qquad a_m = \frac{2n}{\pi m}\sin\frac{m}{2n},$$

and the full series equals δ_n except at the two edges. The difference $\delta_n - S_{n,M}$ has the cosine coefficients a_m for $m > M$ and no others, so Parseval's identity on $(-\pi,\pi)$ and $|a_m| \le 2n/(\pi m)$ give

$$\|\delta_n - S_{n,M}\|_2^2 = \int_{-\pi}^{\pi}(\delta_n - S_{n,M})^2\,dx = \pi\sum_{m>M} a_m^2 \le \frac{4n^2}{\pi}\sum_{m>M}\frac{1}{m^2} \longrightarrow 0 \quad (M\to\infty).$$

Parseval's identity (19.1.13) applies here although the series of the discontinuous δ_n does not converge uniformly: by the L^2 completeness of the Fourier basis, noted in 19.1.13(c), it holds for every square-integrable function. If $|f| \le B$, then $f \in L^2[-\pi,\pi]$ with $\|f\|_2 \le B\sqrt{2\pi}$, and the Cauchy–Schwarz inequality gives

$$\left|\int_{-\pi}^{\pi} f\,(S_{n,M} - \delta_n)\,dx\right| \le \|f\|_2\,\|S_{n,M} - \delta_n\|_2 \longrightarrow 0.$$

Consequently the inner limit is the action of the pulse itself; with $x = u/n$,

$$I_n = \lim_{M\to\infty}\int_{-\pi}^{\pi} f(x)S_{n,M}(x)\,dx = n\int_{-1/(2n)}^{1/(2n)} f(x)\,dx = \int_{-1/2}^{1/2} f(u/n)\,du.$$

Given $\epsilon > 0$, continuity at zero provides $\eta > 0$ such that $|f(x) - f(0)| < \epsilon$ when $|x| < \eta$. Choose $n > 1/(2\eta)$. Then throughout the last integral $|u/n| \le 1/(2n) < \eta$. Since the u-interval has length one, $f(0) = \int_{-1/2}^{1/2} f(0)\,du$ and

$$|I_n - f(0)| \le \int_{-1/2}^{1/2} |f(u/n) - f(0)|\,du \le \epsilon.$$

Hence

$$\boxed{\lim_{n\to\infty}\lim_{M\to\infty}\int_{-\pi}^{\pi} f(x)S_{n,M}(x)\,dx = f(0).}$$

The shrinking-interval step uses only the values of f near zero, so $\int f\delta_n\,dx \to f(0)$ holds for every integrable f continuous at zero. The order of the limits matters: letting $n\to\infty$ first in the coefficients would leave $\int_{-\pi}^{\pi} f(x)\big[1/(2\pi)+\pi^{-1}\sum_{m=1}^{M}\cos mx\big]dx$, which is the Mth Fourier partial sum of f at $x=0$, and that need not converge even for a continuous f (du Bois-Reymond constructed a continuous periodic function whose Fourier series diverges at a point, which a shift moves to $x=0$).

19.2.17

(a) Show that the Dirac delta function $\delta(x-a)$, expanded in a Fourier sine series in the half-interval $(0,L)$ $(0<a<L)$ is given by

$$\delta(x-a)=\frac{2}{L}\sum_{n=1}^{\infty}\sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}.$$

Note that this series actually describes $-\delta(x+a)+\delta(x-a)$ in the interval $(-L,L)$.

(b) By integrating both sides of the preceding equation from 0 to x, show that the cosine expansion of the square wave

$$f(x)=\begin{cases}0, & 0\le x<a\\ 1, & a<x<L,\end{cases}$$

is

$$f(x)=\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L}-\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L}\cos\frac{n\pi x}{L},$$

for $0\le x<L$.

(c) Show that the term $\dfrac{2}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L}$ is the average of $f(x)$ on $(0,L)$.

The delta expansion is distributional. At $x=a$ the integrated Fourier series has the midpoint value $1/2$.

Solution.

(a) The sine functions are orthogonal on the half interval $(0,L)$, with squared norm $L/2$ (δ_{nm} is the Kronecker delta):

$$\int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\,dx=\frac{L}{2}\delta_{nm}.$$

This follows by writing the product as half the difference of cosines,

$$\sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}=\frac{1}{2}\Big[\cos\frac{(n-m)\pi x}{L}-\cos\frac{(n+m)\pi x}{L}\Big]:$$

every $\cos(j\pi x/L)$ with $j\neq 0$ integrates to zero over $(0,L)$, and for $n=m$ the constant term $\frac{1}{2}$ integrates to $L/2$. Multiplying $\sum_n b_n\sin(n\pi x/L)$ by $\sin(m\pi x/L)$ and integrating over $(0,L)$ therefore isolates $(L/2)b_m$, and the sifting property $(0<a<L)$ gives the delta coefficient

$$b_n=\frac{2}{L}\int_0^L\delta(x-a)\sin\frac{n\pi x}{L}\,dx=\frac{2}{L}\sin\frac{n\pi a}{L},$$

which gives the requested expansion

$$\boxed{\delta(x-a) = \frac{2}{L}\sum_{n=1}^{\infty} \sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}, \qquad 0 < x, a < L.}$$

The equality is distributional. Each $\sin(n\pi x/L)$ is odd and has period $2L$, so on $(-L, L)$ the series represents the odd extension $\delta(x-a) - \delta(x+a)$, repeated with period $2L$.

(b) For $0 \le x < L$, $x \ne a$, integrating $\delta(t-a)$ from zero to x gives the step $H(x-a)$: zero for $x < a$, one for $x > a$. For each mode,

$$\int_0^x \sin\frac{n\pi t}{L}\,dt = \frac{L}{n\pi}\left(1 - \cos\frac{n\pi x}{L}\right),$$

and multiplying by the coefficient $(2/L)\sin(n\pi a/L)$ gives $(2/(n\pi))\sin(n\pi a/L)\,[1 - \cos(n\pi x/L)]$. Therefore

$$\boxed{H(x-a) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)}{n} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)}{n}\cos\frac{n\pi x}{L}.}$$

The termwise integration of the delta series is formal, since that series does not converge as a function. The result is nevertheless exact: the boxed series is the Fourier cosine series of the step on $(0, L)$, computed directly in 19.2.18 (its constant term is $1 - a/L$ by (c)), so by Dirichlet's theorem it converges to $H(x-a)$ for $x \ne a$ and to the midpoint $1/2$ at the jump $x = a$. At the endpoints, $x = 0$ makes the two sums cancel, giving 0; at $x = L$, $\cos n\pi = (-1)^n$ leaves

$$\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)}{n}\left[1 - (-1)^n\right] = \frac{4}{\pi}\sum_{n\ \text{odd}}\frac{\sin(n\pi a/L)}{n} = 1,$$

because the square-wave series of 19.2.4 gives $\sum_{n\ \text{odd}}\sin(n\theta)/n = \pi/4$ for $0 < \theta < \pi$.

(c) Put $\theta = \pi a/L \in (0, \pi)$ in 19.1.5:

$$\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin n\theta}{n} = \frac{2}{\pi}\frac{\pi - \theta}{2} = 1 - \frac{a}{L}.$$

Independently the step average is $L^{-1}\int_0^L f(x)\,dx = L^{-1}\int_a^L 1\,dx = 1 - a/L$. Thus the first sum is exactly the required average.

19.2.18 Verify the Fourier cosine expansion of the square wave, Exercise 19.2.17(b), by direct calculation of the Fourier coefficients.

Solution.

For a half-interval cosine expansion, write $f(x) = a_0/2 + \sum_{n\ge1} a_n\cos(n\pi x/L)$. On $(0, L)$ the functions $1, \cos(\pi x/L), \cos(2\pi x/L), \ldots$ are orthogonal, with $\int_0^L 1\,dx = L$ and $\int_0^L \cos^2(n\pi x/L)\,dx = L/2$ for $n \ge 1$. Multiplying the expansion by 1 or by $\cos(n\pi x/L)$ and integrating over $(0, L)$ therefore gives $a_n = (2/L)\int_0^L f(x)\cos(n\pi x/L)\,dx$ for all $n \ge 0$. Since $f = 0$ on $(0, a)$, only (a, L) contributes:

$$a_0 = \frac{2}{L}\int_a^L 1\,dx = 2\left(1 - \frac{a}{L}\right),$$

and, for $n \geq 1$, using $\sin n\pi = 0$,

$$a_n = \frac{2}{L}\int_a^L \cos\frac{n\pi x}{L}\,dx = \frac{2}{n\pi}\left[\sin\frac{n\pi x}{L}\right]_a^L = -\frac{2}{n\pi}\sin\frac{n\pi a}{L}.$$

Hence, with $a_0/2 = 1 - a/L$,

$$\boxed{f(x) = 1 - \frac{a}{L} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)}{n}\cos\frac{n\pi x}{L}.}$$

By 19.2.17(c), $(2/\pi)\sum_n \sin(n\pi a/L)/n = 1 - a/L$, so the constant terms agree and the two expansions coincide term by term. The even extension of f is periodic with period $2L$; it is continuous at $0, L$ and jumps at $\pm a$. Thus, by Dirichlet's theorem, the series has the specified step values away from the jumps and the midpoint $1/2$ there.

19.2.19

(a) A string is clamped at both ends $x = 0$ and $x = L$. Assuming small-amplitude vibrations, we find that the amplitude $y(x,t)$ satisfies the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}.$$

Here v is the wave velocity. The string is set in vibration by a sharp blow at $x = a$. Hence we have

$$y(x,0) = 0, \qquad \frac{\partial y(x,t)}{\partial t} = Lv_0\delta(x-a) \quad \text{at } t = 0.$$

The constant L is included to compensate for the dimensions (inverse length) of $\delta(x-a)$. With $\delta(x-a)$ given by Exercise 19.2.17(a), solve the wave equation subject to these initial conditions.

$$\textit{ANS. } y(x,t) = \frac{2v_0 L}{\pi v}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}\sin\frac{n\pi v t}{L}.$$

(b) Show that the transverse velocity of the string $\partial y(x,t)/\partial t$ is given by

$$\frac{\partial y(x,t)}{\partial t} = 2v_0\sum_{n=1}^{\infty}\sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}\cos\frac{n\pi v t}{L}.$$

Take $L, v > 0$ and $0 < a < L$. Interpret the impulsive initial velocity and its propagated series as distributions.

Solution.

(a) Put $k_n = n\pi/L$ and $\Omega_n = vk_n$. The fixed-end condition selects the sine modes, so write $y(x,t) = \sum_{n\geq 1} q_n(t)\sin k_n x$. Since $(\sin k_n x)'' = -k_n^2 \sin k_n x$, substitution into $v^2 y_{xx} = y_{tt}$ gives $\sum_n (q_n'' + \Omega_n^2 q_n)\sin k_n x = 0$. Multiplying by $\sin k_m x$ and integrating over $(0, L)$ (sine orthogonality, as in 19.2.17(a)) isolates each mode:

$$q_n'' + \Omega_n^2 q_n = 0, \qquad q_n(t) = A_n\cos\Omega_n t + B_n\sin\Omega_n t.$$

The zero initial displacement $y(x,0)=\sum_n A_n \sin k_n x = 0$ forces $A_n = 0$. The sine coefficient of the initial velocity is

$$q_n'(0) = \frac{2}{L}\int_0^L Lv_0\delta(x-a)\sin k_n x\,dx = 2v_0 \sin k_n a.$$

Since $q_n'(0) = \Omega_n B_n$,

$$B_n = \frac{2v_0 \sin k_n a}{\Omega_n} = \frac{2v_0 L}{\pi v}\,\frac{\sin(n\pi a/L)}{n},$$

and we obtain

$$\boxed{y(x,t) = \frac{2v_0 L}{\pi v}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)\sin(n\pi x/L)\sin(n\pi v t/L)}{n}.}$$

Every mode solves the wave equation and vanishes at both fixed ends.

(b) Differentiating $\sin\Omega_n t$ gives $\Omega_n\cos\Omega_n t$, and the factor $\Omega_n = n\pi v/L$ cancels the denominator and prefactor, $(2v_0L/(\pi v))(1/n)(n\pi v/L) = 2v_0$:

$$\boxed{y_t(x,t) = 2v_0\sum_{n=1}^{\infty}\sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}\cos\frac{n\pi v t}{L}.}$$

At $t=0$ the series is $Lv_0\cdot(2/L)\sum_n \sin(n\pi a/L)\sin(n\pi x/L)$, which by the sine closure relation of 19.2.17(a) is exactly $Lv_0\delta(x-a)$. The ideal impulse makes the velocity a distribution. The last series is not asserted to converge as an ordinary velocity function; differentiation and initial conditions are interpreted distributionally.

For an independent propagation description, extend the initial velocity oddly and periodically:

$$G(s) = Lv_0\sum_{j\in\mathbb{Z}}[\delta(s-a-2jL)-\delta(s+a-2jL)].$$

With zero initial displacement, d'Alembert's formula gives

$$y(x,t) = \frac{1}{2v}\int_{x-vt}^{x+vt} G(s)\,ds, \qquad y_t(x,t) = \frac{1}{2}[G(x+vt)+G(x-vt)],$$

the second by differentiating the limits of integration. G is odd about $s=0$ and, being odd with period $2L$, also about $s=L$, so its integrals over $(-vt,vt)$ and $(L-vt,L+vt)$ vanish: $y(0,t)=y(L,t)=0$, and the clamps hold. For $vt<\min(a,L-a)$ and $0<x<L$, the only source that can lie in $(x-vt,x+vt)$ is the one at $s=a$ (every other source lies at $s\le -a$ or $s\ge 2L-a$, outside the interval, because $x-vt>-vt>-a$ and $x+vt<L+vt<2L-a$), and it does exactly when $|x-a|<vt$. Thus, before either pulse reaches an end, $y = Lv_0/(2v)$ where $|x-a|<vt$, and zero outside; the moving fronts and subsequent reflected images explain the singular velocity. This confirms the Fourier solution: the sine coefficients of this plateau are

$$\frac{2}{L}\,\frac{Lv_0}{2v}\int_{a-vt}^{a+vt}\sin k_n x\,dx = \frac{v_0}{vk_n}\big[\cos k_n(a-vt)-\cos k_n(a+vt)\big] = \frac{2v_0L}{\pi v}\,\frac{\sin k_n a}{n}\,\sin\Omega_n t,$$

the last step by $\cos(A-B)-\cos(A+B) = 2\sin A\sin B$ (with $A=k_n a$, $B=k_n vt=\Omega_n t$) and $1/(vk_n) = L/(n\pi v)$. These are exactly the coefficients $B_n\sin\Omega_n t$ found in (a).

19.2.20 A string, clamped at $x = 0$ and at $x = L$, is vibrating freely. Its motion is described by the wave equation

$$\frac{\partial^2 u(x,t)}{\partial t^2} = v^2 \frac{\partial^2 u(x,t)}{\partial x^2}.$$

Assume a Fourier expansion of the form

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \sin \frac{n\pi x}{L}$$

and determine the coefficients $b_n(t)$. The initial conditions are

$$u(x,0) = f(x) \qquad \text{and} \qquad \frac{\partial}{\partial t} u(x,0) = g(x).$$

Note. This is only half the conventional Fourier orthogonality integral interval. However, as long as only the sines are included here, the Sturm–Liouville boundary conditions are still satisfied and the functions are orthogonal.

$$\textit{ANS.}\ \ b_n(t) = A_n \cos \frac{n\pi v t}{L} + B_n \sin \frac{n\pi v t}{L},$$
$$A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \quad B_n = \frac{2}{n\pi v} \int_0^L g(x) \sin \frac{n\pi x}{L}\, dx.$$

Take $L, v > 0$ and real initial data. Because the modal series is differentiated termwise, it gives a classical solution only for data smooth enough and compatible with the clamps; for $f \in H_0^1(0,L)$ and $g \in L^2(0,L)$ (finite initial energy) it gives the finite-energy solution.

Solution.

Here $H_0^1(0,L)$ means that f and its weak first derivative are square integrable and f has zero endpoint values. Thus the assumptions make the initial displacement-gradient and velocity energies finite. Let $k_n = n\pi/L$ and $\Omega_n = v k_n$, with $L, v > 0$. Substitute the sine expansion into the wave equation:

$$\sum_{n\ge 1} [b_n''(t) + \Omega_n^2 b_n(t)] \sin k_n x = 0.$$

Multiplying by $\sin k_m x$ and integrating from 0 to L isolates $(L/2)[b_m'' + \Omega_m^2 b_m] = 0$. Thus

$$b_n(t) = A_n \cos \Omega_n t + B_n \sin \Omega_n t.$$

At zero, $b_n(0) = A_n$ and $b_n'(0) = \Omega_n B_n$, so the initial conditions read

$$\sum_{n\ge 1} A_n \sin k_n x = f(x), \qquad \sum_{n\ge 1} \Omega_n B_n \sin k_n x = g(x).$$

Projecting each onto its sine mode (multiply by $\sin k_m x$ and integrate over $(0,L)$) gives $(L/2)A_m = \int_0^L f \sin k_m x\, dx$ and $(L/2)\Omega_m B_m = \int_0^L g \sin k_m x\, dx$, that is,

$$\boxed{A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \qquad B_n = \frac{2}{n\pi v} \int_0^L g(x) \sin \frac{n\pi x}{L}\, dx.}$$

Together with the displayed time dependence these determine the solution. The factor $2/(n\pi v)$ is $(2/L)/\Omega_n = (2/L)\, L/(n\pi v)$. The cosine time factor carries the initial displacement, and the sine time factor carries the initial velocity.

For a classical solution the initial data must be regular enough for the differentiations and compatible with the clamps (in particular $f(0) = f(L) = 0$, and $g(0) = g(L) = 0$ when these traces exist). For less regular data the modal formula is interpreted in the corresponding weak sense. For example $f \in H_0^1(0, L)$ and $g \in L^2(0, L)$ give a finite-energy solution. Since $u_t = \sum_n b_n' \sin k_n x$ and $u_x = \sum_n k_n b_n \cos k_n x$, and the sines and the cosines are each orthogonal on $(0, L)$ with squared norms $L/2$, we have $\int_0^L u_t^2\,dx = (L/2)\sum_n (b_n')^2$ and $\int_0^L u_x^2\,dx = (L/2)\sum_n k_n^2 b_n^2$. With $v^2 k_n^2 = \Omega_n^2$, the energy

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + v^2 u_x^2)\,dx = \frac{L}{4}\sum_{n\ge 1}[(b_n')^2 + \Omega_n^2 b_n^2]$$

is constant, since each summand has derivative $2b_n'(b_n'' + \Omega_n^2 b_n) = 0$.

19.2.21

(a) Let us continue the vibrating string problem in Exercise 19.2.20. We assume now that the presence of a resisting medium will damp the vibrations according to the equation

$$\frac{\partial^2 u(x,t)}{\partial t^2} = v^2 \frac{\partial^2 u(x,t)}{\partial x^2} - k\frac{\partial u(x,t)}{\partial t}.$$

Introduce a Fourier expansion

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \sin \frac{n\pi x}{L}$$

and again determine the coefficients $b_n(t)$. Take the initial and boundary conditions to be the same as in Exercise 19.2.20. Assume the damping to be small. Since damping is small or large only relative to each mode's frequency $n\pi v/L$, take $k \ge 0$ and first consider each mode for which $k < 2n\pi v/L$.

(b) Repeat, but assume the damping to be large. That is, treat the modes with $k > 2n\pi v/L$, and include the critical case $k = 2n\pi v/L$; explain why finite k cannot overdamp every mode.

ANS. (a) $b_n(t) = e^{-kt/2}\big[A_n \cos \omega_n t + B_n \sin \omega_n t\big]$, $\omega_n^2 = \left(\frac{n\pi v}{L}\right)^2 - \left(\frac{k}{2}\right)^2 > 0$,

$$A_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx, \quad B_n = \frac{2}{\omega_n L}\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx + \frac{k}{2\omega_n}A_n.$$

(b) $b_n(t) = e^{-kt/2}\big[A_n \cosh \sigma_n t + B_n \sinh \sigma_n t\big]$, $\sigma_n^2 = \left(\frac{k}{2}\right)^2 - \left(\frac{n\pi v}{L}\right)^2 > 0$,

$$A_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx, \quad B_n = \frac{2}{\sigma_n L}\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx + \frac{k}{2\sigma_n}A_n.$$

Solution.

Assume $k \ge 0$, write $\Omega_n = n\pi v/L$, and let the initial sine coefficients be

$$F_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx, \qquad G_n = \frac{2}{L}\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx.$$

Substituting the expansion into the damped wave equation gives $\sum_n (b_n'' + k b_n' + \Omega_n^2 b_n)\sin(n\pi x/L) = 0$, and projecting onto each sine mode as in 19.2.20 gives

$$b_n'' + k b_n' + \Omega_n^2 b_n = 0, \qquad b_n(0) = F_n, \quad b_n'(0) = G_n.$$

To remove the first derivative put $b_n = e^{-kt/2} z_n$. Then

$$b_n' = e^{-kt/2}(z_n' - kz_n/2), \quad b_n'' = e^{-kt/2}(z_n'' - kz_n' + k^2 z_n/4),$$

so that

$$\begin{aligned} b_n'' + kb_n' + \Omega_n^2 b_n &= e^{-kt/2}\Big[z_n'' - kz_n' + \frac{k^2}{4} z_n + kz_n' - \frac{k^2}{2} z_n + \Omega_n^2 z_n\Big] \\ &= e^{-kt/2}\Big[z_n'' + \Big(\Omega_n^2 - \frac{k^2}{4}\Big) z_n\Big]. \end{aligned}$$

At $t = 0$, $z_n(0) = b_n(0)$ and, from the expression for b_n', $z_n'(0) = b_n'(0) + kb_n(0)/2$. Hence

$$z_n'' + (\Omega_n^2 - k^2/4) z_n = 0, \quad z_n(0) = F_n, \quad z_n'(0) = G_n + kF_n/2.$$

(a) Underdamped modes. If $\Omega_n > k/2$, define $\omega_n = \sqrt{\Omega_n^2 - k^2/4} > 0$. Then $z_n = C_n \cos \omega_n t + D_n \sin \omega_n t$, and the initial conditions give $C_n = z_n(0) = F_n$ and $\omega_n D_n = z_n'(0) = G_n + kF_n/2$, so

$$\boxed{b_n = e^{-kt/2}\left[F_n \cos \omega_n t + \frac{G_n + kF_n/2}{\omega_n} \sin \omega_n t\right].}$$

This is the printed formula with $A_n = F_n$ and $B_n = (G_n + kA_n/2)/\omega_n$. The textbook's answer to (a) omits the square on $n\pi v/L$ in ω_n^2; the equation for z_n shows that the correct formula is $\omega_n^2 = (n\pi v/L)^2 - (k/2)^2$, as used in the answer above.

(b) Overdamped modes. If $0 < \Omega_n < k/2$, define $\sigma_n = \sqrt{k^2/4 - \Omega_n^2} > 0$. Now $z_n = C_n \cosh \sigma_n t + D_n \sinh \sigma_n t$ with $C_n = F_n$ and $\sigma_n D_n = G_n + kF_n/2$, so

$$\boxed{b_n = e^{-kt/2}\left[F_n \cosh \sigma_n t + \frac{G_n + kF_n/2}{\sigma_n} \sinh \sigma_n t\right].}$$

This is the printed formula with $A_n = F_n$ and $B_n = (G_n + kA_n/2)/\sigma_n$. The two decay exponents $-k/2 \pm \sigma_n$ are negative, since $0 < \sigma_n < k/2$.

Damping is classified mode by mode. No finite k overdamps all infinitely many modes, since Ω_n grows without bound. The critical case $\Omega_n = k/2$, omitted from the textbook's question, also needs a separate formula. Here $z_n'' = 0$, so $z_n = F_n + (G_n + kF_n/2)t$ and

$$\boxed{b_n = e^{-kt/2}[F_n + (G_n + kF_n/2)t].}$$

This is also the continuous $\omega_n \to 0$ or $\sigma_n \to 0$ limit, since $\sin(\omega_n t)/\omega_n \to t$, $\sinh(\sigma_n t)/\sigma_n \to t$ and the cosine and hyperbolic cosine tend to 1. All modes are underdamped if $k < 2\pi v/L$, because then $\Omega_n \geq \Omega_1 = \pi v/L > k/2$; otherwise modes with $n < kL/(2\pi v)$ are overdamped, equality is critical, and the larger indices remain underdamped. As a check, let $E = \frac{1}{2}\int_0^L (u_t^2 + v^2 u_x^2)\, dx$. Differentiating and integrating the second term by parts gives

$$\begin{aligned} E' &= \int_0^L (u_t u_{tt} + v^2 u_x u_{xt})\, dx \\ &= v^2 [u_x u_t]_0^L + \int_0^L u_t (u_{tt} - v^2 u_{xx})\, dx \\ &= -k \int_0^L u_t^2\, dx \leq 0 \qquad (k \geq 0). \end{aligned}$$

The boundary term vanishes because the clamped displacement is constant in time, hence $u_t = 0$ at both ends, and the last line uses the equation of motion, $u_{tt} - v^2 u_{xx} = -k u_t$. For smooth data this is a direct calculation; for finite-energy data it follows by limits of finite modal sums. It is consistent with the time dependence of every mode: as in 19.2.20, $E = (L/4)\sum_n [(b_n')^2 + \Omega_n^2 b_n^2]$, and the mode equation $b_n'' + \Omega_n^2 b_n = -k b_n'$ gives each summand the derivative $2b_n'(b_n'' + \Omega_n^2 b_n) = -2k(b_n')^2$, so $E' = -(kL/2)\sum_n (b_n')^2 = -k\int_0^L u_t^2\,dx$, as found above.

19.3 Gibbs Phenomenon

19.3.1 With the partial-sum summation techniques of this section, show that at a discontinuity in $f(x)$ the Fourier series for $f(x)$ takes on the arithmetic mean of the right- and left-hand limits:

$$\lim_{r\to\infty} s_r(x_0) = \tfrac{1}{2}\big[f(x_0+0) + f(x_0-0)\big].$$

In evaluating $\lim_{r\to\infty} s_r(x_0)$, you may find it convenient to identify part of the integrand as a Dirac delta function. Assume f is 2π-periodic and of bounded variation on a period. The left side is the Fourier-series limit, regardless of the assigned value of $f(x_0)$.

Solution.

The conclusion concerns the Fourier-series value, which need not equal an arbitrarily assigned $f(x_0)$. As in the chapter's Dirichlet conditions, assume a piecewise regular periodic function; bounded variation suffices. The existence of a jump alone is not a general pointwise-convergence hypothesis. The textbook's Eq. (19.29) also omits n from the exponential; the correct finite kernel uses $e^{in(x-t)}$, as in Eq. (19.30). Write N for the textbook's number r of harmonics, so that s_N is its s_r. The variation of a function φ on an interval J is $\operatorname{Var}_J \varphi = \sup \sum_{i=1}^{k} |\varphi(t_i) - \varphi(t_{i-1})|$, the supremum over all finite chains $t_0 < t_1 < \cdots < t_k$ in J, and f has bounded variation when its variation over a period is finite. Such an f is a difference of two increasing functions (Jordan decomposition), so the one-sided limits $f(x_0 \pm 0)$ exist. Let

$$m = \frac{f(x_0+0) + f(x_0-0)}{2}, \qquad D_N(t) = \frac{\sin((N+1/2)t)}{\sin(t/2)}.$$

Inserting the coefficient integrals into $s_N(x) = a_0/2 + \sum_{n=1}^{N}(a_n \cos nx + b_n \sin nx)$ and using $\cos nt \cos nx + \sin nt \sin nx = \cos n(t-x)$ gives the finite partial-sum formula

$$s_N(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\Big[1 + 2\sum_{n=1}^{N}\cos n(t-x)\Big]dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t) D_N(t-x)\,dt,$$

because the finite geometric sum is

$$1 + 2\sum_{n=1}^{N}\cos nu = \sum_{n=-N}^{N} e^{inu} = \frac{e^{-iNu}\big(e^{i(2N+1)u} - 1\big)}{e^{iu} - 1} = \frac{e^{i(N+1/2)u} - e^{-i(N+1/2)u}}{e^{iu/2} - e^{-iu/2}} = D_N(u).$$

Put $t = x_0 + s$; the integrand has period 2π in s, so s may still run over $(-\pi, \pi)$. On $(-\pi, 0)$ replace s by $-s$, using that D_N is even, and rename s as t:

$$s_N(x_0) = \frac{1}{2\pi}\int_0^{\pi} [f(x_0+t) + f(x_0-t)] D_N(t)\,dt.$$

Because $D_N = 1 + 2\sum_{j=1}^{N} \cos jt$ and $\int_0^\pi \cos jt\,dt = \sin j\pi/j = 0$, the integral of D_N on $(0,\pi)$ is exactly π, so $m = (2\pi)^{-1}\int_0^\pi 2m\,D_N(t)\,dt$. Subtracting m therefore gives

$$s_N(x_0) - m = \frac{1}{2\pi}\int_0^\pi g(t)D_N(t)\,dt, \quad g(t) = f(x_0+t) + f(x_0-t) - 2m, \quad g(0+) = 0.$$

We prove that the last integral tends to zero, including its potentially singular neighborhood of zero.

Put $M = N + 1/2$ and $A_N(u) = \int_0^u D_N(t)\,dt$ for $0 \le u \le \pi$. Split

$$\frac{1}{\sin(t/2)} = \frac{2}{t} + q(t), \qquad A_N(u) = 2\int_0^{Mu} \frac{\sin v}{v}\,dv + \int_0^u q(t)\sin Mt\,dt,$$

the first integral by the substitution $v = Mt$. Here $q(t) = t/12 + O(t^3)$ at zero, so q is continuous on $[0,\pi]$ and integrable. The sine integral is uniformly bounded: below $v = 1$, $|\sin v/v| \le 1$ makes its absolute integral at most one; beyond one, integration by parts gives

$$\int_1^z \frac{\sin v}{v}\,dv = \Big[-\frac{\cos v}{v}\Big]_1^z - \int_1^z \frac{\cos v}{v^2}\,dv,$$

whose absolute value is at most $\cos 1 + 1/z + \int_1^z v^{-2}\,dv = 1 + \cos 1 < 2$. Hence $|\int_0^z \sin v/v\,dv| \le 3$ for every $z \ge 0$, and consequently $|A_N(u)| \le C = 6 + \int_0^\pi |q(t)|\,dt$, independently of N, u.

Since g has bounded variation (the variation of a sum is at most the sum of the variations) and $g(0+) = 0$, choose a small $\delta > 0$ for which both $|g(\delta)|$ and its variation on $(0,\delta]$ are small. Since $dA_N = D_N\,dt$ and the boundary term at zero vanishes ($A_N(0) = 0$, $g(0+) = 0$), Stieltjes integration by parts gives

$$\left|\int_0^\delta g(t)D_N(t)\,dt\right| = \left|g(\delta)A_N(\delta) - \int_{(0,\delta]} A_N(t)\,dg(t)\right| \le C\big(|g(\delta)| + \operatorname{Var}_{(0,\delta]} g\big).$$

The last step uses $|A_N| \le C$: every Riemann–Stieltjes sum $\sum_i A_N(\xi_i)[g(t_i) - g(t_{i-1})]$ of $\int A_N\,dg$ is at most $C\sum_i |g(t_i) - g(t_{i-1})| \le C\operatorname{Var}_{(0,\delta]} g$ in absolute value. The variation $\operatorname{Var}_{(0,\delta]} g$ can only decrease as $\delta \downarrow 0$, and its limit is zero: if it stayed above some $\eta > 0$, we could take a chain in $(0,\pi]$ with sum above $\eta/2$, starting at some $t_0 > 0$, then such a chain in $(0, t_0/2]$, and so on; joined, K of these chains form one chain in $(0,\pi]$ with sum above $K\eta/2$, impossible for large K because g has bounded variation. Also $g(\delta) \to 0$. Thus the bound tends to zero as $\delta \downarrow 0$, uniformly in N. For fixed δ, the remaining integrand on $[\delta,\pi]$ has the form $h(t)\sin Mt$ with $h(t) = g(t)/\sin(t/2)$, which is bounded by $\sup|g|/\sin(\delta/2)$ and Riemann integrable (monotone functions are, g is a combination of them, and $1/\sin(t/2)$ is continuous there). Its integral tends to zero as $M \to \infty$ by the Riemann–Lebesgue lemma: given $\epsilon > 0$, take a step function s with $\int_\delta^\pi |h - s|\,dt < \epsilon$ (a lower Darboux step function of h will do). Each of the finitely many sine integrals of s is $O(1/M)$, since $|\int_\alpha^\beta \sin Mt\,dt| \le 2/M$, while $|\int_\delta^\pi (h-s)\sin Mt\,dt| < \epsilon$ for every M; hence $\limsup_{M\to\infty} |\int_\delta^\pi h(t)\sin Mt\,dt| \le \epsilon$ for every $\epsilon > 0$. Combining the two parts, for every such δ

$$\limsup_{N\to\infty} |s_N(x_0) - m| \le \frac{C}{2\pi}\big(|g(\delta)| + \operatorname{Var}_{(0,\delta]} g\big),$$

and the right side tends to zero as $\delta \downarrow 0$. This proves

$$\boxed{\lim_{N\to\infty} s_N(x_0) = \frac{f(x_0+0) + f(x_0-0)}{2}.}$$

This is the precise form of the hint: on $(0,\pi)$ the kernel $D_N/(2\pi)$ has total weight $1/2$ and acts in the limit as half a Dirac delta function at $t = 0$ on $f(x_0+t) + f(x_0-t)$. The kernel's symmetry is what assigns equal weight to the two one-sided limits.

19.3.2 Determine the partial sum, s_n, of the series in Eq. (19.33) by using

(a) $\dfrac{\sin mx}{m} = \displaystyle\int_0^x \cos my\, dy,$

(b) $\displaystyle\sum_{p=1}^{n} \cos(2p-1)y = \frac{\sin 2ny}{2\sin y}.$

Compare the peak limit with Eqs. (19.38) and (19.41) (the textbook cites Eq. (19.40) here; see the solution).

Solution.

Equation (19.33) represents levels $\pm h/2$. Its first n nonzero harmonics give

$$s_n(x) = \frac{2h}{\pi} \sum_{p=1}^{n} \frac{\sin((2p-1)x)}{2p-1}.$$

By hint (a) with $m = 2p-1$, each term is $\sin((2p-1)x)/(2p-1) = \int_0^x \cos((2p-1)y)\, dy$. There are only finitely many terms, so the finite sum and the integral may be interchanged:

$$s_n(x) = \frac{2h}{\pi} \int_0^x \sum_{p=1}^{n} \cos((2p-1)y)\, dy.$$

The identity in hint (b) follows by telescoping: the product formula $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ with $A = y$, $B = (2p-1)y$ gives

$$2\sin y\cos((2p-1)y) = \sin 2py - \sin(2p-2)y,$$

so the sum over $p = 1, \ldots, n$ collapses to $\sin 2ny - \sin 0 = \sin 2ny$. Consequently, since $(2h/\pi)/2 = h/\pi$, the exact partial sum is

$$\boxed{s_n(x) = \frac{h}{\pi} \int_0^x \frac{\sin 2ny}{\sin y}\, dy.}$$

The apparent singularities at multiples of π are removable because the quotient equals twice a finite cosine sum. For $h > 0$, its first positive maximum occurs at $x_n = \pi/(2n)$: by the fundamental theorem of calculus the derivative is $(h/\pi)\sin 2nx/\sin x$, positive for $0 < x < \pi/(2n)$ and negative for $\pi/(2n) < x < \pi/n$. Changing variables $t = 2ny$ ($dy = dt/(2n)$, and $y = x_n$ becomes $t = \pi$) at that maximum gives

$$s_n(x_n) = \frac{h}{\pi} \int_0^{\pi} \frac{\sin t}{2n\sin(t/(2n))}\, dt = \frac{h}{\pi} \int_0^{\pi} \frac{\sin t}{t} \frac{t/(2n)}{\sin(t/(2n))}\, dt \longrightarrow \frac{h}{\pi} \int_0^{\pi} \frac{\sin t}{t}\, dt.$$

The limit holds because $u/\sin u$ increases on $(0, \pi/2)$ (its derivative is $(\sin u - u\cos u)/\sin^2 u$, and $\sin u - u\cos u$ vanishes at $u = 0$ and has derivative $u\sin u > 0$), so for $0 < t \le \pi$ the last factor lies between 1 and $(\pi/(2n))/\sin(\pi/(2n))$, which tends to 1; the factor therefore tends to 1 uniformly, and $|\sin t/t| \le 1$. More generally, at $x = z/(2n)$ for fixed z the limiting profile is $(h/\pi)\int_0^z \sin t/t\, dt$.

Eq. (19.40), cited at the end of the textbook's question, is a tail-integral identity, not a partial-sum formula. The exact expression above agrees with the small-x approximation in (19.37) and the peak limit in (19.38). Using (19.40) together with $\int_0^{\infty} \sin t/t\, dt = \pi/2$ gives the peak integral in (19.41). Thus the limiting peak is $(h/\pi)\int_0^{\pi} \sin t/t\, dt = (h/2)\,(2/\pi)\int_0^{\pi} \sin t/t\, dt = (h/2)I$, with $I = 1.178979744\ldots$ computed in 19.3.3. Its overshoot above the level $h/2$ is $(h/2)(I-1) \approx 0.08949h$: about 8.949% of the jump height h, or 17.898% of the positive level $h/2$.

19.3.3

(a) Calculate the value of the Gibbs phenomenon integral

$$I = \frac{2}{\pi}\int_0^{\pi} \frac{\sin t}{t}\,dt$$

by numerical quadrature accurate to 12 significant figures.

(b) Check your result by (1) expanding the integrand as a series, (2) integrating term by term, and (3) evaluating the integrated series. This calls for double-precision calculation.

ANS. $I = 1.178979744472$.

Solution.

(a) Define the removable endpoint by $(\sin t/t)|_{t=0} = 1$. Map the n-point Gauss–Legendre rule to $[0,\pi]$: with $t = \pi(1+x)/2$ the prefactor $2/\pi$ cancels $dt = (\pi/2)\,dx$, so with the Legendre nodes x_j (the zeros of P_n) and weights $w_j = 2/[(1-x_j^2)P_n'(x_j)^2]$ on $[-1,1]$,

$$I = \int_{-1}^{1} \frac{\sin t}{t}\,dx \approx Q_n = \sum_{j=1}^{n} w_j\,\frac{\sin t_j}{t_j}, \qquad t_j = \frac{\pi}{2}(1+x_j).$$

Because the integrand is entire, the error $Q_n - I$ falls very fast with n: it is -2.9×10^{-12} for $n=6$, 7.5×10^{-15} for $n=7$, -1.5×10^{-17} for $n=8$ and 1.3×10^{-38} for $n=15$. Thus $n=7$ in ordinary double precision already gives the requested twelve significant figures, and $n=20$ in 65-digit arithmetic gives

$$\boxed{I = 1.17897974447216727023202884582 4\ldots.}$$

Adaptive tanh–sinh quadrature at the same precision gives the same digits. To twelve significant figures the result is $\boxed{1.17897974447}$; the textbook's 1.178979744472 agrees when rounded to twelve decimal places.

(b) The entire power series for the integrand is

$$\frac{\sin t}{t} = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k+1)!}.$$

On $[0,\pi]$ its terms are bounded by $\pi^{2k}/(2k+1)!$, the terms of a convergent series, so by the Weierstrass M-test it converges uniformly there and may be integrated termwise. With $\int_0^{\pi} t^{2k}\,dt = \pi^{2k+1}/(2k+1)$,

$$\begin{aligned} I &= \frac{2}{\pi}\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!}\,\frac{\pi^{2k+1}}{2k+1} \\ &= \boxed{2\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{(2k+1)(2k+1)!}}. \end{aligned}$$

Let $T_k = 2\pi^{2k}/[(2k+1)(2k+1)!]$. Since $(2k+3)! = (2k+3)(2k+2)(2k+1)!$, successive magnitudes satisfy

$$\frac{T_{k+1}}{T_k} = \frac{\pi^2(2k+1)\,(2k+1)!}{(2k+3)\,(2k+3)!} = \frac{\pi^2(2k+1)}{(2k+3)^2(2k+2)} < 1 \qquad (k \ge 0).$$

To verify the inequality for every k, put $r_k = (2k+1)/[(2k+3)^2(2k+2)]$. Over the common denominator, using $2k+2 = 2(k+1)$ and $2k+4 = 2(k+2)$, the numerator of $r_k - r_{k+1}$ is $(2k+1)(k+2)(2k+5)^2 - (k+1)(2k+3)^3$, so

$$r_k - r_{k+1} = \frac{16k^3 + 68k^2 + 84k + 23}{2(k+1)(k+2)(2k+3)^2(2k+5)^2} > 0 \qquad (k \geq 0).$$

Thus the ratios decrease, and the largest is $\pi^2 r_0 = \pi^2/18 < 1$. The magnitudes T_k therefore decrease monotonically to zero, and by the alternating-series (Leibniz) test the remainder after N terms is at most T_N. Direct evaluation in high-precision arithmetic yields

N	sum of terms $k = 0, \ldots, N-1$	bound T_N
4	1.1735737377034521530	5.811×10^{-3}
8	1.1789797154459948275	2.978×10^{-8}
12	1.1789797444721629210	4.406×10^{-15}
16	1.1789797444721672702	5.658×10^{-23}

These bounds concern the mathematical series truncation; arithmetic roundoff must be controlled separately. Summing the sixteen terms using ordinary double-precision arithmetic and compensated summation gives 1.178979744472167, independently meeting the requested twelve-digit accuracy. High-precision series evaluation and both quadrature methods provide the additional cross-checks.

20 Chapter 20. Integral Transforms

Problems

20.2 Fourier Transform

20.2.1

(a) Show that $g(-\omega) = g^*(\omega)$ is a necessary and sufficient condition for $f(x)$ to be real.

(b) Show that $g(-\omega) = -g^*(\omega)$ is a necessary and sufficient condition for $f(x)$ to be pure imaginary.

Note. The condition of part (a) is used in the development of the dispersion relations of Section 12.8.

Use Fourier inversion in the usual L^1 or L^2 setting, and interpret reality almost everywhere; continuous representatives give pointwise statements.

Solution.

Use the convention

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{i\omega x} dx, \qquad f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{-i\omega x} d\omega.$$

Conjugating the first formula, with $(e^{i\omega x})^* = e^{-i\omega x}$, shows

$$g^*(\omega) = \frac{1}{\sqrt{2\pi}} \int f^*(x) e^{-i\omega x} dx = \mathcal{F}[f^*](-\omega).$$

(a) If $f = f^*$, the right side is $\mathcal{F}[f](-\omega) = g(-\omega)$, so $g(-\omega) = g^*(\omega)$. Conversely, if $g(-\omega) = g^*(\omega)$, conjugate the inverse transform and put $u = -\omega$ (reversing the limits absorbs the sign of $d\omega = -du$):

$$\begin{aligned} f^*(x) &= \frac{1}{\sqrt{2\pi}} \int g^*(\omega) e^{i\omega x} d\omega \\ &= \frac{1}{\sqrt{2\pi}} \int g(-\omega) e^{i\omega x} d\omega = \frac{1}{\sqrt{2\pi}} \int g(u) e^{-iux} du = f(x). \end{aligned}$$

(b) Purely imaginary means $f^* = -f$. The conjugated forward formula then gives $g^*(\omega) = \mathcal{F}[-f](-\omega) = -g(-\omega)$, i.e., $g(-\omega) = -g^*(\omega)$. Conversely, if $g^*(\omega) = -g(-\omega)$, the steps of the preceding inverse calculation (conjugation, the hypothesis, then $u = -\omega$) give $f^*(x) = -(2\pi)^{-1/2} \int g(-\omega) e^{i\omega x} d\omega = -f(x)$. For L^1 or L^2 transforms the recovered function is determined almost everywhere; these reality assertions have that meaning. They hold at every point for continuous representatives. Changing a function at an isolated point cannot be detected by its Fourier transform.

20.2.2 The function

$$f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

is a symmetrical finite step function.

(a) Find $g_c(\omega)$, Fourier cosine transform of $f(x)$.

(b) Taking the inverse cosine transform, show that

$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}\,d\omega.$$

(c) From part (b) show that

$$\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}\,d\omega = \begin{cases} 0, & |x|>1,\\ \dfrac{\pi}{4}, & |x|=1,\\ \dfrac{\pi}{2}, & |x|<1. \end{cases}$$

Solution.

(a) The unitary cosine transform is

$$g_c(\omega) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos\omega x\,dx = \sqrt{\frac{2}{\pi}}\left[\frac{\sin\omega x}{\omega}\right]_0^1 = \boxed{\sqrt{\frac{2}{\pi}}\frac{\sin\omega}{\omega}}.$$

At zero its continuous value is $\sqrt{2/\pi}$. (b) Substitution in the inverse cosine transform gives

$$f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty g_c(\omega)\cos\omega x\,d\omega = \frac{2}{\pi}\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}d\omega.$$

At a jump, inversion gives the mean of the two one-sided limits, so its value at $x=\pm1$ is $1/2$.

(c) By (b), the integral equals $(\pi/2)f(x)$: it is $\pi/2$ for $|x|<1$ and 0 for $|x|>1$, and the midpoint value $1/2$ at $x=\pm1$ gives $(\pi/2)(1/2)=\pi/4$. To confirm these values, including the edge value, evaluate the integral directly. For $\epsilon>0$ first note that

$$D_\epsilon(b) = \int_0^\infty e^{-\epsilon u}\frac{\sin bu}{u}du, \qquad D'_\epsilon(b) = \int_0^\infty e^{-\epsilon u}\cos bu\,du = \frac{\epsilon}{\epsilon^2+b^2}.$$

Differentiation under the integral sign is allowed because the b-derivative $e^{-\epsilon u}\cos bu$ of the integrand is bounded by the integrable $e^{-\epsilon u}$; the last integral is C of Exercise 20.2.3 with $a=\epsilon$, $\omega=b$. Since $D_\epsilon(0)=0$, integration in b gives

$$D_\epsilon(b) = \int_0^b \frac{\epsilon\,ds}{\epsilon^2+s^2} = \arctan\frac{b}{\epsilon}.$$

As $\epsilon\downarrow 0$, $b/\epsilon\to\pm\infty$ for $b\gtrless 0$. Thus $D(b)=\lim_{\epsilon\downarrow0}D_\epsilon(b)=(\pi/2)\operatorname{sgn}b$ for $b\neq0$, while $D(0)=0$. The improper integral $\int_0^\infty(\sin bu/u)\,du$ converges by Dirichlet's test, so Abel's theorem for integrals makes it equal to this limit $D(b)$: if $\int_0^\infty h\,du$ converges, then $\int_0^\infty e^{-\epsilon u}h\,du\to\int_0^\infty h\,du$ as $\epsilon\downarrow0$. To prove it, put $H(u)=\int_u^\infty h\,dv$; then $H'=-h$, $H(0)=\int_0^\infty h\,du$, and H is continuous with $H(u)\to0$ as $u\to\infty$, hence bounded. Integration by parts gives

$$\begin{aligned}\int_0^\infty e^{-\epsilon u}h\,du &= -\left[e^{-\epsilon u}H\right]_0^\infty - \epsilon\int_0^\infty e^{-\epsilon u}H\,du\\ &= H(0) - \epsilon\int_0^\infty e^{-\epsilon u}H\,du.\end{aligned}$$

Splitting the last integral at any $U > 0$ bounds the last term by $\epsilon U \sup|H| + \sup_{u\ge U}|H(u)|$, which is arbitrarily small when U is chosen large and then ϵ small. Now use

$$\sin\omega\cos\omega x = \tfrac{1}{2}\{\sin[(1+x)\omega] + \sin[(1-x)\omega]\}.$$

Dividing by ω and integrating term by term,

$$\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}d\omega = \tfrac{1}{2}[D(1+x)+D(1-x)] = \frac{\pi}{4}[\operatorname{sgn}(1+x)+\operatorname{sgn}(1-x)],$$

with $\operatorname{sgn}0 = 0$. For $|x|<1$ both signs are positive; for $|x|>1$ they cancel; at $|x|=1$ just one is positive. These are exactly the three stated values.

20.2.3

(a) Show that the Fourier sine and cosine transforms of e^{-at} are

$$g_s(\omega) = \sqrt{\frac{2}{\pi}}\,\frac{\omega}{\omega^2+a^2}, \qquad g_c(\omega) = \sqrt{\frac{2}{\pi}}\,\frac{a}{\omega^2+a^2}.$$

Hint. Each of the transforms can be related to the other by integration by parts.

(b) Show that

$$\int_0^\infty \frac{\omega\sin\omega x}{\omega^2+a^2}\,d\omega = \frac{\pi}{2}\,e^{-ax}, \qquad x>0,$$
$$\int_0^\infty \frac{\cos\omega x}{\omega^2+a^2}\,d\omega = \frac{\pi}{2a}\,e^{-ax}, \qquad x>0.$$

These results can also be obtained by contour integration (Exercise 11.8.12).

Assume $a > 0$.

Solution.

Take $a > 0$ for the ordinary exponentially decaying transforms and the answers as printed.

(a) Put

$$C = \int_0^\infty e^{-at}\cos\omega t\,dt, \qquad S = \int_0^\infty e^{-at}\sin\omega t\,dt.$$

Integrate by parts using an antiderivative $-e^{-at}/a$ of the exponential. The upper-limit terms vanish because $a > 0$, so

$$C = \left[-\frac{e^{-at}}{a}\cos\omega t\right]_0^\infty - \frac{\omega}{a}\int_0^\infty e^{-at}\sin\omega t\,dt = \frac{1}{a} - \frac{\omega}{a}S,$$
$$S = \left[-\frac{e^{-at}}{a}\sin\omega t\right]_0^\infty + \frac{\omega}{a}\int_0^\infty e^{-at}\cos\omega t\,dt = \frac{\omega}{a}C.$$

Substitute the second into the first and multiply by a^2: $a^2C = a - \omega^2 C$, hence

$$C = \frac{a}{a^2+\omega^2}, \qquad S = \frac{\omega}{a^2+\omega^2}.$$

Multiplication by $\sqrt{2/\pi}$ gives g_c and g_s. (b) For $x > 0$, the inversion formulas $f(x) = \sqrt{2/\pi}\int_0^\infty g_c(\omega)\cos\omega x\,d\omega$ and $f(x) = \sqrt{2/\pi}\int_0^\infty g_s(\omega)\sin\omega x\,d\omega$, with $\sqrt{2/\pi}\sqrt{2/\pi} = 2/\pi$, therefore read

$$e^{-ax} = \frac{2a}{\pi}\int_0^\infty \frac{\cos\omega x}{a^2+\omega^2}d\omega, \qquad e^{-ax} = \frac{2}{\pi}\int_0^\infty \frac{\omega\sin\omega x}{a^2+\omega^2}d\omega.$$

Multiplying the first by $\pi/(2a)$ and the second by $\pi/2$ gives the two integrals of (b). The cosine integral is absolutely convergent. The sine integral converges conditionally for $x > 0$; it follows from inversion of the odd extension of e^{-at} on $t > 0$. At $x = 0$ that odd extension has a jump and its inversion value is zero, so the first formula in (b) must not be extended there by substitution.

20.2.4 Find the Fourier transform of the triangular pulse (Fig. 20.6),

$$f(x) = \begin{cases} h(1-a|x|), & |x| < 1/a, \\ 0, & |x| > 1/a. \end{cases}$$

Note. This function provides another delta sequence with $h = a$ and $a \to \infty$.
Assume $a > 0$, so the pulse has positive width.

Solution.

Take $a > 0$, so that the pulse has the stated positive width. Evenness eliminates the sine part:

$$g(\omega) = \frac{2h}{\sqrt{2\pi}}\int_0^{1/a}(1-ax)\cos\omega x\,dx.$$

For $\omega \neq 0$, use $\int\cos\omega x\,dx = \sin\omega x/\omega$ and, integrating by parts,

$$\int x\cos\omega x\,dx = \frac{x\sin\omega x}{\omega} + \frac{\cos\omega x}{\omega^2}.$$

Between the limits 0 and $1/a$ the terms proportional to $\sin(\omega/a)$ cancel; then $1-\cos\theta = 2\sin^2(\theta/2)$ gives

$$\begin{aligned}\int_0^{1/a}(1-ax)\cos\omega x\,dx &= \frac{\sin(\omega/a)}{\omega} - a\left[\frac{\sin(\omega/a)}{a\omega} + \frac{\cos(\omega/a)-1}{\omega^2}\right] \\ &= \frac{a[1-\cos(\omega/a)]}{\omega^2} = \frac{2a\sin^2(\omega/2a)}{\omega^2} \\ &= \frac{1}{2a}\left[\frac{\sin(\omega/2a)}{\omega/2a}\right]^2.\end{aligned}$$

Multiplying by $2h/\sqrt{2\pi}$,

$$\boxed{g(\omega) = \frac{h}{a\sqrt{2\pi}}\left[\frac{\sin(\omega/2a)}{\omega/2a}\right]^2}, \qquad g(0) = \frac{h}{a\sqrt{2\pi}}.$$

The area of the pulse, a triangle of base $2/a$ and height h, is h/a, and $g(0)$ is $(2\pi)^{-1/2}$ times this area. With $h = a$ the area is one; substitute $u = ax$, $dx = du/a$, in the action of $f_a(x) = a(1-a|x|)$ on a continuous test function η:

$$\int f_a(x)\eta(x)dx = \int_{-1}^{1}(1-|u|)\eta(u/a)du \longrightarrow \eta(0)\int_{-1}^{1}(1-|u|)du = \eta(0),$$

because $\eta(u/a) \to \eta(0)$ uniformly for $|u| \le 1$ by the continuity of η at 0. This proves the delta-sequence claim, not merely a pointwise concentration of the graph. Correspondingly, for $h = a$ the transform $g(\omega) = (2\pi)^{-1/2}[\sin(\omega/2a)/(\omega/2a)]^2$ tends, for every ω, to $(2\pi)^{-1/2}$, the transform of $\delta(x)$.

20.2.5 Consider the sequence

$$\delta_n(x) = \begin{cases} n, & |x| < 1/2n, \\ 0, & |x| > 1/2n. \end{cases}$$

This is Eq. (1.152). Express $\delta_n(x)$ as a Fourier integral (via the Fourier integral theorem, inverse transform, etc.). Finally, show that we may write

$$\delta(x) = \lim_{n\to\infty} \delta_n(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ikx}\,dk.$$

The limit and the final Fourier integral are distributional identities, tested against smooth rapidly decreasing functions.

Solution.

The forward transform is

$$\begin{aligned} g_n(k) &= \frac{n}{\sqrt{2\pi}}\int_{-1/(2n)}^{1/(2n)} e^{ikx}dx = \frac{n}{\sqrt{2\pi}}\,\frac{e^{ik/2n} - e^{-ik/2n}}{ik} \\ &= \frac{n}{\sqrt{2\pi}}\,\frac{2\sin(k/2n)}{k} = \frac{1}{\sqrt{2\pi}}\frac{\sin(k/2n)}{k/2n}. \end{aligned}$$

Fourier inversion, $\delta_n(x) = (2\pi)^{-1/2}\int g_n(k)e^{-ikx}dk$, consequently gives

$$\boxed{\delta_n(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\sin(k/2n)}{k/2n}e^{-ikx}dk},$$

with the midpoint value $n/2$ at either edge of the rectangle. Those two point values do not affect any integral. For a smooth rapidly decreasing test function η (real or complex), the interval $|x| < 1/(2n)$ on which $\delta_n = n$ has length $1/n$, so $\eta(0) = n\int_{-1/(2n)}^{1/(2n)}\eta(0)\,dx$ and

$$\begin{aligned} \left|\int \delta_n(x)\eta(x)dx - \eta(0)\right| &= \left|n\int_{-1/(2n)}^{1/(2n)}[\eta(x) - \eta(0)]\,dx\right| \\ &\le \max_{|x|\le 1/(2n)}|\eta(x) - \eta(0)| \longrightarrow 0 \end{aligned}$$

by the continuity of η at 0. To take the same limit in the Fourier integral, let $\tilde\eta(k) = (2\pi)^{-1/2}\int\eta(x)e^{ikx}dx$; it is integrable, and $\eta(x) = (2\pi)^{-1/2}\int\tilde\eta(k)e^{-ikx}dk$. Insert this in $\int\delta_n\eta\,dx$ and interchange the absolutely convergent integrations. Since g_n is even, $\int\delta_n(x)e^{-ikx}dx = \sqrt{2\pi}\,g_n(-k) = \sin(k/2n)/(k/2n)$, so

$$\int\delta_n(x)\eta(x)dx = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{\sin(k/2n)}{k/2n}\,\tilde\eta(k)\,dk \longrightarrow \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\tilde\eta(k)\,dk = \eta(0).$$

The limit follows from dominated convergence: the multiplier tends to one, and $|\sin u/u| \le 1$ for real u. The last step is the inversion formula at $x = 0$. Written out, the limit is $(2\pi)^{-1}\int dk\int\eta(x)e^{ikx}dx = (2\pi)^{-1}\int dk\int\eta(x)e^{-ikx}dx$ $(k\to -k)$, the action on η of the kernel $(2\pi)^{-1}\int e^{-ikx}dk$. Hence the limiting identity is

$$\boxed{\delta(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-ikx}dk}.$$

This last integral is a distributional Fourier identity, not a convergent ordinary integral or a pointwise limit of functions at $x = 0$. The test-function argument supplies its precise meaning.

20.2.6 Using the sequence

$$\delta_n(x) = \frac{n}{\sqrt{\pi}} \exp(-n^2 x^2),$$

show that

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx}\, dk.$$

Hint. Remember that $\delta(x)$ is defined in terms of its behavior as part of an integrand. Interpret the final Fourier integral as a distribution, rather than an ordinary convergent integral.

Solution.

The exponent completes to a square as

$$-n^2x^2 + ikx = -n^2\left(x - \frac{ik}{2n^2}\right)^2 - \frac{k^2}{4n^2}.$$

Set $z = x - ik/(2n^2)$; as x runs over $\mathbb{R}$, z runs along the horizontal line $\Im z = -k/(2n^2)$. This line can be shifted back to the real axis: $e^{-n^2z^2}$ is entire, so by Cauchy's theorem its integral around the rectangle with vertical sides $z = \pm R + iv$ (v between 0 and $-k/(2n^2)$) vanishes, and on those vertical sides the modulus is $|e^{-n^2(\pm R+iv)^2}| = e^{-n^2R^2+n^2v^2}$, with v in a fixed bounded interval, so those integrals tend to zero. Consequently the remaining Gaussian integral is $\int_{\mathbb{R}} e^{-n^2z^2}dz = \sqrt{\pi}/n$, and

$$\begin{aligned} g_n(k) &= \frac{1}{\sqrt{2\pi}} \frac{n}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-n^2x^2+ikx} dx = \frac{1}{\sqrt{2\pi}} \frac{n}{\sqrt{\pi}}\, e^{-k^2/(4n^2)} \frac{\sqrt{\pi}}{n} \\ &= \frac{1}{\sqrt{2\pi}} e^{-k^2/(4n^2)}. \end{aligned}$$

The inverse $\delta_n(x) = (2\pi)^{-1/2} \int g_n(k) e^{-ikx} dk$ is therefore the explicit regularized identity

$$\frac{n}{\sqrt{\pi}} e^{-n^2x^2} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-k^2/(4n^2)} e^{-ikx} dk.$$

For a bounded continuous test function η, put $u = nx$, $dx = du/n$:

$$\int \frac{n}{\sqrt{\pi}} e^{-n^2x^2} \eta(x) dx = \frac{1}{\sqrt{\pi}} \int e^{-u^2} \eta(u/n) du \longrightarrow \frac{\eta(0)}{\sqrt{\pi}} \int e^{-u^2} du = \eta(0).$$

Dominated convergence applies, with the integrable majorant $e^{-u^2} \sup|\eta|$. For a Schwartz test function η, with transform $\tilde{\eta}$, the pairing argument of Exercise 20.2.5 gives likewise

$$\int \frac{n}{\sqrt{\pi}} e^{-n^2x^2} \eta(x) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-k^2/(4n^2)} \tilde{\eta}(k)\, dk \longrightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{\eta}(k)\, dk = \eta(0),$$

because $0 < e^{-k^2/(4n^2)} \leq 1$, the multiplier tends to one, and $\tilde{\eta}$ is integrable. As there, the limit is the action of the kernel $(2\pi)^{-1} \int e^{-ikx} dk$ on η. This gives the required delta identity with the factor $1/(2\pi)$. The unregularized exponential integral is understood in this weak, distributional sense; removing the Gaussian does not make it an ordinary improper integral.

20.2.7 The formula

$$\delta(t-x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{i\omega(t-x)}\,d\omega=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{i\omega t}e^{-i\omega x}\,d\omega$$

can be identified as the continuum limit of an eigenfunction expansion. Derive sine and cosine representations of $\delta(t-x)$ that are comparable to the exponential representation just given.

ANS. $\dfrac{2}{\pi}\displaystyle\int_0^{\infty}\sin\omega t\sin\omega x\,d\omega,\quad \frac{2}{\pi}\int_0^{\infty}\cos\omega t\cos\omega x\,d\omega.$

The two half-line answers above mean distributions for $x,t>0$. On the full line they require reflected delta terms, as derived in the solution.

Solution.

On a finite interval $(0,L)$ the orthonormal Dirichlet eigenfunctions are $\sqrt{2/L}\sin(n\pi x/L)$, $n\ge1$, and the orthonormal Neumann eigenfunctions are $1/\sqrt{L}$ and $\sqrt{2/L}\cos(n\pi x/L)$, $n\ge1$; indeed $\int_0^L\sin^2(n\pi x/L)dx=\int_0^L\cos^2(n\pi x/L)dx=L/2$. With φ_n denoting either orthonormal set, the completeness relations $\sum_n\varphi_n(t)\varphi_n(x)=\delta(t-x)$ give, for interior x,t,

$$\delta(t-x)=\frac{2}{L}\sum_{n=1}^{\infty}\sin\frac{n\pi t}{L}\sin\frac{n\pi x}{L},$$

$$\delta(t-x)=\frac{1}{L}+\frac{2}{L}\sum_{n=1}^{\infty}\cos\frac{n\pi t}{L}\cos\frac{n\pi x}{L}.$$

Set $\omega_n=n\pi/L$, so $\Delta\omega=\omega_{n+1}-\omega_n=\pi/L$. The factor $2/L=(2/\pi)\Delta\omega$ turns the sums into the Riemann sums $(2/\pi)\sum_n\sin\omega_nt\sin\omega_nx\,\Delta\omega$ and $(2/\pi)\sum_n\cos\omega_nt\cos\omega_nx\,\Delta\omega$, which become integrals over $0<\omega<\infty$ as $L\to\infty$; the constant Neumann term $1/L$ vanishes. In the weak sense this gives the two quoted kernels.

Their domains can be checked directly from the product-to-sum formulas $\sin\omega t\sin\omega x=\frac12[\cos\omega(t-x)-\cos\omega(t+x)]$ and $\cos\omega t\cos\omega x=\frac12[\cos\omega(t-x)+\cos\omega(t+x)]$:

$$\frac{2}{\pi}\int_0^{\infty}\sin\omega t\sin\omega x\,d\omega=\delta(t-x)-\delta(t+x),$$
$$\frac{2}{\pi}\int_0^{\infty}\cos\omega t\cos\omega x\,d\omega=\delta(t-x)+\delta(t+x).$$

Here $\delta(u)=\pi^{-1}\int_0^{\infty}\cos\omega u\,d\omega$, which is the given exponential representation with $t-x$ replaced by u: in $(2\pi)^{-1}\int_{-\infty}^{\infty}(\cos\omega u+i\sin\omega u)\,d\omega$ the odd sine term integrates to zero, and the even cosine term gives twice its integral over $(0,\infty)$. The stated sine and cosine answers represent $\delta(t-x)$ on the positive half-line, with $x,t>0$. On the whole real line they contain the reflected source $\mp\delta(t+x)$ shown above. Endpoint values at zero require the corresponding half-line boundary convention. For fixed $x>0$ the reflected source $\delta(t+x)$ is concentrated at $t=-x<0$, outside the support of any test function supported inside $(0,\infty)$, so it contributes nothing there; this proves exactly the needed positive-half-line identities.

20.2.8 In a resonant cavity, an electromagnetic oscillation of frequency ω_0 dies out as

$$A(t)=\begin{cases}A_0e^{-\omega_0t/2Q}e^{-i\omega_0t}, & t>0,\\ 0, & t<0.\end{cases}$$

The parameter Q is a measure of the ratio of stored energy to energy loss per cycle. Calculate the frequency distribution of the oscillation, $a^*(\omega)a(\omega)$, where $a(\omega)$ is the Fourier transform of $A(t)$.

Note. The larger Q is, the sharper your resonance line will be.

$$ANS.\ \ a^*(\omega)a(\omega) = \frac{|A_0|^2}{2\pi}\,\frac{1}{(\omega-\omega_0)^2+(\omega_0/2Q)^2}.$$

Take $\omega_0 > 0$, $Q > 0$. The amplitude A_0 may be complex, so its squared modulus is required in the answer.

Solution.

Put $\gamma = \omega_0/(2Q) > 0$, the amplitude decay rate. In the direct transformation of the causal signal the kernel $e^{i\omega t}$ combines with $e^{-\gamma t - i\omega_0 t}$ into a single exponential, and its value at the upper limit vanishes because $|e^{-[\gamma - i(\omega-\omega_0)]t}| = e^{-\gamma t} \to 0$:

$$\begin{aligned} a(\omega) &= \frac{A_0}{\sqrt{2\pi}}\int_0^\infty e^{-[\gamma - i(\omega-\omega_0)]t}dt = \frac{A_0}{\sqrt{2\pi}}\left[-\frac{e^{-[\gamma-i(\omega-\omega_0)]t}}{\gamma - i(\omega-\omega_0)}\right]_0^\infty \\ &= \frac{A_0}{\sqrt{2\pi}}\frac{1}{\gamma - i(\omega-\omega_0)}. \end{aligned}$$

The conjugate $a^*(\omega)$ has the denominator $\gamma + i(\omega-\omega_0)$, and the product of the two denominators is $\gamma^2 + (\omega-\omega_0)^2$. Hence

$$\boxed{|a(\omega)|^2 = \frac{|A_0|^2}{2\pi}\,\frac{1}{(\omega-\omega_0)^2+(\omega_0/2Q)^2}.}$$

The printed numerator A_0^2 presumes a real initial amplitude; for a complex amplitude it is $|A_0|^2$. Decay requires $\omega_0/Q > 0$, conventionally $\omega_0 > 0$, $Q > 0$. The peak is at ω_0, where the last factor equals $1/\gamma^2$. Half its height, $1/(2\gamma^2)$, occurs where $(\omega-\omega_0)^2 = \gamma^2$, i.e., at $|\omega-\omega_0| = \gamma$, so the full width at half maximum is $2\gamma = \omega_0/Q$. Increasing Q narrows the line. As a normalization check (Parseval's theorem), use $\int_{-\infty}^{\infty} d\omega/[(\omega-\omega_0)^2+\gamma^2] = \pi/\gamma$ and $\int_0^\infty e^{-2\gamma t}dt = 1/(2\gamma)$:

$$\int_{-\infty}^{\infty}|a(\omega)|^2 d\omega = \frac{|A_0|^2}{2\pi}\,\frac{\pi}{\gamma} = \frac{|A_0|^2}{2\gamma} = |A_0|^2\int_0^\infty e^{-2\gamma t}dt = \int_0^\infty |A(t)|^2 dt.$$

20.2.9 Prove that

$$\frac{\hbar}{2\pi i}\int_{-\infty}^{\infty}\frac{e^{-i\omega t}\,d\omega}{E_0 - i\Gamma/2 - \hbar\omega} = \begin{cases} \exp\left(-\dfrac{\Gamma t}{2\hbar}\right)\exp\left(-i\dfrac{E_0 t}{\hbar}\right), & t > 0, \\ 0, & t < 0. \end{cases}$$

This Fourier integral appears in a variety of problems in quantum mechanics: barrier penetration, scattering, time-dependent perturbation theory, and so on.

(The formula as given assumes that $\Gamma > 0$.)

Hint. Try contour integration.

Solution.

Let

$$\omega_p = \frac{E_0 - i\Gamma/2}{\hbar}, \qquad \hbar > 0, \quad E_0 \in \mathbb{R}, \quad \Gamma > 0.$$

Since $E_0 - i\Gamma/2 - \hbar\omega = -\hbar(\omega - \omega_p)$, the factor $\hbar$ cancels and the integral is

$$I(t) = -\frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\omega - \omega_p} d\omega.$$

The pole lies below the real axis: $\Im\omega_p = -\Gamma/(2\hbar) < 0$. If $t > 0$, write $\omega = u + iv$; the modulus of the exponential is $|e^{-iut}e^{vt}| = e^{vt}$, which decays for $v < 0$, so close the contour in the lower half-plane. The closure is clockwise, and, because $1/(\omega - \omega_p) \to 0$ uniformly on large arcs, Jordan's lemma makes its arc integral vanish. The residue of $e^{-i\omega t}/(\omega - \omega_p)$ at the simple pole is $e^{-i\omega_p t}$; consequently

$$I(t) = -\frac{1}{2\pi i}(-2\pi i)e^{-i\omega_p t} = e^{-i(E_0 - i\Gamma/2)t/\hbar} = e^{-\Gamma t/(2\hbar)}e^{-iE_0 t/\hbar}.$$

If $t < 0$, the factor e^{vt} decays for $v > 0$, so close above instead. There is no pole inside, so the integral is zero. This proves both cases, including the sign from the contour orientation. The Instructor's Manual, p. 334, writes $e^{+i\omega_p t}$ in its residue and final step. The exponent must be negative, as the original integrand shows; the positive sign would produce growth for $t > 0$. The published condition $\Gamma > 0$ is necessary for the causal contour choice. At $t = 0$ the symmetric principal-value inversion gives $1/2$, the mean across the jump. Indeed, $\omega - \omega_p$ stays in the upper half-plane, so along the real axis the principal $\ln(\omega - \omega_p)$ is continuous and $\arg(\omega - \omega_p)$ decreases from π to 0, while $\ln|R - \omega_p| - \ln|-R - \omega_p| \to 0$ as $R \to \infty$; hence

$$I(0) = -\frac{1}{2\pi i}\lim_{R\to\infty}\left[\ln(\omega - \omega_p)\right]_{-R}^{R} = -\frac{-i\pi}{2\pi i} = \frac{1}{2}.$$

The problem itself states only $t \neq 0$. For $\Gamma = 0$, a pole prescription would have to be supplied before the real-axis integral is defined.

20.2.10 Verify that the following are Fourier integral transforms of one another:

(a) $\left\{\begin{array}{ll} \sqrt{\dfrac{2}{\pi}} \cdot \dfrac{1}{\sqrt{a^2 - x^2}}, & |x| < a, \\ 0, & |x| > a, \end{array}\right\}$ and $J_0(ay)$,

(b) $\left\{\begin{array}{ll} 0, & |x| < a, \\ -\sqrt{\dfrac{2}{\pi}}\dfrac{1}{\sqrt{x^2 - a^2}}, & |x| > a, \end{array}\right\}$ and $Y_0(a|y|)$,

(c) $\sqrt{\dfrac{\pi}{2}}\dfrac{1}{\sqrt{x^2 + a^2}}$ and $K_0(a|y|)$.

(d) Can you suggest why $I_0(ay)$ is not included in this list?

(It is helpful to use the integral representation $J_0(x) = \dfrac{2}{\pi}\displaystyle\int_0^1 \frac{\cos xt}{\sqrt{1 - t^2}}\, dt$.)

Hint. J_0, Y_0, and K_0 may be transformed most easily by using an exponential representation, reversing the order of integration, and employing the Dirac delta function exponential representation, Eq. (20.20). These cases can be treated equally well as Fourier cosine transforms.

Take $a > 0$. In (b),(c), ordinary transform integrals are asserted for $y \neq 0$; the identities also have a tempered-distribution interpretation.

Solution.

Take $a > 0$ for these transform pairs. Since all the displayed functions are even, the full unitary transform is $\sqrt{2/\pi}\int_0^\infty f(x)\cos(xy)dx$: in $(2\pi)^{-1/2}\int f(x)(\cos xy + i\sin xy)\,dx$ the odd sine part integrates to zero, and the even cosine part gives twice its integral over $(0,\infty)$.

(a) For the compactly supported member the prefactors give $\sqrt{2/\pi}\sqrt{2/\pi} = 2/\pi$, so

$$\begin{aligned}\mathcal{F}f(y) &= \frac{2}{\pi}\int_0^a \frac{\cos(xy)}{\sqrt{a^2-x^2}}dx \\ &= \frac{2}{\pi}\int_0^1 \frac{\cos(ayu)}{\sqrt{1-u^2}}du = J_0(ay).\end{aligned}$$

The substitution $x = au$ gives $dx = a\,du$ and $\sqrt{a^2-x^2} = a\sqrt{1-u^2}$, so both factors of a cancel. The last equality is the Poisson representation stated in the published erratum.

(b) With the corrected square root, the prefactors give $-2/\pi$. Since $\cos(xy) = \cos(x|y|)$, the substitution $x = au$ ($dx = a\,du$, $\sqrt{x^2-a^2} = a\sqrt{u^2-1}$) gives

$$\mathcal{F}f(y) = -\frac{2}{\pi}\int_a^\infty \frac{\cos(xy)}{\sqrt{x^2-a^2}}dx = -\frac{2}{\pi}\int_1^\infty \frac{\cos(a|y|u)}{\sqrt{u^2-1}}du = Y_0(a|y|), \quad y \neq 0.$$

The last equality is the real oscillatory representation $Y_0(z) = -(2/\pi)\int_1^\infty \cos(zt)(t^2-1)^{-1/2}dt$, $z > 0$, derived in Exercise 14.4.8(b), with $z = a|y|$. Its endpoint singularity is integrable and its infinite tail is conditionally convergent for $y \neq 0$.

(c) The prefactors now multiply to one, $\sqrt{2/\pi}\sqrt{\pi/2} = 1$, giving

$$\mathcal{F}f(y) = \int_0^\infty \frac{\cos(xy)}{\sqrt{x^2+a^2}}dx = K_0(a|y|), \quad y \neq 0.$$

To prove the last equality, write Exercise 14.5.13, with its y replaced by a, in the unitary normalization:

$$\frac{1}{\sqrt{x^2+a^2}} = \frac{2}{\pi}\int_0^\infty K_0(aq)\cos(xq)dq = \sqrt{\frac{2}{\pi}}\int_0^\infty \left[\sqrt{\frac{2}{\pi}}K_0(aq)\right]\cos(xq)dq.$$

Thus $1/\sqrt{x^2+a^2}$ is the cosine transform of $\sqrt{2/\pi}K_0(aq)$, which is integrable on $(0,\infty)$ (its singularity at $q = 0$ is only logarithmic) and smooth for $q > 0$. The cosine transform is its own inverse, so for $q > 0$

$$\sqrt{\frac{2}{\pi}}\int_0^\infty \frac{\cos(xq)}{\sqrt{x^2+a^2}}dx = \sqrt{\frac{2}{\pi}}K_0(aq).$$

At $q = |y|$, with $\cos(xy) = \cos(x|y|)$, this is precisely the asserted full-transform relation. Applying the full transform twice reflects the original function, $\mathcal{F}^2 f(x) = f(-x)$: by the inversion formula, $(2\pi)^{-1/2}\int g(y)e^{ixy}dy$ is the inverse transform of g evaluated at $-x$, and for $g = \mathcal{F}f$ it is $f(-x)$. All three members here are even, so this also verifies the reverse directions. The logarithmic singularities in (b) and (c) at $y = 0$ are not finite point values. Their transform identities hold as tempered distributions and as the indicated ordinary improper integrals away from zero; values at the support edges in (a),(b) are likewise singular.

(d) For $a > 0$, the asymptotic form $I_0(z) \sim e^z/\sqrt{2\pi z}$ as $z \to +\infty$ (Section 14.6), together with the evenness of I_0 (its series contains only even powers of z), gives $I_0(ay) \sim e^{a|y|}/\sqrt{2\pi a|y|}$ as $|y| \to \infty$. It therefore has neither an ordinary Fourier integral nor a Fourier transform as a tempered distribution represented by this function. This exponential growth is the obstruction to an analogous pair in the present Fourier setting. At $a = 0$ the constant $I_0(0) = 1$ instead has the distributional transform $\sqrt{2\pi}\delta$, a separate degenerate case.

20.2.11 Show that the left-hand expression is the forward Fourier transform of the right-hand expression, and find the forward transform of the left-hand expression:

$$i^n J_n(t) \qquad \text{and} \qquad \begin{cases} \sqrt{\dfrac{2}{\pi}}\, T_n(x)(1-x^2)^{-1/2}, & |x|<1, \\ 0, & |x|>1. \end{cases}$$

$T_n(x)$ is the nth-order Chebyshev polynomial.

(The left-hand-side quantities are the transforms of their right-hand-side counterparts, but the right-hand-side quantities are $(-1)^n$ times the transforms of the left-hand-side expressions.)

Hint. With $T_n(\cos\theta) = \cos n\theta$, the transform of $T_n(x)(1-x^2)^{-1/2}$ leads to an integral representation of $J_n(t)$.

Solution.

Denote the compactly supported right-hand expression by $f_n(x)$. Its transform, with prefactor $(2\pi)^{-1/2}\sqrt{2/\pi} = 1/\pi$, is

$$\mathcal{F}f_n(t) = \frac{1}{\pi}\int_{-1}^{1} \frac{T_n(x)e^{itx}}{\sqrt{1-x^2}}\,dx.$$

Set $x = \cos\theta$, where $0 \le \theta \le \pi$; then $\sqrt{1-x^2} = \sin\theta$ and $T_n(x) = \cos n\theta$. Reversing the limits removes the sign of $dx = -\sin\theta\,d\theta$, and $\sin\theta$ cancels:

$$\mathcal{F}f_n(t) = \frac{1}{\pi}\int_0^{\pi} e^{it\cos\theta}\cos(n\theta)\,d\theta.$$

For integer $n \ge 0$ the integral is half the full-circle integral, because reflection $\theta \mapsto 2\pi - \theta$ leaves both factors unchanged. The sine part of $e^{in\theta}$ integrates to zero under the same reflection, because $\sin n\theta$ changes sign. To evaluate the full-circle integral, use the Jacobi–Anger expansion of Exercise 14.1.5,

$$e^{it\cos\theta} = \sum_{m=-\infty}^{\infty} i^m J_m(t) e^{im\theta}.$$

Multiply by $e^{in\theta}$ and integrate term by term; as $(2\pi)^{-1}\int_0^{2\pi} e^{i(m+n)\theta}d\theta = \delta_{m,-n}$, only $m = -n$ survives. With $J_{-n} = (-1)^n J_n$ and $i^{-n}(-1)^n = i^n$,

$$\frac{1}{\pi}\int_0^{\pi} e^{it\cos\theta}\cos n\theta\,d\theta = \frac{1}{2\pi}\int_0^{2\pi} e^{it\cos\theta + in\theta}d\theta = i^{-n}J_{-n}(t) = \boxed{i^n J_n(t)}.$$

To check the direction of the transform, apply it again and use $\mathcal{F}^2 f(x) = f(-x)$ (Exercise 20.2.10):

$$\mathcal{F}[i^n J_n](x) = f_n(-x) = (-1)^n f_n(x),$$

since $T_n(-x) = (-1)^n T_n(x)$. Thus the inverse transform of $i^n J_n$ is f_n, while its forward transform is $(-1)^n f_n$. This proves, and explains, the published sign correction. The square-root singularities are integrable; reverse inversion can be read in the tempered-distribution sense at the singular edges.

20.2.12 Show that the Fourier exponential transform of

$$f(\mu) = \begin{cases} P_n(\mu), & |\mu| \le 1 \\ 0, & |\mu| > 1 \end{cases}$$

is $i^n\sqrt{2/\pi}\, j_n(\omega)$, where ω is the transform variable (the book prints $(2i^n/2\pi)\, j_n(kr)$). Here $P_n(\mu)$ is a Legendre polynomial and j_n is a spherical Bessel function.

Solution.

The forward transform, with variable ω, is

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} P_n(\mu) e^{i\omega\mu} d\mu.$$

Use Rodrigues' formula $P_n(\mu) = (2^n n!)^{-1} d^n(\mu^2-1)^n/d\mu^n$. Each integration by parts moves one derivative from $(\mu^2-1)^n$ onto $e^{i\omega\mu}$, contributing a factor -1 and, from $de^{i\omega\mu}/d\mu = i\omega e^{i\omega\mu}$, a factor $i\omega$. There are no endpoint terms: the polynomial $(\mu^2-1)^n = (\mu-1)^n(\mu+1)^n$ has zeros of order n at $\mu = \pm 1$, so it and its first $n-1$ derivatives vanish at both endpoints. After n steps the factor is $(-1)^n(i\omega)^n = (-i\omega)^n$; in the second line below, $(\mu^2-1)^n = (-1)^n(1-\mu^2)^n$ and $(-1)^n(-i\omega)^n = i^n\omega^n$. Hence

$$\begin{aligned} g(\omega) &= \frac{(-i\omega)^n}{2^n n!\sqrt{2\pi}} \int_{-1}^{1} (\mu^2-1)^n e^{i\omega\mu} d\mu \\ &= \frac{i^n\omega^n}{2^n n!\sqrt{2\pi}} \int_{-1}^{1} (1-\mu^2)^n e^{i\omega\mu} d\mu. \end{aligned}$$

The spherical Poisson representation of Exercise 14.7.4 is

$$j_n(\omega) = \frac{\omega^n}{2^{n+1} n!} \int_{-1}^{1} (1-\mu^2)^n e^{i\omega\mu} d\mu.$$

Multiplied by $2i^n/\sqrt{2\pi}$, its right side becomes the second line for $g(\omega)$, since $2/2^{n+1} = 1/2^n$. Hence

$$\boxed{g(\omega) = \frac{2i^n}{\sqrt{2\pi}} j_n(\omega) = i^n\sqrt{\frac{2}{\pi}}\, j_n(\omega).}$$

No division by ω occurs, so the result holds at $\omega = 0$ as well: there both sides equal $\sqrt{2/\pi}\,\delta_{n0}$, since $\int_{-1}^{1} P_n\, d\mu = 2\delta_{n0}$ and $j_n(0) = \delta_{n0}$. This derivation verifies both the scaling and the transform variable in the marked correction; kr is only a possible later choice of ω.

20.2.13

(a) Show that $f(x) = x^{-1/2}$ is a *self-reciprocal* under both Fourier cosine and sine transforms; that is,

$$\sqrt{\frac{2}{\pi}} \int_0^\infty x^{-1/2} \cos xt\, dx = t^{-1/2},$$
$$\sqrt{\frac{2}{\pi}} \int_0^\infty x^{-1/2} \sin xt\, dx = t^{-1/2}.$$

(b) Use the preceding results to evaluate the Fresnel integrals

$$\int_0^\infty \cos\left(y^2\right) dy \quad \text{and} \quad \int_0^\infty \sin\left(y^2\right) dy.$$

In part (a), take $t > 0$.

Solution.

The differential in the second integral must be dx: the textbook, p. 979, prints ds. Take $t > 0$; both displayed integrals then converge conditionally. (a) Introduce positive damping before taking the square-root limit:

$$I_\epsilon(t) = \int_0^\infty x^{-1/2} e^{-(\epsilon - it)x} dx = \Gamma(1/2)(\epsilon - it)^{-1/2}.$$

The gamma integral applies because $\Re(\epsilon - it) > 0$: for real $s > 0$ the substitution $u = sx$ gives $\int_0^\infty x^{-1/2} e^{-sx} dx = s^{-1/2} \int_0^\infty u^{-1/2} e^{-u} du = \Gamma(1/2) s^{-1/2}$, with $\Gamma(1/2) = \sqrt{\pi}$, and both sides are analytic in s for $\Re s > 0$, so the identity persists there with the principal branch of $s^{-1/2}$. As $\epsilon \downarrow 0$, $|\epsilon - it| \to t$ and, on the principal branch, $\arg(\epsilon - it) \to -\pi/2$, so $(\epsilon - it)^{-1/2} \to t^{-1/2} e^{i\pi/4}$ and, with $e^{i\pi/4} = (1+i)/\sqrt{2}$,

$$I(t) = \lim_{\epsilon \downarrow 0} I_\epsilon(t) = \sqrt{\pi}\, t^{-1/2} e^{i\pi/4} = \sqrt{\frac{\pi}{2}}\, t^{-1/2}(1+i).$$

The ordinary integrals converge at zero by the exponent $-1/2$ and at infinity by Dirichlet's test, so Abel's theorem for integrals (Exercise 20.2.2(c), here with $h(x) = x^{-1/2} e^{ixt}$) identifies this limit with their values: $I(t) = \int_0^\infty x^{-1/2} \cos xt\, dx + i \int_0^\infty x^{-1/2} \sin xt\, dx$. Equating real and imaginary parts, each of these integrals is $\sqrt{\pi/2}\, t^{-1/2}$; multiplying by $\sqrt{2/\pi}$ proves both self-reciprocal identities. The Instructor's Manual, p. 336, has two further slips: the change $u = -izt$ (its integration variable z is our x) maps the positive real z axis to the negative imaginary u axis, and its final factor $t^{1/2}$ must be $t^{-1/2}$. The damped gamma calculation above fixes both the phase and the scaling without an ambiguous contour rotation.

(b) For the Fresnel integrals take $t = 1$ in $\int_0^\infty x^{-1/2} \cos xt\, dx = \int_0^\infty x^{-1/2} \sin xt\, dx = \sqrt{\pi/2}\, t^{-1/2}$ and set $x = y^2$. Then $dx = 2y\, dy$ and $x^{-1/2} dx = 2dy$, with the limits $0, \infty$ unchanged, giving

$$2\int_0^\infty \cos(y^2) dy = 2\int_0^\infty \sin(y^2) dy = \sqrt{\frac{\pi}{2}}.$$

Thus each is $\boxed{\sqrt{\pi/8}}$. The substitution is first made at a finite upper limit, followed by the convergent improper limit; it does not require absolute convergence.

20.2.14 Show that $\left[\frac{1}{r^2}\right]^T(\mathbf{k}) = \left(\frac{\pi}{2}\right)^{1/2} \frac{1}{k}$.

Here $k = |\mathbf{k}| > 0$. The equality extends distributionally, with a singular zero-frequency value.

Solution.

Let $k = |\mathbf{k}| > 0$ and use a polar axis parallel to $\mathbf{k}$. To make the large-radius integration explicit, first multiply r^{-2} by $e^{-\epsilon r}$, $\epsilon > 0$. Since $d^3 r = r^2 dr\, d\Omega$, the radial singularity cancels against the volume factor. With $d\Omega = \sin\theta\, d\theta\, d\phi$ and $u = \cos\theta$, the angular integration is

$$\int d\Omega\, e^{ikr\cos\theta} = 2\pi \int_{-1}^{1} e^{ikru} du = 2\pi \frac{e^{ikr} - e^{-ikr}}{ikr} = 4\pi \frac{\sin kr}{kr}.$$

Thus

$$\mathcal{F}\left[\frac{e^{-\epsilon r}}{r^2}\right](\mathbf{k}) = \frac{4\pi}{(2\pi)^{3/2}k}\int_0^\infty e^{-\epsilon r}\frac{\sin kr}{r}dr$$
$$= \sqrt{\frac{2}{\pi}}\frac{1}{k}\arctan\frac{k}{\epsilon}.$$

Here $4\pi/(2\pi)^{3/2} = \sqrt{2/\pi}$, and the last integral is $D_\epsilon(k) = \arctan(k/\epsilon)$ of Exercise 20.2.2 (with $b = k$, $u = r$). As $\epsilon \downarrow 0$, $\arctan(k/\epsilon) \to \pi/2$ and $\sqrt{2/\pi}\,\pi/2 = \sqrt{\pi/2}$. The limit also holds after pairing with a Schwartz test function $\eta(\mathbf{k})$. Put $\tilde{\eta}(\mathbf{r}) = (2\pi)^{-3/2}\int \eta(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}}d^3k$, again a Schwartz function. Since η and $e^{-\epsilon r}r^{-2}$ are integrable ($\int e^{-\epsilon r}r^{-2}d^3r = 4\pi\int_0^\infty e^{-\epsilon r}dr = 4\pi/\epsilon$), the order of integration may be interchanged:

$$\int \mathcal{F}\Big[\frac{e^{-\epsilon r}}{r^2}\Big](\mathbf{k})\,\eta(\mathbf{k})\,d^3k = \int \frac{e^{-\epsilon r}}{r^2}\,\tilde{\eta}(\mathbf{r})\,d^3r.$$

As $\epsilon \downarrow 0$, dominated convergence takes the left side to $\int \sqrt{\pi/2}\,k^{-1}\eta\,d^3k$ (majorant $\sqrt{\pi/2}\,k^{-1}|\eta|$, since $0 < \arctan(k/\epsilon) < \pi/2$) and the right side to $\int r^{-2}\tilde{\eta}\,d^3r$ (majorant $r^{-2}|\tilde{\eta}|$). Both majorants are integrable, because k^{-1} and r^{-2} are locally integrable in three dimensions ($d^3k = k^2dk\,d\Omega$, $d^3r = r^2dr\,d\Omega$) and η, $\tilde{\eta}$ decrease rapidly. The transform of the distribution r^{-2} is defined by this pairing rule, $\int \mathcal{F}[r^{-2}]\,\eta\,d^3k = \int r^{-2}\tilde{\eta}\,d^3r$, so the equality of the two limits for every η states that

$$\boxed{\mathcal{F}[r^{-2}](\mathbf{k}) = \sqrt{\frac{\pi}{2}}\frac{1}{k}}.$$

Here k is the nonnegative magnitude of the wave vector. The formula is for $k > 0$, or is a tempered-distribution identity on all of three-dimensional momentum space. Its value at $k = 0$ is singular, and the original spatial integral is not absolutely convergent at infinity.

20.2.15 The Fourier transform formulas for a function of two variables are

$$F(u,v) = \frac{1}{2\pi}\iint f(x,y)e^{i(ux+vy)}\,dx\,dy,$$
$$f(x,y) = \frac{1}{2\pi}\iint F(u,v)e^{-i(ux+vy)}\,du\,dv,$$

where the integrations are over the entire xy or uv plane. For $f(x,y) = f\big([x^2+y^2]^{1/2}\big) = f(r)$, show that the zero-order Hankel transforms

$$F(\rho) = \int_0^\infty rf(r)J_0(\rho r)\,dr,$$
$$f(r) = \int_0^\infty \rho F(\rho)J_0(\rho r)\,d\rho,$$

are a special case of the Fourier transforms.

Note. This technique may be generalized to derive the Hankel transforms of order $\nu = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$. See the two texts by Sneddon (Additional Readings). It might also be noted that the Hankel transforms of half-integral orders $\nu = \pm\frac{1}{2}$ reduce to Fourier sine and cosine transforms.

Solution.

Put $x = r\cos\theta$, $y = r\sin\theta$, and write $u = \rho\cos\phi$, $v = \rho\sin\phi$. Then $ux + vy = \rho r(\cos\theta\cos\phi + \sin\theta\sin\phi) = \rho r\cos(\theta - \phi)$ and $dx\,dy = r\,dr\,d\theta$. The forward transform is consequently

$$\begin{aligned} F(u,v) &= \frac{1}{2\pi}\int_0^\infty rf(r)dr\int_0^{2\pi} e^{i\rho r\cos(\theta-\phi)}d\theta \\ &= \int_0^\infty rf(r)J_0(\rho r)dr. \end{aligned}$$

The angular integral is $2\pi J_0(\rho r)$, independent of ϕ: the integrand has period 2π in θ, so the shift $\theta - \phi \to \theta$ does not change the integral, and $(2\pi)^{-1}\int_0^{2\pi} e^{iz\cos\theta}d\theta = J_0(z)$ is the case $n = 0$ of the integral evaluated in Exercise 20.2.11. This proves that the transform is radial: $F = F(\rho)$. For the inverse, use $du\,dv = \rho\,d\rho\,d\phi$:

$$\begin{aligned} f(x,y) &= \frac{1}{2\pi}\int_0^\infty \rho F(\rho)d\rho\int_0^{2\pi} e^{-i\rho r\cos(\theta-\phi)}d\phi \\ &= \int_0^\infty \rho F(\rho)J_0(\rho r)d\rho, \end{aligned}$$

because the same steps give $2\pi J_0(-\rho r)$ for the angular integral and J_0 is even. This is the claimed inverse Hankel transform. The factors r and ρ are the two polar Jacobians; the Fourier factors $1/(2\pi)$ cancel the angular integrals. For radial Schwartz functions these operations (polar coordinates and the order of integration) are justified directly. By Parseval's theorem, $\iint |f|^2dx\,dy = \iint |F|^2du\,dv$; in polar coordinates both sides carry the factor 2π of the angular integration, so $\int_0^\infty |f(r)|^2 r\,dr = \int_0^\infty |F(\rho)|^2\rho\,d\rho$. The Hankel pair therefore extends, as the Fourier transform does, to all radial f with $\int_0^\infty |f(r)|^2 r\,dr < \infty$.

20.2.16 Show that the 3-D Fourier exponential transform of a radially symmetric function may be rewritten as a Fourier sine transform:

$$\frac{1}{(2\pi)^{3/2}}\int_{\mathbb{R}^3} f(r)e^{i\mathbf{k}\cdot\mathbf{r}}\,d^3r = \frac{1}{k}\sqrt{\frac{2}{\pi}}\int_0^\infty rf(r)\sin kr\,dr.$$

Take $k = |\mathbf{k}| > 0$, and use a continuous limit at $k = 0$ when it exists.

Solution.

Choose the z axis along $\mathbf{k}$, with $k = |\mathbf{k}| > 0$, so that $\mathbf{k}\cdot\mathbf{r} = kr\cos\theta$. In spherical coordinates,

$$\begin{aligned} \mathcal{F}f(\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}}\int_0^\infty r^2 f(r)dr\int_0^{2\pi} d\phi\int_0^\pi e^{ikr\cos\theta}\sin\theta\,d\theta \\ &= \frac{2\pi}{(2\pi)^{3/2}}\int_0^\infty r^2 f(r)dr\int_{-1}^1 e^{ikru}du \\ &= \frac{4\pi}{(2\pi)^{3/2}}\int_0^\infty r^2 f(r)\frac{\sin kr}{kr}dr \\ &= \boxed{\frac{1}{k}\sqrt{\frac{2}{\pi}}\int_0^\infty rf(r)\sin kr\,dr}. \end{aligned}$$

In the second line $u = \cos\theta$, $du = -\sin\theta\,d\theta$; in the third, $\int_{-1}^1 e^{ikru}du = (e^{ikr} - e^{-ikr})/(ikr) = 2\sin kr/(kr)$; in the last, $4\pi/(2\pi)^{3/2} = \sqrt{2/\pi}$ and $r^2/(kr) = r/k$. Thus $k\mathcal{F}f(k)$ is the unitary sine

transform of $rf(r)$. The marked volume element must be d^3r over all space, as in the published correction. If $\int_0^\infty r^2|f(r)|dr < \infty$, the zero-frequency limit follows from the third line, since $\sin(kr)/(kr) \to 1$ with $|\sin(kr)/(kr)| \le 1$ (dominated convergence):

$$\mathcal{F}f(0) = \sqrt{\frac{2}{\pi}} \int_0^\infty r^2 f(r) dr.$$

Consequently the displayed $1/k$ does not create a singularity when this finite limit exists.

20.3 Properties of Fourier Transforms

20.3.1 Write the 1-D equivalents of the equations for translation, scale change, sign change, and complex conjugation that were given for 3-D transforms in Eqs. (20.48) to (20.51).

Solution.

Write $g(\omega) = \mathcal{F}f(\omega)$ with the positive forward exponential of Eq. (20.10). The four one-dimensional identities are

$$\begin{aligned}
\mathcal{F}[f(x-X)](\omega) &= e^{i\omega X} g(\omega), \\
\mathcal{F}[f(\alpha x)](\omega) &= \frac{1}{|\alpha|} g(\omega/\alpha), \quad \alpha \in \mathbb{R} \setminus \{0\}, \\
\mathcal{F}[f(-x)](\omega) &= g(-\omega), \\
\mathcal{F}[f^*(-x)](\omega) &= g^*(\omega).
\end{aligned}$$

For translation set $u = x - X$, so $dx = du$ and $e^{i\omega x} = e^{i\omega X} e^{i\omega u}$; the factor $e^{i\omega X}$ comes out of the integral. For scaling set $u = \alpha x$, so $dx = du/\alpha$ and $e^{i\omega x} = e^{i(\omega/\alpha)u}$. When $\alpha < 0$ the integration limits reverse, and restoring their order changes $1/\alpha$ into $-1/\alpha = 1/|\alpha|$; in either case

$$\mathcal{F}[f(\alpha x)](\omega) = \frac{1}{|\alpha|} \frac{1}{\sqrt{2\pi}} \int f(u) e^{i(\omega/\alpha)u} du = \frac{1}{|\alpha|} g(\omega/\alpha).$$

The third identity is the second at $\alpha = -1$. For the fourth, set $u = -x$ (the reversal of the limits cancels the sign of $dx = -du$) and obtain

$$\frac{1}{\sqrt{2\pi}} \int f^*(-x) e^{i\omega x} dx = \frac{1}{\sqrt{2\pi}} \int f^*(u) e^{-i\omega u} du = \left[\frac{1}{\sqrt{2\pi}} \int f(u) e^{i\omega u} du \right]^* = g^*(\omega).$$

Equivalently, applying the third identity to $f^*(-x)$, $\mathcal{F}[f^*](\omega) = g^*(-\omega)$. A scale-change factor without an absolute value presumes a positive scale. The formula above covers either sign; $\alpha = 0$ is not an invertible change of variables. The one-dimensional translation line of the Instructor's Manual, p. 337, prints $e^{i\omega t}$ for a shift $f(t-a)$. The phase must contain the fixed displacement, $e^{i\omega a}$, not the integration variable.

20.3.2

(a) Show that by replacement of $\mathbf{r}$ by $\mathbf{r} - \mathbf{R}$ in the formula for the Fourier transform of $f(\mathbf{r})$, one can derive the translation formula, Eq. (20.48).

(b) Using methods similar to those for part (a), establish the formulas for scale change, sign change, and complex conjugation, Eqs. (20.49) to (20.51).

Solution.

Let $g(\mathbf{k}) = (2\pi)^{-3/2}\int f(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}d^3r$. (a) Put $\mathbf{u} = \mathbf{r} - \mathbf{R}$. Translation has unit Jacobian, and the exponential separates:

$$\begin{aligned}\mathcal{F}[f(\mathbf{r}-\mathbf{R})](\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}}\int f(\mathbf{u})e^{i\mathbf{k}\cdot(\mathbf{u}+\mathbf{R})}d^3u \\ &= \boxed{e^{i\mathbf{k}\cdot\mathbf{R}}g(\mathbf{k})}.\end{aligned}$$

(b) For real nonzero α, set $\mathbf{u} = \alpha\mathbf{r}$. The volume element is $d^3r = |\alpha|^{-3}d^3u$ (the Jacobian of $\mathbf{r} = \mathbf{u}/\alpha$ is $|\det(\alpha^{-1}I)| = |\alpha|^{-3}$), and $\mathbf{k}\cdot\mathbf{r} = (\mathbf{k}/\alpha)\cdot\mathbf{u}$, so

$$\begin{aligned}\mathcal{F}[f(\alpha\mathbf{r})](\mathbf{k}) &= \frac{|\alpha|^{-3}}{(2\pi)^{3/2}}\int f(\mathbf{u})e^{i(\mathbf{k}/\alpha)\cdot\mathbf{u}}d^3u \\ &= \boxed{|\alpha|^{-3}g(\mathbf{k}/\alpha)}.\end{aligned}$$

In particular, at $\alpha = -1$ this gives $\mathcal{F}[f(-\mathbf{r})](\mathbf{k}) = g(-\mathbf{k})$. Finally, put $\mathbf{u} = -\mathbf{r}$, again with unit Jacobian, so that $\mathbf{k}\cdot\mathbf{r} = -\mathbf{k}\cdot\mathbf{u}$:

$$\begin{aligned}\mathcal{F}[f^*(-\mathbf{r})](\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}}\int f^*(\mathbf{u})e^{-i\mathbf{k}\cdot\mathbf{u}}d^3u \\ &= \left[\frac{1}{(2\pi)^{3/2}}\int f(\mathbf{u})e^{i\mathbf{k}\cdot\mathbf{u}}d^3u\right]^* = \boxed{g^*(\mathbf{k})}.\end{aligned}$$

The printed α^{-3} in Eq. (20.49) needs $\alpha > 0$. For negative real scaling the correct Jacobian is $|\alpha|^{-3}$, as the sign-change case itself confirms.

20.3.3 Derive Eq. (20.53), the formula for the Fourier transform of $\nabla^2 f(\mathbf{r})$.

Solution.

For a sufficiently decaying differentiable function, integrate by parts in r_j, holding the other coordinates fixed ($d^2r_\perp$ denotes their area element) and using $\partial_j e^{i\mathbf{k}\cdot\mathbf{r}} = ik_j e^{i\mathbf{k}\cdot\mathbf{r}}$:

$$\begin{aligned}\mathcal{F}[\partial_j f](\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}}\int d^2r_\perp\left([fe^{i\mathbf{k}\cdot\mathbf{r}}]_{r_j=-\infty}^{\infty} - \int f\,ik_j e^{i\mathbf{k}\cdot\mathbf{r}}dr_j\right) \\ &= -ik_j g(\mathbf{k}).\end{aligned}$$

The boundary bracket vanishes because $f \to 0$ as $|r_j| \to \infty$. Applying the same calculation to $\partial_j f$ gives $\mathcal{F}[\partial_j^2 f] = -ik_j\mathcal{F}[\partial_j f] = (-ik_j)^2 g = -k_j^2 g$. Summing the three coordinate equations yields

$$\boxed{\mathcal{F}[\nabla^2 f](\mathbf{k}) = -(k_x^2 + k_y^2 + k_z^2)g(\mathbf{k}) = -k^2 g(\mathbf{k})}.$$

Schwartz functions satisfy the required endpoint and interchange conditions. The same identity extends to tempered distributions by defining their derivatives through integration by parts, so singular sources can also be transformed consistently.

20.3.4 Verify Eqs. (20.55) and (20.56), the formulas for the derivatives of 1-D Fourier transforms.

Solution.

Starting from the one-dimensional definition and integrating by parts with $de^{i\omega t}/dt = i\omega e^{i\omega t}$,

$$\begin{aligned}\mathcal{F}[f'](\omega) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f'(t)e^{i\omega t}dt \\ &= \frac{1}{\sqrt{2\pi}}[f(t)e^{i\omega t}]_{-\infty}^{\infty} - i\omega\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)e^{i\omega t}dt \\ &= -i\omega g(\omega),\end{aligned}$$

since the bracket vanishes when $f(t) \to 0$ as $t \to \pm\infty$. For n derivatives repeat the operation, applying it to $f^{(n-1)}, f^{(n-2)}, \ldots$. Each repetition moves one derivative to the exponential and contributes $-i\omega$: $\mathcal{F}[f^{(n)}] = -i\omega\mathcal{F}[f^{(n-1)}] = \cdots$, so

$$\boxed{\mathcal{F}[f^{(n)}](\omega) = (-i\omega)^n g(\omega)}.$$

More explicitly, n integrations by parts on $(-R, R)$ give

$$\begin{aligned}\int_{-R}^{R} f^{(n)}(t)e^{i\omega t}dt = &\sum_{j=0}^{n-1}(-i\omega)^j[f^{(n-1-j)}(t)e^{i\omega t}]_{-R}^{R} \\ &+ (-i\omega)^n\int_{-R}^{R} f(t)e^{i\omega t}dt;\end{aligned}$$

under the usual decaying-function assumptions ($f, \ldots, f^{(n-1)}$ tend to zero at $\pm\infty$, and f and $f^{(n)}$ are integrable) each endpoint term tends to zero as $R \to \infty$. This explains why no initial-value terms occur for this two-sided transform, unlike the one-sided Laplace transform. If f has a jump at t_0, integrating by parts separately on each side of t_0 leaves the extra endpoint term $-[f(t_0^+) - f(t_0^-)]e^{i\omega t_0}/\sqrt{2\pi}$ in the transform of the pointwise derivative, so that derivative alone violates the identity. The identity still holds when f' is the distributional derivative, which contains the term $[f(t_0^+) - f(t_0^-)]\,\delta(t - t_0)$; its transform $[f(t_0^+) - f(t_0^-)]e^{i\omega t_0}/\sqrt{2\pi}$ cancels the endpoint term. Jumps of $f', \ldots, f^{(n-1)}$ are treated in the same way.

20.3.5 Derive the inverse of Eq. (20.56), namely that

$$[t^n f(t)]^T(\omega) = i^{-n}\,\frac{d^n}{d\omega^n}\,g(\omega).$$

Solution.

Differentiate the forward transform with respect to its frequency argument under the integral sign; since $\partial e^{i\omega t}/\partial\omega = it\,e^{i\omega t}$,

$$g'(\omega) = \frac{1}{\sqrt{2\pi}}\int f(t)\frac{\partial}{\partial\omega}e^{i\omega t}dt = \frac{i}{\sqrt{2\pi}}\int tf(t)e^{i\omega t}dt = i\,\mathcal{F}[tf](\omega).$$

After n derivatives, each bringing down another factor it,

$$g^{(n)}(\omega) = \frac{1}{\sqrt{2\pi}}\int (it)^n f(t)e^{i\omega t}dt = i^n\mathcal{F}[t^n f](\omega).$$

Dividing by i^n proves

$$\boxed{\mathcal{F}[t^n f](\omega) = i^{-n} g^{(n)}(\omega)}.$$

For example, $t^n f \in L^1$ justifies these differentiations by dominated convergence, with lower moments controlled by $f \in L^1$ and the highest moment. Schwartz functions satisfy all such conditions. This is the duality between multiplication in direct space and differentiation in frequency space.

20.3.6 The 1-D neutron diffusion equation with a (plane) source is

$$-D\,\frac{d^2\varphi(x)}{dx^2} + K^2 D\varphi(x) = Q\delta(x),$$

where $\varphi(x)$ is the neutron flux, $Q\delta(x)$ is the (plane) source at $x = 0$, and D and K^2 are constants. Apply a Fourier transform. Solve the equation in transform space. Transform your solution back into x-space.

$$ANS.\ \ \varphi(x) = \frac{Q}{2KD}\, e^{-|Kx|}.$$

For the displayed answer assume $D > 0$, $K > 0$, and seek the whole-line solution bounded at both infinities. These conditions select the decaying solution.

Solution.

For the decaying whole-line solution take $D > 0$ and $K^2 > 0$, and put $\kappa = \sqrt{K^2} > 0$. The printed coefficient $1/K$ presumes $K > 0$. Decay, or boundedness at both infinities, must be part of the boundary specification. Write $\Phi = \mathcal{F}\varphi$. The delta transform is $\mathcal{F}[\delta](\omega) = (2\pi)^{-1/2} \int \delta(x) e^{i\omega x} dx = 1/\sqrt{2\pi}$, and $\mathcal{F}[\varphi''] = (-i\omega)^2 \Phi = -\omega^2 \Phi$ by Exercise 20.3.4 with $n = 2$. Transforming the equation term by term gives $-D(-\omega^2\Phi) + \kappa^2 D\Phi = Q/\sqrt{2\pi}$, i.e.,

$$D(\omega^2 + \kappa^2)\Phi(\omega) = \frac{Q}{\sqrt{2\pi}}, \qquad \Phi(\omega) = \frac{Q}{D\sqrt{2\pi}(\omega^2 + \kappa^2)}.$$

Invert with $\varphi(x) = (2\pi)^{-1/2} \int \Phi(\omega) e^{-i\omega x} d\omega$. Because $1/(\omega^2 + \kappa^2)$ is even, the odd part $-i \sin \omega x$ of $e^{-i\omega x}$ integrates to zero and the cosine part gives twice its integral over $(0, \infty)$. The second formula of Exercise 20.2.3(b), with $a = \kappa$ and x replaced by $|x|$ (the cosine integral is even in x, and at $x = 0$ it equals $\int_0^\infty d\omega/(\omega^2 + \kappa^2) = \pi/(2\kappa)$), then gives $\int_0^\infty \cos \omega x\, (\omega^2 + \kappa^2)^{-1} d\omega = (\pi/2\kappa) e^{-\kappa|x|}$:

$$\begin{aligned} \varphi(x) &= \frac{Q}{2\pi D} \int_{-\infty}^{\infty} \frac{e^{-i\omega x}}{\omega^2 + \kappa^2} d\omega \\ &= \frac{Q}{\pi D} \int_0^\infty \frac{\cos \omega x}{\omega^2 + \kappa^2} d\omega = \boxed{\frac{Q}{2\kappa D} e^{-\kappa|x|}}. \end{aligned}$$

As a check, away from zero $\varphi'' = \kappa^2 \varphi$. For $x \gtrless 0$, $\varphi' = \mp \kappa \varphi$, so $\varphi'(0^\pm) = \mp Q/(2D)$ and the derivative jump is

$$\varphi'(0^+) - \varphi'(0^-) = -\frac{Q}{2D} - \frac{Q}{2D} = -\frac{Q}{D}.$$

As a distribution, therefore, $\varphi'' = \kappa^2 \varphi + [\varphi'(0^+) - \varphi'(0^-)]\delta(x) = \kappa^2 \varphi - (Q/D)\delta(x)$, so $-D\varphi'' + \kappa^2 D\varphi = Q\delta(x)$ exactly. A homogeneous addition is $Ae^{\kappa x} + Be^{-\kappa x}$; boundedness at both infinities forces $A = B = 0$. For $K = 0$ and nonzero Q, the formal fundamental solution is $-Q|x|/(2D)$ plus an affine function (since $|x|'' = 2\delta(x)$), and no solution decays at both ends. Substituting $K = 0$ in the printed formula is invalid.

20.4 Fourier Convolution Theorem

20.4.1 Work out the convolution equation corresponding to Eq. (20.71) for

(a) Fourier sine transforms

$$\frac{1}{2}\int_0^\infty g(y)\big[f(y+x)+f(y-x)\big]\,dy=\int_0^\infty F_s(s)G_s(s)\cos sx\,ds,$$

where f and g are odd functions.

(b) Fourier cosine transforms

$$\frac{1}{2}\int_0^\infty g(y)\big[f(y+x)+f(x-y)\big]\,dy=\int_0^\infty F_c(s)G_c(s)\cos sx\,ds,$$

where f and g are even functions.

Solution.

Use the unitary half-line transforms, for example $F_s(s)=\sqrt{2/\pi}\int_0^\infty f(u)\sin(su)du$, with the identical normalization in the inverse.

(a) Extend f oddly. Its inverse sine representation

$$f(u)=\sqrt{\frac{2}{\pi}}\int_0^\infty F_s(s)\sin(su)\,ds$$

then holds for all real u, since both sides are odd in u. Apply it at $u=y+x$ and $u=y-x$, and use the identity $\sin(A+B)+\sin(A-B)=2\sin A\cos B$ with $A=sy$, $B=sx$:

$$f(y+x)+f(y-x)=2\sqrt{\frac{2}{\pi}}\int_0^\infty F_s(s)\sin(sy)\cos(sx)ds.$$

Multiply by $g(y)/2$, integrate on $y>0$, and interchange the order of integration:

$$\begin{aligned}\frac{1}{2}\int_0^\infty g(y)[f(y+x)+f(y-x)]dy&=\sqrt{\frac{2}{\pi}}\int_0^\infty F_s(s)\cos(sx)\left[\int_0^\infty g(y)\sin(sy)dy\right]ds\\&=\int_0^\infty F_s(s)G_s(s)\cos(sx)ds.\end{aligned}$$

By the definition of G_s the bracket is $\sqrt{\pi/2}\,G_s(s)$, and the factors $\sqrt{2/\pi}$ and $\sqrt{\pi/2}$ cancel.

(b) Extend f evenly; then $f(u)=\sqrt{2/\pi}\int_0^\infty F_c(s)\cos(su)\,ds$ for all real u. Since $f(x-y)=f(y-x)$, the identity $\cos(A+B)+\cos(A-B)=2\cos A\cos B$, again with $A=sy$, $B=sx$, gives

$$f(y+x)+f(x-y)=2\sqrt{\frac{2}{\pi}}\int_0^\infty F_c(s)\cos(sy)\cos(sx)ds.$$

The same multiplication and integration replace the inner integral $\int_0^\infty g(y)\cos(sy)dy$ by $\sqrt{\pi/2}\,G_c(s)$, giving exactly the second formula. The calculation is justified first for sufficiently regular decaying transforms, and then by the L^2 pairing limit where appropriate; odd/even extensions are essential for arguments that become negative. The Instructor's Manual's intermediate two-delta kernels on p. 339, which arise when both transforms on the right-hand side are written as integrals and the s

integral is done first, omit reflected sources. For example, the sine kernel on the full range of its arguments is

$$\int_0^\infty \sin(sy)\sin(st)\cos(sx)ds = \frac{\pi}{4}\big[\delta(t-y-x)+\delta(t-y+x) \\ -\delta(t+y+x)-\delta(t+y-x)\big].$$

This follows by applying the product identity and the reflected-source formula in Exercise 20.2.7. Indeed, $\sin(sy)\sin(st) = \frac{1}{2}[\cos s(t-y) - \cos s(t+y)]$, and multiplying each cosine by $\cos(sx)$ with $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta)+\cos(\alpha-\beta)]$ turns the integrand into

$$\tfrac{1}{4}\big[\cos s(t-y+x)+\cos s(t-y-x)-\cos s(t+y+x)-\cos s(t+y-x)\big];$$

each term is then integrated with $\int_0^\infty \cos(su)\,ds = \pi\delta(u)$ (Exercise 20.2.7). The cosine kernel has the same four delta arguments with all plus signs. The reflected terms cannot all be discarded on $t, y > 0$ when x is arbitrary; the odd/even extensions above include them. Since $F_s(s)G_s(s)$ is $2/\pi$ times the integral of $f(t)g(y)\sin(st)\sin(sy)$ over $t, y > 0$, inserting the sine kernel and doing the t integral, in which each delta contributes only where its root t is positive (θ is the unit step function), gives

$$\int_0^\infty F_sG_s\cos(sx)\,ds = \frac{1}{2}\int_0^\infty g(y)\big[f(y+x)\theta(y+x)+f(y-x)\theta(y-x) \\ -f(-y-x)\theta(-y-x)-f(x-y)\theta(x-y)\big]dy.$$

For odd f, $-f(-y-x) = f(y+x)$ and $-f(x-y) = f(y-x)$, so, since $\theta(u)+\theta(-u) = 1$, the bracket is $f(y+x)+f(y-x)$, which gives (a); without the two reflected terms the contributions from $y+x<0$ and from $y<x$ would be lost. The cosine kernel, with all plus signs and even f, gives $f(y+x)+f(x-y)$ in the same way.

20.4.2 Show that for both Fourier sine and Fourier cosine transforms Parseval's relation has the form

$$\int_0^\infty F(t)G(t)\,dt = \int_0^\infty f(y)g(y)\,dy.$$

Solution.

Set $x = 0$ in each of the two formulas just proved. For the sine version the bracket is $f(y)+f(y) = 2f(y)$. For the cosine version it is $f(y)+f(-y) = 2f(y)$ because the extension is even. On either right-hand side $\cos(sx) = 1$. Thus in both cases

$$\boxed{\int_0^\infty f(y)g(y)dy = \int_0^\infty F(s)G(s)ds}.$$

The same result follows by inserting the inverse transform of f into the left-hand side and identifying the transform of g; for the sine case,

$$\int_0^\infty f(y)g(y)dy = \int_0^\infty g(y)\sqrt{\frac{2}{\pi}}\int_0^\infty F(s)\sin(sy)\,ds\,dy \\ = \int_0^\infty F(s)\left[\sqrt{\frac{2}{\pi}}\int_0^\infty g(y)\sin(sy)\,dy\right]ds = \int_0^\infty F(s)G(s)\,ds,$$

and the cosine case is identical with $\cos(sy)$; all normalization factors cancel. For L^2 functions, Cauchy–Schwarz controls both pairings and permits approximation by regular functions. This displayed version is bilinear and is valid also for complex functions. To obtain the norm form set $g = \overline{f}$: the sine and cosine kernels are real, so $G = \overline{F}$, and $\int|f|^2 = \int|F|^2$.

20.4.3

(a) A rectangular pulse is described by

$$f(x) = \begin{cases} 1, & |x| < a, \\ 0, & |x| > a. \end{cases}$$

Show that the Fourier exponential transform is

$$F(t) = \sqrt{\frac{2}{\pi}}\,\frac{\sin at}{t}.$$

This is the single-slit diffraction problem of physical optics. The slit is described by $f(x)$. The diffraction pattern amplitude is given by the Fourier transform $F(t)$.

(b) Use the Parseval relation to evaluate

$$\int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2}\,dt.$$

This integral may also be evaluated by using the calculus of residues (Exercise 11.8.9).

ANS. (b) π.

Take $a > 0$.

Solution.

(a) For $a > 0$, direct integration gives

$$F(t) = \frac{1}{\sqrt{2\pi}}\int_{-a}^{a} e^{itx}dx = \frac{1}{\sqrt{2\pi}}\frac{e^{iat} - e^{-iat}}{it}$$
$$= \frac{1}{\sqrt{2\pi}}\frac{2\sin at}{t} = \boxed{\sqrt{\frac{2}{\pi}}\frac{\sin at}{t}},$$

using $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$ and $2/\sqrt{2\pi} = \sqrt{2/\pi}$. At $t = 0$ take the limit $F(0) = a\sqrt{2/\pi}$, since $\sin at/t \to a$. (b) Parseval's relation in the unitary form $\int |f(x)|^2 dx = \int |F(t)|^2 dt$ gives

$$2a = \int_{-a}^{a} 1\,dx = \int_{-\infty}^{\infty} |F(t)|^2 dt = \frac{2}{\pi}\int_{-\infty}^{\infty} \frac{\sin^2 at}{t^2}dt.$$

Set $u = at$, so $dt = du/a$ and $t^{-2} = a^2/u^2$:

$$2a = \frac{2a}{\pi}\int_{-\infty}^{\infty} \frac{\sin^2 u}{u^2}du.$$

Canceling $2a$ gives the required value $\boxed{\pi}$. The apparent singularity at zero is removable and the tails are bounded by u^{-2}, so the final integral is absolutely convergent.

20.4.4 Solve Poisson's equation, $\nabla^2\psi(\mathbf{r}) = -\rho(\mathbf{r})/\varepsilon_0$, by the following sequence of operations:

(a) Take the Fourier transform of both sides of this equation. Solve for the Fourier transform of $\psi(\mathbf{r})$.

(b) Carry out the Fourier inverse transform.

Seek the free-space solution that vanishes at infinity, for sufficiently localized charge. Without this boundary condition, harmonic additions are possible.

Solution.

(a) Let $\widehat{\psi}$ and $\widehat{\rho}$ denote unitary three-dimensional transforms. By Exercise 20.3.3, $\mathcal{F}[\nabla^2\psi] = -k^2\widehat{\psi}$, so the transformed equation is

$$-k^2\widehat{\psi}(\mathbf{k}) = -\frac{\widehat{\rho}(\mathbf{k})}{\varepsilon_0}, \qquad \widehat{\psi}(\mathbf{k}) = \frac{\widehat{\rho}(\mathbf{k})}{\varepsilon_0 k^2}.$$

(b) To invert the multiplier, first evaluate the absolutely convergent regulated radial transform, $\eta > 0$, with the radial formula of Exercise 20.2.16 (here $rf(r) = e^{-\eta r}$) and the integral S of Exercise 20.2.3 (with $a = \eta$, $\omega = k$):

$$\mathcal{F}\left[\frac{e^{-\eta r}}{r}\right](\mathbf{k}) = \sqrt{\frac{2}{\pi}}\frac{1}{k}\int_0^\infty e^{-\eta r}\sin kr\, dr = \sqrt{\frac{2}{\pi}}\frac{1}{k}\frac{k}{k^2+\eta^2}$$
$$= \sqrt{\frac{2}{\pi}}\frac{1}{k^2+\eta^2}.$$

Letting $\eta \downarrow 0$ in distributions gives $\mathcal{F}[1/r] = \sqrt{2/\pi}/k^2$, so by Fourier inversion $\mathcal{F}^{-1}[\sqrt{2/\pi}/k^2] = 1/r$: the inverse transform of k^{-2} is $\sqrt{\pi/2}/r$. For our normalization, if $A = \mathcal{F}a$ and $B = \mathcal{F}b$, the convolution theorem reads

$$\mathcal{F}^{-1}[AB](\mathbf{r}) = (2\pi)^{-3/2}\int a(\mathbf{u})b(\mathbf{r}-\mathbf{u})d^3u.$$

Indeed, transforming the right-hand side and substituting $\mathbf{r} = \mathbf{u} + \mathbf{v}$ at fixed $\mathbf{u}$ factorizes it:

$$\frac{1}{(2\pi)^3}\int a(\mathbf{u})e^{i\mathbf{k}\cdot\mathbf{u}}d^3u\int b(\mathbf{v})e^{i\mathbf{k}\cdot\mathbf{v}}d^3v = \frac{(2\pi)^{3/2}A\,(2\pi)^{3/2}B}{(2\pi)^3} = AB.$$

Use $a = \rho$ and $b = \sqrt{\pi/2}/r$, so that $B = k^{-2}$:

$$\psi(\mathbf{r}) = \frac{1}{\varepsilon_0}\mathcal{F}^{-1}\left[\frac{\widehat{\rho}}{k^2}\right] = \frac{1}{\varepsilon_0}\frac{\sqrt{\pi/2}}{(2\pi)^{3/2}}\int\frac{\rho(\mathbf{u})}{|\mathbf{r}-\mathbf{u}|}d^3u.$$

Since $\sqrt{\pi/2}/(2\pi)^{3/2} = \sqrt{\pi}/(\sqrt{2}\cdot 2\sqrt{2}\,\pi^{3/2}) = 1/(4\pi)$,

$$\boxed{\psi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\mathbf{u})}{|\mathbf{r}-\mathbf{u}|}d^3u}.$$

This is the free-space solution with vanishing potential at infinity for suitably localized charge. Without a boundary condition, one may add any harmonic function; division by k^2 does not establish uniqueness at the zero mode. For smooth localized charge the Coulomb singularity is locally integrable, and $\nabla^2(1/r) = -4\pi\delta^{(3)}$ verifies the source normalization of the answer.

20.4.5

(a) Given $f(x) = 1 - |x/2|$ for $-2 \le x \le 2$, with $f(x) = 0$ elsewhere, show that the Fourier transform of $f(x)$ is

$$F(t) = \sqrt{\frac{2}{\pi}} \left(\frac{\sin t}{t} \right)^2 .$$

(b) Using the Parseval relation, evaluate

$$\int_{-\infty}^{\infty} \left(\frac{\sin t}{t} \right)^4 dt.$$

ANS. (b) $\dfrac{2\pi}{3}$.

Solution.

(a) Let $q(x)$ equal one on $[-1, 1]$ and zero elsewhere. In this solution $*$ denotes the unnormalized convolution integral, so the Fourier convolution theorem includes its factor $\sqrt{2\pi}$. In $(q * q)(x) = \int q(y) q(x - y) dy$ the integrand is one exactly when $y \in [-1, 1]$ and $y \in [x - 1, x + 1]$. For $0 \le x \le 2$ the overlap of the two intervals is $[x - 1, 1]$, of length $2 - x$; for $-2 \le x \le 0$ it is $[-1, x + 1]$, of length $2 + x$; for $|x| > 2$ the intervals do not overlap. Hence $(q * q)(x) = 2 - |x|$ for $|x| \le 2$ and zero outside, and $f = (q * q)/2$. From Exercise 20.4.3 with $a = 1$, $Q(t) = \sqrt{2/\pi} \sin t/t$. The unitary convolution theorem reads $\mathcal{F}[q * q] = \sqrt{2\pi}\, Q^2$: substituting $x = y + z$ in the transform of $q * q$ gives $(2\pi)^{-1/2} \int q(y) e^{ity} dy \int q(z) e^{itz} dz = (2\pi)^{-1/2} (\sqrt{2\pi}\, Q)^2$. Therefore

$$F(t) = \frac{1}{2} \sqrt{2\pi}\, Q(t)^2 = \frac{1}{2} \sqrt{2\pi}\, \frac{2}{\pi} \left(\frac{\sin t}{t} \right)^2 = \boxed{\sqrt{\frac{2}{\pi}} \left(\frac{\sin t}{t} \right)^2}.$$

(b) Since f is even, its direct-space squared norm is twice the integral over $0 \le x \le 2$; there put $u = 1 - x/2$, $dx = -2\, du$, so u runs from 1 to 0, and reversing the limits absorbs the minus sign:

$$\int |f(x)|^2 dx = 2 \int_0^2 (1 - x/2)^2 dx = 4 \int_0^1 u^2 du = \frac{4}{3}.$$

In transform space it is

$$\int |F(t)|^2 dt = \frac{2}{\pi} \int_{-\infty}^{\infty} \left(\frac{\sin t}{t} \right)^4 dt.$$

Equating the norms yields the value $\boxed{(\pi/2)(4/3) = 2\pi/3}$. The Instructor's Manual's Parseval line on p. 341 prints $(\sin t/t)^2$ where it must print $(\sin t/t)^4$: the already squared amplitude $F(t)$ is squared again to form $|F(t)|^2$.

20.4.6 With $F(t)$ and $G(t)$ the Fourier transforms of $f(x)$ and $g(x)$, respectively, show that

$$\int_{-\infty}^{\infty} \big| f(x) - g(x) \big|^2 dx = \int_{-\infty}^{\infty} \big| F(t) - G(t) \big|^2 dt.$$

If $g(x)$ is an approximation to $f(x)$, the preceding relation indicates that the mean square deviation in t-space is equal to the mean square deviation in x-space.

Assume $f, g \in L^2$, or more generally that the difference and its transform are defined in L^2.

Solution.

Let $h = f - g$. Linearity gives $\mathcal{F}h = F - G$. Parseval's norm identity applied to h is

$$\int |h(x)|^2 dx = \int |\mathcal{F}h(t)|^2 dt,$$

which is exactly the claimed equation. To display the complex cross terms explicitly,

$$|f - g|^2 = |f|^2 + |g|^2 - f\overline{g} - \overline{f}g.$$

The unitary inner-product identity $\int u\overline{v}\, dx = \int U\overline{V}\, dt$ sends each of these four integrals respectively to the integrals of $|F|^2$, $|G|^2$, $F\overline{G}$, and $\overline{F}G$, and with the same signs $|F|^2+|G|^2-F\overline{G}-\overline{F}G = |F-G|^2$. The finite mean-square statement requires $f - g \in L^2$ and its L^2 transform; for example $f, g \in L^2$ suffice. It is an equality of integrated squared errors, not pointwise errors at corresponding coordinates.

20.4.7 Use the Parseval relation to evaluate

$$\text{(a)} \int_{-\infty}^{\infty} \frac{d\omega}{(\omega^2 + a^2)^2}, \qquad \text{(b)} \int_{-\infty}^{\infty} \frac{\omega^2\, d\omega}{(\omega^2 + a^2)^2}.$$

Hint. Compare Exercise 20.2.3.

ANS. (a) $\dfrac{\pi}{2a^3}$, (b) $\dfrac{\pi}{2a}$.

The displayed answers use $a > 0$; for nonzero real a replace their a by $|a|$.

Solution.

Take $a > 0$ and apply the half-line Parseval identities to $f(x) = e^{-ax}$. Its direct squared norm is

$$\int_0^{\infty} e^{-2ax} dx = \frac{1}{2a}.$$

(a) The cosine transform of Exercise 20.2.3 is $g_c(\omega) = \sqrt{2/\pi}\, a/(\omega^2 + a^2)$, so Parseval's relation of Exercise 20.4.2, applied with both functions equal to f (so that both transforms are g_c), gives

$$\frac{1}{2a} = \int_0^{\infty} g_c(\omega)^2 d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{a^2}{(\omega^2 + a^2)^2} d\omega.$$

Dividing by $2a^2/\pi$ gives $\int_0^{\infty} d\omega/(\omega^2 + a^2)^2 = \pi/(4a^3)$. The integrand is even, so doubling the half-line integral gives

$$\boxed{\int_{-\infty}^{\infty} \frac{d\omega}{(\omega^2 + a^2)^2} = \frac{\pi}{2a^3}}.$$

(b) Using the sine transform $g_s(\omega) = \sqrt{2/\pi}\, \omega/(\omega^2 + a^2)$ instead,

$$\frac{1}{2a} = \int_0^{\infty} g_s(\omega)^2 d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{\omega^2}{(\omega^2 + a^2)^2} d\omega,$$

so the half-line integral is $\pi/(4a)$ and, the integrand being even, the whole-line answer is $\boxed{\pi/(2a)}$. If a is allowed to be negative, the integrands are unchanged and the answers contain $|a|^3$ and $|a|$, respectively. At $a = 0$ both integrals diverge. The printed forms use $a > 0$.

20.4.8 The nuclear form factor $F(\mathbf{k})$ and the charge distribution $\rho(\mathbf{r})$ are 3-D Fourier transforms of each other:

$$F(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int \rho(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}}\, d^3r.$$

If the measured form factor is

$$F(\mathbf{k}) = (2\pi)^{-3/2} \left(1 + \frac{k^2}{a^2}\right)^{-1},$$

find the corresponding charge distribution.

$$ANS.\ \ \rho(r) = \frac{a^2}{4\pi}\,\frac{e^{-ar}}{r}.$$

The parameter a in the displayed answer is the positive scale, $a > 0$.

Solution.

Assume $a > 0$ first. Rewrite the measured transform as

$$F(\mathbf{k}) = \frac{a^2}{(2\pi)^{3/2}} \frac{1}{k^2 + a^2}.$$

The radial formula of Exercise 20.2.16, with $rf(r) = e^{-ar}$ and $\int_0^\infty e^{-ar} \sin kr\, dr = k/(k^2 + a^2)$ (the integral S of Exercise 20.2.3 with $\omega = k$), gives

$$\mathcal{F}\left[\frac{e^{-ar}}{r}\right](\mathbf{k}) = \sqrt{\frac{2}{\pi}} \frac{1}{k} \int_0^\infty e^{-ar} \sin kr\, dr = \sqrt{\frac{2}{\pi}} \frac{1}{k^2 + a^2}.$$

Therefore multiplication by $a^2/(4\pi)$ produces the given F, because $(4\pi)^{-1}\sqrt{2/\pi} = (2\pi)^{-3/2}$. Since a function is determined by its Fourier transform (the inversion theorem), the charge distribution is

$$\boxed{\rho(r) = \frac{a^2}{4\pi}\,\frac{e^{-ar}}{r}}.$$

Its total charge is

$$4\pi \int_0^\infty r^2 \rho(r) dr = a^2 \int_0^\infty r e^{-ar} dr = a^2 \cdot \frac{1}{a^2} = 1,$$

in agreement with $\int \rho\, d^3r = (2\pi)^{3/2} F(0) = 1$ (put $\mathbf{k} = 0$ in the definition of F). The $1/r$ origin singularity is integrable with the volume element. The scale in the printed decaying exponential is the positive root. For nonzero real a of either sign use $e^{-|a|r}$; the form factor only determines a^2.

20.4.9 Using convolution methods, find an integral whose value is the electrostatic interaction energy between a charge distribution $\rho(\mathbf{r} - \mathbf{A})$ and a unit point charge at $\mathbf{C}$.

Solution.

Let $H(\mathbf{r}) = 1/|\mathbf{r}|$, write $(\rho * H)(\mathbf{v}) = \int \rho(\mathbf{u}) H(\mathbf{v} - \mathbf{u}) d^3u$ for the unnormalized convolution, and use SI normalization, so the potential of unit charge is $H/(4\pi\varepsilon_0)$. The potential at $\mathbf{C}$ of the translated distribution is its interaction energy with the unit point charge:

$$U = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r} - \mathbf{A})}{|\mathbf{C} - \mathbf{r}|} d^3r.$$

Putting $\mathbf{u} = \mathbf{r} - \mathbf{A}$ identifies a convolution:

$$\boxed{U = \frac{1}{4\pi\varepsilon_0}(\rho * H)(\mathbf{C} - \mathbf{A}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\mathbf{u})}{|\mathbf{C} - \mathbf{A} - \mathbf{u}|}d^3u}.$$

If $R(\mathbf{k}) = \mathcal{F}\rho(\mathbf{k})$, then $\mathcal{F}H = \sqrt{2/\pi}/k^2$ (Exercise 20.4.4), and the convolution theorem of Exercise 20.4.4, in the form $\mathcal{F}(\rho * H) = (2\pi)^{3/2}R\,\mathcal{F}H$, gives $\mathcal{F}(\rho * H) = 4\pi R/k^2$, because $(2\pi)^{3/2}\sqrt{2/\pi} = 2^{3/2}\pi^{3/2}2^{1/2}\pi^{-1/2} = 4\pi$. Inverting,

$$(\rho * H)(\mathbf{v}) = \frac{1}{(2\pi)^{3/2}}\int \frac{4\pi R(\mathbf{k})}{k^2}e^{-i\mathbf{k}\cdot\mathbf{v}}d^3k,$$

and multiplying by $1/(4\pi\varepsilon_0)$ at $\mathbf{v} = \mathbf{C} - \mathbf{A}$ yields the equivalent transform-space integral

$$\boxed{U = \frac{1}{\varepsilon_0(2\pi)^{3/2}}\int \frac{R(\mathbf{k})}{k^2}e^{-i\mathbf{k}\cdot(\mathbf{C}-\mathbf{A})}d^3k}.$$

Both formulas show that only the relative displacement enters. For real charge the frequency-domain integrand has conjugate symmetry ($R(-\mathbf{k}) = R(\mathbf{k})^*$, so its value at $-\mathbf{k}$ is the complex conjugate of its value at $\mathbf{k}$), so U is real. These formulas assume the potential integral exists; point-charge singularities are understood through the corresponding distributions. There is no factor $1/2$ in the mutual energy of these two specified charge distributions.

20.4.10 With $\psi(\mathbf{r})$ a wave function in ordinary space and $\varphi(\mathbf{p})$ the corresponding momentum function, show that

(a) $$\frac{1}{(2\pi\hbar)^{3/2}}\int \mathbf{r}\psi(\mathbf{r})e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}\,d^3r = i\hbar\nabla_p\,\varphi(\mathbf{p}),$$

(b) $$\frac{1}{(2\pi\hbar)^{3/2}}\int \mathbf{r}^2\psi(\mathbf{r})e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}\,d^3r = \left(i\hbar\nabla_p\right)^2\varphi(\mathbf{p}).$$

(This exercise assumes that the units and scaling of the momentum wave function correspond to the formula $\varphi(\mathbf{p}) = \dfrac{1}{(2\pi\hbar)^{3/2}}\displaystyle\int \psi(\mathbf{r})e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}\,d^3r$.)

Note. ∇_p is the gradient in momentum space:

$$\hat{\mathbf{e}}_x\frac{\partial}{\partial p_x} + \hat{\mathbf{e}}_y\frac{\partial}{\partial p_y} + \hat{\mathbf{e}}_z\frac{\partial}{\partial p_z}.$$

These results may be extended to any positive integer power of $\mathbf{r}$ on the appropriate moment domain, and to an analytic function only when its Maclaurin series can be interchanged with the transform for the wave function in use.

Solution.

Use precisely the momentum-space convention stated in the erratum, whose forward exponential has a negative sign:

$$\varphi(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}}\int \psi(\mathbf{r})e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}d^3r.$$

(a) Differentiation of the exponential gives

$$\nabla_p e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} = -\frac{i\mathbf{r}}{\hbar}e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}.$$

Move the derivative under the integral and multiply by $i\hbar$; since $i\hbar(-i\mathbf{r}/\hbar) = \mathbf{r}$,

$$\boxed{i\hbar\nabla_p\varphi = \frac{1}{(2\pi\hbar)^{3/2}}\int \mathbf{r}\psi(\mathbf{r})e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}d^3r}.$$

(b) Apply another component derivative: differentiating twice with respect to p_j brings down $(-ir_j/\hbar)^2 = -r_j^2/\hbar^2$. Summing over $j = x, y, z$,

$$\nabla_p^2 e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} = -\frac{r_x^2 + r_y^2 + r_z^2}{\hbar^2}e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}.$$

Since $(i\hbar\nabla_p)^2 = -\hbar^2\nabla_p^2$ and $-\hbar^2(-r^2/\hbar^2) = r^2$, differentiating under the integral proves

$$\boxed{(i\hbar\nabla_p)^2\varphi = \frac{1}{(2\pi\hbar)^{3/2}}\int r^2\psi(\mathbf{r})e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}d^3r}.$$

The exponent printed in part (b) omits the factor i: it reads $-\mathbf{r}\cdot\mathbf{p}/\hbar$. The Fourier kernel must be $e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}$ in both parts. The same argument works for every polynomial in the coordinate components. Appropriate integrable moments of ψ justify ordinary differentiation; otherwise operator domains or a well-defined distributional interpretation must be specified.

20.4.11 The ordinary space wave function $\psi(\mathbf{r}, t)$ satisfies the time-dependent Schrödinger equation,

$$i\hbar\frac{\partial\psi(\mathbf{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r})\psi.$$

Show that the corresponding time-dependent momentum wave function satisfies the analogous equation

$$i\hbar\frac{\partial\varphi(\mathbf{p}, t)}{\partial t} = \frac{p^2}{2m}\varphi + V(i\hbar\nabla_p)\varphi.$$

Note. Assume that $V(\mathbf{r})$ may be expressed by a Maclaurin series whose termwise transformation is justified on the chosen wave function (a polynomial potential suffices), and use Exercise 20.4.10. $V(i\hbar\nabla_p)$ is the same function of the variable $i\hbar\nabla_p$ that $V(\mathbf{r})$ is of the variable $\mathbf{r}$.

Solution.

Transform with the negative-exponent momentum convention of Exercise 20.4.10; for any function χ write

$$\mathcal{F}_p[\chi](\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}}\int \chi(\mathbf{r})e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}d^3r,$$

so that $\varphi = \mathcal{F}_p[\psi]$. The time derivative commutes with the spatial transform, producing $i\hbar\partial_t\varphi$. For the Laplacian, integrate by parts in x; the boundary terms vanish because $\psi \to 0$ at infinity, and $\partial_x e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} = -(ip_x/\hbar)e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}$, so

$$\mathcal{F}_p\Big[\frac{\partial\psi}{\partial x}\Big] = -\frac{1}{(2\pi\hbar)^{3/2}}\int \psi\,\frac{\partial}{\partial x}e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}d^3r = \frac{ip_x}{\hbar}\varphi.$$

Applying this twice in each coordinate and summing $(ip_j/\hbar)^2 = -p_j^2/\hbar^2$ over $j = x, y, z$ gives

$$\mathcal{F}_p[\nabla_r^2\psi] = -\frac{p^2}{\hbar^2}\varphi,$$

so the kinetic term becomes $-(\hbar^2/2m)(-p^2/\hbar^2)\varphi = p^2\varphi/(2m)$. Write a convergent coordinate expansion with multi-index $\alpha = (\alpha_x, \alpha_y, \alpha_z)$, $\mathbf{r}^\alpha = x^{\alpha_x} y^{\alpha_y} z^{\alpha_z}$ and $|\alpha| = \alpha_x + \alpha_y + \alpha_z$:

$$V(\mathbf{r}) = \sum_\alpha v_\alpha \mathbf{r}^\alpha.$$

Exercise 20.4.10(a), applied component by component, gives $\mathcal{F}_p[x\chi] = i\hbar\,\partial_{p_x}\mathcal{F}_p[\chi]$ for any admissible χ; applying it to $\chi = \psi, x\psi, x^2\psi, \ldots$ and likewise for y and z,

$$\mathcal{F}_p[\mathbf{r}^\alpha \psi] = (i\hbar)^{|\alpha|} \partial_{p_x}^{\alpha_x} \partial_{p_y}^{\alpha_y} \partial_{p_z}^{\alpha_z} \varphi \equiv (i\hbar\nabla_p)^\alpha \varphi.$$

Consequently

$$\mathcal{F}_p[V\psi] = \sum_\alpha v_\alpha (i\hbar\nabla_p)^\alpha \varphi = V(i\hbar\nabla_p)\varphi.$$

Combining the terms gives

$$\boxed{i\hbar \frac{\partial \varphi}{\partial t} = \frac{p^2}{2m}\varphi + V(i\hbar\nabla_p)\varphi}.$$

For a nonpolynomial potential, a local Maclaurin series alone does not justify interchange with an integral over all space. Its action on the chosen wave function must converge and permit that interchange. Polynomial potentials satisfy the algebraic step without an infinite-series issue. When a general potential is treated by its transform instead, the rigorous replacement is the momentum convolution

$$\mathcal{F}_p[V\psi](\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int \widetilde{V}(\mathbf{p} - \mathbf{q})\varphi(\mathbf{q}) d^3q,$$

where $\widetilde{V} = \mathcal{F}_p V$ with the same convention; it follows by inserting $\psi(\mathbf{r}) = (2\pi\hbar)^{-3/2} \int \varphi(\mathbf{q}) e^{i\mathbf{r}\cdot\mathbf{q}/\hbar} d^3q$ into $\mathcal{F}_p[V\psi]$ and doing the $\mathbf{r}$ integral first, which produces $\widetilde{V}(\mathbf{p} - \mathbf{q})$. This holds whenever the convolution, possibly distributional, is defined. The differential expression is its polynomial or convergent-series form.

20.5 Signal-Processing Applications

20.5.1 Find the transfer function $\varphi(\omega)$ for the circuit shown in the left panel of Fig. 20.10. Is this a high-pass, a low-pass, or a more complicated filter?

Solution.

The left circuit has a series resistor R and a capacitor C across the output terminals. With the $e^{i\omega t}$ phasor convention used in Section 20.5, $Z_C = 1/(i\omega C)$. The current is $I = V_{\text{in}}/(R + Z_C)$, and $V_{\text{out}} = IZ_C$. Therefore, dividing numerator and denominator by Z_C and using $R/Z_C = i\omega RC$,

$$\boxed{\varphi(\omega) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{Z_C}{R + Z_C} = \frac{1}{1 + i\omega RC}}.$$

For positive R, C, since $|1 + i\omega RC|^2 = 1 + \omega^2 R^2 C^2$,

$$|\varphi|^2 = \frac{1}{1 + \omega^2 R^2 C^2}, \qquad \arg\varphi = -\tan^{-1}(\omega RC) \quad (\omega \geq 0).$$

The gain decreases from 1 at zero frequency to 0 at high frequency: this is a low-pass filter. The power gain is $1/2$ at $\omega_c = 1/(RC)$. These expressions assume an unloaded output, as in the drawing. The sign of $i\omega$ follows the section's phasor convention; the opposite time convention conjugates the transfer function.

20.5.2 Find the transfer function $\varphi(\omega)$ for the circuit shown in the right panel of Fig. 20.10.

Hint. The potential difference across an inductor is given by $L\,dI/dt$.

Solution.

The intended reference is the right panel of Fig. 20.10, which contains a series inductor L and an output resistor R. Figure 20.11 has a single, different circuit. For a current $Ie^{i\omega t}$, the voltage across the inductor is $L\,dI/dt = i\omega LI$. Kirchhoff's voltage law and the output voltage are consequently

$$V_{\text{in}} = (R + i\omega L)I, \qquad V_{\text{out}} = RI.$$

Dividing gives

$$\boxed{\varphi(\omega) = \frac{R}{R + i\omega L} = \frac{1}{1 + i\omega L/R}}.$$

Its power gain is $|\varphi|^2 = [1 + (\omega L/R)^2]^{-1}$, which decreases from 1 at $\omega = 0$ to 0 as $\omega \to \infty$ and equals $1/2$ at $\omega L/R = 1$; so it is a low-pass filter with half-power frequency R/L, for positive R, L.

20.5.3 Find the transfer function $\varphi(\omega)$ for the circuit shown in Fig. 20.11. This is a band-pass filter.

Hint. Assume the currents in the various parts of the circuit to have the values shown in the figure.

Solution.

Use the currents specified in Fig. 20.11: I_1 flows through R_1, I_2 through the series combination R_2, C_2, and $I_1 + I_2$ through C_1. Put $p = i\omega$. The two node voltages satisfy

$$V_1 = R_1 I_1 = \left(R_2 + \frac{1}{pC_2}\right) I_2, \quad V_{\text{out}} = \frac{I_2}{pC_2}, \quad V_{\text{in}} = V_1 + \frac{I_1 + I_2}{pC_1}.$$

The middle relation gives $I_2 = pC_2 V_{\text{out}}$, and the first then gives

$$V_1 = (1 + pR_2C_2)V_{\text{out}}, \qquad I_1 = \frac{1 + pR_2C_2}{R_1} V_{\text{out}}.$$

Hence $I_1 + I_2 = [(1 + pR_2C_2)/R_1 + pC_2]V_{\text{out}}$. Substitute V_1 and $I_1 + I_2$ into the input relation and multiply by pR_1C_1:

$$pR_1C_1V_{\text{in}} = \big[pR_1C_1(1 + pR_2C_2) + 1 + pR_2C_2 + pR_1C_2\big]V_{\text{out}}.$$

The bracket expands to $1 + p(R_1C_1 + R_1C_2 + R_2C_2) + p^2R_1R_2C_1C_2$, and $p^2 = -\omega^2$. Thus

$$\boxed{\varphi(\omega) = \frac{i\omega R_1C_1}{1 + i\omega(R_1C_1 + R_1C_2 + R_2C_2) - \omega^2 R_1R_2C_1C_2}}.$$

To check the filter type, write $A = R_1C_1$, $B = R_1R_2C_1C_2$, and $D = R_1C_1 + R_1C_2 + R_2C_2$, all positive. The denominator of φ is $(1 - B\omega^2) + iD\omega$, so

$$|\varphi|^2 = \frac{A^2\omega^2}{(1 - B\omega^2)^2 + D^2\omega^2} = \frac{A^2}{\omega^{-2} + B^2\omega^2 + D^2 - 2B},$$

where the second form divides numerator and denominator by ω^2 and expands $(1 - B\omega^2)^2 = 1 - 2B\omega^2 + B^2\omega^4$. It vanishes as $\omega \to 0$ and $\omega \to \infty$. Its maximum occurs where the denominator is smallest. By the inequality of arithmetic and geometric means, $\omega^{-2} + B^2\omega^2 \geq 2\sqrt{\omega^{-2}B^2\omega^2} = 2B$, with equality exactly when $\omega^{-2} = B^2\omega^2$ (equivalently $-2\omega^{-3} + 2B^2\omega = 0$), namely $\omega = 1/\sqrt{B}$. There the denominator is $2B + D^2 - 2B = D^2$, so the maximum gain is $|\varphi| = A/D$. It therefore passes an intermediate band of frequencies.

20.5.4 The circuit elements for Exercise 20.5.3 correspond to the successive transfer functions shown in Fig. 20.12. Explain why the transfer function for this exercise is uniformly approximated by the product of the individual unloaded transfer functions when $R_2 \gg R_1$. Explain the loading correction. The condition $R_2 \gg R_1$ is sufficient for a uniformly accurate product approximation but, contrary to the textbook's word "only", it is not necessary.

Solution.

Taken separately with unloaded outputs, the high-pass and low-pass stages have transfer functions

$$\varphi_1 = \frac{pR_1C_1}{1+pR_1C_1}, \qquad \varphi_2 = \frac{1}{1+pR_2C_2}, \qquad p = i\omega.$$

Their product is

$$\varphi_1\varphi_2 = \frac{pR_1C_1}{1+p(R_1C_1+R_2C_2)+p^2R_1R_2C_1C_2}.$$

The actual denominator found in Exercise 20.5.3 contains the additional term pR_1C_2. This appears because the second stage draws current from the first: the current through C_1 is $I_1 + I_2$, not I_1. Quantitatively,

$$\frac{I_2}{I_1} = \frac{R_1}{R_2 + 1/(i\omega C_2)}, \qquad \left|\frac{I_2}{I_1}\right| = \frac{R_1}{\sqrt{R_2^2 + 1/(\omega^2C_2^2)}} \le \frac{R_1}{R_2}.$$

Hence $R_2 \gg R_1$ makes the load current small at every real frequency. More explicitly, let $D_0 = (1+pR_1C_1)(1+pR_2C_2) = 1 + p(R_1C_1+R_2C_2) + p^2R_1R_2C_1C_2$, so that $\varphi_1\varphi_2 = pR_1C_1/D_0$. The denominator of φ is $D_0 + pR_1C_2 = D_0(1+\eta)$, hence

$$\varphi = \frac{\varphi_1\varphi_2}{1+\eta}, \qquad \eta = \frac{pR_1C_2}{D_0}, \qquad |\eta| \le R_1/R_2.$$

Indeed,

$$|\eta| = \frac{\omega R_1C_2}{\sqrt{1+\omega^2R_1^2C_1^2}\sqrt{1+\omega^2R_2^2C_2^2}} \le \frac{R_1}{R_2} \quad (\omega \ge 0),$$

since the first square root is at least 1 and the second at least ωR_2C_2 (the case $\omega = 0$ has $\eta = 0$). For $\delta = R_1/R_2 < 1$, the relative discrepancy, wherever the product is nonzero, therefore satisfies

$$\left|\frac{\varphi}{\varphi_1\varphi_2} - 1\right| = \left|\frac{1}{1+\eta} - 1\right| = \frac{|\eta|}{|1+\eta|} \le \frac{|\eta|}{1-|\eta|} \le \frac{\delta}{1-\delta},$$

by the reverse triangle inequality $|1+\eta| \ge 1 - |\eta|$ and because $x/(1-x)$ increases on $0 \le x < 1$. The quoted limit is a sufficient condition for a uniformly accurate product approximation, not a necessary one. At sufficiently low and high frequencies the loading is small in any case; an ideal buffer between stages makes the product exact. For finite positive elements in the unbuffered circuit, the product is not an identity in frequency. The exact size of the correction shows that $R_2 \gg R_1$ is not necessary. With $a = R_1C_1$ and $b = R_2C_2$, dividing the numerator and denominator of $|\eta|^2$ by ω^2 and expanding $(1+\omega^2a^2)(1+\omega^2b^2)$ gives

$$|\eta|^2 = \frac{R_1^2C_2^2}{\omega^{-2} + a^2 + b^2 + a^2b^2\omega^2} \le \frac{R_1^2C_2^2}{(a+b)^2},$$

since $\omega^{-2} + a^2b^2\omega^2 \ge 2ab$ (the inequality of Exercise 20.5.3, where $B = ab$), with equality at the passband center $\omega = 1/\sqrt{ab}$. Thus $|\eta| \to 0$ as $\omega \to 0$ or $\omega \to \infty$, and

$$\max_\omega |\eta| = \frac{R_1C_2}{R_1C_1 + R_2C_2} < \frac{R_1C_2}{R_2C_2} = \frac{R_1}{R_2}.$$

The product is therefore uniformly accurate whenever $R_1C_2 \ll R_1C_1 + R_2C_2$, for example also when $C_2 \ll C_1$ with comparable resistances.

20.6 Discrete Fourier Transform

20.6.1 Derive the trigonometric forms of discrete orthogonality corresponding to Eq. (20.120):

$$\sum_{k=0}^{N-1} \cos(2\pi pk/N)\sin(2\pi qk/N) = 0$$

$$\sum_{k=0}^{N-1} \cos(2\pi pk/N)\cos(2\pi qk/N) = \begin{cases} N, & p = q = (0 \text{ or } N/2); \\ N/2, & (p+q = 0 \text{ or } N) \text{ or } p = q \text{ but not both}; \\ 0, & \text{otherwise.} \end{cases}$$

$$\sum_{k=0}^{N-1} \sin(2\pi pk/N)\sin(2\pi qk/N) = \begin{cases} N/2, & p = q \text{ and } p+q \neq (0 \text{ or } N); \\ -N/2, & p \neq q \text{ and } p+q = N; \\ 0, & \text{otherwise.} \end{cases}$$

Note. If N is odd, p and q will never have the value $N/2$.
Hint. Consider the use of trigonometric identities such as

$$\sin A \cos B = \frac{1}{2}\big[\sin(A+B) + \sin(A-B)\big].$$

Take integer indices $0 \le p, q < N$ and a positive integer N in these piecewise formulas.

Solution.

For integers m define the discrete selector

$$\Delta_N(m) = \begin{cases} 1, & m \equiv 0 \pmod N, \\ 0, & \text{otherwise.} \end{cases}$$

With $r = e^{2\pi i m/N}$ the finite geometric series gives

$$\sum_{k=0}^{N-1} e^{2\pi imk/N} = \sum_{k=0}^{N-1} r^k = N\Delta_N(m).$$

Indeed, if $m \equiv 0 \pmod N$, every term is $r^k = 1$; otherwise $r \neq 1$ and the sum is $(1-r^N)/(1-r) = 0$, because $r^N = e^{2\pi im} = 1$. Its imaginary part $\sum_k \sin(2\pi mk/N)$ is therefore zero and its real part $\sum_k \cos(2\pi mk/N)$ is $N\Delta_N(m)$. With $\theta_k = 2\pi k/N$, the product identities $\cos A \sin B = \frac{1}{2}[\sin(B+A) + \sin(B-A)]$ and $\cos A \cos B,\ \sin A \sin B = \frac{1}{2}[\cos(A-B) \pm \cos(A+B)]$ then give

$$\begin{aligned} \sum_k \cos(p\theta_k)\sin(q\theta_k) &= \frac{1}{2}\sum_k \{\sin[(q+p)\theta_k] + \sin[(q-p)\theta_k]\} = 0, \\ \sum_k \cos(p\theta_k)\cos(q\theta_k) &= \frac{1}{2}\sum_k \{\cos[(p-q)\theta_k] + \cos[(p+q)\theta_k]\} \\ &= \frac{N}{2}\{\Delta_N(p-q) + \Delta_N(p+q)\}, \\ \sum_k \sin(p\theta_k)\sin(q\theta_k) &= \frac{1}{2}\sum_k \{\cos[(p-q)\theta_k] - \cos[(p+q)\theta_k]\} \\ &= \frac{N}{2}\{\Delta_N(p-q) - \Delta_N(p+q)\}. \end{aligned}$$

These formulas hold for all integer p, q and positive integer N. The displayed piecewise formulas in the question use the representative range $0 \le p, q < N$. Without that range, congruences modulo N must replace the equalities. In this range $|p-q| < N$, so $\Delta_N(p-q) = 1$ exactly when $p = q$. Also $0 \le p+q \le 2N-2 < 2N$, so $\Delta_N(p+q) = 1$ exactly when $p+q = 0$ or N. Both selectors equal 1 for $p = q = 0$, and also for $p = q = N/2$ when N is even. The cosine sum is then N, whereas the sine sum is zero. When exactly one selector is 1, the cosine sum is $N/2$; the sine sum is $N/2$ for $p = q$ and $-N/2$ for $p+q = N$ with $p \neq q$. When neither selector is 1, both sums vanish. These cases prove the corrected formulas, including $N = 1$ and $N = 2$. The red cases in the statement include the pairs with $p+q = N$, $p \neq q$, for which $\Delta_N(p+q) = 1$: for example, $N = 4$, $p = 1$, $q = 3$ gives the cosine sum $2 = N/2$ and the sine sum $-2 = -N/2$, although $p \neq q$. The middle cosine case is written with $p+q = 0$ or N, so that $p = q = 0$, where both of its conditions hold, falls under the first case only; with $p+q = N$ alone the first two cases would overlap at $p = q = 0$. The final product-to-sum line in the Instructor's Manual's proof on p. 344 has a cosine–cosine product on its left. It must be a sine–sine product to agree with the difference of cosines on its right. The selector derivation above gives that identity directly.

20.6.2 Show in detail how to go from

$$F_p = \frac{1}{N^{1/2}} \sum_{k=0}^{N-1} f_k e^{2\pi i p k/N} \qquad \text{to} \qquad f_k = \frac{1}{N^{1/2}} \sum_{p=0}^{N-1} F_p e^{-2\pi i p k/N}.$$

Solution.

Insert the defining expression for F_p into the proposed inverse, using a different dummy index j:

$$\begin{aligned} \frac{1}{\sqrt{N}} \sum_{p=0}^{N-1} F_p e^{-2\pi i p k/N} &= \frac{1}{N} \sum_{p=0}^{N-1} \sum_{j=0}^{N-1} f_j e^{2\pi i p j/N} e^{-2\pi i p k/N} \\ &= \frac{1}{N} \sum_{j=0}^{N-1} f_j \sum_{p=0}^{N-1} e^{2\pi i p (j-k)/N}. \end{aligned}$$

The inner sum is $N\Delta_N(j-k)$ by the finite geometric series of Exercise 20.6.1 (with $m = j-k$). For $0 \le j, k < N$ we have $|j-k| < N$, so the only surviving index is $j = k$, and its factor N cancels the $1/N$. Thus the expression is f_k, which proves

$$\boxed{f_k = \frac{1}{\sqrt{N}} \sum_{p=0}^{N-1} F_p e^{-2\pi i p k/N}}.$$

All sums are finite, so their interchange requires no convergence assumption. The factors $1/N$ in both exponents are essential: without them every exponential is 1 for integer indices, and the transform cannot distinguish the input entries.

20.6.3 The N-membered point sets f_k and F_p are discrete Fourier transforms of each other. Derive the following symmetry relations:

(a) If f_k is real, F_p is Hermitian symmetric; that is, $F_p = F^*_{N-p}$.

(b) If f_k is pure imaginary, then $F_p = -F^*_{N-p}$.

Note. The symmetry of part (a) is an illustration of aliasing. If F_p describes an amplitude at a frequency proportional to p, we necessarily predict an equal amplitude at the frequency proportional to $N-p$.

Indices are understood modulo N, in particular $F_N = F_0$. Equal paired amplitudes in the note mean equal magnitudes, with conjugate phases.

Solution.

Regard the frequency index modulo N, so $F_N = F_0$. Complex conjugation gives

$$\begin{aligned} F^*_{N-p} &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f^*_k e^{-2\pi i(N-p)k/N} \\ &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f^*_k e^{-2\pi ik} e^{2\pi ipk/N} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f^*_k e^{2\pi ipk/N}, \end{aligned}$$

since $e^{-2\pi ik} = 1$ for integer k. If f_k is real, $f^*_k = f_k$, so $F^*_{N-p} = F_p$. If f_k is purely imaginary, $f^*_k = -f_k$, so $F^*_{N-p} = -F_p$. These are precisely (a) and (b). For a real input, the zero-frequency coefficient and, for even N, the coefficient at $p = N/2$ are real because they are their own conjugates. For a purely imaginary input those same coefficients are purely imaginary. In general the paired coefficients have equal magnitudes; their phases are related by conjugation (and by an extra minus sign in part (b)). The index $N-p$ represents the same sampled frequency as $-p$.

20.7 Laplace Transforms

20.7.1 Prove that

$$\lim_{s\to\infty} sf(s) = \lim_{t\to+0} F(t).$$

Hint. Assume that $F(t)$ can be expressed as $F(t) = \sum_{n=0}^{\infty} a_n t^n$.

Let $s \to +\infty$ along the real axis. Assume a finite $F(0+)$ and an exponential bound $|F(t)| \le Me^{ct}$; these hypotheses suffice without an unjustified termwise integration.

Solution.

Take $s \to +\infty$ along the positive real axis. A sufficient hypothesis is that F has the finite right limit $F(0+)$, is locally integrable, and has an exponential bound $|F(t)| \le Me^{ct}$ for $t \ge 0$. In the defining integral put $u = st$, so $dt = du/s$ and, for $s > 0$, the limits stay 0 and ∞:

$$sf(s) = s\int_0^\infty e^{-st}F(t)dt = \int_0^\infty e^{-u}F(u/s)du.$$

For each fixed $u > 0$, $u/s \to 0^+$ and so $F(u/s) \to F(0+)$. For sufficiently large s, $c/s \le 1/2$, and $|e^{-u}F(u/s)| \le Me^{-u}e^{cu/s} \le Me^{-u/2}$, which is integrable (if $c \le 0$, the bound Me^{-u} holds for every $s > 0$). Dominated convergence therefore gives

$$\lim_{s\to+\infty} sf(s) = F(0+)\int_0^\infty e^{-u}du = \boxed{F(0+)}.$$

The hint also describes the coefficients: if a Taylor expansion can be integrated termwise, then, with $\int_0^\infty e^{-st}t^n dt = \Gamma(n+1)/s^{n+1}$ (substitute $u = st$ as above; see also Exercise 20.7.7),

$$f(s) = \sum_{n\ge0} \frac{a_n\Gamma(n+1)}{s^{n+1}}, \qquad sf(s) = a_0 + \frac{a_1}{s} + \frac{2a_2}{s^2} + \cdots.$$

Even when this infinite series is only asymptotic, the direct limit proof above remains valid and identifies $a_0 = F(0+)$.

20.7.2 Show that

$$\frac{1}{\pi}\lim_{s\to 0} L\{\cos(xt)\} = \delta(x).$$

This formula leaves the transform variable, the approach to zero, and the meaning of convergence unspecified; the published errata therefore say to disregard it. Prove instead the following precise replacement:

$$\frac{1}{\pi}\lim_{s\downarrow 0} L_t\{\cos xt\}(s) = \delta(x) \quad \text{as distributions in } x.$$

Solution.

The published errata withdraw the statement as ill-defined. It has a useful, precise replacement: let s approach zero through positive real values, take the Laplace transform in t, and interpret convergence as convergence of distributions in x. For $s > 0$ and real x, write $\cos xt = \Re e^{ixt}$; direct integration of $e^{-(s-ix)t}$ over $0 < t < \infty$ gives $1/(s - ix)$, so

$$L_t\{\cos xt\}(s) = \Re\frac{1}{s-ix} = \Re\frac{s+ix}{s^2+x^2} = \frac{s}{s^2+x^2}.$$

The resulting functions do not converge to an ordinary function representing a delta: they approach zero at every $x \neq 0$ and diverge at $x = 0$. To prove the correct weak limit, take a smooth test function ϕ with compact support and substitute $x = su$, so $dx = s\,du$ and $s^2 + x^2 = s^2(1 + u^2)$:

$$\int_{-\infty}^{\infty} \frac{s\phi(x)}{\pi(s^2+x^2)}dx = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\phi(su)}{1+u^2}du.$$

The integrand is bounded by the integrable $\|\phi\|_\infty/(1+u^2)$, and $\phi(su) \to \phi(0)$ for each u by continuity. Consequently, by dominated convergence, the limit is

$$\frac{\phi(0)}{\pi}[\tan^{-1} u]_{-\infty}^{\infty} = \phi(0).$$

Thus the repaired assertion is

$$\boxed{\lim_{s\downarrow 0}\frac{1}{\pi}L_t\{\cos xt\}(s) = \delta(x) \quad \text{in the distributional sense.}}$$

The one-sided real approach is part of the assertion; an unspecified complex limit is not the prescription proved here.

20.7.3 Verify that

$$L\left\{\frac{\cos at - \cos bt}{b^2 - a^2}\right\} = \frac{s}{(s^2+a^2)(s^2+b^2)}, \qquad a^2 \neq b^2.$$

Solution.

For real a, b and $\Re s > 0$, write $\cos at = \frac{1}{2}(e^{iat} + e^{-iat})$ and use $L\{e^{\pm iat}\} = \int_0^\infty e^{-(s\mp ia)t}dt = 1/(s \mp ia)$:

$$L\{\cos at\} = \frac{1}{2}\left(\frac{1}{s-ia} + \frac{1}{s+ia}\right) = \frac{1}{2}\,\frac{(s+ia)+(s-ia)}{s^2+a^2} = \frac{s}{s^2+a^2}.$$

Linearity and a common denominator now yield

$$L\left\{\frac{\cos at-\cos bt}{b^2-a^2}\right\}=\frac{s}{b^2-a^2}\left(\frac{1}{s^2+a^2}-\frac{1}{s^2+b^2}\right)$$
$$=\frac{s}{b^2-a^2}\frac{(s^2+b^2)-(s^2+a^2)}{(s^2+a^2)(s^2+b^2)}$$
$$=\boxed{\frac{s}{(s^2+a^2)(s^2+b^2)}}.$$

The stipulated $a^2\neq b^2$ permits the cancellation. For complex frequencies the same algebra applies in a common half-plane of convergence.

20.7.4 Using partial fraction expansions, show that

(a) $L^{-1}\left\{\dfrac{1}{(s+a)(s+b)}\right\}=\dfrac{e^{-at}-e^{-bt}}{b-a},\quad a\neq b.$

(b) $L^{-1}\left\{\dfrac{s}{(s+a)(s+b)}\right\}=\dfrac{ae^{-at}-be^{-bt}}{a-b},\quad a\neq b.$

Solution.

(a) Set $1/[(s+a)(s+b)]=A/(s+a)+B/(s+b)$. After multiplication by the denominator, $1=A(s+b)+B(s+a)$. Setting $s=-a$ gives $1=A(b-a)$ and setting $s=-b$ gives $1=B(a-b)$, so $A=1/(b-a)$ and $B=1/(a-b)=-1/(b-a)$. Since $L\{e^{-at}\}=1/(s+a)$,

$$\boxed{L^{-1}\left\{\frac{1}{(s+a)(s+b)}\right\}=\frac{e^{-at}-e^{-bt}}{b-a}}.$$

(b) Now $s=A(s+b)+B(s+a)$, so the same substitutions give $-a=A(b-a)$ and $-b=B(a-b)$, i.e., $A=a/(a-b)$, $B=-b/(a-b)$. Hence

$$\boxed{L^{-1}\left\{\frac{s}{(s+a)(s+b)}\right\}=\frac{ae^{-at}-be^{-bt}}{a-b}}.$$

The defining transforms converge for $\Re s>\max(-\Re a,-\Re b)$. Although excluded in the question, the equal-parameter limits are regular. By l'Hôpital's rule in b (differentiate numerator and denominator with respect to b, then let $b\to a$),

$$\lim_{b\to a}\frac{e^{-at}-e^{-bt}}{b-a}=\lim_{b\to a}\frac{te^{-bt}}{1}=te^{-at},$$
$$\lim_{b\to a}\frac{ae^{-at}-be^{-bt}}{a-b}=\lim_{b\to a}\frac{-e^{-bt}+bte^{-bt}}{-1}=(1-at)e^{-at};$$

these are the inverse transforms of $1/(s+a)^2$ and $s/(s+a)^2$.

20.7.5 Using partial fraction expansions, show that for $a^2 \neq b^2$,

(a) $L^{-1}\left\{\dfrac{1}{(s^2+a^2)(s^2+b^2)}\right\} = -\dfrac{1}{a^2-b^2}\left(\dfrac{\sin at}{a} - \dfrac{\sin bt}{b}\right)$.

(b) $L^{-1}\left\{\dfrac{s^2}{(s^2+a^2)(s^2+b^2)}\right\} = \dfrac{1}{a^2-b^2}\{a\sin at - b\sin bt\}$.

For real a, b, use $\sin(at)/a \to t$ if $a = 0$, and the corresponding limit if $b = 0$.

Solution.

For real parameters use $L\{\sin at/a\} = 1/(s^2+a^2)$ in $\Re s > 0$. Write the transforms as $A/(s^2+a^2) + B/(s^2+b^2)$ and multiply by $(s^2+a^2)(s^2+b^2)$. (a) Equating coefficients of s^2 and s^0 in

$$1 = A(s^2+b^2) + B(s^2+a^2)$$

gives $A+B = 0$, $Ab^2 + Ba^2 = 1$. Substituting $B = -A$ into the second equation gives $A(b^2-a^2) = 1$. Thus $A = 1/(b^2-a^2)$ and $B = -A$, and inversion gives

$$\boxed{\frac{1}{b^2-a^2}\left(\frac{\sin at}{a} - \frac{\sin bt}{b}\right)} = -\frac{1}{a^2-b^2}\left(\frac{\sin at}{a} - \frac{\sin bt}{b}\right),$$

the printed form. (b) Equating coefficients in $s^2 = A(s^2+b^2) + B(s^2+a^2)$ gives $A + B = 1$, $Ab^2 + Ba^2 = 0$. Substituting $B = 1 - A$ into the second equation gives $A(b^2-a^2) + a^2 = 0$, so $A = a^2/(a^2-b^2)$ and $B = 1 - A = -b^2/(a^2-b^2)$. Inverting $A/(s^2+a^2) + B/(s^2+b^2)$ gives $A\sin(at)/a + B\sin(bt)/b$, in which $a^2\sin(at)/a = a\sin at$ and $b^2\sin(bt)/b = b\sin bt$. Therefore

$$\boxed{L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\} = \frac{a\sin at - b\sin bt}{a^2-b^2}}.$$

If either parameter is zero, the quotient $\sin(at)/a$ in part (a) means its continuous limit t. The condition $a^2 \neq b^2$ alone does not exclude that case.

20.7.6 Show that

(a) $\displaystyle\int_0^\infty \frac{\cos s}{s^\nu}\,ds = \frac{\pi}{2\Gamma(\nu)\cos(\nu\pi/2)}, \qquad 0 < \nu < 1.$

(b) $\displaystyle\int_0^\infty \frac{\sin s}{s^\nu}\,ds = \frac{\pi}{2\Gamma(\nu)\sin(\nu\pi/2)}, \qquad 0 < \nu < 2.$

Why is ν restricted to $(0,1)$ for (a), to $(0,2)$ for (b)? These integrals may be interpreted as Fourier transforms of $s^{-\nu}$ and as Mellin transforms of $\sin s$ and $\cos s$.

Hint. Replace $s^{-\nu}$ by a Laplace transform integral: $L\{t^{\nu-1}/\Gamma(\nu)\}$. Then integrate with respect to s. The resulting integral can be treated as a beta function (Section 13.3).

Solution.

To avoid confusing the transform parameter with the integration variable, write the latter as x. For $\nu > 0$, substitution $v = ux$ into the gamma integral gives

$$x^{-\nu} = \frac{1}{\Gamma(\nu)}\int_0^\infty u^{\nu-1}e^{-ux}\,du.$$

Insert a convergence factor $e^{-\epsilon x}$, $\epsilon > 0$, before changing the order of integration: let

$$C_\epsilon = \int_0^\infty e^{-\epsilon x} x^{-\nu} \cos x \, dx, \qquad S_\epsilon = \int_0^\infty e^{-\epsilon x} x^{-\nu} \sin x \, dx,$$

and substitute the representation of $x^{-\nu}$. In the ranges stated in the question the resulting double integral is absolutely integrable: after integrating its absolute value in u, it is $\int_0^\infty e^{-\epsilon x} x^{-\nu} |\cos x| dx$ or its sine counterpart, finite because $|\cos x| \le 1$ and $|\sin x| \le x$ near $x = 0$. Hence the order may be interchanged, and the inner x integrals are those of Exercise 20.2.3 with $a = u + \epsilon$, $\omega = 1$: $\int_0^\infty e^{-(u+\epsilon)x} \cos x \, dx = (u+\epsilon)/[(u+\epsilon)^2 + 1]$ and $\int_0^\infty e^{-(u+\epsilon)x} \sin x \, dx = 1/[(u+\epsilon)^2 + 1]$. Thus

$$C_\epsilon = \frac{1}{\Gamma(\nu)} \int_0^\infty \frac{u^{\nu-1}(u+\epsilon)}{(u+\epsilon)^2 + 1} du,$$
$$S_\epsilon = \frac{1}{\Gamma(\nu)} \int_0^\infty \frac{u^{\nu-1}}{(u+\epsilon)^2 + 1} du.$$

For $0 < \epsilon \le 1$, on $0 < u < 1$ both integrands are bounded by $2u^{\nu-1}$, since $(u+\epsilon)/[(u+\epsilon)^2+1] \le u + \epsilon < 2$ and $1/[(u+\epsilon)^2+1] \le 1$. For $u \ge 1$, since $(u+\epsilon)/[(u+\epsilon)^2+1] < 1/(u+\epsilon) \le 1/u$ and $1/[(u+\epsilon)^2+1] < 1/u^2$, the bounds are $u^{\nu-2}$ and $u^{\nu-3}$, respectively. These dominating functions are integrable because $\nu > 0$ (at $u = 0$) and $\nu < 1$, respectively $\nu < 2$ (at $u = \infty$). Dominated convergence as $\epsilon \downarrow 0$ gives

$$C_0 = \frac{1}{\Gamma(\nu)} \int_0^\infty \frac{u^\nu}{1+u^2} du, \qquad S_0 = \frac{1}{\Gamma(\nu)} \int_0^\infty \frac{u^{\nu-1}}{1+u^2} du.$$

For $0 < q < 2$, put $v = u^2$, so $u^{q-1} du = \frac{1}{2} v^{q/2-1} dv$, and use the beta integral $\int_0^\infty v^{\alpha-1}(1+v)^{-\alpha-\beta} dv = B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ (Section 13.3) with $\alpha = q/2$, $\beta = 1 - q/2$, followed by the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$:

$$\int_0^\infty \frac{u^{q-1}}{1+u^2} du = \frac{1}{2} \int_0^\infty \frac{v^{q/2-1}}{1+v} dv = \frac{1}{2} \Gamma(q/2)\Gamma(1-q/2) = \frac{\pi}{2\sin(\pi q/2)}.$$

Take $q = \nu + 1$ for the cosine, with $\sin[\pi(\nu+1)/2] = \cos(\pi\nu/2)$, and $q = \nu$ for the sine; the condition $0 < q < 2$ is then $-1 < \nu < 1$, respectively $0 < \nu < 2$:

$$\boxed{C_0 = \frac{\pi}{2\Gamma(\nu)\cos(\pi\nu/2)}}, \quad 0 < \nu < 1; \qquad \boxed{S_0 = \frac{\pi}{2\Gamma(\nu)\sin(\pi\nu/2)}}, \quad 0 < \nu < 2.$$

At the origin the cosine integrand behaves as $x^{-\nu}$, requiring $\nu < 1$, whereas the sine integrand behaves as $x^{1-\nu}$, requiring $\nu < 2$. At infinity Dirichlet's test applies for $\nu > 0$; for $\nu \le 0$ the ordinary oscillatory improper integral fails to converge. Abel's theorem for integrals (Exercise 20.2.2(c), with $h(x) = x^{-\nu}\cos x$ or $x^{-\nu}\sin x$) therefore identifies the regulated limits $C_0 = \lim_{\epsilon\downarrow 0} C_\epsilon$ and $S_0 = \lim_{\epsilon\downarrow 0} S_\epsilon$ above with the ordinary improper integrals in precisely the stated ranges. The gamma notation in the corrected statement accommodates noninteger ν.

20.7.7 A function $F(t)$ can be expanded in a Maclaurin series,

$$F(t) = \sum_{n=0}^\infty a_n t^n.$$

If this series represents F on the half-line and termwise Laplace integration is justified, then

$$L\{F(t)\} = \int_0^\infty e^{-st} \sum_{n=0}^\infty a_n t^n \, dt = \sum_{n=0}^\infty a_n \int_0^\infty e^{-st} t^n \, dt.$$

Show that $f(s)$, the Laplace transform of $F(t)$, contains no powers of s greater than s^{-1}. Check your result by calculating $L\{\delta(t)\}$, and comment on this fiasco.

For the impulse comparison use the causal convention in which a delta at $t = 0$ has full unit mass.

Solution.

Whenever termwise integration is justified, $u = st$ (so $dt = du/s$ and $t^n = u^n/s^n$) gives

$$\int_0^\infty e^{-st} t^n dt = \frac{1}{s^{n+1}} \int_0^\infty e^{-u} u^n du = \frac{n!}{s^{n+1}}, \qquad s > 0,$$

since $\int_0^\infty e^{-u} u^n du = \Gamma(n+1) = n!$. Thus

$$\boxed{f(s) = \sum_{n=0}^\infty \frac{a_n n!}{s^{n+1}}},$$

which contains only $s^{-1}, s^{-2}, \ldots$. Existence of a local Maclaurin series does not by itself justify the displayed interchange over the entire half-line. For example, $F(t) = 1/(1+t)$ has a Maclaurin series near zero and a Laplace transform for $s > 0$, but the resulting formal series $\sum_{n\geq 0}(-1)^n n!/s^{n+1}$ diverges for every finite s. It is an asymptotic expansion at infinity, not a convergent identity. (The Maclaurin series $\sum(-1)^n t^n$ converges only for $|t| < 1$, and in the formal series the ratio of consecutive terms has modulus $(n+1)/s \to \infty$, so its terms do not even tend to zero.) With the causal Laplace convention in which a delta impulse at 0 has unit mass, the distributional pairing gives

$$L\{\delta(t)\}(s) = \langle \delta, e^{-st} \rangle = e^{-s\cdot 0} = 1.$$

This $1 = s^0$ is a higher power than s^{-1}, but it is no contradiction: a delta distribution is not an ordinary function with a Maclaurin series. Its transform lies outside the hypotheses of the argument. The same distinction permits $L\{\delta^{(n)}\} = s^n$ for causal distributions, since $\langle \delta^{(n)}, e^{-st} \rangle = (-1)^n \, d^n e^{-st}/dt^n \big|_{t=0} = (-1)^n (-s)^n = s^n$. The convention at the lower endpoint must be specified: integrating a symmetric regularization over $[0, \infty)$ assigns half the mass, whereas the causal 0^- convention used here assigns the full mass.

20.7.8 Show that the Laplace transform of the confluent hypergeometric function $M(a, c, x)$ is

$$L\big\{M(a,c,x)\big\} = \frac{1}{s} \, {}_2F_1\left(a, 1; c; \frac{1}{s}\right).$$

For the ordinary series definitions take $c \notin \{0, -1, -2, \ldots\}$ and $\Re s > 1$; exceptional terminating cases may admit larger domains.

Solution.

Use the defining confluent hypergeometric series

$$M(a,c,t) = \sum_{n=0}^\infty \frac{(a)_n}{(c)_n} \frac{t^n}{n!}.$$

Each term is integrated with $\int_0^\infty e^{-st}t^n dt = n!/s^{n+1}$, $\Re s > 0$ (Exercise 20.7.7 for real s; both sides are analytic in s for $\Re s > 0$, so the identity extends to that half-plane). For $\Re s > 1$ the series may be integrated absolutely: since $|e^{-st}| = e^{-t\Re s}$, the sum of the integrals of the absolute terms is

$$\sum_{n=0}^{\infty} \left|\frac{(a)_n}{(c)_n}\right| \frac{1}{n!}\frac{n!}{(\Re s)^{n+1}} = \frac{1}{\Re s}\sum_{n=0}^{\infty}\left|\frac{(a)_n}{(c)_n}\right|(\Re s)^{-n},$$

which converges by the ratio test (or terminates): the ratio of consecutive terms is $|(a+n)/(c+n)|/\Re s \to 1/\Re s < 1$. Hence, by dominated convergence (every partial sum is bounded in modulus by the sum of the absolute terms, whose integral, by monotone convergence, is the finite sum above), termwise integration is allowed and

$$\begin{aligned} L\{M(a,c,t)\}(s) &= \sum_{n=0}^{\infty}\frac{(a)_n}{(c)_n n!}\frac{n!}{s^{n+1}} \\ &= \frac{1}{s}\sum_{n=0}^{\infty}\frac{(a)_n(1)_n}{(c)_n n!}\left(\frac{1}{s}\right)^n = \boxed{\frac{1}{s}\,{}_2F_1\left(a,1;c;\frac{1}{s}\right)}, \end{aligned}$$

since $(1)_n = n!$. The ordinary definitions require $c \notin \{0,-1,-2,\ldots\}$, unless a specified terminating or limiting case removes the singularity. The half-plane $\Re s > 1$ is sufficient for the proof; continuation beyond it must stay within the transform's convergence domain and use consistent analytic branches. Terminating cases can have a larger domain.

20.8 Properties of Laplace Transforms

20.8.1 Use the expression for the transform of a second derivative to obtain the transform of $\cos kt$.

Solution.

Put $F(t) = \cos kt$ and $f(s) = L\{F\}(s)$. The differential identity $F'' = -k^2F$ and initial values $F(0) = 1$, $F'(0) = 0$ give

$$L\{F''\} = s^2 f(s) - sF(0) - F'(0) = s^2 f(s) - s = -k^2 f(s),$$

where the last equality transforms $F'' = -k^2F$ by linearity. The derivative formula applies for $\Re s > 0$, since F and $F' = -k\sin kt$ are continuous and bounded. Collecting the f terms gives $(s^2+k^2)f(s) = s$. For $\Re s > 0$ the factor $s^2 + k^2$ does not vanish (its zeros $\pm ik$ lie on the imaginary axis), so

$$\boxed{L\{\cos kt\}(s) = \frac{s}{s^2+k^2}}, \qquad \Re s > 0 \quad (k \in \mathbb{R}).$$

For $k = 0$ this reduces to $L\{1\} = 1/s$, as required.

20.8.2 A mass m is attached to one end of an unstretched spring, spring constant k (Fig. 20.17). Starting at time $t = 0$, the free end of the spring experiences a constant acceleration a, away from the mass. Using Laplace transforms,

(a) find the position x of m as a function of time.

(b) determine the limiting form of $x(t)$ for small t.

ANS. (a) $x = \dfrac{1}{2}at^2 - \dfrac{a}{\omega^2}(1-\cos\omega t)$, $\omega^2 = \dfrac{k}{m}$,
(b) $x = \dfrac{a\omega^2}{4!}\,t^4$, $\omega t \ll 1$.

Use the initially resting preparation $x(0) = x'(0) = 0$, with the driven end initially at rest and $m, k > 0$; the textbook states no initial conditions, and its answer (a) presumes these.

Solution.

(a) Measure x from the initial mass position, positive in the direction of the acceleration a, and subtract the unstretched spring length from the driven endpoint's position. The latter coordinate is $y(t) = at^2/2$, so the spring extension is $y - x$. Newton's law is

$$mx'' = k(y-x), \qquad x'' + \omega^2 x = \frac{a\omega^2 t^2}{2}, \qquad \omega^2 = k/m.$$

The mass is initially at rest at $x = 0$, so with $\widehat{x} = L\{x\}$ we have $L\{x''\} = s^2\widehat{x} - sx(0) - x'(0) = s^2\widehat{x}$; together with $L\{t^2/2\} = 1/s^3$, the transformed equation is

$$(s^2+\omega^2)\widehat{x} = \frac{a\omega^2}{s^3}, \qquad \widehat{x} = \frac{a\omega^2}{s^3(s^2+\omega^2)} = \frac{a}{s^3} - \frac{a}{s(s^2+\omega^2)}.$$

The last step writes the numerator as $\omega^2 = (s^2+\omega^2) - s^2$. The same splitting gives $1/[s(s^2+\omega^2)] = \omega^{-2}[1/s - s/(s^2+\omega^2)]$, and the pairs $L\{t^2/2\} = 1/s^3$, $L\{1\} = 1/s$, $L\{\cos\omega t\} = s/(s^2+\omega^2)$ then invert $\widehat{x}$ term by term:

$$\boxed{x(t) = \frac{1}{2}at^2 - \frac{a}{\omega^2}(1-\cos\omega t)}.$$

(b) Expanding the cosine shows explicitly the cancellation of the quadratic terms:

$$x(t) = \frac{1}{2}at^2 - \frac{a}{\omega^2}\left(\frac{\omega^2t^2}{2} - \frac{\omega^4t^4}{24} + \frac{\omega^6t^6}{720} - \cdots\right) = \boxed{\frac{a\omega^2t^4}{24} + O(t^6)}.$$

The small-time condition is $|\omega t| \ll 1$. Differentiating the exact answer gives $x'' = a(1-\cos\omega t)$, which agrees with $\omega^2(y-x)$ and verifies the equation of motion.

20.8.3 Radioactive nuclei decay according to the law

$$\frac{dN}{dt} = -\lambda N,$$

with N the concentration of a given nuclide and λ its particular decay constant. This equation may be interpreted as stating that the rate of decay is proportional to the number of these radioactive nuclei present. They all decay independently.

Consider now a radioactive series of n different nuclides, with Nuclide 1 decaying into Nuclide 2, Nuclide 2 into Nuclide 3, etc., until reaching Nuclide n, which is stable. The concentrations of the various nuclides satisfy the system of ODEs

$$\frac{dN_1}{dt} = -\lambda_1 N_1, \qquad \frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2, \qquad \cdots, \qquad \frac{dN_n}{dt} = \lambda_{n-1}N_{n-1}.$$

(a) For the case $n = 3$ find $N_1(t)$, $N_2(t)$, and $N_3(t)$, with $N_1(0) = N_0$ and $N_2(0) = N_3(0) = 0$.

(b) Find an approximate expression for N_2 and N_3, valid for small t when $\lambda_1 \approx \lambda_2$.

(c) Find approximate expressions for N_2 and N_3, valid for large t, when (1) $\lambda_1 \gg \lambda_2$, (2) $\lambda_1 \ll \lambda_2$.

ANS. (a) $N_1(t) = N_0 e^{-\lambda_1 t}$, $\quad N_2(t) = N_0 \dfrac{\lambda_1}{\lambda_2 - \lambda_1}\big(e^{-\lambda_1 t} - e^{-\lambda_2 t}\big)$,

$$N_3(t) = N_0 \left(1 - \frac{\lambda_2}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{\lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t}\right).$$

(b) $N_2 \approx N_0 \lambda_1 t$, $\quad N_3 \approx \dfrac{N_0}{2} \lambda_1 \lambda_2 t^2$.

(c) (1) $N_2 \approx N_0 e^{-\lambda_2 t}$

$N_3 \approx N_0\big(1 - e^{-\lambda_2 t}\big)$, $\quad \lambda_1 t \gg 1$.

(2) $N_2 \approx N_0(\lambda_1/\lambda_2) e^{-\lambda_1 t}$,

$N_3 \approx N_0\big(1 - e^{-\lambda_1 t}\big)$, $\quad \lambda_2 t \gg 1$.

The quotient answers in part (a) are written for distinct rates; at equal rates take the continuous limits derived below.

Solution.

(a) Denote Laplace transforms by hats. By $L\{N_i'\} = s\widehat{N}_i - N_i(0)$, the initial values and the three rate equations give

$$(s + \lambda_1)\widehat{N}_1 = N_0, \quad (s + \lambda_2)\widehat{N}_2 = \lambda_1 \widehat{N}_1, \quad s\widehat{N}_3 = \lambda_2 \widehat{N}_2.$$

Hence $\widehat{N}_1 = N_0/(s + \lambda_1)$ and $\widehat{N}_2 = N_0\lambda_1/[(s + \lambda_1)(s + \lambda_2)]$. For $\lambda_1 \neq \lambda_2$ the partial fractions $1/[(s + \lambda_1)(s + \lambda_2)] = [1/(s + \lambda_1) - 1/(s + \lambda_2)]/(\lambda_2 - \lambda_1)$ and the pair $L\{e^{-\lambda t}\} = 1/(s + \lambda)$ give

$$\begin{aligned} N_1(t) &= N_0 e^{-\lambda_1 t}, \\ \widehat{N}_2 &= \frac{N_0 \lambda_1}{\lambda_2 - \lambda_1}\left(\frac{1}{s + \lambda_1} - \frac{1}{s + \lambda_2}\right), \\ N_2(t) &= \frac{N_0 \lambda_1}{\lambda_2 - \lambda_1}(e^{-\lambda_1 t} - e^{-\lambda_2 t}). \end{aligned}$$

Adding the three original equations gives $(N_1 + N_2 + N_3)' = 0$, so their sum remains N_0. Subtraction therefore gives

$$N_3 = N_0 - N_1 - N_2 = N_0 \left[1 - \left(1 + \frac{\lambda_1}{\lambda_2 - \lambda_1}\right) e^{-\lambda_1 t} + \frac{\lambda_1 e^{-\lambda_2 t}}{\lambda_2 - \lambda_1}\right],$$

and $1 + \lambda_1/(\lambda_2 - \lambda_1) = \lambda_2/(\lambda_2 - \lambda_1)$, so

$$\boxed{N_3(t) = N_0 \left[1 - \frac{\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t}}{\lambda_2 - \lambda_1}\right]}.$$

The quotient formulas presume $\lambda_1 \neq \lambda_2$. At the removable equal-rate case $\lambda_1 = \lambda_2 = \lambda$, $N_2 = N_0 \lambda t e^{-\lambda t}$ and $N_3 = N_0[1 - (1 + \lambda t)e^{-\lambda t}]$. They follow either from $\widehat{N}_2 = N_0\lambda/(s + \lambda)^2$ and $L\{te^{-\lambda t}\} = 1/(s + \lambda)^2$, or from the limit $(e^{-\lambda_1 t} - e^{-\lambda_2 t})/(\lambda_2 - \lambda_1) \to -\partial_\lambda e^{-\lambda t} = te^{-\lambda t}$ of the exponential difference; then $N_3 = N_0 - N_1 - N_2$.

(b) Insert $e^{-\lambda t} = 1 - \lambda t + \lambda^2 t^2/2 - \lambda^3 t^3/6 + O(t^4)$ into the formulas of (a):

$$\begin{aligned} e^{-\lambda_1 t} - e^{-\lambda_2 t} &= (\lambda_2 - \lambda_1)t - \frac{\lambda_2^2 - \lambda_1^2}{2} t^2 + O(t^3), \\ \lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t} &= (\lambda_2 - \lambda_1)\left[1 - \frac{\lambda_1 \lambda_2}{2} t^2 + \frac{\lambda_1 \lambda_2 (\lambda_1 + \lambda_2)}{6} t^3\right] + O(t^4); \end{aligned}$$

in the second line the terms linear in t cancel. Dividing by $\lambda_2-\lambda_1$, with $(\lambda_2^2-\lambda_1^2)/(\lambda_2-\lambda_1)=\lambda_1+\lambda_2$, gives

$$N_2 = N_0\lambda_1\left[t - \frac{\lambda_1+\lambda_2}{2}t^2 + O(t^3)\right],$$
$$N_3 = N_0\lambda_1\lambda_2\left[\frac{t^2}{2} - \frac{\lambda_1+\lambda_2}{6}t^3 + O(t^4)\right].$$

The leading terms are those stated. They hold when $\max(\lambda_1,\lambda_2)t \ll 1$, including equal or nearly equal rates; closeness of the rates is not needed for this small-time expansion. Indeed, Taylor's theorem and the rate equations at $t=0$ give the same leading terms for all rates: $N_2(0)=0$ and $N_2'(0)=\lambda_1N_1(0)=\lambda_1N_0$, while $N_3(0)=0$, $N_3'(0)=\lambda_2N_2(0)=0$ and $N_3''(0)=\lambda_2N_2'(0)=\lambda_1\lambda_2N_0$.

(c1) If $\lambda_1 \gg \lambda_2$ and $\lambda_1 t \gg 1$, then $e^{-\lambda_1 t}=e^{-(\lambda_1-\lambda_2)t}e^{-\lambda_2 t}$ is negligible beside $e^{-\lambda_2 t}$, because $(\lambda_1-\lambda_2)t \simeq \lambda_1 t \gg 1$. Discarding it in the exact formulas of (a) leaves

$$N_2 \simeq N_0\frac{\lambda_1}{\lambda_1-\lambda_2}e^{-\lambda_2 t}, \quad N_3 \simeq N_0\left[1-\frac{\lambda_1}{\lambda_1-\lambda_2}e^{-\lambda_2 t}\right].$$

Since $\lambda_1/(\lambda_1-\lambda_2) = 1/(1-\lambda_2/\lambda_1) = 1+O(\lambda_2/\lambda_1)$, these become $N_0e^{-\lambda_2 t}$ and $N_0(1-e^{-\lambda_2 t})$. The positive exponent printed for N_2 in this case must be negative. A positive exponent would violate both the rate equation and $0 \le N_2 \le N_0$. The final N_3 stays accurate even while N_3 itself is still small. By $N_3 = N_0-N_1-N_2$ and part (a), it exceeds the exact N_3 by $N_1+N_2-N_0e^{-\lambda_2 t} = N_0(e^{-\lambda_2 t}-e^{-\lambda_1 t})[\lambda_1/(\lambda_1-\lambda_2)-1]$, that is, by $N_0\lambda_2(e^{-\lambda_2 t}-e^{-\lambda_1 t})/(\lambda_1-\lambda_2)$. Relative to $N_0(1-e^{-\lambda_2 t})$ this is at most $[\lambda_2/(\lambda_1-\lambda_2)]/(e^{\lambda_2 t}-1) \le 1/[(\lambda_1-\lambda_2)t] \ll 1$, by $e^x-1 \ge x$.

(c2) If $\lambda_1 \ll \lambda_2$ and $\lambda_2 t \gg 1$, discard $e^{-\lambda_2 t}$ instead, by the same argument with the rates interchanged:

$$N_2 \simeq N_0\frac{\lambda_1}{\lambda_2-\lambda_1}e^{-\lambda_1 t} \simeq N_0\frac{\lambda_1}{\lambda_2}e^{-\lambda_1 t},$$
$$N_3 \simeq N_0\left[1-\frac{\lambda_2}{\lambda_2-\lambda_1}e^{-\lambda_1 t}\right] \simeq N_0(1-e^{-\lambda_1 t}).$$

The last approximations keep the leading terms in the small ratio λ_1/λ_2, using $\lambda_2-\lambda_1=\lambda_2(1-\lambda_1/\lambda_2)$. Since $N_3 = N_0-N_1-N_2$ exactly, the final form of N_3 drops precisely $N_2 \simeq N_0(\lambda_1/\lambda_2)e^{-\lambda_1 t}$. Relative to $N_0(1-e^{-\lambda_1 t})$ this is $(\lambda_1/\lambda_2)/(e^{\lambda_1 t}-1) \le 1/(\lambda_2 t) \ll 1$, by $e^x-1 \ge x$; so the approximation stays accurate even while N_3 itself is still small.

20.8.4 The rate of formation of an isotope in a nuclear reactor is given by

$$\frac{dN_2}{dt} = \varphi\big[\sigma_1N_1(0) - \sigma_2N_2(t)\big] - \lambda_2N_2(t).$$

Here $N_1(0)$ is the concentration of the original isotope (assumed constant), and N_2 is that of the newly formed isotope. The first two terms on the right-hand side describe the production and destruction of the new isotope via neutron absorption; φ is the neutron flux (units $\mathrm{cm^{-2}\,s^{-1}}$); σ_1 and σ_2 (units $\mathrm{cm^2}$) are neutron absorption cross sections. The final term describes the radioactive decay of the new isotope, with decay constant λ_2.

(a) Find the concentration N_2 of the new isotope as a function of time.

(b) For original isotope ^{153}Eu, $\sigma_1 = 400$ barns $= 400\times10^{-24}$ $\mathrm{cm^2}$, $\sigma_2 = 1000$ barns $= 1000\times10^{-24}$ $\mathrm{cm^2}$, and $\lambda_2 = 1.4\times10^{-9}$ $\mathrm{s^{-1}}$. If $N_1(0)=10^{20}$ and $\varphi = 10^9$ $\mathrm{cm^{-2}\,s^{-1}}$, find N_2, the concentration of ^{154}Eu, after 1 year of continuous irradiation. Is the assumption that N_1 is constant justified?

In (a), retain $N_2(0) = N_{20}$ as an initial datum; for the numerical irradiation in (b), take $N_{20} = 0$.

Solution.

(a) Set $A = \varphi\sigma_1 N_1(0)$ and $b = \varphi\sigma_2 + \lambda_2$. The equation is $N_2' + bN_2 = A$. With $N_2(0) = N_{20}$, the transforms $L\{N_2'\} = s\widehat{N}_2 - N_{20}$ and $L\{A\} = A/s$ give

$$(s+b)\widehat{N}_2 = N_{20} + A/s, \quad \widehat{N}_2 = \frac{N_{20}}{s+b} + \frac{A}{b}\left(\frac{1}{s} - \frac{1}{s+b}\right),$$

by the partial fractions $1/[s(s+b)] = [1/s - 1/(s+b)]/b$. Inverting with $L\{1\} = 1/s$ and $L\{e^{-bt}\} = 1/(s+b)$,

$$\boxed{N_2(t) = N_{20}e^{-bt} + \frac{A}{b}(1 - e^{-bt})}.$$

An initial concentration is needed. For irradiation of the original isotope with none of the new isotope initially present, use $N_{20} = 0$; the formula above also covers a nonzero initial inventory. If $b = 0$, the continuous limit is $N_{20} + At$, since $(1 - e^{-bt})/b \to t$ as $b \to 0$.

(b) For the stated data,

$$A = (10^9)(400 \times 10^{-24})(10^{20}) = 4.00 \times 10^7 \text{ s}^{-1},$$
$$b = (10^9)(1000 \times 10^{-24}) + 1.4 \times 10^{-9} = 1.401 \times 10^{-9} \text{ s}^{-1}.$$

Taking one year to mean 365 days gives $T = 365 \cdot 86\,400 = 31\,536\,000$ s. Then $bT = (1.401 \times 10^{-9})(31\,536\,000) = 0.0441819$, so $1 - e^{-bT} = 0.0432201$, while $A/b = 2.85510 \times 10^{16}$. For $N_{20} = 0$,

$$N_2(T) = \frac{A}{b}\left(1 - e^{-bT}\right) = (2.85510 \times 10^{16})(0.0432201) = \boxed{1.2340 \times 10^{15}}.$$

To the precision of the given data (two significant figures in λ_2) this is 1.2×10^{15}. The parent ^{153}Eu is stable, so it is consumed only by neutron absorption: $N_1' = -\varphi\sigma_1 N_1$ and $N_1(T) = N_1(0)e^{-\varphi\sigma_1 T}$. The depletion rate is $\varphi\sigma_1 = 4.00 \times 10^{-13}$ s^{-1} and $\varphi\sigma_1 T = 1.26144 \times 10^{-5}$, so the fraction consumed is

$$1 - e^{-\varphi\sigma_1 T} = 1.26143 \times 10^{-5} = 0.00126143\%.$$

Thus the assumption is justified: treating N_1 as constant is accurate to about thirteen parts per million over this interval. The calculation uses the same number or concentration units as $N_1(0)$.

20.8.5 In a nuclear reactor ^{135}Xe is formed as both a direct fission product of ^{235}U and by decay of ^{135}I (another fission product), half-life 6.7 hours. The half-life of ^{135}Xe is 9.2 hours. Because ^{135}Xe strongly absorbs thermal neutrons, thereby "poisoning" the nuclear reactor, its concentration is a matter of great interest. The relevant equations are

$$\frac{dN_{\mathrm{I}}}{dt} = \varphi\gamma_{\mathrm{I}}(\sigma_f N_{\mathrm{U}}) - \lambda_{\mathrm{I}} N_{\mathrm{I}},$$
$$\frac{dN_{\mathrm{Xe}}}{dt} = \varphi\big[\gamma_{\mathrm{Xe}}(\sigma_f N_{\mathrm{U}}) - \sigma_{\mathrm{Xe}} N_{\mathrm{Xe}}\big] + \lambda_{\mathrm{I}} N_{\mathrm{I}} - \lambda_{\mathrm{Xe}} N_{\mathrm{Xe}}.$$

Here N_{I}, N_{Xe}, N_{U} are the concentrations of ^{135}I, ^{135}Xe, ^{235}U, with N_{U} assumed to be constant. The neutron flux φ in the reactor causes fission of ^{235}U with cross section σ_f and removes ^{135}Xe by neutron absorption with cross section $\sigma_{\mathrm{Xe}} = 3.5 \times 10^6$ barns $= 3.5 \times 10^{-18}$ cm^2. Neutron absorption by ^{135}I is negligible. The yield of ^{135}I and ^{135}Xe per fission are, respectively, $\gamma_{\mathrm{I}} = 0.060$ and $\gamma_{\mathrm{Xe}} = 0.003$.

(a) Find $N_{\mathrm{Xe}}(t)$ in terms of neutron flux φ and the product $\sigma_f N_{\mathrm{U}}$.

(b) Find $N_{\mathrm{Xe}}(t \to \infty)$.

(c) After N_{Xe} has reached equilibrium, the reactor is shut down: $\varphi = 0$. Find $N_{\mathrm{Xe}}(t)$ following shutdown. Notice the increase in N_{Xe}, which may for a few hours interfere with starting the reactor up again. The increase is not automatic: determine when N_{Xe} increases immediately after shutdown; this depends on the pre-shutdown flux.

Hint. The half-life $t_{1/2}$ of a radioactive isotope is the time required for decay of half of the nuclides in a sample. For a decay rate $dN/dt = -\lambda N$, the half-life has the value $t_{1/2} = \ln 2/\lambda$, so λ can be computed as $\lambda = \ln 2/t_{1/2} = 0.693/t_{1/2}$.

In (a), retain initial inventories $N_{\mathrm{I}}(0) = I_0$ and $N_{\mathrm{Xe}}(0) = X_0$, then specialize to $I_0 = X_0 = 0$.

Solution.

Keep the neutron flux and the uranium concentration constant during operation, and abbreviate

$$P = \varphi\sigma_f N_{\mathrm{U}}, \quad \alpha = \lambda_{\mathrm{I}}, \quad \beta = \lambda_{\mathrm{Xe}}, \quad \mu = \beta + \varphi\sigma_{\mathrm{Xe}}, \quad I = N_{\mathrm{I}}, \quad X = N_{\mathrm{Xe}}.$$

The system becomes $I' + \alpha I = P\gamma_{\mathrm{I}}$ and $X' + \mu X = P\gamma_{\mathrm{Xe}} + \alpha I$. The textbook's part (a) did not specify initial inventories. We first keep $I(0) = I_0$, $X(0) = X_0$ arbitrary, and then give the usual initially clean case $I_0 = X_0 = 0$. The equilibrium values, obtained by setting both derivatives to zero ($\alpha I_e = P\gamma_{\mathrm{I}}$ and $\mu X_e = P\gamma_{\mathrm{Xe}} + \alpha I_e = P(\gamma_{\mathrm{Xe}} + \gamma_{\mathrm{I}})$), are

$$I_e = \frac{P\gamma_{\mathrm{I}}}{\alpha}, \qquad X_e = \frac{P(\gamma_{\mathrm{I}} + \gamma_{\mathrm{Xe}})}{\mu}.$$

(a) With $L\{I'\} = s\widehat{I} - I_0$, the first equation transforms into $(s+\alpha)\widehat{I} = I_0 + P\gamma_{\mathrm{I}}/s = I_0 + \alpha I_e/s$. The partial fractions $\alpha/[s(s+\alpha)] = 1/s - 1/(s+\alpha)$ then give

$$\widehat{I} = \frac{I_e}{s} + \frac{I_0 - I_e}{s+\alpha}, \qquad I(t) = I_e + (I_0 - I_e)e^{-\alpha t}.$$

Next subtract $\mu X_e = P\gamma_{\mathrm{Xe}} + \alpha I_e$ from both sides of the second equation:

$$(X - X_e)' + \mu(X - X_e) = \alpha(I - I_e) = \alpha(I_0 - I_e)e^{-\alpha t}.$$

Since $(X - X_e)(0) = X_0 - X_e$, its transform is

$$(s+\mu)L\{X - X_e\} = X_0 - X_e + \frac{\alpha(I_0 - I_e)}{s+\alpha}.$$

Divide by $s+\mu$, use $1/[(s+\alpha)(s+\mu)] = [1/(s+\alpha) - 1/(s+\mu)]/(\mu - \alpha)$ (for $\mu \neq \alpha$), and invert to obtain

$$\boxed{X(t) = X_e + (X_0 - X_e)e^{-\mu t} + \frac{\alpha(I_0 - I_e)}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})}.$$

In particular, for zero initial inventories ($I_0 = X_0 = 0$, with $\alpha I_e = P\gamma_{\mathrm{I}}$),

$$\boxed{X(t) = \frac{P(\gamma_{\mathrm{I}} + \gamma_{\mathrm{Xe}})}{\mu}(1 - e^{-\mu t}) - \frac{P\gamma_{\mathrm{I}}}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})}.$$

When $\mu = \alpha$, replace $(e^{-\alpha t} - e^{-\mu t})/(\mu - \alpha)$ by $te^{-\alpha t}$. The populations remain finite at this coincident rate. This is the limit of the quotient as $\mu \to \alpha$, namely $-\partial_\mu e^{-\mu t} = te^{-\mu t}$ at $\mu = \alpha$. For example, in the initially clean case this gives $X = X_e(1 - e^{-\alpha t}) - P\gamma_{\mathrm{I}} te^{-\alpha t}$.

(b) Since the rates α and μ are positive, all exponentials in (a) decay, so $X \to X_e$:

$$\boxed{N_{\mathrm{Xe}}(\infty) = \frac{\varphi(\gamma_{\mathrm{I}} + \gamma_{\mathrm{Xe}})\sigma_f N_{\mathrm{U}}}{\lambda_{\mathrm{Xe}} + \varphi\sigma_{\mathrm{Xe}}}}.$$

(c) Reset $t = 0$ at shutdown. The initial values are now I_e, X_e, but production and neutron absorption cease: switching off the flux removes the terms $P\gamma_{\mathrm{I}}$, $P\gamma_{\mathrm{Xe}}$ and $\varphi\sigma_{\mathrm{Xe}}X$, so the equations become $I' = -\alpha I$ and $X' = -\beta X + \alpha I$. Below, φ, P and μ keep their pre-shutdown values, which determine I_e and X_e. Thus $I = I_e e^{-\alpha t}$, and the second equation, with $X(0) = X_e$, transforms into

$$(s + \beta)\widehat{X} = X_e + \frac{\alpha I_e}{s + \alpha}.$$

Here $\alpha > \beta$, because the iodine half-life is the shorter. Dividing by $s + \beta$ and inverting with $1/[(s + \alpha)(s + \beta)] = [1/(s + \beta) - 1/(s + \alpha)]/(\alpha - \beta)$ gives

$$\boxed{X(t) = X_e e^{-\beta t} + \frac{\alpha I_e}{\alpha - \beta}(e^{-\beta t} - e^{-\alpha t})}.$$

In the Instructor's Manual's shutdown formula the coefficient of the exponential difference uses $N_{\mathrm{Xe}}(0)$; it must use $N_{\mathrm{I}}(0) = I_e$. The source of xenon after shutdown is αI, as the differential equation directly shows. The initial slope follows from the shutdown equation at $t = 0$, $X'(0) = \alpha I_e - \beta X_e$. With $\alpha I_e = P\gamma_{\mathrm{I}}$, the value of X_e, and $\mu - \beta = \varphi\sigma_{\mathrm{Xe}}$,

$$\begin{aligned} X'(0) = P\gamma_{\mathrm{I}} - \beta X_e &= \frac{P[\gamma_{\mathrm{I}}\mu - \beta(\gamma_{\mathrm{I}} + \gamma_{\mathrm{Xe}})]}{\mu} \\ &= \frac{P(\gamma_{\mathrm{I}}\varphi\sigma_{\mathrm{Xe}} - \beta\gamma_{\mathrm{Xe}})}{\mu}. \end{aligned}$$

A short-term increase is conditional, not automatic: for a positive pre-shutdown flux it occurs exactly when $\varphi\sigma_{\mathrm{Xe}} > \beta\gamma_{\mathrm{Xe}}/\gamma_{\mathrm{I}}$. Here $\alpha = \ln 2/(6.7 \cdot 3600) = 2.87374 \times 10^{-5}\ \mathrm{s}^{-1}$ and $\beta = \ln 2/(9.2 \cdot 3600) = 2.09284 \times 10^{-5}\ \mathrm{s}^{-1}$. Since $\gamma_{\mathrm{Xe}}/\gamma_{\mathrm{I}} = 0.003/0.060 = 1/20$, the threshold is

$$\varphi > \frac{\beta}{20\sigma_{\mathrm{Xe}}} = 2.98977 \times 10^{11}\ \mathrm{cm}^{-2}\,\mathrm{s}^{-1}.$$

For an explicit peak time put $A = \alpha I_e/(\alpha - \beta) > 0$. Then $X = (X_e + A)e^{-\beta t} - Ae^{-\alpha t}$, and $X' = -\beta(X_e + A)e^{-\beta t} + \alpha A e^{-\alpha t} = 0$ gives $e^{(\alpha - \beta)t} = \alpha A/[\beta(X_e + A)]$, that is,

$$t_{\mathrm{peak}} = \frac{1}{\alpha - \beta} \ln \frac{\alpha A}{\beta(X_e + A)}.$$

This is positive exactly when $\alpha A > \beta(X_e + A)$, i.e. $(\alpha - \beta)A = \alpha I_e > \beta X_e$, which is the preceding increase condition $X'(0) > 0$. Because $e^{\beta t}X' = -\beta(X_e + A) + \alpha A e^{-(\alpha - \beta)t}$ decreases in t, X then has a single maximum, at t_{peak}; otherwise $X' < 0$ for all $t > 0$ and the maximum after shutdown is at $t = 0$. Power reactors operate at thermal fluxes of about 10^{13} to $10^{14}\ \mathrm{cm}^{-2}\,\mathrm{s}^{-1}$, some 30 to 300 times this threshold, so the increase noted in the textbook is the usual case. For example, at $\varphi = 10^{13}\ \mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ ($\mu = 5.59 \times 10^{-5}\ \mathrm{s}^{-1}$; P cancels from the ratios) the peak-time formula gives $t_{\mathrm{peak}} = 6.4$ h with $X(t_{\mathrm{peak}}) = 1.31X_e$, and solving $X(t) = X_e$ numerically shows that X falls back to X_e about 17 h after shutdown.

20.8.6 Solve Eq. (20.160), which describes a damped simple harmonic oscillator, for $X(0) = X_0$, $X'(0) = 0$, and

(a) $b^2 = 4mk$ (critically damped),

(b) $b^2 > 4mk$ (overdamped).

ANS. (a) $X(t) = X_0 e^{-(b/2m)t}\left(1 + \dfrac{b}{2m}t\right)$.

Solution.

The equation is $mX'' + bX' + kX = 0$. Put $\alpha = b/(2m)$ and $\omega_0^2 = k/m$, and divide by m: $X'' + 2\alpha X' + \omega_0^2 X = 0$. With $X(0) = X_0$ and $X'(0) = 0$, the derivative formulas give $L\{X''\} = s^2\widehat{X} - sX_0$ and $L\{X'\} = s\widehat{X} - X_0$, so

$$(s^2 + 2\alpha s + \omega_0^2)\widehat{X} = (s + 2\alpha)X_0, \quad \widehat{X} = X_0\frac{(s+\alpha)+\alpha}{(s+\alpha)^2 + \omega_0^2 - \alpha^2},$$

where the denominator has been completed to a square, $s^2 + 2\alpha s + \omega_0^2 = (s+\alpha)^2 + \omega_0^2 - \alpha^2$. Each case is inverted with the shift theorem $L\{e^{-\alpha t}F(t)\} = f(s+\alpha)$.

(a) Critical damping means $\omega_0^2 = \alpha^2$, so the pairs $L\{1\} = 1/s$ and $L\{t\} = 1/s^2$, shifted, give

$$\widehat{X} = X_0\left(\frac{1}{s+\alpha} + \frac{\alpha}{(s+\alpha)^2}\right), \qquad \boxed{X = X_0 e^{-\alpha t}(1 + \alpha t)}.$$

(b) In the overdamped case define $\kappa = \sqrt{\alpha^2 - \omega_0^2} > 0$, so the denominator is $(s+\alpha)^2 - \kappa^2$. The hyperbolic pairs $L\{\cosh \kappa t\} = s/(s^2 - \kappa^2)$ and $L\{\sinh \kappa t\} = \kappa/(s^2 - \kappa^2)$, shifted by $s \mapsto s + \alpha$, give

$$\widehat{X} = X_0\left[\frac{s+\alpha}{(s+\alpha)^2 - \kappa^2} + \frac{\alpha}{\kappa}\,\frac{\kappa}{(s+\alpha)^2 - \kappa^2}\right],$$

$$\boxed{X = X_0 e^{-\alpha t}\left(\cosh \kappa t + \frac{\alpha}{\kappa}\sinh \kappa t\right)}.$$

Writing cosh and sinh in exponentials, this is

$$\frac{X_0}{2}\left[\left(1 + \frac{\alpha}{\kappa}\right)e^{(-\alpha+\kappa)t} + \left(1 - \frac{\alpha}{\kappa}\right)e^{(-\alpha-\kappa)t}\right].$$

Both exponents $-\alpha \pm \kappa$ are the roots of the characteristic equation $r^2 + 2\alpha r + \omega_0^2 = 0$; the coefficients sum to X_0, and the initial derivative is $(X_0/2)[(1 + \alpha/\kappa)(\kappa - \alpha) - (1 - \alpha/\kappa)(\kappa + \alpha)] = (X_0/2)[(\kappa - \alpha^2/\kappa) - (\kappa - \alpha^2/\kappa)] = 0$. Taking $\kappa \to 0$ gives the critical result because $\sinh \kappa t/\kappa \to t$.

20.8.7 Again solve Eq. (20.160), which describes a damped simple harmonic oscillator, but this time for $X(0) = 0$, $X'(0) = v_0$, and

(a) $b^2 < 4mk$ (underdamped),

(b) $b^2 = 4mk$ (critically damped),

(c) $b^2 > 4mk$ (overdamped).

ANS. (a) $X(t) = \dfrac{v_0}{\omega_1}\, e^{-(b/2m)t} \sin \omega_1 t$,
(b) $X(t) = v_0 t e^{-(b/2m)t}$.

Solution.

With $\alpha = b/(2m)$, $\omega_0^2 = k/m$, and initial data $X(0) = 0$, $X'(0) = v_0$, we have $L\{X''\} = s^2 \widehat{X} - v_0$ and $L\{X'\} = s\widehat{X}$, so transformation of the equation (divided by m) gives

$$(s^2 + 2\alpha s + \omega_0^2)\widehat{X} = v_0, \qquad \widehat{X} = \frac{v_0}{(s+\alpha)^2 + \omega_0^2 - \alpha^2}.$$

(a) If $\omega_1 = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{k/m - b^2/(4m^2)} > 0$, then $\widehat{X} = (v_0/\omega_1)\,\omega_1/[(s+\alpha)^2 + \omega_1^2]$, and the shifted sine transform $L\{e^{-\alpha t}\sin \omega_1 t\} = \omega_1/[(s+\alpha)^2 + \omega_1^2]$ gives

$$\boxed{X(t) = \frac{v_0}{\omega_1} e^{-\alpha t} \sin \omega_1 t}.$$

(b) At $\omega_0^2 = \alpha^2$ the transform is $v_0/(s+\alpha)^2$, whose inverse, by the shifted pair $L\{te^{-\alpha t}\} = 1/(s+\alpha)^2$, is

$$\boxed{X(t) = v_0 t e^{-\alpha t}}.$$

(c) If $\kappa = \sqrt{\alpha^2 - \omega_0^2} > 0$, the denominator factors as $(s+\alpha)^2 - \kappa^2 = (s + \alpha - \kappa)(s + \alpha + \kappa)$, and partial fractions give

$$\widehat{X} = \frac{v_0}{2\kappa}\left(\frac{1}{s+\alpha-\kappa} - \frac{1}{s+\alpha+\kappa}\right),$$

so

$$X(t) = \frac{v_0}{2\kappa}\left(e^{-(\alpha-\kappa)t} - e^{-(\alpha+\kappa)t}\right) = \boxed{\frac{v_0}{\kappa} e^{-\alpha t} \sinh \kappa t}.$$

All three expressions have $X(0) = 0$, $X'(0) = v_0$; for example, in (a) $X' = v_0 e^{-\alpha t}[\cos \omega_1 t - (\alpha/\omega_1)\sin \omega_1 t]$. The critical formula is the continuous limit of both others, using $\sin z/z \to 1$ and $\sinh z/z \to 1$.

20.8.8 The motion of a body falling in a resisting medium may be described by

$$m\,\frac{d^2X(t)}{dt^2} = mg - b\,\frac{dX(t)}{dt}$$

when the retarding force is proportional to the velocity. Find $X(t)$ and $dX(t)/dt$ for the initial conditions

$$X(0) = \left.\frac{dX}{dt}\right|_{t=0} = 0.$$

Solution.

Choose the downward direction positive and set $\Gamma = b/m > 0$. The equation is $X'' + \Gamma X' = g$. With both initial values zero, $L\{X''\} = s^2\widehat{X}$, $L\{X'\} = s\widehat{X}$, and $L\{g\} = g/s$, so

$$(s^2 + \Gamma s)\widehat{X} = g/s, \qquad \widehat{X} = \frac{g}{s^2(s+\Gamma)}.$$

The partial fractions are

$$\frac{1}{s^2(s+\Gamma)} = \frac{1}{\Gamma s^2} - \frac{1}{\Gamma^2 s} + \frac{1}{\Gamma^2(s+\Gamma)}.$$

Multiplication by $s^2(s+\Gamma)$ verifies the identity: its right numerator is $(s+\Gamma)/\Gamma - s(s+\Gamma)/\Gamma^2 + s^2/\Gamma^2 = 1$. Inverting term by term with $L^{-1}\{1/s^2\} = t$, $L^{-1}\{1/s\} = 1$, and $L^{-1}\{1/(s+\Gamma)\} = e^{-\Gamma t}$,

$$X(t) = g\left[\frac{t}{\Gamma} - \frac{1}{\Gamma^2} + \frac{e^{-\Gamma t}}{\Gamma^2}\right] = \frac{g}{\Gamma}t - \frac{g}{\Gamma^2}(1 - e^{-\Gamma t}),$$

and differentiation gives $X' = (g/\Gamma)(1 - e^{-\Gamma t})$. With $\Gamma = b/m$,

$$\boxed{X(t) = \frac{mg}{b}t - \frac{m^2 g}{b^2}(1 - e^{-bt/m})}, \qquad \boxed{X'(t) = \frac{mg}{b}(1 - e^{-bt/m})}.$$

The velocity approaches the terminal value mg/b, since $e^{-bt/m} \to 0$. Expanding $e^{-\Gamma t} = 1 - \Gamma t + \Gamma^2 t^2/2 - \Gamma^3 t^3/6 + \cdots$ gives

$$X = \frac{g}{\Gamma}t - \frac{g}{\Gamma^2}\left(\Gamma t - \frac{\Gamma^2 t^2}{2} + \frac{\Gamma^3 t^3}{6} - \cdots\right) = \frac{gt^2}{2} - \frac{g\Gamma t^3}{6} + \cdots,$$

and similarly $X' = gt - g\Gamma t^2/2 + \cdots$. Hence the zero-drag limits are $X \to gt^2/2$ and $X' \to gt$: the apparent singularities as $b \to 0$ cancel.

20.8.9 Ringing circuit. In certain electronic devices, resistance, inductance, and capacitance are placed in a circuit as shown in Fig. 20.18. A constant voltage is maintained across the capacitance, keeping it charged. At time $t = 0$ the circuit is disconnected from the voltage source. Find the voltages across each of the elements R, L, and C as a function of time. Assume R to be small.

Hint. By Kirchhoff's laws

$$I_{RL} + I_C = 0 \quad \text{and} \quad E_R + E_L = E_C,$$

where

$$E_R = I_{RL}R, \qquad E_L = L\,\frac{dI_{RL}}{dt}, \qquad E_C = \frac{q_0}{C} + \frac{1}{C}\int_0^t I_C\,dt,$$

q_0 = initial charge of capacitor.

Take $R, L, C > 0$ and $R^2 < 4L/C$. Keep the initial branch current $I_{RL}(0) = I_0$ explicit; if the source was connected long enough for DC steady state, use $I_0 = q_0/(CR)$.

Solution.

The drawing has a series R, L branch in parallel with C. Define $V_0 = q_0/C$, write $I = I_{RL}$ for the current through that branch and $I_0 = I(0)$, and use the voltage polarities in the hint. Kirchhoff's laws, with the hint's E_C differentiated ($E_C' = I_C/C$), imply

$$I_C = -I, \qquad E_C' = -I/C, \qquad LI' + RI = E_C.$$

Substituting $I = -CE_C'$ into the last relation gives

$$LCE_C'' + RCE_C' + E_C = 0, \qquad E_C(0) = V_0, \quad E_C'(0) = -I_0/C.$$

Put $\alpha = R/(2L)$ and $\omega = \sqrt{1/(LC) - \alpha^2} > 0$; "small R" means $R^2 < 4L/C$. With a hat denoting the Laplace transform, the derivative formulas give the transforms $s^2\widehat{E}_C - sV_0 + I_0/C$ of E_C'' and $s\widehat{E}_C - V_0$ of E_C', so transformation and division by LC (note $R/L = 2\alpha$) give

$$\left(s^2 + 2\alpha s + \frac{1}{LC}\right)\widehat{E}_C = (s + 2\alpha)V_0 - \frac{I_0}{C},$$

and $s^2 + 2\alpha s + 1/(LC) = (s+\alpha)^2 + \omega^2$. Hence

$$\widehat{E}_C(s) = \frac{(s+2\alpha)V_0 - I_0/C}{(s+\alpha)^2+\omega^2} = V_0\frac{s+\alpha}{(s+\alpha)^2+\omega^2} + \frac{\alpha V_0 - I_0/C}{(s+\alpha)^2+\omega^2}.$$

Since $(s+\alpha)/[(s+\alpha)^2+\omega^2]$ and $\omega/[(s+\alpha)^2+\omega^2]$ are the transforms of $e^{-\alpha t}\cos\omega t$ and $e^{-\alpha t}\sin\omega t$ (shift theorem), inversion gives

$$\boxed{E_C = e^{-\alpha t}\left[V_0\cos\omega t + \frac{\alpha V_0 - I_0/C}{\omega}\sin\omega t\right]}.$$

Write $B = (\alpha V_0 - I_0/C)/\omega$ for the sine coefficient. The product rule gives

$$E_C' = e^{-\alpha t}\big[(\omega B - \alpha V_0)\cos\omega t - (\alpha B + \omega V_0)\sin\omega t\big],$$

where $\omega B - \alpha V_0 = -I_0/C$ and, using $\omega^2 + \alpha^2 = 1/(LC)$, $\alpha B + \omega V_0 = [(\alpha^2+\omega^2)V_0 - \alpha I_0/C]/\omega = [V_0/(LC) - \alpha I_0/C]/\omega$. Multiplying by $-C$, the cosine coefficient of $I = -CE_C'$ is I_0, while its sine coefficient is $(V_0/L - \alpha I_0)/\omega$. Thus

$$I = e^{-\alpha t}\left[I_0\cos\omega t + \frac{V_0/L - \alpha I_0}{\omega}\sin\omega t\right],$$

$$\boxed{E_R = Re^{-\alpha t}\left[I_0\cos\omega t + \frac{V_0/L - \alpha I_0}{\omega}\sin\omega t\right]},$$

$$\boxed{E_L = e^{-\alpha t}\left[(V_0 - RI_0)\cos\omega t + \frac{-\alpha V_0 + (R\alpha - 1/C)I_0}{\omega}\sin\omega t\right]}.$$

The last expression is $E_C - E_R$ (Kirchhoff's voltage law): its cosine coefficient is $V_0 - RI_0$, and its sine coefficient is $[\alpha V_0 - I_0/C - R(V_0/L - \alpha I_0)]/\omega = [-\alpha V_0 + (R\alpha - 1/C)I_0]/\omega$, because $RV_0/L = 2\alpha V_0$. It also equals LI'. These formulas display both initial energy stores explicitly.

For the circuit exactly as drawn, if the constant source has been connected long enough to establish a DC steady state, the inductor has zero DC voltage but the resistor does not:

$$\boxed{I_0 = V_0/R}.$$

Substitution gives, since then $I_0/C = V_0/(RC)$, $R(V_0/L - \alpha I_0) = (2\alpha - \alpha)V_0 = \alpha V_0$, $V_0 - RI_0 = 0$, and $-\alpha V_0 + (R\alpha - 1/C)I_0 = -V_0/(RC)$,

$$E_C = V_0 e^{-\alpha t}\left[\cos\omega t + \frac{\alpha - 1/(RC)}{\omega}\sin\omega t\right],$$
$$E_R = V_0 e^{-\alpha t}\left[\cos\omega t + \frac{\alpha}{\omega}\sin\omega t\right],$$
$$E_L = -\frac{V_0}{RC\omega}e^{-\alpha t}\sin\omega t.$$

These have $E_R(0) = E_C(0) = V_0$, $E_L(0) = 0$, as a prior DC steady state requires. The Instructor's Manual instead uses damping $1/(2RC)$ and $E_C(0) = q_0/C = 0$. Those are not the equations or initial state of the charged-capacitor circuit in Fig. 20.18. Its damping is $R/(2L)$. An ideal inductor in series with R does not short the capacitor. If a different switching preparation leaves the inductor initially unenergized, set $I_0 = 0$ in the general formulas; this is an additional initial condition, not the DC steady state of the displayed circuit. In that case $E_C = V_0 e^{-\alpha t}(\cos\omega t + \alpha\sin\omega t/\omega)$, $E_R = RV_0 e^{-\alpha t}\sin\omega t/(L\omega)$, and $E_L = V_0 e^{-\alpha t}(\cos\omega t - \alpha\sin\omega t/\omega)$. The $R = 0$ idealization admits this separately specified initial state, but no finite steady DC current V_0/R.

20.8.10 With $J_0(t)$ expressed as a contour integral, apply the Laplace transform operation, reverse the order of integration, and thus show that

$$L\{J_0(t)\} = (s^2+1)^{-1/2}, \quad \text{for } s > 0.$$

Solution.

The generating function $\exp[(t/2)(z - z^{-1})] = \sum_n J_n(t) z^n$ (Section 14.1) is a Laurent series in z, so the coefficient formula for its z^0 term gives the contour representation

$$J_0(t) = \frac{1}{2\pi i} \oint_{|z|=1} \exp\left[\frac{t}{2}(z - z^{-1})\right] \frac{dz}{z},$$

with counterclockwise orientation. On this contour $z = e^{i\theta}$ and $z - z^{-1} = 2i \sin\theta$, so the exponential has modulus 1. For real $s > 0$, multiplication by e^{-st} makes the double integral absolutely convergent (the integrand is bounded by e^{-st}), and interchange is valid. The inner integral is elementary, because the real part of $s - (z - z^{-1})/2 = s - i\sin\theta$ is $s > 0$:

$$\begin{aligned} L\{J_0\}(s) &= \frac{1}{2\pi i} \oint \frac{dz}{z} \int_0^\infty e^{-t[s-(z-z^{-1})/2]} dt = \frac{1}{2\pi i} \oint \frac{dz}{z[s - (z - z^{-1})/2]} \\ &= \frac{1}{2\pi i} \oint \frac{2\,dz}{1 + 2sz - z^2}, \end{aligned}$$

since $z[s - (z - z^{-1})/2] = (2sz - z^2 + 1)/2$. Let $D = \sqrt{s^2+1} > 0$. The denominator vanishes at the roots $z_\pm = s \pm D$ of $z^2 - 2sz - 1 = 0$. Here $z_+ > 1$ and, since $(D - s)(D + s) = D^2 - s^2 = 1$, $|z_-| = D - s = 1/(D + s) < 1$, so just z_- lies inside. At this simple pole the residue is the numerator divided by the derivative $2s - 2z$ of the denominator:

$$\frac{2}{2s - 2z_-} = \frac{2}{2D} = \frac{1}{D}.$$

The residue theorem therefore gives

$$\boxed{L\{J_0(t)\}(s) = \frac{1}{\sqrt{s^2+1}}}, \qquad s > 0.$$

The transform extends analytically to $\Re s > 0$ with the branch whose square root is positive on the positive real axis. Indeed, for $s = x + iy$ with $x > 0$, $s^2 + 1 = x^2 - y^2 + 1 + 2ixy$ is real only when $y = 0$, and then it is positive; so the principal root $\sqrt{s^2+1}$ is analytic in $\Re s > 0$. The Laplace integral of the bounded function J_0 is analytic there too, and the two agree for $s > 0$, hence throughout $\Re s > 0$ by the identity theorem.

20.8.11 Develop the Laplace transform of $J_n(t)$ from $L\{J_0(t)\}$ by using the Bessel function recurrence relations.

Hint. Here is a chance to use mathematical induction (Section 1.4).

The formula derived by the indicated induction is for integer $n \geq 0$.

Solution.

Write $g_n(s) = L\{J_n(t)\}(s)$, $D = \sqrt{s^2+1}$, and $r = D - s$ for $s > 0$. Exercise 20.8.10 gives $g_0 = 1/D$. Since $J_0' = -J_1$ and $J_0(0) = 1$, the derivative formula $L\{F'\} = sL\{F\} - F(0)$ gives

$$g_1 = -L\{J_0'\} = -[sg_0 - J_0(0)] = 1 - s/D = (D - s)/D = r/D.$$

For integer $n \geq 1$, $2J_n' = J_{n-1} - J_{n+1}$ and $J_n(0) = 0$, so $L\{J_n'\} = sg_n$; transforming the recurrence gives

$$2sg_n = g_{n-1} - g_{n+1}, \qquad g_{n+1} = g_{n-1} - 2sg_n.$$

Assume $g_{n-1} = r^{n-1}/D$ and $g_n = r^n/D$ for some $n \geq 1$; this holds for $n = 1$ by the two results above. Then

$$g_{n+1} = \frac{r^{n-1}}{D} - 2s\frac{r^n}{D} = \frac{r^{n-1}}{D}(1 - 2sr).$$

The identity (with $D^2 = s^2 + 1$)

$$r^2 = (D - s)^2 = D^2 - 2sD + s^2 = 1 + 2s^2 - 2sD = 1 - 2s(D - s) = 1 - 2sr$$

makes this $r^{n-1}r^2/D = r^{n+1}/D$. Induction establishes

$$\boxed{L\{J_n(t)\}(s) = \frac{(\sqrt{s^2+1} - s)^n}{\sqrt{s^2+1}}}, \qquad n = 0, 1, 2, \ldots.$$

For negative integer order use $J_{-n} = (-1)^n J_n$; the displayed power formula is intended for nonnegative order. Analytic continuation to $\Re s > 0$ uses the branch specified in Exercise 20.8.10.

20.8.12 A calculation of the magnetic field of a circular current loop in circular cylindrical coordinates leads to the integral

$$\int_0^\infty e^{-kz} k J_1(ka)\, dk, \qquad a > 0, \quad \Re z > 0.$$

Show that this integral is equal to $a/(z^2 + a^2)^{3/2}$.

Solution.

For an ordinary improper integral the boundary $\Re z = 0$, which the textbook's printed condition $\Re(z) \geq 0$ includes, must be excluded when $a > 0$: the stated integrand has oscillations with amplitude growing like $\sqrt{k}$. The formula holds ordinarily for $\Re z > 0$ and has boundary values in an Abel or distributional sense. Scale Exercise 20.8.10: for $a > 0$ and $\Re z > 0$, substitute $t = ka$ ($dk = dt/a$) and use that result, continued analytically to $s = z/a$ with $\Re s > 0$:

$$\int_0^\infty e^{-kz} J_0(ka) dk = \frac{1}{a}\int_0^\infty e^{-(z/a)t} J_0(t) dt = \frac{1}{a}\frac{1}{\sqrt{(z/a)^2 + 1}} = \frac{1}{\sqrt{z^2 + a^2}}.$$

By the chain rule and $J_0' = -J_1$, the a-derivative of the integrand is $e^{-kz} k J_0'(ka) = -e^{-kz} k J_1(ka)$. Since $|J_1(x)| \leq 1$ for real x (Exercise 14.1.1), it is bounded by the integrable $ke^{-k\Re z}$, so differentiation under the integral sign with respect to a is allowed:

$$-\int_0^\infty e^{-kz} k J_1(ka) dk = \frac{\partial}{\partial a}(z^2 + a^2)^{-1/2} = -a(z^2 + a^2)^{-3/2}.$$

Therefore

$$\boxed{\int_0^\infty e^{-kz} k J_1(ka) dk = \frac{a}{(z^2 + a^2)^{3/2}}}, \qquad \Re z > 0.$$

The square root is chosen positive for real $z > 0$ and continued throughout the right half-plane. To see the boundary issue directly, at $z = 0$, $kJ_1(ka) \sim \sqrt{2k/(\pi a)}\cos(ka - 3\pi/4)$; its oscillatory primitive does not have a limit. Nevertheless inserting $e^{-\epsilon k}$, that is, taking $z = \epsilon > 0$ in the boxed

formula, and letting $\epsilon \downarrow 0$ gives the Abel value $a/(a^2)^{3/2} = 1/a^2$. For $z = iy$, the analogous limiting formula is singular at $y = \pm a$; it is not a finite ordinary value there. The Instructor's Manual, p. 351, first integrates $\int_0^\infty e^{-kz} J_1(ka)\, dk$ by parts, with $J_1(ka) = -a^{-1}\, dJ_0(ka)/dk$, and then applies $-\partial/\partial z$. In that intermediate integral it prints $K_1(ka)$ where $J_1(ka)$ is required, and it replaces the boundary term $[e^{-kz} J_0(ka)]_0^\infty = -1$ by $+1$. The resulting constant, $-a^{-1}$ times this term, is independent of z and disappears on differentiation with respect to z, but the intermediate equality is incorrect: the integral equals $[1 - z/\sqrt{z^2 + a^2}]/a$.

20.8.13 Show that

$$L\{I_0(at)\} = (s^2 - a^2)^{-1/2}, \qquad a \in \mathbb{R}, \quad s > |a|.$$

Solution.

For real a use the angular representation (Exercise 14.5.14 with $n = 0$; see also Exercise 14.6.3)

$$I_0(at) = \frac{1}{\pi} \int_0^\pi e^{at \cos\theta} d\theta.$$

If a is allowed to be negative, the sufficient real convergence condition is $s > |a|$, not merely $s > a$. For a positive scale a, the textbook's condition $s > a$ is correct. For $s > |a|$ the integrand of the double integral is bounded by $e^{-st} e^{|a|t} = e^{-(s-|a|)t}$, so it converges absolutely and the order of integration may be exchanged. The inner integral is $\int_0^\infty e^{-(s - a\cos\theta)t} dt = 1/(s - a\cos\theta)$, so

$$L\{I_0(at)\}(s) = \frac{1}{\pi} \int_0^\pi \frac{d\theta}{s - a\cos\theta}.$$

Set $u = \tan(\theta/2)$, so $\cos\theta = (1 - u^2)/(1 + u^2)$ and $d\theta = 2du/(1 + u^2)$; the range $0 \le \theta < \pi$ becomes $0 \le u < \infty$, and $s - a\cos\theta = [(s - a) + (s + a)u^2]/(1 + u^2)$. Then

$$\begin{aligned} L\{I_0(at)\} &= \frac{2}{\pi} \int_0^\infty \frac{du}{(s - a) + (s + a)u^2} \\ &= \frac{2}{\pi\sqrt{(s - a)(s + a)}} \left[\tan^{-1}\left(u \sqrt{\frac{s + a}{s - a}} \right) \right]_0^\infty \\ &= \frac{2}{\pi\sqrt{s^2 - a^2}} \cdot \frac{\pi}{2} = \boxed{\frac{1}{\sqrt{s^2 - a^2}}}. \end{aligned}$$

The second line uses $\int du/(p + qu^2) = (pq)^{-1/2} \tan^{-1}(u\sqrt{q/p})$ with $p = s - a > 0$ and $q = s + a > 0$. For complex s in $\Re s > |a|$, use the branch positive on $s > |a|$, equivalently the branch asymptotic to s at positive infinity in the denominator.

20.8.14 Verify the following Laplace transforms:

(a) $L\{j_0(at)\} = L\left\{\dfrac{\sin at}{at}\right\} = \dfrac{1}{a} \cot^{-1}\left(\dfrac{s}{a}\right)$,

(b) $L\{y_0(at)\}$ does not exist.

(c) $L\{i_0(at)\} = L\left\{\dfrac{\sinh at}{at}\right\} = \dfrac{1}{2a} \ln \dfrac{s + a}{s - a} = \dfrac{1}{a} \coth^{-1}\left(\dfrac{s}{a}\right)$,

(d) $L\{k_0(at)\}$ does not exist.

Use $a > 0$, with real $s > 0$ in (a) and $s > a$ in (c), and the real inverse-function branches there.

Solution.

Take $a > 0$; use real $s > 0$ for (a) and $s > a$ for (c). (a) Since $L\{\sin at\}(u) = a/(u^2 + a^2)$, integrate the transform parameter from s to infinity, inserting the defining integral and using $\int_s^\infty e^{-ut}du = e^{-st}/t$:

$$\int_s^\infty \frac{a}{u^2 + a^2}du = \int_0^\infty \sin(at)\left(\int_s^\infty e^{-ut}du\right)dt = \int_0^\infty e^{-st}\frac{\sin at}{t}dt.$$

Absolute integrability after the factor e^{-st} justifies the interchange, because $|\sin at|/t \le a$. The left side is $[\tan^{-1}(u/a)]_s^\infty = \pi/2 - \tan^{-1}(s/a)$, and $\pi/2 - \tan^{-1} x = \cot^{-1} x$ for $x > 0$. Since $j_0(at) = \sin(at)/(at)$, dividing by a yields

$$\boxed{L\{j_0(at)\} = \frac{1}{a}\left[\frac{\pi}{2} - \tan^{-1}(s/a)\right] = \frac{1}{a}\cot^{-1}(s/a)}.$$

(b) The spherical Neumann function is $y_0(at) = -\cos(at)/(at) = -1/(at) + O(t)$ near zero, since $\cos at = 1 + O(t^2)$. Thus its Laplace integral contains a nonzero multiple of $\int_0^\epsilon dt/t$ and diverges for every s.

(c) In the convergent half-plane $u > a$, $L\{\sinh at\}(u) = a/(u^2 - a^2)$. The same integration, now justified by $0 \le \sinh(at)/t \le a\cosh at \le ae^{at}$ and $s > a$, followed by division by a (since $i_0(at) = \sinh(at)/(at)$), and the partial fractions $1/(u^2 - a^2) = [1/(u - a) - 1/(u + a)]/(2a)$, give

$$\begin{aligned} L\{i_0(at)\} &= \int_s^\infty \frac{du}{u^2 - a^2} = \frac{1}{2a}\left[\ln\frac{u - a}{u + a}\right]_s^\infty \\ &= \frac{1}{2a}\left[0 - \ln\frac{s - a}{s + a}\right] = \boxed{\frac{1}{2a}\ln\frac{s + a}{s - a} = \frac{1}{a}\coth^{-1}(s/a)}. \end{aligned}$$

The two logarithms $\ln(u - a)$ and $\ln(u + a)$ must be kept together at the infinite endpoint, where $(u - a)/(u + a) \to 1$; the individual logarithms do not have finite limits. The last form uses $\coth^{-1} x = \frac{1}{2}\ln[(x+1)/(x-1)]$ for $x > 1$, with $x = s/a$. The Instructor's Manual's final logarithmic expression for (c), on p. 351, omits the factor $1/a$. Its coefficient must be $1/(2a)$, as in the textbook answer and the partial fractions shown here.

(d) In the book's spherical normalization $k_0(at) = e^{-at}/(at)$. It also has a nonintegrable $1/(at)$ singularity at zero, so its ordinary Laplace transform does not exist. The real inverse-trigonometric and inverse-hyperbolic forms are written for the positive scale and the domains specified above; the $a = 0$ limits of (a) and (c) are $1/s$. Negative a leaves $j_0(at)$ and $i_0(at)$ unchanged (they are even in a). Both forms in (c) are even in a too and remain valid for $s > |a|$. In (a), however, $\cot^{-1}(s/a)$ at a negative argument depends on the branch (values in $(0, \pi)$ give the wrong sign), so use $\tan^{-1}(a/s)/a$, which is even in a and valid for $s > 0$.

20.8.15 Develop a Laplace transform solution of Laguerre's equation,

$$tF''(t) + (1 - t)F'(t) + nF(t) = 0.$$

Note that you need a derivative of a transform and a transform of derivatives. Go as far as you can with a general value of n; then (and only then) set $n = 0$.

Find the solution regular at the origin, with finite $F(0) = C$ and $F'(0)$. A logarithmically singular second solution is outside these initial-value transform rules.

Solution.

Seek the solution regular at $t=0$, with finite $F(0)=C$ and $F'(0)$. Write $f=L\{F\}$. The derivative formulas and the derivative of a transform, $L\{tG(t)\}=-\frac{d}{ds}L\{G\}$, give the needed transforms

$$L\{F'\}=sf-C, \qquad L\{F''\}=s^2f-sC-F'(0),$$
$$L\{tF'\}=-\frac{d}{ds}(sf-C)=-f-sf', \quad L\{tF''\}=-\frac{d}{ds}(s^2f-sC-F'(0))=-2sf-s^2f'+C.$$

In the sum $L\{tF''\}+L\{F'\}-L\{tF'\}+nf$ the f' terms give $(-s^2+s)f'$, the f terms give $(-2s+s+1+n)f$, and the constants $C-C$ cancel, so the transformed equation is

$$s(1-s)f'+(n+1-s)f=0.$$

For a nonzero solution, divide by $s(1-s)f$ and decompose in partial fractions; covering up s at $s=0$ and $1-s$ at $s=1$ gives $(n+1-s)/[s(1-s)]=(n+1)/s+n/(1-s)$, so

$$\frac{f'}{f}=-\frac{n+1-s}{s(1-s)}=-\frac{n+1}{s}+\frac{n}{s-1}.$$

Integrating each logarithmic derivative gives

$$\ln f=\ln A+n\ln(s-1)-(n+1)\ln s, \qquad f(s)=A\frac{(s-1)^n}{s^{n+1}}.$$

For general n, take the powers positive for real $s>1$ and continue them analytically into the half-plane $\Re s>1$. The initial-value theorem (Exercise 20.7.1) $\lim_{s\to\infty}sf(s)=F(0)$ gives $A=C$, because $sf(s)=A(1-1/s)^n\to A$. In this half-plane $|1/s|\le 1/\Re s<1$, so expand by the generalized binomial theorem, $(1-1/s)^n=\sum_j\binom{n}{j}(-1/s)^j$, where $(-1)^j\binom{n}{j}=(-n)(-n+1)\cdots(-n+j-1)/j!=(-n)_j/j!$:

$$f(s)=\frac{C}{s}\left(1-\frac{1}{s}\right)^n=C\sum_{j=0}^{\infty}\frac{(-n)_j}{j!s^{j+1}}.$$

Since $L\{t^j\}=j!/s^{j+1}$, inverting term by term identifies the regular solution

$$\boxed{F(t)=C\sum_{j=0}^{\infty}\frac{(-n)_j}{(j!)^2}t^j=C\,{}_1F_1(-n;1;t)}.$$

The termwise inversion is justified by Exercise 20.7.8 with $a=-n$, $c=1$: for $\Re s>1$ it gives $L\{M(-n,1,t)\}=s^{-1}{}_2F_1(-n,1;1;1/s)=s^{-1}(1-1/s)^n$, because ${}_2F_1(-n,1;1;x)=\sum_j(-n)_jx^j/j!=(1-x)^n$; by uniqueness of the inverse transform, $F=C\,M(-n,1,t)=C\,{}_1F_1(-n;1;t)$. If n is a nonnegative integer, the series terminates and $F=CL_n(t)$ (the coefficients $(-n)_j/(j!)^2=(-1)^j\binom{n}{j}/j!$ are those of the series (18.53) of L_n, in the ascending form derived in Exercise 18.3.1), with $f=C(s-1)^n/s^{n+1}$ already valid for $\Re s>0$. Only now set $n=0$: $f=C/s$, so

$$\boxed{F(t)=C, \qquad L_0(t)=1 \text{ when } C=1}.$$

This is the regular solution selected by the finite initial values used in transforming derivatives. The second local solution is logarithmically singular at $t=0$ and is not included by that derivation. At $n=0$, direct integration gives $F'=Be^t/t$; its integral is singular unless $B=0$.

20.8.16 Show that the Laplace transform of the Laguerre polynomial $L_n(at)$ is given by

$$L\{L_n(at)\} = \frac{(s-a)^n}{s^{n+1}}, \qquad s > 0.$$

Solution.

For integer $n \geq 0$, use the finite polynomial expansion (Eq. (18.53), in the ascending form derived in Exercise 18.3.1, where $n!/[(n-j)!(j!)^2] = \binom{n}{j}/j!$)

$$L_n(x) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{x^j}{j!}.$$

There is no infinite-series interchange: transform its $n+1$ terms directly in $\Re s > 0$, with $L\{(at)^j\} = a^j j!/s^{j+1}$, and sum by the binomial theorem:

$$\begin{aligned} L\{L_n(at)\}(s) &= \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{a^j}{j!} \frac{j!}{s^{j+1}} \\ &= \frac{1}{s} \sum_{j=0}^{n} \binom{n}{j} \left(-\frac{a}{s}\right)^j = \frac{1}{s}\left(1 - \frac{a}{s}\right)^n \\ &= \boxed{\frac{(s-a)^n}{s^{n+1}}}. \end{aligned}$$

This also includes $n = 0$ and $a = 0$. Because the integrand is an exponential times a polynomial, no condition $s > a$ is needed here; any real $s > 0$ suffices.

20.8.17 Show that

$$L\{E_1(t)\} = \frac{1}{s}\ln(s+1), \qquad s > 0,$$

where

$$E_1(t) = \int_t^\infty \frac{e^{-\tau}}{\tau}\, d\tau = \int_1^\infty \frac{e^{-xt}}{x}\, dx.$$

$E_1(t)$ is the exponential integral function, first encountered in this book in Table 1.2.

Solution.

For real $s > 0$, the given representation has a nonnegative integrand, so Tonelli's theorem permits the interchange:

$$\begin{aligned} L\{E_1\}(s) &= \int_0^\infty e^{-st} \int_1^\infty \frac{e^{-xt}}{x}\, dx\, dt \\ &= \int_1^\infty \frac{dx}{x} \int_0^\infty e^{-(s+x)t} dt = \int_1^\infty \frac{dx}{x(s+x)}. \end{aligned}$$

Use $1/[x(s+x)] = s^{-1}[1/x - 1/(x+s)]$ and a common upper cutoff R:

$$\int_1^R \frac{dx}{x(s+x)} = \frac{1}{s}\big[\ln R - \ln(R+s) + \ln(1+s)\big].$$

The first two logarithms approach 0 together, since $\ln R - \ln(R+s) = -\ln(1+s/R) \to 0$ as $R \to \infty$. Therefore

$$\boxed{L\{E_1(t)\}(s) = \frac{\ln(1+s)}{s}}.$$

The apparent singularity at $s = 0$ is removable: since $\ln(1+s) = s - s^2/2 + \cdots$, the limit is 1. This is consistent with

$$\int_0^\infty E_1(t)dt = \int_1^\infty \frac{dx}{x}\int_0^\infty e^{-xt}dt = \int_1^\infty \frac{dx}{x^2} = 1,$$

obtained by reversing the order in its defining integral.

20.8.18

(a) From Eq. (20.184) show that

$$\int_0^\infty f(x)\,dx = \int_0^\infty \frac{F(t)}{t}\,dt,$$

provided the integrals exist.

(b) From the preceding result show that

$$\int_0^\infty \frac{\sin t}{t}\,dt = \frac{\pi}{2},$$

in agreement with Eqs. (20.146) and (11.107).

In (a), justify the interchange, for example by $\int_0^\infty |F(t)|\,dt/t < \infty$; in (b), use an Abel limiting argument for the conditionally convergent integral.

Solution.

(a) Let $f(x) = \int_0^\infty e^{-xt}F(t)dt$. An interchange of improper integrals requires a convergence justification. A simple sufficient condition for the direct calculation is $\int_0^\infty |F(t)|\,dt/t < \infty$; conditionally convergent cases need a limiting argument, as in part (b). Under this absolute condition the double integral of $e^{-xt}|F(t)|$ over $x, t > 0$ equals $\int_0^\infty |F(t)|\,dt/t < \infty$, so Fubini's theorem gives

$$\begin{aligned}\int_0^\infty f(x)dx &= \int_0^\infty F(t)\left(\int_0^\infty e^{-xt}dx\right)dt \\ &= \boxed{\int_0^\infty \frac{F(t)}{t}\,dt},\end{aligned}$$

because the inner integral is $1/t$ for $t > 0$. Equivalently, Eq. (20.184), $L\{F(t)/t\}(s) = \int_s^\infty f(x)dx$ (the same interchange with $\int_s^\infty e^{-xt}dx = e^{-st}/t$), yields the same relation as $s \downarrow 0$. Under the absolute condition this limit is justified on both sides: the left side tends to $\int_0^\infty F(t)\,dt/t$ by dominated convergence (dominating function $|F(t)|/t$), and the right side to $\int_0^\infty f(x)dx$ because $\int_0^\infty |f(x)|dx \le \int_0^\infty |F(t)|\,dt/t < \infty$.

(b) For $F(t) = \sin t$, $f(x) = 1/(1+x^2)$ for $x > 0$. Insert a positive lower cutoff s in the x integration. The double integral of $e^{-xt}|\sin t|$ over $x > s$, $t > 0$ is $\int_0^\infty e^{-st}|\sin t|\,dt/t \le \int_0^\infty e^{-st}dt = 1/s$, since $|\sin t| \le t$; this absolute integrability permits the interchange of (a), now with $\int_s^\infty e^{-xt}dx = e^{-st}/t$:

$$\int_0^\infty e^{-st}\frac{\sin t}{t}dt = \int_s^\infty \frac{dx}{1+x^2} = \frac{\pi}{2} - \tan^{-1} s.$$

The unregulated sine integral exists: its integrand tends to 1 as $t \to 0$, and integration by parts bounds its tail,

$$\int_R^\infty \frac{\sin t}{t}\,dt = \frac{\cos R}{R} - \int_R^\infty \frac{\cos t}{t^2}dt, \qquad \left|\int_R^\infty \frac{\sin t}{t}\,dt\right| \le \frac{1}{R} + \int_R^\infty \frac{dt}{t^2} = \frac{2}{R}.$$

Abel's theorem for integrals (Exercise 20.2.2(c)) thus lets $s \downarrow 0$ in the regulated integral without changing its improper value, and $\tan^{-1} s \to 0$. Consequently

$$\boxed{\int_0^\infty \frac{\sin t}{t}dt = \frac{\pi}{2}}.$$

This proof uses a regulator precisely because the absolute condition of part (a) fails for $|\sin t|/t$.

20.8.19

(a) Show that

$$L\left\{\frac{\sin kt}{t}\right\} = \cot^{-1}\left(\frac{s}{k}\right).$$

(b) Using this result (with $k = 1$), prove that

$$L\{\mathrm{si}(t)\} = -\frac{1}{s}\tan^{-1} s,$$

where

$$\mathrm{si}(t) = -\int_t^\infty \frac{\sin x}{x}\,dx, \qquad \text{the sine integral.}$$

The real inverse-cotangent expression in (a) assumes $k > 0$, $s > 0$, and values in $(0, \pi/2)$. Use $s > 0$ also in (b).

Solution.

(a) For real $s > 0$, integrate the sine transform $L\{\sin kt\}(u) = k/(u^2 + k^2)$ in its transform parameter, by the interchange of Exercise 20.8.14(a):

$$L\left\{\frac{\sin kt}{t}\right\}(s) = \int_s^\infty \frac{k}{u^2 + k^2}du = \boxed{\tan^{-1}(k/s)}.$$

For $k > 0$ the integral is $[\tan^{-1}(u/k)]_s^\infty = \pi/2 - \tan^{-1}(s/k)$, which equals $\tan^{-1}(k/s)$ by $\tan^{-1} x + \tan^{-1}(1/x) = \pi/2$ for $x > 0$; for $k < 0$ the same steps hold with $-\pi/2$ in place of $\pi/2$, and for $k = 0$ both sides vanish. For $k > 0$ this is $\cot^{-1}(s/k)$ with the usual value in $(0, \pi/2)$. The printed inverse-cotangent form presumes $k > 0$ and this branch. The arctangent form extends continuously to $k = 0$ and all real k.

(b) From the defining tail integral, $\mathrm{si}'(t) = \sin t/t$ (differentiate at the lower limit) and $\mathrm{si}(0+) = -\int_0^\infty (\sin x/x)\,dx = -\pi/2$ by Exercise 20.8.18(b). If $h(s) = L\{\mathrm{si}\}(s)$, the derivative formula $L\{\mathrm{si}'\} = sh(s) - \mathrm{si}(0+)$ (si is continuous and bounded, with a finite limit at 0) and part (a) with $k = 1$ give

$$sh(s) - \mathrm{si}(0+) = sh(s) + \pi/2 = \tan^{-1}(1/s).$$

For $s > 0$, $\tan^{-1}(1/s) = \pi/2 - \tan^{-1} s$, so $sh(s) = -\tan^{-1} s$ and

$$\boxed{L\{\mathrm{si}(t)\}(s) = -\frac{\tan^{-1} s}{s}}.$$

The finite value at the origin and the decaying oscillatory tail of si ensure convergence in $\Re s > 0$; analytic continuation there uses the corresponding logarithmic branch of the arctangent.

20.8.20 If $F(t)$ is periodic (Fig. 20.19) with a period a so that $F(t+a) = F(t)$ for all $t \geq 0$, show that

$$L\{F(t)\} = \frac{1}{1-e^{-as}} \int_0^a e^{-st} F(t)\, dt.$$

Note that the integration is now over only the first period of $F(t)$.

Assume $a > 0$, that F is integrable over a period, and $\Re s > 0$.

Solution.

Assume $a > 0$ and F integrable over one period. For $\Re s > 0$ split the half-line into successive periods, putting $t = na + u$ $(0 \leq u \leq a)$ in the nth interval, so that $e^{-st} = e^{-nas}e^{-su}$, and $F(na+u) = F(u)$ by n applications of $F(t+a) = F(t)$:

$$\begin{aligned} L\{F\}(s) &= \sum_{n=0}^{\infty} \int_{na}^{(n+1)a} e^{-st} F(t) dt \\ &= \sum_{n=0}^{\infty} e^{-nas} \int_0^a e^{-su} F(na+u) du \\ &= \left(\sum_{n=0}^{\infty} e^{-nas} \right) \int_0^a e^{-su} F(u) du. \end{aligned}$$

The geometric series converges because $|e^{-as}| = e^{-a\Re s} < 1$, and its sum is $\sum_{n\geq 0}(e^{-as})^n = 1/(1-e^{-as})$. It also bounds the absolute integrals, $\sum_n \int_{na}^{(n+1)a} |e^{-st}F(t)|dt \leq \int_0^a |F(u)|du/(1-e^{-a\Re s}) < \infty$, justifying the splitting and summation. Hence

$$\boxed{L\{F(t)\}(s) = \frac{\displaystyle\int_0^a e^{-st} F(t) dt}{1-e^{-as}}}, \qquad \Re s > 0.$$

Values assigned at isolated period boundaries do not affect the transform.

20.8.21 Find the Laplace transform of the square wave (period a) defined by

$$F(t) = \begin{cases} 1, & 0 < t < a/2, \\ 0, & a/2 < t < a. \end{cases}$$

ANS. $f(s) = \dfrac{1}{s} \dfrac{1-e^{-as/2}}{1-e^{-as}}$.

Take $a > 0$ and $\Re s > 0$.

Solution.

Apply Exercise 20.8.20. Since $F = 0$ on $(a/2, a)$, only the first half of a period contributes to the numerator:

$$\int_0^a e^{-st} F(t) dt = \int_0^{a/2} e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_0^{a/2} = \frac{1-e^{-as/2}}{s}.$$

Dividing by $1-e^{-as}$ gives

$$\boxed{f(s) = \frac{1-e^{-as/2}}{s(1-e^{-as})} = \frac{1}{s(1+e^{-as/2})}}, \qquad \Re s > 0.$$

The simplification follows from $1 - e^{-as} = (1 - e^{-as/2})(1 + e^{-as/2})$. For real $s \to \infty$, $sf(s) = 1/(1 + e^{-as/2}) \to 1$, which matches $F(0+)$ (the initial-value theorem, Exercise 20.7.1). For $s \downarrow 0$, $sf(s) \to 1/(1 + 1) = 1/2$, the average value over a period; this is not a pointwise final value of the oscillating square wave.

20.8.22 Show that

(a) $L\{\cosh at \cos at\} = \dfrac{s^3}{s^4 + 4a^4}$,

(b) $L\{\cosh at \sin at\} = \dfrac{as^2 + 2a^3}{s^4 + 4a^4}$,

(c) $L\{\sinh at \cos at\} = \dfrac{as^2 - 2a^3}{s^4 + 4a^4}$,

(d) $L\{\sinh at \sin at\} = \dfrac{2a^2 s}{s^4 + 4a^4}$.

Take $a \in \mathbb{R}$ and $\Re s > |a|$.

Solution.

Take real a and $\Re s > |a|$. Define

$$D_- = (s - a)^2 + a^2 = s^2 - 2as + 2a^2, \qquad D_+ = (s + a)^2 + a^2 = s^2 + 2as + 2a^2.$$

The substitution (frequency-shifting) theorem $L\{e^{bt}F(t)\} = f(s - b)$, applied with $b = \pm a$ to $L\{\cos at\} = s/(s^2 + a^2)$ and $L\{\sin at\} = a/(s^2 + a^2)$, gives, since $(s \mp a)^2 + a^2 = D_\mp$,

$$L\{e^{\pm at} \cos at\} = \frac{s \mp a}{D_\mp}, \qquad L\{e^{\pm at} \sin at\} = \frac{a}{D_\mp}.$$

Also $D_\mp = (s^2 + 2a^2) \mp 2as$, so

$$D_- D_+ = (s^2 + 2a^2)^2 - (2as)^2 = s^4 + 4a^4, \qquad D_+ + D_- = 2s^2 + 4a^2, \qquad D_+ - D_- = 4as.$$

Using $\cosh at = (e^{at} + e^{-at})/2$ and $\sinh at = (e^{at} - e^{-at})/2$, each transform is half the sum or difference of two shifted transforms, brought to the common denominator $D_- D_+$:

$$\text{(a)} \quad L\{\cosh at \cos at\} = \frac{(s - a)D_+ + (s + a)D_-}{2D_- D_+} = \boxed{\frac{s^3}{s^4 + 4a^4}},$$

$$\text{(b)} \quad L\{\cosh at \sin at\} = \frac{a(D_+ + D_-)}{2D_- D_+} = \boxed{\frac{as^2 + 2a^3}{s^4 + 4a^4}},$$

$$\text{(c)} \quad L\{\sinh at \cos at\} = \frac{(s - a)D_+ - (s + a)D_-}{2D_- D_+} = \boxed{\frac{as^2 - 2a^3}{s^4 + 4a^4}},$$

$$\text{(d)} \quad L\{\sinh at \sin at\} = \frac{a(D_+ - D_-)}{2D_- D_+} = \boxed{\frac{2a^2 s}{s^4 + 4a^4}}.$$

In (b) and (d) the numerators are $a(2s^2 + 4a^2)$ and $a \cdot 4as$, which halve to the boxed ones. For the two less immediate numerators, direct expansion gives $(s - a)D_+ = s^3 + as^2 - 2a^3$ and $(s + a)D_- = s^3 - as^2 + 2a^3$; adding gives $2s^3$ and subtracting gives $2as^2 - 4a^3$, which proves the simplifications in (a) and (c).

20.8.23 Show that

(a) $L^{-1}\left\{(s^2+a^2)^{-2}\right\} = \dfrac{1}{2a^3}\sin at - \dfrac{t}{2a^2}\cos at,$

(b) $L^{-1}\left\{s(s^2+a^2)^{-2}\right\} = \dfrac{t}{2a}\sin at,$

(c) $L^{-1}\left\{s^2(s^2+a^2)^{-2}\right\} = \dfrac{1}{2a}\sin at + \dfrac{t}{2}\cos at,$

(d) $L^{-1}\left\{s^3(s^2+a^2)^{-2}\right\} = \cos at - \dfrac{at}{2}\sin at.$

Take real $a \neq 0$ in the displayed quotients and use their continuous limits at $a = 0$.

Solution.

Begin with $L\{\sin at/a\} = 1/(s^2+a^2)$ and $L\{\cos at\} = s/(s^2+a^2)$ for real $a \neq 0$, $\Re s > 0$. (a) Differentiate the first pair with respect to a and multiply by $-1/(2a)$. Differentiation under the transform integral is allowed because the a-derivative of the original, $\partial_a(\sin at/a) = t\cos at/a - \sin at/a^2$, is bounded by $t/|a| + 1/a^2$; and $\partial_a(s^2+a^2)^{-1} = -2a(s^2+a^2)^{-2}$. Hence

$$-\frac{1}{2a}\frac{\partial}{\partial a}\frac{1}{s^2+a^2} = \frac{1}{(s^2+a^2)^2}, \qquad -\frac{1}{2a}\frac{\partial}{\partial a}\frac{\sin at}{a} = \boxed{\frac{\sin at}{2a^3} - \frac{t\cos at}{2a^2}}.$$

(b) Since $L\{tF\} = -dL\{F\}/ds$, take $F = \sin at/a$, whose transform is $1/(s^2+a^2)$; then $\frac{1}{2}tF$ has the transform

$$-\frac{1}{2}\frac{d}{ds}\frac{1}{s^2+a^2} = \frac{s}{(s^2+a^2)^2}, \qquad \boxed{L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\} = \frac{t}{2a}\sin at}.$$

(c) Split the numerator as $s^2 = (s^2+a^2) - a^2$ and use (a):

$$\frac{s^2}{(s^2+a^2)^2} = \frac{1}{s^2+a^2} - \frac{a^2}{(s^2+a^2)^2}.$$

Its inverse is $\sin at/a - a^2[\sin at/(2a^3) - t\cos at/(2a^2)] = \sin at/a - \sin at/(2a) + (t/2)\cos at$, or

$$\boxed{\frac{\sin at}{2a} + \frac{t}{2}\cos at}.$$

(d) Similarly, from $s^3 = s(s^2+a^2) - a^2 s$, $s^3/(s^2+a^2)^2 = s/(s^2+a^2) - a^2 s/(s^2+a^2)^2$. The pair $L\{\cos at\} = s/(s^2+a^2)$ and part (b) then give $\cos at - a^2\, t\sin at/(2a)$, that is,

$$\boxed{\cos at - \frac{at}{2}\sin at}.$$

At $a = 0$, interpret all quotient formulas by their continuous limits. The four answers reduce to $t^3/6$, $t^2/2$, t, and 1, the inverses of s^{-4} through s^{-1}. For (a), $\sin at - at\cos at = a^3t^3/3 + O(a^5)$ by the Maclaurin series, so $(\sin at - at\cos at)/(2a^3) \to t^3/6$; the other three limits follow from $\sin at/a \to t$ and $\cos at \to 1$.

20.8.24 Show that

$$L\{(t^2-k^2)^{-1/2}u(t-k)\} = K_0(ks).$$

(Here $u(t-k)$ is the unit step function.) This clarification is from the published errata, which list it under the misprinted number 20.8.34.

Hint. Try transforming an integral representation of $K_0(ks)$ into the Laplace transform integral. Take $k>0$ and $\Re s>0$; the step restricts the real square root to $t>k$.

Solution.

Take $k>0$ and $\Re s>0$. The lower endpoint $t=k$ then has an integrable square-root singularity. At $k=0$ the kernel is $1/t$ and the ordinary transform diverges. The unit step $u(t-k)$ (0 for $t<k$, 1 for $t>k$) restricts the transform to

$$L\{(t^2-k^2)^{-1/2}u(t-k)\} = \int_k^\infty \frac{e^{-st}}{\sqrt{t^2-k^2}}dt.$$

Set $t=k\cosh v$, $v\geq 0$; the limits $t=k$ and $t\to\infty$ become $v=0$ and $v\to\infty$. Since $dt = k\sinh v\,dv$ and, by $\cosh^2 v - 1 = \sinh^2 v$ with $\sinh v \geq 0$, $\sqrt{t^2-k^2} = k\sinh v$, the Jacobian cancels the denominator:

$$\int_k^\infty \frac{e^{-st}}{\sqrt{t^2-k^2}}dt = \int_0^\infty e^{-ks\cosh v}dv = \boxed{K_0(ks)}.$$

The last expression is the standard hyperbolic integral for K_0 on $\Re(ks)>0$ (Exercise 14.5.15, or Exercise 14.6.3 with $\nu=0$). The value assigned to $u(0)$ does not change this integrable transform.

20.9 Laplace Convolution Theorem

20.9.1 From the convolution theorem show that

$$\frac{1}{s}f(s) = L\left\{\int_0^t F(x)\,dx\right\},$$

where $f(s)=L\{F(t)\}$.

Solution.

In the one-sided Laplace convolution, choose $G(t)=1$ for $t\geq 0$. Its transform is $g(s) = \int_0^\infty e^{-st}dt = 1/s$ for $\Re s>0$. Since $G(t-x)=1$ for $0\leq x\leq t$, the definition gives

$$(F*G)(t) = \int_0^t F(x)G(t-x)dx = \int_0^t F(x)dx.$$

The convolution theorem, $L\{F*G\} = f(s)g(s)$, therefore yields

$$\boxed{L\left\{\int_0^t F(x)dx\right\}(s) = f(s)g(s) = \frac{f(s)}{s}}.$$

For locally integrable F of exponential order this holds in the common half-plane where $\Re s>0$ and the integral defining $f(s)$ converges. As a check, the primitive $P(t)=\int_0^t F(x)dx$ has $P'=F$ and $P(0)=0$, so the derivative formula $L\{P'\} = sL\{P\} - P(0)$ gives $f(s) = sL\{P\}$, which is the same result.

20.9.2 If $F(t) = t^a$ and $G(t) = t^b$, $a > -1$, $b > -1$,

(a) Show that the convolution $F * G$ is given by

$$F * G = t^{a+b+1} \int_0^1 y^a (1-y)^b \, dy.$$

(b) By using the convolution theorem, show that

$$\int_0^1 y^a (1-y)^b \, dy = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)} = B(a+1, b+1),$$

where B is the beta function.

Solution.

(a) For $t > 0$, substitute $x = ty$ into the convolution. Then $dx = t\,dy$, $x^a = t^a y^a$, $(t-x)^b = t^b(1-y)^b$, and the limits $x = 0, t$ become $y = 0, 1$:

$$(F * G)(t) = \int_0^t x^a (t-x)^b dx = t^{a+b+1} \int_0^1 y^a (1-y)^b dy.$$

The endpoint conditions $a, b > -1$ make this integral finite: near $y = 0$ the integrand behaves as y^a, and near $y = 1$ as $(1-y)^b$.

(b) Call the integral of (a) C. For $\nu > -1$ and $s > 0$, the substitution $u = st$ in the gamma integral gives

$$L\{t^\nu\} = \int_0^\infty e^{-st} t^\nu dt = \frac{1}{s^{\nu+1}} \int_0^\infty e^{-u} u^\nu du = \frac{\Gamma(\nu+1)}{s^{\nu+1}}.$$

With $\nu = a + b + 1 > -1$, $\nu = a$ and $\nu = b$ in turn,

$$L\{F * G\} = C\frac{\Gamma(a+b+2)}{s^{a+b+2}}, \qquad f(s) = \frac{\Gamma(a+1)}{s^{a+1}}, \qquad g(s) = \frac{\Gamma(b+1)}{s^{b+1}}.$$

The convolution theorem $L\{F*G\} = fg$ now reads $C\,\Gamma(a+b+2)s^{-(a+b+2)} = \Gamma(a+1)\Gamma(b+1)s^{-(a+b+2)}$. Cancelling $s^{-(a+b+2)}$ and dividing by $\Gamma(a+b+2)$ yields

$$\boxed{\int_0^1 y^a (1-y)^b dy = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)} = B(a+1, b+1).}$$

The textbook writes this ratio as $a!\,b!/(a+b+1)!$; since $a, b > -1$ need not be integers, the factorials mean gamma functions, $a! = \Gamma(a+1)$, which is the form displayed above.

20.9.3 Using the convolution integral, calculate

$$L^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\}, \qquad a^2 \neq b^2.$$

Solution.

Factor the transform into $[1/(s^2+a^2)]\,[s/(s^2+b^2)]$. For real b and real $a \neq 0$, the factors are the transforms of $\sin(at)/a$ and $\cos bt$, so the convolution theorem gives

$$H(t) = \frac{1}{a} \int_0^t \sin(ax) \cos[b(t-x)] dx.$$

Use $\sin u\cos v = [\sin(u+v)+\sin(u-v)]/2$ with $u = ax$, $v = b(t-x)$, so that $u+v = (a-b)x+bt$ and $u-v = (a+b)x-bt$. At $x = t$ both arguments equal at; at $x = 0$ they are $\pm bt$, and $\cos(-bt) = \cos bt$. Hence

$$\begin{aligned} H(t) &= \frac{1}{2a}\int_0^t \{\sin[(a-b)x+bt] + \sin[(a+b)x-bt]\}dx \\ &= \frac{1}{2a}\left[-\frac{\cos[(a-b)x+bt]}{a-b} - \frac{\cos[(a+b)x-bt]}{a+b}\right]_{x=0}^{x=t} \\ &= \frac{1}{2a}\left[\frac{\cos bt-\cos at}{a-b} + \frac{\cos bt-\cos at}{a+b}\right] \\ &= \frac{\cos bt-\cos at}{2a}\,\frac{(a+b)+(a-b)}{a^2-b^2} \\ &= \boxed{\frac{\cos at-\cos bt}{b^2-a^2}}. \end{aligned}$$

The assumption $a^2 \neq b^2$ makes $a\pm b$ nonzero. If $a = 0$ (allowed when $b \neq 0$), the first factor is $1/s^2$, the transform of x rather than $\sin(ax)/a$. Then $H(t) = \int_0^t x\cos[b(t-x)]dx$. The substitution $w = t-x$ ($dx = -dw$; the limits $x = 0, t$ become $w = t, 0$) gives $H(t) = \int_0^t (t-w)\cos bw\,dw$. Integrate by parts with $\sin(bw)/b$ as the antiderivative of $\cos bw$: the boundary term $[(t-w)\sin(bw)/b]_0^t$ vanishes at both ends, and since $d(t-w) = -dw$ what remains is $H(t) = \frac{1}{b}\int_0^t \sin bw\,dw = (1-\cos bt)/b^2$, which is the boxed formula at $a = 0$. This agrees with the independently transformed result of Exercise 20.7.3.

20.9.4 An undamped oscillator is driven by a force $F_0\sin\omega t$. Find the displacement $X(t)$ as a function of time, subject to initial conditions $X(0) = X'(0) = 0$. For $\omega^2 \neq \omega_0^2$, the solution is a linear combination of two simple harmonic motions, one with the frequency of the driving force and one with the frequency $\omega_0 > 0$ of the free oscillator. Find the resonant limit separately.

$$\textit{ANS. } X(t) = \frac{F_0/m}{\omega^2-\omega_0^2}\left(\frac{\omega}{\omega_0}\sin\omega_0 t - \sin\omega t\right).$$

Solution.

Let $\omega_0^2 = k/m > 0$. The equation $mX'' + kX = F_0\sin\omega t$ and the initial data, divided by m, are

$$X'' + \omega_0^2 X = \frac{F_0}{m}\sin\omega t, \qquad X(0) = X'(0) = 0.$$

With $L\{X''\} = s^2\widehat{X} - sX(0) - X'(0) = s^2\widehat{X}$ and $L\{\sin\omega t\} = \omega/(s^2+\omega^2)$, transformation gives $(s^2+\omega_0^2)\widehat{X} = (F_0/m)\,\omega/(s^2+\omega^2)$, so

$$\widehat{X}(s) = \frac{F_0}{m}\,\frac{\omega}{(s^2+\omega^2)(s^2+\omega_0^2)}.$$

Equivalently, since $1/(s^2+\omega_0^2)$ and $\omega/(s^2+\omega^2)$ are the transforms of $\sin(\omega_0 t)/\omega_0$ and $\sin\omega t$, the convolution theorem gives the response

$$X(t) = \frac{F_0}{m\omega_0}\int_0^t \sin[\omega_0(t-u)]\sin(\omega u)du.$$

Partial fractions evaluate it more quickly. For $\omega^2 \neq \omega_0^2$, $1/(s^2+\omega^2) - 1/(s^2+\omega_0^2) = (\omega_0^2-\omega^2)/[(s^2+\omega^2)(s^2+\omega_0^2)]$, so

$$\widehat{X} = \frac{F_0\omega}{m(\omega_0^2-\omega^2)}\left(\frac{1}{s^2+\omega^2} - \frac{1}{s^2+\omega_0^2}\right).$$

Inverting the two terms by $L^{-1}\{1/(s^2+c^2)\} = \sin(ct)/c$ yields

$$X(t) = \frac{F_0\omega}{m(\omega_0^2-\omega^2)}\left(\frac{\sin\omega t}{\omega} - \frac{\sin\omega_0 t}{\omega_0}\right),$$

that is, after moving ω into the bracket and changing the sign of both factors,

$$\boxed{X(t) = \frac{F_0/m}{\omega^2-\omega_0^2}\left(\frac{\omega}{\omega_0}\sin\omega_0 t - \sin\omega t\right).}$$

Its value vanishes at $t=0$, and so does its derivative, which is proportional to $(\omega/\omega_0)\omega_0 - \omega = 0$ there: the two sine slopes cancel. The printed answer requires a separate resonant limit. At $\omega = \omega_0$ it is a removable 0/0 form and cannot be used as written. Apply l'Hôpital's rule to the ratio as $\omega \to \omega_0$; for every t the numerator $(\omega/\omega_0)\sin\omega_0 t - \sin\omega t$ and the denominator $\omega^2 - \omega_0^2$ both vanish at $\omega = \omega_0$. Their ω-derivatives are $\sin(\omega_0 t)/\omega_0 - t\cos(\omega t)$ and 2ω, whose ratio at $\omega = \omega_0$ is $(\sin\omega_0 t - \omega_0 t\cos\omega_0 t)/(2\omega_0^2)$. Thus

$$\boxed{X_{\text{res}}(t) = \frac{F_0}{2m\omega_0^2}(\sin\omega_0 t - \omega_0 t\cos\omega_0 t).}$$

This limit is indeed the solution at resonance. For $\omega = \omega_0$ the transform is $\widehat{X} = (F_0/m)\,\omega_0/(s^2+\omega_0^2)^2$, while the rule $L\{tG(t)\} = -g'(s)$ for $g = L\{G\}$ gives $L\{t\cos\omega_0 t\} = -\frac{d}{ds}\frac{s}{s^2+\omega_0^2} = \frac{s^2-\omega_0^2}{(s^2+\omega_0^2)^2}$ and hence

$$L\{\sin\omega_0 t - \omega_0 t\cos\omega_0 t\} = \frac{\omega_0(s^2+\omega_0^2) - \omega_0(s^2-\omega_0^2)}{(s^2+\omega_0^2)^2} = \frac{2\omega_0^3}{(s^2+\omega_0^2)^2},$$

so $X_{\text{res}} = (F_0/m)\,\omega_0 \cdot (\sin\omega_0 t - \omega_0 t\cos\omega_0 t)/(2\omega_0^3)$, as above. The growing $t\cos\omega_0 t$ term is the resonant response; it is no longer a sum of two bounded oscillations at distinct frequencies. For $\omega = 0$ the driving force is identically zero and the solution is $X = 0$.

20.10 Inverse Laplace Transform

20.10.1 Derive the Bromwich integral from Cauchy's integral formula.

Hint. Apply the inverse transform L^{-1} to

$$f(s) = \frac{1}{2\pi i}\lim_{\alpha\to\infty}\int_{\beta-i\alpha}^{\beta+i\alpha}\frac{f(z)}{s-z}\,dz,$$

where $f(z)$ is analytic for $\Re z \geq \beta$.

Assume decay sufficient to make the right closing arc vanish and the usual Laplace inversion hypotheses; analyticity alone does not imply the identity in the hint. Take $\Re s > \beta$.

Solution.

Choose β to the right of all singularities of the Laplace transform, and fix $\Re s > \beta$. Close the upward vertical segment $\beta - iR$ to $\beta + iR$ by a semicircle to its right. This closed contour is clockwise, and it encloses s once $R > |s-\beta|$. The kernel $f(z)/(s-z) = -f(z)/(z-s)$ has residue $-f(s)$ at $z = s$, so Cauchy's formula with this orientation gives

$$\int_{\beta-iR}^{\beta+iR}\frac{f(z)}{s-z}dz + \int_{\text{right arc}}\frac{f(z)}{s-z}dz = -2\pi i[-f(s)] = 2\pi i f(s).$$

If, for example, $|f(z)| \le M|z|^{-\epsilon}$ on these arcs for some $\epsilon > 0$, the arc integral is $O(R^{-\epsilon})$: the arc $z = \beta + Re^{i\theta}$, $|\theta| \le \pi/2$, has length πR, and on it $|s-z| \ge R - |s-\beta| \ge R/2$ once $R \ge 2|s-\beta|$. Since also $|z| \ge R - |\beta|$ on the arc, the arc integral is at most $\pi R \cdot M(R-|\beta|)^{-\epsilon} \cdot (2/R) = 2\pi M(R-|\beta|)^{-\epsilon}$. Letting $R \to \infty$ then gives the identity in the hint. For each z on the line $\Re z = \beta$,

$$\frac{1}{s-z} = \int_0^\infty e^{-st} e^{zt}\, dt, \qquad L_s^{-1}\{(s-z)^{-1}\}(t) = e^{zt},$$

since $|e^{-(s-z)t}| = e^{-(\Re s - \beta)t}$ decays. Insert this integral into the Cauchy representation and interchange the z and t integrations (formally; see the remarks below):

$$\begin{aligned} f(s) &= \frac{1}{2\pi i} \lim_{R\to\infty} \int_{\beta - iR}^{\beta + iR} f(z) \left[\int_0^\infty e^{-st} e^{zt}\, dt \right] dz \\ &= \int_0^\infty e^{-st} \left[\frac{1}{2\pi i} \lim_{R\to\infty} \int_{\beta-iR}^{\beta+iR} e^{zt} f(z)\, dz \right] dt. \end{aligned}$$

Thus f is the Laplace transform of the last bracket, and applying L^{-1} (uniqueness of the inverse transform) gives

$$\boxed{F(t) = \frac{1}{2\pi i} \lim_{R\to\infty} \int_{\beta-iR}^{\beta+iR} e^{zt} f(z)\, dz}, \qquad t > 0,$$

which is the Bromwich inversion formula at a continuity point of F. Analyticity alone is not enough to discard the arc or interchange a limiting inverse transform: $f(z) \equiv 1$ is analytic, yet for real $s > \beta$ (put $z = \beta + i\omega$, $c = s - \beta$, so $1/(s-z) = (c + i\omega)/(c^2 + \omega^2)$) the right side of the hint is $(2\pi)^{-1} \lim_{\alpha\to\infty} \int_{-\alpha}^{\alpha} (c + i\omega)\, d\omega/(c^2 + \omega^2) = (2\pi)^{-1} \int_{-\infty}^{\infty} c\, d\omega/(c^2 + \omega^2) = 1/2$, not 1 (the odd imaginary part cancels between $\pm\omega$). Decay on the closing contours and inversion hypotheses are required. A standard sufficient setting is a causal, piecewise smooth function of exponential order, with the integral interpreted by symmetric truncation (or Abel regularization). At a jump it recovers the mean of the two one-sided limits. Writing $z = \beta + i\omega$ turns this into the Fourier inversion of $e^{-\beta t} F(t)$ extended by zero to $t < 0$; Exercise 20.10.2 supplies a regularized kernel proof of the limiting step.

20.10.2 Starting with

$$\frac{1}{2\pi i} \int_{\beta - i\infty}^{\beta + i\infty} e^{st} f(s)\, ds,$$

show that by introducing

$$f(s) = \int_0^\infty e^{-sz} F(z)\, dz$$

we can convert our integral into the Fourier representation of a Dirac delta function. From this derive the inverse Laplace transform.

Use a Bromwich line to the right of the abscissa of absolute convergence, and recover $F(t)$ at positive continuity points; interpret the delta integral distributionally.

Solution.

Parameterize the line by $s = \beta + i\omega$, so $ds/(2\pi i) = d\omega/(2\pi)$ and $e^{st} e^{-sz} = e^{\beta(t-z)} e^{i\omega(t-z)}$. On inserting the Laplace integral and interchanging the integrations, the proposed inverse becomes

formally

$$\frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{st}f(s)ds = \int_0^\infty F(z)e^{\beta(t-z)}\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\omega(t-z)}d\omega\right]dz$$
$$= \int_0^\infty F(z)e^{\beta(t-z)}\delta(t-z)dz = F(t), \qquad t>0,$$

by the Fourier representation $\delta(u) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\omega u}d\omega$ and the sifting property; the factor $e^{\beta(t-z)}$ equals 1 at $z=t$.

Here is a precise justification of the delta. Multiply the frequency integrand by $e^{-\epsilon\omega^2}$, with $\epsilon>0$. Completing the square, $-\epsilon\omega^2+i\omega u = -\epsilon[\omega - iu/(2\epsilon)]^2 - u^2/(4\epsilon)$, and $\int_{-\infty}^{\infty} e^{-\epsilon v^2}dv = \sqrt{\pi/\epsilon}$ (the shift of the line of integration is allowed by Cauchy's theorem) give the Fourier integral

$$K_\epsilon(u) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\epsilon\omega^2+i\omega u}d\omega = \frac{e^{-u^2/(4\epsilon)}}{\sqrt{4\pi\epsilon}}.$$

It is positive, even and decreasing in $|u|$, and has mass 1 (substitute $u=2\sqrt{\epsilon}\,v$); for fixed $\rho>0$ both $K_\epsilon(\rho)$ and $\int_{|u|\ge\rho}K_\epsilon(u)du$ tend to zero as $\epsilon\to0$. Let $H(z)=e^{-\beta z}F(z)$ for $z\ge0$ and $H(z)=0$ for $z<0$, with $H\in L^1$ (β beyond the abscissa of absolute convergence). On the line, since $e^{s(t-z)} = e^{\beta t}e^{i\omega(t-z)}e^{-\beta z}$,

$$e^{st}f(s) = \int_0^\infty e^{s(t-z)}F(z)dz = e^{\beta t}\int_{-\infty}^{\infty} e^{i\omega(t-z)}H(z)dz.$$

Since $e^{-\epsilon\omega^2}|H(z)|$ is integrable over the (ω,z) plane, Fubini's theorem justifies interchanging the ω and z integrations in the regularized inverse; the inner ω integral is then $K_\epsilon(t-z)$:

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\epsilon\omega^2}e^{st}f(s)\,d\omega = e^{\beta t}\int_{-\infty}^{\infty} K_\epsilon(t-z)H(z)dz.$$

Let $t>0$ be a continuity point of F, hence of H. Given $\eta>0$, choose $\rho>0$ with $|H(z)-H(t)|<\eta$ for $|z-t|<\rho$. Because K_ϵ has mass 1, the difference below equals $\int_{-\infty}^{\infty}K_\epsilon(t-z)[H(z)-H(t)]dz$. On $|z-t|<\rho$ the bracket is less than η in absolute value and K_ϵ integrates to at most 1; on $|z-t|\ge\rho$ use $|H(z)-H(t)|\le|H(z)|+|H(t)|$ and, in the $|H(z)|$ term, $K_\epsilon(t-z)\le K_\epsilon(\rho)$. Hence

$$\left|\int_{-\infty}^{\infty}K_\epsilon(t-z)H(z)dz - H(t)\right| \le \eta + K_\epsilon(\rho)\int_{-\infty}^{\infty}|H|\,dz + |H(t)|\int_{|u|\ge\rho}K_\epsilon(u)du.$$

The last two terms tend to zero as $\epsilon\to0$, and η is arbitrary. Hence the regularized inverse tends to $e^{\beta t}H(t)=F(t)$ at every positive continuity point. This proves

$$\boxed{F(t) = \frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{st}f(s)ds}, \qquad t>0,$$

with the integral understood as the limit $\epsilon\to0$ of its regularized form. The unsmoothed frequency integral is a distribution, not an absolutely convergent integral. At an interior jump symmetric inversion gives $[F(t-)+F(t+)]/2$; at $t=0$ it gives $F(0+)/2$ for the causal extension. For $t<0$ it gives zero.

20.10.3 Derive the Laplace transformation convolution theorem by use of the Bromwich integral.

Solution.

Let $f = L\{F\}$ and $g = L\{G\}$, with both functions extended by zero for negative arguments. In a common half-plane of convergence choose a Bromwich line $\Re s = \beta$. Substitute the defining integral for g into the inverse of the product, combine $e^{st}e^{-su} = e^{s(t-u)}$, and interchange the integrations:

$$\begin{aligned} L^{-1}\{fg\}(t) &= \frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{st} f(s)\left[\int_0^\infty e^{-su}G(u)du\right] ds \\ &= \int_0^\infty G(u)\left[\frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{s(t-u)}f(s)ds\right] du. \end{aligned}$$

The bracket is the Bromwich integral of Exercise 20.10.2 evaluated at the time $t-u$. It therefore equals $F(t-u)$ for $u < t$ and vanishes for $u > t$, where $t-u < 0$: at a time $\tau = t-u < 0$ the function H of Exercise 20.10.2 (built from F) vanishes on a whole neighborhood of τ, so the argument given there yields the limit $e^{\beta\tau}H(\tau) = 0$. For piecewise continuous F the bracket can differ from $F(t-u)$ only at isolated values of u (at $u = t$ it equals $F(0+)/2$, and at a jump of F the mean of the one-sided limits, by Exercise 20.10.2), and such values do not change the u integral. Hence for $t \geq 0$ the u integral stops at $u = t$, and the result is

$$\boxed{L^{-1}\{f(s)g(s)\}(t) = \int_0^t F(t-u)G(u)du}.$$

Changing variables $x = t-u$ ($du = -dx$; the limits $u = 0, t$ become $x = t, 0$) also gives $\int_0^t F(x)G(t-x)dx$, the Laplace convolution. For a direct rigorous interchange, it suffices that the vertical integral of $|f(\beta+i\omega)|$ and $\int_0^\infty e^{-\beta u}|G(u)|du$ are finite: on the line $|e^{s(t-u)}f(s)G(u)| = e^{\beta t}|f(\beta+i\omega)|\,e^{-\beta u}|G(u)|$, so the double integral converges absolutely and Fubini's theorem applies. For the usual broader class of locally integrable functions of exponential order, take β beyond the abscissas of absolute convergence of f and g, and insert the Gaussian regulator $e^{-\epsilon\omega^2}$ of Exercise 20.10.2. Since $|f(\beta+i\omega)| \leq \int_0^\infty e^{-\beta z}|F(z)|dz$, the double integral then converges absolutely and may be interchanged, and by Exercise 20.10.2 the regularized bracket is $e^{\beta(t-u)}\int_{-\infty}^\infty K_\epsilon(t-u-z)H_F(z)dz$. Here $H_F(z) = e^{-\beta z}F(z)$ (the H of that exercise) and $H_G(u) = e^{-\beta u}G(u)$ for nonnegative arguments, and both vanish for negative ones. Since $G(u)e^{\beta(t-u)} = e^{\beta t}H_G(u)$, the regularized inverse of fg is the left side below; substituting $z = v-u$ and interchanging once more (allowed because $K_\epsilon \leq 1/\sqrt{4\pi\epsilon}$ and H_F, H_G are integrable) gives the right side:

$$\begin{aligned} &e^{\beta t}\int_{-\infty}^\infty H_G(u)\int_{-\infty}^\infty K_\epsilon(t-u-z)H_F(z)\,dz\,du \\ &\qquad = e^{\beta t}\int_{-\infty}^\infty K_\epsilon(t-v)\left[\int_{-\infty}^\infty H_F(v-u)H_G(u)\,du\right] dv. \end{aligned}$$

For $v > 0$ the bracket is $\int_0^v e^{-\beta(v-u)}F(v-u)e^{-\beta u}G(u)du = e^{-\beta v}(F*G)(v)$, and for $v < 0$ it is zero. The approximate-identity argument of Exercise 20.10.2, applied to this function in place of H (it is integrable, with integral of its modulus at most $\int|H_F|\,dz\int|H_G|\,du$), now gives $e^{\beta t}e^{-\beta t}(F*G)(t) = (F*G)(t)$ at continuity points of the convolution.

20.10.4 Find

$$L^{-1}\left\{\frac{s}{s^2-k^2}\right\}$$

(a) by a partial fraction expansion.

(b) Repeat, using the Bromwich integral.

Solution.

(a) For $k \neq 0$, since $(s+k)+(s-k)=2s$,

$$\frac{s}{s^2-k^2}=\frac{1}{2}\left(\frac{1}{s-k}+\frac{1}{s+k}\right).$$

Thus, by $L\{e^{\pm kt}\}=1/(s\mp k)$,

$$\boxed{L^{-1}\left\{\frac{s}{s^2-k^2}\right\}=\frac{e^{kt}+e^{-kt}}{2}=\cosh kt}.$$

(b) Take $\beta > |k|$ for real k. For $t>0$, close the Bromwich segment from $\beta-iT$ to $\beta+iT$ by the arc of the circle $|s|=R=\sqrt{\beta^2+T^2}$ to its left; the closed contour is positively oriented. The following arc estimate is used throughout this section. Suppose the factor $\varphi(s)$ multiplying e^{st} satisfies $|\varphi(s)|\le M_R$ on the arc, with $M_R\to 0$ as $R\to\infty$. On the left semicircle $s=Re^{i\theta}$, $\pi/2\le\theta\le 3\pi/2$, put $\theta=\pi/2+\psi$ with $0\le\psi\le\pi$; then $|ds|=R\,d\psi$ and $|e^{st}|=e^{tR\cos\theta}=e^{-tR\sin\psi}$. Since $\sin(\pi-\psi)=\sin\psi$, the halves $\psi\le\pi/2$ and $\psi\ge\pi/2$ contribute equally, and Jordan's inequality $\sin\psi\ge 2\psi/\pi$ on $[0,\pi/2]$ gives

$$\begin{aligned}\int |e^{st}|\,|ds| &= 2R\int_0^{\pi/2} e^{-tR\sin\psi}d\psi \le 2R\int_0^{\pi/2} e^{-2tR\psi/\pi}d\psi\\ &= \frac{\pi}{t}\left(1-e^{-tR}\right)<\frac{\pi}{t}.\end{aligned}$$

The two remaining pieces (present for $\beta>0$), where $0<\Re s<\beta$ and hence $|e^{st}|\le e^{\beta t}$, run between the angles $\pm\arccos(\beta/R)$ of $\beta\pm iT$ and $\pm\pi/2$. Each therefore has length $R[\pi/2-\arccos(\beta/R)]=R\arcsin(\beta/R)\le\pi\beta/2$, since Jordan's inequality at the angle $\arcsin(\beta/R)\in[0,\pi/2]$ reads $\beta/R\ge(2/\pi)\arcsin(\beta/R)$. Hence the arc contributes at most $M_R(\pi/t+\pi\beta e^{\beta t})\to 0$. Here $\varphi(s)=s/(s^2-k^2)$ and $|s^2-k^2|\ge R^2-k^2$, so $M_R=R/(R^2-k^2)$. The Bromwich integral therefore equals the sum of the residues at the simple poles $s=\pm k$, each found by cancelling its factor $s\mp k$ and setting $s=\pm k$:

$$\operatorname*{Res}_{s=k}\frac{se^{st}}{(s-k)(s+k)}=\frac{ke^{kt}}{2k}=\frac{e^{kt}}{2},$$

$$\operatorname*{Res}_{s=-k}\frac{se^{st}}{(s-k)(s+k)}=\frac{-ke^{-kt}}{-2k}=\frac{e^{-kt}}{2}.$$

Their sum $(e^{kt}+e^{-kt})/2$ reproduces $\cosh kt$. At $k=0$ the transform reduces directly to $1/s$, with inverse 1, also the continuous limit of the answer. At $t=0$ the answer takes the right-hand value $F(0+)=\cosh 0=1$. The symmetric Bromwich integral itself, for which the arc estimate (with its factor π/t) is unavailable at $t=0$, gives $F(0+)/2=\frac{1}{2}$ there, as noted in Exercise 20.10.2. Indeed, in $s/(s^2-k^2)=1/s+k^2/[s(s^2-k^2)]$ the first term gives $\frac{1}{2\pi}\int_{-T}^{T}\frac{d\omega}{\beta+i\omega}=\frac{1}{\pi}\arctan\frac{T}{\beta}\to\frac{1}{2}$, while the second is $O(s^{-3})$, so its arc vanishes even at $t=0$ and its residues -1, $\frac{1}{2}$, $\frac{1}{2}$ at $s=0,k,-k$ cancel.

20.10.5 Find

$$L^{-1}\left\{\frac{k^2}{s(s^2+k^2)}\right\}$$

(a) by using a partial fraction expansion.

(b) Repeat using the convolution theorem.

(c) Repeat using the Bromwich integral.

ANS. $F(t) = 1 - \cos kt.$

Solution.

Assume real $k \neq 0$ during the algebra. (a) Writing the numerator as $k^2 = (s^2+k^2) - s^2$ gives the decomposition

$$\frac{k^2}{s(s^2+k^2)} = \frac{1}{s} - \frac{s}{s^2+k^2},$$

and the pairs $L\{1\} = 1/s$, $L\{\cos kt\} = s/(s^2+k^2)$ give $F(t) = 1 - \cos kt$.

(b) Factor the transform as $[1/s]\,[k^2/(s^2+k^2)]$. The inverse factors are 1 and $k\sin kt$, so convolution gives

$$F(t) = \int_0^t 1 \cdot k\sin(ku)du = [-\cos(ku)]_0^t = 1 - \cos kt.$$

(c) For a Bromwich line to the right of 0, close to the left for $t > 0$. The rational factor is $O(s^{-3})$: on $|s| = R > |k|$ it is at most $M_R = k^2/[R(R^2-k^2)]$, so the closing arc vanishes by the arc estimate of Exercise 20.10.4(b). At the three simple poles, cancelling each pole factor and evaluating the rest there, with $(ik)(2ik) = (-ik)(-2ik) = -2k^2$,

$$\begin{aligned}
\operatorname*{Res}_{s=0} \frac{k^2e^{st}}{s(s-ik)(s+ik)} &= \frac{k^2}{(-ik)(ik)} = 1,\\
\operatorname*{Res}_{s=ik} \frac{k^2e^{st}}{s(s-ik)(s+ik)} &= \frac{k^2e^{ikt}}{(ik)(2ik)} = -\frac{1}{2}e^{ikt},\\
\operatorname*{Res}_{s=-ik} \frac{k^2e^{st}}{s(s-ik)(s+ik)} &= \frac{k^2e^{-ikt}}{(-ik)(-2ik)} = -\frac{1}{2}e^{-ikt}.
\end{aligned}$$

Summing gives

$$\boxed{F(t) = 1 - \frac{1}{2}(e^{ikt} + e^{-ikt}) = 1 - \cos kt}.$$

At $k = 0$ the transform and its inverse are both zero, which agrees with the continuous limit. All three methods give $F(0) = 0$.

20.10.6 Use the Bromwich integral to find the function whose transform is $f(s) = s^{-1/2}$. Note that $f(s)$ has a branch point at $s = 0$. The negative x-axis may be taken as a cut line. See Fig. 20.23.

Hint. A portion of the path needed to close the contour will yield nonzero contributions to the contour integral. These will need to be taken into account to get the proper value for the Bromwich integral.

ANS. $F(t) = (\pi t)^{-1/2}.$

Solution.

Choose the principal branch $s^{-1/2} = |s|^{-1/2}e^{-i\arg(s)/2}$ with $-\pi < \arg(s) < \pi$. The Bromwich line lies at $\Re s = \beta > 0$, to the right of the branch point. For $t > 0$, close the Bromwich segment to

the left as in Exercise 20.10.4(b), but leave out the cut: the closed contour follows the arc down to the upper bank of the cut, runs along the upper bank from $-R$ to $-\rho$, circles the origin clockwise on $|s| = \rho$, returns along the lower bank from $-\rho$ to $-R$, and finishes on the arc. It encloses no singularity, so its integral vanishes by Cauchy's theorem. On the arc $|f| = R^{-1/2} \to 0$, so the arc estimate of Exercise 20.10.4(b) applies; on the small circle (length $2\pi\rho$) $|e^{st}f| \le e^{\rho t}\rho^{-1/2}$, so it contributes $O(\sqrt{\rho})$ as $\rho \to 0$. Hence the Bromwich integral equals the reversed bank integrals: along the lower bank from $-\infty$ to 0 and along the upper bank from 0 to $-\infty$, a counterclockwise path around the cut. There are no poles; the two banks supply the entire inverse. For $r > 0$, $\arg s \to \pm\pi$ on the two banks, so

$$f(-r + i0) = r^{-1/2}e^{-i\pi/2} = -ir^{-1/2}, \qquad f(-r - i0) = r^{-1/2}e^{i\pi/2} = +ir^{-1/2}.$$

With $s = -r$, $ds = -dr$, the upper bank (r from 0 to ∞) contributes $-\int_0^\infty e^{-rt}f(-r + i0)dr$ and the lower bank (r from ∞ to 0) contributes $+\int_0^\infty e^{-rt}f(-r - i0)dr$. Their sum is therefore

$$\int_0^\infty e^{-rt}\big[f(-r - i0) - f(-r + i0)\big]dr = 2i\int_0^\infty e^{-rt}r^{-1/2}dr.$$

Dividing by $2\pi i$ and substituting $u = rt$ (so that $r^{-1/2}dr = t^{-1/2}u^{-1/2}du$) gives, with $\Gamma(1/2) = \sqrt{\pi}$,

$$F(t) = \frac{1}{\pi}\int_0^\infty e^{-rt}r^{-1/2}dr = \frac{\Gamma(1/2)}{\pi\sqrt{t}} = \boxed{\frac{1}{\sqrt{\pi t}}}, \qquad t > 0.$$

Its integrable singularity at zero is compatible with a Laplace transform: direct substitution gives $\int_0^\infty e^{-st}(\pi t)^{-1/2}dt = s^{-1/2}$ for real $s > 0$, fixing the branch and normalization independently.

20.10.7 Show that

$$L^{-1}\big\{(s^2 + 1)^{-1/2}\big\} = J_0(t)$$

by evaluation of the Bromwich integral.

Hint. Convert your Bromwich integral into an integral representation of $J_0(t)$. Figure 20.24 shows a possible contour.

Solution.

Take a cut along the segment from $-i$ to i and choose $D(s) = \sqrt{s^2 + 1}$ analytic outside this segment with $D(s) \sim s$ at infinity and $D(s) > 0$ for positive real s. For $s = x + iy$ with $x > 0$, the number $s^2 + 1 = x^2 - y^2 + 1 + 2ixy$ never lies on $(-\infty, 0]$ (a real value needs $xy = 0$, so $y = 0$, and then $s^2 + 1 = x^2 + 1 > 0$). Hence the principal root $\sqrt{s^2 + 1}$ is analytic on $\Re s > 0$; its ratio to D is a continuous function with values ± 1 on this connected half-plane, so it is constant, and it equals 1 for $s > 0$. Thus D is the principal root there. Letting $x \to 0+$ gives, on the right bank $s = iy + 0$, $-1 < y < 1$,

$$D(iy + 0) = +\sqrt{1 - y^2}.$$

On the left bank its sign is reversed: $-D(-s)$ is also analytic outside the cut, squares to $s^2 + 1$ and behaves as s at infinity, so $D(-s) = -D(s)$ and $D(iy - 0) = -D(-iy + 0) = -\sqrt{1 - y^2}$. The Bromwich line lies at $\Re s = \beta > 0$, to the right of the cut. For $t > 0$, close its segment by the arc of Exercise 20.10.4(b). On $|s| = R > 1$, $|1/D(s)| \le (R^2 - 1)^{-1/2} \to 0$, so the arc contributes nothing in the limit. The closed contour encloses the cut, and by Cauchy's theorem it may be shrunk onto a counterclockwise contour hugging the cut: upward along the right bank, around a small circle about i, downward along the left bank, and around a small circle about $-i$. The endpoint singularities are

inverse square roots ($|D(s)|^2 = |s-i|\,|s+i|$), so the circles, of radius ρ and length $2\pi\rho$, contribute $O(\sqrt{\rho})$. With $s = iy$, $ds = i\,dy$, and $1/D_{\text{right}} - 1/D_{\text{left}} = 2/\sqrt{1-y^2}$,

$$F(t) = \frac{1}{2\pi i}\int_{-i}^{i} e^{st}\left[\frac{1}{D(s)_{\text{right}}} - \frac{1}{D(s)_{\text{left}}}\right] ds = \frac{1}{2\pi i}\int_{-1}^{1} e^{iyt}\frac{2\,i\,dy}{\sqrt{1-y^2}}$$
$$= \frac{1}{\pi}\int_{-1}^{1}\frac{e^{iyt}}{\sqrt{1-y^2}}dy = \frac{2}{\pi}\int_{0}^{1}\frac{\cos(yt)}{\sqrt{1-y^2}}dy.$$

The sine part of e^{iyt} cancels by oddness. Finally $y = \cos\theta$ converts the symmetric integral $\frac{1}{\pi}\int_{-1}^{1}\cos(yt)(1-y^2)^{-1/2}dy$: here $dy = -\sin\theta\,d\theta$, $\sqrt{1-y^2} = \sin\theta \geq 0$ on $[0,\pi]$, and $y = -1, 1$ correspond to $\theta = \pi, 0$, so the reversed limits absorb the minus sign. By the integral representation of Exercise 14.1.17(b) with $\nu = 0$,

$$F(t) = \frac{1}{\pi}\int_{0}^{\pi}\cos(t\cos\theta)d\theta = \boxed{J_0(t)}.$$

This also has the right-hand initial value $J_0(0) = 1$, in agreement with the initial-value relation $F(0+) = \lim_{s\to\infty} sf(s) = \lim_{s\to\infty} s/\sqrt{s^2+1} = 1$. The branch must be continued through the exterior of the chosen cut; taking an unrelated square-root sign on the left would spoil the jump and the factor of two.

20.10.8 Evaluate the inverse Laplace transform

$$L^{-1}\{(s^2-a^2)^{-1/2}\}$$

by each of the following methods:

(a) Expansion in a series and term-by-term inversion.

(b) Direct evaluation of the Bromwich integral.

(c) Change of variable in the Bromwich integral: $s = (a/2)(z + z^{-1})$.

Solution.

Take $a > 0$ and the branch of $\sqrt{s^2-a^2}$ asymptotic to s, with cut $[-a, a]$: $D(s) = \sqrt{s-a}\,\sqrt{s+a}$ with principal square roots. The first factor changes sign across $(-\infty, a)$ and the second across $(-\infty, -a)$; on $(-\infty, -a)$ the two sign changes cancel, so D is analytic outside $[-a, a]$, and $D(s) = s\sqrt{1-a^2/s^2} \sim s$. Write $f = 1/D$. The Bromwich line lies at $\Re s = \beta > a$.

(a) For $|s| > a$ put $w = a^2/s^2$, so $|w| < 1$. The binomial series $(1-w)^{-1/2} = \sum_n \binom{-1/2}{n}(-w)^n$ has the coefficients

$$(-1)^n\binom{-1/2}{n} = \frac{1\cdot 3\cdots(2n-1)}{2^n n!} = \frac{(2n)!}{4^n(n!)^2},$$

since $1\cdot 3\cdots(2n-1) = (2n)!/(2^n n!)$. Hence on this branch

$$(s^2-a^2)^{-1/2} = \frac{1}{s}(1-w)^{-1/2} = \frac{1}{s}\sum_{n=0}^{\infty}\frac{(2n)!}{4^n(n!)^2}\frac{a^{2n}}{s^{2n}}.$$

The inverse of s^{-2n-1} is $t^{2n}/(2n)!$, which cancels the $(2n)!$, and $a^{2n}t^{2n}/4^n = (at/2)^{2n}$. Thus, by the power series of I_ν at $\nu = 0$,

$$F(t) = \sum_{n=0}^{\infty}\frac{(at/2)^{2n}}{(n!)^2} = \boxed{I_0(at)}.$$

The resulting series converges for every finite t. To justify the termwise inversion, transform it back. For real $s > a$ every term $e^{-st}(at/2)^{2n}/(n!)^2$ is nonnegative, so by monotone convergence the series may be integrated term by term:

$$L\{I_0(at)\}(s) = \sum_{n=0}^{\infty} \frac{a^{2n}}{4^n (n!)^2} \frac{(2n)!}{s^{2n+1}} = (s^2 - a^2)^{-1/2},$$

the convergent binomial expansion again. By uniqueness of the inverse transform, $F = I_0(at)$; no uniform convergence on a Bromwich line is needed.

(b) For $-a < x < a$, $\sqrt{s+a} \to \sqrt{a+x} > 0$ on both banks, while $\arg(s-a) \to \pm\pi$ gives $\sqrt{s-a} \to \pm i\sqrt{a-x}$ on the upper and lower bank. Hence $D(x \pm i0) = \pm i\sqrt{a^2 - x^2}$, and the bank values of f are

$$f(x + i0) = -\frac{i}{\sqrt{a^2 - x^2}}, \qquad f(x - i0) = +\frac{i}{\sqrt{a^2 - x^2}}.$$

For $t > 0$, close the Bromwich segment by the arc of Exercise 20.10.4(b). On $|s| = R > a$, $|f| \le (R^2 - a^2)^{-1/2} \to 0$, so the arc drops out, and by Cauchy's theorem the closed contour may be shrunk onto a counterclockwise contour around the cut. Its upper bank runs from a to $-a$, and its lower bank from $-a$ to a. The endpoint circles vanish as the square root of their radius, since $|D(s)|^2 = |s-a|\,|s+a|$ (as in Exercise 20.10.7). Reversing the limits of the upper-bank integral gives

$$\begin{aligned} F(t) &= \frac{1}{2\pi i} \int_{-a}^{a} e^{xt} \big[f(x - i0) - f(x + i0) \big] dx = \frac{1}{2\pi i} \int_{-a}^{a} \frac{2i\, e^{xt}}{\sqrt{a^2 - x^2}} dx \\ &= \frac{1}{\pi} \int_{-a}^{a} \frac{e^{xt}}{\sqrt{a^2 - x^2}} dx = \frac{1}{\pi} \int_0^{\pi} e^{at\cos\theta} d\theta = I_0(at). \end{aligned}$$

In the substitution $x = a\cos\theta$, $dx = -a\sin\theta\, d\theta$ and $\sqrt{a^2 - x^2} = a\sin\theta$ for $0 \le \theta \le \pi$; the limits $x = -a, a$ become $\theta = \pi, 0$, and the reversed limits cancel the sign in dx. The last equality is the integral representation of Exercise 14.5.14 with $n = 0$.

(c) Set $s = (a/2)(z + z^{-1})$, taking the exterior inverse $z = (s + \sqrt{s^2 - a^2})/a$, for which $z \sim 2s/a$ at infinity. Since $s^2 - a^2 = \frac{a^2}{4}(z^2 + 2 + z^{-2} - 4) = \frac{a^2}{4}(z - z^{-1})^2$ and $ds = \frac{a}{2}(1 - z^{-2})dz$,

$$\sqrt{s^2 - a^2} = \frac{a}{2}(z - z^{-1}), \qquad \frac{ds}{\sqrt{s^2 - a^2}} = \frac{(1 - z^{-2})dz}{z - z^{-1}} = \frac{dz}{z};$$

the sign of the root is fixed by $\sqrt{s^2 - a^2} \sim s \sim az/2$ at infinity, and $z - z^{-1} = z(1 - z^{-2})$. Deform the s contour around its cut as in (b). For $x = a\cos\theta$, $0 < \theta < \pi$, the bank values $D(x \pm i0)$ of (b) give $z = (x \pm i\sqrt{a^2 - x^2})/a = e^{\pm i\theta}$. So the upper bank ($x$ from a to $-a$, θ from 0 to π) maps to the upper unit semicircle from 1 to -1, and the lower bank to the lower semicircle from -1 to 1. Their union is one counterclockwise unit circle, giving

$$F(t) = \frac{1}{2\pi i} \oint_{|z|=1} \exp\left[\frac{at}{2}(z + z^{-1}) \right] \frac{dz}{z},$$

which is Exercise 14.5.3(a) with $n = 0$ and $x = at$. Directly, expand the two exponentials on this fixed circle:

$$\frac{1}{z} \sum_{n,m \ge 0} \frac{(at/2)^{n+m}}{n!m!} z^{n-m}.$$

On $|z| = 1$ the double series is dominated by $\sum_{n,m}(at/2)^{n+m}/(n!m!) = e^{at}$, so it may be integrated term by term, and $\oint z^{n-m-1}dz$ is $2\pi i$ for $n = m$ and 0 otherwise. Only $n = m$ contributes to the coefficient of z^{-1}, so the residue is $\sum_{n\ge0}(at/2)^{2n}/(n!)^2 = I_0(at)$. For $at \neq 0$, the singularity at $z = 0$ is essential, not the simple pole called for in the Instructor's Manual; the residue calculation still applies. Also, the image of the vertical Bromwich line $\Re s = \beta$ under this map is a curve in the z plane, not a vertical line, so it may not simply be replaced by one; mapping the already deformed cut contour, as above, avoids the issue. The answer is even in a, so real negative a gives the same result with $|a|$. At $a = 0$ the transform is $1/s$ and its inverse is 1.

20.10.9 Show that

$$L^{-1}\left\{\frac{\ln s}{s}\right\} = -\ln t - \gamma,$$

where $\gamma = 0.5772\ldots$ is the Euler–Mascheroni constant.

Solution.

For real $q > 0$ and $s > 0$, the gamma substitution $u = st$ gives

$$L\left\{\frac{t^{q-1}}{\Gamma(q)}\right\}(s) = \frac{1}{\Gamma(q)}\int_0^\infty e^{-st}t^{q-1}dt = \frac{s^{-q}}{\Gamma(q)}\int_0^\infty e^{-u}u^{q-1}du = s^{-q}.$$

Differentiate with respect to q near $q = 1$. For $|q-1| \le \frac12$ the q-derivative of the integrand, $e^{-st}t^{q-1}\ln t$, is bounded by $e^{-st}(t^{-1/2} + t^{1/2})|\ln t|$, which is integrable on $(0,\infty)$ (exponential decay at infinity, an integrable singularity at zero), so differentiation inside the transform is justified. With $\partial_q t^{q-1} = t^{q-1}\ln t$, $\partial_q[1/\Gamma(q)] = -\Gamma'(q)/\Gamma(q)^2$ and $\partial_q s^{-q} = -(\ln s)s^{-q}$,

$$L\left\{\frac{t^{q-1}}{\Gamma(q)}[\ln t - \psi(q)]\right\}(s) = -(\ln s)s^{-q}, \qquad \psi(q) = \frac{\Gamma'(q)}{\Gamma(q)}.$$

Since $\Gamma(1) = 1$ and $\psi(1) = -\gamma$ (Section 13.2), setting $q = 1$ gives $L\{\ln t + \gamma\} = -(\ln s)/s$, and multiplying by -1 yields

$$\boxed{L^{-1}\left\{\frac{\ln s}{s}\right\}(t) = -\ln t - \gamma}, \qquad t > 0.$$

For complex s in $\Re s > 0$, take the branch of $\ln s$ that is real on $s > 0$; both sides are analytic in $\Re s > 0$, so the identity extends there. The logarithmic singularity of the answer at $t = 0$ is integrable, so it causes no failure of the ordinary Laplace transform.

20.10.10 Evaluate the Bromwich integral for

$$f(s) = \frac{s}{(s^2 + a^2)^2}.$$

Solution.

For real $a \neq 0$, take the Bromwich line at $\Re s = \beta > 0$, to the right of the poles $\pm ia$, and close it to the left when $t > 0$. The integrand

$$\frac{se^{st}}{(s - ia)^2(s + ia)^2}$$

has double poles at $s = \pm ia$ and a rational factor $O(s^{-3})$: on $|s| = R > |a|$ it is at most $R/(R^2-a^2)^2$. The closing arc therefore vanishes by the arc estimate of Exercise 20.10.4(b). At a double pole s_0, the residue of $\varphi(s)/(s-s_0)^2$, with φ analytic at s_0, is $\varphi'(s_0)$. At $s = ia$ take $\varphi(s) = se^{st}/(s+ia)^2$; the product and quotient rules give

$$\varphi'(s) = e^{st}\left[\frac{1+st}{(s+ia)^2} - \frac{2s}{(s+ia)^3}\right].$$

With $(2ia)^2 = -4a^2$ and $(2ia)^3 = -8ia^3$,

$$\begin{aligned} R_+ = \varphi'(ia) &= e^{iat}\left[\frac{1+iat}{(2ia)^2} - \frac{2ia}{(2ia)^3}\right] \\ &= e^{iat}\left[-\frac{1+iat}{4a^2} + \frac{1}{4a^2}\right] = -\frac{it}{4a}e^{iat}. \end{aligned}$$

Similarly, at $s = -ia$ the factor is $se^{st}/(s-ia)^2$, with $(s-ia)^2 = -4a^2$ and $(s-ia)^3 = 8ia^3$ there:

$$R_- = e^{-iat}\left[\frac{1-iat}{-4a^2} - \frac{-2ia}{8ia^3}\right] = e^{-iat}\left[-\frac{1-iat}{4a^2} + \frac{1}{4a^2}\right] = \frac{it}{4a}e^{-iat}.$$

Summing, with $e^{-iat} - e^{iat} = -2i\sin at$,

$$F(t) = R_+ + R_- = \frac{it}{4a}(e^{-iat} - e^{iat}) = \boxed{\frac{t}{2a}\sin at}.$$

As a check, the rule $L\{tG(t)\} = -g'(s)$ for $g = L\{G\}$, with $G = \sin at$ and $g = a/(s^2+a^2)$, gives $L\{t\sin at\} = 2as/(s^2+a^2)^2$, which is $2a$ times the given transform. At $a = 0$ the two double poles merge and the reduced transform is s^{-3}; its inverse is $t^2/2$, the continuous limit of the formula above.

20.10.11 Heaviside expansion theorem. If the transform $f(s)$ may be written as a ratio

$$f(s) = \frac{g(s)}{h(s)},$$

where $g(s)$ and $h(s)$ are analytic functions, with $h(s)$ having simple, isolated zeros at $s = s_i$, and with a vanishing left closing-contour contribution and a convergent residue sum, show that

$$F(t) = L^{-1}\left\{\frac{g(s)}{h(s)}\right\} = \sum_i \frac{g(s_i)}{h'(s_i)}\, e^{s_i t}.$$

Hint. See Exercise 11.6.3.

A sufficient elementary setting is a strictly proper rational transform with finitely many simple poles, for $t > 0$. The unqualified analytic-function assertion in the textbook is false, as the counterexample below shows.

Solution.

At a simple zero s_i of h,

$$h(s) = h'(s_i)(s - s_i) + O((s-s_i)^2), \qquad h'(s_i) \neq 0.$$

The residue of the inverse-transform integrand is therefore, since $(s-s_i)/h(s) = (s-s_i)/[h(s)-h(s_i)] \to 1/h'(s_i)$ (this is Exercise 11.6.3 with $f_1 = e^{st}g$, $f_2 = h$),

$$\operatorname*{Res}_{s=s_i} \frac{e^{st}g(s)}{h(s)} = \lim_{s\to s_i} \frac{(s-s_i)e^{st}g(s)}{h(s)} = \frac{g(s_i)}{h'(s_i)} e^{s_i t}.$$

Place the Bromwich line to the right of all the s_i, and close it to the left by contours on which the contribution vanishes in the limit. If the residue sums inside these contours converge, the residue theorem applied to each closed contour gives in the limit

$$\boxed{F(t) = \sum_i \frac{g(s_i)}{h'(s_i)} e^{s_i t}}, \qquad t > 0.$$

A particularly simple sufficient case is a strictly proper rational function g/h with finitely many simple poles: then $|g/h| \le M/|s|$ on large circles, so the arc estimate of Exercise 20.10.4(b) applies, and one large circle encloses all the poles. A zero of h at which g also vanishes is a removable singularity of g/h; its residue is zero, and so is its term $g(s_i)/h'(s_i)$. Analytic g, h and simple zeros of h alone do not imply the stated expansion: one must also control the closing contour, and an infinite residue sum needs its own convergence prescription. For a concrete counterexample satisfying the printed analytic conditions, choose $g(s) = e^{-s}$, $h(s) = s+1$. Its only pole is -1, where $g(-1)/h'(-1) = e$, so the proposed unrestricted sum would be $e \cdot e^{-t} = e^{1-t}$. But direct transformation, with $v = t-1$, gives

$$\int_1^\infty e^{-st}e^{-(t-1)}dt = e^{-s}\int_0^\infty e^{-(s+1)v}dv = \frac{e^{-s}}{s+1},$$

whose causal inverse is $e^{-(t-1)}u(t-1)$: it equals e^{1-t} for $t > 1$ but is zero for $0 < t < 1$. For these times the integrand $e^{st}g/h = e^{s(t-1)}/(s+1)$ has $|e^{s(t-1)}| = e^{(t-1)\Re s}$, which grows as $\Re s \to -\infty$ on a left closing arc, explaining exactly why the omitted hypothesis fails. A polynomial part of a rational transform likewise requires impulses at zero rather than this ordinary exponential sum.

20.10.12 Using the Bromwich integral, invert $f(s) = s^{-2}e^{-ks}$, $k \ge 0$. Express $F(t) = L^{-1}\{f(s)\}$ in terms of the (shifted) unit step function $u(t-k)$.

ANS. $F(t) = (t-k)u(t-k)$.

Solution.

The restriction $k \ge 0$ is needed: for $k < 0$ the ramp $(t-k)u(t-k)$ does not have the transform $s^{-2}e^{-ks}$, as the check at the end shows. Since $e^{st}e^{-ks} = e^{s(t-k)}$, the inversion integral is

$$F(t) = \frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} \frac{e^{s(t-k)}}{s^2} ds, \qquad \beta > 0.$$

Close the segment from $\beta - iT$ to $\beta + iT$ by an arc of the circle $|s| = R = \sqrt{\beta^2+T^2}$. If $t < k$, close to the right ($\Re s \ge \beta$), where $|e^{s(t-k)}| = e^{(t-k)\Re s} \le e^{(t-k)\beta}$. No poles are enclosed, so $F(t) = 0$. If $t > k$, close to the left ($\Re s \le \beta$), where again $|e^{s(t-k)}| \le e^{(t-k)\beta}$, and enclose the double pole at $s = 0$. Its residue is the derivative of $e^{s(t-k)}$ at $s = 0$ (as for the double poles of Exercise 20.10.10):

$$\left.\frac{d}{ds}e^{s(t-k)}\right|_{s=0} = t-k.$$

Both arcs have length at most $2\pi R$, and $|s^{-2}| = R^{-2}$ on them, so they contribute at most $2\pi e^{(t-k)\beta}/R \to 0$. At $t = k$ the value is zero, either by continuity or because $1/s^2$ has zero residue at $s = 0$ (the same arc bound applies). Thus

$$\boxed{F(t) = (t-k)u(t-k)}.$$

A forward check with $v = t - k$ gives

$$\int_k^\infty e^{-st}(t-k)dt = e^{-ks}\int_0^\infty e^{-sv}v\,dv = \frac{e^{-ks}}{s^2}.$$

If $k < 0$, the ramp equals $t-k$ for all $t \geq 0$, and its one-sided transform is instead $\int_0^\infty e^{-st}(t-k)dt = s^{-2} - k/s \neq s^{-2}e^{-ks}$; this confirms the need for the nonnegative delay in the one-sided formula.

20.10.13 You have a Laplace transform:

$$f(s) = \frac{1}{(s+a)(s+b)}, \qquad a \neq b.$$

Invert this transform by each of three methods:

(a) Partial fractions and use of tables,

(b) Convolution theorem,

(c) Bromwich integral.

$$\textit{ANS. } F(t) = \frac{e^{-bt} - e^{-at}}{a-b}, \quad a \neq b.$$

Solution.

(a) Since $(s+a) - (s+b) = a - b$, write

$$\frac{1}{(s+a)(s+b)} = \frac{1}{a-b}\left(\frac{1}{s+b} - \frac{1}{s+a}\right).$$

The exponential transform pair $L\{e^{-ct}\} = 1/(s+c)$ gives

$$F(t) = \frac{e^{-bt} - e^{-at}}{a-b}.$$

(b) The inverse factors are e^{-at} and e^{-bt}. Their convolution, with $e^{-au}e^{-b(t-u)} = e^{-bt}e^{-(a-b)u}$ and $e^{-bt}e^{-(a-b)t} = e^{-at}$, is

$$\begin{aligned} F(t) &= \int_0^t e^{-au}e^{-b(t-u)}du = e^{-bt}\int_0^t e^{-(a-b)u}du \\ &= e^{-bt}\frac{1 - e^{-(a-b)t}}{a-b} = \frac{e^{-bt} - e^{-at}}{a-b}. \end{aligned}$$

(c) Choose $\beta > \max(-\Re a, -\Re b)$ and close the Bromwich line to the left for $t > 0$. The rational factor is $O(s^{-2})$, so the arc vanishes by the arc estimate of Exercise 20.10.4(b). The residues at the simple poles, found by cancelling each pole factor and setting s to the pole, are

$$\operatorname*{Res}_{s=-a}\frac{e^{st}}{(s+a)(s+b)} = \frac{e^{-at}}{b-a}, \qquad \operatorname*{Res}_{s=-b}\frac{e^{st}}{(s+a)(s+b)} = \frac{e^{-bt}}{a-b}.$$

Their sum gives the same result:

$$\boxed{F(t) = \frac{e^{-bt} - e^{-at}}{a - b}}.$$

Its initial value is zero and its initial slope is $F'(0) = (-b + a)/(a - b) = 1$. The excluded coincident case has the regular limit

$$\frac{e^{-bt} - e^{-at}}{a - b} = e^{-at}\,\frac{e^{(a-b)t} - 1}{a - b} \to te^{-at} \qquad (b \to a),$$

since $(e^{\epsilon t} - 1)/\epsilon \to t$ as $\epsilon = a - b \to 0$; it is also obtained from the double pole of $e^{st}/(s + a)^2$, whose residue is $\frac{d}{ds}e^{st}\big|_{s=-a} = te^{-at}$.

21 Chapter 21. Integral Equations

Problems

21.1 Introduction

21.1.1 Starting with the ODE, integrate twice and derive the Volterra integral equation corresponding to

(a) $y''(x) - y(x) = 0; \quad y(0) = 0, \quad y'(0) = 1.$

$$\textit{ANS. } y = \int_0^x (x-t)y(t)\,dt + x.$$

(b) $y''(x) - y(x) = 0; \quad y(0) = 1, \quad y'(0) = -1.$

$$\textit{ANS. } y = \int_0^x (x-t)y(t)\,dt - x + 1.$$

Check your results with Eq. (21.11).

Solution.

Integrating $y''(t) = y(t)$ from $t = 0$ to $t = u$ gives

$$y'(u) = y'(0) + \int_0^u y(t)\,dt.$$

Integrating this from $u = 0$ to $u = x$ gives

$$y(x) = y(0) + xy'(0) + \int_0^x du \int_0^u y(t)\,dt.$$

For $x \geq 0$ the triangular integration region is $0 \leq t \leq u \leq x$. Reverse the order:

$$\int_0^x y(t) \left(\int_t^x du \right) dt = \int_0^x (x-t)y(t)\,dt.$$

For $x < 0$ the identity $\int_0^x du \int_0^u y(t)\,dt = \int_0^x (x-t)y(t)\,dt$ still holds: both sides vanish at $x = 0$, and both have x-derivative $\int_0^x y(t)\,dt$ (the left side by the fundamental theorem of calculus, the right side by Leibniz's rule, whose endpoint term $(x-x)y(x)$ vanishes), so they agree for every x. Thus the general initial-value formula is

$$y(x) = y(0) + xy'(0) + \int_0^x (x-t)y(t)\,dt.$$

Inserting the two pairs of initial data yields, respectively,

$$\boxed{y(x) = x + \int_0^x (x-t)y(t)\,dt}, \qquad \boxed{y(x) = 1 - x + \int_0^x (x-t)y(t)\,dt}.$$

To check with Eq. (21.11), write the ODE as $y'' + A(x)y' + B(x)y = g(x)$ with initial point a: here $A = 0$, $B = -1$, $g = 0$, $a = 0$. Hence

$$K(x,t) = (t-x)[B(t) - A'(t)] - A(t) = (t-x)(-1) = x - t,$$
$$f(x) = \int_0^x (x-t)g(t)\,dt + [A(0)y(0) + y'(0)](x-0) + y(0) = y(0) + xy'(0),$$

exactly as above.

21.1.2 Starting with the given answers of Exercise 21.1.1, differentiate and recover the original ODEs and the boundary conditions.

Solution.

Write either answer as

$$y(x) = a + bx + \int_0^x (x - t)y(t)\,dt,$$

where $(a, b) = (0, 1)$ or $(1, -1)$. Leibniz's rule includes an upper-endpoint term, which vanishes here:

$$y'(x) = b + (x - x)y(x) + \int_0^x \frac{\partial}{\partial x}\big[(x - t)y(t)\big]dt = b + \int_0^x y(t)\,dt.$$

Differentiate once more; by the fundamental theorem of calculus, $\frac{d}{dx}\int_0^x y(t)\,dt = y(x)$, so

$$\boxed{y''(x) = y(x)}.$$

At $x = 0$ both integrals have zero length, so

$$y(0) = a, \qquad y'(0) = b.$$

For $(a, b) = (0, 1)$ this is $y(0) = 0$, $y'(0) = 1$, and for $(a, b) = (1, -1)$ it is $y(0) = 1$, $y'(0) = -1$: precisely the two original initial conditions. The differentiability used here follows from the integral equation itself: for continuous y its right-hand side is continuously differentiable, hence so is y, and then $y' = b + \int_0^x y\,dt$ is differentiable again.

21.1.3 Given

$$\varphi(x) = x - \int_0^x (t - x)\varphi(t)\,dt,$$

solve this integral equation by converting it to an ODE (plus boundary conditions) and solving the ODE (by inspection).

Solution.

First account for the sign, $-(t - x) = x - t$:

$$\varphi(x) = x + \int_0^x (x - t)\varphi(t)\,dt.$$

Differentiating by Leibniz's rule, the upper-endpoint term $(x-x)\varphi(x)$ is zero; a second differentiation uses the fundamental theorem of calculus:

$$\varphi'(x) = 1 + \int_0^x \varphi(t)\,dt, \qquad \varphi''(x) = \varphi(x).$$

At $x = 0$ the integrals in the equation and in φ' have zero length, so $\varphi(0) = 0$, $\varphi'(0) = 1$. The general solution is $Ae^x + Be^{-x}$, and therefore

$$A + B = 0, \qquad A - B = 1, \qquad A = \frac{1}{2}, \quad B = -\frac{1}{2}.$$

Hence $\boxed{\varphi(x) = \sinh x}$. Directly, integrating by parts with $u = x - t$, $dv = \sinh t\,dt$, $v = \cosh t$, $du = -dt$,

$$\int_0^x (x - t)\sinh t\,dt = [(x - t)\cosh t]_0^x + \int_0^x \cosh t\,dt = -x + \sinh x,$$

so substitution gives $x + (-x + \sinh x) = \sinh x$.

21.1.4 Show that the homogeneous Volterra equation of the second kind

$$\psi(x) = \lambda \int_0^x K(x,t)\psi(t)\,dt$$

has no solution (apart from the trivial solution $\psi = 0$). Assume that ψ is continuous and K is continuous on every finite triangle $0 \le t \le x \le L$.

Hint. On a fixed triangle, set $M = \max|K|$, $C = \max|\psi|$, and iterate the integral bound to obtain $|\psi(x)| \le C(|\lambda|Mx)^n/n!$.

Solution.

Use the usual continuous-kernel setting on a finite triangle $0 \le t \le x \le L$. Put

$$M = \max_{0\le t\le x\le L} |K(x,t)|, \qquad C = \max_{0\le x\le L} |\psi(x)|.$$

For $0 \le x \le L$ the equation gives

$$|\psi(x)| \le |\lambda| \int_0^x |K(x,t)|\,|\psi(t)|\,dt \le |\lambda| \int_0^x MC\,dt = C|\lambda|Mx.$$

Suppose $|\psi(t)| \le C(|\lambda|Mt)^n/n!$. Substituting this bound into the same estimate gives

$$|\psi(x)| \le |\lambda|M \int_0^x C\frac{(|\lambda|Mt)^n}{n!}\,dt = C\frac{(|\lambda|Mx)^{n+1}}{(n+1)!}.$$

Starting with $n = 0$, induction therefore proves this bound for every integer $n \ge 0$. For fixed x, the ratio of successive bounds is $|\lambda|Mx/(n+1) \to 0$, so the bound tends to zero. Thus

$$\boxed{\psi(x) = 0 \quad (0 \le x \le L)}$$

for every finite λ. Taking arbitrary finite L covers an unbounded interval whenever the kernel is locally bounded and continuous.

The textbook's hint instead expands ψ in a Maclaurin series. If K is smooth, put $k_j(x) = \partial_x^j K(x,t)\big|_{t=x}$. Each differentiation of $\int_0^x \partial_x^j K(x,t)\psi(t)\,dt$ by Leibniz's rule adds the endpoint term $k_j(x)\psi(x)$, so induction on n gives

$$\psi^{(n)}(x) = \lambda \int_0^x \partial_x^n K(x,t)\psi(t)\,dt + \lambda \sum_{j=0}^{n-1} \frac{d^{n-1-j}}{dx^{n-1-j}}\big[k_j(x)\psi(x)\big].$$

At $x = 0$ the integral has zero length, and by the product rule the sum contains only $\psi(0), \psi'(0), \dots, \psi^{(n-1)}(0)$. Since $\psi(0) = 0$ (zero-length integral), induction on n makes every $\psi^{(n)}(0)$ zero. A vanishing Maclaurin series proves the conclusion only if ψ is analytic, not merely indefinitely differentiable. For example, the smooth extension of e^{-1/x^2} by zero at $x = 0$ has all derivatives zero there but is not the zero function. The bounded-kernel estimate above supplies the missing uniqueness argument without assuming analyticity.

21.2 Some Special Methods

21.2.1 The kernel of a Fredholm equation of the second kind,

$$\varphi(x) = f(x) + \lambda \int_{-\infty}^{\infty} K(x,t)\varphi(t)\,dt,$$

is of the form $k(x-t)$.[6] Assuming that the required transforms exist, show that

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{F(t)e^{-ixt}}{1 - \sqrt{2\pi}\,\lambda K(t)}\,dt.$$

$F(t)$ and $K(t)$ are the Fourier transforms of $f(x)$ and $k(x)$, respectively. Assume also that the denominator has no real zeros and that the quotient in the displayed formula admits Fourier inversion.

Solution.

Use the textbook's symmetric Fourier convention

$$\widehat{g}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(x)e^{i\omega x}dx, \qquad g(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \widehat{g}(\omega)e^{-i\omega x}d\omega.$$

Let $\Phi = \widehat{\varphi}$, $F = \widehat{f}$, $K = \widehat{k}$ (the transform variable is called ω here and t in the statement), and write $(k * \varphi)(x) = \int_{\mathbb{R}} k(x-t)\varphi(t)\,dt$ for the integral in the equation (no factor $1/\sqrt{2\pi}$). To keep track of normalization, set $u = x - t$ for fixed t in its transform, so that $dx = du$ and $e^{i\omega x} = e^{i\omega u}e^{i\omega t}$:

$$\begin{aligned}
\widehat{k * \varphi}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}}\int_{\mathbb{R}} k(x-t)\varphi(t)e^{i\omega x}\,dt\,dx \\
&= \frac{1}{\sqrt{2\pi}} \left(\int_{\mathbb{R}} k(u)e^{i\omega u}du\right)\left(\int_{\mathbb{R}} \varphi(t)e^{i\omega t}dt\right) \\
&= \frac{1}{\sqrt{2\pi}} \left[\sqrt{2\pi}K(\omega)\right]\left[\sqrt{2\pi}\Phi(\omega)\right] = \sqrt{2\pi}K(\omega)\Phi(\omega).
\end{aligned}$$

Transforming each term of the integral equation therefore gives $\Phi = F + \sqrt{2\pi}\lambda K\Phi$, that is,

$$[1 - \sqrt{2\pi}\lambda K(\omega)]\Phi(\omega) = F(\omega).$$

Divide by the bracket and apply the inverse transform to obtain

$$\boxed{\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{F(\omega)e^{-i\omega x}}{1 - \sqrt{2\pi}\lambda K(\omega)}\,d\omega}.$$

Existence of the transforms of f and k alone does not justify division: the quotient must belong to the intended inverse-transform class. Zeros of the denominator require compatibility with F and may permit homogeneous components supported on those zeros in a distributional setting. The displayed formula applies when these obstructions are absent. For an example of such a homogeneous component, suppose $\sqrt{2\pi}\lambda K(\omega_0) = 1$ at a real ω_0. Then $e^{-i\omega_0 x}$ solves the homogeneous equation: with $u = x - t$,

$$\lambda \int_{\mathbb{R}} k(x-t)e^{-i\omega_0 t}dt = \lambda e^{-i\omega_0 x} \int_{\mathbb{R}} k(u)e^{i\omega_0 u}du = \sqrt{2\pi}\lambda K(\omega_0)e^{-i\omega_0 x} = e^{-i\omega_0 x}.$$

Its transform $\sqrt{2\pi}\,\delta(\omega - \omega_0)$ is supported at that zero, and it may be added to any solution φ.

[6] This kernel and a range $0 \le x < \infty$ are the characteristics of integral equations of the Wiener–Hopf type. Details will be found in Chapter 8 of Morse and Feshbach (1953); see the Additional Readings.

21.2.2

(a) The kernel of a Volterra equation of the first kind,

$$f(x) = \int_0^x K(x,t)\varphi(t)\,dt,$$

has the form $k(x-t)$. Assuming that the required transforms exist, show that

$$\varphi(x) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} \frac{F(s)}{K(s)}\,e^{xs}\,ds,$$

where $F(s)$ and $K(s)$ are, respectively, the Laplace transforms of $f(x)$ and $k(x)$.

(b) In terms of the notation of part (a), show that the Volterra equation of the second kind,

$$\varphi(x) = f(x) + \lambda\int_0^x K(x,t)\varphi(t)\,dt,$$

has solution

$$\varphi(x) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} \frac{F(s)}{1-\lambda K(s)}\,e^{xs}\,ds.$$

Assume each quotient is a Laplace transform for which the Bromwich inversion is valid. Choose γ in its right half-plane of analyticity, to the right of its singularities.

Solution.

For $x \geq 0$ use

$$F(s) = \int_0^\infty e^{-sx}f(x)dx, \quad K(s) = \int_0^\infty e^{-sx}k(x)dx, \quad \Phi(s) = \int_0^\infty e^{-sx}\varphi(x)dx.$$

In a right half-plane where the integrals can be interchanged, reverse the order on the region $0 \leq t \leq x < \infty$ (that is, $t \geq 0$, $x \geq t$) and then put $u = x - t$, $e^{-sx} = e^{-su}e^{-st}$:

$$\begin{aligned}\int_0^\infty e^{-sx}\int_0^x k(x-t)\varphi(t)dt\,dx &= \int_0^\infty \varphi(t)\left[\int_t^\infty e^{-sx}k(x-t)\,dx\right]dt \\ &= \int_0^\infty \varphi(t)e^{-st}\left[\int_0^\infty e^{-su}k(u)\,du\right]dt = K(s)\Phi(s).\end{aligned}$$

(a) Transforming the first-kind equation gives $F = K\Phi$, hence

$$\Phi(s) = \frac{F(s)}{K(s)}, \qquad \boxed{\varphi(x) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty}\frac{F(s)}{K(s)}e^{sx}ds}.$$

(b) Transforming the second-kind equation gives $\Phi = F + \lambda K\Phi$, i.e. $(1-\lambda K)\Phi = F$, so

$$\Phi(s) = \frac{F(s)}{1-\lambda K(s)}, \qquad \boxed{\varphi(x) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty}\frac{F(s)}{1-\lambda K(s)}e^{sx}ds}.$$

In each case the quotient must be an admissible Laplace transform; its singularities determine the inversion line $\Re s = \gamma$. A zero of the divisor ($K(s)$ in (a), $1-\lambda K(s)$ in (b)) that is not cancelled by a zero of $F(s)$ is such a singularity, since the quotient is unbounded near it. When classical inversion hypotheses hold, choose γ to the right of all quotient singularities and use the ordinary Bromwich inverse.

21.2.3 Using the Laplace transform solution (Exercise 21.2.2), solve

(a) $\varphi(x) = x + \int_0^x (t-x)\varphi(t)\,dt.$

ANS. $\varphi(x) = \sin x.$

(b) $\varphi(x) = x - \int_0^x (t-x)\varphi(t)\,dt.$

ANS. $\varphi(x) = \sinh x.$

Check your results by substituting back into the original integral equations.

Solution.

Both equations have the form of Exercise 21.2.2(b) with $\lambda = 1$ and $f(x) = x$. In (a) the kernel is $t - x = -(x-t)$, so $k(u) = -u$; in (b) it is $-(t-x) = x - t$, so $k(u) = u$. Since $\mathcal{L}\{x\} = 1/s^2$, the transforms are $F = s^{-2}$ and $K = \mp s^{-2}$:

	transformed equation	$\Phi(s)$
(a)	$\Phi = s^{-2} - s^{-2}\Phi$	$(s^2+1)^{-1}$
(b)	$\Phi = s^{-2} + s^{-2}\Phi$	$(s^2-1)^{-1}$

(multiply each transformed equation by s^2: $(s^2 \pm 1)\Phi = 1$). By the standard transforms $\mathcal{L}\{\sin x\} = 1/(s^2+1)$ and $\mathcal{L}\{\sinh x\} = 1/(s^2-1)$, the inverses are

$$\boxed{\varphi_a(x) = \sin x, \qquad \varphi_b(x) = \sinh x}.$$

For an explicit substitution check, integration by parts (differentiate $x - t$, integrate $\sin t$ or $\sinh t$) gives

$$\int_0^x (x-t)\sin t\,dt = [-(x-t)\cos t]_0^x - \int_0^x \cos t\,dt = x - \sin x,$$

and

$$\int_0^x (x-t)\sinh t\,dt = [(x-t)\cosh t]_0^x + \int_0^x \cosh t\,dt = \sinh x - x.$$

Since $\int_0^x (t-x)\varphi\,dt = -\int_0^x (x-t)\varphi\,dt$, equation (a) therefore reads $x - (x - \sin x) = \sin x$; (b) reads $x + (\sinh x - x) = \sinh x$.

21.2.4 Reformulate the equations of Example 21.2.1 for integrals on the range $(0, \infty)$ using Fourier cosine transforms. Take the unknown and convolution kernel to be even functions on the full line, and retain the reflected kernel in the half-line equation.

Solution.

Take k and φ even on the full line. Their convolution $f(x) = \int_{\mathbb{R}} k(x-t)\varphi(t)\,dt$ is even as well: replacing t by $-t$ in $f(-x)$ and using $k(-y) = k(y)$, $\varphi(-t) = \varphi(t)$ gives back $f(x)$. Split the full-line equation of Example 21.2.1 into positive and negative t, and change the negative half-line by $t \mapsto -t$:

$$\int_{-\infty}^0 k(x-t)\varphi(t)\,dt = \int_0^\infty k(x+t)\varphi(-t)\,dt = \int_0^\infty k(x+t)\varphi(t)\,dt.$$

Hence, for $x \geq 0$,

$$\boxed{f(x) = \int_0^\infty [k(x-t) + k(x+t)]\varphi(t)\,dt}.$$

A half-line cosine formulation requires these even extensions and the reflected kernel $k(x+t)$. Simply replacing the full-line limits by $0, \infty$ while keeping only $k(x-t)$ does not yield the same convolution theorem.

Define the self-inverse cosine transform

$$G_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty g(x) \cos(\omega x) dx.$$

For even g, this equals its symmetric full-line Fourier transform, because $g(x)\sin(\omega x)$ is odd and $g(x)\cos(\omega x)$ is even:

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(x) e^{i\omega x} dx = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(x) \cos(\omega x) dx = \frac{2}{\sqrt{2\pi}} \int_0^\infty g(x) \cos(\omega x) dx = G_c(\omega).$$

Thus the equations of the example become

$$F_c(\omega) = \sqrt{2\pi} K_c(\omega) \Phi_c(\omega), \qquad \Phi_c(\omega) = \frac{F_c(\omega)}{\sqrt{2\pi} K_c(\omega)}.$$

Since G_c is even in ω, the same argument applied to the inverse transform gives

$$g(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty G_c(\omega) \cos(\omega x) d\omega,$$

so the cosine transform is its own inverse. For completeness, inversion of the first relation gives

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \sqrt{2\pi} K_c(\omega) \Phi_c(\omega) \cos(\omega x) d\omega = 2 \int_0^\infty K_c(\omega) \Phi_c(\omega) \cos(\omega x) d\omega.$$

Inverting the second, with $\sqrt{2/\pi}/\sqrt{2\pi} = 1/\pi$, gives the desired half-line solution:

$$\boxed{\varphi(x) = \frac{1}{\pi} \int_0^\infty \frac{F_c(\omega)}{K_c(\omega)} \cos(\omega x) d\omega}.$$

The quotient and inversion need the same existence conditions as in Exercise 21.2.1. If the half-line equation is instead written with one half of the reflected sum, $f(x) = \frac{1}{2}\int_0^\infty [k(x-t)+k(x+t)]\varphi(t)\,dt$, then f is one half of the full-line convolution, so $F_c = \sqrt{\pi/2}\, K_c \Phi_c$, and the factor $1/\pi$ in the solution becomes $\sqrt{2/\pi}/\sqrt{\pi/2} = 2/\pi$.

21.2.5 Given the Fredholm integral equation,

$$e^{-x^2} = \int_{-\infty}^\infty e^{-(x-t)^2} \varphi(t)\, dt,$$

apply the Fourier convolution technique of Example 21.2.1 to solve for $\varphi(t)$ as a tempered distribution.

Solution.

The symmetric Fourier transform of e^{-x^2} follows by completing the square, $-x^2 + i\omega x = -(x - i\omega/2)^2 - \omega^2/4$, and using $\int_{\mathbb{R}} e^{-(x-i\omega/2)^2} dx = \sqrt{\pi}$ (the Gaussian integral with its contour shifted parallel to the real axis):

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2 + i\omega x} dx = \frac{\sqrt{\pi}}{\sqrt{2\pi}} e^{-\omega^2/4} = \frac{1}{\sqrt{2}} e^{-\omega^2/4}.$$

Here $f(x) = e^{-x^2}$ and the kernel is $k(x-t)$ with $k(u) = e^{-u^2}$, so $F = K = e^{-\omega^2/4}/\sqrt{2}$. The convolution theorem $F = \sqrt{2\pi} K\Phi$ therefore gives

$$\frac{1}{\sqrt{2}} e^{-\omega^2/4} = \sqrt{2\pi}\left(\frac{1}{\sqrt{2}} e^{-\omega^2/4}\right)\Phi(\omega).$$

The Gaussian factor never vanishes, so $\Phi(\omega) = 1/\sqrt{2\pi}$. But this is precisely the Fourier transform of the unit point mass at zero, $(2\pi)^{-1/2}\int_{\mathbb{R}} \delta(t) e^{i\omega t} dt = 1/\sqrt{2\pi}$; equivalently, the inverse transform is

$$\varphi(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-i\omega t} d\omega = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\omega t} d\omega,$$

the integral representation of the delta function:

$$\boxed{\varphi(t) = \delta(t)}.$$

Indeed, the original equation is checked without transforms:

$$\int_{\mathbb{R}} e^{-(x-t)^2} \delta(t)\, dt = e^{-x^2}.$$

The answer is a distribution. There is no $L^1(\mathbb{R})$ solution, since its Fourier transform would tend to zero at infinity; nor an $L^2(\mathbb{R})$ solution, since a nonzero constant transform is not square integrable. In the tempered-distribution class the nonzero Gaussian multiplier makes the solution unique: the transform Ψ of the difference of two solutions satisfies $e^{-\omega^2/4}\Psi = 0$, so $\langle \Psi, \chi\rangle = \langle e^{-\omega^2/4}\Psi,\, e^{\omega^2/4}\chi\rangle = 0$ for every smooth compactly supported χ (since $e^{\omega^2/4}\chi$ is again such a function), and therefore $\Psi = 0$.

21.2.6 Solve Abel's equation,

$$f(x) = \int_0^x \frac{\varphi(t)}{(x-t)^\alpha}\, dt, \qquad 0 < \alpha < 1,$$

by the following method:

(a) Multiply both sides by $(z-x)^{\alpha-1}$ and integrate with respect to x over the range $0 \le x \le z$.

(b) Reverse the order of integration and evaluate the integral on the right-hand side (with respect to x) by recognizing it as a beta function.

Note.
$$\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^\alpha} = B(1-\alpha,\alpha) = \Gamma(\alpha)\Gamma(1-\alpha) = \frac{\pi}{\sin \pi\alpha}.$$

Solution.

For locally integrable φ, the singular kernels are integrable when $0 < \alpha < 1$.
(a) Multiply the equation by $(z-x)^{\alpha-1}$ and integrate over $0 \le x \le z$:

$$I(z) := \int_0^z (z-x)^{\alpha-1} f(x) dx = \int_0^z dx \int_0^x (z-x)^{\alpha-1}(x-t)^{-\alpha}\varphi(t) dt.$$

(b) The integration region is the triangle $0 \le t \le x \le z$. Absolute integrability on this triangle follows from the beta integral below applied to $|\varphi|$. Reversing the order (t from 0 to z, then x from t to z),

$$I(z) = \int_0^z \varphi(t) \left[\int_t^z (z-x)^{\alpha-1}(x-t)^{-\alpha} dx\right] dt.$$

Put $x = t + (z-t)u$, so $dx = (z-t)du$, $z - x = (z-t)(1-u)$, $x - t = (z-t)u$, and $x = t, z$ correspond to $u = 0, 1$. The powers of $z - t$ cancel, since $(\alpha - 1) - \alpha + 1 = 0$:

$$\begin{aligned}\int_t^z (z-x)^{\alpha-1}(x-t)^{-\alpha}dx &= \int_0^1 (1-u)^{\alpha-1}u^{-\alpha}du \\ &= B(1-\alpha,\alpha) = \frac{\Gamma(1-\alpha)\Gamma(\alpha)}{\Gamma(1)} = \frac{\pi}{\sin\pi\alpha},\end{aligned}$$

by the reflection formula $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin\pi\alpha$. Hence

$$I(z) = \frac{\pi}{\sin\pi\alpha}\int_0^z \varphi(t)dt.$$

Differentiating with respect to z (fundamental theorem of calculus) and solving for φ,

$$\boxed{\varphi(z) = \frac{\sin\pi\alpha}{\pi}\frac{d}{dz}\int_0^z (z-x)^{\alpha-1}f(x)dx}.$$

For merely locally integrable φ, $I(z)$ is $\pi/\sin\pi\alpha$ times the indefinite integral $\int_0^z \varphi(t)\,dt$, so by the Lebesgue differentiation theorem $I'(z) = (\pi/\sin\pi\alpha)\varphi(z)$ for almost every z.

If f is continuously differentiable up to zero, remove the apparent singular endpoint before differentiating. Integrate by parts, differentiating $f(x)$ and integrating $(z-x)^{\alpha-1}$ to $-(z-x)^\alpha/\alpha$; the boundary term vanishes at $x = z$ because $\alpha > 0$:

$$I(z) = \left[-\frac{f(x)(z-x)^\alpha}{\alpha}\right]_0^z + \frac{1}{\alpha}\int_0^z (z-x)^\alpha f'(x)dx = \frac{f(0)z^\alpha}{\alpha} + \frac{1}{\alpha}\int_0^z (z-x)^\alpha f'(x)dx.$$

Now differentiate: $\frac{d}{dz}\, f(0)z^\alpha/\alpha = f(0)z^{\alpha-1}$, and by Leibniz's rule the upper-endpoint term $(z - z)^\alpha f'(z)/\alpha$ vanishes while $\partial_z(z-x)^\alpha/\alpha = (z-x)^{\alpha-1}$. Thus

$$\boxed{\varphi(z) = \frac{\sin\pi\alpha}{\pi}\left[f(0)z^{\alpha-1} + \int_0^z (z-x)^{\alpha-1}f'(x)dx\right]}.$$

This also proves sufficiency for such f: convolving either term with $x^{-\alpha}$ and using the same beta integral yields respectively $f(0)$ and $\int_0^x f'(y)dy = f(x) - f(0)$. Indeed, with $t = xu$,

$$\int_0^x (x-t)^{-\alpha}t^{\alpha-1}dt = \int_0^1 (1-u)^{-\alpha}u^{\alpha-1}du = B(\alpha,1-\alpha) = \frac{\pi}{\sin\pi\alpha},$$

and, reversing the order of integration as above,

$$\int_0^x \frac{dt}{(x-t)^\alpha}\int_0^t (t-y)^{\alpha-1}f'(y)\,dy = \int_0^x f'(y)\left[\int_y^x \frac{(t-y)^{\alpha-1}}{(x-t)^\alpha}dt\right]dy = \frac{\pi}{\sin\pi\alpha}\int_0^x f'(y)\,dy,$$

because the inner integral is the first one shifted: with $t = y+(x-y)u$, $dt = (x-y)du$, $t-y = (x-y)u$, $x-t = (x-y)(1-u)$, the powers of $x-y$ cancel and it again equals $\int_0^1 u^{\alpha-1}(1-u)^{-\alpha}du = \pi/\sin\pi\alpha$. Multiplying the first result by $f(0)\sin\pi\alpha/\pi$ and the second by $\sin\pi\alpha/\pi$, and adding, the two terms give $f(0) + f(x) - f(0) = f(x)$.

21.2.7 Given the generalized Abel equation with $f(x) = 1$,

$$1 = \int_0^x \frac{\varphi(t)}{(x-t)^\alpha}\, dt, \qquad 0 < \alpha < 1, \quad x > 0,$$

solve for $\varphi(t)$ and verify that $\varphi(t)$ is a solution of the given equation.

$$\textit{ANS.}\ \ \varphi(t) = \frac{\sin \pi\alpha}{\pi}\, t^{\alpha-1}.$$

Solution.

Set $f = 1$ in the first inversion formula of Exercise 21.2.6:

$$\int_0^x (x-t)^{\alpha-1} dt = \left[-\frac{(x-t)^\alpha}{\alpha} \right]_0^x = \frac{x^\alpha}{\alpha},$$

so, since $\frac{d}{dx}(x^\alpha/\alpha) = x^{\alpha-1}$,

$$\boxed{\varphi(x) = \frac{\sin \pi\alpha}{\pi} x^{\alpha-1}}, \qquad x > 0.$$

(The second formula of Exercise 21.2.6, with $f(0) = 1$ and $f' = 0$, gives the same result.) Direct substitution, with $t = xu$, $dt = x\, du$, $t^{\alpha-1} = x^{\alpha-1}u^{\alpha-1}$, $(x-t)^{-\alpha} = x^{-\alpha}(1-u)^{-\alpha}$, yields

$$\begin{aligned} \int_0^x \frac{\varphi(t)}{(x-t)^\alpha} dt &= \frac{\sin \pi\alpha}{\pi} x^{\alpha-1-\alpha+1} \int_0^1 u^{\alpha-1}(1-u)^{-\alpha} du \\ &= \frac{\sin \pi\alpha}{\pi} B(\alpha, 1-\alpha) = \frac{\sin \pi\alpha}{\pi} \Gamma(\alpha)\Gamma(1-\alpha) = \frac{\sin \pi\alpha}{\pi} \frac{\pi}{\sin \pi\alpha} = 1. \end{aligned}$$

Both endpoint singularities in the u integral are integrable. The equation is valid for $x > 0$, with right-hand limit one as $x \downarrow 0$. It cannot also be imposed as an ordinary zero-length integral at $x = 0$, which is zero; the solution itself is singular there.

21.2.8 A Fredholm equation of the first kind has a kernel $e^{-(x-t)^2}$:

$$f(x) = \int_{-\infty}^{\infty} e^{-(x-t)^2} \varphi(t)\, dt.$$

Assume a solution exists in $L^2(\mathbb{R}, e^{-t^2} dt)$. Show that the solution is the following series, converging in that weighted norm:

$$\varphi(x) = \frac{1}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{2^n n!}\, H_n(x),$$

in which $H_n(x)$ is an nth-order Hermite polynomial.

Solution.

Use the physicists' Hermite generating function:

$$e^{-(x-t)^2} = e^{-t^2} e^{2tx - x^2} = e^{-t^2} \sum_{n=0}^{\infty} H_n(t) \frac{x^n}{n!}.$$

Taking the nth derivative with respect to x at zero in the integral equation (only the x^n term of the series survives, giving $H_n(t)$; differentiation under the integral is justified below) gives

$$f^{(n)}(0) = \int_{\mathbb{R}} e^{-t^2} H_n(t) \varphi(t) dt.$$

Now write $\varphi = \sum_n a_n H_n$. Hermite orthogonality is

$$\int_{\mathbb{R}} e^{-t^2} H_n(t) H_m(t) dt = \sqrt{\pi}\, 2^n n!\, \delta_{nm}.$$

Multiplying the expansion by $e^{-t^2} H_n$ and integrating therefore gives

$$a_n = \frac{1}{\sqrt{\pi}\, 2^n n!} \int_{\mathbb{R}} e^{-t^2} H_n(t) \varphi(t) dt = \frac{f^{(n)}(0)}{\sqrt{\pi}\, 2^n n!}.$$

Consequently

$$\boxed{\varphi(x) = \frac{1}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{2^n n!} H_n(x)}.$$

The textbook's unrestricted statement needs a function-space and convergence qualification. If $\varphi \in L^2(\mathbb{R}, e^{-t^2} dt)$, the formula is its convergent weighted-L^2 Hermite expansion; it is not automatically a pointwise series for arbitrary f. Here is the convergence justification. Write $z = a + ib$ and $\|\varphi\|_w^2 = \int e^{-t^2} |\varphi(t)|^2 dt$. Since $\Re(z-t)^2 = (a-t)^2 - b^2$,

$$e^{t^2} \left| e^{-(z-t)^2} \right|^2 = e^{t^2 - 2(a-t)^2 + 2b^2} = e^{-t^2 + 4at - 2\Re(z^2)}, \qquad \Re(z^2) = a^2 - b^2.$$

Cauchy–Schwarz, applied to the product of $e^{-t^2/2}\varphi(t)$ and $e^{t^2/2} e^{-(z-t)^2}$, and $\int_{\mathbb{R}} e^{-t^2+4at} dt = \sqrt{\pi}\, e^{4a^2}$ (complete the square) give

$$\begin{aligned} \left| \int_{\mathbb{R}} e^{-(z-t)^2} \varphi(t) dt \right| &\le \|\varphi\|_w \left(\int_{\mathbb{R}} e^{-t^2+4at-2\Re(z^2)} dt \right)^{1/2} = \|\varphi\|_w \left(\sqrt{\pi}\, e^{2a^2+2b^2} \right)^{1/2} \\ &= \pi^{1/4} e^{|z|^2} \|\varphi\|_w. \end{aligned}$$

Derivatives of the Gaussian add only polynomial factors in $t - z$. On a compact set of z, these are still dominated by an integrable Gaussian times a polynomial after the same Cauchy–Schwarz estimate. Differentiation under the integral is therefore valid to every order, locally uniformly, and the convolution is entire. Parseval's identity for the Hermite basis gives

$$\|\varphi\|_w^2 = \sum_{n=0}^{\infty} |a_n|^2 \sqrt{\pi}\, 2^n n!.$$

Substituting the coefficients found above, $|a_n|^2 \sqrt{\pi}\, 2^n n! = |f^{(n)}(0)|^2 \sqrt{\pi}\, 2^n n! / (\pi\, 4^n (n!)^2)$, gives the necessary condition

$$\sum_{n=0}^{\infty} \frac{|f^{(n)}(0)|^2}{\sqrt{\pi}\, 2^n n!} < \infty.$$

Conversely, if f is entire and this condition holds, the orthogonal series defines a weighted-L^2 function. By the moment formula above, its Gaussian convolution has nth derivative $\sqrt{\pi}\, 2^n n!\, a_n = f^{(n)}(0)$ at zero, i.e. the same Taylor coefficients as f; being entire, it equals f.

The limitation is already visible in Exercise 21.2.5: $f(x) = e^{-x^2}$ has distributional solution δ. Setting $t = 0$ in the generating function gives $e^{-x^2} = \sum_n H_n(0) x^n / n!$, so $f^{(n)}(0) = H_n(0)$; comparing with $e^{-x^2} = \sum_j (-1)^j x^{2j} / j!$, both vanish for odd n and $f^{(2j)}(0) = H_{2j}(0) = (-1)^j (2j)! / j!$. At $x = 0$ the proposed series therefore has the even terms

$$\frac{1}{\sqrt{\pi}} \frac{f^{(2j)}(0) H_{2j}(0)}{2^{2j} (2j)!} = \frac{1}{\sqrt{\pi}} \frac{[(2j)!]^2}{(j!)^2\, 4^j (2j)!} = \frac{1}{\sqrt{\pi}} \frac{\binom{2j}{j}}{4^j} \sim \frac{1}{\pi \sqrt{j}},$$

where Stirling's formula gives $\binom{2j}{j} \sim 4^j / \sqrt{\pi j}$. By comparison with $\sum_j j^{-1/2}$ the positive series diverges, so that example cannot be interpreted as an ordinary pointwise Hermite solution.

21.2.9 Solve the integral equation

$$f(x) = \int_{-1}^{1} \frac{\varphi(t)}{(1 - 2xt + x^2)^{1/2}}\, dt, \qquad -1 \le x \le 1,$$

for the unknown function $\varphi(t)$, with $s = 0, 1, 2, \ldots$, if

(a) $f(x) = x^{2s}$,

(b) $f(x) = x^{2s+1}$.

ANS. (a) $\varphi(t) = \dfrac{4s+1}{2}\, P_{2s}(t)$, (b) $\varphi(t) = \dfrac{4s+3}{2}\, P_{2s+1}(t)$.

Solution.

Let $N = 2s$ in (a) and $N = 2s + 1$ in (b), with $s = 0, 1, \ldots$. For $|x| < 1$, the generating function converges uniformly in $t \in [-1, 1]$ (Weierstrass M-test: $|P_n(t)| \le 1$ for $-1 \le t \le 1$, so $|P_n(t)x^n| \le |x|^n$ there):

$$(1 - 2xt + x^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(t) x^n.$$

Integrating termwise against φ,

$$f(x) = \sum_{n=0}^{\infty} x^n \int_{-1}^{1} P_n(t)\varphi(t)dt, \qquad |x| < 1.$$

Comparing the coefficients of this power series with those of $f(x) = x^N$ (power series coefficients are unique), the equation requires

$$\int_{-1}^{1} P_n(t)\varphi(t)dt = \delta_{nN}.$$

If $\varphi = \sum_m a_m P_m$ in $L^2[-1, 1]$, orthogonality, $\int_{-1}^{1} P_n P_m\, dt = 2\delta_{nm}/(2n+1)$, gives

$$\frac{2a_n}{2n+1} = \delta_{nN}, \qquad a_n = \frac{2n+1}{2}\delta_{nN}.$$

Thus, with $2N + 1 = 4s + 1$ in (a) and $4s + 3$ in (b),

$$\boxed{\varphi_a(t) = \frac{4s+1}{2} P_{2s}(t)}, \qquad \boxed{\varphi_b(t) = \frac{4s+3}{2} P_{2s+1}(t)}.$$

Substitution gives

$$\frac{2N+1}{2} \sum_{n=0}^{\infty} x^n \int_{-1}^{1} P_n(t) P_N(t) dt = \frac{2N+1}{2}\, \frac{2}{2N+1}\, x^N = x^N,$$

which checks both formulas. Completeness of the Legendre polynomials proves uniqueness in L^2: the difference of two solutions has all Legendre moments zero, hence vanishes.

The endpoint values follow by a limit of these finite-polynomial solutions. For $1/2 \le x < 1$,

$$1 - 2xt + x^2 = (1 - x)^2 + 2x(1 - t) \ge 2x(1 - t) \ge 1 - t.$$

Since the numerator $\varphi(t)$ is bounded, the integrand $\varphi(t)(1 - 2xt + x^2)^{-1/2}$ is consequently dominated in absolute value, uniformly in x, by $\max|\varphi|\,(1 - t)^{-1/2}$, which is integrable on $[-1, 1]$. Dominated convergence as $x \to 1^-$ gives $f(1) = \lim_{x\to 1^-} x^N = 1$. Reflection ($t \to -t$, with $\varphi(-t) = (-1)^N \varphi(t)$) gives $f(-1) = (-1)^N$, so the solutions hold at both stated endpoints as improper integrals.

21.2.10 Find the eigenvalues and eigenfunctions of

$$\varphi(x) = \lambda \int_{-1}^{1} (t - x)\varphi(t)\, dt.$$

Solution.

The image of the kernel is linear in x:

$$\int_{-1}^{1} (t - x)\varphi(t)dt = \int_{-1}^{1} t\varphi(t)dt - x\int_{-1}^{1} \varphi(t)dt.$$

Every nonzero solution must therefore have the form $A + Bx$, with $A = \lambda \int_{-1}^{1} t\varphi(t)dt$ and $B = -\lambda \int_{-1}^{1} \varphi(t)dt$. The two required moments are (odd integrands drop out)

$$\int_{-1}^{1} (A + Bt)dt = 2A, \qquad \int_{-1}^{1} t(A + Bt)dt = \frac{2B}{3}.$$

Equating constant and linear coefficients gives

$$A = \frac{2\lambda}{3}B, \qquad B = -2\lambda A.$$

A nonzero solution has $A \neq 0$, since $A = 0$ would force $B = -2\lambda A = 0$. Eliminating B yields $A = \frac{2\lambda}{3}(-2\lambda A) = -\frac{4}{3}\lambda^2 A$, so

$$1 = -\frac{4}{3}\lambda^2, \qquad \lambda^2 = -\frac{3}{4}, \qquad \boxed{\lambda_{\pm} = \pm\frac{i\sqrt{3}}{2}}.$$

Choose $A = 1$, so $B = -2\lambda_{\pm} = \mp i\sqrt{3}$:

$$\boxed{\varphi_+(x) = 1 - i\sqrt{3}\, x, \qquad \varphi_-(x) = 1 + i\sqrt{3}\, x}.$$

Their nonzero scalar multiples give the respective one-dimensional eigenspaces. No real λ permits a nontrivial solution. The imaginary characteristic values are consistent with the real antisymmetric kernel $K(x, t) = -K(t, x)$; the self-adjoint reality theorem does not apply.

21.2.11 Find the eigenvalues and eigenfunctions of

$$\varphi(x) = \lambda \int_{0}^{2\pi} \cos(x - t)\varphi(t)\, dt.$$

ANS. $\lambda_1 = \lambda_2 = \dfrac{1}{\pi}, \qquad \varphi(x) = A\cos x + B\sin x.$

Solution.

Use

$$\cos(x - t) = \cos x \cos t + \sin x \sin t.$$

Hence

$$\int_{0}^{2\pi} \cos(x - t)\varphi(t)\, dt = \cos x \int_{0}^{2\pi} \cos t\, \varphi(t)\, dt + \sin x \int_{0}^{2\pi} \sin t\, \varphi(t)\, dt,$$

a combination of $\cos x$ and $\sin x$ with constant coefficients. Thus every nonzero solution is $\varphi(x) = A\cos x + B\sin x$. Over $[0, 2\pi]$,

$$\int_0^{2\pi} \cos^2 t\, dt = \pi, \quad \int_0^{2\pi} \sin^2 t\, dt = \pi, \quad \int_0^{2\pi} \sin t \cos t\, dt = 0.$$

Inserting $\varphi(t) = A\cos t + B\sin t$, the two coefficient integrals are πA and πB, so the integral is $\pi A\cos x + \pi B\sin x = \pi\varphi(x)$. The equation becomes $(1 - \lambda\pi)\varphi = 0$, so

$$\boxed{\lambda = \frac{1}{\pi}, \qquad \varphi(x) = A\cos x + B\sin x, \quad (A, B) \neq (0, 0)}.$$

The characteristic value has a two-dimensional eigenspace. A function orthogonal to both sine and cosine is annihilated by the integral operator; it does not solve $\varphi = \lambda K\varphi$ nontrivially for any finite λ.

21.2.12 Find the eigenvalues and eigenfunctions of

$$y(x) = \lambda \int_{-1}^{1} (x - t)^2 y(t)\, dt.$$

Hint. This problem may be treated by the separable-kernel method or by a Legendre expansion.

Solution.

Expanding $(x - t)^2 = x^2 - 2xt + t^2$ and writing $m_k = \int_{-1}^{1} t^k y(t)dt$,

$$\int_{-1}^{1} (x - t)^2 y(t)\, dt = m_2 - 2m_1 x + m_0 x^2,$$

a quadratic polynomial in x. So every nonzero solution has the form $y(x) = A + Bx + Cx^2$. Its three moments are (odd powers of t integrate to zero)

$$m_0 = 2A + \frac{2C}{3}, \quad m_1 = \frac{2B}{3}, \quad m_2 = \frac{2A}{3} + \frac{2C}{5}.$$

Equating the coefficients of $1, x, x^2$ in $y = \lambda(m_2 - 2m_1 x + m_0 x^2)$ gives $A = \lambda m_2$, $B = -2\lambda m_1$, $C = \lambda m_0$, that is,

$$\begin{pmatrix} A \\ B \\ C \end{pmatrix} = \lambda \begin{pmatrix} 2/3 & 0 & 2/5 \\ 0 & -4/3 & 0 \\ 2 & 0 & 2/3 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}.$$

The odd equation is $B = -4\lambda B/3$; a solution with $B \neq 0$ requires $1 = -4\lambda/3$, giving

$$\boxed{\lambda_0 = -\frac{3}{4}, \qquad y_0(x) = x}.$$

For the even sector (A, C), put $\mu = 1/\lambda$, so that the first and last rows read $\mu A = \frac{2}{3}A + \frac{2}{5}C$ and $\mu C = 2A + \frac{2}{3}C$. A nonzero (A, C) requires the determinant

$$\begin{vmatrix} 2/3 - \mu & 2/5 \\ 2 & 2/3 - \mu \end{vmatrix} = (2/3 - \mu)^2 - \frac{4}{5} = 0,$$

hence $2/3-\mu=\mp 2/\sqrt{5}$ and $\mu_\pm = 2/3 \pm 2/\sqrt{5}$. Taking $C=1$, the last row says $2A+2/3=\mu_\pm$, so $A=(\mu_\pm - 2/3)/2 = \pm 1/\sqrt{5}$. The first row then holds as well: $(\mu_\pm - \frac{2}{3})A = (\pm\frac{2}{\sqrt{5}})(\pm\frac{1}{\sqrt{5}}) = \frac{2}{5}$. The eigenvalues are $\lambda_\pm = 1/\mu_\pm$; rationalizing,

$$\frac{1}{\frac{2}{3}\pm\frac{2}{\sqrt{5}}} = \frac{3\sqrt{5}}{2\sqrt{5}\pm 6} = \frac{3\sqrt{5}\,(2\sqrt{5}\mp 6)}{20-36} = \frac{30\mp 18\sqrt{5}}{-16} = \frac{-15\pm 9\sqrt{5}}{8}.$$

Therefore

$$\boxed{\begin{aligned} \lambda_+ &= \frac{-15+9\sqrt{5}}{8}, \quad y_+(x) = x^2 + \frac{1}{\sqrt{5}}, \\ \lambda_- &= \frac{-15-9\sqrt{5}}{8}, \quad y_-(x) = x^2 - \frac{1}{\sqrt{5}}. \end{aligned}}$$

All three eigenpairs satisfy every row of the matrix equation. Since the image of the integral operator contains only quadratic polynomials, the calculation exhausts all nonzero finite-characteristic-value solutions.

In the Legendre method of the hint, $x^2 = \frac{1}{3}P_0 + \frac{2}{3}P_2(x)$ gives

$$(x-t)^2 = \frac{2}{3} + \frac{2}{3}\big[P_2(x)+P_2(t)\big] - 2P_1(x)P_1(t).$$

For even $y = a_0P_0 + a_2P_2$, orthogonality ($\int_{-1}^1 P_n^2 dt = 2/(2n+1)$) gives

$$\int_{-1}^1 (x-t)^2 y(t)\,dt = \frac{4a_0}{3} + \frac{4a_2}{15} + \frac{4a_0}{3}P_2(x),$$

and comparing the coefficients of P_2 and P_0 in $y = \lambda\int_{-1}^1 (x-t)^2 y(t)\,dt$,

$$a_2 = \frac{4\lambda}{3}a_0, \qquad a_0 = \lambda\left(\frac{4a_0}{3} + \frac{4a_2}{15}\right).$$

With $a_0=1$, $y_\pm$ are proportional to $P_0 + (4\lambda_\pm/3)P_2$, and the second equation becomes $1 = \lambda\big(\frac{4}{3} + \frac{16\lambda}{45}\big)$, i.e. $16\lambda^2 + 60\lambda - 45 = 0$, whose roots $(-60\pm 36\sqrt{5})/32$ are again $\lambda_\pm$. The odd function P_1 gives $\int_{-1}^1 (x-t)^2 P_1(t)\,dt = -\frac{4}{3}P_1(x)$, i.e. λ_0. To compare with the monomial form: since $P_2 = (3x^2-1)/2$, $P_0 + (4\lambda_\pm/3)P_2$ has constant term $1 - 2\lambda_\pm/3$ and x^2 coefficient $2\lambda_\pm$, whose ratio is $1/(2\lambda_\pm) - 1/3 = \mu_\pm/2 - 1/3 = \pm 1/\sqrt{5}$, as for $y_\pm$.

21.2.13 Use the separable-kernel technique to show that

$$\psi(x) = \lambda\int_0^\pi \cos x \sin t\,\psi(t)\,dt$$

has no solution (apart from $\psi = 0$). Explain this result in terms of separability and symmetry.

Solution.

Put

$$c = \int_0^\pi \sin t\,\psi(t) dt.$$

Separability forces $\psi(x) = \lambda c\cos x$. Substituting this form into the definition of c,

$$c = \lambda c\int_0^\pi \sin t\cos t\,dt = \frac{\lambda c}{2}\,[\sin^2 t]_0^\pi = 0.$$

Then $\psi(x) = \lambda c \cos x$ vanishes identically: $\boxed{\psi = 0}$ for every finite λ.

The cancellation also follows from reflection about $\pi/2$: $\sin(\pi - t) = \sin t$ and $\cos(\pi - t) = -\cos t$, so the product has zero integral. In operator language, K has rank one but

$$K^2 f(x) = \cos x \left(\int_0^\pi \sin t \cos t \, dt \right) \left(\int_0^\pi \sin u \, f(u) du \right) = 0.$$

The operator itself is nonzero, since $K(\sin t) = (\pi/2) \cos x$. Its kernel is not symmetric under $x \leftrightarrow t$, so a nonzero nilpotent operator is allowed. A nonzero real symmetric rank-one kernel $c\,u(x)u(t)$ would instead have nonzero operator eigenvalue $c \int u^2$, and hence a nontrivial finite characteristic value. (For complex u the qualification matters: $u(x) = e^{2ix}$ gives $\int_0^\pi u^2 dx = 0$, and the symmetric kernel $u(x)u(t)$ is again nilpotent.)

21.2.14 Given

$$\varphi(x) = \lambda \int_0^1 (1 + xt)\varphi(t) \, dt,$$

solve for the eigenvalues and the eigenfunctions by the separable-kernel technique.

Solution.

Introduce the two moments

$$c_0 = \int_0^1 \varphi(t) dt, \qquad c_1 = \int_0^1 t\varphi(t) dt.$$

Since $1 + xt$ splits as $1 \cdot 1 + x \cdot t$, the equation is $\varphi(x) = \lambda(c_0 + x c_1)$. Inserting it into those moment definitions, with $\int_0^1 dt = 1$, $\int_0^1 t \, dt = \frac{1}{2}$, $\int_0^1 t^2 dt = \frac{1}{3}$, gives

$$c_0 = \lambda \left(c_0 + \frac{c_1}{2} \right), \qquad c_1 = \lambda \left(\frac{c_0}{2} + \frac{c_1}{3} \right).$$

A nonzero moment vector requires

$$0 = \begin{vmatrix} 1 - \lambda & -\lambda/2 \\ -\lambda/2 & 1 - \lambda/3 \end{vmatrix} = (1 - \lambda)(1 - \lambda/3) - \lambda^2/4$$
$$= 1 - \frac{\lambda}{3} - \lambda + \frac{\lambda^2}{3} - \frac{\lambda^2}{4} = 1 - \frac{4}{3}\lambda + \frac{1}{12}\lambda^2.$$

Multiplying by 12, $\lambda^2 - 16\lambda + 12 = 0$, and the quadratic formula, $\lambda = 8 \mp \sqrt{64 - 12} = 8 \mp \sqrt{52}$, gives

$$\boxed{\lambda_\pm = 8 \mp 2\sqrt{13}}.$$

The constant term of a nonzero eigenfunction cannot vanish: otherwise $\varphi = Bx$ would require $0 = \lambda c_0 = \lambda B/2$, forcing $B = 0$. Normalize that term to one, writing $\varphi_\pm = 1 + b_\pm x$. Then $c_0 = 1 + b/2$, and the constant equation $1 = \lambda(1 + b/2)$ yields $b = 2/\lambda - 2$. Rationalizing,

$$\frac{2}{\lambda_\pm} = \frac{2}{8 \mp 2\sqrt{13}} = \frac{2(8 \pm 2\sqrt{13})}{64 - 52} = \frac{4 \pm \sqrt{13}}{3}, \qquad b_\pm = \frac{4 \pm \sqrt{13}}{3} - 2,$$

hence

$$\boxed{\varphi_\pm(x) = 1 + \frac{-2 \pm \sqrt{13}}{3} x}.$$

Numerically the pairs are

$$(0.7888974491,\ 1+0.5351837585x) \quad \text{and} \quad (15.2111025509,\ 1-1.8685170918x).$$

The x coefficient of the equation, $b=\lambda c_1=\lambda\left(\frac{1}{2}+\frac{b}{3}\right)$, holds as well. With $b=2/\lambda-2$ its right side is

$$\frac{\lambda}{2}+\frac{\lambda}{3}\left(\frac{2}{\lambda}-2\right)=\frac{2}{3}-\frac{\lambda}{6},$$

and $2/\lambda-2=2/3-\lambda/6$ becomes, after multiplication by 6λ, $12-12\lambda=4\lambda-\lambda^2$, i.e. $\lambda^2-16\lambda+12=0$, which $\lambda_\pm$ satisfy. Since the image of K is two-dimensional, these are all the nontrivial eigenspaces.

21.2.15 Knowing the form of the solutions of an integral equation can be a great advantage. For

$$\varphi(x)=\lambda\int_0^1(1+xt)\varphi(t)\,dt,$$

assume $\varphi(x)$ to have the form $1+bx$. Substitute into the integral equation. Integrate and solve for b and λ.

Solution.

Substitute the proposed function before making any spectral assumption. Since $(1+xt)(1+bt)=1+bt+xt+bxt^2$, integrating term by term gives

$$\int_0^1(1+xt)(1+bt)dt=1+\frac{b}{2}+x\left(\frac{1}{2}+\frac{b}{3}\right).$$

Equating coefficients in $1+bx=\lambda[1+b/2+x(1/2+b/3)]$ gives

$$1=\lambda(1+b/2), \qquad b=\lambda(1/2+b/3).$$

The first equation rules out $b=-2$. Eliminating λ (multiply the second equation by $1+b/2$ and use the first),

$$b(1+b/2)=1/2+b/3, \quad \frac{b^2}{2}+\frac{2b}{3}-\frac{1}{2}=0, \quad 3b^2+4b-3=0.$$

The quadratic formula therefore gives

$$b_\pm=\frac{-4\pm\sqrt{16+36}}{6}=\frac{-4\pm2\sqrt{13}}{6}=\frac{-2\pm\sqrt{13}}{3}.$$

Finally, $1+b_\pm/2=(4\pm\sqrt{13})/6$, and rationalizing with $(4\pm\sqrt{13})(4\mp\sqrt{13})=16-13=3$,

$$\lambda_\pm=\frac{1}{1+b_\pm/2}=\frac{6}{4\pm\sqrt{13}}=\frac{6(4\mp\sqrt{13})}{3}=\boxed{8\mp2\sqrt{13}}.$$

Thus $\boxed{\varphi_\pm(x)=1+b_\pm x}$, with arbitrary nonzero overall scale. The ansatz loses no eigenfunctions: the integral's image is linear, and an eigenfunction with zero constant term must vanish, as the constant coefficient equation shows.

21.2.16 The equation

$$f(x) = \int_a^b K(x,t)\varphi(t)\,dt$$

has a degenerate kernel $K(x,t) = \sum_{i=1}^{n} M_i(x)N_i(t)$.

(a) Show that this integral equation has no solution unless $f(x)$ can be written as

$$f(x) = \sum_{i=1}^{n} f_i M_i(x),$$

where the f_i are constants.

(b) Show that to any solution $\varphi(x)$ we may add $\psi(x)$, provided that $\psi(x)$ has zero bilinear moments against every $N_i(x)$:

$$\int_a^b N_i(x)\psi(x)\,dx = 0 \qquad \text{for all } i.$$

Solution.

(a) The finite sum may be brought outside the integral:

$$\int_a^b K(x,t)\varphi(t)dt = \sum_{i=1}^{n} M_i(x) \underbrace{\int_a^b N_i(t)\varphi(t)dt}_{c_i}.$$

Each c_i is independent of x. Hence the equation requires

$$\boxed{f \in \operatorname{span}\{M_1,\dots,M_n\}}, \qquad f(x) = \sum_i c_i M_i(x),$$

i.e. the stated form with constants $f_i = c_i = \int_a^b N_i\varphi\,dt$. This is a necessary condition; to obtain a solution, the moment equations $\int_a^b N_i\varphi\,dt = f_i$ must also be compatible.

(b) If every specified moment of ψ is zero, then

$$K(\varphi+\psi)(x) = K\varphi(x) + \sum_i M_i(x)\int_a^b N_i(t)\psi(t)dt = f(x) + 0.$$

Thus every such $\boxed{\varphi+\psi}$ is another solution.

The Instructor's Manual adds that ψ must itself be a linear combination of the M_i. That extra restriction is false for this first-kind equation. Only the image f, not the unknown or its nullspace additions, is constrained to that span. For a direct counterexample, take $K(x,t) = 1$ on $[0,1]$, $f = 1$, $\varphi = 1$, and $\psi(t) = t - \frac{1}{2}$. Then

$$\int_0^1 \psi(t)dt = 0, \qquad \int_0^1 [1+\psi(t)]dt = 1,$$

although ψ is not in the span of the constant $M_1 = 1$. The displayed conditions $\int_a^b N_i\psi\,dx = 0$ are exactly what the calculation uses. They are called bilinear because no complex conjugate is taken: for complex N_i they differ from Hermitian orthogonality, $\int_a^b N_i^*\psi\,dx = 0$, which would not make the sum $\sum_i M_i\int_a^b N_i\psi\,dt$ vanish. For example, on $[0,2\pi]$, $N_1 = e^{it}$ and $\psi = e^{-it}$ give $\int_0^{2\pi} N_1^*\psi\,dt = \int_0^{2\pi} e^{-2it}dt = 0$ but $\int_0^{2\pi} N_1\psi\,dt = 2\pi$.

21.2.17 A Kirchhoff diffraction theory analysis of a laser leads to the integral equation

$$v(\mathbf{r}_2) = \gamma \iint K(\mathbf{r}_1, \mathbf{r}_2) v(\mathbf{r}_1)\, dA.$$

The unknown, $v(\mathbf{r}_1)$, gives the geometric distribution of the radiation field over one mirror surface; the range of integration is over the surface of that mirror. For square confocal spherical mirrors, the integral equation becomes

$$v(x_2, y_2) = \frac{-i\gamma e^{ikb}}{\lambda b} \int_{-a}^{a}\int_{-a}^{a} e^{-(ik/b)(x_1x_2+y_1y_2)}\, v(x_1, y_1)\, dx_1\, dy_1,$$

in which b is the centerline distance between the laser mirrors. This can be put in a somewhat simpler form by the substitutions

$$\frac{kx_i^2}{b} = \xi_i^2, \qquad \frac{ky_i^2}{b} = \eta_i^2, \qquad \text{and} \qquad \frac{ka^2}{b} = \frac{2\pi a^2}{\lambda b} = \alpha^2.$$

(a) Show that the variables separate and we get two integral equations.

(b) Show that the new limits, $\pm\alpha$, may be approximated by $\pm\infty$ for fixed low-order modes when $a^2 \gg \lambda b$. Explain why $a \gg \lambda$ alone is insufficient.

(c) Solve the resulting integral equations.

Solution.

(a) Choose signed scaled coordinates

$$\xi = \sqrt{k/b}\, x, \qquad \eta = \sqrt{k/b}\, y, \qquad \alpha = a\sqrt{k/b}.$$

This choice retains the signs that were suppressed by the squared relations. The Jacobian is $dx_1 dy_1 = (b/k) d\xi_1 d\eta_1$, the exponent becomes $(k/b)(x_1x_2 + y_1y_2) = \xi_1\xi_2 + \eta_1\eta_2$, the limits $\pm a$ become $\pm\alpha$, and, since $k\lambda = 2\pi$, the prefactor becomes

$$\frac{-i\gamma e^{ikb}}{\lambda b} \cdot \frac{b}{k} = \frac{-i\gamma e^{ikb}}{k\lambda} = \frac{-i\gamma e^{ikb}}{2\pi}.$$

The equation becomes

$$v(\xi_2, \eta_2) = \frac{-i\gamma e^{ikb}}{2\pi} \int_{-\alpha}^{\alpha}\int_{-\alpha}^{\alpha} e^{-i(\xi_1\xi_2+\eta_1\eta_2)} v(\xi_1, \eta_1) d\xi_1 d\eta_1.$$

Define

$$(\mathcal{F}_\alpha h)(u) = \frac{1}{\sqrt{2\pi}} \int_{-\alpha}^{\alpha} e^{-ius} h(s) ds.$$

For a separated mode $v(\xi, \eta) = X(\xi)Y(\eta)$ the exponential factorizes, $e^{-i(\xi_1\xi_2+\eta_1\eta_2)} = e^{-i\xi_1\xi_2} e^{-i\eta_1\eta_2}$, so the double integral is the product $\sqrt{2\pi}(\mathcal{F}_\alpha X)(\xi_2) \cdot \sqrt{2\pi}(\mathcal{F}_\alpha Y)(\eta_2)$ and the factor 2π cancels. The equation is

$$XY = -i\gamma e^{ikb} (\mathcal{F}_\alpha X)(\mathcal{F}_\alpha Y).$$

Where $X(\xi)Y(\eta) \neq 0$, divide by XY:

$$\frac{(\mathcal{F}_\alpha X)(\xi)}{X(\xi)} \cdot \frac{(\mathcal{F}_\alpha Y)(\eta)}{Y(\eta)} = \frac{1}{-i\gamma e^{ikb}}.$$

The first quotient depends only on ξ and the second only on η; holding η fixed and varying ξ shows that the first is a constant ρ, and likewise the second is a constant σ. Thus the two one-dimensional equations and their consistency condition are

$$\boxed{\mathcal{F}_\alpha X = \rho X, \qquad \mathcal{F}_\alpha Y = \sigma Y, \qquad 1 = -i\gamma e^{ikb}\rho\sigma}.$$

These identities also make sense at isolated zeros of X, Y by continuity, without dividing there.

(b) The textbook's condition $a \gg \lambda$ alone does not imply that α is large: the mirror spacing b also enters. The needed large-aperture condition for fixed low-order modes is $a^2 \gg \lambda b$, equivalently $\alpha^2 = 2\pi a^2/(\lambda b) \gg 1$. For example, $a = 10^3\lambda$ and $b = 10^9\lambda$ give $\alpha^2 = 2\pi \cdot 10^6\lambda^2/(10^9\lambda^2) = 2\pi/1000$, much less than one despite $a \gg \lambda$.

Let $\mathcal{F}$ denote $\mathcal{F}_\alpha$ with $\alpha = \infty$, the full Fourier transform with kernel e^{-ius}. The replacement by infinite limits is an approximation for modes concentrated well inside $(-\alpha, \alpha)$. Since $|e^{-ius}| = 1$, its pointwise error obeys

$$|(\mathcal{F} - \mathcal{F}_\alpha)h(u)| = \frac{1}{\sqrt{2\pi}}\left|\int_{|s|>\alpha} e^{-ius}h(s)ds\right| \le \frac{1}{\sqrt{2\pi}}\int_{|s|>\alpha}|h(s)|ds.$$

For a fixed Hermite–Gaussian mode the tail is a polynomial times $e^{-s^2/2}$ and becomes exponentially small as $\alpha \to \infty$. For the ground Gaussian $h_0(s) = \pi^{-1/4}e^{-s^2/2}$, at $u = 0$,

$$\mathcal{F}h_0(0) = \frac{\pi^{-1/4}}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-s^2/2}ds = \pi^{-1/4}, \qquad (\mathcal{F} - \mathcal{F}_\alpha)h_0(0) = \frac{2\pi^{-1/4}}{\sqrt{2\pi}}\int_\alpha^\infty e^{-s^2/2}ds = \pi^{-1/4}\operatorname{erfc}\frac{\alpha}{\sqrt{2}},$$

using $\int_\alpha^\infty e^{-s^2/2}ds = \sqrt{\pi/2}\,\operatorname{erfc}(\alpha/\sqrt{2})$ (put $s = \sqrt{2}\,w$ and use $\operatorname{erfc} z = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-w^2}dw$). So the relative error at $u = 0$ is exactly $\operatorname{erfc}(\alpha/\sqrt{2})$. Higher orders require the aperture to exceed their characteristic width as well.

(c) Now solve the infinite-limit equations. The normalized Hermite functions are

$$h_n(u) = \frac{H_n(u)e^{-u^2/2}}{\pi^{1/4}\sqrt{2^n n!}}.$$

The Gaussian transform gives $\mathcal{F}h_0 = h_0$: completing the square, $-s^2/2 - ius = -(s+iu)^2/2 - u^2/2$, and $\int_{\mathbb{R}} e^{-(s+iu)^2/2}ds = \sqrt{2\pi}$, so

$$\mathcal{F}h_0(u) = \frac{\pi^{-1/4}e^{-u^2/2}}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-(s+iu)^2/2}ds = \pi^{-1/4}e^{-u^2/2} = h_0(u).$$

To obtain all higher orders, use the raising operator $R = (u - d/du)/\sqrt{2}$ (applied to functions of the variable u, or of s inside $\mathcal{F}$). The Hermite recurrences $H_n' = 2nH_{n-1}$ and $H_{n+1} = 2uH_n - 2nH_{n-1}$ give $H_{n+1} = 2uH_n - H_n'$. Write $h_n = c_nH_ne^{-u^2/2}$, where $c_n = (\pi^{1/4}\sqrt{2^n n!})^{-1}$; by the product rule $h_n' = c_n(H_n' - uH_n)e^{-u^2/2}$, so

$$uh_n - h_n' = c_n(2uH_n - H_n')e^{-u^2/2} = c_nH_{n+1}e^{-u^2/2} = \frac{c_n}{c_{n+1}}\,h_{n+1} = \sqrt{2(n+1)}\,h_{n+1},$$

since $c_n/c_{n+1} = \sqrt{2^{n+1}(n+1)!/(2^n n!)} = \sqrt{2(n+1)}$. Dividing by $\sqrt{2}$ gives $Rh_n = \sqrt{n+1}\,h_{n+1}$, i.e. $h_n = R^n h_0/\sqrt{n!}$. Differentiation under the integral and integration by parts (boundary terms vanish for decaying h) give

$$\frac{d}{du}\mathcal{F}h = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}}(-is)e^{-ius}h(s)ds = -i\mathcal{F}(sh), \qquad \mathcal{F}(h') = -\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}}(-iu)e^{-ius}h(s)ds,$$

that is,

$$\mathcal{F}(sh) = i\frac{d}{du}\mathcal{F}h, \qquad \mathcal{F}(h') = iu\,\mathcal{F}h.$$

Therefore

$$\mathcal{F}(Rh) = \frac{1}{\sqrt{2}}\big[\mathcal{F}(sh) - \mathcal{F}(h')\big] = \frac{i}{\sqrt{2}}\left(\frac{d}{du} - u\right)\mathcal{F}h = -i\,R\mathcal{F}h,$$

and induction, $\mathcal{F}h_{n+1} = \mathcal{F}(Rh_n)/\sqrt{n+1} = -i\,R(-i)^n h_n/\sqrt{n+1} = (-i)^{n+1}h_{n+1}$, yields

$$\boxed{\mathcal{F}h_n = (-i)^n h_n}.$$

A separated two-dimensional mode $X = h_n$, $Y = h_m$ thus has $\rho = (-i)^n$, $\sigma = (-i)^m$, and the consistency condition $1 = -i\gamma e^{ikb}(-i)^{n+m}$ gives $\gamma = e^{-ikb}/(-i)^{n+m+1} = i^{\,n+m+1}e^{-ikb}$ (since $1/(-i) = i$):

$$\boxed{v_{nm}(\xi,\eta) = C\,h_n(\xi)h_m(\eta), \qquad \gamma_{nm} = i^{\,n+m+1}e^{-ikb}}.$$

Indeed $-i\gamma_{nm}e^{ikb}(-i)^{n+m} = 1$. In physical coordinates the shape is proportional to

$$H_n\Big(\sqrt{k/b}\,x\Big)\,H_m\Big(\sqrt{k/b}\,y\Big)\,e^{-k(x^2+y^2)/(2b)}.$$

Hermite functions form a complete $L^2(\mathbb{R})$ basis. The full one-dimensional eigenspace for phase $(-i)^r$ consists of convergent combinations of h_n with $n \equiv r \pmod 4$. Likewise, two-dimensional combinations with fixed $n+m$ mod 4 share the same γ. These are the infinite-aperture solutions; for finite α the truncated transform must be solved instead, and the tail bound quantifies the stated approximation for a chosen mode.

21.3 Neumann Series

21.3.1 Using the Neumann series, solve

(a) $\varphi(x) = 1 - 2\displaystyle\int_0^x t\varphi(t)\,dt,$

ANS. (a) $\varphi(x) = e^{-x^2}$.

(b) $\varphi(x) = x + \displaystyle\int_0^x (t-x)\varphi(t)\,dt,$

(c) $\varphi(x) = x - \displaystyle\int_0^x (t-x)\varphi(t)\,dt.$

Solution.

(a) Define $(Tg)(x) = -2\int_0^x t\,g(t)dt$, so the equation is $\varphi = 1 + T\varphi$. Iterating $\varphi_{k+1} = 1 + T\varphi_k$ from $\varphi_0 = 1$ gives the Neumann series $\sum_{n\ge 0} T^n 1$. We find its individual terms by induction: if $T^n 1 = (-1)^n x^{2n}/n!$, then

$$T^{n+1}1 = -2\int_0^x t\,\frac{(-1)^n t^{2n}}{n!}dt = -\frac{2(-1)^n}{n!}\,\frac{x^{2n+2}}{2n+2} = \frac{(-1)^{n+1}x^{2n+2}}{(n+1)!},$$

since $(2n+2)\,n! = 2\,(n+1)!$. The initial case is $T^0 1 = 1$, so, by the exponential series with argument $-x^2$,

$$\boxed{\varphi_a(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} = e^{-x^2}}.$$

Substitution gives, since $-2te^{-t^2} = \frac{d}{dt}e^{-t^2}$, $1 - 2\int_0^x te^{-t^2}dt = 1 + \left[e^{-t^2}\right]_0^x = 1 + (e^{-x^2} - 1) = e^{-x^2}$.

(b) Put $(Jg)(x) = \int_0^x (x-t)g(t)dt$. Since $t - x = -(x-t)$, the equation is $\varphi = x - J\varphi$. The elementary monomial formula is

$$\begin{aligned} J(t^p)(x) &= x\int_0^x t^p\,dt - \int_0^x t^{p+1}dt = x\frac{x^{p+1}}{p+1} - \frac{x^{p+2}}{p+2} \\ &= \frac{x^{p+2}}{(p+1)(p+2)}, \qquad p > -1. \end{aligned}$$

With $p = 2n+1$ it maps $x^{2n+1}/(2n+1)!$ to $x^{2n+3}/[(2n+1)!\,(2n+2)(2n+3)] = x^{2n+3}/(2n+3)!$. Starting with $J^0 x = x$, induction therefore gives $J^n x = x^{2n+1}/(2n+1)!$. The Neumann series of $\varphi = x - J\varphi$ is $\sum_n (-J)^n x = \sum_n (-1)^n J^n x$. Hence

$$\boxed{\varphi_b(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sin x}.$$

(c) Here $-(t-x) = x - t$, so the equation is $\varphi = x + J\varphi$; its Neumann series $\sum_n J^n x$ has the same terms with positive signs:

$$\boxed{\varphi_c(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = \sinh x}.$$

The factorials give uniform convergence on every bounded interval $|x| \le R$ (Weierstrass M-test with the bounds $R^{2n}/n!$ and $R^{2n+1}/(2n+1)!$), justifying termwise integration when T or J is applied to the sums. Direct substitutions for (b),(c) were evaluated explicitly in Exercise 21.2.3.

21.3.2 Solve

$$\psi(x) = x + \int_0^1 (1 + xt)\psi(t)\,dt.$$

Investigate the following methods:

(a) Show that the ordinary Neumann series diverges, so it cannot supply the solution here.

(b) Solve by the separable-kernel technique.

(c) Solve by educated guessing.

Solution.

Let $(Kg)(x) = \int_0^1 (1 + xt)g(t)dt$.

(a) The ordinary Neumann series prescribed by $\psi = x + K\psi$ is

$$x + Kx + K^2 x + \cdots .$$

For $g(x) = A + Bx$, since $\int_0^1 (A + Bt)dt = A + \frac{B}{2}$ and $\int_0^1 t(A + Bt)dt = \frac{A}{2} + \frac{B}{3}$,

$$Kg(x) = A + \frac{B}{2} + x\left(\frac{A}{2} + \frac{B}{3}\right),$$

so (with $A = 0$, $B = 1$, and then $A = \frac{1}{2}$, $B = \frac{1}{3}$)

$$Kx = \frac{1}{2} + \frac{x}{3}, \qquad K^2x = \frac{1}{2} + \frac{1}{6} + x\left(\frac{1}{4} + \frac{1}{9}\right) = \frac{2}{3} + \frac{13x}{36}.$$

Write $K^n x = A_n + B_n x$. Its coefficients obey

$$A_{n+1} = A_n + \frac{B_n}{2}, \qquad B_{n+1} = \frac{A_n}{2} + \frac{B_n}{3}, \quad (A_0, B_0) = (0, 1).$$

The recurrence has nonnegative coefficients, so by induction all $A_n, B_n \geq 0$. Then $A_{n+1} - A_n = B_n/2 \geq 0$, so $A_n \geq A_1 = 1/2$ for every $n \geq 1$. Thus $K^n x \geq A_n \geq 1/2$ on $[0, 1]$ for $n \geq 1$: the terms do not even tend to zero, and the series diverges.

The textbook's part (a) asks for the solution by the Neumann series, but for the equation as printed that series diverges, as just shown, although the solution exists (parts (b) and (c)). The Instructor's Manual's answer -2 does not address this obstruction. Introducing a parameter z locates the obstruction. Since the range of K consists of linear functions, any solution of $\psi_z = x + zK\psi_z$ has the form $\psi_z = A + Bx$, and by the formula for Kg its coefficient vector satisfies

$$\begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + zM \begin{pmatrix} A \\ B \end{pmatrix}, \qquad M = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}.$$

Here $D(z) = \det(I - zM) = (1 - z)(1 - \frac{z}{3}) - \frac{z^2}{4} = 1 - \frac{4}{3}z + \frac{1}{12}z^2$, and Cramer's rule gives $A = z/(2D)$, $B = (1 - z)/D$:

$$\psi_z(x) = \frac{z/2 + (1 - z)x}{1 - \frac{4}{3}z + \frac{1}{12}z^2}.$$

Since $(A_n, B_n)^T = M^n(0, 1)^T$, the expansion $(I - zM)^{-1} = \sum_n z^n M^n$ shows that the Taylor series of ψ_z in z is the Neumann series $\sum_n z^n K^n x$. The zeros of D are the roots $z = 8 \mp 2\sqrt{13}$ of $z^2 - 16z + 12 = 0$ (the characteristic values of Exercise 21.2.14), and the numerator $z/2$ of A does not vanish there, so A has a pole at each. The Taylor series of a function about $z = 0$ converges in the largest open disk about 0 in which the function is analytic and diverges outside it (it cannot converge beyond a pole, where the function is unbounded). So the radius of convergence is the distance $8 - 2\sqrt{13} \approx 0.789 < 1$ to the nearer pole, and the series diverges at $z = 1$. Evaluating the rational continuation at $z = 1$, where $D(1) = 1 - \frac{4}{3} + \frac{1}{12} = -\frac{1}{4}$, gives $\psi_1 = \frac{1}{2}/(-\frac{1}{4}) = -2$, but that continuation is not the sum of the divergent series at $z = 1$.

(b) Since $1 + xt = 1 \cdot 1 + x \cdot t$, the integral term is $c_0 + c_1 x$ with $c_0 = \int_0^1 \psi(t)dt$ and $c_1 = \int_0^1 t\psi(t)dt$. Separability therefore forces $\psi = A + Bx$ with $A = c_0$, $B = 1 + c_1$. By the moments computed in (a), $c_0 = A + \frac{B}{2}$ and $c_1 = \frac{A}{2} + \frac{B}{3}$, so comparing coefficients gives

$$A = A + \frac{B}{2}, \qquad B = 1 + \frac{A}{2} + \frac{B}{3}.$$

The first equation gives $B = 0$ and the second then gives $A = -2$. Thus $\boxed{\psi(x) = -2}$.

(c) Guess a constant c. Since $\int_0^1 (1 + xt)dt = 1 + \frac{x}{2}$, substitution produces

$$c = x + c\int_0^1 (1 + xt)dt = x + c\left(1 + \frac{x}{2}\right) = c + x(1 + c/2).$$

The coefficient of x vanishes exactly when $c = -2$. Finally, $x + \int_0^1 (1 + xt)(-2)dt = x - 2 - x = -2$, an explicit check of the solution of the original equation.

21.3.3 Solve

$$\varphi(x) = 1 + \lambda^2 \int_0^x (x-t)\varphi(t)\,dt$$

by each of the following methods:

(a) Reduction to an ODE (find the boundary conditions),

(b) The Neumann series,

(c) The use of Laplace transforms.

ANS. $\varphi(x) = \cosh \lambda x$.

Solution.

(a) Differentiate the equation with the Leibniz rule $\frac{d}{dx}\int_0^x g(x,t)dt = g(x,x) + \int_0^x \partial_x g(x,t)dt$; the endpoint term $(x-x)\varphi(x)$ is zero:

$$\varphi'(x) = \lambda^2 \int_0^x \varphi(t)dt, \qquad \varphi''(x) = \lambda^2 \varphi(x).$$

Setting $x = 0$ in the equation and in φ' (both integrals vanish) gives the initial data $\varphi(0) = 1$, $\varphi'(0) = 0$. For $\lambda \neq 0$, write $\varphi = Ae^{\lambda x} + Be^{-\lambda x}$. Then $A + B = 1$ and $\lambda(A - B) = 0$, so $A = B = 1/2$. For $\lambda = 0$ the original equation directly gives $\varphi = 1$. In both cases

$$\boxed{\varphi(x) = \cosh(\lambda x)}.$$

(b) With J from Exercise 21.3.1, the equation is $\varphi = 1 + \lambda^2 J\varphi$. The monomial formula of that exercise with $p = 2n$ gives

$$J\left(\frac{t^{2n}}{(2n)!}\right) = \frac{x^{2n+2}}{(2n)!\,(2n+1)(2n+2)} = \frac{x^{2n+2}}{(2n+2)!},$$

so, starting with $J^0 1 = 1$, induction gives $J^n 1 = x^{2n}/(2n)!$, and the Neumann series is

$$\sum_{n=0}^{\infty} \lambda^{2n} J^n 1 = \sum_{n=0}^{\infty} \frac{(\lambda x)^{2n}}{(2n)!} = \cosh(\lambda x).$$

It converges uniformly on every finite x interval for every fixed complex λ.

(c) Since $\mathcal{L}\{1\} = 1/s$ and $\mathcal{L}\{x\} = 1/s^2$, the convolution theorem turns $\int_0^x (x-t)\varphi(t)dt$ into $\Phi(s)/s^2$, so

$$\Phi(s) = \frac{1}{s} + \frac{\lambda^2}{s^2}\Phi(s), \qquad \Phi(s) = \frac{s}{s^2 - \lambda^2}$$

(multiply by s^2: $(s^2 - \lambda^2)\Phi = s$). For $\lambda \neq 0$,

$$\frac{s}{s^2 - \lambda^2} = \frac{1}{2}\left(\frac{1}{s-\lambda} + \frac{1}{s+\lambda}\right),$$

whose inverse, by $\mathcal{L}\{e^{ax}\} = 1/(s-a)$, is $\frac{1}{2}(e^{\lambda x} + e^{-\lambda x}) = \cosh(\lambda x)$. At $\lambda = 0$ the transform is $1/s$, the transform of $1 = \cosh 0$. As a direct check, for $\lambda \neq 0$ integration by parts (differentiate $x - t$, integrate $\cosh \lambda t$) gives

$$\begin{aligned}\int_0^x (x-t)\cosh(\lambda t)dt &= \left[\frac{(x-t)\sinh \lambda t}{\lambda}\right]_0^x + \frac{1}{\lambda}\int_0^x \sinh \lambda t\,dt \\ &= \frac{\cosh(\lambda x) - 1}{\lambda^2},\end{aligned}$$

so the right side of the equation is $1 + (\cosh \lambda x - 1) = \cosh \lambda x$. As $\lambda \to 0$ the integral tends to $x^2/2 = \int_0^x (x-t)dt$, consistent with the zero-parameter case.

21.3.4

(a) In Eq. (21.59), take $V = V_0$, independent of t. Without using Eq. (21.60), show that Eq. (21.59) leads directly to

$$U(t - t_0) = \exp\left[-\frac{i}{\hbar}(t - t_0)V_0\right].$$

(b) Repeat for Eq. (21.60) without using Eq. (21.59).

For the power-series derivations, take V_0 to be a scalar, matrix, or bounded operator. State the qualification for an unbounded self-adjoint operator.

Solution.

(a) With $V(t) = V_0$, Eq. (21.59) is

$$U(t, t_0) = I - \frac{i}{\hbar}\int_{t_0}^{t} V_0 U(s, t_0)\, ds.$$

Differentiating with respect to t (fundamental theorem of calculus), and setting $t = t_0$, where the integral vanishes, gives

$$\partial_t U = -\frac{i}{\hbar}V_0 U, \qquad U(t_0, t_0) = I.$$

Put $\tau = t - t_0$ and $U = \sum_{n\geq 0} C_n \tau^n$, so that $\partial_t U = \sum_{n\geq 0}(n+1)C_{n+1}\tau^n$. Equating powers gives

$$C_0 = I, \qquad (n+1)C_{n+1} = -\frac{i}{\hbar}V_0 C_n.$$

Induction yields $C_n = (-iV_0/\hbar)^n/n!$. This series converges for every τ (its terms are bounded in norm by $(\|V_0\|\,|\tau|/\hbar)^n/n!$) and solves the initial-value problem. It is the only solution: for any solution U, the product $W = e^{i\tau V_0/\hbar}U$ satisfies, since $\partial_t e^{i\tau V_0/\hbar} = e^{i\tau V_0/\hbar}\,\frac{i}{\hbar}V_0$ (differentiate the exponential series termwise),

$$\partial_t W = e^{i\tau V_0/\hbar}\Big(\frac{i}{\hbar}V_0 U + \partial_t U\Big) = 0,$$

so $W = W(t_0) = I$. Multiplying on the left by $e^{-i\tau V_0/\hbar}$, the inverse of $e^{i\tau V_0/\hbar}$ (the exponents commute), gives $U = e^{-i\tau V_0/\hbar} = \sum_n C_n \tau^n$. Thus

$$\boxed{U(t, t_0) = \exp[-i(t - t_0)V_0/\hbar]}.$$

It depends only on the time difference because the constant-coefficient initial-value solution just derived has that form.

(b) Independently, Eq. (21.60) is the ordered integral series

$$U = I + \sum_{n=1}^{\infty}\left(-\frac{i}{\hbar}\right)^n \int_{t_0}^{t} ds_1 \int_{t_0}^{s_1} ds_2 \cdots \int_{t_0}^{s_{n-1}} ds_n\, V(s_1)V(s_2)\cdots V(s_n).$$

For constant V_0 the product is V_0^n, which comes out of the integrals. The remaining scalar simplex integral $S_n(\tau)$, a function of $\tau = t - t_0$, has $S_0 = 1$, and in S_{n+1} the inner n integrals (over $s_2, \ldots, s_{n+1}$) form $S_n(s_1 - t_0)$. With $u = s_1 - t_0$ and the induction hypothesis $S_n(u) = u^n/n!$,

$$S_{n+1}(\tau) = \int_0^{\tau} S_n(u)\, du = \int_0^{\tau} \frac{u^n}{n!}\, du = \frac{\tau^{n+1}}{(n+1)!}.$$

Substituting $S_n(\tau) = \tau^n/n!$,

$$U = I + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar}\right)^n V_0^n \frac{\tau^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i\tau V_0}{\hbar}\right)^n,$$

precisely the same exponential, without using Eq. (21.59). Oriented integrals give the formula also for $\tau < 0$. For a scalar, matrix, or bounded operator V_0 both series converge on finite time intervals. For an unbounded self-adjoint quantum operator, the exponential is defined by its spectral calculus; an operator power series on every state cannot be assumed without domain and convergence conditions.

21.4 Hilbert–Schmidt Theory

21.4.1 In the Fredholm equation

$$\varphi(x) = \lambda \int_a^b K(x,t)\varphi(t)\,dt,$$

assume that the kernel $K(x,t)$ is self-adjoint or Hermitian:

$$K(x,t) = K^*(t,x).$$

Assume a Hilbert–Schmidt kernel, $\int_a^b \int_a^b |K(x,t)|^2\,dx\,dt < \infty$, and nonzero L^2 eigenfunctions. Extend the analysis of the present section to show that

(a) the eigenfunctions are orthogonal, in the sense that

$$\int_a^b \varphi_m^*(x)\varphi_n(x)\,dx = 0, \qquad m \neq n \ (\lambda_m \neq \lambda_n).$$

(b) the eigenvalues are real.

Solution.

Let $\langle u, v\rangle = \int_a^b u(x)^* v(x) dx$ and $(Kg)(x) = \int_a^b K(x,t)g(t)dt$, and assume the kernel defines a self-adjoint operator on this L^2 space (a Hermitian Hilbert–Schmidt kernel is sufficient). The Hilbert–Schmidt condition in the statement is what makes the double integral below absolutely convergent for L^2 functions (by the Cauchy–Schwarz inequality), so that the order of integration may be interchanged (Fubini's theorem). Interchanging it and using $K(x,t) = K^*(t,x)$ gives the double-integral identity

$$\begin{aligned}\langle \varphi_m, K\varphi_n\rangle &= \int_a^b \int_a^b \varphi_m(x)^* K(x,t)\varphi_n(t)\,dt\,dx \\ &= \int_a^b \left[\int_a^b K(t,x)\varphi_m(x)dx\right]^* \varphi_n(t)dt = \langle K\varphi_m, \varphi_n\rangle.\end{aligned}$$

A nonzero eigenfunction in the convention $\varphi_n = \lambda_n K\varphi_n$ has $\lambda_n \neq 0$ (otherwise $\varphi_n = 0$), so $K\varphi_n = \varphi_n/\lambda_n$. Inserting this on both sides of the identity, and taking the factor $1/\lambda_m$ out of the first (conjugate-linear) slot as $1/\lambda_m^*$,

$$\frac{1}{\lambda_n}\langle \varphi_m, \varphi_n\rangle = \frac{1}{\lambda_m^*}\langle \varphi_m, \varphi_n\rangle,$$

or, multiplying by $\lambda_m^* \lambda_n$,

$$(\lambda_m^* - \lambda_n)\langle \varphi_m, \varphi_n \rangle = 0.$$

Take $m = n$. Since $\|\varphi_n\|^2 > 0$, this gives $\boxed{\lambda_n^* = \lambda_n}$, proving (b). Now for $\lambda_m \neq \lambda_n$ the factor $\lambda_m^* - \lambda_n = \lambda_m - \lambda_n$ (by (b)) is nonzero, so

$$\boxed{\langle \varphi_m, \varphi_n \rangle = 0},$$

which proves (a). Within a degenerate eigenspace orthogonality need not hold automatically, but Gram–Schmidt produces an orthonormal basis without changing its eigenvalue.

21.4.2

(a) Show that the eigenfunctions of Exercise 21.2.12 are orthogonal.

(b) Show that the eigenfunctions of Exercise 21.2.14 are orthogonal.

Solution.

Both kernels are real and symmetric, and in each case the eigenvalues are distinct, so Exercise 21.4.1 predicts the result; here it is checked directly.

(a) Use the exact eigenfunctions found in Exercise 21.2.12:

$$y_0 = x, \qquad y_+ = x^2 + \frac{1}{\sqrt{5}}, \qquad y_- = x^2 - \frac{1}{\sqrt{5}}.$$

The products $y_0 y_\pm = x^3 \pm x/\sqrt{5}$ are odd, so $\int_{-1}^{1} y_0 y_\pm dx = 0$. For the two even functions, $y_+ y_- = x^4 - \frac{1}{5}$ and

$$\int_{-1}^{1} y_+ y_- dx = \int_{-1}^{1} \left(x^4 - \frac{1}{5} \right) dx = \frac{2}{5} - \frac{2}{5} = 0.$$

All three distinct eigenfunctions are therefore mutually orthogonal.

(b) Write $\varphi_\pm = 1 + b_\pm x$, where $b_\pm = (-2 \pm \sqrt{13})/3$ (Exercise 21.2.14). Their sum and product are

$$b_+ + b_- = -\frac{4}{3}, \qquad b_+ b_- = \frac{(-2)^2 - 13}{9} = -1.$$

It follows that

$$\begin{aligned} \int_0^1 \varphi_+ \varphi_- dx &= \int_0^1 \left[1 + (b_+ + b_-)x + b_+ b_- x^2 \right] dx \\ &= 1 + \frac{b_+ + b_-}{2} + \frac{b_+ b_-}{3} = 1 - \frac{2}{3} - \frac{1}{3} = \boxed{0}. \end{aligned}$$

Using the exact coefficients is necessary to see exact zero; rounded decimal eigenfunctions only approximate this cancellation.

21.4.3 Use the Hilbert–Schmidt method to solve the inhomogeneous integral equation

$$\varphi(x) = x + \frac{1}{2} \int_{-1}^{1} (t + x)\varphi(t)\, dt.$$

The corresponding homogeneous integral equation was treated in Example 21.2.3.

Note. The application of the Hilbert–Schmidt technique here is somewhat like using a shotgun to kill a mosquito, especially when the equation can be solved quickly by expanding in Legendre polynomials.

Solution.

For $K(x,t) = t + x$, choose normalized eigenfunctions

$$u_\pm(x) = \frac{1}{2}(1 \pm \sqrt{3}\,x).$$

Their norms are one, because the odd term of $(1 \pm \sqrt{3}\,x)^2 = 1 \pm 2\sqrt{3}\,x + 3x^2$ integrates to zero and $\int_{-1}^{1}(1+3x^2)dx/4 = (2+2)/4 = 1$; their mutual inner product is $\int_{-1}^{1}(1-3x^2)dx/4 = 0$. Evaluate the kernel action; the odd integrands t and $\pm\sqrt{3}\,xt$ integrate to zero, leaving

$$\begin{aligned} Ku_\pm(x) &= \int_{-1}^{1}(t+x)\frac{1 \pm \sqrt{3}\,t}{2}dt = \frac{1}{2}\left(\pm\sqrt{3}\int_{-1}^{1}t^2dt + x\int_{-1}^{1}dt\right) \\ &= x \pm \frac{1}{\sqrt{3}} = \pm\frac{2}{\sqrt{3}}u_\pm(x). \end{aligned}$$

Thus $u_\pm = \lambda_\pm K u_\pm$ requires $\lambda_\pm(\pm 2/\sqrt{3}) = 1$: the characteristic values are $\lambda_\pm = \pm\sqrt{3}/2$, as in Example 21.2.3.

The inhomogeneous term $f(x) = x$ has coefficients

$$b_\pm = \langle u_\pm, x\rangle = \pm\frac{\sqrt{3}}{2}\int_{-1}^{1}x^2dx = \pm\frac{\sqrt{3}}{2}\frac{2}{3} = \pm\frac{1}{\sqrt{3}}.$$

Since $x = (u_+ - u_-)/\sqrt{3}$ and every $K\varphi$ is a combination of 1 and x, the solution of $\varphi = x + \lambda K\varphi$ lies in the span of $u_\pm$: $\varphi = a_+u_+ + a_-u_-$. Projecting the equation onto $u_\pm$ and using $\langle u_\pm, K\varphi\rangle = \langle Ku_\pm, \varphi\rangle = a_\pm/\lambda_\pm$ (K is real and symmetric) gives $a_\pm = b_\pm + \lambda a_\pm/\lambda_\pm$. For the requested coupling $\lambda = 1/2$, this Hilbert–Schmidt projection gives

$$a_\pm = \frac{b_\pm}{1 - \lambda/\lambda_\pm}, \qquad \frac{\lambda}{\lambda_\pm} = \pm\frac{1}{\sqrt{3}}.$$

Specifically,

$$a_+ = \frac{1/\sqrt{3}}{1 - 1/\sqrt{3}} = \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2}, \qquad a_- = \frac{-1/\sqrt{3}}{1 + 1/\sqrt{3}} = -\frac{1}{\sqrt{3}+1} = \frac{1-\sqrt{3}}{2}.$$

Then $a_+ + a_- = 1$ and $a_+ - a_- = \sqrt{3}$. Reconstructing,

$$\varphi = a_+u_+ + a_-u_- = \frac{a_+ + a_-}{2} + \frac{\sqrt{3}(a_+ - a_-)}{2}x = \frac{1}{2} + \frac{3}{2}x = \boxed{\frac{1+3x}{2}}.$$

Directly $\int_{-1}^{1}\varphi(t)dt = 1$ and $\int_{-1}^{1}t\varphi(t)dt = 1$, so the right side of the original equation is $x + (1+x)/2 = (1+3x)/2$, as required.

21.4.4 The Fredholm integral equation

$$\varphi(x) = \lambda \int_0^\infty e^{-xt}\varphi(t)\,dt, \qquad x > 0,$$

has an infinite number of solutions, of which one is

$$\varphi(x) = x^{-1/2}, \qquad \lambda = \pi^{-1/2}.$$

Verify that this is a solution and that it is not normalizable.

Note. These are improper-integral solutions, not normalizable Hilbert–Schmidt eigenfunctions. The kernel fails the Hilbert–Schmidt integrability condition; an infinite interval alone would not establish that failure. Its Taylor series cannot be integrated term by term against the displayed power solution.

Solution.

The restriction $x > 0$ in the statement is needed because for $x \le 0$ the integrand $e^{-xt}t^{-1/2}$ is at least $t^{-1/2}$, whose integral over $(0,\infty)$ diverges. For $x > 0$, set $u = xt$, so $dt = du/x$ and $t^{-1/2} = x^{1/2}u^{-1/2}$. Then, with $\int_0^\infty e^{-u}u^{-1/2}du = \Gamma(\frac{1}{2}) = \sqrt{\pi}$,

$$\int_0^\infty e^{-xt}t^{-1/2}dt = x^{-1/2}\int_0^\infty e^{-u}u^{-1/2}du = \sqrt{\pi}\,x^{-1/2}.$$

Multiplication by $\lambda = \pi^{-1/2}$ gives $\boxed{\varphi(x) = x^{-1/2}}$, verifying the equation.

Its truncated squared norm is

$$\int_\epsilon^R |\varphi(x)|^2 dx = \int_\epsilon^R \frac{dx}{x} = \ln R - \ln \epsilon.$$

It diverges as $R \to \infty$ and as $\epsilon \downarrow 0$. No nonzero constant multiple is normalizable on $(0,\infty)$.

This is a pointwise improper-integral solution for $x > 0$, not an L^2 eigenfunction to which the chapter's compact Hilbert–Schmidt conclusions automatically apply. As the Note states, the kernel also fails the Hilbert–Schmidt condition: since $\int_0^\infty e^{-2xt}dx = 1/(2t)$ for $t > 0$,

$$\int_0^\infty\int_0^\infty |e^{-xt}|^2 dx\,dt = \int_0^\infty \frac{dt}{2t} = \infty,$$

the last integral diverging logarithmically at both ends. Thus the kernel is not Hilbert–Schmidt. The infinite interval alone does not cause this: the kernel e^{-x-t} on the same quarter plane has $\int_0^\infty\int_0^\infty e^{-2x-2t}dx\,dt = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4} < \infty$, so it is Hilbert–Schmidt. Finally, the Taylor series of e^{-xt} cannot be integrated term by term against $t^{-1/2}$: the coefficient of $(-x)^m/m!$ would contain $\int_0^\infty t^{m-1/2}dt$, which diverges for every integer $m \ge 0$.

The statement about infinitely many improper-integral solutions can also be verified explicitly. The same substitution $u = xt$ gives, for $s > 0$ and $x > 0$, the gamma integral $\int_0^\infty e^{-xt}t^{s-1}dt = \Gamma(s)x^{-s}$. For $0 < a < 1$, the choices $s = 1-a$ and $s = a$ give, for the integral operator $(Kg)(x) = \int_0^\infty e^{-xt}g(t)\,dt$ of this equation,

$$K(t^{-a})(x) = \Gamma(1-a)x^{a-1}, \qquad K(t^{a-1})(x) = \Gamma(a)x^{-a}.$$

Put $A = \sqrt{\Gamma(a)}$, $B = \sqrt{\Gamma(1-a)}$, and $\mu = AB$. Since $\Gamma(a) = A^2$ and $\Gamma(1-a) = B^2$,

$$K(At^{-a} \pm Bt^{a-1})(x) = AB^2x^{a-1} \pm BA^2x^{-a} = \pm\mu(Ax^{-a} \pm Bx^{a-1}).$$

So $\varphi = \lambda K\varphi$ holds for these combinations when $\lambda(\pm\mu) = 1$. Therefore

$$\varphi_{a,\pm}(x) = \sqrt{\Gamma(a)}x^{-a} \pm \sqrt{\Gamma(1-a)}x^{a-1}, \qquad \lambda_{a,\pm} = \pm\frac{1}{\mu} = \pm\sqrt{\frac{\sin\pi a}{\pi}},$$

where the reflection formula $\Gamma(a)\Gamma(1-a) = \pi/\sin\pi a$ gives $\mu^2 = \pi/\sin\pi a$. The minus combination at $a = 1/2$ is zero and must be excluded; the plus combination there is a nonzero multiple of the requested $x^{-1/2}$. These power combinations belong to the improper-integral solution class, not to the normalizable L^2 eigenfunctions. For $a = \frac{1}{2}$ this is the computation above. For $a \neq \frac{1}{2}$ the two terms are nonzero multiples of the distinct powers x^{-a} and $x^{-(1-a)}$; near $x = 0$ the one with the larger power $p = \max(a, 1-a) > \frac{1}{2}$ of $1/x$ dominates, so $|\varphi_{a,\pm}|^2$ is asymptotic to a positive multiple of x^{-2p}, and $\int_0^1 x^{-2p}dx$ diverges because $2p > 1$. (Similarly, near ∞ the smaller power $\min(a, 1-a) < \frac{1}{2}$ dominates, and the integral diverges there as well.)

21.4.5 Given

$$y(x) = x + \lambda\int_0^1 xt\,y(t)\,dt :$$

(a) Determine $y(x)$ as a Neumann series.

(b) Find the range of λ for which your Neumann series solution is convergent. Compare with the value obtained from

$$|\lambda| \cdot |K|_{\max} < 1.$$

(c) Find the eigenvalue and the eigenfunction of the corresponding homogeneous integral equation.

(d) By the separable-kernel method show that the solution is

$$y(x) = \frac{3x}{3-\lambda}, \qquad \lambda \neq 3.$$

(e) Find $y(x)$ by the Hilbert–Schmidt method. Also decide whether a solution exists at $\lambda = 3$.

Solution.

Let $(Kg)(x) = x\int_0^1 t\,g(t)dt$. Its action on the inhomogeneous term is $Kx = x\int_0^1 t^2dt = x/3$, so $K(x/3^n) = x/3^{n+1}$ and induction gives $K^n x = x/3^n$.

(a) The Neumann series $\sum_n \lambda^n K^n x$ is therefore

$$\boxed{y(x) = x\sum_{n=0}^{\infty}(\lambda/3)^n}.$$

(b) Apart from the identically zero value at $x = 0$, this geometric series converges precisely when $\boxed{|\lambda| < 3}$. For $|\lambda| = 3$ its terms $(\lambda/3)^n x$ have the constant modulus x, so they do not tend to zero, and for larger modulus they grow. On $[0,1]^2$, $|K|_{\max} = \max xt = 1$ and the interval length is one, so the stated sufficient bound only gives $|\lambda| < 1$. It is sufficient, not necessary.

(c) A nonzero homogeneous solution $y = \lambda x\int_0^1 ty(t)dt$ is a multiple of x. Putting $y = Cx$ gives $Cx = \lambda Cx/3$, hence

$$\boxed{\lambda_1 = 3, \qquad y_1(x) = Cx, \quad C \neq 0}.$$

(d) For the inhomogeneous equation put $c = \int_0^1 ty(t)dt$, so $y = x(1 + \lambda c)$. Taking its moment, $c = (1 + \lambda c)\int_0^1 t^2 dt$, that is,

$$c = \frac{1 + \lambda c}{3}, \qquad (3 - \lambda)c = 1.$$

For $\lambda \neq 3$,

$$\boxed{y(x) = x\left(1 + \frac{\lambda}{3 - \lambda}\right) = \frac{3x}{3 - \lambda}}.$$

The formula excludes $\lambda = 3$. At that value the moment equation becomes $0 = 1$, so the inhomogeneous equation has no solution. The rational solution at other λ exists even when its Neumann series diverges.

(e) Normalize the homogeneous eigenfunction: $u(x) = \sqrt{3}\,x$, since $\int_0^1 u^2 dx = 3\int_0^1 x^2 dx = 1$. The coefficient of $f = x$ is $b = \int_0^1 u(x)x\,dx = \sqrt{3}/3 = 1/\sqrt{3}$. Since $f = u/\sqrt{3}$ lies in the span of u, and so does every Ky, the solution is $y = au$. Projecting $y = x + \lambda Ky$ onto u and using $\langle u, Ky\rangle = \langle Ku, y\rangle = a/\lambda_1$ gives $a = b + \lambda a/\lambda_1$, i.e. $(\lambda_1 - \lambda)a = b\lambda_1$. For $\lambda \neq \lambda_1 = 3$, Hilbert–Schmidt projection thus gives

$$y(x) = \frac{b\lambda_1}{\lambda_1 - \lambda}u(x) = \frac{(1/\sqrt{3})3}{3 - \lambda}\sqrt{3}\,x = \frac{3x}{3 - \lambda}.$$

At $\lambda = \lambda_1 = 3$ the projected equation reads $0 \cdot a = 3b = \sqrt{3} \neq 0$, because the overlap b is nonzero: there is no solution, in agreement with (d).

21.4.6 In Exercise 21.2.11 it was found that the integral equation

$$\varphi(x) = \lambda \int_0^{2\pi} \cos(x - t)\varphi(t)\,dt$$

had (unnormalized) eigenfunctions $\cos x$ and $\sin x$, both with eigenvalue $\lambda_i = 1/\pi$. Show that the kernel of this integral equation has an expansion of the form

$$K(x, t) = \sum_{n=1}^{2} \frac{\varphi_n(x)\varphi_n(t)}{\lambda_n},$$

where φ_n in this expansion are the unit-norm versions of the stated eigenfunctions.

Solution.

The squared norms of the stated functions are (Exercise 21.2.11; $\cos^2 x, \sin^2 x = (1 \pm \cos 2x)/2$, and $\cos 2x$ integrates to zero over a period)

$$\int_0^{2\pi} \cos^2 x\,dx = \int_0^{2\pi} \sin^2 x\,dx = \pi,$$

and their mutual integral is $\int_0^{2\pi} \sin x \cos x\,dx = \frac{1}{2}\int_0^{2\pi} \sin 2x\,dx = 0$. Therefore use

$$u_1(x) = \frac{\cos x}{\sqrt{\pi}}, \qquad u_2(x) = \frac{\sin x}{\sqrt{\pi}}.$$

With $\lambda_1 = \lambda_2 = 1/\pi$ and the addition theorem for the cosine,

$$\sum_{n=1}^{2} \frac{u_n(x)u_n(t)}{\lambda_n} = \pi\left[\frac{\cos x \cos t}{\pi} + \frac{\sin x \sin t}{\pi}\right]$$

$$= \cos x \cos t + \sin x \sin t = \boxed{\cos(x - t) = K(x, t)}.$$

The functions used in the displayed kernel expansion must be normalized. Substituting the initially stated unnormalized $\cos x, \sin x$ would instead give $\pi K(x,t)$. Equivalently, retain unnormalized functions but divide each term by its squared norm as well as by its characteristic value.

21.4.7 The integral equation

$$\varphi(x) = \lambda \int_0^1 (1 + xt)\varphi(t)\, dt$$

has eigenvalues $\lambda_{1,2} = 8 \mp 2\sqrt{13}$ (approximately 0.7889 and 15.211). The corresponding unnormalized eigenfunctions are $\varphi_{1,2}(x) = 1 + (-2 \pm \sqrt{13})x/3$, approximately $1 + 0.5352x$ and $1 - 1.8685x$.

(a) Show that these eigenfunctions are orthogonal over the interval $[0, 1]$.

(b) Normalize the eigenfunctions to unity.

(c) Using the normalized functions from part (b), show that

$$K(x,t) = \frac{\varphi_1(x)\varphi_1(t)}{\lambda_1} + \frac{\varphi_2(x)\varphi_2(t)}{\lambda_2}.$$

ANS. (b) $\varphi_1(x) = 0.7831 + 0.4191x$,
$\varphi_2(x) = 1.8403 - 3.4387x$.

Solution.

Use the exact values behind the displayed decimals:

$$b_\pm = \frac{-2 \pm \sqrt{13}}{3}, \qquad \lambda_\pm = 8 \mp 2\sqrt{13}, \qquad v_\pm(x) = 1 + b_\pm x.$$

The textbook's decimal eigenpairs are approximations, so exact orthogonality and exact kernel reconstruction use these radical values. In part (c) the eigenfunctions must be the unit-norm ones of part (b): with the unnormalized $v_\pm$ the same sum is $\frac{4}{3} + \frac{5}{9}(x + t) + \frac{16}{27}xt$ (see the end of the solution), not $K(x,t)$.

(a) As in Exercise 21.4.2(b), since $b_+ + b_- = -4/3$ and $b_+ b_- = -1$,

$$\int_0^1 v_+(x)v_-(x)dx = \int_0^1 \left[1 + (b_+ + b_-)x + b_+ b_- x^2\right]dx = 1 + \frac{b_+ + b_-}{2} + \frac{b_+ b_-}{3} = 0.$$

(b) Their squared norms are

$$N_\pm^2 = \int_0^1 (1 + 2b_\pm x + b_\pm^2 x^2)dx = 1 + b_\pm + \frac{b_\pm^2}{3}.$$

Since $3b_\pm + 2 = \pm\sqrt{13}$, squaring gives the root equation $9b_\pm^2 + 12b_\pm - 9 = 0$, i.e. $3b_\pm^2 + 4b_\pm - 3 = 0$, so $b_\pm^2 = 1 - \frac{4}{3}b_\pm$. This reduces $N_\pm^2$ to

$$N_\pm^2 = 1 + b_\pm + \frac{1}{3} - \frac{4b_\pm}{9} = \frac{4}{3} + \frac{5b_\pm}{9} = \frac{36 + 5(-2 \pm \sqrt{13})}{27} = \frac{26 \pm 5\sqrt{13}}{27}.$$

Thus the normalized functions are

$$\boxed{u_\pm(x) = \sqrt{\frac{27}{26 \pm 5\sqrt{13}}}\left(1 + \frac{-2 \pm \sqrt{13}}{3}x\right).}$$

Their numerical forms (u_+ and u_- are the answer's φ_1 and φ_2) are

$$u_+(x) = 0.7831024897 + 0.4191037337x, \qquad u_-(x) = 1.8403126068 - 3.4386555600x.$$

In φ_2 the textbook prints 3.4386, which is what normalizing its rounded eigenfunction $1 - 1.8685x$ gives $(1.840326 - 3.438649x)$; the exact 3.43865556 . . . rounds to 3.4387.

(c) Put $A_\pm = 1/(\lambda_\pm N_\pm^2)$. Multiplying out,

$$\lambda_\pm N_\pm^2 = \frac{(8 \mp 2\sqrt{13})(26 \pm 5\sqrt{13})}{27} = \frac{78 \mp 12\sqrt{13}}{27} = \frac{26 \mp 4\sqrt{13}}{9},$$

and rationalizing, with $26^2 - 16 \cdot 13 = 468 = 9 \cdot 52$,

$$A_\pm = \frac{9(26 \pm 4\sqrt{13})}{468} = \frac{26 \pm 4\sqrt{13}}{52} = \frac{1}{2} \pm \frac{1}{\sqrt{13}}.$$

Each kernel term is $u_\pm(x)u_\pm(t)/\lambda_\pm = A_\pm(1 + b_\pm x)(1 + b_\pm t) = A_\pm\big[1 + b_\pm(x + t) + b_\pm^2 xt\big]$, so in their sum the constant, $x + t$, and xt coefficients are respectively $\sum A_\pm$, $\sum A_\pm b_\pm$, and $\sum A_\pm b_\pm^2$. Evaluate each, using $b_+ - b_- = 2\sqrt{13}/3$:

$$\sum A_\pm = 1, \qquad \sum A_\pm b_\pm = \frac{1}{2}(b_+ + b_-) + \frac{b_+ - b_-}{\sqrt{13}} = \frac{1}{2}\left(-\frac{4}{3}\right) + \frac{1}{\sqrt{13}}\frac{2\sqrt{13}}{3} = 0,$$

and, using $b_\pm^2 = 1 - \frac{4}{3}b_\pm$,

$$\sum A_\pm b_\pm^2 = 1 - \frac{4}{3}(0) = 1.$$

Consequently

$$\boxed{\frac{u_+(x)u_+(t)}{\lambda_+} + \frac{u_-(x)u_-(t)}{\lambda_-} = 1 + xt}.$$

With the unnormalized $v_\pm$ in place of $u_\pm$, the three coefficients would instead be $\sum 1/\lambda_\pm$, $\sum b_\pm/\lambda_\pm$, and $\sum b_\pm^2/\lambda_\pm$. Since $1/\lambda_\pm = (8 \pm 2\sqrt{13})/(64 - 52) = (4 \pm \sqrt{13})/6$,

$$\sum \frac{1}{\lambda_\pm} = \frac{8}{6} = \frac{4}{3}, \qquad \sum \frac{b_\pm}{\lambda_\pm} = \sum \frac{(-2 \pm \sqrt{13})(4 \pm \sqrt{13})}{18} = \sum \frac{5 \pm 2\sqrt{13}}{18} = \frac{5}{9},$$

and $\sum b_\pm^2/\lambda_\pm = \frac{4}{3} - \frac{4}{3} \cdot \frac{5}{9} = \frac{16}{27}$ by $b_\pm^2 = 1 - \frac{4}{3}b_\pm$. This gives the sum $\frac{4}{3} + \frac{5}{9}(x + t) + \frac{16}{27}xt$ quoted at the beginning of the solution.

21.4.8 An alternate form of the solution to the inhomogeneous integral equation, Eq. (21.81), is

$$\varphi(x) = f_0(x) + \sum_{i=1}^{\infty} \frac{b_i \lambda_i}{\lambda_i - \lambda}\, \varphi_i(x).$$

Here φ_i are orthonormal eigenfunctions for the nonzero operator eigenvalues, $b_i = \langle \varphi_i, f \rangle$, f_0 is the projection of f onto $\ker K$, and $\lambda \neq \lambda_i$. The textbook's pure sum applies when $f_0 = 0$.

(a) Derive this form without using Eq. (21.89).

(b) Show that this form and Eq. (21.89) are equivalent.

Solution.

Let K be the self-adjoint compact integral operator in the chapter, and let its normalized nonzero-eigenvalue functions satisfy $K\varphi_i = \varphi_i/\lambda_i$.

(a) By the Hilbert–Schmidt (spectral) theorem for a compact self-adjoint operator, L^2 is the orthogonal sum of $\ker K$ and the closed span of the φ_i. Decompose both functions accordingly into their nullspace components and their components along the nonzero-eigenvalue eigenfunctions:

$$f = f_0 + \sum_i b_i\varphi_i, \qquad \varphi = \varphi_0 + \sum_i a_i\varphi_i.$$

The projection of $K\varphi$ onto $\ker K$ is zero, since K is self-adjoint: for $h \in \ker K$, $\langle h, K\varphi\rangle = \langle Kh, \varphi\rangle = 0$. Projecting the integral equation onto that nullspace gives $\varphi_0 = f_0$. Project $\varphi = f + \lambda K\varphi$ onto φ_i, using $\langle \varphi_i, K\varphi\rangle = \langle K\varphi_i, \varphi\rangle = a_i/\lambda_i$ (λ_i is real by Exercise 21.4.1):

$$a_i = b_i + \frac{\lambda}{\lambda_i}a_i, \qquad (\lambda_i - \lambda)a_i = b_i\lambda_i.$$

For $\lambda \neq \lambda_i$ this gives

$$\boxed{\varphi(x) = f_0(x) + \sum_i \frac{b_i\lambda_i}{\lambda_i - \lambda}\varphi_i(x)},$$

without using Eq. (21.89).

(b) Split the scalar factor:

$$\frac{\lambda_i}{\lambda_i - \lambda} = 1 + \frac{\lambda}{\lambda_i - \lambda}.$$

Summing the first terms gives $\sum_i b_i\varphi_i = f - f_0$; adding the explicit nullspace term restores f, so

$$\varphi(x) = f(x) + \lambda\sum_i \frac{\varphi_i(x)}{\lambda_i - \lambda}\underbrace{\int_a^b \varphi_i(t)^* f(t)dt}_{b_i}.$$

This is Eq. (21.89), with the conjugate included for a complex orthonormal basis.

The textbook's pure eigenfunction sum assumes that f has no component in the nullspace of K. Nonzero-eigenvalue eigenfunctions need not span the whole L^2 space, for example for the finite-rank kernels in this chapter. In general an additional unchanged nullspace component is required. Thus Eq. (21.89) already retains f_0 in its explicit f term, while the textbook's pure sum requires the extra condition $f_0 = 0$.

For a concrete check, take $K(x,t) = xt$ on $[0,1]$, $f = 1$, and $\lambda = 1$. Here $u = \sqrt{3}x$ satisfies $Ku = x\int_0^1 \sqrt{3}\,t^2dt = u/3$, so $\lambda_1 = 3$; $b = \int_0^1 \sqrt{3}\,x\,dx = \sqrt{3}/2$, and $f_0 = f - bu = 1 - 3x/2$. The corrected formula gives, with $\frac{3b}{3-1}u = \frac{3\sqrt{3}}{4}\sqrt{3}\,x = \frac{9}{4}x$,

$$\varphi = f_0 + \frac{3b}{3-1}u = 1 - \frac{3}{2}x + \frac{9}{4}x = \boxed{1 + \frac{3}{4}x}.$$

Indeed $1 + x\int_0^1 t(1 + 3t/4)dt = 1 + x(\frac{1}{2} + \frac{1}{4}) = 1 + 3x/4$. The sum without f_0 would give $9x/4$, which fails the equation: its right side is again $1 + 3x/4$.

For a compact self-adjoint operator the nonzero operator eigenvalues $\mu_i = 1/\lambda_i$ can accumulate only at zero. For a fixed finite λ, all but finitely many satisfy $|\lambda\mu_i| < 1/2$, so their multipliers $\lambda_i/(\lambda_i - \lambda) = 1/(1 - \lambda\mu_i)$ obey $|1/(1 - \lambda\mu_i)| \leq 1/(1 - |\lambda\mu_i|) \leq 2$. The finitely many remaining

denominators are nonzero away from resonance and therefore have a positive minimum modulus. Hence all multipliers are bounded by some M, so $|a_i| \le M|b_i|$, and Parseval's identity in the closed span of the φ_i gives $\sum_i |a_i|^2 \le M^2 \sum_i |b_i|^2 = M^2 \|f - f_0\|^2 < \infty$: the solution expansion converges in L^2 whenever $f \in L^2$. At $\lambda = \lambda_j$, the projected equation instead requires $b_j = 0$ for every eigenfunction in that eigenspace; if so, its solution coefficients are free, and otherwise there is no solution.

22 Chapter 22. Calculus of Variations

Problems

22.1 Euler Equation

22.1.1 For $dy/dx \equiv y_x \neq 0$, show the equivalence of the two forms of Euler's equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y_x} = 0$$

and

$$\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y_x \frac{\partial f}{\partial y_x}\right) = 0.$$

Solution.

The two partial derivatives in the original textbook statements are interchanged. The correct equivalent forms are

$$\boxed{\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y_x} = 0}, \qquad \boxed{\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y_x \frac{\partial f}{\partial y_x}\right) = 0}.$$

To prove the equivalence, put $p = y_x$, so $p' = y_{xx}$, and regard $f = f(y, p, x)$. Subscripts denote its partial derivatives, evaluated on the curve $y(x)$; d/dx is the total derivative along the curve. The chain rule gives

$$\frac{df}{dx} = f_x + f_y p + f_p p'.$$

By the product rule, $d(pf_p)/dx = p' f_p + p\, df_p/dx$. Therefore

$$\begin{aligned}\frac{d}{dx}(f - pf_p) &= f_x + f_y p + f_p p' - p' f_p - p\frac{df_p}{dx} \\ &= f_x + p\left(f_y - \frac{df_p}{dx}\right),\end{aligned}$$

and consequently

$$f_x - \frac{d}{dx}(f - pf_p) = -p\left(f_y - \frac{df_p}{dx}\right).$$

The second corrected expression is $-p$ times the first. For $p \neq 0$, either one vanishes exactly when the other does. At $p = 0$ the second can vanish without implying the first: for $f = y^2$ and $y \equiv 1$, $f_x - d(f - pf_p)/dx = -d(1)/dx = 0$, whereas $f_y - df_p/dx = 2$.

The printed versions are not generally equivalent. For $f = y$ and $y(x) = 2x$ we have $f_x = f_p = 0$, $f_y = 1$ and $f - pf_p = y = 2x$, so the first printed expression is $f_x - df_p/dx = 0$, whereas the second is $f_y - d(f - pf_p)/dx = 1 - 2 = -1$. This also shows why the stated nonzero-slope condition does not repair the swapped derivatives.

22.1.2 Derive Euler's equation by expanding the integrand of

$$J(\alpha) = \int_{x_1}^{x_2} f\big(y(x,\alpha), y_x(x,\alpha), x\big)\, dx$$

in powers of α. Keep both endpoint values fixed and assume f is C^3 near the smooth trial curves, so that the expansion through order α^2 has an $O(\alpha^3)$ remainder.

Note. The stationary condition is $\partial J(\alpha)/\partial\alpha = 0$, evaluated at $\alpha = 0$. The terms quadratic in α may be useful in establishing the nature of the stationary solution (maximum, minimum, or saddle point).

Solution.

Choose $y(x,\alpha) = y(x) + \alpha\eta(x)$ with $\eta(x_1) = \eta(x_2) = 0$, so $y_x(x,\alpha) = y'(x) + \alpha\eta'(x)$. Taylor expansion about $\alpha = 0$ gives

$$\begin{aligned} f(y + \alpha\eta, y' + \alpha\eta', x) = f &+ \alpha(f_y\eta + f_{y_x}\eta') \\ &+ \frac{\alpha^2}{2}\left(f_{yy}\eta^2 + 2f_{yy_x}\eta\eta' + f_{y_xy_x}(\eta')^2\right) + O(\alpha^3). \end{aligned}$$

All derivatives on the right are evaluated on the unperturbed curve. Integrating over $[x_1, x_2]$ (the remainder is $O(\alpha^3)$ uniformly in x because f is C^3 and η, η' are bounded) gives

$$J(\alpha) = J(0) + \alpha J'(0) + \frac{\alpha^2}{2}J''(0) + O(\alpha^3),$$

where $J'(0)$ and $J''(0)$ are the integrals of the coefficients of α and $\alpha^2/2$ above. Integrating the linear term by parts, $\int f_{y_x}\eta' dx = [f_{y_x}\eta] - \int (df_{y_x}/dx)\eta\, dx$, so

$$\begin{aligned} J'(0) &= \int_{x_1}^{x_2} (f_y\eta + f_{y_x}\eta')dx \\ &= [f_{y_x}\eta]_{x_1}^{x_2} + \int_{x_1}^{x_2} \left(f_y - \frac{d}{dx}f_{y_x}\right)\eta\, dx. \end{aligned}$$

The boundary term is zero because $\eta(x_1) = \eta(x_2) = 0$. Stationarity, $J'(0) = 0$, must hold for every such η. The bracket is continuous; if it were nonzero at an interior point, it would keep one sign on a small interval there, and a nonnegative $\eta \neq 0$ vanishing outside that interval would give $J'(0) \neq 0$. This fundamental lemma of the calculus of variations gives

$$\boxed{f_y - \frac{d}{dx}f_{y_x} = 0}.$$

The coefficient of α^2 is one half of

$$J''(0) = \int_{x_1}^{x_2} \left(f_{yy}\eta^2 + 2f_{yy_x}\eta\eta' + f_{y_xy_x}(\eta')^2\right) dx.$$

If $J''(0) > 0$ for every admissible $\eta \neq 0$, then $J(\alpha) > J(0)$ for all small $\alpha \neq 0$ in every such family (a minimum along each family); $J''(0) < 0$ for every $\eta \neq 0$ gives a maximum in the same sense, and values of both signs give a saddle point. Thus the sign must be tested over all admissible variations; the vanishing first variation alone does not classify the extremum.

22.1.3 Find the Euler equation corresponding to Eq. (22.14) if $f = f(y_{xx}, y_x, y, x)$, assuming that y and y_x have fixed values at the endpoints of their interval of definition.

$$ANS.\ \frac{d^2}{dx^2}\left(\frac{\partial f}{\partial y_{xx}}\right) - \frac{d}{dx}\left(\frac{\partial f}{\partial y_x}\right) + \frac{\partial f}{\partial y} = 0.$$

Solution.

Write $y_\alpha = y + \alpha\eta$, so $y'_\alpha = y' + \alpha\eta'$ and $y''_\alpha = y'' + \alpha\eta''$. Fixing both y and y' at each endpoint requires $\eta = \eta' = 0$ there. Differentiating $J(\alpha) = \int f(y''_\alpha, y'_\alpha, y_\alpha, x)\,dx$ under the integral sign by the chain rule, the first variation $\delta J = J'(0)$ is

$$\delta J = \int_{x_1}^{x_2} (f_y\eta + f_{y_x}\eta' + f_{y_{xx}}\eta'')dx.$$

The first derivative term gives

$$\int f_{y_x}\eta' dx = [f_{y_x}\eta] - \int \frac{d}{dx} f_{y_x}\,\eta\,dx.$$

For the second derivative term, integrate twice:

$$\int f_{y_{xx}}\eta'' dx = [f_{y_{xx}}\eta'] - \int \frac{d}{dx} f_{y_{xx}}\eta' dx = [f_{y_{xx}}\eta' - (df_{y_{xx}}/dx)\eta] + \int \frac{d^2}{dx^2} f_{y_{xx}}\eta\,dx.$$

Every displayed boundary term vanishes, since each contains η or η' at x_1 or x_2. Thus

$$\delta J = \int_{x_1}^{x_2} \left[f_y - \frac{d}{dx} f_{y_x} + \frac{d^2}{dx^2} f_{y_{xx}}\right]\eta\,dx.$$

This vanishes for every such η, so the fundamental lemma (Exercise 22.1.2) yields

$$\boxed{\frac{d^2}{dx^2}\frac{\partial f}{\partial y_{xx}} - \frac{d}{dx}\frac{\partial f}{\partial y_x} + \frac{\partial f}{\partial y} = 0}.$$

22.1.4 The integrand $f(y, y_x, x)$ of Eq. (22.2) has the form

$$f(y, y_x, x) = f_1(x, y) + f_2(x, y)y_x.$$

(a) Show that the Euler equation leads to

$$\frac{\partial f_1}{\partial y} - \frac{\partial f_2}{\partial x} = 0.$$

(b) For C^1 functions f_1, f_2, distinguish this equality along one stationary curve from an identity on a simply connected region. Under which condition is J independent of the path?

Solution.

(a) For $f = f_1(x, y) + f_2(x, y)y'$,

$$f_y = f_{1y} + f_{2y}y', \qquad f_{y'} = f_2, \qquad \frac{d}{dx} f_{y'} = f_{2x} + f_{2y}y'.$$

The y' terms cancel in Euler's equation, leaving

$$\boxed{f_{1y} - f_{2x} = 0 \quad \text{along a stationary curve}}.$$

(b) Since $dy = y'dx$ along the path, the integral is the line integral

$$J = \int (f_1\,dx + f_2\,dy).$$

If the equality $f_{1y} = f_{2x}$ holds identically on a simply connected region, this differential form is exact. Indeed, by Green's theorem $\oint (f_1\,dx + f_2\,dy) = \iint (f_{2x} - f_{1y})\,dx\,dy = 0$ around every simple closed curve in the region (its interior lies in the region), so $F(x,y) = \int_{(x_1,y_1)}^{(x,y)} (f_1\,dx + f_2\,dy)$ does not depend on the path of integration, and $F_x = f_1$, $F_y = f_2$. Along any path $y(x)$, $dF(x,y(x))/dx = F_x + F_y y' = f_1 + f_2 y' = f$. Then

$$\boxed{J = F(x_2,y_2) - F(x_1,y_1)}$$

is independent of the path in that region. Conversely, if J has the same value on all paths joining any two points of the region, every path is stationary, and (a) gives $f_{1y} = f_{2x}$ along every curve, hence identically. So J is path-independent exactly when $f_{1y} = f_{2x}$ holds throughout the (simply connected) region.

An Euler equation satisfied only on one curve does not imply path independence throughout the region. The identity on the region, together with an appropriate topology, is needed for that conclusion. For example, $f_1 = y^2$, $f_2 = 0$ gives $f_{1y} - f_{2x} = 2y$, which vanishes only on the stationary curve $y = 0$. With both endpoints on the axis, $\int y^2 dx$ is 0 on $y = 0$ but positive on every other path, so it changes when the interior path changes. This example also anticipates the correction required in the next exercise.

22.1.5 Show that the condition that $J = \int f(x,y)\,dx$ has a stationary value

(a) leads to $f_y(x,y(x)) = 0$ along the stationary curve and

(b) imposes no separate condition $f_x = 0$.

Dependence on derivatives of y is not essential for a meaningful variational problem. Give a smooth minimizing solution for an integrand that depends on y.

Note. The situation will change when constraints are introduced (compare to Exercise 22.4.6).

Solution.

There is no y' dependence, so for $y + \alpha\eta$ with $\eta(x_1) = \eta(x_2) = 0$ the first variation is simply

$$\delta J = \int_{x_1}^{x_2} f_y(x,y(x))\eta(x)\,dx.$$

It must vanish for every such η, so by the fundamental lemma (Exercise 22.1.2), or by Euler's equation with $f_{y'} = 0$, the correct stationarity condition is the algebraic equation

$$\boxed{f_y(x,y(x)) = 0 \quad \text{for every interior } x}.$$

This is a condition along the selected curve. It does not force $f(x,y)$ to be independent of y, and absence of derivative dependence does not rule out meaningful, smooth minimizing solutions. The

printed part (a), which concludes that $f(x, y)$ is independent of y, and the printed remark that there is then no continuous, differentiable solution, are both false.

For an explicit smooth counterexample take

$$f(x, y) = (y - \sin x)^2$$

and fix the endpoint values to those of $\sin x$. Then $f_y = 2(y - \sin x)$ vanishes only for $y = \sin x$, which also meets the endpoint values, so the stationary function is $\boxed{y(x) = \sin x}$. Moreover,

$$J[y] = \int_{x_1}^{x_2} (y - \sin x)^2 dx \geq 0, \qquad J[\sin x] = 0,$$

which proves a global minimum despite explicit dependence on y.

If f is independent of y, every path has the same J. If f_y never vanishes, no stationary path exists. Otherwise, smooth root branches of $f_y = 0$ may furnish stationary paths, subject to the endpoint conditions. Regarding (b), there is no additional equation $f_x = 0$: x is the independent variable and only y is varied. The second form of Euler's equation (Exercise 22.1.1) adds nothing either: with $f_{y'} = 0$ it reads

$$f_x - \frac{df}{dx} = f_x - (f_x + f_y y') = -f_y y' = 0,$$

which already follows from $f_y = 0$. But x can affect the selected curve through $f_y(x, y) = 0$, as the counterexample demonstrates.

22.1.6 A soap film stretched between two rings of unit radius centered at $\pm x_0$ will have its closest approach to the x-axis at $x = 0$, with the distance from the axis given by c_1, with x_0 and c_1 related by Eq. (22.26) or Eq. (22.29).

(a) Show that dc_1/dx_0 becomes infinite when $x_0 \sinh(x_0/c_1) = 1$, indicating that the soap film becomes unstable if x_0 is increased beyond the value satisfying this condition.

(b) Show that the condition of part (a) is equivalent to

$$\frac{x_0}{c_1} = \coth \frac{x_0}{c_1}.$$

(c) Solve the transcendental equation of part (b) to obtain the critical value of x_0/c_1 and show that the separate values of x_0 and c_1 are then approximately $x_0 \approx 0.6627$ and $c_1 \approx 0.5524$.

Solution.

Put $c = c_1 > 0$ and $u = x_0/c$. The catenoid $y = c\cosh(x/c)$ must have radius 1 at $x = \pm x_0$, so the boundary equation is $c\cosh u = 1$.

(a) Write $c' = dc/dx_0$. Since $u = x_0/c$, the quotient rule gives $du/dx_0 = 1/c - x_0 c'/c^2$. Implicit differentiation of $c\cosh u = 1$ with respect to x_0 therefore gives

$$c' \cosh u + c \sinh u \left(\frac{1}{c} - \frac{x_0 c'}{c^2} \right) = 0,$$

that is, $c' \cosh u + \sinh u - u c' \sinh u = 0$; hence

$$c'(\cosh u - u \sinh u) = -\sinh u.$$

Multiply the numerator and denominator of $c' = -\sinh u/(\cosh u - u\sinh u)$ by c and use $c\cosh u = 1$, $cu = x_0$:

$$\boxed{\frac{dc}{dx_0} = -\frac{c\sinh u}{1 - x_0\sinh u}}.$$

The numerator is nonzero for separated rings ($x_0 > 0$, so $u > 0$). Thus the derivative diverges at $x_0\sinh u = 1$.

(b) Divide that condition by $c\cosh u = 1$; since $x_0/c = u$,

$$\frac{x_0\sinh u}{c\cosh u} = u\tanh u = 1, \qquad \boxed{u = \coth u}.$$

Equivalently, the boundary equation parametrizes all solutions by $u > 0$:

$$c = \operatorname{sech} u, \qquad x_0 = cu = u\operatorname{sech} u.$$

By the product rule,

$$\frac{dx_0}{du} = \operatorname{sech} u - u\operatorname{sech} u\tanh u = \operatorname{sech} u\,(1 - u\tanh u).$$

Because $d(u\tanh u)/du = \tanh u + u\operatorname{sech}^2 u > 0$, the product $u\tanh u$ increases strictly from zero to infinity. So $x_0(u)$ increases while $u\tanh u < 1$ and decreases afterwards: there is one critical point, a maximum, at the root u_c of $u = \coth u$. There $dc/dx_0 = (dc/du)/(dx_0/du)$ is infinite, since $dc/du = -\operatorname{sech} u\tanh u \neq 0$, in agreement with (a). Each x_0 below the maximum is reached twice: on the shallow branch ($u < u_c$, larger neck radius c) and on the deep branch ($u > u_c$). Beyond the maximum no positive-neck catenoid satisfies the two boundary conditions.

(c) Newton iteration for $F(u) = u\tanh u - 1$, with $F'(u) = \tanh u + u\operatorname{sech}^2 u$, is

$$u_{j+1} = u_j - \frac{u_j\tanh u_j - 1}{\tanh u_j + u_j\operatorname{sech}^2 u_j}.$$

Starting from the guess 1.2, two steps already give all the digits shown below; then $c_c = \operatorname{sech} u_c$ (from $c\cosh u = 1$) and $x_{0c} = u_c c_c$:

$$\boxed{u_c = 1.19967864025773, \quad c_c = 0.552434124530883, \quad x_{0c} = 0.662743419349182}.$$

The Instructor's Manual's $c_c = 0.5523434$ is a misprint; substituting $c_c = \operatorname{sech} u_c$ gives the value above, which agrees with the textbook's rounded 0.5524.

The stability threshold can also be seen in the second variation $J''(0)$ of Exercise 22.1.2, written $\delta^2 J$, for the area integrand without its factor 2π, $f = y\sqrt{1 + y'^2}$. Its second partial derivatives with respect to y and y' are

$$f_{yy} = 0, \qquad f_{yy'} = \frac{y'}{\sqrt{1 + y'^2}}, \qquad f_{y'y'} = \frac{y}{(1 + y'^2)^{3/2}}.$$

On the catenoid, $y = c\cosh(x/c)$, $y' = \sinh(x/c)$ and $\sqrt{1 + y'^2} = \cosh(x/c)$, so $f_{yy'} = \tanh(x/c)$ and $f_{y'y'} = c\operatorname{sech}^2(x/c)$. Thus, for a fixed-endpoint variation η,

$$\delta^2 J = \int_{-x_0}^{x_0} \left[2\tanh(x/c)\eta\eta_x + c\operatorname{sech}^2(x/c)\eta_x^2\right] dx.$$

The first term is $\tanh(x/c)(\eta^2)_x$. Integration by parts has zero endpoint term, because $\eta(\pm x_0)=0$, and leaves $-\int[d\tanh(x/c)/dx]\eta^2dx=-c^{-1}\int\operatorname{sech}^2(x/c)\eta^2dx$. Now let $u=x/c$ run over $[-u_0,u_0]$, where $u_0=x_0/c$ is the endpoint value called u above. Then $dx=c\,du$ and $\eta_x=\eta_u/c$ turn $c\operatorname{sech}^2(x/c)\eta_x^2dx$ into $\operatorname{sech}^2u\,\eta_u^2du$ and $c^{-1}\operatorname{sech}^2(x/c)\eta^2dx$ into $\operatorname{sech}^2u\,\eta^2du$, so

$$\delta^2J=\int_{-u_0}^{u_0}\operatorname{sech}^2u\,(\eta_u^2-\eta^2)du.$$

Put $p(u)=\operatorname{sech}^2u$ and

$$w(u)=\cosh u-u\sinh u=\cosh u\,(1-u\tanh u),$$

which is $\partial y/\partial c=\cosh(x/c)-(x/c)\sinh(x/c)$ for the family $y=c\cosh(x/c)$ at fixed x, written in terms of $u=x/c$. Since $w'=-u\cosh u$, we have $pw'=-u\operatorname{sech}u$ and

$$(pw')'=-\operatorname{sech}u+u\operatorname{sech}u\tanh u=-pw,$$

that is, $(pw')'+pw=0$. For $u_0<u_c$ (the shallow branch), $|u|\tanh|u|<1$ on $[-u_0,u_0]$, so $w>0$ there. Write $\eta=wv$; then $v=\eta/w$ is smooth, $v(\pm u_0)=0$, and $\eta'=w'v+wv'$. Expanding the square,

$$p(\eta'^2-\eta^2)=pw^2v'^2+2pww'vv'+(pw'^2-pw^2)v^2=pw^2v'^2+(pww'v^2)',$$

because $(pww'v^2)'=(pw')'wv^2+pw'^2v^2+2pww'vv'$ and $(pw')'w=-pw^2$. The boundary term $[pww'v^2]_{-u_0}^{u_0}$ is zero because $v(\pm u_0)=0$. Consequently

$$\delta^2J=\int_{-u_0}^{u_0}\operatorname{sech}^2u\,w^2(v')^2du>0$$

for a nonzero fixed-endpoint variation: $v'\equiv0$ together with $v(\pm u_0)=0$ would give $v\equiv0$, hence $\eta\equiv0$. At $u_0=u_c$, $w(\pm u_c)=\cosh u_c(1-u_c\tanh u_c)=0$, so $\eta=w$ is itself an admissible variation, and integration by parts with $(pw')'=-pw$ gives $\delta^2J[w]=[pww']_{-u_c}^{u_c}-\int w[(pw')'+pw]\,du=0$. This identifies the loss of strict stability on the shallow branch.

22.1.7 A soap film is stretched across the space between two rings of unit radius centered at $\pm x_0$ on the x-axis and perpendicular to the x-axis. Using the solution developed in Example 22.1.3, set up the transcendental equations for the condition that x_0 is such that the area of the curved surface of rotation equals the area of the two rings (Goldschmidt discontinuous solution). Solve for x_0.

Solution.

For $y=c\cosh(x/c)$ between $x=\pm x_0$, put $u=x_0/c$. The endpoint condition is $c\cosh u=1$. Since $y'=\sinh(x/c)$, $\sqrt{1+y'^2}=\cosh(x/c)$ and $y\sqrt{1+y'^2}=c\cosh^2(x/c)$. The integrand is even; with $v=x/c$, $dx=c\,dv$, and $\cosh^2v=(1+\cosh2v)/2$, the full surface area is

$$\begin{aligned}A&=2\pi\int_{-x_0}^{x_0}y\sqrt{1+y'^2}dx=4\pi c^2\int_0^u\cosh^2v\,dv\\&=2\pi c^2\left[v+\tfrac12\sinh2v\right]_0^u=\pi c^2[\sinh(2u)+2u].\end{aligned}$$

The two unit disks have area 2π. Equating the areas, dividing by π, and using $c^2=\operatorname{sech}^2u$ gives

$$\sinh(2u)+2u=2\cosh^2u=1+\cosh(2u).$$

Since $\sinh 2u - \cosh 2u = -e^{-2u}$, this is the simpler single equation

$$\boxed{2u - 1 = e^{-2u}}.$$

Its left-minus-right side $g(u) = 2u - 1 - e^{-2u}$ has $g(0) = -2$, $g(u) \to \infty$ as $u \to \infty$, and $g'(u) = 2 + 2e^{-2u} > 0$, so it has exactly one positive root. Since $g(0.6) = -0.101 < 0 < g(0.7) = 0.153$, bisection on $(0.6, 0.7)$, or Newton iteration $u_{j+1} = u_j - g(u_j)/g'(u_j)$ from the starting value 0.6, gives

$$u = 0.639232271380537, \qquad c = \operatorname{sech} u = 0.825517453652511$$

(both sides of the boxed equation then equal 0.278464542761). Thus the required half-separation is

$$\boxed{x_0 = cu = u \operatorname{sech} u = 0.527697396962572}.$$

The full ring separation is $2x_0 = 1.05539479392514$. The root is below $u_c = 1.19967\ldots$ from Exercise 22.1.6, so the equality occurs on the shallow catenoid branch. With $c = \operatorname{sech} u$ the area is $A = 2\pi(\tanh u + u \operatorname{sech}^2 u)$, and $dA/du = 4\pi \operatorname{sech}^2 u\,(1 - u \tanh u) > 0$ on that branch, as is dx_0/du. Hence the shallow catenoid has less area than the two disks for $x_0 < 0.527697\ldots$ and more for $0.527697\ldots < x_0 \le x_{0c}$, where the Goldschmidt solution has less.

22.1.8 In Example 22.1.1, expand $J[y(x,\alpha)] - J[y(x,0)]$ in powers of α. The term linear in α leads to the Euler equation and to the straight-line solution, Eq. (22.16). Investigate the α^2 term and show that the stationary value of J, the straight-line distance, is a minimum.

Solution.

Let $y_*(x) = y_1 + m(x - x_1)$, where $m = (y_2 - y_1)/(x_2 - x_1)$ and $x_2 > x_1$. For $y_\alpha = y_* + \alpha\eta$, with $\eta(x_1) = \eta(x_2) = 0$,

$$J[y_\alpha] = \int_{x_1}^{x_2} \sqrt{1 + (m + \alpha\eta')^2}\, dx.$$

The first two derivatives of $g(p) = \sqrt{1+p^2}$ are $g'(p) = p/\sqrt{1+p^2}$ and $g''(p) = (1+p^2)^{-3/2}$. Taylor's theorem at $p = m$ gives, at each x,

$$g(m + \alpha\eta') = g(m) + g'(m)\,\alpha\eta' + \tfrac{1}{2}g''(m)\,\alpha^2(\eta')^2 + O(\alpha^3),$$

with a remainder uniform in x, because $g'''(p) = -3p(1+p^2)^{-5/2}$ and η' are bounded. Integrating over $[x_1, x_2]$,

$$J[y_\alpha] - J[y_*] = \alpha \frac{m}{\sqrt{1+m^2}} \int_{x_1}^{x_2} \eta' dx + \frac{\alpha^2}{2(1+m^2)^{3/2}} \int_{x_1}^{x_2} (\eta')^2 dx + O(\alpha^3).$$

The linear integral is $\eta(x_2) - \eta(x_1) = 0$. The quadratic coefficient is strictly positive for any nonzero admissible η: zero would require $\eta' = 0$, and the endpoints would then require $\eta = 0$. Thus the straight line is a strict local minimum.

It is also the global minimum. Since $g'' > 0$, g is strictly convex, so its graph lies above each tangent line: $g(m + t) \ge g(m) + g'(m)t$, with equality only for $t = 0$. With $t = \eta'(x)$,

$$g(m + \eta') \ge g(m) + g'(m)\eta'.$$

After integration the last term again vanishes, so $J[y_* + \eta] \ge J[y_*]$, with equality only if $\eta' \equiv 0$, that is, only for $\eta = 0$.

22.1.9 Take $x_1 < x_2$, real-valued paths, and fixed endpoint values. In (b), allow endpoint jumps, and compare with the continuous class.

(a) Show that the integral

$$J = \int_{x_1}^{x_2} f(y, y_x, x)\, dx, \quad \text{with} \quad f = y(x),$$

has no extreme values.

(b) If $f(y, y_x, x) = y^2(x)$, find a discontinuous solution similar to the Goldschmidt solution for the soap-film problem.

Solution.

(a) For $f = y$, Euler's equation $f_y - d(f_{y_x})/dx = 0$ reads $1 = 0$, so no curve is stationary. Directly: let η be a smooth function of compact support in (x_1, x_2) with $\int \eta\, dx \neq 0$. Then

$$J[y + \alpha\eta] - J[y] = \alpha \int_{x_1}^{x_2} \eta(x) dx.$$

Arbitrarily small positive and negative α change J in opposite directions. Thus no admissible curve is even a local extremum. Taking $|\alpha| \to \infty$ also shows that J is unbounded above and below.

(b) For $f = y^2$, nonnegativity gives $J \geq 0$. Euler's equation is $f_y = 2y = 0$ (compare Exercise 22.1.5), which forces $y = 0$ at every interior point; nonzero endpoint values can then be met only by jumps at the ends, just as the Goldschmidt solution joins the two disks by the axis. In a class allowing endpoint jumps, set

$$y(x) = 0 \quad (x_1 < x < x_2),$$

while assigning the prescribed values at the two endpoints. Changing isolated points does not change the integral, so this discontinuous path has $\boxed{J_{\min} = 0}$.

If continuity is required and either endpoint value is nonzero, this is an infimum, not an attained minimum. If both endpoint values are zero, the ordinary smooth function $y = 0$ is already the minimum. To see the infimum explicitly, let the endpoint values be A, B, and let $0 < \epsilon < (x_2 - x_1)/2$. Let y_ϵ fall linearly from A to zero on $[x_1, x_1 + \epsilon]$, vanish on $[x_1 + \epsilon, x_2 - \epsilon]$, and rise linearly from zero to B on $[x_2 - \epsilon, x_2]$ (round off the corners if a smooth path is wanted). Since $|y_\epsilon| \leq |A|$ on the first layer and $|y_\epsilon| \leq |B|$ on the last,

$$0 \leq J[y_\epsilon] \leq \epsilon(A^2 + B^2) \longrightarrow 0.$$

On the other hand, a continuous function with $y(x_1) = A \neq 0$ satisfies $|y| > |A|/2$ on some interval $[x_1, x_1 + \delta]$, so $J \geq \delta A^2/4 > 0$; likewise at x_2. Hence no continuous path attains the value 0.

22.1.10 Fermat's principle of optics states that a light ray in a medium for which n is the (position-dependent) index of refraction will follow the path $y(x)$ for which

$$\int_{x_1, y_1}^{x_2, y_2} n(y, x)\, ds$$

is stationary. For $y_2 = y_1 = 1$, $-x_1 = x_2 = 1$, find the ray path and determine whether the minimum is attained if

(a) $n = e^y$,

(b) $n = a(y - y_0)$, $\quad y > y_0$, with $a > 0$ and $y_0 < 1$.

Solution.

Use $ds = \sqrt{1 + y'^2}\,dx$, so $f = n(y)\sqrt{1 + y'^2}$ and $f_{y'} = n(y)y'/\sqrt{1 + y'^2}$. Since neither index depends explicitly on x, the Beltrami first integral $f - y' f_{y'} = C$ (the second form of Euler's equation in Exercise 22.1.1 with $f_x = 0$) holds:

$$f - y' f_{y'} = n(y)\,\frac{(1 + y'^2) - y'^2}{\sqrt{1 + y'^2}} = \frac{n(y)}{\sqrt{1 + y'^2}} = C.$$

Conversely, a solution of this first integral satisfies Euler's equation wherever $y' \neq 0$ (Exercise 22.1.1), but not necessarily where $y' = 0$. On an interval where y is constant, $f_{y'} = 0$ and Euler's expression $f_y - df_{y'}/dx$ reduces to $f_y = n'(y)$, which is $e^y > 0$ in (a) and $a > 0$ in (b). So a ray contains no horizontal segment. In particular, the constant path $y \equiv 1$ satisfies the first integral, with $C = n(1)$, and meets both endpoint conditions, but it is not a ray. The rays found below have $y' = 0$ only at $x = 0$, so Euler's equation holds for $x \neq 0$ and, by continuity, at $x = 0$ as well.

(a) For $n = e^y$ this gives $y'^2 = e^{2y}/C^2 - 1$, with $C > 0$. Separate variables and put $t = Ce^{-y} \in (0, 1]$, so that $\sqrt{e^{2y}/C^2 - 1} = \sqrt{1 - t^2}/t$ and $dt = -t\,dy$:

$$\pm dx = \frac{dy}{\sqrt{e^{2y}/C^2 - 1}} = \frac{t\,dy}{\sqrt{1 - t^2}} = -\frac{dt}{\sqrt{1 - t^2}} = d(\arccos t).$$

Hence $\arccos(Ce^{-y}) = \pm(x - b)$, that is, $e^y = C\sec(x - b)$ on a branch where the cosine is positive. Indeed,

$$y = \ln C - \ln\cos(x - b), \quad y' = \tan(x - b),$$

so $\sqrt{1 + y'^2} = \sec(x - b)$ and $e^y/\sqrt{1 + y'^2} = C$. Equal endpoint values at $x = \pm 1$ require $\cos(1 - b) = \cos(1 + b)$, i.e., $2\sin 1 \sin b = 0$, so $b = k\pi$; since $\cos(x - k\pi) = (-1)^k \cos x$ must be positive on all of $[-1, 1]$, the branch covering the entire interval has $\cos(x - b) = \cos x$, i.e., $b = 0$. The condition $y(1) = 1$, i.e. $e = C\sec 1$, gives $C = e\cos 1$. Thus

$$\boxed{y(x) = 1 + \ln\cos 1 - \ln\cos x, \qquad -1 \le x \le 1}.$$

On it $e^y = e\cos 1 \sec x$ and $\sqrt{1 + y'^2} = \sec x$, so the optical length is

$$J = e\cos 1 \int_{-1}^{1} \sec^2 x\,dx = 2e\cos 1 \tan 1 = 2e\sin 1.$$

This is a global minimum, not just a stationary path. Set

$$U = e^y \cos x, \qquad V = e^y \sin x,$$

polar coordinates with radius e^y and angle $x \in [-1, 1]$, so the map $(x, y) \mapsto (U, V)$ is one-to-one. Since $dU = e^y(\cos x\,dy - \sin x\,dx)$ and $dV = e^y(\sin x\,dy + \cos x\,dx)$, the cross terms cancel and $dU^2 + dV^2 = e^{2y}(dx^2 + dy^2)$. The optical length $\int e^y\,ds$ is therefore ordinary Euclidean length in the (U, V) plane. The endpoints are $(e\cos 1, \pm e\sin 1)$, and their straight segment has $U = e\cos 1$, precisely the curve above (on it $e^y \cos x = C = e\cos 1$). Its length is $2e\sin 1$, the lower bound for any joining path; every other path has a different image, which is strictly longer than the segment.

The Instructor's Manual's inverse formula uses $1+\pi^2/16$ where $\sec^2 1$ is required. At $y=1$ it gives $x=\arctan(\pi/4)$ instead of $x=1$, so it fails the stated endpoint condition.

(b) A positive index requires $a>0$ and the endpoints require $H:=1-y_0>0$. Put $u(x)=y(x)-y_0$, so $n=au$ and $u'=y'$. The first integral, divided by a, is

$$\frac{u}{\sqrt{1+u'^2}}=c>0, \qquad u'^2=\frac{u^2}{c^2}-1.$$

Separate variables on a rising part ($u'>0$), using $\int du/\sqrt{u^2-c^2}=\operatorname{arcosh}(u/c)$:

$$dx=\frac{c\,du}{\sqrt{u^2-c^2}}, \qquad x-b=c\,\operatorname{arcosh}(u/c).$$

A falling part gives $x-b=-c\operatorname{arcosh}(u/c)$; both halves are $u=c\cosh[(x-b)/c]$. Equal endpoint heights require $\cosh[(1-b)/c]=\cosh[(1+b)/c]$, i.e. $|1-b|=|1+b|$, so the horizontal shift is $b=0$:

$$\boxed{y(x)=y_0+c\cosh(x/c), \qquad c\cosh(1/c)=H}.$$

With $w=1/c$, the boundary equation reads $\cosh w/w=H$, or

$$\frac{1}{H}=w\operatorname{sech} w.$$

Exercise 22.1.6 shows that the maximum of the right side is 0.662743419349182, at $w=u_c$. Consequently regular stationary curves exist only for

$$H\ge\frac{1}{0.662743419349182}=1.50887956153832, \quad \text{i.e.} \quad y_0\le -0.50887956153832.$$

Above that threshold there are two branches. Scaling both coordinates by H, $X=x/H$ and $Y=u/H$ (so $dY/dX=u'$), gives

$$\int_{-1}^{1}u\sqrt{1+u'^2}\,dx=H^2\int_{-1/H}^{1/H}Y\sqrt{1+Y'^2}\,dX, \qquad Y(\pm 1/H)=1:$$

the catenoid problem of Exercise 22.1.6 with unit-ring half-separation $1/H$. The scaled catenary $Y=(c/H)\cosh[X/(c/H)]$ has neck c/H, so its endpoint parameter (half-separation over neck radius, called u_0 in Exercise 22.1.6) is $(1/H)/(c/H)=w$. Hence the smaller-w, larger-c branch ($w<u_c$) has positive second variation and is the locally minimizing one.

Part (b) has no single smooth minimizing ray for arbitrary $y_0<1$. Also, Fermat's variational statement is generally stationarity; an available stationary ray need not be the absolute minimum. The distinction can be quantified here. On the catenary $u\sqrt{1+u'^2}=c\cosh^2(x/c)$, so, with $s=x/c$ and $\int_0^w\cosh^2 s\,ds=\frac12 w+\frac14\sinh 2w$, its optical length is

$$J_{\text{cat}}=2ac^2\int_0^w\cosh^2 s\,ds=ac^2\left[w+\frac12\sinh(2w)\right],$$

and since $c^2/H^2=\operatorname{sech}^2 w$,

$$\frac{J_{\text{cat}}}{aH^2}=\operatorname{sech}^2 w\,(w+\sinh w\cosh w)=w\operatorname{sech}^2 w+\tanh w.$$

To see the competing boundary cost, descend vertically from H to a small $\delta > 0$, travel the horizontal distance 2 at that height, and ascend to H. On the vertical segments $ds = |du|$. This piecewise path has

$$J_\delta = a\left(2\int_\delta^H u\,du + 2\delta\right) = a(H^2 - \delta^2 + 2\delta) \longrightarrow aH^2.$$

Smooth positive graphs approach the same limit. Let u fall smoothly and monotonically from H to δ on $[-1, -1+\epsilon]$, equal δ on $[-1+\epsilon, 1-\epsilon]$, and rise likewise back to H on $[1-\epsilon, 1]$. Then $\int_{-1}^1 u\,dx \le 2\epsilon H + 2\delta$, since $u \le H$ on the two layers. Moreover $\sqrt{1+u'^2} \le 1 + |u'|$, and on each monotone layer $\int u|u'|\,dx = \int_\delta^H u\,du = \frac{1}{2}(H^2 - \delta^2)$, so

$$\frac{J}{a} \le \int_{-1}^1 u\,dx + \int_{-1}^1 u|u'|\,dx \le 2\epsilon H + 2\delta + H^2 - \delta^2 \longrightarrow H^2 \quad (\epsilon, \delta \to 0).$$

Equality of the catenary cost with this limit, $w\operatorname{sech}^2 w + \tanh w = 1$, becomes after multiplication by $2\cosh^2 w$ the equation $2w + \sinh 2w = 2\cosh^2 w$ of Exercise 22.1.7, with the same root:

$$2w - 1 = e^{-2w}, \qquad w = 0.639232271380537.$$

On the shallow branch the ratio $w\operatorname{sech}^2 w + \tanh w$ has derivative $2\operatorname{sech}^2 w\,(1 - w\tanh w) > 0$, while $H = \cosh w/w$ decreases as w increases. So the ratio is below 1 exactly when $w < 0.639232\ldots$, i.e., when H exceeds its value $\cosh w/w = 1/(w\operatorname{sech} w) = 1/0.527697396962572$ at that root (there $w\operatorname{sech} w$ is the half-separation x_0 of Exercise 22.1.7): the shallow catenary beats that boundary limit only for

$$H > 1.89502545541442.$$

For a direct global proof, let $d = \min u > 0$ on a competing path. Cauchy–Schwarz applied to the vectors $(d, \sqrt{u^2 - d^2})$ and $(1, |u'|)$, of lengths u and $\sqrt{1+u'^2}$, gives

$$u\sqrt{1+u'^2} \ge d + \sqrt{u^2 - d^2}\,|u'|.$$

The path descends from H to d and returns to H. On a descending or ascending stretch $|u'|\,dx = |du|$, so each of the two stretches contributes at least $\int_d^H \sqrt{u^2 - d^2}\,du$. With the antiderivative $\frac{1}{2}u\sqrt{u^2 - d^2} - \frac{1}{2}d^2 \operatorname{arcosh}(u/d)$, integration over the horizontal span of length 2 yields

$$\frac{J}{a} \ge 2d + 2\int_d^H \sqrt{u^2 - d^2}\,du =: B(d) = 2d + H\sqrt{H^2 - d^2} - d^2 \operatorname{arcosh}(H/d).$$

For a path with additional rises and falls, the absolute-value integral is only larger. Differentiation gives

$$B'(d) = 2 - 2d\operatorname{arcosh}(H/d), \qquad B(0^+) = H^2 :$$

the d-derivative $-Hd/\sqrt{H^2 - d^2}$ of $H\sqrt{H^2 - d^2}$ cancels the term $+Hd/\sqrt{H^2 - d^2}$ that comes from differentiating $\operatorname{arcosh}(H/d)$ in $-d^2\operatorname{arcosh}(H/d)$, and $d^2\operatorname{arcosh}(H/d) \to 0$ as $d \to 0^+$ because $\operatorname{arcosh}(H/d) \approx \ln(2H/d)$.

The product $d\operatorname{arcosh}(H/d)$ vanishes at both ends $d = 0, H$ and has a single maximum: putting $d = H\operatorname{sech} w$ reduces it to $Hw\operatorname{sech} w$, whose maximum is $0.662743\ldots H$. If $H \le 1.50887956\ldots$, then $B' \ge 0$ on $(0, H)$, with at most one zero, so B increases and $B > H^2$ throughout. For larger H there are two critical points: $B' > 0$ near $d = 0$, $B' < 0$ between them (where $d\operatorname{arcosh}(H/d) > 1$), and $B' > 0$ beyond. So the smaller d is a local maximum of B and the larger d is its local minimum. At both critical points $d\operatorname{arcosh}(H/d) = 1$, i.e., $H/d = \cosh(1/d)$: this is the catenary endpoint

equation $c\cosh(1/c) = H$ with $d = c$, and $w = \operatorname{arcosh}(H/d) = 1/d$. The two roots of $Hw\operatorname{sech} w = 1$ lie on either side of u_c (Exercise 22.1.6), and $d = H\operatorname{sech} w$ decreases as w increases. So the larger root d, the minimum of B, has $w < u_c$: it is the neck c of the shallow catenary. Equality in the preceding bound holds for that catenary, since there $c + c\sinh^2(x/c) = c\cosh^2(x/c) = u\sqrt{1+u'^2}$. The endpoint $d = H$ cannot minimize B, since $B'(H^-) = 2$. Thus, for $H > 1.50887956\ldots$, the global lower bound is the smaller of the shallow-catenary cost and H^2. The former is at most H^2, and is then attained by the catenary, precisely for $H \geq 1.8950254554144\ldots$; at equality it ties the boundary limit. For smaller H, every $d > 0$ gives a strictly larger cost than H^2, so no smooth positive-index global minimizer exists. The infimum aH^2 is approached by the endpoint-layer paths described above.

In particular, the Instructor's Manual's claim of a global catenary minimum at $y_0 = -0.70$ is incorrect: $H = 1.70$ lies below the cost-crossing value 1.8950254554.

22.1.11 A particle moves, starting at rest, from point A on the surface of the Earth to point B (also on the surface) by sliding frictionlessly through a tunnel. Find the differential equation satisfied by the path if the transit time is to be a minimum. Assume the Earth to be a nonrotating sphere of uniform density. First take distinct nonantipodal endpoints, so $0 < r_0 < R$; treat the antipodal diameter as a limit.

Hint. The gain in kinetic energy of a particle of mass m a distance $r < R$ from the center of the Earth, with R the Earth's radius, is $\frac{1}{2}mg(R^2 - r^2)/R$, where g is the gravitational acceleration at the Earth's surface. It is convenient to describe the path of the particle (in the plane through A, B, and the center of the Earth) by plane polar coordinates (r, θ), with A at $(R, -\varphi)$ and B at (R, φ).

ANS. Letting r_0 be the minimum value of r (reached at $\theta = 0$), Eq. (22.21) yields $r_\theta^2 = \dfrac{r^2R^2(r^2 - r_0^2)}{r_0^2(R^2 - r^2)}$ (the constant in that equation has the value such that $r_\theta = 0$ at $\theta = 0$).

The solution for the path is a hypocycloid, generated by a circle of radius $\frac{1}{2}(R - r_0)$ rolling inside the circle of radius R. You might like to show that the transit time is

$$t = \pi\frac{(R^2 - r_0^2)^{1/2}}{(Rg)^{1/2}}.$$

For details see P. W. Cooper, *Am. J. Phys.* **34**: 68 (1966); G. Veneziano, et al., **34**: 701 (1966).

Solution.

Inside a uniform sphere only the mass within radius r, the fraction r^3/R^3 of the total, attracts the particle (Newton's shell theorem: the uniform shell outside radius r exerts no net force, and the inner ball attracts as if its mass were at the center), so the surface force mg is multiplied by $(r^3/R^3)(R^2/r^2) = r/R$: the inward gravitational force is $F_r = -mgr/R$. Thus

$$U(r) - U(R) = -\int_R^r F_r\,dr' = \frac{mg}{R}\int_R^r r'\,dr' = \frac{mg}{2R}(r^2 - R^2).$$

The positive quantity in the printed hint is the *decrease* in potential energy from the surface, and hence the kinetic energy gained. It is not the potential energy itself with surface value zero: taken as $U(r)$, it would give the outward force $-dU/dr = +mgr/R$. Energy conservation for release from rest gives

$$\frac{1}{2}mv^2 = U(R) - U(r) = \frac{mg}{2R}(R^2 - r^2), \qquad v = \sqrt{\frac{g}{R}}\sqrt{R^2 - r^2}.$$

Writing the path as $r(\theta)$, the arc length is $ds = \sqrt{dr^2 + r^2 d\theta^2} = \sqrt{r_\theta^2 + r^2}\, d\theta$, and its travel time $T = \int ds/v$ is

$$T = \sqrt{\frac{R}{g}} \int_{-\varphi}^{\varphi} \frac{\sqrt{r_\theta^2 + r^2}}{\sqrt{R^2 - r^2}}\, d\theta.$$

The integrand has no explicit θ dependence. With $p = r_\theta$, omit the constant $\sqrt{R/g}$ and write

$$f = \frac{\sqrt{r^2 + p^2}}{\sqrt{R^2 - r^2}}, \qquad f_p = \frac{p}{\sqrt{R^2 - r^2}\sqrt{r^2 + p^2}}.$$

The Beltrami first integral (Exercise 22.1.1 with θ in place of x) says that

$$f - p f_p = \frac{(r^2 + p^2) - p^2}{\sqrt{R^2 - r^2}\sqrt{r^2 + p^2}} = \frac{r^2}{\sqrt{R^2 - r^2}\sqrt{r^2 + p^2}}$$

is constant. At the minimum radius $r = r_0 > 0$, reached at $\theta = 0$ by symmetry, $r(\theta)$ is stationary, so $p = 0$ there, and this constant is $r_0^2/(r_0\sqrt{R^2 - r_0^2}) = r_0/\sqrt{R^2 - r_0^2}$. Squaring and rearranging,

$$r^4(R^2 - r_0^2) = r_0^2(R^2 - r^2)(r^2 + p^2),$$

so that

$$p^2 = \frac{r^4(R^2 - r_0^2)}{r_0^2(R^2 - r^2)} - r^2 = \frac{r^2[r^2R^2 - r^2r_0^2 - r_0^2R^2 + r_0^2r^2]}{r_0^2(R^2 - r^2)},$$

$$\boxed{r_\theta^2 = \frac{r^2R^2(r^2 - r_0^2)}{r_0^2(R^2 - r^2)}}.$$

To verify the stated geometric identification, put $\rho = (R - r_0)/2$ and consider the hypocycloid

$$z(u) = (R - \rho)e^{iu} + \rho\, e^{-i(R-\rho)u/\rho}.$$

It is generated by a radius-ρ circle rolling within the radius-R circle: its center $(R - \rho)e^{iu}$ turns through the angle u, and rolling without slipping turns the tracing point through $-(R - \rho)u/\rho$ relative to the center. Set $\delta = Ru/\rho$, and note $u + (R - \rho)u/\rho = \delta$. Then

$$\begin{aligned} r^2 = |z|^2 &= (R - \rho)^2 + \rho^2 + 2\rho(R - \rho)\cos\delta \\ &= R^2 - 2\rho(R - \rho)(1 - \cos\delta) = R^2 - 4\rho(R - \rho)\sin^2(\delta/2), \end{aligned}$$

and differentiating, $2r\, dr/du = -2\rho(R - \rho)\sin\delta\,(R/\rho)$. For the polar angle $\theta = \arg z$, write $z = re^{i\theta}$; then $z' = dz/du = (r' + ir\theta')e^{i\theta}$ and $\bar{z}\, z' = rr' + ir^2\theta'$, so $d\theta/du = \operatorname{Im}(\bar{z}\, z')/|z|^2$. Here $z' = i(R - \rho)(e^{iu} - e^{-i(R-\rho)u/\rho})$. Multiplying out,

$$\bar{z}\, z' = i(R - \rho)\left[(R - \rho)(1 - e^{-i\delta}) + \rho(e^{i\delta} - 1)\right],$$

so $\operatorname{Im}(\bar{z}\, z') = (R - \rho)(R - 2\rho)(1 - \cos\delta)$, and $R - 2\rho = r_0$. Hence

$$\frac{dr}{du} = -\frac{R(R - \rho)\sin\delta}{r}, \qquad \frac{d\theta}{du} = \frac{(R - \rho)r_0(1 - \cos\delta)}{r^2}.$$

Their ratio is $dr/d\theta = -(Rr/r_0)\sin\delta/(1-\cos\delta) = -(Rr/r_0)\cot(\delta/2)$. Since $4\rho(R-\rho) = (R-r_0)(R+r_0) = R^2 - r_0^2$, the expression for r^2 gives $\sin^2(\delta/2) = (R^2-r^2)/(R^2-r_0^2)$ and $\cos^2(\delta/2) = (r^2-r_0^2)/(R^2-r_0^2)$. Therefore

$$\left(\frac{dr}{d\theta}\right)^2 = \frac{R^2r^2}{r_0^2}\cot^2(\delta/2) = \frac{R^2r^2(r^2-r_0^2)}{r_0^2(R^2-r^2)},$$

the boxed equation. The radius is r_0 at $\delta = \pi$ and R at the cusps $\delta = 0, 2\pi$. Rotating this arc puts its minimum radius at $\theta = 0$ and its two surface cusps at $\pm\varphi$.

The transit time can be evaluated without integrating the path explicitly. On either half the boxed equation gives $|d\theta/dr|$, and $ds/dr = \sqrt{1 + r^2(d\theta/dr)^2}$, where $R^2(r^2-r_0^2)+r_0^2(R^2-r^2) = r^2(R^2-r_0^2)$:

$$\left|\frac{d\theta}{dr}\right| = \frac{r_0\sqrt{R^2-r^2}}{Rr\sqrt{r^2-r_0^2}}, \qquad \frac{ds}{dr} = \frac{r\sqrt{R^2-r_0^2}}{R\sqrt{r^2-r_0^2}}.$$

Dividing ds by $v = \sqrt{g/R}\sqrt{R^2-r^2}$ and doubling for the two halves, with $\Delta = R^2 - r_0^2$,

$$T = 2\sqrt{\frac{R}{g}}\frac{\sqrt{\Delta}}{R}\int_{r_0}^{R}\frac{r\,dr}{\sqrt{r^2-r_0^2}\sqrt{R^2-r^2}}.$$

Set $r^2 = r_0^2 + \Delta\sin^2\psi$. Then $r\,dr = \Delta\sin\psi\cos\psi\,d\psi$, $r^2 - r_0^2 = \Delta\sin^2\psi$ and $R^2 - r^2 = \Delta\cos^2\psi$, so the integrand becomes $d\psi$, and the limits r_0, R become $0, \pi/2$. Hence $T = 2\sqrt{R/g}\,(\sqrt{\Delta}/R)(\pi/2)$, that is,

$$\boxed{T = \pi\sqrt{\frac{R^2-r_0^2}{Rg}}}.$$

The same substitution in the angular integral $\varphi = \int_{r_0}^{R}|d\theta/dr|\,dr$ turns the integrand into $(r_0/R)\Delta\cos^2\psi\,d\psi/r^2$, where $r^2 = r_0^2\cos^2\psi + R^2\sin^2\psi$ and $\Delta\cos^2\psi = R^2 - r^2$. Hence

$$\varphi = \frac{r_0}{R}\left[R^2\int_0^{\pi/2}\frac{d\psi}{r_0^2\cos^2\psi + R^2\sin^2\psi} - \frac{\pi}{2}\right] = \frac{\pi}{2}\left(1 - \frac{r_0}{R}\right).$$

Here the substitution $t = \tan\psi$ turns the integral into $\int_0^\infty dt/(r_0^2 + R^2t^2) = \pi/(2r_0R)$. As a check, one arch of the hypocycloid spans the arc $2\pi\rho$ of the large circle, i.e., the angle $2\varphi = 2\pi\rho/R = \pi(1-r_0/R)$. The antipodal case is the limiting diameter $r_0 = 0$, with $T = \pi\sqrt{R/g}$; the displayed polar ODE assumes $r_0 > 0$.

22.1.12 A ray of light follows a straight-line path in a first homogeneous medium, is refracted at an interface, and then follows a new straight-line path in the second medium. See Fig. 22.7. Use Fermat's principle of optics to derive Snell's law of refraction:

$$n_1\sin\theta_1 = n_2\sin\theta_2.$$

Hint. Keep the points (x_1, y_1) and (x_2, y_2) fixed and vary x_0 to satisfy Fermat's principle.

Note. This is not an Euler equation problem, because the light path is not differentiable at x_0.

Solution.

Take the interface to be $y = 0$, with fixed endpoints (x_1, y_1) above it and (x_2, y_2) below it. The crossing point is $(x_0, 0)$. In a medium of index n light travels at c/n, with c the vacuum light speed,

so each straight segment of length L takes the time nL/c, and the travel time is $J(x_0)/c$ with the optical path length

$$J(x_0) = n_1\sqrt{(x_0 - x_1)^2 + y_1^2} + n_2\sqrt{(x_2 - x_0)^2 + y_2^2}.$$

Differentiate only the crossing position:

$$J'(x_0) = n_1\frac{x_0 - x_1}{\sqrt{(x_0 - x_1)^2 + y_1^2}} - n_2\frac{x_2 - x_0}{\sqrt{(x_2 - x_0)^2 + y_2^2}}.$$

The interface normal is the y direction. The incident segment has horizontal extent $x_0 - x_1$ and length $\sqrt{(x_0 - x_1)^2 + y_1^2}$, so the first ratio is $\sin\theta_1$, with θ_1 the angle between the incident ray and the normal; likewise the second ratio is $\sin\theta_2$ for the refracted ray (both sines signed, positive when the ray advances toward $+x$). Thus $J'(x_0) = 0$ gives

$$\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}.$$

For positive indices and endpoints off the interface, differentiating $s/\sqrt{s^2 + y^2}$ gives $y^2/(s^2 + y^2)^{3/2}$, so

$$J''(x_0) = \frac{n_1 y_1^2}{[(x_0 - x_1)^2 + y_1^2]^{3/2}} + \frac{n_2 y_2^2}{[(x_2 - x_0)^2 + y_2^2]^{3/2}} > 0.$$

Hence J' increases strictly from $-(n_1 + n_2)$ as $x_0 \to -\infty$ to $n_1 + n_2$ as $x_0 \to +\infty$; it has exactly one zero, where the strictly convex J has its minimum. This proves the stationary crossing is the unique minimum for these fixed endpoints. No differentiability across the corner was assumed.

22.1.13 A second soap-film configuration for the unit-radius rings at $x = \pm x_0$ consists of a circular disk, radius a, in the $x = 0$ plane and two catenoids of revolution, one joining the disk and each ring. One catenoid may be described by

$$y = c_1 \cosh\left(\frac{x}{c_1} + c_3\right).$$

(a) Impose boundary conditions at $x = 0$ and $x = x_0$.

(b) Require that the catenoids form an angle of 120° where they join the central disk. Express this third boundary condition mathematically. For films of equal surface tension and a freely moving junction, this is the equilibrium condition.

(c) Show that the total area of catenoids plus central disk is then

$$A = \pi c_1^2\left[\sinh\left(\frac{2x_0}{c_1} + 2c_3\right) + \frac{2x_0}{c_1}\right].$$

Note. Stability depends on the branch and the allowed variations; see Exercise 22.1.14. The area is larger than that of the *shallow* simple catenoid whenever both exist.

ANS. (a) $\begin{cases} 1 = c_1 \cosh\left(\dfrac{x_0}{c_1} + c_3\right) \\ a = c_1 \cosh c_3 \end{cases}$ (b) $\dfrac{dy}{dx} = \tan 30^\circ = \sinh c_3.$

Solution.

(a) Write $c = c_1$. The right-hand catenoid $y = c\cosh(x/c + c_3)$, $0 \le x \le x_0$, has radius a where it meets the disk ($x = 0$) and radius 1 at the ring ($x = x_0$), so the boundary equations are

$$\boxed{a = c\cosh c_3, \qquad 1 = c\cosh(x_0/c + c_3)}.$$

(b) Work in a meridian plane through the axis, where the three films appear as curves meeting at $(0, a)$. At the junction, the central disk points inward in the negative radial direction, with unit tangent $(0, -1)$. The outward right-catenoid unit tangent is $(1, y')/\sqrt{1 + y'^2}$. A 120° angle therefore requires

$$-\frac{y'}{\sqrt{1 + y'^2}} = \cos 120^\circ = -\frac{1}{2},$$

that is, $4y'^2 = 1 + y'^2$, which gives $y' = 1/\sqrt{3} = \tan 30^\circ$. By reflection the left catenoid has unit tangent $(-1, y')/\sqrt{1 + y'^2}$; its angle with the right one has cosine $(y'^2 - 1)/(y'^2 + 1) = -1/2$, so all three angles are then 120°. Since $y' = \sinh(x/c + c_3)$, at $x = 0$ this is $\sinh c_3 = 1/\sqrt{3}$. Then $\cosh c_3 = \sqrt{1 + 1/3} = 2/\sqrt{3}$, and $\operatorname{arsinh} s = \ln(s + \sqrt{1 + s^2})$ gives $c_3 = \ln(1/\sqrt{3} + 2/\sqrt{3}) = \ln\sqrt{3}$:

$$\boxed{\sinh c_3 = \frac{1}{\sqrt{3}}, \qquad c_3 = \operatorname{arsinh}(1/\sqrt{3}) = \tfrac{1}{2}\ln 3, \qquad a = c\cosh c_3 = \frac{2c}{\sqrt{3}}}.$$

The same angle follows by varying the junction radius. In the reflection-symmetric class the area is $A = 4\pi\int_0^{x_0} y\sqrt{1 + y'^2}\,dx + \pi y(0)^2$. Vary $y \to y + \alpha\eta$ with $\eta(x_0) = 0$ and $\eta(0)$ free, so that the junction radius changes by $\delta a = \alpha\eta(0)$. For $f = y\sqrt{1 + y'^2}$, $f_{y'} = yy'/\sqrt{1 + y'^2}$, and the integration by parts of Exercise 22.1.2 gives, to first order in α,

$$\delta\int_0^{x_0} f\,dx = \alpha\big[f_{y'}\eta\big]_0^{x_0} + \alpha\int_0^{x_0}\left(f_y - \frac{d}{dx}f_{y'}\right)\eta\,dx.$$

The integral vanishes by the interior Euler equation, the upper boundary term vanishes because $\eta(x_0) = 0$, and the lower one is $-\alpha f_{y'}(0)\eta(0) = -a\,y'(0)\,\delta a/\sqrt{1 + y'(0)^2}$, which enters A with the factor 4π. The disk contributes $\delta(\pi a^2) = 2\pi a\,\delta a$. So the first variation is

$$\delta A = \left[-4\pi a\frac{y'(0)}{\sqrt{1 + y'(0)^2}} + 2\pi a\right]\delta a.$$

Since $a > 0$ and δa is free, this again gives $y'(0)/\sqrt{1 + y'(0)^2} = 1/2$. Thus the angle is necessary for this free-junction equilibrium, although one can join mathematical surface pieces at other prescribed angles.

(c) Put $v = x_0/c + c_3$. There are two catenoids and one disk. On each catenoid $\sqrt{1 + y'^2} = \cosh(x/c + c_3)$, so $y\sqrt{1 + y'^2} = c\cosh^2(x/c + c_3)$. Substitute $u = x/c + c_3$, $dx = c\,du$ (limits c_3 and v), and use $\int\cosh^2 u\,du = \frac{1}{2}u + \frac{1}{4}\sinh 2u$ and $a = c\cosh c_3$. Their total area is

$$\begin{aligned} A &= 4\pi\int_0^{x_0} y\sqrt{1 + y'^2}\,dx + \pi a^2 \\ &= 4\pi c^2\int_{c_3}^{v}\cosh^2 u\,du + \pi c^2\cosh^2 c_3 \\ &= \pi c^2[2(v - c_3) + \sinh 2v - \sinh 2c_3 + \cosh^2 c_3]. \end{aligned}$$

The angle condition gives $\cosh c_3 = 2/\sqrt{3}$, so $\sinh 2c_3 = 2\sinh c_3 \cosh c_3 = 2 \cdot \frac{1}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}} = \frac{4}{3} = \cosh^2 c_3$. These last terms cancel, and $2(v - c_3) = 2x_0/c$, $2v = 2x_0/c + 2c_3$:

$$\boxed{A = \pi c^2 \left[\sinh\left(\frac{2x_0}{c} + 2c_3 \right) + \frac{2x_0}{c} \right].}$$

The printed expression is missing the overall factor π. It equals A/π, not the total physical area. The surface-of-revolution factor $2\pi y\, ds$ and the disk area πa^2 fix the normalization.

22.1.14 For the soap film described in Exercise 22.1.13, find (numerically) the maximum value of x_0.

Note. This calls for a calculator with hyperbolic functions or a table of hyperbolic cotangents.

ANS. $x_{0\,\max} = 0.4078$.

Solution.

Retain the angle condition from Exercise 22.1.13: $c_3 = \frac{1}{2}\ln 3$. Write $v = x_0/c + c_3$. The ring condition $1 = c\cosh v$ of Exercise 22.1.13(a) gives $c = \operatorname{sech} v$, and then $x_0 = c(v - c_3)$:

$$v = \frac{x_0}{c} + c_3, \qquad c = \operatorname{sech} v, \qquad x_0 = (v - c_3)\operatorname{sech} v, \quad v > c_3.$$

Differentiate by the product rule, using $(\operatorname{sech} v)' = -\operatorname{sech} v \tanh v$:

$$\frac{dx_0}{dv} = \operatorname{sech} v[1 - (v - c_3)\tanh v].$$

Thus the maximum satisfies

$$\boxed{v - c_3 = \coth v}.$$

The function $v - c_3 - \coth v$ has derivative $1 + \operatorname{csch}^2 v > 0$, equals $-\coth c_3 = -2$ at $v = c_3$ (because $\tanh c_3 = \sinh c_3/\cosh c_3 = 1/2$), and tends to infinity. It has exactly one root. Since $1 - (v - c_3)\tanh v = \tanh v\,[\coth v - (v - c_3)]$, x_0 increases before the root and decreases after it. Moreover x_0 tends to zero at both ends of the parameter range ($x_0 = 0$ at $v = c_3$, and $x_0 < v\operatorname{sech} v \to 0$ as $v \to \infty$), so this root is the global maximum.

Newton's update is

$$v_{j+1} = v_j - \frac{v_j - c_3 - \coth v_j}{1 + \operatorname{csch}^2 v_j}.$$

Starting with $v_0 = 1.6$, three steps give all the digits shown:

$$v = 1.62926936306647, \qquad c = \operatorname{sech} v = 0.377627833193942,$$

and hence

$$\boxed{x_{0\,\max} = (v - c_3)\operatorname{sech} v = 0.407824170219077}.$$

This rounds to the quoted answer 0.4078.

The two branches must not both be called stable. To check this, allow reflection-symmetric, axisymmetric variations of the two catenoids, with the junction remaining in $x = 0$ and its radius free. Set $u = x/c + c_3$ and $p(u) = \operatorname{sech}^2 u$. For $y \to y + \alpha\eta$ on the right catenoid (the left one is its mirror image), the second variation of $4\pi \int_0^{x_0} y\sqrt{1 + y'^2}\, dx$ is, with the second derivatives of $f = y\sqrt{1 + y'^2}$ from Exercise 22.1.6 (now $f_{yy'} = \tanh u$, $f_{y'y'} = c\operatorname{sech}^2 u$),

$$4\pi \int_0^{x_0} \left[2\tanh u\, \eta\eta_x + c\operatorname{sech}^2 u\, \eta_x^2 \right] dx.$$

With $dx = c\,du$ and $\eta_x = \eta_u/c$, the mixed term becomes $4\pi\int_{c_3}^{v}\tanh u\,(\eta^2)_u\,du$ and the last term $4\pi\int_{c_3}^{v} p\,\eta_u^2\,du$. Integrating the former by parts, with $(\tanh u)' = p$ and $\eta(v) = 0$, the catenoids contribute

$$4\pi\left[\int_{c_3}^{v} p\,(\eta_u^2 - \eta^2)\,du - \tanh(c_3)\,\eta(c_3)^2\right], \qquad \tanh c_3 = \tfrac{1}{2}.$$

The disk area $\pi(a + \alpha\eta(c_3))^2$ has second α-derivative $2\pi\eta(c_3)^2 = 4\pi\cdot\frac{1}{2}\eta(c_3)^2$. The boundary terms cancel, so

$$Q[\eta] = \frac{\delta^2 A}{4\pi} = \int_{c_3}^{v} p(u)(\eta'^2 - \eta^2)\,du, \qquad \eta(v) = 0, \quad \eta(c_3) \text{ free.}$$

The function

$$w(u) = \cosh u - (u - c_3)\sinh u,$$

which is $\partial y/\partial c$ for the family $y = c\cosh(x/c + c_3)$ at fixed x and c_3, has $w' = -(u - c_3)\cosh u$. Hence $w'(c_3) = 0$, $pw' = -(u - c_3)\operatorname{sech} u$, and $(pw')' = -\operatorname{sech} u + (u - c_3)\operatorname{sech} u\tanh u = -pw$, so $(pw')' + pw = 0$. Since $w = \sinh u\,[\coth u - (u - c_3)]$ and the bracket decreases strictly from $\coth c_3 = 2$, the first zero of w is $v_c = 1.629269\ldots$, the root just found. For $v < v_c$, $w > 0$ on $[c_3, v]$; putting $\eta = wz$ gives the same square identity as in Exercise 22.1.6, $p(\eta'^2 - \eta^2) = pw^2 z'^2 + (pww'z^2)'$. The boundary term $[pww'z^2]_{c_3}^{v}$ vanishes at $u = v$ because $z(v) = \eta(v)/w(v) = 0$, and at $u = c_3$ because $w'(c_3) = 0$. So

$$Q[\eta] = \int_{c_3}^{v} pw^2 z'^2\,du > 0 \qquad (\eta \not\equiv 0),$$

since $z' \equiv 0$ and $z(v) = 0$ would force $\eta \equiv 0$. To pass from this quadratic test to a local minimum, recall that the first variation vanishes for these variations, by the interior Euler equation and the 120° junction condition (Exercise 22.1.13(b)). So $A[y + \eta] - A[y] = \frac{1}{2}\delta^2 A + R = 2\pi Q[\eta] + R$, with R the Taylor remainder. Next, note that $z(v) = 0$ implies $z(u) = -\int_u^v z'(s)ds$, so $z(u)^2 \le (v - c_3)\int_{c_3}^{v} z'^2 ds$ by Cauchy–Schwarz, and $\int z^2 \le (v - c_3)^2\int z'^2$. On $[c_3, v]$ put $m = \min pw^2 > 0$ and $M = \max(w^2 + 2w'^2, 2w^2)$. Since $\eta^2 + \eta'^2 = w^2 z^2 + (w'z + wz')^2 \le M(z^2 + z'^2)$,

$$\int(\eta^2 + \eta'^2)du \le M[1 + (v - c_3)^2]\int z'^2 du \le \frac{M[1 + (v - c_3)^2]}{m}\,Q[\eta],$$

that is, $Q[\eta] \ge \kappa\int(\eta^2 + \eta'^2)du$ with $\kappa = m/\{M[1 + (v - c_3)^2]\} > 0$. For $\eta \not\equiv 0$ with sufficiently small $\|\eta\|_{C^1}$, the remainder satisfies $|R| < 2\pi\kappa\int(\eta^2 + \eta'^2)du \le 2\pi Q[\eta]$, by uniform continuity of the integrand's Hessian (the disk term is exactly quadratic), so $A[y + \eta] > A[y]$. This proves strict local minimality on the shallow branch in the stated class. At $v = v_c$, $\eta = w$ is a zero mode; the second variation alone does not classify higher orders.

For $v > v_c$, extend w by zero from v_c to v, calling the result η_0. It is an admissible piecewise smooth variation ($\eta_0(v) = 0$), and integration by parts on $[c_3, v_c]$ gives $Q[\eta_0] = [pww']_{c_3}^{v_c} - \int_{c_3}^{v_c} w[(pw')' + pw]\,du = 0$, because $w(v_c) = 0 = w'(c_3)$. Choose a smooth h with $h(v) = 0$ and $h(v_c) > 0$. Integration by parts on $[c_3, v_c]$ gives

$$Q[\eta_0, h] := \int_{c_3}^{v} p(\eta_0' h' - \eta_0 h)\,du = \big[pw'h\big]_{c_3}^{v_c} - \int_{c_3}^{v_c}[(pw')' + pw]h\,du = p(v_c)w'(v_c)h(v_c) < 0,$$

since $w'(c_3) = 0$ and $w'(v_c) = -(v_c - c_3)\cosh v_c < 0$. Therefore $Q[\eta_0 + \epsilon h] = Q[\eta_0] + 2\epsilon Q[\eta_0, h] + \epsilon^2 Q[h] < 0$ for small positive ϵ. Smoothing the corner preserves this strict inequality, because Q changes little when η and η' change little in mean square. The deep branch is unstable even in this restricted class.

The area comparison in Exercise 22.1.13 also needs its branch specified. Write the simple catenoid of Exercise 22.1.6 as $x_0 = \mu \operatorname{sech}\mu$, where μ is the parameter called u there: the ratio of x_0 to that catenoid's own neck radius $\operatorname{sech}\mu$, not to the three-film c above. By Exercise 22.1.7 its area is $A_s = \pi \operatorname{sech}^2\mu\,(2\sinh\mu\cosh\mu + 2\mu) = 2\pi(\tanh\mu + \mu\operatorname{sech}^2\mu)$. Take its shallow root $\mu < u_c$ from Exercise 22.1.6. Let A_t denote the three-film area; Exercise 22.1.13(c) with $c = \operatorname{sech} v$ and $2x_0/c = 2(v - c_3)$ gives $A_t = 2\pi[\tanh v + (v - c_3)\operatorname{sech}^2 v]$. Differentiating,

$$\frac{dA_s}{d\mu} = 4\pi \operatorname{sech}^2\mu\,(1 - \mu\tanh\mu), \qquad \frac{dA_t}{dv} = 4\pi \operatorname{sech}^2 v\,[1 - (v - c_3)\tanh v],$$

and dividing by $dx_0/d\mu$ (Exercise 22.1.6(b)) and by dx_0/dv (above) gives, along either family,

$$\frac{dA_t}{dx_0} = 4\pi \operatorname{sech} v, \qquad \frac{dA_s}{dx_0} = 4\pi \operatorname{sech}\mu.$$

At a common separation, $\mu < v$: for $v \le u_c$, use the increasing function $\mu\operatorname{sech}\mu$ on $(0, u_c]$ and $\mu\operatorname{sech}\mu = x_0 = (v - c_3)\operatorname{sech} v < v\operatorname{sech} v$; for $v > u_c$ it is immediate, since $\mu < u_c$. As sech decreases, $d(A_t - A_s)/dx_0 = 4\pi(\operatorname{sech} v - \operatorname{sech}\mu) < 0$: $A_t - A_s$ decreases with x_0 on the shallow three-film branch. At its fold, where $v_c - c_3 = \coth v_c$ and $\tanh v + \coth v\operatorname{sech}^2 v = (\sinh^2 v + 1)/(\sinh v\cosh v) = \coth v$,

$$A_t = 2\pi\left[\tanh v_c + \coth v_c \operatorname{sech}^2 v_c\right] = 2\pi\coth v_c > 2\pi.$$

Meanwhile $x_{0\max} < 1/2 < 0.527697\ldots$, the area crossing in Exercise 22.1.7; since A_s increases with x_0 on the shallow branch and equals 2π at the crossing, $A_s < 2\pi$ throughout this range. The bound $x_0 < 1/2$, i.e. $\frac{1}{2}\cosh v - v + c_3 > 0$, follows without rounding from $\cosh v \ge 1 + v^2/2$ and $c_3 = \frac{1}{2}\ln 3 > \frac{1}{2}$ (as $3 > e$):

$$\tfrac{1}{2}\cosh v - v + c_3 \ge \tfrac{1}{2} + \tfrac{1}{4}v^2 - v + c_3 = \tfrac{1}{4}(v - 2)^2 + c_3 - \tfrac{1}{2} > 0.$$

The shallow-branch difference is therefore positive even at its minimum, the fold. On the deep three-film branch, $v - c_3 > \coth v$, so $A_t > 2\pi\coth v > 2\pi > A_s$ directly. This proves the stated comparison for both three-film branches. It is not a comparison with the deep simple catenoid.

22.1.15 Find the curve of quickest descent from $(0, 0)$ to (x_0, y_0) for a particle that, starting from rest, slides under gravity and without friction. Assume $x_0, y_0 > 0$ with y measured downward. For the special endpoint ratio $x_0/y_0 = \pi/2$, show that the ratio of times taken by the particle along a straight line joining the two points compared to along the curve of quickest descent is $(1 + 4/\pi^2)^{1/2}$.

Hint. Take y to increase downwards. Apply Eq. (22.21) to obtain $y_x^2 = (1 - c^2 y)/c^2 y$, where c is an integration constant. It is helpful to make the substitution $c^2 y = \sin^2(\varphi/2)$ and take $(x_0, y_0) = (\pi/2c^2,\ 1/c^2)$.

Solution.

Take y positive downward. Conservation of energy, $\frac{1}{2}mv^2 = mgy$ for a start from rest at $y = 0$, gives $v = \sqrt{2gy}$, so $T = \int ds/v$ with $ds = \sqrt{1 + y'^2}\,dx$ is

$$T[y] = \frac{1}{\sqrt{2g}} \int \frac{\sqrt{1 + y'^2}}{\sqrt{y}}\,dx.$$

For $f = \sqrt{1 + y'^2}/\sqrt{y}$, $f_{y'} = y'/(\sqrt{y}\sqrt{1 + y'^2})$ and $f - y' f_{y'} = [(1 + y'^2) - y'^2]/(\sqrt{y}\sqrt{1 + y'^2})$. Since f does not depend on x, this is constant (Eq. (22.21); Exercise 22.1.1 with $f_x = 0$). The Beltrami first integral, with its constant written $c > 0$, is therefore

$$\frac{1}{\sqrt{y}\sqrt{1 + y'^2}} = c, \qquad y'^2 = \frac{1 - c^2 y}{c^2 y},$$

the second form following from $y(1+y'^2)=1/c^2$. Put $c^2y=\sin^2(\varphi/2)$. Then $1-c^2y=\cos^2(\varphi/2)$, so $y'^2=\cot^2(\varphi/2)$ and $y'=\cot(\varphi/2)$: positive while the particle descends ($0<\varphi<\pi$) and negative beyond the lowest point. Also

$$y=\frac{1-\cos\varphi}{2c^2},\qquad \frac{dy}{d\varphi}=\frac{\sin\varphi}{2c^2},\qquad \frac{dx}{d\varphi}=\frac{dy/d\varphi}{y'}=\frac{\sin\varphi\tan(\varphi/2)}{2c^2}=\frac{1-\cos\varphi}{2c^2},$$

using $\sin\varphi\tan(\varphi/2)=2\sin^2(\varphi/2)=1-\cos\varphi$. The start $(0,0)$ has $y=0$, i.e. $\varphi=0$. Choosing $x=0$ at $\varphi=0$ and integrating gives the cycloid

$$\boxed{x=\frac{\varphi-\sin\varphi}{2c^2},\qquad y=\frac{1-\cos\varphi}{2c^2}}.$$

For arbitrary $x_0,y_0>0$, choose $0<\varphi_0<2\pi$ from

$$\frac{x_0}{y_0}=\frac{\varphi_0-\sin\varphi_0}{1-\cos\varphi_0},\qquad \frac{1}{2c^2}=\frac{y_0}{1-\cos\varphi_0}.$$

The ratio increases from zero to infinity on this interval. Near $\varphi=0$ it behaves like $(\varphi^3/6)/(\varphi^2/2)=\varphi/3$, and as $\varphi\to 2\pi$ its denominator tends to zero while its numerator tends to 2π. By the quotient rule, with $(1-\cos\varphi)^2+\sin^2\varphi=2(1-\cos\varphi)$, its derivative is $[2(1-\cos\varphi)-\varphi\sin\varphi]/(1-\cos\varphi)^2$, and the numerator is positive. For $0<\varphi<\pi$ it equals

$$4\sin^2\frac{\varphi}{2}-2\varphi\sin\frac{\varphi}{2}\cos\frac{\varphi}{2}=2\sin\frac{\varphi}{2}\cos\frac{\varphi}{2}\left(2\tan\frac{\varphi}{2}-\varphi\right)>0,$$

because $\tan t>t$ for $0<t<\pi/2$; for $\pi\le\varphi<2\pi$, $\sin\varphi\le 0$ and the sign is immediate. Hence φ_0 is unique, and then c follows from the second equation.

With $A=1/(2c^2)$, $x=A(\varphi-\sin\varphi)$ and $y=A(1-\cos\varphi)$, so $(dx/d\varphi)^2+(dy/d\varphi)^2=A^2[(1-\cos\varphi)^2+\sin^2\varphi]=4A^2\sin^2(\varphi/2)$ and $2gy=4gA\sin^2(\varphi/2)$:

$$ds=2A\sin(\varphi/2)d\varphi,\qquad v=2\sqrt{gA}\sin(\varphi/2),$$

so $dt=ds/v=\sqrt{A/g}\,d\varphi$ and

$$T_{\text{cyc}}=\varphi_0\sqrt{A/g}.$$

One can verify the minimizing character by a calibration. Regard $S(x,y)=\varphi\sqrt{A/g}$ as a function defined by the first-arch endpoint equations $x=A(\varphi-\sin\varphi)$, $y=A(1-\cos\varphi)$, $A>0$, $0<\varphi<2\pi$. By the monotonicity just shown, each point with $x,y>0$ has exactly one such (A,φ), and the Jacobian $\partial(x,y)/\partial(A,\varphi)=-A[2(1-\cos\varphi)-\varphi\sin\varphi]$ is nonzero, so S is smooth there. The differentials of the endpoint equations are

$$dx=(\varphi-\sin\varphi)\,dA+A(1-\cos\varphi)\,d\varphi,\qquad dy=(1-\cos\varphi)\,dA+A\sin\varphi\,d\varphi,$$

whereas

$$dS=\frac{\varphi\,dA}{2\sqrt{gA}}+\sqrt{\frac{A}{g}}\,d\varphi.$$

Equating the coefficients of $dA,d\varphi$ in $dS=S_x dx+S_y dy$ gives two linear equations for S_x,S_y, solved by

$$S_x=\frac{1}{2\sqrt{gA}},\qquad S_y=\frac{\cot(\varphi/2)}{2\sqrt{gA}}.$$

Indeed, use $(1-\cos\varphi)\cot(\varphi/2)=\sin\varphi$ in the first coefficient and $\sin\varphi\cot(\varphi/2)=1+\cos\varphi$ in the second. Consequently, since $2gy=4gA\sin^2(\varphi/2)$,

$$|\nabla S|^2=\frac{1+\cot^2(\varphi/2)}{4gA}=\frac{\csc^2(\varphi/2)}{4gA}=\frac{1}{2gy}.$$

Also $S\to 0$ as $(x,y)\to(0,0)$, by the following two bounds. When $\varphi\le\pi$, eliminating A with $y=2A\sin^2(\varphi/2)$ gives

$$S=\frac{\varphi}{\sin(\varphi/2)}\sqrt{\frac{y}{2g}}\le\pi\sqrt{\frac{y}{2g}},$$

because $\sin(\varphi/2)\ge\varphi/\pi$: the sine is concave on $[0,\pi/2]$, so there its graph lies above the chord $2t/\pi$ from $(0,0)$ to $(\pi/2,1)$; take $t=\varphi/2$. When $\pi\le\varphi<2\pi$, $\sin\varphi\le 0$ gives $x=A(\varphi-\sin\varphi)\ge\pi A$, so $S=\varphi\sqrt{A/g}<2\pi\sqrt{x/(\pi g)}=2\sqrt{\pi x/g}$.

For any joining path in $y>0$, Cauchy–Schwarz gives $dS=\nabla S\cdot d\mathbf{r}\le|\nabla S|\,ds=ds/\sqrt{2gy}=dt$; integrating from the initial limiting point, where $S=0$, proves $T\ge S(x_0,y_0)=T_{\text{cyc}}$. Equality holds along the cycloid, whose tangent $(dx,dy)\propto(1-\cos\varphi,\sin\varphi)\propto(1,\cot(\varphi/2))$ is parallel to ∇S. Reflection handles paths to negative x: replacing x by $|x|$ changes neither the arc length nor the speed.

The fixed numerical time ratio in the question applies to the special endpoint in the hint, not to arbitrary (x_0,y_0). For that endpoint, $x_0/y_0=\pi/2$ is the value of the ratio at $\varphi_0=\pi$, so $\varphi_0=\pi$ and $A=y_0/2$: indeed $x_0=\pi/(2c^2)$, $y_0=1/c^2$, and $T_{\text{cyc}}=\pi\sqrt{A/g}=\pi/(c\sqrt{2g})$. The straight segment has length

$$L=\sqrt{x_0^2+y_0^2}=\frac{1}{c^2}\sqrt{1+\pi^2/4}$$

and constant tangential acceleration gy_0/L (g times the sine of its inclination). Starting from rest, $L=\frac{1}{2}(gy_0/L)T_{\text{line}}^2$, so, with $\sqrt{2/(gy_0)}=c\sqrt{2/g}$,

$$T_{\text{line}}=\sqrt{\frac{2L^2}{gy_0}}=\frac{1}{c}\sqrt{\frac{2}{g}}\sqrt{1+\frac{\pi^2}{4}}.$$

Therefore, since $\sqrt{2/g}\,\sqrt{2g}=2$,

$$\boxed{\frac{T_{\text{line}}}{T_{\text{cyc}}}=\frac{2}{\pi}\sqrt{1+\frac{\pi^2}{4}}=\sqrt{1+\frac{4}{\pi^2}}}\approx 1.185447.$$

22.2 More General Variations

22.2.1 Take $m>0$ and fixed positions at fixed times $t_1<t_2$, since Hamilton's principle compares only paths with the same endpoint data.

(a) Develop the equations of motion corresponding to $L=\frac{1}{2}m(\dot{x}^2+\dot{y}^2)$.

(b) In what sense do your solutions minimize the integral $\displaystyle\int_{t_1}^{t_2}L\,dt$?

Compare the result for your solution with $x =$ constant, $y =$ constant.

Solution.

(a) Since L is independent of x, y,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = m\ddot{x} = 0, \qquad m\ddot{y} = 0.$$

Integrating twice, $x = A + Bt$, $y = C + Dt$. For fixed endpoint positions $\mathbf{r}_1, \mathbf{r}_2$ at times $t_1 < t_2$, the conditions $x(t_1) = x_1$, $x(t_2) = x_2$ give $B = (x_2 - x_1)/(t_2 - t_1)$ and $A = x_1 - Bt_1$, and likewise for C, D. Thus the unique path is

$$\boxed{\mathbf{r}_*(t) = \mathbf{r}_1 + \frac{t - t_1}{t_2 - t_1}(\mathbf{r}_2 - \mathbf{r}_1)},$$

uniform straight-line motion with the constant velocity $\dot{\mathbf{r}}_* = (\mathbf{r}_2 - \mathbf{r}_1)/(t_2 - t_1)$.

(b) Write another admissible path as $\mathbf{r} = \mathbf{r}_* + \boldsymbol{\eta}$, with $\boldsymbol{\eta}(t_1) = \boldsymbol{\eta}(t_2) = 0$. Since $|\dot{\mathbf{r}}|^2 = |\dot{\mathbf{r}}_*|^2 + 2\dot{\mathbf{r}}_* \cdot \dot{\boldsymbol{\eta}} + |\dot{\boldsymbol{\eta}}|^2$ and $\dot{\mathbf{r}}_*$ is constant, its action is

$$S[\mathbf{r}] = S[\mathbf{r}_*] + m\dot{\mathbf{r}}_* \cdot \int_{t_1}^{t_2} \dot{\boldsymbol{\eta}}\, dt + \frac{m}{2}\int_{t_1}^{t_2} |\dot{\boldsymbol{\eta}}|^2 dt.$$

The middle term is zero because $\int_{t_1}^{t_2} \dot{\boldsymbol{\eta}}\, dt = \boldsymbol{\eta}(t_2) - \boldsymbol{\eta}(t_1) = 0$. Therefore, with $S[\mathbf{r}_*] = \frac{m}{2}|\dot{\mathbf{r}}_*|^2(t_2 - t_1)$,

$$\boxed{S[\mathbf{r}] - S[\mathbf{r}_*] = \frac{m}{2}\int_{t_1}^{t_2} |\dot{\boldsymbol{\eta}}|^2 dt \geq 0}, \qquad S[\mathbf{r}_*] = \frac{m|\mathbf{r}_2 - \mathbf{r}_1|^2}{2(t_2 - t_1)}.$$

Equality requires $\dot{\boldsymbol{\eta}} \equiv 0$, i.e. $\boldsymbol{\eta} \equiv \boldsymbol{\eta}(t_1) = 0$: the same path. So the solutions of (a) give the smallest action among all paths with the same endpoint positions at the same times (a strict global minimum). A constant path has zero action, less than $S[\mathbf{r}_*]$ when $\mathbf{r}_1 \neq \mathbf{r}_2$, but it is an admissible competitor only if the prescribed endpoint positions coincide; in that case it is precisely the minimizing solution.

22.2.2 From the Lagrangian equations of motion, Eq. (22.38), show that a system in stable equilibrium has a minimum potential energy. Assume a natural mechanical Lagrangian with positive-definite kinetic energy and a nondegenerate equilibrium.

Solution.

For a natural mechanical system take $T = \frac{1}{2}M_{ij}(q)\dot{q}_i\dot{q}_j$ (summed over repeated indices) with symmetric positive-definite mass matrix, and $L = T - V(q)$. Then $\partial L/\partial \dot{q}_i = M_{ij}\dot{q}_j$ and $\partial L/\partial q_i = \frac{1}{2}(\partial_i M_{jk})\dot{q}_j\dot{q}_k - \partial_i V$, so the Lagrange equations read

$$M_{ij}\ddot{q}_j + (\partial_k M_{ij})\dot{q}_k\dot{q}_j - \tfrac{1}{2}(\partial_i M_{jk})\dot{q}_j\dot{q}_k + \partial_i V = 0.$$

At a rest equilibrium $q = q_0$, $\dot{q} = \ddot{q} = 0$, and they reduce to

$$\boxed{\frac{\partial V}{\partial q_i}(q_0) = 0}.$$

Write $q = q_0 + \eta$ and $K_{ij} = \partial_i\partial_j V(q_0)$. To first order in η and its derivatives, the velocity terms (quadratic) drop out, $M(q)\ddot{\eta}$ becomes $M(q_0)\ddot{\eta}$, and $\partial_i V(q_0 + \eta) = K_{ij}\eta_j + O(|\eta|^2)$; thus

$$M_0\ddot{\eta} + K\eta = 0, \qquad M_0 = M(q_0).$$

Let $M_0^{1/2}$ be the symmetric positive-definite square root of M_0 (write $M_0 = U\,\mathrm{diag}(\mu_j)U^T$ with U orthogonal and all $\mu_j > 0$, and set $M_0^{1/2} = U\,\mathrm{diag}(\sqrt{\mu_j})U^T$), and $M_0^{-1/2}$ its inverse. Put $\eta = M_0^{-1/2}\xi$ and multiply on the left by $M_0^{-1/2}$; since $M_0^{-1/2}M_0M_0^{-1/2} = I$, $\ddot{\xi} + M_0^{-1/2}KM_0^{-1/2}\xi = 0$. The real symmetric matrix $M_0^{-1/2}KM_0^{-1/2}$ equals $O\,\mathrm{diag}(\omega_j^2)O^T$ with O orthogonal, so the normal coordinates $Q = O^T\xi$ satisfy

$$\ddot{Q}_j + \omega_j^2 Q_j = 0.$$

A negative $\omega_j^2 = -\kappa^2$ yields $Q_j = Ae^{\kappa t} + Be^{-\kappa t}$ and hence instability under arbitrarily small perturbations (by Lyapunov's linearization theorem, this growing mode makes the equilibrium of the full nonlinear equations unstable as well). In the nondegenerate stable case no ω_j^2 is negative (stability) or zero (nondegeneracy: $\det K \neq 0$, and $\prod_j \omega_j^2 = \det(M_0^{-1/2}KM_0^{-1/2}) = \det K/\det M_0$), so all $\omega_j^2 > 0$, and K is positive definite: for $\eta \neq 0$,

$$\eta^T K\eta = \xi^T M_0^{-1/2}KM_0^{-1/2}\xi = \sum_j \omega_j^2 Q_j^2 > 0.$$

Taylor expansion about q_0, where $\nabla V = 0$, gives

$$V(q_0+\eta) = V(q_0) + \frac{1}{2}\eta^T K\eta + O(|\eta|^3) > V(q_0)$$

for sufficiently small nonzero η: if $k_{\min} > 0$ is the smallest eigenvalue of K and the remainder is at most $C|\eta|^3$, then $\frac{1}{2}\eta^T K\eta \geq \frac{1}{2}k_{\min}|\eta|^2 > C|\eta|^3$ for $0 < |\eta| < k_{\min}/(2C)$. Thus the equilibrium has a strict local minimum of potential energy.

The usual conclusion assumes a natural mechanical system with positive kinetic energy. Linear stability alone gives only a positive-semidefinite Hessian when zero modes occur; higher-order terms then decide the result. Lagrange's equations do not turn stationarity by itself into a minimum. For example, $V(q) = q^4$ has a stable minimum despite $V''(0) = 0$, whereas $V(q) = -q^4$ has the same zero Hessian and is unstable. Conversely (the Lagrange–Dirichlet theorem), a strict potential well together with positive kinetic energy confines sufficiently small-energy motions. Since $V > V(q_0)$ on a small sphere $|q - q_0| = \rho$, which is compact, $V \geq V(q_0) + \epsilon$ there for some $\epsilon > 0$. A motion starting inside the sphere with total energy $T + V < V(q_0) + \epsilon$ keeps $V(q(t)) < V(q_0) + \epsilon$, by energy conservation (Exercise 22.2.7) and $T \geq 0$, so it never reaches the sphere. This is the energy reason for stable small oscillations about a minimum.

22.2.3 Write out the Lagrangian equations of motion of a particle in spherical coordinates for potential V equal to a constant. Identify the terms corresponding to (a) centrifugal force and (b) Coriolis force. Write the coordinate equations on $r > 0$, $0 < \theta < \pi$, where the spherical coordinates are nonsingular.

Solution.

In spherical coordinates, with $V = V_0$,

$$L = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2\right) - V_0.$$

For r the derivatives are $L_{\dot{r}} = m\dot{r}$ and $L_r = mr(\dot{\theta}^2 + \sin^2\theta\dot{\varphi}^2)$. For θ they are $L_{\dot{\theta}} = mr^2\dot{\theta}$ and $L_\theta = mr^2\sin\theta\cos\theta\dot{\varphi}^2$. For φ, $L_\varphi = 0$ and $L_{\dot{\varphi}} = mr^2\sin^2\theta\dot{\varphi}$. By the product rule,

$$\frac{d}{dt}(mr^2\dot{\theta}) = mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta},$$

$$\frac{d}{dt}(mr^2\sin^2\theta\,\dot{\varphi}) = mr^2\sin^2\theta\,\ddot{\varphi} + 2mr\dot{r}\sin^2\theta\,\dot{\varphi} + 2mr^2\sin\theta\cos\theta\,\dot{\theta}\dot{\varphi}.$$

Setting $dL_{\dot q}/dt = L_q$ for $q = r, \theta, \varphi$ and dividing by m, mr and $mr\sin\theta$, respectively, gives

$$\boxed{\begin{aligned} \ddot r &= r\dot\theta^2 + r\sin^2\theta\,\dot\varphi^2, \\ r\ddot\theta &= r\sin\theta\cos\theta\,\dot\varphi^2 - 2\dot r\dot\theta, \\ r\sin\theta\,\ddot\varphi &= -2\dot r\sin\theta\,\dot\varphi - 2r\cos\theta\,\dot\theta\dot\varphi. \end{aligned}}$$

Equivalently, since $L_\varphi = 0$, $mr^2\sin^2\theta\dot\varphi$ is conserved.

(a) The squared-angular-speed terms on the right in the first two equations, multiplied by m, are the centrifugal terms of the moving spherical frame: $mr\dot\theta^2$ comes from the turning in θ, and the centrifugal force $m\rho\dot\varphi^2$ of the rotation about the z axis ($\rho = r\sin\theta$), directed along $\mathbf{e}_\rho = \sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta$, has the components $mr\sin^2\theta\,\dot\varphi^2$ and $mr\sin\theta\cos\theta\,\dot\varphi^2$. (b) The products with the factor two, each a radial or polar velocity times an angular velocity, are the Coriolis-type terms: $-2m\dot r\dot\theta$, $-2m\dot r\sin\theta\dot\varphi$, and $-2mr\cos\theta\dot\theta\dot\varphi$. There is no physical external force for constant V; these terms account for the changing coordinate basis. Moving them to the left gives the components a_r, a_θ, a_φ of the acceleration $\ddot{\mathbf{r}}$ along $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi$, so the equations state $\ddot{\mathbf{r}} = 0$.

22.2.4 The spherical pendulum consists of a mass on a wire of length l, free to move in polar angle θ and azimuth angle φ (Fig. 22.8). Measure θ from the downward vertical, which fixes the sign of the gravitational term.

(a) Set up the Lagrangian for this physical system.

(b) Develop the Lagrangian equations of motion.

Solution.

(a) As in Fig. 22.8, measure θ from the downward vertical. Relative to the pivot the mass is at $\mathbf{r} = l(\sin\theta\cos\varphi, \sin\theta\sin\varphi, -\cos\theta)$, so its height is $z = -l\cos\theta$. Differentiating, $\dot{\mathbf{r}} = l\dot\theta\,(\cos\theta\cos\varphi, \cos\theta\sin\varphi, \sin\theta) + l\sin\theta\,\dot\varphi\,(-\sin\varphi, \cos\varphi, 0)$; the two vectors are orthogonal unit vectors, so the speed squared is $l^2(\dot\theta^2 + \sin^2\theta\dot\varphi^2)$. With $V = mgz = -mgl\cos\theta$, apart from an irrelevant constant,

$$\boxed{L = \frac{ml^2}{2}(\dot\theta^2 + \sin^2\theta\dot\varphi^2) + mgl\cos\theta}.$$

(b) The polar derivatives are

$$L_{\dot\theta} = ml^2\dot\theta, \qquad L_\theta = ml^2\sin\theta\cos\theta\dot\varphi^2 - mgl\sin\theta.$$

Their Euler equation $ml^2\ddot\theta - L_\theta = 0$, divided by ml^2, is

$$\boxed{\ddot\theta - \sin\theta\cos\theta\,\dot\varphi^2 + \frac{g}{l}\sin\theta = 0}.$$

The azimuth is cyclic ($L_\varphi = 0$), so

$$\boxed{\frac{d}{dt}(ml^2\sin^2\theta\,\dot\varphi) = 0}.$$

By the product rule the latter is $ml^2(\sin^2\theta\,\ddot\varphi + 2\sin\theta\cos\theta\,\dot\theta\dot\varphi) = 0$; where $\sin\theta \neq 0$, division by $ml^2\sin^2\theta$ gives

$$\ddot\varphi + 2\cot\theta\,\dot\theta\dot\varphi = 0.$$

The conserved momentum is the vertical angular momentum, $m(x\dot y - y\dot x) = ml^2\sin^2\theta\,\dot\varphi$. At a polar coordinate singularity the conserved-momentum form is the appropriate one; division by $\sin\theta$ is unnecessary.

22.2.5 Show that the Lagrangian

$$L = m_0 c^2 \left(1 - \sqrt{1 - \frac{v^2}{c^2}}\right) - V(r)$$

leads to a relativistic form of Newton's second law of motion,

$$\frac{d}{dt}\left(\frac{m_0 v_i}{\sqrt{1 - v^2/c^2}}\right) = F_i,$$

in which the force components are $F_i = -\partial V/\partial x_i$. Assume $m_0 > 0$ and $v < c$, so that the square root is real.

Solution.

Let $v^2 = \sum_j \dot{x}_j^2 < c^2$. Since $\partial v^2/\partial \dot{x}_i = 2\dot{x}_i$, the chain rule gives

$$\frac{\partial L}{\partial \dot{x}_i} = -m_0 c^2 \frac{1}{2\sqrt{1 - v^2/c^2}}\left(-\frac{2\dot{x}_i}{c^2}\right) = \frac{m_0 v_i}{\sqrt{1 - v^2/c^2}}.$$

The constant rest-energy term has zero derivative, and the spatial derivative is $\partial L/\partial x_i = -\partial_i V$. Consequently the Lagrange equation $\frac{d}{dt}(\partial L/\partial \dot{x}_i) = \partial L/\partial x_i$ is

$$\boxed{\frac{d}{dt}\left(\frac{m_0 v_i}{\sqrt{1 - v^2/c^2}}\right) = -\partial_i V = F_i}.$$

Write $\gamma = (1 - v^2/c^2)^{-1/2}$. Since $d(v^2)/dt = 2\mathbf{v}\cdot\mathbf{a}$,

$$\dot{\gamma} = -\frac{1}{2}\Big(1 - \frac{v^2}{c^2}\Big)^{-3/2}\Big(-\frac{2\mathbf{v}\cdot\mathbf{a}}{c^2}\Big) = \frac{\gamma^3(\mathbf{v}\cdot\mathbf{a})}{c^2},$$

and the product rule, $d(m_0\gamma v_i)/dt = m_0\gamma a_i + m_0\dot{\gamma} v_i$, turns the same equation into

$$m_0\gamma a_i + \frac{m_0\gamma^3}{c^2} v_i(\mathbf{v}\cdot\mathbf{a}) = F_i.$$

In the limit $v/c \to 0$, $\gamma \to 1$ and the second term vanishes, so this reduces to $m_0 a_i = F_i$.

22.2.6 The Lagrangian for a particle with charge q in an electromagnetic field described by scalar potential φ and vector potential $\mathbf{A}$ is

$$L = \frac{1}{2}mv^2 - q\varphi + q\mathbf{A}\cdot\mathbf{v}.$$

Find the equation of motion of the charged particle.

Hint. $(d/dt)A_j = \partial A_j/\partial t + \sum_i (\partial A_j/\partial x_i)\dot{x}_i$. The dependence of the force fields $\mathbf{E}$ and $\mathbf{B}$ on the potentials φ and $\mathbf{A}$ is developed in Section 3.9; see in particular Eq. (3.108).

ANS. $m\ddot{x}_i = q[\mathbf{E} + \mathbf{v}\times\mathbf{B}]_i$.

Solution.

Treat the potentials as functions of both position and time. The canonical momentum and spatial derivative are

$$\frac{\partial L}{\partial \dot{x}_i} = m\dot{x}_i + qA_i, \qquad \frac{\partial L}{\partial x_i} = -q\,\partial_i\varphi + q\sum_j \dot{x}_j \partial_i A_j.$$

Along the trajectory, by the hint (the chain rule),

$$\frac{dA_i}{dt} = \partial_t A_i + \sum_j \dot{x}_j \partial_j A_i.$$

The Lagrange equation $\frac{d}{dt}(\partial L/\partial \dot{x}_i) = \partial L/\partial x_i$ is therefore

$$m\ddot{x}_i + q\,\partial_t A_i + q\sum_j \dot{x}_j \partial_j A_i = -q\,\partial_i \varphi + q\sum_j \dot{x}_j \partial_i A_j,$$

that is,

$$m\ddot{x}_i = q(-\partial_i \varphi - \partial_t A_i) + q\sum_j \dot{x}_j(\partial_i A_j - \partial_j A_i).$$

Define $\mathbf{E} = -\nabla\varphi - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$ (Eq. (3.108)). For example, at $i = x$ the $j = x$ term of the last sum vanishes, and with $B_z = \partial_x A_y - \partial_y A_x$ and $B_y = \partial_z A_x - \partial_x A_z$ the sum is

$$v_y(\partial_x A_y - \partial_y A_x) + v_z(\partial_x A_z - \partial_z A_x) = v_y B_z - v_z B_y = (\mathbf{v} \times \mathbf{B})_x.$$

The other two components follow by cyclic permutation. Thus

$$\boxed{m\ddot{\mathbf{r}} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})}.$$

The derivatives acting on the position-dependent vector potential are essential; omitting them would omit the magnetic force.

22.2.7 Consider a system in which the Lagrangian is given by

$$L(q_i, \dot{q}_i) = T(q_i, \dot{q}_i) - V(q_i),$$

where q_i and $\dot{q}_i$ represent sets of variables. The potential energy V is independent of velocity and neither T nor V has any explicit time dependence.

(a) Show that

$$\frac{d}{dt}\left(\sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L\right) = 0.$$

(b) The constant quantity

$$\sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L$$

defines the Hamiltonian H. Show that under the preceding assumed conditions, H satisfies $H = T + V$, and is therefore the total energy.

Note. The kinetic energy T is a homogeneous quadratic function of the $\dot{q}_i$.

Solution.

(a) Put $p_j = \partial L/\partial \dot{q}_j$ and $H = \sum_j \dot{q}_j p_j - L$. By the product rule, and the chain rule for $L(q, \dot{q}, t)$, its full derivative is

$$\begin{aligned}\frac{dH}{dt} &= \sum_j (\ddot{q}_j p_j + \dot{q}_j \dot{p}_j) - \sum_j \left(L_{q_j}\dot{q}_j + L_{\dot{q}_j}\ddot{q}_j\right) - L_t \\ &= \sum_j \dot{q}_j(\dot{p}_j - L_{q_j}) - L_t,\end{aligned}$$

where the $\ddot{q}_j$ terms cancel because $p_j = L_{\dot{q}_j}$. The Lagrange equations $\dot{p}_j = L_{q_j}$ make the sum zero, and the stated absence of explicit time dependence makes $L_t = 0$. Hence

$$\boxed{\frac{dH}{dt} = 0}.$$

(b) Use the note's homogeneous quadratic kinetic energy:

$$T = \frac{1}{2}\sum_{i,j} M_{ij}(q)\dot{q}_i\dot{q}_j, \qquad M_{ij} = M_{ji}.$$

Then $\partial T/\partial \dot{q}_j = \sum_i M_{ji}\dot{q}_i$ (the two terms from the product $\dot{q}_i\dot{q}_j$ are equal because $M_{ij} = M_{ji}$), so

$$\sum_j \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = \sum_{i,j} M_{ij}\dot{q}_i\dot{q}_j = 2T,$$

which is Euler's theorem for a function homogeneous of degree two. Since V has no velocity dependence, $L_{\dot{q}_j} = T_{\dot{q}_j}$, and therefore

$$\boxed{H = 2T - (T - V) = T + V}.$$

The quadratic homogeneity is used only in (b); conservation of the Legendre-transform expression in (a) does not require it. In (b) it is essential: if $T = T_2 + T_1 + T_0$ with parts of degree two, one and zero in the velocities, the same computation gives $\sum_j \dot{q}_j T_{\dot{q}_j} = 2T_2 + T_1$ and $H = T_2 - T_0 + V$, which differs from $T + V$ unless $T_1 = T_0 = 0$; hence the word "homogeneous" in the note.

22.2.8 The Lagrangian for a vibrating string (small-amplitude vibrations) is

$$L = \int \left(\frac{1}{2}\rho u_t^2 - \frac{1}{2}\tau u_x^2\right) dx,$$

where ρ is the (constant) linear mass density and τ is the (constant) tension. The x-integration is over the length of the string. Show that application of Hamilton's principle to the Lagrangian density (the integrand), now with two independent variables, leads to the classical wave equation,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\rho}{\tau}\frac{\partial^2 u}{\partial t^2}.$$

Assume $\rho, \tau > 0$ and variations vanishing on the spatial and temporal boundaries, so that the boundary terms of the variation vanish.

Solution.

Hamilton's principle applies to the action

$$S[u] = \int_{t_1}^{t_2} dt \int_{x_1}^{x_2} dx \left(\frac{1}{2}\rho u_t^2 - \frac{1}{2}\tau u_x^2\right).$$

Let $\delta u = \eta$ vanish at the fixed spatial and temporal boundaries. Replacing u by $u + \alpha\eta$ and differentiating at $\alpha = 0$ (for instance $\frac{1}{2}\rho(u_t + \alpha\eta_t)^2$ gives $\rho u_t\eta_t$), the first variation is

$$\delta S = \iint (\rho u_t\eta_t - \tau u_x\eta_x)\, dx\, dt.$$

Integrate the first term by parts in t and the second in x:

$$\int_{t_1}^{t_2} \rho u_t \eta_t \, dt = \big[\rho u_t \eta\big]_{t_1}^{t_2} - \int_{t_1}^{t_2} \rho u_{tt} \eta \, dt,$$
$$\int_{x_1}^{x_2} \tau u_x \eta_x \, dx = \big[\tau u_x \eta\big]_{x_1}^{x_2} - \int_{x_1}^{x_2} \tau u_{xx} \eta \, dx.$$

The bracketed boundary terms vanish because $\eta = 0$ at $t = t_1, t_2$ and at $x = x_1, x_2$, so

$$\delta S = \iint (-\rho u_{tt} + \tau u_{xx}) \eta \, dx \, dt.$$

This vanishes for every such η, so the fundamental lemma (now in two variables) gives $\tau u_{xx} - \rho u_{tt} = 0$, i.e.

$$\boxed{u_{xx} = \frac{\rho}{\tau} u_{tt}},$$

with wave speed $\sqrt{\tau/\rho}$ for positive tension and density. The constancy of ρ, τ permits moving them outside the derivatives.

22.2.9 Show that the stationary value of the total energy of the electrostatic field of Example 22.2.3 is a minimum. The region is charge-free and connected, the boundary potential is fixed, and the permittivity is a positive constant; the proof below uses each of these conditions.

Hint. Investigate the α^2 terms of J.

Solution.

Let φ_* be the harmonic potential satisfying the fixed boundary values in the charge-free region Ω: $\nabla^2 \varphi_* = 0$. For any admissible variation $\varphi = \varphi_* + \alpha\eta$ with $\eta = 0$ on the boundary, the physical field energy is

$$U[\varphi] = \frac{\varepsilon}{2} \int_\Omega |\nabla \varphi|^2 dV.$$

Since $|\nabla\varphi_* + \alpha\nabla\eta|^2 = |\nabla\varphi_*|^2 + 2\alpha \nabla\varphi_* \cdot \nabla\eta + \alpha^2 |\nabla\eta|^2$, the expansion is exact (the functional is quadratic):

$$U[\varphi_* + \alpha\eta] - U[\varphi_*] = \alpha\varepsilon \int_\Omega \nabla\varphi_* \cdot \nabla\eta \, dV + \frac{\alpha^2 \varepsilon}{2} \int_\Omega |\nabla\eta|^2 dV.$$

By the divergence theorem applied to $\eta \nabla\varphi_*$, whose divergence is $\nabla\eta \cdot \nabla\varphi_* + \eta \nabla^2 \varphi_*$, the cross integral is

$$\int_\Omega \nabla\varphi_* \cdot \nabla\eta \, dV = \int_{\partial\Omega} \eta \, \partial_n \varphi_* \, dS - \int_\Omega \eta \, \nabla^2 \varphi_* \, dV = 0,$$

because $\eta = 0$ on $\partial\Omega$ and $\nabla^2\varphi_* = 0$ in Ω. Thus the first-order term vanishes and only the α^2 term of the hint remains:

$$\boxed{U[\varphi_* + \alpha\eta] - U[\varphi_*] = \frac{\alpha^2 \varepsilon}{2} \int_\Omega |\nabla\eta|^2 dV \geq 0}.$$

For $\varepsilon > 0$ and $\alpha \neq 0$, equality requires $\nabla\eta = 0$, so η is constant on the connected region; by continuity the constant is its boundary value 0, i.e. $\eta = 0$. This proves a strict global minimum. The textbook's J omits only the positive constant $\varepsilon/2$, so the conclusion is identical.

22.2.10 The Lagrangian (per unit volume) of an electromagnetic field with a charge density ρ and current density $\mathbf{J}$ is given by

$$L = \frac{1}{2}\left(\varepsilon_0 E^2 - \frac{1}{\mu_0}B^2\right) - \rho\varphi + \mathbf{J}\cdot\mathbf{A}.$$

Show that Lagrange's equations lead to two of Maxwell's equations. (The remaining two are a consequence of the definition of $\mathbf{E}$ and $\mathbf{B}$ in terms of $\mathbf{A}$ and φ.)

Hint. Take φ and the components of $\mathbf{A}$ as dependent variables; and x, y, z, and t as independent variables. $\mathbf{E}$ and $\mathbf{B}$ are given in terms of $\mathbf{A}$ and φ by Eq. (3.108). Use prescribed sources satisfying $\partial_t\rho + \nabla\cdot\mathbf{J} = 0$ and variations vanishing on the spacetime boundary.

Solution.

Use

$$\mathbf{E} = -\nabla\varphi - \partial_t\mathbf{A}, \qquad \mathbf{B} = \nabla\times\mathbf{A},$$

with prescribed sources $\rho, \mathbf{J}$. The action is the spacetime integral of the given density. Its variation is

$$\delta S = \int dt\, d^3x \left[\varepsilon_0\mathbf{E}\cdot\delta\mathbf{E} - \frac{1}{\mu_0}\mathbf{B}\cdot\delta\mathbf{B} - \rho\,\delta\varphi + \mathbf{J}\cdot\delta\mathbf{A}\right].$$

First vary only φ. Then $\delta\mathbf{E} = -\nabla\delta\varphi$ and $\delta\mathbf{B} = 0$. Since $-\mathbf{E}\cdot\nabla\delta\varphi = -\nabla\cdot(\delta\varphi\,\mathbf{E}) + \delta\varphi\,\nabla\cdot\mathbf{E}$ and, by the divergence theorem, the total divergence integrates to a surface term that vanishes with $\delta\varphi$, spatial integration by parts gives

$$\delta_\varphi S = \int(\varepsilon_0\nabla\cdot\mathbf{E} - \rho)\delta\varphi\, dt\, d^3x.$$

This vanishes for arbitrary interior $\delta\varphi$. Hence

$$\boxed{\nabla\cdot\mathbf{E} = \rho/\varepsilon_0}.$$

In component Euler notation, this uses $L_\varphi = -\rho$ and $L_{\partial_i\varphi} = -\varepsilon_0 E_i$.

Next vary only $\mathbf{A}$. Now $\delta\mathbf{E} = -\partial_t\delta\mathbf{A}$ and $\delta\mathbf{B} = \nabla\times\delta\mathbf{A}$. Integrating by parts in t, with no endpoint terms because $\delta\mathbf{A} = 0$ at the initial and final times, the time term becomes

$$-\int\varepsilon_0\mathbf{E}\cdot\partial_t\delta\mathbf{A} = \int\varepsilon_0\partial_t\mathbf{E}\cdot\delta\mathbf{A}.$$

For the magnetic term, the identity

$$\nabla\cdot(\delta\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\delta\mathbf{A}) - \delta\mathbf{A}\cdot(\nabla\times\mathbf{B})$$

and the divergence theorem (the surface term vanishes with $\delta\mathbf{A}$) show that its volume integral $-\mu_0^{-1}\int\mathbf{B}\cdot(\nabla\times\delta\mathbf{A})$ becomes $-\mu_0^{-1}\int(\nabla\times\mathbf{B})\cdot\delta\mathbf{A}$. Thus

$$\delta_\mathbf{A} S = \int\left(\mathbf{J} + \varepsilon_0\partial_t\mathbf{E} - \mu_0^{-1}\nabla\times\mathbf{B}\right)\cdot\delta\mathbf{A}\, dt\, d^3x,$$

and the three independent components of $\delta\mathbf{A}$ therefore give

$$\mathbf{J} + \varepsilon_0\partial_t\mathbf{E} - \mu_0^{-1}\nabla\times\mathbf{B} = 0,$$

or

$$\boxed{\nabla\times\mathbf{B} = \mu_0\mathbf{J} + \mu_0\varepsilon_0\partial_t\mathbf{E}}.$$

Finally the definitions themselves give $\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$ and, since $\nabla \times \nabla\varphi = 0$, $\nabla \times \mathbf{E} = -\partial_t(\nabla \times \mathbf{A}) = -\partial_t \mathbf{B}$, the remaining two Maxwell equations. The divergence of the second boxed equation, with $\nabla \cdot (\nabla \times \mathbf{B}) = 0$ and the first boxed equation, gives $0 = \mu_0(\nabla \cdot \mathbf{J} + \partial_t \rho)$: the field equations can hold only for sources obeying the continuity equation, which is why the statement assumes it.

22.3 Constrained Minima/Maxima

22.3.0 The following problems are to be solved by using Lagrangian multipliers.

Solution.

This entry is the textbook's instruction for Exercises 22.3.1–22.3.7. For an objective $F(q)$ constrained by $G(q) = C$, a permissible displacement satisfies $\nabla G \cdot \delta q = 0$ (if $\nabla G \neq 0$, every such tangent vector is, by the implicit function theorem, the velocity of a curve lying on $G = C$). At a constrained stationary point, $\nabla F \cdot \delta q = 0$ for every such tangent displacement. If $\nabla G \neq 0$, write $\nabla F = \lambda \nabla G + \mathbf{w}$ with $\lambda = \nabla F \cdot \nabla G / |\nabla G|^2$, so that $\mathbf{w} \cdot \nabla G = 0$; taking $\delta q = \mathbf{w}$ gives $\nabla F \cdot \mathbf{w} = |\mathbf{w}|^2 = 0$. The two normal vectors are therefore proportional:

$$\boxed{\nabla F - \lambda \nabla G = 0, \qquad G = C}.$$

For several independent constraints, ∇F is a linear combination of their gradients. The multiplier equations find stationary candidates; one must still examine boundary behavior or use an inequality to decide whether each candidate is a maximum, minimum, or neither. The following solutions include that classification.

22.3.1 The ground-state energy of a quantum particle of mass m in a pillbox (right-circular cylinder) is given by

$$E = \frac{\hbar^2}{2m}\left(\frac{(2.4048)^2}{R^2} + \frac{\pi^2}{H^2}\right),$$

in which R is the radius and H is the height of the pillbox. Find the ratio of R to H that will minimize the energy for a fixed positive volume. Here 2.4048 is the first positive zero $j_{0,1} = 2.404825\ldots$ of J_0, rounded as stated; the solution uses the stated value.

Solution.

Put $j = 2.4048$, retain the positive constant $\hbar^2/(2m)$ outside the minimization, and impose $\pi R^2 H = V$. For

$$\mathcal{F} = \frac{j^2}{R^2} + \frac{\pi^2}{H^2} + \lambda(\pi R^2 H - V),$$

the two multiplier equations are

$$-\frac{2j^2}{R^3} + 2\lambda\pi RH = 0, \qquad -\frac{2\pi^2}{H^3} + \lambda\pi R^2 = 0.$$

Solving each equation for λ and equating the results,

$$\lambda = \frac{j^2}{\pi R^4 H} = \frac{2\pi^2}{\pi R^2 H^3} = \frac{2\pi}{R^2 H^3}, \qquad j^2 H^2 = 2\pi^2 R^2,$$

the last after multiplying the equality by $\pi R^4 H^3$. Thus $(R/H)^2 = j^2/(2\pi^2)$, and since $R, H > 0$ the requested ratio is

$$\boxed{\frac{R}{H} = \frac{2.4048}{\pi\sqrt{2}} = 0.541270169273\ldots}.$$

To classify it, eliminate $H = V/(\pi R^2)$, so that $\pi^2/H^2 = \pi^4 R^4/V^2$:

$$\frac{2mE}{\hbar^2} = f(R) = \frac{j^2}{R^2} + \frac{\pi^4 R^4}{V^2}.$$

This diverges as $R \downarrow 0$ or $R \to \infty$, and its derivative

$$f'(R) = -\frac{2j^2}{R^3} + \frac{4\pi^4 R^3}{V^2}$$

vanishes only at $R^6 = j^2 V^2/(2\pi^4)$. There $(R/H)^2 = (\pi R^3/V)^2 = \pi^2 R^6/V^2 = j^2/(2\pi^2)$, the multiplier ratio. Thus the stationary ratio gives the unique global minimum. (With the unrounded zero $j_{0,1} = 2.4048255576\ldots$ the ratio would be $0.5412759\ldots$.) The Instructor's Manual replaces 2.4048 by 2.048 in both its derivative and its final ratio; the textbook's coefficient is 2.4048, which is used throughout above.

22.3.2 Under the textbook's stipulated postal rule, the total length plus girth may not exceed 36 inches. Using Lagrange multipliers, find the dimensions of the rectangular parallelepiped of maximum volume subject to this constraint.

Solution.

The textbook presents this 36-inch length-plus-girth limit as an actual postal regulation; the statement qualifies it as a stipulation because real postal size limits change over time and need not have this form, and only the mathematical constraint is used. Let the length be L and the two transverse widths be w, h. A maximum uses the whole allowance: otherwise increasing L raises the volume while remaining admissible. Thus

$$V = Lwh, \qquad L + 2w + 2h = 36.$$

For $\mathcal{F} = Lwh - \lambda(L + 2w + 2h - 36)$, the derivatives with respect to L, w, h give

$$wh = \lambda, \qquad Lh = 2\lambda, \qquad Lw = 2\lambda.$$

Substituting $\lambda = wh$ gives $Lh = 2wh$ and $Lw = 2wh$. All dimensions are positive, so dividing by h and by w gives $L = 2w$ and $L = 2h$. The constraint becomes $L + L + L = 36$, hence

$$\boxed{L = 12 \text{ in}, \qquad w = h = 6 \text{ in}, \qquad V_{\max} = 432 \text{ in}^3}.$$

For a global check, the three positive numbers $L, 2w, 2h$ have fixed sum 36, and their product is $4V$. The arithmetic–geometric mean inequality $L \cdot 2w \cdot 2h \le \big((L + 2w + 2h)/3\big)^3$ gives

$$4V \le (36/3)^3 = 1728, \qquad V \le 432 \text{ in}^3,$$

with equality exactly at $L = 2w = 2h$. The identified length is also the longest side, consistently with the length-plus-girth convention.

22.3.3 A thermal nuclear reactor is subject to the constraint

$$\varphi(a, b, c) = \left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2 + \left(\frac{\pi}{c}\right)^2 = B^2, \quad \text{a constant},$$

where the reactor is a rectangular parallelepiped of sides a, b, and c. Take $a, b, c, B > 0$. Find the ratios of a, b, and c that minimize the reactor volume. Show also that no maximum exists.

ANS. $a = b = c$, cube.

Solution.

For positive side lengths, extremizing abc is equivalent to extremizing $\ln(abc)$, since ln is increasing. Apply a multiplier to

$$\mathcal{F} = \ln a + \ln b + \ln c + \lambda \left[\pi^2(a^{-2} + b^{-2} + c^{-2}) - B^2\right].$$

The three equations are

$$\frac{1}{a} - \frac{2\lambda\pi^2}{a^3} = 0, \qquad \frac{1}{b} - \frac{2\lambda\pi^2}{b^3} = 0, \qquad \frac{1}{c} - \frac{2\lambda\pi^2}{c^3} = 0.$$

Multiplying them by a^3, b^3, c^3 gives $a^2 = b^2 = c^2 = 2\lambda\pi^2$, and positivity yields $a = b = c$. The constraint then reads $3\pi^2/a^2 = B^2$, so

$$a = b = c = \frac{\sqrt{3}\pi}{B}.$$

The cube is the *minimum*-volume reactor under this constraint. The printed request for a maximum is false: the volume has no finite upper bound. To prove both statements, set $u = a^{-2}$, $v = b^{-2}$, $w = c^{-2}$, so $u + v + w = B^2/\pi^2$. The arithmetic–geometric mean inequality $uvw \le \big((u + v + w)/3\big)^3$ and $uvw = (abc)^{-2}$ give

$$uvw \le \left(\frac{B^2}{3\pi^2}\right)^3, \qquad abc = (uvw)^{-1/2} \ge \left(\frac{3\pi^2}{B^2}\right)^{3/2} = \boxed{\frac{3\sqrt{3}\pi^3}{B^3}}.$$

Equality occurs only at $u = v = w$, i.e. the cube, whose volume $(\sqrt{3}\pi/B)^3$ is this bound.

For an explicit unbounded family, take

$$a = b = L, \qquad c = \frac{\pi}{\sqrt{B^2 - 2\pi^2/L^2}}, \qquad L > \frac{\sqrt{2}\pi}{B},$$

so that $(\pi/c)^2 = B^2 - 2\pi^2/L^2$. It satisfies the constraint exactly, while $abc = \pi L^2/\sqrt{B^2 - 2\pi^2/L^2} \to \infty$ as $L \to \infty$ (asymptotically $\pi L^2/B$). This settles the printed maximum question as well as identifying the correctly classified stationary cube.

22.3.4 For a lens of focal length f, the object distance p and the image distance q are related by $1/p + 1/q = 1/f$. Find the minimum object-image distance $(p + q)$ for fixed f. Assume real object and image (p and q both positive).

Solution.

The assumptions $p, q > 0$ imply $1/f = 1/p + 1/q > 0$, so $f > 0$. Use

$$\mathcal{F} = p + q + \lambda\left(\frac{1}{p} + \frac{1}{q} - \frac{1}{f}\right).$$

Differentiation gives $1 - \lambda/p^2 = 0$ and $1 - \lambda/q^2 = 0$. Thus $p^2 = q^2 = \lambda$, so $p = q$ (both are positive). The lens equation $2/p = 1/f$ then gives $p = q = 2f$ and

$$\boxed{(p + q)_{\min} = 4f}.$$

For a global proof, put $S = p + q$. Since $1/p + 1/q = (p + q)/(pq)$, the constraint implies $pq = fS$. Since $S^2 - 4pq = (p - q)^2 \ge 0$,

$$S^2 \ge 4pq = 4fS.$$

Divide by the positive S to obtain $S \ge 4f$, with equality only for the multiplier solution.

22.3.5 You have an ellipse $(x/a)^2 + (y/b)^2 = 1$. Find the inscribed rectangle of maximum area. Show that the ratio of the area of the maximum-area rectangle to the area of the ellipse is $2/\pi = 0.6366$.

Solution.

First take the rectangle's vertices to be $(\pm x, \pm y)$ with $x, y > 0$. Its area is $A = 4xy$ and its constraint is $x^2/a^2 + y^2/b^2 = 1$. For

$$\mathcal{F} = 4xy - \lambda(x^2/a^2 + y^2/b^2 - 1),$$

the equations are

$$4y = \frac{2\lambda x}{a^2}, \qquad 4x = \frac{2\lambda y}{b^2}.$$

Multiplying respectively by x, y gives $4xy = 2\lambda x^2/a^2 = 2\lambda y^2/b^2$; here $\lambda \neq 0$ because $xy > 0$, so $x^2/a^2 = y^2/b^2$. By the constraint each is therefore $1/2$, so

$$\boxed{x = \frac{a}{\sqrt{2}}, \quad y = \frac{b}{\sqrt{2}}, \quad A_{\max} = 4 \cdot \frac{a}{\sqrt{2}} \cdot \frac{b}{\sqrt{2}} = 2ab}.$$

The side lengths are $\sqrt{2}a, \sqrt{2}b$ and

$$\boxed{\frac{A_{\max}}{A_{\text{ellipse}}} = \frac{2ab}{\pi ab} = \frac{2}{\pi} = 0.6366197723\ldots}.$$

That this candidate is the maximum follows from the bound below.

This is also the maximum if the rectangle's orientation is unrestricted. Let its center be h, its orthonormal directions be e_1, e_2, and its half-lengths be u, v, so that its vertices are $h + \sigma_1 u e_1 + \sigma_2 v e_2$ with $\sigma_1, \sigma_2 = \pm 1$. With the ellipse matrix $Q = \operatorname{diag}(a^{-2}, b^{-2})$, a point p lies in the closed elliptical region exactly when $p^T Q p \leq 1$, so each vertex satisfies this inequality. Expanding,

$$\begin{aligned}(h + \sigma_1 u e_1 + \sigma_2 v e_2)^T Q (h + \sigma_1 u e_1 + \sigma_2 v e_2) = h^T Q h + u^2 q_1 + v^2 q_2 + 2\sigma_1 u\, h^T Q e_1 \\ + 2\sigma_2 v\, h^T Q e_2 + 2\sigma_1\sigma_2 uv\, e_1^T Q e_2,\end{aligned}$$

with $q_i = e_i^T Q e_i$. The last three terms average to zero over the four sign choices, so averaging the four vertex inequalities gives

$$h^T Q h + u^2 q_1 + v^2 q_2 \leq 1.$$

Dropping $h^T Q h \geq 0$ and using $u^2 q_1 + v^2 q_2 \geq 2uv\sqrt{q_1 q_2}$ (arithmetic–geometric mean) gives $2uv\sqrt{q_1 q_2} \leq 1$. In the orthonormal basis e_1, e_2 the matrix of Q has entries q_1, q_2 and $q_{12} = e_1^T Q e_2$, and its determinant is unchanged, so $q_1 q_2 = \det Q + q_{12}^2 \geq 1/(a^2 b^2)$. Hence $2uv/(ab) \leq 1$, i.e. the area $4uv \leq 2ab$. The aligned multiplier solution attains this bound, proving the global claim.

22.3.6 A rectangular parallelepiped is inscribed in an ellipsoid of semiaxes a, b, and c. Maximize the volume of the inscribed rectangular parallelepiped. Show that the ratio of the maximum volume to the volume of the ellipsoid is $2/(\pi\sqrt{3}) \approx 0.367$.

Solution.

For an aligned box let the positive half-lengths be x, y, z. Maximize $V = 8xyz$ subject to $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$. The multiplier equations give

$$8yz = \frac{2\lambda x}{a^2}, \quad 8xz = \frac{2\lambda y}{b^2}, \quad 8xy = \frac{2\lambda z}{c^2}.$$

Multiplying by x, y, z respectively makes all left sides equal: $8xyz = 2\lambda x^2/a^2 = 2\lambda y^2/b^2 = 2\lambda z^2/c^2$, with $\lambda \neq 0$ because $xyz > 0$. Thus $x^2/a^2 = y^2/b^2 = z^2/c^2$, and the constraint makes each $1/3$, giving

$$\boxed{x = \frac{a}{\sqrt{3}}, \quad y = \frac{b}{\sqrt{3}}, \quad z = \frac{c}{\sqrt{3}}, \qquad V_{\max} = \frac{8abc}{3\sqrt{3}}},$$

since $(1/\sqrt{3})^3 = 1/(3\sqrt{3})$. Dividing by the ellipsoid's $4\pi abc/3$ yields

$$\boxed{\frac{V_{\max}}{V_{\text{ellipsoid}}} = \frac{8abc}{3\sqrt{3}} \cdot \frac{3}{4\pi abc} = \frac{2}{\pi\sqrt{3}} = 0.3675525969\ldots}.$$

The inline ratio must be read as $2/(\pi\sqrt{3})$; $(2/\pi)\sqrt{3}$ would exceed one and would not match the stated decimal approximation.

For completeness, the same bound covers arbitrary orientations and centers; it also shows that the multiplier candidate is the maximum. With $Q = \operatorname{diag}(a^{-2}, b^{-2}, c^{-2})$, center h, orthonormal box directions e_i and half-lengths u_i, the vertices are $h + \sum_i \sigma_i u_i e_i$ with $\sigma_i = \pm 1$. As in Exercise 22.3.5, the cross terms in their values of $p^T Q p$ are odd in some σ_i and average to zero, so the average of the eight vertex inequalities gives $h^T Q h + \sum_i q_i u_i^2 \leq 1$, where $q_i = e_i^T Q e_i$; the nonnegative center term can be discarded. The arithmetic–geometric mean bound then gives $\prod_i (q_i u_i^2) \leq \left(\frac{1}{3}\sum_i q_i u_i^2\right)^3 \leq 1/27$. The Gram matrix $G_{ij} = e_i^T Q e_j$ of the vectors $Q^{1/2} e_i$ is $E^T Q E$, with E the orthogonal matrix of columns e_i, so $\det G = \det Q$ and $G_{ii} = q_i$. The Gram determinant (Hadamard) inequality $\det G \leq \prod_i G_{ii}$ (the squared volume spanned by the vectors $Q^{1/2} e_i$ is at most the product of their squared lengths) therefore gives $\prod_i q_i \geq \det Q = 1/(a^2 b^2 c^2)$. Hence

$$8u_1u_2u_3 \leq \frac{8}{\sqrt{27}\sqrt{q_1q_2q_3}} \leq \frac{8abc}{\sqrt{27}} = \frac{8abc}{3\sqrt{3}}.$$

The aligned box attains equality.

22.3.7 Find the maximum value of the directional derivative of $\varphi(x, y, z)$, a real function differentiable at a fixed point, for which the following formula holds,

$$\frac{d\varphi}{ds} = \frac{\partial\varphi}{\partial x}\cos\alpha + \frac{\partial\varphi}{\partial y}\cos\beta + \frac{\partial\varphi}{\partial z}\cos\gamma,$$

subject to the constraint,

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$

Solution.

At the point in question write $\mathbf{g} = \nabla\varphi$ and $\mathbf{n} = (\cos\alpha, \cos\beta, \cos\gamma)$. Maximize $D = \mathbf{g}\cdot\mathbf{n}$ subject to $\mathbf{n}\cdot\mathbf{n} = 1$. The multiplier equations for $D - \lambda(\mathbf{n}\cdot\mathbf{n} - 1)$ are

$$g_i - 2\lambda n_i = 0.$$

If $\mathbf{g} \neq 0$, then $\lambda \neq 0$ and $\mathbf{n} = \mathbf{g}/(2\lambda)$; normalization gives $4\lambda^2 = |\mathbf{g}|^2$, i.e. $\lambda = \pm|\mathbf{g}|/2$, and hence

$$\mathbf{n}_\pm = \pm\frac{\mathbf{g}}{|\mathbf{g}|}, \qquad D_\pm = \pm|\mathbf{g}|.$$

The maximum is therefore

$$\boxed{\left(\frac{d\varphi}{ds}\right)_{\max} = |\nabla\varphi|, \qquad \mathbf{n}_{\max} = \frac{\nabla\varphi}{|\nabla\varphi|}}.$$

Cauchy–Schwarz proves the global bound $\mathbf{g}\cdot\mathbf{n} \leq |\mathbf{g}||\mathbf{n}| = |\mathbf{g}|$. If $\nabla\varphi = 0$, every unit direction gives zero, so every direction is a maximizer and no preferred direction is defined.

22.4 Variation with Constraints

22.4.1 A particle of mass m is on a frictionless horizontal surface. In terms of plane polar coordinates (r, θ), it is constrained to move so that $\theta = \omega t$ (accomplished by pushing it with a rotating radial arm against which it can slide frictionlessly). With the initial conditions

$$t = 0, \qquad r = r_0, \qquad \dot{r} = 0 :$$

(a) Find the radial position as a function of time.

ANS. $r(t) = r_0 \cosh \omega t.$

(b) Find the force exerted on the particle by the constraint.

ANS. $F^{(c)} = 2m\dot{r}\omega = 2mr_0\omega^2 \sinh \omega t.$

Solution.

Use polar coordinates with the positive tangential direction $\mathbf{e}_\theta$. Before imposing the constraint,

$$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2), \qquad \phi = \theta - \omega t = 0.$$

With the convention $L + \lambda\phi$, the Lagrange equations are $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \lambda\frac{\partial \phi}{\partial q}$; since $\partial\phi/\partial r = 0$ and $\partial\phi/\partial\theta = 1$, they are

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 = 0, \qquad \frac{d}{dt}(mr^2\dot{\theta}) = \lambda.$$

The multiplier in the second equation is a torque, since its coordinate is an angle.

(a) Substitute $\dot{\theta} = \omega$ into the first equation:

$$\ddot{r} - \omega^2 r = 0, \qquad r = A\cosh(\omega t) + B\sinh(\omega t).$$

The initial conditions give $A = r(0) = r_0$ and $\omega B = \dot{r}(0) = 0$, so $B = 0$ (for $\omega \neq 0$), and

$$\boxed{r(t) = r_0 \cosh(\omega t)}, \qquad \dot{r} = r_0\omega \sinh(\omega t).$$

The $\omega = 0$ limit is $r = r_0$.

(b) The frictionless arm pushes perpendicular to itself, along $\mathbf{e}_\theta$. The generalized force conjugate to θ is $\mathbf{F}^{(c)} \cdot \partial\mathbf{r}/\partial\theta = rF_\theta$, because $\partial\mathbf{r}/\partial\theta = r\mathbf{e}_\theta$; so $rF_\theta = \lambda$. With $\dot{\theta} = \omega$ constant,

$$\lambda = \omega\frac{d}{dt}(mr^2) = 2m\omega r\dot{r}, \qquad \boxed{\mathbf{F}^{(c)} = \frac{\lambda}{r}\,\mathbf{e}_\theta = 2m\omega\dot{r}\,\mathbf{e}_\theta = 2mr_0\omega^2 \sinh(\omega t)\,\mathbf{e}_\theta}.$$

There is no radial applied force: the radial acceleration component $\ddot{r} - r\omega^2$ vanishes. As an energy check, with $\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\omega\,\mathbf{e}_\theta$ and $\ddot{r} = \omega^2 r$,

$$\frac{d}{dt}\frac{m}{2}(\dot{r}^2 + r^2\omega^2) = m\dot{r}(\ddot{r} + r\omega^2) = 2m\omega^2 r\dot{r} = \mathbf{F}^{(c)} \cdot \mathbf{v}.$$

Thus the motor turning the arm supplies the increasing kinetic energy.

22.4.2 A point mass m is moving on a flat, horizontal, frictionless plane. The mass is constrained by a string to move radially inward at a constant rate. Using plane polar coordinates (r, θ), $r = r_0 - kt$, with $m, r_0, k > 0$ and $0 \le t < r_0/k$, so that $r > 0$:

(a) Set up the Lagrangian.

(b) Obtain the constrained Lagrange equations.

(c) Solve the θ-dependent Lagrange equation to obtain $\omega(t)$, the angular velocity. What is the physical significance of the constant of integration that you get from your "free" integration?

(d) Using the $\omega(t)$ from part (c), solve the r-dependent (constrained) Lagrange equation to obtain $\lambda(t)$. In other words, explain what is happening to the force of constraint as $r \to 0$.

Solution.

(a) Take r and θ as independent coordinates initially:

$$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2), \qquad \phi(r,t) = r - r_0 + kt = 0.$$

Here $k > 0$, and the prescribed motion applies for $0 \le t < r_0/k$.

(b) Using $L + \lambda\phi$, with $\partial\phi/\partial r = 1$ and $\partial\phi/\partial\theta = 0$, the Lagrange equations $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \lambda\frac{\partial \phi}{\partial q}$ give

$$\boxed{m\ddot{r} - mr\dot{\theta}^2 = \lambda}, \qquad \boxed{\frac{d}{dt}(mr^2\dot{\theta}) = 0}, \qquad r = r_0 - kt.$$

Because $\partial\phi/\partial r = 1$, λ is the signed radial force, positive outward. The physical string tension is $-\lambda$.

(c) Integrating the cyclic-coordinate equation once,

$$mr^2\dot{\theta} = \ell, \qquad \boxed{\omega(t) = \dot{\theta} = \frac{\ell}{m(r_0 - kt)^2} = \omega_0\frac{r_0^2}{(r_0 - kt)^2}}.$$

The constant $\ell = mr_0^2\omega_0$, with $\omega_0 = \omega(0)$, is the conserved angular momentum about the hole through which the string is drawn: the constraint force is radial and therefore has zero torque about that point. The Instructor's Manual prints the third power in the denominator of $\omega(t)$. It must be the second power: the printed expression would make $mr^2\omega = \ell/r$ vary with time. A second integration, if desired, gives, since $d(r_0 - k\tau)^{-1}/d\tau = k(r_0 - k\tau)^{-2}$,

$$\theta(t) - \theta_0 = \frac{\ell}{m}\int_0^t \frac{d\tau}{(r_0 - k\tau)^2} = \frac{\ell}{mk}\left(\frac{1}{r_0 - kt} - \frac{1}{r_0}\right).$$

(d) The angular velocity is found in part (c), hence the corrected part reference in the statement. Since $r = r_0 - kt$ is linear in t, $\ddot{r} = 0$, and with $\dot{\theta} = \omega = \ell/(mr^2)$ the radial equation becomes

$$\boxed{\lambda(t) = -mr\omega^2 = -mr\frac{\ell^2}{m^2r^4} = -\frac{\ell^2}{m(r_0 - kt)^3}}, \qquad T_{\text{string}} = -\lambda = \frac{\ell^2}{mr^3}.$$

For $\ell \neq 0$, the tension tends to $+\infty$ as $r \to 0^+$. Maintaining the specified radial speed therefore requires an unbounded pulling force in that limit. The divergence requires nonzero angular momentum. If $\ell = 0$, then $\omega = 0$ and $\lambda = 0$ throughout the prescribed inward motion; the endpoint $r = 0$ itself is a polar-coordinate singularity.

22.4.3 A flexible cable is suspended from two fixed points. The length of the cable is fixed. Find the curve that will minimize the total gravitational potential energy of the cable. Assume a uniform cable in uniform gravity, supports at different horizontal positions, and a cable length greater than the straight-line distance between the supports.

ANS. Hyperbolic cosine.

Solution.

The hyperbolic-cosine answer assumes a uniform cable in a uniform gravitational field. A nonuniform mass per unit length generally produces a different curve. Let the constant mass per unit length be ρ, let y increase upward, and suppose the endpoints have $x_1 < x_2$. For a graph $y(x)$,

$$U = \rho g \int_{x_1}^{x_2} y\sqrt{1+y'^2}\,dx, \qquad L_0 = \int_{x_1}^{x_2} \sqrt{1+y'^2}\,dx.$$

Introduce a constant multiplier λ and vary

$$\int_{x_1}^{x_2} F\,dx, \qquad F = (\rho g y + \lambda)\sqrt{1+y'^2}.$$

There is no explicit x dependence, so by the second form of Euler's equation (Exercise 22.1.1, with $F_x = 0$) the Beltrami expression $F - y'F_{y'}$ is constant. With $F_{y'} = (\rho g y + \lambda)y'/\sqrt{1+y'^2}$ this first integral is

$$(\rho g y + \lambda)\sqrt{1+y'^2} - y'\frac{(\rho g y + \lambda)y'}{\sqrt{1+y'^2}} = \frac{\rho g y + \lambda}{\sqrt{1+y'^2}} = \rho g c,$$

where the constant has been written as $\rho g c$. For the hanging solution $c > 0$. Put $Y = y + \lambda/(\rho g)$, so that $\rho g y + \lambda = \rho g Y$:

$$Y = c\sqrt{1+Y'^2}, \qquad Y'^2 = \frac{Y^2}{c^2} - 1, \qquad \frac{dY}{\sqrt{Y^2 - c^2}} = \pm\frac{dx}{c}.$$

Since $\int dY/\sqrt{Y^2 - c^2} = \operatorname{arcosh}(Y/c)$ for $Y \geq c$, integration gives $\operatorname{arcosh}(Y/c) = \pm(x-b)/c$ with a constant b. Both signs, i.e. the two sides of the lowest point $x = b$, give $Y = c\cosh((x-b)/c)$ because cosh is even; thus

$$\boxed{y(x) = c\cosh\frac{x-b}{c} + d}, \qquad d = -\frac{\lambda}{\rho g}.$$

The three constants are determined by the two endpoint values and, since $\sqrt{1+y'^2} = \sqrt{1+\sinh^2((x-b)/c)} = \cosh((x-b)/c)$,

$$L_0 = \int_{x_1}^{x_2} \cosh\frac{x-b}{c}\,dx = c\left[\sinh\frac{x_2-b}{c} - \sinh\frac{x_1-b}{c}\right].$$

The Instructor's Manual repeats x_2 in both terms of this length formula. The second evaluation is at x_1; otherwise the displayed length would be zero. These three conditions do fix the constants. Let the supports be (x_1, y_1) and (x_2, y_2), and write $\Delta x = x_2 - x_1 > 0$, $\Delta y = y_2 - y_1$ and $u_k = (x_k - b)/c$, so that $u_2 - u_1 = \Delta x/c$, $\Delta y = c(\cosh u_2 - \cosh u_1)$ and $L_0 = c(\sinh u_2 - \sinh u_1)$. Expanding the squares and using $\cosh^2 u - \sinh^2 u = 1$, $\cosh u_2 \cosh u_1 - \sinh u_2 \sinh u_1 = \cosh(u_2 - u_1)$ and $\cosh z - 1 = 2\sinh^2(z/2)$,

$$L_0^2 - \Delta y^2 = 2c^2\big[\cosh(u_2 - u_1) - 1\big] = 4c^2\sinh^2\frac{\Delta x}{2c},$$

that is, $\sqrt{L_0^2 - \Delta y^2} = \Delta x\,(\sinh\tau)/\tau$ with $\tau = \Delta x/(2c)$. Since $(\sinh\tau)/\tau$ has derivative $(\tau\cosh\tau - \sinh\tau)/\tau^2 > 0$ (because $\tanh\tau < \tau$), it increases strictly from 1 (as $\tau \to 0^+$) to ∞, and τ decreases as c increases, so exactly one $c > 0$ solves this when $L_0^2 > \Delta x^2 + \Delta y^2$, i.e. when the cable is longer than the straight-line distance between the supports, as the statement assumes. Next, $L_0 + \Delta y = c(e^{u_2} - e^{u_1})$ and $L_0 - \Delta y = c(e^{-u_1} - e^{-u_2}) = e^{-u_1-u_2}(L_0 + \Delta y)$, so

$$u_1 + u_2 = \frac{x_1 + x_2 - 2b}{c} = \ln\frac{L_0 + \Delta y}{L_0 - \Delta y}$$

fixes b (both $L_0 \pm \Delta y > 0$ because $L_0 > |\Delta y|$), and $y(x_1) = y_1$ then fixes d. These steps reverse: for the constants found, the catenary's own length ℓ and rise δ satisfy the same identities, so $\ell \pm \delta$ (both positive) have the product and the ratio of $L_0 \pm \Delta y$, hence equal them, and all three conditions hold.

To verify that the hanging solution minimizes energy, rather than merely makes it stationary, parameterize it by arc length $s \in [0, L_0]$. Since $ds/dx = \cosh((x-b)/c)$, there is a constant s_0 (the parameter value of the point $x = b$) such that

$$s - s_0 = c\sinh\frac{x-b}{c}.$$

Define the tension field $\mathbf{T}(s) = \rho g\big(c,\ s - s_0\big)$. The catenary's unit tangent is $(dx/ds, dy/ds) = (1, \sinh((x-b)/c))/\cosh((x-b)/c)$, and $c\cosh((x-b)/c) = \sqrt{c^2 + (s-s_0)^2} = |\mathbf{T}|/(\rho g)$, so

$$\mathbf{r}_*'(s) = \frac{(c,\ s - s_0)}{\sqrt{c^2 + (s-s_0)^2}} = \frac{\mathbf{T}(s)}{|\mathbf{T}(s)|}, \qquad \mathbf{T}\cdot\mathbf{r}_*' = |\mathbf{T}|.$$

Also $\mathbf{T}' = \rho g\,\mathbf{e}_y$, so the energy of any curve parameterized by arc length is $U[\mathbf{r}] = \rho g\int_0^{L_0} y\,ds = \int_0^{L_0}\mathbf{T}'\cdot\mathbf{r}\,ds$. Any other cable of the same length with the same endpoints has an arc-length parameterization $\mathbf{r}(s)$ with $|\mathbf{r}'| = 1$, $\mathbf{r}(0) = \mathbf{r}_*(0)$ and $\mathbf{r}(L_0) = \mathbf{r}_*(L_0)$. Integration by parts gives

$$\begin{aligned} U[\mathbf{r}] - U[\mathbf{r}_*] &= \int_0^{L_0}\mathbf{T}'\cdot(\mathbf{r} - \mathbf{r}_*)\,ds \\ &= \big[\mathbf{T}\cdot(\mathbf{r} - \mathbf{r}_*)\big]_0^{L_0} - \int_0^{L_0}\mathbf{T}\cdot(\mathbf{r}' - \mathbf{r}_*')\,ds \\ &= \int_0^{L_0}\big(|\mathbf{T}| - \mathbf{T}\cdot\mathbf{r}'\big)\,ds \geq 0, \end{aligned}$$

where the endpoint term vanishes because the endpoints coincide, and the last inequality is Cauchy–Schwarz, $\mathbf{T}\cdot\mathbf{r}' \leq |\mathbf{T}||\mathbf{r}'| = |\mathbf{T}|$. Equality requires $\mathbf{r}' = \mathbf{T}/|\mathbf{T}| = \mathbf{r}_*'$ everywhere ($\mathbf{T} \neq 0$ since $c > 0$), hence the same curve with the fixed endpoints. A curved cable needs L_0 greater than the straight-line distance between the endpoints; equality gives the taut straight segment, and a shorter cable cannot join the endpoints. For vertically aligned supports the smooth graph assumption fails. In the ideal zero-thickness model a slack cable has a limiting vertical fold. If its endpoint heights are y_1, y_2 and $L_0 > |y_2 - y_1|$, its lowest point is

$$y_b = \frac{y_1 + y_2 - L_0}{2},$$

since $(y_1 - y_b) + (y_2 - y_b) = L_0$. The same argument applies with horizontal tension zero and $\mathbf{T} = \rho g(s - s_0)\mathbf{e}_y$: the cable goes downward ($\mathbf{r}_*' = -\mathbf{e}_y$) for $s < s_0 = y_1 - y_b$ and upward ($\mathbf{r}_*' = \mathbf{e}_y$) for $s > s_0$, so again $\mathbf{T}\cdot\mathbf{r}_*' = \rho g|s - s_0| = |\mathbf{T}|$ and $\mathbf{T}' = \rho g\,\mathbf{e}_y$.

22.4.4 A fixed volume of water is rotating in a cylinder with constant angular velocity ω. Find the curve of the water surface that will minimize the total potential energy of the water in the combined gravitational-centrifugal force field. Assume constant density, uniform gravity, a flat-bottomed cylinder, no overflow, and negligible surface tension. Allow a dry central region if required by nonnegative water depth.

ANS. Parabola.

Solution.

In the Instructor's Manual's energy integral the centrifugal term is missing the square on ω. Also, its equation $2gry - \omega^2 r^3 - 2\lambda r = 0$ gives $y = (\omega^2 r^2 + 2\lambda)/(2g)$, with a plus sign for its stated multiplier convention. The derivation below uses $U + \lambda V$. Work in the frame rotating with the water. The potential per unit mass for gravity plus the centrifugal force is

$$\Phi(r, z) = gz - \frac{1}{2}\omega^2 r^2,$$

since $-\nabla\Phi = -g\mathbf{e}_z + \omega^2 r\mathbf{e}_r$. Let the cylinder have radius R, a flat bottom $z = 0$, and water height $h(r)$. The density ρ is constant. With volume element $2\pi r\, dr\, dz$, integrating through the depth, $\int_0^h (gz - \frac{1}{2}\omega^2 r^2)\, dz = \frac{g}{2}h^2 - \frac{\omega^2 r^2}{2}h$, gives

$$U[h] = 2\pi\rho \int_0^R \left(\frac{g}{2}h^2 - \frac{\omega^2 r^2}{2}h\right) r\, dr, \qquad V = 2\pi \int_0^R h(r) r\, dr.$$

Under $h \to h + \delta h$ the first variation of $U + \lambda V$ is $2\pi \int_0^R \big[\rho(gh - \omega^2 r^2/2) + \lambda\big]\delta h\, r\, dr$. Where $h > 0$, δh is arbitrary, so stationarity gives the pointwise equation

$$\rho\left(gh - \frac{\omega^2 r^2}{2}\right) + \lambda = 0.$$

Consequently, with $h_0 = -\lambda/(\rho g)$, the free surface is

$$\boxed{z = h(r) = h_0 + \frac{\omega^2 r^2}{2g}},$$

a paraboloid of revolution, whose section through the axis is a parabola. When the bottom is completely wet, the volume fixes

$$V = 2\pi \int_0^R \Big(h_0 + \frac{\omega^2 r^2}{2g}\Big) r\, dr = \pi R^2 h_0 + \frac{\pi\omega^2 R^4}{4g}, \qquad \boxed{h_0 = \frac{V}{\pi R^2} - \frac{\omega^2 R^2}{4g}}.$$

For a volume-preserving perturbation η, since $\frac{g}{2}(h + \eta)^2 - \frac{g}{2}h^2 = gh\eta + \frac{g}{2}\eta^2$,

$$U[h + \eta] - U[h] = 2\pi \int_0^R \rho\Big(gh - \frac{\omega^2 r^2}{2}\Big)\eta\, r\, dr + \frac{\rho g}{2}\int_{\text{bottom}} \eta^2\, dA,$$

with $dA = 2\pi r\, dr$. The first integral is $-\lambda \cdot 2\pi \int_0^R \eta\, r\, dr = 0$, because $\rho(gh - \omega^2 r^2/2) = -\lambda$ is constant and the volume is unchanged. Hence

$$U[h + \eta] - U[h] = \frac{\rho g}{2}\int_{\text{bottom}} \eta^2\, dA \geq 0.$$

This proves the minimum.

The full-disk formula requires $h_0 \ge 0$ and no overflow. If it predicts a negative depth, a dry central region appears. The physical minimizer is $h(r) = \max\{0, C + \omega^2 r^2/(2g)\}$, with C chosen from the volume; the wetted free surface remains parabolic. Explicitly, if its inner wet radius is r_d, then $C = -\omega^2 r_d^2/(2g)$ and $h = \omega^2(r^2 - r_d^2)/(2g)$ on $r_d \le r \le R$, and direct integration,

$$V = \frac{\pi\omega^2}{g}\int_{r_d}^{R}(r^2 - r_d^2)\, r\, dr = \frac{\pi\omega^2}{g}\Big[\frac{r^4}{4} - \frac{r_d^2 r^2}{2}\Big]_{r_d}^{R},$$

gives, since the bracket equals $R^4/4 - r_d^2R^2/2 + r_d^4/4 = (R^2 - r_d^2)^2/4$,

$$V = \frac{\pi\omega^2(R^2 - r_d^2)^2}{4g}, \qquad r_d^2 = R^2 - 2\sqrt{\frac{gV}{\pi\omega^2}},$$

taking the positive square root because $r_d < R$. Thus $0 < r_d < R$ exactly when $V < \pi\omega^2 R^4/(4g)$, which by the full-disk formula for h_0 is exactly the case $h_0 < 0$; for $V \ge \pi\omega^2 R^4/(4g)$ the bottom is fully wet and $C = h_0$. The same expansion proves the minimum with the inequality $h \ge 0$ imposed. Put $\mu(r) = \rho(gh - \omega^2 r^2/2)$: it equals $\rho g C$ on the wet annulus, and $-\rho\omega^2 r^2/2 \ge -\rho\omega^2 r_d^2/2 = \rho g C$ on the dry disk $r < r_d$. For any admissible $\tilde h \ge 0$ with the same volume, write $\eta = \tilde h - h$. As above, $U[\tilde h] - U[h] = \int_{\text{bottom}}[\mu\eta + \frac{\rho g}{2}\eta^2]\, dA$. Since the volumes agree, $\int \rho g C\, \eta\, dA = 0$; and $\mu - \rho g C$ vanishes except on the dry disk, where $h = 0$ and $\eta = \tilde h$. Hence

$$U[\tilde h] - U[h] = \int_{\text{bottom}}\Big[(\mu - \rho g C)\,\tilde h + \frac{\rho g}{2}\eta^2\Big] dA \ge 0.$$

22.4.5 Consider simple closed regular plane curves enclosing positive area, so that Green's theorem gives the enclosed area and the curve has a periodic arc-length parameterization.

(a) Show that for a fixed-length perimeter the plane figure with maximum area is a circle.

(b) Show that for a fixed planar area the boundary with minimum perimeter is a circle.

Hint. The geometric radius of curvature R is given by

$$R = \frac{(r^2 + r_\theta^2)^{3/2}}{\left|r r_{\theta\theta} - 2r_\theta^2 - r^2\right|}.$$

Note. The problems of this section, variation subject to constraints, are often called *isoperimetric*. The term arose from problems of maximizing area subject to a fixed perimeter, as in part (a) of this problem.

Solution.

Let a regular simple closed curve be positively oriented and write $\mathbf{r}(u) = (x(u), y(u))$, $v = \sqrt{x'^2 + y'^2}$. Its area (by Green's theorem) and its length are

$$A = \frac{1}{2}\oint (xy' - yx')\, du, \qquad L = \oint v\, du.$$

Both parts lead to the same stationary curves: vary $A + \lambda L$ in part (a), or $L + \mu A = \mu(A + \mu^{-1}L)$ in part (b), with a constant multiplier. For $A + \lambda L$ the integrand is $F = \frac{1}{2}(xy' - yx') + \lambda v$, with $F_x = y'/2$ and $F_{x'} = -y/2 + \lambda x'/v$, so the x Euler equation is

$$\frac{y'}{2} - \frac{d}{du}\left(-\frac{y}{2} + \lambda\frac{x'}{v}\right) = 0, \quad \text{so} \quad \lambda\frac{d}{du}\frac{x'}{v} = y'.$$

The y equation, with $F_y = -x'/2$ and $F_{y'} = x/2 + \lambda y'/v$, similarly gives

$$-\frac{x'}{2} - \frac{d}{du}\left(\frac{x}{2} + \lambda\frac{y'}{v}\right) = 0, \quad \text{so} \quad \lambda\frac{d}{du}\frac{y'}{v} = -x'.$$

Here $\lambda \neq 0$, since otherwise $x' = y' = 0$. Use arc length s, so $x_s^2 + y_s^2 = 1$; then $x'/v = x_s$, $y'/v = y_s$, $x' = vx_s$, $y' = vy_s$ and $d/du = v\,d/ds$, and the equations become $\lambda x_{ss} = y_s$, $\lambda y_{ss} = -x_s$. Integrating these equations once,

$$x + \lambda y_s = C_1, \qquad y - \lambda x_s = C_2.$$

Squaring and adding, with $x_s^2 + y_s^2 = 1$, yields

$$(x - C_1)^2 + (y - C_2)^2 = \lambda^2.$$

Thus every regular extremal is a circle, of radius $|\lambda|$; equivalently, its curvature $|\mathbf{r}_{ss}| = 1/|\lambda|$ is constant, which is the property the hint is meant to establish.

It remains to prove the requested global maximum and minimum. Translate the curve so $\int_0^L \mathbf{r}(s)\,ds = 0$; this changes neither A nor L. For a mean-zero L-periodic function f, the Fourier expansion $f = \sum_{n\geq 1}[a_n\cos(2\pi ns/L) + b_n\sin(2\pi ns/L)]$ has no constant term, and Parseval's theorem for f and f' gives

$$\int_0^L f'^2\,ds = \frac{L}{2}\sum_{n\geq 1}\left(\frac{2\pi n}{L}\right)^2(a_n^2 + b_n^2) \geq \left(\frac{2\pi}{L}\right)^2\frac{L}{2}\sum_{n\geq 1}(a_n^2 + b_n^2) = \left(\frac{2\pi}{L}\right)^2\int_0^L f^2\,ds.$$

Apply this to both coordinates, whose means are now zero. Because $|\mathbf{r}_s| = 1$,

$$\int_0^L |\mathbf{r}|^2\,ds \leq \frac{L^2}{4\pi^2}\int_0^L |\mathbf{r}_s|^2\,ds = \frac{L^3}{4\pi^2}.$$

Since $xy_s - yx_s = \mathbf{r}\cdot(y_s, -x_s)$ and $|(y_s, -x_s)| = |\mathbf{r}_s|$, the Cauchy–Schwarz inequality, applied pointwise and then to the integral, gives the isoperimetric inequality:

$$2A = \int_0^L (xy_s - yx_s)\,ds \leq \left(\int_0^L |\mathbf{r}|^2\,ds\right)^{1/2}\left(\int_0^L |\mathbf{r}_s|^2\,ds\right)^{1/2} \leq \left(\frac{L^3}{4\pi^2}\right)^{1/2} L^{1/2} = \frac{L^2}{2\pi}.$$

Equality holds for a circle of radius $L/(2\pi)$ centered at the origin, for which $\mathbf{r}_s$ is a constant multiple of the 90° rotation of $\mathbf{r}$: then $\int|\mathbf{r}|^2 ds = L^3/(4\pi^2)$ and $2A = L^2/(2\pi)$. Conversely, equality in $2A \leq L^2/(2\pi)$ requires equality in the Fourier estimate, i.e. $a_n = b_n = 0$ for $n \geq 2$ in both coordinates, so $\mathbf{r} = \mathbf{P}\cos\psi + \mathbf{Q}\sin\psi$ with $\psi = 2\pi s/L$ and constant vectors $\mathbf{P}, \mathbf{Q}$. Then $|\mathbf{r}_s|^2 = (2\pi/L)^2\big(|\mathbf{Q}|^2\cos^2\psi + |\mathbf{P}|^2\sin^2\psi - 2\mathbf{P}\cdot\mathbf{Q}\sin\psi\cos\psi\big) = 1$ for all ψ forces (take $\psi = 0$, $\pi/2$, $\pi/4$) $|\mathbf{Q}| = |\mathbf{P}| = L/(2\pi)$ and $\mathbf{P}\cdot\mathbf{Q} = 0$: a circle. So the circle is the only curve attaining the bound. Therefore, for part (a), and for part (b) by solving $A \leq L^2/(4\pi)$ for L,

$$\boxed{A_{\max} = \frac{L^2}{4\pi} \quad \text{at fixed } L}, \qquad \boxed{L_{\min} = 2\sqrt{\pi A} \quad \text{at fixed } A},$$

both attained only by the circle. The textbook hint's expression is a signed radius of curvature with the opposite sign to the usual counterclockwise convention: for $r = \text{constant} > 0$ it gives $R = -r$. A geometric radius requires the absolute value of its denominator. Its sign does not affect the conclusion that constant curvature gives a circle.

22.4.6 Show that requiring J, given by

$$J = \int_a^b \int_a^b K(x,t)\varphi(x)\varphi(t)\,dx\,dt,$$

to have a stationary value subject to the normalizing condition

$$\int_a^b \varphi^2(x)\,dx = 1$$

leads to a Hilbert–Schmidt integral equation of the form given in Eq. (21.64), provided the corresponding integral-operator eigenvalue is nonzero. State separately what happens in the nullspace.

Note. The kernel $K(x,t)$ is symmetric. Take real $\varphi \in L^2[a,b]$ and a real $K \in L^2([a,b]^2)$, so the quadratic functional is well defined.

Solution.

Assume real functions and a real symmetric kernel, $K(x,t) = K(t,x)$, on $[a,b]$, with integrals for which the following variations are valid. Define

$$(T\varphi)(x) = \int_a^b K(x,t)\varphi(t)\,dt, \qquad \mathcal{A}[\varphi] = J[\varphi] - \mu\left(\int_a^b \varphi^2\,dx - 1\right).$$

Under $\varphi \mapsto \varphi + \epsilon\eta$ the product $\varphi(x)\varphi(t)$ acquires the first-order term $\epsilon[\eta(x)\varphi(t) + \varphi(x)\eta(t)]$, so

$$\begin{aligned}\left.\frac{dJ}{d\epsilon}\right|_0 &= \int_a^b \int_a^b K(x,t)[\eta(x)\varphi(t) + \varphi(x)\eta(t)]\,dx\,dt \\ &= 2\int_a^b \eta(x)(T\varphi)(x)\,dx.\end{aligned}$$

In the second term, interchange the names x, t and use the symmetry of K:

$$\iint K(x,t)\varphi(x)\eta(t)\,dx\,dt = \iint K(t,x)\varphi(t)\eta(x)\,dx\,dt = \int_a^b \eta\,T\varphi\,dx,$$

which equals the first term. The normalization contributes $\frac{d}{d\epsilon}\int(\varphi+\epsilon\eta)^2 dx\big|_0 = 2\int\varphi\eta\,dx$. Hence

$$\delta\mathcal{A} = 2\int_a^b \eta(x)\big[(T\varphi)(x) - \mu\varphi(x)\big]\,dx = 0.$$

Arbitrariness of η (take $\eta = T\varphi - \mu\varphi$, which makes the integral $2\int(T\varphi-\mu\varphi)^2 dx$) gives

$$\boxed{\int_a^b K(x,t)\varphi(t)\,dt = \mu\varphi(x)}.$$

Multiplying by $\varphi(x)$ and integrating over x shows that $J = \int\varphi\,T\varphi\,dx = \mu\int\varphi^2 dx = \mu$. For $\mu \neq 0$, set $\lambda = 1/\mu$:

$$\boxed{\varphi(x) = \lambda\int_a^b K(x,t)\varphi(t)\,dt}.$$

The textbook's reference to Eq. (20.64) should be Eq. (21.64), the homogeneous Hilbert–Schmidt equation. Moreover, the reciprocal form assumes $\mu \neq 0$. A normalized element of the kernel of T is also stationary, with $J = \mu = 0$, and cannot be represented by a finite reciprocal λ. For instance, $K = 1$ on $[0,1]^2$ and $\varphi(x) = \sqrt{12}(x - \tfrac{1}{2})$ give $\int_0^1 \varphi^2 dx = 12\int_0^1 (x-\tfrac{1}{2})^2 dx = 12\cdot\tfrac{1}{12} = 1$ and $(T\varphi)(x) = \int_0^1 \varphi(t)\,dt = \sqrt{12}\,\big[\tfrac{t^2}{2} - \tfrac{t}{2}\big]_0^1 = 0$. The undivided eigenvalue equation above includes this case, with $\mu = 0$.

22.4.7 An unknown function satisfies the differential equation

$$y'' + \left(\frac{\pi}{2}\right)^2 y = 0$$

and the boundary conditions $y'(0) = 0$, $y(1) = 0$, with $y(0) = 1$ fixing the normalization. Regard this as the ground-state problem for $-y'' = \lambda y$, whose exact eigenvalue is $(\pi/2)^2$.

(a) Calculate the approximation $\lambda = F[y_{\text{trial}}]$ for $y_{\text{trial}} = 1 - x^2$.

(b) Compare with the exact eigenvalue.

ANS. (a) $\lambda = 2.5$. (b) $\lambda/\lambda_{\text{exact}} = 1.013$.

Solution.

The nonhomogeneous values $y(0) = 1$, $y(1) = 0$ alone do not define a Sturm–Liouville eigenvalue problem with variable λ. The intended Rayleigh problem has $y'(0) = 0$, $y(1) = 0$, with $y(0) = 1$ fixing the scale of the eigenfunction. The given exact solution and both trial families in their optimum ranges obey that additional Neumann condition. The quotient to be used is

$$F[y] = \frac{N}{D} = \frac{\int_0^1 y'^2\,dx}{\int_0^1 y^2\,dx}, \qquad y(1) = 0.$$

Since $\delta F = (\delta N - F\,\delta D)/D$, F is stationary exactly when $\delta N - \lambda\,\delta D = 0$ with $\lambda = F[y]$. For $y \to y + \epsilon\eta$ with $\eta(1) = 0$ ($\eta(0)$ free: $F[cy] = F[y]$ for every constant $c \neq 0$, so dividing $y + \epsilon\eta$ by $1 + \epsilon\eta(0)$ restores $y(0) = 1$ without changing F), $\delta N = 2\int_0^1 y'\eta'\,dx = 2[y'\eta]_0^1 - 2\int_0^1 y''\eta\,dx$ and $\delta D = 2\int_0^1 y\eta\,dx$, so the condition is

$$2[y'\eta]_0^1 - 2\int_0^1 (y'' + \lambda y)\eta\,dx = 0.$$

The endpoint 1 has $\eta(1) = 0$, so the boundary term is $-2y'(0)\eta(0)$; for arbitrary $\eta(0)$ the free endpoint 0 requires $y'(0) = 0$ (a natural boundary condition). The equation is $-y'' = \lambda y$.

(a) For $y_{\text{trial}} = 1 - x^2$, $y' = -2x$ and

$$\int_0^1 y'^2\,dx = 4\int_0^1 x^2\,dx = \frac{4}{3},$$

and

$$\int_0^1 y^2\,dx = \int_0^1 (1 - 2x^2 + x^4)\,dx = 1 - \frac{2}{3} + \frac{1}{5} = \frac{8}{15}.$$

Consequently

$$\boxed{\lambda_{\text{trial}} = \frac{4/3}{8/15} = \frac{5}{2} = 2.5}.$$

(b) For $\lambda > 0$ the general solution of $-y'' = \lambda y$ is $A\cos(\sqrt{\lambda}x) + B\sin(\sqrt{\lambda}x)$; $y'(0) = B\sqrt{\lambda} = 0$ gives $y = A\cos(\sqrt{\lambda}x)$, and $y(1) = 0$ gives $\cos\sqrt{\lambda} = 0$, i.e. $\sqrt{\lambda} = \pi/2, 3\pi/2, \ldots$ (for $\lambda \leq 0$ only $y \equiv 0$ satisfies both conditions). The lowest eigenvalue and its normalized-at-zero eigenfunction are

$$\lambda_{\text{exact}} = \frac{\pi^2}{4}, \qquad y_{\text{exact}} = \cos\frac{\pi x}{2}.$$

Thus

$$\boxed{\frac{\lambda_{\text{trial}}}{\lambda_{\text{exact}}} = \frac{5/2}{\pi^2/4} = \frac{10}{\pi^2} = 1.013211836\ldots},$$

an excess of approximately 1.32118%.

22.4.8 In Exercise 22.4.7 use a trial function $y = 1 - x^n$, with $n > 1/2$ for finite energy.

(a) Find the value of n that will minimize $F[y_{\text{trial}}]$.

(b) Show that the optimum value of n drives the ratio $\lambda/\lambda_{\text{exact}}$ down to 1.003.

ANS. (a) $n = 1.7247$.

Solution.

Use the Rayleigh quotient and boundary interpretation established in Exercise 22.4.7. The trial $y = 1 - x^n$ has $y' = -nx^{n-1}$ and finite derivative energy precisely for $n > \frac{1}{2}$, because $\int_0^1 x^{2n-2}\,dx$ converges exactly when $2n - 2 > -1$, and then

$$N = \int_0^1 y'^2\,dx = n^2 \int_0^1 x^{2n-2}\,dx = \frac{n^2}{2n-1}.$$

Its squared norm is

$$\begin{aligned} D &= \int_0^1 (1 - 2x^n + x^{2n})\,dx \\ &= 1 - \frac{2}{n+1} + \frac{1}{2n+1} \\ &= \frac{(n+1)(2n+1) - 2(2n+1) + (n+1)}{(n+1)(2n+1)} = \frac{2n^2}{(n+1)(2n+1)}, \end{aligned}$$

since the numerator is $(2n^2 + 3n + 1) - (4n + 2) + (n + 1) = 2n^2$. Therefore

$$F(n) = \frac{N}{D} = \frac{n^2}{2n-1} \cdot \frac{(n+1)(2n+1)}{2n^2} = \frac{(n+1)(2n+1)}{2(2n-1)}.$$

Write $F = f/g$ with $f = 2n^2 + 3n + 1$ and $g = 4n - 2$, so that $f' = 4n + 3$ and $g' = 4$. The quotient rule $F' = (f'g - fg')/g^2$ and $(4n+3)(4n-2) - 4(2n^2+3n+1) = (16n^2 + 4n - 6) - (8n^2 + 12n + 4) = 8n^2 - 8n - 10$ give

$$F'(n) = \frac{(4n+3)(4n-2) - 4(2n^2+3n+1)}{(4n-2)^2} = \frac{2(4n^2 - 4n - 5)}{4(2n-1)^2} = \frac{4n^2 - 4n - 5}{2(2n-1)^2}.$$

The critical points solve $4n^2 - 4n - 5 = 0$, i.e. $n = (4 \pm \sqrt{16 + 80})/8 = (1 \pm \sqrt{6})/2$, so the admissible one is

$$\boxed{n_{\text{opt}} = \frac{1 + \sqrt{6}}{2} = 1.724744871\ldots}.$$

The other root is negative. The numerator $4n^2 - 4n - 5$ is negative between the two roots and positive beyond n_{opt}, so the derivative changes from negative to positive at n_{opt}; and $F(n) \to \infty$ at both ends of $(\frac{1}{2}, \infty)$, so this is the global minimum in the trial family. At n_{opt}, $n + 1 = (3 + \sqrt{6})/2$, $2n + 1 = 2 + \sqrt{6}$ and $2n - 1 = \sqrt{6}$, so substitution yields

$$F_{\min} = \frac{(3 + \sqrt{6})(2 + \sqrt{6})}{4\sqrt{6}} = \frac{12 + 5\sqrt{6}}{4\sqrt{6}} = \frac{5 + 2\sqrt{6}}{4}, \qquad \boxed{\frac{F_{\min}}{\pi^2/4} = \frac{5 + 2\sqrt{6}}{\pi^2} = 1.002976318\ldots}.$$

The energy error is therefore approximately 0.297632%.

The variational domain here is $n > \frac{1}{2}$, not merely $n > 0$: smaller positive n satisfy the endpoint values but have infinite derivative energy. The quadratic-form quotient $\int y'^2 / \int y^2$ remains the appropriate expression throughout this domain; replacing it by $-\int yy'' / \int y^2$ without checking the boundary term is invalid for some of these trials. For example, $n = 1$ ($y = 1 - x$) gives $\int_0^1 y'^2 dx = 1$ but $-\int_0^1 yy'' dx = 0$; the difference is the boundary term $[yy']_0^1 = -y(0)y'(0) = 1$. At the optimum $n > 1$, $y'(0) = 0$, so that boundary term vanishes.

22.4.9 A quantum mechanical particle in a sphere (Example 14.7.1) satisfies

$$\nabla^2 \psi + k^2 \psi = 0,$$

with $k^2 = 2mE/\hbar^2$. The boundary condition is that $\psi = 0$ at $r = a$, where a is the radius of the sphere. For the ground state [where $\psi = \psi(r)$], try an approximate wave function,

$$\psi_a(r) = 1 - \left(\frac{r}{a}\right)^2,$$

and calculate an approximate eigenvalue k_a^2. Take $a > 0$ and impose regularity at the origin, which excludes the singular solutions $\cos(kr)/r$.

Hint. To determine $p(r)$ and $w(r)$, put your equation in self-adjoint form (in spherical polar coordinates).

ANS. $k_a^2 = \dfrac{10.5}{a^2}$, $\quad k_{\text{exact}}^2 = \dfrac{\pi^2}{a^2}$.

Solution.

For a radial function, the differential equation is

$$\frac{1}{r^2}\frac{d}{dr}(r^2\psi') + k^2\psi = 0, \quad \text{or} \quad (r^2\psi')' + k^2 r^2 \psi = 0.$$

Thus the self-adjoint coefficients are $p(r) = r^2$, $w(r) = r^2$, $q(r) = 0$. Multiplication by ψ and integration over $[0, a]$, with $\int_0^a \psi(r^2\psi')'\, dr = [r^2\psi\psi']_0^a - \int_0^a r^2\psi'^2\, dr$ (integration by parts), gives

$$k^2 = \frac{\int_0^a r^2\psi'^2\, dr}{\int_0^a r^2\psi^2\, dr}.$$

The boundary term $[r^2\psi\psi']_0^a$ vanishes at a because $\psi(a) = 0$, and at the origin for the regular radial solution. For a trial function this quotient is the approximate eigenvalue.

For $\psi_{\text{trial}} = 1 - r^2/a^2$, $\psi' = -2r/a^2$ and

$$N = \int_0^a r^2 \frac{4r^2}{a^4}\, dr = \frac{4}{a^4}\int_0^a r^4\, dr = \frac{4a}{5},$$

whereas

$$\begin{aligned} D &= \int_0^a \left(r^2 - \frac{2r^4}{a^2} + \frac{r^6}{a^4}\right) dr \\ &= a^3\left(\frac{1}{3} - \frac{2}{5} + \frac{1}{7}\right) = \frac{8a^3}{105}. \end{aligned}$$

Therefore

$$\boxed{k_{\text{trial}}^2 = \frac{4a/5}{8a^3/105} = \frac{4 \cdot 105}{5 \cdot 8\, a^2} = \frac{21}{2a^2} = \frac{10.5}{a^2}}.$$

For the exact solution put $\psi = u/r$; then $r^2\psi' = ru' - u$ and $(r^2\psi')' = ru''$, so the radial equation becomes $u'' + k^2 u = 0$, with $u = A\sin(kr) + B\cos(kr)$. Regularity at the origin requires $B = 0$, so the exact regular solution is proportional to $\sin(kr)/(kr)$. The condition $\psi(a) = 0$ requires $\sin(ka) = 0$, and the smallest positive zero is $ka = \pi$, so

$$k^2_{\text{exact}} = \frac{\pi^2}{a^2}, \qquad \frac{k^2_{\text{trial}}}{k^2_{\text{exact}}} = \frac{21}{2\pi^2} = 1.063872428\ldots.$$

The resulting energy is approximately 6.38724% above the exact ground-state value, since $E = \hbar^2 k^2/(2m)$.

22.4.10 The wave equation for a quantum mechanical oscillator may be written as

$$\frac{d^2\psi(x)}{dx^2} + (\lambda - x^2)\psi(x) = 0,$$

with $\lambda = 1$ for the ground state; see Eq. (18.17). Take

$$\psi_{\text{trial}} = \begin{cases} 1 - \dfrac{x^2}{a^2}, & x^2 \le a^2 \\ 0, & x^2 > a^2 \end{cases}$$

for the ground-state wave function (with a^2 an adjustable parameter) and calculate the corresponding ground-state energy. How much error do you have? Take $a > 0$ and use the Rayleigh quotient with kinetic term $\int |\psi'|^2 dx$; the trial has derivative jumps at its cutoffs.

Note. Your parabola is really not a very good approximation to a Gaussian exponential. What improvements can you suggest?

Solution.

The dimensionless Hamiltonian is $H = -d^2/dx^2 + x^2$, so its Rayleigh quotient is

$$\lambda[\psi] = \frac{\int_{\mathbb{R}} (\psi'^2 + x^2\psi^2)\, dx}{\int_{\mathbb{R}} \psi^2\, dx}.$$

The stated trial is continuous, but its derivative jumps at $\pm a$. It belongs to the quadratic-form domain, and the quotient above is valid. Treating its second derivative as an ordinary function across those joins would omit delta-function terms. For this trial, however, those terms carry the factor $\psi(\pm a) = 0$ and vanish, so $-\int_{-a}^{a} \psi\psi''\, dx = (2/a^2)\int_{-a}^{a} \psi\, dx = 8/(3a)$ equals the kinetic term N_{kin} found below, and both forms give the same $\lambda(a)$. All three integrals reduce to $[-a, a]$:

$$\begin{aligned} D &= 2\int_0^a \left(1 - \frac{2x^2}{a^2} + \frac{x^4}{a^4}\right) dx = 2a\left(1 - \frac{2}{3} + \frac{1}{5}\right) = \frac{16a}{15}, \\ N_{\text{kin}} &= 2\int_0^a \frac{4x^2}{a^4}\, dx = \frac{8}{3a}, \\ N_{\text{pot}} &= 2\int_0^a \left(x^2 - \frac{2x^4}{a^2} + \frac{x^6}{a^4}\right) dx = 2a^3\left(\frac{1}{3} - \frac{2}{5} + \frac{1}{7}\right) = \frac{16a^3}{105}. \end{aligned}$$

Hence

$$\lambda(a) = \frac{N_{\text{kin}} + N_{\text{pot}}}{D} = \frac{8}{3a}\cdot\frac{15}{16a} + \frac{16a^3}{105}\cdot\frac{15}{16a} = \frac{5}{2a^2} + \frac{a^2}{7}.$$

Put $A = a^2 > 0$, so $\lambda = 5/(2A) + A/7$. Then

$$\frac{d\lambda}{dA} = -\frac{5}{2A^2} + \frac{1}{7} = 0 \quad\Longrightarrow\quad A^2 = \frac{35}{2}, \qquad \boxed{a = \left(\frac{35}{2}\right)^{1/4}}.$$

The second derivative $5/A^3$ is positive. At the minimum the two energy terms are equal, $5/(2A) = A/7$ being the condition $A^2 = 35/2$, so $\lambda_{\min} = 2A/7 = \frac{2}{7}\sqrt{35/2}$:

$$\boxed{\lambda_{\min} = 2\sqrt{\frac{5}{14}} = \sqrt{\frac{10}{7}} = 1.195228609\ldots}.$$

Thus the relative error is

$$\boxed{100(\lambda_{\min} - 1)\% = 19.5228609\ldots\%}.$$

In conventional oscillator units, $E_{\text{trial}} = (\hbar\omega/2)\sqrt{10/7}$. The Instructor's Manual's final value $2\sqrt{10/7}$ has an extra factor of two. Substitution into its own expression $5/(2a^2) + a^2/7$ gives $\sqrt{10/7}$.

One improvement is to make both the value and slope vanish at the cutoff:

$$\psi = (1 - x^2/a^2)^2 \quad (|x| \le a), \qquad \psi = 0 \quad (|x| > a).$$

Writing $t = x/a$, so $dx = a\,dt$ and $\psi' = a^{-1}d(1-t^2)^2/dt = -4t(1-t^2)/a$, the three integrals are $D = a\int_{-1}^{1}(1-t^2)^4 dt$, $N_{\text{kin}} = a^{-1}\int_{-1}^{1} 16t^2(1-t^2)^2 dt$ and $N_{\text{pot}} = a^3\int_{-1}^{1} t^2(1-t^2)^4 dt$. Expanding the even polynomials and integrating term by term, the needed polynomial integrals are

$$\int_{-1}^{1}(1-t^2)^4 dt = \frac{256}{315}, \quad \int_{-1}^{1} 16t^2(1-t^2)^2 dt = \frac{256}{105}, \quad \int_{-1}^{1} t^2(1-t^2)^4 dt = \frac{256}{3465}.$$

For instance, the first is $2\int_0^1 (1 - 4t^2 + 6t^4 - 4t^6 + t^8)\,dt = 2\big(1 - \frac{4}{3} + \frac{6}{5} - \frac{4}{7} + \frac{1}{9}\big) = \frac{256}{315}$. They give

$$\lambda(a) = \frac{(256/105)a^{-1} + (256/3465)a^3}{(256/315)a} = \frac{3}{a^2} + \frac{a^2}{11}.$$

Minimizing in $A = a^2$ as before, $-3/A^2 + 1/11 = 0$, so

$$a^4 = 33, \qquad \lambda_{\min} = \frac{3}{A} + \frac{A}{11} = \frac{2\sqrt{33}}{11} = 2\sqrt{\frac{3}{11}} = 1.044465935\ldots,$$

only 4.44659% high. A still better family is the Gaussian $\psi = e^{-\beta x^2/2}$, $\beta > 0$. Since $\psi^2 = e^{-\beta x^2}$,

$$\langle x^2 \rangle = \frac{\int x^2 e^{-\beta x^2} dx}{\int e^{-\beta x^2} dx} = \frac{\sqrt{\pi}/(2\beta^{3/2})}{\sqrt{\pi/\beta}} = \frac{1}{2\beta},$$

where the numerator follows by differentiating $\int e^{-\beta x^2} dx = \sqrt{\pi/\beta}$ with respect to β, and $\psi' = -\beta x \psi$ gives $\int \psi'^2 dx / \int \psi^2 dx = \beta^2 \langle x^2 \rangle = \beta/2$. Hence

$$\lambda(\beta) = \frac{\beta}{2} + \frac{1}{2\beta} \ge 1,$$

by the arithmetic–geometric mean inequality, with equality at $\beta = 1$, which recovers the exact ground state.

22.4.11 The Schrödinger equation for a central potential may be written as

$$\mathcal{L}u(r) + \frac{\hbar^2 l(l+1)}{2Mr^2}\,u(r) = Eu(r).$$

The $l(l+1)$ term, the angular momentum barrier, comes from splitting off the angular dependence. Compare Eq. (9.80) (divide that equation through by $-r^2$). Use the Rayleigh–Ritz technique to show that $E > E_0$ for $l > 0$ and normalizable bound states, where E_0 is the lowest energy eigenvalue of $\mathcal{L}u_0 = E_0 u_0$ corresponding to $l = 0$. This means that the ground state will have $l = 0$, zero angular momentum. Assume a self-adjoint radial Hamiltonian bounded below, with the usual common quadratic-form domain.

Hint. For a discrete orthonormal eigenbasis, expand $u(r) = \sum_{i=0}^{\infty} c_i u_i(r)$, where $\mathcal{L}u_i = E_i u_i$ and $E_i \geq E_0$. Do not require $c_0 = 1$; a spectral integral handles any continuous part of the spectrum.

Solution.

The strict inequality is for $l > 0$ and a normalizable bound eigenstate, with E_0 the lowest eigenvalue of the $l = 0$ radial operator. For $l = 0$ the ground state has equality. Normalizability matters: if the $l = 0$ operator has no bound state, its spectrum begins at the continuum threshold (0 for a potential vanishing at infinity), and so does that for $l > 0$, because the centrifugal term only raises energies and also vanishes at infinity; the two spectra then have the same bottom, so no strict inequality holds. Normalize u by $\int_0^\infty |u|^2 dr = 1$ and assume it belongs to the quadratic-form domain of the radial Hamiltonian, including the centrifugal term. For form-domain functions, $\langle u, \mathcal{L}u\rangle$ below denotes the associated quadratic form; it agrees with the operator inner product when u is in the operator domain. The variational lower bound for $\mathcal{L}$ is

$$\langle u, \mathcal{L}u\rangle \geq E_0 \langle u, u\rangle = E_0.$$

For the discrete expansion in the hint, this follows directly from $u = \sum_{i\geq 0} c_i u_i$ in an orthonormal eigenbasis: by Parseval's relation and $\mathcal{L}u_i = E_i u_i$, $\langle u_i, u_j\rangle = \delta_{ij}$,

$$\sum_i |c_i|^2 = 1, \qquad \langle u, \mathcal{L}u\rangle = \sum_i |c_i|^2 E_i = E_0 + \sum_{i\geq 1} |c_i|^2 (E_i - E_0) \geq E_0,$$

where $E_0 = E_0 \sum_i |c_i|^2$ has been added and subtracted and each $E_i - E_0 \geq 0$. There is no need to assume $c_0 \neq 0$ or to divide by it. If there is continuous spectrum, the sum is replaced by the spectral integral $\langle u, \mathcal{L}u\rangle = \int E\, d\mu_u(E)$, in which the nonnegative spectral weight $d\mu_u$ of u lives on the spectrum of $\mathcal{L}$ and has total weight $\int d\mu_u = 1$. This integral is $\geq E_0 \int d\mu_u = E_0$ provided no part of the spectrum lies below E_0, i.e. E_0 is the bottom of the spectrum, as it is for a bound ground state below the continuum threshold.

Now let u be, in addition, a bound eigenfunction of the radial equation with $l > 0$ and energy E. Multiply the radial equation by u^*, integrate over $0 < r < \infty$ and use $\int_0^\infty |u|^2 dr = 1$:

$$E = \langle u, \mathcal{L}u\rangle + \frac{\hbar^2 l(l+1)}{2M} \int_0^\infty \frac{|u(r)|^2}{r^2}\, dr.$$

For a nonzero normalizable state the last integral is strictly positive: its integrand is nonnegative, and $r^{-2} > 0$ at every finite $r > 0$, so it could vanish only if $u = 0$ almost everywhere. For $l > 0$ its coefficient is also positive. With $\langle u, \mathcal{L}u\rangle \geq E_0$, thus

$$\boxed{E > E_0 \qquad (l > 0)}.$$

Every bound state with nonzero angular momentum lies above the $l = 0$ ground state. Consequently, if a bound ground state exists for the central-potential problem, it has $l = 0$.

23 Chapter 23. Probability and Statistics

Problems

Conventions for the card and dice exercises: use a standard 52-card deck without jokers and uniform draws, reshuffling after a replacement. Dice rolls and coin tosses are independent, and a fair die has six equally likely faces.

23.1 Probability: Definitions, Simple Properties

23.1.1 A card is drawn from a shuffled deck. (a) What is the probability that it is black, (b) a red nine, (c) or a queen of spades?

Solution.

Use the standard 52-card deck, with four suits of 13 cards and no jokers. Every card has probability $1/52$. There are 26 black cards, two red nines (hearts and diamonds), and one queen of spades. Thus

$$\boxed{\text{(a) } P(\text{black}) = \frac{26}{52} = \frac{1}{2}, \quad \text{(b) } P(\text{red nine}) = \frac{2}{52} = \frac{1}{26}, \quad \text{(c) } P(Q\spadesuit) = \frac{1}{52}}.$$

These are three separate questions; their events need not be disjoint.

23.1.2 Find the probability of drawing two kings from a shuffled deck of cards (a) if the first card is put back before the second is drawn, and (b) if the first card is not put back after being drawn.

Solution.

For a standard 52-card deck there are four kings. Use $P(K_1 \cap K_2) = P(K_1)P(K_2 \mid K_1)$. (a) Replacement restores all 52 cards and all four kings, so

$$\boxed{P = \frac{4}{52}\frac{4}{52} = \frac{1}{169}}.$$

(b) Without replacement a first king leaves three kings among 51 cards:

$$\boxed{P = \frac{4}{52}\frac{3}{51} = \frac{1}{221}}.$$

Equivalently, the unordered count is $\binom{4}{2}/\binom{52}{2} = 6/1326 = 1/221$.

23.1.3 When two fair dice are thrown, what is the probability that their sum is (a) less than 4, or (b) greater than or equal to 4 but less than 6?

Solution.

Interpret the textbook's "observed number" as the *sum* $S = D_1 + D_2$, as in Example 23.2.1; this is the reading marked in the statement. The 36 ordered pairs (D_1, D_2) are equiprobable. (a) $S < 4$ consists of $(1,1), (1,2), (2,1)$, hence

$$\boxed{P(S < 4) = 3/36 = 1/12}.$$

(b) $4 \le S < 6$ means $S = 4$ or 5. Sum 4 has the three pairs $(1,3), (2,2), (3,1)$; sum 5 has the four pairs $(1,4), (2,3), (3,2), (4,1)$. Therefore

$$\boxed{P(4 \le S < 6) = (3+4)/36 = 7/36}.$$

23.1.4 Rolling three fair dice, what is the probability of obtaining six points?

Solution.

There are $6^3 = 216$ equiprobable ordered triples. Sum 6 has the unordered patterns $(1,1,4)$, $(1,2,3)$ and $(2,2,2)$, with respectively $3!/2! = 3$, $3! = 6$ and 1 distinct orderings. Consequently

$$\boxed{P(D_1 + D_2 + D_3 = 6) = \frac{3+6+1}{216} = \frac{5}{108}}.$$

Alternatively, subtract 1 from each die; the nonnegative triples summing to 3 number $\binom{3+2}{2} = 10$ (the separator count of Exercise 23.1.6 with $N = k = 3$), and none violates the upper bound 5.

23.1.5 Determine the probability $P(A \cap B \cap C)$ in terms of $P(A)$, $P(B)$, $P(C)$, $P(A \cup B)$, $P(A \cup C)$, $P(B \cup C)$, and $P(A \cup B \cup C)$.

Solution.

For two events, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$, and likewise for the pairs A, C and B, C. Adding the three identities,

$$\begin{aligned} P(A \cap B) + P(A \cap C) + P(B \cap C) = {} & 2[P(A) + P(B) + P(C)] \\ & - P(A \cup B) - P(A \cup C) - P(B \cup C). \end{aligned}$$

Substitute this sum into the three-event inclusion–exclusion formula of Exercise 23.1.7:

$$\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) \\ & - [2P(A) + 2P(B) + 2P(C) \\ & \quad - P(A \cup B) - P(A \cup C) - P(B \cup C)] + P(A \cap B \cap C). \end{aligned}$$

Solving for $P(A \cap B \cap C)$ gives $P(A \cup B \cup C) - [P(A) + P(B) + P(C)]$ plus the bracketed expression; the single-event terms combine as $-P(A) + 2P(A) = P(A)$, and likewise for B and C, so

$$\boxed{\begin{aligned} P(A \cap B \cap C) = {} & P(A \cup B \cup C) + P(A) + P(B) + P(C) \\ & - P(A \cup B) - P(A \cup C) - P(B \cup C) \end{aligned}}.$$

The Instructor's Manual's conditional-probability product is valid when its conditional probabilities are defined, but does not express the answer in terms of the seven probabilities requested here.

23.1.6 Determine directly or by mathematical induction (Section 1.4) the probability of a distribution of N (Maxwell–Boltzmann) particles in k boxes with N_1 in Box 1, N_2 in Box 2, ..., N_k in the kth box for any numbers $N_j \geq 1$ with $N_1 + N_2 + \cdots + N_k = N$, $k < N$. Repeat this for Fermi–Dirac and Bose–Einstein particles. Treat each box as one single-particle state. Use independent uniform box choices for Maxwell–Boltzmann particles and uniform allowed occupation states for Bose–Einstein particles; compare with conditioning on all boxes being nonempty. For Fermi–Dirac particles, explain why these parameters do not define a normalized ensemble.

Solution.

A probability requires a specified ensemble and its weights. For this exercise, use independent uniform box choices for classical particles, and uniformly weighted allowed occupation states for identical particles. The Instructor's Manual lists incorrect counts instead of normalized probabilities. With $N > k$, the specified Fermi ensemble has no allowed state at all.

For Maxwell–Boltzmann particles label the particles $1, \dots, N$. Each particle can go into any of the k boxes, so there are k^N assignments. To obtain the stated occupation vector, choose the N_1 labels in Box 1, then N_2 of the remaining $N - N_1$ labels for Box 2, and so on:

$$\binom{N}{N_1}\binom{N-N_1}{N_2}\cdots\binom{N-N_1-\cdots-N_{k-1}}{N_k} = \frac{N!}{N_1!\cdots N_k!}.$$

Written with factorials, the factor $(N - N_1)!$ in the first denominator cancels the numerator of the second binomial, and so on; the last denominator contains $(N - N_1 - \cdots - N_k)! = 0! = 1$. Every assignment has probability k^{-N}, so

$$\boxed{P_{\mathrm{MB}}(N_1, \dots, N_k) = \frac{N!}{k^N \prod_j N_j!}}.$$

This is the probability of that positive occupation vector $\boldsymbol{N} = (N_1, \dots, N_k)$ in the *unrestricted* ensemble, which also permits empty boxes. If one conditions on all boxes being nonempty, the denominator k^N must instead be the number of onto assignments,

$$S_{N,k} = \sum_{r=0}^{k}(-1)^r\binom{k}{r}(k-r)^N, \qquad P_{\mathrm{MB}}(\boldsymbol{N} \mid \text{nonempty}) = \frac{N!}{S_{N,k}\prod_j N_j!}.$$

Here $(k - r)^N$ assignments leave a given set of r boxes empty, since each particle must then choose among the other $k - r$ boxes. Let E_i be the event that Box i is empty. As in Exercise 23.1.7, multiplying out a product of indicators gives the general inclusion–exclusion identity

$$1_{\text{nonempty}} = \prod_{i=1}^{k}(1 - 1_{E_i}) = \sum_{r=0}^{k}(-1)^r \sum_{|R|=r} 1_{\bigcap_{i\in R} E_i},$$

where R runs over the $\binom{k}{r}$ sets of r boxes (the $r = 0$ term is 1). Each intersection $\bigcap_{i\in R} E_i$ contains $(k - r)^N$ assignments, so summing over all k^N assignments gives $S_{N,k}$. The conditional probability is $P_{\mathrm{MB}}(\boldsymbol{N})/P(\text{nonempty})$ with $P(\text{nonempty}) = S_{N,k}/k^N$.

For Bose–Einstein particles each occupation vector is one state. Write an occupation vector as a row of N indistinguishable particles and $k - 1$ separators between consecutive boxes. Choosing which $k - 1$ of the $N + k - 1$ positions hold separators shows that the number of nonnegative occupation vectors is $\binom{N+k-1}{k-1}$. Thus, under the stated uniform-state convention,

$$\boxed{P_{\mathrm{BE}}(\boldsymbol{N}) = \binom{N+k-1}{k-1}^{-1}}.$$

If nonempty boxes are an imposed condition, first put one particle in each box and distribute the remaining $N - k$ freely; the same count with $N \to N - k$ gives $\binom{N-k+k-1}{k-1} = \binom{N-1}{k-1}$ states, and the probability is its reciprocal instead.

For Fermi–Dirac particles a single box can contain only 0 or 1 particle. For $0 \le N \le k$ there are $\binom{k}{N}$ states, each with probability $\binom{k}{N}^{-1}$ under uniform weighting. Here $N > k$, so at least one $N_j > 1$ (pigeonhole principle): its multiplicity is zero and the entire allowed sample space is empty. There is *no normalized FD probability distribution* for these parameters. If positivity is retained but $N = k$ is allowed, the sole state is $(1, \dots, 1)$ with probability 1. Physical energy-dependent ensembles would require additional weights.

23.1.7 Show that $P(A\cup B\cup C)=P(A)+P(B)+P(C)-P(A\cap B)-P(A\cap C)-P(B\cap C)+P(A\cap B\cap C)$.

Solution.

For any outcome, put $a=1_A$, $b=1_B$, $c=1_C$, where the indicator 1_E is 1 if the outcome lies in E and 0 otherwise. The product $(1-a)(1-b)(1-c)$ is 1 exactly when the outcome lies in none of the three events, so the indicator of membership in at least one event is

$$1_{A\cup B\cup C}=1-(1-a)(1-b)(1-c)=a+b+c-ab-ac-bc+abc,$$

where $(1-a)(1-b)(1-c)=1-a-b-c+ab+ac+bc-abc$ was multiplied out. Take expectations and use $\langle 1_E\rangle=1\cdot P(E)+0\cdot[1-P(E)]=P(E)$, together with $1_A1_B=1_{A\cap B}$ (a product of indicators is 1 only when every factor is 1) and likewise $1_A1_B1_C=1_{A\cap B\cap C}$. This gives

$$\boxed{\begin{aligned}P(A\cup B\cup C)=P(A)+P(B)+P(C)-P(A\cap B)\\ -P(A\cap C)-P(B\cap C)+P(A\cap B\cap C)\end{aligned}}.$$

No independence or disjointness assumption is needed.

23.1.8 Determine the probability that a positive integer $n\le 100$, chosen uniformly, is divisible by a fixed prime number $p\le 100$. Verify your result for $p=3,5,7$.

Solution.

Choose n uniformly from $1,\dots,100$ and fix the prime p. Its multiples in this set are $p,2p,\dots,\lfloor 100/p\rfloor p$; therefore

$$\boxed{P(p\mid n)=\frac{\lfloor 100/p\rfloor}{100}}.$$

For $p=3,5,7$ the lists of multiples end at $99=3\cdot 33$, $100=5\cdot 20$ and $98=7\cdot 14$ (the next multiples, 102, 105 and 105, exceed 100), so the answers are respectively $33/100$, $20/100$, and $14/100$. Since $100/p-1<\lfloor 100/p\rfloor\le 100/p$, the approximation $1/p$ has error

$$0\le\frac{1}{p}-P(p\mid n)<\frac{1}{100}.$$

Here the notation $p\mid n$ means divisibility, not conditional probability. If "a prime" meant *some* prime rather than a fixed p, every $n=2,\dots,100$ qualifies (it has a prime factor $p\le n\le 100$), and the different question has answer $99/100$.

23.1.9 Put two particles obeying Maxwell–Boltzmann (Fermi–Dirac, or Bose–Einstein) statistics in three boxes. How many ways of doing so are there in each case? Each box represents a single-particle state; use the counting conventions of Exercise 23.1.6, now with $N=2$, $k=3$.

Solution.

For Maxwell–Boltzmann statistics label the particles a,b. Each chooses any of three boxes, so the ordered box pair has $3^2=9$ possibilities. For Fermi–Dirac statistics choose two distinct boxes for the two indistinguishable particles: $\binom{3}{2}=3$ states,

$$(1,1,0),\ (1,0,1),\ (0,1,1).$$

For Bose–Einstein statistics also allow the three doubly occupied states $(2,0,0),(0,2,0),(0,0,2)$. Thus

$$\boxed{W_{\mathrm{MB}}=9,\qquad W_{\mathrm{FD}}=3,\qquad W_{\mathrm{BE}}=6}.$$

The BE count also equals $\binom{2+3-1}{2}=6$.

23.2 Random Variables

23.2.1 Verify the probabilities for the outcomes of the two-card draws in Example 23.2.4, and by direct computation of the mean and variance check the results given in that example.

Solution.

Example 23.2.4 draws two *distinct* cards from the four cards numbered 1 through 4, without replacement. The six equiprobable unordered draws and their sums are

draw	$\{1,2\}$	$\{1,3\}$	$\{1,4\}$	$\{2,3\}$	$\{2,4\}$	$\{3,4\}$
X	3	4	5	5	6	7

Each draw has probability $1/6$, hence $P(3) = P(4) = P(6) = P(7) = 1/6$ and $P(5) = 2/6 = 1/3$. Direct summation gives

$$\langle X\rangle = \frac{3+4+5+5+6+7}{6} = 5, \qquad \langle X^2\rangle = \frac{9+16+25+25+36+49}{6} = \frac{80}{3}.$$

Either subtract $\langle X\rangle^2 = 25$, which gives $80/3 - 25 = 5/3$, or evaluate the centered squares $(X-5)^2$ directly:

$$\boxed{\operatorname{Var} X = \frac{(-2)^2 + (-1)^2 + 0^2 + 0^2 + 1^2 + 2^2}{6} = \frac{5}{3}}.$$

These probabilities and the values $\langle X\rangle = 5$, $\sigma^2 = 5/3$ agree with those given in Example 23.2.4, which the direct computation thus confirms.

23.2.2 Show that adding a constant c to a random variable X changes the expectation value $\langle X\rangle$ by that same constant but not the variance. Show also that multiplying a random variable by a real constant c multiplies the mean by c and the variance by c^2. Show that the random variable $X - \langle X\rangle$ has mean value zero. Assume X is real with a finite second moment.

Solution.

Multiplication by a real constant c multiplies the mean by c, the variance by c^2, and the standard deviation by $|c|$. The textbook's claim of a factor c for the variance is false. Write $\mu = \langle X\rangle$ and assume a finite second moment. Linearity of expectation and $\langle 1\rangle = 1$ give

$$\langle X + c\rangle = \mu + c, \qquad \operatorname{Var}(X+c) = \langle (X + c - \mu - c)^2\rangle = \langle (X-\mu)^2\rangle = \operatorname{Var} X.$$

Likewise

$$\langle cX\rangle = c\mu, \qquad \operatorname{Var}(cX) = \langle (cX - c\mu)^2\rangle = c^2\langle (X-\mu)^2\rangle = c^2 \operatorname{Var} X.$$

These formulas also include $c = 0$ and negative c. Finally $\boxed{\langle X - \langle X\rangle\rangle = \mu - \mu = 0}$. For a quick contradiction to the printed variance rule, $c = -1$ leaves every squared deviation unchanged, rather than making a positive variance negative.

23.2.3 Using the definition given in Eq. (23.27) for the variance σ^2 of a continuous random variable, show that

$$\sigma^2 = \langle X^2\rangle - \langle X\rangle^2.$$

Assume a finite second moment.

Solution.

Let $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x)\,dx = 1$, and suppose $\int x^2 f(x)\,dx < \infty$. Writing $\mu = \langle X \rangle$, expand before integrating:

$$\begin{aligned}\sigma^2 &= \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx \\ &= \int x^2 f(x)\,dx - 2\mu \int x f(x)\,dx + \mu^2 \int f(x)\,dx \\ &= \langle X^2 \rangle - 2\mu^2 + \mu^2 = \boxed{\langle X^2 \rangle - \langle X \rangle^2}.\end{aligned}$$

All undisplayed limits in the second line are $-\infty, \infty$.

23.2.4 A velocity $v_j = x_j/t_j$ is measured by recording the distances x_j at the corresponding times t_j. Show that $\bar{x}/\bar{t}$ is a good approximation for the average velocity v, provided that all the errors are small: $|x_j - \bar{x}| \ll |\bar{x}|$ and $|t_j - \bar{t}| \ll |\bar{t}|$. Here, for N measurements, the average velocity is taken to be the arithmetic mean $v = \bar{v} = N^{-1} \sum_{j=1}^{N} x_j/t_j$ of the individual velocities, and $\bar{x}, \bar{t} \neq 0$.

Solution.

Let there be N measurements, $\bar{v} = N^{-1} \sum_j x_j/t_j$, and write $x_j = \bar{x}(1+a_j)$, $t_j = \bar{t}(1+b_j)$, i.e., $a_j = (x_j - \bar{x})/\bar{x}$ and $b_j = (t_j - \bar{t})/\bar{t}$. The stated relative-error hypotheses require $\bar{x} \neq 0$, $\bar{t} \neq 0$; also $\sum a_j = \sum b_j = 0$, because $\sum_j x_j = N\bar{x}$ and $\sum_j t_j = N\bar{t}$. An exact algebraic identity is

$$\frac{1+a_j}{1+b_j} = 1 + a_j - b_j + \frac{b_j^2 - a_j b_j}{1+b_j};$$

to check it, multiply by $1+b_j$: the right side becomes $(1+a_j-b_j)(1+b_j) + b_j^2 - a_j b_j = 1 + a_j$. Since $v_j = (\bar{x}/\bar{t})(1+a_j)/(1+b_j)$, multiply the identity by $\bar{x}/\bar{t}$ and average over j. The two linear terms average to zero, leaving

$$\boxed{\bar{v} = \frac{\bar{x}}{\bar{t}} \left[1 + \frac{1}{N} \sum_j \frac{b_j^2 - a_j b_j}{1+b_j} \right]}.$$

If $|a_j| \leq \epsilon_x$ and $|b_j| \leq \epsilon_t < 1$, then each numerator obeys $|b_j^2 - a_j b_j| \leq |b_j|(|b_j| + |a_j|) \leq \epsilon_t(\epsilon_t + \epsilon_x)$ and each denominator $|1+b_j| \geq 1 - \epsilon_t$, so

$$\left| \frac{\bar{v}}{\bar{x}/\bar{t}} - 1 \right| \leq \frac{\epsilon_t^2 + \epsilon_x \epsilon_t}{1 - \epsilon_t}.$$

Thus the relative difference is second order in the small relative errors. For its leading term replace $1/(1+b_j)$ by 1; this changes each summand by $(b_j^2 - a_j b_j) b_j/(1+b_j) = O(\epsilon^3)$, where $\epsilon = \max(\epsilon_x, \epsilon_t)$. Then insert $b_j^2 = (t_j - \bar{t})^2/\bar{t}^2$ and $a_j b_j = (x_j - \bar{x})(t_j - \bar{t})/(\bar{x}\bar{t})$; with overbars denoting averages over j, the dimensional form is

$$\bar{v} = \frac{\bar{x}}{\bar{t}} + \frac{\bar{x}}{\bar{t}^3}\overline{(t-\bar{t})^2} - \frac{1}{\bar{t}^2}\overline{(x-\bar{x})(t-\bar{t})} + O(\epsilon^3)\frac{\bar{x}}{\bar{t}}.$$

No statistical independence of the distance and time errors was used.

23.2.5 Redefine the random variable Y in Example 23.2.5 as the number of fours through nines. Then determine the correlation of the X and Y random variables for the drawing of two cards (with replacement, as in the example).

Solution.

As in Example 23.2.5, X is the number of honors (10, J, Q, K, A) among the two cards, while Y now counts the fours through nines. In one draw, the disjoint categories are honors, fours through nines, and twos or threes, comprising 5, 6 and 2 of the 13 ranks. Their probabilities are $p = 5/13$, $q = 6/13$, $r = 2/13$. For two independent draws,

$$P(X = x, Y = y) = \frac{2!}{x!y!(2-x-y)!} p^x q^y r^{2-x-y}, \quad x, y \geq 0,\ x + y \leq 2.$$

The multinomial coefficient counts the orders of the two draws; for example $169P(1,1) = 2 \cdot 5 \cdot 6 = 60$ and $169P(0,0) = 2^2 = 4$. The complete joint table is

$169P(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	4	24	36
$x=1$	20	60	0
$x=2$	25	0	0

It sums to 169. Directly from its rows and columns,

$$\begin{aligned}
\langle X \rangle &= (20 + 60 + 2 \cdot 25)/169 = 10/13, \\
\langle X^2 \rangle &= (20 + 60 + 4 \cdot 25)/169 = 180/169, \\
\langle Y \rangle &= (24 + 60 + 2 \cdot 36)/169 = 12/13, \\
\langle Y^2 \rangle &= (24 + 60 + 4 \cdot 36)/169 = 228/169.
\end{aligned}$$

Only $(x, y) = (1, 1)$ contributes to XY, so $\langle XY \rangle = 60/169$. Consequently, from $\operatorname{Var} X = \langle X^2 \rangle - \langle X \rangle^2$ and $\operatorname{cov}(X, Y) = \langle XY \rangle - \langle X \rangle \langle Y \rangle$,

$$\operatorname{Var} X = \frac{180 - 100}{169} = \frac{80}{169}, \quad \operatorname{Var} Y = \frac{228 - 144}{169} = \frac{84}{169}, \quad \operatorname{cov}(X, Y) = \frac{60 - 120}{169} = -\frac{60}{169}.$$

In $\rho_{XY} = \operatorname{cov}(X, Y)/(\sigma_X \sigma_Y)$ the factors $1/169$ cancel, and $60^2/(80 \cdot 84) = 3600/6720 = 15/28$, so

$$\boxed{\rho_{XY} = \frac{-60}{\sqrt{80 \cdot 84}} = -\sqrt{\frac{15}{28}} \simeq -0.731925}.$$

The draws are independent, but the two category counts are negatively correlated because they compete for the same two positions.

23.2.6 The probability that a particle of an ideal gas travels a distance x between collisions is proportional to $e^{-x/f} dx$, where f is the constant mean free path. Verify that f is the average distance between collisions, and determine the probability of a free path of length $l \geq 3f$. Take $f > 0$ and distances $x \geq 0$.

Solution.

Distances satisfy $x \geq 0$ and $f > 0$. Normalize $p(x) = Ce^{-x/f}$, using $\int_0^\infty e^{-x/f}\, dx = [-fe^{-x/f}]_0^\infty = f$:

$$1 = C \int_0^\infty e^{-x/f}\, dx = Cf, \qquad p(x) = f^{-1} e^{-x/f}.$$

Substitute $x = fu$, $dx = f\,du$ (the limits $0, \infty$ are unchanged), so that $xp(x)\,dx = fue^{-u}\,du$. Integration by parts, with $ue^{-u} \to 0$ as $u \to \infty$, gives

$$\begin{aligned}\langle X\rangle &= \int_0^\infty xp(x)\,dx = f\int_0^\infty ue^{-u}\,du \\ &= f\left([-ue^{-u}]_0^\infty + \int_0^\infty e^{-u}\,du\right) = f(0+1) = \boxed{f}.\end{aligned}$$

The Instructor's Manual drops the factor f in its middle integral after rescaling. The dimensionless integral is $\int_0^\infty ue^{-u}du = 1$; the physical mean is f times that integral. Since the endpoint $x = 3f$ itself has probability zero, $\geq$ and $>$ give the same requested tail:

$$\boxed{P(X \geq 3f) = \int_{3f}^\infty \frac{e^{-x/f}}{f}\,dx = \left[-e^{-x/f}\right]_{3f}^\infty = e^{-3} \simeq 0.0497871}.$$

23.2.7 Determine the probability density for a particle in simple harmonic motion in the interval $-A \leq x \leq A$. For $A > 0$, observe the particle at a uniformly random time in a full period. State the degenerate result when $A = 0$.

Hint. The probability that the particle is between x and $x + dx$ is proportional to the time it takes to travel across the interval.

Solution.

Assume observation at a uniformly random time in a full period, with $A > 0$ and $\omega > 0$. For $x = A\cos\omega t$ (a phase constant does not matter), $dx/dt = -A\omega\sin\omega t$ and $A^2\sin^2\omega t = A^2 - x^2$, so

$$\left|\frac{dx}{dt}\right| = \omega\sqrt{A^2 - x^2}, \qquad T = \frac{2\pi}{\omega}.$$

Each interior interval $[x, x + dx]$ is crossed twice per period, once in each direction, each time in $dt = dx/|dx/dt|$. Therefore its fraction of the total time is, with $2/(T\omega) = 1/\pi$,

$$p(x)\,dx = \frac{2}{T}\frac{dx}{\omega\sqrt{A^2 - x^2}}, \qquad \boxed{p(x) = \frac{1}{\pi\sqrt{A^2 - x^2}} \quad (|x| < A)},$$

and $p = 0$ outside. The endpoint divergences are integrable:

$$\int_{-A}^{A} p(x)\,dx = \frac{1}{\pi}[\arcsin(x/A)]_{-A}^{A} = \frac{1}{\pi}\left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right] = 1.$$

There are no point masses at the turning points. For $A = 0$ the distribution degenerates to a unit point mass at $x = 0$.

23.3 Binomial Distribution

23.3.1 Show that the variable $X = x$, defined as the number of heads in n fair coin tosses, is a random variable and determine its probability distribution. Describe the sample space. What are its mean value, the variance, and the standard deviation? Plot the probability function $P(x) = [n!/x!(n-x)!]2^{-n}$ for $n = 10, 20$, and 30 using graphics software.

Solution.

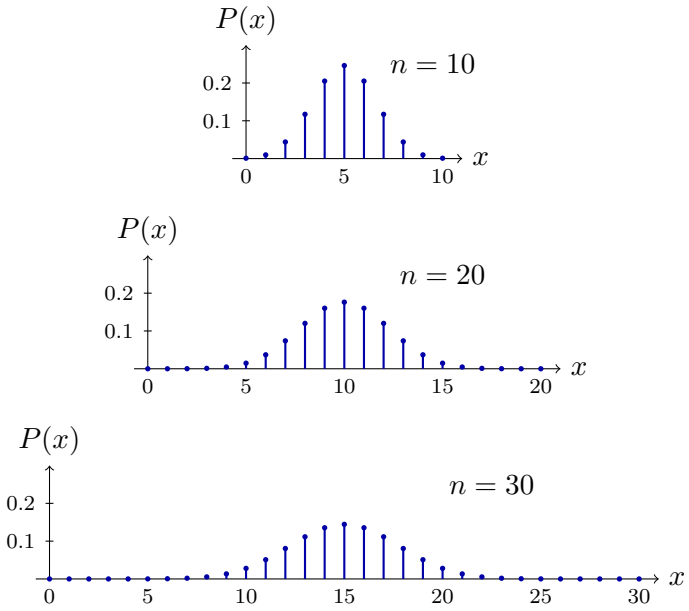
Use independent fair tosses. The elementary sample space is $\Omega = \{H, T\}^n$, with each ordered string having probability 2^{-n}. The map $X(\omega) = \sum_{j=1}^{n} 1_{\{\omega_j = H\}}$ is an integer-valued random variable on this finite space, taking values $0, \ldots, n$. Choosing the x head positions gives exactly $\binom{n}{x}$ strings, hence

$$\boxed{P(X = x) = \binom{n}{x} 2^{-n}, \quad 0 \le x \le n}.$$

The sum of these probabilities is $2^{-n}(1+1)^n = 1$. For each indicator $I_j = 1_{\{\omega_j = H\}}$, $\langle I_j \rangle = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{2}$ and, since $I_j^2 = I_j$, $\operatorname{Var} I_j = \langle I_j^2 \rangle - \langle I_j \rangle^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. Linearity gives $\langle X \rangle = \sum_j \langle I_j \rangle$, and independence makes the cross covariances $\operatorname{cov}(I_i, I_j)$, $i \neq j$, zero, so $\operatorname{Var} X = \sum_j \operatorname{Var} I_j$. Hence

$$\boxed{\langle X \rangle = n/2, \qquad \operatorname{Var} X = n/4, \qquad \sigma_X = \sqrt{n}/2}.$$

The requested plots below show the probability *mass* at each integer, using exactly the displayed finite formula. The horizontal axis has the same scale in all three plots.



23.3.2 Plot the binomial probability function for the probabilities $p = 1/6$, $q = 5/6$, and $n = 6$ throws of a die.

Solution.

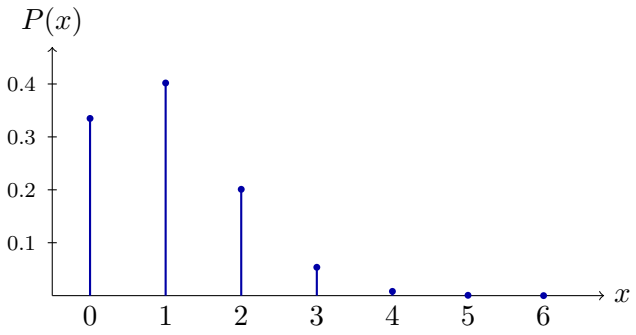
Define a success as a specified face, say a six, on each independent fair-die throw. The number X of successes in six throws has

$$P(X = x) = \binom{6}{x} (1/6)^x (5/6)^{6-x} = \frac{\binom{6}{x} 5^{6-x}}{46656}, \quad x = 0, \ldots, 6.$$

All values, and thus the plot without interpolation, are

x	0	1	2	3	4	5	6
$46656P(x)$	15625	18750	9375	2500	375	30	1

The numerator sum is 46656. The mean is $np = 6 \cdot \frac{1}{6} = 1$, and the largest mass is at $x = 1$, the mode $\lfloor (n+1)p \rfloor = \lfloor 7/6 \rfloor = 1$ of Exercise 23.3.5.



23.3.3 A hardware company knows that the probability of mass-producing nails includes a small probability $p = 0.03$ of defective nails (usually without a sharp tip). What is the probability of finding more than two defective nails in its commercial box of 100 nails? Assume independent defects with the same probability for each nail.

Solution.

Assume nail defects are independent, with common probability 0.03. Then X, the number of defective nails, is binomial with $n = 100$. Use the complement of $X = 0, 1, 2$, with $\binom{100}{2} = 100 \cdot 99/2 = 4950$:

$$\begin{aligned} P(X > 2) &= 1 - \sum_{j=0}^{2} \binom{100}{j} (0.03)^j (0.97)^{100-j} \\ &= 1 - (0.97)^{100} - 100(0.03)(0.97)^{99} - 4950(0.03)^2(0.97)^{98} \\ &= 1 - 0.0475525 - 0.1470696 - 0.2251530 \\ &= \boxed{0.5802249170 \simeq 58.02\%}. \end{aligned}$$

The three subtracted terms are $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, rounded to seven decimals; the boxed value uses their unrounded sum 0.4197750830. The independence assumption matters: a defect rate alone would not determine the box probability if production defects were correlated.

23.3.4 Four cards are drawn from a shuffled bridge deck. What is the probability that they are all red? That they are all hearts? That they are honors? Compare the probabilities when each card is put back at a random place before drawing the next card, with the probabilities when the cards are not replaced in the deck.

Solution.

There are 26 red cards, 13 hearts and 20 honors (10, J, Q, K, A in each suit). If a category contains K cards, replacement gives four independent successes with probability $(K/52)^4$. Without replacement the conditional success probabilities successively decrease:

$$P_K = \frac{K}{52}\frac{K-1}{51}\frac{K-2}{50}\frac{K-3}{49} = \frac{K(K-1)(K-2)(K-3)/4!}{52 \cdot 51 \cdot 50 \cdot 49/4!} = \frac{\binom{K}{4}}{\binom{52}{4}}.$$

With $\binom{52}{4} = 270725$, $\binom{26}{4} = 14950$, $\binom{13}{4} = 715$ and $\binom{20}{4} = 4845$, the results are

category	with replacement	without replacement
red	$(1/2)^4 = 1/16 = 0.0625$	$14950/270725 = 46/833 \approx 0.0552$
hearts	$(1/4)^4 = 1/256 \approx 0.00391$	$715/270725 = 11/4165 \approx 0.00264$
honors	$(5/13)^4 = 625/28561 \approx 0.0219$	$4845/270725 = 57/3185 \approx 0.0179$

For $0 < K < 52$ and $j = 1, 2, 3$, $(K-j)/(52-j) < K/52$, since $52(K-j) < K(52-j)$ reduces to $jK < 52j$, i.e., to $K < 52$. Thus every factor of P_K after the first is smaller than the corresponding factor $K/52$ with replacement, and all three probabilities are strictly smaller without replacement.

23.3.5 Show that for the binomial distribution of Eq. (23.42), the mean value of x is np; find the exact integer mode or modes.

Solution.

The textbook asks to show that the most probable value of x is np. The mean is np, but the exact most probable integer need not be np. The Instructor's Manual differentiates the moment-generating function, which computes the mean and does not prove the printed mode claim. To establish the mean, write $X = I_1 + \cdots + I_n$, where each success indicator has $P(I_j = 1) = p$ and $P(I_j = 0) = 1 - p$. Then

$$\langle I_j \rangle = 1 \cdot p + 0 \cdot (1-p) = p, \qquad \boxed{\langle X \rangle = \sum_{j=1}^{n} \langle I_j \rangle = np}.$$

To find the mode for $0 < p < 1$, write $q = 1 - p$ and compare consecutive positive masses $P(x) = \binom{n}{x} p^x q^{n-x}$, $x = 0, \ldots, n-1$. Since $(x+1)! = (x+1)\,x!$ and $(n-x)! = (n-x)\,(n-x-1)!$,

$$\frac{P(x+1)}{P(x)} = \frac{\binom{n}{x+1}}{\binom{n}{x}} \frac{p}{q} = \frac{x!\,(n-x)!}{(x+1)!\,(n-x-1)!} \frac{p}{q} = \frac{n-x}{x+1} \frac{p}{q}.$$

It exceeds 1 exactly when

$$(n-x)p > (x+1)(1-p) \quad \Longleftrightarrow \quad np - xp > x + 1 - xp - p \quad \Longleftrightarrow \quad x + 1 < (n+1)p,$$

and it equals 1 exactly when $x + 1 = (n+1)p$. If $(n+1)p$ is not an integer, put $m = \lfloor (n+1)p \rfloor$: then $P(x+1) > P(x)$ for $x + 1 \le m$ and $P(x+1) < P(x)$ for $x + 1 \ge m + 1$, so the sequence increases up to $x = m$ and decreases thereafter. If $(n+1)p = k$ is an integer, then $1 \le k \le n$ because $0 < (n+1)p < n + 1$; the sequence increases up to $x = k - 1$, $P(k) = P(k-1)$, and it decreases after $x = k$, so these are the two maxima. Hence

$$\boxed{\begin{cases} \text{unique mode } \lfloor (n+1)p \rfloor, & (n+1)p \notin \mathbb{Z}, \\ \text{modes } k-1, k, & (n+1)p = k \in \{1, \ldots, n\} \end{cases}}.$$

In both cases every mode m obeys $(n+1)p - 1 \le m \le (n+1)p$, i.e., $np - q \le m \le np + p$: the most probable value lies within distance $\max(p, q) < 1$ of the mean np, which is the sense in which the printed claim holds. The endpoint cases $p = 0, 1$ have sole modes $0, n$, respectively; $n = 0$ is the point mass at 0. For example, $n = 3$, $p = 1/2$ has $(n+1)p = 2$, mean 1.5, but modes 1 and 2. For large n, a continuous central approximation places the peak near np; it is not the exact integer-mode statement.

23.4 Poisson Distribution

23.4.1 Radioactive decays for long-lived isotopes are governed by the Poisson distribution. In a Rutherford–Geiger experiment, the numbers of emitted α particles are counted in each of $n = 2608$ time intervals of 7.5 seconds each. In Table 23.1 n_i is the number of time intervals in which i particles were emitted. Determine the average number λ of particles emitted per time interval, and compare the n_i of Table 23.1 with np_i computed from the Poisson distribution with mean value λ.

Table 23.1 Data for Exercise 23.4.1

i	0	1	2	3	4	5	6	7	8	9	10
n_i	57	203	383	525	532	408	273	139	45	27	16

Solution.

The total number N of recorded intervals (the n of the statement) and the total particle count S are

$$N = \sum_i n_i = 57 + 203 + 383 + 525 + 532 + 408 + 273 + 139 + 45 + 27 + 16 = 2608,$$
$$S = \sum_i i\, n_i = 203 + 766 + 1575 + 2128 + 2040 + 1638 + 973 + 360 + 243 + 160 = 10086.$$

Thus the fitted mean per 7.5-second interval is

$$\boxed{\widehat{\lambda} = S/N = 10086/2608 = 3.8673312883}.$$

Use $E_0 = Ne^{-\widehat{\lambda}} = 2608 \times 0.0209141 = 54.544$, followed by the exact recurrence $E_{i+1} = \widehat{\lambda} E_i/(i+1)$, which follows from $p_{i+1}/p_i = \widehat{\lambda}/(i+1)$ for $p_i = e^{-\widehat{\lambda}}\widehat{\lambda}^i/i!$, to compute $E_i = Np_i$.

i	observed n_i	Poisson expected Np_i
0	57	54.544
1	203	210.940
2	383	407.887
3	525	525.811
4	532	508.372
5	408	393.208
6	273	253.444
7	139	140.022
8	45	67.689
9	27	29.086
10	16	11.249
> 10	0	5.749

The final expected tail is $2608 - \sum_{i=0}^{10} E_i$; it is required because the Poisson model has unbounded support. Observed values fluctuate around their expectations rather than agreeing termwise; the natural scale is the standard deviation of n_i. In the model each of the N intervals independently contains i particles with probability p_i, so n_i is binomial with mean Np_i and standard deviation $\sqrt{Np_i(1-p_i)}$; since $p_i \le p_3 \approx 0.2016$ for every i, this lies between $0.89\sqrt{Np_i}$ and $\sqrt{Np_i}$, the Poisson value of Exercise 23.4.2. For $i \le 10$ the differences $n_i - Np_i$ stay within $1.5\sqrt{Np_i}$ except at $i = 8$,

where $45 - 67.689 = -22.689 \approx -2.76\sqrt{67.689}$. If, as in the original Rutherford–Geiger data, the last column counts all intervals with $i \geq 10$ (there 10, 4, 0, 1 and 1 intervals had $i = 10, \ldots, 14$), then S is slightly underestimated (the original total is 10097, so $\widehat{\lambda} = 10097/2608 = 3.8715$) and the last two rows combine to 16 observed against $11.249 + 5.749 \approx 17.0$ expected. The Instructor's Manual's entry 55.7 at $i = 0$ is inconsistent with the stated data and fitted mean; the direct value is 54.544. Its displayed formula for np_i also omits the factor n on the right: $p_i = e^{-\lambda}\lambda^i/i!$, whereas the expected count is np_i.

23.4.2 Derive the standard deviation of a Poisson distribution of mean value μ.

Solution.

For $\mu > 0$, the Poisson mass is $p_n = e^{-\mu}\mu^n/n!$, $n = 0, 1, \ldots$. In the two sums below the terms $n = 0$ (respectively $n = 0, 1$) vanish. Use $n/n! = 1/(n-1)!$ and $n(n-1)/n! = 1/(n-2)!$, shift the summation index to $m = n - 1$ (respectively $m = n - 2$), and use $\sum_{m\geq 0} \mu^m/m! = e^{\mu}$:

$$\langle X \rangle = e^{-\mu} \sum_{n=1}^{\infty} \frac{n\mu^n}{n!} = \mu e^{-\mu} \sum_{m=0}^{\infty} \frac{\mu^m}{m!} = \mu,$$

$$\langle X(X-1) \rangle = e^{-\mu} \sum_{n=2}^{\infty} \frac{n(n-1)\mu^n}{n!} = \mu^2 e^{-\mu} \sum_{m=0}^{\infty} \frac{\mu^m}{m!} = \mu^2.$$

Since $X^2 = X(X-1) + X$, $\langle X^2 \rangle = \mu^2 + \mu$, and $\operatorname{Var} X = \langle X^2 \rangle - \langle X \rangle^2 = \mu^2 + \mu - \mu^2$; that is,

$$\boxed{\operatorname{Var} X = \mu, \qquad \sigma_X = \sqrt{\mu}}.$$

At $\mu = 0$ the distribution is concentrated at 0 and the same result holds.

23.4.3 The number of α-particles emitted by the decay of a radium sample is counted per minute for 40 hours. The total number is 5000. How many 1-minute intervals are expected with (a) 2, and (b) 5 α-particles? Use a constant-rate Poisson model with its rate estimated from the observed total.

Solution.

There are $40 \cdot 60 = 2400$ one-minute intervals. Their fitted mean is $\mu = 5000/2400 = 25/12$ decays per minute. The expected number of intervals containing exactly k decays is

$$E_k = 2400 P(X = k) = 2400 e^{-25/12} \frac{(25/12)^k}{k!}.$$

With $e^{-25/12} = 0.12451447$, the single-interval probabilities are $P(X = 2) = e^{-25/12}(25/12)^2/2! = 0.2702137$ and $P(X = 5) = e^{-25/12}(25/12)^5/5! = 0.0407223$. Multiplying by 2400,

$$\boxed{E_2 = 648.513 \simeq 649, \qquad E_5 = 97.734 \simeq 98 \text{ intervals}}.$$

These are expectations under the fitted Poisson model, not guaranteed integer observations. The Instructor's Manual calls the 2400 intervals seconds rather than minutes and stops at single-interval probabilities. The question asks for the corresponding expected numbers of intervals.

23.4.4 For a radioactive sample, 10 decays are counted on average in 100 seconds. Use the Poisson distribution to estimate the probability of counting 3 decays in 10 seconds.

Solution.

The rate is $10/100 = 0.1$ decay per second, so a 10-second interval has Poisson mean $\mu = 1$. Therefore

$$\boxed{P(X = 3) = e^{-1}\frac{1^3}{3!} = \frac{1}{6e} = 0.0613132402 \simeq 6.13\%}.$$

The time rescaling changes the mean, not the factorial in the mass formula.

23.4.5 ^{238}U has a half-life of 4.51×10^9 years. Its decay series ends with the stable lead isotope ^{206}Pb. The ratio of the number of ^{206}Pb to ^{238}U atoms in a rock sample is measured as 0.0058. Estimate the age of the rock assuming that all the lead in the rock is from the initial decay of the ^{238}U, which determines the rate of the entire decay process, because the subsequent steps take place far more rapidly. Assume a closed sample, with no uranium or lead gained or lost since formation and no initial lead.

Hint. This is not a Poisson distribution problem, but is an application of the decay law $N(t) = Ne^{-\lambda t}$, where λ, the decay constant, is related to the half-life T by $T = \ln 2/\lambda$.

ANS. 3.8×10^7 years.

Solution.

Let U_0 be the initial uranium count and $U(t) = U_0 e^{-\lambda t}$ the remaining count. Under the stated closed-sample, zero-initial-lead and rapid-daughter-decay assumptions, $L(t) = U_0 - U(t)$. Thus the measured ratio is

$$R = \frac{L(t)}{U(t)} = \frac{U_0}{U(t)} - 1 = e^{\lambda t} - 1.$$

Solve $e^{\lambda t} = 1 + R$ for t, and use the *given* half-life T through $1/\lambda = T/\ln 2 = 6.50655463 \times 10^9$ years; with $\ln(1.0058) = 0.00578324476$,

$$\boxed{t = \frac{\ln(1+R)}{\lambda} = \frac{4.51 \times 10^9}{\ln 2}\ln(1.0058) = 3.7628998 \times 10^7 \text{ years} \simeq 3.8 \times 10^7 \text{ years}}.$$

One uranium decay ultimately yields one lead atom in this accounting; the intermediate decay steps are not additional initial uranium atoms.

23.4.6 The probability of hitting a target in one shot is known to be 20%. If five shots are fired independently, what is the probability of striking the target at least once?

Solution.

The probability that one shot misses is $1 - 0.2 = 0.8$. By independence, all five miss with probability $0.8^5 = 0.32768$. Its complement is

$$\boxed{P(\text{at least one hit}) = 1 - 0.8^5 = 0.67232 = 67.232\%}.$$

This is an exact binomial complement; a Poisson approximation is unnecessary.

23.5 Gauss' Normal Distribution

23.5.1 Show that the variance of the normal distribution given by Eq. (23.56) is σ^2, the symbol in that equation. Take $\sigma > 0$.

Solution.

For $\sigma > 0$, write the density explicitly:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}.$$

Substitute $z = (x-\mu)/\sigma$, i.e., $x = \mu + \sigma z$, $dx = \sigma\, dz$, with the limits $-\infty, \infty$ unchanged since $\sigma > 0$; then $f(x)\,dx = (2\pi)^{-1/2} e^{-z^2/2}\, dz$, and the Gaussian integral $\int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \sqrt{2\pi}$ gives the normalization $(2\pi)^{-1/2}\int_{-\infty}^{\infty} e^{-z^2/2}\, dz = 1$. Since $ze^{-z^2/2}$ is odd and integrable, $\langle X - \mu\rangle = \sigma(2\pi)^{-1/2}\int_{-\infty}^{\infty} ze^{-z^2/2}\, dz = 0$, so the mean really is μ. For the variance, $(x-\mu)^2 = \sigma^2 z^2$; integrate by parts with $u = z$ and $dv = ze^{-z^2/2}\, dz$, so that $v = -e^{-z^2/2}$ because $d(e^{-z^2/2}) = -ze^{-z^2/2}\, dz$:

$$\begin{aligned}
\operatorname{Var} X &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \left(\left[-ze^{-z^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-z^2/2}\, dz \right) \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \left(0 + \sqrt{2\pi}\right) = \boxed{\sigma^2}.
\end{aligned}$$

The boundary term is zero because the Gaussian dominates the linear factor: $ze^{-z^2/2} \to 0$ as $z \to \pm\infty$.

23.5.2 Show that Eq. (23.62) can be obtained by manipulation of the formula Eq. (23.61) for $W(v)$. Use $0 < p < 1$, $q = 1 - p$, $n > 0$, and $-pn < v < qn$; both formulas represent the same Stirling approximation.

Solution.

Let $0 < p < 1$, $q = 1 - p$, and $s = pn + v$, so that $n - s = n - pn - v = qn - v$. The Stirling approximation in Eq. (23.61), obtained from $m! \approx \sqrt{2\pi}\, m^{m+1/2} e^{-m}$ for $m = n, s, n-s$ in $\binom{n}{s} p^s q^{n-s}$, is

$$W(v) = \frac{p^s q^{n-s} n^{n+1/2} e^{-n+s+(n-s)}}{\sqrt{2\pi}(pn+v)^{s+1/2}(qn-v)^{n-s+1/2}}.$$

The exponential is $e^0 = 1$. Factor pn and qn out of the two denominator powers:

$$(pn+v)^{s+1/2} = (pn)^{s+1/2}(1 + v/(pn))^{s+1/2},$$

$$(qn-v)^{n-s+1/2} = (qn)^{n-s+1/2}(1 - v/(qn))^{n-s+1/2}.$$

The remaining constant quotient, with $(s+\tfrac{1}{2}) + (n-s+\tfrac{1}{2}) = n+1$, is

$$\frac{p^s q^{n-s} n^{n+1/2}}{p^{s+1/2} q^{n-s+1/2} n^{(s+1/2)+(n-s+1/2)}} = \frac{n^{n+1/2}}{p^{1/2} q^{1/2} n^{n+1}} = p^{-1/2} q^{-1/2} n^{-1/2}.$$

Therefore, writing the exponents as $s+\frac{1}{2}=pn+v+\frac{1}{2}$ and $n-s+\frac{1}{2}=qn-v+\frac{1}{2}$,

$$\boxed{W(v)=\frac{(1+v/(pn))^{-(pn+v+1/2)}(1-v/(qn))^{-(qn-v+1/2)}}{\sqrt{2\pi pqn}}}.$$

The rearrangement is exact for the Stirling expression; its identification with the original binomial mass is asymptotic, requiring both s and $n-s$ large. For real logarithms also require $-pn<v<qn$.

23.5.3 With $W(v)$ the expression in Eq. (23.62), show that the expansion of $\ln W(v)$ in powers of v leads to Eq. (23.63). Take $p,q>0$ fixed, $p+q=1$, and $|v|<\min(pn,qn)$. For the Gaussian limit use the central scaling $v=O(\sqrt{n})$.

Solution.

Take the logarithm of Eq. (23.62), the result of Exercise 23.5.2:

$$\begin{aligned}\ln W(v)=\ &-\tfrac{1}{2}\ln(2\pi pqn)-(pn+v+\tfrac{1}{2})\ln(1+v/(pn))\\ &-(qn-v+\tfrac{1}{2})\ln(1-v/(qn)).\end{aligned}$$

For $|v|<\min(pn,qn)$ use $\ln(1+u)=u-u^2/2+O(u^3)$ with $u=v/(pn)$, $|u|<1$. Multiplying out,

$$\Big(pn+v+\frac{1}{2}\Big)\Big(u-\frac{u^2}{2}\Big)=v-\frac{v^2}{2pn}+\frac{v^2}{pn}+\frac{v}{2pn}-\frac{v^2}{4p^2n^2}-\frac{v^3}{2p^2n^2},$$

the terms coming from pnu, $-pnu^2/2$, vu, $u/2$, $-u^2/4$ and $-vu^2/2$; the remainder contributes $(pn+v+\frac{1}{2})O(u^3)=O(v^3)$. Since $-v^2/(2pn)+v^2/(pn)=v^2/(2pn)$, the negative of this product gives the first line below. The second line follows from the first by $p\to q$, $v\to -v$, which maps $pn+v+\frac{1}{2}$ to $qn-v+\frac{1}{2}$ and $1+v/(pn)$ to $1-v/(qn)$. Keeping all terms through v^2, including those from the half-power factors,

$$\begin{aligned}-(pn+v+\tfrac{1}{2})\ln(1+v/(pn))&=-v-\frac{v}{2pn}-\frac{v^2}{2pn}+\frac{v^2}{4p^2n^2}+O(v^3),\\ -(qn-v+\tfrac{1}{2})\ln(1-v/(qn))&=v+\frac{v}{2qn}-\frac{v^2}{2qn}+\frac{v^2}{4q^2n^2}+O(v^3).\end{aligned}$$

The $-v$ and v cancel. Collecting the rest yields Eq. (23.63):

$$\boxed{\begin{aligned}\ln W(v)=\ &-\tfrac{1}{2}\ln(2\pi pqn)-\frac{v}{2n}\left(\frac{1}{p}-\frac{1}{q}\right)\\ &-\frac{v^2}{2n}\left(\frac{1}{p}+\frac{1}{q}\right)+\frac{v^2}{4n^2}\left(\frac{1}{p^2}+\frac{1}{q^2}\right)+O(v^3)\end{aligned}}.$$

Here $O(v^3)$ first denotes a Taylor remainder for fixed n,p,q. For the limiting argument hold $p,q>0$ fixed and take $v=O(\sqrt{n})$. Then $u=v/(pn)=O(n^{-1/2})$. In the two-line expansion before the box, the first line omits $v^3/(2p^2n^2)$ (the negative of the last term of the product) and the remainder $(pn+v+\frac{1}{2})O(u^3)=O(|v|^3/n^2)+O(v^4/n^3)+O(|v|^3/n^3)$, where $v^4/n^3\le p\,|v|^3/n^2$ because $|v|<pn$; by $p\to q$, the second line omits terms of the same orders. The remainder is then $O(|v|^3/n^2+|v|^3/n^3)=O(n^{-1/2})$, while the displayed linear term, $O(|v|/n)=O(n^{-1/2})$, and the last quadratic term, $O(v^2/n^2)=O(n^{-1})$, also vanish as $n\to\infty$. Since $1/p+1/q=(q+p)/(pq)=1/(pq)$, the surviving form is

$$W(v)\sim\frac{1}{\sqrt{2\pi npq}}e^{-v^2/(2npq)}.$$

This central scaling does not assert a uniformly accurate Gaussian approximation all the way to the extreme binomial tails.

23.5.4 What is the probability for a normally distributed random variable to differ by more than 4σ from its mean value? Compare your result with the corresponding one from Chebyshev's inequality. Explain the difference in your own words. Assume $\sigma > 0$.

Solution.

Standardize $Z = (X - \mu)/\sigma$, for $\sigma > 0$; then $|X - \mu| > 4\sigma$ means $|Z| > 4$, and by the symmetry of the standard normal density the tails $Z > 4$ and $Z < -4$ have equal probability. Substitute $z = \sqrt{2}\,u$, $dz = \sqrt{2}\,du$, with new lower limit $4/\sqrt{2} = 2\sqrt{2}$, and use $\operatorname{erfc} x = (2/\sqrt{\pi}) \int_x^\infty e^{-u^2}\,du$:

$$P(|X - \mu| > 4\sigma) = \frac{2}{\sqrt{2\pi}} \int_4^\infty e^{-z^2/2}\,dz = \frac{2}{\sqrt{\pi}} \int_{2\sqrt{2}}^\infty e^{-u^2}\,du$$
$$= \operatorname{erfc}(2\sqrt{2}) = \boxed{6.33424837 \times 10^{-5}}.$$

This is about 0.00633425%. Chebyshev's inequality, $P(|X - \mu| \geq k\sigma) \leq 1/k^2$, with $k = 4$ gives only

$$P(|X - \mu| > 4\sigma) \leq \frac{1}{4^2} = 0.0625 = 6.25\%.$$

The bound is $0.0625/(6.33425 \times 10^{-5}) \approx 987$ times the exact Gaussian tail. Chebyshev uses only the variance and applies to every distribution with that variance; the Gaussian computation uses its entire density, including its rapidly decaying tails. A general upper bound need not be attained by this particular density. The Instructor's Manual's trailing approximation $6.3 \times 10^{-5} \approx 10^{-3}$ is numerically incorrect.

23.5.5 An instructor grades a final exam of a large undergraduate class, obtaining the mean value of points m and the variance σ^2. Assuming a normal distribution for the random variable M of points, he defines a grade F when $M < m - 3\sigma/2$, D when $m - 3\sigma/2 < M < m - \sigma/2$, C when $m - \sigma/2 < M < m + \sigma/2$, B when $m + \sigma/2 < M < m + 3\sigma/2$, and A when $M > m + 3\sigma/2$. What is the percentage of As, Fs; Bs, Ds; and Cs? Redesign the cutoffs so that there are equal percentages of As and Fs (5%), 25% Bs and Ds, and 40% Cs. Assume $\sigma > 0$.

Solution.

Use the corrected notation: m is the mean, M the random score (the cutoffs compare a student's score M with the class mean m, so the two need different symbols), and $Z = (M - m)/\sigma$ has standard normal CDF

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2}\,du = \tfrac{1}{2}[1 + \operatorname{erf}(z/\sqrt{2})].$$

The original cutoffs $m \pm \sigma/2$, $m \pm 3\sigma/2$ correspond to $Z = \pm 0.5$, ± 1.5. By the symmetry $\Phi(-z) = 1 - \Phi(z)$, $P(F) = \Phi(-1.5) = 1 - \Phi(1.5)$, $P(D) = \Phi(-0.5) - \Phi(-1.5) = \Phi(1.5) - \Phi(0.5)$ and $\Phi(0.5) - \Phi(-0.5) = 2\Phi(0.5) - 1$. With $\Phi(1.5) = 0.93319279873$ and $\Phi(0.5) = 0.69146246127$ the cutoffs give

$$P(A) = P(F) = 1 - \Phi(1.5) = 0.0668072013,$$
$$P(B) = P(D) = \Phi(1.5) - \Phi(0.5) = 0.2417303375,$$
$$P(C) = \Phi(0.5) - \Phi(-0.5) = 0.3829249225.$$

Thus the percentages are $\boxed{6.68072\%,\ 24.17303\%,\ 38.29249\%}$ for each extreme grade, each adjacent grade, and C, respectively. They sum as $2(6.68072) + 2(24.17303) + 38.29249 = 100\%$, up to rounding.

For the redesign let the positive thresholds be $a > b > 0$, with the boundaries at $Z = \pm a, \pm b$ (symmetric, so that A and F, and B and D, get equal shares). The outer 5% requires $P(A) = 1 - \Phi(a) = 0.05$, i.e., $\Phi(a) = 0.95$, and the next 25% requires $P(B) = \Phi(a) - \Phi(b) = 0.25$, hence $\Phi(b) = 0.70$. Solving these numerically (from a table of Φ, or by Newton's method with $\Phi'(z) = e^{-z^2/2}/\sqrt{2\pi}$) gives

$$a = \Phi^{-1}(0.95) = 1.644853627, \qquad b = \Phi^{-1}(0.70) = 0.524400513.$$

The new four score boundaries, in order, are

$$\boxed{m - 1.644853627\sigma, \quad m - 0.524400513\sigma, \quad m + 0.524400513\sigma, \quad m + 1.644853627\sigma}.$$

They yield 5%, 25%, 40%, 25%, 5%, since $P(C) = \Phi(b) - \Phi(-b) = 2\Phi(b) - 1 = 0.40$. Any assignment of an exact boundary to one adjacent grade gives the same probabilities because a continuous normal variable has no point masses.

23.6 Transformations of Random Variables

23.6.1 Let $X_1, X_2, \ldots, X_n$ be independent normal random variables with the same mean $\bar{x}$ and variance σ^2. Show that

$$\frac{\sum_i X_i/n - \bar{x}}{\sigma/\sqrt{n}}$$

is normal with mean zero and variance 1. Take integer $n \geq 1$ and $\sigma > 0$.

Solution.

Take $n \geq 1$ and $\sigma > 0$. To track the distribution, use the characteristic function, not just its first two moments. For a centered normal variable of variance σ^2, with density $f(x) = e^{-x^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$, write $\phi(t) = \langle e^{itX} \rangle = \int e^{itx} f(x)\, dx$. Since $f'(x) = -xf(x)/\sigma^2$, i.e. $xf(x) = -\sigma^2 f'(x)$, differentiation under the integral sign (allowed because $\int |x| f(x)\, dx < \infty$) gives

$$\phi'(t) = i \int x e^{itx} f(x)\, dx = -i\sigma^2 \int e^{itx} f'(x)\, dx.$$

Integration by parts gives

$$\int e^{itx} f'(x)\, dx = \Big[e^{itx} f(x) \Big]_{-\infty}^{\infty} - it \int e^{itx} f(x)\, dx = -it\phi(t),$$

since the boundary term vanishes: $|e^{itx} f(x)| = f(x) \to 0$ as $x \to \pm\infty$. Hence $\phi'(t) = -i\sigma^2(-it)\phi(t) = -\sigma^2 t \phi(t)$, so that $\frac{d}{dt}\big[e^{\sigma^2 t^2/2}\phi(t)\big] = e^{\sigma^2 t^2/2}\big[\phi'(t) + \sigma^2 t \phi(t)\big] = 0$. With $\phi(0) = 1$ this yields $\phi(t) = e^{-\sigma^2 t^2/2}$. Translation by $\bar{x}$ multiplies the characteristic function by $e^{it\bar{x}}$, because $\langle e^{it(\bar{x}+X)} \rangle = e^{it\bar{x}} \langle e^{itX} \rangle$; hence $\phi_{X_j}(t) = e^{it\bar{x} - \sigma^2 t^2/2}$. Now let $\overline{X} = n^{-1} \sum_{j=1}^{n} X_j$. Then $e^{it\overline{X}} = \prod_j e^{i(t/n)X_j}$, and for independent variables the expectation of such a product is the product of the expectations, as in Exercise 23.6.4: against the product density $\prod_j p_{X_j}(x_j)$, the n-fold integral of $\prod_j e^{i(t/n)x_j}$, absolutely convergent because each factor has modulus 1, separates into n one-dimensional integrals. Therefore

$$\phi_{\overline{X}}(t) = \prod_{j=1}^{n} \phi_{X_j}(t/n) = \Big[e^{it\bar{x}/n - \sigma^2 t^2/(2n^2)} \Big]^n = e^{it\bar{x} - \sigma^2 t^2/(2n)}.$$

This is the characteristic function of the normal distribution with mean $\bar{x}$ and variance σ^2/n. A characteristic function determines its distribution uniquely (it is $\sqrt{2\pi}$ times the Fourier transform of the density, which Fourier inversion recovers), so $\overline{X}$ is normal with mean $\bar{x}$ and variance σ^2/n. The requested variable is $Z = (\overline{X} - \bar{x})/(\sigma/\sqrt{n}) = c\overline{X} + d$ with $c = \sqrt{n}/\sigma$ and $d = -\bar{x}\sqrt{n}/\sigma$, and the rule $\phi_{cY+d}(t) = \langle e^{it(cY+d)} \rangle = e^{itd}\phi_Y(ct)$ gives

$$\phi_Z(t) = e^{-it\bar{x}\sqrt{n}/\sigma}\, \phi_{\overline{X}}\left(\frac{t\sqrt{n}}{\sigma}\right) = e^{-it\bar{x}\sqrt{n}/\sigma}\, e^{it\bar{x}\sqrt{n}/\sigma - t^2/2} = e^{-t^2/2},$$

the characteristic function of the standard normal law; thus

$$\boxed{Z \sim N(0,1)}.$$

The corrected denominator is $\sigma/\sqrt{n}$. The printed $\sigma\sqrt{n}$ would give variance $1/n^2$ instead. For $\sigma = 0$ all observations are constant and this standardization is undefined, so positivity is necessary.

23.6.2 If the random variable X is normal with mean value 29 and standard deviation 3, what can you say about the distributions of $2X - 1$ and $3X + 2$?

Solution.

For $Y = aX + b$ with $a \neq 0$ the map $x \mapsto ax + b$ is one-to-one, so $f_Y(y) = f_X(x)\,|dx/dy|$ with $x = (y - b)/a$ and $|dx/dy| = 1/|a|$. With $f_X(x) = e^{-(x-29)^2/18}/(3\sqrt{2\pi})$ and $((y - b)/a - 29)^2 = (y - b - 29a)^2/a^2$,

$$f_Y(y) = \frac{1}{|a|}\frac{1}{3\sqrt{2\pi}} \exp\left[-\frac{((y-b)/a - 29)^2}{18}\right] = \frac{1}{3|a|\sqrt{2\pi}} \exp\left[-\frac{(y - (29a + b))^2}{2(3a)^2}\right].$$

This is the normal density with mean $29a + b$ and standard deviation $3|a|$. For $(a, b) = (2, -1)$ and $(3, 2)$ it identifies both full distributions:

$$\boxed{2X - 1 \sim N(57, 36), \quad \sigma = 6; \qquad 3X + 2 \sim N(89, 81), \quad \sigma = 9}.$$

The second argument of $N(\mu, \tau^2)$ here denotes variance. The Instructor's Manual multiplies variance by a and reports $3\sqrt{2}$ and $3\sqrt{3}$ as standard deviations. The correct variance factor is a^2, so the standard-deviation factor is $|a|$.

23.6.3 For a normal distribution of mean value m and variance σ^2, find the distance r such that half the area under the bell shape is between $m - r$ and $m + r$. Assume $\sigma > 0$ and seek $r \geq 0$.

Solution.

For $\sigma > 0$, standardize $z = (x - m)/\sigma$; with the standard normal distribution function Φ this gives $P(m - r \leq X \leq m + r) = \Phi(r/\sigma) - \Phi(-r/\sigma)$. The symmetry of the bell shape gives $\Phi(-z) = 1 - \Phi(z)$, and $\Phi(z) = \frac{1}{2}\big[1 + \operatorname{erf}(z/\sqrt{2})\big]$, so

$$\frac{1}{2} = P(m - r \leq X \leq m + r) = 2\Phi(r/\sigma) - 1 = \operatorname{erf}\left(\frac{r}{\sigma\sqrt{2}}\right).$$

Hence $\Phi(r/\sigma) = 3/4$, and strict monotonicity of Φ gives the unique nonnegative answer (the quantile is read from a table of Φ, or computed by Newton's method as in Exercise 23.5.5)

$$\boxed{r = \sigma\Phi^{-1}(3/4) = \sigma\sqrt{2}\,\operatorname{erf}^{-1}(1/2) = 0.6744897502\sigma}.$$

This interval contains the central half of the probability mass.

23.6.4 If $\langle X\rangle$, $\langle Y\rangle$ are the average values of two independent random variables X, Y, what is the expectation value of the product XY? Assume finite absolute first moments.

Solution.

Assume finite absolute first moments. Independence means that the joint law of (X,Y) is the product of the two marginal laws; with densities, $p(x,y)=p_X(x)p_Y(y)$. For the nonnegative function $|xy|$, Tonelli's theorem then gives $\langle|XY|\rangle=\langle|X|\rangle\langle|Y|\rangle<\infty$, so by Fubini's theorem the double integral may be computed as iterated integrals and separated. When densities exist, this reads

$$\boxed{\begin{aligned}\langle XY\rangle &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\,p_X(x)p_Y(y)\,dx\,dy \\ &= \left(\int xp_X(x)\,dx\right)\left(\int yp_Y(y)\,dy\right)=\langle X\rangle\langle Y\rangle\end{aligned}}.$$

For general laws the same calculation uses their product probability measure; discrete laws give sums. The Instructor's Manual omits the factors xy, x, and y in its three integrals. Without them those integrals are normalization integrals equal to 1, not the stated expectations. It follows in particular that $\operatorname{cov}(X,Y)=\langle XY\rangle-\langle X\rangle\langle Y\rangle=0$ when second moments exist.

23.6.5 If X and Y are two independent random variables with different probability densities and the function $f(x,y)$ has derivatives of any order, explain why the two means alone do not determine $\langle f(X,Y)\rangle$. Derive a finite Taylor expansion in central moments, including its remainder, and state conditions for an infinite moment expansion. Give the covariance and correlation of X,Y; also derive the corresponding formulas for transformed variables $G=g(X,Y)$ and $H=h(X,Y)$ when the required moments exist.

Solution.

The two means alone do not determine $\langle f(X,Y)\rangle$. Higher moments are needed. Existence of derivatives of every order also does not justify an infinite Taylor expansion or its termwise expectation. The Instructor's Manual's weighted integral is not the ordinary covariance of the independent variables.

Let $\mu=\langle X\rangle$, $\nu=\langle Y\rangle$, $U=X-\mu$, $V=Y-\nu$, and define central moments $M_r=\langle U^r\rangle$, $N_s=\langle V^s\rangle$ when finite. Then $M_0=N_0=1$, $M_1=N_1=0$, $M_2=\sigma_X^2$, $N_2=\sigma_Y^2$, and independence implies $\langle U^rV^s\rangle=M_rN_s$ (Exercise 23.6.4 applied to U^r and V^s). The always valid starting point, when $\langle|f(X,Y)|\rangle<\infty$, is

$$\langle f(X,Y)\rangle=\iint f(x,y)p_X(x)p_Y(y)\,dx\,dy.$$

For a finite Taylor order k, put $F(t)=f(\mu+tU,\nu+tV)$, so that $F(0)=f(\mu,\nu)$ and $F(1)=f(X,Y)$, and apply the one-variable Taylor theorem with integral remainder at $t=0$, evaluated at $t=1$:

$$F(1)=\sum_{j=0}^{k}\frac{F^{(j)}(0)}{j!}+\frac{1}{k!}\int_0^1(1-t)^kF^{(k+1)}(t)\,dt.$$

By the chain rule, with U,V fixed, each t-derivative acts on f as the operator $U\partial_x+V\partial_y$, so $F^{(j)}(t)=(U\partial_x+V\partial_y)^jf(\mu+tU,\nu+tV)$. The two partial derivatives commute, so the binomial theorem gives, with $f_{rs}=\partial_x^r\partial_y^sf$,

$$F^{(j)}(t)=\sum_{r+s=j}\frac{j!}{r!s!}U^rV^sf_{rs}(\mu+tU,\nu+tV).$$

Dividing $F^{(j)}(0)$ by $j!$ gives the following coefficients:

$$f(X,Y) = \sum_{r+s\le k} \frac{f_{rs}(\mu,\nu)}{r!s!} U^r V^s + R_k,$$

$$R_k = \frac{1}{k!}\int_0^1 (1-t)^k (U\partial_x + V\partial_y)^{k+1} f(\mu+tU, \nu+tV)\, dt.$$

Assuming integrability of these terms, taking expectations with $\langle U^r V^s\rangle = M_r N_s$ gives the rigorous finite form

$$\boxed{\langle f(X,Y)\rangle = \sum_{r+s\le k} \frac{f_{rs}(\mu,\nu)}{r!s!} M_r N_s + \langle R_k\rangle}.$$

For example, through second order ($k=2$), the two first-order terms contain M_1 or N_1 and vanish, and so does the mixed term $f_{11}M_1N_1$; with $M_2 = \sigma_X^2$ and $N_2 = \sigma_Y^2$,

$$\langle f(X,Y)\rangle = f(\mu,\nu) + \tfrac{1}{2} f_{20}(\mu,\nu)\sigma_X^2 + \tfrac{1}{2} f_{02}(\mu,\nu)\sigma_Y^2 + \langle R_2\rangle.$$

If the Taylor series equals f on the support and is integrable term by term (for example, $\sum_{r,s} |f_{rs}(\mu,\nu)|\langle |U|^r\rangle\langle |V|^s\rangle/(r!s!) < \infty$, which allows the expectation and the double sum to be interchanged), the limit is

$$\langle f(X,Y)\rangle = \sum_{r,s\ge 0} \frac{f_{rs}(\mu,\nu)}{r!s!} M_r N_s.$$

Two examples show that both caveats of the erratum above are needed. For a counterexample to determination by the means, take $f(x,y) = x^2$, independent Y uniform on $[-3,3]$, and compare X uniform on $[-1,1]$ with X uniform on $[-2,2]$. All means are zero, but for X uniform on $[-a,a]$, $\langle X^2\rangle = \int_{-a}^{a} x^2\, dx/(2a) = a^2/3$, so the expectations are $1/3$ and $4/3$. For a counterexample to smoothness sufficing, take the smooth function $f(x,y) = e^{-1/x^2}$ for $x \neq 0$, extended by 0 at $x=0$; it does not depend on y, so only its x-derivatives can be nonzero. By induction, for $x \neq 0$ its nth derivative has the form $P_n(1/x)e^{-1/x^2}$ with a polynomial P_n: $P_0 = 1$, and

$$\frac{d}{dx}\Big[P_n(1/x)e^{-1/x^2}\Big] = \Big[2x^{-3}P_n(1/x) - x^{-2}P_n'(1/x)\Big]e^{-1/x^2},$$

so $P_{n+1}(u) = 2u^3 P_n(u) - u^2 P_n'(u)$. Each $f^{(n)}(x)$ tends to 0 as $x \to 0$, because with $q = 1/x^2 \to \infty$ each of its terms obeys $|x|^{-m}e^{-1/x^2} = q^{m/2}e^{-q} \to 0$. The difference quotient $[f^{(n)}(x) - 0]/x$ at $x = 0$ has the same form, so every derivative at $(0,0)$ exists and is zero: the Taylor series at $(0,0)$ vanishes identically, and the infinite expansion would give 0. For centered independent normal X, Y, however, the expectation is strictly positive, because $f > 0$ except on the line $x = 0$, which has probability zero.

For the variables actually named in the question,

$$\boxed{\operatorname{cov}(X,Y) = \langle UV\rangle = M_1 N_1 = 0, \qquad \rho_{XY} = 0}$$

provided both variances are finite and nonzero; correlation is undefined if either variance is zero. For the transformed variables $G = g(X,Y)$ and $H = h(X,Y)$, write their Taylor coefficients as $g_{rs}/(r!s!)$ and $h_{uv}/(u!v!)$, all evaluated at (μ,ν). Under convergence and interchange conditions for the product as well, multiplying the two series and using $\langle U^{r+u}V^{s+v}\rangle = M_{r+u}N_{s+v}$ gives

$$\langle GH\rangle = \sum_{r,s,u,v\ge 0} \frac{g_{rs}h_{uv}}{r!s!u!v!} M_{r+u}N_{s+v}, \qquad \langle G\rangle\langle H\rangle = \sum_{r,s,u,v\ge 0} \frac{g_{rs}h_{uv}}{r!s!u!v!} M_r N_s M_u N_v.$$

Subtracting,

$$\boxed{\operatorname{cov}(G,H)=\sum_{r,s,u,v\ge 0}\frac{g_{rs}h_{uv}}{r!s!u!v!}\left[M_{r+u}N_{s+v}-M_rN_sM_uN_v\right]}.$$

Setting $h=g$ gives the variance. The correlation is this covariance divided by $\sqrt{\operatorname{Var}G\operatorname{Var}H}$, when positive. For small fluctuations keep the lowest order. The bracket vanishes when $(r,s)=(0,0)$ or $(u,v)=(0,0)$, because $M_0=N_0=1$. The lowest-order surviving terms are therefore the second-order ones with $r+s=u+v=1$: the bracket is $M_2=\sigma_X^2$ for $(r,s)=(u,v)=(1,0)$, $N_2=\sigma_Y^2$ for $(r,s)=(u,v)=(0,1)$, and $M_1N_1-M_1N_0M_0N_1=0$ for the two mixed choices. With $g_x=g_{10}$, $g_y=g_{01}$, and likewise for h,

$$\operatorname{cov}(G,H)\simeq g_xh_x\sigma_X^2+g_yh_y\sigma_Y^2.$$

With $h=g$ this approximation gives $\operatorname{Var}G\simeq g_x^2\sigma_X^2+g_y^2\sigma_Y^2$, and with $g=h$ it gives $\operatorname{Var}H\simeq h_x^2\sigma_X^2+h_y^2\sigma_Y^2$; so to the same order, when both are positive,

$$\rho_{GH}\simeq\frac{g_xh_x\sigma_X^2+g_yh_y\sigma_Y^2}{\sqrt{(g_x^2\sigma_X^2+g_y^2\sigma_Y^2)(h_x^2\sigma_X^2+h_y^2\sigma_Y^2)}}.$$

Finally the Instructor's Manual's object $\langle UVf(X,Y)\rangle$ would instead expand as

$$\langle UVf(X,Y)\rangle=\sum_{r,s\ge0}\frac{f_{rs}}{r!s!}\langle U^{r+1}V^{s+1}\rangle=\sum_{r,s\ge0}\frac{f_{rs}}{r!s!}M_{r+1}N_{s+1}.$$

Its terms with $r=0$ or $s=0$ contain M_1 or N_1, respectively, and vanish, so its leading term is $f_{11}M_2N_2=f_{11}\sigma_X^2\sigma_Y^2$. It is a weighted mixed moment, not a covariance or a normalized correlation of X,Y.

23.6.6 Let $f(x,y)$ be the joint probability density of two random variables X, Y. Find the variance $\sigma^2(aX+bY)$, where a, b are constants. What happens when X, Y are independent? Take real a,b,X,Y and finite second moments.

Solution.

Set $\mu=\langle X\rangle$, $\nu=\langle Y\rangle$ and assume finite second moments. Since $\langle aX+bY\rangle=a\mu+b\nu$,

$$\begin{aligned}\operatorname{Var}(aX+bY)&=\iint[a(x-\mu)+b(y-\nu)]^2f(x,y)\,dx\,dy\\&=a^2\iint(x-\mu)^2f\,dx\,dy+b^2\iint(y-\nu)^2f\,dx\,dy\\&\quad+2ab\iint(x-\mu)(y-\nu)f\,dx\,dy\\&=\boxed{a^2\sigma_X^2+b^2\sigma_Y^2+2ab\operatorname{cov}(X,Y)}.\end{aligned}$$

All integrations are over $\mathbb{R}^2$. Integrating f over y gives the marginal density $p_X(x)=\int f(x,y)\,dy$ of X, so the first integral is $\int(x-\mu)^2p_X(x)\,dx=\sigma_X^2$; likewise the second is σ_Y^2, and the third is the definition of $\operatorname{cov}(X,Y)$. For independent X,Y, $f(x,y)=p_X(x)p_Y(y)$ makes the final integral the product of two zero centered means, $\int(x-\mu)p_X(x)\,dx\int(y-\nu)p_Y(y)\,dy=0\cdot0$. Thus $\boxed{\operatorname{Var}(aX+bY)=a^2\sigma_X^2+b^2\sigma_Y^2}$ in that case.

23.6.7 Obtain an addition theorem for the distribution of a random variable $Y = X_1 + X_2$ where X_1 and X_2 are independent Gauss normal random variables with different mean values μ_j and variances σ_j^2.

ANS. Y is normal with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$.

Solution.

The stated answer requires independence, as in the addition theorem being used. Normal marginal distributions alone are insufficient. The Instructor's Manual's final exponential also omits a factor 1/2. Let $X_j \sim N(\mu_j, \sigma_j^2)$ independently. Their characteristic functions, derived in Exercise 23.6.1, multiply by independence (Exercise 23.6.4):

$$\begin{aligned}\phi_Y(t) &= \langle e^{it(X_1+X_2)}\rangle = \langle e^{itX_1}\rangle\langle e^{itX_2}\rangle \\ &= e^{it\mu_1-\sigma_1^2t^2/2}e^{it\mu_2-\sigma_2^2t^2/2} = e^{it(\mu_1+\mu_2)-(\sigma_1^2+\sigma_2^2)t^2/2}.\end{aligned}$$

This is the characteristic function of the normal law with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$, and a characteristic function identifies the complete distribution (Exercise 23.6.1):

$$\boxed{Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)}.$$

One can also check it directly by convolution: for independent variables the density of the sum is $f_Y(y) = \int f_{X_1}(x) f_{X_2}(y - x)\, dx$. With $S = \sigma_1^2 + \sigma_2^2$ and $h = y - \mu_1 - \mu_2$, set $u = x - \mu_1$, so that $y - x - \mu_2 = h - u$ and

$$f_Y(y) = \frac{1}{2\pi\sigma_1\sigma_2}\int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{u^2}{\sigma_1^2} + \frac{(h-u)^2}{\sigma_2^2}\right)\right] du.$$

Expanding $(h - u)^2$, the expression in parentheses has u^2-coefficient $1/\sigma_1^2 + 1/\sigma_2^2 = S/(\sigma_1^2\sigma_2^2)$, u-coefficient $-2h/\sigma_2^2 = -2\,[S/(\sigma_1^2\sigma_2^2)]\,(\sigma_1^2 h/S)$, and constant term h^2/σ_2^2. Completing the square subtracts $[S/(\sigma_1^2\sigma_2^2)](\sigma_1^2 h/S)^2 = \sigma_1^2 h^2/(\sigma_2^2 S)$ from that constant, leaving $h^2(S - \sigma_1^2)/(\sigma_2^2 S) = h^2/S$ because $S - \sigma_1^2 = \sigma_2^2$:

$$\frac{u^2}{\sigma_1^2} + \frac{(h-u)^2}{\sigma_2^2} = \frac{S}{\sigma_1^2\sigma_2^2}\left(u - \frac{\sigma_1^2 h}{S}\right)^2 + \frac{h^2}{S}.$$

The remaining Gaussian integral is $\int \exp[-S(u - \sigma_1^2 h/S)^2/(2\sigma_1^2\sigma_2^2)]\, du = \sqrt{2\pi\sigma_1^2\sigma_2^2/S}$, so the convolution integral is

$$f_Y(y) = \frac{e^{-h^2/(2S)}}{2\pi\sigma_1\sigma_2}\sqrt{\frac{2\pi\sigma_1^2\sigma_2^2}{S}} = \frac{e^{-h^2/(2S)}}{\sqrt{2\pi S}}.$$

This calculation assumes positive variances; a zero variance just adds a constant, and both zero give a point mass. Without independence, take $X_1 = Z$, $X_2 = 1 + 2Z$ for $Z \sim N(0, 1)$. Then $X_1 + X_2 = 1 + 3Z$ has variance 9, whereas the printed sum of marginal variances is $1 + 4 = 5$. For jointly normal dependent variables the correct variance adds $2\,\mathrm{cov}(X_1, X_2)$ (Exercise 23.6.6); here $\mathrm{cov}(Z, 1 + 2Z) = 2\langle Z^2\rangle = 2$ and $5 + 2 \cdot 2 = 9$.

23.6.8 Show that the Fourier transform of the gamma-distribution probability density, Eq. (23.83), has the functional form given in Eq. (23.84). Take $p, \sigma > 0$ and use the book's positive-exponent, $1/\sqrt{2\pi}$ Fourier convention.

Solution.

For $p > 0$, $\sigma > 0$, Eq. (23.83) is

$$g(y) = \frac{y^{p-1}e^{-y/(2\sigma^2)}}{(2\sigma^2)^p\Gamma(p)}, \quad y > 0,$$

and zero elsewhere. In the book's positive-exponent Fourier convention, $\widehat{g}(t) = (2\pi)^{-1/2} \int g(y) e^{ity}\, dy$; combining the two exponentials, $e^{-y/(2\sigma^2)} e^{ity} = e^{-ay}$, gives

$$\widehat{g}(t) = \frac{1}{\sqrt{2\pi}(2\sigma^2)^p \Gamma(p)} \int_0^\infty y^{p-1} e^{-ay}\, dy, \qquad a = \frac{1}{2\sigma^2} - it.$$

For positive real a, the substitution $u = ay$ (limits unchanged, $dy = du/a$) gives

$$\int_0^\infty y^{p-1} e^{-ay}\, dy = a^{-p} \int_0^\infty u^{p-1} e^{-u}\, du = \Gamma(p) a^{-p}.$$

Both sides are holomorphic in $\Re a > 0$. On the right, $a^{-p} = e^{-p \operatorname{Log} a}$ with the principal logarithm, which is real for $a > 0$. On the left, $|e^{-ay}| = e^{-y \Re a}$, so on each half-plane $\Re a \geq \delta > 0$ the integrand $y^{p-1} e^{-ay}$ and its a-derivative $-y^p e^{-ay}$ are bounded in modulus by $y^{p-1} e^{-\delta y}$ and $y^p e^{-\delta y}$, which are integrable over $y > 0$ and independent of a; hence the integral may be differentiated with respect to a under the integral sign. The identity theorem extends the formula throughout that half-plane, which contains every a above for real t, since $\Re a = 1/(2\sigma^2) > 0$. Substitution, with $(2\sigma^2)^p a^p = (2\sigma^2 a)^p$ (valid for principal powers because $2\sigma^2 > 0$) and $2\sigma^2 a = 1 - 2i\sigma^2 t$, gives

$$\boxed{\widehat{g}(t) = \frac{1}{\sqrt{2\pi}} (1 - 2i\sigma^2 t)^{-p}}.$$

At $t = 0$ it is $1/\sqrt{2\pi}$, confirming normalization. The characteristic function is $\phi(t) = \langle e^{itY} \rangle = \sqrt{2\pi}\, \widehat{g}(t) = (1 - 2i\sigma^2 t)^{-p}$; it has value 1 at 0, and since $\phi'(t) = 2ip\sigma^2 (1 - 2i\sigma^2 t)^{-p-1}$ while $\phi'(0) = i\langle Y \rangle$, it gives the mean $\langle Y \rangle = 2p\sigma^2$.

23.7 Statistics

23.7.1 Let ΔA be the error of a measurement of A, etc. Use first-order propagation of uncorrelated errors at nonzero nominal values to show that

$$\left(\frac{\sigma(C)}{C}\right)^2 \simeq \left(\frac{\sigma(A)}{A}\right)^2 + \left(\frac{\sigma(B)}{B}\right)^2$$

holds for the product $C = AB$ and the ratio $C = A/B$.

Solution.

The stated rule is first-order error propagation for uncorrelated errors, evaluated at nonzero nominal A, B. Correlations add a cross term, and nonlinear propagation need not satisfy this rule exactly. Write the errors as δA, δB, with zero means. For the product, $C + \delta C = (A + \delta A)(B + \delta B) = AB + B\delta A + A\delta B + \delta A\, \delta B$; dropping the second-order term $\delta A\, \delta B$ (first-order Taylor expansion) and dividing by $C = AB$ gives

$$\delta C = B\delta A + A\delta B, \qquad \frac{\delta C}{C} = \frac{\delta A}{A} + \frac{\delta B}{B}.$$

For the ratio, the two derivatives are $\partial_A C = 1/B$ and $\partial_B C = -A/B^2$, so first-order expansion, followed by division by $C = A/B$, gives

$$\delta C = \frac{\delta A}{B} - \frac{A\delta B}{B^2}, \qquad \frac{\delta C}{C} = \frac{\delta A}{A} - \frac{\delta B}{B}.$$

Squaring gives $(\delta C/C)^2 = (\delta A/A)^2 + (\delta B/B)^2 \pm 2\,\delta A\,\delta B/(AB)$. Average, with $\sigma_A^2 = \langle \delta A^2\rangle$, $\sigma_B^2 = \langle \delta B^2\rangle$, and $\operatorname{cov}(A,B) = \langle \delta A\,\delta B\rangle$:

$$\frac{\sigma_C^2}{C^2} \simeq \frac{\sigma_A^2}{A^2} + \frac{\sigma_B^2}{B^2} \pm \frac{2\operatorname{cov}(A,B)}{AB},$$

where $+$ is for the product and $-$ for the ratio. With zero covariance,

$$\boxed{\left(\frac{\sigma_C}{C}\right)^2 \simeq \left(\frac{\sigma_A}{A}\right)^2 + \left(\frac{\sigma_B}{B}\right)^2}.$$

To see the scope of the approximation, take independent random A, B with nonzero means μ_A, μ_B. Then $\langle AB\rangle = \mu_A\mu_B$ and $\langle A^2B^2\rangle = \langle A^2\rangle\langle B^2\rangle$ (Exercise 23.6.4), with $\langle A^2\rangle = \mu_A^2 + \sigma_A^2$ and $\langle B^2\rangle = \mu_B^2 + \sigma_B^2$, so the exact product formula is

$$\begin{aligned}\operatorname{Var}(AB) &= \langle A^2\rangle\langle B^2\rangle - \mu_A^2\mu_B^2 \\ &= (\mu_A^2 + \sigma_A^2)(\mu_B^2 + \sigma_B^2) - \mu_A^2\mu_B^2 \\ &= \mu_B^2\sigma_A^2 + \mu_A^2\sigma_B^2 + \sigma_A^2\sigma_B^2.\end{aligned}$$

Dividing by $(\mu_A\mu_B)^2$,

$$\frac{\operatorname{Var}(AB)}{(\mu_A\mu_B)^2} = \frac{\sigma_A^2}{\mu_A^2} + \frac{\sigma_B^2}{\mu_B^2} + \frac{\sigma_A^2\sigma_B^2}{\mu_A^2\mu_B^2}.$$

First-order propagation omits the last, higher-order term. For a ratio it additionally requires fluctuations small enough that the denominator stays away from zero in the regime being approximated.

23.7.2 Find the mean value and standard deviation of the sample of measurements $x_1 = 6.0$, $x_2 = 6.5$, $x_3 = 5.9$, $x_4 = 6.1$, $x_5 = 6.2$. If the point $x_6 = 6.1$ is added to the sample, how does the change affect the mean value and standard deviation? Use the sample standard deviation with divisor $N - 1$; state separately the empirical-distribution convention with divisor N.

Solution.

Use the sample standard deviation of Eq. (23.95), with divisor $N - 1$. For the five given measurements,

$$\bar{x}_5 = \frac{6.0 + 6.5 + 5.9 + 6.1 + 6.2}{5} = \frac{30.7}{5} = 6.14.$$

The deviations $x_j - \bar{x}_5$ are $-.14$, $.36$, $-.24$, $-.04$, $.06$, so the squared-deviation sum is

$$\begin{aligned}S_5 &= (-.14)^2 + (.36)^2 + (-.24)^2 + (-.04)^2 + (.06)^2 \\ &= .0196 + .1296 + .0576 + .0016 + .0036 = .212.\end{aligned}$$

Thus

$$\boxed{\bar{x}_5 = 6.14, \quad s_5^2 = .212/4 = .053, \quad s_5 = .230217289}.$$

Adding $x_6 = 6.1$ gives $\bar{x}_6 = (30.7 + 6.1)/6 = 36.8/6 = 92/15 = 6.133333333$. To compute its spread exactly, note $92/15 = 184/30$, so $x_j - \bar{x}_6 = (30x_j - 184)/30$ with $30x_j = 180, 195, 177, 183, 186, 183$. The six deviations are therefore

$$\left(-\frac{4}{30}, \frac{11}{30}, -\frac{7}{30}, -\frac{1}{30}, \frac{2}{30}, -\frac{1}{30}\right).$$

They sum to zero and their squared sum is $S_6 = (16+121+49+1+4+1)/900 = 192/900 = 16/75$. Therefore

$$\boxed{s_6^2 = \frac{16/75}{5} = \frac{16}{375}, \qquad s_6 = .206559112}.$$

The mean decreases by $6.14 - 92/15 = 921/150 - 920/150 = 1/150 = 0.006666667$, and the sample standard deviation decreases by $.230217289 - .206559112 \approx 0.0236582$. If one instead describes the empirical distribution with divisor N as in Eq. (23.23), its standard deviations are $\sqrt{.212/5} = .205912603$ and $\sqrt{(16/75)/6} = .188561808$; these are different conventions. The Instructor's Manual omits the original $x_4 = 6.1$, computing four then five observations. The exercise supplies five then six.

23.7.3 Carry out a χ^2 analysis of the fit corresponding to Fig. 23.10 using the same points as in Example 23.7.2, but with the errors now associated with the t_i rather than the y_i. Keep the fit through the origin. Treat the ordinates as exact and the quoted uncertainties as known standard deviations of independent normal time errors.

Solution.

The textbook's Example 23.7.2 has $y_3 = 2.7$. The Instructor's Manual substitutes 3.0 and consequently obtains a different fit. Use the textbook data throughout. With the errors now assigned to time, the exact ordinates are $y = (0.8, 1.5, 2.7)$ and the measured times are $t = (1, 2, 3)$, with independent standard deviations $\sigma = (0.1, 0.05, 0.2)$. The physical constraint remains that the line passes through the origin. Fit $t = by$ by minimizing

$$\chi^2(b) = \frac{(1-.8b)^2}{.1^2} + \frac{(2-1.5b)^2}{.05^2} + \frac{(3-2.7b)^2}{.2^2}.$$

Differentiate and set the result to zero; the condition is linear in b:

$$0 = -2\sum_j \frac{y_j(t_j - by_j)}{\sigma_j^2}, \qquad \widehat{b} = \frac{N}{D}, \quad N = \sum_j \frac{y_j t_j}{\sigma_j^2}, \quad D = \sum_j \frac{y_j^2}{\sigma_j^2}.$$

With $1/\sigma_j^2 = 100, 400, 25$, the individual terms give

$$N = .8 \cdot 100 + 3 \cdot 400 + 8.1 \cdot 25 = 80 + 1200 + 202.5 = 1482.5,$$

$$D = .64 \cdot 100 + 2.25 \cdot 400 + 7.29 \cdot 25 = 64 + 900 + 182.25 = 1146.25.$$

Since $d^2\chi^2/db^2 = 2D > 0$, this is the unique minimum. Multiplying N and D by 4 gives $\widehat{b} = 5930/4585$; cancelling the common factor 5, and inverting $t = \widehat{b}y$ for the fitted line,

$$\boxed{\widehat{b} = \frac{1186}{917} = 1.2933478735, \qquad y = \frac{917}{1186}t = 0.7731871838t}.$$

For its uncertainty, differentiate the estimator $\widehat{b} = N/D$ with respect to each time; only N depends on the t_j, so $\partial\widehat{b}/\partial t_j = y_j/(D\sigma_j^2)$. Because the estimator is linear, independent-error propagation is exact:

$$\operatorname{Var}\widehat{b} = \sum_j \frac{y_j^2\sigma_j^2}{D^2\sigma_j^4} = \frac{1}{D^2}\sum_j \frac{y_j^2}{\sigma_j^2} = \frac{1}{D} = \frac{4}{4585}, \qquad \boxed{\sigma_b = 0.0295365880}.$$

Write $T = \sum_j t_j^2/\sigma_j^2 = 100 + 1600 + 225 = 1925$. Expanding the squares gives $\chi^2(b) = T - 2bN + b^2 D$, and completing the square in b gives

$$\chi^2(b) = \chi^2_{\min} + D(b - \widehat{b})^2, \qquad \chi^2_{\min} = T - \frac{N^2}{D},$$

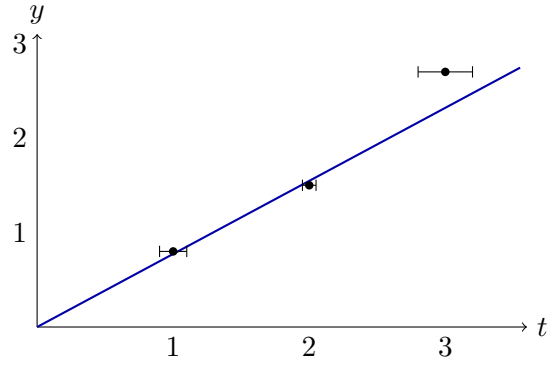
where, since $N^2/D = N\widehat{b} = 1482.5 \cdot 1186/917 = 1758245/917$ and $1925 = 1765225/917$,

$$\chi^2_{\min} = 1925 - \frac{1482.5^2}{1146.25} = 1925 - \frac{1758245}{917} = \frac{6980}{917} = 7.6117775354.$$

For independent normal measurement errors with the specified variances, write $z_j = t_j/\sigma_j$, $v_j = y_j/\sigma_j$, so that $\chi^2(b) = \|z - bv\|^2$. If the model holds with true slope b_0, then $z = b_0 v + e$, where the standardized errors $e_j = (t_j - b_0 y_j)/\sigma_j$ are independent $N(0,1)$. Minimizing $\|z - bv\|^2$ projects z onto the one-dimensional span of v; since $b_0 v$ lies in that span, the residual $z - \widehat{b}v$ is the component of e orthogonal to v. An orthogonal rotation $w = Oe$ taking v to the first coordinate axis leaves the components w_1, w_2, w_3 independent with unit variance, because the joint density of e, proportional to $\exp(-\|e\|^2/2)$, is unchanged by rotations. The residual squared norm is the sum $w_2^2 + w_3^2$ of the squares of the other two coordinates, so $\chi^2_{\min}$ has the χ^2 distribution with $3 - 1 = 2$ degrees of freedom. That density is $\frac{1}{2}e^{-x/2}$ for $x > 0$: a sum of ν squares of independent $N(0,1)$ variables has the density of Eq. (23.83) with $\sigma = 1$ and $p = \nu/2$, here $p = 1$. Directly, polar coordinates in the (w_2, w_3)-plane give $P(w_2^2 + w_3^2 \ge x) = \frac{1}{2\pi}\int_0^{2\pi}\int_{\sqrt{x}}^{\infty} e^{-r^2/2}\, r\, dr\, d\varphi = e^{-x/2}$, whose negative x-derivative is $\frac{1}{2}e^{-x/2}$. The upper-tail probability is therefore

$$\boxed{P(\chi^2_2 \ge \chi^2_{\min}) = \int_{\chi^2_{\min}}^{\infty} \tfrac{1}{2}e^{-x/2}\, dx = e^{-\chi^2_{\min}/2} = 0.0222394227}.$$

Thus a residual at least this large occurs about 2.22% of the time under this model and the stated errors. The fit has reduced $\chi^2 = \chi^2_{\min}/2 = 3.805888768$; this alone does not identify which model assumption is responsible for the discrepancy. The figure shows the three data points with their horizontal (time) error bars and the fitted line $y = 0.7731871838\, t$.



If a 95% confidence interval for the slope is also desired: with known normal measurement errors, $\widehat{b} = N/D = \sum_j y_j t_j/(D\sigma_j^2)$ is a linear combination of the independent normal t_j, hence exactly normal (each term is normal by the rule $\phi_{cY+d}(t) = e^{itd}\phi_Y(ct)$ of Exercise 23.6.1, and a sum of three independent normal variables is normal by Exercise 23.6.7, applied twice). Since $\langle t_j \rangle = b_0 y_j$, its mean is $\sum_j y_j \langle t_j \rangle/(D\sigma_j^2) = b_0 \sum_j y_j^2/(D\sigma_j^2) = b_0$, and its standard deviation is σ_b. Thus $P(|\widehat{b} - b_0| \le c\,\sigma_b) = 2\Phi(c) - 1 = 0.95$ for $c = \Phi^{-1}(0.975) = 1.959963985$, and with $1.959963985\,\sigma_b = 0.0578906488$,

$$b_0 \in \widehat{b} \pm 1.959963985\,\sigma_b = [1.235457225, 1.351238522].$$

Taking reciprocals reverses the positive endpoints, giving

$$\frac{dy}{dt} = \frac{1}{b_0} \in \left[\frac{1}{1.351238522}, \frac{1}{1.235457225}\right] = [0.74006179, 0.80941693].$$

Here σ_b already is the estimator's standard deviation; it must not be divided by $\sqrt{3}$. With known individual variances there is no additional Student-t factor either.

23.7.4 Using the data from Exercise 23.7.2 (including the point x_6), find the 90% and 95% confidence intervals for the mean of the x_i. Assume independent observations from one normal population with unknown mean and variance.

Solution.

Assume the six measurements are an independent sample from a normal population with unknown mean μ and unknown variance σ^2. Exercise 23.7.2 gives

$$N = 6, \quad \bar{x} = \frac{92}{15}, \quad s^2 = \frac{1}{N-1}\sum_{j=1}^{N}(x_j - \bar{x})^2 = \frac{16}{375}.$$

Let $X_1, \dots, X_N$ be the random observations, with sample mean $\overline{X} = N^{-1}\sum_j X_j$ and sample variance $S^2 = (N-1)^{-1}\sum_j (X_j - \overline{X})^2$; their observed values are $\bar{x}$ and s^2. The mean of N independent observations has standard deviation $\sigma/\sqrt{N}$ (Exercise 23.6.1); replacing the unknown σ by s gives the standard error of the mean

$$\frac{s}{\sqrt{N}} = \sqrt{\frac{16}{375 \cdot 6}} = \sqrt{\frac{8}{1125}} = 0.0843274043.$$

To distinguish sample size from degrees of freedom, standardize the six observations by their true μ, σ, $z_j = (X_j - \mu)/\sigma$, and rotate the vector z orthogonally, $w = Oz$, with an orthogonal matrix O whose first row is $(1, \dots, 1)/\sqrt{N}$ (complete it to an orthonormal basis, e.g. by the Gram–Schmidt process). The density of z is proportional to $\exp(-\sum z_j^2/2)$. Since $\sum_j w_j^2 = \sum_j z_j^2$ and $|\det O| = 1$, the density of w is proportional to $\exp(-\sum w_j^2/2)$, which factors over the new coordinates: $w_1, \dots, w_N$ are independent $N(0,1)$ variables. Now $w_1 = \sum_j z_j/\sqrt{N} = \sqrt{N}\,\bar{z}$, and expanding the square with $\sum_j z_j = N\bar{z}$ gives

$$\sum_j (z_j - \bar{z})^2 = \sum_j z_j^2 - N\bar{z}^2 = \sum_{j=2}^{N} w_j^2.$$

Since $\bar{z} = (\overline{X} - \mu)/\sigma$ and $z_j - \bar{z} = (X_j - \overline{X})/\sigma$, this shows that

$$Z = \frac{\sqrt{N}(\overline{X} - \mu)}{\sigma} = w_1 \sim N(0,1), \qquad Q = \frac{(N-1)S^2}{\sigma^2} = \sum_{j=2}^{N} w_j^2 \sim \chi^2_{N-1}$$

are independent: the residual squared norm uses the other five coordinates. By definition, the ratio of a standard normal variable to the square root of an independent χ^2_ν/ν variable has Student's t distribution with ν degrees of freedom. Here $\nu = N - 1 = 5$ and $\sqrt{Q/(N-1)} = S/\sigma$, so σ cancels:

$$T = \frac{Z}{\sqrt{Q/(N-1)}} = \frac{\overline{X} - \mu}{S/\sqrt{N}} \sim t_5.$$

Let c_p have $P(T \le c_p) = p$. By symmetry, $P(-c_p \le T \le c_p) = 2p-1$. Multiplying these inequalities by $S/\sqrt{N} > 0$ and solving them for μ,

$$-c_p \le \frac{\overline{X}-\mu}{S/\sqrt{N}} \le c_p \iff \overline{X} - c_p\frac{S}{\sqrt{N}} \le \mu \le \overline{X} + c_p\frac{S}{\sqrt{N}},$$

which gives the confidence interval $\bar{x} \pm c_p s/\sqrt{N}$. For 90% and 95% use respectively $p = .95, .975$:

$$c_{.95} = 2.0150483733, \qquad c_{.975} = 2.5705818356.$$

They can be obtained by integrating the normalized density of T, Student's density with ν degrees of freedom,

$$f_\nu(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})}\Big(1+\frac{t^2}{\nu}\Big)^{-(\nu+1)/2},$$

which for $\nu = 5$ (with $\Gamma(3) = 2$ and $\Gamma(5/2) = 3\sqrt{\pi}/4$) is

$$f_5(t) = \frac{\Gamma(3)}{\sqrt{5\pi}\Gamma(5/2)}(1+t^2/5)^{-3} = \frac{8}{3\pi\sqrt{5}}(1+t^2/5)^{-3}$$

and solving $\int_{-\infty}^{c_p} f_5(t)\,dt = p$. The substitution $t = \sqrt{5}\tan\theta$, with $1 + t^2/5 = \sec^2\theta$ and $dt = \sqrt{5}\sec^2\theta\,d\theta$, turns $f_5(t)\,dt$ into $\frac{8}{3\pi}\cos^4\theta\,d\theta$, and $t \to -\infty$ corresponds to $\theta \to -\pi/2$. Applying $\cos^2 x = \frac{1}{2}(1+\cos 2x)$ twice gives $\cos^4\theta = \frac{1}{4}(1+\cos 2\theta)^2 = \frac{3}{8} + \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta$, with antiderivative $\frac{3}{8}\theta + \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta$, which equals $-3\pi/16$ at $\theta = -\pi/2$. Multiplying by $\frac{8}{3\pi}$, with $\frac{8}{3\pi}\cdot\frac{3\pi}{16} = \frac{1}{2}$,

$$P(T \le c) = \frac{1}{2} + \frac{1}{\pi}\left[\theta + \frac{2}{3}\sin 2\theta + \frac{1}{12}\sin 4\theta\right], \qquad \theta = \arctan\frac{c}{\sqrt{5}}.$$

Newton's method, $c \to c - [P(T \le c) - p]/f_5(c)$, started at $c = 2$ and $c = 2.5$, reproduces the two values above to ten decimals in three iterations. Multiplication by the standard error gives the half-widths $2.0150483733 \times 0.0843274043 = 0.169923799$ and $2.5705818356 \times 0.0843274043 = 0.216770494$; with the endpoints rounded to seven decimals,

$$\boxed{\begin{array}{ll} 90\%: & 6.133333333 \pm 0.169923799 = [5.9634095, 6.3032571], \\ 95\%: & 6.133333333 \pm 0.216770494 = [5.9165628, 6.3501038] \end{array}}.$$

The sample size is 6; the Student distribution has 5 degrees of freedom. The Instructor's Manual's use of $\sqrt{5}$ in the standard error confuses these quantities and makes both intervals too wide. The stated coverage is for repeated sampling under the normal-sample assumptions, rather than a probability distribution assigned to the fixed unknown population mean.